A Biopolymer Transistor: Electrical Amplification by Microtubules

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ABSTRACT Microtubules (MTs) are important cytoskeletal structures engaged in a number of specific cellular activities, including vesicular traffic, cell cyto-architecture and motility, cell division, and information processing within neuronal processes. MTs have also been implicated in higher neuronal functions, including memory and the emergence of ''consciousness''. How MTs handle and process electrical information, however, is heretofore unknown. Here we show new electrodynamic properties of MTs. Isolated, taxol-stabilized MTs behave as biomolecular transistors capable of amplifying electrical information. Electrical amplification by MTs can lead to the enhancement of dynamic information, and processivity in neurons can be conceptualized as an ''ionic-based'' transistor, which may affect, among other known functions, neuronal computational capabilities.

INTRODUCTION

MTs are long hollow cylinders made up of guanosine-triphosphate (GTP)-dependent $\alpha\beta$-tubulin dimer assemblies (1). MTs have outer diameters of ~25 nm and inner diameters of 15 nm, with lengths that reach several micrometers. Recently, it has become apparent that neurons may utilize MTs in cognitive processing. MT-associated proteins, including tau and MAP2, have been implicated in such neuronal processes as learning and memory (2–4). MTs are also linked to the regulation of a number of ion channels, thus contributing to the electrical activity of excitable cells (5). Thus, MTs may play a heretofore unknown role in the processing of electrical signals within the neuron. Morphologically speaking, the elongating processes that sprout from the neuronal soma include a single axon and a variable number of dendrites; both take cable-like properties and contain abundant MTs. Dendrites, in particular, are responsible for both synaptic integration and synaptic plasticity, essential functions of neuronal activity (6,7). It is increasingly clear that cytoskeletal structures play a central role in both the process of ramification and more specialized neuronal functions, including neuronal processivity, long-term potentiation, long-term depression, and memory formation (8,9). Cytoskeletal alterations may reflect changes of neural circuits in response to learning and experience, and they must involve highly dynamic regulatory mechanisms. How this cytoskeletal control of neuronal plasticity takes form remains ill defined. Interestingly, a century ago Cajal proposed that the ''cytoskeleton contain a system for the conduction of the nerve impulse'' (10). To elucidate the electrodynamic properties of MTs and gain insight into the regulatory role that MTs play in dendritic activity, we assessed whether MTs are capable of processing electrical information.

METHODS

Preparation of isolated MTs

Tubulin was polymerized with a combined GTP-taxol ''hybrid'' protocol, following loosely general guidelines as described in Mitchinson Lab's online protocols (http://mitchison.med.harvard.edu/protocols/). Briefly, all reactions were conducted in BRB80 solution, containing 80 mM PIPES (1,4-piperazinediethanesulfonic acid), 1 mM $MgCl_2$, 1 mM EGTA, and pH 6.8 with KOH. In some experiments 10 μl glycerol was added before mixing the tubulin. An aliquot of tubulin in solution (catalog No. T238, Cytoskeleton, Denver, CO) was mixed with 1 mM dithiothreitol (DTT) and 1 mM GTP in an ice bed and incubated for 5 min. Taxol (paclitaxel, Sigma; St. Louis, MO) was then added in steps, first from a 1 μM and then 10 μM solutions. Each stepwise addition of taxol was followed by incubation for 20–30 min in a water bath at 37°C. Taxol was kept in a 10-mM stock in anhydrous dimethylsulfoxide (DMSO). In some instances, the tubulin-containing solution was mixed with BRB80 solution (×2) and added 2 mM DTT, 2 mM GTP, and 20% DMSO, followed by incubation at 37°C for 20–30 min. Polymerized tubulin was centrifuged in a microfuge for 30 min at 14,000 rpm. Two-thirds of the supernatant was discarded, and the remainder constituted the MT sample, which was further stabilized by addition of 1 mM GTP and 1 μM taxol. MTs were visualized under phase contrast microscopy (IMT-2, Olympus, Tokyo, Japan; ×60) to ascertain the lack of nucleated centers induced by higher concentrations of taxol. MTs were stable at room temperature for up to 2 days.

Measurement of electrical signals in isolated MTs

Electrical data were collected with a modified dual ''patch-clamp'' setup as previously reported (11). Briefly, the electrical setup consisted of two independent patch-clamp amplifiers, PC501 and PC501A (Warner Instruments, Hamden, CT), with 1-GΩ and 10-GΩ feedback resistors, respectively, one to stimulate the isolated MT (stimulus, s) and the other to collect current signals (collection, c). Electrical signals were simultaneously collected by respective analog inputs of an analog-to-digital (A/D) converter (TL-1 DMA Interface, Axon Instruments, Foster City, CA). Voltage stimulus protocols were constructed in Clampex 5.5.1 (Axon Instruments) run from a computer. The sampling intervals varied depending on the protocol, such that in most cases a 100-ms signal length contained at least 400 samples. Offset tip potentials within the range of a few millivolts were

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compensated immediately before data acquisition. Further zeroing of the offset signal was conducted offline during data processing. Signals were Gaussian filtered at a cutoff frequency of 2.2 kHz. Pipettes were made from borosilicate capillaries (1.2-mm external diameter) and were pulled with tip diameters below 3 μm. The pipettes contained and were immersed in a solution containing in mM: 135 KCl, 5 NaCl, 0.8 $MgCl_2$, 1.2 $CaCl_2$, 10 Hepes. Occasionally, the pipette tip contained a 1:10 dilution of the taxol solution to help attach the MT. The pipettes were connected to the electrodes and head stage of their respective patch-clamp amplifiers. Agar bridges containing 3% agar in 150 mM KCl solution grounded both amplifiers to the chamber. The circuit was closed by two AgCl-plated Ag ground electrodes in solutions far away (1–2 cm) from the pipette tips.

Data analysis

Data were acquired at 2 KHz and further filtered offline for display and analysis when required. Noise analysis was conducted as Lorentzian spectral analysis with a subroutine of Axograph 4.0 (Axon Instruments) from unfiltered files where the variance versus frequency was plotted for paired experiments before and after MT attachment was obtained in the experiment. The method for deriving the amplification is different for the square pulse and for the triangle wave. In the case of the square pulse, we evaluated the average amplitude in solution only and with MT attached for the same conditions. The amplification is simply the ratio. In the case of triangle pulses, a different strategy was used, using the slope of the linear regression of the MT-attached data as a function of the solution-only data at the pulse times. Data are expressed as n = number of files averaged ± SE, where t-test was used to assess statistical significance at $p < 0.05$.

RESULTS

MTs were formed and stabilized as indicated in Methods. Isolated MTs were visualized and identified under phase contrast microscopy and connected to the tips of two patch-clamp amplifiers (Fig. 1 *a*), such that electrical stimulation could be applied to one of them, whereas the elicited signal could be collected with the other one (Fig. 1 *b*). MTs were electrically stimulated by applying 5–10-ms input voltage pulses with amplitudes in the range of ±200 mV. The resulting electrical signals were obtained at the opposite end of an MT, 20–50 μm away with a pipette also connected to a patch amplifier, which was kept ''floating'' at 0 mV. We observed two remarkable findings. First, coupling of the stimulus pipette to an MT increased the overall conductance of the pipette by >300%. The stimulus pipette resistance in KCl saline (135 mM) was 21.6 ± 0.6 MΩ (n = 16). The coupling ratio between pipettes in solution was 41% ± 12% before attachment to an MT. Thus, the collection pipette ''read'' ~40% of the stimulus in solution. The evoked current increased from 1.91 ± 0.13 nA (n = 16) to 2.78 ± 0.17 nA (n = 35) after attachment (Fig. 2 *a*). The signals reaching the collection site were in 30/35 of the cases higher than those obtained in free solution, indicating that the MT amplified both the electrical pulse injected at the stimulus site and the collection site as well (Fig. 2 *a*). Transfer amplification ratios were up to 2.35 with an average of 1.69 ± 0.06 (n = 30, Fig. 2 *c*). Thus, MTs improve electrical connectivity between two locations in saline solution. The currents measured at the collection site were linearly dependent on the stimulus pipette input voltage, indicating a strictly inverse ohmic response, i.e., linear amplification (Fig. 2 *b*). The MT conductances reached up to 9 nS, much higher than that expected from channel conductances (5–200 pS, Fig. 2 *b*, *inset*). The electrical amplifying effect of the MT was observed in either direction (stimulus by either

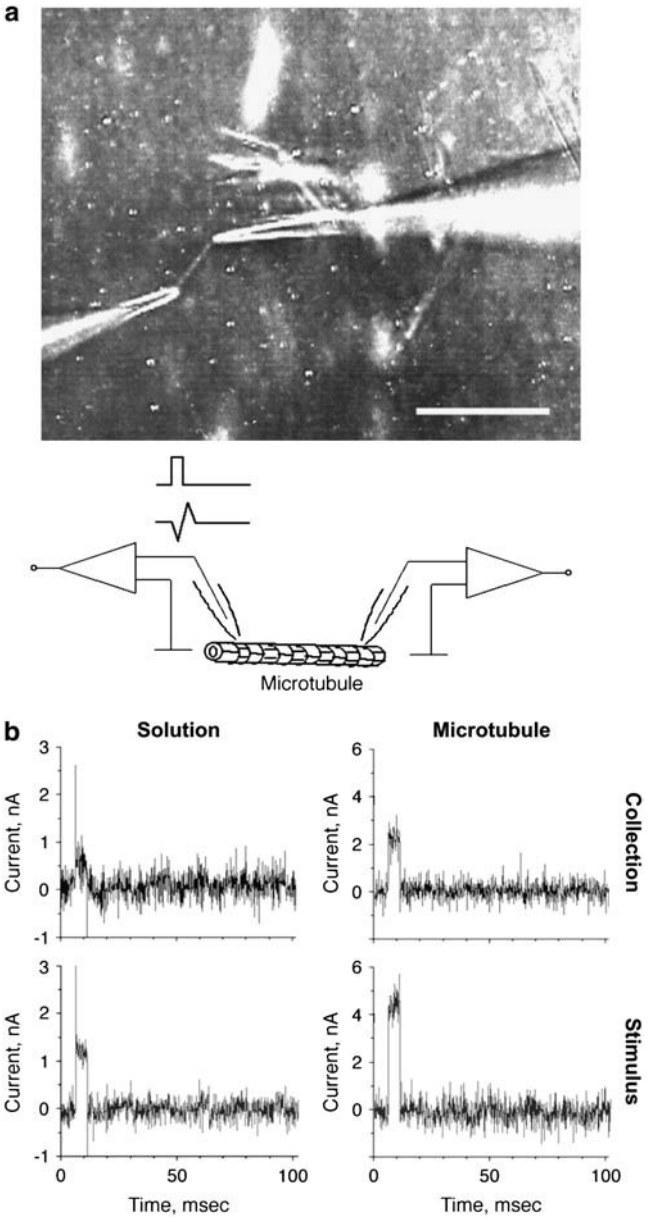

FIGURE 1 Electrical setup. (*a*) (*Top*) A free-floating MT was ''connected'' to two patch pipettes. Horizontal bar is 50 μm. (*Bottom*) Electrical signals were applied to one end and collected from the other end with two patch-clamp amplifiers. (*b*) Voltage pulse (+100 mV) was first applied to the ''stimulus'' pipette in saline solution. The recorded current from stimulus pipette (*left bottom*) and electrical coupling by the ''collection'' pipette (*left top*) were obtained. A similar approach was repeated after MT attachment. Currents were higher after attachment to an MT (*right*), compared to saline solution (*left*). Representative tracings from 16 experiments are shown.

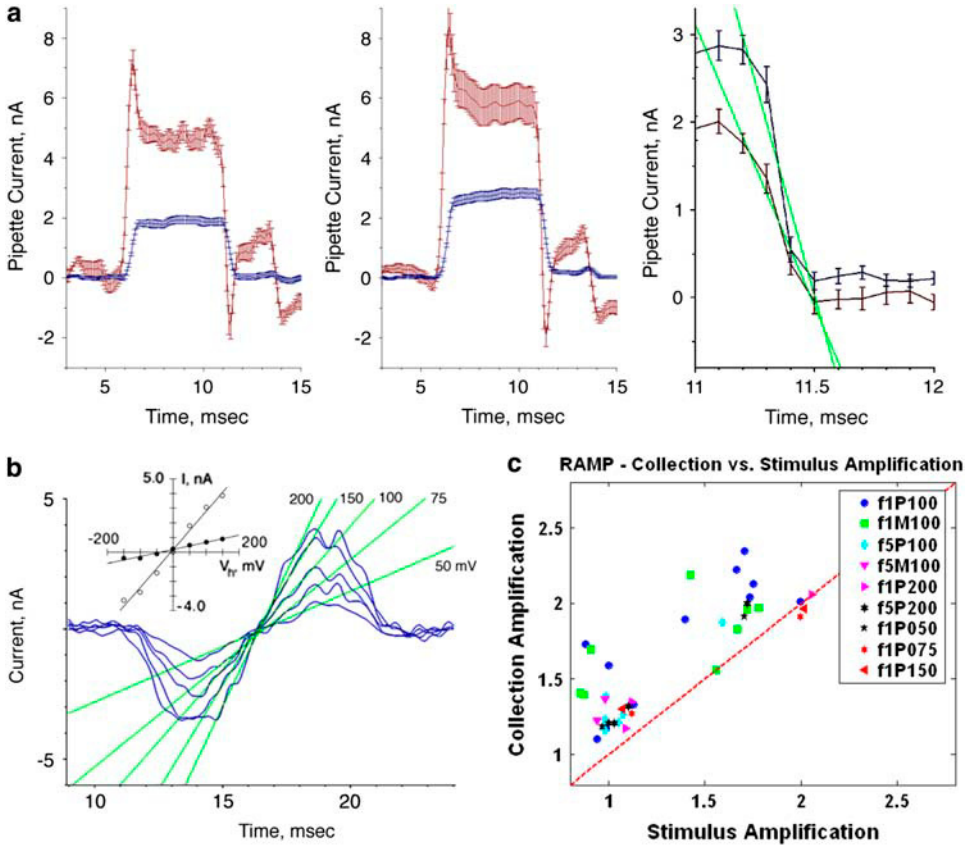

FIGURE 2 Pulse stimulation of isolated MTs. (*a*) Average electric pulses (100 mV, 5 ms) for the ''stimulus'' (*red*) and the ''collection'' (*blue*) sites. Data on the left indicate the signal before attachment. Values are the mean ± SE for five pipettes with similar electrical properties. The decay response of the MT-connected tip at the end of the electrical pulse was ~40 μs faster after connection (*right*). (*b*) MT-attached signals on the collection site for 5-ms ramps at different voltages. Fitted linear slopes (*green*) indicate a linear response. (*Inset*) Current-to-voltage relationship for a connected MT (*open circles*) compared to free solution (*solid circles*). The conductance is linear. The signals increased in average by 1.69 at an average distance of 35 μm. (*c*) Correlation of amplification ratios between the stimulus and collection sites. Identity line (*red*) indicates no amplification by the connected MT. Amplification factors are shown for either 10 ms (f1) or 5 ms (f5) ramps for positive (P) and negative (M) potentials (last three digits).

amplifier). To further prove the linear response in amplification as well as the speed of the electrical amplifying phenomenon, voltage ramps were also applied within the range of ±100 mV at two different rates (100 V/s and 20 V/s, Fig. 2 *b*). The stimulated currents at either rate were again linear, showing a remarkable reproducibility within the range of voltages studied. The amplification ratio remained constant within the range of voltages and rates tested, and the noise in the signal was, as expected, increased as well. The fastest ramps tested showed, however, a 40-μs (or shorter) delay between the stimulus and collection sites, which was consistent with the shift in voltage decay after pulse stimulation (Fig. 2 *a*, *right*). This suggests a lower bound for the transfer rate of the electrical signal of the order of 1.0 m/s. In two out of seven experiments, electrical amplification further increased by addition of GTP (1 mM) to the bath solution (data not shown).

DISCUSSION

Our findings demonstrate that electrical amplification by MTs is equivalent to the polymer's ability to act as a biomolecular transistor at its core. Based on a minimalistic model, both a constant electrical polarization and an active component are required. Despite current interest and extensive literature on the properties and functions of MTs, practically nothing is known about the ionic properties of the MTs intrapolymeric compartment. However, molecular dynamics simulation of tubulin structure (12,13) indicates a strong negative surface charge distribution of the order of $20e^-$ per monomer, distributed more on the outer surface than in the inner core with ratio of ~2:1. Our findings support these theoretical results manifested in the model via a constant (permanent) electric polarization, which follows localized Nernst potentials arising from asymmetries in the ionic distributions between the intra- and extra-MT environments. This polarization is modulated by electrical stimulation such that the forward-reverse biased junctions of an intramolecular transistor (Fig. 3) creates a proper MT-adjacent ionic cloud environment, which allows amplification of axially transferred signals. The proposed model implies that intrinsic semiconductive like properties of the structured tubulin dimers are such that an effective transistor is being formed (14) whose gating ability to modulate localized charges may help amplify axial ionic movements. Recent studies aimed at solving the Poisson-Boltzmann equation for a section of an MT in ionic conditions similar to those reported here (i.e., 150 mM saline) (12) suggest a highly ordered charge distribution. Positive charges are ''sandwiched'' within a two-layered structure of negative charges facing either side of the MT. There also is an asymmetrical distribution of charges on the plus-minus

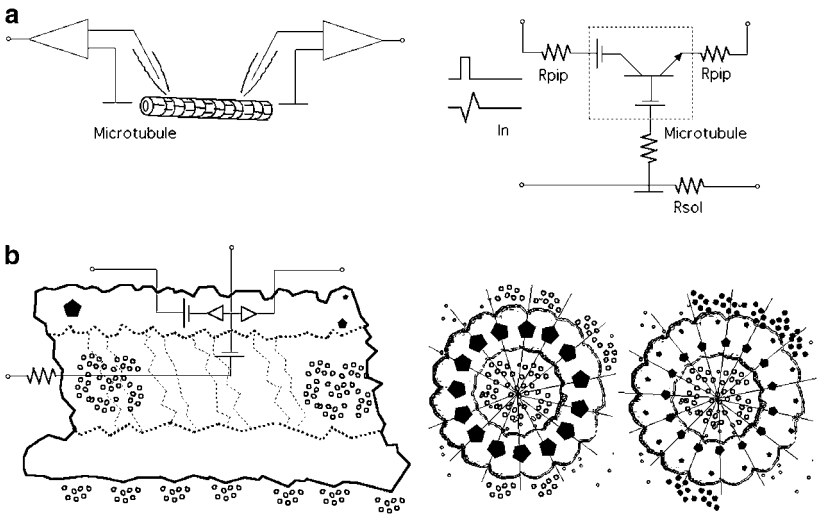

FIGURE 3 Electrical model of the MT. (*a*) Effective electrical model of the MT (*right*) is consistent with an electrical amplifier, including a molecular ''biotransistor'' and energy sources in the form of batteries to polarizing the amplifier. (*b*) Relation of electrical model to the electrostatic properties of an MT for both the plus and minus ends, potential isocontours, reproduced from Baker et al. (11). Electrostatic profile of a cross section of the MT (*right*) shows periodic distribution of positive charges (*black*) on the surface of the electronegative MT. Periodically distributed charges (*large* and *small pentagons*) maintain a ''band junction'' in the surrounding counterions. This is based on an electric potential difference in the MT wall (*open circles* and *solid circles* for cations and anions, respectively).

orientation of the MT. However, to better understand the dynamical properties of the electric field around the MT, it is imperative to apply a method that incorporates the diffusive part of the system. A possible candidate is the Poisson-Nernst-Planck (PNP) theory in which both the electrical potential generated by fixed charges and the flux generated by the moving charges due to potential and concentration gradients are taken into account. This method has been applied successfully to describe ion channels (15), and it would be interesting to test the applicability of this method to a system as large as the MT.

An electrical amplifier by a cytoskeletal biopolymer such as the MT provides a relevant and important key component to an electrodynamic network in conjunction with electrical coupling by current generators (i.e., MT-regulated channels (5)) and MT-coupled intracellular transmission lines (F-actin) (11,16). It is therefore plausible that electrical amplification by MTs may play an important role in processing electrical information in neuronal function. Our data further indicate that this phenomenon is accomplished with speed at least as high as those reported for cable properties in neurites (17).

MT networks (MTNs) play relevant roles in neuron formation and function (18–20). Interconnected bundles of MTs, for example, are present in the axon, the axon hillock, and the dendritic shaft, all regions where ion channels are found. In particular, dendritic MTs are arranged in grid-like networks of mixed polarity interconnected by MAP2s. Thus, we envision a mechanism in which MT-ion channel interactions may regulate synaptic plasticity by the MT's ability to allow spatially oriented intracellular electrical signals in connection with actin filaments also able to transmit electrical information (11,16). According to this hypothesis, postsynaptic electrical signals elicit ion waves along the associated actin filaments at the synaptic spine that propagate to the MTN, where they serve as input signals. The MTN, operating as a large high dimensional ''state machine'', evolves these input states, e.g., by supporting nonlinear wave collisions. The output from the MTN is the state of the system that is being ''read'' (sensed) to propagate and electrically stimulate remote voltage-sensitive ion channels. Thus, our findings provide several advantages in the context of neural function. Cable theory analysis of dendrites (21,22) has challenged the simple integrate and fire models, suggesting that dendrites impose a heavy conductance load on the soma, acting as low-pass filters of postsynaptic potentials, hence changing the response to synaptic activities (22). The dendritic electrical properties are dynamically changed by modulation of voltage-gated ion channels (6,22,23) and by changes in cytoskeletal structures (25–27). Thus, MT electrical amplification may be central to revised models of neuronal adaptability (23,28,29) providing renewed support to nonlinear models of neuronal activity.

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A FRÖHLICH INTERACTION OF HUMAN ERYTHROCYTES

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Experiments on human red cells provide evidence for an interaction compatible with the quantum mechanical theory of Fröhlich.

Fröhlich has a theory [1,2] that long-range coherent interactions, known to exist in the solid state, should occur in living cells in a unique form. Necessary conditions of the theory [2] are (a) a cell membrane potential difference, (b) a supply of metabolic energy, and (c) organisation, a type of order which, according to Fröhlich "... exists in a well-working machine in contrast to a broken-down one [2]". In work on human red cell rouleau formation [3–6] we noticed a small attraction between erythrocytes in plasma priorto the actual contact which precedes doublet formation. When we (a) reduce the membrane potential near to zero or (b) deplete the cells of ATP or (c) derange the cellular organisation with glutaraldehyde, the attraction can no longer be detected.

Membrane potentials act across very short distances (≈ 10 nm) which implies that the membrane is strongly polarised electrically. Vibrations of membrane molecules will be dipolar, representing an electric wave of half wavelength equal to the thickness of the membrane [2]. If the velocity of mechanical membrane waves is assumed to be 10^3 ms^{-1}, the frequency of such oscillations will be of the order of 10^{11} Hz. There is evidence for such oscillations in microorganisms [2].

Fröhlich considers two basic collective excitations. The first, quasi-static, is of metastable states of very high electric polarisation. He then introduces metabolic energy which can excite coherent vibrations. If the supply of energy to the modes of a band of electric polarisation waves exceeds a minimum value, the energy may be channelled into a single mode, giving rise to a strong coherent oscillation extending throughout the cell. Collective vibrations are quantised (phonons) and obey Bose–Einstein statistics. Fröhlich likens the excitation of a single mode to the Einstein condensation of a Bose gas [7], a condensation in momentum space, not in coordinate space as in liquefaction. In the quasi-static case interactions between highly polarised cell constituents will be repulsive and of short range. Fröhlich posits quite a different situation at $\approx 10^{11}$ Hz when ionic screening is only significant at distances of micrometres. Moreover interactions may be attractive or repulsive as conditions change. Cooper [8] has applied the theory to the movement of macromolecules in the complex processes of cell mitosis.

In our experiments, heparinised blood at a haematocrit of $\approx 0.5\%$ in a haemacytometer chamber is photographed every 15 s with a time-lapse 16 mm ciné-microscope for 3 h. About 200 erythrocytes are in view. In an early frame of the developed film, pairs of cells less than a cell diameter apart (<8 μm) are identified and their subsequent motion is followed until a doublet forms or until the end of the film. Isolated cells execute brownian movement. If the motion of pairs is also random, simple geometry predicts a maximum probability of the pair making contact of 0.5 if the initial separation is zero, to 0.15 at an initial separation of 8 μm. The average probability is about 0.3. It will be seen from table 1 that in six experiments on four subjects and involving 212 pairs, the fraction of pairs which collided was 0.85 ± 0.06.

The pH of the plasma was reduced with 0.14 M HCl

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Table 1
Fraction of pairs of cells initially <8 μm apart which collide within 3 h.

Subject	Pairs	Collisions	Fraction
SR	15	14	0.93
SR	56	46	0.82
SR	21	18	0.86
DLF	63	47	0.75
LSS	23	20	0.87
REL	34	30	0.88
	212 total	175 total	0.85 mean (sd ±0.06)

in 0.14 M NaCl and restored to normal with 0.14 M NaOH. At pH = 6.3 the membrane potential is close to zero [9,10]. Pair collision is reduced to the random value but is restored to a non-random value on returning the same cells to pH = 7.6 (table 2). Ionophore A23187 induces rapid entry of Mg^{++} and depolarisation of membrane potential [11]. Cells were washed in standard phosphate-buffered saline (PBS) containing 10 μM A23187 and then suspended in plasma to which had been added 10 mM $MgCl_2$, 10 μM EGTA and 10 μM A23187 [11]. Pair collision is reduced to just within the random range by this procedure (table 2). Cells were metabolically depleted by washing in PBS followed by incubation in PBS for 22 h at 37°C [12,13]. Such cells also give a result in the random range and the same cells are restored nearly to the mean fraction for normal cells by incubation for a further 3 h in 30 mM adenosine (table 2). Glutaraldehyde crosslinks protein and so "disorganises" the cell [14]. Cells were washed twice in PBS containing 0.04% glutaraldehyde, resuspended in the same solution and shaken gently for 24 h at 22.0°C. The mean fraction is well within the theoretical random range (table 2) but it is likely that in these embalmed cells the membrane potential and metabolic mechanisms were abolished. Latex microspheres suspended in plasma were used as controls (table 2).

Brownian motion is monitored in each experiment by measuring the X coordinate against time of ten or more cells which remain isolated (>8 μm) throughout 3 h. From theory [15]

$$\langle X^2 \rangle = 2kTt\mu^{-1},$$

where μ is the drag coefficient. Correlation coefficients averaged 0.94 ± 0.03 (for microspheres r = 0.96) but μ (including that for microspheres) was consistently higher than the value calculated from Stoke's law or from the rate of free fall of erythrocytes [16]. So cells, normal and experimentally modified, and artificial particles when resting on a surface, experience a drag greater than that of the suspending medium.

By simple experiments we have demonstrated a hitherto unknown interaction of human erythrocytes and tested it against the Fröhlich theory. The most satisfactory of our tests is the reversible disappearance of the interaction by abolition of the membrane potential. Metabolic depletion does more than cut off the supply of energy. Although reversible it modifies structure [17] and results in ion shifts [18]. The zeta potential, however, is unchanged [12]. We measured intra- and extra-cellular chloride before and after metabolic depletion and, from the Nernst equation [10], calculated the membrane potential. It dropped from 15.0 mV to 7.1 mV and so might have fallen beneath some threshold for the Fröhlich effect. Glutaraldehyde

Table 2
Fraction of pairs of erythrocytes initially < 8 μm apart (< 3.4 μm for microspheres) which collide within 3 h.

	No. of experiments	Mean fraction
cells at pH 6.3	3	0.33
cells restored to pH 7.6	3	0.76
ionophore-treated cells	2	0.48
metabolically depleted cells	2	0.39
cells repleted with adenosine	1	0.75
glutaraldehyde fixed cells	5	0.37
3.4 μm latex microspheres	1	0.24

has a rather drastic effect on structure although some enzyme activity remains [14]. The theory regards organisation, membrane potential and energy supply as separate but the three entities are likely to be functionally interwoven and may prove difficult to isolate. Fröhlich forces will be universal in living systems. We offer our preliminary findings to enable other workers to check and extend them.

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ORIGINAL PAPER

A nonlinear model of ionic wave propagation along microtubules

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Abstract Microtubules (MTs) are important cytoskeletal polymers engaged in a number of specific cellular activities including the traffic of organelles using motor proteins, cellular architecture and motility, cell division and a possible participation in information processing within neuronal functioning. How MTs operate and process electrical information is still largely unknown. In this paper we investigate the conditions enabling MTs to act as electrical transmission lines for ion flows along their lengths. We introduce a model in which each tubulin dimer is viewed as an electric element with a capacitive, inductive and resistive characteristics arising due to polyelectrolyte nature of MTs. Based on Kirchhoff's laws taken in the continuum limit, a nonlinear partial differential equation is derived and analyzed. We demonstrate that it can be used to describe the electrostatic potential coupled to the propagating localized ionic waves.

Keywords Microtubule · Polyelectrolyte · Nonlinear transmission line · Intrinsic electric fields · Ionic waves

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Introduction

The cytoskeleton is a major component of all living cells. It is made up of three different types of filamentous structures, including actin-based filaments, intermediate filaments (e.g. keratin) and tubulin based microtubules (MTs). All of them are organized into networks, which are interconnected through numerous proteins, with specific roles to play in the functioning of the cell. MTs are hollow cylinders with outer diameter of approximately 25 nm. Their lengths vary but commonly reach 5–10 μm. They are comprised of exclusively 13 parallel protofilaments in vivo. These protofilaments are strongly bound internally and are connected via weaker lateral bonds to form a sheet that is wrapped up into a tube following the nucleation process (Chrétien et al. 1995) (see Fig. 1).

Each tubulin monomer of the MT lattice has a short C-terminal helix H12 followed by a highly acidic amino acid sequence projecting out of the MT outer surface called a tubulin tail (TT). The TTs are short polypeptide sequences of 4–5 nm length when completely outstretched, protruding from the MT surface into the solution. They are essential for MT interactions with MT associated proteins and motor proteins.

Jimenez et al. (1999) determined the helicity of α (404–451) and β (394–445) tubulin C-terminal recombinant peptides with the use of NMR spectroscopy. The α-tubulin TT amino acid sequence is EEVGADSVEGEGEEEGEEY. The α-TT is 19 amino acids long and possesses ten negatively charged residues. The situation in the β-tubulin C-terminal domain is even more interesting. Jimenez et al. found a helix there, which is nine amino acids longer than in α-tubulin. This suggests an extension into the protein, supporting the possibility of a functional coil-to-helix transition at the TT zone. The β-tubulin TT helix H12 is formed by

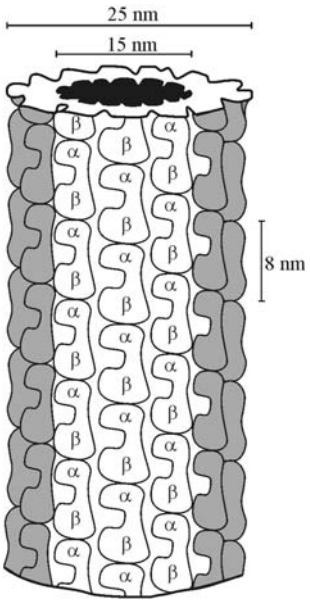

Fig. 1 Sketch of a MT cylinder with characteristic dimensions and α–β dimers clearly discerned

amino acid residues (408–431) but it seems that the reversible transition between coil and the helix comprising the last nine amino acid residues (423–431) from β-TT helix (with sequence QQYQDATAD) could either increase or decrease the length of H12, at the same time changing the β-TT length, as has been recently demonstrated using molecular dynamics simulations on human beta tubulin C-termini by Luchko et al. (2008). The corresponding residues (432–445) contain the 14 amino acids EQGEFEEEEGE DEA with nine negatively charged residues (E, A). At neutral pH the negative charge on a TT causes it to remain extended due to electrostatic repulsion within the tail (Tuszynski et al. 2005). Under more acidic conditions the negative charge of TTs is partially neutralized by associated counter-ions condensed on them. The effect is to allow the β-TTs to acquire a more compact form by folding (see Fig. 3). This is probably the largest structural effect in MTs which occurs due to change in the cell's pH. This fact will be of great importance in our model, with respect to the variable effective electrical capacitance of MTs.

The detailed charge distribution on the tubulin surface was calculated by Tuszynski et al. (2005), leading to the determination of an electrostatic potential around tubulin within a MT. It exhibits non-uniformity along the MT radius with a peak and a trough corresponding to every protofilament and the area between them, respectively. This fact plays a crucial role in our modeling of MTs as a "cable" conducting 13 parallel currents comprised of ionic flows. Keeping in mind that MTs are mostly negatively charged, especially on their outer surface, they can be referred to as polyelectrolyte polymers. This is based on the fact that numerous amino acids forming tubulin have many negatively charged residues under physiological conditions, especially those located on TTs. Since a sufficiently high surface charge density is present on the MT surface, Manning's theory (1969) can be applied to determine the conditions for the solution's positive counter-ions to "condense" on the surface of a MT. Thus, every MT should attract positive counter-ions near its surface, while negative ions of the cytosol are repelled such that a cylindrical depletion area around a MT is created. The thickness of this depleted area is defined by the so-called Bjerrum length, which is the distance from the MT surface at which the Coulomb energy of the screened surface charges equals the thermal energy:

$$\frac{e^2}{4\pi\varepsilon_0\varepsilon\ell_B} = k_B T. \tag{1.1}$$

Here, e is the charge of an electron, ε_0 the permittivity of vacuum, ε the relative permittivity of cytosol, and k_B is Boltzmann's constant. At a physiological temperature of 310 K and taking $\varepsilon = 80$, it is easy to find from Eq. 1.1 that:

$$\ell_B = 6.7 \times 10^{-10}\,\text{m}. \tag{1.2}$$

Counter-ion condensation along MTs occurs when the average distance between the intrinsic negative surface charges is smaller than ℓ_B.

The idea to examine MTs in the context of polyelectrolyte features was inspired by the experimental and theoretical results obtained earlier for actin filaments (Lin and Cantiello 1993; Tuszynski et al. 2004). From Figs. 2 and 3 it is clear that the surface of a MT filament is very rough due to the presence of TTs. We assume that the flow

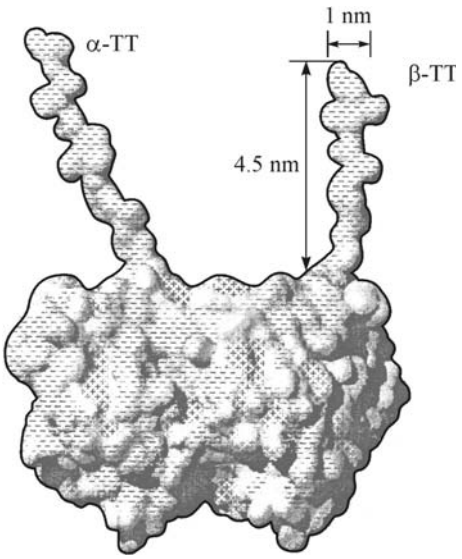

Fig. 2 Tubulin dimer charge distribution according to Tuszynski et al. (2005), with TTs and characteristic dimensions

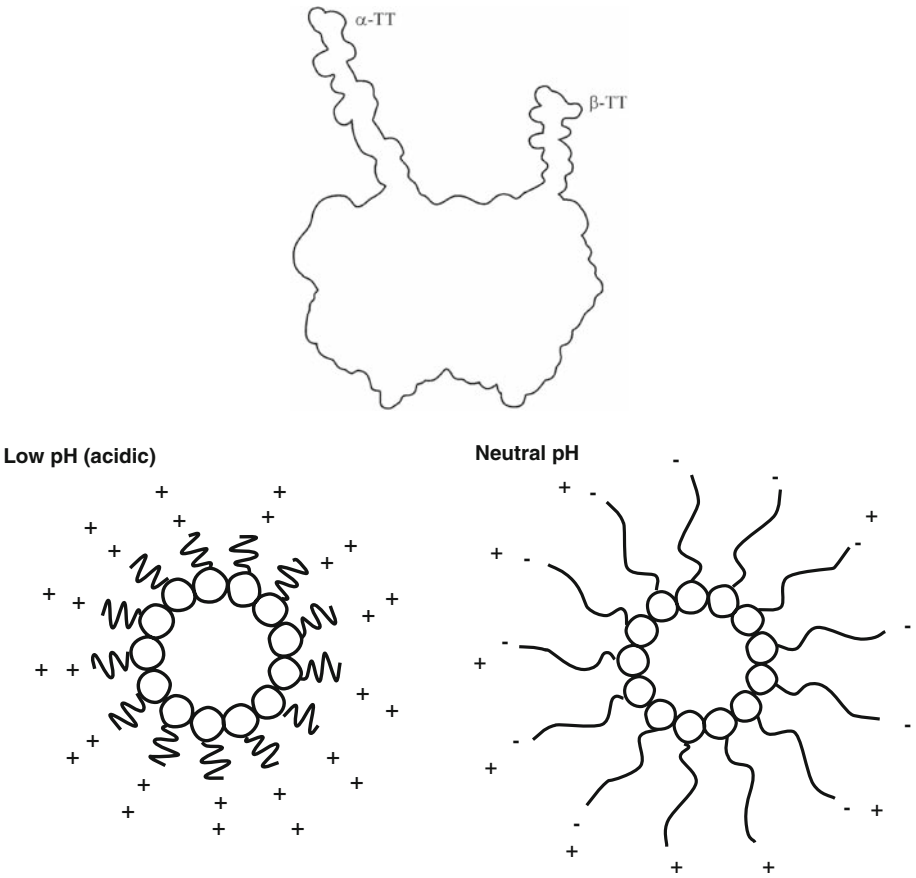

Fig. 3 *Top* shrinking of β-TT as a result of increased counterion condensation. It brings about the variable capacity of a MT as a whole. *Bottom* the conformation of TTs at (*left*) low pH and (*right*) high pH values

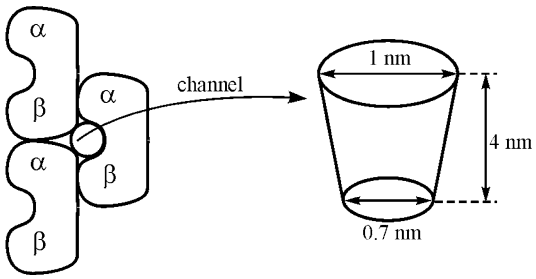

Fig. 4 Sketch of the channel connecting inner and outer surfaces of a MT

of ions along MTs, caused by the concentration gradient and by intrinsic electric fields created by the protein's charge distribution, should be mostly channeled through potential valleys (troughs) running parallel to each of the 13 protofilaments. The walls of valleys include TTs whose flexibility could significantly affect ionic current flows. In addition to the presence of TTs, it is important to note that the globular shape of tubulin dimers results in the creation of small pores representing channels that connect the inner and the outer surface of a MT (see Fig. 4).

These channels could be thought of as candidates for additional leakage of ions which would then be able to rectify currents along the depleted (Bjerrum) cylindrical layer around a MT. Recent experimental evidence (Priel et al. 2006) indicates that MTs could act as biopolymer transistors clearly amplifying input electrical signals. It is our objective in this paper to propose a model which provides an explanation of these peculiar properties of MTs and their possible role in information processing. We will follow the same theoretical approach first outlined in Tuszynski et al. (2004) for actin filaments, but it will be applied to MTs with special attention placed on their specific structural features.

The paper is organized in the following way: in Sect. "The model of MT as a nonlinear electric transmission line" we introduce the model of a MT as a nonlinear electric transmission line segmenting it into elementary rings. In Sect. "Modeling MT as nonlinear RLC transmission line" we derive and analyze a nonlinear differential equation for the local electrostatic potential coupled to the ionic wave propagating along a MT. Section "The impact of the cell's intrinsic electric fields on the dynamics of AKs in MTs" demonstrates the possible role of cell's intrinsic electric fields in the dynamics of localized ionic waves propagating along MTs. Section "Conclusion and discussion" presents a discussion and conclusions regarding this important physical mechanism in terms of the cell's electrical activity brought about by the presence of MTs and their effects on ions.

The model of MT as a nonlinear electric transmission line

We start from the fact that a MT with condensed counter-ions is separated from the rest of the ions in the bulk solution by a depleted layer which plays the role of a dielectric medium located between MTs and the bulk solution. This depleted layer provides both a resistive and capacitive component for the electrical characteristics of protofilaments that make up the MT. The injected ions are expected to flow at a radial distance from the surface of the MT with the thickness of ℓ_B. We will also estimate the inductive component of ionic waves due to 13 parallel currents and show it to be negligible compared to the other impedance components.

The estimation of elementary capacitance of a MT ring

In a continuum model we treat the MT as a polyelectrolyte represented by a volume which is bounded by a charged cylindrical surface in contact with an electrolyte (cytosol) whose dielectric constant is assumed to be time-invariant and uniform. In this case Poisson's equation is an appropriate physical description and it has the form:

$$\nabla^2 V = -\frac{\rho}{\varepsilon_0 \varepsilon}, \tag{2.1}$$

where ρ is the volume charge density at any point in the cytosol and ε is its relative dielectric constant. The charge density ρ results from the presence of counter-ions and co-ions. These are assumed to satisfy a Boltzmann distribution function. This means that in a solution containing n_s dissociated molecules of salt (for example NaCl) the number n_1 of counter-ions (Na^+) with charge $z_1 = e$ at a distance r from the MT cylinder axis is given by the expression:

$$n_1(r) = n_s \exp\left(\frac{eV}{k_B T}\right), \tag{2.2}$$

since z_1 and V have opposite signs. The number of negative (Cl^-) ions is:

$$n_2(r) = n_s \exp\left(-\frac{eV}{k_B T}\right). \tag{2.3}$$

This implies the total charge density given by:

$$\begin{aligned}\rho(r) &= n_s e\left[-\exp\left(\frac{eV}{k_B T}\right) + \exp\left(-\frac{eV}{k_B T}\right)\right] \\ &= -2n_s e \sinh\left(\frac{eV}{k_B T}\right).\end{aligned} \tag{2.4}$$

Using this result in Eq. 2.1 yields the Poisson–Boltzmann equation (PBE):

$$\nabla^2 V = \frac{2n_s e}{\varepsilon_0 \varepsilon} \sinh\left(\frac{eV}{k_B T}\right). \tag{2.5}$$

This nonlinear differential equation can be brought into a more compact form by introducing two dimensionless variables:

$$\psi = \frac{eV}{k_B T} \quad \text{and} \quad x = \frac{r}{\mathcal{D}}, \tag{2.6}$$

where the Debye length $\mathcal{D}$ is defined as:

$$\mathcal{D} = \sqrt{\frac{\varepsilon_0 \varepsilon k_B T}{2n_s e^2}}. \tag{2.7}$$

It gives:

$$\nabla^2 \psi(x) = \sinh[\psi(x)]. \tag{2.8}$$

In the Debye–Hückel approximation it is assumed that $eV \ll k_B T$ or $\psi \ll 1$, so that Eq. 2.8 can be linearized by the approximation $\sinh[\psi(x)] \approx \psi(x)$. Thus PBE in cylindrical coordinates now reads:

$$\frac{1}{x}\frac{\mathrm{d}}{\mathrm{d}x}\left(x\frac{\mathrm{d}\psi}{\mathrm{d}x}\right) = \psi \quad \text{or} \quad \frac{\mathrm{d}^2\psi}{\mathrm{d}x^2} + \frac{1}{x}\frac{\mathrm{d}\psi}{\mathrm{d}x} - \psi = 0. \tag{2.9}$$

On the surface of the MT cylinder we have: $x_0 = R/\mathcal{D}$ and $\psi_0 = eV_0/k_B T$, where R = 12.5 nm. The solution of PBE, Eq. 2.9 is a modified Bessel function of zero order denoted by $K_0(x)$, so we have:

$$\psi(x) = cK_0(x) \tag{2.10}$$

where c is the constant of integration. The above Bessel function does not oscillate and behaves asymptotically like $\exp[-x]$ as $x \to \infty$. From Eq. 2.10 one has:

$$\frac{\mathrm{d}\psi}{\mathrm{d}x} = -cK_1(x), \tag{2.11}$$

where $K_1(x)$ is the modified Bessel function of the first order. Using the relationship for the electric field:

$$E = -\frac{k_B T \mathrm{d}V}{e\mathcal{D}\ \mathrm{d}x}$$

we have:

$$-x_0\left(\frac{\mathrm{d}\psi}{\mathrm{d}x}\right)_{x=x_0} = \frac{e^2}{4\pi\varepsilon_0 \varepsilon k_B T g} = \xi; \quad g = \frac{\mathcal{L}}{N} \tag{2.12}$$

where ξ represents a scaled charge parameter of a MT, $\mathcal{L}$ is the length of a MT and Ne is the total surface charge of the same MT, Fig. 5.

Returning to the variable r, the above equations yield:

$$\psi(r) = \frac{2\xi\mathcal{D}K_0\left(\frac{r}{\mathcal{D}}\right)}{RK_1\left(\frac{r}{\mathcal{D}}\right)}. \tag{2.13}$$

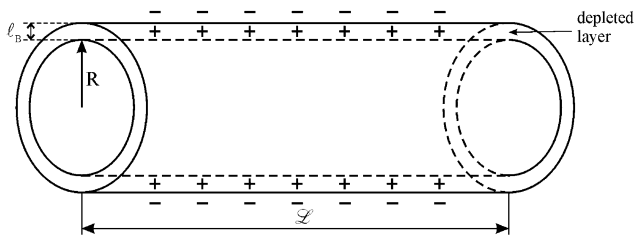

Fig. 5 MT cylinder with a depleted layer with the thickness equal to Bjerrum length ℓ_B

This equation in the case of low ionic strengths $(r/\mathcal{D} \to 0)$ behaves like $-\ln\left(\frac{r}{\mathcal{D}}\right)$, and hence we obtain the solution of PBE as:

$$V(r) = -\frac{eN}{2\pi\varepsilon_0\varepsilon\mathcal{L}}(\ln r - \ln \mathcal{D}) = -\frac{Q}{2\pi\varepsilon_0\varepsilon\mathcal{L}}\ln\left(\frac{r}{\mathcal{D}}\right). \quad (2.14)$$

The first term corresponds to the potential due to the surface charge of a MT and the second term $(\ln \mathcal{D})$ to the potential due to the counter-ions. The potential difference between the MT surface cylinder and the outer cylinder of the depleted zone at a distance ℓ_B is therefore given by:

$$\Delta V = -\frac{Q}{2\pi\varepsilon_0\varepsilon\mathcal{L}}\ln\left(1+\frac{\ell_B}{R}\right). \quad (2.15)$$

The capacitance of a MT is defined by $C = \frac{Q}{|\Delta V|}$, giving the total value for a MT:

$$C = \frac{2\pi\varepsilon_0\varepsilon\mathcal{L}}{\ln\left(1+\frac{\ell_B}{R}\right)} \quad (2.16)$$

Taking as an elementary capacitor C_0 a ring of 13 tubulin dimers with the length $\ell = 8 \times 10^{-9}$ m, and using $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m, and $\varepsilon = 80$, with the above values for ℓ_B and R, we find the first part of the ring's capacitance from Eq. 2.16:

$$C_0^{(1)} = 6.8 \times 10^{-16}\,\mathrm{F} = 0.68\,\mathrm{fF}.$$

In the same way, considering an extended TT as a cylinder with length $\ell_C = 4.5$ nm and radius $R_C = 0.5$ nm we calculate an additional capacity $C_0^{(2)}$ for $n = 13 \times 2 = 26$ TTs contained in the same elementary ring composed of 13 dimers as above (see Fig. 6).

Using the same formula, Eq. 2.16, we obtain:

$$C_0^{(2)} = \frac{26 \times 3.14 \times 80 \times 8.85 \times 10^{-12} \times 4.5 \times 10^{-9}}{\ln\left(1+\frac{0.67}{0.5}\right)}\,\mathrm{F}$$
$$= 0.64 \times 10^{-15}\,\mathrm{F} = 0.64\,\mathrm{fF}.$$

Thus, treating the two contributing capacitances to be connected in parallel, the total maximal capacity of the elementary ring is found to be:

$$C_0 = C_0^{(1)} + C_0^{(2)} = 1.32 \times 10^{-15}\,\mathrm{F}. \quad (2.17)$$

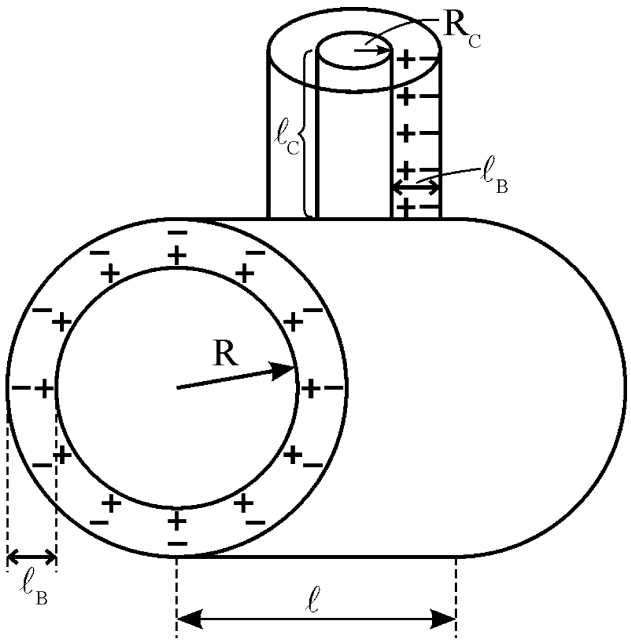

Fig. 6 Elementary ring of a MT with characteristic cylindrical capacitors and corresponding dimensions

We have already stressed that the increase of negative voltage should increase the concentration of positive counter-ions. These ions additionally condense on the surface of a MT and on TTs. This causes β-TTs to shrink diminishing their effective length ℓ_C and thus the capacity $C_0^{(2)}$. We may initially at least assume that this effect is linear with respect to the applied voltage. This observation is of crucial importance to the development of an appropriate model of MTs as nonlinear transmission lines for ionic currents.

The estimation of effective inductance of a MT ring

We start from the assumption that a current I along the MT can be conceived as a sum of 13 thin parallel wires along every protofilament (see Fig. 7).

If we calculate the resulting magnetic field in the location $R < r < R + \ell_B$, we have to add the 13 components together. Due to the radial symmetry present, we found that the total flux of the magnetic field through area $A = \ell \cdot \ell_B$ is given as follows:

$$\Phi = \frac{\mu_0 \ell I_0}{\pi}\left[\frac{3\ell_B}{R} + \frac{1}{2}\ln\left(1+\frac{\ell_B}{R_d}\right)\right],$$

where I_0 stands for the ionic current along a single protofilament and R_d is the radius of a protofilament (tubulin dimer). Using Faraday's law, we readily find the induced voltage as:

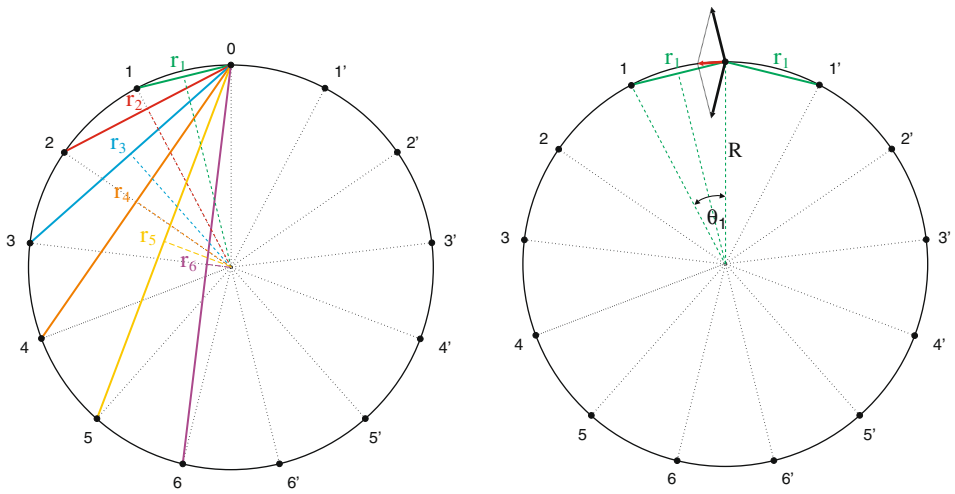

Fig. 7 The diagram for estimation of flux of magnetic field around a MT brought about by 13 currents I_0 along protofilaments

H/m=henry/meter= magnetic permeability

$$\mathcal{E} = -\frac{\mathrm{d}\Phi}{\mathrm{d}t} = -L\frac{\mathrm{d}I_0}{\mathrm{d}t} = -\frac{\mu_0 \ell}{\pi}\left[\frac{3\ell_\mathrm{B}}{R} + \frac{1}{2}\ln\left(1 + \frac{\ell_\mathrm{B}}{R_\mathrm{d}}\right)\right]\frac{\mathrm{d}I_0}{\mathrm{d}t},$$

which brings about the inductance of a MT ring:

$$L = \frac{\mu_0 \ell}{\pi}\left[\frac{3\ell_\mathrm{B}}{R} + \frac{1}{2}\ln\left(1 + \frac{\ell_\mathrm{B}}{R_\mathrm{d}}\right)\right]. \tag{2.18}$$

Inserting the set of parameters: $\ell = 8$ nm, $R = 12.5$ nm, $R_\mathrm{d} = 4$ nm, $\ell_\mathrm{B} = 0.67$ nm, $\mu_0 = 4\pi \times 10^{-7}$ H/m, we obtain:

$$L = 8 \times 10^{-16}\,\mathrm{H} = 0.8\,\mathrm{fH}. \tag{2.19}$$

Modeling the longitudinal and transverse resistance components

Recently published accurate measurements of individual MTs using the electro-orientational method performed in vitro by Minoura and Muto (2006) produced the value of MT conductivity:

$$\sigma = (0.15 \pm 0.01)\,\mathrm{S/m}. \tag{2.20}$$

If we now take that the resistivity within the depleted cylindrical shell around a MT with thickness ℓ_B is homogeneous, the longitudinal resistance of the elementary ring with length $\ell = 8 \times 10^{-9}$ m and area $A_1 = 2\pi R\,\ell_\mathrm{B}$ (see Fig. 8) is expected to be:

$$R_1 = \frac{1}{\sigma}\frac{\ell}{2\pi R \ell_\mathrm{B}} = \frac{8 \times 10^{-9}}{0.15 \times 2\pi \times 12.5 \times 10^{-9} \times 0.67 \times 10^{-9}}\,\Omega = 10^9\,\Omega. \tag{2.21}$$

The transversal resistance of the same ring R_2 is calculated by the same formula, but the length of each resistor is now ℓ_B and the area $A_2 = 2\pi R\,\ell$, giving:

$$R_2 = 7 \times 10^6\,\Omega. \tag{2.22}$$

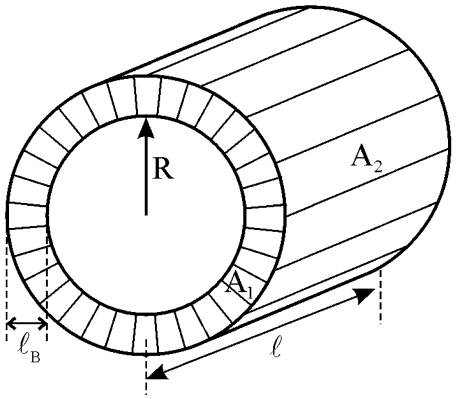

Fig. 8 Elementary ring of a MT with characteristic areas A_1 and A_2 pertaining to resistivities R_1 and R_2

Modeling MT as nonlinear RLC transmission line

On the basis of the above estimates for the resistive components of the elementary ring of a MT we are now able to develop and analyze an electrical model by applying Kirchhoff's laws to the series of elementary rings envisaged to represent coupled electrical circuits. We expect a potential difference between the MT charged surface with condensed counter-ions and ions located along the coaxial cylinder at a distance of one Bjerrum length away. These "Bjerrum ions" are responsible for the creation of 13 time-dependent currents which generate the inductance L, Eq. 2.19. In series with L one should insert a resistive component R_1, Eq. 2.21 (see Fig. 9).

In addition to the above components there is also a cylindrical capacitance C_0, Eq. 2.17 connected in series with transversal resistivity R_2, Eq. 2.22, acting between the "Bjerrum ions" and the surface of a MT. We have already

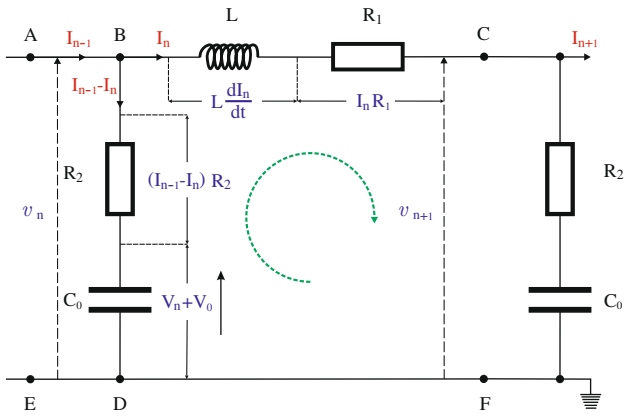

Fig. 9 An effective circuit diagram for the nth elementary ring between the doted lines

pointed out that the capacitance C_0 should change with an increasing concentration of counter-ions due to the flexibility of β-TTs. It implies that the charge on this elementary capacitor varies in a nonlinear way with voltage, much like the charge-voltage relation for a reverse-biased diode junction (Wang et al. 1999). Thus we take for the nth MT ring:

$$Q_n = C_0\left(V_n - bV_n^2\right), \tag{3.1}$$

where b is expected to be small ($bV_n \ll 1$) since the change of the area of β-TT is small compared with the outer surface of the corresponding dimer. From Kirchhoff's laws, on the basis of Fig. 9, if I_n is the current through the inductance L and the resistor R_1, and I_{n-1} is the other current flowing along AB, the current through line DB should be $I_n - I_{n-1}$. Thus for the section BC of the nth MT ring we obtain:

$$v_n - v_{n+1} = L\frac{\mathrm{d}I_n}{\mathrm{d}t} + I_nR_1, \tag{3.2}$$

where v_n and v_{n+1} are the voltages across AE and CF, respectively. Similarly, if the voltage across the variable capacitor $C_0\left(V_n - bV_n^2\right)$ is $V_n + V_0$, where V_0 is the bias voltage of the capacitor, we have:

$$v_n = R_2(I_{n-1} - I_n) + V_0 + V_n. \tag{3.3}$$

Moreover, the current through the section BD should be found as the rate of change of Q_n:

$$I_{n-1} - I_n = \frac{\mathrm{d}Q_n}{\mathrm{d}t}. \tag{3.4}$$

From Eq. 3.2 we have

$$L\frac{\mathrm{d}I_{n-1}}{\mathrm{d}t} = v_{n-1} - v_n - I_{n-1}R_1 \quad \text{and}$$
$$L\frac{\mathrm{d}I_n}{\mathrm{d}t} = v_n - v_{n+1} - I_nR_1$$

and including it in Eq. 3.4 we find

$$L\frac{\mathrm{d}^2Q_n}{\mathrm{d}t^2} = v_{n+1} + v_{n-1} - 2v_n + R_1(I_n - I_{n-1}). \tag{3.5}$$

Substituting v_{n+1}, v_n and v_{n-1} from Eqs. 3.3 to 3.5 one obtains

$$\begin{aligned} L\frac{\mathrm{d}^2}{\mathrm{d}t^2}\left[C_0\left(V_n - bV_n^2\right)\right] &= V_{n+1} + V_{n-1} - 2V_n \\ &- R_1C_0\frac{\mathrm{d}}{\mathrm{d}t}\left(V_n - bV_n^2\right) \\ &- R_2C_0\left\{2\frac{\mathrm{d}}{\mathrm{d}t}\left(V_n - bV_n^2\right) - \frac{\mathrm{d}}{\mathrm{d}t}\left(V_{n+1} - bV_{n+1}^2\right)\right. \\ &\left. - \frac{\mathrm{d}}{\mathrm{d}t}\left(V_{n-1} - bV_{n-1}^2\right)\right\}. \end{aligned} \tag{3.6}$$

Bearing in mind that voltage V_n changes gradually from ring n to its neighbors, we can safely expand V_n in a continuum approximation using a Taylor expansion in terms of a small spatial parameter ℓ (the length of a dimer):

$$\begin{aligned} V_{n\pm1} = V &\pm \ell\left(\frac{\partial V}{\partial x}\right) + \frac{\ell^2}{2!}\left(\frac{\partial^2 V}{\partial x^2}\right) \\ &\pm \frac{\ell^3}{3!}\left(\frac{\partial^3 V}{\partial x^3}\right) + \frac{\ell^4}{4!}\left(\frac{\partial^4 V}{\partial x^4}\right) \pm \cdots. \end{aligned} \tag{3.7}$$

We discarded terms on the order of ℓ^5 and ℓ^6 since $\ell^5 \sim 0.33 \times 10^{-40}$ m^5 is a very small value. The above expansion leads to the following relation:

$$V_{n+1} - 2V_n + V_{n-1} = \ell^2\left(\frac{\partial^2 V}{\partial x^2}\right) + \frac{\ell^4}{12}\left(\frac{\partial^4 V}{\partial x^4}\right). \tag{3.8}$$

Inserting Eqs. 3.7 and 3.8 into Eq. 3.6 yields:

$$\begin{aligned} LC_0\frac{\partial^2}{\partial t^2}\left(V - bV^2\right) &= \ell^2\left(\frac{\partial^2 V}{\partial x^2}\right) + \frac{\ell^4}{12}\left(\frac{\partial^4 V}{\partial x^4}\right) \\ &- R_1C_0\frac{\partial}{\partial t}\left(V - bV^2\right) + R_2C_0\frac{\partial}{\partial t}\left[\ell^2\left(\frac{\partial^2 V}{\partial x^2}\right) + \frac{\ell^4}{12}\left(\frac{\partial^4 V}{\partial x^4}\right)\right] \\ &- R_2C_0b\frac{\partial}{\partial t}\left[2\ell^2\left(\frac{\partial^2 V}{\partial x^2}\right) + 2\ell^2\left(\frac{\partial V}{\partial x}\right)^2 + \frac{\ell^4}{6}V\left(\frac{\partial^4 V}{\partial x^4}\right)\right. \\ &\left. + \frac{2}{3}\ell^4\left(\frac{\partial V}{\partial x}\right)\left(\frac{\partial^3 V}{\partial x^3}\right) + \frac{\ell^4}{2}\left(\frac{\partial^2 V}{\partial x^2}\right)^4\right]. \end{aligned} \tag{3.9}$$

The above derivation closely parallels an analogous derivation for actin filaments whose details can be found in Tuszynski et al. (2004).

Considering that the time variations of the local voltage V are small compared to the constant background voltage V_0, we can safely assume that the time derivative is of the order of the small parameter ε as well as the nonlinear

Capital V is voltage and small v is velocity

voltage terms bV^2 are of the order ε^2. Thus Eq. 3.9 can be vastly reduced by collecting only the leading terms:

$$LC_0\frac{\partial^2 V}{\partial t^2} = \ell^2\left(\frac{\partial^2 V}{\partial x^2}\right) + R_2C_0\frac{\partial}{\partial t}\left[\ell^2\left(\frac{\partial^2 V}{\partial x^2}\right)\right] - R_1C_0\frac{\partial V}{\partial t} + R_1C_0 2bV\frac{\partial V}{\partial t}. \tag{3.10}$$

Taking into account the values of L and C_0, Eqs. 2.19 and 2.17, it is clear that the factor $LC_0 \sim 10^{-30}$ s is exceedingly small and hence the second derivative term can be safely ignored. Moreover, since $R_1 \gg R_2$, the term proportional to the factor $R_2\ C_0$ can also be discarded bringing Eq. 3.10 into its final form:

$$\ell^2\frac{\partial^2 V}{\partial x^2} - R_1C_0\frac{\partial V}{\partial t} + R_1C_0 2bV\frac{\partial V}{\partial t} = 0. \tag{3.11}$$

As a first step we can linearize this equation taking $b \to 0$, which yields $b \to 0$, thus getting

$$\frac{\ell^2}{R_1C_0}\frac{\partial^2 V}{\partial x^2} = \frac{\partial V}{\partial t}. \tag{3.12}$$

This simply represents Fick's law of diffusion with a coefficient of diffusion

$$D = \frac{\ell^2}{R_1C_0} = 0.8\times 10^{-10}\,\mathrm{m^2/s},$$

a value that, for example, falls within the experimentally observed range (Albriton et al. 1992) for calcium waves in intracellular calcium dynamics from endoplasmic reticulum.

The spatio-temporal voltage function in Eq. 3.12 follows the diffusion of ions along the MT. The solution of this equation is a well-known Gaussian wave packet that spreads with time according to:

$$V(x,t) = V_0\left(\frac{R_1C_0}{\pi\ell^2 t}\right)^{\frac{1}{2}}\exp\left\{-\left[\frac{x}{2\left(\frac{\ell^2 t}{R_1C_0}\right)^{\frac{1}{2}}}\right]^2\right\}. \tag{3.13}$$

We now analyze Eq. 3.10 by introducing a set of dimensionless variables as follows:

$$\left.\begin{aligned} &\frac{x}{\ell} = \eta; \quad \frac{t}{R_1C_0} = \tau; \quad \xi = \eta - v_0\tau \\ &V_0 = \frac{v}{v_0}; \quad v_0 = \frac{\ell}{R_1C_0}; \quad W = \frac{V}{V_0} \end{aligned}\right\} \tag{3.14}$$

Here, v is the actual propagation velocity of ionic waves while v_0 is the maximum value of the propagation velocity in a given medium.

The characteristic velocity of ionic waves is found to be

$$v_0 = \frac{8\times 10^{-9}}{10^9\times 1.32\times 10^{-15}}\,(\mathrm{m/s}) = 6\,(\mathrm{mm/s}), \tag{3.15}$$

and is two orders of magnitude greater than the velocity of calcium waves in bulk cytoplasm (Wang et al. 2004). However, since $v < v_0$ ($V_0 < 1$) this appears to be a reasonably accurate estimate which is likely to correspond to the velocity of MT-guided ionic waves in vivo. In terms of the new variables in Eq. 3.14, the nonlinear partial differential equation of (3.11) now becomes an ordinary differential equation which reads:

$$\frac{\mathrm{d}^2 W}{\mathrm{d}\xi^2} + v_0\frac{\mathrm{d}W}{\mathrm{d}\xi} - 2v_0V_0bW\frac{\mathrm{d}W}{\mathrm{d}\xi} = 0. \tag{3.16}$$

Integrating it once we easily find that:

$$\frac{\mathrm{d}W}{\mathrm{d}\xi} + v_0W - v_0V_0bW^2 = c_1. \tag{3.17}$$

Interestingly, the same form of a differential equation can be found to describe the thermophoresis of biopolymers in solution (Braun and Libchaber 2004) where temperature changes linearly with distance ($T = T_0 + \alpha x$)

$$j = -D\frac{\mathrm{d}c}{\mathrm{d}x} - D_{\mathrm{T}}c(1-c)\frac{\mathrm{d}T}{\mathrm{d}x} = \mathrm{const}, \tag{3.18}$$

where $c(x)$ is the polymer concentration function, D is the diffusion coefficient, D_{T} is the thermal diffusion coefficient, and j is the steady-state current density of the polymer. This is the basis for a thermal force scenario proposing that life might have originated in pores, driven by gradients of temperature and electrical potential at hydrothermal mounds developed over submarine seepages. Moreover, if $c_1 = 0$ in Eq. 3.17, it reduces to the same mathematical form describing the logistic equation of population growth (Strogatz 1994).

In order to determine c_1 in Eq. 3.17 we impose the natural initial condition of the type

$$\frac{\mathrm{d}W}{\mathrm{d}\xi} = -A < 0, \quad \text{for} \quad W = 0.$$

Separating variables in Eq. 3.17 and factorizing it, we have

$$\int\frac{\mathrm{d}W}{W - W_1} - \int\frac{\mathrm{d}W}{W - W_2} = \alpha\xi + \ln c_2, \tag{3.19}$$

where

$$W_{1,2} = \frac{1}{2bV_0}\left[1 \pm \left(1 + 4\frac{AV_0b}{v_0}\right)^{\frac{1}{2}}\right] \quad \text{and} \quad \alpha = (W_1 - W_2)V_0v_0b > 0. \tag{3.20}$$

The integration of Eq. 3.19 with the initial conditions: $\xi = 0 \Rightarrow W = 0$, yields:

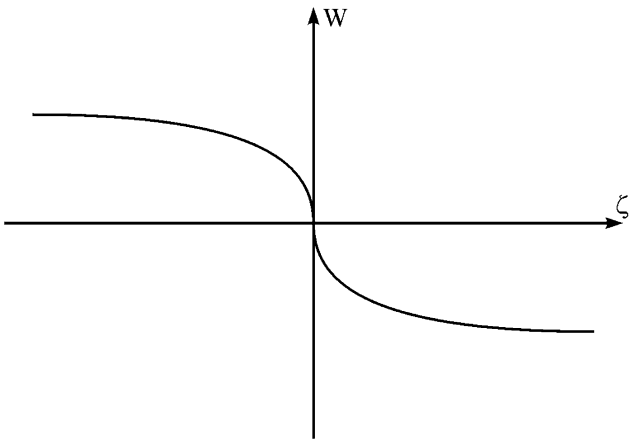

Fig. 10 The anti-kink profile of ionic potential propagating with small velocity along MT

$$W(\xi) = W_1\left[\frac{1-\exp(\alpha\xi)}{1-\frac{W_1}{W_2}\exp(\alpha\xi)}\right]. \tag{3.21}$$

Let us consider the interesting case of slow waves with velocities v_0 obeying the inequality

$$\frac{4AV_0b}{v_0} \gg 1, \tag{3.22}$$

which implies that Eq. 3.21 becomes

$$W_0(\xi) = -\left(\frac{A}{bV_0v_0}\right)^{\frac{1}{2}}\tanh(\alpha\xi), \quad \alpha = 2(AbV_0v_0)^{\frac{1}{2}}. \tag{3.23}$$

This is a symmetric anti-kink (AK) represented graphically in Fig. 10.

In order to examine Eq. 3.17 with respect to its stability we start from a slightly distorted trial solution of the form

$$W = W_0 + \delta; \quad \delta \ll W_0.$$

Inserting this trial solution into Eq. 3.17 we readily obtain

$$\frac{d^2\delta}{d\xi^2} + v_0(1-bV_0)\frac{d\delta}{d\xi} - \left(\frac{dW_0}{d\xi}\right)bV_0\delta = 0. \tag{3.24}$$

Since the inequalities

$$bV_0 \ll 1; \quad \frac{dW_0}{d\xi} < 0$$

always hold, this clearly indicates that the AK of Eq. 3.23 represents a robust stable front of propagating ionic wave as shown in Fig. 10.

The impact of the cell's intrinsic electric fields on the dynamics of AKs in MTs

Satarić et al. (1993, 2007) and Satarić and Tuszynski (2003, 2005) developed the ferroelectric model of MTs with oppositely charged ends as giant dipoles arising as a consequence of the fact that every tubulin dimer possesses an intrinsic electric dipole (see Fig. 11) that becomes compensated by the dipoles of its neighbors only partially.

These ferroelectric features of MTs lead to the conclusion that every MT has an intrinsic, almost constant, electric field E whose magnitude depends on the environmental pH conditions and the MT length. This field gives rise to an additional potential V_{int}. In this context the dimensionless Eq. 3.16 cab be analyzed numerically by taking into account V_{int}.

If we judiciously choose the set of parameters:

$$v_0 = 0.1, \quad v_0bV_0 = 0.05, \quad \frac{V_{int}}{V_0} = 0.4,$$

we have the resultant equation in the form

$$\frac{d^2W}{d\xi^2} + 0.1\frac{dW}{d\xi} - 0.05W\frac{dW}{d\xi} + 0.1 = 0. \tag{4.1}$$

Taking the initial conditions as

$$\frac{dW}{d\xi}(0) = -0.5, \quad W(0) = 0$$

we calculate the trajectory for the center of the AK as a function of the dimensionless time represented graphically in Fig. 12.

This figure shows that the AK is initially driven by acceleration and after $\tau = 50 (t \sim 0.5\,\mu s)$ it attains a constant terminal velocity $V_0 = 0.1$ ($v = 0.1 \cdot v_0 \approx 600$ μm/s) which is reasonably close to the experimentally determined values for waves of Ca^{2+} ions.

We further assume that certain intrinsic alternating electric fields (IAEF) can be elicited in the cell's compartments. It was already suggested earlier that such fields can originate from the oriented (vicinal) water molecules which form an electric dipole field occurring on either side

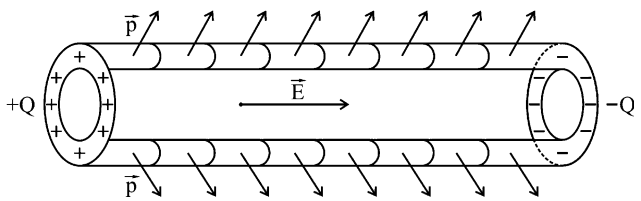

Fig. 11 Ferroelectric features of a MT. The charged tips bring about a constant intrinsic electric field E

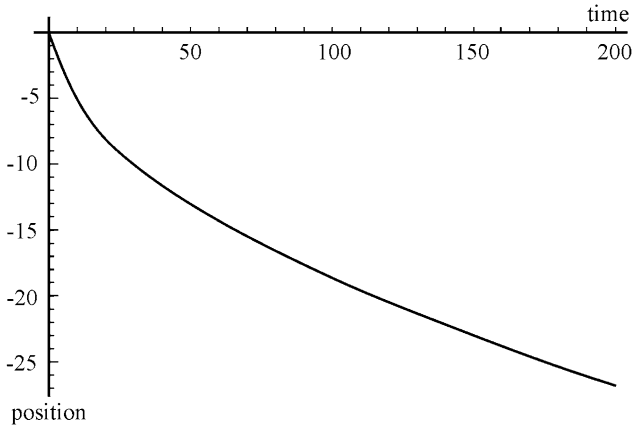

Fig. 12 The trajectory of the center of AK under the action of constant intrinsic electric field, Eq. 4.1

of the cell membrane. Del Giudice et al. (1988) have proposed that electromagnetic fields arising from the coherent oscillations of the above-mentioned water molecules represent signals comparable in size with the dimensions of the cell's MTs. It is potentially of major significance that MTs themselves may be a source of oscillating electric fields in living cells as discussed by Pokorny et al. (2005). We can safely assume that the associated dimensionless driving force has a wavelength greater than the average MT length and it harmonically depends on the dimensionless time τ as follows

$$V_{\mathrm{WF}}(\tau) = A_0 \cos(\Omega\tau + \varphi_0), \tag{4.2}$$

where A_0 is the amplitude, Ω is the frequency and φ_0 is the initial phase of the potential of the water field (WF). Using the same parameters for the homogeneous part of Eq. 4.1 as those used before, and taking $A_0 = 0.5$, $\Omega = 0.5$ and $\varphi_0 = 0$, we have

$$\frac{\mathrm{d}^2 W}{\mathrm{d}\xi^2} + 0.1\frac{\mathrm{d}W}{\mathrm{d}\xi} - 0.05W\frac{\mathrm{d}W}{\mathrm{d}\xi} + 0.1 = 0.5\cos(0.5\tau). \tag{4.3}$$

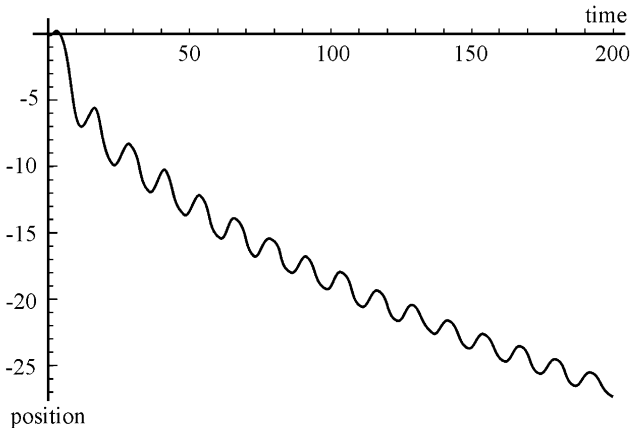

Fig. 13 The trajectory of the center of AK subjected to constant and harmonically changeable intrinsic electric fields, Eq. 4.3

A numerical solution of the above equation produces a trajectory of the corresponding AK as shown in Fig. 13.

We have again found similar characteristics of the unidirectional AK motion with the same average velocity but now superimposed with harmonic oscillations around the center of the AK.

Conclusion and discussion

A new model for ionic currents along MTs has been presented in this paper with a strong emphasis on the concept of a MT as a polyelectrolyte. The background for the model is the molecular structure and geometry of a MT and its interaction with solvent ions. In this context the MT cylinder with brush-like TTs surrounded by ions from cytosol is represented as an electrical transmission line with R, C and L elements whose values have been estimated using the available experimental data and known geometry. The important feature of the nonlinearity of MT capacitance has arisen naturally from the structure of β-TTs. As a result, the nonlinear second order differential equation for the electric potential propagating along the MT was evaluated. The traveling wave solution of this equation, Eq. 3.16, is a localized front profile of the AK type. This front resembles a so-called "nano-tsunami" and it propagates with a velocity which has been estimated to be reasonably close to the experimental findings, for example for waves of Ca^{2+} ions elicited by endosome reticulum compartments. These AKs appear to be stable and robust against perturbations. The remaining part of the paper has been dedicated to the possible impact of the cell's endogenous (intrinsic) electric fields on the dynamics of ionic AKs. It appears that the velocity and direction of propagation of an AK could be tuned by changing the parameters of the above-mentioned fields (i.e. their amplitude, frequency and phase). It is our hope that the model developed here will provide a better insight into the still unknown roles of MTs in the cell's information processing capabilities.

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A quantum mechanical interaction of human erythrocytes[1]

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Theory predicts that the membrane potential will polarize membrane molecules and cause them to vibrate coherently at a frequency of $\sim 10^{11}$ Hz. If the supply of metabolic energy exceeds a minimum value, membrane phonons may condense their momentum into a single "giant" vibrational mode. At 10^{11} Hz ionic screening is small up to distances of approximately a micrometre, so forces of a range several orders of magnitude longer than chemical forces can arise. These forces may be attractive or repulsive depending on frequency. They should occur in every metabolically active membrane and may control macromolecular transport and enzyme–substrate interactions. We find that normal human erythrocytes in plasma form rouleaux faster than Brownian motion predicts. When cells are fixed in glutaraldehyde or are metabolically depleted, or if the membrane potential is brought to zero, the rate of aggregation agrees with Brownian theory. When the metabolically depleted cells are revived or if the membrane potential is restored, then the interaction returns.

ROWLANDS, S., L. S. SEWCHAND et E. G. ENNS. 1982. A quantum mechanical interaction of human erythrocytes. Can. J. Physiol. Pharmacol. **60**: 52–59.

La théorie veut que le potentiel de membrane polarise les molécules membraneuses et les fait vibrer cohéremment à une fréquence d'environ 10^{11} Hz. Si l'apport d'énergie métabolique dépasse une valeur minimale, les phonons membraneux peuvent concentrer leur momentum en un seul mode de vibration. A 10^{11} Hz, l'interaction ionique est faible jusqu'à des distances de quelques micromètres, de sorte que des forces de plusieurs ordres de grandeur plus élevées que les forces chimiques peuvent se développer. Ces forces peuvent être attractives ou répulsives suivant la fréquence. Elles devraient apparaître dans chaque membrane métaboliquement active et peuvent contrôler le transport macromoléculaire et les interactions enzyme–substrat. Nous constatons que les érythrocytes humains normaux contenus dans le plasma forment des rouleaux plus rapidement que ne le prévoit le mouvement brownien. Quand les cellules sont fixées dans le glutaraldéhyde ou appauvries métaboliquement ou encore si le potentiel de membrane est placé à zéro, le taux d'aggrégation concorde avec la théorie brownienne. Quand les cellules appauvries métaboliquement sont revivifiées ou lorsque le potentiel de membrane est rétabli, alors l'interaction se normalise.

[Traduit par le journal]

Introduction

For over a decade Fröhlich (1968) has been developing the concept of a unique collective interaction of the molecules comprising cell membranes. Collective interactions are well known in the physics of the condensed state of matter and are responsible for such subtleties as super-conductivity and the bizarre behaviour of liquid helium II. The manifestations of Fröhlich's theory should arise at normal temperatures and should occur in all living systems. In particular the theory predicts the existence of very long-range forces (~ 1 μm) which are highly specific in that an organelle or a cell could recognize another at a distance and be attracted or repelled. Fröhlich (1980) recently reviewed the theory and the evidence in its support.

Since the membrane potential acts across a membrane thickness of the order of 10 nm, the electric field is very great ($\sim 10^7$ V·m^{-1}), and membrane molecules will be strongly polarized in this field. Thermal vibration of the dipoles in the membrane correspond to a polarization wave of half wavelength roughly equal to the membrane thickness. If the velocity of mechanical waves in the membrane has the reasonable value of 10^3 m·s^{-1}, the vibrations will have a frequency of $\sim 10^{11}$ Hz. Resonant absorption in tissue cultures and in microorganisms in the neighbourhood of this frequency comprises the main evidence for the theory (Fröhlich 1980). Since polar molecules interact electrically, the vibrations of the membrane constituents will be col-

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lective and the quantum-mechanical properties of these collective interactions will be apparent and important at 10^{11} Hz. The name phonon is given to this aspect of the vibration by analogy with the photon, the corresponding "quasi-particle" of the electromagnetic waves. In a strong electric field, conformational change in macromolecules and the readjustment of counter ions can result in minima appearing in the relationshp between polarization and electric field. If these minima are above the energy of the ground state, they will correspond with metastable polar states; if lower than the ground state, the system will be ferroelectric. Even if the metastable states have a short life their occupancy may remain high by the exchange of thermal phonons with the surroundings, which is a heat bath in the thermodynamic sense. If energy, say from cell metabolism, is now supplied to the modes of such a system of collective polar excitations, a new process arises with the exchange of the difference in energy of two quanta with the heat bath. If this process dominates single-phonon exchange, the supply of energy will become concentrated into the mode of highest wave number and constitute a giant vibration. Phonons have zero spin and obey Bose–Einstein statistics. The mathematical formulation of the accumulation of energy into a single zero wave-number mode is similar to that of the Einstein condensation of a Bose gas (London 1950; Kittel 1958), a condensation in momentum space, not in coordinate space which is the case for the condensation of a common gas. Liquid helium II is presumed to result from such a quantum condensation (London 1950).

Fröhlich's theory as it stands is quite general and makes no predictions about specific cell systems. The important conclusion is that forces, of a range several orders of magnitude longer than London–van der Waals' forces should exist between macromolecular systems within the cell and between cells, perhaps over distances of micrometres. The forces may be attractive or replusive depending on the natural frequencies of the interacting systems and the dispersion of the dielectric constant, as a function of frequency, of the medium in which the systems are suspended. The necessary features of an entity which might be expected to manifest a Fröhlich effect are (*a*) an organization of molecules into some system, (*b*) a membrane potential to polarize the array of molecules in the system, and (*c*) a supply of metabolic energy. In the course of experimentation on red-cell rouleau formation (Rowlands and Skibo 1972; Sewchand and Canham 1979) we noticed a small attraction and a "jiggling" of the membranes just prior to the contact which precedes doublet formation. When we upset the organization of the membrane or remove the supply of energy or alter the membrane potential, this attraction is affected. An account of our early results has been published (Rowlands et al. 1981).

Methods

Venous blood was drawn on the morning of an experiment and immediately mixed with heparin (14 USP units/mL of blood). The plasma was separated by centrifugation and filtered through a 0.3-μm Millipore filter. Red cells, normal or experimentally modified, were added to the plasma at a concentration (v/v) of 0.5%. The low haematocrit blood was stirred rapidly with a magnetic stirrer to break up rouleaux and quickly transferred to a siliconized (Dow Corning, Midland, MI, Sylgard 184 or R4-3117) haemacytometer chamber (Spencer Bright-Line, American Optical Corporation, Buffalo, NY) using a 30-gauge needle to ensure high shear rate and so minimize prior rouleau formation. The chamber was sealed with silicone grease to prevent evaporation and photographed with a time-lapse cinécamera on an Ortholux microscope (Wild-Leitz, Rockleigh, NJ) at about 60× magnification. An exposure was taken every 15 s for 3 h. About 150–200 cells were in view.

Glutaraldehyde-fixed cells

Red cells were washed twice in phosphate-buffered saline (PBS) containing 0.04% glutaraldehyde (PBS composition: 150 m*M* NaCl, 7.3 m*M* Na_2HPO_4, 2.8 m*M* NaH_2PO_4, 2.2 m*M* $CaCl_2$, 4.6 m*M* $MgCl_2$, 310 mosmol, pH 7.4 at 22°C), suspended in the same solution, and shaken gently for 24 h at 22°C. They were stored at 4°C until use. Cells so treated retain normal shape, area, and volume (R. E. Lovlin, unpublished observations).

Microspheres

Latex microspheres 3.4 μm in diameter (Coulter Electronics, Hialeah, FL) were added to plasma at a concentration (v/v) of about 0.2%, shaken, and used immediately. Polystyrene DVB microspheres 5.9 and 8.0 μm in diameter (Duke Scientific, Palo Alto, CA) stuck to the floor of the haemacytometer chamber when in plasma. They were therefore suspended in soapy water at a similar concentration.

Variation of membrane potential

The pH of the cell suspension was measured (PHM 64 Radiometer, Copenhagen, electrode GK2421C) and then lowered by adding 0.14 *M* HCl in 0.14 *M* NaCl (so keeping $[Na^+]$ constant) or increased by adding 0.14 *M* NaOH (Funder and Wieth 1966; Lassen et al. 1980). The membrane potential was calculated from $[Cl_i^-]/[Cl_o^-]$ using the Nernst equation. $[Cl^-]$ was measured with a Corning chloride meter (model 920 M, Corning, Medfield, MA).

Abolition of membrane potential by ionophore A23187

Flatman and Lew (1980) used this ionophore to study magnesium buffering in the human red cell and showed that entry of magnesium depolarized the membrane. We used their method to achieve an approximately zero potential in the following way. The cells were washed twice in PBS containing 10 μ*M* A23187 and then were suspended in plasma to which had been added 10 m*M* $MgCl_2$, 10 μ*M* EGTA, and 10 μ*M* A23187. The pretreatment of the cells with the ionophore was necessary because it is readily adsorbed by albumin (Sarkadi et al. 1976).

Metabolic depletion and repletion

Cells were washed in PBS, resuspended in PBS, and incu-

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bated at 37°C for 22 h. They were repleted by adding 30 m*M* adenosine to the suspending medium and incubating for a further 3 h (Weed et al. 1969; Meiselman and Baker 1977).

Viscosity

The viscosity of plasma ($\eta \sim 1.7 \times 10^{-3}$ Pa·s) for each subject and of the soapy water ($\eta = 1.06 \times 10^{-3}$ Pa·s) for the microspheres was measured with a cone and plate viscometer (Synchro-Lectric, Brookfield, Stoughton, MA).

Analysis of results

The phenomenon is a coagulation of particles undergoing Brownian movement on which is superimposed a specific interaction at a distance. Smoluchowski (1917) developed a theory for the coagulation of the particles of colloidal suspensions. The theory is reviewed in a shortened form by Chandrasekhar (1943) and experimental tests are described by Swift and Friedlander (1964). It is assumed that the diffusion coefficient of the individual particles is given by the Stokes–Einstein relation

$$[1] \quad D = \frac{kT}{6\pi\eta r}$$

where k is Boltzmann's constant, T absolute temperature, η the viscosity of the suspending medium, and r the particle radius. In the theory, Smoluchowski incorporates the possibility of a zone of attraction, radius $Z > r$, around each particle, which ceases to exist if $Z = r$, and if $Z < r$ a repulsive force is implied. Chandrasekhar (1943, equation 461) gives an expression for the variation in the total number of particles as a function of time. We also adopt Swift and Friedlander's (1964) simplifying assumption that all particles have the same radius (and the implication that all zones of attraction have the same radius $Z \neq r$). If, then, N is the total number of particles (single and multiple) the kinetics of coagulation is given by

$$[2] \quad \frac{1}{N_t} - \frac{1}{N_0} = \frac{4kTZ}{3\eta r} t$$

Equation 2 is for a three-dimensional process and would apply if the cells had the same density as plasma and so were distributed randomly in depth as well as in the other two dimensions of the chamber. Cells sediment, however, so the process is compressed into two dimensions and the rate constant will be higher by a factor of 1.5. Equation 2 becomes

$$[3] \quad \frac{1}{N_t} - \frac{1}{N_0} = \frac{2kTZ}{\eta r} t$$

which predicts a linear relation between N_t^{-1} and t.

The developed cinéfilm was projected with a photo-optical data analyser (model 224A, L-W International, Woodland Hills, CA), and in the earliest frame at which the cells had settled, the total number of cells and rouleaux (N_0) was counted in an area defined by the haemacytometer grid. The count was repeated every fifth frame (75 s) initially, and at increasing intervals up to 20 frames (300 s). The regression of N^{-1} on t was computed.

Einstein's (1926) theory of Brownian motion gives the mean square deviation in two dimensions as a function of time

$$[4] \quad \langle U^2 \rangle = 4kTt\mu^{-1}$$

where μ is Stokes' drag coefficient for spheres

$$[5] \quad \mu = 6\pi\eta r$$

which is also the denominator in [1]. To check that isolated cells were executing Brownian motion, U was measured against t for each of 30 cells which did not come within 6 μm of another during the first 15 min of each experiment. Beyond 15 min fewer than 30 cells fulfilled this criterion. The correlation of $\langle U^2 \rangle$ with t and the value of μ were calculated.

Results

Drag coefficients

The interaction occurs against a background of Brownian movement and, in differentiating between the two, the observations on the drag coefficient are pivotal. If the measured value of plasma viscosity is inserted into Stokes' Eq. 5, the drag coefficient for spheres of the same diameter as cells is 1.28×10^{-7} N·s·m^{-1}. Canham et al. (1971) observed the free fall of erythrocytes in Ringer's solution. Allowing for the different viscosity of plasma, we calculate from their published data $\mu = 1.49 \times 10^{-7}$ N·s·m^{-1}. The cells were tumbling as they fell so Perrin's correction to Stokes' formula for oblate spheroids (Tanford 1961) is appropriate, bringing the theoretical value up to $\mu = 1.54 \times 10^{-7}$ N·s·m^{-1}. Deformability of cells could well account for the difference in the last two figures. These two values are substantially less than the experimental values (Table 1). The results for microspheres give an explanation. For the small microspheres the drag coefficient by Stokes' Eq. 5 is 0.57×10^{-7} N·s·m^{-1} and the measured value 0.62×10^{-7} N·s·m^{-1}, 1.1 times higher. On the cinéfilm these particles are seen to be going in and out of focus, rising and falling continually and so spending a negligible fraction of time in contact with the floor of the chamber. The larger spheres have a specific gravity of $\rho = 1.05$ and are suspended in soapy water ($\rho = 1$). They remain in focus and are therefore close to the surface. For 5.9-μm spheres, the Stokes' $\mu = 0.59 \times 10^{-7}$ N·s·m^{-1} ($\eta = 1.06 \times 10^{-3}$ Pa·s) whereas the experimental value is 1.8 times higher at 1.06×10^{-7} N·s·m^{-1}. Similarly for 8.0-μm spheres the factor is 2.1. If we compare the mean μ for cells (Table 1) with the Stokes' μ for oblate spheroids of the same major diameter the factor is 1.78. We infer that the juxtaposition of either cells or large microspheres with the floor of the chamber introduces an additional hydrodynamic drag.

Since [2] and thus [3] are derived from [1] in which the denominator is the Stokes' coefficient, the rate constant in [3] has to be divided by the ratio of the experimentally determined to the theoretical drag coefficient (Table 1). In this way, for each experiment, a

TABLE 1. Drag coefficients

	$\mu \times 10^7$ N·s·m^{-1}	Experimental value/ Stokes' value for oblate spheroids
Normal cells	2.80±0.66[a]	1.82±0.43[a]
Cells fixed in glutaraldehyde	2.96±0.68	1.92±0.44
Cells at pH 6.3	2.78±0.76	1.80±0.50
Cells restored to pH ~7.6	2.57±0.83	1.67±0.54
Ionophore-treated cells	3.39±0.14	2.21±0.09
Metabolically depleted cells	2.18±0.34	1.42±0.23
Cells repleted with adenosine	2.68±0.74	1.74±0.48
Mean value for cells	2.75±0.10[b]	1.78±0.06[b]
Microspheres 3.4 μm	0.62	1.1[d]
Microspheres 5.9 μm	1.06[c]	1.8[d]
Microspheres 8.0 μm	1.67[c]	2.1[d]

[a]Standard deviation.
[b]Standard error of the mean.
[c]The larger microspheres in plasma stick to the floor of the chamber. These experiments were done in soapy water ($\eta = 1.06 \times 10^{-3}$ Pa·s).
[d]The Stokes' drag coefficient for spheres was used in this calculation.

"Brownian motion rate constant" (no interaction; $Z = r$) is calculated. The values are in the neighbourhood of 2.6×10^{-18} m^3·s^{-1}. For convenience we use an "interaction coefficient" Ξ which has a magnitude of unity when representing the "Brownian motion rate constant". Slopes of the graph N_t^{-1} against t are reported in terms of Ξ and from [3]

$$[6] \quad \Xi = Z/r$$

To get a measurement of the accuracy of the determination of the drag coefficient, U ([4]) was measured in seven experiments on the same blood. The standard error of the mean value of $\langle U^2 \rangle$ was 19% which sets a lower limit to the accuracy of the experiments as presently performed. Correlation coefficients for the regression of $\langle U^2 \rangle$ on t averaged 0.94 and for N_t^{-1} on t averaged 0.98.

Normal cells, glutaraldehyde-fixed cells, and microspheres

In Table 2 the present results and those derived by an earlier method (Rowlands et al. 1981) are displayed together. The simple geometrical analysis, which we used previously, predicted that if there was no interaction the mean probability that pairs of cells, initially less than 8 μm apart, would collide in 3 h was 0.3. The theory used in analyzing the present results has ample experimental support (Swift and Friedlander 1964) from the coagulation of hydrosols. This theory predicts an attractive interaction if $\Xi > 1$. The figures in column two, by having the same trend as those in column one, confirm our previous conclusions though the mathematical relationship between the two sets of figures is nonlinear, as might be expected.

The interaction coefficient Ξ for normal cells is three times greater than the theoretical value for neutral particles. The experimental value of $\Xi = 2.9$ taken together with a probability by the earlier method of 0.85 (value for no interaction ~0.3) is our evidence for an attraction at a distance between normal erythrocytes. Cells fixed in glutaraldehyde and the microspheres have significantly lower values for Ξ close to unity. Glutaraldehyde-fixed cells have a negative surface charge which is at least as large as that of normal cells (Vassar et al. 1972) but it will be neutralized by counter ions up to very short distances of approach (~1 nm) so an electrostatic repulsion is unlikely to be detected in the present experiments.

Variation in pH of the suspending medium

At pH 6.3 the membrane potential is close to zero (Funder and Wieth 1966), there should be no Fröhlich-type interaction, and Ξ should be equal to unity, which it is. At pH 6.3 the membrane potential calculated from the chloride ratio was 1.5 mV as against that in normal cells of 16 mV. Restoring pH to 7.6 raises Ξ significantly above unity and the membrane potential to 12 mV.

Ionophore-treated cells

Reducing the membrane potential by the procedure

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TABLE 2. Interaction coefficients Ξ

	Probability[a]	Ξ[b]	No. of experiments	No. of subjects
Normal cells	0.85	2.92±1.12[c]	15	9
Cells fixed in glutaraldehyde	0.46	0.92±0.46	9	5
Cells at pH 6.3	0.40	1.07±0.41	10	5
Cells restored to pH ~7.6	0.71	2.57±0.99	9	3
Ionophore-treated cells	0.48	1.69±0.04	2	2
Metabolically depleted cells	0.44	0.43±0.11	4	2
Cells repleted with adenosine	0.87	3.26±0.68	3	2
Microspheres 3.4 μm	0.24	0.72	1	—
Microspheres 5.9 μm	—	0.93	1	—
Microspheres 8.0 μm	—	1.15	1	—

[a]Probability of pairs of cells colliding in 3 h (Rowlands et al. 1981). Value for no interaction ~0.3.
[b]Interaction coefficient as defined in text. $\Xi > 1$ attraction.
[c]Standard deviation: $p < 0.001$ normal versus glutaraldehyde cells; normal versus pH 6.3 cells; pH 6.3 versus pH 7.6 cells; normal versus metabolically depleted cells; $p < 0.003$ normal versus ionophore-treated cells.

of Flatman and Lew (1980) also reduced Ξ significantly. There were only two experiments so further work is needed to discover if the difference between the two methods of changing membrane potential is real. There are some difficulties common to both the ionophore and the metabolic depletion experiments which will be explained in the Discussion.

Metabolic depletion and repletion

Depleting the cells of ATP caused the largest decrease in Ξ into a range suggestive of a repulsion. Repletion with adenosine reversed the alteration completely. Metabolic depletion reduced the calculated membrane potential from 15.0 to 7.1 mV so the marked change might be the result of a combination of two effects which may be difficult to separate (see Discussion).

Range of the attraction

From [3] and the definition of Ξ, it follows that the range of attraction $Z = \Xi r$ where r is the radius of the cell. For normal cells $\Xi = 3$ and therefore $Z = 12$ μm. The implication is that cells separated beyond this zone execute Brownian movement. On closer approach, pairs of cells are, on the average, committed to make contact and to form a rouleau. Energetic fluctuations of Brownian movement can of course drive some "committed" pairs apart.

Discussion

Substantial experimental support for Fröhlich's theory would impose radical changes on physiological thought. The evidence which Fröhlich recently reviewed (Fröhlich 1980) has made little impact. The frequency of the predicted oscillations poses technically difficult problems and the results, while indicative of vibrational modes, offer no evidence for the broader implication of the theory. The experiments reported in this paper were designed to test three major aspects of the theory, all of which are necessary features (see Introduction).

The membrane potential

When the pH of the suspending medium is reduced to 6.3 and the chloride determinations indicate that the membrane potential is close to zero, the interaction coefficient is that of a random interaction between cells. In normal cells the interaction coefficient is three times greater and if the pH is restored from 6.3 to 7.6, the interaction returns. In addition to having a membrane potential, erythrocytes have a negative surface charge which keeps cells apart when they are suspended in artificial media devoid of macromolecules. This charge varies with pH and the isoelectric point is in the neighbourhood of pH 2 (Ponder 1948; Bateman and Zellner 1956; Heard and Seaman 1960). The effect, if any, of a reduction to pH 6.3 will therefore be to reduce the surface charge and so act in the opposite direction to the change we have observed. For reasons already given (see Results) an electrostatic repulsion would have a very short range. There was no change in plasma viscosity with change in pH.

Metabolic depletion

Whereas, in the pH experiments, cells which touch proceed to rouleau formation and so make the analysis unequivocal, crenation and sphering of metabolically depleted cells introduce a technical difficulty. Such

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cells adhere but they do not form rouleaux and it is a matter of judgement to decide, at the low magnification and resolution perforce used, the instant of contact. When the cellular concentration of ATP falls to about half the normal value, crenation occurs quite abruptly (Nakao 1974) but it is difficult to predict just how long the incubation should last to avoid this transition. In one of our experiments, however, less than 10% of the cells were crenated and the rest were forming rouleaux. The interaction coefficient was close to that in the other three experiments. This obviates the objection that it might be crenation *per se* which led to the low value of Ξ in metabolic depletion.

Nevertheless, the technique may have significant effects in addition to that of cutting off the supply of metabolic energy and so account for the value of $\Xi < 1$ which implies a repulsion, small enough to be overcome by Brownian motion. We know that the membrane potential is halved to 7 mV. There are also ion shifts (Clemens et al. 1980) but the zeta potential and thus the surface charge is unaffected (Weed et al. 1969). Once again there was no change in plasma viscosity.

Metabolic depletion modifies membrane structure in other ways than crenation (Lutz 1978). Indeed, Glansdorff and Prigogine (1971) assert that a supply of metabolic energy is a fundamental thermodynamic necessity, not only for the function of a cellular membrane but also for its stability and for the assembly of its structure.

Glutaraldehyde-fixed cells

Fröhlich's last major requirement is for molecular organization. A great deal is known about the structure of the cellular membrane (Marchesi et al. 1976), the functions of receptors (Edelman 1976), the assembly of proteins (Wickner 1980), and the transport of various substances (Lassen et al. 1980). But of a membrane regarded by Fröhlich as an intricate machine, needing a power supply and undoubtedly subject to control of its function as an entity, we know as yet nothing that enables us to interfere other than destructively. Almost certainly glutaraldehyde, which cross-links proteins and aminophospholipids (Carvalho et al. 1980), wreaks havoc. Some enzyme and receptor activity remains and our method preserves the shape, volume, and area of the cells, but undoubtedly the supply of energy and the membrane potential are lost. So at present we regard this type of experiment as a control along with experiments using microspheres.

Ionophore-treated cells

The treatment of erythrocytes with ionophore A23187 in the manner described caused morphological changes, largely sphering, in a minority of the cells. This led to difficulties similar to those described for crenated cells after metabolic depletion. Nevertheless, reduction in the membrane potential by this method reduced the interaction substantially and so confirms the necessity of a membrane potential to the interaction.

General discussion

All the described experiments were done in plasma and there is no detectable attraction between cells suspended in artificial media devoid of macromolecules. Preliminary experiments using PBS containing polyvinylpyrrolidone or Dextran 70, both rouleau-inducing agents (Sewchand and Canham 1979; Sewchand and Bruckschwaiger 1980), suggest that these macromolecules can transmit the interaction. Experiments are in progress to dissect plasma to discover which component is the transmitting medium, but we are as yet unable to suggest what is being transmitted, be they phonons, some other quasiparticle for which there are many candidates, or quantized polar waves. Exchange of phonons between superconducting electrons overcomes electrical repulsion with an attractive force (Fröhlich 1950 cited by Rose-Innes and Rhoderick 1978) so perhaps an analogous process is more than abolishing the Coulomb repulsion between erythrocytes.

The Fröhlich theory predicts that polarized cell membranes may be triggered into coherent, cooperative oscillations if supplied with energy above a minimum rate. The laser is the best known example of such a process but the phonon-maser (Chiao et al. 1964) is the closest comparison we know. The work presented in this paper demonstrates a hitherto unknown interaction of human erythrocytes and we have tested it against Fröhlich's theory. The theory demands a strong transmembrane electric field. When it is abolished experimentally, the interaction disappears, only to reappear when the field is restored. Cutting off ATP also abolishes the interaction but we are in doubt about the integrity of the subtleties of membrane structure as a result, and certainly the electric field is halved. The theory deals with organization, electric field, and energy supply as separate entities but the three are likely to be functionally inseparable. It is an obvious suggestion that the electric field not only polarizes membrane molecules but also is the ultimate supply of energy to the vibrations. The mechanism is there in the work of Mitchell (1976). He uses the term "proticity" to refer to the flow of protons, as electricity refers to the flow of electrons, and describes a number of proticity generators in biological cells. "A generator of proticity plugged through the membrane at any point can act as a source of protonic power than can be drawn upon by a suitable consumer unit plugged through the membrane at any other point, even when the producer and consumer units are quite far apart on a molecular scale . . .

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the use of proticity for power transmission in biology is unique in its ubiquitous occurrence and wide range of application." It may therefore be impossible to dissociate, unequivocally, membrane potential difference from the supply of energy.

This interaction of human erythrocytes has features in common with Fröhlich's theoretical predictions. One prediction which is not tested in the present work is that of specificity. The interaction should be attractive, repulsive, or null depending on the relative frequencies or phases of the vibrations of the neighbouring membranes. Evidence of specificity can be deduced from the experiments of Sewchand and Canham (1976) on rouleau formation in mixtures of erythrocytes from different mammalian species. In such interspecies rouleaux, the cells can be recognized by differences in size and they are seen not to be distributed at random within a rouleau. Overbeek (1952) reviewed theoretical and experimental work on the aggregation of polydispersed colloids and showed that the *least* likely collision in Brownian movement is between particles of the *same* size. Yet, in the interspecies experiments, cells of the same size and so of the same species show a preference to adhere to each other. In 10 pairings of species, the only exception was the rat–mouse combination, though the difference in cell size is larger than in some pairs which did show a species preference. As they belong to the same biological family, perhaps all murine erythrocytes are indistinguishable in respect to a Fröhlich interaction. Whether the interspecies interaction is a repulsion, a null interaction, or a smaller attraction than that between cells of the same species cannot be deduced from Sewchand and Canham's (1976) results, but further experiments using methods similar to those described in this paper should answer that question.

It is interesting to note that the history of this line of thought goes back at least 30 years. Fritz London (1950) asking if the study of liquid helium II was of any practical use, wrote "... certain actions between macromolecules in biochemistry could not be understood unless they could be conceived as conditioned by some quantum mechanism involving the system as a whole ...". More recently Glansdorff and Prigogine (1971) wrote "... [There] exist new dynamic states of matter induced by a flow of free energy far from equilibrium ... governed by a new physical chemistry on a supermolecular level, while all the laws referring to the molecular level remain essentially unchanged ...".

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A Review of the Applications of Solid State Physics Concepts to Biological Systems*†

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ABSTRACT. The evidence for solid state physical processes in diverse biological systems is reviewed. Semiconduction of electrons across the enzyme particles as the rate-limiting process in cytochrome oxidase is evidenced by the peculiar kinetic patterns of this enzyme and by microwave Hall effect measurements. PN junction conduction of electrons is suggested by kinetics of photobiological free radicals in eye and photosynthesis. Superconduction at physiological temperatures may be involved in growth and nerve. Phonons and polarons seem likely to be involved in mitochondrial phosphorylation. Piezoelectricity and pyroelectricity may be involved in growth and nerve. Infrared electromagnetic waves may transmit energy in lipid bilayers of nerve and mitochondria. Complexed sodium and potassium ions in structured cell water may be analogous to valence band electrons in a semiconductor, and the free cations may be considered analogous to conduction band electrons. Ionic processes in cell water therefore resemble electronic conduction processes in solid semiconductors, which leads to kinetic predictions in agreement with experiment. The future of solid state biology depends on the development of new experimental methods able to measure solid state physical properties in biological materials which are non-crystalline, impure, particulate, and wet.

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*Dedicated to Felix Gutmann, who stimulated me to write this, and whose book on organic solid state physics provides a foundation for biological solid state physics.

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I. Introduction

The physics and chemistry of the *liquid* state has been widely applied to biological systems. However in addition to liquids, cells contain *solids* (particles and membranes), so that it is reasonable to expect that events of biological importance might occur in the solid parts of cells, and that these events should be governed by the physics of solids. Only recently, and only in a few laboratories, has serious attention been given to this possibility.

Szent-Györgyi[1] in 1941 suggested a role for semiconduction in biology, but the high activation energies of semiconduction in many biological solids measured during subsequent years made this suggestion seem unlikely. However, a particular protein (cytochrome oxidase), which gives kinetic evidence of semiconduction, was measured to have a much lower semiconduction activation energy, approximately equal to the activation energy of the enzymatic reaction catalyzed by this protein, which supports the concept of a semiconductive mechanism. Evidence for other sorts of solid state physical processes in cells (P-N junction conduction, superconduction, piezoelectricity, infrared electromagnetic wave transmission) has also been developed in recent years.

The only previous comprehensive reviews of solid state biology were those of Rosenberg and Postow[2] in 1969 and Cope[3] in 1970. Much has been done since then. The present review will update the 1970 review of Cope.[3] We shall describe in considerable detail work appearing since 1970, but earlier work will be described less completely, and the previous review[3] should be consulted if more details are needed regarding earlier studies. For general background in theory and experiment of organic semiconduction, the books by Gutman and Lyons[4] and Boguslavskii and Vannekov[5] may be consulted.

Solid state biology has been approached in two different ways from two opposite ends of the problem. The two approaches are described in partial separation in the next two sections of this review.

First, purified organic solids, extracted from living animals or plants, have been tested for various solid state physical properties such as semiconduction, superconduction and piezoelectricity. These and other solid state physical phenomena have indeed been observed in purified biological solids. One does not immediately know, however, whether the solid state physical properties of isolated, purified biological solids necessarily play any role in the function of living systems. They might not.

Second, solid state biology has been approached from the other end. Some observed properties of *organized living systems* are not easy to explain in terms of chemical processes in aqueous solutions, but may be derived readily from hypotheses of solid state physical processes in cells. One may then use the observed characteristics of organized living systems to deduce the quantitative properties of solid state physical phenomena which may cause them. One may then compare these deductions with the observed solid state physical properties of purified components of these complex organized living systems.

II. Solid State Physical Phenomena in Components of Living Cells

A. Semiconduction. In 1941, Szent-Györgyi[1] hypothesized that semiconduction of electrons in solid proteins or DNA might play a role in biological function. This idea led many investigators to measure the temperature dependence of electron conductivity for dried, compressed pellets of many components and extracts of biological systems, such as proteins or cell fragments. In general, the data did indeed conform to the semiconduction equation

$$\sigma = \sigma_0 \exp(-E_a/kT) \tag{1}$$

where σ = measured conductivity, E_a is activation energy of semiconduction, k is Boltzmann constant, T is absolute temperature and σ_0 is a constant. However, the value of E_a was usually in the region of 1.0 ev, so that conductivity at physiological temperature was very low, perhaps too low to

serve any useful biological function, and the 1.0 ev was much higher than the observed activation energies of most biological processes.[6] For these reasons, by the middle 1950's, most investigators concluded that semiconduction probably played no role in biology.

Several important points had however been overlooked.

First, the possible influence of water was overlooked. Measurements of semiconduction had been made only in *dried* systems. Living systems function only when wet (80 percent water). Water as an impurity at the surface of tiny biological particles or membranes might change electron conductivity by many orders of magnitude, as impurities adsorbed on thin film inorganic semiconductors are often observed to do. This effect of water adsorbed on a thin film semiconductor is used for humidity measurement in a commercially available humidity meter. The experiments of Rosenberg[7,8] suggest that the presence of water may indeed cause significant changes in conductivity and activation energy of semiconduction of proteins. A pulsed electron beam method with wet protein films in Russia seems to show the same result.[5,9]

Second, it was overlooked that a few proteins might exist which had unusual structures, specially adapted for high electron conductivities and low semiconduction activation energies. Cope[10] showed that the enzyme cytochrome oxidase had kinetic properties which suggested that it was rate-limited by solid state conduction in this particular protein (which has a specific and complicated structure quite different from most proteins). Then Straub[11,12] measured semiconduction activation energy in this particular protein, cytochrome oxidase, and found it to be approximately 0.3 ev (3 times lower than with most proteins, the high values of which were confirmed by Straub). It therefore seems that although most proteins do indeed have very high values of semiconduction activation energy which make them poorly suited to perform biological functions by semiconduction, there is at least one specialized protein with a low activation energy well suited to the biochemical role which it seems to play.

Third, it was overlooked that electron conduction *within* individual solid particles might be quite different from conduction across interfaces from particle to particle, which may be what one measures when one puts electrodes across a compressed mass of thousands of microscopic particles. To measure electron conductivity *within* individual tiny biological particles, a microwave Hall effect technique was developed by Trukhan[13,14] in Russia, and applied to various biological particles by various investigators.[14-22] The results seem to show that electron conductivities within individual biological particles may be much higher than previously deduced from conductivities of compressed masses of particles, and that cytochrome oxidase activity is positively correlated with microwave Hall effect mobility of electrons.[19,20]

It therefore seems likely that semiconduction within solid biological particles may indeed play a role in biological function in at least a few important biological systems, especially in the enzyme cytochrome oxidase. For more details and references to the earlier work, the previous review[3] may be consulted.*

B. PN Junction or Activation Energy Barrier Conduction. Conduction across an activation energy barrier, such as at a PN solid-state junction or liquid-solid junction of an electrode in an aqueous solution commonly shows a exponential or logarithmic voltage-current equation.

In exponential form

$$I = I_0 \exp(V/kT) \tag{2}$$

is known as the diode equation. In logarithmic form

$$V = a + b \log(I) \tag{3}$$

is known as the Tafel equation of electrodes, where V is voltage, I is current, k is the Boltzmann constant, T is absolute temperature and a and b are constants.

Liquid-solid, and solid-solid interfaces are common in biological systems. Therefore, this type of conduction might be expected in biological systems. Its existence has been deduced from behavior of organized systems as described in Section III of this review.

C. Phonons and Polarons. It has been proposed by Straub[11,23,24] that phonons (packets of lattice vibrational energy) may be involved in transmission of energy from oxidative reactions to phosphorylative reactions in solid protein systems such as membranes of mitochondria. Destruction of the phonons in the solid protein by thermalization may be prevented by the hydrophobic bonding of a layer of lipid separating the protein from the surrounding water.[24] The mitochondrial processes of respiratory control, reversed electron flow, uncoupling, and utilization of two electron transfer to accomplish one bond formation seem to be compatable with the phonon hypothesis.[11,23-26]

Phonon coupling between a pair of electrons to form a superconductive Cooper pair may also occur in biological systems at physiological temperatures. Biological superconduction will be discussed in the next section of this review.

A problem with the concept of phonons in mitochondria is probable lack of mobility, because these phonons would need to have an energy of approximately 0.39 ev (9.0 kcal/mole) which is the energy required for phosphorylation of ADP (the chief phosphorylation reaction carried out by mitochondria). Phonons of this energy are *optical* phonons (wavelength

*Good experimental evidence for electronic conduction in lipid membranes[167-169] and across the shells of crabs[170-172] is also available.

3.24 microns in the infrared band). Optical phonons are generally immobile, so that transport of energy by phonons from sites of oxidation to sites of phosphorylation might be expected to be slow.[25] However, the structure and geometry of the lipid bilayer of mitochondria seems well adapted to carry infrared electromagnetic waves (photons), which could serve as a rapid and efficient method to carry the energy of the phonons throughout the mitochondrion.[25] This mechanism is discussed in Section IIIC of this review.

That polarons rather than phonons may be the vibrational packets in mitochondria is suggested by the reversal at low temperatures of the enzyme activity vs (1/T) curve of cytochrome oxidase measured by DeVault and Chance[27] as analyzed by Cope and Straub.[12] Such a reversal is predicted from the concept that the charge carriers in solid cytochrome oxidase are polarons, according to the theory of Holstein.[28,29] A polaron is an electron or hole tied to a phonon. The probability of polaron conduction in organic semiconductors has been discussed by Siebrand[30] and by Gutman and Lyons.[4]

It has been suggested by Kemeny and Rosenberg[159] that polarons may be the charge carriers in proteins and other organic semiconductors, and may be responsible for the observed compensation law in these substances.

Volkenstein[160] has proposed a state in a protein which he calls a conformon, which he claims should be different from a polaron because of the non-regularity of the atomic lattice, resulting in lack of mobility and short lifetime. This has been studied further by others.[165-166]

It has been proposed on theoretical grounds that melanin (the black pigment found in skin, eye, and in parts of the brain and ear) may be a particularly efficient converter of photons to phonons and of the reverse process.[161-162] Evidence suggesting the phonon to photon conversion by melanin was the observation that ultrasound (10^4 Hz) was more damaging to cells containing melanin than to similar cells without melanin, correlated with previous observations that photons of high intensity light are more damaging to cells if melanin is present.[163-164] At lower intensities, however, melanin protects cells against damage by light.[162]

D. **Superconduction.** Superconduction is the conduction of electrons without the generation of heat, and hence with an electrical resistance equal to zero. Until recently, superconduction had been observed only in certain metals and only at a temperature colder than approximately 20° Kelvin. Theorists have long predicted that superconduction might occur at room temperature in organic solids. The development of new techniques of very high sensitivity has now led to the experimental detection of superconductive transitions in organic solids (various cholates) at temperatures as high as

277°K. Indirect evidence suggests that superconduction plays a controlling role in various functions of living systems at physiological temperatures.

A comprehensive mathematical theory (the BCS theory) of superconduction has been developed which explains and predicts many (but not all) of the experimentally observed electrical and magnetic properties of superconductors[31] and properties of electron tunneling across junctions between superconductors.[32,33] An extension of the BCS theory by Little[34,35] predicted that superconduction should occur in certain types of organic polymers *at room temperatures and above.* There is much controversy among theoretical physicists regarding the validity of this prediction.[36] All attempts to synthesize Little's superconductive polymers have so far failed, which may be due to inadequacies of the attempts.

Pauling[37] and London[38] claim that electron currents around benzene rings in aromatic compounds are analogous to superconductive currents in metals. This claim is disputed by Musher.[39,40] However, much quantitive data on magnitudes of chemical shifts of observed nuclear magnetic resonance spectra of compounds containing benzene rings is explainable in terms of superconductive ring currents.[41-45]

High temperature superconduction has also been predicted in a thin metal film adjacent to a dielectric layer, where the interaction between the two electrons of the pair occurs through the dielectric instead of within the conductor.[36,46-48] All attempts to manufacture such superconductive bilayers or sandwiches have failed, perhaps because the correct method was not tried.

Experimental evidence of superconduction in organic solids at high temperatures has been obtained very recently by Halpern and Wolf.[49-51] These findings required the use of new instrumental methods for the measurement of extremely small changes in magnetic susceptibility.[49-51,177] Transitions (abrupt changes) in susceptibility as a function of temperature and of applied magnetic field were observed in various cholates (4 rings with various side chains).[49-51] The rise of susceptibility is predicated on the concept that transition from normal to superconducting regions would initiate the Meissner effect which would cause internal magnetic field changes to be less than those imposed externally. X-ray diffraction patterns were observed to be unchanged at temperatures across the transition temperature, which showed that no change occured in the in the structure of the atomic lattice.[52] Therefore, the transition must be due to *electronic* rearrangements. A change between superconducting and normal conducting state was the most obvious possibility, and was consistent with the experimental data.[49-51,177] Transition temperatures observed for the cholates were as high as 277°K. Other types of organic compounds studied did not show evidence of superconductivity.[50,51] Cholates showed superconductive phenomena only when crystallized in

certain ways.[50-51,177] The heights of the observed transitions were small, which indicated that only a small fraction of cholate molecules were in the superconducting state at any given time.

Extrapolation[52] of the experimental data on cholates[49-51,177] suggests that cholesterol (a major constituent of nerve and of other tissues) may superconduct at physiological temperatures and above, so that cholesterol might be a site of superconduction in living systems.

Another experimental observation of probable high temperature organic superconduction is the recent data of Antonowicz.[54] He observed fluctuations in current through thin carbon films at room temperature as a function of magnetic field at very low fields (0.5 to 2.0 Gauss), with a pattern like that seen with the Josephson effect for two-electron tunneling at superconducting junctions. No other known physical phenomenon seems adequate to explain this experimental finding.

There is evidence suggesting that high electric fields may promote superconduction in organic fields at high temperatures, and that this might play a role in nerve conduction.[53] The room temperature Josephson effect of Antonowicz[54] in carbon films required a 2 volt electric potential across a carbon film that was presumably only a few molecules thick, so that the electric field would be very high. Magnetic fields of 2000-3000 Gauss cause changes in photocurrent in organic solids (anthracene and tetracene) at room temperature but only if the voltage is sufficiently high to be above the ohmic conduction region and into the square law conduction region.[55] This probably implies charge injection into the organic crystal.[53,55] The magnetic effects in both the above sets of experiments, as well as the square law conduction, may be understood from the hypothesis that high voltage electron injection into organic solids may produce superconductive Cooper pair charge carriers.[53] It is reasonable to consider that this superconductive process might occur in the thin lipid layers of nerve in response to the high electric fields which occur there,[53] especially since extrapolation of the data on cholates suggests that cholesterol (which occurs in large quantities in nerve) should superconduct at physiological temperatures.[52]

The evidence for high temperature superconduction in organic solids raises the possibility that superconduction in solid portions of cells might play a role in living systems. Three lines of indirect evidence suggest that this is so.

First, that bulk superconduction may play a biological role is suggested by the abrupt changes in rates of metabolism of growing tissues induced by magnetic fields of 50 to 150 Gauss,[56-58] although lower fields produced no effect, the higher fields produced no additional effect. A superconductive transition is the most obvious possible cause of a discontinuous effect produced by a magnetic field.[56]

Second, a role in biology for *two*-electron superconductive tunneling (Josephson effect) is suggested by the demonstration tnat numorous living systems (birds, insects, snails, and possibly man) can detect very low magnetic fields (0.5 to 1.0 Gauss).[59,60] No physical phenomenon other than the Josephson effect seems to have sufficiently high magnetic sensitivity to provide the physical basis for this observed biological magneto-detection.[59,60] In addition, the rates of some biological processes are observed to show *negative* temperature coefficients, which cannot be explained by ordinary chemical and physical rate processes, which all show *positive* temperature coefficients. However, Josephson superconductive tunneling has a *negative* temperature coefficient and hence could provide a physical basis for these observed biological phenomena.[56]

Third, a controlling role for *single*-electron superconductive tunneling in living systems is suggested by the temperature dependences of the rates of certain nerve and growth processes.[56] For the rates of most biological processes, like most chemical processes, the temperature dependence is given by linear plots of log of rate vs 1/T (a linear Arrhenius plot). However, a few biological processes show curved Arrhenius plots, and some of these are described by the equation

$$E_a^2 = aT + b \tag{4}$$

where E_a is the activation energy, measured from the slope of the Arrhenius plot, T is temperature in degrees Kelvin and a and b are constants. This type of temperature dependence is predicted for and observed in *single*-electron superconductive tunneling.[56] Therefore, one may surmise that these biological processes (all involving either nerve or growth) are rate-limited by single-electron superconductive tunneling.[56]

piezoelectricity in microtubules

E. **Piezoelectricity and Pyroelectricity.** A piezoelectric solid is usually defined as one which generates a voltage proportional to an applied mechanical force, due to mechanical distortion or displacement of electric dipoles in the lattice.[61] Piezoelectricity has been observed directly in bone[62,63] and wood[64] by measuring electrical potentials induced by bending or compression.

Biological materials which are available only in powdered form cannot be tested for piezoelectricity by the *direct* method used for bone and wood. However, an *indirect* method capable of measuring weak piezoelectricity in powders is available and has been applied to dry biological powders. This method was developed by Giebe and Sheibe[65] and has been improved by others.[66-70] It uses the principle of the vacuum tube crystal oscillator, which has for many years been used as a source of frequency-stable radio-frequency voltage. In this circuit, a piezoelectric quartz crystal, cut to a size and shape to allow mechanical vibration resonance at the desired radiofrequency, is placed between metal electrodes, which are connected

to a vacuum tube feedback circuit. The entire electro-mechanical system then goes into oscillation, which generates a sinusoidal voltage whose frequency is determined by the mechanical properties of the piezoelectric crystal.

Because electrical oscillation in this type of circuit depends on the piezoelectricity of the crystal, the electrical oscillations can be used to detect piezoelectricity in the crystal. The same may be done if a compacted powder is used instead of a single crystal.

To obtain highest sensitivity of detection of weak piezoelectricity in powdered materials, special superregenerative oscillator circuits have been developed in which the amplitude of oscillation is decreased at frequencies at which piezoelectric resonance occurs in the powder.[65-70] These circuits have been used to measure the piezoelectric spectrum of biological materials as a function of frequency and temperature by Duchesne and Monfils[69,71-73] and others.[70,74-77]

Duchesne and Monfils[69,71-73] observed that the positions of the peaks in the piezoelectric spectrum showed a temperature dependence which could be described by the equation

$$\Delta = \frac{-1}{f}\frac{df}{dT} \tag{4A}$$

where df/dT is the observed change in the frequency (f) of the piezoelectric absorption peak per unit change in temperature (T), f is the approximate frequency of the peak, and the calculated constant (Δ) turns out to have approximately the same value for all peaks at all frequencies for any particular type of powdered sample. Δ is therefore a measurable piezoelectric constant characteristic of each organic or biological powder. Duchesne *et al*[69,71-73] believe that the constant Δ is related to other physical parameters of the solid by equation

$$\Delta = \alpha\gamma \tag{4B}$$

where α is the thermal expansion coefficient of the solid sample, and γ is the Grüneisen constant,[78,79] which is a dimensionless measure of the change in entropy with volume of the crystal.

Measured values of Δ for various biological and non-biological powders are given in Table 1.

We should like to know whether the observed piezoelectric properties of these biological materials play any role in function of living systems. They might or might not.

The characteristics of several biological processes suggest that piezoelectric phenomena do indeed perform biological functions.

First, it is observed that the architecture of bone continuously adapts itself to the mechanical loads placed upon it.[80] Growth of bone is

TABLE 1

Piezoelectric Temperature Coefficient of Various Substances

Sample	Δ ($\times 10^{-6}$) $(°C)^{-1}$	Reference
Quartz	6	77
Zinc sulfide	46	77
Potassium sodium tartrate	400	77
1-Chloramphenicol	100	71
DNA (thymus)	50	73
DNA (sperm)	56	73
DNA (yeast)	18	77
RNA (yeast)	6	77
Myosin	85	73
Actomyosin	100	73
Collagen	47	73

The constant Δ is characteristic of the material and is calculated by equation 4A from measurements of the frequency shifts of piezoelectric resonance absorption peaks as a function of temperature, which are measured as described in the text.

experimentally observed to be controllable by applied electric potentials.[81,83-88] Piezoelectric potentials of bone, generated in response to mechanical stress on the bone, may then serve to control the growth of bone to adapt it to the stresses placed upon it.[82]

Second, growth of tissue is stimulated by implantation of piezoelectric dusts.[89-91]

Third, the temperature dependence of the resting potential of nerve is observed to conform to the temperature dependence of piezoelectric resonance.[92] Petrov[93] has theorized that a membrane potential may be dependent on piezoelectric (flexoelectric) properties of liquid crystalline lipids. Piezoelectric properties of liquid crystals have been studied theoretically[94-97] and experimentally.[98,99] Therefore, piezoelectricity may be involved in nerve function.

Fourth, some of the observed properties of muscle stretch receptors may be derived from the hypothesis that piezoelectric elastic polymers are involved in detection of muscle stretch.[100]

Piezoelectric materials are usually also pyroelectric, which means that an electric potential is generated by the application of heat or cold. Pyroelectricity has been measured directly in tendon and bone,[149-155] meninges,[156] spinal cord,[149] and nerves.[156] Also, it is observed that the electrical responses of temperature sense organs of living systems show behavior which is typical of pyroelectric materials.[100] Application of a square pulse of increased temperature causes transient nerve responses of opposite sign at the two ends of the temperature pulse[100] (Fig. 1). This suggests that pyroelectricity does indeed play a role in biological temperature detection.

F. Amorphous Semiconductor Switching. McGinness *et al*[157] have shown that melanins (from melanoma tumors or synthetic melanins) switch from low current to high current when voltage is increased past a certain threshold voltage, which resembles the switching behavior of inorganic amorphous semiconductors (Ovshinsky effect). The increase from low to high current is 10 to 1000 times.

The mechanism of this effect in inorganic materials is in dispute, so that extrapolations of mechanism from inorganics to melanin have not been made. Some information regarding the mechanism of switching in melanin has been deduced from the experiments of McGinness *et al.*[157]

First, the current in the on-state is mostly electronic rather than ionic, because weight loss of sample which would occur if water were lost by electrolysis is negligible.[157]

Second, even though conduction is electronic, water is necessary for switching, because thoroughly dried melanin does not switch, but switching ability is recovered when the melanin is slightly hydrated.[157] Perhaps, water alters electronic conductivity of the melanin by lowering the activation energy for conduction of electrons between sites, analogous to the mechanism proposed by Rosenberg[8] for proteins. This requirement for water for switching to the high conductivity state may be related to the requirement for water for reversibility of electron transfer measured by generation and decay of photogenerated free radicals in eye melanin.[158]

The voltage threshold for switching for hydrated melanin is approximately 500 volts/cm which is much lower than the 5×10^5 volts/cm observed for cytochrome C, which is comparable to the values of voltage threshold observed for inorganics.[157] Various other organic materials of biological origin were tested and found not to show switching behavior.[157]

It would be interesting to know whether switching behavior in melanin plays any role in biological switching processes in regions where melanin occurs, such as skin, retina, brain, or ear.

III. Solid State Processes in Integrated Biological Systems

A. Growth. It is experimentally observable that a living bone grows continuously in a way that adapts it to withstand the stresses to which it is subjected.[80] The boney struts on the inside of a long bone (e.g. the leg bone) are located along the lines of stress. During life, a bone may change its structure to increase or decrease its strength to adapt to the loads habitually placed upon it. For example, a longshoreman's bones

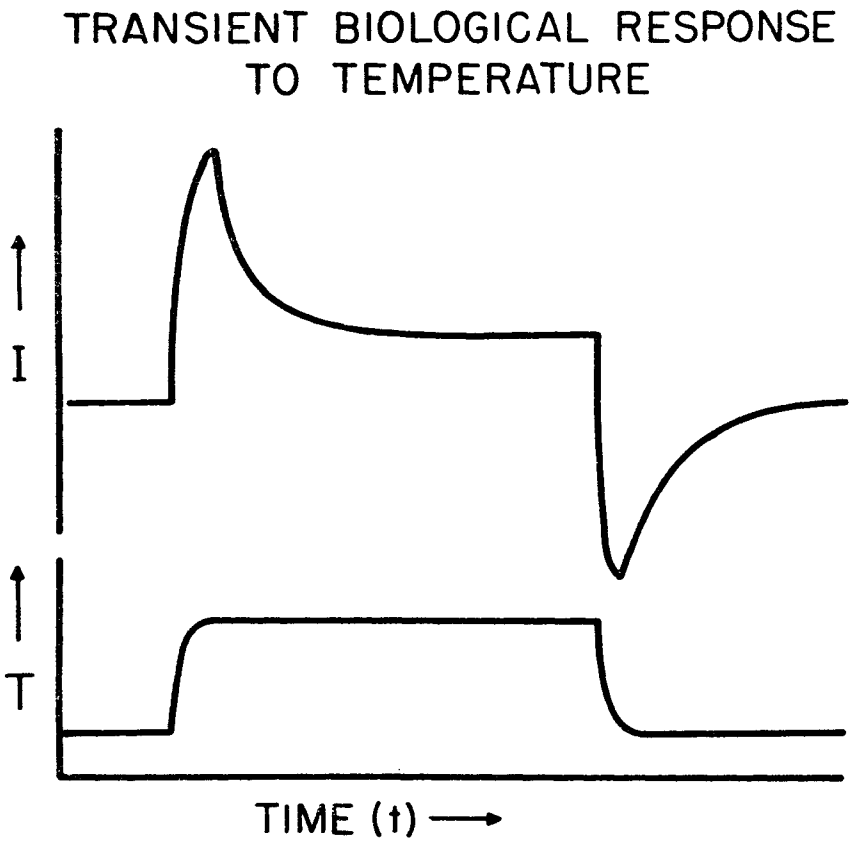

Fig. 1. Transient Biological Response to Temperature Pulse. Lower curve shows a pulse of temperature (T) applied to biological system during the passage of time (t). Upper curve sketches the form of a typical pattern of biological response (I), with transients of opposite sign at the start and end of the temperature pulse, and a steady state response after the initial transient has faded away.[100]

will become much stronger than average, while a person who remains in bed for some months will develop lighter bones. There must therefore exist some sort of system which controls the growth of bones in response to applied stress. It has been observed that application of a D.C. electric potential to the end of a broken bone causes growth.[81,83-88] It has also been observed that bones are piezoelectric, i.e. bending a bone generates an electrical potential.[62,63] It was therefore suggested by Becker, Bassett and Bachman[82] that these two phenomena working together might serve as a closed-loop feedback system by which the body controls the growth of bones in response to the applied stresses.

An additional indication of a relation of piezoelectricity to growth is the observation that implantation of peisoelectric dusts stimulates tissue growth.[89-91]

That superconductivity participates in cell growth is suggested by two lines of evidence. First, various growth processes show a temperature dependence described by a linear relationship between temperature and the square of the activation energy.[56] Such behavior is predicted and observed for single electron superconductive tunneling.[56] Second, the incidence of cancer throughout the world is a periodic function of earth's magnetic field for very low fields.[101] Such behavior is characteristic of two electron Josephson superconductive tunneling.[101]

B. Photobiological Systems (Eye and Photosynthesis). When light falls on a biological system, it may kick an electron up out of its resting position into some other state. One may know that this has happened by the observation by electron spin resonance spectroscopy of generation of free radicals (unpaired electrons) when the biological system is illuminated. This can be observed with illuminated particles (melanin granules) from the eye[102] and with photosynthetic particles (chromatophores) from bacteria.[103] Because decay of the free radical concentration after the end of illumination is slow (several seconds), there must be a barrier of some kind between the illuminated and resting positions of electrons.

Because these biological systems are solid particles floating in watery solution, it is reasonable to suppose that the laws of solid state physics apply within the particle, and that the particle surface acts as a solid-liquid interface, to which electrode physics should apply.

A simple hypothesis for such a system as to why return of illuminated electrons to equilibrium might be slow is that light has kicked the electrons across an activation energy barrier at the particle surface. Activation energy barriers at liquid-solid interfaces are commonly observed[105-107] in electrode physics, and give rise to a voltage-current relation across the interface of the form

$$V = b - \frac{RT}{Fa} \log_e(-i) \tag{5}$$

where V is voltage across the interface, i is current across the interface, T is absolute temperature, R is the gas constant, and a and b are constants. In electrode physics, this relationship is known as the Tafel equation,[104,105] but the same relation (written in exponential form) is seen in solid state physics at solid-solid interfaces, at which it is known as the diode equation. The use of equation 5 for electron conduction across interfaces is well justified on the basis of experimental observation. It also has a theoretical

basis, because it may be derived from the concept of electron conduction across an activation energy barrier at an interface by charge carriers having a Boltzmann distribution of energies and constant charge carrier concentration.[3,105]

We might like to test the hypothesis of an interfacial activation energy barrier in a particulate photobiological system by testing for conformity to equation 5. Unfortunately, a *direct* test is impossible because the eye melanin particles and photosynthetic particle are so small that one cannot make electrical contacts to the two sides of the interface in order to measure the voltage-current curve. An indirect test however, is possible because equation 5 leads to a prediction regarding the shape of the decay of concentration of unpaired electrons vs time after the end of illumination, which can be measured by electron spin resonance.

We may derive the concentration vs time curve as follows:

Let us consider a liquid-solid interface at a particle surface, across which electron current flow is described by equation 5. In the solution outside the particle, let there be an ion (a small molecular weight substance) which can exist in either the oxidized or reduced state, depending on whether it has gained or lost an electron from the nearby solid surface. This reaction may be written

$$x_{ox} + \epsilon^- \rightleftharpoons x_r$$

where x_{ox} and x_r are the oxidized and reduced forms of substrate molecule X respectively, and ϵ^- represents an electron, which may be rapidly and reversibly exchanged between the molecule X and the solid surface of the biological particle. According to theory and experiment of electrode physics, the particle surface then experiences an electrical potential *(V)* given by the familiar electrode equation

$$V = V_0 + (RT/F) \log_e (x_{ox}/x_r) \tag{6}$$

where x_{ox} and x_r are concentrations of oxidized and reduced forms respectively of substate molecule X, V_0 is a constant, and the other symbols are constants as previously defined. Because the sum of oxidized plus reduced forms of X-substrate is always constant $(= C_x)$, we may write

$$x_{ox} + x_r = C_x \tag{7}$$

Combining equations 6 and 7, we obtain

$$V = V_0 - (RT/F) \log_e \left[x_r/(C_x - x_r) \right] \tag{8}$$

If as assumed, light kicks electrons across the particle surface, it disturbs the equilibrium between the two sides of the surface and sets up an electrical potential difference across the surface. Decay of photogenerated unpaired electrons then consists of conduction of electrons back across the

surface in accord with the conduction equation (equation 5), driven by the potential difference of equation 8 for the potential on the liquid side of the surface minus a constant potential which we shall assume for simplicity on the solid side of the surface (this merely assumes an electron concentration in the solid that is large compared to the number of electrons transferred). *The surface reaction described above is equivalent to an electric circuit consisting of a diode connected across a battery.*

We may then equate the driving potential across the interface (given by equation 8 minus a constant) to the voltage drop across the interface given by equation 5 to yield

$$-\frac{RT}{F}\log\left[x_r/(C_x - x_r)\right] = \frac{RT}{Fa}\log(-i) + B \tag{9}$$

where B is a new constant. Because the conduction of one mole of electrons across the particle surface reduces (or oxidized) one mole of substrate molecules, we may write

$$i = F\,dx_r/dt \tag{10}$$

where t is time. Substituting equation 10 in equation 9, we obtain

$$-\frac{RT}{F}\log\left[x_r/(C_x - x_r)\right] = \frac{RT}{Fa}\log\left[-F\frac{dx_r}{dt}\right] + B \tag{11}$$

which is easily converted to

$$-\frac{dx}{dt} = \frac{1}{Fg}\left[\frac{x}{C_x - x_r}\right]^a \tag{12}$$

where g is a new constant. If a = 1, as is theoretically the case for an ideal electrode surface,[108] and if reaction velocity (v) is defined as $v = dx/dt$, then equation 12 may be written in the form

$$\frac{1}{v} = \frac{Fg}{1}\left[\frac{C_x}{x_r} - 1\right] \tag{13}$$

which may be compared with the classical rate equation of Michaelis-Menten for enzyme kinetics

$$\frac{1}{v} = \frac{1}{V}\left[\frac{K_s}{s} + 1\right] \tag{14}$$

where s is substrate concentration and V and K_s are constants.

The rate equation of classical enzyme kinetics (equation 14) is derived from the hypothesis of reactants in free solution obeying the mass action principle, while equation 13 was derived from solid state and electrode physics concepts. Equation 13 and 14 both predict linearity of a double

reciprocal plot of reaction velocity vs substrate concentration, but differ in that they predict opposite signs for the intercepts on the 1/v − axis.

The relevance of the solid state activation energy barrier equation (equation 13) to biology is supported by the fact that decay of photo-generated free radicals in eye melanin particles (Fig. 2) conforms to equation 13 and so does the decay of photoconductivity (Fig. 3) in invertebrate nerve.

Equation 12 for activation barrier kinetics may be put into another form, which is sometimes more convenient for application to experimental data, and which shows an important correlation with experimental solid state physics. This is true because the function log $[x_r/(C_x - x_r)]$ may be adequately approximated by a linear function of x_r which may be shown in the following way: Let us define a new variable λ equal to that fraction of total substrate concentration which is present in the reduced form. Hence, λ is defined by

$$\lambda = x_r/C_x \tag{15}$$

A graph of (Fig. 4) of the function $log_{10}[\lambda/(1-\lambda)]$ *vs* λ shows that $log_{10}[\lambda/(1-\lambda)]$ can be expressed fairly accurately as a linear function of λ for values of λ between approximately 0.15 and 0.85. By measuring the slope and intercept of the straight line approximation, we may write the approximate equation

$$log_{10}\left[\frac{\lambda}{1-\lambda}\right] = 2\lambda - 1 \tag{16}$$

The right hand side of equation 16 is actually the first two terms of the Taylor series expansion of the left hand side. Substitution of equation 15 in equation 16 and conversion to *log* to the base *e* gives

$$log_e\left[\frac{x_r}{C_x - x_r}\right] = \frac{4.6x_r}{C_x} - 2.3 \tag{17}$$

Equation 17 may be written in the form

$$\left[\frac{x_r}{C_x - x_r}\right] = exp\left[2.3\left(\frac{2x_r}{C_x} - 1\right)\right] \tag{18}$$

Equation 18 may be substituted in equation 12 to give

$$-\frac{dx_r}{dt} = me^{nx_r} \tag{19}$$

where m and n are new constants defined by

$$m = \frac{A'}{Fg}\ exp(2.3a) \tag{20}$$

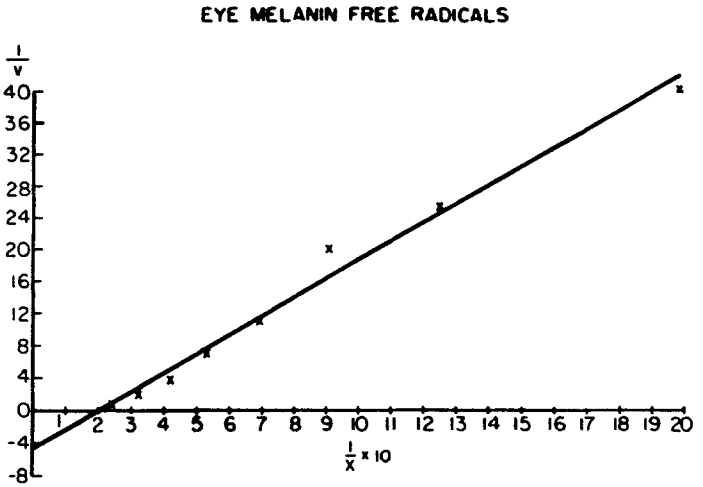

Fig. 2. Double Reciprocal Plot for Decay of Eye Melanin Free Radicals. x = substrate (free radical) concentration. v = reaction velocity = $-dx/dt$. This graph shows a *negative* intercept on the 1/v-axis, which may be contrasted with the *positive* intercept seen in the usual Lineweaver-Burk plot of data from an enzyme reaction conforming to Michaelis-Menten kinetics. Experimental data was obtained from electron spin resonance analysis of eye melanin particles in aqueous suspension at end of illumination with visible light.[109]

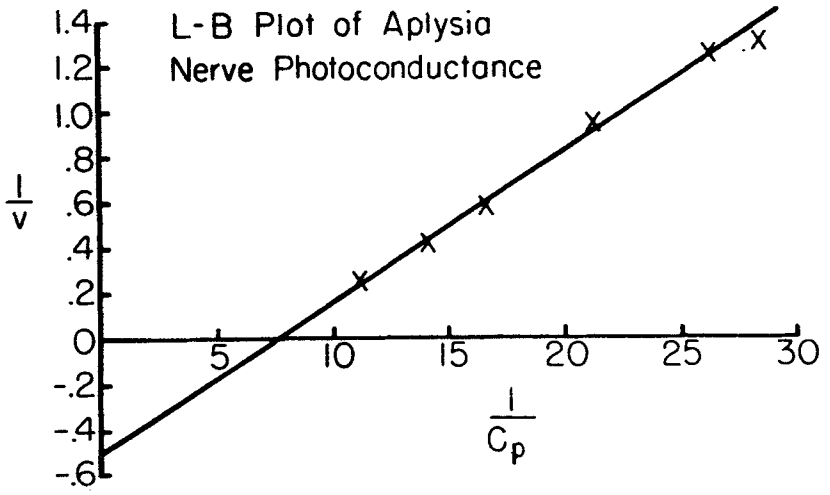

Fig. 3. Double Reciprocal Plot for Decay of Photoconductivity in Nerve. C_p = photoconductance in $(M\ \Omega)^{-1}$ of membrane of *Aplysia* neuron; $v = dC_p/dt$, where t is time. Experimental data were taken from figure 9 of Chalazonitis,[118] in which the data points represent microelectrode measurements of electrical resistance across the membrane of the photo-active neuron of *Aplysia* at various times after the end of exposure to 18 seconds of light with a wavelength of 470-510 $m\mu$. Calculations are taken from Cope.[119]

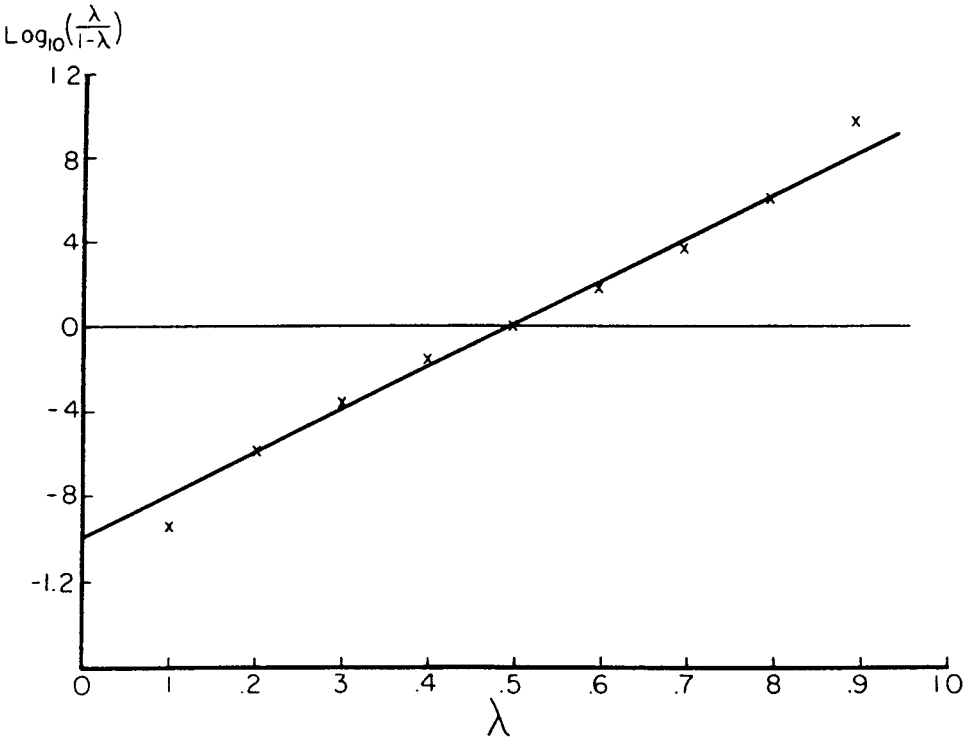

Fig. 4. Plot λ vs. the log function. The accuracy of this linear approximation is studied in the appendix of Reference 3.

and

$$n = \frac{4.6a}{C_x} \tag{21}$$

Equation 19 is called the Roginsky-Zeldovich equation or the Elovich equation.[109] It is named after its discoverers who observed that it described data on oxygen absorption and charge decay at surfaces of inorganic semiconductors. The applicability of the Elovich equation to experimental data of surface physics of semiconductors, such as Ge and Si, has been confirmed repeatedly.[109] The frequency of experimental observation of this equation suggested that it was a manifestation of fundamental physico-chemical principles of surfaces, probably connected with electron transport processes. A variety of derivations of the Elovich equation have been

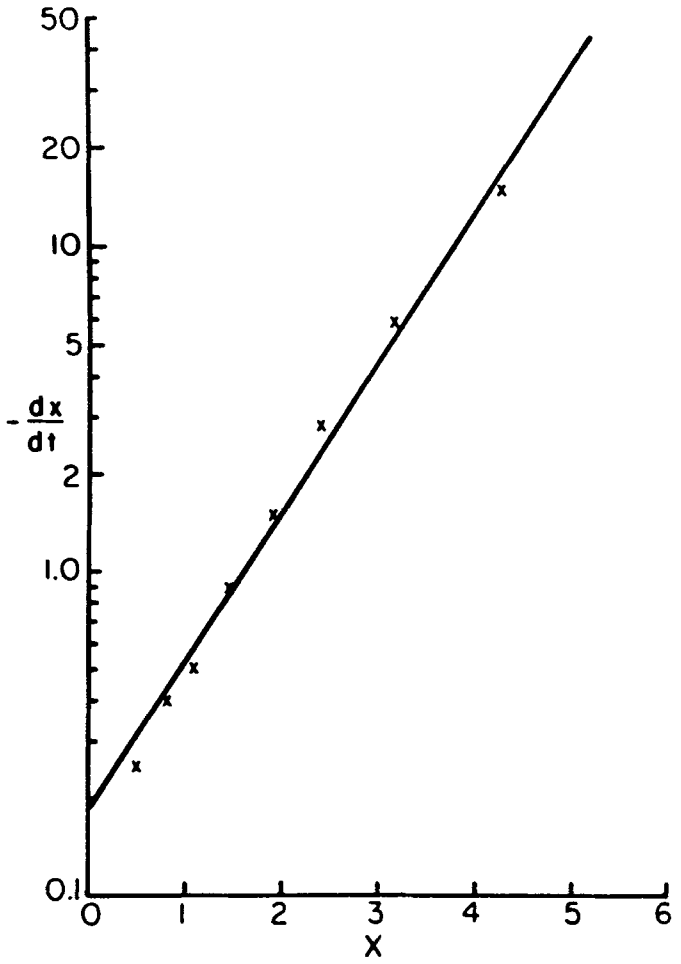

Fig. 5. Test of Eye Melanin Free Radical Decay Data for Conformity to Elovich Equation. Graph of log of reaction rate (−dx/dt) vs free radical concentration (x) for the experimental data of Fig. 2. Linearity of this graph indicates conformity of this data to the Elovich equation (equation 19).

proposed, but all seem to be based on postulates of doubtful validity.[109] The derivation presented here represents a new approach, which is based on postulates that have a good basis in experiment and theory.[109]

After the Elovich equation had been derived by Cope[109] from solid state activation barrier theory, its relevance to biology was confirmed by showing that it described considerable biological data. The Elovich equation describes well the decay of free radicals in eye melanin particles (Fig. 5) and the decay of photoconductivity in nerve (Fig. 6). It also described the decay of free radicals in photosynthetic particles,[110] and the decay of molecules responsible for delayed emission of light from green leaves.[111] All of these biological systems contain particles or membranes

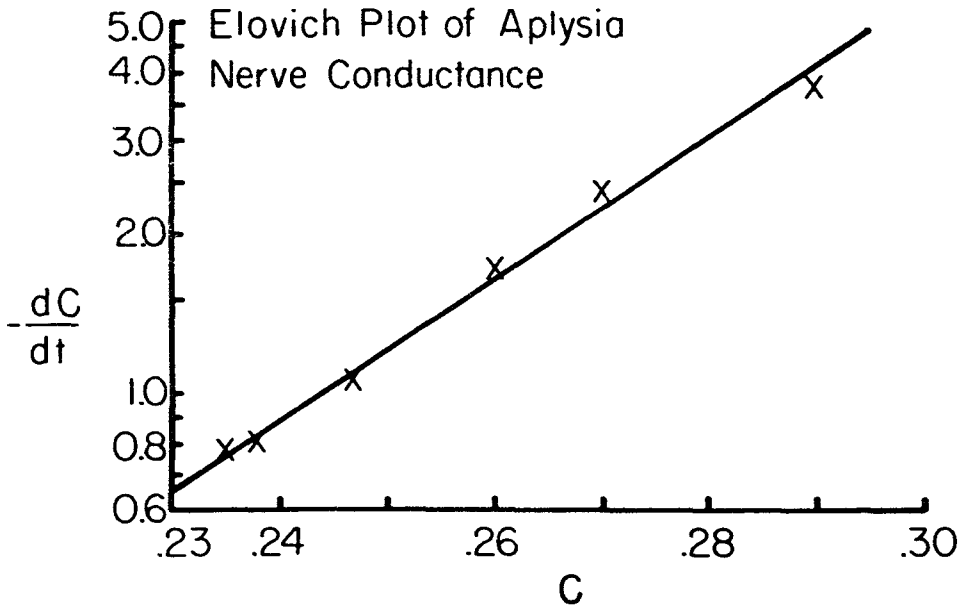

Fig. 6. Test of Nerve Photoconductivitv Decay Data for Conformity to the Elovich Equation. Graph of log of rate of change of nerve membrane conductivity (−dC/dt) vs conductivity (C) for the experimental data of Fig. 3 on *Aplysia* nerve membrane, using the data of Chalazonitis[118] as calculated by Cope.[119] Linearity of this graph indicates conformity of this data to the Elovich equation (equation 19).

as required by the activation overvoltage derivation of the Elovich equation. Support for the concept that solid state electron conduction may be involved in the eye melanin reaction derives from the measurement of relatively high electron mobilities in eye melanin by the new experimental technique of microwave Hall effect.[15]

In addition, the temperature dependence of the Elovich equation has been derived from activation overvoltage theory, and has been found to agree well with experiment, both for inorganic solid systems and for biological systems.[112]

C. Enzymes (Cytochrome Oxidase). Enzymes are the catalysts in cells with cause the chemical reactions necessary for life to procede at rapid rates. Without enzymes these chemical reactions would occur too slowly to support life. Because of their importance to life, the structure of enzymes, and the kinetics and mechanisms of chemical reactions catalyzed by enzymes, have been studied extensively by biochemists and physical chemists for many years.

The classical approach to chemical kinetics is the mass action theory. This is based on the hypothesis that all reacting molecules float free in solution, which implies that the rate of reaction is proportional to the rate of collision between reactants, which is proportional to concentrations of reactants. Most kinetic theories of enzyme action are merely elaborations of this simple idea. The Michaelis-Menten theory of enzyme kinetics is a well known form of the mass action treatment of enzyme kinetics, which has been found to describe much experimental data on many enzymes. For many of these cases, the enzymes and their substrates probably do float free in solution in cell water, so that the basic hypothesis of the mass action theory is in accord with reality.

Some enzymes are bound in or on particles or membranes in the cell. Despite this violation of the mass action hypothesis, biochemists have nevertheless attempted to use mass action theory to analyze such enzymes. One such enzyme is cytochrome oxidase, which in the cell exists as a part of the mitochondrial membrane, which can by appropriate treatment be cracked up into fragments of molecular weight 500,000 or more which retain the ability to catalyze the reduction of O_2 by the reduced form of the protein cytochrome c. The membrane fragment called cytochrome oxidase contains protein, unsaturated lipids, at least two types of haeme-iron groups, and copper. Purifications which remove any of these substances cause loss of catalytic activity of the enzyme.

Despite the fact that cytochrome oxidase is part of a membrane and hence violates the free solution hypothesis of mass action theory, biochemists have nevertheless tried to analyze its kinetics on the basis of mass action theory. The results have been poor. Cytochrome oxidase does *not* conform to the Michaelis-Menten pattern. Instead, it shows first order kinetics, with the peculiar feature that larger initial amounts of reduced cytochrome c substrate decrease the first order rate constant in a hyperbolic pattern. Such a result is difficult to derive from mass action theory. It can be done by assuming a four step reaction, *if the rate constants of the first and last step are assumed to be equal.*[113] The latter is possible, but seems rather unlikely.

Persistence by biochemists in attempts to apply mass action theory to membrane-bound enzymes which violate its postulates is psychologically understandable because of the great success of mass action theory in dealing with soluble enzymes. Reason dictates, however, that when reality conflicts with the postulates of theory and when predictions of theory are difficult to reconcile with experiment, then a different theoretical approach should be sought. This has been done by Cope[10,12,114] in order to deal with the kinetics of redox reactions in biological particles and membranes using realistic postulates. No longer does one impose the restrictive hypothesis of mass action theory that all processes must occur

in free solution. One concedes that the catalytic system contains solid particles and membranes, and that therefore catalysis may involve electron conduction through these solids in accord with the laws of solid state physics, and may involve electron conduction across the liquid-solid interfaces in accord with the behavior of electrodes. This approach leads in a direct manner to kinetic predictions that describe accurately and in considerable detail the kinetic experimental behavior of various particulate and membranous biological processes, including that of cytochrome oxidase.

The purpose of a kinetic theory of enzymes is to predict correctly the substrate concentration vs time curve for the chemical reaction when catalyzed by the enzyme (concentration vs time is the most easily measured characteristic of an enzymatic process). The derivation should be based on physical and chemical principles appropriate to the enzyme structure and conditions.

We shall now derive the substrate concentration vs time curve for a redox reaction catalyzed by an enzyme which is assumed to be a solid particle across which electrons can be conducted in response to a potential difference generated by electrode redox potentials of substrates on the opposite ends of the conductive particle (Fig. 7).* An equivalent electrical circuit would be a resistor connected across a battery. This model seems appropriate for an enzyme such as cytochrome oxidase which when studied in the test tube is a high-molecular weight fragment of a membrane. In life, cytochrome oxidase exists as part of the membrane of the mitochondrion. When in this larger and more complex solid system, some additional solid state processes may influence the cytochrome oxidase reaction, as will be discussed later in this section. The derivation, which will now be presented in simplified form, can be found in complete detail elsewhere.[10,12,114]

Let us postulate an enzymatic particle (or membrane fragment) as shown in Fig. 7. Let there be two sites on the particle at which small molecular weight substrate molecules can exchange electrons with the particle. The two substrates (X and Y) can each exist in an oxidased or reduced form. We assume enzymatic specificity of site for substrate: in other words, the X-site on the particle can exchange electrons only with X-substrate, and the Y-site can exchange electrons only with the Y-site.

Let us now consider the system shown in Fig. 7 from the point of view of solid state theory. The X-site and Y-site may be considered as two electrodes immersed in two solutions with different equilibrium electrode potentials. When the two electrode sites are connected by a wire, or by the internal resistance of the enzyme particle, a current of electrons will flow, which will be limited by the electrical resistance across the particle. This solid state model of an enzyme is electrically equivalent to a resistor

*Digby[170] independently proposed that this same mechanism operates across the shells of crabs and shrimps. The same mechanism had been proposed earlier by Lund[173-175] and by Jahn.[176]

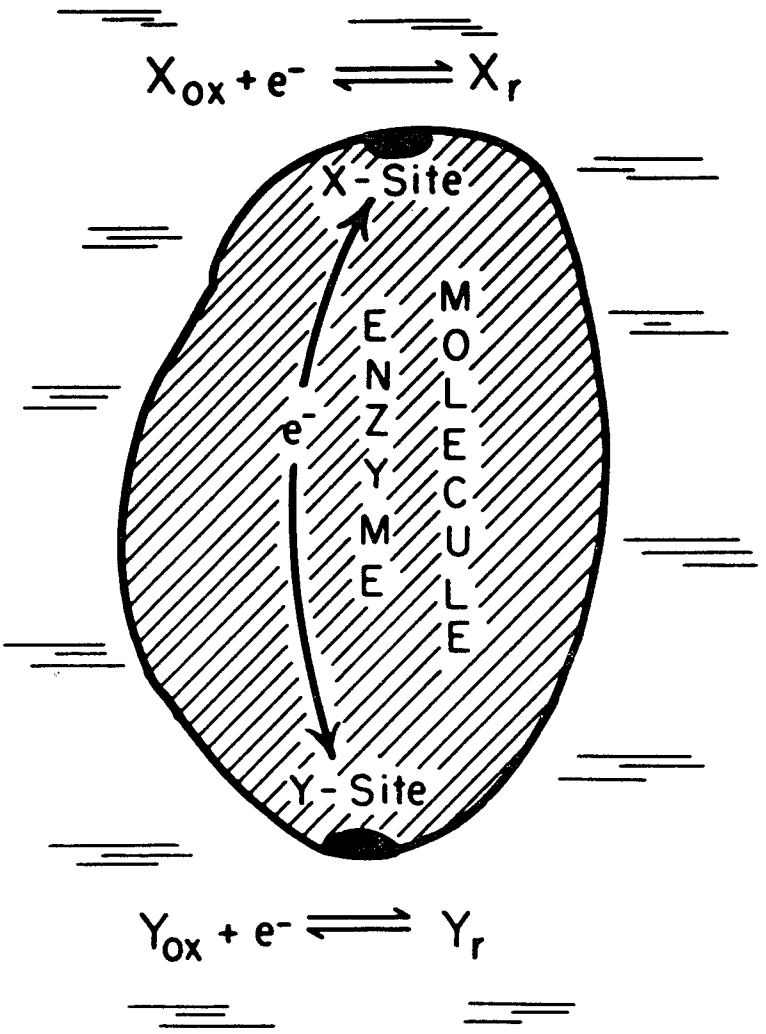

Fig. 7. Solid State and Electrode Model of the Cytochrome Oxidase Enzyme Particle. X-site on the particle acts like an electrode immersed in a solution of redox substrate X, so that the X-site develops on equilibrium electrode potential governed by the percent reduction of substrate X. The X-site is assumed to have enzymatic specificity for reaction with X-substrate. In other words, X-site can exchange electrons freely with X-substrate but will not react with Y-substrate. The Y-site, at the other end of the enzyme particle acts like a second electrode immersed in a solution of a second redox substrate Y, and develops an equilibrium electrode potential governed by percent reduction of Y. The direct reaction of X- and Y-substrates in solution is assumed to be insignificantly slow. The difference in electrode potentials at the X- and Y-sites causes a current of electrons to flow through the conductive solid enzyme particle. The rate of current flow, and hence the rate of oxidation of X-substrate by Y-substrate, is governed by the over-voltage phenomena at the X- and Y-sites, and by the electrical resistance across the solid enzyme particle.

connected across a tiny battery. To develop the theory of this behavior, we shall express the site (or electrode) potentials and the voltage-current equation in terms of substrate concentrations and time.

First, let us calculate the potential difference across the particle (Fig. 7), which we shall call the cell potential (V_{cell}). This is merely the difference in equilibrium electrode potentials experienced by the two sites at the two ends of the particle. Hence:

$$V_{cell} = V_x - V_y \tag{22}$$

where V_x and V_y are the electrode potentials at the X- and Y-sites respectively. Equation 22 may be put in terms of substrate concentration by the use of the familiar electrode equation

$$V_x = V_x^0 + (RT/F) \log_e (x_{ox}/x_r) \tag{23}$$

where x_{ox} and x_r are the concentrations of the oxidized and reduced forms of X-substrate, R is the gas constant, T is absolute temperature, F is the Faraday and V_x is a constant. Because the sum of oxidized plus reduced forms of X-substrate is always constant $(= C_x)$, we may write

$$x_{ox} + x_r = C_x \tag{24}$$

We shall assume for simplicity that Y-substrate is present in excess, so that the redox potential at the Y-site on the particle is constant $(= Y)$. We may then combine equations 22, 23, and 24 to obtain the potential difference across the particle (V_{cell}) (Fig. 7) in terms of concentration of reduced substrate as follows:

$$V_{cell} = V_0 - (RT/F) \log \left[x_r/(C_x - x_r) \right] \tag{25}$$

where V_0 is a new constant $(= V_x^0 - Y)$.

The potential difference across the enzyme particle causes a flow of electron current across the particle, like that which would occur through a resistor connected across a pair of electrodes in a battery. If we assume that current flow *(i)* across the enzyme particle is governed by Ohm's law, as is true for most organic semiconductors, we may write

$$V_{cell} = ir \tag{26}$$

where r is the electrical resistance across the enzyme particle.

Because the conduction of one mole of electrons across the enzyme particle reduces (or oxidizes) one mole of substrate, we may write

$$i = F\, dx_r/dt \tag{27}$$

We may now derive the differential equation for substrate reduction by substituting equations 25 and 27 in equation 26 to obtain an equation of the form

$$-\frac{dx_r}{dt} = A \log \left[\frac{x_r}{C_x - x_r}\right] - B \tag{28}$$

where x_r is concentration of the reduced form of X-substrate, t is time, C_x is total concentration of oxidized plus reduced forms of X-substrate, and A and B are constants.

Equation 28 is the basic differential rate equation describing the kinetics of a particulate enzyme that is rate-limited by ohmic conduction of electrons across the particle. Unfortunately, because of the log term, equation 28 cannot be integrated by any exact method. We may, however, test experimental data for ohmic conduction kinetics by plotting a graph of (dx_r/dt) vs log $[x_r/(C_x - x_r)]$, although this is rather inconvenient. Linearity of such a plot would indicate conformity of the data to equation 28.

Fortunately, equation 28 may be converted to an approximate form of reasonable accuracy, which is easy to integrate and easy to use for the analysis of experimental data. We may substitute the approximate equation 17 in equation 28 to give a new differential rate equation for a particulate electron conduction equation having the simple form

$$-\frac{dx_r}{dt} = \left[\frac{\alpha}{C_x}\right] x_r + \delta \tag{29}$$

where α and δ are constants. This approximation[10] is moderately accurate from 15 percent to 85 percent reduction of substrate (Fig. 4).

Equation 29 was derived on the assumption that ir is the only voltage drop in the equivalent circuit of Fig. 7. If we make our picture more realistic by assuming that a diffusion overvoltage at the X-site also exists due to slow diffusion of large substrate molecules and that Y-substrate is also significantly reduced during the course of the reaction, then the differential equation of reaction kinetics remains of the same general form as equation 29, but with some extra constants and becomes

$$-\frac{dx_r}{dt} = \left[\frac{\alpha}{C_x + \beta} + \gamma\right] x_r + \epsilon \tag{30}$$

where β is a positive constant that is a function of the diffusion constant of substrate x_r, γ is a positive constant and ϵ is a constant. Hence, like equation 29 for the simpler model, the improved theory predicts a hyperbolic relationship between the first order rate constant (k') and C_x, but of a more complicated form as follows:

$$k' = \frac{\alpha}{C_x + \beta} + \gamma \tag{31}$$

or

$$(k' - \gamma)\cdot(C_x + \beta) = \alpha \qquad (32)$$

Equations 29 and 30 predict the kinetics of substrate reduction to be expected from an enzyme particle which is rate limited by ohmic conduction of electrons across the solid particle without or with a diffusion overvoltage at the particle surface. The kinetics are predicted to be of first order with respect to the concentration of reduced substrate (x_r). The important and unusual special feature of the prediction is that the first order rate constant (k′) should decrease if larger *total* (oxidized plus reduced) concentrations (C_x) of substrate are used, and the relationship is predicted to be hyperbolic of the form given by equation 32.

A kinetic pattern of exactly this form has been observed experimentally with the enzyme cytochrome oxidase when catalyzing the oxidation of reduced cytochrome c by O_2 and has also been observed for the enzymes cytochrome peroxidase[10] and pyruvate carboxylase.[114] To account for this unusual behavior in terms of classical mass action mechanisms, one would have to postulate some sort of substrate inhibition, or a complicated sequence of reactions.[113]

In addition to predicting the kinetic pattern of the cytochrome oxidase reaction, the solid state theory predicts the experimentally observed temperature dependence of this enzyme.[12] The theory given here predicts that the activation energy of the functioning cytochrome oxidase enzyme should equal the semiconduction activation energy for the solid protein of the enzyme particle.[12] Solid state physical measurements made recently on purified dried cytochrome oxidase do indeed agree with this prediction, and show that cytochrome oxidase is unique among proteins thus far studied, in that it has a semiconduction activation energy three times lower than other proteins.[11,12] Additional support for the concept that cytochrome oxidase activity involves solid state electron conduction is provided by measurements of relatively high electron mobilities in mitochondria by two new experimental techniques (pulsed electron beam[5,9] and microwave Hall effect[13-22]).

When the cytochrome oxidase reaction is studied at low oxygen concentrations, the rate equation of the reaction is observed to change from first to second order, the oxygen concentration shows an inhibitory effect on the second order rate constant of hyperbolic form.[115] These experimental findings are quantitatively understandable from the same concept of a solid state electron conduction enzyme, with the additional features of space change injection of electrons (yielding square law instead of ohmic conduction) and trapping.[116]

According to intuition, and also according to detailed theory,[12] if cytochrome oxidase functions by a solid state mechanism, then the

activation energy of the cytochrome oxidase reaction should approximately equal the activation energy for semiconduction in dried cytochrome oxidase, as measured by solid state physical techniques. For various simple proteins, semiconduction activation energies as measured by numerous investigators had always been found to be in the region of 1.0 eV. Since the cytochrome oxidase enzymatic reaction had been observed to have an activation energy of 0.3 eV or lower, a semiconductive mechanism for enzyme action had been thought unlikely. The possibility was overlooked that cytochrome oxidase protein might have a different activation energy of semiconduction from most other proteins. When at long last it was measured,[11,12] the semiconduction activation energy of dried cytochrome oxidase was found to be approximately 0.3 eV, which supports the solid state physical concept of the cytochrome oxidase mechanism.

Direct experimental measurements of electron conductivity under physiological conditions in cytochrome oxidase and in other systems that show solid state kinetics would be desirable to test theoretical deductions. Such measurements present great technical difficulties. The conductivity and electron mobility techniques generally used in inorganic solid state physics require samples that are single crystals yet are big enough to allow the attachment of electrodes, purity to parts per million, absence of water, and ohmic contacts between electrode and sample. Biological samples usually satisfy none of these criteria. Therefore, conventional solid state physical techniques have proved useful only to a limited extent. New techniques to avoid the problems inherent in biological samples are under development and have yielded some interesting preliminary data.

The only study by conventional solid-state techniques that has proved illuminating is the measurement by Straub[11,12] of the activation energy of semiconduction of purified, dried cytochrome oxidase. This showed that cytochrome oxidase was unique among the proteins thus far studied in having an activation energy threefold lower than others, in agreement with the predictions of solid state kinetic theory. With regard to new techniques, preliminary results are available from a pulsed electron beam method developed in Russia,[5,9] which showed relatively high electron mobilities in mitochondria. This result, which agrees with theoretical expectations, has been confirmed in preliminary measurements with a new microwave Hall-effect technique by Eley and Pethig.[18] The method was invented in Russia by Trukhan[13,14] and improved in England by Eley and Pethig.[16,22] More recently, Eley *et al*[19,20] showed that cytochrome oxidase activity is positively correlated with microwave Hall mobility of electrons, which strongly supports the concept that cytochrome oxidase enzyme activity is rate-limited by solid state electron conduction.

In conclusion, the evidence in support of a solid state mechanism for cytochrome oxidase now seems fairly strong.

The theory just presented describes well the experimental behavior of cytochrome oxidase in reactions in the test tube. In life, however, cytochrome oxidase exists as a part of a much larger solid membrane system, which performs at least one important biochemical function in addition to what is done by purified cytochrome oxidase in the test tube. In life, cytochrome oxidase exists as part of the membrane of mitochondria (small granules present in every cell which are barely visible under the light microscope). Cytochrome oxidase participates in the oxidative or electron transport process of the mitochondria. In mitochondria which are undamaged, the energy from oxidation is converted into high energy phosphate bond energy by phosphorylation of adenosine dephosphate (ADP) to adenosine triphosphate (ATP) which is the immediate source of energy for most body functions (muscle contraction, nerve firing, secretion of acid in the stomach, etc.).

The mechanism of coupling between oxidation and phosphorylation in mitochondria is not understood by biochemists. Various biochemists present various explanations with an air of confidence, but the various explanations are different, so that at least some of them must be wrong.

If one concedes that the *oxidative* part of the oxidative phosphorylation of cytochrome oxidase might occur in a solid, then one may ask whether *phosphorylation* and the coupling process might also occur by solid state mechanisms. Preliminary theoretical considerations suggest that this is indeed possible. No experimental tests of this hypothesis have yet been made.

It was suggested by Straub[11,23,24] that phonons (packets of lattice vibrational energy) might be the immediate source of energy for the phosphorylation in the mitochondrial solid. It was then suggested by Cope and Straub[12] that the coupling process was a polaron (a combination of a phonon and an electron), which could be conducted across the mitochondrial solid and would give up its phonon for a phosphorylation. The evidence that suggested polaron conduction was derived from an analysis in terms of solid state kinetic theory of the temperature dependence of the cytochrome oxidase reaction as measured by deVault and Chance.[27]

If the carrier of the energy for phosphorylation is the phonon, we may inquire about the mobility of this carrier for transport of the energy through the mitochondrial solid. We must first note that the free energy of hydrolysis of adenosine triphosphate is 9.0 kcal/M or 0.39 eV, which must be the energy of phosphorylation of ADP. Hence, the phosphorylation phonon must carry at least this much energy, which is equivalent to a photon of light of 3.24μ wavelength, which is in the infrared (IR) range. Phonons of this energy range are localized vibrations (optical phonons) and do not significantly propagate through the solid, in contrast to lower energy

phonons (acoustic phonons) which move freely. It therefore seems unlikely that phonon movement can play a significant role in dissemination of energy through the lattice of the solid mitochondrial membrane. If the solid state carrier of energy should be an exciton, this might be either free or localized.[25]

A different method of dissemination of an energy packet of this size is suggested by the dimensions of the mitochondrion (3.3μ long by 0.5μ radius for rat liver). Because the length of the mitochondrion is approximately one wavelength of this phonon energy packet, Cope[25] considered the possibility that the mitochondrion might act as a resonant cavity for IR electromagnetic waves, which could serve as a rapid and efficient method for dissemination of IR energy throughout the mitochondrion from generation sites to utilization sites. The mitochondrion would then be a small scale analog of a microwave cavity made from a shorted section of waveguide, as used in radar systems.

A serious objection to this idea is the fact that water, which is the principle constituent of the mitochondrial sap, absorbs IR energy strongly, so that IR radiation within the mitochondrion would be mostly absorbed by water and wasted, instead of being transmitted.

A variation of the above mechanism would allow the mitochondrion to avoid the IR absorption by water.[25] Suppose that the IR radiation does not pass into the aqueous sap but is contained within the lipid bilayer of the mitochondrial membrane. Suppose that IR electromagnetic waves are generated at the surface of the lipid bilayer and kept within the lipid bilayer by internal reflections from the adjacent layers of protein. The mitochondrion would then be analogous to a coaxial transmission line where the lipid bilayer is the dielectric and the adjacent protein layers are analogous to the metal surfaces of the central wire and of the outer metal sheath.[25] This coaxial transmission line, when shorted at both ends, forms a coaxial cavity which is resonant for any integral multiple of a half wavelength (the multiple is 2 for the mitochondrion for 3.24μ wavelength energy packet). The composition and geometrical structure of the mitochondrial lipid bilayer minimize absorption of IR waves by the lipid dielectric, so that this could be a highly efficient method for energy propagation throughout the mitochondrial membrane.

The first reason for expecting low absorption, and hence low wastage of IR energy within the lipid bilayer of the mitochondrial membrane results from its chemical composition.[25] Analysis of mitochondrial lipids reveals that they consist mostly of compounds with very low IR absorption, including 33 percent saturated straight-chains (16-18 carbons), and 52 percent straight chain (18-20 carbons) with 2 or 3 double bonds. Indeed, the IR absorption of saturated straight-chain hydrocarbons is so low that such compounds (Nujol) are commonly used to make slurries of crystals for IR spectroscopy. The only significant IR absorption in saturated

straight chains is the C—H stretching frequency, which is located near 3.3μ. In unsaturated carbon chains, there is also an IR absorption band due to the C=C stretching frequency, which is very weak, although it becomes more intense when conjugation occurs with multiple double bonds.

Even though IR absorption in the mitochondrial lipid bilayer must already be low because of the chemical composition of the lipid, it must be reduced lower by the particular geometrical arrangement of lipid molecules relative to the geometrical pattern of the electric vector, which is produced in a coaxial cavity of this type.[25]

The electric field configuration to be expected in the IR coaxial cavity consisting of the mitochondrial membrane may be deduced easily from the theory of coaxial transmission lines. For a coaxial transmission line, at a wavelength (λ) longer than the cutoff wavelength (λ_c), only the principle transverse electromagnetic (TEM) mode can be propagated, where λ_c is given approximately by

$$\lambda_c = \sqrt{\epsilon}\,\pi\,(r_1 + r_2) \tag{33}$$

where r_1 and r_2 are the inner and outer radii of the coaxial transmission line, and ϵ is the dielectric constant.[25] For the rat liver mitochondrion, radius $\cong 0.5\mu$ and the thickness of the lipid bilayer dielectric is relatively small (60 Å = $6\times10^{-3}\mu$), so that $r_1 \cong 0.5\mu$. If we take the dielectric constant (ϵ) of the lipid bilayer as one, then we may substitute the numbers in equation 33 to obtain the cutoff wavelength for the mitochondrion $\lambda_c = 3.0\mu$. Because the wavelength of the energy packet for phosphorylation is 3.24μ which is longer than λ_c, only the principle TEM mode can be propagated in the mitochondrial membrane.[25] Higher order modes with other field configurations cannot exist.

The field configuration for the principle TEM mode in a coaxial transmission line has the electric vector (E) directed radially and the magnetic (H) directed circumferentially, which must therefore be the situation the mitochondrial membrane. According to the Danielli model of the lipid bilayer membrane, the carbon chains of the lipid molecules are directed perpendicular to the membrane surface, so that in the mitochondrial membrane, the carbon chains of the lipids are directed radially, which is parallel to the direction of the E vector. The C—H bonds are perpendicular to the chains and are therefore perpendicular to the E vector. As mentioned before, the principle IR absorption of straight carbon chain hydrocarbon molecules is the C—H stretching frequency. It has been shown by both theory and experiment that IR absorption by a molecular group is a minimum (which may be zero) when the E vector is perpendicular to the direction of the molecular vibration. It is therefore to be expected that the absorption of IR energy by the lipid bilayer of the mitochondrial membrane which already must be low because of the chemical composition, must be even lower because the C—H bonds are perpendicular to E. This expected

low IR absorption implies that dissemination of IR phosphorylating photons by the coaxial resonant cavity mechanism may be extremely efficient, because of very low losses expected from absorption in the lipid dielectric.[25]

In summary, it seems likely that the mitochondrial enzyme system, which operates in an organized solid membrane, functions by an integrated set of solid state processes. Oxidation by cytochrome oxidase occurs by solid state conduction of polarons, which yield phonons which perform phosphorylation of ADP to ATP. The phonon energy is disseminated rapidly and with low loss throughout the entire mitochondria membrane by infrared electromagnetic waves in the lipid bilayer of the mitochondrial membrane, so that the mitochondrian as a whole acts as an infrared resonant coaxial cavity.

D. Nerve. Nerve fibers consist of a solid matrix of protein and lipid with intersitial water containing various small ions, such as sodium (Na), potassium (K), and chloride (Cl). When a nerve fires, a complicated set of movements of these ions occurs through the water in the nerve. Extensive measurements of these ion movements have been made, and much of the experimental data has been put together (i.e. the experimental data has been curve fitted) by a complicated set of equations called the Hodgkin-Huxley theory.[117]

The Hodgkin-Huxley theory is not an explanation of nerve conduction in terms of the principles of physical chemistry. Rather, it is merely an ingenious curve fit of much experimental data on ion flows in nerve water. It ignores completely the possibility of solid state events within the solid protein-lipid matrix of the nerve.

Evidence is now available for several different solid state physical processes in the solid matrix of nerve. These include electron conduction over activation energy barriers, superconduction, piezoelectric phenomena, and transmission of infrared electromagnetic waves in the lipid bilayer. It seems reasonable to suppose that these solid state processes take part in transmission of the nervous impulse down the nerve fiber, and are integrated with the ionic events which take place in the nerve water.

Electron conduction across activation energy barriers in nerve is evidenced by the experimental observation of Chalazonitis[118] as analyzed by Cope[119] that decay of photoconductivity in pigmented invertebrate nerve conforms to the Elovich equation, the derivation and significance of which was discussed in Section IIIB of this review.

A role for superconduction in nerve function is suggested by several types of evidence. These include (a) the temperature dependence of certain neural processes is characteristic of single electron superconductive tunneling,[56] (b) certain other neural processes show *negative* temperature dependences, which are difficult to explain except from the hypothesis of two electron superconductive tunneling,[56] (c) numerous behavioral effects

on organisms are caused by magnetic fields smaller than 2 Gauss, for which no physical mechanism of adequate sensitivity is known, except the superconductive Josephson effect,[59,60] (d) dilute superconductivity has been observed experimentally[49-51] in various bile acids at temperatures as high as 277°K, and extrapolation of this data leads to the prediction[52] the cholesterol should superconduct as physiological temperatures and above. Cholesterol is a major constituent of the solid matrix of nerve. In addition, solid state magnetic phenomena suggestive of superconduction have been observed experimentally in the presence of high electric fields.[53] Nerve possesses thin lipid layers (containing cholesterol) in which high electric fields exist. The superconductivity predicted for cholesterol at zero electric field might be enhanced by these high fields,[53] so that superconduction could become of significant magnitude and play a role in nerve function.

Piezoelectricity and pyroelectricity in nerve are suggested by the following evidence. The temperature dependence of the resting membrane potential of an invertebrate nerve cell shows a temperature dependence for which log of the activation energy is a linear function of temperature, which conforms to theory of peizoelectric resonance.[92] A square pulse of increased temperature often produces in nerve systems two response transients of opposite sign at the start and end of the temperature pulse (Fig. 1). This is characteristic of a pyroelectric process.[100] It is seen in the fish thermoreceptor, which also shows proportionality between the time derivative of response and temperature, which is predicted from pyroelectric theory.[100] The muscle stretch receptor shows linearity of neural response with the logarithm of applied mechanical force. This is predicted from the concept that the neural response is governed by a piezoelectric effect in an elastic polymer macromolecule.[100] In addition, nerve is observed to be pyroelectric.[156]

Infrared (IR) transmission of electromagnetic energy by a coaxial transmission line mechanism in nerve fibers (axons) seems possible.[25] Like in mitochondria, the probable immobility of IR phonons will limit the transfer of energy along the axon, where high mobility would be useful to move energy along the great distances of axon length. Like in mitochondria, the cylindrical lipid bilayer of the axon membrane seems well constructed to serve as a coaxial transmission line for rapid and efficient transfer of IR electromagnetic waves along the axon within the lipid dielectric.[25] A bundle of axons in the nerve could then be regarded as a coaxial IR fiber optic system. Supporting this concept is experimental observation of emission of IR radiation from nerve during excitation.[120]

Some nerve fibers are larger in diameter than mitochondria, and consequently will have longer cutoff wavelengths and lower impedances than mitochondria. For an A-nerve fiber, such as commonly found in a mammalian motor nerve, a radius of 10μ might be typical, from which

one may calculate using equation 33 that the cutoff wavelength (λ_c) will be 60μ. Hence, in a typical motor nerve axon for wavelengths over most of the IR range, higher order modes will carry a significant fraction of the total IR energy, so that longitudinal transmission of IR energy will be less efficient than in mitochondria or in smaller nerve fibers, and dielectric losses may be somewhat higher because E is not always perpendicular to the C—H bonds of the lipid chains. In a C-nerve fiber, such as one might find in the sympathetic system, a radius of 0.25μ might be found, which substituted in equation 33 leads to a cutoff wavelength (λ_c) of 1.5μ, which makes for efficient longitudinal IR transmission and low dielectric losses.[25]

Impedance of a coaxial transmission line is given by (if resistive losses are small)[25]

$$Z = 138 \sqrt{\mu_r/\epsilon}\, \log_{10} (r_2/r_1) \tag{34}$$

where Z is impedance in ohms, r_1 and r_2 are inner and outer radii of the coaxial line, μ_r is permeability (= 1 except for ferroelectrics), and ϵ is dielectric constant of the lipid bilayer. The impedance (Z) for IR transmission will be different for different nerve fibers because of differences in radius, although the thickness of the dielectric lipid layer is probably much the same in all nerves and in mitochondria, and will here be considered a constant (= 0.006μ) as for mitochondria.[25] For a C-nerve fiber, with an inner radius (r_1) of 0.25μ, using equation 34, we may calculate impedance (Z) of 1.4Ω. However, for a A-nerve fiber with a radius of 10μ, we calculate[25] impedance (Z) of 0.036 Ω.

Because for different nerve fibers, radius varies markedly but thickness of the dielectric lipid bilayer remains approximately constant, it is instructive to show the simple approximate relationship between the radius of a nerve and its impedance for transmission of IR waves.[25] We use the approximate equation (derived by Clark[121] and Cope[122])

$$\log_e \left[\frac{\chi + \Delta}{\chi} \right] \cong \left[\frac{\Delta}{\chi + \Delta} \right] \cong \left[\frac{\Delta}{\chi} \right] \tag{35}$$

where $\chi > 0$ and $\Delta \ll \chi$. If we let χ = r (inner radius) $\cong$ r (mean radius of nerve fiber) and Δ = thickness of dielectric lipid bilayer, using 0.434 for the ratio of $\log_{10} x$ over $\log_e x$, we may substitute equation 35 in equation 34 and obtain

$$Z \cong 60 \sqrt{\mu_r/\epsilon} \left[\frac{\Delta}{r} \right] \tag{36}$$

where Z is axon impedance for IR transmission in ohms and the other symbols are as previously defined. This shows that, approximately, the axon impedance is inversely proportional to its radius.

The impedances for coaxial IR transmission for nerve fibers (1.4 Ω for C-fibers and 0.036 Ω for A-fibers) are much lower than the 75 Ω which is usual in man-made coaxial transmission lines for radio frequency waves. Most efficient energy transfer between generator, line and load occurs when impedance of the line equals that of generator and load. One may speculate that therefore the generators and absorbing systems for IR energy in nerve should have low load impedances to match the nerve transmission line, and might be of fundamentally different character than the relatively high impedance devices ordinarily used in man-made systems. Pertinent to the possible nature of the low impedance nerve generator and loads is the evidence for superconductive phenomena in nerve at physiological temperatures as discussed in previous paragraphs of this section. Superconductive junctions (Josephson junctions) have been fabricated which will generate and detect electromagnetic energy in the microwave and IR range of wavelengths.[123,124] These superconductive generating and absorbing systems have been observed[32,125] to have impedances in the range of 1 Ω. The expected low impedance of a superconductive IR generator or load in nerve might therefore be a reasonably good match to the calculated low impedance of an IR coaxial nerve transmission line.

Larger nerve fibers have lower impedances, with impedances inversely proportional to radius (see equation 36). Therefore, for efficient IR energy transfer, impedances of generators and loads should be correspondingly lower in larger nerve fibers. One might speculate[25] that if the generators and loads were uniformly distributed as sites over the circumferential area of the nerve with the same density in all nerve fibers, the number of sites per unit length of nerve would be proportional to radius, and the reciprocal of the total impedance of the unit length would be proportional to the sum of reciprocals of impedances of individual sites, whose number is proportional to circumferential area. Therefore, total impedance of generator or load sites per unit length of nerve would be inversely proportional to nerve radius, so that an equally good match of generator or load to transmission impedance could be obtained for nerves of all diameters using the same type and density of generator and load sites for all nerves. The hypothesis just expressed that sites of IR generation and absorption might be continuously distributed along the surface of the nerve lipid bilayer implies the possibility of repeated absorption and re-emission of the energy by the conductive walls (protein) as the IR energy is transmitted along the length of the nerve fiber within the dielectric lipid bilayer.[25]

IV. Application of Solid State Physical Concepts to Ionic Phenomena in Cell Water

We may consider the cell to resemble a sponge, consisting of a solid skeleton of membranes and particles constructed of lipids and proteins, in

which electron currents may flow, governed by the laws of solid state physics.[92] The interstices of the sponge are filled with water, in which reside dissolved proteins and small ions. NMR evidence[126-128] indicates clearly that cell water possesses more structure than liquid water, and that much of the Na^+ and K^+ in the cell is not free in aqueous solution, but is associated with charged sites on macromolecules.[129-133] The quantity of Na^+ and K^+ dissolved in tissue water is low, because of the structuring of the water. One may then make an analogy between the aqueous portion of the cell and a solid semiconductor. One may regard complexed Na^+ and K^+ as analogous to valence band electrons in a semiconductor, and free cations as analogous to conduction band electrons.[134,135] The complexed ions exchange rapidly with, but are only slightly soluble in, the structured water. Conduction, diffusion, and potential equations for ionic processes in cell water therefore resemble those for electronic conduction in a semiconductor solid. Experimentally useful predictions regarding ion transport have been obtained from this approach.[134,135]

An ionic form of solid state activation energy barrier kinetics has been developed by Cope[135,135] which predicts that Na^+ and K^+ leakage from cells should obey the Elovich equation, which agrees with experimental observation. This theory requires that there be an activation barrier to ion conduction at the cell surface. The cell surface must resemble a liquid-solid interface like an electrode surface. Since the extracellular space is presumably liquid, the intracellular phase must have solid-like properties if the theory is correct. Specifically, it is necessary to assume that cell water has more structure than liquid water and consequently a low solubility for Na^+ and K^+, and that Na^+ and K^+ exist in cells mostly associated with macromolecules rather than in free solutions. During the last few years, excellent evidence for cell water structuring from nuclear magnetic resonance (NMR)[126-128] and from water absorption studies[136,137] has become available and this has become the basis for an NMR method of cancer diagnosis.[138,139] It has been shown that the metabolism of the cell produces much less energy than is required to operate the Na^+ and K^+ pumps that are required to maintain ionic concentrations in the cell for the liquid state model of the cell.[140-143] Therefore, the liquid-state-free-cation model of the cell is thermodynamically impossible, but the solid state, or ion exchange resin, or structured water and associated cation model of the cell is compatible with all the evidence and with thermodynamics. An entire issue of the Annals of the N.Y. Academy of Sciences[144] was recently devoted to this problem.

Therefore, the weight of the experimental evidence supports strongly the analogy of the cell surface to a liquid-solid interface. Therefore, the application of solid state physical theory to the kinetics of ion transport

across the cell surface seems appropriate. Cope's solid state theory of cell water and ions is probably equivalent to a simplified form of the association-induction hypothesis of Ling[141,145] and the ion exchange theory of Damadian.[146,147]

V. The Future

New and better experimental techniques are needed for solid state biology. Biological processes function only when the system is complex, impure, noncrystalline and wet. The conventional experimental techniques of solid state physics are not usable under these conditions.

New experimental techniques suitable for solid state biology are beginning to be developed. Measurements of electron mobilities within wet protein particles by Hall effect at microwave frequencies have begun.[13-22,148] A pulsed electron beam method for measurement of electron mobilities in wet proteins has been developed in Russia.[5,9] Methods for measurement of dilute superconductivity, applicable to organic solids[49-51,177] and perhaps to biological systems have been developed.

When the experimental techniques of solid state physical biology become as well developed as those of liquid state biology, the author expects that solid state processes will be found to be equally as important for biological functions as liquid state processes.

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A REVIEW OF THE FERROELECTRIC MODEL OF MICROTUBULES

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It has been 25 years since the piezoelectric properties of the microtubule were first discussed. Subsequent to the discovery of its piezoelectric properties, its consequences have been observed in experiments which have shown microtubules to align under the influence of electric fields. In this regard, our investigations have sought to determine in which situations the ferroelectric properties are important for the function of the microtubule, and in addition, how the microtubule may be affected by the transient endogenous electric fields of living cells. We report a summary of recent findings and discuss their implications.

1. INTRODUCTION

Microtubules have recently generated much interest well beyond the biological community[1-3] because they demonstrate how physical interactions are important within living systems. The microtubule is a polymer of a protein known as tubulin which is common to most animal, plant and protist cells. Microtubules are one of three main structures which form the cytoskeleton that supports the cell. Microtubules perform many functions related to both their structural strength and flexibility. Other than serving as a cellular backbone, they act as a railroad for the transportation of cellular materials. For this purpose the orientation of the microtubule is essential since the motor proteins which carry materials toward or away from the cell body may travel in only one direction along a microtubule. Hence, the permanent dipole moment of the tubulin molecules may be important in the guidance of molecules, known as motor proteins, that *walk* along the microtubule surface. More recently, microtubules have been implicated in

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simple signalling within the cell. In protists, which are single-cell organisms, information processing cannot occur through any sort of multicellular neural networking, microtubules have been proposed as the most likely organelle to accomplish this task. These curious possibilities are explained in this review from the standpoint of ferroelectrics.

The piezoelectric properties of the microtubule were first discovered by Athenstädt, who was searching for piezoelectric effects in living systems.[4] The measurement of its piezoelectric properties and subsequent classification as a piezoelectric polymer declared the existence of a permanent electric dipole on the microtubule. Since the microtubule is composed of many identical subunits ($\alpha-\beta$ tubulin dimers), each much itself possess a permanent dipole. This has been confirmed by the experiments of Vassilev *et al.*[5] and subsequently by Kaimanovich *et al.* [6, 7] and White *et al.* [8] which have shown that arrays of microtubules may be oriented by electric fields and have gone on to speculate that intracellular electric fields may induce microtubule orientation. Further experiments have demonstrated that the tubulin dimer of which the microtubule is composed has two different conformational states within the microtubule.[9] When guanosine triphosphate (GTP) hydrolysis occurs, the length of the tubulin dimer is reduced from 8.3 nm to 8.0 nm. With any atomic reorganization such as that involved in a conformational change of a molecule, the electrons must also be redistributed. Given that the surface of tubulin is strongly negatively charged at cellular pH and that its size changes by a significant amount on the scale of electronic interactions, this structural compression must result in a change of the dipole moment. This electron redistribution has also spurred interest in charge conduction along microtubules.[10] However, the length scale of tubulin implies that microtubules lie in the gray area between quantum and classical physics. To the best of our knowledge, the first theoretical model of ferroelectricity in microtubules has been proposed by Satarić *et al.*[11] and it was based on an effective field approximation in the continuum limit giving rise to travelling ferroelectric domain walls.

Our present model of ferroelectric phases in microtubules is simply a classical Ising model where: (a) the interaction constants are determined by classical dipole–dipole electromagnetic forces; (b) the 2D lattice is periodic in one direction so as to form a tube; and (c) the dipole moment has only two possible orientations, though they are not precisely opposite to each other. The last property shall be important when interactions between the dipole of tubulin and transient endogenous electric fields are considered; the dipole moment must have a radial component so that the dipole interaction with the external field is non-zero.

Some of the interest in microtubule ferroelectric properties has been sparked by their unusual assembly properties, their special arrangement in cells of various types, their involvement with cell division and their suspected role in some cellular signalling. We elaborate upon these examples of microtubule structure and function.

- *In vitro*, microtubules experience a form of stochastic assembly known as dynamic instability.[12] Typically, individual microtubules are either undergoing slow steady growth or extremely rapid shortening, known as a catastrophe event. The entire pool of tubulin in the cell, however, exhibits smooth oscillations in the quantity of tubulin which is assembled. Electromagnetic interactions resulting from dipoles on assembled microtubules and on free tubulin dimers have been proposed as one coupling mechanism to describe the transition from individual to ensemble dynamics. In addition, the catastrophe event may be explained. The idea is that the dipole interactions induce a conformation with some *fault* energy. At some point, the fault energy is sufficient to induce a catastrophe event.
- The arrangement of microtubules in cells is always such that the negative end is closest to the center of the cell and the positive end extends toward the cell membrane, or down the long extension of a nerve cell known as an axon.[13] The positive end of the microtubule should not be confused as denoting the direction in which a microtubule's dipole is pointing since at this point the direction has not been experimentally determined. However, this convention is consistent with the biological literature on the subject and the plus end indicates the orientation of the structure. Perhaps 5% or fewer microtubules will typically be improperly oriented. Given that cell membranes have a resting potential caused by ionic gradients across the membrane, it is unlikely that the electrical potential drops precisely to zero immediately inside the membrane. There is a small residual electric field on the order of $1-10\,\mathrm{V/m}$, which one suspects aids in the efficient alignment of microtubule orientation within the cell. This alignment is crucial to motor proteins which travel along microtubules and specifically carry material toward or away from the centrosome which is located close to the cell nucleus.
- In cells which are undergoing division, microtubules align themselves from the two poles of the cells and extend toward the metaphase plate which will ultimately divide the two daughter cells. Pairs of chromosomes are held together along the plate with two microtubules, one anchored near each pole. The pattern of microtubules is highly reminiscent of the

field lines between a positive and negative charge. This has reinforced the speculation that electromagnetic effects are at work, given that it is known that microtubules may be ordered by electric and magnetic fields [14] and has driven investigators to study charge flow in the cell membrane before and during cell division. More intriguing than the array of microtubules which forms is the ability of the microtubules to synchronize chromosome separation and subsequently initiate the segregation process as the microtubules themselves disassemble. It seems necessary for the microtubules to interpret or carry some signal in this process since a signal carried by a chemical messenger, through diffusion, could not arrive at the chromosomes with the perfect synchrony that is observed. If the signal is mechanical, then the structure which supplies the push or pull on the microtubules must be found, an alternative explanation in terms of piezoelectric or ferroelectric coupling seems more reasonable.

Finally, it has recently been shown that microtubules and associated members of the cytoskeleton play a role in intracellular signalling. [15, 16] The mechanism of signal transduction is not understood at present and may again be intimately associated with the ferroelectric properties of the polymer.

2. THE MICROTUBULE LATTICE

The lattice of the microtubule is helical. Within cells, it is most commonly composed of 13 columns, known as protofilaments. The outer ring of protofilaments is 25 nm in diameter, the inner ring 15 nm (Fig. 1). The interior of a microtubule is hollow and they may have an arbitrary length; however, microtubules with a length of a few microns are not uncommon. Since the length of each dimer is about 8 nm, the typical microtubule has perhaps 100–1000 rows. Each α–β tubulin dimer is composed of two nearly identical monomers, individually known as α and β tubulin. This lattice is arranged such that there is an upward offset between neighbouring protofilaments of 0.92 nm (12 nm/13) when moving around the microtubule in a left-handed sense. Since the monomer spacing along a protofilament is 4 nm, this represents a rise of 3 monomers for each loop of the helix. However, there are two types of lattice which have been described in the literature, known as the A lattice and the B lattice. If one follows the helix with the small 0.92 nm offset (known as the $\bar{3}$ start helix), the monomers alternate (α–β–α–β–$\cdots$) in the A lattice but remain the same in the B lattice (α–α–α–α–$\cdots$), except at the seam, where there is an α–β link.

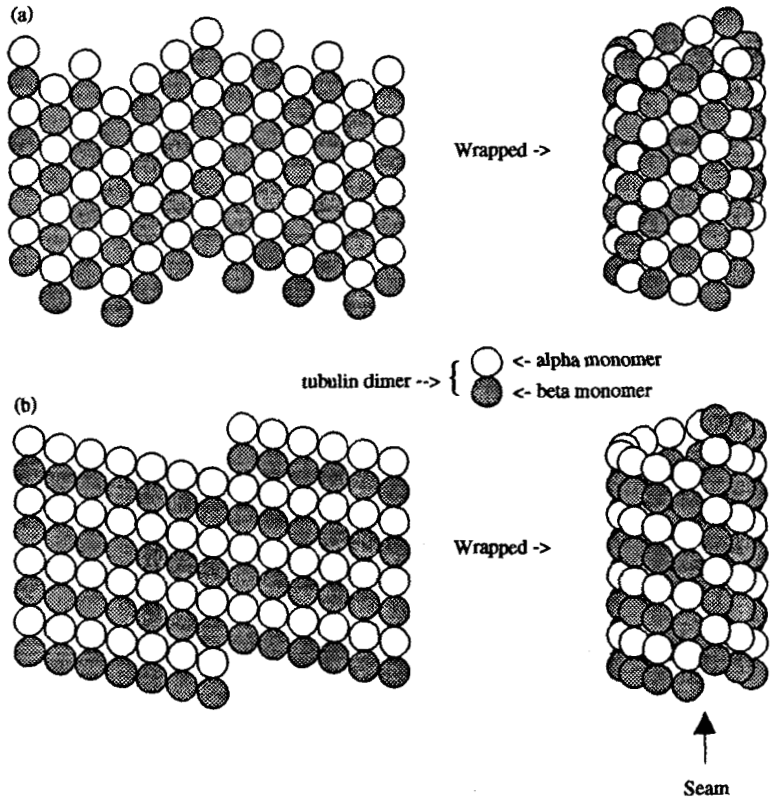

FIGURE 1 The two possible microtubule lattices are depicted. Typical microtubules are much longer than these which are shown with a length of only 4 dimers along each of the 13 protofilaments (columns). (a) The A lattice, which has helical symmetry. (b) The B lattice, which has a seam where a line following the α-monomers around the microtubule meets a β-monomer. The left side depicts unwrapped (non-physical) microtubules; the figures resulting from our simulations are given in a similar manner.

Thus the A lattice has complete helical symmetry which the B lattice does not possess. So, the monomer lattices are identical but the dimer lattices are quite different. Since the dimer is the fundamental unit from which microtubules are constructed, the dipole is associated with each dimer and the lattice change means that interactions are quite different between the two geometries.

Our model of ferroelectric effects in microtubules was inspired by the cellular automaton model of Hameroff *et al.*[2] Within our model, there is no net charge on each dimer but there is a dipole associated with each dimer,

and consequently the overall structure has an electric polarization. One orientation of the dipole points along the protofilament axis, this is the up state (Fig. 2). We have some freedom to choose the direction of the dipole in the down state since the electric dipole has not been measured in each of tubulin's conformational states. We have chosen the down state's orientation in a direction which is roughly opposite to the direction of the up state but which is 29° from the vertical and points out radially away from the microtubule. This choice is based on the geometry of microtubule ends that are splaying apart during disassembly. Under these conditions, tubulin adopts a different conformation with curved protofilaments.[17] The results of simulations presented here have been carried out with $\phi = 29°$ down states. Some degree of tilting is necessary if action potentials are to influence the dipole dynamics because the non-tilted dipole is entirely axial and the electric field of the action potential is purely radial.[18] Consequently, there would be no interaction unless one of the states has some radial component since the interaction energy between a dipole and an electric field is simply

$$U_{\text{int}} = -\bar{p} \cdot \bar{E}, \tag{1}$$

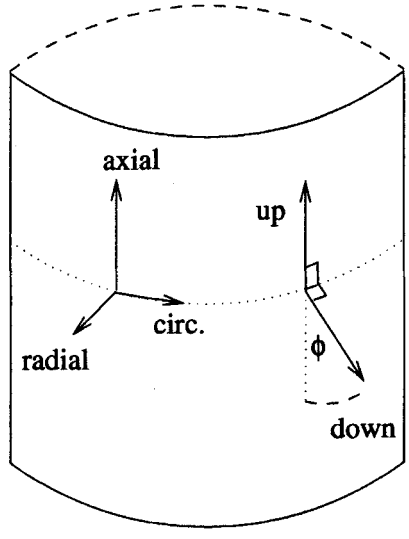

FIGURE 2 This view of the outer surface of a microtubule identifies the directions of fields we consider. The axial or vertical direction in this diagram points away from the cell body. At the right, the up and down directions which were selected for tubulin's dipole moment are shown ($\phi = 29°$).

where $\bar{p}$ is the dipole moment and $\bar{E}$ is the electric field. The interaction energy between two elementary dipoles is given by the following well-known formula: [19]

$$U_{\text{int}} = \frac{1}{4\pi\varepsilon\varepsilon_f} \frac{\bar{p}_1 \cdot \bar{p}_2 - 3\,(\bar{p}_1 \cdot \bar{n})(\bar{p}_2 \cdot \bar{n})}{r^3}, \tag{2}$$

where ε is the relative permittivity of the medium, ε_f is the permittivity of free space, $\bar{n}$ is the normal vector pointing from the position of the first dipole to the second dipole and r is the distance separating the dipoles. Consider the interaction between dipoles of the microtubule lattice: From a purely structural point of view, the lattice type may seem insignificant, as one does not expect that it will change assembly dynamics or elastic properties much. However, from the point of view of dipole interaction, the lattice type is crucial. In the B lattice, dimers are aligned in nearly tangential rows (Fig. 3). As a result, there is a strong interaction between neighboring protofilaments which favours antiparallel dipole orientations. In the A lattice, the neighboring protofilaments are shifted vertically such that identical orientation of the dipoles is favoured. In either case, there is a strong interaction along individual protofilaments which favours similarly oriented dipoles. The effect of the dipole interactions is that the lattice of up and down dipoles will self-organize into an energetically favourable configuration. At high temperatures, the lattice has a random state – each individual dimer can be found in the up or down state in a rather arbitrary fashion. At lower temperatures, the interaction energy dictates the dipole

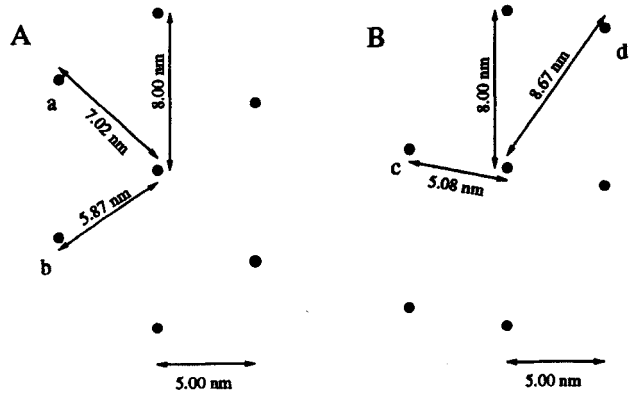

FIGURE 3 View of lattice structures. Note that the A lattice on the left is nearly perfectly triangular while the B lattice on the right is nearly rectangular. Shown as a planar lattice in the diagram, the lattice curls around in a cylindrical fashion to form the hollow microtubule.

arrangement and the lattice becomes ordered. Our task is to investigate this transition in order to determine the likely state of a microtubule under physiological conditions.

The lattice dynamics in our model are simulated using a standard Monte Carlo technique.[20] For each timestep, the energy of the present state and the opposite state are calculated. A change of state is a random event; the probability of its occurrence is determined by the temperature. There are two numbers which are put into the model by hand. One is the relative permittivity of tubulin in cytoplasm, ε and the other is the magnitude of the dipole moment of tubulin. The dielectric constant of dry tubulin or of tubulin in solution has not been found in the literature although some experiments give permittivities in excess of 100 for some other proteins in solution. [21] Our choice of these parameters allows us to compare results with previous modelling attempts. All of this dependence can be accounted for by a single parameter, Q, where

$$Q = p^2/\varepsilon \,. \tag{3}$$

The dynamics of the model are unaffected by the choice of Q but it simply scales with the temperature. Larger values of Q increase interaction strengths and thereby increase the transition temperature. We have chosen $Q = 12 \times 10^{-56} \mathrm{C}^2 \cdot \mathrm{m}^2 \simeq 10800$ (Debye)^2. For a dipole corresponding to an electronic charge, with a charge separation of one dimer unit, 8 nm, $\varepsilon \simeq 13$ given this choice. Such a value for the dipole moment has the same order of magnitude as some other globular proteins.[21] This may be a conservative estimate as Athenstädt has reported the dipole moment of another fibrous molecule, collagen, to be about 15 000 Debye. The important number is not the total dipole but the change in the dipole, consequently the smaller figure that we report seems reasonable. The precise value of the dipole moment depends on the pH of the surrounding medium. The following results are derived from these estimates of the tubulin dimer's dipole moment. We shall comment on how any changes to these parameters affect the results of our simulations.

3. RESULTS

The results presented represent a summary of our work, and the most important features are highlighted. A more extensive look at the microtubule lattices and their ferroelectric ordering may be found in our

previous publications.[22–24] We are plotting an order parameter, P, representing scaled polarization, which is defined over the N lattice sites by

$$P = \frac{1}{N}\sum_{i,j}\sigma_{ij}, \tag{4}$$

where the σ_{ij} is $+1$ or -1 and describes the conformation of the dipole labelled (i, j) of the 2D cylindrical lattice. The results for the 13A lattice and the 13B lattice (as illustrated in Fig. 1) are shown in Figures 4 and 5. In addition, the polarization *versus* temperature graph is displayed for the 13A lattice as a function of an applied external electric field in Figure 6. We have also computed polarization curves in the case of lattice defects, both impurities and microtubules with alternative protofilament numbers (12 and 14 being the most common of such defects).

Along a protofilament, head to tail alignment of dipoles is preferred, so that the positive end of one dipole is next to the negative end of the adjoining dipole. Consequently, dipoles will seek parallel alignment along a protofilament at low temperature when the microtubule lies in its ground state. In the microtubule lattice, each dimer has six nearest neighbours due to the triangular nature of the lattice. Consider the two nearest neighbouring dimers which are located on the protofilament directly to the left of the protofilament of the dimer, labelled a and b (Fig. 3). Relative to the

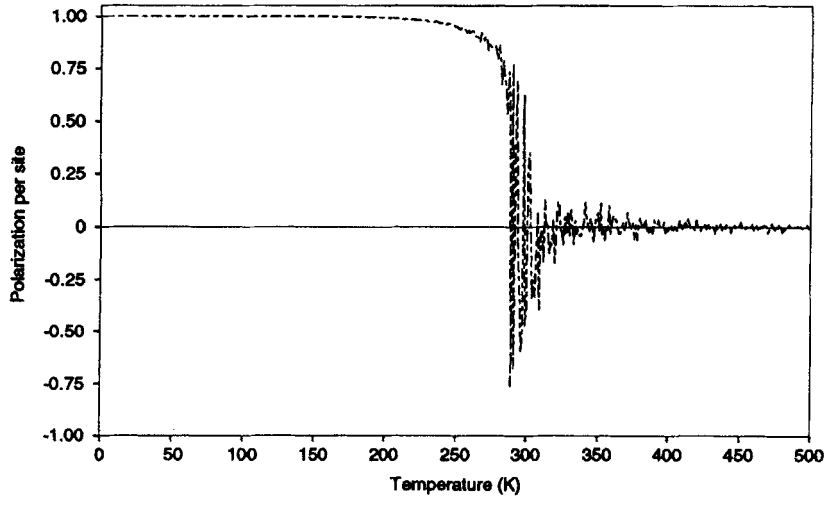

FIGURE 4 Relative electric polarization of the microtubule 13A lattice as a function of temperature, obtained from Monte Carlo modelling.

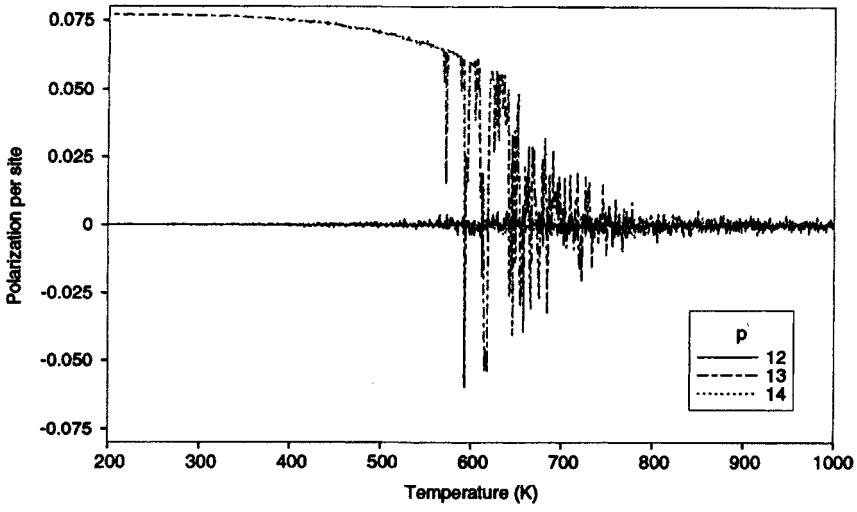

FIGURE 5 Polarization of the microtubule 12B, 13B and 14B lattices as a function of temperature. The 12B and 14B lattices show only random fluctuations about zero relative polarization because the ordered low temperature state consists of antiferroelectric protofilaments. In the 13B lattice, one protofilament of 13 must be unpaired, resulting in a finite polarization.

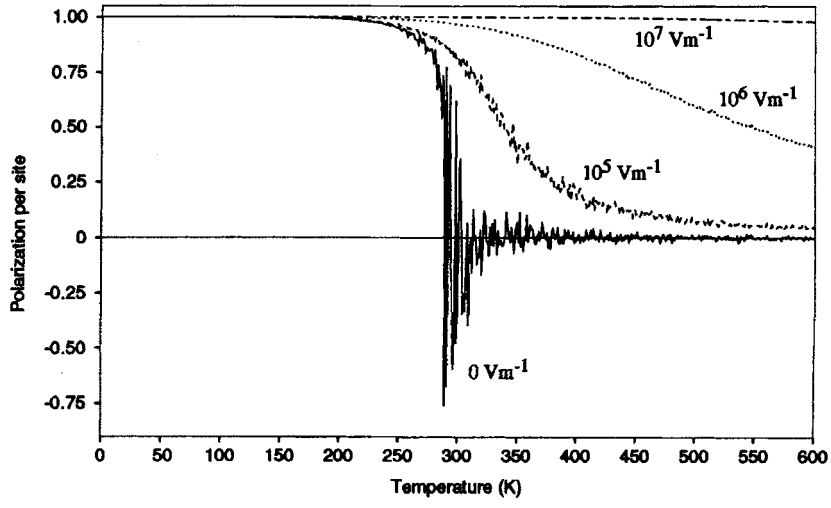

FIGURE 6 Polarization of the microtubule 13A lattice as a function of temperature as an electric field is applied in the direction of the up moment.

interaction along the protofilament, J, there is an interaction of strength 0.35 J along the direction a, and an interaction of strength -0.22 J along the direction b. Both of these are smaller than the interaction along the

protofilament, so indeed parallel alignment along the protofilament is expected. Of the off-protofilament interactions, the larger interaction (a) is ferroelectric so neighbouring protofilaments adopt a configuration of parallel dipoles. Consequently, the ground state consists of all dipoles in parallel alignment (Fig. 8a). In the B lattice, the situation is changed; there is an interaction of strength -1.76 J along the direction c, and an interaction of strength 0.39 J along the direction d (Fig. 3). This direction, c, must have alternating dipoles. Given that dipoles alternate along the helix defined by the d direction, the other two interactions are conflicting but the interaction along the protofilament is larger. As a result, dipoles are still in parallel alignment along the protofilament. The entire lattice now consists of alternating protofilaments (Fig. 8b).

3.1. Lattice Defects

In addition to the effects of lattice type and applied fields upon the ordering of the lattice, there is also the question of how defects in the lattice have an impact upon dipole ordering. Firstly, consider lattice defects where the protofilament number is different than the usual 13. In the case of such a microtubule, there is now a seam for both the A and B lattices. In the 14A lattice, the interaction at the seam is now B-lattice-like. As a result, at the

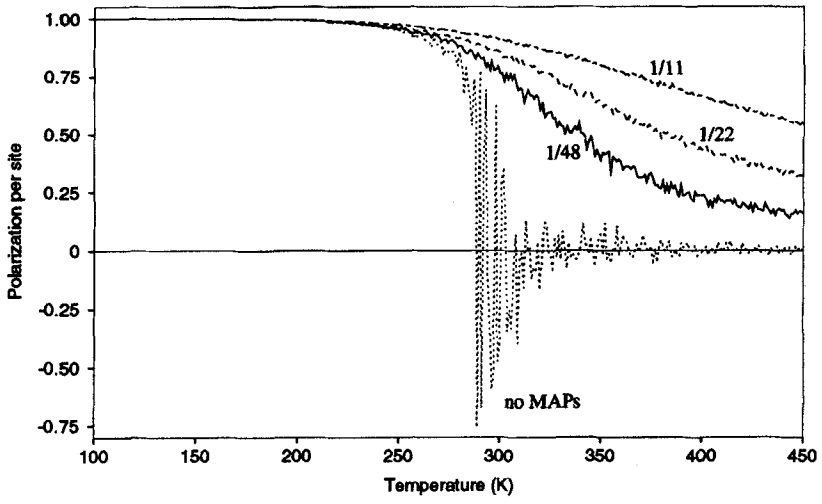

FIGURE 7 Polarization of the microtubule 13A lattice as a function of the ratio of microtubule-associated proteins to tubulin dimers. Ratios taken from Dustin [25] and 1/22 means one in each 22 dimers is attached to a MAP and held in the up conformation.

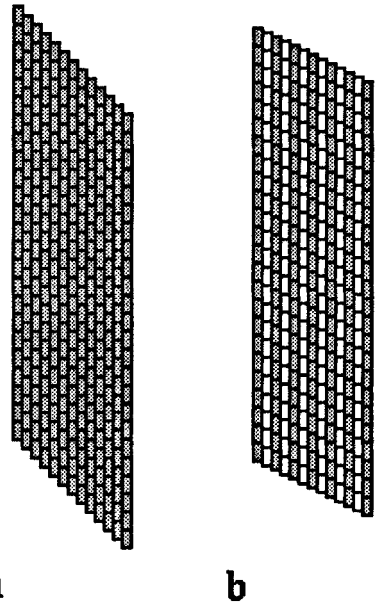

FIGURE 8 (a) Ground state of the microtubule 13A lattice, which is completely ferroelectric. Dimers are illustrated and dark boxes and light boxes indicate states of opposite dipole moments. (b) Ground state of the microtubule 13B lattice, which demonstrates antiferroelectric order between neighbouring protofilaments. Dimers are illustrated as in (a).

seam the ordering between the neighbouring protofilaments favours antiparallel ferroelectric order. These two protofilaments adopt the antiparallel ferroelectric ordering and the remaining protofilaments have some flexibility. That is, a ferroelectric seam must exist on the microtubule due to the dipole frustration. It should be distinguished from the seam we have discussed up until this point, which was structural; its location is immaterial since it does not have an energetic consequence. So, one drawback to such a lattice is the relative lack of order since dipole interactions alone do not make all microtubules with 14 protofilaments and an A lattice alike. Similarly, if we consider the 14B lattice, the bulk of the lattice imposes that neighbouring protofilaments should have alternating dipole orientations; but at the structural seam which closes the microtubule cylinder the preferred interaction is of antiparallel dipole orientations. The ground state occurs when the frustration is localized at the structural seam; however, the net polarization of the microtubule defined by Eq. (4) is zero since the numbers of dimers in the up state and in the down state are equal. Since the dipole of the up state and down state are not exactly opposite, the

microtubule does retain a small polarization. Thus the lattice type and the boundary conditions of the structural seam play an important role in determining the ordered state of the polymer.

We have also studied the impact which microtubule associated proteins (MAPs) have on dipole ordering in microtubules. MAPs attach themselves to microtubules in periodic patterns.[25] Their function is to stabilize the assembly properties of the microtubule. Our hypothesis is that this is accomplished by preventing the conformational change of the tubulin dimer to which it is attached. The net effect of such MAPs has been simulated given this assumption, and the result is qualitatively similar to the application of an electric field (Fig. 7). This may be understood by viewing the fixed conformation dipoles as seeds of order. In the limit where one could connect a MAP to each tubulin dimer in a microtubule, the lattice would be ordered at all temperatures.

One thing which has not yet been experimentally established is the magnitude of the dipole moment on the tubulin dimer. It seems quite likely that it is sufficiently large for dipole interactions to cause ordering of the molecular conformations of the microtubule. Indeed, it is even possible that, due to the shielding of the microtubule by ordered water, the effective temperature experienced by the polymer would be somewhat lower and dipolar effects would be even more important.

Our investigations show that the microtubule is also sensitive to electric fields of the order 10^5 V/m; this may be more usefully understood as 100 mV/μm and is smaller than the typical electric field of 10^7 V/m measured across cellular membranes. If the MAPs function as we have hypothesized, then weaker fields would suffice. The significance of this observation comes mainly from the geometry of microtubules in nerve cells, which pass electrical signals (nerve impulses) along their membranes and give rise to transient electric fields. These short-lived fields may be sufficient to cause a wave of reordering along the microtubule, just as a wake follows a ship. It is essential that the microtubule has a highly ordered resting state if such signals are to be interpreted by receptor proteins at the synaptic junction between two nerve cells. That is to say, a signal could not be distinguished from the background unless the background was relatively smooth. This may indeed be the function of the dipole interactions.[18] Our simulations show signal processing or transmission as possibilities but rule out the storage of information as Hameroff *et al.*[2] have hypothesized. The main problem with information storage through particular patterns of dipole states is the need to shield the system from thermal fluctuations. In our model this means operation below the transition temperature. Below the

transition temperature, signals are only transient because the microtubule must ultimately reside in its ground state.

4. DISCUSSION

The ferroelectric properties of the microtubule seem to be important in determining the orientation of the microtubule, may be important in directing the traffic of motor proteins on their outer surface and are also essential if signals are passed along the microtubule by a wave of conformational change. While the electromagnetic hand may not be required for this direction, it does simplify the explanation of certain signalling properties such as a synchronizing signal prior to chromosome separation or signal propagation along an axonal microtubule. The phase transition from a disordered to an ordered lattice of dipoles is not necessarily observable within a living cell, since one expects that the microtubule would always remain ordered to retain its function. However, the transition temperature does determine the magnitude of the dipole required for electromagnetic interaction to be the dominant influence on lattice dipole dynamics. The larger the dipole, the more important both the dipole–dipole interaction and the interaction with external electric fields.

Currently investigation is underway to examine the conduction properties of the microtubule. Should the microtubule be found to have semiconductor properties, this electron mobility will lend much weight to the ferroelectric explanation of microtubule function.

Acknowledgements

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Amplification of Superpositional Effects Through Electronic–Conformational Interactions

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Abstract—Electronic–conformational interactions allow for a speedup of biomolecular structure formation that derives from the quantum mechanical superposition principle. Changing phase relationships among electronic states of different energy lead to perturbative interactions between non Born–Oppenheimer electrons and atomic nuclei that can have a self-amplifying (positive feedback) character. Conformations that satisfy the conditions for such an interaction are inherently unstable. The term "conformational soliton" refers to nuclear configurational changes that ripple through macromolecular lattices in a manner that preserves informational integrity. The model suggests that such states of ordered conformational motions are generic to polymacromolecular organizations of the type that occur in biological cells.

1. INTRODUCTION

Conformational changes appear to accompany all biomolecular recognition-action processes. An enzyme, for example, may be pictured metaphorically as interacting with a substrate in a lock–key manner; if the shapes fit together a catalytic action occurs. In reality, the shape of the enzyme changes during the course of the reaction and the response is enabled and modulated by a variety of milieu features (including control molecules). The formation of the complex is typically attributed to diffusive motion ("Brownian search") and shape fitting augmented by conformational motion. Recent years have also seen efforts to develop a variety of dynamic pictures in which nonlinear interactions, such as electronic–conformational coupling, play a self-organizing role [1–7].

In previous publications the author has argued that the rates of complex formation and decomposition (the docking and undocking of the interacting molecules) depend on interactions between electronic and nuclear degrees of freedom that are inherently unstable [8–10]. The purpose of the present paper is to show how such interactions can draw on the parallelism inherent in the quantum mechanical wave function to generate conformational changes that propagate over long distances in macromolecular networks in biological cells. Cytoskeleton would be a prominent example; DNA strands and the bilayer membrane itself are other possible media. The term "conformational soliton" is intended to evoke the picture of an electronic-conformational instability that perpetuates itself over substantial distances in a macromolecular network in a way that requires no net dissipation. The term "self-perpetuating conformational change" could also be used, since the propagation processes resulting from electronic–conformational interactions are in many respects quite different from those of conventional solitons.

2. QUALITATIVE PREVIEW

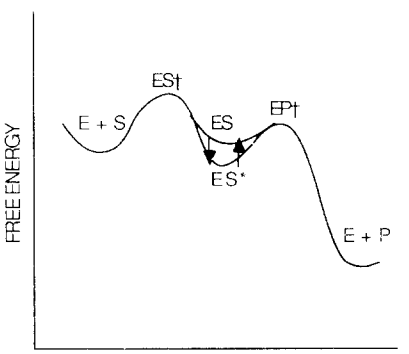
2.1. *Dynamic free energy surfaces*

Let us begin with an informal description of the physical principles underlying docking speed. As noted above diffusion and dynamic shape fitting play a role. The probability that a "molecular lock" and a "molecular key" would find each through random collisions should decrease as the specificity of the fit increases. To achieve both specificity and speed it is necessary that the molecules dynamically draw each other into the required relative orientation. The conformational changes that accompany recognition reflect this requirement. The greater the "drawing in" effect the lower the free energy of the complex. The likelihood of decomposition due to purely thermal agitation must then decrease. This implies that specificity and speed are conflicting requirements in a purely thermal model of enzyme catalysis.

Enzymatic reactions are nevertheless both quite specific and quite fast. This can be achieved if the complex is inherently unstable [3, 11]. The thermodynamic picture is that of a dynamic free energy surface (Fig. 1). The substrate molecule bounces into the enzyme, undergoes a catalytic modification, and then bounces out. A bouncing ball provides a simple example of such a nonthermal motion. The elasticity of the ball allows an exchange between potential and kinetic energy. In the model to be presented the exchange occurs between the electronic and nuclear coordinates. The nuclear system transfers (perturbation) energy to the electronic system, which in turn transfers it back to the nuclear system. In the presence of suitable nonlinearities (with a positive feedback character) the effect is self-amplifying.

2.2. *Quantum speedup principle*

The quantum mechanical superposition principle plays the key role in this exchange. The wave function of the interacting molecules constitutes a set of possible states that interfere with each other. Let us divide the entire system—for example, enzyme and substrate—into electrons, atomic nuclei, and hydrogen bonds (which may either be shared or mobile protons). The de Broglie wavelength of the atomic nuclei is too small for wave properties



$$E + S \rightleftharpoons ES \rightleftharpoons E + P$$

Fig. 1. Unstable free energy surface. E represents enzyme and S represents substrate. ES‡ and EP‡ are activated states. ES represents the enzyme–substrate complex. ES*, which is not explicitly represented in the reaction scheme, is a second form of the complex that has lower energy than ES. The double potential surface indicates that ES and ES* are connected by a nonthermal motion (e.g. an electronic–conformational interaction).

to play an important role. The possible positions of the nuclei converge to a definite value, and consequently the superposition of states is of negligible significance. The "shape state" of the complex is defined in terms of the coordinates of the nuclei, but even if we considered individual nuclei the superposition of states would be of only minor significance. The important point is that definite shape is alien to the superposition principle. Consequently quantum features cannot speed up the docking process if the configuration of the nuclei is the only factor considered, even if dynamic changes in the configuration are allowed.

The situation is entirely different for the electrons, and to some extent for the hydrogen bonds. The possible positions of an electron can differ significantly. When a measurement is made, the electron jumps into one of its states with some probability. The possible states interfere with one another, but the probability of finding the electron in any particular state is independent of time as long as the states are all of the same energy [12]. This corresponds to a stationary state. When states of different energy contribute significantly to the superposition, the probability of finding the electron in any particular state changes, corresponding to time development. The change in probability is due to constructive and destructive interference among the possible states of differing energy (the phase relationships among the different possible states would stay the same if they all had the same energy). The nuclei will be agitated by the resulting variations in the electronic charge distribution. In the model to be developed this agitation contributes to the speed of complex formation. In the presence of suitable nonlinearities the electronic states that contribute to the superposition are further spread out in energy by the motions they induce in the nuclei, leading to a self-amplifying effect that allows the nuclear configuration to jump out of local minima. This speeds the discovery of a unique minimum (or family of minima) in the case of molecular self-assembly and speeds both the formation and decomposition of the complex in the case of enzyme catalysis. Conformational solitons result when the conformation changes triggered in one macromolecule lead to unstable conformational interactions with a neighboring macromolecule.

2.3. *Breakdown of Born–Oppenheimer*

At first glance the above scenario is incompatible with the fact that the energy differences between different electronic states in biological macromolecules appear to be too great to allow for a speedup effect of the type described. Clearly self-assembly and enzyme catalysis occur in the dark. Thus in order for superpositional speedup to occur it would be necessary for the electrons of biological macromolecules to undergo transitions in response to the infrared radiation field. This appears to contradict the fact that the infrared spectra are ordinarily attributed to the vibration and rotation of the atomic nuclei. This is probably why models of the type described have not previously been seriously considered.

The above objection rests on the Born–Oppenheimer approximation, that is, on the assumption that the electrons are tightly enough bound to the atomic nuclei so that the potential due to their charge can be wrapped into the nuclear potential. This is not true for delocalized electrons. Such electrons will experience some motion relative to the nuclei, hence some accelerations that are not rigidly tied to the thermally induced accelerations of the nuclei. Accordingly, such electrons should make a contribution to the infrared radiation normally attributed to the thermal motions of the nuclei that deviates from the contribution they would make if Born–Oppenheimer were strictly accurate. Since the electrons and nuclei form a bound system the electronic levels must at the same time be discrete. Consequently closely spaced transitions must in fact exist. This would be of no significance in a system in which the nuclei are undergoing a periodic motion. Complex formation and

decomposition is a time dependent process that is not periodic, however. The infrared spectra of a complex should change in the course of its self-assembly, since degrees of freedom become constrained in different ways as the complex forms. If the Born–Oppenheimer assumption fails, some component of this change will then be due to closely spaced electronic transitions.

There is now substantial evidence that electrons can tunnel over significant distances in proteins, for example by tunneling through hydrogen bonds [13]. Surface electrons and some highly delocalized bonds or highly polarizable electrons (such as the electrons in the polar head groups of lipids) are probably also attached to the atomic nuclei in a sufficiently loose manner for Born–Oppenheimer to become an inadequate approximation. Paralled spin electrons that ordinarily are well approximated by Born–Oppenheimer could conceivably also be transiently delocalized if the nuclei to which they are attached are forced, in the process of complex formation, to be much closer together than their unconstrained equilibrium separation. The confinement of such electrons would serve to correlate their motions during short periods of downward fluctuation in thermal energy (over longer periods spin flipping through the spin-orbit interaction would be the more likely fate). During these short periods the electrons will behave as a pair with an energy that is lower than the unpaired ground state energy [2, 3, 11]. In this way states of different energy would be available; additionally the correlated motion of the pair would exert a perturbative effect on nuclei.

A more general argument, based on the principle of microscopic reversibility, implies that even mildly delocalized electrons should undergo transitions to different energy states in the course of complex formation. When the nuclear conformation changes, the potential to which electrons are subjected changes; the energy levels of the electrons thus change in the course of complex formation [14]. It might at first be thought that this is consistent with Born–Oppenheimer, since the change in the electronic levels follows the nuclear motion. However, according to the principle of microscopic reversibility the time reversed process should also be possible. In this case the changes in the electronic levels would lead the nuclear motions, and would therefore influence them. In short, if it is possible to follow it is possible to lead. (The effect of the electrons on the nuclei should not be analogized to the effect of a football on the earth, since the electromagnetic forces exerted by the two systems are equal and independent of mass. The difference between the masses of the two systems only controls the speed with which the follower system follows.)

2.4. *Quantum versus thermal fluctuations*

A second objection also suggests itself, namely that the quantum effect would be small as compared to the thermal effect. This would be true on a static potential surface. For a dynamic surface, such as the one schematically pictured in Fig. 1, the effects of thermal and quantum fluctuations are incommensurable. If the electronic–conformational interaction opens up a new pathway on the surface the controlling factor will be the barrier presented by this pathway; the issue is not whether a quantum effect is more effective than thermal agitation for jumping over (or tunneling through) a given barrier at room temperature, but whether it is more effective in allowing for reduction or sidestepping of the barrier. Here is where the nonlinearity of the interaction between the electronic and nuclear wave functions plays the crucial role. The quantum speedup effect outlined above is in a sense a thermal effect, since thermal (infrared) agitation initially spreads out the energies of the possible electronic states. The nonlinear interaction between nuclear conformation and the charge variations which result from the interference of possible

electronic states of different energy amplifies this initial thermal influence, a feature that is uniquely dependent on the superposition principle.

3. FORMAL REPRESENTATION

3.1. *Superposition principle*

The time development of the whole system (complex plus heat bath) is governed by some true time independent potential $V(x)$. Here x schematically denotes coordinates of all relevant particles—electrons, nuclei, and particles of the heat bath. This potential is unknown, and is in any event useless since it provides too much information about the time development of the system (we are not interested in the heat bath). At each stage of development we can suppose that some features of this true potential become relevant. We can denote this time dependent potential by $V(x_e, x_n, t)$, where x_e denotes the coordinates of the electrons of the interacting macromolecules and x_n denotes the coordinates of their nuclei. The time evolution of the potential function means that some information about it is absorbed into the heat bath and represented in terms of a statistical description.

Let us denote the electronic wave function of the complex by $\Psi_e(x_e, t)$, and the nuclear wave function by $\Psi_n(x_n, t)$. For the purposes of eliciting the speedup and instability inherent in electronic–conformational interactions it is sufficient to regard these as notational wave functions—that is, we need not here attempt to represent the detailed character of the electronic or configurational states. Also, we will work with wave functions rather than density matrices; the latter would make it possible to include thermal statistical effects more elegantly, but at the expense of losing focus on the quantum statistical effect that is here to be proposed.

If the potential energies (or Hamiltonians) of the electronic and nuclear systems were additive we could write

$$\Psi(x_e, x_n, t) = \Psi_e(x_e, t)\Psi_n(x_n, t). \tag{1}$$

This is not possible in a system in which the potentials share coordinate dependencies. In ordinary atomic or molecular problems the nuclear wave function, expressed in center of mass coordinates, is considered constant; hence the electronic levels can be static and governed by a fixed potential that is determined by the interaction between electrons and nuclei and between electrons and electrons. This is equivalent to assuming the complete absence of an electronic–conformational interaction. In our case the latter interaction is the main point of interest, and consequently the clean splitting described by equation (1) cannot be strictly accurate since the electronic and nuclear wave functions must mix to some extent.

The time scales of electronic and nuclear transitions are very different, however. Our idealization assumption is that the difference is so great that we can treat the wave functions as being sufficiently separable so we can make a quasi-independence assumption. When the molecules come into contact the wave functions are modified by an interaction term that requires them to be updated. Updating continues until the wave functions converge to mutual consistency (in which case the complex stabilizes) or until they diverge (in which case the complex breaks up). Quasi-independence means that the updating can be done independently—that is, the electronic wave function can evolve for a while, then the nuclear wave function changes, then the electronic wave function evolves again, and so forth. Apart from the updating points the potential function can then with reasonable approximation be split into two parts

$$V(x_e, x_n, t) = V_n(x_e, x_n, t) + V_e(x_e, x_n, t), \tag{2}$$

where V_e denotes the potential controlling the behavior of the electronic system and V_n denotes the potential controlling the behavior of the nuclear system. The reversal of subscripts relative to those of the corresponding wave functions (e.g. V_n is the potential for Ψ_e) is intended to indicate that the positions of nuclei make a more significant contribution to the potential governing the time evolution of the electronic system than do interactions among the electrons and that the distribution of electrons makes a contribution to the potential governing the time evolution of the nuclear configuration that is not entirely attributable to the strong and weak bonding interactions among the nuclei (taken in the sense of the interactions used in a molecular dynamics calculation). The idealization is that the potentials do not significantly depend on the same coordinates between the updates. All the nonlinearity is thus concentrated at the updating points; between these points the time development is linearizable. These are extreme simplifications. If we can elicit a significant electronic–conformational effect under these conditions it is reasonable to assume that the effect would even be stronger if the wave functions are less separable and the nonlinearity operative at all times.

For simplicity suppose that the complex has two accessible families of nuclear configurations, labeled by I and II, and two accessible families of electronic states, also labeled by I and II. Suppose that these families (or conformations) are both separated by potential barriers, at least initially. Also for simplicity we will think in terms of two wave functions, one for all the electrons of the interacting macromolecules and one for all the nuclei. In reality, of course, the potential surface would be much more complicated and the wave functions of each of the interacting molecules would ordinarily be treated as independent of one another to a good approximation; but for the purposes of exhibiting the speedup effect in a formal way it is the interaction between electronic and nuclear wave functions that is the main point of interest.

The electronic wave function can then be expressed as a superposition of the form

$$\Psi_e(x_e, t) = \sum_{e \in \mathrm{I}} a_e(t) e_e(x_e) \exp(-iE_{e(\mathrm{I})} t/h) + \sum_{e \in \mathrm{II}} a_e(t) e_e(x_e) \exp(-iE_{e(\mathrm{II})} t/h) \qquad (3)$$

where the e_e are eigenfunctions of the electronic system that are most naturally associated with nuclear conformation I, the e_e are eigenfunctions most naturally associated with nuclear conformation II, the a_e and a_e are expansion coefficients, and we assume that the wave function can be split into time independent and time dependent parts. The $E_{e(\mathrm{I})}$ and the $E_{e(\mathrm{II})}$ are the electronic energies associated with the eigenfunctions e_e and e_e.

Any complete set of orthonormal basis functions could be used to express $\Psi_e(x_e, t)$. The choice of eigenfunctions allows for the most natural association between individual basis functions and particular conformations. Since the potential is time dependent it is necessary to choose the set of eigenfunctions at some particular point in time and in a way that allows for the associations noted above. A natural choice is the initial point in time, assuming the true potential is operative. Alternatively, we can regard the electronic eigenfunctions as defined over the initial potential $V_e(x_e, x_n; t = 0)$ and the nuclear eigenfunctions as defined over the initial potential $V_n(x_e, x_n; t = 0)$. In the present case we are considering a coupled double well potential. Consequently the electronic eigenfunctions are determined by the double well potential representing the two possible nuclear conformations, and similarly the nuclear eigenfunctions would be determined by the double well representing the two possible families of electronic states. These are of course purely conceptual definitions. It would never be possible to actually ascertain the mathematical form of these eigenfunctions starting with the double surfaces. But in principle the eigenfunctions exist and this is sufficient for the present purposes.

Some of our "notational eigenfunctions" might not be assignable in an unambiguous way to either conformation I or conformation II. This class of eigenfunctions would contribute to speedup, due to the fact that they extend over a larger region of the potential surface. If this class is ignored we can think of the superposition in a more practical way as built up from eigenfunctions calculated in the individual wells. For the purpose of demonstrating the speedup effect we can ignore the ambiguous eigenfunctions, since their presence would in fact enhance the effect.

The wave function for the collection of nuclei may, in accordance with the above conventions, be expressed as

$$\Psi_n(x_n, t) = \sum_{r\varepsilon \mathrm{I}} b_r(t) n_r(x_n) \exp(-iE_{n(r)}t/\hbar) + \sum_{s\varepsilon \mathrm{II}} b_s(t) n_s(x_n) \exp(-iE_{n(s)}t/\hbar). \tag{4}$$

The n_r are the eigenfunctions of the nuclear system when it is in conformation I, the n_s are the eigenfunctions when it is in conformation II, and the $E_{n(r)}$ and $E_{n(s)}$ are the energies associated with these eigenfunctions. However, the nuclear system, because of its large mass, should with high probability be assigned to a definite state. Thus if it is initially in conformation I all the b_s can for all practical purposes be set to zero. To make matters as unfavorable as possible to the speedup effect we can also assume that the nuclear configuration is initially in a single eigenstate, so that all the b_r apart from one can initially be set to zero. Practically speaking this is reasonable since the average values associated with the different n_r should peak very sharply due to the large mass.

For the electrons it is not possible to ignore superposition effects. An electron, because of its small mass, might with reasonable probability be found in a location more naturally associated with II than with I even if the nuclei are in conformation I. To make matters as unfavorable to the speedup effect as possible, let us assume that initially all the a_s are negligible and that the electronic system is initially in a single eigenstate, so that only one of the a_r is initially nonnegligible.

If some of the electrons are not tightly bound to the nuclei it will be possible for perturbations, arising for example from the radiation field, to induce transitions to eigenfunctions with energies close to that of the energy of the initial eigenfunction. The probability distribution of electronic coordinates will then be given by

$$p(x_e, t) = \sum_r a_r^*(t) a_r(t) e_r^*(x_e) e_r(x_e) + \sum_r \sum_{r'} a_r^*(t) a_{r'}(t) e_r^*(x_e) e_{r'}(x_e) \exp\left(\frac{(-iE_{e(r)} - iE_{e(r')})t}{\hbar}\right), \tag{5}$$

where $r \neq r'$. This explicitly demonstrates the statement made earlier that the electronic charge density will change with time due to interference among the eigenfunctions of different energies, since these are associated with different frequencies of vibration. Our simplifying assumption that the wave function can be split into time independent and time dependent parts serves to elicit this well known quantum mechanical feature in an explicit manner; but in fact the assumption is more restrictive than necessary.

Equation (5) can be used to give more precise meaning to the phrase "parallelism of the quantum mechanical wave function". The structure of the wave function is determined by two factors: the choice of the basis functions and the superposition of these basis functions. The contribution of these two factors to the "possibility space" of a system cannot be separated. Even a single basis function, say an eigenfunction, will allow for penetration into the classically forbidden region of a barrier. It might therefore appear that the quantum speedup effect could occur without interference between states of different energy. This is not the case, however, since interference inevitably occurs if the time

evolution of the system is considered. Suppose, for example, that an electron initially has a high amplitude to be in a particular potential well. With time the amplitude for being in a second well increases. The energy of the electron at the later time will be smaller, since the space in which it can reasonably be found will be larger. (This can be seen from the position–momentum uncertainty principle. An increase in the spatial range of the electron allows for a decrease in the uncertainty of its momentum, and therefore for a decrease in the kinetic energy fluctuations it exhibits.) Consequently the time evolution will involve interference among states of different energy regardless of the initial choice of basis functions. The treatment that follows will emphasize superposition, since interference is definitive of time evolution. However, the intimate connection between the choice of basis functions and their superposition must be kept in mind, since the interference pattern, and all physical properties, are independent of any acceptable choice.

3.2. *Self-consistent field*

The electronic and nuclear wave functions (equations 3 and 4) must be consistent with each other. This suggests a Hartree-type self-consistent field approach. Ordinarily this is viewed as an approximation procedure. For example, the electronic wave function of a multi-electron atom is approximated by starting with an initial trial wave function and potential and iterating until convergence is achieved. The steps toward convergence are ordinarily not viewed as representing the self-organization of the multi-electron atom, and accordingly the trial wave function need not represent any physical reality; questions about how such atoms organize themselves are not ordinarily considered. The objective is to obtain a description of a final, static system of electrons which presumably forms so rapidly and in a way so controlled by energy that questions about the formation process are not of major phenomenological interest. The situation is completely different for the formation of macromolecular complexes, such as the enzyme–substrate complex. The self-organization and self-disorganization of the complex, and the specificity of these processes, are the phenomenological features of greatest importance.

The idea, then, is to use a Hartree-like self-consistency scheme to represent, in a conceptual way, the evolution of the complex to a self-consistent state (which may ultimately be the dissociated state if no self-consistent complex is possible). Could it be that inconsistencies are possible in nature? The answer is certainly yes. A room controlled by a thermostat exhibits an inconsistency between the state of the room (ambient temperature) and the state of the thermostat (curvature of a piece of metal). If the relation between these two states is one of negative feedback the ambient temperature converges to a constant value, with some slight oscillation about this value. If the relation is one of positive feedback the temperature diverges. From the point of view of nonlinear dynamics the state of the room and the state of the thermostat are of course inconsistent in a consistent way. Our hypothesis is that the state of the electronic and nuclear systems can be inconsistent in just this fashion, that is, can be in a relation of either positive or negative feedback.

From the assumption (equation 2) that the potentials can be split in an additive manner we can express the electronic and nuclear wave functions as solutions of the Schrödinger equations

$$-\sum_i \frac{\hbar^2}{2m_i}\nabla_i^2\Psi_e(x_i, t) + V_e(x_i, x_n, t)\Psi_e(x_i, t) = i\hbar\frac{\partial\Psi_e(x_i, t)}{\partial t} \tag{6a}$$

$$-\sum_n \frac{\hbar^2}{2M_n}\nabla_n^2\Psi_n(x_n, t) + V_n(x_i, x_n, t)\Psi_n(x_n, t) = i\hbar\frac{\partial\Psi_n(x_n, t)}{\partial t} \tag{6b}$$

Here m_e is the mass of the electron, ∇_j^2 is the Laplacian operator of the jth electron, M_k is the mass of the kth nucleus, and ∇_k^2 is the Laplacian operator of the kth nucleus.

The complex must form from a large fraction of the initial conditions that correspond to initial contacts. The initial potentials governing the electronic and nuclear systems, since they are time dependent, would in fact then not be independent of the initial conditions. Let us suppose some initial arbitrary contact between the interacting molecules. We suppose that it is possible in principle to construct initial potentials corresponding to this initial arrangement. The initial electronic and nuclear wave functions are in principle given by the equations (6a) and (6b). The average positions of the electrons are then obtained from equation (5), and the average positions of the nuclei may be obtained analogously (except that interference effects should not be noticeable). The first cycle is completed by using these new average positions to update the potentials. The process is repeated until convergence is achieved (corresponding to mutually consistent electronic and nuclear wave functions) or until the system diverges to a self-consistent state. Divergence means jumping from one metastable state to another, leading eventually to a self-consistent complex. It is entirely possible that no self-consistent complex exists, apart from the dissociated state. Two interacting molecules could in this case jump, through a series of divergent processes, to a unique minimum that is suited to a particular catalytic function and then diverge to a dissociated state.

The updating of the potentials may be expressed notationally by writing

$$V_e(x_e, x_n, t) = V_e(\Psi_e(x_e, t), \Psi_n(x_n, t)) \tag{7a}$$

$$V_n(x_e, x_n, t) = V_n(\Psi_e(x_e, t), \Psi_n(x_n, t)), \tag{7b}$$

where the functional notation expresses the fact that V_e and V_n depend on features of both the electronic and nuclear wave functions. As noted above the updated functional forms utilize the probabilities

$$p(x_e, t) = \Psi_e^*(x_e, t)\Psi_e(x_e, t) \tag{8a}$$

and

$$p(x_n, t) = \Psi_n^*(x_n, t)\Psi_n(x_n, t). \tag{8b}$$

But of course some criteria would have to be used for obtaining adequate potentials from these probabilities; for example, interactions may be considered irrelevant if the likely coordinate values are such that the particles are more than some specified distance apart, or some averaging may be done beyond a certain distance. Updating the potentials in this manner does not represent any process inherent in the phenomena—in particular it does not imply that wave function reduction is occurring. New potentials could, with some less intuitive procedure, be obtained without calculating probabilities from the wave function. To construct a theoretical demonstration of the speedup effect it is only necessary to assume that relevant potentials exist in principle.

To formally express the scheme it is convenient to write the Schrödinger equations (6a) and (6b) in the abbreviated form:

$$\Psi(x, t + \tau_b) = \Psi(V(x, t + \tau_a), t + \tau_b). \tag{9}$$

Here $\Psi(x, t + \tau_b)$ is the (electronic or nuclear) wave function at time $t + \tau_b$, $t + \tau_a$ (with $\tau_b > \tau_a$) is the time at which the (nuclear or electronic) potential $V(x, t + \tau_a)$ is turned on, and $\tau_b - \tau_a$ is the time interval over which this potential remains applicable. The assumption that time dependent perturbation theory is adequate for dealing with any changes in the potential functions during the intervals means that the potentials take the

form

$$V(x, t) = V'(x) + v(x, t), \tag{10}$$

in a piecewise continuous manner, where $V'(x)$ is a time independent unperturbed potential applicable during the interval $\tau_{ik} - \tau_{ii}$ and $v(x, t)$ is a small time dependent perturbation that develops over this interval. Written explicitly in terms of what we have called the potentials for the electronic and nuclear subsystems

$$V_e(x_e, x_n, t) = V'_e(x_e, x_n) + v_e(x_e, x_n, t) \tag{11a}$$

$$V_n(x_e, x_n, t) = V'_n(x_e, x_n) + v_n(x_e, x_n, t). \tag{11b}$$

This assumption, as noted above, is more restrictive than necessary, but it allows us to separate the solution of the Schrödinger equations for $V'(x)$ into time independent and time dependent components

$$\Psi'(x, t) = \psi'(x)\phi(t), \tag{12}$$

in a piecewise continuous manner and therefore to represent the vibratory terms in equations (3) and (4) in an explicit manner. The $\Psi(x, t)$ can then be written in terms of $\Psi'(x, t)$ with time varying coefficients. (Note: we have here put the prime on the unperturbed potential so as to avoid carrying it through in the discussion to follow.)

Let the time interval $t_{i+1} - t_i$ denote the ith cycle of nuclear–electronic interactions. The electronic system is pictured as undergoing an evolution over some time interval τ and then as remaining constant while the nuclear system follows over a time interval $\tau_{i+1,0} - \tau_{i,i+1}$. The subscript $(i0)$ associates these intervals with the start time of the cycle. The time evolution of the complex may be described by an iterative (recursive) process of the following form:

$$\Psi_e(x_e, t_i + \tau_{i,i+1}) = \Psi_e(V_e(\Psi_e(x_e, t_i), \Psi_n(x_n, t_i)), t = t_i + \tau_{i,i+1}) \tag{13a}$$

$$\Psi_n(x_n, t_i + \tau_{i,i+1}) = \Psi_n(V_n(\Psi_e(x_e, t_i + \tau_{i,i+1}), \Psi_n(x_n, t_i)), t = t_i + \tau_{i,i+1}) \tag{13b}$$

In each interval the potential comprises a time independent part and a small time dependent perturbation, in accordance with equations (11a) or (11b). The sequence of cycles is initially entered by setting the potentials at time t_{i0}, which subsumes initial wave function Ψ_e and Ψ_n at time t_{i0}. These potentials are used to solve for Ψ_n at time $t_i + \tau_{i,i+1}$. The time index $i = 0$ is then updated to $i = 1$, $\Psi_e(x_e, t_i)$ and $\Psi_n(x_n, t_i)$ in equation (11a) are set equal to $\Psi_e(x_e, t_i + \tau_{i,i0})$ and $\Psi_n(x_n, t_i + \tau_{i,i0})$, respectively, and the cycle thereby re-entered. The potential V_e is updated and used to solve for $\Psi_e(x_e, t_i + \tau_{i,i+1})$, and so on.

The term "solving" could be replaced by "evolving in accordance with" since we are interested in a conceptual representation of the complex formation and decomposition process. A bona fide calculation scheme would have to introduce auxiliary procedures for approximating the wave functions and potentials. For example, the initial potentials and wave functions at time t_{i0} should represent the true situation. In principle these potentials and wave functions exist, but in practice they would have to be calculated or in some way measured; this auxiliary calculation process would have no necessary relationship to the self-organizing dynamics of the molecules.

In a practical calculation specific reference to different time intervals between significant changes in either the electronic or nuclear state might be dispensed with, since there would be no way of knowing what these time intervals are in advance without some kind of look-ahead procedure (e.g. using the change in the probability distribution of the electrons as a criterion). The intervals could be made arbitrarily small, so that one could simply switch back and forth between equations (6a) and (6b). However, these intervals serve to

highlight the feature that the nuclear motion is slower than the electronic motion. The electrons adjust rapidly to a change in the nuclear coordinates (in which case $\tau_{e(u)}$ is small), but they can also exert a leading effect due to the charge variations accompanying interference. A more descriptive notational picture might therefore utilize two electronic time constants—a fast one for the rapid readjustment to the change in the nuclear configuration and a slow one for the interference effect [8, 10]. In the present formalism the evolution from $\Psi_e(x_e, t_u)$ to $\Psi_e(x_e, t_u + \tau_{e(u)})$ on re-entering equation (13a) represents both rapid readjustment and the slower developing interference effect.

In reality the division into two interacting subsystems, electrons and nuclei, is a great oversimplification. The hydrogen bonds should be treated as a subsystem in their own right. As noted earlier, the de Broglie wavelength is large enough to allow for some tunneling motions. At the same time hydrogen bonds serve as preferred pathways for electron tunneling. The tunneling electrons are highly delocalized and therefore good candidates for exhibiting interference effects among possible states that are accessible through infrared perturbation. The hydrogen bonds are the nuclei that will be most agitated by the consequent variations in the charge distribution, and therefore the first nuclei to amplify the perturbation to the electronic system. As the variations in the charge distribution increase, the heavier nuclei will be increasingly affected, further contributing to the amplification effect. The possible sequences of events are much more complicated than in the two component schemes described by equations (13a, b). The scheme would obviously be even more complicated if the particular orbital characteristics of the various electrons were taken into account.

3.3. *Processing the wave function*

The iteration scheme (equations 13a, b) processes the superpositions of electronic and nuclear states (equations 3 and 4). Let us consider what can happen as these superpositions are iterated. Recall that the possible nuclear states are so closely spaced in energy that the fact of superposition can be ignored in favor of a classical description. The worst case assumption is that the electronic system is in a single eigenstate associated with nuclear conformation I. Conceivably the electronic wave function is consistent with the nuclear wave function. Conformation I will then be metastable, and could for all practical purposes be stable. For example, we can suppose that II is a lower energy state, but that the barrier between I and II is sufficiently high that thermal fluctuations are very unlikely to trigger a transition; quantum fluctuations in the static case would even be more unlikely to trigger a transition.

Suppose, however, that the perturbation to the electronic system induces transitions to other electronic states. The magnitude of the initial nonnegligible expansion coefficient will decrease, and the magnitude of one or more other expansion coefficients will increase. The nuclei will be agitated by the changing phase relationships among electronic states, since more than one state now has a nonnegligible probability amplitude. The initial perturbation may be from the radiation field, but the resulting agitation of the nuclei serves as an additional source of perturbation. This means that the expansion coefficients of the electronic subsystem will have a functional dependence on the expansion coefficients of the nuclear system, since both contribute to V_n.

Let us trace through some possible "execution sequences" of the wave functions in a more formal way. The original electronic wave function is determined, according to our maximally unfavorable assumption, by a single expansion coefficient $a_r(t_0)$. Due to infrared perturbations and variations of the potential due to thermal motions of the nuclei this is converted to a superposition with some set of nonnegligible expansion coefficients:

$$a_i(t_0) \rightarrow \{a_i(t_0 + \tau_{i(i)}),\ a_{i'}(t_0 + \tau_{i(i')}),\ a_{i''}(t_0 + \tau_{i(i'')}) \ldots\} \tag{14a}$$

Here i, i', i'' ... all belong to the index set of conformation I. The number of elements in this set is at least two, since by assumption one or more electrons are not closely tied to the nuclei. Interference phenomena, including charge variations, thus appear. The wave function of the nuclei is determined by an equivalence class of eigenfunctions that are hardly distinguishable from the macroscopic point of view. To keep the notation simple we will ignore this distinction and, as previously indicated, denote the nonnegligible expansion coefficient by a single expansion coefficient. Because of the electronic charge variations the initial nonnegligible equivalence class $b_i(t_0)$ undergoes the transition

$$b_i(t_0) \rightarrow b_{i'}(t_0 + \tau_{i(i')}). \tag{14b}$$

The wave function defined by the new equivalence class of coefficients $b_{i'}$ corresponds to a new configuration of the nuclei, which we suppose (so as not to beg the question) is still assignable to conformation I. The electronic system undergoes transitions to new states in response to the nuclear motions:

$$\{a_i(t_0 + \tau_{i(i)}),\ a_{i'}(t_0 + \tau_{i(i')}),\ a_{i''}(t_0 + \tau_{i(i'')}) \ldots\}$$

$$\rightarrow \{a_{i'}(t_0 + \tau_{i'(i')}),\ a_{i''}(t_0 + \tau_{i'(i'')}),\ a_{i'''}(t_0 + \tau_{i'(i''')}) \ldots\} \tag{14c}$$

This corresponds to re-entry into the second cycle, and therefore involves a fast readjustment of the electrons to the new nuclear potential ($V_{i'}$ at time $t_0 + \tau_{i(i')}$) and also the slower interference effect. The latter effect now has three sources: the infrared radiation field and thermal motions of the nuclei, as before, and in addition the agitation of the nuclei that results from the interference itself. This additional source of perturbation remains after the fast readjustment unless it is assumed that all the transitions go directly to a single electronic state. In reality of course the agitation of the nuclei begins at time t_0 and acts reciprocally on the electrons on a continuous time scale. All this is represented (conceptually) in the updated $V_{i'}$ at $t_0 + \tau_{i(i')}$, or equivalently at time t_1. The time $t_0 + \tau_{i(i'')}$ and time $t_1 + \tau_{i'(i'')}$ sets might be entirely disjoint, or there may be some overlap.

Completing the $n = 1$ cycle

$$b_{i'}(t_0 + \tau_{i(i')}) \rightarrow b_{i'}(t_0 + \tau_{i(i')}) \tag{14d}$$

and entering the $n = 2$ cycle

$$\{a_{i'}(t_0 + \tau_{i(i')}),\ a_{i''}(t_0 + \tau_{i(i'')}),\ a_{i'''}(t_0 + \tau_{i(i''')}) \ldots\}$$

$$\rightarrow \{a_{i'}(t_1 + \tau_{i'(i')}),\ a_{i''}(t_1 + \tau_{i'(i'')}),\ a_{i'''}(t_1 + \tau_{i'(i''')}) \ldots\} \tag{14e}$$

The wave function defined by the equivalence class of coefficients $b_{i'}$ might still be assignable to conformation I, but conceivably they are assignable to II. If so the $t_1 + \tau_{i'(i'')}$ set should also be predominated by coefficients belonging to conformation II, since the electronic system should make a fast readjustment to the new conformation. The molecule will then be locked in conformation II, assuming that this can eventually evolve to a self-consistent state or to a cycle of quasi-consistent states all of which belong to conformation II. If II does not admit such a state or set of states the complex will eventually switch back to conformation I (since the example has been unrealistically restricted to only two conformations).

Feedback between a quantum and a classical system is the important feature illustrated by the above sequence. *The superposition of possible electronic states serves as the channel through which motions of the nuclear system are fed back into the nuclear system.* If the feedback is positive the motions of the nuclei will be self-amplifying. The complex will exhibit increasingly agitated motion, and then undergo a sudden significant change of form

If the feedback is negative the electronic and nuclear system evolve to mutual consistency. Negative here means that the charge variations resulting from inteference push the nuclei back towards their initial arrangement. The nuclei will then oscillate around some equilibrium configuration. The vast majority of molecules fall to a state of lowest energy that is for all practical purposes completely consistent and time independent. From our point of view the feedback loops are present but of negligible importance in these cases. They are unmasked only in those molecules in which the nuclear motions are intricately constrained in three dimensions (conformation is important) and in which electrons are present that can move somewhat independently of the nuclei.

From equations (6a, b) and (7a, b) it can be seen that the potentials are, through the updating procedure, functionally dependent on each other. This is an alternative way of looking at the time dependence of the expansion coefficients and the coupling of these coefficients.

3.4. *Connection to fluctuation-dissipation*

Fluctuation is necessary to induce transitions of the electronic system to states of different energy. The time development of a perturbed molecule would then ordinarily involve dissipation of energy into the heat bath in accordance with the equipartition of energy. The nonlinearity of the electronic–conformational interaction allows this dissipation to be focused into selected degrees of freedom of the nuclear configuration. The self-amplifying (positive feedback) form of the interaction is driven by fluctuation in a focused way over an extended period of time. The agitation of the nuclei can serve to bounce energy back to the electronic system in a selective way, but the self-amplification would die out in the absence of heat bath fluctuation.

Fluctuation-dissipation is concomitant to all chemical reactions, since fluctuation is the source of collisions that drive reactions over activation barriers. No macroscopic work is performed. What is controlled, as in all catalytic processes, is the rate at which work is performed and the selectivity of the pathway.

4. COUPLED INSTABILITIES

So far we have pictured the complexing of two molecules, at least one of which must be a macromolecule. The formation of the complex is described by a dynamic free energy surface. The parallelism inherent in the quantum mechanical wave function (i.e. in the superposition of electronic states) speeds the search of this potential surface, and also greatly increases the likelihood of finding a unique minimum that is independent of the initial conditions. It does so by agitating the nuclei in a positive feedback fashion, thereby opening up new pathways for formation or decomposition, or lowering or narrowing existing barriers. If the complex that is formed is self-consistent, or if it allows for cycling among a family of conformationally equivalent states, we have self-assembly. The folding of an individual protein fits into this situation. The folded protein maybe viewed as self-organizing from the interactions among the different regions of a single covalently bonded sequence of amino acids. Enzyme–substrate interactions correspond to the situation of an unstable complex. If the enzyme recognizes the substrate it makes or breaks a particular covalent bond and then ejects it; if it does not recognize the substrate it dissociates without taking an action. Conformational switching in response to control molecules (allosteric control) can be interpreted in a similar manner.

Let us now turn to supermacromolecular complexes. We can think of these as highly structured arrangements of macromolecules that self-assemble in jigsaw puzzle fashion.

Such polymacromolecular constructions are ubiquitous in biology. They come in the form of extensive sheets (membranes), localized but highly intricate constructions (chloroplasts, mitochrondria, ribosomes), wide ranging but highly structured networks (the cytoskeleton), and linearly oriented architectures (chromosomes). These structures exhibit motions that help to choreograph the flow of components in the cell at all levels, ranging from directed movement of electrons in chloroplasts and mitochondria to the micromuscular activities of the cytoskeleton. In all likelihood they also serve as channels for signal flows that coordinate distant events.

For simplicity we will picture these polymacromolecular constructions as lattices, with each lattice point occupied by a macromolecule. Each molecule potentially interacts with any of its immediate neighbors. The interaction can be viewed in analogy to the interaction between an enzyme and substrate, except that in this case we are not concerned with the making or breaking of a covalent bond. The "substrate" acts solely as a control molecule that switches the enzyme into a particular conformation. The macromolecules of the lattice can thus act as control molecules relative to one or more of their neighbors and can act as controlled molecules relative to one or more neighbors.

Figure 2 illustrates in schematic fashion how interactions of the type proposed in the previous section could lead to a propagating conformational change in a one-dimensional lattice built from a single type of macromolecule. Each molecule has two conformations available to it, I and II, as in our previous discussion. Conformation I is represented by a left handed arrow and conformation II by a right handed arrow. Conformation I is self-consistent (stable) except when it complexes with a conformation II molecule, in which case it undergoes a transition to conformation II (complexing is represented by two arrowheads pointing towards each other). Similarly conformation II is stable except when it complexes with conformation I, in which case it reverts to conformation I. The initial conformational switch is triggered by an externally imposed input on the far left hand side of the chain. The conformational change propagates through the chain until it reaches the right hand side, where it triggers an effector molecule into an active state.

To the above lattice we have attributed properties inherited from the superposition-amplification model described in the previous section and characteristic of the biological examples mentioned above. The subsumed conformational changes are significant, they have a high degree of directional specificity, and the rate of propagation is high. The changes can propagate over substantial distances in a homogeneous lattice in a manner that maintains their original form. In an inhomogeneous lattice the conformation change would

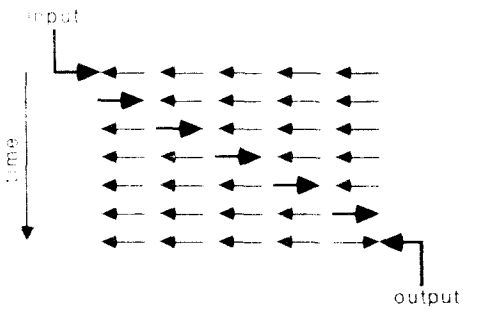


Fig. 2. Schematic representation of a conformational soliton propagating through a one-dimensional polymacromolecular lattice. The first frame (top line) represents the presence of an external input that triggers the initial switch in conformation. Each subsequent frame represents the migration of the switch to the next lattice position. The last frame (bottom line) represents the triggering of an output action.

itself change in character depending on the particularities of the neighboring molecules, but could do so without losing its informational integrity.

The above properties are incompatible from a classical point of view, but are generic to superposition–amplification models since in these the time development is generated by inherent instabilities. In a classical model purely thermal fluctuations could lead to switching between significantly different conformational states, but the potential barriers would have to be low and consequently the directional specificity with which the conformation changes ripple through the lattice could not be high. If the potential barriers were high enough to confer high selectivity on the propagation patterns, the rate of propagation would be low. In the absence of high specificity and high speed the spatial range over which it would be possible to maintain informational coherence would be compromised. But according to our superposition-amplification model the chance that a macromolecular aggregate exhibits these properties simultaneously becomes more rather than less likely as the architectural complexity of the aggregate increases, since the chance of satisfying self-consistency conditions decreases as the number of constraints that have to be satisfied increases.

5. CONCLUDING REMARKS

The new feature of our model is the manner in which the linear superposition principle is embedded in a nonlinear interaction. The nonlinear interaction between the electrons and nuclei constitutes a feedback loop, but one that cannot be treated in its entirety as if all elements are describable by classical variables. The rich structure of the quantum mechanical wave function is drawn into the loop and hence contributes to specificity and speed.

The nonlinearity is here built on top of linear quantum mechanics, not built into it in a fundamental way. It enters in the same way as it would if Newton's equations were used—by focusing attention on major components of a system and ignoring details of the heat bath. Our analysis begins with the standard Schrödinger equation. In complete microscopic detail the admissable solutions would be confined to unitary transformations (translations and rotations in Hilbert space, no stretching or folding). The time dependence of the potentials in our model and the treatment of the nuclear configuration in classical terms rests on the departure from a completely detailed description. The time development of the system of interest—the interacting molecules *sans* heat bath—is no longer confined to unitary transformations. Exotic dynamic phenomena can occur that would appear to be excluded by the Schrödinger equation.

Sensitive dependence on initial conditions suggestive of chaos is an example. Divergence of trajectories (stretching) can be physically interpreted in terms of the spreading out of the electronic wave function. The consequent interference induces an irregular motion of the nuclei. This is the phase of the electronic–conformational interaction in which the electrons lead. Confinement of trajectories to some domain (folding) can be interpreted in terms of the adjustment of the electronic levels to changes in the nuclear configuration. This is the phase of the interaction in which the electrons follow. If the number of nonnegligible eigenfunctions is very small the electronic structure will approximate time independence; if it is very large an ergodic, statistically regular regime will be approached. Between these extremes the nuclei should undergo a series of configurational changes that are highly irregular and initial condition sensitive. If folding (confinement) dominates the series will belong to a single conformation. This is the negative feedback situation. If stretching (divergence) dominates jumping to a new conformation (basin of attraction) will occur. This is the positive feedback situation.

The model makes a number of experimental predictions of a general nature. Sustained low frequency radiation should affect the rate of complex formation by inducing closely spaced transitions among possible electronic states. It should be possible to introduce color groups into proteins in such a way that folding and complex formation are disrupted in the presence of visible light, but not in the dark. It should be possible to alter the rate of complex formation by replacing hydrogen atoms by deuterium or tritium; the effect would be pronounced for those hydrogen bonds that provide preferred pathways for electron tunneling. If confinement of parallel spin electrons can augment superpositional speedup by inducing transient delocalization this should be detectable as a short-lived alteration of a spin resonance signal prior to a shift in conformation.

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AN IMPROVED NANOSCALE TRANSMISSION LINE MODEL OF MICROTUBULE: THE EFFECT OF NONLINEARITY ON THE PROPAGATION OF ELECTRICAL SIGNALS

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Abstract. *In what manner the microtubules, cytoskeletal nanotubes, handle and process electrical signals is still uncompleted puzzle. These bio–macromolecules have highly charged surfaces that enable them to conduct electric signals. In the context of electrodynamic properties of microtubule, the paper proposes an improved electrical model for divalent ions (Ca^{2+} and Mg^{2+}) based on the cylindrical structure of microtubule with nano–pores in its wall. Relying on our earlier ideas, we represent this protein–based nanotube with the surrounding ions as biomolecular nonlinear transmission line with corresponding nanoscale electric elements in it. One of the key aspects is the nonlinearity of associated capacitance due to the effect of shrinking/stretching and oscillation of C–terminal tails. Accordingly, a characteristic voltage equation of electrical model of microtubule and influence of capacitance nonlinearity on the propagation of electrical pulses are numerically analyzed here.*

Key words: *microtubule, electrodynamic properties, nonlinear transmission line, voltage equation, soliton wave*

1. INTRODUCTION

Biological systems at the nanoscale are rich in electrical activity. Besides the well–known pathways of biochemical regulation, there exist additional pathways of biological communication governed by electrical signals [1]. Recently, *in silico* demonstration has shown that the microtubules (MTs), essential cellular biopolymers, are the source of electromagnetic fields in the form of the electric pulses, playing an important role in the intracellular signaling and information processing [2]. MTs are self–assembling protein mesostructures made up of electrically polar tubulin heterodimer subunits, α– and β–tubulin monomers. Each dimer, 8 nm in length, possesses two highly flexible C–terminal tail regions, by one of each monomer, which can extend up to 4.5 nm from the surface, see Fig. 1a. *In vivo* MT is generally made up of 13 parallel, loosely connected chains, or

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protofilaments, formed by polymerization of the tubulin heterodimer molecules so as to result in a helical structure. Cryo–electron microscopic analyses indicate that the single MT resembles a hollow tube with its external and internal diameters of 25 nm and 15nm, respectively, see Fig. 1b, while its length typically varies in the range of few tens of nanometers to several micrometers. Due to the shape and spacing of the tubulin heterodimers, the MT possesses small, nanometer size pores between the outer environment and the inner MT's lumen. Two different types of these nano–pores exist within MT's wall.

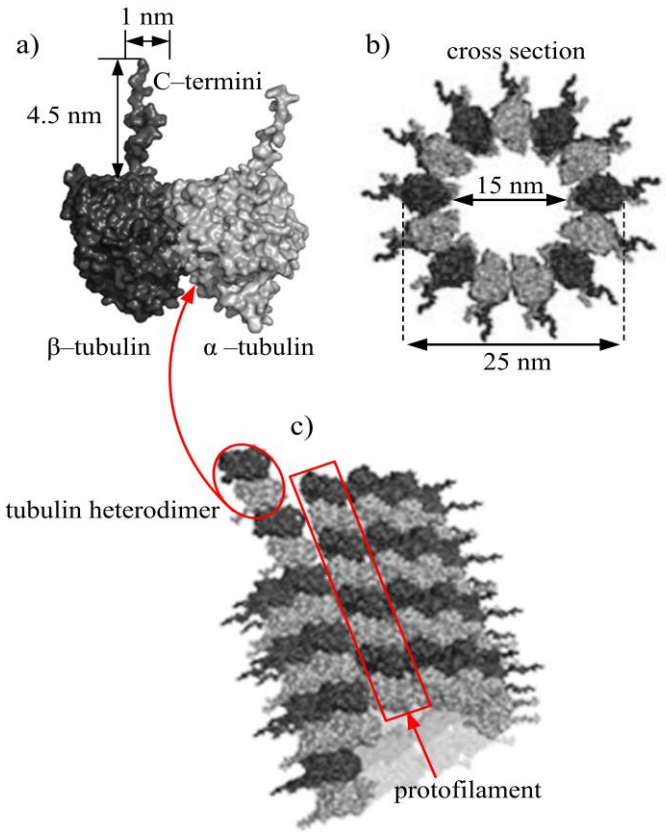

Fig. 1 a) The topology of tubulin heterodimer with dimensions of the C– terminal tail regions b) Cross–sectional view of microtubule c) Microtubule tube–like structure with marked protofilament

MT functionalization plays a critical role in making the jump from *in vivo* intracellular transport to *in vitro* nanoscale device applications [3]. The potential of this protein structure for application in novel bio–inspired nanoelectronic components was recently demonstrated [4]. In addition, the process of formation of MTs by polymerization of αβ–tubulin heterodimers, which can be controlled by physical (temperature) and chemical (pH, concentration of protein and ions), can result to the production of closely or widely spaced MTs, centers, sheets, rings and other structures [5], thus facilitating fabrication of

nanowires, nodes and networks in the future for bionanotechnological applications. Interestingly, MTs are self–assembling protein–based nanotubes with mechanical behavior similar to carbon nanotubes despite their very different chemical composition: proteins and non–covalent interactions in the case of MTs; carbon and covalent bonds in the case of carbon nanotubes [6].

In view of the key role of MTs in intracellular electrodynamic signaling and information processing, the objective of this study is to provide a framework for the analysis of propagation of electric pulses along this biological transmission line. More specifically, for this purpose we improve our previous electrical model [7] based on polyelectrolyte concept applied to the molecular structure and geometry of these nanotubes. The initial motivation for the model of MT as electrical transmission line was the experimental evidence [8], which revealed that these biopolymers are good conductors of electrical signals at nano–level. Here, we will focus on the specific importance of this mechanism for intracellular transportation of divalent ions, in the first place Ca^{2+} and Mg^{2+} as the most important ions in human body. The particular attention will be paid to the role of nonlinearity parameter that arises from structure and dynamics of C– terminal tails.

2. Electrodynamic Properties and Characterization of MT

The conductance of MTs is the result of their electrostatic and structural properties. On the basis of detailed molecular dynamics computations, it was recently demonstrated that the outer surface of tubulin heterodimer is mostly electrostatically negative charged, whereby a specifically large negative charge is located on the C–terminal tail regions [9]. As a result, overall surface of MT is predominantly negative charged, as one can see from Fig. 2, while the ends of this nanotube possess a different amount of net charge. Accordingly, it can be readily concluded that the MT supports an intrinsic electric field [5]. The average linear charge density of single MT is estimated to be 52.2 electronic charges (e) per dimer or approximately 85 e/nm [10]. Also, recent experimental observation was showed that the MT conductivity in a solution is on the order of 10 nS [8], indicating a high level of ionic conductivity along this biopolymer.

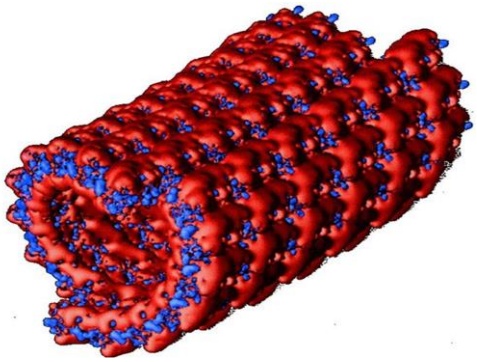

Fig. 2 Charge distribution on the surface of microtubule (color red indicates negative charges, blue positive charges and white neutral regions) [9]

In the context of the above–mentioned findings, it is possible to consider MT like one–dimensional polymer that behaves as highly charged polyelectrolyte [7, 10]. In a solution, positive ions are attracted to the negative charged surface of the MT, while negative ions are repelled, creating a positive condensed ionic cloud (CIC) localized around the MT's landscape. In the presence of an applied voltage, the loosely held CIC is free to migrate along MT, creating an ionic flow and an associated electrical current [11]. In accordance with Manning's condensation theory of polyelectrolytes [12], negative ions in solution are repelled at the distance called the Bjerrum length (l_B), which is defined by the balance between electrostatic attraction energy of the ions and the pertaining thermal energy:

$$l_B = \frac{ze^2}{4\pi\,\varepsilon_r\varepsilon_0 k_B T}. \tag{1}$$

The particular emphasis is paid on the divalent ions (z = 2) such as Ca^{2+} and Mg^{2+}, which are primary attracted by the MT's surface. Their flows around MTs play fundamental roles in auditory processes as well as in the cardiac muscles action. For these important ions in human body, the width of such "depleted layer" sandwiched between two charged regions is l_B = 1.34 nm at physiological temperature T = 310 K. Other parameters used in the estimation of Bjerrum length are the elementary electric charge e = 1.6×10^{-19} C, the dielectric constant of cytosol ε_r = 80, the permittivity of vacuum ε_0 = 8.85×10^{-12} F/m and Boltzmann's constant k_B = 1.38×10^{-23} J/K.

Based on Manning's approach, the thickness of CIC λ around the rod–like filamentous polyon of radius r is given by the following expression:

$$\lambda = \mathcal{A}\left(rl_{Db}\right)^{1/2};\quad \mathcal{A}<1,\ \ l_{Db}^{-1} = \left(8\pi\, nl_B\right)^{1/2}. \tag{2}$$

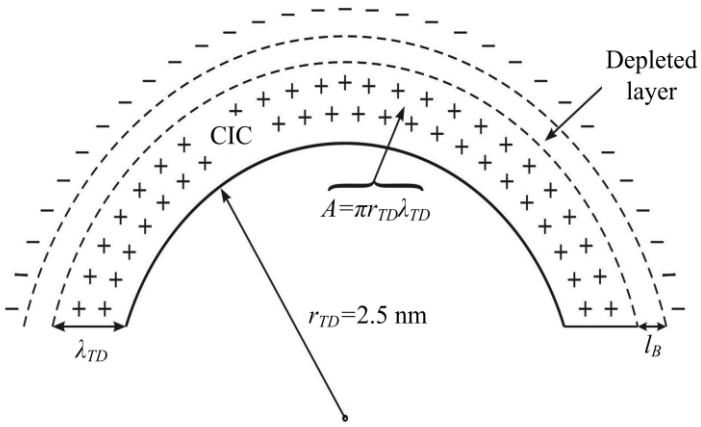

Fig. 3 Schematic illustration of a tubulin dimer with its condensed ionic cloud (CIC), depleted layer and the geometry of "coaxial cable" with the dimensions. *A* is the cross section area of a CIC around tubulin dimer.

Using the above equation, the corresponding values for tubulin heterodimer (TD) and C– terminal tails (CT) are λ_{TD} = 2.96 nm and λ_{CT} = 0.92 nm, respectively, for equilibrium ionic concentration $n = 1.5\times10^{23}$ m^{-3} and parameter $\mathcal{A}$ = 0.5. The estimated depleted layer between CIC and repelled negative ions (anions) plays the role of a dielectric medium between charged plates in "coaxial cable", see Fig. 3. It provides resistive and capacitive components for the behavior of the tubulin heterodimer that make up the MT.

2.1. Electrical parameters of the MT model

Bearing in mind the CIC around the MT's cylinder, this protein–based nanotube may act as biological "electrical wire" which can be modeled as nonlinear transmission line [7, 11]. In this section, we describe the nature and estimate the value of electrical components at nanoscale that make up transmission line model that mimics the behavior of MT in solution. Due to symmetry of this biopolymer, it is plausible to consider just one of thirteen MT's protofilaments and to introduce the so called elementary electric unit, which is a single tubulin dimer with its two C–terminal tails and two different types of nano–pores in MT's wall.

In order to estimate the total static capacitance of elementary electric unit for the case of divalent ions, we firstly consider the contribution of a tubulin heterodimer as half–cylindrical capacitor which has the value

$$C_{TD} = \frac{\pi\varepsilon_r\varepsilon_0 l}{\ln\left(1+\frac{l_B}{R}\right)} = 0.75\times10^{-16}\ \text{F}. \tag{3}$$

Here l = 8 nm is the length of tubulin dimer, and the parameter $R = r_{TD} + \lambda_{TD}$ = 5.46 nm represents the outer radius of respective CIC, see Fig. 3, while the other parameters are already mentioned and used. In a similar way, viewing an extended C–terminal tail as a smaller cylinder with $r = r_{CT} + \lambda_{CT}$ = 1.42 nm, one obtains its capacitance as follows

$$C_{CT} = \frac{2\pi\varepsilon_r\varepsilon_0 l^{\text{eff}}}{\ln\left(1+\frac{l_B}{r}\right)} = 0.14\times10^{-16}\ \text{F}, \tag{4}$$

where $l^{\text{eff}} = l_{CT} - \lambda_{TD}$ = 1.54 nm is the corresponding effective length of the part of C–terminal not plunged in λ_{TD}. Since each tubulin heterodimer has two C–terminal tails, and keeping in mind that the capacitances of tubulin dimer and C–terminal tails are in parallel arrangement [13], it implies that total maximal static capacitance of the elementary electric unit is readily estimated as the simple sum of above determined components:

$$C_0 = C_{TD} + 2\times C_{CT} = 1.03\times10^{-16}\ \text{F}. \tag{5}$$

It is evident that the capacitance in our model represents the charge distribution in the region from CIC to approximately one l_B away perpendicularly to the surface of the MT. Since the ions in this layer are assumed to be condensed, the charge of this capacitor will vary in a nonlinear way with voltage as follows:

$$Q_n = C_0[1-\Gamma\Omega(t-t_0)-bv_n]v_n \ . \tag{6}$$

The non–stationary term $\Gamma\Omega(t–t_0)$ reflects the fact that the presence of extra ions injected within the ionic cloud affects the local concentration during the time compared with the characteristic charging time of localized C–terminal's capacitor. The nonlinear parameter b represents the change of capacitance of elementary electric unit with an increasing concentration of condensed cations due to the flexibility of the C–terminal tails (their shrinking/stretching). This expression (6) was derived and explained in detail in the previous version of electrical model of MT [7].

In order to estimate the conductance of nano–pores, we rely on the detailed atomic–scale *in silico* calculations using 3D Brownian dynamics [14]. In accordance with these numerical results, the conductance of both nano–pores is estimated to be G = 10.7 nS, which is determined as a sum of pretty different components reflecting difference in two types of nano–pores [7, 13]. Using the same numerical calculation, the resistance of the parallel flow of ions along elementary electric unit is $R = 6.2\times10^7$ Ω, neglecting the ionic current which leaks through the depleted layer. Finally, in the earlier simplified version of the electrical model of MT [11], we found that corresponding inductance of elementary electric unit is small enough that its role can be safely ignored in this improved model.

3. ELECTRICAL MODEL OF MT AS BIOMOLECULAR NONLINEAR TRANSMISSION LINE

On the basis of above estimations for the components of elementary electric unit of the MT, we are now in the position to establish the corresponding electrical model. Electric circuit that simulates single MT's protofilament is presented in Fig. 4.

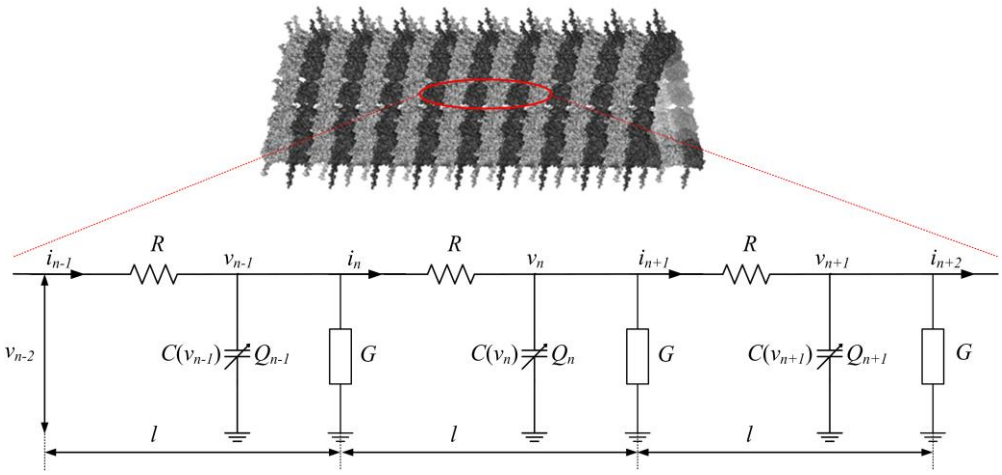

Fig. 4 A scheme describing the corresponding electrical model of MT

In order to derive the characteristic voltage equation of model, we begin by introducing the discrete potential in one section of the MT where Kirchhoff's laws for the currents and voltages read as follows

$$i_n - i_{n+1} = \frac{\partial Q_n}{\partial t} + G v_n , \qquad v_{n-1} - v_n = R i_n . \tag{7}$$

We then introduce new function $u_n(x, t)$ unifying voltage and its accompanying current along a MT as follows

$$u_n(x,t) = Z^{1/2} i_n = Z^{-1/2} v_n, \tag{8}$$

where Z is characteristic impedance of the elementary electric unit corresponding to characteristic frequency $\omega = 2\pi/(RC_0)$

$$Z = R + \left(\frac{G^{-1}}{C_0^2 \omega^2 G^{-2} + 1} \right) \left(1 + \frac{C_0^2 \omega^2}{G^2} \right)^{1/2} . \tag{9}$$

This expression for Z is much more accurate compared to the previous that was used in [7]. Assuming that a large number of tubulin heterodimer along the protofilament of MT, the next step is to go over to the continuum approximation with respect to space variable x and as a result we get the following voltage equation of established model of MT

$$\frac{\partial^3 u}{\partial \xi^3} + 3\left(\frac{ZC_0 s}{T} - 2 \right) \frac{\partial u}{\partial \tau} + 3ZC_0 \Gamma\Omega \left(\xi - \xi_0 \right) \frac{\partial u}{\partial \xi} + 6 \frac{Z^{3/2} b C_0 s}{T} \left(u \frac{\partial u}{\partial \xi} \right) + 3(ZG + Z^{-1} R - ZC_0 \Gamma\Omega) u = 0. \tag{10}$$

Here, the first term is dispersive one arising from diffusive spreading of ions within the pulse. The second one with negative sign ($ZC_0 s < 2$) resembles the corresponding time dependent term in Fick's diffusion law. Third term reflects the non–stationary change of capacity due to the ratchet mechanism. Fourth term represents nonlinearity that competes with dispersion. The key point here is that the last term accounts the competition between energy expenditure due to ohmic losses and energy supply by non–local ratchet mechanism. This term can be discarded as being very small due to the balance of above competition. In characteristic voltage equation (10), the characteristic charging (discharging) time of the elementary capacitor through the elementary resistance is given by $T = RC_0$, while the dimensionless variables of space ξ, time τ and velocity s are defined as follows:

$$\xi = \frac{x}{l} - \tau , \quad \tau = s \frac{t}{T} , \quad s = \frac{\mathrm{v}}{\mathrm{v}_0} , \quad \mathrm{v}_0 = \frac{l}{T} , \tag{11}$$

where parameter $\mathrm{v}_0 = l/T$ represents the characteristic cut–off velocity, and l is the length of one tubulin heterodimer.

3.1. Numerical analysis of the voltage equation

Since one of the key aspects of electrical model of MT as transmission line is the nonlinearity of capacitance of elementary electric unit, the characteristic voltage equation and effect of capacitance nonlinearity on the propagation of ionic pulses along MT are numerically analyzed. Numerical simulations are based on a finite–difference time–domain method applied to the differential equation (10) governing the voltages of proposed nonlinear circuit [15]. A circuit simulator is implemented in *Matlab* software package.

In Fig. 6 is shown the numerical results of voltage equation $u(\xi, \tau)$ for estimated values of electric components of the established model and different values of capacitance nonlinearity.

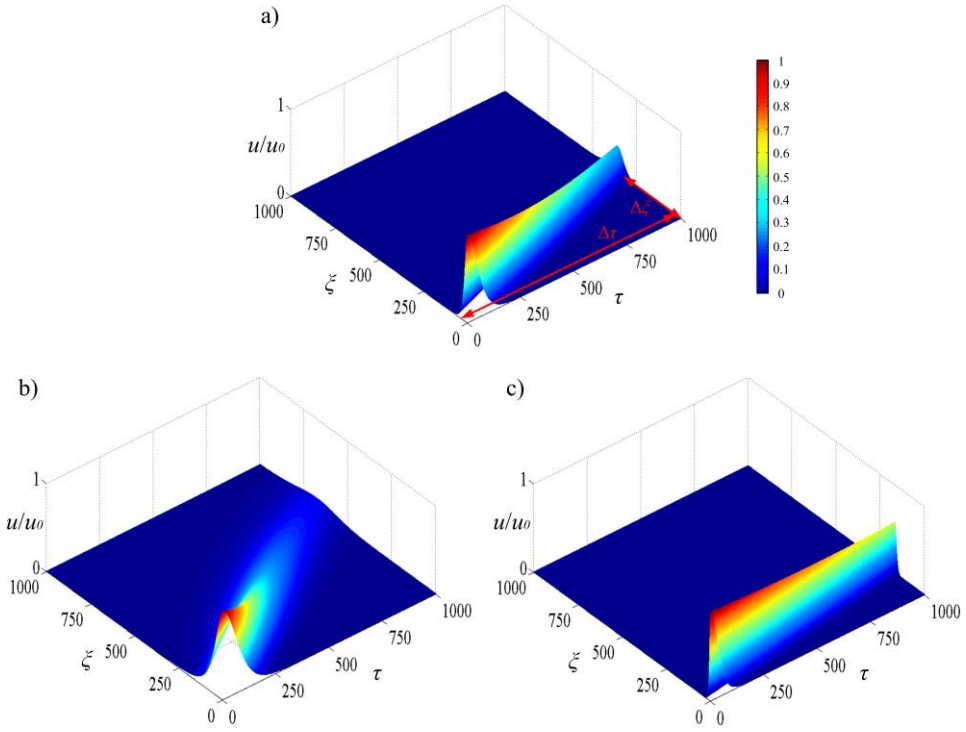

Fig. 5 Numerical solution of the voltage equation $u(\xi, \tau)$ for the set of estimated electric components of the established model and a) capacitance nonlinearity of 0.05 V^{-1}, b) lower value of capacitance nonlinearity of 0.01 V^{-1} and c) higher value of capacitance nonlinearity of 0.1 V^{-1}

For reasonable value of the nonlinearity parameter of 0.05 V^{-1}, the pulse appears in the form of soliton–like wave exhibiting the slowly decaying amplitude and an almost constant velocity of propagation. This wave possesses the energy losses due to ohmic resistance, but it preserves the stable localized form. Using the values of time and space units, the average velocity of localized voltage pulse can be estimated from the obtained graphic shown in Fig. 5a as follows:

$$\langle \mathrm{v} \rangle = \frac{\Delta\xi}{\Delta\tau} = \frac{280l}{1000T_{\mathrm{CT}}} \cong 3\frac{\mathrm{cm}}{\mathrm{s}}, \tag{12}$$

where l is length of elementary electric unit that is one tubulin heterodimer. The parameter T_{CT} represents a swing period of C–terminal tail which is given by the following expression [10]

$$T_{\mathrm{CT}} = 6\pi \frac{ZC_0}{ZG + Z^{-1}R} \cong 0.72 \times 10^{-7}\,\mathrm{s}. \tag{13}$$

The obtained propagation velocity of voltage pulse is the order of several cm/s that is close to experimental findings [8] and it depends on the electrical papameters of established circuit model of MT. It is also possible to estimate that the pulse width is of the order of 10 elementary units. This fact supports the validity of continuum approximation. The range of this soliton–like localized pulse is about $280 \times l \approx 2.24$ μm, which is of the order of cell's diameter.

In the case of lower value of nonlinearity parameter of 0.01 V^{-1} presented in Fig. 9b, the amplitude of pulse decreases significantly faster so that over about 350 elementary units it becomes negligible. The graphic shows the deceleration of the soliton–like pulse along its path. For higher value of capacitance nonlinearity of 0.1 V^{-1}, the voltage pulse exhibits not only a higher localization, but also a slower decay of its amplitude and greater robustness compared to the previous two cases. Also, Fig. 5c indicates that the propagation velocity is lower than the average velocity of 3 cm/s for the case of nonlinearity parameter of 0.05 V^{-1}. Overall, the performed numerical analysis of the voltage equation demonstrates that the role of capacitance nonlinearity is of decisive importance for the stability and localized character of Ca^{2+} and Mg^{2+} pulses along MTs.

4. Discussion and Conclusion

In this paper, we have developed an improved electrical model of MT that provides qualitative framework for the observations of propagation of Ca^{2+} and Mg^{2+} signals along this charged polar biopolymer. In summary, the geometric symmetry of MT provided the opportunity that one MT's protofilament can be approximately seen as a "coaxial cable" with the "charged plates" composed of counterions of solution with a depleted layer of water molecules in between. It enabled that the series of tubulin heterodimers with pertaining C– terminal tails could be observed as series of identical elementary electric units with estimated capacitance and resistance, including the conductance of two different types of nano–pores existing in MT's wall. The background for the proposed model is the molecular structure of MT and its polyelectrolyte properties.

Numerical analysis of characteristic voltage equation of model showed that the nonlinearity of capacitance, arising from structure and dynamics of C– terminal tails, is essential important parameter for the character propagation of Ca^{2+} and Mg^{2+} signals along MT. For reasonable value of the nonlinearity parameter of 0.05 V^{-1}, numerical results indicated stable and localized soliton–like pulses which propagate with a velocity value close to experimental findings and cross the distance of cell's size despite ohmic dissipation. Velocities of these pulses are of the order of several cm/s, depending on the amount of the injected ions and valence of the ions. Under physiological conditions, this effect is much faster than the process of propagation of the ions by pure diffusion (a few μm/s). Interestingly, if we compare the estimated velocity with the drift velocity of electrons in a typical semiconductor, we get the same order of magnitude at current density of order of A/mm^2. However, the velocity of establishing a local field around the protein biopolymers is much lower due to heaviness of the ions, which are much more massive than electrons.

Finally, we believe that our predictions may have important consequences for understanding of MT's ability to conduct electrical signals, which may affect the neuronal computational capabilities, among other known functions. Also, our findings could encourage experimentalists to conduct more subtle assays in an attempt to elucidate this important aspect of cellular activities. On the other hand, the much–needed experimental validation of presented results would undoubtedly lead to exciting new opportunities in the development of biological nanoscale electronic applications.

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Anharmonic Oscillator. II. A Study of Perturbation Theory in Large Order

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This paper is concerned with the nature of perturbation theory in very high order. Specifically, we study the Rayleigh-Schrödinger expansion of the energy eigenvalues of the anharmonic oscillator. We have developed two independent mathematical techniques (WKB analysis and difference-equation methods) for determining the large-n behavior of A_n^K, the nth Rayleigh-Schrödinger coefficient for the Kth energy level. We are not concerned here with placing bounds on the growth of A_n^K as n, the order of perturbation theory, gets large. Rather, we consider the more delicate problem of determining the precise asymptotic behavior of A_n^K as $n \to \infty$ for both the Wick-ordered and non-Wick-ordered oscillators. Our results are in exact agreement with numerical fits obtained from computer studies of the anharmonic oscillator to order 150 in perturbation theory.

I. INTRODUCTION

This paper reports an intensive study of the behavior of perturbation theory in very high order for the anharmonic oscillator.[1] We give a detailed treatment of the new mathematical techniques we have developed to obtain this behavior and we discuss the related physical ideas.

The quantum anharmonic oscillator is described by the time-independent Schrödinger equation

$$\left[\frac{d^2}{dx^2}+\frac{x^2}{4}+\frac{\lambda x^4}{4}-E(\lambda)\right]\psi(x)=0 \tag{1.1}$$

with the associated boundary condition

$$\lim_{|x|\to\infty}\psi(x)=0\,. \tag{1.2}$$

The boundary condition selects out a discrete set of energy eigenvalues $E^K(\lambda)$, $K=0,1,2,\ldots$, whose perturbation expansions take the form

$$E^K(\lambda)\sim K+\tfrac{1}{2}+\sum_{n=1}^{\infty}A_n^K\lambda^n\,. \tag{1.3}$$

We refer to Eq. (1.3) as the Rayleigh-Schrödinger series and A_n^K as a Rayleigh-Schrödinger coefficient. We investigate the behavior of A_n^K as n, the order of perturbation theory, becomes large.

The anharmonic oscillator is also expressible as a $(\lambda\phi^4)_1$ quantum field theory.[2] The Hamiltonian for the theory may be written as

$$H=\tfrac{1}{2}\dot{\phi}^2+\tfrac{1}{2}m^2\phi^2+\lambda\phi^4\,, \tag{1.4}$$

where

$$[\phi,\dot{\phi}]=i\,. \tag{1.5}$$

The energy levels $E^K(\lambda)$ are the eigenvalues of H:

$$H|K\rangle=E^K(\lambda)|K\rangle\,. \tag{1.6}$$

For all m, the field-theoretic perturbation series takes the form

$$E^K(\lambda)\sim m\left(K+\tfrac{1}{2}+\sum_{n=1}^{\infty}A_n^K(\lambda m^{-3})^n\right). \tag{1.7}$$

One can recover Eq. (1.3) by setting $m=1$ in Eq. (1.7).

Viewing the anharmonic oscillator as a quantum field theory, we can express A_n^K in terms of Feynman diagrams.[3] A_n^0 is the sum of all connected diagrams having n vertices and no external legs. A_n^K is the K-particle pole of the $2K$-point Green's function. Thus, the problem of determining the large-n behavior of A_n^K corresponds to studying diagrams with many vertices. There has already been an intensive study of the properties of many-vertex diagrams in higher-dimensional quantum field theory[4] as well as in the anharmonic oscillator.[5] However, these studies give only bounds on the growth of the Rayleigh-Schrödinger coefficients and do not reveal much about the mathematical and physical nature of these theories. In this paper we completely by-pass the problem of obtaining bounds. Instead, we solve here the much more formidable problem of determining the precise asymptotic behavior of A_n^K for large n.

This paper will conform to the following outline: In Sec. II we use dispersion techniques to derive an exact relation between the nth Rayleigh-Schrödinger coefficient and the lifetimes of the unstable states of a negatively coupled anharmonic oscilla-

tor ($\lambda = -\epsilon$, $\epsilon > 0$). Section III contains a WKB derivation of approximate lifetimes for small ϵ. In Sec. IV we combine the results of Secs. II and III to find the leading behavior of A_n^K for large n. In Sec. V we discuss the relation between the WKB analysis of Sec. III and other WKB results we obtained in a previous paper. Then we compute the corrections (of order n^{-1}) to the leading behavior of A_n^K given in Sec. IV. A completely independent derivation of the results of Sec. IV for the two lowest energy levels is given in Sec. VI. This derivation uses difference-equation and generating-function methods. Finally, in Sec. VII we define and discuss the large-order perturbation behavior of the Wick-ordered version of the Hamiltonian in Eq. (1.4). The Appendix gives a short and physically intuitive derivation of the material in Sec. III.

The results of this paper may be summarized in two equations. For large n,

$$A_n^K = \frac{(-1)^{n+1}12^K\sqrt{6}}{K!\pi^{3/2}}\,3^n\Gamma(n+K+\tfrac{1}{2}) \times\left[1-\frac{1}{n}\left(\frac{95}{72}+\frac{29K}{12}+\frac{17K^2}{12}\right)+O(n^{-2})\right] \tag{1.8}$$

and

$$\frac{A_n^K}{B_n^K} = e^{-3}\left(1+\frac{1+3K}{n}+\frac{\frac{71}{6}+31K+23K^2}{4n^2}+O(1/n^3)\right), \tag{1.9}$$

where B_n^K is a Rayleigh-Schrödinger coefficient for the Wick-ordered perturbation series. These results have been verified: On a computer we have carried out perturbation theory to order $n = 150$ (the limiting factor is the amount of available core memory), performed numerical fits,[6] and obtained spectacular agreement.

In this paper we have established an easily generalized mathematical framework for determining the large-order behavior of perturbation theory for more complicated models. Three such models have already been successfully investigated: The anharmonic oscillator with a λx^{2N} perturbation,[7] the anharmonic oscillator with an arbitrary polynomial interaction of the form[8]

$$\lambda(x^{2n}+ax^{2N-2}+bx^{2N-4}+cx^{2N-6}+\cdots),$$

and the N-mode problem (N-coupled anharmonic oscillators).[9] This last problem involves applying WKB methods to partial differential equations and is thus far more difficult and profound than the problem considered here. We hope to extend these methods to infinite-mode problems and thereby obtain results for higher-dimensional quantum field theories.

II. DISPERSION RELATION FOR $E^K(\lambda)$

In this section we use a once-subtracted dispersion integral to establish an exact relation between A_n^K, the nth Rayleigh-Schrödinger coefficient for the Kth energy level, and the lifetime of the Kth (unstable) state of a negatively-coupled anharmonic oscillator. This relation follows from three rigorous properties of $E^K(\lambda)$ in the cut λ plane (cut along the real λ axis from $-\infty$ to 0): (a) $|E(\lambda)| \sim |\lambda|^{1/3}$ as $|\lambda|\to\infty$; (b) $E(\lambda)$ is analytic; and (c) the series in Eq. (1.3) is asymptotic. Property (a) is a consequence of a simple scaling argument known as the Symanzik transformation applied to Eq. (1.1).[10] Property (b) is a profound result obtained by Loeffel and Martin.[11] Property (c) was established by Loeffel, Martin, Simon, and Wightman.[12]

Properties (b) and (c) were originally established heuristically by us.[13] They follow directly from a study of Eqs. (1.1) and (1.2) in the complex x plane. The boundary condition Eq. (1.2) holds in a pair of sectors of angular opening $\frac{1}{3}\pi$ centered about the real x axis. As λ rotates into the complex λ plane ($\arg\lambda$ increases from 0), the sectors rotate clockwise. (A full description of the rotating sectors is given in BW.) The distant turning points[2] at $x_1 \sim \pm i\lambda^{-1/2}$ rotate clockwise and faster than the sectors rotate. The turning points enter the sector when $\arg\lambda = \pi$ (λ is negative). When $\arg\lambda < \pi$ WKB methods[2] give the asymptotic series in Eq. (1.3). When $\arg\lambda > \pi$ the WKB results are completely different[2] and predict nonanalyticity (branch points at which level crossing occurs) of $E^K(\lambda)$ in the λ plane. Thus, at $\arg\lambda = \pi$ (and $-\pi$), the distant turning point which enters the sector marks the edge of the region in the λ plane where the series in Eq. (1.3) is asymptotic and where the nonanalyticity of $E(\lambda)$ appears. While not rigorous, these are strong arguments for the truth of properties (b) and (c).

We proceed to derive the dispersion relation. Properties (a), (b), and (c) motivate the definition

$$F^K(\lambda) \equiv [E^K(\lambda) - K - \tfrac{1}{2}]\lambda^{-1}. \tag{2.1}$$

In the cut λ plane, $F^K(\lambda)$ is analytic and

$$\lim_{|\lambda|\to 0} \lambda F(\lambda) = 0, \tag{2.2}$$

$$|F^K(\lambda)| \sim |\lambda|^{-2/3} \text{ as } |\lambda|\to\infty. \tag{2.3}$$

Equation (2.2) is a consequence of property (c).

Since $F^K(\lambda)$ is analytic in the cut λ plane, we use the Cauchy theorem to write

$$F^K(\lambda) = \frac{1}{2\pi i}\oint_C \frac{F^K(x)}{x-\lambda}\,dx. \tag{2.4}$$

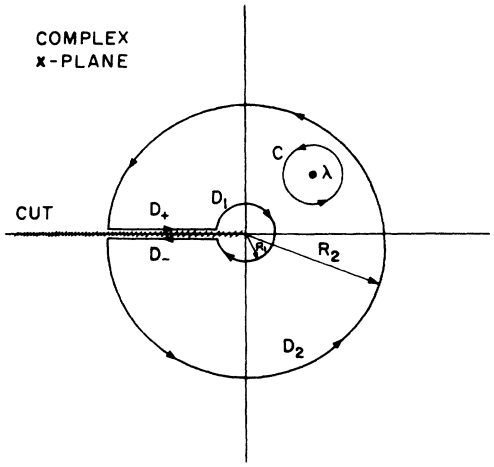

FIG. 1. Change of contour in the x plane used to derive the dispersion relation in Eq. (2.5).

The contour C may be shifted to the contour $D = D_+ + D_- + D_1 + D_2$ (see Fig. 1). Letting $R_1 \to 0$ and $R_2 \to \infty$, the contributions from the curves D_1 and D_2 vanish as a consequence of Eq. (2.2) and (2.3). Thus,

$$F^K(\lambda) = \frac{1}{2\pi i}\int_{-\infty}^{0} \frac{D^K(x)}{x-\lambda}dx\,, \tag{2.5}$$

where

$$D^K(x) \equiv \lim_{\epsilon\to 0}[F^K(x+i\epsilon) - F^K(x-i\epsilon)]\,. \tag{2.6}$$

Equation (2.5) is the once-subtracted dispersion relation we have sought.

Finally, we insert the identity

$$\frac{1}{x-\lambda} = \sum_{n=0}^{\infty}\frac{\lambda^n}{x^{n+1}}$$

into Eq. (2.5), compare the resulting series in powers of λ with that in Eq. (1.3), and obtain

$$A_n^K = \frac{1}{2\pi i}\int_{-\infty}^{0} D^K(x)x^{-n}\,dx\,. \tag{2.7}$$

Equation (2.7) is an exact and rigorous result.[14]

Physically, the relation in Eq. (2.7) may be understood by comparing two potentials with positive and negative anharmonic terms:

$$V_1(x) = \frac{x^2}{4} + \frac{\lambda x^4}{4}, \quad \lambda > 0$$

$$V_2(x) = \frac{x^2}{4} - \frac{\epsilon x^4}{4}, \quad \epsilon > 0\,.$$

These potentials are shown on Fig. 2. The first potential has bound states. The second potential has only unstable states (having complex energy). If ϵ is very small, the real part of the energy of an unstable state of V_2 is approximately equal to the energy of the corresponding bound state of V_1.

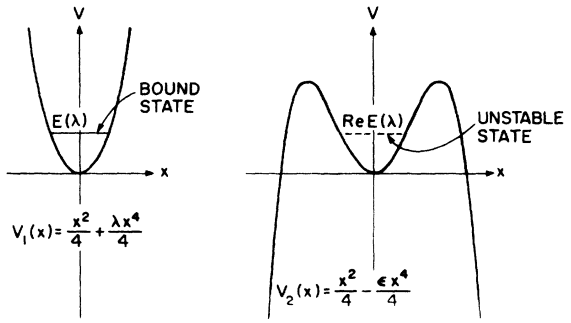

FIG. 2. Comparison of potentials having positive and negative anharmonic terms.

$\mathrm{Im}E(\epsilon)$ is very small and is related to the lifetime of the state which is very long. Equation (2.7) relates the Rayleigh-Schrödinger coefficients of the perturbation expansion of $E(\lambda)$ to the inverse moments of $\mathrm{Im}E(\epsilon)$.

Equation (2.7) is the basis for most of the approximations used in this paper. When n is very large, the contribution to the integral in Eq. (2.7) comes entirely from the region near $x = 0$. Thus, it is necessary to compute approximately $\mathrm{Im}E(\epsilon)$ for small ϵ. This will be done in the next section using WKB techniques.

III. WKB ZEROTH ORDER AT $\arg\lambda = \pi$

In the previous section it was shown that the large-order behavior of the perturbation series for $E^K(\lambda)$ depends on the imaginary part of the energy for negative coupling constant. In this section we use WKB techniques to calculate $\mathrm{Im}E^K(\epsilon)$ to lowest order in powers of ϵ, where

$$\epsilon = -\lambda > 0\,. \tag{3.1}$$

Until now, WKB approximations to $E(\lambda)$ have been given for every value of $\arg\lambda$ except for $\arg\lambda = \pi$.[2] Well-known technical problems, such as the phenomenon of subdominance, associated with transmission through a potential barrier,[15] make this calculation an order of magnitude more difficult than all previous ones. However, we will show how, with extreme care, one may avoid the ambiguities of this transmission problem.

$E^K(\epsilon)$ is defined by the differential equation

$$\left[-\frac{d^2}{dx^2} + \frac{x^2}{4} - \epsilon\frac{x^4}{4} - E^K(\epsilon)\right]\Phi(x) = 0\,. \tag{3.2}$$

Equation (3.2) describes a particle in a potential well in a slightly unstable state (very long half-life). As ϵ becomes smaller, the lifetime of the state increases because the particle must pass through a higher potential barrier to escape. Thus, $\mathrm{Im}E^K(\epsilon)$, which goes as the reciprocal of the life-

time, is extremely small.

The wave function $\Phi(x)$ satisfies two boundary conditions. At the origin,

$$\Phi(0)=0 \quad \text{for odd-parity states}, \tag{3.3a}$$

$$\Phi'(0)=0 \quad \text{for even-parity states}. \tag{3.3b}$$

At $x=+\infty$, the boundary conditions are somewhat complicated because at first glance they seem to be undefined. It would appear that any linear combination of outgoing waves and incoming waves, $\exp(\pm i\epsilon^{1/2}x^3/6)$, would suffice. However, we recall that the analytic continuation of the energy levels into the complex λ plane is accomplished by simultaneously rotating x into the complex x plane.[2] When $\arg\lambda=\pi$, the sector in which the boundary condition, $\lim_{|x|\to\infty}\psi(x)=0$, applies is given by $-\frac{1}{3}\pi<\arg(\pm x)<0$. Thus it is necessary to pick that asymptotic behavior which vanishes exponentially if the argument of x lies between 0° and −60°. Hence, $\Phi(x)$ must obey the boundary condition

$$\Phi(x)\sim\left(\frac{\text{const}}{x}\right)e^{-i\epsilon^{1/2}x^3/6} \quad \text{as} \quad x\to+\infty. \tag{3.4}$$

For small ϵ the real part of $E^K(\epsilon)$ is approximately $K+\frac{1}{2}$ while the imaginary part is very small. Thus, to prevent $\mathrm{Im}E^K(\epsilon)$ from being eclipsed, it is necessary to separate Eq. (3.2) into its real and imaginary parts. For this purpose we let

$$E^K(\epsilon)=E_1^K(\epsilon)+iE_2^K(\epsilon)$$
$$\sim K+\tfrac{1}{2}+iE_2^K(\epsilon), \tag{3.5a}$$

$$\Phi(x)=\Phi_1(x)+i\Phi_2(x). \tag{3.5b}$$

Furthermore, for sufficiently large x in the range $0<x<\epsilon^{-1/2}$, $\Phi(x)$ consists of an increasing and a decreasing part (see Fig. 3). We choose the phase of $\Phi(x)$, without loss of generality, such that the *decreasing part is purely real.* [We will see that the decreasing part behaves like $D_K(x)$ in regions A and B, below.] From these considerations we have

$$\left(-\frac{d^2}{dx^2}+\frac{x^2}{4}-\epsilon\frac{x^4}{4}-K-\frac{1}{2}\right)\Phi_1(x)=0 \tag{3.6a}$$

and

$$\left(\frac{d^2}{dx^2}+\frac{x^2}{4}-\epsilon\frac{x^4}{4}-K-\frac{1}{2}\right)\Phi_2(x)=E_2\Phi_1(x). \tag{3.6b}$$

In Eq. (3.6a) we have neglected $E_2\Phi_2$ compared with $E_1\Phi_1$ (see Fig. 3).

Our procedure will be to solve for Φ_1 approximately using Eq. (3.6a) and then to use this result to solve Eq. (3.6b) for Φ_2 and ultimately E_2. Our approximation scheme is based on dividing the x axis into four regions (see Fig. 3): region A, where $x\gtrsim(4K+2)^{1/2}=O(1)$; region B, where $x\gg(4K+2)^{1/2}$ but $x\ll\epsilon^{-1/2}$; region C, where $x\lesssim\epsilon^{-1/2}$; region D, where $x\sim\epsilon^{-1/2}$.

A. Φ_1 in Region A

In region A, we approximate Eq. (3.6a) by

$$\left(-\frac{d^2}{dx^2}+\frac{x^2}{4}-K-\frac{1}{2}\right)\Phi_{1A}=0, \tag{3.7}$$

to lowest order in ϵ. The solution to Eq. (3.7) which decreases for increasing x (see Fig. 3) is[16]

$$\Phi_{1A}=D_K(x), \tag{3.8}$$

where we have chosen the normalization of Φ_{1A} without loss of generality. Note that[16]

$$D_K(x)=2^{-K/2}e^{-x^2/4}H_K(x/\sqrt{2}) \tag{3.9}$$

satisfies Eq. (3.3). When x is large[16] (x approaches region B)

$$\Phi_{1A}\sim x^K e^{-x^2/4}. \tag{3.10}$$

B. Φ_2 in Region B

In regions B and C we approximate Φ_1 using lowest-order WKB:

$$\Phi_{1\,\mathrm{WKB}}=C_1(x^2-\epsilon x^4-4K-2)^{-1/4}$$
$$\times\exp\left(-\tfrac{1}{2}\int_{x_0}^{x}(t^2-\epsilon t^4-4K-2)^{1/2}\,dt\right), \tag{3.11}$$

where

$$x_0=(4K+2)^{1/2}. \tag{3.12}$$

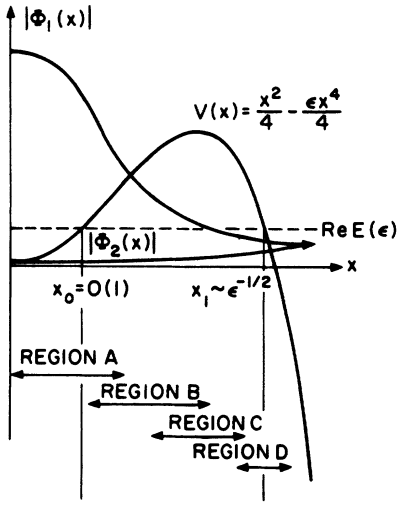

FIG. 3. Relative sizes of $|\Phi_1(x)|$ and $|\Phi_2(x)|$, which are exponentially decreasing and increasing functions of x. Regions A, B, C, and D and turning points at $x=x_0$ and $x=x_1$ are indicated.

In Eq. (3.11) we have only taken the *decreasing* exponential term (see Fig. 3). We will use the increasing exponential when we approximate Φ_2.

When x is near region B, we perform the integral in Eq. (3.11) approximately and get

$$\Phi_{1B} \sim C_1 e^{-x^2/4} x^K \exp\{\tfrac{1}{2}(K+\tfrac{1}{2})[1-\ln(K+\tfrac{1}{2})]\}. \tag{3.13}$$

Comparing Eqs. (3.10) and (3.13) gives

$$C_1 = \exp\{\tfrac{1}{2}(K+\tfrac{1}{2})[\ln(K+\tfrac{1}{2})-1]\}. \tag{3.14}$$

C. Φ_1 in Region C

When x is near the distant turning point at x_1,

$$x_1 \sim \frac{1}{\sqrt{\epsilon}}, \tag{3.15}$$

Equation (3.11) gives[17]

$$\Phi_{1C}(x) \sim (2x_1 - 2x)^{-1/4}\epsilon^{1/8} \times \exp[\tfrac{1}{3}\sqrt{2}\,\epsilon^{-1/4}(x_1-x)^{3/2} - 1/6\epsilon - (\tfrac{1}{2}K+\tfrac{1}{4})\ln(\tfrac{1}{4}\epsilon)], \tag{3.16}$$

where we have used Eq. (3.14).

D. Φ_1 and Φ_2 in Region D

In region D we substitute

$$r = (x_1 - x)(2\sqrt{\epsilon})^{-1/3}$$

into Eq. (3.2). Note that in region D we treat Φ_1 and Φ_2 at the same time – they are the same order of magnitude (see Fig. 3). In terms of r, Eq. (3.2) is approximately

$$\left(\frac{d^2}{dr^2} - r\right)\Phi_D(r) = 0, \tag{3.17}$$

which is the Airy equation.[18] Two linearly independent solutions to Eq. (3.17) are $\mathrm{Ai}(r)$ and $\mathrm{Bi}(r)$. Thus,

$$\Phi_D(x) = D_1\,\mathrm{Ai}(r) + D_2\,\mathrm{Bi}(r). \tag{3.18}$$

When we allow $x \to +\infty$, that is, $r \to -\infty$, we consult the formulas[19]

$$\mathrm{Ai}(-\rho) \sim \pi^{-1/2}\rho^{-1/4}\sin(\tfrac{2}{3}\rho^{3/2}+\tfrac{1}{4}\pi), \tag{3.19a}$$

$$\mathrm{Bi}(-\rho) \sim \pi^{-1/2}\rho^{-1/4}\cos(\tfrac{2}{3}\rho^{3/2}+\tfrac{1}{4}\pi), \tag{3.19b}$$

as $\rho \to +\infty$. However, Φ_D must have only decreasing phase as $x \to \infty$ [Eq. (3.4)]. From this requirement we obtain

$$D_1 = -iD_2 \equiv -iD. \tag{3.20}$$

Thus,

$$\Phi_D(x) = D[-i\,\mathrm{Ai}(r) + \mathrm{Bi}(r)]. \tag{3.21}$$

When $x \to 0$, $r \to +\infty$, and we consult the formulas[20]

$$\mathrm{Ai}(\rho) \sim \tfrac{1}{2}\pi^{-1/2}\rho^{-1/4}\exp(-\tfrac{2}{3}\rho^{3/2}), \tag{3.22a}$$

$$\mathrm{Bi}(\rho) \sim \pi^{-1/2}\rho^{-1/4}\exp(\tfrac{2}{3}\rho^{3/2}), \tag{3.22b}$$

as $\rho \to +\infty$. Recall that Φ_1, the real part of Φ, is decreasing as $x: 0 \to x_1$. We thus take D real and identify the decreasing and increasing solutions with the real and imaginary solutions:

$$\Phi_{1D}(x) = D\,\mathrm{Bi}(r), \tag{3.23a}$$

$$\Phi_{2D}(x) = D\,\mathrm{Ai}(r). \tag{3.23b}$$

Moreover, Eqs. (3.23a), (3.22b), and (3.16) enable us to match Φ_{1C} and Φ_{1D} asymptotically and thus to determine that

$$D = \epsilon^{1/12}\pi^{1/2}2^{-1/3}\exp[-1/6\epsilon - \tfrac{1}{2}(K+\tfrac{1}{2})\ln(\tfrac{1}{4}\epsilon)]. \tag{3.24}$$

Now that D is known, we easily calculate the asymptotic behavior of Φ_2 in region D for large r (that is, for $x < x_1$):

$$\Phi_{2D} \sim -\epsilon^{1/8}2^{-5/4}(x_1-x)^{-1/4} \times \exp[-\tfrac{1}{3}\sqrt{2}\,\epsilon^{-1/4}(x_1-x)^{3/2} - 1/6\epsilon - (\tfrac{1}{2}K+\tfrac{1}{4})\ln(\tfrac{1}{4}\epsilon)]. \tag{3.25}$$

E. Φ_2 in Region C

In regions B and C we neglect $E_2\Phi_1$ compared with $E_1\Phi_2$ in Eq. (3.6b). Thus, the WKB approximation to Φ_2 that applies in these regions is the same as that given for Φ_2 in Eq. (3.11). However, we choose a new constant C_2 and change the sign in the exponential because Φ_2 is an increasing function (see Fig. 3):

$$\Phi_{2\mathrm{WKB}} = C_2(x^2 - \epsilon x^4 - 4K - 2)^{-1/4}\exp\left(\tfrac{1}{2}\int_{x_0}^{x}(t^2 - \epsilon t^4 - 4K - 2)\,dt\right). \tag{3.26}$$

In region C, we approximate Eq. (3.26) by

$$\Phi_{2C} \sim C_2\epsilon^{1/8}(2x_1 - 2x)^{-1/4}\exp\{-\tfrac{1}{3}\sqrt{2}\,\epsilon^{-1/4}(x_1-x)^{3/2} + 1/6\epsilon - (\tfrac{1}{2}K+\tfrac{1}{4})[1 - \ln(\tfrac{1}{4}\epsilon) - \ln(K+\tfrac{1}{2})]\}. \tag{3.27}$$

Comparing this result with Eq. (3.25) gives

$$C_2 = -\tfrac{1}{2}e^{-1/3\epsilon}\exp\{-(K+\tfrac{1}{2})[\ln(\tfrac{1}{4}\epsilon) - \tfrac{1}{2} + \tfrac{1}{2}\ln(K+\tfrac{1}{2})]\}. \tag{3.28}$$

F. Φ_2 in Region B

Now that C_2 is known, we use Eq. (3.26) again to determine the asymptotic behavior of Φ_2 in region B. We find that

$$\Phi_{2B} \sim -\tfrac{1}{2} e^{-1/3\epsilon} e^{-(K+1/2)\ln(\epsilon/4)+x^2/4} x^{-K-1}. \quad (3.29)$$

G. Φ_2 in Region A

In region A, we must not neglect $E_2\Phi_1$ compared with $E_1\Phi_2$. However, we do neglect $\frac{1}{4}\epsilon x^4$ compared with $\frac{1}{4}x^2$. Thus, we must solve the differential equation

$$(-d^2/dx^2 + \tfrac{1}{4}x^2 - K - \tfrac{1}{2})\Phi_{2A}(x) = E_2\Phi_{1A}(x), \quad (3.30)$$

for the eigenvalue E_2. The boundary conditions on Φ_{2A} at $x=0$ are those given in Eq. (3.3). The large-x behavior of Φ_{2A} is given in Eq. (3.29).

To solve Eq. (3.30) we reduce it to a homogeneous equation as follows: Recall that E_2 is very small, and that Φ_{2A} is exponentially small compared to Φ_1 in region A. Then consider a perturbation expansion in powers of E_1 of the form

$$\Phi(x) = \Phi_{1A} + E_2\left(\frac{\Phi_{2A}}{E_2}\right) + \cdots. \quad (3.31)$$

$\phi(x)$ solves the equation

$$(-d^2/dx^2 - K - \tfrac{1}{2} - E_2 + \tfrac{1}{4}x^2)\phi(x) = 0. \quad (3.32)$$

To verify Eq. (3.32), observe that to lowest order in E_2, it reduces to Eq. (3.7); to next order in E_2 it is precisely Eq. (3.30).

The even- (and odd-) parity solutions of Eq. (3.32) [those obeying Eq. (3.3)] are[16]

$$\phi(x) = \tfrac{1}{2}[D_{K+E_2}(x) + (-1)^K D_{K+E_2}(-x)]. \quad (3.33)$$

The factor of $\frac{1}{2}$ comes from comparing Eq. (3.33) with the result for lowest order in E_2. That is, if we let $E_2=0$ in Eq. (3.33), the expression reduces to $\phi(x)|_{E_2=0} = \Phi_{1A}(x) = D_K(x)$, which agrees with Eq. (3.9).

To next order in E_2 we get

$$\frac{\Phi_{2A}}{E_2} = \frac{1}{2}\frac{\partial}{\partial E_2}[D_{K+E_2}(x) + (-1)^K D_{K+E_2}(-x)]\Big|_{E_2=0}. \quad (3.34)$$

E_2 will be determined by demanding that as $x \to \infty$, Eqs. (3.34) and (3.29) agree. The dominant large-x behavior of Φ_{2A}, which comes from the second term in Eq. (3.34), is[21]

$$\Phi_{2A}(x) = \left[\frac{E_2}{2}\frac{\partial}{\partial E_2}\ \frac{(2\pi)^{1/2}}{\Gamma(-K-E_2)} \times e^{(K+E_2)\pi i} x^{-(K+E_2+1)} e^{x^2/4}\right]\Big|_{E_2=0}. \quad (3.35)$$

Demanding that Eqs. (3.35) and (3.29) agree to lowest order in ϵ and E_2 gives the result

$$E_2 = \frac{4^{K+1/2}}{K!\sqrt{2\pi}} e^{-1/3\epsilon}\epsilon^{-K-1/2}. \quad (3.36)$$

Equation (3.36) is the result we have sought. Note that E_2 is exponentially small, which justifies *a posteriori* our neglecting of $E_2\Phi_1$ compared with $E_1\Phi_2$ in part E of this calculation. In the next section we use Eq. (3.36) and the dispersion integral of Sec. II to determine the large-order behavior of perturbation theory. In the appendix we give a very short and intuitive derivation of Eq. (3.36).

IV. LOWEST-ORDER WKB RESULTS

In this section we show how to use Eq. (3.36), the result of the lowest-order WKB calculation, to obtain the dominant large-n behavior of the Rayleigh-Schrödinger coefficients.

First, we compute $D^K(x)$ by combining Eqs. (2.1) and (2.6) with (3.6). We find that to lowest order in x (x is small and negative)

$$D^K(x) = -\frac{4^{K+1}i}{K!(2\pi)^{1/2}} e^{1/3x}(-x)^{-K-3/2}. \quad (4.1)$$

To compute the discontinuity D in Eq. (4.1), it is necessary to use a well-known property of $E^K(\lambda)$, namely,[2]

$$E^K(\lambda) = E^K{}^*(\lambda^*). \quad (4.2)$$

Next we substitute Eq. (4.1) into Eq. (2.7) and obtain for large n

$$A_n^K = \frac{-4^K\sqrt{2}}{K!\pi^{3/2}}\int_{-\infty}^{0} dx\, x^{-n} e^{1/3x}(-x)^{-K-3/2}. \quad (4.3)$$

Performing the integral in Eq. (4.3) exactly gives the leading large-n behavior of A_n (see Ref. 22):

$$A_n^K = \frac{(-1)^{n+1}12^K\sqrt{6}}{K!\pi^{3/2}} 3^n\,\Gamma(n+K+\tfrac{1}{2})[1+O(1/n)]. \quad (4.4)$$

We have indicated that the corrections to Eq. (4.4) are of order n^{-1}. This is verified in the next section, where we derive the next-higher-order WKB results. We emphasize here that *lowest-order WKB is sufficient to give the precise leading behavior of A_n^K for large n.* Successively higher-order WKB approximations to $\mathrm{Im}E^K(\lambda)$ do not change the results in Eq. (4.4); rather, they give the successive coefficients of inverse powers of n which multiply the leading behavior in Eq. (4.4). Equation (4.4) is the first term of a series in powers of n^{-1} (which may be asymptotic).

V. FIRST-ORDER WKB RESULTS

In this section we obtain the $O(n^{-1})$ corrections to Eq. (4.4). We could, of course, accomplish this task by carrying out the WKB calculations of Sec. III to order ϵ, plugging those results into Eq. (2.7), and evaluating the integral. Instead of presenting this extremely tedious and unenlightening calculation, we will rely on previous WKB calculations,[2] and, with a minimum of work, determine the $O(n^{-1})$ corrections. Our procedure will be to (A) compare the results of Sec. III with previous zeroth-order WKB calculations, and (B) use this comparison to find the first-order WKB result corresponding to Eq. (3.36).

A. Discussion of Zeroth-Order WKB

In Ref. 2 the approximate λ dependence of the energy levels $E^K(\lambda)$ was discovered through the use of WKB techniques. When $|\lambda|$ is small and $|\arg\lambda| < \pi$,

$$E^K(\lambda) = K + \tfrac{1}{2} + O(\lambda). \tag{5.1}$$

Thus, as $\arg\lambda$ increases from 0 to π, the energy eigenvalues stay very near the points $K+\frac{1}{2}$ on the positive real axis. As $\arg\lambda$ increases from π to $3\pi/2$, the eigenvalues move abruptly to the ImE axis. These trajectories in the complex E plane are governed by[2]

$$\frac{\Gamma(\frac{1}{4}+\frac{1}{2}E^K)}{\Gamma(\frac{1}{4}-\frac{1}{2}E^K)} = \exp\left(\frac{1}{3\lambda}+\frac{5\pi i}{4}-E^K\ln(\tfrac{1}{2}i\lambda)\right), \tag{5.2a}$$

for even-parity energy levels (K even), and

$$\frac{\Gamma(\frac{3}{4}+\frac{1}{2}E^K)}{\Gamma(\frac{3}{4}-\frac{1}{2}E^K)} = \exp\left(\frac{1}{3\lambda}-\frac{5\pi i}{4}-E^K\ln(\tfrac{1}{2}i\lambda)\right), \tag{5.2b}$$

for odd-parity energy levels.

As we emphasized in Sec. II, Eq. (5.1) is valid when $|\arg\lambda| < \pi$, and Eq. (5.2) is valid when $\pi < |\arg\lambda| \leq 2\pi$. When $|\arg\lambda| = \pi$, the point at which the distant turning point enters the sector in the complex x plane where the boundary condition $\lim_{|x|\to\infty}\psi(x) = 0$ holds, we have from Sec. III

$$\mathrm{Re}\,E^K(\lambda) = K + \tfrac{1}{2} + O(\lambda) \tag{5.3a}$$

and

$$\mathrm{Im}E^K(\lambda) = \pm\frac{4^{K+1/2}}{K!(2\pi)^{1/2}}\,e^{1/3\lambda}(-\lambda)^{-K-1/2}[1+O(\lambda)], \tag{5.3b}$$

where the $\pm$ corresponds to $\arg\lambda = \pm\pi$.

We raise the following question: What do we get from Eqs. (5.1) and (5.2) if they are assumed to be valid at $\arg\lambda = \pm\pi$? For this purpose, we must consider the real part and the imaginary part of $E^K(\lambda)$ separately because, as seen from Eq. (5.3), for $|\arg\lambda| = \pi$, the imaginary part is exponentially small compared with the real part. Consider the real part first. Equation (5.3a) agrees trivially with Eq. (5.1) for $|\arg\lambda| = \pi$. Furthermore, for $\arg\lambda = \pi$, the right-hand sides of Eqs. (5.2a) and (5.2b) are exponentially small and hence may be neglected. Accordingly, we get Eq. (5.3a) once again from Eqs. (5.2a) and (5.2b).

It is much more interesting to compare the imaginary parts. If Eq. (5.1) is assumed to hold for $|\arg\lambda| = \pi$, then, since the coefficient of every power of λ is real, we must conclude that

$$\mathrm{Im}E^K(\lambda) = 0. \tag{5.4}$$

More care must be taken with Eq. (5.2). Consider first the even-parity energy levels,

$$E^K = E^{2J} = 2J + \tfrac{1}{2} + iE_2, \tag{5.5}$$

where E_2 is assumed to be exponentially small (of order $e^{1/\lambda}$, for λ negative real). We approximate the left-hand side of Eq. (5.2a) as follows[23]:

$$\frac{\Gamma(\frac{1}{4}+\frac{1}{2}E^K)}{\Gamma(\frac{1}{4}-\frac{1}{2}E^K)} \sim \frac{\Gamma(J+\frac{1}{2})}{\Gamma(-J-iE_2/2)} \sim \frac{\Gamma(J+\frac{1}{2})\Gamma(J+1)\sin(\pi J+\pi iE_2/2)}{-\pi}$$

$$\sim i\sqrt{\pi}\,2^{-2J-1}E_2(-1)^{J+1}(2J)!\,. \tag{5.6}$$

The right-hand side of Eq. (5.2a) is approximately

$$e^{1/3\lambda}e^{5\pi i/4}(-\tfrac{1}{2}\lambda)^{-E^K}e^{-E^K\ln(-i)} \sim ie^{1/3\lambda}e^{5\pi i/4}(-\tfrac{1}{2}\lambda)^{-2J-1/2}e^{(2J+1/2)\ln i} \sim e^{1/3\lambda}(-1)^{J+1}(-\tfrac{1}{2}\lambda)^{-2J-1/2}, \tag{5.7}$$

where we have chosen that branch of the ln function for which $\ln i = \frac{1}{2}i\pi$. Combining Eqs. (5.6) and (5.7) gives

$$E_2 \sim \frac{\sqrt{2}\,4^{2J+1/2}}{\sqrt{\pi}\,(2J)!}\,e^{1/3\lambda}(-\lambda)^{-2J-1/2}. \tag{5.8}$$

By an analogous procedure, we have

$$E_2 \sim \frac{\sqrt{2}\,4^{2J+3/2}}{\sqrt{\pi}\,(2J+1)!}\,e^{1/3\lambda}(-\lambda)^{-2J-3/2}, \tag{5.9}$$

for odd-parity energy levels of the form

$$E^K = E^{2J+1} = 2J + \tfrac{3}{2} + iE_2.$$

Equations (5.8) and (5.9) verify the assumption that E_2 is exponentially small.

Finally we combine Eqs. (5.8) and (5.9) into one:

$$\mathrm{Im}E^K(\lambda)\sim\frac{\sqrt{2}\,4^{K+1/2}}{\sqrt{\pi}\,K!}\,e^{1/3\lambda}(-\lambda)^{-K-1/2}. \tag{5.10}$$

Again we observe a disagreement with Eq. (5.3b) – the imaginary part of the energy in Eq. (5.10) is too large by a factor of 2. Specifically, this factor arises from the equations for the asymptotic behavior of the Airy functions [see Eqs. (3.19) and (3.22)]. However, factors of 2 appear quite generally in Stokes' related discontinuous phenomena. In our case the discontinuity occurs when the distant turning point crosses a Stokes line of the asymptotic expansion of $\psi(x)$ for large x. The Stokes line is the edge of the sector in which $\psi(x)\to 0$ for large x.

A simple example illustrating this discontinuous phenomenon is a contour integral in the complex plane where the contour is shifted so that it passes through a simple pole. When the pole lies on the contour there is a discontinuity in the value of the integral which may be obtained by the principal-parts method. If we average the integrals for the pole lying inside and outside the contour, we obtain the principal-parts result.

Keeping these comments in mind, we are now not surprised to observe that Eq. (5.3b) is precisely the average of Eqs. (5.4) and (5.10). This averaging phenomenon is quite general. We will use it in Sec. V B to determine $E^K(\lambda)$, $\lambda<0$, to first order in λ by averaging the first-order versions of Eqs. (5.4) and (5.10).

Finally, we point out one further way of understanding this averaging. With reference to Sec. II, suppose we take the cut not along the $\arg\lambda=180°$ direction, but at $\arg\lambda=179°$. Then the discontinuity is given by

$$E^K(\arg\lambda=179°)-E^K(\arg\lambda=-181°).$$

The first term can be computed from Eq. (5.1) and the second from Eq. (5.2). Some contributions such as $K+\frac{1}{2}$ cancel and we get the sum of Eqs. (5.4) and (5.10) after analytically continuing the discontinuity to 180°.

B. First-Order WKB Results

The first-order analog of Eq. (5.1) is

$$E^K(\lambda)=K+\tfrac{1}{2}+\tfrac{3}{4}(2K^2+2K+1)\lambda+O(\lambda^2). \tag{5.11}$$

Equation (5.11) follows from a straightforward application of Rayleigh-Schrödinger perturbation theory to Eqs. (1.1) and (1.2). The first-order analog of Eq. (5.2a) is[24]

$$\frac{\Gamma(\frac{1}{2}+\frac{1}{2}\nu)}{\Gamma(-\frac{1}{2}\nu)}=\exp\left(\frac{1}{3\lambda}+\frac{5\pi i}{4}-(\nu+\tfrac{1}{2})\ln(\tfrac{1}{2}i\lambda)+\lambda A+\lambda BE^2\right), \tag{5.12}$$

where

$$\nu\equiv E-\tfrac{1}{2}-3\lambda(E^2+\tfrac{1}{4})/2. \tag{5.13}$$

The values of A and B in Ref. 2 are incorrect. Their correct values are

$$A=\tfrac{67}{48},\quad B=\tfrac{17}{4}. \tag{5.14}$$

Equations (5.13) and (5.1) imply that for the Jth even-parity energy level,

$$\nu=2J+\eta, \tag{5.15}$$

where η is at most of order λ. We insert this result into Eq. (5.12) and find that (as in Sec. V A) the left-hand side reduces to η times a constant while the right-hand side is of order $e^{1/3\lambda}$. Thus, for negative real λ, η *is smaller than any power of* λ. Noting this result, we return to Eq. (5.13) and iterate it once to get an approximate expression for $E^K(\lambda)=E^{2J}(\lambda)$:

$$E^{2J}=2J+\tfrac{1}{2}+\tfrac{3}{4}\lambda(8J^2+4J+1)+\eta[1+\tfrac{3}{2}\lambda(1+4J)]. \tag{5.16}$$

Equation (5.16), the first-order version of Eq. (5.4), contains λ twice. Apparently, the real and the imaginary parts of E (η is imaginary, as will be seen shortly) express their higher-order corrections in separate and independent power series in λ.

Next, we use Eqs. (5.12), (5.15), and (5.16) to evaluate η. The left-hand side of Eq. (5.12) reduces to

$$\sqrt{\pi}\,2^{-2J-1}(-1)^{J+1}(2J)!\,\eta, \tag{5.17}$$

following the procedure of Eq. (5.5). The right-hand side of Eq. (5.12) is approximated by

$$e^{1/3\lambda}e^{5\pi i/4}[1+\lambda(A+BE^2)]\,e^{-(2J+1/2)\ln(i\lambda/2)}\sim e^{1/3\lambda}i[1+\lambda(A+BE^2)](-\tfrac{1}{2}\lambda)^{-2J-1/2}(-1)^{J+1}. \tag{5.18}$$

Combining Eqs. (5.17) and (5.18) gives

$$\eta=\frac{i\sqrt{2}\,4^{2J+1/2}}{\sqrt{\pi}\,(2J)!}\,e^{1/3\lambda}(-\lambda)^{-2J-1/2}[1+\lambda(A+BE^2)]. \tag{5.19}$$

Next, we combine Eq. (5.19) with (5.16):

$$E^{2J} = 2J + \tfrac{1}{2} + \tfrac{3}{4}\lambda(8J^2 + 4J + 1) + \frac{i\sqrt{2}\,4^{2J+1/2}}{\sqrt{\pi}\,(2J)!}\, e^{1/3\lambda}(-\lambda)^{-2J-1/2}\{1 + \lambda[\tfrac{3}{2} + 6J + A + B(2J + \tfrac{1}{2})^2]\}. \tag{5.20}$$

Finally, we replace $2J$ in Eq. (5.20) with K and have for all energy levels

$$E^K = K + \tfrac{1}{2} + \tfrac{3}{4}\lambda(2K^2 + 2K + 1) + \frac{i\sqrt{2}\,4^{K+1/2}}{\sqrt{\pi}\,K!}\, e^{1/3\lambda}(-\lambda)^{-K-1/2}\{1 + \lambda[\tfrac{3}{2} + 3K - A - B(K + \tfrac{1}{2})^2]\}. \tag{5.21}$$

The imaginary part of Eq. (5.21) is the first-order generalization of Eq. (5.10). Note that the real parts of Eqs. (5.21) and (5.11) are identical and that only Eq. (5.21) has a nonzero imaginary part (as was true of the zeroth-order counterparts of these equations).

As in Sec. V A we can now *average* the expressions in Eqs. (5.11) and (5.21) to obtain the true value of $E^K(\lambda)$ for $\lambda < 0$. This true value differs from Eq. (5.21) in that the $\sqrt{2}$ is in the denominator. From this result we obtain the discontinuity which was defined in Eq. (2.6):

$$D^K(x) = -\frac{4^{K+1} i}{K!(2\pi)^{1/2}}\, e^{1/3x}(-x)^{-K-3/2} \times [1 + x(\tfrac{95}{24} + \tfrac{29}{4}K + \tfrac{17}{4}K^2)], \tag{5.22}$$

where we have substituted the values of A and B given in Eq. (5.14).

We can now compute the $O(n^{-1})$ corrections to Eq. (4.4). Following the procedure of Sec. IV, we insert Eq. (5.22) into Eq. (2.7), perform the integral, and obtain for large n

$$A_n^K = \frac{(-1)^{n+1}12^K\sqrt{6}}{K!\,\pi^{3/2}}\, 3^n\Gamma(n + K + \tfrac{1}{2}) \times\left(1 - \frac{1}{n}(\tfrac{95}{72} + \tfrac{29}{12}K + \tfrac{17}{12}K^2) + O(n^{-2})\right). \tag{5.23}$$

This is the desired result which was stated in Sec. I. It is a generalization to all K of Eq. (4) in Ref. 1.

We conclude with two remarks. First, recall that in Ref. 2 there is a numerical fit of the first 75 Rayleigh-Schrödinger coefficients for the ground-state energy ($K = 0$).[25] There, the coefficient of n^{-1} is given as

$$-1.319\,453.$$

The decimal equivalent of $-\frac{95}{72}$, the exact answer in Eq. (5.23), is

$$-1.319\,444\cdots.$$

This agreement strongly testifies to the power of WKB analysis.

Second, recall that as pointed out in Sec. IV, A_n^K will in general be given by the result in Eq. (4.4) [and Eq. (5.23)] followed by a series in inverse powers of n. [This must be so because the corrections to the discontinuity $D^K(x)$, in Eq. (5.22), for example, are a power series in x.] The series for A_n^0 will thus take the form

$$A_n^0 \sim (-1)^{n+1}\frac{\sqrt{6}}{\pi^{3/2}}\Gamma(n + \tfrac{1}{2})3^n\left(1 - \sum_{j=1}^{\infty}\frac{a_j}{n^j}\right). \tag{5.24}$$

We would not be surprised if the series in Eq. (5.24) is asymptotic. In fact, we would not even be surprised to find that the coefficients a_j are all positive, rational fractions. We have calculated numerically some of the coefficients a_j and have obtained the following results[26]:

$$a_1 = \tfrac{95}{72},$$

$$a_2 = \tfrac{20\,099}{10\,368}\ (= 1.938\,560\,9_{567\,901\,234}),$$

$$a_3 = \tfrac{15\,422\,651}{2\,198\,748}.$$

These values for a_1, a_2, and a_3 were of course originally computed as decimals; afterward, we guessed the above fractional representations of which we are extremely confident. Note that the denominators are all products of powers of 2 and 3.

We have also computed these additional coefficients:

$$a_4 = 40.118\,943\,1,$$

$$a_5 = 305.5223,$$

$$a_6 = 2808.09,$$

$$a_7 = 0.2995 \times 10^5,$$

$$a_8 = 0.365 \times 10^6,$$

$$a_9 = 0.44 \times 10^7,$$

$$a_{10} = 1 \times 10^8.$$

It is not yet possible to determine the fractional representations for the above coefficients because they are not known to a sufficient number of significant figures.

VI. INDEPENDENT DERIVATION

We exhibit below a new and independent derivation of Eq. (4.4) for the ground-state ($K = 0$) energy. At the end of this section, we generalize these results to the first energy level. Our methods here are far more direct, although somewhat more involved, than those of Secs. II, III, and IV. We will find a *linear* difference equation which for large n asymptotically approximates the nonlinear differ-

ence equation [Eq. (6.3)] which generates nth-order perturbation theory. We then solve this linear difference equation. Our methods are nonrigorous; nonetheless, we feel that this analysis is extremely worthwhile because it clarifies the rather subtle and intricate perturbative structure of the anharmonic oscillator model.

We obtain the nonlinear difference equation which generates the ground-state energy perturbation series by substituting

$$E^0(\lambda) \sim \tfrac{1}{2} - \sum_{n=1}^{\infty} (-\lambda)^n C_n \tag{6.1}$$

and

$$\psi^0(x) \sim e^{-x^2/4}\left(1 + \sum_{n=1}^{\infty} (-\lambda)^n \sum_{j=1}^{2n} (\tfrac{1}{2}x^2)^j C_{n,j}\right) \tag{6.2}$$

into the differential equation in Eq. (1.1). Collecting powers of x and λ gives the desired difference equation[27]:

$$2jC_{n,j} = (j+1)(2j+1)C_{n,j+1} + C_{n-1,j-2} - \sum_{p=1}^{n-1} C_{p,1}C_{n-p,j}, \tag{6.3}$$

where $C_n = C_{n,1}$ and $C_{0,0} = 1$. The C_n are related to the Rayleigh-Schrödinger coefficients A_n^0 by the formula

$$A_n^0 = (-1)^{n+1} C_n . \tag{6.4}$$

The minus signs in Eqs. (6.1) and (6.2) are chosen so that the $C_{n,j}$ are all positive.

Figure 4 illustrates schematically how Eq. (6.3) iterates. The dots represent the nonzero entries in the matrix $C_{n,j}$. The dot in the circle represents the value of $C_{n,j}$ to be computed, that is, the left-hand side of Eq. (6.3). The underlined dots are the values of $C_{n,j}$ on the right-hand side of Eq. (6.3).

A glance at Fig. 4 indicates several general properties of Eq. (6.3). Each $C_{n,j}$ depends on all numbers above and to the right of it. Thus, the matrix in Fig. 4 must be filled out from right to left for fixed n; after each row is completed, n, the order of perturbation theory, is increased by unity. As is typical in perturbation theory, each new order is increasingly difficult to compute because it depends on *all* previous orders. Note, however, that it is easy to find exact formulas for the entries on the right-hand side of the matrix:

$$C_{n,2n} = \frac{1}{4^n n!}, \tag{6.5a}$$

$$C_{n,2n-1} = \frac{4n+5}{3\,4^n (n-1)!}, \tag{6.5b}$$

$$C_{n,2n-2} = \frac{16n^2+64n+87}{18\,4^n (n-2)!}. \tag{6.5c}$$

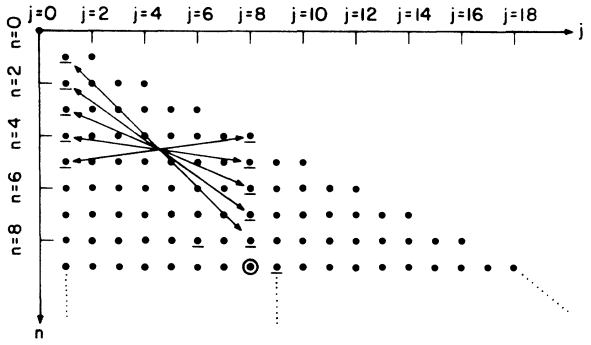

FIG. 4. Schematic representation of Eq. (6.3). The dots represent the nonzero values of $C_{n,j}$. The dot in the circle is the value of $C_{n,j}$ to be computed; the underlined dots are the values of $C_{n,j}$ which Eq. (6.3) requires for this computation. The arrows indicate those pairs of $C_{n,j}$ which are multiplied together.

Our problem is to find the large-n behavior of $C_{n,1}$, the entries in the first column of the matrix.

A. Linearization of Eq. (6.3)

The positivity of $C_{n,j}$ implies that the summation on the right-hand side of Eq. (6.3) is small compared with the other two terms. This suggests that Eq. (6.3) may be replaced by a sequence of *linear* equations, the Nth equation having the summation in

$$-\sum_{p=1}^{n-1} C_{p,1}C_{n-p,j}$$

replaced by $\sum_{p=1}^{N}$ and $\sum_{p=n-N}^{n-1}$, where $N=0$ means no sum is performed, and where the actual numerical values of $C_{p,j}$, $1 \le p \le N$ are calculated and then inserted into the summation. One would expect that as N increases one obtains a better approximation to $C_{n,1}$. This has been verified to order 150 on a computer where we found that

$$C_{n,1}/C_{n,1}(N=0) \sim 1 + O(1/n),$$

$$C_{n,1}/C_{n,1}(N=1) \sim 1 + O(1/n^2),$$

$$C_{n,1}/C_{n,1}(N=2) \sim 1 + O(1/n^3), \quad \text{etc.}$$

We now demonstrate analytically that the $N=0$ linear equation

$$2jC_{n,j} = (j+1)(2j+1)C_{n,j+1} + C_{n-1,j-2}, \quad C_{0,0} = 1 \tag{6.6}$$

gives the same leading behavior of $C_{n,1}$ for large n as in Eq. (4.4). The $N \ge 1$ equations have still not been solved analytically.

B. Approximation of the Linear Equation

Equation (6.6) may be further approximated. Detailed inspection of computer output shows that for large n,

$$(j+1)(2j+1)C_{n,j+1} \gg C_{n-1,j-2}, \quad \text{for } j=O(1) \tag{6.7a}$$

$$(j+1)(2j+1)C_{n,j+1} \ll C_{n-1,j-2}, \quad \text{for } j=O(2n). \tag{6.7b}$$

Equation (6.6) suggests the substitution

$$D_{n,j}=j\Gamma(j+\tfrac{1}{2})C_{n,j}, \tag{6.8}$$

which gives

$$D_{n,j}=D_{n,j+1}+\left(\frac{j}{2}+\frac{3}{8j-16}\right)D_{n-1,j-2}. \tag{6.9}$$

For large n, the term $3(8j-16)^{-1}$ can be dropped because for large j it is small compared with $\frac{1}{2}j$ and for small j Eq. (6.7a) shows that contribution from the $D_{n-1,j-2}$ term is negligible. Thus, we approximate Eq. (6.9) by

$$E_{n,j}=E_{n,j+1}+\tfrac{1}{2}jE_{n-1,j-2}. \tag{6.10}$$

However, this approximation is only good for large n. Furthermore, Eq. (6.10) is homogeneous (multiples of solutions are solutions). Therefore, we calculate the initial value $E_{0,0}$ by demanding that $E_{n,j}$ approximate $D_{n,j}$ for large n. It is simplest to require that

$$\lim_{n\to\infty}\frac{E_{n,2n}}{D_{n,2n}}=1. \tag{6.11}$$

$D_{n,2n}$ is obtained from Eqs. (6.5a) and (6.8). It is easy to see that $E_{n,2n}=E_{0,0}n!$. Thus,

$$E_{0,0}=\lim_{n\to\infty}\frac{2n\Gamma(2n+\frac{1}{2})}{4^n(n!)^2}=(2/\pi)^{1/2}. \tag{6.12}$$

Having established Eq. (6.12), we expect that for large n,

$$C_{n,1}=\frac{D_{n,1}}{\Gamma(\frac{3}{2})}$$

$$=\frac{2}{\sqrt{\pi}}D_{n,1}\sim\frac{2}{\sqrt{\pi}}E_{n,1}. \tag{6.13}$$

Finally, for convenience we define a new function $F_{n,j}$ which is a multiple of $E_{n,j}$ and obeys the equations

$$F_{0,0}=1, \tag{6.14a}$$

$$F_{n,j}=F_{n,j+1}+\tfrac{1}{2}jF_{n-1,j-2}, \tag{6.14b}$$

$$C_{n,1}\sim\frac{2\sqrt{2}}{\pi}F_{n,1}. \tag{6.14c}$$

C. Conversion to a Partial Differential Equation

To solve Eq. (6.14) we define the generating functions

$$F(x)=\sum_{n=1}^{\infty}\frac{F_{n,1}x^{n-1}}{(n-1)!}, \tag{6.15a}$$

$$G(x,y)=\sum_{n=1}^{\infty}\frac{x^n}{n!}\sum_{j=1}^{2n}y^{j+2}F_{n,j}, \tag{6.15b}$$

where

$$G(x,0)=G(0,y)=0. \tag{6.16}$$

Then we multiply Eq. (6.14b) by y^{j+2} and sum over j, multiply by $x^{n-1}/(n-1)!$ and sum over n, and use Eq. (6.15) to convert the result into the partial differential equation

$$\frac{\partial}{\partial x}G(x,y)+\frac{y^4}{2(1-y)}\frac{\partial}{\partial y}G(x,y)=\frac{y^3}{y-1}[y^2-F(x)]. \tag{6.17}$$

Note that unknowns appear on *both* sides of this equation

To solve Eq. (6.17) we introduce a new variable ξ, where

$$\frac{dy}{d\xi}=\frac{y^4}{2(1-y)}. \tag{6.18}$$

Integrating Eq. (6.18) gives

$$\xi(y)=\frac{3y-2}{3y^3}. \tag{6.19}$$

$\xi(y)$ is plotted on Fig. 5. Note that for $\xi>0$, $\xi(y)$ has a two-valued inverse. The inverse functions $g_1(\xi)$ and $g_2(\xi)$ are plotted on Fig. 6. In terms of x and ξ the solution of Eq. (6.17) is

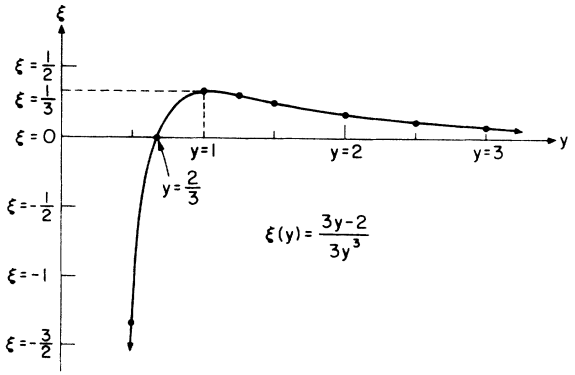

FIG. 5. Graph of $\xi(y)=(3y-2)/3y^3$ showing the two-valued inverse of $\xi(y)$ for $\xi>0$.

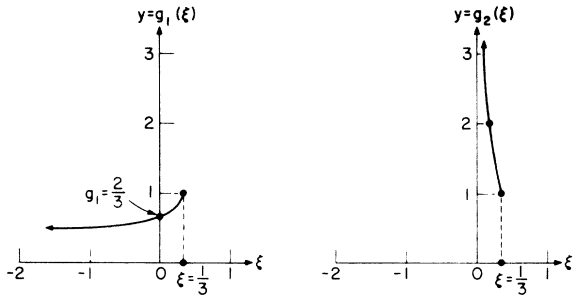

FIG. 6. Graph of the two inverse functions $y = g_1(\xi)$ and $y = g_2(\xi)$ which solve the equation $\xi(y) = (3y-2)/3y^3$.

$$G_{1,2}(x,\xi) = \int_0^x dx' \, \frac{g_{1,2}{}^3(\xi - x + x')}{1 - g_{1,2}(\xi - x + x')} \times [F(x') - g_{1,2}{}^2(\xi - x + x')], \tag{6.20}$$

where

$$G_{1,2}(x,\xi) = G(x, g_{1,2}(\xi)). \tag{6.21}$$

Note that Eq. (6.16) allows us to eliminate an arbitrary solution $V(x-\xi)$ of the homogeneous equation from Eq. (6.20).

The integral in Eq. (6.20) can be simplified by using Eq. (6.18) as follows:

$$-\int_0^x dx' \, \frac{g_{1,2}{}^5(\xi - x + x')}{1 - g_{1,2}(\xi - x + x')} = -2\int_0^x dx' \, g_{1,2}(\xi - x + x') \, \frac{dg_{1,2}(\xi - x + x')}{dx'} = -g_{1,2}{}^2(\xi) + g_{1,2}{}^2(\xi - x). \tag{6.22}$$

D. Conversion to an Integral Equation for $F(x)$

From Fig. 6 observe that $g_1(\frac{1}{3}) = g_2(\frac{1}{3})$. Hence, using Eq. (6.21), we find that

$$G_1(x, \tfrac{1}{3}) = G_2(x, \tfrac{1}{3}). \tag{6.23}$$

Combining Eqs. (6.20) and (6.23) then eliminates all reference to G and gives the following singular integral equation for F:

$$\int_0^x dx' F(x') \left[\frac{g_1{}^3(\frac{1}{3} - x + x')}{1 - g_1(\frac{1}{3} - x + x')} - \frac{g_2{}^3(\frac{1}{3} - x + x')}{1 - g_2(\frac{1}{3} - x + x')} \right] = g_2{}^2(\tfrac{1}{3} - x) - g_1{}^2(\tfrac{1}{3} - x). \tag{6.24}$$

E. Approximate Solution of the Integral Equation

Figure 6 shows that for small ξ, $g_2(\xi)$ is singular while $g_1(\xi)$ is finite. Indeed, Eq. (6.19) implies that for small ϵ,

$$g_1(\epsilon) \sim \tfrac{2}{3}, \tag{6.25a}$$

$$g_2(\epsilon) \sim \epsilon^{-1/2}. \tag{6.25b}$$

Equation (6.25) indicates that as x approaches $\frac{1}{3}$, the right-hand side of Eq. (6.24) becomes singular. We are thus motivated to identify the leading singularity in $F(x)$ which causes the equivalent divergence of the left-hand side. To this end, we substitute

$$x = \tfrac{1}{3} - \epsilon, \quad x' = \tfrac{1}{3} - \epsilon' \tag{6.26}$$

into Eq. (6.24). The resulting approximate integral equation is

$$\int_\epsilon^{1/3} d\epsilon' \, F(\tfrac{1}{3} - \epsilon') \left[\frac{g_1{}^3(\frac{1}{3} + \epsilon - \epsilon')}{1 - g_1(\frac{1}{3} + \epsilon - \epsilon')} - \frac{g_2{}^3(\frac{1}{3} + \epsilon - \epsilon')}{1 - g_2(\frac{1}{3} + \epsilon - \epsilon')} \right] \sim \frac{1}{\epsilon}. \tag{6.27}$$

Next we approximate the expression in square brackets in Eq. (6.27). Figure 6 shows that $g_{1,2}(\frac{1}{3}) = 1$. When ξ is near $\frac{1}{3}$ (y near unity) we can replace the differential equation in Eq. (6.18) by

$$d\xi \sim 2(1-y)\,dy. \tag{6.28}$$

Equation (6.28) integrates to

$$\tfrac{1}{3} - \xi = (1-y)^2. \tag{6.29}$$

Thus, using Fig. 6 to guide the choice of sign,

$$1 - g_1(\tfrac{1}{3} + \epsilon - \epsilon') \sim (\epsilon' - \epsilon)^{1/2}, \tag{6.30a}$$

$$1 - g_2(\tfrac{1}{3} + \epsilon - \epsilon') \sim -(\epsilon' - \epsilon)^{1/2}. \tag{6.30b}$$

From Eq. (6.30), we conclude that Eq. (6.27) may be further approximated by

$$\int_\epsilon^{1/3} d\epsilon' \, F(\tfrac{1}{3} - \epsilon') \frac{2}{(\epsilon' - \epsilon)^{1/2}} \sim \frac{1}{\epsilon}. \tag{6.31}$$

The leading singularity in F which causes the divergence of the right-hand side for small ϵ clearly has the form

$$F(\tfrac{1}{3} - \epsilon') \sim \frac{a}{\epsilon'^k}. \tag{6.32}$$

The substitution $\epsilon' = \epsilon/z$ yields $k = \frac{3}{2}$ and the equation

$$\int_{3\epsilon}^1 \frac{dz}{(1-z)^{1/2}} = \frac{1}{2a}.$$

Letting $\epsilon \to 0$ gives $a = \frac{1}{4}$. Hence, the leading singularity of $F(x)$ is given by

$$F(x) \sim \tfrac{1}{4}(\tfrac{1}{3} - x)^{-3/2}. \tag{6.33}$$

F. Recovery of $C_{n,1}$ for Large n

The actual behavior of $F(x)$ as given by Eq. (6.24) has the form

$$F(x) \sim \tfrac{1}{4}(\tfrac{1}{3}-x)^{-3/2} + \alpha(\tfrac{1}{3}-x)^{-1/2} + \beta(\tfrac{1}{3}-x)^{1/2} + \cdots . \tag{6.34}$$

However, when $F(x)$ is expanded into a power series in x, the *dominant* contribution to the coefficient of the x^n term for large n comes from the *most* singular term in Eq. (6.34). Thus, the analysis we have carried out is just sufficient to obtain the leading behavior of C_n for large n. Hence, using the binomial theorem to get the nth term in the expansion of $\frac{1}{4}(\frac{1}{3}-x)^{-3/2}$ and combining this result with Eqs. (6.15a) and (6.14c) gives the desired result

$$C_{n,1} \sim \frac{\sqrt{6}}{\pi^{3/2}}\, 3^n\, \Gamma(n+\tfrac{1}{2}) \tag{6.35}$$

or

$$A_n^0 \sim (-1)^{n+1} \frac{\sqrt{6}}{\pi^{3/2}}\, 3^n\, \Gamma(n+\tfrac{1}{2}), \tag{6.36}$$

which agrees with the result in Eq. (4.4) when $K=0$.

G. Discussion

We discuss below one subtle aspect of this derivation. At first glance it might appear that the delicate and intricate use of the multivaluedness of y as a function of ξ to eliminate the unknown generating functions G_1 and G_2 in Sec. V D was not necessary. It would seem that a more straightforward approach would use the relation

$$F(x) \equiv 1 + \frac{1}{2} \frac{\partial G(x,y)}{\partial y}\bigg|_{y=1}. \tag{6.37}$$

This identity may be proved by differentiating G in Eq. (6.15b) and letting $y=1$. The result is

$$1 + \frac{1}{2}\frac{\partial G}{\partial y}\bigg|_{y=1} = 1 + \sum_{n=1}^{\infty} \frac{x^n}{n!} \sum_{j=1}^{2n} \frac{j+2}{2} F_{n,j}$$

$$= 1 + \sum_{n=1}^{\infty} \frac{x^n}{n!} \sum_{j=3}^{2n+2} (F_{n+1,j} - F_{n+1,j+1}), \tag{6.38}$$

where we have shifted the j index and substituted Eq. (6.14b). The j summation in Eq. (6.38) collapses. Comparing with Eq. (6.15a) verifies Eq. (6.37).

Converting the identity to one involving the x and ξ variables gives

$$F(x) \equiv 1 + \lim_{\xi \to 1/3} \left(\pm(\tfrac{1}{3}-\xi)^{1/2} \frac{\partial G_{1,2}(x,\xi)}{\partial \xi} \right). \tag{6.39}$$

It would now appear that applying Eq. (6.39) to Eq. (6.20) is a far simpler way to eliminate the unknowns $G_{1,2}$ in favor of F. In fact, this procedure leads to a tautology – the resulting equation for F reduces to an identity. The apparently new and independent condition in Eq. (6.39) is *already* contained in Eq. (6.20). Thus, we are *forced* to follow the procedure of Sec. V D.

H. First Energy Level ($K=1$)

It is easy to repeat these techniques for the first energy level. We list below some intermediate $K=1$ results, which correspond with the $K=0$ results given above.

The linearized difference equation which corresponds with Eq. (6.6) is

$$2j\, C_{n,j} = (j+1)(2j+3) C_{n,j+1} + C_{n-1,j-2}, \tag{6.40}$$

where $C_{0,0}=1$ and

$$A_n^1 = 3(-1)^{n+1} C_{n,1}. \tag{6.41}$$

To derive Eq. (6.40) we have dropped the convolution term

$$-3 \sum_{p=1}^{n-1} C_{p,j}\, C_{n-p,1}$$

from the right-hand side. The substitution

$$D_{n,j} = j\Gamma(j+\tfrac{3}{2}) C_{n,j}$$

[see Eq. (6.8)] and some approximations give

$$F_{n,j} = F_{n,j+1} + \tfrac{1}{2}(j+2) F_{n-1,j-2}, \tag{6.42}$$

where $F_{0,0}=1$ and

$$A_n^1 = (-1)^{n+1} \frac{8\sqrt{2}}{\pi} F_{n,1}$$

[see Eq. (6.14)].

Instead of solving Eq. (6.42) let us consider a one-parameter family of such equations. Let M be arbitrary and $F_{0,0}=1$. Then we solve

$$F_{n,j} = F_{n,j+1} + \tfrac{1}{2}(j+M) F_{n-1,j-2} \tag{6.43}$$

by introducing the generating functions

$$F(x) = \sum_{n=1}^{\infty} \frac{x^{n-1}}{(n-1)!} F_{n,1} \tag{6.44a}$$

and

$$G(x,y) = \sum_{n=1}^{\infty} \frac{x^n}{n!} \sum_{j=1}^{2n} y^{j+M+2} F_{n,j}. \tag{6.44b}$$

The analog of Eq. (6.17) is

$$\frac{\partial}{\partial x} G(x,y) + \frac{y^4}{2(1-y)} \frac{\partial}{\partial y} G(x,y) = \frac{y^{M+3}}{y-1} \left(\frac{M+2}{2} y^2 - F(x) \right). \tag{6.45}$$

Following the analysis of Secs. V C–V D, we obtain

$$F(x)\sim\frac{\Gamma[\frac{1}{2}(M+3)]}{2\sqrt{\pi}\,\Gamma[\frac{1}{2}(M+2)]}\,(\tfrac{1}{3}-x)^{-(M+3)/2}, \tag{6.46}$$

which is the generalization of Eq. (6.33). Finally, we repeat the work of Sec. V F and find that

$$F_{n,1}=\frac{3^{n+(M+1)/2}\Gamma[n+\frac{1}{2}(M+1)]}{2\sqrt{\pi}\,\Gamma[\frac{1}{2}(M+2)]}. \tag{6.47}$$

The particular case $M=2$ in Eq. (6.42) reduces Eq. (6.47) to

$$A_n^1\sim(-1)^{n+1}\frac{12\sqrt{6}}{\pi^{3/2}}\,3^n\Gamma(n+\tfrac{3}{2}), \tag{6.48}$$

which is precisely the case $K=1$ in Eq. (4.4).

VII. LARGE-ORDER BEHAVIOR OF WICK-ORDERED PERTURBATION THEORY

It is *necessary* to Wick-order the Hamiltonian of a multidimensional field theory. Wick-ordering removes the divergence associated with the zero-point energies of the infinitely many degrees of freedom. Correspondingly, in perturbation theory, Wick-ordering removes those (divergent) diagrams having self-loops (lines with both ends connected to the same vertex). Because the anharmonic oscillator is a field theory having only *one* degree of freedom,[28] it is *not necessary* to Wick-order the Hamiltonian H in Eq. (1.4). In addition, the non-Wick-ordered diagrams in this field theory are finite and need not be eliminated. Nevertheless, to make a stronger connection with higher-dimensional field theory, we investigate the large-order behavior of the Wick-ordered perturbation series.

We begin by Wick-ordering the Hamiltonian in Eq. (1.4). To do this, we express the fields in the interaction picture in terms of creation and annihilation operators:

$$\phi=(2m)^{-1/2}(ae^{-imt}+a^\dagger e^{imt}), \tag{7.1a}$$

$$\dot\phi=(\tfrac{1}{2}m)^{1/2}(-iae^{-imt}+ia^\dagger e^{imt}). \tag{7.1b}$$

Then, in terms of a and $a^\dagger$, H is

$$H=m(a^\dagger a+\tfrac{1}{2})+g(2m)^{-2}(a+a^\dagger)^4. \tag{7.2}$$

The standard way to Wick-order H in Eq. (7.2) is to place all creation operators to the left of all annihilation operators.

To elucidate this definition we use the commutation relation

$$[a,a^\dagger]=1. \tag{7.3}$$

Equation (7.3) allows us to express Wick-ordered quantities simply in terms of non-Wick-ordered ones, to wit:

$$:\phi^2:=\phi^2-\frac{1}{2m}, \tag{7.4a}$$

$$:\phi^4:=\phi^4-\frac{3}{m}\,\phi^2+\frac{3}{4m^2}, \tag{7.4b}$$

$$:\tfrac{1}{2}\dot\phi^2+\tfrac{1}{2}m^2\phi^2:=\tfrac{1}{2}\dot\phi^2+\tfrac{1}{2}m^2\phi^2-\tfrac{1}{2}m. \tag{7.4c}$$

We now use Eq. (7.4) to rewrite the Wick-ordered Hamiltonian $:H:$ as a *new non-Wick-ordered* Hamiltonian:

$$:H:=\tfrac{1}{2}\dot\phi^2+\tfrac{1}{2}m^2\phi^2-\tfrac{1}{2}m+\lambda\phi^4-\left(\frac{6\lambda}{m^3}\right)\frac{m^2\phi^2}{2}+\frac{3\lambda}{4m^2}. \tag{7.5}$$

Comparing Eq. (7.5) with Eq. (1.4) shows that Wick-ordering is equivalent to the replacement

$$m^2\to m^2(1-6\lambda/m^3), \tag{7.6a}$$

$$E\to E-\tfrac{1}{2}m+\tfrac{3}{4}m\lambda/m^3. \tag{7.6b}$$

From Eqs. (7.6) and (1.7), it follows that we can immediately express the perturbation series for the Kth energy level of the Wick-ordered Hamiltonian in terms of the Rayleigh-Schrödinger coefficients for the non-Wick-ordered Hamiltonian:

$$E^K_{\rm Wick}\sim-\tfrac{1}{2}m+\tfrac{3}{4}m\Lambda+m(K+\tfrac{1}{2})(1-6\Lambda)^{1/2}+m\sum_{n=1}^{\infty}A_n^K\Lambda^n(1-6\Lambda)^{1/2-3n/2}, \tag{7.7}$$

where $\Lambda=\lambda m^{-3}$.

Using the binomial expansion,

$$(1-6\Lambda)^{1/2-3n/2}=\sum_{j=0}^{\infty}\frac{(6\Lambda)^j\Gamma(\frac{3}{2}n-\frac{1}{2}+j)}{j!\,\Gamma(\frac{3}{2}n-\frac{1}{2})}, \tag{7.8}$$

we simplify Eq. (7.7) to read[29]

$$E^K_{\rm Wick}\sim mK+\tfrac{3}{2}K(K-1)m\Lambda+m\sum_{n=2}^{\infty}\Lambda^nB_n, \tag{7.9}$$

where

$$B_n^K=\sum_{j=0}^{n}\frac{A^K_{n-j}6^j\Gamma(\frac{3}{2}n-\frac{1}{2}j-\frac{1}{2})}{j!\,\Gamma(\frac{3}{2}n-\frac{3}{2}j-\frac{1}{2})}. \tag{7.10}$$

[In Eq. (7.10), $A_0^K=K+\frac{1}{2}$.] Equations (7.9) and (7.10) are exact.

We proceed to compute the large-order behavior of B_n^K. To do this, we observe that for large n, the dominant contribution to the sum in Eq. (7.10) comes from the small-j terms. We thus approximate[30]

$$\frac{\Gamma(\frac{3}{2}n-\frac{1}{2}j-\frac{1}{2})}{\Gamma(\frac{3}{2}n-\frac{3}{2}j-\frac{1}{2})}\sim(\tfrac{3}{2}n)^j\left(1-\frac{2j(j+1)}{3n}+\frac{j(j-1)(12j+11)(j+1)}{54n^2}+O(1/n^3)\right). \tag{7.11}$$

As will be seen, it is most productive to compute the large-n behavior of the ratio A_n^K/B_n^K. Equations (7.10) and (7.11) give

$$\frac{B_n^K}{A_n^K} \sim \sum_{j=0}^{N} \frac{(9n)^j}{j!} \frac{A_{n-j}^K}{A_n^K} \left(1 - \frac{2j(j+1)}{3n} + \frac{j(j-1)(12j+11)(j+1)}{54n^2} + O(1/n^3)\right) . \tag{7.12}$$

Next, we refer to Eq. (5.23) and determine[30] that for large n

$$\frac{A_{n-j}^K}{A_n^K} \sim \frac{1}{(-3n)^j} \left(1 + \frac{j^2 - 2jK}{2n} + \frac{-j(\frac{95}{3} + 58K + 34K^2) + j(j+1)(12K^2 - 12Kj + 3j^2 + j - 1)}{24n^2} + O(1/n^3)\right) . \tag{7.13}$$

Note that even though we have only determined the large-n behavior of A_n^K up to terms of order $1/n^2$ we can still calculate the ratio in Eq. (7.13) up to terms of order $1/n^3$. This is the reason why we have chosen to approximate the ratio.

Finally, we combine Eqs. (7.12) and (7.13) and allow the summation index to run from zero to ∞ to obtain

$$\frac{B_n^K}{A_n^K} \sim \sum_{j=0}^{\infty} \frac{(-3)^j}{j!} \left(1 - \frac{j^2 + 4j + 6Kj}{6n} + j\,\frac{3j^3 + 8j^2 - 48j - 338 - 522K - 306K^2 + 36Kj(j+1) + 108K^2(j+1)}{216n^2} + O(1/n^3)\right) . \tag{7.14}$$

By repeatedly shifting the summation index, we do the indicated sums in Eq. (7.14) exactly. We obtain

$$\frac{B_n^K}{A_n^K} = e^{-3}\left(1 + \frac{1+3K}{n} + \frac{\frac{71}{6} + 31K + 23K^2}{4n^2} + O(1/n^3)\right) . \tag{7.15}$$

This is the desired result which was also given in Sec. I.

To compare this prediction with our numerical results, we let $K=0$ in Eq. (7.15), take the reciprocal, and get[31]

$$\frac{A_n^0}{B_n^0} = e^3\left(1 - \frac{1}{n} - \frac{47}{24n^2} + O(1/n^3)\right) . \tag{7.16}$$

We have calculated the ratio A_n^0/B_n^0 to 74th order in perturbation theory on a computer and find that

$$\left(\frac{A_{74}^0}{B_{74}^0}\right)_{\text{computer}} = 19.806\,55 . \tag{7.17}$$

The agreement between the theoretical result in Eq. (7.16) and the computer output in Eq. (7.17) is most impressive:

$$e^3 = 20.085\,537 , \tag{7.18a}$$

$$e^3(1 - 74^{-1}) = 19.814\,111 , \tag{7.18b}$$

$$e^3[1 - 74^{-1} - 47 \cdot 24^{-1} \cdot 74^{-2}] = 19.806\,928 . \tag{7.18c}$$

Despite the excellent results of Eq. (7.15) we point out that our derivation does *not* generalize to Wick-ordered $(\lambda\phi^{2N})$ field theories with $N>2$. For these theories there is no replacement equivalent to that in Eq. (7.6). However, the general problem of finding the large-n behavior of the ratio of the Rayleigh-Schrödinger coefficients in the perturbation series for the energy levels of the $(\lambda\phi^{2N})_1$ and $(:\lambda\phi^{2N}:)_1$ theories has now been solved and will be published elsewhere.[8]

APPENDIX

We give below a derivation of Eq. (3.36) which is less careful than that given in Sec. III, but which clarifies the physical ideas and minimizes the mathematical detail. In Sec. III we took pains to show how to deal with the subdominant contribution to the wave function [the exponentially decreasing $\Phi_2(x)$]. Now that $\Phi_2(x)$ is comfortably under control, we cavalierly rederive $\mathrm{Im}E(\lambda)$ without regard to subdominance. In a future paper, where we generalize to the much more complicated and difficult problem of many coupled oscillators, we will follow the techniques of this appendix rather than those of Sec. III.

First, we use a well-known relation[32] between the imaginary part of the energy of an unstable state of a real potential and the probability current $J(x)$:

$$\mathrm{Im}E = E_2 = J(x) \Big/ \int_0^x \Phi^*(x')\Phi(x')dx' , \tag{A1}$$

where

$$J(x) \equiv \frac{1}{2i}\left(-\Phi^*(x)\frac{d}{dx}\Phi(x) + \Phi(x)\frac{d}{dx}\Phi^*(x)\right) . \tag{A2}$$

Equation (A1) is trivially derived by multiplying Eq. (3.2) by $\Phi^*(x)$, subtracting the complex conjugate of this result, integrating from $-x$ to x, where $x > x_1$, and using the invariance of the potential under reflections $x \to -x$. (We integrate past x_1, the distant turning point, so that there is no point between x and ∞ where the current will be reflected back toward $x=0$. Thus the amount

of outward probability current flow is a measure of the decay rate of the unbound state.) Note that by conservation of probability current the right-hand side of Eq. (A1) does not depend on x.

Second, we approximate the denominator of Eq. (A1) as follows: For $x<x_1$, we can replace Φ in the integral with $\mathrm{Re}\Phi$ because $\mathrm{Im}\Phi$ is small, as was shown in Sec. III. Moreover, we can approximate $\mathrm{Re}\Phi$ by $\Phi_{1A}=D_K(x)$ [see Eq. (3.8)], because $\mathrm{Re}\Phi$ and Φ_{1A} differ only in the shape of their exponentially small tails. Finally, we extend the range of integration to all positive x. Thus,[33]

$$\int_0^x \Phi^*(x')\Phi(x')dx' \sim \int_0^\infty [D_K(x')]^2 dx' = (\tfrac{1}{2}\pi)^{1/2} K! \,. \tag{A3}$$

Third, we evaluate $J(x)$ in Eq. (A2). We will do this in two distinct ways.

Method 1

We follow the approach of Sec. III until Eqs. (3.21), (3.22), and (3.24) have been established. These equations together imply the following behavior for the wave function $\Phi(x)$ when $x>x_1$:

$$\Phi(x)\sim 2^{-1/4}\epsilon^{1/8}(x-x_1)^{-1/4}\exp[-\tfrac{1}{4}i\pi-(6\epsilon)^{-1}-\tfrac{1}{2}(K+\tfrac{1}{2})\ln\tfrac{1}{4}\epsilon-\tfrac{1}{3}i\sqrt{2}\epsilon^{-1/4}(x-x_1)^{3/2}]. \tag{A4}$$

The expression in Eq. (A4) is then inserted in Eq. (A2) giving

$$J(x)\sim \tfrac{1}{2}e^{-1/3\epsilon}\left(\frac{4}{\epsilon}\right)^{K+1/2}. \tag{A5}$$

Method 2

Here we use a trick to circumvent the asymptotic matching at the distant turning point x_1. We thus reduce the problem to just *one* asymptotic connection. We follow Sec. III until Eqs. (3.11), (3.12), and (3.14) have been established. Equation (3.11) is the WKB approximation to Φ in region B:

$$\Phi(x)\sim C_1(x^2-\epsilon x^4-4K-2)^{-1/4} \times\exp\left(-\tfrac{1}{2}\int_{x_0}^{x}(x'^2-\epsilon x'^4-4K-2)^{1/2}dx'\right). \tag{A6}$$

x_0 and C_1 are given in Eqs. (3.12) and (3.14). C_1 is determined by asymptotically matching $\Phi_{\mathrm{WKB}}(x)$ in Eq. (A6) to $D_K(x)$ at the boundary of regions A and B, as was done in Sec. III B. We would now like to allow x, the upper endpoint of the WKB integral, to be larger than x_1. But, of course, this expression for the WKB wave function is no longer valid as we pass the turning point at x_1. To avoid this difficulty we approach the point x along a path which goes around x_1: The path goes up the real axis, circles x_1 counterclockwise, and continues up the real axis until it reaches x. Along this contour, WKB gives a *good* approximation to Φ.

Fortunately, the *integral* in the WKB wave function depends only on the endpoint. Thus, for simplicity it may be taken entirely along the real axis. We then break the integral into two parts: (a) the portion below x_1 which is real and factors out of $J(x)$ as a constant, and (b) the portion above x_1 which is imaginary. To leading order [neglecting $O(x^{-2})<O(\epsilon)$ compared with $O(1)$], $J(x)$ is a real constant which we easily simplify to

$$J(x)=\tfrac{1}{2}C_1{}^2\exp\left(-\int_{x_0}^{x_1}(x'^2-\epsilon x'^4-4K-2)^{1/2}dx'\right). \tag{A7}$$

We evaluate the integral in Eq. (A7) (Ref. 17) to obtain the same result as in Eq. (A5).

Finally, we combine Eqs. (A3) and (A5) as indicated in Eq. (A 1) and get

$$\mathrm{Im}E=E_2 = \frac{4^{K+1/2}}{K!(2\pi)^{1/2}}e^{-1/3\epsilon-K-1/2}, \tag{A8}$$

which agrees with Eq. (3.36).

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[1]This paper is a sequel to C. M. Bender and T. T. Wu, Phys. Rev. Letters 27, 461 (1971).

[2]The equivalence is demonstrated in C. M. Bender and T. T. Wu, Phys. Rev. 184, 1231 (1969), Appendix A. This paper is referred to hereafter as BW.

[3]The Feynman rules are given in BW, Appendix B.

[4]A. M. Jaffe, Commun. Math. Phys. 1, 127 (1965). This paper gives references to the works of C. Hurst, W. Thirring, A. Petermann, and E. R. Caianiello, A. Campolattaro, and M. Marinaro.

[5]BW, Appendix C.

[6]For a description of this numerical analysis see BW,

Appendix E.

[7]Most of the results are given in Ref. 1. A complete treatment is given in C. M. Bender and T. T. Wu (unpublished).

[8]T. I. Banks and C. M. Bender, J. Math. Phys. 13, 1320 (1972).

[9]T. I. Banks, C. M. Bender, and T. T. Wu (unpublished).

[10]See B. Simon, Ann. Phys. (N.Y.) 58, 79 (1970), for discussion.

[11]J. J. Loeffel and A. Martin, CERN Report No. CERN-TH-1167, 1971 (unpublished).

[12]J. J. Loeffel, A. Martin, B. Simon, and A. S. Wightman, Phys. Letters 30B, 656 (1969).

[13]C. M. Bender and T. T. Wu, Phys. Rev. Letters 21, 406 (1968), and BW.

[14]An independent derivation of Eq. (7) may be found in the paper by Simon (Ref. 10).

[15]See N. Fröman and P. O. Fröman, *JWKB Approximation* (North-Holland, Amsterdam, 1965). The nature of the difficulty is that one of the two solutions of the Schrödinger equation becomes exponentially small ("subdominant," or "recessive") compared with the other in the region of the potential barrier and must be disregarded when one uses matched asymptotic expansions. The associated loss of information makes it very difficult to obtain the secular equation for the eigenvalues.

[16]*Higher Transcendental Functions* (Bateman Manuscript Project), edited by A. Erdélyi (McGraw-Hill, New York, 1953), Vol. 2, pp. 116–123. Hereafter, we label this reference BMP.

[17]See Ref. 2 for the relevant mathematical techniques for approximating the WKB integral.

[18]M. Abramowitz and I. A. Stegun, *Handbook of Mathematical Functions* (National Bureau of Standards, Washington, D. C., 1964), Eq. 10.4.1. Hereafter, we label this reference NBS.

[19]NBS, Eqs. 10.4.60 and 10.4.64.

[20]NBS, Eqs. 10.4.59 and 10.4.63.

[21]BMP, Vol. 2, p. 123, Eq. (2).

[22]There is a misprint in Eq. (3) of Ref. 1 which becomes evident when it is compared with Eq. (4.4) of this paper. In Eq. (3) the curly brackets should be raised to the $nN - n + K + \frac{1}{2}$ power (the K was omitted).

[23]See BMP, Vol. 1, Sec. (1.2), Eqs. (5) and (15).

[24]See Ref. 2, Appendix F.

[25]See Ref. 2, Appendix E, Eq. (E8a). There is a misprint in Eq. (E8): All decimals should be preceeded by a $(-)$ sign.

[26]We are most grateful to Ken Anderson for his assistance in this difficult computer calculation. The numerical fits were obtained from a 36-place (4-fold precision) calculation of the ground state energy to 150th order in perturbation theory.

[27]This difference equation is also derived in Ref. 2, Sec. II.

[28]See Ref. 2, Appendixes A and B.

[29]To compute the coefficient of A in Eq. (7.9) it is necessary to know A_1^K. See Eq. (5.11).

[30]See NBS, p. 257, Eq. 6.1.47.

[31]There is a misprint in Ref. 1, Eq. (18). The coefficient of the n^{-2} term was given incorrectly as $-41/24$; it should be $-47/24$ as in Eq. (7.16).

[32]We credit Dr. George Zipfel as the first to use Eq. (A1) in connection with the anharmonic oscillator.

[33]See BMP, Vol. 2, Sec. (8.3), Eq. (23). There is a factor of 2 error in BMP which has been corrected in Eq. (A3).

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Anharmonic Oscillator

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We consider the anharmonic oscillator defined by the differential equation $(-d^2/dx^2+\frac{1}{4}x^2+\frac{1}{4}\lambda x^4)\Phi(x)=E(\lambda)\Phi(x)$ and the boundary condition $\lim_{x\to\pm\infty}\Phi(x)=0$. This model is interesting because the perturbation series for the ground-state energy diverges. To investigate the reason for this divergence, we analytically continue the energy levels of the Hamiltonian H into the complex λ plane. Using WKB techniques, we find that the energy levels as a function of λ, or more generally of λ^α, have an infinite number of branch points with a limit point at $\lambda=0$. Thus, the origin is not an isolated singularity. Level crossing occurs at each branch point. If we choose $\alpha=\frac{1}{3}$, the resolvent $(z-H)^{-1}$ has no branch cut. However, for all z it has an infinite sequence of poles which have a limit point at the origin. The anharmonic oscillator is of particular interest to field theoreticians because it is a model of $\lambda\varphi^4$ field theory in one-dimensional space-time. The unusual and unexpected properties exhibited by this model may give some indication of the analytic structure of a more realistic field theory.

I. INTRODUCTION

IN this paper, we discuss the familiar anharmonic oscillator.[1] We are concerned with the eigenvalue problem defined by the differential equation

$$(-d^2/dx^2+\tfrac{1}{4}x^2+\tfrac{1}{4}\lambda x^4)\Phi(x)=E(\lambda)\Phi(x) \quad (1.1)$$

and the associated boundary condition

$$\lim_{x\to\pm\infty}\Phi(x)=0. \quad (1.2)$$

We propose to answer the following questions within the context of the anharmonic oscillator:

(i) Is the perturbation series for the ground-state energy, which is a power series in the coupling constant λ, convergent for any $\lambda\neq 0$?

(ii) If not, does the ground-state energy, considered as a function of complex λ or more generally as a function of some fractional power α of λ, have an isolated singularity at $\lambda=0$?

(iii) Is the resolvent $(z-H)^{-1}$, considered as a function of λ^α for fixed z, analytic at $\lambda=0$? If not, is the point $\lambda=0$ an isolated singularity?

The answer to all these questions is *no*. More precisely, the energy levels, which are originally defined only for positive values of λ, can be analytically continued into the complex λ plane. This analytic continuation has an infinite number of branch points, which have a limit point at the origin $\lambda=0$. Moreover, level crossing occurs at each branch point. If α is chosen to be $\frac{1}{3}$, the resolvent has no branch cut. However, for all z, $(z-H)^{-1}$ has an infinite number of poles which have a limit point at the origin. These qualitative analytic properties of the energy levels and of the resolvent were totally unexpected and are, we feel, most unusual and exciting.

This paper will adhere to the following outline: In Sec. II (and Appendices A–E), we inspect in great detail the properties of the perturbation series for the ground-state energy in our model. However, this direct inspection gives no clue as to the nature of the singularities in the λ plane which cause this divergence. To

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[1] The techniques used in this paper to analyze the anharmonic oscillator were first introduced by C. M. Bender and T. T. Wu, Phys. Rev. Letters **21**, 406 (1968).

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understand these singularities better, in Sec. III, we analytically continue the energy levels into the λ plane and derive some exact properties of this continuation. In particular, we derive an exact condition which gives the locations of the singularities in the λ plane.

To apply this condition we must know the wave function in coordinate space. To calculate the wave function approximately, in Sec. IV, we present a technique using zeroth-order WKB methods in the complex plane. (In Appendix F, we make the calculation of the wave function more precise by using first-order WKB techniques.)

In Sec. V, we use the results of Sec. IV to determine the singularities of the resolvent. In Sec. VI (and Appendix G), we combine the results of Secs. III and IV to find the approximate locations of the singularities in the analytic continuation of the energy levels. Section VI contains an exhaustive qualitative description of the analytic continuation of the energy levels $E(\lambda)$. The major result of this paper is that the energy levels of the anharmonic oscillator for a given positive real λ are the positive real values of $E(\lambda)$ on each of the infinite number of branches of a Riemann surface. Each energy level corresponds to a sheet of this Riemann surface.

Although it is not necessary to understand any field theory to read this paper (aside from Appendices A–C), it is important to point out that the anharmonic oscillator is a simple model field theory in one-dimensional space-time. This field theory is defined by the Hamiltonian

$$H=\tfrac{1}{2}\dot{\varphi}^2+\tfrac{1}{2}m^2\varphi^2+\lambda\varphi^4 \tag{1.3}$$

and the commutation relations

$$[\varphi,\dot{\varphi}]=i. \tag{1.4}$$

[In Appendix A, we prove rigorously that Eqs. (1.1) and (1.2) are consequences of Eqs. (1.3) and (1.4). In fact, Eqs. (1.1) and (1.2) are the coordinate representation of H.]

Since this field-theory model involves no space dimensions, there are no asymptotic states and therefore no particle scattering. Also, there is only one degree of freedom. In fact, all this theory describes is a universe which sits at one point and oscillates. Nevertheless, this model has value to the field theoretician. It enables one to conjecture with slight supporting evidence[2] that many of the fascinating properties we have observed are also present in a more realistic field theory. This model may give us some idea of the vast underlying complexity of such a theory.

The model was originally chosen for study because it had a well-defined but divergent perturbation series. We believe that the divergence of perturbation series in field theory was first discussed by Dyson.[3] Since then, many papers have been written on this topic. A particularly interesting investigation was carried out by Jaffe.[4] Jaffe proved that in two-dimensional space-time the perturbation series for Green's functions in self-interacting boson field theories diverge even though these series exist term by term. However, as far as we know, no techniques have been discovered to elucidate the singularities in field theories with divergent perturbation series. In this paper, we develop for the first time methods to describe the actual singularity structure of a particular field theory.

Having achieved the results of this paper, we hope in the future to apply the new analytical tools of Secs. III–VI to other models of increasing complexity in order to better understand how singularities depend upon the form of interaction. Additional zero-dimensional bose field theories which ought to be considered are self-interactions such as φ^6, φ^{2N}, and $\cos\varphi$ and couplings such as $\varphi_1^2\varphi_2^2$. Hopefully, these WKB techniques will be advanced to the point where they may be used to analyze higher-dimensional theories.

II. GROUND-STATE-ENERGY PERTURBATION SERIES

The perturbation series for the ground-state energy $E_0(\lambda)$ in this model is a power series in the coupling constant λ. This power series takes the form

$$E_0(\lambda)=\tfrac{1}{2}m+\sum_{n=1}^{\infty} mA_n(\lambda/m^3)^n. \tag{2.1}$$

Two methods for calculating the A_n in this equation are discussed below. Each method reveals some important properties of Eq. (2.1).

Method 1. In general, in a field theory the perturbation series for the ground-state energy is the sum of all connected Feynman diagrams having no external legs. Hence, Feynman diagrams can be used to calculate the A_n. (As an illustration of Feynman-diagram techniques, we calculate A_1, A_2, and A_3 in Appendix B.)

Moreover, an integral representation for these Feynman diagrams provides upper and lower bounds on the terms A_n in the perturbation series. We outline below the procedure for estimating these upper and lower bounds. (Detailed proofs for the following statements are given in Appendix C.)

(i) From the form of the Feynman integral representation, it follows that in the perturbation series all diagrams with the same number of vertices add in phase.

(ii) The integral representation places *uniform* bounds on the contribution of each diagram having n vertices.

[2] A. M. Jaffe, in his thesis [Princeton University, 1965 (unpublished)], was able to analytically continue the resolvent for $\lambda\varphi^4$ field theories of any space dimension into the cut λ plane. Jaffe did not predict the properties of the resolvent when $|\arg\lambda|\geq 180°$ (off the first sheet) but, as far as he did predict, his work agrees with ours.

[3] F. J. Dyson, Phys. Rev. **85**, 631 (1952).

[4] A. M. Jaffe, Commun. Math. Phys. **1**, 127 (1965). This paper contains a very complete list of references.

(iii) The maximum and minimum number of diagrams with n vertices is found to be $(2n-1)!!8^n$ and $(n-1)!3^{-n}$.

Combining the above three results implies that $A_n(-1)^{n+1}$ are positive numbers bounded below and above by

$$A\Gamma(n)B^n<A_n(-1)^{n+1}<C\Gamma(\tfrac{5}{2}n)D^n, \quad (2.2)$$

where A, B, C, and D are positive constants.[5] Equation (2.2) implies that, although the perturbation series [Eq. (2.1)] is finite in every order, the ground-state energy is not an analytic function of λ^α about $\lambda=0$ for any α.

Method 2. The A_n can also be calculated from a difference equation. To derive this difference equation we use the differential equation [Eq. (1.1)] which is

$$(-d^2/dx^2+\tfrac{1}{4}x^2+\tfrac{1}{4}\lambda x^4)\Phi(x)=E(\lambda)\Phi(x), \quad (2.3)$$

and the boundary conditions [Eq. (1.2)]

$$\lim_{x\to\pm\infty}\Phi(x)=0. \quad (2.4)$$

In Eq. (2.3), we substitute

$$\Phi(x)=\sum_{n=0}^{\infty}\lambda^n e^{-x^2/4}B_n(x), \quad (2.5)$$

where $B_0=1$ and $B_n(x)$ are the polynomials in x to be determined for $n=1, 2, 3, \cdots$.

We then combine Eqs. (2.1), (2.3), and (2.5) to get

$$\sum_{n=0}^{\infty}\lambda^n[xB_n'(x)-B_n''(x)+\tfrac{1}{4}x^4B_{n-1}(x)]$$
$$=(\sum_{n=1}^{\infty}\lambda^nA_n)(\sum_{m=0}^{\infty}\lambda^mB_m(x)). \quad (2.6)$$

Finally, we let $x=\sqrt{2}y$ and

$$B_i(y)=\sum_{j=1}^{2i}y^{2j}B_{i,j}(-1)^i. \quad (2.7)$$

The result is

$$2jB_{i,j}=(j+1)(2j+1)B_{i,j+1}+B_{i-1,j-2}$$
$$-\sum_{p=1}^{i-1}B_{i-p,1}B_{p,j}. \quad (2.8)$$

Equation (2.8) is the desired difference equation because the numbers $B_{i,j}$ are related to the A_n by the simple equation

$$A_n=-B_{n,1}. \quad (2.9)$$

[5] If the Hamiltonian were Wick-ordered, there would be fewer diagrams of order n because a line could not have both ends connected to the same vertex. Thus, the terms in the Wick-ordered perturbation series would be slightly smaller than those of our non-Wick-ordered series. However, the estimate [Eq. (2.2)] would hold for both perturbation series.

Equation (2.8) was used on a computer to calculate the first 75 terms in the ground-state-energy perturbation series [Eq. (2.1)].[6] From these results we have determined that their detailed asymptotic growth is

$$A_n\sim(-1)^{n+1}(6/\pi^3)^{1/2}\Gamma(n+\tfrac{1}{2})3^n. \quad (2.10)$$

[In Appendix D we discuss the details of this computer calculation and list the first 75 terms in the perturbation series to 12-place accuracy. In Appendix E, we present the calculation of Eq. (2.10).]

Numerical investigation of the difference equation [Eq. (2.8)] brings to light the following exact additional properties of the A_n which are not immediately evident from the Feynman-diagram treatment of the perturbation series:

(i) All A_n are rational fractions.

(ii) If we define

$$I_n\equiv(-1)^{n+1}4^nA_n, \quad (2.11)$$

then the I_n are all positive *integers*.

(iii) Every integer I_n is divisible by 3.

We list below the first nine terms in the perturbation series in exact fraction form to illustrate the above characteristics of the A_n.

$$A_1=\tfrac{3}{4}, \quad (2.12a)$$
$$A_2=-21/8, \quad (2.12b)$$
$$A_3=333/16, \quad (2.12c)$$
$$A_4=-30\,885/128, \quad (2.12d)$$
$$A_5=916\,731/256, \quad (2.12e)$$
$$A_6=-65\,518\,401/1024, \quad (2.12f)$$
$$A_7=2\,723\,294\,673/2048, \quad (2.12g)$$
$$A_8=-1\,030\,495\,099\,053/32\,768, \quad (2.12h)$$
$$A_9=54\,626\,982\,511\,455/65\,536. \quad (2.12i)$$

III. ANALYTIC CONTINUATION OF ENERGY LEVELS

In Sec. II and associated appendices, we studied the properties of the ground-state-energy perturbation series [Eq. (2.1)]. However, this study did not reveal the reason for the divergence of the series. In other words, the singularity structure of $E_0(\lambda)$ in the complex λ plane is as yet unknown. In this section, we define the analytic continuation of the energy levels, which are the eigenvalues of the differential equation [Eq. (1.1)], into the λ plane. Then we present some heuristic derivations of some properties of this continuation.

[6] One author (CMB) wishes to acknowledge partial financial support for this calculation from Dr. B. M. McCoy.

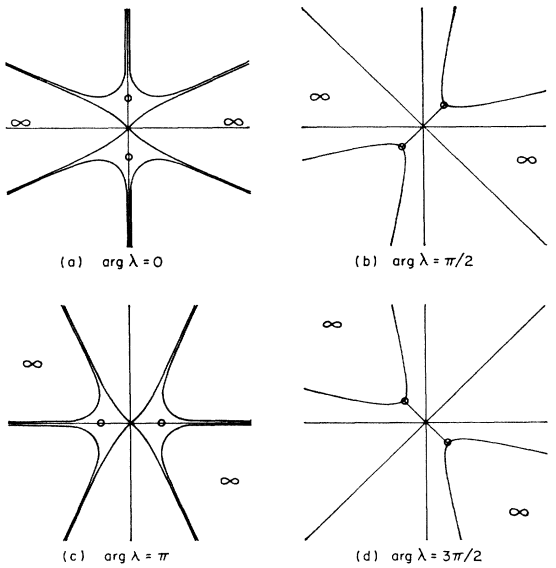

FIG. 1. Curves in the complex x plane where $\mathrm{Re}\lambda^{-1}[(1+\lambda x^2)^{1/2}-1]=0$ for various values of $\arg\lambda$. The circles denote the turning points at $x=\pm i\lambda^{-1/2}$. The symbol ∞ indicates the sectors in which the boundary condition [Eq. (3.4)] applies.

We shall need only Eqs. (1.1) and (1.2) which are

$$(-d^2/dx^2+\tfrac{1}{4}x^2+\tfrac{1}{4}\lambda x^4)\Phi(x)=E(\lambda)\Phi(x) \tag{3.1}$$

and

$$\lim_{x\to\pm\infty}\Phi(x)=0. \tag{3.2}$$

Equations (3.1) and (3.2) yield the asymptotic behavior of $\Phi(x)$ for large $|x|$ along the real axis:

$$\Phi(x)\sim\exp(-\tfrac{1}{6}\lambda^{1/2}|x|^3). \tag{3.3}$$

Equation (3.3) implies that Eq. (3.2), the boundary condition, holds not only on the real axis but also for complex x in the sector $|\arg x|<\frac{1}{6}\pi$.

Until now λ has been positive. When λ is complex, $\Phi(x)$ must satisfy the general boundary condition

$$\lim_{|x|\to\infty}\Phi(x)=0 \quad \text{when } |\arg(\pm x)+(\tfrac{1}{6}\arg\lambda)|<\tfrac{1}{6}\pi, \tag{3.4}$$

which follows from Eqs. (3.2) and (3.3).

We can now continue $E(\lambda)$ into the λ plane. We *define* the analytic continuation $E(\lambda)$ by the differential equation [Eq. (3.1)] in which $\Phi(x)$ satisfies the boundary condition in Eq. (3.4).

We shall use Eqs. (3.1) and (3.4) to find some exact properties of the analytic continuation of $E(\lambda)$:

Property (*i*). We may analytically continue $E(\lambda)$ to all points λ (except $\lambda=0$) where $\int_C\Phi^2(x,\lambda)dx\neq 0$. The integral is taken along an appropriate path C in the complex x plane.

Proof. Normalize $\Phi(x)$ by choosing

$$\Phi(0,\lambda)=1 \text{ for an even-parity wave function} \tag{3.5a}$$

and

$$(d/dx)\Phi(0,\lambda)=1 \text{ for an odd-parity wave function.} \tag{3.5b}$$

Vary λ by

$$\lambda\to\lambda+\delta\lambda. \tag{3.6}$$

Then,

$$E(\lambda)\to E(\lambda)+\delta E(\lambda) \tag{3.7}$$

and

$$\Phi(x,\lambda)\to\Phi(x,\lambda)+\delta\Phi(x,\lambda). \tag{3.8}$$

$\delta\Phi$ has the property that

$$\delta\Phi(0,\lambda)=0 \quad \text{or} \quad \delta\Phi'(0,\lambda)=0$$

for even or odd parity, respectively. $\delta\Phi(x,\lambda)$ obeys the same boundary conditions at ∞ as $\Phi(x,\lambda)$ does, namely, Eq. (3.4).

Substituting Eqs. (3.6), (3.7), and (3.8) into Eq. (3.1) gives

$$(d^2/dx^2-\tfrac{1}{4}x^2-\tfrac{1}{4}\lambda x^4)\delta\Phi(x,\lambda)-\tfrac{1}{4}\delta\lambda x^4\Phi(x,\lambda) =-E(\lambda)\delta\Phi(x,\lambda)+\delta E(\lambda)\Phi(x,\lambda). \tag{3.9}$$

Multiply Eq. (3.9) by $\Phi(x,\lambda)$. Then integrate by parts along a path C in the complex x plane such that each end of the path goes to ∞ in a sector in which the boundary condition [Eq. (3.4)] applies. [See Fig. 1 for the location of the sectors in which Eq. (3.4) applies for various choices of $\arg\lambda$.] The result of this manipulation is

$$0=\tfrac{1}{4}\delta\lambda\int_C x^4\Phi^2(x,\lambda)dx-\delta E(\lambda)\int_C\Phi^2(x,\lambda)dx. \tag{3.10}$$

Hence, $E(\lambda)$ is analytic (its derivative with respect to complex λ exists) at all points except those for which

$$\int_C\Phi^2(x,\lambda)dx=0. \tag{3.11}$$

Equation (3.11) will be used in Sec. VI as a condition for determining the location of the singularities of $E(\lambda)$.

Property (*ii*). At points where $E(\lambda)$ is analytic, there is no degeneracy in the energy.

Proof. Suppose $\Phi_1(x,\lambda)$ and $\Phi_2(x,\lambda)$ are two linearly-independent wave-function solutions to Eq. (3.1) each associated with energy E. We find that the derivative of the Wronskian $W'[\Phi_1(x),\Phi_2(x)]$ vanishes. Hence the Wronskian W is a constant. Because Φ_1 and Φ_2 obey the boundary condition [Eq. (3.4)], the constant must be 0. A vanishing Wronskian implies that Φ_1 and Φ_2 are not linearly-independent. Hence there is no degeneracy.

Property (*iii*). If there is a point λ_0 in the λ plane where $E_1(\lambda)\to E_2(\lambda)$ as $\lambda\to\lambda_0$, then $E_1(\lambda)$ is singular at $\lambda=\lambda_0$.

Proof. If $\Phi_i(x,\lambda)$ are the wave functions associated with $E_i(\lambda)$ for $i=1$ and 2, then because of property (ii)

$\Phi_1(x,\lambda) \to \Phi_2(x,\lambda)$ as $\lambda \to \lambda_0$. Consider the equation

$$\Phi_1(x,\lambda)(-d^2/dx^2+\tfrac{1}{4}x^2+\tfrac{1}{4}\lambda x^4)\Phi_2(x,\lambda) = E_2\Phi_1(x,\lambda)\Phi_2(x,\lambda). \quad (3.12)$$

Integrating Eq. (3.12) over the contour C in the complex x plane and integrating by parts gives

$$(E_1-E_2)\int_C \Phi_1(x,\lambda)\Phi_2(x,\lambda)dx=0. \quad (3.13)$$

Thus, when $\lambda \neq \lambda_0$,

$$\int_C \Phi_1(x,\lambda)\Phi_2(x,\lambda)dx=0. \quad (3.14)$$

Letting $\lambda \to \lambda_0$ in Eq. (3.14) gives

$$\int_C \Phi_2^2(x,\lambda)dx=0, \quad (3.15)$$

which is just Eq. (3.11). As a result of property (i), $E_1(\lambda)$ is singular at $\lambda=\lambda_0$. Thus, level crossing only occurs at singularity points of $E(\lambda)$.

Property (iv). The singularities of $E(\lambda)$ are of the square-root type for $\lambda \neq 0$.

Proof. Suppose $E(\lambda)$ has a singularity at $\lambda=\lambda_0\neq 0$. Let

$$\epsilon=(\lambda-\lambda_0)^{1/2}. \quad (3.16)$$

Then,

$$(-d^2/dx^2+\tfrac{1}{4}x^2+\tfrac{1}{4}\lambda_0 x^4+\tfrac{1}{4}\epsilon^2 x^4)\Phi(x,\epsilon)=E(\epsilon)\Phi(x,\epsilon). \quad (3.17)$$

Expand

$$\Phi(x,\epsilon)=\Phi_0(x)+\epsilon\Phi_1(x)+\epsilon^2\Phi_2(x)+\cdots \quad (3.18)$$

and

$$E(\epsilon)=\alpha+\epsilon\beta+\epsilon^2\gamma+\cdots. \quad (3.19)$$

If we let $\Theta(x)=-d^2/dx^2+\frac{1}{4}x^2+\frac{1}{4}\lambda_0 x^4$, then

$$\Theta(x)\Phi_0(x)=\alpha\Phi_0(x), \quad (3.20a)$$

$$\Theta(x)\Phi_1(x)=\alpha\Phi_1(x)+\beta\Phi_0(x), \quad (3.20b)$$

and

$$\Theta(x)\Phi_2(x)+\tfrac{1}{4}x^4\Phi_0(x)=\alpha\Phi_2(x)+\beta\Phi_1(x)+\gamma\Phi_0(x). \quad (3.20c)$$

Equation (3.20b) gives no useful information, but multiplying Eq. (3.20c) by $\Phi_0(x)$ and integrating gives

$$\beta=\int_C \tfrac{1}{4}x^4\Phi_0^2(x)dx \Big/ \int_C \Phi_1(x)\Phi_0(x)dx. \quad (3.21)$$

β exists if the denominator in Eq. (3.21) does not vanish[7]; that is,

$$\int_C \Phi_1(x)\Phi_0(x)dx\neq 0. \quad (3.22)$$

[7] To zeroth order in the WKB expansion discussed in Secs. IV and VI, it can be shown that for those points which satisfy Eq. (3.11), Eq. (3.22) is indeed true. If Eq. (3.22) were not true, we would have to reexpand $E(\lambda)$ and $\Phi(x,\lambda)$ in terms of $(\lambda-\lambda_0)^{1/3}=\epsilon$. Physically, this would mean that three energy levels cross at λ_0 instead of two as is the case. One would expect intuitively that the chance of three levels crossing at the same point λ_0 is small.

An argument similar to the one given in the proof of property (i) implies that λ_0 is a square-root-type singularity point of $E(\lambda)$. Since a square-root singularity has associated with it a double Riemann sheet, energy-level crossing must take place at all points $\lambda_0\neq 0$ satisfying Eq. (3.11).

IV. WKB SOLUTION

In this section, we shall approximate the wave function $\Phi(x)$ using zeroth-order WKB techniques.[8] We only consider the case where $|\lambda|$ is small. We will need Eqs. (1.1) and (1.2) which are

$$(-d^2/dx^2+\tfrac{1}{4}x^2+\tfrac{1}{4}\lambda x^4)\Phi(x)=E(\lambda)\Phi(x) \quad (4.1)$$

and

$$\lim_{x\to\pm\infty}\Phi(x)=0. \quad (4.2)$$

The zeroth-order WKB solution to Eq. (4.1) is[8]

$$\Phi=\text{const}\times\mu^{-1/2}\exp\left(\pm\int^x \mu(x)dx\right), \quad (4.3)$$

where

$$\mu(x)=[-E(\lambda)+\tfrac{1}{4}x^2+\tfrac{1}{4}\lambda x^4]^{1/2}. \quad (4.4)$$

For large $|x|$, Eq. (4.3), the zeroth-order WKB solution, becomes

$$\Phi(x)\sim\text{const}\times(x^2+\lambda x^4)^{-1/4}\times\exp\{(-6\lambda)^{-1}[(1+\lambda x^2)^{3/2}-1]\}. \quad (4.5)$$

Given $\arg\lambda$, we use Eq. (4.5) to locate in the complex x plane the turning points and the sectors in which the general boundary condition [Eq. (3.4)] applies. The turning points are at $\pm i\lambda^{-1/2}$. The boundaries of the above sectors are curves given by

$$\text{Re}\{\lambda^{-1}[(1+\lambda x^2)^{3/2}-1]\}=0.$$

The sectors and turning points are plotted in Fig. 1 for $\arg\lambda=0$, $\frac{1}{2}\pi$, π, and $\frac{3}{2}\pi$.

We now begin the WKB analysis. Near the origin $x=0$, the differential equation [Eq. (4.1)] becomes

$$(d^2/dx^2+E-\tfrac{1}{4}x^2)\Phi(x)=0, \quad (4.6)$$

which is the defining equation for the parabolic cylinder function $D_{E-\frac{1}{2}}(x)$.[9] The correct physical solution to Eq. (4.6) is

$$\Phi(x)=C[D_{E-\frac{1}{2}}(x)\pm D_{E-\frac{1}{2}}(-x)] \quad (4.7)$$

[8] For a review of zeroth-order WKB techniques, see A. Messiah, *Quantum Mechanics* (John Wiley & Sons, Inc., New York, 1961), Vol. 1, pp. 232–233. For a discussion of WKB techniques in the complex plane see T. T. Wu, Phys. Rev. **143**, 1110 (1966).

[9] A. Erdélyi, W. Magnus, F. Oberhettinger, and F. G. Tricomi, *Bateman Manuscript Project* (McGraw-Hill Book Co., New York, 1953), Vol. 2, pp. 116–123. In the future we will refer to this reference as BMP.

for even- or odd-parity wave functions. C is a multiplicative constant.

When the phase of λ is less than 270°, we can asymptotically connect the parabolic cylinder function in Eq. (4.7) to the WKB solution in Eq. (4.5). The connection restricts the possible values of E to

$$E=2n+\tfrac{1}{2} \quad \text{for even parity} \tag{4.8a}$$

and

$$E=2n+\tfrac{3}{2} \quad \text{for odd parity}, \tag{4.8b}$$

where $n=0, 1, 2, \cdots$.

However, Eq. (4.7) cannot be directly connected to Eq. (4.5) when the phase of λ is near 270° because a turning point lies in the path of the connection [see Fig. 1(d)]. Instead, we use the following procedure:

We define new variables $r\equiv xe^{i\pi/4}$, $\rho\equiv\lambda e^{-3\pi i/2}$, and $\epsilon\equiv 4iE$. In terms of these variables, Eq. (4.1) becomes

$$[d^2/dr^2+\tfrac{1}{4}(-\epsilon+r^2-\rho r^4)]\Phi(r)=0. \tag{4.9}$$

We assume that $|\lambda|$ is so small that $|\rho\epsilon|\ll 1$.

We define

$$r_0\equiv\{[1-(1-4\rho\epsilon)^{1/2}](2\rho)^{-1}\}^{1/2}\sim\epsilon^{1/2} \tag{4.10a}$$

and

$$r_1\equiv\{[1+(1-4\rho\epsilon)^{1/2}](2\rho)^{-1}\}^{1/2}\sim\rho^{-1/2}. \tag{4.10b}$$

r_1 is approximately the position of the turning point. $|\rho\epsilon|\ll 1$ implies that $|r_0|\ll|r_1|$.

Next, we identify four regions in the complex x plane: region A in which r is near the origin, $0<|r|\ll|r_0|\ll|r_1|$; region B in which $|r|\ll|r_1|$ but $|r|\gg|r_0|$; region C in which r is near the turning point, $|r|\lesssim|r_1|$ but $|r|\gg|r_0|$; and region D in which r is at the turning point, $|r|\sim|r_1|$.

In order to carry out the WKB procedure when $\arg\lambda$ is near 270°, approximate solutions to Eq. (4.9) must be found in regions A, B, C, and D. Then these solutions must be connected asymptotically at the boundaries of the regions.

Region A. In region A, Eq. (4.9) becomes approximately (that is, to zeroth order in λ)

$$(d^2/dr^2-\tfrac{1}{4}\epsilon-\tfrac{1}{4}r^2)\Phi_A(r)=0. \tag{4.11}$$

Equation (4.11) is the transformed defining equation for a parabolic cylinder function.[9] The even-parity solution to this equation is

$$\Phi_A(r)=C[D_{-\frac{1}{2}-i\epsilon/4}(re^{-i\pi/4})+D_{-\frac{1}{2}-i\epsilon/4}(-re^{-i\pi/4})]. \tag{4.12}$$

(In this section, we will treat even-parity wave functions in detail; only the final results will be given for odd-parity solutions.)

To compute the asymptotic expansion of Φ_A in region A, we need the leading term in the asymptotic expansion of $D_\nu(z)$,[9] which is

$$D_\nu(z)\sim z^\nu e^{-\frac{1}{4}z^2} \quad \text{for } -\tfrac{3}{4}\pi<\arg z<\tfrac{3}{4}\pi \tag{4.13a}$$

and

$$D_\nu(z)\sim z^\nu e^{-\frac{1}{4}z^2}-[(2\pi)^{1/2}/\Gamma(-\nu)]e^{\nu\pi i}z^{-\nu-1}e^{\frac{1}{4}z^2}$$

$$\text{for } \tfrac{1}{4}\pi<\arg z<(5/4)\pi. \tag{4.13b}$$

Equation (4.13) yields the asymptotic behavior of $\Phi_A(r)$ in region A for large $|r|$; that is, for r near region B. After much algebraic manipulation we have the desired asymptotic expansion:

$$\Phi_A(r)\sim\frac{C_1}{\sqrt{r}}\left\{\frac{1}{\Gamma(\frac{1}{4}-\frac{1}{8}i\epsilon)}\exp[\tfrac{1}{4}i(r^2-\epsilon\ln r+\tfrac{1}{2}\epsilon\ln 2-\tfrac{1}{2}\pi)]\right.$$

$$\left.+\frac{1}{\Gamma(\frac{1}{4}+\frac{1}{8}i\epsilon)}\exp[-\tfrac{1}{4}i(r^2-\epsilon\ln r+\tfrac{1}{2}\epsilon\ln 2-\tfrac{1}{2}\pi)]\right\}, \tag{4.14}$$

where

$$C_1=\frac{C2\pi e^{\pi\epsilon/16}2^{-i\epsilon/8}}{\Gamma(\frac{3}{4}+\frac{1}{8}i\epsilon)}. \tag{4.15}$$

Region B. In regions B and C, we need the zeroth-order WKB solution[8] to Eq. (4.9) which is

$$\Phi_{\mathrm{WKB}}(x)=(-\epsilon+r^2-\rho r^4)^{-1/4}$$

$$\times\left\{C_2\exp\left[\tfrac{1}{2}i\int_{r_0}^{r}(-\epsilon+r^2-\rho r^4)^{1/2}dr\right]\right.$$

$$\left.+C_3\exp\left[-\tfrac{1}{2}i\int_{r_0}^{r}(-\epsilon+r^2-\rho r^4)^{1/2}dr\right]\right\}. \tag{4.16}$$

In Eq. (4.16), we choose to integrate from the reference point r_0. We are free to choose the reference point, but this choice must be maintained for the WKB solution in region C as well as in region B.

For r in region B, the choice of r_0 and r_1 in Eq. (4.10) allows us to simplify Eq. (4.16). To zeroth order in ρ,

$$(-\epsilon+r^2-\rho r^4)\sim r^2-\epsilon. \tag{4.17}$$

Using Eq. (4.17), we easily evaluate the integrals and expand the expression in front of the curly brackets in Eq. (4.16). The result is

$$\Phi_B(x)\sim(1/\sqrt{r})\{C_2\exp[\tfrac{1}{4}i(r^2-\epsilon\ln r-\tfrac{1}{2}\epsilon+\epsilon\ln(\tfrac{1}{2}\sqrt{\epsilon}))]$$
$$+C_3\exp[-\tfrac{1}{4}i(r^2-\epsilon\ln r-\tfrac{1}{2}\epsilon+\epsilon\ln(\tfrac{1}{2}\sqrt{\epsilon}))]\}. \tag{4.18}$$

Region C. In region C, we need the zeroth-order WKB solution [Eq. (4.16)] to Eq. (4.9). The WKB solution in region C has the same constants C_2 and C_3 as the solution in region B. However, the integrals for region C are more difficult to perform than those for region B because of the approximations associated with region C which are $|r_0|\ll|r|\lesssim|r_1|$. We use the following

decomposition to evaluate the integrals:

$$\tfrac{1}{2}i\int_{r_0}^{r} dr(-\epsilon+r^2-\rho r^4)^{1/2} = -\tfrac{1}{2}i\rho^{1/2}\int_{r}^{r_1}[(r^2-r_0^2)(r_1^2-r^2)]^{1/2}dr + \tfrac{1}{2}i\rho^{1/2}\int_{r_0}^{r_1}[(r^2-r_0^2)(r_1^2-r^2)]^{1/2}dr. \quad (4.19)$$

To do the first integral, we introduce a change of variables

$$R=r_1-r. \quad (4.20)$$

Then, the integral becomes

$$-\tfrac{1}{2}i\rho^{1/2}(2r_1)^{1/2}(r_1^2-r_0^2)^{1/2}\int_0^R R^{1/2}dR \sim -i3^{-1}2^{1/2}\rho^{-1/4}R^{3/2}. \quad (4.21)$$

In the second integral (the definite integral), we let $K^2=1-r_0^2r_1^{-2}$. Then,

$$\tfrac{1}{2}i\rho^{1/2}\int_{r_0}^{r_1}[(r^2-r_0^2)(r_1^2-r^2)]^{1/2}dr = \tfrac{1}{6}ir_1^3[(1+r_0^2r_1^{-2})E(k)-2r_0^2r_1^{-2}K(k)]. \quad (4.22)$$

$E(k)$ and $K(k)$ are the standard elliptic integrals.[10] Expanding $E(k)$ and $K(k)$ for k near 1 gives[11]

$$E(k)\sim 1+\frac{r_0^2}{4r_1^2}\left(4\ln 2-1-2\ln\frac{r_0}{r_1}\right) \quad (4.23a)$$

and

$$K(k)\sim\ln(4r_1/r_0). \quad (4.23b)$$

Thus,

$$\tfrac{1}{2}i\rho^{1/2}\int_{r_0}^{r_1} dr[(r^2-r_0^2)(r_1^2-r^2)]^{1/2} \sim\frac{i}{6\rho}-\tfrac{1}{8}i\epsilon\left[\ln\frac{1}{\rho\epsilon}+4\ln 2+1\right]. \quad (4.24)$$

We also note that

$$(-\epsilon+r^2-\rho r^4)^{-1/4}\sim\rho^{-1/4}(r_1^2-r_0^2)^{-1/4}(2r_1)^{-1/4}R^{-1/4} \sim 2^{-1/4}\rho^{1/8}R^{-1/4}. \quad (4.25)$$

Equations (4.19)–(4.25) imply that near the turning point the WKB solution [Eq. (4.16)] in region C is asymptotically

$$\Phi_C(r)\sim 2^{-1/4}\rho^{1/8}R^{-1/4}\left\{C_2\exp\left[-\tfrac{1}{3}i\sqrt{2}\rho^{-1/4}R^{3/2}+\frac{i}{6\rho}-\tfrac{1}{8}i\epsilon\left(\ln\frac{1}{\rho\epsilon}+4\ln 2+1\right)\right] +C_3\exp\left[+\tfrac{1}{3}i\sqrt{2}\rho^{-1/4}R^{3/2}-\frac{i}{6\rho}+\tfrac{1}{8}i\epsilon\left(\ln\frac{1}{\rho\epsilon}+4\ln 2+1\right)\right]\right\}. \quad (4.26)$$

Region D. In region D, we use Eq. (4.20) to change variables from r to R in the differential equation [Eq. (4.9)] and find that to zeroth order in ρ

$$(d^2/dR^2+\tfrac{1}{2}R\rho^{-1/2})\Phi_D(R)=0. \quad (4.27)$$

Equation (4.27) is an Airy differential equation.[12] The solution to Eq. (4.27) is

$$\Phi_D(R)=D(\tfrac{1}{3}y)^{1/2}K_{1/3}(2y^{3/2}), \quad (4.28)$$

where

$$y=-(18\sqrt{\rho})^{-1/3}R. \quad (4.29)$$

There is only one undetermined constant D in Eq. (4.28) because only the $K_{1/3}$ function obeys the general boundary conditions at ∞ [Eq. (3.4)].

To determine the behavior of $K_{1/3}(2y^{3/2})$ near the turning point, we note that[13]

$$K_{1/3}(2e^{3\pi i/2}(-y)^{3/2})=-\tfrac{1}{2}i\pi\{e^{i\pi/6}H_{1/3}^{(1)}(2(-y)^{3/2}) +e^{-i\pi/6}H_{1/3}^{(2)}(2y^{3/2})\}. \quad (4.30)$$

Then we use the asymptotic expansions[14] for $H^{(1)}$ and $H^{(2)}$ and find that in region D near the turning point

$$\Phi_D\sim C_4R^{-1/4}[\exp(\tfrac{1}{3}i\sqrt{2}\rho^{-1/4}R^{3/2}-\tfrac{1}{4}i\pi) +\exp(-\tfrac{1}{3}i\sqrt{2}\rho^{-1/4}R^{3/2}+\tfrac{1}{4}i\pi)], \quad (4.31)$$

where

$$C_4=D\pi^{1/2}3^{-1/3}2^{-11/12}\rho^{1/24}. \quad (4.32)$$

We have thus completed the determination of the asymptotic behavior of Φ in regions A, B, C, and D. Next we must connect Φ across regions A and B and across C and D. (Φ in region B is already connected to Φ in region C because the same WKB solution is used in both regions.) Connecting Eq. (4.14) in region A to Eq. (4.18) in region B and Eq. (4.26) in region C to Eq. (4.31) in region D gives two independent determinations of the ratio C_2/C_3. These are, respectively,

$$\frac{C_2}{C_3}=\frac{\Gamma(\frac{1}{4}+\frac{1}{8}i\epsilon)}{\Gamma(\frac{1}{4}-\frac{1}{8}i\epsilon)}\exp\left[i\left(\frac{\epsilon}{2}+\frac{\epsilon}{2}\ln\frac{8}{\epsilon}-\frac{\pi}{2}\right)\right] \quad (4.33a)$$

and

$$\frac{C_2}{C_3}=\exp\left[i\left(\frac{\pi}{2}-\frac{1}{3\rho}+\frac{\epsilon}{4}\ln\frac{16}{\rho\epsilon}+\frac{\epsilon}{4}\right)\right]. \quad (4.33b)$$

[10] BMP, Vol. 2, pp. 317–318.

[11] To expand $E(k)$ and $K(k)$ near $k=1$, we use

$$E(k)=\tfrac{1}{2}\pi\,{}_2F_1(-\tfrac{1}{2},\tfrac{1}{2};1;k^2) \quad \text{and} \quad K(k)=\tfrac{1}{2}\pi\,{}_2F_1(\tfrac{1}{2},\tfrac{1}{2};1;k^2).$$

[See BMP, Vol. 2, p. 318, Eqs. (5) and (6).] Then we consult BMP, Vol. 1, p. 110, Eq. (12), where we find the appropriate expansion for the hypergeometric functions. These expansions may be simplified by using BMP, Vol. 1, pp. 1–20.

[12] BMP, Vol. 2, p. 22.

[13] BMP, Vol. 2, p. 5, Eq. (15) and p. 80, Eq. (43).

[14] BMP, Vol. 2, p. 85, Eqs. (1) and (2).

Combining Eqs. (4.33a) and (4.33b) gives a single transcendental equation which contains ρ and ϵ only.

$$\frac{\Gamma(\frac{1}{4}-\frac{1}{8}i\epsilon)}{\Gamma(\frac{1}{4}+\frac{1}{8}i\epsilon)}=\exp\left[\frac{5i\pi}{4}+\frac{i}{3\rho}+\frac{i\epsilon}{4}\ln\frac{\rho}{2}\right]. \quad (4.34)$$

Because Φ_D goes exponentially to 0 at ∞ as required by Eq. (4.5), we have connected the origin to ∞ via regions A, B, C, and D. We have found that Eq. (4.34) is the one requirement which must be obeyed for this connection to be valid. One should note that since Eq. (4.9) is a second-order differential equation, there are two undetermined constants in each region A, B, C, and D. An equation such as Eq. (4.34) is posible only because parity considerations restrict Φ_A to one constant C_1 and the boundary conditions [Eq. (3.4)] at ∞ limit Φ_D to one constant C_4.

If we rewrite Eq. (4.34) in terms of E instead of ϵ, we get

$$\frac{\Gamma(\frac{1}{4}+\frac{1}{2}E)}{\Gamma(\frac{1}{4}-\frac{1}{2}E)}=\exp\left(\frac{i}{3\rho}+\frac{5\pi i}{4}-E\ln\frac{\rho}{2}\right). \quad (4.35a)$$

For odd-parity wave functions the equivalent result is

$$\frac{\Gamma(\frac{3}{4}+\frac{1}{2}E)}{\Gamma(\frac{3}{4}-\frac{1}{2}E)}=\exp\left(\frac{i}{3\rho}-\frac{5\pi i}{4}-E\ln\frac{\rho}{2}\right). \quad (4.35b)$$

We may regard these results as approximate equations to zeroth order in λ relating the low-lying energy levels to complex λ near $\arg\lambda=270°$.

Appendix F contains a WKB calculation similar to the calculation in this section but to first order in λ. The result of Appendix F is more precise than Eq. (4.35), but it reduces to Eq. (4.35) when terms to first order in λ are neglected.

V. RESOLVENT

As a simple application of Eq. (4.35), we calculate in this section the locations of the singularities of the resolvent of the Hamiltonian H [Eq. (1.3)]. The resolventof the Hamiltonian is conventionally defined as $(z-H)^{-1}$. We will prove that $(z-H)^{-1}$ has poles when $\arg\lambda$ is near 270°.

It is clear that $(z-H)^{-1}$ has a pole in the λ plane whenever the following equations are satisfied:

$$\frac{\Gamma(\frac{1}{4}+\frac{1}{2}z)}{\Gamma(\frac{1}{4}-\frac{1}{2}z)}=e^{5\pi i/4}e^{i/3\rho}e^{-z\ln(\rho/2)} \quad (5.1a)$$

for even parity;

$$\frac{\Gamma(\frac{3}{4}+\frac{1}{2}z)}{\Gamma(\frac{3}{4}-\frac{1}{2}z)}=e^{-5\pi i/4}e^{i/3\rho}e^{-z\ln(\rho/2)} \quad (5.1b)$$

for odd parity.

We solve each of the above equations separately for ρ by taking the logarithm of both sides. Given z, it is always possible to choose a sufficiently small $|\rho|$ so that the $i/3\rho$ term dominates the $z\ln(2/\rho)$ term. Thus, the resolvent has poles at

$$\lambda=e^{3\pi i/2}\left[-3i\ln\frac{\Gamma(\frac{1}{4}+\frac{1}{2}z)}{\Gamma(\frac{1}{4}-\frac{1}{2}z)}+6\pi N+\tfrac{1}{4}\pi\right]^{-1}, \quad (5.2a)$$

provided that z is not an even integer $+\frac{1}{2}$, and at

$$\lambda=e^{3\pi i/2}\left[-3i\ln\frac{\Gamma(\frac{3}{4}+\frac{1}{2}z)}{\Gamma(\frac{3}{4}-\frac{1}{2}z)}+6\pi N-\tfrac{1}{4}\pi\right]^{-1}, \quad (5.2b)$$

provided that z is not an odd integer $+\frac{1}{2}$. In Eq. (5.2), N is any sufficiently large positive integer.

Equation (5.2) implies that for both parities $(z-H)^{-1}$ has an infinite sequence of poles approaching the origin. These correspond to an infinite sequence of choices for large positive N.

That the resolvent has poles was a surprise to us and, as far as we know, this had not even been conjectured. Jaffe[2] has proved that for negative z the resolvent is analytic in the cut λ plane, the cut extending from the origin to $-\infty$ along the negative real axis. This is entirely consistent with our results which maintain that no singularities appear in the resolvent until the phase of the coupling constant reaches nearly $\pm 270°$. Since the analytic continuation of the resolvent disappears through Jaffe's cut at $\arg\lambda=180°$, Jaffe observed no singularities.

VI. QUALITATIVE DESCRIPTION OF $E(\lambda)$

In this section, we use the results of Secs. III and IV to discover the properties of the analytic continuation of the energy levels $E(\lambda)$ as a function of complex λ. For purposes of clarity, this section is divided into three parts. Part A gives a calculation of the locations in the λ plane of the branch points of $E(\lambda)$. Part B describes in detail the paths traced out by the energy levels in the complex E plane as λ moves about the complex λ plane. Part C contains a discussion of level crossing.

A. Locations of Branch Points

We find the coordinates in the λ plane of the branch points of $E(\lambda)$ by solving a pair of simultaneous equations. The first of these is Eq. (4.35) which resulted from matching the zeroth-order WKB solution across the boundaries of regions A, B, C, and D, as was done in Sec. IV. We repeat Eq. (4.35) here:

$$\frac{\Gamma(\frac{1}{4}+\frac{1}{2}E)}{\Gamma(\frac{1}{4}-\frac{1}{2}E)}=\exp\left(\frac{i}{3\rho}+\frac{5\pi i}{4}-E\ln\frac{\rho}{2}\right)$$

for even parity, (6.1a)

$$\frac{\Gamma(\frac{3}{4}+\frac{1}{2}E)}{\Gamma(\frac{3}{4}-\frac{1}{2}E)}=\exp\left(\frac{i}{3\rho}-\frac{5\pi i}{4}-E\ln\frac{\rho}{2}\right)$$

for odd parity. (6.1b)

The second of the simultaneous equations is obtained by explicitly performing the integral in Eq. (3.11) which is

$$0=\int_C dx\,\Phi^2(x,\lambda). \tag{6.2}$$

Equation (6.2) is an exact condition which is satisfied by those λ and only those λ which are branch points of $E(\lambda)$. Now that the approximate WKB wave function is known from Sec. IV, it is possible to evaluate this integral and in turn to use the result to find the branch points. The evaluation is very lengthy and is done in Appendix G. The result of this calculation is

$$0=-\psi(E+\tfrac{1}{2})+\frac{\pi}{2}\cot\left(\frac{1}{4}-\frac{E}{2}\right)\pi+\ln\frac{4}{\rho}$$

for even parity (6.3a)

and

$$0=-\psi(E+\tfrac{1}{2})+\frac{\pi}{2}\cot\left(\frac{3}{4}-\frac{E}{2}\right)\pi+\ln\frac{4}{\rho}$$

for odd parity, (6.3b)

where ψ is the logarithmic derivative of the Γ function. Equation (6.3) now embodies the condition for λ to be a singularity of $E(\lambda)$. Because Eqs. (6.3) and (6.1) are both correct to zeroth order in λ and since these are the pair of simultaneous equations that will be solved to find the singularities of $E(\lambda)$, the positions of the singularities will be known to zeroth order in λ.

Equations (6.1) and (6.3) must be put into a form favorable for simultaneous solution. To do so we use the following procedure:

(i) In this calculation, we consider $|\rho|$ to be small and argρ to be near 0° (because argλ is near 270°). We may express this by

$$\mathrm{Re}(1/\rho)=O(\Lambda) \tag{6.4a}$$

and

$$\mathrm{Im}(1/\rho)=O(\ln\Lambda), \tag{6.4b}$$

where Λ is very large.

(ii) We also consider only low-lying energy levels (that is, levels near the ground state) because the derivation of Eq. (6.1) in Sec. IV is valid only for such levels. We may express this by

$$|E|=O(1). \tag{6.5}$$

(iii) Define η by

$$E=2n+\tfrac{1}{2}+\eta \quad \text{for even parity} \tag{6.6a}$$

and

$$E=2n+\tfrac{3}{2}+\eta \quad \text{for odd parity}, \tag{6.6b}$$

where $n=0, 1, 2, \cdots$.

(iv) Take the logarithm of both sides of Eq. (6.1), substitute Eq. (6.6), and apply the Legendre duplication formula[15] to get

$$\ln\Gamma(-n-\tfrac{1}{2}\eta)-\ln\Gamma(n+\tfrac{1}{2}\eta)+\ln\Gamma(2n+\eta)+\ln(2\sqrt{\pi})-(2n+\eta)\ln 2=\frac{i}{3\rho}+\frac{5\pi i}{4}-E\ln\frac{\rho}{2}-2\pi Ni \tag{6.7a}$$

and

$$\ln\Gamma(-n-\tfrac{1}{2}\eta)-\ln\Gamma(n+\tfrac{1}{2}\eta)+\ln\Gamma(2n+\eta)+\ln(2\sqrt{\pi})-(2n+\eta)\ln 2+\ln(\tfrac{1}{2}+n+\tfrac{1}{2}\eta)=\frac{i}{3\rho}-\frac{5\pi i}{4}-E\ln\frac{\rho}{2}-2\pi Ni. \tag{6.7b}$$

N in this equation is an integer referring to the multiple-valued logarithmic function.

(v) Expand the lnΓ functions using[16]

$$\ln\Gamma(1+z)=-\gamma z+\sum_{m=2}^{\infty}(-1)^m\xi(m)(z^m/m). \tag{6.8}$$

The result is

$$\ln(\tfrac{1}{2}\eta)+\ln(2n)!+\tfrac{1}{2}\ln(2\pi)+\eta\left(-\gamma+\sum_{k=1}^{2n}\frac{1}{k}\right)+\sum_{m=2}^{\infty}\frac{(-1)^m\eta^m}{m}\left\{\xi(m)-\xi(m)\left[\frac{1}{2^m}+\frac{1}{(-2)^m}\right]-\sum_{k=1}^{2n}\frac{1}{k^m}\right\}=\frac{i}{3\rho}+\frac{\pi i}{4}-(n\pi i)-2\pi Ni-(2n+\eta+\tfrac{1}{2})\ln\frac{\rho}{4} \tag{6.9a}$$

and

$$\ln(\tfrac{1}{2}\eta)+\ln(2n+1)!+\tfrac{1}{2}\ln(2\pi)+\eta\left(-\gamma+\sum_{k=1}^{2n+1}\frac{1}{k}\right)+\sum_{m=2}^{\infty}\frac{(-1)^m\eta^m}{m}\left\{\xi(m)-\xi(m)\left[\frac{1}{(2)^m}+\frac{1}{(-2)^m}\right]-\sum_{k=1}^{2n+1}\frac{1}{k^m}\right\}=\frac{i}{3\rho}+\frac{\pi i}{4}-(n\pi i)-2\pi(N+1)i-(2n+\eta+\tfrac{3}{2})\ln\frac{\rho}{4}. \tag{6.9b}$$

(vi) Substitute Eq. (6.6) into Eq. (6.3) and expand the function using[17]

$$\psi(1+z)=-\gamma+\sum_{m=2}^{\infty}(-1)^m\xi(m)z^{m-1}. \tag{6.10}$$

The result is

$$\frac{1}{\eta}+\left(-\gamma+\sum_{k=1}^{2n}\frac{1}{k}\right)+\sum_{m=2}^{\infty}(-1)^m\eta^{m-1}\left[-\sum_{k=1}^{2n}\frac{1}{k^m}+\xi(m)-\left(\frac{1}{2^m}+\frac{1}{(-2)^m}\right)\xi(m)\right]=-\ln(\tfrac{1}{4}\rho) \tag{6.11a}$$

[15] BMP, Vol. 1, p. 5, Eq. (15).
[16] BMP, Vol. 1, p. 45, Eq. (2). $\xi(m)$ is the Riemann zeta function.
[17] BMP, Vol. 1, p. 45, Eq. (5).

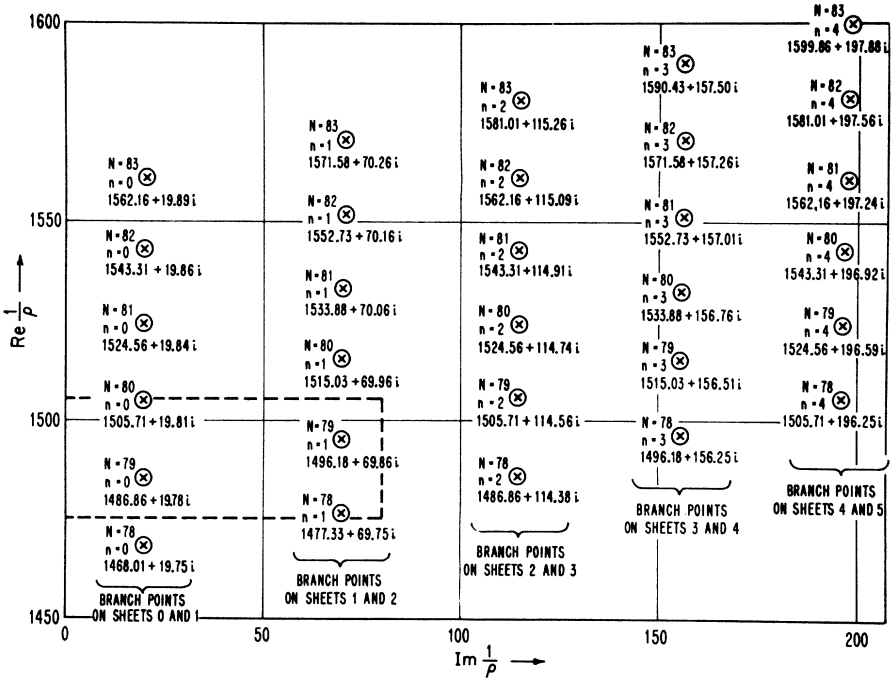

FIG. 2. A portion of the $1/\rho$ plane on which are plotted the even-parity branch points. The branch points are labeled by the integers n ranging from 0 through 4 and N ranging from 78 to 83. Next to each branch point are its coordinates correct to the nearest hundredth. The region of the graph surrounded by the dashed line is enlarged in Fig. 4 and will be discussed in connection with level crossing.

and

$$\frac{1}{\eta}+\left(-\gamma+\sum_{k=1}^{2n+1}\frac{1}{k}\right)+\sum_{m=2}^{\infty}(-1)^m\eta^{m-1}\left[-\sum_{k=1}^{2n+1}\frac{1}{k^m}+\xi(m)-\left(\frac{1}{2^m}+\frac{1}{(-2)^m}\right)\xi(m)\right]=-\ln(\tfrac{1}{4}\rho). \quad (6.11b)$$

Equations (6.9) and (6.11) are now in a form amenable to approximate simultaneous solution. The expansions in Eqs. (6.9) and (6.11) are power series in η and are most susceptible to techniques of approximation if $|\eta|\ll 1$. η is indeed small as will be seen. N is a large integer.

The simultaneous solution consists of using Eqs. (6.9) and (6.11) alternatively to provide successively more accurate approximations to $1/\rho$ and $1/\eta$. First, we use Eq. (6.9). To zeroth order in η, the locations of the singularities in the analytic continuation $E(\lambda)$ are approximately

$$1/\rho=6\pi N \quad (6.12a)$$

and

$$1/\rho=6\pi(n+1). \quad (6.12b)$$

Next we use Eqs. (6.12) and (6.11) to find $1/\eta$ to zeroth order in η:

$$\frac{1}{\eta}=\ln(24\pi N)+\gamma-\sum_{k=1}^{2n}\frac{1}{k} \quad (6.13a)$$

and

$$\frac{1}{\eta}=\ln[24\pi(N+1)]+\gamma-\sum_{k=1}^{2n+1}\frac{1}{k}. \quad (6.13b)$$

Remembering that N is large, we see that Eq. (6.13) verifies the earlier assumption that $|\eta|$ is small. We use Eqs. (6.13) and (6.9) and iterate once more to find the locations of the singularities of $E(\lambda)$ to first order in η:

$$1/\rho=6\pi N+3\pi n-3\pi/4-3i[\ln(2n)!+\tfrac{1}{2}\ln(\pi/2)-\ln\ln N-(2n+\tfrac{1}{2})\ln(24\pi N)-1] \quad (6.14a)$$

and

$$1/\rho=6\pi(N+1)+3\pi n+3\pi/4-3i\{\ln(2n+1)!+\tfrac{1}{2}\ln(\pi/2)-\ln\ln N-(2n+\tfrac{3}{2})\times\ln[24\pi(N+1)]-1\}. \quad (6.14b)$$

Equation (6.14) is the equation we have sought. It lists the coordinates of the branch points of $E(\lambda)$ parametrized by two integers N (large) and n (small).

Before discussing this equation, it is appropriate to make some observations here on the structure of Eqs. (6.9) and (6.11), the simultaneous equations that were used to derive Eq. (6.14). We note that Eq. (6.11) is the derivative of Eq. (6.9) with respect to η. This is true because Eq. (6.3) is the logarithmic derivative of Eq. (6.1) with respect to E. However, the significance of this relationship is not obvious.

An explanation for this relationship comes from a broader interpretation of Eqs. (4.35) than we have chosen to use. We might have assumed that Eqs. (4.35) were more than just equations relating E and λ for small λ. We might have treated them as approximate (to zeroth order in λ) analytic continuations of E as a function of λ. Based on this assumption, the condition for locating those points in the λ plane where the energy

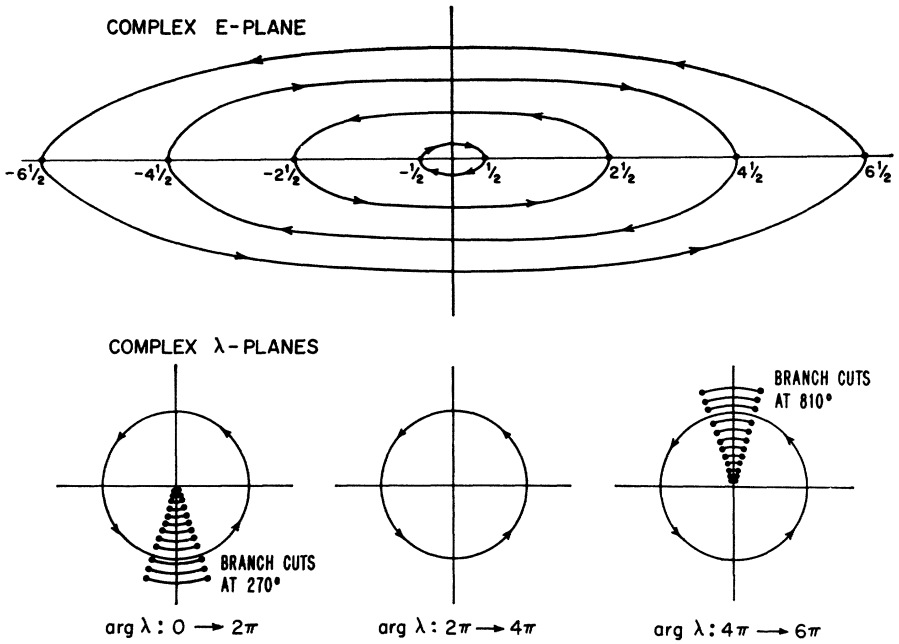

FIG. 3. A rough picture of a tour of the λ plane with the associated image of $E(\lambda)$ in the E plane. $|\lambda|$ is held fixed while $\arg\lambda$ ranges from 0 to 6π. The λ plane is drawn as three planes to show the full rotation of $\arg\lambda$. The branch points and arc-shaped branch lines are indicated schematically. The graph gives the paths of the even-parity energy levels only. The odd-parity levels behave similarly.

is twofold degenerate would be that the derivative of Eq. (6.9) with respect to E (or η) must be a valid equation. Because it gives the correct answer, this argument would have constituted a "derivation" of Eq. (6.11) and would have been a short cut past Appendix G.

We did not use the above argument on grounds of rigor. It is not clear what is meant by an "approximate zeroth-order analytic continuation." It is difficult to understand how one could approximate an exact concept like analyticity by zeroth-order techniques. The laborious method of getting Eq. (6.11) by deriving and evaluating $\int_C \Phi^2(x)dx$ in Sec. III and Appendix G was presented in this paper to avoid these mathematical imprecisions.

Having made this point we return to analyze Eq. (6.14). We note that as the integer N takes on a sequence of values approaching ∞, the associated sequence of branch points in the λ plane given by Eq. (6.14) begins at a finite distance from the origin and has a limit point at the origin. (Remember that $\rho=\lambda e^{-3\pi i/2}$.) That this sequence of branch points has a limit point at the origin explains the divergence of the perturbation series discussed earlier in Sec. II. Note that the origin in the λ plane is not an isolated singularity of $E(\lambda)$.

It is useful to plot the results of Eq. (6.14) on a graph. Figure 2 shows a portion of the $1/\rho$ plane on which are plotted the even-parity branch points labeled by the integer n ranging from 0 through 4 and N ranging from 78 to 83. Next to each branch point are its coordinates correct to two decimal places. The region of the graph surrounded by the dashed line is blown up in Fig. 4 and will be discussed in detail in Part C of this section in connection with level crossing.

It is clear from Fig. 2 that the branch points lie in a sequence of sequences. Each sequence of branch points is labeled by n and each sequence approaches the origin. A glance at Fig. 4 shows that the first ($n=0$) sequence of points lies at an angle of about $\frac{3}{4}°$ less than 270° in the λ plane. The next ($n=1$) sequence lies approximately $2\frac{1}{2}°$ less than 270°. Each successive sequence is rotated by a small angle toward the real axis in the λ plane.

However, we suspect that as n increases, these sequences bunch up before they reach 180° and that no branch points have a complex argument of less than 180°. If there were such branch points then, by the argument of Sec. V, the resolvent of the Hamiltonian would have poles in the cut λ plane. Jaffe[2] proved that this is not so. Unfortunately, Eq. (6.14) only holds for small n so this bunching phenomenon cannot be illustrated graphically.[18]

B. Paths in E Plane

Having found the branch points of $E(\lambda)$, we can give a qualitative picture of the complex function $E(\lambda)$. We will do this by tracing the paths of the energy levels in the complex E plane as λ moves about the complex λ plane.

We begin our analysis by observing that both differential equations [Eqs. (4.1) and (4.9)] are real in

[18] In Sec. VI C, it will be shown that the WKB solution [Eq. (6.1)] is valid in two semidisk-shaped regions of the λ plane bounded by $\arg\lambda=270°\pm90°$ and $\arg\lambda=810°\pm90°$. In the rest of the λ plane, the WKB solution is superseded by Eq. (4.8). Since the existence of branch points is a phenomenon associated only with Eq. (6.1), branch points must lie throughout the region in which Eq. (6.1) holds. Therefore, we expect that the bunching phenomenon occurs at the boundaries of the region; that is, at $\arg\lambda=180°$, 360°, 720°, and 900°.

their respective variables. From this arise some symmetry properties of $E(\lambda)$ which are

$$E(\lambda)=E^*(\lambda^*)=-E^*(e^{3\pi i}\lambda^*)=-E(e^{3\pi i}\lambda). \quad (6.15)$$

We also note that since the Hamiltonian [Eq. (1.3)] is Hermitian and positive definite, the energy levels lie along the positive real axis in the energy plane when λ is positive real. Similarly, when $\arg\lambda=270°$, Eq. (4.9) indicates that the energy levels are pure imaginary. At $\arg\lambda=270°$, the energy levels lie on both the positive and negative imaginary axes.

Equation (4.8) indicates that for fixed $|\lambda|$ the energy levels lie near the real axis in the energy plane as $\arg\lambda$ increases from 0 until nearly 270°. However, for $\arg\lambda$ in the neighborhood of 270° the full energy spectrum moves rapidly toward the imaginary axis.

Combining the symmetry properties [Eq. (6.1)] with the results of the previous two paragraphs enables us to predict the paths of the energy levels in the complex E plane when $|\lambda|$ is held fixed and $\arg\lambda$ is varied. (These paths are given in Fig. 3.) We predict that:

(i) When $\arg\lambda=0$, the energy spectrum lies along the positive real axis.

(ii) As $\arg\lambda$ increases toward $\frac{1}{4}(6\pi)$, each energy level traces out a path in the complex E plane which approaches the imaginary axis. The energy levels lie along the imaginary axis at $\arg\lambda=\frac{1}{4}(6\pi)$.

(iii) At $\arg\lambda=\frac{1}{2}(6\pi)$, the energy levels lie on the negative real axis at points equal to the negative of their original values for $\arg\lambda=0$.

(iv) As $\arg\lambda$ increases from $\frac{1}{2}(6\pi)$ to 6π, each energy level traces a path reflectionally symmetric about the origin to the path described above. At $\arg\lambda=6\pi$ the energy levels return to the values on the real axis that they had for $\arg\lambda=0$.

Thus, as λ goes in a circle three times about the origin in the λ plane, starting at the real axis, each energy level in the E plane traces a closed path starting and ending on the positive real axis. Moreover, it will be shown in Part C of this section that the paths in the complex E plane are concentric and that the directions of successive concentric paths alternate between clockwise and counterclockwise.

Because the energy levels return to their original values when $\arg\lambda$ is increased by 6π, in addition to the branch-point singularities discussed in Part A of this section there is also a $\lambda^{1/3}$ singularity at the origin. However, the divergence of perturbation theory discussed in Sec. II is in no way related to this singularity. If it were, we could then expand $E(\lambda)$ in powers of $\lambda^{1/3}$ and get a convergent power series. But there is no convergent power-series representation of $E(\lambda)$ about the origin in any fractional power of λ. Perturbation theory diverges because of the sequences of branch points discussed in Part A of this section.

How do these branch points affect the curves in the E plane in Fig. 3? Surprisingly, the answer to this question is "*not at all.*" To understand why this is we argue as follows:

(i) As proved in Sec. III, each of these singularities is a square-root type. (In the neighborhood of each branch point, there is a double-sheeted Riemann surface.) Level crossing (twofold degeneracy) must occur at each branch point. A branch line is connected to each singularity.

(ii) The symmetry properties listed in Eq. (6.15) imply that there is another set of branch points symmetrically reflected about 270° in the λ plane. Moreover, all these singularities (those in Fig. 2 and those reflected through 270°) also must occur at −270° (which is the same as 810° because of the cube-root singularity at the origin).

(iii) The branch lines join the branch points on either side of 270° and on either side of 810° pairwise; that is, each point is joined to its reflected image. These branch cuts are shaped like circular arcs and are very short because the branch points lie a short distance from either side of 270° or 810°. The cuts form two sequences of concentric arcs with each sequence of arcs approaching the origin. These cuts are schematically displayed on Fig. 3.

(iv) In the circular tour of the λ plane shown on Fig. 3 it is clear that none of the branch cuts was crossed because $|\lambda|$ was held fixed.

This reasoning explains why the branch points could not affect the curves in the E plane in Fig. 3. In addition, it shows that the presence of the branch points can never be detected as long as $|\lambda|$ is held fixed.

Therefore, to exhibit the delicate intricacy of this theory caused by the array of cuts in Fig. 3, we must make a new tour of the λ plane in which we cross the branch cuts. We still increase $\arg\lambda$ from 0 to 6π. However, when $\arg\lambda$ is near 270° (or 810°) we vary $|\lambda|$ so that the path in the λ plane crosses a branch line. If we follow an energy level as it traces a path in the complex E plane, we find that it returns to the positive real axis when $\arg\lambda=6\pi$ but as a different energy level from the original. This is what is meant by level crossing.

A complete description of level crossing is given in Part C of this section. Nevertheless, it should now be clear that the energy levels are completely interrelated by the structure of the branch cuts. That is, one can get to any given energy level from any other energy level by continuing around the appropriate branch points.

As a corollary of this, once one knows the ground-state energy of the anharmonic oscillator, one knows all the energy levels by analytic continuation. Or, in other words, *the physical energy levels of the anharmonic oscillator for a given positive real* λ *are the positive real values of* $E(\lambda)$ *on each of the infinite number of branches of a Riemann surface. Each energy level corresponds to a sheet of this Riemann surface.* This is the major result of our paper.

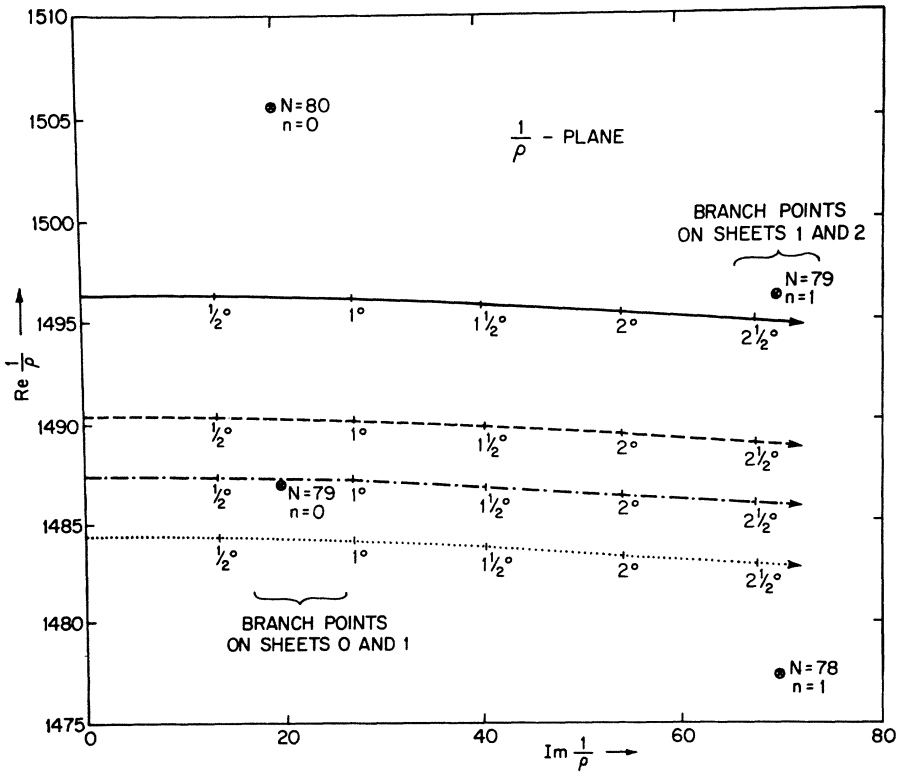

FIG. 4. An enlargement of a small portion of the $1/\rho$ plane of Fig. 2. The $(N=78, n=1)$, $(N=79, n=0)$, $(N=79, n=1)$, and $(N=78, n=1)$ branch points are precisely located. Four short circular arcs of unequal radii are drawn and the angular rotation along each arc is given in degrees. The arcs are indicated by a solid line, dashed line, dash-dotted line, and dotted line. The radii of these arcs to the nearest hundredth are 1496.50, 1490.50, 1487.50, and 1484.50, respectively. The images of these arcs in the E plane are plotted on Fig. 5 in connection with level crossing.

C. Level Crossing

We have shown that level crossing takes place at the branch points in the λ plane and we know the positions of these branch points. But the details of level crossing are not yet fully explained. In particular, in this part of Sec. VI we will answer these questions: (a) Which levels cross at which branch points? (b) What does level crossing look like graphically?

To give a precise answer to these questions, we carry out an intensive program of numerical calculation which enables us to construct Figs. 4 and 5. Figures 4 and 5 will give a beautiful pictorial answer to the questions we have posed. The numerical calculation involves the following procedure:

(i) We take a sequence of circular paths of varying radii in the $1/\rho$ plane. (This is equivalent to and easier to handle numerically than a sequence of circles in the λ plane.) The image of each of these circles in the E plane is a set of energy-level curves like those crudely drawn in Fig. 3. As in Fig. 3, we will consider the even-parity case only. (The odd-parity case is qualitatively similar.)

(ii) By symmetry [see Eq. (6.15) and the accompanying discussion] we need only to vary $\arg\lambda$ from 0° to 270°. Moreover, until $\arg\lambda$ is very nearly 270°, the energy levels stay extremely close to the values given in Eq. (4.8a). Therefore, we only need to consider the portions of the energy-level curves for λ near 270° (that is, where $1/\rho$ is approximately real).

(iii) We choose the sequence of radii in the $1/\rho$ plane so that a branch point is crossed. We then use Eq. (6.1a) to plot points accurately in the E plane and expect to observe the process of level crossing. We choose four values for the radii of the curves in the $1/\rho$ plane. To the nearest hundredth, these values of $|1/\rho|$ are 1496.50, 1490.50, 1487.50, and 1484.50. These curves are drawn on Fig. 4. (Figure 4 is an enlarged version of a part of Fig. 2. That portion which is enlarged is indicated on Fig. 2 by a dashed line.) The branch point that is crossed is the one labeled $N=79$, $n=0$.

(iv) We then choose points along the curves in Fig. 4 and solve the transcendental equation [Eq. (6.1)] numerically for complex E to the nearest hundredth. To solve Eq. (6.1a) numerically we take the logarithm of both sides. This gives

$$\ln\Gamma(\tfrac{1}{4}+\tfrac{1}{2}E)-\ln\Gamma(\tfrac{1}{4}-\tfrac{1}{2}E) = \frac{i}{3\rho}+\frac{5\pi i}{4}+E\ln\frac{2}{\rho}+2\pi iM. \quad (6.16)$$

M is an integer referring to the many-valued logarithmic function. M occurs for the same reason that N did in Eq. (6.7).

(v) We choose the branch line for the logarithmic function in Eq. (6.16) to lie along the negative real axis. We then break up Eq. (6.16) into its real and imaginary parts and, given $1/\rho$, solve the resulting

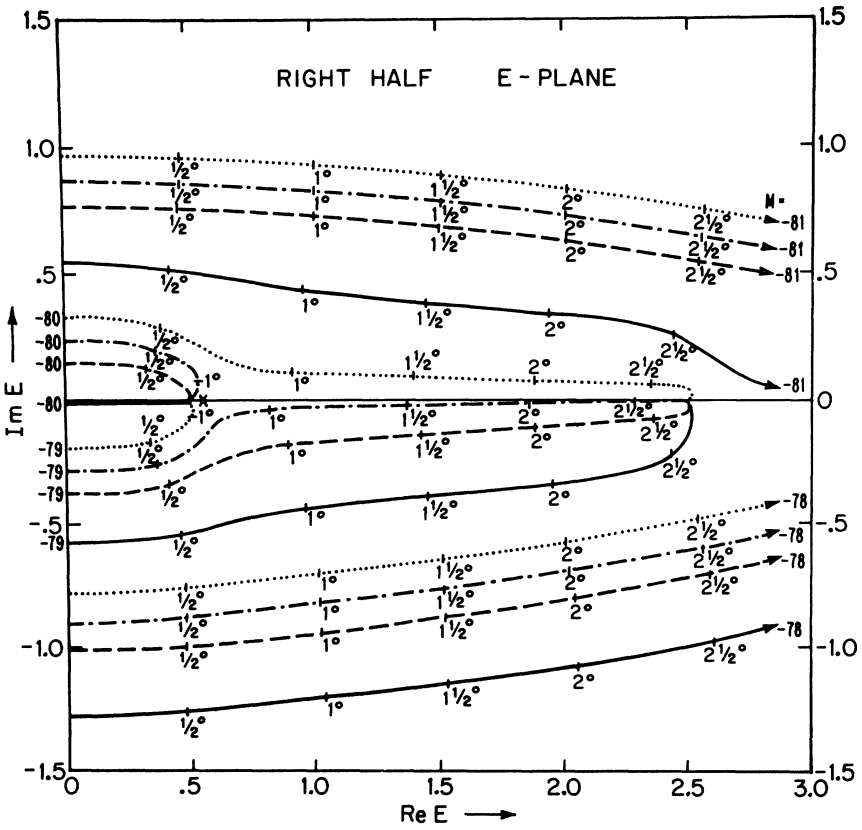

FIG. 5. The complex E plane, showing to the nearest hundredth the images of the circular arcs in the $1/\rho$ plane of Fig. 4. These curves exhibit level crossing. The precise point where this level crossing occurs is marked with an ×. × is the image of the branch point ($N=79$, $n=0$). The energy-level curves shown correspond with the four lowest energy levels. The curves are labeled by their M values [see Eq. (6.16)] and are represented by dotted, dot-dashed, dashed, and solid lines to correspond with the values of $|1/\rho|$ on Fig. 4 (1484.50, 1487.50, 1490.50, and 1496.50). The numbers in degrees along each curve indicate the rotation in the $1/\rho$ plane of the preimage points along the curves on Fig. 4.

pair of simultaneous transcendental equations for E. The solution is a painstaking trial-and-error process involving tables of $\ln\Gamma(z)$ functions. Fortunately, this mammoth task is tractable because $\arg(1/\rho)$ is small, ranging from 0 to approximately 3° (see Fig. 4). Several hundred values for E have been plotted on Fig. 5 and the energy-level curves have been drawn and labeled by their M values. We have plotted the energy-level curves connected to the four lowest energy levels.

This completes the description of the construction of Figs. 4 and 5. We list below our observations and conclusions with regard to these graphs.

As $|1/\rho|$ gets smaller, we observe that the points where the energy-level curves meet the imaginary axis in the E plane move upward. {This fact can be proved from the differential equation [Eq. (4.9)] independently of the numerical calculations involved in the graphing of Fig. 5.} Therefore, as $|1/\rho|$ decreases, either we will find that (a) level crossing occurs (that is, the curves associated with a given M value become associated with different energy values along the real E axis) or else that (b) the curves all remain attached to the same energy values on the real E axis and stretch higher and higher up the imaginary axis. The enormous distortion of the curves in the latter possibility (b) makes the former (a) a good probability.

Level crossing is indeed found on inspection of Fig. 5. (For each value of M the curves on Fig. 5 are represented in order of decreasing $|1/\rho|$ by solid lines, dashed lines, dash-dotted lines, and dotted lines.) We observe the following example of level crossing.

The solid $M=-79$ line is connected to $E=2\frac{1}{2}$ along the real axis. There is just a hint of inflection as this curve passes below $E=\frac{1}{2}$. Its companion, the solid $M=-80$ line, goes directly to $E=\frac{1}{2}$ and the dash-dotted line inflects strongly. Meanwhile, the companion $M=-80$ lines belly increasingly outward above $E=\frac{1}{2}$ to meet the inflecting $M=-79$ lines. As the curves in the $1/\rho$ plane pass through the $N=79$, $n=0$ branch point, the energy levels flip and the dotted $M=-79$ line is suddenly and discontinuously connected to $E=\frac{1}{2}$ and the companion $M=-80$ line to $E=2\frac{1}{2}$. *This is a precise pictorial description of level crossing.*

It is interesting to determine where the curves connecting $E=\frac{1}{2}$ and $E=2\frac{1}{2}$ cross. To determine this crossing point we have calculated the image of the branch point $N=79$, $n=0$ in the E plane. The crossing point is labeled on Fig. 5 by an ×.

In addition to the above level crossing we also observe an incipient level crossing at $E=2\frac{1}{2}$. As we look at the lines in the opposite order (dotted, dot-dashed, dashed, and solid), we see on Fig. 4 that a level crossing is about to take place at the $N=79$, $n=1$

branch point. The solid lines $M=-81$ and $M=-79$ are ready to cross near $E=2\frac{1}{2}$.

As $|\lambda|$ increases and the circle in the λ plane passes through the appropriate branch points, each pair of adjacent levels flips repeatedly. *Only adjacent levels can cross.* For example, the zeroth and first energy levels cross at $n=0$ branch points, the $n=1$ and $n=2$ energy levels cross at $n=1$ branch points, and so on. In general, *the nth sequence of branch points is common to and lies only on the nth and (n+1)st sheets of the Riemann surface of* $E(\lambda)$. (Remember that the nth sheet is the sheet which contains the nth energy level for real positive λ.) This is equivalent to saying that *the nth and (n+1)st energy levels can only cross at the nth sequence of branch points.*

Figure 5 evokes an additional, unexpected, and most welcome observation. Recall that near $\lambda=270°$ the WKB equation [Eq. (6.1a)] describes the paths of the energy levels in the E plane. Near $\arg\lambda=0$, Eq. (4.8a) describes these paths. Therefore, at some value of $\arg\lambda$ between 270° and 0°, Eq. (6.1a) is no longer accurate and is superseded by Eq. (4.8a). The angle at which this changeover occurs defines the angular opening about 270° (and 810°) in which the WKB solution is valid. The numerical investigations used to plot Fig. 5 give answers to two important questions: (a) At what angle does the changeover occur? (b) How accurate is the WKB result?

To answer these questions, we notice that the curves on Fig. 5 all approach the values predicted by Eq. (4.8a) which are $E=\frac{1}{2}, 2\frac{1}{2}, 4\frac{1}{2}, \cdots$. However, at some angle in the $1/\rho$ plane the WKB result [Eq. (6.1a)] no longer directs the energy-level curves toward these values. At that angle the curves begin to veer away. Therefore, the angle at which the curves are closest to the points $E=\frac{1}{2}, 2\frac{1}{2}, 4\frac{1}{2}, \cdots$ is the answer to question (a). Furthermore, the *distance* of closest approach of the WKB curves to these points is a useful measure of how accurate the WKB solution is and will answer question (b). The WKB result is accurate if the distance of closest approach is small.

It is easy to calculate the changeover angle and the distance of closest approach. We expand $\ln\Gamma(\frac{1}{4}-\frac{1}{2}E)$ in a Taylor series[16] about $E=\frac{1}{2}, 2\frac{1}{2}, 4\frac{1}{2}, \cdots$, and insert the series into Eq. (6.16). We determine from the resulting equation that $\arg(1/\rho)=90°$ is the condition for closest approach. That is, the WKB result applies in an angular opening of 90° on either side of $\arg\lambda=270°$ (and 810°). This answers question (a). This angular opening is much wider than originally anticipated and shows that the WKB solution is valid in a very large region of the λ plane.

Furthermore, the changeover angle at $\arg\lambda=180°$ ties in beautifully with three earlier results of this paper, First, Fig. 1(c) in Sec. IV shows that it is exactly at $\arg\lambda=180°$ that the WKB turning point enters the sector in which the general boundary condition [Eq. (3.4)] applies. This is just the point at which we would expect Eq. (6.1a), the WKB equation, to become valid. Second, a changeover angle of 180° indicates that the bunching phenomenon discussed in Part A of Sec. VI takes place at 180°.[18] Third, when combined with the arguments of Sec. V, a changeover angle of 180° implies that the poles of the resolvent occur when $\arg\lambda$ is larger than 180°. This is consistent with Jaffe's findings.[2]

A short numerical calculation reveals that the distance of closest approach to $E=\frac{1}{2}$ is the amazingly small number $\frac{1}{3}\times10^{-153}$. This distance is much smaller than originally anticipated. It is gratifying to observe that, in answer to question (b), the WKB solution is extremely accurate.

ACKNOWLEDGMENTS

We wish to thank Professor A. M. Jaffe for many interesting discussions. We are most grateful to Mrs. Carl M. Bender for her careful editorial reading of this paper.

APPENDIX A

In this appendix, we use field theory to derive the coordinate representation given by Eqs. (1.1) and (1.2) which are

$$(-d^2/dx^2+\tfrac{1}{4}x^2+\tfrac{1}{4}\lambda x^4)\Phi(x)=E(\lambda)\Phi(x) \tag{A1}$$

and

$$\lim_{x\to\pm\infty}\Phi(x)=0. \tag{A2}$$

A quick derivation follows from substituting

$$\varphi=2^{-1/2}x \quad \text{and} \quad \dot{\varphi}=-i2^{1/2}d/dx \tag{A3}$$

into the Hamiltonian H [Eq. (1.3)] which is

$$H=\tfrac{1}{2}\dot{\varphi}^2+\tfrac{1}{2}m^2\varphi^2+\lambda\varphi^4. \tag{A4}$$

We choose $m=1$ without loss of generality. Then the wave function $\Phi(x)$ satisfies the differential equation [Eq. (A1)]. Equation (A2) results from the requirement that $\Phi(x)$ be normalizable.

A more rigorous derivation follows from the assumption that in Fock space the energy eigenstates of H are normalizable.

In the interaction picture, the representation of the field φ in terms of creation and annihilation operators on Fock space is

$$\varphi=[1/(2m)^{1/2}](ae^{-imt}+a^\dagger e^{imt}), \tag{A5a}$$

$$\dot{\varphi}=(\tfrac{1}{2}m)^{1/2}(-iae^{-imt}+ia^\dagger e^{imt}), \tag{A5b}$$

$$[a,a^\dagger]=1. \tag{A5c}$$

In terms of the creation and annihilation operators, the Hamiltonian is

$$H=m(a^\dagger a+\tfrac{1}{2})+(g/4m^2)(a+a^\dagger)^4. \tag{A6}$$

Let $|E\rangle$ be an eigenstate of H of energy E. Then

$$H|E\rangle=E|E\rangle. \tag{A7}$$

The Fock representation consists of the states

$$|n\rangle=[1/\sqrt{(n!)}](a^\dagger)^n|0\rangle, \quad n=0, 1, 2, \cdots \tag{A8a}$$

where

$$\langle n|m\rangle=\delta_{nm}. \tag{A8b}$$

Assuming that $\{|n\rangle\}$ are a complete set of states,

$$|E\rangle=\sum_{n=0}^{\infty} a_n|n\rangle\sqrt{(n!)}. \tag{A9}$$

Equations (A6)-(A9) can be combined to give a difference equation that the $\{a_n\}$ satisfy:

$$\begin{aligned}Ea_n=m(n+\tfrac{1}{2})a_n+(g/4m^2)[&a_{n-4}+4(n-2)a_{n-2}\\&+6n(n-1)a_n+4n(n+1)(n+2)a_{n+2}\\&+(n+1)(n+2)(n+3)(n+4)a_{n+4}].\end{aligned} \tag{A10}$$

Furthermore, since $|E\rangle$ is assumed normalizable,

$$\sum_{n=0}^{\infty} |a_n|^2 n!<\infty. \tag{A11}$$

From Eq. (A11) it follows that as $n\to\infty$, $|a_n|\to 0$ faster than $1/\sqrt{(n!)}$. Thus, if we define the generating function $f(z)$ by

$$f(z)=\sum_{n=0}^{\infty} a_n z^n, \tag{A12}$$

then $f(z)$ is entire.

The difference equation [Eq. (A10)] becomes a differential equation satisfied by $f(z)$:

$$Ef(z)=\frac{g}{4m^2}\left(z+\frac{d}{dz}\right)^4 f(z)+mz\frac{d}{dz}f(z)+\tfrac{1}{2}mf(z). \tag{A13}$$

The differential equation [Eq. (A13)] and the side condition that $f(z)$ is entire embody the assumption that $|E\rangle$ is an eigenstate of H of finite norm and that $\{|n\rangle\}$ are a complete set of states.

Now let

$$f(z)=e^{-z^2/2}h(z). \tag{A14}$$

Then,

$$\begin{aligned}Eh(z)=&(g/4m^2)h''''(z)\\&+m[zh'(z)-z^2h(z)]+\tfrac{1}{2}mh(z).\end{aligned} \tag{A15}$$

Next we show that $h(z)$ must go to 0 rapidly along the imaginary axis. To do this, we apply the Cauchy formula to Eq. (A12):

$$a_n=\frac{1}{2\pi i}\oint\frac{f(z)}{z^{n+1}}dz=\frac{1}{2\pi i}\oint\frac{e^{-z^2/2}g(z)}{z^{n+1}}. \tag{A16}$$

Suppose that $g(z)$ did not go to 0 along the imaginary axis. In fact, suppose for simplicity that $g(z)$ were constant. Then, since the major contribution to this integral would come from the region where the path of integration crosses the imaginary axis, we calculate (using the method of steepest descent) that $a_n\sim A/\sqrt{n}$ for large n. This contradicts Eq. (A11) and the entirety of $f(z)$. Hence, $h(z)$ must go to 0 along the imaginary axis.

To see how fast $h(z)\to 0$, we consider the differential equation for $h(z)$ [Eq. (A15)]. By systematically investigating pairs of terms, we find that the dominant behavior at large z is given by

$$(g/4m^2)h''''(z)-mz^2h(z)=0. \tag{A17}$$

Thus,

$$h(z)\sim\exp[C\tfrac{2}{3}(4m^2/g)^{1/4}z^{3/2}], \tag{A18}$$

where $C=1, -1, i, -i$. Since there are four solutions to the fourth-order differential equation [Eq. (A17)], we have them all. Combining Eq. (A18) with our knowledge that $h(z)$ goes to 0 along the imaginary axis, we learn that $h(z)$ goes to 0 along the imaginary axis extremely rapidly. Hence, $h(z)$ may be Fourier-transformed along the imaginary axis. We thus define (for real x)

$$h(z)=\int_{-\infty}^{\infty} H(x)e^{zx}dx. \tag{A19}$$

Fourier-transforming Eq. (A15) according to Eq. (A19) gives

$$\begin{aligned}&mH''(x)+mxH'(x)+(E+\tfrac{1}{2}m)H(x)\\&\qquad-(g/4m^2)x^4H(x)=0.\end{aligned} \tag{A20}$$

Finally, we let

$$H(x)=e^{-x^2/4}\Phi(x). \tag{A21}$$

This substitution [Eq. (A21)] simplifies (A20) to

$$m\Phi''(x)+E\Phi(x)-\tfrac{1}{4}mx^2\Phi(x)-(g/4m^2)x^4\Phi(x)=0. \tag{A22}$$

Equation (A22) implies that the dominant behavior of Φ for large x is $e^{|x|^3}$ or $e^{-|x|^3}$. The former is not possible because the inverse transform would not exist and $h(z)$ would be undefined. Hence, $\Phi(x)\to 0$ as $|x|\to\infty$.

We set $m=1$ as we did earlier in this appendix and Eq. (A22) becomes

$$-\Phi''(x)+\tfrac{1}{4}x^2\Phi(x)+\tfrac{1}{4}\lambda x^4\Phi(x)=E\Phi(x), \tag{A23}$$

and

$$\lim_{|x|\to\infty}\Phi(x)=0. \tag{A24}$$

Equations (A23) and (A24) are precisely Eqs. (A1) and (A2). We have thus rigorously derived the coordinate representation and the boundary condition satisfied by the wave function Φ.

This connection between field theory and the wave function in coordinate space will be used in Sec. II and Appendices B and C.

APPENDIX B

In this appendix, we use Feynman rules to calculate A_1, A_2, and A_3. These are the first three terms beyond the constant zero-point energy term in the perturbation series for the ground-state energy [Eq. (2.1)]. The Feynman rules for the anharmonic oscillator considered as a field theory are

$$(E^2-m^2+i\epsilon)^{-1} \quad \text{for a propagator,} \tag{B1a}$$

$$24\lambda \quad \text{for a vertex,} \tag{B1b}$$

$$i(2\pi)^{-1}\int_{-\infty}^{\infty} dE \quad \text{for a loop integration.} \tag{B1c}$$

The symmetry numbers are the same as in the (3+1)-dimensional theory.[19]

The nth term in the perturbation series for the ground-state energy is the sum of all connected n-vertex Feynman diagrams with no external legs. (Each diagram is of course multiplied by its appropriate symmetry number.) Figure 6 shows all such diagrams of order $n=1$, 2, and 3 with their symmetry numbers.[20]

We find the contribution of each of the diagrams by applying the Feynman rules [Eq. (B1)] and by evaluating the resulting simple contour integrals. This calculation yields

$$\begin{aligned}
&(1a)=(24)\tfrac{1}{4}(\lambda/m^3)m,\\
&(2a)=(24)^2(-1/16)(\lambda/m^3)^2m,\\
&(2b)=(24)^2(-1/32)(\lambda/m^3)^2m,\\
&(3a)=(24)^3(3/128)(\lambda/m^3)^3m,\\
&(3b)=(24)^3(5/512)(\lambda/m^3)^3m,\\
&(3c)=(24)^3(3/512)(\lambda/m^3)^3m,\\
&(3d)=(24)^3(1/64)(\lambda/m^3)^3m.
\end{aligned} \tag{B2}$$

Next, we combine the symmetry numbers listed on Fig. 6 with Eq. (B2) and find that[21]

$$A_1=(24/4)\times\tfrac{1}{8}=\tfrac{3}{4}, \tag{B3a}$$

$$A_2=-\frac{(24)^2}{16\times16}-\frac{(24)^2}{32\times48}=-\frac{21}{8}, \tag{B3b}$$

[19] T. T. Wu, Phys. Rev. **125**, 1436 (1962).

[20] The interaction we are discussing is not Wick-ordered. If it were Wick-ordered, only diagrams (2b) and (3c) would contribute to the ground-state-energy perturbation series. All other diagrams would be excluded because they have at least one line emerging from and returning to the same vertex.

[21] In the definition of the ground-state-energy perturbation series [Eq. (2.1)], which is

$$E_0(\lambda)=\tfrac{1}{2}m+\sum_{n=1}^{\infty}(\lambda/m^3)^n mA_n,$$

the factors of m and $(\lambda/m^3)^n$ have been isolated from the A_n.

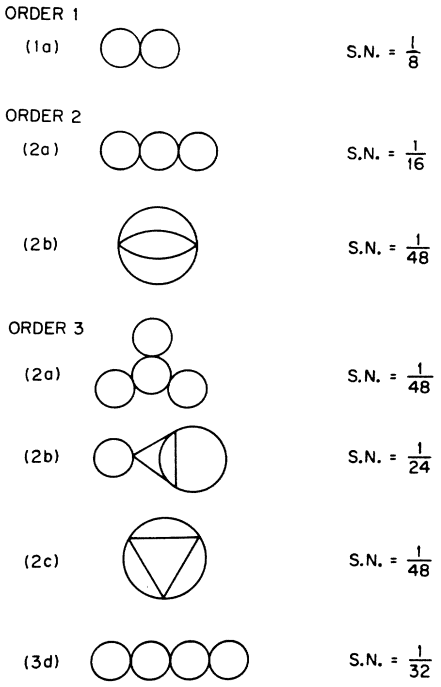

FIG. 6. All connected Feynman diagrams of order $n=1$, 2, and 3 with no external legs. To the right of each diagram is its symmetry number (S.N.).

$$A_3=\frac{(24)^3\times3}{128\times48}+\frac{(24)^3\times5}{512\times24}+\frac{(24)^3\times3}{512\times48}+\frac{(24)^3}{64\times32}=\frac{333}{16}. \tag{B3c}$$

The results in Eqs. (B3a)–(B3c) agree with Eqs. (2.12a)–(2.12c). This agreement illustrates the equivalence of the Feynman-diagram and difference-equation techniques for computing the numbers A_n.

APPENDIX C

Equation (2.2) gave without proof upper and lower bounds on $(-1)^{n+1}A_n$,[22] the terms in the perturbation series for the ground-state energy [Eq. (2.1)]. In this appendix, we rigorously derive these bounds using a Feynman-integral representation.

To develop the notation of the Feynman-integral representation, we restate here the ground-state-energy perturbation series [Eq. (2.1)]:

$$E_0(\lambda)=\tfrac{1}{2}m+\sum_{n=1}^{\infty} m(\lambda/m^3)^n A_n. \tag{C1}$$

A_n is the sum of all connected diagrams of order n (diagrams with n vertices) with no external legs.

[22] We find upper and lower bounds on $(-1)^{n+1}A_n$ because $(-1)^{n+1}A_n$ are positive quantities. This is evident from Eq. (C3).

Rewriting the perturbation series [Eq. (C1)] in terms of individual diagrams gives

$$E_0(\lambda)=\tfrac{1}{2}m+\sum_{n=1}^{\infty} m(\lambda/m^3)^n\sum I, \tag{C2}$$

where the summation $\sum I$ is over all connected diagrams of order n with no external legs.

I is the numerical value of each diagram. In terms of a Feynman-integral representation for non-Wick-ordered connected diagrams with no external legs, I is given by

$$I=(\text{symmetry number})(\tfrac{1}{2})^w(24)^n(-1)^{n+1}\times(1/4\pi)^{(n+1-w)/2}\Gamma(\tfrac{3}{2}n-\tfrac{1}{2}w-\tfrac{1}{2})\times\int_0^\infty\prod_{i=1}^{N}\frac{dx_i}{\sqrt{x_i}}\{\delta(1-\sum_{i=1}^{N}x_i)/[D(1/x_i)]^{1/2}\}. \tag{C3}$$

In Eq. (C3), (i) n=order of diagram=number of vertices. In Eq. (C3), $n\geq 2$. For $n=1$, A_1 is calculated in Appendix B. (ii) w=number of lines returning to the same vertex from which they emerged (w would be zero if the interaction were Wick ordered). (iii) N=number of lines$=2n$. (iv) x_i=the Feynman parameters. (v) A skeleton of a diagram is a simply connected subdiagram. $D(1/x_i)$=the sum over all skeletons of a diagram of

$$\left(\prod_{i=1}^{N}\frac{1}{x_i}\right)^{-1}.$$

The product is taken over all lines in the skeleton.[23]

We use Eq. (C3) to establish bounds on $(-1)^{n+1}A_n$ by proving a sequence of theorems.

Theorem 1. Let

$$L\equiv\int_0^\infty\prod_{i=1}^{N}\frac{dx_i}{\sqrt{x_i}}\{\delta(1-\sum_{i=1}^{N}x_i)/[D(1/x_i)]^{1/2}\}. \tag{C4}$$

Then, (i) L is uniformly bounded above and (ii) L is uniformly bounded below. (A uniform bound on diagrams means that the bound depends only on the number of vertices and not on the individual diagrams.)

Proof of part (i). We substitute $y=\sqrt{x_i}$ into the definition of L [Eq. (C4)] and get

$$L=\int_0^\infty\prod_{i=1}^{N}2dy_i\frac{\delta(1-\sum y_i^2)}{[D(1/y_i^2)]^{1/2}}. \tag{C5}$$

Let K_N be the surface area of a unit sphere in N dimensions. Then,

$$L\leq K_N\sup\frac{1}{[D(1/y_i^2)]^{1/2}}\Big|_{\Sigma y_i^2=1}. \tag{C6}$$

We evaluate the factors in Eq. (C6) separately.

[23] Precise definitions of "skeleton" and "$D(1/x_i)$" are given in Jaffe's paper. See Ref. 4.

By a straightforward evaluation of the surface-area integral, we get for the first factor

$$K_N=2\pi^{\frac{1}{2}N}/\Gamma(\tfrac{1}{2}N). \tag{C7}$$

In order to find the maximum of $[D(1/y_i^2)]^{-1/2}|_{\Sigma y_i^2=1}$, we note that $0\leq y_i^2\leq 1$ for each $1\leq i\leq N$. We can overestimate a maximum by letting every $y_i=1$. Then

$$\frac{1}{[D(1/y_i^2)]^{1/2}}\Big|_{\Sigma y_i^2=1}\leq\frac{1}{\sqrt{D(1)}}\leq\frac{1}{\sqrt{(\text{number of skeletons})}}. \tag{C8}$$

Clearly, the smallest number of skeletons a graph could have is 1. Hence,

$$\sup\frac{1}{[D(1/y_i^2)]^{1/2}}\Big|_{\Sigma y_i^2=1}=1. \tag{C9}$$

Combining Eqs. (C6), (C7), and (C9) and using $N=2n$ yields

$$L\leq 2\pi^n/\Gamma(n). \tag{C10}$$

This is the desired estimate for Theorem 1, part (i).

Proof of part (ii).[24] We make the substitution $\alpha_i=1/x_i$ in Eq. (C4) and integrate over the δ function. The result is that

$$L=\int_0^\infty\cdots\int_0^\infty\prod_{i=2}^{N}\frac{d\alpha_i}{\alpha_i^{3/2}}\alpha_1^{1/2}\frac{1}{[D(\alpha_i)]^{1/2}}\Big|_{\Sigma(1/\alpha_i)=1}. \tag{C11}$$

Hence,

$$L\geq\int_{2N}^{3N}\cdots\int_{2N}^{3N}\prod_{i=2}^{N}\frac{d\alpha_i}{\alpha_i^{3/2}}\alpha_1^{1/2}\frac{1}{[D(\alpha_i)]^{1/2}}\Big|_{\Sigma(1/\alpha_i)=1}. \tag{C12}$$

So

$$L\geq\left\{\int_{2N}^{3N}\prod_{i=2}^{N}\frac{d\alpha_i}{\alpha_i^{3/2}}\right\}\{\inf(\sqrt{\alpha_1})\,|\,{}_{2N\leq\alpha_i\leq 3N}\}\times\left\{\inf\frac{1}{[D(\alpha_i)]^{1/2}}\Big|_{\Sigma(1/\alpha_i)=1,\ 2N\leq\alpha_i\leq 3N}\right\}. \tag{C13}$$

We treat each of the factors in (C13) separately.

We evaluate the first factor exactly:

$$\int_{2N}^{3N}\prod_{i=2}^{N}\frac{d\alpha_i}{\alpha_i^{3/2}}=\left(\frac{2\sqrt{3}-2\sqrt{2}}{(6N)^{1/2}}\right)^{N-1}\geq N^{(1-N)/2}(\tfrac{1}{10})^{N-1}. \tag{C14}$$

The smallest possible value for the second factor is obtained when $\alpha_i=3N$. Then,

$$\inf(\sqrt{\alpha_1})=\left(\frac{3}{2+1/N}\right)^{1/2}\geq 1. \tag{C15}$$

[24] The proof of Theorem 2, part (ii) closely follows the proof given by Jaffe. See Ref. 4. However, Jaffe's other arguments do not apply here because of counting and duplication problems associated with diagrams having no external legs.

The minimum value for the third term in Eq. (C13) is found by computing the maximum value for $D(\alpha_i)\,|_{\Sigma\alpha_i=1,\ 2N\le\alpha_i\le 3N}$. But

$$\sup D(\alpha_i) = (\text{max number of skeletons}) \times (\text{max value for each skeleton}). \quad \text{(C16)}$$

An overestimation of the maximum number of skeletons a graph of n vertices has is 4^{n-1}. Also, the maximum value for each skeleton is achieved when all the $\alpha_i = 3N$. Putting this into Eq. (C16) yields

$$\sup D(\alpha_i) = 4^{n-1}(3N)^{n-1}. \quad \text{(C17)}$$

Hence,

$$\inf\left\{\frac{1}{[D(\alpha_i)]^{1/2}}\bigg|_{\Sigma(1/\alpha_i)=1,\ 2N\le\alpha_i\le 3N}\right\} = (12N)^{(1-n)/2}. \quad \text{(C18)}$$

We combine Eqs. (C14), (C15), and (C18) with Eq. (C13) to get

$$L \ge (N^{(1-N)/2})(\tfrac{1}{10})^{N-1}(12N)^{(1-n)/2}. \quad \text{(C19)}$$

Substituting $N = 2n$,

$$L \ge (2n)^{(1-2n)/2}(10)^{1-2n}(24n)^{(1-n)/2}. \quad \text{(C20)}$$

This is the desired estimate for Theorem 1, part (ii). Note that the bounds given in Eq. (C10) and in Eq. (C20) are uniform. That is, they depend on n but not on the individual diagrams.

Theorem 2. The symmetry number for diagrams is uniformly bounded (i) above and (ii) below.

Proof of part (i). Since all symmetry numbers are less than 1, an upper bound on symmetry numbers for all diagrams is 1.

Proof of part (ii). A small symmetry number corresponds to a diagram having much symmetry. The class of connected diagrams having the greatest degree of symmetry per number of vertices is shown in Fig. 7. If n is the number of vertices, the symmetry numbers for diagrams belonging to this infinite class is exactly

$$\text{S.N.} = 1/(8\times 6^{n/2}). \quad \text{(C21)}$$

Equation (C21) serves as a lower bound on all symmetry numbers. This completes Theorem 2.

Combining the results of Theorems 1 and 2 places upper and lower bounds on $(-1)^{n+1}I$. Those bounds are

$$(-1)^{n+1}I \le (\tfrac{1}{2})^w(24)^n\left(\frac{1}{(4\pi)^{1/2}}\right)^{n+1-w} \times\Gamma(\tfrac{3}{2}n-\tfrac{1}{2}w-\tfrac{1}{2})\frac{2\pi^n}{\Gamma(n)} \quad \text{(C22)}$$

and

$$(-1)^{n+1}I \ge \frac{1}{8}\frac{1}{6^{n/2}}(\tfrac{1}{2})^w(24)^n\left(\frac{1}{(4\pi)^{1/2}}\right)^{n+1-w} \times\Gamma(\tfrac{3}{2}n-\tfrac{1}{2}w-\tfrac{1}{2})(2n)^{(1-2n)/2}\times 10^{1-2n}(24n)^{(1-n)/2}. \quad \text{(C23)}$$

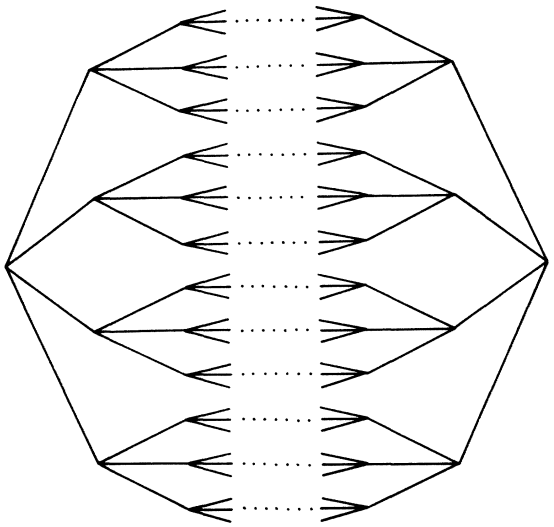

FIG. 7. The infinite class of diagrams having maximum symmetry per number of vertices. These diagrams have the smallest symmetry numbers for diagrams with a comparable number of vertices. All diagrams in this class start from a single vertex, each of whose four lines branches repeatedly and symmetrically into three new lines; after an arbitrary number of branchings, the lines symmetrically recombine into a single vertex.

To find bounds on $(-1)^{n+1}A_n$, we must make the estimates for $(-1)^{n+1}I$ independent of w. Clearly, for a connected diagram of n vertices, w can range from 0 to n. Therefore, in Eq. (C22) we take $w=0$ and in Eq. (C23) we take $w=n$. After simplification, the estimates [Eqs. (C22) and (C23)] become

$$(-1)^{n+1}I \le \frac{1}{\sqrt{\pi}}(12\sqrt{\pi})^n\frac{\Gamma(\frac{3}{2}n-\frac{1}{2})}{\Gamma(n)} \quad \text{(C24)}$$

and

$$(-1)^{n+1}I \ge \frac{5}{2}\left(\frac{3}{\pi}\right)^{1/2}\left(\frac{1}{200}\right)^n\Gamma(n-\tfrac{1}{2})n^{(2-3n)/2}. \quad \text{(C25)}$$

We now return to the integral representation [Eq. (C3)] and observe that for a given order n the contributions from all diagrams add in phase. Therefore, upper and lower bounds on $(-1)^{n+1}A_n$, the terms in the perturbation series, may be obtained once we place bounds on the number of diagrams in each order.

Theorem 3. The number of diagrams of order n is (i) bounded above by $(4n-1)!!$ and (ii) bounded below by $n!/3^n$.

Proof of part (i). We establish an upper bound on the number of diagrams by constructing them as follows.

We begin with n vertices. None of the vertices are connected, so each has four free lines emerging from it. There are $4n$ free lines.

We pick any line and connect it to one of the other $4n-1$ lines. There are now $4n-2$ lines. We pick and connect another line in one of $4n-3$ ways. We continue until there are no free lines. Therefore, there are at most $(4n-1)!!$ diagrams.

Proof of part (ii). In order to establish a lower bound on the number of diagrams, we establish a lower bound on the number of diagrams belonging to a particular class. We consider only those diagrams in which all the vertices lie on a continuous closed line (a circular path). We construct such a class of diagrams as follows.

Draw a circle and place all the n vertices on it. The circle uses up two of the four lines at each vertex. This leaves two free (unconnected) lines extending from every vertex. Now start with any vertex and pick one of the two free lines emerging from it. Join this line to a line coming from any of the n vertices (including itself). Connect the last free line coming from this new vertex to one of the remaining $n-1$ vertices. Continue this process, going sequentially from vertex to vertex until all pairs of free lines are used up. Since we use both lines at each vertex one right after the other, the number of available vertices keeps decreasing by one. Of course, at any time the process may close on itself if the free line is connected to the last line available at the original vertex. If this happens, we simply start over at any other free vertex, which will have two available lines, and continue.

Clearly, there are $n!$ ways to carry out the above process, but there will be many duplications. To avoid these duplications, we divide by three at each vertex.

We conclude that there are at least $n!/3^n$ diagrams. This establishes Theorem 3.

Combining the results of Theorem 3 with Eqs. (C24) and (C25) places upper and lower bounds on $(-1)^{n+1}A_n$:

$$(-1)^{n+1}A_n \leq \frac{(4n-1)!!\Gamma(\frac{3}{2}n-\frac{1}{2})}{\Gamma(n)} \frac{(12\sqrt{\pi})^n}{\sqrt{\pi}} \quad \text{(C26)}$$

and

$$(-1)^{n+1}A_n \geq \frac{n!\Gamma(n-\frac{1}{2})}{n^{(3n-2)/2}}(600)^{-n}\frac{5}{2}\left(\frac{3}{\pi}\right)^{1/2}. \quad \text{(C27)}$$

Simplifying Eqs. (C26) and (C27) yields

$$AB^n\Gamma(\tfrac{1}{2}n) \leq (-1)^{n+1}A_n \leq CD^n\Gamma(\tfrac{5}{2}n). \quad \text{(C28)}$$

Equation (C28) is the final result. Note that it agrees with the bounds in Eq. (2.2).

APPENDIX D

In this appendix, we discuss the details of the computer calculation of the first 75 terms in the ground-state-energy perturbation series [Eq. (2.1)] which is

$$E_0(\lambda) = \tfrac{1}{2}m + \sum_{n=1}^{\infty} mA_n(\lambda/m^3)^n. \quad \text{(D1)}$$

In the calculation, the difference equation [Eq. (2.8)] which is

$$2jB_{i,j} = (j+1)(2j+1)B_{i,j+1} + B_{i-1,j-2} - \sum_{p=1}^{i-1}(B_{i-p,1})(B_{p,j}) \quad \text{(D2)}$$

and Eq. (2.9) which is

$$A_i = -B_{i,1} \quad \text{(D3)}$$

were used on an IBM 7094 in double-precision mode. (The results of this computer calculation are listed in Table I. The absolute value of A_n is listed to the right of the order n. The number following the comma is the power of 10 multiplying the decimal.)

The computer program had three interesting features.

(i) To calculate A_n it was necessary to know all $B_{i,j}$ for $j=1, 2, \cdots, 2n$ and $i=1, \cdots, n$. Because $B_{i,j}$ has two indices, the limiting factor in this calculation was not time but memory capacity. Seventy-five was the maximum number of terms in the perturbation series

TABLE I. The absolute values of the first 75 terms A_n in the perturbation series for the ground-state energy [Eq. (2.1)]. The number following the comma is the power of 10 multiplying the decimal. The signs of A_n alternate beginning with plus, i.e., $A_n = (-)^{n-1}|A_n|$.

n	$\lvert A_n\rvert$		n	$\lvert A_n\rvert$	
1	0.750000000000,	0	39	0.559961162001,	64
2	0.262500000000,	1	40	0.664186377517,	66
3	0.208125000000,	2	41	0.807715625319,	68
4	0.241289062500,	3	42	0.100646863922,	71
5	0.358098046875,	4	43	0.128429408835,	73
6	0.639828134766,	5	44	0.167730490075,	75
7	0.132973372705,	7	45	0.224085871222,	77
8	0.314482146928,	8	46	0.306093131234,	79
9	0.833541603263,	9	47	0.427287814796,	81
10	0.244789407028,	11	48	0.609277658120,	83
11	0.789333316003,	12	49	0.887045954492,	85
12	0.277387769635,	14	50	0.131804184041,	88
13	0.105564665831,	16	51	0.199796510697,	90
14	0.432681068354,	17	52	0.308853507485,	92
15	0.190081719760,	19	53	0.486698406263,	94
16	0.891210175364,	20	54	0.781543229859,	96
17	0.444255088999,	22	55	0.127844051221,	99
18	0.234646430681,	24	56	0.212959346691,	101
19	0.130915026105,	26	57	0.361127834038,	103
20	0.769399985382,	27	58	0.623214611884,	105
21	0.475124077343,	29	59	0.109419743747,	108
22	0.307579295113,	31	60	0.195392812615,	110
23	0.208301009434,	33	61	0.354775753462,	112
24	0.147290492138,	35	62	0.654807077714,	114
25	0.108552296005,	37	63	0.122820914441,	117
26	0.832483627531,	38	64	0.234056056576,	119
27	0.663329371112,	40	65	0.453052532884,	121
28	0.548392431330,	42	66	0.890541554378,	123
29	0.469782420784,	44	67	0.177719768349,	126
30	0.416502699806,	46	68	0.359993943833,	128
31	0.381734895502,	48	69	0.740009711275,	130
32	0.361299554016,	50	70	0.154336989562,	133
33	0.352778001982,	52	71	0.326515091790,	135
34	0.355023393768,	54	72	0.700567351807,	137
35	0.367917476478,	56	73	0.152414090414,	140
36	0.392301600257,	58	74	0.336160314738,	142
37	0.430055097492,	60	75	0.751507982642,	144
38	0.484327278330,	62			

which could be computed without using additional memory space on disks or tapes.

(ii) The numbers $B_{i,j}$ grow exponentially large or small with i for j near 1 or near $2i$. As a result, the capacity of the 7094 to manipulate these large and small exponents in floating point would be exceeded before $i=20$. Therefore, it was necessary to scale the $B_{i,j}$. One particularly useful scaling is

$$C_{i,j}=(1/100^{i-j})B_{i,j}. \tag{D4}$$

Equation (D4) is adequate for $n\lesssim 100$.

(iii) The structure of the difference equation [Eq. (D2)] indicated that a huge number of manipulations would have to be performed in which numbers with greatly differing exponents would be added together. This would cause round-off errors which on a computer are cumulative. Therefore, although the calculation was done to 16 decimal places (double-precision mode), one must take care to determine the number of decimal places which are really significant.

If we assume that the round-off errors are purely random (not systematic), then the cumulative error grows as the square root of the number of operations. Because there are approximately 10^8 operations performed in the calculation of A_{75}, we expect that the error is about 10^4 in 10^{16}. That is, the last four decimal places (of the sixteen) are random.

We can check that there are indeed 12 significant figures in the table of A_n as follows: Divide out the dominant asymptotic behavior of the A_n [see Eq. (2.10)] and make a table of the resulting numbers Q_n:

$$Q_n\equiv\frac{A_n}{(\sqrt{6})\pi^{-3/2}\Gamma(n+\frac{1}{2})3^n}. \tag{D5}$$

Then compute a table of the successive differences:

$$Q_n{}^{(1)}\equiv Q_n-Q_{n-1}, \tag{D6a}$$

$$Q_n{}^{(2)}\equiv Q_n{}^{(1)}-Q_{n-1}{}^{(1)}, \tag{D6b}$$

and so on.

The $Q_n{}^{(i)}$ are regularly monotonic as i increases until they become random. By inspection we find that $Q^{(9)}$ is random and, as expected, it is nonzero in the last four decimal places (of the sixteen). This verifies that the Q_n and hence the A_n are correct to 12 places.

APPENDIX E

There is no known mathematically rigorous proof of the detailed asymptotic growth of the A_n as given in Eq. (2.10). However, in this appendix, we justify this equation using numerical methods. The raw data used in this appendix are contained in Table I.

We approach the problem by finding the rough growth of A_n. To do this, we construct a table of the ratios

$$R_n=A_{n+1}/A_n \tag{E1}$$

from the values of A_n listed in Table I. As $n\to 75$, we observe that to a good approximation $R_n\to 3(n+\frac{1}{2})$. This indicates that A_n grows as $3^n\Gamma(n+\frac{1}{2})$.

Next we try to fit A_n very accurately by

$$A_n\sim 3^n\Gamma(n+\tfrac{1}{2})\left(a_0+\frac{a_1}{n}+\frac{a_2}{n^2}+\frac{a_3}{n^3}+\cdots\right). \tag{E2}$$

[The $\sim$ notation in Eq. (E2) is justified because the term in square brackets is an asymptotic (nonconvergent) series.] Note that it would not be possible to fit A_n by a series such as that in Eq. (E2) if A_n did not grow as $3^n\Gamma(n+\frac{1}{2})$; that is, if the rough growth of A_n were, say, $(2.9)^n\Gamma(n+\frac{1}{2})$ or $3^n\Gamma(n+0.6)$.

To evaluate the a_i in Eq. (E2), we use Table I to calculate

$$P_n=A_n/3^n\Gamma(n+\tfrac{1}{2}). \tag{E3}$$

Then we terminate the series in Eq. (E2) at a_k and solve for a_i, $i=0, 1, \cdots, k$, in the resulting set of simultaneous equations:

$$P_n=a_0+\frac{a_1}{n}+\frac{a_2}{n^2}+\cdots+\frac{a_k}{n^k}, \tag{E4}$$

where $n=75-k, 75-k+1, \cdots, 75$.

We solve for a_0 in Eq. (E4) by multiplying each equation by n^k, taking successive differences, and using the beautiful identity:

$$\sum_{i=0}^{k}(n-i)^k\binom{k}{i}(-1)^i=k!. \tag{E5}$$

a_0 may thus be given exactly in closed form by

$$a_0=\sum_{i=0}^{k}(-1)^iP_{75-i}\binom{k}{i}(75-i)^k\Big/k!. \tag{E6}$$

Once a_0 is known, we can reduce Eq. (E4) to a new set of equations in which P_n is replaced by $(P_n-a_0)n$ and k by $k-1$. We solve as before for the new a_0 which is the old a_1. Repeating this process k times gives all a_i.

If we terminate the series at a_5, we find that to four parts in 10^9

$$a_0=(\sqrt{6})/\pi^{3/2}. \tag{E7}$$

We also find that

$$a_1/a_0=1.3194528, \tag{E8a}$$

$$a_2/a_0=1.9366649, \tag{E8b}$$

$$a_3/a_0=7.1004516, \tag{E8c}$$

$$a_4/a_0=27.445970, \tag{E8d}$$

$$a_5/a_0=665.57303. \tag{E8e}$$

Apparently, as i increases, the growth of the numbers a_i is so rapid that when $n\sim 75$, the term a_6/n^6 is larger than a_5/n^5. This suggests that the asymptotic series

in Eq. (E4) is an optimal approximation of P_n when it is terminated at a_5/n^5. (More terms could be included, giving a more accurate approximation to P_n, only if n were larger.)

Since we keep six terms in the asymptotic series, we expect that P_n should be represented accurately and uniformly to a few parts in 10^9. This turns out to be the case for n near 75 down to well below 50. Because a_0 is $(\sqrt{6})\pi^{-3/2}$ to four parts in a billion and since the magnitude of this error is exactly what we would expect, we assert that a_0 *is* $(\sqrt{6})\pi^{-3/2}$.

Combining this value for a_0 with Eq. (E2) gives Eq. (2.10) which is the result we had set out to derive in this appendix, namely,

$$A_n \sim 3^n \Gamma(n+\tfrac{1}{2})(\sqrt{6})\pi^{-3/2}. \tag{E9}$$

One might note that although a_0 is known to four parts in 10^9, a_i/a_0 in Eq. (E8) is known less and less accurately as i increases. a_1/a_0 is accurate to seven or eight places; a_2/a_0 is accurate to five or six places; and a_5/a_0 is accurate to one place at most.

To increase the accuracy of a_i, A_n would have to be calculated for n much larger than 75. As is always the case with asymptotic series, nothing is gained by calculating A_n more precisely for n near 75.

Aside from simplicity, there is no compelling reason why P_n should be expanded into an asymptotic series of the form in Eq. (E2). One might prefer to use the more general series

$$P_n \sim \alpha_0 + \frac{\alpha_1}{(n+\epsilon)} + \frac{\alpha_2}{(n+\epsilon)^2} + \frac{\alpha_3}{(n+\epsilon)^3} + \cdots, \tag{E10}$$

where ϵ is any number which is small compared to n. Then a simple formula relates the a_i to the α_i:

$$\alpha_0 = a_0, \tag{E11a}$$

$$\alpha_1 = a_1, \tag{E11b}$$

$$\alpha_2 = a_2 + \epsilon a_1, \tag{E11c}$$

$$\alpha_3 = a_3 + 2\epsilon a_2 + \epsilon^2 a_1, \tag{E11d}$$

$$\alpha_4 = a_4 + 3\epsilon a_3 + 3\epsilon^2 a_2 + \epsilon^3 a_1, \tag{E11e}$$

and, in general,

$$\alpha_n = \sum_{i=1}^{n} a_i \epsilon^{n-i} \binom{n-1}{i-1}. \tag{E11f}$$

This completes our remarks on these numerical calculations. However, before concluding this appendix, we will discuss an example of how investigations of this sort might lead to a better understanding of the singularity structure of the ground-state energy $E_0(\lambda)$ in the complex plane.

Example. As a crude illustration of how one might recover the analytic properties of $E_0(\lambda)$ from Eq. (E2), we neglect a_i for $i \geq 1$. Then Eqs. (E2) and (2.1) give

$$E_0(\lambda) \sim \sum_{n=0}^{\infty} \lambda^n (-1)^n (\sqrt{6})\pi^{-3/2}\Gamma(n+\tfrac{1}{2})3^n. \tag{E12}$$

Then, ignoring the possibility that two unequal functions can have the same asymptotic expansion, we substitute the expression

$$\Gamma(n+\tfrac{1}{2}) = \int_0^\infty e^{-t} t^{n-\frac{1}{2}} dt \tag{E13}$$

into Eq. (E12). Interchanging summation and integration, summing the series, and letting $s=\sqrt{t}$ gives a convergent integral representation for E_0:

$$E_0(\lambda) \cong \frac{2\sqrt{6}}{\pi^{3/2}} \int_0^\infty \frac{e^{-s^2}}{1+3\lambda s^2} ds. \tag{E14}$$

For λ real and positive, this integral can be evaluated explicitly with the result that

$$E_0(\lambda) \cong (2/\pi\lambda)^{1/2} \exp(1/3\lambda)\ \mathrm{Erfc}[(3\lambda)^{-1/2}], \tag{E15}$$

where

$$\mathrm{Erfc}(x) = \frac{2}{\sqrt{\pi}} \int_0^\infty e^{-t^2} dt. \tag{E16}$$

More importantly, the dispersion-type integral in Eq. (E14) defines a nontrivial singularity structure in the λ plane—a cut along the negative real λ axis. The jump across the cut is easily calculated. Unfortunately, this result does not come close to giving the true singularity structure of $E_0(\lambda)$ as derived in Secs. III, IV, and VI of this paper. Nevertheless, it is hoped that future calculations of this sort will lead to positive results.

APPENDIX F

In this appendix, we present the WKB calculation of Eq. (4.35) to first order in λ. A successful calculation would be a significant result because:

(i) Our continued ability to match the asymptotic expansions across the boundaries in regions A, B, C, and D would give added assurance that the application of WKB techniques to the problem of locating singularities in $E(\lambda)$ is valid.

(ii) It is most important to make sure that the first-order result we obtain here differs from the zeroth-order result only by terms of order λ. This would imply that the error in Eq. (4.35) is indeed small, as is tacitly assumed in Secs. IV, V, and VI.

(iii) As a byproduct, this calculation would define the boundaries of regions A, B, C, and D more precisely than the zeroth-order calculation does. This added precision would result from the greater number of terms that would be matched across the boundaries.

In this appendix, we will follow the same procedure as in the zeroth-order calculation in Sec. IV. However, we keep terms of order λ as well as of order 1. We neglect terms of order λ^2 or smaller. We again use the differential equation [Eq. (4.9)] which is

$$[d^2/dr^2+\tfrac{1}{4}(-\epsilon+r^2-\rho r^4)]\Phi(r)=0, \tag{F1}$$

and recall the definitions $\rho=\lambda e^{-3\pi i/2}$, $\epsilon=4iE$, and $r=xe^{i\pi/4}$.

We define the four regions A, B, C, and D as in Sec. IV. That is, in region A, $0<|r|\ll|r_0|$; in region B, $|r_0|\ll|r|\ll|r_1|$; in region C, $|r_0|\ll|r|\lesssim|r_1|$; and in region D, $|r|\sim|r_1|$.

Also, for the first-order calculation r_0 and r_1 must be given more exactly than for the zeroth-order calculation. From Eq. (4.10), we have

$$r_0=\{[1-(1-4\rho\epsilon)^{1/2}](2\rho)^{-1}\}^{1/2}\sim(\sqrt{\epsilon})(1+\tfrac{1}{2}\rho\epsilon) \tag{F2a}$$

and

$$r_1=\{[1+(1-4\rho\epsilon)^{1/2}](2\rho)^{-1}\}^{1/2}\sim(1/\sqrt{\rho})(1-\tfrac{1}{2}\rho\epsilon). \tag{F2b}$$

As in Sec. IV, we consider separately the regions A, B, C, and D.

Region A. We solve Eq. (F1) to first order in λ. To do this for region A, we find a transformation which to first order in λ turns Eq. (F1) into a parabolic cylinder function differential equation.[25] The transformation which works is

$$w=x+\lambda(\tfrac{1}{8}x^3+\tfrac{3}{4}Ex). \tag{F3}$$

Then, the even-parity solution to (D1) is

$$\Phi_A=C(1-\tfrac{3}{16}\lambda x^2)[D_\nu(w)+D_\nu(-w)], \tag{F4}$$

where

$$\nu=E-\tfrac{1}{2}-\tfrac{3}{2}\lambda(E^2+\tfrac{1}{4}). \tag{F5}$$

[In this appendix, we do the WKB calculation for even-parity wave functions only. The differences for odd parity involve the algebraic manipulations in region A.]

We compute the asymptotic expansion of Φ_A in region A for r near region B. The first two terms in the asymptotic expansion[26] for $D_\nu(w)$ are

$$D_\nu(w)\sim w^\nu e^{-\frac{1}{4}w^2}\left[1+\frac{\nu(1-\nu)}{2w^2}\right] \quad \text{for } -\tfrac{3}{4}\pi<\arg w<\tfrac{3}{4}\pi \tag{F6a}$$

and

$$D_\nu(w)\sim w^\nu e^{-\frac{1}{4}w^2}\left[1+\frac{\nu(1-\nu)}{2w^2}\right]-\frac{(2\pi)^{1/2}}{\Gamma(-\nu)}\times e^{\nu\pi i}w^{-\nu-1}e^{\frac{1}{4}w^2}\left[1+\frac{(1+\nu)(2+\nu)}{2w^2}\right] \quad \text{for } \tfrac{1}{4}\pi<\arg w<(5/4)\pi. \tag{F6b}$$

Equations (F2)–(F6) are used to expand Φ_A into its asymptotic form. We expand to first order in powers of ρ and r_0^2/r^2. We keep terms of the form ρ and r_0^2/r^2 and neglect terms of the form ρ^2, $\rho r_0^2/r^2$, and r_0^4/r^4. These approximations are justified because they give the correct asymptotic connection with Φ_B. As we had expected, these approximations define the extent of region A more clearly than the zeroth-order calculation did.

After considerable simplification,[27] the asymptotic expansion for Φ_A is

$$\Phi_A\sim C2\pi 2^{\frac{1}{2}\nu+\frac{1}{4}}e^{\pi i/8}e^{\pi i\nu/4}\left(1+\frac{3\rho r^2}{16}\right)\left(1+\frac{\epsilon}{4r^2}+\frac{\rho r^2}{16}+\frac{17\rho\epsilon}{64}\right)\left[1-\frac{\rho\epsilon}{32}\left(\frac{37}{8}+\frac{\epsilon^2}{32}\right)\right]$$
$$\times\left\{\frac{r^\nu}{\Gamma(\frac{1}{2}+\frac{1}{2}\nu)}\exp\left[\frac{ir^2}{4}\left(1-\frac{3\rho\epsilon}{8}-\frac{\rho r^2}{4}\right)-\frac{i\pi}{8}\right]\exp\left[\frac{i}{4}\left(\frac{3\rho\epsilon^2}{16}+\frac{\rho\epsilon r^2}{8}+\frac{\epsilon^2}{8r^2}-\frac{2}{2r^2}\right)\right]2^{-\frac{1}{2}\nu-\frac{1}{4}}\right.$$
$$\left.+\frac{r^{-\nu-1}}{\Gamma(-\frac{1}{2}\nu)}\exp\left[\frac{-ir^2}{4}\left(1-\frac{3\rho\epsilon}{8}-\frac{\rho r^2}{4}\right)+\frac{i\pi}{8}\right]\exp\left[-\frac{i}{4}\left(\frac{3\rho\epsilon^2}{16}+\frac{\rho\epsilon r^2}{8}+\frac{\epsilon^2}{8r^2}-\frac{3}{2r^2}\right)\right]2^{\frac{1}{2}\nu+\frac{1}{4}}\right\}. \tag{F7}$$

This is the desired result for region A.

Region B. The first-order WKB solution[28] to the differential equation [Eq. (F1)] is

$$\Phi_{\mathrm{WKB}}=\frac{\sqrt{\mu}}{(1-\frac{1}{8}\mu'^2+\frac{1}{4}\mu''\mu)^{1/2}}\left(C_2\exp\left\{i\left[\int^r\left(\frac{1}{\mu}-\frac{\mu'^2}{8\mu}\right)dr+\frac{\mu'}{4}\right]\right\}C_3\exp\left\{-i\left[\int^r\left(\frac{1}{\mu}-\frac{\mu'^2}{8\mu}\right)dr+\frac{\mu'}{4}\right]\right\}\right), \tag{F8}$$

[25] BMP, Vol. 2, p. 116, Eq. (1).

[25] BMP, Vol. 2, pp. 116–123. There is a serious misprint not on the errata sheet which must be corrected before one can do the first-order WKB calculation. Equation (2), p. 123, should read

$$D_\nu(z)=z^\nu e^{-z^2/4}\left[\sum_{n=0}^{N}\frac{(-\frac{1}{2}\nu)_n(\frac{1}{2}-\frac{1}{2}\nu)_n}{n!(-z^2/2)^n}+O|z^2|^{-N-1}\right]-\frac{(2\pi)^{1/2}}{\Gamma(-z)}e^{\nu\pi i}z^{-\nu-1}e^{z^2/4}\left[\sum_{n=0}^{N}\frac{(1+\frac{1}{2}\nu)_n(\frac{1}{2}+\frac{1}{2}\nu)_n}{n!(z^2/2)^n}+O|z^2|^{-N-1}\right] \text{ for } \pi/4<\arg z<5\pi/4.$$

Equation (3), p. 123, has a similar misprint but it is not needed in this paper. The equation is derived correctly in E. T. Whittaker and G. N. Watson, *A Course of Modern Analysis* (Cambridge University Press, Cambridge, 1952).

[27] BMP, Vol. 1, pp. 1–5.

[28] See Ref. 8. For first-order WKB techniques, see A. Messiah, pp. 233–234.

where

$$\mu=2(-\epsilon+r^2-\rho r^4)^{-1/2}. \tag{F9}$$

We would encounter a serious difficulty with this first-order calculation if we were to proceed as we did in Sec. IV. If we fixed the constants C_2 and C_3 in Eq. (F8) by defining the lower endpoint (the reference point) in the integrals to be r_0, we would find that the integrals become divergent regardless of the upper endpoint. The same would be true if we picked r_1 to be the reference point. In order to avoid this difficulty, we carry out the following procedure.

We treat $-\frac{1}{8}\int dr\,\mu'^2/\mu$ as an indefinite integral and integrate by parts repeatedly. This isolates the divergences at $r=r_0$ and r_1. After some tedious algebra, the result is

$$\begin{aligned}-\frac{1}{8}\int^r dr\frac{\mu'^2}{\mu}=&-\frac{1}{6}\left(\rho r^3-\frac{r}{2}\right)(-\epsilon+r^2-\rho r^4)^{-3/2}\\&-\frac{1}{12\epsilon(1-4\rho\epsilon)}(-\epsilon+r^2-\rho r^4)^{-1/2}\\&\times[\rho r^3(1-12\rho\epsilon)-r(1-8\rho\epsilon)]\\&+\frac{\rho}{12\epsilon(1-4\rho\epsilon)}\int_{r_0}^r\frac{4\epsilon+(1-12\rho\epsilon)r^2}{(-\epsilon+r^2-\rho r^4)^{1/2}}dr+\text{const}.\end{aligned} \tag{F10}$$

In Eq. (F10),

$$\epsilon=r_1^2r_0^2/(r_1^2+r_0^2), \tag{F11}$$

$$\rho=(r_1^2+r_0^2)^{-1}. \tag{F12}$$

Equation (F10) is an exact equation. Equations (F11) and (F12) are also exact and follow from the definitions of r_0 and r_1 [Eq. (4.10)].

We determine the constant in Eq. (F10) to first order by evaluating the integral $-\frac{1}{8}\int^r dr\,\mu'^2/\mu$ using a second method. Using Eq. (F9), we have

$$\begin{aligned}\int^r-\frac{1}{8}\frac{\mu'^2}{\mu}dr&\\=-\int^r dr(-\epsilon+r^2-\rho r^4)^{-5/2}(\rho r^3-\tfrac{1}{2}r)^2.&\end{aligned} \tag{F13}$$

Expanding the right-hand side of Eq. (F13) in region B to first order, we find that

$$\begin{aligned}\int^r-\frac{1}{8}\frac{\mu'^2}{\mu}dr\sim-\frac{1}{4}\int^r dr\left(r^{-3}+\frac{5}{2}\epsilon r^{-5}-\frac{3}{2}\rho r^{-1}-\frac{5}{4}r^{-3}\rho\epsilon\right)&\\\sim\frac{1}{8r^2}+\frac{3\rho}{8}\ln\frac{2r}{r_0}.&\end{aligned} \tag{F14}$$

As in region A, the first-order asymptotic expansion in region B means that we expand in powers of ρ and and r_0^2/r^2. We keep terms of the form ρ and r_0^2/r^2 and neglect terms of the form r_0^4/r^4, ρ/r^2, and ρ^2.

Finally, we return to Eq. (F10) and approximate that equation to first order in region B with the result that

$$\int^r-\frac{1}{8}\frac{\mu'^2}{\mu}dr\sim\text{const}+\frac{1}{8r^2}+\frac{3\rho}{8}\ln\frac{2r}{r_0}+\frac{1}{12\epsilon}-\frac{3\rho}{8}. \tag{F15}$$

Comparing Eqs. (F14) and (F15) determines that the constant in Eq. (F13) gives

$$\text{const}=3\rho/8-1/12\epsilon. \tag{F16}$$

We have thus avoided the divergences which come from the endpoints in the evaluation of $-\frac{1}{8}\int dr\,\mu'^2/\mu$. As a result, Eq. (F10) with the constant of integration given in Eq. (F16) is a well-defined and consistent integral evaluation valid in both regions B and C. Hence, the constants C_2 and C_3 are fixed.

We now calculate to first order the asymptotic expansion of Φ_{WKB} [Eq. (F8)] in region B.

First, we expand $\frac{1}{4}\mu'$.

$$\tfrac{1}{4}\mu'=(-\epsilon+r^2-\rho r^4)^{-3/2}(\rho r^3-\tfrac{1}{2}r) \tag{F17}$$

$$\sim-\frac{1}{2r^2}+\frac{\rho}{4}. \tag{F18}$$

Second, we expand $(1-\frac{1}{8}\mu'^2+\frac{1}{4}\mu\mu'')^{-1/2}$ to first order and get simply

$$(1-\tfrac{1}{8}\mu'^2+\tfrac{1}{4}\mu\mu'')^{-1/2}\sim1. \tag{F19}$$

Third, we expand $\sqrt{\mu}$. To first order,

$$\sqrt{\mu}\sim\left(\frac{2}{r}\right)^{1/2}\left(1+\frac{3}{16}\rho r^2\right)\left(1+\frac{\epsilon}{4r^2}+\frac{\rho r^2}{16}+\frac{17\rho\epsilon}{64}\right). \tag{F20}$$

Fourth, we must evaluate $\int_{r_0}^r(1/\mu)dr$. Note that there is no difficulty with endpoint divergences in this integral. Therefore, we treat it just as we did in Sec. IV except that we retain first-order terms. From Eq. (F9), we have

$$\int_{r_0}^r\frac{1}{\mu}dr=\frac{1}{2}\int_{r_0}^r(-\epsilon+r^2-\rho r^4)^{1/2}dr. \tag{F21}$$

We expand the integral in Eq. (F21) and get

$$\int_{r_0}^r\frac{1}{\mu}dr\sim\frac{\sqrt{\rho}}{2}\int_{r_0}^r dr(r^2-r_0^2)^{1/2}r_1-\left(\frac{r^2}{2r_1}\right). \tag{F22}$$

Substituting $r=r_0\cosh\Theta$ into Eq. (F22) and integrating gives

$$\begin{aligned}\int_{r_0}^r\frac{1}{\mu}dr\sim\frac{1}{4}\Big\{&-\epsilon\left(1+\frac{3}{8}\rho\epsilon\right)\ln\frac{2r}{r_0}+r^2\left(1-\frac{1}{4}\rho\epsilon\right)\\&-\frac{r^4\rho}{4}-\frac{1}{2}\epsilon\left(1+\frac{9}{16}\rho\epsilon\right)+\frac{\epsilon^2}{8r^2}\Big\}.\end{aligned} \tag{F23}$$

We combine Eqs. (F14), (F18)–(F20), and (F23). These five results are needed to give us the asymptotic expansion of Φ_{WKB} (F8) in region B:

$$\Phi_B\sim\left(\frac{2}{r}\right)^{1/2}\left(1+\frac{3}{16}\rho r^2\right)\left(1+\frac{\epsilon}{4r^2}+\frac{\rho r^2}{16}+\frac{17\rho\epsilon}{64}\right)$$
$$\times\Bigg(C_2\exp\Bigg\{i\Bigg[\frac{1}{4}\Bigg(-\epsilon(1+\tfrac{3}{8}\rho\epsilon)\ln\frac{2r}{r_0}+r^2(1-\tfrac{1}{4}\rho\epsilon)-\tfrac{1}{4}r^4\rho$$
$$-\tfrac{1}{2}\epsilon\left(1+\frac{9}{16}\rho\epsilon\right)+\frac{\epsilon^2}{8r^2}\Bigg)-\frac{1}{2r^2}+\frac{\rho}{4}+\frac{1}{8r^2}+\frac{3\rho}{8}\ln\frac{2r}{r_0}\Bigg]\Bigg\}$$
$$+C_3\exp\{-i[\cdots]\}\Bigg). \quad \text{(F24)}$$

Region C. We calculate to first order the asymptotic expansion of Φ_{WKB} [Eq. (F8)] in region C. For this purpose it is convenient to use a new variable R:

$$R\equiv r_1-r_0. \quad \text{(F25)}$$

Expansions in region C are done in powers of ρ and R/r_1. For first order we keep terms of the form ρ and R/r_1 and neglect terms of the form R^2/r_1^2, $R\rho/r_1$, and ρ^2.

The components of the asymptotic expansion of Φ_{WKB} [Eq. (F8)] are computed in turn:

First,

$$\mu^{1/2}\sim\frac{2^{1/4}\rho^{1/8}}{R^{1/4}}\left[1+\frac{\rho}{4}\left(\frac{5}{2}\epsilon+\frac{5}{2}\frac{R}{\rho^{1/2}}\right)\right]. \quad \text{(F26)}$$

Second, as in region B,

$$(1-\tfrac{1}{8}\mu'^2+\tfrac{1}{4}\mu''\mu)^{-1/2}\sim 1. \quad \text{(F27)}$$

Third,

$$\tfrac{1}{4}\mu'\sim\frac{R^{-3/2}\rho^{1/4}}{24\sqrt{2}}\left(6+\frac{15}{2}\rho\epsilon-\frac{15}{2}R\sqrt{\rho}\right). \quad \text{(F28)}$$

Fourth, we decompose the integral $\int_{r_0}^{r}(1/\mu)dr$ as follows:

$$\int_{r_0}^{r}\frac{1}{\mu}dr=\int_{r_0}^{r_1}\frac{1}{\mu}dr-\int_{r}^{r_1}\frac{1}{\mu}dr. \quad \text{(F29)}$$

The first integral in Eq. (F29) will be expressed later in terms of elliptic integrals [see Eq. (F47)]. For now we label it P:

$$P=\int_{r_0}^{r_1}\frac{1}{\mu}dr. \quad \text{(F30)}$$

The second integral in Eq. (F29) may be approximated easily, with the result that

$$-\int_{r}^{r_1}\frac{1}{\mu}dr\sim-\tfrac{1}{2}\sqrt{3}\rho^{-1/4}R^{3/2}\left(1-\frac{5}{4}\rho\epsilon-\frac{3}{4}(\sqrt{\rho})R\right). \quad \text{(F31)}$$

Fifth, we use Eq. (F10) to approximate $\int-\frac{1}{8}(\mu'^2/\mu)dr$. We decompose the integral in Eq. (F10) using the same method as in Eq. (F29) and define the definite integral as Q:

$$Q=\int_{r_0}^{r_1}\frac{\rho dr}{12\epsilon(1-4\rho\epsilon)}\frac{4\epsilon+(1-12\rho\epsilon)r^2}{(-\epsilon+r^2-\rho r^4)^{1/2}}. \quad \text{(F32)}$$

Then, after some heavy algebra,

$$\int^{r}-\frac{1}{8}\frac{\mu'^2}{\mu}dr=\frac{3\rho}{8}-\frac{1}{12\epsilon}+\frac{10R^{-3/2}\rho^{1/4}R\sqrt{\rho}}{24\sqrt{2}}$$
$$-\frac{\rho^{1/4}R^{-3/2}}{24\sqrt{2}}\left(1+\frac{5}{4}\rho\epsilon-\frac{5}{4}(\sqrt{\rho})R\right)+Q. \quad \text{(F33)}$$

We have now determined Φ_C, the asymptotic expansion of Φ_{WKB} in region C:

$$\Phi_C\sim\frac{2^{1/4}\rho^{1/8}}{R^{1/4}}\left(1+\frac{5}{8}\rho\epsilon\right)\left[1+\frac{5}{8}(\sqrt{\rho})R\right]$$
$$\times\Bigg(C_2\exp\Bigg\{i\Bigg[\frac{R^{-3/2}\rho^{1/4}}{24\sqrt{2}}\left(5+\frac{25}{4}\rho\epsilon+\frac{15}{4}R\sqrt{\rho}\right)+P$$
$$+Q+\tfrac{3}{8}\rho-\frac{1}{12\epsilon}-\tfrac{1}{3}\sqrt{2}\rho^{-1/4}R^{3/2}[1-\tfrac{3}{2}(\tfrac{5}{6}\rho\epsilon+\tfrac{1}{2}\rho^{1/2}R)]\Bigg]\Bigg\}$$
$$+C_3\exp\{-i[\cdots]\}\Bigg), \quad \text{(F34)}$$

where P and Q are defined in Eqs. (F30) and (F32), respectively.

Region D. In region D, we substitute Eq. (F25) into the differential equation (F1) and transform the equation to

$$0=\left\{\frac{d^2}{dR^2}+\tfrac{1}{2}\rho^{-1/2}R\left[1-\rho\left(\frac{5}{2}\epsilon+\frac{5}{2}\frac{R}{\rho^{1/2}}\right)\right]\right\}\Phi(R). \quad \text{(F35)}$$

Equation (F35) must be solved to first order and Φ_D is its solution. We solve Eq. (F35) by finding a transformation of variable which to first order changes Eq. (F35) into an Airy equation.[29] The correct change of variables is

$$R=\rho^{1/2}[z+\rho(\tfrac{3}{2}+\tfrac{5}{6}\epsilon z+\tfrac{1}{2}z^2)]. \quad \text{(F36)}$$

Then,

$$\Phi_D=\{\exp\tfrac{1}{2}\rho[z-\rho(\tfrac{5}{6}\epsilon+\tfrac{1}{2}z^2)]\}\chi. \quad \text{(F37)}$$

χ in Eq. (F37) satisfies this Airy equation to first order:

$$\{d^2/dz^2+\tfrac{1}{2}\rho z\}\chi=0. \quad \text{(F38)}$$

The solution to the Airy equation [Eq. (F38)] is

$$\chi(z)=[-\tfrac{1}{3}(18\rho^{-1})^{-1/3}z]^{1/2}$$
$$\times K_{1/3}(2(-z)^{3/2}(18\rho^{-1})^{-1/2}). \quad \text{(F39)}$$

[29] BMP, Vol. 2, Sec. 7.3.7, p. 22.

As in Sec. IV, to calculate the asymptotic behavior of Φ_D we need to use Eq. (4.31) which is[30]

$$K_{1/3}(e^{3\pi i/2}x)=-\tfrac{1}{2}i\pi[e^{i\pi/6}H_{1/3}^{(1)}(x)+e^{-i\pi/6}H_{1/3}^{(2)}(x)]. \quad \text{(F40)}$$

Also, we need the asymptotic expansions of $H_{1/3}^{(1)}$ and $H_{1/3}^{(2)}$ to first order[31]:

$$H_{1/3}^{(1)}(x)\sim\frac{\sqrt{2}}{\sqrt{(\pi x)}}e^{i(x-5\pi/12)}\left(1+\frac{5}{i72x}\right) \quad \text{(F41a)}$$

and

$$H_{1/3}^{(2)}(x)\sim\frac{\sqrt{2}}{\sqrt{(\pi x)}}e^{-i(x-5\pi/12)}\left(1-\frac{5}{72ix}\right). \quad \text{(F41b)}$$

We combine Eqs. (F36), (F37), (F39), (F40), and (F41), and after some heavy algebra, we have the asymptotic expansion of Φ_D:

$$\Phi_D(R)\sim\frac{2^{-11/12}\pi^{1/2}3^{-1/3}\rho^{1/24}D}{R^{1/4}}\left(1+\frac{5\rho\epsilon}{24}+\frac{5R\sqrt{\rho}}{8}\right)\times\{\exp\{i[\tfrac{1}{3}\sqrt{2}\rho^{-1/4}R^{3/2}[1-\tfrac{3}{2}(\tfrac{5}{8}\rho\epsilon+\tfrac{1}{2}R\sqrt{\rho})]-\tfrac{1}{4}\pi-(5/24\sqrt{2})\rho^{1/4}R^{-3/2}[1+\tfrac{3}{2}(\tfrac{5}{6}\rho\epsilon+\tfrac{1}{2}R\sqrt{\rho})]\}+\exp\{-i[\cdots]\}\}. \quad \text{(F42)}$$

This completes the calculation of the asymptotic expansions for Φ_A, Φ_B, Φ_C, and Φ_D. It is necessary to asymptotically connect Φ_A with Φ_B and Φ_C with Φ_D as was done in the zeroth-order calculation in Sec. IV.

First, we match Φ_A in Eq. (F7) with Φ_B in Eq. (F24) and determine the ratio of C_2 to C_3:

$$\frac{C_2}{C_3}=(\tfrac{1}{8}r_0^2)^{\nu+1/2}e^{-i\pi/4}\frac{\Gamma(-\frac{1}{2}\nu)}{\Gamma(\frac{1}{2}+\frac{1}{2}\nu)}e^{i\epsilon/4}\times\exp\left[i\rho\left(1+\frac{15}{64}\epsilon^2\right)\right], \quad \text{(F43)}$$

where ν is given in Eq. (F5).

Second, we match Φ_C in Eq. (F34) with Φ_D in Eq. (F42) and, once again, determine the ratio of C_2 to C_3:

$$\frac{C_2}{C_3}=e^{i\pi/2}\exp\left[-2i\left(P+Q+\frac{3\rho}{8}-\frac{1}{12\epsilon}\right)\right], \quad \text{(F44)}$$

where P and Q are given in Eqs. (F30) and (F32), respectively.

It only remains to calculate the integrals P and Q. Both integrals may be treated by substituting

$$z^2=(r_1^2-r^2)/(r_1^2-r_0^2). \quad \text{(F45)}$$

After some algebra both P and Q are found to be linear combinations of $K(k)$ and $E(k)$. These are the complete elliptic integrals of the first and second kind.[32] $K(k)$ and $E(k)$ are defined by

$$K(k)\equiv\int_0^1\frac{dz}{(1-z^2)^{1/2}(1-k^2z^2)^{1/2}} \quad \text{(F46a)}$$

and

$$E(k)\equiv\int_0^1\frac{dz(1-k^2z^2)^{1/2}}{(1-z^2)^{1/2}}, \quad \text{(F46b)}$$

where in our case

$$k^2\equiv 1-r_0^2/r_1^2. \quad \text{(F46c)}$$

Using Eq. (F46), P and Q may be written as

$$P=\frac{(\sqrt{\rho})r_1^3}{2}\left[\frac{2-k^2}{3}E(k)+2\left(\frac{k^2-1}{3}\right)K(k)\right] \quad \text{(F47)}$$

and

$$Q=\frac{\sqrt{\rho}}{12r_1\epsilon(1-4\rho\epsilon)}[4\epsilon K(k)+(1-12\rho\epsilon)r_1^2E(k)]. \quad \text{(F48)}$$

We expand $K(k)$ and $E(k)$ to first order for k near 1 by means of the following formulas[33]:

$$K(k)=\tfrac{1}{2}\pi\,{}_2F_1(\tfrac{1}{2},\tfrac{1}{2};1;k^2) \quad \text{(F49a)}$$

and

$$E(k)=\tfrac{1}{2}\pi\,{}_2F_1(-\tfrac{1}{2},\tfrac{1}{2};1;k^2). \quad \text{(F49b)}$$

We then expand the hypergeometric functions in Eq. (F49) to first order for k near 1[34]:

$$K(k)\sim\ln\frac{4r_1}{r_0}+\frac{r_0^2}{4r_1^2}\left(\ln\frac{4r_1}{r_0}-1\right) \quad \text{(F50a)}$$

and

$$E(k)\sim 1+\frac{r_0^2}{4r_1^2}\left(2\ln\frac{4r_1}{r_0}-1\right)+\frac{3}{32}\frac{r_0^4}{r_1^4}\left(2\ln\frac{4r_1}{r_0}-\frac{13}{6}\right). \quad \text{(F50b)}$$

Equation (F2) allows us to write Eq. (F50) in terms of ρ and ϵ:

$$K(k)\sim\left(\ln\frac{4}{\sqrt{(\rho\epsilon)}}\right)\left(1+\frac{\rho\epsilon}{4}\right)-\frac{5\rho\epsilon}{4} \quad \text{(F51a)}$$

and

$$E(k)\sim 1-\frac{\rho\epsilon}{4}-\frac{77}{64}\rho^2\epsilon^2+\rho\epsilon\left(\frac{1}{2}+\frac{19}{16}\rho\epsilon\right)\ln\left(\frac{4}{\sqrt{(\rho\epsilon)}}\right). \quad \text{(F51b)}$$

Finally, we combine Eqs. (F47), (F48), and (F51) to

[30] BMP, Vol. 2, p. 5, Eq. (15) and p. 80, Eq. (43).
[31] BMP, Vol. 2, p. 85, Eqs. (1) and (2).
[32] BMP, Vol. 2, p. 317, Eqs. (1) and (2).
[33] BMP, Vol. 2, p. 318, Eqs. (5) and (6).
[34] For the expansion of $K(k)$ and $E(k)$ see Ref. 11.

get the first-order evaluation of P and Q:

$$P\sim-\frac{\epsilon}{4}\left(\ln\frac{4}{\sqrt{(\rho\epsilon)}}\right)\left(1+\frac{3\rho\epsilon}{8}\right)+\frac{1}{6\rho}-\frac{\epsilon}{8}+\frac{17}{128}\rho\epsilon^2 \quad \text{(F52)}$$

and

$$Q\sim\frac{1}{12\epsilon}-\frac{35\rho}{48}+\frac{3\rho}{8}\left(\ln\frac{4}{\sqrt{(\rho\epsilon)}}\right). \quad \text{(F53)}$$

Combining Eqs. (F52) and (F53) with (F44), we have

$$\frac{C_2}{C_3}=e^{i\pi/2}\exp\left\{i\left[\frac{\epsilon}{4}-\frac{1}{3\rho}+\rho\left(\frac{17}{24}-\frac{17}{64}\epsilon^2\right)+\left(\ln\frac{16}{\rho\epsilon}\right)\left(\frac{\epsilon}{4}-\frac{3\rho}{8}+\frac{3\rho\epsilon^2}{32}\right)\right]\right\}. \quad \text{(F54)}$$

We have thus calculated two independent expressions for C_2/C_3. Setting these expressions in Eqs. (F43) and (F54) equal gives an even-parity version of Eq. (4.35) correct to first order instead of zeroth order as previously computed in Sec. IV. Thus,

$$\frac{\Gamma(\frac{1}{2}+\frac{1}{2}\nu)}{\Gamma(-\frac{1}{2}\nu)}=\exp\left[(\nu+\tfrac{1}{2})\ln\frac{2}{\rho}+i\left(\frac{5\pi}{4}+\frac{1}{3\rho}\right)+i\rho\left(\frac{7}{24}-\frac{61E^2}{4}\right)\right], \quad \text{(F55)}$$

where

$$\nu=E-\tfrac{1}{2}+\tfrac{3}{2}i\rho(E^2+\tfrac{1}{4}).$$

This is the result we have sought. It is reassuring to observe that Eq. (F55) reduces to the zeroth-order calculation [Eq. (4.35)] if we neglect $\frac{3}{2}i\rho(E^2+\frac{1}{4})$ compared to $(E-\frac{1}{2})$ in ν and $\rho(7/24+E^2 61/4)$ compared to $5\pi/4$.

Considering the zeroth- and first-order results, it is tempting to conjecture that to all orders the WKB result corresponding to Eq. (F55) takes the form

$$\frac{\Gamma(\frac{1}{2}+\frac{1}{2}\nu(\rho))}{\Gamma(-\frac{1}{2}\nu(\rho))}=\exp\left\{[\nu(\rho)+\tfrac{1}{2}]\ln\frac{2}{\rho}+i\left(\frac{5\pi}{4}+\frac{1}{3\rho}\right)+i\alpha(\rho)\right\}, \quad \text{(F56)}$$

where $\nu(\rho)$ and $\alpha(\rho)$ are power series in ρ:

$$\nu(\rho)=\sum_{n=0}^{\infty}\nu_n\rho^n, \quad \text{(F57)}$$

with $\nu_0=E-\frac{1}{2}$ and $\nu_1=\frac{3}{2}i(E^2+\frac{1}{4})$;

$$\alpha(\rho)=\sum_{n=1}^{\infty}\alpha_n\rho^n, \quad \text{(F58)}$$

with $\alpha_1=7/24-61E^2/4$.

APPENDIX G

In Sec. VI, the result of performing the integral $\int c\Phi_{\mathrm{WKB}}^2(z)dz$ was stated in Eq. (6.3). In this appendix, we demonstrate explicitly the evaluation of this integral which we call J. We will calculate J only for even-parity wave functions Φ. The odd-parity case is similar and presents no new difficulties.

We chose the contour C to be the straight line $z=re^{-i\pi/4}$, where r runs from $-\infty$ to ∞. Because the WKB solutions have definite parity about $r=0$, we simplify J to

$$J=2e^{-i\pi/4}\int_{r=0}^{\infty}dr\,\Phi_{\mathrm{WKB}}^2(re^{-i\pi/4}). \quad \text{(G1)}$$

In Eq. (G1) we break the region of integration into four pieces:

(I) $0\leq r\leq\bar{r}_0$,

(II) $\bar{r}_0\leq r\leq\bar{r}_1$,

(III) $\bar{r}_1\leq r\leq r_2$,

(IV) $r_2\leq r$,

where

$$r_0<\bar{r}_0\ll\bar{r}_1<r_1<r_2. \quad \text{(G2)}$$

The $<$ sign in Eq. (G2) means that we disregard r_0 compared to $\bar{r}_0$ to zeroth order.

Φ_{WKB} is approximated by Φ_A [Eq. (4.12)] in region I, by the oscillating WKB solution [Eq. (4.16)] in region II, by Φ_D [Eq. (4.28)] in region III, and by the exponentially decreasing WKB solution [Eq. (4.5)] in region IV. The integrals in each of these regions must be treated separately using

$$J=J_{\mathrm{I}}+J_{\mathrm{II}}+J_{\mathrm{III}}+J_{\mathrm{IV}}. \quad \text{(G3)}$$

Region I. In region I, we evaluate

$$J_{\mathrm{I}}=2e^{-i\pi/4}\int_{r=0}^{\bar{r}_0}dr\,\Phi_A^2(re^{-i\pi/4}). \quad \text{(G4)}$$

Inserting Eq. (4.12), the definition of Φ_A, into Eq. (G4) we get

$$J_{\mathrm{I}}=2e^{-i\pi/4}C^2\int_{r=0}^{\bar{r}_0}[D_\nu(re^{-i\pi/4})+D_\nu(-re^{-i\pi/4})]^2dr, \quad \text{(G5)}$$

where $\nu=-\frac{1}{2}-\frac{1}{4}i\epsilon$ and $\epsilon=4iE$.

We expand the integrand in Eq. (G5) using identities of the parabolic cylinder functions[35] so that

$$J_{\mathrm{I}}=2e^{-i\pi/4}C^2\left[(1-e^{2\nu\pi i})\int_{r=0}^{\bar{r}_0}D_\nu^2(re^{-i\pi/4})dr+\left(\frac{\sqrt{(2\pi)}}{\Gamma(-\nu)}\right)^2e^{\pi i(\nu+1)}\int_{r=0}^{\bar{r}_0}D_{-\nu-1}^2(re^{i\pi/4})dr+2(1+e^{\pi i\nu})\int_{r=0}^{\bar{r}_0}D_\nu(re^{-i\pi/4})D_\nu(-re^{-i\pi/4})dr\right]. \quad \text{(G6)}$$

[35] BMP, Vol. 2, p. 117, Eq. (7).

We evaluate the integrals in Eq. (G6) by introducing a new function F into Eq. (G4) as follows:

$$J_{\mathrm{I}}=2e^{-i\pi/4}\left\{\int_{r=0}^{\bar{r}_0} dr[\Phi_A{}^2(re^{-i\pi/4})-F(r)] + \int_{r=0}^{\bar{r}_0}\right\} dr\, F(r). \quad \text{(G7)}$$

$F(r)$ is the zeroth-order asymptotic approximation to those terms in $\Phi_A{}^2(re^{-i\pi/4})$ which would have divergent tails if integrated to $r=\infty$. This definition of $F(r)$ implies that a good approximation to J_{I} is

$$J_{\mathrm{I}}\sim 2e^{-i\pi/4}\left\{\int_{r=0}^{\infty} dr[\Phi_A{}^2(re^{-i\pi/4})-F(r)] + \int_{r=0}^{\bar{r}_0} dr\, F(r)\right\}. \quad \text{(G8)}$$

All the integrations in Eq. (G8) can be done easily.

Explicitly these integrals are

$$J_{\mathrm{I}}\sim 2e^{-i\pi/4}C^2\left\{(1-e^{2\nu\pi i})\int_{r=0}^{\infty} D_\nu{}^2(re^{-i\pi/4})dr + \left(\frac{\sqrt{(2\pi)}}{\Gamma(-\nu)}\right)^2 e^{\pi i(\nu+1)}\int_{r=0}^{\infty} D_{-\nu-1}{}^2(e^{i\pi/4}r)dr + 2(1+e^{\pi i\nu}) \times\int_{r=0}^{\infty} dr[D_\nu(re^{-i\pi/4})D_\nu(-re^{-i\pi/4})-F(r)] + \int_{r=0}^{\bar{r}_0} F(r)dr\right\}. \quad \text{(G9)}$$

Asymptotically, $D_\nu(re^{-i\pi/4})D_\nu(-re^{-i\pi/4})$ behaves as $[\sqrt{(2\pi)}/\Gamma(-\nu)]r^{-1}e^{i\pi/4}$ for large r.[36] However, if we were to choose the counterterm $F(r)$ to have this form, we would introduce a spurious pole at $r=0$. Therefore, we pick a form for $F(r)$ with the same asymptotic behavior at $r=\infty$ but with no pole in the region of integration. We thus define

$$F(r)\equiv\frac{1}{r+1}\frac{\sqrt{(2\pi)}}{\Gamma(-\nu)}e^{i\pi/4}. \quad \text{(G10)}$$

After rotating the contours in the complex plane by $\pi/4$ and $-\pi/4$, the first two integrals in Eq. (G9) become known forms.[37] In order to perform the third integral in Eq. (G9) which is

$$\int_{r=0}^{\infty}\left[D_\nu(re^{-i\pi/4})D_\nu(-re^{-i\pi/4}) - \frac{\sqrt{(2\pi)}}{\Gamma(-\nu)}(r+1)^{-1}e^{i\pi/4}\right]dr, \quad \text{(G11)}$$

we introduce a small positive ϵ into Eq. (G11) as follows:

$$\lim_{\epsilon\to 0}\int_{r=0}^{\infty}\left[D_{\nu-\epsilon}(re^{-i\pi/4})D_\nu(-re^{-i\pi/4}) - \frac{\sqrt{(2\pi)}}{\Gamma(-\nu)}\frac{1}{(r+1)^{\epsilon+1}}e^{i\pi/4}\right]dr. \quad \text{(G12)}$$

After rotating the contour in Eq. (G12) by $\pi/4$, the integrals become known forms.[38] We evaluate the integrals in Eq. (G12) and take the limit as $\epsilon\to 0$. After considerable simplification[39] the final result of this evaluation is

$$J_{\mathrm{I}}\sim\frac{4C^2\sqrt{(2\pi)}\cos(\frac{1}{2}\nu\pi)e^{\nu\pi i/2}}{\Gamma(-\nu)} \times[2\ln\bar{r}_0-\psi(E+\tfrac{1}{2})+\tfrac{1}{2}\pi\cot(\tfrac{1}{4}-\tfrac{1}{2}E)\pi], \quad \text{(G13)}$$

where $\psi(x)\equiv d\ln\Gamma(x)/dx$.

Region II. In region II, we evaluate

$$J_{\mathrm{II}}\sim 2e^{-i\pi/4}\int_{r=\bar{r}_0}^{\bar{r}_1} dr(-\epsilon+r^2-\rho r^4)^{-1/2}2C_2C_3. \quad \text{(G14)}$$

To get this equation, we have substituted Eq. (4.16) into Eq. (G1) and squared the expression. In squaring, we have disregarded two small oscillatory terms because their contribution to the integral could only become significant if r were near r_0 or r_1. The choice of $\bar{r}_0$ and $\bar{r}_1$ as endpoints of the integration region prevents r from getting near r_0 or r_1.

To do the integral in Eq. (G14) we decompose the region of integration into three pieces:

$$J_{\mathrm{II}}\sim 4C_2C_3e^{-i\pi/4}\left(\int_{r=r_0}^{r_1}-\int_{r=r_0}^{\bar{r}_0}-\int_{r=\bar{r}_1}^{r_1}\right) \times dr(-\epsilon+r^2-\rho r^4)^{-1/2}. \quad \text{(G15)}$$

By substituting $x^2=(r_1{}^2-r^2)/(r_1{}^2-r_0{}^2)$ and $k^2=1-r_0{}^2/r_1{}^2$, the first integral in Eq. (G13) is shown to be $K(k)$, the elliptic integral of the first kind.[40] The last two integrals are approximated by factoring $(-\epsilon+r^2-\rho r^4)=\rho(r^2-r_0{}^2)(r_1{}^2-r^2)$. Combining these results gives

$$J_{\mathrm{II}}\sim 4C_2C_3e^{-i\pi/4}\left[-\ln\frac{2\bar{r}_0}{r_0}-\left(2-\frac{2\bar{r}_1}{r_1}\right)^{1/2} + \frac{1}{r_1\sqrt{\rho}}K(k)\right]. \quad \text{(G16)}$$

We now expand $K(k)$ for k near 1 [see Eq. (4.23b) for

[36] BMP, Vol. 2, pp. 122–123, Eqs. (1) and (2).
[37] BMP, Vol. 2, p. 122, Eq. (22).
[38] BMP, Vol. 2, p. 122, Eq. (21).
[39] BMP, Vol. 1, pp. 1–5 and 16.
[40] BMP, Vol 2, p. 317, Eq. (1).

this expansion] and find that the contribution from region II is

$$J_{\rm II}\sim 4C_2C_3e^{-i\pi/4}[\ln(2/\bar{r}_0\sqrt{\rho})-\rho^{1/4}\sqrt{2}(r_1-\bar{r}_1)^{1/2}]. \quad \text{(G17)}$$

Regions III and IV. Using Eqs. (4.20), (4.28), and (4.29), we can write down the integral contribution from region III:

$$J_{\rm III}=2e^{-i\pi/4}\int_{r=\bar{r}_1}^{r_2} dr\frac{D^2}{3}$$
$$\times K_{1/3}{}^2[2(18\sqrt{\rho})^{-1/2}(r-r_1)^{3/2}](18\sqrt{\rho})^{-1/3}(r-r_1). \quad \text{(G18)}$$

We notice that the contribution from region IV can be included in Eq. (G18) by increasing the upper limit of integration from r_2 to ∞. This is true because the $K_{1/3}$ Bessel function for region III and the WKB solution for region IV [Eq. (4.5)] have similar exponentially decreasing tails. Hence,

$$J_{\rm III}+J_{\rm IV}=2e^{-i\pi/4}\int_{r=\bar{r}_1}^{\infty} dr\frac{D^2}{3}$$
$$\times K_{1/3}{}^2(2(18\sqrt{\rho})^{-1/2}(r-r_1)^{3/2})(18\sqrt{\rho})^{-1/3}(r-r_1). \quad \text{(G19)}$$

We simplify Eq. (G19) by substituting $s=r-r_1$:

$$J_{\rm III}+J_{\rm IV}=\frac{2e^{-i\pi/4}D^2(18\sqrt{\rho})^{-1/3}}{3}$$
$$\times\int_{s=\bar{r}_1-r_1}^{\infty} sds\, K_{1/3}{}^2(2(18\sqrt{\rho})^{-1/2}s^{3/2}). \quad \text{(G20)}$$

Next, as we did in region I, we introduce a counterterm $G(s)$ in Eq. (G21) and extend the region of integration in Eq. (G22). This procedure facilitates the evaluation of the integrals.

$$J_{\rm III}+J_{\rm IV}=\tfrac{2}{3}e^{-i\pi/4}D^2(18\sqrt{\rho})^{-1/3}$$
$$\times\left\{\int_{s=0}^{\infty} sds\, K_{1/3}{}^2(2(18\sqrt{\rho})^{-1/2}s^{3/2})\right.$$
$$+\int_{s=\bar{r}_1-r_1}^{0} ds[sK_{1/3}{}^2(2(18\sqrt{\rho})^{-1/2}s^{3/2})-G(s)]$$
$$\left.+\int_{s=\bar{r}_1-r_1}^{0} G(s)ds\right\}. \quad \text{(G21)}$$

This may be approximated by

$$J_{\rm III}+J_{\rm IV}\sim\tfrac{2}{3}e^{-i\pi/4}D^2(18\sqrt{\rho})^{-1/3}$$
$$\times\left\{\int_{s=0}^{\infty} sds\, K_{1/3}{}^2(2(18\sqrt{\rho})^{-1/2}s^{3/2})\right.$$
$$+\int_{s=-\infty}^{0} ds[sK_{1/3}{}^2(2(18\sqrt{\rho})^{-1/2}s^{3/2})-G(s)]$$
$$\left.+\int_{s=\bar{r}_1-r_1}^{0} G(s)ds\right\}. \quad \text{(G22)}$$

Equation (G22) is a good approximation to Eq. (G21) because $G(s)$ is chosen to behave asymptotically as $K_{1/3}{}^2$ for s near $-\infty$.

We reduce the first integral Eq. (G22) to a known form by substituting

$$x=2(18\sqrt{\rho})^{-1/2}s^{3/2}, \quad \text{(G23)}$$

and by using[41]

$$\int_0^{\infty} K_{1/3}{}^2(x)x^{1/3}dx=2^{-5/3}3\Gamma^2(\tfrac{2}{3}). \quad \text{(G24)}$$

We simplify the second integral in Eq. (G22) by substituting Eq. (G23) again and by observing that[42]

$$K_{1/3}(xe^{\frac{3}{2}\pi i})=(-\pi i/\sqrt{3})[J_{1/3}(x)+J_{-1/3}(x)]. \quad \text{(G25)}$$

Three Bessel function terms arise from plugging the square of Eq. (G25) into the second term in Eq. (G22). The techniques for integrating only one of these three terms will be presented here because the other two terms can be integrated similarly. Investigation of Eq. (G25) shows that in terms of the variable x the counterterm G is proportional to $x^{-2/3}$. As in region I, a limiting procedure must be used to do the counterterm integration as follows.

We insert a positive ϵ and the first of the three terms becomes

$$\lim_{\epsilon\to 0}\int_0^{\infty}\left[J_{1/3}(x)J_{1/3}(x(1+\epsilon))x^{1/3}\right.$$
$$\left.-\frac{(\cos x\epsilon)x^{-2/3}}{(1+\epsilon)^{1/2}\pi}\right]dx. \quad \text{(G26)}$$

The first integral in Eq. (G26) is a Weber-Schafheitlin integral.[43] The second may be evaluated by changing the contour. This gives a Γ function. The results are

$$\lim_{\epsilon\to 0}\left(2^{1/3}(1+\epsilon)^{-5/3}\frac{F(1,\frac{2}{3};\frac{4}{3};(1+\epsilon)^{-2})}{\Gamma(\frac{4}{3})\Gamma(\frac{1}{3})}\right.$$
$$\left.-\frac{\Gamma(\frac{1}{3})\epsilon^{-1/3}\cos\frac{1}{6}\pi}{\pi}\right). \quad \text{(G27)}$$

We transform[44] the hypergeometric function in Eq. (G27) as follows:

$$F(1,\tfrac{2}{3};\tfrac{4}{3};(1+\epsilon)^{-2})=\frac{\Gamma(\frac{4}{3})\Gamma(-\frac{1}{3})}{\Gamma(\frac{1}{3})\Gamma(\frac{2}{3})}$$
$$\times(1+\epsilon)^2F(1,\tfrac{1}{3};\tfrac{4}{3};-2\epsilon-\epsilon^2)+\frac{\Gamma(\frac{4}{3})\Gamma(\frac{1}{3})}{\Gamma(\frac{2}{3})}(1+\epsilon)^{2/3}$$
$$\times\left[1-\frac{1}{(1+\epsilon)^2}\right]^{-1/3}F(\tfrac{1}{3},0;\tfrac{2}{3};-2\epsilon-\epsilon^2). \quad \text{(G28)}$$

[41] BMP, Vol. 2, p. 93, Eq. (36).
[42] BMP, Vol. 2, p. 5, Eq. (15), p. 80, Eq. (43), and p. 4, Eqs. (5) and (6).
[43] BMP, Vol. 2, Sec. 7.7.4, pp. 51–52.
[44] BMP, Vol. 1, p. 109, Eq. (4).

We simplify[39] Eq. (G28) and insert it into Eq. (G27). We can then take the limit as $\epsilon \to 0$ because the terms containing $\epsilon^{-1/3}$ cancel. The result is that

$$\lim_{\epsilon\to 0}\int_0^\infty dx\left(J_{1/3}(x)J_{1/3}((\epsilon+1)x)x^{1/3}-\frac{(\cos x\epsilon)x^{-2/3}}{\pi x(1+\epsilon)^{1/2}}\right)=-\frac{3\times 2^{1/3}}{\Gamma^2(\frac{1}{3})}. \quad \text{(G29)}$$

The second term $(J_{1/3}J_{-1/3})$ and the third term $(J_{-1/3}J_{-1/3})$ each contribute 0 to the second term in Eq. (G22).

Thus, the first two integrals in Eq. (G22) cancel. After substituting for G, we are left with the third term in Eq. (G22) which is

$$J_{\mathrm{III}}+J_{\mathrm{IV}}\sim -ie^{-i\pi/4}\frac{2D^2}{3}\left(\frac{\sqrt{\rho}}{3}\right)^{1/3}\pi\int_{s=\bar{r}_1-r_1}^{0}x^{-2/3}dx, \quad \text{(G30)}$$

where $x=2(18\sqrt{\rho})^{-1/2}s^{3/2}$. We evaluate Eq. (G30) and find that

$$J_{\mathrm{III}}+J_{\mathrm{IV}}\sim \pi D^2 e^{-i\pi/4}2^{7/6}\rho^{1/12}3^{-2/3}(r_1-\bar{r}_1)^{1/2}. \quad \text{(G31)}$$

We have now evaluated all of the integrals in Eq. (G1). Finally, we must relate the constants C, C_1, C_2, C_3, C_4, and D in Eqs. (G13), (G17), and (G31). To do so we use Eqs. (4.14)–(41.6), (4.32), (4.33), and (4.35) and show algebraically that

$$C^2=\frac{\Gamma^2(\frac{3}{4}-\frac{1}{2}E)e^{-\pi iE/2}2^{-E}C_1^2}{(2\pi)^2}, \quad \text{(G32a)}$$

$$C_2C_3=\frac{C_1^2}{\Gamma(\frac{1}{4}-\frac{1}{2}E)\Gamma(\frac{1}{4}+\frac{1}{2}E)}, \quad \text{(G32b)}$$

$$D^2=\frac{\rho^{1/6}2^{4/3}3^{2/3}C_1^2}{\pi\Gamma(\frac{1}{4}+\frac{1}{2}E)\Gamma(\frac{1}{4}-\frac{1}{2}E)}. \quad \text{(G32c)}$$

We may pick an arbitrary value for C_1^2 because we are free to choose the over-all normalization of the original WKB wave function. We let

$$C_1^2=\tfrac{1}{2}e^{i\pi/4}\Gamma(\tfrac{1}{4}+\tfrac{1}{2}E)\Gamma(\tfrac{1}{4}-\tfrac{1}{2}E). \quad \text{(G33)}$$

Substituting Eq. (G33) into Eq. (G32) and simplifying,[39] the results are

$$C^2=\frac{\Gamma(-\nu)e^{-\pi iE/2}(\frac{1}{2}\pi)^{1/2}e^{i\pi/4}}{4\pi\cos\frac{1}{2}\pi\nu}, \quad \text{(G34a)}$$

$$C_2C_3=\tfrac{1}{2}e^{i\pi/4}, \quad \text{(G34b)}$$

$$D^2=(18\sqrt{\rho})^{1/3}e^{i\pi/4}/\pi. \quad \text{(G34c)}$$

Combining Eq. (G34) with Eqs. (G13), (G17), and (G31) shows that the contributions to J are

$$J_{\mathrm{I}}\sim 2\ln\bar{r}_0-\psi(E+\tfrac{1}{2})+\tfrac{1}{2}\pi\cot(\tfrac{1}{4}-\tfrac{1}{2}E)\pi, \quad \text{(G35a)}$$

$$J_{\mathrm{II}}\sim -2\ln\bar{r}_0+\ln(4/\rho)-2\rho^{1/4}\sqrt{2}(r_1-\bar{r}_1)^{1/2}, \quad \text{(G35b)}$$

$$J_{\mathrm{III}}+J_{\mathrm{IV}}\sim 2\sqrt{2}\rho^{1/4}(r_1-\bar{r}_1)^{1/2}. \quad \text{(G35c)}$$

Referring to Eq. (G3), we add together the three expressions in Eq. (G35) to find the total contribution to J which is

$$J=-\psi(E+\tfrac{1}{2})+\tfrac{1}{2}\pi\cot(\tfrac{1}{4}-\tfrac{1}{2}E)\pi+\ln(4/\rho). \quad \text{(G36)}$$

Note that $\bar{r}_0$ and $\bar{r}_1$ have dropped out of this final expression. This completes the evaluation of J for even parity.

For odd parity the result is

$$J=-\psi(\tfrac{1}{2}+E)+\tfrac{1}{2}\pi\cot(\tfrac{3}{4}-\tfrac{1}{2}E)\pi+\ln(4/\rho). \quad \text{(G37)}$$

The only differences between the even- and odd-parity derivations occur in the algebraic manipulations in region I.

Automaton Model of Dynamic Organization in Microtubules

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Microtubules (MT) participate in a wide range of organized dynamical activities that apparently involve information processing. Cellular automata are dynamical systems that can generate and process patterns and information by way of rules based on lattice neighbor interactions.[1-3] Lattice structure and apparent information processing capabilities have suggested an MT-automaton behavior based on coherent dipole oscillations within MT subunits.[4-6]

Various models of dynamic activities in biomolecules predict functional organization and regulation. Among these, Frohlich has theorized that nonlinear dipole excitations among subunits of biomolecular arrays result in coherent oscillations (10^{-10} to 10^{-11} sec fluctuations), lower frequency metastable states, long-range cooperativity and order, and coupling of protein conformational states to dipole oscillations within protein hydrophobic regions.[7,8] Individual subunit conformation within a lattice array would thus depend on various factors (GTP, Ca^{++}, primary structure, lattice neighbor electrostatic interactions) influencing oscillation phase. Based on MT geometry, we have calculated MT-lattice neighbor electrostatic influences on dipole oscillations as rules for an MT-automaton computer simulation (FIGURE 1). Dimer states at each generation can be determined by neighbor states at the previous generation.

$$\text{state} = \alpha, \text{ if } \sum_{i=1}^{n} f(y) > 0, \text{ state} = \beta, \text{ if } \sum_{i=1}^{n} f(y) < 0,$$

where n = 7 as the number of neighbors, and f(y) the force from the "*i*th" neighbor in the y direction.

$$f(y) \propto \frac{\sin\theta}{r^2}, \sin\theta = \frac{y}{r}; f(y) \propto \frac{y}{r^3};$$

thus states may be determined by $\Sigma_{i=1}^{n}\, y/r^3$ and relative values calculated from known MT structure.

FIGURE 2 shows one sequence from an "opened" MT automaton simulation in which 13 MT protofilaments are arranged horizontally. Each generation represents one Frohlich oscillation period (~ one nanosecond). "a" represents an alpha-tubulin dipole oscillation phase, whereas dots connected by lines represent beta-tubulin oscillation phases. Generations 22, 28, 34, and 40 from a particular sequence

demonstrate stable patterns (during even-numbered generations) and a traveling kink pattern (~80 meters/sec), which leaves an altered "wake" (? memory). These traveling patterns are known as gliders in automaton nomenclature and have been shown by von Neumann to be capable of universal computing.

Stable and moving patterns of an MT-subunit oscillation phase could function as specific sites of ion or protein binding and transport, orchestration of biomolecular activities, as well as stored information or memory. Interaction and interference of these patterns could process information as in computers and provide "real time" biological information processing. Although the MT-automaton rules and patterns described here are somewhat arbitrary, they demonstrate that MT-automaton activities are feasible and are of potential importance in dynamic organization.

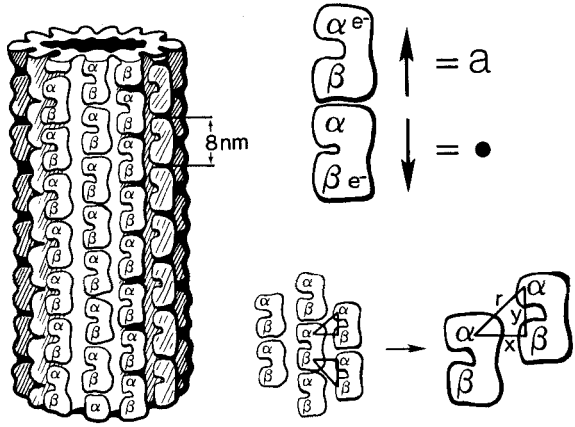

FIGURE 1. Left: MT structure from X-ray diffraction crystallography. Right top: MT-tubulin dimer subunits comprised of α and β monomers. Electron occupancy of either monomer may correlate with coherent dipole oscillations and are represented as "a" for α-monomer electron occupancy and dots connected by lines for β-monomer electron occupancy. Center bottom and right: hexagonal relations lead to neighbor rules based on lattice distances and electrostatic interactions. For neighbors shown (lower right), x = 5 nm, y = 4 nm, r = 6.4 nm.

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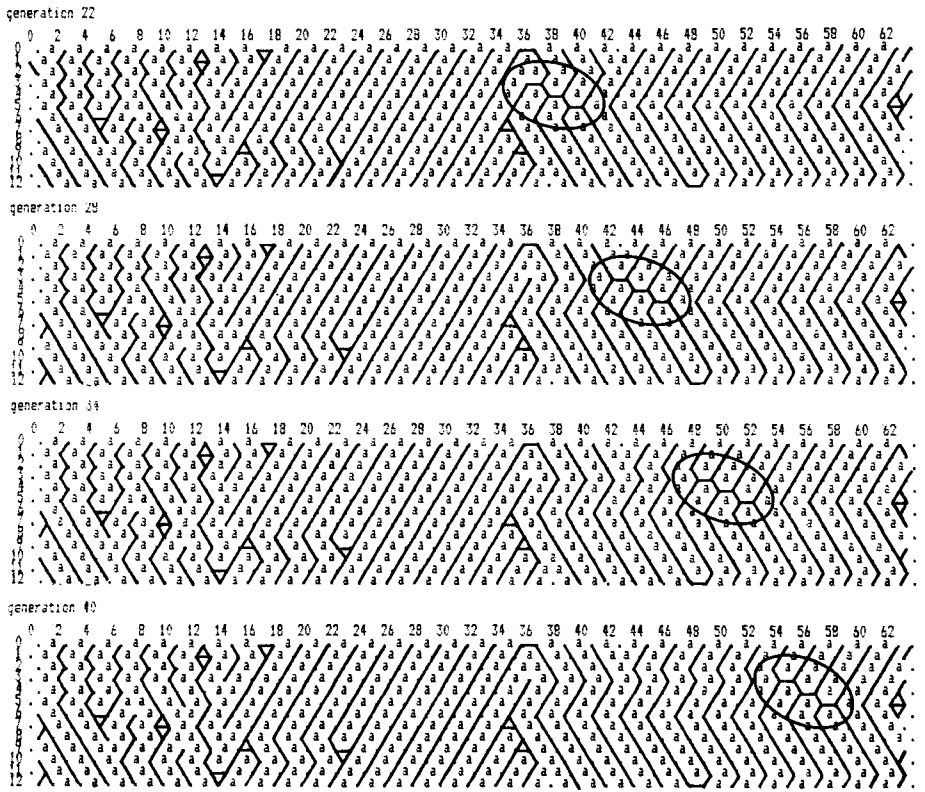

FIGURE 2. Generations 22, 28, 34, and 40 from an MT-automaton simulation. Stable patterns have evolved during even-numbered generations. A traveling "glider" has left an altered "wake," which may be likened to a computer operation.

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BIOLOGICAL BOSE CONDENSATION AND THE TIME THRESHOLD FOR BIOLOGICAL EFFECTS ☆

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Fröhlich's model of Bose condensation in biological systems is analyzed using finite temperature Green's function techniques. The appropriate biological system excitation lifetime is calculated and represents the time needed to produce the condensed state. The possible relevance of this calculated lifetime to experiments concerning the microwave irradiation of biological organisms is discussed.

The dipole–dipole interaction within biological structure first suggested by Fröhlich [1–6] has stimulated many recent theoretical studies [7–12, 19–21]. Also some recent experimental reports [13–18] have appeared to conform in several aspects to the predictions of the Fröhlich model. The purpose of this paper is to extend the theoretical analysis utilizing a Green's function technique and to discuss the results in relation to biological experimentation that has been performed up to the present.

Fröhlich's model has been particularly useful in understanding recent experiments revealing the effects of low level microwaves on various biological systems. One important general characteristic that is reported in these experiments is the existence of a time threshold for the initiation of the differing biological effects. Sevastyanova and Vilenskaya [13] performed an experiment concerning the simultaneous irradiation of mice with X-rays and microwaves where they counted surviving bone marrow cells. The time threshold they determined before a protective effect occurred with simultaneous irradiation was 30 minutes of microwave exposure. Smolyanskaya and Vilenskaya [18] observed the effects of low level millimeter waves on the induction coefficient of colicin synthesis. Again, it was found that a minimum irradiation time is needed to produce measurable biological effects. The time threshold for this experiment was of the order of 30 minutes. One must keep in mind when considering these time effects that the threshold will depend on the particular biological activity being monitored and will also be temperature dependent. An example of the latter is clearly observed by analyzing Smolyanskaya and Vilenskaya's experiment. Irradiation for 30 minutes at $T = 20^{\circ}\mathrm{C}$ produced no change in the rate of colicin synthesis, whereas at $T = 37^{\circ}\mathrm{C}$, colicin synthesis was induced after 30 minutes of irradiation time. Therefore, it is expected that the analysis of the time threshold will not be a trivial matter. Both the detailed microscopic structure of the biological system and its thermodynamic behavior must be considered to provide a full understanding of the specific time thresholds encountered for different biosystems.

It is the authors' present intention to obtain an approximation for a segment of this time threshold using the Fröhlich model. In applying the Fröhlich hypothesis, one sees that the biological effects cannot begin until the condensed phase occurs. How long this condensation has to be maintained in order to produce biological effects is undetermined as yet. Most generally, the time needed to initiate biological effects will consist of two parts, τ_1 and τ_2. τ_1 represents the time from the beginning of irradiation to the onset of condensation. τ_2 represents the time elapsed from the on-

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set of the condensed phase to the production of the actual biological effects. Thus, the total time, τ, elapsed before biological effects occur from the initiation of irradiation, is given by $\tau = \tau_1 + \tau_2$. In the following analysis, we calculated τ_1 to the lowest order in coupling strength using a Green's function method. An explicit form for τ_2 requires an in-depth biochemical analysis of the particular system under scrutiny and is beyond the scope of the present study. Undoubtedly, τ_2 represents an important part and should be studied in the future.

The hamiltonian of the biological system is:

$$H = \sum_p \omega_p a_p^\dagger a_p + \sum_{p'} \Omega_{p'} b_{p'}^\dagger b_{p'} + \sum_q \bar{\Omega}_q p_q^\dagger p_q + \sum_{p,p',p''} (\lambda a_p b_{p'}^\dagger b_{p''} + \lambda^* a_p^\dagger b_{p'} b_{p''}^\dagger) + \tfrac{1}{2} \sum_{p,p',p''} (\alpha a_p b_{p'}^\dagger b_{p''}^\dagger + \alpha^* a_p^\dagger b_{p'} b_{p''}) + \sum_{p,p',p''} (\chi a_p a_{p'}^\dagger b_{p''}^\dagger + \chi^* a_p^\dagger a_{p'} b_{p''}) + \tfrac{1}{2} \sum_{p,p',p''} (\beta a_p a_{p'} b_{p''}^\dagger + \beta^* a_p^\dagger a_{p'}^\dagger b_{p''}) + \sum_{p,q} (\xi p_q a_p^\dagger + \xi^* p_q^\dagger a_p). \quad (1)$$

All notations and the meanings ot the various coupling constants and operators are the same as those given in ref. [8]. For simplicity in the calculation, the excitations will be regarded as phonons for both the biological system and the heat bath. However, the heat bath quanta may also consist of fermions [8]. It should be noted that the lifetime (and thus self-energy) of interest is due to the terms with coupling constants χ and χ^* because these terms are responsible for creating the condensed phase. The contributions from the other terms that involve the simultaneous creation or destruction of two quanta within the biological system (i.e., those terms with the β and β^* coupling constants) are negligible compared to the above as shown in ref. [8]. The rest of the terms in the hamiltonian do not contribute to the condensation to the lowest order in coupling constant and therefore are not relevant to the present calculation.

Using finite temperature Green's function techniques the self-energy, $\Sigma(p, \omega_p)$, for the biological system phonon associated with operators a_p and $a_p^\dagger$ was calculated yielding:

$$\Sigma(p, \omega_p) = |\chi|^2 \sum_{p'} \left[\frac{1 + n(\Omega_{p'}) + n(\omega_{p-p'})}{\omega_p - \Omega_{p'} - \omega_{p-p'} + i\delta} + \frac{n(\Omega_{p'}) - n(\omega_{p-p'})}{\omega_p + \Omega_{p'} - \omega_{p-p'} + i\delta} \right], \quad (2)$$

where $n(\omega) = [\exp(\omega/kT) - 1]^{-1}$, Ω_p is the energy of a phonon with momentum p in the heat bath and ω_p is the energy of a phonon with momentum p in the biological system. Since at room temperature, these energies are much smaller than kT, $n(\omega_p)$ and $n(\Omega_{p'})$ are much larger than 1. Also $\Omega_{p'} = \pm(\omega_p - \omega_{p-p'})$ which is smaller than the width of the narrow energy band of the biological system under consideration which in turn is smaller than a typical energy in the set of vibrational modes (i.e., $\Omega_{p'} \ll \omega_p$). Thus, $n(\Omega_{p'}) \gg n(\omega_{p-p'})$. Therefore, the self-energy can be approximately written as:

$$\Sigma(p, \omega_p) = \frac{|\chi|^2 kT}{4\pi^2 \hbar^3} \int_{-1}^{1} dx \int_0^\infty \frac{p'^2 dp'}{\Omega_{p'}} \times \left[\frac{1}{\omega_p - \Omega_{p'} - \omega_{p-p'} + i\delta} + \frac{1}{\omega_p + \Omega_{p'} - \omega_{p-p'} + i\delta} \right], \quad (3)$$

where the sum over heat bath momenta has been converted to an integration. Using the Debye model, for the excitations in the heat bath, $\Omega_{p'} = v'p'$, where v' = the velocity of sound in the heat bath. The form of the dispersion relation for the biological system is unknown and depends in an intricate manner upon the biological structure. To obtain an approximate result, however, it is presumed that $\omega_p - \omega_{p-p'} = \pm\gamma\Delta$ where Δ = the width of the energy band (which is assumed narrow) and γ is a positive quantity less than 1. The value of γ depends on the form of the dispersion relation for the biological system.

Using the relation:

$$\tau_1 = \hbar/|\mathrm{Im}\,\Sigma|, \quad (4)$$

and the imaginary part of the self-energy in eq. (3):

$$\mathrm{Im}\,\Sigma \doteq -|\chi|^2 kT\gamma\Delta/\pi v'^3\hbar^3, \quad (5)$$

we have:

$$\tau_1 \doteq \hbar^4\pi v'^3/|\chi|^2 kT\gamma\Delta, \quad (6)$$

which is inversely proportional to T and to the square of the coupling constant. As seen from eq. (6), τ_1 decreases with increasing temperature. Thus, as the temperature is raised, the condensed phase should occur sooner also causing the biological effects to begin sooner. This agrees, at least qualitatively, with the example stated previously from Smolyanskaya and Vilenskaya's experiment where changing T from 20°C to 37°C brought the biological effects into play holding the irradiation time fixed at 30 minutes.

τ_1 given in eq. (6) represents the irradiation time needed to produce the condensation and is the minimum time needed to produce biological effects. In order to apply this equation to specific circumstances, model studies and more experiments need to be performed to obtain expressions for the coupling constants and the widths of the energy bands. Given this information, a more complete understanding of the Fröhlich model and all its implications can be obtained.

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BOSE CONDENSATION IN BIOSYSTEMS

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A microscopic approach to Fröhlich's theory of Bose condensation in biological systems is discussed. The conclusions from this microscopic analysis offer further support for Fröhlich's hypothesis.

The possible importance of vibrational modes in the frequency range of 10^{11}–$10^{12}\,\mathrm{sec}^{-1}$ of biological macromolecules in stimulating important life processes was first suggested by Fröhlich [1–5]. Recently, several experimental results have been published which tend to support Fröhlich's theory [6–10].

Generally, biological systems possess constituents that are capable of electric dipolar oscillations. Long range Coulomb interaction between these dipoles establishes a branch or branches of longitudinal electric modes. Fröhlich has suggested if energy is supplied locally to one constituent (or removed from it), this energy variation will be shared by all the constituents within the branch of vibrational modes due to the long range coulomb interaction. He further concluded if energy is supplied beyond a certain critical rate, the energy redistribution among the dipolar modes will lead to its concentration in the lowest energy state (i.e. Bose–Einstein condensation).

Fröhlich's mathematical analysis was based upon the utilization of rate equations. We have approached the theoretical problem using microscopic techniques and the interaction Hamiltonian. As Fröhlich, we suppose that the biological system consists of the following:

(i) oscillating segments of giant dipoles occurring along macromolecules,

(ii) the remainder of the biosystem which forms a heat bath,

(iii) an external energy source which couples to the oscillating units.

The interaction between oscillating units produces a narrow band of frequencies, $\omega_i\,(\omega_0 \leqslant \omega_i \leqslant \omega_{\max})$ representing the normal electromagnetic modes. Assigned to each mode are creation and destruction operators, a_i^+ and a_i. The vibrational modes of the heat bath, Ω_k, are associated with creation and destruction operators, b_k^+ and b_k. Similarly, the external energy source has frequencies, $\bar{\Omega}_p$, and creation and destruction operators P_p^+ and P_p associated with it. Thus, the Hamiltonian of the biosystem can be given as:

$$H = \sum_i \hbar\omega_i a_i^+ a_i + \sum_k \hbar\Omega_k b_k^+ b_k + \sum_p \hbar\bar{\Omega}_p P_p^+ P_p + \hbar\sum_{i,k}(\lambda b_k^+ a_i + \lambda^* b_k a_i^+)$$

$$+ \tfrac{1}{2}\hbar \sum_{i,j,k}{}' (\chi a_i^+ a_i b_k^+ + \chi^* a_j a_i^+ b_k) + \hbar\sum_{p,i}(\xi P_p a_i^+ + \xi^* P_p^+ a_i)\,, \tag{1}$$

where λ, χ and ξ are coupling constants for the one quantum process, the two quantum process and the energy supply to the oscillating constituents respectively. Using the commutation relations the rate of change of ith mode quanta is

$$\dot{n}_i = (\mathrm{i}\hbar)^{-1}[n_i, H]\,, \tag{2}$$

where $n_i = a_i^+ a_i$.

Using the commutation relations for the various creation and destruction operators, we found the expectation value of $\dot{n}_i$ ($\langle\dot{n}_i\rangle = \langle\psi(t)|\dot{n}_i|\psi(t)\rangle$) to be:

$$\langle \dot{n}_i \rangle = S - \phi(T, \omega_i)[\langle n_i \rangle \exp(\hbar\omega_i/kT) - (1 + \langle n_i \rangle)] - \sum_j \Lambda(T, \omega_i, \omega_j)[\langle n_i \rangle (1 + \langle n_j \rangle) \exp(\hbar(\omega_i - \omega_j)/kT) - \langle n_j \rangle (1 + \langle n_i \rangle)], \tag{3}$$

where the angular brackets refer to thermal averaging in the grand ensemble appropriate to the entire system. $|\psi(t)\rangle$ is the state vector of the whole system at time t. Since $\langle \dot{n}_i \rangle$ cannot be solved exactly, perturbation theory was utilized to obtain it to lowest order in coupling strength. For stationarity, $\langle \dot{n}_i \rangle = 0$. As a result, we found the average number of ith mode quanta to be:

$$\langle n_i \rangle = \left[1 + \frac{S}{\phi(T, \omega_i) + \Sigma_j \Lambda(T, \omega_i, \omega_j)\langle n_j \rangle}\right] \{\exp[\hbar\omega_i - \mu)/kT] - 1\}^{-1}, \tag{4}$$

where

$$\exp(-\mu/kT) = \frac{1 - \Sigma_j \Lambda(T, \omega_i, \omega_j)(\langle n_j \rangle - (1 + \langle n_j \rangle) \exp(-\hbar\omega_j/kT)}{\phi + \Sigma_j \Lambda \langle n_j \rangle}, \tag{5}$$

$$S = 4\pi |\xi|^2 \sum_i \langle p_i^+ p_i \rangle, \tag{6}$$

$$\phi(T, \omega_i) = 4\pi |\lambda|^2 \exp(-\hbar\omega_i/kT)[1 + N(\hbar\omega_i)], \tag{7}$$

and

$$\Lambda(T, \omega_i, \omega_j) = \pi |\chi|^2 \exp[-\hbar(\omega_i - \omega_j)/kT] \times \begin{cases} [N(\hbar\omega_j - \hbar\omega_i)], & \omega_j > \omega_i, \\ [1 + N(\hbar\omega_i - \hbar\omega_j)], & \omega_i > \omega_j, \end{cases} \tag{8}$$

where $N(\hbar\omega)$ is the number of excitations with energy $\hbar\omega$ in the heat bath.

Eqs. (4) and (5) are of exactly the same form as those derived by Fröhlich. Bose–Einstein condensation into the lowest energy state occurs when μ approaches $\hbar\omega_0$. This takes place when the energy supplied exceeds a certain critical rate as shown by Fröhlich. Fröhlich formed the ansätze in his rate equations by requiring a Bose–Einstein distribution for thermal equilibrium when no external power is supplied. Otherwise, the form of the rate equations were rather arbitrary. However, by approaching Fröhlich's theory from a macroscopic point of view, we have arrived at the same conclusions as Fröhlich did and removed the arbitrariness. Our results, therefore, further strengthen the possibility that such a phenomenon as Bose condensation in biological systems exists.

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BOSE CONDENSATION OF STRONGLY EXCITED LONGITUDINAL ELECTRIC MODES

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If the energy fed into the branch of longitudinal electric modes exceeds a critical rate then, under stationary conditions, the excitation energy is channelled into the mode with lowest frequency in a manner typical for Bose condensation.

In the introductory talk to the recent conference on Theoretical Physics and Biology [1] I have conjectured that in living systems certain coherent longitudinal electric modes are strongly excited, and stabilized by deformations arising from non linear effects. This conjecture was stimulated by the extraordinary dielectric properties of cells and of some of the relevant giant molecules in conjecture with the requirement of a non-obvious long-range order (coherence) in systems which are relatively stable but not in all respects in thermal equilibrium. It is the purpose of the present note to show that under very general circumstances the energy fed into the branch of longitudinal electric modes gets channelled into a single mode in a manner which closely resembles the condensation phenomenon in a Bose gas.

Assume the system to consist of Z components suspended in a heat bath, and consider each component capable of dipolar electric oscillations. Such a component may be a repeated dipolar part of a giant molecule (e.g. H-bond) or it may be a larger unit. The long range interaction of these components then leads to the establishment of Z modes, usually in a relatively narrow frequency range, say

$$\omega_1 \leqslant \omega \leqslant \omega_2 . \tag{1}$$

Assume that energy is supplied locally to each component at a rate s, so that it may be considered supplied to each mode of the branch at the same rate s. Interaction with the heat bath then leads to emission of energy until a steady state has been reached. This interaction with the heat bath must depend on the excitation of the modes; processes of first and second order in the number of excited quanta will be considered. The condition for stationarity then requires for each l,

$$s = L_l , \tag{2}$$

where the rate of loss of energy L_l from mode ω_l (containing n_l quanta) is assumed of the form

$$L_l = \phi(T)(n_l \exp(\hbar\omega_l/kT) - (1+n_l)) + \tag{3}$$
$$+ \chi(T) \sum_j (n_l(1+n_j) \exp\{\hbar(\omega_l - \omega_j)/kT\} - n_j(1+n_l)) .$$

The interaction terms $\phi(T) > 0$ and $\chi(T) > 0$ with the heat bath will in general depend on temperature T. Clearly L_l vanishes in thermal equilibrium when n_l is a Planck distribution n_{lT}. With the notation $N = \sum n_j$ one finds from eqs. (2) and (3)

$$n_l = \left(1 + \frac{s}{\phi + \chi N}\right) \frac{1}{\exp\{(\hbar\omega_l - \mu)/kT\} - 1} , \tag{4}$$

which requires $\hbar\omega_1 > \mu \geqslant 0$ to satisfy $n_l \geqslant 0$. Also

$$\exp(-\mu/kT) = \frac{\phi + \chi \sum (1+n_j) \exp(-\hbar\omega_j/kT)}{\phi + \chi N} = \tag{5}$$
$$= 1 - \frac{\chi}{\phi + N\chi} \frac{S(T)}{\phi} > 0 ,$$

where

$$S(T) = \phi \sum \{n_j - (1+n_j) \exp(-\hbar\omega_j/kT)\} . \tag{6}$$

Clearly Bose condensation into the lowest mode ω_1 will take place when μ approaches $\hbar\omega_1$ very closely. One would expect this to occur when s becomes sufficiently large. We shall demonstrate this explicitly for the high temperature case,

$$kT \gg \hbar\omega_1 . \tag{7}$$

In this approximation one finds from (4)

$$n_l = \left(1 + \frac{s}{\phi + \chi N}\right) \frac{kT}{\hbar\omega_l - \mu} + \ldots . \tag{8}$$

Let $D(\omega)d\omega$ be the number of modes in a range $d\omega$. Then if no condensation has taken place $D(\omega_l)\, n_l$ can be considered a continuous function and one finds (N_n denotes the total number of quanta in this case)

$$N_n = \left(1 + \frac{s}{\phi + \chi N_n}\right) kT \int_{\omega_1}^{\omega_2} \frac{D d\omega}{\hbar\omega - \mu} < \left(1 + \frac{s}{\phi + \chi N_n}\right) N_0 \qquad (9)$$

and hence in the absence of condensation when $N = N_n$ one has

$$\frac{N(\phi + \chi N)}{\phi + \chi N + s} < N_0 = \frac{kT}{\hbar} \int_{\omega_1}^{\omega_2} \frac{D d\omega}{\omega - \omega_1} \,. \qquad (10)$$

Now N can also be estimated from the generally valid eq. (6), which yields

$$S(T) = \phi \frac{\hbar\bar{\omega}}{kT} (N - N_T) \,, \qquad (11)$$

where $\omega_1 \leqslant \bar{\omega} \leqslant \omega_2$, and N_T is the number of quanta in thermal equilibrium. Also in this approximation $S(T) \approx Zs$. It follows from eq. (11) that N is a linear function of s because $\bar{\omega}$ depends only weakly on s. Hence a value of s_0 of s exists above which (10) can no longer be fulfilled. The situation then closely resembles that in the Bose condensation of ideal gases. This implies that a continuous distribution $n_l D(\omega_l)$ can no longer accommodate the total number of quanta. $\mu/\hbar$ then takes a value below, but so close to the lowest frequency ω_1 that condensation into ω_1 takes place. The total number of quanta in this case becomes $N = N_n + kT/(\hbar\omega_1 - \mu)$.

Details depend, of course, strongly on the particular model. It can be seen, however, that in general N_0 will be larger than the number of thermal quanta by a factor of the order $\omega_1/(\omega_2 - \omega_1)$, but from (10) it follows that the critical number of quanta at which condensation starts is less than N_0. Finally it should be mentioned that N, according to eq. (11) is independent of χ, but χ is very relevant in the condensation condition.

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CRYSTALLINE FIELD EFFECTS AND FIRST ORDER PHASE TRANSITION IN $ErCo_2$

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The temperature dependence of the nuclear hyperfine splitting of ^{166}Er has been measured in the magnetically ordered $ErCo_2$ compound, showing a first order magnetic phase transition. The measurements indicate a crystalline field interaction which is large compared to the exchange interaction.

Measurements of the dependence upon temperature of the nuclear magnetic and quadrupole hyperfine splitting provide information on the influence of the crystalline electric field on the energy levels of ions embedded in a magnetically ordered metal. We report here the observation of a crystalline field interaction within the $J = \frac{15}{2}$ ground multiplet of Er^{3+} in the intermetallic Laves compound $ErCo_2$, which produces an over-all splitting of the multiplet which is large compared to the exchange splitting of about $100^{\circ}K$ of the lowest $|\pm\frac{15}{2}\rangle$ multiplet.

The hyperfine splitting was observed in the recoilless gamma resonance spectrum of the 80.6 keV transition of ^{166}Er in $ErCo_2$ as absorber material where the Er environment has cubic symmetry. The measurements employed sources of $HoAl_2$ which were kept at T =

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Bose–Einstein Condensation in Biological Systems

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Biological systems contain dipolar elements which are capable of electric oscillations at certain specific frequencies. Longitudinal electric modes in a frequency range of 10^{11}–10^{12} s^{-1} are typical. Fröhlich has suggested that if energy is supplied above a critical rate to the branch or branches of electric modes, Bose–Einstein condensation into the lowest energy state occurs. This phenomenon possibly represents a channel for various life processes such as cell division and protein synthesis.

Fröhlich's model is analyzed utilizing microscopic techniques, and it is subsequently concluded that Bose condensation in biosystems may occur under certain conditions.

1. Introduction

The possible participation of the vibrational modes of macromolecules in biological processes has been discussed repeatedly (Volkenshtein, 1964; Khurgin, 1967). It is well known that biological systems contain components capable of electric dipolar oscillations. For example, cell membranes possess a thickness on the order of 10^{-6} cm which maintains a strong dipolar layer. Local vibrations within the membrane could cause positively and negatively charged segments to vibrate against each other yielding an electric dipole oscillation. Similarly, biological macromolecules, e.g. proteins, DNA, etc. have marked dipolar characteristics due primarily to the presence of hydrogen bonds. The interaction between these giant dipoles through long range Coulomb forces establishes a branch or several branches of longitudinal electric modes in a frequency range of 10^{11}–10^{12} s^{-1}.

Recently, H. Fröhlich (Fröhlich, 1968*a*,*b*, 1970, 1972, 1975), has suggested that if energy is supplied locally to one component (or removed from there), then, due to the long range Coulomb interaction, this energy change will be shared by all the components comprising the branch of electric modes. If the rate of energy supply is sufficiently large, the redistribution of the energy between the various degrees of freedom will lead to its concentration in the

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lowest energy mode. This phenomenon is known in physics as Bose–Einstein condensation. This coherent oscillation of electric modes might serve not only as a method for efficient energy storage, but also as a channel for specific bioprocesses.

The biological implications of Bose–Einstein Condensation are numerous. With the onset of condensation, a strong coherent electromagnetic oscillation is established in the system, and the energy channeled into the lowest energy mode could possibly disrupt the cell membrane. Thus, this effect might be one of the energetic mechanisms crucial to the production of cell division. A Soviet investigator (Devyatkov, 1974) has reported enhanced cell division in Rhodotorula and Candida cultures after irradiation by electromagnetic waves in the frequency range appropriate to the Fröhlich model. Similarly, the energy stored by biological Bose–Condensation could be directed to other life sustaining activities such as protein synthesis.

Besides the possible important participation of Bose–Einstein Condensation in the processes of cell division and protein synthesis, a variety of other biological effects could perhaps be attributed to this phenomenon. Recently, several investigators (Kiselev & Zalyuboskaya, 1974; Sevastyanova & Vilenskaya, 1974; Smolyanskaya & Vilenskaya, 1974) have reported interesting results concerning the microwave irradiation of various biological objects in the frequency range 10^{11}–10^{12} s^{-1}. Assuredly, the specific effects represent very complex biological processes. However, they exhibit two pronounced features: (i) they depend very strongly on frequency exhibiting typical resonance behavior, (ii) there exists a critical power supply threshold below which no effect is observed and above which there is a sharp increase in the biological effects which is essentially independent of the power supply. These features strongly support Fröhlich's Bose condensation theory as applied to biological systems. The complexity and diversity of the experimental results may reflect the possibility that different biological effects occur depending upon which dipolar elements are participating in the Bose condensation. For example, the vibrations of one type of macromolecule within the biological system would most likely produce different effects than a different type of macromolecule or the vibrations of the cell membrane itself.

Fröhlich arrived at his conclusions on the basis of rate equations. The general forms of the rate equations were dictated by requiring a Bose–Einstein distribution for thermal equilibrium when there is no power supplied. Beyond this, their forms were somewhat arbitrary. In this paper, the interaction Hamiltonian is utilized. It is subsequently shown that Fröhlich's ansätze in the rate equations are correct, and that there is a Bose–Einstein Condensation.

2. Theory

Similar to Fröhlich, it is presumed that the biological system consists of three parts: (i) oscillating units of giant dipoles occurring approximately periodically along the length of the macromolecules, (ii) the rest of the biosystem constituting a heat bath, (iii) an external energy source which couples to the oscillatory units. The interaction between the oscillatory units will produce a narrow band of frequencies, ω_i ($\omega_0 \leq \omega_i \leq \omega_{\max}$), corresponding to the normal modes of the electromagnetic oscillations.

To each mode, ω_i, a creation operator, a_i^+, and a destruction operator, a_i, are assigned. The normal modes of oscillation will then interact with the rest of the biological system (the heat bath) which is represented by a set of independent excitation frequencies, Ω_k, associated with creation and destruction operators, b_k^+ and b_k respectively. Furthermore, the external energy source, associated with creation and destruction operators P_p^+ and P_p respectively and frequency, $\overline{\Omega}_p$, feeds into the electromagnetic oscillating units and provides impetus for initiation of the biological effects. The Hamiltonian of the biological system can be written as:

$$\begin{aligned} H = & \sum_i \hbar\omega_i a_i^+ a_i + \sum_k \hbar\Omega_k b_k^+ b_k \\ & + \sum_p \hbar\overline{\Omega}_p P_p^+ P_p \\ & + \sum_{i,k} \lambda b_k^+ a_i + \lambda^* b_k a_i^+ \\ & + \sum_{i,j,k} (\chi a_j^+ a_i b_k^+ + \chi^* a_j a_i^+ b_k) \\ & + \sum_{p,i} (\xi P_p a_i^+ + \xi^* P_p^+ a_i) \end{aligned} \tag{1}$$

where λ, χ, and ξ are coupling constants for the one quantum process, two quanta process, and the energy supply to the oscillating units respectively. The electromagnetic oscillations are bosons. Yet the excitations within the heat bath can be either bosons, e.g. phonons, or fermions, e.g. electrons.

The operators a_i^+, a_i, b_k^+, b_k, P_p^+ and P_p satisfy the commutation relations:

$$\begin{aligned} & a_i a_j^+ - a_j^+ a_i = \delta_{ij} \\ & a_i^+ a_j^+ - a_j^+ a_i^+ = a_i a_j - a_j a_i = 0 \\ & b_k b_l^+ \pm b_l^+ b_k = \delta_{kl} \end{aligned} \tag{2}$$

($-$ sign for Bosons and $+$ sign for fermions)

$$\begin{aligned} & b_k b_l \pm b_l b_k = b_k^+ b_l^+ \pm b_l^+ b_k^+ = 0 \\ & P_p P_{p'}^+ - P_{p'}^+ P_p = \delta_{pp'} \end{aligned}$$

The operators corresponding to different excitations commute with one another.

The rate of change of the number of quanta in the ith mode is given by:

$$\dot{n}_i = \frac{1}{i\hbar}[n_i, H] \tag{3}$$

with $n_i = a_i^+ a_i$.

Using the commutation relations in equation (2), equation (3) becomes:

$$\begin{aligned}\dot{n}_i = & -\frac{i}{\hbar}\sum_{i,k}(\lambda^* b_k a_i^+ - \lambda b_k^+ a_i) \\ & -\frac{i}{\hbar}\sum_{j,k}\{\chi(a_i^+ a_j b_k^+ - a_j^+ a_i b_k^+) + \chi^*(a_i^+ a_j b_k - a_j^+ a_i b_k)\} \\ & -\frac{i}{\hbar}\sum_{p,i}(\xi P_p a_i^+ - \xi^* P_p^+ a_i)\end{aligned} \tag{4}$$

The expectation value of $\dot{n}_i$ is given by:

$$\langle \dot{n}_i \rangle = \langle \psi(t) | \dot{n}_i | \psi(t) \rangle \tag{5}$$

where the angular brackets refer to thermal averaging in the grand ensemble describing the whole system, and $|\psi(t)\rangle$ is the state vector of the whole system at time t. Since equation (5) cannot be evaluated exactly, perturbation theory is utilized to solve it to the lowest order in coupling strength. To this end, it is presupposed that the interactions between the electromagnetic oscillations of the system with the heat bath and energy source are turned on adiabatically from the past. Then, if $|\psi_0\rangle$ is the exact eigenstate of the non-interaction system with the Hamiltonian

$$\begin{aligned}H_0 = & \sum_i \hbar\omega_i a_i^+ a_i + \sum_k \hbar\Omega_k b_k^+ b_k \\ & + \sum_P \hbar\bar{\Omega}_p P_p^+ P_p,\end{aligned}$$

it is then possible to write the state vector at time t, to the lowest order in coupling constants as:

$$\begin{aligned}|\psi(t)\rangle = e^{-iH_0 t/\hbar} \Bigg[& 1 - \sum_{i,k} \frac{\lambda b_k^+ a_i \, e^{i(\Omega_k - \omega_i)t}}{(\omega_i - \Omega_k) + i\delta} \\ & - \sum_{i,k} \frac{\lambda^* b_k a_i^+ \, e^{-i(\Omega_k - \omega_i)t}}{(\omega_i - \Omega_k) - i\delta} \\ & - \sum_{i,j,k} \frac{\chi a_j^+ a_i b_k^+ \, e^{i(\Omega_k + \omega_j - \omega_i)t}}{(\Omega_k + \omega_j - \omega_i) + i\delta} \\ & - \sum_{i,j,k} \frac{\chi^* a_j a_i^+ b_k \, e^{-(\Omega_k - \omega_j + \omega_i)t}}{(\Omega_k - \omega_j + \omega_i) - i\delta} \\ & - \sum_{p,i} \frac{\xi P_p a_i^+ \, e^{i(\omega_i - \bar{\Omega}_p)t}}{(\omega_i - \bar{\Omega}_p) - i\delta} \\ & - \sum_{p,i} \frac{\xi^* P_p^+ a_i \, e^{-i(\omega_i - \quad p)t}}{(\omega_i - \bar{\Omega}_p) + i\delta} \Bigg] \; |\psi_0\rangle\end{aligned} \tag{6}$$

Where δ is an infinitesimally small positive value, changing the summations over Ω_k and $\bar{\Omega}_p$ to integrations and using appropriate contours yields:

$$\begin{aligned}|\psi(t)\rangle = e^{-iH_0t/\hbar}\Big[1 &- 2i\pi\sum_i(\lambda b_i^+ a_i - \lambda^* b_i a_i^+) \\ &- 2i\pi\sum_{i,j}(\chi a_j^+ a_i b_{i-j}^+ + \chi^* a_j a_i^+ b_{j-i}) \\ &- 2i\pi\sum_i(\xi P_i a_i^+ + \xi^* P_i^+ a_i)\Big]\ |\psi_0\rangle \end{aligned} \tag{7}$$

Substituting equation (7) into equation (5) and using the commutation relations, one obtains:

$$\begin{aligned}\langle\dot{n}_i\rangle = S &- \phi(T,\omega_i)[\langle n_i\rangle e^{\hbar\omega_i/kT} - (1+\langle n_i\rangle)] \\ &- \sum_j \Lambda(T,\omega_i,\omega_j)[\langle n_i\rangle(1+\langle n_j\rangle)\, e^{[\hbar(\omega_i-\omega_j)]/kT} \\ &\qquad - \langle n_j\rangle(1+\langle n_i\rangle)]\end{aligned} \tag{8}$$

where $\langle n_i\rangle$ is the thermal average of the number of ith quanta and $S = 4\pi/\hbar|\xi|^2\sum_i\langle P_i^+ P_i\rangle_0$ is the number of quanta supplied to the ith mode from the external source. $\phi(T,\omega)$ and $\Lambda(T,\omega_i,\omega_j)$ are given by:

$$\begin{aligned}\phi(T,\omega_i) &= 4\pi/\hbar|\lambda|^2\, e^{-(\hbar\omega_i/kT)}\ [1\pm N(\hbar\omega_i)] \\ \Lambda(T,\omega_i,\omega_j) &= 2\pi/\hbar|\chi|^2\, e^{-[\hbar(\omega_i-\omega_j)]/kT} \\ &\qquad \begin{cases}[N(\hbar\omega_j-\hbar\omega_i)], & \text{for } \omega_j > \omega_i \\ [1\pm N(\hbar\omega_i-\hbar\omega_j)], & \text{for } \omega_i > \omega_j\end{cases}\end{aligned} \tag{9}$$

where $N(\hbar\omega)$ is the number of excitations with energy $\hbar\omega$ in the heat-bath and the plus or minus sign corresponds to bosons or fermions respectively. Note that the 2nd and 3rd terms in equation (8) possess exactly the same forms as Fröhlich's ansätze for loss rates of ith quanta in one and two quanta processes respectively.

The condition for stationarity requires for each oscillator $\langle\dot{n}_i\rangle = 0$. From this, it is found from equation (8) that:

$$\langle n_i\rangle = \left[1 + \frac{S}{\phi(T,\omega_i) + \sum_j \Lambda(T,\omega_i,\omega_j)\langle n_j\rangle}\right] e^{[(\hbar\omega_i-\mu)/kT]^{-1}} - 1, \tag{10}$$

and

$$e^{-(\mu/kT)} = 1 - \frac{\sum_j \Lambda(T,\omega_i,\omega_j)[\langle n_j\rangle - (1+\langle n_j\rangle)\, e^{-(\hbar\omega_j/kT)}]}{\phi + \sum_j \Lambda\langle n_j\rangle} \leq 1. \tag{11}$$

The inequality of equation (11) together with the requirement that $\langle n_i\rangle \geq 0$ dictates that:

$$\hbar\omega_0 \geq \mu \geq 0.$$

3. Conclusion

Equations (10) and (11) are precisely the same equations obtained by Fröhlich. Clearly, from equation (10) one sees that Bose Condensation into the lowest energy mode occurs when μ is approximately equal to $\hbar\omega_0$. As demonstrated by Fröhlich, this takes place when the energy supply becomes sufficiently large and exceeds a certain specific rate which depends on the coupling strength of the heat bath. If there is no energy supplied, $s = 0$, $\mu = 0$, and equation (10) becomes the thermal equilibrium distribution as required. Also it is interesting to take note that if there are no two quanta processes, i.e. $\chi = 0$, then $\mu = 0$, and no Bose Condensation occurs even if energy is supplied externally.

By approaching this theoretical problem using microscopic techniques, a better understanding of the possible existence of Bose–Einstein Condensation in biological systems has been obtained. This model, therefore, is a useful tool for gaining insight into intricate biological structures via straightforward physical concepts. However, the explicit understanding of the complex biochemical aspects of this problem and how they relate to specific life processes is yet to be developed.

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CANONICAL TRANSFORMATIONS IN QUANTUM FIELD THEORY AND SOLITONS

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In a sine-Gordon model we discuss a transformation which can be viewed as a quantum image of the Bäcklund transformation. The soliton is described in terms of condensation of quanta; their confinement is ascribed to the non-implementability of unitary transformations among unitarily inequivalent representations of canonical commutators and to the related topological conservation law.

1. Introduction

In many-body physics, as well as in high-energy particle physics, much attention has been devoted to the study of soliton-like solutions to non-linear evolution equations [1, 15]. Such solutions could be indeed useful in the description of "extended objects" such as dislocations in crystals, vortices in superconductivity or, in high-energy physics, monopoles in non-abelian gauge theories, bag models for elementary particles, etc. The quantum origin of these extended objects requires, however, an understanding of the soliton-like solutions in terms of basic quantum constituents.

It is known that for classical non-linear evolution equations the Bäcklund transformation is a canonical one which keeps the hamiltonian form invariant [2].

In the quantum field theory an operator has been constructed generating a canonical transformation among Heisenberg fields and leading to a coherent state representation for a soliton solution [3]. In the present work we consider a quantum sine-Gordon $1+1$ dimensional model. The condition that the vacuum expectation value (v.e.v.) of the Heisenberg scalar field reproduces the static soliton solution is found to be satisfied when the LSZ asymptotic quantum field (constituent) undergoes a canonical transformation.

The same condition leads us to consider the tree approximation only.

It will be seen that although the transformation for the asymptotic field cannot be defined for $x_1 \to -\infty$ (or $x_1 \to +\infty$), the transformation for the Heisenberg field

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can be formally extended to the full x_1 axis. This fact expresses a well-known property characterizing the classical behaviour of the soliton, namely its interpolation between degenerate vacua. In sect. 3 we see that this is related, on the other hand, to the existence of a topological conservation law and to the stability of the soliton.

By successive canonical transformations of the LSZ asymptotic field, n-soliton solutions can be generated; they appear thus as the "quantum image" of the Bäcklund transformations. Their generator is considered in sect. 3, where the soliton solution is represented in terms of coherent state; the relation to the coherent representation of ref. [3] is also discussed.

Finally let us observe that the reconstruction of the soliton in terms of the v.e.v. of a quantum field has been presented in refs. [4–6]. There the tree approximation has also been used and a canonical transformation ("the boson transformation") for the asymptotic field has been assumed. Canonical transformations from the free field base have been also used in ref. [18].

2. Canonical transformations and the soliton solution

We first introduce the quantum field theory framework. Let us consider the 1 + 1 dimensional quantum sine-Gordon model whose Heisenberg field equation is given by

$$\partial^2\hat{\phi}(x)+\frac{m^3}{g}\sin g\frac{\hat{\phi}(x)}{m}=0, \qquad x\equiv(x_0,x_1), \qquad \partial^2\equiv\partial_0^2-\partial_1^2. \tag{1}$$

$\hat{\phi}(x)$ is a Heisenberg scalar field. Eq. (1) can be written, upon expansion of the sine, as

$$(-\partial^2-m^2)\hat{\phi}(x)=-\frac{m^3}{g}\sum_{k=1}^{\infty}\frac{(-1)^{k+1}}{(2k+1)!}\left(\frac{g}{m}\right)^{2k+1}\hat{\phi}(x)^{2k+1}, \tag{2}$$

i.e., with obvious notation,

$$\Lambda(\partial)\hat{\phi}(x)=J\left[\hat{\phi}(x)\right]. \tag{3}$$

In the previous equations m is the physical mass. Let $\hat{\phi}^{\mathrm{in}}(x)$ denote the LSZ asymptotic in-field (equivalently we could work with the out-field $\hat{\phi}^{\mathrm{out}}(x)$):

$$\Lambda(\partial)\hat{\phi}^{\mathrm{in}}(x)=0. \tag{4}$$

Our quantum field theory is completely specified only when the Hilbert space on which the field operators are to be realized is given. We choose as our Hilbert space

the Fock space for the in-field $\hat{\phi}^{\text{in}}(x)$ denoted by $\{|0\rangle\}$, with the vacuum state $|0\rangle$:

$$\hat{\phi}^{\text{in}}(x)^{(+)}|0\rangle = 0. \tag{5}$$

$\hat{\phi}^{\text{in}}(x)^{(+)}$ denotes the annihilation operator of $\hat{\phi}^{\text{in}}(x)$. Also we assume that

$$\langle 0|\hat{\phi}(x)|0\rangle = 0. \tag{6}$$

The Heisenberg field operates on the states of $\{|0\rangle\}$ when expressed in terms of $\hat{\phi}^{\text{in}}(x)$. In other words a solution of the dynamics in terms of $\hat{\phi}^{\text{in}}$ is determined when the "dynamical map" [7]

$$\langle a|\hat{\phi}(x)|b\rangle = \langle a|F\left[\hat{\phi}^{\text{in}}(x)\right]|b\rangle \tag{7}$$

is found. Here F denotes a functional of normal products of $\hat{\phi}^{\text{in}}$ and is determined by the dynamics; $|a\rangle$ and $|b\rangle$ belong to $\{|0\rangle\}$. It is important to note that $|a\rangle$ and $|b\rangle$ are wave-packet states since $\{|0\rangle\}$ is the space of physical states.

Let us observe that eq. (3) leads to the Yang-Feldman equation

$$\langle a|\hat{\phi}(x)|b\rangle = \langle a|\left[\hat{\phi}^{\text{in}}(x) - i\int \mathrm{d}^2 y \Delta_{\text{R}}(x-y)\right] J\left[\hat{\phi}(y)\right]|b\rangle, \tag{8}$$

where $\Delta_{\text{R}}(x-y)$ is the retarded Green function:

$$\Lambda(\partial)\Delta_{\text{R}}(x-y) = i\delta^{(2)}(x-y). \tag{9}$$

Let us now turn our attention to the classical Euler equation framework. The static Euler equation is

$$\Lambda_{\text{s}}(\partial)\phi_{\text{c}}^{(1)}(x_1) = J\left[\phi_{\text{c}}^{(1)}(x_1)\right] \tag{10}$$

where $\Lambda_{\text{s}}(\partial) \equiv \partial_1^2 - m^2$ and J has the same form as in eq. (3). $\phi_{\text{c}}^{(1)}(x_1)$ is the c-number function for the single-soliton solution:

$$\phi_{\text{c}}^{(1)}(x) = -\frac{4m}{g}\,\text{tg}^{-1}\text{e}^{-m(x_1-a)}. \tag{11}$$

Let $f(x_1)$ denote a solution of

$$\Lambda_{\text{s}}(\partial)f(x_1) = 0; \tag{12}$$

e.g.

$$f(x_1) = \text{const}\cdot\text{e}^{-mx_1}. \tag{13}$$

We do not choose as $f(x_1)$ a superposition of e^{-mx_1} and e^{+mx_1} since it would be divergent both as $x_1 \to -\infty$ and as $x_1 \to +\infty$. When the constant in (13) is chosen to be $-4(m/g)e^{ma}$, $\phi_c^{(1)}(x_1)$ can be written as

$$\phi_c^{(1)}(x_1) = f(x_1) + \int dy_1 K(x_1 - y_1) J\left[\phi_c^{(1)}(y_1)\right] \tag{14}$$

$$= f(x_1) - \frac{g^2}{3!} \int dy_1 K(x_1 - y_1) f^3(y_1) + \cdots, \tag{15}$$

with

$$\Lambda_s(\partial) K(x_1 - y_1) = \delta(x_1 - y_1). \tag{16}$$

The choice (13) and the convergence of the integrals in (15) require that

$$K(x_1 - y_1) = -\theta(y_1 - x_1) \frac{1}{m} \sinh m(x_1 - y_1). \tag{17}$$

Our assumption is now that a correspondence can be established between the quantum level and the classical level. To exhibit such a correspondence let us assume that there exists an invariant transformation B for the static Heisenberg field equation:

$$\hat{\phi}^{(1)}(x) = B\hat{\phi}(x), \tag{18}$$

$$\Lambda(\partial)\hat{\phi}^{(1)}(x) = J\left[\hat{\phi}^{(1)}(x)\right]. \tag{19}$$

We want to relate the v.e.v. of $\hat{\phi}^{(1)}$ to $\phi_c^{(1)}$. Since the Heisenberg field operates on the Fock space through the physical particle field $\hat{\phi}^{in}(x)$, we have to find a transformation β,

$$\hat{\phi}^{in}(x) \to \hat{\phi}^{in\prime}(x) = \beta\hat{\phi}^{in}(x), \tag{20}$$

such that, as an effect of the non-linearity of the dynamical map eq. (7), it "dynamically rearranges" [8, 9] as

$$\langle a | B\hat{\phi}(x) | b \rangle = \langle a | F\left[\beta\hat{\phi}^{in}(x)\right] | b \rangle . \tag{21}$$

Furthermore, we require that β is a canonical transformation for $\hat{\phi}^{in}(x)$, i.e. it preserves the canonical commutation relation (c.c.r.)

$$\left[\hat{\phi}^{in}(x), \dot{\hat{\phi}}^{in}(y)\right]_{x_0 = y_0} = i\delta(x_1 - y_1), \qquad \left[\hat{\phi}^{in}(x), \hat{\phi}^{in}(y)\right]_{x_0 = y_0} = 0, \quad \text{etc.} \tag{22}$$

and is an invariant transformation for the free field equation (4):

$$\Lambda(\partial)\hat{\phi}^{\mathrm{in}\prime}(x)=0. \tag{23}$$

The Yang-Feldman equation for $\hat{\phi}^{(1)}$ is

$$\langle 0|\hat{\phi}^{(1)}(x)|0\rangle=\langle 0|\hat{\phi}^{\mathrm{in}\prime}(x)|0\rangle-i\int \mathrm{d}^2y\Delta_{\mathrm{R}}(x-y)\langle 0|J\big[\hat{\phi}^{(1)}(y)\big]|0\rangle\,. \tag{24}$$

Observe now that

$$\langle 0|J\big[\hat{\psi}(x)\big]|0\rangle=J\big[\langle 0|\hat{\psi}(x)|0\rangle\big]+\mathrm{O}(h), \tag{25}$$

where $\mathrm{O}(h)$ denotes contribution of the order of h, the Planck constant, due to the contraction of in fields which create loop diagrams. Thus

$$\begin{aligned}\lim_{h\to 0}\langle 0|\hat{\phi}^{(1)}(x)|0\rangle &= \langle 0|\hat{\phi}^{\mathrm{in}\prime}(x)|0\rangle-i\int \mathrm{d}^2y\Delta_{\mathrm{R}}(x-y)J\Big[\lim_{h\to 0}\langle 0|\hat{\phi}^{(1)}(y)|0\rangle\Big]\\ &= \langle 0|\hat{\phi}^{\mathrm{in}\prime}(x)|0\rangle-i\int \mathrm{d}^2y\Delta_{\mathrm{R}}(x-y)\big[\langle 0|\hat{\phi}^{\mathrm{in}\prime}(y)|0\rangle\big]^3+\cdots.\end{aligned} \tag{26}$$

If the v.e.v. of $\hat{\phi}^{\mathrm{in}\prime}(x)$ is time independent, when the static case is considered, Δ_{R} is replaced by $K(x_1-y_1)$ and d^2y by $i\mathrm{d}y_1$. Therefore, we can put

$$\phi_{\mathrm{c}}^{(1)}(x_1)=\lim_{h\to 0}\langle 0|\hat{\phi}^{(1)}(x)|0\rangle=\lim_{h\to 0}\langle 0|B\hat{\phi}(x)|0\rangle\,. \tag{27}$$

The limit $h\to 0$ suggests use of the tree approximation when explicit computation is to be carried out [4–6]*. To find β we compare eq. (15) with eq. (26) for the static case:

$$\begin{aligned}&f(x_1)-\frac{g^2}{3!}\int \mathrm{d}y_1K(x_1-y_1)f^3(y_1)+\cdots\\ &\quad=\langle 0|\hat{\phi}^{\mathrm{in}\prime}(x)|0\rangle-\frac{g^2}{3!}\int \mathrm{d}y_1K(x_1-y_1)\big(\langle 0|\hat{\phi}^{\mathrm{in}\prime}(y)|0\rangle\big)^3+\cdots,\end{aligned} \tag{28}$$

* It could be observed that in refs. [4–6] use of the tree expansion from the beginning is equivalent to finding just a perturbative solution of the classical equation. However it can be shown (cf. refs. [12, 13]) that quantum corrections, including renormalization effects, can be calculated from the dynamical map in the tree approximation. These problems are extensively analyzed in refs. [12–14] where interaction between quanta and static soliton are also studied.

so that

$$\langle 0|\hat{\phi}^{\mathrm{in}\prime}(x)|0\rangle = f(x_1). \tag{29}$$

In conclusion

$$\hat{\phi}^{\mathrm{in}\prime}(x) = \beta\hat{\phi}^{\mathrm{in}}(x) = \hat{\phi}^{\mathrm{in}}(x) + f(x_1). \tag{30}$$

It should be observed, however, that $\hat{\phi}^{\mathrm{in}\prime}(x)$ is *not* well definite as $x_1 \to -\infty$; so we restrict eq. (30) to the domain $x_1 > 0$. Nevertheless, the B transformation can be formally extended to the full x_1 axis since $\phi_c^{(1)}(x_1)$ [cf. eq. (27)] is everywhere well behaved*. This is related to the interpolating character of the soliton solution between degenerate vacua:

$${}_{2\pi}\langle 0|\hat{\phi}(x)|0\rangle_{2\pi} = -2\pi\frac{m}{g} \underset{-\infty \leftarrow x_1}{\leftarrow} \phi_c^{(1)}(x_1) \underset{x_1 \to +\infty}{\rightarrow} 0 = {}_0\langle 0|\hat{\phi}(x)|0\rangle_0, \tag{31}$$

which expresses, equivalently, the existence of a topological conservation law (topological charge $Q = \phi_c^{(1)}(x_1 = +\infty) - \phi_c^{(1)}(x_1 = -\infty)$). $|0\rangle_0$ denotes the Fock vacuum introduced in eq. (5). The state $|0\rangle_{2\pi}$ denotes the degenerate vacuum for which

$${}_{2\pi}\langle 0|\hat{\phi}(x)|0\rangle_{2\pi} = -2\pi\frac{m}{g}. \tag{32}$$

The related space $\{|0\rangle_{2\pi}\}$ is one of the infinitely many unitarily inequivalent representations of the c.c.r. allowed in our theory in the infinite volume limit ($x_1 \to \pm\infty$) [11]. Due to the unitary inequivalence, there does not exist a well defined unitary operator generating the transformation $|0\rangle_0 \to |0\rangle_{2\pi}$ or vice versa. Nevertheless, the interpolation between expectation values of $\hat{\phi}$ in $|0\rangle_0$ and $|0\rangle_{2\pi}$, as expressed in eq. (31), is characteristic of the non-trivial topology of the soliton $\phi_c^{(1)}(x)$. This is a manifestation of the classical nature of the soliton solution: it allows the interpolation between unitarily inequivalent representations. At the quantum level, such an interpolation is strictly forbidden since quantization requires a unique definite choice of the representation**.

Let us remark that when $\hat{\phi}$ is written through the dynamical map in terms of $\hat{\phi}^{\mathrm{in}}$, B does not preserve the canonical commutators [12–14]

$$\left[\hat{\phi}(x), \dot{\hat{\phi}}(y)\right]_{x_0=y_0} = i\delta(x_1 - y_1), \qquad \left[\hat{\phi}(x), \hat{\phi}(y)\right]_{x_0=y_0} = 0, \qquad \text{etc.}$$

* It can be proved that if $\hat{\psi}(x) = \Psi(x, \hat{\psi}^{\mathrm{in}}(x))$ satisfies the Heisenberg equation, then $\hat{\psi}'(x) = \Psi(x, \hat{\psi}^{\mathrm{in}} + f)$ satisfies the same Heisenberg equation, with $f(x)$ the solution of the free field equation for field $\hat{\psi}^{\mathrm{in}}$; in our case, cf. eq. (19). This is the content of the boson transformation theorem, see ref. [10].

** In problems involving order parameters $\alpha_i = {}_{\alpha_i}\langle 0|\hat{\phi}(x)|0\rangle_{\alpha_i}$, $i = 1,2\ldots$ where $|0\rangle_{\alpha_i}$ denote inequivalent degenerate vacua, soliton solutions can be relevant in the study of transitions between different α_i.

To reintegrate them (which may be interpreted as completeness relations) the inclusion of bound state modes and the introduction of a "quantum coordinate" (collective coordinate) for the soliton are required. When fluctuations of this coordinate are negligible with respect to the size of the soliton, it behaves as a classical object; otherwise it behaves as a quantum soliton. In ref. [14] an extensive discussion of these problems can be found.

Since the above considerations are independent of the particular form of J in eq. (3), they also apply to a scalar $1+1$ dimensional $\lambda\phi^4$ theory.

In the case of the sine-Gordon model one can consider a n-soliton solution $\phi_c^{(n)}(x_0, x_1)$ with $n > 1$, which is characterized by n parameters [1, 15]. We have to boost $f(x_1)$ given in eq. (13):

$$x_1 \to x_1' = \gamma(x_1 - ux_0) + \gamma u \frac{\Delta}{2m}$$

$$c = 1, \qquad 0 < u < 1, \qquad \gamma = (1-u^2)^{-1/2}, \qquad \tfrac{1}{2}\Delta = \frac{1}{u\gamma}\log u,$$

$$f(x_1) = \text{const} \cdot e^{-mx_1} \to \text{const} \cdot e^{-m\gamma x_1 + m\gamma u x_0 - \Delta u\gamma/2} = f(x_0, x_1) = f(x_1'). \quad (33)$$

Here we are partially using notations and conventions of ref. [15]. Of course $f(x_1')$ solves the free field equation (4) due to its Lorentz invariance. As we did for the one-soliton case, we can write in this case

$$\phi_c^{(n)}(x_1') = f^{(n)}(x_1') + \int dy_1' K(x_1' - y_1') J\left[\phi_c^{(n)}(y_1')\right],$$

where $f^{(n)}(x_1')$ is a solution of the free field equation (4) of the general form

$$f^{(n)}(x_1') = f^{(n)}(x_0, x_1) = \sum_{i=1}^{n} f_i(x_1'). \quad (34)$$

f_i are of the form (33) and characterized by the velocity parameters u_i.

We assume now that a correspondence can be established between the quantum level and the classical level such that

$$\lim_{h \to 0} \langle 0|\hat{\phi}^{(n)}(x_0, x_1)|0\rangle = \phi_c^{(n)}(x_0, x_1), \quad (35)$$

with $\hat{\phi}^{(n)}$ solution of the Heisenberg field equation. Then we look for a canonical transformation $\beta^{(n)}$ for $\hat{\phi}^{\text{in}}$ such that

$$\langle 0|\hat{\phi}^{(n)}(x_0, x_1)|0\rangle = \langle 0|B^{(n)}\hat{\phi}(x_0, x_1)|0\rangle = \langle 0|F\left[\beta^{(n)}\hat{\phi}^{\text{in}}(x_0, x_1)\right]|0\rangle, \quad (36)$$

and such that it is an invariant transformation for the free field equation (4). We can

repeat the procedure followed in the one-soliton case by changing the label (1) into (n) everywhere. Under our assumptions, we thus find $\beta^{(n)}$:

$$\hat{\phi}^{\text{in}} \to \hat{\phi}^{\text{in}\prime} = \hat{\phi}^{\text{in}} + f^{(n)} = \hat{\phi}^{\text{in}} + \sum_{i=1}^{n} f_i \tag{37}$$

as the transformation which induces $B^{(n)}$

$$\hat{\phi} \to \hat{\phi}^{(n)} = B^{(n)}\hat{\phi}. \tag{38}$$

Like eq. (30), eq. (37) must be restricted to a convenient x_1 domain to make $\hat{\phi}^{\text{in}\prime}(x)$ a well-defined field operator. We can also think of (37) as

$$\hat{\phi}^{\text{in}} \overset{\beta}{\to} \hat{\phi}^{\text{in}} + f_1 \overset{\beta}{\to} \hat{\phi}^{\text{in}} + f_1 + f_2 \to \cdots, \tag{39a}$$

in such a way that

$$\hat{\phi} \overset{B}{\to} \hat{\phi}^{(1)} \overset{B}{\to} \hat{\phi}^{(2)} \to \cdots. \tag{39b}$$

In this sense we refer to (38) [or to (39)] as to the "quantum image" of the Bäcklund transformation. Eq. (37) suggests that the soliton can be viewed as an extended object created by the condensation of $\hat{\phi}^{\text{in}}$ quanta (constituents) controlled by the f shift. By using $f^{(n)}$ as given in eq. (34) with $n = 2$, an explicit reconstruction of the 2-soliton case is presented in ref. [5]; there the n-soliton case, $n > 2$, is also discussed.

3. The coherent state representation and the confinement of the constituents

To study the coherent state representation, we introduce the generator of the transformation (30); it is given by

$$D^{\text{in}} = -\int_{-\infty}^{+\infty} \mathrm{d}y_1 g(y_1)\dot{\hat{\phi}}^{\text{in}}(y), \tag{40}$$

with $g(x_1) = \theta(x_1)f(x_1)$ and $\theta(x_1)$ the step function. Use of the c.c.r. shows indeed that

$$\hat{\phi}^{\text{in}\prime}(x) = \mathrm{e}^{-iD^{\text{in}}}\hat{\phi}^{\text{in}}(x)\mathrm{e}^{iD^{\text{in}}}. \tag{41}$$

Thus, eq. (29) gives

$$\langle 0|\mathrm{e}^{-iD^{\text{in}}}\hat{\phi}^{\text{in}}(x)\mathrm{e}^{iD^{\text{in}}}|0\rangle = \langle f|\hat{\phi}^{\text{in}}(x)|f\rangle = f(x_1), \tag{42}$$

where we have put

$$|f\rangle = e^{iD^{in}}|0\rangle \tag{43}$$

and a restriction to a convenient x_1 domain must be introduced to make (41) well defined. Due to the isometry of the transformation induced by (40), it is

$$\langle f|f\rangle = 1. \tag{44}$$

The state (43) is thus a coherent state for $\hat{\phi}^{in}(x)$ [16]:

$$\hat{a}^{in}_k|f\rangle = \alpha(k)|f\rangle, \tag{45}$$

where $\hat{a}^{in}_k$ denotes the annihilation operator of $\hat{\phi}^{in}$ and the c-number function $\alpha(k)$ is related (in a convenient x_1 domain) to the $f(x_1)$ Fourier amplitude. By using the dynamical map (7) and eq. (42) we have

$$\langle f|\hat{\phi}(x)|f\rangle = F[f(x_1)],$$

$$\langle f|J[F(\hat{\phi}^{in}(x))]|f\rangle = J[F(f(x_1))]. \tag{46}$$

Thus $\langle f|\hat{\phi}(x)|f\rangle$ solves the static Euler equation

$$\Lambda_s(\partial)\langle f|\hat{\phi}(x)|f\rangle = J[\langle f|\hat{\phi}(x)|f\rangle],$$

$$\langle f|\hat{\phi}(x)|f\rangle = \phi^{(1)}_c(x_1). \tag{47}$$

Again we stress that, although (42) is restricted to $x_1 > 0$ only, eq. (47) can also be formally extended to $x_1 < 0$, which expresses the interpolating character of the soliton between degenerate vacua [cf. eq. (31)].

Let us note that $|f\rangle$ is not the soliton coherent state for $\hat{\phi}(x)$ as usually given in the literature (cf. refs. [3, 17, 18]), i.e.

$$|\phi^{(1)}_c\rangle = e^{iD}|0\rangle \tag{48}$$

with

$$D = -\int dy_1 \phi^{(1)}_c(y_1)\dot{\hat{\phi}}(y), \tag{49}$$

which is indeed different from D^{in}. Since $|0\rangle$ is the free field vacuum state, to compute the r.h.s. of (48), the dynamical map for $\hat{\phi}$ must be used to express D in terms of $\hat{\phi}^{in}$. In this way the non-linearity of the dynamics is taken into account. It

should be noticed, however, that in the infinite volume limit ($x_1 \to \pm\infty$), since the soliton solution goes to a constant ($\phi_c^{(1)}(x_1 = \pm\infty) = \text{const}^{(\pm)}$) a generator such as the one in eq. (49), cannot be safely defined for the transformation

$$\hat{\phi}(x) \to \hat{\phi}(x) + \text{const}^{(\pm)}, \qquad \text{at } x_1 \to \pm\infty \text{ and } \text{const}^{(+)} \neq \text{const}^{(-)} \tag{50}$$

as is well known [19]. This fact expresses the stability of the soliton state against decay into a finite number of constituents $\hat{\phi}^{\text{in}}(x)$. By following ref. [18] indeed, we can evaluate the overlap between the soliton state $|\phi_c^{(1)}\rangle$ at $x_0 = \pm\infty$ and a state of a finite number of constituents. We compute (cf. ref. [18], sect. 3)

$$\langle 0|\phi_c^{(1)}\rangle\big|_{x_0=\pm\infty} = \exp\left[-\tfrac{1}{2}\int \mathrm{d}k\,|\tilde{\phi}_c^{(1)}|^2\,\omega_k\right], \tag{51}$$

where eq. (49) and the asymptotic field for $\hat{\phi}(x)$ have been used. In eq. (51) $\tilde{\phi}_c^{(1)}$ is the Fourier amplitude of $\phi_c^{(1)}(x_1)$ and $\omega_k = (m^2 + k^2)^{1/2}$. Since

$$\infty = \int |\phi_c^{(1)}|^2\,\mathrm{d}x_1 = \int |\tilde{\phi}_c^{(1)}|^2\,\mathrm{d}k < \int \omega_k |\tilde{\phi}_c^{(1)}|^2\,\mathrm{d}k,$$

the resulting transition probability is zero. We thus conclude that the confinement of the constituents of an extended object finds its origin in the fact that the transformation

$$\hat{\phi}^{\text{in (out)}} \to \hat{\phi}^{\text{in (out)}\prime} = \hat{\phi}^{\text{in (out)}} + \text{const}^{(\pm)} \tag{52}$$

is not unitarily implementable. To implement (52) an infinite amount of energy would be required, because it is not an invariant transformation for the free field equation (4). As in ref. [18], this can be expressed by saying that the existence of a topological conservation law [topological charge $Q = \phi_c^{(1)}(x_1 = +\infty) - \phi_c^{(1)}(x_1 = -\infty)$] corresponds to the soliton being a stable bound state of infinite constituents.

4. Conclusion

We have discussed the transformations which the Heisenberg and asymptotic fields undergo when one looks at the classical soliton from the point of view of quantum basic constituents*. We have considered the 1 + 1 dimensional sine-Gordon model.

Suppose that there exists a quantum field operator $\hat{\phi}^{(1)}(x)$ which solves the static Heisenberg equation, and whose v.e.v. in the $h \to 0$ limit is given by the static soliton

* For an interesting example of condensation generating solitons see ref. [20]. See also refs. [18, 21] for solitons in terms of free fields.

$\phi_c^{(1)}(x_1)$. Suppose also that there exists a transformation B such that $\hat{\phi}^{(1)}(x) = B\hat{\phi}(x)$, with $\hat{\phi}(x)$ a solution of the Heisenberg equation with zero v.e.v.

When $\hat{\phi}(x)$ is expressed in terms of the LSZ asymptotic in-field we have that the transformation B is induced by

$$\hat{\phi}^{\text{in}}(x) \overset{\beta}{\to} \hat{\phi}^{\text{in}}(x) + f(x_1),$$

with $f(x_1) = \text{const} \cdot e^{-mx_1}$, $x_1 > 0$ solving the free field equation. It is interesting to note that although β is definite for $x_1 > 0$ only, B can be formally extended to $x_1 < 0$ also. We have observed that this is a manifestation of the soliton interpolation between degenerate vacua expectation values of $\hat{\phi}$:

$$_{2\pi}\langle 0|\hat{\phi}(x)|0\rangle_{2\pi} = -2\pi\frac{m}{g} \underset{-\infty \leftarrow x_1}{\leftarrow} \phi_c^{(1)}(x_1) \underset{x_1 \to +\infty}{\to} 0 = {}_0\langle 0|\hat{\phi}(x)|0\rangle_0,$$

which expresses the non-trivial topology of the soliton ($\phi_c^{(1)}(x_1 \to +\infty) - \phi_c^{(1)}(x_1 = -\infty) \neq 0$). $|0\rangle_{2\pi}$ and $|0\rangle_0$ belong to representations of the free field c.c.r. which are unitarily inequivalent in the infinite volume limit ($x_1 \to \pm\infty$) [11]. Since quantization requires a unique definite choice of the representation, we have ascribed this interpolation property to the fact that the soliton is the solution to the classical Euler equation. A discussion of a coherent state representation has been presented in sect. 3 where the interpolating character of the soliton has been related to the existence of a topological conservation law and to the "confinement" of the constituents (stability of the soliton, see ref. [18]). In the infinite volume limit the "confinement" has indeed its origin in the non-implementability of transformation among unitarily inequivalent representations.

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CELLULAR AUTOMATA IN CYTOSKELETAL LATTICES

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Cellular automata (CA) activities could mediate biological regulation and information processing via nonlinear electrodynamic effects in cytoskeletal lattice arrays. Frohlich coherent oscillations and other nonlinear mechanisms may effect discrete 10^{-10} to 10^{-11} s interval events which result in dynamic patterns in biolattices such as cylindrical protein polymers: microtubules (MT). Structural geometry and electrostatic forces of MT subunit dipole oscillations suggest neighbor rules among the hexagonally packed protein subunits. Computer simulations using these suggested rules and MT structural geometry demonstrate CA activities including dynamical and stable self-organizing patterns, oscillators, and traveling "gliders". CA activities in MT and other cytoskeletal lattices may have important biological regulatory functions.

1. Introduction

Cellular automata (CA) are dynamical systems which can generate and process patterns and information via lattice neighbor interactions [1–3]. Various biomolecular systems have lattice structures and participate in organized dynamical functions. These structures include membrane lipids and proteins, extracellular glycoproteins, DNA, and the cytoskeleton: microtrabecular lattice, protein filaments and microtubules (MT) [4–7] (fig. 1). Coherent dipole oscillations resulting in long-range cooperativity, as well as other nonlinear electrodynamic effects have been theorized to regulate activities in these structures [8–9]. Coherent oscillations among biolattice grid subunits could mediate information processing via CA-like activities. This paper considers the possibilities and potential significance of CA-like activities in a particular biolattice: cytoskeletal MT.

2. Coherent excitations in biological systems

Frohlich has theorized that non-linear dipole excitations among subunits of biomolecular arrays will result in coherent oscillations (10^{-10} to 10^{-11} s fluctuations), lower frequency metastable states, and long range order [8–9]. Frohlich also theorized that protein conformational states are coupled to electron dipole oscillations within protein hydrophobic regions [8]. Conformational coupling to coherent electronic excitation by individual sub-

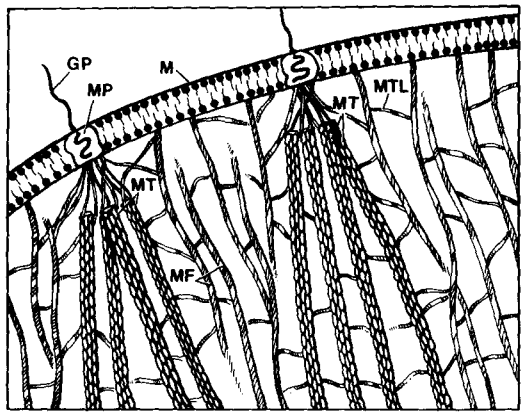

Fig. 1. Schematic representation of cytoskeletal lattice structures. M: cell membrane, MP: membrane protein (ion channel, receptor etc.), GP: glycoproteins which extend into extracellular space, MT: microtubules, MF: microfilaments including actin, intermediate, and neurofilaments, MTL: microtrabecular lattice. MT, MF, and MTL comprise the cytoskeletal lattice which interconnects and organizes cellular structures and activities [6].

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units in biomolecular arrays would apparently depend on various (genetic and environmental) biochemical factors as well as neighbor interactions. Neighbor interactions in a protein polymer can be mediated by factors which include electrostatic dipole interactions, water and calcium ion binding, conformational changes, binding of various substances including GTP and associated proteins, intrinsic subunit factors, or solitons [10, 11]. Such influences could alter the oscillatory phase of particular subunits in a coherent biomolecular array, resulting in dynamic patterns and CA-like activities.

Dynamic and stable patterns of excitation and conformational states in biomolecular structures could have direct, profound effects on regulation and organization of cellular processes. Patterns of resonant excitation and conformational states in various structures could mediate protein binding sites, ion fluxes, transport of biomolecules, growth, organization of cellular activities, and information processing. CA are thus theoretically possible and of potential significance in biological systems.

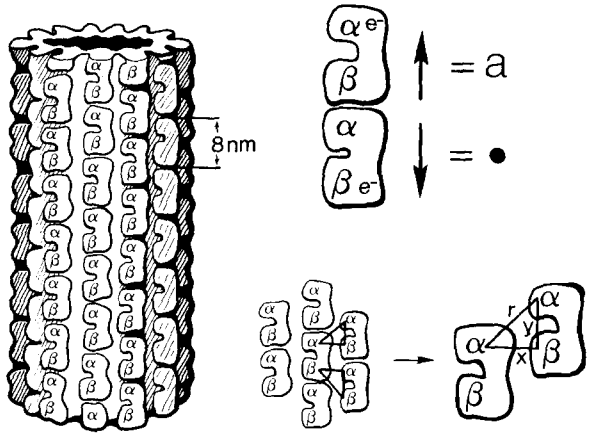

Fig. 2. Left: microtubule (MT) structure from X-ray diffraction crystallography [15]. Right top: MT tubulin dimer subunits comprised of α and β monomers. Electron occupancy of either monomer may correlate with coherent dipole oscillations and are represented [figs. 3–6] as "a" for α monomer electron occupancy and "." for β monomer electron occupancy. Center bottom and right: hexagonal relations lead to neighbor rules based on lattice distances. For neighbors shown (lower right), $x = 5$ nm, $y = 4$ nm, $r = 6.4$ nm.

3. Cytoskeletal lattices

Cytoskeletal lattices including protein polymers, microtubules (MT), actin, intermediate filaments, and the microtrabecular lattice interconnect and organize nearly all cell components and dynamic activities (fig. 1). Intracellular movement and coordination, growth, mitosis, transport and turnover of membrane proteins are accomplished by cytoskeletal lattices and attached proteins. MT are cylindrical polymers which comprise cilia, mitotic spindles, other organelles, and are the major cellular architectural elements [6, 12]. MT are intimately involved in dynamic biological activities, but mechanisms of "real time" regulation and control by MT or other structures are unknown [13]. The particular structure and geometry of MT will be considered as potential CA lattices.

Most MT are assemblies of 13 longitudinal protofilaments which are each a series of polar, tubulin dimers (fig. 2). A dimer consists of two slightly different 55,000 molecular weight monomers (α and β tubulin) which are each 4 nm in diameter. Hexagonal packing of the dimers results in 5 and 8 start helical patterns [14–16]. MT outer diameters are 25–30 nm and inner diameters are 14–15 nm [17]. MT outer surfaces are surrounded by a "clear zone" of several nm which apparently represents electronegative charges on tubulin [18]. Functional MT lengths may range from hundreds of nm to micrometers and meters in some mammalian neurons [19, 20]. Contractile or enzymatic proteins may be attached at specific dimer sites resulting in various helical patterns [16]. Various proteins are transported by MT at rates from one to 400 millimeters/day [21, 22]. Also binding reversibly at specific dimer sites are GTP molecules, hydrolysis of which can yield phosphate bond energies capable of oscillations or soliton formation [23]. Thus programming modes by genetic or environmental factors can introduce solitons or oscillatory phase shifts at specific lattice sites.

Table I

Neighbor position (fig. 2)	Central dimer α		Central dimer β	
	Neighbor α	Neighbor β	Neighbor α	Neighbor β
North	+0.063	+0.254	+0.028	+0.063
Northeast	+0.063	+0.007	+0.040	+0.063
Southeast	−0.061	−0.037	−0.007	−0.061
South	−0.063	−0.028	−0.254	−0.063
Southwest	−0.063	−0.040	−0.007	−0.063
Northwest	+0.061	+0.007	+0.037	+0.061

Configuration of tubulin dimer subunits may favor dipole oscillations between α and β states (fig. 2). In the context of Frohlich coherent excitation, various factors could modulate oscillatory phase of individual lattice subunits. (See section on Coherent excitations in biological systems.) The most predictable and consistent of these factors, electrostatic forces due to MT subunit neighbor dipoles, will be considered as the rules for CA activity in the structural geometry of MT lattices.

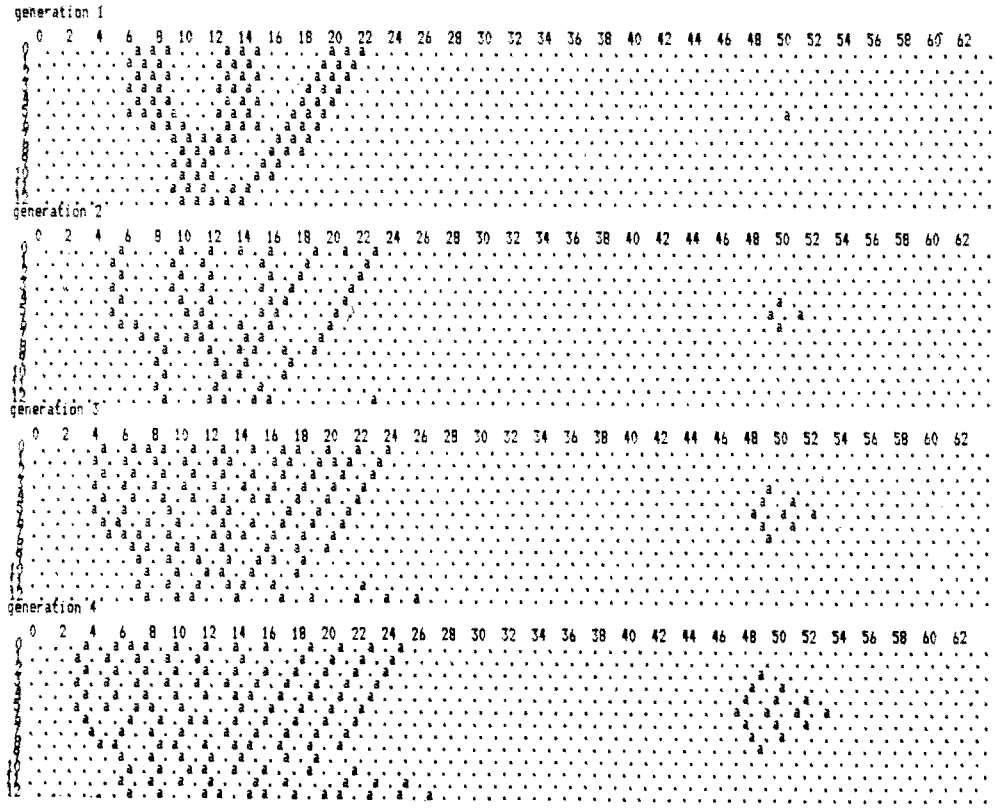

Fig. 3. Unfolded MT lattice with 13 protofilaments (0–12) on vertical axis, and "north" neighbors and dimer orientation on horizontal axis. An organized (arbitrary) pattern of α states starts on the left side and a single α state originates on the right. In generations 1–4 (generation numbers are on top of accompanying lattice) both the pattern and single α state develop and grow according to rules described in text.

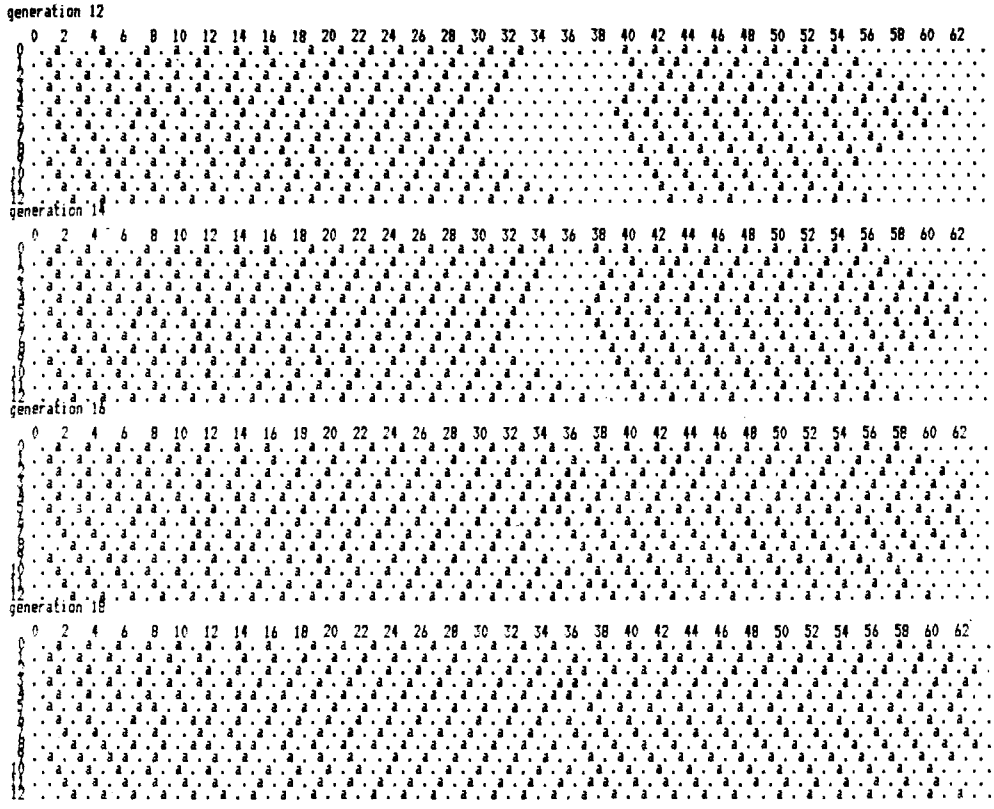

Fig. 4. Generations 12, 14, 16, and 18 of the same sequence as fig. 3. The two patterns have evolved and begin to interact.

4. CA rules for MT lattices

Frohlich's theory predicts 10^{-10} to 10^{-11} s coherent dipole oscillations of MT subunit dimers. If dipole polarities may exist in either the α and β orientation (fig. 2) then the state of each dimer at each oscillation may depend largely on neighbor dipole electrostatic forces.

Relative contributions of neighbor dipoles will consist of the y-component of the electrostatic force (along the axis of the dimers and protofilaments). The state of a dimer in the subsequent 10^{-10} s generation may thus be expressed:

$$\text{state} = \alpha, \quad \text{if } \sum_{i=1}^{n} f(y) > 0,$$

where n is the number of neighbors ($n = 7$);

$$\text{state} = \beta, \quad \text{if } \sum_{i=1}^{n} f(y) < 0,$$

$f(y)$ is the force from the "ith" neighbor in the y direction,

$$f(y) \propto \frac{\sin\theta}{r^2}, \quad \sin\theta = \frac{y}{r}; \quad f(y) \propto \frac{y}{r^3}.$$

x, y, and r are distances and θ the angle between neighboring monomers (fig. 2).

Thus states may be determined by $\Sigma_{i=1}^{n} y/r^3$ and relative values calculated from known MT structural geometry; see table I.

5. Computer simulation of CA patterns in MT lattices

Using the described rules, the following computer simulations were done. The cylindrical MT structures are represented as unfolded, horizontal ("North" neighbors are now "East") planes. α states are shown as "a", and β states by dots (figs. 2–6). The number of each successive generation indicating an oscillation period of 10^{-10} to 10^{-11} s is shown at the top of each lattice. In fig. 3, an organized pattern of "a" states starts on the left side, and a single, isolated "a" state starts on the right side. In generations 1–4 both the pattern and single state develop. Fig. 4 demonstrates generations 12, 14, 16 and 18 in which the 2 patterns begin to interact. In figs. 5 and 6, the β dots have been connected to enhance contrast. Fig. 5 demonstrates generations 22, 28, 34, and 40 from the same sequence. Even numbered stable patterns have evolved. The right side reveals a traveling pattern moving at maximum velocity (8 nm/nanosecond if based on 10^{-11} s oscillations) and leaving an altered wake. In fig. 6 generations 44, 46, 47, and 48 demonstrate the traveling pattern, which may be likened to a "glider" [3], Frohlich metastable state, or soliton reaching the end of the display. A single odd numbered generation (47) is a stable oscillator which includes a noticeable crossing pattern. Thus CA rules based

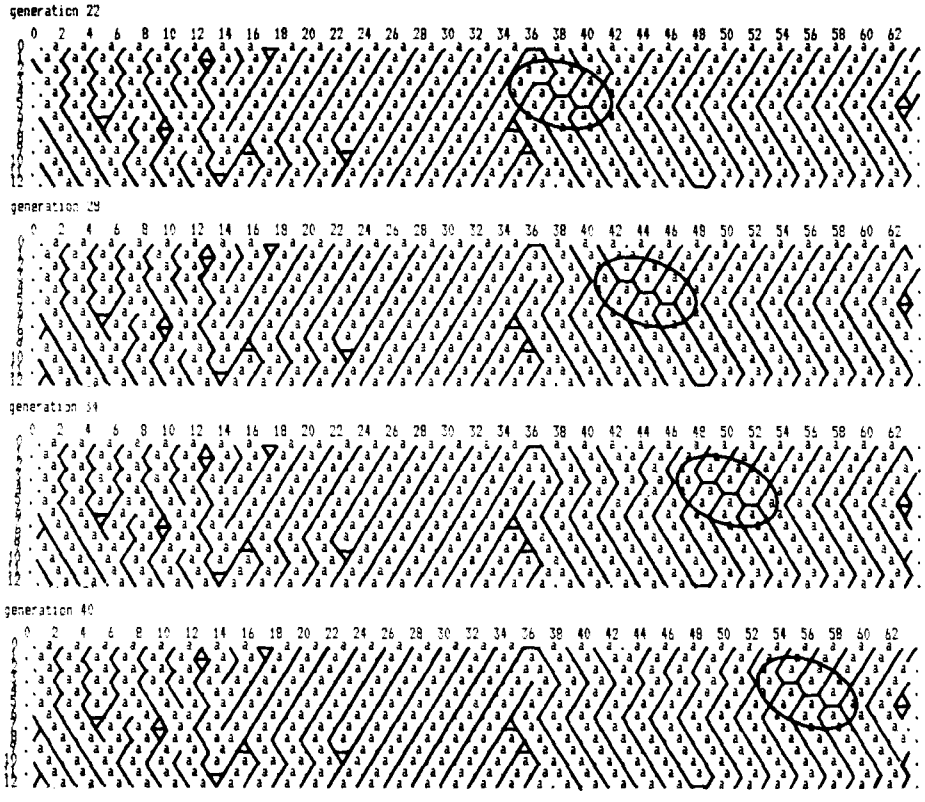

Fig. 5. Generations 22, 28, 34, and 40 from the same sequence. In figs. 5 and 6, β states represented by dots have been connected by lines to enhance contrast. Stable patterns have evolved (during even numbered generations) except for a traveling pattern which may be likened to a "glider", and the "glider's" altered "wake".

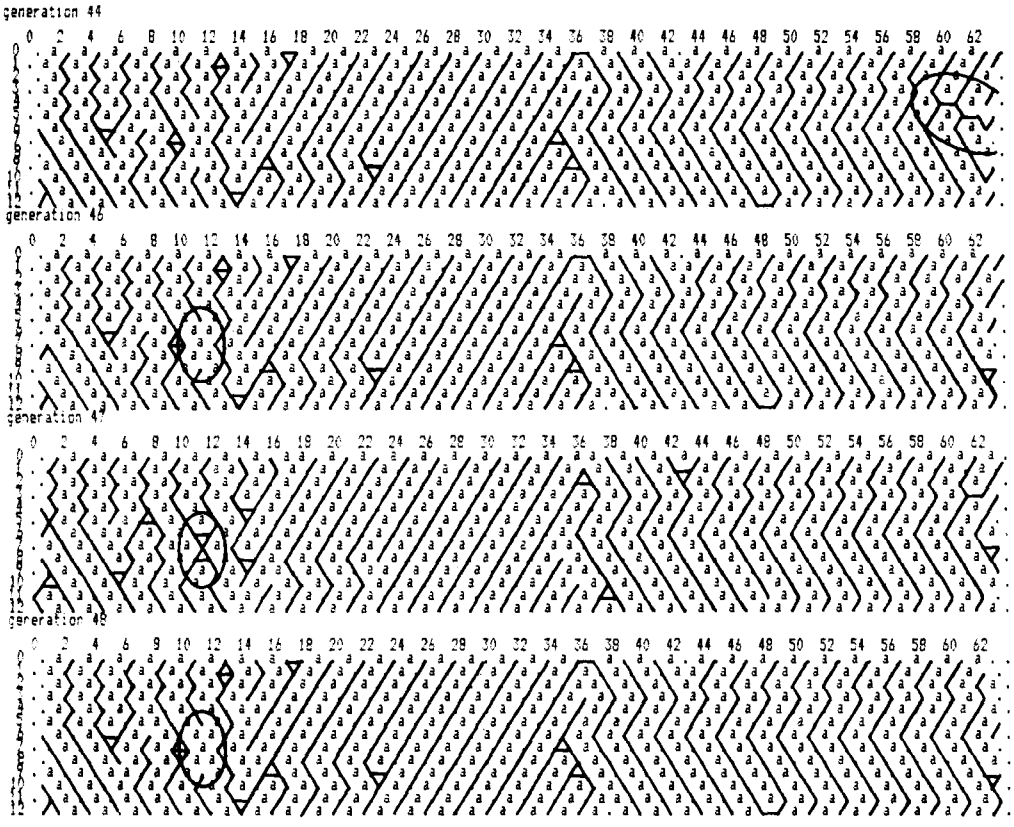

Fig. 6. Generations 44, 46, 47, and 48 demonstrate the "glider" leaving the display. Odd numbered generation 47 reveals a stable oscillator patterns which includes a noticeable crossing pattern.

on electrostatic dipole interactions in cytoskeletal geometries can result in stable patterns, oscillators, and traveling patterns ("gliders").

6. Conclusion

Stable and moving patterns of MT subunit oscillation phase could function as specific sites of protein binding and transport, orchestration of biomolecular activities, as well as stored information or memory. Interaction and interference of these patterns including "gliders" could process information as in computers. Although the CA rules and patterns described are somewhat arbitrary, they demonstrate that CA-like activities in cytoskeletal lattices are thus feasible and of potentially great importance.

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Ch'i: A Neural Hologram? Microtubules, Biohology, and Acupuncture

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Abstract: Structure, occurrence, and functions of microtubules are reviewed. The theoretical principles of biohology are outlined and the neuronal microtubular system is offered as an anatomical substrate for such a mechanism. This entails microtubules acting as coherency inducing, dielectric waveguides and resonators for atmospheric ultraviolet radiation which is refracted by the stratum corneum and other tissues. The proposed corporeal energy and its pathways are correlated with Ch'i, the ancient Chinese life energy and medium of acupuncture.

MICROTUBULES ARE CYLINDRICAL cytoplasmic inhabitants whose existence was unknown prior to the advent of the electron microscope. Even then they were only occasionally observed until 1963, when glutaraldehyde gained prominence as a fixative. Osmium tetroxide, the previously used agent is now known to commonly dissolve microtubules. Since then their presence has been noted in so many plant and animal cells that they are now considered universal cytoplasmic components(1,2).

Microtubules (MT's) are hollow cylinders and seem to be a uniform size although their diameters are variously listed from 200 to 270 angstroms (A). This range is thought to be due to variations among investigators since the measurement is usually constant from each laboratory over a wide range of cells and MT's studied. The walls of the cylinders are 50 to 70 A thick and they may be as long as a neuron. The walls seem to be composed of about 13 subfilaments which are either arrayed parallel or in a mild helix. These subfilaments appear to be strands of globular protein units, roughly 50 A x 60 A x 55 A. This molecular length coincides with observed striations at 80 A intervals in negatively stained samples, the slight pitch to the striations suggesting a helical arrangement of the subfilaments(3). The globular protein units ("tubulin") are a labile pool which may polymerize and dissolve repeatedly, even in vitro(4). Hemoglobin (A,S, or C) will form MT's under gel condi-

tions(5,6) so the labile pool may vary among tissues. MT's are dissolved by low temperature, increased hydrostatic pressure, and certain drugs, and may re-form upon return to prior conditions(7,8). Their origins and terminations are rarely seen but have been described as being embedded in a dendritic cortex or, if loose in the cytoplasm, capped by a dense blob(9). A second set of MT's, the neurofilaments, similar in structure but only 100 A in diameter is prevalent among MT's in neurons, and a third, smaller set, the "microfibrils" resides in astrocytes(10).

MT's may occur in several patterns. In cilia, flagella, and centrioles they are in couplets and arranged in a circle in the 9&2 or 9&0 formations that are familiar to all biologists. In mitosis, microtubular spindle fibers form between the centrioles and guide the chromosomes to their proper poles. They then dissolve, their constituent protein becoming available for subsequent MT formation(11). This may be to establish or support any cell asymmetry or to direct (in conjunction with a sol-gel transformation) a transient cytoplasmic rearrangement such as cyclosis or pseudopod formation(12). In permanently asymmetric cells, MT's seem to direct cytoplasmic flow and then to act as a cytoskeleton to maintain cell shape. This can occur longitudinally, as in axons, dendrites, glial cells, and spermatazoa, or circumferentially, as in platelets and discoidal nucleated erythrocytes. Muscular sarcotubules are morphologically indistinguishable from MT's.

How MT's function remains a mystery. It seems they give direction and guidance to cytoplasmic flow and they may participate in a propulsion mechanism. An associated contractile protein or ATPase substance has been found on several occasions(13, 14,15). In axons, peristaltic contractions of the overlying Schwann cells, or activity of extra-microtubular finger like projections have been credited with propelling axoplasm parallel to and among MT's(16). This flow achieves great velocity relative to the size of the organelles carried and the diameter of the axon. In this instance MT's may only be passive guides but they undoubtedly play a role in the development and orientation of the axon along distances up to meters. In megakaryocytes, MT's align to form cytoplasmic projections which break off, creating platelets. They then arrange circumferentially and participate in a spiral contraction during the second phase of platelet aggregation or clot retraction(17.) MT's have been observed outlining and establishing the cleavage furrow following mitosis, thus effecting daughter cell shape before cell division has concluded(18,19). Regardless of the motile force, MT's may be seen to orient, shape, and direct cytoplasmic movement and cell polarity, and subsequently to serve a cytoskeletal role. MT's are known to be strong and rigid enough to provide such support(20), but how they can function in a guidance capacity is as yet unanswered.

Perhaps MT's work like radar? Klystron tubes which are used in radar and other electronic devices utilize cavity resonators(21). These are geometric waveguides in which a specific electromagnetic wave may resonate and dwell, not unlike air in a tuned organ pipe. Cylinders are a particularly good shape for this purpose, and the radius of a cylindrical cavity resonator is related to the frequency of the resonant energy by the equation: $\gamma = 1.19\ c\ /\ 2\ \pi\ a$

where c is the *speed of light*, *a is the cavity* radius, and γ the resonant frequency(22). The function is independent of cylindrical length. In the radio-wave and micro-wave regions, a is in millimeters and cavity resonators (i.e., Klystron tubes)

are a convenient size. If a is set equal to the radius of microtubules, γ is on the order of 10^{16} cycles per second. This energy is in the far ultra-violet (UV) range, beyond the visible spectrum and the upper limit of the narrow window of solar and celestial nonionizing radiation that our atmosphere admits. The equation is given for a vacuum and an intra-microtubular dielectric (as well as skin and other intervening tissue) would alter the propagative properties according to the refractive index of that dielectric. The stratum corneum of the skin is known to have powerful refractile properties due to the abundance of birefringent keratin polymers(23). Thus the velocity and wave length of refracted penetrating radiation are decreased (24,25). Longer wave length energy readily available in our atmosphere (mid to near UV), upon attenuation and refraction, would be available to MT's, tuned to the new refracted wave length, in neurons and other cells. It would be difficult to measure the attenuation of light which is refracted subcutaneously to the specific resonant wave length of MT's (i.e., without constructing resonators the size of MT's). Permeating, or ambient light in the visible and UV ranges has been measured however, and there is some evidence of a physiological role for such energy.

Ganong and co-workers(26) measured extra ocular sunlight penetration into the mammalian brain. Stereotactically placed photoreceptors recorded intensities of 10^{-3} lumens in a sheep's hypothalamus when surface intensity was 0.4 lumens, with a logarithmic diminution. The most light permeable areas were lateral to the orbits, the temporal poles and hippocampus receiving maximum intensity. Light penetration remained constant for about 30 minutes after the animal's demise, at which time tissue opacity rose sharply. To the pineal gland has been attributed regulation of circadian rhythms, although no optic-pineal pathway has been demonstrated(27). (The pineal gland, along with retinal rod cells and hair cells of the inner ear are rich in "sensory" cilia, nine MT couplets with no central pair(28,29,30)). Cyclical, light dependent functions independent of the visual apparatus have been demonstrated in new born rats(31), and continuous photic stimulation by hypothalamic fiber optic implants have been shown to elicit neuroendocrine responses(32). Whether MT's act as inherent fiber optic waveguides remains to be seen.

Apparently analogous and homologous tubular dielectric waveguide structures have been well studied in the antenna sensillae of certain moths and provide a theoretical basis for electromagnetic communication among them, particularly between mating pairs(33). The waveguides are of a diameter considerably greater than MT's and the energy received and transmitted is consequently in the longer wave length infra-red and micro-wave regions. Similarly, EEG evoked responses have been obtained from a boa constrictor following infra-red and micro-wave stimulations, and potentially dangerous, internal human micro-wave sensors have been postulated(34).

The fate of the proposed intra-microtubular wave energy at synaptic junctions throughout the nervous system might include components of: transmission to post synaptic MT's (refracted to fit 100 A MT's ?), reflection to promote standing waves, and dissipation into surrounding tissue. Dissipation at MT terminals near synapses and along MT lengths would seem to be a physiological condition in surrounding tissue. A metabolic role might be surmised, and indeed, mitochondria have been observed aggregating at nodes of Ranvier(35). The physical properties of wave-

guides and resonators include absorption of kinetic energy in linear and angular fashions(36). MT orientation, and hence cell orientation can then theoretically be viewed as reflecting the flux in the resonant wave length (phototropism ?). Colchicine induced dissolution of MT mitotic spindles has reportedly been reversed with low intensity UV light(37). Aberration of endogenous resonant wave fields could retard spindle formation and could conceivably play a role in teratogenic or malignant change.

The observed striations at 80 A intervals along MT's correspond to one half of the resonant wave length. This could serve to foster coherent waves within MT's by diffraction. Assuming the 80 A intervals represent the globular protein units (as they are known to in hemoglobin gels) the waves which strike the dense protein would be reflected and remain in the tubule while those striking the gap areas would be dissipated into the cytoplasm. The intra-tubular wave energy, including reflection from MT endings would be "in phase" or coherent. (A well known technological source of coherent photic energy is the laser.) The dissipated cytoplasmic energy would also be coherent due to diffraction by the MT wall. Dissipated coherent energy from among various MT's would interfere in the cytoplasm. An interference pattern from among two or more coherent sources is, by definition, a hologram.

The 1971 Nobel prize in Physics was awarded to Dennis Gabor for his invention (in 1947) and development of holography. A laboratory curiosity at first, holography became important with the advent of the laser in the 1950's. Briefly, holography employs coherent light to record a three dimensional image of an object. One method utilizes a partially transmitting mirror by which a laser simultaneously illuminates a photographic plate directly, and an object which reflects the light onto the plate secondarily. The plate records an interference pattern between the two impingent wave trains. This pattern, the hologram looks like a garbled mess, but it contains all the information in three dimensions of the object expressed as the phase differences between the two impingent, originally coherent, wave trains. If the hologram is reilluminated by the laser source, the object appears due to cancellation of "non object"(38). A detailed treatment of holography is beyond the scope of this paper, but several properties of holograms are particularly interesting and may prove pertinent. One is that holograms have an enormous capacity to store information in a small area, although it is necessary to reilluminate that area to translate that information. A second property is that all the information in a given hologram is contained in any one, small discreet portion of it, although not as sharp or well focused as the entire hologram. These properties have prompted certain analogies:

> "Indeed it has been suggested that the human brain may store information in this way. It has always been a great puzzle how the vast amount of information stored in the human memory can be contained in the volume of the brain and the holographic method might well be the way it is done."
>
> Ronald Brown
> "Lasers: Tools of Modern Technology"(39)

> "For some years now this property of holograms has attracted the interest of neurophysiologists who were puzzled by the difficulty in locating the 'engram'

in the human or animal memory. As is well known, especially since the famous experiments of Lashley, large parts of the brain can be destroyed without wiping out a learned pattern of behavior. This has led to speculation that the brain may contain a holographic mechanism."

Dennis Gabor, 1969(40)

It has been suggested that bats, dolphins, and whales employ an ultra-sonic holographic system of information processing(41). Several scientists, including Dr. Gabor have created mathematical expressions for theoretical holographic systems of temporal recall(42,43,44,45). A psychiatrist, Dr. H.C. Tien has devised a holographic cybernetic model which relates consciousness, personality, time, memory, and acupuncture(46,47). Dr. Karl H. Pribram has proposed a model of brain function with two basic processes: spatially organized states and operations on that state by pulsed neural transmissions(48). He as well as Dr. Philip R. Westlake(49), suggest holography, expressed in Fourier transforms, to be the stratum of that spatially organized state.

But as yet this is all conjecture, for as Dr. Gabor has remarked:

"I am inclined to believe that there exists an abstract mathematical similarity but I am rather skeptical regarding the existence of waves or even of tuned resonators in the brain(50).

But we now know the neurons of the brain and nervous system are packed with MT's which are at least the right size, shape, and configuration to be tuned resonators. If such electromagnetic waves do exist within neuronal MT's, they would seem to be in equilibrium with, or at least affected by the state of surface membrane polarization and ionic flow. Neuronal transmission might then result in an increased dissipation of wave energy by a transient ionic effect upon MT structure (vice versa ?). This interaction fits well with the holographic models, the net coherent wave fields providing a spatially organized state within neuronal groups. Recent evidence has shown MT structure and function to be sensitive to lidocaine and the volatile anesthetics(51) and MT's "may be involved in impulse propagation in ways as yet completely unsuspected"(52).

Subtle magnetic fields have been detected outside the human head which fluctuate directly with alpha rhythms(53), and neuronal generated magnetic fields (cat sciatic nerve) have been measured and analyzed(54). The heart possess a magnetic vector (55,56) and electric fields (i.e., EEG, EKG) are well recognized. These are presently ascribed solely to ionic dipoles. Completely unexplained however, is the phenomenon of Kirlian photography(57,58). This technique utilizes high frequency (radio wave region) oscillating electric fields to visualize on film emanations from living organisms, as well as certain metals and crystals. Lensless photography may suggest a holographic mechanism and coherent interference may be the nature of these emanations. In human subjects, densest emanations of energy originate from acupuncture points(59).

The ancient Chinese described empirically a flow of energy through the body(60,61). The nature of the energy and the anatomical pathways through which

it flows have been unexplained, and consequently denied, by Western science. It is suggested here that Ch'i, the Chinese life energy is in fact interfering, coherent photic energy from the sun and stars which is refracted by the stratum corneum and resonant within microtubules. Tissue pathways of least resistivity to this energy have been described by the Chinese in terms of meridians. These appear to reflect embryological axes of development and, like the axes of cleavage and gastrulation(62) may be determined by MT's in the forms of neurons and non-neuronal cytoskeletal architecture. These seem to manifest primarily in the small, unmyelinated fibers mediating pain and autonomic function, both afferent and efferent. They may be embraced in a peripheral nerve, autonomic innervation of a blood vessel, or a tissue axis with no obvious neuronal axis. The relationship between the meridians and the organ systems they represent by the Chinese categorization lies presumably in the autonomic ganglia. Acupuncture therapy may be viewed as correcting an energy excess or deficiency in a given meridian system, perhaps by allowing equilibrium to be reached among the meridians or across the stratum corneum.

By holographic theory any arbitrary neuronal group, a spatially organized state, should contain the "engram" of the whole body, as measured by modulation of a pulsed function. This might hold diagnostic value and indeed, Chinese acupuncturists have claimed to do just this for thousands of years. Carefully palpating three fingerbreadths of the radial artery, subtle modulation of the pulse character by the neuronal group of autonomic innervation of this segment of the vessel is said to be diagnostic(63,54). What may be the fetal engram can be construed in the human outer ear (inverted) in which ectoderm, mesoderm, and endoderm analogues have distinct innervations(65). The many acupuncture points on the outer ear correspond to their respective body sites by this analogy(66).

Neuronal dissipation of the postulated energy might be dampened by myelin, a recognized insulator (i.e., saltatory conduction). Accordingly small unmyelinated fibers and unmyelinated ganglia would provide most efficient absorption and dissipation of energy and would be the mediators of acupuncture therapy. Myelin itself, however, might be of great significance. Myelin lamella formation, which seems to correlate with learning and development in any given fiber tract(67), consists of concentric spiral wrappings around a neuronal fiber of a lipoprotein blanket. The spacing between the lamellae is constant and corresponds to the intramicrotubular diameter. The insulating effect of myelin might include absorption and resonance into the spiral. A possible analogy is the conch shell, in which sound waves that were present during lamella formation may be reproduced later by a non specific wave. Memory, learning, and development may lie in the resonance of myelin lamellae.

Is there a microtubular forest in the neuronal tree? Did Gabor invent holography, or discover it? Physiology may follow engineering. It is reported that Casper and Klug derived their model of viral structure after studying Buckminster Fuller's geodesic domes(68). Is it that unlikely that we may holographically utilize the energy of the universe so directly, that we may, in fact be part and parcel of that energy? One need only study Eastern philosophy and Maxwell's equations to appreciate that man may have described electromagnetism no less than electromagnetism described man.

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Coherence in the Cytoskeleton: Implications for Biological Information Processing

S. R. HAMEROFF [1]

1 Introduction

Fröhlich's model of coherent excitations in biological systems can provide mechanisms for long-range order and cooperativity, factors useful for biomolecular communication and information processing. Fröhlich's coherence is based on oscillating dipoles in a voltage field, and has been applied primarily to biological membranes (Fröhlich 1970, 1975, 1986). Relatively overlooked, the interiors of living cells contain parallel networks of dynamic protein filamentous polymers which organize and regulate intracellular activities and whose properties strongly suggest communication and intelligence. These networks are collectively termed the cytoskeleton because they were originally thought to provide merely structural bone-like support to living cells. It is now recognized that the cytoskeleton is a dynamic information-processing system capable of organizing cell movement, division, growth and behavior. Thus the cytoskeleton may be described as the nervous system within all cells, ranging from single cell organisms like amoeba and paramecium, to nerve cells (neurons) within the human brain. In neurons, the cytoskeleton accounts for formation and rearrangements of neuronal form and synaptic connections, factors implicated in wide ranges of cognitive functions including neural networking, learning, memory, and consciousness. Components of the cytoskeleton (microtubules, actin and intermediate filaments, microtrabecular lattice) are oriented assemblies of "polar" subunits and have been accordingly described as electrets. The characteristics of polar electrets have been considered sufficient to support Fröhlich-type coherent excitations in the cytoskeleton (Del Giudice et al. 1983c). The presence of coherent excitations within the cytoskeleton would provide an intracellular dimension of molecular cognition. Several models of information representation and processing could take advantage of coherent "clocking" mechanisms in the cytoskeleton. These include parallel switching (neural network) matrices, reaction-diffusion patterns, cellular automata, nonlinear wave propagation, and coherent interference (holography). This chapter will review characteristics of the cytoskeleton, application of Fröhlich coherence mechanisms to cytoskeletal components, and theories of cytoskeletal information processing for which coherent oscillations may be important (Hameroff 1987).

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2 The Cytoskeleton

Living organisms are collective assemblies of cells which contain collective assemblies of organized material called protoplasm. In turn, protoplasm consists of membranes, organelles, nuclei and the bulk interior medium of living cells: cytoplasm. Dynamic rearrangements of cytoplasm within living cells account for their changing shape, repositioning of internal organelles, and in many cases, movement from one place to another. We now know that the cytoskeleton, a dynamic network of filamentous proteins, is responsible for cytoplasmic organization (Fig. 1).

2.1 The Nature of Cytoplasm

Nineteenth century microscopists recognized that cytoplasm had structural, contractile, motile, elastic and rigid properties which permitted elaborate and dynamic forms. They concluded that cytoplasm is not merely a liquid nor an emulsion nor an aqueous suspension of life-bearing granules. Rather, cytoplasmic architecture and contractility could be explained by the proposal of a mesh-like reticular or fibrous substructure whose interstices were filled with fluid (Burnside 1974). Biologists began to consider cytoplasm as a gel in which rod-shaped filaments formed cross-linkages. Gel aptly describes the mechanical properties of cytoplasm: an elastic intermeshing of linear crystalline units giving elasticity and rigidity to a fluid while allowing it to flow.

Development of the electron microscope through the 1960's initially did not illuminate the substructures of cytoplasm. Portions of cells which were optically empty by light microscopy persisted in being empty in electron micrographs and the cell was perceived by many to be a "bag of watery enzymes". Ironically, the fixative then used in electron microscopy, osmium tetroxide, had been dissolving filamentous elements so that their presence was observed only sporadically. Later, with the advent of glutaraldehyde fixation, delicate tubular structures were found to be present in virtually all cell types and they came to be called microtubules. Subsequent characterization of actin, intermediate filaments, microtrabecular lattice and the structure of centrioles led to the recognition that cell interiors were comprised of dynamic networks of connecting filaments and brought the cytoskeleton out of the closet.

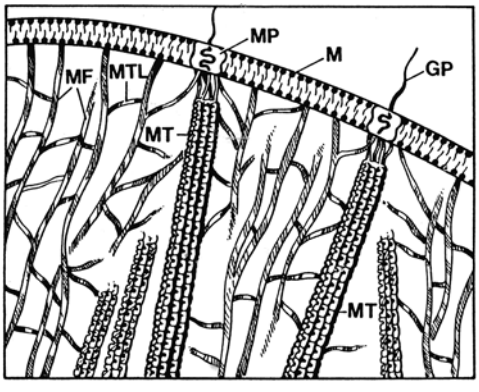

Fig. 1. Schematic of cellular cytoskeleton/membrane. *M* cell membrane; *MP* membrane protein; *GP* glycoprotein extending into extra-cellular space; *MT* microtubules; *MF* microfilaments (actin filaments or intermediate filaments); *MTL* microtrabecular lattice. Cytoskeletal proteins which connect MT and membrane proteins include spectrin, fodrin, ankyrin, and others. (Hameroff 1987)

2.2 Microtubules

Soifer (1986):

"When microtubules are required by a cell for a particular function, microtubules assemble in the appropriate part of the cell, with the necessary orientation. As microtubules are no longer needed, they depolymerize."

The most visible and widely studied cytoskeletal elements are microtubules (MT), slender cylinders intimately involved in the real time execution of important cell functions. Microtubules (MT) are hollow cylinders about 25 nanometers (1 nm = 10 Å = 10^{-9} m) in diameter whose walls are polymerized arrays of protein subunits. Their lengths may range from tens of nanometers during early assembly, to possibly meters in nerve axons within large animals. The protein subunits assemble in longitudinal strings called protofilaments; 13 parallel protofilaments laterally align to form the hollow tubules. The subunits are barbell-shaped proteins (dimers) which in turn consist of two globular proteins (monomers) known as alpha and beta tubulin (Fig. 2). Alpha and beta tubulin monomers are similar, but slightly different molecules with identical orientation within protofilaments and tubule walls. Each monomer consists of about 500 amino acids, weighs about 55 kilodaltons, and has a polarity or charge orientation. MT which grow from cell centers have a plus end (beta tubulin) which extends outward from the cell center (centrosome) into the cell periphery. The minus end (alpha tubulin) remains anchored to a microtubule organizing center (MTOC) within the centrosome. Each dimer, as well as each MT, appears to have an electrical polarity or dipole, with the negative end oriented towards the alpha monomer and cell center, and the positive end towards the beta monomer and cell periphery. The dimers are held together by relatively weak Van der Waals hydrophobic forces such as dipole coupling. Dimer neighbors form hexagonal lattices with a "leftward" tilt and several helical patterns occur in the relations among dimers (Dustin 1984).

MT from different life forms have marked similarities, but subtle differences. Comparison of MT from nerve cells of earthworms and mammals shows that the more

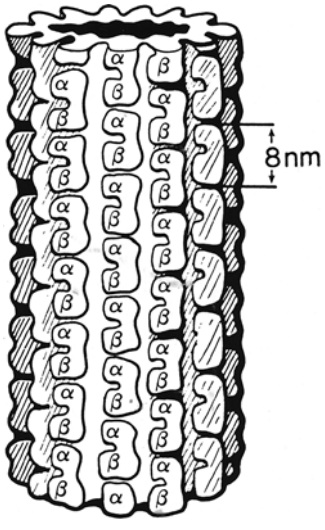

Fig. 2. Microtubules are cylinders whose walls are 13 protofilaments, each a string of 8-nm tubulin dimers. Alpha and beta tubulin monomers form the dimers. Each dimer has six neighbors. (Hameroff 1987)

primitive worm MT are more variable in geometric structure with MT ranging from 9 to 11 protofilaments, whereas mammalian MT are generally more consistent with 13. Tubulins from among different species including mammals and plants bind to common antibodies, and tubulins from different species may coassemble into hybrid MT. Despite these common traits, the diversity of tubulin gene expression has proved far greater than imagined years ago. Analysis of tubulin by amino acid sequencing and advanced electrophoretic techniques has shown that different alpha and beta tubulins exist concurrently, with the greatest diversity shown by beta tubulin. For example, Lee et al. (1986) have shown that as many as 11 different tubulin forms exist in rat thyroid microtubules and 17 different forms exist in rat brain microtubules. Thus alpha and beta tubulin are families of "isozymes," each of which may have specific functions or binding of microtubule-associated proteins (MAPs). Another tubulin variable, detyrosination, occurs in the cytoplasm subsequent to DNA transcription. Detyrosination is the removal of the terminal amino acid, tyrosine, from the polypeptide chain which comprises beta tubulin. Removal of tyrosine exposes an acidic amino acid, glutamate. Local factors in the cytoplasm beyond genetic programming determine whether or not individual tubulin subunits are "tyrosinated" or "glutamated." Thus both genetic and nongenetic coding mechanisms exist for the structure of tubulin subunits assembled within MT.

Since early electron microscopy studies, microtubules have been invariably described as being surrounded by a "clear zone" which gives the impression of a halo around them when they are viewed in cross-section. A 5-10 nanometer distance from the surface of MT is free of cytoplasmic ground substance or any other material normally seen elsewhere throughout the cell. Stebbings and Hunt (1982) have studied the "clear zone" and point out that MT surfaces are strongly "anionic" since tubulin is an acidic protein due to its high content of acidic amino acids such as glutamate and aspartate. These amino acids give up positively charged hydrogen ions to solution, leaving MT with excess electrons. Stebbings and Hunt propose that anionic, or electronegative fields at MT surfaces can explain the clear zones as well as the staining of MT by positively charged dyes, binding to MT of positively charged proteins, cations such as calcium, metals and other compounds. Electronegative fields surrounding MT may be heterogeneous or patterned due to detyrosinated subunits (exposing anionic glutamate), or other factors and act as excitable ionic charge layers (Debye layers) which are also thought to occur immediately adjacent to cell membranes (Green and Triffet 1985). Excitable clear zone charge layers and their ordered water and ions next to MT could facilitate collective communicative mechanisms occurring due to coherent excitations within the cytoskeleton.

The question of the hollow core within MT is even more mysterious; it too appears devoid of ground substance. It is unknown whether the interiors of MT are also electronegative zones, or perhaps positive ones which would create voltage gradients across MT walls. Del Giudice et al. (1986) have suggested electromagnetic focusing and possibly superconductivity within microtubule cores. Insulated from aqueous surroundings and held together by water-excluding hydrophobic forces, MT and the rest of the cytoskeleton comprise a solid state network within living cells.

What are the functions of microtubules? They do provide skeletal support, being the most rigid structures in most cells. To establish the pattern of the cytoskeleton

and the form and function of living cells, MT assemble from subunits at the proper time, place, and direction. Biochemical energy in the form of GTP plays an important role in MT assembly (and ATP for actin polymerization). Tubulin subunits bound to GTP add on to polymerizing MT; however, energy is not utilized via hydrolysis until after the GTP tubulin is incorporated. Phosphate bond hydrolysis energy from GTP and ATP may be providing energy to cytoskeletal structures for coherent excitations and/or lattice solitons, polarization waves etc.

MT are often anchored and guided by MT-organizing centers (MTOC) containing centrioles. Centrioles are cylinders comprised of nine MT triplets which exist in perpendicularly oriented pairs. Centrioles provide guidance and directional nagivation for nearly all cytoplasmic movements including cell division, or mitosis (Fig. 3). Once in place, MTOC and MT direct and participate in movement of cytoplasm, organelles and materials, growth, reproduction, synaptic plasticity, and nearly all examples of dynamic cytoplasmic activity.

The full range of MT function is achieved by the actions of various proteins which may bind in precise fashions at specific patterns of tubulin dimers in MT lattices. These *microtubule-associated proteins* (MAPs) include electromechanical enzymes which generate force and movement, communicative crossbridges to other cytoskeletal filaments and organelles, MAPs which enhance MT assembly and a class of MAPs within neurons whose functions are not understood. In many cases, the attachment patterns of MAPs to MT lattice walls have a precise geometrical configuration which appears related to the function of the MAP-MT complex (Fig. 4). Motion producing MAP "arms" which consume ATP hydrolysis energy to generate force are analogous to myosin bridges of skeletal muscle. Moving arms attached to MT are made of pro-

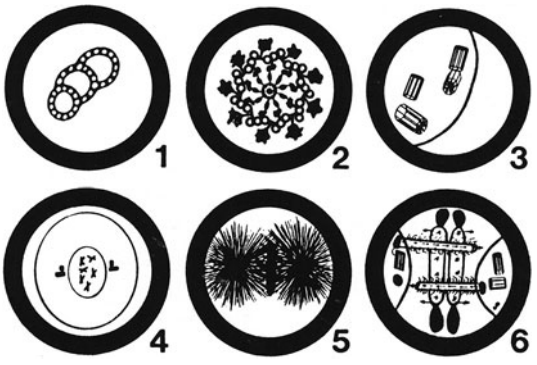

Fig. 3

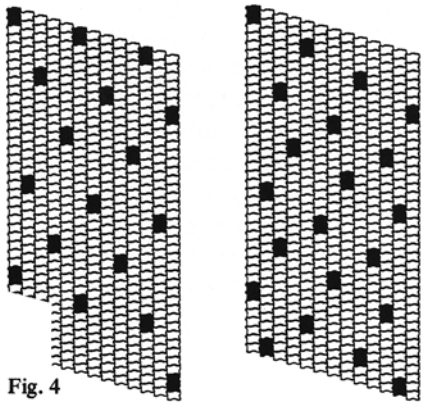

Fig. 4

Fig. 3. Centrioles in cell division. *1* Cross-section of centriole microtubule triplet. *2* Cross section of a centriole with 9 microtubule triplets, 9 satellite bodies, and central "pinwheel" structure. *3* Centriole pairs near cell nucleus, prior to cell division. *4* Centriole pairs have separated and migrated; chromosomes ready for separation. *5* Mitotic spindles, composed of microtubules, have formed from centriole centers (MTOCs) with chromosomes in middle. *6* Centriole anchored microtubules separating a pair of duplicate chromosomes. (Paul Jablonka in: Hameroff 1987)

Fig. 4. Patterns of MAP attachment to microtubules observed by electron microscopy. (Burns 1978; Kim et al. 1986; and others, from Hameroff 1987)

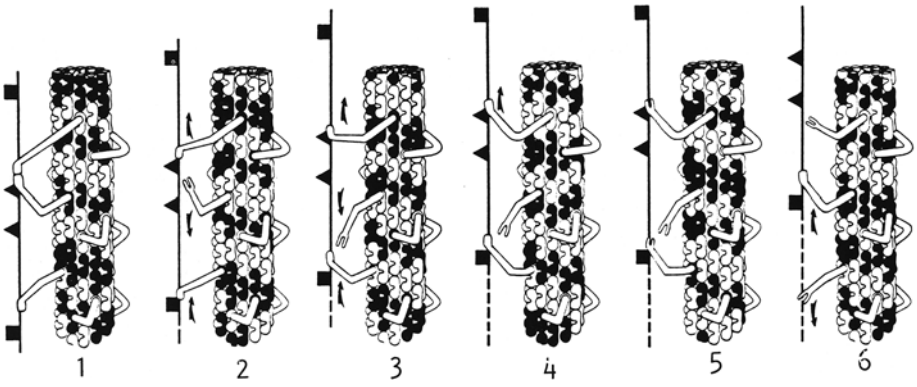

Fig. 5. Axoplasmic transport occurs by coordinated activities of microtubule attached sidearm, contractile proteins ("dynein") which cooperatively pass material in a "bucket brigade." The orchestration mechanism is unknown, but shown here as the consequence of signaling by "soliton" waves of microtubule subunit conformational states. By Fred Anderson

teins called kinesin and dynein. Dynein arms on MT which contract in organized sequences to produce collective movements were first described and characterized in cilia and flagella by (Gibbons 1968). The arms reside at periodic intervals along the outer MT within cilia and flagella and contract conformationally due to hydrolysis of ATP. Many MT-related activities generate force, locomotion and movement of vesicles and other material by a similar mechanism; axoplasmic transport is one well-studied example (Lasek 1981; Ochs 1982). Enzymes, neurotransmitter precursors, and organelles synthesized in neuronal cell bodies are transported along the lengths of axons by contractile dynein arms attached to axonal MT which act collectively to pass material in "bucket brigade" axoplasmic transport. The contractility mechanism is understood; however, the orchestration, signaling and coordination of the dynein MAPs appears to require some form of MT information processing (Fig. 5).

MT also participate in sensory perception of the cell's external environments. Many sensory receptors are modified cilia, assemblies of microtubules similar in structure to centrioles. Sensory transduction, centriole-based guidance and alignment, and axoplasmic transport are "intelligent" MT activities which are vital to biological growth, embryological development, secretion, synapse formation and many other important biological functions. MT are the scaffolding, conveyor belts, nagivators, and computers of living cells.

2.3 Intermediate Filaments, Actin and the Microtrabecular Lattice

The major filamentous components of the cytoskeleton are MT, actin filaments, intermediate filaments (IF) and a class of delicate interconnecting fibrils called the microtrabecular lattice (MTL). Actin is the most versatile component of the cytoskeleton. In conjunction with other proteins, it can polymerize in string-like filaments, form dynamic branching nanoscale meshworks or geodesic polyhedrons. Even more evanescent than labile MT, assembly of actin and associated proteins creates transient con-

figurations of cytoplasm for specific purposes. Dynamic cytoplasmic expressions such as contractile actin-based rings which divide the cytoplasm in cell division, probing amoeboid lamellipodia, and dendritic spines and synapses in neurons may be the roots of intelligence.

Interaction with a variety of other proteins (actin-binding proteins) unleash actin's full capabilities. When crosslinked by proteins such as fimbrin, actin can form rigid bundles which provide structural support. When associated with myosin (a mechano-enzyme), useful muscle-like contraction occurs. Also important are proteins such as talin, spectrin, vinculin, ankyrin and fodrin which connect the cytoskeleton with membrane proteins, and calmodulin which mediates effects of calcium ion fluxes on the cytoskeleton. Proteins such as alpha actinin, troponin, and filamin link actin in networks which cause a gelatinous consistency to cytoplasm – a "gel" (Fig. 6). In the presence of calcium ions and actin-fragmenting proteins such as villin and gelsolin, the gel network liquefies to a solution – "sol." Other actin regulatory proteins include actin capping proteins, which stabilize polymerized actin and promote a gel condition, and profilin, which binds actin subunits, prevents polymerization and maintains sol conditions (Satir 1984). Actin gel polymers, like microtubules, are structurally coherent assemblies which consume sufficient phosphate bond hydrolysis energy (ATP) to fuel coherent oscillations, depolarization waves or solitons-phenomena which would help explain the complex and dynamic functions of actin gels.

The "unknown" members of the cytoskeleton, intermediate filaments represent the most nebulous and chemically variable subgroup among the cytoskeleton. Lazarides (1980) has shown that different types associate with specific types of cells. For example, subunit structure defines five major classes of intermediate filaments: (1) keratin (tonofilaments), which are found in epithelial cells, (2) desmin filaments, predominantly found in smooth, skeletal and cardiac muscle cells, (3) vimentin filaments, found in mesenchymal cells, (4) neurofilaments, found in neurons, and (5) glial filaments, found in glial cells. Often two or more of these classes co-exist in the same cell.

MT and intermediate filaments are not the finest texture of cytoplasmic organization. Smaller, more delicate structures made of actin branch and interconnect in gel state networks which comprise the microtrabecular lattice, or ground substance of living material. Techniques in electron microscopy developed by Porter and colleagues at the University of Colorado (Porter and Tucker 1981) led to observation of an ir-

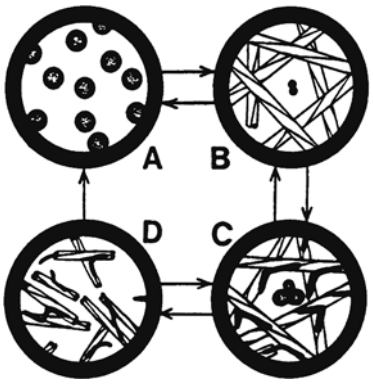

Fig. 6. Components of the cytomatrix. Assembly of actin and related proteins form cytoplasmic "gel" states. *A* monomeric actin; *B* assembled actin filaments which are helical pairs of monomeric strands; center-cross section of actin filament; *C* the presence of filamin (dark strands) causes actin cross-linking and dense "gel state;" *D* calcium and/or actin fragmenting proteins cause liquification to a "sol state." (By Paul Jablonka, from Hameroff 1987)

regular three-dimensional "trabecular" lattice of slender strands throughout the cytoplasm, interconnecting nearly everything in the cell. Slender strands of the microtrabecular lattice range in diameter from 4 to 10 nanometers with lengths of 10 to 100 nanometers and divide the ground substance of cytoplasm into two phases, a protein-rich polymerized phase and a water-rich fluid phase that fills the intratrabecular spaces.

MT, actin, intermediate filaments and the MTL comprise a dynamic cytoskeletal, solid state network within cytoplasm. Gershon et al. (1985) found that the cytoskeleton/MTL comprised from 16 to 21% of cytoplasmic volume with a surface area from 69,000 to 91,000 square microns per cell (69 to 91 billion nm^2). Biologist James Clegg (1981) has accrued evidence regarding the role of water surrounding cytoskeletal structures. Using neutron diffraction and other techniques, Clegg has found that water adjacent to the cytoskeleton is "ordered," that is aligned with polar bonds on the protein surfaces. Thus a layer of ordered water extends at least 3 nm from the billions of square nanometers of solid state surface area within each cell. This ordered water may be dipole coupled to coherent dynamics of the solid state, inhibit the thermal dissipation of protein oscillation energy from within the solid state, and shield calcium and other ions from random solid state interactions. Far from being "bags of watery enzymes," the interiors of living cell are highly ordered networks of dynamic, structurally coherent polymers.

3 Protein Conformational Dynamics

Proteins are the structural and organizational elements of the living state. Their essential functions are intrinsically linked to their structure and dynamic switching among different conformational states. Proteins such as enzymes, ion channels, receptors, cytoskeletal filaments, muscle myosin and hemoglobin undergo important conformational changes in response to a variety of stimuli. Coherent conformational states among cytoskeletal subunits may serve to represent information, exert control over routine biological functions and provide a basis for biomolecular communication.

3.1 Protein Conformation

Most protein molecules are able to shift reversibly between several different but related stable conformations due to several alternative sets of hydrogen bonds, disulfide bridges and hydrophobic Van der Waals forces of about equal energy among their constituent amino acid side chains. Thus step-like, nonlinear jumping occurs between specific conformations so that only a discrete number of alternative conformations exist for a given protein, and random switching from conformation to conformation is limited. Fröhlich (1975) proposed that dipole oscillations within water-excluding hydrophobic protein pockets were coupled to the protein's conformational state.

Different frequencies of conformational changes coexist cooperatively in the same protein (Table 1). The vast majority of processes of biological interest are in the time scale greater than 10^{-9} s, or one ns, which also lies in the collective mode realm for protein dynamics (Karplus and McCammon 1983). Protein conformational changes in the nanosecond time frame are coupled to stimuli by three fundamental features:

Table 1. Protein conformational dynamics – motions of globular proteins at physiological temperature. (After Karplus and McCammon 1983; from Hameroff 1987)

Amplitude of motions	0.001 to 10 nm
Energy	0.1 to 100 kcal
Time range	10^{-15} s (fs) to 10^{3} s (many minutes)
Time for collective functional steps	10^{-9} s (ns)
Types of motion	Local atom fluctuations
	Side chain oscillations
	Displacements of loops, arms, helices, domains and subunits
	Collective elastic body modes, coupled atom fluctuations, solitons and other nonlinear motions, coherent excitations

hydrophobic interactions, charge redistribution and hydrogen bonding. Appropriate stimuli including neurotransmitters, voltage alterations, hormones, calcium and other ions, and enzyme substrates can induce conformational changes within specific proteins (Fig. 7). Conformational transduction is the sensitive link in the mechanism of anesthetic gas molecules, and indicates that cognitive functions relating to consciousness depend in some way on protein conformational regulation. Dynamic conformational changes of proteins are the dynamics of living organisms.

3.2 Hydrophobic Interactions

The polymerization of cytoskeletal polymers and other biomolecules appears to flow against the tide of order proceeding to disorder dictated by the second law of thermodynamics. This apparent paradox is explained by the hydrophobic forces which bind protein assemblies including cytoskeletal polymers and which exclude ordered water from their protein sites. Thus the net effect is order proceeding to disorder because of the hydrophobic exclusion and disordering of previously ordered water.

The attractive forces which bind hydrophobic groups are called Van der Waals forces after the Dutch chemist who described them in 1873 (Baron 1961). At that time it had been experimentally observed that gas molecules failed to follow behavior predicted by the "ideal gas laws" regarding pressure, temperature and volume relationships. Van der Waals attributed this deviation to the volume occupied by the gas molecules and by attractive forces among the gas molecules. These same attractive forces are vital to the assembly of organic molecules, including protein assemblies and can explain cooperativity and coherent interaction among polymer subunits.

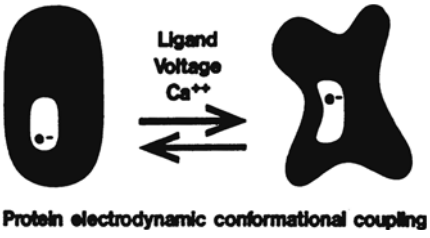

Fig. 7. Protein switching between two different conformational states induced by binding of ligand, calcium ion, or voltage change. One mechanism proposed by Fröhlich is that dipole oscillations (e^-) within hydrophobic regions of proteins may be a trigger, or switch for the entire protein. (Hameroff 1987)

Hydrophobic Van der Waals forces which are essential to the assembly and function of important biomolecules consist of dipole-dipole attraction, induction effect, and London dispersion forces.

Dipole-dipole attractions occur among molecules with permanent dipole moments. Alignments in which attractive, low energy arrangements occur are favored and net attraction between two polar molecules can result if their dipoles are properly configured. The induction effect occurs when a permanent dipole in one molecule can polarize and distort electrons in a nearby molecule so that dipole interactions are attractive. London dispersion forces explain why all molecules, even those without intrinsic dipoles, attract each other. Recognized by F. London in 1930 (Barrow 1961), the effect depends on quantum mechanical motion of electrons. Electrons in atoms without permanent dipole moments (and "shared" electrons in molecules) have, on the average, a zero dipole. However, dynamic instantaneous dipoles can be recognized which can induce dipoles in neighboring polarizable atoms or molecules. The strength of London forces can be significant only when two or more atoms or molecules are very close together (Barrow 1961). By dynamically coupling dipoles and conformational states among neighboring proteins, Van der Waals hydrophobic interactions can result in cooperativity and coherent oscillations of protein assemblies.

3.3 Electret, Piezo, and Pyroelectric Effects

Assemblies whose microscopic subunits and macroscopic whole both possess permanent electric dipoles are known as electrets. They exhibit properties known as piezoelectricity, which may help explain biological activities. In these crystals, dipolar elementary subunits are arranged in such a way that all the positive dipole ends point in one direction and all the negative dipole ends are oriented in the opposite direction. In microtubules, the positive ends of tubulin dimer subunits point away from microtubule organizing centers (MTOC) toward the cell periphery, and the negative ends point towards MTOC. Electrets can store charge and polarization and have now been identified in a variety of nonbiological materials such as ionic crystals, molecular solids, polymers, glasses, ice, liquid crystals and ceramics. Biological tissues demonstrating electret properties include bone, blood vessel wall materials, keratin, cellulose, collagen, gelatin, artificial polypeptides, keratin, DNA, cellulose and microtubules (Athenstaedt 1974). An electret effect accounts for specific properties such as anti-blood clotting in biomaterials and the non-stickiness of Teflon. Sources of polarization or charge storage in macromolecules are dipoles, ionic space charges, or ordered surface water.

Electret materials are piezoelectric (Gubkin and Sovokin 1960). Piezoelectric materials change their shape or conformation in response to electrical stimuli, and change their electrical state in response to mechanical stimuli. Koppenol (1980) has shown that the dipole moment orientation of an enzyme changes in concert with its functional activity. Electrets are also pyroelectric, in that any change in temperature alters the electrical and conformational characteristics of the molecule. The requisites for such behavior are permanent dipole moments in molecules bound in parallel alignment. Microtubules and other cytoskeletal structures appear to be appropriately configured electret, pyroelectric, piezoelectric devices.

The electret state within bone has been well studied and is able to store large amounts of polarization of the order of 10^{-8} coulombs/cm^2 (Mascarenhas 1974, 1975). The limit of charge separation (equivalent to maximal information density) has been calculated (Gutman 1986) to be about 10^{17} electronic charges/cm^3, while there may be about 10^{21} total molecules/cm^3. One cm^3 of densely arrayed parallel microtubules 100 nm apart contains about 10^{17} tubulin subunits and therefore may contain 10^{17} dipoles equivalent to the maximal density of charge! Electret and related properties can impart interesting and potentially useful coherent properties to biomolecules including cytoskeletal polymers.

3.4 Coherent Excitations (Fröhlich)

Proteins are vibrant, dynamic structures in physiological conditions. A variety of recent techniques (nuclear magnetic resonance, X-ray diffraction, fluorescence depolarization, infrared spectroscopy, Raman and Brillouin laser scattering) have shown that proteins and their component parts undergo conformational motions over a range of time scales from femtoseconds (10^{-15} s) to many minutes. The most significant conformational vibrations are suggested by Harvard's Karplus and McCammon (1983) to be in the middle of this range: nanoseconds. Such fluctuations are appropriate for conformational motion of globular proteins (4 to 10 nm diameter), consistent with enzymatic reaction rates and coupled modes like solitons.

Collective cooperativity of dynamic conformational states have been elegantly woven in a theory of coherent protein excitations by Professor Herbert Fröhlich, who has come to several profound conclusions. One is that changes in protein conformation in the nanosecond time scale may be triggered by a charge redistribution such as a dipole oscillation within hydrophobic regions of proteins (Fröhlich 1975). Another Fröhlich (1970) concept is that a set of proteins connected in a common voltage gradient field (membrane or polymer electret) would oscillate coherently at nanosecond periodicity if energy such as biochemical ATP or GTP hydrolysis were supplied. Far-reaching biological consequences may be expected from such coherent excitations, particularly when applied to the cytoskeleton.

Biomolecular excited states with very high dipole moments may, according to Fröhlich, become stabilized (metastable states) through internal and external deformations and through displacement of counter ions like calcium. Metastable states, which can correlate with functional conformations, are thus collective effects involving the molecule and its surroundings. A molecule might be lifted into a metastable state through the action of electric fields, binding of ligands or neurotransmitters, or effects of neighbor proteins. Thus rapid, nanosecond oscillations may become "locked" in specific modes which correspond to useful conformations of a protein. For example, an ion channel, receptor, enzyme, or tubulin subunit may stay in metastable conformational states for relatively long periods, on the order of milliseconds.

Expanding on Fröhlich's work, Wu and Austin (1978) concluded that coherently oscillating dipoles within a narrow band of resonance frequencies with large enough coupling constants may be expected to cause strong long-range (about 1 μ) attractive forces among dipoles. In a dense microtubule array, 1 μ^3 (one billion nm^3) would encompass about 160,000 tubulin subunits – an array sufficiently large for collective effects.

In a further extension of Fröhlich's work, a group of scientists from the University of Milan (Del Giudice et al. 1986) have viewed living matter as a sea of electric dipoles. The Milan group sees order in living systems to be induced by reduction of tridimensional symmetry to a rotational alignment along filamentous electrets such as cytoskeletal structures. In their view, quantum field theory and the Goldstone theorem require that the symmetry breaking (Bose condensation) results in long range interactions among system components (dipoles) conveyed by massless particle/waves (Goldstone bosons). The Milan group argues that the energy required to generate massless bosons is invested in the electret states of biomolecules and correlated fluctuations of their surrounding water and ions.

Celaschi and Mascarenhas (1977) showed that electret activation energy of biomolecules (0.2-0.4 eV) is equivalent to the hydrolysis of one ATP or GTP molecule and what Soviet scientist A.S. Davydov predicted for initiation of biological solitons. Consequently solitons, massless bosons, and Fröhlich's coherent polarization waves may be synonymous.

Pursuing their quantum field approach, the Milan group arrived at an astounding concept of self-focusing of electromagnetic energy within cytoskeletal filaments. Electromagnetic energy exceeding a threshold originating within, or penetrating into, cytoplasm would be confined inside filaments whose diameters depended on the original symmetry breaking (Bose condensation) of ordered dipoles. Any electric disturbance produced by thermally fluctuating dipoles or by any other source would be confined inside filamentous regions whose diameter depends on the polarization density, or ordering of biological water. The Milan group calculated a self focusing diameter of about 15 nm (precisely the inner diameter of microtubules) and argues the cytoskeleton is the material consequence of dynamic self-focusing of polarization waves in the cytoplasm. The Milan group concludes that focusing occurs in cytoplasm of living cells due to the spatial coherence and ordering imparted by cytoskeletal electret behavior.

4 Coherence and Models of Cytoskeletal Information Processing

Cooperative, collective effects of coherent protein conformational oscillations are a likely substrate for biological intelligence ranging from cytoplasmic probing to human consciousness. The activities, functions, and structures of microtubules and other cytoskeletal components have led at least a dozen author groups to publish theoretical models of information processing within MT and the cytoskeleton. The concepts include passive MT signal transduction, descriptive patterns among MT subunit states, dynamic cooperative automaton effects among coherent oscillations of microtubule subunits, and cytoplasmic/cytoskeletal sol-gel field effects utilizing holographic imagery. Coherent oscillations of cytoskeletal protein subunits such as tubulin are relevant to several of these communicative modes.

Experimental findings which support the notion of cytoskeletal communication include electrical signaling along microtubules (Vassilev et al. 1985), fluorescence resonance transfer among MT subunits and membranes (Becker et al. 1975), parallel alignment of MT in applied electric and magnetic fields (Vassilev et al. 1982) and in-

volvement of intraneuronal MT in nerve membrane excitability and synaptic transmission (Matsumoto and Sakai 1979; Alvarez and Ramirez 1979). The following models of cytoskeletal information processing are consistent with these findings and would be strengthened by coherent protein oscillations.

4.1 MT Sensory Transduction (Atema)

Biologist Jelle Atema (1973) proposed that sensory cilia transduced environmental information to the rest of the cell by propagated conformational changes in the microtubule subunits which constituted these cilia. Atema reviewed conformational changes in MT tubulin subunit dimers observed by a number of authors, and concluded that sequences of subunit conformational changes are likely to occur in microtubules coupled to wavelike or whiplike motions of cilia and flagella. Atema's microtubule theory assumed that distortion of tubulin conformation by any number of sources was sufficient to propagate a conformational wave. Thus light energy, chemical bond energy, and mechanical forces could be transduced by sensory cilia.

Atema's view of MT information processing was an all-or-none propagation of conformational changes along tubulin protofilaments. His was the first theory to look beyond the global behavior of cilia to consider conformational effects in tubulin components. Subsequent theories became more elaborate to consider localized analog functions, switching, and collective neighbor interactions among MT subunits.

4.2 MT Mechano-Ionic Transducers (Moran and Varela)

Expanding on Atema's suggestion, Harvard biologists Moran and Varela (1971) suggested that compression or bending of MT would cause release and flux of bound ions from MT subunits (16 calcium ions per tubulin). Moran and Varela saw these ionic currents capable of generating membrane depolarization much like the coordinated release of calcium ion waves by muscle cell sarcoplasmic reticulum triggers actin-myosin contractile shortening and like the calcium waves thought to regulate bending and waving of cilia and flagella as well as calcium wave coupling to sol-gel state in the cytoskeletal ground substance. Moran and Varela's contribution was to observe that MT could release ions such as calcium in a controlled and modifiable manner useful for intracellular communication.

4.3 Cytomolecular Computing (Conrad and Liberman)

Wayne State University computer scientist Michael Conrad and Soviet information scientist E.A. Liberman (Conrad and Liberman 1982) have viewed a molecular analog within the cytoskeleton as a representation of the external world.

Conrad (1985) notes that

"highly parallel signal processing and vibratory behavior on the part of microtubules and other cytoskeletal elements could play a significant role".

4.4 MT Signal Processing (DeBrabander)

Cell biologist Marc DeBrabander and colleagues (Geuens et al. 1987) have developed a method of labeling individual tubulin subunits within MT which are either tyrosinated or glutamated. Tyrosine, the terminal amino acid in tubulin, may be removed from polymerized MT to expose the more acidic glutamate as the new terminal amino acid. Thus all tubulins within polymerized MT are either tyrosinated, or glutamated which may serve as a convenient programming or memory function. DeBrabander and colleagues developed a double labeling technique in which immunogold particles bind to tubulin subunits. Large immunogold particles (10 nm) identify glutamated tubulin and smaller particles (5 nm) bind to tyrosinated tubulin. Patterns of tyrosinated/glutamated tubulin within MT show heterogenous distributions, and suggest the potential for a coding mechanism. MAP attachment, tubulin conformation, calcium binding and other factors could be coupled to MT function via such patterns. DeBrabander and colleagues have provided the first direct evidence of modifiable patterns of tubulin variability in intact microtubules. Consequently, MT appear capable of signal processing in addition to signal transduction.

4.5 Cytoskeletal String Processors (Barnett)

Brooklyn College computer and information scientist Michael P. Barnett has pursued the design of computer components suited to molecular scale devices and artificial intelligence (AI). Unlike most AI researchers, Barnett scoured the subcellular biological realm in search of molecular scale information processing concepts and "discovered" cytoskeletal microtubules and neurofilaments! He has proposed information representation as patterns in the subunits of cytoskeletal polymer subunits which may operate like information strings analogous to word processors.

In Barnett's conceptualization (Fig. 8), information strings move from right to left along processing channels which run parallel to one-dimensional memory channels in which character strings can be stored. String transformers can perform global replacements on sequences of characters like common word-processors. Barnett's model assumes the existence of processing channels (MT) along which strings of information can move, and memory channels (neurofilaments) which consist of a succession of locations, each of which can hold a single character. Parallel array and lateral interconnectedness of MT and neurofilaments could qualify these cytoskeletal elements as string processors, assuming that information may be represented in the polymers. Barnett's model is thus compatible and complementary with other models of conformational patterns within MT and the cytoskeleton. The actual implementation of Barnett's string processing would be greatly facilitated by clocking, or coherent oscillations of subunits in both MT and neurofilaments.

4.6 Microtubule "Gradions" (Roth, Pihlaja, Shigenaka)

Roth et al. (1970), and Roth and Pihlaja (1977) have considered information processing in biomolecular assemblies. Citing cooperative effects among adjacent proteins or protein subunits, they proposed that conformational gradients in microtubules and other protein arrays represent information by patterns of conformational states (gradions) among near neighbors in protein lattices.

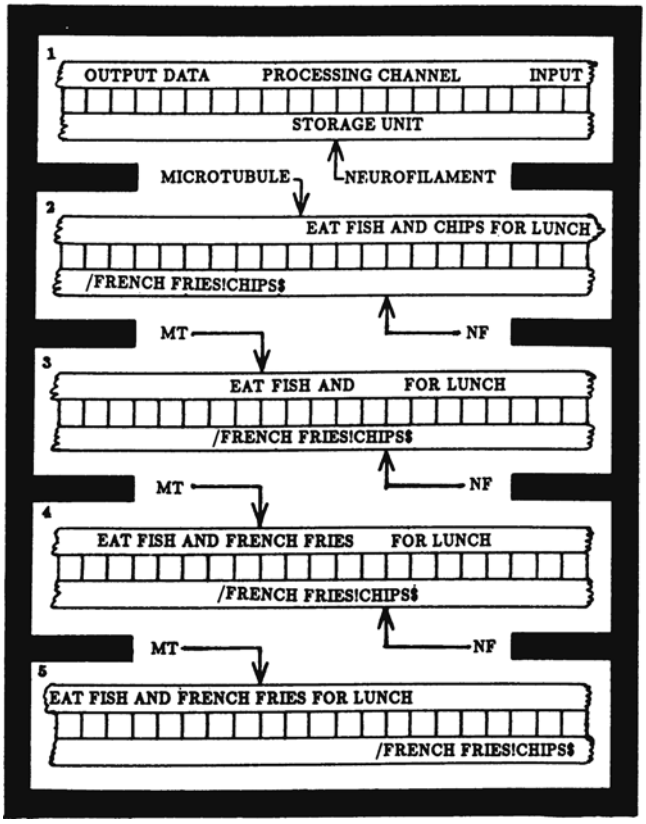

Fig. 8. Barnett's string processor model of microtubule-neurofilament memory storage. *1* a processing channel (microtubule-MT) is laterally connected to a parallel storage unit (neurofilament-NF); *2* information traverses the processing channel from right to left; on the storage channel, synonym pairs move from left to right; *3* as the word "chips" moves by its own self in storage, it is deleted; *4* "chips" is replaced by its synonym "french fries", and *5* the replacement is complete. (By Paul Jablonka, from Hameroff 1987)

Conformational states of individual tubulin subunits are thought to be controlled by factors which include ligand induced conformation, tubulin isozyme, detyrosination, or binding of MAPs including inter-microtubule bridges. Discrete gradion fields within microtubule surfaces were defined as the neighborhood areas around inter-microtubule MAP bridges. Roth, Pihlaja and Shigenaka assumed five possible conformational states for each dimer determined by inter-MT linkage sites and their cooperative allosteric effects, as well as dimer binding by a number of different native substrates or foreign molecules. The pattern of tubulin dimer conformational states in a region of MT lattice governed by attachment of inter-MT linkages (or other MAPs) is defined as a gradion, or conformational field which includes about seventeen dimers (Fig. 9). Consequently, 5^{17} different gradion patterns could exist if the conformation of each subunit were independent of all others. Such deterministic patterns would be generally useful when coupled to coherent dynamics in the cytoskeleton.

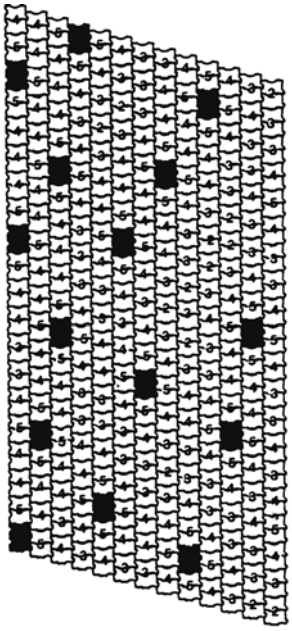

Fig. 9. "Gradions" within microtubule lattice. Dark shaded dimers are MAP attachment patterns; numbered dimers are determined by proximity to MAP attachment sites. MAP binding and other "gradionators" are thought to induce coded patterns representing information. (By Paul Jablonka from Roth 1970)

4.7 Gyroscopic Centrioles (Bornens)

French molecular biologist Michel Bornens (1979) argues for a dynamic stability based on organizational properties of centrioles (Fig. 3) and the cytoskeleton. Specifically, Bornens proposes that centrioles are animated by rapid oscillatory rotation about their longitudinal axis, which results in a dynamic stability and inertia analogous to a spinning top or gyroscope.

The rate of rotation required for gyroscopic inertia has been calculated as 2.3 million revolutions per minute (Albrecht-Buehler 1981). To account for such rapid activity, Bornens suggests a submicroscopic mechanism "allowing more independence of the centriole with respect to surrounding material." Factors which could support this contention include ordered water coupled to centriolar oscillation, some unknown property of the "pericentriolar material," ionic charge layer, or even superconductivity as suggested by Fröhlich and Del Giudice's group. Rapid rotatory oscillations in protein assemblies would be consistent with the findings of Michels et al. (1985), who found collective energy absorption in the range of 10^7 per second by protein assemblies such as virus coats. Bornens also suggests that ATP-generated centriolar rotation triggers propagating impulses along microtubules by transitory contact/stimulus of the nine rotating MT doublet/triplets in the centriole wall with their surrounding satellite bodies. This would lead to coherent rhythmic signals throughout the cytoskeleton with a frequency nine times greater than that of the centriole's rotation. Occurrence of coherent, rhythmic signals throughout the cytoskeleton would be mechanisms close to the nature of life itself.

4.8 Centriole-MT Signaling (Albrecht-Buehler)

Northwestern University biologist Guenter Albrecht-Buehler has considered the general question of intelligence in cytoplasm as well as two specific models of cytoskeletal information processing: centriole signal detection, and propagation of MT impulses.

Albrecht-Buehler (1985) has described a mechanism for impulse propagation along microtubules. He considers that each MT protofilament is a chain of alpha-beta tubulin dimers: AB,AB,AB,...,AB. Within the wall of a microtubule, each monomer is in contact with other monomers of the same protofilament and with those of adjacent protofilaments. Each monomer is consequently subject to attractive Van der Waals forces from surrounding tubulin monomers which hold together the protofilaments and MT cylinders. Albrecht-Buehler proposed that each of these interactions must weaken the A-B dimer bond; consequently the wall of a microtubule exists in a "state of resonance as to the relative strengths of the intermonomeric bonds. For example, (A-B)(A-B)...(A-B)(A-B) could resonate with (A)(B-A)(B-A)...(B-A)(B). Such a resonating chain could propagate information at close to the speed of light".

Albrecht-Buehler is suggesting a coherent communicative resonance among protein conformational states within MT assemblies.

4.9 Dynamic Tensegrity (Heidemann and Jarosch)

Buckminister Fuller proposed an interesting architecture he called tensegrity, constructed from an assembly of tension and compression members (Fuller 1975). The compression members of solid struts are isolated from each other, held together by the tension members. Joshi et al. (1985) have shown that cytoplasm has both compressive and tensile elements. Semi-rigid microtubules are under compression, presumably due to tension generated by actin filaments and actin based microtrabecular lattice. In general, MT do not contact each other so that the self-supporting capability of cytoplasm may indeed stem from tensegrity.

Robert Jarosch (1986a,b) has published a series of papers describing actin-MT interactions which suggest that (1) contractile actin filaments are spirally wound around microtubules, (2) coordinated contraction of the actin filaments imparts a rotational torque to MT, somewhat like a spinning top, (3) actin filaments wound in opposite directions on the same MT can cause rotational oscillations of the MT. These two models fit together to provide a picture of a dynamic cytoplasmic tensegrity network in which the cytoskeletal elements may be coherently twisting back and forth. Perturbation of any part of such a tensegrity network could have dynamic consequences throughout its domain. Transient changes in tension, compression, or oscillatory rhythm caused by a variety of factors would be transmitted and possibly amplified throughout the cytoskeleton.

4.10 Dynamic MT Probing (Kirschner and Mitchison)

Strong evidence supports the concept of dynamic instability in microtubule assembly. Kirschner and Mitchison (1986) ask: "how can a peripheral clue lead to reorganization deep within a cell?" One possibility is that a signal is relayed to the microtubule-organizing center leading to a change in its structure, orientation, and directed nuclea-

tion of microtubules. This would be consistent with a primary role for information integration and decision-making within the MTOC. A simpler non-hierarchical idea is that a signal at the periphery affects distribution directly. Since the whole cytoskeletal array is very dynamic, it would only be necessary to stabilize or destabilize a particular subset of microtubules for the entire cytoskeleton to rapidly transform.

4.11 Sphere Packing Screw Symmetry (Koruga)

There are 32 possible symmetry arrangements of packed spheres in a cylindrical crystal. Erickson (1973) used hexagonal packing of protein monomers to explain the form and patterns of viruses, flagella and microtubules. Djuro Koruga (1986) of the University of Belgrade's Molecular Machines Research Unit has analyzed the symmetry laws which describe cylindrical sphere packing and the structure of microtubules.

Koruga (1986):

"The sixfold symmetry and dimer configuration lead to screw symmetry on the cylinder: a domain may repeat by translocating it in a spiral fashion on the cylinder. From coding theory, the symmetry laws of tubulin subunits suggest that 13 protofilaments are optimal for the best known binary error correcting codes with 64 code words. Symmetry theory further suggests that a code must contain about 24 monomer subunits or 12 dimers".

Koruga's symmetry arguments may be compared with the gradion concept of Roth, Pihlaja, and Shigenaka in which a field of about 17 monomers is thought to represent a basic information unit. Koruga concludes that microtubule symmetry and coherent structure are optimal for information processing.

4.12 Cytoskeletal Self-Focusing (Del Giudice)

Starting from Fröhlich's model of coherent excitations, Del Giudice and colleagues (Del Giudice et al. 1983a,b, 1986; Del Giudice 1986) have used quantum field theory to describe the electret state of biological systems (ordered water surrounding linear biomolecules) and determined that there exists a strong likelihood for the propagation of particle-like waves in biomolecules. Further, the ordering of water should lead to self-focusing of electromagnetic energy into filamentous beams excluded by the ordered symmetry. For ordered cytoplasm, they calculate the diameter for the confinement and propagation of particle-like waves (massless bosons, polarization waves, or solitons) in biomolecules to be about 15 nanometers, exactly the inner diameter of microtubules.

Cytoskeletal polymers may thus be capable of capturing and utilizing ambient or biologically generated electromagnetic energy. One possible example is infrared energy which is routinely generated by dipoles in biological molecules. This energy is generally believed to be dissipated into heat within the aqueous cytoplasm, however self-focusing within a solid-state cytoskeleton could utilize this energy productively in a communicative medium.

4.13 MT Automata, Holography (Hameroff, Watt, Smith)

The self-focusing of electromagnetic energy described by the Milan group is thought to occur by an electret induced increase in the refractive index of cytoplasm. A similar concept was proposed (Hameroff 1974) in which microtubules were thought to act like dielectric waveguides for electromagnetic photons. It was also proposed that the periodic array of MT subunits leaked or diffracted energy with 8 nanometer periodicity, resulting in a source of coherent energy (or calcium ions) from MT. Cytoplasmic interference of the coherent sources from among multiple MT could lead to holographic imaging in cytoplasm. Coupling of calcium concentrations to cytoplasmic sol-gel states could "hardwire" holographic interference patterns into the microtrabecular lattice. In parallel arrays of MT within nerve fibers, graded potentials or traveling action potentials were suggested to activate planes of cytoplasm perpendicular to the long axis of the MT and nerve fibers (Fig. 10). These traveling planes may be likened to image screens as in TV sets. In a TV picture tube, the screen is motionless and electron beams move to create a pattern by their intersection with the screen. Perhaps imaging within neurons occurs on travelling screens generated by action potentials moving through parallel MT arrays. The content of such images would depend on programming mechanisms in the conformation of tubulin subunits which comprise the MT walls, and which would update with each successive action potential. Hameroff and Watt (1982) described a method of MT tubulin programming in which charge carriers (calcium ions, electrosolitons) or conformational waves such as phonons or solitons were steered through MT lattices by genetically or cytoplasmically programmed tubulins and specific MAP-binding sites.

Hameroff et al. (1984, 1986) have utilized principles of cellular automata to explain information processing in MT. Cellular automata are dynamical systems which can generate and process patterns and information, and are capable of computing. Cellular automata require a lattice structure of like neighbors with discrete states and neighbor rules, and a universal clock to which all neighbors are timed. Adopting Fröhlich's model of coherent nanosecond dipole oscillations coupled to conformational

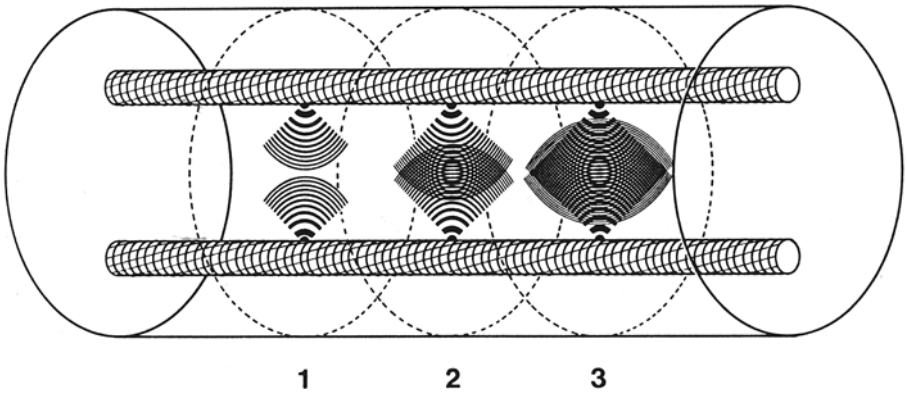

Fig. 10. Interference patterns in cytoplasm caused by coherent waves (e.g. Ca^{2+}, sol-gel state, MTL) generated by dynamic activities in microtubules may be a basis for holographic information imagery. (Hameroff 1987)

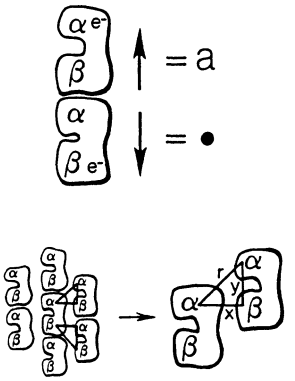

Fig. 11. *Top* MT-tubulin dimer subunits comprised of α and β monomers. Relative electron occupancy of either monomer may correlate with coherent dipole oscillations in the nanosecond time domain. *Bottom* screw symmetry hexagonal packing leads to unequal neighbor rules based on lattice distances and Van der Waals dipole interactions. For dimers shown at right, x = 5 nm, y = 4 nm, r = 6.4 nm. The relative strength of each neighbor dimer dipole state is y/r^3

states as a clocking mechanism, the authors calculated MT lattice neighbor Van der Waals dipole interactions as neighbor rules for an MT-automaton computer simulation. Each tubulin dimer was considered in one of two possible states at each nanosecond generation. The two states were related to Fröhlich's concept of dipole oscillation so that the dipole can be oriented either toward the alpha tubulin end (a, Fig. 11) or toward the beta tubulin end (represented by a dot in Fig. 11). The polarity and electret behavior of MT indicate that in the resting state, tubulin dimer dipoles should be oriented toward the beta monomer.

Dimer states at each "clock tick," or generation were determined by neighbor states at the previous generation.

$$\text{state} = \alpha, \text{ if } \sum_{i=1}^{n} f(y) > 0, \text{ state} = \beta, \text{ if } \sum_{i=1}^{n} f(y) < 0 ,$$

where n = 7 as the number of neighbors, and f(y) the force from the *i*th neighbor in the y direction.

$$f(y) \propto \frac{\sin\theta}{r^2}, \sin\theta = \frac{y}{r}; f(y) \propto \frac{y}{r^3} .$$

The dipole state of any particular dimer at each clock tick thus depends on the summation of the dimer's neighbor dipole states (including its own) at the previous clock tick. The neighbor influences are unequal because of the screw symmetry of the MT lattice. Distant dimers (more than one neighbor away) would be expected to have little influence because of the dropoff in force intensity (y/r^3) with distance. However, collective influences from many like oriented dimers could augment long-range cooperativity. Using only near neighbor influences, computer simulation of an MT automaton yields interesting patterns and behavior of dipole/conformational states. These include both stable and traveling interactive patterns capable of computing and regulation of cytoskeletal activities. For example, Fig. 12 shows such patterns traveling through MT lattices, leaving altered wakes, or memory. Assuming nanosecond generations, these traveling patterns would travel at 8 nm per ns (8 m/s), a velocity consistent with propagating action potentials, or solitons. Variability in individual tubulin dimer isozymes, ligand binding, or MAP attachments could program and read out information in routine cellular functions. As one example, propagation

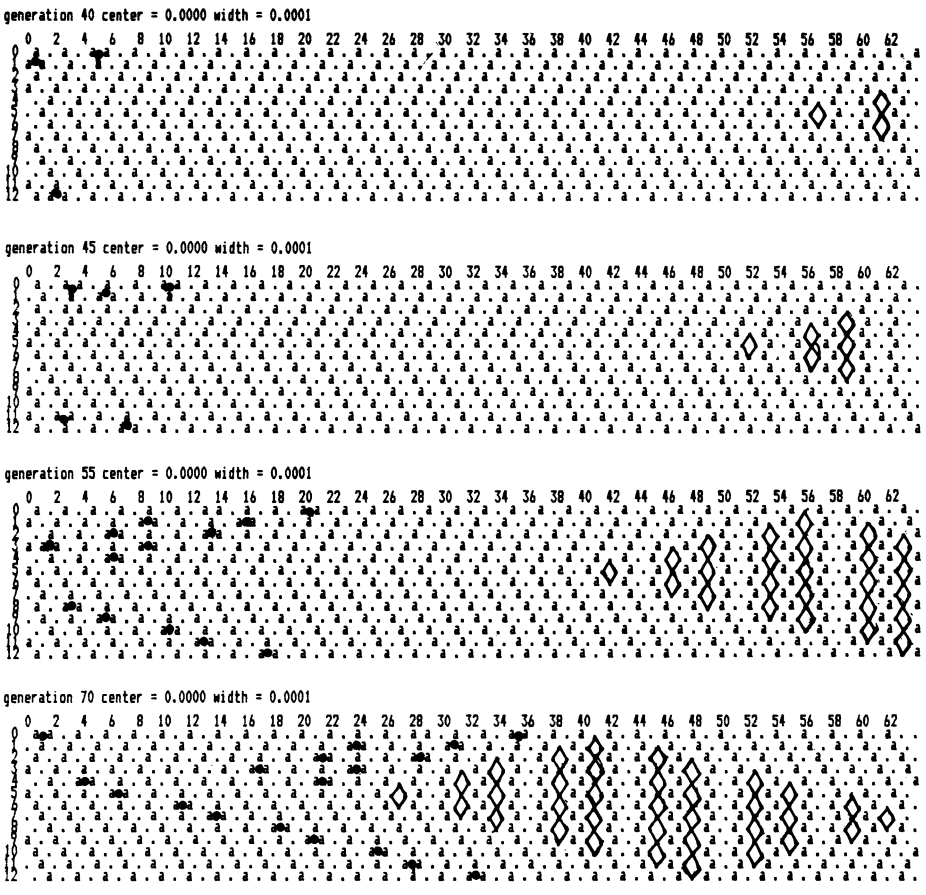

Fig. 12. 30 "nanoseconds" of computer simulation of MT cellular automaton based on Fröhlich coherent oscillations and tubulin subunit neighbor rules based on dipole-dipole interactions. Thirteen protofilaments are arranged horizontally, 64 rows are shown vertically yielding a flat MT lattice. "Alpha" states of tubulin (Fig. 11) are shown by "a," beta states by a *dot*. Four contiguous beta states are highlighted by *diamond patterns*. Four contiguous alpha states by *darkened dots*. The stimulation sequence (generations 40, 45, 55 and 70) shows movement of patterns at velocity of 8 nm/ns (8 m/s). (Hameroff et al. 1984)

of MT conformational patterns could coordinate the activities of contractile MAP sidearms in axoplasmic transport (Fig. 5). In a general sense, MT automata may be the information substrate for biological activities ranging from ciliary bending to human consciousness. Specific automata patterns distributed throughout wide volumes of cytoskeletal arrays within the brain could lead to cooperative resonance and collective effects resulting in images, thoughts, or ideas similar to coherence-induced phase transitions in certain metals yielding emergent collective properties such as superconductivity.

5 The Cytoskeletal Connection

The current prevalent model of brain function and artificial intelligence is the neural network, a collective dynamical system whose output is determined by the states of component neuronal synapses. In turn, the state of each synapse is determined by other neurons, modulated by the interneuronal cytoskeletons. In turn, the cytoskeleton is a collective dynamical system whose net activity is determined by the states of its component subunits. Their states, in turn, are determined by another level of organization including cytoplasmic factors, ordered water and ions, genetic isozymes of individual subunits, other cytoskeletal elements such as the microtrabecular lattice, and cooperative effects – coherence – from among collective assemblies of subunits. Cognitive processes which have been ascribed to neural net concepts may be more accurately interpreted as fractal hierarchies of dynamical systems which are highly parallel, highly interconnected, and of increasing capacity as they become more microscopic. Coherent oscillations of these subunits would permit collective effects to occur at each level and at successively more macroscopic levels with a net effect of consciousness.

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Coherent Electric Vibrations in Biological Systems and the Cancer Problem

H. FRÖHLICH

Abstract—**The existence of coherent electric vibrations in the 5×10^{10}–3×10^{12} Hz region has now been established experimentally for biologically active microorganisms. It is shown that such vibrations might play a decisive role in the control of growth of tissues and, hence, in the cancer problem.**

I. Survey

SEVERAL YEARS AGO, it had been proposed [1] that electric vibrations with frequencies of the order of 10^{11}–10^{12} Hz should be excited coherently in active biological materials through metabolic processes. The given frequency range is to be considered as a very rough order of magnitude. It thus borders closely on the now accessible microwave range. Excitations of the proposed type could have far-reaching biological consequences, for they would lead to selective long-range interactions which probably are required to control growth in normal tissue and whose absence might lead to cancer.

Verification of the theoretical conjecture posed considerable experimental difficulties. During recent months, however, conclusive experimental evidence for their existence has been obtained in bacteria [2], [3] and in yeast cultures [4]. This latter followed and extended earlier work [5] on the action of coherent millimeter waves on biological systems for which it was found difficult to obtain detailed experimental information.

Existence of such extraordinary physical properties in biological systems must be the result of long evolutionary processes. It may be hoped, therefore, that such vibrations exist not only in the monocellular systems for which their existence has been verified, but also in higher systems and particularly in tissues.

While the stimulus for this work has arisen from theoretical considerations, it must be emphasized that the role of the theory must first be the introduction of relevant concepts—coherent electric vibrations in the present case. Model calculations are useful (Section IV) in that they may further specify such concepts. More detailed predictions, however, require experimental corroboration. For, in contrast to the physics of materials where, from their structure, many properties can be derived theoretically through the theory of linear response, biological systems are largely characterized through nonlinear response.

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II. The Model and Its General Consequences

Membranes of biological systems usually have a thickness of about 10^{-6} cm across which an electric potential difference of the order of 100 mV is maintained. An enormous electric field of the order of 10^5 V/cm thus acts on the membrane which, therefore, is strongly polarized electrically. The only generally accepted role of this field refers to nerve conduction, but membranes of other than nerve cells also maintain such a field. A possible role of such fields may refer to proteins dissolved in a membrane which, under the influence of the field, may undergo configurational changes, particularly if they possess metastable states with high electric dipole moments as has been discussed on other occasions [6]–[9].

For the present purpose, it will be noted that a section of the membrane oscillating (with displacement) perpendicular to the surfaces represents an electric wave with half-wavelength of the width of the membrane, and (in view of its polarization) an oscillating electric dipole. Assuming elastic properties corresponding to a velocity of sound of 10^5 cm/s this yields a frequency of one-half 10^{11} Hz. Proteins built into the membrane and possessing a high dipole moment may, of course, also oscillate, probably with higher frequency. Sections of giant molecules such as DNA or larger complexes might oscillate with lower or higher frequencies. All this assumes that the damping of such oscillations is not unduly high. In each such case, harmonics which no longer represent dipolar oscillations—with appropriately higher frequencies—should exist.

Oscillating polar systems of this type interact with each other, and their resulting oscillations can usually be presented in terms of normal modes. It will now be assumed that the lowest of these modes is excited very strongly (coherently) through the supply of energy from metabolic processes. In general, this will involve nonlinear features. A mathematical treatment for a possible model will be presented in Section IV, but much more complex features than these considered may be at work. The existence of such vibrations, however, will have consequences which

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are far reaching and largely independent of the manner in which they are excited, provided they involve oscillating electric dipoles.

The most important of these consequences refers to a long-range attractive selective interaction [8], [10]. It should be noticed first that, in biological systems, static charges are usually screened at relatively short distances in contrast to electric vibrations in the frequency region of 10^{11} Hz which have a range much larger than the dimensions of a cell. Consider now two such oscillating systems with frequencies ω_1 and ω_2, respectively, at a distance R, large compared with their linear dimensions. In this case, dipolar terms in the interaction only need be considered, leading to normal mode frequencies $\omega_\pm$

$$\omega_\pm^2 = \frac{1}{2}(\omega_1^2+\omega_2^2)\left[1 \pm \left\{\left(\frac{\omega_1^2-\omega_2^2}{\omega_1^2+\omega_2^2}\right)^2 + \frac{M^2}{R^6}\right\}^{1/2}\right]. \quad (1)$$

Here, for given displacements of the oscillators, M/R^3 measures the ratio of the interaction energy between the dipoles and of their individual potential energies. Thus as long as the { } bracket is small compared to unity,

$$\omega_\pm = \left(\frac{\omega_1^2+\omega_2^2}{2}\right)^{1/2}\left[1 \pm \frac{1}{2}\left\{\left(\frac{\omega_1^2-\omega_2^2}{\omega_1^2+\omega_2^2}\right)^2 + \frac{M^2}{R^6}\right\}^{1/2} - \frac{1}{8}\left\{\left(\frac{\omega_1^2-\omega_2^2}{\omega_1^2+\omega_2^2}\right)^2 + \frac{M^2}{R^6}\right\} + \cdots\right]. \quad (2)$$

The energy of interaction between the two systems is now obtained by subtracting from the free energy at distance R the free energy as $R \to \infty$. In thermal equilibrium, both ω_+ and ω_- contribute, and the result is the well-known van der Waals interaction $\alpha 1/R^6$, whether or not $\omega_1 = \omega_2$. In our case, when ω_- is very strongly excited, say with $n_- \gg 1$ quanta, the thus defined interaction energy I becomes (let $\omega_1 \geqslant \omega_2$)

$$-I \simeq \hbar n_- \left(\frac{\omega 2_1+\omega_2^2}{2}\right)^{1/2}\left[\frac{1}{2}\left\{\left(\frac{\omega_1^2-\omega_2^2}{\omega_1^2+\omega_2^2}\right)^2 + \frac{M^2}{R^6}\right\}^{1/2} - \frac{1}{2}\frac{\omega_1^2-\omega_2^2}{\omega_1^2+\omega_2^2} - \frac{1}{8}\frac{M^2}{R^6} + \cdots\right]. \quad (3)$$

When $\omega_1 \simeq \omega_2$, an attractive interaction $\alpha 1/R^3$ thus arises in contrast to the case $\omega_1 \neq \omega_2$ when the attraction is again $\alpha 1/R^6$ at large distances.

Furthermore, since the magnitude of the interaction depends on the strength of excitation, such interaction will be switched on or off depending on whether or not metabolic energy is supplied. This might apply to the attraction of an enzyme and a substrate at certain stages of cell development [6].

The case of a whole assembly of such pumped equal systems which should be of particular interest for the control of tissue growth has not yet been treated in detail but general theorems concerning normal modes do, of course, apply although such theorems usually refer to regular arrangements of the individual oscillating systems, frequently though not necessarily with nearest neighbor interaction only. In our case, e.g., in an assembly of cells pumped metabolically to oscillate with equal frequency, interaction with more distant neighbors will be relevant and, in the lowest normal mode, each individual oscillator will exhibit a definite phase relation to the phase of the mode so as to make the total (negative) interaction energy a minimum. This will cause certain contractions and in this way, will very likely involve nonlinear terms. The total interaction energy may thus be expected to become size and shape dependent.

An individual cell in the neighborhood of this assembly may then be expected to be attracted to it much more strongly when its frequency is the same as that of the assembly than when it is different. Assuming that this frequency is different for differently differentiated cells [11], this would explain the great attraction that equally differentiated cells exert on each other.

The possibility of switching on or off the collective mode through metabolic processes might give rise to changes in the configuration of larger regions such as required prior to cell division. The supply of metabolic energy, in turn, would have to be regulated so as to be available at the correct time. A whole temporal succession would thus arise.

Models for such processes could be treated mathematically in some detail. At the present stage of development, it seemed to be more important to find conclusive experimental proof for the existence of the conjectured vibrations.

III. Experimental Evidence

If vibrational states with frequencies of the order one-half 10^{11}–10^{12} Hz (or higher) are strongly excited by metabolic processes, then corresponding Raman lines should be found when the system is metabolically active but not when it is nonactive. The excitation, moreover, should be much stronger than thermal excitation.

An alternative approach would be to attempt interfering with the biological processes by exciting the systems from outside.

A. Raman Effect

Careful investigation of the Raman spectrum of an enzyme (lysozyme) in aqueous solutions yields broad bands in the region of 100 cm^{-1}(3×10^{12} Hz) [12]. Such broadening indeed should be expected from the high friction with water.

The Raman spectrum of resting (not metabolizing) bacterial cells was found to be of a similar nature [2]. If synchronized cells (i.e., they are all in the same stage of development) are supplied with an appropriate nutrient (which on its own also shows a broadband), then a number of relatively sharp strong Raman lines appear in a

range between 7×10^{10} and 5×10^{12} Hz. They thus fall into the expected membrane frequency region and may well be due to the predicted excitations. In principle, however, they might also arise from intermediate chemical products which for unknown reasons are not subject to the friction with water. The crucial experiment consists in the measurement of the ratio R of the intensities of anti-Stokes and Stokes lines. If the lines (frequency ν) arise from an intermediate chemical, then this ratio should be

$$R_0=\exp(-h\nu/kT) \tag{4}$$

for their excitation would, in a very short time, reach the thermal equilibrium value (temperature $T\simeq300$ K). If the excitation arises from metabolic pumping, however, and is very large compared with thermal excitation, then $R_0\simeq1$ must be expected.

Experiments on active synchronized E coli B bacteria conclusively proved the existence of strong excitations [3]. A Raman line which, during the life cycle of the bacteria, shifts from 118 to 125 cm^{-1} (3.75×10^{12} Hz) showed $R_0=1.01\pm0.09$ for 118 cm^{-1} and $R_0=0.93\pm0.13$ for 125 cm^{-1}, while the respective thermal equilibrium values (3.1) are $R_0=0.57$ and $R_0=0.55$. When the thermal value of R_0 is larger, such tests would, of course, require much higher accuracy.

Experiments to date showed that a multiplet near 40 cm^{-1} appeared during a short period in the cell cycle only; many more such temporary lines may be anticipated since nonsynchronized cells show many more lines though they are of much lower intensity than those of synchronized cells. The hope thus exists that these excitations may be attributed to definite cell activities as suggested in Section II.

Lines also seem to exist near 5 cm^{-1} (1.5×10^{11} Hz) or possibly lower, though this was beyond the available resolving power.

B. Action of Coherent Millimeter Waves

Already several years ago, it had been observed that electromagnetic (EM) radiation of 7.1 and 7.3×10^{10} Hz depresses the growth of E coli bacteria [13]. These findings were confirmed recently [14].

A series of most startling results on the influence of coherent millimeter waves on various properties of biological materials was presented in the U.S.S.R. Academy of Sciences [5]. According to this report, irradiation in the range of 5–8 mm ($4–6\times10^{10}$ Hz) can influence certain properties of biological objects ranging from microorganisms to higher organisms. It is claimed that in almost all cases, sharp frequency resonances in the biological effects are found while only optically broadbands have been observed. Also, the dependence on intensities exhibits threshold properties. A critical value S_0 of the rate of energy supply exists such that no effect is observed when $S<S_0$, while for $S>S_0$ the effect is nearly independent of S. These findings confirm our general conjecture of the biological importance of frequency selective vibrations [15].

Details of these experiments are not available, and it was decided, therefore, to repeat one of the experiments with the greatest care aiming at the highest possible accuracy [4]. The influence of the radiation on the rate of growth of yeast was selected where, in [5], results at four frequencies had been reported: one showing an increase and three showing a depression of this rate. The culture was different in the two cases so that one could not expect exactly the same results. The reproducibility of the measured rate was ±3 percent; the long-time frequency stability was ±3 MHz (a single rate determination takes about 10 hours). Yeast was suspended in a stirred aqueous nutrient medium and the growth was measured optically. A frequency range between 4.14 and 4.19×10^{10} Hz was investigated. About 40 values with 20 control (absence of microwaves) were measured. Power (a few milliwatts) and temperature in and near the sample were monitored. The finite penetration depth of the waves in water implied that at most 20 percent of the suspension was illuminated at a time. The results show five minima with growth rates between 70 and 85 percent of the control within 4.65 and 4.79×10^{10} Hz, separated by maxima of rates between 100 and 115 percent of the control, and no noticeable effect outside this range. Thus the effect is largely negative and exhibits extremely sharp biological frequency resonances. The dependence on the intensity of the radiation has not yet been investigated.

These findings strongly support the conjecture that coherent electric vibrations play an important role in biological activities. The extremely narrow frequency resonances pose most interesting problems whose answers demand further investigation. A model for the predominantly negative effects might arise from the assumption that the active biological system excites certain frequencies in terms of limit cycles. Application of radiation at slightly different frequencies might then destroy the original limit cycle and, hence, reduce the biological activity connected with it.

In conclusion, we have evidence for the biological importance of vibrations in the region of 5×10^{10} Hz from experiments with coherent millimeter waves. The strongest evidence from the Raman effect is at about 3×10^{12} Hz, but it extends to lower frequencies, possibly to about 10^{11} Hz. The interconnection of these various frequencies is unknown at present.

IV. A Model Calculation

The original proposal of coherent excitation of electric vibrations (polarization waves) through supply of metabolic energy was supported by the calculation of a model [13] which can be generalized in the following manner. Consider a band of z polarization waves with frequencies $\omega_k(k=1,2,\cdots,z;\omega_1<\omega_2\cdots<\omega_z)$ in strong interaction with a heat bath at temperature T, referring thus to the interaction of our vibrations with the rest of the system. In lowest order, this can be described in terms of emission of quanta $\hbar\omega_k$ from the polarization waves into the heat bath and of the corresponding absorption. To obtain interest-

ing effects, it will be necessary to consider the lowest nonlinear processes involving two quanta. Suppose that energy is supplied to the k mode at the rate S_k. Then, if this mode contains n_k quanta $\hbar\omega_k$, the following kinetic equation holds:

$$\dot{n}_k = S_k - \phi\left(n_k \exp^{(\beta\hbar\omega_k)} - (n_k+1)\right) - \chi\sum_l \left\{n_k(1+n_e)\exp^{(\beta\hbar\omega_k)} - n_e(1+n_k)\exp^{(\beta\hbar\omega_e)}\right\} - \lambda\sum_e \left\{\dot{n}_k n_e \exp^{(\beta\hbar(\omega_k+\omega_e))} - (n_k+1)(n_e+1)\right\}. \quad (5)$$

Here $\beta = 1/kT$ and the exponential factors arise from the condition of detail balancing. The transition probabilities ϕ, χ, λ are averages depending on temperature and a possible dependence on frequency is neglected. They are largely connected with the interchanged energy with nearly free ions and dipoles in the cell media.

Under stationary conditions then, $n_k = 0$, so that

$$\sum S_k = \phi\sum_k \left(n_k \exp^{(\beta\hbar\omega_k)} - (n_k+1)\right) + \lambda\sum_k\sum_l \left\{n_k n_e \exp^{(\beta\hbar(\omega_k+\omega_e))} - (n_k+1)(n_e+1)\right\} = \sum_k \left(n_k \exp^{(\beta\hbar\omega_k)} - (n_k+1)\right) \cdot \left\{\phi + \lambda\sum_e \left(n_e \exp^{(\beta\hbar\omega_e)} + (n_e+1)\right)\right\} \quad (6)$$

and

$$n_k = \left(1 + \frac{S_k}{\phi + \chi\Sigma n_e \exp^{(\beta\hbar\omega_e)} + \lambda\Sigma(n_e+1)}\right)\frac{1}{\exp^{(\beta\hbar\omega_e-\mu)}-1} \quad (7)$$

where

$$1 - e^{-\beta\mu} = \frac{(\chi-\lambda)\Sigma\left(n_e \exp^{(\beta\hbar\omega_e)} - (n_e+1)\right)}{\phi + \chi\Sigma n_e \exp^{(\beta\hbar\omega_e)} + \lambda\Sigma(n_e+1)}. \quad (8)$$

Thus unless $\chi = \lambda$ (requiring $\mu = 0$), ω_k represents a modified Bose distribution with chemical potential μ.

Now, for most more specific models, $\lambda \ll \chi$ holds ([16]). If this inequality holds to the extent that λ is negligible, then the above equations reduce to

$$\Sigma S_k = \phi\Sigma\left(n_k \exp^{(\beta\hbar\omega_k)} - (n_k+1)\right) \quad (9)$$

$$n_k = \left(1 + \frac{S_k}{\Sigma S_e}\frac{\phi}{\chi}\left(1 - e^{-\beta\mu}\right)\right)\frac{1}{\exp^{(\beta(\hbar\omega_k-\mu))}-1} \quad (10)$$

and μ satisfies

$$\hbar\omega_1 > \mu \geqslant 0. \quad (11)$$

The upper limit $\hbar\omega_1$ follows from the condition $n_k \geqslant 0$.Thus from (10) and (11)

$$\Sigma n_k \leqslant \left(1 + \frac{S_0}{\Sigma S_e}\frac{\phi}{\chi}\left(1 - \exp^{(-\beta\hbar\omega_1)}\right) \perp \Sigma \frac{1}{\exp^{(\beta(\hbar\omega_k-\mu))}-1}\right) \quad (12)$$

where S_0 is the largest of the S_k.

Now, from (9) it follows that Σn_k must increase proportionally to ΣS_k if this quantity is sufficiently large. This can be achieved only if μ comes very close to $\hbar\omega_1$ so that n_1 becomes much larger than the other n_k.

Thus the pumped system exhibits a phenomenon very well known for massive particles in thermal equilibrium: Einstein's condensation of a Bose gas; but, while in this case, the corresponding order arises when the temperature is decreased below a critical value, our case requires that the energy supply is raised above a critical value.

It should be emphasized that these results rest on a large magnitude of the nonlinear χ interaction. Other terms such as nonlinear direct interaction between the polarization waves could have been introduced. Such terms would tend to increase the temperature of the polarization waves above that of the heat bath and thus counteract Bose condensation.

Calculations of this type thus demonstrate the possibility of Bose condensation, i.e., of strong excitation of the lowest mode, provided certain material conditions are satisfied. This, one would conclude, has arisen through long evolutionary processes.

Even so, it seems difficult to expect excitations as narrow as those found experimentally and reported in Section III. Quite possibly, a second stabilizing agent will have to be found. If, for instance, the hypothesis of a limit cycle mentioned in Section III would hold, then the amplitude x would have to satisfy an equation of the type

$$\ddot{x} + \omega_1^2 x + \gamma\dot{x}(x^2 - a^2) = 0 \quad (13)$$

in the absence of an external field; γ and a^2 are constants. The missing agent would then replace ordinary friction terms $\alpha\dot{x}$ by the appropriate nonlinear terms which depending on the amplitude can extract or supply energy to the oscillator.

V. Coherent Vibrations and the Cancer Problem

While bacterial cells divide as long as nutrient and space are available, cell division in normal tissue is subjected to a control which is absent in cancer. It is required, therefore, to find the origin of this control.

We shall show that, in principle, this control could rest on the coherent electric vibrations discussed in the previous sections. In microorganisms, these vibrations have been found to exhibit quite extraordinary properties with regard to the sharpness of frequency resonances (Section III). It will be assumed then that they also exist in active normal tissue, though it will be for future experiments to verify this.

Suppose the coherent vibration in the assembly of cells forming a tissue to be excited as discussed in Section II. This requires each cell to oscillate with appropriate phase and with correct frequency. Changes in a cell that alter this frequency lead to the loss of most of its (negative) interaction energy with the coherent mode. Such changes can, therefore, occur only if sufficient energy is supplied. Now cell division may at certain stages lead to alteration of the relevant frequency. This step is, therefore, con-

trolled energetically by the interaction of the coherently excited mode with the particular cell. The magnitude of the required energy will depend on the strength of excitation of the mode. The availability of the required energy will depend on the available amount of nutrient though, to some extent this should also influence the strength of excitation.

Clearly, the oscillating system of cells exhibits a certain stability. Not only does it attract cells in the neighborhood if they have the correct frequency (Section II), but also an influence will be exerted on cells that for some reason have lost the correct frequency to reinstall it, for this gains free energy.

Now a cell that has lost the correct frequency will no longer contribute to the collective vibration and, hence, weaken it so that the potential barrier controlling cell division in other cells is reduced; but this will be a small effect. If a number of cells, however, will be so affected, then the strength of the collective vibrations will decrease more significantly, and a critical number of such cells should thus exist at which the control becomes ineffective. A phase transition into an uncontrolled state will thus take place.

Clearly, ideas of this type can give rise to many different specific situations in which the coherent vibration in conjunction with the available nutrient controls cell division. A more specific model [18] would, e.g., assume that the DNA protein complex of the cell nucleus of a differentiated cell is responsible for the vibration. Prior to cell division, this complex must unfold and would then loose the correct frequency which, as shown above, would require an appropriate supply of energy which may only rarely be available. Cancer would arise when, from external reasons, the frequency of a sufficient number of cells is changed. This might be caused through the invasion of the nucleus by foreign molecules. It might also arise through the transfer of a sufficient number of electrons to the conduction band of proteins, e.g., by sufficiently short-wave ultraviolet light. For the arising plasma oscillation would interact with our mode and alter its frequency [1].

VI. Proposed Experiments

The most important task should be experimental establishment of the conjectured vibrations in active (growing) tissues. Naturally, one would use the methods that have been successful in the cases of bacteria and other microorganisms, i.e., Raman effect and use of coherent millimeter waves to influence properties. In normal growth, one would expect sharp lines with some of the frequencies depending on the nature of differentiation. In cancer growth, some of these lines would either be absent or broadened.

If monolayers of cells are used, then the strength of the collective mode will not necessarily be the same as in bulk because of the contributions of other layers. The observed contact inhibition in normal tissue, i.e., the absence of growth in three dimensions, should then be understood in terms of the control by the collective mode which is absent in cancer tissues which do not exhibit contact inhibition. Another factor in such investigations must also be the amount of nutrient available. A certain amount of nutrient will be required to establish our mode and, hence, the energy barrier which a normal cell will have to overcome in cell division. The energy available to overcome this barrier might, however, also depend on the amount of available nutrient so that a fairly complex situation could be expected.

Spectroscopy with coherent millimeter waves might also be successful although it must be realized that individual cells will most likely oscillate with phase shifts such that the whole system will only be very weakly optically active. This also implies that the excited coherent vibration will emit only minute amounts of EM radiation.

In conclusion, it is hoped that coherent electric vibrations in the 5×10^{10}–3×10^{12} Hz region whose existence has been established in microorganisms also exist in growing tissues and play an important role in the control of growth. If the low frequency side, i.e., the range of millimeter waves is relevant, and if this relevance shows resonant effects similar to those discussed in Section III, then one should expect important nonthermal effects of millimeter waves of correct frequencies on cancer tissues.

Note Added in Proof

Recently S. J. Webb, R. Lee, and M. E. Stoneham (*J. Quantum Chem: Quantum Biology*, vol. 4, pp. 277–284, 1977) have demonstrated the existence of several (Stokes-) Raman lines in the region of 50–200 cm^{-1}. These were found to be broadened and split in the case of mammary carcinoma. If confirmed, and if strong anti-stokes lines also exist, then this finding is in accordance with the suggestion (1) that cancer cells contain unattached electrons such that their plasma frequency is close to the frequency of the coherent vibrations discussed in the present paper; interaction between the two vibrations will then lead to a splitting of lines.

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8

Coherent Excitation in Active Biological Systems

H. Fröhlich

ABSTRACT: The principal types of coherent excitations: metastable state with high electric dipole moment, and coherent high-frequency electric vibrations are derived from simple models. Far-reaching consequences for biological systems are discussed. The conjecture is supported by a great variety of experiments.

1. General

From the point of view of atomic and molecular physics (and chemistry), biological materials are extremely complex and complicated systems. Yet they function in a most systematic way. Molecular biology succeeded in establishing the structure of the relevant giant molecules. Thus, e.g., it was found that enzymes possess an active site where the relevant processes can take place according to the laws of chemistry. Yet, this fails to explain their enormous catalytic power. Clearly, function must be treated in terms of dynamic, rather than static, properties. As a next step one may be tempted to introduce the known interaction between the atoms and ions of the system and, based on the known structure, calculate their dynamical properties. Such a procedure would be relevant in the range of linear displacements from a rigid structure like a solid. Enzymes, however, do not form such a rigid structure. Attempts to carry out the appropriate calculation have been discussed recently.[1] Even for small enzymes they

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require quite considerable computer time. Such endeavors, thus, while useful as a demonstration of the importance of dynamic properties, will not result in a general theory of the relevant processes, if such a theory exists.

One might then try an approach that emphasizes the common features of various systems—enzymes in the above example—and treat them in a general way thus creating a framework into which the individual cases will fit. To take an example from pure physics, consider the dynamic properties of fluids. Here we do not proceed by first considering the structure of a particular fluid, a most formidable task, and then, based on this knowledge, investigate the dynamic properties. If this were possible then we would obtain the motion of individual atoms or molecules which constitute the fluid, a result which is not very interesting. What is interesting is the space-time dependence of the mean density, $\rho(x, t)$ and of the mean current density, $j(x, t)$. Both are connected by a set of differential equations, the Navier–Stokes equations, from which their space-time dependence can be derived in terms of certain parameters like compressibility, frictional constants, which define the particular fluid. These Navier–Stokes equations can be derived exactly from the basic multiparticle equation, the Schrödinger equation,[2] not by solving it in terms of the multiparticle coordinates, but rather by first expressing ρ and j in terms of individual particle properties, and then finding the differential equations satisfied by them, which essentially arise as symmetry properties of the multiparticle system.

The success of this procedure rests in the definition of the two multiparticle quantities ρ and j, which describe the essential properties of hydrodynamic flow. Derivation of the actual values of the various parameters entering the equations of flow would require, of course, deeper investigation of the basic multiparticle equation. Could we not apply a corresponding method in the treatment of the dynamic properties of biological systems? The answer is no, not for the time being, for we do not yet know the quantities that would take a role corresponding to ρ and j in the previous case, though later in this chapter we shall make proposals in this respect and discuss their experimental verification. The difficulty arises from the enormous number of states of which excited multiparticle systems, like the biological ones, are capable and which make a systematic investigation and inspection impossible. Thus the lowest energy state of a system of N particles, the ground state, usually—though not always—is a crystal. Small displacements of the particles from their equilibrium positions can be treated in a systematic way as linear excitations. In biological systems, however, we are very far from these states, in the realm of nonlinear excitation. The number of states then rises exponentially with the number N of particles—in some particular case say as 10^N. Quite clearly it is impossible, even with a supercomputer, to evaluate and

investigate all these states. Hence we must find a way of deciding on "interesting" properties, usually connected with some kind of order. This order need not, however, be of a spatial nature. In a machine, for instance, this order consists in its way of functioning. Normally we treat this with the laws of macrophysics, which, in turn, follow from the laws of microphysics.

Early microbiologists were tempted to consider spatial order as the only order existing in physics. Thus, after no spatial order had been found in the arrangement of amino acids in an enzyme, Monod[3] says that knowing the arrangement of 199 amino acids would not permit us to predict the 200th. He concludes that the arrangements are random. Actually, however, the whole chain must function as an enzyme; replacing the "correct" 200th amino acid by a "wrong one" disturbs this function.

In looking to physics for guidance to find a concept capable of describing biological "order" we are led to the concept of coherence, which applies to cases as different as superconductors, superfluids, lasers, and masers. Roughly speaking it implies that if a certain property is known at a space-time region near (x, t) it is also determined at another, (x', t'). Thus in a coherent wave, for instance, amplitude and phase at (x, t) determine their values at another, (x', t'). Considering now biological systems, periodic processes are very common among them with periods ranging from that of life cycles to very high-frequency electric vibrations. In contrast to most physical periodic processes, periods of biological processes often vary from case to case. Thus the periods of the life cycles of cells grown under exactly the same conditions vary within certain limits, i.e., each cell exhibits a certain amount of personality. Also, if an apparent cause for a certain biological process is removed, another takes over because biological processes are frequently multicausal.

We shall presently describe the derivation of certain coherent excitations on the basis of definite physical models. Such developments show that random energy supplied to a certain system need not lead to heating but may result in the excitation of ordered (coherent) states. This need not imply that other models could not lead to similar excitations, i.e., that a multicausal situation exists—in other words that experimental verification of the excitation need not necessarily be considered as proof of a particular model. Clearly a situation then arises which requires close collaboration between theory and experiment. Thus, e.g., theory has predicted certain coherent excitations and experiment verifies their existance. This presents a development from the point of view of physics. From the point of view of biology, however, we may ask a question that is prohibited in physics: what is the purpose of these excitations? Evidence will be presented later in this chapter for the first (physical) stage, but the second, biological question has hardly been touched yet. Its solution will, of course, lead back again to physics, i.e., a certain process has certain consequences.

If these consequences are vital biologically then multicausal possibilities must be considered again, i.e., what other processes could have the same consequences.

2. Coherent Excitations

In looking for a physical characteristic common to most biological systems, one is at once led to their dielectric properties.[4] Most startling among these is the high electric field of about 10^5 V/cm present in biological membranes. This field acts in the range of nonlinear response. Materials inside membranes are, therefore, strongly, nonlinearly polarized. Furthermore, the many ions contained in biomolecules make an essential contribution to their stabilization through their interaction[5] as well as through their interaction with the surrounding medium, e.g., water.[6] On the other end biological systems are capable of reacting to extremely weak external impulses as reported in Section II of Ref. 7. Hence, by appropriate arrangements and excitations, biological systems may form most sensitive instruments that in normal technology would be devised in completely different ways with very different materials.

As a first case of coherent excitation, a metastable state with very high electric polarization (ferroelectric state) will be shown to exist in materials with high polarizability, combined with elastic softness, as discussed in Section III B of Ref. 7, and Ref. 8. Such excitation is coherent as the polarization at a point x determines the value at another, x'. The metastability of this excitation arises from the shape dependence of the electrostatic energy of a material. Thus, e.g., the ratio of the energies of a slab and that of a sphere both having equal volume and polarization is $3\varepsilon/(\varepsilon+2)$, where ε is the dielectric constant (cf. p. 168 of Ref. 9). To the electrostatic energy must be added the elastic energy arising from a deformation of the material.

Thus consider a sphere with polarization P and electrostatic energy $\frac{1}{2}\gamma P^2$. If deformed into an ellipsoid with excentricity η, this energy is changed by terms proportional to η, and higher powers, as the electrostatic energy does not have an *extremum* for the sphere, $\eta = 0$. The elastic energy arising from the displacement will, however, have no linear term in η but be proportional to η^2, as elastically the sphere is the equilibrium state. The total energy U, therefore, for small η will have the form

$$U = \tfrac{1}{2}\gamma P^2(1 + c_1\eta + \tfrac{1}{2}c_2\eta^2) + \tfrac{1}{4}C\eta^2 \tag{1}$$

where $C > 0$ and $\gamma > 0$, and the C_2 term has been inserted as it might influence the C term. Minimizing with regard to η yields

$$\eta = -c_1\gamma P^2/(C + c_2\gamma P^2) \tag{2}$$

and hence

$$U = \tfrac{1}{2}\gamma P^2 - (c_1 \tfrac{1}{2}\gamma P^2)^2/(C + c_2 \gamma P^2) \tag{3}$$

Thus U rises more slowly than it would without the deformation ($\eta = 0$), and it reaches a maximum beyond which it decreases with increasing P^2, provided $c_2 > 0$, and $2c_2 < c_1^2$. For still larger P^2, higher-order P^2 terms will be relevant, so that U goes through a minimum. Should that be at negative U, then this represents the ground state and the material is ferroelectric. Otherwise this minimum represents a metastable state with very high polarization, a quasi-ferro-electric state.

On a more formal basis we may consider a homogeneous polarization field P, energy density $\frac{1}{2}\omega^2 P^2$, interaction with an elastic field with potential energy density $\frac{1}{2}s^2\rho^2$ for longitudinal elastic displacements yielding a density ρ. Their interaction must be proportional to ρP^2, to provide the effect discussed above.

The total potential energy has then the form

$$W = \tfrac{1}{2}\omega^2 P^2 + (\tfrac{1}{2}s^2\rho^2 + cP^2\rho)(1 - d^2P^2) \tag{4}$$

where c and d are constants and the d^2P^2 term has been included to take care of nonlinearity required to stabilize the material. Minimizing with regard to ρ yields

$$\rho = -\frac{c}{s^2} P^2 \tag{5}$$

and hence

$$W = \frac{1}{2}\omega^2 P^2 - \frac{1}{2}\frac{c^2}{s^2} P^4 + \frac{1}{2}\frac{c^2 d^2}{s^2} P^6 \tag{6}$$

Thus this potential energy increases proportional to P^2 for small P^2, and proportional to P^6 for large P^2. In between it has a maximum, followed by a minimum provided a parameter λ satisfies

$$\lambda = 3s^2 d^2 \omega^2 / c^2 < 1 \tag{6a}$$

and this minimum represents a metastable state when $\frac{3}{4} < \lambda < 1$, while for $\lambda < \frac{3}{4}$ this highly polar state is the ground state. This confirms our previous results, and (6a), moreover, indicates the conditions the material must satisfy. They are, in particular, elastic softness (small s) and high polarizability (small ω) in conjunction with large coupling between the fields (large c^2).

Existence of highly polar metastable states should thus be a general property of biological materials. One would expect this to hold in less sim-

plified models, introducing particular structures together with anisotropy. One would expect then to find more than one metastable state. A small generalization of the above is possible, in fact, if the various parameters s, ω, c slowly depend on space coordinates, provided derivatives of P and s need not be introduced into the expression for the potential energy.

Most generally, of course, small excitations in materials with definite molecular structure are described through frequency–wave number bands, which can be classified as polar or nonpolar. Introduction of nonlinear features, as required here, must then be investigated for various particular cases, though our above treatment should always be correct as a limiting case. From the structural point of view, metastable excited states would then be classified as arising from conformational changes.

A candidate frequently mentioned in this respect is the displacement of H^+ in an H bond (e.g., Refs. 10–12 and related papers), for which also evidence from an analysis of dielectric properties in the millimeter region is available.[13,14] Displacement of various H^+ ions does, of course, yield electric polarization, though this need not be homogeneous. It is important, however, to realize that this polarization necessarily is connected with electric polarization of the rest of these molecules, which in turn leads to elastic displacements in it, as follows from our general model. Within this frame, polarization of the H bond represents a subsystem which enhances the stability of the metastable state. Detailed calculations have not been carried out yet. P= polarization state

So far we have considered the potential energy W of our system only, and found it to have a minimum at $P=0$ and at $P=\pm P_0$ if P_0^2 is the value of P^2 at which the metastable state occurs. The system can of course oscillate harmonically if the displacement from $P=0$ or $P=\pm P_0$ is small. More generally, however, the system will carry out anharmonic oscillations. In fact it is shown in Ref. 8 that to a good approximation P satisfies the equation of motion

$$\ddot{P}+\frac{\partial V(P)}{\partial P}=0 \tag{7}$$

where

$$V=A+\tfrac{1}{2}BP^2+CP^4 \tag{8}$$

i.e., as long as P^4 can be neglected, the oscillations are harmonic. Yet even then the system possesses highly nonlinear properties as A, B, C are not constant but depend on the time average p of P^2,

$$p=\overline{P^2} \tag{9}$$

by

$$A = \frac{1}{2}\frac{c^2}{s^2}p, \qquad B = \omega^2 - 2\frac{c^2}{s^2}p - \frac{d^2c^2}{s^2}p^2, \qquad C = \frac{c^2d^2}{s^2}p \tag{10}$$

Thus each amplitude of P defines its own potential V. Clearly for small amplitudes when p can be neglected in B, and the P^4 terms are also negligible, the frequency is ω. For growing amplitude, but when P^4 is still negligible, the frequency of oscillations is softened, i.e., B becomes smaller until finally it changes sign; V then has a maximum at $P=0$ and a minimum further outside.

Near the metastable state, P may, of course carry out small, harmonic oscillations again, but with a frequency different from ω.

The coherent excitation considered so far is essentially static; its establishment requires a supply of energy larger than the energy of the metastable state above the round state $P=0$. A dynamic, very different type of coherent excitation may exist whose maintenance requires a continuous supply of energy at a rate larger than a critical s_0. The model is again based on polar oscillations of a large biological system, forming a band of modes with frequencies ω_1, ω_2,..., ω_z. Metabolic energy is considered to be supplied at a rate s_j to mode ω_j. This systems of polar modes is considered to interact with a heat bath which is kept at a constant temperature T. In cells, this will largely consist of cell water with its ions. Linear interaction will then impose a Planck distribution for the excitation of the polar modes. This interaction is described in terms of emission and absorption of single quanta $h\omega_j$ between the modes and the heat bath. When nonlinear interactions are taken into account, however, then energy exchange in terms of two quanta will occur. Most important among these are simultaneous emissions and absorption of quanta $h\omega_j$ and $h\omega_l$. If n_j represents the number of quanta by which mode ω_j is excited, then the following equation arises for the rate of change, n_j (cf. Section III C of Ref. 7):

$$\begin{aligned}\dot{n}_j = s_j - \phi_j(n_j e^{\beta\omega_j} - (1+n_j)) \\ - \sum_l \chi_{jl}[n_j(1+n_j)\, e^{\beta\omega_j} - n_l(1+nj)\, e^{\beta\omega_l}]\end{aligned} \tag{11}$$

where $\beta = h/kT$, and the ϕ and χ are transition probabilities. The ratio of absorption and emission terms, proportional to n and $1+n$, respectively, follows from the principle of detailed balance. Note that after further summation over j, the χ terms vanish so that in the stationary case when $\dot{n}_j$ vanishes, the total number $\sum n_j$ of quanta that arises from the total rate of supply $\sum s_j$ is determined by the first-order terms only through the ϕ. The

actual distribution will, however, be largerly determined by the χ terms, provided that they are relatively large. In the stationary case then they must vanish so that

$$n_j = \frac{1}{\exp[\beta(\omega_j - \mu)] - 1} \tag{12}$$

follows where μ must be determined from the already known total number of quanta $\sum n_j$. When this number exceeds a critical one, i.e., when $\sum s_j$ exceeds s_0, then μ must approach the lowest ω_1, yielding a large n_1, as in Bose condensation. This frequency is then very strongly, i.e., coherently, excited. Detailed considerations[7] show that in certain circumstances other frequencies might be strongly excited.

To understand the possibility of this type of coherent excitation, the strong interaction with the heat bath must be emphasized. It endeavors to impose its temperature on the distribution which is demonstrated by the form (12), which represents a Bose distribution at temperature T. But while in a Bose gas the number of particles is fixed so that the Einstein condensation arises only when the temperature is sufficiently lowered, in our case the temperature is fixed, but the number of particles (quanta) increases with increasing rate of energy supply.

It must also be emphasized that this type of coherent excitation arises from the strong nonlinear coupling with the heat bath (χ terms). Other terms could be introduced, in principle, which would counteract this effect. Thus a very strong nonlinear interaction among the quanta of the polar modes, without the use of the heat bath, would simply tend to increase the temperature. We thus have found a particular example of the two different ways energy supply to a system may act, resulting in either an increase in temperature, or in the creation of a new order, that may be dynamic. Our considerations have emphasized the second possibility. Experiment must decide whether biological systems make use of it.

Finally, in this section, possible frequencies for the excitation of coherent vibrations should be considered. Originally special reference was made to membrane oscillations as their high polarization will then yield coherent electric oscillations (cf. Ref. 7, with earlier literature). Since the thickness of a membrane is about 10^{-6} cm and its elastic constant is equivalent to a velocity of sound of about 10^5 cm/s, a frequency of the order of 10^{11} Hz was expected, corresponding to electromagnetic waves in the millimeter region. When based on proteins or DNA, however, both higher and lower frequencies may be expected.

It must be emphasized at this point that most of the polar modes of a system interact only weakly with electromagnetic waves of the same frequency, as usually the wave lengths do not match. Thus, e.g., excitation of

sections of the outer membranes of a cell may result in a very high multipole electric oscillation, and hence in small interactions with electromagnetic waves of the same frequency.

3. *Consequences*

The two coherent excitations, metastable highly polar states, and coherent electric vibrations, may have far-reaching consequences arising from the forces exerted by the respective fields which are absent in the nonexcited states. The electric field in the metastable state is static; it thus acts inside and near the excited system, while at larger distances it is screened by the ions in the surrounding medium. For molecules, e.g., proteins, surrounded by cell water, the screening distance naturally is much shorter than for proteins dissolved in membranes. In the case of excitation of coherent vibrations, on the other hand, screening is negligible once the frequency exceeds a critical one.

Forces arising from the excitation of the metastable state may then be employed to reduce activation energies in enzyme processes.[15,4] The system enzyme + substrate will change with the motion of ions in it and, hopefully, the direction and magnitude of the internal field will change as required by the process. The whole system, enzyme + substrate, while active, possesses a high electric dipole moment which, in principle, could be measured by standard dielectric methods. Such measurements have not been attempted yet; they face difficulties arising from the presence of unattached ions in the medium in which the enzyme reaction occurs.

For a protein dissolved in a membrane the possibility arises that it may act as a switch element. For if μ is the dipole moment in the metastable state, and F is the electric field in the membrane, then if the directions of μ and F are appropriate the interaction lowers the energy by $(\mu - \mu_0)F$, where μ_0 is the dipole moment in the ground state. If this energy equals the energy of excitation of the metastable state, then the system is degenerate with two equal energy states. They then can be switched by the addition of relatively weak fields.

It should also be noticed that in proteins, aligned in a membrane, the excitation energy may be transferred from a particular excited protein to its neighbor, thus providing transport of energy that is nearly loss free.

Solitons, furthermore, represent a particular case of excitation to a metastable state.[16,17] In the Davydov soliton, in particular, stretching of the C = 0 bond forms the polar excitation, stabilized by the surroundings which play the part of the elastic field. This soliton, too, provides a loss-free transport of energy.

Excitation of a metastable state requires a certain minimum amount of energy. Excitation of the coherent polar mode, on the other hand, requires supply of a minimum rate of energy, s_0; its maintenance thus requires biological pumping. Its excitation results in a long-range, frequency-selective interaction that may lead to attraction between two, or many molecules, provided they oscillate with equal frequency. In particular cases this may also result in selective repulsion. In the case of the enzyme–substrate system, in the first stage the (frequency) correct substrate is attracted to the enzyme (long-range selectivity); in the second stage the usual short-range chemical processes take place—an example of a multicausal process.

As indicated in connection with Eq. (11), the energy supply s_j may occur at any, or all, frequencies of the polar modes. The subsequent excitation of a particular mode, e.g., ω_1, then requires some time even when the total rate of supply exceeds s_0. The system thus possesses storing ability. One would expect, however, that this time is particularly short if the energy is supplied at the frequency of the mode that will be excited coherently. Furthermore, arising from the considerations of mode softening, this mode, when excited, may be detached from the band and hence provide a very frequency-sensitive target for further energy supply. Detailed calculations on this question have not been carried out yet.

Consider now biological processes that make use of coherent excitation of a particular polar mode, requiring pumping at a rate $\sum s_j > s_0$. Let (cf. Section IV C of Ref. 7) t_1 be the time required for this process; then, we expect it to be the faster the larger $\sum s_j$; but it will never occur ($t_1 = \infty$) as long as $\sum s_j < s_0$. Thus if a and n are positive numbers we may assume

$$t_1 = a \Big/ \left(\sum s_j - s_0 \right)^n, \qquad \sum s_j \geqslant s \tag{13}$$

If we now consider a complex biological process consisting of many steps, only one of which depends on the excitation, i.e., on s_j, then if t_2 is the time required for all these s_j-independent processes, the total time for the process is

$$t = t_1 + t_2 \tag{14}$$

i.e., the total rate is

$$\frac{1}{t} = \frac{(\sum s_j - s_0)^n}{\alpha + t_2 (\sum s_j - s_0)^n}, \qquad \sum s_j > s_0 \tag{15}$$

It thus depends in a steplike manner on the rate of supply $\sum s_j$.

We must assume now that the biological system is controlled in such a manner that the rate of energy supply is switched on at a period that integrates the particular process considered here with the whole biological development. External supply of energy, e.g., by microwaves at a rate say s_e, will then influence the rate $1/t$ and hence interfere with the general development. The effect of even a small outside s_e can be very big at a stage when the biological $\sum s_j$ is close to s_0, because replacing $\sum s_j$ by $\sum s_j + s_e$ in Eq. (15) is equivalent to replacing s_0 by $s_0 - s_e$, i.e., it shifts the steplike curve Eq. (15). At particular times in the biological development, small s_e hence may cause large effects, in particular when s_e is supplied at the "correct" frequency, which involves little time for the reaching of the required quasistationary distribution of the excitation of modes.

In considering now candidates that may act as carriers for the electric vibrations arising in the coherent excitation of a particular mode, recent results on the structure of cells must be presented. It has been found[18] that an intricate network of protein structures known as the cytoskeleton exists, among which the so-called microtrabecular lattice extends from the cell surface throughout the cytoplasm. It has also been proposed[18] that enzymes do not float in the cytoplasm but are attached to this lattice, which may act as carrier of the coherent electric vibrations that provide frequency-selective interaction between appropriately excited molecules. Supporting this conjecture is the conclusion that "microtubules $\cdots$ form a vectorial framework upon which some other force generating component is organized."[19] This also is supported by theoretical investigations on self-focusing.[20] To avoid misunderstandings it should be repeated again that while the relevant frequencies might be in the range of electromagnetic millimeter waves, interaction with such radiation will be weak as the wavelengths will not be in the millimeter region, but probably will be less than the cross section of the tubules, possibly of the order of molecular distances.

It will now be shown that combination of our two coherent excitations may result in low-frequency periodic enzyme reactions giving rise to corresponding electric vibrations—another coherent electric vibration (Ref. 21 and Section IV E of Ref. 7). Consider an enzyme process consisting of the interaction of a substrate and an excited enzyme and assume the energy thus made available to excite a nonexcited enzyme. Then if N, Z, S are the respective numbers of excited enzymes, nonexcited enzymes, and substrates, the rate of change of N is proportional to NZS with a rate constant α. Also if there exists a spontaneous decay of excitation of an enzyme, rate constant β, then

$$\frac{dN}{dt} = \alpha NZS - \beta N \tag{16}$$

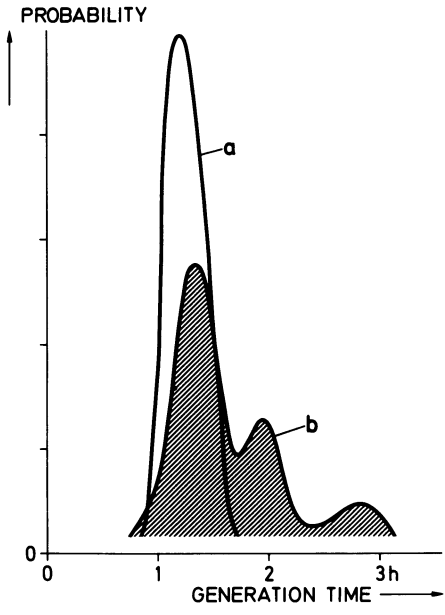

FIGURE 1. Probability of time of cell cycle when irradiated (a) with 83800 MHz or without irradiation, (b) with 83835 MHz (W. Grundler, personal communication).

and the number of substrates decreases at the rate αNZS. Consider now that during this process substrates are coherently excited to a frequency that attracts other substrates to the region (rate constant γ); then

$$\frac{dS}{dt} = -\alpha NZS + \gamma S \tag{17}$$

Consider the case that Z is so large that it can be treated as time independent; then time-independent solutions N_0, S_0 exist,

$$N_0 = \gamma/\alpha Z, \qquad S_0 = \beta/\alpha Z \tag{18}$$

Hence if

$$N = N_0 + v, \qquad S = S_0 + \sigma \tag{19}$$

then

$$\dot{v} = \gamma\sigma + \alpha Z\sigma v, \qquad \dot{\sigma} = -\beta v - \alpha Z\sigma v \tag{20}$$

These are the well-known Lotka–Volterra equations that have periodic solutions. In the limit that σ and v are small so that the σv term can be neglected, $(\beta\gamma)^{1/2}$ represents the frequency of oscillation, which thus is determined by the rate γ of spontaneous decay of enzyme excitation, and

by the rate β of attraction of substrate molecules. According to our first type of coherent excitation, the excited enzyme carries a large electric dipole moment, which thus also will oscillate with the frequency $(\beta\gamma)^{1/2}$. Outside electric observation will normally integrate the contribution of all enzymes and thus will also depend on their spatial disposition.

This scheme was originally derived for brain waves, and extended by taking into account a possible short-range interaction between excited enzymes, which then tends to organize them into a ferroelectric system. This results in an extension of Eqs. (19) and (20) into a complicated set of equations whose solutions are limit cycles; this means that the system will then tend to oscillate with a fixed amplitude.

This case has been investigated in detail[22] and it has been shown that a weak, appropriate, external perturbation may cause the collapse of the limit cycle and hence lead to the liberation of the considerable energy stored in it. This was intended as a model to demonstrate how weak electric perturbations can cause large effects in the brain (e.g., a nerve impulse).

The considerations given here are of a very general nature, and modifications of the theory can easily be performed without altering the principal result of coherent excitations. One will expect the excitation of a polar mode to be relevant whenever long-range coordination is required, as in cell division, tissue growth, and the cancer problem; cf. Section IV D of Ref. 7 and Ref. 23.

The theoretical derivation of the existence of a coherently excited polar mode following from (11) can also be phrased in a field theoretical model, or in a more exact way be supported by numerical work as quoted in Ref. 7, as well as by more recent work—the present chapter is not meant to provide a complete review. It is of importance, however, that other models involving nonlinear coupling of oscillations can lead to coherent excitation of a single mode.[24,25]

It has also been suggested that the assembly of biological oscillators may be modeled as parameter tunnel diode memory system with couplings. The frequencies of such a system can include values very much lower than the "design" frequency, which thus supports our prediction of mode softening.[26]

4. *Experiments*

A great variety of experiments support the conjecture of coherent excitations in biological systems. The complexity of biological materials might permit different interpretations in a number of these experiments. The possibility of understanding them from a common point of view gives strength, however, to our interpretation. In fact the conclusions of a recent

meeting[27] were "that coherent excitation had now been shown to be an intimate feature of biological activity; further progress would rely on detailed studies...." The proceedings of this meeting were published in the "Green Book."[28]

A number of these experiments require subtle preparation of the biological material, which is not always described in detail and hence may lead to failure of reproducibility. As a first example consider the Raman spectrum of life algae. If certain frequencies are nonthermally excited, as required in coherent excitation, then the ratio of anti-Stokes to Stokes lines should be larger than in thermal equilibrium. This was actually found.[29] A repetition of the experiment showed no such effect.[30] The authors forgot, however, to supply CO_2, which is required to maintain the biological activity, giving rise to the excitation.

A great number of investigations on the Raman effect in active bacteria have been carried out.[31] To be active, considerable dilation must be maintained, and it has been estimated that an enhancement of excitation by a factor of 10^5 is required to obtain sufficient intensity to be noticiable.[32] Such enhancement cannot be expected for a single cell. A number of cells must, therefore, be assumed to oscillate coherently because the scattering intensity then is proportional to the square of the number of scattering units. While thus, seeing the lines at all already provides an indication for coherent excitation, measurement of the ratio R anti-Stokes: Stokes intensity, carried out in one case, gives a more direct proof. Thus at about 120 cm^{-7}, in *E. coli* B bacteria, $R \simeq 1$ was measured while in thermal equilibrium $R \simeq 0.55$. Note that $R = 1$ is the largest possible value.[33] To maintain the required synchronization is a task that so far has not been properly documented. Adverse influences may arise from the laser beam, or from turbulance that may develop in a flow instrumentation, used to avoid destructive action of the laser beam on bacteria.

To avoid some of these difficulties a special method of synchronization has been used, whereby appropriate nutrients are provided, suddenly, after a period of starvation.[34] It was found that a nutrient line at 980 cm^{-1} is highly excited, the enhancement above thermal being by a factor between 8 and 15; and it showed a slow periodicity of about 10 min. For a tentative interpretation we assume that nutrient molecules in or near an active cell (the measurements integrate excited and nonexcited molecules) are very highly excited and attract outside nutrient molecules to the cell, or to a particular region in it. This would then correspond to the term γS in Eq. (17). The periodicity, then, might follow in a similar way. By further investigation of the activity of this system, it should be possible to separate the diffusion part of the molecules from the region in which they are subjected to forces, and hence, possibly, determine the relevant biochemical process.

While this separation of movement due to forces, from diffusion, remains to be carried out, a very thorough separation of this type has been performed in connection with the formation of so-called rouleaux of red blood cells, i.e., stacks of cells that form in their plasma.[35,36] It had been observed that when such cells approach to a certain distance they seem to rush towards each other. Analysis with the help of the Smoluchowski theory of diffusion revealed, quantitatively, the existence of a long-range attractive force. Its properties were found to agree with the requirements of the theory of long-range interaction arising from coherent excitation of electric vibrations in the membrane. Thus on removal of the membrane potential, the interaction disappears, but is restored when the potential is restored. Similarly it is found that the interaction disappears when the metabolism, required to maintain the excitation [terms s_j in Eq. (11)] is inhibited. Furthermore the interaction is specific in that a mixture of red blood cells from different animals segregate into rouleaux of equal origin.

It was also found that the interaction is carried by long molecules which, inside cells, corresponds to the role attributed[18,20] to the microtrabecular lattice as discussed in Section 3. Furthermore it has been shown[37] that albumin, a globular protein, does not transmit the interaction at physiological concentrations. Albumin can, however, be converted into a transmitter by a treatment that unravels the tertiary structure and allows it to polymerize.

The frequencies for the excitation discussed here have not been measured directly, but the involvement of the membranes suggests a region near 10^{11} Hz, though both lower frequencies (mode softening) and higher frequencies (proteins dissolved in the membrane) may be contemplated. In addition, much lower frequencies arise from periodic enzyme reactions, discussed in Eqs. (16)–(20). Among other experiments that support the existence of coherent excitation[38] are the attraction of particles of a dielectric powder to active cells,[39] by the method of "microdielectrophoresis." Investigations have been carried out on a great variety of living cells. They show that an oscillating electric field with minimum frequency of about 10 kHz exists near the surfaces of living but not of dead cells. This can be suppressed by metabolic inhibitors and is maximal (order 100 V/cm) near mitosis (in yeast).

Since the dielectric particles used in these experiments are much smaller than the cells to which they are attracted, they do not give much information about the relative phase on various sections of the cell surface. Thus a dipolar oscillation would interact strongly with electromagnetic waves of the same frequency, and the field will then decrease slowly with the distance from the cell, in contrast to the case of a high multipole whose electric field will be significant near the cell surface only. Nevertheless, within this region it will give rise to a secondary magnetic field, which

might result in the attraction of small magnetic particles, as has been observed recently.[40] While thus observation of emission of electromagnetic radiation would present a useful identification, lack of such observation does not imply absence of electric vibrations; for this might require measurement instruments to be placed very close to the cell surface.

An enormous literature has accumulated on the action of microwaves on biological systems, much of it, however, at single frequencies or single intensities only, which makes it difficult to analyze the results. An exception is formed by investigation of the action of coherent millimeter waves on biological systems. These wavelengths are strongly absorbed by cell water, and great care has been taken, therefore, to eliminate thermal effects by using low intensities. In this case direct action of the electromagnetic field is excluded as this would require intensities many millions of times stronger than those used in the relevant experiments.[41] Early experiments are discussed in Section V B of Ref. 7. Their results are best summarized as follows[42]: "(a) the effect of irradiation depends strongly on the frequency of the microwaves; (b) in certain microwave-power ranges, the effect of exposure depends weakly on variation of the power through several orders of magnitude; (c) the effects are observed to depend significantly on the time of irradiation."

These results seem to confirm all theoretical predictions first made in 1968[43]: As shown in previous sections, the excitation of coherent membrane oscillations should be in the 10^{11} Hz region at very sharp frequencies; the response should be steplike as a function of the supplied energy [cf. Eq. (15)], and the system has storing capability. However, the supplied energy, in the theory, is largely of metabolic nature—this has to be so as the energy supplied externally alone would be much too small to cause big effects. It has been pointed out, however, in the discussion following Eq. (15) that at an appropriate instant of the development of the biological system, a small external energy supply s_e may shift the position of the step in the response function, i.e., it will act strongly at a very sensitive stage of development. This may be compared to a phase transition in the development of the whole biological system; the externally supplied energy then changes its timing of this process relative to other processes, and the consequences for the whole system may be specific in particular cases.

It seems obvious that such subtle interference might be influenced by the natural fluctuations present in highly excited systems, like biological ones. This is in particular so as it has been shown that noise may cause phase transitions in nonlinear systems.[44] It might be suggested, therefore, that details of the experimentation are relevant for the outcome of the investigation.

Consider now, in particular, the sharp resonances in the rate of growth of yeast under the influence of microwaves in the 42-GHz

region.[45,46] A great number of very sharp maxima and minima in the rate of growth versus frequency curve was found. The experiment emphasizes high-frequency stability and provides the use of two very different electrodes to supply the radiation to the yeast cultures that grow in liquid suspension. Cross correlation between the two geometries in the range between 41700 and 41800 MHz reveals a central line of about 10 MHz width with three side bands on each side, separated by about 18 MHz. The effects are relatively small, about $\pm 10\%$, but very well resolved and reproducible. Another feature, apart from the extremely sharp resonances, lies in the absence of a difference in magnitude between the two electrodes, although their surfaces differ by a factor of 10, and the radiation penetrates from the surface only a fraction of a millimeter. The suspension is constantly stirred, and it can be concluded that a single yeast cell is at most affected once during its life cycle, and then only in a particular state of development.

Clearly one would expect such extraordinary effects to arise from important, highly sensitive properties which will not reveal themselves in population effects, though these have the property of high reproducibility.

An investigation of the time of a life cycle of individual cells, and the influence of millimeter waves on it, was therefore undertaken. Frequencies in the 83-GHz region were used—for organizational reasons different from those of the source used in the experiments described above. The cell cycle time was not influenced by radiation of 83800 MHz, but very strongly so by the neighboring 83835 MHz, as shown in Figure 1. Remarkably, however, about half of the cells are not affected, whereas the others are affected very strongly. This leaves the question open whether cells show a "personality" by responding to one or another frequency, or whether a secondary effect, like a particular fluctuation at a critical time, is required to produce the effect, as suggested above.

Further experimentation should result in a more detailed knowledge of the particular cell processes involved. The sharpness of the resonances makes it quite clear, already at the present stage, that coherent excitations must be involved, and the fine structure of 18 MHz might arise from periodic enzyme reactions, Eq. (20).

While the above results are highly reproducible, it is not yet quite clear which particular items in the instrumentation, and the handling of the biological materials, are vital. Thus, at the same laboratory, a similar instrumentation has been produced which shows no effect. Up to the time of writing, the reason for this had not yet been discovered.

Other investigations in the field are published in Ref. 28. It should be expected that multicellular systems might show a larger effect. In fact[47] the growth of cress roots can be completely stopped within 100 seconds after irradiation with low-intensity 56-GHz microwaves, provided the

polarization is parallel to the root. From the theoretical discussion it can be concluded that the process, which owing to the effect of external energy supply is started at an inappropriate time, completely disrupts the growth. Further biological and physical investigations are required to find this process. Clearly it must, at its own time, make use of coherent excitation.

Other biological systems exist, however, whose dynamic properties are much better known than those of cress root. Among these, properties of the membranes systems of electron transport phosphorylation have been studied from the point of view of bioenergies.[48] This requires energy transport over relatively long distances, and other concepts that can all be supplied by various coherent excitations. Measurements to confirm this are now planned.

Finally, to return to the metastable highly polar state of individual molecules, it should be possible, in sufficiently strong external fields, to lift the molecules into this state. Transitions into a metastable state were actually discovered when strong fields were applied to so-called Langmuir–Blodget multilayers and the low-frequency dielectric response was then measured.[49] Haemoglobin and some other molecules were investigated. When the applied field reached about 10^5 V/cm, characteristic changes in the dielectric response were observed. They are maintained even when they are remeasured in low fields, i.e., the layers have been transferred into a metastable state. This state persists for up to six weeks. The gradual reversion to the initial characteristics can be accelerated by heating for short periods to about 40°C. It is to be expected that, owing to the presence of water and ions, the high polarization is screened. The fact, however, that the metastable state is reached through the application of very high fields indicates its basic high polarity.

Another support for the existence of a highly polar metastable state arises in the so-called electret method. Here the material is polarized in strong fields, and the polarization is frozen in by cooling to very low temperatures. Gradual increase in temperature then releases polarization at characteristic temperatures. In our metastable state, the release should begin at temperatures above physiological ones. Both investigated enzymes, trypsin and urease, show this effect,[50] though the polarizing field amounted to only 25000 V/cm.

In a very different kind of experiment it was found that laser irradiation increases the activity of chymotrypsin by a factor of up to 3.[51] This was investigated as arising from lifting the enzyme into the active, metastable state through the Raman effect. Detailed investigations of the dependence on the density of molecules and on laser power support this interpretation. For other possibilities see Ref. 52.

In summary it is seen that a variety of different experiments support the conjecture of two types of coherent excitation: a metastable state with

large electric polarization, and existance of coherent electric vibrations. It should be emphasized again that the theoretical models show that such excitations should be possible. They are not unique; other models might achieve similar results. In fact if, as we think, biological activity makes great use of these excitations, the manner of establishing them need not be unique.

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Comments on the Bose Condensation of Phonons in the Biological Systems

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Abstract

Following the conjecture of Fröhlich that the Bose condensation of phonons is the dominant biological effect, Wu and Austin derived from the Fröhlich Hamiltonian an expression for the relaxation time which is inversely proportional to the temperature by using the technique of Green's function. In this paper, we have generalized the Wu and Austin expression by improving their approximations with the superconducting canonical transformation and obtained the relaxation time depending exponentially on the inverse temperature. Moreover, we have checked our expression by using a different method based on the "continued fraction expansion" and also employed this technique to derive the dielectric constant for this system.

Introduction

Fröhlich [1–6] proposed that the phonon condensation may be possible in the biological systems because of their extraordinary dielectric properties. The dipoles of the biosystem generate longitudinal coupled oscillations to produce a narrow phonon energy band. Fröhlich [1–6] showed from the rate equation that such a phonon system will undergo Bose Condensation if it is coupled to a thermal bath system and an energy pump with a pumping rate exceeding a critical value. Wu and Austin [7] showed that this phenomenon of Bose condensation in the biosystem can be demonstrated from the fundamental quantum mechanical arguments. Furthermore, Wu and Austin [8] derived an expression for the relaxation time in terms of the phonon-bath coupling coefficient and estimated the numerical value for it.

In this paper we have derived an expression for the dielectric constant in terms of the same phonon-bath coupling coefficient. The calculations have been based on the "continued fraction expansion" technique developed by one of the authors (Paul [9]), and provided an estimate of this coefficient from the available experiment data.

Theory

Wu and Austin [7] assumed for the biosystem the following Hamiltonian of the

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superconductors introduced by Fröhlich [10]:

$$H = \sum_i \hbar\omega_i a_i^\dagger a_i + \sum_k \hbar\Omega_k b_k^\dagger b_k + \sum_p \hbar\Omega_p P_p^\dagger P_p + \sum_{ijk} \hbar(\chi a_i^\dagger a_j b_k^\dagger + \chi^* a_j a_i^\dagger b_k) + \sum_{pi} (\xi P_p a_i^\dagger + \xi^* P_p^\dagger a_i), \quad (1)$$

where each of the normal modes of the oscillation of the biosystem is represented by the creation and the destruction operators $a_a^\dagger$ and a_i, respectively. Similar operators $(b_k^\dagger, b_k)$ and $(P_p^\dagger, P_p)$ are assigned for the rest of the biosystem which are the bath and the pump, respectively. The χ is the coupling coefficient which is responsible for the interaction between the phonons and the bath.

Rate of change of the number of quanta in the ith mode N_i is given by the Heisenberg equation

$$\frac{dN_i}{dt} = \frac{1}{i\hbar}[N_i, H]; \quad (2)$$

and the expectation value of Eq. (2) is

$$\left\langle \frac{dN_i}{dt} \right\rangle = \left\langle \psi(t) \middle| \frac{dN_i}{dt} \middle| \psi(t) \right\rangle. \quad (3)$$

Using the perturbation method and the commutation relationship of the boson operators, Wu and Austin [7] derived the rate equation given by

$$\left\langle \frac{dN_i}{dt} \right\rangle = S_i - \Phi(T,\omega_i)[\langle N_i \rangle e^{\beta\omega_i} - (1 + \langle N_i \rangle)] - \sum_j \Lambda(T,\omega_i,\omega_j)[\langle N_i \rangle (1 + \langle N_j \rangle) e^{\beta\omega_i} - \langle N_j \rangle (1 + \langle N_j \rangle) e^{\beta\omega_j}], \quad (4)$$

where we have $\beta = \hbar/kT$, S is the rate of supply of energy from external source to the ith mode, and Φ depends on the coupling coefficients other than χ.

For the stationary state, we have

$$\left\langle \frac{dN_i}{dt} \right\rangle = 0. \quad (5)$$

Using Eqs. (4) and (5) one may solve for the average occupation number of each phonon state and obtain an expression of the form

$$\langle N_i \rangle \approx (e^{\beta(\omega_i - \mu_i)} - 1)^{-1}, \quad (6)$$

where μ_i is defined as

$$e^{-\beta\mu_i} = (\Phi(T,\omega_i) + \sum_j \Lambda(T,\omega_i,\omega_j)(1 + \langle N_j \rangle)) \times (\Phi(T,\omega_i) + \sum_j \Lambda(T,\omega_i,\omega_j)\langle N_j \rangle e^{\beta\omega_j})^{-1} \quad (7)$$

and

$$\Lambda(T,\omega_i,\omega_j) = (2\pi/\hbar)|\chi|^2 e^{-\beta(\omega_i-\omega_j)} \tag{8}$$

So if we have $\chi = 0$, $\mu_i = 0$, no condensation can take place.

Bose condensation is possible, however, if and only if

$$\omega_i > \mu_i \geq 0. \tag{9}$$

From the rate equation it is obvious that the Bose condensation in the biosystem depends upon the coupling coefficient χ. Wu and Austin [11] have shown that this coupling coefficient dominates the other ones that appear in Eq. (1). Therefore, they employed Green's function technique [8] to derive an expression for the relaxation time τ_1 in terms of χ only

$$\tau_1 = \hbar^3\pi\nu'^3/|\chi|^2 kT\gamma\Delta, \tag{10}$$

where ν' is the speed of sound, k is the Boltzman constant, T is the absolute temperature, Δ is the width of the energy band, and γ is a positive quantity less than 1. From Eq. (10) we estimated a numerical value for $|\chi|$ as 4×10^{-2} by using $\tau_1 \approx 10^{-10}$ s, $\nu' \approx 300$ m/s and $\gamma\Delta \approx 10^6$ as given in Ref. 8.

Measurement of χ from the Dielectric Constant

Since oscillators involved are dipolar in nature it is easy to see that they should leave a remarkable effect upon the dielectric properties of the system. If a medium composed of dipolar entities is subjected to a rapidly changing electromagnetic field the dipoles will oscillate with the field but if they are coupled to a bath, then the response will be modified by this coupling. This modification will appear in the form of a phase lag which is related to the imaginary part of the dielectric constant $\epsilon''(\omega_0)$ and therefore $\epsilon''(\omega_0)$ is related to the parameter χ. The precise details of this calculation are based upon the continued fraction technique developed by Paul [9] involving a two-body Green's function. These details will be published elsewhere but for the present we are providing a simplified version of the final equation in order to estimate χ from experimental measurement of the dielectric constant.

The expression for the dielectric constant derived by us is given by

$$\epsilon''(\omega_0) = \frac{4\pi^3\hbar^2\rho|\mu|^2\omega_0(N_i - N_j)^2}{|\chi|^2 kT\Gamma(\omega_0)}, \tag{12}$$

where ρ is the density of the material, $|\mu|$ is the mean dipole moment, ω_0 is the resonant frequency, and $\Gamma(\omega_0)$ is the linewidth. Since the relaxation time is inversely proportional to $\Gamma(\omega_0)$ and in our derivations the latter contains exponentials of $(-h\omega_0/kT)$, thus the relaxation time will be proportional to the exponent of temperature's inverse.

Discussion and Results

Supekar and Bandopathyaya [12] provided some experimental data for cholesterol

at and around two resonant frequencies which are given below in the SI units

$$\rho = 7.5 \text{ kg/m}^2, \quad T = 300°\text{K},$$
$$\omega_{01} = 0.63 \text{ MHz}, \quad \Gamma(\omega_{01}) = 0.92 \text{ MHz},$$
$$\epsilon''(\omega_{01}) = 0.270 \text{ F/m},$$
$$\omega_{02} = 2.19 \text{ MHz}, \quad \Gamma(\omega_{02}) = 3.16 \text{ MHz},$$
$$\epsilon''(\omega_{02}) = 0.265 \text{ F/m}.$$

Using the above values, we obtained from our equation (12) $|\chi/\hbar| = 5 \times 10^{-3}$ (experimental accuracy is 2%) for both resonant frequencies. This shows that the coupling coefficient indeed does not depend on the frequency. This value of $|\chi/\hbar|$ agrees satisfactorily with the value found from formula (10) of Wu and Austin which is $|\chi/\hbar| \approx 4 \times 10^{-2}$.

Detailed measurement of the dielectric constant of the biosystem in microwave frequencies are in progress in our EPR laboratory.

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COMPUTATIONAL CONNECTIONISM WITHIN NEURONS: A MODEL OF CYTOSKELETAL AUTOMATA SUBSERVING NEURAL NETWORKS

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"Neural network" models of brain function assume neurons and their synaptic connections to be the fundamental units of information processing, somewhat like switches within computers. However, neurons and synapses are extremely complex and resemble entire computers rather than switches. The interiors of the neurons (and other eucaryotic cells) are now known to contain highly ordered parallel networks of filamentous protein polymers collectively termed the cytoskeleton. Originally assumed to provide merely structural "bone-like" support, cytoskeletal structures such as microtubules are now recognized to organize cell interiors dynamically. The cytoskeleton is the internal communication network for the eucaryotic cell, both by means of simple transport and by means of coordinating extremely complicated events like cell division, growth and differentiation. The cytoskeleton may therefore be viewed as the cell's "nervous system". Consequently the neuronal cytoskeleton may be involved in molecular level information processing which subserves higher, collective neuronal functions ultimately relating to cognition. Numerous models of information processing within the cytoskeleton (in particular, microtubules) have been proposed. We have utilized cellular automata as a means to model and demonstrate the potential for information processing in cytoskeletal microtubules. In this paper, we extend previous work and simulate associative learning in a cytoskeletal network as well as assembly and disassembly of microtubules. We also discuss possible relevance and implications of cytoskeletal information processing to cognition.

1. Cognition: emergent computation in the brain

1.1. Connectionism

The use of connectionist models of functional brain organization has advanced understanding of cognition and linked neuroscience to computer science. "Neural networks", which approximate cortical neurons as parallel processors with variable lateral interconnections, may be simulated on computers and can provide a working model of some aspects of brain function [32, 41]. A key concept relating neural network dynamics to neuroscience was introduced by Hebb [40]. He hypothesized that spatially organized neural networks called "cell assemblies" function as reverberatory circuits which constituted elements of thought. Gazzaniga's [30] "modules", Minsky's [55] "agents", Freeman's [26] "cartels" and other conceptualized functional entities are examples of more recent, comparable proposals for anatomic neural networks. In Hebb's view, an individual neuron could participate in many cell assemblies just as an individual member of society may participate in many social groups. By strengthening or reinforcing repeatedly used connections ("synaptic plasticity"), Hebb suggested that recognition, learning and problem solving occurred

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through lowered thresholds of specific loops. By assigning energy levels to threshold loop patterns ("landscapes"), mathematical solutions could be applied to neural net configurations [41].

There are, however, at least two inconsistencies in the analogy between neural nets and brain function. The first is that neurons and their synapses are extremely complex and by themselves resemble computers more than fundamental switches. Accordingly, further attempts to approach mechanisms of brain cognition need to consider dynamic regulatory activities within each neuron. The second is that a number of important artificial neural networks require "back propagation": tuning of internal parameters from output conditions. Both of these inconsistencies may be resolved by consideration of the cytoskeleton, an intracellular parallel network of protein polymers which regulate synapses and performs other important functions.

We contend that rudimentary information processing occurs within and among intraneuronal cytoskeletal elements such as microtubules (MTs) (fig. 1). By subserving synaptic connectionism, such information processing can provide a lower level in a hierarchy of cognitive processes from whose highest level emerges consciousness. Further, retrograde patterns travelling through the cytoskeleton could serve a role analogous to back-propagation in some artificial neural networks. In sections 4 and 5, we describe simulation of such retrograde patterns travelling through microtubules.

In the cytoskeleton, filamentous bridges among parallel MTs and/or neurofilaments could serve functions comparable to the recognition automata of Reeke and Edelman [61]. To take maximal advantage of parallelism, these authors have modelled two parallel automata which communicate laterally and have distinct and complementary personalities. One model automaton ("Darwin") is highly analytical, keyed to recognizing edges, dimensions, orientation, color, intensity, etc. The other ("Wallace") is more "gestalt" and attempts to merely categorize objects into preconceived

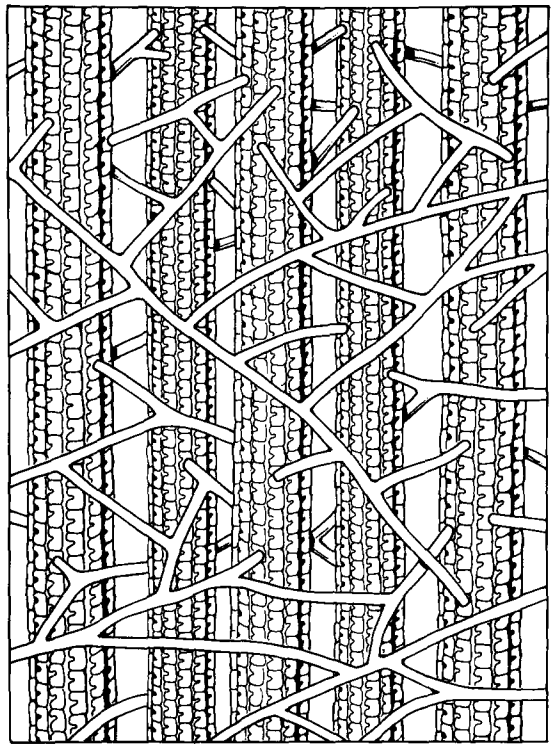

Fig. 1. Connectionist network of five parallel arrayed microtubules (MTs) interconnected by lateral crosslink filaments. Drawn by Fred Anderson from electron micrograph of neuronal dendrite cytoplasm (Aoki and Siekevitz [5]).

classifications. Lateral communications between the two parallel automata resolve conflicting output and form an associative memory. In the cytoskeleton, filamentous bridges among parallel MTs and/or neurofilaments could serve comparable functions.

Multi-level nets are consistent with connectionist brain models in which neural net "assemblies", "modules", etc. fit in an overall cognitive hierarchy of parallel layers of brain/mind organization [13, 20]. The highest cognitive level in such models is global brain function which correlates with awareness, thought, or "consciousness" (although some would argue for even higher-level social consciousness among people, political parties, societies, etc. [42, 59]). The second level appears to be comprised of anatomically and functionally recognizable brain systems and centers (i.e. respiratory

center, satiety center, etc. [66]). At intermediate levels, maps such as the motor and sensory homunculi represent the body and outside world. Finally, the lowest level is thought to be the neural synaptic network, module or cartel which may operate cooperatively by utilizing dense interconnectedness, parallelism, associative memory and learning due to synaptic plasticity.

The lowest level of brain/mind organizational hierarchy, and potential correlate of the binary switch in computers, is generally considered to be the neuronal synapse. However, we suggest that information processing in the cytoskeleton could provide a basement level to the cognitive hierarchy. The cytoskeleton is a parallel, interconnected network of microtubules, actin, and intermediate filaments which also connects to membrane proteins, cell nuclei, centrioles and other structures. Capabilities for dynamic organization coupled with the cytoskeleton's grid-like lattice structure have prompted at least a dozen author groups to model microtubules and other cytoskeletal structures as information processing devices [35]. In neurons, synaptic connectionism and regulation (plasticity) depend on cytoskeletal functions as do other cognitive processes. Information representation, transduction, translocation (including retrograde signaling suitable for "back-propagation") and computation in the cytoskeleton may function just below synaptic neural networks in a cognitive hierarchy.

The next sections will describe: (1.2) evidence linking the cytoskeleton to computation, (2) a biochemical overview of microtubules and other cytoskeletal elements, and (3) models of cytoskeletal information processing. In section 3.3, our previously published model of molecular automata in microtubules [39] is reviewed. Section 4 describes a network model of connected microtubule automata capable of learning and association and section 5 covers an assembly/disassembly microtubule automata model relevant to cell division, growth and plasticity. Section 6 concludes with a discussion of the potential implications of cytoskeletal computation.

1.2. Cytoskeleton and computation

The concept of cytoskeletal computation is based on the observation that cytoplasmic interiors of living cells, particularly nerve cells, are not watery soups. The current picture of cell interiors is one of a highly organized network in which cell water is bound and governed by cytoskeletal structures rather than a freely diffusing solution of macromolecules. Two lines of evidence support this concept. The first is the cytoskeleton which interpenetrates the cytoplasm and defines its architecture and function. Some cytoskeletal components are in polymerization equilibrium, interconverting cytoplasm between "sol" (liquid, solution) and "gel" (gelatinous, viscous) states. This "sol/gel" equilibrium (mediated by Ca^{2+} and Mg^{2+} effects on cytoskeletal actin and other proteins) indicate that the interiors of living cells operate very close to the solid–liquid phase transition. Theoretical findings predict optimal computational capabilities for distributed dynamical systems near this phase transition [47, 58].

Another line of evidence supporting the possibility of a computational cytoplasm includes data from nuclear magnetic resonance (NMR), neutron diffraction and other techniques which show a high degree of bound water in the cytoplasm. Water molecules abutting cytoskeletal (and other) cytoplasmic surfaces are attracted to these surfaces and are restricted in their motions. As a result, a greater proportion has four (rather than three or less) hydrogen bonds with their neighbors. Such water is called "vicinal" water, and has unusual properties compared to "bulk" water: lower density, greater heat capacity and greater viscosity. Further, such vicinal water may cooperatively oscillate with coherent excitations in cytoskeletal proteins (section 3.2).

Further circumstantial evidence for cytoskeletal computation and communication stems from the spatial distribution of discrete sites/states in the cytoskeleton. For example, subunits in microtubules have a "density" of about 10^{17} per cm^3, very close to the theoretical limit for charge sepa-

ration [34]. Thus, cytoskeletal arrays have maximal density for information storage via charge, and the capacity for dynamically coupling that information to mechanical and chemical events via conformational states of proteins (i.e. Fröhlich's mechanism of dipole excitations coupled to conformational state).

The dynamic complexity of neurons and their synaptic connections suggest that they utilize "lower" layers which may include dendritic processing, branch point conductance and molecular dynamics [14, 63]. Lynch and Baudry [51] have studied long-term potentiation ("LTP") of synaptic efficacy in glutamate–NMDA hippocampal neurons, a correlative model of learning. They find that LTP depends on rearrangement of subsynaptic cytoskeleton – a dynamic network of parallel arrayed, interconnected proteins including microtubules, actin, and intermediate filaments. Desmond and Levy [21] have studied dendritic spine structural changes during a synaptic learning paradigm. They find mechanical shape changes in spines mediated by cytoskeletal actin connected to microtubules in dendrites. Aoki and Siekevitz [5] have studied dendritic spine synaptic plasticity in development and find that synaptic down-regulation occurs in conjunction with depolymerization of microtubules in the main dendrite. They find also that signaling and regulation for dendritic spine synapse function depends on phosphorylation of a cytoskeletal protein: MAP2. Microtubule associated proteins (MAPs) attach to and connect microtubules to other microtubules, actin, intermediate filaments, organelles, membrane proteins and cell nucleus. MAP2 is found only in neuronal dendrites, and consumes a large proportion of phosphorylation energy. MAP2 phosphorylation is regulated by cyclic AMP-dependent protein kinase and calcium–calmodulin protein kinase. Theurkauf and Vallee [68] have found MAP2 to be "the major substrate for endogenous cyclic AMP-dependent phosphorylation in cytosolic brain tissue", and concluded that MAP2 phosphorylation "may be an important reaction in response to neurotransmitter stimulation". Thus, numerous mechanisms exist for coupling membrane synaptic events and cytoskeletal activities.

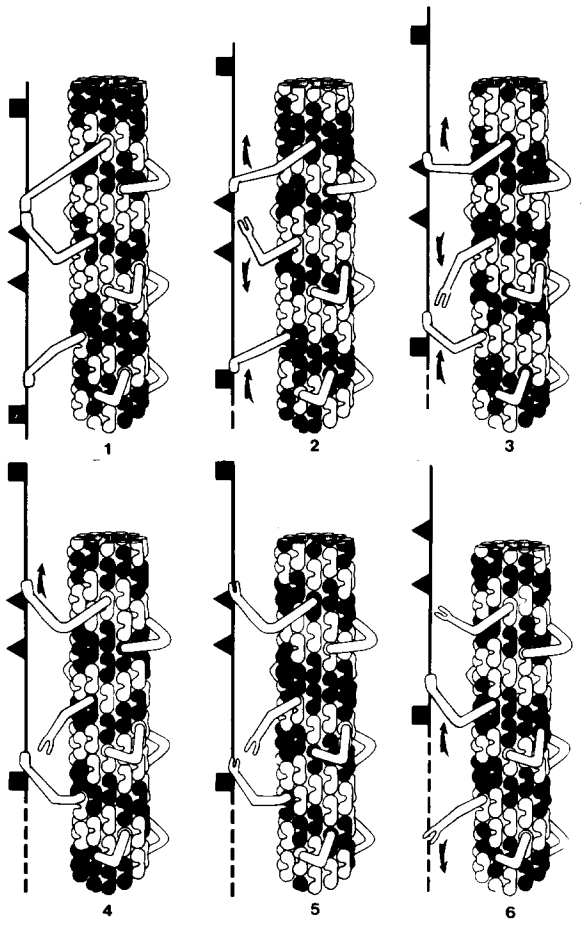

Fig. 2. Axoplasmic transport by coordinated activities of microtubule attached contratile (MAP) proteins ("dynein"). The orchestration mechanism is unknown, but shown here as the consequence of traveling conformational patterns on the MTs. By Fred Anderson.

Maintenance and modulation of synaptic function also depend on axoplasmic transport, an MT based conveyor system which supplies structural components to synapses [48, 57]. Contractile "sidearm" MAPs attached at specific sites on microtubules pass materials in a cooperative bucket brigade which achieves a velocity of 400 mm/day (fig. 2). The contractile sidearms hydrolyze ATP for energy, but the mechanism of orchestration and signaling is unknown. Material can travel in

opposite directions along a single microtubule. Other organized activities in cytoskeletal structures help control the molecular machinery of cell division, growth, differentiation, formation of synapses and dendritic spines, as well as relatively complex functions within single cell organisms like amoeba and paramecium which occur without benefit of brain, neuron, or synapse [22, 65].

Evidence for direct cytoskeletal involvement in cognitive processes comes from Mileusnic et al. [54], who have correlated production of MT subunit protein ("tubulin") and microtubule activities with peak learning, memory and experience in baby chick brain. Cronly-Dillon and co-workers [16] have shown that when baby rats first open their eyes, neurons in visual cortex begin producing vast quantities of tubulin. When the rats are 35 days old and the critical learning phase is over, tubulin production is drastically reduced. Other studies have shown that cytoskeletal proteins directly link to nerve membrane ion channels and receptors [46, 70] and that the intra-neuronal cytoskeleton is linked to nerve membrane excitability and synaptic transmission [2, 53]. Direct evidence for signal propagation in MT has been generated by Vassilev et al. [71], who suspended parallel excitable membranes in ionic solution. Only when two membranes were connected by MTs, excitation in one membrane provoked excitation in the other. The authors suggested that similar communication signals occurred routinely within the cytoskeleton. Also, Becker et al. [10] studied energy resonance transfer among fluorescent groups separately attached to different MT subunits or to membranes. They showed energy resonance transfer occurs both among MT subunits and among MT subunits and membrane proteins.

The intra-neuronal, sub-synaptic cytoskeleton is important to neural function including synaptic plasticity. In assessing cognition, low-level processing may be important. Because the cytoskeleton comprises a connectionist network within each branching neuron, it may be described as a cytoskeletal forest within each dendritic tree.

2. Microtubules and the cytoskeleton

2.1. Cytoplasm

Neurons and other cells are comprised of protoplasm, which in turn consists of membranes, organelles, nuclei, and the bulk interior medium of living cells: cytoplasm. Nineteenth century light microscopists described cytoplasm as containing or consisting of "reticular threads", "alveolar foam", or "watery soup". Development of the electron microscope through the 1960's did not initially illuminate the substructure of cytoplasm because the commonly used fixative (osmium tetroxide) was dissolving fine structure and the cell was still often perceived to be a "bag of watery enzymes".

In the early 1970's, with the event of glutaraldehyde fixation, delicate tubular filamentous structures were found in virtually all cell types and they came to be called microtubules. Originally thought to provide merely structural, or bone-like support, MTs and other filamentous structures such as actin, intermediate filaments and centrioles were collectively termed the "cytoskeleton". Recognition that complex dynamic activities of microtubules and other cytoskeletal elements were essential for organization, movement and growth of cellular cytoplasm finally brought the cytoskeleton out of the closet.

2.2. Cytoskeletal networks

Bulk cytoplasm contains networks of individual microtubules, arrayed in parallel and interconnected by filamentous strands. Other interconnecting networks of smaller filamentous proteins (actin, intermediate filaments, microtrabecular lattice, etc.) intersperse with MTs to form a dynamic gel whose activities in all types of cells (i.e. mitosis, growth and differentation, locomotion, food ingestion or phagocytosis, synapse modulation, dendritic spine formation, cytoplasmic movement, neurotransmitter release, etc.) are essential to the living state [18]. Orientation and directional guid-

ance of these cytoskeletal functions depend on centrioles: assemblies of nine pairs or triplets of microtubules arranged in replicating super-cylinders always found in pairs oriented perpendicular to each other. To initiate cell mitosis, centriole pairs migrate to opposite poles of the cell from where MT "mitotic spindles" separate chromosomes and establish orientation and architecture for the next generation cells [17, 19].

2.3. Microtubules

Of the various filamentous structures which comprise the cytoskeleton, microtubules are the most prominent, best characterized and appear best suited for dynamic information processing [22, 65].

MT are hollow cylinders 25 nm in diameter whose lengths may span meters in some mammalian neurons. MT cylinder walls are assemblies of 13 longitudinal protofilaments which are each a series of subunit proteins known as tubulin (fig. 3). Each tubulin subunit is a polar, 8 nm dimer which consists of two slightly different classes of 4 nm, 55 kilodalton monomers known as α and β tubulin. Genes for α and β tubulin are complex, multi-gene families which give rise to varying tubulin isozymes. During evolution some tubulins have been highly conserved for basic cell functions. However, extensive microtubule heterogeneity exists in complex cells due to genetic diversity, expression, post-translational modifications, MAPs, and assembly patterns [24, 31]. For example, two-dimensional gel electrophoresis has shown 17 different β tubulin isozymes exist in mammalian brain MTs, whereas fewer exist in other tissues [49].

The tubulin dimer subunits within MTs are arranged in a hexagonal lattice which is slightly twisted, resulting in differing neighbor relationships among each subunit and its six nearest neighbors (fig. 4).

MT self-assembly and disassembly are dynamic, complex processes which depend on various factors including temperature and calcium ion concentration. Oriented by centrioles, MT polymerization determines the architecture and form of cells which can quickly change by MT depolymerization and reassembly in another direction [45]. GTP, an energy-providing analog of ATP, binds to polymerizing tubulin; GTP hydrolysis energy is subsequently delivered to assembled MTs. MTs whose ends are GTP tubulin are stable and will continue to grow. After the GTP is hydrolyzed to GDP, GDP tubulin is exposed at MT ends and unless stabilized by MAPs, centrioles or other structures, MTs rapidly disassemble with their component subunits being utilized by assembling MTs. This dynamic instability results in a "selectionist" activity in which MT networks can probe or retreat in cellular appendages including dendritic synapses. The precise mechanism of consumption of GTP hydrolysis energy in MTs is not understood, although one possible utilization is the production of solitons or coherent lattice excitations as proposed by Fröhlich [27–29]. It is known that GTP tubulin and GDP tubulin have different conformational states [8].

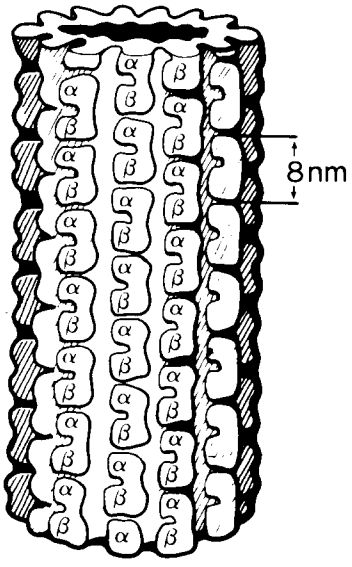

Fig. 3. Microtubules (MTs) are cylinders whose walls are 13 protofilaments, each a string of 8 nm tubulin dimers. Alpha and beta tubulin monomers form the dimers; each dimer has 6 neighbors. Drawn by Fred Anderson from X-ray crystallographic data of Amos and Klug [4].

When viewed in cross section by electron microscopy, MT outer surfaces are surrounded by a "clear zone" of several nm, which apparently represents an electronegative field due to excess electrons in tubulin and may also serve to organize cytoplasmic water and enzymes [67]. MTs, as well as their individual dimers, have dipoles with negative charges localized toward α monomers [17]. Thus MTs are "electrets": oriented assemblies of dipoles which are predicted to have piezoelectric properties [7, 52]. Contractile or enzymatic MAP proteins may be attached to MTs at specific dimer sites and MAP attachments can result in various helical patterns on MT surface lattices [12]. Contractile MAPs participate in axoplasmic transport while other MAPs form bridges which laterally connect specific subunits on parallel arrayed MTs.

3. Models of microtubular information processing

3.1. Cytoskeletal signal processing

The lattice polymer structure of MTs have suggested capabilities for information processing, and numerous author groups (some of which are mentioned below) have published theoretical models of MT/cytoskeletal information processing.

Atema [6] proposed that cellular sensory transduction occurred by propagated conformational changes in MT subunits. His view was an "all or none" propagation in which entire MTs were essentially trigger switches.

Barnett [9] proposed that filamentous cytoskeletal structures operated like information strings analogous to word processors. He proposed that MTs are processing channels along which strings of information can move, and that neurofilaments (intermediate filaments within neurons) are parallel arrayed memory channels. While not specifying the mechanism of pattern representation within MTs or neurofilaments, Barnett's model emphasizes the potential utility of parallel arrays of laterally interconnected molecular structures.

A model which does address patterns within MTs was proposed by Roth and Pihlaja [62]. They suggested conformational states of tubulin subunits within MTs were regulated by binding of inter-MT linkages of MAPs. They assumed five possible conformational states for each tubulin subunit (based on drug binding studies) could be induced by proximity of MAP binding and cooperative allosteric effects.

Hameroff and Watt [36] proposed that conformational states of tubulin within MTs were coupled to charge or dipole states, and interacted cooperatively with neighbor tubulin states. Such interactions were thought to be based on "molecular automata" in which the conformational states of individual tubulin dimer subunits represent "bits" of information subject to dynamic influence by neighboring subunits [36–39, 64]. The control of protein conformation is, by itself, an interesting enigma. Before considering molecular automata as a specific model of MT information processing, some aspects of protein conformational regulation theory will be mentioned.

3.2. Coherent protein dipole excitations

Proteins are vibrant, dynamic structures in physiological conditions. A variety of recent techniques have shown that proteins and their component parts undergo conformational motions over a range of time scales from 10^{-15} s to many minutes with the functionally most significant conformational vibrations appearing in the "nanosecond" range of 10^{-9}–10^{-11} s. Biologically relevant motions of globular proteins in this time scale are thought to include collective elastic modes, e.g. solitons and other nonlinear motions, and coherent excitations [43, 44, 60]. These motions are global changes in protein conformation rather than more rapid thermal fluctuations of side chains or local regions. Such global protein conformations appear suitable for computation: finite states which can be influenced by dynamic neighbor interactions.

need to include these

Collective conformational states near the nanosecond time domain have been woven into a theory of coherent protein excitations by Fröhlich [27–29]. Changes in protein conformation, according to Fröhlich [28], are triggered by charge redistributions such as dipole oscillations within specific hydrophobic regions of proteins. Another concept of Fröhlich [27] is that a set of proteins connected in a common physical structure and electromagnetic field such as within a polarized membrane (or polymer electret like a microtubule) may be excited coherently if chemical energy such as ATP or GTP were supplied. Coherent excitation frequencies of the order of 10^9–10^{11} Hz are deduced by Fröhlich, who cites as evidence sharp windows of sensitivity to electromagnetic energy in this region by a variety of biological systems [33]. Circumstantial evidence for Fröhlich's activities in microtubules include the apparent electronegative field surrounding MTs, and the unexplained consumption of GTP hydrolysis energy by MT lattices. Other aspects of Fröhlich's model include metastable states (longer-lived conformational state patterns stabilized by local factors) and polarization waves, i.e. travelling regions of conformational states slightly out of phase with the majority of coherently excited states.

3.3. *Molecular automata in microtubules*

Conrad and coworkers [15] introduced the idea of "molecular" automata within neurons as a basis for intracellular information processing related to cognition. We believe molecular automata activities in cytoskeletal structures such as microtubules could explain much of their organizational activities.

introduction for automaton behavior

Automaton behavior including dynamic patterns and capabilities for computation require a lattice whose subunits can exist in two or more states at discrete time steps, and transition rules which determine those states among lattice neighbor subunits. Fröhlich's [27] model of coherent excitations and cooperative coupling among proteins arrayed in an electromagnetic field may be applied to tubulin subunit dimers within MTs to serve as a "clocking" mechanism and define discrete generations for automaton behavior. We can obtain an estimate for the clocking frequency assuming one coherent "sound" wave across the MT diameter ($\Phi \simeq 25$ nm). A crude calculation, assuming that $V_{\text{sound}} \simeq 10^3$ m/s, yields a clocking frequency of approximately 4×10^{11} Hz for one wave across the MT diameter. This is in the predicted range and correlates with a clocking generation time of 2.5×10^{-11} s. However, faster and slower automata are possible, depending on the type of MT excitation being considered.

Orientation or phase of any tubulin subunit dimer relative to the coherently oscillating majority at any excitation period would depend on factors such as initial conformational states, binding of water, ions, or MAPs, bridges to other MTs, tyrosinated/glutamated state, energy-providing phosphate nucleotides (i.e. GTP) and associated proteins, genetically determined subunit factors and electrostatic dipole interactions among neighboring subunits. Many of these factors may be considered programming modes (either hereditary or environmental) whose net effects would serve to alter phases of particular subunits in coherently excited arrays. To simulate MT molecular automata, we will focus solely on electrostatic dipole interactions among MT subunits.

Our derivation of automaton neighborhoods and neighbor dynamics for computer simulation are illustrated in figs. 4 and 5. The MT cylinder (25 nm in diameter) has a circumference of 13 tubulin subunit dimers and the pitch pattern (angle from horizontal) of the leftward helix is one and one-half dimers. For automaton simulation we consider seven-member neighborhoods of tubulin dimers: a central dimer surrounded by a tilted hexagon of six neighbor dimers. Taking the cylinder axis of the MTs to define the y-axis, the dimer neighborhood is defined as follows (fig. 4): "C" is the center dimer and "N" (north) is the nearest-neighbor dimer in the positive direction of the y-axis. Similarly, dimers labeled "NE" (northeast), "SE" (southeast), "S" (south), "SW" (southwest),

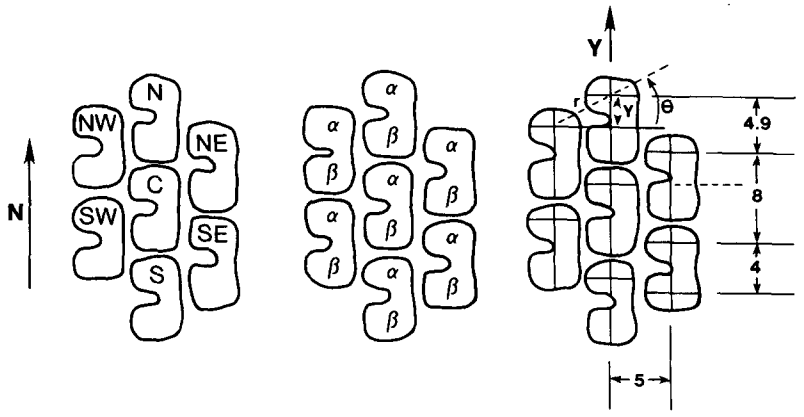

Fig. 4. Microtubule automata (MTA) neighborhood. Left: Definition of neighborhood dimers. Center: α and β monomers within each dimer are labelled. Right: Distances (in nm) and orientation among lattice neighbors. Interaction forces are calculated using $y_i = r_i \sin\theta$.

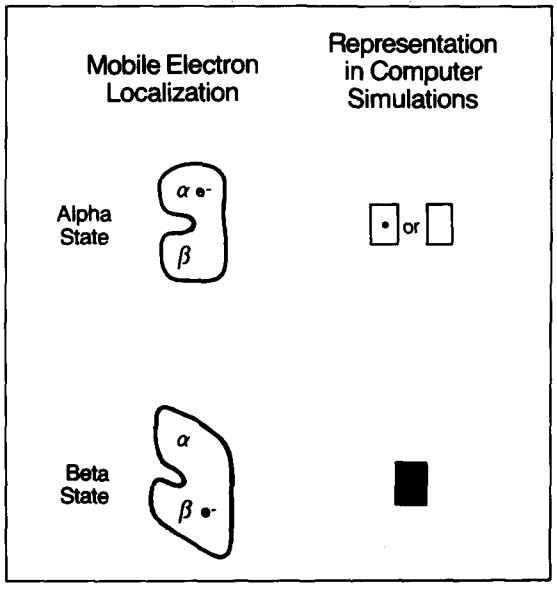

Fig. 5. Switching between conformational states in microtubule (MT) dimers. Top row: α states, bottom row: β states. Left: dimer conformation coupled to mobile electron localization, Right: representation of states in computer simulation.

and "NW" (northwest) are appropriately oriented around the "C" dimer. Within each dimer the α monomer is north of its β partner. The helical twist yielding an offset along the y-axis for NW, SW, NE, and SE neighbors leads to neighbor dimer distances and dipole interaction forces which differ among the six neighbors surrounding the C dimer.

To calculate the neighbor dipole interaction forces as a basis for automaton transition rules, each dimer may be viewed as having a mobile electron shared by the two monomers (fig. 5). At each time step the electron's average position is considered to be oriented either more toward the α monomer ("alpha state") or more toward the β monomer ("beta state") with associated changes in dimer conformation. Experimental evidence [69] suggests that a conformational shift of 29° from vertical may occur. Because MT net dipoles are negative toward the α ends, oscillations need not necessarily change direction totally to effect a conformational change. For example, a relative dipole reorientation toward the α monomer may be sufficient to induce a tubulin dimer conformational state. For our calculations, however, we have assumed that electrons localize in the centers of the two monomers.

Considering the MT lattice geometry, the y-component of the resulting electrostatic force acting on an electron in a central dimer can then be calculated as:

$$f_{\text{net}} = \frac{e^2}{4\pi\epsilon} \sum_{i=1}^{6} \frac{y_i}{r_i^3}, \qquad (1)$$

Table 1
Relative forces ($= -1000y/r^3$) for neighbor configurations. Net forces are summation of six neighbors.

Neighbor position	Central dimer α		Central dimer β	
	neighbor α	neighbor β	neighbor α	neighbor β
north	+15.625	+62.500	+6.944	+15.625
northeast	+15.205	−7.022	+9.635	+15.205
southeast	−14.250	−8.338	−7.022	−14.250
south	−15.625	−6.944	−62.500	−15.625
southwest	−15.205	−9.635	+7.022	−15.205
northwest	+14.250	+7.022	+8.338	+14.250

where y_i and r_i are defined as illustrated in fig. 4, e is the electron charge, and ϵ is the average permittivity for MT proteins, typically 10 times the vacuum permittivity [50]. We have assumed that only the y-component of the interaction forces are effective and have neglected any net force around the MT circumference.

Relative neighbor dipole coupling forces calculated from eq. (1) and MT geometry are shown in table 1. These relative forces, which we use for transition rules, may be multiplied by 2.3×10^{-14} N to obtain calculated absolute values of intersubunit forces.

To simulate MT automata, we represent MT structure as a two-dimensional grid in which the cylindrical MT has been fileted open and flattened (figs. 6 and 7). The grid consists of MT subunit dimer loci which can exist in either an α state (blank or blank with dot) or β state (solid black). At each time step (generation), resulting neighbor forces are calculated for each dimer in the grid. The extent to which each dimer is influenced by neighbor forces acting upon its mobile electron is represented by a "threshold" parameter. The higher the threshold, the greater are the summated neighbor forces necessary to induce a transition. For example, a threshold of ± 9.000 means that net neighbor forces greater than $+9.0 \times 2.3 \times 10^{-14}$ N will induce an α state, and negative forces of less than $-9.0 \times 2.3 \times 10^{-14}$ N will induce a β state.

Biological factors which might affect the threshold include temperature, pH, voltage gradients, ionic concentration, genetically determined vari-

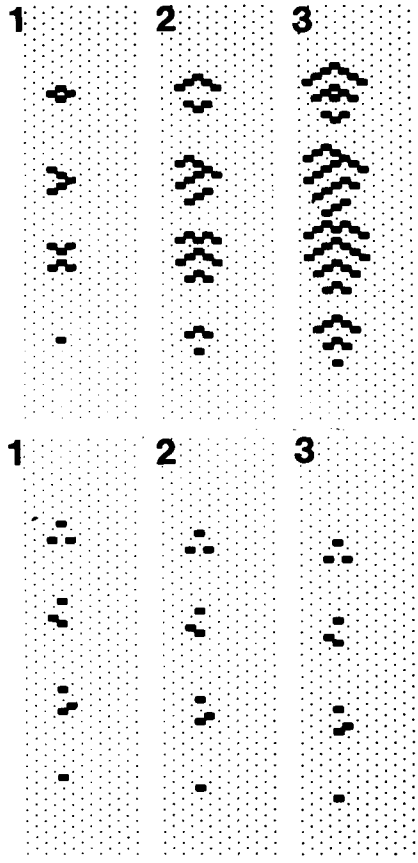

Fig. 6. Some virtual microtubule automata existing for two different threshold values. Top: objects in left panel are labelled (top to bottom) as diamond blinker, triangle glider, spider glider and dot glider. The middle and right sided panels in the top show the next two generations at threshold ± 1.000. The objects either blink or move and leave "wakes" which lead to travelling wave structures. Bottom: the first three generations of some other gliders with threshold ± 9.000. At this higher threshold, the gliders travel without creating a wake or travelling waves.

ability in individual dimers, and binding of molecules including MAPs and/or drugs to dimer subunits. In nerve cells, travelling membrane depolarizations could induce transient waves of lowered threshold along parallel arrayed MTs. Consequently, the frequency of depolarization for a particular nerve cell would directly influence the elaboration of patterns within that nerve cell's MT automata.

In the MT automata of Smith, Watt, and Hameroff [64] and Hameroff, Smith and Watt [37, 38] boundary conditions were chosen in which reflection of automata patterns occurred at each end of simulated MTs. In more recent studies using toroidal boundary conditions [39], we have investigated effects of varying thresholds on MT automata behavior as well as the effects of asymmetrical thresholds in which α and β transition thresholds differ. The simulations have demonstrated behaviors of both a general nature (e.g. virtual automata, gliders and blinkers) similar to behavior in other automata systems, and of a specific nature (e.g. perturbation stable gliders, wedge patterns, NE–SW orientations) derived from unique MT properties. Threshold-dependent behaviors include gliders (dot, spider, bus and giant gliders), travelling and standing wave patterns, blinkers, linearly growing patterns ("bean sprouts"), bidirectional gliders, and frozen patterns. The bifurcations found in the automata dynamics are summarized in fig. 8.

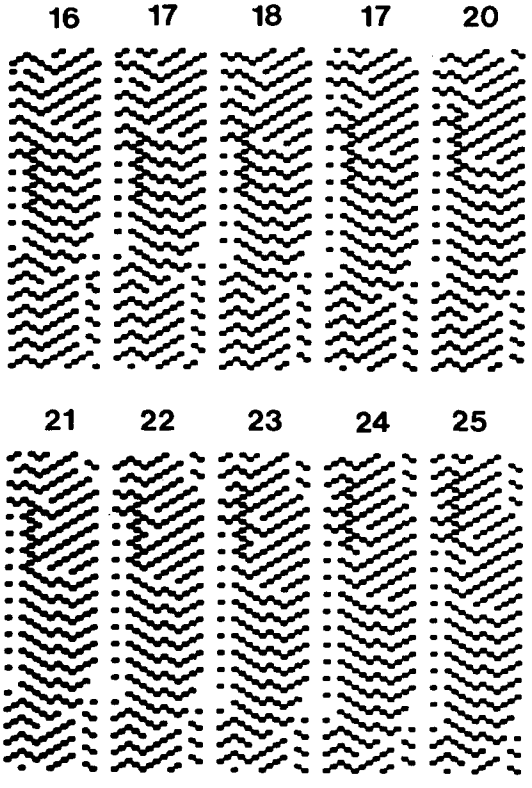

Fig. 7. Time steps 16 through 25 of a microtubule automata simulation with asymmetric threshold, demonstrating localized bilateral information transformation. The initial condition was few randomly scattered α seeds on a β background. The threshold from α to β is −20.000. From β to α the threshold is 2.000. Note that as the β bus (black line of dimers) moves north a new α glider (white line of dimers) is developed in the western region of the neighborhood. This new glider continues to expand until generation 22 when it is "released" from the β bus and starts to travel south together with the other α glider. This dynamical phenomenon may account for the observed bilateral transport on MTs as well as processes related to "back-propagation" in neural networks.

Having demonstrated the potential capability of MT automata for information processing in single MTs, in the present study we have begun to model simple MT networks as well as MT assembly and disassembly.

4. A cytoskeletal connectionist network

The largest portion of free energy used in neuronal dendrite cytoplasm is consumed by MAP2, which actively interconnects dendrite MTs and other structures [68]. A number of empirical studies have also shown that cytoskeletal structures within dendrites change during learning. The relation between cognitive structural changes within neurons and among neurons is enigmatic: which is cause and which is effect? In "true" neural networks (i.e. the brain), the formation of new connections, as well as the breakdown of old ones, is presumably a function of both internal neuronal states (configuration and states of the cytoskeleton, membranes, membrane proteins, second messengers, etc.), as well as signals received from other neutrons. Thus, a multi-level hierarchy of information processing is suggested in biological

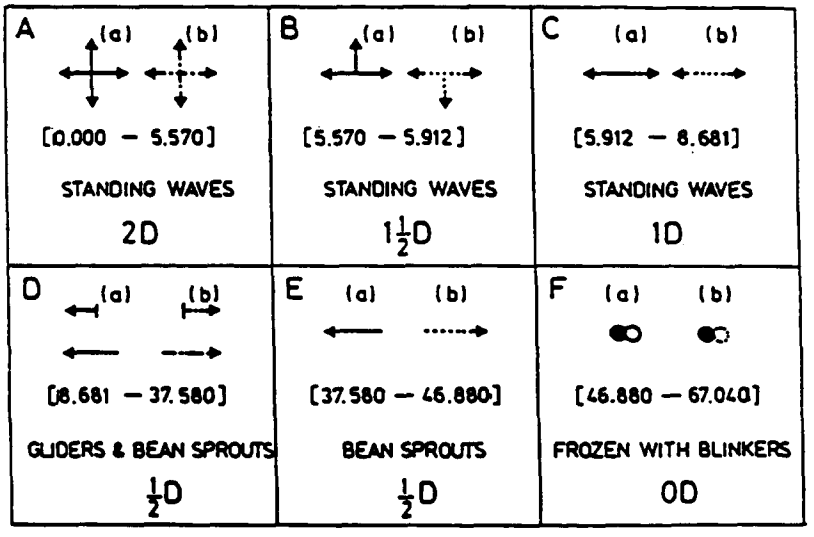

Fig. 8. Bifurcation table for toroidal microtubule automata dynamics with symmetrical dipole coupling thresholds. The upper items in each sector (A)–(F) indicate the dominant global behavior for: (a, solid arrows) β seeds on an α background, and (b, dashed arrows) α seeds induced on a β background. Below the dominant behaviors are the threshold intervals, how we describe each type of dynamic phenomena, and the dimensionality of the growth patterns.

neural networks. To explore this possibility, we have devised an adaptive network based on cytoskeletal structure and function.

Present technologies cannot yet address nanoscale actions of individual protein subunits (although scanning tunnelling microscopy and related techniques may soon be capable). Consequently, there are currently insufficient data to directly test or accurately model such actions and their collective properties.

To merely explore some possibilities, we have used what is known about cytoskeletal structure to serve as the basic architecture for a new connectionist computer. The result is an adaptive dynamical system having certain similarities with the cytoskeleton.

Parallel arrays of MT automata (MTA) are interconnected via MAP-like connections enabling different MTA to communicate with one another. The MAP connections are themselves modeled to be simple automata (MAPA) capable of transmission of signals (sequences of α and β states) from one MTA to another. In the simplest case, this transmission is unidirectional, only transferring signals from a predefined input area on one MTA to a predefined output area on another. Consequently, any tubulin dimer state (α or β) from the MAPA origin on the "input" MTA1 is transmitted to the MAPA target area on the "output" MTA2 (fig. 9).

Conditions which can contribute to dynamic behavior of the MTA network (MTAN) system include MTA thresholds as well as number, attachment locations, and directionality of MAPA. By defining inputs and outputs for each MTA (fig. 10), together with a recursive procedure for updating subunit states, we are able to construct a simple adaptive network.

To model a computationally efficient connectionist network via a feedback between connection topology, internal automata dynamics and automata rules, we use an optimization process often used by Nature: *variation* combined with *selection*. Several systems compete in parallel with respect to a given set of tasks. At each step in the optimization process the most efficient system is selected as the "mother system". At the next step in the optimization process a population of other systems is created as "daughters" or random offspring from the mother system. When one of the offspring begins to perform better than the original system, this offspring is turned into the mother

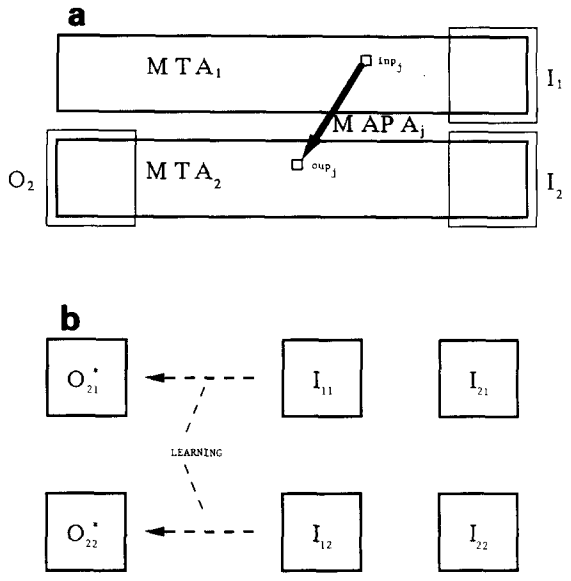

Fig. 9. (a) The simple microtubule automata network, the MTA net, consists of two toroidal MTA of length l dimers, coupled via microtubule associated protein automata, MAPA. The output for the MTA net is defined at the left end of the lower MTA indicated by O_2, and it is taken when the system is updated $l-(13+1)$ times after the input patterns are set at locations I_1 and I_2 at time 0. Input and output areas occupy a matrix area of $13 \times 13 = 169$ dimers. The jth MAPA transmits the dimer states from the input area inp_j at the upper MTA to the output area oup_j at the lower MTA. For each pair of input patterns, I_1 and I_2 at the upper and lower MTA, there exists a desired output pattern O_2^* at the lower MTA. (b) By means of the learning algorithm, the mappings $(I_{11}, I_{21}) \rightarrow O_{21}^*$ and $(I_{12}, I_{22}) \rightarrow O_{22}^*$ are generated.

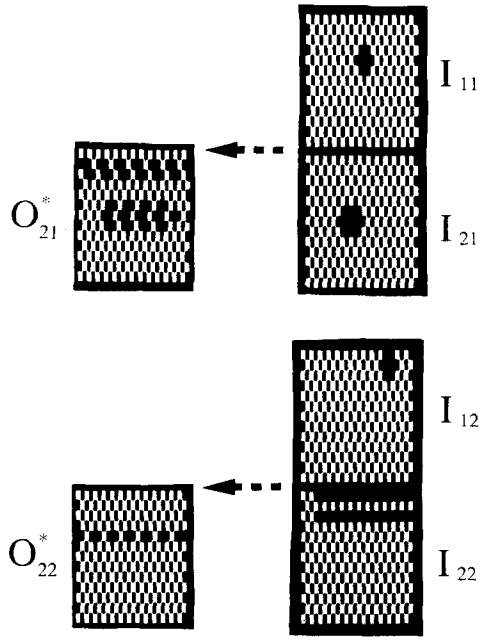

Fig. 10. A MTA network with start configurations (I_{11}, I_{21}) and (I_{12}, I_{22}) and corresponding desired output patterns O_{21}^* and O_{22}^*. These two maps have been chosen from the learnable set for the MTA network. The structure of this set is as yet unknown. The present realization of the MTA network is not able to satisfy every input–output pattern combination.

system as the next step in the optimization process. In this way it is possible to evolve efficient networks with respect to different computational tasks.

In our initial efforts to model a cytoskeletal network, two parallel MTA are capable of being interconnected by MAPA at any subunit loci. Each MTA has an input area at the right end, and output area at the left. Output area states are compared to desired output states using their Hamming distance. For the system to recognize inputs to produce desired outputs, we have random MAPA connections between the two MTA. The learning algorithm for our network is represented both by the threshold parameter for the individual MTA and by the interconnections between the different MTA. A detailed description of the learning algorithm is given in the appendix.

The MTAN obtains truly distributed computation since it is only governed by low-level, neighborhood interactions mediated by intermolecular forces. Such a spatial locality is usually not found in other connectionist models. Another important difference is that the learning algorithm for the MTAN does not use a local learning rule as most artificial neural networks do; rather, MTAN uses an *evolutionary search* to accomplish learning. However, our current MTAN system needs input/output learning in the same sense as most artificial neural net systems do.

In the following example we have fixed the threshold parameters for the upper and the lower MTA at ± 5.900 and ± 9.000 respectively. The corresponding input–output maps used for learning are shown in fig. 10. The connection topology

$C_j = C_2$ for the first accepted input–output map is shown in fig. 11 together with the dynamics for the MTA net at time 0, 43, and 66. The Hamming distance between the desired and the actual map is $H_d = 0$. In fig. 12 the connection topology $C_k = C_4$ for the second input–output map is shown together with the dynamics at time 0, 23, and 66 ($H_d = 0$). It turns out that C_4 also satisfies the first input–output map. We had to redo pass (5) and (6) in the learning algorithm a number of times before the system found a correct C_k satisfying both the maps $(I_{11}, I_{21}) \rightarrow O_{21}^*$ and $(I_{12}, I_{22}) \rightarrow O_{22}^*$.

In the situations where the two pattern combinations can be memorized, there always seem to be several C_k's satisfying the two maps. However, the present realization of the network is not able to memorize every pattern combination at the same time. Some maps cannot coexist.

The MTA net is also able to "associate" or generalize patterns. In figs. 13a and 13b the system tries to associate or recognize the first input pattern, and in 13c and 13d the system tries to associate the second pattern. Note that H_d may be quite high due to a simple horizontal offset, even though the patterns seem very similar. The basis for this associative ability is the existence of relatively perturbation-stable virtual automata in this system. The dynamics of the system is more

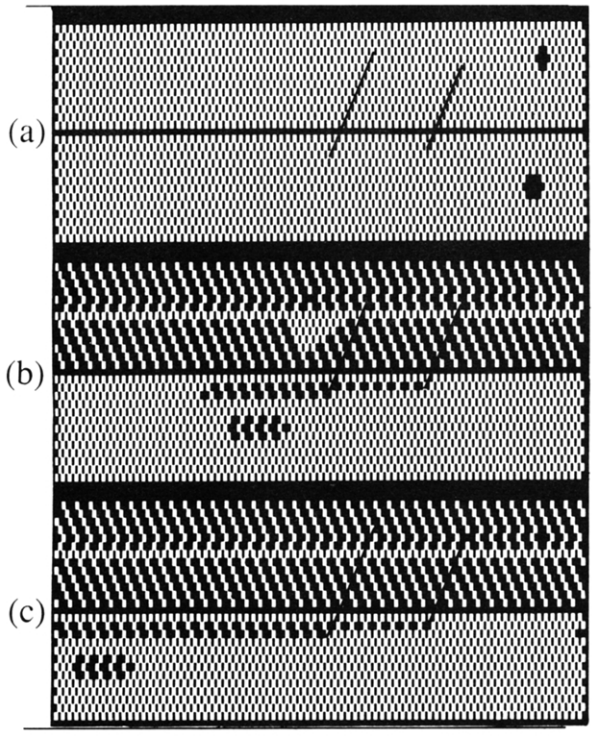

Fig. 11. MTA net at stage (3) in the learning process. The parameters for this net are $l = 80$ and $\text{inp}_j = \text{oup}_j = 1$. The connection topology $C_j = C_2$, with MAPA at the dimer locations $(60,6) \rightarrow (55,2)$ and $(47,4) \rightarrow (41,3)$, turns out to satisfy the first input–output map. The Hamming distance $H_d = |O_{21}^* - O_{21}| = 0$. The dynamics is shown at time: (a) 0, (b) 43, and (c) 66. The thresholds for the upper and lower MTA are ± 5.900 and ± 9.000 respectively.

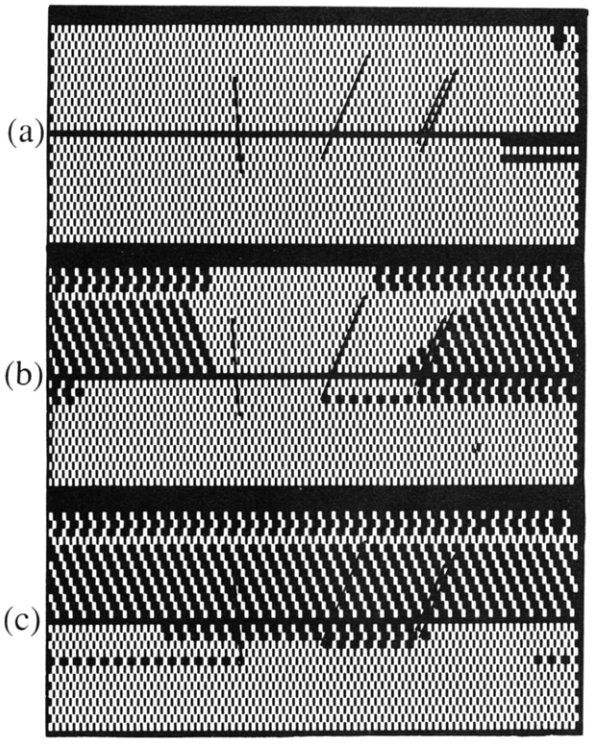

Fig. 12. The same MTA net as shown in fig. 11 at stage (5) in the learning process. $C_k = C_4$ is here given by: $(60,6) \rightarrow (55,2)$, $(47,4) \rightarrow (41,3)$, $(59,7) \rightarrow (55,1)$, and $(28,7) \rightarrow (29,5)$. The dynamics is shown at time: (a) 0, (b) 26, and (c) 66. It can be shown that C_4 both satisfies the first and the second input–output map ($H_d = 0$). However, pass (5) and (6) had to be repeated a number of times, with $D = 10$, before a correct connection topology was found. The thresholds for the upper and lower MTA are still ± 5.900 and ± 9.000 respectively.

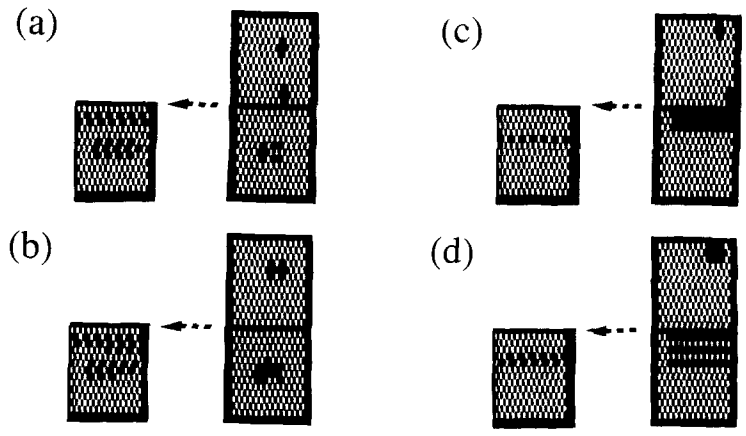

Fig. 13. Association of patterns in the MTA net with connection topology C_4. (a) and (b) are related to the first map, and (c) and (d) are related to the second map. In (a) $H_d = 26$ and in (b) $H_d = 6$ although (a) looks more right than (b). (c) is perfectly associated. In (d) we see a less correctly associated pattern, $H_d = 26$. Compare with fig. 10.

resistant to perturbations in the longitudinal (north–south) MTA direction than in the east–west direction. This property appears due to chosen threshold values which mainly support communication along the north–south axis on the MTA (see the bifurcation table in fig. 8). The reason for the preferred north–south communication for higher thresholds is due to the geometry of the MT lattice. The east–west interactions between most of the dipole configurations are simply cut off for higher thresholds. For example, table 1 shows relative forces for the 24 configuration pairs occurring. None of the pairwise east–west interactions exceed an absolute value of the relative force larger than $15.205 \times 2.3 \times 10^{-14}$ N. The pairwise north–south interactions have absolute values up to $62.500 \times 2.3 \times 10^{-14}$ N.

Selection of specific asymmetric threshold values which differ between the two connected MTA as well as selection of different background states in the two connected MTA can lead to significant behavior. For example, a larger absolute positive threshold in MTA1 and a larger absolute negative threshold in MTA2 cause north → south direction gliders in MTA1 which induce, by MAPA connections, south → north gliders in MTA2. Another example of retrograde information transport is a system with MAPA connections between MTA in different ground states. This is possible since β-gliders always travel south on an α background and α-gliders always travel north on a β background (for details on the dynamics in a single MTA, see ref. [39]). Such retrograde signaling could implement "back-propagation" within neural net neurons. MTs from separate dendritic branches merge and interconnect in the dendrite; thus dendritic MTA nets can merge information from multiple dendritic branch synapses. "Up-regulation" due to increased activity in one synaptic area could thereby "down-regulate" relatively inactive synapses on separate dendritic branches.

Although our system presently is able to adapt to certain dynamics, the learning process is not optimal for several reasons. These include the basis on which the daughter systems are both constructed and chosen, and because of an inefficient use of the MTA patterns. Also, the Hamming distance is probably not the best index for selection of a connection topology because two identical patterns shifted just for one dimer often result in a huge Hamming distance. This behavior also limits the associative properties.

Several potential improvements could be implemented in the MTA net system. The resolution of

different maps on the same MTA net would increase significantly if the system were able to use "coded" MAPA. A coded MAPA could be an automaton which only transmits if a certain α/β sequence ("an initiator pattern") passes the input area for the MAPA. Similarly, the MAPA should stop the transmission whenever a certain stop pattern passes the input area; thus the system would have properties very similar to the DNA transcription system. Alternatively, the MAPA could be coded to transmit a signal averaged over a certain area in the input zone. Another possible improvement which could reflect true biological activity involves MAPA connection patterns. Rather than random connection topologies, MTA patterns could determine where the MAPA connections should be located. For example, connection sites could be determined by the location of a persistent diamond blinker or some other pattern occupying roughly $2 \times 2 = 4$ dimer subunits. These possibilities have yet not been tested, but indicate a potential for adaptation and learning in structures based on cytoskeletal networks. Future work will explore these areas.

Assembly and disassembly of MTs, MAPs, centrioles and other cytoskeletal structures determine cell division, differentiation, movement, transport, and (in neurons) synaptic connectivity. Coupling between MTA dynamics and MT assembly/disassembly could explain the orchestration of these elegant and delicate functions. In section 5, we investigate this area.

5. Assembly and disassembly of microtubules

Experimental results of studies on assembly and disassembly of MTs both in vivo and in vitro are summarized in ref. [11]. In order to grow, the microtubule must have dimers containing exchangable GTP at the end. Without this GTP "cap", depolymerization occurs unless the MT is stabilized by other factors such as binding to other cytoskeletal structures. Thus, free MTs tend to either grow or shrink. These results support the dynamic instability model of MT assembly and disassembly [45]. In a simple cellular automaton model for such a process, each boundary site can be considered to be in one of five different states: as polymerized tubulin dimers with either GTP or GDP, as free tubulin with either GTP or GDP, or as water. The possible transitions are shown in fig. 14.

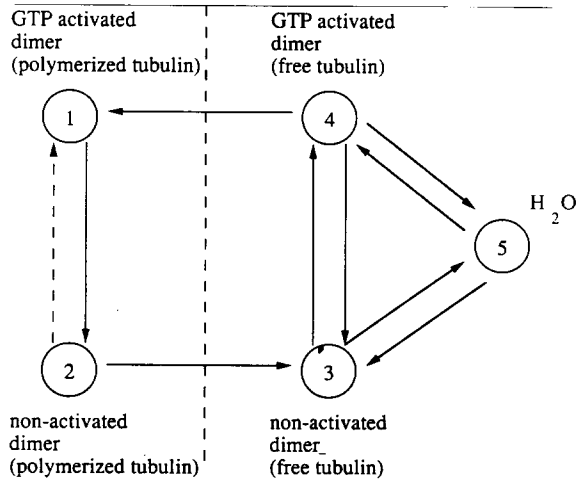

Fig. 14. State diagram for a simple model of assembly/disassembly of microtubules. State 1 is GTP activated tubulin dimer in polymerized form (microtubule geometry), state 2 is microtubule tubulin is non-activated form (GDP), state 3 is non-activated free dimer (in solution), state 4 is GTP activated free dimer, and state 5 is water. The transitions are discussed in the text. Note that this model is formulated partly as a Markov chain, and partly as a cellular automaton.

In the present study, we are concerned with the elongation of already existing tubules, since formation of initial fragments is governed by complex nucleation processes and interactions with centrioles.

We assume that the GTP hydrolysis to form GDP tubulin occurs through a process $(1 \rightarrow 2)$ described by some probability P_{12} at each update of the system (fig. 14). The disassembly $(2 \rightarrow 3)$ of the MTs occurs through dissolving of the non-activated dimers. This process is assumed to depend on the concentration of activated free tubulin [11]: a polymerized boundary dimer will

get free if a major part (three or more) of its neighbors is of the non-GTP free form (including water). Thus, the lower the GTP-activated free dimer concentration, the faster the disassembly. In this model both the concentration of the activated GTP tubulin and the non-activated GDP tubulin are determined by a stochastic process involving internal transitions between the three states: 3, 4, and 5. The higher the overall probability for the presence of an activated dimer among the free states, the more likely is a polymerization process in the system. Polymerization ($4 \rightarrow 1$) takes place if an activated free dimer is aligned with at least two activated polymerized dimers. A necessary condition for polymerization is that the fractional occupation time for activated dimer among the free states is significantly higher than the decay probability P_{12} for the activated microtubule dimer. The transition $2 \rightarrow 1$ is not modeled at this point, but will be discussed later.

Simulations of microtubule assembly and disassembly with the simple model are shown in fig. 15. The clocking frequency in this model is significantly lower than the assumed clocking frequency for signal propagation on the microtubule (10^9–10^{11} Hz). Elongation of existing dentrite MTs in vivo may result in a MT growth of 1 mm/day corresponding to 1 nm/s (or one dimer row per 8 s), although in vitro assembly under physiological conditions may elongate at least two orders of magnitude faster. The simulation in figs. 15d–15f shows an elongation rate of approximately 6 times 8 nm for each 50 generations. This corresponds to approximately 1 nm/s. Therefore the updating of the assembly/disassembly model does not demand nearly the same frequency as the signal propagation model does (10^9–10^{11} Hz). In the present simulation we update with a real time of only 1 Hz.

If specific dynamics on the MTA are able to activate dimers at certain locations, for instance at the free ends (the $2 \rightarrow 1$ transition), there exists an internal mechanism for determining whether a microtubule in the network should start growing. Introducing to the simple cytoskeletal model from

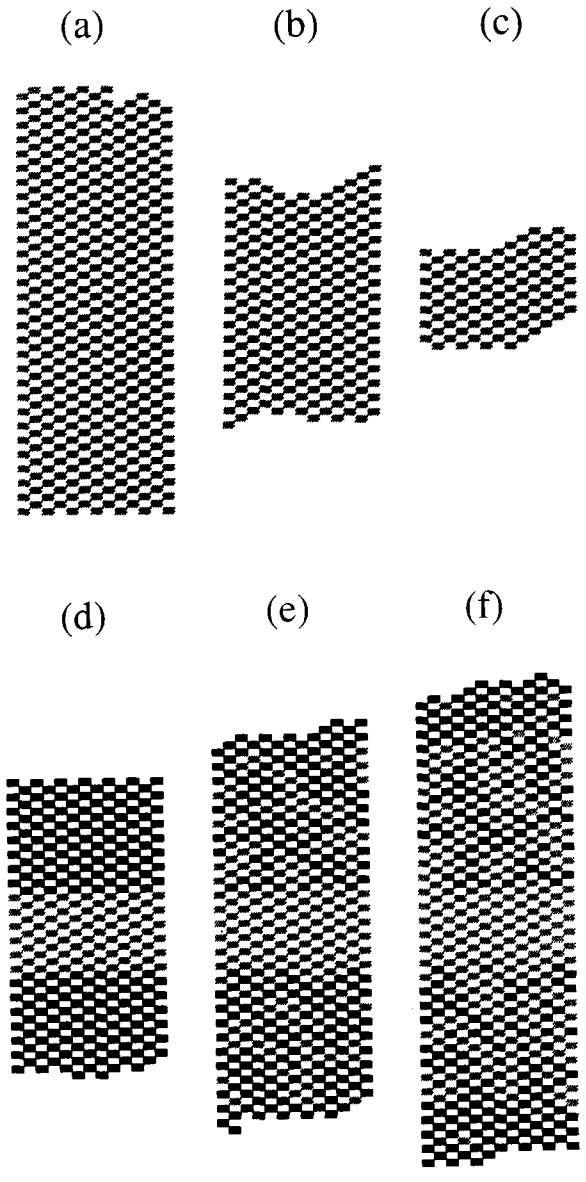

Fig. 15. (a)–(c) disassembly of a microtubule shown at time 0, 21, and 49, with hydrolysis probability $P_{12} = 0.05$ per generation, and an overall probability of 0.091 that the free states are visited by an activated dimer during the automaton generation time. Non-activated microtubule tubulin dimers (state 2) are represented by shaded squares, whereas the free tubulin and water are blank (states 3, 4, and 5). (d)–(f) Assembly of a microtubule with GTP activated caps (state 1 shown as dark squares), shown at time 0, 20, and 50, with hydrolysis $P_{12} = 0.05$ per generation, and an overall probability of 0.987 that the free states are visited by an activated dimer during the automaton generation time.

section 4, such a mechanism adds another feedback between local dynamics and meta-dynamics of our network. Relevant to section 4, selecting the sites of MAPA connections by means of MTA dynamics would also be consistent with this type of feedback.

In vivo assembly/disassembly of MTs may actually be controlled by signals transmitted on the cytoskeleton.

6. Discussion: connectionism in the cytoskeleton

Deciphering of the genetic code in DNA and RNA has led to many technological applications. The existence of a "real-time" processing code in the cytoskeleton could be comparably useful because of the important functions which could be controlled or regulated (mitosis, differentiation, aging, growth, immune function, etc.).

In the neuronal cytoskeleton, MTA can be relevant to brain/mind cognitive function. For example, MT automata may be utilized for guidance and movement of cells and cell processes (axon growth cones, dendritic spines) during morphogenesis of the brain's network architecture. In learning, axoplasmic transport and other mechanisms orchestrated by MT automata could regulate synaptic plasticity. Threshold-dependent bidirectional MT automata patterns indicate that a "back-propagation" type of synaptic regulation could occur because rapid information transport in directions opposite to the nerve impulse is possible. Combined with possible retrograde information flow across synapses ("trophism", pre-synaptic modulation) the cytoskeleton could support "neural-net" type phenomena in the brain. Beyond these morphological roles, MT automata could themselves directly participate an imprinting, pattern recognition, memory and other cognitive functions. MT automaton gliders (8 to 800 m/s) may exist as traveling waves, coupled to nerve membrane depolarizations. Coupling between membrane events and MT automaton behaviors may occur via ionic fluxes or direct connection through specialized sub-membrane cytoskeletal proteins such as ankyrin, fodrin and spectrin to the cytoskeleton. This coupling can lead to a view of the brain as a hierarchy of networks with a cytoskeletal basement level. MT automata would then comprise the "fine grain" of cognition, the "roots of consciousness".

Relegating cognition function to the level of protein conformational dynamics is supported by efforts to understand the mechanism of general anesthesia. Anesthetic gas molecules erase consciousness (while sparing other brain functions like breathing) by inhibiting conformational switching in classes of proteins such as receptors, ion channels, enzymes and cytoskeletal subunits [25]. The anesthetic molecules are thought to bind in hydrophobic regions within these proteins: the same sites proposed by Fröhlich to house dipole oscillations. Anesthetics such as halothane have also been shown (at relatively high concentrations) to cause depolymerization of microtubules [1]. Anesthesia (the reversible cessation of consciousness) may be due to interruption of communicative interactions within and among cytoskeletal networks and their membrane connections.

We have demonstrated that a population of simple networks of paired MTA is able to learn two different patterns by using an evolutionary mechanism as well as associate nearby patterns. However, many improvements may be implemented in the model which can (a) more closely resemble biological activities, and (b) increase MTA network performance. Some of these are discussed in section 4.

We have described the cytoskeleton as a connectionist information network. As such, it may be compared with other connectionist systems like artificial neural networks, classifier systems, etc. Farmer (these Proceedings [23]) has described a group of parameters into which all such systems may be fit for comparison and clarity. These parameters include nodes, states, connections, parameters, interaction rules, learning algorithm and metadynamics. In table 2, we have catalogued

Table 2
Farmer's Rosetta Stone [23] for adaptive dynamic systems applied to microtubule automata (MTA).

	Single MT	MTA networks
nodes	cellular automata cells; MTA subunits	cellular automata networks; MTA-MAPA
states	α and β state patterns	α and β state patterns
connections	lattice geometry	MAPA and other connections
parameters	local thresholds	coding in MAPA connections
interaction rules	local dipole coupling forces	MAPA connections
learning algorithm	input/output learning; selection of the best in a population (selectionist learning)	input/output learning; selection of the best in a population (selectionist learning)
metadynamics	local threshold changes; metastable states	threshold change in MTA-MAPA loci and assembly/disassembly

both a single MT automaton and the adaptive MT automaton network according to Farmer's connectionist classifications.

Connectionist networks within single elements of larger connectionist networks suggest a self-similar "fractal-like" relationship. Information capacity of a high-resolution scale within a self-similar structure is far greater than the capacity at larger scales. Roboticist Hans Moravec [56] has attempted to compare the computing power of a computer and of the human brain. Considering the number of next states available per time in binary digits, a microcomputer has a capacity of about 10^6 bits per second. Moravec calculates brain "computing" power by assuming 40 billion neurons, each with perhaps one thousand synaptic connections which can change states hundreds of times per second, resulting in 4×10^{15} bits per second. We estimate approximately 10^{14} MT tubulin subunits in a human brain based on MTs spaced 100 nm apart and 40% of brain volume being neuronal cytoplasm. In our model of MT automata these 10^{14} brain MT subunits can change states 10^9 to 10^{11} times per second leading to a total brain capacity of 10^{23} to 10^{25} bits per second! Such a huge capacity could provide a high resolution to the grain of cognition, permit massive parallelism and redundancy, and account for cytoplasmic "housekeeping" chores not directly related to brain/mind function. Still, is such a huge capacity necessary to explain cognitive function? That question is presently unanswerable. However, we surmise that the highest level of brain/mind cognitive functions are collective emergent effects whose fine grain may exist within the cytoskeleton.

We conclude with possible applications of MTA molecular cognition. Deciphering real-time MTA codes may permit interfacing to emerging technologies such as spinoffs of scanning tunneling microscopy. These techniques should allow nanometer scale monitoring and programming of dynamic actions of individual protein subunits within assemblies such as MTs. Combining ge-

netic engineering of MT tubulin subunits with real-time programming could lead to replicating MTA with complex, purposeful activities. Perhaps akin to programmable viruses, mobile MTA might be capable of intracellular repair with profound medical applications.

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We appreciate the discussions we had with Kuni Kaneko on how to design the input and the output for the adaptive microtubular automaton network.

Appendix. The learning algorithm

The process of teaching the MTA net to map two different sets of input to two different output patterns consists of six steps.

(1) Primarily depending on the shape of the desired output pattern, the thresholds for both MTA are set manually. This step demands certain knowledge about what kind of dynamics is possible for each value of the threshold parameter (recall fig. 8), and is not yet endogenous in the algorithm.

(2) A single random MAPA connection is placed at D successive MTA nets. Each of these MTA nets are then simulated for $l-(13+1)$ generations, at which time the Hamming distance, H_d, between the actual output O_{21}, and the desired output pattern, O_{21}^*, is calculated. The MAPA connection pattern which resulted in the smallest H_d with the two first input patterns (I_{11}, I_{21}) is kept for the next steps. This new connection topology is denoted by C_1.

(3) With C_1 as the basic (mother) system, D new random MAPA connections are tested in D different (daughter) systems. Again the system with the smallest $H_d = |O_{21}^* - O_{21}|$ is chosen. This new connection topology is denoted by C_2. (3) is then repeated until an acceptable H_d is reached. The connection topology for the new system is denoted by C_j.

(4) I_{11} and I_{21} are now replaced with I_{12} and I_{22}, still conserving C_j.

(5) Additional MAPA to C_j as in step (3) which is repeated until an acceptable $H_d = |O_{22}^* - O_{22}|$ is reached for a new C_k.

(6) The map $(I_{11}, I_{21}) \rightarrow O_{21}^*$ is then tested with C_k. If $|O_{21}^* - O_{21}|$ still is acceptable, the sequence is finished satisfactorily. If not, the sequence is either started all over again at (1) or repeated from (5).

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BIOELECTROCHEMISTRY AND BIOENERGETICS

Conditions for coherent vibrations in the cytoskeleton

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Abstract

Mechanism of organization and order in biological systems is not satisfactorily explained yet. Biophysical theories predicted that coherent endogenous electric field of high frequency can play a significant role in organization. The polarity of vibration structures, spectral energy transfer caused by nonlinearities, and energy supply can lead to energy condensation and excitation of coherent states. These conditions are satisfied in the polymer structures of the cytoskeleton, in particular, in the microtubules as follows from experimental findings. Nonetheless, experimental verification of energy condensation and coherent vibrations in the cytoskeleton is still missing. © 1999 Elsevier Science S.A. All rights reserved.

Keywords: Biological order; Cytoskeleton; Microtubules; Actin filaments; Coherence; Fröhlich coherence

1. Introduction

Biological activity of living systems manifests a high degree of cooperation and organization. An inconceivable number of various macromolecules, polymers, and organelles interact, join together in chemical reactions, and form larger structures whose properties cannot be predicted from the characteristics of the components. All the processes of biochemical, biophysical, and biological nature are coordinated simultaneously and successively in time but the mechanisms of order and self-organization are largely unknown.

Living matter on the cellular, subcellular and molecular levels does not have spatial order typical for symmetrical structures of inorganic crystalline substances. The 'translation symmetry' of microtubules and actin filaments seems to be exceptional. Coordination of biochemical and biophysical processes, dynamic organization and coherence is fundamental to the self-organization of biological systems [1–3]. Self-organization principles are important for evolution and for existence of living matter.

The idea that self-assembly and order are based on biophysical mechanisms using energy is now generally accepted [4,5]. Nevertheless, the concrete mechanism seems still to be rather obscure. The biological matter has extraordinary polar and polarization properties which suggests that endogenous electric and electromagnetic fields can play a fundamental role in organization and order. Supply of metabolic energy is an important feature which together with nonlinear properties enable explanation of coherent states in biological systems as was done, e.g., by Fröhlich [6–9]. Experimental research on cytoplasmic structures disclosed a fundamental role of cytoskeleton in biological activity of cells. We will compare the properties of the cytoskeleton in living cells with the theoretical requirements for excitation of coherent states.

2. Fundamentals of the Fröhlich theory

In the original work, Fröhlich postulated that long-range quantum mechanical phase correlations exist in biological systems, that the majority of the degrees of freedom behave like being close to thermal equilibrium, that a few modes of motion are very strongly excited and remain far from thermal equilibrium, and that the strong polar character of biological objects suggest longitudinal oscillations as stabilising modes [6]. He deeply analyzed the physical properties of biological systems and developed theory of energy condensation [7–9]. In short, Fröhlich assumed that some modes of ubiquitous thermal vibrations in polar

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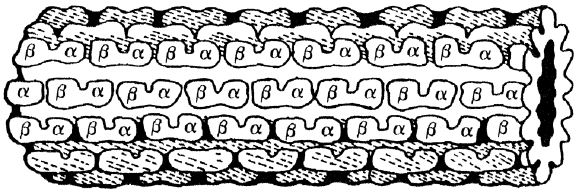

Fig. 1. A schematic figure of a microtubule with tubulin heterodimer molecules in parallel rows (protofilaments). After Refs. [10,12].

systems are excited far from thermodynamic equilibrium. The Fröhlich theory of coherent vibration states has three foundation stones: polar properties of the biological structures, energy transfer between vibration modes (spectral energy transfer), and energy supply to the system.

Long-range interactions within the vibration system as well as between the vibration system and its surroundings (i.e., the heat bath) depend on the electric nature of the vibrations that generate the oscillating electric field. The system may consist of N identical polar oscillators interacting with each other which results in a band of normal modes with frequencies $\nu_1 < \nu_2 \ldots < \nu_N$. Fröhlich postulated the rate equation in the form:

$$\dot{n}_i = s_i - \Phi_i\left[n_i \exp(\beta\nu_i) - (n_i + 1)\right] - \sum_j \chi_{ij}\left[n_i(n_j + 1)\exp(\beta\nu_i) - (n_i + 1)\, n_j \exp(\beta\nu_j)\right],$$

where $\dot{n}_i$ is the time derivative of the occupation number n_i of the normal mode with frequency ν_i, s_i is the number of energy quanta supplied from metabolic sources in the unit time, and Φ_i and χ_{ij} are the linear and the nonlinear transition probability of energy transfer, respectively. The nonlinear interaction makes possible energy transfer between two normal modes with different frequencies in the vibration system with participation of the heat bath. Energy transfer of this type is a fundamental property of the Fröhlich system and a necessary condition for energy condensation. If energy is not supplied (i.e., $s_i = 0$) the system is in thermodynamic equilibrium. In this case, the third term on the right hand side of the rate equation (the χ term) equals zero as well as the second term (the Φ term) and, therefore, $\dot{n}_i = 0$. The steady state solution of the rate equation (for $s_i \neq 0$), given by the relation:

$$n_i = A\frac{1}{\exp\left[\beta(\nu_i - \mu)\right] - 1},$$

where A is the amplitude and $\beta = h/kT$ has the usual meaning, shows that if the excitation potential μ depending on the excitation is high enough, the mode with the lowest frequency is highly excited.

The Fröhlich theory shows the possibility of energy condensation and coherence in the momentum space.

3. Electric properties of the cytoskeleton

The cytoskeleton is an important polymer structure for organization of the cell. The cytoskeleton is composed of intermediate filaments, actin filaments, microtubules, and associated proteins. The microtubules have a fundamental role in organization of the cytoskeleton. The microtubule is a cylindrical hollow tube with the outer diameter of about 25 μm composed of 13 (or 14) protofilaments which are one dimensional chains of heterodimer tubulin molecules (Fig. 1). Description of the cytoskeleton may be found, e.g., in Refs. [4,10–22]. The tubulin molecule is formed by two globular proteins—α- and β-tubulins. The heterodimer molecules can have two conformation states—nontilted and tilted by 29° from the microtubule axis (Fig. 2). The heterodimer molecule is an electric dipole. The microtubules are polar structures with the plus and the minus end embedded in the centrosome—a center of the cytoskeleton. Polarity of the microtubules are closely connected with their function. Fig. 3 shows positions of heterodimers on the wall of a microtubule (a), orientation of the dipole moments in the ferroelectric state—nontilted dimers (b), and the dipole moments of a structure with nontilted and tilted dimers (c). The function of the microtubules depends on phosphorylation and dephosphorylation of microtubule associated tau proteins [16].

4. Nonlinear properties

Vibrations in polar systems excite polarization waves that generate electromagnetic field (with the dominant near zone field electric component) around the vibration structure [5,23]. Energy is exchanged between the near zone field and the vibration structure. (If the vibrations are periodic the energy exchange is periodic too.) A part of the energy may be lost in the ambient medium. Conductivity losses cause ordinary damping represented by the linear transition probability (Φ) in the Fröhlich rate equation. Another type of interaction with the surroundings may exist. A photon of the field can excite a valence electron which loses a part of the energy in a collision (without emission) and then the electron reemit the remaining energy. The collision can increase the energy of the electron too. The emitted quantum of energy differs from the absorbed quantum. This mechanism enables energy trans-

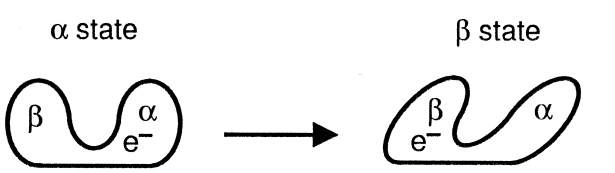

Fig. 2. Conformation change of a tubulin heterodimer from the α state to the β state. The symbol e^- indicates the position of mobile electron charge.

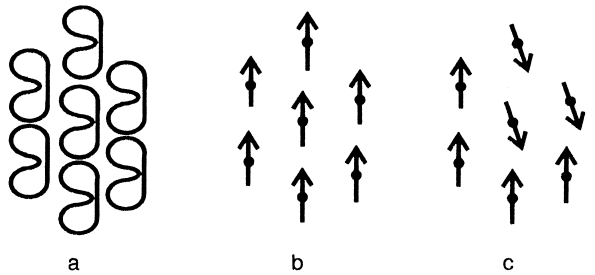

Fig. 3. Position of heterodimers on the wall of a microtubule (a), ferroelectric state of a part of the wall with nontilted heterodimers (b), and dipole moments of a structure with nontilted and tilted heterodimers (c).

fer between modes with different frequencies. Therefore, the polar vibrations in microtubules and actin filaments may be involved in the Fröhlich nonlinear energy transfer.

The microtubule structure is very likely nonlinear by itself. The binding energy of the tubulins in a heterodimer is very likely larger than that between two heterodimers. The elastic force constants along the microtubule chain are not equal and, therefore, the potential as a function of distance of the tubulin 'particle' is asymmetric with respect to the equilibrium position. The nonlinear systems can generate higher harmonic components and components with combination frequencies and enable spectral energy transfer too.

5. Vibrations in microtubules and energy supply

The heterodimers in a protofilament form a one dimensional chain. The globular tubulins in the chain may be considered as mass units with translation symmetry (Fig. 1). These mass units have internal vibrations (exerted by atoms and large parts of tubulins) as well as external vibrations in the chain (i.e., vibrations exerted by each tubulin as a whole). A similar rationale may be applied to the actin filaments. We have analyzed the external vibrations in the protofilaments and in the actin filaments using a method similar to that employed in solid state physics [5,24]. Frequency of vibrations have two branches: the optical and the acoustical branch. The highest frequencies of external vibrations are in the range 0.1–1 GHz [5,23,25]. The frequencies of internal vibrations may be much higher (in the submillimeter and far infrared regions).

The dynamics of the microtubules is very likely one of their most prominent features. The microtubules are capable of rapid polymerization and depolymerization which enables exchange of the subunits between soluble and polymer pool. A schematic picture of a microtubule structure with centrosome in the center is shown in Fig. 4. A microtubule grows towards the cell periphery and then shrinks back towards the centrosome. It may shrink partially and grow again or it may disappear completely to be replaced be new microtubules. This process is called the dynamic instability. The treadmilling (i.e., addition of the tubulin subunits at the plus end and removal of the tubulin subunits at the minus end) appears if both the ends of a microtubule are exposed. After polymerization the guanosine triphosphate (GTP) bound to the β-tubulin is hydrolyzed to the guanosine diphosphate (GDP). Depolymerization and repolymerization is a continual process in animal cells. A similar process of hydrolyzation of adenosine triphosphate (ATP) exists in actin filaments which are important parts of the cytoskeleton too.

Due to depolymerization and repolymerization, the majority of the heterodimer subunits (which are energetically exhausted) is replaced by the energy-rich subunits within about 20 min. The exchange of subunits is much faster in the M phase of the cell cycle. This exchange supplies the standard amount of energy. The dynamic instability provides additional energy supply above the standard value.

We will assess the amount of energy supplied by hydrolysis of the GTP's (attached to the β-tubulins) to the GDP's. The measurements of the hydrolysis of the GTP analogue guanylyl-(a,b)-methylene-diphosphonate (GMPCPP) in solution, in tubulin heterodimers, and in tubulin subunits in the microtubule led to a conclusion that the difference in the free energy was -5.18 kcal/mole, -3.79 kcal/mole, and $+0.7$ kcal/mole [19,26,27]. The reaction with the positive difference of the free energy is thermodynamically unfavorable. Nevertheless, this reaction will ultimately be favorable in the cell. As inorganic phosphate is buffered in the cytosol at approximately 1 mM, then for the equilibrium constant of the reaction smaller than 1, the energy released is -3.38 kcal/mole [27]. A part of the energy is expended on dissociation of the inorganic phosphate but a considerable part is stored in the microtubule lattice (1.7 kcal/mole).

6. Energy usage

Usage of the hydolysis energy is not well understood. The energy can change bond strengths [13,28] or excite

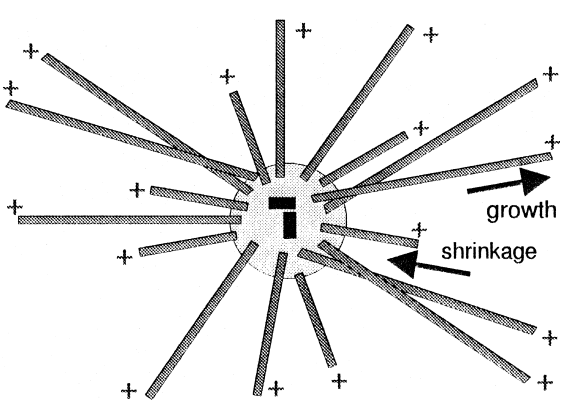

Fig. 4. A centrosome with attached microtubules. The arrows indicate growth and shinkage in the microtubule array.

Table 1
Theoretical requirements and cytoskeleton properties

Theoretical requirements	Properties of the cytoskeleton
Polar systems	Tubulin herodimers and actin molecules are dipoles
Long-range interaction	Interaction by the electric field
Spectral energy transfer by	Mediated by
(a) nonlinear interaction with a heat bath	(a) a near zone electric field
(b) nonlinear intrinsic properties	(b) nonlinear oscillations in asymmetric potential valleys
Energy supply	Hydrolysis of
	(a) GTP in microtubules
	(b) ATP in actin filaments
Consequences	
Energy condensation	Not experimentally proved
Coherence	

soliton or soliton-like waves [11,28]. It can be used for conformation changes of tubulin heterodimers (or for imparting energy via tubulin conformation changes) [12] and for kink wave reorientation of heterodimer dipoles [11,13]. It can excite coherent lattice vibrations by the Fröhlich mechanism [13,28] and/or by the intrinsic nonlinear mechanism. Real physical mechanisms of energy transformation are irreversible and a part of the energy is always transferred to vibrations. Even if the energy levels of two conformation states of a tubulin subunit are different (e.g., if the level of the β state is higher than that of the α state) after transition to the higher energy level, a part of the energy which is used to surmount the potential barrier between the conformation states can be transformed into vibration energy. (The opposite relaxation process should convert energy to vibrations too.) Creation of the tense state with altered forces between the subunits in the microtubule has to supply energy into the lattice. Therefore, we may assume that energy is stored in the microtubule lattice not only in the conformation changes and in the tense states of the lattice but also in the vibration states [5,25].

The energy stored in the microtubule can do work in the cell but a precise manner in which the energy is used is still not understood. We assume that the energy can excite the lattice vibrations and polarization waves, which may be coherent on account of the Fröhlich mechanism or of the intrinsic nonlinear mechanism. The majority of the energy stored in the microtubules may be converted into the vibrations. Therefore, the vibrations are excited and can be far from thermodynamic equilibrium. The electric field generated by the polar vibrations can do work in the surroundings of the microtubule. In the vicinity of the microtubule (e.g., at a distance of about 10 nm) the intensity of the electric field is about 10^5 V/cm (the vacuum value) [5,23].

We assessed the amount of energy which might be supplied to the cytoskeleton vibrations from the hydrolysis of GTP. If the energy supplied by the hydrolysis of GTP is 1.7 kcal/mole, then the energy per one entity is about three times greater than its energy in thermodynamic equilibrium in the frequency range from 10^7 to 10^{11} Hz. The standard amount of energy is supplied within about 20 min when the microtubule pool of heterodimers is exchanged. Due to dynamic instability, the total amount of energy supplied to the microtubule structure in this time interval is considerably greater.

7. Theoretical requirements and microtubule properties

Table 1 shows comparison of theoretical requirements for creation of coherent states and properties of the microtubules. The microtubules satisfy the conditions for energy condensation and coherence. The measured electromagnetic activity of living cells (observed, for instance, in yeast cells—*Saccharomyces cerevisiae* [29,30]) may be assigned to microtubules. Further measurements have to be carried out to prove or disprove coherence of the emitted electromagnetic field.

8. Conclusions

Microtubules are polar—their heterodimer tubulin subunits are electric dipoles whose near-zone field mediates interaction with the ambient medium. The microtubule structure is nonlinear. Energy supply is provided by the hydrolysis of GTP to GDP. The microtubules in the cytoskeleton satisfy the requirements for excitation of coherent vibrations but at present experimental proof of coherent states is missing. We assume that electromagnetic activity of living cells originates in the cytoskeleton and in particular in the microtubules.

Acknowledgements

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Conduction pathways in microtubules, biological quantum computation, and consciousness

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Abstract

Technological computation is entering the quantum realm, focusing attention on biomolecular information processing systems such as proteins, as presaged by the work of Michael Conrad. Protein conformational dynamics and pharmacological evidence suggest that protein conformational states—fundamental information units ('bits') in biological systems—are governed by quantum events, and are thus perhaps akin to quantum bits ('qubits') as utilized in quantum computation. 'Real time' dynamic activities within cells are regulated by the cell cytoskeleton, particularly microtubules (MTs) which are cylindrical lattice polymers of the protein tubulin. Recent evidence shows signaling, communication and conductivity in MTs, and theoretical models have predicted both classical and quantum information processing in MTs. In this paper we show conduction pathways for electron mobility and possible quantum tunneling and superconductivity among aromatic amino acids in tubulins. The pathways within tubulin match helical patterns in the microtubule lattice structure, which lend themselves to topological quantum effects resistant to decoherence. The Penrose–Hameroff 'Orch OR' model of consciousness is reviewed as an example of the possible utility of quantum computation in MTs. © 2002 Elsevier Science Ireland Ltd. All rights reserved.

Keywords: Microtubules; Quantum computation; Consciousness; Protein conformation; Biolomolecular computation

1. Introduction: proteins, anesthesia and quantum computation

1.1. Moore's law and quantum biology

Technology is approaching the limit of classical computing through the operation of Moore's Law, which states that the number of transistors or fundamental switches that can be fabricated on a silicon integrated circuit doubles every 18–24 months. This amazing trend has been miniaturizing microelectronics for close to four decades, and today the smallest available silicon chips contain up to 100 million transistors on a few square centimeters of wafer. This translates into dimensions on the order of 200 nm or less per fundamental switch. As further miniaturization occurs, switching processes enter the quantum world, with extremely useful potential advantages inherent in quantum computation.

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Quantum computers are proposed to take advantage of the bizarre quantum property of superposition, in which particles may be in two or more states or locations simultaneously. Whereas conventional computers typically utilize binary 'bits' in which information is represented by particles in classical states of either 1 or 0, quantum computers utilize quantum superpositions of both 1 and 0 ('qubits'). While in superposition qubits interact with other qubits by another bizarre quantum feature, nonlocal entanglement, and ultimately each qubit 'collapses' or 'reduces' to classical states of either 1 or 0. The set of post-reduction classical bits is the solution, or output. Quantum computers will in principle solve at least certain types of difficult problems (e.g. searches, factoring large numbers into primes) with unprecedented speed, and other unique applications are likely to emerge. Quantum computation promises to be a dominant form of information technology.

Miniaturization also brings information technology into the biological realm. Moore's law predicts that in 10 years fundamental switches should reach a scale of about 2 nm roughly the size of proteins, which can act as conformational switches. Biology has had 4 billion years to evolve the most efficient form of information processing. It should not be surprising to find that protein conformational switching is regulated by quantum computation.

1.2. Protein conformational dynamics

Protein function—such as enzymatic catalysis, ion channel opening, cell movement and information processing—depends on regulated changes in protein shape, or conformation.

Individual proteins are synthesized as linear chains of amino acids which 'fold' into 3-D conformation. The precise folding depends on attractive and repulsive forces among various amino acid side groups, and a current view is that many possible intermediate conformations precede the final one (Baldwin, 1994). Predicting final 3-D folded shape using computer simulation has proven difficult if not impossible. This conundrum is known as the 'protein folding problem' and so far appears to be 'NP complete': the answer can be calculated in theory, but the space and time required of any classical computer is prohibitive. Perhaps protein folding is a quantum computation?

The main driving force in protein folding occurs as uncharged non-polar amino acid groups join together, repelled by solvent water. These 'hydrophobic' groups attract each other by van der Waals forces and bury themselves within the protein interior. Intra-protein hydrophobic pockets result, composed of side groups of non-polar (but polarizable) amino acids such as leucine, isoleucine, phenylalanine, tryptophan, tyrosine and valine. Hydrophobic pockets range in size from single non-polar amino acids, up to groups of them occupying $\sim$400 cubic Å, or 0.4 cubic nm (roughly 1/30–1/250 the total volume of a single protein). The physical solvent which most closely resembles hydrophobic pockets is olive oil, a medium in which anesthetic gases readily bind. Within hydrophobic pockets, quantum mechanical activities regulate protein conformational changes.

Proteins in a living state are dynamical, with transitions occurring at many scales. However, conformational transitions in which proteins move globally and upon which protein function generally depends occur in the ns (10^{-9} s) to 10 ps (10^{-11} s) time scale (Karplus and McCammon, 1983). Proteins are also only marginally stable. A protein of 100 amino acids is stable against denaturation by only $\sim$40 kJ mol^{-1} whereas thousands of kJ mol^{-1} are available in a protein from side group interactions including van der Waals forces. Consequently protein conformation is a 'delicate balance among powerful countervailing forces' (Voet and Voet, 1995).

1.3. Quantum effects in proteins

Can electron movements influence protein conformation? The Born–Oppenheimer approximation assumes that electrons cannot significantly influence the position of atomic nuclei (and hence the position of the atom) because the electron mass is so small compared with protons and neutrons (like a soccer ball to the earth). However, as Michael Conrad pointed out (Conrad,

1994) the Born–Oppenheimer approximation does not apply because electron charge is equivalent to proton charge. Quantum processes such as electron tunneling, delocalization, and superposition could, therefore, couple to nuclear locations and influence conformational changes, particularly if the quantum processes are collective and self-organized. Proteins may be designed to amplify these quantum processes. Within proteins ideal sites for electron delocalization are hydrophobic pockets (sites of anesthetic effect), in particular those containing aromatic rings like that of tryptophan.

Electron movements and dipoles are among several forces operating among amino acid side groups within a protein. These include charged interactions such as ionic forces and hydrogen bonds, as well as interactions between dipoles—separated charges in electrically neutral groups. Dipole–dipole interactions are known as van der Waals forces and include three types:

1. permanent dipole–permanent dipole.
2. Permanent dipole–induced dipole.
3. Induced dipole–induced dipole.

Type 3 induced dipole–induced dipole interactions are the weakest but most purely non-polar. They are known as London dispersion forces and they ensue from the fact that atoms and molecules which are electrically neutral and spherically symmetrical nevertheless have instantaneous electric dipoles due to asymmetry in their electron distribution ('electron cloud'). The electric field from each fluctuating dipole in an electron cloud couples to others in electron clouds of adjacent non-polar amino acid side groups. The couplings are quite delicate (40 times weaker than hydrogen bonds) but are numerous and influential. The London force attraction between any two atoms is usually less than a few kiloJoules, however, thousands occur in each protein. Electron clouds in aromatic ring structures (i.e. tryptophan, tyrosine, phenylalanine, histidine) are larger and more complex, and may give rise to cooperatively organized London forces particularly suited to governing protein conformational states. Due to inherent uncertainty in electron localization, London forces are quantum effects which may couple to 'zero point fluctuations' of the quantum vacuum (London, 1937; Milloni, 1994).

1.4. Quantum neuropharmacology: anesthesia and psychedelia

Quantum dipole oscillations within hydrophobic pockets were first proposed by Fröhlich (1968) to regulate protein conformation and engage in macroscopic coherence. In a series of proposals Conrad (e.g. 1994) showed how electron superpositions could influence nuclear movement (conformation), and suggested that quantum superposition of various possible protein conformations occurs before one is selected. Roitberg et al. (1995) showed functional protein vibrations which depend on quantum effects centered in two hydrophobic phenylalanine residues, and Tejada et al. (1996) have evidence to suggest quantum coherent states exist in the protein ferritin. And recently Matsuno (2001) has claimed to observe magnetic quantum coherence in actin, a main component of the contractile apparatus in muscle cells, and of the cytoskeleton in all cells.

Close examination of effects of general anesthetics (which erase consciousness) and hallucinogens (which are said to expand consciousness) reveal that both act in opposite ways on delicate quantum effects.

The mechanism of general anesthesia supports a role for quantum London forces in the phenomenon of consciousness. A century ago Meyer and Overton (working independently in Germany and England, respectively) showed a remarkable correlation between anesthetic potency and solubility in a particular lipid-like environment. Although these results were originally taken to imply that the anesthetics acted in lipids within the cell membrane, studies in the past decades (e.g. Wulf and Featherstone, 1957; Franks and Lieb, 1982, 1984, 1985, 1994; Halsey, 1989 and others) conclude that anesthetic gas molecules act in hydrophobic (lipid-like, water-excluding) regions within critical target proteins. The solubility/binding occurs by weak van der Waals London forces between the anesthetic and non-polar amino acid groups (the same type of endogenous interactions occurring between non-polar amino acid groups in the absence of anesthetics). Critical brain proteins affected by anesthetics include receptors for neurotransmit-

ters such as gamma amino butyric acid (GABA), acetylcholine, serotonin and glycine, as well as certain second messenger proteins, enzymes and tubulin which comprise microtubules (MTs) (Franks and Lieb, 1982, 1984, 1985). What do anesthetics do at their site of action?

Franks and Lieb (1994) suggested that anesthetics act simply by following the Meyer–Overton correlation: their mere presence in hydrophobic pockets prevents conformational switching. However, a variety of molecules, which follow the Meyer–Overton correlation and occupy the same pockets are nonanesthetic, or even convulsant (Fang et al., 1996). The mere presence of molecules in hydrophobic pockets may be insufficient to explain anesthesia.

Another view is that anesthetics somehow disrupt van der Waals London force interactions normally occurring in the critical hydrophobic pockets. Quantum superposition requires electron mobility—electron pairs must be relatively free to roam among allowed orbitals. Evidence shows that anesthetics retard electron mobility—the movement of free electrons in a corona discharge is inhibited by anesthetics (Hameroff and Watt, 1983). By forming their own London force attractions in hydrophobic pockets, anesthetics may inhibit electron mobility required for protein dynamics, quantum superposition and consciousness. Nonanesthetics may be understood as occupying hydrophobic pockets without altering electron mobility, and convulsants as forming cooperative van der Waals interactions, which promote excessive electron mobility and protein dynamics in excitatory proteins.

Another class of drugs, the hallucinogenic ('psychedelic') tryptamine, ergoline and phenylethylamine derivatives bind and act in hydrophobic pockets within serotonin receptors and elsewhere. For example the hallucinogens LSD (an ergoline) and DMT (a tryptamine) are based on indole rings, exactly like that in tryptophan. Nichols et al. (1977) showed that these psychedelic drugs bind in hydrophobic pockets of less than 0.6 nm (6 Å) length. Kang and Green (1970), Snyder and Merrill (1965) measured the capacity of a series of psychedelic drug molecules to donate electron orbital resonance energy. In both studies, the drug's electron resonance donation is correlated with psychedelic potency.

Taken together with dependence of protein conformational regulation on quantum van der Waals forces, the anesthetic and psychedelic studies suggest that (1) consciousness depends on quantum processes in hydrophobic pockets; (2) these quantum processes are inhibited by anesthetics which impair electron mobility in van der Waals London forces. The same processes are enhanced (hallucinations, but also enlightenment) in the presence of psychedelic drugs, e.g. those with indole rings donating electron resonance energy to indole rings in tryptophan within the hydrophobic pocket, forming a collective quantum state. Consciousness depends on quantum states of electrons within hydrophobic pockets in a class of brain proteins.

With conformational states controlled by a quantum process, proteins may thus be viewed as quantum bits, or 'qubits'. For quantum computation, qubits can be arranged in a geometrical lattice (Lloyd, 1993). In biological systems the protein tubulin is arranged in a particular geometrical lattice in MTs whose functions appear to include organization, communication and information processing. Are MTs quantum computers?

2. Information processing in microtubules

2.1. Microtubules and the cytoskeleton

Interiors of living cells are functionally organized by webs of protein polymers known as the cytoskeleton (Fig. 2). Major components of the cytoskeleton are MTs, self-assembling hollow crystalline cylinders whose walls are hexagonal lattices of subunit proteins known as tubulin (Fig. 3). Other major cytoskeletal components include actin, intermediate filaments, and centrioles, MT-based organelles, which orient the cell and guide cell movement and division.

MTs are essential for a variety of biological functions including cell movement, cell division (mitosis) and establishment and maintenance of cell form and function. In neurons (which lack centrioles), MTs self-assemble to extend axons

and dendrites and form synaptic connections. Microtubule-associated proteins (MAPs) interconnect MTs to form MT-MAP networks, which define cell architecture and function. Once established, MT-MAP networks maintain and regulate synaptic strengths responsible for learning and cognitive functions. MTs interact with membrane structures and activities (e.g. in actin-based dendritic spines) by linking proteins (e.g. fodrin, ankyrin) and 'second messenger' chemical signals. Woolf (1997) has shown that activation of neuronal acetylcholine receptors causes changes in MT-MAP connections. (For a more complete description of the role of MTs and other cytoskeletal structures in cognitive functions see Dayhoff et al., 1994; Hameroff and Penrose, 1996a; Hameroff, 1994). MTs have traditionally been considered as purely structural components, however, recent evidence has demonstrated MT mechanical signaling and communication functions (see Tables 1 and 2).

How might MTs signal and process information? Tubulin can undergo several types of conformational changes (e.g. Engelborghs, 1992; Cianci et al., 1986). In one example of tubulin conformational change observed in single protofilament chains, one monomer can shift 27° from the dimer's vertical axis (Melki et al., 1989). Whether that degree of mechanical deformation occurs in tubulin within intact MTs is unknown; neighbor tubulins in the MT lattice might be expected to constrain movement. However, cooperativity among MT subunit tubulins bound 'loosely' in the MT lattice by hydrophobic forces could coordinate conformational changes, and support propagation of wave-like vibrational sig-

Table 1
Experimental evidence for signaling in microtubules

Author(s), year	Signaling mechanism observed
Vassilev et al., 1985	MT bridge between two membranes conveys depolarization
Vernon and Woolley, 1995	Propagation of 'active zone' along flagellar MTs
Maniotis et al., 1997	Mechanical signals from membrane proteins through MTs to cell nucleus
Glanz, 1997	Force transduction through cytoskeleton

Table 2
Theoretical models of microtubule information processing

Author(s), year	Theoretical model
Sherrington, 1953	'Cyto-skeleton' as protozoan nervous system
Moran and Varela, 1971	MTs as mechano-ionic transducers
Atema, 1973	Propagating tubulin conformational changes in cilia and flagella
Hameroff, 1974	MT holography
DeBrabander, 1975	MTs as signal transducers between membrane and nucleus
Roth and Pihlaja, 1977	Information in tubulin conformational gradient within MTs MTs as signal transducers between membrane and nucleus
Bornens, 1979	Centrioles as gyroscopic gravity sensors
Albrecht-Buehler, 1981	MT-centriole signalling by infra-red photons
Conrad and Lieberman, 1982	MTs as vibrational computers
Hameroff and Watt, 1982	MT computation
Del Giudice et al., 1982	Energy self-focusing in hollow MT cores
Hameroff et al., 1984	MT molecular automata
Joshi et al., 1985	MT tensegrity signalling
Rasmussen et al., 1990	MT molecular automata; MT automata networks
Sataric et al., 1992	Solitons in MTs
Tuszynski et al., 1995	Ferroelectric, spin glass behavior
Penrose and Hameroff, 1995	MT quantum computation
Kirschner and Mitchison, 1986	MT probing/signalling
Puck and Krystosek, 1992	Phosphorylation/dephosphorylation along MT subunits
Wang and Ingber, 1994	Tensegrity signalling
Chou et al., 1994	Soliton propagation in MTs
Jibu and Yasue 1995	MT super-radiance, self-induced transparency
Mavromatos and Nanopoulos, 1997	Quantum computation in MTs with string theory collapse

nals. MTs are also ferroelectric with electron and proton conduction, which may couple by phonons to conformational vibrations. The crystal-like MT structure makes them attractive candidates for quantum coherent excitations, e.g. in the gigaHz range by a mechanism suggested by Fröhlich ('pumped phonons' Fröhlich, 1968, 1970, 1975, 1975 cf. Penrose and Onsager, 1956).

One particular model of MT information processing potential utilizes Fröhlich excitations of tubulin subunits within MTs to support computation and information processing (e.g. Rasmussen et al., 1990). The coherent excitations are proposed to 'clock' computational transitions occurring among neighboring tubulins acting as 'cells' as in molecular scale 'cellular automata'. Dipole coupling mediates logical interactions among neighboring tubulins, resulting in self-organizing patterns capable of information processing, memory and learning (Fig. 4).

In a series of simulations (e.g. Hameroff et al., 1984; Rasmussen et al., 1990) Fröhlich's excitations were used as a clocking mechanism and electrostatic dipole coupling forces as 'transition rules' for cellular automata behavior by dynamic conformational states of tubulins within MTs. In Fig. 3, each tubulin dimer conformation is ruled by a single London force electron pair coupling. For automata behavior, the dipole strength of each dimer is coupled to its six surrounding tubulin neighbors at each 'Fröhlich coherent' time step (e.g. 10^{-9}–10^{-11} s). The net electrostatic force F_{net} from the six surrounding neighbors acting on each tubulin can then be calculated as:

$$f_{net} = \frac{e^2}{4\pi\varepsilon} \sum_{i=1}^{6} \frac{Y_i}{r_i^3}$$

where y_i and r_i are inter-tubulin distances, e is the electron charge, and ε is the average protein permittivity. MT automata simulations (Fig. 5) show conformational pattern behaviors including standing waves, oscillators and gliders traveling one dimer length (8 nm) per time step (10^{-9}–10^{-11} s) for a velocity range of 8–800 m s^{-1}, consistent with the velocity of propagating nerve action potentials.

MT automata patterns can thus represent and process information through each cell; gliders may convey signals which regulate synaptic strengths, represent binding sites for MAPs (and thus neuronal and synaptic connectionist architecture) or material to be transported. Information could become 'hardened' in MTs by tubulin modifications or stored in neurofilaments via MAPs.

2.2. Tryptophan and histidine locations in tubulin

In conventional computers the 'currency' of information in semiconductors is electrons. Microtubule automata could utilize Fröhlich phonons for information currency, although other quasi-particles such as solitons, excitons, instantons, anyons, as well as photons, electrons and protons could serve. Electron propagation would be extremely useful, as conformational energy (e.g. in the form of phonons) could be coupled to electrons and both classical and quantum processes could be supported. However, proteins have never been considered especially conductive, nor in most cases semiconductive (of course neither has DNA which has recently been shown to be highly conductive, and perhaps superconductive (Barton, 1998; Kasumov et al., 2000)).

Electron (or proton) transfer within and among specific amino acid residues within proteins may mediate protein function. For example catalysis in the enzyme class 1 ribonuclease reductase utilizes electron or proton transfer over 3.5 nm from a tyrosine to a cysteine. Photoactivation in *E. coli* DNA photolyase enzyme involves electron 'hopping' or tunneling along a chain of three separate tryptophans (Aubert et al., 2000). Electron tunneling over significant distances within proteins has now been shown experimentally, and there is evidence of electron transfer from intra-protein tryptophan to DNA (Wagenknecht et al., 2000).

The amino acids tryptophan and tyrosine are convenient depots for electron hopping or tunneling because of the high polarizability of their ring structure. The 'aromatic' amino acids such as tyrosine, tryptophan, phenylalanine and histidine have residues with resonant ring structures in which electrons are mobile and delocalizable.

Tyrosine, phenylalanine and histidine have single rings; each tyrosine and phenylalanine have six-carbon benzene-like rings, and histidine has a

ring with three carbons and two nitrogens. Tryptophan has a particular double ring (an 'indole ring')—a six carbon ring conjoined with a five ring with one nitrogen and four carbons. The extra lines between the atoms in the rings in Fig. 1 signify shared, delocalized, or resonant electrons. Thus tryptophan has the greatest electron resonance (and thus is the most fluorescent), and histidine the lowest electron mobility, and least fluorescent. Becker et al. (1975) showed fluorescent resonance energy transfer (non-radiative photon exchange) between tryptophan and other aromatics in adjacent tubulins in MTs, and between MTs and membranes. Tryptophan is the most highly suited amino acid for transiting electrons and exchanging photons. In this study, we have begun to examine and map intra-tubulin

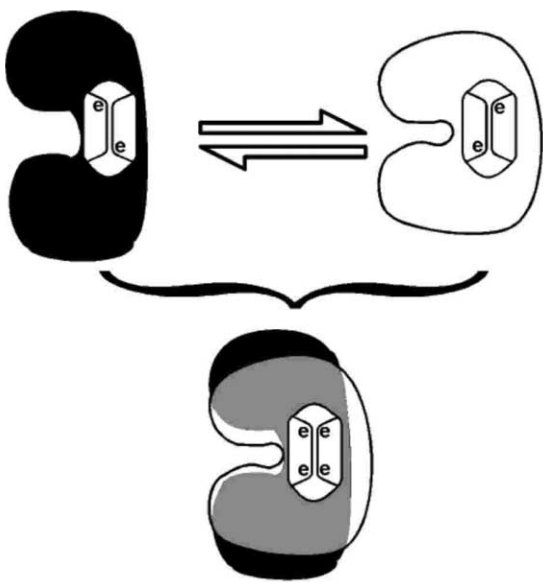

Fig. 1. Model of protein conformational switching. Tubulin is depicted as an example, and for simplicity one pair of electrons coupled by London forces is shown in a single hydrophobic pocket. Top: coupled electrons in one configuration in a single hydrophobic pocket correspond to 'open' (black) conformation. In the opposite London force electron coupling the protein is 'closed' (white). Bottom: since London forces are quantum mechanical, the electron pair may occupy both states, and the protein exist in a quantum superposition of both open and closed conformations. By going from bottom (quantum superposition) to top (one particular conformation) the protein functions as a qubit.

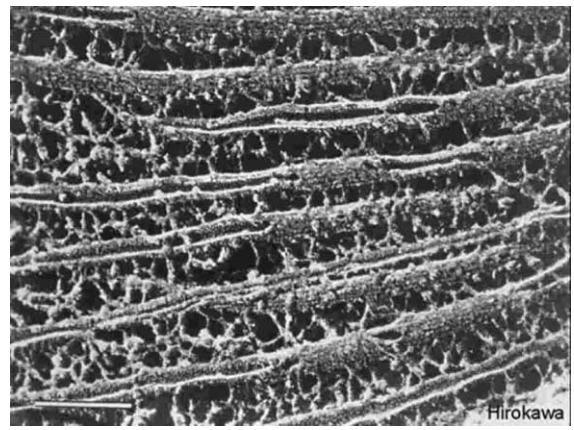

Fig. 2. The neuronal cytoskeleton. Immunoelectron micrograph of dendritic MTs interconnected by dendrite-specific MAPs. Some MTs have been sheared, revealing internal hollow core. The granular 'corn-cob' surface of MTs is barely evident to close inspection. Scale bar, lower left: 100 nms. With permission from Hirokawa, 1991.

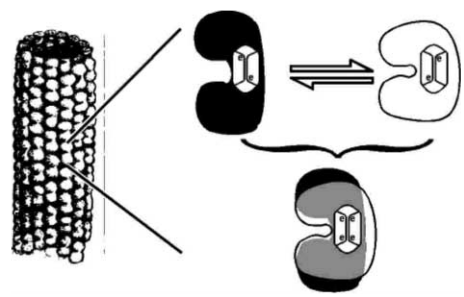

Fig. 3. Left: microtubule, a cylindrical lattice of tubulin proteins. Right (Fig. 1): coupled to position of a pair of quantum coupled electrons in an internal hydrophobic pocket, each tubulin may occupy two classical conformations (top) or exist in quantum superposition of both conformational states (bottom). A tubulin may thus act as a classical bit (top) or as a quantum bit, or 'qubit'.

locations of the aromatic amino acids tryptophan and histidine.

Tubulin dimers are 8 nm in length, and 4 nm wide (and 4 nm 'deep'). The peanut-shaped dimer is composed of two roughly equal monomers, alpha tubulin and beta tubulin each approximately $4 \times 4 \times 4$ nm. The distances between the tryptophans is thus no more than 2 nm. The 3-D crystallographic structure of tubulin was solved in 1998 by Nogales, Wolf and Downing (Nogales et

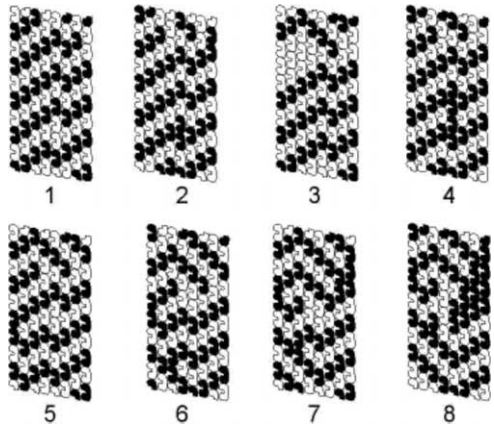

Fig. 4. Microtubule automaton simulation (from Rasmussen et al., 1990). Black and white tubulins correspond to states shown in Fig. 4. Eight ns time steps of a segment (eight of 13 protofilaments) of one microtubule are shown in 'classical computing' mode in which patterns ('gliders') move, evolve, interact and lead to emergence of new patterns.

al., 1998). Recently Alex Nip (now at University de Montreal) and Jack Tuszynski (University of Alberta, Starlab.org) simulated the locations of tryptophans and other aromatics in tubulin. Those results for tryptophan and histidine in 2-D projection are shown here. Tryptophan locations in the tubulin dimer are shown in Figs. 6–8, and histidine locations in Figs. 9–11.

Fig. 6 shows the 'front view'of tryptophan locations, looking at the tubulin dimer as looking at the microtubule from the outside, e.g. as in Fig. 3. However, the front view gives only a 2-D picture. We gain 3-D perspective by also looking at 'side' (within the MT wall) and 'top' (down the protofilament). There are eight tryptophans, four in the alpha (top) monomer, and four in the beta (bottom) monomer. Two tryptophans (one from each monomer) are in the neck, or hinge of the dimer.

The side view indicates that two of the tryptophans are alone in the extended portion of each monomer, separated from each other by exactly 4 nm. The other tryptophans are close to the inner surface and arrayed vertically. The top view shows that the tryptophans are arrayed vertically along three specific axes within the dimer.

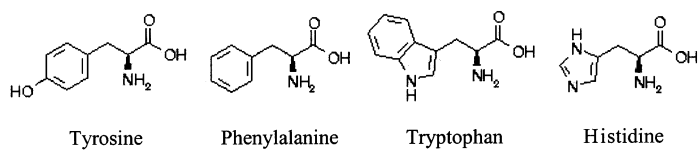

Fig. 5. The 'aromatic' amino acids contain ring structures with resonant (shared) electrons.

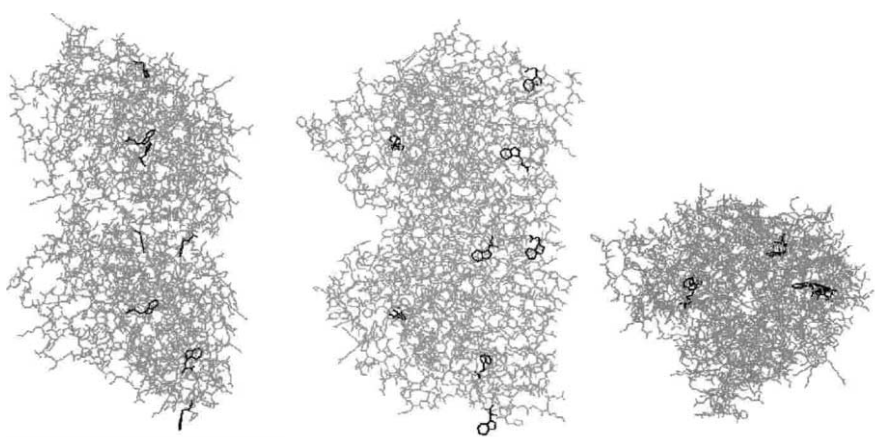

Fig. 6. Tryptophans (black) in the tubulin dimer. Left: front view (looking at the outer MT surface), Middle: side view (from within the MT wall), Right: top view (looking down on the MT protofilament).

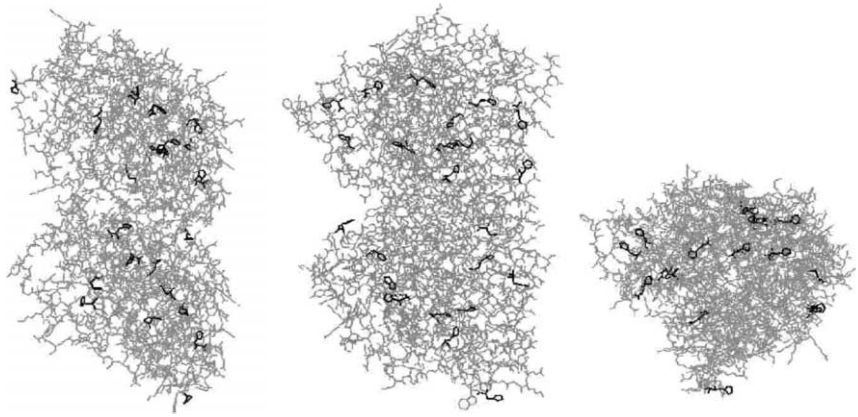

Fig. 7. Histidines (black) in the tubulin dimer. Left: front view, Middle: side view, Right: top view.

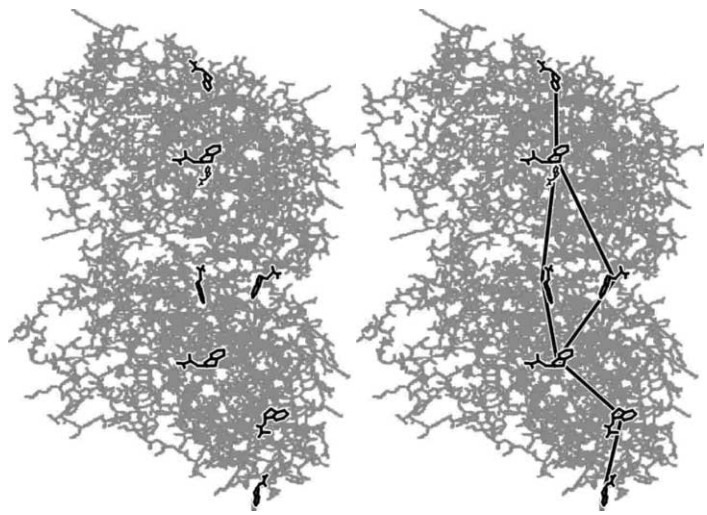

Fig. 8. Tryptophans in the tubulin dimer. Left: trypyophan locations in front view. Right: tryptophans interconnected by estimated shortest paths (2-D approximation). Typical separations are 2 nm, close enough for 'through-bond' hopping. The path separates near the cleft, or hinge of the dimer.

Histidines are more numerous than tryptophans, 22 per dimer, and their arrangement is more complex (Fig. 7).

We next consider possible electron pathways between tryptophans and histidines within tubulin.

2.3. Conduction pathways within tubulin

Conventional wisdom indicates that electron tunneling or hopping in proteins is only possible over distances under 1 nm. This is the 'Foerster distance' (maximum length of an excitation to travel). However the Foerster distance pertains to free hopping via an inert medium like an ionic solution. Within proteins electron movements may be facilitated by 'through bond hopping' over distances of 3 nm or more. Furthermore, if there are sufficient available electrons to fill half or more of the available sites, then conditions can exist within proteins at physiological temperature for (semi)conductivity comparable with silicon or

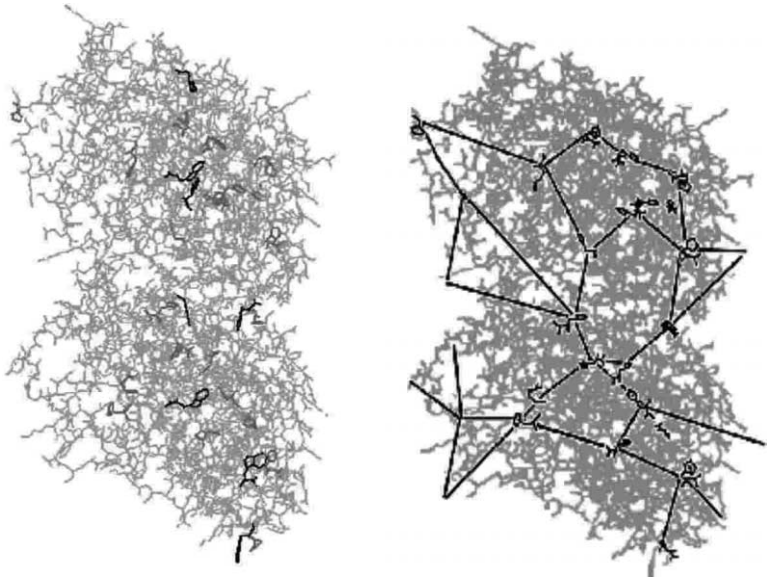

Fig. 9. Histidines in the tubulin dimer. Left: histidine locations in front view. Right: Histidines interconnected by estimated shortest paths. Typical separations are 1 nm.

even semi-metals (e.g. Brown, 1999). With dynamic water ordered at the protein surface, conductivity may be even further enhanced, and proton conduction can also occur.

As described earlier, in some enzymes electron hopping between amino acid residues may span 3.5 nm or more, and electron hopping along 3 tryptophans is known to occur in some enzymes. Based on a qualitative 2-D approximation, Fig. 8 estimates shortest direct paths between tryptophans in the tubulin dimer.

The tryptophan paths are mostly vertical, along the protofilament axis. The path separates near the cleft, or hinge of the tubulin. Occupation of one path versus the other path may control the dimer conformation, for example bending at the hinge.

In Fig. 9, we examine possible paths between histidines.

Estimation of shortest paths between histidines shows multiple loops and alternate possible routes. The average distance between histidines is about 1 nm.

Electron conduction between aromatic tryptophan and histidine amino acid residues is within range of known conduction mechanisms in proteins (tunneling, 'hopping', semiconduction).

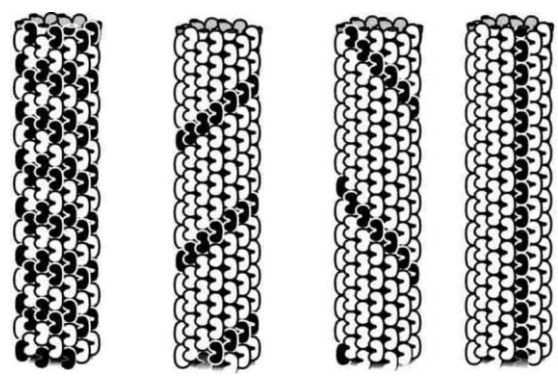

Fig. 10. Helical winding patterns of tubulin subunits within MTs. The windings repeat on any given protofilament according to the Fibonacci series: from left to right, 3, 5, 8 and 13 step repeats. The 3-step repeat is based on monomers, whereas the 5 and 8 step repeats are based on dimmrs. The 13-step repeat is simply the direct path along the vertical protofilaments. These patterns form the basis for attachment patterns of MAPs.

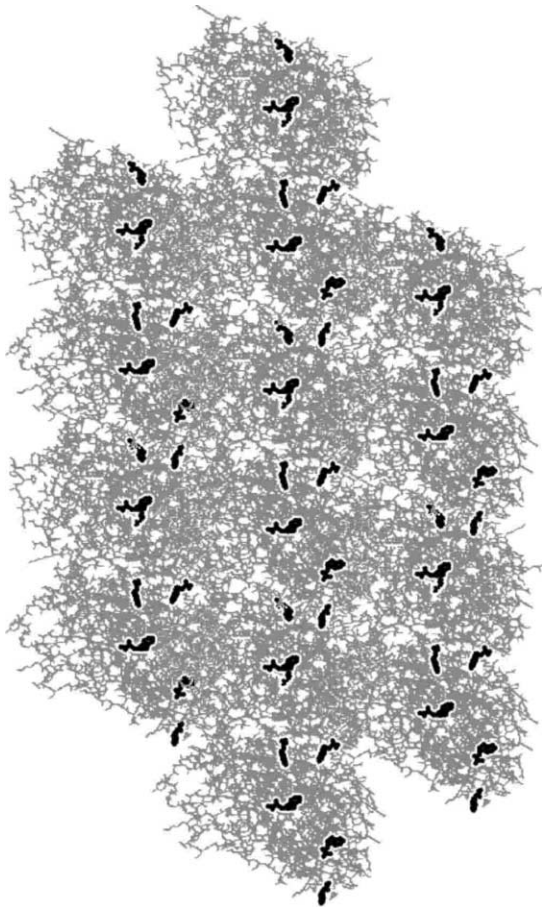

Fig. 11. A lattice neighborhood of seven tubulin dimers as found in a microtubule with tryptophans highlighted in black.

2.4. Inter-tubulin conduction pathways in microtubules

Tubulin dimers are arranged not only in vertical protofilament chains, but form a particular skewed hexagonal lattice whose crystal structure gives rise to helical winding patterns with regular repeat intervals. Increasingly steep winding patterns complete one cycle around the cylinder at a certain number of tubulins above where the cycle started (Fig. 10). These numbers, 3, 5, 8, 13, 21 etc. follow the well known 'Fibonacci series' found throughout nature. The repeating patterns also determine binding sites of MAPs'.

A neighborhood of seven tubulin dimers with tryptophans represented in black is shown in Fig. 11.

If we then show the intra-tubulin tryptophan paths in the lattice format by connecting nearest tryptophans (as in Fig. 12), we see that the tryptophan paths extend vertically along each protofilament.

Tryptophans seem oriented vertically whereas histidine locations suggest helical paths. When added to tryptophan, all possible structural winding pathways appear available (Figs. 13 and 14). In Section 4, we will discuss the possible implications for topological quantum error correction in microtubular computation.

3. Quantum computation in microtubules

3.1. Penrose–Hameroff 'Orch OR' model

Quantum theory and consciousness have been intertwined since the days of Schrödinger's cat when conscious observation was believed to cause collapse of the wave function. More recently, consciousness has itself been viewed as a collapse process, analogous to quantum computation. Penrose (1989, 1994) correlated the multiple possibilities of quantum superposition with multiple sub-conscious, or pre-conscious possibilities 'collapsing' to distinct choices or perceptions (cf. Stapp, 1993). In his early work in this area, Penrose suggested quantum superpositions of neurons both firing and not firing, collapsing or reducing (by his quantum gravity 'objective reduction' process, OR) to either firing or not firing. Thus the neuron was acting as qubit. However, for a variety of reasons MTs within each neuron were deemed better candidates for biological quantum computation, with individual tubulins within MTs acting as qubits.

The Penrose–Hameroff 'Orch OR' model portrays consciousness as quantum computation in MTs which collapse or reduce by an objective factor related to quantum gravity. For detailed explanations see Penrose and Hameroff, 1995; Hameroff and Penrose, 1996a,b; Hameroff, 1998, etc. The basic ideas are these:

- Conformational states of individual tubulins within neuronal MTs are determined by quantum mechanical London forces within the tubulin interiors which can induce conformational quantum superposition (Fig. 3).
- While in superposition, tubulins communicate/compute with entangled tubulins in the same microtubule, and in other MTs in the same neuron.
- Quantum states in MTs in any given neuron may extend to MTs in neighboring neurons, and through macroscopic regions of brain via tunneling through gap junctions (see Section 3.3, Fig. 16).
- Quantum states of tubulin/MTs are isolated/protected from environmental decoherence by biological mechanisms which include phases of actin gelation, ordered water, coherent pumping and topological quantum error correction (see Section 3.2).
- Microtubule quantum computations/superpositions are tuned or 'orchestrated' by MAPs during a classical, liquid phase which alternates (e.g. at 40 Hz) with a quantum, solid state phase.
- Following periods of pre-conscious quantum computation (e.g. on the order of tens to hundreds of milliseconds) tubulin superpositions reduce or collapse by Penrose quantum gravity 'objective reduction' (OR Fig. 15). The classical output states which result from the OR process are chosen non-algorithmically ('non-com-

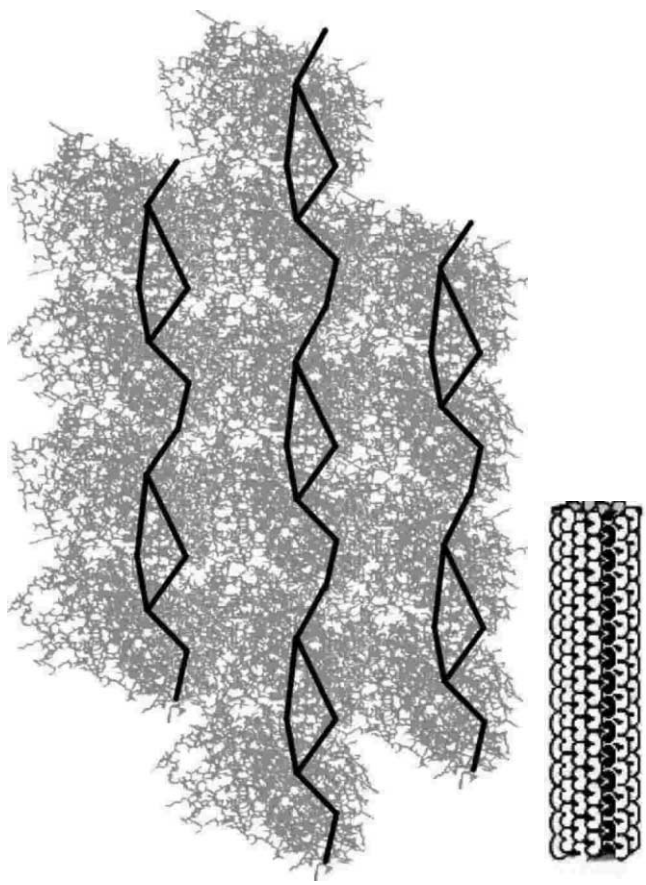

Fig. 12. Left: a lattice of seven tubulin dimers as found in the microtubule lattice. Black lines connect tryptophans. Right: vertical path along protofilament ('13 start' winding pattern) which corresponds with tryptophan conduction path.

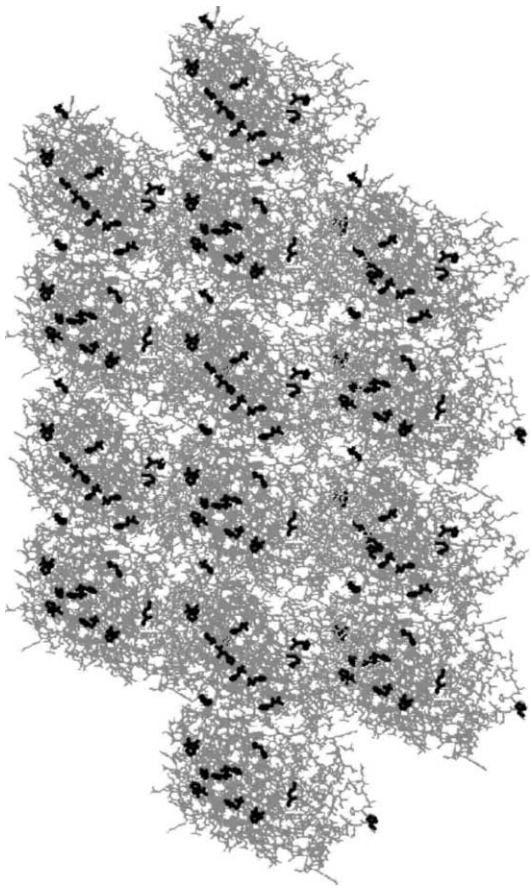

Fig. 13. A lattice neighborhood of seven tubulin dimers as found in a microtubule with histidines highlighted in black.

putably') and then govern neurophysiological events by binding of MAPs, regulating synapses and membrane functions etc.

- The reduction or 'self-collapse' in the orchestrated objective reduction 'Orch OR' model is suggested to be a 'conscious moment', linked to Penrose's quantum gravity mechanism which ties the process to fundamental spacetime geometry. This connection enables a panprotopsychist approach to the 'hard problem' of subjective experience (Chalmers, 1996a).

Other quantum models related to consciousness include those of Marshall (1989), Beck and Eccles (1992), Stapp (1993), Jibu and Yasue (1995).

Quantum models have potential explanatory value for the enigmatic features of consciousness (Table 3), but face at least two conceptual obstacles: (1) the apparent likelihood of rapid 'decoherence' (loss of quantum state) due to environmental thermal interactions in the seemingly-too-warm brain (Tegmark, 2000; Seife, 2000); and (2) the question of how a quantum state or field located within neurons might extend across membranes and anatomical regions to approach 'brain-wide' proportions. The next section deals with these issues.

3.2. Quantum states in the brain? Decoherence and biological feasibility

Quantum computing surpasses classical computing in certain critical functions (e.g. Grover's quantum search algorithm) and would be of extreme benefit to biological organisms for survival and adaptation. Billions of years of evolution may have solved the problems of decoherence and spatiotemporal spread. A number of mechanisms to prevent environmental decoherence have been suggested, specifically for quantum computation in MTs. These include (1) coherent pumping of the environment; (2) screening due to counterion Debye double layers surrounding MTs; (3) screening by actin gelation and ordered water; (4) quantum error correction; (5) topological effects of the microtubule cylindrical lattice. Recent calculations of protein decoherence times indicate quantum superpositions may indeed survive for neurophysiological time durations (Hagan et al., 2000), and brain imaging by 'quantum coherence MRI' utilizes quantum couplings of proton spins in proteins and water to give a neuroanatomical correlate of consciousness (Richter et al., 2000; Rizi et al., 2000). This quantum coherence is a MRI-induced artifact, but shows that quantum coherence of some sort can indeed occur in the brain.

Technological quantum computation became feasible with the advent of quantum error correction codes. This means that algorithms run on the quantum computer which detect and correct errors (random localized decoherence) before they destroy the global quantum state. In some cases topological structure of the quantum computer enhances the error correction. For example

toroidal surfaces (e.g. Kitaev, 1997) may have global, topological degrees of freedom which are protected from local errors and decoherence. Topological quantum computation and error correction have been suggested to occur in MTs by Porter (2001) and the helical windings discussed in Section 2 may correspond to quantum-computational 'basis states' distinguished by 'winding number' (Figs. 12 and 14).

3.3. Spatio-temporal spread of quantum states in the brain

Regarding spatial extension of a quantum (or 'quantum-like' (John, 2001)) field throughout the brain, a possible solution may be gap junctions-window-like 'electrical' connections between cells including neurons (e.g. Dermietzel and Spray, 1993). Gap junctions are more primitive and less

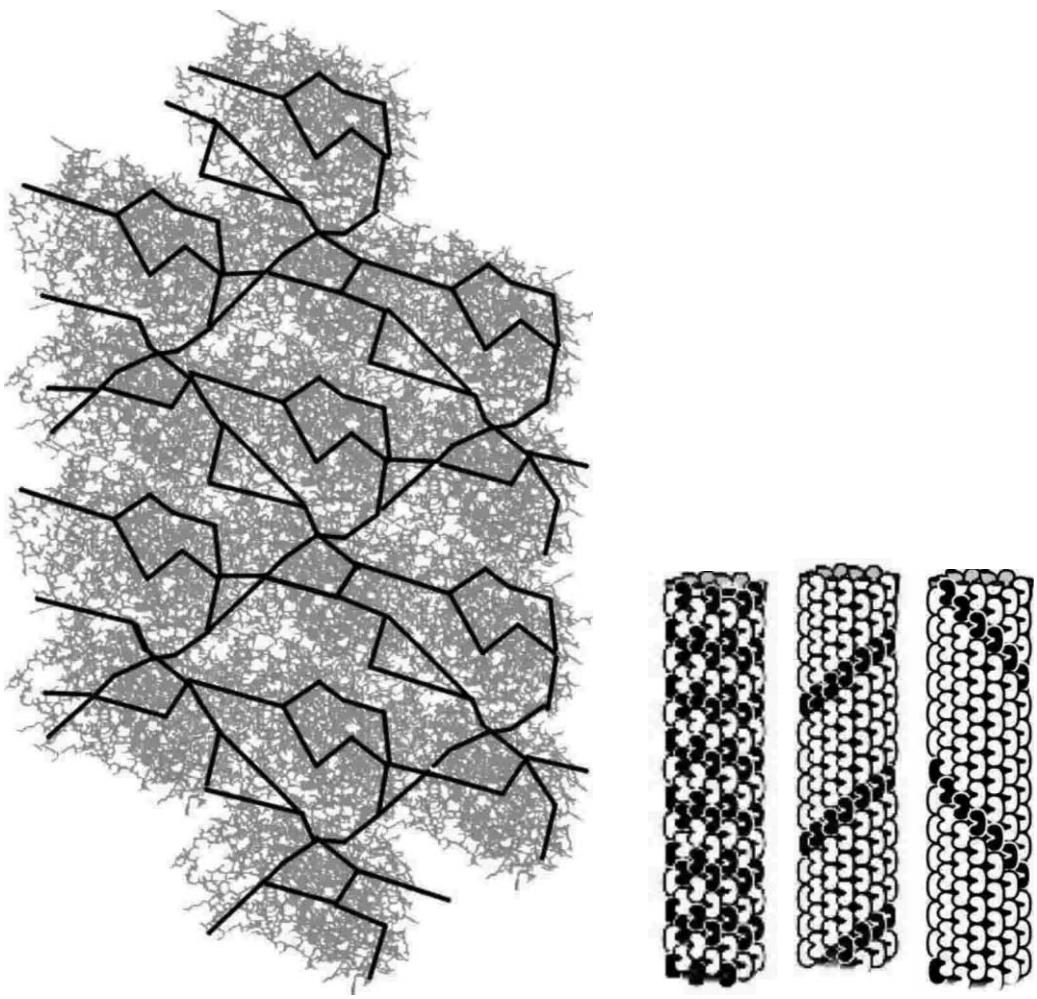

Fig. 14. Left: a lattice of seven tubulin dimers as found in the microtubule lattice. Black lines connect histidines. Right: 3, 5 and 8 start helical winding patterns in MTs. Histidine pathways in the lattice may be seen to follow these winding patterns.

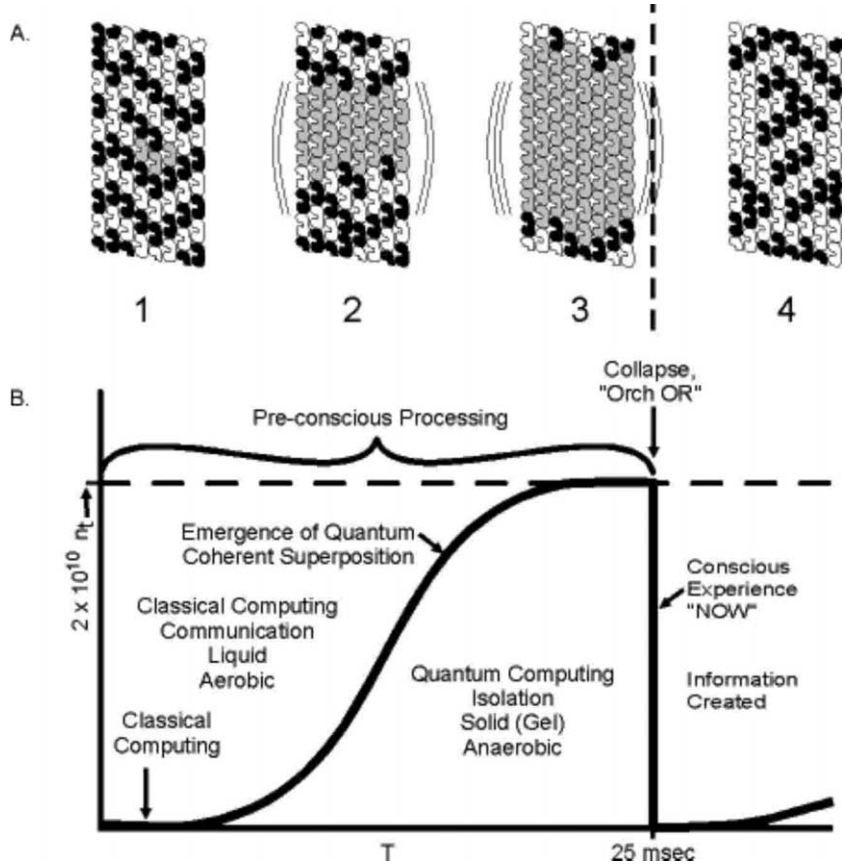

Fig. 15. An Orch OR event. (a) Microtubule simulation in which classical computing (step 1) leads to emergence of quantum coherent superposition and quantum computing (steps 2, 3) in certain (gray) tubulins. Step 3 (in coherence with other microtubule tublins) meets critical threshold related to quantum gravity for selfcollapse (Orch OR). A conscious event (Orch OR) occurs in the step 3 to 4 transition. Tubulin states in step 4 are noncomputably chosen in the collapse, and evolve by classical computing to regulate neural function. (b) Schematic graph of proposed quantum coherence (number of tubulins) emerging versus time in MTs. Area under curve connects superposed mass energy E with collapse time T in accordance with $E = h - \mathrm{bar}/T$. E may be expressed as N_t, the number of tubulins whose mass separation (and separation of underlying space time) for time T will selfcollapse. For $T = 25$ ms (e.g. 40 Hz oscillations), $N_t = 2 \times 10^{10}$ tubulins.

numerous connections than chemical synapses, and occur between dendrites, axons, cell bodies and/or glial cells. Dendritic–dendritic gap junctions in particular have been implicated in the mediation of conscious processes (Pribram, 1991; Eccles, 1992). Cell interiors (cytoplasm) are continuous through gap junctions so that cells connected by gap junctions have actually one complex interior. Quantum states isolated in one cell interior may thus extend to neighboring cells by quantum tunneling of electrons across the 4 nm gap junction. Specific intracellular organelles have been discovered in dendrites, immediately adjacent to dendritic–dendritic gap junctions. These are layers of membrane covering a mitochondrion, and are called 'dendritic lamellar bodies' (De Zeeuw et al., 1995). The dendritic lamellar bodies are tethered to small cytoskeletal proteins anchored to MTs, and it is suggested that the mitochondria within the bodies provide free electrons for tunneling, forming a tunneling diode pair or Josephson junction between cells (Fig. 16). As few as three gap junction connections per cortical neuron (with perhaps thousands of chem-

ical synapses) to neighboring neurons and glia which in turn have gap junction connections elsewhere may permit spread of cytoplasmic quantum states throughout significant regions of the brain, weaving a widespread syncytium whose unified interior hosts a unified quantum state or field (Hameroff and Penrose, 1996a,b; Woolf and Hameroff, 2000).

Kandel et al. (1991) remarked that neurons connected by gap junctions fire synchronously,

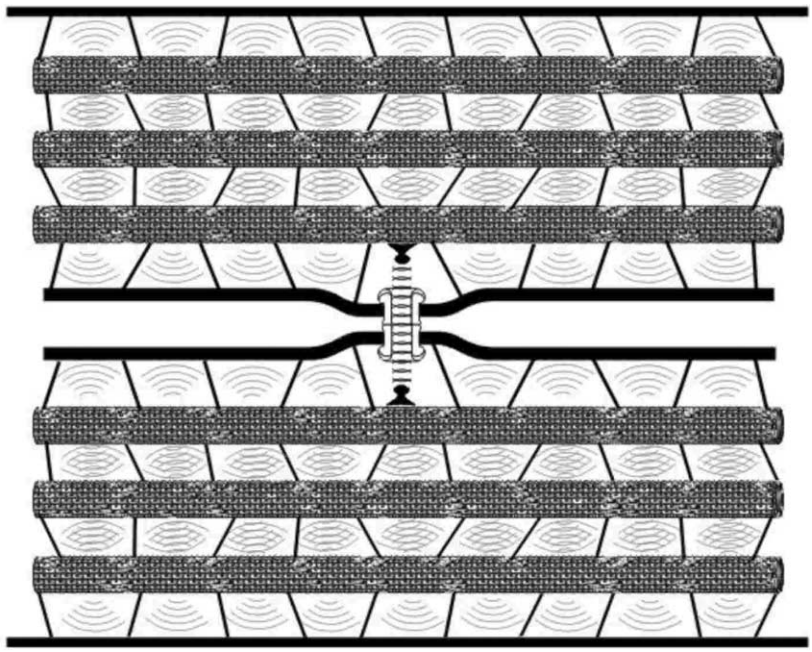

Fig. 16. Schematic representation of a gap junction connecting two dendrites in which MTs are in quantum superposition/quantum computation 'tuned' by interconnecting MAP proteins as suggested in the Penrose–Hameroff Orch OR model. On either side of the gap junction, dendritic lamellar bodies (DLBs) containing mitochondria may act as tunneling diodes to convey the quantum state between the dendrites.

Table 3
Enigmatic features of consciousness and possible solutions via the Orch OR model

Enigmatic feature of consciousness	Classical explanations	Possible Orch OR solutions
Essential nature of human experience ('qualia', the 'hard problem')	'Emergence': conscious experience emerges from computational complexity	Pan-protopsychist philosophy; qualia are fundamental features of spacetime geometry, accessed by quantum gravity OR
Unitary sense: 'binding problem'	Temporal synchrony e.g. coherent 40 Hz	(1) Non-local quantum coherence; indivisible macroscopic quantum state (e.g. Bose-Einstein condensate); (2) Instantaneous self-collapse of superpositioned states (Orch OR)
Transition from pre-conscious processes to consciousness	No apparent threshold	Collapse, or reduction of the quantum wave function, a la quantum computation
(Apparent) 'free will'	None	Non-computability in quantum gravity OR

Table 4
Potential single-body approaches to microtubule electronics

Single body theory
Diffusion equations for single mobile electrons
Microtubule as solenoid
Crystal field theory (microtubule as paracrystal, tubulin dimer as unit cell)

Table 5
Potential many body theory approaches to microtubule electronics

Many body theory
2-D models from condensed matter physics (e.g. chiral spin liquid, Bonesteel, 2000)
Quantum-classical phase transition in quantum computers (Aharonov, 1999)
Topological quantum computation and error correction (Kitaev, 1997)
Josephson-junction arrays (Ivanov et al., 2001)
Band theory (Brown and Tuszynski, 2001)

behaving like 'one giant neuron', and E.R. John (2001) has suggested that gap junction-connected neurons ('hyper-neurons') mediate zero phase lag coherence. Dendritic lamellar bodies are associated with synchronously firing neurons (De Zeeuw et al., 1997) and several studies (Galarreta and Hestrin, 1999; Gibson et al., 1999; Velasquez and Carlen, 2000) implicate gap junction-connected interneurons in the mediation of coherent ('40 Hz') oscillations. These gap junction-connected interneurons form 'dual' connections (gap junctions and GABAergic chemical synapses) with pyramidal cells and other cortical neurons. GABA inhibition could quiet membrane activities, avoiding decoherence to enable quantum states in neuronal cytoplasmic interiors to develop and spread among many gap junction-linked cells across wide areas of the brain. Thus gap junction-connected coherent 40 Hz neurons may support spatially extended quantum states.

4. Conclusion

Scientists speak of 'levels of theory', a hierarchy of models ranked by physical detail. The ultimate level would be a 'Theory of Everything'. One possible over-arching approach (suggested by Roger Penrose) attributes quantum state reduction to gravity. The Orch OR model asserts that this ultimate level is relevant to the physics of consciousness.

The discovery of the aromatic lattice within the microtubule has no immediately discernible implications at this ultimate level, but it suggests a host of new possibilities at simpler levels of theory (one-body quantum models, many-body quantum models, and quantum-field models). For example, in the Orch OR model it is assumed that neighboring dimers become entangled by the dipole–dipole interaction, which appears in the classical cellular automaton models of the microtubule. But it may be that mobile electrons following these newly recognized pathways contribute more to the entanglement process; tubulin dimers may become entangled indirectly, by way of interaction with strongly correlated mobile electrons.

Tables 4 and 5 list a few theoretical approaches to motion of electrons within the lattice of aromatic amino acids. These and others need to be explored and their predictions calculated (Brown and Tuszynski (2001) is a promising starting point). The structures of other biomolecules must also be examined, in order to show whether these pathways are unique to tubulin or found in many proteins. Logical candidates for examination include actin and other cytoskeletal proteins, the bacterial protein FtsZ, which is related to tubulin, and microtubule associated proteins such as MAP-2, MAP-4 and tau (Eva Nogales, personal communication).

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Conformational Analysis of the Carboxy-Terminal Tails of Human β-Tubulin Isotypes

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ABSTRACT Several isotypes of the structural protein tubulin have been characterized. Their expression offers a plausible explanation for differences regarding microtubule function. Although sequence variation between tubulin isotypes occurs throughout the entire protein, it is the extreme carboxy-terminal tails (CTTs) that exhibit the greatest concentration of differences. In humans, the CTTs range in length from 9 to 25 residues and because of a considerable number of glutamic acid residues, contain over 1/3 of tubulin's total electrostatic charge. The CTTs are believed to be highly disordered and their precise function has yet to be determined. However, their absence has been shown to result in altered microtubule stability and a reduction in the interaction with several microtubule-associated proteins (MAPs). To characterize the role that CTTs play in microtubule function, we examined the global conformational differences within a set of nine human β-tubulin isotypes using replica exchange molecular dynamics simulations. Through the analysis of the resulting configuration ensembles, we quantified differences such as the CTTs sequence influence on overall flexibility and average secondary structure. Although only minor variations between each CTT were observed, we suggest that these differences may be significant enough to affect interactions with MAPs, thereby influencing important properties such as microtubule assembly and stability.

INTRODUCTION

Microtubules (MTs) are hollow cylinders constructed from linear chains of the protein tubulin. Tubulin is highly conserved throughout the entire eukaryotic kingdom, and the two main classes, α/β, are expressed from multiple genes, producing several isotypes with seemingly identical functions (1–3). At the cellular level, it has been hypothesized that MT stability is regulated by subtle variations observed between α and β isotypes (4–6). There is ~80%–95% sequence identity between isotypes; however, the extreme carboxy-terminal tails (CTTs) exhibit considerable differences, having only 50%–60% identity in this region (7,8). Early phylogenetic comparisons of the vertebrate β-tubulin families identified the CTTs, along with an internal variable domain, as the primary isotype defining features of the β-tubulin protein (9).

The importance of the CTTs has been demonstrated through their removal using limited proteolytic cleavage by substilisin. After cleavage, the critical tubulin concentration required for polymerization was found to be ~50 times lower than that for intact tubulin (10). This rate was shown to decrease further through the addition of MT-stabilizing compounds such as paclitaxel (11). Proteolyzed tubulin also exhibits altered protofilament-bending, resulting in the formation of sheets, bundles of twisted filaments, rings, unstructured aggregates, or MTs with reversed polarity (10,12). Finally, the presence of the highly charged CTTs is thought to obstruct tubulin/tubulin interactions, regulating the rate and conformation of MT assembly through unfavorable electrostatic interactions (13). This hypothesis is supported by the observation that divalent cations, or the substitution of acidic residues, results in an increased rate of MT assembly and a decreased rate of MT disassembly (14,15).

In addition to tubulin interactions within the MT itself, the functional stability of MTs in vivo has been shown to depend on interactions with MT-associated proteins (MAPs) through interactions with the CTTs. For example, the correct assembly of the kinetochore and the integrity of the mitotic spindle is dependent on the ability of the Dam1 complex to bind the CTTs of β-tubulin (16). Interactions between the CTTs and MAP2 or MAP τ have also been shown to result in the stabilization of MT structures in vivo (17,18). Interactions between kinesin and MTs may also be significantly modulated through direct contacts with the CTT (19) or through the indirect modulation of kinesin's ability to bind ATP (20). Recent evidence also suggests that the CTTs may play a role in apoptosis, as interactions between MTs and the proapoptotic and antiapoptotic members of the Bcl-2 family are also affected by the presence of the CTT regions of α- and β-tubulin (21), an observation that may offer a plausible explanation for the previously observed interactions between Bcl-2 and paclitaxel (22).

Although the presence of intact CTTs is seemingly essential for proper MT assembly and function, little is known about their structure and how this may affect interactions with other proteins. Several early studies have examined possible conformations of the CTTs using NMR and circular dichroism (CD) spectroscopy and demonstrated that there was

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inherent disorder within the CTTs (23,24). Although a region of increased helicity toward the amino terminus was identified, a finding that was subsequently confirmed by electron crystallographic analysis, this has provided little additional information about CTT structure, as the last 10 residues of α-tubulin and the last 18 residues of β-tubulin were not visualized due to lack of density (25). A possible explanation for this lack of density is the nonhomogenous presence of tubulin isotypes in the MT preparations used in the crystallographic analysis. However, homogenous samples of $\alpha\beta$II and $\alpha\beta$III tubulin failed to improve the quality of density maps, and Nogales et al. (25) concluded that their lack of resolution was indeed due to disorder in this region.

Fortunately, molecular modeling provides us with the unique ability to examine conformations of the CTTs at a level of detail experimental analysis is unable to yet provide. However, even modeling has its limitations in this regard, where a persistent problem with low temperature protein-folding simulations is that of obtaining adequate sampling. This problem exists because simulations generally become trapped in one of a large number of local energy minima. Several generalized ensemble algorithms, based on non-Boltzmann probability weight factors, are capable of overcoming this problem by introducing a random walk in energy space (26). However, it is often not a trivial matter to determine the non-Boltzmann weight factors, and for this reason we chose replica exchange molecular dynamics (REMD) as a method to examine possible conformations of the tubulin CTTs (27,28). REMD utilizes a large number of parallel simulations at different temperatures, with exchanges between trajectories attempted periodically using Metropolis criteria. All replicas of the system then perform a random walk through temperature space and, as a consequence, also through energy space. A replica may therefore overcome an energy barrier through exchange with replicas at higher temperatures. This allows configurations to be sampled at a given temperature on timescales not otherwise possible while still maintaining a thermodynamically consistent ensemble of configurations. Here, we discuss the creation of nine models of human β-tubulin CTT peptides using REMD and their analysis for relative conformational flexibility within the ensemble.

METHODS AND MATERIALS

Peptide construction and system configuration

Based on our previous examination of β-tubulin isotypes, we chose a set of nine peptides corresponding to the consensus CTT sequences of βI (GI:34222261), βII (GI:68299771 and GI:42476191), βIII (GI:50592995), βIVa (GI:21361321), βIVb (GI:68051719), βV (GI:14210535), βVI (GI:41152077), βVII (TUBB4Q) (GI:55770867), and βVIII (TUBB8) (GI:42558278) (Table 1) (8). Accession numbers correspond to Entrez Nucleotide expressed mRNA sequence IDs contained only within the human genome. Using the crystallographic structure of tubulin (Protein Data Bank (PDB) ID: 1JFF) as a guide, we selected the conserved DA[TK]A motif that is positioned at the extreme end of helix H12 as the initiation point for the construction of all the peptides (29). This position marks the beginning of the structurally undefined region of the CTT in the structure and the domain examined using NMR and CD spectroscopy (23,24). All nine CTTs were prepared identically in an extended conformation, having the N-terminal charge neutralized by capping with an acetyl group using the PyMol v0.99 residue and fragment builder facility (30). The conventional protonation state at pH 7.0 was used for all residues, with a focus on His, which was protonated on the ε nitrogen throughout.

TABLE 1 ClustalW multiple sequence alignment of the CTT sequences

Isotype	Sequence	Length	Charge
I	DATAEEEED-FGEEAEEEA	18	−12
II	DATADEQGE-FEEEEGEDEA	19	−12
III	DATAEEEGEMYEDDEEESEAQGPK	24	−12
IVa	DATAEEGEF-EEEAEEEVA	18	−11
IVb	DATAEEEGE-FEEEAEEEVA	19	−12
V	DATANDGEEAFEDEEEEIDG	20	−12
VI	DAKAVLEED-EEVTEEAEMEPEDKGH	25	−11
VII	DATAEGGGV	9	−3
VIII	DATAEEEED-EEYAEEEVA	18	−12

The length and net charge at pH 7.0 was determined for each peptide, including the C-terminal cap.

Parameterization and model preparation

All calculations utilized the AMBER99 force field (31,32), which was applied using PDB2GMX in GROMACS 3.3 (33). Each peptide was placed in a rhombic-dodecahedron unit cell consisting of ~2800 TIP3P waters (34). Sodium and chlorine counterions were added to the most energetically favorable locations (as determined using GENION) such that the net charge of the system was neutralized and a final ion concentration of 100 mM was established. For all molecular dynamics (MD) and REMD calculations, rigid bonds were maintained using the LINCS algorithm for the peptide and SETTLE for waters. Particle Mesh Ewald (PME) was used for electrostatic interactions with a cutoff of 0.8 nm and a Lennard-Jones cutoff of 1.0 nm. Constant pressure was maintained by allowing the box size to fluctuate isotropically. Each system was minimized using a low-memory Broyden-Fletcher-Goldfarb-Shanno minimizer in GMXRUN until machine precision was achieved. This was followed by 20 ps of heating and 1.1 ns of equilibration.

REMD

To achieve a transition probability of 0.3, 43 target temperatures between 273 and 382 K were selected. Each replica was then heated/cooled to its target temperature over 50 ps and simulated without exchange for 500 ps, followed by 500 ps of REMD. Production REMD runs consisted of 10 ns of dynamics for each replica with exchanges attempted every 5 ps. Energy and conformational snapshots were saved every 1 and 10 ps, respectively. For each isotype, this produced an ensemble of 43,000 conformations and an aggregate simulation time of 430 ns over all temperatures.

Cluster analysis

A cluster analysis of the resulting REMD conformations was used to determine preferred conformations and relative populations of each peptide. The GROMOS clustering algorithm (39), implemented in G_CLUSTER, was used for this purpose. All Cα atoms were root mean-square deviation (RMSD) fit to the extended starting structure, and an RMSD cutoff was set for the Cα atoms; for each structure from the ensemble, all structures within

this cutoff were assigned as neighbors. The structure with the most neighbors was then identified as the center of a cluster and was removed from the pool with all its neighbors. This process was repeated until all structures had been assigned to a cluster. The RMSD cutoff was chosen such that the largest cluster contained 50% plus 1% of the structures.

Principal component analysis

As MD simulations tend to produce immense quantities of data, principal component analysis (PCA) is a powerful mathematical tool used to detect correlations in MD trajectories (40,41). To perform PCA, all of the conformations from the REMD ensemble were RMSD least squares fit to the reference structures, effectively removing all rotations and translations. Then, for non-mass-weighted PCA, the covariance matrix can be calculated as

$$\sigma_{ij} = \langle (r_i - \langle r_i \rangle)(r_j - \langle r_j \rangle) \rangle,$$

where $r_1 \ldots r_{3N}$ are the Cartesian coordinates of the N atoms used in the analysis and $\langle \ldots \rangle$ denotes the ensemble average. The resulting matrix can then be diagonalized, and the resulting $3N$-dimensional eigenvectors, $\mathbf{v}_i$, are organized in descending order of eigenvalues, λ_i. Eigenvalues represent the variance along their associated eigenvector, and the larger the eigenvalue the more significant the correlated motion.

A PCA of all nine isotypes of β-tubulin was performed at 311 K using the positions of all Cα atoms. To mimic the CTTs bound at their N-terminus, the backbone atoms of the first three amino acids of the ensemble were RMSD fit to the reference structure for the isotype at 311 K, producing an average structure (Fig. 1). This was essential for the physiological relevance of the PCA calculation and the considerable motion of the CTTs. Eigenvalues of different isotypes cannot be directly compared, as different numbers of atoms were used in the covariant analysis. This can be overcome by normalizing the eigenvalues by the number of atoms used in the analysis:

$$\sigma_i = \sqrt{\frac{\lambda_i}{N}},$$

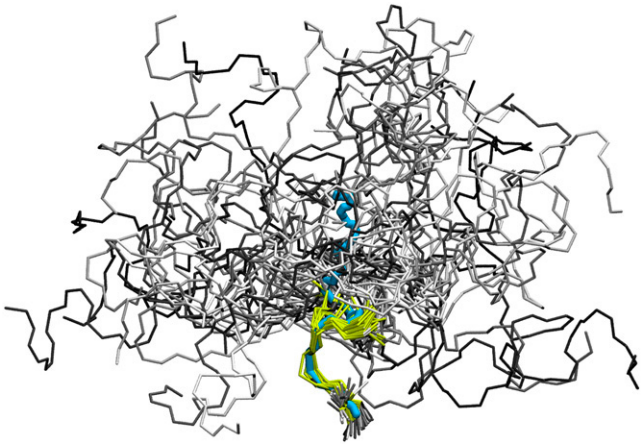

FIGURE 1 Human β-tubulin CTT sequence alignment and consensus. Illustration of reference and average structures used for PCA. The N, C, and Cα atoms from the three N-terminal residues of each conformation from the βI ensemble were RMSD fit to a representative structure backbone. The backbone atoms of 50 randomly selected conformers of the ensemble are shown in black, gray, and white to delineate conformers in no specific order. The first four residues of each peptide are shown in yellow. The average structure, used for subsequent PCA calculations, is shown in light blue. Image produced using VMD (63).

where σ_i is the root normalized eigenvalue representing the standard deviation along the ith eigenvector.

Sequence alignments

Sequence alignments of CTTs were performed with the default values using the European Bioinformatics Institute Clustal server (http://www.ebi.ac.uk/clustalw/), with the exception that the extended gap penalty was increased to a value of 0.5 (Table 1).

Motif identification

Structural motifs within the CTT ensemble were identified by performing pairwise alignment of each CTT sequence using the ClustalW alignment facility in MacVector (MacVector, Cary, NC). Motifs were identified as those sequences containing at least four consecutive identical residues. A structural similarity search was then performed using only sequence runs of four or more identical or similar residues. Structural similarity was determined by calculating the ρ_{sc} between all pairs of conformations from the ensembles of each isotype to create a 1001 $\times$ 1001 matrix. Here, ρ is defined as

$$\rho(A,B) = \frac{2 \times \mathrm{RMSD}(A,B)}{\left(R_{gyr}^2(A) + R_{gyr}^2(B) - \mathrm{RMSD}^2(A,B)\right)^{1/2}},$$

where A and B are structures with an equal number of points, R_{gyr} is the radius of gyration, and RMSD stands for the RMSD of the two structures. To obtain ρ_{sc}, we translate the centroid of each structure to the origin and rotate it such that the principal axes of inertia lie on the coordinate axes. Then, for both structures, we satisfy the condition that $R_{gyr}=1$ by scaling each axis independently such that each contributes equally to the radius of gyration. The ρ_{sc} provides a length-independent measure of conformational similarity of two structures (42). Structures with a ρ_{sc} of 0 are considered identical; those with a factor of 2 are considered maximally dissimilar. Mirror images have a value of $\sqrt{2}$, whereas structures with a factor of 0.3–0.5 indicate visual similarity. The fraction of conformational pairs with a ρ_{sc} of <0.3 is the fraction of time that regions within the two isotypes are structurally similar. See Table 3 for regions that showed higher than 50% ρ_{sc} similarity between isotypes.

RESULTS

Amino acid composition of CTT consensus sequences

The absolute length of each human β-tubulin CTT ranges from 9 to 25 residues; however, their typical length falls between 18 and 20 residues (Table 1). The exceptions are βVII, which contains only nine residues, and βIII/βVI, which have 24 and 25 residues, respectively. Although the overall amino acid composition of each CTT is quite similar, there are some notable factors to consider. When we examine all nine human CTT sequences in aggregate, the residue that occurs most frequently is Glu (42%). The occurrence of all other residues drops significantly from this value: Ala (18%), Asp (12%), Gly (8%), Thr (5%), Val (4%), and Phe (3%). The βVIII CTT contains no Gly, the βIVa/b CTTs have a proportionally low Asp content, and, finally, the βVII CTT has a reduced Glu content, which may simply be a result of its length. Interestingly, the remainder of the amino acids

occurs exclusively in the βIII (Lys, Met, Pro, Gln, Ser, and Tyr), βV (Ile, Asn), and βVI (His, Lys, Leu, and Pro) CTTs at frequencies of 1% or less. Finally, Cys, Trp, and Arg are absent from all the CTT sequences.

REMD and completeness of sampling

A notable exclusion from these simulations is the tubulin protein itself, a factor that will undoubtedly influence the results as they have been explained here. The decision to exclude the bulk tubulin was made due to its large system size and the computational resources that would be required to explore the conformational space of an entire tubulin dimer. Therefore, simulations of only the CTTs become a problem of protein folding, and adequate sampling is critical to the proper interpretation of the results. To increase sampling efficiency at different temperatures, we used replica exchange MD (REMD), which does not provide information on the dynamics of the system. However, as our ultimate interest is in the overall conformations of the CTTs, the loss of dynamics data was an acceptable compromise to gain increased sampling.

The completeness of sampling can be determined by calculating the normalized overlaps between two different parts of an MD trajectory. Such overlaps can indicate whether or not both parts are spanning over the conformational space equally and not diffusing to new parts (43–45). Subspace overlap between two sets of n orthonormal vectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$ and $\mathbf{w}_1, \ldots, \mathbf{w}_n$ is defined as

$$\text{overlap}(\mathbf{v}, \mathbf{w}) \equiv \frac{1}{n} \sum_{i=1,j=1}^{n,n} (\mathbf{v} \times \mathbf{w})^2.$$

When the overlap has a value of 1, the sets $\mathbf{v}$ and $\mathbf{w}$ can be considered to span the same subspace. This measure, however, does not account for the magnitudes of the eigenvalues, meaning that differences between all eigenvectors contribute equally. Furthermore, when two or more eigenvalues are equal, the corresponding eigenvectors are random, causing a random variation in the subspace overlap number. An alternative method, suggested by Hess (45), is the normalized overlap, defined as

$$d = \sqrt{\text{tr}\left(\left(\sqrt{C_1} - \sqrt{C_2}\right)^2\right)}$$

$$\text{normalized overlap}(C_1, C_2) = 1 - \frac{d}{\sqrt{\text{tr}\left(C_1\right) + \text{tr}\left(C_2\right)}},$$

where d is the difference between the covariant matrices, C_1 and C_2, and tr is the trace. Here, if the overlap is 0, then the two sets are considered to be orthogonal, whereas an overlap of 1 indicates that the matrices are identical. Covariant analysis of the trajectories from each isotype at 311 K, chronologically divided into thirds, was performed using the same procedure used for the PCA. The subspace and normalized overlaps calculated between each of these thirds are reported in Table 2. The high overlap between the thirds indicates that each part of the simulation is sampling approximately the same conformational space, and it is unlikely that there are unexplored regions missed earlier in the run. Although not a guarantee of complete equilibrium sampling, we concluded that the overlap using both of the above mentioned methods is acceptable and that adequate sampling within all of these systems has been obtained.

TABLE 2 PCA overlap of CTT peptides

PCA overlap at 311 K for N-terminus fit						
	Subspace overlap			Normalized overlap		
Isotype	1st vs. 2nd	1st vs. 3rd	2nd vs. 3rd	1st vs. 2nd	1st vs. 3rd	2nd vs. 3rd
I	0.93	0.95	0.95	0.86	0.84	0.86
II	0.90	0.87	0.92	0.84	0.77	0.78
III	0.94	0.91	0.95	0.82	0.78	0.86
IVA	0.92	0.84	0.91	0.77	0.62	0.75
IVB	0.93	0.89	0.89	0.78	0.76	0.83
V	0.92	0.89	0.89	0.75	0.78	0.76
VI	0.88	0.88	0.94	0.72	0.71	0.84
VII	0.96	0.91	0.98	0.70	0.56	0.76
VIII	0.92	0.86	0.94	0.79	0.76	0.85

The subspace overlap consists of the first 10 eigenvectors only. For both overlap calculations, the ensemble was divided into thirds and all are compared to each other.

Clustering and secondary structure

Clustering each CTT peptide can provide a representation of probable folded conformations; however, a bell-shaped ρ_{sc} distribution about a representative structure demonstrates that there is actually no native folded conformation for most of the isotypes (Fig. 2). The only exception was the βVII CTT, which exhibited a significant population of folded structures with a $\rho_{sc} < 0.5$ and a second population of unfolded structures with a $\rho_{sc} > 0.5$. These results recapitulate previous observations that the CTTs are extremely flexible, and any structures within them are transient in nature and cannot be captured by the clustering methodology used in this study. Therefore, we felt that a more appropriate analysis was to consider the ensemble average of secondary structures as calculated from the entire 10 ns of ensemble conformations at 311 K by STRIDE (46) (Fig. 3). These results demonstrated that, whereas the ρ_{sc} distribution showed no native folded conformations, many of the CTTs contain regions that are either α- or 3–10-helical at least 40% of the time.

Motif identification

Having established the presence of transient secondary structures within the ensembles, we performed pairwise alignments of all the CTT sequences to identify potential sequence

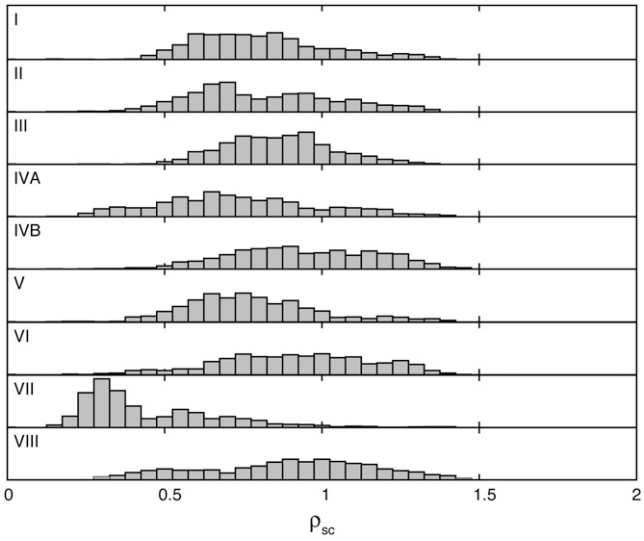

FIGURE 2 ρ_{sc} distributions of CTTs about representative structures. The large spread in the histograms of the $\boldsymbol{\rho_{sc}}$ distributions for each CTT indicates the lack of a single folded conformation. A distinct population of 67% conformers have a $\boldsymbol{\rho_{sc}} < 0.5$ for the βVII isotype, suggesting a distinct folded conformation for this CTT.

motifs with which to correlate our modeling results. A ρ_{sc} matrix was then calculated for all the conformations across each isotype at 311 K. Only those motifs having a high fraction of low ρ_{sc} were determined to be significantly similar (ρ_{sc} of 0.3 or less) (Table 3). Through this analysis, we identified two motifs that showed both sequence and conformational similarities. To visually compare the motifs, the conformations of the aligned subsequences were clustered as described in Materials and Methods. Illustrated in Fig. 4 are RMSD alignments of representative structures (those with the most neighbors) from each isotype containing the respective motif. The first of these motifs was identified as a probable casein kinase-2 (CK-2) binding motif at the N-terminal end of the peptide (Fig. 4 *A*). This motif was also independently identified using a Prosite search for motifs within each of the CTT sequences (47) (not shown). The second motif was determined to correspond to a previously identified MAP2 binding motif found within α-tubulin (Fig. 4 *B*) (48). We should note that although a common conformation for each of the motifs across isotypes has been identified, these are not stable folds; and depending on the isotype and residues included in the search, anywhere between 1% and 69% of the ensemble structures have a $\rho_{sc} < 0.3$ when compared to the motif conformation.

PCA

Not only is the secondary structure of each CTT significant, but their flexibility is also of critical importance when attempting to understand how they are able to conform when interacting with MAPs. Because of the complex interplay of many factors, sophisticated statistical analysis of conformational ensembles, such as PCA, appears to be the most appropriate computational tool to characterize the flexibility. PCA was performed on the ensemble at 311 K, with the backbone atoms of the first three residues RMSD fit to a reference structure (Fig. 1). Resulting eigenvectors were ordered by descending eigenvalues, which represent the variance of the motion along the principal components. In Fig. 5, the six largest root normalized eigenvalues are shown for each isotype. Except for βVII and βIVa, to a lesser extent, three components can be identified with prominent eigenvalues of comparable magnitude, which is significantly higher than the magnitude of other eigenvalues. The three components with the largest eigenvalues represent correlated motions of the peptide fragments with the most significant standard deviations of the motion along the corresponding orthogonal directions. Since the standard deviations shown in Fig. 5 were normalized for the number of residues, they can be employed as a universal measure for comparison of conformational motions in different CTT peptides.

Two-dimensional projections of the REMD conformation ensemble

A more detailed representation of the conformational motions in peptides is provided by projections of the ensemble of conformations onto the planes spanned by the most important principal components (Fig. 6). Here, the spatial distributions of occupancies of the various conformational states are shown over the planes spanned by the first and second and by the first and third principal components. A representative isotype, βV, illustrates that both distributions are smooth and without evidence of considerable clustering. Similar behavior exists in all isoforms considered except for βIVa and βVII, which show significant clustering of conformational occupancy, with maxima at (0.1,0) and (−0.2,0), respectively. The widths of the distributions shown in Fig. 6 are not directly dependent on the length of the peptide, as the eigenvalues have been normalized by the number of residues. The reduced width of βVII, therefore, indicates its reduced mobility. The bin size of the histogram has been chosen in proportion to the width of the distribution; so the maxima seen for βIVa and βVII are not artifacts of the sampling but represent regions of increased occupancy. Since the occupancy of a conformational state is inversely proportional to the free energy of that state (49,50), the lack of clustering in the other isotypes suggests that the global minima are broad or that there are generally no significant local minima. In contrast, βVII has a significant basin of attraction, indicative of a folded conformation. This observation is consistent with the preceding cluster analysis, which identified a well-defined folded state in the βVII isoform only. Although less defined, the occupancy distribution for βIVa also exhibits a preferred folded conformation, demonstrating that the PCA methodology is more sensitive to transient structural motifs than the

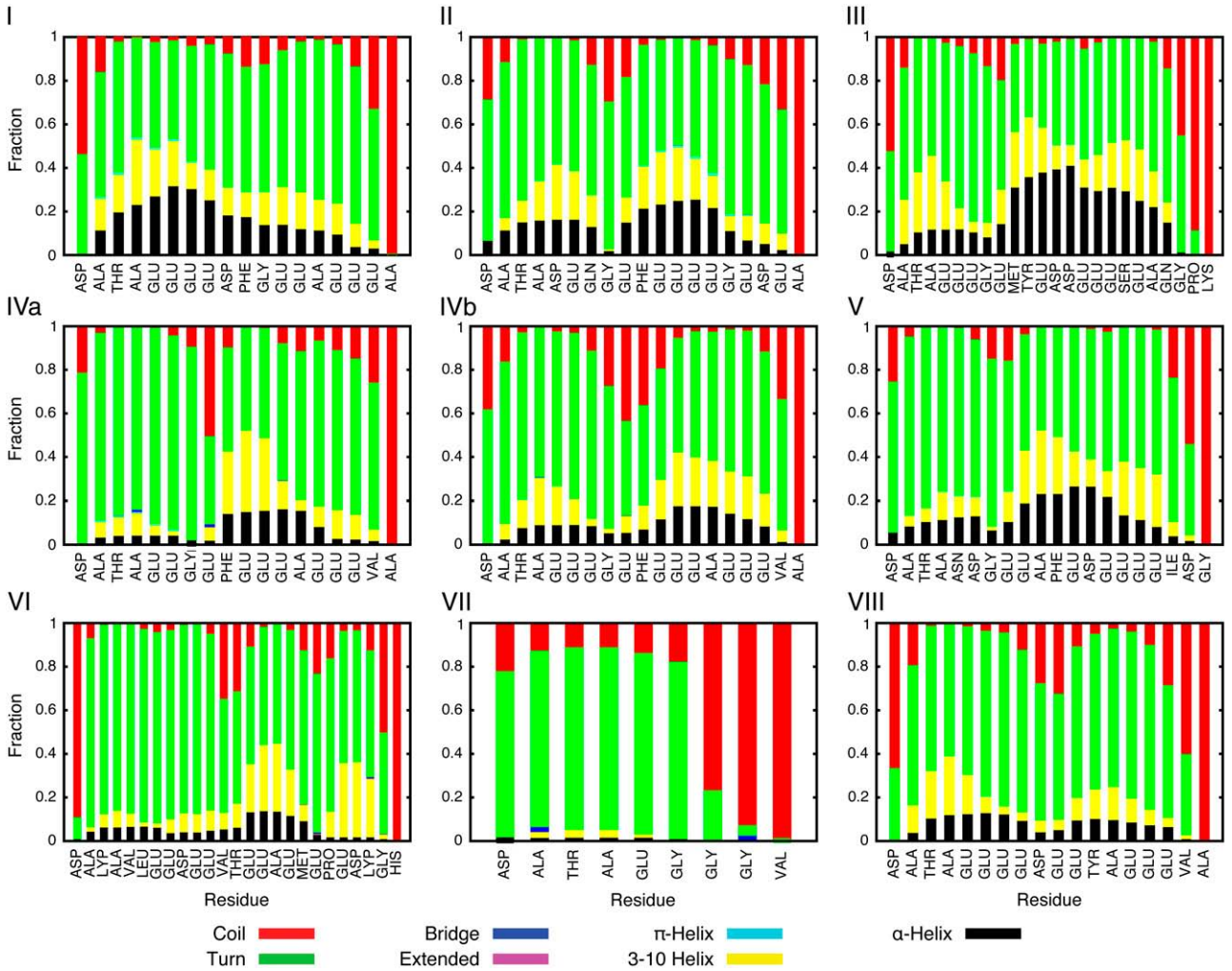

FIGURE 3 Time average of each type of secondary structure for each CTT. The total fractional secondary structure content was determined as an average over the entire 10 ns MD simulation at 311 K. The fractional time averages of each type of secondary structure are stacked to sum up to one. Although experimental studies of tubulin suggest that the CTTs are unstructured, these results suggest that many of the CTTs contain a significant amount of transient helix.

clustering analysis. One more noteworthy feature is that the two-dimensional distributions over the first and second components are largely similar to those over the first and third components, suggesting a significant level of symmetry among principal components.

Relative CTT flexibility

In addition to information regarding the motion of each CTT peptide, PCA provides the ability to compare relative flexibilities. As we studied only the nine human β-tubulin isotypes, it is not possible to comment on the flexibility of the CTT peptides in an absolute sense. However, there have been a few studies discussing the flexibility of small peptides. Although they use a different approach to calculating the flexibility, Ma et al. made a survey of 28 short peptides and determined that the native helical structures of the peptides were more flexible than random or disordered conformations, in agreement with our observations (Fig. 7 *B*) (51). Unfortunately, their methodology, calculating the vibrational free energy, does not allow comparison of flexibility between peptides. Here, as a measure of the relative flexibility of each peptide, the eigenvalues of principal components were compared. To characterize the flexibility, we employed the value $\sigma_{12} = \sqrt{\sigma_1^2+\sigma_2^2}$, where σ_1 and σ_2 are the normalized eigenvalues for the first and the second principal components, respectively. The magnitude of the PCA eigenvalue parameter σ_{12} can be viewed as the mean square width of the two-dimensional REMD configuration ensembles shown in Fig. 6, and thus σ_{12} can be directly interpreted as the flexibility of the peptide associated with its major correlated motions. Although the CTTs are characterized by three principal components (Fig. 5), the considerable symmetry between the second and third component demonstrated in Fig. 6 allows the use of only two components out of three to quantify the flexibility of CTTs. The correlations between the normalized distance (the distance between the N- and C-termini Cαs, divided by the number of residues),

TABLE 3 Sequence motif identification

CK2 Motif						
Isotype	Range	Sequence	Isotype	Range	Sequence	Similarity
III	1–9	DATAEEEGE	IVb	1–9	DATAEEEGE	72.56
II	1–8	DATADEQG	III	1–8	DATAEEEG	76.64
II	1–8	DATADEQG	IVb	1–8	DATAEEEG	62.97
II	1–5	DATAD	VII	1–5	DATAE	61.4
IVa	1–5	DATAE	VII	1–5	DATAE	64
II	1–4	DATA	V	1–4	DATA	64.1
IVb	1–4	DATA	V	1–4	DATA	62.34
V	1–4	DATA	VI	1–4	DAKA	51.26
V	1–4	DATA	VII	1–4	DATA	67.19
V	1–4	DATA	VIII	1–4	DATA	52.9
VI	1–4	DATA	VII	1–4	DATA	52.7
MAP2 Motif						
Isotype	Range	Sequence	Isotype	Range	Sequence	Similarity
II	10–14	FEEEE	III	11–15	YEDDE	60.89
II	10–14	FEEEE	V	11–15	FEDEE	71.61
IVa	11–14	EEAE	VI	14–17	EEAE	71.87
I	12–15	EEAE	VI	14–17	EEAE	78.71
IVb	12–15	EEAE	VI	14–17	EEAE	80.93
I	15–18	EEEA	III	15–18	EEES	67.62
IVa	14–17	EEEV	V	15–18	EEEI	69.12
IVb	15–18	EEEV	V	15–18	EEEI	85.56
V	15–18	EEEI	VIII	14–17	EEEV	55.65

Pairwise alignments of each CTT sequence identified several similar regions between peptides. Structural similarity between sequences was determined by calculating the ρ_{sc} between all pairs of conformations. Those pairs with high conformational and sequence similarity were used to characterize the motif conformations for the CK-2 and MAP2 domains (Fig. 4).

the time-averaged helical content, and the average clustering parameter $\langle \rho_{sc} \rangle$ with the value σ_{12}, which we use as the major measure of flexibility, can be seen in Fig. 7.

DISCUSSION

The CTT's involvement in MT regulation is critical; it is essential to understand the conformational differences adopted by different tubulin isotypes. We suggest that the sequence variability within the CTTs may have arisen as a mechanism to conserve the overall tubulin structure to maintain proper MT assembly, while still providing a flexible, solvent-exposed region of increased variability for interactions with proteins that affect MT function (see alignment in Table 1). This is both an attractive and reasonable hypothesis, as the role of intrinsically disordered proteins in protein interactions has been implicated as a mechanism to enhance specificity (52). The following discussion addresses three main points about the CTTs and their interactions with MAPs: first, the relative flexibilities of each of the CTT peptides; second, the presence of transient structure within these peptides; and finally, how this may influence protein interactions. We will discuss this possibility further when examining CTT flexibility in the context of MAP interactions, in particular interactions with the CK-2 and MAP2 binding motifs that we identified in this work.

CTT flexibility and secondary structure

Accurate flexibility and secondary structure measurements are critical when the goal is to propose mechanisms for MAP interactions with the large number of possible CTT conformations on the MT surface. The overall distribution of amino acids within the CTT will obviously have an effect on their

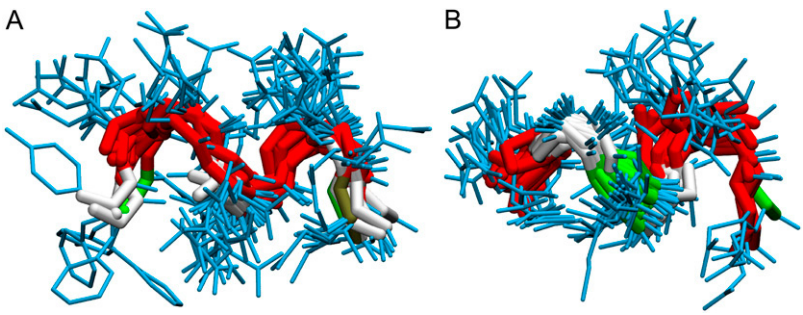

FIGURE 4 Sequence motif identification. High scoring sequence pairs from Table 3 were RMSD fit to illustrate the overall fold and position of side chains in the structural motifs. In both images, red residues are acidic, white residues are nonpolar, green residues are polar, and all side chains are blue. Image created using VMD (63).

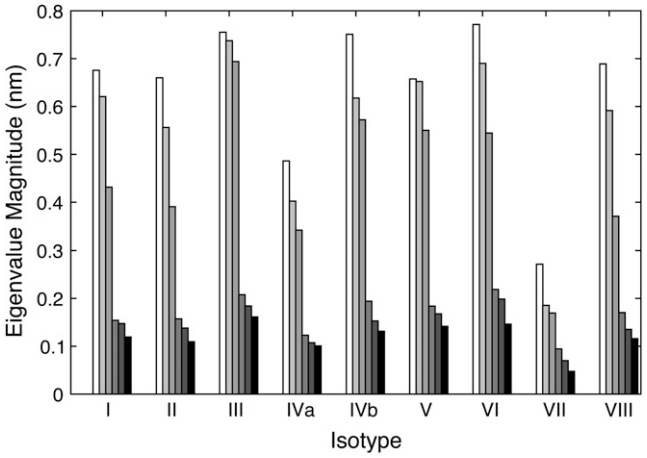

FIGURE 5 Root mean-square normalized eigenvalues for each isotype. The magnitudes of the first six root normalized eigenvalues (*white* to *black*) are shown for each isotype and represent the standard deviation of the motion along the eigenvector normalized for the overall length of the peptide. The relative size of the root-normalized eigenvalues indicates the relative flexibilities of the individual peptides.

flexibility and result in secondary structure. For example, the overall helical propensity of βII, βIII, βIVa, βIVb, and βV is interrupted by the presence of Gly (Fig. 3). Additionally, regions with uninterrupted stretches of Glu also tend to produce regions with greater helical propensity, providing an argument for the prevalence of this residue within the CTT sequences. This observation can be compared with the results of Roe et al. for simulations performed on decaalanine (53). Although shorter than all but the βVII peptide, the overall α- and 3–10-helical content within decaalanine was similar to the helical content observed here. However, through the analysis of the distribution of secondary structure along the CTT sequences, we note that our results differ from that of Roe et al. in that the helical content of the CTTs drops near the middle of the sequence for most isotypes whereas for decaalanine the central residues have a maximal amount of helix.

Three characteristics emerge which are representative of the behaviors of CTTs: the end-to-end distance, normalized by the number of residues, the PCA-based eigenvalues, and the averaged helical content (Fig. 7, *A* and *B*). To test the applicability of PCA as a measure of relative flexibility, we compared PCA-based flexibilities with the average parameter of structural similarity $\langle\rho_{sc}\rangle$, which we computed from the distributions in Fig. 2 (Fig. 7 *C*). Although less detailed than PCA, the distributions of conformations over the parameter ρ_{sc} provide an alternative statistical characterization of the CTTs configurations, and thus they can be expected to correlate with the PCA-based results. The correlation between the PCA-based flexibilities and $\langle\rho_{sc}\rangle$ is more pronounced than any other correlation that we investigated. Most of the CTTs show a clear proportionality between σ_{12} and $\langle\rho_{sc}\rangle$, and the variability of $\langle\rho_{sc}\rangle$ that corresponds to similar σ_{12} is $<15\%$. This similarity of results from two fundamentally different statistical tools demonstrates that these statistical methodologies are the best suited to characterize CTTs' flexibility. Interestingly, rather than the expected exponential decay, the magnitudes of the first three eigenvalues are of similar magnitude and significantly larger than the remaining eigenvalues, indicating an isotropy of occupancies in the three-dimensional space (Fig. 5).

More simply, the CTTs are flexible enough for the ensemble of their configurations to be spherically symmetric with respect to an immobilized base of amino acids (Fig. 1). Since these three eigenvalues are significantly larger than all the others, three-dimensional symmetry is likely reached through a few highly flexible bonds within the peptides that allow rotations of large angles, thereby generating nearly spherically symmetric distributions of occupancies. In Fig. 6 this is illustrated by a comparison of the histogram for βV, which is representative of a significant level of symmetry typical for most isotypes, with the less symmetric ones for

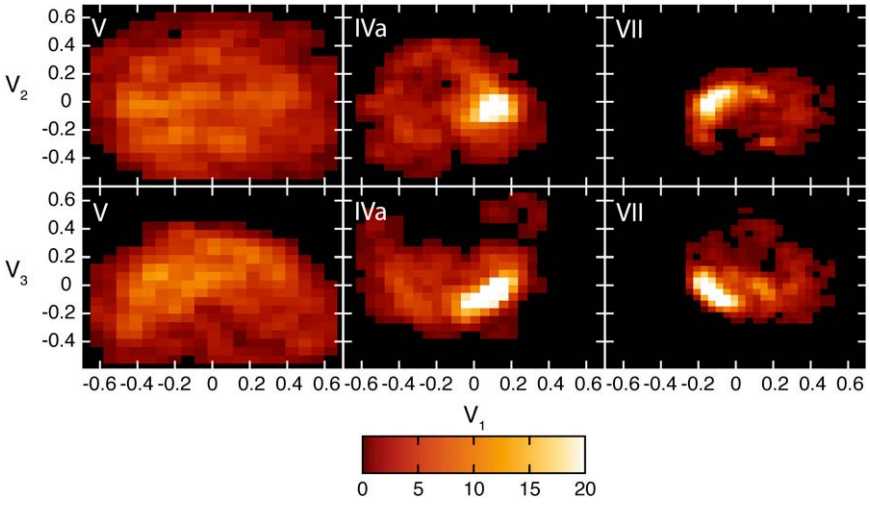

FIGURE 6 Projections of the ensemble of conformations onto the planes of the three most important principal components. The first and second principal components (*upper row*) and the first and third principal components (*lower row*) are plotted on the *x* and *y* axes, respectively, for βV, βIVa, and βVII. The histograms represent the occupancies of the corresponding conformation states, with lighter colors indicating more frequently visited areas. The same coloring scheme is used for all isotypes and is capped at 20 counts. βV is typical of the remaining isotypes in the width of the distribution and lack of a significant maxima.

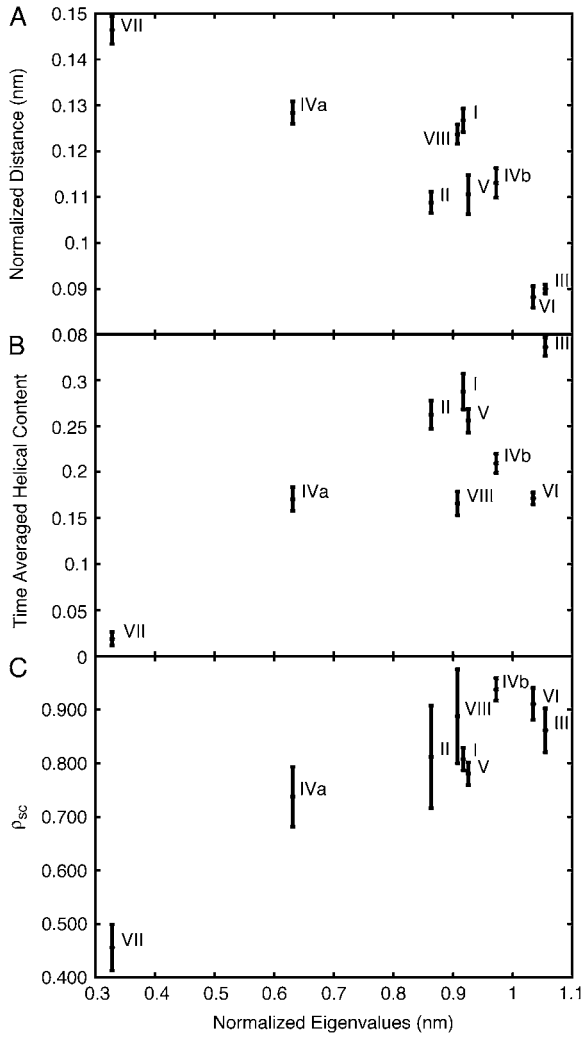

FIGURE 7 Peptide's flexibility indicators. The end-to-end distance normalized by the number of residues (*A*), time average helicity (*B*), and $\langle \rho_{sc} \rangle$ (*C*) of each CTT peptide at 311 K plotted as functions of σ_{12}. The plots demonstrate the correlation between each of the above values and the major flexibility indicator, σ_{12}. In *b*, βVI can be excluded from the trend due to the presence of a Pro, which disrupts the secondary structure but contributes to the flexibility of the peptide. The error in all three plots is the standard error calculated using box averaging (64).

βIVa and βVII. As the magnitudes of the eigenvalues shift to an exponential decay, spherical symmetry is lost and correlated motions begin to appear. At the same time, an individual CTT may preserve relatively stable motifs contained within some secondary structure. In terms of the PCA results presented in Fig. 5, these motifs are represented by the fourth, fifth, and higher eigenvalues. The fact that these components, although significantly less pronounced than the first three, still have considerable nonzero eigenvalues indicates that the conformational motifs within the CTTs are of transient nature and subject to variability over the trajectory. This is consistent with our observation of a significant amount of transient helicity (Fig. 3), indicated by a substantial amount of 3–10-helix.

Considering the PCA eigenvalue parameter σ_{12} as the measure of flexibility, it is evident that the βIII and βVI CTTs are the most flexible of all CTTs. These are the longest two sequences, which may account for some of their flexibility, but interestingly they also have the shortest end-to-end distance per residue. The cause of these shared structural properties differs in the two cases: whereas the βIII CTT has among the highest degree of secondary structure, the βVI CTT has among the lowest. The βVI CTT differs from all the other fragments, as it contains a Pro near the C-terminus of the peptide. This Pro breaks the helical structure of the fragment and, at the same time, does not facilitate an extended conformation. This accounts for the lack of secondary structure but is in contrast to βIII, which lies predominantly in the α-helical region of the Φ/Ψ distribution. When accounting for the lone Pro and omitting the data for βVI, the correlation between helicity and flexibility becomes greater. The βII, βIVb, and βV CTTs show the same spring-like flexibility of βIII, exhibiting a similar helical content and moderate end-to-end distance per residue. These CTTs also contain Gly at the same position that breaks the helical content of the fragment. In contrast, the βIVa and βVII CTTs displayed the least amount of flexibility. In the case of βIVa, one reason for this reduction in flexibility may be the early occurrence of Gly in its sequence, which stiffens this region of the peptide by decreasing the helicity. This decrease in helicity in the first seven residues, compared to βIVb, can be seen in Fig. 3. βIVa also contained the most stable DA[TK]AEE motif (data not shown), which resulted in the largest normalized distance of all the common isotypes.

The observation that peptides containing a significant amount of transient helical conformations are more flexible than an extended one may seem counterintuitive. The correlation between the normalized distance and flexibility in Fig. 7 *A* is most easily understood in terms of entropy. Take the following example: the freely jointed, or ideal chain, polymer model states that compact forms of the polymer contain more states that are accessible (i.e., have greater entropy) than do extended states (54). Thus, an entropic force drives the polymer or polypeptide into compact conformations. A greater number of accessible states suggests that the chain is more flexible and that extended states are more rigid. Although the ideal chain has no internal energy and is free to collapse, a peptide fragment does contain internal energy that can stiffen the peptide. The source of the force that causes the peptides to be in a rigid, extended state is the peptide backbone. Individual residues that induce a compact conformation through helical propensity or residues that induce a turn (such as proline) provide flexibility to the structure of the peptide while reducing the normalized distance. However, there is not a complete one-to-one correspondence between the flexibility and the normalized distance. For example, βI and βVIII have a normalized distance comparable to βIVa but a σ_{12} closer to the more compact βV. Although the normalized distance of the peptide is a strong indicator of flexibility,

the details of the compact structure, such as the degree of helicity, also contribute. We are not the first to observe this phenomenon in peptides; Ma et al. calculated the vibrational free energy of 28 short peptides in native helical and random extend states and determined that the native helical structures of the peptides were more flexible in all cases (51).

It is becoming clear that each β-tubulin isotype has a unique pattern of expression ranging from specific for βII, βIII, βIVa, and βVI to constitutive for βI and βIV (56,57). The βI isotype is the most commonly expressed in humans and, as such, is also the most common isotype found in cancer cells (58). Our observations also suggest that the CTT of this isotype shows ''typical'' behavior in our analysis. It is difficult to correlate our results to biological function; however, a clear pattern illustrating βIII as a statistical outlier has emerged. The βIII CTT was one of the most flexible peptides and contained a large amount of transient secondary structure. This is interesting, as βIII-tubulin has been observed at increased levels in human tumors and implicated in the development of drug resistance to standard chemotherapy treatments (59–61). In addition to the altered expression of βIII, the alteration of tubulin regulatory proteins such as MAP4 has also been implicated in changes to MT dynamics and the development of drug resistance (62). Because several MAPs have now been shown to interact directly with the CTTs (16–19,21), the differences that we observed here may have a significant impact on their affinity to the surface of an MT.

Motifs and MAP interactions

Using the conformational similarities for each CTT peptide, in combination with sequence comparisons, we identified two distinct transient structural motifs (Table 3). Several similar regions were observed as significant conformations throughout the trajectory files, indicating that they may be common motifs that could potentially be recognized by MAPs. The first motif was identified as a putative CK-2 phosphorylation site for which the consensus motif has been identified as [ST]XX[DE] (47). All of the CTTs, with the exception of βVI and βVII, were shown to contain this motif within the conserved DA[TK]A sequence. The second motif identified was the MAP2 binding site, for which the consensus sequence has previously been characterized as EEAEEE (48). Only the CTT of βI, βIVa, and βIVb strictly contain this motif; however, most of the other CTTs also contain a similar motif (Table 3). The conformations identified for βI, βIVa, βIVb, and βVI most commonly form a transient helix that is between four and five residues in length. In βVIII, there seems to be an increasing amount of unstructured loop associated with this region, which is most likely a result of the displacement of the amino end due to the presence of a bulky Tyr residue. Both the CK-2 and MAP2 motifs display a significantly higher degree of helical propensity when compared to the entire ensemble of CTT conformations (Fig. 3). Interestingly, with the exception of βVI and βVII, the increased helical propensity of the proposed CK-2 domain suggests that the H12 helix may extend several additional residues farther than those that have already been observed in the crystal structures of tubulin.

Finally, the motifs identified here are not stable structures but rather commonly occurring, transient conformations. For example, in the ''stiffest'' isotype, βIVa, the CK-2 DATA motif is occupied 69% of the time whereas the MAP2 motif EEEV is occupied 17% of the time (Table 3). The two motifs are simultaneously folded in the ensemble of CTT trajectories only 12% of the time. As the sequences contained within the two motifs are extended to include DATAEE and AEEEVA, respectively, this drops to 2% (Table 3). Rather, the shared similarity of the folded states indicates that these conformations are individually inducible, particularly when binding to their respective substrates. As we are dealing with a subset of all CTTs, we were unable to determine accurately if any additional structural or sequence motifs exist in the tails. A more accurate determination of CTT motifs would require a larger survey of CTTs across all identified α- and β-tubulin isotypes (see Table 1).

CONCLUSIONS

The role of the unstructured tubulin CTTs, particularly concerning MT stability and interactions with MAPs, is a question that has yet to be conclusively answered. Here we have shown, using a combination of REMD and PCA, that although the experimental structures of tubulin suggest the CTTs are unstructured, many contain a significant amount of transient conformations that are linked by a few highly rotatable bonds. We believe that the role of the transient secondary structure, coupled with increased CTT flexibility, may be twofold. First, global flexibility coupled with weak helical tendencies could provide a mechanism that allows the CTTs to search conformational space and hence easily bind to other proteins. This could provide a convenient mechanism for MAPs to dock to the MT surface. Second, increased flexibility could also enhance binding affinities to MAPs as a result of their ability to conform to a specific binding site. The presence of well-defined structural motifs within the CTTs may, consequently, play an important role in binding specificity, where the presence of smaller domains provides a common structural signal for binding, while the global flexibility of the CTT is maintained.

The role of β-tubulin CTTs in MT function is appealing, since subtle differences in a small stretch of amino acids could have profound consequences on MAP interactions. Unfortunately, it is extremely challenging at this time to elicit tight correlations between the results presented here and their biological consequences, as there is very little experimental data regarding CTTs explicitly in the context of tubulin isotypes. As interest in the properties of tubulin isotypes and their potential role in targeted chemotherapy

treatments increases, we anticipate a significant body of data to become available soon. As a result, a clear understanding of CTT structure and function may provide the means by which we can begin to rationally design and develop novel drugs or peptide mimetics that target CTT binding sites, providing better selectivity for chemotherapies.

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CONSCIOUS EVENTS AS ORCHESTRATED SPACE–TIME SELECTIONS

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Abstract: What *is* consciousness? Some philosophers have contended that 'qualia', or an experiential medium from which consciousness is derived, exists as a fundamental component of reality. Whitehead, for example, described the universe as being comprised of 'occasions of experience'. To examine this possibility scientifically, the very nature of physical reality must be re-examined. We must come to terms with the physics of space–time — as is described by Einstein's general theory of relativity — and its relation to the fundamental theory of matter — as described by quantum theory. This leads us to employ a new physics of *objective reduction*: '**OR**' which appeals to a form of 'quantum gravity' to provide a useful description of fundamental processes at the quantum/classical borderline (Penrose, 1994; 1996). Within the **OR** scheme, we consider that consciousness occurs if an appropriately organized system is able to develop and maintain quantum coherent superposition until a specific 'objective' criterion (a threshold related to quantum gravity) is reached; the coherent system then self-reduces (objective reduction: **OR**). We contend that this type of objective self-collapse introduces non-computability, an essential feature of consciousness. **OR** is taken as an instantaneous event — the climax of a self-organizing process in fundamental space–time — and a candidate for a conscious Whitehead-like 'occasion' of experience. How could an **OR** process occur in the brain, be coupled to neural activities, and account for other features of consciousness? We nominate an **OR** process with the requisite characteristics to be occurring in cytoskeletal microtubules within the brain's neurons (Penrose and Hameroff, 1995; Hameroff and Penrose, 1995; 1996).

In this model, quantum-superposed states develop in microtubule subunit proteins ('tubulins'), remain coherent, and recruit more superposed tubulins until a mass–time–energy threshold (related to quantum gravity) is reached. At that stage, self-collapse, or objective reduction (**OR**) abruptly occurs. We equate the pre-reduction, coherent superposition ('quantum computing') phase with pre-conscious processes, and each instantaneous (and non-computable) **OR**, or self-collapse, with a discrete conscious event. Sequences of **OR** events give rise to a 'stream' of consciousness. Microtubule-associated proteins can 'tune' the quantum oscillations of the coherent superposed states; the **OR** is thus self-organized, or 'orchestrated' ('**Orch OR**'). Each **Orch OR** event selects (non-computably) microtubule subunit states which regulate synaptic/neural functions using classical signalling.

The quantum gravity threshold for self-collapse is relevant to consciousness, according to our arguments, because macroscopic superposed quantum states each have their own space–time geometries (Penrose, 1994; 1996). These geometries are also superposed, and in some way 'separated', but when sufficiently separated, the superposition of space–time geometries becomes signifcantly unstable, and reduce to a single universe state. Quantum gravity determines the scale of the instability; we contend that the actual choice of state made by Nature is non-computable. Thus each **Orch OR** event is a self-selection of space–time geometry, coupled to the brain through microtubules and other biomolecules.

If conscious experience is intimately connected with the very physics underlying space–time structure, then **Orch OR** in microtubules indeed provides us with a completely new and uniquely promising perspective on the hard problem of consciousness.

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Introduction: Self-Selection in an Experiential Medium?

The 'hard problem' of incorporating the phenomenon of consciousness into a scientific world-view involves finding scientific explanations of qualia, or the subjective experience of mental states (Chalmers, 1995; 1996). On this, reductionist science is still at sea. Why do we have an inner life, and what exactly is it?

One set of philosophical positions, addressing the hard problem, views consciousness as a fundamental component of physical reality. For example an extreme view — 'panpsychism' — is that consciousness is a quality of all matter: atoms and their subatomic components having elements of consciousness (e.g. Spinoza, 1677; Rensch, 1960). 'Mentalists' such as Leibniz and Whitehead (e.g. 1929) contended that systems ordinarily considered to be physical are constructed in some sense from mental entities. Bertrand Russell (1954) described 'neutral monism' in which a common underlying entity, neither physical nor mental, gave rise to both. Recently Stubenberg (1996) has claimed that qualia are that common entity. In monistic idealism, matter and mind arise from consciousness — the fundamental constituent of reality (e.g. Goswami, 1993). Wheeler (1990) has suggested that information is fundamental to the physics of the universe. From this, Chalmers (1995; 1996) proposes a double-aspect theory in which information has both physical and experiential aspects.

Among these positions, the philosophy of Alfred North Whitehead (1929; 1933) may be most directly applicable. Whitehead describes the ultimate concrete entities in the cosmos as being actual 'occasions of experience', each bearing a quality akin to 'feeling'. Whitehead construes 'experience' broadly — in a manner consistent with panpsychism — so that even 'temporal events in the career of an electron have a kind of 'protomentality'. Whitehead's view may be considered to differ from panpsychism, however, in that his discrete 'occasions of experience' can be taken to be related to 'quantum events' (Shimony, 1993). In the standard descriptions of quantum mechanics, randomness occurs in the events described as quantum state reductions — these being events which appear to take place when a quantum-level process gets magnified to a macroscopic scale.

Quantum state reduction (here denoted by the letter **R**; cf. Penrose 1989, 1994) is the random procedure that is adopted by physicists in their descriptions of the quantum measurement process. It is still a highly controversial matter whether **R** is to be taken as a 'real' physical process, or whether it is some kind of illusion and not to be regarded as a fundamental ingredient of the behaviour of Nature. Our position is to take **R** to be indeed real — or, rather to regard it as a close approximation to an objectively real process **OR** (objective reduction), which is to be a non-computable process instead of merely a random one (see Penrose 1989; 1994). In almost all physical situations, **OR** would come about in situations in which the random effects of the environment dominate, so **OR** would be virtually indistinguishable from the random **R** procedure that is normally adopted by quantum theorists. However, when the quantum system under consideration remains coherent and well isolated from its environment, then it becomes possible for its state to collapse spontaneously, in accordance with the **OR** scheme we adopt, and to behave in non-computable rather than random ways. Moreover, this **OR** scheme intimately involves the geometry of the physical universe at its deepest levels.

Our viewpoint is to regard experiential phenomena as also inseparable from the physical universe, and in fact to be deeply connected with the very laws which govern the physical universe. The connection is so deep, however, that we perceive only glimmerings of it in our present day physics. One of these glimmerings, we contend, is a necessary non-computability in conscious thought processes; and we argue that this

non-computable = describes a problem that is not solvable

non-computability must also be inherent in the phenomenon of quantum state *self*-reduction — the 'objective reduction' (**OR**) referred to above. This is the main thread of argument in *Shadows of the Mind* (Penrose, 1994). The argument that conscious thought, whatever other attributes it may also have, is non-computable (as follows most powerfully from certain deductions from Gödel's incompleteness theorem) grabs hold of one tiny but extremely valuable point. This means that at least some conscious states cannot be derived from previous states by an algorithmic process — a property which distinguishes human (and other animal) minds from computers. Noncomputability *per se* does not directly address the 'hard problem' of the nature of experience, but it is a clue to the kind of physical activity that lies behind it. This points to **OR**, an underlying physical action of a completely different character from that which would appear to underlie nonconscious activity. Following this clue with sensitivity and patience should ultimately lead to real progress towards understanding mental phenomena in their inward manifestations as well as outward.

In the **OR** description, consciousness takes place if an organized quantum system is able to isolate and sustain coherent superpositions until its quantum gravity threshold for space–time separation is met; it then *self*-reduces (non-computably). For consciousness to occur, *self*-reduction is essential, as opposed to the reduction being triggered by the system's random environment. (In the latter case, the reduction would itself be effectively random and would lack useful non-computability, being unsuitable for direct involvement in consciousness.) We take the *self*-reduction to be an instantaneous event — the climax of a *self*-organizing process fundamental to the structuring of space–time — and apparently consistent with a Whitehead 'occasion of experience'.

As **OR** could, in principle, occur ubiquitously within many types of inanimate media, it may seem to imply a form of 'panpsychism' (in which individual electrons, for example, possess an experiential quality). However according to the principles of **OR** (as expounded in Penrose, 1994; 1996), a single superposed electron would spontaneously reduce its state (assuming it could maintain isolation) only once in a period longer than the present age of the universe. Only large collections of particles acting coherently in a single macroscopic quantum state could possibly sustain isolation and support coherent superposition in a timeframe brief enough to be relevant to our consciousness. Thus, only very special circumstances could support consciousness:

1) High degree of coherence of a quantum state — a collective mass of particles in superposition for a time period long enough to reach threshold, and brief enough to be useful in thought processes.

2) Ability for the **OR** process to be at least transiently isolated from a 'noisy' environment until the spontaneous state reduction takes place. This isolation is required so that reduction is not simply random. Mass movement in the environment which entangles with the quantum state would effect a random (not non-computable) reduction.

3) Cascades of **OR**s to give a 'stream' of consciousness, huge numbers of **OR** events taking place during the course of a lifetime.

By reaching the quantum gravity threshold, each **OR** event has a fundamental bearing on space–time geometry. One could say that a cascade of **OR** events charts an actual course of physical space–time geometry selections.

It may seem surprising that quantum gravity effects could plausibly have relevance at the physical scales relevant to brain processes. For quantum gravity is normally viewed

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as having only absurdly tiny influences at ordinary dimensions. However, we shall show later that this is not the case, and the scales determined by basic quantum gravity principles are indeed those that are relevant for conscious brain processes.

We must ask how such an **OR** process could actually occur in the brain? How could it be coupled to neural activities at a high rate of information exchange; how could it account for preconscious to conscious transitions, have spatial and temporal binding, and both simultaneity and time flow?

We here nominate an **OR** process with the requisite characteristics to be occurring in cytoskeletal microtubules within the brain's neurons. In our model, microtubule-associated proteins 'tune' the quantum oscillations leading to **OR**; we thus term the process 'orchestrated objective reduction' (**Orch OR**).

it is natural and intuitive to think of every object being in its own eigenstate. This is another way of saying that every object appears to have a definite position, a definite momentum, a definite measured value, and a definite time of occurrence.

Space–time: Quantum Theory and Einstein's Gravity

Quantum theory describes the extraordinary behaviour of the matter and energy which comprise our universe at a fundamental level. At the root of quantum theory is the wave/particle duality of atoms, molecules, and their constituent particles. A quantum system such as an atom or subatomic particle which remains isolated from its environment behaves as a 'wave of possibilities' and exists in a coherent complex-number valued 'superposition' of many possible states. The behaviour of such wave-like, quantum-level objects can be satisfactorily described in terms of a state vector which evolves deterministically according to the Schrödinger equation (unitary evolution), denoted by **U**.

Somehow, quantum microlevel superpositions lead to unsuperposed stable structures in our macro-world. In a transition known as wave function collapse, or state reduction (**R**), the quantum wave of alternative possibilities reduces to a single macroscopic reality — an 'eigenstate' of some appropriate operator. (This would be just one out of many possible alternative eigenstates relevant to the quantum operator.) This process is invoked in the description of a macroscopic measurement, when effects are magnified from the small, quantum scale to the large, classical scale.

According to conventional quantum theory (as part of the standard 'Copenhagen interpretation'), each choice of eigenstate is entirely random, weighted according to a probability value that can be calculated from the previous state according to the precise procedures of the quantum formalism. This probabilistic ingredient was a feature with which Einstein, among others, expressed displeasure: 'You believe in a God who plays dice, and I in complete law and order.'[1] Penrose (1989; 1994) has contended that, at a deeper level of description, the choices may more accurately arise as the result of some presently unknown 'non-computational' mathematical/physical (i.e. 'Platonic realm') theory, that is they cannot be deduced algorithmically. Penrose argues that such non-computability is essential to consciousness, because (at least some) conscious mental activity is unattainable by computers.

It can be argued that present-day physics has no clear explanation for the cause and occurrence of wave function collapse **R**. Experimental and theoretical evidence through the 1930's led quantum physicists (such as Schrödinger, Heisenberg, Dirac, von Neumann, and others) to postulate that quantum-coherent superpositions persist indefinitely in time, and would, in principle be maintained from the micro to macro levels. Or perhaps they would persist until conscious observation collapses, or reduces, the wave function (subjective reduction, or '**SR**'). Accordingly, even macroscopic objects, if unobserved, could remain superposed. To illustrate the apparent absurdity of this notion, Erwin

[1] From a letter to Max Born.

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Schrödinger (e.g. 1935) described his now-famous 'cat in a box' being simultaneously *both* dead and alive until the box was opened and the cat observed.

As a counter to this unsettling prospect, various new physical schemes for collapse according to objective criteria (objective reduction — '**OR**') have recently been proposed. According to such a scheme, the growth and persistence of superposed states could reach a critical threshold, at which collapse, or **OR** rapidly occurs (e.g. Pearle, 1989; Ghirardi *et al.*, 1986). Some of these schemes are based specifically on gravitational effects mediating **OR** (e.g. Károlyházy, 1966; Károlyházy *et al.*, 1986; Diósi, 1989; Ghirardi *et al.*, 1990; Penrose, 1989; 1994; Pearle and Squires, 1995; Percival, 1995). Table 1 categorizes types of reduction.

Context	**Cause of Collapse (Reduction)**	**Description**	**Acronym**
Quantum coherent superposition	No collapse	Evolution of wave function (Schrödinger equation)	**U**
Conventional quantum theory (Copenhagen interpretation)	Environmental entanglement, Measurement, Conscious observation	Reduction; Subjective reduction	**R** **SR**
New physics (Penrose,1994)	Self-collapse — quantum gravity induced (Penrose, Diosi, etc.)	Objective reduction	**OR**
Consciousness (present paper)	Self-collapse, quantum gravity threshold in microtubules orchestrated by MAPs etc.	Orchestrated objective reduction	**Orch OR**

Table 1: Description of wave function collapse

The physical phenomenon of gravity, described to a high degree of accuracy by Isaac Newton's mathematics in 1687, has played a key role in scientific understanding. However, in 1915, Einstein created a major revolution in our scientific world-view. According to Einstein's theory, gravity plays a unique role in physics for several reasons (cf. Penrose, 1994). Most particularly, these are:

1) Gravity is the only physical quality which influences causal relationships between space–time events.
2) Gravitational force has no local reality, as it can be eliminated by a change in space–time coordinates; instead, gravitational tidal effects provide a *curvature* for the very *space–time* in which all other particles and forces are contained.

It follows from this that gravity cannot be regarded as some kind of 'emergent phenomenon', secondary to other physical effects, but must be a 'fundamental component' of physical reality.

There are strong arguments (e.g. Penrose, 1987; 1996) to suggest that the appropriate union of general relativity (Einstein's theory of gravity) with quantum mechanics — a union often referred to as 'quantum gravity' — will lead to a significant change in *both* quantum theory and general relativity, and, when the correct theory is found, will yield a

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profoundly *new* understanding of physical reality. And although gravitational *forces* between objects are exceedingly weak (feebler than, for example, electrical forces by some 40 orders of magnitude), there are significant reasons for believing that gravity has a fundamental influence on the behaviour of quantum systems as they evolve from the micro to the macrolevels. The appropriate union of quantum gravity with biology, or at least with advanced biological nervous systems, may yield a profoundly new understanding of consciousness.

Curved Space–time Superpositions and Objective Reduction ('OR')

According to modern accepted physical pictures, reality is rooted in 3-dimensional space and a 1-dimensional time, combined together into a 4-dimensional space–time. This space–time is slightly curved, in accordance with Einstein's general theory of relativity, in a way which encodes the gravitational fields of all distributions of mass density. Each mass density effects a space–time curvature, albeit tiny.

This is the standard picture according to *classical* physics. On the other hand, when *quantum* systems have been considered by physicists, this mass-induced tiny curvature in the structure of space–time has been almost invariably ignored, gravitational effects having been assumed to be totally insignificant for normal problems in which quantum theory is important. Surprising as it may seem, however, such tiny differences in space–time structure *can* have large effects: for they entail subtle but fundamental influences on the very rules of quantum mechanics.

Superposed quantum states for which the respective mass distributions differ significantly from one another will have space–time geometries which correspondingly differ. Thus, according to standard quantum theory, the superposed state would have to involve a quantum superposition of these differing space–times. In the absence of a coherent theory of quantum gravity there is no accepted way of handling such a superposition. Indeed, the basic principles of Einstein's general relativity begin to come into profound conflict with those of quantum mechanics (cf. Penrose, 1996). Nevertheless, various tentative procedures have been put forward in attempts to describe such a superposition. Of particular relevance to our present proposals are the suggestions of certain authors (e.g. Károlyházy, 1966; 1974; Károlyházy *et al.*, 1986; Kibble, 1981; Diósi, 1989; Ghirardi *et al.*, 1990; Pearle and Squires, 1995; Percival, 1995; Penrose, 1993; 1994; 1996) that it is at this point that an objective quantum state reduction (**OR**) ought to occur, and the rate or timescale of this process can be calculated from basic quantum gravity considerations. These particular proposals differ in certain detailed respects, and for definiteness we shall follow the specific suggestions made in Penrose (1994; 1996). Accordingly, the quantum superposition of significantly differing space–times is unstable, with a life-time given by that timescale. Such a superposed state will decay — or 'reduce' — into a single universe state, which is one or the other of the space–time geometries involved in that superposition.

Whereas such an **OR** action is not a generally recognized part of the normal quantum-mechanical procedures, there is no plausible or clearcut alternative that standard quantum theory has to offer. This **OR** procedure avoids the need for 'multiple universes' (cf. Everett, 1957; Wheeler, 1957, for example). There is no agreement, among quantum gravity experts, about how else to address this problem. For the purposes of the present article, it will be assumed that a gravitationally induced **OR** action is indeed the correct resolution of this fundamental conundrum.

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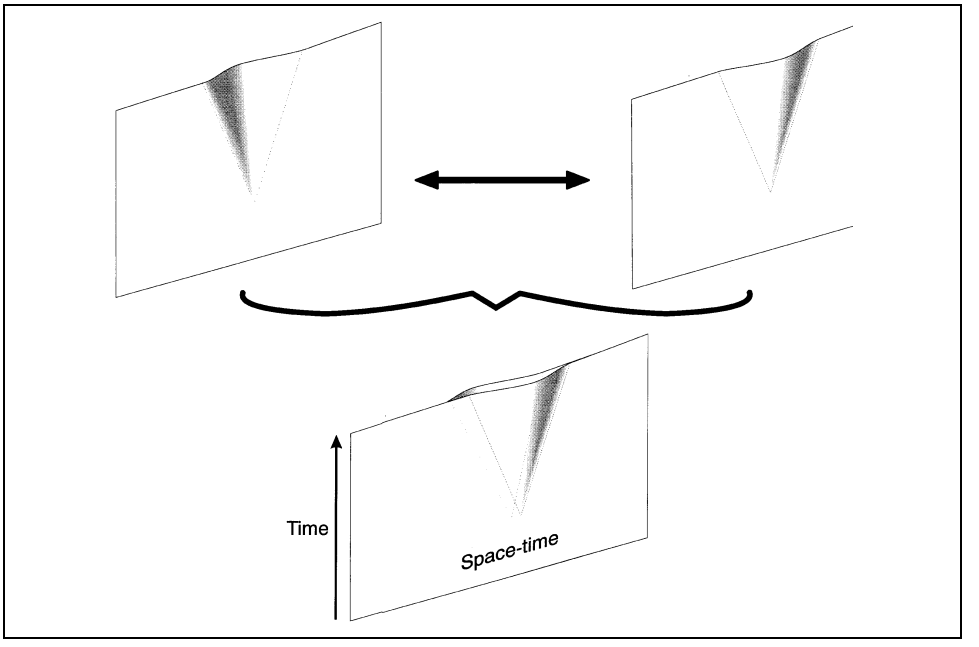

Figure 1. Quantum coherent superposition represented as a separation of space–time. In the lowest of the three diagrams, a bifurcating space–time is depicted as the union ('glued together version'), of the two alternative space–time histories that are depicted at the top of the Figure. The bifurcating space–time diagram illustrates two alternative mass distributions actually in quantum superposition, whereas the top two diagrams illustrate the two individual alternatives which take part in the superposition (adapted from Penrose, 1994, p. 338).

Figure 1 (adapted from Penrose, 1994, p. 338) schematically illustrates the way in which space time structure can be affected when two macroscopically different mass distributions take part in a quantum superposition. Each mass distribution gives rise to a separate space–time, the two differing slightly in their curvatures. So long as the two distributions remain in quantum superposition, we must consider that the two space–times remain in superposition. Since, according to the principles of general relativity, there is no natural way to identify the points of one space–time with corresponding points of the other, we have to consider the two as separated from one another in some sense, resulting in a kind of 'blister' where the space–time bifurcates.

A bifurcating space–time is depicted in the lowest of the three diagrams, this being the union ('glued together version') of the two alternative space–time histories that are depicted at the top of Figure 1. The initial part of each space–time is at the lower end of each individual space–time diagram. The bottom space–time diagram (the bifurcating one) illustrates two alternative mass distributions actually in quantum superposition, whereas the top two illustrate the two individual alternatives which take part in the superposition. The combined space–time describes a superposition in which the alternative locations of a mass move gradually away from each other as we proceed in the upward direction in the diagram. Quantum mechanically (i.e. so long as **OR** has not taken place), we must think of the 'physical reality' of this situation as being illustrated as an actual superposition of these two slightly differing space–time manifolds, as indicated in the bottom diagram. As soon as **OR** has occurred, one of the two individual space–times takes over, as depicted as one of the two sheets of the bifurcation. For clarity only, the bifurcating parts of these two sheets are illustrated as being one convex and the other

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concave. Of course there is additional artistic licence involved in drawing the space–time sheets as 2-dimensional, whereas the actual space–time constituents are 4-dimensional. Moreover, there is no significance to be attached to the imagined '3-dimensional space' within which the space–time sheets seem to be residing. There is no 'actual' higher dimensional space there, the 'intrinsic geometry' of the bifurcating space–time being all that has physical significance. When the 'separation' of the two space–time sheets reaches a critical amount, one of the two sheets 'dies' — in accordance with the **OR** criterion — the other being the one that persists in physical reality. The quantum state thus reduces (**OR**), by choosing between either the 'concave' or 'convex' space–time of Figure 1.

It should be made clear that this measure of separation is only very schematically illustrated as the 'distance' between the two sheets in the lower diagram in Figure 1. As remarked above, there is no physically existing 'ambient higher-dimensional space' inside which the two sheets reside. The degree of separation between the space–time sheets is a more abstract mathematical thing; it would be more appropriately described in terms of a *symplectic measure* on the space of 4-dimensional metrics (cf. Penrose, 1993) — but the details (and difficulties) of this will not be important for us here. It may be noted, however, that this separation is a space–time separation, not just a spatial one. Thus the *time* of separation contributes as well as the spatial displacement. Roughly speaking, it is the product of the temporal separation $\boldsymbol{T}$ with the spatial separation $\boldsymbol{S}$ that measures the overall degree of separation, and **OR** takes place when this overall separation reaches the critical amount. [This critical amount would be of the order of unity, in absolute units, for which the Planck-Dirac constant □□ (= h/2π), the gravitational constant G, and the velocity of light c, all take the value unity; cf. Penrose, 1994, pp. 337–9.] Thus for small $\boldsymbol{S}$, the lifetime $\boldsymbol{T}$ of the superposed state will be large; on the other hand, if $\boldsymbol{S}$ is large,then $\boldsymbol{T}$ will be small. To calculate $\boldsymbol{S}$, we compute (in the Newtonian limit of weak gravitational fields) the gravitational self-energy $\boldsymbol{E}$ of the difference between the mass distributions of the two superposed states. (That is, one mass distribution counts positively and the other, negatively; see Penrose, 1994; 1996.) The quantity $\boldsymbol{S}$ is then given, in absolute units, by:

ħ = h/(2π)

$$\boldsymbol{S} = \boldsymbol{E}$$

Thus, restoring standard units,

$$\boldsymbol{T} = \square\square\boldsymbol{E}^{-1}$$

Schematically, since $\boldsymbol{S}$ represents three dimensions of displacement rather than the one dimension involved in $\boldsymbol{T}$, we can imagine that this displacement is shared equally between each of these three dimensions of space — and this is what has been depicted in Figure 3 (below). However, it should be emphasized that this is for pictorial purposes only, the appropriate rule being the one given above. These two equations relate the mass distribution, time of coherence, and space–time separation for a given **OR** event. If, as some philosophers contend, experience is contained in space–time, **OR** events are *self*-organizing processes in that experiential medium, and a candidate for consciousness.

But where in the brain, and how, could coherent superposition and **OR** occur? A number of sites and various types of quantum interactions have been proposed. We strongly favour microtubules as an important ingredient; however various organelles and biomolecular structures including clathrins, myelin (glial cells), presynaptic vesicular grids (Beck and Eccles, 1992) and neural membrane proteins (Marshall, 1989) might also participate.

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Microtubules

Properties of brain structures suitable for quantum coherent superposition, **OR** and relevant to consciousness might include: 1) high prevalence, 2) functional importance (for example regulating neural connectivity and synaptic function), 3) periodic, crystal-like lattice dipole structure with longrange order, 4) ability to be transiently isolated from external interaction/observation, 5) functionally coupled to quantum-level events, 6) hollow, cylindrical (possible wave-guide), and 7) suitable for information processing. Membranes, membrane proteins, synapses, DNA and other types of structures have some, but not all, of these characteristics. Cytoskeletal microtubules appear to qualify in all respects.

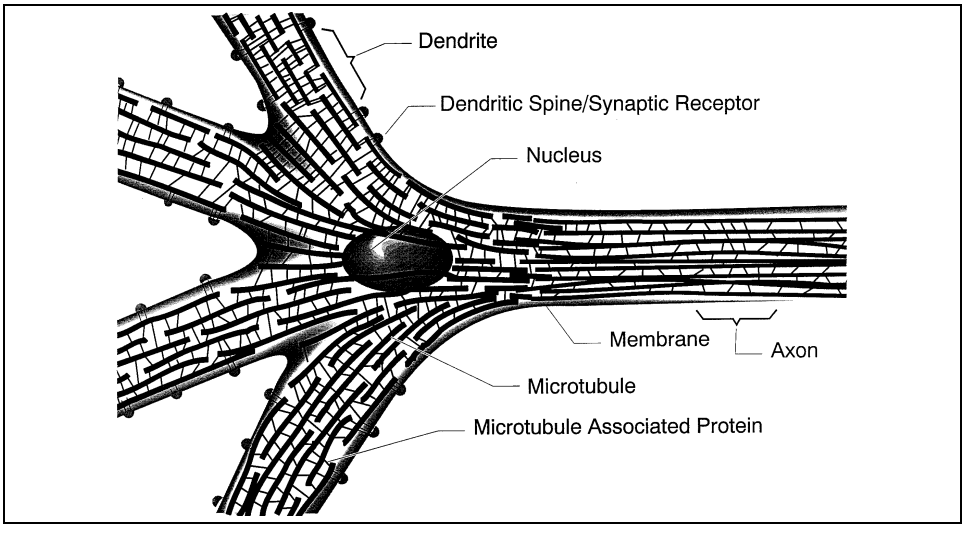

Figure 2. Schematic of central region of neuron (distal axon and dendrites not shown), showing parallel arrayed microtubules interconnected by MAPs. Microtubules in axons are lengthy and continuous, whereas in dendrites they are interrupted and of mixed polarity. Linking proteins connect microtubules to membrane proteins including receptors on dendritic spines.

Interiors of living cells, including the brain's neurons, are spatially and dynamically organized by *self*-assembling protein networks: the cytoskeleton. Within neurons, the cytoskeleton establishes neuronal form, and maintains and regulates synaptic connections. Its major components are microtubules, hollow cylindrical polymers of individual proteins known as tubulin. Microtubules ('MTs') are interconnected by linking proteins (microtubule-associated proteins: 'MAPs') to other microtubules and cell structures to form cytoskeletal lattice networks (Figure 2).

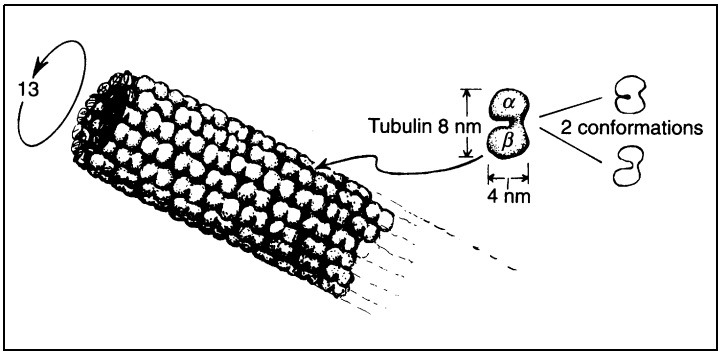

Figure 3.

Microtubule structure: a hollow tube of 25 nanometers diameter, consisting of 13 columns of tubulin dimers. Each tubulin molecule is capable of (at least), two conformations. (Reprinted with permission from Penrose, 1994, p. 359.)

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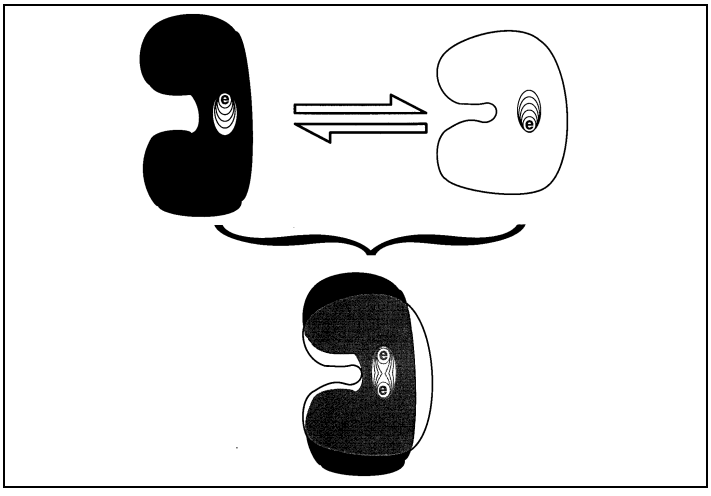

Figure 4.

Top:
Two states of tubulin in which a *single* quantum event (electron localization), within a central hydrophobic pocket is coupled to a *global* protein conformation. Switching between the two states can occur on the order of nanoseconds to picoseconds.

Bottom:
Tubulin in quantum coherent superposition.

MTs are hollow cylinders 25 nanometers (nm) in diameter whose lengths vary and may be quite long within some nerve axons. MT cylinder walls are comprised of 13 longitudinal protofilaments which are each a series of subunit proteins known as tubulin (Figure 3). Each tubulin subunit is a polar, 8 nm dimer which consists of two slightly different 4 nm monomers (α and β tubulin — Figure 4). Tubulin dimers are dipoles, with surplus negative charges localized toward monomers (DeBrabander, 1982), and within MTs are arranged in a hexagonal lattice which is slightly twisted, resulting in helical pathways which repeat every 3, 5, 8 and other numbers of rows. Traditionally viewed as the cell's 'bonelike' scaffolding, microtubules and other cytoskeletal structures also appear to fill communicative and information processing roles. Numerous types of studies link the cytoskeleton to cognitive processes (for review, cf. Hameroff and Penrose, 1996). Theoretical models and simulations suggest how conformational states of tubulins within microtubule lattices can interact with neighbouring tubulins to represent, propagate and process information as in molecular-level 'cellular automata', or 'spinglass' type computing systems (Figure 5; e.g. Hameroff and Watt, 1982; Rasmussen

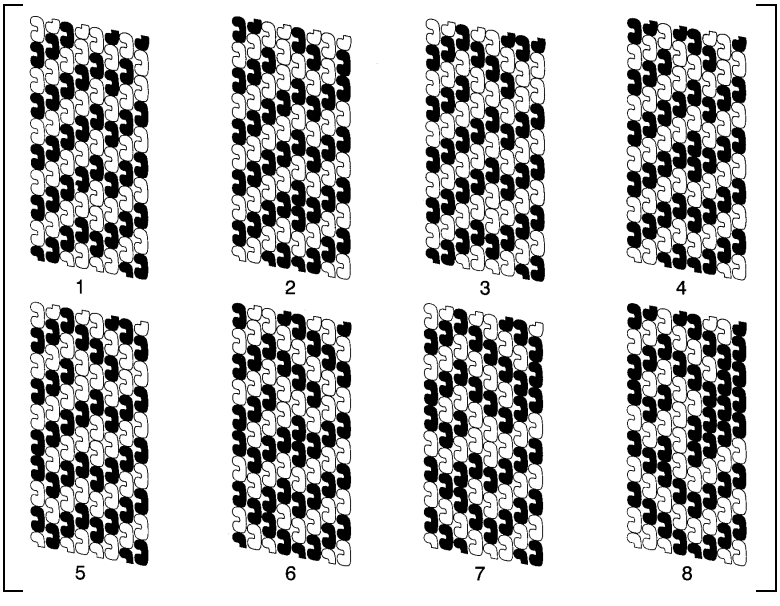

Figure 5. Microtubule automaton simulation (from Rasmussen *et al*., 1990). Black and white tubulins correspond to states shown in Figure 2. Eight nanosecond time steps of a segment of one microtubule are shown in 'classical computing' mode in which patterns move, evolve, interact and lead to emergence of new patterns.

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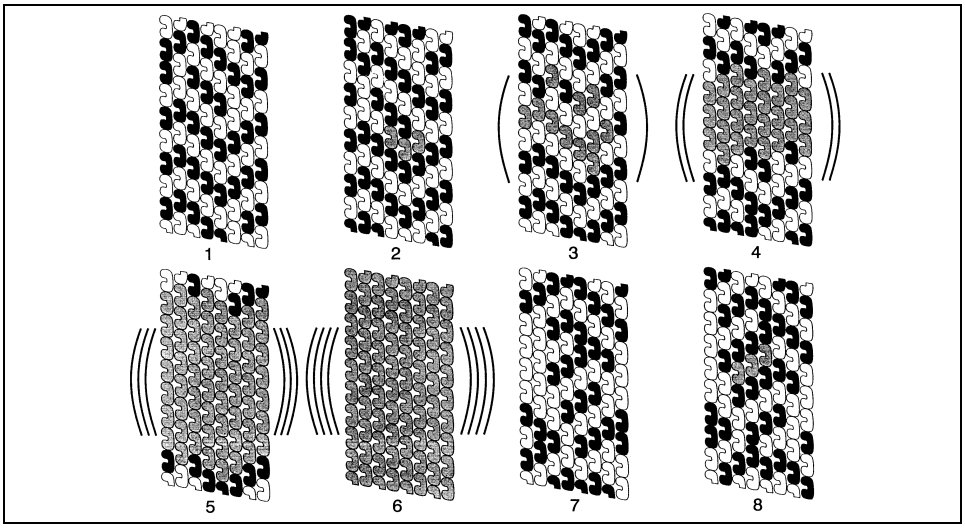

Figure 6. Microtubule automaton sequence simulation in which classical computing (step 1), leads to emergence of quantum coherent superposition (steps 2–6), in certain (grey) tubulins due to pattern resonance. Step 6 (in coherence with other microtubule tubulins) meets critical threshold related to quantum gravity for self–collapse (**Orch OR**). Consciousness (**Orch OR**) occurs in the step 6 to 7 transition. Step 7 represents the eigenstate of mass distribution of the collapse which evolves by classical computing automata to regulate neural function. Quantum coherence begins to re–emerge in step 8.

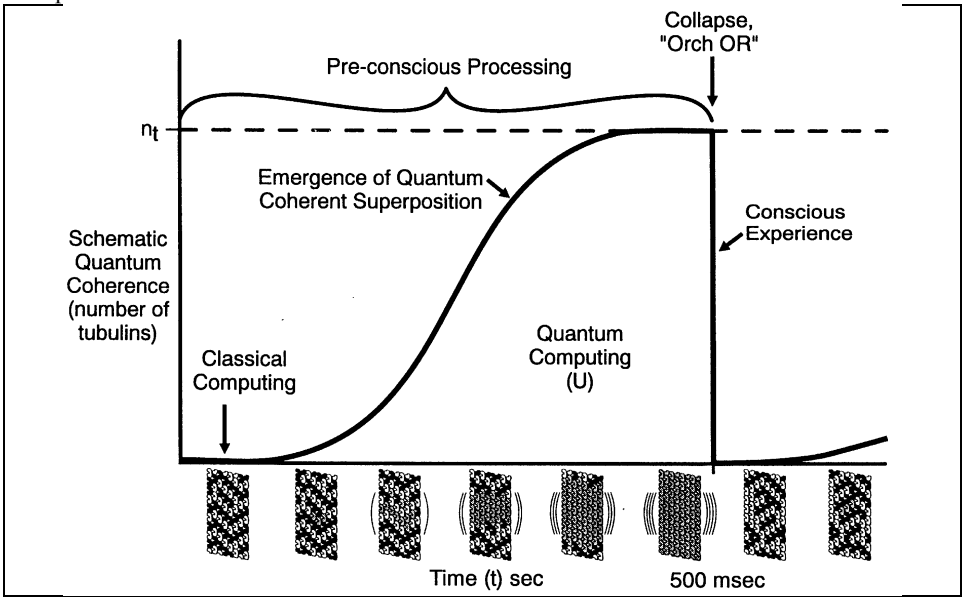

Figure 7. Schematic graph of proposed quantum coherence (number of tubulins), emerging vs time in microtubules. 500 milliseconds is time for pre–conscious processing (e.g. Libet *et al.*, 1979). Area under curve connects mass–energy differences with collapse time in accordance with gravitational **OR**. This degree of coherent superposition of differing space–time geometries leads to abrupt quantum → classical reduction ('self–collapse' or 'orchestrated objective reduction: **Orch OR**').

et al., 1990; Tuszy□ski *et al.*, 1995). In Hameroff and Penrose (1996, and in summary form Penrose and Hameroff, 1995), we present a model linking microtubules to consciousness, using quantum theory as viewed in the particular 'realistic' way referred to

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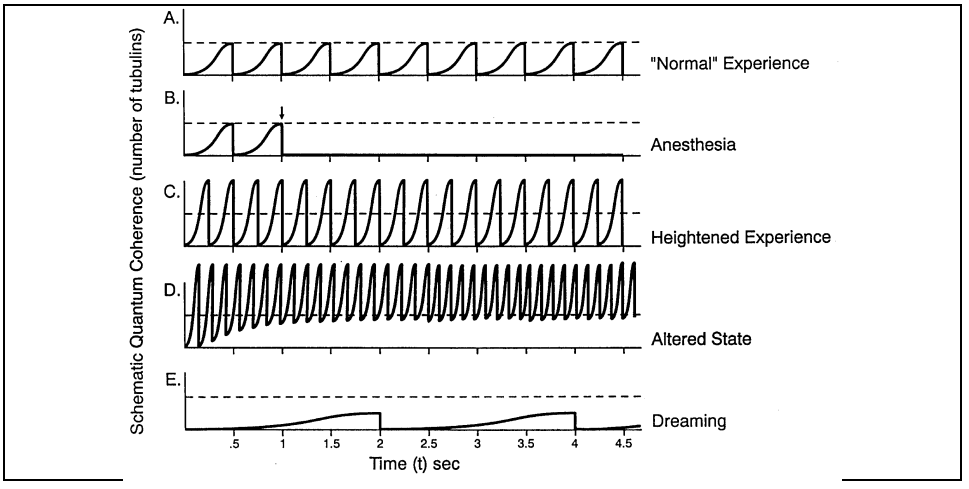

Figure 8. Quantum coherence in microtubules schematically graphed on longer time scale for five different states related to consciousness. Area under each curve equivalent in all cases. A. Normal experience: as in Figure 8. B. Anaesthesia: anaesthetics bind in hydrophobic pockets and prevent quantum delocalizability and coherent superposition (e.g. Louria and Hameroff, 1996). C. Heightened Experience: increased sensory experience input (for example), increases rate of emergence of quantum coherent superposition. **Orch Or** threshold is reached faster (e.g. 250 msec), and **Orch Or** frequency is doubled. D. Altered State: even greater rate of emergence of quantum coherence due to sensory input and other factors promoting quantum state (e.g. meditation, psychedelic drug, etc.). Predisposition to quantum state results in baseline shift and only partial collapse so that conscious experience merges with normally sub–conscious quantum computing mode. E. Dreaming: prolonged quantum coherence time. (Highly schematic.)

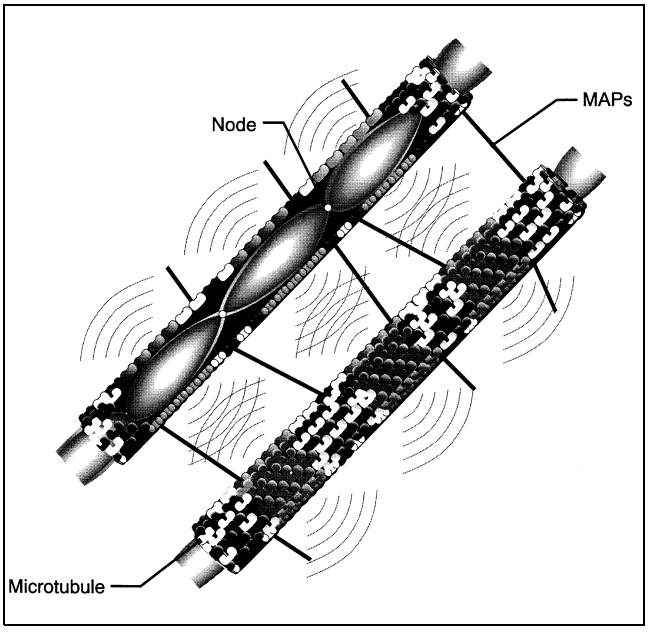

Figure 9. Quantum coherence in microtubules. Having emerged from resonance in classical automaton patterns, quantum coherence non–locally links superpositioned tubulins (grey), within and among microtubules. Upper microtubule: cutaway view shows coherent photons generated by quantum ordering of water on tubulin surfaces, propagating in microtubule waveguide. MAP (microtubule-associated protein), attachments breach isolation and prevent quantum coherence; MAP attachment sites thus act as 'nodes' which tune and orchestrate quantum oscillations and set possibilities and probabilities for collapse outcomes ('orchestrated objective reduction': **Orch OR**).

above, and as described in *Shadows of the Mind* (Penrose, 1994). In our model, quantum coherence emerges, and is isolated, in brain microtubules until the differences in mass–energy distribution among superposed tubulin states reach the above threshold of instability, related to quantum gravity (Figure 6). The resultant *self*-collapse (**OR**), considered

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to be a time-irreversible process, creates an instantaneous 'now' event in our model. Sequences of such events create a flow of time and consciousness (Figures 7 and 8).

We envisage that attachments of MAPs on microtubules 'tune' quantum oscillations, and 'orchestrate' possible collapse outcomes (Figure 9). Thus we term the particular *self*-organizing **OR** occurring in MAP-connected microtubules, and relevant to consciousness, orchestrated objective reduction ('**Orch OR**'). **Orch OR** events are thus *self*-selecting processes in fundamental space–time geometry. If experience is truly a component of fundamental space–time, **Orch OR** may indeed begin to to address the 'hard problem' of consciousness.

Summary of the 'Orch OR' Model for Consciousness

The full details of this model are given in Hameroff and Penrose (1996). The picture we are putting forth involves the following ingredients:

(1) Aspects of quantum theory (e.g. quantum coherence) and of the suggested physical phenomenon of quantum wave function '*self*-collapse' (objective reduction: **OR** — Penrose, 1994; 1996) are essential for consciousness, and occur in cytoskeletal microtubules (MTs) and other structures within each of the brain's neurons.

(2) Conformational states of MT subunits (tubulins) are coupled to internal quantum events, and cooperatively interact with other tubulins in both classical and quantum computation (Hameroff *et al.*, 1992; Rasmussen *et al.*, 1990 — Figures 4, 5 and 6).

(3) Quantum coherence occurs among tubulins in MTs, pumped by thermal and biochemical energies (perhaps in the manner proposed by Fröhlich, 1968; 1970; 1975). Evidence for some kind of coherent excitation in proteins has recently been reported by Vos *et al.* (1993).

It is also considered that water at MT surfaces is 'ordered' — dynamically coupled to the protein surface. Water ordering within the hollow MT core (acting something like a quantum waveguide) may result in quantum coherent photons (as suggested by the phenomena of 'super-radiance'and '*self*-induced transparency' — Jibu *et al.*, 1994; 1995). We require that coherence be sustained (protected from environmental interaction) for up to hundreds of milliseconds by isolation (a) within hollow MT cores, (b) within tubulin hydrophobic pockets, (c) by coherently ordered water, (d) sol-gel layering (Hameroff and Penrose, 1996). Feasibility of quantum coherence in the seemingly noisy, chaotic cell environment is supported by the observation that quantum spins from biochemical radical pairs which become separated retain their correlation in cytoplasm (Walleczek, 1995).

(4) During preconscious processing, quantum coherent superposition/computation occurs in MT tubulins and continues until the mass-distribution difference among the separated states of tubulins reaches a threshold related to quantum gravity. Self-collapse (**OR**) then occurs (Figures 6 & 7).

(5) The **OR** *self*-collapse process results in classical 'outcome states' of MT tubulins which then implement neurophysiological functions. According to certain ideas for **OR** (Penrose, 1994), the outcome states are 'non-computable'; that is they cannot be determined algorithmically from the tubulin states at the beginning of the quantum computation.

(6) Possibilities and probabilities for post**OR** tubulin states are influenced by factors including initial tubulin states, and attachments of microtubule-associated proteins (MAPs) acting as 'nodes' which tune and 'orchestrate' the quantum oscillations (Figure 9). We thus term the *self*-tuning **OR** process in microtubules 'orchestrated objective reduction' — **Orch OR**.

(7) According to the arguments for **OR** put forth in Penrose (1994), superposed states each have their own space–time geometries. When the degree of coherent mass–energy

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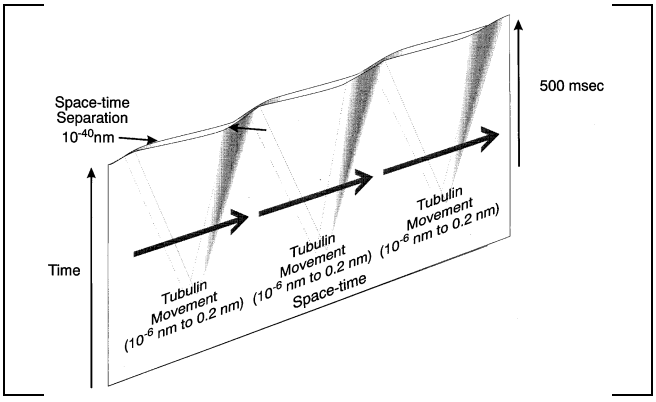

Figure 10.

Schematic space–time separation illustration of three superposed tubulins. The space–time differences are very tiny in ordinary terms (10^{-40} nm), but relatively large mass movements (e.g. hundreds of tubulin conformations, each moving from 10^{-6} nm to 0.2 nm), indeed have precisely such very tiny effects on the space–time curvature.

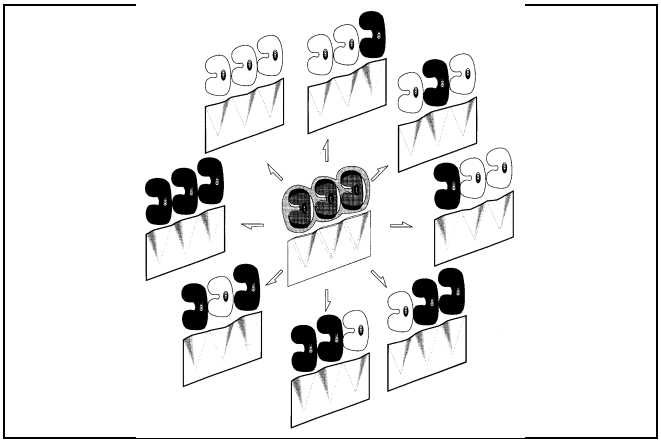

Figure 11.

Centre:
Three superposed tubulins (e.g. Figure 4), with corresponding schematic space–time separation illustrations (Figures 1 and 10). Surrounding the superposed tubulins are the eight possible post–reduction 'eigenstates' for tubulin conformation, and corresponding space–time geometry.

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difference leads to sufficient separation of space–time geometry, the system must choose and decay (reduce, collapse) to a single universe state. Thus **Orch OR** involves *self*-selections in fundamental space–time geometry (Figures 10 & 11).

(8) To quantify the **Orch OR** process, in the case of a pair of roughly equally superposed states, each of which has a reasonably well-defined mass distribution, we calculate the gravitational *self*-energy $\boldsymbol{E}$ of the difference between these two mass distributions, and then obtain the approximate lifetime $\boldsymbol{T}$ for the superposition to decay into one state or the other by the formula $\boldsymbol{T} = \hbar/\boldsymbol{E}$. Here, $\hbar$ is Planck's constant over 2π. We call $\boldsymbol{T}$ the coherence time for the superposition (how long the coherence is sustained). If we assume a coherence time $\boldsymbol{T}$ = 500 msec (shown by Libet *et al*., 1979, and others to be a relevant time for preconscious processing), we calculate $\boldsymbol{E}$, and determine the number of MT tubulins whose coherent superposition for 500 msec will elicit **Orch OR**. This turns out to be about 10^9 tubulins.

(9) A typical brain neuron has roughly 10^7 tubulins (Yu and Baas, 1994). If, say, 10 % of tubulins within each neuron are involved in the quantum coherent state, then roughly 10^3 (one thousand) neurons would be required to sustain coherence for 500 msec, at which time the quantum gravity threshold is reached and **Orch OR** then occurs.

(10) We consider each *self*-organized **Orch OR** as a single conscious event; cascades of such events would constitute a 'stream' of consciousness. If we assume some form of excitatory input (e.g. you are threatened, or enchanted) in which quantum coherence emerges faster, then, for example, 10^{10} coherent tubulins could **Orch OR** after 50 msec,

or 10^{11} after 5 msec (e.g. Figure 8c). Turning to see a bengal tiger in your face might perhaps elicit 10^{12} in 0.5 msec, or more tubulins, faster. A slow emergence of coherence (your forgotten phone bill) may require longer times. A single electron would require more than the age of the universe.

(11) Quantum states are non-local (because of quantum entanglement — or 'Einstein-Podolsky-Rosen' (EPR) effects), so that the entire non-localized state reduces all at once. This can happen if the mass movement that induces collapse takes place in a small region encompassed by the state, or if it takes place uniformly over a large region. Thus, each instantaneous **Orch OR** could 'bind' various superpositions which may have evolved in separated spatial distributions and even over different time scales, but whose net displacement *self*-energy reaches threshold at a particular moment. Information is bound into an instantaneous event (a 'conscious now'). Cascades of **Orch OR**s could then represent our familiar 'stream of consciousness', and create a 'forward' flow of time (Aharonov and Vaidman, 1990; Elitzur, 1996; Tollaksen, 1996).

It may be interesting to compare our considerations with subjective viewpoints that have been expressed with regard to the nature of the progression of conscious experience. For example, support for consciousness consisting of sequences of individual, discrete events is found in Buddhism; trained meditators describe distinct 'flickerings' in their experience of reality (Tart, 1995). Buddhist texts portray consciousness as 'momentary collections of mental phenomena', and as 'distinct, unconnected and impermanent moments which perish as soon as they arise'. Each conscious moment successively becomes, exists, and disappears — its existence is instantaneous, with no duration in time, as a point has no length. Our normal perceptions, of course, are seemingly continuous, presumably as we perceive 'movies' as continuous despite their actual makeup being a series of frames. Some Buddhist writings even quantify the frequency of conscious moments. For example the Sarvaastivaadins (von Rospatt, 1995) described 6,480,000 'moments' in 24 hours (an average of one 'moment' per 13.3 msec), while other Buddhist writings decribe one moment per 0.13 msec (Conze, 1988), and some Chinese Buddhism as one 'thought' per 20 msec. These accounts, including variations in frequency, seem to be consistent with our proposed **Orch OR** events. For example a 13.3 msec preconscious interval would correspond with an **Orch OR** involving 4×10^{10} coherent tubulins, a 0.13 msec interval would correspond with 4×10^{12} coherent tubulins, and a 20 msec interval with 2.5×10^{10} coherent tubulins. Thus Buddhist 'moments of experience', Whitehead 'occasions of experience', and our proposed **Orch OR** events seem to correspond tolerably well with one another.

The **Orch OR** model thus appears to accommodate some important features of consciousness:

1) control/regulation of neural action
2) pre-conscious to conscious transition
3) non-computability
4) causality
5) binding of various (time scale and spatial) superpositions into instantaneous 'now'
6) a 'flow' of time, and
7) a connection to fundamental space–time geometry in which experience may be based.

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Conclusion: What Is It Like To Be a Worm?

The **Orch OR** model has the implication that an organism able to sustain quantum coherence among, for example, 10^9 tubulins for 500 msec might be capable of having conscious experience. More tubulins coherent for a briefer period, or fewer for a longer period ($\boldsymbol{E} = \square / \boldsymbol{T}$) will also have conscious events. Human brains appear capable of, for example, 10^{12} tubulin, 0.5 msec 'bengal tiger experiences', but what about simpler organisms?

From an evolutionary standpoint, introduction of a dynamically functional cytoskeleton (perhaps symbiotically from spirochetes, e.g. Margulis, 1975) greatly enhanced eukaryotic cells by providing cell movement, internal organization, separation of chromosomes and numerous other functions. As cells became more specialized with extensions like axopods and eventually neural processes, increasingly larger cytoskeletal arrays providing transport and motility may have developed quantum coherence via the Fröhlich mechanism as a by-product of their functional coordination.

Another possible scenario for emergence of quantum coherence leading to **Orch OR** and conscious events is 'cellular vision'. Albrecht-Buehler (1992) has observed that single cells utilize their cytoskeletons in 'cellular vision' — detection, orientation and directional response to beams of red/infrared light. Jibu *et al.* (1995) argue that this process requires quantum coherence in microtubules and ordered water, and Hagan (1995) suggests the quantum effects/cellular vision provided an evolutionary advantage for cytoskeletal arrays capable of quantum coherence. For whatever reason quantum coherence emerged, one could then suppose that, one day, an organism achieved sufficient microtubule quantum coherence to elicit **Orch OR**, and had a 'conscious' experience.

At what level of evolutionary development might this primitive consciousness have emerged? A single cell organism like *Paramecium* is extremely clever, and utilizes its cytoskeleton extensively. Could a paramecium be conscious? Assuming a single paramecium contains, like each neuronal cell, 10^7 tubulins, then for a paramecium to elicit **Orch OR**, 100% of its tubulins would need to remain in quantum coherent superposition for nearly a minute. This seems unlikely.

Consider the nematode worm *C elegans*. It's 302 neuron nervous system is completely mapped. Could *C elegans* support **Orch OR**? With 3×10^9 tubulins, *C elegans* would require roughly one third of its tubulins to sustain quantum coherent superposition for 500 msec. This seems unlikely, but not altogether impossible. If not *C elegans*, then perhaps *Aplysia* with a thousand neurons, or some higher organism. **Orch OR** provides a theoretical framework to entertain such possibilities.

Would a primitive **Orch OR** experience be anything like ours? If *C elegans* were able to *self*-collapse, what would it be like to be a worm? (cf. Nagel, 1974). A single, 10^9 tubulin, 500 msec **Orch OR** in *C elegans* should be equal in gravitational *self*-energy terms (and thus perhaps, experiential intensity) to one of our 'everyday experiences'. A major difference is that we would have many **Orch OR** events sequentially (up to, say, 50 per second) whereas *C elegans* could generate, at most, 2 per second. *C elegans* would also presumably lack extensive memory and associations, and have poor sensory data, but nonetheless, by our criteria a 10^9 tubulin, 500 msec **Orch OR** in *C elegans* could be a conscious experience: a mere smudge of known reality, the next space–time move.

Consciousness has an important place in the universe. **Orch OR** in microtubules is a model depicting consciousness as sequences of non-computable *self*-selections in fundamental space time geometry. If experience is a quality of space–time, then **Orch OR** indeed begins to address the 'hard problem' of consciousness in a serious way.

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Contractils: Quantum Mechanical Fibrils

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ABSTRACT: When an erythrocyte rouleau is pulled apart by micromanipulation the cells remain attached to each other by long fibrils which may reach lengths of over 100 μm. The tension and length of the fibrils are drastically reduced when the cellular membrane potential is abolished or when the cells are depleted of ATP. It is inferred that the fibrils consist of macromolecules and that the tension is largely derived from a quantum mechanical long-range interaction which accords with the theory of Fröhlich.

INTRODUCTION

HERBERT FRÖHLICH'S THEORY (1968, 1980) of a phonon-maser-like phenomenon in cell membranes predicts intercellular forces of range several orders of magnitude longer than London-van der Waals' force. Interactions between human erythrocytes at a range of several micrometers have recently been demonstrated by deviation from Brownian movement (Rowlands et al. 1981, 1982a, 1982b). Essential to Frohlich's hypothesis are (a) an organized array of molecules in a membrane, (b) a membrane potential to polarize membrane molecules and (c) a supply of metabolic energy which, if it exceeds a minimum value, will trigger membrane vibrations into a coherent polar oscillatory mode ($\nu \approx 10^{11}$Hz), a process analogous with the transition to coherence in a maser or in a laser. When the membrane of the erythrocyte is disorganized with glutaraldehyde the deviations from Brownian movement disappear and they disappear *reversibly* when the membrane potential is reduced to zero or if the cells are deprived of ATP (Rowlands et al. 1981, 1982a, 1982b). The presence of macromolecules in the medium suspending the erythrocytes is essential to the interaction (Rowlands et al. 1982b: Sewchand et al. 1982b) and although the necessary and sufficent properties of the macromolecules have yet to be delineated they need to have an extended chain structure.

In the experiments described by Rowlands et al. the Fröhlich interaction is superimposed on Brownian movement and the two motions are comparable in magnitude. When erythrocytes in Brownian motion collide they adhere and quite quickly (in the time frame of the experiment) slide face to face. Further cells and groups of cells adhere and so columns of erythrocytes develop, which is the well-known process of erythrocyte rouleau formation. In the present work rouleaux are pulled apart by micromanipulation, In the process long contractile fibrils (contractils) form between the cells. The properties of these contractils vary when membrane function is interfered with by the methods (a), (b) and (c) above and these variations provide additional evidence for a Fröhlich mechanism in human erythrocytes.

METHODS

Fresh heparinized blood with the haematocrit adjusted to about 1% was placed on a large cover slip on an inverted microscope (Leitz Diavert, Rockleigh, NJ). In some experiments a tiny chamber was constructed on the cover slip with pieces of microscope slides and of cover slips. Micropipettes could be inserted into a specimen of blood in the chamber and then the whole preparation sealed with medicinal paraffin. These latter experiments showed that the observed phenomena were not a consequence of the specimen drying out. In both types of experiment the end cells of a rouleau were held by gentle suction on each of two diametrically opposed micropipettes of internal diameter <3 μm. Canham and Burton (1968) showed that human erythrocytes have a "minimum cylindrical diameter" of 3 μm. That is, a cell will deform into a cylinder, with hemispherical ends, of this diameter without change of volume or of surface area. With a pipette of internal diameter <3 μm gentle suction pulls part of the cell into the pipette and thus the cell is firmly held. If the pipettes are now separated, so as to exert traction on the rouleau, the cells do not fall apart but remain linked by fibrils of length 15 μm or more (Fig. 1). If the negative pressure in one pipette is cut off, or if a fibril breaks, the remaining fibrils contract, pulling the cells through the plasma at velocities of the order of tens of micrometers per second. Observations were recorded with a still camera, a television camera, and tape recorder, or with a 16mm. cine-camera.

Experimental modification of the cells. The methods used were those fully described in Rowlands et al. (1982b). In brief, the membrane potential was brought close to zero by reducing the pH of plasma to 6.3 and restored by returning it to pH = 7.4. (Funder and Wieth, 1966; Lassen et al. 1980). The membrane potential was calculated from measurements of the chloride ratio $[Cl_i^-/Cl_o^-]$ using the Nernst equation. The other (nonreversible) method of reducing the membrane potential involved the use of the ionophore A23187 and the procedure of Flatman and Lew (1980). Metabolic depletion was achieved by incubation for 24 hr at 37°C and cells were repleted by further incubation after adding 30 mM adenosine (Meiselman and Baker 1977). The membrane was disorganized by a 24 hr incubation in 0.04g% glutaraldehyde (Rowlands et al. 1982b). In whatever manner the cells were modified, they were resuspended in their own plasma, except in the following experiments.

Mixed species experiments. Rouleaux composed of a mixture of cells from man and rabbit were examined. The cells cannot be suspended in human plasma because of immunological reactions. The suspending liquid was 0.4g% polyvinylpyrrolidone (molecular weight 360,000) in standard phosphate-buffered saline. Human serum albumin (0.2g%) was added to prevent crenation of the cells. Polyvinylpyrrolidone is a rouleau-inducing agent and it also transmits the long-range Fröhlich interaction between human erythrocytes (Sewchand et al. 1982b).

Scanning Electron Microscopy. The contractils are very thin and we have not succeeded, with the light microscope, in making a picture which demonstrates contractils unequivocally. The two younger authors of this paper see contractils by direct vision, but the old one never does. We therefore

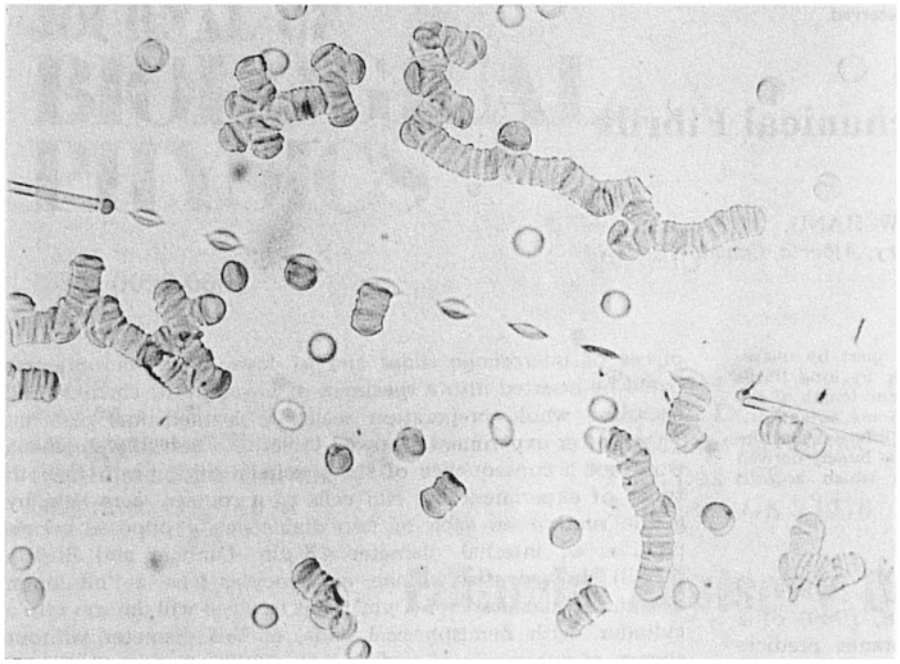

Fig. 1 A branched rouleau has been extended between the micropipettes seen at extreme left and right. The internal diameter of the pipettes is less than the cellular minimum cylindrical diameter (see text) and so the end cells are firmly held by negative pressure. Branches of the original rouleau are seen still attached to cells three, four, and eight of the chain. Human erythrocytes in their own plasma. Red cell diameter ≈ 8 μm.

embarked on scanning electron microscopy. In the end success was achieved but the technique demands the utmost delicacy. Contractils were pulled in an open chamber which could be perfused. With a satisfactory chain of cells in view through the light microscope, glutaraldehyde (0.5g% in phosphate-buffered saline) slowly replaced the rouleau-inducing fluid. This was followed by flow of distilled water (40 ml) to remove all the glutaraldehyde solution. The water was sucked out of the chamber and the specimen was allowed to dry overnight, still on the light microscope with micromanipulators and pipettes in place. If the specimen looked satisfactory the next day, the micropipettes were rigidly glued to the floor of the chamber with a drop or two of epoxy cement. When the cement was hard the outer parts of the pipettes were sheared off with heavy metal cutters. The flow chamber could now be taken from the light microscope. The glass floor of the chamber was removed and cut around the tips of the attached pipettes to a size to fit a metal sample holder. The rest was the standard technique of scanning electron microscopy. The thickness of the gold coating, according to the machine calibration, was 80 nanometres.

RESULTS

The chain of cells in Fig. 1 was drawn from a large branched rouleau such as seen elsewhere in the field of view. That is why small rouleaux are attached to some of the cells in the chain. The gaps in the chain are on the average 10 μm long. These are by no means the longest contractils we have seen. The maximum length of a contractil before breaking varies from rouleau to rouleau and from day to day in the blood of the same donor.

Glutaraldehyde-fixed cells. Our method leaves the shape, area, and volume of the cells unaffected. Such cells are rigid and cannot be drawn into a pipette. Rouleaux do not form. Occasionally cells will stick together and they can be separated by shear from the tips of the pipettes. There is no sign of a contractil. Once separated the cells fall apart.

Ionophore-treated (A23187) cells. Two experiments were done with identical results. Rouleaux formed without delay and their appearance was normal; Yet no contractils were seen. As soon as the cell at one end of a rouleau was sucked into the first pipette, the cell came away. The cells could be picked off one by one from a rouleau with a single pipette.

Variation of membrane potential by varying pH (four experiments). At pH = 6.3 when the membrane potential is within 1.5 mV of zero (Rowlands et al. 1982b) some rouleaux fell apart on manipulation but others formed short (<4 μm) contractils which gave the appearance of being weak, perhaps because further extension led to rupture. No long contractils were formed. Restoring the pH to 7.4 gave contractils with most rouleaux and they were consistently longer (>6 μm) than at pH = 6.3.

Metabolic depletion and repletion (four experiments). The majority of metabolically depleted cells are crenated. Crenated cells do not form rouleaux but they often stick together. They can be sucked easily into a micropipette and on manipulation adhering cells immediately fall apart with no sign of a contractil. A minority of cells are normal in appearance and some of these form rouleaux. Some of the rouleaux fall apart on micromanipulation: others demonstrated contractils but always they were short (<4 μm) as in the pH = 6.3 low membrane potential experiments. When the cells were repleted with adenosine crenation disappeared and rouleaux formed normally. Contractils reappeared and they were >8 μm in length but, as when the membrane potential was restored, fully normal contractils were rarely seen.

Interspecies Experiments. Rouleaux which form from mixtures of erythrocytes from different species of mammals are not random in their grouping of cells. (Sewchand and Canham 1976). This non-randomness is further evidence for a Fröhlich interaction between erythrocytes (Rowlands et al. 1982a,b). A human-rabbit mixed rouleau comprises short runs of cells of the two species and the cells can be identified by size. On micromanipulation such a mixed rouleau always separates at a junction between a rabbit cell and a human cell. There is sometimes a short contractil at the break. But when a mixed rouleau is stripped until the remaining rouleau is all rabbit or all human and then pulled apart, normal contractils occur.

Scanning Electron Microscopy. In dozens of attempts a break always occurred in all but the shortest contractils subsequent to removal from the light microscope. The contractil shown in Fig. 2 is no exception. Such breaks can occur in any

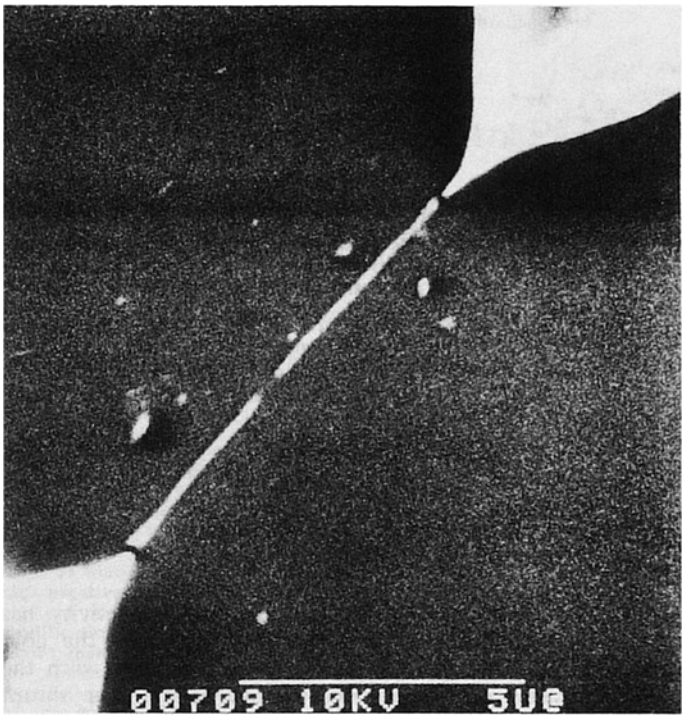

Fig. 2 Scanning electron microscope picture. See text for technique. The break in the middle of the contractil is an artefact. The breaks where the contractils begin to widen and merge with the contours of the cells are always present. The line above the identification marks is 5 μm long.

position along the fibre. Quite consistently, however, two additional and very short discontinuities are seen at the ends of each contractil just as it begins to widen and merge with the contours of the two cells to which it is attached. Since contractils and cells, after fixation, were washed in distilled water and then dried, the contractils as prepared for scanning electron microscopy, are stretched in air. Two breaks would cause them to fall and be displaced from alignment with the pointed aspects of the cells. Such displacement is never seen. This finding is considered in the Discussion.

Velocity of recoil. If a contractil breaks or if the negative pressure in one micropipette is released the stretched chain of cells recoils back into a rouleau. The maximum velocity of recoil measured by frame analysis of a cine-film was 180 $\mu m.s^{-1}$. The length of the contractil at this velocity was 18 μm. By observing Brownian movement of erythrocytes resting on horizontal surfaces we have made extensive measurements on drag coefficients (Rowlands et al. 1982a,b; Sewchand et al. 1982a). If we assume for the purposes of an order of magnitude calculation that the cell was traveling at its terminal velocity under the force (F) from the contractil then

$$F = \nu\mu$$

where ν is the velocity and μ is the drag coefficient. Such a calculation gives a value of the force of 10^{-6} N. If, from the width of the image of a contractil by scanning electron microscopy is subtracted the thickness of the gold shadow, the contractil itself comes out to be of width 0.2 μm (which explains the difficulty of resolution by light miscroscopy). From this a very rough estimate of the cross-sectional area and so of the stress in a contractil can be calculated giving 10^7 N/m^2. This must be regarded as an upper limit.

DISCUSSION

Coherent quantum mechanical excitations in cell membranes as postulated by Fröhlich provide a mechanism for many cell processes which await explanation (Fröhlich 1977; Cooper 1979). There is evidence, from Raman spectroscopy, of vibrations of the predicted frequency. Irradiation with microwaves of microorganisms and of cells in tissue culture demonstrates resonances as frequency is varied. Fröhlich (1980) has reviewed this evidence. But the important prediction of the theory is for the existence of ultra-long-range specific interactions for which evidence is only recently beginning to accumulate (Pohl 1980,1981a,b,c; Rowlands et al. 1981, 1982a,b; Sewchand et al. 1982b). In brief Pohl has shown that particles of high dielectric constant will adhere to cells more readily than particles of low dielectric constant. Rowlands et al. have demonstrated an interaction between erythrocytes in their own plasma at a range of several micrometers. The interaction requires a membrane potential and a supply of energy.

The Rowlands et al. interaction takes place *before* cellular contact. Contractils arise *after* contact when the cells are pulled apart by micromanipulation. Certain macromolecules are essential to both processes and the inference is that the Fröhlich interaction of erythrocytes is transmitted by these macromolecules (Rowlands et al. 1982b; Sewchand et al. 1982b). Solutions of macromolecules can be pulled out into thin threads (e.g. polyethylene oxide) and it is no surprise that erythrocytes, adhering by virtue of suspension in a macromolecular solution, maintain adherence by thin threads when pulled apart. It is therefore necessary to review the evidence that the threads are powered and that the power is compatible with the predictions of Fröhlich's theory.

Glutaraldehyde-fixed cells. Although suspended in the same medium, fixed cells do not form contractils. They are, however, much more rigid than normal cells and so their area of contact is small. Therefore the area from which macromolecules could be recruited into a thread is much smaller than for normal cells. Thus the absence of contractils with fixed cells might merely be an area effect.

Ionophore-treated cells. These form normal-looking rouleaux which fall apart on micromanipulation. They are not so flexible as normal cells but they have a much larger area of contact than gluraraldehyde-fixed cells, as large as for normal cells. But reduced flexibility could prevent the macromolecules holding the cells together from being drawn towards the final point of membrane contact as the cells come apart.

Change of membrane potential by change in pH. At pH = 6.3 when the membrane potential was close to zero, rouleaux occur and sometimes these formed short, weak threads. The cells were of normal flexibility, the area of contact was maximal, and so these threads were probably a passive drawing out of the macromolecules which had held the membranes together. They never exceeded 4 μm in length and they broke within a few seconds. With normal cells the threads are very much longer, at least 10 μm and sometimes over 100 μm in length. Moreover threads from normal cells continue to exist for half an hour or more prior to fixation in the scanning electron microscope experiments. Restoring pH to 7.4 and so the membrane potential nearly to normal (Rowlands et al. 1982b) partially restored the contractils. They were longer and longer lasting than at pH = 6.3 but we never observed the very long contractils seen with normal cells.

Metabolic depletion and repletion. No contractils formed when the depleted cells were crenated and they reappeared when the metabolism was restored with adenosine. A minority of depleted cells have normal appearance, but formed rouleaux and weak contractils. These cells may not have been depleted as much as those which crenated.

Interspecies experiments. The nonrandomness of the arrangement of the cells in mixed rouleaux has been adduced as

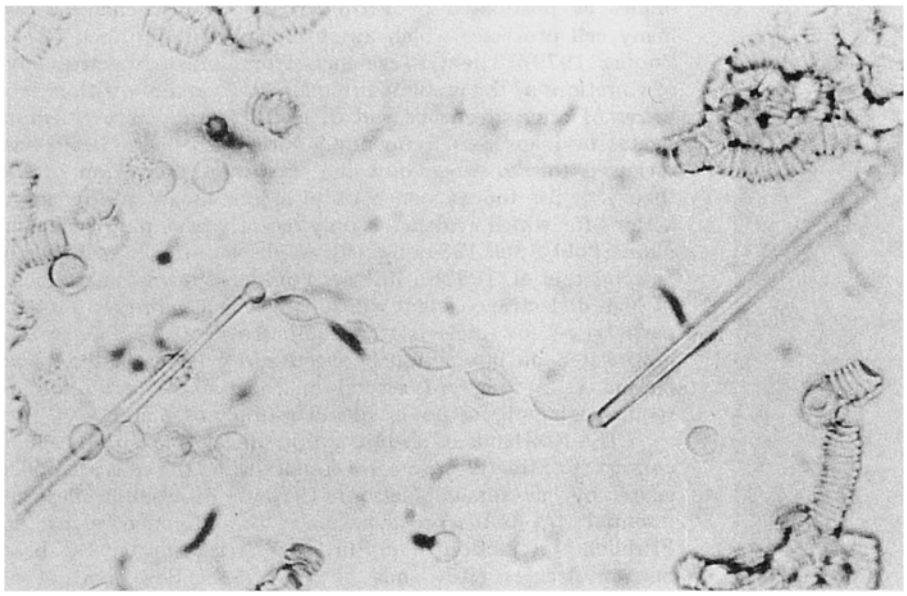

Fig. 3 The pipettes are offset and the axis of the contractils is seen to be radial with respect to the spherical external part of each of the two cells held in the micropipettes. As the extent of the offset is varied the contractils always point towards the centres of the spheres. Human erythrocytes in their own plasma. Red cell diameter ≈ 8 μm.

evidence for specificity in the interaction of erythrocytes (Rowlands et al. 1982a,b). The present experiments clearly demonstrate an adhesiveness between cells of different species which is lower than that between cells of the same species.

General Discussion. The experiments described in this paper are technically difficult and the results are qualitative. Partly this is a consequence of the technique but mainly because there is considerable variation in the properties of the cells in a single sample from an individual. There is for instance a very wide variation in the permeability of single cells to various non-electrolytes (Jay and Rowlands 1975). One "defective" cell in a rouleau would vitiate contractil formation. Such defective cells might be ripe for destruction in the spleen. Nevertheless the properties of these threads vary regularly in the manner expected if their tension is derived, in the main, from the living state of the erythrocytes. There are two further important pieces of evidence. When one contractil breaks the remaining cells are pulled together at great speed by the remaining contractils back into a rouleau. If the thread was just visco-elastic, it would shorten, but not down to a zero length, so rapidly bringing the membranes again into contact. Moreover a passive thread would always attach at the point on the membrane from which it originated. Reference to Fig. 3 shows that the line of attachment of the contractil to the spherical portion of the end cells held in the micropipettes is radial, directed towards the center of the sphere. As the pipettes are moved out of and into alignment, with the tips kept at a fixed distance apart, the attachment moves over the spherical surfaces, always directed towards the center of the sphere without hysteresis. If the pressure in the pipette is released slightly so that the sphere sags, then a point appears at the attachment which also moves freely with the contractil as its orientation is changed.

We conclude that the observed fibrils have an active contraction powered by the cells and that the contraction is consistent with the interaction postulated by Fröhlich.

An unequivocal picture of a contractil by light microscopy was not obtained. What two out of three of the authors saw might have been an optical illusion because the line of a contractil is clearly oriented by the direction of the points on the parent cells. Scanning electron microscopy proved the existence of a physical thread, which is composed of macromolecules. The constant gaps in the contractils which occur just as they merge into the cell outline are difficult to explain. By the time the gold, which is what is seen, falls on the contractils they and the cells have been fixed in glutataldehyde, washed in distilled water and air dried overnight, so all quantum activity has ceased. There must still be continuity otherwise the gold threads would have been displaced from alignment with the points of the cells. Clearly there must be a rather abrupt thinning of the contractil close to the pointed distortion of the cell membrane. Rowlands et al. (1982b) have suggested that the interaction is transmitted by quantum quasi-particles which may be tunnelling at this point between the membrane and the contractil.

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COOPERATIVE BEHAVIOR IN BIOLOGICAL SYSTEMS

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Fröhlich's model of Bose condensation in biological systems is discussed. A criticism of the theory levied by Lifshits is examined utilizing microscopic techniques. It is subsequently concluded that the criticism is invalid.

Cooperative behavior in biological systems. The possible participation of the vibrations of the dipolar segments of biological macromolecules in fundamental biological processes has been discussed repeatedly [1–10]. Fröhlich [1–5] proposed that such coherent behavior is feasible if energy is supplied to the dipolar constituents above a certain critical rate which causes the oscillating subunits to condense into the lowest energy mode in a manner characteristic of Bose condensation. Fröhlich's analysis was based upon rate equations whereby each dipolar segment was considered to be suspended in a heat bath.

Subsequently, other investigators [6–10] approached the theoretical problem from differing standpoints. Lifshits [6] concluded that the Fröhlich model was erroneous because omitted in the rate equations is a "two quantum" term which, when included, obliterates the Bose condensation effect.

The purpose of the present paper is to illustrate that Lifshits' argument is incorrect, and that Bose condensation does occur even when the Lifshits term is included in the calculation.

According to Fröhlich, the biological system is presumed to consist of (i) oscillating units of giant dipoles (e.g., sections of the membrane of cells or parts of H-bonds within biological macromolecules), (ii) the remainder of the biological system which formed the heat bath, and (iii) an external energy source which couples to the vibratory units. Long-range Coulomb interaction between the oscillatory units produces a narrow band of frequencies, $\omega_i(\omega_0 \leqslant \omega_i \leqslant \omega_{\max})$, corresponding to the normal modes of the electromagnetic oscillations. Utilizing these assumptions, Fröhlich deduced that the average rate of change of ith mode quanta is [2]

$$\langle \dot{n}_i \rangle = S - \phi[\langle n_i \rangle \exp(\hbar\omega_i/kT) - (1 + \langle n_i \rangle)] - \chi \sum_j [\langle n_i \rangle (1 + \langle n_j \rangle) \exp[\hbar(\omega_i - \omega_j)/kT] - \langle n_j \rangle (1 + \langle n_i \rangle)], \tag{1}$$

where S is the rate of energy supplied, and ϕ and χ are coupling constants to the heat bath for the one- and two-quantum processes respectively.

Lifshits claims that included in the interaction hamiltonian should be "two-quantum" terms which allow the simultaneous creation and destruction of the quanta (i) and (j). Thus, according to Lifshits, a modification of the Fröhlich formula is required which yields for the average rate of change of ith mode quanta [6]:

$$\langle \dot{n}_i \rangle = S - \phi[\langle n_i \rangle \exp(\hbar\omega_i/kT) - (1 + \langle n_i \rangle)] - \sum_j \chi_j [\langle n_i \rangle (1 + \langle n_j \rangle) \exp(\hbar\omega_i/kT) \tag{2}$$

$$- \langle n_j \rangle (1 + \langle n_i \rangle) \exp(\hbar\omega_j/kT) - (1 + \langle n_i \rangle)(1 + \langle n_j \rangle) + \langle n_i \rangle \langle n_j \rangle \exp[\hbar(\omega_i + \omega_j)/kT]]$$

$$= S - B_i[\langle n_i \rangle \exp(\hbar\omega_i/kT) - (1 + \langle n_i \rangle)], \tag{3}$$

where $\chi_j = \chi \exp(-\hbar\omega_j/kT)$, $B_i = \Sigma_j \chi_j[1 + \langle n_j\rangle + \langle n_j\rangle \exp(\hbar\omega_j/kT)] + \phi$. Therefore, the two-quantum term is of the same form as the one-quantum term and no Bose condensation occurs.

To confirm the weakness of Lifshits' conclusions, the problem must be treated microscopically. Assigning creation and destruction operators to each mode associated with the oscillating subunits (a_i^+, a_i), heat bath (b_k^+, b_k), and the energy source (p_p^+, p_p), it is possible to write the hamiltonian of the biological system as:

$$H = \sum_i \hbar\omega_i a_i^+ a_i + \sum_k \hbar\Omega_k b_k^+ b_k + \sum_p \hbar\bar{\Omega}_p p_p^+ p_p + \sum_{i,j,k} (\lambda a_i b_j^+ b_k + \lambda^* a_i^+ b_j b_k^+)$$
$$+ \sum_{i,j,k}' (\chi a_j^+ a_i b_k^+ + \chi^* a_j a_i^+ b_k) + \frac{1}{2}\sum_{i,j,k} (\beta a_j a_i b_k^+ + \beta^* a_j^+ a_i^+ b_k) + \sum_{p,i} (\xi p_p a_i^+ + \xi^* p_p^+ a_i), \tag{4}$$

where Ω_k and $\bar{\Omega}_p$ are the frequencies associated with excitations in the heat bath and energy source respectively and λ, χ, and ξ are coupling constants for the one-quantum process, for the Fröhlich two-quantum process, and for the energy supply to the oscillating units respectively.

β is the coupling constant for the simultaneous creation and destruction two-quantum terms suggested by Lifshits. Using this hamiltonian and perturbation theory [10] yielded for the $\langle \dot{n}_i\rangle$, the average rate of change of quanta with frequency ω_i in the biological system:

$$\langle \dot{n}_i\rangle = S - \phi(T,\omega_i)[\langle n_i\rangle \exp(\hbar\omega_i/kT) - (1 + \langle n_i\rangle)] - \sum_j \Lambda(T,\omega_i,\omega_j)[\langle n_i\rangle(1+\langle n_j\rangle)\exp(\hbar(\omega_i-\omega_j)/kT) - \langle n_j\rangle(1+\langle n_i\rangle)]$$
$$- \sum_j \Gamma(T,\omega_i,\omega_j)[\langle n_i\rangle\langle n_j\rangle \exp(\hbar(\omega_i+\omega_j)/kT) - (1+\langle n_i\rangle)(1+\langle n_j\rangle)],$$

where

$$\phi(T,\omega_i) = \frac{4\pi|\lambda|^2}{\hbar}\sum_j N(\hbar\Omega_j)N(\hbar\omega_i + \hbar\Omega_j)\exp(\hbar\Omega_j/kT),$$
$$\Lambda(T,\omega_i,\omega_j) = \frac{2\pi|\chi|^2}{\hbar}\times\begin{cases} 1 + N(\hbar\omega_j - \hbar\omega_i), & \text{for } \omega_j > \omega_i, \\ N(\hbar\omega_i - \hbar\omega_j), & \text{for } \omega_j < \omega_i, \end{cases} \tag{6}$$
$$S = \frac{4\pi|\xi|^2}{\hbar}\sum_i \langle p_i^+ p_i\rangle_0, \qquad \Gamma(T,\omega_i,\omega_j) = \frac{2\pi|\beta|^2}{\hbar}N(\hbar\omega_i + \hbar\omega_j),$$

and $N(\hbar\omega)$ is the number of excitations with energy $\hbar\omega$ in the heat bath.

If $\Lambda(T,\omega_i,\omega_j)\exp(-\hbar\omega_j/kT) = \Gamma(T,\omega_i,\omega_j)$ as Lifshits presumes, then eq. (5) can be reduced to Lifshits' form. However, as seen in eq. (6), allowing equality of $\Lambda\exp(-\hbar\omega_j/kT)$ and Γ is incorrect. Nonetheless, presupposing that the coupling constants are in the same order of magnitude for each type of two-quantum interaction, the coefficient, $\Lambda(T,\omega_i,\omega_j)$, is much larger than the coefficient, $\Gamma(T,\omega_i,\omega_j)$. This can be demonstrated readily for biological systems where ω_i is of the order of 10^{11} sec^{-1}. At room temperature $\hbar\omega_i/kT \doteq 10^{-2}$, thus Γ and Λ can be approximately written as:

$$\Gamma \doteq (2\pi|\beta|^2 kT/\hbar^2)(\omega_i + \omega_j)^{-1} \quad \text{and} \quad \Lambda \doteq (2\pi|\chi|^2 kT/\hbar)(|\omega_j - \omega_i|)^{-1}. \tag{7a, b}$$

For the Γ term, since $\omega_i \doteq \omega_j$ due to the very narrow range of normal frequencies, let $\omega_i \doteq \omega_j = \bar{\omega}$. For the Λ term, let $|\omega_j - \omega_i| = \Delta\omega$, the width of the frequency band. Thus eqs. (7) become:

$$\Gamma \doteq (\pi|\beta|^2/\hbar)(kT/\hbar\bar{\omega}) \quad \text{and} \quad \Lambda \doteq (2\pi|\chi|^2/\hbar)(kT/\hbar\Delta\omega). \tag{8a, b}$$

Since the range of normal frequencies of the oscillations is very small, $\Delta\omega \ll \bar{\omega}$. Therefore, the terms in eq. (8b) are much larger than those in eq. (8a). The simultaneous creation and destruction of quanta proposed by Lifshits can be regarded as negligible. Thus, the predominant "two-quantum" process is precisely the one postulated by

Fröhlich, and the addition of the Lifshits term does not cancel out the Bose condensation phenomenon predicted by Fröhlich.

As Fröhlich has pointed out to us, oscillations as introduced in refs. [7–9] and by the above authors form only a part of the degrees of freedom of the heat bath. Very important for the relevant energy transfer are non-oscillating systems, e.g. free ions, mobile ionic groups (dipoles), and certain low-lying electronic levels. For all of these it can be shown in a trivial way that Lifshits' contention is incorrect. In fact some of them may permit excitations in the energy range $(0, \hbar(\omega_{max} - \omega_0))$ but not in the range $(2\hbar\omega_0, 2\hbar\omega_{max})$, thus making two-phonon absorption or emission impossible.

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Coupled Anharmonic Oscillators. I. Equal-Mass Case

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This is the first in a series of papers on the large-order behavior of perturbation theory for coupled anharmonic oscillators. We exploit previously published dispersion techniques to convert the calculation of perturbation theory in large order into a barrier-penetration problem. We then introduce new semiclassical methods for describing tunneling through nonspherically symmetric, N-dimensional potentials. To illustrate our new methods, we calculate the large-order behavior of perturbation theory for a simple system of two equal-mass oscillators with quartic coupling. Our predictions are in complete agreement with computer calculations. We then extend our results to oscillators with x^{2N} coupling, N-oscillator systems, and some infinite-oscillator systems.

I. INTRODUCTION

In a recent paper Adler[1] argues that α, the physical charge on the electron, is an essential singularity of the Gell-Mann–Low function. Since the location of an essential singularity cannot be affected by the low-order terms in a perturbation expansion, an asymptotic study of perturbation theory for quantum electrodynamics in extremely large order seems indicated.

There has already been much work on the large-order behavior of perturbation series in quantum

field theory.[2] However, these papers are concerned with determining whether diagrammatic perturbation theory converges or diverges and are satisfied with order-by-order bounds on the terms in the perturbation series. What is required now is a precise analytic study of the behavior of perturbation theory in large order and not just upper and lower bounds.

Our general approach to perturbation theory in large order is guided by recent successful studies (which we review later) of perturbation theory for the anharmonic oscillator. We hope to extend and apply this work to the perturbation series encountered in quantum field theory. A $(\phi^4)_2$ quantum field theory is a good model with which to begin because, unlike four-dimensional quantum electrodynamics, its perturbation series are finite order by order. Moreover, Glimm and Jaffe[3] have shown that, with spatial and ultraviolet cutoffs, this field theory is equivalent to a finite system of coupled anharmonic oscillators (modes). Our goal is to find the large-order behavior of perturbation theory for this system as a function of the two cutoffs, remove the ultraviolet and spatial cutoffs, and, assuming that this limit exists (we see no reason why it should not), thereby obtain the answer. We have not solved this problem, but we do report major progress toward that goal. We have developed new mathematical techniques for investigating the large-order behavior of perturbation theory for coupled oscillator systems. Specifically, we have developed very general *multidimensional* semiclassical techniques[4] for treating nonspherically symmetric potentials such as those encountered in the cutoff $(\phi^4)_2$ model.

The purpose of the first paper in this series is to explain our new methods in detail in the limited context of equal-mass oscillators. (Our second paper deals with the more difficult problem of unequal-mass oscillators.) We begin by solving the two-mode oscillator defined by

$$\left[-\frac{\partial^2}{\partial x^2}-\frac{\partial^2}{\partial y^2}+\tfrac{1}{4}x^2+\tfrac{1}{4}y^2 +\lambda(ax^4+by^4+2cx^2y^2)/4-E(\lambda)\right]\psi(x,y)=0, \tag{1.1}$$

where

$$\lim_{|\vec{r}|\to\infty}\psi(x,y)=0, \quad \vec{r}=(x,y). \tag{1.2}$$

We assume that $a>0$, $b>0$, and $c\geq -\sqrt{ab}$, making the potential bounded below. Then it is possible to satisfy Eq. (1.2) and ψ is a bound-state wave function with $E(\lambda)$ its associated energy eigenvalue. The perturbation series for the ground-state energy is given by[5]

$$E(\lambda)=1+\sum_{n=1}^{\infty}A_n\lambda^n. \tag{1.3}$$

Our goal is to predict the behavior of A_n for large n as a function of a, b, and c. We solve this problem in Secs. II–VI. The solution is a discontinuous function of a, b, and c, and the discontinuities have an interesting physical explanation. In Sec. VII we hope to whet the reader's appetite by giving a partial solution of the N-mode oscillator defined by

$$\left[\sum_{i=1}^{N}\left(-\frac{\partial^2}{\partial x_i^2}+\tfrac{1}{4}x_i^2\right) +\tfrac{1}{4}\lambda\sum_{i,j=1}^{N}a_{ij}x_i^2x_j^2-E(\lambda)\right]\psi(x_i)=0. \tag{1.4}$$

In addition, we exhibit some matrices $\{a_{ij}\}$ for which the limit $N\to\infty$ may be taken, and explicitly compute the large-order behavior of the ground-state energy perturbation series for those infinite systems.

Before discussing more fully the content and results of this paper, we briefly review what is known about the one-oscillator problem so that this paper will be reasonably self-contained. In early computer studies[6,7] of the ground-state energy of the anharmonic oscillator, an extraordinarily simple behavior was observed for perturbation theory in large order. In particular, for the anharmonic oscillator defined by

$$[-d^2/dx^2+x^2/4+\lambda x^4/4-E(\lambda)]\psi(x)=0 \tag{1.5}$$

and

$$\lim_{|x|\to\infty}\psi(x)=0, \tag{1.6}$$

and the ground-state energy perturbation series defined by

$$E(\lambda)=\tfrac{1}{2}+\sum_{n=1}^{\infty}A_n\lambda^n, \tag{1.7}$$

a numerical fit to A_n for large n was found to be

$$A_n=-\frac{\sqrt{6}}{\pi^{3/2}}(-3)^n\,\Gamma(n+\tfrac{1}{2})\left[1+O\left(\frac{1}{n}\right)\right]. \tag{1.8}$$

Later, methods were developed that explain this behavior analytically.[8,9] Indeed, two completely independent techniques were developed which predict formulas similar to Eq. (1.8) for *all* energy levels of *all* anharmonic oscillators having λx^{2N} interactions. Analogous results for anharmonic oscillators with arbitrary polynomial interactions were also obtained and the effect of Wick ordering on perturbation theory in large order was determined.[10] Moreover, higher-order corrections [$O(1/n)$ terms] were also calculated and these also agree with computer calculations.[8,9]

The anharmonic oscillator suggests how perturbation theory might behave in a $(\phi^4)_2$ field theory because it is also a quantum field theory. Equations (1.5) and (1.6) can be reinterpreted as describing a ϕ^4 field theory in one-dimensional space-time.[11] For more complicated systems such as those defined in Eqs. (1.1) and (1.2) and in Eq. (1.4), A_n behaves for large n very much like the A_n in Eq. (1.8). We would not be very surprised to find that the ground-state energy perturbation series for a $(\phi^4)_2$ field theory behaves similarly.

We now give a more detailed outline of this paper. Section II employs dispersion techniques to reduce the problem of finding the leading large-n behavior of A_n to that of solving a quantum-mechanical barrier-penetration problem. The barrier penetration is then treated semiclassically. In Secs. III and IV we develop those semiclassical methods needed to solve Eqs. (1.1) and (1.2) for the special case where $a=b=1$. In Sec. III we solve the problem approximately using geometrical optics, and in Sec. IV we find the precise leading behavior of A_n for large n using physical optics. The notions of geometrical and physical optics are discussed in detail. The results are as follows: For $-1 \leq c < 1$,

$$A_n \sim -[48c/\sin(\pi\nu)]^{1/2}(-3)^n \Gamma(n+\tfrac{1}{2})/\pi, \tag{1.9}$$

where $\nu(\nu+1)=2c$; for $c>1$,

$$A_n \sim -[48\bar{c}/\sin(\pi\bar{\nu})]^{1/2}[(-3c-6)/2]^n \Gamma(n+\tfrac{1}{2})/\pi, \tag{1.10}$$

where $\bar{\nu}(\bar{\nu}+1)=2\bar{c}=(3-c)/(1+c)$; and for $c=1$,

$$A_n \sim 2(-3)^{n+1}\Gamma(n+1)/\pi. \tag{1.11}$$

The discontinuity at $c=1$ is caused by a kind of constructive interference.

The calculation for the special spherically symmetric case ($c=1$) is done in Appendix A. Appendix A actually treats the more general case of N oscillators with spherically symmetric coupling. In Appendix B we repeat the calculations of Sec. IV for a system of two coupled oscillators with $2N$ order rather than quartic coupling.

The predictions in Eqs. (1.9)–(1.11) are compared with computer calculations in Sec. V and we observe spectacular agreement. Section VI generalizes the results in Eqs. (1.9)–(1.11) to the case where a and b are not 1. Again the theoretical and numerical calculations agree. Finally, in Sec. VII we investigate Eq. (1.4). In Appendix C we discuss some properties of equal-mass N-oscillator systems that are more general than those in Eq. (1.4).

II. DISPERSION RELATION

In this section we reduce the problem of finding the large-order behavior of the Rayleigh-Schrödinger coefficients to a semiclassical barrier-penetration problem. This reduction is necessary because, as we explain below, a direct analysis of the difference equation which generates the perturbation series is too difficult.

In general, the Rayleigh-Schrödinger coefficients [such as those defined in Eq. (1.3)] for a perturbation problem in quantum mechanics may be computed from a recursion relation which gives the nth coefficient in terms of its predecessors and integrals over all the lower-order contributions to the wave function. The special form of the unperturbed wave function for the harmonic oscillator and our choice of a polynomial perturbing potential [see Eqs. (1.1) and (1.4)] enable us to reduce the recursion relation to a single nonlinear partial difference equation. This difference equation, which we derive and discuss in detail in Secs. V and VI, may be solved on a computer.

The corresponding but much simpler difference equation for a single anharmonic oscillator can actually be solved approximately.[8,9] However, for a system of coupled anharmonic oscillators, the indices in the difference equation proliferate and the relevant approximation techniques become unwieldy, even for the two-mode oscillator. With this difficulty in mind, we have developed in a previous paper a rather indirect, but physically intuitive, method for computing the large-n behavior of A_n which generalizes immediately to the many-coupled oscillator problem. This method rests on Eq. (2.4).

Equation (2.4) may be derived by assuming a once-subtracted dispersion relation for $E(\lambda)$:

$$F(\lambda) = \frac{1}{2\pi i}\int_{-\infty}^{0} \frac{D(x)}{x-\lambda}\,dx, \tag{2.1}$$

where

$$F(\lambda) \equiv [E(\lambda)-E(0)]/\lambda \tag{2.2}$$

and

$$D(\lambda) \equiv \lim_{\epsilon\to 0}[F(\lambda+i\epsilon)-F(\lambda-i\epsilon)]. \tag{2.3}$$

From Eq. (2.1) we obtain an exact expression for the nth Rayleigh-Schrödinger coefficient after expanding $(x-\lambda)^{-1}$ and comparing with Eq. (1.3). We find that

$$A_n = \frac{1}{2\pi i}\int_{-\infty}^{0} D(x)x^{-n}\,dx. \tag{2.4}$$

The proof of Eq. (2.1) for the one-mode oscillator is outlined in Sec. II of BW II. Although there is

as yet no proof, it is intuitively reasonable that Eq. (2.1) is valid for the N-mode oscillator. However, a proof would require that $E(\lambda)$ be analytic in the cut λ plane. Analyticity has only been proved[12] in the region shown in Fig. 1. Fortunately, analyticity in this region is sufficient to rigorously establish Eq. (2.4) for large n.[12]

Equation (2.4) may be understood as follows. When λ is negative, the perturbing potential in Eq. (1.1) becomes repulsive. The states that were found for $\lambda > 0$ now decay. Decaying states have complex eigenvalues and the imaginary part of the eigenvalue is inversely proportional to the lifetime of the state. Equation (2.4) relates A_n to the inverse moments of the imaginary part of $E(\lambda)$ and thus to the lifetime of the decaying state. For large n, the integral is dominated by small negative values of x. Thus, the large-order behavior of A_n is related to the imaginary part of $E(\lambda)$ for small negative coupling constant.

It is interesting to note that if the perturbing potential is dominated by the unperturbed potential when λ gets small enough, all states will be bound for such values of λ. We then have $\mathrm{Im}E(\lambda) = 0$ for $|\lambda|$ sufficiently small. In this case, Eq. (2.4) tells us that $\lim_{n\to\infty} A_n = 0$, and the perturbation series converges. Thus, Eq. (2.4) quantifies Dyson's[13] old argument about the divergence of perturbation theory. We emphasize the similarity between the above discussion and that given by Adler in Sec. V A of Ref. 1.

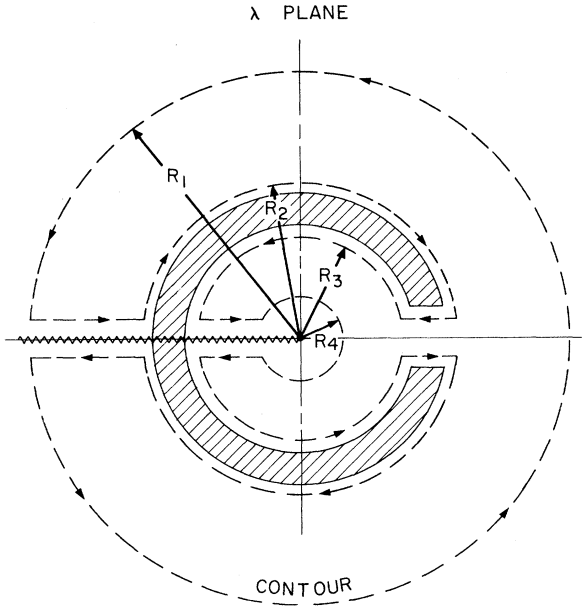

FIG. 1. The domain of analyticity of $E(\lambda)$ rigorously established by Simon. $E(\lambda)$ is analytic except on the cut along the negative real axis and in the shaded region. The contour appropriate for proving Eq. (2.4) is indicated by a dashed line.

Equation (2.4) relates the computation of A_n to the familiar quantum-mechanical barrier-penetration problems as follows. Generalizing the arguments of the Appendix in Ref. 9 we can express $\mathrm{Im}E(\lambda)$ as

$$\mathrm{Im}E = \frac{\int_{S_V} \vec{\mathbf{J}} \cdot d\vec{\mathbf{S}}}{\int_V \psi^* \psi \, dV}, \tag{2.5}$$

where $\vec{\mathbf{J}}$ is the probability current,

$$\vec{\mathbf{J}} \equiv \tfrac{1}{2} i(\psi^* \vec{\nabla} \psi - \psi \vec{\nabla} \psi^*), \tag{2.6}$$

and S is the boundary of volume V. We take S to lie outside the barrier so that there are no points outside of S where the probability current can reflect back toward the origin. Thus $\vec{\mathbf{J}}$ measures decay of the state by expressing the flow of probability density out to $|\vec{\mathbf{r}}| = \infty$.

Equations (2.4) and (2.5) relate A_n to the perturbed wave function for small negative λ.

III. GEOMETRICAL OPTICS

A. Discussion of Method of Solution

In Sec. II we reduced the calculation of the large-order behavior of perturbation theory to the familiar quantum-mechanical problem of tunneling through a nonspherical N-dimensional potential barrier. We propose a method for solving this problem which rests on a simple physical picture. A particle in an unstable state centered at the origin will ultimately penetrate the barrier and escape to infinity. The total amplitude for escape is the sum of the amplitudes over all possible paths of escape. We will show that there exist most probable escape paths (MPEP's) and that the relative amplitude to escape along other paths is exponentially small. The dominant contribution to the escape amplitude comes from regions, which we call tubes, surrounding the MPEP's. The probability current is negligible outside of these tubes during tunneling and flows outward in narrow beams. We will show that for a system without a rotational symmetry the number of tubes is finite and that they are well separated. We will then use semiclassical (WKB-related) methods to approximate the solutions to the Schrödinger equation [Eq. (1.1)] within these tubes.

We have introduced the notion of a tube in order to reduce our nonspherical multidimensional problem to one which is approximately one-dimensional. It is natural to try to solve a tunneling problem using WKB techniques.[14] There has been much work on the problem of WKB approximations to many-dimensional systems.[14] The zeroth-order WKB

equation for the phase S of a wave function with energy E is[15]

$$(\vec{\nabla}S)^2 = V - E\,. \tag{3.1}$$

This is just the Hamilton-Jacobi equation for a classical system with Hamiltonian $\vec{p}^2 + V$. In one dimension it reduces to $(dS/dx)^2 = V - E$, whose solution is $S = \pm\int (V-E)^{1/2}$. For the general multidimensional case it is a nonlinear partial differential equation. Of course, if the Hamiltonian has a continuous symmetry, Eq. (3.1) will be separable. However, Eq. (3.1) is nontrivial in general. The new multidimensional techniques which we have discovered simplify the problem of solving Eq. (3.1) because now we need to solve it only in a small, approximately one-dimensional region. Our technique is expressly designed to deal with problems which do *not* have continuous symmetries, and is thus *complementary* to the separation of variables idea.

We briefly review the path-integral formalism. The amplitude for a particle of energy E to take a particular path P in a potential V is

$$\frac{1}{N}\exp\left[-\int_P (V-E)^{1/2}\right], \tag{3.2}$$

where $\int_P (E-V)^{1/2}$ is the classical action and the normalization factor N depends only on the end points of the path. The total amplitude is just the sum of Eq. (3.2) over all paths P.

In the tunneling region, $(V-E)$ is positive, and the amplitude is exponentially damped. Therefore, the dominant contribution to the amplitude comes from regions near the paths which minimize the action integral and thus satisfy

$$\delta\int (V-E)^{1/2} = 0\,. \tag{3.3}$$

The Euler-Lagrange equations following from Eq. (3.3) are[16]

$$2(V-E)\frac{d^2x_i}{ds^2} + \frac{dx_i}{ds}\left[\sum_j \frac{dx_j}{ds}\frac{dV}{dx_j}\right] = \frac{\partial V}{\partial x_i}\,, \tag{3.4}$$

where s is the path length. All solutions of Eq. (3.4) are local stationary points of the action. However, we are interested in the *global* minima. This will eliminate all except a discrete[17] set of paths which are just the MPEP's.

Once we have found a set of MPEP's, we must find approximate solutions to the Schrödinger equation along these trajectories. As in any semiclassical or ray description of a wave phenomenon, we must distinguish two levels of approximation. At the first level, called geometrical optics or the eikonal approximation, the phase of the wave function is approximated by a line integral along the trajectory, while its amplitude is assumed to be constant. This is just zeroth-order WKB. The second level, called physical optics or first-order WKB, takes into account the variation of the amplitude and the spread of the wave function into the region around the trajectory. Thus, physical optics is characterized by a set of tubes through which most of the probability current flows.

These two levels of approximation are clearly distinguished in our results for the large-order behavior of perturbation theory. We find that in general for large n

$$A_n \sim KL^n\,\Gamma(Mn+J)[\,1+O(1/n)]\,. \tag{3.5}$$

The constants L, M, and J are determined by geometrical optics alone. Physical optics is needed to find the value of K.

B. Determination of Most Probable Escape Paths

The program we outlined in part A of this section for finding the MPEP's is of course very difficult. It involves actually finding closed-form solutions to Eq. (3.4) and explicitly selecting those solutions which minimize $\int (V-E)^{1/2}$. Fortunately, in many cases, a heuristic argument enables us to guess the most probable paths without solving Eq. (3.4) and these turn out to be straight lines. In fact, it is generally true that the MPEP's for the equal-mass oscillators defined in Eq. (1.4) are straight lines. The more difficult problem of unequal-mass oscillators, which have curved MPEP's, will be discussed in the next paper of this series.[18] It is easy to show that the straight MPEP's satisfy Eq. (3.4), but we have no way of proving that they are global minima of the action. The only convincing evidence we have for this is the excellent agreement of our results with our computer calculations.

In this section we use geometrical optics to treat the special case of Eq. (1.1) for which $a = b = 1$. This simplifies the notation without obscuring any of the important features of the problem. In Sec. IV we use physical optics to treat this same case. Equation (1.1) is solved in general in Sec. VI.

We expect a straight MPEP to satisfy certain reasonable criteria. It should be a "path of least resistance" to tunneling and thus should pass through a saddle point of the potential $V = \frac{1}{4}x^2 + \frac{1}{4}y^2 - \frac{1}{4}\epsilon(x^4 + y^4 + 2cx^2y^2)$. The saddle point should be oriented along the path (which is a radial line). A saddle point of V satisfies the equations

$$\begin{aligned}\frac{\partial V}{\partial x} &= \tfrac{1}{2}x - \epsilon(x^3 + cxy^2) = 0\,,\\ \frac{\partial V}{\partial y} &= \tfrac{1}{2}y - \epsilon(y^3 + cx^2y) = 0\,.\end{aligned} \tag{3.6}$$

Equations (3.6) have nine solutions, namely,

$$(x, y) = (0, 0),$$
$$(\pm(2\epsilon)^{-1/2}, 0), \quad (0, \pm(2\epsilon)^{-1/2}), \tag{3.7}$$

and

$$\pm([2(c+1)\epsilon]^{-1/2}, \pm[2(c+1)\epsilon]^{-1/2}).$$

To identify those critical points in Eq. (3.7) which are radially oriented saddle points, we compute the Hessian matrix H (matrix of second partial derivatives). We demand that H have one positive and one negative eigenvalue at the critical point and further require that the eigenvector having negative eigenvalue must lie along the radial line connecting the critical point with the origin. We find that

$$H = \begin{bmatrix} \frac{1}{2} - \epsilon(3x^2 + cy^2) & -2c\epsilon xy \\ -2c\epsilon xy & \frac{1}{2} - \epsilon(3y^2 + cx^2) \end{bmatrix}. \tag{3.8}$$

It follows that $(0, 0)$ is not a saddle point, that $(\pm(2\epsilon)^{-1/2}, 0)$ and $(0, \pm(2\epsilon)^{-1/2})$ are acceptable saddle points when $-1 \leq c < 1$, and $\pm([2(c+1)\epsilon]^{-1/2}, \pm[2(c+1)\epsilon]^{-1/2})$ are acceptable saddle points when $c > 1$. When $c = 1$, there are no saddle points. This is the spherically symmetric case where all straight-line paths contribute equally to the amplitude and it is treated separately in Appendix A.

It is now easy to show that radial lines through the saddle points are solutions of Eq. (3.4).[19] Equation (3.4) reduces to

$$\frac{dx}{ds}\left(\frac{dx}{ds}\frac{\partial V}{\partial x} + \frac{dy}{ds}\frac{\partial V}{\partial y}\right) = \frac{\partial V}{\partial x},$$
$$\frac{dy}{ds}\left(\frac{dx}{ds}\frac{\partial V}{\partial x} + \frac{dy}{ds}\frac{\partial V}{\partial y}\right) = \frac{\partial V}{\partial y}, \tag{3.9}$$

because our straight-line paths have the property that

$$\frac{d^2x}{ds^2} = \frac{d^2y}{ds^2} = 0.$$

Equation (3.9) is satisfied by $(dx/ds, dy/ds) = (0, \pm 1), (\pm 1, 0)$, and $(\pm\sqrt{2}/2, \pm\sqrt{2}/2)$ because of three properties of V:

$$\left.\frac{\partial V}{\partial y}\right|_{y=0} = 0,$$
$$\left.\frac{\partial V}{\partial x}\right|_{x=0} = 0, \tag{3.10}$$
$$\frac{\partial V}{\partial x} = \pm\frac{\partial V}{\partial y}, \quad \text{when } x = \pm y.$$

C. Geometrical Optics

We outline here a brief and heuristic treatment. A careful and mathematically detailed approach is given in Sec. IV. We follow method 2 of the Appendix of BW II. Up to multiplicative constants the wave function $\psi(x, y)$ in the tunneling region on the MPEP is given by

$$e^{-\int_{s_0}^{(x,y)}(V-E)^{1/2}\,ds}.$$

We are ignoring all paths except the four MPEP's. E is the unperturbed value of the energy, namely, unity. The integral is taken along the MPEP from the inner turning point $s_0 = O(1)$ (solution of $V - E = 0$) to the argument of $\psi(x, y)$.

We are interested in computing the current flowing out to infinity. Since the distant turning point $s_1 = O(\epsilon^{-1/2})$ is the last place where the current can be reflected back toward the origin, we compute the current at a point just beyond s_1. The total current J is the sum of the currents along each MPEP. J is proportional to

$$\exp\left[-2\int_{s_0}^{s_1}(V-E)^{1/2}ds\right]$$

for each MPEP. This reduces to

$$\exp\left\{-2\int_{s_0}^{s_1}[\tfrac{1}{4}s^2 - \tfrac{1}{4}\epsilon s^4\alpha - 1]^{1/2}ds\right\},$$

where $\alpha = \cos^4\theta + \sin^4\theta + 2c\sin^2\theta\cos^2\theta$, and θ is the angle between the MPEP and the x axis. Computing the above integral approximately gives

$$J \propto \epsilon^{-1/2}\, e^{-1/(3\epsilon\alpha)}$$

for each MPEP. From Eqs. (2.4) and (2.5) we thus obtain

$$A_n \propto \int_{-\infty}^{0} d\epsilon(-\epsilon)^{-n-3/2}\, e^{1/(3\epsilon\alpha)}$$
$$\propto \Gamma(n+\tfrac{1}{2})(-3\alpha)^n. \tag{3.11}$$

When $-1 \leq c < 1$, the MPEP's are along the axes and $\alpha = 1$. When $c > 1$, the MPEP's are at 45° to the axes and $\alpha = \frac{1}{2}(c+1)$. Thus, for large n,

$$A_n \propto (-3)^n\,\Gamma(n+\tfrac{1}{2}), \quad -1 \leq c < 1,$$

and (3.12)

$$A_n \propto [-3(c+1)/2]^n\,\Gamma(n+\tfrac{1}{2}), \quad c > 1.$$

Equation (3.12) is continuous in c at $c = 1$.

Equation (3.12) clearly illustrates the phenomenon of *decoupling* that takes place in the large-order behavior of perturbation theory. When the coupling of the oscillators is strong enough ($c > 1$), A_n depends on the coupling term. But when $-1 \leq c < 1$, the system seems to behave as if the oscillators were completely uncoupled. Actually, when $c < 1$ the multiplicative constant K, which we will determine in Sec. IV, still depends on c. Nevertheless, the decoupling of the *dominant* behavior of A_n for large n is quite remarkable and is typical of the simplification that we observe in large order.

IV. PHYSICAL OPTICS

In this section we use physical optics to approximate the imaginary part of the ground-state energy for the system

$$\left\{-\frac{\partial^2}{\partial x^2}-\frac{\partial^2}{\partial y^2}+\tfrac{1}{4}x^2+\tfrac{1}{4}y^2 -\tfrac{1}{4}\epsilon(x^4+y^4+2cx^2y^2)-E\right\}\psi=0. \quad (4.1)$$

We will solve the problem explicitly only for $-1 \leq c<1$. When $c>1$ we use the following symmetry transformation to reduce the problem back to the $c<1$ case[20]:

$$x\to\bar{x}=\frac{x+y}{\sqrt{2}}, \qquad y\to\bar{y}=\frac{x-y}{\sqrt{2}}. \quad (4.2)$$

Equation (4.2) converts Eq. (4.1) into

$$\left\{-\frac{\partial^2}{\partial\bar{x}^2}-\frac{\partial^2}{\partial\bar{y}^2}+\tfrac{1}{4}\bar{x}^2+\tfrac{1}{4}\bar{y}^2 -\tfrac{1}{4}\epsilon\left(\frac{1+c}{2}\right)\left(\bar{x}^4+\bar{y}^4+2\frac{3-c}{1+c}\bar{x}^2\bar{y}^2\right)-E\right\}\psi=0. \quad (4.3)$$

We then make the additional transformations

$$c\to\bar{c}=(3-c)/(1+c), \qquad \epsilon\to\bar{\epsilon}=\epsilon(1+c)/2, \quad (4.4)$$

and observe that $c>1$ implies that $|\bar{c}|<1$.

A. The Physical-Optics Approximation

We will solve Eq. (4.1) in a tube of thickness $O(1)$ surrounding the positive x axis. (The x axis is an MPEP when $|c|<1$.) To do so, we break the tube into two regions: Region I, where $y=O(1)$ and $x<\epsilon^{-1/4}$, and Region II, where $y=O(1)$ and $\epsilon^{-1/6}<x\ll\epsilon^{-1/2}$. Notice that the regions overlap.

In Region I we approximate Eq. (4.1) by

$$\left[-\frac{\partial^2}{\partial x^2}-\frac{\partial^2}{\partial y^2}+\tfrac{1}{4}x^2+\tfrac{1}{4}y^2-1\right]\psi=0, \quad (4.5)$$

whose solution is the unperturbed ground-state wave function

$$\psi_{\mathrm{I}}=e^{-(x^2+y^2)/4}. \quad (4.6)$$

We have freely chosen the normalization of ψ_{I}.

In Region II, we approximate Eq. (4.1) by

$$\left[-\frac{\partial^2}{\partial x^2}-\frac{\partial^2}{\partial y^2}+\tfrac{1}{4}x^2+\tfrac{1}{4}y^2-\tfrac{1}{4}\epsilon(x^4+2cx^2y^2)-1\right]\psi=0, \quad (4.7)$$

where we have neglected ϵy^4 compared with y^2. As in BW II, we seek a solution which is exponentially decaying with increasing x in the tunneling region. Thus, we factor off a decreasing WKB-type function of the x variable:

$$\psi=\chi(\tfrac{1}{4}x^2-\tfrac{1}{4}\epsilon x^4-\tfrac{1}{2})^{-1/4} \times\exp\left[-\int_{\sqrt{2}}^{x}(\tfrac{1}{4}t^2-\tfrac{1}{4}\epsilon t^4-\tfrac{1}{2})^{1/2}dt\right]. \quad (4.8)$$

We have chosen the lower end point of the integral arbitrarily. The new equation satisfied by χ is

$$(x^2-\epsilon x^4)^{1/2}\chi_x-\chi_{yy}+(\tfrac{1}{4}y^2-\tfrac{1}{2}-\tfrac{1}{2}\epsilon cx^2y^2)\chi=0. \quad (4.9)$$

The change of scale

$$\epsilon x^2=z^2 \quad (4.10)$$

eliminates all reference to ϵ from Eq. (4.9) and gives

$$z(1-z^2)^{1/2}\chi_z-\chi_{yy}+(\tfrac{1}{4}y^2-\tfrac{1}{2}-\tfrac{1}{2}cy^2z^2)\chi=0. \quad (4.11)$$

In Eq. (4.8) we factored off the rapidly changing geometrical optics behavior. Equation (4.11) contains the next-order correction to this behavior which we have referred to as physical optics; that is, Eq. (4.11), when solved, will provide the multiplicative constants that were missing in Eq. (3.12). However, there are no further approximations to be made because all quantities in Eq. (4.11) are of order unity. Equation (4.11) must now be solved exactly.

The change of variables

$$w=(1-z^2)^{1/2} \quad (4.12)$$

is useful because the resulting equation,

$$(w^2-1)\chi_w-\chi_{yy}+[\tfrac{1}{4}y^2-\tfrac{1}{2}-\tfrac{1}{2}cy^2(1-w^2)]\chi=0, \quad (4.13)$$

no longer contains a square-root term.

One strategy for solving Eq. (4.13) is to transform the dependent variable so that a Fourier transform in the y variable gives a (hopefully soluble) first-order partial differential equation. Of course, an immediate Fourier transform of Eq. (4.13) is useless because of the y^2 term. We are thus led to the substitution

$$\chi=e^{-y^2f(w)/4}A. \quad (4.14)$$

The undetermined function $f(w)$ will be chosen to eliminate the y^2 term from the differential equation for A. It will then be profitable to Fourier transform that equation because the highest power of y will be one. The above constraint on $f(w)$ takes the form of a Riccati equation:

$$-(w^2-1)f'(w)-f^2(w)+1-2c+2cw^2=0. \quad (4.15)$$

When Eq. (4.15) is satisfied, the equation for A simplifies to

$$(w^2-1)A_w - A_{yy} + yf(w)A_y + \tfrac{1}{2}[f(w)-1]A = 0. \tag{4.16}$$

First we solve Eq. (4.15). A standard substitution which linearizes the Riccati equation is

$$f(w) = (w^2-1)u'(w)/u(w). \tag{4.17}$$

We obtain

$$(1-w^2)u'' - 2wu' + u\left(2c - \frac{1}{1-w^2}\right) = 0. \tag{4.18}$$

We gratefully recognize that Eq. (4.18) is the associated Legendre equation.[21] Solutions to this equation are

$$u(w) = P_\nu^\mu(w),\ Q_\nu^\mu(w), \tag{4.19}$$

where

$$\nu(\nu+1) = 2c, \quad \mu^2 = 1. \tag{4.20}$$

For definiteness we choose[22]

$$u(w) = P_\nu^{-1}(w). \tag{4.21}$$

Next, we return to Eq. (4.16) and complete its solution. It is clear that the strategy of the substitution in Eq. (4.14) has succeeded; that is, if we Fourier transform in the y variable, the resulting equation will be first order and should yield to the method of characteristics. However, we are fortunate that there is an even simpler approach. We change to new independent variables

$$(w, y) \to (w,\ s = y/u(w)). \tag{4.22}$$

In terms of these variables, Eq. (4.16) becomes

$$(w^2-1)u^2(w)A_w + \tfrac{1}{2}u^2(w)[f(w)-1]A = A_{ss}, \tag{4.23}$$

which is separable.

We now argue that the separation constant for Eq. (4.23) is 0. To justify this contention explicitly we separate

$$A(w, s) = B(w)C(s). \tag{4.24}$$

For separation constant α^2, the equation for $C(s)$ is $C''(s) = \alpha C(s)$, whose solution is $C(s) \propto \cosh(\alpha s) = \cosh[\alpha y/u(w)]$. Here we have kept only the even solution in y because only an even solution can be matched to $\psi_{\rm I}$ in Eq. (4.6). However, $u(w)$ in Eq. (4.21) vanishes [see Eq. (4.29)] at $w = 1$ and $w \lesssim 1$ is in the overlap of Regions I and II [see Eqs. (4.10) and (4.12)]. Therefore, there is no asymptotic match of ψ across Regions I and II unless, of course, $\alpha^2 = 0$.

Having shown that $\alpha^2 = 0$ and thus that $C(s)$ is constant, it is straightforward to solve for $B(w)$. We obtain (omitting a multiplicative constant)

$$B(w) = u^{-1/2}[(1-w)/(1+w)]^{1/4}. \tag{4.25}$$

Note that as $w \to 1$, $B(w) \to$ a finite constant [see Eq. (4.31)]. Thus, it is possible to match $\psi_{\rm I}$ and $\psi_{\rm II}$ asymptotically in the overlap region.

We have now solved Eq. (4.1) in Region II up to an over-all multiplicative constant β. Our final result is

$$\psi_{\rm II} = \beta(\tfrac{1}{4}x^2 - \tfrac{1}{4}\epsilon x^4 - \tfrac{1}{2})^{-1/4} \times \exp\left[-\int_{\sqrt{2}}^{x} (\tfrac{1}{4}t^2 - \tfrac{1}{4}\epsilon t^4 - \tfrac{1}{2})^{1/2} dt\right] \times \exp[-y^2 f(w)/4][u(w)]^{-1/2} \times [(1-w)/(1+w)]^{1/4}. \tag{4.26}$$

It is easy to identify the physical meaning of the three types of terms in Eq. (4.26). There is a rapidly varying term from geometrical optics and several slowly varying terms that do not depend on y. These describe the amplitude along the MPEP. Finally, the term $\exp[-y^2 f(w)/4]$ describes the falloff of probability current in the tube surrounding the x axis. A quick calculation shows that as x approaches the turning line [the line along which $V(x, y) - E = 0$] at the end of the tunnel near $x = \epsilon^{-1/2}$, w approaches 0, and, for positive c, the tube gradually widens. At the turning line the tube flares out like the bell of a trumpet. When $-1 \leq c < 0$, the tube narrows as $w \to 0$. When $c = 0$, the thickness of the tube is constant along its length.

It might appear that our solution Eq. (4.26) is the result of an amazing sequence of lucky substitutions whose application is rather limited. Actually these techniques immediately generalize to *all* straight-line path problems [see Secs. V–VII and Appendix B]. Moreover, when we study an arbitrary curved-path problem in our next paper[23] we show that factoring off the geometrical optics behavior always leads to a Riccati equation whose solution is related to the thickness of the tube. So, on the contrary, the substitutions we have made are both natural and general.

B. Asymptotic Matching of Regions I and II

We now determine β by requiring that $\psi_{\rm I}$ in Eq. (4.6) and $\psi_{\rm II}$ in Eq. (4.26) become asymptotically equal in the overlap of Regions I and II. In this overlap region, x is large compared with 1 but small compared with $\epsilon^{-1/2}$. Thus, we approximate

$$(\tfrac{1}{4}x^2 - \tfrac{1}{4}\epsilon x^4 - \tfrac{1}{2})^{-1/4} \sim (\tfrac{1}{2}x)^{-1/2}, \tag{4.27}$$

$$-\int_{\sqrt{2}}^{x}(\tfrac{1}{4}t^2-\tfrac{1}{4}\epsilon t^4-\tfrac{1}{2})^{1/2}dt \sim -\tfrac{1}{2}\int_{\sqrt{2}}^{x}(t^2-2)^{1/2}dt-\tfrac{1}{4}x^2+\tfrac{1}{2}\ln x+\tfrac{1}{4}+O\left(\frac{1}{x^2}\right). \tag{4.28}$$

Also, in the overlap of Regions I and II, $w\sim 1$. Thus, we use[24]

$$u(w)=P_\nu^{-1}(w)\sim 2^{-1/2}(1-w)^{1/2}, \quad \text{when } w\sim 1. \tag{4.29}$$

Equation (4.29) implies that

$$f(w)=\frac{(w^2-1)u'(w)}{u(w)}\sim 1, \tag{4.30}$$

$$u^{-1/2}\left(\frac{1-w}{1+w}\right)^{1/4}\sim 1. \tag{4.31}$$

Combining Eqs. (4.26)–(4.31) gives

$$\psi_{II}\sim \beta e^{-(x^2+y^2)/4}e^{1/4\sqrt{2}}. \tag{4.32}$$

Thus, comparing Eqs. (4.32) and Eq. (4.6) gives

$$\beta=2^{-1/2}e^{-1/4}. \tag{4.33}$$

Now ψ in Region II is completely determined relative to the normalization of ψ in Region I.

C. Determination of the Probability Current $J(x)$

In the Appendix of BW II we developed a trick (referred to there as Method 2) for evaluating the probability current for values of x further from the origin than the turning line without ever doing turning-point analysis. Without further explanation we use this technique to obtain the magnitude of the probability current emerging from the end of the tube along the positive x axis:

$$J(y)=\beta^2 e^{-y^2 f(0)/2}u^{-1}(0)\times\exp\left[-\int_{\sqrt{2}}^{x_1}(t^2-\epsilon t^4-2)^{1/2}dt\right]. \tag{4.34}$$

x is the distant zero of the integrand. Note that J is a function of y only.

The evaluation of the integral is given in BW II as

$$\exp\left[-\int_{\sqrt{2}}^{x_1}(t^2-\epsilon t^4-2)^{1/2}dt\right]\sim e^{-1/3\epsilon}2(e/\epsilon)^{1/2}. \tag{4.35}$$

To compute $u(0)$ and $f(0)$ we use the formulas[25]

$$u(0)=P_\nu^{-1}(0)=\tfrac{1}{2}\pi^{-1/2}\sin(\tfrac{1}{2}\pi\nu)\Gamma(\tfrac{1}{2}\pi\nu)/\Gamma(\tfrac{3}{2}+\tfrac{1}{2}\nu),$$
$$u'(0)=\frac{d}{dw}P_\nu^{-1}(w)\Big|_{w=0} =-\pi^{-1/2}\cos(\tfrac{1}{2}\pi\nu)\,\Gamma(\tfrac{1}{2}+\tfrac{1}{2}\nu)/\Gamma(1+\tfrac{1}{2}\nu). \tag{4.36}$$

From Eqs. (4.17) and (4.36) we have

$$\frac{f(0)}{2}=\frac{\cos(\frac{1}{2}\pi\nu)\,\Gamma(\frac{1}{2}+\frac{1}{2}\nu)\Gamma(\frac{3}{2}+\frac{1}{2}\nu)}{\sin(\frac{1}{2}\pi\nu)\,\Gamma(\frac{1}{2}\nu)\Gamma(1+\frac{1}{2}\nu)}. \tag{4.37}$$

Finally, we combine Eqs. (4.33)–(4.35) and obtain

$$J(y)=e^{-1/3\epsilon}\epsilon^{-1/2}\exp[-y^2 f(0)/2]/u(0), \tag{4.38}$$

with $u(0)$ and $f(0)/2$ given above.

D. Computation of ImE

To calculate ImE, we use Eq. (2.5) and proceed to evaluate the integrals in the denominator and numerator in turn.

The integral in the denominator is done by replacing ψ with ψ_I in Eq. (4.6) and allowing V, the region of integration, to be unbounded. This is a good approximation because the dominant contribution comes from Region I. We obtain

$$\int_V \psi^*\psi\,dV\sim 2\pi. \tag{4.39}$$

The integral in the numerator is a surface integral which reduces to an integral over y. We allow the end points of the integral to be unbounded, use Eq. (4.38), and find that

$$\int_{S_V}\vec{J}\cdot d\vec{S}\sim\frac{e^{-1/3\epsilon}\epsilon^{-1/2}}{u(0)}\int_{-\infty}^{\infty}dy\,e^{-y^2 f(0)/2} \sim e^{-1/3\epsilon}\left[\frac{2\pi}{\epsilon f(0)u^2(0)}\right]^{1/2}. \tag{4.40}$$

After using Eqs. (4.36) and (4.37), Eq. (4.40) simplifies drastically. From this result and Eqs. (2.5) and (4.39) we have finally

$$\text{Im}E=2\sqrt{2}\,e^{-1/3\epsilon}[\nu(\nu+1)/\epsilon\sin(\pi\nu)]^{1/2}. \tag{4.41}$$

We have multiplied by an extra factor of 4 to obtain Eq. (4.41) because the contributions to the integral for ImE come from four equal tubes. As much current flows out along the positive x axis as along the negative x axis and symmetrically along the y axis in both directions.

E. Perturbation Theory in Large Order

From Eqs. (2.4) and (4.41), we have for large n

$$A_n=\left[\frac{\nu(\nu+1)\pi}{\sin(\pi\nu)}\right]^{1/2}\times\frac{2\sqrt{2}}{\pi^{3/2}}\int_0^\infty d\epsilon\,e^{-1/3\epsilon}\epsilon^{-n-3/2}(-1)^{n+1} =2\left[\frac{2c\pi}{\sin(\pi\nu)}\right]^{1/2}\frac{\sqrt{6}}{\pi^{3/2}}(-1)^{n+1}3^n\Gamma(n+\tfrac{1}{2}). \tag{4.42a}$$

Equation (4.42a) is valid for $c<1$, $\nu(\nu+1)=2c$.

When $c>1$, we use Eqs. (1.3) and (4.4) to obtain

$$A_n \sim 2\left[\frac{2\bar{c}\pi}{\sin(\pi\bar{\nu})}\right]^{1/2}\frac{\sqrt{6}}{\pi^{3/2}}(-1)^{n+1}\left(\frac{3c+6}{2}\right)^n\Gamma(n+\tfrac{1}{2}), \tag{4.42b}$$

where $\bar{\nu}(\bar{\nu}+1)=2\bar{c}=(3-c)/(1+c)$.

For $c=1$, we cite the result in Eq. (A18), to wit:

$$A_n \sim (-1)^{n+1}\frac{6}{\pi}3^n\Gamma(n+1). \tag{4.42c}$$

F. Discussion

We can immediately verify Eq. (4.42a) for the case $c=0\Rightarrow\nu=0$. As $\nu\to 0$, the quantity in square brackets approaches unity and we obtain exactly twice the result in Eq. (1.8). This is because now we have two uncoupled anharmonic oscillators.

The other and more interesting limit to investigate is $c\to 1\Rightarrow\nu\to 1$. Now the term in square brackets blows up. This singularity corresponds with the onset of spherical symmetry. Recall that our analytical procedures necessarily break down for that case because there are no isolated MPEP's—all radial paths are equally probable. The sudden increase from four to an infinite number of MPEP's allows the probability current to escape to infinity faster, reducing the lifetime of the unstable state. We thus observe a constructive interference phenomenon which causes an enhancement in the rate of divergence of perturbation theory—A_n in Eq. (4.42c) is larger than A_n in Eqs. (4.42a) and (4.42b) by a factor of $n^{1/2}$.

One encounters many similar phenomena in optics. Consider, for example, a light beam parallel to the z axis and incident upon a flat elliptical plate centered about and perpendicular to the z axis. We use ray tracing to determine the amplitude of the scattered wave at a point on the z axis behind the ellipse. Only two rays scattering off the edge of the ellipse, namely, those at the ends of the minor axes, contribute appreciably to the amplitude. However, when the lengths of the major and minor axes become equal, the scattering amplitude suffers a discontinuous jump because of constructive interference. All rays scattering off the edge of the now circular disk contribute equally to the scattered wave.

We describe the numerical verification of Eq. (4.42) in Sec. V.

V. COMPARISON WITH NUMERICAL DATA

In this section we verify the theoretical predictions in Eqs. (4.42a) and (4.42b). We have run a computer program which calculates A_n to 55th order in perturbation theory by solving a difference equation. It then fits (to six significant figures) the raw Rayleigh-Schrödinger coefficients to a formula similar to but slightly less general than that in Eq. (3.5):

$$A_n = -\frac{\sqrt{6}}{\pi^{3/2}}\alpha(-3\beta)^n\Gamma(n+\tfrac{1}{2})\left(1+\frac{\gamma_1}{n}+\frac{\gamma_2}{n^2}+\frac{\gamma_3}{n^3}+\frac{\gamma_4}{n^4}\right). \tag{5.1}$$

The relevant numerical techniques are discussed in BW I, Appendixes D and E.

The actual difference equation that was solved is

$$\begin{aligned}(2i+2j)C_{n,i,j} &= (i+1)(2i+1)C_{n,i+1,j}\\ &\quad +(j+1)(2j+1)C_{n,i,j+1}\\ &\quad +aC_{n-1,i-2,j}+bC_{n-1,i,j-2}\\ &\quad +2cC_{n-1,i-1,j-1}-\sum_{k=1}^{n-1}D_{n-k}C_{k,i,j},\end{aligned} \tag{5.2}$$

where

$$C_{n,1,0}+C_{n,0,1}=D_n=(-1)^{n+1}A_n, \tag{5.3}$$

and

$$C_{0,0,0}=1,$$

$$C_{n,0,0}=0 \quad \text{for } n>0,$$

$$C_{n,i,j}=0 \quad \text{for } i+j>2n,$$

$$C_{n,i,j}=0 \quad \text{for } i<0 \text{ or } j<0.$$

Equation (5.2) is derived by substituting

$$E(\lambda)=1-\sum_{n=1}^{\infty}D_n(-\lambda)^n, \tag{5.4}$$

$$\psi(x)=e^{-(x^2+y^2)/4}\left\{1+\sum_{n=1}^{\infty}B_n(x,y)(-\lambda)^n\right\}, \tag{5.5}$$

$$B_n(x,y)=\sum_{i,j=0}^{2n}(\tfrac{1}{2}x^2)^i(\tfrac{1}{2}y^2)^jC_{n,i,j} \tag{5.6}$$

into Eq. (1.1) and collecting powers of $(\frac{1}{2}x^2)^i$, $(\frac{1}{2}y^2)^j$, and λ^n.

The numerical predictions for α and β for various values of a, b, and c are given in Table I. The predictions in Eq. (4.42) are as follows: For $-1\leq c<1$, $a=b=1$,

$$\begin{aligned}\alpha &= \{-8\pi c/\cos[\tfrac{1}{2}\pi(1+8c)^{1/2}]\}^{1/2},\\ \beta &= 1,\end{aligned} \tag{5.7}$$

where we have eliminated ν in favor of c using Eq. (4.20). For $c>1$, $a=b=1$,

$$\begin{aligned}\alpha &= \left\{\frac{8\pi(c-3)}{(1+c)\cos[\pi(25-7c)^{1/2}(4+4c)^{-1/2}]}\right\}^{1/2},\\ \beta &= (c+1)/2,\end{aligned} \tag{5.8}$$

where we have used Eq. (4.4). For $a=b=1$ the values of α and β in Table I agree to six places with

TABLE I. Numerical values of α and β in Eq. (5.1) for various values of a, b, and c in Eq. (1.1). The theoretical predictions in Eqs. (4.42a) and (4.42b) and the numerical calculations of α and β agree to six figures, which was the available limit of computer accuracy. Some values of α were not computed.

a	b	c	α	β
1	1	−1	···	1.000 00
1	1	−0.5	···	1.000 00
1	1	−0.25	1.582 42	1.000 00
1	1	$-\frac{1}{6}$	1.705 41	1.000 00
1	1	−0.05	1.903 78	1.000 00
1	1	−0.005	1.990 02	1.000 00
1	1	0	2.000 00	1.000 00
1	1	0.005	2.010 02	1.000 00
1	1	0.05	2.104 10	1.000 00
1	1	0.1	···	1.000 00
1	1	$\frac{1}{6}$	2.383 99	1.000 00
1	1	0.25	2.623 72	1.000 00
1	1	$\frac{1}{3}$	2.907 40	1.000 00
1	1	0.5	3.672 06	1.000 00
1	1	1.5	4.338 36	1.250 00
1	1	2	2.907 40	1.500 00
1	1	2.5	2.322 11	1.750 00
1	1	3	2.000 00	2.000 00
1	1	5	1.472 28	3.000 00
1	1	33	···	17.0000
1	2	0.25	···	2.000 00
1	2	1	···	2.000 00
1	2	5	···	3.285 74
1	3	0.25	···	3.000 00
1	3	1	···	3.000 00
1	3	5	···	3.666 67
1	5	0.25	···	5.000 00
1	5	1	···	5.000 00

the expressions in Eqs. (5.7) and (5.8). Note that when the argument of the cosine function becomes imaginary, cos is replaced by cosh. The function in the curly brackets is always positive.

We have done one further and rather amusing numerical calculation which does not appear in Table I. We computed A_n for $c=-5$. This problem has no apparent physical significance because the Hamiltonian is not bounded below, and therefore has no discrete eigenvalues. Nevertheless, the perturbation series is still well defined and we found that $\beta=-2.000\,00$. This result agrees with β in Eq. (5.8).

In Sec. VI we obtain theoretical values for α and β when $a, b \neq 1$ and compare these with the data in Table I.

VI. THE CASE $a, b \neq 1$

The generalization of the discussion of Secs. III and IV to the case where $a, b \neq 1$ in Eq. (1.1) is entirely straightforward. As before, we find the saddle points of V by solving

$$\frac{\partial V}{\partial x}=\tfrac{1}{2}x-\epsilon(ax^3+cxy^2)=0,$$

$$\frac{\partial V}{\partial y}=\tfrac{1}{2}y-\epsilon(by^3+cx^2y)=0$$

and requiring that the Hessian matrix

$$H=\begin{bmatrix} \frac{1}{2}-\epsilon(3ax^2+cy^2) & -2cxy\epsilon \\ -2cxy\epsilon & \frac{1}{2}-\epsilon(3by^2+cx^2) \end{bmatrix}$$

have negative determinant.

The critical points are $(0,0)$, $(0,\pm(2b\epsilon)^{-1/2})$, $(\pm(2a\epsilon)^{-1/2},0)$, and $\pm[2\epsilon(c^2-ab)]^{-1/2}((c-b)^{1/2}, \pm(c-a)^{1/2})$. $(0,\pm(2b\epsilon)^{-1/2})$ are saddle points if $c/b<1$ and $(\pm(2a\epsilon)^{-1/2},0)$ are saddle points if $c/a<1$. If $c>\max(a,b)$ then the off-axis critical points are saddle points. (Recall that for the Hamiltonian to be bounded below we must have $a\geq 0$, $b\geq 0$, and $c\geq -\sqrt{ab}$.) All of the saddle points are radially directed.

As in Sec. IV, the off-axis case can be reduced to the case $c/a<1$ by a rotation. To simplify the algebra we introduce the following notation:

$$D=ab-c^2=\mathrm{Det}\begin{vmatrix} a & c \\ c & b \end{vmatrix}, \tag{6.1a}$$

$$D_1=b-c=\mathrm{Det}\begin{vmatrix} 1 & c \\ 1 & b \end{vmatrix}, \tag{6.1b}$$

$$D_2=a-c=\mathrm{Det}\begin{vmatrix} a & 1 \\ c & 1 \end{vmatrix}, \tag{6.1c}$$

$$S=D_1+D_2=a+b-2c. \tag{6.1d}$$

Then a suitable rotation is

$$\begin{aligned} x&=(D_1/S)^{1/2}\bar{x}\pm(D_2/S)^{1/2}\bar{y}, \\ y&=\mp(D_2/S)^{1/2}\bar{x}+(D_1/S)^{1/2}\bar{y}. \end{aligned} \tag{6.2}$$

In terms of the new variables the potential is

$$V=\tfrac{1}{4}(\bar{x}^2+\bar{y}^2)-\tfrac{1}{4}\epsilon(\bar{a}\bar{x}^4+\bar{b}\bar{y}^4+2\bar{c}\,\bar{x}^2\bar{y}^2), \tag{6.3}$$

where

$$\begin{aligned} \bar{a}&=D/S, \\ \bar{b}&=[(D_1-D_2)^2+D]/S, \\ \bar{c}&=3D/S-2c. \end{aligned} \tag{6.4}$$

Observe that

$$\bar{c}-\bar{a}=2D/S-2c=\frac{2D_1D_2}{S}<0,$$

when $a>0$, $b>0$, and $c>\max(a,b)$.

Without loss of generality, then, we assume $c<a$. We will solve the Schrödinger equation in a narrow tube surrounding the x axis. In Region I where $0\leq x<\epsilon^{-1/4}$ we have

$$\psi_{\mathrm{I}}\sim e^{-(x^2+y^2)/4}. \tag{6.5}$$

In Region II, $\epsilon^{-1/8} < x < \epsilon^{-1/2}$ and we can approximate the differential equation by

$$\left[-\frac{\partial^2}{\partial x^2} - \frac{\partial^2}{\partial y^2} + \tfrac{1}{4}x^2 - \tfrac{1}{4}\epsilon(ax^4 + 2cx^2y^2) - E \right] \psi_{\rm II} = 0 . \tag{6.6}$$

The substitutions

$$\begin{aligned} a\epsilon &\to \epsilon , \\ c/a &\to c \end{aligned} \tag{6.7}$$

reduce Eq. (6.6) to Eq. (4.7), which has already been solved.

We therefore immediately deduce that the large-order behavior of perturbation theory is

$$A_n = -\frac{\sqrt{6}}{\pi^{3/2}} \alpha (-3\beta)^n \, \Gamma(n + \tfrac{1}{2}) , \tag{6.8}$$

where

$$\begin{aligned} \alpha &= \left\{ \frac{8\pi(2ac + 2bc - 3ab - c^2)}{(ab - c^2)\cos[\tfrac{1}{2}\pi(25 - 16c(a+b-2c)/(ab-c^2))^{1/2}]} \right\}^{1/2} , \\ \beta &= \frac{ab - c^2}{a + b - 2c} , \end{aligned} \tag{6.11}$$

where $a > 0$, $b > 0$, $c > \max(a, b)$.

The results in Eqs. (6.9)–(6.11) agree to six places with the numbers in Table I.

$$\begin{aligned} \alpha &= \left\{ \frac{-2\pi c}{a \cos[\tfrac{1}{2}\pi(1 + 8c/a)^{1/2}]} \right\}^{1/2} , \\ \beta &= a , \end{aligned} \tag{6.9}$$

for the case $a > b > 0$, $a > c \geq -\sqrt{ab}$.

There is a factor of 2 missing from the expression for α in Eq. (6.9) relative to Eq. (5.7) because the contribution from the tube along the y axis is negligible when $a > b$. When $b > a$, we have similar results:

$$\begin{aligned} \alpha &= \left\{ \frac{-2\pi c}{b \cos[\tfrac{1}{2}\pi(1 + 8c/b)^{1/2}]} \right\}^{1/2} , \\ \beta &= b , \end{aligned} \tag{6.10}$$

where $b > a > 0$, $b > c \geq -\sqrt{ab}$.

For the off-axis case we use Eq. (6.4) to deduce that

VII. GENERALIZATION TO N DIMENSIONS

It is natural to try to extend the techniques we have developed for two-mode oscillators to more complicated problems. In this section we will show that such an extension is possible for a large class of N-mode equal-mass oscillator systems. Our aim will be to present a brief overview of what can be accomplished in N dimensions, and we will not dwell on algebraic or numerical details.

We begin by studying systems having potentials of the form

$$V = \sum_{i=1}^{N} \tfrac{1}{4} x_i^{\,2} + \tfrac{1}{4}\lambda \sum_{i,j=1}^{N} a_{ij} \, x_i^{\,2} x_j^{\,2} , \tag{7.1}$$

where a_{ij} is a real symmetric matrix. The Hamiltonian in (7.1) must be bounded below, and therefore a must satisfy

$$\sum_{i,j} a_{ij} \, y_i \, y_j \geq 0 \text{ for } y_i \geq 0 . \tag{7.2}$$

When $\lambda = -\epsilon (\epsilon > 0)$ the critical points of V are the solutions of

$$\begin{aligned} 0 &= \frac{\partial V}{\partial x_i} \\ &= \tfrac{1}{2} x_i - \epsilon x_i \sum_{j=1}^{N} a_{ij} \, x_j^{\,2} . \end{aligned} \tag{7.3}$$

Thus

$$x_i = 0 ,$$

or

$$\frac{1}{2\epsilon} = \sum_{j=1}^{N} a_{ij} \, x_j^{\,2} . \tag{7.4}$$

We will first consider the case where all $x_i \neq 0$. Then Eq. (7.4) has a solution if a is nonsingular. We define D_i to be the determinant of the matrix obtained from a by replacing each element of the ith column by one. Then $\sum_j a_{ij} D_j = \det(a)$ for all i, and

$$x_j^{\,2} = \frac{1}{2\epsilon} \frac{D_j}{\det(a)} . \tag{7.5}$$

If this is to correspond to a point in real space we must have

$$\frac{D_i}{\det(a)} > 0 , \quad 1 \leq i \leq N . \tag{7.6}$$

Of course there are 2^{N-1} vectors which satisfy Eq. (7.5) because we can choose the sign of each component of x_i independently.

The Hessian matrix at the critical point is

$$H_{ij} = \tfrac{1}{2}\delta_{ij} - \epsilon \delta_{ij} \sum_k a_{ik} \, x_k^{\,2} - 2\epsilon x_i \, x_j a_{ij} . \tag{7.7}$$

Using Eq. (7.5) we can rewrite this as

$$H_{ij} = -2\epsilon x_i x_j a_{ij}$$

$$= -\frac{\sigma_i \sigma_j}{\det(a)} (D_i D_j)^{1/2} a_{ij}, \tag{7.8}$$

where σ_i is the sign of x_i. Note that $D_i D_j$ is always positive [see Eq. (7.6)]. Equations (7.3) and (7.8) imply that

$$\sum_j H_{ij} \sigma_j (D_j)^{1/2} = -\sigma_i (D_i)^{1/2}, \tag{7.9}$$

which means that the radial line through each critical point is one of the critical point's principal axes. Furthermore, the minus sign in Eq. (7.9) implies that along this radial line, V has a maximum at the critical point. Thus if all the other eigenvalues of H at the critical point are positive, we have a radially directed saddle point.

A necessary condition for H to have one negative and $N-1$ positive eigenvalues is

$$\det H < 0. \tag{7.10}$$

But,

$$\det H = [-\det(a)]^{-N} \det[\sigma_i \sigma_j (D_i D_j)^{1/2} a_{ij}]$$

$$= -[-\det(a)]^{-N-1} \prod_{i=1}^{N} D_i.$$

Using Eq. (7.6) we see that

$$\det H < 0 \Longleftrightarrow (-1)^N \det(a) < 0. \tag{7.11}$$

In two dimensions, Eq. (7.11) is indeed satisfied by the off-axis saddle points that we discussed in Sec. VI. In fact this condition and Eq. (7.6) imply that $c > \max(a, b)$, which we have shown to be a necessary and *sufficient* condition for an off-axis saddle point in two dimensions. In higher dimensions however, we can have $\det H < 0$ without having one negative and $N-1$ positive eigenvalues. It is difficult to give a simple necessary and sufficient condition for saddle points in the general N-dimensional case.

If several of the x_i are zero the procedure for finding a saddle point is slightly more complicated. We choose to label the axes so that the first M x_i are zero. Then, the condition for a critical point becomes

$$x_i = 0, \quad i = 1, \ldots, M$$

$$\frac{1}{2\epsilon} = \sum_{j=M+1}^{N} a_{ij} x_j^2, \quad i = M+1, \ldots, N. \tag{7.12}$$

The discussion proceeds as before in the subspace of nonzero components. We find that the Hessian matrix is given by

$$\begin{bmatrix} \frac{1}{2} - \epsilon \sum_{k=M+1}^{N} a_{1k} x_k^2 & \cdots & 0 & 0 \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \cdots & \frac{1}{2} - \epsilon \sum_{k=M+1}^{N} a_{Mk} x_k^2 & 0 \\ 0 & \cdots & 0 & -2\epsilon x_i x_j a_{ij} \end{bmatrix},$$

where the x_i's are the solutions of Eq. (7.12). Thus, in addition to the usual conditions on the N-M dimensional matrix $x_i x_j a_{ij}$, we must have

$$\frac{1}{2} - \epsilon \sum_{M+1}^{N} a_{ik} x_k^2 > 0, \quad i = 1, \ldots, M.$$

The special case of an on-axis critical point, where all but one of the x_i vanish, is important because any other configuration can be reduced to this one by a rotation. Here the condition for a saddle point becomes

$$\frac{1}{2} - \epsilon a_{iN} x_N^2 > 0, \quad i = 1, \ldots, N-1$$

where

$$x_N^2 = \frac{1}{2\epsilon a_{NN}}. \tag{7.13}$$

It is now easy to generalize the arguments of Secs. IV and VI to compute the contribution to the large-order behavior of perturbation theory from this saddle point. We observe that in the tube where $\epsilon^{-1/6} < x_N < \epsilon^{-1/2}$, $x_i = O(1)$, $i \neq N$, we can make the approximation

$$\frac{1}{4}\epsilon \sum_{i,j} a_{ij} x_i^2 x_j^2 \sim \frac{1}{4}\epsilon x_N^2 \sum_{j=1}^{N-1} a_{Nj} x_j^2 + \frac{1}{4}\epsilon a_{NN} x_N^4.$$

Following the procedure of Sec. IV, we factor a rapidly varying WKB function of the x_N variable [see Eq. (4.8)] out of the wave function and neglect all terms in the resulting equation which vanish as $\epsilon \to 0$. After a simple change of variables [see Eq. (4.12)], we obtain a partial differential equation which must be solved exactly:

$$(w^2 - 1)\chi_w + \sum_{i=1}^{N-1} \left[-\frac{\partial^2}{\partial x_i^2} + \frac{1}{4} x_i^2 - \frac{1}{2} - \frac{a_{iN}}{2a_{NN}} x_i^2 (1 - w^2) \right] \chi = 0.$$

The ansatz

$$\chi = A\exp\left[-\tfrac{1}{4}\sum_{i=1}^{N-1} f_i(w)x_i^2\right]$$

generates $N-1$ Riccati equations whose solutions govern the thickness of the tube of probability current in the directions perpendicular to the MPEP. Then the change of variables

$$w = w\,,$$

$$s_i = x_i/u_i(w)\,, \quad 1 \leq i \leq N-1$$

with

$$f_i = (w^2-1)u_i'(w)/u_i(w)$$

reduces the equation for A to one that is separable:

$$(w^2-1)A_w + \tfrac{1}{2}\sum_{i=1}^{N-1}[f_i(w)-1]A = \sum_{i=1}^{N-1}\frac{1}{u_i^2(w)}\frac{\partial^2}{\partial x_i^2}A\,.$$

Finally, we require that the wave function in the tunneling region match to a harmonic-oscillator wave function (the solution of the Schrödinger equation near the origin). As in Sec. IV, this implies that A is a function only of w. It is then easy to determine the probability current and evaluate the dispersion integral. The resulting contribution to the large-order behavior of perturbation theory is

$$A_n = -\left\{\prod_{i=1}^{N-1}\left[\frac{8\pi a_{iN}}{a_{NN}\sin(\pi\nu_i)}\right]^{1/2}\right\} \times\sqrt{6}\,\pi^{-3/2}(-3a_{NN})^n\,\Gamma(n+\tfrac{1}{2})\,, \tag{7.14}$$

where $\nu_i(\nu_i+1) = 2a_{iN}/a_{NN}$.

This expression will be equal to the true large-order behavior if the x_N axis is the MPEP. However, as we mentioned above, we can use Eq. (7.14) even if the dominant saddle point does not lie on this axis. As an example, let us consider the case of a dominant critical point whose coordinates are nonvanishing. Then the following rotation will align the radial line through the saddle point with the $\bar{x}_N$ axis:

$$x_i = \sum_{j=1}^{N} R_{ij}\bar{x}_j\,, \tag{7.15}$$

$$R = \begin{bmatrix} a_{11} & \cdots & a_{N-1\,1} & \left(D_1/\sum_{i=1}^{N} D_i\right)^{1/2} \\ \vdots & & \vdots & \vdots \\ a_{1N} & \cdots & a_{N-1\,N} & \left(D_N/\sum_{i=1}^{N} D_i\right)^{1/2} \end{bmatrix}.$$

We choose the first $N-1$ column vectors of R to lie along the other principal axes of the saddle point. In the new coordinate system V will no longer have the simple form in Eq. (7.1). It will contain terms like $\bar{x}_i^3\bar{x}_j$. However, it is easy to see that there are no terms of the form $\bar{x}_N^3\bar{x}_j$ $(j\neq N)$ or $\bar{x}_N^2\bar{x}_i\bar{x}_j$ $(i,j\neq N)$. Such terms would give a nonvanishing contribution to $\partial^2 V/\partial\bar{x}_N\partial\bar{x}_j$ or to $\partial^2 V/\partial\bar{x}_i\partial\bar{x}_j$ at the critical point. (The coordinates of the critical point are $\bar{x}_j = 0$, $1\leq j\leq N-1$, $\bar{x}_N\neq 0$.) Thus V has the form

$$V = \sum_{i=1}^{N}\tfrac{1}{4}\bar{x}_i^2 - \tfrac{1}{4}\epsilon\left(\bar{a}_{NN}\bar{x}_N^4 + \sum_{i=1}^{N-1}\bar{a}_{iN}x_i^2x_N^2 + b\right),$$

where b depends at most linearly on $\bar{x}_N$. In the tube where $\bar{x}_i = O(1)$, $1\leq i\leq N-1$, and $\epsilon^{-1/6}\leq\bar{x}_N\leq\epsilon^{-1/2}$, we can clearly approximate V by neglecting b entirely.

We then use Eq. (7.14) to compute the large-order behavior of perturbation theory as before. The computation of $\bar{a}_{iN}$ for all i is a tedious algebraic problem. Still it is easy to find the value of $\bar{a}_{NN}$, the coefficient of $\bar{x}_N^4$ in $\sum_{i,j}a_{ij}x_i^2x_j^2$. Using Eq. (7.15) it is

$$\begin{aligned}\bar{a}_{NN} &= \sum_{i,j} a_{ij}\frac{D_iD_j}{(\sum_j D_j)^2} \\ &= \frac{1}{(\sum_j D_j)^2}\sum_j D_j\det(a) \\ &= \frac{\det(a)}{\sum_j D_j}\,.\end{aligned}$$

Thus, the large-order behavior of perturbation theory is

$$A_n \sim -\sqrt{6}\,\pi^{-3/2}2^{N-1}K_N\left[\frac{-3\det(a)}{\sum_j D_j}\right]^n\Gamma(n+\tfrac{1}{2})\,. \tag{7.16}$$

The constant K_N may be determined from Eq. (7.14) once we have computed $\bar{a}_{iN}$ for $i\neq N$. The factor 2^{N-1} reflects the possible choices of the sign of x_j in Eq. (7.5). Similar formulas exist for the case where some of the x_i vanish at the dominant saddle point.

Equation (7.14) may also be used to find the large-order behavior of the perturbation series for systems having an infinite number of degrees of freedom. As an example, consider the sequence of potentials

$$V^{(N)} = \tfrac{1}{4}\sum_{i=1}^{N}x_i^2 - \tfrac{1}{4}\epsilon\sum_{\substack{i,j=1\\ i\neq j}}^{N}x_i^2x_j^2\,. \tag{7.17}$$

The critical points of $V^{(N)}$ are given by

$$0 = \frac{\partial V^{(N)}}{\partial x_i} = \tfrac{1}{2}x_i - \tfrac{1}{4}\epsilon x_i\sum_{j\neq i}x_j^2\,, \tag{7.18}$$

whose solutions are

$$x_i = 0 \text{ or } x_i{}^2 = [2\epsilon(M-1)]^{-1}. \tag{7.19}$$

M is the number of nonzero x_i. Note that M can never be one, so Eq. (7.19) always makes sense. The Hessian matrix is

$$H_{ij}^{(N)} = -\frac{\delta_{ij}}{2(M-1)} - 2\epsilon x_i x_j + 3\epsilon\delta_{ij}(x_i{}^2). \tag{7.20}$$

Observe from Eq. (7.20) that we cannot have a saddle point if any of the x_i vanish because Eq. (7.13) is violated. Hence, all $x_i \neq 0$ and

$$H_{ij}^{(N)} = \frac{\delta_{ij}}{2(N-1)} - \frac{\sigma_i\sigma_j}{(N-1)},$$

where σ_i is the sign of x_i. The eigenvalues of this matrix are $\frac{1}{2} - N/(N-1)$, which is negative, and $2/(N-1)$, the latter having multiplicity $N-1$. Therefore, we have a saddle point. There is a different saddle point for each of the 2^{N-1} choices of sign for x_i.

To determine the large-order behavior of perturbation theory for this system, we must compute $D_i^{(N)}$ and $\det[a^{(N)}]$ [see Eq. (7.16)]. The matrix $a^{(N)}$ is given by

$$\begin{bmatrix} 0 & 1 & 1 & 1 & \cdots \\ 1 & 0 & 1 & 1 & \cdots \\ 1 & 1 & 0 & 1 & \cdots \\ 1 & 1 & 1 & 0 & \\ \cdots & \cdots & \cdots & & \ddots \end{bmatrix}.$$

Thus,

$$\det[a^{(N)}] = (N-1)(-1)^{N-1}$$

and

$$D_i^{(N)} = (-1)^{N-1}, \quad 1 \leq i \leq N.$$

We can also compute $\bar{a}_{iN}^{(N)}$ because the potential is so symmetric. We find that

$$\begin{aligned} \bar{a}_{iN}^{(N)} &= \frac{N-3}{N}, \quad 1 \leq i \leq N-1 \\ \bar{a}_{NN}^{(N)} &= \frac{N-1}{N}. \end{aligned} \tag{7.21}$$

Therefore, from Eqs. (7.14) and (7.21), the large-order behavior of perturbation theory is

$$A_n^{(N)} = -\sqrt{6}\,\pi^{-3/2}\left[\frac{-8\pi(N-3)/(N-1)}{\cos\frac{3}{2}\pi(1-16/9(N-1))^{1/2}}\right]^{(N-1)/2}\left(-3\frac{N-1}{N}\right)^n\Gamma(n+\tfrac{1}{2}). \tag{7.22}$$

Now consider the limit as $N\to\infty$. This limit defines an infinite-mode oscillator system which, strictly speaking, is some nonlocal field theory. The leading contribution to Eq. (7.22) which comes from geometrical optics remains finite in this limit:

$$\left(-3\frac{N-1}{N}\right)^n\Gamma(n+\tfrac{1}{2}) \to (-3)^n\Gamma(n+\tfrac{1}{2}).$$

However, the constant from physical optics blows up. The divergence of this constant derives from two sources. The factor $(\sqrt{8\pi})^{N-1}$ occurs in any N-mode problem in which no axis passes through the dominant saddle point. The vanishing of the cosine term as $N\to\infty$ is a more singular divergence of the form $N^{N/2}$. It reflects the disappearance of the saddle point. The extreme symmetry of the potential makes the saddle become flat as N becomes large. This kind of symmetry is not present in potentials arising from $(\phi^N)_2$ quantum field theories.

It is amusing that we can eliminate this divergence by a mass renormalization. We will argue that by adding a lower-order mass term to the potential we can ensure that the $N\to\infty$ limit of $A_n^{(N)}$ exists. Consider the effect of adding a term of the form

$$\epsilon R^{(N)}\sum_{i=1}^{N}\tfrac{1}{4}x_i{}^2.$$

Because this term is at most $O(1)$ in the tunneling region, it cannot affect the determination of the MPEP. This term is merely a correction of $O(\epsilon)$ to the mass, and thus can actually be viewed as a mass renormalization. Following the scaling arguments of Sec. VII, Ref. 9, we find that this term contributes an over-all multiplicative constant (independent of n) to the large-order growth of perturbation theory, namely,

$$\exp\left[\frac{-NR^{(N)}}{2(N-1)}\right].$$

On the other hand, we easily determine from Eq. (7.22) that the two large-N divergences which we discussed give precisely

$$[6(N-3)]^{(N-1)/2}.$$

Hence, if we choose

$$R^{(N)} = \frac{(N-1)^2}{N}\ln(6N-18), \tag{7.23}$$

then the limit $N\to\infty$ may be taken. We obtain the remarkable result that the large-order behavior of the Rayleigh-Schrödinger coefficients A_n for the

ground-state energy perturbation series of the infinite-mode system described by the potential

$$V=\lim_{N\to\infty}\left\{\tfrac{1}{4}[1-\lambda R^{(N)}]\sum_{i=1}^{N}x_i{}^2+\tfrac{1}{4}\lambda\sum_{\substack{i,j=1\\ i\neq j}}^{\infty}x_i{}^2x_j{}^2\right\} \tag{7.24}$$

is

$$A_n\sim-\sqrt{6}\,\pi^{-3/2}(-3)^n\,\Gamma(n+\tfrac{1}{2}), \tag{7.25}$$

where $N\to\infty$ in such a way that λN is small. This is precisely the result in Eq. (1.8) for the one-mode anharmonic oscillator. Of course, it can be argued that the choice of the potential in Eq. (7.24) is somewhat contrived. Nonetheless, we feel that Eq. (7.25) is just one more example of the extraordinary simplification that takes place in the large-order limit of perturbation theory.

APPENDIX A: OSCILLATORS WITH SPHERICAL SYMMETRY

Here we investigate the ground-state-energy perturbation series for spherically coupled oscillators. This special case obtains when we choose $c=1$ in Eq. (1.4). However, a spherically symmetric configuration of oscillators is so easy to treat that we immediately generalize from the two-mode problem of Eq. (1.4) to the N-mode problem, which we define by the equation

$$\left\{\sum_{i=1}^{N}\left(-\frac{\partial^2}{\partial x_i{}^2}+\tfrac{1}{4}x_i{}^2\right)+\tfrac{1}{4}\lambda\left(\sum_{i=1}^{N}x_i{}^2\right)^2-E(\lambda)\right\}\psi(x_i)=0, \tag{A1}$$

where $\lim_{|x_i|\to\infty}\psi=0$.

We use spherical symmetry to transform Eq. (A1) to N-dimensional spherical coordinates. Moreover, we seek a wave function ψ which depends only on r, the radial coordinate, because the ground-state wave function has no angular dependence. We thus reduce Eq. (A1) to the ordinary differential equation

$$\left[-\frac{d^2}{dr^2}-\frac{N-1}{r}\frac{d}{dr}+\tfrac{1}{4}r^2+\tfrac{1}{4}\lambda r^4-E(\lambda)\right]\psi(r)=0, \tag{A2}$$

where

$$r^2=\sum_{i=1}^{N}x_i{}^2.$$

Our problem is to compute the eigenvalue $E(\lambda)$ perturbatively in large order. We expand $E(\lambda)$ into the perturbation series

$$E(\lambda)=\tfrac{1}{2}N-\sum_{n=1}^{\infty}(-\lambda)^nC_n. \tag{A3}$$

We could solve this problem by removing the first derivative term from Eq. (A2) by making a suitable transformation and then using WKB in the same manner as in the body of this paper. However, we prefer a much simpler approach. We will convert Eq. (A2) into a partial difference equation which has already been solved asymptotically in BW II.

To transform Eq. (A2) into a partial difference equation, we substitute the expression

$$\psi(r)=e^{-r^2/4}\left\{1+\sum_{n=1}^{\infty}(-\lambda)^n\sum_{j=1}^{2n}(\tfrac{1}{2}r^2)^jC_{n,j}\right\} \tag{A4}$$

and Eq. (A3) into Eq. (A2) and collect powers of $r^2/2$ and $-\lambda$. The coefficient of $(\frac{1}{2}r^2)^j$ and $(-\lambda)^n$ is the desired difference equation:

$$2jC_{n,j}=(j+1)(2j+N)C_{n,j+1}+C_{n-1,j-2}-\sum_{p=1}^{n-1}C_{p,1}C_{n-p,j}, \tag{A5}$$

with initial value $C_{0,0}=1$ and boundary condition $C_{n,j}\neq0$ for $n>1$ and $1\leq j\leq 2n$; $C_{n,j}=0$ otherwise. C_n is related to $C_{n,j}$ by

$$C_n=NC_{n,1}. \tag{A6}$$

Following Sec. VI A of BW II, we approximate Eq. (A5) by dropping the nonlinear convolution term. As was argued there, the neglected term does not affect the leading asymptotic behavior of $C_{n,1}$ for large n. Thus, the equation to be solved is

$$2jC_{n,j}=(j+1)(2j+N)C_{n,j+1}+C_{n-1,j-2}. \tag{A7}$$

We put Eq. (A7) into a more useful form by substituting

$$C_{n,j}=D_{n,j}/[j\Gamma(j+\tfrac{1}{2}N)]. \tag{A8}$$

The equation satisfied by $D_{n,j}$ is

$$D_{n,j}=D_{n,j+1}+\frac{(j+\frac{1}{2}N-1)(j+\frac{1}{2}N-2)}{2(j-2)}D_{n-1,j-2}. \tag{A9}$$

Next, we replace Eq. (A9) by a new approximate equation satisfied by a new dependent variable $E_{n,j}$:

$$E_{n,j}=E_{n,j+1}+\tfrac{1}{2}(j+N-1)E_{n-1,j-2}. \tag{A10}$$

Equation (A10) is derived by approximating the coefficient of $D_{n-1,j-2}$ in Eq. (A9) for large j, keeping terms of orders j and 1 and neglecting terms of order j^{-1}.

We must introduce an extra condition which fixes the multiplicative scale of $E_{n,j}$ because Eq. (A10) is homogeneous; to wit, we require that

$$\lim_{n\to\infty}E_{n,2n}/D_{n,2n}=1. \tag{A11}$$

From Eq. (A10) we easily deduce that

$$E_{n,2n} = E_{0,0}\Gamma(n+\tfrac{1}{2}(N+1))/\Gamma(\tfrac{1}{2}(N+1)). \quad \text{(A12)}$$

$E_{0,0}$ is the multiplicative factor which adjusts the scale of $E_{n,j}$. Also, from Eqs. (A7) and (A8) we have

$$D_{n,2n} = 2n\Gamma(2n+\tfrac{1}{2}N)4^{-n}/n! \,. \quad \text{(A13)}$$

Combining Eqs. (A11)–(A13) gives

$$E_{0,0} = \Gamma(\tfrac{1}{2}(N+1))2^{N/2}\pi^{-1/2}. \quad \text{(A14)}$$

Finally, we recall that Sec. VI H of BW II gives a complete treatment of the asymptotic behavior of solutions for difference equations like that in Eq. (A10). It is shown there that for large n,

$$E_{n,1} \sim E_{0,0}\frac{3^{n+N/2}\Gamma(n+\tfrac{1}{2}N)}{2\sqrt{\pi}\,\Gamma(\tfrac{1}{2}(N+1))}. \quad \text{(A15)}$$

Thus, we combine Eqs. (A6), (A8), and (A15) to obtain

$$C_n \sim \Gamma(n+\tfrac{1}{2}N)3^n 6^{N/2}\pi^{-1}/\Gamma(\tfrac{1}{2}N). \quad \text{(A16)}$$

This is the precise leading asymptotic behavior of C_n for large n and is the general result we have sought.

Two special cases of this equation are noteworthy. For the one-mode oscillator ($N=1$), the coefficients of the ground-state-energy perturbation series grow like

$$C_n \sim 6^{1/2}\pi^{-3/2}3^n\,\Gamma(n+\tfrac{1}{2}), \quad \text{(A17)}$$

which agrees with Eq. (1.8).

Second, the coefficients for the two-mode oscillator ($N=2$) diverge like

$$C_n \sim \frac{2}{\pi}\,3^{n+1}n! \,, \quad \text{(A18)}$$

as in Eqs. (1.11) and (4.42c).

Observe also that the rate of divergence of perturbation theory increases with increasing N: $\Gamma(n+\frac{1}{2}N) \sim \Gamma(n)n^{N/2}$. This is a phenomenon characteristic only of spherically symmetrically coupled oscillators. As is shown in Sec. IV, it results physically from a kind of constructive interference.

APPENDIX B: NONQUARTIC COUPLING

To demonstrate that the analytical methods of Sec. IV are easily generalized to more difficult problems, we treat the two-mode, nonquartically coupled oscillator. This system is defined by the equation

$$\left[-\frac{\partial^2}{\partial x^2}-\frac{\partial^2}{\partial y^2}+\tfrac{1}{4}x^2+\tfrac{1}{4}y^2+\lambda 2^{-N}(x^{2N}+cx^{2N-2}y^2+dx^{2N-4}y^4+\cdots+dx^4y^{2N-4}+cx^2y^{2N-2}+y^{2N})-E(\lambda)\right]\psi=0. \quad \text{(B1)}$$

Corresponding to Eq. (4.1) we have

$$\left[-\frac{\partial^2}{\partial x^2}-\frac{\partial^2}{\partial y^2}+\tfrac{1}{4}x^2+\tfrac{1}{4}y^2-\epsilon^{N-1}2^{-N}(x^{2N}+cx^{2N-2}y^2+\cdots)-1\right]\psi=0. \quad \text{(B2)}$$

As in Sec. IV, we solve Eq. (B2) in a tube surrounding the positive x axis. In Region I we obtain the same result as in Eq. (4.6) and in Region II we make a substitution similar to that in Eq. (4.8):

$$\psi = \chi(\tfrac{1}{4}x^2-\epsilon^{N-1}2^{-N}x^{2N}-\tfrac{1}{2})^{-1/4}\exp\left[-\int_{\sqrt{2}}^{x}(\tfrac{1}{4}t^2-\epsilon^{N-1}2^{-N}t^{2N}-\tfrac{1}{2})^{1/2}dt\right]. \quad \text{(B3)}$$

In the resulting differential equation for χ we let

$$\epsilon x^2 = z^2 \quad \text{(B4)}$$

and obtain

$$z(1-2^{2-N}z^{2N-2})^{1/2}\chi_z - \chi_{yy} + \left(\tfrac{1}{4}y^2-\frac{c}{2^N}z^{2N-2}y^2-\tfrac{1}{2}\right)\chi=0. \quad \text{(B5)}$$

Equation (B5), like Eq. (4.11), must be solved *exactly*—there are no further approximations.

To simplify Eq. (B5), we let

$$\omega = (1-2^{2-N}z^{2N-2})^{1/2}, \quad \text{(B6)}$$

which gives

$$(N-1)(\omega^2-1)\chi_\omega - \chi_{yy} + [\tfrac{1}{4}y^2-\tfrac{1}{2}-\tfrac{1}{4}c(1-\omega^2)y^2]\chi=0. \quad \text{(B7)}$$

The substitution

$$\chi = e^{-y^2 f(\omega)/4}A \quad \text{(B8)}$$

leads to a Riccati equation that is almost identical to that in Eq. (4.15):

$$-(N-1)(\omega^2-1)f'(\omega)-f^2(\omega)+1-c+c\omega^2=0. \quad \text{(B9)}$$

To linearize Eq. (B9), we let

$$f(\omega) = (N-1)(\omega^2-1)u'(\omega)/u(\omega). \quad \text{(B10)}$$

As in Sec. IV, $u(\omega)$ satisfies a Legendre equation:

$$u(\omega)=P_\nu^\mu(\omega), \tag{B11}$$

where $\mu=-1/(N-1)$ and $\nu(\nu+1)=c/(N-1)^2$.

When Eq. (B9) is satisfied, A solves a much simpler equation than Eq. (B7), namely,

$$(N-1)(\omega^2-1)A_\omega-A_{yy}+yf(\omega)A_y+\tfrac{1}{2}[f(\omega)-1]A=0. \tag{B12}$$

The substitution $(\omega, y)\to(\omega, s=y/u(\omega))$ reduces Eq. (B12) to one that is separable:

$$(N-1)(\omega^2-1)u^2(\omega)A_\omega+\tfrac{1}{2}u^2(\omega)[f(\omega)-1]A=A_{ss}. \tag{B13}$$

The argument given in Sec. IV that the separation constant for Eq. (4.23) vanishes also holds for Eq. (B13). It follows that, up to an arbitrary multiplicative constant β,

$$A=\beta[u(\omega)]^{-1/2}[(1-\omega)/(1+\omega)]^{1/(4N-4)}. \tag{B14}$$

This completes the solution of Eq. (B2).

We determine β by asymptotically matching ψ in Region II to ψ in Region I. In the matching region, $\omega\sim1$, and we have[24]

$$P_\nu^{-1/(N-1)}(\omega)\sim(\tfrac{1}{2}-\tfrac{1}{2}\omega)^{(N-1)/2}/\Gamma(N/(N-1)). \tag{B15}$$

From Eq. (B15) we have for $\omega\sim1$

$$A\sim\beta\Gamma^{1/2}(N/(N-1)),\qquad f(\omega)\sim1. \tag{B16}$$

In the matching region we also have formulas similar to those in Eqs. (4.27) and (4.28) from which we conclude that [see Eq. (B3)]

$$\psi/\chi\sim2^{1/2}e^{1/4}e^{-x^2/4}. \tag{B17}$$

From Eqs. (B16) and (B17) and Eq. (4.6), we obtain the generalization of Eq. (4.33):

$$\beta=e^{-1/4}\{2\Gamma(N/(N-1))\}^{-1/2}. \tag{B18}$$

Continuing to follow Sec. IV, we use the formulas[25]

$$P_\nu^\mu(0)=2^\mu\pi^{-1/2}\cos\frac{\pi(\nu+\mu)}{2}\,\frac{\Gamma(\frac{1}{2}+\frac{1}{2}\nu+\frac{1}{2}\mu)}{\Gamma(1+\frac{1}{2}\nu-\frac{1}{2}\mu)}, \tag{B19}$$

$$\frac{d}{dx}P_\nu^\mu(x)\Big|_{x=0}=2^{\mu+1}\pi^{-1/2}\times\sin\frac{\pi(\nu+\mu)}{2}\,\frac{\Gamma(1+\frac{1}{2}\nu+\frac{1}{2}\mu)}{\Gamma(\frac{1}{2}+\frac{1}{2}\nu-\frac{1}{2}\mu)}, \tag{B20}$$

from which it follows that

$$f(0)=2(1-N)\times\frac{\sin[\pi(\nu+\mu)/2]\Gamma(1+\frac{1}{2}\nu+\frac{1}{2}\mu)\Gamma(1+\frac{1}{2}\nu-\frac{1}{2}\mu)}{\cos[\pi(\mu+\nu)/2]\Gamma(\frac{1}{2}+\frac{1}{2}\nu-\frac{1}{2}\mu)\Gamma(\frac{1}{2}+\frac{1}{2}\nu+\frac{1}{2}\mu)}. \tag{B21}$$

Equations (B19) and (B21) are now used to determine $J(x)$, and then ImE, and finally the behavior of A_n for large n. We jump immediately to the final answer because the calculation is relatively easy and the techniques are described in Sec. IV. Our result is that for large n,

$$A_n\sim-(2K)\frac{N-1}{\pi^{3/2}}(-2)^{-n}\Gamma(nN-n+\tfrac{1}{2})\times\left[\frac{\Gamma(2N/(N-1))}{\Gamma^2(N/(N-1))}\right]^{nN-n+1/2}, \tag{B22}$$

where

$$K=\left[\frac{\Gamma(1+\nu-\mu)\Gamma(-\nu-\mu)}{\Gamma(1-\mu)\Gamma(-\mu)}\right]^{1/2}. \tag{B23}$$

Finally, we check (B22) and (B23) in three important limiting cases. First, we let $\nu\to0$. In this limit K in Eq. (B23) is 1. Thus, A_n in Eq. (B22) is exactly *twice* the theoretical prediction for the large-order behavior of perturbation theory for the x^{2N} oscillator.[8] This is reasonable because when $\nu=0$, $c=0$, and we have two noninteracting x^{2N} oscillators.

Second, we pick $N=2$, when $\mu=-1$. In this case

$$K=[\Gamma(2+\nu)\Gamma(1-\nu)]^{1/2},$$

which further simplifies to

$$K=[\nu(\nu+1)\pi/\sin(\pi\nu)].$$

Here we recover the same result as in Eq. (4.42a).

Third, we expect to find a singularity in K when Eq. (B1) has spherical symmetry because of constructive interference. Hence we investigate the limit $C\to N$. Solving $\nu(\nu+1)=N/(N-1)^2$ gives

$$\nu_{1,2}=1/(N-1),\ -N/(N-1).$$

But, $\mu=-1/(N-1)$, so

$$u+\nu_1=0,\quad 1+\nu_2-\mu=0.$$

Thus, one of the two Γ functions in Eq. (B23) blows up and K is indeed singular.

We conclude that the results in Eqs. (B22) and (B23) are quite reasonable and attest to the generalizability of the methods introduced in Sec. IV.

APPENDIX C

Our discussion of N-dimensional oscillators in Sec. VII was incomplete because we did not show that straight lines through radially directed saddle points were solutions of the classical orbit equa-

tions, and we did not give criteria for the existence of saddle points. The purpose of this appendix is to remedy these omissions.

Consider an N-dimensional potential of the form

$$V=\sum_i \tfrac{1}{4}x_i{}^2+\frac{\lambda}{2M}\sum_{i_1,\ldots,i_{2M}} A_{i_1\cdots i_{2M}}x_{i_1}\cdots x_{i_{2M}}. \tag{C1}$$

The interaction term is the most general $2M$th-order homogeneous polynomial, subject to the restriction that V be bounded below for $\lambda>0$. Then we have the following theorem: If $\lambda=-\epsilon$, $\epsilon>0$, then the radial line through any critical point of V is a solution of the classical equations [Eq. (3.4)]. Furthermore, this radial line is a principal axis of the critical point and V reaches a maximum at the critical point along this line.

Proof: Let $a=(a_1,\ldots,a_n)$ be a critical point of V. Then,

$$\left.\frac{\partial V}{\partial x_i}\right|_{x_i=a_i}=\tfrac{1}{2}a_i-\epsilon\sum_{i_2,\ldots,i_{2M}} A_{ii_2\cdots i_{2M}}a_{i_2}\cdots a_{i_{2M}}=0. \tag{C2}$$

The radial line through a_i is

$$x_i(s)=\frac{a_i}{|a|}s. \tag{C3}$$

Using Eqs. (C2) and (C3) we can evaluate the expressions in Eq. (3.4) in terms of s:

$$\begin{aligned}V_iV(x_i(s))&=\frac{a_i}{2|a|s}-\epsilon\sum_{i_2,\ldots,i_{2M}} A_{ii_2\cdots i_{2M}}\left(\frac{s}{|a|}\right)^{2M-1}\\&=\frac{a_is}{2|a|s}-\frac{a_i}{s}\left(\frac{s}{|a|}\right)^{2M-1}\end{aligned} \tag{C4}$$

and

$$\sum_i x_i'(s)\nabla_i x_i(s)=\tfrac{1}{2}s-\frac{s^{2M-1}}{|a|^{2M-2}}. \tag{C5}$$

Thus

$$x_i'(s)\sum_j x_j'(s)\nabla_j x_i(s)=\nabla_i V(x_i(s)),$$

and $x_i(s)$ satisfies Eq. (3.4), the equation of motion.

Next, to show that the radial line $x_i(s)$ is a principal axis of the critical point, we compute the Hessian matrix H_{ij}:

$$\begin{aligned}H_{ij}&=\frac{\partial^2 V}{\partial x_i\,\partial x_j}(a)\\&=\tfrac{1}{2}\delta_{ij}-\epsilon^{(2M-1)}\sum_{i_3,\ldots,i_{2M}} A_{iji_3\cdots i_{2M}}a_{i_3}\cdots a_{i_{2M}}.\end{aligned} \tag{C6}$$

Thus,

$$\sum_j H_{ij}x_j(s)=\frac{sa_i}{|a|}(1-M);$$

that is, $x_i(s)$ is an eigenvector of H with eigenvalue $1-M$ and by definition it is a principal axis of the critical point. Because $1-M$ is less than 0, the critical point is a maximum along the x_i direction. This completes the proof.

Next we will show that for almost all values of the parameters of V in Eq. (C1) there is at least one radially directed saddle point.

Let us first review the results that we obtained in two dimensions. For the potential $V=\frac{1}{4}(x^2+y^2)-\frac{1}{4}\epsilon(ax^4+2cx^2y^2+by^4)$, there are three distinct possibilities:

$$c\neq a,\quad c\neq b, \tag{C7a}$$

$$a=b=c, \tag{C7b}$$

$$a=c>b. \tag{C7c}$$

In the first case we found that V always has a radial saddle point. The second case is spherically symmetric and is treated by separation of variables in Appendix A. The third case was not treated in Sec. VI, but it is easy to see that there is a degenerate critical point along the x axis (the Hessian has a zero eigenvalue). We will see that a similar trichotomy occurs for the general potential in Eq. (C1).

Let us consider an arbitrary unit vector $\hat{u}$ and study the variation of V along the radial line through $\hat{u}$:

$$V=\tfrac{1}{4}r^2-\frac{\epsilon}{2M}r^{2M}T(\hat{u}), \tag{C8}$$

where $x_j=r\hat{u}_j$ and

$$T(\hat{u})=\sum_{i_1,\ldots,i_{2M}} A_{i_1\cdots i_{2M}}\hat{u}_{i_1}\cdots\hat{u}_{i_{2M}}.$$

Then

$$\frac{\partial V}{\partial r}=\tfrac{1}{2}r-\epsilon r^{2M-1}T(\hat{u}), \tag{C9}$$

and

$$\frac{\partial^2 V}{\partial r^2}=\tfrac{1}{2}-\epsilon(2M-1)r^{2M-2}T(\hat{u}). \tag{C10}$$

Thus V has a maximum on this radial line at

$$r=\left(\frac{1}{2\epsilon T(\hat{u})}\right)^{1/(2M-2)} \tag{C11}$$

Equation (C11) defines an $N-1$ dimensional hypersurface B. It is easy to show that V attains a minimum on B. If $T(u)$ does not vanish anywhere, then the hypersurface is compact and, because V is continuous, a minimum exists. If T does vanish,

it can only do so for a finite number of vectors $\hat{u}$. Therefore consider a small cone of solid angles around each such $\hat{u}$. These cones will intersect B in a finite number of small patches, and the complement of these patches B' is compact. Thus V has a minimum on B'. Since we can choose the opening angle of the cones to be as small as we wish, the point where V has a minimum is separated from the patches by a finite distance. (Otherwise the minimum would lie along one of the directions where T vanishes; but this would be absurd because $V \to +\infty$ in such a direction.)

There are now two possibilities. Either the minimum of V is isolated or else V attains its minimum on some connected subset of V. The latter case corresponds to Eq. (C7b) and is exceptional in the sense that we can make the minimum isolated by changing the interaction term $A_{i_1 \cdots i_{2M}}$ infinitesimally and we will therefore not consider it further.

If the minimum is isolated, then we have a radially directed saddle point because, by construction, V has a maximum along the radial direction and a minimum in all other directions. As we have shown above, the radial direction is one of the principal axes of the saddle point. The saddle point may of course be degenerate as in Eq. (C7c). (The WKB methods which we have introduced depend on the existence of a nondegenerate saddle point.) However, since degeneracy is also extremely exceptional, there is almost always a saddle point.

We conclude finally that the straight-line WKB methods which we have introduced in this paper are applicable to almost all potentials of the form in Eq. (C1).

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[2]See A. M. Jaffe, Commun. Math. Phys. 1, 127 (1965). This paper gives references to the works of C. Hurst, W. Thirring, A. Petermann, and E. R. Caianiello, A. Campolattaro, and M. Marinaro.

[3]See, for example, A. M. Jaffe, in *Proceedings of the Conference on the Mathematical Theory of Elementary Particles*, edited by R. Goodman and I. Segal (M.I.T. Press, Cambridge, Mass., 1966), p. 45.

[4]Adler also suggests the possibility of using semiclassical methods to determine α. See Ref. 1.

[5]The coefficients A_n are called Rayleigh-Schrödinger coefficients and Eq. (1.3) is called a Rayleigh-Schrödinger perturbation series.

[6]C. M. Bender and T. T. Wu, Phys. Rev. Lett. 21, 406 (1968).

[7]C. M. Bender and T. T. Wu, Phys. Rev. 184, 1231 (1969), Appendix E. This paper is referred to hereafter as BWI.

[8]C. M. Bender and T. T. Wu, Phys. Rev. Lett. 27, 461 (1971).

[9]C. M. Bender and T T. Wu, Phys. Rev. D 7, 1620 (1973). This paper is referred to hereafter as BWII.

[10]T. I. Banks and C. M. Bender, J. Math. Phys. 13, 1320 (1972).

[11]The equivalence is demonstrated in BWI, Appendix A. In addition to the usual quantum-mechanical perturbative methods, Feynman rules and diagrams can be used to calculate the terms in the series in Eq. (1.7). A_n is the sum of all connected bubble diagrams having n vertices (see BWI, Appendix B). The topology of diagrams for ϕ^4 field theories in any space-time dimension is the same.

[12]To prove Eq. (2.4) we use two rigorous results of B. Simon as given in Ann. Phys. (N.Y.) 58, 79 (1970). Simon proved that $E(\lambda)$ is analytic in the domain shown in Fig. 1 and that the Rayleigh-Schrödinger series is asymptotic in the cut plane for sufficiently small λ. We can use the Cauchy theorem to write

$$F(\lambda) = \frac{1}{2\pi i} \oint_C \frac{F(z)}{z-\lambda}\, dz,$$

where C is the dashed contour shown in Fig. 1. A simple scaling argument (the Symanzik transformation) shows that $F(\lambda) \sim \lambda^{-2/3}$ as $|\lambda| \to \infty$. Thus as $R_1 \to \infty$, the contribution from the large circle vanishes. Furthermore, since the Rayleigh-Schrödinger series is asymptotic, the integral along the small circle $\to 0$ as $R_4 \to 0$. Expanding the denominator as in the text gives the formula

$$A_n = \frac{1}{2\pi i} \int_{-\infty}^{-R_2} D(x) x^{-n}\, dx + \frac{1}{2\pi i} \int_B F(z) z^{-n}\, dz + \frac{1}{2\pi i} \int_{-R_3}^{0} D(x) x^{-n}\, dx,$$

where B is the contour around the boundary of the open annulus. We now let n get large and observe that the first two integrals contributing to A_n are overwhelmed by the third, assuming that $D(x) \neq 0$ for $-R_3 < x < 0$. This establishes Eq. (2.4) for large n.

[13]F. J. Dyson, Phys. Rev. 85, 631 (1952).

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[15]See, for example, L. I. Schiff, *Quantum Mechanics* (McGraw-Hill, New York, 1949), p. 179.

[16]Equation (3.4) has a classical analogy. It is formally identical to classical equations for the orbit of a particle in the potential V. Of course, for classical motion to occur we must have $E > V$, while in the quantum-mechanical tunneling problem $E < V$. Thus our most probable escape paths are analytic continuations of the classical orbits.

[17]In a problem with a continuous symmetry, such as spherical symmetry, the set of MPEP's will of course be continuous. This happens in our problem only when $c = 1$. When $c \neq 1$ (and $a = b = 1$) we have four paths because of reflection symmetry in x and y.

[18]T. Banks and C. M. Bender, following paper, Phys. Rev. D8, 3366 (1973).

[19]It is also easy to show that these are the only straight-line solutions of Eq. (3.4).

[20]This transformation is of course motivated by the direction of the MPEP's for $c > 1$ (they meet the x axis at 45° angles). See Sec. III.

[21]See the Bateman Manuscript Project, *Higher Transcendental Functions*, edited by A. Erdélyi (McGraw-Hill, New York, 1953), Vol. 1, p. 121, Eq. (1). This work is referred to hereafter as BMP.

[22]We are of course free to choose any linear combination of $P_\nu^{-1}(w)$ and $P_\nu^1(w)$. It is most convenient and simplest to make the choice in Eq. (4.21) because it is easy to argue that α^2, the separation constant for Eq. (4.23), vanishes. For choices other than that in Eq. (4.21) there may be an integral over separation constants.

[23]See Ref. 18, Sec. II.

[24]BMP, Vol. 1, p. 163, Eq. (8).

[25]BMP, Vol. 1, p. 145, Eqs. (20) and (23). There is an error in Eq. (23) which we have corrected.

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Coupled Anharmonic Oscillators. II. Unequal-Mass Case

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We develop a general formalism for calculating the large-order behavior of perturbation theory for quantized systems of unequal-mass coupled anharmonic oscillators. Our technique is based on a generalization of the semiclassical approximation which was used to study equal-mass oscillators in the first paper of this series. The unequal-mass problem is much more difficult because the path which minimizes the classical action is not a straight line. Assuming that this tunneling path is known, we derive a general expression for the physical-optics approximation to the wave function of a tunneling particle. This derivation rests on the construction of a WKB approximation in curved space. We thus completely reduce the general quantum problem to the much simpler classical one of determining the path. Then we present a perturbation scheme for finding the classical path for systems of oscillators whose masses only differ by a small amount. Finally, we illustrate our techniques by solving a two-mode unequal-mass oscillator and comparing these results with a computer calculation. Our theoretical predictions and numerical calculations agree.

I. INTRODUCTION

In the first paper of this series,[1] we developed a method for computing the large-order behavior of the perturbation series for the ground-state energy of a system of N equal-mass coupled anharmonic oscillators. The method employs an extension of the semiclassical (WKB) approximation to multidimensional systems. However, before we can make the WKB approximation we must find the particular solutions of the classical orbit equations which make the action integral $\int (V-E)^{1/2}$ both a local and global minimum.[2] In general, it is very difficult to solve the classical equations, but in BBW I it was shown that for equal-mass oscillators the MPEP's are straight lines. In the present paper we will investigate the more difficult problem of computing the large-order behavior of perturbation theory for systems whose MPEP's are *not* straight lines. To demonstrate that our techniques work in such situations, we completely reduce the calculation of the large-order behavior of perturbation theory to the classical task of finding the MPEP.

Our interest in curved-path problems is not merely academic, for we are ultimately interested in quantum field theoretic perturbation series. The $(\phi^4)_N$ field theory with spatial and ultraviolet cutoffs is equivalent[3] to a system of unequal mass anharmonic oscillators. The "masses" in this case are just the energies of the modes of the field. It therefore seems likely that any attempt

Dielectric constant of biological systems

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In this paper we follow the Fröhlich model of long-range coherence in biological systems and use the field-theoretic Hamiltonian of Wu and Austin. We calculate the complex dielectric constant of the metabolically active dipolar parts in the cell. The actual computation proceeds in two stages. In the first stage the imaginary part of the dielectric constant is obtained by the method of continued fractions. In the second stage the real part of the dielectric constant is calculated by an application of the Kramers-Kronig theorem. The main objective of the work presented in this paper is the real part of the dielectric constant. We also discuss the contributions from water which constitutes the majority of the cell content. Our results confirm the predictions of Fröhlich, and we give analytical expressions which can be utilized in the Fröhlich potential.

I. THE FRÖHLICH MODEL

In a series of papers Fröhlich points out certain physically intriguing features of biological membranes:

(i) the various components possess a high degree of organization and cooperation;
(ii) there exists a strong transmembrane electric field;
(iii) the many types of dipoles present in a membrane are capable of high-frequency vibrations (10^{11}–10^{12} Hz);
(iv) in order to support metabolic activity the energy supply must exceed a critical value.

Based on these precepts Fröhlich[1–7] developed a mathematical model in which a chain of coupled dipole moments interact with their environment consisting of a heat bath and an energy supply. To illustrate the principles involved let us consider just two neighboring dipolar molecules which vibrate initially at frequencies ω_1 and ω_2, respectively. Then, due to their coupling, modified frequencies of dipole vibrations, ω_+ and ω_-, arise:

$$\omega_\pm = \left[\tfrac{1}{2}(\omega_1^2+\omega_2^2) \pm \left[\tfrac{1}{4}(\omega_1^2-\omega_2^2)^2 + \frac{\beta_0^4}{(\epsilon'_\pm)^2} \right]^{1/2} \right]^{1/2}, \quad (1)$$

where $\beta_0^2=\gamma e^2\sqrt{z_1 z_2}/MR^3$ and γ is a size- and shape-dependent coupling parameter, z_1 and z_2 are the numbers of elementary charges within each of the molecules, R is the equilibrium chain spacing, M is the molecule's mass, and $\epsilon'_\pm \equiv \epsilon'(\omega_\pm)$ is the real part of the complex dielectric constant of the system. The interaction energy

$$\Delta E = \frac{\hbar}{2}[(\omega_+ + \omega_-) - (\omega_1+\omega_2)] \quad (2)$$

is calculated by expanding the external square root in Eq. (1) as

$$\Delta E \cong \frac{\hbar}{2}\left[[2(\omega_1^2+\omega_2^2)]^{1/2} - (\omega_1+\omega_2) + \frac{\left[(\omega_1^2-\omega_2^2)^2 + \frac{4\beta_0^4}{(\epsilon'_+)^2} \right]^{1/2} - \left[(\omega_1^2-\omega_2^2)^2 + \frac{4\beta_0^4}{(\epsilon'_-)^2} \right]^{1/2}}{[8(\omega_1^2+\omega_2^2)]^{1/2}} \right]. \quad (3)$$

It is only when the two original frequencies are close to resonance, i.e., for $|\omega_1^2-\omega_2^2| \ll 4\beta_0^2/\epsilon'_\pm$, that we can expand the internal square root in a series in which the first term will be proportional to R^{-3}. Otherwise, the suitable expansion contains terms falling off as R^{-6}, R^{-12}, etc. At resonance then ($\omega_1=\omega_2=\bar{\omega}$)

$$\Delta E = \frac{A}{R^3} - \frac{B}{R^6}. \quad (4)$$

Here B is the van der Waals constant and A is the long-range constant given by

$$A = \left[\frac{\hbar e^2 z \gamma}{\sqrt{8}\bar{\omega} M} \right] \left[\frac{1}{\epsilon'(\omega_+)} - \frac{1}{\epsilon'(\omega_-)} \right], \quad (5)$$

where $z=z_1=z_2$. The constant A vanishes when either

(i) ω_1 and ω_2 differ significantly or

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(ii) $\epsilon'(\omega_+)=\epsilon'(\omega_-)$, i.e., the medium is frequency independent in the region considered.

Moreover, A depends on the rate of energy supply S through the value of the resonance frequency $\bar{\omega}$. It is clear that by considering the energy supply time dependent, the harmonic oscillators driven by it will attain the frequency of the driving force. In addition, A is size and shape dependent as a function of γ. If ω_- (ω_+) is excited, then the interaction is a selective attraction (repulsion). All these properties make this mechanism very appealing from the point of view of growth- and division-control effects in biology.

The only unknown parameter in Eq. (5) has been the dielectric constant $\epsilon'(\bar{\omega})$. This paper is devoted to solving this problem.

II. THE WU-AUSTIN HAMILTONIAN

Wu and Austin[8–11] proposed the following second-quantized Hamiltonian as a model Hamiltonian of the Fröhlich theory:

$$H=\sum_i \tilde{\omega}_i a_i^\dagger a_i+\sum_k \Omega_k b_k^\dagger b_k+\sum_l \theta_l P_l^\dagger P_l+\tfrac{1}{2}\sum_{i,j,k}(\chi a_i^\dagger a_j b_k^\dagger+\chi^* a_j a_i^\dagger b_k)+\sum_{i,k}(\lambda b_k a_i^\dagger+\lambda^* b_k^\dagger a_i)+\sum_{i,l}(\xi P_l a_i^\dagger+\xi^* P_l^\dagger a_i)\,. \tag{6}$$

Here $(a_i^\dagger,a_i)$, $(b_k^\dagger,b_k)$, and $(P_l^\dagger,P_l)$ are the dipole, heat-bath, and energy-pump creation and annihilation operators, respectively. The first three terms in Eq. (6) describe the three constituent parts of the system in the harmonic approximation and the remaining terms relate to the interactions between the subsystems. The commutation relations for the operators involved in Eq. (6) are those of the Bose-Einstein statistics:

$$[a_i,a_j^\dagger]=[b_i,b_j^\dagger]=[P_i,P_j^\dagger]=\delta_{ij}\,, \tag{7}$$

$$[a_i,a_j]=[b_i,b_j]=[P_i,P_j]=0\,, \tag{8}$$

$$[a_i^\dagger,a_j^\dagger]=[b_i^\dagger,b_j^\dagger]=[P_i^\dagger,P_j^\dagger]=0\,. \tag{9}$$

All operators of different types commute. For the present purposes, the Wu-Austin Hamiltonian is approximated by excluding the effects of the pump and linear dipole—heat-bath coupling. These changes, however, can be accommodated by modifying the dipole frequencies from $\tilde{\omega}_i$ to ω_i. Following Wu and Austin,[10] the effective Hamiltonian is adopted in the form

$$H=H_0+H_1\,, \tag{10}$$

where H_0 is the unperturbed cell Hamiltonian

$$H_0=\sum_i \omega_i a_i^\dagger a_i \tag{11}$$

and the interaction operator H_1 contains only the dominant, second-order dipole—heat-bath coupling

$$H_1=\tfrac{1}{2}\sum_{i,j,k}(\chi a_i^\dagger a_j b_k^\dagger+\chi^* a_j a_i^\dagger b_k)\,. \tag{12}$$

A similar Hamiltonian was used by Mishra *et al.*[12] to calculate the interaction potential between two collections of bosons. The effective Hamiltonian, Eq. (10), will be later used to evaluate the dielectric constant of the biological cell.

III. THE COMPLEX DIELECTRIC CONSTANT

In this section we outline a well-established[13] approach to the frequency-dependent complex dielectric constant. We have

$$\epsilon(\omega)=\epsilon'(\omega)-i\epsilon''(\omega)\,, \tag{13}$$

where $\epsilon'(\omega)$ and $\epsilon''(\omega)$ are the real and imaginary parts, respectively. The Kramers-Kronig[14] theorem relates the two parts through

$$\epsilon'(\omega)=1+\frac{1}{\pi}\mathrm{P}\int_{-\infty}^{\infty}\frac{\epsilon''(\nu)}{\nu-\omega}d\nu \tag{14}$$

and

$$\epsilon''(\omega)=-\frac{1}{\pi}\mathrm{P}\int_{-\infty}^{\infty}\frac{\epsilon'(\nu)}{\nu-\omega}d\nu\,, \tag{15}$$

where $\epsilon_\infty=\epsilon(\infty)$ and P denotes taking the principal part of the integral. Moreover, the imaginary part of $\epsilon(\omega)$ is proportional to the band-shape function $I(\omega)$ which is an observable quantity

$$\epsilon''(\omega)=\frac{4\pi^2(1-e^{-\beta\hbar\omega})}{V\hbar}I(\omega)\,, \tag{16}$$

where V is the volume, $\hbar$ is Planck's constant, and $\beta=(kT)^{-1}$ with k the Boltzmann constant and T the absolute temperature. The band-shape function, in turn, is derived from the dipole-dipole autocorrelation function by the Fourier transformation

$$I(\omega)=\frac{1}{2\pi}\int_{-\infty}^{\infty}e^{-i\omega t}\langle\hat{\vec{\mathrm{M}}}\cdot\hat{\vec{\mathrm{M}}}(t)\rangle dt\,. \tag{17}$$

Here, $\vec{\mathrm{M}}$ is the total-dipole-moment operator of the sample:

$$\hat{\vec{\mathrm{M}}}=\sum_{i=1}^{N}\hat{\vec{\mu}}_i \tag{18}$$

and t is time. The angular brackets $\langle\ \rangle$ imply a thermal average of the operators within, i.e.,

$$\langle\hat{\vec{\mathrm{M}}}\cdot\hat{\vec{\mathrm{M}}}(t)\rangle\equiv\mathrm{Tr}\left[\left(\frac{e^{-\beta H}}{Z}\right)\hat{\vec{\mathrm{M}}}\cdot\hat{\vec{\mathrm{M}}}(t)\right]\,, \tag{19}$$

where H is the Hamiltonian of the system and Z is the canonical partition function. The time development of the operator $\hat{\vec{\mathrm{M}}}(t)$ is given by

$$\hat{\vec{\mathrm{M}}}(t)=e^{i\hat{L}t}\hat{\vec{\mathrm{M}}}\,, \tag{20}$$

where $\hat{L}$ is the Liouville operator defined by

$$\hat{L}\hat{\vec{M}} \equiv \frac{1}{\hbar}[H,\hat{\vec{M}}] . \tag{21}$$

Furthermore, the average $\langle \hat{\vec{M}} \cdot \hat{\vec{M}}(t) \rangle$ is related to the causal Green's function $\bar{G}(t)$ since

$$\bar{G}(t) \equiv -i\Theta(t)\langle \hat{\vec{M}}(t) \cdot \hat{\vec{M}} \rangle - i\Theta(-t)\langle \hat{\vec{M}} \cdot \hat{\vec{M}}(t) \rangle , \tag{22}$$

where $\Theta(t)$ is the Heaviside step function. Denoting the Fourier transform of $\bar{G}(t)$ by $\bar{G}(\omega)$ we arrive at

$$I(\omega) = \frac{-1}{1-e^{-\beta\hbar\omega}} \mathrm{Im}\bar{G}(\omega) . \tag{23}$$

Defining the binary product of two arbitrary operators $\hat{B}$ and $\hat{C}$ as

$$(\hat{B}^{\dagger} | \hat{C}) \equiv \mathrm{Tr}\left[\left[\frac{e^{-\beta H}}{Z} \right] [\hat{B},\hat{C}] \right] \tag{24}$$

and employing the complex frequency variable $z = i\omega \pm \eta$, $\bar{G}(z)$ acquires a very compact form:

$$\frac{2\pi}{i}\bar{G}(z) = \left(\hat{\vec{M}}^{\dagger} \left| \frac{1}{z - iL} \right| \hat{\vec{M}} \right) . \tag{25}$$

The operator $\hat{\vec{M}}$ can be written in a second-quantized form as

$$\hat{\vec{M}} = \sum_{j,l} \langle j | \hat{\vec{\mu}} | l \rangle a_j^{\dagger} a_l , \tag{26}$$

where the creation and annihilation operators $a_j^{\dagger}$ and a_l span the complete and orthonormal space of many-body states $| n_1, n_2, \ldots \rangle$:

$$a_i^{\dagger} | n_1, n_2, \ldots, n_i, \ldots \rangle = \sqrt{n_i + 1} | n_1, n_2, \ldots, n_i + 1, \ldots \rangle , \tag{27}$$

$$a_i | n_1, n_2, \ldots, n_i, \ldots \rangle = \sqrt{n_i} | n_1, n_2, \ldots, n_i - 1, \ldots \rangle , \tag{28}$$

where n_i is the occupation number in the ith state ϕ_i. Substituting Eq. (26) into Eq. (25) yields

$$\bar{G}(z) = \sum_{j,l,m,n} \langle j | \hat{\vec{\mu}} | l \rangle \langle m | \hat{\vec{\mu}} | n \rangle G_{jlmn}(z) , \tag{29}$$

where $G_{jlmn}(z)$ is a Green's function in terms of the field operators alone,

$$\frac{2\pi}{i} G_{jlmn}(z) \equiv \left(F_{jl} \left| \frac{1}{z - i\hat{L}} \right| F_{mn} \right) , \tag{30}$$

where $F_{ij} \equiv a_i^{\dagger} a_j$. Equation (30) can be equivalently written in a matrix notation as

$$\frac{2\pi}{i} \underline{\underline{G}}(z) = \left(\underline{\underline{F}} \left| \frac{1}{z - i\hat{L}} \right| \underline{\underline{F}} \right) . \tag{31}$$

By introducing a projection operator P,

$$P \equiv | \underline{\underline{F}}) \cdot \underline{\underline{\alpha}} \cdot (\underline{\underline{F}} | , \tag{32}$$

where $\underline{\underline{\alpha}} = (\underline{\underline{F}} | \underline{\underline{F}})^{-1}$ and P's complement $Q \equiv 1 - P$ and following the prescription due to Mori,[15] the Green's function, Eq. (31), can be converted into a continued fraction expression

$$\frac{2\pi}{i} \underline{\underline{G}}(z) = \cfrac{(\underline{\underline{F}} | \underline{\underline{F}})}{z\underline{\underline{1}} - (\underline{\underline{F}} | i\hat{L} | \underline{\underline{F}}) \cdot \underline{\underline{\alpha}} + \cfrac{(\underline{\underline{G}} | \underline{\underline{G}}) \cdot \underline{\underline{\alpha}}}{z\underline{\underline{1}} - (\underline{\underline{G}} | iL_1 | \underline{\underline{G}}) \cdot \underline{\underline{\beta}} + \cdots}} , \tag{33}$$

where the following symbols have been used:

$$| \underline{\underline{G}}) \equiv QiL | \underline{\underline{F}}) , \tag{34}$$

$$\hat{L}_1 \equiv Q\hat{L}Q , \tag{35}$$

$$\underline{\underline{\beta}} \equiv (\underline{\underline{G}} | \underline{\underline{G}})^{-1} , \tag{36}$$

and $\underline{\underline{1}}$ is an identity matrix.

Equation (33) is the central equation of this paper. Its Fourier transform will be used to calculate the band-shape function $I(\omega)$ via Eq. (23). This, in turn, will enable us to evaluate both imaginary and real parts of the dielectric constant, using Eqs. (16) and (14), respectively. However, the continued fraction expression, Eq. (33), is just another way of formally representing the Green's function $\underline{\underline{G}}(z)$. It is an exact result which, unless further simplifications are given, must not be viewed as a solution to the problem. In Sec. IV we employ an effective Hamiltonian, Eq. (10), and carry out approximate calculations of the continued fraction expression, Eq. (33).

IV. APPROXIMATE CALCULATIONS OF THE GREEN'S FUNCTION

In order to calculate the Green's function of the Hamiltonian, Eq. (10), we rewrite Eq. (33) as

$$\frac{2\pi}{i} \underline{\underline{G}}(z) = [z\underline{\underline{1}} - (\underline{\underline{F}} | i\hat{L} | \underline{\underline{F}}) \cdot \underline{\underline{\alpha}} - \underline{\underline{\sigma}}(z)]^{-1} \cdot (\underline{\underline{F}} | \underline{\underline{F}}) , \tag{37}$$

where

$$\underline{\underline{\sigma}}(z) \equiv -\left(\underline{\underline{G}} \left| \frac{1}{z - i\hat{L}_1} \right| \underline{\underline{G}} \right) \cdot \underline{\underline{\alpha}} \tag{38}$$

and $\underline{\underline{\sigma}}(z)$ must be approximated. The details of our calculations involved in Eq. (37) are presented in Appendix A and here we just summarize the results of subsequent stages of these calculations.

(i) Matrix $(\underline{\underline{F}} | \underline{\underline{F}})$ is given by

$$(F_{jl} | F_{mn}) = (n_l - n_m)\delta_{j,m}\delta_{l,n} , \tag{39}$$

where n_i denotes the number of dipolar phonons in the state i

$$n_i = [\exp(\omega_i / kT) - 1]^{-1} \tag{40}$$

according to the Bose-Einstein statistics.

(ii) Matrix $\underline{\underline{\alpha}}$ is the inverse of $(\underline{\underline{F}} | \underline{\underline{F}})$. Since the latter matrix is diagonal, so is $\underline{\underline{\alpha}}$:

$$\alpha_{jlmn} = (n_l - n_m)^{-1}\delta_{j,m}\delta_{l,n} . \tag{41}$$

(iii) Matrix $(\underline{F}\,|\,i\hat{L}\,|\,\underline{F})$ contains the following elements:

$$(F_{jl}\,|\,i\hat{L}\,|\,F_{mn})=\frac{i}{\hbar}(\omega_m-\omega_n)(n_l-n_m)\delta_{j,m}\delta_{l,n}\ . \tag{42}$$

(iv) Matrix $(\underline{G}\,|\,(z-i\hat{L}_1)^{-1}\,|\,\underline{G})$ equals $(\underline{F}\,|\,i\hat{L}Q\times(z-i\hat{L}_1)^{-1}Qi\hat{L}\,|\,\underline{F})$. However, its elements cannot be obtained exactly. The approximate calculations carried out in Appendix A yield

$$\begin{aligned}\left[F_{jl}\left|i\hat{L}Q\frac{1}{z-i\hat{L}_1}Qi\hat{L}\right|F_{mn}\right]=\frac{|\chi|^2}{\hbar}\Bigg[&\sum_{p,r}\frac{\rho_r(n_l-n_p)-\bar{n}_l n_p}{z+i\hbar^{-1}\omega_{p,l}-i\hbar^{-1}\Omega_r}\delta_{l,n}-\sum_r\frac{\rho_r(n_l-n_m)-\bar{n}_l n_m}{z+i\hbar^{-1}\omega_{m,l}-i\hbar^{-1}\Omega_r}\\&-\sum_r\frac{\rho_r(n_n-n_j)-\bar{n}_n n_j}{z+i\hbar^{-1}\omega_{j,n}-i\hbar^{-1}\Omega_r}+\sum_{p,r}\frac{\rho_r(n_p-n_m)-\bar{n}_p n_m}{z+i\hbar^{-1}\omega_{m,p}-i\hbar^{-1}\Omega_r}\delta_{j,m}\\&+\sum_{p,r}\frac{\bar{\rho}_r(n_l-n_p)-\bar{n}_l n_p}{z+i\hbar^{-1}\omega_{p,l}+i\hbar^{-1}\Omega_r}\delta_{l,n}-\sum_r\frac{\bar{\rho}_r(n_l-n_m)+\bar{n}_l n_m}{z+i\hbar^{-1}\omega_{m,l}+i\hbar^{-1}\Omega_r}\\&-\sum_r\frac{\bar{\rho}_r(n_n-n_j)+\bar{n}_n n_j}{z+i\hbar^{-1}\omega_{j,n}+i\hbar^{-1}\Omega_r}+\sum_{p,r}\frac{\rho_r(n_p-n_m)+\bar{n}_p n_m}{z+i\hbar^{-1}\omega_{m,p}+i\hbar^{-1}\Omega_r}\delta_{j,m}\Bigg]\ .\end{aligned} \tag{43}$$

The following symbols have been used:

$$\omega_{i,j}\equiv\omega_i-\omega_j\ , \tag{44}$$

$$\bar{n}_i\equiv 1+n_i\ , \tag{45}$$

$$\rho_i\equiv[\exp(\Omega_i/kT)-1]^{-1}\ , \tag{46}$$

$$\bar{\rho}_i\equiv 1+\rho_i\ . \tag{47}$$

At this point we have calculated all the matrices that are required in Eq. (37), but we still must carry out a matrix inversion. This inversion is preceded by deriving a Dyson equation which is then solved by a variational technique. The details of this computation are given in Appendix B. Based on this, the approximate Green's function is

$$\begin{aligned}\frac{2\pi}{i}G_{jlmn}(z)=\Bigg[z-i\hbar^{-1}\omega_{j,l}-\frac{|\chi|^2}{\hbar^2(n_m-n_n)}\Bigg[&\sum_{p,r}\frac{\rho_r(n_l-n_p)-\bar{n}_l n_p}{z+i\hbar^{-1}\omega_{p,l}-i\hbar^{-1}\Omega_r}-\sum_p\frac{\bar{\rho}_p(n_l-n_m)-\bar{n}_l n_m}{z+i\hbar^{-1}\omega_{m,l}-i\hbar^{-1}\Omega_p}\\&-\sum_p\frac{\bar{\rho}_p(n_n-n_j)-\bar{n}_n n_j}{z+i\hbar^{-1}\omega_{j,n}-i\hbar^{-1}\Omega_p}+\sum_{p,r}\frac{\rho_r(n_p-n_m)-\bar{n}_p n_m}{z+i\hbar^{-1}\omega_{m,p}-i\hbar^{-1}\Omega_r}\\&+\sum_{p,r}\frac{\rho_r(n_l-n_p)-\bar{n}_l n_p}{z+i\hbar^{-1}\omega_{p,l}+i\hbar^{-1}\Omega_r}-\sum_p\frac{\bar{\rho}_p(n_l-n_m)+\bar{n}_l n_m}{z+i\hbar^{-1}\omega_{m,l}+i\hbar^{-1}\Omega_p}\\&-\sum_p\frac{\bar{\rho}_p(n_n-n_j)+\bar{n}_n n_j}{z+i\hbar^{-1}\omega_{j,n}+i\hbar^{-1}\Omega_p}\\&+\sum_{p,r}\frac{\rho_r(n_p-n_m)+\bar{n}_p n_m}{z+i\hbar^{-1}\omega_{m,p}+i\hbar^{-1}\Omega_r}\Bigg]\Bigg]^{-1}(n_n-n_m)\delta_{j,m}\delta_{l,n}\ .\end{aligned} \tag{48}$$

This complicated equation can be written in a compact form:

$$\frac{2\pi}{i}G_{jlmn}(z)\equiv\left[z-ih^{-1}\omega_{j,l}-\frac{|\chi|^2}{\hbar^2(n_m-n_n)}\sum_{Q=1}^{8}\sigma^{(Q)}_{jlmn}(z)\right]^{-1}(n_l-n_j)\delta_{j,m}\delta_{l,n}\ . \tag{49}$$

The precise definition of the eight quantities $\sigma^{(Q)}_{jlmn}(z)$ can be obtained by comparing the right-hand sides of Eqs. (48) and (49).

The Green's function given by Eq. (49) depends upon the complex frequency variable z. The corresponding function dependent upon the real frequency ω is obtained by the method outlined in Appendix C. The final form of the required Green's function is

$$G_{jlmn}(\omega)=\frac{1}{2\pi}\frac{(n_l-n_j)\delta_{j,m}\delta_{l,n}}{\omega-\hbar^{-1}\omega_{j,l}\pm[i\,|\chi|^2/\pi\hbar^2(n_l-n_j)]\Gamma_{jl}(\omega)}\ . \tag{50}$$

The function $\Gamma_{jl}(\omega)$ is the frequency-dependent linewidth function. Note that the line shift function $S(\omega)$ has been

neglected since the resonance frequencies are usually determined from experiment anyhow. The form of the linewidth function will be discussed in Sec. V.

V. CALCULATIONS OF THE COMPLEX DIELECTRIC CONSTANT

Having found the explicit form of the frequency-dependent correlation function, Eq. (50), we use Eqs. (23) and (16) to obtain the imaginary (absorptive) part of $\epsilon(\omega)$ and then, applying the Kramers-Kronig formula, Eq. (14), the real (nonabsorptive) part of $\epsilon(\omega)$ will be calculated. We start with the imaginary part of $G_{jlmn}(\omega)$,

$$\mathrm{Im}G_{jlmn}(\omega)=\frac{1}{2\pi}\frac{(|\chi|^2/\pi\hbar^2)\Gamma_{jl}(\omega)\delta_{j,m}\delta_{l,n}}{(\omega-\hbar^{-1}\omega_{j,l})^2+[|\chi|^4/\pi\hbar^2(n_l-n_j)^2]\Gamma_{j,l}^2(\omega)} . \tag{51}$$

Assuming the dipole moment matrix elements in the phonon states in Eq. (26), $\langle i|\hat{\vec{\mu}}|j\rangle$, to be real quantities, we obtain

$$\begin{aligned}\mathrm{Im}\bar{G}(\omega)&=\sum_{j,l,m,n}\langle j|\hat{\vec{\mu}}|l\rangle\langle m|\hat{\vec{\mu}}|n\rangle\mathrm{Im}G_{jlmn}(\omega)\\&=\sum_{j,l}|\langle j|\hat{\vec{\mu}}|l\rangle|^2\frac{(|\chi|^2/\pi\hbar^2)\Gamma_{jl}(\omega)}{(\omega-\hbar^{-1}\omega_{j,l})^2+[|\chi|^4/\pi^2\hbar^4(n_l-n_j)]\Gamma_{jl}^2(\omega)} .\end{aligned} \tag{52}$$

This, together with Eqs. (23) and (16), is used to evaluate the imaginary part of the dielectric constant:

$$\epsilon''(\omega)=\frac{4\pi^2(1-e^{-\beta\hbar\omega})}{V\hbar(1+e^{-\beta\hbar\omega})}\sum_{j,l}|\langle j|\hat{\vec{\mu}}|l\rangle|^2\frac{(|\chi|^2/\pi\hbar^2)\Gamma_{jl}(\omega)}{(\omega-\hbar^{-1}\omega_{j,l})^2+[|\chi|^4/\pi^2\hbar^4(n_l-n_j)^2]\Gamma_{jl}^2(\omega)} . \tag{53}$$

If the imaginary part of the dielectric constant is to have any practical value, an explicit form for the quantity $\Gamma_{jl}(\omega)$ must be obtained. The mathematical details have been relegated to Appendix D and we outline only the most important points here.

(i) If, in Eq. (48), the coupling constant χ between the dipolar and the bath phonons is allowed to vanish, a very simple formula is obtained:

$$\frac{2\pi}{i}G_{jlmn}(z)\cong\frac{(n_n-n_m)\delta_{j,m}\delta_{l,n}}{z-i\hbar^{-1}\omega_{j,l}} . \tag{54}$$

Now, applying the principles discussed in Appendix C, we find

$$\mathrm{Im}G_{jlmn}(\omega)=\pi(n_n-n_m)\hbar\delta_{j,m}\delta_{l,n}\delta(\hbar\omega-(\omega_j-\omega_l)) . \tag{55}$$

Thus the absorptive part of the dielectric constant calculated from this equation displays a sharp (narrow) line corresponding to a dipolar frequency transition given by $\omega_l\rightarrow\omega_j$, where $\omega_j>\omega_l$. An absorption line will be observed at an experimental frequency $\omega=\hbar^{-1}(\omega_j-\omega_l)$.

(ii) As the effects of the bath phonons are introduced ($\chi\neq0$), the situation described in (i) is altered as both shifting and broadening of the absorption line is observed. Focusing on the transitions between the modes j and l, two types of terms in Eq. (48) can be distinguished.

(a) The bath directly influences the modes j and l:

$$\text{Bath}\rightarrow\begin{cases}\text{mode } j\\ \text{mode } l\end{cases} .$$

This coupling scheme corresponds to those terms in Eq. (48) that contain a single summation over p.

(b) The bath influences a dipolar mode other than the modes j and l and the result of this is fed into the modes j and l:

$$\text{Bath}\rightarrow\text{intermediate dipolar mode}\rightarrow\begin{cases}\text{mode } j\\ \text{mode } l\end{cases} .$$

This coupling scheme corresponds to all those terms in Eq. (48) that contain a double summation over p and r.

(iii) The summations involved in terms of category (a) can be easily carried out but some difficulty is associated with the terms of category (b). To simplify, it is assumed that there exists only one dipolar mode k other than modes j and l participating in processes (b) and that the bath phonons are acoustic phonons with sound velocity v. The function $\Gamma_{jl}(\omega)$ is now evaluated (see Appendix D for details) as

$$\begin{aligned}\Gamma_{jl}(\omega)=\frac{V\hbar}{2\pi^2v^3}\{&w_1^2(n_l-n_k)[\rho(w_1)+\rho(-w_1)]\\&+w_2^2(n_k-n_j)[\rho(w_2)+\rho(-w_2)]\\&-2\bar{w}_1^2(n_l-n_j)[\bar{\rho}(\bar{w}_1)+\bar{\rho}(-\bar{w}_1)]\} ,\end{aligned} \tag{56}$$

where

$$w_1\equiv\hbar\omega+\omega_{k,l} , \tag{57}$$

$$w_2\equiv\hbar\omega+\omega_{j,k} , \tag{58}$$

$$\bar{w}_1\equiv\hbar\omega+\omega_{j,l} . \tag{59}$$

We now calculate the real part of the dielectric constant using the Kramers-Kronig formula, Eq. (14), and the expression for $\epsilon''(\omega)$, Eq. (53). In order to simplify the result we make the following two approximations:

(i) the resonance approximation, $\omega\cong\hbar^{-1}\omega_{j,l}$, and
(ii) the high-temperature limit, $kT\gg\hbar\omega$.

The necessary integration is carried out with the aid of a contour in the complex plane (see Appendix E). The final expression for the real part of the dielectric constant is obtained as

$$\epsilon'(\omega)=1+\frac{2\pi}{VkT}\sum_{j,l}|\langle j|\hat{\vec{\mu}}|l\rangle|^2(n_l-n_j)\frac{\left[\hbar^{-2}\omega_{j,l}^2+\left[\frac{|\chi|^4}{\pi^2\hbar^4(n_l-n_j)^2}\right]\Gamma_{jl}^2(\hbar^{-1}\omega_{j,l})-\hbar^{-1}\omega_{j,l}\omega\right]}{\left[(\omega-\hbar^{-1}\omega_{j,l})^2+\frac{|\chi|^4}{\pi^2\hbar^4(n_l-n_j)^2}\Gamma_{jl}^2(\hbar^{-1}\omega_{j,l})\right]}. \tag{60}$$

VI. DISCUSSION

In our Hamiltonian the coupling constant χ is an unknown parameter. The equations that have been derived for ϵ' and ϵ'', which are measurable quantities, should provide a means for estimating this parameter. Since both Eqs. (53) and (60) are fairly complicated, we introduce two additional assumptions:

(i) the resonance condition, $\omega=\omega_0\cong\hbar^{-1}\omega_{j,l}$, and
(ii) the high-temperature limit, $\exp(-\beta\hbar\omega)\cong 1-\beta\hbar\omega$.

As a result, an approximate expression for $\epsilon''(\omega)$ is obtained:

$$\epsilon''(\omega_0)\cong\frac{2\pi^3\hbar^2\omega_0}{VkT|\chi|^2}\sum_{j,l}\frac{|\langle j|\vec{\mu}|l\rangle|^2}{\Gamma_{jl}(\omega_0)}(n_l-n_j)^2. \tag{61}$$

Furthermore, we approximate the remaining summation by

$$\sum_{j,l}\frac{|\langle j|\vec{\mu}|l\rangle|^2}{\Gamma_{jl}(\omega_0)}(n_l-n_j)^2\cong\frac{N|\vec{\mu}|^2}{\Gamma(\omega_0)}, \tag{62}$$

where N is the total number of phonons and $|\vec{\mu}|$ is the mean dipole moment of the sample. Hence, the final form of $\epsilon''(\omega)$ is

$$\epsilon''(\omega_0)\cong\frac{2\pi^3\hbar^2\rho|\vec{\mu}|^2\omega_0}{|\chi|^2kT\Gamma(\omega_0)}, \tag{63}$$

where ρ is the density of the material.

Similarly, the real part of the dielectric constant, $\epsilon'(\omega)$ of Eq. (60), can be evaluated close to the resonance, i.e., for $\omega=\omega_0\pm\delta\omega$, as

$$\epsilon'(\omega_0\pm\delta\omega)=1+\frac{2\pi}{kT}\rho|\vec{\mu}|^2\frac{1\mp\omega_0\Delta\,\delta\omega}{1-\Delta(\delta\omega)^2}, \tag{64}$$

where $\Delta\cong\pi^2\hbar^4N/|\chi|^4\Gamma^2(\omega_0)$. Therefore, $\epsilon'(\omega)$ is of an asymmetric shape whose value at ω_0 is equal to the static value

$$\epsilon'(\omega_0)=1+\frac{2\pi}{kT}\rho|\vec{\mu}|^2=\epsilon'(0) \tag{65}$$

and $\epsilon'(\pm\infty)=1$. Equation (65) resembles the Debye equation in the case of ϵ resulting from the existence of permanent dipole moments. Within this approximation the real part, $\epsilon'(\omega_0)$, is strongly frequency dependent, confirming the assumption of Fröhlich. We can therefore conclude that the point dipoles do contribute to the long-range attraction predicted by Fröhlich. The imaginary part, $\epsilon''(\omega_0)$, is proportional to ω_0 and inversely proportional to the coupling constant $|\chi|^2$. The real part becomes frequency independent as $|\chi|\to 0$. Both ϵ' and ϵ'' decrease with the absolute temperature and increase with the density of the substance.

Our calculations of the complex dielectric constant of a biological cell have been based on a number of assumptions which have to be scrutinized.

(i) The omission of energy-pump and linear cell—heat-bath coupling can be justified as causing a constant shift of eigenfrequencies $\omega_i\to\tilde{\omega}_i$ which are not known beforehand.

(ii) Similar argument applies to neglecting the line-shift function $S(\omega)$.

(iii) The continued fraction expression, Eq. (48), is approximate. However, it is still complicated enough for our purposes.

(iv) In dealing with the heat-bath modes we have adopted continuum approximation and Debye model, both realistic in view of the large size of the heat bath.

(v) The high-temperature limit seems appropriate as $T\cong 300$ K is a typical temperature necessary for life processes.

(vi) Finally, taking only one excited mode k into consideration in Eq. (56) is yet to be verified but it seems compatible with the tenets of the Fröhlich model.

It should be realized that the derived complex dielectric constant, Eqs. (53) and (60), covers the giant dipoles present in the cell membrane. It does not include the bulk of a cell which is free and bound water. This fact must be accounted for when attempting to correlate our theory with experiment. Unfortunately, no satisfactory theory on the role of water content in the dielectric properties of organic materials exists. It is well known[16] that water affects the value of ϵ'' more profoundly than ϵ'. For pure water ϵ'' reaches 30 while dry organic material has ϵ'' in the range 10^{-3}—10^{-1} (microwave frequencies considered). Since moist materials achieve ϵ'' in the order of 1, both water and organic molecules seem to be involved strongly in dielectric processes.

Based on the excellent treatise on aqueous dielectrics by Hasted,[16] we write the expression for the dielectric constant of an entire tissue approximately as

$$\epsilon=v_m\epsilon_m+v_b\epsilon_b+(1-v_m-v_b)\epsilon_w, \tag{66}$$

where v_m and v_b are the volume fractions of the membrane dipolar complexes and bound water, respectively, and ϵ_m,ϵ_b are the corresponding dielectric constants. The values of v_m and v_b vary from substance to substance, but for muscle tissue,[16] for example, $v_m=25\%$. The dielectric constant of free water ϵ_w can be assumed to obey the Debye-Drude equations

$$\epsilon'(\omega)=\epsilon_\infty+\frac{\epsilon_s-\epsilon_\infty}{1+\omega^2\tau^2}, \tag{67}$$

$$\epsilon''(\omega)=\frac{(\epsilon_s-\epsilon_\infty)\omega\tau}{1+\omega^2\tau^2}, \tag{68}$$

where $\epsilon_s=\epsilon(0)$, $\epsilon_\infty=\epsilon(\infty)$, and τ is the single relaxation time. At 20°C, the most appropriate temperature for considering living organisms, the free-water parameters, are[16]

$$\epsilon_s\cong 78.5,\quad \epsilon_\infty\cong 4.5,\quad \tau=0.9\times 10^{-11}\text{ s}.$$

The dipole moment of H_2O molecules is $|\vec{\mu}|=1.84$ D whereas for various protein molecules it ranges between 150 and 400 D depending on the type of the protein and the pH of the electrolyte.

The dielectric properties of bound water are not known in detail but are presumed to be between those of ice and liquid water. Its volume fraction v_b is much smaller than that of free water and it relaxes at much lower frequencies with almost no contribution to the dielectric loss at microwave frequencies.

Finally, the dielectric constant of the membrane dipolar complexes ϵ_m is given in Eqs. (53) and (60) or their estimates, Eqs. (63) and (65). We believe that it should be possible to experimentally detect these unusual dielectric properties of biological substances. We expect that in the 10^{10}–10^{12} Hz region there should be resonance peaks clearly distinguishable from the water background. After all, the membrane content in the tissue is by no means negligible. At this stage of development we are not prepared to offer any numerical values, which is simply due to the scarcity of relevant experimental data. Nevertheless we recognize the importance of this question and will deal with it in our future work.

APPENDIX A

(i) In order to calculate the matrix elements of $(\underline{F}\,|\,\underline{F})$ we evaluate the commutator

$$[F_{ik},F_{jl}]=\delta_{kj}a_i^\dagger a_l-\delta_{il}a_j^\dagger a_k \tag{A1}$$

and use Eq. (24) to find

$$(F_{jl}\,|\,F_{mn})=(n_l-n_m)\delta_{j,m}\delta_{l,n}\ , \tag{A2}$$

where we have approximated H by H_0.

(ii) The calculations of $(\underline{F}\,|\,i\hat{L}\,|\,\underline{F})$ require applying the Liouville operator $\hat{L}$ of Eq. (21) to the element F_{mn}:

$$i\hat{L}F_{mn}=\frac{i}{\hbar}\left[(\omega_m-\omega_n)a_m^\dagger a_n+\sum_{p,r}\chi^*(a_p^\dagger a_n-a_n^\dagger a_p)b_r+\sum_{p,r}\chi(a_p^\dagger a_n-a_m^\dagger a_p)b_r^\dagger\right]. \tag{A3}$$

Taking the inner product of the "ket" produced from this result with the "bra" $(F_{jl}|$ yields

$$(F_{jl}\,|\,i\hat{L}\,|\,F_{mn})=\frac{i}{\hbar}(\omega_m-\omega)(n_l-n_m)\delta_{j,m}\delta_{l,n}\ . \tag{A4}$$

(iii) The matrix $(\underline{G}\,|\,(z-i\hat{L}_1)^{-1}\,|\,\underline{G})$ is calculated in several steps.

(a) Using $Q=1-P$ and Eq. (32) for P, we write Q explicitly as

$$Q=1-\sum_{q,r}|\,a_q^\dagger a_r)\frac{1}{(n_r-n_q)}(a_q^\dagger a_r|\ . \tag{A5}$$

(b) The definition of $|\,\underline{G})$, Eq. (34), applied in conjunction with Eq. (A5) gives

$$|\,G_{mn})=\frac{i}{\hbar}\sum_{p,r}|\,(\chi^*b_r+\chi b_r^\dagger)(F_{pn}-F_{mp}))\ . \tag{A6}$$

(c) Similarly, utilizing Eq. (A5) in the definition of $\hat{L}_1$, Eq. (35) yields

$$i\hat{L}_1=i\hat{L}-\sum_{s,t}i\hat{L}\,|\,a_s^\dagger a_t)\frac{1}{n_t-n_s}(a_s^\dagger a_t|-\sum_{q,r}|\,a_q^\dagger a_r)\frac{1}{(n_r-n_q)}(a_q^\dagger a_r\,|\,i\hat{L}+\sum_{q,r,s,t}|\,a_q^\dagger a_r)\frac{1}{(n_r-n_q)}(a_q^\dagger a_r\,|\,i\hat{L}\,|\,a_s^\dagger a_t)\frac{1}{(n_t-n_s)}(a_s^\dagger a_t|\ . \tag{A7}$$

(d) Making use of Eq. (A6) we obtain

$$\left[G_{jl}\left|\frac{1}{z-i\hat{L}_1}\right|G_{mn}\right]=\left[-\frac{i}{\hbar}\right]\left[\frac{i}{\hbar}\right]\sum_{p,q,r,s}\left[(\chi^*b_s+\chi b_s^\dagger)(F_{rl}-F_{jr})\left|\frac{1}{z-i\hat{L}_1}\right|(\chi^*b_q+\chi b_q^\dagger)(F_{pn}-F_{mp})\right] \tag{A8}$$

and upon expanding the right-hand side, eight terms arise, each of which is of the form

$$\left[b_iF_{jk}\left|\frac{1}{z-i\hat{L}_1}\right|b_lF_{mn}\right]\equiv G^{(1)}_{ijklmn}(z)\ , \tag{A9}$$

$$\left[b_i^\dagger F_{jk}\left|\frac{1}{z-i\hat{L}_1}\right|b_l^\dagger F_{mn}\right]\equiv G^{(2)}_{ijklmn}(z)\ . \tag{A10}$$

These new Green's functions involve the new variables bF and $b^\dagger F$ which implies that as we proceed further down the continued fraction, the bath variables begin to make their appearance. It is convenient to introduce the vectors $\underline{A}$ and $\underline{B}$ of which bF and $b^\dagger F$ are the components. Thus we have $b_iF_{jk}\in\underline{A}$ and $b_i^\dagger F_{jk}\in\underline{B}$. Hence the right-hand sides of Eqs. (A9) and (A10) are the elements of the following matrices: $G^{(1)}_{ijklm}(z)\in\underline{\underline{G}}^{(1)}(z)$, $G^{(2)}_{ijklmn}(z)\in\underline{\underline{G}}^{(2)}(z)$. Formally, these new Green's functions resemble Eq. (37) since

$$\underline{\underline{G}}^{(1)}(z)=[z\underline{\underline{1}}-(\underline{A}\,|\,i\hat{L}_1\,|\,\underline{A})-\underline{\underline{\Sigma}}^{(1)}(z)]^{-1}\cdot(\underline{A}\,|\,\underline{A}) \tag{A11}$$

and

$$\underline{\underline{G}}^{(2)}(z)=[z\underline{\underline{1}}-(\underline{B}\,|\,i\hat{L}_1\,|\,\underline{B})-\underline{\underline{\Sigma}}^{(2)}(z)]^{-1}\cdot(\underline{B}\,|\,\underline{B})\ . \tag{A12}$$

Here, the two functions $\underline{\underline{\Sigma}}^{(1)}(z)$ and $\underline{\underline{\Sigma}}^{(2)}(z)$ represent the third stage in the continued fraction development and are set equal to zero as an approximation:

$$\underline{\underline{\Sigma}}^{(1)}(z)=\underline{\underline{\Sigma}}^{(2)}(z)=0 . \tag{A13}$$

With this approximation Eqs. (A11) and (A12) may be written as

$$\underline{\underline{G}}^{(1)}(z)\simeq(\underline{A}\mid\underline{A})\cdot[\underline{\underline{\Delta}}^{(1)}(z)]^{-1}\cdot(\underline{A}\mid\underline{A}) , \tag{A14}$$

$$\underline{\underline{G}}^{(2)}(z)\simeq(\underline{B}\mid\underline{B})\cdot[\underline{\underline{\Delta}}^{(2)}(z)]^{-1}\cdot(\underline{B}\mid\underline{B}) , \tag{A15}$$

where

$$\underline{\underline{\Delta}}^{(1)}(z)\equiv[z(\underline{A}\mid\underline{A})-(\underline{A}\mid i\hat{L}_1\mid\underline{A})] , \tag{A16}$$

$$\underline{\underline{\Delta}}^{(2)}(z)\equiv[z(\underline{B}\mid\underline{B})-(\underline{B}\mid i\hat{L}_1\mid\underline{B})] . \tag{A17}$$

The final forms of the matrix elements are written as follows:

$$G^{(1)}_{ijklmn}(z)=\frac{\rho_i(n_k-n_m)-\bar{n}_k n_m}{z+i\hbar^{-1}\omega_{m,n}-i\hbar^{-1}\Omega_1}\delta_{il}\delta_{jm}\delta_{kn} , \tag{A18}$$

$$G^{(2)}_{ijklmn}(z)=\frac{\bar{\rho}_i(n_k-n_m)+\bar{n}_k n_m}{z+i\hbar^{-1}\omega_{m,n}+i\hbar^{-1}\Omega_1}\delta_{il}\delta_{jm}\delta_{kn} . \tag{A19}$$

The various quantities appearing in these equations are defined in Sec. IV [see Eqs. (40) and (44)—(47)].

APPENDIX B

In this appendix we calculate the inverse required in Eq. (37) using a variational technique.[15] For convenience we extract from Eq. (37) the part that has to be inverted and represent it by the matrix $\underline{\underline{g}}(z)$,

$$\underline{\underline{g}}(z)\equiv[z\underline{\underline{1}}-(\underline{F}\mid i\hat{L}\mid\underline{F})\cdot\underline{\underline{\alpha}}-\underline{\underline{\sigma}}(z)]^{-1} . \tag{B1}$$

We first convert Eq. (B1) into a Dyson equation defining a zero-order diagonal Green's function:

$$\underline{\underline{g}}_0^{-1}(z)=z\underline{\underline{1}}-(\underline{F}\mid i\hat{L}\mid\underline{F})\cdot\underline{\underline{\alpha}} . \tag{B2}$$

Therefore

$$\underline{\underline{g}}^{-1}(z)=\underline{\underline{g}}_0^{-1}(z)-\underline{\underline{\sigma}}(z) . \tag{B3}$$

Multiplying both sides of this equation by $\underline{\underline{g}}(z)$ and then from the left by $\underline{\underline{g}}_0(z)$ and rearranging we obtain the desired Dyson equation

$$\underline{\underline{g}}(z)=\underline{\underline{g}}_0(z)+\underline{\underline{g}}_0(z)\cdot\underline{\underline{\sigma}}(z)\cdot\underline{\underline{g}}(z) . \tag{B4}$$

Since $\underline{\underline{\sigma}}$ is large we use a variational approach to solve the Dyson equation instead of the iterative method which employs

$$\underline{\underline{g}}(z)=\underline{\underline{g}}_0(z)+\underline{\underline{g}}_0(z)\cdot\underline{\underline{\sigma}}(z)\cdot\underline{\underline{g}}_0(z)+\underline{\underline{g}}_0(z)\cdot\underline{\underline{\sigma}}(z)\cdot\underline{\underline{g}}_0(z)\cdot\underline{\underline{\sigma}}(z)\cdot\underline{\underline{g}}_0(z)+\cdots . \tag{B5}$$

Dyson's equation can be written as a functional-matrix equation:

$$\underline{\underline{F}}(\underline{\underline{g}})\equiv\underline{\underline{F}}=\underline{\underline{g}}_0+\underline{\underline{g}}_0\cdot\underline{\underline{\sigma}}\cdot\underline{\underline{g}}-\underline{\underline{g}}=\underline{\underline{0}} \tag{B6}$$

and the problem is to obtain a solution of this equation for the unknown matrix $\underline{\underline{g}}$. If we start with a trial solution $\underline{\underline{g}}^t$ and expand $\underline{\underline{F}}$ about this solution and retain only the first-order terms then we will obtain

$$\underline{\underline{F}}(\underline{\underline{g}})=\underline{\underline{F}}(\underline{\underline{g}}^t)+\left[\frac{\delta\underline{\underline{F}}(\underline{\underline{g}})}{\delta\underline{\underline{g}}}\right]_{\underline{\underline{g}}^t}\cdot(\underline{\underline{g}}-\underline{\underline{g}}^t)=\underline{\underline{0}} . \tag{B7}$$

Equation (B7) can be immediately solved to give an approximate result $\underline{\underline{g}}^a$,

$$\underline{\underline{g}}^a=\underline{\underline{g}}^t-\left[\frac{\delta\underline{\underline{F}}}{\delta\underline{\underline{g}}}\right]^{-1}_{\underline{\underline{g}}^t}\cdot\underline{\underline{F}}(\underline{\underline{g}}^t) . \tag{B8}$$

The matrix inverse required in Eq. (B8) can be easily found:

$$(\delta\underline{\underline{F}}\mid\delta\underline{\underline{q}})^{-1}=-(\underline{\underline{1}}+\underline{\underline{g}}\cdot\underline{\underline{\sigma}}) . \tag{B9}$$

Since Eq. (B9) contains the unknown $\underline{\underline{g}}$ which must be replaced by $\underline{\underline{g}}^t$ that appears in Eq. (B8), we replace it by another trial function $\underline{\underline{g}}^\tau$ and then substitute the result in Eq. (B8), thereby producing another approximation which we call $\underline{\underline{g}}^v$:

$$\underline{\underline{g}}^v=\underline{\underline{g}}_0+\underline{\underline{g}}_0\cdot\underline{\underline{\sigma}}\cdot\underline{\underline{g}}^t+\underline{\underline{g}}^\tau\cdot\underline{\underline{\sigma}}\cdot\underline{\underline{g}}_0+\underline{\underline{g}}^\tau\cdot\underline{\underline{R}}\cdot\underline{\underline{g}}^t , \tag{B10}$$

where $\underline{\underline{R}}\equiv\underline{\underline{\sigma}}\cdot\underline{\underline{g}}_0\cdot\underline{\underline{\sigma}}-\underline{\underline{\sigma}}$.

The matrix $\underline{\underline{g}}^v$ is stationary with respect to variations in both $\underline{\underline{g}}^t$ and $\underline{\underline{g}}^\tau$ about the exact solution $\underline{\underline{g}}$. A very convenient set of trial functions for the problem may be generated by taking linear combinations of the terms arising in the iterative expansion of the Dyson equation, Eq. (B6):

$$\underline{\underline{g}}^t=\sum_{L=1}^{N}c_L\underline{\underline{T}}^{(L)},\quad \underline{\underline{g}}^\tau=\sum_{L=1}^{N}d_L\underline{\underline{T}}^{(L)} , \tag{B11}$$

where

$$\begin{aligned}&\underline{\underline{T}}^{(1)}=\underline{\underline{g}}_0 ,\\ &\underline{\underline{T}}^{(2)}=\underline{\underline{g}}_0\cdot\underline{\underline{\sigma}}\cdot\underline{\underline{g}}_0 ,\\ &\underline{\underline{T}}^{(3)}=\underline{\underline{g}}_0\cdot\underline{\underline{\sigma}}\cdot\underline{\underline{g}}_0\cdot\sigma\cdot\underline{\underline{g}}_0 ,\\ &\cdots .\end{aligned} \tag{B12}$$

As a first approximation we select one-parameter trial functions:

$$\underline{\underline{g}}^t=c_1\underline{\underline{g}}_0,\quad \underline{\underline{g}}^\tau=d_1\underline{\underline{g}}_0 . \tag{B13}$$

Inserting these trial functions into Eq. (B10), minimizing with respect to c_1 and d_1, and then solving the two linear equations for c_1 and d_1 we obtain the "best" matrix $\underline{\underline{g}}$. A typical element is

$$\underline{\underline{g}}^{(z)}_{jlmn}=\frac{\delta_{j,m}\delta_{l,n}}{z-i\hbar^{-1}\omega_{jl}-\sigma_{jlmn}(z)} . \tag{B14}$$

The function $\underline{\underline{\sigma}}$ is defined in Eq. (38) and requires an explicit form for $(\underline{G}\mid(z-i\hat{L}_1)^{-1}\mid\underline{G})$ and $\underline{\underline{\alpha}}$. These two are defined in Eqs. (43) and (41), respectively. Thus the explicit form of $\underline{\underline{\sigma}}$ can be obtained and from it the matrix $\underline{\underline{g}}$ via Eq. (B14).

APPENDIX C

In this appendix we present the necessary details to obtain the imaginary part of the Green's function $\underline{\underline{G}}(z)$ for real frequencies. For simplicity Eq. (48) can be symbolically rewritten as

$$2\pi G_{jlmn}(i\omega\pm\eta):=\left[\omega\mp i\eta-A_1+\frac{A_2}{\omega\mp i\eta+A_3}\right]^{-1}, \tag{C1}$$

where we have used $z=i\omega\pm\eta$. Since the quantity A_3 is real, we must use the standard equation to properly deal with singularities:

$$\frac{1}{x\pm i\eta}=\mathrm{P}\left[\frac{1}{x}\right]\mp i\pi\delta(x)\ . \tag{C2}$$

Then Eq. (C1) becomes

$$\tilde{G}_{jlmn}(\omega\mp i\eta):=\frac{1}{2\pi}\left[\omega\mp i\eta-A_1+A_2\mathrm{P}\left[\frac{1}{\omega+A_3}\right]\pm i\pi A_2\delta(\omega+A_3)\right]^{-1}, \tag{C3}$$

where we have used the definition $\tilde{G}(i\omega\pm\eta)=G(i(\omega\mp i\eta))\equiv\tilde{G}(\omega\mp i\eta)$. Since the equation contains a finite imaginary part in the form of the term $i\pi A_2\delta(\omega+A_3)$ we can now set $\eta=0$ without impunity as there is no danger of encountering singularities for real ω. We now have

$$\tilde{G}_{jlmn}(\omega):=\frac{1}{2\pi}[\omega-A_1+S(\omega)\pm i\Gamma(\omega)]^{-1}\ . \tag{C4}$$

Here, $S(\omega)=A_2\mathrm{P}((\omega+A_3)^{-1})$ becomes a frequency-dependent line shift and $\Gamma(\omega)=\pi A_2\delta(\omega+A_3)$ the frequency-dependent line broadening. The imaginary part of Eq. (C4) is readily taken to give

$$\mathrm{Im}\,\tilde{G}_{jlmn}(\omega):=\frac{1}{2\pi}\frac{\Gamma(\omega)}{[\omega-A_1+S(\omega)]^2+\Gamma^2(\omega)}\ . \tag{C5}$$

In all future development we will call $\tilde{G}(\omega)$ as simply $G(\omega)$.

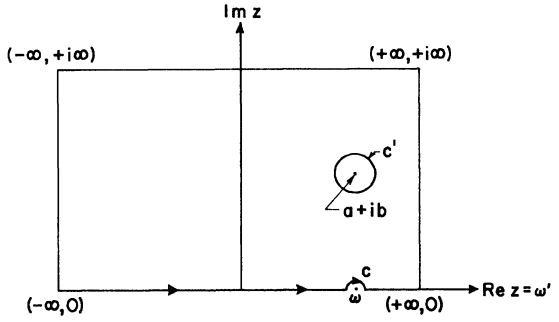

FIG. 1. Details of the contour used in Eq. (E6).

APPENDIX D

The calculation of the linewidth function consists, essentially, of computing the summations contained in the eight terms in the bracket of Eq. (48) for real values of the frequency. Since the calculations with the eight different terms are repetitive, it is sufficient to consider just one term, e.g., $\sigma^{(1)}$, for illustrating the technique.

We begin with

$$\sigma^{(1)}_{jlmn}(z)=\sum_{p,r}\frac{\rho_r(n_l-n_p)-\bar{n}_l n_p}{z+i\hbar^{-1}\omega_{p,l}-i\hbar^{-1}\Omega_r}\ . \tag{D1}$$

Employing the discussion of Appendix C we obtain for real frequencies

$$\sigma^{(1)}_{jlmn}(\omega)=\sum_{p,r}P\left[\frac{\rho_r(n_l-n_p)-\bar{n}_l n_p}{\omega+\hbar^{-1}\omega_{p,l}-\hbar^{-1}\Omega_r}\right]\pm i\pi\sum_{p,r}[\rho_r(n_l-n_p)-\bar{n}_l n_p]\delta(\omega+\hbar^{-1}\omega_{p,l}-\hbar^{-1}\Omega_r)$$
$$\equiv S^{(1)}_{jlmn}(\omega)\pm i\pi\Gamma_{jlmn}(\omega)\ . \tag{D2}$$

Here, $S^{(1)}(\omega)$ is the contribution to the line shift and $\Gamma^{(1)}(\omega)$ is the contribution to the linewidth from $\sigma^{(1)}$ and consequently $S(\omega)=\sum_{Q=1}^{8}S^{(Q)}(\omega)$ and $\Gamma(\omega)=\sum_{Q=1}^{8}\Gamma^{(Q)}(\omega)$. The line shift will be ignored in the present calculations as the resonance frequencies are not known *a priori* but are equated to the experimental values.

To evaluate the linewidth we convert the summation over r into an integral since the heat bath is much larger than the cell itself:

$$\Gamma^{(1)}_{jlmn}(\omega)=\hbar\delta_{j,m}\delta_{l,n}\sum_{p}\int dr\,w(r)[\rho(\Omega_r)(n_l-n_p)-\bar{n}_l n_p]\delta(\hbar\omega+\omega_p-\omega_l-\Omega_r)\ , \tag{D3}$$

where $w(r)$ is a density of phonon states for which we use the Debye model,

$$w(r)=\frac{V\hbar}{2\pi^2}r^2\ . \tag{D4}$$

The quantity Ω_r is the bath phonon frequency,

$$\Omega_r=vr\ , \tag{D5}$$

where v is the speed of sound in the bath. Inserting Eqs. (D4) and (D5) into Eq. (D3) gives

$$\Gamma^{(1)}_{jlmn}(\omega)=\frac{V\hbar}{2\pi^2 v^3}\sum_p(\hbar\omega+\omega_p-\omega_l)^2[\rho(\hbar\omega+\omega_p-\omega_l)(n_l-n_p)-\bar{n}_l n_p]\delta_{j,m}\delta_{l,n}\ . \tag{D6}$$

From the discussion given in Sec. V we can see that since $\sigma^{(1)}$ is a term of category (b), the final summation in Eq. (D6) is approximated by a single term $p=k$ and

$$\Gamma^{(1)}_{jlmn}(\omega)=\frac{V\hbar}{2\pi^2 v^3}(\hbar\omega+\omega_k-\omega_l)^2[\rho(\hbar\omega+\omega_k-\omega_l)(n_l-n_k)-\bar{n}_l n_k]\delta_{j,m}\delta_{l,n}\ . \tag{D7}$$

APPENDIX E

In this appendix we present some of the mathematical details required for the calculation of the principal-value integration appearing in Eq. (14). In the high-temperature limit ($\beta\to 0$) the ratio of the two exponential factors in Eq. (53) simplifies to

$$\epsilon''(\omega)=\frac{2\pi^2\omega}{VkT}\sum_{j,l}\frac{|\langle j|\hat{\vec{\mu}}|l\rangle|^2(|\chi|^2/\pi\hbar^2)\Gamma_{jl}(\hbar^{-1}\omega_{j,l})}{(\omega-\hbar^{-1}\omega_{j,l})^2+[|\chi|^4/\pi^2\hbar^4(n_l-n_j)^2]\Gamma^2_{jl}(\hbar^{-1}\omega_{j,l})}\ . \tag{E1}$$

Substituting Eq. (E1) into Eq. (14) we obtain

$$\epsilon'(\omega)=1+\frac{2|\chi|^2}{VkT\hbar^2}\sum_{j,l}|\langle j|\hat{\vec{\mu}}|l\rangle|^2\Gamma_{jl}(\hbar^{-1}\omega_{j,l})\mathrm{P}\int_{-\infty}^{\infty}d\nu\frac{\nu}{\{(\nu-\hbar^{-1}\omega_{j,l})^2+[|\chi|^4\Gamma^2_{jl}(\hbar^{-1}\omega_{j,l})/\pi^2\hbar^4(n_l-n_j)^2]\}(\nu-\omega)}\ . \tag{E2}$$

We define

$$a\equiv\hbar^{-1}\omega_{j,l}\ , \tag{E3}$$

$$b^2\equiv\frac{|\chi|^4}{\pi^2\hbar^4(n_l-n_j)^2}\Gamma^2_{jl}(\hbar^{-1}\omega_{j,l})\ . \tag{E4}$$

The principal-value integral can be given in a shorthand form by

$$I\equiv\mathrm{P}\int_{-\infty}^{\infty}d\nu\frac{\nu}{[(\nu-a)^2+b^2](\nu-\omega)}\ . \tag{E5}$$

Instead of I as defined by Eq. (E5) we consider another integral J,

$$J\equiv\int dz\frac{z}{(z-a+ib)(z-a-ib)(z-\omega)}\ . \tag{E6}$$

Here, the contour of integration over the variable z is selected according to Fig. 1. It is easy to show that the integrand vanishes along the contour except along the lines $-\infty<\chi\le\omega-i\eta$ and $\omega+i\eta\le\chi\le+\infty$ and the infinitesimally small semicircular contour C. Since the integrand is analytic over the entire upper half of the complex plane, with the exception of the point $z=a+ib$, the integral can be evaluated by shrinking the large, almost rectangular contour to an infinitesimally small circular contour around the simple pole at $(a+ib)$. Thus

$$\mathrm{P}\int_{-\infty}^{\infty}dx\frac{x}{(x-a+ib)(x-a-ib)(x-\omega)}+\int_C dz\frac{z}{(z-a+ib)(z-a-ib)(z-\omega)}=\oint_{C'}dz\frac{z}{(z-a+ib)(z-a-ib)(z-\omega)}\ . \tag{E7}$$

The right-hand side is evaluated by applying Cauchy's theorem as $\pi(a+ib)/b(a+ib-\omega)$. The second term in the left-hand side is calculated by changing to an angular variable θ, $z=\omega+\eta e^{i\theta}$, where θ extends from π to 0 and η is the radius of the semicircle C. Upon integration over θ we get for this term $-\pi\omega i/[(\omega-a+ib)(\omega-a-ib)]$. Collecting all the results yields

$$I=\frac{\pi(a^2+b^2-a\omega)}{b[(\omega-a)^2+b^2]}\ . \tag{E8}$$

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Dielectric Measurement of Individual Microtubules Using the Electroorientation Method

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ABSTRACT Little is known about the electrostatic/dynamic properties of microtubules, which are considered to underlie their electrostatic interactions with various proteins such as motor proteins, microtubule-associated proteins, and microtubules themselves (lateral association of microtubules). To measure the dielectric properties of microtubules, we developed an experiment system in which the electroorientation of microtubules was observed under a dark-field microscope. Upon application of an alternating electric field (0.5–1.9 $\times$ 10^5 V/m, 10 kHz–3 MHz), the microtubules were oriented parallel to the field line in a few seconds because of the dipole moment induced along their long axes. The process of this orientation was analyzed based on a dielectric ellipsoid model, and the conductivity and dielectric constant of each microtubule were calculated. The analyses revealed that the microtubules were highly conductive, which is consistent with the counterion polarization model—counterions bound to highly negatively charged microtubules can move along the long axis, and this mobility might be the origin of the high conductivity. Our experiment system provides a useful tool to quantitatively evaluate the polyelectrolyte nature of microtubules, thus paving the way for future studies aiming to understand the physicochemical mechanism underlying the electrostatic interactions of microtubules with various proteins.

INTRODUCTION

Microtubules of eukaryotic cells play an important role in cellular organization, intracellular transport, and cell motility. In these cellular processes, microtubules serve as molecular tracks for motor proteins such as kinesin and dynein (1) and interact with a variety of microtubule-associated proteins (MAPs) (2).

Electrostatic interaction is thought to play an essential role in mediating these intermolecular interactions of the microtubules. For instance, the processive movement of the motor protein kinesin is modulated by its electrostatic interaction with the microtubule; the distance traveled by the kinesin along the microtubule is dramatically reduced by treatment with high salt concentration or by proteolytic removal of the acidic C-terminal tails of tubulin (3,4). This indicates that this part of the microtubule may serve as an electrostatic tether that keeps the kinesin molecule close to the microtubule surface. A similar decrease was observed in the distance traveled by cytoplasmic dynein by the proteolytic removal of the C-terminal region of tubulin (5). The electrostatic interaction is also implicated in the lateral association of microtubules, which leads to the formation of a variety of cellular architectures composed of microtubule bundles. In this context, the ability of microtubules to form bundles is affected by the presence of ions (6) or MAPs (7,8) in solution. The morphology of the bundle appears to depend on the species of ions (9) or the number of charges included in the microtubule-binding domain of the MAPs (10). These results indicate that both ions and MAPs may serve as cationic ligands that neutralize the negative charges of microtubules (9,10), although the formulation for these observations in terms of electrostatic interaction has not yet been presented.

Despite the potential significance of electrostatic interactions in mediating and regulating their intermolecular interactions, little is known about the electrostatic/dynamic properties of microtubules (11,12). This situation contrasts with the case of other biomolecular filaments such as DNA and actin; the polyelectrolyte nature of these polymers has been extensively characterized, and theoretical studies to explain the observed phenomena are underway (13–19).

To characterize the electrostatic/dynamic properties of microtubules, we performed dielectric measurements by using the electroorientation method (20,21). An alternating electric field was applied to microtubules, inducing dipoles and aligning them parallel to the direction of the applied field. By analysis of this electroorientation process, the microtubule conductivity was calculated to be 1.5 $\pm$ 0.1 mS/m, which is $>$15 times that of the surrounding medium. This high conductivity is probably due to counterion polarization (22–24). Since the microtubule has a large negative potential (12), counterions bind loosely and nonspecifically to it, whereas their diffusional motion along the long axis is unconstrained. Thus, the high conductivity is mainly attributed to the longitudinal movement of these counterions in response to the applied field. The polyelectrolyte nature of microtubules characterized in this study might define the physicochemical basis for nonspecific electrostatic interactions between microtubules and various proteins.

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MATERIALS AND METHODS

Preparation of proteins

Tubulin was prepared from porcine brain by temperature-dependent cycles of polymerization and depolymerization (25). Tubulin depleted of MAPs was obtained by performing the second cycle of polymerization in a buffer containing 0.5 M piperazine-1,4-bis(2-ethanesulfonic acid) (PIPES) (pH 6.8) and 10% dimethylsulfoxide (DMSO) (26). Microtubules with lengths of 3–20 μm were polymerized using a short polymer segment as a seed for polymerization (27) in BRB80 buffer (80 mM 2-morpholinoethanesulfonic acid (MES) (pH 6.8), 2 mM $MgCl_2$, and 1 mM EGTA). The polymerized microtubules were stabilized by 0.2 mM taxol (T7402, Sigma, St. Louis, MO) and centrifuged at 250,000 $\times$ g for 12 min through glycerol cushions (40% v/v glycerol in the BRB80 buffer). The resulting pellet was resuspended in the BRB80 buffer containing 0.2 mM taxol at a final protein concentration of 15 mg/ml; it was stored on ice until use.

For the preparation of subtilisin-treated microtubules (s-MTs), taxol-stabilized microtubules in the BRB80 buffer (3 mg/ml) were incubated with subtilisin (P-5380, Sigma) at a molar ratio of 1:1 (tubulin dimer/subtilisin) at 37°C for 2 h (28). The reaction was stopped by 2 mM phenylmethylsulfonyl fluoride, and the mixture was centrifuged through a glycerol cushion to separate the microtubules from the enzyme and C-terminal fragments. The pellet was then resuspended in the BRB80 buffer containing 0.2 mM taxol. The digestion of both α- and β-tubulin was confirmed by SDS polyacrylamide gel electrophoresis.

To determine the site of subtilisin cleavage, amino acid sequencing with liquid chromatography/mass spectrometry (LC/MS; LCQ-DEACA XP, ThermoQuest, San Jose, CA) was carried out on subtilisin-treated tubulin. LC/MS analysis of the subtilisin-treated α-tubulin detected two cleavage sites—at positions Asp^{438} and Glu^{443}. The cleavage site in β-tubulin was not uniquely determined because of its multiple isoforms; the possible cleavage site is at position Ala^{430}, Asp^{431}, Glu^{432}, Gln^{433}, or Gly^{434}.

Buffer condition

A low-salt buffer (LS buffer) composed of 1 mM MES (pH 6.7), 10 μM $MgCl_2$, and 80 μM EGTA, supplemented with 10 μM taxol in DMSO, was used for electroorientation. The final concentration of DMSO was 1%. The LS buffer had a free Mg^{2+} concentration of 9.95 μM, as calculated by the method of Goldstein (29). The conductivity of this buffer solution was 9.7 mS/m, as measured with a conductivity meter (Horiba, Kyoto, Japan). The microtubules were stable in this solution for at least 3 h; all experiments were completed within 1 h of the microtubules being transferred to the LS buffer. EGTA was excluded from the LS buffer in electroorientation experiments for testing the effect of ion species.

Application of electric fields and observation of electroorientation

The orientation chamber with dimensions 6 mm $\times$ 6 mm $\times$ 0.15 mm was composed of a glass slide and a coverslip that were affixed to each other by means of double-sticky tape (Fig. 1 *A*). A pair of transparent electrodes was microfabricated on the upper surface of the glass slide; a 1500-Å-thick indium-tin-oxide (ITO)-precoated glass surface (Sanyo Shinku, Osaka, Japan) was etched by photolithography to form a pair of parallel electrodes with a spacing of 100 $\pm$ 1 μm. Each of these ITO electrodes was connected to a coaxial cable by means of a flexible phosphor bronze plate.

For the orientation experiment, a 7-μl microtubule solution, diluted to a final concentration of $\sim$1 μg/ml with the LS buffer, was first applied in the open orientation chamber; a coverslip was then placed on the top of the chamber to complete the ceiling. The motion of individual microtubules was observed under a dark-field microscope (Olympus, Tokyo, Japan) with a 50×, 0.5–0.9 NA oil-immersion objective lens (Olympus) and 100 W mercury-lamp illumination (OSRAM, Munich, Germany). Dark-field images were projected onto an electron-bombarded charge-coupled device camera (C7190 Hamamatsu Photonics, Hamamatsu, Japan) and stored using a videocassette recorder.

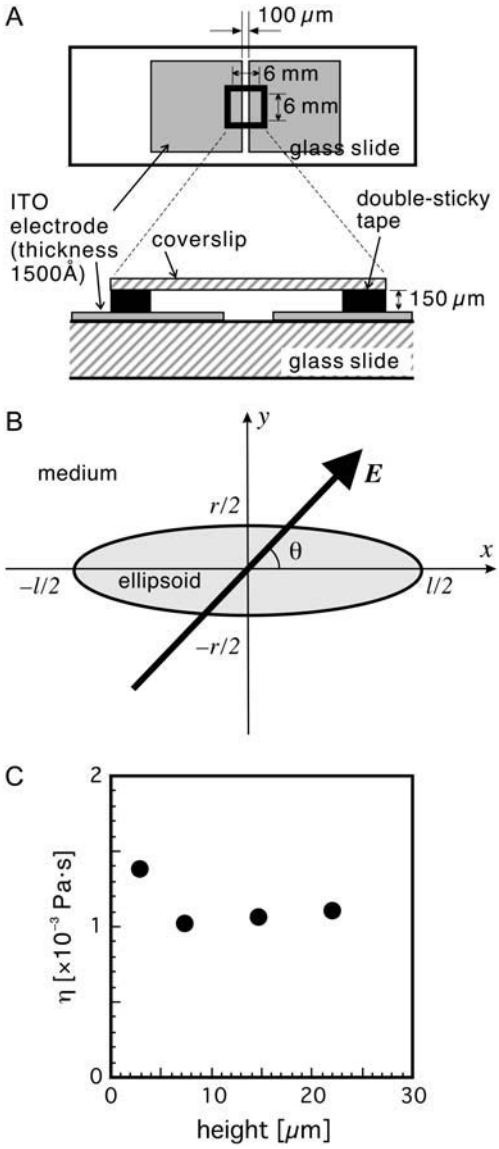

FIGURE 1 (*A*) Top and side views of the electroorientation chamber. Side view is vertically expanded to emphasize the structure. (*B*) Diagram illustrating a dielectric ellipsoid with its long axis x placed at angle θ to the external field E. The z axis is perpendicular to the page. (*C*) Viscosity in the chamber is plotted as a function of the distance from the bottom of the electroorientation chamber. The rotational diffusion constant of microtubule D_r was measured from the mean square rotation angle of the microtubule $\langle\theta^2\rangle$, and η was calculated from $D_r = kT/\pi\eta l^3 C_r = \langle\theta^2\rangle/2t$, where C_r is a form factor (Eq. 5) (55).

To apply the AC field to the microtubule solution, sinusoidal voltage in the frequency range of 1 kHz–10 MHz was generated by a function generator (WAVE Factory 1952, NF Co., Yokohama, Japan) and amplified up to 22 V rms (at 1 MHz) by a power amplifier (High Speed Power Amplifier 4055, NF Co.). The applied sinusoidal voltage was monitored using a digital oscilloscope.

The voltage applied to the electrode may get damped due to the resistance, capacitance, and polarization of the electrodes. Therefore, the voltage actually applied in the microtubule solution was estimated by the method of

Schwan (30), i.e., by using an electronic circuit simulation program (Circuit Maker, Altium, Sydney, Australia). In this simulation, the electrode resistance (1 Ω) was taken into account and coaxial cables were modeled as ideal transmission lines. The capacitance and resistance between the electrodes were measured using an impedance meter (Hioki E. E. Co., Ueda, Japan). The voltage drop caused by the impedance of the electrodes was thus estimated to be <1% at frequencies >10 kHz. The voltage drop increased at frequencies <10 kHz, and was estimated to be 6% at 1 kHz.

As the electric field was not uniformly generated near the edge of the ITO electrodes, dielectrophoresis of the microtubules was inevitable in that area (20). Thus, to avoid the effect of dielectrophoresis, the electroorientation of only the microtubules located in the central region of the chamber (>30 μm from the edges of both the electrodes) was analyzed. Experiments were performed at 25 ± 0.5°C.

Electroorientation model and calculation of microtubule polarization

Since the orientation is caused by an electric-field-induced torque acting on a microtubule filament, the orientation process was modeled as described below, following the equivalent dipole moment method (31,32).

The microtubule is considered as an ellipsoid with rotational symmetry of length l and radius r (Fig. 1 *B*). When the ellipsoid is placed with its long axis at angle θ to the electric field $\mathbf{E}^*$, the electrostatic torque exerted on the ellipsoid is calculated by

$$\mathbf{T}_e = 1/2\mathrm{Re}[\mathbf{p}^* \times \mathbf{E}^*], \tag{1}$$

where $\mathbf{p}^*$ is the dipole moment induced in the ellipsoid and Re[ ] denotes the real part. If we take the coordinate x in the direction of the longest axis of the ellipsoid and coordinate y perpendicular to it (both x and y are in the plane parallel to the direction of applied electric field), the orientation of the ellipsoid is caused by the torque about the z axis, which is perpendicular to the direction of electric field. This torque is thus expressed as

$$T_{ez} = 1/2\alpha E^2 \cos\theta \sin\theta, \tag{2}$$

where

$$\alpha = \mathrm{Re}[a_x^* - a_y^*] \tag{3}$$

is the effective polarization coefficient of the ellipsoid, $E = |\mathbf{E}^*|$, and a_x^* and a_y^* are the polarizability of the ellipsoid in the x and y directions, respectively.

On the other hand, the viscous countertorque T_v experienced by the ellipsoid is given by

$$T_v = \pi l^3 \eta C_r \frac{d\theta}{dt}, \tag{4}$$

where $\eta = 1.0 \times 10^{-3}$ Pa·s (see Fig. 1 *C*) is the viscosity of the medium, and C_r is a form factor. Assuming that $l \gg r$, C_r can be expressed as

$$C_r = \frac{1}{3(\ln(l/r) - 1/2)}, \tag{5}$$

where r is 12.5 nm for the microtubule.

The electrostatic torque should be balanced with the viscous countertorque by equating Eqs. 2 and 4, thus resulting in the following equation of motion:

$$\tau \frac{d\theta}{dt} = -\cos\theta \sin\theta, \tag{6}$$

where τ is the time constant of the orientation expressed as

$$\tau = \frac{2\pi l^2 \eta C_r}{(\alpha/l) E^2}. \tag{7}$$

Eq. 6 can be solved analytically for the case in which the field is applied stepwise with the initial condition $\theta = \theta_0$ at $t = 0$. The solution is

$$\ln(\tan\theta) = \ln(\tan\theta_0) - t/\tau. \tag{8}$$

Based on these equations, the time constant of the orientation, τ, was calculated from the orientation angle of the ellipsoid measured as a function of time (Eq. 8), and the effective polarization coefficient of the microtubule, α, was obtained (Eq. 7).

Calculation of microtubule conductivity

The effective polarization coefficient $\alpha = \mathrm{Re}[a_x^* - a_y^*]$ is actually a parameter related to both the conductivity and dielectric constant of an ellipsoid. For a high aspect ratio ellipsoid such as a microtubule, a_x^* and a_y^* can be approximated by the following equations (32,33):

$$\begin{aligned} \alpha_x^* &= \varepsilon_0 \varepsilon_1 V(\varepsilon_2^* - \varepsilon_1^*)/\varepsilon_1^* \\ \alpha_y^* &= \varepsilon_0 \varepsilon_1 V(\varepsilon_2^* - \varepsilon_1^*)/(\varepsilon_2^* + \varepsilon_1^*), \end{aligned} \tag{9}$$

where

$$\varepsilon_j^* = \varepsilon_j - i\sigma_j/2\pi f \varepsilon_0. \tag{10}$$

ε_j^* is the complex dielectric constant ($j = 1$ for the surrounding medium and 2 for the microtubule), ε_j is the dielectric constant, σ_j is the conductivity, ε_0 is the permittivity of free space, V is the volume of the ellipsoid, and f is the frequency of the applied field. The dielectric constant of water at 25°C, ε_1, is 78.5. According to these equations, σ_2 and ε_2 can be obtained by fitting the spectrum of α/l.

In the actual experiment, since the electroorientation was performed in a solution with low ionic strength, a cloud of counterions surrounding the microtubule may have significantly affected the orientation. Therefore, to calculate the conductivity, the entire microtubule including the surrounding counterion cloud was regarded as a polar entity of an ellipsoid, the radius of which was represented by the sum of the radius of the microtubule (12.5 nm) and the Debye length in the LS buffer ($\kappa^{-1} = (\sum_i \rho_i e^2 z_i^2/\varepsilon\varepsilon_0 kT)^{-1/2} = 9.1$ nm (34), where ρ_i is the concentration of ions in m^{-3}, z_i is the valence of ions, i represents each ion species, e is the elementary charge, k is the Boltzmann constant, and $T = 298$ K is the temperature).

Viscosity in the orientation chamber

The viscosity is assumed to be constant in the model. However, it is not homogeneous in the orientation chamber, but increases near the glass surface due to the wall effect (35). To analyze the electroorientation with our model, the orientation must be measured at positions sufficiently distant from the glass surface.

To determine the height appropriate for measurement, the viscous coefficient η was calculated from the rotational diffusion constant of the microtubules, D_r, at various heights above the glass surface (Fig. 1 *C*). The height of each microtubule from the bottom surface of the glass was determined by calibrating the focal plane of the microscope using the fine adjustment knob. The result revealed that the viscosity was constant at a height of 7–22 μm above the glass surface. Since the focal depth of the microscope was <6 μm, the electroorientation experiment was conducted on a focal plane 15 μm above the bottom of the chamber.

Temperature control

To accurately calculate the viscosity, a constant temperature should be maintained during electroorientation. However, the temperature may increase with the application of the electric field because of excessive Joule heating.

Therefore, we monitored the temperature of the solution using particles of stearic acid butyl ester (S-5001, Sigma) as a temperature indicator. Stearic acid butyl ester is a colorless wax with a melting point of 26.8°C. To monitor the temperature, particles of this wax (10–20 μm in size) were placed between the electrodes in the LS buffer, and the temperature of the stage was initially set to 24.0 ± 0.2°C. Then, upon application of an alternating electric field of 1.9×10^5 V/m (1 MHz), the time required to melt the particles was measured. Continuous application of the electric field for a period of 4 s resulted in melting of the particles (i.e., the temperature increased by ~3°C), whereas the repetitive application of the electric field for a period of 3 s with an interval of 7 s left the particles intact.

Based on these results, for the electroorientation conducted at 25.0 ± 0.5°C, we applied an electric field of $<1.9 \times 10^5$ V/m for a period <3 s with an interval >15 s. Under this condition, the increase in temperature of the solution during measurement should be <3°C.

Calculation of net charge density

The net charges of each α- and β-tubulin subunit at various pH values were calculated by the method of Skoog and Wichman (36), where the number of dissociated charges is obtained by the sum of the charges dissociated from the amino acid side chains and the terminal amino acids. In the calculation, the amino acid sequences of porcine α- and β-tubulin subunits were used (37,38). One molecule each of Mg^{2+} and GTP was included in α-tubulin, whereas one molecule of GDP was included in β-tubulin (39). The net charge of the tubulin dimer was then obtained by summing the net charge of each subunit. The formation of intradimer salt bridges was not taken into consideration in this summation (40) because the dissociated charges that participate in the salt bridge should cancel each other and therefore would not significantly affect the net charge of the dimer. The net charge of the tubulin dimer thus calculated was comparable to the values obtained by molecular dynamics simulation based on the crystal structure of tubulin (41).

RESULTS AND DISCUSSION

Observation and quantitative analysis of electroorientation

It has been demonstrated that in response to the applied electric field, parallel arrays of microtubules are formed due to the dipole moment induced along the long axis of the microtubules ((42), K. Sakata, M. Kurachi, and H. Tashiro, RIKEN, personal communication, 1994). We first attempted a reproduction of their results.

When the microtubules in the LS buffer solution (ionic strength 1.11 mM) were initially observed using a dark-field microscope, they exhibited Brownian motion in the chamber, being random in orientation (Fig. 2 *A*, *left*). Upon application of an alternating electric field (9×10^4 V/m, 1 MHz), the microtubules were oriented parallel to the field line in a few seconds (Fig. 2 *A*, *right*, and *B*; see Supplementary Material for video). The electroorientation occurred at a field strength $>5 \times 10^4$ V/m and in the frequency range of 10 kHz–5 MHz. At frequencies <10 kHz, the electroorientation was accompanied by a rapid flow of microtubules due to the convection of the solution. This convection can be

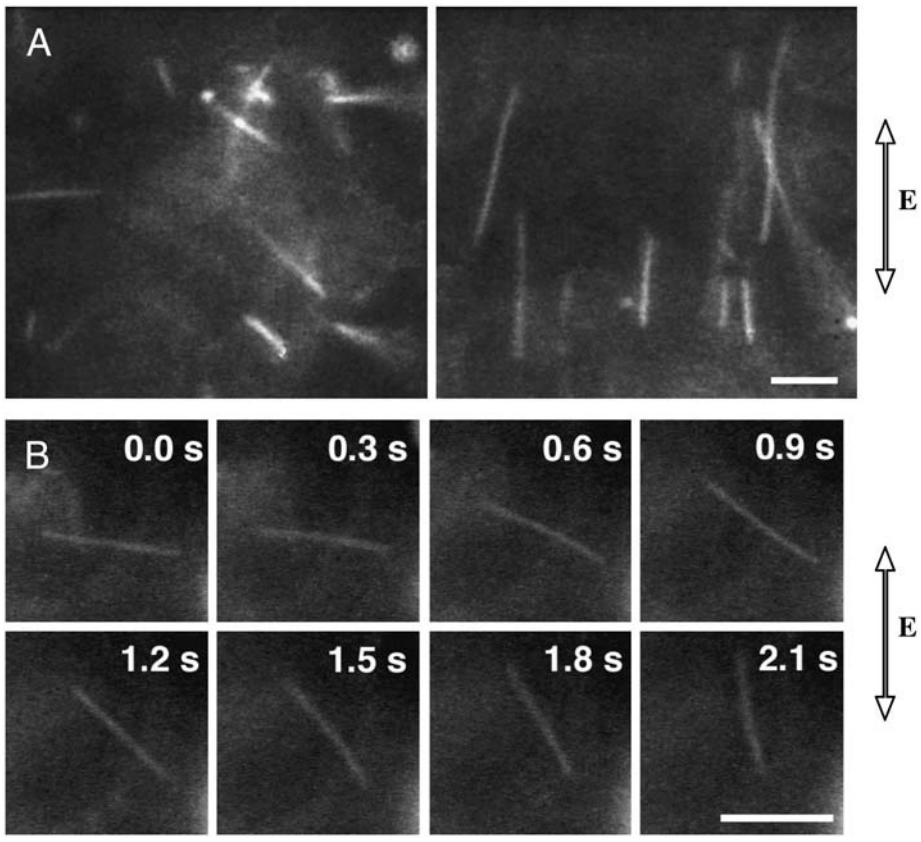

FIGURE 2 Orientation of microtubules at $E = 9 \times 10^4$ V/m, frequency 1 MHz. (*A*, *Left*) Orientation of microtubules is random in the absence of the electric field. (*Right*) Upon application of the electric field, the microtubules are oriented parallel to the direction of the field line. (*B*) Sequential images of a single microtubule after the onset of the electric field taken at intervals of 0.3 s. Arrows indicate the direction of the applied electric field. Scale bar, 10 μm.

attributed to the ionic current in the solution, as the flow became more obvious when the electroorientation was performed in a solution with higher ionic strength. To avoid convection, a frequency of 1 MHz and a field strength just sufficient to orient the microtubules was used in the following experiments, unless otherwise stated.

The process of electroorientation was analyzed by modeling the microtubule as a dielectric ellipsoid on which electrostatic and viscous countertorques were balanced (see Materials and Methods for details). From the video record of the electroorientation of a single microtubule, the angle of the microtubule long axis against the direction of the field line, θ, was measured frame by frame, and $\ln(\tan\theta)$ was plotted as a function of time (Fig. 3 *A*). As expected from Eq. 8 of the model, $\ln(\tan\theta)$ decreased linearly with time, and the slope gives the inverse of the orientation time constant, $1/\tau$.

When τ was calculated for microtubules with various lengths, it appeared to be proportional to the square of the microtubule lengths, l^2 (Fig. 3 *B*). This result is consistent with the model (Eq. 7), given that the dielectric property of microtubule is homogeneous along its length (α/l = constant). The model also predicts that τ normalized by l^2 should be proportional to E^{-2}, where E is the field strength (Eq. 7). This relationship was confirmed with data measured at various field strengths (0.53–1.9 $\times 10^5$ V/m) (Fig. 3 *C*). Together, these results were consistent with the dielectric ellipsoid model; therefore, the effective polarization coefficient of the microtubule, α, could be reliably deduced from our measurement. The effective polarization coefficient normalized for the microtubule length, α/l, was calculated to be $6.3 \pm 0.3 \times 10^{-24}$ Cm/V at 1 MHz.

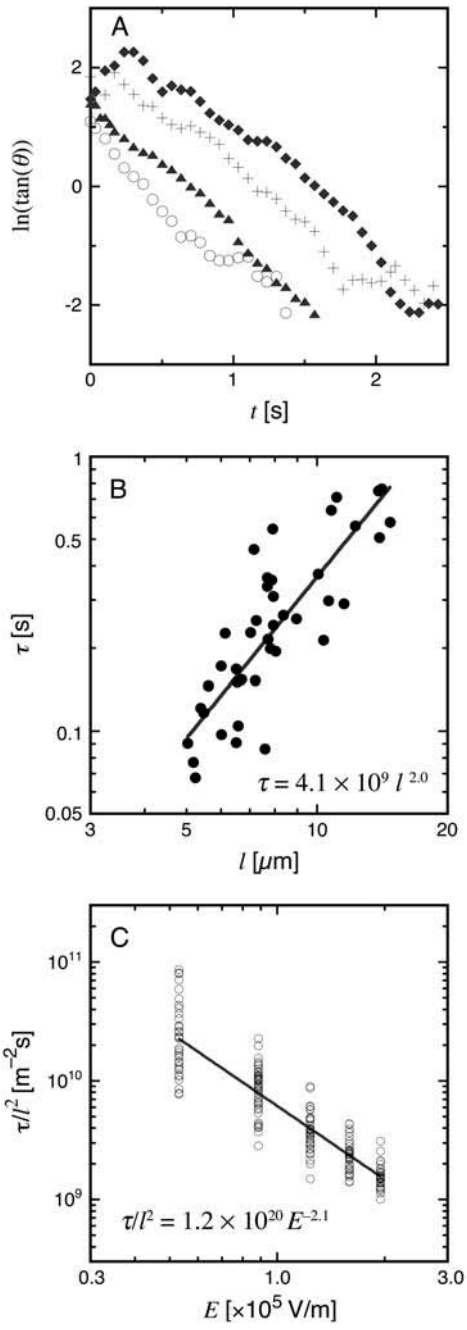

FIGURE 3 Quantitative analysis of microtubule electroorientation. (*A*) $\ln(\tan(\theta))$ plotted against time. Since the taxol-stabilized microtubule is rigid enough, θ can be uniquely determined at each time instant (see Gittes et al. (56)). Four representative examples for microtubules with lengths of 8–12 μm are shown. $E = 1.2 \times 10^5$ V/m, frequency 1 MHz. (*B*) Electroorientation time constant τ plotted as a function of the microtubule length l. τ is proportional to the square of the filament length (line represents the best fit: $\tau = 4.1 \times 10^9\ l^{2.0}$). (*C*) Time constant normalized for the square of microtubule length, τ/l^2, plotted as a function of the field strength E. Line represents the best fit: $\tau/l^2 = 1.2 \times 10^{20}\ E^{-2.1}$.

Dielectric properties of microtubules

The effective polarization coefficient $\alpha = \mathrm{Re}[a_x^* - a_y^*]$ is actually a parameter related to both the conductivity σ_2 and dielectric constant ε_2 of a microtubule. It is possible to approximate σ_2 and ε_2 if the frequency dependence of α is known (see Materials and Methods for details).

Fig. 4 *A* shows the spectrum of α/l measured in the frequency range of 10 kHz–7 MHz. Since the dielectric property of the microtubule was homogeneous along its length, the effective polarization constant was normalized by the microtubule length for the purpose of comparison between the frequencies. The magnitude of α/l was almost constant in the frequency range of 30–700 kHz. In the higher frequency range, α/l gradually declined with an increase in the frequency, and the orientation of the microtubule was not observed at frequencies >7 MHz. The gradual decline of α/l is not an artifact due to the voltage drop at high frequencies (see Materials and Methods for details). In contrast, at frequencies <30 kHz, the electroorientation was accompanied by fluid convection, and α/l measured at 10 kHz showed a large deviation.

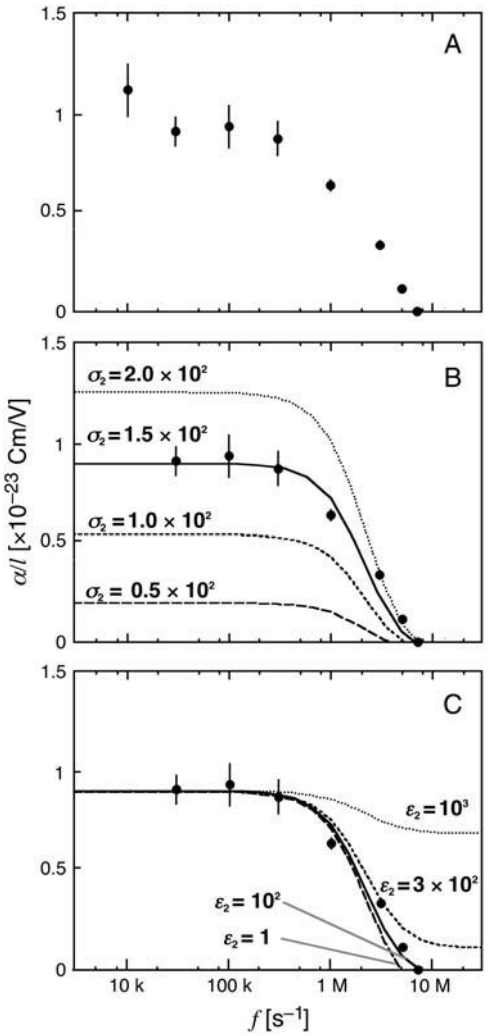

FIGURE 4 Spectrum of the effective polarization coefficient. (*A*) The effective polarization coefficient normalized for the microtubule length, α/l, was measured as a function of the frequency of the electric field, f. The field strength is just sufficient to orient the microtubules; $E = 0.71$ (<700 kHz), 0.88 (700 kHz), 1.2 (1 MHz), and 1.7 (>1 MHz) $\times$ 10^5 V/m. Scale bar represents the standard error for 15–20 measurements. (*B*) α/l simulated by Eqs. 3, 9, and 10, assuming $\varepsilon_2 = 10$ and $\sigma_2 = 0.5$, 1.0, 1.5, and 2.0 $\times$ 10^2 mS/m, respectively. The data obtained at 10 kHz was excluded in this simulation. (*C*) α/l simulated by the same equations, assuming $\sigma_2 = 1.5 \times 10^2$ mS/m and $\varepsilon_2 = 1$, 100, 300, and 1000. In both *B* and *C*, dots represent the measured values.

Fitting the spectrum by Eqs. 3, 9, and 10 using the nonlinear least-squares Marquardt-Levenberg method (with Maxima and GNUPLOT), σ_2 was estimated to be $1.5 \pm 0.1 \times 10^2$ mS/m (mean $\pm$ SD of the fit) (Fig. 4 *B*). ε_2 was estimated to be <100, but the exact value was not determined. As shown in the simulations, the spectrum is insensitive to the variation of ε_2 when ε_2 is <100 (Fig. 4 *C*).

The microtubule conductivity of 1.5×10^2 mS/m is ~15 times that of the surrounding medium. Such a high conductivity can be attributed to counterion polarization (22,24,43). A microtubule is a highly negatively charged polymer and is therefore expected to attract counterions from the solution. These counterions are loosely bound to the microtubule, yet they are mobile along its length. In response to the applied electric field, these counterions may move along the long axis of microtubule, resulting in a high microtubule conductivity.

If this interpretation is correct, according to the counterion condensation theory, the conductivity of microtubules should be affected by both their charge density and the counterion species (44,45). To validate this, the following experiments were conducted. Since the spectrum analysis showed that measurement at one frequency is enough to calculate the microtubule conductivity (Fig. 4), the measurements were performed at 1 MHz.

Effect of ion valence on polarization

The counterion polarization hypothesis predicts that the conductivity of polyelectrolytes might depend on the ion species in the solution. On the other hand, if the electro-orientation is caused by the electronic polarization of the tubulin molecule itself, the conductivity might not depend on the ion species in the solution.

To examine the influence of ion species on the conductivity of microtubules, the electroorientation was examined using various species of monovalent/multivalent ions (Fig. 5). Electroorientation of the microtubules was measured in a solution containing 0.90–1.00 mM MES (pH 6.8), 10 μM taxol, and 10 μM of one of the following salts: KCl, NaCl, $ZnCl_2$, $MgCl_2$, $[Co(NH_3)_6]Cl_3$, or $AlCl_3$. Although the microtubules become unstable in the absence of EGTA and Mg^{2+} (46), the filaments remained intact for several hours provided 10 μM taxol was included in the solution. The result showed that in the presence of monovalent ions such as K^+ or Na^+, the microtubule conductivity was ~20% higher than that measured with divalent cations. In contrast, in the presence of trivalent cations, the conductivity was ~75% of that measured with divalent cations.

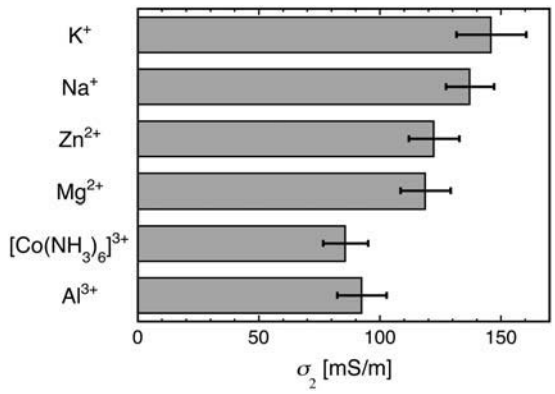

FIGURE 5 Effect of ion species on microtubule conductivity. Conductivity of the solution was adjusted to 9.5 $\pm$ 0.2 mS/m by controlling the concentration of the MES buffer. $E = 1.2 \times 10^5$ V/m. Error bar represents the standard error for 17–26 measurements.

The dependence of microtubule conductivity on ion species might be consistent with the counterion polarization hypothesis. In comparison with cations with lower valences, cations with higher valences are expected to be more strongly bound to the negatively charged surface of the microtubules due to Coulomb interaction (23,24). With an increase in the number of valences, the fraction of counterions being trapped in the so-called Stern layer would increase (34,47), and these trapped counterions should have less mobility compared to the counterions in the diffuse electric double layer (48). Consequently, the conductivity of microtubules in a solution containing ions with higher valences is expected to be lower than that in a solution containing ions with lower valences.

Effect of charge density on polarization

According to the counterion condensation theory, the formation of counterion clouds depends on the linear charge density of the polyelectrolyte. If the linear charge density of the polyelectrolyte is greater than the critical value ($q_{\text{crit}} = 1/z\lambda_B = 4\pi\varepsilon_0\varepsilon kT/e^2$, where ε is the relative dielectric constant of the polyelectrolyte, ε_0 is the permittivity of vacuum, kT is the thermal energy, e is the elementary charge, z is the valence of the counterion, and λ_B is the so-called ''Bjerrum length''), the counterions will condense around it (24,43). In such a situation, the number of mobile counterions on the polyelectrolyte is expected to be a linear function of its charge density.

To examine the actual relationship between the conductivity and charge density of microtubules, we analyzed the effect of pH on microtubule polarization. By changing the pH of the solution, we varied the degree of dissociation of the charges in tubulin and accordingly modulated the charge density of the microtubules. The result showed that the microtubule conductivity increased as the pH of the solution became farther from the isoelectric point of the microtubules (p$I \approx 4.2$ (49)) (Fig. 6 *A*). We were unable to measure the electroorientation below pH 5.5 because of microtubule depolymerization.

The microtubule conductivities were replotted as a function of the number of net charges per tubulin dimer under each pH condition, which was calculated based on the amino acid composition of tubulin (Fig. 6 *B*). The conductivity appeared to increase linearly with the number of net negative charges, with the x-intercept at 20 e per tubulin dimer. This result is consistent with the counterion condensation theory, and the threshold of net negative charges required for electroorientation (20 e per tubulin dimer) may correspond to the critical charge density for counterion condensation.

Baker et al. (12) recently calculated the electrostatic potential for a microtubule structure based on the Poisson-Boltzmann equation. Their result demonstrated that the ribs of a strong negative potential field are formed along the ridges of protofilaments, each intervened by small patches of a positive potential field. This suggests that counterion condensation occurs along the ridges of individual protofilaments rather than occurring homogeneously along the entire surface of the microtubule filament. In keeping with this idea, the threshold of net negative charges derived from our measurement—20 e per tubulin dimer—corresponds to a linear charge density of 2.5 e/nm along a protofilament, which is of the same order as the critical charge density theoretically predicted for a linear polyelectrolyte ($q_{\text{crit}} = 1.4$

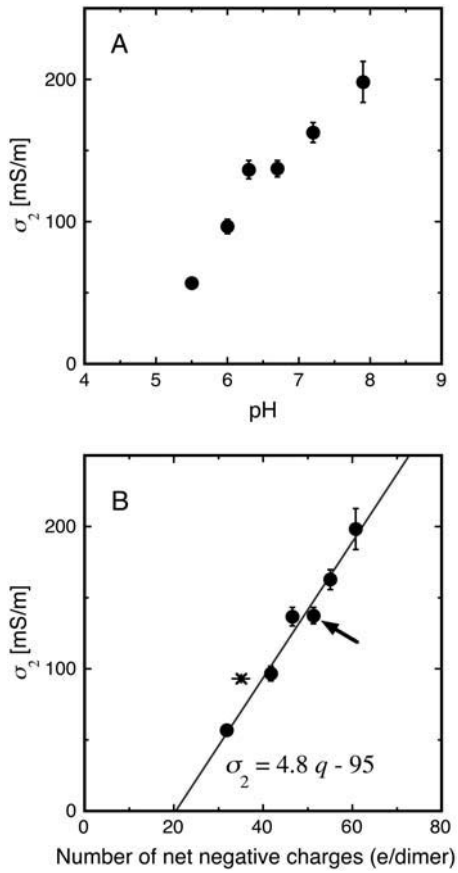

FIGURE 6 Effect of charge density on microtubule conductivity. (*A*) Conductivity of microtubules (σ_2) measured as a function of pH. Either MES (pH 5–7) or HEPES (pH 7–8) was used to adjust the pH of the medium, whereas the conductivity of the medium was kept constant by controlling the concentration of the buffer solution (9.7 ± 0.3 mS/m). $E = 1.2–1.7 \times 10^5$ V/m. Scale bar represents the standard error for 30–40 measurements. (*B*) Conductivity of microtubules replotted as a function of the number of net charges per tubulin dimer. The arrow indicates the data point measured for intact microtubules at a pH of 6.7, whereas the cross represents the conductivity of the s-MTs measured at the same pH. Since the subtilisin cleavage site was not unique, a range of all plausible charge densities is represented by a horizontal bar (see Materials and Methods for details). The line represents the best fit ($\sigma_2 = 4.8q–95$). It should be noted that our calculation of the net charges (*abscissa*) includes all the dissociated charges in the tubulin dimer and could thus overestimate the surface charges involved in the counterion condensation. Among the dissociated charges included in our calculation, those of some amino acids that are not directly exposed on the microtubule surface may or may not contribute to the counterion condensation.

e/nm for monovalent counterions and 0.7 *e*/nm for divalent counterions (see 24,43)).

Contribution of the C-terminus of tubulin to counterion polarization

In both tubulin α- and β-subunits, negative charges are highly concentrated in the C-terminal region. In each subunit, the disordered C-terminal region of a tubulin emerging from $\alpha-$helix 12, composed of only ~20 amino acid residues (a.a.), contains ~50% of the net negative charges (Table 1) (11). To assess the contribution of this highly acidic region to the conductivity of the microtubule, we examined the electroorientation of s-MTs, in which most of the C-terminal region, <1.5 kDa in size, was deleted from each subunit (50). The result showed that the conductivity of these s-MTs was ~60% that of the intact microtubules; this is consistent with the number of net negative charges expected for s-MTs (Table 2).

It has been observed that the decrease in the number of negative charges in microtubules—whether caused by the protease digestion or a change in the pH—resulted in a reduced microtubule conductivity. As shown in Fig. 6 *B*, the data points fall on the same regression line, represents a linear relationship between the charge density and conductivity of microtubules. These results indicate that all the charges in microtubules equally contribute to the microtubule conductivity.

Implication of nonspecific electrostatic interactions based on the polyelectrolyte nature of microtubules

Our analyses revealed that the microtubules had a high conductivity, which is probably due to counterion polarization. Counterions are trapped loosely and nonspecifically in the electrostatic field surrounding the microtubules, and they exhibited a behavior typical of counterions in polyelectrolyte solutions. The result was obtained under a low ionic strength condition (~1 mM); therefore, it is not immediately applicable to the physiological ionic condition (>0.1 M).

Nevertheless, under the physiological ionic condition, the fraction of the microtubule surface charges associated with the counterions would be same as that measured under a low ionic strength condition because the number of condensed counterions is determined solely by the charge density of the microtubule. Our result showed a linear relationship between the microtubule conductivity and the charge density with the *x*-intercept at 20 *e* per tubulin dimer (Fig. 6). The result indicates that each protofilament can be approximately considered as a linear polyelectrolyte with the *x*-intercept corresponding to the critical charge density for counterion condensation (24). Thus, based on the counterion condensation theory (23), the fraction of microtubule surface charge compensated by the condensed counterions can be determined by the equation $\theta = 1 - b/z\lambda_B$, where λ_B is the Bjerrum length, z is the valence of the counterion, and b is the average charge spacing of the protofilament. At pH 6.8, b is calculated to be 0.15 nm (which is the size of the tubulin dimer, 8 nm, divided by its charge density, 52.2 *e*); hence, θ is expected to be 79% in a solution of monovalent cations. It may reach 89% in the presence of sufficient divalent cations. The counterion condensation described by this equation is insensitive to the salt concentration in the solution (24,51). In short, at physiological pH and ionic strength, the majority of the microtubule surface charges are expected to be associated with the counterions.

The polyelectrolyte nature of microtubules may underlie the physicochemical mechanism by which microtubules interact with various microtubule-binding proteins. Analogous to the behavior of counterions, some of these binding interactions might be mediated by weak, nonspecific interactions in which the proteins behave simply as polyvalent cations attracted to the negatively charged surface of the microtubules. The interaction does not require the complementary surface of each protein (lock and key mechanism). The stability of the interaction is determined by the charge density of the microtubules and the net positive charges included at the binding site of the proteins. Such nonspecific electrostatic interaction might be involved, for example, in the processive movement of motor proteins such as kinesin and dynein along the microtubule filament (3–5). The weak electrostatic interaction may be advantageous because it may

TABLE 1 Distribution of charges in tubulin (pH 6.7)

	α-tubulin			β-tubulin		
	Intact	w/o C-terminus	C-terminus*	Intact	w/o C-terminus	C-terminus*
Number of residues	451	434	17	445	424	21
Acidic/basic†	65/40	57/40	8/0	62/37	51/37	11/0
Number of net negative charges‡	24.8	16.7	8.1	26.1	17.2	8.9
Relative charge/residue§	5.5	3.8	47.6	5.9	4.1	42.4

*C-terminal tail region that has no secondary-structure assignment (54).
†Number of acidic (Asp and Glu) and basic amino acids (Lys and Arg) in the amino acid sequence.
‡The number of net negative charges of intact tubulin, as well as that of tubulin without the C-terminus, was calculated by the method of Skoog and Wichman (36). The number of net charges in the C-terminus was calculated by subtracting the latter from the former value.
§Calculated as 100 × (number of net negative charges/number of residues).

TABLE 2 Conductivity and net negative charges of intact and subtilisin-treated microtubules at pH 6.7

	Intact MT	s-MT*
Number of residues	896	868–877†
Number of net negative charges	50.9	34.2–38.4
$\delta_2 \times 10^2$ mS/m	1.5 ± 0.07	0.96 ± 0.06

*Subtilisin-treated microtubules.
†The site of subtilisin digestion was estimated by LC/MS analysis (see Materials and Methods for details).

allow the motors some freedom of motion along the longitudinal axes of microtubules during the weak binding state of the motors.

The stability of the nonspecific protein-microtubule interaction might be affected not only by the electrostatic attraction between the binding pair, but also by the entropy gained by the release of monovalent or divalent cations condensed on the microtubule surface (52,53). As described above, the majority of the microtubule surface charges are associated with counterions; therefore, the binding of cationic proteins would involve the release of counterions into solution, which is accompanied by an entropic gain. Accordingly, the thermal fluctuations of the condensed counterion density along the microtubules may affect/modulate the interaction of cationic proteins with the microtubule.

Regarding protein-protein interaction, we have largely focused on the short-range interaction, which is dependent on the specific, complementary structure of each protein. However, the importance of the weak electrostatic binding interaction, which is probably effective over a long range, has not been recognized until recently (18,34). A better understanding of the latter type of interaction may ultimately lend a new perspective to the function of intermolecular interactions mediated by cytoskeletal filaments. The experiment system developed in this study provides a useful tool to evaluate the polyelectrolyte nature of microtubules. Since the method requires only a small amount of sample ($\sim$1 μg), it is applicable to isotypically pure microtubules obtained by genetic engineering (Katsuki et al., unpublished). A combination of dielectric measurements and mutational analyses may facilitate future studies aiming to understand the fundamental physicochemical mechanisms underlying the intermolecular interactions of microtubules.

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Dielectric polarization, electrical conduction, information processing and quantum computation in microtubules. Are they plausible?

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The multitude and diversity of functions performed by the cytoskeleton of eukaryotic cells poses a major scientific puzzle. Microtubules, which are the main components of the cytoskeleton, are engaged in such important activities as material transport, cell motility, cell division, signal transduction and possibly information processing within axons of nerve cells. The latter aspect has been recently brought to the forefront of consciousness studies by R. Penrose and S. Hameroff. In this paper we discuss the potential of microtubules as information processing units from a physical standpoint. In particular, since electric dipoles are a characteristic property of protein molecules, and as such may undergo various ordering phase transitions, it can be expected that microtubules support the existence of various ferroelectrically ordered states. It is also argued that the piezoelectric effect may link electric and elastic properties of a dielectric polymer system and hence explain a number of experimental observations both *in vitro* and *in vivo*. Furthermore, preliminary results are shown of recent quantum mechanical calculations for the electrical conduction properties of microtubule protofilaments which support the assertion that the latter may function under certain conditions very much like semiconducting devices. Indirect evidence links the cytoskeleton with information processing and cognitive function. We discuss some plausible ways in which axonal microtubules can be involved in the functioning of the brain. Finally, we present a hierarchy of energy values for a number of physical and chemical interactions of interest and discuss their relative importance.

Keywords: microtubules; quantum computing; electrical conduction; neurotransmitter release; signal propagation; dielectric polarization

1. What are microtubules and what do they do in the cell?

The physical and chemical design underlying the complex multifunctional behaviour of the cell's cytoskeleton is one of the most intriguing unsolved problems of molecular biophysics. The cytoskeleton comprises actin, intermediate filaments, microtubules and centrioles (which, in turn, also contain bundles of microtubules). Microtubules (MTs) are ubiquitous in eukaryotic cell biology and they perform an astounding array of functions in addition to sustaining the structural architecture of the cytoplasm (Dustin 1984). These are (a) material transport of organelles via motor proteins; (b)

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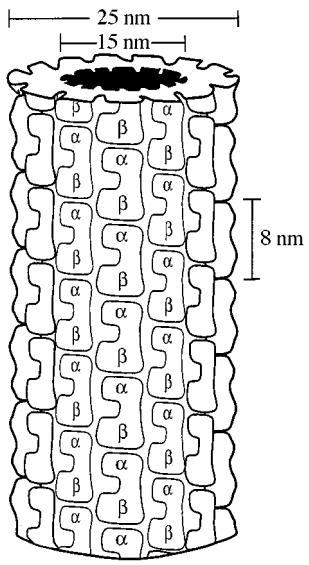

Figure 1. A typical microtubule consisting of 13 protofilaments (vertical columns) Dimensions are indicated. Note the hollow interior and helical packing of tubulin dimers.

cell motility through cilia and flagella; (c) chromosome separation during mitosis; (d) signal transduction within axonal MTs; and most recently discovered (e) a role in the communication between the exterior of the cell and the nucleus (Glanz 1997; Maniotis *et al*. 1997).

Microtubules are hollow cylinders with a 25 nm outer and 15 nm inner diameter (see figure 1) which are formed in the process of polymerization of tubulin protein. Each tubulin molecule is a dimer composed of two highly homologous monomers called α-tubulin and β-tubulin, respectively. The length of each dimer is approximately 8 nm, the width 5 nm and the thickness also 5 nm. These dimers assemble themselves into protofilaments which number 13 almost without exception *in vivo*, although *in vitro* experiments indicate various deviations from the rule to be quite common.

In non-dividing cells, almost all MTs emanate from a centrosome but their individual length constantly varies in a process referred to as *dynamic instability*. In a dividing cell, the above phenomenon is used to form mitotic spindles, which grow at first randomly from the two oppositely positioned centrosomes to eventually anchor themselves to chromosomes and pull the latter apart in the final mitotic phase. During the stable state of a cell, MTs serve as tracks for the transport of vesicles and organelles mediated by various types of motor proteins. In neurons, vesicles are moved along MTs towards synapses by motor proteins.

Two distinct crystal lattices have been seen to form the MT structure and have become known as the A-lattice and the B-lattice (Chrétien & Wade 1991; Amos 1995). The A-lattice with 13 protofilaments is the only lattice without a seam where protofilaments run parallel to the MT axis (figure 2). The assembly and disassembly processes of MTs have been extensively studied over the past two decades both *in vivo* and *in vitro* (Mitchison & Kirshner 1984; Horio & Hotani 1986; Bayley *et al*. 1990; Mandelkow & Mandelkow 1995). The development of a single microtubule passes

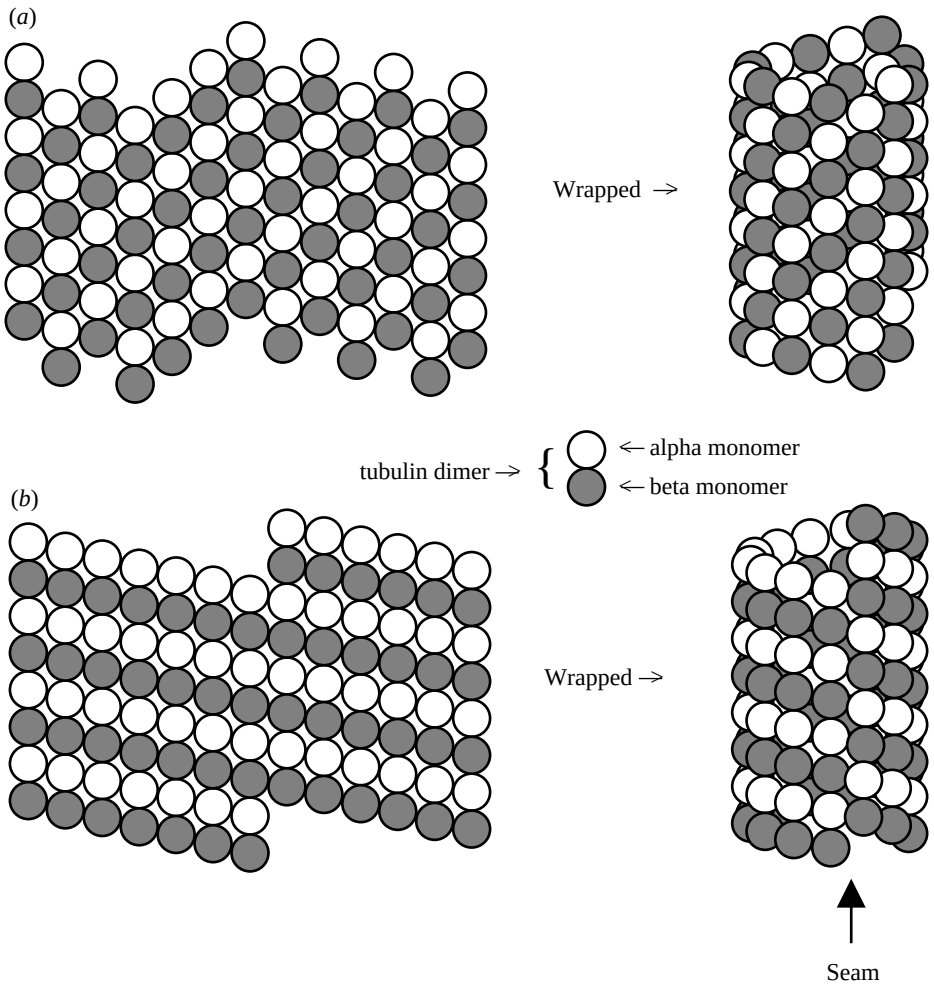

Figure 2. The A and B microtubule lattices: (*a*) the A-lattice exhibits perfect helical symmetry; (*b*) the B-lattice has a structural discontinuity known as a seam.

through several stages starting with a nucleation phase from seed oligomers or microtubule organizing centres (MTOCs) and requiring the presence of γ-tubulin (Sunkel *et al*. 1995). This slow, gradual process eventually reaches an asymptotic density of MT ends and is followed by an almost continuous growth stage which is randomly interrupted by a sudden and catastrophic disassembly followed by another regrowth stage. The ratio of guanosine 5′-triphosphate (GTP) to guanosine 5′-diphosphate (GDP) concentrations in solution determines whether or not and at what rate MTs elongate from a steady-state condition (Engelborghs & Van Houtte 1981). The modelling of MT assembly and disassembly processes *in vitro* as well as *in vivo* through the use of the sophisticated master-equation formalism based on the known chemical reactions and their rates has provided a very accurate and reliable analysis (Flyvbjerg *et al*. 1994). It should be emphasized in this context that MT assembly is a non-equilibrium phenomenon driven by the supply of energy-giving GTP molecules

since only the GTP-tubulin complex is capable of binding to a microtubule. Furthermore, Raman spectroscopy studies revealed that following binding to an MT, GTP attached to the exchangeable site on a tubulin dimer rapidly hydrolyses to GDP, imparting its energy to an unknown destination on the MT lattice. This picture is believed to apply to all but the top layer (or two) of tubulin dimers which have been speculated to form a so-called lateral cap that is required to prevent disassembly.

It is also noteworthy that 'the geometry' of growing MTs (straight protofilaments) differs from that of shrinking ones (curved protofilament ends). This is a direct result of the conformational change that tubulin undergoes upon GTP hydrolysis (Howard & Timasheff 1986). The breaking of lateral bonds first is also consistent with close to a two order of magnitude difference between the elasticity moduli along the axis versus about the circumference in a stable MT; the former being greater.

Finally, it should be mentioned that various types of defects are not uncommon, especially *in vitro*. The number of protofilaments may vary between 12 and as many as 17. MTs often exhibit internal vacancies, breakage points and line defects (Mandelkow & Mandelkow 1994). *In vitro* assembly initially involves a two-dimensional sheet which ultimately closes to form a cylinder and may continue growing as a sheet with the seam closing up behind the growth front. Some researchers claim that the closing of the sheet acts as a trigger for disassembly (Chrétien *et al*. 1995). These problems are almost non-existent *in vivo* where MTOCs provide a template for growth of MTs and MAPs provide additional stabilizing agents (Amos 1995).

We should also add that MTs are highly sensitive to the ionic environment, temperature, pH levels and a number of specific chemicals which bind to their surfaces. For example, Ca^{2+} ions promote disassembly of MTs by enhancing the catastrophe rates (O'Brien *et al*. 1997). The dynamics of MT polymerization are dramatically altered through the use of the drugs: vinblastine (Dhamodharan *et al*. 1995), taxol (Klauber *et al*. 1997) and colchicine (Skoutias & Wilson 1992). An increase in the value of pH of the solution causes an increase in the critical concentration of tubulin for MT formation (Audenaert *et al*. 1989). It has been demonstrated recently that high osmotic pressure in the order of MPa reduces the rate of MT assembly (Barthou *et al*. 1997). All of the above findings are of potential importance in devising new cancer therapies due to the crucial involvement of MTs in the cell division process.

Finally, in this introductory section we should comment on the structure of the constituent protein, tubulin, which has just been resolved at an atomic level (Downing 1998). The development of tubulin appears to be an early evolutionary achievement and a relatively small diversity in the structural isotypes in various species has been found. In a given organism, isotypes of α- and β-tubulin have been found to correlate to some degree with the function performed by the corresponding microtubule, although in some cases isotypes may be interchangeable (Joshi & Cleveland 1990). There are known to be six isotypes of α- and seven of β-tubulin in humans. What complicates the issue of linking structure with function is the plethora of post-translational modifications of tubulin which include phosphorylation, acetylation, detyrosination and glutamylation. It is interesting to note that the greatest divergence among tubulin isotypes is localized to a variable domain called the C-terminus, which is a quasi-linear protrusion from the protein surface containing at least 16 residues (Downing 1998) and possessing a net charge which depends on the solution's pH (Sackett 1997). We believe that this is closely linked to the dipolar properties of microtubules and their role in MTs as carriers of signals. In the next

section we take a closer look at the possibility of ordered dipole phases decorating the surface of a microtubule.

2. Can microtubules be ferroelastic?

Since MTs are protein polymers, there is reason to believe that they are dipolar in character and consequently should be sensitive to electric fields and their gradients. Calculations of electric multipole contributions for a number of proteins have been performed, and the dipole moments found for globular proteins similar in size to tubulin would appear to indicate the existence of a dipole moment in the order of 10^{-27} Cm (Nakamura & Wada 1985; Schlecht 1969). The dipole moment is directly related to each of the peptide groups. Residues of the peptide groups may be either protonated or deprotonated giving rise to a possible net charge on the helix.

According to Sackett (1997) the net charge on a tubulin dimer is pH dependent, being positive at low pH values and highly negative at high pH values. At physiological pH, eight negative charges are expected to exist, which reside mainly in the C-terminus region. Of course, in solution these charges are surrounded by counterions forming a dipolar zone whose thickness corresponds to the Bjerrum length and is of the order of 1 nm at room temperature. If this picture is even approximately correct, it would offer some very attractive possibilities, such as: dipolar orientation predominantly along the protofilament axis, relatively easy to create dipole oscillations, sensitivity to the pH and contents of the solution, cooperative effects, sensitivity to electric fields and their gradients, coupling between elastic deformation and the dipole moment (the piezoelectric effect), etc. Many of the above effects have been experimentally observed and we briefly list them below.

It was suggested a long time ago that bioelectric fields may act to orientate various growth-guiding filaments in the cell (Jaffe & Nuccitelli 1977). Indeed, rearrangement of the cytoskeleton and nuclear transfer was seen in *Tetrahymena thermophila* cells fused by electric fields in the range 300–900 V cm^{-1}. Pronounced effects of electric fields in the 10^3 V m^{-1} range on MTs and actin stress fibres in the fibroblast were observed where filament elongation took place perpendicular to the field gradient and a weakening parallel to it (Harris *et al*. 1990). A possible explanation involves Ca^{2+} influx charges which have been known to be sensitive to low-frequency (3–75 Hz) and low-amplitude electric fields ($E > 10^{-2}$ V m^{-1}) (Berg 1995). Unfortunately, there is only scant evidence indicating direct response of MTs to both electric and magnetic fields. Vassilev *et al*. (1982) observed alignment of MTs in parallel arrays due to the application of both electric and magnetic fields. Electric field intensities used were relatively low, in the range of 20–50 V m^{-1} and were of a pulsed shape. The magnetic field intensity applied was 0.02 T. Similarly, DC electric fields of magnitude 20 V m^{-1} applied in regenerating *Mongeotia* protoplasts resulted in the ordering of MTs (White *et al*. 1990).

If MTs are capable of ordering their dipoles into a ferroelectric state, as we try to argue in this section, this would not be an unusual situation among biopolymers. Indeed, the electret state of most hydrated biopolymers has been found to be a general property of these systems (Fukada 1974). Moreover, bound water was found to strongly contribute to dielectric polarization of biomolecules in solution. Nonlinear effects in transport and polarization storage were discovered which depend on hydration. The nonlinear bioelectret may stimulate ferroelectric hysteresis curves which

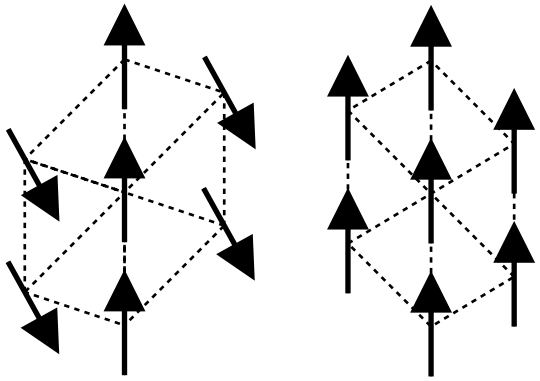

Figure 3. The nearest neighbours of each dimer are arranged as shown in the following diagram. Along the protofilament, parallel alignment is preferred so that the negative heads are close to the positive tails. Between protofilaments, the preferred orientation is determined by the vertical offset between the neighbouring protofilaments.

introduce both memory and irreversibility to the behaviour of these systems (Mascarenhas 1974). Fukada (1974) reported that DNA and RNA are bioelectrets in their hydrated state with a polarization storage as high as $10^{-4}\ \mathrm{C\,m^{-2}}$ at fields as low as $100\ \mathrm{V\,m^{-1}}$. Polarization storage varies drastically with the hydration level, indicating the role of bound water in creating an ordered dielectric state of the complex.

The dielectric constant has a nonlinear dependence upon the protein concentration; this has been linked to the existence of local order in protein molecular assemblies. This is indicative of positional ordering of elongated molecules with permanent dipole moments and it could lead to liquid crystalline properties as well as to molecular electric fields of the order of 10^4–$10^5\ \mathrm{V\,m^{-1}}$ in solutions with protein concentrations of the order of $100\ \mathrm{mg\,mL^{-1}}$ (Post 1995).

In our modelling of dipolar properties of MTs our working assumption was that each tubulin dimer possesses a dipole moment $\boldsymbol{p} = Q \cdot \boldsymbol{d}$, where Q is the net charge capable of transfer over distance d. We have also assumed that the projection of the dipole moment on the protofilament axis can only be $+p$ or $-p$ due to strong anisotropy which constrains the mobile charge to only two locations in the dimer. We chose these locations to correspond to the central region in an α- and β-monomer, respectively (Tuszyński *et al*. 1995). The interaction energy between the neighbouring tubulin dimers, labelled i and j, was taken to be of the standard dipole–dipole form:

$$E_{ij} = \frac{1}{4\pi\epsilon\epsilon_0}\frac{3\cos^2\theta - 1}{r_{ij}^3}p^2, \tag{2.1}$$

where ϵ_0 is the permittivity of the vacuum, ϵ is the dielectric constant of the medium, r_{ij} is the distance between sites i and j and θ is the angle between the dipole axis and the direction joining the two neighbouring dipoles (see figure 3).

As stated earlier, we assume that the electric dipole of the tubulin dimer has two possible orientations depending on the conformational state of the molecule. One orientation of the dipole points along the protofilament axis; this is the *up* state. We have some freedom to choose the direction of the dipole in the *down* state since the electric dipole has not been measured in each of tubulin's conformational states. We

have chosen a direction which is roughly opposite to the direction of the *up* state but which is 29° from the vertical and points out away from the MT. This choice is based on the geometry of MT ends which are splaying apart and of tubulin oligomer rings. Our simulations have been carried out with both the tilted and non-tilted down states for comparison. Note that some degree of tilting is necessary for axonal MTs if action potentials are to influence the dipole dynamics.

However, from the dipole interaction point of view, the lattice type is crucial. In the B-lattice, dimers are aligned in nearly horizontal rows. As a result, there is a strong interaction between neighbouring dimers which favours opposite dipole orientations. In the A-lattice, the neighbouring dipole is shifted vertically such that identical orientation of the dipoles is favoured. In either case, there is a strong interaction along individual protofilaments which favours similarly orientated dipoles. The effect of the dipole interactions is that the lattice of *up* and *down* dipoles will self-organize into an energetically favourable configuration. It can be easily predicted that at high temperatures, the lattice has a random state—each individual dimer can be found in the *up* or the *down* state in a rather arbitrary fashion. At lower temperatures, the interaction energy dictates the dipole arrangement and the lattice becomes ordered. The key question is in what temperature range is the the resultant phase transition from dipolar order to disorder to be expected? The answer crucially depends on the choice of as yet largely unknown physical parameter values.

There are two numbers which are put into the model by hand. One is the relative permittivity of tubulin, or more precisely that of tubulin in cytosol, ϵ. As an approximation, we are using the value 10. The dynamics of the model are unaffected by the choice of ϵ but it scales with temperature. Larger values of ϵ reduce interaction strengths and thereby act like a temperature increase. The other number which is put in by hand is the dipole strength of the conformational states of tubulin. We have chosen the dipole magnitude, p, by considering the corresponding dipole charge to be an elementary charge in magnitude and the charge separation to be 4 nm which is the monomer spacing. This gives a value of $p = 192$ Debye which is comparable to the measured dipole moments of some other protein molecules.

The simulations show that a highly ordered phase exists for both the A- and B-lattices at low temperatures. Whether the MT is ordered at physiological temperature depends critically upon p and ϵ since these parameters determine the transition temperature from disorder to order. In the MT A-lattice with 13 protofilaments, we have found a transition temperature of 105 K and in the MT 13B-lattice, the corresponding transition temperature is 218 K. In the case where the *down* state is directly opposite to the *up* state (the non-tilted case), the corresponding transition temperatures are somewhat higher: 160 K for 13A and 248 K for 13B. This shows that the existence of a transition to order is quite robust but that the transition temperature is sensitive to the specific choice of the *down* dipole. All of these temperatures are below human physiological temperature (310 K, 37 °C) but all have been derived based upon our estimates of the dipole moment, p, and the electric permittivity, ϵ:

$$T_{\mathrm{c}} = T_{\mathrm{c}}^{0}\left(\frac{p}{p_0}\right)^{2}\left(\frac{\epsilon_0}{\epsilon}\right). \tag{2.2}$$

This equation demonstrates the scaling of the transition temperature, T_{c}, in a MT which is not subject to electric fields. The values p_0 and ϵ_0 are the values which we have estimated, while p and ϵ are the actual values of the dipole moment and

permittivity of tubulin. The scaling is more complicated in the presence of electric fields. The transition temperature scales directly with the square of p, and inversely with ϵ. Hence, if the dipoles are twice as large as our estimate then the transition temperature would be raised by a factor of four. Should the permittivity be half as large as estimated, the transition temperature would double. The main result is that the MT could easily exist in an ordered state at physiological temperatures without the application of external fields provided $p^2/\epsilon_{\mathrm{r}} > 10\,800\,\mathrm{Debye}^2$. Our original estimate of $p^2/\epsilon_{\mathrm{r}}$ was about $3600\,\mathrm{Debye}^2$ which comes within a factor of three of what the model requires. We are hopeful that an experimental value for the dipole moment of tubulin will soon be available or a better estimate arrived at using the latest atomic resolution map of tubulin (Downing 1998).

For modelling purposes, in both lattice types the effective Hamiltonian has been represented as a two-dimensional Ising model (Yeomans 1992) on an appropriate triangular lattice whose interaction constants have been calculated based on the geometry of the lattice chosen:

$$H = -\sum_{\langle nn \rangle} J_{ij} S_i^z S_j^z, \tag{2.3}$$

where S_i^z are dipole projection values on the z-axis which can be either $+1$ or -1. The application of an axial electric field, E, along the length of the MT may serve to order the MT at higher temperatures than without an external field and the effective Hamiltonian becomes

$$H = -\sum_{\langle nn \rangle} J_{ij} S_i^z S_j^z - p \sum_i E S_i^z, \tag{2.4}$$

where p is the magnitude of the dipole moment.

Our simulations suggest that for the 13A-lattice, the electric field starts to be effective at the range 10^4–$10^5\ \mathrm{V\ m^{-1}}$. This is because at these fields the potential energy $-pE$ becomes of the order of magnitude of the dipole–dipole interaction energies. These values of the electric fields are at least three orders of magnitude larger than those reported by Vassilev *et al.* (1982) as being used in their experiments on aligning MT assemblies. On the other hand, such strong electric fields may be supported across cell membranes. While the effect of the electric field is quite dramatic in the MT 13A-lattice, for the 13B-lattice the effect is greatly reduced because the orientation of the ground state is not ferroelectric. The dipolar ordering is also affected by the binding of microtubule-associated proteins (MAPs). The influence of MAPs was studied for three different superhelical patterns that attached MAPs formed on a MT. An assumption was made that MAPs prevent the dipoles on the tubulin dimers to which they bind from flipping. MAPs enhance the number of dipoles that point in the same direction as the dipole on the MAP tubulin.

Of the three types of cytoskeletal polymers (actin filaments, intermediate filaments and microtubules), it is the two polar structures which participate in material transport (including axonal transport) and cell motility through the use of their respective motor proteins. It is tantalizing to speculate that it might be this polarization which is responsible for guiding the motor proteins, kinesin and dynein, which travel along the MT in opposite directions (Audenaert *et al.* 1989; Barthou *et al.* 1997). It would immediately explain why the transport would be so efficient because collisions would not occur. The motor proteins would simply have to bind preferentially to a particular conformational state of tubulin. The kinesin and dynein protein motors may

be sensitive to the dipole moments of the tubulin molecules on which they step. An ordered lattice would allow for the efficient transport of material along the lattice. For example, if the lattice were ordered in the same way as the MT 13B-lattice, where alternating protofilaments have opposite dipole orientations, one motor might follow the first, third, fifth, etc., protofilaments while the other would follow the second, fourth, sixth, etc., protofilaments.

3. Can axonal MTs propagate secondary signals?

Neurons have a specialized structure and function. Many of the resources of the cell are spread out in the axons which may project for distances greater than 1 m from the cell body. The cytoplasm within the axon is distinct from that in the cell body. It is generally free of organelles, and filled with microtubules and neurofilaments. These filamentous proteins are arranged parallel to the axon, each neuronal MT being typically about 100 μm long, thus spanning more than 10^5 tubulin subunits. The network of cytoskeletal tubes is interconnected by high molecular weight proteins known as MAPs. Their precise function is not understood but tubulin dimers coassembled with MAPs *in vitro* are polymerized more easily and are more stable than MTs assembled from tubulin in the absence of MAPs (Mandelkow & Mandelkow 1995). The MTs of the axon have uniform polarity and lie with their positive ends distal to the cell body (Baas *et al*. 1988; Baas & Black 1990). Once the axonal MTs have been assembled, they are post-translationally modified and their properties are changed. These changes to the MTs cause them to become more stable (Li & Black 1996; Pryer *et al*. 1992). The post-translational post-MT assembly change has also been frequently studied in the last decade. The transition from newly synthesized tubulin to detyrosinated tubulin within a MT can be used to estimate the age of an individual MT.

Greater stability allows the neuronal MTs to participate in other cell activities. Information processing (Hameroff 1987; Tuszyński *et al*. 1995) and energy transport (Satarić *et al*. 1993) have been proposed as secondary MT functions. Consider the central theme of biology in which DNA codes for functional proteins. Now, not only is the tubulin which makes up neuronal MTs specific to humans, but it is also specific to nerve cells and is known to be post-translationally modified. The highly specialized nature of the functional proteins suggests that it has been selected to perform a very specific function which we conjecture may be signal transduction.

Hameroff *et al*. (1988) have introduced a cellular automaton model where a discrete charge is associated with each tubulin dimer and can either be localized in the top monomer (α state) or bottom monomer (β state). The MT lattice of dimers in their α or β states has then been considered as a binary biological computer. The Hameroff model (Hameroff *et al*. 1988) found that signals introduced onto the MT 13A-lattice would propagate along protofilaments. The propagation was either bidirectional or unidirectional depending upon the choice of flipping force thresholds and whether they were symmetric or not. The model also admitted the possibility of signals which periodically flipped back and forth between α and β states but did not move along the protofilaments. The moving patterns of defects were named gliders and the non-propagating defects were called blinkers. The other feature of the Hameroff model was its ability to filter input signals. Some would propagate, others would be modified and would subsequently propagate, and still others would be annihilated. Thus, the

cellular automaton model accepted certain patterns and rejected others. The first modification introduced in our work (Brown & Tuszyński 1997) was simply to replace the discrete charges with dipoles as motivated by our earlier discussion in this paper. Some arbitrary torque threshold was required akin to the force threshold of the original model. Despite the relative weakness of dipole interactions, most of the original features of signal transduction were still present in the modified model. The one difference was a shift in the direction of signal propagation from the N–S direction (along the protofilament) to the NW–SE direction (around the helix).

The lattice dynamics were simulated using a standard Monte Carlo technique (Binder 1988). For a given time-step, the energy of the present state and of the opposite state were calculated. Whether a change of state occurs is a random event whose probability is determined by the availability of stored lattice energy, the temperature, and the threshold to reaction. An important additional feature of the model is the removal of the unphysical flip-flop of states which plagued our early attempts at modelling this system. This is removed by forbidding nearest neighbours from both changing states in the same time-step. Thus, when the dipole interaction favours paired dipoles and the current state of the neighbouring dipoles is *up–down*, in the next time-step they may be *up–down* (unchanged), *up–up* or *down–down* but not *down–up*. When a dipole flips, the change in conformational energy is either removed from or added to the lattice at that location. After the evolution step, energy diffuses to neighbouring sites. Our simulations have been carried out with a diffusion constant which is isotropic and small enough that several time-steps are required for localized energy to dissipate. Energy is conserved by the lattice in our current model; none of the energy is returned to the surrounding medium. It is important to note in this context, that the conformational state of tubulin is not coupled to GTP hydrolysis. These two events are separate but GTP hydrolysis can easily induce the conformational change. Some authors have directly linked these two events and on occasion have proposed the opposite causality, that the conformational change of tubulin induces GTP hydrolysis (Chrétien *et al*. 1995).

Consider the interaction of a MT lattice with an action potential, in which a signal can be identified as a particular sequence of antialigned dipoles on an otherwise well-ordered lattice. Associated with this signal or defect is some additional energy which is recovered when the antialigned dipole falls back to its original configuration (see figure 4). The energy may diffuse in all directions so with six nearest neighbours a single defect is not likely to cause neighbouring dipoles to flip since they receive only about a sixth of the required energy. However, a larger defect such as a group of three or more dimers might successfully maintain its integrity. This has been observed in our model; larger defects have larger lifetimes.

The interesting question is: how does the MT respond to the presence of defects on the ordered lattice? A defect could arise by GTP hydrolysis to GDP at the exchangeable site upon addition of an additional tubulin subunit or by the less frequent hydrolysis of GTP at the non-exchangeable site (Semenov 1996). In either case, the energy released might go into changing the conformation of the molecule and its electric dipole. We no longer observe the smooth propagation of signals along the MT unless some additional mechanism is added to the model. There is nothing to direct the propagation of the defect so it takes a random walk about the MT and dissipates its energy. The efficient propagation of signals could be restored by several mechanisms. Such mechanisms include (i) the application of an external

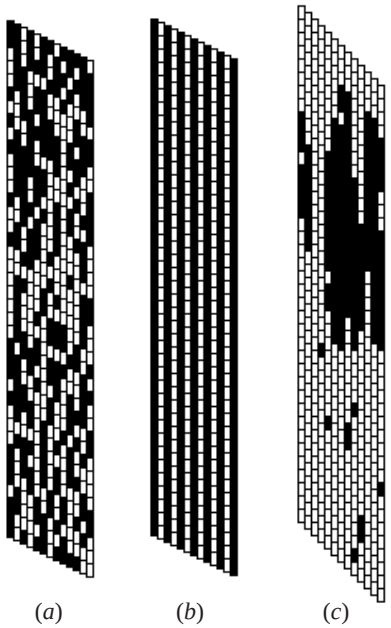

Figure 4. Portions of three MTs are shown. Light boxes represent the 'up' state and dark boxes the 'down' state: (*a*) above the critical temperature of the B-lattice there is disorder; (*b*) below the critical temperature there is spontaneous ordering; (*c*) below the critical temperature of the A-lattice, a large defect (signal) has been introduced. Smaller thermal defects also dot the structure.

field which would bias signal propagation; (ii) an asymmetry in the dipole structure which simply makes it more favourable to propagate in a particular direction; or (iii) mechanical stress if dipoles are coupled to a lattice distortion. The second mechanism could be the result of the bonding between tubulin dimers. If the energy deposited by a dipole flip is comparable with the vibrational energy of a particular bond, it is most likely that this energy would be propagated in that direction. The third mechanism would be the result of a piezoelectric effect, which will be discussed in the next section. There could also be some sort of refractory period which prevents the retrograde propagation of this signal. Since external fields are known to act upon MTs in neurons, the study of these fields and their interaction garners our attention.

As expected, application of a large electric field along the MT causes nearly all dipoles to orientate themselves in the direction which most closely follows that of the field. Thus, a wave of dipole flips is induced along the MT as the field is translated along the MT. This is similar to what happens as an action potential moves along an axon (figure 5). The field is felt by MTs in the vicinity of the cell membrane. Suppose the field is orientated in a direction which favours an alternate ordering for the lattice, such as is the case in MTs, these dipoles will reorientate themselves. The field acts like a pump and stores energy in the dimers. Once the field has passed, these dimers may return to their original configuration and release their stored energy. A weaker field does not actually create defects by changing the orientation of dipoles in the ground state, but may act as a bias and direct the movement of existing defects.

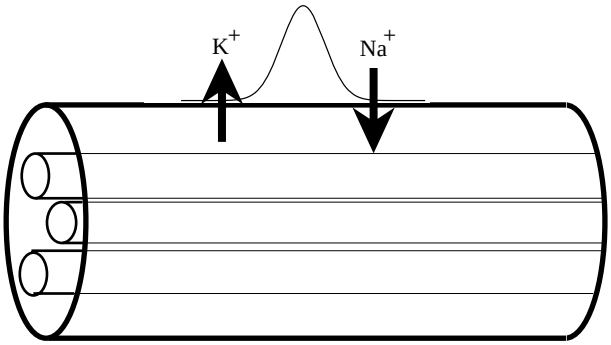

Figure 5. An action potential moving along an axon. There is an electric field caused by the potential difference and a magnetic field caused by the moving charges.

Our simulations have shown this exact behaviour. When the strength of the interacting action potential is large, such as for those MTs in close proximity to the cell membrane, a wave of structural deformation travels parallel to the action potential along the MT. As the wave reaches the MT end, its effects are unclear. Once the MT relaxes to its ground-state structure, energy should once again be available for cellular use if the structural change at the MT terminus is coupled to another cellular structure via a MAP. It might also simply serve as a cellular signal indicating when motor proteins should be activated, MT assembly instructions or any other host of functions associated with the MT terminal which could involve signalling another MT or some part of the axon terminal depending on the location of the individual MAP coupled to an MT.

Once dipole defects have been induced upon the MT lattice, they may propagate along the length of the MT. While some nonlinear effect or external guidance seems to be required, processing of these signals cannot be ruled out. In particular, signal transduction from the distal end of the axon towards the cell body is at least theoretically possible—although unlikely in our opinion. Transduction along the length of the MT is simpler to explain from the proximal end to the distal end though because of the existence of action potentials and the need to explain how a signal may be passed from one MT to another.

Our simulations place a firm limit on the minimum strength of the dipoles required for self-organization. The required dipole strength of about 320 Debye is slightly larger than the value which we estimated for tubulin (as discussed in § 2). In addition, if the angle of the dipole directions of each of tubulin's conformational states is smaller than predicted, an even larger dipole strength would be required. Should the value for tubulin be smaller than this figure, information processing must be ruled out. Energy might still be passed along due to the passage of an action potential, but it would be marked by the highly ordered configuration of the signal upon the random background. The quantity of energy associated with such an entropic change is much smaller than other values mentioned, about $0.06\,\mathrm{kcal\,mol^{-1}}$.

On the topic of information storage in MTs, the simple introduction of thermal energy has removed the possibility of information storage in MT. The energy involved in flipping the conformation of the dimer would have to be significantly larger for

such a form of storage to be possible; not to mention the need for some kind of mechanism which would preserve the integrity of the information.

It is clear that electromagnetic properties are important in cell biology. In addition, the intrinsic polarity of MTs seems to be very important. If it were not so, one would expect the MTs within the axon to be randomly orientated. The question which might be asked is: what is the specific usefulness of a signal which is coupled to the axonal action potential and which travels along MT protofilaments inside the axon? We shall return to provide a plausible answer in the next section.

There has been additional speculation about the role of MAPs in this discussion. Certain MAPs are specific to neuronal MTs (Hirokawa 1991) and their periodic patterns of attachment are somewhat of a paradox (Dustin 1984; Hameroff 1987). Models have attempted to confirm their organizing effect; it seems that due to the interconnections and intraconnections which MAPs make between neuronal MTs, the possibility of parallel information processing is facilitated. In particular, it has been ascertained that the content of an especially important MAP, τ, increases progressively with distance to the soma (Black *et al*. 1996), indicating that its function may not only be a greater stability and assembly but perhaps enhancement of signal transfer properties. We have not pursued this sort of examination to this point; however, if even the most elementary features of MT cellular automata are experimentally corroborated, the importance of electric properties within cells will be clearly demonstrated.

4. Can signal transduction be piezoelectric?

A piezoelectric solid is usually defined as one which generates a voltage proportional to an applied mechanical force, either due to mechanical distortion or displacement of electric dipoles in its crystal lattice. Biological piezoelectricity has been observed directly in bone, DNA, RNA, myosin and collagen (Cope 1975). Piezoelectric materials are usually also pyroelectric, which means that an electric potential is generated by the application of a temperature gradient. Pyroelectricity has been directly observed and measured in tendon, bone and nerve cells. Athenstaedt (1974) was the first to describe microtubules as piezoelectric biopolymers. Piezoelectric properties of biological polymers have been reviewed by Fukada (1983), indicating that almost all biologically relevant macromolecules exhibit piezoelectricity. Included as an example of piezoelectricity was the mechanism of flagellar beating. The pyroelectric effect has also been listed as common among proteins.

Trying to link piezoelectricity with the properties of MTs, the natural starting point is to recall changes taking place in the tubulin dimer upon assembly. GTP bound to tubulin hydrolyses rapidly into GDP, imparting energy to the MT lattice. Furthermore, comparison studies of GTP and its non-hydrolysable analogue, guanylyl-(a,b)-methylene-diphosphonate (GMPCPP), by Hyman *et al*. (1995) indicate that hydrolysis results in an anisotropic deformation of the protein affected such that the non-hydrolysed monomer length is increased by approximately 1.5 Å with the opposite effect taking place in the perpendicular direction.

At neutral pH, the hydrolysis of GTP in solution proceeds according to the chemical reaction

$$\mathrm{GTP}^{4-} + \mathrm{H_20} \rightarrow \mathrm{GDP}^{3-} + \mathrm{HPO}_4^{2-} + \mathrm{H}^{+} + \Delta E, \tag{4.1}$$

where $\Delta E = 8.7\,\mathrm{kcal\,mol^{-1}}$. It is also worth pointing out that the release of a proton in this reaction may play an important role in both dielectric and conductive properties of the cytoplasm.

It appears that the amount of hydrolysis energy changes drastically when GTP is bound to either a single tubulin or a whole microtubule. This was studied by Caplow *et al.* (1994) using GMPCPP, which stabilizes microtubules. In solution, the hydrolysis of GMPCPP releases $5.18\,\mathrm{kcal\,mol^{-1}}$, which is reduced to $3.79\,\mathrm{kcal\,mol^{-1}}$ for GMPCPP attached to a tubulin dimer and to only $0.90\,\mathrm{kcal\,mol^{-1}}$ when GMPCPP is embedded in a microtubule structure. The difference in these numbers is believed to indicate the amount of energy stored in the microtubule structure. However, the last figure also indicates that upon assembly an excess of energy is imparted to the tubulin subunits without an apparent designation. It has been suggested that this excess amount of energy could be used to change protein bond strengths, or cause conformational changes of tubulin, or even be stored as fault energy of the lattice which may be released later through a collapse catastrophe. This energy can conceivably be harnessed in the form of mechanical work, for example, by coupling disassembly to the vesicle or chromosome movement or even providing the coordinated behaviour of the ciliary motor proteins.

Several groups of researchers have developed nonlinear models of energy propagation along a microtubular protofilament. Chou *et al.* (1994) showed that kinks and pulses excited by the energy freed in the GTP hydrolysis can propagate along MTs due to elastic coupling between tubulin dimers. Another model of solitary wave formation was proposed by Sataric *et al.* (1993) and further developed by Trpišova & Tuszyński (1997). The working assumption here is that tubulin dimers possess dipole moments which are spontaneously ordered at physiological temperature. Consequently, solitary waves carrying the free portion of the hydrolysis energy correspond to kink-type domain walls connecting two regions of the protofilament with opposite electric polarizations.

The physical model is based on the lattice Hamiltonian

$$H = \sum_{n=1}^{N}\left[\tfrac{1}{2}M\left(\frac{\mathrm{d}u_n}{\mathrm{d}t}\right)^2 + \tfrac{1}{2}K(u_{n+1}-u_n)^2 - (\tfrac{1}{2}\alpha_2 u_n^2 - \tfrac{1}{4}\alpha_4 u_n^4) - cu_n\right], \qquad (4.2)$$

where u_n represents the projection on the protofilament axis of the dimer's conformational change (coupled to the dipole moment) and the various other terms represent the kinetic energy, the elastic restoring energy and the effective nonlinear potential, while the last term is responsible for the effect of external (electric or pressure) fields on the system of coupled dipoles placed on tubulin molecules, each of mass M.

Furthermore, cellular MTs are embedded in cytosol, which is a liquid ion-filled solution which interacts with the MTs both via electric forces and by providing a viscous medium which damps the motion of tubulin dimers. This latter effect can be modelled by including a frictional force $F_{\mathrm{v}} = -\gamma\partial u_n/\partial t$ in the equation of motion. Taking the continuum limit such that $u_n(t) \to u(x,t)$ yields the following equation of motion for the displacement variable:

$$M\frac{\partial^2 u(x,t)}{\partial t^2} - KR_0^2\frac{\partial^2 u}{\partial x^2} - \alpha_2 u + \alpha_4 u^3 + \gamma\frac{\partial u}{\partial t} - q_{\mathrm{eff}}E = 0. \qquad (4.3)$$

Somewhat surprisingly, an analytic kink-type solution has been found in the form

$$u(\xi) = u_2 + \frac{u_1 - u_2}{1 + \exp[(u_1 - u_2)\xi/\sqrt{2}]}, \tag{4.4}$$

where u_1 and u_2 are the real roots of the cubic polynomial in equation (4.3) and ξ is the moving coordinate $\xi = x - vt$. The sound velocity of MTs was estimated elsewhere as $v = 610\,\mathrm{m\,s^{-1}}$ (Sirenko *et al.* 1996). The damping coefficient γ was estimated to be $\gamma = 5.2 \times 10^{-11}\,\mathrm{kg\,s^{-1}}$ (Trpišova & Tuszyński 1997). Consequently, it was determined that for sufficiently small electric fields E, the velocity of kink propagation, v, is proportional to the electric field strength E and it ranges from $10^{-2}\,\mathrm{m\,s^{-1}}$ for $E \sim 2 \times 10^5\,\mathrm{V\,m^{-1}}$ to $10^{-4}\,\mathrm{m\,s^{-1}}$ for $E \sim 10^2\,\mathrm{V\,m^{-1}}$. These numbers appear to be on the high side of the known MT-associated processes (assembly, disassembly speeds or motor protein velocities) but one should also be aware of the great sensitivity of v to the dielectric constant of the medium and to the presence of impurity potentials along the path of a moving kink which was studied numerically. In fact, for large enough potential wells or bumps, kink motion can be stopped altogether (Trpišova & Tuszyński 1997). The question arises as to what happens to the energy transported by a kink excitation. We may speculate that it is transferred to another MT via an attached MAP or that it could play an important role in the movement of a motor protein. Another possibility is that it creates a domain wall between two portions of a MT protofilament in which tubulin dimers are in different dipolar and conformational states. This could be an important factor in developing microscopic models of dynamic instability.

Models have also been developed which explicitly account for the coupling between tubulin dipoles and the conformational (elastic) degrees of freedom (Trpišova & Tuszyński 1997). A phenomenological Landau–Ginzburg free-energy expansion has been formulated for the entire stable MT in terms of the total dielectric polarization P and the mechanical stress σ which is coupled to P via the piezoelectric effect. The free energy then takes the form

$$F = F_0 + \tfrac{1}{2}AP^2 - \tfrac{1}{4}BP^4 + \tfrac{1}{6}CP^6 - \tfrac{1}{2}e\sigma P^2 + \tfrac{1}{2}D(\partial P/\partial x)^2 - \tfrac{1}{2}s_0\sigma^2. \tag{4.5}$$

The time-evolution of the order parameter (net polarization P) is given by the time-dependent Landau–Ginzburg equation

$$\frac{\partial P}{\partial t} = -\Gamma\frac{\delta F}{\delta P}, \tag{4.6}$$

where Γ is the phenomenological Landau–Khalatnikov damping coefficient providing a time-scale for relaxation processes. In our case this equation takes the form

$$\frac{\partial P}{\partial t} = \Gamma(AP - BP^3 + CP^5 - e\sigma P) + \Gamma D\frac{\partial^2 P}{\partial x^2}, \tag{4.7}$$

and it is coupled to the stress–stain relationship

$$\epsilon = \frac{\partial u}{\partial x} = -\frac{\partial F}{\partial \sigma} = \tfrac{1}{2}eP^2 + s_0\sigma. \tag{4.8}$$

These two equations can be effectively decoupled by taking note of the fact that ϵ also satisfies the wave equation

$$\rho\frac{\partial^2 \epsilon}{\partial t^2} = \frac{\partial^2 \sigma}{\partial x^2}, \tag{4.9}$$

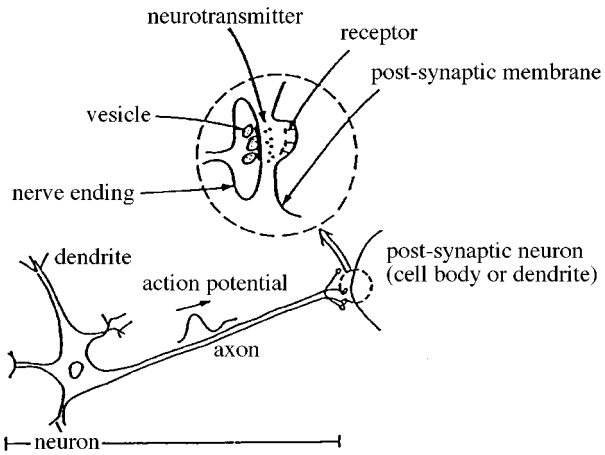

Figure 6. Schematic diagram of a neuron. Inset shows a neuron–neuron synapse.

which results in an effective equation for travelling polarization waves:

$$\Gamma D\frac{\mathrm{d}^2P}{\mathrm{d}\xi^2} + v\frac{\mathrm{d}P}{\mathrm{d}\xi} - \Gamma(AP - \tilde{B}P^3 + CP^5) = 0, \tag{4.10}$$

where $\xi = x - vt$ and $\tilde{B}$ is a 'dressed' expansion coefficient. This model was also studied numerically (Trpišova & Tuszyński 1997). The polarization wave describes a localized nucleation which moves with a constant velocity along a protofilament. Above the transition temperature for the paraelectric phase to the ferroelectric phase transition, which could be in the physiological range for most of electric field values, collisions of coupled $P - \sigma$ waves with an impurity potential result in the formation of a constant stress along the MT. This could lead to a build-up of fault energy and a subsequent disassembly. This model has also been analysed in the presence of an external pressure applied to a MT. Several effects have been seen in these latter simulations: (a) a decrease in the kink's propagation velocity down to the $\mu\mathrm{m\ s}^{-1}$ range when hydrostatic pressures are applied; (b) a dielectric phase transition which may be induced by applying pressure to the microtubule. Some numerical estimates were made and compared to the observed values of Young's modulus, Y, in the protofilament direction where $Y = 0.5 \times 10^9\ \mathrm{N\ m}^{-2}$ has been reported (Mickey & Howard 1995). Our estimate of the longitudinal stresses which accompany the propagation of a kink along the protofilament is $\sigma = 10^6\ \mathrm{N\ m}^{-2}$.

We conclude that nonlinear models of energy transfer along microtubules provide a conceptual link between the excess energy released in GTP hydrolysis directly after tubulin assembly and the various processes associated with the functions of MTs, especially motor protein motion, chromosome segregation forces and disassembly velocities.

In the context of neuronal MT functioning, the piezoelectric properties of MTs may offer a very attractive connection between the internal structure of the axon and its function, namely synaptic communication (see figure 6). The first link that may be established has to do with the control mechanism of exocytosis of neurotransmitters. Segel & Parnas (1991) argued convincingly that the Ca^{2+} wave in the extracellular medium is insufficient to explain quantitatively the sensitivity of the neurotransmitter release mechanism to depolarization at the release site. They pro-

posed the existence of an additional polarization mechanism which, in our opinion, may well be the coupled polarization–stress wave propagating along the MT that accompanies the action potential motion.

The other possible link involves the architecture of neurons which number 10^{10} in the brain and, through sensory stimulation, each of them develops an extensive dendrite tree with a signal target made up of 10^3–10^5 other neurons. These nearest-neighbour neurons are connected via synapses mentioned above. In some neurons over 99% of their surface area is accounted for in the dendritic tree. The functioning of the dendritic trees consumes up to 60% of the brain's energy (Mel 1994). How can individual synaptic connections within such a tree be maintained or abandoned depending on their usefulness or lack thereof, respectively? As argued earlier, when the action potential travels along the axon it affects the MTs dipoles such that the GTP energy that MTs constantly receive from the cell is smoothly transferred from one MT end to the other. We found that the propagation of dipole kinks is feasible even at velocities in the 1–20 $\mathrm{m\,s^{-1}}$ range, roughly corresponding to the action potential velocity of propagation. We also stressed here that there is electroconformational coupling in MTs. That means that stable MTs have straight protofilaments (tense). However, unstable MTs have protofilaments which curve away from the axis (relaxed). Instability comes about when too much energy is delivered to MTs and it is stored in them without an outlet. For example, it can get stored in the form of mechanical fault energy, causing crystal lattice defects. Therefore, when action potential signals are very infrequent, MTs may have difficulty getting rid of excess energy, so they become curved or full of defects. This would eventually lead to several possible modes of MT degeneration: (a) falling apart of the middle part if, for example, MAPs are also weakened. This could be the case with Alzheimer's disease where τ protein does not clamp onto the scaffold structure of MT bundles. The consequence would be that axons themselves would be affected and synaptic connections destroyed. (b) If they don't fall apart, they may still experience negative effects due to piezoelectricity. The MTs would, for example, build unidirectional stress that could curve their axis away from the straight line of the synaptic connection. The result could be the two ends on the opposite sides of the synapse moving away from each other. They may eventually find other new partners with which to pair up within the dendritic tree. The latter possibility would explain why synaptic connections are eliminated if they are not used sufficiently.

5. Can MTs conduct electrical currents?

Conjectures regarding the possibility of electrical conduction in biological polymers were originated by Szent-Györgyi in the 1940s and elaborated on later by Pullman & Pullman (1962) who pointed to the role of delocalized π-electrons in biochemical compounds. They predicted the existence of both extrinsic and intrinsic semiconducting properties of protein polymers. Pethig (1982) performed band-structure calculations for some simple protein polymers and found the band gap to be of the order of 6.0 eV with a band width of approximately 1.5 eV, indicating that they would most likely be insulators unless disorder in the structure was introduced to drastically reduce the energy gap.

Table 1. *Potential functions of molecular complexes as bioelectronic materials*

function	molecular complexes
wiring	polyene antibiotics conductive polymers
storage	bacteriorhodopsin reaction centres (PS II) cytochromes blue proteins ferritin collagen, DNA
gates and switches	bacteriorhodopsin photosynthetic systems cell receptors ATPase
input/output devices	photosensitive proteins enzymes receptors metal–protein complexes

Table 2. *Abundance of n-type and p-type amino acids in α- and β-tubulin*

amino acid	electronic type	% in α-tu	% in β-tu
Asp	*n*	10.3	11.1
Lys	*p*	3.9	3.8

A general discussion regarding the potential for bioelectronic devices based on molecular complexes predicts a range of fascinating possibilities, as shown in table 1 following Tsong (1989) and Wangermann (1980).

Based on table 1, one might expect MTs under suitable conditions to provide wiring for the transmission of electrical pulses (e.g. inside axons or as mitotic spindle apparatus) or perform input/output functions using electromagnetic signals if they can be photosensitive under special circumstances (as is believed to be the case in centrioles and in the phototaxis mechanism of cilia). A recent paper (Insinna *et al*. 1996) has given an exhaustive account of the accumulated evidence for the participation of MTs in electrical conduction within the cell via electron or proton transfer mechanisms which can be coupled with GTP hydrolysis. In fact, Koruga & Simić-Krstic (1990) pointed out that semiconducting properties of MTs are to be expected simply on the basis of the tubulin's constitution from the linear *n*-type (Asp) or *p*-type (Lys) amino acids. This is shown in table 2 and would indicate that there is approximately three times more *n*-type than *p*-type amino acids based on the number of amino acid residues and, hence, one expects an excess of mobile electrons to exist.

Using the above premise we have undertaken to examine the issue of conductivity in a theoretical detail. For comparison purposes, we assumed the presence of just

two conduction electrons for each tubulin dimer, leading to the conduction electron density of $n_0 = 10^{19}$ cm^{-3}. This may be an overestimation, since in comparison with other materials, this value is higher than typical semiconductor concentrations, usually 10^{13}–10^{17} cm^{-3}, but is much less than typical metals which have a conduction electron density of 10^{22}–10^{23} cm^{-3}. However, the electrons are expected to have relatively low mobilities unless they are excited from the ground-state configuration. The excitation energy is a function of both the number of electrons within the protofilament and its size. Suppose that the mobility of excited electrons is at least similar to the mobility of electrons within semiconductors such as Si, where the electron mobility, μ_e, is about 1300 cm^2 V^{-1} s^{-1}. The conductivity, σ, is given in a semiclassical model by

$$\sigma = ne\mu_e, \tag{5.1}$$

where n is the conduction electron density and e is the electron charge. The electron density is given as

$$n = n_0 e^{-\Delta/kT}, \tag{5.2}$$

where n_0 represents the total number of electrons per cm^3. If we assume that the gap between valence and conduction bands, Δ, is about eV, which is the small barrier height in our model, then at physiological temperature, some of the carriers will be thermally excited and should be available to conduct. Applying equation (5.1), a conductivity of about 10^{-2} is expected for the protofilament:

$$\sigma \sim (10^{25} e^{-0.40/0.026})(1.6 \times 10^{-19})(0.13) \sim 0.04\,\Omega^{-1}\,\mathrm{m}^{-1}. \tag{5.3}$$

However, if the hybridization of electronic orbitals lowers the gap to a value closer to zero, the expected conductivity would be significantly higher:

$$\sigma \sim (10^{25})(1.6 \times 10^{-19})(0.13) \sim 2 \times 10^5\,\Omega^{-1}\,\mathrm{m}^{-1}. \tag{5.4}$$

This is an upper limit on conduction in MTs due to the optimistic choice of electron mobility. This value should be compared to values in excess of 10^8 for copper and other good conductors, 10^7 for lesser conductors and semimetals. Typical semiconductors such as Si and Ge have intrinsic conductivities of 10^{-3} to 10^{+1} at room temperature, but when heavily doped the conductivity may rise to between 10^4 and 10^5.

In subsequent quantum-mechanical calculations each protofilament is pictured as a chain of periodically distributed potential double wells whose two sites represent two binding sites, one each on each monomer between which the electron may hop (figure 7). We can estimate the relevant parameters of the well depth and the well width from kinetic studies and with a knowledge of the molecule's overall geometry. Once the bound states and their energies are known for this system, this knowledge may be applied to a second-quantization scheme. Hopping along the protofilament by electrons presumably has an energy comparable to the binding energy. We have estimated 0.4 eV for intradimer hopping. Thus, a single hydrolysis event is not expected to cause disassembly.

The results for the single-electron eigenstates of a double quantum well may be found in a rather straightforward manner using a transfer matrix method for the eigenfunctions of the time-independent one-dimensional Schrödinger equation:

$$\hat{H}\psi = \frac{\hat{p}^2}{2m}\psi + \hat{V}\psi = E\psi. \tag{5.5}$$

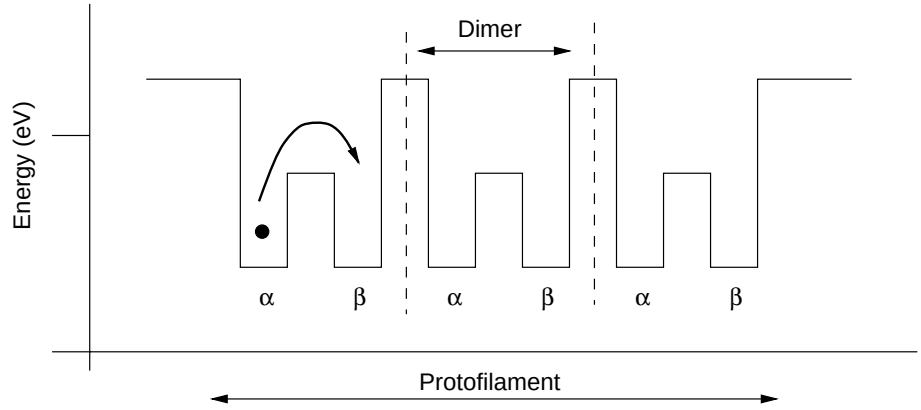

Figure 7. Model of the MT protofilament with a binding site located on each α-monomer and β-monomer.

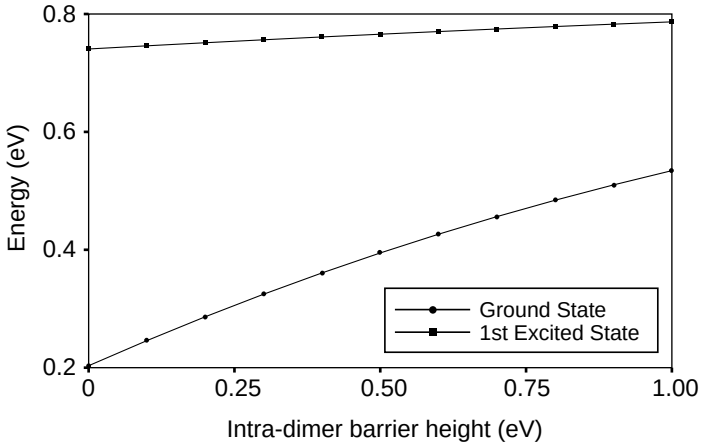

Figure 8. Variation of the bound-state energies of a dimer with the central barrier height.

In figure 8 we show the results of our numerical solutions for the two bound states of a single dimer (ground and excited) as a function of the energy of the central potential barrier. One may then calculate the energy of such states for a sequence of n such dimers. This leads to the development of bands as hybrid orbitals stretch across the polymer. Roughly speaking, one band is of hybridized symmetric orbitals and the high-lying band of hybridized antisymmetric orbitals (figure 9). The separation of the two bands and the spread of each individual band is important. They determine the values of the parameters which shall be used in the tight-binding Hubbard model which shall be used to model the polymer. The Hubbard model is the simplest second-quantized model to capture all of the essential features of a real system of interacting particles (Lieb 1995). In this model, the particles, i.e. electrons, are fundamental excitations described by the Hamiltonian

$$\hat{H} = \sum_{i,\sigma} \epsilon_i \hat{C}^{\dagger}_{i\sigma} \hat{C}_{i\sigma} + \sum_{i \neq j,\sigma} t_{ij} \hat{C}^{\dagger}_{i\sigma} \hat{C}_{j\sigma} + \sum_{i} U_i \hat{C}^{\dagger}_{i\uparrow} \hat{C}_{i\uparrow} \hat{C}^{\dagger}_{i\downarrow} \hat{C}_{i\downarrow}, \tag{5.6}$$

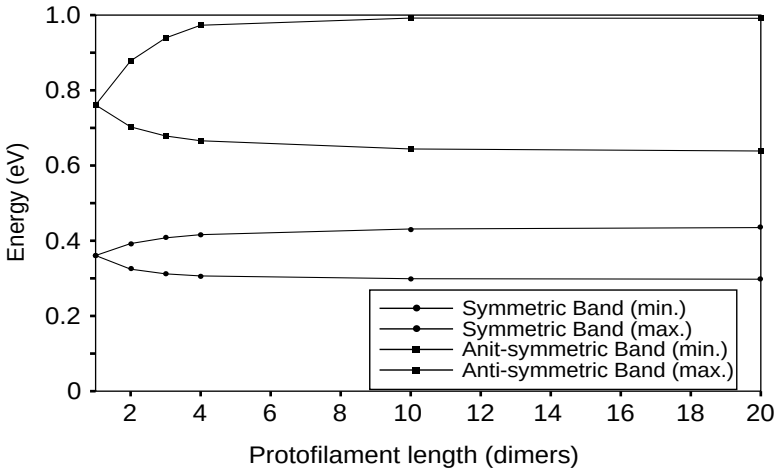

Figure 9. Band formation as the protofilament is lengthened.

where the operators $\hat{C}^{\dagger}_{i\sigma}$ and $\hat{C}_{i\sigma}$ are the usual fermion creation and annihilation operators, and where $i = 1, \ldots, 2n$ represent the two sites on each of the n dimers. The ϵ_i represent the potential energy at each site and the kinetic terms t_{ij} account for hopping from one site to another. We are accounting for nearest-neighbour interactions in the model, but there are two different kinds of interactions depending on whether an electron hops within a dimer, or between dimers. Hence, this kinetic parameter has two different values along the chain: t_{d} or t_{l}, respectively. The final term accounts for electron–electron repulsion now that we have gone to a many-electron system. In our simple model, this is limited to an on-site interaction. To this point, we have made each of the site energies equivalent ($\epsilon_i = \epsilon$); since the Hamiltonian conserves the particle number, this term may be ignored as a trivial constant. We have also made the electron repulsion the same at all sites ($U_i = U$). We expect $U = 2.0\,\mathrm{eV}$, which is the energy associated with two bare electrons lying about 0.7 nm apart. This choice seems reasonable within a well of width 2.0 nm.

On comparison with figure 9, which shows the band formation in the quantum well formulation, we can assign values to t_{d} and t_{l}. The two bands are separated by about 0.40 eV, so $t_{\mathrm{d}} \approx 0.20\,\mathrm{eV}$ and using the bottom band to estimate t_{l}, it widens until it is about 0.10 eV thick, so $t_{\mathrm{l}} \approx 0.05\,\mathrm{eV}$.

Suppose an external field is applied; we want the conductivity that is the response to the total electric field in the solid. This conductivity takes into account all of the currents induced by the external fields which create their own fields. In other words, we are looking for the self-consistent conductivity:

$$J_\alpha(r,t) = \sum_\beta \sigma_{\alpha\beta}(q,\omega) E_\beta \mathrm{e}^{\mathrm{i}qr - \mathrm{i}\omega t}. \tag{5.7}$$

This tensor, $\sigma_{\alpha\beta}$, is the conductivity we are looking for which relates the macroscopic electric field to the currents of the system which may be measured.

In terms of the Hamiltonian's eigenstates $|n\rangle$ and frequency dependence, conduc-

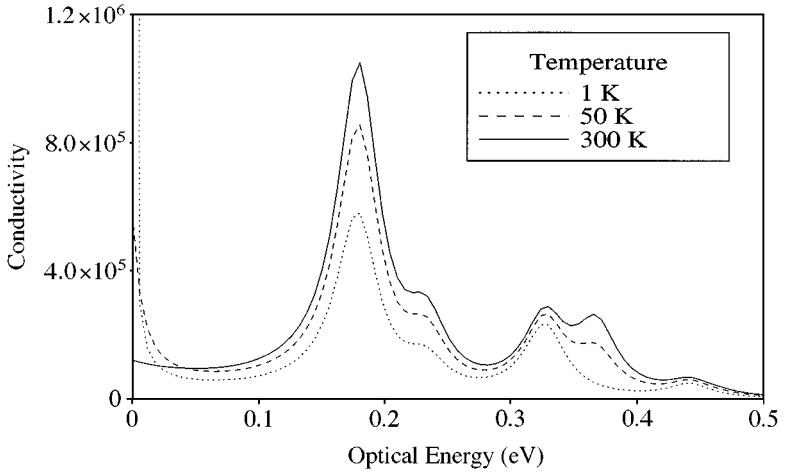

Figure 10. Conductivity is plotted as a function of the optical energy for a short protofilament. The spectrum changes as the temperature is raised. Note the single peak which becomes a double peak close to 0.35 eV.

tivity is expressed as (Mahan 1981)

$$\sigma(\omega) = \frac{\pi}{Z}\frac{1}{\omega}\sum_{m,n}(\mathrm{e}^{-\beta E_m} - \mathrm{e}^{-\beta E_n})|\langle n|j_\alpha|m\rangle|^2\delta(\omega - (E_n - E_m)). \tag{5.8}$$

Modelling the protofilament as a Hubbard chain, the eigenstates are found by exact diagonalization techniques and these are then used as input for the Kubo model of electrical conduction. The δ-functions of the last expression are replaced with Lorentzians of finite width as follows:

$$\delta(\omega - a) \to \frac{1}{\pi}\frac{\epsilon}{(\omega - a)^2 + \epsilon^2}. \tag{5.9}$$

This approximates relaxation processes and the degree of experimental precision. In addition, the temperature which we have chosen, 300 K, is approximately physiological temperature since conductivity is temperature dependent. In the Hubbard chain model of the MT protofilament at 0 K, there is conduction only at the frequencies which are equal to jumps between the ground state and each of the excited states. As the temperature is raised, the excited states may themselves be populated. Consequently, more peaks are observed as transitions between excited states may also contribute to the overall conduction (figure 10). Now, we are interested not only in the peak values of conductivity and how it may vary with the protofilament length, but particularly also in how the conductivity varies with the number of charge carriers. The tubulin which forms protofilaments is composed of two dimers, α and β. Each of these monomers has several isotypes. MTs localized in different places within the body differ in their isotypic composition. The isotypic composition may also change in a given location as a function of the cell cycle (Moskowitz & Oblinger 1995). These isotypes of α- and β-tubulin have different regulatory roles and differ almost superficially along certain side chains. That is, these changes do not seem to affect the MT structure, or cause the assembly properties to change much, but may be of utmost importance to the conducting properties of MTs by introducing electron or hole doping into this picture. For example, distinguishable β-tubulin

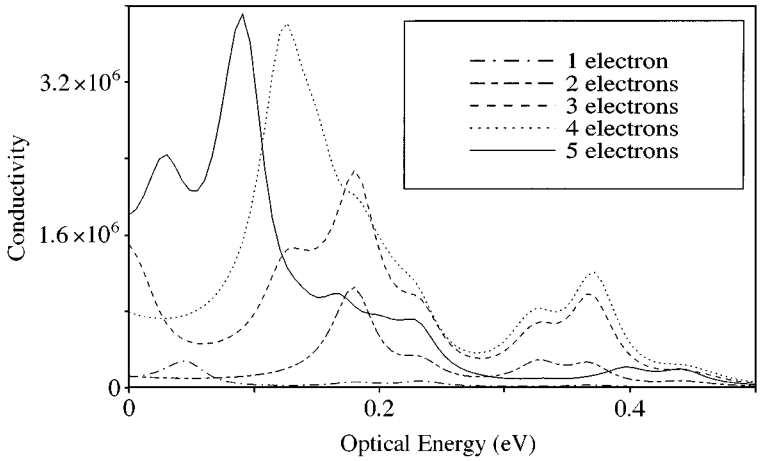

Figure 11. Conductivity is plotted as a function of the carrier concentration.

isotypes have been found to be differentially distributed among the peripheral and central axoplasm. In addition to the interactions these specific isotypes may have with MAPs, certain amino acid residues on these side chains may be hydrolysed or glutamylated to varying extents and this modification may provide additional (or fewer) charge carriers. Consequently, we hope to find the best carrier concentration for conduction which might dictate a specific ratio between the levels of the various tubulin isotypes which should be incorporated into MTs.

Typical MTs in which we are interested span 1000 or more subunits. Such objects which are of the order of a few microns long could be experimentally studied. However, these cannot be tackled theoretically using the present techniques because of computational limitations related to the diagonalization and multiplication of large matrices. As a result, the work is being carried out on short filaments of a few dimers length.

Figure 11 displays a sample of conductivity curves in a protofilament of three dimers with various numbers of electrons. The single-electron case is expected to differ from the others since there is obviously no Coulomb repulsion in such a system and it turns out to have the lowest conductivity. Conductivity seems to increase as electrons are added to the system until it is precisely half full. The situation at half-filling is somewhat of a puzzle but we surmise that the low conduction is the result of more localized electrons at that specific fractional filling. One electron may be confined to each monomer and Coulomb repulsion avoided. However, the movement of any electron ruins this energetically stable state. At other fractional fillings, electron movement is possible without disturbing the energy of the system similarly.

Now, while our results are preliminary to this point, it is interesting to attempt to make predictions for the entire MT. This involves extending the Hubbard model to a second dimension. Hopping is allowed in a second direction about the helix in addition to along the protofilament. This adds several complications; first, there are many more states in a model of this system. Consequently, the matrix which must be diagonalized is much larger. As the density of states is much greater in the MT, one may expect that the conductivity will also be higher—that is in an MT of the

same length as a protofilament. Physically, one might rationalize this simply. There are a larger number of paths available for the electron to move from one end to the other; consequently, one would expect the resistivity to be lower—conductivity, which is the inverse of the resistivity would therefore be larger. In addition, the close spacing of energy levels implies that there should be more peaks at lower energies corresponding to the level spacing. Consequently, conduction along the MT should be greatly improved over protofilament conduction, especially at lower frequencies. The added complication is that conductivity becomes a tensor with the introduction of the additional protofilaments in the MT. In axonal MTs the field is expected to be radial while it would be conduction along the MT which is of interest.

There are also questions of conductivity if the charge carriers are not electrons but hydrogen ions (H^+). Proton hopping may be modelled in a similar way but protons have a much larger mass than do electrons. The higher mass means that energy levels are more closely spaced; again this probably means higher conduction at lower frequencies, but for our choice of parameters for the double quantum well, no bound states exist. Thus, binding of the protons must be somewhat larger. This would therefore have a negative effect on the conductivity. No meaningful statement can be made on protonic versus electronic conduction without further investigation.

6. Can MTs function as quantum computers?

It is a very attractive proposition to consider protein assemblies performing the task of a computer. Protein-based computers could operate 10^3 times faster than semiconductor-based computers. Furthermore, they could be designed one atom at a time, offer parallel processing architecture, three-dimensional data storage and a number of other features. In fact, a case has recently been made (Bray 1995) to view the already existing intracellular assemblies of proteins as computational elements which in essence constitute the cell's equivalent of a nervous system. Many aspects of cell behaviour display capacity for information processing independent of the genome and are controlled by proteins, e.g. motility. Bray (1995) maintains that GTP supply is the key element as it activates proteins while GDP deactivates them, which is consistent with our earlier discussions. Protein circuits in cells include hundreds of different channels, receptors and kinases which together monitor and respond to internal and external environmental stimuli. Can MTs be viewed as part of such circuitry? If so, how can their functioning as information processors be accomplished?

Some evidence links the cytoskeleton with information processing and cognitive function. Production of tubulin, the MT subunit and MT activities is correlated with peak learning, memory and experience in baby chick brains. When baby rats begin their critical learning phase for the visual system (when they first open their eyes), neurons in the visual cortex begin producing vast quantities of tubulin (Cronly-Dillon 1974). Tubulin production is drastically reduced when the critical learning phase is over (when the rats are 35 days old). Bensimon & Chernat (1991) found that selective destruction of brain MTs by the drug colchicine caused cognitive defects in learning and memory which mimic the clinical symptoms of Alzheimer's disease, in which the cytoskeleton becomes entangled. Further suggestion for cytoskeleton computation and/or information storage stems from the spatial distribution of discrete sites (or states) in the cytoskeleton. For example, tubulin subunits in closely arrayed MTs have a density of about $10^{17}\,\mathrm{cm}^{-3}$, which is very close to the theoretical limit for

Table 3. *Characteristic physical quantities describing MTs and their environment.*

tubulin dimer mass	110 kDa $\simeq 10^{-21}$ kg
dielectric constant	5–10
mobile charge	1–2e^- $\simeq 10^{-19}$ C
electric field strengths	
in membrane	10^7 V m^{-1}
in cytoplasm	10^{-1}–10^1 V m^{-1}
temperature	300 K
velocity	
phonon	600 m s^{-1}
tubulin in cytoplasm	1 μm min^{-1}
charge in axon	1–2 m s^{-1}
motor protein	0.5–2 μm s^{-1}
light wavelength (λ)	400–700 nm

charge separation (Gutmann 1986). Thus the cytoskeleton polymers have maximal density for information representation by charge, and the capacity for dynamically coupling that information to mechanical energy and chemical events via cooperative dipole states.

Recently, Hameroff & Penrose (1996) advanced the hypothesis which in a nutshell states that MTs are biological quantum computers providing a molecular substrate for the emergence of consciousness. In their model, called orchestrated objective reduction, or Orch OR for short, MTs' tubulin dimers are viewed as existing in a quantum superposition state which may self-collapse (hence the term objective reduction) upon reaching a quantum gravity threshold. The resultant wavefunction collapse event was linked by these authors (Hameroff & Penrose 1996) with numerous synaptic/neural functions and, in particular, with the formation of a thought process. While the Orch OR hypothesis cannot be proved or disproved at present, we will try to assess its validity within the context of the physical properties of MTs presented earlier in the paper.

In this connection, it is worthwhile recalling several thoughts on the topic of physics, computation and biology verbalized not too long ago by Hopfield (1991). First, he stated that: 'the human brain is the most mysterious and complex of these biological computational systems'—which would perhaps give legitimacy to searches for very exotic explanations of its functioning. However, he also said that 'contrary to the expectations of a long history of ill-prepared physicists approaching biology, there is absolutely no indication that quantum mechanics plays a significant role in biology'. He emphasized the role of complexity in biological systems where a change in a meaningful bit of information will have an effect on macroscopic behaviour of a living system. This is not so in physical systems.

Returning now to the issue of MTs, we wish to summarize the key physical parameters describing these protein polymers and their cellular environment (cytosol). These are shown in table 3. Based on these numbers and our earlier discussions throughout this paper we can put together a summary of energy-scale estimates for the processes that have been either observed or postulated to play a role in the functioning of MTs. This is summarized in table 4. For better clarity, this information is illustrated graphically in figure 12. What are the general conclusions we can draw regarding these processes? First of all, for all intents and puposes, gravitational effects

Table 4. *Estimates of energy scales*

chemical (hydrolysis)	
free GTP	0.2 eV
Tu-GTP	0.12 eV
MT-GTP	0.03 eV
electrical (qE)	10^{-4}–10^{+1} eV
dipole–dipole ($d^2/\epsilon r^3$)	10^{-2}–10^{-1} eV
thermal (kT)	*ca.* 0.02 eV
phonon ($\hbar\omega$)	*ca.* 10^{-4} eV
$\omega = \sqrt{K/M}$	
light (hc/λ)	*ca.* 1–2 eV
kinetic ($mv^2/2$)	
motor protein	3×10^{-15} eV
electron	*ca.* 10^{-10} eV
gravitational ($GM_{\mathrm{Tu}}^2/r_{\mathrm{Tu}}$)	10^{-27} eV
mechanical conformation ($kx^2/2$)	
$E_{\parallel}$	0.3 eV
$E_{\perp}$	0.03 eV

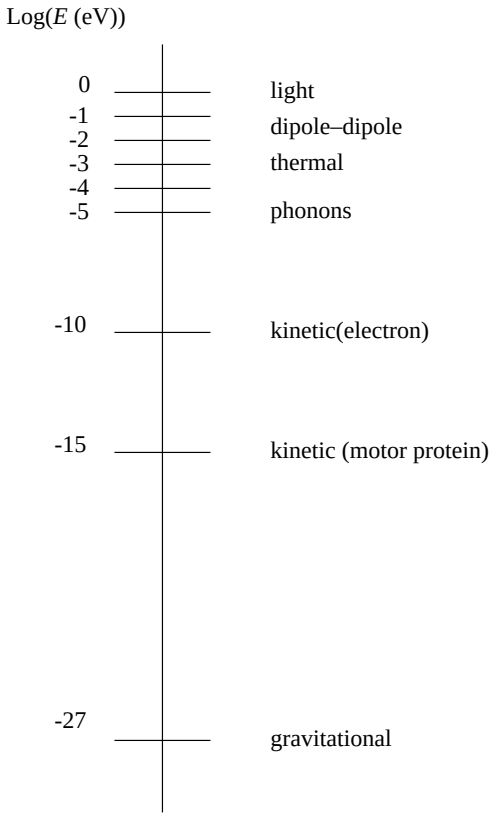

Figure 12. Graphical display of relative energies of MT-associated processes.

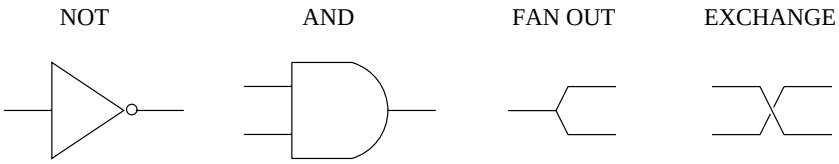

Figure 13. Basic function elements for a quantum computer.

should be entirely overshadowed by the remaining processes. Secondly, it is likely that electrons and motor proteins are guided by electric fields in order to overcome thermal noise. There is also substantial scope for cross-effects between different types of interactions, e.g. (a) photoconduction; (b) mechano-chemical coupling; (c) chemically triggered dipole reversals; (d) electromechanical; and (e) a possible induction of magnetic fields by electric currents. Turning now specifically to the Orch OR model of Hameroff & Penrose (1996), the chief difficulties of reconciling their proposal with our analysis can be listed as follows.

(i) The size of the tubulin protein is probably too large to make quantum effects easily sustainable.

(ii) Conformational effects are expected to involve distances *ca.* 10Å, which, again, are larger than those called for in the Orch OR model.

(iii) Physiological (room) temperature requirements make it extremely difficult to defend the use of the quantum regime due to the persistence of thermal noise.

(iv) A possible effect mediating against thermal fluctuations might be some form of screening of tubulin from the cell's 'noisy' environment. As discussed earlier, MTs are extremely sensitive to their environment, especially ionic and microelement concentrations. Hence, we doubt that MTs can be shielded from their environment.

(v) It appears that the two (or possibly more) conformational states of tubulin are separated by a sizeable potential barrier which again requires an external stimulus (such as GTP hydrolysis) to overcome it.

(vi) The calculations of the 500 ms preconscious processing time may be directly related to the action potential's travel time along the axon plus the refractory lag time in synaptic transmission, rather than to the quantum collapse time.

We, therefore, believe that the Orch OR model is not tenable for MT functioning. Is there still another possibility, that MTs are quantum computational devices? Feynmann (1986) presented a very lucid analysis of a quantum-mechanical computer which basically deals with a physical system composed of two-state units. The four key operations are shown in figure 13. Feynmann translated the action of these logical elements into the language of second quantization and found corresponding Hamiltonian terms. Recently, an application of these concepts to a system of quantum dots was made (Brum & Hawrylak 1997) and it was demonstrated that a quantum exclusive-OR gate can be made. The Hamiltonian relevant in this case was of the Hubbard type and it represents the same class of problems as the electronic conduction model in a MT presented in § 5. The full set of quantum functions shown in figure 13 can be realized through the coupling provided by MAPs with the proviso that there exists a mechanism maintaining the two-level structure of mobile electronic states.

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Discussion

P. Marcer (*BCS Cybernetic Machine Group, Keynsham, UK*). I believe that no one, with the exception of Professor Zoller, has touched on the fact that computations must concern UNIT WIRE and EXCHANGE primitives as well as ANDs and ORs (as was pointed out by Feynman), or their quantum equivalents. Perhaps, therefore, this is a basic function of the microtubule, particularly as it has been shown that such tubules have two fundamental phases, one conductive and one semiconductive. This would support the Penrose–Hameroff consciousness postulate, which states that this phase transition is pertinent to allowing the transmitted signals to rise to a higher 'quantum' level of awareness via conduction rather than semiconduction until the brain becomes consciously rather than unconsciously aware. The ubiquitous nature of microtubules would also be consonant with this function as a unit wire, as is the case with current technology.

J. A. Tuszyński. I agree with the first part of the statement. We do indeed maintain that microtubules can be suitable semiconducting devices. As much as we have tried to open-mindedly examine, however, we have not found any empirical basis for viewing microtubules quantum mechanically. They certainly play a significant role in the functioning of the brain, but linking them with the emergence of consciousness is in our view somewhat premature.

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BIOELECTROCHEMISTRY AND BIOENERGETICS

Electric field around microtubules

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Abstract

Living cells are organized by the cytoskeleton with a fundamental role of microtubules. The mechanisms of organization are largely unknown. We analyze the vibrations in the microtubules which are polar and are accompanied by polarization waves. Oscillating electric field generated around microtubules can be as high as 10^5 Vm^{-1} and may have an important role in information system and mass transport in living cells. Energy from hydrolysis of guanosine triphosphate (GTP) stored in microtubules can excite the vibrations above thermodynamic equilibrium level. © 1998 Elsevier Science S.A. All rights reserved.

Keywords: Molecular biophysics; Vibration states in microtubules; Bioelectric phenomena; Electric field of microtubules; Hydrolysis of GTP; Condensation of energy

1. Introduction

Living cells are structurally and dynamically organized by protein polymer network which is called cytoskeleton. The cytoskeleton is a highly dynamic structure which reorganizes continually as the cell changes its shape, divides, and responds to its environment. The essential data and description of the cytoskeleton, its components, and function are given in the book by Alberts et al. [1]. The cytoskeleton is composed of intermediate filaments, of actin filaments (which are also called microfilaments), and of microtubules. The filaments and the microtubules are connected to a large variety of accessory proteins and form a three dimensional network in the cell. The cytoskeleton is connected to transmembrane proteins. The cytoskeleton enables transport of molecules, structures, and organelles (by means of motor proteins), receives signals from cellular environment through the membrane proteins, and has an important role in the mitotic spindle.

The microtubules are the main organizers of the cytoskeleton. They are formed by cylindrical structures (hollow tubes with the outer and the inner diameters of about 25 nm and 17 nm, respectively) composed of 13 (or 14) protofilaments which are one dimensional chains of heterodimer tubulin molecules. The structure of the microtubules was disclosed by Amos and Klug [2]—Fig. 1a. A tubulin molecule is formed by two globular proteins—α- and β-tubulins; each of them has the relative molecular mass of about 55,000 ('molecular weight' in daltons) but the masses are not equal. Conformation of the heterodimer molecule can be changed from the α state (non tilted) to the β state (tilted by 29° from the microtubule axis)—Fig. 1b. The heterodimer molecule is an electric dipole [3–5] (18 Ca^{2+} ions are bound in the β-monomer). Therefore, the microtubules are polar structures with the plus end (fast growing) and the minus end (slow growing) embedded in the centrosome—a centre of the cytoskeleton. Polarity of the microtubules are closely connected with their function [6,7]. The function of the microtubules can be regulated by phosphorylation and dephosphorylation of microtubule-associated tau protein [8].

The dynamics of the microtubules is one of the most prominent features. The microtubules are capable of rapid polymerization and depolymerization [9] which enables exchange of the subunits between soluble and polymer pools. The microtubules grow toward the cell periphery and then shrink back toward the centrosome. They may shrink partially and grow again or they may disappear to be replaced by new microtubules. This process is called the dynamic instability. The treadmilling (i.e. addition of the tubulin subunits at the plus end and removal of the tubulin subunits at the minus end at an identical rate)

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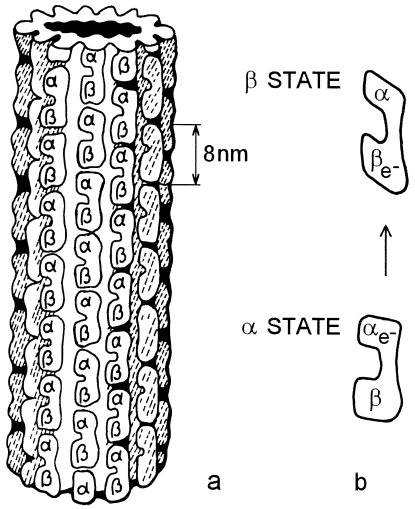

Fig. 1. A schematic picture of a microtubule (a), and of conformation change of a tubulin heterodimer from the α to the β state (b). After Refs. [2,4].

appears if both the ends of a microtubule are exposed [1]. Newly polymerized subunits (heterodimer molecules) contain energy rich nucleotide guanosine triphosphate (GTP). After polymerization the guanosine triphospate molecules bound to the β-tubulins are hydrolyzed to the guanosine diphosphate (GDP). The significance of the GTP in microtubule dynamics is analysed in Refs. [10–12]. The growth of a microtubule can continue if an energy rich cap at its plus end is maintained, i.e. if the GTP's in the cap subunits are not hydrolyzed to the GDP's. The microtubule dynamics may depend on kinetochores too [13].

The microtubules depolymerize and repolymerize continually in animal cells. The majority of the microtubules can incorporate new subunits within 20–30 min although complete exchange of the subunits in a cell can be achieved within 1 h [14]. The half-life of an individual microtubule is of about 10 min [1]. Depolymerization removes the subunits with the GDP and repolymerization adds the energy rich subunits with the GTP to a microtubule. The hydrolysis energy is not used for polymerization. The majority of the hydrolysis energy is stored in the microtubule [15]. The mechanism of depolymerization and repolymerization provides continual supply of energy into the microtubule structures in a cell and signifies that the total energy storage is renewed within one hour.

The mechanisms of the cytoskeleton activity are still not well understood. The cytoskeleton exerts forces and generates movements without any major chemical changes [1]. A physical mechanism is probably responsible for the cytoskeleton function. Vibrations in the microtubules can be one of possible mechanisms as follows from the analysis given in Ref. [16]. As the tubulin heterodimers are polar the vibrations generate an oscillating electric field. The vibrations can be excited by the energy released from the hydrolysis of the GTP. The excited vibrations may be far from thermodynamic equilibrium and need not be of random nature. We will analyze the properties of the electric field generated by the microtubules and its possible effects on ordering and on the mass transport in living cells.

2. Vibrations in the microtubules

A part of the microtubule composed of protofilaments is shown in Fig. 1a. Fig. 1b shows the tubulin heterodimer molecule in the α and in the β states. The globular tubulins in a protofilament may be considered as mass units in a chain with translation symmetry. These units have internal vibrations (exerted by atoms and by parts of the tubulins) and external vibrations in the chain (i.e. the vibrations exerted by each tubulin as a whole). On account of translation symmetry the external vibrations may be analysed using a method which is similar to that employed in the solid state physics [17]. Nevertheless, we have to point out certain differences between crystalline solids with translation symmetry and the microtubules. The 'diameters' of atoms in solids are of about 0.1 nm and the distances between their centres are about twice or three times greater. The 'diameters' of the tubulins are of about 4 nm; they are only slightly smaller than the distances between their centres of mass. Atoms in a lattice are not rigid but in comparison with them the tubulin globules are highly elastic particles. The external vibrations will be analyzed without including dynamic deformations of tubulin globules.

Fig. 2 shows a scheme of the one-dimensional chain of particles. The circles denote the centres of mass of tubulins in equilibrium positions, the black spots show the theoretical equidistant control points (the distance between them is a), Δ_1, Δ_2 are distances of tubulins from the corresponding control points, m_1, m_2 are masses of tubulins, and f_1, f_2 are the elastic force constants which need not be equal.

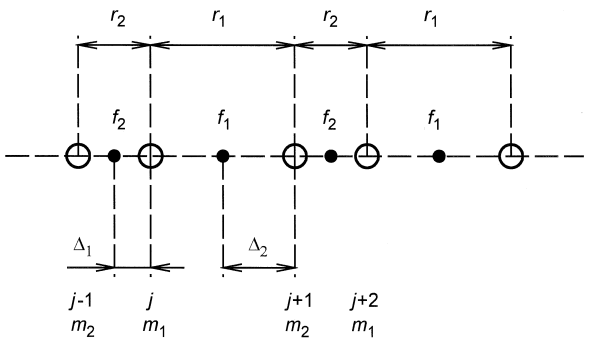

Fig. 2. Translation symmetry in a protofilament.

The interactions between the closest neighbours will be assumed. The dispersion relation is given by [16]

$$\omega_{\pm}^2 = \frac{(f_1+f_2)(m_1+m_2)}{2m_1m_2} \pm \sqrt{\left[\frac{(f_1+f_2)(m_1+m_2)}{2m_1m_2}\right]^2 - 4\frac{f_1f_2}{m_1m_2}\sin^2 ka} \quad (1)$$

where k is the absolute value of the wave vector and ω_+ and ω_- denote the frequencies of the optical and of the acoustical branches of vibrations, respectively. In the acoustical branch the neighbouring particles (tubulins) move in the same direction and in the optical branch the particles move in opposite directions under condition that the centre of gravity remains at rest [17]. The term vibration mode is used for the waves propagating in both directions along the chain with the same absolute value k of the wave vector.

The protofilaments in a microtubule are bound and build up a wall of a cylinder. The vibrations in particular protofilaments are not independent and form a common vibration system in a microtubule. As a microtubule is composed of 13 protofilaments we assume, that a microtubule 'particle' has 13 times greater mass than a protofilament 'particle' (this assumption may be considered as a zero order approximation). The elastic force constants in isolated parts of the protofilaments (oligomers) and in the microtubules may be different. Frequency vs. absolute value of the wave vector evaluated from Eq. (1) is shown in Fig. 3a for $f_1 = f_2$ and in Fig. 3b for $f_1 = 10f_2$ [16]. The elastic force constants are parameters of the curves. We used several values for the elastic force constants as they can vary in a wide range of values. We will not discuss here the dependence of the elastic force constants on the viscoelastic properties of the cytoskeleton including the entanglements, cross-linkings, and lengths of microtubule sections and filaments as it is beyond the scope of this paper. The reader may find it e.g. in Ref. [16]. We only note that they may be determined from the elastic shear moduli and Young's moduli. Fig. 3a,b shows strong dependence of the frequency of the vibrations on the elastic force constants [16].

The microtubule chain is composed of discrete mass units which we denote microtubule 'particles' and, therefore, the frequency spectrum of the vibrations is discrete. The boundary conditions enable us to determine the absolute values of the wave vector k. If we assume that the tubulins at the end of the microtubule are at rest we get [17]

$$k = \frac{\pi}{Na}g \quad (2)$$

where $N+1$ is the number of microtubule 'particles' in the chain, g is an integer $(1 \le g \le N-1)$, and the length

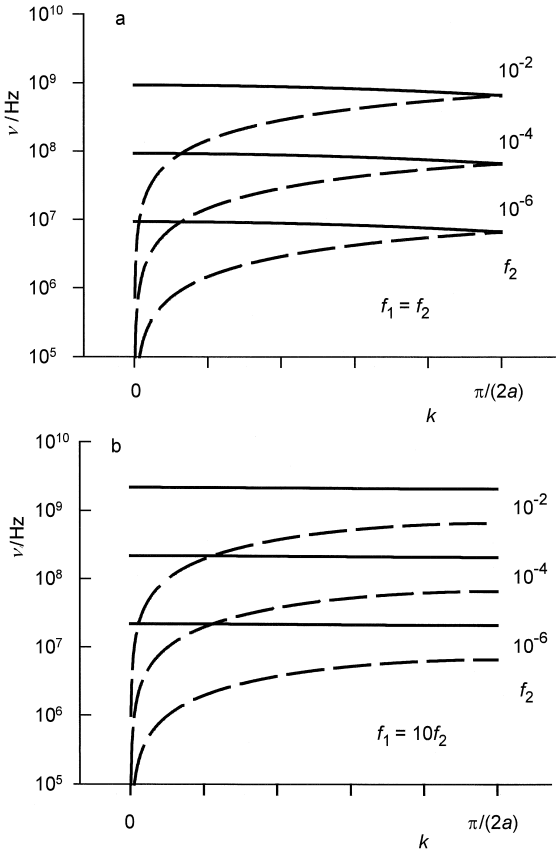

Fig. 3. Dispersion relation for a microtubule (frequency vs. wave vector). The full lines and the dashed lines denote the optical and the acoustical branches, respectively. Elastic force constants are assessed $f_1 = f_2$ (a), $f_1 = 10f_2$ (b). After Ref. [16].

of the microtubule is $L_m = Na$. The number of modes in one branch of vibrations is equal to the number of mobile particles (in the Brillouin zone). We will assess the lowest frequency of the acoustical branch. For a microtubule the lowest frequency for $\nu \neq 0$ of the acoustical branch is in the band of 5–50 MHz if the length of a microtubule is 10 μm $(= \lambda/2)$ and velocity of the acoustical waves is in the range from 10^2–10^3 ms^{-1}.

If the elastic force constants in a heterodimer and between two heterodimers are not equal (i.e. $f_1 \neq f_2$) the potential of 'the tubulin particle' as a function of distance is asymmetric with respect to the equilibrium position. Therefore, the system is non-linear and higher harmonic components (i.e. $\omega = n\omega_j$) and components with combination frequencies (i.e. $\omega = n\omega_j + m\omega_k + \ldots$) are generated ($n$, m, ... are integers). The dispersion relation (1) does not contain these harmonic and combination terms.

The biological reality can certainly yield more complicated patterns of vibrations in the microtubules than those derived in this paper as a number of simplifications was used. For example the internal vibrations were neglected and simple rigid structures of the tubulins were assumed. The internal vibrations are combined with the external

vibrations giving rise to complicated spectral patterns. On account of small mass and great values of the elastic force constants the frequencies of the internal vibrations are much higher than those of the external vibrations. The frequency spectrum can contain frequency lines in the millimeter and in the far infrared regions too.

3. Energy supply to the microtubules

When a microtubule grows tubulin heterodimers add to the free plus end and the GTP's attached to the β-tubulins are hydrolyzed to the GDP's [1,10–12,15]. The rate of polymerization may be greater than the rate of hydrolysis and an energy rich cap can be formed at the end of the microtubule. This energy rich cap is hydrolyzed later. The measurements of hydrolysis of the GTP analogue guanylyl-(a,b)-methylene-diphosphonate (GMPCPP) in solution, in tubulin heterodimers, and in tubulin subunits in the microtubules led to a conclusion that the free energy was -5.18, -3.79, and -0.9 kcal mole^{-1} (-21.7, -15.9, and -3.8 kJ mole^{-1}), i.e. 0.22, 0.16, and 0.04 eV per one entity, respectively [15]. Therefore, energy of about 4 kcal mole^{-1} (17 kJ mole^{-1}) was assumed to be potentially available from the hydrolysis of the GTP. A part of the energy is expended on dissociation of the inorganic phosphate P_i but the majority of it is stored in the microtubule lattice [15]. A remeasurement of the dissociation rate constant allowed better estimation of the free energy for hydrolysis and release of P_i which is $+0.7$ kcal mole^{-1} ($+2.9$ kJ mole^{-1}) [18] rather than -0.9 kcal mole^{-1} as was reported in Ref. [15]. The energy stored in the microtubule lattice is assessed to be 1.7 kcal mole^{-1} (7.1 kJ mole^{-1}). We have to emphasize that the measurements of the energy available from hydrolysis were carried out on the GTP analogue GMPCPP.

The experimental findings given e.g. in Refs. [15,18] show that energy from hydrolysis is stored in the microtubules. The form in which the energy is stored in the microtubule lattice is a special issue. Mandelkow et al. [11] assume that the tubulin–GDP subunit prefers a conformation which leads to coiling of the protofilaments and as a consequence energy may be stored in the form of a tense state of the microtubule lattice. Caplow et al. [15] come to the conclusion that energy is stored in protein conformation and presumably as a repulsive force between subunits. A similar mechanism of energy storage in the microtubules is given in Ref. [4], i.e. that energy from the hydrolysis of the GTP to the GDP is imparted into the microtubule lattice via tubulin conformation changes. Real physical mechanisms of energy transformation are irreversible and a part of the energy is always transferred to the random heat vibrations. Even if the energy levels of two conformation states of a tubulin subunit are different (e.g. if the level of the β state is higher than that of the α state) after the transition to the higher energy level a part of the energy which is used to surmount the potential barrier between the conformation states can be transformed into vibration energy. The opposite relaxation process and restoration of the α conformation state should convert energy to the microtubule vibrations too. Creation of the tense state with altered forces between the subunits in the microtubule has to supply energy into the lattice. Therefore, we may assume that energy is stored in the microtubule lattice not only in the conformation changes and in the tense state of the lattice but also in the vibration states. Besides energy stored in the cytoskeleton during polymerization a part of the energy from the ATP hydrolysis supplied to motor proteins moving along a microtubule [1] can be transferred to the microtubule as random excitations of the vibrations.

The energy stored in the microtubule can do work in the cell [15] but a precise manner in which the energy is utilized is still not understood [19]. As the cytoskeleton is not subjected to any major chemical changes [1] the energy has to be used in some biophysical processes. The hydrolysis energy can change bond strength [19] and excite solitons or soliton like waves [3,19]. As it causes the conformation changes of the tubulin heterodimers kink wave reorientation of the heterodimer dipoles can be excited in the microtubule [3]. It can excite the lattice vibrations and polarization waves, which may be coherent on account of the Fröhlich mechanism [19] or the intrinsic non-linear mechanism. (The Fröhlich mechanism is described e.g. in Refs. [20–22]). Regardless of the primary mechanisms of the energy storage at a later stage at least a part of it is always supplied to the vibrations in the microtubules. Nevertheless, the majority of the energy stored in the microtubules may be converted into the vibrations. Therefore, the vibrations are excited and can be far from thermodynamic equilibrium. We suggest that the mechanism of the coherent polar vibrations in microtubules utilizing the stored energy is a tool for the microtubule organization function. The electric fields generated by the polar vibrations can do work in the surroundings of the microtubules. A mechanism of this type (large scale quantum coherent phenomena in microtubules) is suggested and supported by theoretical analysis e.g. in Refs. [23,24]. The mechanism discussed e.g. in Ref. [23] assumes interaction between coherent modes in the microtubule with water in its surroundings, especially in the microtubule cavity. Experimental data necessary for assessment of the energy transfer between the microtubule and its ambient medium and of damping of the vibrations are still missing.

We assess the amount of energy which might be supplied to the cytoskeleton vibrations from the hydrolysis of the GTP. If the energy supplied by the hydrolysis of the GTP is 4 kcal mole^{-1} (17 kJ mole^{-1}) then energy per one entity is about 7 times greater than energy in thermodynamic equilibrium in the frequency range from 10^7 to 10^{11} Hz. On account of non-linear mechanism the energy may be accumulated in certain vibration modes.

4. Oscillating electric field

The thermal vibrations are omnipresent and biological systems are no exception. The typical feature of the vibrations in thermodynamic equilibrium is their randomness. Using classical concepts we may state that the 'amplitude' and the 'phase' have no coherent components. The tubulin heterodimers are polar and the vibrations create oscillating dipoles or oscillating multipoles. The polar vibrations generate an electric field. We will adopt a linear model of a microtubule with a length of $L_m = 3$ μm where each heterodimer carries an oscillating electric dipole $p = bel_h$ (e is the elementary charge, $l_h = 8$ nm, and $b = b(x)$ where x is the distance along the microtubule axis). Except for b the dipole moments are identical. The frequency of vibrations $\nu = 100$ MHz and the relative permittivity of the ambient medium $\varepsilon' = 81$. We evaluate the intensity of the electric field along the lines parallel with the microtubule axis.

The intensity of the electric field for random orientations of the dipole moments is given in Fig. 4. (We note that this case may correspond to the thermal vibrations.) E_r and E_x shown in Fig. 4a,b are the absolute values of the intensity in the r and x directions (where r denotes the radial direction perpendicular to the microtubule axis),

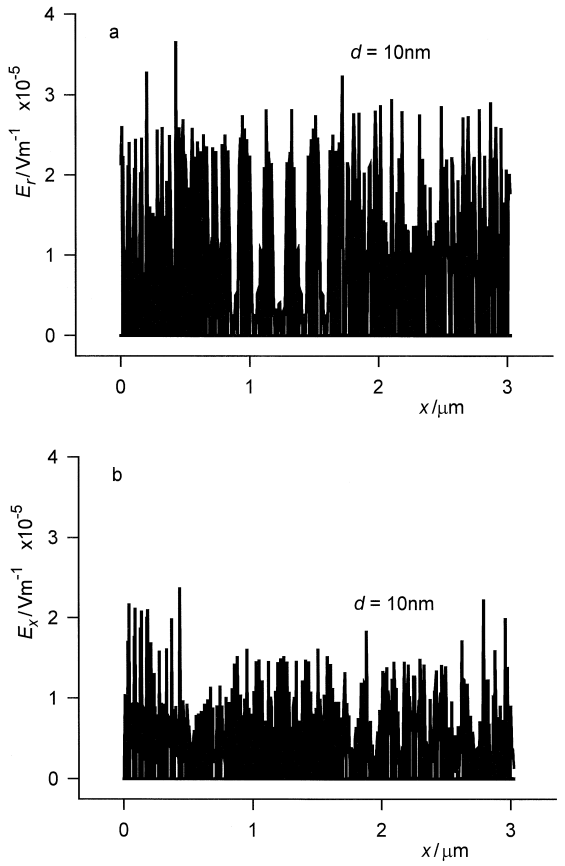

Fig. 4. The thermal vibrations: The absolute value of the intensity of the electric field E_r (a) and E_x (b) as a function of x.

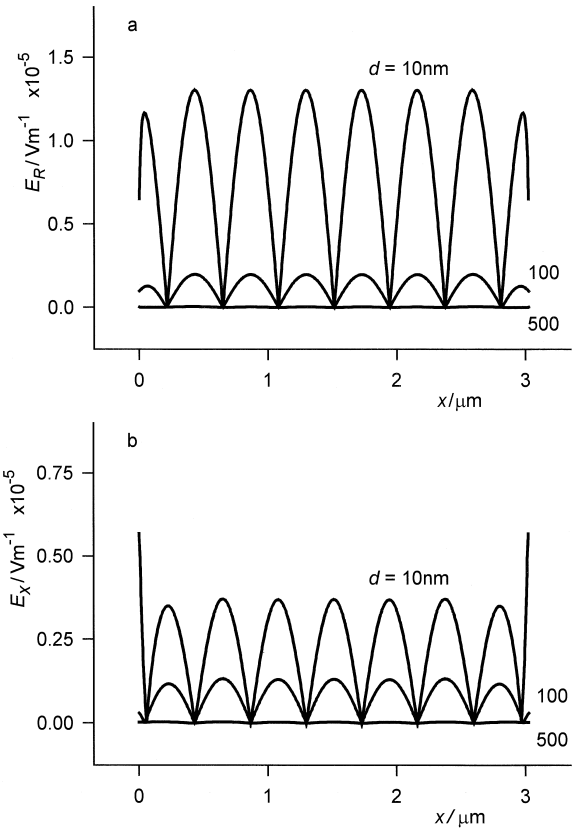

Fig. 5. The coherent vibrations of a vibration mode: E_r (a) and E_x (b) as a function of x.

respectively. The distance d from the 'surface' of the model microtubule is 10 nm. The greatest absolute value of the intensity is of the order of 10^5 Vm^{-1}. At a distance 0.1 and 0.5 μm the peak values are about 10^2 and 1 V m^{-1}, respectively. We used the generator of random numbers to determine the orientations of dipole moments in particular heterodimers ($b = \pm 1$). The intensity of the electric field has random fluctuations in space and time and can decrease to very small values. In the random case the space–time distribution of the dipole moments should correspond to the maximum value of entropy (on account of the Second Law of Thermodynamics).

In thermodynamic equilibrium the Planck's Law determines the distribution of energy in the normal modes. A vibration mode is equivalent to the harmonic oscillator which is subjected to random disturbances. The resulting vibrations are given by superposition of vibrations of all the modes. Nevertheless, under certain conditions the excited vibrations can be far from thermodynamic equilibrium with an important coherent component. In a coherent state certain mode (or modes) is (are) excited; its (their) energy is far from thermodynamic equilibrium value. An example is shown in Fig. 5. Fig. 5a,b shows E_r and E_x, respectively, of the vibration mode $k = 7\pi/L_m$ as a function of x. The distance d is 10 nm, 0.1, and 0.5 μm. The vibrations generate a dynamic periodic potential around

the microtubule. The k value may be considerably greater than that used in Fig. 5 and the period of the standing waves may be comparable in size to the heterodimers. The intensity of the electric field depends on the excitation of vibrations in the microtubule.

5. Discussion

The potential energy U of a dipole in the external electric field E is given by

$$U = -p \cdot E \tag{3}$$

where p is the dipole moment. If $E = 10^6$ V m^{-1}, $p = 10^{-28}$ Cm then the absolute value of U is of the order of 10^{-22} J, i.e. about one order smaller than the energy kT.

The electric field exerts forces on charges, dipoles and multipoles, and even on neutral molecules and particles on account of dielectrophoretic effect. For instance the force exerted on a dipole by the generated field is equal to the gradient of the scalar product

$$F = \nabla(p \cdot E). \tag{4}$$

If the intensity of the electric field decreases from 10^6 to 10^5 V m^{-1} within a distance of 10 nm then for a dipole moment $p = 10^{-28}$ Cm parallel to the intensity of the electric field the force is of the order of 10^{-14} N. The force acts in the direction of the gradient if the dipole is free, i.e. if it is capable of the flip–flop reorientation of the dipole moment.

The inhomogeneous electric field can exert forces on and cause translation motion of neutral dielectric particles. The particle is polarized and the field acts on the induced dipole. The relation for the dielectrophoretic force can be written in the form [25,26]

$$F \propto V(\varepsilon_2' - \varepsilon_1')\varepsilon_0 \nabla(E^2) \tag{5}$$

where ε_2' (ε_1') is the relative permittivity of the particle (of the ambient medium), ε_0 is the permittivity of free space, and V is the volume of the particle. If $\varepsilon_2' > \varepsilon_1'$ the particle is attracted in the direction of the gradient of the intensity of the electric field. If $\varepsilon_2' < \varepsilon_1'$ the force acts in the opposite direction (i.e. the particle is expelled from the field). If the difference of $\varepsilon_2' - \varepsilon_1' \approx 70$, $V = 10^{-27}$ m^3 we can derive for the same electric field as in the case of the dipole that F is of the order 10^{-16} N.

If the vibrations in particular parts of the source structure are random then the generated electric field has the same random properties. If the vibrations are coherent the generated electric field is coherent too. The oscillating electric field in biological systems may have an important role in cell signaling system. Motion of matter and its transport may be caused or organized by the electric oscillating field generated by the cytoskeleton (especially the long distance transport). The endogenous electric field generated by the microtubules can be an organizing factor in living cells.

6. Conclusions

The thermal vibrations are an omnipresent phenomenon existing in inorganic as well as in organic and biological matter. A method similar to that employed in the solid state physics can be used to analyse vibrations in the microtubules. The optical and the acoustical branches of vibrations can exist in microtubules. As the tubulin heterodimer molecules are polar the vibrations generate an oscillating electric field. Generation of the electric field by the microtubules was not analysed yet.

The energy supplied from the hydrolysis of the GTP's to the GDP's can cause excitations of the vibrations to a state far from thermodynamic equilibrium. The generated electric field can have a strong effect on the long distance mass transport and on the organization of chemical reactions in living cells.

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Electric field generated by axial longitudinal vibration modes of microtubule

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ABSTRACT

Microtubules are electrically polar structures fulfilling prerequisites for generation of oscillatory electric field in the kHz to GHz region. Energy supply for excitation of elasto-electrical vibrations in microtubules may be provided from GTP-hydrolysis; motor protein–microtubule interactions; and energy efflux from mitochondria. We calculated electric field generated by axial longitudinal vibration modes of microtubules for random, and coherent excitation. In case of coherent excitation of vibrations, the electric field intensity is highest at the end of microtubule. The dielectrophoretic force exerted by electric field on the surrounding molecules will influence the kinetics of microtubule polymerization via change in the probability of the transport of charge and mass particles. The electric field generated by vibrations of electrically polar cellular structures is expected to play an important role in biological self-organization.
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1. Introduction

Cytoskeleton, which consists of microtubules, actin filaments, and intermediate filaments, is considered to be an organizing structure of the eukaryotic cell (Alberts et al., 2008). Microtubules (MTs) are of special interest since they manifest several features that distinguish them from other subcomponents of the cytoskeleton. MTs, which consist usually of 13 protofilaments (PT), resemble hollow tubes with inner and outer diameter of 17 nm and 25 nm, respectively (Fig. 1a). The building subunits of MT are the tubulin heterodimers composed of α- and β-tubulin (Fig. 2). The heterodimers have high electric dipole moment of over 1000 Debye (10^{-26} C m) (Tuszyński et al., 2002; Mershin et al., 2004); they are responsible for the high electrical polarity of MTs. In the interphase of the cell cycle, microtubules are radially organized with their chemical minus ends embedded in a centrosome (organizing center), which is located in the center of the cell near the cell nucleus. There are several hundreds of MTs in a cell, depending on the organism and the cell type. In the interphase microtubules undergo dynamic growth (polymerization) and shrinkage (depolymerization), so-called "dynamic instability" (Fig. 1b). During the existence of mitotic spindle, MTs are subject to treadmilling, i.e. they polymerize at the plus end and depolymerize at the minus end (chemical plus and minus).

1.1. Microtubule Vibrations

Elastic and vibrational properties of MTs have been studied by several authors. Sirenko et al. (1996) analyzed vibrations of microtubules in a fluid modeled as elastic cylindrical shell with isotropic mechanical properties. Maximum frequency of tens of GHz corresponded to nonradiative elastic waves localized at the MT wall.

Pokorný et al. (1997) analyzed the vibrations of MTs modeled as one dimensional chain of mass particles with translation symmetry. Vibrations of cellular structures have been considered to be viscously overdamped in cytosol and thus not realistic (Foster and Baish, 2000; Adair, 2002). However, most of the water in the cell has different properties than bulk water (Preparata, 1995; Del Giudice et al., 2000, 2005; Zhadin and Giuliani, 2006; Zhadin et al., 2007) due to the molecular crowding and consequent large hydration surfaces. If the slip layer condition for MT longitudinal vibrations is taken into account, vibrations may be excited (Pokorný, 2004, 2003, 2005).

Portet et al. (2005) describe the microscopic dynamical properties of MTs using a discrete model based on a lattice of dimers. It was pointed out that various experimental methods yielded wide range of values of Young's and shear modulus between 1 Pa (Janmey et al., 1991; Sato et al., 1988) up to 1 GPa (Portet et al., 2005). This would be explainable by the fact that mechanical properties of microtubules are anisotropic and orthotropic.[1] This was showed by Kis et al. (2002) who obtained the most accurate values for both the Young's ($E > 100$ MPa) and shear modulus ($G = 1.4 \pm 0.4$ MPa).

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[1] Orthotropic–anisotropic in the directions perpendicular to each other.

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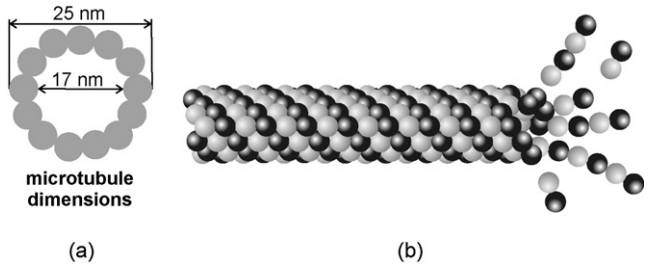

Fig. 1. (a) Cross section of microtubule. (b) Detail of chemical plus end of a microtubule undergoing dynamic instability.

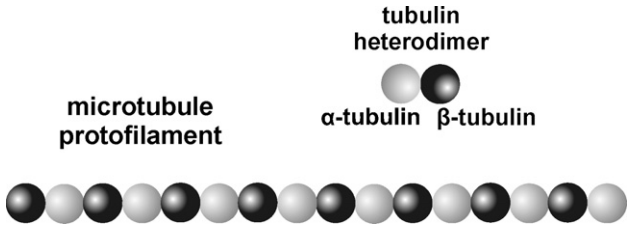

Fig. 2. Structural subunits of microtubules: protofilament composed of tubulin heterodimers.

The oscillation modes involving a change in the diameter of the MTs strongly depend on the shear modulus (Kasas et al., 2004).

Tuszyński et al. (2005) reviewed characteristic elastic properties of microtubules. Wang et al. (2006) and Wang and Zhang (2008) developed orthotropic elastic shell model to study microtubule vibrations and determined more realistic frequencies of radial and circumferential modes than that provided by Sirenko et al. (1996) which was confirmed by Qian et al. (2007).

It can be seen from above given references that frequencies of MT vibration modes may span over broad spectral range from kHz to GHz. Due to non-linearities and to mechanical anisotropy of MTs, frequency mixing and energy transfer between the modes, one may expect various frequencies of oscillations.

1.2. Energy Supply to Microtubules

Excitation of mechanical vibrations in MTs, which are electrically polar, would give rise to electrical oscillations. To excite and sustain vibrations the energy needs to be supplied. In the case when at least a part of energy supplied to MTs excites elastic vibrations of electrically polar structures, generated oscillatory electric field has the same frequency as that of elastic MT vibrations. We analyzed three possible sources of energy supply to MTs (Jelínek et al., 2009):

- Energy released from hydrolysis of guanosine triphosphate (GTP) in processes of dynamic instability of MTs (Fig. 2) is the energy supply dependent on internal MT process. β-Tubulin bound GTP is hydrolyzed to guanosine diphosphate (GDP) after polymerization of tubulin heterodimer. In the interphase the half lifetime of individual unstable MTs (subject to dynamic instability) is about 5–10 min (Alberts et al., 2008). Energy from hydrolysis of GTP stored in MT lattice is estimated to be 1.7 kcal/mol[2] (Caplow et al., 1994; Caplow, 1995; Caplow and Shanks, 1996). The corresponding power input to MT lattice due to GTP hydrolysis is about 1.8×10^{-14} W/mm of MT (if we accept 5 min as a half life of a single MT) which range roughly corresponds to the previous estimate (Pokorný, 1999; Pokorný and Wu, 1998). Assuming 100 μm total length of MT structure in a cell (the length is assumed for yeast cell (King and Hyams, 1982; Winey et al., 1995; Sobel, 1997)), one obtains the power of about 3×10^{-18} W in whole microtubular network. In the M phase, the dynamics of polymerization and depolymerization is increased about 20-fold, for instance by treadmilling in the metaphase.
- Energy transferred from moving of motor proteins and their interaction with MTs may also contribute to excitation of vibrations in MTs (Fig. 3). Motor protein dynein moves towards the cell center (minus end of a MT) and kinesin moves towards the cell periphery (plus end of a MT). One ATP (adenosine triphosphate) molecule is hydrolyzed for each step of motor protein. One step of motor protein is ca. 8 nm long, corresponding to axial dimensions of tubulin heterodimer. Energy released during ATP hydrolysis under cellular ATP concentrations (1–10 mM) is ca. 14 kcal/mol[3] which corresponds to activation energy of motor proteins (Kawaguchi and Ishiwata, 2001, 2000; Pelling et al., 2004). Motor proteins move with a velocity ca. 0.1–2 μm/s (Alberts et al., 2008). Assuming four motor proteins on each of 100 MTs in the cell, each motor protein moving with average velocity of 800 nm/s, power exerted by all motor proteins in a cell is about 4×10^{-15} W. Assuming that 1–25% of this power excites vibrations in MTs, it corresponds to 4×10^{-17}–10^{-15} W, respectively.
- Energy released from mitochondria as "wasted" energy in the course of citric acid cycle may be the most significant energy source for excitations of MT vibrations.

  Whole metabolic turnover of *Saccharomyces cerevisiae* cell is 0.1 pW (10^{-13} W) (Lamprecht, 1980). Mitochondria are considered "power plants of the cell" generating ATP by means of energy from pyruvate and fatty acid. Less than 50% of energy is stored in ATP. The rest of the energy ("wasted" energy), i.e., more than 50%, is released to surroundings of mitochondria in the form of heat and emitted radiation (Fig. 4). Corresponding wasted power may be 0.05 pW. Since mitochondria are located near microtubules, part of the "wasted" power (we may assume 20%, i.e., 10^{-14} W) may be involved in excitation of the vibrations in MTs.

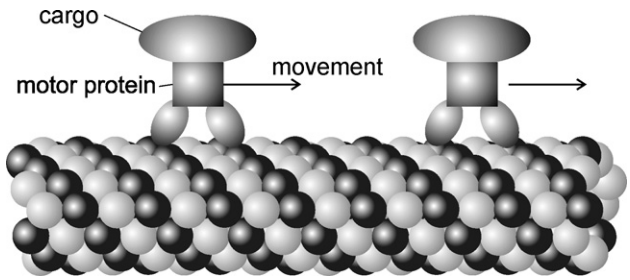

Fig. 3. Movement of the motor proteins on the microtubule, cargo not in scale.

2. Method and Parameters Used for Calculation of Electric Field Around MT

First assessment of oscillating electric field around MTs has been done by Pokorný et al. (1997). Intensity of the electric field was evaluated along the lines parallel with the MT axis up to the distance of 500 nm. Their model included microtubules with heterodimers with random or coherent oscillating dipoles. Frequency of coherent excitations has been chosen as 100 MHz.

We made a preliminary assessment of MT electric field with different calculation method in Havelka and Cifra (2009a), Havelka (2008), Havelka and Cifra (2009b), but we used a higher value of dipole moment than corresponds to the real state. The complete

[2] 1.7 kcal/mol = 7.1 kJ/mol = 118×10^{-22} J = 0.074 eV.

[3] 58.2 kJ/mol = 0.97562×10^{-19} J = 0.609 eV.

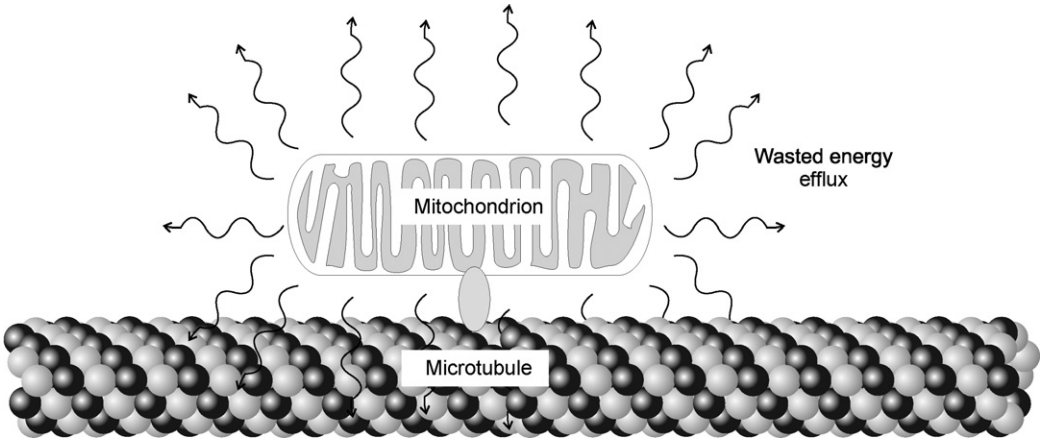

Fig. 4. Microtubule bathed in energy efflux from a mitochondrion. The mitochondrion not in scale to microtubule.

and more realistic results are presented in this paper. We analyze electric field generated by MT longitudinal axial vibration modes. We also study the spatial distribution of electric field of MT since this has not been performed before.

Due to axial longitudinal vibrations in the MT lattice an electric dipole moment will have an oscillating component in the direction parallel to MT axis. Oscillating dipole with the dimensions much smaller than wavelength of generated electric field can be represented by elementary oscillating electric dipole (Hertz dipole).

2.1. Components of the Electric and the Magnetic Fields of the Elementary Oscillating Electric Dipole

We use standard relations for the field of oscillating electric dipole in the form as provided in Novotný (1997):

$$\vec{H}_\phi = -\frac{Id}{4\pi}k^2 \sin\theta \left(\frac{1}{jkr} + \frac{1}{(jkr)^2}\right) e^{-jkr}\vec{\phi}_0 \quad (1)$$

$$\vec{E}_r = -\frac{Id}{4\pi}Zk^2 2\cos\theta \left(\frac{1}{(jkr)^2} + \frac{1}{(jkr)^3}\right) e^{-jkr}\vec{r}_0 \quad (2)$$

$$\vec{E}_\theta = -\frac{Id}{4\pi}Zk^2 \sin\theta \left(\frac{1}{jkr} + \frac{1}{(jkr)^2} + \frac{1}{(jkr)^3}\right) e^{-jkr}\vec{\theta}_0 \quad (3)$$

H is magnetic field intensity; E is electric field intensity; I is equivalent current; d is length of the dipole; k is propagation constant; Z is wave impedance; j is imaginary unit and coordinate symbols are according to Fig. 5. None of these equations is dependent on ϕ and all equations include:

$$Id = j\omega p \quad (4)$$

ω is angular frequency, and p is dipole moment ($Id = (\partial Q/\partial t)d = j\omega Qd$, $Qd = p$, Q is charge).

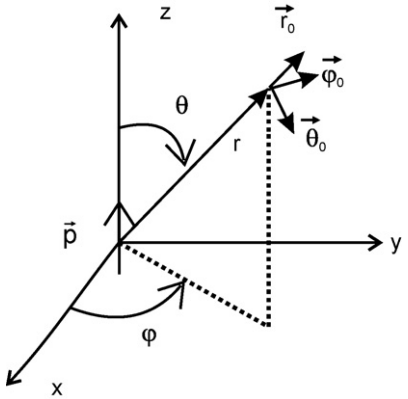

Fig. 5. A dipole in the Cartesian and spherical coordinate system.

2.2. Description of the Electric Field Around a Microtubule

Electric field around MT can be calculated as a vectorial sum of elementary electric components generated by individual tubulin heterodimer dipoles. Frequency of the oscillations of individual dipoles has been set to the 1 GHz which falls into the spectrum predicted by Sirenko et al. (1996), Wang et al. (2006) and Wang and Zhang (2008) and also fits almost exactly the frequency of optical branches of vibration modes of microtubule calculated by Pokorný and Wu (1998) and Pokorný et al. (1997).

The near field around electric dipole has the electric component much stronger that the magnetic one. We concentrated therefore on the electric field component.

We have rewritten the Eqs. (2) and (3) for electric field in Cartesian coordinates which can be used for arbitrarily placed and tilted dipole in lossy medium. The procedure is described thoroughly in Cifra (2009). All calculations and figures have been made in MatLab.

2.3. Shift of Protofilaments

Although the MTs are tube-like structures, their wall does not have cylindrical symmetry. This is due to longitudinal shift of protofilaments in MT lattice. Accordingly, lattice with longitudinal shift of PTs simulates more realistic organization of protofilaments in the MT. This gives us more accurate information about the spatial distribution of electric field intensity of the around MT. We have chosen the configuration in A-lattice (Song and Mandelkow, 1993, 1995; Kikkawa et al., 1994; Amos, 1995), which is a lattice with a longitudinal shift of protofilaments of 4.92 nm (Table 1 and Fig. 6).

2.4. Types of Excitation and Calculation Parameters

Geometry and orientation of tubulin dipole moments in MT is approximately known. However, the elasto-electrical waves in

Table 1
Longitudinal shift of PT in A-lattice.

Number of PT	2	3	4	5	6	7
Shift to 1st PT (nm)	−4.92	−1.84	−6.76	−3.68	−0.6	2.48
Number of PT	8	9	10	11	12	13
Shift to 1st PT (nm)	−2.44	0.64	−4.28	−1.2	−6.12	−3.04

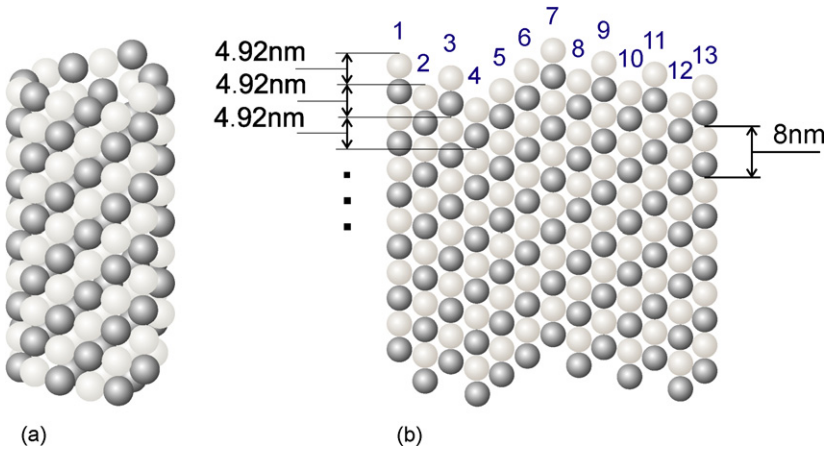

Fig. 6. (a) Microtubule with longitudinal shift of PT of 4.92 nm. (b) Shell of MT with longitudinal shift of protofilament.

microtubules also determine the efficiency, spatial distribution and dynamics of generated electric oscillations. There may be acoustical and optical modes of axial longitudinal oscillations. In optical modes, neighbor particles move in opposite direction. In acoustical modes, neighbor particles move in the same direction (except the end of Brillouin zone where the k-vector is large).

In the case of optical mode a component of dipole moment which is oriented along the axis would oscillate and generate electric oscillations.

Vibration modes which displace whole dipole along its axis would be significantly less efficient in generation of electric oscillations. This is due to the fact that opposite charges near each other and oscillating along the same axis with the same phase and frequency will behave as antiphase dipoles which effectively cancel each other field.

As shown in Tuszyński et al. (2002) and Mershin et al. (2004) a significant part of the tubulin dipole moment is oriented radially from MT axis. Certain kind of vibration modes involve radial oscillation of MT wall. Such kind of modes would involve displacement of rather large mass of surrounding cytosol, hence they would undergo stronger damping compared to the longitudinal optical oscillations in the direction along MT axis.

Circumferential elastic vibrations may be also connected with generation of oscillating electric field due to oscillatory changes of the distance between electric charges. Frequency of the modes depends on Young's and shear modulus. As the moduli exhibit large differences in magnitudes in different directions the frequencies of oscillations are corresponding.

Based on the discussion above, we have focused our analysis on the longitudinal optical oscillations of MT. Frequency of oscillations was chosen to be 1 GHz. It corresponds to calculations of MT vibrations of Pokorný and Wang. We estimate the oscillatory displacement of the charge in tubulin dipole moment along the MT axis to be around 0.1 nm for the vibrations in GHz region.[4] The static dipole moment (p_s) in the direction of the MT axis is accepted to be about 330 D (Tuszyński et al., 2002; Mershin et al., 2004). Length of the tubulin heterodimer is 8 nm, so the effective oscillatory component of the dipole p_a at the frequency of 1 GHz may be estimated as $p_a = (0.1/8)p_s = 4.125\,\mathrm{D} = 1.376 \times 10^{-29}\,\mathrm{C\,m}$.

Excitation of vibrations in our calculations was based on following models:

- Random excitation–random phase of oscillations was approximated by modulation of dipole moment of each of dipoles by function $\sin(2\pi n)$, n = random number from interval $\langle 0, 1\rangle$.
- Coherent excitation – modes with wavenumber k (k-modes) – longitudinal waves on the MT have been approximated by modulation of dipole moment of the individual heterodimers along the MT axis by function $a + b \times \sin(k_0 nz)$, z-axis corresponds to MT axis. Physically, these vibration modes belong to optical branch where neighbor particles (here α- and β-tubulin monomers in protofilaments) oscillate against each other. This corresponds to our approximation of heterodimer as an oscillating dipole. Number of mode n is provided instead of wavenumber k to indicate number of maxima. Note that direction of electric field vector is opposite in neighboring maxima as given by $\sin(k_0 nz)$ modulating function. In a special case all the dipoles may oscillate with the same phase (corresponds to $k = 0$ optical branch of longitudinal lattice vibration).

Dielectric parameters of the surrounding medium have been chosen based on experimentally obtained results from Asami et al. (1976) and Asami and Yonezawa (1996). Compared to the cited literature, we use slightly different value of the cytoplasm relative permittivity; instead 50 (Asami et al., 1976) we use 60. Slightly higher permittivity used corresponds physically to higher water content assumed (Table 2).

3. Results

Colormap figures depict MT electric field spatial distribution in the plane which contains the MT axis. White lines in the figures are contours of the MT wall. The arrows in the colormap figures denote the direction of the electric field vector. Note that neither arrow density nor arrow length is related to intensity. For greater dynamic range common logarithm of electric intensity is used for all

Table 2
Fixed parameters of calculations.

Frequency of vibrations	1 GHz
Number of protofilaments in MT	13
Number of heterodimers in each protofilament	50
Oscillatory component of dipole moment	$4.125\,\mathrm{D} = 1.376 \times 10^{-29}\,\mathrm{C\,m}$
Relative permittivity of the medium	60
Conductivity of the medium	1 S/m

[4] This estimate is based on measurements of Pelling et al. (2004) and Pelling et al. (2005) who found oscillations of yeast cell membrane (likely excited by cytoskeleton) in the frequency range about 1 kHz and amplitudes about 1–10 nm. Microtubules associated with motor proteins are also able to vibrate at frequencies of hundreds of kHz and amplitudes up to 10 nm in cilia and flagella (Kamimura and Kamiya, 1989, 1992; Sakakibara et al., 2004).

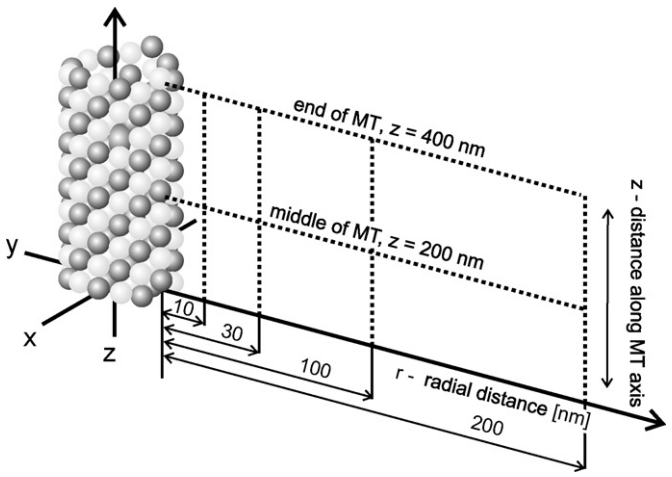

Fig. 7. The lines along which the electric field intensity has been plotted.

graphs and figures, e.g. color on the colorbar denoted 4 corresponds to intensity 10^4 V/m.

Electric field intensity dependence on distance radially from MT axis and along MT axis in selected distances from MT axis has been plotted in separate figures for simpler comparison. See Fig. 7 for the clarification of the plotting geometry.

3.1. Coherent Excitation

First we studied the simple model of the special case of coherent excitation. The spatial distribution of electric field around MT in this case is depicted in Fig. 8.

Coherently excited (at $k = 0$) MT behaves like a giant dipole having maxima of electric field intensity at its ends. This is of no wonder since it consists of chains of in phase oscillating dipoles with the same orientation of dipole moments.

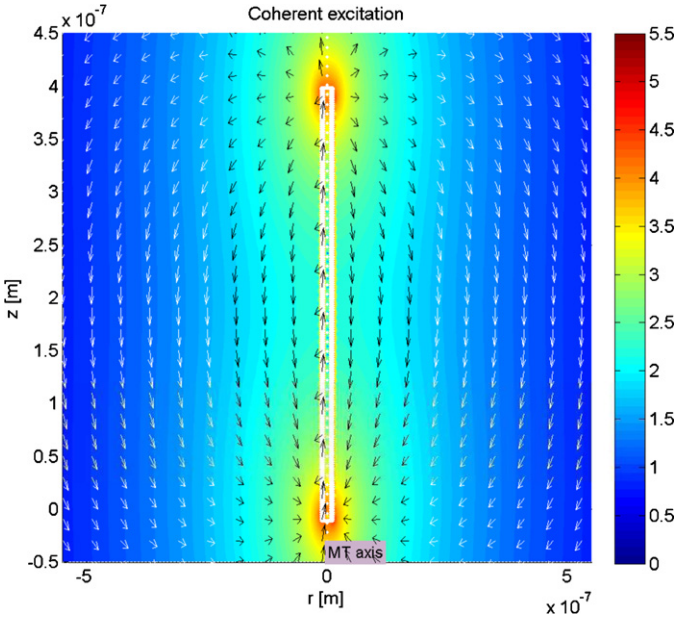

Fig. 8. Electric field around microtubule with coherent excitation ($k = 0$). Color coded electric field magnitude is in log scale (log(V/m)). Arrows represent electric field vector.

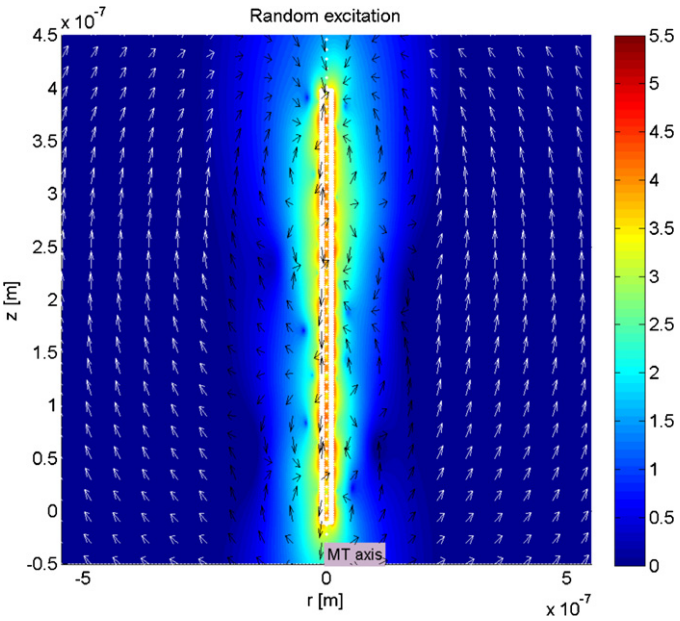

Fig. 9. Electric field around microtubule with random excitation. Color coded electric field magnitude is in log scale (log(V/m)). Arrows represent electric field vector.

3.2. Random Excitation

Spatial distribution of electric field generated by random (thermal) excitation of MT is given in Fig. 9.

The spatial distribution depicted in the Fig. 9 is just one example which results from the certain random number sequence. Naturally, the spatial distribution of random excitation can take various forms and often exhibits local minima.

3.3. Comparison of Coherent and Random Excitation

Comparison of coherent ($k = 0$) and random excitation of electric field is provided in Fig. 10.

We clearly see the difference between the field generated by random and in phase (coherent) oscillations. Although electric field intensity is higher in the middle of the MT axis ($z = 200$ nm) at the surface of MT for random excitation, it falls off faster with the dis-

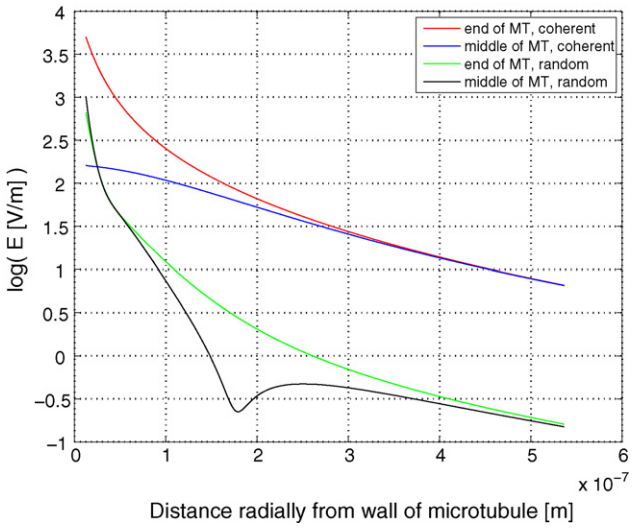

Fig. 10. Electric field intensity radial dependence from the wall of microtubule for the coherent and random excitation. From the end of MT ($z = 400$ nm) and from the middle of MT ($z = 200$ nm).

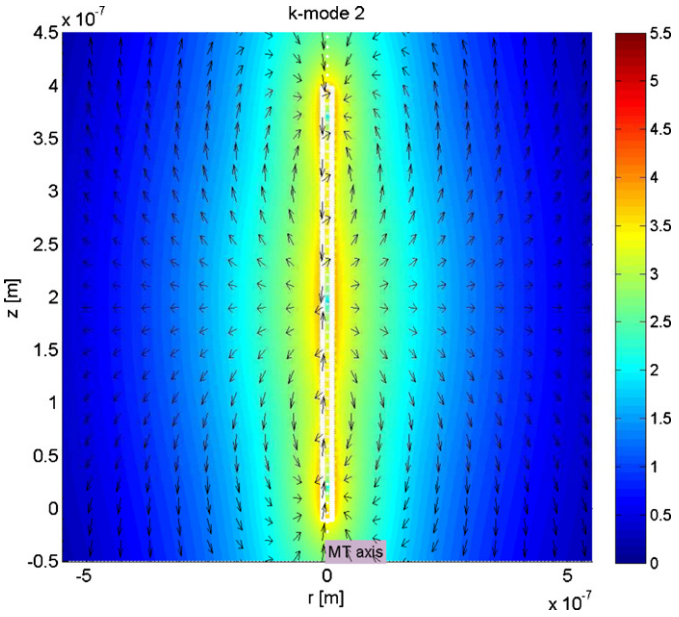

Fig. 11. Electric field distribution around MT of *k*-mode 2. Color coded electric field magnitude is in log scale (log(V/m)). Arrows represent electric field vector.

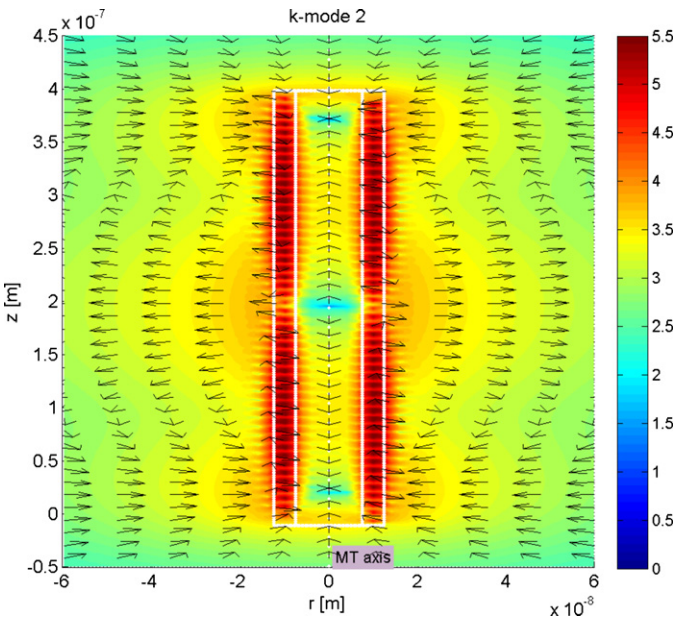

Fig. 12. Electric field distribution around MT of *k*-mode 2, zoomed in radial axis. Color coded electric field magnitude is in log scale (log(V/m)). Arrows represent electric field vector.

tance compared to the in phase (coherent) excitation (Fig. 10). The electric field intensity is higher at the ends of MT ($z = 400$ nm and 0 nm[5]) for the in phase excitation compared to the random one (Fig. 10). There is a difference by one and a half orders of magnitude between the electric field intensity of randomly and in phase excited vibrations in the distance larger than ca. 100 nm.

3.4. Coherent Excitation of Higher k-modes

Normal vibration modes (termed as *k*-modes) can be excited in microtubules. We primarily chose to study *k*-modes number $n = 2$, 5 and 50 as the representative ones. We depict the spatial distribution of the *k*-mode 2 in Fig. 11 and zoomed in spatial distribution in 12.

Two maxima in the MT lattice are clearly discernible in Fig. 12 whereas in Fig. 11 they are not. Therefore we chose to provide only zoomed in figures when displaying spatial distribution of electric field for higher ($n > 0$) modes. Note that abscissa and ordinate have different scales in zoomed in figures; the *r*-axis (ordinate) is 10-fold stretched to resolve the details of electric field in direction radially from MT axis.

The zoomed in spatial distribution of electric field around MT of *k*-modes 5 and 50 is depicted in Figs. 13 and 14, respectively. Electric field of higher modes is depicted in figures in supplementary material.

We observed that every half-wave of *k*-mode of longitudinal lattice vibration produces effectively a *k*-mode "dipole" field[6] the maxima of which are located on the poles of the "dipole". Moreover, *k*-mode "dipoles" are oriented to each other with the same pole. This can be seen also from the orientation of electric field vectors (see, e.g. Fig. 12 where two *k*-mode "dipolar" fields are discernible in each intersection of the MT wall; the plus poles are in the middle of the MT wall, i.e. the electric field vector is oriented radially from the plus poles of the "dipole"). The effective number of maxima in the field intensity up to the distance of 10 nm from the MT wall (or more, depending on the mode number) for the *k*-modes $n > 0$ is actually $n + 1$ since each pole of *k*-mode "dipole" contributes by one maximum. The number of maxima further changes in greater distance.

Higher the mode, the total dipole moment of whole MT is split into many small *k*-mode "dipoles" with opposite directions. The MT obtains multipolar *k*-mode character with rising mode number. Multipolar character of field source distribution explains the steeper decrease of intensity from the MT for higher mode numbers (e.g. see Fig. 15).

We provide figures depicting electric field intensity radial dependence from MT end and middle of MT for *k*-mode 2, 5, 10, 20 and 50 in Figs. 15 and 16.

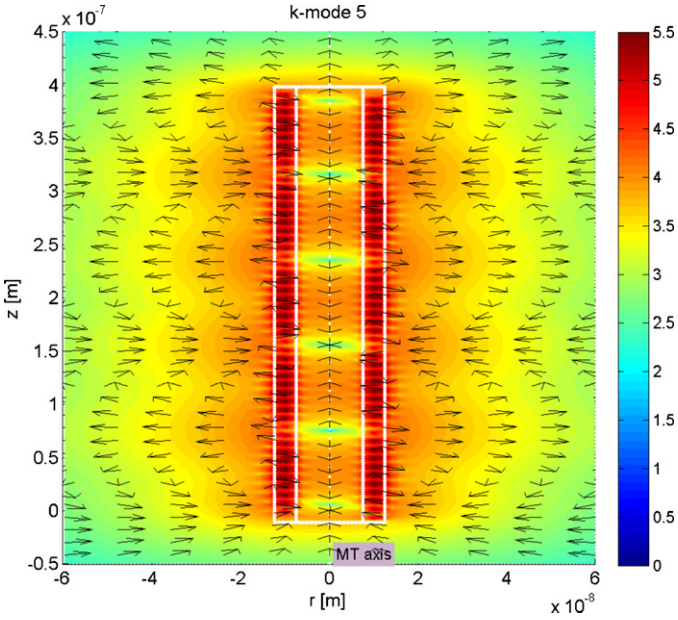

Fig. 13. Electric field distribution around MT of *k*-mode 5, zoomed in radial axis. Color coded electric field magnitude is in log scale (log(V/m)). Arrows represent electric field vector.

[5] Values are not depicted for $z = 0$ nm but they are essentially the same as for $z = 400$ nm due to the symmetry.

[6] We use term "*k*-mode 'dipole'" to describe an effective "dipole" field structure which appears in the MT wall due to every half wave of MT vibration. Number of mode indicates the number of *k*-mode "dipoles".

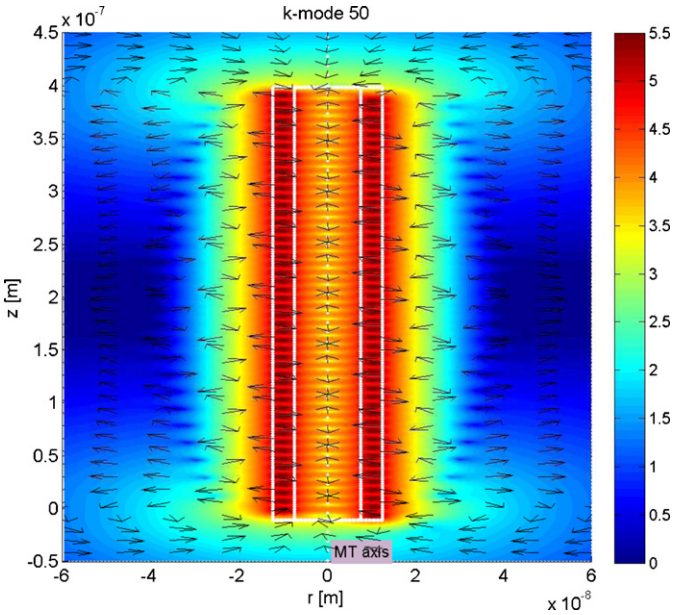

Fig. 14. Electric field distribution around MT of mode 50, zoomed in radial axis. Color coded electric field magnitude is in log scale (log(V/m)). Arrows represent electric field vector.

At the close distance from the MT, for certain regions, there is a specific k-mode which has highest intensity, see Figs. 15 and 16. We see that different k-modes manifest different rate of decrease of intensity with increasing distance radially from MT wall. This resembles multipolar behavior of intensity dependence on distance: higher the multipole degree (k-mode number of MT vibrations) steeper the decrease of intensity from the multipolar source.

Following this line of thought, it is interesting to observe the profile of the intensity for various distances r from the microtubule. In Figs. 17, 18 and 19 we provide the profiles of intensity parallel to MT axis for k-mode 2, 5 and 50, respectively.

4. Discussion

First the frequency dependence will be discussed. For the frequencies up to high THz region, we can work with the microtubule generated field as near field in the distances we treat here (up

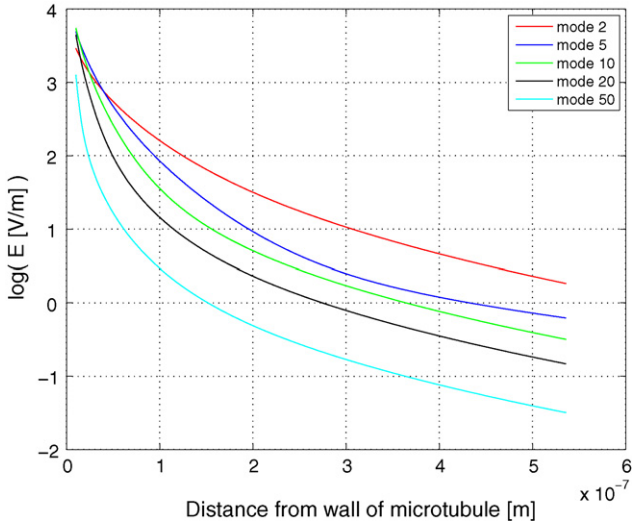

Fig. 15. Electric field intensity radial dependence from the wall of microtubule for the k-modes 2, 5, 10, 20 and 50. From the end of MT ($z = 400$ nm).

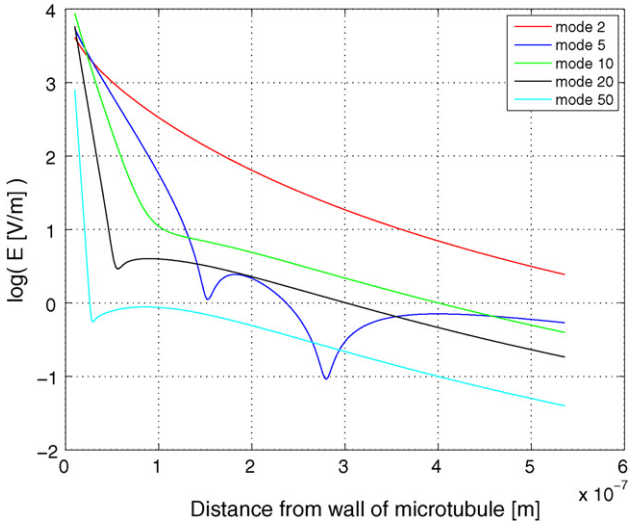

Fig. 16. Electric field intensity radial dependence from the wall of microtubule for the k-modes 2, 5, 10, 20 and 50. From the middle of MT ($z = 200$ nm).

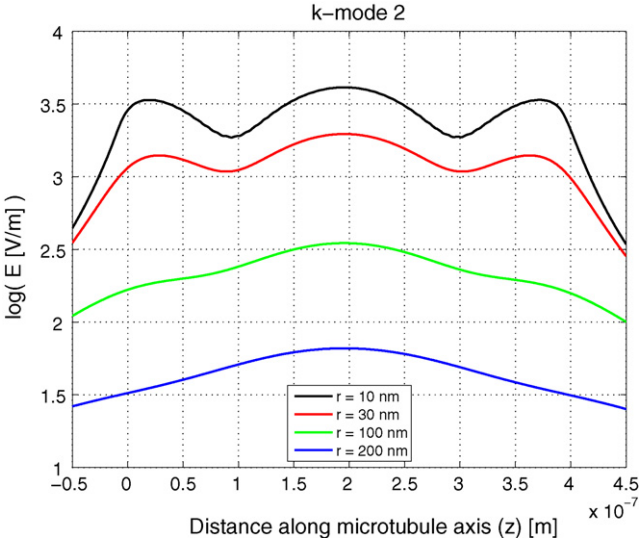

Fig. 17. Electric field intensity along the lines parallel to microtubule axis at the distance r from microtubule wall, mode 2. Microtubule is located from $z = 0$ to 400 nm.

to 1 μm from MT). So there is basically no difference in the field geometry for various frequencies.[7] With changing frequency the magnitude of the intensity may change due to permittivity and conductivity relaxations. Increasing frequency has an effect on effective current created by dipole oscillations ($Id = \omega p$), so the generated intensity would be higher if the amplitude of the vibrations does not decrease.

Discussing dipole moment separately; the generated intensity increases linearly with increasing dipole moment p magnitude. Amount of power in MT vibrations will influence effective value of dipole moment oscillating component p_a. Effective oscillating dipole moment would be increased if more power is supplied into the vibrations. If the amount of the oscillating charge is fixed,

[7] Actually, there would be frequency dependence of the field geometry since wavenumber of vibration modes is bound to frequency of oscillations by dispersion relations. However, in the case of optical branch of longitudinal oscillations the dispersion relation is rather flat, i.e. the function of frequency on wavenumber is almost constant.

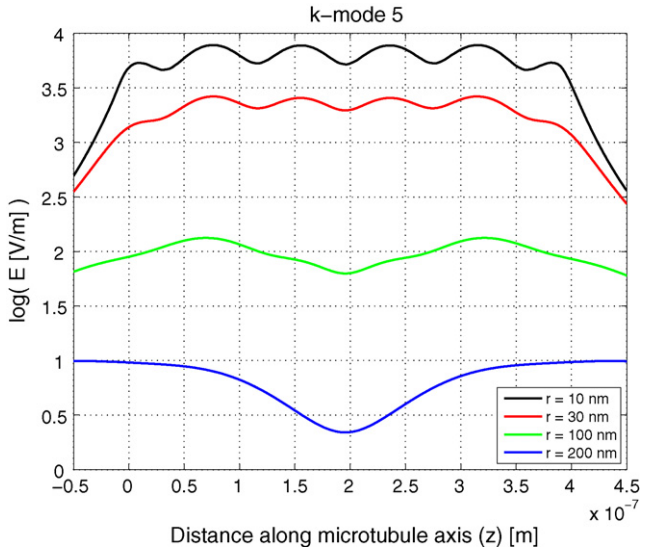

Fig. 18. Electric field intensity along the lines parallel to microtubule axis at the distance r radially from microtubule axis, mode 5. Microtubule is located from $z = 0$ to 400 nm.

increased dipole moment would correspond to the greater oscillating distance. However, for different frequencies, there is different amount of charge available for oscillations, given by the mobility of charge.

In a special case of strong energy condensation in certain microtubule vibration modes, electric field generated at certain frequency could be higher by several orders of magnitude and could exert force at the correspondingly greater distance. This kind of phenomenon may be transient only. Nevertheless, it may be involved in cellular interactions, if not by force interaction then by information transfer.

Only optical branches of vibrations can contribute significantly to electric field generation since in these kind of modes neighbor particles (monomers of tubulin) oscillate against each other. Actually, transverse (Kittel, 1976) and longitudinal (Berreman, 1963) optical modes in ionic crystals are responsible for electromagnetic field absorption. We can see the reciprocity of electromagnetic field absorption and generation in these processes. Acoustic longitudinal branch could also effectively contribute to the electric field generation in the MT near the limit of the Brillouin zone, where the k vector is already large (vibration wavelength short) so that particles in the lattice oscillate against each other.

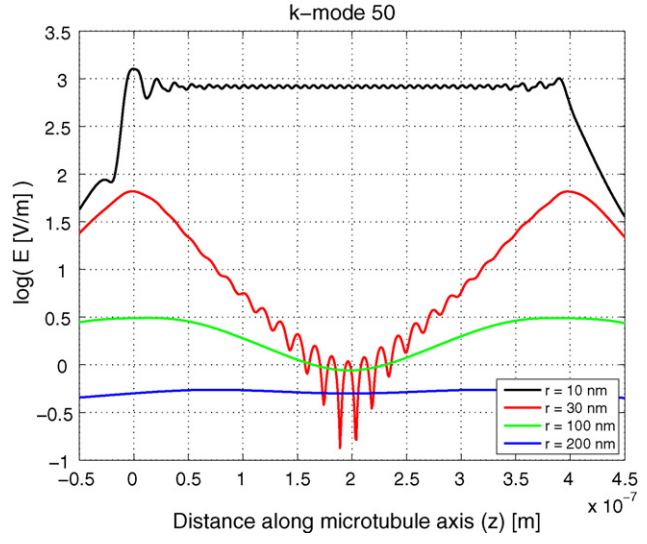

Fig. 19. Electric field intensity along the lines parallel to microtubule axis at the distance r from microtubule, mode 50. Microtubule is located from $z = 0$ to 400 nm. The slight asymmetry seen at the $r = 10$ nm is due to protofilament shift.

Microtubule associated proteins (MAPs) may also influence MT vibrations acting like a nodes imposing minimum of vibration amplitude (adding the mass to the binding point). This will influence generated field geometry, frequency of vibrations and the transfer of vibrations to other structures bound to the other end of MAPs.

In case of excitation of higher vibration modes, there will be regions of electric field with spatially oscillating direction of electric field vector. The dense distribution of regions with opposite field vector direction renders a problem for direct measurement of electric oscillations. Point detector needs to be used to resolve spatial regions of opposite direction of electric field vector.

We also studied the impact of the MT model with real lattice (protofilaments shifted to each other), which corresponds to realistic model of MT lattice compared to nonshifted models of MT (see supplementary material). For the low mode numbers ($n < N/2$), where N is the number of tubulin heterodimers in the MT, there was a small difference in the electric field around MT between the model with protofilaments (lattice) shifted and not shifted. However, the differences become significant for the growing mode number. The break in the symmetry introduced by protofilament shift changes the field geometry. The difference between the models with shifted and not shifted lattice are most pronounced for the higher mode number ($n > N/2$).

Vibrations of microtubules or of other electrically polar cellular structures may be the underlying mechanism generating the electric oscillations which were observed around living cells in kHz to MHz region (Pohl et al., 1981; Phillips et al., 1987; Hölzel, 2001; Pokorný et al., 2001; Jelínek et al., 2009).

We calculated and compared behavior of the electric field generated by random and coherent oscillations of microtubules. While random oscillations occur naturally without any special assumptions, the process where coherence of oscillations arise needs to be touched. Fröhlich's theory (Fröhlich, 1968a,b, 1980) describes a system of oscillators with energy supply, linear and nonlinear coupling with heat bath. If sufficient energy supply is provided to this system, condensation of energy occurs in lowest mode leading to its coherent excitation. Thus, coherent oscillations in microtubules described in this paper can be based on Fröhlich's theory. Numerous works have been inspired by Fröhlich's theory, for review see, e.g. Pokorný and Wu (1998), Hyland and Rowlands (2006) and Cifra (2009). Recently, Reimers et al. (2009) and McKemmish et al. (2009) presented analysis of Penrose–Hameroff orchestrated objective reduction (Orch OR) model of brain activity (Penrose and Hameroff, 1995; Hameroff, 1998; Hagan et al., 2002) with conclusion, that the model is founded on incorrect assumptions, in particular on Fröhlich's coherence, the essential requirement for Orch OR function. McKemmish et al. (2009) claimed that Fröhlich's coherence time is not long enough and that biochemical energy is too small to facilitate coherent condensation which follows from analysis in McKemmish et al. (2009). However, as described closer in comment by Pokomý (submitted for publication), authors of (McKemmish et al., 2009; Reimers et al., 2009) did not take into account the main sources of energy in the cell ("wasted" energy from mitochondria – see Section 1.2 this paper) and the fact that coherence may be strongly raised by

1. Weak damping due to organization of water (Fuchs et al., 2007, 2008, 2009; Giuliani et al., 2009) in strong static electric field around mitochondria (Tyner et al., 2007).

2. Interactions between individual Fröhlich's systems (Pokomý, submitted for publication).
3. Higher order nonlinear terms (Pokorný, 1982; Pokorný et al., 1986a,b).

If Fröhlich's theory of coherent excitations is reconsidered in the light of the facts in comment of Pokomý (submitted for publication), the disclaim of the Orch OR proposal of the brain function by Reimers et al. (2009) on the basis of insufficient coherence (McKemmish et al., 2009) does not seem to be justified.

5. Conclusions

Based on previous works (Pokorný and Wu, 1998; Pokorný, 1999; Pokorný et al., 1997, 1998; Havelka and Cifra, 2009a; Havelka, 2008), we have analyzed electric oscillations excited by vibrations of microtubule. Our model corresponds to optical branches of longitudinal vibrations.

The coherent vibrations of MT generate greater electric field in greater distance than random vibrations.

It was also found that the lattice shift of protofilaments has increasingly significant influence on the field geometry for the lattice vibration modes which wavelength is shorter than two tubulin heterodimer dipoles. This suggests necessity to model higher vibrations modes taking into account lattice shift, which is the real case.

More realistic oscillatory component of dipole moment based on experimental values of static dipole moment and amplitude of mechanical vibrations has been estimated than in previous works (Pokorný, 1999; Havelka and Cifra, 2009a).[8] Using more realistic dipole moments electric field intensity around microtubules is about two orders of magnitude lower than calculated in previous works (Pokorný and Wu, 1998; Pokorný, 1999; Pokorný et al., 1997, 1998; Havelka and Cifra, 2009a; Havelka, 2008). Nevertheless, the electric field generated by microtubules may contribute to self-organization of biological processes also under these conditions.

Acknowledgments

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.biosystems.2010.02.007.

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18

Electronic and Ionic Conductivities of Microtubules and Actin Filaments, Their Consequences for Cell Signaling and Applications to Bioelectronics

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18.1 Introduction

The prevailing view of neural information processing is based on passive properties of the membrane derived from the application of linear cable theory to dendrites, e.g., [114]. Recent studies, however, have suggested nonlinear models that accommodate new experimental evidence that appears to be inconsistent with this classical view. In particular, conventional theories do not satisfactorily explain: (a) significant fluctuations of synaptic efficacy over short periods of time in response to a recent burst of activation, e.g., [54], (b) a variety of active ion-channels capable of affecting the local membrane electric properties, and (c) nonlinear responses localized to specific dendritic sites, pointing to highly specialized mechanisms correlated with specific inputs. On the other hand, nonlinearities inherent in the new models, e.g., [74,133], give rise to a wider repertoire of (computational) capabilities, such as multiplication, fast correlation, etc. Moreover, experimental and theoretical support for conductive properties of protein filaments and the ionic clouds around them is becoming available, which indicates a far more complex picture of the signaling and information processing phenomena in neurons and other cells.

In this chapter, we suggest that functional electrodynamic interactions between cytoskeletal structures and ion-channels are central to the neural information processing mechanism. These interactions are supported by long-range ionic wave propagation along microtubule networks and actin filaments, and exhibit subcellular control of ionic channel activity, hence impacting the computational capabilities of the whole neural function. Cytoskeletal biopolymers, including actin filaments (AFs) and microtubules (MTs), constitute the backbone of wave propagation, and in turn interact with membrane components to modulate synaptic connections and membrane channels. Indeed, only recently clear functional interactions between these cytoskeletal structures have become apparent. Association of MTs with AFs in neuronal filopodia appears to guide microtubule growth, and plays a key role in neurite initiation [28]. This is further evidenced by the presence in neurons of proteins that interact with both MTs and F-actin, and proteins that can mediate signaling between both types of filaments. This is likely used to control microtubular invasion. The microtubule associated proteins MAP1B and MAP2, for example, are known to interact with actin in vitro, e.g., [146]. Cross-linking, MAP2 and/or MAP1B is highly probable to associate with both types of filaments contributing to the guidance of MTs along AF bundles. Extensive evidence exists confirming a direct interaction between AFs and ion channels, and a regulatory functional role associated with actin. Thus it is clear that the cytoskeleton has a direct connection to membrane components, in particular channels and synapses.

In the following, we demonstrate how each individual cytoskeletal component is capable of supporting both ionic and electronic wave propagation. In particular, we present a molecular dynamics model of the dendritic microtubule network (MTN) where arrays of MTs are interconnected by MAP2s. In this model, ionic waves propagate along the MTN and interact with C-termini of MTs to generate collective modes of behavior. Also in this chapter, a biophysical model of nonlinear ionic wave propagation along AFs is presented, supported by experimental evidence. Finally, we describe a model for a direct regulation of ion-channels and synaptic strength by AFs and networks of MTs that control the electrical response of the dendrite in particular, and the neuron in general. We begin this chapter by providing a general overview of the structure and function of a nerve cell, emphasizing the role of cytoskeletal filaments, ion channels, and their interactions.

18.2 The Neuron

The neuron is the quintessential communicating cell. A great variety of neural subtypes can be found interconnected to one another in complex electrochemical circuits via multiple synaptic connections. Thus, the amount of information processing by a neural circuit depends on the number of such connections or synapses. Neurons are highly polarized cells, whose level of morphological complexity increases during maturation into distinct subcellular domains. A typical neuron possesses three functional domains. These are the cell body or *soma* containing the nucleus and all major cytoplasmic organelles, a single axon, which extends away from the soma and takes cable like properties, and a variable number of dendrites, different

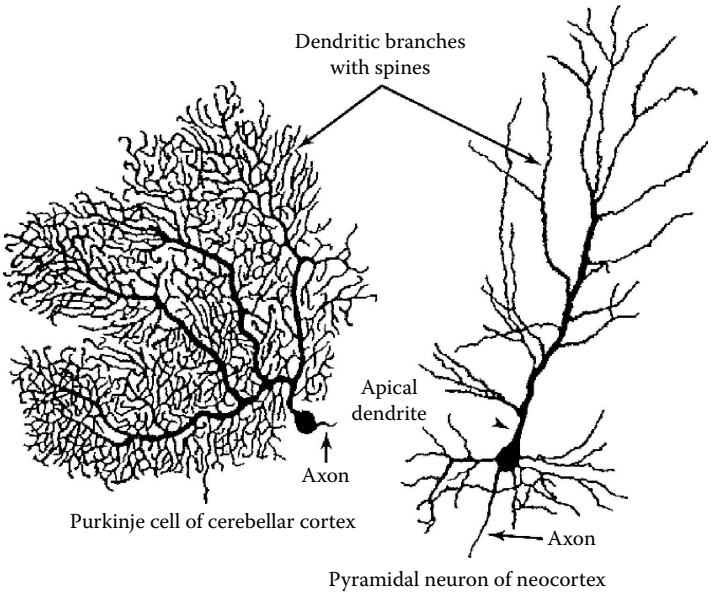

FIGURE 18.1 Two prominent examples of nerve cells.

in shape and complexity, which emanate from the soma. The axonal terminal region, where the synapse contacts it with other cells, displays a wide range of morphological specialization (see Figure 18.1).

Two major types of synapses can be found: (a) asymmetric, responsible for the transmission of excitatory impulses, and (b) symmetrical, present in inhibitory synapses. Turning morphology into a functional feature, the soma and dendritic tree are the major domains of receptive inputs. Thus, both the dendritic arborization and the axonal ramifications confer a high level of sub-cellular specificity in the localization of particular synaptic contacts on a given neuron. The extent to which a neuron may be interconnected largely depends on the three-dimensional spreading of the dendritic tree. Dendrites contain a large number of dendritic spines, which are small sac-like organelles, projecting from the dendritic trunk. Dendritic spines are more abundant in highly arborized cells such as pyramidal neurons, and scarcer in lowly interconnected smaller-sized dendritic tree inter-neurons.

Spines are the dendritic regions where most excitatory inputs are found, such that each spine contains one asymmetric synapse. The number of excitatory inputs can be correlated with the number of spines present in the dendrite. Dendritic spines are complex saccular enlargements, containing ribosomes, and cytoskeletal structures, including actin, and α- and β-tubulin. Electrically speaking, a single neuronal circuit is closed such that afferent inputs reaching the soma and the dendritic tree are transferred to the axon at the other pole of the cell which, in turn, is responsible for transmitting efferent neural information. Both the morphology of the axon and its course through the nervous system are correlated with the type of information processed by a particular neuron, and by its interconnectivity with other neurons. While the axon leaves the soma with a small swelling known as the axon hillock, some neurons protrude the axonal projection from the main dendrite instead. Cytoskeletal components are highly relevant in the structure/function of the neuron. The axonal hillock contains large "parallel" bundles of MTs. Axons are also located with other cytoskeletal structures, such as neurofilaments, which are more abundant in axons than in dendrites. The dynamics in cytoskeletal structures is central to our contention that information can be processed and "delivered" to the synaptic function by changes in the cytoskeletal structures.

Dendrites are the principal element responsible for synaptic integration and for the changes in synaptic strengths that take place as a function of neuronal activity. While dendrites are the principal sites for excitatory synaptic input, little is still known about their function. The activity patterns inherent to

the dendritic tree-like structures, such as in integration of synaptic inputs, are likely based in the wide diversity of shapes and sizes of the dendritic arborizations [116]. Some dendrites are unipolar, while others are multipolar; some have many orders of branching, while others have only one or two; and some branch primarily in two dimensions, while others have complex three-dimensional structures. The size and complexity of dendritic trees increase during development, which has been associated with the ability of the neural system to organize and process information [49]. The majority of synapses, both excitatory and inhibitory, terminate on dendrites, so it has long been assumed that dendrites somehow integrate the numerous inputs to produce single electrical outputs. It is increasingly clear that the morphological functional properties of dendrites are central to its integrative function. Branching in this context is essential to the increasingly complex ability of the neuron to respond to developmental complexity. Nerve branching and complexity are controlled by both external and internal cues. Of current interest to this are substrate and electromagnetic fields, while also recognized are the chemical gradients involved in local adaptation. Highly relevant to adaptation is cytoskeletal dynamics, as the dendritic cytoskeleton plays a central role in the process of ramification, filopodia formation, and more specialized neuronal activity such as long-term potential (LTP) and long-term depression (LTD) circuit dynamics, including memory formation.

The electrotonic or passive properties of dendrites became of great importance to our understanding of dendritic integration of synaptic signals. Early hypotheses were supported by work on motor neurons [34] and sensory receptors [38,67]. In that action, potentials were initiated in the axon hillock region as a result of the graded or algebraic summation of EPSPs and IPSPs occurring in various parts of the neuron. Because of the relatively short space constant calculated for motor neurons [18], it was predicted that excitatory post-synaptic potentials from synapses on the distal portions of dendrites would be ineffective in firing a neuron. Rall [115] revolutionized the field of dendrite physiology by using cable theory for the study of neurons and their dendrites. Action potentials initiated in dendrites by synaptic stimulation were later observed propagating in both directions at velocities of around 0.3 m/s. [7,21,43]. Direct intracellular recordings from dendrites of cerebellar Purkinje cells [84], cortical neurons [57], and hippocampal neurons [153] confirmed that action potentials could be initiated and propagated in dendrites. Very little is known, however, about the active, or voltage-dependent, properties of dendrites, and many questions remain unanswered. Such as the types of ion channels present in dendrites, location, and how active properties alter synaptic integration.

The firing of action potentials elicits characteristic M-shaped Ca^{2+} profiles across the neuron. The Ca^{2+} rise is small in the soma, highest in the proximal apical and basal dendrites, and small again in the distal apical dendrites [62,100,118–120]. The spatial distribution of Ca^{2+} channels explains the distinct Ca^{2+} rise in various cell locations but rather reflects the propagation into the dendrites of trains of action potentials. Imaging with Na^{+}-sensitive dyes has yielded similar patterns, suggesting that Ca^{2+} influx into the dendrites is driven by Na^{+} influx into the dendrites which is driven by Na^{+}-dependent action potentials [62]. Ca^{2+}-dependent action potentials have been demonstrated in hippocampal neurons and dendrites [131]. Most of the action potential induced Ca^{2+} entry is mediated by voltage-gated Ca^{2+} channels [100]. Multiple types of voltage-gated Ca^{2+} channels have been demonstrated in hippocampal neurons [36,40,68,101]. Based on the addition of specific Ca^{2+} channel blockers, the presence of L-, N-, P-, Q-, R-, and T-type Ca^{2+} channels has been reported [16]. This channel distribution may contribute to spike-triggered Ca^{2+} entry into the soma and apical dendrites. Both low-voltage activated (LVA) channels and at least one high-voltage activated (HVA) channel seem to be most predominant in regions of the dendrite at least a 100 mm from the soma. Experimental results strongly suggest a heterogeneous distribution of different types of voltage-gated Ca^{2+} channels within the soma and dendrites of hippocampal neurons [152]. The presence of Na^{+} channels in distal dendrites helps explain the back-propagation of action potentials well into the dendritic tree [88,89,138,139,141]. Action potential propagation in dendrites is rather low, and susceptible to the effects of cable filtering.

A most prominent feature of dendritic ion channel regulation is the potential boosting and enhancement of distal synaptic events. Voltage-gated Na^{+} and Ca^{2+} channels may play an important role in amplifying the magnitude of EPSPs towards the soma. However, channel regulation may also be important for

dendritic interactions in the immediate vicinity of the synaptic input. Voltage-gated channels may alter the local input resistance and time constant, which in turn would influence both spatial and temporal summation of EPSPs and IPSPs, thus conveying highly nonlinear interactions; signal amplification [64,137]. Thus, the nonlinear properties of dendrites may serve as electronic devices. In addition to providing an output, the action potentials elicited by synaptic input, whether initiated at the local synaptic sites or at the axon hillock and back-propagated into the dendrites, would essentially reset the dendritic membrane potential. Action potentials might also provide feedback or an associative signal to other synapses active just before or just after the action potential.

Back-propagating action potentials may also provide feedback to dendritic synapses after a neuronal output has occurred. Ca^{2+} signals are implicated in post-synaptically-induced synaptic plasticity such as short-term potentiation (STP) [92], LTP, and LTD [8,15]. Actin has long been postulated to play important roles in neuronal morphogenesis. High concentrations of actin have been found in the leading edge of the growth cones of developing neurons, and in synapses of mature neurons. Actin filament dynamics also plays a role in a number of related phenomena, including axon initiation, growth, guidance, and branching, in the morphogenesis and stability of dendrites and dendritic spines in synapse formation and stability. Neurons conduct two types of cytoplasmic growth processes in axons and dendrites, both of which are functionally and morphologically distinct, but are developmentally led from the growth cone. Growth cones are composed of finger-like filopodia and veil-like lamellipodia that help the growing processes explore their environment. Axons travel long distances, and upon reaching their targets, produce terminal branches where the growth cones are converted into pre-synaptic terminals. Dendrites usually do not extend such long distances, but often branch extensively, giving rise to dendritic trees. Upon proper contact with axons, post-synaptic dendritic specializations create functional synapses by opposition with the pre-synaptic terminals of axons. These are localized to small protrusions on their dendritic trees called dendritic spines, which are the site of most excitatory synapses. In the mammalian brain these multiple synaptic connections are believed to play important roles in learning and memory. Thus, neural morphogenesis requires changes in both the underlying cytoskeleton and specific interactions with the plasma membrane. To generate a membrane protrusive structure such as an axon, coordinated changes in actin and microtubular organization are required. Growth cones of developing neurons, for example, rapidly extend and retract filopodia by concentred changing in cytoskeletal dynamics, namely bundled AFs in filopodia, and composed of a cross-linked actin network in lamellipodia [80,81,155]. In this section, we have outlined the main features of signaling taking place within a neuron. Major emphasis was placed on the role played by the cytoskeleton in the neuron's structure, growth, and internal signaling. The next section discusses the main biophysical properties of the cytoskeleton.

18.3 The Cytoskeleton

The cytoskeleton is a major component of all living cells. It is made up of three different types of filamental structures, including actin-based microfilaments (MFs), intermediate filaments (IFs) (e.g., neurofilaments, keratin), and tubulin-based microtubules (MTs). All of them are organized into networks, which are interconnected through numerous particular proteins, and which have specific roles to play in the functioning of the cell. The cytoskeletal networks are mainly involved in the organization of different directed movements in cell migration, cell division, or in the internal transport of materials. Polymerization of MFs is responsible for cell migration and for the remodeling of the leading edge of cells. Molecular motors are protein complexes associated with the cytoskeleton and drive organelles along MTs and MFs in a "vectorial" transport.

MTs are some of the most basic and important cytoskeletal elements in the morphological scheme of neurons. As for the case of cytochalasins in actin structures, the addition of microtubular modifying agents was first used to assess the role of MTs in axon elongation. Microtubular disrupting agents affect axons first and have no early effects on growth cones and filopodia, suggesting that MTs provide support for the axon. Conversely, microtubular invasion may be a critical element in the formation of neurite projections. Initially, expanded lamellipodia first undergo segmentation at certain spots where MTs accumulate in an

ordered array [27,144,156] and gradually "migrate away" from the cell body. These lamellipodia transform into nascent growth cones. Neurite elongation takes place as the MTs in the shaft become compressed into a narrower bundle. In cultured hippocampal and cortical neurons, this process most likely implies the coordinated control of the major cytoskeletal components, including actin, MTs, and associated proteins.

Thus, MTs are essential for neurite development. One particular aspect of microtubular involvement in neurite outgrowth is vesicular transport. Most membrane insertion occurs at the growth cone of axons [22,24]. MAP-stabilized MTs, form parallel or antiparallel arrays, which might act as compression resistant struts inside neurites [61]. Because MTs constantly invade the actin cytoskeleton in lamellipodia of epithelial [125] and neuroblastoma cells [27], such a mechanical role for MTs could also be important during de novo neurite initiation. Neurite-like protrusions, for example, can be induced in non-neuronal cells by the exogenous expression of stabilizing proteins such as MAP2 and tau [72,79,157]. Spontaneous neurite initiation is dependent on both the presence and the dynamic properties of MTs.

A number of proteins associate with MTs, including MT motors, kinases, so-called "structural" MAPs, which alter microtubule structure, and specialized MAPs, which bind to microtubule plus or minus ends. Members of the MAP2/tau family have long been proposed to be important regulators of neurite behavior [10,48,151]. MAP2 and tau stabilize MTs by reducing catastrophe and promoting rescues, leading to prolonged growth periods and thus enhanced net microtubule accumulation [32,45,75]. MAP2 and tau stabilize MTs by binding along the outer ridges of protofilaments [2].

F-actin and MTs cytoskeletal networks are typically thought to fulfill separate, independent cellular roles. Highly dynamic actin networks are known for their role in cell spreading and contraction. The more stable MT cytoskeleton is known for its importance in cell division and organelle trafficking. However, recent studies provide important roles for the actin cytoskeleton in cell division and trafficking and for MTs in the generation and plasticity of cellular morphology. A direct physical association between both cytoskeletons has been suggested, because MTs often preferentially grow along actin bundles and transiently target actin-rich adhesion complexes.

Association of MTs with actin cables in filopodia appears to guide microtubule growth along the most efficient path anti-parallel to retrograde flow [130]. Thus, specific coupling between MTs and actin bundles presumably promotes MT advance. Proteins that are able to interact with both MTs and F-actin, or proteins that can mediate signaling between both cytoskeletons likely control the regulation of MT invasion. This is further strengthened by specific microtubule association of the microtubule-associated proteins MAP1B and MAP2, both known to interact with actin in vitro [20,109,128,134,146]. Furthermore, MAP2 binds to F-actin and efficiently induces actin bundle formation [51,129]. It is likely that by crosslinking, MAP2 and/or MAP1B associated with both cytoskeletons could be involved in the guidance of MTs along actin filament bundles. Alternatively, MAPs could shuttle from MTs to actin and could alter F-actin behavior by actively crosslinking AFs.

The effect of changes in the cytoskeleton on ion channel activity has been the focus of recent attention. Most studies have employed drugs that selectively stabilize or destabilize either AFs or MTs, with resulting effects on specific ion channel activity, either as changes in whole cell conductance or single channel activity by the patch clamping techniques. Although direct binding of specific cytoskeletal proteins to individual channel proteins remains to be demonstrated, recent work in which exogenous purified actin has been added to the cytoplasmic side of channels excised from the cell membrane has in numerous cases directly confirmed an effect of cytoskeletal filaments. A body of earlier evidence suggested that various cytoskeletal components, including actin and actin-associated proteins, anchor, co-localize, and regulate both the spatial stability as well as the function of ion transport proteins. It is possible that actin either binds directly to the channels, or that actin may first interact with actin-binding proteins, which in turn regulate (by binding or other indirect interaction) ion channel function. Independent studies demonstrated that membrane-resident CFTR is functionally regulated by AF organization, which may also involve a direct interaction with actin. Thus, ion channel may be controlled by direct and indirect cytoskeletal interactions.

Before we present an integrated view of the conductive properties of the active and passive structural blocks of the cytoskeleton, we should review the current state of knowledge regarding their conductive properties.

18.4 Overview of Biological Conductivity

Cope [19] has presented a detailed review of the applications of solid-state physics concepts to biological systems. He divided the discussion into seven groups. The first involves semiconduction of electrons across enzyme particles as a rate-limiting process in cytochrome oxidase, evidenced by kinetic patters of enzymes and microwave Hall measurements. Second, pn-junction conduction electrons were suggested by kinetics of photobiological free radicals in the eye and in photosynthesis. Their I-V characteristics conform to the diode equation:

$$I = I_0 exp(V/kT) \tag{18.1}$$

Third, there may be a connection between superconduction and growth in nerves. Fourthly, phonons and polarons are involved in mitochondrial phosphorylation. Next, piezoelectricity and pyroelectricity may be involved in the growth of nerves and bone structures. Then, infrared electromagnetic waves may transmit energy in lipid bilayers of nerves and mitochondria. Finally, complex sodium and potassium ions in structured cell water may be analogous to valence band electrons in semiconductors, free cations being analogous to conduction electrons. The ionic process in cell water resembles electronic conduction in seminconductors. For dried proteins or DNA, Szent-Gyorgi[142] noticed that their conductivities obeyed the relationship:

$$\sigma = \sigma_0 exp(-E_a/kT) \tag{18.2}$$

With activation energies E_a on the order of 1 eV. The presence of water may cause considerable changes in conductivities, (for example, in cytochrome oxidase this may be reduced to 0.3 eV).

Adessi *et al.* [1] have provided a theoretical treatment of transmission through a poly(g)DNA molecule. They found that there was a modification of the rise of a B-DNA form, which can induce a shift of the conduction channel towards a valence one. They found that deformation of the backbone of the molecule has a significance on the hole transport. Furthermore, the presence of ionic species (e.g., Na) in the surrounding of the molecule can create new conduction channels. In photo-excitation experiments, hole transport occurs through coherent transport at short distances and long-range incoherent or hopping transport at large distances. Fink *et al.* [39] studied the metallic behavior in λ DNA using an electron projection technique. Superconducting behavior has been observed by Kasumov *et al.* [66]. Porath *et al.* [111] observed semiconductor behavior with a poly(g)-poly(c)DNA molecule. A scanning force microscope has been used by de Pablo *et al.* [26] to observe insulating behavior. This paper models a single poly(g) DNA lying between electrodes, and transmission was computed with a Landauer–Buettiker formalism based on Green's functions. The electronic properties of O_2-doped DNA have been studied by Mehrez *et al.* [96], who suggested that DNA conduction is affected by O_2-doping, in particular hole doping. However, ab initio atomic structure calculations of O_2-doped DNA indicate negligible charge transfer. The absence of dc conductivity in λ DNA has been investigated by de Pablo *et al.* [26]. First principles electronic structure calculations indicate a minimum DNA resistance of 10^{16} Ω per molecule, and a minimum resistivity of 10^4 Ωm. However, lower energy electron bombardment may dramatically increase conductivity and with electron energies between 50 and 200 eV may lead to metallic conduction with a resistance of 2×10^8 Ω. Fink *et al.* [39] discussing DNA and conducting electrons stated that I-V characteristics of DNA indicate ohmic-like behavior with a resistance of 2 $M\Omega$ for a 600 nm-long DNA rope corresponding to $\rho = m\Omega$ cm. These authors speculate that DNA may be semiconducting, insulating, or metallic simultaneously depending on the arrangement of molecules in the structure. Endres *et al.* [37] discussed the quest for high-conductance DNA saying that DNA conduction is an unsolved problem first enunciated by Eley and Spivey [35]. Experimental outcomes range from insulating to semiconducting to even superconducting, and the uncertainties depend on: (a) contacts between the electrodes and the DNA and (b) differences in the molecules (sequence, length, arrangement, preparation) within the DNA. It appears that drying DNA can lead to conformations with localized electronic high states, although hole doping of the backbone by counter-ions might also be possible. Wet DNA may support electrical current due to solvent impurity

states in the large high-Π^* energy gap. Davies and Inglesfield [25] discussed the embedding method for conductance of DNA calculations. It is believed that the pathway for transport in DNA runs through the bases in the center of the double helix molecule. The dominant transport mechanism for DNA appears to be coherent hopping. Calculations show extreme sensitivity to the choice of contact orbitals.

Marechal [95] discussed the transfer of protons as a third fundamental property of hydrogen bonds. The importance of the cooperative resonance types in cyclic structures is stressed. Proton transfers along H-bonds have been discussed as a mechanism of vital importance for chains, helices, and in cyclic structures such as rings. Consta and Kapral [17] discussed proton transfer in mesoscopic molecular systems. Proton transfer rates and mechanics are studied theoretically with solvent molecules in solution. The proton transfer occurs as a result of orientational fluctuations on the cluster's surface. The environment influences solvent effects and reaction rates. The rate constant was established to be 1/59 ps at 260 K and for the bond length fixed at 2 Å. Morowitz [102] discussed proton semiconductors and energy transduction in biological systems. He presented the possibility of protochemistry paralleling electrochemistry. ATP synthesis was discussed as coupled to proton transport assuming a gated protein semiconductor across a membrane. Protein conduction in biological structures involves the water of hydration and protein molecules.

Ripoll *et al.* [121] investigated ionic condensation and signal transduction. Ion condensation on biopolymers at a critical value of the charge density may lead to intra-cellular signaling due to the diffusion of condensed counter-ions in the near region along cytoskeletal filaments. A feedback mechanism was proposed between condensation/decondensation of calcium and the activation of calcium-dependent enzymes. This is shown to create coherent patters of protein phosphorylation. It was postulated that ion condensation operates in signal transduction. In the case of polymer ionics [117], the conductivity depends on temperature via the formula:

$$\sigma T = Aexp[-B/(T - T_o)] \tag{18.3}$$

indicating activated hopping. The concept of the Fermi level in solution has been extended by Bockris and Khan [9] through the introduction of a reversible potential for a redox couple.

Unfortunately, measurements of conductivity of biopolymers (including MTs) are fraught with difficulties due to their structural heterogeneity, strong dependence on the environmental conditions (pH, ionic concentration, temperature, etc.), and the inherent liquid state of the sample. Nonetheless, attempts have been made recently to measure resistance values of MTs, both intrinsically and as ionic conduction cables. Fritzsche *et al.* [41] made electrical contacts to single MTs following dry-etching of a substrate containing gold microelectrodes. Their results indicate intrinsic resistance of a 12 micron-long MT to be in the range of 500 megaohms, giving a value of resistivity of approximately 40 mega-ohms/micron in their dry state. The same group performed measurements on gold-coated MTs [42] which were covered with a 30-nm layer of gold following sputtering. The resistance of the thus metallized MTs was estimated to be below 50 ohms, (i.e., originated entirely due to the metallic coating). Minoura and Muto [99] measured the ionic conductivity of MTs using an electro-orientation method under an alternating electric field. The ionic MT conductivity was calculated as $1.5 + / - 0.5$ mS/m, which is approximately 15 times greater than that of the buffer solution. This was attributed to the counter-ion polarization and is consistent with another recent study of ionic conductivity along MTs. Furthermore, Goddard and Whittier [47] reported their measurements of RF reflectance spectroscopy of samples containing the buffer solution, free tubulin in buffer, MTs in buffer, and finally, MTs with MAPs in buffer. The sample size in each case was 5 ml. The concentration of tubulin was 5 mg/ml and in the last case the concentration of MAP2 and the tau protein was 0.3 mg/ml. The average DC resistance reported by these authors was (a) 0.999 $k\Omega$ (buffer), (b) 0.424 $k\Omega$ (tubulin), (c) 0.883 $k\Omega$ (MTs), and (d) 0.836 $k\Omega$ (MTs + MAPs). It's not straightforward to translate these results into resistivity of MTs without making assumptions about their geometrical arrangement and connectivity. However, assuming all tubulin being polymerized in case (c) and a uniform distribution of MTs forming a combination of parallel and series networks, one can find the resistance of a 10 μm–long MT, forming a basic electrical element in such a circuit, to have approximately an 8 $M\Omega$ resistance. This compares very favorably to an early theoretical estimate of MT conductivity based on the Hubbard model with electron hopping between tubulin monomers. This model predicted the resistance of a 1-μm MT

to be in the range of 200 $k\Omega$, hence a 10-μm MT would be expected to have an intrinsic resistance of 2 $M\Omega$, within the same order of magnitude as the result reported by Goddard and Whittier [47]. While direct experimental analysis of intrinsic conductivity of the various cytoskeletal components is still nebulous, theoretical modeling of these effects is very well developed. The next section presents our current understanding of the intrinsic conductivity of MTs.

18.5 Intrinsic Electronic Conductivity of Microtubules

Microtubules (MTs) are ubiquitous in eukaryotic cell biology. Their primary role is to serve as the cellular scaffold and act as a corridor for the transport of vesicles through the cell by motor proteins. The interest in MTs stems from their involvement in a number of crucial cellular processes including mitosis and, more recently, the discovery of their role in the communication between the exterior of the cell and the nucleus [46,94]. The special arrangement of MTs just prior to chromosone separation makes them an ideal candidate for mediating a signal of some sort, which would cause the coordinated breakage of sister chromatids. In the axons of nerve cells, the MTs are arranged in bundles, parallel to the axon [56]. These groups of MTs span the entire length of the axon and might serve to carry signals from the cell body to the nerve terminus or vice versa. No mechanism for this sort of direct feedback is known. We demonstrate here that the MT may be able to conduct a current along its length through a process of electron hopping between binding sites on each dimer.

The MT is made up of a protein called tubulin. Each subunit contains α-tubulin and β-tubulin monomers joined together to form a dimer. The dimers are connected end to end to form a protofilament. These protofilaments, usually 13 in number, are wrapped up to form a hollow tube, the MT. The dimers are themselves 5 nm in diameter and have a length of about 8 nm. A MT has an outer diameter of 25 nm and an inner diameter of 15 nm (see Figure 18.2).

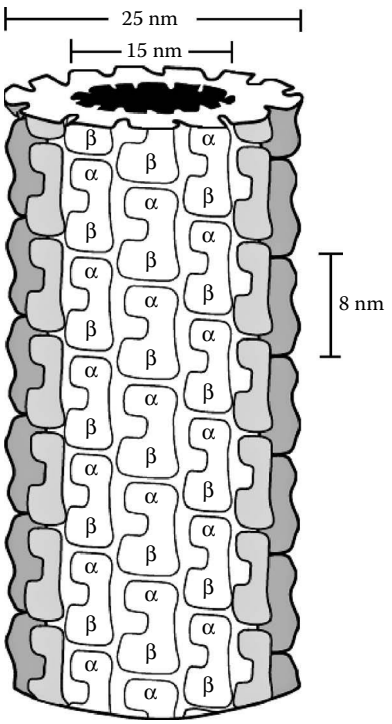

FIGURE 18.2 Structure of a typical microtubule demonstrating how it is formed by a special arrangement of tubulin dimers.

The tubulin dimer of which the MT is composed has at least two different conformational states [58]. That is, while the amino-acid chain of the protein remains fixed, its three-dimensional shape may change. It is believed that these conformational changes are the result of hydrolysis and an associated charge movement from one binding site within the molecule to another. It is precisely this charge movement that may lead to the conduction through the polymer.

One may take a semi-classical view of the movement of electrons along the protofilament. We ignore the fact that the protofilament is nearly one-dimensional and consider its *bulk* properties. Assuming two conduction electrons for each tubulin dimer, the conduction electron density is

$$\begin{aligned} n_0 &= 2/(8nm \times 5nm \times 5nm) \\ &= 1.0 \times 10^{19}/cm^3. \end{aligned} \tag{18.4}$$

In comparison with other materials, this value is slightly higher than typical semi-conductor concentrations, usually $10^{13} - 10^{17}/cm^3$, but is much less than typical metals, which have a conduction electron density of $10^{22} - 10^{23}/cm^3$. However, the electrons are expected to have relatively low mobilities unless they are excited from the ground state configuration. The excitation energy is a function of both the number of electrons within the protofilament and its size. Suppose that the mobility of excited electrons is at least similar to the mobility of electrons within semiconductors, such as Si where the electron mobility, μ_e, is about 1300 $cm^2\ V^{-1}\ s^{-1}$. The electrical conductivity σ is given in a semi-classical model by the well-known formula

$$\sigma = ne\mu_e \tag{18.5}$$

where e is the electron charge and n is the conduction electron density given by

$$n = n_0 e^{-\Delta/\kappa T} \tag{18.6}$$

where n_0 represents the total number of electrons. If we assume that the gap between valence and conduction bands, Δ, is about 0.40 eV, then at physiological temperature, some of the carriers will be excited and should be available to conduct. Applying the preceding formula (18.6), a conductivity of about 10^{-2} (in SI units) is expected for the protofilament.

$$\sigma \sim (10^{25} e^{-0.40/0/026})(1.6 \times 10^{-19})(0.13) \sim 0.04\Omega^{-1} m^{-1} \tag{18.7}$$

However, if the hybridization of electronic orbitals lowers the gap to a value closer to zero, the expected conductivity would be significantly higher:

$$\sigma \sim (10^{25})(1.6 \times 10^{-19})(0.13) \sim 2 \times 10^5 \Omega^{-1} m^{-1} \tag{18.8}$$

These values should be compared to values in excess of 10^8 for copper and other good conductors, and 10^7 for lesser conductors and semi-metals. Typical semi-conductors such as Si and Ge have intrinsic conductivities of $10^{-3} - 10^{+1}$ at room temperature, but when heavily doped, the conductivity may rise to between 10^4 and 10^5.

In order to model the MT as a semiconductor, the individual tubulin dimer must first be physically analyzed. The $\alpha\beta$-tubulin dimer is comprised of about 890 amino acids which represents about 13000 nuclei. A gross simplification which we introduce is to treat each monomer as having an effective site energy for electronic binding. Fluctuations of the site energy on scales less than the monomer spacing are therefore washed out. It is convenient to model this system as two quantum wells separated by a potential barrier. The two wells represent two binding sites for an electron, which may hop between the monomers, resulting in the change of molecular conformation. We can estimate the relevant parameters of the well depth and the well width from kinetic studies and with a knowledge of the molecule's overall geometry. Once the bound states and their energies are known for this system, this knowledge may be applied to a

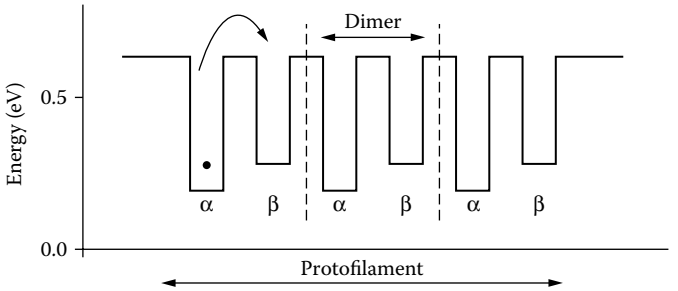

FIGURE 18.3 A one-dimensional chain of quantum wells representing the energy profile along a MT protofilament where the α and β monomers have alternating site energies.

second-quantization scheme. This procedure allows us to make progress without specifically requiring us to solve for the wavefunction of the system.

A 1D chain of quantum double wells is pictured in Figure 18.3. For simplicity, we are initially making each binding site have the same energy. If this is related to GTP hydrolysis, they may differ by about 0.17 eV (4 kcal/mol) [59]. This is roughly the quantity of energy stored in the lattice when GTP is hydrolyzed to GDP [13,60]. The barrier height between dimers has been estimated from kinetic data. The bond strength must be somewhat larger than the hydrolysis energy delivered to the lattice or else hydrolysis of the dimers would cause immediate disassembly. Hopping along the protofilament by electrons presumably has an energy comparable to this binding energy. We have estimated 0.4 eV for intra-dimer hopping and 1.0 eV for the inter-dimer hopping. What may be varied in the model, is the central barrier which must be overcome for each tubulin dimer to change conformation. If it is nearly as large as the barrier to electron movement between dimers, then the protofilament will look like a chain of single quantum wells rather than double quantum wells because the difference between the α and β monomers will not be distinguished. However, if the central barrier is too small, the chain will again look like a series of single quantum wells since the effect of the barrier would be negligible. The depth of the deeper well, f_1, has been selected as 0.4 eV initially. Given the value, we assumed a 25% difference in well depths. If f_1 is much larger, the dimer nature of the protofilament is lost because the wells are effectively the same depth.

The results for the single electron eigenstates of a double quantum-well may be found using a transfer matrix method for the time-independent 1D Schrodinger equation in each interval in which the piece-wise constant potential is defined as

$$\hat{H}\psi = \frac{\hat{p}^2}{2m}\psi + \hat{V}\psi = E\psi \tag{18.9}$$

The oscillatory solution

$$\psi(x) = A\cos(kx) + B\sin(kx) \tag{18.10}$$

applies to $E > V$ where

$$k = a\sqrt{2m(E-V)}/\hbar \tag{18.11}$$

while for $E < V$, the solutions are exponentially damped

$$\psi(x) = Ae^{\kappa x} + Be^{-\kappa x} \tag{18.12}$$

where

$$k = \sqrt{2m(V-E)}/\hbar \tag{18.13}$$

Now, matching conditions for the wave function and its derivative are applied at each interface. This gives a 2×2 matrix relating the coefficients in the wave function expansion according to

$$\begin{pmatrix} A_2 \\ B_2 \end{pmatrix} = \begin{pmatrix} {}^{2,1}M_{11} & {}^{2,1}M_{12} \\ {}^{2,1}M_{21} & {}^{2,1}M_{22} \end{pmatrix} \begin{pmatrix} A_1 \\ B_1 \end{pmatrix} \tag{18.14}$$

In the preceding equation, the superscript to the left of the M denotes which coefficients are being linked. The subscripts indicate the matrix element. In an iterated fashion, one eventually arrives at the following:

$$\begin{pmatrix} A_n \\ B_n \end{pmatrix} = \begin{pmatrix} {}^{n,1}M_{11} & {}^{n,1}M_{12} \\ {}^{n,1}M_{21} & {}^{n,1}M_{22} \end{pmatrix} \begin{pmatrix} A_1 \\ B_1 \end{pmatrix} \tag{18.15}$$

Now, for bound states $A_1 = B_n = 0$ so that the wavefunction decays outside the region of the quantum wells:

$$B_n = 0 \quad \Rightarrow \quad {}^{n,1}M_{22} = 0 \tag{18.16}$$

Given the choices for our quantum double well (width, $a = 2$ nm; and height, $V = 1.0$ eV), one finds that there are two bound states for this dimer problem. As the number of dimers is increased, one resorts to numerical analysis. In Figure 18.4, the energy of the two bound states is plotted as the energy of the central barrier is varied. The lowest energy state is the symmetric wavefunction, which increases in energy sharply as the central barrier is increased. The second bound state is the antisymmetric wavefunction it has a node within the barrier, hence its energy increases only marginally.

The energy calculation for a sequence of n such dimers leads to the development of bands as hybrid orbitals develop across the polymer. Roughly speaking, one band is of hybridized symmetric orbitals and the higher lying band of hybridized antisymmetric orbitals. The separation of the two bands and the spread of each individual band determine the values of the parameters which shall be used in the tight-binding Hubbard model used to model the polymer.

The second-quantized Hamiltonian for a many-electron multidimer system becomes

$$\hat{H} = \sum_{i,\sigma} \epsilon_i \hat{c}^{\dagger}_{i\sigma} \hat{c}_{i\sigma} + \sum_{i \neq j,\sigma} t_{ij} \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} \tag{18.17}$$

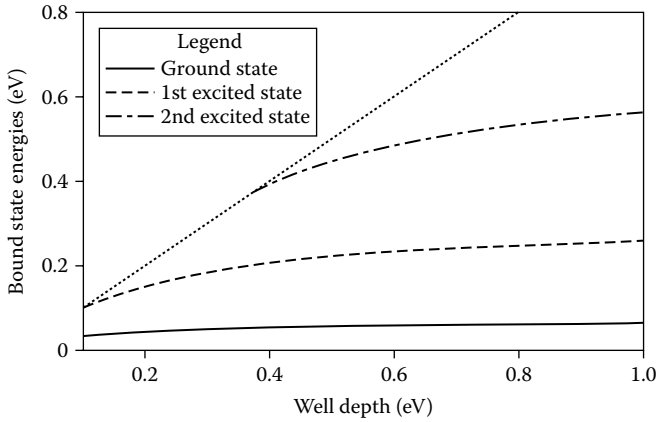

FIGURE 18.4 Variation of the bound state energies with the depth of the double quantum wells, $f_1 = f_2$. The straight line indicates where the bound state energy is equal to the well depth.

where the operators $\hat{c}^{\dagger}_{i\sigma}$ and $\hat{c}_{i\sigma}$ are the usual fermion creation and annihilation operators, and where i runs from 1 to $2n$ representing the two sites on each of the n dimers. The ϵ_i represent the potential energy at each site and the kinetic terms t_{ij} account for hopping from one site to another. There are two different kinds of interactions depending on whether an electron hops within a dimer, or between dimers. Hence, this kinetic parameter has two different values along the chain. Since there are no interactions between the states with $+\frac{1}{2}$ spin and those with $-\frac{1}{2}$ spin, the state space may be divided into two identical subspaces labeled by the electron spin. In each of these subspaces, the matrix corresponding to this chain of dimers is tri-diagonal and may be diagonalized to determine the itinerant eigenstates of the system. The coefficient t_d which represents the movement of an electron within a dimer causes the eigenstates to be split into two bands by an amount, $2t_d$. Within each of these bands, the coefficient t_l of the inter-dimer hopping, causes these two single electron bands to broaden each to about $2t_l$. Now, although the preceding description is qualitative, clearly whenever $t_d \sim t_l$, these bands will overlap and transitions between the states will be important.

Considering the band formation in the quantum well formulation, we can assign values to t_d and t_l. The two bands are separated by about 0.40 eV, so $t_d \sim 0.20$ eV, and using the bottom band to estimate t_l it widens until it is about 0.10 eV thick, so $t_l \sim 0.05$ eV.

The Hubbard model is the simplest second quantized model to capture all of the essential features of a real system of interacting particles [82]. In this model, electrons are the fundamental excitations. All states may be built up from the vacuum using the fermion creation operators. Once the number of sites is selected, n, and the number of electrons is determined, e, there are

$$\binom{2n}{e} = \frac{2n!}{e!(2n-e)!} \tag{18.18}$$

distinct electronic states. These states are built up by applying in normal order each of the combinations of e-electron states.

The second quantized Hamiltonian for the Hubbard model with electron–electron interactions is conventionally written as

$$\hat{H} = \sum_{i,\sigma} \epsilon_i \hat{c}^{\dagger}_{i\sigma} \hat{c}_{i\sigma} + \sum_{i \neq j,\sigma} t_{ij} \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} + \sum_{i} U_i \hat{c}^{\dagger}_{i\uparrow} \hat{c}_{i\uparrow} \hat{c}^{\dagger}_{i\downarrow} \hat{c}_{i\downarrow} \tag{18.19}$$

The last term accounts for electron–electron repulsion now that we have gone to a many electron system. In our simple model, this is limited to an onsite interaction. To this point, we have made each of the site energies equivalent ($\epsilon_i = \epsilon, \forall i$). In what follows, $U = 2.0$ eV which is the energy associated with two bare electrons lying about 0.7 nm apart. This choice seems reasonable within a well of width 2.0 nm. We try to solve the problem by considering the electron–electron term as a perturbation to study how it affects the electronic states of the dimer. The Hamiltonian in the absence of the final U-term may be diagonalized by the following unitary transformation:

$$\alpha^{\dagger}_{1'\sigma} = \frac{1}{\sqrt{2}}(c^{\dagger}_{1\sigma} + c^{\dagger}_{2\sigma}) \tag{18.20}$$

$$\alpha^{\dagger}_{1'\sigma} = \frac{1}{2\sqrt{2}}(c^{\dagger}_{1\sigma} - c^{\dagger}_{2\sigma}) \tag{18.21}$$

The linear combination of operators is simple because we have made the site energies ϵ_1 and ϵ_2 equal. These new operators are easily shown to obey the usual anticommutation relations for fermions and are said to be in the itinerant or distributed electron basis. The Hamiltonian may now be rewritten in terms of these new operators, as

$$\hat{H} = \sum_{\sigma}(\epsilon + t)\alpha^{\dagger}_{1\sigma}\alpha_{1\sigma} + \sum_{\sigma}(\epsilon - t)\alpha^{\dagger}_{2\sigma}\alpha_{2\sigma} + \frac{1}{2}\sum_{\sigma \neq \sigma'} V_{ijlm}\alpha^{\dagger}_{i\sigma}\alpha^{\dagger}_{j\sigma'}\alpha_{l\sigma'}\alpha_{m\sigma} \tag{18.22}$$

the vacuum state is represented by $|\Omega>$. Acting with our Hamiltonian on this basis gives us the matrix form of the Hamiltonian:

$$\hat{H} = \begin{pmatrix} 2(\epsilon+t)+\frac{(U_1+U_2)}{4} & 0 & \frac{(U_1-U_2)}{4} & \frac{(U_2-U_1)}{4} & 0 & \frac{(U_1+U_2)}{4} \\ 0 & 2\epsilon & 0 & 0 & 0 & 0 \\ \frac{(U_1-U_2)}{4} & 0 & 2\epsilon+\frac{(U_1+U_2)}{4} & -\frac{(U_1+U_2)}{4} & 0 & \frac{(U_1-U_2)}{4} \\ \frac{(U_2-U_1)}{4} & 0 & -\frac{(U_1+U_2)}{4} & 2\epsilon+\frac{(U_1+U_2)}{4} & 0 & \frac{(U_2-U_1)}{4} \\ 0 & 0 & 0 & 0 & 2\epsilon & 0 \\ \frac{(U_1+U_2)}{4} & 0 & \frac{(U_1-U_2)}{4} & \frac{(U_2-U_1)}{4} & 0 & 2(\epsilon-t)+\frac{(U_1+U_2)}{4} \end{pmatrix}$$

The Hamiltonian is then diagnolized in order to determine its eigenstates. In the absence of the Coulomb repulsion ($U = 0$), the electrons are localized on the $2'$ site with energy $2(\epsilon - t)$, the gap to the first excited state being $2t$. However, as U is increased, the effect of the Coulomb interaction raises the energy of the ground state relative to the excited singlet states, which do not feel the Coulomb repulsion. Consequently, the gap between the ground state and the first excited state is actually reduced through the introduction, as shown in Figure 18.5. Thus, when conductivity is being considered, it is important to bear in mind that the gap is reduced by the presence of the electron–electron interactions.

To this point, we have made each of the site energies equivalent, ($\epsilon_i = \epsilon$), and the electron repulsion the same at all sites ($U_i = U$). The lone complication lies within the kinetic term, where the hopping parameter will depend on the two sites between which the electron hops. However, since we first consider the MT protofilament, it shall initially be a constant as well, $t = 0.4$ eV, where the choice comes from the separation of the energy bands in the 1D Schrödinger picture.

Since the Hamiltonian conserves both the particle number and the overall spin of each state, we are free to work in subspaces enumerated by the spin of the states it contains. This reduces the problem slightly since it is possible to consider individually the spin $+1$, spin 0, and spin -1 systems separately in a two-electron system. In addition, the symmetry implies that the spin $+1$ and spin -1 systems have the same conductive properties. We also find that the ground state is always a state consisting of the lowest possible total spin.

When the inter-dimer hopping term is small relative to the intra-dimer hopping term $t_{inter} < t_{intra}$, the form of the ground state is independent of the number of dimers. In each case, the ground state is a linear combination of singlet states on each dimer. The singlet states are those where two electrons are

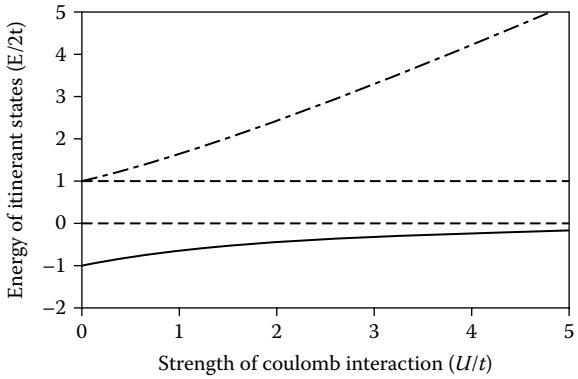

FIGURE 18.5 Energy of the itinerant states as a function of U/t with the energy plotted in units of $2t$.

paired on an individual binding site, or where the state consists of the anti-symmetric combination of an up-electron and a down-electron on neighboring binding sites of the dimer. When the inter-dimer hopping parameter is increased, the density of states becomes more uniform between the ground state and the highest energy eigenstate. With the gap between the ground state and the first excited state of the system becoming diminished. Since the hopping term within a dimer is largest, the system prefers to have two electrons with the same spin within the same dimer in the ground state.

For electronic conductivity, the band formation importantly depends upon the hopping parameter between dimers. When this parameter is small relative to the intra-dimer hopping, $t_{inter}/t_{intra} \sim 0.1$, there remains a rather large gap between the ground state and the first excited state of about $2t_{intra}$ which is reduced as the repulsion term increases. However, when the hopping between dimers is equal to the hopping within dimers, $t_{inter} \sim t_{intra}$, the gap is minimized. In the event that $t_{inter} > t_{intra}$, we would effectively have to relabel the dimers and would then return to the picture where $t_{inter} < t_{intra}$.

Kubo was the first to derive formulae for the electrical conductivity in solids, and his derivation is applied to our case in the present section [77]. In the Coulomb gauge, we write the vector potential for a uniform electric field along the direction of the chain, $A(t)$, as

$$A(t) = \frac{cE(t)}{-i\omega} e^{i\omega t}, \tag{18.23}$$

Adding a field-coupling term to the Hubbard Hamiltonian and using a canonical transformation gives [91]

$$\hat{H}_0' = -t \sum_{i\sigma} [\hat{c}_{i\sigma}^{\dagger} \hat{c}_{i+1\sigma} e^{-ieA(t)/c} + \hat{c}_{i\sigma}^{\dagger} \hat{c}_{i-1\sigma} e^{ieA(t)/c}]. \tag{18.24}$$

where a is the lattice spacing and c is the speed of light. Expanding to second order in the vector potential, we find

$$\hat{H}_A = \frac{A}{c} \hat{j} - e \frac{A^2}{c^2} a^2 \hat{H}_0, \tag{18.25}$$

where

$$\hat{j} = i\mathrm{Eat} \sum_{i\sigma} (\hat{c}_{i\sigma}^{\dagger} \hat{c}_{i+1\sigma} - \hat{c}_{i\sigma}^{\dagger} \hat{c}_{i-1\sigma}). \tag{18.26}$$

So, the energy of the original system is corrected to this order and the interaction term is proportional to the current.

Suppose a time-dependent external electric field is applied to a solid:

$$E_{\alpha}^{\mathrm{ext}}(r,t) = \Xi_{\alpha}^{\mathrm{ext}} e^{iq\cdot r - i\omega t}, \tag{18.27}$$

where α represents the Cartesian coordinate directions. In a linear response theory, the induced current is proportional to the applied electric field:

$$J_{\alpha}(r,t) = \sum_{\beta} \sigma_{\alpha\beta}'(q,\omega) \Xi_{\beta}^{\mathrm{ext}} e^{iq\cdot r - \omega t}, \tag{18.28}$$

where $\sigma_{\alpha\beta}'$ is a parameter relating the observed current density to the applied field's direction, its periodicity, and frequency. However, the symbol appearing in (18.28) is not the conductivity we seek. Rather, we want the conductivity which represents the response to the total electric field in the solid, a quantity that can be measured. This conductivity takes into account all of the currents, induced by the external fields, that

create their own electric fields. Thus, we seek $\sigma_{\alpha\beta}$ that relates the macroscopic electric field to the currents of the system.

$$J_\alpha(r,t) = \sum_\beta \sigma_{\alpha\beta}(q,\omega) E_\beta e^{iq\cdot r - i\omega t} \tag{18.29}$$

$$E_\alpha(r,t) = \Xi_\alpha e^{iq\cdot r - i\omega t} \tag{18.30}$$

$$\sigma_{\alpha\beta} = Re(\sigma_{\alpha\beta}) + i\,Im(\sigma_{\alpha\beta}) \tag{18.31}$$

We write the Hamiltonian of the system as $\hat{H} + \hat{H}'$ where the latter term contains the electric field, which we shall introduce as a time-dependent perturbation. The evolution of the operators is given as follows [30]:

$$\hat{j}(x,t) = e^{i\hat{H}t}\hat{j}(x)e^{-i\hat{H}t}. \tag{18.32}$$

where $\hat{H}$ is the unperturbed Hamiltonian. Thus, the interaction picture is adopted. The Kubo formula for electrical conductivity gives the result in terms of a current–current correlation function:

$$\sigma_{\alpha\beta}(q,w) = \frac{1}{\omega}\int_0^\infty dt e^{i\omega t} < \Omega|\hat{j}^\dagger_\alpha(q,t)\hat{j}_\beta(q,0) - \hat{j}_\beta(q,0)\hat{j}^\dagger_\alpha(q,t)|\Omega > + i\frac{n_0 e^2}{m\omega}\delta_{\alpha\beta} \tag{18.33}$$

where $|\Omega >$ is the ground state of the system, which has not been perturbed by the electric field. The first term in (18.33) is known as the incoherent contribution to the conductivity. Now, we use (18.32) in the preceding equation and use the fact that the basis we are using is that of the eigenstates. Consequently, with proper normalization the new expression for the real part of the conductivity is [90]

$$\sigma(\omega) = \frac{\pi}{Z}\frac{(1-e^{-\beta\omega})}{\omega}\sum_{n,m} e^{-\beta E_m}| < n|\hat{j}_\alpha|m > |^2\delta[\omega - (E_n - E_m)] \tag{18.34}$$

The DC conductivity is calculated by a procedure [90,91], which employs the sum rule

$$\int_0^\infty \sigma'(\omega)d\omega = -\frac{\pi e^2}{2}a^2 < K > \tag{18.35}$$

where

$$\sigma'(\omega) = D\delta(\omega) + \sigma_{kubo}(\omega) \tag{18.36}$$

and $< K >$ is the expectation value of the kinetic energy term of the Hamiltonian. The strategy of computing $\sigma(\omega)$, and subsequently the Drude contribution, at either zero or finite temperature is reasonably straight forward. The first step is to construct the matrix corresponding to the Hamiltonian [Equation (18.19)] in a subspace which is defined by the geometry of the lattice, the number of electrons, and the total electron spin. This leads to serious computational difficulties since for a moderately sized lattice of say ten dimers and eight electrons of arbitrary spin, the corresponding space is about 77 million states. A matrix of dimension 77 million square has about 5.9×10^{15} elements. Obviously, working on a moderately sized lattice, we are restricted to a low concentration of electrons. Due to electron-hole symmetry, we are also able to study the nearly filled situation. It is useful to keep in mind the relationship

$$\lambda = \frac{1240nm \cdot eV}{\text{Energy}}. \tag{18.37}$$

So that in the plots following, the visible light range is between 1.75 eV and 3.10 eV. Energies larger than this range correspond to ultraviolet, and those smaller to infrared wavelengths. Since cells respond to light in the near-infrared [4], and centrioles comprised of MTs are proposed to be the site of light detection [5], absorption in this energy range is of great interest to cell biology. These energies should also be compared

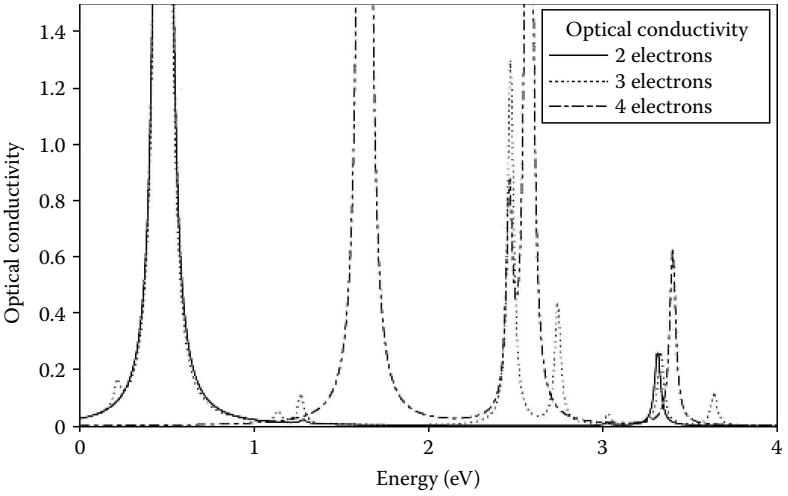

FIGURE 18.6 Optical conductivity of the two dimer system with two, three, and four electrons.

to the energy available from ATP and GTP hydrolysis, which are 0.49 eV and 0.22 eV, respectively. These values provide limits for the interactions of chemical energy in the conduction process.

At zero temperature, the linear polymer results are summarized concisely by saying that there is no DC conductivity that agrees with the previous modeling of Hubbard systems [44].

The following results for the 1D chain of tubulin dimers represent the AC conductivity. There are peaks in $\sigma(\omega)$, also known as the optical conductivity, which correspond to excitations between electronic states of the polymer. Electronic excitation results in electron redistribution and hence the development of a current. We have selected $\epsilon = 0.02$ eV to present our results, which is roughly kT and is smaller than the three energy scales in the Hubbard model, t, U, and $4t^2/U$ [82] so we expect to be able to pick out features corresponding to specific transitions in the predicted optical spectra.

We shall begin with two dimers and $t = 0.4$ eV, $U = 2.0$ eV, and equal site energy on the α and β monomer sites. This system can support up to eight electrons. The results for these systems are presented in Figure 18.6. The predicted spectra for the two and three electron cases are quite similar. The differences include a new but small peak in the three electron spectrum at an energy just below the major peak, and variations in the detail of the absorption band centered about 2.6 eV. However, the qualitative similarity of the two spectra is remarkable in terms of band locations. Contrast these two cases with the four-electron system that corresponds to half-filling. The difference is that the peak in the optical conductivity spectrum increases dramatically from about 0.48 eV to 1.64 eV. This change indicates how the half-filled case is special. The ground state in this system largely consists of electrons spread out to minimize the Coulombic energy. Exciting this system by moving any particular electron will come at the cost of greatly increasing the electron–electron repulsion and, consequently, the location of the required excitation energy corresponds roughly to U.

As the length of the polymer is increased, results remain qualitatively similar. In the three-dimer structure, results are shown for the conductivity in Figure 18.7(a) and for the integrated conductivity in Figure 18.7(b). Once again, the spectra are quite similar for all fillings aside from the half-filled case which consists of six electrons. The shift of the conductivity to higher energies can be clearly seen as the filling fraction increases towards half filling. We now present results where the broadening parameter, ϵ, and the temperature have been varied to see the effect that they have on the results. Figure 18.8 demonstrates that as the width is decreased, the spectrum becomes sharper. Note how the two peaks in this simple spectrum that occur near 2.5 eV are successively blended into a band as the ϵ is increased. Peaks represent transitions from one specific electron configuration to another and are weighted with the associated conductivity.

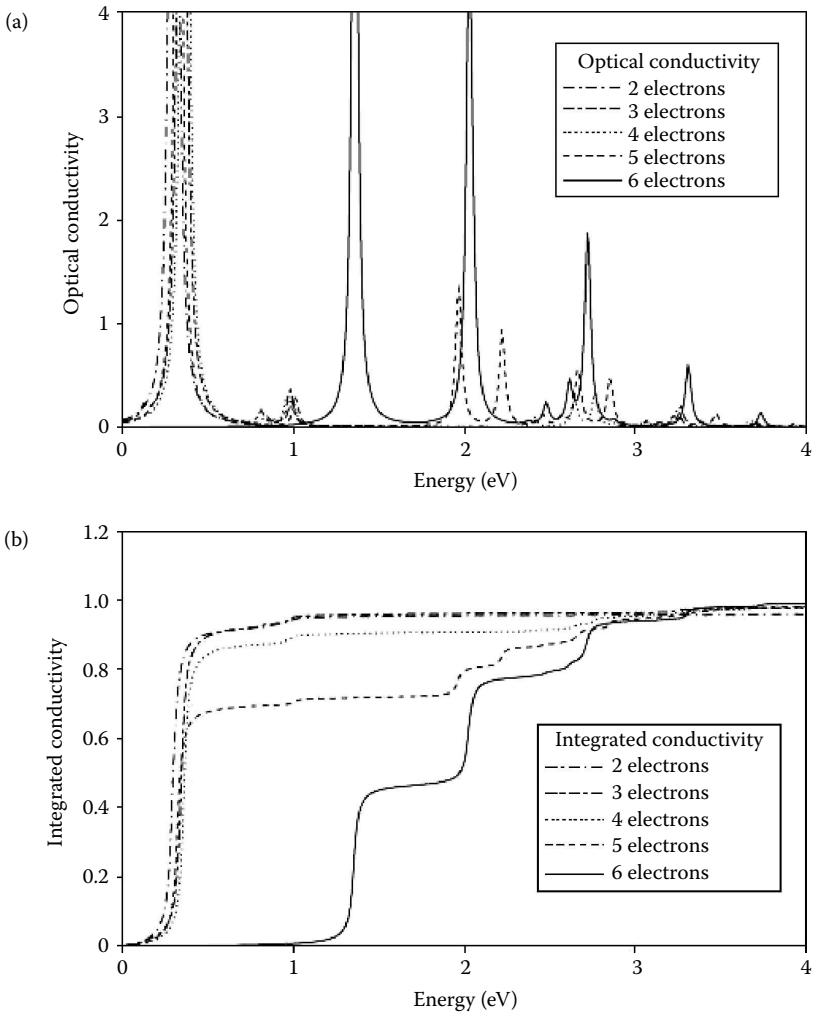

FIGURE 18.7 (a) Optical conductivity of the three-dimer system from two electrons through half filling. (b) Integrated optical conductivity of the three-dimer system shows that indeed the protofilament is an insulator. The gap for excitation of the half-filled polymer is seen easily here.

The effect of changing the temperature is to excite electron configurations other than the ground state. Consequently, transitions from one excited state to another can now contribute to the conductivity. Figure 18.9 shows how this often reduces the gap to conduction: all the major peaks in the 300 K spectrum shift slightly towards lower energies in the 3000 K spectrum, but many more features develop as well as states that previously did not contribute to conductivity may make transitions to excited states. Physiological temperature of 300 K ($kT = 0.026$ eV) is small relative to the system's energy scales, consequently lowering the temperature further causes very little change to the spectrum.

If the α-tubulin and β-tubulin sites have different energies, then the MT A and B lattices can be distinguished. The effect of this change on the optical spectrum seems to be small when individual protofilaments are considered. In addition, the sum rule shows that protofilaments remain insulators. There are a few parameters we can consider individually and which can be compared as the size of the system changes: the threshold to conductivity, and the expectation value of the kinetic energy operator. The threshold to conductivity is simply the location of the first absorption peak in the optical spectrum.

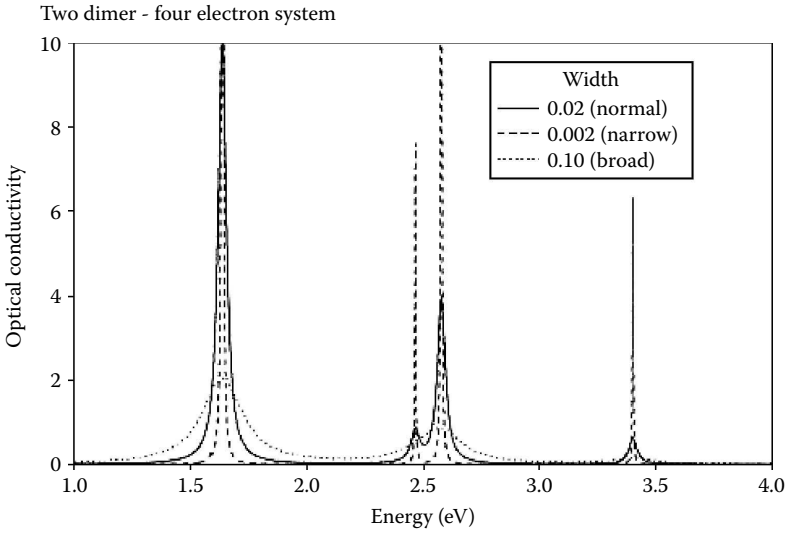

FIGURE 18.8 Optical conductivity spectrum variation with the broadening parameter.

It is a decreasing function of the polymer length. Starting at 0.48 eV for the two-dimer system, it seems to drop a little faster than $1/n$ for systems consisting of n dimers and containing two electrons. The fitted value in Figure 18.10 varies $\alpha 1/n^{1.4}$. Extrapolating to larger values of n, this gap appears certain to fall into the range where these excited states could be thermally excited. The protofilament then develops a finite DC conductivity at finite temperature. It is interesting that although the first peak does move towards zero, the fraction of the conductivity it contains is gradually diminished. Some of the conduction continues to reside in a peak located close to t in the energy spectrum. In the 2D lattice, there is a phase transition in the dimensionality. DC conduction becomes possible even at zero temperature. However, we shall still be concerned about the boundary conditions since calculations are being performed on a small lattice.

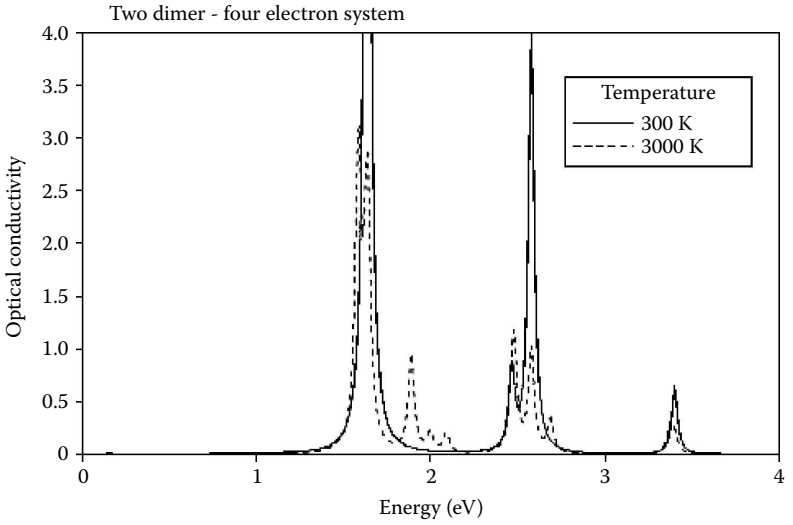

FIGURE 18.9 The variation of the optical conductivity spectrum with temperature is shown.

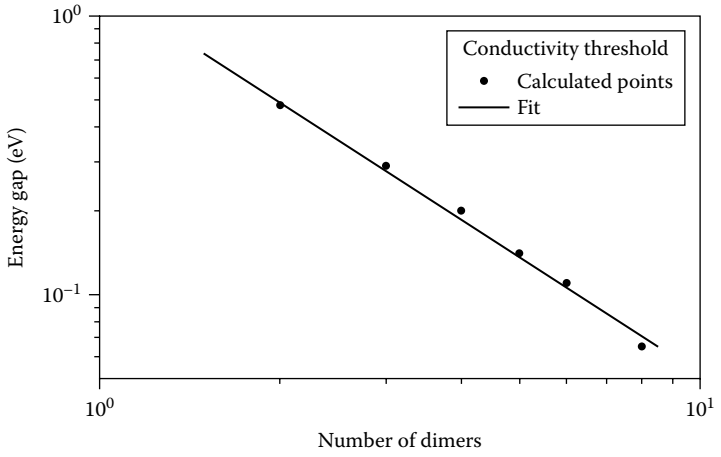

FIGURE 18.10 Threshold to conduction as a function of protofilament length.

Consider the kinetic energy/electron that has been calculated in a Hubbard model with our parameters for t and U and with either one, two or three protofilaments. Figure 18.11 demonstrates how the kinetic energy of each electron approaches the limit of $2t$ or 0.80 eV as the size of the system increases. We can easily see that for more than about 20 electron sites, the average kinetic energy of each electron starts to approach the infinite limit and we expect that the results obtained from our model should converge to the long protofilament limit in a similar manner. In particular, we are considering 28 electron sites in our MT model, which span three protofilaments. While this is not the ideal situation, the boundary conditions are not expected to create huge effects in our results. We are also using open boundary conditions since they minimize the boundary effects when compared with a variety of periodic boundary conditions.

The unit cells depicted in Figure 18.12 are the basis for the calculations. In the A lattice used for the calculations, the α-dimers connect to β-dimers along the $\bar{3}$-start helix, both within the lattice and at the seam when the MT is wrapped up. In the B lattice, the connections are $\alpha - \alpha$ or $\beta - \beta$ along the $\bar{3}$-start helix and never $\alpha - \beta$. For comparison, calculations have also been carried out with a flat tubulin sheet, specifically an A lattice MT lacking periodic boundary conditions in the lateral direction. This simulates the

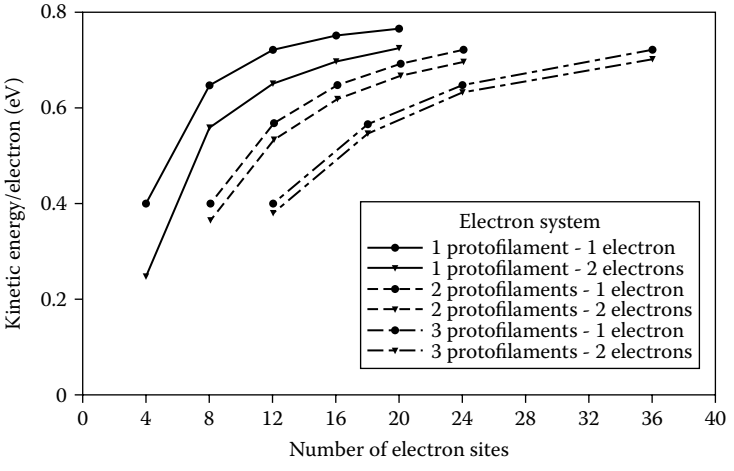

FIGURE 18.11 Kinetic energy per electron as a function of system size.

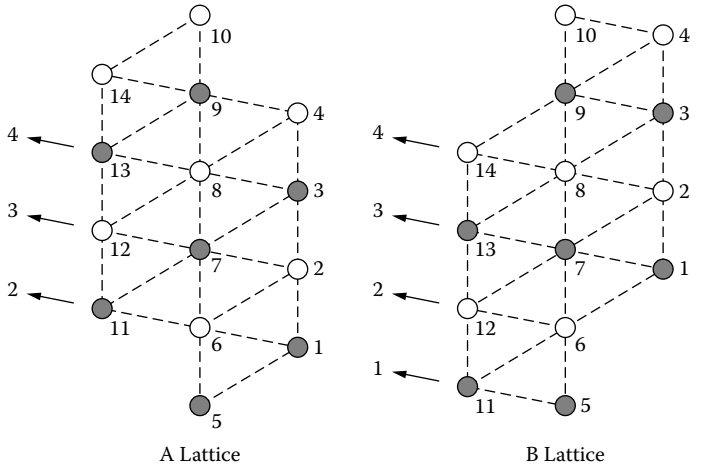

FIGURE 18.12 Lattice unit cells for conductivity calculations. The unit cell on the left corresponds to the MT A lattice, while that on the right corresponds to the MT B lattice.

MT when it has been unwrapped. We recall that the MT lattice is triangular and has three distinct hopping directions. Since protofilaments are strongly bound together along their length but relatively weakly bound to other protofilaments, one also expects the hopping parameter to be much smaller when an electron moves between protofilaments. The calculations have been carried out with the same parameters as in the one-dimensional case, $U = 2.00$ eV, $t = 0.40$ eV along the protofilaments and the site energies have been set equal to zero for all sites. The value of the hopping parameter along the $\bar{3}$-start helix is called t_ℓ (left) and that along the 8-start helix that connects diagonally from a monomer in a direction up and to the right is called t_r (right).

We consider first the results of the MT lattice which has not rolled up to form a tube but is that of a tubulin sheet where $t_r = 0$ and only the value of t_l is varied (see Figure 18.13). Since $t_r = 0$, the nature of the lattice remains rectangular to this point. When $t_l = 0$, this particular system is equivalent to individual protofilaments. The two peaks in the optical spectrum arise from the fact that the protofilaments have different lengths. The absorption peaks of 0.22 eV and 0.40 eV correspond roughly to those of the two-dimer and three-dimer, two electron cases discussed earlier in the chapter. The difference here is that the two electrons are spread over three protofilaments and consequently the effect of the electron repulsion is reduced. However, as t_l is increased from zero, the entire lattice becomes accessible and the protofilament character of the optical spectrum is lost. The two peaks of the t_l spectrum coalesce into a single peak with a large activation energy. A second but smaller absorption peak forms just above 0.50 eV. In the plot of the integrated conductivity, we can see that there is also a smaller peak at 0.16 eV when $t_l = 0.10$ eV. The source of this peak is an absorption for conduction along the direction of t_l. When t_l is increased to 0.40 eV, this peak occurs at 0.61 eV. Thus, this peak seems to occur at roughly $1.5t_l$. The activation energy for conduction along the protofilament axis also increases but is not as sensitive to t_l. Finally, examination of the integrated conductivity demonstrates that the total absorption combines such that the Drude weight remains zero.

The results in Figure 18.14 show a difference between an individual protofilament and a tubulin sheet. The behavior here is inconsistent with that of an insulator since the Drude weight is non-zero. In the optical conductivity, we find that the largest peak occurs at the same location in all directions but with different weightings. The first line traces the results for $t_r = 0.00$ eV and has peaks at 0.38 eV and 0.61 eV. Once the third hopping direction is allowed, the peak absorption is raised slightly to 0.43 eV for $t_r = 0.10$ eV and a smaller peak forms at 0.74 eV. The lower peak consists mainly of hopping along the protofilament while the second peak consists largely of hopping along the t_l direction, which also corresponds to a large

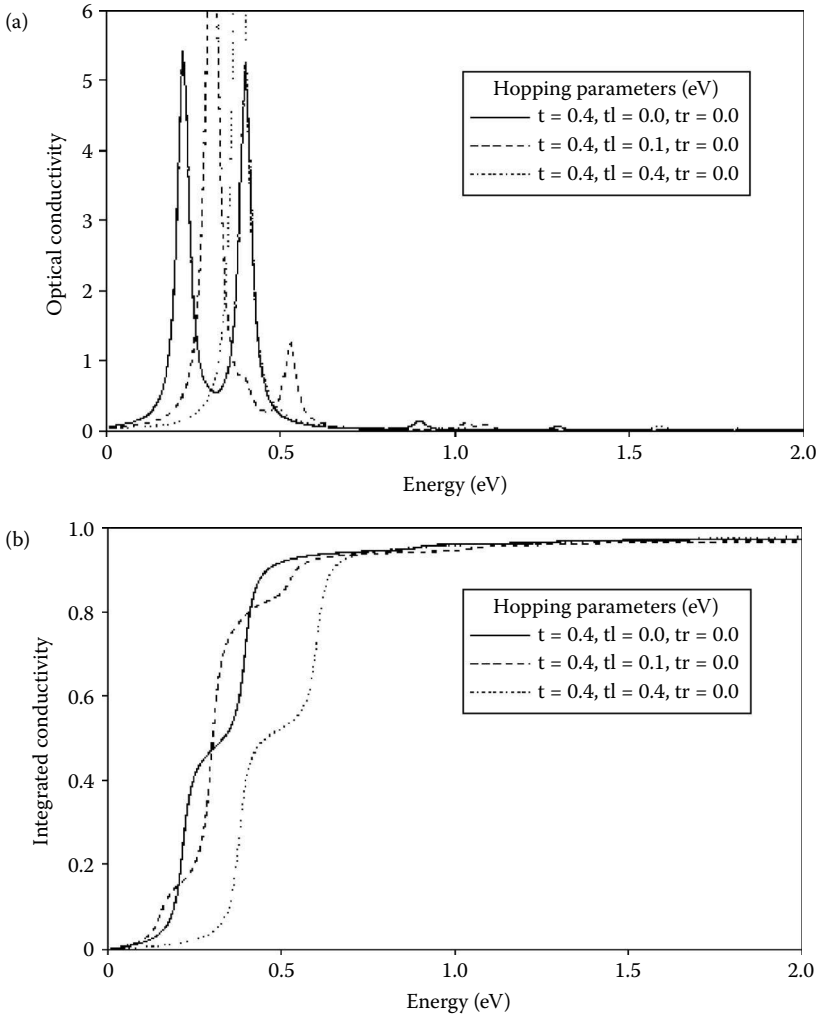

FIGURE 18.13 (a) Optical conductivity along the protofilament axis of the tubulin sheet consisting of a 14-site lattice with two electrons while $t_r = 0$ and t_l is varied. (b) Integrated conductivity of the tubulin sheet with two electrons while $t_r = 0$ and t_l is varied.

hopping parameter. Once t_r is raised to 0.40 eV so that it has the same magnitude as the other hopping parameters, there is only a single peak at 0.70 eV.

The effect of applying the MT A lattice boundary conditions to our system is shown in Figure 18.15, where we see that changing only t_l to a finite quantity while still maintaining t_r as zero is sufficient to result in a non-zero Drude weight. Thus, the periodic boundary conditions in the lateral direction seem to act in a manner similar to the additional hopping directions of the unwrapped lattice. When the kinetic parameters between protofilaments are set to zero, the optical conductivity spectrum shows there are two large absorption peaks at about 0.23 eV and 0.40 eV. As t_l increased, these peaks move higher in energy and the second peak becomes smaller in integrated weight. Eventually, once $t_l = t$, there is a single large peak near 0.47 eV. Introduction of the third hopping direction serves only to raise further the absorption peak, up to 0.70 eV in the case where the hopping parameter is 0.40 eV in all directions.

From Figure 18.15 one can see that more conduction results in the B lattice for a small t_l, such as 0.1 eV, than in the MT A lattice, by comparing it with Figure 18.16. However, even the positioning of absorption

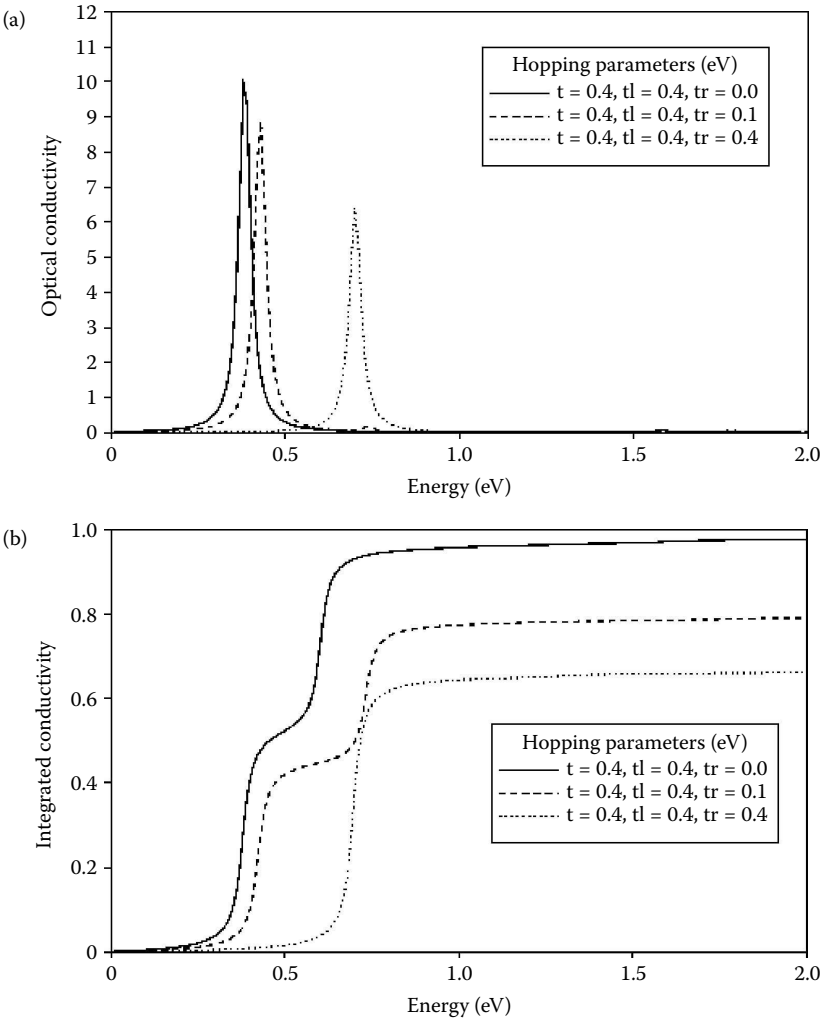

FIGURE 18.14 (a) Optical conductivity along the protofilament axis of the unwrapped 14-site lattice with two electrons while $t_l = 0.4$ and t_r is varied. (b) Integrated conductivity of the unwrapped 14-site lattice with two electrons while $t_l = 0.4$ and t_r is varied.

peaks is similar in the two lattices. Again, it is sufficient for only one of t_l or t_r to be non-zero for the Drude weight to be finite.

Finally, we compare the MT A and B lattices with the unwrapped tubulin sheet as well as with individual protofilaments of tubulin dimers. There is an interesting geometrical interpretation to the results, which shows that the MT A and MT B lattices have identical conduction properties when $t_l = t_r$. This is because in our calculations we have considered all monomers to have the same site energy. Consequently, the B lattice can be viewed as an A lattice with the opposite helicity but the same structure. The sign of the system's helicity does not affect the conduction properties along the major axis. The unwrapped MT lattice is not as good a conductor since some of the hopping freedom has been removed from the lattice but does remain conducting, provided that both t_l and t_r are non-zero.

If we consider the case where one of t_l or t_r is quite small, however, the wrapping of the lattice is much more important. As discussed earlier, the unwrapped tubulin sheet is an insulator while the MT A and B lattices may carry electrons. This observation is especially interesting given that MTs have been observed to zip up and essentially change their structure from a tube to that of a sheet *in vivo*.

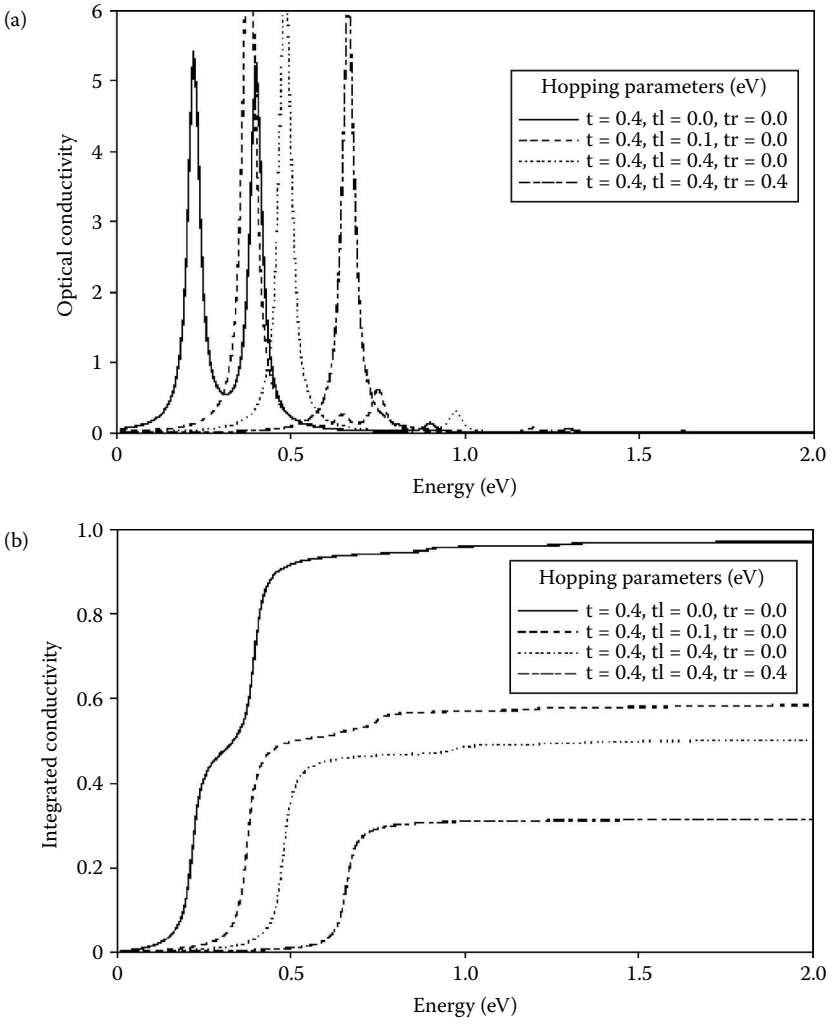

FIGURE 18.15 (a) Optical conductivity along the protofilament direction of the 14-site MT B lattice with two electrons. (b) Integrated conductivity of the 14-site MT B with two electrons shows a non-zero weight for finite t_l.

Our calculations based on the Hubbard model of electron hopping show that the MT is not an insulator, but that instead its conductivity depends on the lattice geometry and whether the MT is wrapped up or not. How well or poorly does one expect the MT to conduct relative to known semi-conductors? Referring back to Equations (18.35) and (18.36), the DC conductivity of the MT is simply given by D as

$$\frac{D}{2} = -\frac{\pi e^2}{2} a^2 < K > - \int_0^\infty \sigma_{kubo}(\omega) d\omega. \tag{18.38}$$

In this equation, the constants can be factored, and what remains is the fraction that can be read from the graphs presented earlier. The Kubo fraction (KF) is the limit of the integral as the frequency tends to infinity.

$$\frac{D}{2} = -\frac{\pi e^2}{2} a^2 < K > (1 - KF). \tag{18.39}$$

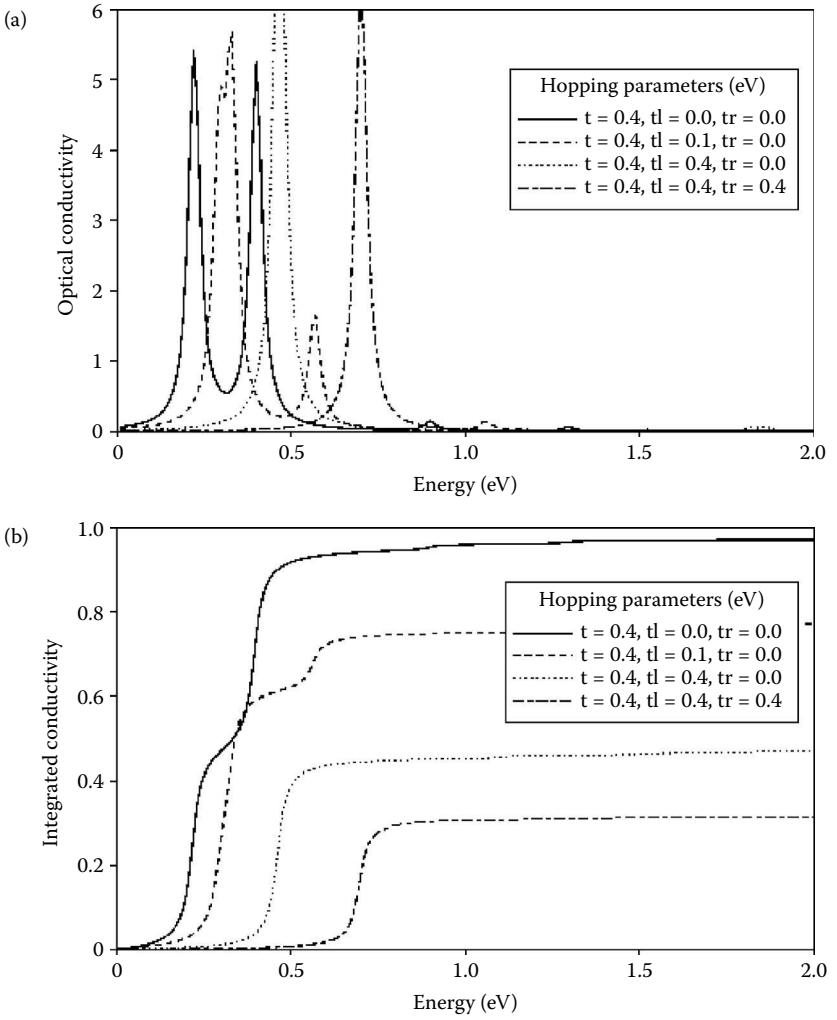

FIGURE 18.16 (a) Optical conductivity along the protofilament direction of the 14-site MT A lattice with two electrons. (b) Integrated conductivity of the 14-site MT A lattice with two electrons shows a non-zero Drude weight for finite t_1.

By returning constants of the system so that the dimensionality of D is correct, we find

$$D = -\frac{\pi n e^2 a^2}{\hbar} \frac{< K >}{t} (1 - KF). \tag{18.40}$$

Finally, we are in a position to predict the conductivity of individual MTs of varying lattice types. For the following calculations, the electron density has been taken to be of two electrons within the lattice of 14 dimers and corresponds to a density of $1.8 \times 10^{18}/cm^3$. Thus, the results must assume that the conductivity remains constant as the length of the polymer is extended, provided the degree of filling remains the same.

The values in Table 18.1 should be compared with the conductivity of metals, such as copper ($6 \times 10^7\, \Omega m^{-1}$) and iron ($1 \times 10^7\, \Omega m^{-1}$). Indeed the values can also be compared to the intrinsic semiconductors germanium ($2.5\, \Omega m^{-1}$) and silicon ($4 \times 10^{-4}\, \Omega m^{-1}$). Given these comparisons, the MT may indeed be quite a good semi-conductor given our assumptions. What is particularly interesting is the

TABLE 18.1 Calculated Conductivity and Resistance of a 1-μm Polymer Composed of Tubulin Dimers

Tubulin Structure	Conductivity ($\Omega^{-1}m^{-1}$)	Resistance
Individual Protofilament	Non-Conducting	
Tubulin sheet ($t_1 = t_r = 0.04\ eV$)	9.4×10^3	$418k\ \Omega$
Tubulin sheet ($t_l = t_r = 0.1\ eV$)	2.0×10^4	$195k\ \Omega$
Tubulin sheet ($t_1 = t_r = 0.4\ eV$)	6.3×10^4	$62k\ \Omega$
MT (A lattice, $t_l = t_r = 0.04\ eV$)	1.8×10^4	$219k\ \Omega$
MT (A lattice, $t_l = t_r = 0.1\ eV$)	4.5×10^4	$87k\ \Omega$
MT (A lattice, $t_l = t_r = 0.4\ eV$)	1.6×10^5	$25k\ \Omega$
MT (B lattice, $t_l = t_r = 0.04\ eV$)	1.9×10^4	$211k\ \Omega$
MT (B lattice, $t_l = t_r = 0.1\ eV$)	4.7×10^4	$84k\ \Omega$
MT (B lattice, $t_l = t_r = 0.4\ eV$)	1.6×10^5	$24k\ \Omega$

way the conduction properties depend on the lattice and the particular boundary conditions. The situation can be compared to carbon nanotubes which have a similar size and structure to MTs. In addition, theoretical consideration of nanotube structure predicts that the conduction properties depend on the boundary conditions. In the case of nanotubes, this means the way in which the graphene sheet is wrapped up to form the nanotube. It is quite interesting that particular sets of boundary conditions produce a semi-conducting nanotube while the appropriate choice of wrapping the nanotube gives rise to metallic conduction [31].

The ability to change from an insulator to a conductor by a simple geometrical change could be biologically relevant even without observing a single electron to be conducted along the MT since it is the conductivity property that affects the way the cell views and responds to external electromagnetic fields. Specifically, the reflectivity of MTs to electromagnetic fields is high when they are conducting, and, consequently, they could act to direct infrared signals to the interior of the centrosome. Since the centrioles are maintained at right angles to each other, the cell would be able to determine the location in latitude and longitude of a light source if the MTs are conducting [4,5]. Peaks in the AC conductivity are all at energies significantly above the thermal activation threshold so no conductivity is expected along a darkened MT, but photoconduction in the infrared-visible range is possible. Therefore, it is possible this photo-activated conduction has functional repercussions. In addition to the intrinsic semiconducting behavior of MTs, and possibly AFs, these protein filaments possess very intensely active cable properties when immersed in an ionic buffer solution. In the next section, we discuss the ionic conductivity effects of AFs.

18.6 Actin Filaments Support Nonlinear Ionic Waves

Several experiments [12,83] indicate the possibility of ionic wave generation along AFs. As the condensed cloud of counter-ions separates the filament core from the rest of the ions in the bulk solution, we expect it to act as a dielectric medium between the two. It has both resistive and capacitive components associated with each monomer that makes up the AF. Ion flow is expected to occur at a radial distance from the surface of the filament approximately equal to the Bjerrum length. An inductive component is proposed to emerge due to the actin's double-stranded helical structure that induces the ionic flow in a solenoidal manner. Due to the presence of the sheath of counter-ions around the AF, these polymers act as biological "electrical wires" [83], and have been modeled as nonlinear inhomogeneous transmission lines propagating nonlinear dispersive solitary waves. Recently, Lader *et al.* [78] applied an input voltage pulse with an amplitude of approximately 200 mV and a duration of 800 μs to an AF, and measured electrical signals at the opposite end of the AF, indicating that AFs support ionic waves in the form of axial nonlinear currents. In an earlier experiment [83], the wave patterns observed in electrically stimulated single AFs were remarkably similar to recorded solitary waveforms for electrically stimulated nonlinear transmission lines [85]. Considering the AFs highly nonlinear complex physical structure and the thermal fluctuations of the counter-ionic

cloud [105], the observation of soliton-like ionic waves is consistent with the idea of AFs functioning as biological transmission lines.

The electro-conductive medium is a condensed cloud of ions surrounding the polymer and separated from it due to the thermal fluctuations in the solution. The distance beyond which thermal fluctuations are stronger than the electrostatic attractions or repulsions between charges in solution is defined as the so-called Bjerrum length, λ_B. With the dielectric constant of the medium denoted by ε, the Bjerrum length is given by

$$\frac{e^2}{4\pi\varepsilon\varepsilon_0\lambda_B} = k_B T \tag{18.41}$$

for a given temperature T in Kelvin. Here e is the electronic charge, ε the dielectric constant, ε_0 the permittivity of the vacuum, and k_B is Boltzmann's constant. For a temperature of 293 K, we find that $k_B = 7.13 \times 10^{-10} m$. Counter-ion condensation occurs when the mean distance between charges, b, is such that $\lambda_B B/b = S > 1$. Each actin monomer carries an excess of 14 negative charges in vacuum, and accounting for events such as protonation of histidines, and assuming there to be 3 histidines per actin monomer, there exist 11 fundamental charges per actin subunit [143]. Assuming an average of 370 monomers per μm, we find there is approximately $4e/nm$. Thus, we expect a linear charge spacing of $b = 2.5 \times 10^{-10} m$ so $S = 2.85$. As the effective charge, q_{eff}, or renormalized rod charge is the bare value divided by S, we find $q_{eff} = 3.93e/monomer$. Consequently, it can be shown that approximately 99% of the counter-ion population is predominantly constrained within a radius of $8nm$ [110] around the polymer's radial axis [158]. Significant ionic movements within this "tightly bound" ionic cloud are therefore allowed along the length of the actin, provided that it is shielded from the bulk solution [104,107].

The physical significance of each of the components of the electrical network, and additional details, are described in [148]. For actin in solution, a key feature is that the positively charged end assembles more quickly than the negatively charged end [135]. This results in an asymmetry in the charges at the ends of the filaments and F-actin's electric polarization. Actin monomers arrange themselves head to head to form actin dimers resulting in an alternating distribution of electric dipole moments along the length of the filament [73]. We assume, therefore, that there is a helical distribution of ions winding around the filament at approximately one Bjerrum length. This corresponds to a solenoid in which a fluctuating current flows as a result of voltage gradient between the two ends.

The capacitive element of the electric circuit is obtained following the observation of oppositely charged layers surrounding the filament surface. We envisage the protein surface's negative charge to be distributed homogeneously on a cylinder defining the filament surface. Furthermore, positive counter-ionic charges in the bulk are expected to form another cylinder at a radius greater than the AF itself, approximately one Bjerrum length, λ_B, away from the actin surface, which includes the condensed ions. The permittivity, ε_0, is given by $\varepsilon = \varepsilon_0\varepsilon_r$ where ε_r is the relative permittivity which we take to be that of water, i.e., $\varepsilon_r = 80$. We take the length of an actin monomer typically as $a = 5.4\ nm$ and the radius of the actin filament, r_{actin}, to be $r_{actin} = 2.5\ nm$ [14]. The next step is to consider a cylindrical Gaussian surface of length a whose radius is r such that $r_{actin} < r < r_{actin} + \lambda_B$.

Applying of Gauss's law for the total charge enclosed in the cylinder, we have the following expression for the capacitance

$$C_0 = \frac{2\pi\varepsilon a}{ln(\frac{r_{actin}+\lambda_B}{r_{actin}})} \tag{18.42}$$

With the parameters given previously, we estimate the capacitance per monomer is $C_0 = 96 \times 10^{-6}\ pF$. The resistive part is obtained from Ohm's law. Taking into account the potential difference and the current I, the magnitude of the resistance, $R = V/I$, for an actin filament is given by

$$R = \frac{\rho ln(\frac{r_{actin}+\lambda_B}{r_{actin}})}{2\pi l} \tag{18.43}$$

where ρ is the resistivity. Typically, for K^+ and Na^+, intracellular ionic concentrations are 0.15 M and 0.02 M, respectively [147]. Kohlrausch's law states that the molar conductance of a salt solution is the sum of the conductivities of the ions comprising the salt solution. Thus

$$\sigma = \Lambda_0^{K^+} c_{K^+} + \Lambda_0^{Na^+} c_{Na^+} = 1.21(\Omega m)^{-1} \tag{18.44}$$

Using this with $\rho = \sigma^{-1}$, the resistance estimate becomes $R = 6.11\ M\Omega$, which is much lower than pure water since $R_{water} = 1.8x10^6\ M\Omega$.

To describe the properties of the whole filament, we simply connect n subcircuits, as described earlier, to obtain an effective resistance, inductance, and capacitance, respectively, such that:

$$R_{eff} = \left(\sum_{i=1}^{n} \frac{1}{R_{2,i}} \right)^{-1} + \sum_{i=1}^{n} R_{1,i} \tag{18.45}$$

$$L_{eff} = \sum_{i=1}^{n} L_i \tag{18.46}$$

and

$$C_{eff} = \sum_{i=1}^{n} C_{0,i} \tag{18.47}$$

where $R_{1,i} = 6.11 \times 10^6\ \Omega$, $R_{2,i} = 0.9x10^6\ \Omega$ such that $R_{1,i} = 7R_{2,i}$. Note that we have used $R_{1,i} = R_1$, $R_{2,i} = R_2$, $L_i = L$, and $C_{0,i} = C_0$. For a 1 μm of the AF, we find

$$R_{eff} = 1.2 \times 10^9\ \Omega \tag{18.48}$$

$$L_{eff} = 340 \times 10^{-12}\ H \tag{18.49}$$

$$C_{eff} = 0.02 \times 10^{-12}\ F \tag{18.50}$$

This solenoidal flow geometry leads to an equivalent electrical element possessing self-inductance. From Faraday's law, we can derive an effective inductance for the actin filament in solution by

$$L = \frac{\mu N^2 A}{l} \tag{18.51}$$

where l is the length of the F-actin and A is the cross-sectional area of the effective coil given by

$$A = \pi (r_{actin} + \lambda_B)^2 \tag{18.52}$$

The number of turns is approximated by simply working out how many ions could be lined up along the length of a monomer. We would then be approximating the helical turns as circular rings lined up along the axis of the F-actin. We also take the hydration shell of the ions into account in our calculation. The hydration shell is then the group of water molecules oriented around an ion. It can be shown that $L = 1.7pH$ for the length of the monomer.

The electrical model of the AF is an application of Kirchhoff's laws to one section of the effective electrical circuit that is coupled to neighboring monomers. Taking the continuum limit [148] for a large number of monomers along an AF, we derive the following equation which describes the spatio-temporal

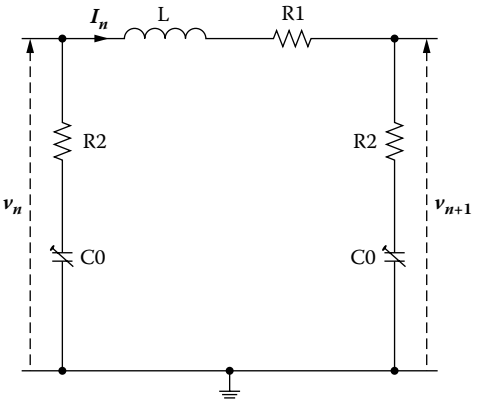

FIGURE 18.17 An equivalent electrical circuit for a segment of an actin filament.

behavior of the potential along the AF:

$$LC_0\frac{\partial^2 V}{\partial t^2} = a^2(\partial_{xx}V) + R_2C_0\frac{\partial[a^2(\partial_{xx}V)]}{\partial t} - R_1C_0\frac{\partial V}{\partial t} + R_1C_0 2bV\frac{\partial V}{\partial t} \tag{18.53}$$

Motivated by this picture, physical properties of the ionic distribution along a short stretch of the polymer (the average pitch 35 to 40 nm) have been modeled by Tuszynski *et al.*, [148] as an electrical circuit with non linear components (see Figure 18.17). The main elements of the circuit are: (a) a non linear capacitor associated with the spatial charge distribution between the ions located in the outer and inner regions of the polymer; (b) an inductance due to the helical geometry of the filament; and (c) a resistor due to the viscosity imposed by the solution.

From Kirchhoff's laws, one can derive an equation governing the propagation of voltage along the filament (see [148]). One of the key aspects in this model is the nonlinearity of the associated capacitance [86,150] that eventually gives rise to the self-focusing of the ionic waves. The equations developed for the model originate from the application of Kirchhoff's laws to the RLC resonant circuit of a model actin monomer in a filament. Perhaps the most important finding is the existence of the traveling wave, which describes a moving transition region between a high and low ionic concentration due to the corresponding inter-monomeric voltage gradient. The velocity of propagation was estimated to range between 1 and 100 ms^{-1} depending on the characteristic properties of the electrical circuit model. It is noteworthy that these values overlap with action potential velocities in excitable tissues [55].

Analysis of the model described earlier reveals the possibility of stationary waves in time that may lead to the establishment of spatial periodic patterns of ionic concentration. As mentioned previously, Lader *et al.* [78] applied an input voltage pulse with an amplitude of approximately 200 mV and a duration of 800 μs to an AF, and measured electrical signals at the opposite end of the AF, indicating that AFs support ionic waves in the form of axial nonlinear currents. In an earlier experiment [83], the wave patterns observed in electrically stimulated single AFs were similar to recorded solitary waveforms for electrically stimulated nonlinear transmission lines [85]. Considering the AFs highly nonlinear physical structure and thermal fluctuations of the counter-ionic cloud [105], the observation of soliton-like ionic waves is consistent with the idea of AFs functioning as biological transmission lines.

This section only provides an indication as to a realistic model of actin that can support soliton-like ionic traveling waves. Modeling relies on data constrained by experimental conditions, and/or assumptions made, including the charge density, which is calculated based on the net surface charges of actin. It should also be considered that soliton velocity is directly proportional to the magnitude of the stimulus, which in a biological setting has not been formally described. Actin interacts with a number of ion channels of different ionic permeability and conductance. Thus, it is expected that channel opening, single channel

currents, and other channel properties, including the resting potential of the cell, may significantly modify the amplitude and velocity of the soliton-supported waves. This should correlate with the velocity of the traveling waves along channel-coupled filaments. Other parameters that may play a role in this type of electrodynamic interaction are the local ionic gradients and the regulatory role of actin binding proteins, which can help "focus" the conductive medium, or otherwise impair wave velocity. The physical significance of each of the components of the electrical network (see Figure 18.17) and additional details, are described in [148]. MTs differ from AFs in several respects. One of the key differences is the presence of peptide chains called C-termini that protrude from their surface. In the following section, we outline their role in ionic wave conductivity along MTs.

18.7 Long-Range Spatio-Temporal Ionic Waves Along Microtubules

MTs are long hollow cylinders made of $\alpha\beta$-tubulin dimers [33], as described in Section 18.5. Recently, it has become apparent that neurons utilize MTs in cognitive processing. Both kinesin and MAP2 that associate with MTs have been implicated in learning and memory [69,154]. Dendritic MTs are implicated in particular, and it is highly probable that the precisely coordinated transport of critical proteins and mRNAs to the post-synaptic density via kinesin along MT tracks in dendrites is necessary for learning, as well as for LTP [70].

The following molecular dynamics (MD) simulation results focus on the C-termini of neighboring tubulins, whose biophysical properties have a significant influence on the transport of material to activated synapses. This affects cytoskeletal signal transduction and processing as well as synaptic functioning related to LTP. Using MD modeling we calculated conformation states of the C-termini protruding from the outer surfaces of MTs and strongly interacting with other proteins, such as MAP2 and kinesin [124]. To elucidate the biophysical properties of C-termini and gain insight into the role they play in the functioning of dendrites, we developed a quantitative computational model based on the currently available biophysical and biochemical data regarding the key macromolecular structures involved, including tubulin, their C-termini, and associated MAP2. In the proposed model of the C-termini microtubular network, the tubulin dimer is considered to be the basic unit. Each dimer is decorated with two C-termini that may either extend outwardly from the surface of the protofilament or bind to it in one of a few possible configurations.

The most dynamic structural elements of the system (i.e., its elastic and electric degrees of freedom) are envisaged as conformational states of the C-termini. Each state of the unbound C-terminus evolves so as to minimize the overall interaction energy of the system. The negatively charged C-termini interact with (a) the dimer's surface, (b) neighboring C-termini and (c) adjacent MAPs. While the surface of the dimer is highly negatively charged overall, it has positive charge regions that attract the C-termini causing them to bend and bind in a "downward" state. The energy difference between the two major metastable states is relatively small, on the order of a few $k_B T$ [112].

To simplify our calculations, we first used a bead-spring model representing the C-terminus as a sequence of beads with flexible connections. We have accounted for the electric field exerted by the dimer, the external field generated by the environment, and various short-range interactions within the simulated C-terminus, including interactions between the beads (i.e., Lennard–Jones potential, angular forces, etc.).

A simplified model of the interaction between MAP2 and its ionic environment via counter-ions was used to investigate the ability of MAP2 to function as a "wave-guide" that transfers the conformational change in a C-terminus state to an adjacent MT (see Figure 18.18). A perturbation applied to the counter-ions at one end of the MAP2 drives them out of equilibrium and initiates a wave that travels along the MAP2 (for details see [112]). Figure 18.18 (right) depicts the main result for a localized perturbation applied for a few picoseconds to the counter-ions near the binding site of MAP2. Wave propagation along the chain of $N = 50$ counter-ions is shown as a counter-ion displacement parallel to MAP2, u_i, where i denotes the i^{th} counter-ion. This perturbation along MAP2 propagates almost as a "kink" whose phase velocity has been found to be $v_{ph} \approx 2nm \cdot ps^{-1}$.

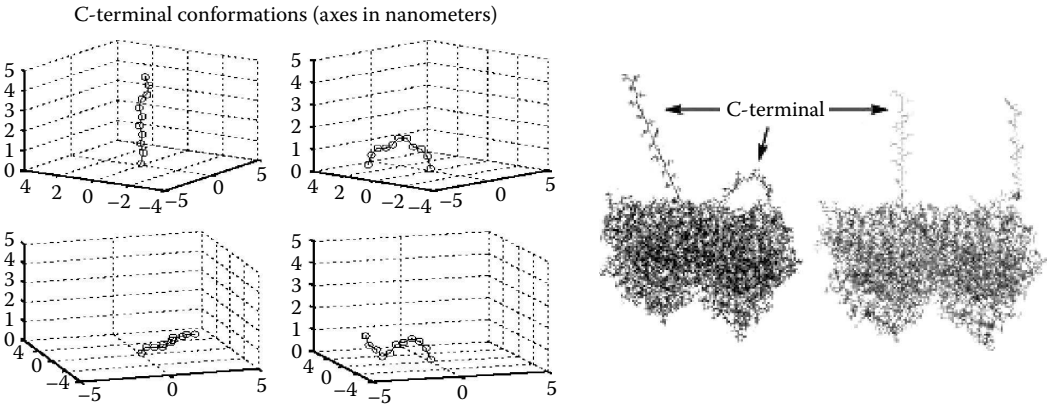

FIGURE 18.18 Examples of the conformational states of the C-termini in a tubulin dimer (right) and results from bead-spring model simulations (left).

Our simulations indicate the ability of an ionic wave to trigger a coupled wave of C-termini state changes from their upright to downward orientations. Four examples of "up" and "down" states of the C-termini obtained through bead-spring model simulations are shown in Figure 18.18 (left) and results from molecular dynamics simulations for an actual tubulin dimer are shown in Figure 18.18 (right). Calculations of the energy minimized positions of the individual beads representing the amino acids of the C-terminus in two equivalent forms reveal that the probability of the down position, which includes all cases of full or partial attachment, is 15%. This means that the system has two major states with a strong bias towards the stretched-up state.

In the remainder of this section, we present an MD simulation study of the interaction between MTs and MAP2, followed by preliminary experimental results of ionic wave conduction by MTs.

A simplified model of the interaction between MAP2 and its ionic environment via counter-ions was used to investigate the ability of MAP2 to function as a "biological wire" that transfers the conformational change in a C-terminus state to an adjacent MT (see Figure 18.19).

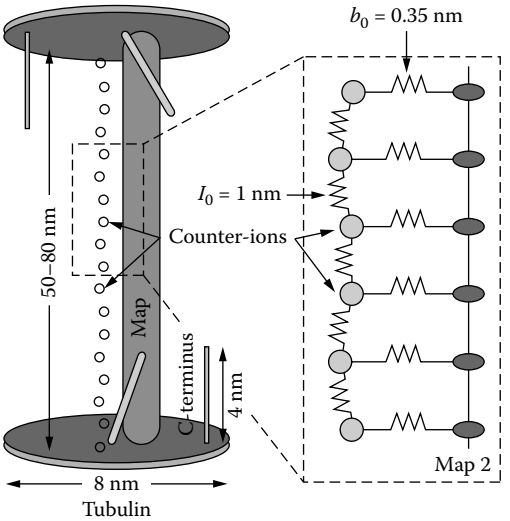

FIGURE 18.19 An illustration of a MT C-terminus conformational change transfer process. Ionic waves travel along an adjacent MAP via local perturbations to the bounded counter-ions.

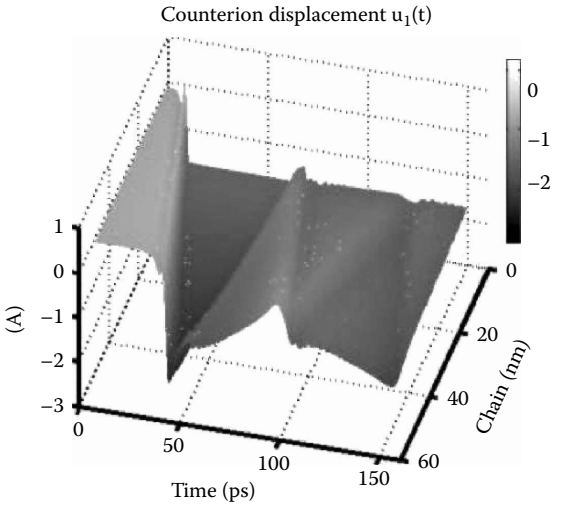

FIGURE 18.20 Propagation of a localized perturbation applied to counter-ions near the binding site of a MAP2.

A perturbation applied to the counter-ions at one end of the MAP2 drives them out of equilibrium and initiates a wave that travels along the MAP2 [112]. Figure 18.20 depicts the main result for a localized perturbation applied for a few picoseconds to the counter-ions near the binding site of a MAP2. Wave propagation along the chain of counter-ions is demonstrated as counter-ions displacement parallel to MAP2.

To validate the proposed model, we have conducted experiments on electro-stimulation of MTs in a solution. A detailed description of the experiment and its results has been published elsewhere [113]. Here we only describe the main aspects of the work on this phenomenon. Figure 18.21 depicts the experimental setup used. Isolated taxol-stabilized MTs have been shown to be able to amplify an electric signal applied to them through a micropipette. The input signal was of 5 to 10 msec duration, with amplitude in the range of $\pm$200 mV. The signal that arrived at the other end of the MT was more than twice as high as the signal recorded in a control experiment where the same two pipettes were immersed in a solution with no MT making contact to them. The calculated conductivity of MTs was found to be on the order of 10 nS, indicating a high level of ionic conductivity along an MT. By comparison, for a typical ion channel, the corresponding value ranges between 5 and 200 pS.

The results of the MD modeling previously described raise the possibility of transmitting electrostatic perturbations collectively among neighboring C-termini, and from C-termini on one MT to those C-termini on another MT via MAP2. The importance of collective conformational states of C-termini and of transmitting perturbations among the C-termini as a novel information processing mechanism operating at a subneuronal level is clear. MAP2 and kinesin bind near to the C-termini on tubulin and the electrostatic properties of C-termini affect this binding [2,145]. Hence the conformational states of the C-termini must

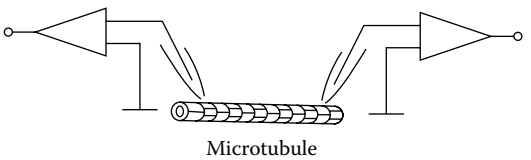

FIGURE 18.21 Schematic representation of the experimental set-up used for the measurements of a MTs ionic conductivity. The MT is attached to two micropipettes that are connected to signal amplifiers.

at some point be taken into account in order to understand neural processing that depends on transport of synaptic proteins inside of neurons. Kim and Lisman [71] have shown that inhibition of MT motor proteins reduces an AMPA receptor-mediated response in hippocampal slice. This means that a labile pool of AMPA-receptors depends on MT dynamics, and MT-bound motors determine the amplitudes of excitatory postsynaptic currents (EPSC). A walking kinesin carries with it a protein or an mRNA molecule. Since kinesin binds to a MT on a C-terminus, as it steps on it, it brings the C-terminus to the MT surface and makes it ineffective in binding for the next kinesin over a period of time that it takes the C-terminus to unbind and protrude outside. From this we can deduce that long stretches of C-termini in the upright position are going to be most efficient at transporting kinesin and kinesin cargo while C-termini that lie flat are expected to be most efficient at detaching kinesin. Thus, in considering the trafficking of many kinesins, collective electrostatic effects of C-termini become crucially important. Collective states that correspond to transport strategies that will send optimal numbers of kinesin molecules to synaptic zones are likely to occur when synapses are activated (e.g., when synapses are generating EPSCs). One type of kinesin cargo associated with learning and memory is the NMDA receptor, which is well known to be associated with LTP. The PSD might be expected to require replenishment of NMDA receptors at a later time than AMPA receptors are replenished (see [71]). Kinesin protein (specifically KIF17) actively transports NMDA receptor 2B subunits (NR2B) to the region of the PSD [52]. Once NR2B is in the vicinity of an active synapse it dissociates from kinesin and then becomes associated with the PSD, presumably using actin transport as an intermediary step. Hence, kinesin-mediated transport of NMDA receptors along MTs has a built-in negative feedback mechanism: whenever too many NMDA receptors are transported to the synaptic site (often located on a spine head), then those NMDA receptors can initiate proteolytic breakdown of MAP2. The latter event would be expected to reduce further transport of NMDA receptors to the synaptic site. The signal transduction molecule CaMKII is also critical for learning and LTP. Similar to the NMDA receptor, CaMKII is transported to active synapses via a kinesin-mediated transport mechanism. However, it appears to be the mRNA for CaMKII to a larger extent than the protein that is transported to spines, since dendrites are enriched with polyribosomes and CaMKII mRNA [140]. Ribosomes are redistributed from dendrites to spines with LTP [106]. Local translation of CaMKIIα in dendrites appears to be necessary for the late phase of LTP: fear conditioning and spatial memory [98]. Moreover, changes in synaptic efficacy are often accompanied by changes in morphology; reorganization of underlying MTs is a fundamental factors for these morphological changes. Having discussed the individual properties and roles played by AFs and MTs in subcellular conduction processes, we now turn to an integrated view of their functioning within a cell. The next several sections discuss this issue in depth.

18.8 Dendritic Cytoskeleton Information Processing Model

Our current hypothesis states that the cytoskeletal biopolymers constitute the backbone for ionic wave propagation that interacts with, and regulates, dendritic membrane components, such as ion channels to effectively control synaptic connections. Figure 18.22 depicts a portion of the dendritic shaft where MTs are decorated by C-termini and interconnected by MAP2 (thick line). Connections between MTs and AFs are shown as well as two types of synaptic bindings. On the upper left side, actin bundles bind to the post-synaptic density (PSD) of a spineless synapse. On the lower right side, a spiny synapse is shown, where actin bundles enter the spine neck and bind to the PSD, which at the other end, is connected to the MTN.

We envision a mechanism in which a direct regulation of ion channels and thus synaptic strength by AFs and associated cytoskeletal structures controls and modifies the electrical response of the neuron. In this picture, MTs arranged in networks of mixed polarity receive signals in the form of electric perturbations, from synapses via AFs connected to MTs by MAP2 [122], or via direct MT connections to postsynaptic density proteins by molecules such as CRIPT [108]. As discussed next, the MTN may be viewed as a high-dimensional dynamic system where the main degrees of freedom are related to the conformational state of the C-termini. The input signals perturb the current state of the system that continues to evolve. Hypothetical integration of the preceding ideas is outlined as follows. Electrical signals arrive at the PSD

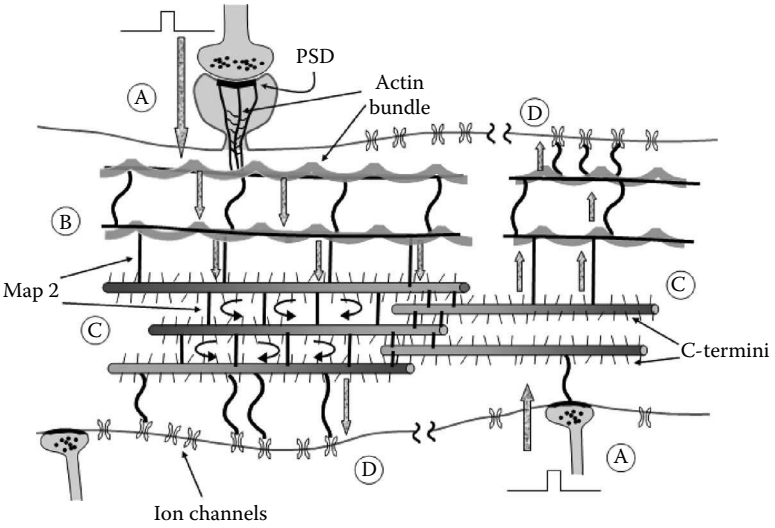

FIGURE 18.22 A scheme of the dendritic shaft with MTs arranged in networks of mixed polarity that receive signals in the form of electric perturbations from synapses via AFs connected to MTs by MAP2, or via direct MT connections to postsynaptic density proteins by molecules such as CRIPT. A spiny synapse is depicted in the bottom right where an actin bundle enters the spine neck.

via synaptic transmission, which in turn elicits ion waves along the associated AFs at the synaptic spine. These dendritic input signals propagate in the form of ionic waves through AFs to the MTN where they serve as input signals. The MTN, operating as a large high-dimensional state machine, evolves these input states, e.g., by dynamically changing C-termini conformation. The output from the MTN is the state of the system that may propagate via AFs to remote ion-channels. These output functions are assumed to regulate the temporal gating state of voltage-sensitive channels. This process subsequently regulates the membrane conductive properties and controls the axon hillock behavior by changing the rate, distribution, and topology of open/close channels. The overall functions of the dendrite and neuron can thus be regulated in this manner.

The attractiveness of the concept that the cytosol, with its cytoskeletal structures, may behave as a large dynamical system is clear since it provides a means for real-time computation without the need for stable attractor states. Moreover, the output is relatively insensitive to small variations in either the MTN (cytoskeletal networks) or the input patterns. It should be noted that the temporal system's state evolves continuously, even without external inputs. Recent perturbations, however, have a long-term effect on the MTN trajectories, i.e., there is a memory effect inherent to this system (not to be mistaken with synaptic LTP, which has a much longer time scale). The output from the MTN may converge at or near ion channels to regulate their temporal behavior. The issue of adaptation requires a feedback mechanism that will, at least locally, enable the change of the output function. In the context of neuronal function, with a focus on processivity, synaptic strengthening, LTP, and memory enhancement, the output function may simply reflect an effect of the MTN on synaptic channel function, such that the desired state of the channel appears to have a higher probability of being open/close upon the presentation of the associated input pattern. One possibility is a Hebbian-based response where more frequent activity of certain subdomains of the MTN output states gives rise to a higher/lower density of AFs connecting to corresponding channels.

We now explain how this integrative view may serve as a regulatory (adaptive) mechanism. The input level denoted by A in Figure 18.22 is associated with external electric perturbations passing through membranes, mainly synaptic inputs arriving from other neurons. These signals arrive at the cytoskeleton and directly affect MTNs, and/or actin filaments (bundles), in spiny synapses. However, the actin cytoskeleton is responsible for the propagation of signals to the MTN (see level B). These propagated signals (level C)

are in turn used as inputs to the MTN, viewed as a dynamic system. We further propose that the MTN generates diverse phase space trajectories in response to different input vectors. The requirements from such a system are not too restrictive since the output-state is not an attractor of the system. In other words, information processing at this level is not necessarily based on attractor dynamics but rather on real-time computations. This proposition relies on the observed ability of the MTN to propagate signals and on the specific topological features of MTNs in dendrites, in which shorter MTs of mixed polarity are interconnected by MAP2s. The output from the MTN would be a function of the evolved state vector in certain areas accessed by actin filaments and/or directly linked to ion channels (see both possibilities at level D). These output signals may modify the temporal channel activity, either by directly arriving from the MTN to the channels, or mediated by actin filaments. Hence, the channel-based synaptic membrane conductance is regulated, in particular in the axon hillock region, which is, in most cases, responsible for the generation of action potentials.

The idea that a nonspecific high-dimensional dynamical system may serve as a reservoir of trajectories in the context of liquid state machines (LSMs), has recently been suggested as an explanation for the existence of microcircuits in the brain [87]. The basic structure of an LSM is an excitable medium (hence "liquid") and an output function that maps the current liquid state. The liquid must be sufficiently complex and dynamic to guarantee a universal computational power. This is sufficient to ensure that different input vectors lead to separate trajectories. A network of spiking neurons and a recurrent neural network has been used as a "liquid", whereas the output (readout) function has been implemented by simple perceptrons, threshold functions, or even linear regression functions. Clearly, simpler readout functions restrict the ability of the whole system to capture complex nonlinear dependencies.

18.9 The Inter-Relation Between the Neural Cytoskeleton and the Membrane

Usually, F-actin and microtubular cytoskeletal networks are thought to fulfill separate, independent cellular roles. Highly dynamic actin networks are known for their role in cell motility, in particular the spreading of the leading edge and contraction. The more stable MT cytoskeleton is best known for its importance in cell division and organelle trafficking. However, recent studies provide a more unified role, ascribing important roles to the actin cytoskeleton in cell division and trafficking and important roles for MTs in the generation and plasticity of cellular morphology. Coordination between the actin- and MT- based cytoskeletons has been recently observed during cellular migration and morphogenesis, processes that share some similarities with neurite initiation [28]. A direct physical association between both cytoskeletons has been suggested, because MTs often preferentially grow along actin bundles and transiently target actin-rich adhesion complexes. In neurons, certain plakins and neuron-specific MAPs, like MAP1B and MAP2, may play a role in linking MTs and AFs, helping in the transition from an undifferentiated state to neurite-bearing morphology.

MAP1B and MAP2 are both known to interact with actin in vitro [20,109,128,134,146]. It is likely that by crosslinking, MAP2 and/or MAP1B associated with both cytoskeletons could be involved in guidance of MTs along AF bundles. Alternatively, MAPs could shuttle from MTs to actin and could alter F-actin behavior by actively crosslinking AFs.

Little is known, however, about interactions between neuronal ion channels and the cytoskeleton. Recently, whole-cell and single channel recordings showed that acute disruption of endogenous AFs with cytochalasin D activated voltage-gated K^+ currents in these cells, which was largely prevented by intracellular perfusion with the AF-stabilizer agent, phalloidin. Interestingly, the direct addition of actin to excised, inside-out patches activated, and/or increased single K^+ channels. Thus, acute changes in actin-based cytoskeleton dynamics regulate voltage-gated ion channel activity in bipolar neurons. This may be indicative of a more general and quite appealing mechanism by which cytoskeletal structures control feedback mechanisms in neuronal channels.

Recent theoretical and experimental studies of the electrical properties of AFs and MTs in solution revealed their capability to act as biopolymer wires [83,148]. This means that these protein polymer

filaments are capable of conducting nonlinear ionic waves and even amplifying the signal with respect to the conducting solution [112]. We conjecture a mechanism in which a direct regulation of ion channels and thus synaptic strength by AFs and associated cytoskeletal structures controls and modifies the electrical response of the neuron. According to this scenario (see Figure 18.22), MTs arranged in networks of mixed polarity receive signals in the form of electric perturbations from synapses via AFs connected to MTs by MAP2 [122], or via direct MT connections to postsynaptic density proteins by molecules such as CRIPT [108]. These signals propagate in the form of ionic waves. Specific physical properties of the propagating ionic waves will be discussed in sections that follow.

18.10 Relationship to Cognitive Functions

Transport of proteins and receptors in neurons is likely to have an electromagnetic basis to the extent that this function is possibly a result of MT computation. The perspective advanced in this chapter is that a specific fingerprint defined by a particular electromagnetic state of a microtubular array potentially corresponds to a unique unit of cognition (e.g., a basic visual parameter). Recently, it has been shown that visual components can be represented musically [23]; hence, the idea that one common type of energy underlies divergent perceptual and cognitive processes appears likely. Activation of one electromagnetic fingerprint could, in turn, activate another electromagnetic fingerprint, irrespective of sensory input. Moreover, the subjective feels of this widespread pattern of electromagnetic energy can be specified according to those key physical properties of MTs that influence the transport of proteins to synapses. Factors influencing kinesin-mediated transport include the protein conformation of tubulin and the nature of the C-termini (see [112,136]).

Not only is the conformational state of tubulin critical to effective transport, motor proteins appear to alter albeit temporarily, the conformation of tubulin. Kinesin binding and that of MAP, tau, significantly alter the direction of the protruding protofilament ridges along MTs, which in turn influences their further binding abilities [127]. More than mere local adaptation to binding, MTs may alter their conformation ahead of kinesin processivity [76], supporting the notion of long-range cooperative effects between tubulin dimers located along longitudinal protofilaments of MTs. These biochemical relationships have consequences for electromagnetic fields. The dipole moment of tubulin depends on its configuration in the MT [53,97]. Thus, electromagnetic fields among MTs could, in theory, be induced or inhibited by synaptic inputs that affect the protein conformation of tubulin directly or through alterations in kinesin or MAP binding. Synaptic effects upon MTs could be mediated through ionic currents, by propagation via actin filaments [148], or by signal transduction cascades resulting in the phosphorylation of MAPs [126].

Due to lengthwise electric dipoles of tubulin dimers, information in the form of traveling waves propagated along microtubular tracks can, in principle, be transmitted between synapses with high fidelity (see [149]). MAP2 bridges keep microtubular arrays within the dendritic core parallel and antiparallel by aligning portions of polarized MTs. The antiparallel alignment of MTs, which specifically occurs in dendrites, would severely attenuate any electromagnetic field generated by MTs. However, during enhanced kinesin-mediated transport, as is likely to occur with heightened synaptic activity, MAP2 bound to the MT would be perturbed and may even temporarily detach from the MT. A similar phenomenon might also occur due to dynein-mediated transport, which occurs largely in the opposite direction to that of kinesin-mediated transport. Assuming that at least some MAP2 stays attached to the antiparallel MTs, keeping the dendritic array intact, any net unidirectional transport along the MT array should increase the strength of the electromagnetic field associated with the fingerprint and should further result in the spread of that electromagnetic field to adjacent MTs. Once a sufficient number of MTs were engaged in dynamically sending and receiving complementary electromagnetic energies, whole neuronal compartments (e.g., dendrites) might be expected to interact. Due to the parallel/antiparallel arrangement of MTs in cortical dendrites and the ability of electromagnetic fields to pass from one dendrite to adjacent dendrites, information could, in principle, pass between neurons when such electromagnetic fields were sufficiently amplified as a result of changes in the binding of MAPs or kinesin.

While it is true that there are some 100 billion neurons and some 100 trillion synapses in the human cerebral cortex, so too does each sensory field afford the luxury of extremely high resolution. The fovea of the retina, for example, contains roughly 1 million receptor cells that relay information to the cerebral cortex, diverging to eventually drive the synaptic activity of at least 1 billion cortical cells. Although the mammalian cortex has many neurons, and even more synapses, its connectivity can be readily grasped by one recurrent theme: repetitive high-resolution topographic representation. Despite this simple organizational plan, cells in cortical areas, in particular cells in higher sensory or association areas, inexplicably show correlated responses during cognitive tasks. Experience alters the basic structure of dendrites, and as a consequence, it should also alter any electromagnetic fields generated by MTs as a result of transport proteins moving along them. As described earlier, a rearrangement of the cytoskeleton occurs during early development, with learning, and with neurodegeneration underlying dementia.

Although models of changes in synaptic efficacy, e.g., long-term potentiation or depression (LTP or LTD), offer great potential as memory mechanisms (see [93]), one often overlooked problem is that large numbers of synapses are affected in concert. If only a few synapses are changed with each memory, then the entire cortical system would have a near unlimited capacity, but if many synapses participate in each memory, as seems to be the case for LTP, then there could be a serious saturation effect. This relates back to the topographic organization of the cerebral cortex. Strictly synaptic models (electrochemical) suggest that complex neural networks increase synaptic weights (i.e., neurons that fire together wire together). Nonetheless, visual cortical regions, for example, do not have massively random interconnections among all parts of the visuotopic map, thereby making many of the changes in synaptic efficacy necessary for encoding complex perceptual features impossible.

The biophysical properties of MTs are just beginning to be understood at molecular and atomic levels, and recent empirical evidence suggests interesting effects occur between MTs under certain experimentally induced conditions. Two groups, one led by Watt Webb at Cornell and another led by Paul Campagnola and William Mohler at the University of Connecticut, observed that MTs give rise to intense second-harmonic generation: a frequency doubling upon exposure to a sapphire laser in the 880-nm range. (Other frequencies were partially effective.) MTs were one of the few biological materials having electric dipoles that constructively interfered with the dipoles of neighboring MTs [11,29]. This occurred for parallel MTs in axons, but not for antiparallel MTs in dendrites. Could such a phenomenon be expected to occur with natural learning or upon exposure to oscillatory input, an LTP-inducing tetanus or various pharmacological agents? To the extent that these induce electromagnetic energy, it is conceivable. In addition to being sensitive to electromagnetic radiation, MTs may themselves produce this kind of energy. Second-harmonic generation by MTs is consistent with their ferroelectric properties. In second-harmonic generation, the most strongly enhanced wavelength is 880-nm, supporting the frequently overlooked reports of electromagnetic signaling by cells. Guenther Albrecht-Buehler [5] observed electromagnetic energy in the near infrared region overlapping the 880 nm value that was generated by centrioles for the purpose of cell-to-cell communication, leading him to suggest that: "... one of the functions of MTs may be to play the role of cellular 'nerves'." Infrared light in this range of wavelengths also induces cell aggregation [6]. Since frequency (and hence wavelength) determines the long-range effect of dielectric polarization of a given MT, one would expect this effect to be length-dependent. The distance between MTs was shown to be critical in second-harmonic generation by MTs.

18.11 The Potential for Bioelectronic Applications and Neuromorphic Computing

Traditionally, biologists viewed the cell as a "fluid mosaic bag" containing enzymes, organelles, etc. Recent work has focused on finding the proteins that come together in this cellular "soup" and proceed to effect a change corresponding to a particular biological function. Through the molecular-level analysis of these protein-protein interactions, recent work in various labs has proceeded to build a model of "systems biology," in which these biochemical reactions that have been individually characterized can be brought together to form a cohesive whole, a synergistic dynamics network. From this knowledge, we hope to extract

useful information that will affect fields as diverse as nanotechnology, biotechnology, and materials science. Inside each cell, there is an elaborate network that serves a multitude of functions. It is an intracellular "highway" that provides a roadway for motor proteins to carry cargo and travel along in their journey to extend axonal length, for example. It controls synapses, and dendrite outgrowth. It is responsible for pulling the chromosomes apart during mitosis, for cell motility, etc.

As man-made engineered systems become more and more miniaturized, the push to build "nano-machinery" is driving physicists and engineers towards biomimetics, (i.e., the branch of engineering that mimics biological materials). This is highly feasible and much progress has been made in this area. However, due to the limitations inherent in trying to build mechanical devices on such a small scale, there is still much to learn. It is conceivable that through our better understanding of the cell's individual functional and structural components we will be able to advance medicine and technology beyond even our wildest dreams. One area that may undergo a major paradigm shift as a result of it is computer engineering.

As we enter into the 21st century, the dawn of the exciting new world of computational biotechnology is upon us and we are poised for a new generation of computers that do not have to be told what to do, but have the power to learn on their own. The development of a biological computer that is fast, small, and evolvable is no longer considered science fiction. Scientists are combining biological materials with the latest silicon-based technology in an effort to give us not only electronic devices that are smaller, but are also more flexible in terms of structure and function.

Ironically, our brain can serve as proof of concept. Many scientists now believe that individual neurons display wiring patterns and communication powers that resemble a human computer, and therefore may provide the framework for the design of a biological computer. The brain is composed of 10 billion neurons, each of which may be communicating with 1000 others. The neuron transmits electric signals along its arms, or 'axons'. Inside each axon a parallel architecture of microtubules is interconnected with other proteins, not unlike the parallel computer's wiring. In fact, the structure of microtubules may have evolved towards optimal computational efficiency. The piezoelectric properties of these protein filaments allow them to bend as a result of electric fields or currents which makes them ideal candidates as regulators of synaptic plasticity, a mechanism that could explain the popular principle of "use it or lose it." Thus, seldom used synaptic connections would be switched off in favor of more active ones. Biologist Guenter Albrecht-Buehler [3] demonstrated that cells perceive their environment through a tiny organelle called the centriole, composed of microtubules that were shown to interact with electromagnetic radiation. To further display that microtubules themselves can be conductive, the German biotechnology group, led by Eberhard Unger [41,42] succesfully conducted required experiments and is now working on building nanoelectronic components using these and other proteins. Simultaneously, ideas of combining biological and silicon-based materials in hybrid arrangements are gaining more and more support for future bioelectronic applications. Work is underway at various labs around the world to develop a biological chip. The main objective is to design nanoelectronic components using hybrid protein–silicon structures with arrays of proteins as biological oscillators that can be stimulated by electrodes or acoustic couplers. This will provide an analog circuit with rich dynamics.

DNA computers are unlikely to become stand alone competitors for electronic computers. But digital memory in the form of DNA and proteins is a real possibility with exquisitely efficient editing machines that navigate through the cell, cutting and pasting molecular data into the stuff of life. Beyond that, the innate intelligence built into DNA molecules could help fabricate tiny, complex structures: in essence, using computer logic not to crunch numbers but to build things, an idea conceived of by Caltech's Winfree and Rothemund [123]. A single test tube of DNA tiles could perform about 10 trillion additions per second (about a million times faster than an electronic computer). Interfacing DNA or protein-based hybrid structures with living cells will enable a host of medical and technological applications including non-invasive diagnostic and therapeutic medical applications and computational devices.

One major component of this framework is a protein-based subcellular structure called a microtubule, to which we have devoted a large portion of this chapter. Microtubules are composed of 13 protofilaments, arranged in a spiral pattern, and each protofilament is composed of subunits of tubulin. Each tubulin possesses one of two conformational states: either both monomers can align, or one monomer can shift

30° relative to the vertical axis. Based on the conformation an individual tubulin exists in at a given moment, the conformation of its nearest neighbors can be altered as well. This conformational change can be transmitted throughout the microtubule in a similar manner as patterns are transmitted through cellular automata programs. In addition, as we argued earlier in this chapter, tubulin can have several electronic states as well as dipolar states, some of which can be conceivably linked to the conformations of C-termini and the counter-ion clouds around them. The idea that we could input an electric, chemical, or mechanical signal and generate another one at the output end of the system brings computing or information processing to mind. We believe that ordered protein arrays can function as a programmable processor or even a computer either inside cells such as neurons or as nano-scale devices.

Nanotechnology could and should take advantage of biology in a completely novel manner, (i.e., instead of building materials that mimic biological components, we could use the components themselves). In the case of microtubules and tubulin, rather than purify it from traditional sources such as cow or pig brains, tubulin can be readily cloned into an expression vector, and purified from bacteria in mass quantities. Then, purified tubulin of a predetermined sequence (and thus with desirable biophysical and biochemical characteristics) can be routinely polymerized to form microtubules with specific structural and functional properties. Following that, the microtubules produced could be stabilized in a polymer gel capable of undergoing a phase transition at a particular value of the control parameter such as temperature, applied pressure, pH, or ionic concentration. This would then trigger a signal that would initiate computational processes carried out by individual microtubules or their assemblies interconnected by MAPs or even connected to actin filaments in architecures that could mimic subneuronal designs. This technology could also be applied to sensors or artifical intelligence systems in building an evolvable computer. The only limitations in this endeavor are those imposed by our imagination and our ability to dream up new applications.

18.12 Discussion

Considering the abundance of MTNs and AFs in axons and dendritic trees, the findings and theoretical models described earlier may have important consequences for our understanding of the signaling and ionic transport at an intracellular level. Extensive new information (see [63]) indicates that AFs are both directly [14] and indirectly linked to ion channels in both excitable and non-excitable tissues, providing a potentially relevant electrical coupling between these current generators (i.e., channels), and intracellular transmission lines (i.e., AFs and MTs). Furthermore, both filaments are crucially involved in cell motility and, in this context, they are known to be able to rearrange their spatial configuration. In nerve cells, AFs are mainly located in the synaptic bouton region, whereas MTs are located in both dendrites and the axon. Again, it would make sense for electrical signals supported by these filaments to help trigger neurotransmitter release through a voltage-modulated membrane deformation leading to exocytosis [132]. Actin is also prominent in post-synaptic dendritic spines, and its dynamics within dendritic spines has been implicated in the post-synaptic response to synaptic transmission. Kaech *et al.*, [65] have shown that general anesthetics inhibit this actin mediated response. Among the functional roles of actin in neurons, we mention in passing glutamate receptor channels, which are implicated in long-term potentiation. It is therefore reasonable to expect ionic wave propagation along AFs and MTs to lead to a broad range of physiological effects.

In summary, this chapter broadly described the physical conditions that enable cytoskeletal polymers such as AFs and MTs to act as electrical transmission lines for ion flows along their lengths and along MAP-mediated interconnections between adjacent filaments. In the case of AFs, we propose a model in which each protein subunit is equivalent to an electric circuit oscillator. The physical parameters used in the model were evaluated based on the molecular properties of the polymer. Using the general conductivity rules that apply to electrical circuits we analyzed the properties of ionic waves that propagate along AFs and compared these values to those observed in earlier experiments. In the context of the role played by MTs in neurons, we described the dynamics of C-termini states. We discussed both individual MTs and their networks, including the interactions with ions and signal transmission via MAPs. Recent experiments

on ionic conductivity along AFs and MTs show the validity of the basic assumptions postulated in our models. In light of these results, we conjectured a new dendritic signaling mechanism that involves ion waves along protein filaments which may travel without significant decay over tens of microns, potentially affecting the function of synapses and ion channels.

We proposed a new model for information processing in dendrites based on electrical signaling involving the cytoskeleton. This model predicts that the dendritic cytoskeleton, including MTs and AFs plays an active role in computations affecting neuronal function. These cytoskeletal filaments are affected by, and regulate, ion channel activity. A molecular dynamics description of the C-termini protruding from the surface of a MT has revealed the existence of several conformational states, which lead to collective dynamical properties of the neuronal cytoskeleton. These collective states of the C-termini on MTs have been shown to have a significant effect on the ionic condensation and ion cloud propagation with physical similarities to those recently found in actin-filaments. We have been able to then provide an integrated view of these phenomena in a bottom-up scheme that demonstrates how ionic wave propagation along cytoskeletal structures may impact channel functions, and thus neuronal computational capabilities.

The possibility of an evolvable, dynamic, and responsive electrical circuitry within the cell provided by actin and MT filaments could be of enormous consequence to our understanding of the way cells operate internally and interact with their environment. In particular, it would cast an entirely new light on cell differentiation, cell division, and cell–cell communication. While an integrated theory of this type of behavior is far from being constructed, its individual elements are gradually taking shape.

Acknowledgments

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18

Electronic and Ionic Conductivities of Microtubules and Actin Filaments, Their Consequences for Cell Signaling and Applications to Bioelectronics

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18.1 Introduction

The prevailing view of neural information processing is based on passive properties of the membrane derived from the application of linear cable theory to dendrites, e.g., [114]. Recent studies, however, have suggested nonlinear models that accommodate new experimental evidence that appears to be inconsistent with this classical view. In particular, conventional theories do not satisfactorily explain: (a) significant fluctuations of synaptic efficacy over short periods of time in response to a recent burst of activation, e.g., [54], (b) a variety of active ion-channels capable of affecting the local membrane electric properties, and (c) nonlinear responses localized to specific dendritic sites, pointing to highly specialized mechanisms correlated with specific inputs. On the other hand, nonlinearities inherent in the new models, e.g., [74,133], give rise to a wider repertoire of (computational) capabilities, such as multiplication, fast correlation, etc. Moreover, experimental and theoretical support for conductive properties of protein filaments and the ionic clouds around them is becoming available, which indicates a far more complex picture of the signaling and information processing phenomena in neurons and other cells.

In this chapter, we suggest that functional electrodynamic interactions between cytoskeletal structures and ion-channels are central to the neural information processing mechanism. These interactions are supported by long-range ionic wave propagation along microtubule networks and actin filaments, and exhibit subcellular control of ionic channel activity, hence impacting the computational capabilities of the whole neural function. Cytoskeletal biopolymers, including actin filaments (AFs) and microtubules (MTs), constitute the backbone of wave propagation, and in turn interact with membrane components to modulate synaptic connections and membrane channels. Indeed, only recently clear functional interactions between these cytoskeletal structures have become apparent. Association of MTs with AFs in neuronal filopodia appears to guide microtubule growth, and plays a key role in neurite initiation [28]. This is further evidenced by the presence in neurons of proteins that interact with both MTs and F-actin, and proteins that can mediate signaling between both types of filaments. This is likely used to control microtubular invasion. The microtubule associated proteins MAP1B and MAP2, for example, are known to interact with actin in vitro, e.g., [146]. Cross-linking, MAP2 and/or MAP1B is highly probable to associate with both types of filaments contributing to the guidance of MTs along AF bundles. Extensive evidence exists confirming a direct interaction between AFs and ion channels, and a regulatory functional role associated with actin. Thus it is clear that the cytoskeleton has a direct connection to membrane components, in particular channels and synapses.

In the following, we demonstrate how each individual cytoskeletal component is capable of supporting both ionic and electronic wave propagation. In particular, we present a molecular dynamics model of the dendritic microtubule network (MTN) where arrays of MTs are interconnected by MAP2s. In this model, ionic waves propagate along the MTN and interact with C-termini of MTs to generate collective modes of behavior. Also in this chapter, a biophysical model of nonlinear ionic wave propagation along AFs is presented, supported by experimental evidence. Finally, we describe a model for a direct regulation of ion-channels and synaptic strength by AFs and networks of MTs that control the electrical response of the dendrite in particular, and the neuron in general. We begin this chapter by providing a general overview of the structure and function of a nerve cell, emphasizing the role of cytoskeletal filaments, ion channels, and their interactions.

18.2 The Neuron

The neuron is the quintessential communicating cell. A great variety of neural subtypes can be found interconnected to one another in complex electrochemical circuits via multiple synaptic connections. Thus, the amount of information processing by a neural circuit depends on the number of such connections or synapses. Neurons are highly polarized cells, whose level of morphological complexity increases during maturation into distinct subcellular domains. A typical neuron possesses three functional domains. These are the cell body or *soma* containing the nucleus and all major cytoplasmic organelles, a single axon, which extends away from the soma and takes cable like properties, and a variable number of dendrites, different

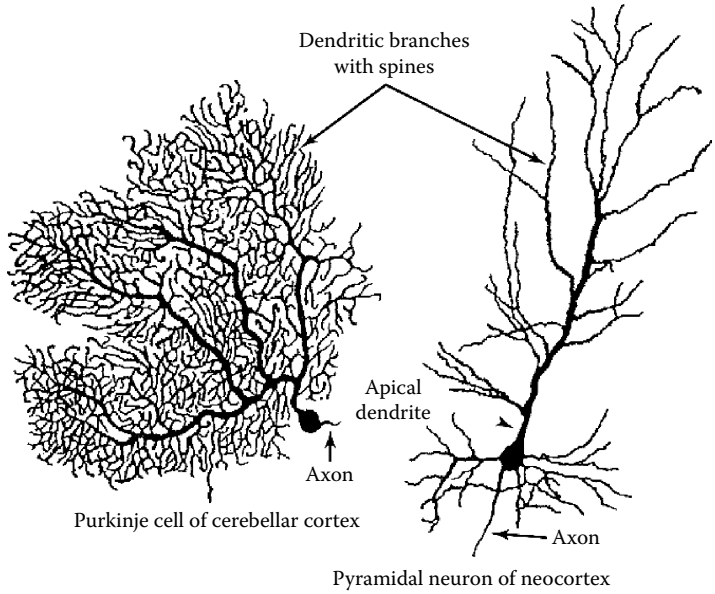

FIGURE 18.1 Two prominent examples of nerve cells.

in shape and complexity, which emanate from the soma. The axonal terminal region, where the synapse contacts it with other cells, displays a wide range of morphological specialization (see Figure 18.1).

Two major types of synapses can be found: (a) asymmetric, responsible for the transmission of excitatory impulses, and (b) symmetrical, present in inhibitory synapses. Turning morphology into a functional feature, the soma and dendritic tree are the major domains of receptive inputs. Thus, both the dendritic arborization and the axonal ramifications confer a high level of sub-cellular specificity in the localization of particular synaptic contacts on a given neuron. The extent to which a neuron may be interconnected largely depends on the three-dimensional spreading of the dendritic tree. Dendrites contain a large number of dendritic spines, which are small sac-like organelles, projecting from the dendritic trunk. Dendritic spines are more abundant in highly arborized cells such as pyramidal neurons, and scarcer in lowly interconnected smaller-sized dendritic tree inter-neurons.

Spines are the dendritic regions where most excitatory inputs are found, such that each spine contains one asymmetric synapse. The number of excitatory inputs can be correlated with the number of spines present in the dendrite. Dendritic spines are complex saccular enlargements, containing ribosomes, and cytoskeletal structures, including actin, and α- and β-tubulin. Electrically speaking, a single neuronal circuit is closed such that afferent inputs reaching the soma and the dendritic tree are transferred to the axon at the other pole of the cell which, in turn, is responsible for transmitting efferent neural information. Both the morphology of the axon and its course through the nervous system are correlated with the type of information processed by a particular neuron, and by its interconnectivity with other neurons. While the axon leaves the soma with a small swelling known as the axon hillock, some neurons protrude the axonal projection from the main dendrite instead. Cytoskeletal components are highly relevant in the structure/function of the neuron. The axonal hillock contains large "parallel" bundles of MTs. Axons are also located with other cytoskeletal structures, such as neurofilaments, which are more abundant in axons than in dendrites. The dynamics in cytoskeletal structures is central to our contention that information can be processed and "delivered" to the synaptic function by changes in the cytoskeletal structures.

Dendrites are the principal element responsible for synaptic integration and for the changes in synaptic strengths that take place as a function of neuronal activity. While dendrites are the principal sites for excitatory synaptic input, little is still known about their function. The activity patterns inherent to

the dendritic tree-like structures, such as in integration of synaptic inputs, are likely based in the wide diversity of shapes and sizes of the dendritic arborizations [116]. Some dendrites are unipolar, while others are multipolar; some have many orders of branching, while others have only one or two; and some branch primarily in two dimensions, while others have complex three-dimensional structures. The size and complexity of dendritic trees increase during development, which has been associated with the ability of the neural system to organize and process information [49]. The majority of synapses, both excitatory and inhibitory, terminate on dendrites, so it has long been assumed that dendrites somehow integrate the numerous inputs to produce single electrical outputs. It is increasingly clear that the morphological functional properties of dendrites are central to its integrative function. Branching in this context is essential to the increasingly complex ability of the neuron to respond to developmental complexity. Nerve branching and complexity are controlled by both external and internal cues. Of current interest to this are substrate and electromagnetic fields, while also recognized are the chemical gradients involved in local adaptation. Highly relevant to adaptation is cytoskeletal dynamics, as the dendritic cytoskeleton plays a central role in the process of ramification, filopodia formation, and more specialized neuronal activity such as long-term potential (LTP) and long-term depression (LTD) circuit dynamics, including memory formation.

The electrotonic or passive properties of dendrites became of great importance to our understanding of dendritic integration of synaptic signals. Early hypotheses were supported by work on motor neurons [34] and sensory receptors [38,67]. In that action, potentials were initiated in the axon hillock region as a result of the graded or algebraic summation of EPSPs and IPSPs occurring in various parts of the neuron. Because of the relatively short space constant calculated for motor neurons [18], it was predicted that excitatory post-synaptic potentials from synapses on the distal portions of dendrites would be ineffective in firing a neuron. Rall [115] revolutionized the field of dendrite physiology by using cable theory for the study of neurons and their dendrites. Action potentials initiated in dendrites by synaptic stimulation were later observed propagating in both directions at velocities of around 0.3 m/s. [7,21,43]. Direct intracellular recordings from dendrites of cerebellar Purkinje cells [84], cortical neurons [57], and hippocampal neurons [153] confirmed that action potentials could be initiated and propagated in dendrites. Very little is known, however, about the active, or voltage-dependent, properties of dendrites, and many questions remain unanswered. Such as the types of ion channels present in dendrites, location, and how active properties alter synaptic integration.

The firing of action potentials elicits characteristic M-shaped Ca^{2+} profiles across the neuron. The Ca^{2+} rise is small in the soma, highest in the proximal apical and basal dendrites, and small again in the distal apical dendrites [62,100,118–120]. The spatial distribution of Ca^{2+} channels explains the distinct Ca^{2+} rise in various cell locations but rather reflects the propagation into the dendrites of trains of action potentials. Imaging with Na^{+}-sensitive dyes has yielded similar patterns, suggesting that Ca^{2+} influx into the dendrites is driven by Na^{+} influx into the dendrites which is driven by Na^{+}-dependent action potentials [62]. Ca^{2+}-dependent action potentials have been demonstrated in hippocampal neurons and dendrites [131]. Most of the action potential induced Ca^{2+} entry is mediated by voltage-gated Ca^{2+} channels [100]. Multiple types of voltage-gated Ca^{2+} channels have been demonstrated in hippocampal neurons [36,40,68,101]. Based on the addition of specific Ca^{2+} channel blockers, the presence of L-, N-, P-, Q-, R-, and T-type Ca^{2+} channels has been reported [16]. This channel distribution may contribute to spike-triggered Ca^{2+} entry into the soma and apical dendrites. Both low-voltage activated (LVA) channels and at least one high-voltage activated (HVA) channel seem to be most predominant in regions of the dendrite at least a 100 mm from the soma. Experimental results strongly suggest a heterogeneous distribution of different types of voltage-gated Ca^{2+} channels within the soma and dendrites of hippocampal neurons [152]. The presence of Na^{+} channels in distal dendrites helps explain the back-propagation of action potentials well into the dendritic tree [88,89,138,139,141]. Action potential propagation in dendrites is rather low, and susceptible to the effects of cable filtering.

A most prominent feature of dendritic ion channel regulation is the potential boosting and enhancement of distal synaptic events. Voltage-gated Na^{+} and Ca^{2+} channels may play an important role in amplifying the magnitude of EPSPs towards the soma. However, channel regulation may also be important for

dendritic interactions in the immediate vicinity of the synaptic input. Voltage-gated channels may alter the local input resistance and time constant, which in turn would influence both spatial and temporal summation of EPSPs and IPSPs, thus conveying highly nonlinear interactions; signal amplification [64,137]. Thus, the nonlinear properties of dendrites may serve as electronic devices. In addition to providing an output, the action potentials elicited by synaptic input, whether initiated at the local synaptic sites or at the axon hillock and back-propagated into the dendrites, would essentially reset the dendritic membrane potential. Action potentials might also provide feedback or an associative signal to other synapses active just before or just after the action potential.

Back-propagating action potentials may also provide feedback to dendritic synapses after a neuronal output has occurred. Ca^{2+} signals are implicated in post-synaptically-induced synaptic plasticity such as short-term potentiation (STP) [92], LTP, and LTD [8,15]. Actin has long been postulated to play important roles in neuronal morphogenesis. High concentrations of actin have been found in the leading edge of the growth cones of developing neurons, and in synapses of mature neurons. Actin filament dynamics also plays a role in a number of related phenomena, including axon initiation, growth, guidance, and branching, in the morphogenesis and stability of dendrites and dendritic spines in synapse formation and stability. Neurons conduct two types of cytoplasmic growth processes in axons and dendrites, both of which are functionally and morphologically distinct, but are developmentally led from the growth cone. Growth cones are composed of finger-like filopodia and veil-like lamellipodia that help the growing processes explore their environment. Axons travel long distances, and upon reaching their targets, produce terminal branches where the growth cones are converted into pre-synaptic terminals. Dendrites usually do not extend such long distances, but often branch extensively, giving rise to dendritic trees. Upon proper contact with axons, post-synaptic dendritic specializations create functional synapses by opposition with the pre-synaptic terminals of axons. These are localized to small protrusions on their dendritic trees called dendritic spines, which are the site of most excitatory synapses. In the mammalian brain these multiple synaptic connections are believed to play important roles in learning and memory. Thus, neural morphogenesis requires changes in both the underlying cytoskeleton and specific interactions with the plasma membrane. To generate a membrane protrusive structure such as an axon, coordinated changes in actin and microtubular organization are required. Growth cones of developing neurons, for example, rapidly extend and retract filopodia by concentred changing in cytoskeletal dynamics, namely bundled AFs in filopodia, and composed of a cross-linked actin network in lamellipodia [80,81,155]. In this section, we have outlined the main features of signaling taking place within a neuron. Major emphasis was placed on the role played by the cytoskeleton in the neuron's structure, growth, and internal signaling. The next section discusses the main biophysical properties of the cytoskeleton.

18.3 The Cytoskeleton

The cytoskeleton is a major component of all living cells. It is made up of three different types of filamental structures, including actin-based microfilaments (MFs), intermediate filaments (IFs) (e.g., neurofilaments, keratin), and tubulin-based microtubules (MTs). All of them are organized into networks, which are interconnected through numerous particular proteins, and which have specific roles to play in the functioning of the cell. The cytoskeletal networks are mainly involved in the organization of different directed movements in cell migration, cell division, or in the internal transport of materials. Polymerization of MFs is responsible for cell migration and for the remodeling of the leading edge of cells. Molecular motors are protein complexes associated with the cytoskeleton and drive organelles along MTs and MFs in a "vectorial" transport.

MTs are some of the most basic and important cytoskeletal elements in the morphological scheme of neurons. As for the case of cytochalasins in actin structures, the addition of microtubular modifying agents was first used to assess the role of MTs in axon elongation. Microtubular disrupting agents affect axons first and have no early effects on growth cones and filopodia, suggesting that MTs provide support for the axon. Conversely, microtubular invasion may be a critical element in the formation of neurite projections. Initially, expanded lamellipodia first undergo segmentation at certain spots where MTs accumulate in an

ordered array [27,144,156] and gradually "migrate away" from the cell body. These lamellipodia transform into nascent growth cones. Neurite elongation takes place as the MTs in the shaft become compressed into a narrower bundle. In cultured hippocampal and cortical neurons, this process most likely implies the coordinated control of the major cytoskeletal components, including actin, MTs, and associated proteins.

Thus, MTs are essential for neurite development. One particular aspect of microtubular involvement in neurite outgrowth is vesicular transport. Most membrane insertion occurs at the growth cone of axons [22,24]. MAP-stabilized MTs, form parallel or antiparallel arrays, which might act as compression resistant struts inside neurites [61]. Because MTs constantly invade the actin cytoskeleton in lamellipodia of epithelial [125] and neuroblastoma cells [27], such a mechanical role for MTs could also be important during de novo neurite initiation. Neurite-like protrusions, for example, can be induced in non-neuronal cells by the exogenous expression of stabilizing proteins such as MAP2 and tau [72,79,157]. Spontaneous neurite initiation is dependent on both the presence and the dynamic properties of MTs.

A number of proteins associate with MTs, including MT motors, kinases, so-called "structural" MAPs, which alter microtubule structure, and specialized MAPs, which bind to microtubule plus or minus ends. Members of the MAP2/tau family have long been proposed to be important regulators of neurite behavior [10,48,151]. MAP2 and tau stabilize MTs by reducing catastrophe and promoting rescues, leading to prolonged growth periods and thus enhanced net microtubule accumulation [32,45,75]. MAP2 and tau stabilize MTs by binding along the outer ridges of protofilaments [2].

F-actin and MTs cytoskeletal networks are typically thought to fulfill separate, independent cellular roles. Highly dynamic actin networks are known for their role in cell spreading and contraction. The more stable MT cytoskeleton is known for its importance in cell division and organelle trafficking. However, recent studies provide important roles for the actin cytoskeleton in cell division and trafficking and for MTs in the generation and plasticity of cellular morphology. A direct physical association between both cytoskeletons has been suggested, because MTs often preferentially grow along actin bundles and transiently target actin-rich adhesion complexes.

Association of MTs with actin cables in filopodia appears to guide microtubule growth along the most efficient path anti-parallel to retrograde flow [130]. Thus, specific coupling between MTs and actin bundles presumably promotes MT advance. Proteins that are able to interact with both MTs and F-actin, or proteins that can mediate signaling between both cytoskeletons likely control the regulation of MT invasion. This is further strengthened by specific microtubule association of the microtubule-associated proteins MAP1B and MAP2, both known to interact with actin in vitro [20,109,128,134,146]. Furthermore, MAP2 binds to F-actin and efficiently induces actin bundle formation [51,129]. It is likely that by crosslinking, MAP2 and/or MAP1B associated with both cytoskeletons could be involved in the guidance of MTs along actin filament bundles. Alternatively, MAPs could shuttle from MTs to actin and could alter F-actin behavior by actively crosslinking AFs.

The effect of changes in the cytoskeleton on ion channel activity has been the focus of recent attention. Most studies have employed drugs that selectively stabilize or destabilize either AFs or MTs, with resulting effects on specific ion channel activity, either as changes in whole cell conductance or single channel activity by the patch clamping techniques. Although direct binding of specific cytoskeletal proteins to individual channel proteins remains to be demonstrated, recent work in which exogenous purified actin has been added to the cytoplasmic side of channels excised from the cell membrane has in numerous cases directly confirmed an effect of cytoskeletal filaments. A body of earlier evidence suggested that various cytoskeletal components, including actin and actin-associated proteins, anchor, co-localize, and regulate both the spatial stability as well as the function of ion transport proteins. It is possible that actin either binds directly to the channels, or that actin may first interact with actin-binding proteins, which in turn regulate (by binding or other indirect interaction) ion channel function. Independent studies demonstrated that membrane-resident CFTR is functionally regulated by AF organization, which may also involve a direct interaction with actin. Thus, ion channel may be controlled by direct and indirect cytoskeletal interactions.

Before we present an integrated view of the conductive properties of the active and passive structural blocks of the cytoskeleton, we should review the current state of knowledge regarding their conductive properties.

18.4 Overview of Biological Conductivity

Cope [19] has presented a detailed review of the applications of solid-state physics concepts to biological systems. He divided the discussion into seven groups. The first involves semiconduction of electrons across enzyme particles as a rate-limiting process in cytochrome oxidase, evidenced by kinetic patters of enzymes and microwave Hall measurements. Second, pn-junction conduction electrons were suggested by kinetics of photobiological free radicals in the eye and in photosynthesis. Their I-V characteristics conform to the diode equation:

$$I = I_0 exp(V/kT) \tag{18.1}$$

Third, there may be a connection between superconduction and growth in nerves. Fourthly, phonons and polarons are involved in mitochondrial phosphorylation. Next, piezoelectricity and pyroelectricity may be involved in the growth of nerves and bone structures. Then, infrared electromagnetic waves may transmit energy in lipid bilayers of nerves and mitochondria. Finally, complex sodium and potassium ions in structured cell water may be analogous to valence band electrons in semiconductors, free cations being analogous to conduction electrons. The ionic process in cell water resembles electronic conduction in seminconductors. For dried proteins or DNA, Szent-Gyorgi[142] noticed that their conductivities obeyed the relationship:

$$\sigma = \sigma_0 exp(-E_a/kT) \tag{18.2}$$

With activation energies E_a on the order of 1 eV. The presence of water may cause considerable changes in conductivities, (for example, in cytochrome oxidase this may be reduced to 0.3 eV).

Adessi *et al.* [1] have provided a theoretical treatment of transmission through a poly(g)DNA molecule. They found that there was a modification of the rise of a B-DNA form, which can induce a shift of the conduction channel towards a valence one. They found that deformation of the backbone of the molecule has a significance on the hole transport. Furthermore, the presence of ionic species (e.g., Na) in the surrounding of the molecule can create new conduction channels. In photo-excitation experiments, hole transport occurs through coherent transport at short distances and long-range incoherent or hopping transport at large distances. Fink *et al.* [39] studied the metallic behavior in λ DNA using an electron projection technique. Superconducting behavior has been observed by Kasumov *et al.* [66]. Porath *et al.* [111] observed semiconductor behavior with a poly(g)-poly(c)DNA molecule. A scanning force microscope has been used by de Pablo *et al.* [26] to observe insulating behavior. This paper models a single poly(g) DNA lying between electrodes, and transmission was computed with a Landauer–Buettiker formalism based on Green's functions. The electronic properties of O_2-doped DNA have been studied by Mehrez *et al.* [96], who suggested that DNA conduction is affected by O_2-doping, in particular hole doping. However, ab initio atomic structure calculations of O_2-doped DNA indicate negligible charge transfer. The absence of dc conductivity in λ DNA has been investigated by de Pablo *et al.* [26]. First principles electronic structure calculations indicate a minimum DNA resistance of 10^{16} Ω per molecule, and a minimum resistivity of 10^4 Ωm. However, lower energy electron bombardment may dramatically increase conductivity and with electron energies between 50 and 200 eV may lead to metallic conduction with a resistance of 2×10^8 Ω. Fink *et al.* [39] discussing DNA and conducting electrons stated that I-V characteristics of DNA indicate ohmic-like behavior with a resistance of 2 $M\Omega$ for a 600 nm-long DNA rope corresponding to $\rho = m\Omega$ cm. These authors speculate that DNA may be semiconducting, insulating, or metallic simultaneously depending on the arrangement of molecules in the structure. Endres *et al.* [37] discussed the quest for high-conductance DNA saying that DNA conduction is an unsolved problem first enunciated by Eley and Spivey [35]. Experimental outcomes range from insulating to semiconducting to even superconducting, and the uncertainties depend on: (a) contacts between the electrodes and the DNA and (b) differences in the molecules (sequence, length, arrangement, preparation) within the DNA. It appears that drying DNA can lead to conformations with localized electronic high states, although hole doping of the backbone by counter-ions might also be possible. Wet DNA may support electrical current due to solvent impurity

states in the large high-Π^* energy gap. Davies and Inglesfield [25] discussed the embedding method for conductance of DNA calculations. It is believed that the pathway for transport in DNA runs through the bases in the center of the double helix molecule. The dominant transport mechanism for DNA appears to be coherent hopping. Calculations show extreme sensitivity to the choice of contact orbitals.

Marechal [95] discussed the transfer of protons as a third fundamental property of hydrogen bonds. The importance of the cooperative resonance types in cyclic structures is stressed. Proton transfers along H-bonds have been discussed as a mechanism of vital importance for chains, helices, and in cyclic structures such as rings. Consta and Kapral [17] discussed proton transfer in mesoscopic molecular systems. Proton transfer rates and mechanics are studied theoretically with solvent molecules in solution. The proton transfer occurs as a result of orientational fluctuations on the cluster's surface. The environment influences solvent effects and reaction rates. The rate constant was established to be 1/59 ps at 260 K and for the bond length fixed at 2Å. Morowitz [102] discussed proton semiconductors and energy transduction in biological systems. He presented the possibility of protochemistry paralleling electrochemistry. ATP synthesis was discussed as coupled to proton transport assuming a gated protein semiconductor across a membrane. Protein conduction in biological structures involves the water of hydration and protein molecules.

Ripoll *et al.* [121] investigated ionic condensation and signal transduction. Ion condensation on biopolymers at a critical value of the charge density may lead to intra-cellular signaling due to the diffusion of condensed counter-ions in the near region along cytoskeletal filaments. A feedback mechanism was proposed between condensation/decondensation of calcium and the activation of calcium-dependent enzymes. This is shown to create coherent patters of protein phosphorylation. It was postulated that ion condensation operates in signal transduction. In the case of polymer ionics [117], the conductivity depends on temperature via the formula:

$$\sigma T = Aexp[-B/(T - T_o)] \tag{18.3}$$

indicating activated hopping. The concept of the Fermi level in solution has been extended by Bockris and Khan [9] through the introduction of a reversible potential for a redox couple.

Unfortunately, measurements of conductivity of biopolymers (including MTs) are fraught with difficulties due to their structural heterogeneity, strong dependence on the environmental conditions (pH, ionic concentration, temperature, etc.), and the inherent liquid state of the sample. Nonetheless, attempts have been made recently to measure resistance values of MTs, both intrinsically and as ionic conduction cables. Fritzsche *et al.* [41] made electrical contacts to single MTs following dry-etching of a substrate containing gold microelectrodes. Their results indicate intrinsic resistance of a 12 micron-long MT to be in the range of 500 megaohms, giving a value of resistivity of approximately 40 mega-ohms/micron in their dry state. The same group performed measurements on gold-coated MTs [42] which were covered with a 30-nm layer of gold following sputtering. The resistance of the thus metallized MTs was estimated to be below 50 ohms, (i.e., originated entirely due to the metallic coating). Minoura and Muto [99] measured the ionic conductivity of MTs using an electro-orientation method under an alternating electric field. The ionic MT conductivity was calculated as $1.5 + / - 0.5$ mS/m, which is approximately 15 times greater than that of the buffer solution. This was attributed to the counter-ion polarization and is consistent with another recent study of ionic conductivity along MTs. Furthermore, Goddard and Whittier [47] reported their measurements of RF reflectance spectroscopy of samples containing the buffer solution, free tubulin in buffer, MTs in buffer, and finally, MTs with MAPs in buffer. The sample size in each case was 5 ml. The concentration of tubulin was 5 mg/ml and in the last case the concentration of MAP2 and the tau protein was 0.3 mg/ml. The average DC resistance reported by these authors was (a) 0.999 $k\Omega$ (buffer), (b) 0.424 $k\Omega$ (tubulin), (c) 0.883 $k\Omega$ (MTs), and (d) 0.836 $k\Omega$ (MTs + MAPs). It's not straightforward to translate these results into resistivity of MTs without making assumptions about their geometrical arrangement and connectivity. However, assuming all tubulin being polymerized in case (c) and a uniform distribution of MTs forming a combination of parallel and series networks, one can find the resistance of a 10 μm–long MT, forming a basic electrical element in such a circuit, to have approximately an 8 $M\Omega$ resistance. This compares very favorably to an early theoretical estimate of MT conductivity based on the Hubbard model with electron hopping between tubulin monomers. This model predicted the resistance of a 1-μm MT

to be in the range of 200 $k\Omega$, hence a 10-μm MT would be expected to have an intrinsic resistance of 2 $M\Omega$, within the same order of magnitude as the result reported by Goddard and Whittier [47]. While direct experimental analysis of intrinsic conductivity of the various cytoskeletal components is still nebulous, theoretical modeling of these effects is very well developed. The next section presents our current understanding of the intrinsic conductivity of MTs.

18.5 Intrinsic Electronic Conductivity of Microtubules

Microtubules (MTs) are ubiquitous in eukaryotic cell biology. Their primary role is to serve as the cellular scaffold and act as a corridor for the transport of vesicles through the cell by motor proteins. The interest in MTs stems from their involvement in a number of crucial cellular processes including mitosis and, more recently, the discovery of their role in the communication between the exterior of the cell and the nucleus [46,94]. The special arrangement of MTs just prior to chromosone separation makes them an ideal candidate for mediating a signal of some sort, which would cause the coordinated breakage of sister chromatids. In the axons of nerve cells, the MTs are arranged in bundles, parallel to the axon [56]. These groups of MTs span the entire length of the axon and might serve to carry signals from the cell body to the nerve terminus or vice versa. No mechanism for this sort of direct feedback is known. We demonstrate here that the MT may be able to conduct a current along its length through a process of electron hopping between binding sites on each dimer.

The MT is made up of a protein called tubulin. Each subunit contains α-tubulin and β-tubulin monomers joined together to form a dimer. The dimers are connected end to end to form a protofilament. These protofilaments, usually 13 in number, are wrapped up to form a hollow tube, the MT. The dimers are themselves 5 nm in diameter and have a length of about 8 nm. A MT has an outer diameter of 25 nm and an inner diameter of 15 nm (see Figure 18.2).

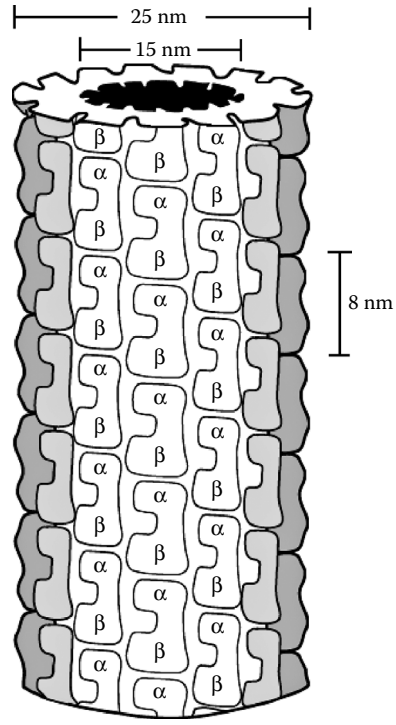

FIGURE 18.2 Structure of a typical microtubule demonstrating how it is formed by a special arrangement of tubulin dimers.

The tubulin dimer of which the MT is composed has at least two different conformational states [58]. That is, while the amino-acid chain of the protein remains fixed, its three-dimensional shape may change. It is believed that these conformational changes are the result of hydrolysis and an associated charge movement from one binding site within the molecule to another. It is precisely this charge movement that may lead to the conduction through the polymer.

One may take a semi-classical view of the movement of electrons along the protofilament. We ignore the fact that the protofilament is nearly one-dimensional and consider its *bulk* properties. Assuming two conduction electrons for each tubulin dimer, the conduction electron density is

$$\begin{aligned} n_0 &= 2/(8nm \times 5nm \times 5nm) \\ &= 1.0 \times 10^{19}/cm^3. \end{aligned} \tag{18.4}$$

In comparison with other materials, this value is slightly higher than typical semi-conductor concentrations, usually $10^{13} - 10^{17}/cm^3$, but is much less than typical metals, which have a conduction electron density of $10^{22} - 10^{23}/cm^3$. However, the electrons are expected to have relatively low mobilities unless they are excited from the ground state configuration. The excitation energy is a function of both the number of electrons within the protofilament and its size. Suppose that the mobility of excited electrons is at least similar to the mobility of electrons within semiconductors, such as Si where the electron mobility, μ_e, is about 1300 $cm^2\ V^{-1}\ s^{-1}$. The electrical conductivity σ is given in a semi-classical model by the well-known formula

$$\sigma = ne\mu_e \tag{18.5}$$

where e is the electron charge and n is the conduction electron density given by

$$n = n_0 e^{-\Delta/\kappa T} \tag{18.6}$$

where n_0 represents the total number of electrons. If we assume that the gap between valence and conduction bands, Δ, is about 0.40 eV, then at physiological temperature, some of the carriers will be excited and should be available to conduct. Applying the preceding formula (18.6), a conductivity of about 10^{-2} (in SI units) is expected for the protofilament.

$$\sigma \sim (10^{25} e^{-0.40/0/026})(1.6 \times 10^{-19})(0.13) \sim 0.04\Omega^{-1} m^{-1} \tag{18.7}$$

However, if the hybridization of electronic orbitals lowers the gap to a value closer to zero, the expected conductivity would be significantly higher:

$$\sigma \sim (10^{25})(1.6 \times 10^{-19})(0.13) \sim 2 \times 10^5 \Omega^{-1} m^{-1} \tag{18.8}$$

These values should be compared to values in excess of 10^8 for copper and other good conductors, and 10^7 for lesser conductors and semi-metals. Typical semi-conductors such as Si and Ge have intrinsic conductivities of $10^{-3} - 10^{+1}$ at room temperature, but when heavily doped, the conductivity may rise to between 10^4 and 10^5.

In order to model the MT as a semiconductor, the individual tubulin dimer must first be physically analyzed. The $\alpha\beta$-tubulin dimer is comprised of about 890 amino acids which represents about 13000 nuclei. A gross simplification which we introduce is to treat each monomer as having an effective site energy for electronic binding. Fluctuations of the site energy on scales less than the monomer spacing are therefore washed out. It is convenient to model this system as two quantum wells separated by a potential barrier. The two wells represent two binding sites for an electron, which may hop between the monomers, resulting in the change of molecular conformation. We can estimate the relevant parameters of the well depth and the well width from kinetic studies and with a knowledge of the molecule's overall geometry. Once the bound states and their energies are known for this system, this knowledge may be applied to a

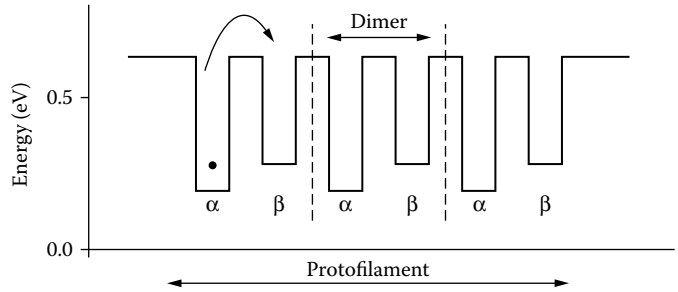

FIGURE 18.3 A one-dimensional chain of quantum wells representing the energy profile along a MT protofilament where the α and β monomers have alternating site energies.

second-quantization scheme. This procedure allows us to make progress without specifically requiring us to solve for the wavefunction of the system.

A 1D chain of quantum double wells is pictured in Figure 18.3. For simplicity, we are initially making each binding site have the same energy. If this is related to GTP hydrolysis, they may differ by about 0.17 eV (4 kcal/mol) [59]. This is roughly the quantity of energy stored in the lattice when GTP is hydrolyzed to GDP [13,60]. The barrier height between dimers has been estimated from kinetic data. The bond strength must be somewhat larger than the hydrolysis energy delivered to the lattice or else hydrolysis of the dimers would cause immediate disassembly. Hopping along the protofilament by electrons presumably has an energy comparable to this binding energy. We have estimated 0.4 eV for intra-dimer hopping and 1.0 eV for the inter-dimer hopping. What may be varied in the model, is the central barrier which must be overcome for each tubulin dimer to change conformation. If it is nearly as large as the barrier to electron movement between dimers, then the protofilament will look like a chain of single quantum wells rather than double quantum wells because the difference between the α and β monomers will not be distinguished. However, if the central barrier is too small, the chain will again look like a series of single quantum wells since the effect of the barrier would be negligible. The depth of the deeper well, f_1, has been selected as 0.4 eV initially. Given the value, we assumed a 25% difference in well depths. If f_1 is much larger, the dimer nature of the protofilament is lost because the wells are effectively the same depth.

The results for the single electron eigenstates of a double quantum-well may be found using a transfer matrix method for the time-independent 1D Schrodinger equation in each interval in which the piece-wise constant potential is defined as

$$\hat{H}\psi = \frac{\hat{p}^2}{2m}\psi + \hat{V}\psi = E\psi \tag{18.9}$$

The oscillatory solution

$$\psi(x) = A\cos(kx) + B\sin(kx) \tag{18.10}$$

applies to $E > V$ where

$$k = a\sqrt{2m(E-V)}/\hbar \tag{18.11}$$

while for $E < V$, the solutions are exponentially damped

$$\psi(x) = Ae^{\kappa x} + Be^{-\kappa x} \tag{18.12}$$

where

$$k = \sqrt{2m(V-E)}/\hbar \tag{18.13}$$

Now, matching conditions for the wave function and its derivative are applied at each interface. This gives a 2×2 matrix relating the coefficients in the wave function expansion according to

$$\begin{pmatrix} A_2 \\ B_2 \end{pmatrix} = \begin{pmatrix} {}^{2,1}M_{11} & {}^{2,1}M_{12} \\ {}^{2,1}M_{21} & {}^{2,1}M_{22} \end{pmatrix} \begin{pmatrix} A_1 \\ B_1 \end{pmatrix} \tag{18.14}$$

In the preceding equation, the superscript to the left of the M denotes which coefficients are being linked. The subscripts indicate the matrix element. In an iterated fashion, one eventually arrives at the following:

$$\begin{pmatrix} A_n \\ B_n \end{pmatrix} = \begin{pmatrix} {}^{n,1}M_{11} & {}^{n,1}M_{12} \\ {}^{n,1}M_{21} & {}^{n,1}M_{22} \end{pmatrix} \begin{pmatrix} A_1 \\ B_1 \end{pmatrix} \tag{18.15}$$

Now, for bound states $A_1 = B_n = 0$ so that the wavefunction decays outside the region of the quantum wells:

$$B_n = 0 \quad \Rightarrow \quad {}^{n,1}M_{22} = 0 \tag{18.16}$$

Given the choices for our quantum double well (width, $a = 2$ nm; and height, $V = 1.0$ eV), one finds that there are two bound states for this dimer problem. As the number of dimers is increased, one resorts to numerical analysis. In Figure 18.4, the energy of the two bound states is plotted as the energy of the central barrier is varied. The lowest energy state is the symmetric wavefunction, which increases in energy sharply as the central barrier is increased. The second bound state is the antisymmetric wavefunction it has a node within the barrier, hence its energy increases only marginally.

The energy calculation for a sequence of n such dimers leads to the development of bands as hybrid orbitals develop across the polymer. Roughly speaking, one band is of hybridized symmetric orbitals and the higher lying band of hybridized antisymmetric orbitals. The separation of the two bands and the spread of each individual band determine the values of the parameters which shall be used in the tight-binding Hubbard model used to model the polymer.

The second-quantized Hamiltonian for a many-electron multidimer system becomes

$$\hat{H} = \sum_{i,\sigma} \epsilon_i \hat{c}^{\dagger}_{i\sigma} \hat{c}_{i\sigma} + \sum_{i \neq j,\sigma} t_{ij} \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} \tag{18.17}$$

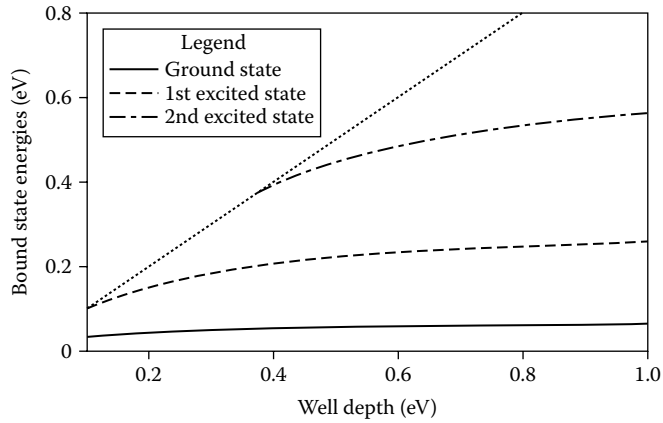

FIGURE 18.4 Variation of the bound state energies with the depth of the double quantum wells, $f_1 = f_2$. The straight line indicates where the bound state energy is equal to the well depth.

where the operators $\hat{c}^{\dagger}_{i\sigma}$ and $\hat{c}_{i\sigma}$ are the usual fermion creation and annihilation operators, and where i runs from 1 to $2n$ representing the two sites on each of the n dimers. The ϵ_i represent the potential energy at each site and the kinetic terms t_{ij} account for hopping from one site to another. There are two different kinds of interactions depending on whether an electron hops within a dimer, or between dimers. Hence, this kinetic parameter has two different values along the chain. Since there are no interactions between the states with $+\frac{1}{2}$ spin and those with $-\frac{1}{2}$ spin, the state space may be divided into two identical subspaces labeled by the electron spin. In each of these subspaces, the matrix corresponding to this chain of dimers is tri-diagonal and may be diagonalized to determine the itinerant eigenstates of the system. The coefficient t_d which represents the movement of an electron within a dimer causes the eigenstates to be split into two bands by an amount, $2t_d$. Within each of these bands, the coefficient t_l of the inter-dimer hopping, causes these two single electron bands to broaden each to about $2t_l$. Now, although the preceding description is qualitative, clearly whenever $t_d \sim t_l$, these bands will overlap and transitions between the states will be important.

Considering the band formation in the quantum well formulation, we can assign values to t_d and t_l. The two bands are separated by about 0.40 eV, so $t_d \sim 0.20$ eV, and using the bottom band to estimate t_l it widens until it is about 0.10 eV thick, so $t_l \sim 0.05$ eV.

The Hubbard model is the simplest second quantized model to capture all of the essential features of a real system of interacting particles [82]. In this model, electrons are the fundamental excitations. All states may be built up from the vacuum using the fermion creation operators. Once the number of sites is selected, n, and the number of electrons is determined, e, there are

$$\binom{2n}{e} = \frac{2n!}{e!(2n-e)!} \tag{18.18}$$

distinct electronic states. These states are built up by applying in normal order each of the combinations of e-electron states.

The second quantized Hamiltonian for the Hubbard model with electron–electron interactions is conventionally written as

$$\hat{H} = \sum_{i,\sigma} \epsilon_i \hat{c}^{\dagger}_{i\sigma} \hat{c}_{i\sigma} + \sum_{i \neq j,\sigma} t_{ij} \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} + \sum_{i} U_i \hat{c}^{\dagger}_{i\uparrow} \hat{c}_{i\uparrow} \hat{c}^{\dagger}_{i\downarrow} \hat{c}_{i\downarrow} \tag{18.19}$$

The last term accounts for electron–electron repulsion now that we have gone to a many electron system. In our simple model, this is limited to an onsite interaction. To this point, we have made each of the site energies equivalent ($\epsilon_i = \epsilon, \forall i$). In what follows, $U = 2.0$ eV which is the energy associated with two bare electrons lying about 0.7 nm apart. This choice seems reasonable within a well of width 2.0 nm. We try to solve the problem by considering the electron–electron term as a perturbation to study how it affects the electronic states of the dimer. The Hamiltonian in the absence of the final U-term may be diagonalized by the following unitary transformation:

$$\alpha^{\dagger}_{1'\sigma} = \frac{1}{\sqrt{2}}(c^{\dagger}_{1\sigma} + c^{\dagger}_{2\sigma}) \tag{18.20}$$

$$\alpha^{\dagger}_{1'\sigma} = \frac{1}{2\sqrt{2}}(c^{\dagger}_{1\sigma} - c^{\dagger}_{2\sigma}) \tag{18.21}$$

The linear combination of operators is simple because we have made the site energies ϵ_1 and ϵ_2 equal. These new operators are easily shown to obey the usual anticommutation relations for fermions and are said to be in the itinerant or distributed electron basis. The Hamiltonian may now be rewritten in terms of these new operators, as

$$\hat{H} = \sum_{\sigma}(\epsilon + t)\alpha^{\dagger}_{1\sigma}\alpha_{1\sigma} + \sum_{\sigma}(\epsilon - t)\alpha^{\dagger}_{2\sigma}\alpha_{2\sigma} + \frac{1}{2}\sum_{\sigma \neq \sigma'} V_{ijlm}\alpha^{\dagger}_{i\sigma}\alpha^{\dagger}_{j\sigma'}\alpha_{l\sigma'}\alpha_{m\sigma} \tag{18.22}$$

the vacuum state is represented by $|\Omega>$. Acting with our Hamiltonian on this basis gives us the matrix form of the Hamiltonian:

$$\hat{H} = \begin{pmatrix} 2(\epsilon+t)+\frac{(U_1+U_2)}{4} & 0 & \frac{(U_1-U_2)}{4} & \frac{(U_2-U_1)}{4} & 0 & \frac{(U_1+U_2)}{4} \\ 0 & 2\epsilon & 0 & 0 & 0 & 0 \\ \frac{(U_1-U_2)}{4} & 0 & 2\epsilon+\frac{(U_1+U_2)}{4} & -\frac{(U_1+U_2)}{4} & 0 & \frac{(U_1-U_2)}{4} \\ \frac{(U_2-U_1)}{4} & 0 & -\frac{(U_1+U_2)}{4} & 2\epsilon+\frac{(U_1+U_2)}{4} & 0 & \frac{(U_2-U_1)}{4} \\ 0 & 0 & 0 & 0 & 2\epsilon & 0 \\ \frac{(U_1+U_2)}{4} & 0 & \frac{(U_1-U_2)}{4} & \frac{(U_2-U_1)}{4} & 0 & 2(\epsilon-t)+\frac{(U_1+U_2)}{4} \end{pmatrix}$$

The Hamiltonian is then diagnolized in order to determine its eigenstates. In the absence of the Coulomb repulsion ($U = 0$), the electrons are localized on the $2'$ site with energy $2(\epsilon - t)$, the gap to the first excited state being $2t$. However, as U is increased, the effect of the Coulomb interaction raises the energy of the ground state relative to the excited singlet states, which do not feel the Coulomb repulsion. Consequently, the gap between the ground state and the first excited state is actually reduced through the introduction, as shown in Figure 18.5. Thus, when conductivity is being considered, it is important to bear in mind that the gap is reduced by the presence of the electron–electron interactions.

To this point, we have made each of the site energies equivalent, ($\epsilon_i = \epsilon$), and the electron repulsion the same at all sites ($U_i = U$). The lone complication lies within the kinetic term, where the hopping parameter will depend on the two sites between which the electron hops. However, since we first consider the MT protofilament, it shall initially be a constant as well, $t = 0.4$ eV, where the choice comes from the separation of the energy bands in the 1D Schrödinger picture.

Since the Hamiltonian conserves both the particle number and the overall spin of each state, we are free to work in subspaces enumerated by the spin of the states it contains. This reduces the problem slightly since it is possible to consider individually the spin $+1$, spin 0, and spin -1 systems separately in a two-electron system. In addition, the symmetry implies that the spin $+1$ and spin -1 systems have the same conductive properties. We also find that the ground state is always a state consisting of the lowest possible total spin.

When the inter-dimer hopping term is small relative to the intra-dimer hopping term $t_{inter} < t_{intra}$, the form of the ground state is independent of the number of dimers. In each case, the ground state is a linear combination of singlet states on each dimer. The singlet states are those where two electrons are

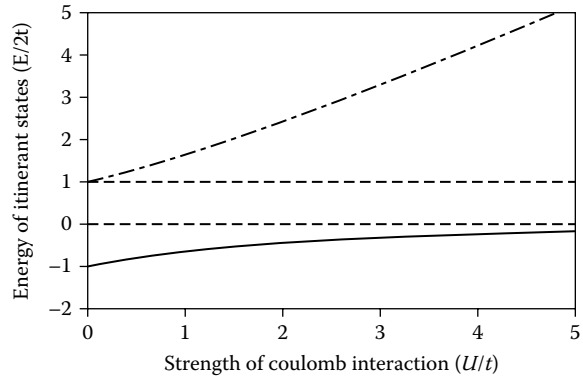

FIGURE 18.5 Energy of the itinerant states as a function of U/t with the energy plotted in units of $2t$.

paired on an individual binding site, or where the state consists of the anti-symmetric combination of an up-electron and a down-electron on neighboring binding sites of the dimer. When the inter-dimer hopping parameter is increased, the density of states becomes more uniform between the ground state and the highest energy eigenstate. With the gap between the ground state and the first excited state of the system becoming diminished. Since the hopping term within a dimer is largest, the system prefers to have two electrons with the same spin within the same dimer in the ground state.

For electronic conductivity, the band formation importantly depends upon the hopping parameter between dimers. When this parameter is small relative to the intra-dimer hopping, $t_{inter}/t_{intra} \sim 0.1$, there remains a rather large gap between the ground state and the first excited state of about $2t_{intra}$ which is reduced as the repulsion term increases. However, when the hopping between dimers is equal to the hopping within dimers, $t_{inter} \sim t_{intra}$, the gap is minimized. In the event that $t_{inter} > t_{intra}$, we would effectively have to relabel the dimers and would then return to the picture where $t_{inter} < t_{intra}$.

Kubo was the first to derive formulae for the electrical conductivity in solids, and his derivation is applied to our case in the present section [77]. In the Coulomb gauge, we write the vector potential for a uniform electric field along the direction of the chain, $A(t)$, as

$$A(t) = \frac{cE(t)}{-i\omega} e^{i\omega t}, \tag{18.23}$$

Adding a field-coupling term to the Hubbard Hamiltonian and using a canonical transformation gives [91]

$$\hat{H}'_0 = -t \sum_{i\sigma} [\hat{c}^\dagger_{i\sigma} \hat{c}_{i+1\sigma} e^{-ieA(t)/c} + \hat{c}^\dagger_{i\sigma} \hat{c}_{i-1\sigma} e^{ieA(t)/c}]. \tag{18.24}$$

where a is the lattice spacing and c is the speed of light. Expanding to second order in the vector potential, we find

$$\hat{H}_A = \frac{A}{c} \hat{j} - e \frac{A^2}{c^2} a^2 \hat{H}_0, \tag{18.25}$$

where

$$\hat{j} = i\mathrm{Eat} \sum_{i\sigma} (\hat{c}^\dagger_{i\sigma} \hat{c}_{i+1\sigma} - \hat{c}^\dagger_{i\sigma} \hat{c}_{i-1\sigma}). \tag{18.26}$$

So, the energy of the original system is corrected to this order and the interaction term is proportional to the current.

Suppose a time-dependent external electric field is applied to a solid:

$$E^{\mathrm{ext}}_\alpha(r,t) = \Xi^{\mathrm{ext}}_\alpha e^{iq\cdot r - i\omega t}, \tag{18.27}$$

where α represents the Cartesian coordinate directions. In a linear response theory, the induced current is proportional to the applied electric field:

$$J_\alpha(r,t) = \sum_\beta \sigma'_{\alpha\beta}(q,\omega) \Xi^{\mathrm{ext}}_\beta e^{iq\cdot r - \omega t}, \tag{18.28}$$

where $\sigma'_{\alpha\beta}$ is a parameter relating the observed current density to the applied field's direction, its periodicity, and frequency. However, the symbol appearing in (18.28) is not the conductivity we seek. Rather, we want the conductivity which represents the response to the total electric field in the solid, a quantity that can be measured. This conductivity takes into account all of the currents, induced by the external fields, that

create their own electric fields. Thus, we seek $\sigma_{\alpha\beta}$ that relates the macroscopic electric field to the currents of the system.

$$J_\alpha(r,t) = \sum_\beta \sigma_{\alpha\beta}(q,\omega) E_\beta e^{iq\cdot r - i\omega t} \tag{18.29}$$

$$E_\alpha(r,t) = \Xi_\alpha e^{iq\cdot r - i\omega t} \tag{18.30}$$

$$\sigma_{\alpha\beta} = Re(\sigma_{\alpha\beta}) + iIm(\sigma_{\alpha\beta}) \tag{18.31}$$

We write the Hamiltonian of the system as $\hat{H} + \hat{H}'$ where the latter term contains the electric field, which we shall introduce as a time-dependent perturbation. The evolution of the operators is given as follows [30]:

$$\hat{j}(x,t) = e^{i\hat{H}t}\hat{j}(x)e^{-i\hat{H}t}. \tag{18.32}$$

where $\hat{H}$ is the unperturbed Hamiltonian. Thus, the interaction picture is adopted. The Kubo formula for electrical conductivity gives the result in terms of a current–current correlation function:

$$\sigma_{\alpha\beta}(q,w) = \frac{1}{\omega}\int_0^\infty dt e^{i\omega t} < \Omega|\hat{j}^\dagger_\alpha(q,t)\hat{j}_\beta(q,0) - \hat{j}_\beta(q,0)\hat{j}^\dagger_\alpha(q,t)|\Omega > + i\frac{n_0 e^2}{m\omega}\delta_{\alpha\beta} \tag{18.33}$$

where $|\Omega>$ is the ground state of the system, which has not been perturbed by the electric field. The first term in (18.33) is known as the incoherent contribution to the conductivity. Now, we use (18.32) in the preceding equation and use the fact that the basis we are using is that of the eigenstates. Consequently, with proper normalization the new expression for the real part of the conductivity is [90]

$$\sigma(\omega) = \frac{\pi}{Z}\frac{(1-e^{-\beta\omega})}{\omega}\sum_{n,m} e^{-\beta E_m}| < n|\hat{j}_\alpha|m > |^2\delta[\omega - (E_n - E_m)] \tag{18.34}$$

The DC conductivity is calculated by a procedure [90,91], which employs the sum rule

$$\int_0^\infty \sigma'(\omega)d\omega = -\frac{\pi e^2}{2}a^2 < K > \tag{18.35}$$

where

$$\sigma'(\omega) = D\delta(\omega) + \sigma_{kubo}(\omega) \tag{18.36}$$

and $< K >$ is the expectation value of the kinetic energy term of the Hamiltonian. The strategy of computing $\sigma(\omega)$, and subsequently the Drude contribution, at either zero or finite temperature is reasonably straight forward. The first step is to construct the matrix corresponding to the Hamiltonian [Equation (18.19)] in a subspace which is defined by the geometry of the lattice, the number of electrons, and the total electron spin. This leads to serious computational difficulties since for a moderately sized lattice of say ten dimers and eight electrons of arbitrary spin, the corresponding space is about 77 million states. A matrix of dimension 77 million square has about 5.9×10^{15} elements. Obviously, working on a moderately sized lattice, we are restricted to a low concentration of electrons. Due to electron-hole symmetry, we are also able to study the nearly filled situation. It is useful to keep in mind the relationship

$$\lambda = \frac{1240nm\cdot eV}{\text{Energy}}. \tag{18.37}$$

So that in the plots following, the visible light range is between 1.75 eV and 3.10 eV. Energies larger than this range correspond to ultraviolet, and those smaller to infrared wavelengths. Since cells respond to light in the near-infrared [4], and centrioles comprised of MTs are proposed to be the site of light detection [5], absorption in this energy range is of great interest to cell biology. These energies should also be compared

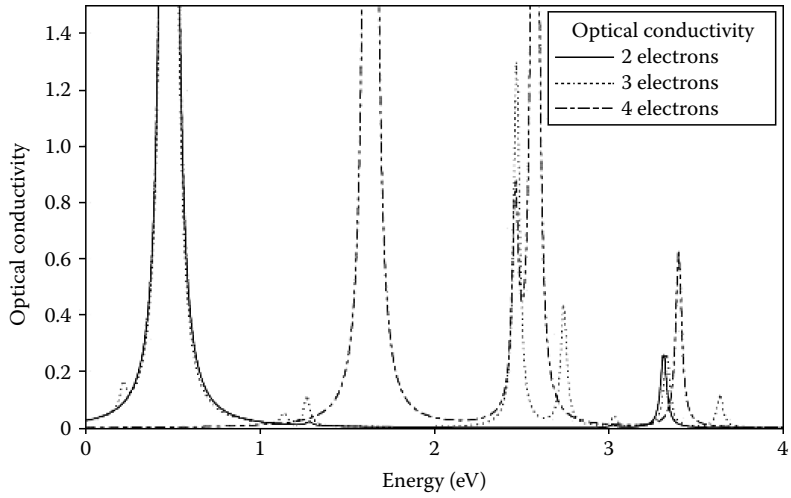

FIGURE 18.6 Optical conductivity of the two dimer system with two, three, and four electrons.

to the energy available from ATP and GTP hydrolysis, which are 0.49 eV and 0.22 eV, respectively. These values provide limits for the interactions of chemical energy in the conduction process.

At zero temperature, the linear polymer results are summarized concisely by saying that there is no DC conductivity that agrees with the previous modeling of Hubbard systems [44].

The following results for the 1D chain of tubulin dimers represent the AC conductivity. There are peaks in $\sigma(\omega)$, also known as the optical conductivity, which correspond to excitations between electronic states of the polymer. Electronic excitation results in electron redistribution and hence the development of a current. We have selected $\epsilon = 0.02$ eV to present our results, which is roughly kT and is smaller than the three energy scales in the Hubbard model, t, U, and $4t^2/U$ [82] so we expect to be able to pick out features corresponding to specific transitions in the predicted optical spectra.

We shall begin with two dimers and $t = 0.4$ eV, $U = 2.0$ eV, and equal site energy on the α and β monomer sites. This system can support up to eight electrons. The results for these systems are presented in Figure 18.6. The predicted spectra for the two and three electron cases are quite similar. The differences include a new but small peak in the three electron spectrum at an energy just below the major peak, and variations in the detail of the absorption band centered about 2.6 eV. However, the qualitative similarity of the two spectra is remarkable in terms of band locations. Contrast these two cases with the four-electron system that corresponds to half-filling. The difference is that the peak in the optical conductivity spectrum increases dramatically from about 0.48 eV to 1.64 eV. This change indicates how the half-filled case is special. The ground state in this system largely consists of electrons spread out to minimize the Coulombic energy. Exciting this system by moving any particular electron will come at the cost of greatly increasing the electron–electron repulsion and, consequently, the location of the required excitation energy corresponds roughly to U.

As the length of the polymer is increased, results remain qualitatively similar. In the three-dimer structure, results are shown for the conductivity in Figure 18.7(a) and for the integrated conductivity in Figure 18.7(b). Once again, the spectra are quite similar for all fillings aside from the half-filled case which consists of six electrons. The shift of the conductivity to higher energies can be clearly seen as the filling fraction increases towards half filling. We now present results where the broadening parameter, ϵ, and the temperature have been varied to see the effect that they have on the results. Figure 18.8 demonstrates that as the width is decreased, the spectrum becomes sharper. Note how the two peaks in this simple spectrum that occur near 2.5 eV are successively blended into a band as the ϵ is increased. Peaks represent transitions from one specific electron configuration to another and are weighted with the associated conductivity.

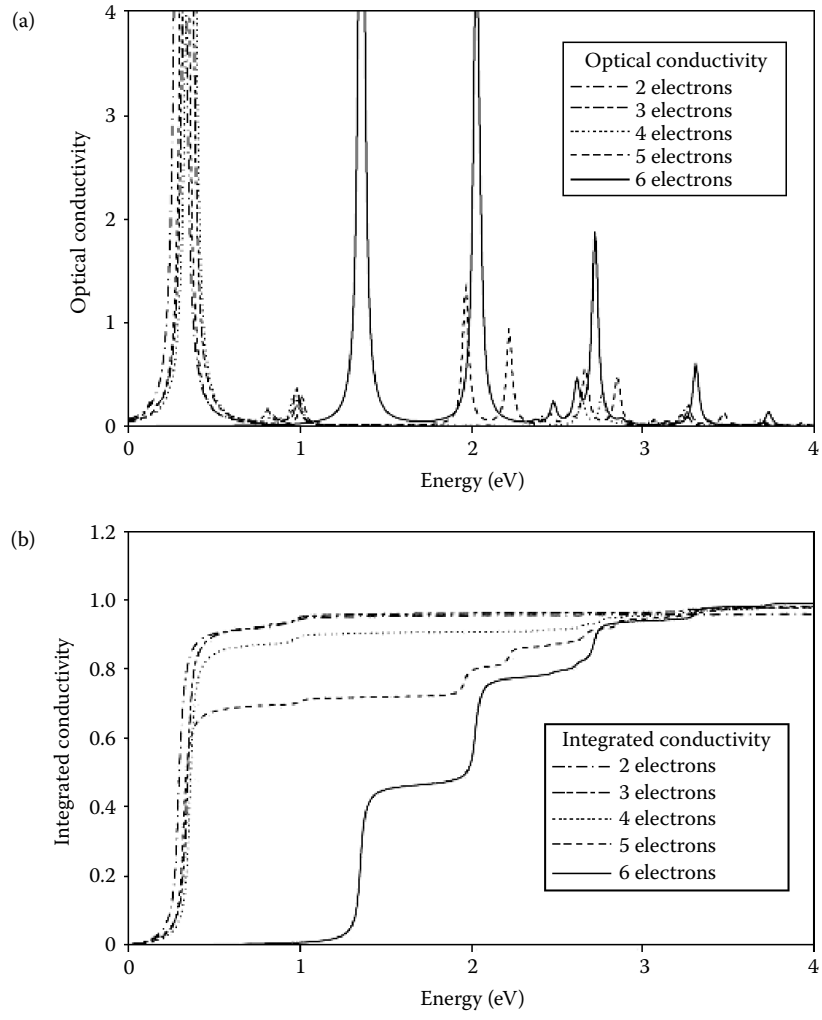

FIGURE 18.7 (a) Optical conductivity of the three-dimer system from two electrons through half filling. (b) Integrated optical conductivity of the three-dimer system shows that indeed the protofilament is an insulator. The gap for excitation of the half-filled polymer is seen easily here.

The effect of changing the temperature is to excite electron configurations other than the ground state. Consequently, transitions from one excited state to another can now contribute to the conductivity. Figure 18.9 shows how this often reduces the gap to conduction: all the major peaks in the 300 K spectrum shift slightly towards lower energies in the 3000 K spectrum, but many more features develop as well as states that previously did not contribute to conductivity may make transitions to excited states. Physiological temperature of 300 K ($kT = 0.026$ eV) is small relative to the system's energy scales, consequently lowering the temperature further causes very little change to the spectrum.

If the α-tubulin and β-tubulin sites have different energies, then the MT A and B lattices can be distinguished. The effect of this change on the optical spectrum seems to be small when individual protofilaments are considered. In addition, the sum rule shows that protofilaments remain insulators. There are a few parameters we can consider individually and which can be compared as the size of the system changes: the threshold to conductivity, and the expectation value of the kinetic energy operator. The threshold to conductivity is simply the location of the first absorption peak in the optical spectrum.

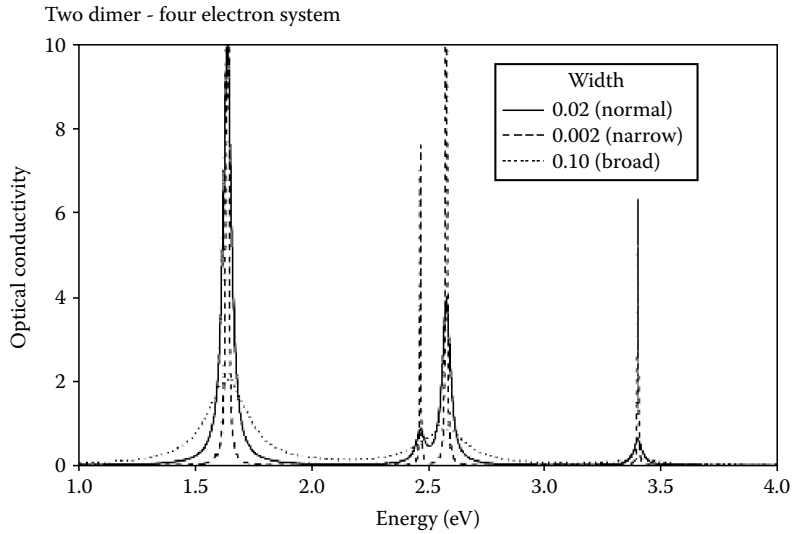

FIGURE 18.8 Optical conductivity spectrum variation with the broadening parameter.

It is a decreasing function of the polymer length. Starting at 0.48 eV for the two-dimer system, it seems to drop a little faster than $1/n$ for systems consisting of n dimers and containing two electrons. The fitted value in Figure 18.10 varies $\alpha 1/n^{1.4}$. Extrapolating to larger values of n, this gap appears certain to fall into the range where these excited states could be thermally excited. The protofilament then develops a finite DC conductivity at finite temperature. It is interesting that although the first peak does move towards zero, the fraction of the conductivity it contains is gradually diminished. Some of the conduction continues to reside in a peak located close to t in the energy spectrum. In the 2D lattice, there is a phase transition in the dimensionality. DC conduction becomes possible even at zero temperature. However, we shall still be concerned about the boundary conditions since calculations are being performed on a small lattice.

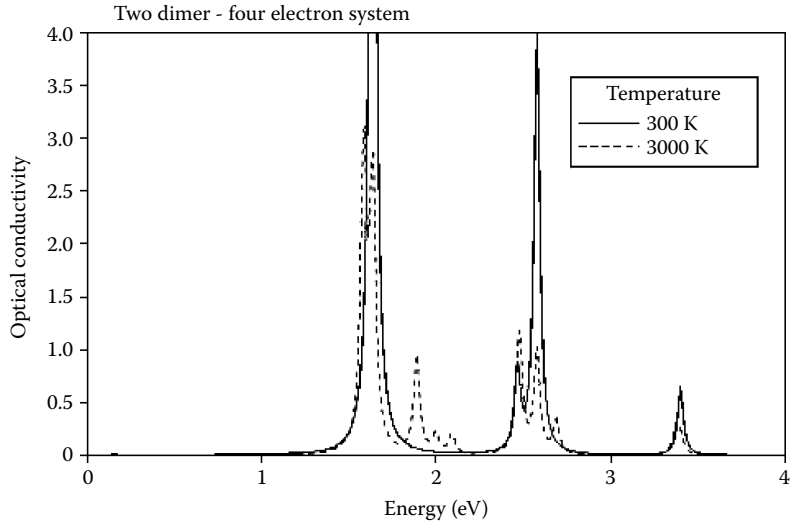

FIGURE 18.9 The variation of the optical conductivity spectrum with temperature is shown.

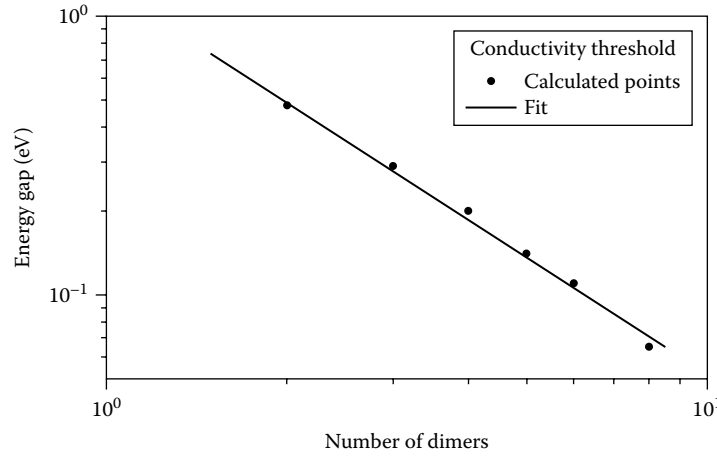

FIGURE 18.10 Threshold to conduction as a function of protofilament length.

Consider the kinetic energy/electron that has been calculated in a Hubbard model with our parameters for t and U and with either one, two or three protofilaments. Figure 18.11 demonstrates how the kinetic energy of each electron approaches the limit of $2t$ or 0.80 eV as the size of the system increases. We can easily see that for more than about 20 electron sites, the average kinetic energy of each electron starts to approach the infinite limit and we expect that the results obtained from our model should converge to the long protofilament limit in a similar manner. In particular, we are considering 28 electron sites in our MT model, which span three protofilaments. While this is not the ideal situation, the boundary conditions are not expected to create huge effects in our results. We are also using open boundary conditions since they minimize the boundary effects when compared with a variety of periodic boundary conditions.

The unit cells depicted in Figure 18.12 are the basis for the calculations. In the A lattice used for the calculations, the α-dimers connect to β-dimers along the $\bar{3}$-start helix, both within the lattice and at the seam when the MT is wrapped up. In the B lattice, the connections are $\alpha - \alpha$ or $\beta - \beta$ along the $\bar{3}$-start helix and never $\alpha - \beta$. For comparison, calculations have also been carried out with a flat tubulin sheet, specifically an A lattice MT lacking periodic boundary conditions in the lateral direction. This simulates the

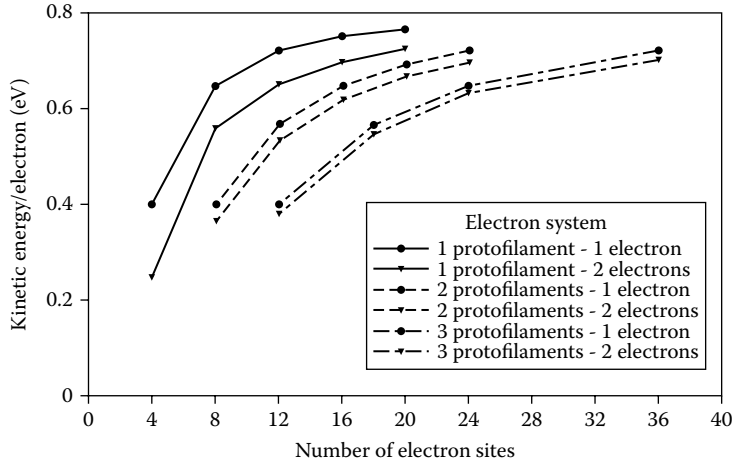

FIGURE 18.11 Kinetic energy per electron as a function of system size.

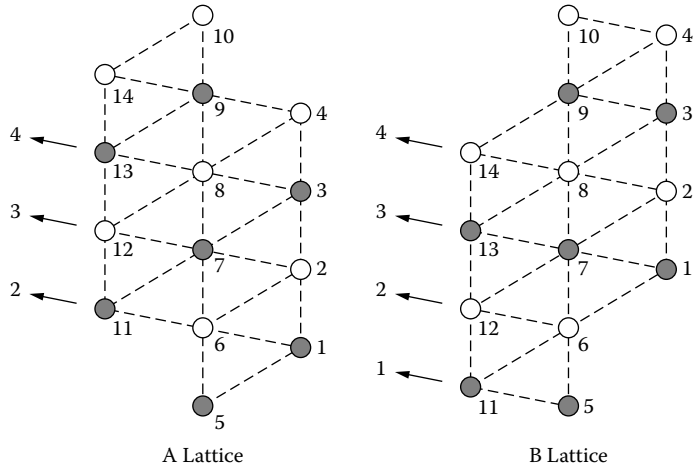

FIGURE 18.12 Lattice unit cells for conductivity calculations. The unit cell on the left corresponds to the MT A lattice, while that on the right corresponds to the MT B lattice.

MT when it has been unwrapped. We recall that the MT lattice is triangular and has three distinct hopping directions. Since protofilaments are strongly bound together along their length but relatively weakly bound to other protofilaments, one also expects the hopping parameter to be much smaller when an electron moves between protofilaments. The calculations have been carried out with the same parameters as in the one-dimensional case, $U = 2.00$ eV, $t = 0.40$ eV along the protofilaments and the site energies have been set equal to zero for all sites. The value of the hopping parameter along the $\bar{3}$-start helix is called t_ℓ (left) and that along the 8-start helix that connects diagonally from a monomer in a direction up and to the right is called t_r (right).

We consider first the results of the MT lattice which has not rolled up to form a tube but is that of a tubulin sheet where $t_r = 0$ and only the value of t_l is varied (see Figure 18.13). Since $t_r = 0$, the nature of the lattice remains rectangular to this point. When $t_l = 0$, this particular system is equivalent to individual protofilaments. The two peaks in the optical spectrum arise from the fact that the protofilaments have different lengths. The absorption peaks of 0.22 eV and 0.40 eV correspond roughly to those of the two-dimer and three-dimer, two electron cases discussed earlier in the chapter. The difference here is that the two electrons are spread over three protofilaments and consequently the effect of the electron repulsion is reduced. However, as t_l is increased from zero, the entire lattice becomes accessible and the protofilament character of the optical spectrum is lost. The two peaks of the t_l spectrum coalesce into a single peak with a large activation energy. A second but smaller absorption peak forms just above 0.50 eV. In the plot of the integrated conductivity, we can see that there is also a smaller peak at 0.16 eV when $t_l = 0.10$ eV. The source of this peak is an absorption for conduction along the direction of t_l. When t_l is increased to 0.40 eV, this peak occurs at 0.61 eV. Thus, this peak seems to occur at roughly $1.5t_l$. The activation energy for conduction along the protofilament axis also increases but is not as sensitive to t_l. Finally, examination of the integrated conductivity demonstrates that the total absorption combines such that the Drude weight remains zero.

The results in Figure 18.14 show a difference between an individual protofilament and a tubulin sheet. The behavior here is inconsistent with that of an insulator since the Drude weight is non-zero. In the optical conductivity, we find that the largest peak occurs at the same location in all directions but with different weightings. The first line traces the results for $t_r = 0.00$ eV and has peaks at 0.38 eV and 0.61 eV. Once the third hopping direction is allowed, the peak absorption is raised slightly to 0.43 eV for $t_r = 0.10$ eV and a smaller peak forms at 0.74 eV. The lower peak consists mainly of hopping along the protofilament while the second peak consists largely of hopping along the t_l direction, which also corresponds to a large

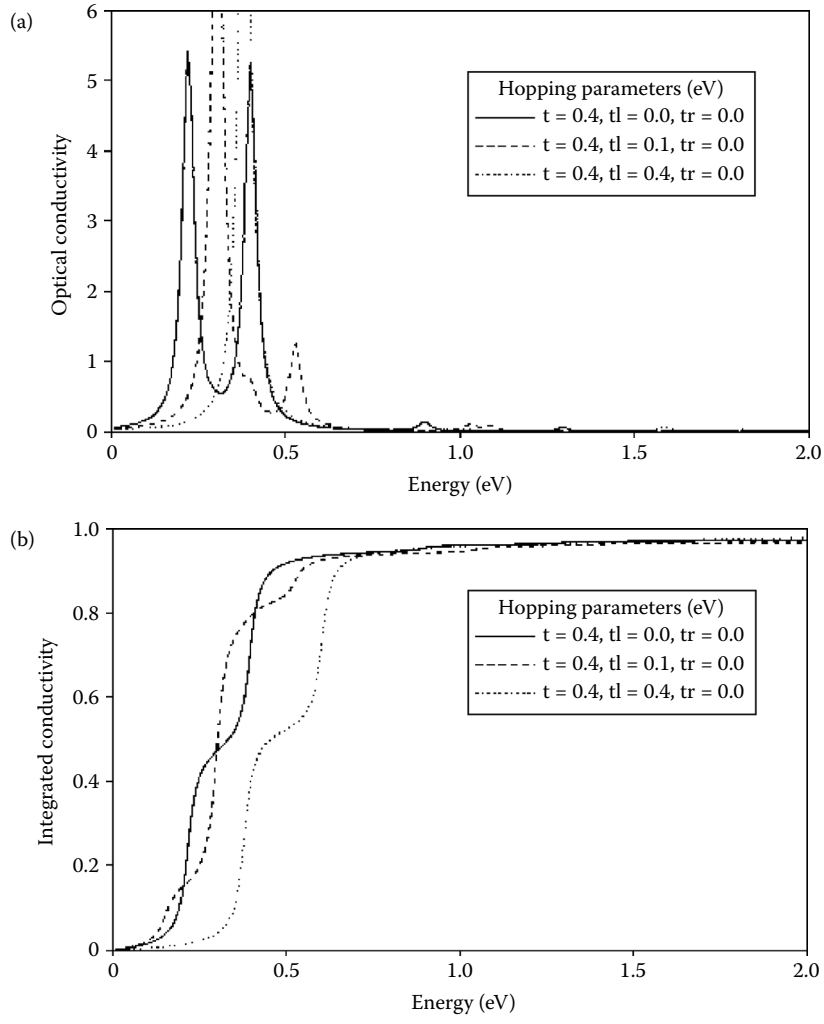

FIGURE 18.13 (a) Optical conductivity along the protofilament axis of the tubulin sheet consisting of a 14-site lattice with two electrons while $t_r = 0$ and t_l is varied. (b) Integrated conductivity of the tubulin sheet with two electrons while $t_r = 0$ and t_l is varied.

hopping parameter. Once t_r is raised to 0.40 eV so that it has the same magnitude as the other hopping parameters, there is only a single peak at 0.70 eV.

The effect of applying the MT A lattice boundary conditions to our system is shown in Figure 18.15, where we see that changing only t_l to a finite quantity while still maintaining t_r as zero is sufficient to result in a non-zero Drude weight. Thus, the periodic boundary conditions in the lateral direction seem to act in a manner similar to the additional hopping directions of the unwrapped lattice. When the kinetic parameters between protofilaments are set to zero, the optical conductivity spectrum shows there are two large absorption peaks at about 0.23 eV and 0.40 eV. As t_l increased, these peaks move higher in energy and the second peak becomes smaller in integrated weight. Eventually, once $t_l = t$, there is a single large peak near 0.47 eV. Introduction of the third hopping direction serves only to raise further the absorption peak, up to 0.70 eV in the case where the hopping parameter is 0.40 eV in all directions.

From Figure 18.15 one can see that more conduction results in the B lattice for a small t_l, such as 0.1 eV, than in the MT A lattice, by comparing it with Figure 18.16. However, even the positioning of absorption

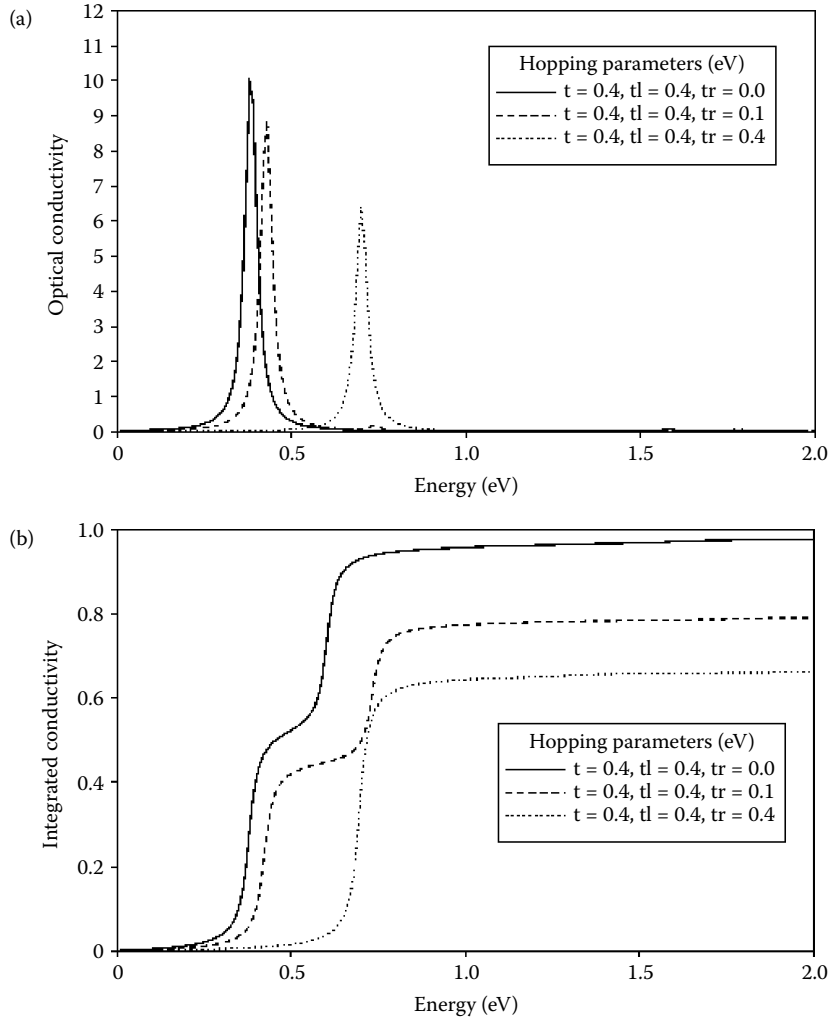

FIGURE 18.14 (a) Optical conductivity along the protofilament axis of the unwrapped 14-site lattice with two electrons while $t_l = 0.4$ and t_r is varied. (b) Integrated conductivity of the unwrapped 14-site lattice with two electrons while $t_l = 0.4$ and t_r is varied.

peaks is similar in the two lattices. Again, it is sufficient for only one of t_l or t_r to be non-zero for the Drude weight to be finite.

Finally, we compare the MT A and B lattices with the unwrapped tubulin sheet as well as with individual protofilaments of tubulin dimers. There is an interesting geometrical interpretation to the results, which shows that the MT A and MT B lattices have identical conduction properties when $t_l = t_r$. This is because in our calculations we have considered all monomers to have the same site energy. Consequently, the B lattice can be viewed as an A lattice with the opposite helicity but the same structure. The sign of the system's helicity does not affect the conduction properties along the major axis. The unwrapped MT lattice is not as good a conductor since some of the hopping freedom has been removed from the lattice but does remain conducting, provided that both t_l and t_r are non-zero.

If we consider the case where one of t_l or t_r is quite small, however, the wrapping of the lattice is much more important. As discussed earlier, the unwrapped tubulin sheet is an insulator while the MT A and B lattices may carry electrons. This observation is especially interesting given that MTs have been observed to zip up and essentially change their structure from a tube to that of a sheet *in vivo*.

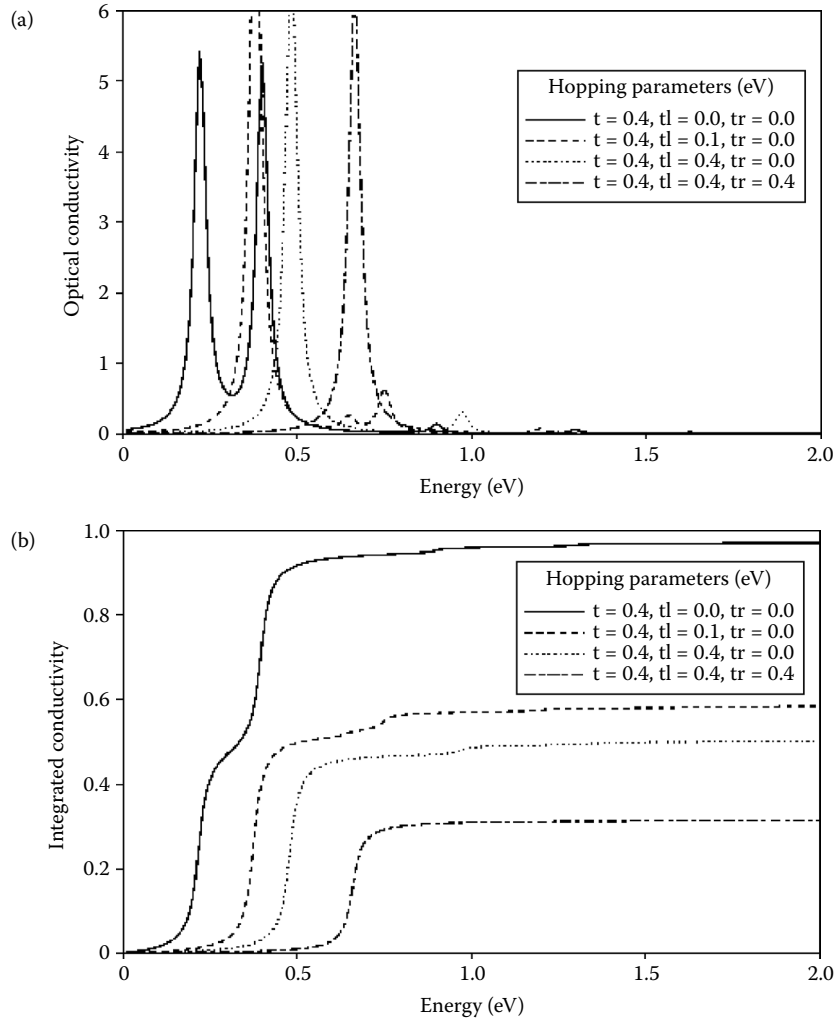

FIGURE 18.15 (a) Optical conductivity along the protofilament direction of the 14-site MT B lattice with two electrons. (b) Integrated conductivity of the 14-site MT B with two electrons shows a non-zero weight for finite t_l.

Our calculations based on the Hubbard model of electron hopping show that the MT is not an insulator, but that instead its conductivity depends on the lattice geometry and whether the MT is wrapped up or not. How well or poorly does one expect the MT to conduct relative to known semi-conductors? Referring back to Equations (18.35) and (18.36), the DC conductivity of the MT is simply given by D as

$$\frac{D}{2} = -\frac{\pi e^2}{2} a^2 < K > - \int_0^\infty \sigma_{kubo}(\omega) d\omega. \tag{18.38}$$

In this equation, the constants can be factored, and what remains is the fraction that can be read from the graphs presented earlier. The Kubo fraction (KF) is the limit of the integral as the frequency tends to infinity.

$$\frac{D}{2} = -\frac{\pi e^2}{2} a^2 < K > (1 - KF). \tag{18.39}$$

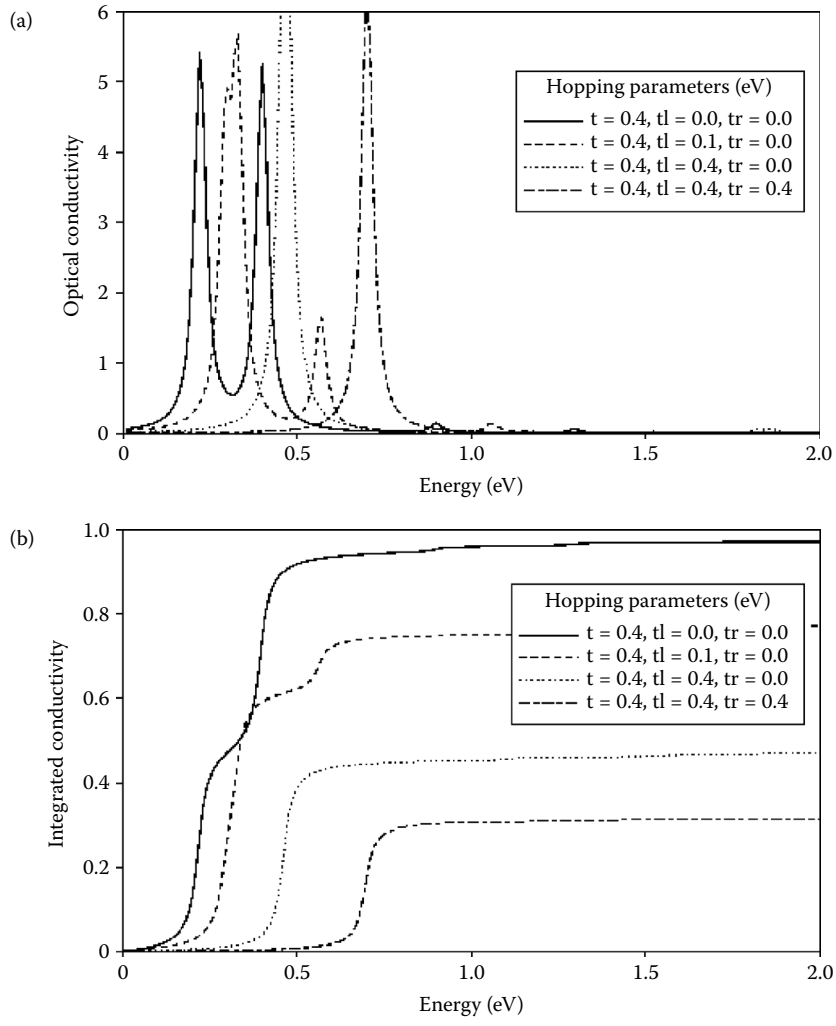

FIGURE 18.16 (a) Optical conductivity along the protofilament direction of the 14-site MT A lattice with two electrons. (b) Integrated conductivity of the 14-site MT A lattice with two electrons shows a non-zero Drude weight for finite t_1.

By returning constants of the system so that the dimensionality of D is correct, we find

$$D = -\frac{\pi n e^2 a^2}{\hbar} \frac{< K >}{t} (1 - KF). \tag{18.40}$$

Finally, we are in a position to predict the conductivity of individual MTs of varying lattice types. For the following calculations, the electron density has been taken to be of two electrons within the lattice of 14 dimers and corresponds to a density of $1.8 \times 10^{18}/cm^3$. Thus, the results must assume that the conductivity remains constant as the length of the polymer is extended, provided the degree of filling remains the same.

The values in Table 18.1 should be compared with the conductivity of metals, such as copper ($6 \times 10^7\ \Omega m^{-1}$) and iron ($1 \times 10^7\ \Omega m^{-1}$). Indeed the values can also be compared to the intrinsic semiconductors germanium ($2.5\ \Omega m^{-1}$) and silicon ($4 \times 10^{-4}\ \Omega m^{-1}$). Given these comparisons, the MT may indeed be quite a good semi-conductor given our assumptions. What is particularly interesting is the

TABLE 18.1 Calculated Conductivity and Resistance of a 1-μm Polymer Composed of Tubulin Dimers

Tubulin Structure	Conductivity ($\Omega^{-1}m^{-1}$)	Resistance
Individual Protofilament	Non-Conducting	
Tubulin sheet ($t_1 = t_r = 0.04\ eV$)	9.4×10^3	$418k\,\Omega$
Tubulin sheet ($t_l = t_r = 0.1\ eV$)	2.0×10^4	$195k\,\Omega$
Tubulin sheet ($t_1 = t_r = 0.4\ eV$)	6.3×10^4	$62k\,\Omega$
MT (A lattice, $t_l = t_r = 0.04\ eV$)	1.8×10^4	$219k\,\Omega$
MT (A lattice, $t_l = t_r = 0.1\ eV$)	4.5×10^4	$87k\,\Omega$
MT (A lattice, $t_l = t_r = 0.4\ eV$)	1.6×10^5	$25k\,\Omega$
MT (B lattice, $t_l = t_r = 0.04\ eV$)	1.9×10^4	$211k\,\Omega$
MT (B lattice, $t_l = t_r = 0.1\ eV$)	4.7×10^4	$84k\,\Omega$
MT (B lattice, $t_l = t_r = 0.4\ eV$)	1.6×10^5	$24k\,\Omega$

way the conduction properties depend on the lattice and the particular boundary conditions. The situation can be compared to carbon nanotubes which have a similar size and structure to MTs. In addition, theoretical consideration of nanotube structure predicts that the conduction properties depend on the boundary conditions. In the case of nanotubes, this means the way in which the graphene sheet is wrapped up to form the nanotube. It is quite interesting that particular sets of boundary conditions produce a semi-conducting nanotube while the appropriate choice of wrapping the nanotube gives rise to metallic conduction [31].

The ability to change from an insulator to a conductor by a simple geometrical change could be biologically relevant even without observing a single electron to be conducted along the MT since it is the conductivity property that affects the way the cell views and responds to external electromagnetic fields. Specifically, the reflectivity of MTs to electromagnetic fields is high when they are conducting, and, consequently, they could act to direct infrared signals to the interior of the centrosome. Since the centrioles are maintained at right angles to each other, the cell would be able to determine the location in latitude and longitude of a light source if the MTs are conducting [4,5]. Peaks in the AC conductivity are all at energies significantly above the thermal activation threshold so no conductivity is expected along a darkened MT, but photoconduction in the infrared-visible range is possible. Therefore, it is possible this photo-activated conduction has functional repercussions. In addition to the intrinsic semiconducting behavior of MTs, and possibly AFs, these protein filaments possess very intensely active cable properties when immersed in an ionic buffer solution. In the next section, we discuss the ionic conductivity effects of AFs.

18.6 Actin Filaments Support Nonlinear Ionic Waves

Several experiments [12,83] indicate the possibility of ionic wave generation along AFs. As the condensed cloud of counter-ions separates the filament core from the rest of the ions in the bulk solution, we expect it to act as a dielectric medium between the two. It has both resistive and capacitive components associated with each monomer that makes up the AF. Ion flow is expected to occur at a radial distance from the surface of the filament approximately equal to the Bjerrum length. An inductive component is proposed to emerge due to the actin's double-stranded helical structure that induces the ionic flow in a solenoidal manner. Due to the presence of the sheath of counter-ions around the AF, these polymers act as biological "electrical wires" [83], and have been modeled as nonlinear inhomogeneous transmission lines propagating nonlinear dispersive solitary waves. Recently, Lader *et al.* [78] applied an input voltage pulse with an amplitude of approximately 200 mV and a duration of 800 μs to an AF, and measured electrical signals at the opposite end of the AF, indicating that AFs support ionic waves in the form of axial nonlinear currents. In an earlier experiment [83], the wave patterns observed in electrically stimulated single AFs were remarkably similar to recorded solitary waveforms for electrically stimulated nonlinear transmission lines [85]. Considering the AFs highly nonlinear complex physical structure and the thermal fluctuations of the counter-ionic

cloud [105], the observation of soliton-like ionic waves is consistent with the idea of AFs functioning as biological transmission lines.

The electro-conductive medium is a condensed cloud of ions surrounding the polymer and separated from it due to the thermal fluctuations in the solution. The distance beyond which thermal fluctuations are stronger than the electrostatic attractions or repulsions between charges in solution is defined as the so-called Bjerrum length, λ_B. With the dielectric constant of the medium denoted by ε, the Bjerrum length is given by

$$\frac{e^2}{4\pi\varepsilon\varepsilon_0\lambda_B} = k_B T \tag{18.41}$$

for a given temperature T in Kelvin. Here e is the electronic charge, ε the dielectric constant, ε_0 the permittivity of the vacuum, and k_B is Boltzmann's constant. For a temperature of 293 K, we find that $k_B = 7.13 \times 10^{-10} m$. Counter-ion condensation occurs when the mean distance between charges, b, is such that $\lambda_B B/b = S > 1$. Each actin monomer carries an excess of 14 negative charges in vacuum, and accounting for events such as protonation of histidines, and assuming there to be 3 histidines per actin monomer, there exist 11 fundamental charges per actin subunit [143]. Assuming an average of 370 monomers per μm, we find there is approximately $4e/nm$. Thus, we expect a linear charge spacing of $b = 2.5 \times 10^{-10} m$ so $S = 2.85$. As the effective charge, q_{eff}, or renormalized rod charge is the bare value divided by S, we find $q_{eff} = 3.93e/monomer$. Consequently, it can be shown that approximately 99% of the counter-ion population is predominantly constrained within a radius of $8nm$ [110] around the polymer's radial axis [158]. Significant ionic movements within this "tightly bound" ionic cloud are therefore allowed along the length of the actin, provided that it is shielded from the bulk solution [104,107].

The physical significance of each of the components of the electrical network, and additional details, are described in [148]. For actin in solution, a key feature is that the positively charged end assembles more quickly than the negatively charged end [135]. This results in an asymmetry in the charges at the ends of the filaments and F-actin's electric polarization. Actin monomers arrange themselves head to head to form actin dimers resulting in an alternating distribution of electric dipole moments along the length of the filament [73]. We assume, therefore, that there is a helical distribution of ions winding around the filament at approximately one Bjerrum length. This corresponds to a solenoid in which a fluctuating current flows as a result of voltage gradient between the two ends.

The capacitive element of the electric circuit is obtained following the observation of oppositely charged layers surrounding the filament surface. We envisage the protein surface's negative charge to be distributed homogeneously on a cylinder defining the filament surface. Furthermore, positive counter-ionic charges in the bulk are expected to form another cylinder at a radius greater than the AF itself, approximately one Bjerrum length, λ_B, away from the actin surface, which includes the condensed ions. The permittivity, ε_0, is given by $\varepsilon = \varepsilon_0\varepsilon_r$ where ε_r is the relative permittivity which we take to be that of water, i.e., $\varepsilon_r = 80$. We take the length of an actin monomer typically as $a = 5.4\ nm$ and the radius of the actin filament, r_{actin}, to be $r_{actin} = 2.5\ nm$ [14]. The next step is to consider a cylindrical Gaussian surface of length a whose radius is r such that $r_{actin} < r < r_{actin} + \lambda_B$.

Applying of Gauss's law for the total charge enclosed in the cylinder, we have the following expression for the capacitance

$$C_0 = \frac{2\pi\varepsilon a}{ln(\frac{r_{actin}+\lambda_B}{r_{actin}})} \tag{18.42}$$

With the parameters given previously, we estimate the capacitance per monomer is $C_0 = 96 \times 10^{-6}\ pF$. The resistive part is obtained from Ohm's law. Taking into account the potential difference and the current I, the magnitude of the resistance, $R = V/I$, for an actin filament is given by

$$R = \frac{\rho ln(\frac{r_{actin}+\lambda_B}{r_{actin}})}{2\pi l} \tag{18.43}$$

where ρ is the resistivity. Typically, for K^+ and Na^+, intracellular ionic concentrations are 0.15 M and 0.02 M, respectively [147]. Kohlrausch's law states that the molar conductance of a salt solution is the sum of the conductivities of the ions comprising the salt solution. Thus

$$\sigma = \Lambda_0^{K^+} c_{K^+} + \Lambda_0^{Na^+} c_{Na^+} = 1.21(\Omega m)^{-1} \tag{18.44}$$

Using this with $\rho = \sigma^{-1}$, the resistance estimate becomes $R = 6.11\ M\Omega$, which is much lower than pure water since $R_{water} = 1.8x10^6\ M\Omega$.

To describe the properties of the whole filament, we simply connect n subcircuits, as described earlier, to obtain an effective resistance, inductance, and capacitance, respectively, such that:

$$R_{eff} = \left(\sum_{i=1}^{n} \frac{1}{R_{2,i}} \right)^{-1} + \sum_{i=1}^{n} R_{1,i} \tag{18.45}$$

$$L_{eff} = \sum_{i=1}^{n} L_i \tag{18.46}$$

and

$$C_{eff} = \sum_{i=1}^{n} C_{0,i} \tag{18.47}$$

where $R_{1,i} = 6.11 \times 10^6\ \Omega$, $R_{2,i} = 0.9x10^6\ \Omega$ such that $R_{1,i} = 7R_{2,i}$. Note that we have used $R_{1,i} = R_1$, $R_{2,i} = R_2$, $L_i = L$, and $C_{0,i} = C_0$. For a 1 μm of the AF, we find

$$R_{eff} = 1.2 \times 10^9\ \Omega \tag{18.48}$$

$$L_{eff} = 340 \times 10^{-12}\ H \tag{18.49}$$

$$C_{eff} = 0.02 \times 10^{-12}\ F \tag{18.50}$$

This solenoidal flow geometry leads to an equivalent electrical element possessing self-inductance. From Faraday's law, we can derive an effective inductance for the actin filament in solution by

$$L = \frac{\mu N^2 A}{l} \tag{18.51}$$

where l is the length of the F-actin and A is the cross-sectional area of the effective coil given by

$$A = \pi (r_{actin} + \lambda_B)^2 \tag{18.52}$$

The number of turns is approximated by simply working out how many ions could be lined up along the length of a monomer. We would then be approximating the helical turns as circular rings lined up along the axis of the F-actin. We also take the hydration shell of the ions into account in our calculation. The hydration shell is then the group of water molecules oriented around an ion. It can be shown that $L = 1.7pH$ for the length of the monomer.

The electrical model of the AF is an application of Kirchhoff's laws to one section of the effective electrical circuit that is coupled to neighboring monomers. Taking the continuum limit [148] for a large number of monomers along an AF, we derive the following equation which describes the spatio-temporal

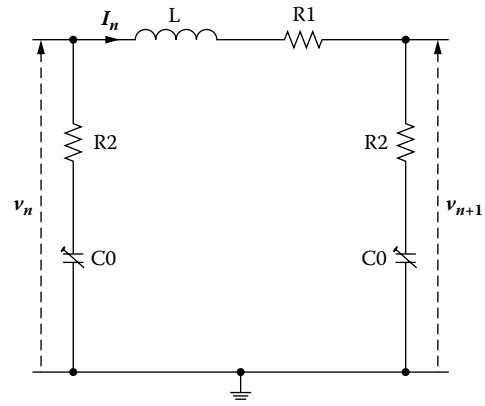

FIGURE 18.17 An equivalent electrical circuit for a segment of an actin filament.

behavior of the potential along the AF:

$$LC_0\frac{\partial^2 V}{\partial t^2} = a^2(\partial_{xx}V) + R_2C_0\frac{\partial[a^2(\partial_{xx}V)]}{\partial t} - R_1C_0\frac{\partial V}{\partial t} + R_1C_0 2bV\frac{\partial V}{\partial t} \tag{18.53}$$

Motivated by this picture, physical properties of the ionic distribution along a short stretch of the polymer (the average pitch 35 to 40 nm) have been modeled by Tuszynski *et al.*, [148] as an electrical circuit with non linear components (see Figure 18.17). The main elements of the circuit are: (a) a non linear capacitor associated with the spatial charge distribution between the ions located in the outer and inner regions of the polymer; (b) an inductance due to the helical geometry of the filament; and (c) a resistor due to the viscosity imposed by the solution.

From Kirchhoff's laws, one can derive an equation governing the propagation of voltage along the filament (see [148]). One of the key aspects in this model is the nonlinearity of the associated capacitance [86,150] that eventually gives rise to the self-focusing of the ionic waves. The equations developed for the model originate from the application of Kirchhoff's laws to the RLC resonant circuit of a model actin monomer in a filament. Perhaps the most important finding is the existence of the traveling wave, which describes a moving transition region between a high and low ionic concentration due to the corresponding inter-monomeric voltage gradient. The velocity of propagation was estimated to range between 1 and 100 ms^{-1} depending on the characteristic properties of the electrical circuit model. It is noteworthy that these values overlap with action potential velocities in excitable tissues [55].

Analysis of the model described earlier reveals the possibility of stationary waves in time that may lead to the establishment of spatial periodic patterns of ionic concentration. As mentioned previously, Lader *et al.* [78] applied an input voltage pulse with an amplitude of approximately 200 mV and a duration of 800 μs to an AF, and measured electrical signals at the opposite end of the AF, indicating that AFs support ionic waves in the form of axial nonlinear currents. In an earlier experiment [83], the wave patterns observed in electrically stimulated single AFs were similar to recorded solitary waveforms for electrically stimulated nonlinear transmission lines [85]. Considering the AFs highly nonlinear physical structure and thermal fluctuations of the counter-ionic cloud [105], the observation of soliton-like ionic waves is consistent with the idea of AFs functioning as biological transmission lines.

This section only provides an indication as to a realistic model of actin that can support soliton-like ionic traveling waves. Modeling relies on data constrained by experimental conditions, and/or assumptions made, including the charge density, which is calculated based on the net surface charges of actin. It should also be considered that soliton velocity is directly proportional to the magnitude of the stimulus, which in a biological setting has not been formally described. Actin interacts with a number of ion channels of different ionic permeability and conductance. Thus, it is expected that channel opening, single channel

currents, and other channel properties, including the resting potential of the cell, may significantly modify the amplitude and velocity of the soliton-supported waves. This should correlate with the velocity of the traveling waves along channel-coupled filaments. Other parameters that may play a role in this type of electrodynamic interaction are the local ionic gradients and the regulatory role of actin binding proteins, which can help "focus" the conductive medium, or otherwise impair wave velocity. The physical significance of each of the components of the electrical network (see Figure 18.17) and additional details, are described in [148]. MTs differ from AFs in several respects. One of the key differences is the presence of peptide chains called C-termini that protrude from their surface. In the following section, we outline their role in ionic wave conductivity along MTs.

18.7 Long-Range Spatio-Temporal Ionic Waves Along Microtubules

MTs are long hollow cylinders made of $\alpha\beta$-tubulin dimers [33], as described in Section 18.5. Recently, it has become apparent that neurons utilize MTs in cognitive processing. Both kinesin and MAP2 that associate with MTs have been implicated in learning and memory [69,154]. Dendritic MTs are implicated in particular, and it is highly probable that the precisely coordinated transport of critical proteins and mRNAs to the post-synaptic density via kinesin along MT tracks in dendrites is necessary for learning, as well as for LTP [70].

The following molecular dynamics (MD) simulation results focus on the C-termini of neighboring tubulins, whose biophysical properties have a significant influence on the transport of material to activated synapses. This affects cytoskeletal signal transduction and processing as well as synaptic functioning related to LTP. Using MD modeling we calculated conformation states of the C-termini protruding from the outer surfaces of MTs and strongly interacting with other proteins, such as MAP2 and kinesin [124]. To elucidate the biophysical properties of C-termini and gain insight into the role they play in the functioning of dendrites, we developed a quantitative computational model based on the currently available biophysical and biochemical data regarding the key macromolecular structures involved, including tubulin, their C-termini, and associated MAP2. In the proposed model of the C-termini microtubular network, the tubulin dimer is considered to be the basic unit. Each dimer is decorated with two C-termini that may either extend outwardly from the surface of the protofilament or bind to it in one of a few possible configurations.

The most dynamic structural elements of the system (i.e., its elastic and electric degrees of freedom) are envisaged as conformational states of the C-termini. Each state of the unbound C-terminus evolves so as to minimize the overall interaction energy of the system. The negatively charged C-termini interact with (a) the dimer's surface, (b) neighboring C-termini and (c) adjacent MAPs. While the surface of the dimer is highly negatively charged overall, it has positive charge regions that attract the C-termini causing them to bend and bind in a "downward" state. The energy difference between the two major metastable states is relatively small, on the order of a few $k_B T$ [112].

To simplify our calculations, we first used a bead-spring model representing the C-terminus as a sequence of beads with flexible connections. We have accounted for the electric field exerted by the dimer, the external field generated by the environment, and various short-range interactions within the simulated C-terminus, including interactions between the beads (i.e., Lennard–Jones potential, angular forces, etc.).

A simplified model of the interaction between MAP2 and its ionic environment via counter-ions was used to investigate the ability of MAP2 to function as a "wave-guide" that transfers the conformational change in a C-terminus state to an adjacent MT (see Figure 18.18). A perturbation applied to the counter-ions at one end of the MAP2 drives them out of equilibrium and initiates a wave that travels along the MAP2 (for details see [112]). Figure 18.18 (right) depicts the main result for a localized perturbation applied for a few picoseconds to the counter-ions near the binding site of MAP2. Wave propagation along the chain of $N = 50$ counter-ions is shown as a counter-ion displacement parallel to MAP2, u_i, where i denotes the i^{th} counter-ion. This perturbation along MAP2 propagates almost as a "kink" whose phase velocity has been found to be $v_{ph} \approx 2nm \cdot ps^{-1}$.

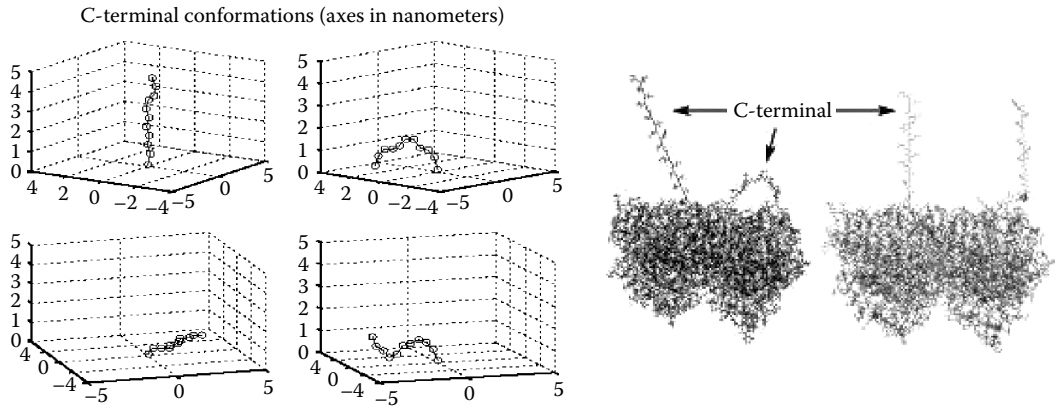

FIGURE 18.18 Examples of the conformational states of the C-termini in a tubulin dimer (right) and results from bead-spring model simulations (left).

Our simulations indicate the ability of an ionic wave to trigger a coupled wave of C-termini state changes from their upright to downward orientations. Four examples of "up" and "down" states of the C-termini obtained through bead-spring model simulations are shown in Figure 18.18 (left) and results from molecular dynamics simulations for an actual tubulin dimer are shown in Figure 18.18 (right). Calculations of the energy minimized positions of the individual beads representing the amino acids of the C-terminus in two equivalent forms reveal that the probability of the down position, which includes all cases of full or partial attachment, is 15%. This means that the system has two major states with a strong bias towards the stretched-up state.

In the remainder of this section, we present an MD simulation study of the interaction between MTs and MAP2, followed by preliminary experimental results of ionic wave conduction by MTs.

A simplified model of the interaction between MAP2 and its ionic environment via counter-ions was used to investigate the ability of MAP2 to function as a "biological wire" that transfers the conformational change in a C-terminus state to an adjacent MT (see Figure 18.19).

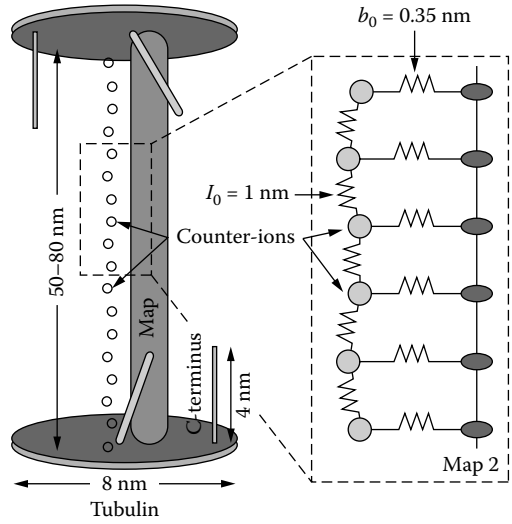

FIGURE 18.19 An illustration of a MT C-terminus conformational change transfer process. Ionic waves travel along an adjacent MAP via local perturbations to the bounded counter-ions.

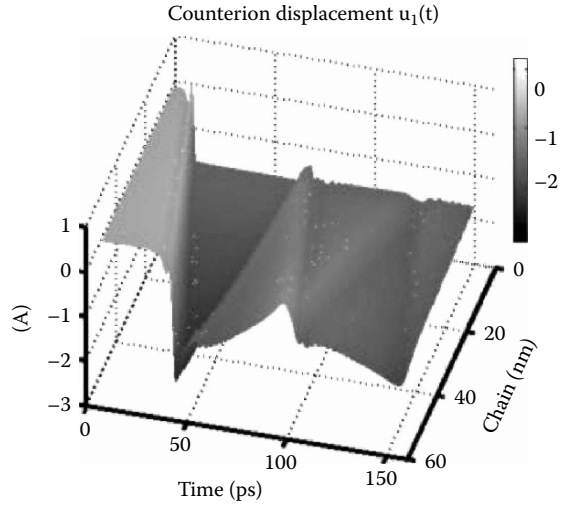

FIGURE 18.20 Propagation of a localized perturbation applied to counter-ions near the binding site of a MAP2.

A perturbation applied to the counter-ions at one end of the MAP2 drives them out of equilibrium and initiates a wave that travels along the MAP2 [112]. Figure 18.20 depicts the main result for a localized perturbation applied for a few picoseconds to the counter-ions near the binding site of a MAP2. Wave propagation along the chain of counter-ions is demonstrated as counter-ions displacement parallel to MAP2.

To validate the proposed model, we have conducted experiments on electro-stimulation of MTs in a solution. A detailed description of the experiment and its results has been published elsewhere [113]. Here we only describe the main aspects of the work on this phenomenon. Figure 18.21 depicts the experimental setup used. Isolated taxol-stabilized MTs have been shown to be able to amplify an electric signal applied to them through a micropipette. The input signal was of 5 to 10 msec duration, with amplitude in the range of $\pm$200 mV. The signal that arrived at the other end of the MT was more than twice as high as the signal recorded in a control experiment where the same two pipettes were immersed in a solution with no MT making contact to them. The calculated conductivity of MTs was found to be on the order of 10 nS, indicating a high level of ionic conductivity along an MT. By comparison, for a typical ion channel, the corresponding value ranges between 5 and 200 pS.

The results of the MD modeling previously described raise the possibility of transmitting electrostatic perturbations collectively among neighboring C-termini, and from C-termini on one MT to those C-termini on another MT via MAP2. The importance of collective conformational states of C-termini and of transmitting perturbations among the C-termini as a novel information processing mechanism operating at a subneuronal level is clear. MAP2 and kinesin bind near to the C-termini on tubulin and the electrostatic properties of C-termini affect this binding [2,145]. Hence the conformational states of the C-termini must

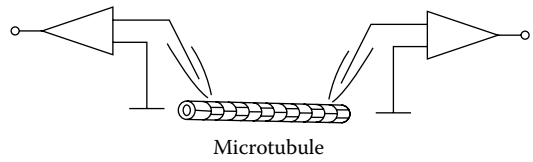

FIGURE 18.21 Schematic representation of the experimental set-up used for the measurements of a MTs ionic conductivity. The MT is attached to two micropipettes that are connected to signal amplifiers.

at some point be taken into account in order to understand neural processing that depends on transport of synaptic proteins inside of neurons. Kim and Lisman [71] have shown that inhibition of MT motor proteins reduces an AMPA receptor-mediated response in hippocampal slice. This means that a labile pool of AMPA-receptors depends on MT dynamics, and MT-bound motors determine the amplitudes of excitatory postsynaptic currents (EPSC). A walking kinesin carries with it a protein or an mRNA molecule. Since kinesin binds to a MT on a C-terminus, as it steps on it, it brings the C-terminus to the MT surface and makes it ineffective in binding for the next kinesin over a period of time that it takes the C-terminus to unbind and protrude outside. From this we can deduce that long stretches of C-termini in the upright position are going to be most efficient at transporting kinesin and kinesin cargo while C-termini that lie flat are expected to be most efficient at detaching kinesin. Thus, in considering the trafficking of many kinesins, collective electrostatic effects of C-termini become crucially important. Collective states that correspond to transport strategies that will send optimal numbers of kinesin molecules to synaptic zones are likely to occur when synapses are activated (e.g., when synapses are generating EPSCs). One type of kinesin cargo associated with learning and memory is the NMDA receptor, which is well known to be associated with LTP. The PSD might be expected to require replenishment of NMDA receptors at a later time than AMPA receptors are replenished (see [71]). Kinesin protein (specifically KIF17) actively transports NMDA receptor 2B subunits (NR2B) to the region of the PSD [52]. Once NR2B is in the vicinity of an active synapse it dissociates from kinesin and then becomes associated with the PSD, presumably using actin transport as an intermediary step. Hence, kinesin-mediated transport of NMDA receptors along MTs has a built-in negative feedback mechanism: whenever too many NMDA receptors are transported to the synaptic site (often located on a spine head), then those NMDA receptors can initiate proteolytic breakdown of MAP2. The latter event would be expected to reduce further transport of NMDA receptors to the synaptic site. The signal transduction molecule CaMKII is also critical for learning and LTP. Similar to the NMDA receptor, CaMKII is transported to active synapses via a kinesin-mediated transport mechanism. However, it appears to be the mRNA for CaMKII to a larger extent than the protein that is transported to spines, since dendrites are enriched with polyribosomes and CaMKII mRNA [140]. Ribosomes are redistributed from dendrites to spines with LTP [106]. Local translation of CaMKIIα in dendrites appears to be necessary for the late phase of LTP: fear conditioning and spatial memory [98]. Moreover, changes in synaptic efficacy are often accompanied by changes in morphology; reorganization of underlying MTs is a fundamental factors for these morphological changes. Having discussed the individual properties and roles played by AFs and MTs in subcellular conduction processes, we now turn to an integrated view of their functioning within a cell. The next several sections discuss this issue in depth.

18.8 Dendritic Cytoskeleton Information Processing Model

Our current hypothesis states that the cytoskeletal biopolymers constitute the backbone for ionic wave propagation that interacts with, and regulates, dendritic membrane components, such as ion channels to effectively control synaptic connections. Figure 18.22 depicts a portion of the dendritic shaft where MTs are decorated by C-termini and interconnected by MAP2 (thick line). Connections between MTs and AFs are shown as well as two types of synaptic bindings. On the upper left side, actin bundles bind to the post-synaptic density (PSD) of a spineless synapse. On the lower right side, a spiny synapse is shown, where actin bundles enter the spine neck and bind to the PSD, which at the other end, is connected to the MTN.

We envision a mechanism in which a direct regulation of ion channels and thus synaptic strength by AFs and associated cytoskeletal structures controls and modifies the electrical response of the neuron. In this picture, MTs arranged in networks of mixed polarity receive signals in the form of electric perturbations, from synapses via AFs connected to MTs by MAP2 [122], or via direct MT connections to postsynaptic density proteins by molecules such as CRIPT [108]. As discussed next, the MTN may be viewed as a high-dimensional dynamic system where the main degrees of freedom are related to the conformational state of the C-termini. The input signals perturb the current state of the system that continues to evolve. Hypothetical integration of the preceding ideas is outlined as follows. Electrical signals arrive at the PSD

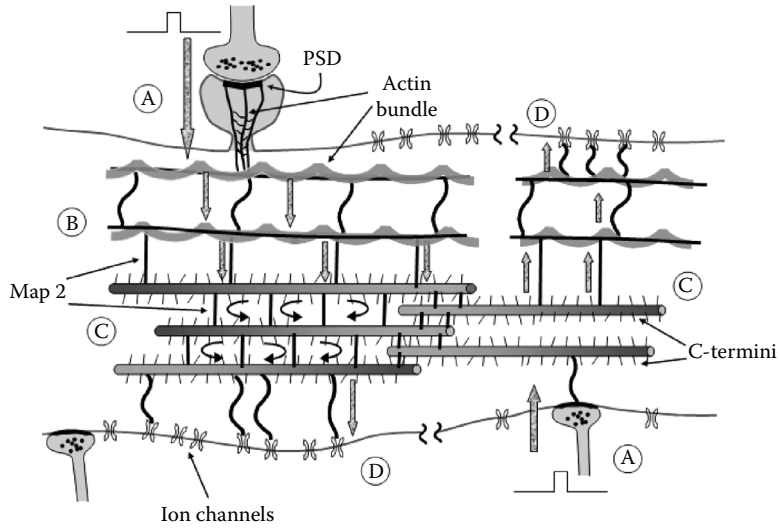

FIGURE 18.22 A scheme of the dendritic shaft with MTs arranged in networks of mixed polarity that receive signals in the form of electric perturbations from synapses via AFs connected to MTs by MAP2, or via direct MT connections to postsynaptic density proteins by molecules such as CRIPT. A spiny synapse is depicted in the bottom right where an actin bundle enters the spine neck.

via synaptic transmission, which in turn elicits ion waves along the associated AFs at the synaptic spine. These dendritic input signals propagate in the form of ionic waves through AFs to the MTN where they serve as input signals. The MTN, operating as a large high-dimensional state machine, evolves these input states, e.g., by dynamically changing C-termini conformation. The output from the MTN is the state of the system that may propagate via AFs to remote ion-channels. These output functions are assumed to regulate the temporal gating state of voltage-sensitive channels. This process subsequently regulates the membrane conductive properties and controls the axon hillock behavior by changing the rate, distribution, and topology of open/close channels. The overall functions of the dendrite and neuron can thus be regulated in this manner.

The attractiveness of the concept that the cytosol, with its cytoskeletal structures, may behave as a large dynamical system is clear since it provides a means for real-time computation without the need for stable attractor states. Moreover, the output is relatively insensitive to small variations in either the MTN (cytoskeletal networks) or the input patterns. It should be noted that the temporal system's state evolves continuously, even without external inputs. Recent perturbations, however, have a long-term effect on the MTN trajectories, i.e., there is a memory effect inherent to this system (not to be mistaken with synaptic LTP, which has a much longer time scale). The output from the MTN may converge at or near ion channels to regulate their temporal behavior. The issue of adaptation requires a feedback mechanism that will, at least locally, enable the change of the output function. In the context of neuronal function, with a focus on processivity, synaptic strengthening, LTP, and memory enhancement, the output function may simply reflect an effect of the MTN on synaptic channel function, such that the desired state of the channel appears to have a higher probability of being open/close upon the presentation of the associated input pattern. One possibility is a Hebbian-based response where more frequent activity of certain subdomains of the MTN output states gives rise to a higher/lower density of AFs connecting to corresponding channels.

We now explain how this integrative view may serve as a regulatory (adaptive) mechanism. The input level denoted by A in Figure 18.22 is associated with external electric perturbations passing through membranes, mainly synaptic inputs arriving from other neurons. These signals arrive at the cytoskeleton and directly affect MTNs, and/or actin filaments (bundles), in spiny synapses. However, the actin cytoskeleton is responsible for the propagation of signals to the MTN (see level B). These propagated signals (level C)

are in turn used as inputs to the MTN, viewed as a dynamic system. We further propose that the MTN generates diverse phase space trajectories in response to different input vectors. The requirements from such a system are not too restrictive since the output-state is not an attractor of the system. In other words, information processing at this level is not necessarily based on attractor dynamics but rather on real-time computations. This proposition relies on the observed ability of the MTN to propagate signals and on the specific topological features of MTNs in dendrites, in which shorter MTs of mixed polarity are interconnected by MAP2s. The output from the MTN would be a function of the evolved state vector in certain areas accessed by actin filaments and/or directly linked to ion channels (see both possibilities at level D). These output signals may modify the temporal channel activity, either by directly arriving from the MTN to the channels, or mediated by actin filaments. Hence, the channel-based synaptic membrane conductance is regulated, in particular in the axon hillock region, which is, in most cases, responsible for the generation of action potentials.

The idea that a nonspecific high-dimensional dynamical system may serve as a reservoir of trajectories in the context of liquid state machines (LSMs), has recently been suggested as an explanation for the existence of microcircuits in the brain [87]. The basic structure of an LSM is an excitable medium (hence "liquid") and an output function that maps the current liquid state. The liquid must be sufficiently complex and dynamic to guarantee a universal computational power. This is sufficient to ensure that different input vectors lead to separate trajectories. A network of spiking neurons and a recurrent neural network has been used as a "liquid", whereas the output (readout) function has been implemented by simple perceptrons, threshold functions, or even linear regression functions. Clearly, simpler readout functions restrict the ability of the whole system to capture complex nonlinear dependencies.

18.9 The Inter-Relation Between the Neural Cytoskeleton and the Membrane

Usually, F-actin and microtubular cytoskeletal networks are thought to fulfill separate, independent cellular roles. Highly dynamic actin networks are known for their role in cell motility, in particular the spreading of the leading edge and contraction. The more stable MT cytoskeleton is best known for its importance in cell division and organelle trafficking. However, recent studies provide a more unified role, ascribing important roles to the actin cytoskeleton in cell division and trafficking and important roles for MTs in the generation and plasticity of cellular morphology. Coordination between the actin- and MT- based cytoskeletons has been recently observed during cellular migration and morphogenesis, processes that share some similarities with neurite initiation [28]. A direct physical association between both cytoskeletons has been suggested, because MTs often preferentially grow along actin bundles and transiently target actin-rich adhesion complexes. In neurons, certain plakins and neuron-specific MAPs, like MAP1B and MAP2, may play a role in linking MTs and AFs, helping in the transition from an undifferentiated state to neurite-bearing morphology.

MAP1B and MAP2 are both known to interact with actin in vitro [20,109,128,134,146]. It is likely that by crosslinking, MAP2 and/or MAP1B associated with both cytoskeletons could be involved in guidance of MTs along AF bundles. Alternatively, MAPs could shuttle from MTs to actin and could alter F-actin behavior by actively crosslinking AFs.

Little is known, however, about interactions between neuronal ion channels and the cytoskeleton. Recently, whole-cell and single channel recordings showed that acute disruption of endogenous AFs with cytochalasin D activated voltage-gated K^+ currents in these cells, which was largely prevented by intracellular perfusion with the AF-stabilizer agent, phalloidin. Interestingly, the direct addition of actin to excised, inside-out patches activated, and/or increased single K^+ channels. Thus, acute changes in actin-based cytoskeleton dynamics regulate voltage-gated ion channel activity in bipolar neurons. This may be indicative of a more general and quite appealing mechanism by which cytoskeletal structures control feedback mechanisms in neuronal channels.

Recent theoretical and experimental studies of the electrical properties of AFs and MTs in solution revealed their capability to act as biopolymer wires [83,148]. This means that these protein polymer

filaments are capable of conducting nonlinear ionic waves and even amplifying the signal with respect to the conducting solution [112]. We conjecture a mechanism in which a direct regulation of ion channels and thus synaptic strength by AFs and associated cytoskeletal structures controls and modifies the electrical response of the neuron. According to this scenario (see Figure 18.22), MTs arranged in networks of mixed polarity receive signals in the form of electric perturbations from synapses via AFs connected to MTs by MAP2 [122], or via direct MT connections to postsynaptic density proteins by molecules such as CRIPT [108]. These signals propagate in the form of ionic waves. Specific physical properties of the propagating ionic waves will be discussed in sections that follow.

18.10 Relationship to Cognitive Functions

Transport of proteins and receptors in neurons is likely to have an electromagnetic basis to the extent that this function is possibly a result of MT computation. The perspective advanced in this chapter is that a specific fingerprint defined by a particular electromagnetic state of a microtubular array potentially corresponds to a unique unit of cognition (e.g., a basic visual parameter). Recently, it has been shown that visual components can be represented musically [23]; hence, the idea that one common type of energy underlies divergent perceptual and cognitive processes appears likely. Activation of one electromagnetic fingerprint could, in turn, activate another electromagnetic fingerprint, irrespective of sensory input. Moreover, the subjective feels of this widespread pattern of electromagnetic energy can be specified according to those key physical properties of MTs that influence the transport of proteins to synapses. Factors influencing kinesin-mediated transport include the protein conformation of tubulin and the nature of the C-termini (see [112,136]).

Not only is the conformational state of tubulin critical to effective transport, motor proteins appear to alter albeit temporarily, the conformation of tubulin. Kinesin binding and that of MAP, tau, significantly alter the direction of the protruding protofilament ridges along MTs, which in turn influences their further binding abilities [127]. More than mere local adaptation to binding, MTs may alter their conformation ahead of kinesin processivity [76], supporting the notion of long-range cooperative effects between tubulin dimers located along longitudinal protofilaments of MTs. These biochemical relationships have consequences for electromagnetic fields. The dipole moment of tubulin depends on its configuration in the MT [53,97]. Thus, electromagnetic fields among MTs could, in theory, be induced or inhibited by synaptic inputs that affect the protein conformation of tubulin directly or through alterations in kinesin or MAP binding. Synaptic effects upon MTs could be mediated through ionic currents, by propagation via actin filaments [148], or by signal transduction cascades resulting in the phosphorylation of MAPs [126].

Due to lengthwise electric dipoles of tubulin dimers, information in the form of traveling waves propagated along microtubular tracks can, in principle, be transmitted between synapses with high fidelity (see [149]). MAP2 bridges keep microtubular arrays within the dendritic core parallel and antiparallel by aligning portions of polarized MTs. The antiparallel alignment of MTs, which specifically occurs in dendrites, would severely attenuate any electromagnetic field generated by MTs. However, during enhanced kinesin-mediated transport, as is likely to occur with heightened synaptic activity, MAP2 bound to the MT would be perturbed and may even temporarily detach from the MT. A similar phenomenon might also occur due to dynein-mediated transport, which occurs largely in the opposite direction to that of kinesin-mediated transport. Assuming that at least some MAP2 stays attached to the antiparallel MTs, keeping the dendritic array intact, any net unidirectional transport along the MT array should increase the strength of the electromagnetic field associated with the fingerprint and should further result in the spread of that electromagnetic field to adjacent MTs. Once a sufficient number of MTs were engaged in dynamically sending and receiving complementary electromagnetic energies, whole neuronal compartments (e.g., dendrites) might be expected to interact. Due to the parallel/antiparallel arrangement of MTs in cortical dendrites and the ability of electromagnetic fields to pass from one dendrite to adjacent dendrites, information could, in principle, pass between neurons when such electromagnetic fields were sufficiently amplified as a result of changes in the binding of MAPs or kinesin.

While it is true that there are some 100 billion neurons and some 100 trillion synapses in the human cerebral cortex, so too does each sensory field afford the luxury of extremely high resolution. The fovea of the retina, for example, contains roughly 1 million receptor cells that relay information to the cerebral cortex, diverging to eventually drive the synaptic activity of at least 1 billion cortical cells. Although the mammalian cortex has many neurons, and even more synapses, its connectivity can be readily grasped by one recurrent theme: repetitive high-resolution topographic representation. Despite this simple organizational plan, cells in cortical areas, in particular cells in higher sensory or association areas, inexplicably show correlated responses during cognitive tasks. Experience alters the basic structure of dendrites, and as a consequence, it should also alter any electromagnetic fields generated by MTs as a result of transport proteins moving along them. As described earlier, a rearrangement of the cytoskeleton occurs during early development, with learning, and with neurodegeneration underlying dementia.

Although models of changes in synaptic efficacy, e.g., long-term potentiation or depression (LTP or LTD), offer great potential as memory mechanisms (see [93]), one often overlooked problem is that large numbers of synapses are affected in concert. If only a few synapses are changed with each memory, then the entire cortical system would have a near unlimited capacity, but if many synapses participate in each memory, as seems to be the case for LTP, then there could be a serious saturation effect. This relates back to the topographic organization of the cerebral cortex. Strictly synaptic models (electrochemical) suggest that complex neural networks increase synaptic weights (i.e., neurons that fire together wire together). Nonetheless, visual cortical regions, for example, do not have massively random interconnections among all parts of the visuotopic map, thereby making many of the changes in synaptic efficacy necessary for encoding complex perceptual features impossible.

The biophysical properties of MTs are just beginning to be understood at molecular and atomic levels, and recent empirical evidence suggests interesting effects occur between MTs under certain experimentally induced conditions. Two groups, one led by Watt Webb at Cornell and another led by Paul Campagnola and William Mohler at the University of Connecticut, observed that MTs give rise to intense second-harmonic generation: a frequency doubling upon exposure to a sapphire laser in the 880-nm range. (Other frequencies were partially effective.) MTs were one of the few biological materials having electric dipoles that constructively interfered with the dipoles of neighboring MTs [11,29]. This occurred for parallel MTs in axons, but not for antiparallel MTs in dendrites. Could such a phenomenon be expected to occur with natural learning or upon exposure to oscillatory input, an LTP-inducing tetanus or various pharmacological agents? To the extent that these induce electromagnetic energy, it is conceivable. In addition to being sensitive to electromagnetic radiation, MTs may themselves produce this kind of energy. Second-harmonic generation by MTs is consistent with their ferroelectric properties. In second-harmonic generation, the most strongly enhanced wavelength is 880-nm, supporting the frequently overlooked reports of electromagnetic signaling by cells. Guenther Albrecht-Buehler [5] observed electromagnetic energy in the near infrared region overlapping the 880 nm value that was generated by centrioles for the purpose of cell-to-cell communication, leading him to suggest that: "... one of the functions of MTs may be to play the role of cellular 'nerves'." Infrared light in this range of wavelengths also induces cell aggregation [6]. Since frequency (and hence wavelength) determines the long-range effect of dielectric polarization of a given MT, one would expect this effect to be length-dependent. The distance between MTs was shown to be critical in second-harmonic generation by MTs.

18.11 The Potential for Bioelectronic Applications and Neuromorphic Computing

Traditionally, biologists viewed the cell as a "fluid mosaic bag" containing enzymes, organelles, etc. Recent work has focused on finding the proteins that come together in this cellular "soup" and proceed to effect a change corresponding to a particular biological function. Through the molecular-level analysis of these protein-protein interactions, recent work in various labs has proceeded to build a model of "systems biology," in which these biochemical reactions that have been individually characterized can be brought together to form a cohesive whole, a synergistic dynamics network. From this knowledge, we hope to extract

useful information that will affect fields as diverse as nanotechnology, biotechnology, and materials science. Inside each cell, there is an elaborate network that serves a multitude of functions. It is an intracellular "highway" that provides a roadway for motor proteins to carry cargo and travel along in their journey to extend axonal length, for example. It controls synapses, and dendrite outgrowth. It is responsible for pulling the chromosomes apart during mitosis, for cell motility, etc.

As man-made engineered systems become more and more miniaturized, the push to build "nano-machinery" is driving physicists and engineers towards biomimetics, (i.e., the branch of engineering that mimics biological materials). This is highly feasible and much progress has been made in this area. However, due to the limitations inherent in trying to build mechanical devices on such a small scale, there is still much to learn. It is conceivable that through our better understanding of the cell's individual functional and structural components we will be able to advance medicine and technology beyond even our wildest dreams. One area that may undergo a major paradigm shift as a result of it is computer engineering.

As we enter into the 21st century, the dawn of the exciting new world of computational biotechnology is upon us and we are poised for a new generation of computers that do not have to be told what to do, but have the power to learn on their own. The development of a biological computer that is fast, small, and evolvable is no longer considered science fiction. Scientists are combining biological materials with the latest silicon-based technology in an effort to give us not only electronic devices that are smaller, but are also more flexible in terms of structure and function.

Ironically, our brain can serve as proof of concept. Many scientists now believe that individual neurons display wiring patterns and communication powers that resemble a human computer, and therefore may provide the framework for the design of a biological computer. The brain is composed of 10 billion neurons, each of which may be communicating with 1000 others. The neuron transmits electric signals along its arms, or 'axons'. Inside each axon a parallel architecture of microtubules is interconnected with other proteins, not unlike the parallel computer's wiring. In fact, the structure of microtubules may have evolved towards optimal computational efficiency. The piezoelectric properties of these protein filaments allow them to bend as a result of electric fields or currents which makes them ideal candidates as regulators of synaptic plasticity, a mechanism that could explain the popular principle of "use it or lose it." Thus, seldom used synaptic connections would be switched off in favor of more active ones. Biologist Guenter Albrecht-Buehler [3] demonstrated that cells perceive their environment through a tiny organelle called the centriole, composed of microtubules that were shown to interact with electromagnetic radiation. To further display that microtubules themselves can be conductive, the German biotechnology group, led by Eberhard Unger [41,42] succesfully conducted required experiments and is now working on building nanoelectronic components using these and other proteins. Simultaneously, ideas of combining biological and silicon-based materials in hybrid arrangements are gaining more and more support for future bioelectronic applications. Work is underway at various labs around the world to develop a biological chip. The main objective is to design nanoelectronic components using hybrid protein–silicon structures with arrays of proteins as biological oscillators that can be stimulated by electrodes or acoustic couplers. This will provide an analog circuit with rich dynamics.

DNA computers are unlikely to become stand alone competitors for electronic computers. But digital memory in the form of DNA and proteins is a real possibility with exquisitely efficient editing machines that navigate through the cell, cutting and pasting molecular data into the stuff of life. Beyond that, the innate intelligence built into DNA molecules could help fabricate tiny, complex structures: in essence, using computer logic not to crunch numbers but to build things, an idea conceived of by Caltech's Winfree and Rothemund [123]. A single test tube of DNA tiles could perform about 10 trillion additions per second (about a million times faster than an electronic computer). Interfacing DNA or protein-based hybrid structures with living cells will enable a host of medical and technological applications including non-invasive diagnostic and therapeutic medical applications and computational devices.

One major component of this framework is a protein-based subcellular structure called a microtubule, to which we have devoted a large portion of this chapter. Microtubules are composed of 13 protofilaments, arranged in a spiral pattern, and each protofilament is composed of subunits of tubulin. Each tubulin possesses one of two conformational states: either both monomers can align, or one monomer can shift

30° relative to the vertical axis. Based on the conformation an individual tubulin exists in at a given moment, the conformation of its nearest neighbors can be altered as well. This conformational change can be transmitted throughout the microtubule in a similar manner as patterns are transmitted through cellular automata programs. In addition, as we argued earlier in this chapter, tubulin can have several electronic states as well as dipolar states, some of which can be conceivably linked to the conformations of C-termini and the counter-ion clouds around them. The idea that we could input an electric, chemical, or mechanical signal and generate another one at the output end of the system brings computing or information processing to mind. We believe that ordered protein arrays can function as a programmable processor or even a computer either inside cells such as neurons or as nano-scale devices.

Nanotechnology could and should take advantage of biology in a completely novel manner, (i.e., instead of building materials that mimic biological components, we could use the components themselves). In the case of microtubules and tubulin, rather than purify it from traditional sources such as cow or pig brains, tubulin can be readily cloned into an expression vector, and purified from bacteria in mass quantities. Then, purified tubulin of a predetermined sequence (and thus with desirable biophysical and biochemical characteristics) can be routinely polymerized to form microtubules with specific structural and functional properties. Following that, the microtubules produced could be stabilized in a polymer gel capable of undergoing a phase transition at a particular value of the control parameter such as temperature, applied pressure, pH, or ionic concentration. This would then trigger a signal that would initiate computational processes carried out by individual microtubules or their assemblies interconnected by MAPs or even connected to actin filaments in architecures that could mimic subneuronal designs. This technology could also be applied to sensors or artifical intelligence systems in building an evolvable computer. The only limitations in this endeavor are those imposed by our imagination and our ability to dream up new applications.

18.12 Discussion

Considering the abundance of MTNs and AFs in axons and dendritic trees, the findings and theoretical models described earlier may have important consequences for our understanding of the signaling and ionic transport at an intracellular level. Extensive new information (see [63]) indicates that AFs are both directly [14] and indirectly linked to ion channels in both excitable and non-excitable tissues, providing a potentially relevant electrical coupling between these current generators (i.e., channels), and intracellular transmission lines (i.e., AFs and MTs). Furthermore, both filaments are crucially involved in cell motility and, in this context, they are known to be able to rearrange their spatial configuration. In nerve cells, AFs are mainly located in the synaptic bouton region, whereas MTs are located in both dendrites and the axon. Again, it would make sense for electrical signals supported by these filaments to help trigger neurotransmitter release through a voltage-modulated membrane deformation leading to exocytosis [132]. Actin is also prominent in post-synaptic dendritic spines, and its dynamics within dendritic spines has been implicated in the post-synaptic response to synaptic transmission. Kaech *et al.*, [65] have shown that general anesthetics inhibit this actin mediated response. Among the functional roles of actin in neurons, we mention in passing glutamate receptor channels, which are implicated in long-term potentiation. It is therefore reasonable to expect ionic wave propagation along AFs and MTs to lead to a broad range of physiological effects.

In summary, this chapter broadly described the physical conditions that enable cytoskeletal polymers such as AFs and MTs to act as electrical transmission lines for ion flows along their lengths and along MAP-mediated interconnections between adjacent filaments. In the case of AFs, we propose a model in which each protein subunit is equivalent to an electric circuit oscillator. The physical parameters used in the model were evaluated based on the molecular properties of the polymer. Using the general conductivity rules that apply to electrical circuits we analyzed the properties of ionic waves that propagate along AFs and compared these values to those observed in earlier experiments. In the context of the role played by MTs in neurons, we described the dynamics of C-termini states. We discussed both individual MTs and their networks, including the interactions with ions and signal transmission via MAPs. Recent experiments

on ionic conductivity along AFs and MTs show the validity of the basic assumptions postulated in our models. In light of these results, we conjectured a new dendritic signaling mechanism that involves ion waves along protein filaments which may travel without significant decay over tens of microns, potentially affecting the function of synapses and ion channels.

We proposed a new model for information processing in dendrites based on electrical signaling involving the cytoskeleton. This model predicts that the dendritic cytoskeleton, including MTs and AFs plays an active role in computations affecting neuronal function. These cytoskeletal filaments are affected by, and regulate, ion channel activity. A molecular dynamics description of the C-termini protruding from the surface of a MT has revealed the existence of several conformational states, which lead to collective dynamical properties of the neuronal cytoskeleton. These collective states of the C-termini on MTs have been shown to have a significant effect on the ionic condensation and ion cloud propagation with physical similarities to those recently found in actin-filaments. We have been able to then provide an integrated view of these phenomena in a bottom-up scheme that demonstrates how ionic wave propagation along cytoskeletal structures may impact channel functions, and thus neuronal computational capabilities.

The possibility of an evolvable, dynamic, and responsive electrical circuitry within the cell provided by actin and MT filaments could be of enormous consequence to our understanding of the way cells operate internally and interact with their environment. In particular, it would cast an entirely new light on cell differentiation, cell division, and cell–cell communication. While an integrated theory of this type of behavior is far from being constructed, its individual elements are gradually taking shape.

Acknowledgments

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Electrostatics of nanosystems: Application to microtubules and the ribosome

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Evaluation of the electrostatic properties of biomolecules has become a standard practice in molecular biophysics. Foremost among the models used to elucidate the electrostatic potential is the Poisson-Boltzmann equation; however, existing methods for solving this equation have limited the scope of accurate electrostatic calculations to relatively small biomolecular systems. Here we present the application of numerical methods to enable the trivially parallel solution of the Poisson-Boltzmann equation for supramolecular structures that are orders of magnitude larger in size. As a demonstration of this methodology, electrostatic potentials have been calculated for large microtubule and ribosome structures. The results point to the likely role of electrostatics in a variety of activities of these structures.

The importance of electrostatic modeling to biophysics is well established; electrostatics have been shown to influence various aspects of nearly all biochemical reactions. Advances in NMR, x-ray, and cryo-electron microscopy techniques for structure elucidation have drastically increased the size and number of biomolecules and molecular complexes for which coordinates are available. However, although the biophysical community continues to examine macromolecular systems of increasing scale, the computational evaluation of electrostatic properties for these systems is limited by methodology that can handle only relatively small systems, typically consisting of fewer than 50,000 atoms. Despite these limitations, such computational methods have been immensely useful in analyses of the stability, dynamics, and association of proteins, nucleic acids, and their ligands (1–3). Here we describe algorithms that open the way to similar analyses of much larger subcellular structures.

One of the most widespread models for the evaluation of electrostatic properties is the Poisson-Boltzmann equation (PBE) (4, 5)

$$-\nabla\cdot\varepsilon(x)\nabla\phi(x) + \bar{\kappa}^2(x)\sinh\phi(x) = f(x), \qquad [1]$$

a second-order nonlinear elliptic partial differential equation that relates the electrostatic potential (ϕ) to the dielectric properties of the solute and solvent (ε), the ionic strength of the solution and the accessibility of ions to the solute interior ($\bar{\kappa}^2$), and the distribution of solute atomic partial charges (f). To expedite solution of the equation, this nonlinear PBE is often approximated by the linearized PBE (LPBE) by assuming $sinh\phi(x) \approx \phi(x)$. Several numerical techniques have been used to solve the nonlinear PBE and LPBE, including boundary element (6–8), finite element (9–11), and finite difference (12–14) algorithms. However, despite the variety of solution methods, none of these techniques has been satisfactorily applied to large molecular structures at the scales currently accessible to modern biophysical methods. To accommodate arbitrarily large biomolecules, algorithms for solving the PBE must be both efficient and amenable to implementation on a parallel platform in a scalable fashion, requirements that current methods have been unable to satisfy. Although boundary element LPBE solvers provide an efficient representation of the problem domain, they are not useful for the nonlinear problem and have not been applied to the PBE on parallel platforms. Similarly, adaptive finite element methods have shown some success in parallel evaluation of both the LPBE and nonlinear PBE (15), but limitations in current solver technology and difficulty with efficient representation of the biomolecular data prohibits their practical application to large biomolecular systems. Finally, unlike the boundary and finite element techniques, finite difference methods have the advantage of very efficient multilevel solvers (12, 16) and applicability to both the linear and nonlinear forms of the PBE; however, existing parallel finite difference algorithms often require costly interprocessor communication that limits both the nature and scale of their execution on parallel platforms (17–21) [see especially Van de Velde (19) for reviews of the various methods].

Multigrid Solution Through Parallel Focusing

Recently, a new algorithm (Bank-Holst) was described for the parallel adaptive finite element solution of elliptic partial differential equations with negligible interprocess communication (22). Unlike finite difference methods, which use a fixed resolution of the problem domain, adaptive finite element techniques generate very accurate solutions of these equations by locally enriching the basis set in regions of high error through refinement of the domain discretization. The first step in the Bank-Holst parallel finite element algorithm is the solution of the equation over the entire problem domain using a coarse resolution basis set by each processor. This solution is then used, in conjunction with an *a posteriori* error estimator, to partition the problem domain into P subdomains, which are assigned to the P processors of a parallel computer. The algorithm is load balanced by equidistributing *a posteriori* error estimates across the subdomains. Each processor then solves the partial differential equation over the global mesh, but confines its adaptive refinement to the local subdomain and a small surrounding overlap region. This procedure results in a very accurate representation of the solution over the local subdomain. After all processors have completed their local adaptive solution of the equation, a global solution is constructed by the piecewise assembly of the solutions in each subdomain. It can be rigorously shown that the piecewise-assembled solution is as accurate as the solution to the problem on a single global mesh (23). Because each processor performs all of its computational work independently, the Bank-Holst algorithm requires very little interprocess communication and exhibits excellent parallel scaling. Unfortunately, although this algorithm works well for adaptive techniques, such as finite elements, it is not directly applicable to the fixed-resolution finite difference methods that currently offer the most efficient solution of the PBE for large molecular systems. However, by combining the Bank-Holst method with

Abbreviations: PBE, Poisson-Boltzmann equation; LPBE, linearized PBE; NPACI, National Partnership for Advanced Computational Infrastructure.

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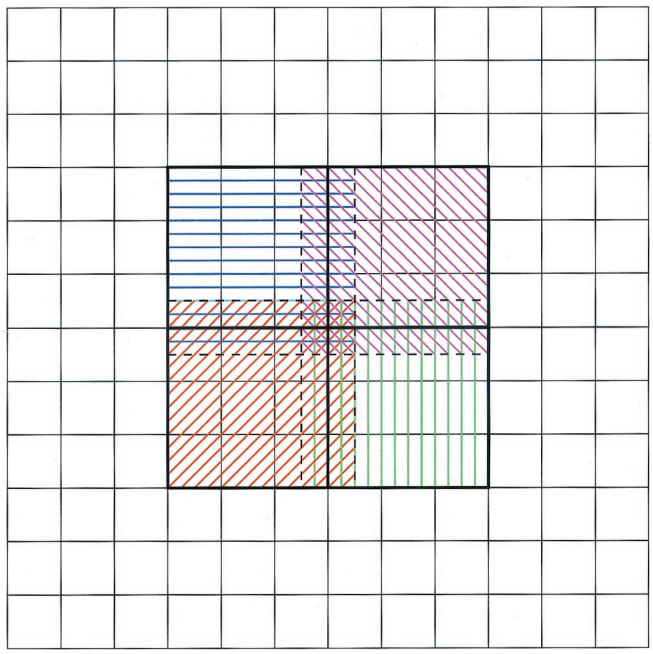

Fig. 1. Parallel focusing subdomains for four processors on simple two-dimensional mesh. Parallel computations occur on subset (hashed areas) of global mesh. Each processor performs calculations over a domain (colored hashed patterns) that combines a unique partition (inside heavy, black lines) with an overlap region (between heavy black lines and dashed lines).

other techniques, we have been able to extend this algorithm to finite difference solvers.

Electrostatic "focusing" is a popular technique in finite difference methods for generating accurate solutions to the PBE in subsets of the problem domain, such as a binding or titratable sites within a protein (4, 14). Like the Bank-Holst method, the first step in electrostatic focusing is the calculation of a low-accuracy solution on a coarse finite difference mesh spanning the entire problem domain. This coarse solution is then used to define the boundary conditions for a much more accurate calculation on a finer discretization of the desired subdomain. As noted previously (22), this focusing technique is superficially related to the Bank-Holst algorithm, where the local enrichment of a coarse, global solution has been replaced by a the solution of a fine, local multigrid problem using the solution from a coarse, global problem for boundary conditions.

We have combined standard focusing techniques and the Bank-Holst algorithm into a "parallel focusing" method for the solution of the PBE. Unlike previous parallel algorithms for solving the PBE, this method has excellent parallel complexity, permitting the treatment of very large biomolecular systems on massively parallel computational platforms. Furthermore, the finite difference discretization on a regular mesh allows for fast solution by certain highly efficient multigrid solvers (12). The algorithm is summarized below:

Given the problem data and P processors of a parallel machine:

(i) Each processor $i = 1...,P$ (a) obtains a coarse solution of the PBE over the global domain, (b) heuristically subdivides the global domain into P subdomains ($\Omega_1...,\Omega_P$), which are assigned to processors 1...,P, (c) assigns boundary conditions to its subdomain Ω_i using the coarse global solution, and (d) solves the PBE on subdomain Ω_i.

(ii) A master processor collects observable data (free energy, etc.) from the other processors and controls I/O.

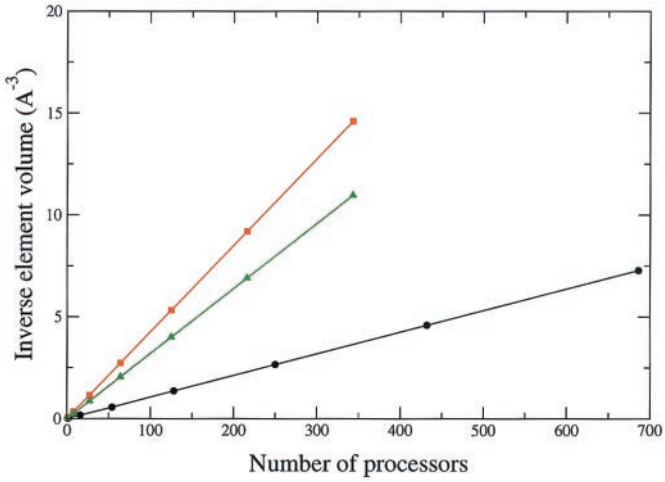

Fig. 2. Scaling of the parallel focusing algorithm to solve the LPBE for the microtubule (black line), 50S ribosome subunit (green line), and 30S ribosome subunit (red line) systems on the NPACI IBM Blue Horizon supercomputer. As discussed in the text, the inverse element volume (a measure of the mesh resolution) is a linear function of the number of processors, $v(P)^{-1} = (h_{xhyhz})^{-1} = cP$, where c = 0.011, 0.043, and 0.032 Å^{-3}/processor for the microtubule, 30S, and 50S systems, respectively.

This parallel focusing algorithm begins with each processor independently solving a coarse global problem. The per-processor subdomains are chosen in a heuristic fashion as outlined in Fig. 1. Like standard focusing calculations, only a subset of the global mesh surrounding the area of interest is used for the parallel calculations. This subset is partitioned into P approximately equal subdomains, which are distributed among the processors. Each processor then performs a fine-scale finite difference calculation over this subdomain and an overlap region that usually spans about 5–10% of the neighboring subdomains. The overlap regions are included to compensate for inaccuracies in the boundary conditions derived from the global coarse solution; however, these regions are not used to assemble the fine-scale global solution and do not contribute to calculations of observables such as forces and energies. After the fine-scale calculations are complete, a master processor accumulates the desired data from the other processes and controls I/O of the solution. Like the Bank-Holst algorithm, each process computes independently on both the global coarse problem and its subdomain. These independent computations result in an algorithm that requires negligible interprocess communication and offers excellent parallel performance.

Because of its low communication requirements, this parallel focusing algorithm can be used on parallel platforms ranging from low-bandwidth networks of workstations to high-performance supercomputers. Because step d of the parallel focusing algorithm typically dominates the overall computation, this algorithm provides excellent time complexity with respect to the number of processors. Most importantly for Poisson-Boltzmann calculations on large molecules, the mesh resolution increases linearly with the number of processors.** This linear behavior has been observed on calculations involving hundreds of processors; Fig. 2 shows the scaling results for calculations on the biomolecular systems discussed below. The parallel focusing algorithm was tested on up to 700 processors and shows linear

**For example, let $v(P) = h_x(P)h_y(P)h_z(P)$ be the volume of a grid element in a P-processor calculation with $100f$% overlap on a $n_x \times n_y \times n_z$ mesh with spacings $h_x(P),h_y(P),h_z(P)$ over a problem domain of volume V. One measure of the mesh resolution is the inverse element volume, which behaves as $v(P)^{-1} \sim (1 - 2f)^3P(n_x - 1)(n_y - 1)(n_x - 1)/V$, which is linear in P.

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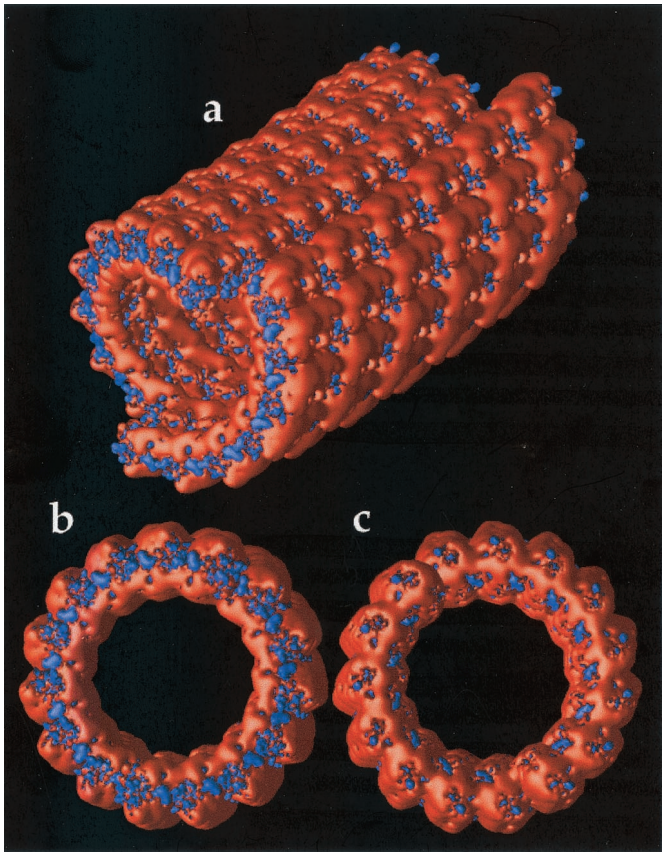

Fig. 3. Electrostatic properties of the microtubule exterior. Potential isocontours are shown at +1 kT/e (blue) and −1 kT/e (red) and obtained by solution of the LPBE at 150 mM ionic strength with a solute dielectric of 2 and a solvent dielectric of 78.5. (*a*) Exterior view of entire microtubule with − end (α tubulin monomer exposed) forward. (*b*) View of − end. (*c*) View of + end of microtubule (β tubulin monomer exposed).

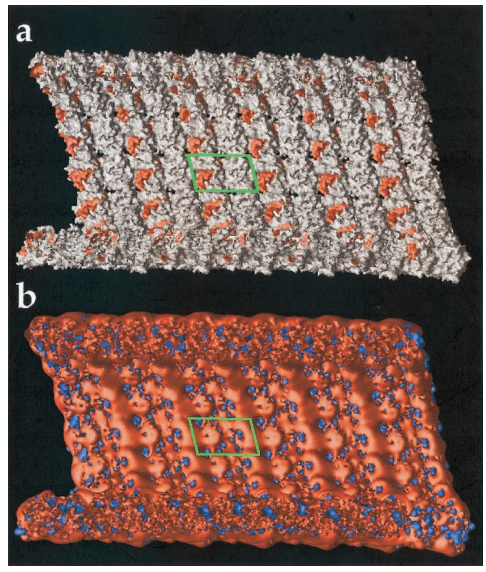

Fig. 4. Electrostatic properties of microtubule interior. Cross-section view with − end to the left and tubulin dimer outlined by green box. (*a*) View of microtubule molecular surface with regions implicated in vinblastine, colchicine, and taxol binding shown in red. (*b*) Same view of microtubule with electrostatic potential isocontours at +1 kT/e (blue) and −1 kT/e (red). Potential calculated as in Fig. 3.

scaling over the entire range. The implications of the improved performance available through this parallel algorithm are discussed in more detail in the next section.

Parallel focusing has been implemented as an extension (23) of the APBS (Adaptive Poisson-Boltzmann Solver) software package (9) by using the PMG multigrid library to assemble and solve the PBE algebraic systems (12, 16). The parallel routines have been implemented by using MALOC, a portable hardware abstraction layer library (23), which allows interprocess communication through a variety of methods.

The following sections describe the application of the parallel focusing techniques to very large biomolecular systems whose electrostatic properties were unattainable by using traditional methods for solving the PBE. Specifically, we examine two important cellular components: a million-atom microtubule fragment and the large and small subunits of the ribosome.

Electrostatic Properties of Microtubules

Within every eukaryotic cell is a complex system of proteins and filaments called the cytoskeleton. The largest structures within the cytoskeleton are microtubules, cylindrical polymers formed from the protein tubulin. Microtubules perform many functions within the cell, from providing basic structure to cell division and transport, and many of these roles involve electrostatic interactions. The recent solution of the structure of tubulin (24) has allowed us to construct a working model of a microtubule containing 90 tubulin dimers. This structure contains more than 1.25 million atoms [the negatively charged C termini of both the α and β subunits were not resolved in the tubulin structure (24) and have not been rebuilt in our microtubule structure], and the structure has dimensions of ≈300 Å × 300 Å × 600 Å. The LPBE was solved at 0.54-Å mesh spacing by using 686 processors of the National Partnership for Advanced Computational Infrastructure (NPACI) Blue Horizon supercomputer. Blue Horizon is a massively parallel IBM RS6000 supercomputer consisting of 1,152 375-MHz Power3 processors distributed among 144 eight-way symmetric multiprocessing nodes. These nodes are connected with the IBM Colony switch, which delivers 350 MB/sec peak message-passing interface performance with 17-ms latency. Following the parallel focusing algorithm, each processor solved the LPBE on a coarse 97^3 global mesh, as well as a finer 97^3 spanning a subset of the overall domain. The global solution was obtained in less than 1 h. This system provides an excellent example of the improved performance available through the parallel focusing method. A sequential PBE solver treating the same system at a similar resolution would discretize the problem on a 1,111 × 555 × 555 mesh, requiring more than 350 times more memory and time to solve.

Although more work is presently underway to elucidate the implications of electrostatics on microtubule structure and function, Fig. 3 shows some early results. The most obvious feature is the overall negative electrostatic potential of the microtubule with smaller regions of positive potential. This large negative potential likely plays a significant role in the interaction with microtubule-associated proteins, motor proteins such as kinesin and dynein, and other microtubules. Fig. 3 *b* and *c* also exhibits dramatic differences between the + end (β tubulin monomer exposed) and − end (α tubulin monomer exposed) of the microtubule, which, as has been shown for actin filaments (3), may contribute to differences in the polymerization properties (25) and relative stabilities of the two ends. Fig. 4 shows a microtubule inner surface in cross-section. The binding regions of compounds such as vinblastine, colchicine, and taxol (26) are shown in Fig. 4*a*, while electrostatic isocontours are shown, for the same view, in Fig. 4*b*. Qualitative examination of the electrostatic data shows interesting variations in the drug binding

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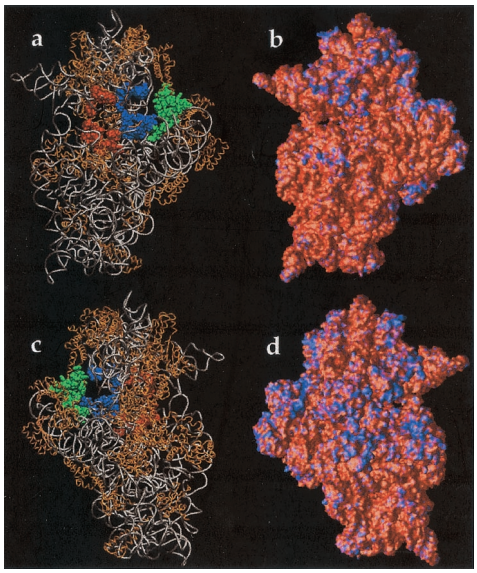

Fig. 5. Electrostatic properties of the 30S ribosomal subunit. Potential obtained by solution of the LPBE at 150 mM ionic strength with a solute dielectric of 2 and a solvent dielectric of 78.5 by using the 30S structure from the 1FJG Protein Data Bank entry (29, 34). (*a*) Front view (face that contacts the 50S subunit) of the 30S backbone with protein shown in gold; nucleic acids shown in silver. Selected components of the A-site (16S residues 525–535, 955–965, 1055–1060, 1490–1493; S12 residues 45–49) are shown in red, components of the P-site (16S residues 789–791, 925–927, 965–967, 1229–1230, 1338–1341, 1399–1403, 1497–1499; S9 residues 124–128; S13 residues 122–126) are shown in blue, and components of the E-site (16S residues 691–695, 792–797; S7 residues 80–90, 150–170; S11 residues 45–60) are shown in green. (*b*) Front view of the electrostatic potential mapped on the 30S molecular surface; a blue color indicates regions of positive potential ($> +2.6$ kT/e) whereas red depicts negative potential (< -2.6 kT/e) values. (*c*) Back view (opposite the 50S-binding face) of the 30S backbone. (*d*) Back view of the electrostatic potential mapped on the 30S molecular surface.

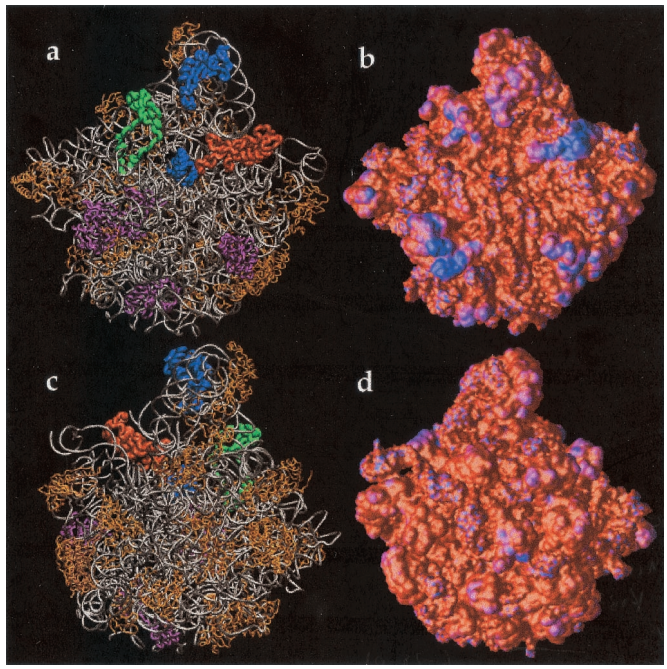

Fig. 6. Electrostatic properties of the 50S ribosomal subunit. Potential obtained by solution of the LPBE at 150 mM ionic strength with a solute dielectric of 2 and a solvent dielectric of 78.5 by using the 50S structure from the 1FFK Protein Data Bank entry (27). (*a*) Front view (face that contacts the 30S subunit) of the 50S backbone with protein component (L10e) of the A-site shown as green tube, protein component (L5) of the P-site shown as gold tube, nucleic P-loop portion of P-site shown as gold spheres, protein component (L44e) of E-site shown as white tube, various protein components (L2, L14, L19) of 50S-30S interface shown in purple. Remaining protein components are shown in gold and nucleic acid components in silver. (*b*) Front view of the electrostatic potential mapped on the 50S molecular surface; a blue color indicates regions of positive potential (>0 kT/e) whereas red depicts negative potential values (< -9.6 kT/e). (*c*) Back view (opposite the 30S-binding face) of the 50S backbone. (*d*) Back view of the electrostatic potential mapped on the 50S molecular surface.

regions; however, much more work needs to be done to quantitatively elucidate contribution of electrostatics to the interaction of these pharmaceuticals with microtubules.

The microtubule also provides a good example for examining the improved performance of this parallel algorithm with respect to sequential solvers.

Electrostatic Properties of the Ribosome

Ribosomes are macromolecular complexes responsible for the translation of the mRNA into protein. These complexes consist of two subunits: the large 50S subunit and the small 30S subunit, both of which are composed of RNA and protein constituents. During translation, the large and small subunits associate to form an active ribosome. Recently, high-resolution structures of the 50S subunit from *Haloarcula marismortui* (27) and the 30S subunit from *Thermus thermophilus* (28, 29) were solved by x-ray crystallography. These structures present a unique opportunity to examine the electrostatic potential of large ribonucleoprotein complexes.

The protonated 30S structure consists of more than 88,000 atoms and roughly spans a 200-Å-long cubic box. Using APBS on 343 processors of the NPACI Blue Horizon, LPBE was solved to give the electrostatic potential of the small ribosomal subunit at 0.41-Å resolution. Calculations also were performed on the protonated 50S subunit structure, which consisted of 94,854 atoms[††] and has similar dimensions to the 30S subunit. The electrostatic potential of the large subunit was obtained at 0.45-Å resolution by using APBS to solve the LPBE on 343 processors of the Blue Horizon. Both systems were solved by using the parallel focusing algorithm: each processor first solved a coarse problem defined by a 97^3 mesh covering the entire problem domain, and the solved the LPBE on a 97^3 mesh covering the particular subset of the global problem. Examination of the electrostatic potential mapped to the molecular surfaces of the 30S (Fig. 5) and 50S (Fig. 6) reveals large areas of negative potential with smaller regions of positive potential corresponding to some of the proteins implicated in binding of the 30S and 50S subunits to form the active ribosomal complex (27, 29–31). Not surprisingly, the potential surface maps between the two subunits exhibit qualitative electrostatic complementarity. Despite the fact that the 30S and 50S structures are from distinct species that thrive in very different environments, such comparisons of electrostatic potential are not unwarranted. Formation of active ribosomal complexes has been observed by using subunits from different species (32, 33), indicating some evolutionary conservation of the elements involved in subunit association.

Fig. 5 *b* and *d* shows the electrostatic potential mapped to the 30S subunit molecular surface. As evidenced by the 30S structure (Fig. 5 *a* and *c*), regions of positive potential typically correspond to protein components of the small subunit or cocrystallized counterions. Of particular interest is the active site of the 30S subunit, shown in Fig. 5 *a* and *b*. There are interesting variations in the electrostatic potential near regions of antibiotic

[††]The 50S Protein Data Bank coordinate file (1FFK) (27) contained only Cα coordinates for the protein constituents. Therefore, each protein residue was simply represented by its Cα atom, which was assigned the total charge of the residue and a radius of 4.0 Å.

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binding (34) and the A, P, and E tRNA binding sites (29, 34). For example, the codon binding sites and "platform region" (protuberance on the right side of Fig. 5 *a* and *b*) are surrounded by regions of positive potential, which could help stabilize complexes of the ribosome with mRNA and tRNA. However, more quantitative calculations will be needed to fully explore these hypotheses.

The electrostatic potential of the 50S subunit is shown in Fig. 6 *b* and *d*. Like the small subunit, the electrostatic surface potential is largely negative, with scattered regions of positive potential typically associated with ribosomal proteins. Some of the most interesting aspects of this data set are the regions of positive potential on the 50S "crown" (see upper protuberance in Fig. 6 *a* and *b*), which correspond to proteins of the large subunit involved in tRNA binding. Specifically, these positive regions are due to proteins L44e, L5, and L10e (see Fig. 6*a*), which have been implicated in tRNA binding to the 50S subunit (35), and contribute large regions of positive potential to the molecular surface in the upper portions of the A, P, and E tRNA binding sites. Furthermore, the P-loop (shown as blue spheres in Fig. 6*a*), an important component of the 50S P-site (35, 36), shows significant positive surface potential due to a nearby Mg^{2+} ion. These areas of positive potential may provide stabilizing interactions with mRNA or tRNA bound to the ribosomal complex during translation. However, as with the 30S calculations, more detailed work is required to accurately resolve the role of electrostatics in tRNA and mRNA binding to the 50S subunit.

Conclusions

We have described the combination of standard finite difference focusing techniques and the Bank-Holst algorithm into a parallel focusing method to facilitate the solution of the PBE for nanoscale systems. Unlike previous multiprocessor algorithms for solving the PBE, this method has excellent parallel complexity that permits the solution of these problems on massively parallel computational platforms. Furthermore, the finite difference discretization on a regular mesh allows for fast solution by highly efficient multigrid solvers. This algorithm has been implemented in the APBS software by using the PMG multigrid solver and tested on the NPACI IBM Blue Horizon supercomputer.

Using these methods for the parallel multigrid solution of elliptic differential equations, the electrostatic properties of very large biomolecular assemblages are now amenable to computation. This technique relies on the efficient solution of the PBE combined with parallel focusing techniques to solve these large problems in a variety of distributed computational environments. Solution of the LPBE for the 1.2 million-atom microtubule system provided electrostatic potential data, which revealed interesting features near drug binding sites and provided possible insight into stability differences at the + and − ends of the microtubule. Such detailed electrostatic information will be central to future studies that examine the possible collective effects involved in the formation of structural defects and the stabilizing effects of taxol binding to the interior of microtubules. Likewise, application of this methodology to ribosome systems elucidated intriguing electrostatic properties of the ribosomal active site, which will provide the starting point for investigation of the stabilization of the tRNA- and mRNA-ribosome complexes during translation and the rational design of novel antibiotics. Finally, the ability to determine the contribution of electrostatics to the forces and energies of nanoscale systems should extend the scale of implicit solvent dynamics methods to much larger macromolecular complexes.

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Electrostatics of nanosystems: Application to microtubules and the ribosome

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Evaluation of the electrostatic properties of biomolecules has become a standard practice in molecular biophysics. Foremost among the models used to elucidate the electrostatic potential is the Poisson-Boltzmann equation; however, existing methods for solving this equation have limited the scope of accurate electrostatic calculations to relatively small biomolecular systems. Here we present the application of numerical methods to enable the trivially parallel solution of the Poisson-Boltzmann equation for supramolecular structures that are orders of magnitude larger in size. As a demonstration of this methodology, electrostatic potentials have been calculated for large microtubule and ribosome structures. The results point to the likely role of electrostatics in a variety of activities of these structures.

The importance of electrostatic modeling to biophysics is well established; electrostatics have been shown to influence various aspects of nearly all biochemical reactions. Advances in NMR, x-ray, and cryo-electron microscopy techniques for structure elucidation have drastically increased the size and number of biomolecules and molecular complexes for which coordinates are available. However, although the biophysical community continues to examine macromolecular systems of increasing scale, the computational evaluation of electrostatic properties for these systems is limited by methodology that can handle only relatively small systems, typically consisting of fewer than 50,000 atoms. Despite these limitations, such computational methods have been immensely useful in analyses of the stability, dynamics, and association of proteins, nucleic acids, and their ligands (1–3). Here we describe algorithms that open the way to similar analyses of much larger subcellular structures.

One of the most widespread models for the evaluation of electrostatic properties is the Poisson-Boltzmann equation (PBE) (4, 5)

$$-\nabla\cdot\varepsilon(x)\nabla\phi(x) + \bar{\kappa}^2(x)\sinh\phi(x) = f(x), \qquad [1]$$

a second-order nonlinear elliptic partial differential equation that relates the electrostatic potential (ϕ) to the dielectric properties of the solute and solvent (ε), the ionic strength of the solution and the accessibility of ions to the solute interior ($\bar{\kappa}^2$), and the distribution of solute atomic partial charges (f). To expedite solution of the equation, this nonlinear PBE is often approximated by the linearized PBE (LPBE) by assuming $sinh\phi(x) \approx \phi(x)$. Several numerical techniques have been used to solve the nonlinear PBE and LPBE, including boundary element (6–8), finite element (9–11), and finite difference (12–14) algorithms. However, despite the variety of solution methods, none of these techniques has been satisfactorily applied to large molecular structures at the scales currently accessible to modern biophysical methods. To accommodate arbitrarily large biomolecules, algorithms for solving the PBE must be both efficient and amenable to implementation on a parallel platform in a scalable fashion, requirements that current methods have been unable to satisfy. Although boundary element LPBE solvers provide an efficient representation of the problem domain, they are not useful for the nonlinear problem and have not been applied to the PBE on parallel platforms. Similarly, adaptive finite element methods have shown some success in parallel evaluation of both the LPBE and nonlinear PBE (15), but limitations in current solver technology and difficulty with efficient representation of the biomolecular data prohibits their practical application to large biomolecular systems. Finally, unlike the boundary and finite element techniques, finite difference methods have the advantage of very efficient multilevel solvers (12, 16) and applicability to both the linear and nonlinear forms of the PBE; however, existing parallel finite difference algorithms often require costly interprocessor communication that limits both the nature and scale of their execution on parallel platforms (17–21) [see especially Van de Velde (19) for reviews of the various methods].

Multigrid Solution Through Parallel Focusing

Recently, a new algorithm (Bank-Holst) was described for the parallel adaptive finite element solution of elliptic partial differential equations with negligible interprocess communication (22). Unlike finite difference methods, which use a fixed resolution of the problem domain, adaptive finite element techniques generate very accurate solutions of these equations by locally enriching the basis set in regions of high error through refinement of the domain discretization. The first step in the Bank-Holst parallel finite element algorithm is the solution of the equation over the entire problem domain using a coarse resolution basis set by each processor. This solution is then used, in conjunction with an *a posteriori* error estimator, to partition the problem domain into P subdomains, which are assigned to the P processors of a parallel computer. The algorithm is load balanced by equidistributing *a posteriori* error estimates across the subdomains. Each processor then solves the partial differential equation over the global mesh, but confines its adaptive refinement to the local subdomain and a small surrounding overlap region. This procedure results in a very accurate representation of the solution over the local subdomain. After all processors have completed their local adaptive solution of the equation, a global solution is constructed by the piecewise assembly of the solutions in each subdomain. It can be rigorously shown that the piecewise-assembled solution is as accurate as the solution to the problem on a single global mesh (23). Because each processor performs all of its computational work independently, the Bank-Holst algorithm requires very little interprocess communication and exhibits excellent parallel scaling. Unfortunately, although this algorithm works well for adaptive techniques, such as finite elements, it is not directly applicable to the fixed-resolution finite difference methods that currently offer the most efficient solution of the PBE for large molecular systems. However, by combining the Bank-Holst method with

Abbreviations: PBE, Poisson-Boltzmann equation; LPBE, linearized PBE; NPACI, National Partnership for Advanced Computational Infrastructure.

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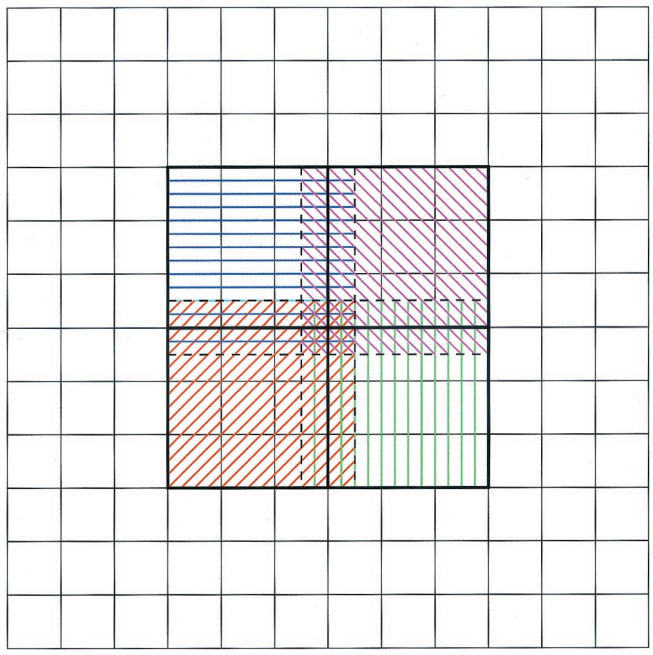

Fig. 1. Parallel focusing subdomains for four processors on simple two-dimensional mesh. Parallel computations occur on subset (hashed areas) of global mesh. Each processor performs calculations over a domain (colored hashed patterns) that combines a unique partition (inside heavy, black lines) with an overlap region (between heavy black lines and dashed lines).

other techniques, we have been able to extend this algorithm to finite difference solvers.

Electrostatic "focusing" is a popular technique in finite difference methods for generating accurate solutions to the PBE in subsets of the problem domain, such as a binding or titratable sites within a protein (4, 14). Like the Bank-Holst method, the first step in electrostatic focusing is the calculation of a low-accuracy solution on a coarse finite difference mesh spanning the entire problem domain. This coarse solution is then used to define the boundary conditions for a much more accurate calculation on a finer discretization of the desired subdomain. As noted previously (22), this focusing technique is superficially related to the Bank-Holst algorithm, where the local enrichment of a coarse, global solution has been replaced by a the solution of a fine, local multigrid problem using the solution from a coarse, global problem for boundary conditions.

We have combined standard focusing techniques and the Bank-Holst algorithm into a "parallel focusing" method for the solution of the PBE. Unlike previous parallel algorithms for solving the PBE, this method has excellent parallel complexity, permitting the treatment of very large biomolecular systems on massively parallel computational platforms. Furthermore, the finite difference discretization on a regular mesh allows for fast solution by certain highly efficient multigrid solvers (12). The algorithm is summarized below:

Given the problem data and P processors of a parallel machine:

(i) Each processor $i = 1...,P$ (a) obtains a coarse solution of the PBE over the global domain, (b) heuristically subdivides the global domain into P subdomains ($\Omega_1...,\Omega_P$), which are assigned to processors 1...,P, (c) assigns boundary conditions to its subdomain Ω_i using the coarse global solution, and (d) solves the PBE on subdomain Ω_i.

(ii) A master processor collects observable data (free energy, etc.) from the other processors and controls I/O.

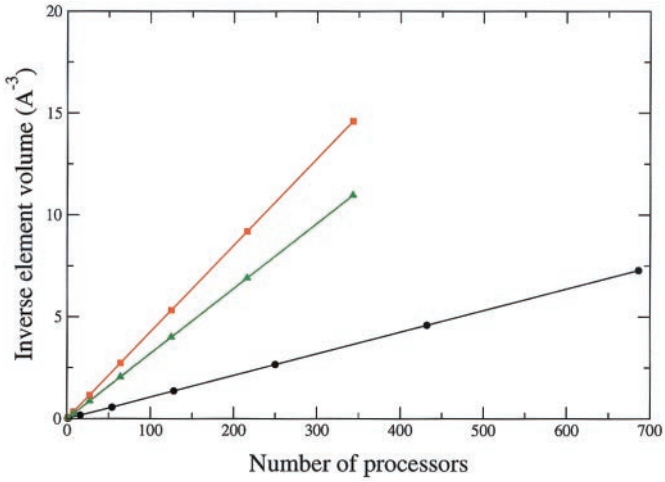

Fig. 2. Scaling of the parallel focusing algorithm to solve the LPBE for the microtubule (black line), 50S ribosome subunit (green line), and 30S ribosome subunit (red line) systems on the NPACI IBM Blue Horizon supercomputer. As discussed in the text, the inverse element volume (a measure of the mesh resolution) is a linear function of the number of processors, $v(P)^{-1} = (h_{xhyhz})^{-1} = cP$, where c = 0.011, 0.043, and 0.032 Å^{-3}/processor for the microtubule, 30S, and 50S systems, respectively.

This parallel focusing algorithm begins with each processor independently solving a coarse global problem. The per-processor subdomains are chosen in a heuristic fashion as outlined in Fig. 1. Like standard focusing calculations, only a subset of the global mesh surrounding the area of interest is used for the parallel calculations. This subset is partitioned into P approximately equal subdomains, which are distributed among the processors. Each processor then performs a fine-scale finite difference calculation over this subdomain and an overlap region that usually spans about 5–10% of the neighboring subdomains. The overlap regions are included to compensate for inaccuracies in the boundary conditions derived from the global coarse solution; however, these regions are not used to assemble the fine-scale global solution and do not contribute to calculations of observables such as forces and energies. After the fine-scale calculations are complete, a master processor accumulates the desired data from the other processes and controls I/O of the solution. Like the Bank-Holst algorithm, each process computes independently on both the global coarse problem and its subdomain. These independent computations result in an algorithm that requires negligible interprocess communication and offers excellent parallel performance.

Because of its low communication requirements, this parallel focusing algorithm can be used on parallel platforms ranging from low-bandwidth networks of workstations to high-performance supercomputers. Because step d of the parallel focusing algorithm typically dominates the overall computation, this algorithm provides excellent time complexity with respect to the number of processors. Most importantly for Poisson-Boltzmann calculations on large molecules, the mesh resolution increases linearly with the number of processors.** This linear behavior has been observed on calculations involving hundreds of processors; Fig. 2 shows the scaling results for calculations on the biomolecular systems discussed below. The parallel focusing algorithm was tested on up to 700 processors and shows linear

**For example, let $v(P) = h_x(P)h_y(P)h_z(P)$ be the volume of a grid element in a P-processor calculation with $100f$% overlap on a $n_x \times n_y \times n_z$ mesh with spacings $h_x(P),h_y(P),h_z(P)$ over a problem domain of volume V. One measure of the mesh resolution is the inverse element volume, which behaves as $v(P)^{-1} \sim (1 - 2f)^3P(n_x - 1)(n_y - 1)(n_x - 1)/V$, which is linear in P.

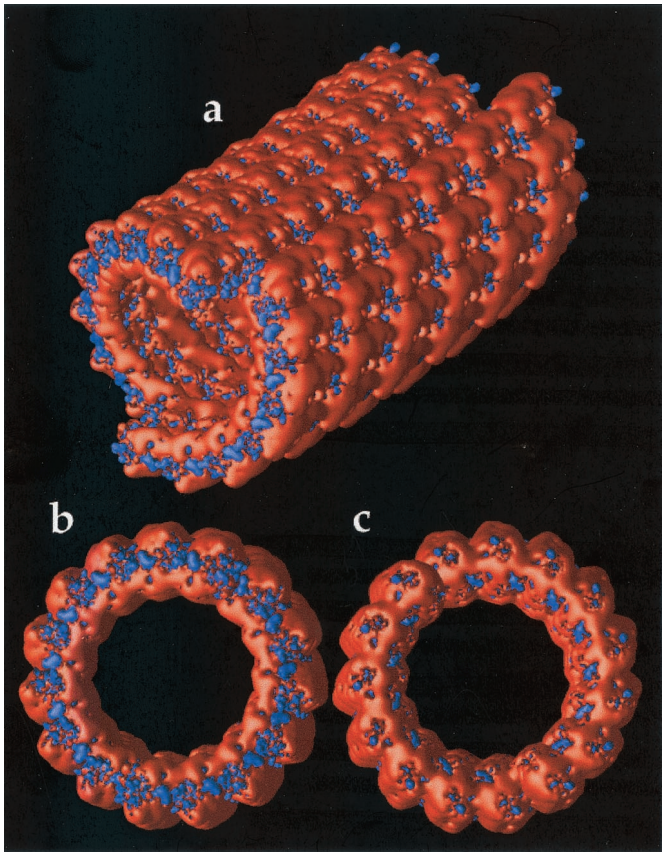

Fig. 3. Electrostatic properties of the microtubule exterior. Potential isocontours are shown at +1 kT/e (blue) and −1 kT/e (red) and obtained by solution of the LPBE at 150 mM ionic strength with a solute dielectric of 2 and a solvent dielectric of 78.5. (*a*) Exterior view of entire microtubule with − end (α tubulin monomer exposed) forward. (*b*) View of − end. (*c*) View of + end of microtubule (β tubulin monomer exposed).

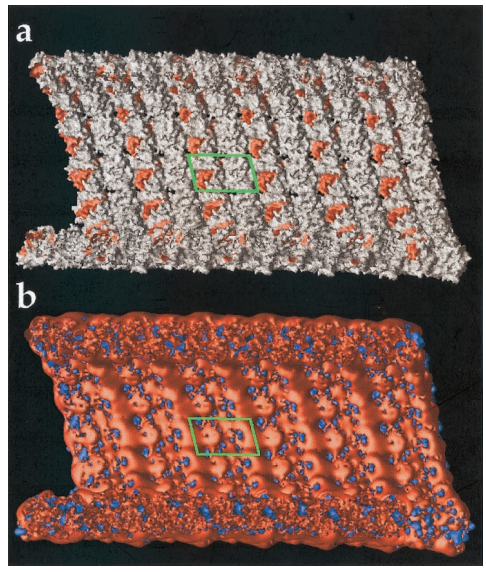

Fig. 4. Electrostatic properties of microtubule interior. Cross-section view with − end to the left and tubulin dimer outlined by green box. (*a*) View of microtubule molecular surface with regions implicated in vinblastine, colchicine, and taxol binding shown in red. (*b*) Same view of microtubule with electrostatic potential isocontours at +1 kT/e (blue) and −1 kT/e (red). Potential calculated as in Fig. 3.

scaling over the entire range. The implications of the improved performance available through this parallel algorithm are discussed in more detail in the next section.

Parallel focusing has been implemented as an extension (23) of the APBS (Adaptive Poisson-Boltzmann Solver) software package (9) by using the PMG multigrid library to assemble and solve the PBE algebraic systems (12, 16). The parallel routines have been implemented by using MALOC, a portable hardware abstraction layer library (23), which allows interprocess communication through a variety of methods.

The following sections describe the application of the parallel focusing techniques to very large biomolecular systems whose electrostatic properties were unattainable by using traditional methods for solving the PBE. Specifically, we examine two important cellular components: a million-atom microtubule fragment and the large and small subunits of the ribosome.

Electrostatic Properties of Microtubules

Within every eukaryotic cell is a complex system of proteins and filaments called the cytoskeleton. The largest structures within the cytoskeleton are microtubules, cylindrical polymers formed from the protein tubulin. Microtubules perform many functions within the cell, from providing basic structure to cell division and transport, and many of these roles involve electrostatic interactions. The recent solution of the structure of tubulin (24) has allowed us to construct a working model of a microtubule containing 90 tubulin dimers. This structure contains more than 1.25 million atoms [the negatively charged C termini of both the α and β subunits were not resolved in the tubulin structure (24) and have not been rebuilt in our microtubule structure], and the structure has dimensions of ≈300 Å × 300 Å × 600 Å. The LPBE was solved at 0.54-Å mesh spacing by using 686 processors of the National Partnership for Advanced Computational Infrastructure (NPACI) Blue Horizon supercomputer. Blue Horizon is a massively parallel IBM RS6000 supercomputer consisting of 1,152 375-MHz Power3 processors distributed among 144 eight-way symmetric multiprocessing nodes. These nodes are connected with the IBM Colony switch, which delivers 350 MB/sec peak message-passing interface performance with 17-ms latency. Following the parallel focusing algorithm, each processor solved the LPBE on a coarse 97^3 global mesh, as well as a finer 97^3 spanning a subset of the overall domain. The global solution was obtained in less than 1 h. This system provides an excellent example of the improved performance available through the parallel focusing method. A sequential PBE solver treating the same system at a similar resolution would discretize the problem on a 1,111 × 555 × 555 mesh, requiring more than 350 times more memory and time to solve.

Although more work is presently underway to elucidate the implications of electrostatics on microtubule structure and function, Fig. 3 shows some early results. The most obvious feature is the overall negative electrostatic potential of the microtubule with smaller regions of positive potential. This large negative potential likely plays a significant role in the interaction with microtubule-associated proteins, motor proteins such as kinesin and dynein, and other microtubules. Fig. 3 *b* and *c* also exhibits dramatic differences between the + end (β tubulin monomer exposed) and − end (α tubulin monomer exposed) of the microtubule, which, as has been shown for actin filaments (3), may contribute to differences in the polymerization properties (25) and relative stabilities of the two ends. Fig. 4 shows a microtubule inner surface in cross-section. The binding regions of compounds such as vinblastine, colchicine, and taxol (26) are shown in Fig. 4*a*, while electrostatic isocontours are shown, for the same view, in Fig. 4*b*. Qualitative examination of the electrostatic data shows interesting variations in the drug binding

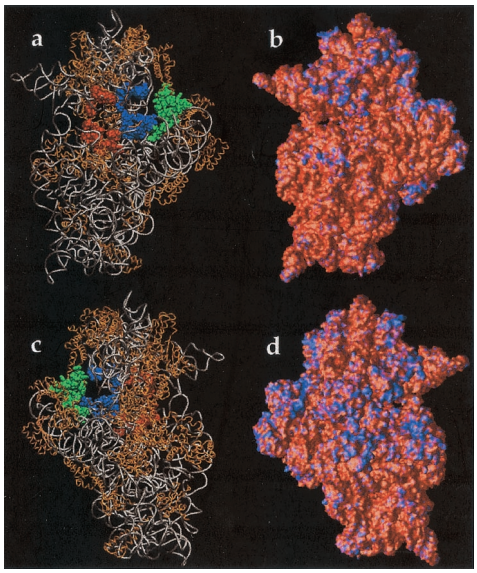

Fig. 5. Electrostatic properties of the 30S ribosomal subunit. Potential obtained by solution of the LPBE at 150 mM ionic strength with a solute dielectric of 2 and a solvent dielectric of 78.5 by using the 30S structure from the 1FJG Protein Data Bank entry (29, 34). (*a*) Front view (face that contacts the 50S subunit) of the 30S backbone with protein shown in gold; nucleic acids shown in silver. Selected components of the A-site (16S residues 525–535, 955–965, 1055–1060, 1490–1493; S12 residues 45–49) are shown in red, components of the P-site (16S residues 789–791, 925–927, 965–967, 1229–1230, 1338–1341, 1399–1403, 1497–1499; S9 residues 124–128; S13 residues 122–126) are shown in blue, and components of the E-site (16S residues 691–695, 792–797; S7 residues 80–90, 150–170; S11 residues 45–60) are shown in green. (*b*) Front view of the electrostatic potential mapped on the 30S molecular surface; a blue color indicates regions of positive potential ($> +2.6$ kT/e) whereas red depicts negative potential (< -2.6 kT/e) values. (*c*) Back view (opposite the 50S-binding face) of the 30S backbone. (*d*) Back view of the electrostatic potential mapped on the 30S molecular surface.

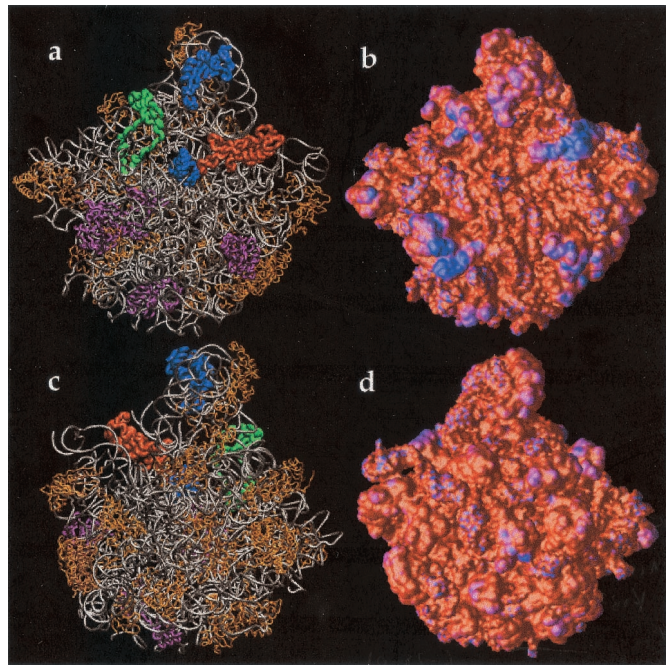

Fig. 6. Electrostatic properties of the 50S ribosomal subunit. Potential obtained by solution of the LPBE at 150 mM ionic strength with a solute dielectric of 2 and a solvent dielectric of 78.5 by using the 50S structure from the 1FFK Protein Data Bank entry (27). (*a*) Front view (face that contacts the 30S subunit) of the 50S backbone with protein component (L10e) of the A-site shown as green tube, protein component (L5) of the P-site shown as gold tube, nucleic P-loop portion of P-site shown as gold spheres, protein component (L44e) of E-site shown as white tube, various protein components (L2, L14, L19) of 50S-30S interface shown in purple. Remaining protein components are shown in gold and nucleic acid components in silver. (*b*) Front view of the electrostatic potential mapped on the 50S molecular surface; a blue color indicates regions of positive potential (>0 kT/e) whereas red depicts negative potential values (< -9.6 kT/e). (*c*) Back view (opposite the 30S-binding face) of the 50S backbone. (*d*) Back view of the electrostatic potential mapped on the 50S molecular surface.

regions; however, much more work needs to be done to quantitatively elucidate contribution of electrostatics to the interaction of these pharmaceuticals with microtubules.

The microtubule also provides a good example for examining the improved performance of this parallel algorithm with respect to sequential solvers.

Electrostatic Properties of the Ribosome

Ribosomes are macromolecular complexes responsible for the translation of the mRNA into protein. These complexes consist of two subunits: the large 50S subunit and the small 30S subunit, both of which are composed of RNA and protein constituents. During translation, the large and small subunits associate to form an active ribosome. Recently, high-resolution structures of the 50S subunit from *Haloarcula marismortui* (27) and the 30S subunit from *Thermus thermophilus* (28, 29) were solved by x-ray crystallography. These structures present a unique opportunity to examine the electrostatic potential of large ribonucleoprotein complexes.

The protonated 30S structure consists of more than 88,000 atoms and roughly spans a 200-Å-long cubic box. Using APBS on 343 processors of the NPACI Blue Horizon, LPBE was solved to give the electrostatic potential of the small ribosomal subunit at 0.41-Å resolution. Calculations also were performed on the protonated 50S subunit structure, which consisted of 94,854 atoms[††] and has similar dimensions to the 30S subunit. The electrostatic potential of the large subunit was obtained at 0.45-Å resolution by using APBS to solve the LPBE on 343 processors of the Blue Horizon. Both systems were solved by using the parallel focusing algorithm: each processor first solved a coarse problem defined by a 97^3 mesh covering the entire problem domain, and the solved the LPBE on a 97^3 mesh covering the particular subset of the global problem. Examination of the electrostatic potential mapped to the molecular surfaces of the 30S (Fig. 5) and 50S (Fig. 6) reveals large areas of negative potential with smaller regions of positive potential corresponding to some of the proteins implicated in binding of the 30S and 50S subunits to form the active ribosomal complex (27, 29–31). Not surprisingly, the potential surface maps between the two subunits exhibit qualitative electrostatic complementarity. Despite the fact that the 30S and 50S structures are from distinct species that thrive in very different environments, such comparisons of electrostatic potential are not unwarranted. Formation of active ribosomal complexes has been observed by using subunits from different species (32, 33), indicating some evolutionary conservation of the elements involved in subunit association.

Fig. 5 *b* and *d* shows the electrostatic potential mapped to the 30S subunit molecular surface. As evidenced by the 30S structure (Fig. 5 *a* and *c*), regions of positive potential typically correspond to protein components of the small subunit or cocrystallized counterions. Of particular interest is the active site of the 30S subunit, shown in Fig. 5 *a* and *b*. There are interesting variations in the electrostatic potential near regions of antibiotic

[††]The 50S Protein Data Bank coordinate file (1FFK) (27) contained only Cα coordinates for the protein constituents. Therefore, each protein residue was simply represented by its Cα atom, which was assigned the total charge of the residue and a radius of 4.0 Å.

binding (34) and the A, P, and E tRNA binding sites (29, 34). For example, the codon binding sites and "platform region" (protuberance on the right side of Fig. 5 *a* and *b*) are surrounded by regions of positive potential, which could help stabilize complexes of the ribosome with mRNA and tRNA. However, more quantitative calculations will be needed to fully explore these hypotheses.

The electrostatic potential of the 50S subunit is shown in Fig. 6 *b* and *d*. Like the small subunit, the electrostatic surface potential is largely negative, with scattered regions of positive potential typically associated with ribosomal proteins. Some of the most interesting aspects of this data set are the regions of positive potential on the 50S "crown" (see upper protuberance in Fig. 6 *a* and *b*), which correspond to proteins of the large subunit involved in tRNA binding. Specifically, these positive regions are due to proteins L44e, L5, and L10e (see Fig. 6*a*), which have been implicated in tRNA binding to the 50S subunit (35), and contribute large regions of positive potential to the molecular surface in the upper portions of the A, P, and E tRNA binding sites. Furthermore, the P-loop (shown as blue spheres in Fig. 6*a*), an important component of the 50S P-site (35, 36), shows significant positive surface potential due to a nearby Mg^{2+} ion. These areas of positive potential may provide stabilizing interactions with mRNA or tRNA bound to the ribosomal complex during translation. However, as with the 30S calculations, more detailed work is required to accurately resolve the role of electrostatics in tRNA and mRNA binding to the 50S subunit.

Conclusions

We have described the combination of standard finite difference focusing techniques and the Bank-Holst algorithm into a parallel focusing method to facilitate the solution of the PBE for nanoscale systems. Unlike previous multiprocessor algorithms for solving the PBE, this method has excellent parallel complexity that permits the solution of these problems on massively parallel computational platforms. Furthermore, the finite difference discretization on a regular mesh allows for fast solution by highly efficient multigrid solvers. This algorithm has been implemented in the APBS software by using the PMG multigrid solver and tested on the NPACI IBM Blue Horizon supercomputer.

Using these methods for the parallel multigrid solution of elliptic differential equations, the electrostatic properties of very large biomolecular assemblages are now amenable to computation. This technique relies on the efficient solution of the PBE combined with parallel focusing techniques to solve these large problems in a variety of distributed computational environments. Solution of the LPBE for the 1.2 million-atom microtubule system provided electrostatic potential data, which revealed interesting features near drug binding sites and provided possible insight into stability differences at the + and − ends of the microtubule. Such detailed electrostatic information will be central to future studies that examine the possible collective effects involved in the formation of structural defects and the stabilizing effects of taxol binding to the interior of microtubules. Likewise, application of this methodology to ribosome systems elucidated intriguing electrostatic properties of the ribosomal active site, which will provide the starting point for investigation of the stabilization of the tRNA- and mRNA-ribosome complexes during translation and the rational design of novel antibiotics. Finally, the ability to determine the contribution of electrostatics to the forces and energies of nanoscale systems should extend the scale of implicit solvent dynamics methods to much larger macromolecular complexes.

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ENHANCED ANTI-STOKES RAMAN SCATTERING FROM LIVING CELLS OF *CHLORELLA PYRENOIDOSA*

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The anti-Stokes/Stokes ratios for living *Chlorella pyrenoidosa* cells at 294 K considerably exceed those given by the thermal population factors $[n/(n+1)]$. We have shown that resonance enhancement in the scattered photons is mainly responsible for this, but cannot rule out that some population enhancement occurs.

Recent studies [1,2] have shown that the investigation of Raman scattering from photosynthetic systems can provide information on the chemical and structural composition of the absorbing pigments, and the molecular surroundings in which they are found. Resonances of the incident laser light with visible absorption bands of these pigments lead to an enhancement of the scattering. The use of such resonances to investigate specific molecules in heterogeneous biological systems is rahter widespread [3].

We have measured Raman spectra of active *Chlorella pyrenoidosa* cells at 294 K in an attempt to clarify questions related to the role of vibrational states in biological membranes as predicted by theoretical considerations [4–7] and as suggested by some experimental observations [8,9]. According to the theory, a non-thermal population of vibrational states can occur in living systems which should lead to enhanced anti-Stokes scattering.

We have observed enhanced anti-Stokes scattering for several modes in *Chlorella pyrenoidosa* but point out that a large part of this arises from resonances in the scattered photon channel which is different for Stokes and anti-Stokes photons. Population factors derived from our experimental data show, however, that we cannot rule out that some population enhancement occurs.

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The starting culture of *Chlorella pyrenoidosa* cells was obtained from D. Fork, Carnegie Institution, Stanford, and grown in nutrient solution as proposed by Lorenzen [10]. A CO_2–air mixture (1.5% CO_2) was bubbled through the cell culture which was illuminated with fluorescent lamps. The Raman measurements were performed with a flow arrangement as described elsewhere [11] to avoid heating the cells with the laser beam and to be able to keep them in a rather adapted state. The cell concentration was 10^7 cells/mℓ with a flow speed of 40 cm/s. White background light was available for illumination of the reservoir in order to investigate a possible relation between Raman intensities and the activity of photosynthesis. All spectra were recorded at $\sim 5\ \mathrm{cm}^{-1}$ resolution using Ar^+-laser excitation, a 1 m-double monochromator and conventional photon counting detection.

The Stokes Raman spectrum of active *Chlorella pyrenoidosa* cells at 294 K is shown in fig. 1 for a laser wavelength of 5145 Å (19 436 cm^{-1}). The main features are a few groups of sharp lines (width $\sim 10\ \mathrm{cm}^{-1}$) between 800 and 1600 cm^{-1} superimposed on a rather structureless background. The anti-Stokes Raman spectrum for this range is shown in fig. 2 for the same laser frequency and power.

The intensity ratios R of the main anti-Stokes lines with respect to their Stokes counterparts ($R = I_{as}^{19436}/I_{s}^{19436}$) deviate considerably from the thermal population factor $[n(\Omega, T)/(n(\Omega, T)+1)]$ which is

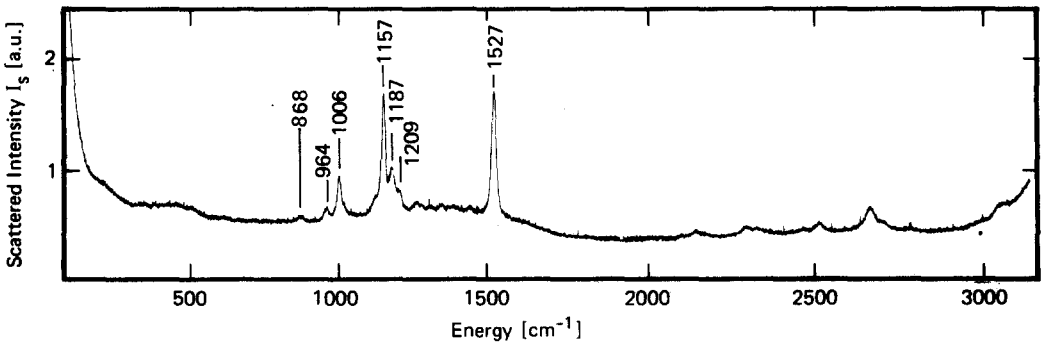

Fig. 1. Stokes Raman spectrum measured at 294 K with active cells of the green alga *Chlorella pyrenoidosa* during irradiation at 5145 Å. The bands at 868 cm^{-1}, 964 cm^{-1}, 1006 cm^{-1}, 1157 cm^{-1}, 1187 cm^{-1}, 1209 cm^{-1} and 1527 cm^{-1} are assigned to vibrational modes of carotenoid molecules [1].

given by a Planck equilibrium (Ω is the vibrational mode frequency, $T = 294$ K, see table 1, column 3). The dependence of the anti-Stokes intensity on laser power shows that local heating by the laser is not responsible for its enhancement.

As the laser wavelength was changed between 5145 Å and 4545 Å substantial changes in the scattered Stokes intensities occurred which overwhelmed the non-resonant ω^4 behaviour (fig. 3), and are attributed resonance with the absorption band of carotenoid pigments.

Furthermore, the influence of additional light (i.e., illumination of the cells in the reservoir with white light) on the intensity of the Stokes lines at 868 cm^{-1}, 1006 cm^{-1} and 1527 cm^{-1} was measured. Our preliminary results show that a reproducible increase in the intensity of those lines occurred which amounted to ~5% and which reached its maximum value approximately 60 s after light was switched on.

We concentrate our attention on the enhanced anti-Stokes/Stokes ratios R of the Raman lines at 868 cm^{-1}, 1006 cm^{-1}, 1157 cm^{-1} and 1527 cm^{-1}. These are assigned to vibrations of carotenoid molecules by Lutz [1].

The intensity ratio R for excitation at laser frequency ω is given by eq. (1):

$$R(\Omega, \omega, T) = \frac{I_{as}^{\omega}}{I_{s}^{\omega}} = a\frac{(x-\Omega)^2}{(x+\Omega)^2} P(\Omega, T), \qquad (1)$$

where a accounts for reabsorption of Stokes and anti-Stokes light, the second factor with $x = \omega - \omega_e$ expresses the resonance term, and P is the population factor which in normal thermal equilibrium is just $n/(n+1)$. Relaxing the assumption of a single resonance does not affect the essentials of our arguments.

Far from resonance both a and Ω can be neglected and R measures the population factor. On the other hand near resonance, and particularly for large Ω,

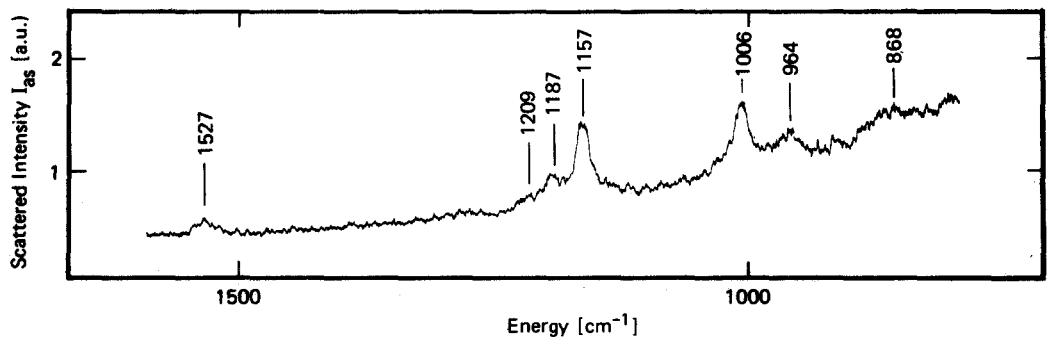

Fig. 2. Anti-Stokes spectrum between 800 and 1600 cm^{-1} measured under the same experimental conditions as the Stokes spectrum shown in fig. 1. Each Stokes shifted line has its counterpart in the anti-Stokes spectrum. The intensity ratio $I_{as}^{19436}/I_{s}^{19436}$ appears to be much larger than given by thermal population factors assuming a Planck equilibrium.

Table 1

Vibrational energy Ω (cm^{-1})	Observed ratio R $I_{as}^{19436}/I_{s}^{19436}$	Anti-Stokes enhancement $R/n(n+1)$	Res. factor $I_{s}^{19436+\Omega}/I_{s}^{19436}$ from fig. 3	$\frac{P}{n/(n+1)}$
868	0.1	10.3	4.0	2.6±0.5
1006	0.060	7.2	5.6	1.3±0.1
1157	0.020	5.3	4.1	1.3±0.1
1527	0.0024	3.5	2.7	1.3±0.2

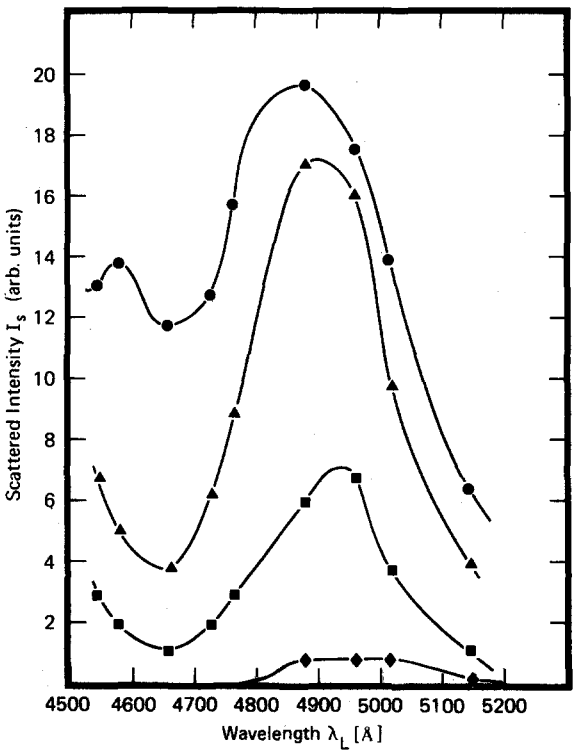

Fig. 3. Intensity dependence of the Stokes shifted bands at 868 cm^{-1} (♦), 1006 cm^{-1} (■), 1157 cm^{-1} (▲) and 1527 cm^{-1} (●) on the wavelength of the incident laser light.

R can be dominated by the frequency-dependent cross section factor. We have attempted to separate this from the population factor P by considering the ratio of Stokes intensity $I_s^{\omega+\Omega}$ (laser at $\omega+\Omega$) and anti-Stokes intensity I_{as}^{ω} (laser at ω). In this configuration it can be shown from the usual equations for Raman intensities that

$$P = I_{as}^{\omega}/I_s^{\omega+\Omega} = R(I_s^{\omega}/I_s^{\omega+\Omega}), \qquad (2)$$

is independent of resonance and reabsorption factors.

Eq. (2) can be rewritten to show the anti-Stokes enhancement as:

$$\frac{R}{n(n+1)} = \frac{I_s^{\omega+\Omega}}{I_s^{\omega}} \frac{P}{n(n+1)} . \qquad (3)$$

Table 1 shows values for the factors in eq. (3), the ratio $I_s^{\omega+\Omega}/I_s^{\omega}$ being obtained from fig. 3.

The population ratio P appears to be enhanced by factors ranging from 1.3 to 2.6 if compared with the thermal population factor $n/(n+1)$ (enhanced population factors have been suggested by refs. [4–7]).

However, the major part of the experimentally measured anti-Stokes enhancement can be explained as due to a resonance with the carotenoid absorption band at ~480 nm. This shows clearly the importance of an analysis of resonance enhancements for investigations of population effects.

We are grateful to H. Fröhlich for suggesting the experiments, and to F. Keilmann for assistance in initial measurements.

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EVIDENCE FOR BOSE CONDENSATION-LIKE EXCITATION OF COHERENT MODES IN BIOLOGICAL SYSTEMS

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Experimental evidence of the action of coherent millimeter electromagnetic radiation on biological systems shows very striking effects which are strongly frequency selective at very low power supply thus indicating resonance effects. The existence of a threshold for the power supply below which no effects are found strongly supports the earlier theoretical prediction that energy supply above a critical rate will lead to excitation of polar coherent modes in the system in a manner comparable to Bose condensation.

Some time ago it was suggested that many biological systems have a branch of polar modes in the $10^{11} - 10^{12}$ s^{-1} frequency region, and that supply of energy to these modes leads to very strong (coherent) excitation of a single mode provided the rate of supply of energy, S, is larger than a critical one, S_0, i.e. $S > S_0$, [1–3]. In the formal treatment the transition from $S < S_0$ to $S > S_0$ shows great similarity with Bose condensation.

The vibrations are considered to involve regions of biological membranes, or substantial sections of giant molecules – such as proteins, DNA, etc. Coherent excitation of these modes could have very far reaching biological influences in particular in the case that the excited mode represents a giant dipole oscillation (polarisation wave with zero wave number, $k = 0$). Amongst the many consequences of such excitations are the establishment of long range and short range selective forces [4] which so far have been invoked to interpret a number of specific biological features [5, 6], but which might in very many cases be the agent which initiates positional or conformational changes from which a variety of biological effects might arise (cf also [7]).

While the vibrations with higher wave numbers may be expected to be largely – though not exclusively – longitudinal no such classification holds for the giant dipole ($k = 0$) oscillation. Furthermore the linear dimension of most of the relevant biological objects (molecules, membranes) is less than the wave length of electromagnetic radiation in the $10^{11} - 10^{12}$ s^{-1} frequency region. It follows then that radiation in resonance with the $k = 0$ frequency would be selectively absorbed by the material. The energy of excitation (cf [3] with minor modifications for the present case) would then be shared with other modes as long as $S < S_0$. Ultimately this would lead to an increase in temperature. For $S > S_0$, however, a single vibrational mode would be strongly (coherently) excited in a manner comparable to a Bose condensation. This mode may – but need not – be the mode $k = 0$ which absorbs the radiation; in this case S_0 would be relatively small.

Recently in a report of a meeting of the USSR Academy of Science (January 1973) [8] several remarkable contributions have demonstrated a striking influence of 6–7 mm coherent electromagnetic radiation on the behaviour of a variety of biological objects. The corresponding frequency of $\frac{1}{2} \times 10^{11}$ s^{-1} is close to the theoretically predicted region for such effects. while the reported biological effects represent without doubt very complex processes they exhibit two very simple common features; (i) they depend very strongly on frequency exhibiting a typical resonance behaviour, and (ii) (cf. fig. 1 of [9] and fig. 3 of [10] there exists a critical power supply $S = S_0$ below which no effects are observed and above which the observed effects are nearly independent of S.

These features strongly support the theoretical prediction of the existence of polar vibrations in the mentioned frequency region and of their biological importance when excited coherently. The crucial point here is the evidence for the existence of a threshold S_0 such that for $S > S_0$ a complete change of behaviour occurs comparable to a phase transition at $S = S_0$.

The theoretical interpretation of these results rests on the proposal that arising from the $k = 0$ resonance absorption the experimentally observed transition at $S = S_0$ consists in the coherent excitation of the polar mode mentioned above which does exhibit the required transition at $S = S_0$. This coherent excitation which is possible only when $S > S_0$ should then initiate the essential first step (e.g. selective attraction of molecules, opening of a section of a DNA complex, etc) in a possibly very complicated sequence of biological events. The time t_1, required for this first step will decrease with increasing coherent excitation, i.e. with S (provided $S > S_0$), but the time t_2 for all the subsequent processes will be independent of S. An S value will thus be reached for which $t_1 \ll t_2$ holds so that for larger S the total rate will become nearly independent of S in agreement with the observations.

It thus appears that the very simple common features discovered in experiments on a variety of biological objects find their natural interpretation in the previous theoretical predictions. It should be emphasised that the new concepts introduced there make use of a greatly simplified model. These concepts are, however, of a very general nature and permit the considerable refinements and modifications of the model which will be required for the detailed description of the observed phenomena. It is very striking to see, however, that complex biological phenomena may contain common features of very great simplicity in terms of the appropriate physical concepts.

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EVIDENCE FOR NON-THERMAL EXCITATION OF ENERGY LEVELS IN ACTIVE BIOLOGICAL SYSTEMS

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The ratio R of the intensities of anti Stokes and Stokes Raman shift lines of 124 cm^{-1} and of 118 cm^{-1} of active *E. coli* B was measured. For an oscillating system in therm equilibrium $R \approx 0.55$ is expected whereas we find $R \approx 1.0$. This shows that this system is excited strongly above thermal excitation in agreement with a previous theoretical conjecture [2–5].

Recent investigation of Raman effect in bacteria has demonstrated [1] that Raman shift lines below 200 cm^{-1} are found in active (metabolizing) synchronised cells whereas broad bands only are exhibited by both resting cells and nutrient solutions. These findings were considered a first step towards confirmation of the theoretical prediction [2–5] that metabolic energy may be used by biological systems to excite coherent vibrations in the region of 10^{11}–10^{12} Hz.

Details of the spectra found in [1] depend on both the type of bacteria and the type of nutrient. To show that these Raman lines represent excitations beyond thermal equilibrium requires the ratio R of the intensities of anti Stokes: Stokes lines (in an oscillating system) to lie close to $R \approx 1$ when the thermal equilibrium value $R = \exp(-h\nu/kT)$ is considerably smaller (ν = Raman shift).

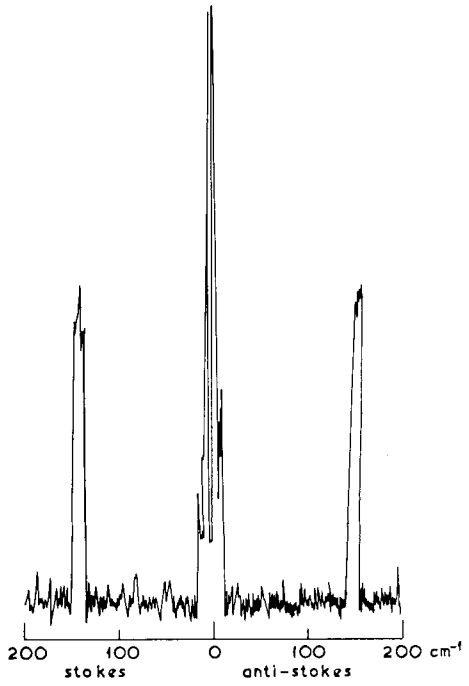
We have, therefore, measured R for active cells of *E. coli* B (glucose nutrient). Experimental procedures were as in [1] except that the sensitivity setting for the Rayleigh lines was reduced by a factor 200 compared with that of the shift lines. It should also be reemphasized that the incubation temperature must be held above 25°C [6].

Figs. 1a, b, c give typical results of spectra taken 40, 50, and 60 minutes after begin of incubation. A total of eight series of such spectra was taken so far. We note that the line near 120 cm^{-1} ranges between 118 cm^{-1} and 125 cm^{-1} depending on the age of the cell. At these frequencies the thermal equilibrium value R lies between 0.55 and 0.57. Experimentally, however, we find $R = 1.01 \pm 0.10$ for 118 cm^{-1} and $R = 0.93 \pm 0.13$ for 125 cm^{-1} after averaging over the eight spectra. These values clearly are far above the thermal equilibrium values and thus constitute conclusive confirmation of the theoretical conjecture.

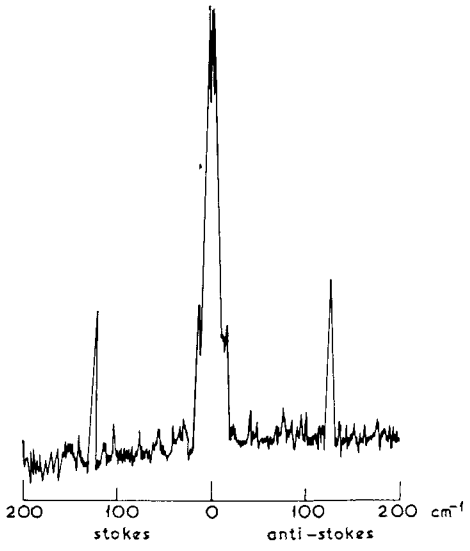
These excitations should be interpreted as arising from certain activities of the cell which change with its age. We note in this context the band near about 40 cm^{-1} which occurs for a short period only.

Finally the lines near 5 cm^{-1} should be noted. They might form a link with biological effects of coherent mm waves (near 2 cm^{-1}) [7].

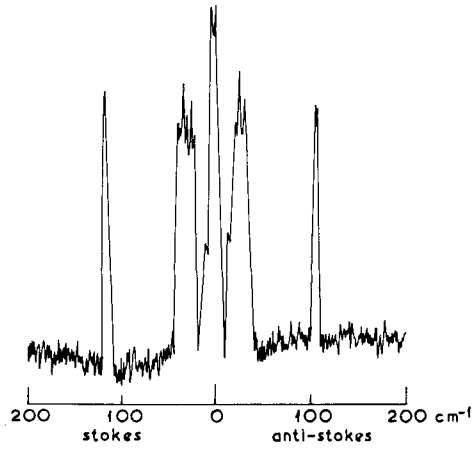
We are grateful to Jane Montgomery for help in the experiments.



(a)



(b)



(c)

Fig. 1a, b, c. Raman spectra of synchronised cells of *E. coli* B taken 40, 50, and 60 minutes after begin of incubation.

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Exact solutions to the time-dependent Landau-Ginzburg model of phase transitions

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Using the Landau-Ginzburg Hamiltonian, which provides a phenomenological description of phase transition, we derive its Lagrangian and subsequently the Lagrange-Euler equations of motion for the order parameter. In the case of one-dimensional systems these equations appear in the form of a nonlinear Schrödinger equation and are solved exactly following the method of Hasse, using the assumption of constant envelope velocity. The resulting time-dependent solutions possess a permanent profile, and their role and range of existence are discussed. The temperature dependence of their velocities and widths is analyzed.

I. INTRODUCTION

This paper is intended to generalize the Landau-Ginzburg model of phase transitions and to investigate the resulting time-dependent solutions of the equation of motion of the order parameter η. We treat the order parameter as a complex time- and space-dependent function and utilize the Hamiltonian approach. The potential function appears as a power series up to order $2n$ in $|\eta|$. The equation of motion also includes a dissipative term. We begin by reviewing the original Landau-Ginzburg approach.

Landau[1] developed a phenomenological theory of second-order phase transitions by expanding the Gibbs free energy G of the system in a series of symmetry invariants of the low-temperature symmetry group g_0 which is also a subgroup of the high-temperature symmetry group g_1,

$$G(T,P)=G_0(T,P)\eta^2+A_4\eta^4+\cdots , \tag{1}$$

where P denotes the external field conjugated with the order parameter η and G_0 is the Gibbs free energy of the high-temperature phase. In order to obtain the desired transition at T_c Landau postulated that

$$A_2(T,P)=a(P)\epsilon , \tag{2}$$

where $\epsilon=(T-T_c)/T_c$ is the normalized temperature. The extremal value of η, $\bar{\eta}$, is found simply by minimizing G,

$$\left[\frac{\partial G}{\partial \eta}\right]_{\eta=\bar{\eta}}=0 ,$$

and using the stability conditions

$$\left[\frac{\partial^2 G}{\partial \eta^2}\right]_{\eta=\bar{\eta}}>0 ,$$

$$\bar{\eta}=\begin{cases}\pm[-(a/2A_4)\epsilon]^{1/2}, & T<T_c\\ 0, & T\geq T_c .\end{cases} \tag{3}$$

It is required that $A_4>0$ and $a>0$ for the low-temperature phase to be ordered.

When the expansion Eq. (1) is carried out to the next order,

$$G(T,P)=G_0(T,P)+a(P)\epsilon\eta^2+A_4\eta^4+A_6\eta^6+\cdots , \tag{4}$$

then $\bar{\eta}$ becomes

$$\bar{\eta}=\begin{cases}\dfrac{\pm[-A_4+(A_4^2-4aA_6\epsilon)^{1/2}]^{1/2}}{3A_6}, & T<T_c^*\\ 0, & T\geq T_c^*\end{cases} \tag{5}$$

where the new transition temperature T_c^* is

$$T_c^*=T_c+(A_4^2/4aA_6) , \tag{6}$$

and the transition may be either of second or first order depending on whether $A_4>0$ or $A_4<0$, respectively.[2]

In order to include spatial inhomogeneities of the order parameter, when these inhomogeneities are slowly varying, Ginzburg[3] added another correction to the original Landau expression,

$$G=G_0+\int(A_2\eta^2+A_4\eta^4+D\,|\,\vec{\nabla}\eta\,|^2)dV , \tag{7}$$

where the sixth-order term of Eq. (4) has been dropped and D is called the dissipation constant. Minimizing this functional with respect to η one obtains[4]

$$2(a\epsilon+2A_4\eta^2-D\nabla^2)\eta=0 , \tag{8}$$

which has two trivial, spatially homogeneous solutions [those of Eq. (5)] and one nontrivial solution (in one-dimensional systems) representing a boundary wall between two possible ordered phases of opposite signs,

$$\eta=\eta_0\tanh[x/\sqrt{2}\zeta(T)] , \tag{9}$$

where $\eta_0=\eta(x\rightarrow\infty)$ and $\zeta^2(T)\equiv D/|A_2|$. Here ζ is the coherence length. The latter solution exists only below T_c and only if $D>0$.

In order to consider time-dependent phenomena one

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uses the Gibbs free energy G from Eq. (7) and arrives at the Langevin[5] equation (without the presence of external forces),

$$\dot{\eta}=(-2)\left[(A_2=2A_4\eta^2)\eta-D\frac{\partial^2\eta}{\partial x^2}\right], \tag{10}$$

where the constants, A_2, A_4, and D should be taken in units of the rate constant as Eq. (10) is a rate equation. This equation has been termed as "rather hopeless"[15] to be solved exactly. Therefore, a usual procedure has been to linearize it and use various approximations to perturb the linearized solution.[6] However, Parliński and Zieliński[7] discussed the first-order structural phase transitions in a recent paper and postulated two types of special solutions to Eq. (10). The Gibbs free energy they used can be represented as

$$G=G_0+\int\left[A_2\eta^2+A_{n+2}\eta^{n+2}+A_{2n+2}\eta^{2n+2}+\frac{D}{2}(\vec{\nabla}\eta)^2\right]dV, \tag{11}$$

where $n=2$ corresponds to the standard Landau functional and $n=1$ may be the case in systems where η is time reversal or space-inversion invariant. They obtained an equation of motion analogous to Eq. (10) and postulated the following solutions (which they checked by substitution). (i) A kink in the form of

$$\eta=\eta_0\{1+\exp[\delta(X-X_0-vt)]\}^{-1/n}, \tag{12}$$

where η_0 corresponds to $\bar{\eta}$ of Eq. (5), δ is the reciprocal width, and v is the velocity. (ii) A kink-antikink couple in the form of

$$\eta=\eta_0(1+\exp\{-\delta[X-X_A(t)]\})^{-1/n}\times(1+\exp\{\delta[X-X_B(t)]\})^{-1/n}. \tag{13}$$

The first solution is interpreted as an interface separating the ordered phase from the disordered one, whereas the other solution describes a nucleation center of the ordered phase. Both solutions have permanent profiles and travel with constant velocities. All of their parameters depend significantly on temperature.

It is our intention to provide a derivation of exact solutions of a generalized time-dependent Landau-Ginzburg model without restricting the order of the transition. Instead of the free Gibbs energy we will employ the Hamiltonian formalism and the order parameter will be a complex function. The resultant equation which replaces Eq. (10) is

$$-D\eta_{xx}+[2A_2+(n+2)A_{n+2}|\eta|^2+(2n+2)A_{2n+2}|\eta|^{2n}]\eta=-i\eta_t. \tag{14}$$

II. EQUATION OF MOTION

In this paper we follow the approach taken by Ma[8] and employ a block Hamiltonian for a system undergoing phase transition. The free energy of the system is therefore not used. The necessary steps leading to the final form of the equation of motion may be conveniently summarized as follows.

(1) We use the continuum approximation and include the possibility of inhomogeneities by writing an appropriate Hamiltonian H as

$$H=\int dx[(m/2)\dot{\eta}(x)^2+A_2|\eta|^2+A_{n+2}|\eta|^{n+2}+A_{2n+2}|\eta|^{2n+2}-(D/2)|\vec{\nabla}\eta|^2]\equiv\int dx\,\mathscr{H}(x), \tag{15}$$

where (x) is the Hamiltonian density.

(2) The Landau condition given by Eq. (2) is adopted.

(3) We invoke the Gaussian approximation with a probability density ρ given by

$$\rho=Z^{-1}\exp[-(\eta-\tilde{\eta})^2/kT]. \tag{16}$$

Here Z is the appropriate normalization constant and $\tilde{\eta}$ is the most probable value of η. Within this approximation $\tilde{\eta}=\bar{n}$.

(4) Assuming that the kinetic energy term in Eq. (15) can be treated as a constant we replace the block Hamiltonian by an approximate equivalent Hamiltonian expressed as

$$H=\int dx[_0+A_2\bar{\eta}^2+A_{n+2}|\bar{\eta}|^{n+2}+A_{2n+2}|\bar{\eta}|^{2n+2}-(D/2)|\vec{\nabla}\bar{\eta}|^2]. \tag{17}$$

Here $\bar{\eta}=\bar{\eta}(x)$.

(5) We express $\bar{\eta}$ as a complex function depending upon both space (x) and time (t),

$$\bar{\eta}=|\eta(x,t)|e^{i\phi(x,t)}\equiv\eta e^{i\phi}, \tag{18}$$

where for the sake of convenience we have also denoted $\eta\equiv|\bar{\eta}|$.

(6) By using the Legendre transformation we convert the Hamiltonian density (x) into a Lagrangian density $L(\eta_x,\eta_t,\eta)$. This transformation is readily carried out as follows:

$$H=L-\pi\eta_t,\quad \pi=\frac{\partial L}{\partial\eta_t}, \tag{19a}$$

$$\eta_t=\frac{\partial\eta}{\partial t},\quad \eta_x=\frac{\partial\eta}{\partial x}. \tag{19b}$$

From Eqs. (19) it is easy to see that π is the momentum density conjugate to η. The Lagrangian density resulting from these calculations has the form

$$L=(i/2)(\eta\eta_t^*-\eta^*\eta_t)(D/2)\eta_x^2-A_2|\eta|^2-A_{n+2}|\eta|^{n+2}-A_{2n+2}|\eta|^{2n+2}-_0. \tag{20}$$

(7) The standard Lagrange-Euler equations of motion may be written as

$$\frac{\partial}{\partial x}\frac{\partial L}{\partial\eta_x}+\frac{\partial}{\partial t}\frac{\partial L}{\partial\eta_t}-\frac{\partial L}{\partial\eta}=0. \tag{21}$$

(8) Substituting Eq. (20) in Eq. (21) we get

$$-D\eta_{xx}+[2A_2+(n+2)A_{n+2}|\eta|^n+(2n+2)A_{2n+2}|\eta|^{2n}]\eta=-i\eta_t. \tag{22}$$

This is our final equation of motion and is an example of the well-known nonlinear Schrödinger equation.[9] Hasse[10] has presented a method which can be adapted for the exact solution of Eq. (22). In the next section we turn our attention to this aspect of the problem.

III. SOLUTION OF THE NONLINEAR SCHRÖDINGER EQUATION

Representing the potential term in Eq. (22) by

$$W(\eta)=2A_2(n+2)A_{n+2}|\eta|^n+(2n+2)A_{2n+2}|\eta|^{2n}, \tag{23}$$

and substituting $\bar{\eta}$ from Eq. (18) into Eq. (22), we obtain

$$i(\eta_t+i\eta\phi_t)=Dn_{xx}+2iD\eta_x\phi_x+iD\eta\phi_{xx}-D\eta\phi_x^2 - W(\eta)\eta . \tag{24}$$

We may now separate the real and imaginary parts as follows.

(i) The imaginary part yields

$$\eta_t=2D\eta_x\phi_x+D\eta\phi_{xx} , \tag{25}$$

which when multiplied through by 2η transforms into

$$\frac{\partial}{\partial t}(\eta^2)=\frac{\partial}{\partial x}(\eta^2)(2D\phi_x)+\eta^2(2D\phi_{xx}) . \tag{26}$$

Introducing the quantum velocity as

$$v=-2D\phi_x \tag{27}$$

and the density as

$$\rho(x,t)=\eta^2(x,t) , \tag{28}$$

we rewrite Eq. (26) in the form of an equation of continuity,

$$\rho_t=-(\rho V)_x . \tag{29}$$

(ii) The real part yields

$$-\eta\phi_t=D\eta_{xx}-D\eta\phi_x^2-W(\eta)\eta , \tag{30}$$

which can be transformed into

$$\phi_t-W(\eta)-(1/4d)v^2+(D/\eta)\eta_{xx}=0 . \tag{31}$$

Integration of Eq. (27) yields

$$\phi(x,t)=(1/2D)\int_{x'=vt}^{x'=x}dx'v(x',t)+f(t) , \tag{32}$$

where the function $f(t)$ is selected in the manner suggested by Hasse[10] such that the expectation value of the Hamiltonian gives the energy of the wave packet. To carry out this program we interpret $(2D)^{-1}$ as the effective mass of the quasiparticle described by the Schrödinger equation, Eq. (22): $m=(2D)^{-1}$, and we write

$$\langle H\rangle=-\langle\phi_t\rangle=mv^2/2=v^2/4D . \tag{33}$$

We now assume v to be constant and set $f(t)=v^2/4Dt$ to obtain

$$\phi(x,t)=(1/2D)v(x-vt)+(v^2/4D)t , \tag{34}$$

which yields $-\phi_t=-\langle\phi_t\rangle=v^2/4D$ as required by Eq. (33). Substituting this value of ϕ_t into Eq. (31) we arrive at

$$\eta_{xx}=[(1/2D)v^2+W(\eta)]\eta/D . \tag{35}$$

Because of the assumed constancy of v, Eq. (35) can be solved exactly upon introducing a new variable: $z\equiv(\eta_x)^2$. As a result we transform Eq. (35) into

$$\frac{dz}{d\eta}=\frac{2}{D}\eta\left[\frac{1}{2D}v^2+W(\eta)\right] , \tag{36}$$

which can be written in the integral form

$$z(\eta)=\frac{2}{D}\int_0^\eta d\eta\eta\left[\frac{1}{2D}v^2+W(\eta)\right] . \tag{37}$$

By substituting the definition of W from Eq. (23) and carrying out a termwise integration over η we obtain

$$z(\eta)=c_1\eta^2+c_2\eta^{n+2}+c_3\eta^{2n+2} , \tag{38}$$

where the constants are

$$c_1=\frac{1}{D}\left[2A_2+\frac{v^2}{2D}\right] , \tag{39}$$

$$c_2=\frac{2A_{n+2}}{D} , \tag{40}$$

$$c_3=\frac{2A_{2n+2}}{D} . \tag{41}$$

We now employ the definition of z to solve for η

$$\pm(x-vt)=\int_{\eta_0}^{\eta}d\eta' z(\eta')^{-1/2} =\int_{\eta_0}^{\eta}d\eta'[\eta'(c_1+c_2\eta'^n+c_3\eta'^{2n})^{1/2}]^{-1} , \tag{42}$$

where $\eta_0=\eta(x=vt)$. Changing the integration variable to $\zeta=\eta^n$, Eq. (42) becomes

$$\pm(x-vt)=\frac{1}{n}\int_{\eta_0^n}^{\eta^n}d\zeta[\zeta(c_1+c_2\zeta+c_3\zeta)^{1/2}]^{-1} , \tag{43}$$

which upon another change of variable, $\tau=1/\zeta$ assumes the form of a standard elliptic integral,

$$\pm(x-vt)=-\frac{1}{n}\int_{\eta_0^{-n}}^{\eta^{-n}}d\tau(C_3+C_2\tau+c_1\tau^2)^{-1/2} , \tag{44}$$

whose value depends on the constants c_1, c_2, and c_3, and another constant defined as

$$\Delta=4c_3c_1-c_2^2=\frac{4}{D^2}\left[2A_{2n+2}\left[2A_2+\frac{v^2}{2D}\right]-A_{n+2}^2\right] . \tag{45}$$

Equation (44) is obtained, after a series of transformations, as a result of using the Hamiltonian, Eq. (15), to derive the equation of motion of η, Eq. (22). The latter equation has been integrated out by assuming the form of η as in Eq. (18) and keeping the velocity v constant. The variable τ is simply $\tau=\eta^{-n}$. The analytical expressions for the integral Eq. (44) are given in Appendix A. Substituting those expressions into Eq. (44) and solving for η we

obtain the following possibilities.

(i) If $c_1>0$ and $\Delta>0$, then

$$\eta(x,t)=\{(1/2c_1)\sqrt{\Delta}\sinh[x_1\mp n\sqrt{c_1}(x-vt)] -c_2/2c_1\}^{-1/n}, \tag{46}$$

where

$$x_1=\sinh^{-1}[(2c_1\eta_0^{-n}+c_2)/\sqrt{\Delta}] . \tag{47}$$

(ii) If $c_1<0$ and $\Delta<0$, then

$$\eta(x,t)=\{(1/2c_1)\sqrt{-\Delta}\sin[x_2\pm n\sqrt{-c_1}(x-vt)] -c_2/2c_1\}^{-1/n}, \tag{48}$$

where

$$X_2=\sin^{-1}[(2c_1\eta_0^{-n}+c_2)/\sqrt{-\Delta}] . \tag{49}$$

(iii) If $c_1>0$ and $\Delta=0$, then

$$\eta(x,t)=\{(1/2c_1)\exp[X_3\pm n\sqrt{c_1}(x-vt)]-c_2/2c_1\}^{-1/n}, \tag{50}$$

where

$$X_3=\ln(2c_1\eta_0^{-n}+c_2) . \tag{51}$$

(iv) If $c_1>0$ and arbitrary Δ, then

$$\eta(x,t)=\left[\frac{1+\Delta}{4c_1}\sinh[X_4\pm n\sqrt{c_1}(x-vt)] +\frac{1-\Delta}{4c_1}\cosh[X_4\pm n\sqrt{c_1}(x-vt)]-\frac{c_2}{2c_1}\right]^{-1/n}, \tag{52}$$

where

$$X_4=\ln\{2[c_1(c_1\eta_0^{-2n}+c_2\eta_0^{-n}+c_3)]^{1/2}+2c_1\eta_0^{-n}+c_2\} . \tag{53}$$

Note that Eq. (52) becomes identical to Eqs. (50) and (46) if $\Delta=0$ and 1, respectively. A special use of this solution with $\Delta=-1$ and $n=4$ is listed in Table 1 of Hasse.[10]

All the solutions given in Eqs. (46), (48), (50), and (52) are essentially waves propagating through the system with a constant velocity v and a permanent profile. These formulas determine the real part of $\bar{\eta}$ [see Eq. (18)] which is related to the envelope wave,[9] whereas the carrier wave is given by $e^{i\phi}$ where ϕ is displayed in Eq. (34). Therefore, each of these time-dependent solutions can, in general, be written as

$$\eta(x,t)=\eta_0 f[\delta_e(\Delta X_e-V_e t)]\exp[i\delta_c(\Delta X_c-v_c t)] , \tag{54}$$

where the index e refers to the envelope and c refers to the carrier properties, η_0 is the amplitude, and f is the function of the profile expressed in (i)–(iv). It is readily seen that

$$\delta_c=\frac{v}{2},\quad v_c=\frac{v}{2},\quad \delta_e=n\sqrt{c_1},\quad v_e=v . \tag{55}$$

It is perhaps worth mentioning that apart from the time-dependent solutions (i)–(iv), Eq. (22) possesses a number of trivial, time- and/or space-independent solutions which have been presented in conjunction with the earlier models. These are the following.

(v) The solution corresponding to the disordered phase (stable for $T\geq T_c^*$),

$$\eta(x,t)=0 . \tag{56}$$

Here

$$\delta_c=v_c=\delta_e=v_e=\eta_0=0 .$$

(iv) The solution corresponding to the homogeneously ordered phase

$$\eta(x,t)=\pm\left[\frac{-(n+2)A_{n+2}+[(n+2)^2A_{n+2}^2-8(2n+2)A_2A_{2n+2}]^{1/2}}{2A_{2n+2}}\right]^{1/n} . \tag{57}$$

This solution exists only when $T<T_c^*$. Here $\delta_c=v_c=v_e=0$; $\delta_e=\infty$ and $\eta_0\neq 0$.

(vii) Provided $T<T_c^*$ and $D>0$, there exist solutions describing a stationary boundary wall between the solutions (vi) with opposite signs,

$$\eta(x,t)=\eta_0\tanh\frac{x}{\sqrt{2}\xi(T)} . \tag{58}$$

Here $\delta_c=v_c=v_e=0$; $\delta_e=[\sqrt{2}\xi(T)]^{-1}$. In the next section we proceed to analyze in detail the properties and existence ranges of the time-dependent solutions (i)–(iv).

IV. DISCUSSION OF THE SOLUTIONS

We now discuss the constraints imposed on the parameters c_1, c_2, and c_3 and through them on the coefficients A_2, A_4, A_6, and D in order for the time-dependent solutions to exist. From the requirements on the signs of c_1 and Δ stated for each of the solutions we easily deduce the following.

(1) Solution (i) requires that $D>0$ (dissipative systems) and the temperature range must be restrained to

$$T>T_c\left[1-\frac{v^2}{4aD}+\frac{A_{n+2}^2}{2A_{n+2}^2}\right]\equiv T_{c_1} . \tag{59}$$

However, T_{c_1} may be made arbitrarily small by the appropriate choice of parameters.

(2) Solution (ii) requires that $D<0$ and $T>T_{c_1}$.

(3) Solution (iii) exists only if $D>0$.

(4) Solution (iv) may exist both if $D<0$ and if $D>0$. In the former case one needs

$$T<T_c\left[1-\frac{v^2}{4aD}\right]\equiv T_{c_2} , \tag{60}$$

and in the latter case one needs $T > T_{c_2}$. From the form of these solutions we note that f contained in Eq. (54) is comprised of a function raised to the $(-1/n)$th power. Therefore, additional restrictions arise if n is even ($n=2$ is the most common case). In this case we require that in the entire range of $x-vt$, the function whose root is taken should be positive, otherwise unphysical divergences appear.

Upon analyzing this condition in all the four cases we find the following additional restrictions that have to be imposed in order for the solutions to attain physical meaning.

(1) Solution (i) requires that $c_2 < 0$ or else there will be at least one singular point. This combined with the previous requirement that $D > 0$ results in $A_{n+2} < 0$ which means that only systems exhibiting first-order transitions may possess this solution.

(2) Similarly, solution (ii) may only exist if $c_2\sqrt{-\Delta} > 1$ which implies that $c_2 > 0$ and $(c_2 - 4c_3c_1)^{1/2}$. However, c_3 is positive and c_1 is negative, and therefore the latter conditions can never be realized and consequently solution (ii) is unphysical.

(3) Solution (iii) exists only if $c_2 < 0$ which is equivalent to $A_{n+2} < 0$. The latter condition is satisfied only by systems undergoing first-order transitions.

(4) Solution (iv) can only be admitted if $c_2 < -2\sqrt{c_1 c_3}$, which in terms of the Hamiltonian constants, is written as

$$\frac{A_{n+2}}{D} < -\frac{1}{|D|}\left[2A_{2n+2}\left[2A_2 + \frac{v^2}{2D}\right]\right]^{1/2}. \quad (61)$$

This inequality is satisfied irrespective of the sign of A_{n+2} and therefore solution (iv) may exist in system exhibiting both first- and second-order transitions. The assumption that $D > 0$ is followed by $A_{n+2} < 0$, and conversely if $D < 0$; then it is required that $A_{n+2} > 0$. We conclude that solution (iv) may exist either in systems exhibiting first-order transitions with $D > 0$ or in systems exhibiting second-order transitions with $D < 0$. The final analysis we make is with respect to the form of the traveling solutions. Using Eq. (54) and Eqs. (46), (48), (50), and (52), we notice the following common features.

(1) The envelope velocity $v_e = v$ is given in Eq. (33) where we adopt

$$\langle H \rangle = A_2\bar{\eta}^2 + A_{n+2}\bar{\eta}^{n+2} + A_{2n+2}\bar{\eta}^{2n+2} \quad (62)$$

along with Eq. (57) for $\bar{\eta}$. Therefore we find that $\bar{\eta} \to 0$ as $T \to T_c^*$ and $\bar{\eta} = 0$ when $T > T_c^*$, $v \to 0$ as $T \to T_c^*$, and $v = 0$ when $T > T_c^*$. The envelope velocity v decreases continuously to zero with temperature if the transition is of second order and has an abrupt discontinuity at T_c^* in the case of a first-order transition. We conclude that all time-dependent solutions become stationary (or vanish altogether) above the transition temperature T_c^*.

(2) In all four cases the envelope width is $\delta_e = n\sqrt{|c_2|}$ and thus $\delta_e \to 0$ as $T \to T_c^*$.

(3) The profile of the time-dependent solutions is preserved throughout their propagation and can be desired as follows. Solution (i) is a kink-antikink couple, solution (ii) is a periodic function, solution (iii) is a kink, and solution (iv) is of an intermediate form between (i) and (iii). Solution (iv) becomes (i) when $\Delta = 1$ and (iii) when $\Delta = 0$. We have depicted these four profiles in Figs. 1–4.

(4) The amplitude for each solution can be written as

$$\eta_0 = (1/2c_1)^{-1/n} \text{ and } \eta_0 \to 0 \text{ as } T \to T_c^*.$$

The properties of solutions (i)–(iv) are summarized in Table I.

Finally, by comparison with the work of Parliński and Zieliński[7] we note that solution (iii) corresponds to their kinklike solution which has been interpreted as an interface between two allowed phases and solution (ii) corresponds to their kink-antikink couple which has been identified with the nucleation center. We emphasize here that we have found that these solutions exist only for first-order phase transitions in a limited temperature range and they require that $D > 0$ which characterizes a dissipative medium. The other two solutions which we include in Table I have not been demonstrated previously. The periodic solution, however, may only exist if n is odd (the order parameters must be invariant under time reversals

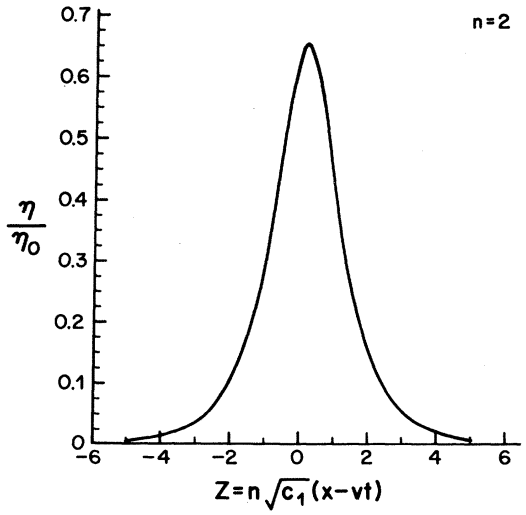

FIG. 1. Profile of solution (i).

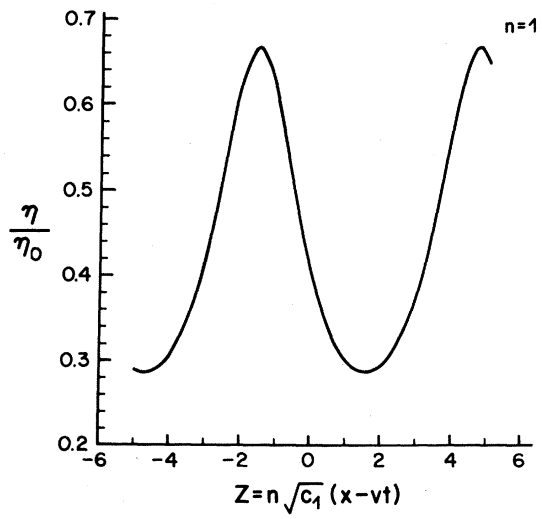

FIG. 2. Profile of solution (ii).

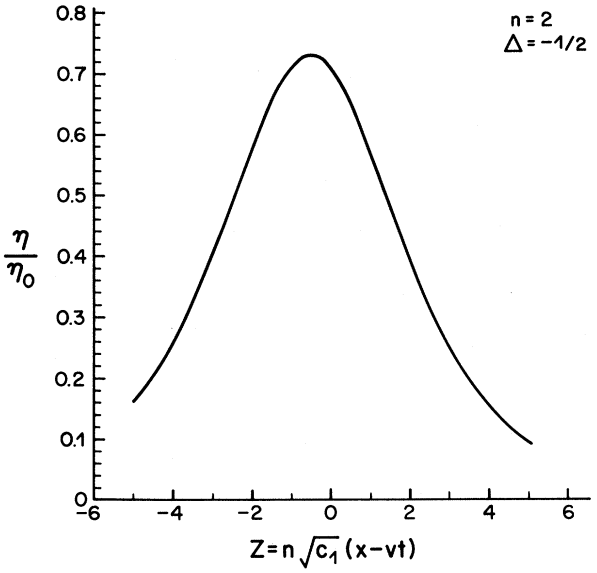

FIG. 3. Profile of solution (iii).

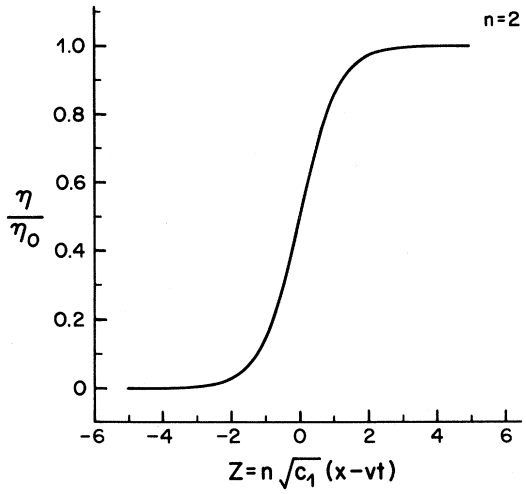

FIG. 4. Profile of solution (iv).

or space inversions). Furthermore, solution (iv) exists both in systems undergoing first- and second-order transitions. Both these new solutions appear if only the appropriate temperature ranges are ensured to coincide with the prescribed sign of the dissipation constant D. We may identify solution (ii) with elementary excitations[11] around the extremal value $\bar{\eta}$ and solution (iv) with local regions of order within the correlation length. Also notice that all the time-dependent solutions can propagate in both directions along the x axis and can have both positive and negative values of the amplitude as long as no external field breaking this symmetry is applied.

The physical meaning of the solutions with $D<0$ is, however, uncertain to the authors, as for the order parameter to be stable against fluctuations it is required that $D>0$. The opposite case can only be taken if a higher-order term proportional to $|\nabla^2\eta|$ is present to ascertain the stability. We intend to examine this question and also investigate the role of external fields in a future paper.

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APPENDIX

From the table of integrals[12] we find that the value of the following elliptic integral,

$$I=\int dx(c_3+c_2x+c_1x^2)^{-1/2}\,, \tag{A1}$$

depends on the sign of c_1 and $\Delta=4c_1c_3-c_2^2$, namely the following four possibilities exist.

(i) If $c_1>0$, then

$$I=(1/\sqrt{c_1})\ln\{2[c_1(c_3+c_2x+c_1x^2)]^{1/2}+2c_1x+c_2\}\,. \tag{A2}$$

TABLE I. Properties of solutions (i)—(iv).

Solution	Profile	Dissipation constant	Temperature	If $n=2$
(i)	Kink-antikink couple	$D>0$	$T_c^*\geq T>T_{c_1}$	First order
(ii)	Periodic function	$D<0$	$T_c^*\geq T>T_{c_1}$	Disappears
(iii)	Kink	$D>0$	$T_c^*\geq T$	First order
(iv)	Kink-antikink couple or kink depending on Δ	$D<0$	$T<T_{c_2}$, $T\leq T_c^*$	Second order
		$D>0$	$T_c^*\geq T>T_{c_2}$	First order

(ii) If $c_1 > 0$ and $\Delta > 0$, then

$$I = (1/\sqrt{c_1})\sinh^{-1}[(2c_1x + c_2)/\sqrt{\Delta}] . \quad \text{(A3)}$$

(iii) If $c_1 < 0$ and $\Delta < 0$, then

$$I = (-1/\sqrt{-c_1})\sin^{-1}[(2c_1x + c_2)/\sqrt{-\Delta}] . \quad \text{(A4)}$$

(iv) If $c_1 > 0$ and $\Delta = 0$, then

$$I = (1/\sqrt{c_1})\ln(2c_1x + c_2) . \quad \text{(A5)}$$

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Excitation of vibrations in microtubules in living cells

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Abstract

Microtubules, which are thought to be the primary organizers of the cytoskeleton, are electrical polar structures with extraordinary elastic deformability at low stress and with energy supply from hydrolysis of guanosine triphosphate (GTP) to guanosine diphosphate (GDP). At least a part of the energy supplied from hydrolysis can excite vibrations. Energy is mainly lost by viscous damping of the surrounding cytosol. Viscous damping is diminished by a slip layer which is formed by an attracted ionic charge layer and by a thin surface layer of the microtubule. Relaxation time caused by viscous damping may be several orders of magnitude greater than period of vibrations at 10 MHz. Energy supplied to the microtubule is of the order of magnitude of 10^{-14} W cm^{-1} (per unit length of the microtubule).
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Keywords: Biological order; Cytoskeleton; Microtubules; Coherence; Fröhlich coherence

1. Introduction

Physical processes in living matter participating in its organization are still not adequately understood. Frauenfelder et al. [1] claim than physical mechanisms of organization are of electromagnetic nature. Fröhlich pointed out that biological systems exhibit relative stability in a way in which some modes of behaviour remain very far from thermal equilibrium, postulated the existence of long range quantum mechanical phase correlations, and very strong excitation of a few modes of motion. Strong polar character of biological objects suggests longitudinal oscillations as stabilising modes [2–4]. The postulated rate equation for the *i*-th vibration mode may be written in a symbolic form

$$\dot{n}_i = s_i - [\dot{n}_i]_\phi - [\dot{n}_i(n_i, n_j)]_\chi$$

where $\dot{n}_i$ is the total rate of change of occupation number n_i (in 1 s), and s_i, $[\dot{n}_i]_\phi$, $[\dot{n}_i(n_i, n_j)]_\chi$ is the rate of change of n_i by energy supply, and by the linear and by the nonlinear interaction with the heat bath, respectively. The linear and the nonlinear term of the interaction with the heat bath describes transfer of one energy quantum between a vibration mode and the heat bath and energy transfer between two vibration modes with participation of the heat bath, respectively. The steady state solution to the rate equation shows that due to nonlinear term the supplied energy is not thermalized but condensed in a vibration mode. Excitation depends on the amount of the energy supplied to the system but not on the manner of its supply [5].

Eucaryotic cells are structurally and dynamically organized by a protein polymer network called the cytoskeleton [6]. The cytoskeleton exerts forces and generates movements without any major chemical changes. The cytoskeleton is a highly dynamic structure which reorganizes continually as the cell changes its shape, divides, and responds to its environment. The fundamental structure of the cytoskeleton formed by the microtubules satisfies the basic requirements for excitation of vibrations and generation of endogenous oscillating electric field [7]. Viscous damping by surrounding cytosol may quench excitation of vibrations. Viscosity effects on vibrations were examined using a model with longitudinal vibrations parallel to the planar interface between the microtubular medium below and the cytosol medium above the interface [8]. Slip boundary conditions make possible excitation of vibrations. Foster and Baish [9] used a cylindrical model of a microtubule but without slip boundary conditions and claimed that damping is so strong that excitation of vibrations is not possible. A cylindrical model of microtubule with the slip boundary conditions used in Refs. [10,11] confirmed the possibility of excitation. We present assessment of excitation in microtubules including viscous damping.

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2. Subunits of the microtubule—electric dipoles

The microtubule is a hollow cylindrical polymer of tubulin heterodimers (Fig. 1a). The outer and the inner diameter is about 25 and 17 nm, respectively. Tubulin heterodimer—composed of α- and β-tubulins—may exist in two conformation states (Fig. 1b): the α state and the β state where the electron negative charge is placed toward the α-tubulin and the β-tubulin, respectively. In the β state, the heterodimer is tilted 29° with respect to the microtubule axis. In the interphase microtubules are organized in a radial structure extending from the centrosome to the periphery of the cell (Fig. 2a). In the M phase, microtubules assemble from the two separated centrosomes to form a mitotic spindle (Fig. 2b).

Each heterodimer binds 18 calcium ions [12–14]. The electric dipole moment is about 1000 Debye (10^{-26} Cm). The induced dipole moment per dimer arising only from the motion of mobile electrons or protons was estimated to be 200–400 Debye [15]. Ionic concentration inside a cell is of about 150 mM of cations and the same concentration of anions [6]. The ions from the cytosol form a charge layer—a cylindrical envelope around the microtubule [15]. Similar situation may be inside the microtubule (Fig. 3) as the liquid medium in the inner circular cavity is very likely to be cytosol. The ionic charge layer, which is schematically shown in Figs. 1 and 3, screens electrical potential generated by the charges in the microtubule [15]. Within a distance of about 2 Debye lengths the static potential is considerably screened. The Debye length is of about 0.8 nm [16]. Due to hydration, the ions are surrounded by bound water molecules up to a distance of 0.4–0.5 nm. Charged proteins of considerably greater size than ionized atoms may be in the ionic layer too. The majority of water molecules in the charge layer are not free and its viscous properties are dependent on hydrated ions. Some ions can penetrate inside the tubulin structure and behave like a part

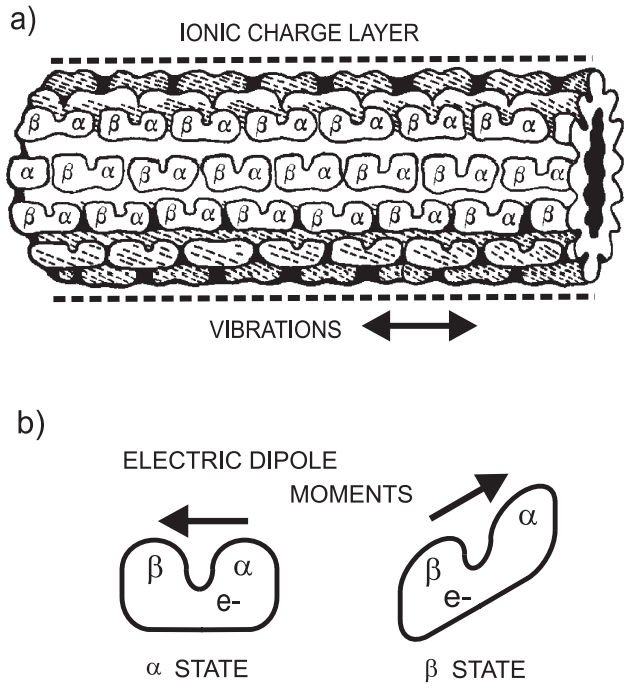

Fig. 1. (a) A schematic picture of a microtubule with tubulin heterodimers in parallel protofilaments and (b) conformation change of a heterodimer from the α to the β state after hydrolysis of GTP (guanosine triphosphate) to GDP (guanosine diphosphate) in the β-tubulin (b). The lines with arrows indicate the dipole moment vectors.

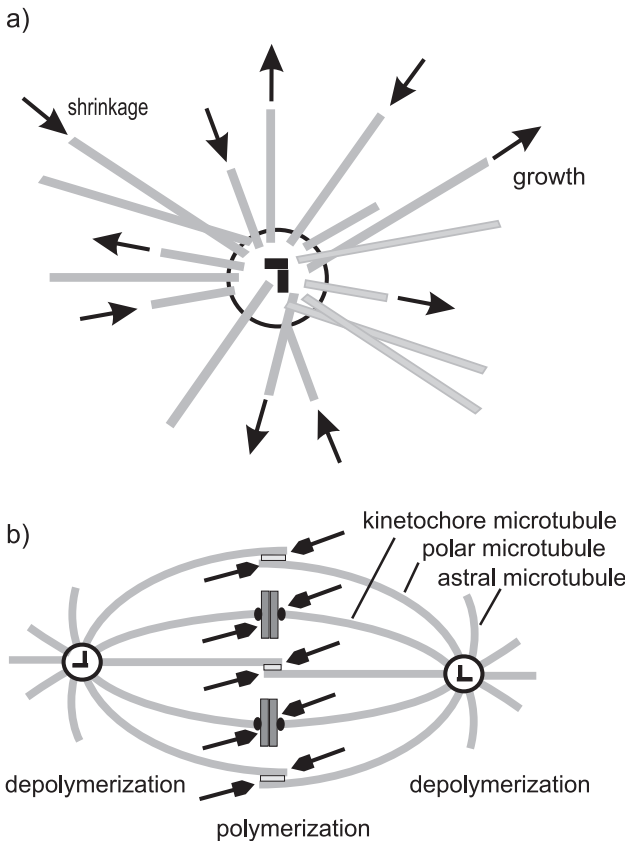

Fig. 2. (a) A centrosome with attached microtubules in the interphase. (b) The mitotic spindle in the M phase. The heterodimers with exhausted energy are exchanged for energy rich ones by the dynamic instability—the growth and shrinkage are indicated by arrows (a) and by the treadmilling—polymerization (indicated by arrows) and depolymerization take place at the center and at the poles of the mitotic spindle, respectively (b).

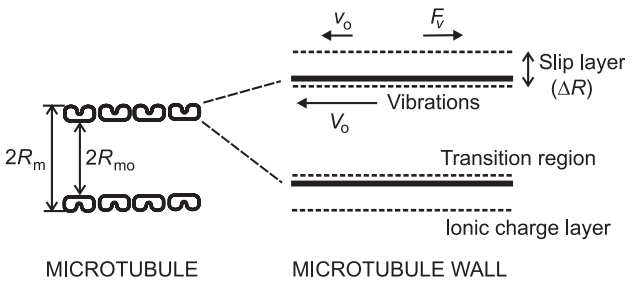

Fig. 3. A schematic picture of the microtubule wall with the slip layers. Each slip layer includes the ionic charge layer and the transition layer in the microtubule.

of the tubulin (as well as water molecules). Ions in the cytosol have small mobility and, therefore, there is slip between the charge layer and the microtubule surface.

We have to mention here that the cytoskeleton is a complicated structure with associated proteins cross-linking its parts, changing elastic properties of the structures and the viscosity effects. There are e.g. microtubule-associated proteins (MAP-1, MAP-2 forming sidearms), tau proteins and motor proteins. Phosphorylation of tau proteins plays very likely a role in microtubule binding and function [17]. Skeletor reticular matrix, spring like lattice around kinetochore microtubules, and protein cross-linked lattice around the spindle poles [18], are formed in the M phase. These structures might diminish viscosity damping but their description and analysis of their effects on excitation of microtubules is outside the scope of this paper.

3. The microtubule—a highly deformable structure

In contrast with actin and intermediate filaments the microtubules can be easily deformed and begin to flow without limit when the strain exceeds 50% (Fig. 4, [6,19]). Dynamic shear modulus G is of 5 N m^{-2} [19,20]. The dynamic shear modulus was measured in medium with ionic concentration similar to that in cytosol [19]. As the thickness of the ionic charge layer at the inner and at the outer surface of a microtubule are comparable with the thickness of the microtubule wall the measured dynamic shear modulus is an average value representing shear properties of the microtubule together with the charge layer. Therefore, as a rough approximation we will use the measured G value of the shear modulus for slip conditions inside the slip layer which consists of the microtubule surface layer (transition layer) and of the charge layer (Fig. 3). Fig. 3 shows similar slip layers on the outer and on the inner side of the microtubule wall. Large deformability is an important property for slip boundary conditions at the microtubule surface.

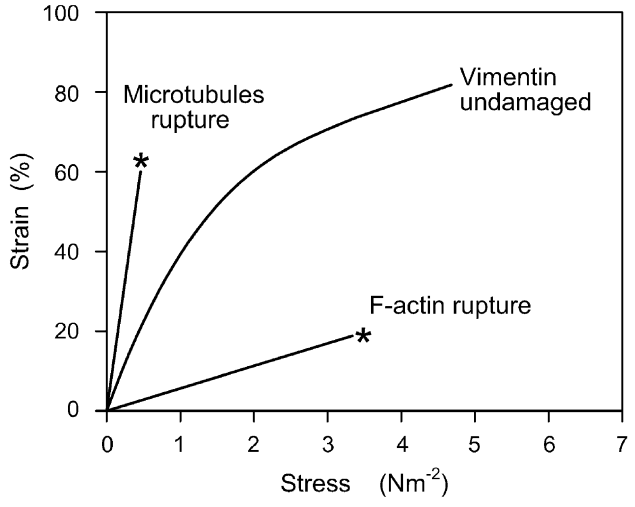

Fig. 4. Strain versus stress of actin, vimentin, and tubulin filaments. The microtubules are capable of large elastic deformation at low stress. After Refs. [6,19].

4. Energy excitation of the microtubular structure

The microtubule is a highly dynamic polymer that continuously exchange subunits (heterodimers) by the mechanism of dynamic instability (growth and shrinkage of the microtubules) and by treadmilling (heterodimers are continually added to one end and lost from the other one) [6,21,22]. After polymerization, guanosine triphosphate (GTP) bound to the β-tubulin in the heterodimer is hydrolyzed to guanosine diphosphate (GDP) and considerable amount of energy is stored in the microtubule [23,24]. Dynamic instability and treadmilling are important processes for energy supply to the microtubule structure.

The average rate of growth of a microtubule may be determined from histograms of lengths published e.g. in Refs. [25,26]. The mean microtubule length in histogram reported by Melki et al. [25] is about 0.5 μm and, therefore, the rate of growth is about 0.5 μm per 20 s. Five percent of heterodimers is incorporated into a microtubule in 1 s. For simplicity, we will assume microtubular structure where microtubules have the total length of 1 cm, and that 5% of it is exchanged in 1 s, i.e. $n = 8 \times 10^5$ of heterodimers may be replaced. From the energy released by hydrolysis of GTP to GDP, the amount $\epsilon = 7.1$ kJ mol^{-1} [23,24] is stored in the microtubule. Energy supplied within 1 s (i.e. power) $P_m = \epsilon n/L \approx \times 10^{-14}$ W cm^{-1} (where L is Avogadro's number). In the M phase of the cell cycle, the rate of exchange of heterodimers is more than one order of magnitude greater than that in the interphase and energy supply corresponds to this rate.

It is known that energy produced during GTP hydrolysis is delivered to microtubules and may be used for increase of tension in the microtubule, may excite solitons, may be transformed into vibrations and excite coherent states of the Fröhlich's type. In the last case, longitudinal polar vibrations (polarization waves) in the direction of the microtubule axis are of particular interest. These vibrations can generate electromagnetic field with dominant electric component around microtubules. Intensity of the electric field in the near vicinity of the outer surface and inside the circular cavity may be high.

Hydrolysis of GTP to GDP takes place at the fast growing end (plus end) below the cap with nonhydrolyzed GTP. The released energy may relax inside a microtubule which is from 1 to 10 μm long within about 1–10 ns for the velocity of vibration modes about 10^3 ms^{-1}. We cannot exclude energy transfer to the whole microtubule structure by this mechanism in a very short time of the order of magnitude of 10^{-8} s.

5. Viscous force at the upper surface of the slip layer

The microtubule may be approximated by a cylindrical layer with circular cross section and with rotation symmetry with respect to the microtubule axis. We will assume longitudinal harmonic vibrations along the axis of the cylinder (but independent of the distance along the axis). Due to viscosity oscillations in the microtubule excite oscillations in the liquid medium (in the cytosol) around the microtubule. The viscous force exerted by Newtonian liquid per unit area of the slip layer is given by

$$F'_v = \eta\left(\frac{\partial v}{\partial r}\right)_{r=R}, \tag{1}$$

where $\eta = \rho\kappa$ is the dynamic viscosity of the cytosol (ρ is the density of mass and κ is the kinematic viscosity), and $v = v_0\exp(i\omega t)$ is the velocity of vibrations in the direction of the microtubule axis (v_0 is the amplitude). In this case, the momentum equation is given by the diffusion equation $\partial v/\partial t = \kappa\nabla v$ [27]. Solution to the momentum equation in cylindrical coordinates can be found using Kelvin's functions [28,29] which can be expressed by Macdonald's functions (K_0, K_1). We get for the viscous force per unit length (shown in Fig. 3) [11]

$$F_v = -\beta v_0\exp(i\omega t) \quad \text{where}$$

$$\beta = \exp\left(i\frac{\pi}{4}\right)\sqrt{\omega\rho\eta}\,\frac{K_1\left(R\sqrt{i\frac{\omega}{\kappa}}\right)}{K_0\left(R\sqrt{i\frac{\omega}{\kappa}}\right)}S_i, \tag{2}$$

where R is the radius of the outer cylindrical surface and $S_i = 2\pi R$ is the area of the cylindrical surface of unit length.

6. Transfer of vibrations across the slip layer

As a first order approximation, we assume that the oscillations in a microtubule act on the damped upper surface of the slip layer by the force [11]

$$F_i = f\,Y_0\exp[i(\omega t + \phi)] \quad \text{where} \quad f = \frac{GS_i}{\Delta R}, \tag{3}$$

Y_0 is the amplitude of oscillations below the transition region in the microtubule, f is the elastic force constant, and ΔR is the thickness of the slip layer. The force F_i acts in the direction of the microtubule axis. Equation of motion for forced vibrations at the upper surface of the slip layer has the form

$$\frac{d^2y}{dt^2} + \frac{\beta}{m}\frac{dy}{dt} + \omega_0^2 y = \frac{F_i}{m} \tag{4}$$

(where ω_0 is the characteristic angular frequency, y is the displacement along the microtubule axis, and m is the mass of the slip layer). Under steady state conditions we get from Eqs. (2)–(4) for $\omega = \omega_0$ [11]

$$\frac{y_0}{Y_0} = \frac{f}{|\beta|\omega} \tag{5}$$

where y_0 is the amplitude of the displacement and $|\beta| = \sqrt{\Re(\beta^2) + \Im(\beta^2)}$. The same relation is valid for v_0/V_0 where V_0 is amplitude of velocity in the microtubule (shown in Fig. 3).

7. Relaxation time and energy losses

The average kinetic energy stored in the microtubule per unit length and the average rate of energy loss per unit length caused by viscous damping (as follows from Eq. (2)) are given by [11]

$$E = \frac{1}{4}\pi(R_m^2 - R_{m0}^2)\,\rho V_0^2 \quad \text{and} \quad E' = -\frac{1}{2}|\beta|v_0^2, \tag{6}$$

respectively. R_m and R_{m0} in Eq. (6) are the outer and the inner radius of the microtubule (Fig. 3). Using Eq. (6) the relaxation time τ and its relative value τ_{rel} (in the number of periods of oscillations) may be evaluated from

$$\tau = \frac{E}{|E'|} = \frac{1}{2}\pi(R_m^2 - R_{m0}^2)\rho\,|\beta|\,\frac{\omega^2}{f^2} \quad \text{and} \quad \tau_{rel} = \frac{\omega\tau}{2\pi}. \tag{7}$$

Fig. 5 shows the relative relaxation time as a function of frequency for $\eta = 5\times10^{-4}$ (the dashed lines) and 10^{-3} Pa s (the full lines). The range of magnitude from 5×10^{-4} to

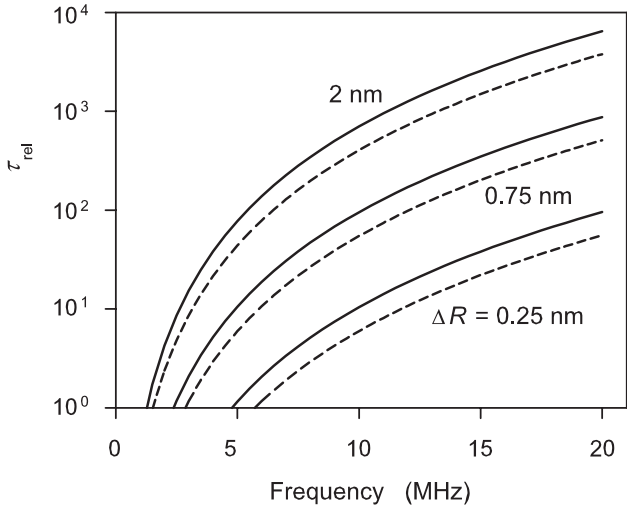

Fig. 5. Relative relaxation time τ_{rel} caused by viscous damping in a microtubule versus frequency. Parameter is the thickness of the slip layer ΔR (0.25, 0.75, and 2 nm). Dynamic viscosity η of the liquid medium is 5×10^{-4} (the dashed lines) and 1×10^{-3} Pa s (the full lines). The shear modulus G is 5 N m^{-2}.

10^{-3} Pa s corresponds to the viscosity of water for temperatures from 55 to 25 °C, respectively (more detailed information about viscosity of cytosol is in the discussion). $R_m = 12.5$ nm, $R_{m0} = 8.5$ nm [6], $R = R_m + \Delta R$, and the shear modulus $G = 5$ N m^{-2} [19]. The relative relaxation time τ_{rel} for $\Delta R = 2$ nm is about 10^3 at 10 MHz. The curves for $\Delta R = 0.25$ nm in Fig. 5 may correspond to the relaxation time for a slip in the transition surface layer of the microtubule. τ_{rel} is greater than 10 at 10 MHz. Assessment of damping for the magnitude of viscosity from 10^{-4} to 10^{-2} Pa s is in Ref. [11].

The average rate of energy loss is given by Eq. (6). We determine v_0 from Eq. (5) for amplitude of oscillations in the microtubule 0.5 nm. At 10 MHz power lost per 1 cm length is about 3.3×10^{-14} W cm^{-1} ($\eta = 5 \times 10^{-4}$ Pa s). This power is of the same order of magnitude as the power P_m supplied by hydrolysis of GTP to GDP.

8. Discussion

Fröhlich postulated that excitation of vibrations, condensation of energy in a vibration mode, and creation of the state far from thermodynamic equilibrium are fundamental physical properties of the living system [2–4]. This paper is a contribution to Ref. [7] in which we concluded that microtubules may satisfy the requirements for the Fröhlich's structure. Theoretical treatment describes a real biological mechanism of energy supply and possibility of energy excitation of a polar structure. Polar structures generate an electromagnetic field with dominant near zone electrical component which can participate for instance in the directional transport of reaction component [30], in orientation of macromolecules and polymer structures, and in polymerization processes.

We will make remarks on energy supply and energy losses conditioning excitation of the Fröhlich's system. The energy supply by hydrolysis of GTP to GDP seems to be sufficient for excitation even if we take into account losses at the outer and at the inner surface of the microtubule. Vibrations in bodies immersed in liquids are strongly damped by viscosity. Description of the properties of cytoplasm, especially of its viscosity, is contained in Ref. [31]. Cytoplasmic viscosity was measured as high as 0.011 Pa s—the value more than 10 times greater than that of water. (About 30% of mass in a mammalian cell is in the solid state form but due to compartmentalization its concentration is not uniform [6].) Other measurements revealed that apparent viscosity did not significantly differ from the viscosity of bulk water. Water is the major part of cytosol (a typical cell contains about 70% of water). Nevertheless, surface associated layers of ordered water up to the thickness of 200 nm may exist. Therefore, properties of water in living cells do not appear to be greatly altered in comparison with pure water but layers of ordered water may exist. Consequently, properties of cytosol around microtubules may differ from those of pure water and low damping of vibrations by viscosity may exist. Viscous losses from the inner surface might be very small as all the liquid medium in the cavity may be in an ordered state. We may conclude that even for viscosity of water excitation of vibrations is possible on account of the slip boundary conditions.

Experimental proof of the postulated theory is rather difficult as we have to measure living cells or subcellular structures of micrometer or submicrometer dimensions. Measurement method of electromagnetic emission from individual cells using special microelectronic sensors is described e.g. in Ref. [32]. Each detection structure in a microelectronic sensor has two electrodes 10 μm wide with a small gap between them. The electromagnetic emission of a sedimented cell at the gap is measured. The measurement results are promising. Nonetheless, the microtubular structures organized inside a cell need not be a source of electromagnetic emission from the cell. Two needle electrodes pricked inside a cell may be more convenient for measurement of the oscillating electric field generated by microtubules.

9. Conclusions

Microtubules satisfy conditions for excitation of vibrations and energy condensation. The damping effects of cytosol viscosity are minimized by slip layer at the boundary between the microtubule and cytosol. Formation of the slip layer depends on large deformability of microtubules at low stress and on the ionic layer at their surface. Energy supplied to the microtubule from the hydrolysis of GTP to GDP may be sufficient for excitation.

Acknowledgements

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Excitation of vibrations in microtubules in living cells

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Abstract

Microtubules, which are thought to be the primary organizers of the cytoskeleton, are electrical polar structures with extraordinary elastic deformability at low stress and with energy supply from hydrolysis of guanosine triphosphate (GTP) to guanosine diphosphate (GDP). At least a part of the energy supplied from hydrolysis can excite vibrations. Energy is mainly lost by viscous damping of the surrounding cytosol. Viscous damping is diminished by a slip layer which is formed by an attracted ionic charge layer and by a thin surface layer of the microtubule. Relaxation time caused by viscous damping may be several orders of magnitude greater than period of vibrations at 10 MHz. Energy supplied to the microtubule is of the order of magnitude of 10^{-14} W cm^{-1} (per unit length of the microtubule).
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where $\dot{n}_i$ is the total rate of change of occupation number n_i (in 1 s), and s_i, $[\dot{n}_i]_\phi$, $[\dot{n}_i(n_i, n_j)]_\chi$ is the rate of change of n_i by energy supply, and by the linear and by the nonlinear interaction with the heat bath, respectively. The linear and the nonlinear term of the interaction with the heat bath describes transfer of one energy quantum between a vibration mode and the heat bath and energy transfer between two vibration modes with participation of the heat bath, respectively. The steady state solution to the rate equation shows that due to nonlinear term the supplied energy is not thermalized but condensed in a vibration mode. Excitation depends on the amount of the energy supplied to the system but not on the manner of its supply [5].

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2. Subunits of the microtubule—electric dipoles

The microtubule is a hollow cylindrical polymer of tubulin heterodimers (Fig. 1a). The outer and the inner diameter is about 25 and 17 nm, respectively. Tubulin heterodimer—composed of α- and β-tubulins—may exist in two conformation states (Fig. 1b): the α state and the β state where the electron negative charge is placed toward the α-tubulin and the β-tubulin, respectively. In the β state, the heterodimer is tilted 29° with respect to the microtubule axis. In the interphase microtubules are organized in a radial structure extending from the centrosome to the periphery of the cell (Fig. 2a). In the M phase, microtubules assemble from the two separated centrosomes to form a mitotic spindle (Fig. 2b).

Each heterodimer binds 18 calcium ions [12–14]. The electric dipole moment is about 1000 Debye (10^{-26} Cm). The induced dipole moment per dimer arising only from the motion of mobile electrons or protons was estimated to be 200–400 Debye [15]. Ionic concentration inside a cell is of about 150 mM of cations and the same concentration of anions [6]. The ions from the cytosol form a charge layer—a cylindrical envelope around the microtubule [15]. Similar situation may be inside the microtubule (Fig. 3) as the liquid medium in the inner circular cavity is very likely to be cytosol. The ionic charge layer, which is schematically shown in Figs. 1 and 3, screens electrical potential generated by the charges in the microtubule [15]. Within a distance of about 2 Debye lengths the static potential is considerably screened. The Debye length is of about 0.8 nm [16]. Due to hydration, the ions are surrounded by bound water molecules up to a distance of 0.4–0.5 nm. Charged proteins of considerably greater size than ionized atoms may be in the ionic layer too. The majority of water molecules in the charge layer are not free and its viscous properties are dependent on hydrated ions. Some ions can penetrate inside the tubulin structure and behave like a part

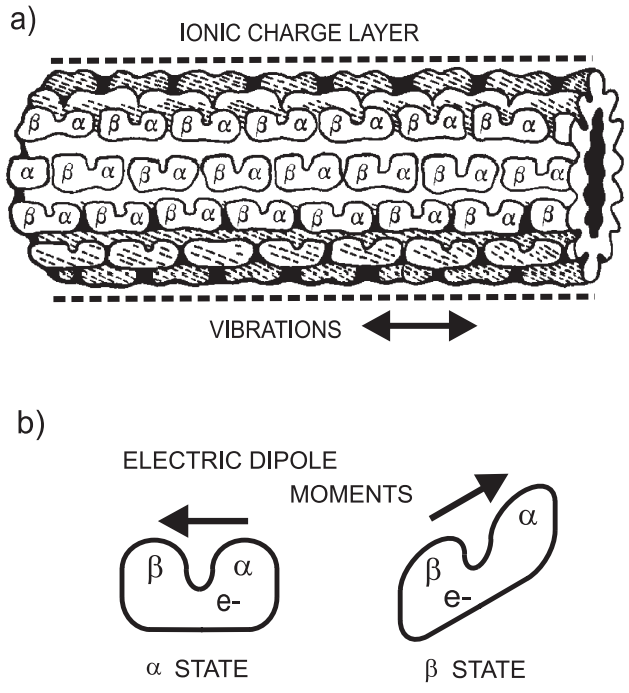

Fig. 1. (a) A schematic picture of a microtubule with tubulin heterodimers in parallel protofilaments and (b) conformation change of a heterodimer from the α to the β state after hydrolysis of GTP (guanosine triphosphate) to GDP (guanosine diphosphate) in the β-tubulin (b). The lines with arrows indicate the dipole moment vectors.

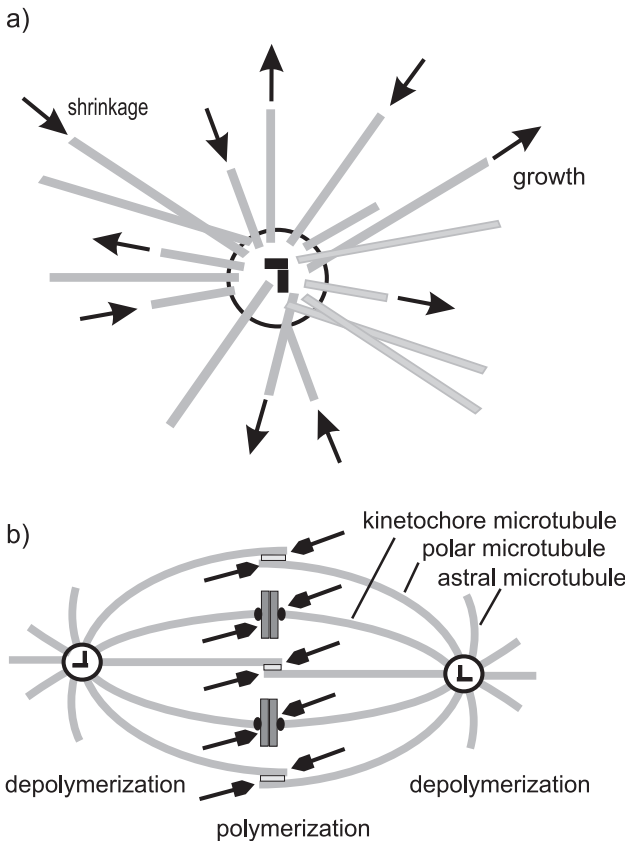

Fig. 2. (a) A centrosome with attached microtubules in the interphase. (b) The mitotic spindle in the M phase. The heterodimers with exhausted energy are exchanged for energy rich ones by the dynamic instability—the growth and shrinkage are indicated by arrows (a) and by the treadmilling—polymerization (indicated by arrows) and depolymerization take place at the center and at the poles of the mitotic spindle, respectively (b).

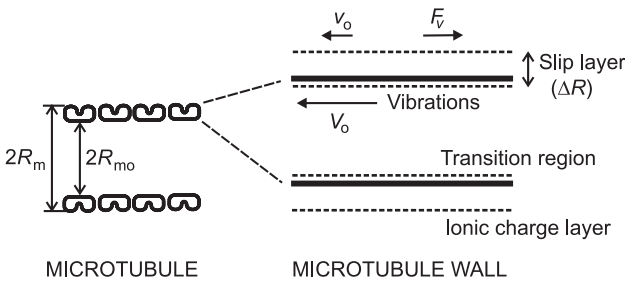

Fig. 3. A schematic picture of the microtubule wall with the slip layers. Each slip layer includes the ionic charge layer and the transition layer in the microtubule.

of the tubulin (as well as water molecules). Ions in the cytosol have small mobility and, therefore, there is slip between the charge layer and the microtubule surface.

We have to mention here that the cytoskeleton is a complicated structure with associated proteins cross-linking its parts, changing elastic properties of the structures and the viscosity effects. There are e.g. microtubule-associated proteins (MAP-1, MAP-2 forming sidearms), tau proteins and motor proteins. Phosphorylation of tau proteins plays very likely a role in microtubule binding and function [17]. Skeletor reticular matrix, spring like lattice around kinetochore microtubules, and protein cross-linked lattice around the spindle poles [18], are formed in the M phase. These structures might diminish viscosity damping but their description and analysis of their effects on excitation of microtubules is outside the scope of this paper.

3. The microtubule—a highly deformable structure

In contrast with actin and intermediate filaments the microtubules can be easily deformed and begin to flow without limit when the strain exceeds 50% (Fig. 4, [6,19]). Dynamic shear modulus G is of 5 N m^{-2} [19,20]. The dynamic shear modulus was measured in medium with ionic concentration similar to that in cytosol [19]. As the thickness of the ionic charge layer at the inner and at the outer surface of a microtubule are comparable with the thickness of the microtubule wall the measured dynamic shear modulus is an average value representing shear properties of the microtubule together with the charge layer. Therefore, as a rough approximation we will use the measured G value of the shear modulus for slip conditions inside the slip layer which consists of the microtubule surface layer (transition layer) and of the charge layer (Fig. 3). Fig. 3 shows similar slip layers on the outer and on the inner side of the microtubule wall. Large deformability is an important property for slip boundary conditions at the microtubule surface.

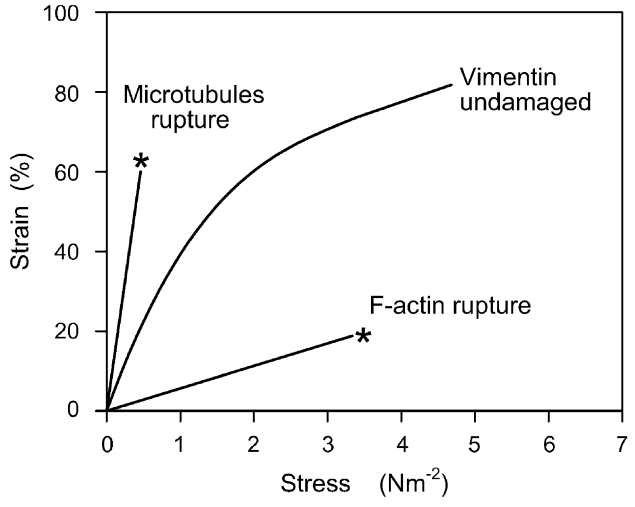

Fig. 4. Strain versus stress of actin, vimentin, and tubulin filaments. The microtubules are capable of large elastic deformation at low stress. After Refs. [6,19].

4. Energy excitation of the microtubular structure

The microtubule is a highly dynamic polymer that continuously exchange subunits (heterodimers) by the mechanism of dynamic instability (growth and shrinkage of the microtubules) and by treadmilling (heterodimers are continually added to one end and lost from the other one) [6,21,22]. After polymerization, guanosine triphosphate (GTP) bound to the β-tubulin in the heterodimer is hydrolyzed to guanosine diphosphate (GDP) and considerable amount of energy is stored in the microtubule [23,24]. Dynamic instability and treadmilling are important processes for energy supply to the microtubule structure.

The average rate of growth of a microtubule may be determined from histograms of lengths published e.g. in Refs. [25,26]. The mean microtubule length in histogram reported by Melki et al. [25] is about 0.5 μm and, therefore, the rate of growth is about 0.5 μm per 20 s. Five percent of heterodimers is incorporated into a microtubule in 1 s. For simplicity, we will assume microtubular structure where microtubules have the total length of 1 cm, and that 5% of it is exchanged in 1 s, i.e. $n = 8 \times 10^5$ of heterodimers may be replaced. From the energy released by hydrolysis of GTP to GDP, the amount $\epsilon = 7.1$ kJ mol^{-1} [23,24] is stored in the microtubule. Energy supplied within 1 s (i.e. power) $P_m = \epsilon n/L \approx \times 10^{-14}$ W cm^{-1} (where L is Avogadro's number). In the M phase of the cell cycle, the rate of exchange of heterodimers is more than one order of magnitude greater than that in the interphase and energy supply corresponds to this rate.

It is known that energy produced during GTP hydrolysis is delivered to microtubules and may be used for increase of tension in the microtubule, may excite solitons, may be transformed into vibrations and excite coherent states of the Fröhlich's type. In the last case, longitudinal polar vibrations (polarization waves) in the direction of the microtubule axis are of particular interest. These vibrations can generate electromagnetic field with dominant electric component around microtubules. Intensity of the electric field in the near vicinity of the outer surface and inside the circular cavity may be high.

Hydrolysis of GTP to GDP takes place at the fast growing end (plus end) below the cap with nonhydrolyzed GTP. The released energy may relax inside a microtubule which is from 1 to 10 μm long within about 1–10 ns for the velocity of vibration modes about 10^3 ms^{-1}. We cannot exclude energy transfer to the whole microtubule structure by this mechanism in a very short time of the order of magnitude of 10^{-8} s.

5. Viscous force at the upper surface of the slip layer

The microtubule may be approximated by a cylindrical layer with circular cross section and with rotation symmetry with respect to the microtubule axis. We will assume longitudinal harmonic vibrations along the axis of the cylinder (but independent of the distance along the axis). Due to viscosity oscillations in the microtubule excite oscillations in the liquid medium (in the cytosol) around the microtubule. The viscous force exerted by Newtonian liquid per unit area of the slip layer is given by

$$F'_v = \eta\left(\frac{\partial v}{\partial r}\right)_{r=R}, \tag{1}$$

where $\eta = \rho\kappa$ is the dynamic viscosity of the cytosol (ρ is the density of mass and κ is the kinematic viscosity), and $v = v_0\exp(i\omega t)$ is the velocity of vibrations in the direction of the microtubule axis (v_0 is the amplitude). In this case, the momentum equation is given by the diffusion equation $\partial v/\partial t = \kappa\nabla v$ [27]. Solution to the momentum equation in cylindrical coordinates can be found using Kelvin's functions [28,29] which can be expressed by Macdonald's functions (K_0, K_1). We get for the viscous force per unit length (shown in Fig. 3) [11]

$$F_v = -\beta v_0\exp(i\omega t) \quad \text{where}$$

$$\beta = \exp\left(i\frac{\pi}{4}\right)\sqrt{\omega\rho\eta}\,\frac{K_1\left(R\sqrt{i\frac{\omega}{\kappa}}\right)}{K_0\left(R\sqrt{i\frac{\omega}{\kappa}}\right)}S_i, \tag{2}$$

where R is the radius of the outer cylindrical surface and $S_i = 2\pi R$ is the area of the cylindrical surface of unit length.

6. Transfer of vibrations across the slip layer

As a first order approximation, we assume that the oscillations in a microtubule act on the damped upper surface of the slip layer by the force [11]

$$F_i = f\,Y_0\exp[i(\omega t + \phi)] \quad \text{where} \quad f = \frac{GS_i}{\Delta R}, \tag{3}$$

Y_0 is the amplitude of oscillations below the transition region in the microtubule, f is the elastic force constant, and ΔR is the thickness of the slip layer. The force F_i acts in the direction of the microtubule axis. Equation of motion for forced vibrations at the upper surface of the slip layer has the form

$$\frac{d^2y}{dt^2} + \frac{\beta}{m}\frac{dy}{dt} + \omega_0^2 y = \frac{F_i}{m} \tag{4}$$

(where ω_0 is the characteristic angular frequency, y is the displacement along the microtubule axis, and m is the mass of the slip layer). Under steady state conditions we get from Eqs. (2)–(4) for $\omega = \omega_0$ [11]

$$\frac{y_0}{Y_0} = \frac{f}{|\beta|\omega} \tag{5}$$

where y_0 is the amplitude of the displacement and $|\beta| = \sqrt{\Re(\beta^2) + \Im(\beta^2)}$. The same relation is valid for v_0/V_0 where V_0 is amplitude of velocity in the microtubule (shown in Fig. 3).

7. Relaxation time and energy losses

The average kinetic energy stored in the microtubule per unit length and the average rate of energy loss per unit length caused by viscous damping (as follows from Eq. (2)) are given by [11]

$$E = \frac{1}{4}\pi(R_m^2 - R_{m0}^2)\,\rho V_0^2 \quad \text{and} \quad E' = -\frac{1}{2}|\beta|v_0^2, \tag{6}$$

respectively. R_m and R_{m0} in Eq. (6) are the outer and the inner radius of the microtubule (Fig. 3). Using Eq. (6) the relaxation time τ and its relative value τ_{rel} (in the number of periods of oscillations) may be evaluated from

$$\tau = \frac{E}{|E'|} = \frac{1}{2}\pi(R_m^2 - R_{m0}^2)\rho\,|\beta|\,\frac{\omega^2}{f^2} \quad \text{and} \quad \tau_{rel} = \frac{\omega\tau}{2\pi}. \tag{7}$$

Fig. 5 shows the relative relaxation time as a function of frequency for $\eta = 5\times10^{-4}$ (the dashed lines) and 10^{-3} Pa s (the full lines). The range of magnitude from 5×10^{-4} to

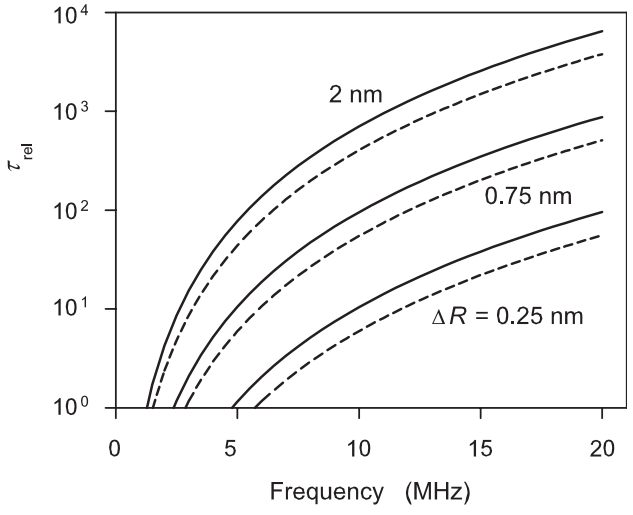

Fig. 5. Relative relaxation time τ_{rel} caused by viscous damping in a microtubule versus frequency. Parameter is the thickness of the slip layer ΔR (0.25, 0.75, and 2 nm). Dynamic viscosity η of the liquid medium is 5×10^{-4} (the dashed lines) and 1×10^{-3} Pa s (the full lines). The shear modulus G is 5 N m^{-2}.

10^{-3} Pa s corresponds to the viscosity of water for temperatures from 55 to 25 °C, respectively (more detailed information about viscosity of cytosol is in the discussion). $R_m = 12.5$ nm, $R_{m0} = 8.5$ nm [6], $R = R_m + \Delta R$, and the shear modulus $G = 5$ N m^{-2} [19]. The relative relaxation time τ_{rel} for $\Delta R = 2$ nm is about 10^3 at 10 MHz. The curves for $\Delta R = 0.25$ nm in Fig. 5 may correspond to the relaxation time for a slip in the transition surface layer of the microtubule. τ_{rel} is greater than 10 at 10 MHz. Assessment of damping for the magnitude of viscosity from 10^{-4} to 10^{-2} Pa s is in Ref. [11].

The average rate of energy loss is given by Eq. (6). We determine v_0 from Eq. (5) for amplitude of oscillations in the microtubule 0.5 nm. At 10 MHz power lost per 1 cm length is about 3.3×10^{-14} W cm^{-1} ($\eta = 5 \times 10^{-4}$ Pa s). This power is of the same order of magnitude as the power P_m supplied by hydrolysis of GTP to GDP.

8. Discussion

Fröhlich postulated that excitation of vibrations, condensation of energy in a vibration mode, and creation of the state far from thermodynamic equilibrium are fundamental physical properties of the living system [2–4]. This paper is a contribution to Ref. [7] in which we concluded that microtubules may satisfy the requirements for the Fröhlich's structure. Theoretical treatment describes a real biological mechanism of energy supply and possibility of energy excitation of a polar structure. Polar structures generate an electromagnetic field with dominant near zone electrical component which can participate for instance in the directional transport of reaction component [30], in orientation of macromolecules and polymer structures, and in polymerization processes.

We will make remarks on energy supply and energy losses conditioning excitation of the Fröhlich's system. The energy supply by hydrolysis of GTP to GDP seems to be sufficient for excitation even if we take into account losses at the outer and at the inner surface of the microtubule. Vibrations in bodies immersed in liquids are strongly damped by viscosity. Description of the properties of cytoplasm, especially of its viscosity, is contained in Ref. [31]. Cytoplasmic viscosity was measured as high as 0.011 Pa s—the value more than 10 times greater than that of water. (About 30% of mass in a mammalian cell is in the solid state form but due to compartmentalization its concentration is not uniform [6].) Other measurements revealed that apparent viscosity did not significantly differ from the viscosity of bulk water. Water is the major part of cytosol (a typical cell contains about 70% of water). Nevertheless, surface associated layers of ordered water up to the thickness of 200 nm may exist. Therefore, properties of water in living cells do not appear to be greatly altered in comparison with pure water but layers of ordered water may exist. Consequently, properties of cytosol around microtubules may differ from those of pure water and low damping of vibrations by viscosity may exist. Viscous losses from the inner surface might be very small as all the liquid medium in the cavity may be in an ordered state. We may conclude that even for viscosity of water excitation of vibrations is possible on account of the slip boundary conditions.

Experimental proof of the postulated theory is rather difficult as we have to measure living cells or subcellular structures of micrometer or submicrometer dimensions. Measurement method of electromagnetic emission from individual cells using special microelectronic sensors is described e.g. in Ref. [32]. Each detection structure in a microelectronic sensor has two electrodes 10 μm wide with a small gap between them. The electromagnetic emission of a sedimented cell at the gap is measured. The measurement results are promising. Nonetheless, the microtubular structures organized inside a cell need not be a source of electromagnetic emission from the cell. Two needle electrodes pricked inside a cell may be more convenient for measurement of the oscillating electric field generated by microtubules.

9. Conclusions

Microtubules satisfy conditions for excitation of vibrations and energy condensation. The damping effects of cytosol viscosity are minimized by slip layer at the boundary between the microtubule and cytosol. Formation of the slip layer depends on large deformability of microtubules at low stress and on the ionic layer at their surface. Energy supplied to the microtubule from the hydrolysis of GTP to GDP may be sufficient for excitation.

Acknowledgements

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Extended objects in quantum systems and soliton solutions

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It is shown that the soliton solutions of the classical Euler equations are closely related to the extended objects created in quantum many-body systems. The general argument is supplemented by a concrete example which shows how the boson transformation applied to a quantum system leads to the static soliton solution in the $(1+1)$-dimensional $\lambda\phi^4$ model.

There has recently been an active interest in the study of extended objects which are described by the soliton solutions of the Euler equations for classical fields (say ϕ).[1] The Ginzburg-Landau equation[2] can also be regarded as a kind of Euler equation in which ϕ is the order parameter. In parallel to the study of soliton solutions of the Euler equations, we have been formulating a systematic method for the construction of extended objects which appear in quantum many-body systems.[3] This method begins with a set of Heisenberg equations for quantized fields (say ψ) and makes use of the so-called boson transformation method[4] to obtain those states in which classically behaving extended objects and quantum modes coexist. This method has been applied to the study of extended objects in various ordered states such as dislocations, grain boundaries, point defects and surface sound waves in crystals[5] and vortices, surface electromagnetic waves, Josephson currents in superconductors, etc.[4,6]

The aim of this paper is to present a brief description of the relation between the above-mentioned two approaches in the study of extended objects. In particular, it will be pointed out that the extended objects constructed by the boson method become the soliton solutions of the Euler equations when the Planck constant, h, is ignored, implying that the soliton solutions can be regarded as the extended objects with quantum origin. We feel that this consideration sheds light on the question about the gap between the classical and quantum physics.

Let us begin with a brief summary of the boson method. Consider a Heisenberg equation

$$\Lambda(\partial)\psi = F[\psi]. \tag{1}$$

To make the essence of the boson method transparent, we first consider a simple case, in which no composite particles appear and the perturbative expansion method is usable. We assume also that ψ is a scalar field. Equation (1) leads to the Yang-Feldman equation

$$\psi = \varphi^{\text{in}} + [\Lambda(\partial)]^{-1}F[\psi], \tag{2}$$

where φ^{in} is a free boson field satisfying

$$\Lambda(\partial)\varphi^{\text{in}} = 0. \tag{3}$$

It is important to note here that the mass in the $\Lambda(\partial)$ is the physical mass and, therefore, that φ^{in} is renormalized (or physical) free field. To understand this, we rewrite (2) in the form

$$\langle a|\psi(x)|b\rangle = \langle a|\varphi^{\text{in}}|b\rangle + (-i)\int d^4y\Delta(x-y)\langle a|F_y[\psi]|b\rangle, \tag{4}$$

where the states $|a\rangle$ and $|b\rangle$ are the ones in the Fock space of φ^{in}, and $F_y[\psi]$ means $F[\psi]$ at the space-time position y. The function $\Delta(x)$ in (4) is the Green's function

$$\Lambda(\partial)\Delta(x) = i\delta(x). \tag{5}$$

The second term on the right-hand side of (4) diverges unless $\langle a|F_y[\psi]|b\rangle$ vanishes in a reasonable manner at the limit $t_y \to \pm\infty$. To avoid this divergence we must choose φ^{in} to be the renormalized one. The above argument implies that we regard the Heisenberg equation as an equation among matrix elements (i.e., the weak relation). All the equations for Heisenberg operators in the following should be understood as weak relations.

When we solve (2) by successive iteration, we are led to the usual perturbative expansion. The result is an expression of ψ in terms of φ^{in},

$$\psi(x) = \psi[x:\varphi^{\text{in}}], \tag{6}$$

which is called the dynamical map.[7]

Let us now introduce a c-number function $f(x)$ which satisfies

$$\Lambda(\partial)f = 0. \tag{7}$$

Then we can generalize the Yang-Feldman equation as

$$\psi = \varphi^{\text{in}} + f + [\Lambda(\partial)]^{-1}F[\psi], \tag{8}$$

which obviously satisfies (1). Solving (8) by successive iteration, we obtain a new solution of (1),

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$$\psi^f(x) = \psi(x:\varphi^{\text{in}} + f). \tag{9}$$

Note that ψ^f is equal to ψ in which φ^{in} is replaced by $\varphi^{\text{in}} + f$ (the boson transformation). The fact that both ψ and ψ^f satisfy the same Heisenberg equation (1) is the content of the boson transformation theorem,

$$\Lambda(\partial)\psi^f = F[\psi^f]. \tag{10}$$

Let us now introduce the c-number field ϕ^f as

$$\phi^f = \langle 0|\psi^f|0\rangle. \tag{11}$$

When we ignore h (the Planck constant), ϕ^f will be denoted by ϕ_0^f,

$$\phi_0^f = \lim_{h\to 0}\phi^f. \tag{12}$$

Note that now the difference between the vacuum expectation value of the product of ψ^f and the product of the vacuum expectation value of ψ^f is due to the contraction of in-fields, which creates the loop diagrams in the course of successive iteration applied to (2). Since the contraction of in-fields creates terms which vanish at $h=0$, we can write

$$\langle 0|F[\psi^f]|0\rangle = F[\phi^f] + O(h), \tag{13}$$

where $O(h)$ stands for those terms which vanish at $h=0$. Thus (8) leads to

$$\phi_0^f = f + [\Lambda(\partial)]^{-1}F[\phi_0^f], \tag{14}$$

which gives the classical Euler equation

$$\Lambda(\partial)\,\phi_0^f = F[\phi_0^f]. \tag{15}$$

The above argument shows that ϕ_0^f is given by tree diagrams only.

Summarizing, when we calculate $\psi(x:\varphi^{\text{in}})$ by means of the tree approximation, the vacuum expectation value of the boson-transformed one (i.e., $\langle 0|\psi^f|0\rangle$) gives ϕ_0^f which satisfies the classical Euler equation.

Let us now illustrate the construction of the soliton solution ϕ_0^f from the boson transformation by using the following model of (1+1) dimensions:

$$(-\partial^2 - \mu^2)\psi(x) = \lambda\psi^3(x). \tag{16}$$

Here $x=(x_0, x_1)$ and $\psi(x)$ is a scalar Heisenberg field. We are particularly interested in finding what choice of the boson transformation function $f(x)$ leads to the static soliton solution.

Using the notation

$$v \equiv \langle 0|\psi(x)|0\rangle, \tag{17}$$

we define the Heisenberg operator $\rho(x)$ by the relation

$$\psi(x) = v + \rho(x). \tag{18}$$

Equation (16) then leads to

$$(-\partial^2 - m^2)\rho(x) = \tfrac{3}{2}mg\rho^2(x) + \tfrac{1}{2}g^2\rho^3(x), \tag{19}$$

and also to $\lambda v^2 = -\mu^2$. Here $g=\sqrt{2\lambda}$ and $m^2 = 2\lambda v^2$. Let ρ^{in} denote the in field which is the asymptotic limit of ρ. We have

$$(-\partial^2 - m^2)\rho^{\text{in}}(x) = 0. \tag{20}$$

In the following we consider the extended objects created by the condensation of ρ^{in}. The tree approximation leads to the following dynamical map for ρ:

$$\rho(x) = \rho^{\text{in}}(x) + \tfrac{3}{2}mg(-i)\int d^2y\Delta(x-y):[\rho^{\text{in}}(y)]^2: + \left\{\tfrac{1}{2}g^2(-i)\int d^2y\Delta(x-y):[\rho^{\text{in}}(y)]^3: + \tfrac{9}{2}m^2g^2(-i)^2\int d^2yd^2z\Delta(x-y)\Delta(y-z):\rho^{\text{in}}(y)[\rho^{\text{in}}(z)]^2:\right\} + \cdots. \tag{21}$$

Here $\Delta(x-y)$ is the Green's function satisfying

$$(-\partial^2 - m^2)\Delta(x-y) = i\delta^{(2)}(x-y). \tag{22}$$

We put (21) in the form

$$\rho(x) = \sum_{n=1}^{\infty}\rho^{(n)}(x), \tag{23}$$

where n denotes the order of the normal products. Then the following relation holds in the tree approximation:

$$\rho^{(n)}(x) = \tfrac{3}{2}mg(-i)\int d^2y\,\Delta(x-y):\sum_{i+j=n}\rho^{(i)}(y)\rho^{(j)}(y): + \tfrac{1}{2}g^2(-i)\int d^2y\,\Delta(x-y):\sum_{i+j+k=n}\rho^{(i)}(y)\rho^{(j)}(y)\rho^{(k)}(y):. \tag{24}$$

We now perform the boson transformation

$$\rho^{\mathrm{in}}(x) \to \rho^{\mathrm{in}}(x) + f(x), \tag{25}$$

where the c-number function f satisfies

$$(-\partial^2 - m^2) f(x) = 0. \tag{26}$$

Denoting the boson-transformed ρ-field operator by ρ^f, we have

$$\phi_0^f(x) = v + \langle 0 | \rho^f(x) | 0 \rangle. \tag{27}$$

We consider the static case. The space coordinates $x_1, y_1, \ldots$ will be simply denoted by $x, y, \ldots$. Then (26) and (27) read

$$\frac{d^2}{dx^2} f(x) = m^2 f(x) \tag{28}$$

and

$$\phi_0^f(x) = v + f(x) + \tfrac{3}{2} mg \int dy\, K(x-y) f^2(y) + \left[\tfrac{1}{2} g^2 \int dy\, K(x-y) f^3(y) + \tfrac{9}{2} m^2 g^2 \int dy\, K(x-y) f(y) \int dz\, K(y-z) f^2(z) \right] + \cdots, \tag{29}$$

where the Green's function $K(x-y)$ is defined by

$$\left(\frac{d^2}{dx^2} - m^2 \right) K(x-y) = \delta(x-y). \tag{30}$$

The recurrence relation (24) becomes

$$\rho_f^{(n)}(x) = \tfrac{3}{2} mg \int dy\, K(x-y) : \sum_{i+j=n} \rho_f^{(i)}(y) \rho_f^{(j)}(y) : + \tfrac{1}{2} g^2 \int dy\, K(x-y) : \sum_{i+j+k=n} \rho_f^{(i)}(y) \rho_f^{(j)}(y) \rho_f^{(k)}(y) :. \tag{31}$$

As a solution of (28), we choose $f(x)$ which diverges at $x = -\infty$ and regular at $x = \infty$; $f(x) = A \exp[-mx]$. Since $f(x)$ diverges at $x = -\infty$, we choose the Green's function K in such a way that $K(x-y) = 0$ for $x > y$:

$$K(x) = -\theta(-x) \frac{1}{m} \sinh mx. \tag{32}$$

Noticing that $\rho_f^{(1)}(x) = f(x)$, we can easily see from (31) that

$$\rho_f^{(n)}(x) = C_n (e^{-mx})^n, \tag{33}$$

and C_n satisfy the recurrence formula

$$C_n = \frac{1}{m^2(n^2-1)} \left(\tfrac{3}{2} mg \sum_{i+j=n} C_i C_j + \tfrac{1}{2} g^2 \sum_{i+j+k=n} C_i C_j C_k \right), \quad (n \geq 2), \tag{34}$$

$C_1 = A$.

Solving (34), together with the relation $v = m/g$, we have

$$C_n = 2v \left(\frac{A}{2v} \right)^n, \tag{35}$$

which leads to

$$\phi_0^f(x) = v + 2v \sum_{n=1}^{\infty} \left(\frac{A}{2v} e^{-mx} \right)^n = v \frac{1 + (A/2v) e^{-mx}}{1 - (A/2v) e^{-mx}}. \tag{36}$$

When A is chosen as

$$A = -2ve^{ma}, \tag{37}$$

the $\phi_0^f(x)$ is obtained as

$$\phi_0^f(x) = v \tanh[(m/2)(x-a)], \tag{38}$$

which is the well-known static solution of the Euler equation

$$(-\partial - \mu^2) \phi_0^f(x) = \lambda [\phi_0^f(x)]^3. \tag{39}$$

When $A = 2ve^{ma}$, $\phi_0^f(x)$ is given by

$$\phi_0^f(x) = v \coth[(m/2)(x-a)], \tag{40}$$

which is another solution of (39).

For the static case, (38) and (40) are the only solutions which satisfy (39) with the condition $\phi_0^f(x) \to v (x \to \infty)$. The choice of $f(x)$ and the Green's function $K(x)$ is restricted from the well-defined property of the integration in the right-hand side

of (29). If $f(x)$ is extended to be time dependent, there are many varieties for the choice of $f(x)$. The relation between the choice of $f(x)$ and the N-soliton solution will be a future problem.

The situation becomes involved when there appear certain bound states which prohibit us from using the successive iteration. Such a situation arises in many practical cases, in which the extended objects are created by condensation of composite Goldstone bosons (such as crystal phonons, magnons, etc.). When we cannot use the successive iteration, proof of the boson transformation theorem becomes very complicated. Since this has been discussed in Ref. 4, it will not be repeated here. In the following we present a sketch of the general structure of the boson method for derivation of the soliton solutions.

Suppose that the Heisenberg equation (1) gives rise to the set of in fields $(\varphi_1^{\text{in}}\,\varphi_2^{\text{in}}\cdots\varphi_n^{\text{in}}\cdots)$. Here $\varphi_\alpha^{\text{in}}$ $(\alpha=1,\ldots,n)$ are the boson in fields and other in fields are fermions. The free field equation for the boson in fields are

$$(-\partial^2-m_\alpha^2)\varphi_\alpha^{\text{in}}=0. \tag{41}$$

Since we require that ψ is realized in the Fock space of the in fields, all the matrix elements of ψ among the vectors in this Fock space should be well defined. This means that ψ can be expressed by a linear combination of normal products of the in fields:

$$\psi(x)=\psi(x;\varphi_\alpha^{\text{in}},\ldots). \tag{42}$$

This is the dynamical map and should be understood as a weak relation. To calculate $F[\psi]$ in the right-hand side of the Heisenberg equation (1), we need to define the products of ψ. The products are defined by the rule which states that one first calculates the products of normal products of the in fields and rearranges them into a linear combination of normal products. The space-time integration should be performed only after the products of normal products are well taken care of. We assume that when certain divergences appear through the course of calculation, they can be eliminated by some regularization procedures in a reasonable manner. Once the products of ψ are determined, all the matrix elements of both sides of the Heisenberg equation (1) can be calculated. A well-known expression for the dynamical map is the Lehmann-Symanzik-Zimmermann (LSZ) formula.

Now perform the substitution called the boson transformation

$$\varphi_\alpha^{\text{in}}(x)\to\varphi_\alpha^{\text{in}}(x)+f_\alpha(x), \tag{43}$$

in which $f_\alpha(x)$ $(\alpha=1,\ldots,n)$ are c-number functions which satisfy the equations for $\varphi_\alpha^{\text{in}}$,

$$(-\partial^2-m_\alpha^2)f_\alpha=0. \tag{44}$$

Then, the boson transformation theorem states that the boson-transformed Heisenberg field

$$\psi^f(x)=\psi(x;\varphi_\alpha^{\text{in}}+f_\alpha,\ldots) \tag{45}$$

also satisfies the Heisenberg equation (1), thus leading to (10). Note that the boson $\varphi_\alpha^{\text{in}}$ in (43) can be either elementary or composite.

The expression (43) for the boson transformation indicates that $f_\alpha(x)$ are created by the condensation of the bosons $\varphi_\alpha^{\text{in}}$. The result of this condensation is the appearance of certain extended objects. Since $f_\alpha(x)$ are c numbers, these extended objects behave classically: Intuitively speaking, the quantum fluctuation becomes much smaller than the macroscopic effect of the condensed bosons. The c-number field ϕ^f is defined by (11). Note that $\phi^f(x)$ describes a classically behaving extended object even when $h\neq 0$. When we ignore h, $\phi^f(x)$ becomes $\phi_0^f(x)$, i.e., Eq. (12). Following the argument which led to (13), we can prove that ϕ_0^f satisfies the classical Euler equation (15).

When the boson which makes the boson transformation happens to be composite, we should formulate the tree approximation in such a way that the internal lines of the tree diagrams include the propagation function of the composite boson. Such a formulation of the tree approximation can be made by means of the Ward-Takahashi relations. Since this kind of tree approximation has been formulated in Ref. 8 in which the itinerant electron ferromagnetism was studied, it will not be repeated here.

Equation (44) for f_α admits various propagating wave solutions. A well-known example of wavelike extended objects in solid state physics is the ultrasonic waves in crystals.

When we consider a static extended object, (44) becomes

$$(\nabla^2-m^2)f_\alpha=0. \tag{46}$$

When $m=0$, this equation has a trivial solution, i.e., $f_\alpha(x)=$constant. When we disregard this trivial solution, Eq. (46) with $m^2\geq 0$ does not admit any solution which is Fourier transformable. We thus conclude that the static objects created by the condensation of bosons carry certain singularities which prohibit the Fourier transform of f_α.

Since $f_\alpha(x)$ for static objects should carry certain singularities which prohibit its Fourier transform, $f_\alpha(x)$ has either a divergent singularity or a topological singularity. Here the divergent singularity means that $f_\alpha(x)$ diverges at $|\vec{x}|=\infty$ at least in certain directions of $\vec{x}$. The topological singularity means that $f_\alpha(x)$ is not single valued.

Since $f_\alpha(x)$ is path dependent when it is not single valued, the topological singularity can be mathematically expressed by the relation

$$G^{\alpha\dagger}_{\mu\nu}(x)\neq 0 \quad \text{for certain } x,\ \mu,\ \nu,\ \text{and } \alpha, \tag{47}$$

where $G^{\alpha\dagger}_{\mu\nu}$ is defined by

$$G^{\alpha\dagger}_{\mu\nu}(x)\equiv[\partial_\mu,\partial_\nu]f_\alpha(x). \tag{48}$$

Furthermore, existence of the path-dependent $f_\alpha(x)$ requires that $\partial_\rho f_\alpha(x)$ should be single valued,

$$[\partial_\mu,\partial_\nu]\partial_\rho f_\alpha(x)=0. \tag{49}$$

Equation (44) together with (48) and (49) leads to

$$\partial_\nu f_\alpha(x)=\frac{1}{\partial^2+m_\alpha^2}\,\partial^\mu G^{\alpha\dagger}_{\mu\nu}(x), \tag{50}$$

where the derivative operator $(1/\partial^2+m_\alpha^2)$ is defined in terms of the Fourier representation as follows:

$$\frac{1}{\partial^2+m_\alpha^2}\,e^{ipx}=-\frac{1}{p^2+m_\alpha^2}\,e^{ipx}. \tag{51}$$

Thus $(1/\partial^2+m_\alpha^2)$ is the Green's function of the Klein-Gordon equation with the mass m_α. Note that $G^{\alpha\dagger}_{\mu\nu}$ is Fourier transformable. Equation (50) leads to $\partial^2 f_\alpha=0$, implying that $m_\alpha=0$. We thus conclude that the extended objects associated with the topological singularity (47) can be created only by the condensation of *massless* bosons. A systematic study of extended objects with topological singularities has been presented in Ref. 3.

We need a special remark when the Heisenberg field ψ is a fermion field, because $\langle 0|\psi^f|0\rangle$ vanishes. In this case we can still construct various bosonlike operators by means of products of an even number of ψ's. When we find a set of bosonlike operators [say $\varphi_a(\psi)$; $a=1,\ldots,l$] which satisfy a closed set of Heisenberg equations, we can write down the classical equations of the same form. Latter classical equations are regarded as the Euler equations, and our consideration in this paper can be applied to these Euler equations. However, it frequently happens that the Heisenberg equations are not closed by a finite number of bosonlike operators; to obtain a closed set of Heisenberg equations for bosonlike operators, we usually need certain approximations. The Gor'kov equation[9] for the order parameter $\Delta(x)$ in the theory of superconductivity is a well-known example of this kind of Euler equation. There the Goldstone boson is a bound state of two electrons. When we denote the boson-transformed electron Heisenberg field by ψ^f, the order parameter $\Delta(x)$ is equal to $\langle 0|\psi^f_\downarrow\psi^f_\uparrow|0\rangle$. When $\Delta(x)$ is very small, the Gor'kov equation becomes the Ginzburg-Landau equation. When $\Delta(x)$ is not small, its Euler equation (i.e., the Gor'kov equation) has an extremely complicated structure. However, we can derive also a classical equation for physical quantities such as the electromagnetic field and current by applying the boson transformation to the dynamical maps of the electromagnetic Heisenberg field and electron Heisenberg field.[4,10] This equation can be used for the study of extended objects in superconductors. The same situation is true in the relativistic superconductorlike models (e.g., the Nambu model).

Let us close this paper with two comments on the boson method. In this paper we considered mostly $\phi^f(x)$ which is the vacuum expectation value of the boson-transformed Heisenberg field $\psi^f(x)$. When we calculate other matrix elements of ψ^f and the boson-transformed S matrix, we can study the reactions among extended objects and quantum particles. Another comment is the following: Since ϕ^f contains $\hbar$, it has quantum effects in it, although it describes a classical object. In other words, ϕ^f is the soliton solution with quantum correction; the quantum correction is given by the loop diagrams.

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Ferroelectric Behavior in Microtubule Dipole Lattices: Implications for Information Processing, Signaling and Assembly/Disassembly

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Cytoskeletal microtubules structurally organize interiors of living eukaryotic cells. As polymers of subunit proteins ("tubulin"), which are each dipoles, microtubules are thus lattices of oriented dipoles. In general, three types of arrangements of dipoles in lattices may occur: (i) random, (ii) ferroelectric (parallel-aligned) and (iii) an intermediate weakly ferroelectric phase, which is length-dependent. Because of involvement in dynamical cell activities (movement, growth, mitosis, differentiation, etc.), models of microtubule signaling and information processing have been proposed. In these, tubulin units are assumed to represent informational "bit states" and to be coupled to intra-tubulin dipoles. In the present paper, we consider microtubules as lattice arrays of coupled local dipole states that interact with their immediate neighbors. Depending on the values of assumed model parameters, the system may exhibit "frustration": conflict in satisfying all dipole couplings. Such systems have properties suitable for efficient information processing and computation. By slightly altering temperature and external field (both within physiological conditions), microtubule dipole lattices may assume a ferroelectric phase with long-range order and alignment with capabilities to propagate kink-like excitations. The ferroelectric phase appears to be optimal for microtubule signaling and assembly/disassembly. Microtubules may organize cell activities by operating in different modes suitable for information processing and computation (intermediate phase) or signaling and assembly/disassembly (ferroelectric phase).

1. Introduction

Interiors of living cells are structurally and dynamically organized by networks of protein polymers called the cytoskeleton. Of these filamentous structures which include actin and intermediate filaments, microtubules (MTs) are the best characterized and appear to be the most fundamental (Dustin, 1984). MTs are hollow cylinders 25 nanometers (nm) in diameter whose lengths may span macroscopic dimensions. MTs are assemblies of 13 longitudinal protofilaments, each of which is a series of subunit proteins known as tubulin (Fig. 1). Each tubulin subunit is a polar, 8 nm-long dimer which consists of two slightly different 4 nm-long monomers, each with a molecular weight of 55 kDa. These two constituent parts are referred to as α and β tubulin. Each dimer has a net mobile electronegative charge due to an abundance of acidic amino acids, and binds 18 calcium ions. The net negative mobile charge is localized predominantly towards the α monomer so that each tubulin dimer bears an oriented dipole. Thus, MTs are an example of an electret substance, i.e. an assembly of oriented dipoles. In this paper we investigate the types of spatial arrangements of these dipoles in detail and their implications on MT behavior.

Conformational states of tubulin dimers present within MTs have been suggested to be coupled to charge or dipolar states, thereby allowing for cooperative interactions with neighboring tubulin dimer sites (Hameroff & Watt, 1982; Hameroff, 1987;

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Rasmussen *et al*., 1990). This coupling would also lead to the presence of piezoelectric properties which are very common in ferroelectrics which, in turn, could prove very important in MT signaling, communication and assembly/disassembly behavior (Athenstaedt, 1974; Margulis *et al*., 1978).

MT associated proteins (MAPs) which may be structural, contractile or enzymatic are attached to MTs at specific tubulin dimer sites and can link MTs with other MTs, other parallel arrayed cytoskeletal elements, membranes and organelles to form networks and scaffolding within living cells. MAP attachment sites may be irregular, or can describe various helical patterns on MT surface lattices (Burns, 1978; Kim *et al*., 1986). The focal point of the cytoskeleton is the centriole, a pair of cylindrical assemblies of nine MT triplets oriented perpendicular to each other. Centrioles organize the array of cytoplasmic MTs during interphase, duplicate at mitosis to nucleate the two poles of the mitotic spindle and establish orientation and architecture for the next generation of cells.

Genes for α and β tubulin are complex, multi-gene families that give rise to varying tubulin isozymes. During evolution, some tubulins have been highly conserved for basic cell functions; however, extensive MT heterogeneity exists in complex cells due to genetic diversity, expression, post-translational modifications, MAPs and assembly patterns. For example, two-dimensional gel electrophoresis has shown 17 different varieties of β tubulin exist in mammalian brain MTs, whereas fewer exist in other tissues (Lee *et al*., 1986). Tubulin structure may also be altered by "post-translational" modification: enzymatic alteration (usually addition or removal of amino acids such as glycosylation or detyrosination) triggered by intracellular events including second messenger activities.

MT self-assembly and disassembly are dynamic, complex processes whose states depend on various factors including temperature and calcium ion concentration. Oriented by centrioles, MT polymerization determines the architecture and form of cells which can quickly change by MT depolymerization and reassembly in another direction. GTP, an energy-providing analog of ATP, binds to free, unpolymerized tubulin; GTP-tubulins then self assemble to form MTs in an entropy driven process. Within assembled MTs, GTP hydrolyzes to GDP, imparting energy into the MT lattice via tubulin conformational changes. MTs whose ends are comprised of GTP-tubulin are stable and will continue to grow. MTs whose ends are comprised of GDP-tubulin are unstable and will depolymerize rapidly. As GTP is hydrolyzed to GDP within MT, GDP-tubulin is exposed at MT ends and, unless stabilized by MAPs, centrioles or other structures, MTs rapidly disassemble. Free GDP-tubulin is reconverted to GTP-tubulin which becomes available for reassembly. *In vitro*, this "dynamic instability" (Kirschner & Michison, 1986; Hotani *et al*., 1992) results in erratic alterations between growing and shrinking phases; *in vivo* "selectionist" activity occurs in which MT networks can adaptively probe or retreat in cellular appendages including axonal and dendritic growth cones and developing synapses (e.g. Reinsch *et al*., 1991; Sabry *et al*., 1991). Recent works have shown that MT spatial structures or patterns may form under the maintenance of a sustained energy source (GTP hydrolysis), suggesting that MTs may be dissipative structures (Tabony & Job, 1990, 1992).

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similar to fractal nature of entropy

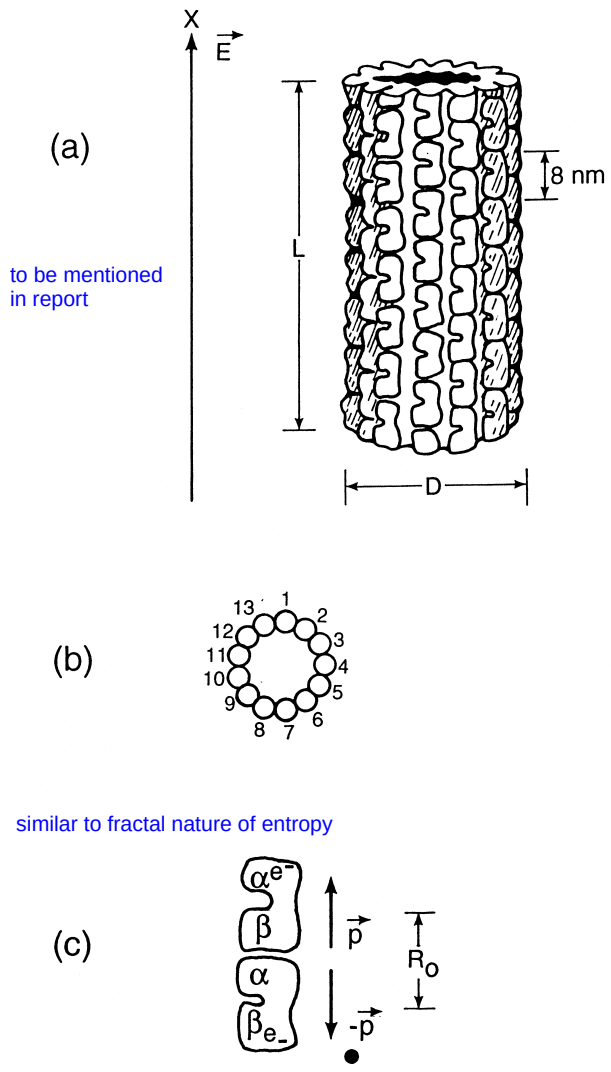

FIG. 1. An illustration of the microtubular arrangement.

Some evidence links the cytoskeleton with information processing and cognitive function. For example, Mileusnic *et al*. (1980) correlated production of MT subunit protein (''tubulin'') and MT activities with peak learning, memory and experience in baby chick brains. Cronley-Dillon *et al*. (1974) showed that when baby rats begin their critical learning phase for the visual system (when they first open their eyes), neurons in the visual cortex begin producing vast quantities of tubulin. Tubulin production is drastically reduced when the critical learning phase is over (when the rats are 35 days old). Moshkov *et al*. (1992) showed structural and chemical changes in goldfish brain neuronal cytoskeleton following sensory stimulation. In gerbils exposed to cerebral ischemia, Kudo *et al*. (1990) correlated the amount of reduction in dendritic MAP-2 with the degree of cognitive impairment. Bensimon & Chernat (1991) found that selective destruction of brain MTs by the drug colchicine caused cognitive defects in learning and memory which mimic the clinical symptoms of Alzheimer's disease, in which the cytoskeleton becomes entangled. Geerts *et al*. (1992) showed that sabeluzole, a memory-enhancing drug, increases fast axoplasmic transport. Matsuyama & Jarvik (1989) have proposed that Alzheimer's is a disease of MTs and MAPs. In specific hippocampal regions of the brains of schizophrenic patients, Arnold *et al*. (1991) found distorted neuronal architecture due to a lack of two MAPs (MAP-2 and MAP-5).

Models of learning in mammalian NMDA (n-methyl-d-aspartate) hippocampal neurons (i.e. long-term potentiation—LTP) involve pre- and post-synaptic enhancement of synaptic function. Silva *et al*. (1992) have shown calcium-calmodulin kinase to be essential for LTP. Aszodi *et al*. (1991) have demonstrated protein kinase A to be involved in learning. Halpain & Greengard (1990) have shown that post-synaptic activation of NMDA receptors induces rapid dephosphorylation of dendrite-specific MAP-2. Aoki & Siekevitz (1988) linked learning to protein kinase C mediated MAP-2 phosphorylation/ dephosphorylation, and Theurkauf & Vallee (1983) demonstrated that MAP-2 phosphorylation/ dephosphorylation consumed a huge proportion of brain biochemical energy. Bigot & Hunt (1990) showed that NMDA and other excitatory amino acid neurotransmitters caused redistribution of intraneuronal MAPs, and Wang & Rasenick (1991) have shown that G proteins directly link with MTs. Surridge & Burns (1992) demonstrated that the lipid second messenger phosphatidylinositol binds specifically to MAP-2, a coupling which may ''constitute a link by which extracellular factors influence the microtubule cytoskeleton''. Kwak & Matus (1988) showed that denervation caused long-lasting changes in MAP distribution, and Desmond & Levy (1988) showed that LTP involved cytoskeletal-mediated shape changes in dendritic spines. Lynch & Baudry (1987) have proposed that proteolytic digestion of the sub-synaptic cytoskeleton, followed by its structural reorganization, correlate with learning, and Friedrich (1990) has formalized a learning model of synaptic cytoskeletal restructuring. Synaptic regulation, both in learning and steady state, also depends on material (enzymes, receptors, neurotransmitters, etc.) synthesized in the cell body and transported along axons and dendrites by contractile MAPs attached to MT (''axoplasmic transport''). Similar mechanisms involving actin, synapsin and other cytoskeletal polymers are involved in neurotransmitter release.

Further suggestion for cytoskeletal computation and/or information storage stems from the spatial distribution of discrete sites (or states) in the cytoskeleton. For example, tubulin subunits in closely arrayed MTs have a density of about 10^{17} cm^{-3}, which is very close to the theoretical limit for charge separation (Gutmann, 1986). Thus cytoskeletal polymers have maximal density for information representation by charge, and the capacity for dynamically coupling that information to mechanical and chemical events via cooperative dipole states.

Previous papers have modeled signaling and information processing in MTs based on coherent phonons, solitons and cellular automata based on dipole coupled conformational states of tubulin.

In this paper we address the question of how some functions of MTs (i.e. assembly/disassembly and proposed signaling and information processing) may be related to the ground state properties of the lattice of dipoles. In general, three kinds of geometrical arrangements of dipoles are found: (i) random where no order in the dipole arrangements exists, (ii) ferroelectric (long-range parallel-aligned order) and (iii) intermediate, weakly ferroelectric where short range order is present. Each of these arrangements may exist under different conditions of temperature, electric field and MT length, and a single MT may switch among the different arrangements. The long-range ordered ferroelectric state appears most suitable for assembly/disassembly and signaling processes while the short-range ordered weakly ferroelectric state is well suited for information processing. In the next two sections we discuss these properties in detail.

2. The Dipolar Lattice and its Phases

Our basic premise physically views the entire MT as a regular array of coupled local dipole states

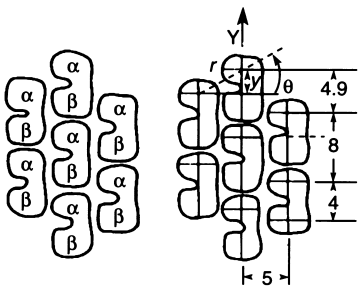

FIG. 2. A tubulin dimer and its nearest neighbors.

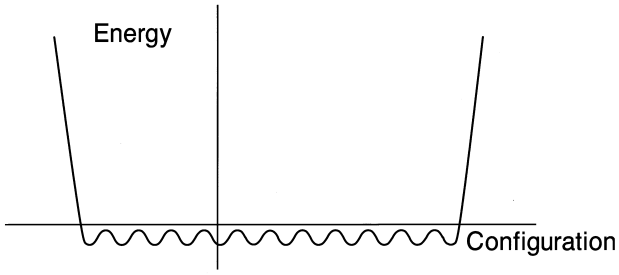

FIG. 4. A schematic representation of the energy dependence on conformational states.

which interact with their immediate neighbors. Although tubulin undergoes conformational changes (e.g. Melki *et al*., 1989), we adopt a simplified view in which elastic degrees of freedom are not explicitly included in the description. We will return to this question in the Discussion.

The starting point in the analysis is to adopt a hexagonal lattice structure with the dimensions and orientations observed by X-ray diffraction of MTs and shown in Fig. 2. Each lattice site is assumed to possess a dipole moment $p = Q \cdot d$ where $Q = 2e$ and $d \cong 4$ nm whose projections on the vertical axis can only be $+p$ or $-p$. The interaction energy between two neighboring lattice sites is

$$H_{ij} = \frac{1}{4\pi\epsilon\epsilon_0} \frac{3\cos^2\theta - 1}{r_{ij}^3} p^2, \tag{1}$$

where ϵ_0 is the vacuum permittivity and r_{ij} is the distance between sites i and j. The angle θ is between the dipole axis and the direction between the two dipoles. Figure 3 illustrates the situation resulting from our calculations.

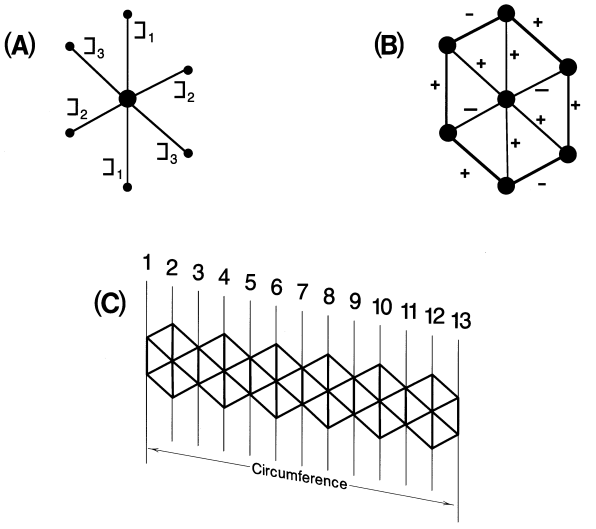

FIG. 3. Exchange constants (a) and their signs (b) within a unit hexagonal cell. (c) A band of hexagons spanning the circumference of a microtubule.

In Fig. 3(b) the signs "+" and "−" refer to dipole–dipole interactions that prefer either a parallel or antiparallel arrangements of dipole moments, respectively. Below, we present the numerical results for the constants J_1, J_2, J_3 and the corresponding angles.

$$J_1 = 5.77 \times 10^{-21}\ \text{J} \quad \theta_1 = 0^\circ$$

$$J_2 = -0.71 \times 10^{-21}\ \text{J} \quad \theta_2 = 58.2^\circ$$

$$J_3 = 3.40 \times 10^{-21}\ \text{J} \quad \theta_3 = 45.6^\circ$$

We can, therefore, map this situation as an anisotropic two-dimensional Ising model on a triangular lattice so that the effective Hamiltonian is

$$H = \underset{\langle nn \rangle}{-\Sigma} J_{ij} S_i^z S_j^z, \tag{2}$$

and the effective spin variable $S_i^z = \pm 1$ denotes the dipole's projection on the vertical axis. The exchange constants J_{ij} take the values J_1, J_2, J_3 depending on the choice of dipole pairs.

What, therefore, appears in Fig. 3(b) is that the system exhibits a certain amount of *frustration* (Suzuki, 1977) in its ground state. This means that for a closed path around the 13 protofilaments of the microtubule, it is impossible to satisfy all bond requirements since there is an odd number of antiparallel bonds. Note that no frustration exists in the direction along the MT axis giving the overall ferroelectric character to the lattice. It will also be seen how this is manifested through size dependence of net polarization.

Small potential barriers separating the various equivalent arrangements of spins (see Fig. 4) play another useful role. Relaxation times are very long for the various accessible states giving them long-term stability. Some other properties of this intermediate, weakly ferroelectric phase are: (i) absence of long-range order between spin sites and (ii) the presence of short-range correlations. This means that the system is very "soft" energetically, permitting a number of spin arrangements which are relatively

stable. However, there is no tendency towards the formation of large correlated spin clusters. This intermediate phase also allows easy formation of local ordered states, each of which carries an information content and is long-lived, i.e. relatively stable over time. On the other hand, it is not difficult to switch from one local state to another through various physical means by overcoming local potential barriers. Furthermore, a complete phase transformation can be achieved by:

(i) an application of an electric field along the axis. We estimate that intracellular fields on the order of $10^4\ \mathrm{Vm}^{-1}$ are sufficient to switch an MT from an intermediate state to a ferroelectric state (F). (Note that fields up to $10^7\ \mathrm{Vm}^{-1}$ are known to exist across cell membranes).

(ii) Raising the temperature above the ferroelectric transition temperature will drive the system to a disordered (paraelectric) phase.

We mentioned above that another ordered phase is favored when a moderate external field is applied (or indeed when the temperature is lowered further). This phase, called a ferroelectric phase, is characterized by long-range order manifested by an alignment of spin directions along the axis. This is shown in Fig. 5. Obviously, from the point of view of optimal information content, this is *not* a very useful phase. However, the ferroelectric phase can play a major role in signaling and the assembly/disassembly processes. This will be discussed at length in the next Section. Before we discuss this aspect, we investigate its general properties. The ferroelectric phase can also be destabilized by temperature when T exceeds T_c. To obtain a crude estimate of T_c we approximated the MT by an infinite triangular lattice with the exchange constants J_1, J_2 and J_3 given above and used the formula (Domb, 1960).

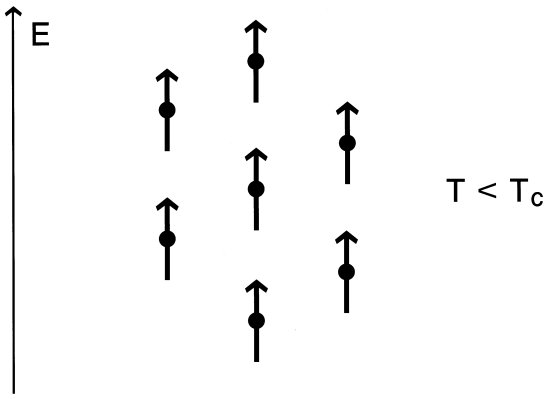

FIG. 5. The dipole orientations in the ferroelectric state.

$$\sinh\left(\frac{2J_1}{K_B T_C}\right)\sinh\left(\frac{2J_2}{K_B T_C}\right) + \sinh\left(\frac{2J_1}{K_B T_C}\right)\sinh\left(\frac{2J_3}{K_B T_C}\right) + \sinh\left(\frac{2J_2}{K_B T_C}\right)\sinh\left(\frac{2J_3}{K_B T_C}\right) = 1. \quad (3)$$

The result we obtained using the above crude estimate for an infinitely long lattice is $T_c \cong 402\ K$ *for the charge of* $q = 2e$, the permittivity $\epsilon = 10$ and the dipole displacement $d = 4$ nm. On the other hand, taking $q = 1e$, $\epsilon = 5$ and $d = 4$ nm yields $T_c = 201$ K. These two very rough estimates of T_c indicate nevertheless that it is feasible to expect an interesting phase behavior in the physiological temperature range around 300 K for a specific choice of model parameters. In addition, it is well known that the size of a lattice affects T_c greatly. In general, for size N, $T_c(N)$ is related to that for an infinite lattice $T_c(\infty)$ through (Binder *et al*., 1985):

$$T_c(N) = T_c(\infty) - \alpha N^{-1/\gamma}, \quad (4)$$

where α is an approximate proportionality coefficient and γ is the correlation length exponent (typically close to 1/2). Thus, for a finite-size lattice, the transition temperature is expected to be less than that for an infinite lattice. In other words, dipole ordering is more likely to occur when a critical size of the MT is achieved. In addition, other important factors may affect the value of T_c and thus provide sensitive control mechanisms. Through a coupling to the elastic degrees of freedom (conformational change), the value of T_c may be shifted. The dielectric constant ϵ may be altered by the presence of water molecules surrounding an MT structure. This would decrease the value of T_c and introduce disorder. Small changes in the angles between the dimer dipoles may remove the frustration mechanism effectively preventing the onset of the intermediate phase. Changes in the opposite direction can enhance frustration favoring the intermediate phase over the F phase and effectively switching from the growth mode of operation to an information processing behavior. Thus symmetry transitions or mechanical forces may effect switching between phases.

Once in the ferroelectric phase, the system is also very sensitive to the frequency of electromagnetic oscillations. It is known that in organic ferroelectric materials a resonant frequency exists W_0 which is inversely proportional to the length, stiffness constant

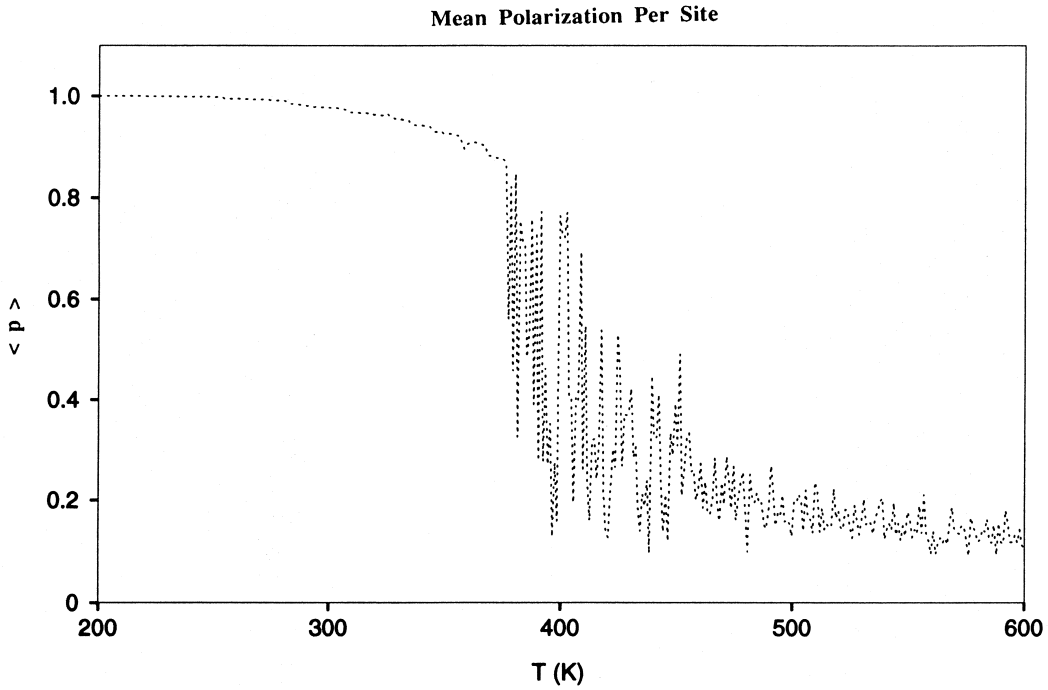

FIG. 6. A plot of the mean polarization per site for a triangular lattice whose size is 13×26.

and the square root of the density (Matthias, 1973). Extrapolating the data given by Matthias (1973), one finds for microtubules a resonant frequency range exceeding 10^8 Hz and thus being rather close to the microwave range predicted by Fröhlich (1980) to play a major role in living cells.

In Figs 6 and 7 we have shown the results of our Monte Carlo computations intended to reveal the size dependence of the phase behavior discussed above. Figure 6 is a plot of the mean polarization per site when the microtubule is composed of only 26 layers (i.e. it is a 13×26 lattice). It is clear that three distinct ranges exist: (i) the low-temperature ferroelectric range, (ii) the intermediate weakly ordered phase range and (iii) the high-temperature paraelectric range. Figure 7 illustrates the same dependence for a microtubule with 5000 layers (i.e. it is a 13×5000 lattice). We conclude that the intermediate weakly ferroelectric phase all but disappeared.

To summarize, in this section we have argued that two ordered phases are likely to govern the behavior of dipole moments in an MT. A ferroelectric phase is characterized by long-range order and has many characteristics useful in energy transfer properties discussed in the next section. An intermediate weakly ordered phase exhibits short-range order but has a multitude of equivalent, long-lived dipolar states. It appears that in the latter phase the system can explore a large subset of the phase space at a low energetical cost unlike in the ferroelectric phase. Having unhindered access to a multitude of conformational states indicates easy transitions from one configuration to another at little or no energy cost. A number of control mechanisms are possible which allow switching from one phase to the other and hence from one mode of operation to the other.

3. Ferroelectricity, Signaling and Assembly/Disassembly

It is well known that ferroelectrics are one of the few types of materials whose description is most adequately provided in terms of mean field models (Kretschmer & Binder, 1979). A typical approach is to postulate a double-well quartic on-site potential $V(S_n)$ that gives the overall effect of the environment on a selected dipole site n. Thus,

$$V(S_n) = -\tfrac{1}{2}AS_n^2 + \tfrac{1}{4}BS_n^4, \tag{5}$$

where S_n is the effective spin variable at site n and the coefficients A and B are model-dependent. It will be subsequently assumed that over long distances the variable S_n changes very gradually and can be treated as a continuous variable.

In the F phase, the MT cylinder taken as a whole represents a giant dipole. When the cross-section of an MT is viewed using electron microscopy, the outer surfaces of an MT are surrounded by a "clear zone" of several nm which apparently represents the oriented molecules of cytoplasmic water and enzymes (Steb-

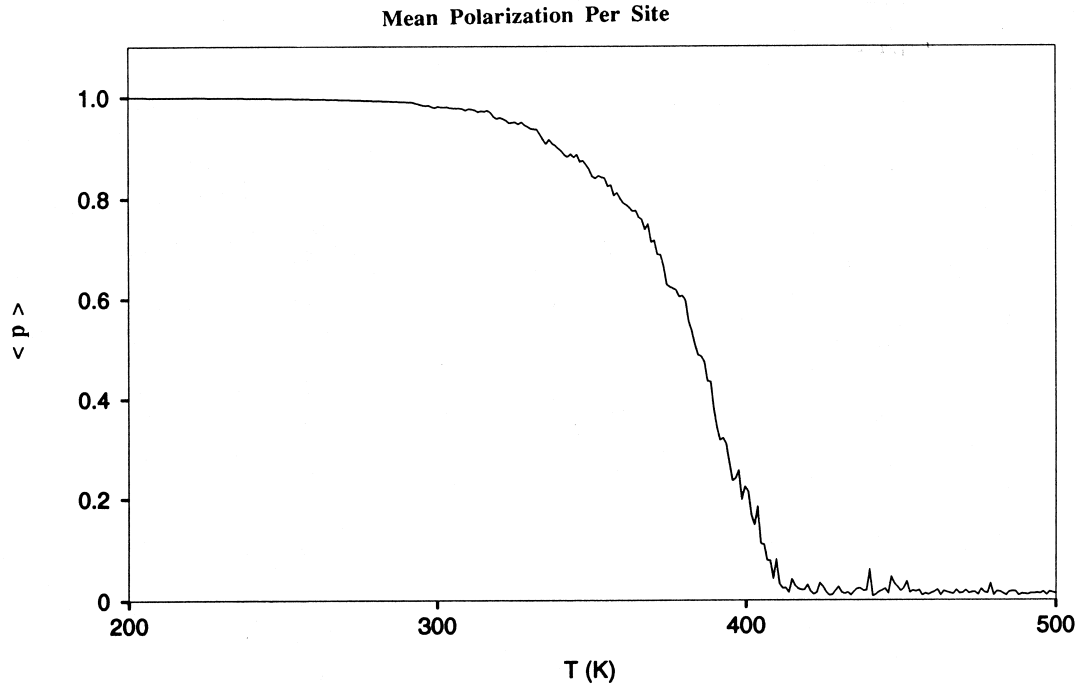

FIG. 7. A plot of the mean polarization per site for a triangular lattice whose size is 13×5000.

bings & Hunt, 1982). This could be explained by the presence of an electric field produced by an MT. Therefore, we assume that together with the polarized water surrounding it, an MT generates a nearly uniform intrinsic electric field parallel to its axis (see Fig. 1). Consequently, the additional potential energy due to this electric field and associated with each dipole is

$$V_{\mathrm{el}} = -CS_n, \quad C = qE, \tag{6}$$

where q denotes the effective charge of a single dimer and E is the magnitude of the intrinsic electric field. Within this picture, the model Hamiltonian for the dipole moments of a microtubule can be written as

$$H = \sum_{i=1}^{n} \left[\frac{1}{2} M \left(\frac{\mathrm{d}S_n}{\mathrm{d}t} \right)^2 + \frac{1}{2} K (S_{n+1} - S_n)^2 - \frac{A}{2} S_n^2 + \frac{B}{4} S_n^4 - CS_n \right]. \tag{7}$$

The first term above represents the kinetic energy associated with the longitudinal displacements of constituent dimers each of which has mass M. The second term arises from the restoring strain forces between adjacent dimers in the protofilament. If the stiffness parameter K is sufficiently large, then it is expected that large-amplitude long-wavelength excitations of the displacement field will be formed manifested by a slowly varying modulation of S_n along the MT axis.

In order to derive a realistic equation of motion for this system, the viscosity of the solvent is indispensable and the associated damping force must be introduced. Assuming for simplicity that the solvent is made up of only water molecules we may infer that water will provide a viscous medium that will damp out vibrations of dimer dipoles. This can be taken into account by adding the viscous force to the equation of motion with

$$F = -\gamma \frac{\mathrm{d}S_n}{\mathrm{d}t},$$

where γ is the damping coefficient.

Based on our assumption that the dipolar oscillations of dimers within an MT form a system that can be classified as displacive ferrodistortive, we can use the continuum approximation to obtain the corresponding equation of motion as

$$M \frac{\mathrm{d}^2 S}{\mathrm{d}t^2} - KR_0^2 \frac{\mathrm{d}^2 S}{\mathrm{d}x^2} - AS + BS^3 + \gamma \frac{\mathrm{d}S}{\mathrm{d}t} - qE = 0. \tag{8}$$

Note that the x-coordinate is along the protofilament axis. Moreover, for longitudinal sound waves the dispersion relation

$$\omega = \sqrt{K/M}$$

can be used to identify the sound velocity with

$$V_0^2 = \omega R_0 = \sqrt{K/M}\, R_0.$$

We now seek solutions of eqn (7) in the traveling-wave form where the moving coordinate ξ is given by

$$\xi = \left[\frac{|A|}{M(V_0^2 - V^2)}\right]^{1/2}(x - Vt) = \alpha(x - Vt), \qquad (9)$$

with V denoting the propagation velocity and the coefficient α is, of course,

$$\alpha = \left[\frac{|A|}{M(V_0^2 - V^2)}\right]^{1/2}$$

Consequently, eqn (8) is reduced to the ordinary differential equation below

$$M\alpha^2(V^2 - V_0^2)S'' - \gamma\alpha V S' - AS + BS^3 - qE = 0, \quad (10)$$

where $S' \equiv \mathrm{d}S/\mathrm{d}\xi$. This is an equation for an anharmonic oscillator with linear friction.

For the sake of convenience we now introduce a normalized displacement field as:

$$\Psi(\xi) = S(\xi)/S_0$$

where the normalizing coefficient

$$S_0 = \sqrt{|A|/B}$$

corresponds to the minimum of the double well potential in eqn (5).

It has been shown by Collins *et al*. (1979) that eqn (10) has a unique bounded solution which is given by the formula (and illustrated graphically in Fig. 8)

$$\Psi(\xi) = a + \frac{b - a}{1 + \exp(\beta\xi)}, \qquad (11)$$

where

$$\beta = (b - a)/\sqrt{2}$$

and the parameters a, b and d satisfy the cubic equation

$$(\Psi - a)(\Psi - b)(\Psi - d) = \Psi^3 - \Psi - \sigma, \qquad (12)$$

where

$$\sigma = q\sqrt{B}|A|^{-3/2}E.$$

Formula (11) describes a kink-like excitation (KLE) which takes the form of a domain wall separating two distinct ferroelectric domains.

It is also important to note that the above kink-like soliton propagates along the protofilament with a fixed velocity

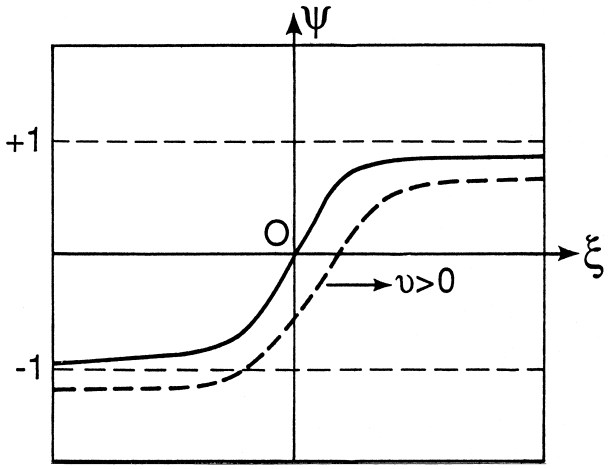

FIG. 8. The kink-like profile to the solution $\Psi(\xi)$ of eqn (11).

$$V = V_0/\left[1 + \frac{2\gamma^2}{9d^2MV_0^2}\right]^{1/2}, \qquad (13)$$

which is *less* than the sound velocity V_0 and decreases with an increase of the friction coefficient γ. It is clear that V depends on the magnitude of the electric field E through the parameter d.

For T well below T_c, V can be approximated by a linear relationship with E

$$V \approx \frac{3V_0}{\gamma|A|}\sqrt{\frac{MB}{2}}\, qE, \qquad (14)$$

where the coefficient of proportionality is called mobility and will be denoted by μ. Using simple arguments from fluid dynamics, μ can be estimated numerically as

$$\mu \approx 2 \times 10^{-5}\ \mathrm{m^2\ V^{-1}\ sec^{-1}}$$

Thus, for $E = 10^5\ \mathrm{V\ m^{-1}}$ we obtain $\mathrm{V} \cong 2\ \mathrm{m\ sec^{-1}}$ which can increase to $10^2\ \mathrm{m\ sec^{-1}}$ close to the MTs ends where E increases dramatically.

Assuming a smooth journey from one end of the protofilament to the other we estimate the average time of propagation $\bar{T}$ for a single KLE to be

$$\bar{T} = \frac{L}{\bar{V}} \approx 5 \times 10^{-7}\ \mathrm{sec}$$

A recent paper (Chou *et al*., 1994) also pointed to the possible role of solitary waves in the dynamics of microtubules.

To summarize, we showed that a unique kink-like excitation exists in the ferroelectric phase which

may propagate along an MT with a velocity that is proportional to the electric field E. KLEs in stable microtubules may constitute signals transducing membrane events via second messengers to other cell regions. Energy for such KLEs may come from GTP hydrolysis or phosphorylation via MAPs. In neurons, KLEs in microtubules may signal error information retrograde (i.e. from axon to dendrite) which would be useful in synaptic regulation and learning as in artificial neural network "back-propagation". In unstable MTs (i.e. whose ends are not stabilized by MAPs or other structures), KLEs can explain the fact that growth rates are different at the two ends of an MT. To this end, we assume that KLE formation is mainly due to the hydrolysis of GTP into GDP so that one act of hydrolysis corresponds to conformational change resulting in the formation of a single KLE. However, a KLE is preferentially oriented towards the direction of the intrinsic electric field. The propagation of a KLE will then distribute the energy of hydrolysis at the preferred end of an MT. This energy may then be used to detach dimers from the MT. This picture is in accordance with a hypothesis put forward by Kirschner & Mitchison (1986) who stated that on-rate (growth) is limited in principle only by the rate of diffusion of the monomer subunits into MT polymers (i.e. by the concentration of the constituent monomers in solution). On the other hand, the off-rate (depolymerization) can be extremely rapid and the net amount of MT polymer formed can be regulated independently by the hydrolysis of GTP.

It is reasonable to expect that at the onset of the MT assembly process, when MTs are very short, the formation of KLEs is very unlikely (or even impossible), since for a short chain the displacement field gradient required would be very high and the associated energy to form it prohibitively large. However, when the length of an MT reaches a threshold value, kink propagation could become a real possibility. Excitation of kinks being related to GTP hydrolysis would become dependent on the GTP concentration. As MTs become longer, KLE propagation accelerates leading to a disassembly at the "-" end. KLE propagation in stable MTs would depend on MAP phosphorylation or cytoplasmic kinase activity triggered by membrane events and second messengers.

4. Summary

We have argued in this paper that the spatial arrangement of dipole moments of a microtubule is crucial to its functioning as a dynamic system. Due to the presence of frustration in the dipole–dipole interactions, an intermediate weakly ferroelectric phase is predicted to arise at low enough temperatures and electric fields as well as for sufficiently short microtubules. Its attractiveness lies in maximal computational capabilities offered by a highly degenerate ground state and by long relaxation times giving relative stability to short-range correlated dipole patterns. Each pattern can be seen as containing binary information. This phase strongly resembles the well-known spin glass phenomenon which has received much attention in frustrated spin systems such as random-anisotropy magnets. Recently, spin-glass models have been applied in connection with biological systems (Stein, 1992).

The other possible ordered state is a ferroelectric phase with an almost perfect alignment of dipole movement along the protofilament axis. It is characterized by long-range order and hence its usefulness for information processing is dubious. However, it eagerly supports the formation of domain walls between the two stable orientations of dipole movements. The application of an external electric field preferentially directs kink-like excitations towards the properly aligned end. These kink-like excitations may be utilized as signaling mechanisms or causing disassembly due to energy release.

We have discussed the various possible control mechanisms (temperature, electric field, mechanical distortion) that could select the MT operating modes among information processing, weakly ordered and signaling and assembly/disassembly (ferroelectric) types.

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Ferroelectric Behavior in Microtubule Dipole Lattices: Implications for Information Processing, Signaling and Assembly/Disassembly

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Cytoskeletal microtubules structurally organize interiors of living eukaryotic cells. As polymers of subunit proteins (''tubulin''), which are each dipoles, microtubules are thus lattices of oriented dipoles. In general, three types of arrangements of dipoles in lattices may occur: (i) random, (ii) ferroelectric (parallel-aligned) and (iii) an intermediate weakly ferroelectric phase, which is length-dependent. Because of involvement in dynamical cell activities (movement, growth, mitosis, differentiation, etc.), models of microtubule signaling and information processing have been proposed. In these, tubulin units are assumed to represent informational ''bit states'' and to be coupled to intra-tubulin dipoles. In the present paper, we consider microtubules as lattice arrays of coupled local dipole states that interact with their immediate neighbors. Depending on the values of assumed model parameters, the system may exhibit ''frustration'': conflict in satisfying all dipole couplings. Such systems have properties suitable for efficient information processing and computation. By slightly altering temperature and external field (both within physiological conditions), microtubule dipole lattices may assume a ferroelectric phase with long-range order and alignment with capabilities to propagate kink-like excitations. The ferroelectric phase appears to be optimal for microtubule signaling and assembly/disassembly. Microtubules may organize cell activities by operating in different modes suitable for information processing and computation (intermediate phase) or signaling and assembly/disassembly (ferroelectric phase).

1. Introduction

Interiors of living cells are structurally and dynamically organized by networks of protein polymers called the cytoskeleton. Of these filamentous structures which include actin and intermediate filaments, microtubules (MTs) are the best characterized and appear to be the most fundamental (Dustin, 1984). MTs are hollow cylinders 25 nanometers (nm) in diameter whose lengths may span macroscopic dimensions. MTs are assemblies of 13 longitudinal protofilaments, each of which is a series of subunit proteins known as tubulin (Fig. 1). Each tubulin subunit is a polar, 8 nm-long dimer which consists of two slightly different 4 nm-long monomers, each with a molecular weight of 55 kDa. These two constituent parts are referred to as α and β tubulin. Each dimer has a net mobile electronegative charge due to an abundance of acidic amino acids, and binds 18 calcium ions. The net negative mobile charge is localized predominantly towards the α monomer so that each tubulin dimer bears an oriented dipole. Thus, MTs are an example of an electret substance, i.e. an assembly of oriented dipoles. In this paper we investigate the types of spatial arrangements of these dipoles in detail and their implications on MT behavior.

Conformational states of tubulin dimers present within MTs have been suggested to be coupled to charge or dipolar states, thereby allowing for cooperative interactions with neighboring tubulin dimer sites (Hameroff & Watt, 1982; Hameroff, 1987;

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Rasmussen *et al*., 1990). This coupling would also lead to the presence of piezoelectric properties which are very common in ferroelectrics which, in turn, could prove very important in MT signaling, communication and assembly/disassembly behavior (Athenstaedt, 1974; Margulis *et al*., 1978).

MT associated proteins (MAPs) which may be structural, contractile or enzymatic are attached to MTs at specific tubulin dimer sites and can link MTs with other MTs, other parallel arrayed cytoskeletal elements, membranes and organelles to form networks and scaffolding within living cells. MAP attachment sites may be irregular, or can describe various helical patterns on MT surface lattices (Burns, 1978; Kim *et al*., 1986). The focal point of the cytoskeleton is the centriole, a pair of cylindrical assemblies of nine MT triplets oriented perpendicular to each other. Centrioles organize the array of cytoplasmic MTs during interphase, duplicate at mitosis to nucleate the two poles of the mitotic spindle and establish orientation and architecture for the next generation of cells.

Genes for α and β tubulin are complex, multi-gene families that give rise to varying tubulin isozymes. During evolution, some tubulins have been highly conserved for basic cell functions; however, extensive MT heterogeneity exists in complex cells due to genetic diversity, expression, post-translational modifications, MAPs and assembly patterns. For example, two-dimensional gel electrophoresis has shown 17 different varieties of β tubulin exist in mammalian brain MTs, whereas fewer exist in other tissues (Lee *et al*., 1986). Tubulin structure may also be altered by "post-translational" modification: enzymatic alteration (usually addition or removal of amino acids such as glycosylation or detyrosination) triggered by intracellular events including second messenger activities.

MT self-assembly and disassembly are dynamic, complex processes whose states depend on various factors including temperature and calcium ion concentration. Oriented by centrioles, MT polymerization determines the architecture and form of cells which can quickly change by MT depolymerization and reassembly in another direction. GTP, an energy-providing analog of ATP, binds to free, unpolymerized tubulin; GTP-tubulins then self assemble to form MTs in an entropy driven process. Within assembled MTs, GTP hydrolyzes to GDP, imparting energy into the MT lattice via tubulin conformational changes. MTs whose ends are comprised of GTP-tubulin are stable and will continue to grow. MTs whose ends are comprised of GDP-tubulin are unstable and will depolymerize rapidly. As GTP is hydrolyzed to GDP within MT, GDP-tubulin is exposed at MT ends and, unless stabilized by MAPs, centrioles or other structures, MTs rapidly disassemble. Free GDP-tubulin is reconverted to GTP-tubulin which becomes available for reassembly. *In vitro*, this "dynamic instability" (Kirschner & Michison, 1986; Hotani *et al*., 1992) results in erratic alterations between growing and shrinking phases; *in vivo* "selectionist" activity occurs in which MT networks can adaptively probe or retreat in cellular appendages including axonal and dendritic growth cones and developing synapses (e.g. Reinsch *et al*., 1991; Sabry *et al*., 1991). Recent works have shown that MT spatial structures or patterns may form under the maintenance of a sustained energy source (GTP hydrolysis), suggesting that MTs may be dissipative structures (Tabony & Job, 1990, 1992).

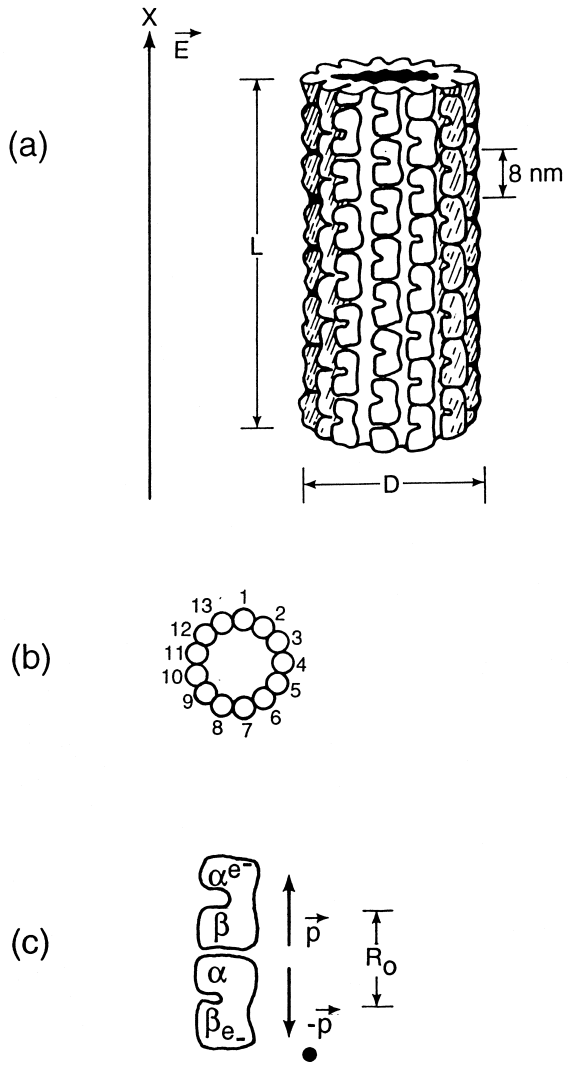

FIG. 1. An illustration of the microtubular arrangement.

Some evidence links the cytoskeleton with information processing and cognitive function. For example, Mileusnic *et al*. (1980) correlated production of MT subunit protein (''tubulin'') and MT activities with peak learning, memory and experience in baby chick brains. Cronley-Dillon *et al*. (1974) showed that when baby rats begin their critical learning phase for the visual system (when they first open their eyes), neurons in the visual cortex begin producing vast quantities of tubulin. Tubulin production is drastically reduced when the critical learning phase is over (when the rats are 35 days old). Moshkov *et al*. (1992) showed structural and chemical changes in goldfish brain neuronal cytoskeleton following sensory stimulation. In gerbils exposed to cerebral ischemia, Kudo *et al*. (1990) correlated the amount of reduction in dendritic MAP-2 with the degree of cognitive impairment. Bensimon & Chernat (1991) found that selective destruction of brain MTs by the drug colchicine caused cognitive defects in learning and memory which mimic the clinical symptoms of Alzheimer's disease, in which the cytoskeleton becomes entangled. Geerts *et al*. (1992) showed that sabeluzole, a memory-enhancing drug, increases fast axoplasmic transport. Matsuyama & Jarvik (1989) have proposed that Alzheimer's is a disease of MTs and MAPs. In specific hippocampal regions of the brains of schizophrenic patients, Arnold *et al*. (1991) found distorted neuronal architecture due to a lack of two MAPs (MAP-2 and MAP-5).

Models of learning in mammalian NMDA (n-methyl-d-aspartate) hippocampal neurons (i.e. long-term potentiation—LTP) involve pre- and post-synaptic enhancement of synaptic function. Silva *et al*. (1992) have shown calcium-calmodulin kinase to be essential for LTP. Aszodi *et al*. (1991) have demonstrated protein kinase A to be involved in learning. Halpain & Greengard (1990) have shown that post-synaptic activation of NMDA receptors induces rapid dephosphorylation of dendrite-specific MAP-2. Aoki & Siekevitz (1988) linked learning to protein kinase C mediated MAP-2 phosphorylation/ dephosphorylation, and Theurkauf & Vallee (1983) demonstrated that MAP-2 phosphorylation/ dephosphorylation consumed a huge proportion of brain biochemical energy. Bigot & Hunt (1990) showed that NMDA and other excitatory amino acid neurotransmitters caused redistribution of intraneuronal MAPs, and Wang & Rasenick (1991) have shown that G proteins directly link with MTs. Surridge & Burns (1992) demonstrated that the lipid second messenger phosphatidylinositol binds specifically to MAP-2, a coupling which may ''constitute a link by which extracellular factors influence the microtubule cytoskeleton''. Kwak & Matus (1988) showed that denervation caused long-lasting changes in MAP distribution, and Desmond & Levy (1988) showed that LTP involved cytoskeletal-mediated shape changes in dendritic spines. Lynch & Baudry (1987) have proposed that proteolytic digestion of the sub-synaptic cytoskeleton, followed by its structural reorganization, correlate with learning, and Friedrich (1990) has formalized a learning model of synaptic cytoskeletal restructuring. Synaptic regulation, both in learning and steady state, also depends on material (enzymes, receptors, neurotransmitters, etc.) synthesized in the cell body and transported along axons and dendrites by contractile MAPs attached to MT (''axoplasmic transport''). Similar mechanisms involving actin, synapsin and other cytoskeletal polymers are involved in neurotransmitter release.

Further suggestion for cytoskeletal computation and/or information storage stems from the spatial distribution of discrete sites (or states) in the cytoskeleton. For example, tubulin subunits in closely arrayed MTs have a density of about 10^{17} cm^{-3}, which is very close to the theoretical limit for charge separation (Gutmann, 1986). Thus cytoskeletal polymers have maximal density for information representation by charge, and the capacity for dynamically coupling that information to mechanical and chemical events via cooperative dipole states.

Previous papers have modeled signaling and information processing in MTs based on coherent phonons, solitons and cellular automata based on dipole coupled conformational states of tubulin.

In this paper we address the question of how some functions of MTs (i.e. assembly/disassembly and proposed signaling and information processing) may be related to the ground state properties of the lattice of dipoles. In general, three kinds of geometrical arrangements of dipoles are found: (i) random where no order in the dipole arrangements exists, (ii) ferroelectric (long-range parallel-aligned order) and (iii) intermediate, weakly ferroelectric where short range order is present. Each of these arrangements may exist under different conditions of temperature, electric field and MT length, and a single MT may switch among the different arrangements. The long-range ordered ferroelectric state appears most suitable for assembly/disassembly and signaling processes while the short-range ordered weakly ferroelectric state is well suited for information processing. In the next two sections we discuss these properties in detail.

2. The Dipolar Lattice and its Phases

Our basic premise physically views the entire MT as a regular array of coupled local dipole states

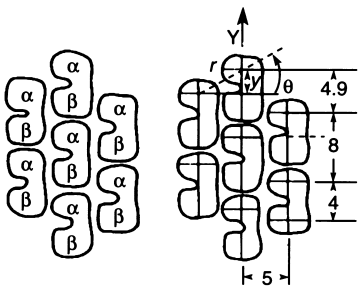

FIG. 2. A tubulin dimer and its nearest neighbors.

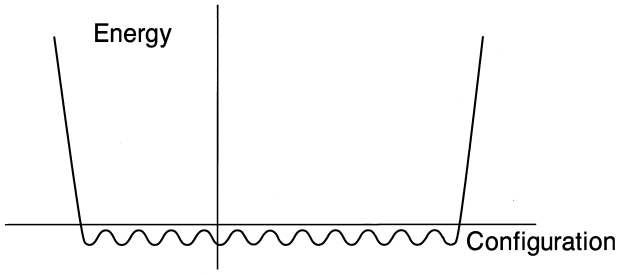

FIG. 4. A schematic representation of the energy dependence on conformational states.

which interact with their immediate neighbors. Although tubulin undergoes conformational changes (e.g. Melki *et al*., 1989), we adopt a simplified view in which elastic degrees of freedom are not explicitly included in the description. We will return to this question in the Discussion.

The starting point in the analysis is to adopt a hexagonal lattice structure with the dimensions and orientations observed by X-ray diffraction of MTs and shown in Fig. 2. Each lattice site is assumed to possess a dipole moment $p = Q \cdot d$ where $Q = 2e$ and $d \cong 4$ nm whose projections on the vertical axis can only be $+p$ or $-p$. The interaction energy between two neighboring lattice sites is

$$H_{ij} = \frac{1}{4\pi\epsilon\epsilon_0} \frac{3\cos^2\theta - 1}{r_{ij}^3} p^2, \tag{1}$$

where ϵ_0 is the vacuum permittivity and r_{ij} is the distance between sites i and j. The angle θ is between the dipole axis and the direction between the two dipoles. Figure 3 illustrates the situation resulting from our calculations.

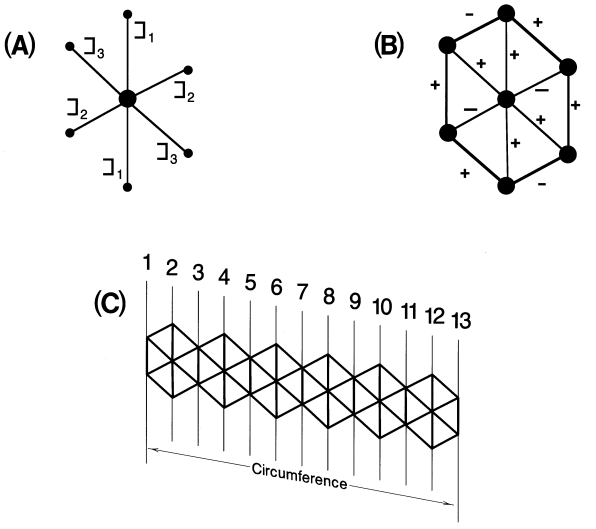

FIG. 3. Exchange constants (a) and their signs (b) within a unit hexagonal cell. (c) A band of hexagons spanning the circumference of a microtubule.

In Fig. 3(b) the signs "+" and "−" refer to dipole–dipole interactions that prefer either a parallel or antiparallel arrangements of dipole moments, respectively. Below, we present the numerical results for the constants J_1, J_2, J_3 and the corresponding angles.

$$J_1 = 5.77 \times 10^{-21}\ \text{J} \quad \theta_1 = 0^\circ$$

$$J_2 = -0.71 \times 10^{-21}\ \text{J} \quad \theta_2 = 58.2^\circ$$

$$J_3 = 3.40 \times 10^{-21}\ \text{J} \quad \theta_3 = 45.6^\circ$$

We can, therefore, map this situation as an anisotropic two-dimensional Ising model on a triangular lattice so that the effective Hamiltonian is

$$H = \underset{\langle nn \rangle}{-\Sigma} J_{ij} S_i^z S_j^z, \tag{2}$$

and the effective spin variable $S_i^z = \pm 1$ denotes the dipole's projection on the vertical axis. The exchange constants J_{ij} take the values J_1, J_2, J_3 depending on the choice of dipole pairs.

What, therefore, appears in Fig. 3(b) is that the system exhibits a certain amount of *frustration* (Suzuki, 1977) in its ground state. This means that for a closed path around the 13 protofilaments of the microtubule, it is impossible to satisfy all bond requirements since there is an odd number of antiparallel bonds. Note that no frustration exists in the direction along the MT axis giving the overall ferroelectric character to the lattice. It will also be seen how this is manifested through size dependence of net polarization.

Small potential barriers separating the various equivalent arrangements of spins (see Fig. 4) play another useful role. Relaxation times are very long for the various accessible states giving them long-term stability. Some other properties of this intermediate, weakly ferroelectric phase are: (i) absence of long-range order between spin sites and (ii) the presence of short-range correlations. This means that the system is very "soft" energetically, permitting a number of spin arrangements which are relatively

stable. However, there is no tendency towards the formation of large correlated spin clusters. This intermediate phase also allows easy formation of local ordered states, each of which carries an information content and is long-lived, i.e. relatively stable over time. On the other hand, it is not difficult to switch from one local state to another through various physical means by overcoming local potential barriers. Furthermore, a complete phase transformation can be achieved by:

(i) an application of an electric field along the axis. We estimate that intracellular fields on the order of $10^4\ \mathrm{Vm}^{-1}$ are sufficient to switch an MT from an intermediate state to a ferroelectric state (F). (Note that fields up to $10^7\ \mathrm{Vm}^{-1}$ are known to exist across cell membranes).

(ii) Raising the temperature above the ferroelectric transition temperature will drive the system to a disordered (paraelectric) phase.

We mentioned above that another ordered phase is favored when a moderate external field is applied (or indeed when the temperature is lowered further). This phase, called a ferroelectric phase, is characterized by long-range order manifested by an alignment of spin directions along the axis. This is shown in Fig. 5. Obviously, from the point of view of optimal information content, this is *not* a very useful phase. However, the ferroelectric phase can play a major role in signaling and the assembly/disassembly processes. This will be discussed at length in the next Section. Before we discuss this aspect, we investigate its general properties. The ferroelectric phase can also be destabilized by temperature when T exceeds T_c. To obtain a crude estimate of T_c we approximated the MT by an infinite triangular lattice with the exchange constants J_1, J_2 and J_3 given above and used the formula (Domb, 1960).

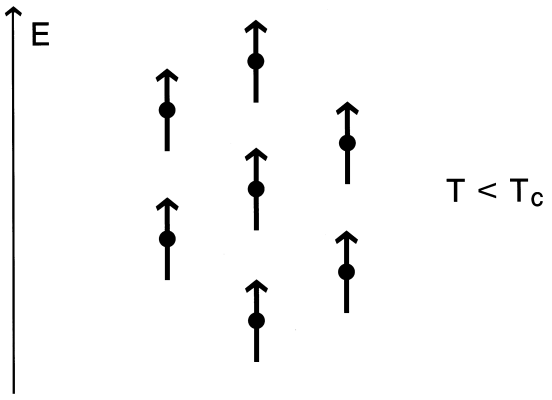

FIG. 5. The dipole orientations in the ferroelectric state.

$$\sinh\left(\frac{2J_1}{K_B T_C}\right)\sinh\left(\frac{2J_2}{K_B T_C}\right) + \sinh\left(\frac{2J_1}{K_B T_C}\right)\sinh\left(\frac{2J_3}{K_B T_C}\right) + \sinh\left(\frac{2J_2}{K_B T_C}\right)\sinh\left(\frac{2J_3}{K_B T_C}\right) = 1. \quad (3)$$

The result we obtained using the above crude estimate for an infinitely long lattice is $T_c \cong 402\ K$ *for the charge of* $q = 2e$, the permittivity $\epsilon = 10$ and the dipole displacement $d = 4$ nm. On the other hand, taking $q = 1e$, $\epsilon = 5$ and $d = 4$ nm yields $T_c = 201$ K. These two very rough estimates of T_c indicate nevertheless that it is feasible to expect an interesting phase behavior in the physiological temperature range around 300 K for a specific choice of model parameters. In addition, it is well known that the size of a lattice affects T_c greatly. In general, for size N, $T_c(N)$ is related to that for an infinite lattice $T_c(\infty)$ through (Binder *et al*., 1985):

$$T_c(N) = T_c(\infty) - \alpha N^{-1/\gamma}, \quad (4)$$

where α is an approximate proportionality coefficient and γ is the correlation length exponent (typically close to 1/2). Thus, for a finite-size lattice, the transition temperature is expected to be less than that for an infinite lattice. In other words, dipole ordering is more likely to occur when a critical size of the MT is achieved. In addition, other important factors may affect the value of T_c and thus provide sensitive control mechanisms. Through a coupling to the elastic degrees of freedom (conformational change), the value of T_c may be shifted. The dielectric constant ϵ may be altered by the presence of water molecules surrounding an MT structure. This would decrease the value of T_c and introduce disorder. Small changes in the angles between the dimer dipoles may remove the frustration mechanism effectively preventing the onset of the intermediate phase. Changes in the opposite direction can enhance frustration favoring the intermediate phase over the F phase and effectively switching from the growth mode of operation to an information processing behavior. Thus symmetry transitions or mechanical forces may effect switching between phases.

Once in the ferroelectric phase, the system is also very sensitive to the frequency of electromagnetic oscillations. It is known that in organic ferroelectric materials a resonant frequency exists W_0 which is inversely proportional to the length, stiffness constant

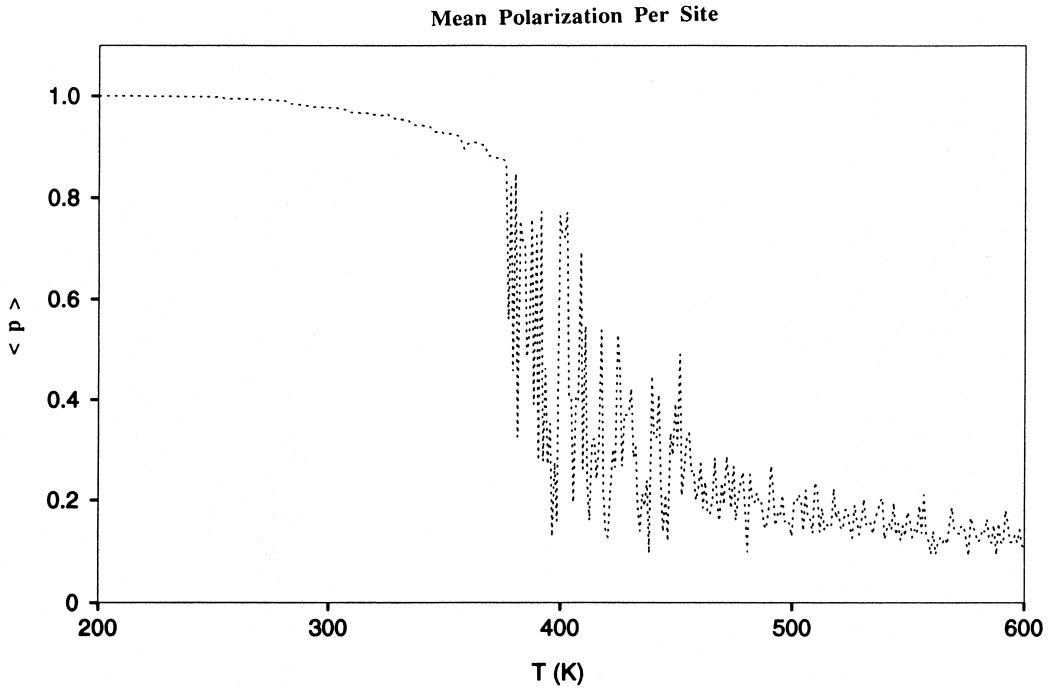

FIG. 6. A plot of the mean polarization per site for a triangular lattice whose size is 13×26.

and the square root of the density (Matthias, 1973). Extrapolating the data given by Matthias (1973), one finds for microtubules a resonant frequency range exceeding 10^8 Hz and thus being rather close to the microwave range predicted by Fröhlich (1980) to play a major role in living cells.

In Figs 6 and 7 we have shown the results of our Monte Carlo computations intended to reveal the size dependence of the phase behavior discussed above. Figure 6 is a plot of the mean polarization per site when the microtubule is composed of only 26 layers (i.e. it is a 13×26 lattice). It is clear that three distinct ranges exist: (i) the low-temperature ferroelectric range, (ii) the intermediate weakly ordered phase range and (iii) the high-temperature paraelectric range. Figure 7 illustrates the same dependence for a microtubule with 5000 layers (i.e. it is a 13×5000 lattice). We conclude that the intermediate weakly ferroelectric phase all but disappeared.

To summarize, in this section we have argued that two ordered phases are likely to govern the behavior of dipole moments in an MT. A ferroelectric phase is characterized by long-range order and has many characteristics useful in energy transfer properties discussed in the next section. An intermediate weakly ordered phase exhibits short-range order but has a multitude of equivalent, long-lived dipolar states. It appears that in the latter phase the system can explore a large subset of the phase space at a low energetical cost unlike in the ferroelectric phase. Having unhindered access to a multitude of conformational states indicates easy transitions from one configuration to another at little or no energy cost. A number of control mechanisms are possible which allow switching from one phase to the other and hence from one mode of operation to the other.

3. Ferroelectricity, Signaling and Assembly/Disassembly

It is well known that ferroelectrics are one of the few types of materials whose description is most adequately provided in terms of mean field models (Kretschmer & Binder, 1979). A typical approach is to postulate a double-well quartic on-site potential $V(S_n)$ that gives the overall effect of the environment on a selected dipole site n. Thus,

$$V(S_n) = -\tfrac{1}{2}AS_n^2 + \tfrac{1}{4}BS_n^4, \tag{5}$$

where S_n is the effective spin variable at site n and the coefficients A and B are model-dependent. It will be subsequently assumed that over long distances the variable S_n changes very gradually and can be treated as a continuous variable.

In the F phase, the MT cylinder taken as a whole represents a giant dipole. When the cross-section of an MT is viewed using electron microscopy, the outer surfaces of an MT are surrounded by a "clear zone" of several nm which apparently represents the oriented molecules of cytoplasmic water and enzymes (Steb-

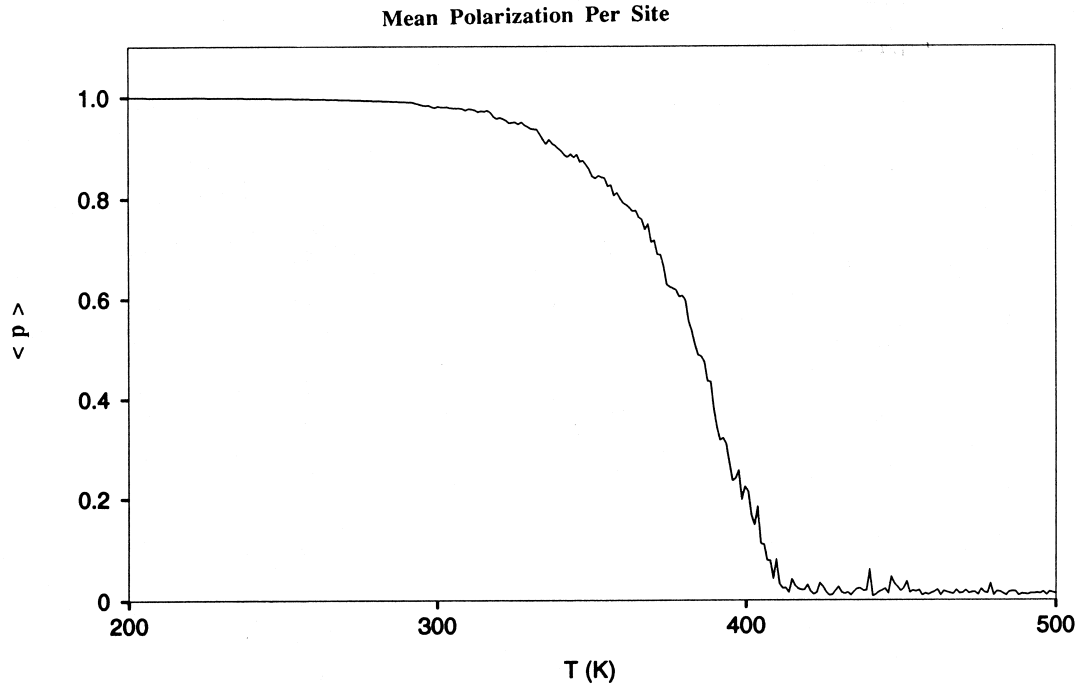

FIG. 7. A plot of the mean polarization per site for a triangular lattice whose size is 13×5000.

bings & Hunt, 1982). This could be explained by the presence of an electric field produced by an MT. Therefore, we assume that together with the polarized water surrounding it, an MT generates a nearly uniform intrinsic electric field parallel to its axis (see Fig. 1). Consequently, the additional potential energy due to this electric field and associated with each dipole is

$$V_{\mathrm{el}} = -CS_n, \quad C = qE, \tag{6}$$

where q denotes the effective charge of a single dimer and E is the magnitude of the intrinsic electric field. Within this picture, the model Hamiltonian for the dipole moments of a microtubule can be written as

$$H = \sum_{i=1}^{n} \left[\frac{1}{2} M \left(\frac{\mathrm{d}S_n}{\mathrm{d}t} \right)^2 + \frac{1}{2} K (S_{n+1} - S_n)^2 - \frac{A}{2} S_n^2 + \frac{B}{4} S_n^4 - CS_n \right]. \tag{7}$$

The first term above represents the kinetic energy associated with the longitudinal displacements of constituent dimers each of which has mass M. The second term arises from the restoring strain forces between adjacent dimers in the protofilament. If the stiffness parameter K is sufficiently large, then it is expected that large-amplitude long-wavelength excitations of the displacement field will be formed manifested by a slowly varying modulation of S_n along the MT axis.

In order to derive a realistic equation of motion for this system, the viscosity of the solvent is indispensable and the associated damping force must be introduced. Assuming for simplicity that the solvent is made up of only water molecules we may infer that water will provide a viscous medium that will damp out vibrations of dimer dipoles. This can be taken into account by adding the viscous force to the equation of motion with

$$F = -\gamma \frac{\mathrm{d}S_n}{\mathrm{d}t},$$

where γ is the damping coefficient.

Based on our assumption that the dipolar oscillations of dimers within an MT form a system that can be classified as displacive ferrodistortive, we can use the continuum approximation to obtain the corresponding equation of motion as

$$M \frac{\mathrm{d}^2 S}{\mathrm{d}t^2} - KR_0^2 \frac{\mathrm{d}^2 S}{\mathrm{d}x^2} - AS + BS^3 + \gamma \frac{\mathrm{d}S}{\mathrm{d}t} - qE = 0. \tag{8}$$

Note that the x-coordinate is along the protofilament axis. Moreover, for longitudinal sound waves the dispersion relation

$$\omega = \sqrt{K/M}$$

can be used to identify the sound velocity with

$$V_0^2 = \omega R_0 = \sqrt{K/M}\, R_0.$$

We now seek solutions of eqn (7) in the traveling-wave form where the moving coordinate ξ is given by

$$\xi = \left[\frac{|A|}{M(V_0^2 - V^2)}\right]^{1/2}(x - Vt) = \alpha(x - Vt), \qquad (9)$$

with V denoting the propagation velocity and the coefficient α is, of course,

$$\alpha = \left[\frac{|A|}{M(V_0^2 - V^2)}\right]^{1/2}$$

Consequently, eqn (8) is reduced to the ordinary differential equation below

$$M\alpha^2(V^2 - V_0^2)S'' - \gamma\alpha V S' - AS + BS^3 - qE = 0, \quad (10)$$

where $S' \equiv \mathrm{d}S/\mathrm{d}\xi$. This is an equation for an anharmonic oscillator with linear friction.

For the sake of convenience we now introduce a normalized displacement field as:

$$\Psi(\xi) = S(\xi)/S_0$$

where the normalizing coefficient

$$S_0 = \sqrt{|A|/B}$$

corresponds to the minimum of the double well potential in eqn (5).

It has been shown by Collins *et al*. (1979) that eqn (10) has a unique bounded solution which is given by the formula (and illustrated graphically in Fig. 8)

$$\Psi(\xi) = a + \frac{b - a}{1 + \exp(\beta\xi)}, \qquad (11)$$

where

$$\beta = (b - a)/\sqrt{2}$$

and the parameters a, b and d satisfy the cubic equation

$$(\Psi - a)(\Psi - b)(\Psi - d) = \Psi^3 - \Psi - \sigma, \qquad (12)$$

where

$$\sigma = q\sqrt{B}|A|^{-3/2}E.$$

Formula (11) describes a kink-like excitation (KLE) which takes the form of a domain wall separating two distinct ferroelectric domains.

It is also important to note that the above kink-like soliton propagates along the protofilament with a fixed velocity

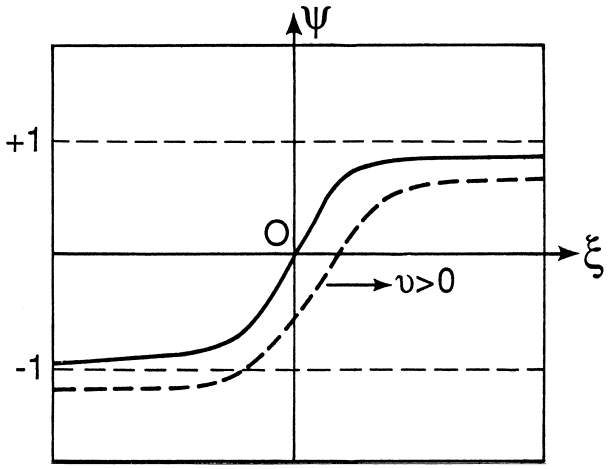

FIG. 8. The kink-like profile to the solution $\Psi(\xi)$ of eqn (11).

$$V = V_0/\left[1 + \frac{2\gamma^2}{9d^2MV_0^2}\right]^{1/2}, \qquad (13)$$

which is *less* than the sound velocity V_0 and decreases with an increase of the friction coefficient γ. It is clear that V depends on the magnitude of the electric field E through the parameter d.

For T well below T_c, V can be approximated by a linear relationship with E

$$V \approx \frac{3V_0}{\gamma|A|}\sqrt{\frac{MB}{2}}\, qE, \qquad (14)$$

where the coefficient of proportionality is called mobility and will be denoted by μ. Using simple arguments from fluid dynamics, μ can be estimated numerically as

$$\mu \approx 2 \times 10^{-5}\ \mathrm{m^2\ V^{-1}\ sec^{-1}}$$

Thus, for $E = 10^5\ \mathrm{V\ m^{-1}}$ we obtain $\mathrm{V} \cong 2\ \mathrm{m\ sec^{-1}}$ which can increase to $10^2\ \mathrm{m\ sec^{-1}}$ close to the MTs ends where E increases dramatically.

Assuming a smooth journey from one end of the protofilament to the other we estimate the average time of propagation $\bar{T}$ for a single KLE to be

$$\bar{T} = \frac{L}{\bar{V}} \approx 5 \times 10^{-7}\ \mathrm{sec}$$

A recent paper (Chou *et al*., 1994) also pointed to the possible role of solitary waves in the dynamics of microtubules.

To summarize, we showed that a unique kink-like excitation exists in the ferroelectric phase which

may propagate along an MT with a velocity that is proportional to the electric field E. KLEs in stable microtubules may constitute signals transducing membrane events via second messengers to other cell regions. Energy for such KLEs may come from GTP hydrolysis or phosphorylation via MAPs. In neurons, KLEs in microtubules may signal error information retrograde (i.e. from axon to dendrite) which would be useful in synaptic regulation and learning as in artificial neural network "back-propagation". In unstable MTs (i.e. whose ends are not stabilized by MAPs or other structures), KLEs can explain the fact that growth rates are different at the two ends of an MT. To this end, we assume that KLE formation is mainly due to the hydrolysis of GTP into GDP so that one act of hydrolysis corresponds to conformational change resulting in the formation of a single KLE. However, a KLE is preferentially oriented towards the direction of the intrinsic electric field. The propagation of a KLE will then distribute the energy of hydrolysis at the preferred end of an MT. This energy may then be used to detach dimers from the MT. This picture is in accordance with a hypothesis put forward by Kirschner & Mitchison (1986) who stated that on-rate (growth) is limited in principle only by the rate of diffusion of the monomer subunits into MT polymers (i.e. by the concentration of the constituent monomers in solution). On the other hand, the off-rate (depolymerization) can be extremely rapid and the net amount of MT polymer formed can be regulated independently by the hydrolysis of GTP.

It is reasonable to expect that at the onset of the MT assembly process, when MTs are very short, the formation of KLEs is very unlikely (or even impossible), since for a short chain the displacement field gradient required would be very high and the associated energy to form it prohibitively large. However, when the length of an MT reaches a threshold value, kink propagation could become a real possibility. Excitation of kinks being related to GTP hydrolysis would become dependent on the GTP concentration. As MTs become longer, KLE propagation accelerates leading to a disassembly at the "-" end. KLE propagation in stable MTs would depend on MAP phosphorylation or cytoplasmic kinase activity triggered by membrane events and second messengers.

4. Summary

We have argued in this paper that the spatial arrangement of dipole moments of a microtubule is crucial to its functioning as a dynamic system. Due to the presence of frustration in the dipole–dipole interactions, an intermediate weakly ferroelectric phase is predicted to arise at low enough temperatures and electric fields as well as for sufficiently short microtubules. Its attractiveness lies in maximal computational capabilities offered by a highly degenerate ground state and by long relaxation times giving relative stability to short-range correlated dipole patterns. Each pattern can be seen as containing binary information. This phase strongly resembles the well-known spin glass phenomenon which has received much attention in frustrated spin systems such as random-anisotropy magnets. Recently, spin-glass models have been applied in connection with biological systems (Stein, 1992).

The other possible ordered state is a ferroelectric phase with an almost perfect alignment of dipole movement along the protofilament axis. It is characterized by long-range order and hence its usefulness for information processing is dubious. However, it eagerly supports the formation of domain walls between the two stable orientations of dipole movements. The application of an external electric field preferentially directs kink-like excitations towards the properly aligned end. These kink-like excitations may be utilized as signaling mechanisms or causing disassembly due to energy release.

We have discussed the various possible control mechanisms (temperature, electric field, mechanical distortion) that could select the MT operating modes among information processing, weakly ordered and signaling and assembly/disassembly (ferroelectric) types.

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FURTHER EVIDENCE FOR A FRÖHLICH INTERACTION OF ERYTHROCYTES

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The aggregation of red blood cells, of a single species and of mixtures of species, deviates from brownian aggregation theory. The deviations are compatible with Fröhlich's theory of a quantum mechanical interaction between living cell membranes.

In a previous paper [1] we showed that a newly discovered attraction between human erythrocytes was compatible with a theory of Fröhlich [2,3] of long-range coherent interactions in living cells. As necessary in the theory the attraction disappeared when the cellular membrane was disorganised (with glutaraldehyde) and disappeared reversibly when the transmembrane potential difference was abolished or when the cells were depleted of their supply of metabolic energy. The interaction is associated with subsequent aggregation or rouleau formation [4–7] and does not occur unless there are rouleau-inducing macromolecules in the suspending medium. In place of the simple geometrical analysis previously used we now employ an adaptation of the Smoluchowski theory of colloid aggregation by brownian movement [8]. This method gives an estimate of the range of the interaction and provides evidence that the interaction is specific, from experiments on the aggregation of mixtures of erythrocytes from different mammalian species.

Chandrasekhar [9] reviewed Smoluchowski's theory and gives an expression for the total number of particles (aggregates and singles) N_t, in a coagulating colloid suspension as a function of time t [9, eq. (461)]

$$N_t^{-1} - N_0^{-1} = (4kTZ/3\eta r)\, t, \qquad (1)$$

where η is the viscosity of the suspending medium, r the average radius of the particles and Z the radius of the hypothetical zone of attraction around each particle. In our experiments the brownian motion of erythrocytes in their own plasma, resting on the floor of a sealed siliconized haemacytometer chamber, is recorded by time-lapse cinémicroscopy [1]. When cells touch they quickly slide together [4,6,7] face to face. So linear aggregates, called rouleaux, form in time in the chamber (fig. 1). The process is similar to the coagulation of a colloid though the "particles" are much larger and the aggregates more regular. Eq. (1) would apply if the cells had the same density as the suspending fluid and so were distributed randomly in depth as well as in the horizontal extent of the chamber. In fact the cells have a slightly higher density and quickly fall to the floor. The three-dimensional process described by eq. (1) is thus compressed into two dimensions. The number of collisions in brownian motion and so the rate constant are accordingly increased by a factor of 3/2 giving

$$N_t^{-1} - N_0^{-1} = (2kTZ/\eta r)\, t\,. \qquad (2)$$

Einstein's theory of brownian motion [10] gives the mean square deviation in two dimensions as a function of time,

$$\langle U^2 \rangle = 4kTt\mu^{-1}\,, \qquad (3)$$

where μ is Stokes' drag coefficient for spheres,

$$\mu = 6\pi\eta r\,. \qquad (4)$$

To check that cells beyond the range of an interaction were executing brownian motion, U was measured for each of 30 cells which remained further than 6 μm

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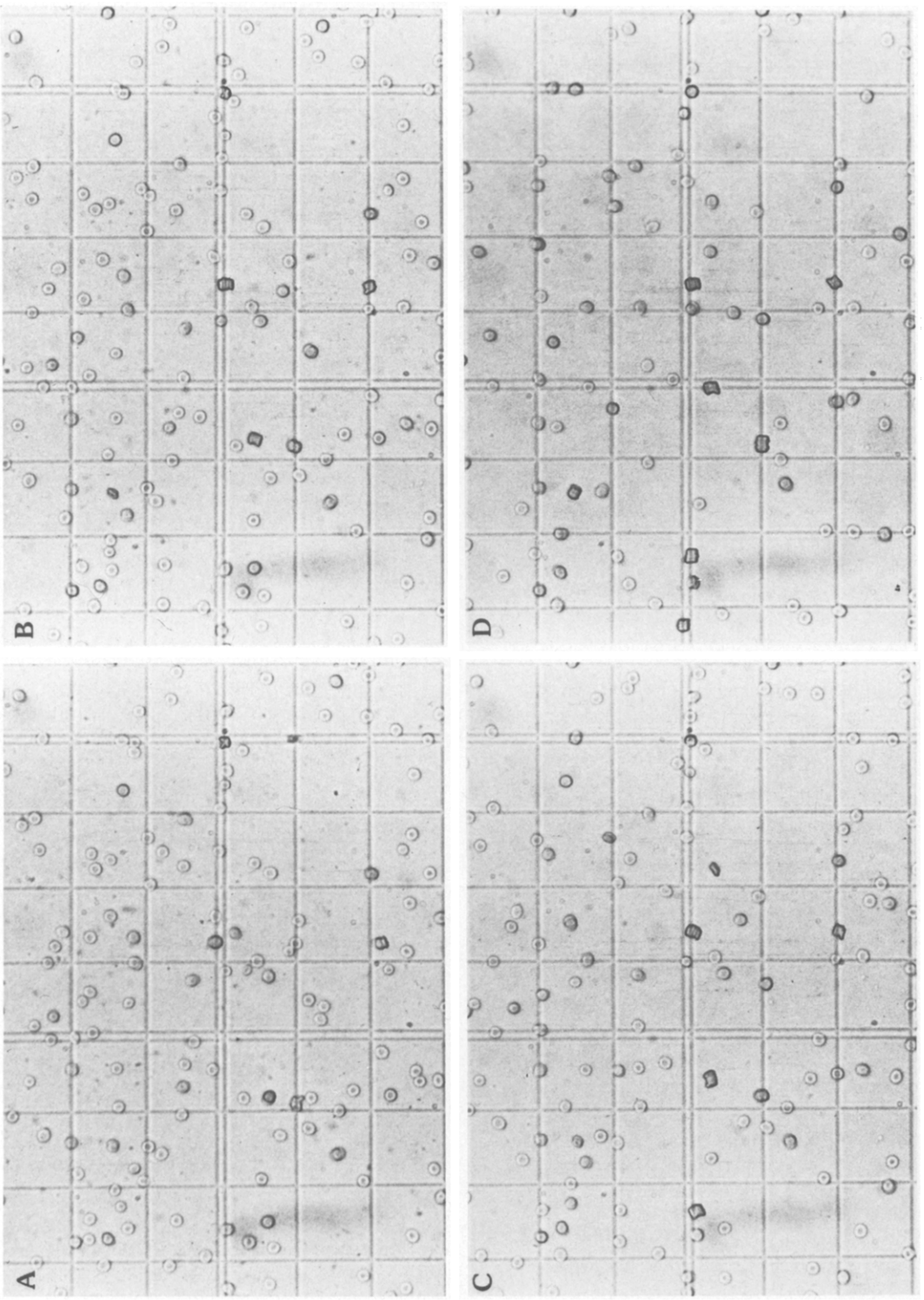

Fig. 1. Normal cells in plasma. Four frames from a ciné film. (A) the cells have just settled and some rouleaux have formed; (B) after 6 min; (C) after 22 min; (D) after 114 min.

Table 1
Drag coefficients.

	$\mu \times 10^7$ (N s m^{-1}) experimental value	experim. value/ Stokes' value, oblate spheroids
normal cells	2.80 ± 0.66 [a]	1.82 ± 0.43 [a]
cells fixed in glutaraldehyde	2.96 ± 0.68	1.92 ± 0.44
cells at pH = 6.3	2.78 ± 0.76	1.80 ± 0.50
cells restored to $pH \approx 7.6$	2.57 ± 0.83	1.67 ± 0.54
ionophore-treated cells	3.39 ± 0.14	2.21 ± 0.09
metabolically depleted cells	2.18 ± 0.34	1.42 ± 0.23
cells repleted with adenosine	2.68 ± 0.74	1.74 ± 0.48
mean value for cells	2.75 ± 0.10 [b]	1.78 ± 0.06 [b]
microspheres 5.8 μm	1.06 [c]	1.8
microspheres 8.0 μm	1.67 [c]	2.1

[a] Standard deviation. [b] Standard error of the mean.
[c] Microspheres in plasma stick to the floor of the chamber. These experiments were done in soapy water (η = 1.06 × 10^{-3} Pa s).

from other cells during the first 15 min of each experiment. Beyond 15 min fewer than 30 cells remained unassociated. For an average plasma viscosity (1.7 × 10^{-3} Pa s) and incorporating Perrin's correction to eq. (4) for oblate spheroids [11] the calculated value for μ = 1.54 × 10^{-7} N s m^{-1}. The experimental values for cells (table 1) are nearly twice as large and a similar discrepancy occurs with microspheres (table 1). We infer that the hydrodynamic drag is substantially increased when cells or microspheres are resting on the floor of the haemacytometer. In deriving eqs. (1) and (2) it is assumed [9] that the diffusion coefficient of the individual particles is given by the Stokes–Einstein relation

$$D = kT/6\pi\eta r\,, \qquad (5)$$

where the Stokes' drag coefficient appears as the denominator. The rate constant in eq. (2) must therefore be reduced by the ratio of the experimentally determined drag to the Stokes' drag, and this ratio is determined for each experiment. In seven experiments conducted on the same blood the standard error of the mean value of $\langle U^2 \rangle$, and so of μ was 19%. This sets a lower limit for the accuracy of the experiments as presently performed.

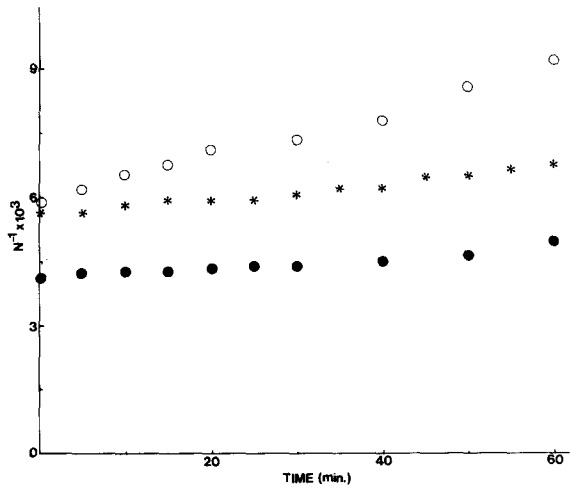

Fig. 2. Plots of the reciprocal of the total number of particles in an area of 10^5 μm^2 against time. ○ normal cells, Ξ = 3.37: ● glutaraldehyde-fixed cells, Ξ = 0.83: * microspheres, Ξ = 1.15.

For a typical experiment and if there is no interaction before cell contact ($Z = r$) the calculated corrected slope of the graph of N_t^{-1} against t will be 2.6 × 10^{-18} m^3 s^{-1}, a value which varies slightly from experiment to experiment. Experimental results are displayed in figs. 2–4 and in column 3 of table 2 as multiples of this base value (interaction coefficient Ξ = Z/r). In this table results obtained by the previous method [1] are displayed

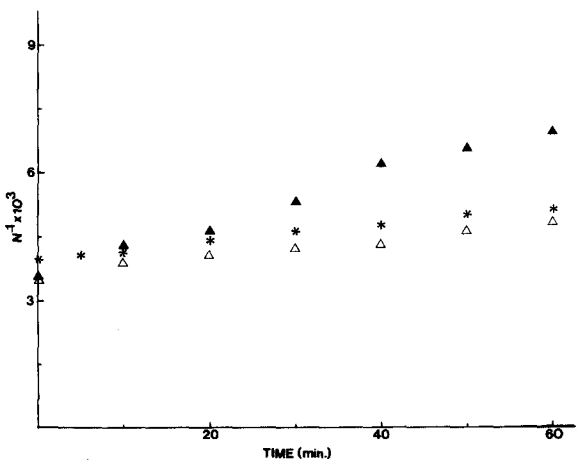

Fig. 3. As in fig. 2. △ membrane potential reduced to 1.5 mV by changing pH of plasma to 6.3, Ξ = 1.28: ▲ membrane potential restored by returning pH to 7.6, Ξ = 3.51: * membrane potential reduced by treatment of cells with ionophore A23187, Ξ = 1.66.

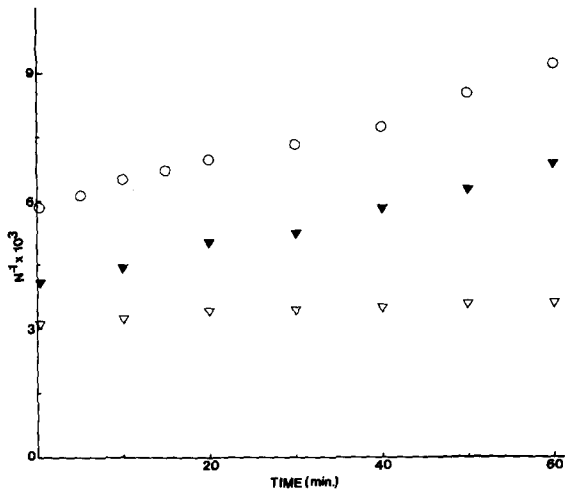

Fig. 4. As in figs. 2 and 3. ▽ cells depleted of metabolic energy, $\Xi = 0.40$: ▼ cells repleted with adenosine, $\Xi = 3.65$: ○ normal cells, $\Xi = 3.37$.

alongside (column 2). By having the same trend the results confirm each other though the relationship between the two sets of figures is non-linear as may be expected.

Fixed cells and microspheres have interaction coefficients close to unity and so, taking account of the added drag, show no significant deviation from brownian movement. For normal cells the interaction coefficient is three times the base value. Abolishing the membrane potential by reducing *pH* to 6.3 abolishes the interaction. Abolishing the membrane potential by the ionophore causes a significant decrease in Ξ but the two experiments suggest a remnant of interaction. Metabolic depletion causes a great fall in Ξ to a value which suggests a repulsion but both of these two last findings need further study for confirmation. Repleting energy-deprived cells and partially restoring the membrane potential after 3 h [1] brings the interaction back to the range of normal cells. Since the interaction coefficient $\Xi = Z/r$ we conclude that the interaction of normal cells extends on the average to a distance of three cell radii.

An assumption of the Smoluchowski theory of colloid aggregation is that all the aggregating particles are of the same size which is a reasonable assumption for the erythrocytes from a single individual. Mixtures of erythrocytes from different species of mammals will also aggregate into rouleaux when suspended in isotonic solutions of certain macromolecules [12]. Plasma cannot be used because of immunological reactions. In such interspecies rouleaux, cell volumes may differ by as much as a factor of two. Overbeek [13], in discussing the aggregation of polydispersed colloids, shows that

Table 2
Interaction coefficients Ξ.

	Probability [a]	Ξ [b]	No. of exp.	No. of subjects
normal cells	0.85	2.92 ± 1.12 [c]	15	9
cells fixed in glutaraldehyde	0.46	0.92 ± 0.46	9	5
cells at *pH* = 6.3	0.40	1.07 ± 0.41	10	5
cells restored to *pH* ≈ 7.6	0.71	2.57 ± 0.99	9	3
ionophore-treated cells	0.48	1.69 ± 0.04	2	2
metabolically depleted cells	0.44	0.43 ± 0.11	4	2
cells repleted with adenosine	0.87	3.26 ± 0.68	3	2
microspheres 5.9 μm		0.93	1	
microspheres 8.0 μm		1.15	1	

[a] Probability of pairs of cells colliding in 3 h [1]. Value for no interaction ≈0.3.
[b] Interaction coefficient as defined in text. $\Xi > 1$ attraction.
[c] Standard deviation: $p < 0.001$ normal versus glutaraldehyde cells; normal versus *pH* = 6.3 cells; *pH* = 6.3 versus *pH* = 7.6 cells; normal versus metabolically depleted cells; $p < 0.003$ normal versus ionophore-treated cells.

the *least* likely collision is between particles of the same size. Yet in the interspecies experiments [12] cells of the *same* size and so of the same species show a preference to adhere to each other. In ten combinations of different species the only exception was a rat–mouse mixture though there is a significant difference in cell size, larger than in some pairs which did show a species preference. Perhaps rats and mice, being of the same biological family, have red cells indistinguishable in respect to a Fröhlich interaction.

The interspecies interaction is contrary to that expected of a passive phenomenon [13]. It gives a different type of evidence from that adduced above for a specific interaction at a distance between mammalian erythrocytes. Further work will reveal whether the interspecies interaction is a repulsion, a null interaction, or an attraction smaller than that between cells of the same species. Brownian motion could overcome a small repulsion allowing rouleaux to form at a slow rate.

During growth, particularly in the embryo, cells appear to have the ability to recognise and to adhere to like cells. Experimentally a random mixture of dispersed cells of embryonic liver and heart will sort itself out. The hepatocytes form a ring around, and an island within, the cardiac myocytes [14]. In synchronized Chinese Hamster Ovary-KI cells, GI cells adhere to each other at twice the rate S cells adhere, while GI cells adhere to S cells at an intermediate rate [15]. In explanation, cell motility and chemical and physicochemical mechanisms are invoked [16]. Mammalian erythrocytes are not motile so their mutual attraction is not chemotactic. The first time they touch they adhere and then quickly slide into the rouleau array [4]. So the process is not one of tentative random collisions followed by immobilisation as chemical binding sites interlock. Yet their interactions are not dissimilar from those reported for other groups of cells [14–16]. This suggests that a Fröhlich effect is playing at least some part in the interaction of a wide range of mammalian cells.

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GENERAL THEORY OF COHERENT EXCITATIONS ON BIOLOGICAL SYSTEMS

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ABSTRACT

Coherent excitations, as first proposed in 1968, can arise in the presence of nonlinear interactions if metabolic energy is supplied at a rate s above a critical s_o. Three types of excitations will be shown to exist: A excitation of a single mode; B excitation of a metastable highly polar state; C excitation of limit cycles or Lotka-Volterra oscillations in complex systems. They may have far reaching consequences on the dynamics of biological systems.

Externally applied radiation may, at certain frequencies and at certain stages of development, act as trigger to biological events. In this case a steplike dependence on the intensity is to be expected.

Relevant experimental evidence will be indicated.

INTRODUCTION

A general theory of biological activity necessarily will be concerned with properties common to most biological materials. From the point of view of physics the extraordinary dielectric properties (1) are outstanding in this respect. Most remarkable amongst these are the very high electric fields of about 10^7 volt/m, usually present in biological membranes, leading to very strong polarization of the materials in them.

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Biologically the term "life-cycle" already indicates a crude periodicity in biological dynamics, though here, each particular case exhibits individuality prohibiting the exact reproducibility of a period.

We shall take a vast step and assume that coherent excitation of electric vibrations in a large range of frequencies play an important role in biological activity. Modern physics offers this type of excitation arising through nonlinear interactions, with the necessary energy supplied, in general, from metabolic processes. Thus while supply of energy to a material usually leads to heating, coherent excitations may arise if certain conditions are satisfied.

This point of view was first presented more than fifteen years ago (2,3). It has led to far reaching consequences which stimulated a great number of experiments (review up to 1979 in (4). In a recent meeting on coherent excitations in biological systems (5,6) the assembled physicists and biologists have arrived at the conclusion that coherent excitation had now been shown to be an intimate feature of biological activity*.

COHERENT EXCITATIONS

Three types of coherent excitation have been proposed:

A. Coherent excitation of a single polar mode.

B. Excitation of a metastable highly polar state; this includes solitons.

C. Vibrations arising from more complex processes and giving rise to limit cycles or Lotka-Volterra oscillations.

All three involve essentially nonlinear processes.

Frequencies of type A processes in membrane vibrations have been estimated in the 10^{11} Hz region; this corresponds to the frequency of millimeter waves. Both, higher and lower frequencies of polar modes in biological molecules, and larger systems, do exist and must be relevant: Such vibrations interact with the

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*Misrepresentation in this subject is fairly common. Thus in a recent book review in Bioelectromagnetics (December 1982) K.R. Foster seems to be quite unaware that experiments on the influence of millimeter waves on the rate of growth of yeast were undertaken to test the theoretical conjectures.

abundant cell water and thus will be strongly damped. Spectroscopy of non-excited systems thus shows broad bands only. Coherent excitation, however, as is well known from the theory of lasers, overcomes the damping and leads to sharp lines.

In the type A model, nonlinear interaction of the polar vibrations with a heat bath under the influence of energy supply at a rate s leads to the transformation of a Planck distribution of excitation at frequency s into a Bose distribution containing a chemical potential μ, i.e. to a replacement of $h\omega$ by $h(\omega-\mu)$. Here increases with increasing s and at a critical value s_o causes Bose condensation, i.e. high excitation of a single mode*.

A metastable highly polar state (B) may arise from the shape dependence of the electrostatic energy of a polarized material (7). Material deformation then must overcome the elastic energy, and the condition for the existence of the state B requires a certain softness which implies the sound velocity c to be below a certain critical value c_o. In extended regions where the sound velocity may vary spatially, those for which $c < c_o$ may exhibit the excitation.

Excitation of type C involves the consequences of A and B and will be discussed below.

Excitation of a metastable state (B) requires a single appropriate energy supply. Excitation of coherent vibrations, however, as mentioned above involves a rate of energy supply s above a critical s_o. When the existing s is just below s_o then small further supply at the rate s_o-s will induce the excitation.

It must also be pointed out that the frequency of the excited vibration need not be exactly reproducible. Assume an ion to be attached to the vibrating region in some cases but not in others. Through electrostatic interaction it may lead to a small shift in frequency through which in course of the nonlinear excitation may be stabilized. Also, nonlinear interaction may induce the frequency shifts which depend on the degree of excitation.

CONSEQUENCES

Coherent excitation of a single polar mode (A) leads to frequency selective long range interaction between systems with

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*Extensive theoretical work in this direction has been carried out by Bhaumik et al., Mills and Wu et al.; quoted in (4). Regrettably, very faulty theoretical work has also been published in this field.

equal frequencies; in biological membranes this will be in the 10^{11} Hz region, but higher as well as lower frequencies, originating in other cell structures, will be expected. Such interaction may lead to selective attraction (or repulsion) of systems, to other control of their activities, and to communication within and between cells. It will be noted that electrostatic long range interaction is screened through counter ions.

Excitation of the metastable polar state (B) may lead to storage and transport of energy. It could represent the active state of enzymes and through the high internal field lead to a reduction of activation energies. It could also be effective in transporting energy from protein to protein. An array of proteins dissolved in a membrane would thus act as the energy transporting network postulated by current thinking (8).

The excitation discussed here may be switched on or off depending on the supply of energy at the appropriate time and place. In general this energy will arise from metabolic processes (or light in case of photosensitivity) and must be regulated by a control system. In the case of excitations A the energy must be supplied to the band of polar modes. According to the theoretical model, the system possesses storage capacity, and a certain time will be required to effect the excitation even if the rate of supply s exceeds the critical one, $s > s_o$. At this stage the system becomes sensitive to interference from the outside. When the relevant band of frequencies is in the microwave, in particular in the millimeter wave region, then low intensity external radiation may cause large biological effects as the switching on of the excitation depends on s in a steplike fashion.

Such effects will be noticeable whenever the externally applied frequency falls into the band of polar modes. When a coherent mode begins to be excited, however, then energy application at its frequency will be more effective. Biological effects dependent on resonance must be expected. The external radiation then acts as a switch. It will be effective at certain periods of the biological development only.

The selective attraction arising in excitation A may play a crucial role in multienzyme processes. Assume enzymes to be spatially organized as discussed by Clegg in (6). Then some of the energy liberated during the enzymatic process may be used to cause excitation A in substrates which then will attract other substrates. This leads to an increase in available substrates and in turn to an increase in the rate of enzymatic processes which decreases the number of substrates. A typical Lotka-Volterra oscillation arises - characteristic of our type C of coherent

oscillations. More complex situations may arise leading to limit cycles as discussed in these proceedings by Kaiser. Further consequences are also discussed by Del Guidice.

Such excitations may also be sensitive to external vibrations which might, by relatively weak supply of energy, liberate the stored energy and thus cause large biological effects.

EXPERIMENTS

Existence of three types of coherent excitations: A) a coherent single mode; B) a polar metastable state; C) a periodic activity arising from A and B, arises as the main conclusion of theoretical considerations. Detailed mechanisms might well differ considerably from the particular theoretical models; they might in fact be different in different cases.

Relevant experiments have been discussed in reference (6). They refer to the attraction of nutrients to a cell - similar to the attraction of a substrate as discussed above. The excitation is measured through non thermal excitation of anti-Stokes lines in Raman effect (Drissler et al. in (6). Rowland (6) gives evidence for the mutual attraction of blood cells satisfying the requirements of the theory.

Several contributions are related to biological effects arising from irradiation by very low intensity milimeter waves. Attempts to interpret these in conventional ways, e.g. in terms of the field turning a protein, are quite out of place as they would require (9) intensities of millions W/cm^2 when experiments use 10 mW/cm^2 or less.

Experiments conducted so far do not permit specification of the particular biological process on which electromagnetic fields act. Although recently important evidence in this direction has been obtained, the experiments will have to be continued to establish this evidence. For example, extremely sharp frequency dependence of the influence of millimeter waves on the rate of growth of yeast have been reported by Grundler et al. (6). They find the step-like intensity dependence required by the theory. Apart from a very sharp resonance in the region of 41700 MHz, the frequency dependence exhibits a fine structure of about twenty minima and maxima, spaced at 7-8 MHz. It is tempting to interpret the basic line in terms of a type A excitation with the low frequency arising from processes of type C which require a type A excitation. With very sensitive instrumentation, carried out in an electrically screened laboratory, C.W. Smith and A.H. Jafary have measured very sharp emission in the 7-8 MHz region from single yeast cells. They occur during the period of cell division only, and last only about 3 minutes. The frequencies are not always

exactly the same, but this should be interpreted in terms of individuality as discussed in reference 2. We should conclude that the excitation is closely connected with the highly organized processes occurring in cell division.

Long range interaction may also be expected relevant for the control of cell division, important in the cancer problem (4) IVD). Experiments on growing tissue cultures would, therefore, be highly desirable. They do require, however, development of new experimental techniques.

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High-frequency electric field and radiation characteristics of cellular microtubule network

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ABSTRACT

Microtubules are important structures in the cytoskeleton, which organizes the cell. Since microtubules are electrically polar, certain microtubule normal vibration modes efficiently generate oscillating electric field. This oscillating field may be important for the intracellular organization and intercellular interaction. There are experiments which indicate electrodynamic activity of variety of cells in the frequency region from kHz to GHz, expecting the microtubules to be the source of this activity. In this paper, results from the calculation of intensity of electric field and of radiated electromagnetic power from the whole cellular microtubule network are presented. The subunits of microtubule (tubulin heterodimers) are approximated by elementary electric dipoles. Mechanical oscillation of microtubule is represented by the spatial function which modulates the dipole moment of subunits. The field around oscillating microtubules is calculated as a vector superposition of contributions from all modulated elementary electric dipoles which comprise the cellular microtubule network. The electromagnetic radiation and field characteristics of the whole cellular microtubule network have not been theoretically analyzed before. For the perspective experimental studies, the results indicate that macroscopic detection system (antenna) is not suitable for measurement of cellular electrodynamic activity in the radiofrequency region since the radiation rate from single cells is very low (lower than 10^{-20} W). Low noise nanoscopic detection methods with high spatial resolution which enable measurement in the cell vicinity are desirable in order to measure cellular electrodynamic activity reliably.

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1. Introduction

Electrodynamic activity of eukaryotic cells is a topic, which is gaining considerable theoretical and experimental support in recent years. In the 1960s Fröhlich postulated and theoretically treated a concept in which he presented the possible existence of the mechanical vibrations of electrically polar structures in living cell causing generation of electric oscillations around and in the cell. Firstly, he proposed electrically polar longitudinal vibrations interacting with an elastic field, giving rise to coherent cellular electric oscillations (Fröhlich, 1968b, 1969). Secondly, he showed that the energy condenses to the lowest frequency mode in his nonlinear model system (Fröhlich, 1968a). Lastly, he estimated that oscillating polar structures are situated in the cellular membrane and their oscillating frequency is in range $10^{11}-10^{12}$ Hz. However, with the advent of deeper understanding of the cellular structure the microtubules emerge as hot candidates for generation of endogenous electrodynamic activity of living cells (Pokorný et al., 1997, 1998).

A possible biological role of the endogenous cellular high-frequency electric field has been analyzed by several authors. Its role has been proposed in long-range protein interactions (Fröhlich, 1972, 1975), in chemical reaction transfer and consequently chemical reaction kinetics (Pokorný and Wu, 1998; Pokorný et al., 2005a, 2005b; Pokorný, 2001), in disrupted cellular organization state such as cancer (Fröhlich, 1978; Pokorný et al., 2008), and in intercellular interactions (Pokorný et al., 1983; Pokorný and Fiala, 1994; Pokorný, 2006; Pohl, 1981, 1980; Rowlands et al., 1981, 1982).

The electrodynamics of cellular structures is inseparably connected to the vibrational dynamics of biomacromolecules. Many marvelous biological functions in proteins and DNA and their profound dynamic mechanisms, such as switch between active and inactive states (Chou, 1984b; Chou et al., 2001; Wang and Chou, 2009), cooperative effects (Chou, 1989), allosteric transition (Chou, 1984a, 1987; Schnell and Chou, 2008), intercalation of drugs into DNA (Chou and Mao, 1988), and assembly of microtubules (Chou et al., 1994), can be revealed by studying their internal motions (Chou, 1988) or terahertz frequency motions. Such inferences have been later observed by NMR (Chou et al., 2001), and applied in medical treatments (Gordon, 2007, 2008;

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Madkan et al., 2009). Recently, investigation into the internal motion in biomacromolecules and its biological functions is deemed as a 'genuinely new frontier in biological physics' (Vermont Photonics Technologies Corp, 2011). Likewise, to really understand the action mechanism of microtubules, we should consider not only its static structure but also its dynamical character. The electrodynamic behavior of microtubules inevitably underlies electrodynamic activity of living cells.

Electric oscillations of living cells have been experimentally investigated by several authors in the frequency range from kHz (Jelínek et al., 2008, 2009) through MHz (Pohl et al., 1981; Hölzel and Lamprecht, 1994; Hölzel, 2001; Pokorný et al., 2001) to the GHz (Jelínek et al., 2007; Cifra, 2009) region. These pioneering experiments suggest cellular electrodynamic activity in mentioned frequency region, but the measurement systems used were of the threshold sensitivity. It was recently explained (Kučera et al., 2010) which technical prerequisites must be fulfilled by the measurement system which is designed to detect cellular electrodynamic activity in kHz–GHz region. One of the biophysical aspects underlying the technical prerequisites for high gain and low noise of the first pre-amplifier is the obviously low power radiated from the cell. However, there are few theoretical calculations which provide quantitative evaluation of radiated power from cell and electric field intensity around the cell in the high-frequency region. Pokorný carried out calculations where the source of MHz–THz cellular electric oscillations were assumed to be membrane proteins (Pokorný and Wu, 1998; Pokorný, 1990, 1987) (some selected results are also in Pokorný et al., 1983, 1996; Pokorný, 1985). Shein and Kharlanov (2007) and Kharlanov (2006) represented the whole membrane by discrete dipole layer and considered acoustoelectric vibrations of the membrane in MHz–GHz region. The analysis of Russian authors was inspired by the earlier works of Devyatkov, Golant and Betskii (summarized in books Betskii et al., 1988; Devyatkov et al., 1991) who proposed that the acoustoelectric vibrations of the cell membrane can generate electric oscillations in the region of 30–300 GHz and made the first estimation of radiated electromagnetic power from single cell. Results from both groups of authors show clearly that single cell radiation under most conditions is of the power well below 10^{-20} W, thus hardly detectable in far field at room temperature if the exact frequency of radiation is not known a priori. As can be seen, all of these works considered only cell membrane as the source of the oscillatory electric field, although modern findings suggest that cytoskeleton, especially microtubules, is also very probable source of endogenous cellular high-frequency electric oscillations (Cifra et al., 2010d). Work presented here is filling the gap by calculating electric field and radiation of the whole cellular microtubule network.

2. Microtubules

Cytoskeleton is not only the mechanical support of the cell, but also the dynamic structure responsible for dividing the cell in two, for cellular motion and selforganization of the cell. Cytoskeleton is composed of three types of filaments. One of them are microtubules. Microtubules resemble hollow rods with inner and outer diameter 17 and 25 nm, respectively. These structures are mostly composed of 13 protofilaments, see Fig. 1. The subunits of protofilaments are tubulin heterodimers, which are constituted from α- and β-tubulin. Tubulin heterodimer is very polar structure with static dipole moment $p_{AXIS,STATIC} = 337$ D $(1.12 \times 10^{-27}$ C m) (Tuszyński et al., 2005) in the direction of the microtubule axis. In most of eukaryotic cells, microtubule nucleating and organizing center is a centrosome, which is a cell organelle near the nucleus. Microtubules grow up from this center through the cytoplasm to the cell membrane and

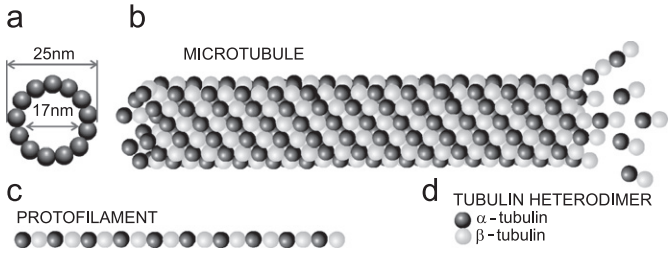

Fig. 1. (a) Cross section of a microtubule. (b) A single microtubule with 13 protofilaments. (c) One protofilament. (d) One tubulin heterodimer.

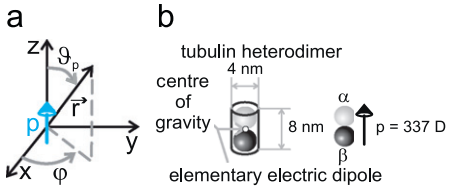

Fig. 2. (a) Dipole moment in the Cartesian coordinate system. (b) Tubulin heterodimer (dimensions and center of gravity).

create intricate network. Together with an extraordinary elasticity, microtubules fulfill all conditions for generation of electric oscillations (Cifra et al., 2010c).

2.1. Electric properties of microtubule

Electric properties of tubulin heterodimer were approximated by an elementary electric dipole (in further text the term "dipole" is used for brevity). Field of such dipole is described by

$$H_\varphi = -\frac{Idl}{4\pi}k^2\sin(\vartheta_p)\left(\frac{1}{jkr}+\frac{1}{(jkr)^2}\right)e^{-jkr} \quad (1)$$

$$E_r = -\frac{Idl}{2\pi}Zk^2\cos(\vartheta_p)\left(\frac{1}{(jkr)^2}+\frac{1}{(jkr)^3}\right)e^{-jkr} \quad (2)$$

$$E_\vartheta = -\frac{Idl}{4\pi}Zk^2\sin(\vartheta_p)\left(\frac{1}{jkr}+\frac{1}{(jkr)^2}+\frac{1}{(jkr)^3}\right)e^{-jkr} \quad (3)$$

where E is the electric field intensity, H is the magnetic field intensity, I is the equivalent current, dl is the length of the dipole, Z is the wave impedance, k is the propagation constant, ω is the angular frequency, p is the dipole moment, j is the imaginary unit ($j^2 = -1$) and other symbols are according to Fig. 2a.

All equations include:

$$Id\vec{l} = j\omega\hat{p} \quad (4)$$

In the near zone the dominant components are with $E \sim 1/(jkr)^3$ and $H \sim 1/(jkr)^2$ and the time-averaged Poynting vector (power density, or Poynting vector in further text, for brevity) composed of these components is zero in lossless medium, because the phase shift between E and H is $\pi/2$. Time-averaged Poynting vector $\vec{S}$ is defined as $\vec{S} = \frac{1}{2}\mathrm{Re}\{\vec{E} \times \vec{H}^*\}$, where Re{} denotes real part of complex number and * denotes complex conjugate.

Every tubulin heterodimer in one microtubule is approximated by elementary electric dipole to obtain the model of electrical properties of microtubule (Fig. 3).

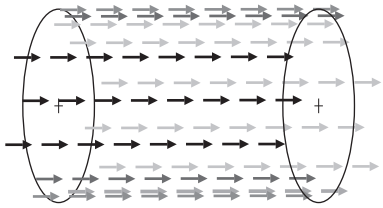

Fig. 3. Model of electrical properties of microtubule. Every arrow represents one dipole.

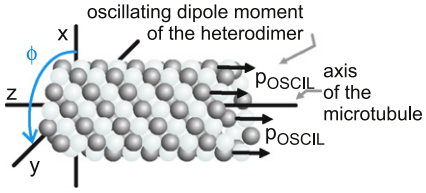

Fig. 4. Location of microtubule in coordinate system and sense of the angles ϑ and ϕ.

3. Method of calculation

Scripts for calculation of electromagnetic field generated by microtubule network were written in Matlab. It had to be dealt with three problems during calculations. The first is the distribution of points on the sphere with the smallest potential energy—to obtain the most symmetric distribution of arbitrary number of microtubules. The second is the rotation of microtubules and then creation of microtubule network. The last problem to deal with was a summation of small and great numerical values. The first problem was analyzed by Bulatov (1996). The solution of problem of microtubule rotation will be described in the following: The microtubule is situated in Cartesian coordinate system (Fig. 4) and coordinates of vector of elementary electric dipoles (both position and direction) are known.[1]

These coordinates were rotated according to the points distributed on the sphere[2] with the smallest potential energy. Two representative models of microtubule network were created in this way. The first model (symmetric) represents the microtubule network of nondividing cell. End points of microtubules' axes are distributed symmetrically and uniformly on the sphere in this model (the axes of microtubules are situated along radius vectors from center of the sphere to end points which are uniformly distributed on the sphere. Uniform distribution was obtained by an algorithm which calculates coordinates of point charges uniformly distributed on sphere—distributed according to their total smallest potential energy) (Fig. 5). This uniform and symmetric distribution of microtubules simulates complete (phase and geometric) symmetry in the cell. The second model (asymmetric) is the model of asymmetric distribution of microtubules (Fig. 6). This model represents microtubule network during mutual cellular interaction or during cellular interaction with solid substrates when cytoskeleton is redistributed. Both models are composed of 100 microtubules, which is reasonable estimate based on electron microscopy studies of eukaryotic cells (Sobel, 1997; Winey et al., 1995; May and Hyams, 1998). Each microtubule in our model is constituted of 1950 heterodimers.[3] We assumed that the ends of microtubules are in direct contact with the cell membrane, i.e. diameter of the model cell is 2.8 μm (0.4 μm diameter of the centrosome $+2\times1.2$ μm of microtubule length).

The problem of summation of small and large numeric values occurred in calculations of power mainly in lossless medium, because the value of field calculated only from near field

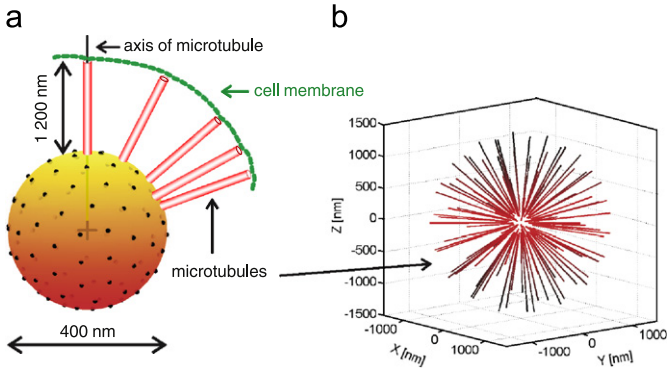

Fig. 5. (a) Model of centrosome with nucleation centers (black points) uniformly and symmetrically distributed on the sphere. (b) The symmetric model of a simple microtubule network with 100 microtubules. The ends of microtubules are distributed according to (a). Each red line represents one microtubule. Centrosome is in no way involved in the generation of mechanical or electric oscillations in the current calculations model, it is depicted here only for clarity and explanation of the geometrical distribution of microtubules. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

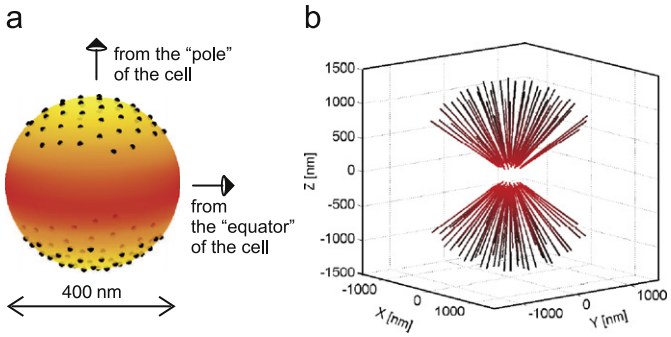

Fig. 6. (a) Model of centrosome with nucleation centers (black points). There are 50 points on each pole. (b) The asymmetric model of a simple microtubule network with 100 microtubules. The ends of microtubules are distributed according to (a). Each red line represents one microtubule. Centrosome is in no way involved in the generation of mechanical or electric oscillations in the current calculations model, it is depicted here only for clarity and explanation of the geometrical distribution of microtubules. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

components is much greater than value of field calculated from far field components. This was solved by decomposition of Eqs. (1)–(3) into parts $1/jkr$, $1/(jkr)^2$ and $1/(jkr)^3$ and by calculation of small and large values separately. All three problems and their solutions were described closely by Havelka (2010).

The calculation is executed by vector summation of contributions from all dipoles in the point of space in which the value of electric and/or magnetic field (in further text the term "point" is used for brevity) has to be calculated and then this procedure is repeated for all points in the chosen set (sets included either plane, coaxial surface of microtubule or spherical surface around

[1] Coordinates of the initial point of dipole vector are in the center of gravity of tubulin heterodimer and coordinates of the endpoint of dipole vector depend on angle ϑ. ϑ is the angle between the axis of dipole vector and the axis of microtubule. Thus, $\vartheta=0$ corresponds to case when the dipole vector is parallel with the microtubule axis. In other words, when we consider longitudinal axial oscillations of microtubule (and heterodimer from which it consists), only component of the dipole vector which oscillates significantly is the one which is parallel with the microtubule axis.

[2] Sphere with radius 200 nm represents simple approximation of spherical surface of centrosome, from where the microtubules grow.

[3] Thirteen protofilaments $\times150$ heterodimers in one protofilament, corresponds to the length of 1.2 μm of a single microtubule.

cell). This method was used for calculation of the power radiated from models and electric intensity in radiant distance from a cell membrane. The effect of the cell membrane is neglected in our calculations. To obtain the radiated power P, numerical integration of Poynting vector (power density) was performed over the sphere concentric to cell center using summation $P = \sum_i^N S_i \cdot A_i$ (N, number of elementary surfaces, A_i, i-th surface with S_i power density flowing through it) which is discrete equivalent of continuous integration $P = \int \vec{S}\, d\vec{A}$, where P is the power flowing through the surface A and S is the Poynting vector (power density). Electric field of the whole microtubule network was calculated along the radius vector, either in the direction from the "pole" or from the "equator" (Fig. 6).

4. Parameters used for calculations

Calculations were performed in frequency regions in which various workers have been attempting to measure electric oscillations of cells (Jelínek et al., 2007, 2008, 2009; Pelling et al., 2005; Pohl et al., 1981; Hölzel and Lamprecht, 1994; Hölzel, 2001; Pokorný et al., 2001; Cifra, 2009). Two different surrounding media were taken into account: lossy and lossless. The parameters of surrounding media are in Table 1 (ε_r and σ have subscript 1 for lossy medium and subscript 2 for lossless medium). Longitudinal shift between neighboring protofilaments is 4.92 nm (corresponds to the microtubule lattice A, Tuszyński et al., 2005) as can be seen in Fig. 4. Longitudinal oscillations of tubulin heterodimers are represented by oscillating dipole moment in the direction of microtubule axis ($p_{AXIS,STATIC} = 337$ D static part of dipole moment). We note for clarity that every tubulin in microtubule is oscillating. This follows from the fact that oscillations of microtubules as wholes are considered in this paper. Amplitude of oscillatory component of the tubulin dipole moment around kHz was estimated based on Pelling et al. (2004, 2005). In this work Pelling measured oscillations of yeast cell wall, likely connected with cytoskeleton oscillations, in the amplitude range 1–10 nm, so the amplitude of 1 nm was accepted. Amplitudes of oscillations for higher frequencies are assumed to be of the level of thermal motion amplitude of proteins (Chou, 1988), which is around 0.1 nm. Thus, taking ratio of the oscillatory amplitude (1 and 0.1 nm) and the length of the dipole (tubulin heterodimer length of 8 nm) amplitude of the dipole oscillatory component were estimated: for kHz $p_{OSCIL1} = \frac{1}{8} p_{AXIS,STATIC}$ and for higher frequencies $p_{OSCIL2} = \frac{0.1}{8} p_{AXIS,STATIC}$.

Oscillations of tubulin heterodimers in one protofilament were approximated by longitudinal oscillations of the chain of rigid particles of two types, optical branch (Kittel, 1976). Optical branch is able to generate and absorb electromagnetic field effectively. Movement of particles (tubulin heterodimers) and corresponding oscillations of the dipole moment are described by spatial modulation function:

$$p_m = p_{OSCIL} \cdot \sin(mv) \tag{5}$$

Table 1
Parameters of surrounding medium around microtubules and cell model for eight frequencies. ε_r and σ have subscript 1 for lossy medium and subscript 2 for lossless medium.

f (Hz)	1 k, 1 M, 10 M, and 100 M	1 G	10 G	42 G	100 G
ε_{r1} (–) (relative permittivity)	81	81	71	20	10
σ_1 (S/m) (conductivity)	1	1.33	17.22	39.67	68.50
ε_{r2} (–) (relative permittivity)	81	81	71	20	10
σ_2 (S/m) (conductivity)	0	0	0	0	0

where p_m is the modulated dipole moment, p_{OSCIL} is the oscillatory component of the dipole moment and mv is basically distance along the microtubule. Modulation function is specific for each mode, minimal mode number is zero, maximal mode number (max) depends on the number of tubulin heterodimers, i.e. length of microtubule. The mode zero represents oscillations of all dipoles in phase and the max mode represents oscillations of neighboring dipoles in antiphase.

For the illustration, modulation function of the 1st protofilament is depicted in Fig. 7 for the mode 3. Modulation functions of other protofilaments (2nd–13th) are axially shifted, according to the position of protofilaments in the real microtubule A lattice. Each circle in Fig. 7 represents the value of dipole moment of one heterodimer. Modulation function represents effect of a vibrational longitudinal wave on dipole moments of individual tubulin heterodimers. This wave can be described as the wave which has the same phase in the cutting plane orthogonal to the axis of the microtubule (Fig. 8—mode 3). Integer values of mode numbers correspond to boundary conditions of rigidly bound ends of microtubule. It is important to note that in our models of microtubule network, phase difference between oscillations of individual microtubules is zero, i.e. microtubules oscillate in phase. The distribution of electric field around microtubule excited by vibration modes min (mode 0) and max (mode 50, in this case) can be seen in Fig. 9. The intensity of

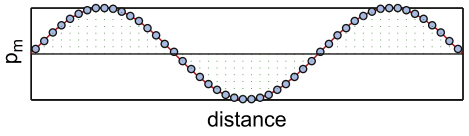

Fig. 7. The modulation function of the 1st protofilament (mode 3).

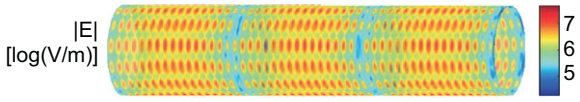

Fig. 8. The intensity of electric field in logarithmic scale on the coaxial surface 1 nm above the microtubule wall (mode 3).

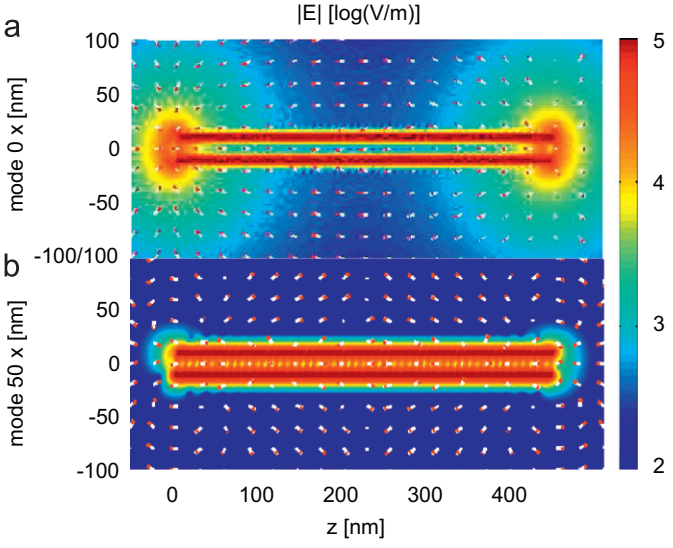

Fig. 9. The intensity of electric field in logarithmic scale in section through axis of the microtubule, which is excited by (a) mode 0 (min mode) and by (b) mode 50 (max mode).

Table 2
Total radiated power from the whole cellular microtubule network for different frequencies of the microtubule oscillations, lossless medium. The radiated power is the same both from symmetric and asymmetric model.

f (Hz)	1 k	1 M	10 M	100 M	1 G	10 G	42 G	100 G
P [log(W)]	−52.22	−40.22	−36.22	−32.22	−28.22	−24.25	−22.03	−20.68

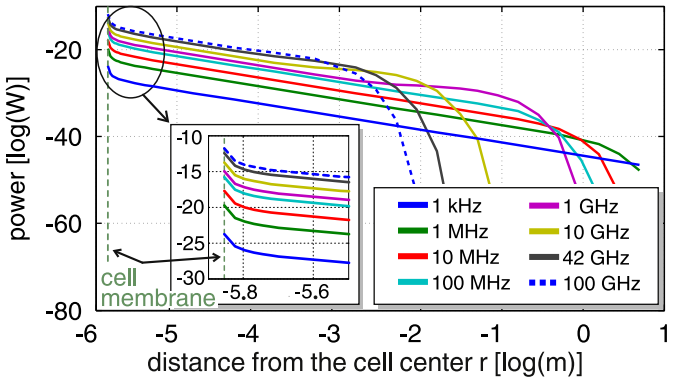

Fig. 10. Power calculated from the cell membrane to the distance of 1 m for eight frequencies (asymmetric model, mode 0 and lossy medium).

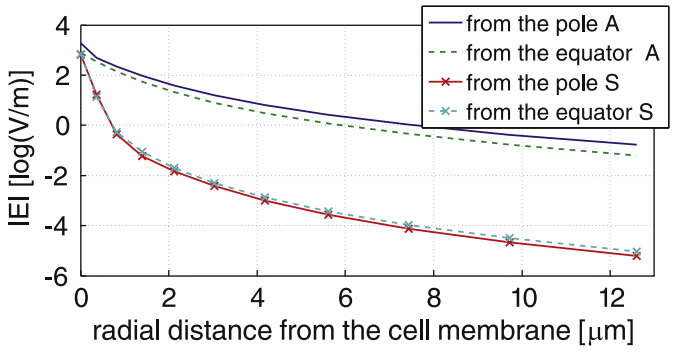

Fig. 11. Radial dependence of the electric intensity from the cell membrane in the direction from the equator (dashed lines) and from the pole (full lines) up to 12.6 μm. A and S stands for the asymmetric and the symmetric model, respectively (mode 0, f=100 GHz, lossy medium).

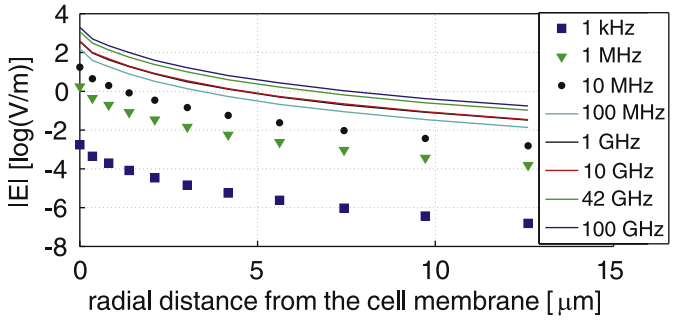

Fig. 12. Radial dependence of the electric intensity from the pole of the cell membrane—comparison of frequencies which are in the legend of the figure in Hz (asymmetric model, mode 0 and lossy medium).

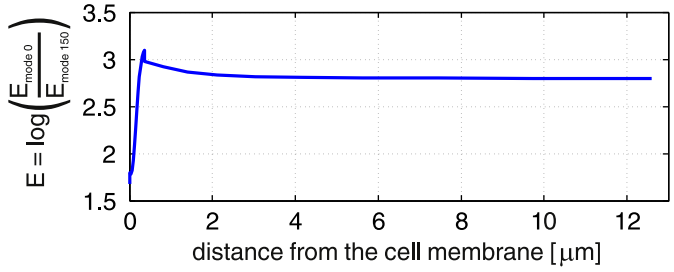

Fig. 13. Logarithm of the ratio of the electric intensity generated by mode 0 and that generated by mode 150 in dependence on radial distance from the pole of the cell membrane ($\log_{10}(E_{mode0}(r)/E_{mode150}(r))$). Electric intensity generated by mode 0 has higher values (asymmetric model, f=100 GHz and lossy medium).

electric field around microtubule generated by mode 50 is smaller by several orders of magnitude compared to electric field intensity generated by mode 0.

5. Results

At first, selected results of calculations of power of asymmetric microtubule network are presented. Eight frequencies (Table 2) and mode 0 were used for excitation of microtubules. The mode 0 is the most effective for generation of electric oscillations. Lossy and lossless medium was considered. The calculated results show that the values of radiated power from microtubule network generated by oscillations of mode min and of mode max are negligibly different (see Figs. 1–4 in supplementary material) and the values of power radiated from asymmetric and symmetric model are the same (see Figs. 5 and 6 in supplementary material). Therefore, from the calculations of radiated power results from asymmetric model only are shown. Poynting vector computed from near-field components ($E\sim 1/(\mathrm{j}kr)^3$ and $H\sim 1/(\mathrm{j}kr)^2$) has zero value in lossless medium. In contrast, the Poynting vector calculated from far field components ($E\sim 1/(\mathrm{j}kr)$ and $H\sim 1/(\mathrm{j}kr)$) is not zero. The values of power radiated from asymmetrical model in lossless medium for all eight frequencies are in Table 2.

The phase shift between near field components of intensity of electric and magnetic field is not $\pi/2$ but $\pi/2+2\phi_k$ in lossy medium (ϕ_k is a phase of propagation constant k of electromagnetic field in lossy medium, in lossless medium $\phi_k=0$). Consequently, the values of Poynting vector calculated from components of near field are not zero and are higher than Poynting vector calculated from other components in lossy medium. On the one hand the values of power near the membrane in lossy medium are higher than in lossless medium but on the other hand the values of power in lossy medium decrease steeply with distance from cell membrane. Fig. 10 shows the dependence of power generated by asymmetric microtubule network in lossy medium for all eight frequencies on radial distance. The value of power is the highest in the proximity of the cell membrane and decreases fast with growing distance.

Further results are from the calculations of intensity of electric field depending on radial distance from the pole or equator of cell (Fig. 6) for both models (Fig. 11). Fig. 12 depicts intensity of electric field depending on the distance from pole of the cell membrane for all eight frequencies. The difference between the electric intensity ΔE generated by mode 0 and that by mode 150 (maximal mode for the 1.2 μm long microtubule) in dependence of radial distance from the pole of asymmetric model in lossy medium and for frequency 100 GHz is shown in Fig. 13. The intensity of electric field on the border (at the cell membrane) of asymmetric model of the microtubule network is depicted in Fig. 14.

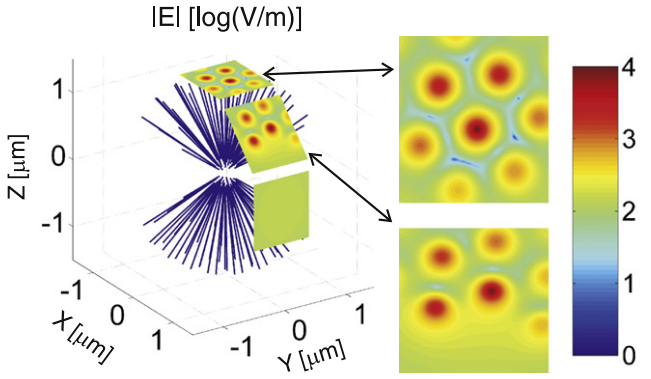

Fig. 14. The electric intensity on the border (at the cell membrane) of asymmetric model of the microtubule (mode 0, f=42 GHz, lossy medium).

6. Discussion

6.1. Approximations and assumptions on the level of the single microtubule

On the level of the single microtubule approximations and assumptions used include:

- integer mode numbers used;
- magnitude of oscillating dipole moment;
- shift of protofilaments in microtubule;
- microtubule protofilament approximated by chain of rigid particles, usage of axial longitudinal vibration modes, optical branch;
- coherent or at least quasi-coherent oscillations of individual dipoles, which require low damping of microtubule vibrations.

All of these points were discussed in the previous works of current authors (Cifra, 2009; Cifra et al., 2010d) where electric field generated by vibrations of single microtubule was already analyzed, so they are touched only briefly here.

Integer values of vibration mode number is related to boundary conditions of rigidly bound ends of microtubule. In real physiological conditions all microtubules grow from centrosome or other microtubule organizing center and can be considered to be bound rigidly (or quasi-rigidly) on the centrosome connected end (minus end). However, not all microtubules are capped and bound on its other end, the plus end. Substantial number of microtubules (exact fraction depends on the cell state and the phase of the cell cycle) undergoes dynamic instability and perpetual growth and shrinkage (Alberts et al., 2008). Vibrations of microtubules with one free end could assume infinite number of noninteger mode numbers.

Value of oscillating dipole moment used in calculations presented here represents the lower boundary of the possible values. This means that in real conditions the oscillating dipole moment can be also higher and consequently the generated electric field intensity may be higher, by approximately one order of magnitude. Axial shift of protofilaments in microtubule which was used in the model presented here corresponds to the real arrangement of protofilaments in microtubule lattice and is called lattice A, as already mentioned in Section 4. Another arrangement of protofilaments also often found in microtubules in vivo is lattice B (Tuszyński et al., 2005).

Vibrations of microtubule protofilament were approximated by vibrations of chains of rigid two-type particles, optical branch. A similar model of microtubule vibrations was used by Pokorný et al. (1997). On the level of the whole microtubule, this corresponds to axial longitudinal vibration modes of microtubules, such as treated by Wang et al. (2006), but is a special case in which neighboring tubulin particles oscillate against each other (optical branch). Vibrations of single microtubules were treated based on tubulin atomic coordinates obtained from crystallography (Deriu et al., 2010) but this approach is not computationally feasible for the whole cellular microtubule network. Therefore the approximation of tubulin as single rigid particles is considered reasonable for the current purpose.

Calculations presented here assume that there are coherent or at least quasi-coherent oscillations of individual dipoles (tubulin heterodimer). Coherent excitation can be provided by external coherent electromagnetic radiation or can be provided endogenously if system of oscillators fulfills conditions such as described by Fröhlich (1968a,b, 1980). Microtubules fulfill necessary conditions of the Fröhlich system (Cifra, 2009). Recently, Reimers et al. (2009) and McKemmish et al. (2009) performed computational studies of Fröhlich system based on using Wu–Austin Hamiltonian. They concluded that condensation of energy in certain modes strong enough to provide coherent excitations (one of the main features of Fröhlich system) is not feasible in biological systems. It is not necessary to go into details here but it has been pointed out (Cifra et al., 2010a,b) that the studies of Reimers and McKemmish treat incomplete Fröhlich system (nonlinear temperature dependence of coupling coefficients is missing) and contains several further flaws. Also another group of authors (Salari et al., 2011) noticed several errors in Reimers' study. Thus, conclusion of Reimers et al. (2009) and McKemmish et al. (2009) that the "…coherent Fröhlich condensates are not biologically feasible,…" cannot be fully accepted and definitely cannot be used for overgeneralization.

6.2. Damping of microtubule vibrations

Regardless of the mechanism for selective excitation of certain microtubule mode, it is necessary that the modes to be excited have low damping so the energy may be stored quasi-coherently in them. This requires that the energy supplied to microtubules is not dissipated rapidly to surrounding medium. Physically the problem is related to the damping of microtubule vibrations. If the microtubule vibrations are treated classically like for a cylinder immersed into bulk water, the result will be that the vibrations will be overdamped (Foster and Baish, 2000), i.e. $Q<1$ (Q, quality factor of oscillations). This may not be the case if one considers ions on the microtubule surface behaving as a slip layer (Pokorný, 2005), especially for longitudinal vibrations where negligible or no water at all will be displaced as for example in the case of standing waves with a zero vibration amplitude at the ends of microtubule. The quality factor (and the lifetime) of the biomolecule vibrations depends also on the organization of the water around the biomolecule surface: the higher the ability of the surface to attract water molecules (i.e. the higher the hydrophilicity of the surface), the higher is the Q (and lifetime) of the vibrations (Liu et al., 2009a,b). The physical mechanism is such that the water due to the electric polarity of its molecules is organized by the electric field of the surface charge of the hydrophilic biomolecule and the water undergoes visco-elastic transition (Van Zandt, 1986, 1987). There is extensive experimental evidence that the water is organized at the hydrophilic surfaces. Several layers of organized water molecules above the biomolecule interface can be detected and visualized by atomic force microscopy (Fukuma, 2010; Kimura et al., 2010). However, ordering of the water is not a matter of only a few molecular layers but can extend to macroscopic distances from strongly hydrophilic interfaces, e.g. few hundreds of micrometers as in the case of exclusion zone found by Zheng and Pollack (2003) and Zheng et al. (2006). These zones carry their name "exclusion zones" after the process they are

visualized with; the zones exclude micrometer sized particles (e.g. latex particles) in such a manner that two distinct water phases (zones) are visible at the hydrophilic interface observed under the microscope. The zone of water adjacent to the surface contains no particles (they are excluded to the outer zone), while beyond rather sharp boundary, particles are present in the second, outer water zone. It was experimentally proved that the organized water in the exclusion zone has different physical properties compared to bulk water including higher viscosity (Pollack et al., 2006), lowered thermal motion of molecules (Zheng et al., 2006), further it exhibits separation of charge, different pH (Chai et al., 2009), different spectroscopic properties (Chai et al., 2008), solvent exclusion (Zheng and Pollack, 2003; Zheng et al., 2006) and that infrared radiation promotes formation of the exclusion zones (Chai et al., 2009), etc. (Pollack et al., 2006, 2009).

Direct evidence of damping can be obtained from spectroscopy. While some spectroscopic and theoretical studies suggest that collective vibrations of proteins are overdamped by the water (Hayward et al., 1993) which is a major constituent of the cytosol, there are indications (Romanovsky et al., 2003) that the broadening of the spectral lines (a physical signature of a damping) may not be solely due to the damping but also due to inhomogeneous broadening of the spectral line—slight difference in the mode frequency due to conformational differences in the protein molecules in the ensemble under study. Actually, nonlinear spectroscopy already brought experimental evidence that lifetime (similar to quality factor Q of the oscillating system—inversely proportional to damping) of certain protein modes can be rather large, over 1000 (Xie et al., 2001). Currently, published data from direct spectroscopic studies on microtubules in MHz–GHz region are very limited. Hameroff et al. (1986) reported 8 GHz acoustic mode of microtubules with Brillouin spectroscopy, from the sole figure they provided the half width of the peak (Δf) is about 2 GHz, giving Q factor ($f/\Delta f$) about 4. Pizzi et al. (2011) measured very weak microwave absorption (0.5 dB) of in vitro microtubules with mean length of 2 μm at 1510 MHz which probably corresponds to some bending modes of microtubules which are overdamped (Samarbakhsh, 2010). Regarding these two past and any future experiments, it needs to be noted that inhomogeneous broadening contributes to broadening of the peak, thus leads to underestimation of the real Q factor. If the spectral resolution of the spectroscopy technique used is low, multiple spectral peaks will be merged into one broad peak, additionally clouding the real Q factor of oscillations. Further, longitudinal axial modes (which are most electromagnetically active in comparison with twisting or bending modes because the "stretching" of the tubulin dipole moment is greatest for longitudinal axial wave on microtubule) which conserve microtubule length will be damped much more weakly since they do not cause any displacement of the surrounding water. It is also important to realize that microtubules are naturally present in a cellular environment where the surrounding water is strongly organized due to numerous interfaces (Pollack, 2001) and by strong static electric field of mitochondria (Tyner et al., 2007). It can be expected that the in vivo damping of microtubule elastic vibrations will be lower than that measured on extracted microtubules. This can be reasonable if the amplitude of oscillations is small and comparable to size of damping units (water molecules or water clusters). In such case, classical hydrodynamic treatment of the damping (Stokes' formula) fails (Mitrofanov et al., 2006) and it underestimates the Q factor of protein oscillations especially in the THz region.

6.3. Related works on microtubule and protein vibrations (low-frequency phonons)

It may be of interest for readers to discuss several related works of Sinkala (2006) and Chou et al. (1994) and their relation to the topic of the current paper. Sinkala analyzed possibility of soliton/exciton transport in proteins which would enable non-dissipative transport of energy and related his model to the model of Resonant Recognition Model of Cosic (1994). The main connection with the current work is that solitons treated in Sinkala's paper are having low-frequency phonon (vibrational) component. In the current work, low frequency (low compared to higher frequencies of the protein vibrations) collective vibrations of the whole microtubules are the basis for the calculation of the distribution of the electric field around them. However, vibrations considered in the current paper are not of the soliton type but of simple standing harmonic wave of longitudinal axial type, optical branch, as described in our earlier paper focused on the field calculation around single microtubules (Cifra et al., 2010d). Chou et al. (1994) has focused on the soliton-like solution of the kink waves generated by the GTP hydrolysis and the possible role of these waves in the regulation of the microtubule polymerization dynamics. In the current paper, we focus on the electric field characteristic of the whole microtubule network. Regarding Davydov's solitons in proteins (namely the α-helix) and its relation to the generation of electromagnetic field we can refer reader to the works of Brizhik (Brizhik and Eremko, 2001, 2003; Brizhik, 2003), where emission of electromagnetic waves due to acceleration of the soliton involving charge transfer has been studied. The connecting work between microtubule soliton propagation and electric field generation seems to be in work of Satarić and Tuszyński. They carried out analysis of propagation of soliton type (kink) of tubulin dipole moment tilt on protofilament, which can have effect on microtubule polymerization kinetics (Satarić et al., 1992), energy transfer along microtubules (Satarić et al., 1993; Trpisová and Tuszyński, 1997), provide a possible elucidation of the unidirectional transport of cargo via motor proteins (Sataric and Tuszyński, 2005; Sataric et al., 2008). To summarize this paragraph regarding microtubules, soliton propagation and intrinsic microtubule electric field, it can be said that another approach to the problem treated in this paper could be based on calculation of electric field around microtubules generated by propagation of solitons on microtubules involving phonon–charge or phonon–dipole interaction.

6.4. Approximations and assumptions on the level of the whole cellular microtubule network

Regarding the model of the whole cellular microtubule network there are following approximations in the models in the study presented here:

- geometry of the models,
- in phase and single frequency oscillations of individual microtubules,
- size of the cell.

Geometry of the symmetric model was chosen to correspond to geometry of nondividing cell (interphase). Asymmetric model cannot be attributed to specific phase of the cell cycle, although it resembles the position of astral microtubules during the cell division. Asymmetric model was constructed in order to study influence of simple asymmetry of cytoskeleton distribution on radiation and electric field characteristic. Current work of the present authors involves modeling of complete mitotic spindle—microtubule formation during the cell division. It should be also noted that the microtubule organizing center (centrosome in many eukaryotic cells) is not always located in the center of the cell, so the microtubules do not grow exactly radially from the cell center.

In the model used in the study presented here the mutual phase difference between oscillations of individual microtubules is zero, i.e. microtubules oscillate in phase. This situation is probably very natural. Oscillations of individual microtubules can be coupled mechanically through the centrosome (or other microtubule organizing center) from which microtubules grow and coupling can be also established via microtubule electrodynamic field. From the time of Huygens it is known and nowadays very well established that coupled oscillators tend to synchronize each other (Bennett et al., 2002) in frequency (see Maffezzoni, 2010 for illustration) under certain conditions. In principle, even if the microtubules are of different length, they can support oscillations of the same frequencies (same modes). The condition of in phase oscillations of microtubules also corresponds to the requirement of minimal energy of the system. The radiation rate is lowest for such configuration since geometrical symmetry together with in phase oscillations causes effective canceling of radiative component of generated electromagnetic field. In the case when the minimum energy state is perturbed, models using random or quasi-random phase distribution of microtubules on the one hand or models using opposite phases of microtubule on opposite hemispheres on the other hand could be used.

Size of the cell (diameter 2.8 μm) used in models presented here corresponds to a rather small cell since typical diameter e.g. of standard eukaryotic model cell, yeast *Saccharomyces Cerevisiae* is 5−10 μm. Nevertheless, the total amount of microtubules used in current model corresponds to amounts present in larger cells. Work in progress includes parametric analysis of influence of number of microtubules and cell sizes on radiated power and electric field intensity at the cell membrane.

6.5. Discussion of results

Intensity of electric field and magnitude of the radiated power are comparable with the results obtained from the calculations using membrane proteins as the sources of the electric oscillations (Pokorný and Wu, 1998). Models presented here give good quantitative estimates of power and intensity of the cellular microtubule network unperturbed and unloaded by the detection electrodes. However, few other biophysical aspects influencing the technical requirements for measurement system (Kučera et al., 2010) could not be covered in current models. Electrodynamic activity probably arises from coupled oscillatory system of polar cellular structures, i.e. cytoskeleton coupled to membrane. Vibrations excited at various points of the membrane (e.g. where the cytoskeleton binds to the cell membrane) would create very complex interference pattern of multiple standing waves on the membrane. In such general case, spatial distribution of electric field intensity associated with the vibrations of the membrane will resemble high order spherical harmonics, having closely spaced regions of opposite orientation of the electric field vector. In order to resolve the spatial minima and maxima, detector electrode needs to have tip with radius of few hundreds of nanometers, ideally units of nanometers. Another aspect is related to the "softness" of the source. Providing the Q factor of oscillating system is high, energy accumulates in the system. Once the system is loaded by the detector and the energy is drained, the effective power supplied to the detector drops to the value of power delivered by the metabolism.

Models presented here do not take into account the effect of the cell membrane. It should be noted that the cell membrane behaves due to its capacitive character as a high pass filter. Therefore electric oscillations up to few MHz (or up to few hundreds of kHz only, depending on the cell type and effective capacitance of its membrane) (Asami and Yonezawa, 1996; Wang et al., 1993) which originate inside the cell cannot pass through the membrane. However, if there are mechanical vibrations present in cytoskeleton, they can be transferred into cell membrane through the cytoskeleton-membrane protein junctions. Consequently, electric oscillations can be generated by vibrations of the polar cell membrane.

7. Conclusion

Electromagnetic radiation and electric field of the whole cellular network consisting of 100 microtubules have been calculated for microtubule oscillations in the frequency region from 1 kHz to 100 GHz. Two basic models have been used for microtubule distribution, symmetric and asymmetric model. Microtubules are uniformly distributed in the symmetric model and they are distributed toward the poles of the cell in the asymmetric model. Total radiated power of the single cell for the lossless medium from the symmetric model is the same as from asymmetric model, being highest (10^{-20} W) for the highest frequency (100 GHz) calculated and lowest (10^{-52} W) for the lowest frequency (1 kHz) calculated. Electric field intensity of the symmetric model decreases steeper with the distance from the cell membrane than that of the asymmetric model, the difference is 4 orders of magnitude at the distance of 10 μm.

From current results, there are serious implications for the measurement system which is aimed to detect cellular electrodynamic activity. It is clear that macroscopic detection of the cellular far field radiation at room temperature with classical antennas is very doubtful. Cellular electrodynamic activity in radiofrequency region will be detectable under most conditions only within 10 μm from the cell membrane, conditions are more favorable when the symmetry (both phase and geometrical symmetry) of the microtubule network is perturbed. Further requirements for detectors, which cannot be quantitatively evaluated from the current models but have been enumerated and reasoned by Kučera et al. (2010), include minimal draining of the energy from biological nanosource (oscillating microtubule) and spatial resolution of hundreds to tens of nanometers. Development of sensors with these parameters is necessary to elucidate the biological role of the cellular electrodynamic activity.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jtbi.2011.07.007.

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ARTICLE

Homology modeling of tubulin: influence predictions for microtubule's biophysical properties

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Abstract Using comparative modeling, we have generated structural models of 475 α and β tubulins. Using these models, we observed a global, structural similarity between the tubulin isotypes. However, a number of subtle differences in the isotypes physical properties, including net electric charges, solvent accessible surface areas, and electric dipole moments were also apparent. In order to examine the roles that these properties may play in microtubule (MT) assembly and stability, we have created a model to evaluate the dipole–dipole interaction energies of varying MT lattice conformations, using human tubulin isotypes as particularly important examples. We conclude that the dipole moments of each tubulin isotype may influence their functional characteristics within the cell, resulting in differences for MT assembly kinetics and stability.

Keywords Tubulin · Isotype · Comparative/homology modeling · Microtubule · Dipole

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Introduction

Microtubules (MTs) are hollow cylinders constructed from linear chains, or protofilaments, of repeating subunits of α and β tubulin. MTs are involved in many cellular processes within eukaryotic organisms, including maintenance of cell morphology and active transport of proteins and organelles throughout the cytoplasm (Pichichero and Avers 1973; Hyams and Stebbings 1979). MTs go through periods of assembly and disassembly, a property known as dynamic instability (Mitchison and Kirschner 1984). Dynamic instability has been extensively investigated and is critical during mitosis when a delicate equilibria of forces are necessary for proper chromosome alignment prior to segregation (Margolis and Wilson 1981; Horio and Hotani 1986; Kirschner and Schulze 1986; Melki et al. 1989; Walker et al. 1991; Mandelkow and Mandelkow 1992). The correct assembly of MTs into the mitotic spindle is essential, as it likely provides the mechanical force for mitotic chromosome segregation (Westermann et al. 2006). The failure of this assembly process generally results in mitotic arrest and for this reason, several chemotherapeutic agents disrupt normal cell division by targeting MTs (Jordan and Wilson 2004).

Most eukaryotic organisms have multiple genes which encode distinct isoforms or isotypes of α and β tubulin. In humans, several isotypes have been identified and characterized (Lu and Luduena 1994; Roach et al. 1998; Luduena 1998). In addition to α and β tubulin, three other tubulin homologues are required for efficient MT assembly in vivo. The first, γ tubulin, is not essential for assembly, yet is found at MT-organizing centers and is thought to aid in initial MT nucleation (Oakley et al. 1990). Two other

homologues, δ and ε tubulin, have uncertain roles in MT assembly, although models have been proposed that describe their possible involvement in MT self-assembly (Inclan and Nogales 2001). At the molecular level, the roles and interactions of tubulin are complex and differ between isotypes. For instance, MT dynamics appears to change significantly with β tubulin isotype composition (Panda et al. 1994). This result implies that if MT assembly/disassembly equilibria are disrupted, cells might respond by producing an appropriate isotype mix to restore normal balance. While the presence of numerous tubulin isotypes, whose differences are often highly conserved in evolution (Ludueña 1998), suggests that they may play specific roles in MT function, there are no precise models to describe differences between them.

A major advance in our understanding of MT structure and function was the solution of tubulin's three-dimensional crystallographic structure (Nogales et al. 1998). Several crystallographic structures of tubulin are now available from the RCSB Protein Data Bank (PDB) and we refer to these structures by their PDB ID (Berman et al. 2000). The first approach to obtain tubulin crystals was taxane-stabilized bovine-brain tubulin (mixed isotypes) in flat Zn^{2+} induced sheets, producing protofilament-like end-to-end α/β dimer repeats. The earliest structure, 1TUB, utilized a drug known as docetaxel to stabilize protofilaments for crystallography (Nogales et al. 1998). Due to difficulties in fitting to the electron density, this structure contains several misalignments and density omissions and was superceded by 1JFF, which utilized paclitaxel, a drug similar to docetaxel, to stabilize the protofilaments (Lowe et al. 2001). Similarly, the structure 1TVK utilized the drug epothilone A, which binds at the paclitaxel/docetazxel binding site (Nettles et al. 2004). One final structure from this group, 1IA0 contains direct copies of the 1TUB monomers which were docked with the monomeric kinesin motor KIF1A (Kikkawa et al. 2001). A second approach to producing tubulin crystals utilizes the stathmin-like domain RB3-SLD, which complexes a pair of α/β dimer repeats. The earliest of these, 1FFX, utilized molecular replacement from the 1TUB data (Gigant et al. 2000). Most recently two higher resolution structures, 1SA0 and 1SA1, were co-crystallized with colchicine and podophyllotoxin, respectively (Ravelli et al. 2004), while both colchicine and vinblastine are included in 1Z2B (Gigant et al. 2005). Other, non-eukaryotic or tubulin-like structures also exist. The structures, 1Z5V and 1Z5W, are of human γ tubulin crystals (Aldaz et al. 2005). Related prokaryotic structures include 2BTO and 2BTQ products of *Prosthectobacter dejongeii* genes BtubA and BtubB (Schlieper et al. 2005). Additionally, several structures of the prokaryotic tubulin homolog FtsZ are also available.

The availability of high quality structural data for tubulin has enabled us to create comparative models of all currently known α and β tubulin isotypes. Comparative modeling uses known protein structures as templates to produce 3D structural models of a target protein with a known, related amino acid sequence. By assuming that parts of the target and templates with similar sequences have similar 3D structures, portions of the template structures may be copied and amino acids substituted, inserted, or deleted to produce models. Any part of the target not obtained from the template structure must be constructed via de novo methods. Finally, the resulting model must then be optimized to a protein-like geometry. This technique has been verified for a small number of homologous protein families and can yield accurate results for template/target pairs with strong sequence identity (Chothia and Lesk 1986).

From such comparative models, we have calculated the energy associated with MT lattice dipole–dipole interactions for all the human tubulin isotypes. Schoutens (2005) estimated that MT dipole–dipole interactions destabilize the MT structure by 50–60% of the elastic energy of the underlying lattice. The strength of these interactions is roughly proportional to the squares of the dipole moments; increased tubulin dipole moments should therefore make MTs less stable. From our estimates we hope to understand how tubulin isotype sequences may affect both MT assembly and stability through basic electrostatic properties of protein.

Materials and methods

Tubulin sequences

Tubulin sequences were obtained from the Uniprot Knowledgebase, consisting of Swiss-Prot release 42.1 and TrEMBL release 25.12 (15 March 2004) (Boeckmann et al. 2003). The entire database was downloaded and sequences annotated with the term ''tubulin'', were extracted, producing an initial set of 2,294 entries. This initial set was then filtered to remove entries labeled as fragmentary, leaving only 652 complete tubulin sequences. These sequences were then aligned and using cluster analysis assembled into a guide tree, or dendrogram using default parameters within Clustal W (version 1.82) (Higgins and Sharp 1988). This step identified sequences that are not tubulin, but unrelated

proteins with tubulin in their name (e.g. tubulin tyrosine ligase or tubulin-folding cofactor D). Since these nontubulin sequences differ significantly from the tubulins, they clustered separately in the dendrogram, providing a clear delineation between 557 ''tubulin-like'' and 95 nontubulin sequences. A second dendrogram guide tree was then produced using only the ''tubulin-like'' sequences resulting in three major branches, one of 190 α tubulin sequences, one of 285 β tubulin sequences, and the third of 82 tubulin sequences that were neither α nor β tubulin.

Template selection

As we were constructing models of eukaryotic α and β tubulins, we rejected γ tubulin and bacterial proteins as structural templates. We chose to use only monomers from the 1JFF and 1SA0 structures as these are high-resolution representatives of the two crystallographic approaches. We also reasoned that all three tubulin dimers within these structures have distinct environments and would produce variation in the models. In the 1SA0 structure, the colchicine molecule is at opposite ends of the two dimers and each dimer has a different interaction with the stathmin-like domain, while the 1JFF dimer is in a different crystal packing. Thus, for α tubulins we used the 1JFF A chain and the 1SA0 A and C chains, while for β tubulins we used the 1JFF B chain and the 1SA0 C and D chains.

Modeling

To build the models, we used Modeller (version 6v2) (Sanchez and Sali 2000). While Modeller provides many options for model refinement and fine-tuning, default parameters and settings were deemed sufficient for generating the models presented here. (See Table 1 for a sample script building sequence P05218 on the 1JFF β monomer.) For each tubulin sequence, three models were built, each using a different one of the three templates (Table 1). To check the correctness of each of the models, some were tested with PROCHECK and WHAT_CHECK (Laskowski et al. 1993; Hooft et al. 1996). Since the results of these programs require human inspection, only a small subset of models was tested in this way. These results suggested to us that the isotype models are likely of similar quality to the input template structures. As a final step, models were re-positioned onto the coordinates of the templates used to create them by RMSD (root-mean-square deviation) fitting over corresponding alpha-carbons.

Generation of human isotype models

Among the 190 α and 285 β tubulin sequences are 22 human sequences. Two are quite different in length from the templates, one short (Q8WU19), and one long (Q8WUL7). The other 20 were used to examine the stability of the modeling technique and accuracy of the models by producing 100 models of each target. As Modeller uses pseudo-random noise in model construction, each differs slightly, providing a sample of 100 Modeller output conformations for these human sequences.

Calculations of the surface areas, charges, and dipole moments

After models were built, surface areas, total charges, and dipole moments were estimated using analysis tools in Gromacs v. 3.3 (Lindahl et al. 2001). Models were converted using pdb2gmx with the GROMOS96 43a1 force field (van Gunsteren 1996), providing atomic charges, sizes, and some hydrogen atom

Table 1 Sample MODELLER routine

```
# comparative modelling by the MODELLER TOP routine MODEL
SET KNOWNS = '1JFFB'                    # code of the templates
INCLUDE                                 # load Modeller defaults from __defs.top
READ_TOPOLOGY FILE = '$(LIB)/top_heav.lib'
# read template sequence
READ_ALIGNMENT FILE = 'structure.ali', ALIGN_CODES = KNOWNS
# add sequence to be modeled, remembering how many templates are already aligned
SET ADD_SEQUENCE = ON, ALIGN_BLOCK = NUMB_OF_SEQUENCES
READ_ALIGNMENT FILE = 'tubulins.ali', ALIGN_CODES = ALIGN_CODES 'P05218'
# align sequence to template
ALIGN2D
# save alignment – read by routine MODEL
WRITE_ALIGNMENT FILE = 'P05218.ali'
# do model construction
CALL ROUTINE = 'model', ALNFILE = 'P05218.ali', SEQUENCE = 'P05218'
STOP
```

locations. As a side effect, the net charge of the sequence is reported, assuming charges of $+1e$ for lysine and arginine, $-1e$ for aspartic and glutamic acids, and no charge for other residues. Using g_sas, Connolly solvent accessible surfaces (SAS) were then calculated (Connolly 1983). Next, editconf was used to shift the co-ordinate origin to the model's geometric center, which is expected to be a good approximation of the monomer's center-of-mass. Dipole moments were then calculated using g_dipoles. These dipole components were then converted to a cylindrical coordinate system based on MT geometry. Each 1SA0 monomer was α-carbon RMSD fit onto the corresponding 1JFF monomer, these transformations taken as conversions between the 1SA0-based co-ordinate space and a 1JFF-based one; 1JFF co-ordinates were converted to the MT-based one by 63° rotation about the protofilament axis, a value chosen with data presented by Li et al. (2002). Each dipole moment was transformed identically to its model's template.

Calculations of the microtubule dipole–dipole interaction energies

The same MT geometry based cylindrical coordinate system, as described above, was used for dipole interaction energy estimates. We defined a direction away from the central cylindrical axis as radial, one parallel to the central axis as longitudinal, and one perpendicular to both of these (around the surface of a cylinder) as tangential. Following the description by Li et al. (2002), we position dipoles at a radius of 112 Å, in 13 parallel protofilament-like rows, at the 13 angular multiples of (360/13)°. Along these rows, dipole values were placed every 81.2 Å

The interaction energy between two dipoles $\vec{p}_1$ and $\vec{p}_2$ separated by $\vec{r}$ is given by:

$$U = \frac{1}{4\pi\varepsilon\varepsilon_0}\left(\frac{\vec{p}_1\cdot\vec{p}_2}{r^3} - 3\frac{(\vec{p}_1\cdot\vec{r})(\vec{p}_2\cdot\vec{r})}{r^5}\right). \quad (1)$$

Each dipole is described as

$$\vec{p} = (p_r, p_\theta, p_z),$$

having a radial, tangential, and longitudinal component. The interaction energy of two dipoles of known positions can then be rewritten:

$$\begin{aligned} 4\pi\varepsilon\varepsilon_0 U = {} & c_1 p_{r1} p_{r2} + c_2 p_{r1} p_{\theta 2} + c_3 p_{r1} p_{z2} \\ & + c_4 p_{\theta 1} p_{r2} + c_5 p_{\theta 1} p_{\theta 2} + c_6 p_{\theta 1} p_{z2}, \\ & + c_7 p_{z1} p_{r2} + c_8 p_{z1} p_{\theta 2} + c_9 p_{z1} p_{z2}, \end{aligned} \quad (2)$$

where the nine c_i are known geometry-dependent constants dependent on $\vec{r}$ and the rotation of coordinate system components between the two dipoles.

To calculate the interaction energy between a reference dipole and those in an adjacent protofilament, as the adjacent protofilament is displaced along its length, we take several dipoles from the adjacent protofilament near the reference and evaluate the c_i of (2) between adjacent dipoles and the reference. Then with all the adjacent dipoles identical, and $4\pi\epsilon\epsilon_0$ uniform, the p are common factors for all interaction pairs, so each of the nine c_i terms can be separately summed over all pairs to obtain a total dipole–dipole energy of the same form as (2), with a new set of c_i.

We evaluated summations over adjacent sections of length $2l + 1$, expecting that as we summed longer sections we would approach a limit, more separated dipole pairs having weaker interactions. We placed the reference dipole at $z = 0$ Å and the adjacent dipoles at $z = (81.2\text{ Å})(i + d)$ where integer $i \in [-l,+l]$, and the displacement, d, varied from 0 to 1. At $d = 0$, we found that a limit is rapidly approached in our calculations, with little difference in the summed c_i for several $l \geq 10$, and those of $l = 10{,}000$.

For $l = 1{,}500$ and d = 0.00, 0.01, 0.02,...,1.00 we calculated the summed values of the nine coefficients c_i and observed near sinusoidal variation with displacement (Fig. 1). Fits to the three lowest frequency Fourier components,

$$c_i(d) = A_i + \sum_{k=1}^{3} B_{ki}\cos kd + C_{ki}\sin kd,$$

give coefficients where either the sine or the cosine series coefficients are large (more than one part in 10^6) relative to those of the other series. In this significant series the second-order terms are all less than 5% of the first-order terms, and the third-order terms less

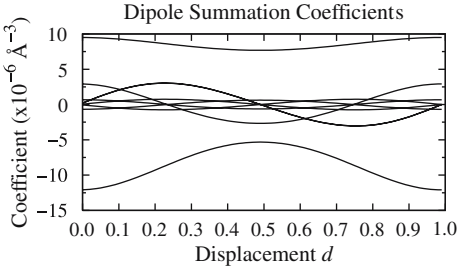

Fig. 1 The nine coefficients c_i as functions of the displacement, d for the case of an adjacent protofilament. Note the high level of symmetry, and that two of the nine curves are identical and superimposed in the plot

than 3% of the second-order terms. In only two of the nine cases are the constant terms significantly nonzero. Thus, we take only the first order and constant terms and obtain the approximations $c_i(d) \approx A_i + B_i \cos d$ or $c_i(d) \approx C_i \sin d$. We note that this approximation implies sinusoidal variation with displacement to the energy in (2).

This calculation can be easily extended to the dipole interactions between a reference dimer and an adjacent protofilament when each monomer has its own dipole moment. Treating the α and β contributions in the two filaments separately gives four cases. In the first, we can place all the α dipoles at the locations discussed above and obtain the same result. The case of the row of β dipoles midway between adjacent α row interacting with the reference α dipole has the same form but an additional one-half displacement is included. Similarly, a reference β dipole may be placed at $z = 40.6$ Å, the same as subtracting one-half from the displacements when interacting with the row of α dipoles. In the β–β interactions case, the two additional displacements cancel out. By summing the contributions of the four cases, a single sum is again obtained. For simplicity, we use only one α and one β monomer in both the adjacent and reference positions. Then by symmetry, many coefficients cancel and the result may be rewritten as

$$4\pi\varepsilon\varepsilon_0 U = F + G\cos d \tag{3}$$

where both F and G have the form

$$\begin{aligned} &e_1(p_{r\alpha}^2 + p_{r\beta}^2) + e_2(p_{\theta\alpha}^2 + p_{\theta\beta}^2) + e_3(p_{z\alpha}^2 + p_{z\beta}^2) \\ &\quad + e_4(p_{r\alpha}p_{r\beta}) + e_5(p_{\theta\alpha}p_{\theta\beta}) + e_6(p_{z\alpha}p_{z\beta}) \end{aligned} \tag{4}$$

for particular values of the geometric constant coefficients e_i.

Results and discussion

Comparative modeling is a theoretical method with predictive power limited by the degree of similarity between target and template structures and input template quality. Quality is further limited by the choice of refinement approximations following model construction. Further constraints on predictive power arise from failures to include environmental factors, such as ligands, ranging from ions to protein complexes, and chemical modifications such as post-translational modification, both of which are relevant in the case of tubulin.

Tubulin sequences

Our set of α and β tubulin sequences includes animal, plant, fungal, and protist tubulins. We believe that this set is likely large enough to well sample the diversity present within the tubulin family. In general, the tubulin sequences modeled were similar to those of the templates. However, some of the tubulin sequences vary greatly in length from the template sequences. It seems likely that some of these sequences may not assemble into MTs, e.g. a γ tubulin sequence. Such cases would be less well modeled using this technique than other sequences. Except for the two human sequences omitted during the human repeat calculations, such possible problems were ignored and all sequences were treated identically.

It should be noted that the assignments generated by the initial clustering did not always agree with the database labels. Alignments and scoring were performed without knowledge of each isotypes cellular function, close similarity between the isotypes occasionally produced ambiguous results. For instance, the Uniprot Knowledgebase sequence Q94EN1 is described as ''Pi-tubulin'' in the database, but clustered within the α tubulin set using the procedure described above.

Template selection

All of the available tubulin structures would likely provide good templates for modeling. All structures of crystallized α and β tubulin are similar, being nearly indistinguishable at 6 Å, even with only 40% amino acid identity between them (Li et al. 2002). Since sequences within the α and β tubulin groups are more similar to sequences within the same group than with those of the other group, it is reasonable to expect that any given sequence likely has a structure very similar to that of other members of its group. An example is the 1TUB and 1JFF structures which are of a porcine sequence, but fit to structural data from inhomogeneous bovine samples. Accordingly, either crystallographic structure can be used as a template to produce models with some confidence. However, each of these template structures contains missing or weakly defined residues. Trivially, neither template contains the initial methionine residue. Additionally, one α tubulin loop is incomplete with residues Q35–L60 missing from the A chain in 1JFF and residues P37–D46 and S38–D46 missing from the A and C chains of 1SA0, respectively. Both of the 1SA0 β monomers, chains B and D lack residues R278–A285 and several side chains within these structures are also truncated. Further, all

template structures lack a 20–30 residue region, commonly referred to as the C-terminal tail (approximately residue 440 to the C-terminus); this region is believed to be highly flexible. Within these missing regions the model is being created with few constraints and less confidence should therefore be placed in the reality of the results.

Comparison of the tubulin models shows that regions found within the templates have well-defined structures, and those without templates are more disordered. The region missing from the 1JFF α tubulin template (Q35–L60) produces a large unstructured loop away from the core of the monomer in many models. The large variability of this region within the human repeats led us to reject the 1JFF α tubulin derived models from further consideration. In the 1SA0-based models, due to the resolved structure of more of the template, this region has a more compact structure, although it is more variable than most other regions in the models. In all the models, loops corresponding to residues 360–371 and 275–286 are also of high variability, owing to unresolved structures in the templates. Such variations within the structure of tubulin could have an effect on dimer–dimer interactions, which may translate into significant changes in assembly/disassembly rates for MTs. These differences are fairly well studied experimentally (Panda et al. 1994) but poorly understood theoretically. It should be noted that the generated C-terminal regions are too flexible and variable for models to represent, at best, more than one of many possible conformations. To what extent the variation in the models is experimental artifact, and to what extent real variation in the tubulin structures is unknown.

Estimation of physical properties

It is possible to estimate various physical characteristics from these models. These properties can then be considered for possible effects of these on tubulin dimer–dimer interactions, and thus MT assembly and stability. One contribution to the interactions between tubulin dimers is the electrostatics of each tubulin monomer, in particular, net charges and dipole moments. The net charge or monopole is perhaps the simplest approximation to the complicated electrostatic field around a protein. Slightly more complex is the electrostatic dipole moment, which has opposing positive and negative directions along a central axis, and is radially symmetric about this axis. Like the monopole, the dipole is an average over a real system and is generally inadequate for representing the complex field near a protein, and is also subject to attenuation by solvent mediated effects, but having both a position and an orientation, a dipole may give more complex interactions. With charges at points in space, we can calculate a dipole moment about a chosen coordinate origin. If a system has zero net charge, this dipole moment is independent of the choice of origin, but is otherwise dependent upon the origin. For protein dipole calculations, the center-of-mass is commonly used.

Solvent accessible surface area

For each tubulin model, we estimated the solvent accessible surface areas and show that they are largely the same for α and β tubulins with mean areas of approximately 220 nm^2, and standard deviations of 10 and 30 nm^2, respectively. This difference in variance is visible where surface areas for α and β tubulins are compared with their estimated dipole moments (Fig. 2). Little correlation between these two properties is evident from these scatter plots. However, there does appear to be an observable difference within the mean dipole between the alpha and beta tubulin.

Electrostatics

Tubulin sequences generally contain more acidic than basic residues and accordingly we report negative electrostatic charges. In an overly simplistic analysis we could expect that these negative charges would repel, thereby preventing MT formation. However, at short ranges charge is partially screened and some close charge pairs can dominate the electrostatic interaction

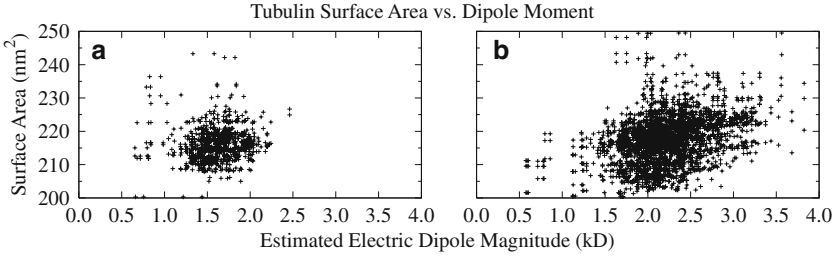

Fig. 2 Comparison between tubulin isotype models solvent accessible surface areas and estimated dipole moment magnitudes. Panel **a** illustrates correlation within α tubulin models. Panel **b** illustrates correlation within β tubulin models

over more distant pairs. The extreme of this occurs between monomers, where contacting regions have interactions strong enough to form a stable bond.

The α tubulins are estimated to have dipole magnitudes about their centers in the range 1,000–2,000 D, while β tubulin magnitudes are generally higher, in the range 2,000–3,000 D (Fig. 3). The β tubulins are, on average, more highly charged with an approximate mean of –24*e*, compared with –22*e* for the α tubulins. The β tubulins also have a slightly broader net charge distribution than α tubulins (standard deviation of 3.0*e* versus 2.6*e*). There appears to be little if any correlation, in the scatter plots, between the dipole moment and the net charge as might be expected since polynomial expansions coefficients such as the electrostatic potential generally have no dependence on one another (Fig. 4).

In this work we give specific values for tubulin charges and dipole moments. It is important to recognize that these values only consider the protein and not effects due to the surrounding solution. A significant solvation effect is on mean protonation state; net tubulin charge becomes more positive as pH decreases. In particular, near pH 5, tubulin is electrically neutral (pI from SWISS-2DPAGE, see Hoogland et al. 2004). However, our electrostatic numbers are derived with a charge model (Gromos96 43a1) with a specific charge for each atom, appropriate near pH 7. Another solvation effect is counter-ion cancellation of the electrostatic field about a charged object. Charge effects only occur when there is some separation of this screening charge and the charge. Trivially, objects in close proximity will exclude some or all of the screen. More generally, this separation will depend on the solution composition. Since the extent of screening will depend on the circumstance we report values for unscreened systems, and as a further simplification use uniform dielectric constants.

Comparing our calculated values to experimentally based values reveals the possible magnitude of these solvent effects. For instance, experiments orienting or dragging MTs with external electric fields have reported per dimer charges equivalent to –0.9*e* and –0.2*e* (Minoura and Muto 2006; Stracke et al. 2002). These results may also be interpreted as a measure of the screening charge pulled away from the tubulin. In a different example, the addition of tubulin to a solution is reported to affect the high-frequency dielectric constant by 0.0050 mL/mg (Schuessler et al. 2003). Using this value in the dilute limit Oncley formula gives a tubulin dipole moment of 78 D for a dimer at 300 K. While our values do not compare well with these numbers, but it must be emphasized that we did not expect them to be considered absolute and are only valid within the results presented here.

Human isotypes

The 100 models of the human tubulin sequences allow the reproducibility of results for an isotype sequence to be assessed. In particular, the dipole moment estimates have roughly a 300 D standard deviation. These values suggest that much of the observed variation between sequences may be attributed to model variability and not intrinsic differences among the sequences. For one of the 55 sequence-template pairs in this repeat set, Modeller stopped prematurely with an error, so 64 of the 5,500 repeat models were not built.

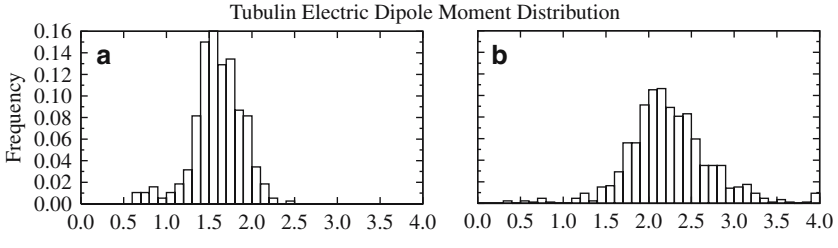

Fig. 3 Distribution of the estimated dipole moment magnitudes for α tubulin (**a**) and β tubulin (**b**)

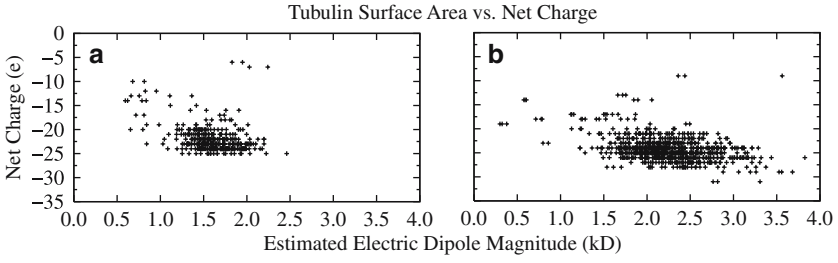

Fig. 4 Comparison between the dipole moments and net charges for α tubulin (**a**) and β tubulin (**b**)

Human microtubule dipole interaction energies

We considered all the α/β pairings of models from our set of replicate human runs, and evaluated all of the dipole moments with (3). We noted G, the sinusoidal magnitude, is typically small compared to F, the offset term, and so we ignored the sinusoidal term. The results show a great deal of similarity, with the five α monomers being largely indistinguishable, as are many of the β monomers; only the Q99867 sequence is significantly different. The distribution of interaction energies for $\epsilon\epsilon_0 = 1$ of each human β tubulin sequence is shown in Fig. 5a. A similar calculation can be done within a single protofilament by replacing a dimer in the ''adjacent'' protofilament with the reference dimer. Displacement makes no sense in this case, avoiding the sinusoidal term (Fig. 5b). Again, only Q99867 stands out as significantly different in this regard.

Summary and conclusions

We have used comparative modeling as a computational tool to generate structural models of 475 α and β tubulin sequences. We note a structural similarity across tubulin isotypes, along with some subtle differences in physical properties, including net electric charges, solvent accessible surface areas, and electric dipole moments. In order to examine the roles these properties may play in MT assembly and stability, we evaluated dipole–dipole interaction energies of MT lattice conformations with various human tubulin isotypes. In this case, we already know that isotype composition affects MT assembly kinetics (Panda et al. 1994). We therefore expected that properties, such as the dipole moment, may influence the functional behavior of tubulin monomers and dimers within the cell, resulting in differences for MT stability and assembly kinetics. It was demonstrated that electrostatics contributes to the energy of relative longitudinal offsets between two protofilaments, helping to shape the energy minima corresponding to the two MT lattice types, especially the A-lattice (Sept et al. 2003).

Further, besides its involvement in MT stability, the dipole moment of tubulin may play a role in other processes involving tubulin, for example, docking of molecules such as kinesin or MT associated proteins to tubulin and in creating the proper steric configuration of a tubulin dimer as it approaches an MT for binding. Interestingly, in an MT, dimers have near parallel dipole moments relative to their nearest neighbors, suggesting energy minimization takes place during MT assembly. However, when all dimer–dimer interactions in a MT are summed, the total is a positive energy leading to MT destabilization. Hence the greater the magnitude of the individual dipole moments (assuming all are identical), the less stable the MT since these interactions will be repulsive, pushing dimers apart. The strength of these interactions is proportional to the squares of the dipole moments; MTs made up of tubulin with larger dipole moments should be more prone to disassembly than those composed of lower dipole moment tubulins. For cells expressing more than one isotype, one can conceive MT dynamics regulated by altering relative amounts of isotypes. However, these effects are moderated by local interactions with more complicated electrostatic fields than the above simplified assembly of dipoles; for details see Baker et al. (2001). Nonetheless, it is possible that the differing kinetic properties of different isotypes are linked to the electrostatic differences.

From the discussion of experimental determinations of electrostatic properties of tubulin and MTs it is clear that the experiment and its precise conditions can ultimately affect the outcome of any measurement. Hence, we require more accurate and systematic analyses under a range of conditions such as pH, ionic concentrations, tubulin concentration, and temperature values. It is, however, clear that electrostatics, and in particular the dipole moment, plays an important role in the interactions of tubulin with itself, with other proteins, molecules, and ions.

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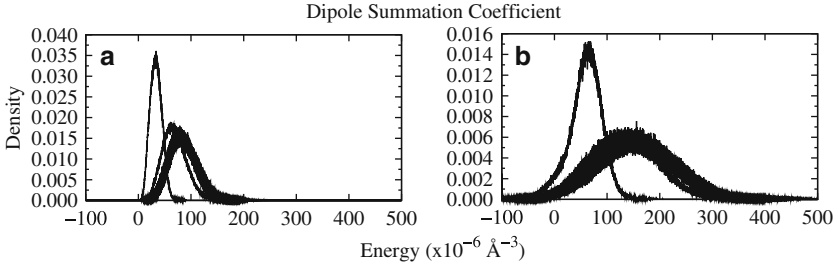

Fig. 5 The per dimer interaction energies within a microtubule for all human tubulin isotypes. Panel **a** illustrates results obtained for adjacent protofilaments within a MT structure. Panel **b** illustrates results for a single protofilament using a reference dimmer only

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Chaos

Ionic wave propagation and collision in an excitable circuit model of microtubules

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Ionic wave propagation and collision in an excitable circuit model of microtubules

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In this paper, we report the propensity to excitability of the internal structure of cellular microtubules, modelled as a relatively large one-dimensional spatial array of electrical units with nonlinear resistive features. We propose a model mimicking the dynamics of a large set of such intracellular dynamical entities as an excitable medium. We show that the behavior of such lattices can be described by a complex Ginzburg-Landau equation, which admits several wave solutions, including the plane waves paradigm. A stability analysis of the plane waves solutions of our dynamical system is conducted both analytically and numerically. It is observed that perturbed plane waves will always evolve toward promoting the generation of localized periodic waves trains. These modes include both stationary and travelling spatial excitations. They encompass, on one hand, localized structures such as solitary waves embracing bright solitons, dark solitons, and bisolitonic impulses with head-on collisions phenomena, and on the other hand, the appearance of both spatially homogeneous and spatially inhomogeneous stationary patterns. This ability exhibited by our array of proteinic elements to display several states of excitability exposes their stunning biological and physical complexity and is of high relevance in the description of the developmental and informative processes occurring on the subcellular scale. *Published by AIP Publishing.*
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The existence of a mechanical signal due to some applied centrifugal force, though marginal, has been suspected to be responsible of the positional information in cortical microtubules. On the same footing, their electrical activities due to a constant state of flux, continually polymerize and depolymerize them (as observed in animal cells), unless stabilized by the binding of guanosine triphosphate to the ends, which inhibits depolymerization. The dynamics and nature of the information processes underlying these functional mechanisms at the subcellular scale are still not completely understood. We have developed an electrical model with nonlinear resistive properties that can model the excitability of these biological systems. This tunable excitable model, where the dispersion, nonlinearity, and dissipation can be easily regulated, demonstrates rich dynamics, and can serve as a tool for studying the physiological complexity of cortical microtubules.

I. INTRODUCTION

During the last decades, quite a sizable amount of mathematical treatment has been performed in the field of complex nonlinear systems as a whole and in the area of nonlinear excitable media, in particular. Typical examples of such systems include chemical excitable milieux, namely, the Belousov-Zhabotinsky reaction[1,2] and the reaction of oxidation of *CO* on platinum catalysts.[3] They equally incorporate biological paradigms such as intracellular waves of calcium induced calcium release, head-on collisions of waves in excitable FitzHugh-Nagumo systems,[4] the electrical membrane excitation in nerve and muscular tissues (relevant in cardiac arrhythmia for instance), and some simple morphogenetic systems,[5] to name just a few.

Nowadays, biological systems have been the core components in the development of nonlinear science due to their outstanding complexity both at small and large scales. Their baroque complexity then requires some strong computational treatments safely guided by mathematical insights, based on simple nonlinear systems, or rationally reduced models. Starting from the ground that biological systems usually emerge as a large collective set of topologically connected (all-to-all, sparsely, or nearest neighbors interactions) biochemical units (cells), their computational modelling, crucial for their quantitative and qualitative dynamical studies, forms a key ingredient of the interpretation of the mathematical and experimental outcomes obtained from their inspections. Structurally speaking, the basic building block of the architecture of any biological system is the cell. It is mainly made up of a substance called the cytoplasm, containing 80% of water in its composition. Throughout the cytoplasm of eukaryotic cells (cells that have membrane-bound organelles such as nucleus and endoplasmic reticulum) the cytoskeleton, which comprises tubular polymers called microtubules, is found.[6,7] Microtubules play some key roles in living organisms. They are triggered by the signalling processes occurring within the cellular milieu. Microtubules

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play some crucial roles in a number of cellular processes and are involved in maintaining the structure of the cell. Together with microfilaments and intermediate filaments, they form the cytoskeleton. They also make up the internal structure of cilia and flagella which play a great role in movement.[7–10] The act of withdrawing the hand when it comes into contact with a hot object, the act of an organism displacing itself to locate food when in need, and many other occurrences are evidence of some degree of information processing and transmission within the internal structure of living beings. These processes usually occur in the form of wave phenomena.[2] Such information or impulse called ionic wave is in the form of an electric field. This ionic wave follows a pattern that sometimes emerge in the form of a localized wave, solution of a widespread class of weakly non-linear dispersive partial differential equations referred to as non-linear localized waves of soliton types.[11,12]

References 13–15 model microtubule as a nanobioelectronic nonlinear circuit possessing the polyelectrolyte features of cylindrical biopolymers, with the assumption that microtubules play a fundamental role in sub-cellular computation and signalling via electronic and protonic conductivity and can then be made useful in hybrid material that offer novel electronic characteristics. Based on these considerations, the nature of the wave solution (or energy) resulting from the model is observed to be a solitonic pulse. Recently, on the same footing, the authors in Ref. 16 investigated the phenomena of wave transmission and amplification in a similar electrical model of microtubules. Therein, a comparison is made between the results obtained on one hand in the absence of a negative nonlinear resistance, and on the other hand, with those obtained in its presence. The signal was found to be decreasing in the former case and to be amplified in the latter.

In this work, considering the strong intrinsic nonlinear character of these biological systems, we consider an electrical model of microtubule with a nonlinear resistance at the level of each electrical unit. This is motivated by the fact that the previous models failed to display some features of excitability as it has been experimentally reported to exist to some extent in microtubules.[17–26] This consideration accounts for the fact that the processes of opposition to the flow of calcium ions within and between microtubules are more than just linear. We believe that the fluctuations in the internal resistance of the dynamical unit, occurring in the course of the information processing mechanisms occurring along the structure of the microtubule, and which depend to a great extent on the content of the microtubule, may impact significantly on the nature of the physical and biochemical solicitations of these biological systems. Therefore, the aim of this work is to investigate under the aforementioned circumstances (conditions), the related liability of these biological systems to display some interesting features of excitable media. The main goal of this exploration is to throw more light on the puzzling processes pertaining in the robust internal functional activities of cells, which have relevance in maintaining homeostasis in living beings, via stable internal operating conditions, crucial for the perpetuation of life.

This article is organized as follows. In Sec. II, we present our electrical model of microtubule (*MT*) with nonlinear resistive elements. This model aims at emulating the dynamics of the voltage stimulus, compelling calcium ions to move within and in between microtubule protofilaments (PF). Subsequently, we derive a discrete differential equation of motion for the voltage stimulus of the calcium ions, following some circuit analysis principles. At this stage, we propose ourselves to solve this equation while seeking for solitary waves solutions. For this purpose, we shall use a perturbation technique in the form of the multiple scales expansion method with semi-discrete approximation. The description of this method is coarsely provided, and its development related to the issue at hand is as well expounded. Afterward, an equation of evolution for the ionic wave amplitude is obtained in the form of a complex Ginzburg-Landau equation (*CGLE*), and the Benjamin-Feir stability analysis of the plane waves solutions of this equation is conducted numerically. In Sec. III, we analytically examine the stability of the plane waves solutions of the *CGLE* based on the Eckhaus stability analysis approach. The outcomes of this analytical inquiry are compared to those obtained numerically in Sec. II in order to check for their consistency. In Sec. IV, based on the Nozaki-Bekki approach, we propose an analytical solution for the solitary wave solution of our *CGLE*, whose characteristics will be analyzed in the entire computational domain. Next, we perform the analytical resolution of our *CGLE* based on the harmonic balance method, in order to work toward the determination of exact travelling symmetric-like ionic waves, solutions of our nonlinear equation. Section V is dedicated to the direct numerical treatment of the main discrete voltage equation of motion, while assuming Ansatz as the exact analytical solution obtained in Sec. IV. Finally, Sec. VI concludes the work.

II. THE ELECTRICAL CIRCUIT MODEL OF MICROTUBULES AND THE WAVE AMPLITUDE EQUATION

A. The model

Microtubules (MTS) are essential components of the cytoskeleton, found throughout the cytoplasm of eukaryotic cells. They are hollow cylinders made up of protofilaments (PF) representing series of proteins known as tubulin dimers. Each PF is assimilated to a succession of N adjacent electrical units. Each unit cell number n of the lattice is constituted of a linear inductance L in series with a linear resistance r, shunted by a non-linear capacitance $C(V_n)$ in parallel with a non-linear resistance R_n.[16] In standard models, the non-linear components are only represented by a conductance and a capacitance.[9,14,27,28] Most mathematical models for transmitters[27,29] add a cubic non-linear resistance for ionic current flow through nano-pores of neurons. Freedman *et al.*[27] observed that the current-voltage characteristic of the MT lumen resistance in the presence of cations can be fitted by a cubic polynomial. Thus, a non-linear resistance was suggested for modelling cation flow through nano-pores of MT.[16] Priel *et al.*,[16,30] in an experiment, isolated

microtubules and infused electrical signals through one end. Overwhelmed by the reception of the emitted signal at the other end of the microtubules, they could ascertain the fact that microtubules can be looked upon and described as biological conducting wires, suitable for electrical pulses transmission. Therefore, microtubules play an important role in electrical activities within cells.

It is noteworthy that excitable systems usually appear as spatially extended systems, which as a result of the application of a super-threshold stimulus, are capable of generating and conducting nonlinear waves of excitations. These regenerative waves travel without decrement as a solitary propagating wave or wave train. In this case, subsequent to the excitation, the medium needs some time to recover its excitability. During this time interval called the refractory period, the medium cannot be excited again. Thus, based on these facts, our proposed electrical model of microtubule should be consistent with these features of excitable media. In an attempt to do so, we suggest below in Fig. 1 an electrical circuit model for the microtubules, with a variety of specifications, including nonlinear resistive elements. In this model, we focus mainly on the voltage stimulus V_n urging cations (calcium ions or currents) within and between microtubule protofilaments.

The mathematical description of the model at hand is based on Kirchoff's network rules. The microtubule, being assimilated to a succession of N adjacent unit cells, depicts a transmission line. This transmission line then portrays one protofilament of a microtubule. Each unit cell number, n, of the lattice is made up of a linear inductance, L, in series with a linear resistance r, shunted by a nonlinear capacitance $C(V_n)$ in parallel with a nonlinear resistance R_n. The proposed circuit model combines two models where on one hand the $C(V_n)$ characteristics is a nonlinear function of the voltage,[15] and on the other hand, the resistance is a nonlinear function of the voltage.[16] The characterizations of the origins of the current, capacity, and resistance are well described in Ref. 15.

However, these models were better suited to explore the ability of microtubules to act as transmission lines supporting wave amplification phenomena. They could not display properties of excitability which have been experimentally shown to exist in microtubules.[17–26] The role played by microtubules in neuronal excitability had been proposed several years ago by Matsumoto and Sakai, based on experiment using squid giant axons which resulted in the direct correlation between excitability and microtubule assembly.[17–19] In fact, the excitability of the axon membrane, once suppressed by intracellular perfusion with solutions unfavourable for microtubule assembly, can be restored by perfusing the axon

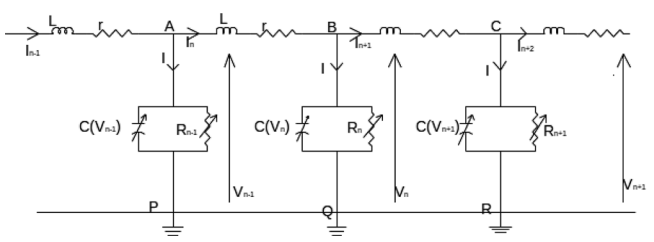

FIG. 1. The electrical circuit model of microtubule.

with a solution containing tubulin and its polymerizing reagents.[19,25] In Ref. 25, Terakawa *et al.*, following experimental observations, suggested that the submembranous cytoskeletons, including microtubules, provide an important mechanical support to the excitable membrane but are not essential elements in channel activities. Also, Ichikawa *et al.* artificially controlled the physiological conditions of axonal microtubules in squid giant axons under an optical microscope by perfusing them intracellularly with artificial solutions and by clamping the membrane voltage to measure membrane currents.[26] Their observations enabled them to conclude that axonal microtubules determine the generation and maintenance of membrane excitability in squid giant axons. In the same line, Braak *et al.*[20] reported from experimental studies that the microtubule-associated protein Tau is implicated in neurodegenerative disorders and described briefly how the Tau protein may modulate neuronal excitability by interacting with the microtubules at the axon initial segment. The authors in Ref. 9 have hypothesized that microtubules may exhibit some features of excitable media as they support the creation and propagation of some sort of solitonic ionic waves along their axis. In the context of their analysis, the origin of the nonlinearity in microtubules stems from the nonlinear capacitance of each tubulin dimer. Similarly, in this work, it is keynote that the polynomial form of the nonlinearity we use here, as shown in Eq. (2), stems from its electrical charge and is well known to be appropriate in excitable neural models,[31] in replacement of the previously used logarithmic form which has been shown to have some limitation in mimicking enough the experimentally observed physiological features of excitability of these cellular systems.

Applying the current rule to the circuit at points A and B of the circuit, and the voltage rule to the loops BCRQB and ABQPA, where the entire transmission line represents a protofilament of a microtubule and V_n denotes the voltage stimulus of the calcium ions (cations), we obtain the equation

$$V_{n+1} - 2V_n + V_{n-1} = L\frac{d^2Q_n}{dt^2} + 3B_1LV_n{}^2\frac{dV_n}{dt} - B_2L\frac{dV_n}{dt} + r\frac{dQ_n}{dt} + rB_1V_n{}^3 - rB_2V_n, \quad (1)$$

where Q_n denotes the quantity of charge that flows through the n_{th} unit cell (i.e., the number of calcium ions driven across the n_{th} unit cell multiplied by the quantity of charge in one calcium ion). $C(V_n)$ denotes the nonlinear capacitance element, which represents quantity of charge urged across the n_{th} unit cell per unit voltage. $C(V_n)$ possesses an electrical charge, Q_n, which takes the form

$$Q_n = C_0(1 - \alpha V_n - \beta V_n{}^2)V_n, \quad (2)$$

obtained by using the form of the charge proposed in Ref. 31.

Here, C_0 is the capacitance in the linear regime and α and β are positive parameters that account for non-linearity in the charge.

The nonlinear resistance effects are characterized by the relation

$$I_{R_n} = f(V_n) = B_1 {V_n}^3 - B_2 V_n, \tag{3}$$

that is, I_{R_n} is the current flowing through the non-linear resistance R_n; B_2 is a positive parameter that accounts for linearity in the current; and B_1 is a positive parameter that accounts for non-linearity in the current.[16] Equation (3) is approximately the I-V relationship of the nonlinear resistance and the nonlinear differential equation [Eq. (1)] becomes

$$\begin{aligned}&\ddot{V}_n - \Omega^2 V_n + (\gamma_0 + \gamma_1 V_n + \gamma_2 V_n^2)\dot{V}_n + (\eta_1 + \delta V_n)\dot{V}_n^2 \\ &\quad + (\eta_1 V_n + \lambda {V_n}^2)\ddot{V}_n + \eta_2 V_n^3 = D_0(V_{n+1} - 2V_n + V_{n-1}),\end{aligned} \tag{4}$$

where

$$\Omega^2 = \frac{rB_2}{LC_0}, \quad \gamma_0 = \frac{r}{L} - \frac{B_2}{C_0}, \quad \gamma_1 = -2\frac{\alpha r}{L}, \quad \gamma_2 = 3\left(\frac{B_1}{C_0} - \frac{\beta r}{L}\right),$$
$$\eta_1 = -2\alpha, \quad \delta = -6\beta, \quad \lambda = -3\beta, \quad \eta_2 = \frac{rB_1}{LC_0}, \quad D_0 = \frac{1}{LC_0}.$$

Equation (4) describes the spatiotemporal evolution of the voltage $V_n(t)$. The terms containing B_1 and B_2 are the contribution of the ionic flow. The terms with r represent the contribution of the linear resistance r. After the equality sign, we have the coupling entity between nearest neighbors $(n-1)$, n, $(n+1)$ of the dimer lattice. The system is dimensionless and it is needless to insist on space and time units. An approximate solution to Eq. (4) can be obtained using perturbation methods. In order to do so, we introduce the variables

$$V_n = \epsilon \phi_n,$$

where $\epsilon \ll 1$.

We consider that γ_0 is a perturbed parameters of order ϵ^2. This perturbation arises from the fact that we are looking for solutions in a weakly dissipative medium. Keeping the first two nonlinear terms of the development, the equation becomes

$$\begin{aligned}&\ddot{\phi}_n - \Omega^2 \phi_n + \epsilon(\epsilon\gamma_0 + \gamma_1 \phi_n + \epsilon\gamma_2 \phi_n^2)\dot{\phi}_n + (\epsilon\eta_1 + \epsilon^2 \delta\phi_n)\dot{\phi}_n^2 \\ &\quad + (\epsilon\eta_1 \phi_n + \epsilon^2 \lambda \phi_n^2)\ddot{\phi}_n + \epsilon^2 \eta_2 \phi_n^3 = D_0(\phi_{n+1} - 2\phi_n + \phi_{n-1}).\end{aligned} \tag{5}$$

The resolution of Eq. (4) reveals the nature of the voltage signal within and between dimer molecules. However, its solution is obtained through the use of Eq. (5).

It is keynote to mention that several authors investigated the mechanical and electrical features of microtubules.[6–10,13–28,30,32–35] Indeed, due to their electrical properties, microtubules are known to play important roles in many cellular functions including motility, mitosis, transport, and the maintenance of cell shape by forming cytoskeletons.[30,32,33] Nerve tissues contain a large amount of tubulin and microtubules, and experimental evidence has been accumulated in support of the view that neuronal microtubules are directly involved in axonal transport.[34,35] Motivated by these facts, Sekulic *et al.*[32] suggested that microtubules (by for instance the collective dynamics of the flexible C-terminal tails decorating its outer surface) could crucially act as a mechanical support of localized waves aimed for communication between microtubules and associated proteins.

B. The wave amplitude equation

At this stage, let us reiterate the fact that our major concern is to investigate for the existence of low-amplitudes non-linear ionic waves in a microtubule. For this purpose, it is convenient to use the multiple scale expansion method which can be applied with a semi-discrete approximation in order to find the solution to our differential equation. The semi-discrete approximation is a perturbation technique in which the carrier waves are kept discrete while the amplitude is treated in the continuum limit. Applying this method allows one to study the modulation of a plane wave caused by nonlinear effects. We proceed by making a change of variables according to the new time and space scales $U_n = \epsilon^n u$ and $T_n = \epsilon^n t$. It should be noted that the principal purpose of using the multiple scale expansion is to find a solution $V(u, t)$ depending on these new sets of variables as a perturbation series of functions. We will consider here that

$$V(u,t) = \sum\nolimits_{n=1}^{\infty} \epsilon^n \phi_n(U_0, U_1, U_2, \ldots, T_0, T_1, T_2, \ldots), \tag{6}$$

where each U_n or T_n is treated as an independent variable. Thus, we also obtain a perturbation series of operators from both independent variables. The temporal derivatives can be expressed as

$$\frac{\partial}{\partial t} = \frac{\partial T_0}{\partial t}\frac{\partial}{\partial T_0} + \frac{\partial T_1}{\partial t}\frac{\partial}{\partial T_1} + \frac{\partial T_2}{\partial t}\frac{\partial}{\partial T_2} + \cdots. \tag{7}$$

Similarly, the spatial derivatives can be expressed as

$$\frac{\partial}{\partial u} = \frac{\partial U_0}{\partial u}\frac{\partial}{\partial U_0} + \frac{\partial U_1}{\partial u}\frac{\partial}{\partial U_1} + \frac{\partial U_2}{\partial u}\frac{\partial}{\partial U_2} + \cdots. \tag{8}$$

An important feature of this method is that the solution of the original problem will only be obtained if the multidimensional space generated by the new set of variables U_n and T_n comes from the physical line.[11] Worth noting is the fact that these auxiliary variables permit us to impose appropriate conditions to the system thus assuring that the asymptotic expansion converges uniformly for small values of ϵ.[39]

In order to solve the equation of motion of the voltage stimulus, we consider the following form of solution (ansatz) for the dynamics of our discrete electrical circuit model of microtubule:

$$\begin{aligned}\phi_n = {} & F_n e^{j\theta_n} + F_n^* e^{-j\theta_n} + \epsilon(G_n + H_n e^{2j\theta_n} + H_n^* e^{-2j\theta_n}) \\ & + O(\epsilon^2),\end{aligned} \tag{9}$$

with $\theta_n = qn - wt$, where q is the normal mode wave vector, n denotes the unit cell number in the dimer lattice, and w is the angular velocity of the wave.

The fact that there are nonlinear terms in Eq. (5) leads one into the prediction that through frequency superpositions, the first harmonics of the wave will contain terms in

$e^{\pm 2j\theta_n}$ as well as terms without any exponential dependence. We will equally consider that the amplitudes F, G, and H change slowly in space and time. This is formalized in a multiple scale expansion in a method referred to as semi-discrete approximation because it preserves a carrier wave which obeys a discrete dispersion relation and only the variation of the amplitude is treated in the continuum limit.

In this semi-discrete approximation, F, G, and H are set to be independent of the "fast" variables t and u. but to depend on the "slow" variables $U_1 = \epsilon u$, $U_2 = \epsilon^2 u$, $T_1 = \epsilon t$, and $T_2 = \epsilon^2 t$. Applying now the continuum limit approximation on these amplitudes will yield $F_n(t)$, $G_n(t)$, and $H_n(t)$ becoming $F(U_1, U_2, T_1, T_2)$, $G(U_1, U_2, T_1, T_2)$, and $H(U_1,U_2,T_1,T_2)$, respectively. $F_{n\pm1}$ is obtained up to ϵ^2 by a Taylor expansion

$$F_{n\pm1} = F \pm \epsilon \frac{\partial F}{\partial U_1} \pm \epsilon^2 \frac{\partial F}{\partial U_2} + \frac{\epsilon^2}{2}\frac{\partial^2 F}{\partial U_1^2} + O(\epsilon^3).$$

Similar equations are written for G_n and H_n. These together with Eq. (9) are inserted into the two equations depicting our model, that is, Eq. (5). We seek relations for which terms in $e^{\pm j\theta_n}$, $e^{\pm 2j\theta_n}$ and terms without exponential dependence cancel out at different orders of ϵ (at different orders of perturbation).

At the zeroth order of perturbation, we obtain the dispersion relation of linear waves of the system described by Eq. (5)

$$w^2 = -\Omega^2 + 4D_0 \sin^2\frac{q}{2}. \tag{10}$$

By definition, the coefficients, B_1, B_2, r, L, and C_0 are positive constants. Therefore, Eq. (10) is valid if the wave number satisfies the condition[16]

$$q \geq q_c = 2\arcsin\sqrt{\frac{\Omega^2}{4D_0}}. \tag{11}$$

It follows from Eq. (11) that the system is a low band pass filter with the allowed band defined by the frequency interval $[0, \omega_c]$, where $\omega_c^2 = -\Omega^2 + 4D_0$ is the square of the upper cut-off frequency.[16] The dispersion relation, Eq. (10), is indeed that of a discrete model and thus meets up with expectations. Figure 2 presents the graph of this dispersion relation. It depicts a scenario whereby there are limited intervals of values of the wave number which are suitable for the existence of linear waves within the medium. These epochs of values of q suitable for existence of linear waves are regularly separated by forbidden intervals of values of the wave number. These forbidden gaps then naturally indicate that linear waves are not accessible for the system for some given values of q.

At the first and second orders of perturbation, we obtain the coefficient

$$G = 0, \tag{12}$$

the relation

$$H = -\left[\frac{\omega D_0(2\omega\eta_1 + j\gamma_1)}{D_0(\Omega^2 + 2\omega^2) + 2(\omega^4 - 2\omega^2\Omega^2 + \Omega^4)}\right]F^2, \tag{13}$$

the group velocity of the ionic waves conveyed by our model

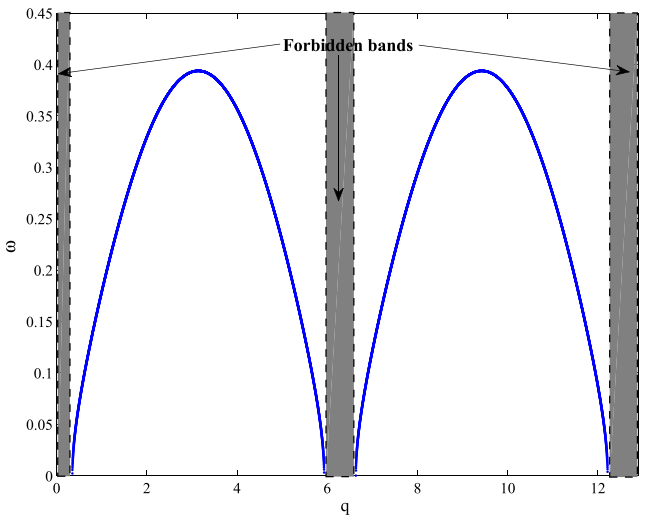

FIG. 2. The dispersion relation of the ionic wave. $D_0 = 0.04$ and $\Omega^2 = 0.0049$.

$$v_g = \frac{D_0 \sin q}{w}, \tag{14}$$

and at higher order, the equation of evolution of the solitonic function

$$j\frac{\partial F}{\partial \tau_2} + \frac{P}{2}\frac{\partial^2 F}{\partial \xi_1^2} + Q|F|^2 F + j\frac{R}{2}F = 0, \tag{15}$$

where we have used the rescaling (change of variables)

$$\xi_n = U_n - v_g T_n, \quad \tau_n = T_n. \tag{16}$$

Equation (15) shows that the evolution of ionic waves in this electrical circuit model of microtubules is described by the Complex Ginzburg-Landau equation (CGLE) where the coefficients of dispersion and dissipation, P and R, respectively, are real, while the coefficient of nonlinearity, Q, is complex. It can be seen that the wave propagates in a frame of reference that travels with velocity, v_g (the group velocity), where

$$P = \frac{D_0 w^2 \cos q - D_0^2 \sin^2 q}{w^3}, \tag{17}$$

$$Q = Q_r + jQ_i, \tag{18}$$

$$R = \gamma_0, \tag{19}$$

where Q_r and Q_i are the real and imaginary parts of the nonlinearity coefficient, respectively, such that

$$Q_r = \frac{\omega D_0}{2}\left(\frac{2\omega^2{\eta_1}^2 - 3{\gamma_1}^2}{D_0(\Omega^2 + 2\omega^2) + 2(\omega^4 - 2\omega^2\Omega^2 + \Omega^4)}\right) - \frac{3\eta_2 + \omega(\delta - 3\lambda)}{2\omega}, \tag{20}$$

$$Q_i = \frac{\omega D_0}{2}\left(\frac{7\omega\gamma_1\eta_1}{D_0(\Omega^2 + 2\omega^2) + 2(\omega^4 - 2\omega^2\Omega^2 + \Omega^4)}\right) + \frac{\gamma_2}{2}. \tag{21}$$

The variation of the coefficient of dispersion P, the real and imaginary parts of the coefficient of nonlinearity Q_r and

Q_i, and the product PQ_r, as functions of the wave number q, are plotted in Fig. 3.

When the dissipation is not considered in some cases, the Complex Ginzburg-Landau equation reduces to the Nonlinear Schrödinger equation.[40] However, that is not the case in this model because if the dissipation is neglected, the nonlinearity coefficient will still have complex and imaginary parts. Therefore, our amplitude equation will always be described by the Complex Ginzburg-Landau equation.

The dispersion coefficient is real. It follows that the modulational instability depends on the sign of PQ_r. According to Benjamin-Feir instability analysis, plane waves are unstable for positive values of PQ_r whilst they are stable for negative values. Note that this stability criterion does not depend on the manner with which the wave propagates.

In Fig. 3(a), it is observed that the dispersion coefficient P is always negative. Also, in Figs. 3(b) and 3(c), we see that the real and imaginary parts of the coefficient of nonlinearity are always negative and positive, respectively, irrespective of the accessible values of q. In accordance with these observations, as expected from the outcomes of Figs. 3(a) and 3(b), Fig. 3(d) illustrates the fact that the product PQ_r is permanently positive. This suggests that in line with the Benjamin-Feir instability criteria, plane waves' solutions of our biological system are unconditionally unstable. As a result of this, our model of microtubule steadily exhibits a tendency to promote the generation of modulated amplitude waves along the structure of the subcellular lattice. Thus, based on these facts, our biological model depicts an environment purposely conducive for the existence of stable solitary waves, as solutions of our dynamical system. The determination of these soliton solutions in this proteinic network model shall be the core of Secs. III–VI. First of all, in order to be relatively complete in this survey, Sec. III will be dedicated to analytically ascertain this startling predilection to excitability by the means of the Eckhaus instability analysis of the plane waves solutions of our *CGLE*.

III. NONLINEAR ANALYSIS OF THE ECKHAUS INSTABILITY OF THE MODEL

The complex Ginzburg-Landau equation that describes our model is given in Eq. (15).

A plane wave is a field of energy that propagates with constant frequency and amplitude. An amplitude modulated wave is a wave whose amplitude varies spatiotemporally, while a phase modulated wave is a wave whose phase changes spatiotemporally. In order to study the stability of our wave amplitude equation, we use the following ansatz for our saturated amplitude modulations:[44,45]

$$F(\xi_1, t) = a(z)e^{j[\phi(z)+\omega_1 t]}, \tag{22}$$

where a and ϕ are real-valued functions of $z = \xi_1 - v_g\tau_2$. Substituting the ansatz in Eq. (22) into Eq. (15), separating the real and imaginary parts and setting $b = a_z$ and $\psi = \phi_z$ yield the set of three coupled nonlinear ordinary differential equations (ODEs)

$$\begin{cases} a_z = b \\ b_z = \dfrac{2\omega}{P}a + \dfrac{2}{P}a\psi + a\psi^2 - \dfrac{2}{P}a^3Q_r \\ \psi_z = \dfrac{2v_g}{P}\dfrac{b}{a} - \dfrac{R}{P} - 2\dfrac{b\psi}{a} - \dfrac{2Q_i}{P}a^2. \end{cases} \tag{23}$$

Therefore, the system of Eq. (23) describes the dynamics of amplitude-modulated waves in our electrical circuit model of microtubules.

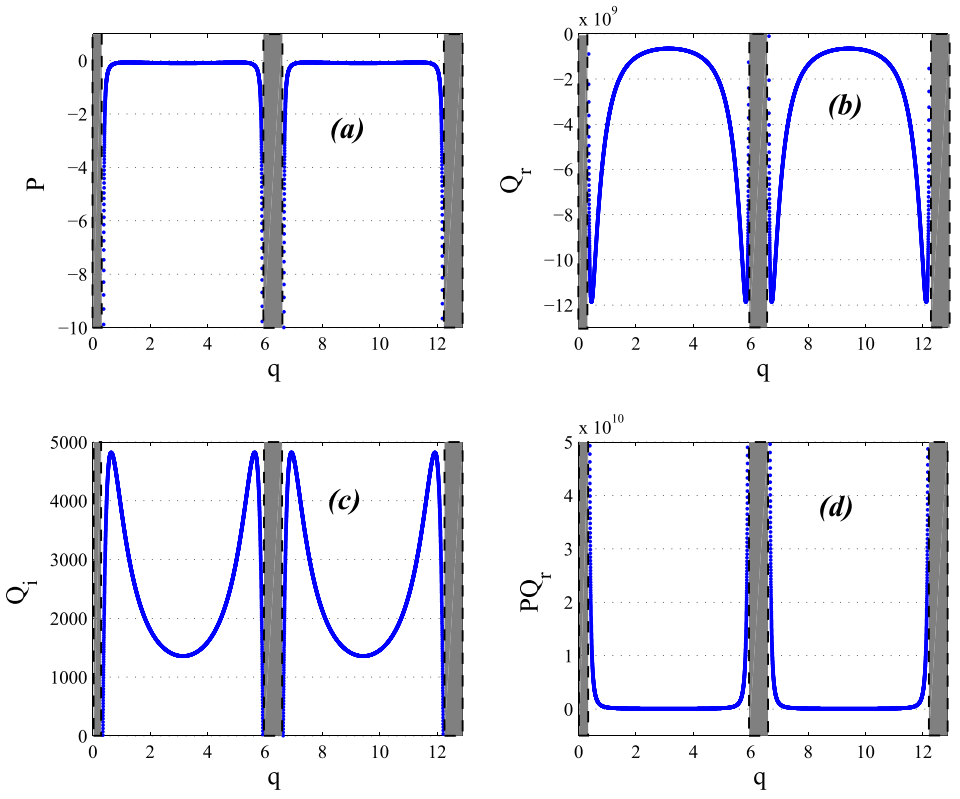

FIG. 3. Variations of the coefficient of dispersion P, the real and imaginary parts of the coefficient of nonlinearity Q_r and Q_i, and the product PQ_r, as functions of the wave number q. The parametric values are $D_0 = 0.04$, $\Omega^2 = 0.0049$, $\eta_1 = 0.09$, $\eta_2 = 0.005$, $\gamma_1 = 40\,000$, $\gamma_2 = 5.0$, $\lambda = 0.05$, and $\delta = 0.8$.

We proceed to perform a stability analysis of the fixed points of our model. Since the equilibria of dynamical systems are not always stable, it behooves us to characterize the stability of the fixed points pertaining to our model due to the fact that stable and unstable equilibria play quite different roles in the dynamics of a system. If the fixed points are stable, then the plane waves become stable; if the fixed points are unstable, the plane waves become unstable. The equilibrium points are determined by setting all gradients to zero, that is, $a_z = b_z = \psi_z = 0$, and, taking only positive square roots, we find that the procedure starts from the fixed point $(a,\ b,\ \psi_z) = (\sqrt{\frac{R}{2Q_i}}, 0, \frac{1}{p}[-1 + \sqrt{1 + P\left(\frac{Q_r}{Q_i}R - 2\omega\right)}])$, which corresponds to a plane wave solution. In a bid to characterize the stability of the fixed point, we obtain the characteristic polynomial

$$a_0\lambda^3 + a_1\lambda^2 + a_2\lambda + a_3 = 0, \tag{24}$$

where

$$\begin{aligned} &a_0 = 1, \quad a_1 = 0, \\ &a_2 = 4\psi^2 + \frac{4\psi}{P} + \frac{6Q_r}{P} - \frac{4v_g\psi}{P} - \frac{4v_g}{P^2} - \psi - \frac{2\psi}{P} - \frac{2\omega}{P}, \\ &a_3 = \frac{4av_g\psi}{P} + \frac{4av_g}{P^2} - 4a\psi^2 - \frac{4a\psi}{P}. \end{aligned} \tag{25}$$

For the fixed point to be stable, all the eigenvalues, λ_k for $k = 1$, 2, and 3 of Eq. (24) must have negative real parts. Using the *Routh-Hurwitz stability criteria*, the necessary and sufficient conditions for stability to occur are

$$a_1 > 0, \quad a_3 > 0, \quad \textit{and} \quad a_1a_2 > a_3. \tag{26}$$

The simultaneous fulfillment of these criteria implies that all the eigenvalues of the characteristic equation have negative real parts, synonym of convergence of our dynamical system, following some perturbation deeds, toward the plane waves solutions. Yet, from Eq. (26), $a_1 = 0$. Then, this implies that the Routh-Hurwitz stability criteria can never be concomitantly fulfilled. Thus, the plane waves solutions of our *CGLE* are always unstable. This fact is in perfect agreement with the results obtained numerically in Fig. 3(d) in Sec. II, using the Benjamin-Feir stability condition. Therefore, our *CGLE* admits modulated amplitude waves as only stable solutions. It then suitably describes the dynamics of our lattice of biochemical units as an excitable medium.

IV. NONLINEAR IONIC WAVE SOLITON SOLUTION

A. The Nozaki-Bekki soliton solution

A form of the soliton solution of the amplitude Eq. (15) can be obtained using the Nozaki and Bekki ansatz[36,38,40,41]

$$F(\xi_n, t) = \frac{F_0 e^{\varphi}}{1 + e^{(\varphi+\varphi^*)^{(1+j\alpha)}}}, \tag{27}$$

where

$$\varphi = q\xi_n - w\tau, \tag{28}$$

$$\alpha = \beta \pm (2 + \beta^2)^{\frac{1}{2}}, \tag{29}$$

$$\beta = \frac{3Q_r}{2Q_i}. \tag{30}$$

Now, rewriting the ansatz in Eq. (27) by considering the real and imaginary parts and transforming its coefficients following the process in Ref. 37, we arrive at the solution for the calcium wave propagation in the microtubule described by

$$\begin{aligned} V_n = {} & \epsilon F_0\left[\frac{e^{\varphi}\cos(\theta - 2\alpha\varphi) + e^{-\varphi}\cos\theta}{(\cosh 2\varphi + \cos 2\alpha\varphi)}\right] + \epsilon^2 F_0^2\left[(x\cos 2\theta\right. \\ & \left. - y\sin 2\theta)\left(\frac{e^{2\varphi}\cos 4\alpha\varphi + 2\cos 2\alpha\varphi + e^{-2\varphi}}{2(\cosh 2\varphi + \cos 2\alpha\varphi)^2}\right)\right] \\ & - \epsilon F_0^2\left[(x\sin 2\theta - y\cos 2\theta)\left(\frac{e^{2\varphi}\sin 4\alpha\varphi - \sin 2\alpha\varphi}{(\cosh 2\varphi + \cos 2\alpha\varphi)^2}\right)\right], \end{aligned} \tag{31}$$

where

$$x = -\frac{2\omega^2\eta_1 D_0}{D_0(\Omega^2 + 2\omega^2) + 2(\omega^4 - 2\omega^2\Omega^2 + \Omega^4)}, \tag{32}$$

$$y = -\frac{\omega\gamma_1 D_0}{D_0(\Omega^2 + 2\omega^2) + 2(\omega^4 - 2\omega^2\Omega^2 + \Omega^4)}. \tag{33}$$

Figure 4 portrays the depiction and behavior this ionic wave in all the computational domains. From Figs. 4(a)–4(c), It is clearly noticed that this analytical solution emerges in the form of a single asymmetric pulse, propagating along the medium. These figures show the three-dimensional spatiotemporal evolution of the travelling ionic impulse [Fig. 4(a)], its upper view [Fig. 4(b)], and its profiles at different times [Fig. 4(c)], respectively. The snapshots of the lattice at several distinct times [as seen in Fig. 4(c)] reveal the fact that the pulse is structurally unstable, as it changes its shape in the course of its propagation through the lattice. Indeed, this asymmetric solution is indicative of the presence of a bright envelope soliton propagating within the one-dimensional spatial array of biochemical units, and whose amplitude is correspondingly depicted in Fig. 4(d) (three-dimensional spatiotemporal evolution), Fig. 4(e) (upper view), and Fig. 4(f) (lattice snapshots at different times).

Equally, it is important to note that the parameter ϵ helps us to differentiate the orders of development of the solitonic pulse. Its effect on the ionic wave can be seen in Fig. 5. This figure represents the snapshots of the lattice structure at the time $t = 46$, for different values of the perturbation, namely, $\epsilon = 0.0001$ [Fig. 5(a)], $\epsilon = 0.001$ [Fig. 5(b)], $\epsilon = 0.01$ [Fig. 5(c)], and $\epsilon = 0.1$ [Fig. 5(d)]. We notice that as the perturbation increases the amplitude of the ionic wave also increases. A similar phenomenon has already been observed in other biological excitable media, namely, in diffusively connected Hindmarsh-Rose neural networks.[37] A generalization of this fact is depicted in Fig. 6.

Thus, the previously obtained outcomes show that information processing and transmission along the structure of a

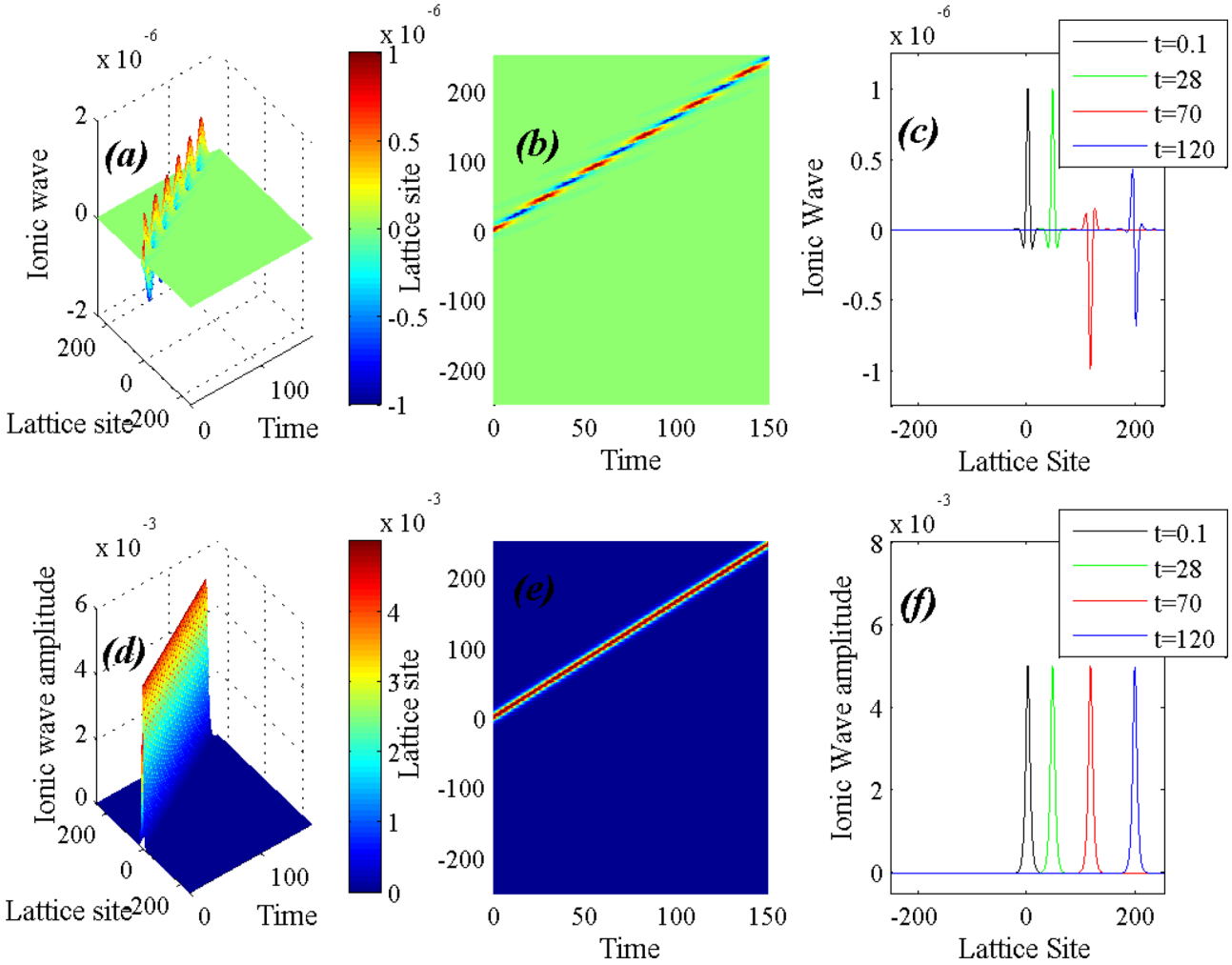

FIG. 4. Spatiotemporal evolution of the ionic wave and its corresponding envelope. The ionic wave is presented in (a) three-dimensional view of the travelling pulse, (b) upper view of the pulse, and (c) ionic wave profiles at different times. The corresponding envelope is depicted in (d) three-dimensional view of the propagating envelope, (e) upper view of the envelope, and (f) envelope profiles at different times. The parametric values are $D_0 = 0.04$, $\Omega^2 = 0.0049$, $\eta_1 = 0.09$, $\eta_2 = 0.005$, $\gamma_1 = 40\,000$, $\gamma_2 = 5.0$, $\lambda = 0.05$, $\delta = 0.8$, $q = 1.5$, $F_0 = 0.01$, and $\epsilon = 0.0001$.

microtubule are achievable by means of localized excitations. These solitary waves, because of their robustness and stability, are ideal tools to secure the transmission processes taking place along some relatively large arrays of biological dynamical systems. Their remarkable robustness is predominantly due to their regenerative feature. Yet, one of the challenges usually faced in communication processes is the ability to concomitantly convey not only safely but also quantitatively, a huge diversity of information. This implies the simultaneous transmission of a huge set of harmonics along our biochemical milieu. A typical means to achieve that is to relay the stimulus in the form of a wave packet. Subsequent to these facts, we propose ourselves to seek for wave packets solutions of our nonlinear equation of motion. In an attempt to meet this target, in Sec. IV B, we use the harmonic balance method to solve our wave amplitude equation and derive the expression of the ionic wave propagating in the model.

B. The symmetric solution using the harmonic balance method

Here, we seek exact solutions of the CGLE that pertain to our model. We use the harmonic balance method with a nonlinear transform to find solutions in terms of Fourier series.[42] This method, which is a technique used in determining the limit cycle of a dynamical system, will be utilized in this section to obtain the exact travelling solution of the equation describing the dynamics or evolution of our model (the CGLE). In this technique, we define and use a nonlinear transform $l(\xi)$ such that

$$\frac{dl}{d\xi} = f, \tag{34}$$

where $f = f(l) = f(l(\xi))$ is a periodic function of period 2π. Trigonometric functions, which are easier to handle, will be used here.

For our system, we choose

$$f(l) = \cos l. \tag{35}$$

Since the equation describing the evolution of our model, the complex Ginzburg-Landau equation (CGLE), is a complex equation, we consider an ansatz (solution) to it of the form:

$$F(\xi, t) = g(\xi) e^{j[s(\xi) + wt]}, \tag{36}$$

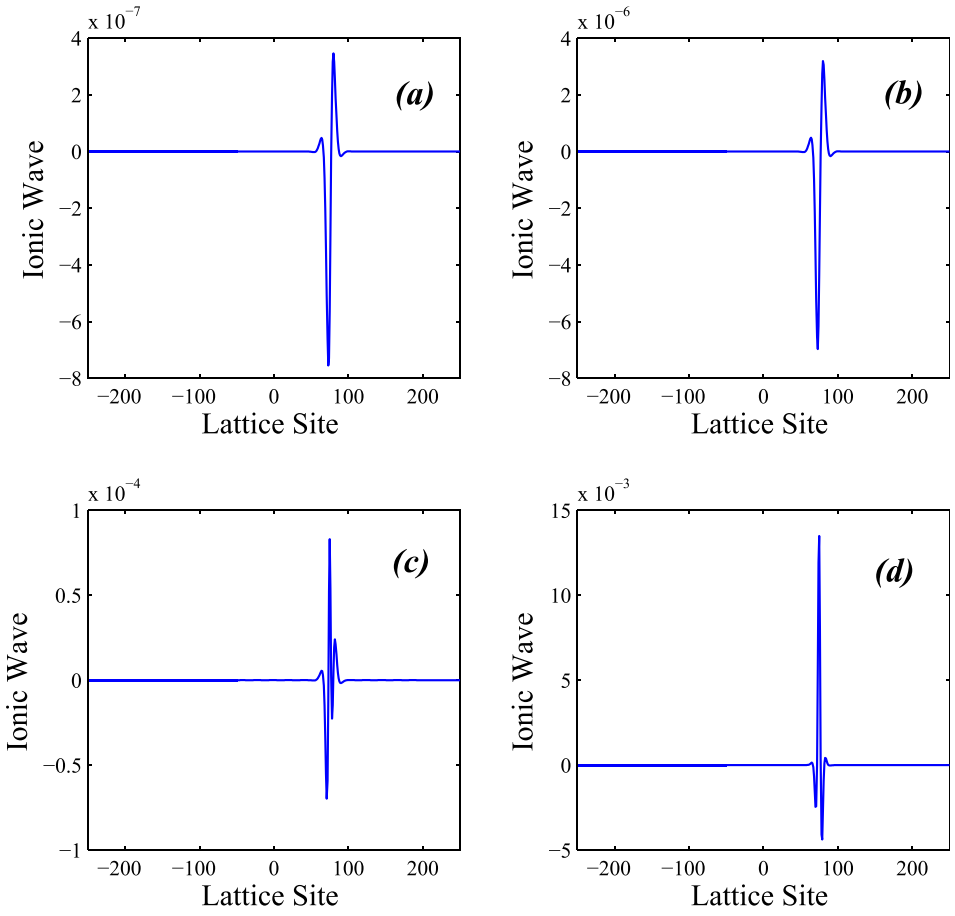

FIG. 5. Effects of small perturbations on the ionic wave. Snapshots of the wave profile taken at $t=46$. (a) $\epsilon=0.0001$, (b) $\epsilon=0.001$, (c) $\epsilon=0.01$, and (d) $\epsilon=0.1$. The parametric values are the same as in Fig. 4.

where $g(\xi)$ and $s(\xi)$ are real functions and $\xi=n_1u+n_2t$, n_1 and n_2 being constants.

Inserting Eq. (36) into Eq. (15) yields, for real and imaginary terms

$$g_{\xi\xi}-gs_\xi^2-n_2n_3gs_\xi+n_4g^3-n_5g=0, \tag{37}$$

$$gs_{\xi\xi}+2g_\xi s_\xi+n_2n_3g_\xi+n_6g^3+n_7g=0, \tag{38}$$

where

$$n_3=\frac{2}{Pn_1^2},\quad n_4=n_3Q_r,\quad n_5=wn_3,\quad n_6=n_3Q_i,\quad n_7=n_3\frac{R}{2}.$$

In order to find travelling wave solutions, we shall use the ansatz (solutions) suggested by Porubov and Velarde[43]

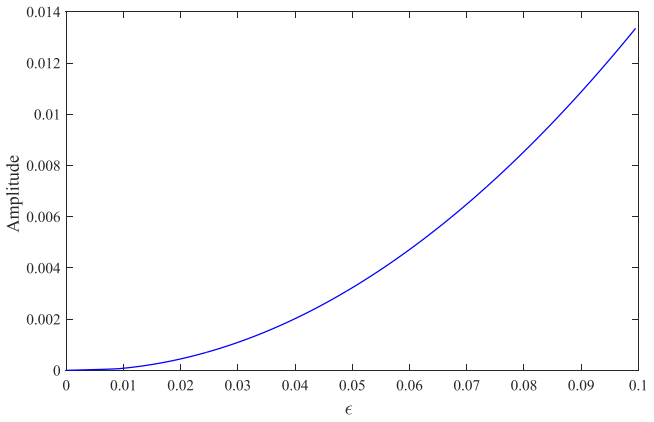

FIG. 6. Plot of the ionic wave amplitude as a function of the amplitude of the perturbation. The parametric values are the same as in Fig. 4.

$$R=g^2, \tag{39}$$

$$A=g^2s_\xi. \tag{40}$$

Inserting these in Eqs. (37) and (38) yield

$$\frac{1}{2}RR_{\xi\xi}-\frac{1}{4}R_\xi^2-A^2-n_2n_3RA+n_4R^3-n_5R^2=0, \tag{41}$$

$$A_\xi+\frac{1}{2}n_2n_3R_\xi+n_6R^2+n_7R=0. \tag{42}$$

The harmonic balance method is carried out by choosing a solution that utilizes terms which enhance a check at different orders of neglected harmonics. Thus, one needs to have an insight into the possible solution of the system in order to apply this method. The idea here is to express solutions in a periodic form at different harmonics. We therefore suppose that

$$R=r_0+s_1\sin l+r_2\cos 2l, \tag{43}$$

$$A=c_0+a_1\sin l+c_2\cos 2l+a_3\sin 3l. \tag{44}$$

In terms of l, we can show that

$$R_\xi=f\frac{dR}{dl},\quad A_\xi=f\frac{dA}{dl},\quad R_{\xi\xi}=f^2\frac{d^2R}{dl^2}+f\frac{df}{dl}\frac{dR}{dl}. \tag{45}$$

Revisiting our nonlinear transform Eq. (34) and Eq. (35), we obtain $d\xi=\frac{dl}{f(l)}\Rightarrow\xi=\int\frac{dl}{\cos l}$, which leads us to the identities

$$\sec h\xi = \cos l, \quad \textit{and} \tag{46}$$

$$\tanh\xi = \sin l. \tag{47}$$

We therefore obtain different equations at the different harmonics $\sin wl$ and $\cos yl$, where w is odd (1, 3, 5, etc.) and y is even (0, 2, 4, 6, etc.). The equations at the different harmonics are then solved to obtain expressions for the required coefficients in Eqs. (43) and (44).

From the above deductions, one obtains the factors of the Porubov and Velarde ansatz solution to the equation that describes our model as

$$R = 2r_2 \sec h^2\xi, \tag{48}$$

$$A = -\left(\frac{1}{2}\eta + \frac{2}{3}m\sin l\right)R - \frac{8}{3}mr_2\sin l, \tag{49}$$

where $n_6 r_2 = m$ and $n_2 n_3 = \eta$.

We therefore obtain the ionic wave or voltage stimulus in our transmission line or model as

$$\begin{aligned} V_n(u,t) = {} & \left(2\epsilon\sqrt{2r_2}\sec h\xi\right)\cos\left[\theta_n - \left(\frac{1}{2}\eta\xi\right)\right. \\ & \left. -\left(\frac{2}{3}m\right)\ln(\cosh\xi) - \left(\frac{1}{3}m\right)\cosh 2\xi + wt\right] \\ & + 4\epsilon^2 r_2 x\sec h^2\xi\cos\left[2\theta_n - \eta\xi - \left(\frac{4}{3}m\right)\ln(\cosh\xi)\right. \\ & \left. -\left(\frac{2}{3}m\right)\cosh 2\xi + 2wt\right] - 4\epsilon^2 r_2 y \\ & \times \sec h^2\xi\sin\left[2\theta_n - \eta\xi - \left(\frac{4}{3}m\right)\ln(\cosh\xi)\right. \\ & \left. -\left(\frac{2}{3}m\right)\cosh 2\xi + 2wt\right], \end{aligned} \tag{50}$$

where

$$x = -\frac{2\omega^2\eta_1 D_0}{D_0(\Omega^2 + 2\omega^2) + 2(\omega^4 - 2\omega^2\Omega^2 + \Omega^4)}, \tag{51}$$

$$y = -\frac{\omega\gamma_1 D_0}{D_0(\Omega^2 + 2\omega^2) + 2(\omega^4 - 2\omega^2\Omega^2 + \Omega^4)}, \tag{52}$$

$$\xi = n_1 u + n_2 t. \tag{53}$$

The voltage stimulus in Eq. (50) is that of the ionic wave propagated in our model of microtubules. This wave has its origin from the drifting of calcium ions from the region where they are more concentrated (region of higher potential) to the region where they are less concentrated (region of lower potential) within a microtubule protofilament between dimer molecules. An electric field is thus built up and is depicted by our transmission line or model.

Figure 7 presents this ionic wave on all the computational domains. It is observed that this analytical solution emerges in the form of a symmetric localized wave packet, travelling along the one-dimensional spatial subcellular network. Figures 7(a)–7(c) show its three-dimensional spatiotemporal evolution, its upper view, and its profiles at three different propagation times, respectively. This ionic wave is indicative of the existence and propagation along the array of biochemical systems of a symmetric bright envelope soliton. It is noteworthy that by definition, a bright soliton is a solution of a partial differential equation that is localized in space with the characteristic that it vanishes at infinity on the spatial domain.[39] Furthermore, in soliton theory, keeping in mind the concept of amplitude modulation of signal, it is well known that a carrier wave (wave front) can change its profile in the course of time since it can possibly be structurally unstable,[37,39,40] while its corresponding envelope preserves its shape. In other words, several profiles of a carrier

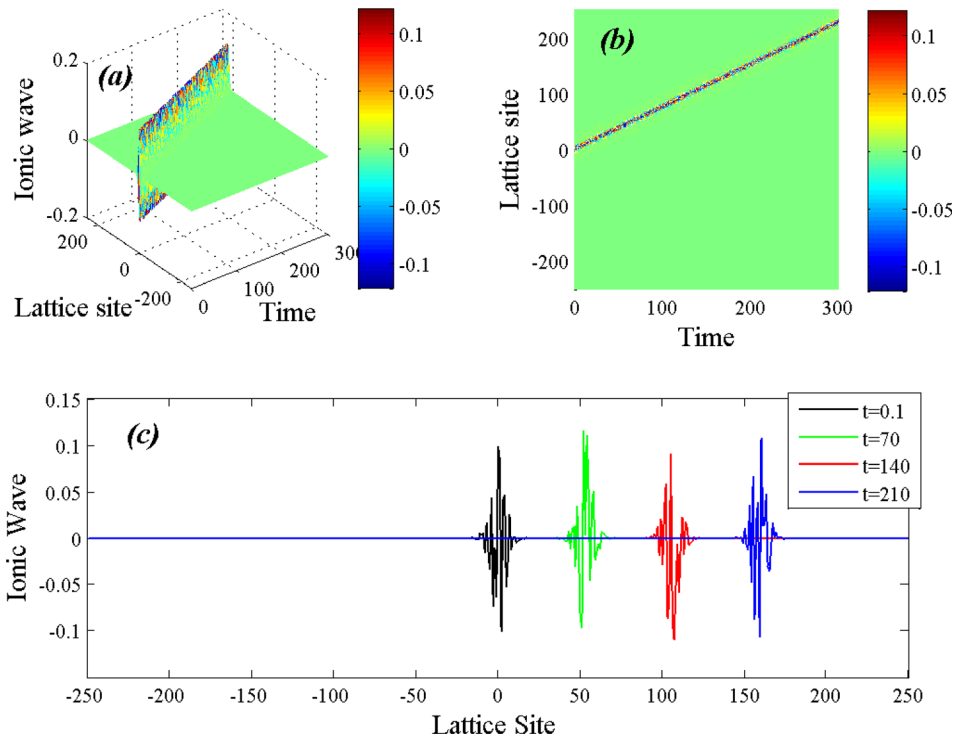

FIG. 7. Spatiotemporal evolution of the ionic wave in all the computational domains: (a) Three-dimensional view of the ionic wave, (b) upper view of the ionic wave, and (c) ionic wave profiles at different times. The signal appears in the form of a nearly symmetric wave packet. The parametric values are $D_0 = 0.04$, $\Omega^2 = 0.0049$, $\eta_1 = 0.09$, $\eta = 1.0$, $\gamma_1 = 40\,000$, $r_2 = 1.0$, $n_1 = 1.0$, $n_2 = 0.069441$, $q = 1.5$, and $\epsilon = 0.001$.

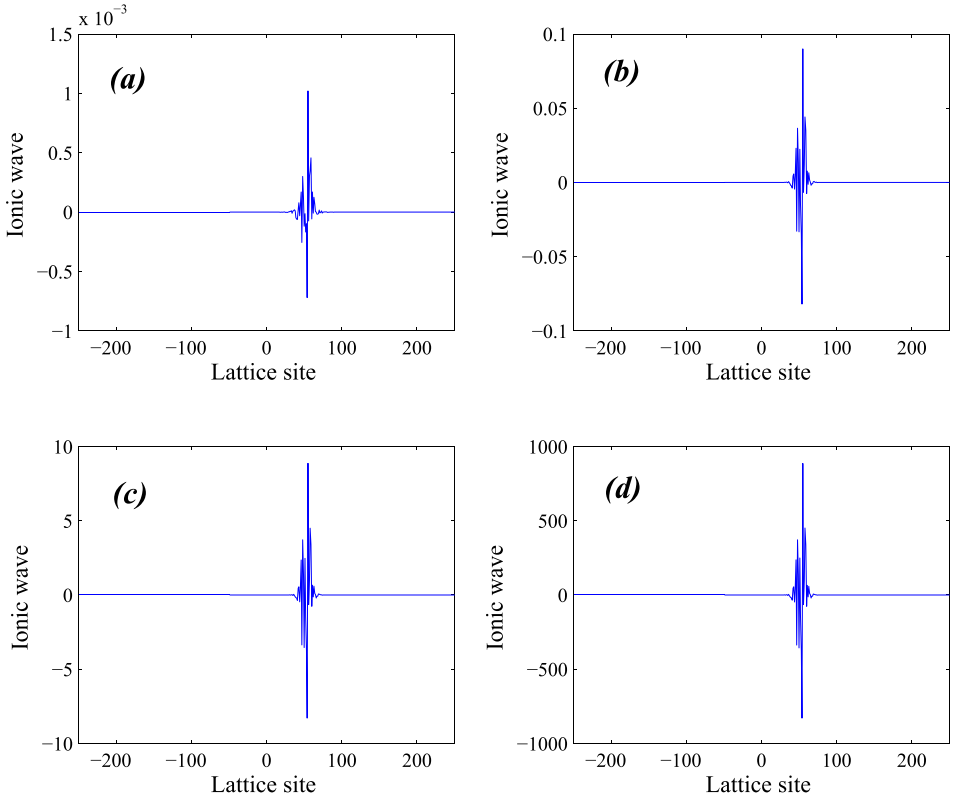

FIG. 8. Effects of small perturbations on the ionic wave. Snapshots of the wave profile taken at $t = 71$. (a) $\epsilon = 0.0001$, (b) $\epsilon = 0.001$, (c) $\epsilon = 0.01$, and (d) $\epsilon = 0.1$. The parametric values are the same as in Fig. 7.

wave can support the same envelope (or amplitude). Indeed, a close look at the wave front in Fig. 7 shows that the wave packet, in the course of its propagation, retains its width and amplitude. Therefore, its corresponding envelope naturally maintains its shape as it moves in the lattice.

The impact of the perturbation on the ionic wave can be seen in Fig. 8. This figure depicts the snapshots of the internal spatial state of the microtubule at the time $t = 71$, for different values of the perturbation, specifically, $\epsilon = 0.0001$ [Fig. 8(a)], $\epsilon = 0.001$ [Fig. 8(b)], $\epsilon = 0.01$ [Fig. 8(c)], and $\epsilon = 0.1$ [Fig. 8(d)]. We notice that as the perturbation increases, the amplitude of the ionic wave is tremendously enhanced. A similar trend has been reported in Sec. III in the case of the asymmetric ionic wave. A generalization of this fact is equally displayed in Fig. 9.

It therefore follows that, for very small perturbations of microtubules, impulses are exchanged between microtubules and equally amid their inner parts in the form of nearly symmetric envelope solitons. These impulses are steadily transmitted at a constant velocity with no change in the topology of the conduction medium. Microtubules play crucial roles in higher neuronal functions including mechanisms of consciousness and memory; and it is well known that some reasonable degree of disturbances may usually occur during the events of loss of consciousness and memory in living beings. In such scenarios, an increase in the amplitude of the wave is noticed. This is as a result of the energy released from the bursting of bubbles that are formed due to the agitations stirred up in the medium by the perturbations.

Unlike the results obtained with the asymmetric solitary wave in Subsection IV A above, Fig. 7(c) shows that this symmetric ionic wave propagates in the form of a train of localized pulses embedding a series of harmonics. Therefore, this solitary wave satisfactorily meets our expectations.

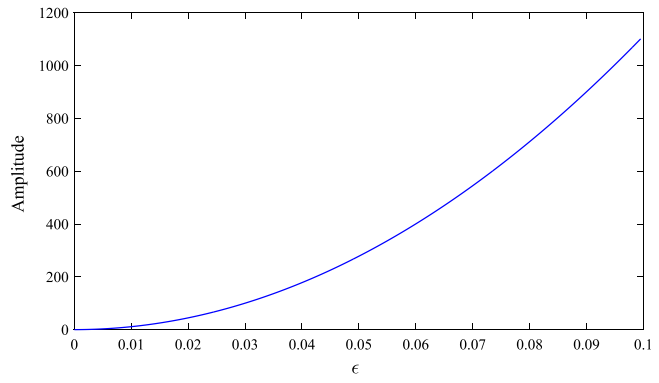

FIG. 9. Plot of the ionic wave amplitude as a function of the amplitude of the perturbation. The parametric values are the same as in Fig. 7.

V. NUMERICAL ANALYSIS OF THE VOLTAGE EQUATION: EXCITABILITY AND COLLISION

This section is solely dedicated to the direct numerical resolution of the original discrete equation of the voltage stimulus obtained in Sec. II and given by Eq. (4). The aim of this study is to check if our model of microtubule, in its full dimension, can equally display the features of excitability by supporting strongly nonlinear localized waves along its structure. For the purpose of this examination, we assume a huge partition of 100 electrical units in a one-dimensional spatial domain [–L, L], with a time step of 0.1. We numerically analyze our second order ordinary differential equation based on the fifth-order Runge-Kutta scheme. We take in this case $L = 50$, with periodic boundary conditions.

This initial value problem will be resolved while assuming as Ansatz, the exact analytical solution obtained in Sec. IV as Eq. (31). An inspection of the collective dynamical behavior of our electrical network enables us to witness several distinct wave phenomena occurring in our subcellular milieu within several parametric conditions.

In the upper part of Fig. 10, it is observed that after sending the impulse in the system, the structure preserves this excitation state permanently, with no event of propagation, for some given set of the system parametric values. This suggests that stationary localized pulses are accessible for our biological system. This intimates the existence of a stationary bright envelope soliton solution of our discrete voltage equation. Figures 10(a)–10(c) show the three-dimensional spatiotemporal display, the upper view, and the profile of this ionic wave at all times, respectively.

Interestingly, a further numerical study in other parametric conditions exhibits scenarios whereby, subsequent to the insertion of the Ansatz, it breaks down into a bi-pulse (that is two pulses) propagating in opposite directions within the lattice. This phenomenon is observed in both the lower part of Fig. 10 and in Figs. 11 and 12, for different lattice settings. Due to the periodic boundary conditions, these two pulses collide at the boundaries of the biological lattice and subsequent to this collision, several physical events may occur. First, Figs. 10(d)–10(f) disclose a situation where the two pulses merge into a non-trivial spatially homogeneous steady state for the biological ensemble. This stationary state is reminiscent of the phenomenon of oscillation quenching (observed in many physical and biological systems), alongside with the concept of synchronization as all the dynamical units states saturate at the same value of the amplitude of the ionic signal (voltage).

Second, in Fig. 11, with some additional boundary conditions, namely, the homogeneous Dirichlet boundary conditions, the two pulses after the collision, coalesce into a stable stationary dark soliton. Figures 11(a)–(11c) portray the three-dimensional spatiotemporal evolution of the bi-pulse propagation and collision, its upper view, and the profiles of the ionic wave at different times, respectively. Figures

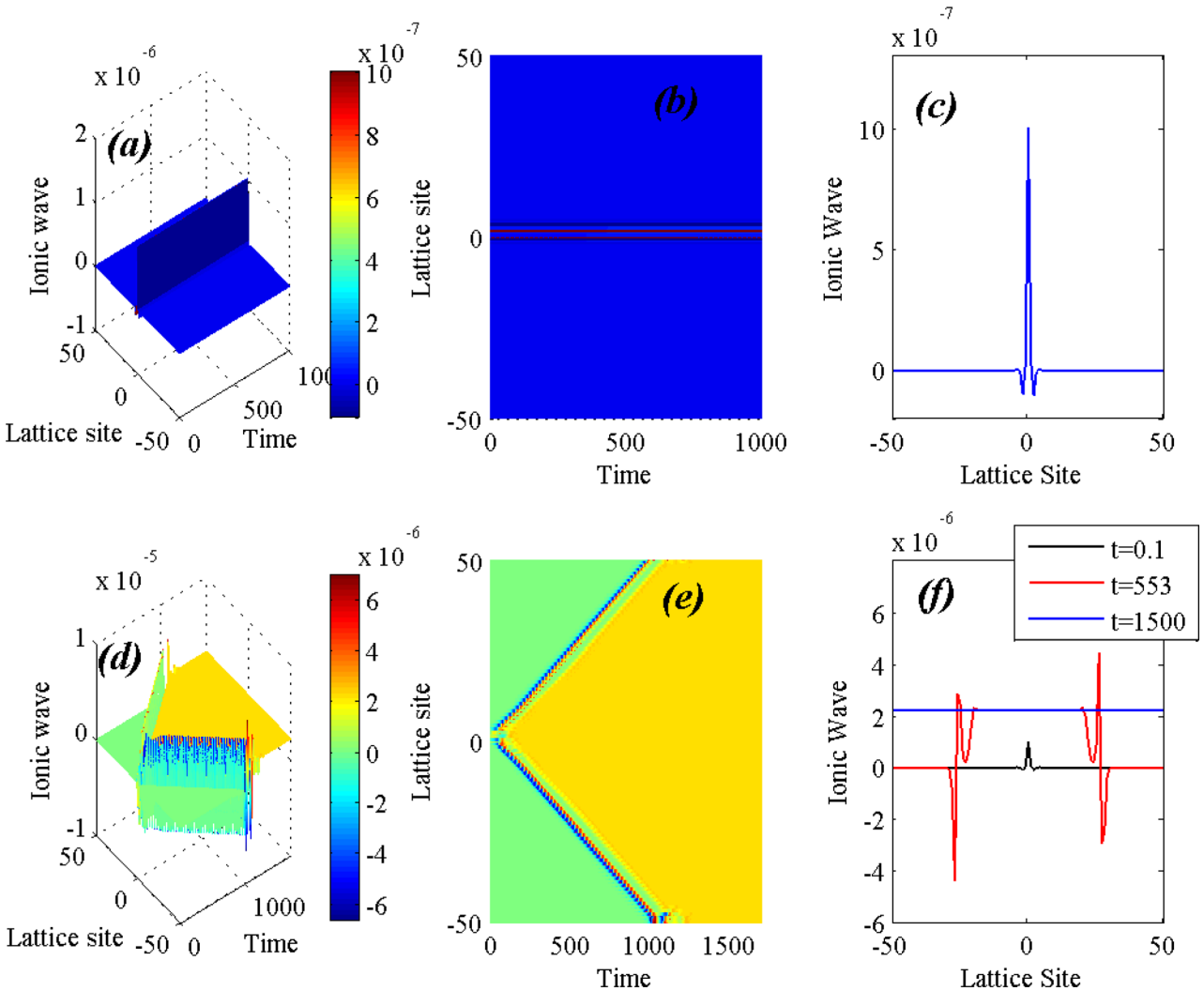

FIG. 10. Ionic waves solutions of the discrete voltage equation (4) in two different parametric conditions. The *upper figures* depict a stationary localized pulse. (a) Three-dimensional view, (b) upper view, and (c) profile at all times. The parametric values are $\gamma_0 = 65.0$, $\Omega^2 = 0.0049$, $\eta_1 = 0.09$, $\eta_2 = 0.005$, $\gamma_1 = 40000$, $\gamma_2 = 5.0$, $\delta = 0.1$, $\lambda = -1.0$, and $D_0 = 0.0024$. The *lower figures* depict a Bi-pulse solution travelling in opposite directions, yielding after collision a non-trivial spatially homogeneous stationary pattern. (d) Three-dimensional view, (e) upper view, and (f) profiles of the wave at different times. The parametric values are $\gamma_0 = 0.0$, $\Omega^2 = 0.0049$, $\eta_1 = 0.09$, $\eta_2 = 10^9$, $\gamma_1 = 40\,000$, $\gamma_2 = 5.0$, $\delta = 0.1$, $\lambda = -1.0$, and $D_0 = 0.0024$.

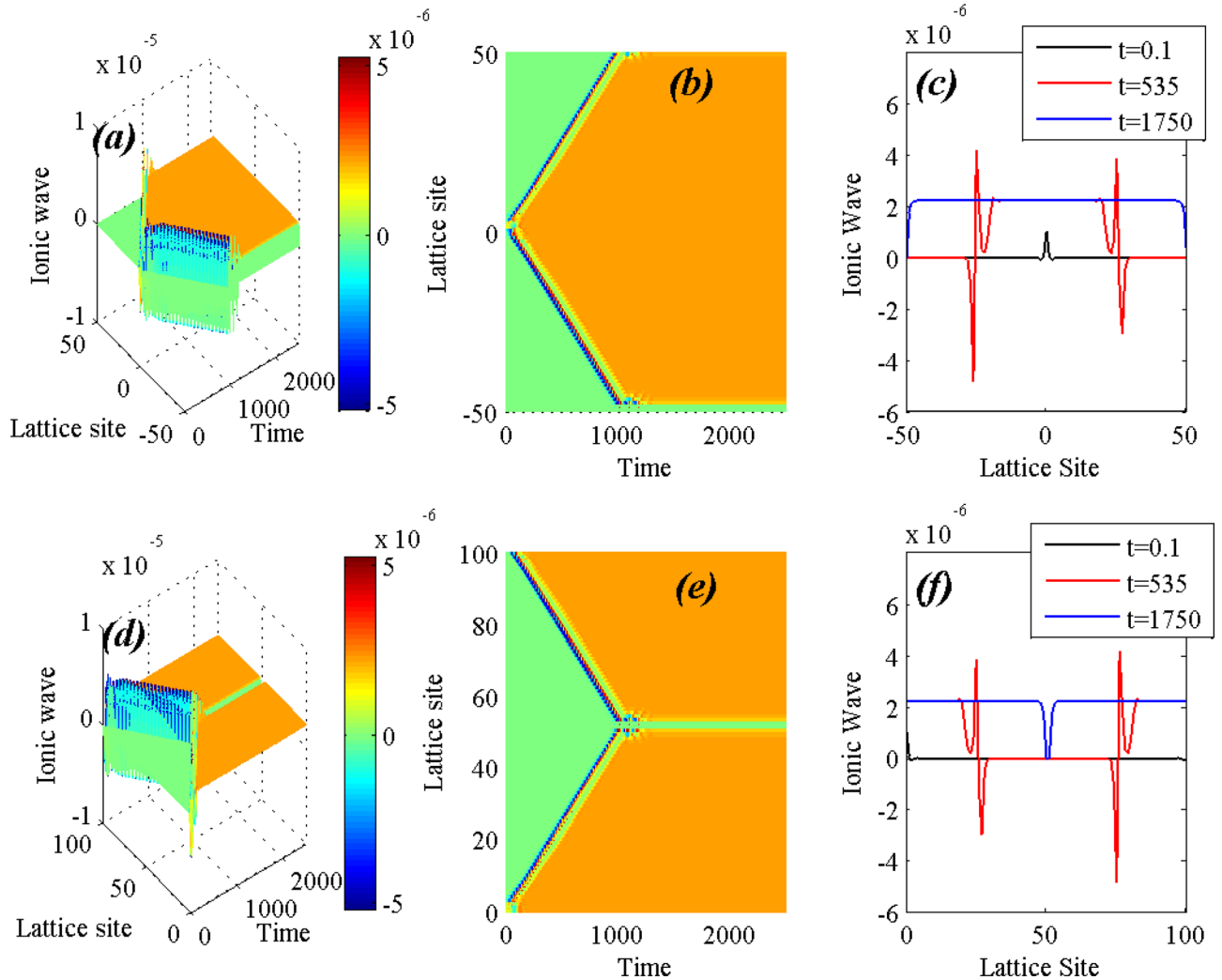

FIG. 11. Ionic waves solutions of the discrete voltage equation (4). The *upper figures* show a Bi-pulse solution travelling in opposite directions, yielding after collision, a stationary dark soliton. (a) Three-dimensional spatiotemporal evolution, (b) upper view, and (c) profiles at three different times. The parametric values are $\gamma_0 = 0.0$, $\Omega^2 = 0.0049$, $\eta_1 = 0.09$, $\eta_2 = 10^9$, $\gamma_1 = 40000$, $\gamma_2 = 5.0$, $\delta = 0.1$, $\lambda = -1.0$, and $D_0 = 0.0024$. The *lower figures* (d), (e), and (f) depict a rotation of the cyclic lattice observed in the upper figure in order to clearly highlight the collision process of the two pulses and the emergence of the dark soliton.

11(d)–11(f) show a rotation of this large cyclic network in order to underscore the formation of the dark soliton in the lattice, after the collision process of the two propagating pulses has taken place.

Third, in other parametric conditions, Fig. 12 shows that the two pulses, as they propagate through the spatial domain, leave behind them a permanent deformation of the lattice that leans toward generating a spatially inhomogeneous stationary pattern, in the form of stripes disseminated all over the spatiotemporal domain. Figures 12(a)–12(e) depict the three-dimensional spatiotemporal evolution of the travelling ionic waves, its upper view, and the profiles of the lattice at four different times, respectively. It is important to note here that after the collision, the two pulses annihilate each other.

Therefore, the aforementioned facts stress the existence, prior to the collision process, of a robust bisoliton solution propagating along the biochemical medium.

Furthermore, in cellular automata, the state of a particular cell in the next time steps depends on the state of the cells around it (that is its neighbors) at current time. The Greenberg-Hasting cellular automaton, for instance, can be conveniently used to model the excitability of heart tissues.[46] In general, it is well known that each cell can be in one of the following states: quiescent (or excitable) state, excited state, or refractory state. In the last case, the cell has recently been excited and is temporally not excitable. This refractory cell possesses a refractory period which is lessened at the next time step, until it reaches the end of the refractory period and becomes excitable once more. Thus, the refractory period is indeed the time interval elapsed between the time step that follows the onset of excitation, and the time step at which the cell returns to a quiescent state, when it becomes excitable again. At this stage, we numerically assess this refractory period in the context of our work. For this purpose, we select a specific cell, namely, the cell at location $n = 25$ and numerically compute its refractory period on the basis of its local dynamics as highlighted in Fig. 13.

Indeed, based on its time series observed in the same parametric conditions as in Fig. 10, we obtain a refractory period of 372.1 time units whose span is indicated by the double arrow (Magenta online). It is noteworthy that in the

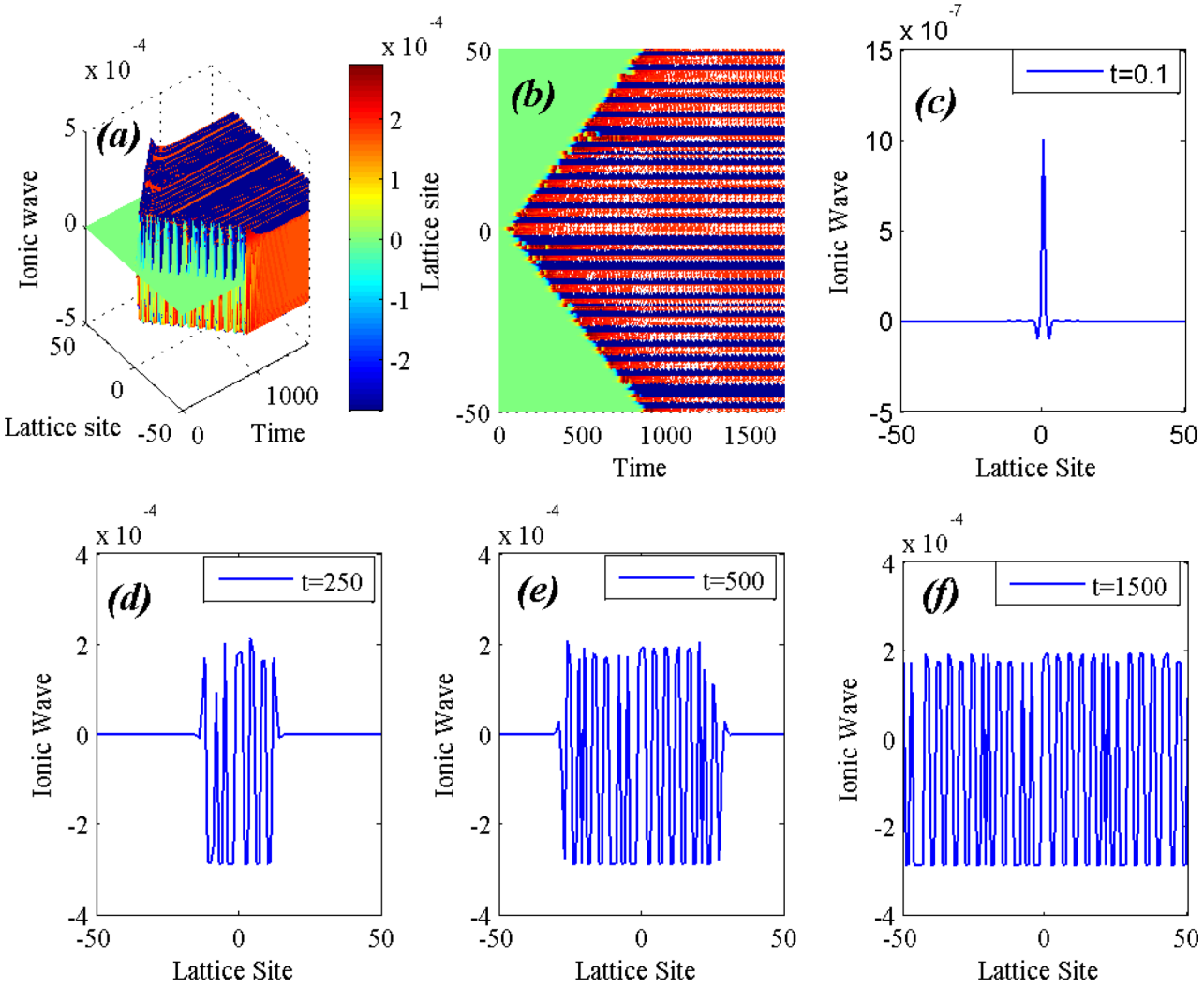

FIG. 12. Ionic waves solutions of the discrete voltage equation (4). Bi-pulse propagation, generating the emergence of a spatially inhomogeneous stationary pattern along the lattice. (a) Three dimensional spatiotemporal evolution of the bi-pulse; (b) upper view of the propagating phenomenon; (c)–(f) snapshots of the structure of the lattice at different times. The parametric values are $\gamma_0 = 65.0$, $\Omega^2 = (2.85)^2$, $\eta_1 = 0.09$, $\eta_2 = 10^8$, $\gamma_1 = 4000$, $\gamma_2 = 5.0$, $\delta = 0.1$, $\lambda = -1.0$, and $D_0 = 0.0024$.

parametric conditions of Fig. 12, this same value of the refractory period was retrieved. This fact appears to be logical due to the fact that we assumed from the onset that our lattice is made up of identical dynamical units. Therefore, based on these facts, the minimum time interval needed for any dynamical unit along our biological lattice to regain its excitability subsequent to the onset of its excitation is about 372.1 time units.

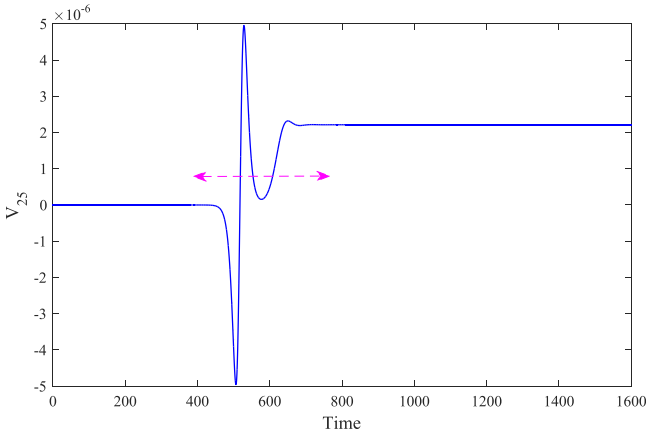

FIG. 13. Local time series of the dynamical unit at location $n = 25$, indicating the refractory period of the lattice found to be approximately 372.1 time units. The parametric conditions are the same as in Fig. 10.

At this point of our numerical analysis, it becomes obvious that our biological system, via its electrical model of microtubule, is resiliently qualified to uphold nonlinear localized waves of excitation. This predisposition to excitability has thoroughly been proven both analytically and numerically.

VI. CONCLUSION

Excitable media are widely spread amid biological systems. They include among others, the heart, the muscular tissues, and the neural networks. In this work, we have studied the capability of the inner part of a microtubule, namely, a protofilament, to behave like an excitable structure. For this purpose, we have proposed an electrical model mimicking the behavior of a microtubule protofilament, with strongly nonlinear resistive elements. Based on this model, we could derive the discrete equation of evolution of the voltage

stimulus urging the calcium ions to flow both within the medium and in between the microtubules. We showed that this equation of motion could be reduced to a *CGLE* describing the dynamics of the amplitude of the ionic wave. An analytical and numerical treatment of this equation indicates that its plane waves solutions are continually unstable, thereby depicting the propensity of our biological milieu to favor the emergence of modulated amplitude waves. These localized modes incorporate both stationary and travelling bright solitons, bi-solitons, and dark solitons. We could equally notice the accessibility of our array of biochemical units to some stationary homogeneous and inhomogeneous spatial patterns.

Microtubule model for excitability was originally proposed for wireless communication in the brain, where microtubules at the axon initial segment may serve as the receiving antennas for converting electromagnetic waves into neuronal excitability.[21] Some investigators hypothesized that reducing the number of microtubules with general anesthetics,[22] colchicine,[23] or other antimitotic drugs[24] suppressed the ability of nerve cells to produce action potentials. In this case, the states of excitability analysed indicate that microtubules possess a high degree of complexity, peculiar to many nonlinear biological systems, and it plays a crucial role in the processes underlying the mechanisms of information processing and transmission in living beings at the subcellular scale.

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Kinklike excitations as an energy-transfer mechanism in microtubules

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A model is presented that is intended to provide a realistic physical picture of the energy-transfer mechanism in cell microtubules. A classical ϕ^4 model in the presence of a constant electric field is used as a conceptual basis. It is demonstrated that kinklike excitations arise as a result of the guanosine 5′-triphosphate (GTP) hydrolysis and that an intrinsic electrical force may cause them to propagate along a microtubule. A discussion is given on the possible effects of these excitations on the dynamics of microtubules.

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GTP binds to tubulin dimers

I. INTRODUCTION

The interiors of biological cells are structurally and dynamically organized through the presence of networks of interconnected protein polymers. These networks are referred to as the cytoskeleton due to their bonelike structure. Of the various filamentary structures which comprise the cytoskeleton, microtubules (MT's) appear to be the most prominent ones. It can safely be said that MT's are ubiquitous through the entire biology [1].

Parallel-arrayed MT's are interconnected by cross-bridging proteins which are called microtubule-associated proteins (MAP's), as shown in Fig. 1. These interconnections, together with MT's, are believed to be responsible for numerous cellular activities such as growth and division, which are essential for the living state. From the characterization of structure and function, MT's appear very well suited for dynamic information processing [1–3].

MT's represent hollow cylinders formed by protofilaments aligned along their axes (see Fig. 2) and whose lengths may span macroscopic dimensions. In vivo, the cylindrical walls of MT's are assemblies of 13 longitudinal protofilaments, each of which is a series of subunit proteins known as tubulin dimers. Each tubulin subunit is a polar, 8-nm dimer which consists of two slightly different 4-nm monomers with molecular weight of 55 kilodaltons. These two constituent parts are usually called α and β tubulin. Each dimer may be physically viewed as an electric dipole $\mathbf{p}$ whose dipolar character originates from the 18 calcium ions bound within each β monomer. An equal number of negative charges required for the electrostatic balance are localized near the neighboring α monomer. Thus, MT's can be identified as an example of electret substances, i.e., oriented assemblies of dipoles. They are predicted to possess piezoelectric properties that may be quite important in their functions, especially in the very interesting assembly and disassembly behavior [4,5].

A very important role is played by guanosine 5′-triphosphate (GTP), an energy-providing analog of adenosine 5′-triphosphate (ATP) which binds to polymerizing tubulin dimers. It is a very appealing mechanism from the biochemical point of view since the order of magnitude of the energy produced is just right. The chemical reaction involved is [6] the hydrolysis of GTP, which is

$$GTP^{4-}+H_2O \rightarrow GDP^{3-}+HPO_4^{2-}+H^+ ,$$

and under normal physiological conditions approximately 10 kcal/mol (0.42 eV/molecule) of free energy is released in this reaction, where GDP denotes guanosine 5′-disphosphate. It is known that the energy produced during GTP hydrolysis is delivered to assembled MT's,

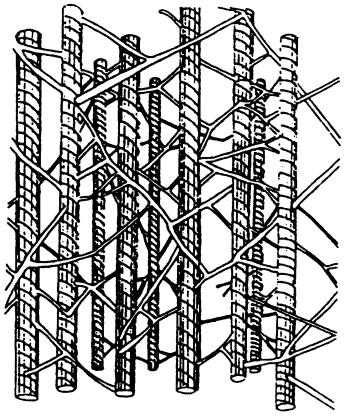

FIG. 1. A parallel microtubule network.

peizoelectric = electricity due to mechanical forces

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although a precise manner in which this energy is utilized is still not understood. One possible mechanism for the utilization of this form of energy is the production of coherent lattice vibrations as proposed by Fröhlich [7] and by the scores of other researchers that followed his idea of biological coherence over the past two decades. Although Fröhlich's theory has never been confirmed by experiment, we believe that the basic mechanism proposed by Fröhlich merits continued interest. Our approach in the present paper will not follow Fröhlich's theory but will be influenced to a certain degree by his ideas.

An interesting suggestion was made by Barnett [8] that filamentary cytoskeletal structures may operate much like information strings in analogy to semiconductor word processors. Barnett conjectured that MT's are indeed processing channels along which strings of information bits can move, transferring messages from place to place. In addition, within this context protofilaments could be envisaged as playing the role of parallel-arrayed memory channels. It was also suggested [2,3,9] that conformational states of tubulin dimers present within MT's may be coupled to charge or dipolar states, thereby allowing for cooperative interactions with neighboring tubulin states. Such interactions were modeled in the framework of molecular automata in which the conformational states of individual tubulin subunits represent bits of information that can change under the dynamic influence of the neighbor subunits.

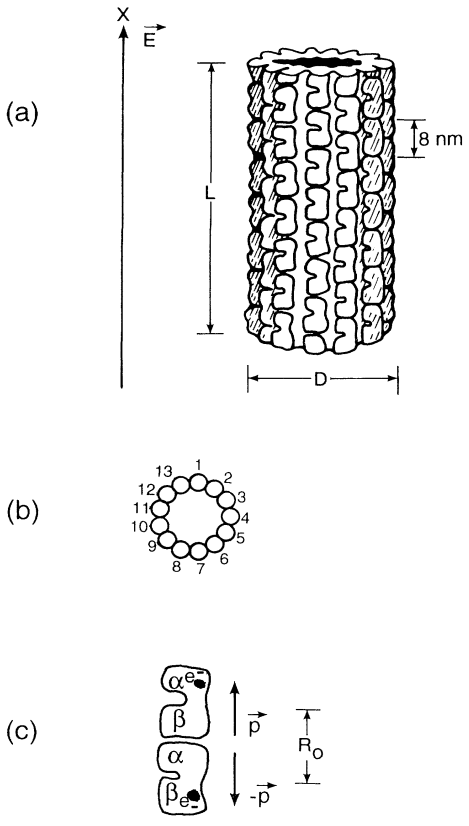


FIG. 2. (a) The structure of a microtubule, (b) its cross section, and (c) two neighboring dimers.

Another intriguing property of MT's is the process of their assembly. Since in solution containing their subunits MT's are not true equilibrium polymers, the process is rather complicated to model. Dark-field microscopy was the technique used to investigate the process experimentally [10]. It was found that when a single MT is monitored, its two ends grow at different rates. The active "+" end grows faster than the inactive "−" end. However, each end independently stops growing in a stochastic manner and then immediately begins to shorten at a high rate. After the shortening period, a microtubule suddenly stops and restarts the growth phase. A very important experimental observation was made that the presence of MAP's significantly suppresses uncorrelated phase conversions and stabilizes MT's in their growth phase.

In this paper we will put forward a theoretical model based on the creation of kinklike excitations in MT's. Their presence will be linked to GTP hydrolysis and the energy released therein. This will offer a plausible explanation of this aspect of the very complex and intriguing MT dynamics. The model proposed in this paper will also provide a starting point for the physical description of a molecular computation system.

II. THE PHYSICAL MODEL

Biological systems in general, and MT's in particular, possess a high degree of order. This is predominantly in the form of functional organization but also, to a large degree, represents spatial arrangement. One basic physical idea concerning the nature of biological order was put forward by Fröhlich [11]. Fröhlich considered an ensemble of molecular dipoles that are capable of high-frequency oscillations that can be subjected to an external electric field and allowed to strongly interact among themselves. He conjectured that, provided the supply of energy to the dipolar system exceeds a critical value, a nonequilibrium metastable state may be formed that is characterized by long-range coherence manifested by a macroscopic occupation of a single mode. An example of such a system could be the dipoles of a cell membrane in the presence of a transmembrane potential [12]. The conditions described above may also apply to the assembly of dimers within a MT which was already described as an electret. The supply of biochemical energy in the form of ATP or GTP hydrolysis may be the source of nonequilibrium excitations.

We believe that the entire MT may be viewed within the context of the Fröhlich model as a regular array of coupled dipole oscillators interacting through resonant (i.e., frequency-specific) long-range forces. Furthermore, coherent vibrations within regions of a MT may take the form of kinklike excitations (KLE's) separating adjacent regions with opposite polarization vectors. The dipole orientation is in the direction of the protofilament axis. It should be mentioned that macroscopic mobile domain structures are of great practical interest due to their potential applications as active components in optical switching and memory devices [13]. It is the number of

KLE's formed and their mobility that determine their most important characteristics, such as the switching time, effective polarization, power dissipation rate, and hysteretic behavior.

In the physical model that is put forward here, the basic argument is that the MT system has a strong uniaxial dielectric anisotropy so that the array of dipole oscillators can be effectively described in terms of only one degree of freedom. In fact, experiments showed [14] that tubulin undergoes a conformational change induced by the GTP-GDP hydrolysis in which one monomer shifts its orientation by 29° from the dimer's vertical axis. Thus, we deduce that the single degree of freedom retained in our description (analogous to Landau's order parameter; see Ref. [15]) is the projection on the MT cylinder's axis of the monomer's displacement from its equilibrium position. The inherent symmetry of a MT enables one to view them as nearly perfect one-dimensional crystals.

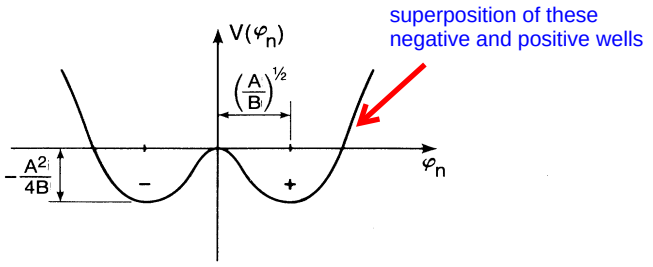
In this paper we intend to describe the nonlinear dynamics of dimer dipoles in one protofilament of a MT in terms of the well-known double-well potential model. This approach has been proven extremely useful in the physics of bistable molecular systems [16]. Its successes in other related areas, for example in the description of dipolar excitations in ferroelectrics [17], are widely acknowledged. The essential argument in favor of introducing the double-well model stems from the fact that the longitudinal projection of the dimer displacement interacts with the rest of the lattice through a mean-field force due to an anharmonic crystal-field potential. A similar inorganic ferroelectric, i.e., antimony sulphoiodide (SbSI), possessing a filamentary structure with parallel filaments [18], has been also analyzed in a similar manner.

The overall effect of the surrounding dipoles on a chosen site n can be qualitatively described by the double-well quartic potential

f(x)=ax^4+bx^3+cx^2+dx+e is a quartic equation

$$V(u_n) = -\tfrac{1}{2}Au_n^2 + \tfrac{1}{4}Bu_n^4 \ , \tag{2.1}$$

where u_n is the longitudinal displacement of the nth dimer, A and B are model parameters such that $B>0$ and is temperature independent, while A is typically a linear function of temperature that may change its sign at an instability temperature T_c, i.e., $A \cong \alpha(T-T_c)$. With $\alpha>0$, below T_c $A>0$ and $u_n=0$ is a maximum of the potential such that $V(0)=0$. Two symmetric local minima exist at $u_n=\pm\sqrt{A/B}$ for which $V_{\min}=-A^2/4B$. This is illustrated in Fig. 3. The anharmonic potential discussed here is meant to approximate the average effect due to the environment when all the neighboring dimers assume their equilibrium positions. As shown in Fig. 4, the mobile electron on each dimer can be localized either more toward the α monomer or more toward the β monomer. The latter possibility is associated with changes in dimer conformation. Experimental evidence indicates [14] that a conformational distortion of 29° from vertical occurs in the β state. We, therefore, identify the variable u_n with the amount of β-state distortion when the latter is projected on the vertical axis.



superposition of these negative and positive wells

FIG. 3. The double-well on-site potential of Eq. (2.1).

This equilibrium being affected by ATP or GTP hydrolysis

In the next step of developing our model, account is taken of the experimental fact that the MT cylinder taken as a whole represents a giant dipole. When the cross section of a MT is viewed using electron microscopy, the outer surfaces of a MT are surrounded by a "clear zone" of several nm which apparently represents the oriented molecules of cytoplasmic water and enzymes [19]. This could be explained by the presence of an electric field produced by a MT. Therefore, we assume that, together with the polarized water surrounding it, a MT generates a nearly uniform intrinsic electric field parallel to its axis (see Fig. 2). Consequently, the additional potential energy due to this electric field and associated with each dipole is

$$V_{\rm el} = -cu_n \ , \quad c=qE \ , \tag{2.2}$$

where q denotes the effective mobile charge of a single dimer and E is the magnitude of the intrinsic electric field.

We are now in a position to write the model Hamiltonian for a microtubule. It can be given as

$$H=\sum_{i=1}^{N}\left[\frac{1}{2}M\left[\frac{du_n}{dt}\right]^2+\tfrac{1}{4}K(u_{n+1}-u_n)^2-\frac{A}{2}u_n^2 + \frac{B}{4}u_n^4-cu_n\right] . \tag{2.3}$$

The first term above represents the kinetic energy associ-

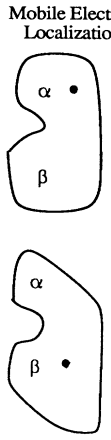


FIG. 4. An illustration of switching between conformational states (α and β) in microtubule dimers.

ated with the longitudinal displacements of constituent dimers, each of which has mass M. The second term arises from the restoring strain forces between adjacent dimers in the protofilament. If the stiffness parameter K is sufficiently large (so that the intersite strain energy exceeds the depth of the on-site potential well: $-A^2/4B$), then it is expected that large-amplitude long-wavelength excitations of the displacement field will be formed. They will be manifested by a slowly varying modulation of u_n along the MT axis.

In order to derive a realistic equation of motion for this system, it is indispensable to include the viscosity of the solvent and introduce the associated damping force. Assuming for simplicity that the solvent is made up of only water molecules, we may infer the following properties [20]:

(i) The dipolar water molecules will have a very significant effect on the long-range electrostatic energy between the dimer dipoles.

(ii) Water will provide a viscous medium that will damp out vibrations of dimer dipoles.

The first effect has already been mentioned, and it can be accounted for by an appropriate choice of the model parameters, especially by an inclusion of the dielectric constant. The second effect can be simply taken into account by adding the viscous force to the equation of motion with

$$F_v=-\gamma\frac{\partial u_n}{\partial t}\ , \tag{2.4}$$

where γ represents the damping coefficient that will be estimated in the next section. We are now in a position to obtain and analyze the relevant equation of motion for the displacement field.

III. THE EQUATION OF MOTION

Based on our assumption that the dipolar oscillations of dimers within a MT form a system that can be classified as displacive ferrodistortive, we can use the continuum approximation whereby

$$u_n(t)\rightarrow u(x,t) \tag{3.1}$$

and

$$u_{n+1}(t)\rightarrow u(x,t)+R_0\frac{\partial u(x,t)}{\partial x}+\tfrac{1}{2}R_0^2\frac{\partial u^2(x,t)}{\partial x^2}+\cdots\ , \tag{3.2}$$

where R_0 represents the equilibrium spacing between adjacent dimers.

An equation of motion for dipolar oscillations can be now derived using the Hamiltonian of Eq. (2.3), applying the continuum limit of Eqs. (3.1) and (3.2) and adding the viscous force of Eq. (2.4) to the thus-obtained Euler-Lagrange equation. We therefore find that

$$M\frac{\partial^2 u}{\partial t^2}-KR_0^2\frac{\partial^2 u}{\partial x^2}-Au+Bu^3+\gamma\frac{\partial u}{\partial t}-qE=0\ . \tag{3.3}$$

Note that the x coordinate is along the protofilament axis. Moreover, for longitudinal sound waves the dispersion relation $\omega=\sqrt{K/M}$ can be used to identify the sound velocity with $v_0=\omega R_0=\sqrt{K/M}R_0$.

We now seek solutions of Eq. (3.3) in the traveling-wave form where the moving coordinate ξ is given by

$$\xi=\left[\frac{|A|}{M(v_0^2-v^2)}\right]^{1/2}(x-vt)=\alpha(x-vt)\ , \tag{3.4}$$

where v denotes the propagation velocity and the coefficient α is, of course,

$$\alpha=\left[\frac{|A|}{M(v_0^2-v^2)}\right]^{1/2}\ . \tag{3.5}$$

Consequently, Eq. (3.3) is reduced to the ordinary differential equation below:

$$M\alpha(v^2-v_0^2)u''-\gamma\alpha u'-Au+Bu^3-qE=0\ , \tag{3.6}$$

where $u'\equiv du/d\xi$. This is an equation for an anharmonic oscillator with linear friction, and a recent paper [21] was devoted to describing is analytical solutions, one of which will be discussed shortly. In addition, Lal [22] discussed a similar problem concerning the existence of kink solutions when friction is present and emphasized its importance in biological applications.

For the sake of convenience we now introduce a normalized displacement field as

$$\psi(\xi)=u(\xi)/\omega\ , \tag{3.7}$$

where the normalizing coefficient $\omega=\sqrt{|A/B|}$ corresponds to the minimum of the double-well potential in Eq. (2.1). The dimensionless equation of motion now takes the form of the ordinary differential equation below:

$$\psi''+\rho\psi'-\psi^3+\psi+\sigma=0\ , \tag{3.8}$$

where we have used the symbols

$$\rho=\gamma v[M(v_0^2-v^2)|A|]^{-1/2} \tag{3.9}$$

and

$$\sigma=q\sqrt{B}\,|A|^{-3/2}E\ . \tag{3.10}$$

It has been shown [17] that Eq. (3.8) has a unique bounded solution which is given by the formula (and illustrated graphically in Fig. 5)

$$\psi(\xi)=a+\frac{b-a}{1+\exp(\beta\xi)}\ , \tag{3.11}$$

where

$$\beta=(b-a)/\sqrt{2} \tag{3.12}$$

and the parameters a, b, and d satisfy the cubic equation

$$(\psi-a)(\psi-b)(\psi-d)=\psi^3-\psi-\sigma\ . \tag{3.13}$$

It is important to note that the above kinklike solution propagates along the protofilament with a fixed velocity

$$v=v_0\left[1+\frac{2\gamma^2}{9d^2Mv_0^2}\right]^{-1/2}\ . \tag{3.14}$$

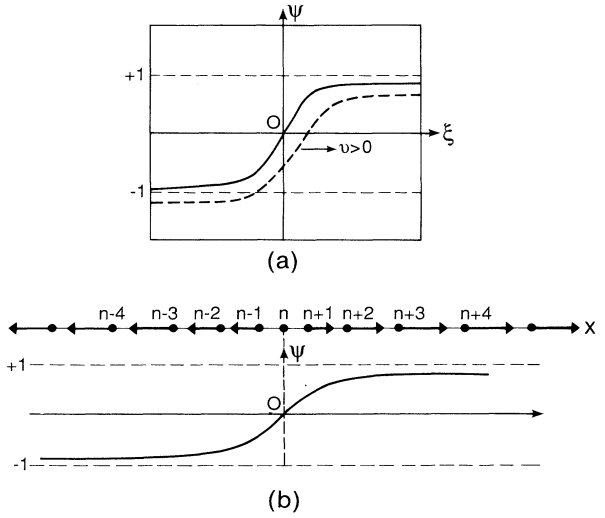


FIG. 5. The form of a kinklike excitation of Eq. (3.11).

This velocity is less than the sound velocity v_0 and it obviously decreases with an increase of the friction coefficient γ. It is clear that v depends on the magnitude of the electric field E through the parameter d. The dependence of d on E is not very simple, since it involves the cubic equation (3.13). However, for $T<T_c$ and large fields E, we know that $d\sim E^{1/3}$, while for small fields $d\sim\psi_0+\lambda E$, where $\psi_0=\pm 1$ is the mean-field solution for $E=0$ and λ is a constant coefficient. In order to gain a more quantitative understanding of this result, we now wish to estimate the magnitude of the intrinsic electric field E in the protofilament. To this end, we assume that the length of a MT, L, is much greater than the diameter D, $L\gg D$. Hence, for points along the protofilament that are sufficiently removed from its ends, the magnitude of the electric field can be approximated simply as

$$E\cong\frac{Q}{4\pi\epsilon_0 r^2}\ , \tag{3.15}$$

where Q represents the effective charge on the ends of the hollow cylinder and r is the distance between the selected point and the end of the protofilament.

Taking, as an example, a moderately long MT consisting of approximately 10^2 dimers gives $L\sim 10^{-6}$ m. Then the effective charge Q consists of 2×13 protofilament end dimers, each of which has a charge of $18\times 2e$ due to the presence of 18 Ca^{2+} ions. Consequently, $Q\cong 26\times 36e\cong 9.4\times 10^2 e$ with $e=1.6\times 10^{-19}$ C. Therefore, the field in the vicinity of the middle point on the protofilament is found to be on the order of $E\sim 4\times 10^6$ V/m. Taking into account the dielectric effects of the surrounding water molecules, this value must be reduced by almost two orders of magnitude to approximately $E\sim 10^5$ V/m. As shown in Fig. 6, the magnitude of the electric field $E(x)$ sharply increases as we approach the end points.

Unfortunately, no reliable experimental data exist at present regarding the values of the potential coefficients

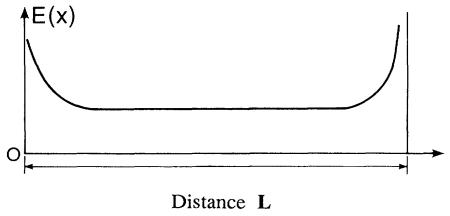


FIG. 6. Plot of the intrinsic electric-field distribution along the MT axis.

A and B or indeed whether the form of our potential is suitable. In principle, at least, neutron-scattering experiments should provide a dynamic structure factor for microtubules which could be used for modeling purposes in a manner similar to that used for ferroelectrics [23]. In the absence of any hard data for MT's we can use some typical values for crystalline ferroelectrics, knowing that they do not vary substantially between different compounds. For example, the well-studied [23] ferroelectric $Pb_5Ge_3O_{11}$ has been demonstrated to support the formation of kinklike domain walls. There, $A(T)=\alpha(T-T_c)$ with $\alpha=10^2$ J m^2 and $B\cong 1.6\times 10^{24}$ J m^{-4}, while $T_c\cong 500$ K. Assuming that the critical temperature T_c for MT's is in the room-temperature range or somewhat above it, we take $T_c\sim 320$ K and estimate that in the room-temperature region

$$A(T=300\text{ K})\cong 200\text{ J m}^{-2}\ ,\quad B\cong 10^{24}\text{ J m}^{-4}\ . \tag{3.16}$$

Thus, with the parameter values obtained in this section we find

$$\sigma\simeq 2\times 10^{-9}E\ . \tag{3.17}$$

It is, therefore, clear that even for very strong electric fields ($E\sim 10^8$ V/m) the inequality $\sigma\ll 1$ holds right up to the protofilament ends. Under these circumstances we can safely approximate the KLE of Eq. (3.11) as

$$\begin{aligned} u(\xi)&=u_0\left[1-\frac{\sigma}{2}-\frac{2}{1+\exp(\sqrt{2}\xi)}\right] \\ &\cong u_0\left[1-\frac{2}{1+\exp(\sqrt{2}\xi)}\right]\ , \end{aligned} \tag{3.18}$$

where the amplitude of the dimer displacement field has the value $u_0=\sqrt{|A|/B}\cong 1.4\times 10^{-11}$ m. Let us now assess other consequences of smallness of the electric field. First, the propagation velocity of a KLE is approximated (for T well below T_c and large values of γ) as

$$v\cong\frac{3v_0}{\gamma|A|}\left[\frac{MB}{2}\right]^{1/2}qE \tag{3.19}$$

and it is generally much smaller than the sound velocity ($v\ll v_0$). Equation (3.19) can be interpreted as a linear response between the propagation velocity v and the magnitude of the electric field E. The coefficient of proportionality in this relationship represents the KLE mobility

$$\mu \cong \frac{3v_0}{\gamma |A|} \left[\frac{MB}{2} \right]^{1/2} q \ . \tag{3.20}$$

In order to evaluate the magnitude of μ we require an estimate of the damping coefficient γ. To do so we will use simple fluid mechanic arguments. First of all, note that each dimer can be roughly viewed as a sphere of mass M. Then, using the well-known result for a sphere of radius R moving through fluid with velocity du/dt, the drag force exerted by the fluid on the sphere is

$$F_d = -6\pi R \eta \frac{du}{dt} = -\gamma \frac{du}{dt} \ , \tag{3.21}$$

where η is the absolute viscosity of the fluid. In our case, the fluid is assumed to be water and its viscosity is temperature dependent. However, at physiological temperatures (~ 300 K) we take

$$\eta \cong 7 \times 10^{-4} \text{ kg m}^{-1} \text{s}^{-1} \ . \tag{3.22}$$

Substituting $R \cong 4 \times 10^{-9}$ m into Eq. (3.21) we obtain

$$\gamma \cong 5.6 \times 10^{-11} \text{ kg s}^{-1} \ . \tag{3.23}$$

The last quantity required in our estimate is the sound velocity v_0. The experimental results [24] of v_0 for DNA produced the value

$$v_0 = 1.7 \times 10^3 \text{ m/s} \ . \tag{3.24}$$

Finally, then, putting $M = (55 \times 10^3) \times (2 \times 10^{-27})$ kg $\cong 1 \times 10^{-22}$ kg and $q = 18 \times 2 \times (1.6 \times 10^{-19})$ C $\cong 6 \times 10^{-18}$ C, Eq. (3.21) yields

$$\mu \cong 2 \times 10^{-5} \text{ m}^2 \text{ V}^{-1} \text{s}^{-1} \ . \tag{3.25}$$

Thus, taking $E = 10^5$ V m^{-1} as a representative average value of the electric field, the propagation velocity of a KLE is on the order of

$$\bar{v} \cong 2 \text{ m/s} \ . \tag{3.26}$$

However, in close proximity of MT ends E may reach 10^7 V m^{-1} and, consequently, the KLE may accelerate up to $v \sim 10^2$ m/s.

Assuming a smooth journey from one end of the protofilament to the other, we estimate the average time of propagation for a single KLE to be

$$\bar{\tau} = L/\bar{v} \cong 5 \times 10^{-7} \text{ s} \ . \tag{3.27}$$

However, increasing the length of the protofilament will increase $\bar{\tau}$ on two accounts: (a) increasing the numerator in Eq. (3.27), and (b) affecting the mean velocity through the dependence of the electric field on L. Indeed, $E \sim L^{-2}$ and hence $\bar{v} \sim L^{-2}$. Consequently,

$$\bar{\tau} \sim L^{+3} \ , \tag{3.28}$$

leading to a rapid increase of $\bar{\tau}$ with L.

While the model presented here is very closely related to the one developed by Collins *et al.* [17] for ferroelectrics, we should mention that the effects of discreteness of the lattice may play an important role. For example, Kimball's analysis [25] of kink dynamics for a discrete lattice indicates that there may exist a threshold value of the external field required to sustain kink motion. Moreover, it appears that fast-propagating kinks are more stable than slow ones. Similar conclusions were obtained by Sánchez and Vázquez [26], who studied the propagation of topological kinks in a stochastically perturbed ϕ^4 model. Their numerical simulations confirmed the conclusion that for all noise strengths studied slow kinks were more affected by the perturbation than fast ones. Finally, in this context, Combs and Yip [27] demonstrated that discreteness of the lattice may be accommodated by continuous equations of the form in Eq. (3.3) by adding higher powers of the time derivative of the field, i.e., terms like $(\partial u/\partial t)^2$. A general discussion on the propagation and stability of kinks and their generalizations in dissipative nonlinear models can be found in Ref. [28].

IV. STATISTICAL PROPERTIES OF THE KLE ENSEMBLE

It is well known that extra energy is required in order to create a kink, since kinks are not solutions of the equation of motion with the lowest energy. Thus, outside the immediate proximity of the critical temperature T_c, kinks are not easily formed. Due to their topological stability, however, they are not readily destroyed once they are formed. An injection of energy into the system in the form of incoherent pulses of electromagnetic radiation is expected to stimulate kink creation. This extra energy can be produced, for example, during the hydrolysis of ATP or GTP, as was already mentioned in this paper. Due to their mutual repulsion over short distances, the number of KLE's formed is limited from above.

In order to formulate a statistical description of an ensemble of KLE's we must first calculate the energy of a single kink. This is found by substituting the displacement field solution of Eq. (3.18) into the model Hamiltonian of Eq. (2.3) so that

$$E = \frac{1}{R_0} \int_{-\infty}^{+\infty} dx \, H(x,t) \ , \tag{4.1}$$

and we find

$$E = \frac{2\sqrt{2}}{3} \frac{A^2}{B} + \frac{\sqrt{2}}{3} K \frac{A}{B} + \frac{1}{2} \left[\frac{4}{3\sqrt{2}} \frac{MA\alpha}{R_0 B} \right] v^2 \ . \tag{4.2}$$

The first term in the formula above represents the well-known binding energy of a KLE. Its magnitude is much greater than that of the remaining two, being on the order of 1 eV. The second term in Eq. (4.2) corresponds to the resonant transfer of energy responsible for the propagation of KLE's along the MT. The third term is the classical kinetic energy of a KLE, which can be expressed as

$$E_k = \tfrac{1}{2} M^* v^2 \ , \tag{4.3}$$

where the effective mass M^* is

$$M^* = \frac{4}{3\sqrt{2}} \frac{MA\alpha}{R_0 B} \ . \tag{4.4}$$

Thus, the total energy can be rewritten more compactly

as

$$E=\Delta+\tfrac{1}{2}M^*v^2 , \tag{4.5}$$

where

$$\Delta=\frac{2\sqrt{2}}{3}\frac{A^2}{B}\left[1+\frac{K}{3A}\right] . \tag{4.6}$$

Using the same set of parameters as before, including the DNA value for K ($=32\ \mathrm{N\,m^{-1}}$), we obtain the following estimates for a MT:

$$\begin{aligned}&\frac{2\sqrt{2}}{3}\frac{A^2}{B}\cong 0.4\times 10^{-19}\ \mathrm{J} ,\\ &\frac{\sqrt{2}}{3}K\frac{A}{B}\cong 0.3\times 10^{-20}\ \mathrm{J} ,\\ &M^*\simeq 5\times 10^{-27}\ \mathrm{kg} .\end{aligned} \tag{4.7}$$

This estimate appears to indicate that thermal energy is insufficient to create a significant number of KLE's. Thus, we must seriously consider the hydrolysis of GTP from the surrounding medium (or injection of energy pulses of another type) as a main potential source of KLE formation. At normal concentrations of attached GTP molecules one expects a few KLE's to be generated so that their mean separation by far exceeds their width. In this regime, KLE's can be viewed as a gas of weakly interacting quasiparticles whose velocities are thermalized.

In the following we provide a more quantitative statistical calculation [29] intended to demonstrate that thermal generation of KLE's is negligibly small. The partition function for a single KLE with a number of available momenta p is

$$Z=\frac{1}{D}\int_0^L dx\int_0^{M^*v_0} dp\,\exp(-\beta E) , \tag{4.8}$$

where D is a normalization constant, $d\Gamma=dx\,dp$ is an element of the phase space, $p=M^*v$ is the momentum of one KLE, and $\beta=(k_BT)^{-1}$, where k_B is the Boltzmann constant and T is the absolute temperature. In order to calculate the mean number of KLE's, we first define the grand partition function Ξ as

$$\Xi\equiv\sum_{i=1}^{\infty}\frac{1}{i!}Z^i\exp(i\beta\mu)=\exp[Z\exp(\beta\mu)] , \tag{4.9}$$

where Z^i is an i-particle partition function and μ is the chemical potential. Then, standard thermodynamic relations give

$$F=-k_BT\ln\Xi , \quad \langle n\rangle=-\frac{\partial F}{\partial\mu} , \tag{4.10}$$

and the mean number of KLE's is found as

$$\langle n\rangle=Z\exp(\beta\mu) . \tag{4.11}$$

The mean density of KLE's in the MT is their number per unit length, i.e.,

$$\begin{aligned}\rho_0&\equiv\frac{\langle n\rangle}{L}\\ &=\frac{1}{D}\left[\frac{M^*}{2\beta}\right]^{1/2}\mathrm{erf}\left[v_0\left[\frac{M^*\beta}{2}\right]^{1/2}\right]\exp\left[\frac{\mu-\Delta}{k_BT}\right] ,\end{aligned} \tag{4.12}$$

where erf( ) denotes the error function which stems from the p integration in Eq. (4.8). However, the most important factor in Eq. (4.12) is $\exp(-\Delta/k_BT)$, since at room temperature $\Delta\simeq 0.2$ eV, $k_BT\cong 0.02$ eV, and the expression for ρ_0 is *reduced* by a factor of e^{10}. This fully supports our earlier predictions and also agrees with the results of Koehler *et al.* [30].

Finally, the thermodynamic equation of state for an ensemble of KLE's on a MT can be found as

$$\lambda L=\left[\frac{Z}{\beta}\right]\exp(\beta\mu)=\langle n\rangle k_BT , \tag{4.13}$$

where λ is the uniaxial stress on the MT. This result takes the form of an ideal-gas equation relating the space occupied by KLE's with the magnitude of the electric field that influences the value of the chemical potential. This is consistent with the emergence of piezoelectric properties of MT's.

V. SUMMARY AND CONCLUSIONS

This paper provided a general biophysical picture regarding the structure and function of microtubules. A physical model has been developed based on nonlinear dipolar excitations. Although this is but a first step in the direction of quantitative modeling of microtubular behavior, we believe that it will be very useful in future investigations. The basis of our model has been the approach due to Collins *et al.* [17] that has proved very useful in ferroelectric and ferroelastic applications. In our model, we have assumed the existence of an intrinsic electric field in MT's and included its interactions with dipolar oscillations in the constituent dimers. Lastly, the presence of the surrounding water molecules has been accounted for by the addition of an effective viscous force that opposes the motion of dimers in an aqueous environment. We have estimated the required model parameters using available experimental data.

Our main result is that a unique kinklike excitation exists which may propagate with a unique velocity that, in the given range of parameters, is proportional to the magnitude of the electric field E. We predict that these excitations may be observed experimentally, for example, through neutron-scattering and polarization experiments. In the past, somewhat similar ferroelectric materials possessing filamentary structure [23] (e.g., Sb-Si) have been precisely analyzed using light-, neutron-, and x-ray-scattering data [31,32].

Although quantitative experimental evidence to support our model appears rather tenuous at present, there are a number of facts that are consistent with the qualitative features of our approach. For example, the observed growth rates of MT's are different at both ends in spite of

an identical concentration of the surrounding monomers. This can be explained on the basis of the following argument. We assume that KLE formation is mainly due to the hydrolysis of GTP into GDP so that one act of hydrolysis corresponds to the conformational change resulting in the formation of a single KLE. However, a KLE is preferentially oriented towards the direction of the intrinsic electric field. The propagation of a KLE will then distribute the energy of hydrolysis at the preferred end of a MT. This energy will then be used to detach dimers from the MT. This picture is in accordance with a hypothesis put forward in a recent paper by Krischner and Mitchison [33], who stated that on-rate (growth) is limited in principle only by the rate of diffusion of the monomer subunits into the MT polymer (i.e., by the concentration of the constituent monomers in solution). On the other hand, the off-rate (depolymerization) can be extremely rapid and the net amount of MT polymer formed can be regulated independently by the hydrolysis of GTP. Our model can be especially useful in the explanation of a peculiar dynamical effect known as "treadmilling." In it, the "+" end grows at the same rate as the "−" end shortens so that the overall length of the MT remains constant under the condition of constant temperature and concentrations. See Fig. 7 for an illustration of this process.

It is reasonable to expect that at the onset of the MT assembly process, when MT's are very short, the formation of KLE's is very unlikely (or even impossible), since for a short chain the displacement field gradient required would be very high and the associated energy to form it prohibitively large. However, when the length of a MT reaches a threshold value, kink formation could become a real possibility. Excitation of kinks, being related to GTP hydrolysis, would become dependent on the GTP concentration. As MT's become longer, KLE propagation becomes an important new factor, and it introduces disassembly at the "−" end, whose rate is proportional to the rate of KLE's arriving per unit time at the "−" end.

It has been emphasized [33] that the existence of a so-called "cap" of dimers containing GTP near the "+" end is a factor stabilizing the growth of GTP. If dimers within this cap do not hydrolyze, the growth of a MT is maintained. However, an intensive hydrolysis of this cap is a trigger mechanism for the disassembly of MT's occurring at *both* ends. We also attribute this instability to the energy transfer through KLE's, since the probability of a growing MT to start shrinking is thought to be governed by the stochastic loss of the fluctuating GTP cap.

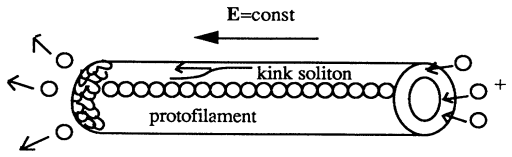


FIG. 7. Schematic representation of the "treadmilling" process involving tubulin dimers of a microtubule which takes place at the (−) end with disassembly and at the (+) end with assembly.

We have seen in our model that the intrinsic electric field governs the rate of propagation of KLE's. By adding an external electric field to the Hamiltonian we can introduce a new control mechanism in the MT dynamics. An applied electric field will result in a faster-moving population of KLE's and thus a greater stability against thermal fluctuations; if, on the other hand, the intrinsic and applied electric fields are oriented in opposite directions, then KLE's motion may be slowed down or stopped altogether. This then can be seen as a basis for treating MT's as artificial information strings. Each KLE within a MT can be viewed as a bit of information whose propagation can be controlled by an external electric field.

Another important experimental fact is that stability of MT's is greatly enhanced by the presence of lateral cross-bridging proteins (MAP's). From the physical point of view, these bridges represent lattice impurities in the MT structure. Interactions between solitonlike pulses and impurities have been studied very extensively in the pst few years. It was demonstrated that impurity modes play an important role in soliton propagation. In particular, kinks may be totally reflected by an attractive impurity if their velocities lie in specific resonance velocity "windows" [34]. Hence, if the velocity of KLE's lies within such a "window," the rate of arrival of KLE's at the "−" end will decrease or even stop as the number of MAP's increases. This mechanism can lead to a significantly reduced rate of MT disassembly. This mechanism together with a control mechanism offered by an external electric field may provide a number of interesting possibilities for information transmission and storage in MT's.

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LASER–RAMAN SPECTROSCOPY OF LIVING CELLS

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Abstract:

Investigations into the laser–Raman shift spectra of bacterial and mammalian cells have revealed that many Raman lines observed at 4–6 K, do not appear in the spectra of cells held at 300 K. At 300 K, Raman activity, at set frequencies, is observed only when the cells are metabolically active; however, the actual live cell spectrum, between 0 and 3400 cm^{-1}, has been found to alter in a specific way with time as the cells' progress through their life cycles. Lines above 300 cm^{-1}, from *in vivo* Raman active states, appear to shift to higher wave numbers whereas those below 300 cm^{-1} seem to shift to lower ones. The transient nature of many shift lines observed and the intensity of them when present in the spectrum indicates that, *in vivo*, a metabolically induced condensation of closely related states occurs at a set time in the life of a living cell. In addition, the calculated ratio between the intensities of Stokes and anti-Stokes lines observed suggests that the metabolically induced "collective" Raman active states are produced, *in vivo*, by non thermal means. It appears, therefore, that the energetics of the well established cell "time clock" may be studied by laser–Raman spectroscopy; moreover, Raman spectroscopy may yield a new type of information regarding the physics of such biological phenomena as nutrition, virus infection and oncogenesis.

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Introduction

Over the past four decades investigations into the working of living cells designed to understand those special properties of matter that constitute life, have been dominated, almost exclusively, by biochemically oriented considerations. Such works, by a large number of dedicated researchers, have (a), increased our knowledge of the chemical composition of life forms; (b), unravelled the chemical structures of some large macromolecules; (c), tracked the pathways by which many components are synthesized and (d), led to the introduction of a mechanism by which the primary structure of proteins *in vivo* may be determined and the required codes handed down from one generation to the next. The sheer number of such investigations and the wealth of information they have provided, however, have tended to overshadow other types of approaches to the problem of life, and to drown the words of caution, issued from time to time, from the voices of those who have walked different paths. It is understandable, therefore, that initial ideas and theory regarding the nature of life were based on chemical concepts and the random collision kinetics of physical chemistry.

In more recent times cell biologists and geneticists have uncovered new facts which reveal the cell to be a highly ordered, dynamic entity in which the chemical events, described by the biochemists, occur in sequences of extreme precision with respect to time, place and accuracy; few mistakes, if any, are made in either of these three parameters. In addition the rates, at which the *in vivo* syntheses of macromolecules occur are several orders of magnitude faster than those performed, in the test tube, by individual components in isolation from the rest of the cell. From the standpoint of statistics and the enormous number of steps performed by a cell, in just one sequence of events, the probability of an error, if it was governed by random collision, would far exceed the small number of errors actually made. Some process exists *in vivo*, therefore, able to direct, in time and space, rapid sequential events with great accuracy, a process accredited by biologists, to the "cell time clock"; giving the phenomenon a name, however, does not provide, in any way, a clue to its mechanism.

The living cell has been found to possess unexpected electrical properties, with membrane potentials 10^7 V/M and dielectric characteristics which alter with time and nutrition, phenomena which have occupied the research efforts of some scientists over many years. It is susceptible also to injury from specific frequencies of electromagnetic radiations of extremely low photon energy, such as microwaves, by mechanisms which appear to be of a non-thermal nature [1, 2, 3, 4]. While the cell is stable and capable of extreme accuracy in its internal manoeuvres, under "normal" thermal conditions, its precision can be seriously impaired by an input of small quantities of electromagnetic energy. The livelihood of the cell thus rests heavily on the specific, yet delicately balanced, physical relationship it creates between its component parts; balances which it is able to generate and maintain sometimes in the presence of excess thermal energy.

The creation of specific states and transitions between them are known vital parts of the life process. Somehow, however, the cell is able to direct, in time and space, the occurrence of these transitions so as to extract configurational energy from its nutrients, with a minimum of loss in the form of heat or the emission of electromagnetic radiation, and produce an entity identical to itself, or one with a limited lifetime but special mission within societies of cells that make up multicellular life forms.

It is clear today that life, even in its simplest form, the cell, does not rest on the independent activities of enzymes, DNA or even membrane potentials but arises from the creation of unique complexes of macromolecules in which the chemical and physical characteristics of the individual molecules are changed and their activities integrated. It is, therefore, a collective phenomenon of enormous complexity as far as the structure and activities of its component parts are concerned, but one

which may be understood more easily perhaps, through the acquisition of knowledge regarding the physics of the phenomenon, as Fröhlich has attempted to do over the years [5, 6]. Much of the enormous effort of biochemists and cell biologists could be wasted, therefore, if physicists do not take up the challenge of ascertaining how the information supplied by these scientists may be collectively viewed in terms of energy, space, time and motion. Any physicist who does try to solve anyone of these problems, however, must keep in mind the limitations of the theories of physics based on "simple" models in which he or she has been schooled. Most, if not all of these amply apply to an individual atom or small collections of them or to collections of identical but small molecules; there is, however, no guarantee that these theories will hold true as far as the life phenomenon is concerned. Indeed, one of the excitements of such explorations is the possibility today of new physical "laws", or phenomena, being discovered via studies of the life process just as they were a century or so ago. Moreover, while it is important that the physicist acquaints himself with the fundamentals and theory of modern biochemistry and cell biology (which may seem, at first, a formidable task, although the essentials, free of intimate details, are not difficult to grasp) it is equally important that he appreciates information acquired much earlier, something which even modern biologists often fail to do. It is essential to recognize, for instance, that ever since chromosomes and the developing embryo were first viewed under a primitive microscope directed and orderly motions of macromolecular complexes and whole cells have been known to be vital parts of the life process; these activities have been given names, such as mitosis and meiosis etc., but the mechanism by which they occur remains neither explained nor covered by popular biological theory. Another feature of the cell is that simply to survive as a life form it requires no use of outside energy, inputs of thermal or electromagnetic energy can be "anti-life" but if these are avoided a cell can retain, for an indefinite period, its ability to renew its life activities if the correct environmental conditions are established. In this state of anabiosis, a condition not understood, the cell possesses, in the form of enzymes, electrical potentials, and so on, all the energy necessary to perform its tasks. It lacks only the ingredients with which to do them. If these are supplied, it simply uses its inherent energy to re-arrange the configurational energy, present in the supplied nutrients, to form a unit just like itself, or to make something other cells require. No one can say how self perpetuating systems, able to re-arrange random energy sources into identical organized systems, evolved, all that can be said today is that they exist and constitute the only method by which energy "packets" with the property of life may be reproduced. Since all the components of "ruptured" cells in a test tube will not reform into cells, it may be said safely that it takes life to create life and it is with life that we must work if we are to understand it.

In the knowledge that the cell is a special form of integrated changes in energy states transformed by metabolic processes into coherent and specific vibrations of large macromolecular complexes and mechanical motions (analogous to the crystals and cogs of a highly accurate watch) some researchers have tried to bring to bear, on the problem of understanding the energy transactions of life, new physical tools such as the laser and Raman spectroscopy. The purpose of this report, therefore, is to describe the difficulties and frustrations encountered in such endeavours and the preliminary successes achieved.

Why laser–Raman spectroscopy?

If life is to be considered a collective phenomenon and to arise from a particular way in which transitions in the states of large macromolecules are integrated then, to understand the life "force", methods must be used which will display not only the energy distribution in isolated molecules but also

how this is affected by the formation of macromolecular complexes and the presence of adjacent ones. In addition, the techniques should permit a determination of those transitions allowed and propagated *in vivo* by living entities. Some form of spectroscopy, therefore, is suggested. Many powerful tools exist such as Nuclear Magnetic (NMR) and Electron Spin Resonance (ESR) but these require, often, the addition to the system of special chemicals toxic to the living cell. The same is true of methods which employ gamma rays, X-rays and ultraviolet light, all have been, and are, invaluable techniques for molecular biology but all are lethal to life processes. To look at life the best techniques are those which do the least harm. Thus visible and infrared light appear to be the preferred probes.

Straight forward absorption spectroscopy in the visible would seem to be insufficient. Only a few biological molecules absorb in the visible and those that do usually have very specialized functions, in vision, photosynthesis and so on, and while vision is a part of some life forms it is not essential to life as such. Photosynthesis in the overall picture is essential to life on earth but it plays no direct role in the organization of animal and other heterotrophic life forms. Thus while valuable information about the above phenomenon has been obtained by such absorption techniques, the procedures have limitations which may prevent them from yielding the type of knowledge sought.

Since it is the allowed transitions in states *in vivo* about which information seems vital, infrared or Raman spectroscopy appear the methods of choice. In accordance with classical electrodynamics any motion of an atomic system involving a change in dipole moment leads to absorption, or emission, of radiation. Any vibrational motion of a molecule may be resolved into a sum of normal vibrations and, as these are simple periodic motions, the frequencies are the ones emitted or absorbed and lie in the near infrared. To be infrared active, the vibrations must be connected with a change in the dipole moment (M), so in the harmonic oscillation approximation only the fundamentals are active. In asymmetric molecules, however, every normal vibration is infrared active, only in symmetrical molecules may the change in dipole moment be zero and thus a motion be infrared inactive.

In order for a fundamental to appear as a Raman shift, the amplitude of M must change. The magnitude of the dipole moment $p = \alpha E$, where E is the electric vector of the incident radiation of frequency ν and α = polarizability. If, therefore, α changes during a vibration with a frequency ν_1, P will change with frequencies $\nu + \nu_1$, $\nu - \nu_1$ and ν. In asymmetrical molecules, all normal vibrations are associated with a periodic change in polarizability and are Raman active. From classical considerations an induced dipole moment will oscillate at frequency ν of the incident radiation (Rayleigh scattering) and at $\nu \pm \nu_1$ (Raman scattering); the vibration ν_1 will appear as a Raman shift, however, only if one of the components of the polarizability α, for the various scattering geometries, differs from zero. From classical theory, therefore, the intensity of the Raman line $\nu - \nu_1$ should equal that of the $\nu + \nu_1$.

From quantum theory the frequencies of infrared absorption and Raman shift bands are determined by the energy difference between the levels of a transition with the selection rule $\Delta\nu = \pm 1$ for each normal vibration. For molecules with a centre of symmetry infrared active transitions can occur only between states of opposite symmetry with respect to the centre of symmetry, i.e. components of M are antisymmetric but Raman active ones can occur only between states of the same symmetry. For all molecules, however, without a centre of symmetry many transitions can be both infrared and Raman active. As far as sublevels are concerned, only those of opposite parity are IR active whereas those of the same parity are Raman active. Rotational transitions follow similar selection rules in the IR, a transition can take place only if there is a permanent dipole moment and positive levels can combine only with negative ones, thus $\Delta J = \pm I$; for Raman active transitions the selection rule is $+ \leftrightarrows +$ and $- \leftrightarrows -$ and $\Delta J = 0 \pm 2$.

From the above it is clear that the wave number of IR and Raman shift bands are equal to the actual

frequencies of the movements, thus from classical and quantum theory the spectra are the same as far as position and occurrence of bands are concerned. There is, however, one important difference, classical theory suggests that the intensity of Raman bands $\nu - \nu_1$ (Stokes lines) and $\nu + \nu_1$ (anti-Stokes lines) are the same, whereas quantum theory predicts that the number of vibrations in the first state of the anti-Stokes line is only $e^{-h\nu/kT}$ times the number of vibrations in the initial state of the Stokes line (e.g. the ground state). Thus anti-Stokes lines are considerably weaker than their Stokes counterpart and are rarely seen at wave numbers above $500\,cm^{-1}$. It is this difference, therefore, that makes Raman spectroscopy more attractive for biological endeavours than infrared.

The large molecules of life systems are, in the main, linear and asymmetrical thus both IR and Raman active. The metabolically induced electronic transitions between them, *in vivo*, moreover, should involve not only changes in dipole moment but in large alterations in their amplitude, and hence polarizability; and, since it is controlled shifts in electron density which seem to control life process, such processes should exhibit Raman activity. A living system, however, it not governed by thermal equilibria as, in order for it to function, it must produce "excited" states at what must be considered to be low temperatures. An amino acid, for instance, may be produced by the chemist on refluxing certain chemicals at temperatures lethal, on microsecond exposures, to all life forms. The cell, however, produces them, at rapid rates, at very low temperatures and it does so by using the "configurational" energy present in its large organic catalysts, namely its enzymes, in place of heat. In general, heat is toxic to life forms and is a waste product of some of their internal processes. The association of even a small molecule, such as H_2O, with a macromolecule can induce enormous shifts in electron cloud densities causing severe alterations in the conformation of the macromolecule. For instance, transitions between the B and A forms of DNA are brought about by the association or disassociation of just a few molecules of H_2O to or from its outer groups [7, 8, 9]. Thus a rhythmic oscillation between the two conformations can be set up physiologically by H_2O exchange with large changes in dipole moments and their amplitudes. For a cell to "live", it has to create unstable states; a cell in thermal equilibrium, therefore, is either dormant, i.e. in an anabiotic state, or dead. The physical mechanism by which life forms perform their activities suggests, therefore, that large rhythmic shifts in the amplitudes of dipole moments of specific frequencies occur which should give rise to Raman shifts of high intensity, moreover, the population of upper states should differ from that arising from normal thermal distribution. Thus, if this is fact, then the ratio between the intensities of Stokes and anti-Stokes lines should differ from the thermal factor and anti-Stokes lines observed at higher Raman shift frequencies. Raman spectroscopy, therefore, appears to offer a greater versatility in the interpretation of the physics of life processes than infrared, although, for a full picture to be seen, both should be performed.

Experimental findings

With the advancements in laser technology laser–Raman spectroscopy has proven a powerful tool for studies in many areas of physics, chemistry and molecular biology, by which new information regarding the structure of small and large molecules has been obtained. The excellent works of experimentalists such as Yu and colleagues on the proteins of the eye [10, 11]; Hartmann [12] and Lord [13] on many proteins and nucleic acids have revealed the spectra of these molecules to be complex, reproducible and specific. Data on the spectra of isolated biomolecules [14–20] and membranes [21–25] now are abundant, but it is not, however, the purpose of this report to describe such information and interested readers are referred to the recent reviews of the subject [14–25]. It was the success of these endeavours,

nevertheless, that led the author to believe that the technique should yield new information about *in-vivo* energy states even though the spectra of whole cells, by expectation, would be extremely complex; work on the subject began, therefore, in our laboratories in 1971.

Stokes spectra

The biological models chosen initially were the microbes *Bacillus megaterium* and *Escherichia coli*, as much is known about their physiology and molecular composition. Following the procedures used by prior investigators with isolated macromolecules the microbial cells were collected and thoroughly washed in 0.85% NaCl (saline) or 0.1 M phosphate buffer (pH 6.9) (Buffer) and then used either as suspensions in saline, Buffer or H_2O or as a wet "paste". The physical equipment consisted of a high resolution Cary 82 spectrometer, a Spectra-Physics argon–ion laser and a PDB 11 minicomputer with a link to an IBM 370. The spectrophotometer possessed a multipass system, by which some ten passes of the laser beam could be focussed on a spot in the sample, and a sample chamber equipped with facilities by which the sample could be maintained at any desired temperature. The available apparatus, therefore, was of a high quality and conducive to immediate success, but this was not the case. For almost a year my colleagues and I worked with "pastes" and cell suspensions varying from 10^3 to 10^9 cells/ml but without success, the only Stokes spectrum obtained, if one could call it a spectrum, was that of water and the suspending medium which, because of its dilute nature produced only a few weak lines (fig. 1). These results were confusing, therefore, and disappointing as, from the standpoint of physics, the spectra seen, or lack of spectra seen, made little sense. The enterprise was given up, therefore, because of the belief that something was wrong with either the equipment or the techniques employed. Repeated publications in the literature of the spectra of large biomolecules prompted, however, a return to the project but again with no success.

It was our custom to discuss our failures each morning over tea and on such a morning, with the usual "Raman" run in operation, we heard a loud whistle from our recorder, a characteristic of it, even though new, when it recorded a strong line. Over the next few minutes it continued to whistle and produce a spectrum the like of which we had not seen before. A hasty reset of the instrument produced a second and a third spectrum then, disappointment, on the fourth time of asking, no lines appeared. Over the next week total failures again were experienced, but, in checking over all procedures, suspending fluids etc. it was discovered, by Mrs. Montgomery, our technician, that on the "successful" morning she had inadvertently placed the cells in a fluid containing glucose; the salts fluid used, therefore, was not a holding one as thought, but one in which the cells were able to grow. This experiment using a growth medium, therefore, was repeated and, up to a point, it was successful.

The cuvettes used were of a special kind in that, for use with the multipass system, set Brewster angles were required and only a 0.5 ml volume of sample could be employed. With any one sample in the cuvette, only two or three spectra with lines could be obtained, after this the recorded spectra became, once again, essentially devoid of lines. In addition, in the two or three runs able to be made and showing lines, the spectra were almost totally different; if aliquots of fresh cells, however, were placed in the cuvettes, renewed Raman activity was seen but again each spectrum obtained differed, in large part, from that of the others. Later experiments revealed that more spectra with lines could be achieved, with a given sample, if the laser power was reduced (but here the lines were much weaker) and also if a given scan of up to 3400 cm^{-1} was conducted as short consecutive scans of 400 cm^{-1} with a 2 minute, dark period, between them. Thus the maximum time the cells could be in a 300 mW laser beam and a spectrum produced was

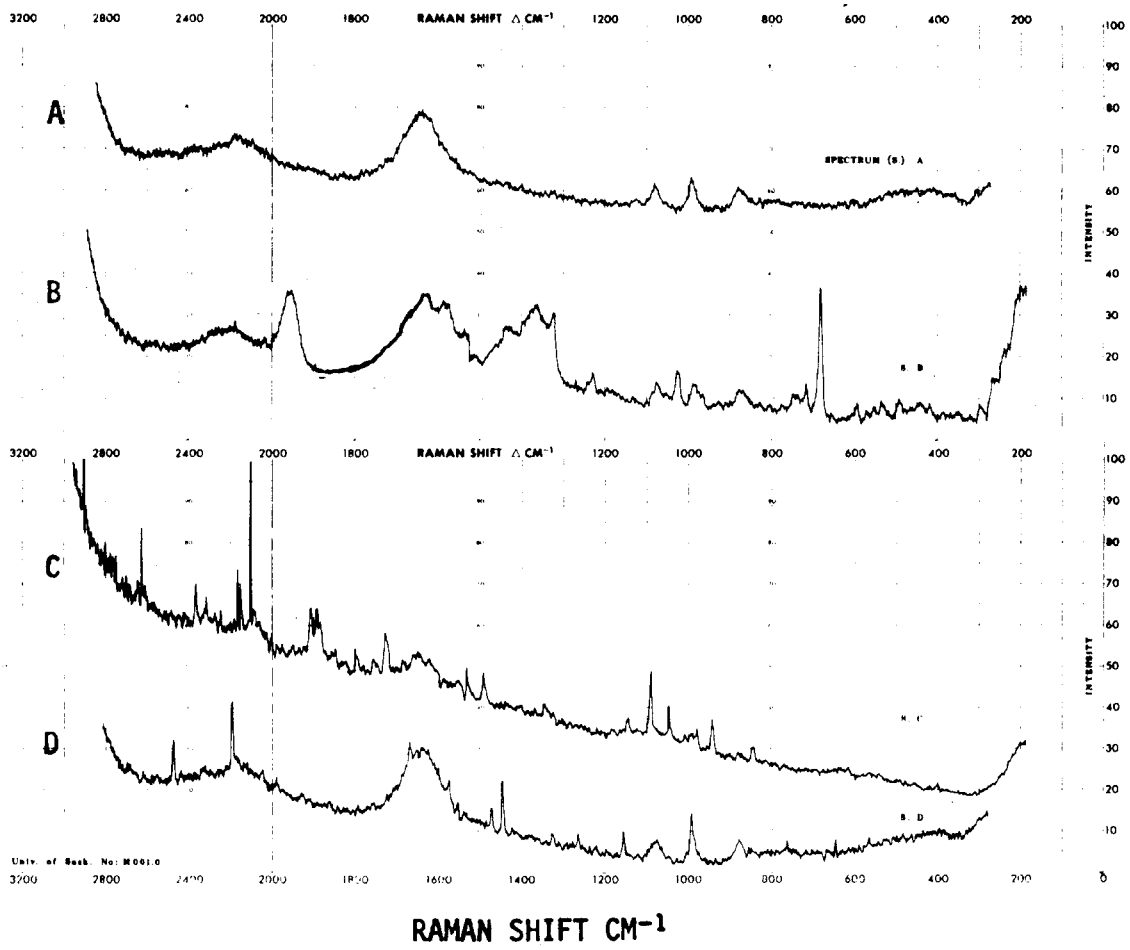

Fig. 1. Examples of initial spectra obtained with washed *B. megaterium* cells in Davis NH_4^+ salts medium. A, without glucose at any incubation time; B, C and D, with glucose at 30, 60 and 90 minutes of incubation respectively. Sensitivity 1×10^4 cps. Band width 5 cm^{-1}; laser 488 nm. Taken from Webb et al. [26].

about 6 minutes. The technique adopted, therefore, was to scan for 3 minutes only and then change to a fresh sample of cells taken from a culture actively growing in a growth medium.

Normal bacteriological growth media, such as the various nutrient broths could not be used because all are highly coloured and hence, in a laser light, they display complex fluorescences. The growth medium of choice, therefore, was the Davis minimal salts medium (MM) which contains only inorganic salts and glucose; if this was used as a 1:3 dilution in H_2O of its normal strength it produced a characteristic Raman shift spectrum with three weak lines at 873 cm^{-1}, 988 cm^{-1} and 1068 cm^{-1} and one stronger peak at 1660 cm^{-1} which was due, in part to water (fig. 1). For cells unable to proliferate in this medium a second one was used which consisted of the MM to which a given amino acid, or group of amino acids, at 1.0 μg/ml was added. At this low concentration no Raman shift lines were produced from these additives and no fluorescence problems were encountered.

Stokes spectra with synchronized populations

Even with the sequential 3 minute scans and the above growth media the spectrum, from 5 to 3200 cm^{-1}, varied from run to run (fig. 1) and, moreover, if the temperature in the sample chamber of the spectrophotometer was held between 15°C and 25°C no lines, except those of the medium were observed. It was clear, by now, that, to obtain Raman shift lines from intact cells, the cells had to be in an active metabolic state which required that they be given a usable carbon (energy) source and environmental temperatures above 25°C. Thus, it was concluded that, if these conditions held, the Raman shift lines observed were due to energy states produced *in vivo* by an active metabolism, and the non-reproducibility of the spectrum was because these states changed, with time, during the lifetime of a given cell. To test this hypothesis synchronized populations of cells next were employed.

Synchronization was achieved by taking cells from a stationary phase culture and first; washing them three times with 0.85% NaCl (saline); second, resuspending them at 10^7 cells/ml in saline and subjecting them to two successive heat and cold shocks, at 40°C and 4°C, for 15 minutes at each temperature; and third, resuspending them at 2×10^7 cells/ml in a fresh medium held at 35°C or 37°C. A second technique was the starvation for 30 minutes, of an auxotrophic strain of *Escherichia coli K12* of a vital nutrient, such as thymine, and then add the nutrient to the medium. Both techniques produced a population of cells in which the metabolic events in each cell were synchronized, to within 3 to 5 minutes, in approximately 70% of the population. After resuspending the synchronized cells in the new medium, new samples were taken every 3 minutes and the shift spectrum between the Rayleigh line and about 3200 cm^{-1} determined. This procedure was carried out only over the first generation (or lag phase) of the cells, which varied from 70 to 180 minutes depending on the growth medium used, at a scan rate of 2.5 cm^{-1}/sec as this gave the best results. Thus a complete spectrum could be obtained every 20 minutes or repeat segments of 450 cm^{-1} run every 3 minutes. For the examination and presentation of the results the abbreviation, MAR, was coined to signify the number of minutes after the cells had been resuspended, in the appropriate growth medium, and at which, a spectral run was started.

The four spectra, obtained over a 90 minute growth period, always differed from one another, only two very broad lines between 1600–1800 cm^{-1} and 2000–2150 cm^{-1} appeared in all but, even here, the fine characteristics of each of these changed (fig. 2). This type of result had been expected, what was not anticipated, however, was the subsequent inability to reproduce, in toto, each of the four spectra in repeat runs: some lines occurred again, others did not and new ones appeared. The reason for this was found to be the inability to reproduce exactly the same characteristics in successive microbial cultures. The lag phase of the cells, after the treatments given, varied by ±5 minutes and apparently this variation, which could not be improved upon, was sufficient to alter the spectral characteristics.

At this time much data had been collected and so some 62 complete spectra were examined; each distinct line seen was tabulated, broad lines of more than 100 cm^{-1} were omitted unless a clear peak appeared on them, as seen in fig. 2A and C at 1225 cm^{-1} or 1620 cm^{-1}. In addition the very narrow "spikes", seen in fig. 1C at 2132 cm^{-1}, were omitted for reasons to be discussed below. When these lines were graphed as a histogram, in terms of the number of times a given line (±2.5 cm^{-1}) was seen, it became clear that the appearance of lines was not random, all the lines occurred over and over again although they were not always present in a given spectrum (fig. 3). The composite, shown in fig. 3, therefore, may be considered as the, in toto, Raman Shift Stokes spectrum of metabolically active *Escherichia coli*. When these lines were histogrammed in terms of the time of their spectral appearance during the life span of the cells they became even more informative. As can be seen from fig. 4 the bulk

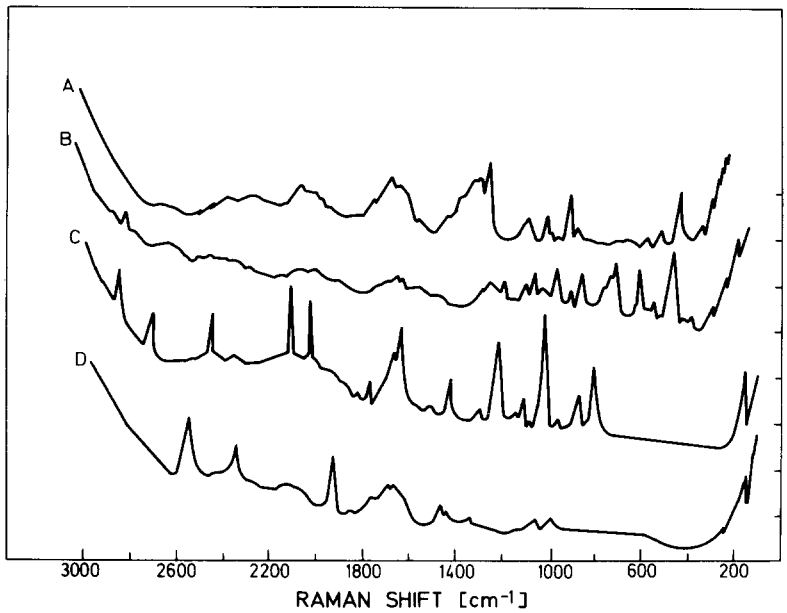

Fig. 2. Examples of spectra obtained with synchronized *Escherichia coli* cells in Davis NH_4^+ salts. A, 0–20 min; B, 20–40 min; C, 40–60 min; D, 65–85 min. Conditions as for fig. 1.

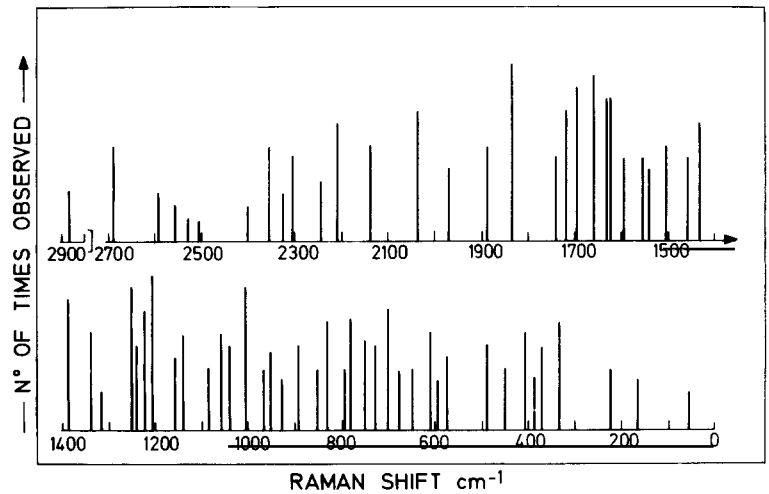

Fig. 3. Composite of all Raman lines observed in the spectra obtained with synchronized populations of *Escherichia coli* cells. Each Raman line seen in a given spectrum, recorded at a set time in the lifetime of the cell, is tabulated along with the number of times each line re-occurred in repeat spectral determinations. The height of the frequency markers, used to scale, the illustration, = two appearances of a line.

of the lines, during the first 30 minutes of the cells' lifetime, occurred between 300 cm^{-1} and 1200 cm^{-1}, from 30 to 60 minutes it was between 800 cm^{-1} and 1900 cm^{-1} and from 60 to 80 minutes between 1400 cm^{-1} and 2600 cm^{-1}. Thus the lines, associated with Raman active metabolic states appeared to shift, with incubation time, to higher wavenumbers; there was, however, an exception to this, the lines below 200 cm^{-1} appeared to move towards lower wavenumbers (both facts apparent from figs. 1, 2, 3, 14). Another interesting finding was that, approximately 10 minutes before cell divisions began, lines at about 2035 cm^{-1} and 2142 cm^{-1} often became very strong (fig. 5).

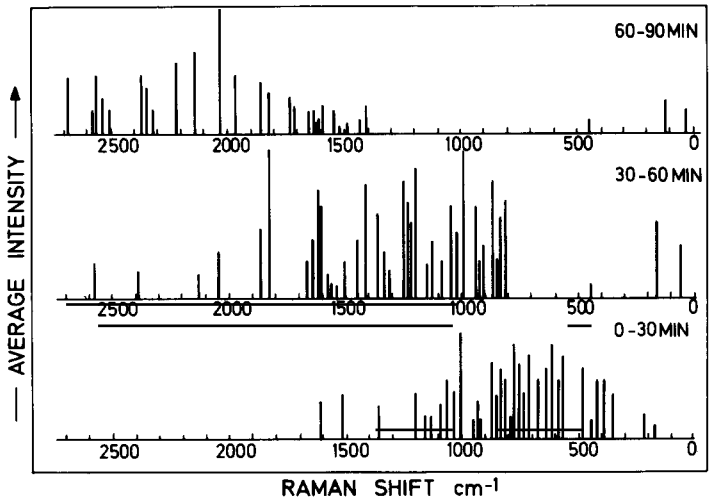

Fig. 4. The timed appearance of Raman lines seen in the spectra of synchronized populations of *Escherichia coli* cells during a 90 minute cell cycle. The times shown represent the time after the cells were placed in a minimal medium and, at which, their Raman shift spectrum was determined. The average intensity of each line is that calculated from two, or more, spectra in which the line appeared and is presented merely to demonstrate the relative intensities of the lines and change in them with time.

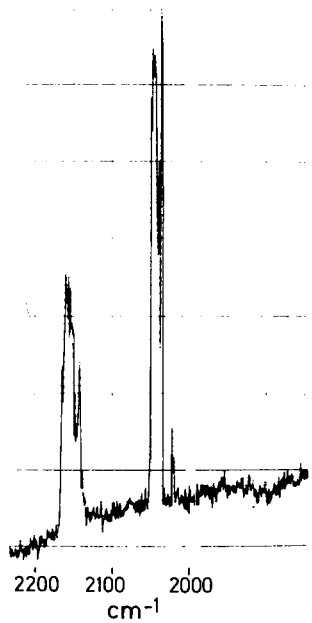

Fig. 5. The strong Raman lines which appeared in the spectrum of *Escherichia coli* just before cell divisions began. Laser 5145 Å, sensitivity 10 K.

Low temperature spectra

The spectra of *Escherichia coli* at 6 K recently obtained is shown in fig. 6. These spectra were produced from a 20 mg, wet weight, paste of 18 hr cells, grown in MM at 33°C, placed on a silvered

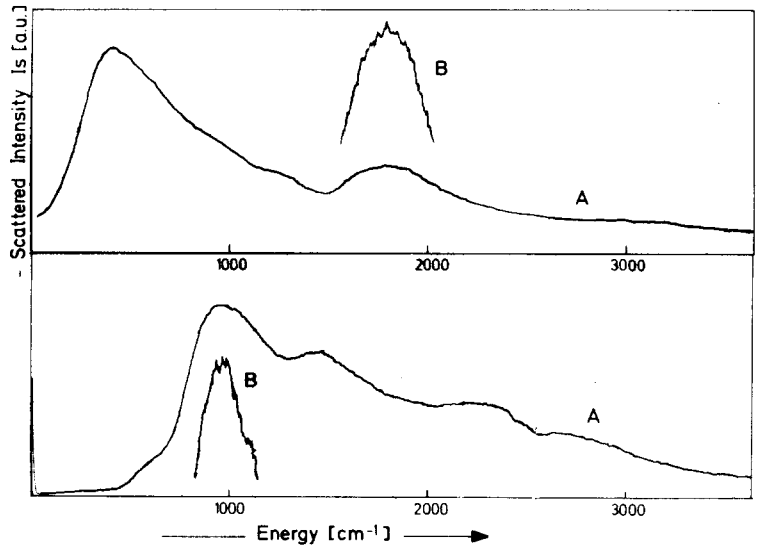

Fig. 6. Low temperature Raman shift spectra of *Escherichia coli* cell pastes obtained at 6 K. Top, laser at 5145 Å; Bottom, laser at 4880 Å. A = sensitivity at 100 K, B = sensitivity at 10 K.

plate. The technical details of the apparatus and procedures may be found in the publication by Drissler and Webb [27]. These spectra are reproduced here for comparison with the lines from time scans of active cells shown in figs. 3 and 4. The low temperature low sensitivity spectrum showed three distinct very broad bands while the high sensitivity scans revealed each of these bands to have a fine structure and thus the broad bands to be a result of the overlapping of many closely related Raman active modes. At low temperatures, this fine structure was completely reproducible. The interesting feature of the low temperature spectra was that the shift frequencies of broad humps appeared to correspond with the peaks in the average intensities and shift frequencies of the lines obtained with active cells (fig. 4). There was, however, one obvious difference in that many more Raman active modes were present in the fine structure of the low temperature humps than in the active cell spectra.

Stokes spectra below 300 cm^{-1}

At this stage of the studies little attention had been paid to shifts below 300 cm^{-1} although it had been noticed that the Rayleigh wings, which were "smooth" with resting cells, always displayed more "noise" when active cells were investigated and, in some cases, definite sharp likes were seen superimposed on these wings. Moreover, the spectra obtained with non-synchronized cells displayed lines below 200 cm^{-1} which were reproducible as far as shift frequencies were concerned but not intensity (fig. 7). These lines below 200 cm^{-1} became of special interest because they did not occur in the published Raman spectra of solutions of isolated macromolecules surveyed by the author, but several groups had reported such lines in the spectra of crystals and other "solid" preparations of natural and synthetic proteins [28, 29, 30]. In our laboratory, moreover, such low frequency lines could not be detected in the spectra of concentrated, but aqueous, lysates of cells produced by bacterophages or chloroform. It seemed, therefore, that the states associated with these lines were peculiar to the intact cell and crystals of biomolecules.

When synchronized cells were studied lines at low frequencies were seen in nearly every spectrum,

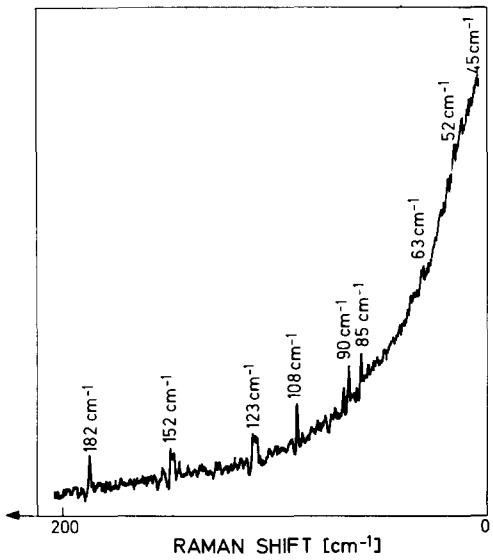

Fig. 7. The Raman shift spectrum, between 30 and 200 cm^{-1} of non-synchronized cultures of *Escherichia coli* in Davis minimal medium. Cell concentrations 5×10^7/ml; laser at 5145 Å and 300 mW; sensitivity 10 K.

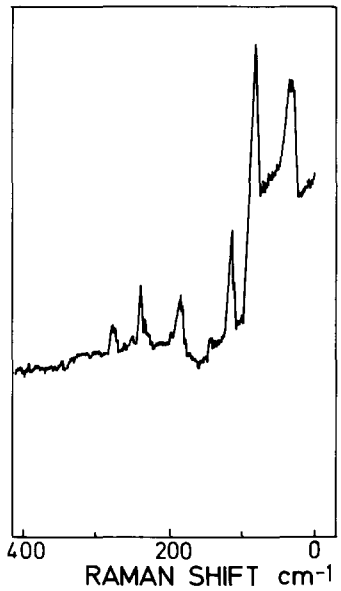

Fig. 8. A low frequency Raman shift spectrum obtained with cultures of *Escherichia coli* in which lines at 45 and 85 cm^{-1} became very strong. Cell concentration 2×10^7/ml; laser at 5145 Å and 300 mW; growth medium Davis minimal medium.

from 0 to 200 cm^{-1}, taken 20 minutes after the start of cell incubation (i.e. 20 MAR). Their intensity varied, however, from appearing as a shoulder on the wings to very strong lines at 45 cm^{-1}, 85 cm^{-1}, 150 cm^{-1} and 186 cm^{-1} (figs. 8, 9). Such lines were seen in the spectra of both *Escherichia coli* and *B. megaterium* and in both cases their exact frequency and detection depended on the MAR time at which the spectrum was obtained. The line at about 150 cm^{-1} appeared to move, as MAR time increased, from 158 cm^{-1}, at the time of its first appearance, to 126 just before cell divisions began. The line at 45 cm^{-1}, which was not detected until after 40 MAR, also moved, but only by a few wavenumbers, to 38 cm^{-1} (see fig. 14). The suggestion that the lines moved came from consecutive scans from 0 to 200 cm^{-1} each taking 80 seconds to run, in which only one predominant line between 126 cm^{-1} and 158 cm^{-1} was seen in any one scan (at least as far as the bacterial cells were concerned) which moved with time a few wavenumbers towards lower frequencies (fig. 10). Such lines at almost the same low wavenumbers have been observed recently in the spectrum of active algal cells and these also seem to change in intensity with time [32]. It is not known at present whether the apparent movement of the line is real or whether it is just another example of the time dependency of given Raman active states which, in this case, arise from a narrow cluster of *in vivo* states.

Time scans

By now at least one thing had been established, this was that the Raman activity of *in vivo* states was a time-dependent phenomenon. It was natural, therefore, that time scanning should follow to ascertain

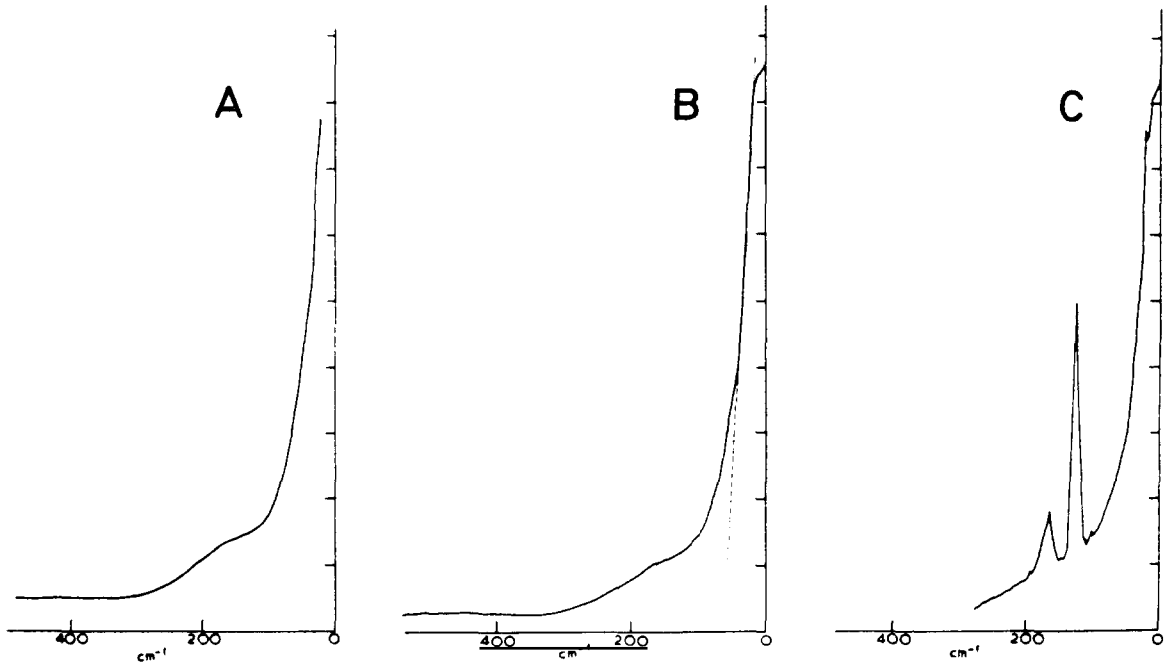

Fig. 9. Low frequency Raman spectra of A, Davis minimal medium; B, resting cells of *Bacillus megaterium*; C, active synchronized cells. Taken from Webb and Stoneham [31].

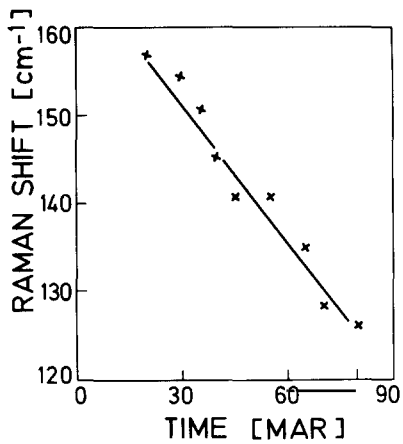

Fig. 10. The change seen in the frequency of a strong line between 126 cm^{-1} and 158 cm^{-1}, as cells of *Escherichia coli* progressed through a 90 minute life cycle.

the time, or times, during one cell cycle at which a given activity occurred. To do this a given shift frequency was chosen, the Raman instrument set on this frequency and a fresh sample of cells placed in the cuvettes every 5 minutes. The results repeated well the observations seen above (fig. 11) in that lines at the lower shift frequencies, but above 300 cm^{-1}, were seen more abundantly early in the life of the cells and a gradual movement to higher frequency shifts took place as the life cycle of the cells progressed. Thus the time phenomenon would have been well established if it was not for a new finding. A given Raman activity at a given frequency certainly occurred at definite times in the life of the cell

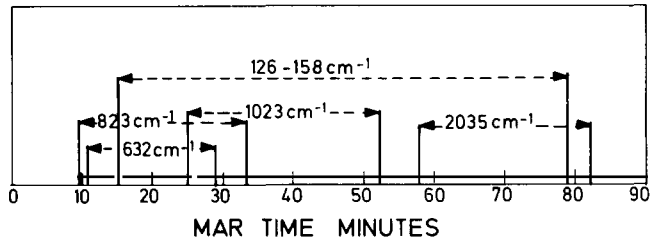

Fig. 11. The times at which set Raman shift lines occurred, in fixed shift frequency time scans, during a 90 minute cycle of synchronized cells of *Escherichia coli* in Davis minimal medium. Cell concentration 3×10^7/ml; laser 5145 Å and 300 mW.

but some of this activity appeared as "bursts" of lines over a 3 to 5 minute period (fig. 12). Since metabolic synchrony could be achieved only within 3–5 minutes a given line should have appeared at a set time and the line traced in amplitude, as a normal Gaussian curve indicating the normal distribution of the effective Raman active state within the 3–5 minute heterogeneity of the population. As seen in fig. 12 this, however, was not the case and so an analysis was made of the intervals of time between the appearance of successive strong lines (Δt) on the spectral traces from many independent scans. The result, histogrammed in fig. 13, revealed that the interval between successive strong lines was not random; Δt's of 15–20 sec and 50–55 sec apparently were the most commonly occurring intervals. The shaded portion of the histogramme represents only the frequency with which the most common time interval occurred when all lines, including very weak ones, seen only in high resolution spectra, were included. It is shown, mainly because the computer analysis revealed its presence but, since many of the lines analysed were extremely weak, it must be viewed with scepticism. The Δt's of the strong lines, which are well under the minimal 3 minute synchrony time, however, seem real and this phenomenon is discussed below.

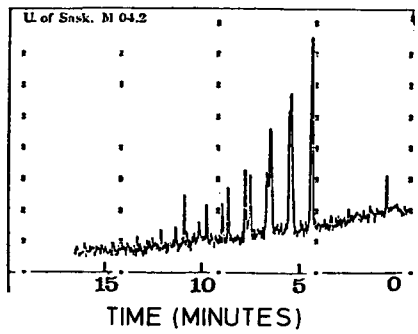

Fig. 12. Bursts of Raman activity which occurred at 1273 cm^{-1} over a 5 minute period and seen in fixed frequency time scans of synchronized cultures of *Escherichia coli*. From Stoneham and Webb [33].

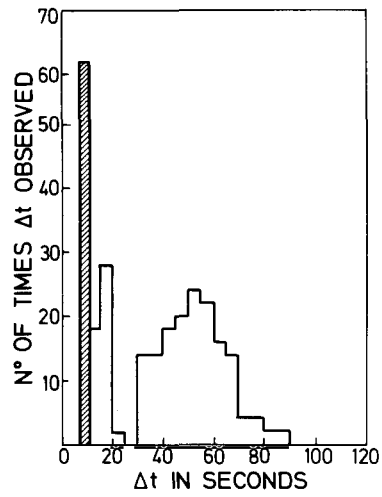

Fig. 13. Histogramme of the intervals of time (Δt) which occurred between the sequential appearance of a line at 1273 cm^{-1} within a 3 to 5 minute period during fixed frequency time scans with synchronized *Escherichia coli* cells.

Anti-Stokes lines

At this stage of the investigations the anti-Stokes spectra had not been explored since it was assumed that the intensity of such lines above $300\,cm^{-1}$ would be too low to be detected. However, the appearance of Stokes lines below $200\,cm^{-1}$ which, at times were of considerable strength, made it reasonable to search for them. Such lines were observed but again their intensity varied from run to run and many appeared as broad bands. At the MAR time when the Stokes line reached its maximum intensity so did the corresponding anti-Stokes line but detailed analysis of their intensity ratio proved difficult because of the need to change the cell sample between successive spectral runs. This required many cultures of cells each started (0 MAR) at various times so that each population could be examined at the same MAR time. The inherent small, but uncontrollable, variation in the time of the cell cycle, from culture to culture, made it impossible to obtain the same absolute intensity of the lines. One of the best sequences observed is shown in fig. 14, but it must be emphasized that this has been selected from many spectra of which most were not of the same quality. To obtain this type of spectrum the sensitivity of the instrument was set at 1×10^5 cps during the first $30\,cm^{-1}$ wavenumbers of the scan past the laser line and then switched through 1×10^4 to 5×10^3 cps. This procedure tended to flatten the wings, result in small "false" lines over the first $35\,cm^{-1}$ and amplify the intensity of real lines between 40 and $60\,cm^{-1}$ which appear on the Rayleigh wings. The line at about $150\,cm^{-1}$, therefore, is a true representation of its Stokes and anti-Stokes intensities, that at $43\,cm^{-1}$, however, must be viewed with caution. This caution kept in mind, it is interesting nevertheless that both Stokes and anti-Stokes at $43\,cm^{-1}$, have almost the same intensity. This type of intensity ratio also held true of the line at $158\,cm^{-1}$ which, as stated above, appeared to move to lower wavenumbers as the MAR time increased. In eight successive spectral runs the anti-Stokes to Stokes ratio of this line was found to vary from a low of 0.73 to a high of 1.08.

As was done with the Stokes lines some time runs were conducted at set anti-Stokes shift frequencies

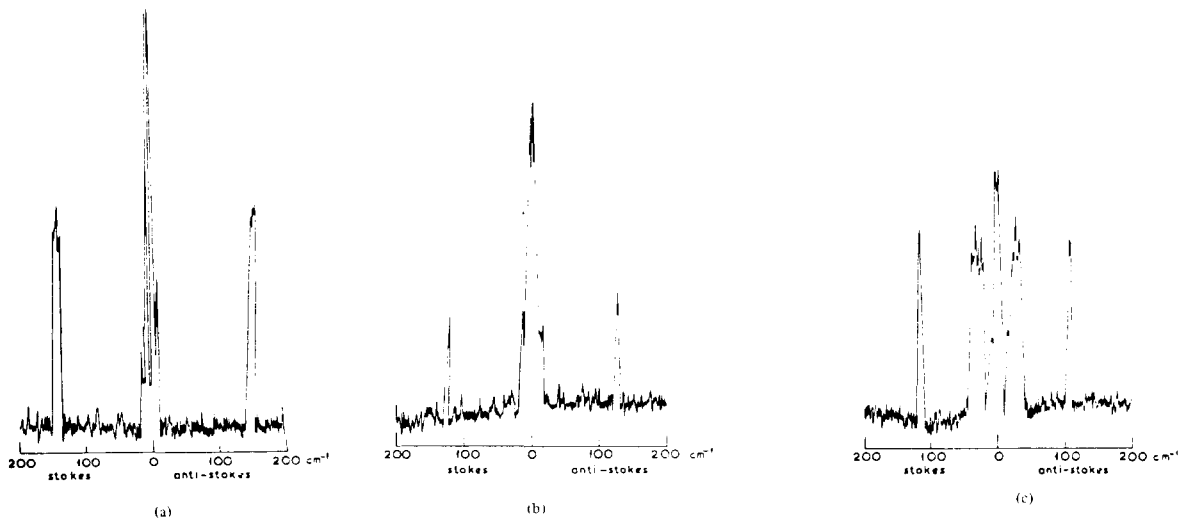

Fig. 14. An example of Stokes and anti-Stokes low frequency spectra of *Escherichia coli* cells obtained at 40, 50 and 60 minutes after resuspension. Taken from Webb et al. [34].

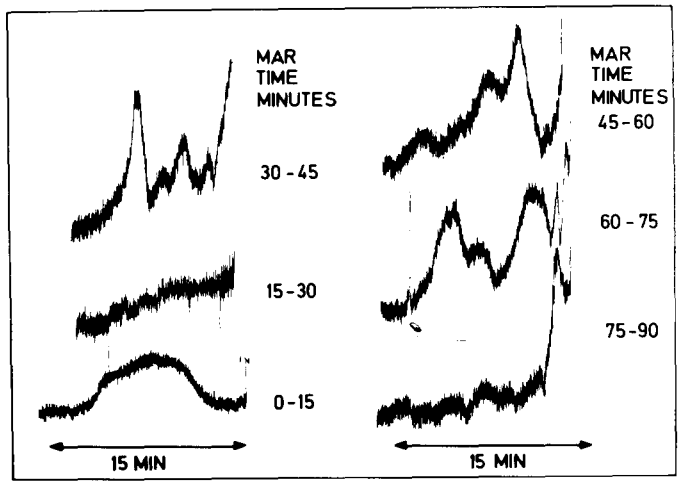

Fig. 15. The changing pattern of anti-Stokes line at 455 cm^{-1} seen during fixed frequency time scans with synchronized cells of *Escherichia coli* MAR TIME = minutes after resuspension over which a 15 minute time scan was conducted. Laser at 5145 Å.

and one line at 455 cm^{-1} so far has been studied. This line was selected because of its relatively low wavenumber and its frequent appearance in normal spectral scans. As can be seen from fig. 15 the characteristics of the line altered considerably with MAR time, also the line did not reappear if a given suspension of cells was left unchanged in the laser light for more than 10–15 minutes, if, at this time, however, a new sample of cells was introduced the line reappeared during the next 15 minute period. Thus this anti-Stokes line behaved in a similar manner to that of many Stokes lines in that its intensity and appearance fluctuated with MAR time and the *in vivo* activity, responsible for the Raman activity, seemingly sensitive to some "bleaching" action of the laser light, if exposure to the light was prolonged. Bleaching was indicated because the length of time, in the laser light, over which the line could be seen, appeared to be inversely proportional to the laser power used.

Cells other than bacteria

In recent times both Stokes and anti-Stokes lines at relatively high shift frequencies have been observed in the spectrum of algal cells exposed for only a millisecond or so to light to activate photosynthesis. The lines, seen at 300 K, however, were those of the carotenoid pigments only, the spectral lines associated with the large chlorophyll molecules were not detected (fig. 16). Chlorophyll lines, however, now have been seen clearly in the spectrum of these algal cells held at low temperatures (fig. 17) and, moreover, in experiments where the temperature has been gradually increased from 6 K to 300 K, these lines have been found to "disappear" from the spectrum at about 273 K, i.e. at the melting point of water. In addition, at 300 K and certain laser frequencies, lines below 200 cm^{-1} which vary greatly in intensity from run to run and with laser wavelength have been observed. A detailed description of these thermal effects will appear shortly in a new publication [37]. The lack of appearance of the chlorophyll lines and indeed other lines, seems to simulate the non appearance of lines in the spectra of resting bacterial cells at 300 K, however, it is possible that some chlorophyll and carotenoid lines overlap.

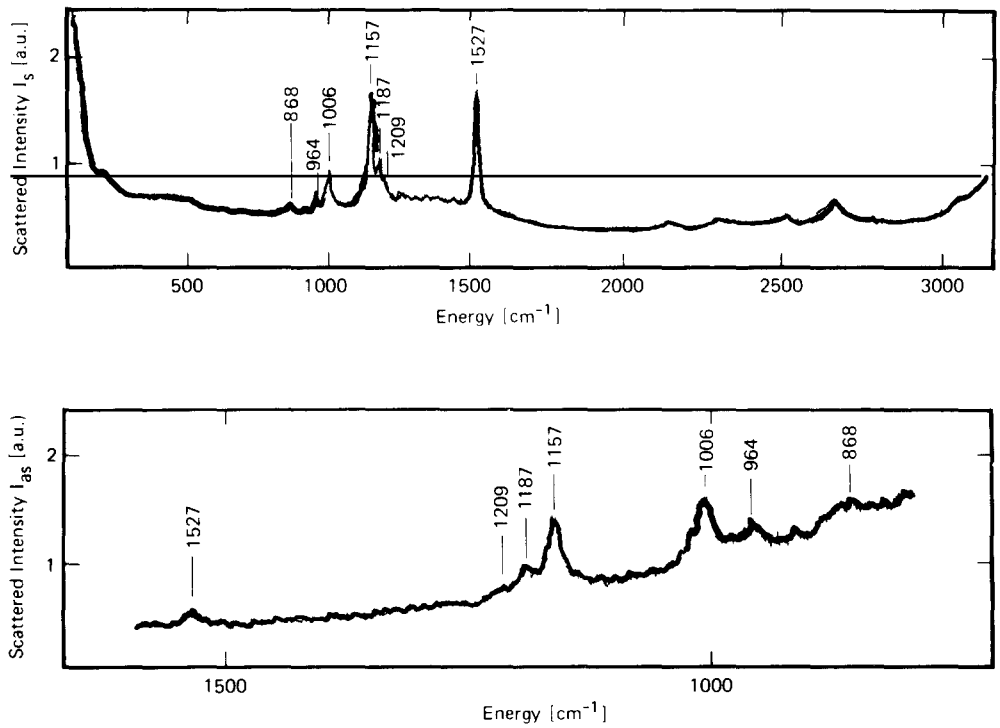

Fig. 16. Raman shift spectra of *Chlorella pyrenoidosa*. Top, Stokes lines; Bottom, anti-Stokes lines. Taken from Drissler and MacFarlane [35]. The Stokes bands shown, have been assigned to vibrational modes of carotenoid molecules by Lutz [34].

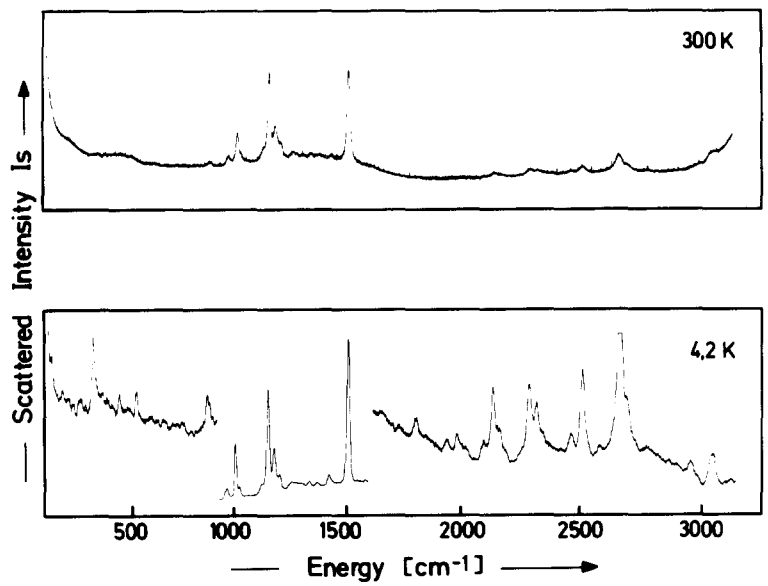

Fig. 17. Stokes spectra of *Chlorella pyrenoidosa* at 300 K (top) and at 4.2 K (bottom). Courtesy of F. Drissler [37].

Normal and tumour animal cells

In fig. 18 are shown some recently determined spectra of "normal" and tumour mammary cells, between 0 and 200 cm^{-1}. These were obtained by taking scraping from the outer regions of freshly removed 2 to 3 diameter mammary carcinomas or pieces of normal tissue [38]. The scrapings, containing pieces of tissue about 1.0 mm in thickness, were washed in Dulbecco's salts solution and then placed in a tissue culture medium similar to that of the commercial 1066 medium. Modifications, however, had to be made in this medium in that dyes and animal serum were omitted because, if present, both produced fluorescences which tended to hinder the characteristics of the spectra obtained. Such cells, if suspended in holding fluids such as Hanks (without dyes) and Dulbecco's basic salt solutions, produced no discernable Raman activity between 0 and 200 cm^{-1} nor did those placed in a tissue culture medium several hours after they had been removed from a patient or subjected previously to "fixing" solutions, only with the fresh tissue was the Raman activity, shown in fig. 17, seen. An interesting feature of these spectra was that the Raman lines from tumour tissue and the "normal" tissue from a diseased patient were broad and double spiked areas whereas the spectral lines from normal tissue, i.e. from a patient found not to be suffering mammary carcinoma, were narrower and sharp. Further studies of this phenomenon seem warranted, therefore, to ascertain whether the pattern of Raman activity seen is characteristic of other tumour cells.

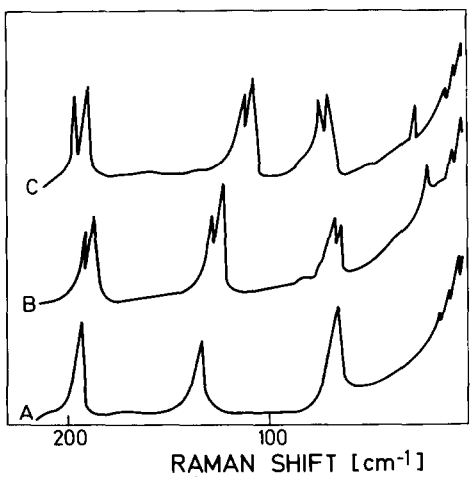

Fig. 18. Low frequency Raman shift spectra of: A, normal mammary tissue; B, "normal" right side tissue; C, left side 2–3 cm diameter mammary carcinoma of a human female. Taken from Webb et al. [38].

Discussion

The studies presented paint a complex picture and pose more questions than answers, a great deal more work will be required, therefore, before any sound conclusions may be made as to what the above findings mean. With this caution in mind, it does appear that at physiological temperatures the resting cell is in a stable condition with few, if any, of its modes Raman active. This stability is thought to be due to the presence and activity of those water molecules which form an intricate lattice *in vivo* in and around macromolecules and their complexes. It may be argued that the Raman lines are not seen in the

spectra of resting cells in suspension is because, at a concentration of 10^7 cells/ml, their total volume is too small. If this was the case, however, no lines would be detected in the spectra of any such suspension but, as demonstrated, the active cell spectra display many lines which often have a very high intensity. If the fine low temperature spectra of *Escherichia coli* is compared with the composite active cell spectra it becomes evident that fewer lines within a given 300 to 400 cm^{-1} band, appear in the active spectra than in the low temperature ones, but because of the small volume of cells used in the former case, the lines in the active spectra are very much stronger. The impression is given then that some form of condensation of closely related states takes place as a result of metabolic activity which gives rise to Raman activity, at a given frequency, of a high amplitude. Thus the Stokes lines in the active cell spectra may arise from a condensation *in vivo* of states excited by metabolic processes as suggested by Fröhlich [5, 6].

An indication that a condensation type of process does take place comes from the ratios of respective Stokes and anti-Stokes lines, and indeed the appearance of anti-Stokes lines above 500 cm^{-1}. Under normal thermal equilibrium conditions the intensity of anti-Stokes lines above 500 cm^{-1} would be too small to be detected while the intensity of those below 200 cm^{-1} would be approximately half that of their Stokes counterpart. As pointed out by Drissler and MacFarlane [35], the intensity ratio R at laser frequency ω is given by:

$$R = \frac{I_{as}^{\omega}}{I_{s}^{\omega}} = A \frac{(\omega - \omega_e - \Omega)^2}{(\omega - \omega_e + \Omega)^2} P[\Omega, T],$$

where A accounts for reabsorption, the second factor, the resonance term (with Ω the vibrational mode frequency) and P the thermal population ratio $(n/(n+1))$ derived from Boltzmann's statistics in which n is the number of vibrators in a particular energy level.

With cells containing a large quantity of molecules which have a strong absorption in the visible at about 480 nm, such as the carotenoids, the anti-Stokes line is nearer resonance with these pigments than is the Stokes line when the laser is set at say 514.5 nm. Near resonance, and in particular for large Ω, R could be influenced, and perhaps dominated, by preferential resonance enhancement in the intensity of the anti-Stokes line due to resonance with the carotenoid absorption band at 480 nm; i.e., mathematically, by the cross sectional factor. In recognition of this the authors [35] attempted to separate resonance factors from the population ratio P by a consideration of the ratio between the Stokes intensity $I_s^{\omega+\Omega}$ (laser at $\omega + \Omega$) and the anti-Stokes intensity I_{as}^{ω} (laser at ω) because under these conditions

$$P = I_{as}^{\omega} / I_{s}^{\omega+\Omega},$$

from which

$$P = \frac{I_{as}^{\omega}}{I_{s}^{\omega+\Omega}} \cdot \frac{I_{s}^{\omega}}{I_{s}^{\omega}} = \frac{I_{as}^{\omega}}{I_{s}^{\omega}} \cdot \frac{I_{s}^{\omega}}{I_{s}^{\omega+\Omega}} = R \frac{I_{s}^{\omega}}{I_{s}^{\omega+\Omega}}.$$

Thus R was measured at a given ω and corrected from a plot of the intensities of the Stokes lines at 868 cm^{-1}, 1157 cm^{-1}, 1006 cm^{-1} and 1527 cm^{-1} found with equal laser power but different wavelengths (fig. 19). With their flow system, in which the cells were exposed to light for only a millisecond or so, they calculated, under these conditions, for the 4 Ω's, population enhancements, over the thermal population factor, of 1.2 to 2.6. They found, however, in preliminary experiments, that, if the cells were exposed to white light for longer periods, the intensity of the Raman lines was increased by approximately 5%; this increase, if proven true would yield a much greater population enhancement

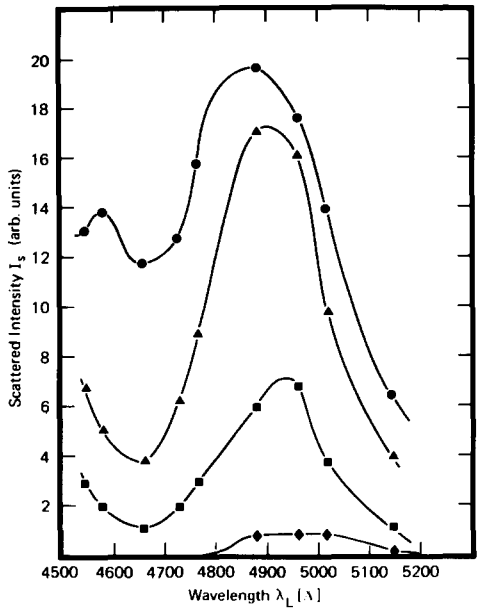

Fig. 19. Intensity dependence of the Stokes shifted bands at 868 cm^{-1} (◆), 1006 cm^{-1} (■), 1157 cm^{-1} (▲) and 1527 cm^{-1} (●) on the wavelength of the incident laser light. Taken from [35].

over the thermal factor. As far as the bacteria studied are concerned the anti-Stokes/Stokes ratio (R) at 150 cm^{-1} had an average value of about 0.8 which, at this frequency, indicates a very large population enhancement over the thermal factor. It is not known, at present, whether resonance enhancements of the above type occur in bacteria able to influence R in the same way. At this time it seems unlikely because they do not possess, in great quantities, the carotenoid pigments present in algal cells and display no absorption in the visible but further work with laser wavelengths other than the 5145 Å and 4880 Å wavelengths used will clarify this point. In any event, in two independent studies the anti-Stokes/Stokes ratios have been found to exceed considerably those predicted by thermal population factors. Thus metabolic processes do appear to excite, and perhaps produce, by non-thermal means specific Raman active modes.

One thing that certainly emerges from the studies presented is the time dependency of the onset of *in vivo* Raman activity at specific frequencies. This suggests that the living cell only allows small "cracks" to occur in its otherwise Raman inactive stable state and somehow controls, in time and frequency, the occurrence of these cracks. It is assumed, therefore, that the great stability and accuracy of function of the living system, which is in a constant state of flux, resides in its ability to confine the occurrence of unstable states, which are required for metabolic activities to take place, to set times in its lifetime. In this way it maintains its dynamic stability by allowing only minute "cracks", or perhaps one crack, to occur at any given time while the rest of the system is held in a stable state. If, as suspected, water molecules are involved in the production of the stable system, then since nearly all *in vivo* metabolic processes involve the addition or subtraction of one or two H_2O molecules to a given molecule, the movement of H_2O at a specific time and place could lead to the "cracks" in the otherwise stable architecture of the cell and lead to Raman activity of the groups with which H_2O directly associates or dissociates. It has been suggested that the known timing and occurrence of events *in vivo* may arise from an overall uniform spiral, or helical motion within the cell which brings together enzymes and their

substrates at specific times. The pitch of this motion, determined by nutritional factors, thus controls when, and if, an event will take place [39, 40, 41, 42]. In recent preliminary studies nutritional factors, both in terms of the type of carbon and nitrogen supplied to the cells, have been found to alter the frequencies and timing of Raman active events but it will take much effort to establish just how each nutritional factor alters the *in vivo* "Raman active time clock".

The Raman lines below 200 cm^{-1} clearly show a different behaviour in time from those above 200 cm^{-1}. First, these lines are present in the spectra over a prolonged period of the cells' life time, even though they may vary in intensity from a mere shoulder on the Rayleigh wings to the strong lines shown in figs. 8, 9, 14 and second, their movement with time appears to be towards lower frequencies. The metabolic process associated with these Raman lines, therefore, seems to be of a different nature from those producing lines at higher shift frequencies. Such low frequency Raman lines have been seen only in the spectra of large crystals of proteins and then it has required special techniques for their presence to be detected. In the living cell enzyme molecules of a given type are known to be clustered together in blocks, thus the lines below 200 cm^{-1} may arise from collective coherent modes peculiar to blocks of proteins, or the cell membrane, which, with time, increase in size as their metabolic synthesis proceeds.

In this report no effort has been made to identify any of the Raman lines with specific metabolic activities, or indeed the particular modes of biosystems. Only in one case has this been attempted by blocking the respiratory chain with carbon monoxide which resulted in severe changes in the characteristics, with time, of a line at 1273 cm^{-1} [33]. To identify each of the lines by the specific inhibition of set metabolic processes obviously will require an enormous amount of work, if the techniques of analysis of many time spectra, used in this study, is employed. What is required is instrumentation capable of displaying simultaneously all the Raman lines present at a given MAR time; such equipment is available but expensive. In addition, there is the danger that the inhibition of one *in vivo* process may affect the behaviour of others, a factor that could lead to falsely identifying a line with a given process. Some method should be found, however, to achieve this goal and perhaps the best one would be to study, as mentioned above, the effects of various types of nutrients on the actual shift frequencies seen and their time of appearance. Much is known about the biochemical and molecular changes induced *in vivo* by nutritional factor and it is possible that this knowledge could be put to use to identify lines.

The "bursts" of lines, seen at set MAR times but occurring within the 3–5 minute synchrony, is a mystery. This pattern could indicate that some sort of metabolic pumping, of modes with frequencies close to the Raman line, occurs over a few seconds which produces a coherent mode of high amplitude. It is difficult to comprehend, however, how this could be seen in the spectra of populations of cells metabolically, "out of phase" by at least 3 minutes. A second possibility is that some sort of resonance enhancement, induced by the laser at the frequencies the phenomenon is seen, takes place and that alternate laser excitation and biologically induced energy transfer produce this type of pattern; perhaps future work will shed some light on the phenomenon.

Time resolved Raman spectroscopy, of time scales from 10^{-1} to 10^{-8} seconds has been achieved with isolated bacteriorhodopsin, a retinal-proteins complex found in some halophilic bacterial cells [43–46]. Such studies have shown changes in the spectrum of the complex within this small timescale, and it may be that the phenomenon has some relationship to "bursts" of lines seen in this study, however, the timescale here is at least one order of magnitude higher.

The very sharp "spikes" with narrow bandwidths, sometimes seen on the tops of weak broad bands, are thought to be monochromatic fluorescences induced by the laser. The emission of virtually monochromatic light by some genera of bacteria and fungi as a result of metabolic activity, is not uncommon. In the present case, however, if the laser light was removed, no such spikes were seen; so they do not appear to arise from the metabolic activities of the cells used in this study.

One other fact that seems to arise from the study presented is that the living cell system appears unique in its ability to control the occurrence of Raman active states. The closest entities to life forms are the viruses and independent studies of aqueous suspension of bacterial [47] and plant [48, 49, 50] viruses have shown them to have a clear reproducible spectra, between 200 cm^{-1} and 3100 cm^{-1} which possess lines of high intensity. These entities are macromolecular complexes of protein and nucleid acid, and much smaller than bacterial cells. The question arises, therefore, as to why viruses display such high Raman activity while resting bacterial cells, in suspension, do not. Locked up in this phenomenon, therefore, may be a key as to why and how viruses are able to infect cells, disrupt and take control of their metabolic activities. Here again studies of the effect of virus infection on the time dependent Raman spectra of cells should prove enlightening and may provide a new tool by which virus infection may be better understood from the standpoint of the physics of the process.

Finally the phenomenon shown as far as the Raman spectra of normal versus tumour cells are concerned [26, 51]. In the one clinical case shown [38], there is a clear difference in the characteristics of the Raman lines below 200 cm^{-1}. It is possible, therefore, that in normal cells degenerate states exist which, as a result of the oncogenic process, becomes resolved and this process produces the broad double spiked peaks seen in the tumour cell spectrum. Should this prove true of other tumour cells, then Raman spectroscopy, and perhaps time dependent spectroscopy, may prove a new method by which tumour, and pretumour, cells may be rapidly identified. Indeed, differences in the Raman spectra of the membranes of such cells already have been reported [52].

The study described, at this stage, has solved very little and has merely confirmed facts cell biologists have uncovered by other means, concerning the existence of an accurate metabolic time clock within living cells. It has introduced nevertheless a different parameter in the form of the physical nature of this clock and a new method by which it may be studied. Clearly, however, the above discussion concerning "why Raman spectroscopy" is grossly inadequate as groundwork for any interpretation to be made here as to the seemingly complex physics governing the observations shown. It is hoped, therefore, that these findings will encourage some theoretical physicists to explore, in detail, their possible meanings. In addition the work seems to have opened a new approach by which a variety of different biological phenomena, such as nutrition, virus infection and oncogenesis may be identified and studied with respect to energetics and time. There is, therefore, much to be done.

Acknowledgements

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Long Range Coherence and the Action of Enzymes

SOME time ago it was proposed that the energy produced in biological activities is partly stored in various materials through excitation of coherent electrical vibrations (polarization waves). If strong enough, such excitations can be stabilized through non-linear effects leading to various types of deformations[1,2]. R. Ferreira (personal communication) has suggested that such considerations might be of importance for an understanding of the action of enzymes. In fact the properties of a model which I have considered recently (unpublished) seems to support this idea.

The model consists of a system of interacting vibrating electric dipoles suspended in a neutral fluid with whose compressional modes they interact in a non-linear way. This system contains the case treated in appendix I of ref. 1. The electric polarization field P of the dipoles will be (but need not be) restricted to one direction. For weak excitation, this system behaves in the usual way, that is, it exhibits polarization waves whose frequency is very insensitive to the wavelength, and whose energy U is proportional to $\overline{P^2}$, the mean value of P^2. For larger excitations, however, the system shows remarkable properties, the most important being the establishment of a metastable state characterized by a non-vanishing mean value P_+ of P, that is, by a permanent mean displacement as it is found in ferroelectric materials. In the present case, however, this state is metastable; it has a higher energy than the undisplaced ground state though it is stable against small displacements. This metastable state is best reached through excitation of coherent modes which have very different characteristics from non-coherent ones. As a consequence of non-linear properties of the model, excitation of a certain amount of incoherent vibrations in the metastable state will lead as an immediate mechanical consequence to excitation of a coherent mode—quite apart from the further arguments for such excitation given in ref. 2.

Assume now that the model may be applicable to enzyme molecules which would possess metastable excited states in which polar groups are stretched so that the molecule may have a very large dipole moment resembling a ferroelectric case, and in which excitations above this state are partly transformed into coherent vibrational modes. An enzyme molecule so excited will have strong long range interaction with other molecules. This interaction will tend to lift other enzyme molecules into the metastable state, the required energy to be provided from the coherently excited modes. The interaction will also lead to attraction of other types of molecules (substrates) to the excited enzyme molecule, especially when they possess a proper frequency close to that of the coherent modes. It is therefore likely that the total energy of the complex of stretched enzyme and substrate is lower than it would be otherwise and this may facilitate the relevant chemical processes in the substrates.

Some of the properties of our model resemble Koshland's hypothesis of conformational changes[4] but they go much further because they imply a stretching of dipoles throughout the large enzyme molecule rather than a deformation near the active site. A principal consequence is the occurrence of the new long range forces which are likely to dominate the dynamic behaviour.

We thus arrive at the following dynamic picture. Assume the first substrate–enzyme complexes to be formed by fluctuations. The energy liberated by the relevant chemical transformations leaves the enzyme molecules coherently excited near their metastable state; return to the ground state would take a relatively long time. Through the long range forces so established they will (*a*) assist other enzyme molecules to transfer to the metastable state and (*b*) attract further substrates. The energy liberated by the chemical processes so acts as a pumping device. At this first stage the number of activated enzyme molecules—and therefore chemical processes—will consequently increase at an accelerating rate, much as in a chain reaction. This will continue until most enzyme molecules are in the metastable state after which the process will continue in a normal way.

These proposals could be tested in three ways. First, by investigation of the build-up of the reaction rate (first stage; chain-like process) as a function of the enzyme density. Second, by direct observation of the excited coherent modes. These modes are optically inactive. Use of Raman effect would, therefore, be appropriate. Through surface effects weak coherent radiation might, however, be observable. Third, by the detection of increase in the static dielectric constant as a result of the expected large increase in the dipole moment of the enzyme molecules when they are transferred to the metastable state.

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G_{M1}-gangliosidosis Types 1 and 2: Enzymatic Differences in Cultured Fibroblasts

G_{M1}-GANGLIOSIDOSIS occurs in two clinical forms[1,2]. Type 1 resembles Hurler's syndrome somatically, but its psychomotor aspect is apparent from birth and progresses rapidly to cause death by 2 years of age. Type 2 presents no Hurler signs and its psychomotor course, though similar to that in type 1, is neither congenital nor is rapidly progressive; death occurs about 5 years of age. In addition to the accumulation of G_{M1}-ganglioside in neural, and to a lesser extent in extraneural, tissues both types share a profound, generalized deficiency of lysosomal β-galactosidase, with storage and excretion of modified keratin sulphates. Although the clinical and biochemical definition of the type 2 form is very recent[3], failure to observe both clinical types within single sibships is sufficient to suggest that each is caused by a different mutant gene. To test this hypothesis we have examined the gangliosides, glycosaminoglycans and residual β-galactosidase activities in fibroblast cell strains derived from skin in the deltoid area of patients with each type of G_{M1}-gangliosidosis. This communication deals only with the last focus of interest.

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Long-Range Coherence in Biological Systems (*).

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1. – Introduction.

The great success of molecular biology arises from the discovery of the detailed structure of biomolecules such as DNA and enzymes. No systematic spacial order was found, and it was concluded that these molecules do not possess a physical order. Physical order, however, does not express itself in spacial order only. Thus superfluid helium, or the electrons in a superconductor exhibit an order that might be termed « motional order ». This type of order is connected with the existence of macroscopic wave functions, which implies the existence of certain quantum phase correlations over macroscopic distances or the existence of coherent states.

Other correlations connected with the phase of oscillating systems rather than with those of macroscopic wave functions are also well known when those systems are lifted from their thermal equilibrium. A well-known example are laser systems which need not exhibit a spacial order as, *e.g.*, in the case of gas lasers.

Many other types of physical order may be imposed on certain systems

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when they are lifted from their thermal-equilibrium state even though they are spacially disordered. This does not hold, however, for all systems, and no general criterium has been found, so far, which when satisfied permits particular systems to exhibit some type of order when energy is supplied to them. Most systems, however, simply raise their temperature.

Biological materials, when they exhibit typical biological properties, must be « fed », *i.e.* energy passes through them. This energy may arise from chemical processes, or it may be supplied by the Sun.

It seems rash, therefore, to conclude from the absence of spacial order —obtained from structural analysis—that no physical order exists in biological systems. On the contrary, it has been suggested several years ago that they belong to those systems in which the supply of energy imposes a characteristic order [1, 2].

In approaching these questions from the point of view of theoretical physics it must be realized that, in contrast to most materials usually treated in physics, biological materials have been selected through long evolutionary processes. As a consequence, they often show quite extraordinary properties, and, rather than ask whether some material properties seem likely, one should ask whether they are possible. Clearly, in the first place, the role of theoretical physics must be to decide what questions should be asked; subsequent developments should then work in close contact with experiments.

It has often been thought that exact knowledge of the atomic structure of a system will provide all answers at least in principle. This is already a basic mistake [3], even in the case of the physics of simple systems. Thus, for example, the above-mentioned concept of macroscopic wave function refers to the whole assembly of particles, *i.e.* it is a collective property which cannot in a simple way be reduced to properties of individual particles.

2. – General.

In looking for a striking material property common to most biological systems one will, no doubt, choose their dielectric properties [1, 2, 4]. Biological systems are, of course, not uniform, but composed of many different such systems. In ordinary physics, one would then extract each subsystem, *e.g.* a certain type of molecule, and determine its dielectric properties, *e.g.* its dipole moment. In solution this dipole moment, say μ_i, would be slightly different from the moment μ_v in vacuum owing to polarization effects. Such a procedure would be highly misleading in the biological case, say for water-soluble proteins. Their conformation is largely dominated by the interaction of their charged groups with the surrounding water and ions which thus form an integral part of the structure and hence of the measured dipole moment. Under these circumstances, very high internal electric fields may exist in such systems.

Most striking, of course, are the fields in cell membranes which have average values of the order of 10^5 V/cm, which in ordinary materials would require great precautions to avoid breakdown. Inhomogeneous regions, *e.g.* through proteins dissolved in the membrane, might lead to much higher local fields. These fields play a basic role in nerve conduction, but here weaker fields could fulfil the same purpose. Their role in the membranes of other cells is unknown.

One notices that the energy difference of a unit electronic charge across the membrane (about 10^{-6} cm) would, at room temperature T, amount to about $3kT$. This suggests that the transport of single charges obtains a significance against thermal noise.

Also in a field $F \sim 10^5$ V/cm the energy μF is larger than kT when μ is the dipole moment of a small protein, say $\mu \sim 300$ Debye units. This suggests that proteins dissolved in membranes are oriented by the field.

In addition, however, the whole membrane system is strongly polarized by the field. As a consequence, an oscillation of a section of the membrane—or of certain proteins dissolved in it—is connected with a corresponding electric vibration, so that the longest electric wave usually yields an electric-dipole vibration. According to one of our main suggestions, such vibrations are very strongly, coherently, excited in biological systems when they are active, *i.e.* when metabolic energy is available.

Now it must be realized that the displacement of a single ion by 10^{-6} cm produces a very large dipole moment of the order of 500 Debye units. It would be wrong, however, to neglect other processes arising from such displacements, for the change of electric field arising from this displacements may substantially alter the polarization in the neighbourhood of the ion (note that the electric field at a $3 \cdot 10^{-7}$ cm distance from the electronic charge is of the order of 10^6 V/cm) and cause other ions to move. Under these circumstances, one would expect such displacements to become stabilized, and our second principal suggestion states that biological systems have metastable states with very high electric polarization.

How can one support theoretically such general statements referring to a variety of complicated materials? It is already clear from the above that we are dealing with strongly nonlinear effects. This provides possibilities only occasionally contemplated in ordinary physics. Our method will be to formulate simple, but general models which do show the required properties. General conditions required in these models then impose corresponding conditions on the respective biological systems. If experiments confirm the theoretical predictions, then we can conclude that the molecular evolution has led to the selection of materials with the required properties. This does not imply that other systems do not exist which do not have the required properties, though from a structural point of view they could be very similar to the systems which

do have these properties. The difference between the two would then rest in their nonlinear behaviour, *i.e.* in properties of certain excited states which would not be revealed in structural analysis.

3. – The vibrational model.

Any system containing charged particles is capable of long-wave electrical vibrations. Such a system need not be homogeneous. One of the characteristics of these vibrations is the existence of a lowest frequency $\omega_1/2\pi$, different from zero. Our system is assumed to possess such modes, $\omega_1, \omega_2, \ldots, \omega_z$. They are considered not to interact with each other, but to be strongly coupled to the surroundings, *i.e.* to the other modes of the systems which will be treated as a heat bath at temperature T. Let n_k be the number of quanta $\hbar\omega_k$ in the mode ω_k. Then, through first-order processes, single quanta $\hbar\omega_k$ may be emitted to or absorbed from the heat bath, so that the rate of change $\dot{n}_{k,1}$ of n_k due to these processes is [2, 5]

$$\dot{n}_{k,1} = -\varphi\big(n_k \exp[\beta\hbar\omega_k] - (n_k + 1)\big)\,, \tag{3.1}$$

where $\beta = 1/kT$, and φ will depend on T and, in general, also on ω_k. For simplicity we shall neglect the ω_k-dependence. The ratio of emission to absorption processes is governed by detailed balance.

Similarly, for second-order processes we find

$$\dot{n}_{k,2} = -\chi \sum_j \big(n_k(1 + n_j) \exp[\beta\hbar\omega_k] - n_j(1 + n_k) \exp[\beta\hbar\omega_j]\big)\,. \tag{3.2}$$

Again, apart from depending on T, the transition probability χ could depend on k and j, but we shall neglect this. This will greatly simplify the calculations, but should have no influence on the main results.

We note that, on this choice,

$$\sum_k \dot{n}_{k,2} = 0\,, \tag{3.3}$$

i.e. such processes lead to interchange of quanta $\hbar\omega_k$ without alteration of their total number.

Now we consider that energy is supplied at the rate s_k to the mode ω_k. We then find, for the total rate of change $\dot{n}_k$,

$$\dot{n}_k = s_k - \varphi\big(n_k \exp[\beta\hbar\omega_k] - (n_k + 1)\big) - \\ - \chi \sum_j \big(n_k(1 + n_j) \exp[\beta\hbar\omega_k] - n_j(1 + n_k) \exp[\beta\hbar\omega_j]\big)\,. \tag{3.4}$$

We wish to solve this equation for the stationary case, *i.e.* when

$$\dot{n}_k = 0 \,. \tag{3.5}$$

We may derive the general behaviour of n_k in a simple manner, though it would be difficult to solve this system of quadratic equations exactly.

Summation over k with the definition

$$S = \sum s_k \tag{3.6}$$

yields

$$S = \varphi \sum_k \left(n_k \exp\left[\beta\hbar\omega_k\right] - (n_k + 1)\right) . \tag{3.7}$$

Introducing the total number of quanta N,

$$N = \sum n_k \,, \tag{3.8}$$

we find with

$$\omega_1 \leqslant \omega_k \leqslant \omega_z \tag{3.9}$$

that $\left(\text{note } \sum_k 1 = z\right)$

$$N\left(\exp\left[\beta\hbar\omega_z\right] - 1\right) > S/\varphi + z > N\left(\exp\left[\beta\hbar\omega_1\right] - 1\right) . \tag{3.10}$$

This means that N must increase proportional to S if the supply S is sufficiently large.

Now define μ by

$$\exp\left[-\beta\mu\right] = \frac{\varphi + \chi \sum (1 + n_j)}{\varphi + \chi \sum n_j \exp\left[\beta k\omega_j\right]} , \tag{3.11}$$

so that

$$1 - \exp\left[-\beta\mu\right] = \frac{\chi \sum \left(n_j \exp\left[\beta\hbar\omega_j\right] - (1 + n_j)\right)}{\varphi + \chi \sum n_j \exp\left[\beta\hbar\omega_j\right]} . \tag{3.12}$$

From (3.7) it then follows that

$$\mu \geqslant 0 \tag{3.13}$$

and

$$1 - \exp\left[-\mu\beta\right] = \frac{\chi}{\varphi} \frac{S}{\varphi + \chi \sum n_j \exp\left[\beta\hbar\omega_j\right]} \geqslant 0 . \tag{3.14}$$

Equation (3.4) with (3.5) can thus be formally solved by

$$n_k = \left(1 + \frac{\varphi}{\chi} \frac{s_k}{S} \left(1 - \exp\left[-\beta\mu\right]\right)\right) \frac{1}{\exp\left[\beta(\hbar\omega_k - \mu)\right] - 1} , \tag{3.15}$$

and $n_k \geqslant 0$ together with (3.13) yields

$$\hbar\omega_1 > \mu \geqslant 0 \,. \tag{3.16}$$

This inequality permits the establishment of an upper limit for N although μ is a complicated function of all the n_k. For (3.15) then yields

$$N = \sum n_k < \left(1 + \frac{s_m(1 - \exp[-\beta\hbar\omega_1])\,\varphi}{\chi S}\right) \sum_k \frac{1}{\exp[\beta(\hbar\omega_k - \mu)] - 1}\,, \tag{3.17}$$

where s_m is the largest of the s_k.

Clearly, if the sum in (3.17) can be replaced by an integral, then for $\mu \sim \hbar\omega_1$ a value independent of the s_k will be reached. Moreover, always $s_m/S \leqslant 1$, so that an upper limit for N independent of S is obtained. This contradicts the conclusion drawn from (3.10), according to which N must increase proportional to S if S is sufficiently large. This situation is the same as normally found in Einstein condensation of a Bose gas. Thus the replacement of the sum by an integral is incorrect; μ takes a value close to $\hbar\omega_1$ such that n_1 becomes very much larger that all the other n_k.

It follows then that a single mode ω_1 gets very strongly (coherently) excited when the energy supply is larger than a critical one.

As special cases we may consider that all s_k are equal. Then

$$s_k/S = 1/z \tag{3.18}$$

and (3.15) becomes

$$n_k = \left(1 + \frac{\varphi}{\chi_z}\left(1 - \exp[-\beta\mu]\right)\right) \frac{1}{\exp[\beta(\hbar\omega_k - \mu)] - 1}\,. \tag{3.19}$$

As another extreme consider that all s_k vanish except for $k = k_0$.

Then

$$s_k = S\delta_{k,k_0}\,, \qquad s_{k_0} = S \tag{3.20}$$

and

$$n_k = \left(1 + \delta_{k,k_0}(1 - \exp[-\beta\mu])\,\frac{\varphi}{\chi}\right) \frac{1}{\exp[\beta(\hbar\omega_k - \mu)] - 1}\,. \tag{3.21}$$

To understand qualitatively these conclusions, we notice first of all that the case $\chi = 0$ leads to a completely trivial result, namely

$$n_k = \left(1 + \frac{s_k}{\varphi}\right) \frac{1}{\exp[\beta\hbar\omega_k] - 1}\,, \qquad \text{if } \chi = 0\,, \tag{3.22}$$

i.e. to a modified thermal-equilibrium Planck distribution. It will be noticed from (3.11) that $\chi = 0$ in fact entails $\mu = 0$ and thus eliminates the occurrence of Bose condensation. Thus the nonlinear terms χ arising from a strong interaction with the heat bath tend to force the system into a thermal-equilibrium distribution making use of a chemical potential μ.

A main difference from ordinary condensation of a Bose gas in thermal equilibrium should also be noted. There the ordered (condensed) state is reached by lowering the temperature. In our case, the ordered state is reached by supplying sufficient external energy to the system. Quite a similar situation prevails with respect to the second of our concepts (sect. **4**).

The above model, naturally, is a very simplified version of any actual situation. It has a very general consequence which might not depend on modifications required in particular cases, namely the coherent excitation of a polar mode when energy supply exceeds a critical magnitude.

Which frequencies should we anticipate? Roughly speaking, if a polarized system of linear dimensions l carries out long-wave elastic vibrations (*e.g.* breathing vibrations), then the electric polarization of the system implies coexistence of an electric-polarization vibration with the same frequency

$$\nu \sim v/l\,, \tag{3.23}$$

where v depends on the elastic constants and can formally be expressed as a velocity of sound. Thus $l = 10^{-6}$ cm and $v = 10^5$ cm/s would yield $\nu = 10^{11}$ Hz, but $l = 3 \cdot 10^{-7}$ cm (small protein) with $v = \frac{1}{3} 10^5$ cm/s yields, of course, the same value. The band of polarization waves ω_k would then consist of the interacting vibrations of such units, and also of higher vibrational states of the same unit. It must be emphasized again that the existence of long polarization waves does not require homogeneous material.

A further condition is the absence of unduly high friction, which probably will mean that surfaces facing water should not move. This does not hold, however, for structured water, which is to be considered as an integral part of the particular biological unit. Most likely, these conditions are satisfied by proteins in or near membranes. Thus, if $\Delta\nu$ is the frequency width, we have in this case

$$\nu \sim 10^{11}\ \mathrm{Hz}\,, \qquad \frac{\Delta\nu}{\nu} \ll 1\,, \qquad s > s_0 \tag{3.24}$$

as condition for the excitation of coherent vibrations. Here s is an appropriate average over the rate of energy supply and s_0 is the threshold value for Bose condensation.

It should be emphasized that other frequencies, both higher and lower, are well feasible. Thus larger units such as DNA-protein complexes might

well possess lower frequencies. Higher ones, on the other hand, may be based on a combination of the various rotational and vibrational subgroups of relevant molecules.

4. – The high-polarization model.

Consider [6] a polarizable material, polarization P. Its mean « self-energy » can then be written as $\frac{1}{2}a^2\overline{P^2}$, where the bar indicates an appropriate average and the positive a^2 depends on the polarizability, including size and shape of the material. These definite size and shape imply the elastic energy of the material to be a minimum relative to small deformations. If these are measured in terms of a deformation parameter η, then on deformation the elastic energy increases by $\frac{1}{2}\eta^2\varepsilon$, where $\varepsilon > 0$ depends on the elastic properties. In the case of a sphere, for instance, the deformation could produce an ellipsoid with eccentricity η. Now the polarizability in general is not an extremum at $\eta = 0$, so that a^2 would have to be replaced by $a^2(1 + c\eta)$, where c may be positive or negative. The total contribution to the free energy thus amounts to

$$U = \tfrac{1}{2}a^2(1 + c\eta)\,\overline{P^2} + \tfrac{1}{2}\eta^2\varepsilon\,. \tag{4.1}$$

Minimizing with regard to η at fixed $\overline{P^2}$ yields

$$\eta = -\frac{1}{2}\frac{ca^2}{\varepsilon}\,\overline{P^2}, \tag{4.2}$$

so that

$$U = \frac{1}{2}a^2\,\overline{P^2}\left(1 - \frac{1}{4}\frac{c^2}{\varepsilon}a^2\,\overline{P^2}\right). \tag{4.3}$$

Thus increasing $\overline{P^2}$ ultimately may lead to decreasing U. At this stage, the linear approximation η and the quadratic one in P will probably have become invalid. Higher-order terms may thus reverse this trend and lead to a minimum of U at a certain $\overline{P^2}$. If U at this value is negative, then the material would acquire a spontaneous dipole moment. If U is positive, then a metastable state with spontaneous polarization exists.

These conclusions are of a very general nature. They follows from the shape dependence of the polarizability and require the material to be sufficiently soft, *i.e.* ε should be sufficiently small.

They can be extended to a dynamical model composed of a polarization field $\boldsymbol{P}$ and an elastic (acoustic) field $\boldsymbol{A}$. Let ω_0 be the circular frequency of long-wave polarization waves in the absence of any interaction with the elastic field, then with such an interaction the density of potential energy has the

form

$$W = \tfrac{1}{2}\omega_0^2 \boldsymbol{P}^2 + \left(\tfrac{1}{2}v^2(\operatorname{div}\boldsymbol{A})^2 + c\boldsymbol{P}^2 \operatorname{div}\boldsymbol{A}\right)(1 - d^2\boldsymbol{P}^2)\,, \tag{4.4}$$

where v is the velocity of sound, d^2P^2 is an *ad hoc* term introduced to achieve stabilization for high P^2.

Note that, for long waves,

$$\varrho = \operatorname{div}\boldsymbol{A} \tag{4.5}$$

may be introduced as change in density. The static case requires minimization of (4.4) with regard to ϱ and P^2, so that for homogeneous displacements

$$\varrho = -\frac{cP^2}{v^2}\,, \tag{4.6}$$

yielding

$$W = \frac{1}{2}\omega_0^2 P^2 - \frac{1}{2}\frac{c^2}{v^2}(1 - d^2P^2)\,. \tag{4.7}$$

Apart from the different notation, this result corresponds to (4.3) except that the d^2P^2 term leads to a stabilization at high P^2.

Now we can introduce the density of kinetic energy

$$K = \tfrac{1}{2}(\dot{\boldsymbol{P}}^2 + \dot{\boldsymbol{A}}^2) \tag{4.8}$$

and hence from the Lagrangian density L obtain the equations of motion in the usual way,

$$\delta\int L\,\mathrm{d}^3\boldsymbol{r}\,\mathrm{d}t = \delta\int (K - W)\,\mathrm{d}^3\boldsymbol{r}\,\mathrm{d}t = 0\,, \tag{4.9}$$

where $\boldsymbol{r}$ is the space vector and t the time. The assumption that acoustic phonons are absent then permits the elimination of the $\boldsymbol{A}$-field and yields an equation of motion of the form

$$\ddot{\boldsymbol{P}} + \frac{\partial V(P)}{\partial \boldsymbol{P}} = 0\,. \tag{4.10}$$

The effective potential $V(P)$ has the form

$$V(P) = A + BP^2 + CP^4\,, \tag{4.11}$$

apparently a harmless anharmonic oscillator. This is misleading, however, for the coefficients A, B, C depend on the mean square amplitude

$$p = \overline{P^2}\,, \tag{4.12}$$

such that

$$A = \frac{1}{2}\frac{c^2}{v^2}p^2, \tag{4.13}$$

$$B = \frac{1}{2}\left(\omega_0^2 - 2\frac{c^2}{v^2}p - \frac{d^2c^2}{v^2}p^2\right), \tag{4.14}$$

$$C = \frac{c^2d^2}{v^2}p\,. \tag{4.15}$$

For small p clearly the CP^4 term is negligible and B becomes $\omega_0^2/2$, so that an harmonic oscillation with frequency ω_0 is obtained.

For increasing p, however, the magnitude of B decreases, *i.e.* the mode softens. For appropriate values of the parameters it is then found that a metastable state with permanent polarization $\pm P_0$ exists and that the system has two branches of oscillation: one around $P = 0$ and another with limited amplitude around $\pm P_0$. In this case, the effective potential $V(P)$ has a maximum at $P = 0$ and two minima at $P = \pm P_0$. The value of P_0 is found by minimizing (4.7), which yields

$$P_0^2 = p_0 = \frac{1}{3d^2}\left(1 + \sqrt{1-\lambda}\right), \tag{4.16}$$

where

$$\lambda = \frac{3v^2d^2\omega_0^2}{c^2}. \tag{4.17}$$

Thus $\lambda < 1$ is a necessary condition for the existence of a P_0, and it can be shown that this state is metastable, *i.e.* has positive energy when

$$\frac{3}{4} < \lambda < 1\,. \tag{4.18}$$

For $\lambda < \frac{3}{4}$ its energy is negative, *i.e.* it becomes the (ferroelectric) ground state.

We notice again that small v^2, *i.e.* softness of the material, favours the establishment of a polarized state. We also note that in many ferroelectric materials the effective potential has a minimum at $P = 0$ at high temperatures. The mode softens as we decrease the temperature towards the Curie point and below we find two minima at $P \pm P_0$.

In our case $V(P)$ has its minimum at $P = 0$ at low energies, but supply of energy may establish the quasi-ferroelectric state with minima at $P = \pm P_0$. Thus here as in the case of sect. **3** a long-range order—coherent vibrations, or permanent polarization—is established through the supply of energy.

Clearly actual biological materials will behave in a much more complex way than discussed in this simple model. We should recall in this case that structured water and various ions are to be considered as an integral part of the system.

This implies considerably higher nonlinear effects than discussed above. While these will have to be treated case by case, it seems that they usually will contribute to the stabilization of our metastable state. Thus a polarized molecule will shift mobile ions so that its energy is lowered; in turn, this will, of course, affect the magnitude of polarization.

5. – General consequences.

The two types of excitation discussed in sect. **3** and **4** can have far-reaching biological consequences. First of all, it should be noticed that they require supply of energy, *i.e.* activation. Both can then lead to forces which were absent before activation, and to storage and transport of energy.

Cell water contains considerable amounts of small ions which tend to screen any static electric charges at short range. Hence a system activated to a high dipole moment (sect. **4**), *e.g.* a protein, may exert considerable forces in water-free regions, but will be screened for long-range interaction. This screening does not hold, however, for an oscillating field at frequencies of the order of 10^{11} Hz (sect. **3**). Thus, if the coherently excited vibration represents an oscillating giant dipole, then long-range forces may be activated. They will be shown to be frequency selective.

Consider two oscillating giant dipoles at distance R with frequencies ω_e and ω_s, respectively, in the absence of interaction [4, 7]. At sufficient distance, their interaction is essentially dipolar, since other interactions decrease much more rapidly with R. The combined system then has two normal modes ω_+, ω_-, where

$$\omega_\pm^2 = \tfrac{1}{2}(\omega_e^2 + \omega_s^2)\left(1 \pm (q^2 + Q^2)^{\frac{1}{2}}\right). \tag{5.1}$$

Here (by assuming $\omega_e^2 \geqslant \omega_s^2$)

$$q = \frac{\omega_e^2 - \omega_s^2}{\omega_e^2 + \omega_s^2} > 0 \tag{5.2}$$

and Q essentially is the ratio of the interaction energy between the two giant dipoles and their internal potential energies.

It thus depends on the distance as $1/R^3$.

In the particular case that the giant-dipole oscillations in the two molecules consist in correlated oscillations of, respectively, z_e and z_s elastically bound ions of charge e and of equal mass M, then

$$Q = \gamma e^2 (z_e z_s)^{\frac{1}{2}} / M R^3 (\omega_e^2 + \omega_s^2), \tag{5.3}$$

where γ is a numerical constant.

Clearly the expression for ω_-^2 holds only when $\omega_-^2 > 0$; at short distance thus mode softening occurs.

The interaction energy I is defined as the difference of the free energy F at distance R from its value at $R \to \infty$. Now consider the case that the state with the lower frequency has been coherently excited (n_- quanta) so that the contribution of ω_+ to F can be neglected. Then

$$I = F(R) - F(\infty) = n_- \hbar(\omega_-(R) - \omega_-(\infty)) = \tag{5.4}$$

$$= \frac{1}{\sqrt{2}} n_- \hbar(\omega_e^2 + \omega_s^2)^{\frac{1}{2}} \left\{(1 - \sqrt{q^2 + Q^2})^{\frac{1}{2}} - (1 - q)^{\frac{1}{2}}\right\}.$$

This energy is always negative, *i.e.* attractive. Furthermore, if $\omega_e \neq \omega_s$, then at sufficient distance $Q^2 < q^2$ and we find, after development,

$$I = -\frac{\hbar n_-}{4\sqrt{2}} (\omega_e^2 + \omega_s^2)^{\frac{1}{2}} \frac{Q^2}{q(1-q)^{\frac{1}{2}}} \propto -\frac{1}{R^6}. \tag{5.5}$$

If, however, $\omega_e^2 \simeq \omega_s^2$ and $Q^2 \gg q^2$, $q^2 \ll 1$, then

$$I \simeq -\frac{\hbar n_-}{\sqrt{2}} (\omega_e^2 + \omega_s^2)^{\frac{1}{2}} Q = -\frac{\hbar n_-}{\sqrt{2}} \frac{\gamma e^2 (z_e z_s)^{\frac{1}{2}}}{M(\omega_e^2 + \omega_s^2)^{\frac{1}{2}}} \frac{1}{R^3} \propto -\frac{1}{R^3}. \tag{5.6}$$

In this case of near resonance, the always attractive interaction has very long range. Roughly, if

$$U = \hbar n_- \omega_e \tag{5.7}$$

is the energy of excitation, using $\omega_e \simeq \omega_s$, we have

$$I \simeq -U \frac{e^2 (z_e z_s)^{\frac{1}{2}}}{M \omega_e^2 R^3}. \tag{5.8}$$

While thus activation of coherent vibrations may lead to long-range, frequency-selective interaction, activation of the polar state leads to short-range nonselective forces which might facilitate transport of ions within the particular material, or across it.

Long-range selective forces may lead to selective transportation of molecules like enzymes and thus initiate a particular trend of chemical events. It might also be relevant for the transportation of larger units.

It has been suggested that these properties are an essential feature in the high catalytic power of enzymes [4, 6, 8]. Thus the high internal fields arising from the strongly polarized state could reduce activation energies, while excitation of coherent modes could be used for the transport of energy.

Independently [9], the same conclusion has been reached from an analysis of a great variety of enzymatic reactions.

In [10] it has been suggested that the selective long-range interaction (sect. **3**) may be responsible for the attraction of a homologous pair of chromosomes in meiosis—an interaction which according to [11] has till then remained « a great mystery ».

A number of other possible consequences has been discussed elsewhere [1, 2, 4, 12].

Selective long-range interaction in conjunction with the existence of highly polar metastable states may also be decisive for the establishment of the well-known low-frequency electric vibrations in the brain, whose biological significance is only now being explored [13]. It has been shown that the long-range coherent interaction (based on the 10^{11} Hz coherent vibrations) could in an enzyme-substrate system lead to collective enzymatic reactions resulting in slowly varying (periodic) excitation and de-excitation of the system of enzymes from the only slightly polar ground state into the highly polar metastable state. Thus a slow chemical oscillation is connected with a corresponding electric vibration. The strong electric interaction between the highly polar states in conjunction with the considerable damping of electric currents then imposes on them a limit cycle condition, which recently has been investigated [14]. This model requires large regions to oscillate coherently as has actually been found [15].

6. – Experimental evidence.

In active biological systems experimental evidence for the existence of coherent electric oscillations in the 10^{11} Hz region would require methods which have not yet been fully developed (cf., however, Note added in proofs). Evidence does exist, however, that radiation in this frequency region may have very strong effects on such systems. The most comprehensive report on such effects has been presented in a meeting of the USSR Academy of Science [16]. Properties of a great variety of objects have been found to be strongly influenced by coherent electromagnetic waves in the $(5 \div 7)$ mm (vacuum) wave-length region. Almost all experimental results have in common:

i) That they very strongly depend on frequency in a region in which the optical constants have the appearance of broad bands; often the biological effects showed typical resonance character.

ii) The dependence on intensity exhibits a threshold below which no biological effects occur and above which they are nearly independent of intensity.

To interpret these results [17], we assume that these biological effects are initiated by a first step which is sensitive to a radiation followed by a chain of other possibly very complicated effects which are insensitive to it. The

first step might be connected with a long-range interaction arising from coherent excitation of a molecule, *e.g.* an enzyme which could be attracted to its substrate. Let the time required for this be $t_1(s)$, where s represents the radiative supply of energy to it. According to sect. **3**, excitation of the coherent mode requires a threshold s_0. With growing s ($> s_0$) the time $t_1(s)$ will decrease, so that the total time for the whole process

$$t = t_1(s) + t_2 \to t_2 \tag{6.1}$$

will approach the s-independent time t_2 required for the s-independent chain of further biological steps. Thus in these experiments we find

$$\nu \sim \frac{1}{2} 10^{11}\,\mathrm{Hz}\,, \qquad \frac{\Delta\nu}{\nu} \ll 1\,, \qquad s > s_0\,, \tag{6.2}$$

as predicted in (3.24).

We note that the resonancelike frequency dependence in a region in which the optical constants show little frequency dependence is the best proof that we are not dealing with thermal effects.

The magnitude of s_0 has been estimated theoretically [18] from calculations using methods of laser physics and found of the order 0.04 mW/cm², which agrees with some of the experimental results (*).

The Russian experiments range from various effects on bacteria, growth of yeast cultures, influence on human and animal cells to effects on various genetic properties. Depending on the type of process stimulated by the radiation, a certain biological activity might be increased or decreased. Independently, on a smaller scale and without extensive frequency resolution, biological effects near $0.7 \cdot 10^{11}$ Hz and $1.4 \cdot 10^{11}$ Hz have been found by other investigators [19, 20].

In some of the Russian experiments [21], the biological resonance has been found as narrow as $2 \cdot 10^8$ Hz. This can be understood on the assumption that the only broadening of the excited giant-dipole vibration arises from its electric coupling to the thin layers of structured water whose dielectric absorption lies in this region [22, 23]. The remarkable absence of other frictional effects reminds one, of course, of the various suggestions of the existence of room temperature superconductive regions in biological systems. So far no definite experimental proof for their existence has been found. These narrow resonances are most likely to occur when the level which is excited by the radiation is identical with the level which is strongly populated in Bose condensation,

(*) It should be mentioned that in [18] the presentation of a quadratic Hamiltonian is misleading, though the results are correct. For, as required, the authors do use the influence of a heat bath which in terms of a Hamiltonian would require at least 4th order terms to obtain their eq. (8*a*)—as can be seen from our eq. (3.2) which is responsible for the occurrence of a chemical potential μ.

i.e. when in (3.21) this is the level K_0. Broader bands must be expected when this is not the case.

In experiments of this type, one must be prepared for unexpected non-linear effects of the imposed radiation. Thus, within the range in which the radiation is absorbed, in the first place coherent excitation of some molecules is expected. If the same type of molecules is present at greater depth, then they in turn may become excited by transfer of the excitation energy from the outside molecules. Thus the excitation may be expected to move beyond the penetration depth of the imposed radiation.

Now, in turning to the excitation of the strongly polar metastable state, one would, of course, attempt to excite this state by bringing the molecule in question into a strong electric field. It should be of interest in this context that investigations on nerve membranes have led to the conclusion that the electric field in the membrane makes it nearly impermeable to Na^+ and K^+ ions, whereas the absence of the field leads to a much higher permeability presumably through proteins dissolved in the membrane; it is suggestive to attribute these two different properties to the two states in the proteins.

More direct evidence for conformational changes of certain molecules in membranes due to changes of the electric field is available [24].

A number of other possibilities has been discussed in [4], including results found with the electret method [25] and with laser-induced stimulation of enzyme activity at very high dilution [26].

Thus, while experiments available at present seem to support the theoretical predictions, in a general way much greater refinement of the experimental methods is desirable to obtain more specific experimental results. This in turn should stimulate theoretical investigations of a more specific nature.

7. – Cancer.

The problem of cancer is one of the control of cell division. While bacterial cells under appropriate conditions keep on dividing, cells in a fully grown differentiated tissue or organ divide relatively rarely, though they divide more rapidly in growing tissues and organs. We shall assume that a controlling agency belonging to the whole organ, *i.e.* a collective property, exists and that individual cells respond to it.

The nuclei of these cells contain a DNA-protein complex, whose detailed structure is not well known, though it seems reasonable to assume that its spacial arrangement differs in different organs and depends on the type of differentiation that has taken place. Now we shall assume that these materials are capable of electric vibrations which can be excited coherently in the sense of our model (cf. sect. **3**). Thus metabolic energy would be used to excite these vibrations at a well-defined frequency (or frequencies) which would be char-

acteristic for the particular organ or tissue. Note that experiments discussed in sect. **6** confirm the existence of such vibrations in some biological materials.

This then implies that a coherent electric vibration carried collectively by the individual nuclei extends through the whole tissue (organ), and that the vibrations of each individual cell respond to it and are thus held in the appropriate phase, such that the total electric energy is a minimum. In general, this vibrating polarization field will form a complicated spacial pattern.

Now, preceding cell division, the DNA-protein complex of the nucleus of a cell must unfold and may then be expected to change some of its physical characteristics, such that it looses its resonance with the collective mode. This implies that its energy of interaction with the collective mode is increased, *i.e.* energy has to be spent to perform this change (apart from changes in the internal energy). This energy may not always be available and, as a consequence, the interaction of an individual cell with the collective mode acts as an inhibition on cell division. The strength of this inhibition depends on the magnitude of excitation of the collective mode. This excitation may well depend on the size of the tissue because of the long range of the interactions as discussed in sect. **5**. The possibility of a size-dependent inhibition on cell division thus exists.

A situation thus prevails which in some respects corresponds to order in physical systems described in terms of an order parameter as, for instance, in lasers (cf. also [27]). In our case, the collective vibration represents the order parameter. It is built up of the contributions of the relevant vibrations of the individual cells, and they in turn obey the instruction of the order parameter, namely not to divide.

Now assume that changes have taken place in a particular cell such that its frequency is altered. This cell no longer contributes to the coherent collective vibration, whose intensity will thus be slightly reduced, nor does this cell obey the instruction of the coherent mode, *i.e.* the inhibition to divide disappears. The slight reduction of intensity of the coherent mode due to the change of property of a single cell should have a minute effect on other cells. A critical intensity should exist, however, below which the control of cell division no longer is effective. It is feasible, therefore, that something like a phase transition takes place from the controlled to the uncontrolled state once a critical number of cells has been affected.

Such an approach requires that—while the controlling vibration is strong enough—a cell in the tissue will be induced to transfer into the correctly vibrating state provided it possesses such a state, for only thus can the system exhibit stability.

This may have two types of consequences:

i) Differentiation may be expected to consist in rearrangements of the DNA-protein complex, such that it obtains relevant frequencies assumed

absent before differentiation. Under the influence of the coherently excited mode(s) of the particular tissue (organ), this transition is energetically favourable and hence is stimulated [28]. The differentiated cell then gets subjected to the control.

ii) Mutations which lead to alterations in the relevant frequency(cies) are energetically less favourable than those which do not alter them and which thus still respond to the control and are, therefore, statistically less likely.

Thus, in our model, the sufficiently excited collective vibration exerts an influence on individual cells which inhibits both cell division and « unfavourable » mutations (which would not react to control) and, at the same time, tends to bring cells which do not respond to the control into states in which they do react to it.

If by external influences some cells are changed such that their relevant frequencies are altered, then *a*) these cells no longer respond to the control and *b*) the strength of the controlling vibration is reduced. Both these effects are rectified by the above-mentioned stabilizing influence provided they are not too large. If, however, the number of altered cells is so large that the excitation of the collective mode is reduced below a critical value, then the control becomes ineffective. As a consequence, all cells divide more frequently, and simultaneously « unfavourable » mutations (having « wrong » frequencies) become possible, *i.e.* a phase transition from order to disorder (cancer) has taken place.

It should be noted here that the frequencies of the changed cells will differ from each other; thus no collective mode with sharp frequency will arise: the ordered state has a sharp frequency, the disordered one a broad frequency band.

Which are the external influences that can lead to a change in frequencies? It has been shown [2] that supply of unattached electrons could have such an influence. Such electrons might occur in the conduction band of various molecules. They are then capable of long-wave electric vibrations, plasma waves, whose frequencies increase proportional to the square root of the electron density. These vibrations will interact with the coherent vibrations of our system and under appropriate circumstances can significantly alter the frequency.

Such electrons might be supplied by excitation through ultraviolet light. Or they might be supplied by foreign molecules which are electron donors. A change of frequency might, of course, also arise from an invasion by foreign molecules into the relevant DNA-protein complex.

Other possibilities may arise, however, if we assume that the frequency sharpness found in some of the experiments discussed in sect. **6** is a feature of the proposed collective vibrations. It has been previously indicated that electric coupling with structured water would account for the measured frequency width indicating absence of all other types of friction. Some structural

changes might be sufficient to eliminate this property and with it the effectiveness of control. The affected cells could then still have frequencies in the region of the nonaffected one, but exhibit greater width.

The model described here can without difficulty be presented in mathematical form by using well-known techniques developed, *e.g.*, in connection with laser physics [29]. Even without doing so, it will be noticed that the term « collective mode » might be given a different physical interpretation from the one suggested here, provided individual cells have selective properties which enable them to react to the collective mode and which could be destroyed by external influences.

If the interpretation of the collective mode as a vibration is correct, then one would wish to know the order of magnitude of relevant frequencies. The 10^{11} Hz region is, of course, a first candidate both for theoretical and experimental reasons (sect. **3** and sect. **6**). It might well be, however, that quite different frequency regions are also of importance, and it will in the first place be a task for experiments to find these frequencies.

Most desirable would, of course, be experiments which measure the excitations of the biological material *in vivo*. For frequencies near or above 10^{12} Hz laser Raman effect would be suitable. For frequencies near or below 10^{11} Hz, this becomes inefficient because of the broadening of the primary line through water. Thus the frequencies for which selective biological effects have been discovered (sect. **6**) are beyond the available range. New methods (corresponding to Brillouin scattering) are being developed, but must still be adapted for use in biological materials.

It should be emphasized in this context that vibrations in this frequency range may very sensitively depend on details of the arrangements and on the conformation of relevant molecules including structured water and ions, which must both be considered an integral part of the system. Investigations on molecules, *e.g.* proteins, outside their natural surroundings may but need not give relevant results. Something similar holds of the measurement of absorption of electromagnetic waves by the biological material. We have mentioned above that the excited collective mode might have a complicated spacial pattern. This will probably not be optically active. Optically active modes are, however, to be expected in the neighbourhood of each « relevant » frequency; they would show up as a band rather than as the relatively sharp line we expect from the collective mode, except in cases in which the spacial pattern is homogeneous in the range of penetration of the electromagnetic wave—as might hold for thin layers of the material. It should be emphasized in this context that some of the experiments mentioned in sect. **6** exhibit extremely sharp frequency response for biological effects, whereas broad bands are found in optical measurements. Nevertheless changes found in the optical spectra would certainly indicate major changes and could thus provide valuable information.

One might in particular wish to measure the changes arising from the excitation of electrons into the conduction bands discussed above. It must be pointed out in this context that plasma waves are not optically active except through the influence of surfaces. The same is likely to hold of the collective mode with which they interact. For optical detection, therefore, one would use biological samples which are small compared with the wavelength of the electromagnetic radiation with the relevant frequency. When using a thin layer, then the electric vector should have a component perpendicular to the surface. Internal surfaces may, however, also be effective which to some extent may relax these conditions.

Finally biological effects of the electromagnetic radiation are most likely to provide significant results. If we can learn from the experiments mentioned in sect. **6**, then we might expect sharp frequency resonances in the biological response. As a consequence, a tunable source will be required.

In the first place, we would expect the naturally excited mode to be in its most favourable state. External excitation of modes with the same frequency, though probably with different spacial pattern, would thus interfere with the efficiency of the collective controlling mode. In a grown material this would then increase the rate of cell division, while in a growing system it should decrease it.

Finally, when under the influence of external agencies a sufficient number of cells have undergone changes in frequency to the extent that the controlling collective mode has become ineffective, then a stronger externally imposed vibration of the correct frequency (though possibly with different spacial pattern) might still be able to restore the original order. This, of course, would be the most startling influence of electromagnetic waves on biological systems.

Note added in proofs on 27/9/1977.

Conclusive evidence for the metabolic excitation of vibrational states in bacteria has now been obtained by Raman effect. WEBB and STONEHAM (*Phys. Lett.*, **60** A, 267 (1977)) show that, while resting bacteria exhibit only bands in the (100÷200) cm^{-1} region, active bacteria show strong lines. Moreover (S. J. WEBB, M. E. STONEHAM and H. FRÖHLICH: *Phys. Lett.*, A, in print), the intensity ratio of anti-Stokes to Stokes lines in thermal equilibrium should be about 0.5, whereas it is found close to 1.0, evidence for strong nonthermal excitation.

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Long-Range Coherence and Energy Storage in Biological Systems

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Abstracts

Biological systems are expected to have a branch of longitudinal electric modes in a frequency region between 10^{11} and 10^{12} sec^{-1}. They are based on the dipolar properties of cell membranes; of certain bonds recurring in giant molecules (such as H bonds) and possibly on pockets of non-localized electrons. In Section 2 it is shown quite generally that if energy is supplied above a certain mean rate to such a branch, then a steady state will be reached in which a single mode of this branch is very strongly excited. The supplied energy is thus not completely thermalized but stored in a highly ordered fashion. This order expresses itself in long-range phase correlations; the phenomenon has considerable similarity with the low-temperature condensation of a Bose gas. General consequences and proposals of experiments are discussed in Section 3.

Dans les systèmes biologiques on peut présumer une branche de modes électriques longitudinaux dans une région de fréquences entre 10^{11} et 10^{12} sec^{-1}. Ceux-ci peuvent être associés à des propriétés dipolaires de membranes cellulaires, de certaines liaisons récurrentes dans les molécules géantes (telles que des liaisons d'hydrogène) et peut-être de poches d'électrons non localisés. Dans le Section 2 on montre d'une façon toute générale, que si l'on fournit de l'énergie à une telle branche à une certaine vitesse, un état stationnaire sera atteint, dans lequel un seul mode de cette branche est très fortement excité. L'énergie ainsi fournie n'est donc pas complètement thermalisée, mais reste emmagasinée d'une manière bien ordonnée. Cet ordre apparaît dans les corrélations de phase à longue portée. Le phénomène a une similarité considérable à la condensation d'un gaz de Bose aux basses températures. Dans le Section 3 on discute des conséquences générales et on propose des expériences.

Man kann erwarten dass biologische Systeme einen Zweig von longitudinalen elektrischen Schwingungen in einem Frequenzbereich zwischen 10^{11} und 10^{12} sec^{-1} haben. Diese sind auf Dipolareigenschaften von Zellmembranen, von gewissen Bindungen in Riesenmolekülen (z. B. Wasserstoffbindungen) und vielleicht von Taschen von nichtlokalisierten Elektronen gegründet. In Sektion 2 wird im allgemeinen gezeigt, dass wenn Energie mit mehr als einer gewissen Durchschnittsrate einem solchen Zweig zugeführt wird, wird ein stationärer Zustand erreicht, in welchem eine einzige Schwingung dieses Zweigs sehr stark angeregt ist. Die zugeführte Energie ist also nicht vollständig thermalisiert, sondern wird in einer höchst geordnetenr Weise gespeichert. Diese Ordnung zeigt sich in Phasenkorrelationen mit langer Reichweite. Das Phänomen hat beträchtlicle Ähnlichkeiten mit der Kondensation eines Bosegases bei tiefen Temperaturen. Allgemeine Folgerungen und Vorschläge von Experimenten werden in Sektion 3 diskutiert.

1. General

The present paper arises from development and clarification of ideas presented by myself [1] in the opening talk of the recent Versailles conference on Theoretical Physics and Biology. In the past decades biology has been highly successful in the correlation of biological properties with the structure and properties of relevant molecules. This stage of development might be compared with the correlation of the physical and chemical properties of materials with the structure and properties of the atoms and molecules of which the material is composed. These properties modified by interaction with their neighbours account for many features of the substance. Some of the most interesting properties in the physics of materials, however, can be understood in terms of the collective (cooperative) behaviour of a great number of these constituents only, usually associated with some kind of long-range order. Most frequently this order is described in terms of a definite spatial pattern. A much more subtle type of long-range order is found in superfluid helium, and in superconductors. This order is connected with the concept of long-range phase correlation (coherence). Superfluids (and superconductors) undergo at very low temperatures a phase transition. The simplest model available for this transition is the Einstein condensation of a Bose gas in which—with decreasing temperature—more and more particles "condense" into a single quantum state.

At first thought one might consider the order exhibited in these low-temperature phenomena to be quite irrelevant for Biology. Stimulated by a different attitude, however, I have been able to show [2] that under appropriate conditions a phenomenon quite similar to Bose condensation may occur in substances which possess longitudinal electric modes. If energy is fed into these modes and thence transferred to other degrees of freedom of the substance then a stationary state will be reached in which the energy content of the electric modes is larger than in thermal equilibrium. This excess energy is found to be channelled into a single mode—exactly as in Bose condensation—provided the energy supply exceeds a critical value. Under these circumstances a random supply of energy is thus not completely thermalized but partly used in maintaining a coherent electric wave in the substance.

The original reason for considering longitudinal electric modes lay in the extraordinary dielectric properties of biological systems. Furthermore living systems though relatively stable are in some respects far from thermal equilibrium; also they may be expected to exhibit some order of a non-trivial nature. From a very general point of view of physics, three observations may thus be made about living systems.

[1]. They are relatively stable yet in some respects far from thermal equilibrium.

[2]. They probably exhibit some kind of order typical for them yet not of an obvious spatial nature.

[3]. Cells and many of the relevant giant molecules have quite extraordinary dielectric properties.

In connection with (3) it should be remembered that cells have a membrane of about 10^{-6} cm thickness which maintains a very strong dipolar layer. In the (from the point of view of physics) complicated shape of a cell surface, local vibrations of a part of the cell membrane are feasible such that the positive and the negative part of a particular section of the membrane vibrate against each other leading to an oscillating electric dipole. Its frequency is of the order 10^{11}–10^{12} sec^{-1} if a sound velocity in the layer of order 10^{5}–10^{6} cm/sec perpendicular to the surface is assumed.

The giant molecules occurring in biological systems also have extraordinary dipolar properties largely in view of the frequently occurring hydrogen bonds. Careri [3] has pointed out that in the α helix they may be expected to exhibit collective dipolar oscillations in the same frequency region.

Other sources for dipolar oscillations in the same frequency region might exist. They would arise, for instance, from the presence of "non-localized" electrons in some regions of a cell.

Dielectric systems of the type described here are capable of longitudinal electric oscillations extending over the whole material. The existence of such oscillations follows directly from Maxwell's equations though little use has been made of this possibility in physics. They can have a certain range of macroscopic wavelengths depending only weakly on frequency.

Certain processes in living systems involving the cell membrane, or the hydrogen bonds of molecules, or other dipolar constituents, may be expected to feed energy locally to the particular dipolar oscillations. The long-range Coulomb interaction then causes this energy to be shared with other dipoles. The situation thus is exactly as assumed in the above-mentioned model, so that the dipoles will tend to oscillate coherently provided the energy supply is sufficiently large compared with the energy loss. Non-linear effects are likely to reduce this loss with increasing excitation and effectively transfer the system into a metastable state in which the energy supplied locally to dipolar constituents is channelled into a single longitudinal mode which exhibits long-range phase correlations.

It should be remarked that the present model fits the general conditions for the stability of "dissipative structures" presented by Prigogine [4] at the Versailles conference.

Furthermore the channelling of randomly supplied energy into a single mode—described above—should be considered as a special case of long-range quantum-mechanical phase correlations as pointed out in [1]. Such phase correlations are a common property of superfluids and superconductors as described by Yang [5]. They also may be expected to arise in certain forms of chemical bond as has been emphasized by Löwdin [6].

2. Coherent Excitations

Consider a suspension of a large number Z of units in a substance which will be treated as a heat bath. Each unit is assumed capable of electric dipole oscillations with a certain frequency ω_0. Long-range Coulomb interaction between these individual oscillators then gives rise to a branch of Z longitudinal electric modes in a frequency range say

$$\omega_1 \leqq \omega \leqq \omega_2 \tag{1}$$

which may be shifted considerably from ω_0 but is relatively narrow. The modes of this branch describe longitudinal electric waves in the material as a whole. Oscillations of single units can then be described by appropriate superposition of these waves. Assume that energy is supplied locally to each unit at a mean rate s. This is equivalent with a supply of energy at the same rate s to each mode. Assume furthermore that each unit can exchange energy locally with the surrounding heat bath; this can again be expressed in terms of an exchange of energy in quanta $\hbar\omega$ with the modes of the branch at a rate which is nearly independent of ω. The net rate of loss L_{1l} of the mode with frequency ω_l (containing n_l quanta) can then be written in the form

$$L_{1l} = \phi(T)(n_l e^{\hbar\omega_l/KT} - (1 + n_l)) \tag{2}$$

where $\phi(T)$ may depend on temperature T. In a higher order individual units could exchange two, or more quanta with the heat bath but never a fraction of a quantum. The long-range Coulomb interaction between units gives rise, however, to the whole range (1) of frequencies. In second order, therefore, absorption of a quantum $\hbar\omega_l$ in conjunction with emission of a quantum $\hbar\omega_j$, or vice versa, permits an exchange of energy between the branch of electric modes, and the heat bath in a range

$$0 < \hbar\,|\omega_l - \omega_j| \leqq \hbar(\omega_2 - \omega_1) \tag{3}$$

The net range of loss L_{2l} of the mode ω_l due to such processes can be written in the form

$$L_{2l} = \chi(T) \sum_j (n_l\,(1 + n_j) e^{\hbar(\omega_e - \omega_j)/KT} - n_j(1 + n_l)) \tag{4}$$

The general forms of (2) and (4) are dictated by the requirement that in the absence of energy supply, $s = 0$, thermal equilibrium ($L_{1l} = 0$, $L_{2l} = 0$) requires a Planck distribution n_{lT} for n_l,

$$n_{lT} = \frac{1}{e^{\hbar\omega_l/KT} - 1} \tag{5}$$

Note that χ may depend on temperature. A slightly more general form of L_{2l} would permit explicit dependence of χ on ω_l and ω_j.

The condition for stationarity requires for each l,

$$s = L_{1l} + L_{2l} \tag{6}$$

With the notations

$$N = \sum n_j\,, \qquad S(T) = s \sum e^{-\hbar\omega_j/KT} \tag{7}$$

one finds

$$S(T) = \phi \sum (n_j - (1 + n_j)e^{-\hbar\omega_j/KT}) \tag{8}$$

independent of χ, and

$$n_l = \left(1 + \frac{s}{\phi + \chi N}\right) \frac{1}{Ae^{\hbar\omega_l/KT} - 1} \tag{9}$$

where

$$A = \frac{\phi + \chi \sum (1 + n_j)e^{-\hbar\omega_j/KT}}{\phi + N\chi} > 0 \tag{10}$$

or making use of (8),

$$A = 1 - \frac{\chi}{\phi + N\chi} \frac{S(T)}{\phi} \leqq 1 \tag{11}$$

Clearly $s = 0$ leads to $A = 1$ and (9) becomes the thermal equilibrium distribution (5). However, $\chi = 0$ also leads to $A = 1$ even if $s \neq 0$ though in this case n_l no longer represents a Planck distribution.

Inequality (11) together with the condition $n_l \geqq 0$ leads to

$$A = e^{-\mu/KT} \quad \text{where} \quad \hbar\omega_1 > \mu \geqq 0 \tag{12}$$

Clearly a Bose type of condensation into the lowest mode ω_1 will take place when μ approaches $\hbar\omega_1$ very closely. We shall show in the high temperature limit

$$kT \gg \hbar\omega_1 \tag{13}$$

that this will occur when s exceeds a critical value s_0. In this approximation one has from (9) and (12),

$$n_l = \left(1 + \frac{s}{\phi + \chi N}\right) \frac{kT}{\hbar w_l - \mu} + \cdots \tag{14}$$

Let $D(\omega)\, d\omega$ be the number of modes in $d\omega$. Then if no condensation has taken place $D(\omega_l)n_l$ may be considered a continuous function. Let N_n be the total

number of quanta in this case. From the inequality of (12) one then finds an upper limit for N_n, for then

$$N_n = \left(1 + \frac{s}{\phi + \chi N_n}\right) kT \int_{\omega_1}^{\omega_2} \frac{D\, d\omega}{\hbar\omega - \mu} < \left(1 + \frac{s}{\phi + \chi N_n}\right) N_0 \tag{15}$$

or

$$\frac{N_n(\phi + \chi N_n)}{\phi + \chi N_n + s} < N_0 \tag{16}$$

where

$$N_0 = \frac{kT}{\hbar} \int_{\omega_1}^{\omega_2} \frac{D\, d\omega}{\omega - \omega_1} \tag{17}$$

Now the total number N of quanta is quite generally determined by s and ϕ, and may well exceed the maximum of "non-condensed" quanta N_n imposed by (16). For an introduction of the excess number m_l of quanta in ω_l over the number in thermal equilibrium,

$$m_l = n_l - n_{l,T}; \qquad \sum m_l = N - N_T \tag{18}$$

where N_T is the total number of quanta in thermal equilibrium, one finds from (8), (5), and (13),

$$S(T) = \phi \sum m_l (1 - e^{-\hbar\omega/KT}) \simeq \phi \sum m_l \frac{\hbar\omega_l}{kT} = \phi \frac{\hbar\bar{\omega}}{kT} (N - N_T) \tag{19}$$

Here $\bar{\omega}$ must have the range

$$\omega_1 \lessapprox \bar{\omega} \lessapprox \omega_2 \tag{20}$$

Also in approximation (13) one has $S(T) \simeq Zs$, so that

$$N = N_T + \frac{sZ}{\phi} \frac{kT}{\hbar\bar{\omega}} \tag{21}$$

Now $\bar{\omega}$ is limited by (20) and may therefore vary only slightly with s. Hence according to (21) N is linearly increasing with s. A value s_0 of s therefore exists above which $N = N_n$ can no longer hold because (16) can no longer be satisfied. In this case

$$N = n_1 + N_n = \frac{kT}{\hbar\omega_1 - \mu} + N_n \tag{22}$$

where μ approaches $\hbar\omega_1$ very closely. Its actual value follows from (21) and (22). Condensation into the mode ω_1 thus takes place.

It is now of great interest to note that apart from the relatively small variations permitted for $\bar{\omega}$, the excess $N - N_T$ is entirely determined by Zs/ϕ. The smaller ϕ, therefore, the smaller is the critical value s_0 required for condensation. Compared with other processes like energy transfer from acoustic (low frequency) vibrations the processes L_1 (cf.(2)) may indeed be expected to be rather slow (i.e. ϕ relatively small) because they must occur in quanta $\hbar\omega$. The processes L_2 (4), on the other hand, which can transfer smaller amounts contribute only in a higher order, $(\hbar\omega/kT)^2$, which is considered negligible in view of (13).

Assume now that condensation into the mode ω_1 has been established, so that a coherent longitudinal electric vibration exists in the substance. Non-linear effects associated with this excitation are then expected to lead to its further stabilization as pointed out for a simple model in Appendix I of [1], though the argument used there is very general. Roughly speaking the free energy associated with the vibration can be decreased if the system is permitted to undergo deformations. If these deformations are measured in terms of parameters, say η, then for small η the elastic energy associated with a deformation will be proportional to η^2 because in the absence of the vibrations the system was elastically in equilibrium. The energy associated with the electric polarization will, however, be linear in η—for small η. Hence a new equilibrium is established in which η is proportional to P^2, the mean square of the polarization. The total free energy then consists of a positive term proportional to P^2, and a negative one proportional to P^4, thus pointing to an instability at sufficiently large P^2. This may indicate a rupture or else stabilization under the influence of higher order terms in η and P^2. In the latter case a metastable state will be reached, and ϕ will then be very small. Further supply of energy s should then lead to instabilities at a later stage.

3. Discussion and Proposal of Experiments

Biological systems contain components capable of electric dipole oscillations. These interact through long-range Coulomb forces and thus establish a branch (or several branches) of longitudinal electric modes probably in a frequency range of 10^{11}–10^{12} sec^{-1}. The relevant components may be connected with certain regions in the cell membrane, with certain chemical bonds in particular with hydrogen bonds, or with regions containing non-localized electrons. If energy is supplied locally to one component (or removed from there) then in view of the long-range interaction this energy change is shared by all the components which establish the branch of electric modes. If the rate of energy supply is sufficiently large then, as shown in Section 2, the energy gets channelled into a single mode which then presents a strongly excited coherent longitudinal electric vibration. Its wavelength depends on details of the geometrical arrangement of the components. A certain fraction of the supplied energy, therefore, is not thermalized but stored in a single mode which then exhibits long-range phase correlations. With respect to the electric modes the system is thus maintained

through a flux of energy in a state of high order far from thermal equilibrium. In this excited state the mean value P^2 of the square of the electric polarization may be large and hence lead to deformations (and stresses) in the material. This in turn may lead to a relative stabilization of the excitation so that a relatively small supply of energy is required to maintain it. Continued energy supply of a higher magnitude must then ultimately lead to an instability.

If established the phenomenon proposed here should have a profound influence on biological properties like cell division which may involve relatively extensive regions. Thus it has been suggested in [1] that the deformations connected with the excitation should stimulate cell division. This stimulus would increase with increasing size of the cell. It would be partly due to the vibrations inside the cell under consideration, but partly it would arise from the electric fields based on the dipolar oscillations in other cells. This latter influence would disappear, however, when cells are densely packed for then deformations in the cell membrane no longer influence the dielectric response. Such an influence of packing has actually been observed; it is known as "contact inhibition".

Formation of certain bonds as required in the synthesis of giant molecules will feed energy into the branch if the particular bonds belong to the components which establish the electric modes. The coherent excitation, on the other hand, presents a reservoir of energy in a highly ordered form available for various uses.

Observations by Szent-Györgyi [7] may be of considerable relevance in this connection. He concludes that "cellular activities, like cell division or protein synthesis are dominated, to a great extent, by the electronic charges of the system", and he indicates that some of these electrons may be in non-localized states. Such electrons will take part in the proposed electric vibrations and reinforce them provided their density is not too high. Hence if cell division or protein synthesis are stimulated by our vibrations then additional (electronic) sources for it will increase these effects. A change must be expected, however, when the electron density becomes so high (larger than $10^{15}\,\mathrm{cm}^{-3}$) that their proper frequency exceeds ω_1. In this case the reaction on the original vibrations can no longer be neglected. This then may lead to a qualitative change in such cell activities as depend on our vibrations. It is suggestive, of course, to propose that this change characterizes cancer which accordingly would be caused by an excessive density of non-localized electrons in some region of a cell (e.g. in certain giant molecules). Carcinogenic molecules must then have the ability of transferring electrons to the relevant regions of a cell. This hypothesis could possibly be checked if ionizing radiation exists which produces "non-localized" electrons.

Such specific questions must, of course, be studied in detail in connection with the particular structures and processes. Most important in the first place, however, would be experimental establishment of the existence of the coherent vibrations in living systems. The magnitude of the electric polarization should be fairly high, but direct measurement seems difficult in view of the expected high frequency. A very interesting suggestion has been made, however, by

Palma [8]. He proposes investigation of the spectral distribution of scattered light of laser beams. One then deals essentially with Raman-like scattering and thus from the frequency shift obtains the frequency of the longitudinal vibrations. One would have to work with low intensities, of course, so as not to kill the organism.

Careri [9] proposes to investigate the infrared properties of biological systems. The oscillations considered here are of course longitudinal, and hence optically not active. Nevertheless they should be expected to produce a certain amount of radiation through surface effects. Also the deformations associated with the establishment of the proposed oscillations would influence the transverse (optical) resonance frequencies.

More indirect methods would work with the proposed biological effects of the electric vibrations. Thus the stimulus on a cell arising from the dipolar oscillations of other cells could be screened with the appropriate use of absorbers or otherwise altered with the use of filters or reflectors.

The proposals discussed in this paper are speculative. The derivation given in Section 2 in conjunction with the striking dielectric properties of biological systems seems to point definitely, however, to the possibility of the storage of supplied energy in a highly ordered form.

Note added in proof

I have now found that absorption of light ($\hbar\omega \gg kT$) may also lead to energy storage in a single state. This may be relevant to photosynthesis.

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Long-Range Coherence in Biological Systems (*).

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1. – Introduction.

The great success of molecular biology arises from the discovery of the detailed structure of biomolecules such as DNA and enzymes. No systematic spacial order was found, and it was concluded that these molecules do not possess a physical order. Physical order, however, does not express itself in spacial order only. Thus superfluid helium, or the electrons in a superconductor exhibit an order that might be termed « motional order ». This type of order is connected with the existence of macroscopic wave functions, which implies the existence of certain quantum phase correlations over macroscopic distances or the existence of coherent states.

Other correlations connected with the phase of oscillating systems rather than with those of macroscopic wave functions are also well known when those systems are lifted from their thermal equilibrium. A well-known example are laser systems which need not exhibit a spacial order as, *e.g.*, in the case of gas lasers.

Many other types of physical order may be imposed on certain systems

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when they are lifted from their thermal-equilibrium state even though they are spacially disordered. This does not hold, however, for all systems, and no general criterium has been found, so far, which when satisfied permits particular systems to exhibit some type of order when energy is supplied to them. Most systems, however, simply raise their temperature.

Biological materials, when they exhibit typical biological properties, must be « fed », *i.e.* energy passes through them. This energy may arise from chemical processes, or it may be supplied by the Sun.

It seems rash, therefore, to conclude from the absence of spacial order —obtained from structural analysis—that no physical order exists in biological systems. On the contrary, it has been suggested several years ago that they belong to those systems in which the supply of energy imposes a characteristic order [1, 2].

In approaching these questions from the point of view of theoretical physics it must be realized that, in contrast to most materials usually treated in physics, biological materials have been selected through long evolutionary processes. As a consequence, they often show quite extraordinary properties, and, rather than ask whether some material properties seem likely, one should ask whether they are possible. Clearly, in the first place, the role of theoretical physics must be to decide what questions should be asked; subsequent developments should then work in close contact with experiments.

It has often been thought that exact knowledge of the atomic structure of a system will provide all answers at least in principle. This is already a basic mistake [3], even in the case of the physics of simple systems. Thus, for example, the above-mentioned concept of macroscopic wave function refers to the whole assembly of particles, *i.e.* it is a collective property which cannot in a simple way be reduced to properties of individual particles.

2. – General.

In looking for a striking material property common to most biological systems one will, no doubt, choose their dielectric properties [1, 2, 4]. Biological systems are, of course, not uniform, but composed of many different such systems. In ordinary physics, one would then extract each subsystem, *e.g.* a certain type of molecule, and determine its dielectric properties, *e.g.* its dipole moment. In solution this dipole moment, say μ_i, would be slightly different from the moment μ_v in vacuum owing to polarization effects. Such a procedure would be highly misleading in the biological case, say for water-soluble proteins. Their conformation is largely dominated by the interaction of their charged groups with the surrounding water and ions which thus form an integral part of the structure and hence of the measured dipole moment. Under these circumstances, very high internal electric fields may exist in such systems.

Most striking, of course, are the fields in cell membranes which have average values of the order of 10^5 V/cm, which in ordinary materials would require great precautions to avoid breakdown. Inhomogeneous regions, *e.g.* through proteins dissolved in the membrane, might lead to much higher local fields. These fields play a basic role in nerve conduction, but here weaker fields could fulfil the same purpose. Their role in the membranes of other cells is unknown.

One notices that the energy difference of a unit electronic charge across the membrane (about 10^{-6} cm) would, at room temperature T, amount to about $3kT$. This suggests that the transport of single charges obtains a significance against thermal noise.

Also in a field $F \sim 10^5$ V/cm the energy μF is larger than kT when μ is the dipole moment of a small protein, say $\mu \sim 300$ Debye units. This suggests that proteins dissolved in membranes are oriented by the field.

In addition, however, the whole membrane system is strongly polarized by the field. As a consequence, an oscillation of a section of the membrane—or of certain proteins dissolved in it—is connected with a corresponding electric vibration, so that the longest electric wave usually yields an electric-dipole vibration. According to one of our main suggestions, such vibrations are very strongly, coherently, excited in biological systems when they are active, *i.e.* when metabolic energy is available.

Now it must be realized that the displacement of a single ion by 10^{-6} cm produces a very large dipole moment of the order of 500 Debye units. It would be wrong, however, to neglect other processes arising from such displacements, for the change of electric field arising from this displacements may substantially alter the polarization in the neighbourhood of the ion (note that the electric field at a $3 \cdot 10^{-7}$ cm distance from the electronic charge is of the order of 10^6 V/cm) and cause other ions to move. Under these circumstances, one would expect such displacements to become stabilized, and our second principal suggestion states that biological systems have metastable states with very high electric polarization.

How can one support theoretically such general statements referring to a variety of complicated materials? It is already clear from the above that we are dealing with strongly nonlinear effects. This provides possibilities only occasionally contemplated in ordinary physics. Our method will be to formulate simple, but general models which do show the required properties. General conditions required in these models then impose corresponding conditions on the respective biological systems. If experiments confirm the theoretical predictions, then we can conclude that the molecular evolution has led to the selection of materials with the required properties. This does not imply that other systems do not exist which do not have the required properties, though from a structural point of view they could be very similar to the systems which

do have these properties. The difference between the two would then rest in their nonlinear behaviour, *i.e.* in properties of certain excited states which would not be revealed in structural analysis.

3. – The vibrational model.

Any system containing charged particles is capable of long-wave electrical vibrations. Such a system need not be homogeneous. One of the characteristics of these vibrations is the existence of a lowest frequency $\omega_1/2\pi$, different from zero. Our system is assumed to possess such modes, $\omega_1, \omega_2, \ldots, \omega_z$. They are considered not to interact with each other, but to be strongly coupled to the surroundings, *i.e.* to the other modes of the systems which will be treated as a heat bath at temperature T. Let n_k be the number of quanta $\hbar\omega_k$ in the mode ω_k. Then, through first-order processes, single quanta $\hbar\omega_k$ may be emitted to or absorbed from the heat bath, so that the rate of change $\dot{n}_{k,1}$ of n_k due to these processes is [2, 5]

$$\dot{n}_{k,1} = -\varphi\big(n_k \exp[\beta\hbar\omega_k] - (n_k + 1)\big)\,, \tag{3.1}$$

where $\beta = 1/kT$, and φ will depend on T and, in general, also on ω_k. For simplicity we shall neglect the ω_k-dependence. The ratio of emission to absorption processes is governed by detailed balance.

Similarly, for second-order processes we find

$$\dot{n}_{k,2} = -\chi \sum_j \big(n_k(1 + n_j)\exp[\beta\hbar\omega_k] - n_j(1 + n_k)\exp[\beta\hbar\omega_j]\big)\,. \tag{3.2}$$

Again, apart from depending on T, the transition probability χ could depend on k and j, but we shall neglect this. This will greatly simplify the calculations, but should have no influence on the main results.

We note that, on this choice,

$$\sum_k \dot{n}_{k,2} = 0\,, \tag{3.3}$$

i.e. such processes lead to interchange of quanta $\hbar\omega_k$ without alteration of their total number.

Now we consider that energy is supplied at the rate s_k to the mode ω_k. We then find, for the total rate of change $\dot{n}_k$,

$$\dot{n}_k = s_k - \varphi\big(n_k \exp[\beta\hbar\omega_k] - (n_k + 1)\big) - \\ - \chi \sum_j \big(n_k(1 + n_j)\exp[\beta\hbar\omega_k] - n_j(1 + n_k)\exp[\beta\hbar\omega_j]\big)\,. \tag{3.4}$$

We wish to solve this equation for the stationary case, *i.e.* when

$$\dot{n}_k = 0 \,. \tag{3.5}$$

We may derive the general behaviour of n_k in a simple manner, though it would be difficult to solve this system of quadratic equations exactly.

Summation over k with the definition

$$S = \sum s_k \tag{3.6}$$

yields

$$S = \varphi \sum_k \left(n_k \exp\left[\beta\hbar\omega_k\right] - (n_k + 1)\right) . \tag{3.7}$$

Introducing the total number of quanta N,

$$N = \sum n_k \,, \tag{3.8}$$

we find with

$$\omega_1 \leqslant \omega_k \leqslant \omega_z \tag{3.9}$$

that $\left(\text{note } \sum_k 1 = z\right)$

$$N\left(\exp\left[\beta\hbar\omega_z\right] - 1\right) > S/\varphi + z > N\left(\exp\left[\beta\hbar\omega_1\right] - 1\right) . \tag{3.10}$$

This means that N must increase proportional to S if the supply S is sufficiently large.

Now define μ by

$$\exp\left[-\beta\mu\right] = \frac{\varphi + \chi \sum (1 + n_j)}{\varphi + \chi \sum n_j \exp\left[\beta k\omega_j\right]} , \tag{3.11}$$

so that

$$1 - \exp\left[-\beta\mu\right] = \frac{\chi \sum \left(n_j \exp\left[\beta\hbar\omega_j\right] - (1 + n_j)\right)}{\varphi + \chi \sum n_j \exp\left[\beta\hbar\omega_j\right]} . \tag{3.12}$$

From (3.7) it then follows that

$$\mu \geqslant 0 \tag{3.13}$$

and

$$1 - \exp\left[-\mu\beta\right] = \frac{\chi}{\varphi} \frac{S}{\varphi + \chi \sum n_j \exp\left[\beta\hbar\omega_j\right]} \geqslant 0 . \tag{3.14}$$

Equation (3.4) with (3.5) can thus be formally solved by

$$n_k = \left(1 + \frac{\varphi}{\chi} \frac{s_k}{S} \left(1 - \exp\left[-\beta\mu\right]\right)\right) \frac{1}{\exp\left[\beta(\hbar\omega_k - \mu)\right] - 1} , \tag{3.15}$$

and $n_k \geqslant 0$ together with (3.13) yields

$$\hbar\omega_1 > \mu \geqslant 0 \,. \tag{3.16}$$

This inequality permits the establishment of an upper limit for N although μ is a complicated function of all the n_k. For (3.15) then yields

$$N = \sum n_k < \left(1 + \frac{s_m(1 - \exp[-\beta\hbar\omega_1])\,\varphi}{\chi S}\right) \sum_k \frac{1}{\exp[\beta(\hbar\omega_k - \mu)] - 1}\,, \tag{3.17}$$

where s_m is the largest of the s_k.

Clearly, if the sum in (3.17) can be replaced by an integral, then for $\mu \sim \hbar\omega_1$ a value independent of the s_k will be reached. Moreover, always $s_m/S \leqslant 1$, so that an upper limit for N independent of S is obtained. This contradicts the conclusion drawn from (3.10), according to which N must increase proportional to S if S is sufficiently large. This situation is the same as normally found in Einstein condensation of a Bose gas. Thus the replacement of the sum by an integral is incorrect; μ takes a value close to $\hbar\omega_1$ such that n_1 becomes very much larger that all the other n_k.

It follows then that a single mode ω_1 gets very strongly (coherently) excited when the energy supply is larger than a critical one.

As special cases we may consider that all s_k are equal. Then

$$s_k/S = 1/z \tag{3.18}$$

and (3.15) becomes

$$n_k = \left(1 + \frac{\varphi}{\chi_z}(1 - \exp[-\beta\mu])\right) \frac{1}{\exp[\beta(\hbar\omega_k - \mu)] - 1}\,. \tag{3.19}$$

As another extreme consider that all s_k vanish except for $k = k_0$.

Then

$$s_k = S\delta_{k,k_0}\,, \qquad s_{k_0} = S \tag{3.20}$$

and

$$n_k = \left(1 + \delta_{k,k_0}(1 - \exp[-\beta\mu])\frac{\varphi}{\chi}\right) \frac{1}{\exp[\beta(\hbar\omega_k - \mu)] - 1}\,. \tag{3.21}$$

To understand qualitatively these conclusions, we notice first of all that the case $\chi = 0$ leads to a completely trivial result, namely

$$n_k = \left(1 + \frac{s_k}{\varphi}\right) \frac{1}{\exp[\beta\hbar\omega_k] - 1}\,, \qquad \text{if } \chi = 0\,, \tag{3.22}$$

i.e. to a modified thermal-equilibrium Planck distribution. It will be noticed from (3.11) that $\chi = 0$ in fact entails $\mu = 0$ and thus eliminates the occurrence of Bose condensation. Thus the nonlinear terms χ arising from a strong interaction with the heat bath tend to force the system into a thermal-equilibrium distribution making use of a chemical potential μ.

A main difference from ordinary condensation of a Bose gas in thermal equilibrium should also be noted. There the ordered (condensed) state is reached by lowering the temperature. In our case, the ordered state is reached by supplying sufficient external energy to the system. Quite a similar situation prevails with respect to the second of our concepts (sect. **4**).

The above model, naturally, is a very simplified version of any actual situation. It has a very general consequence which might not depend on modifications required in particular cases, namely the coherent excitation of a polar mode when energy supply exceeds a critical magnitude.

Which frequencies should we anticipate? Roughly speaking, if a polarized system of linear dimensions l carries out long-wave elastic vibrations (*e.g.* breathing vibrations), then the electric polarization of the system implies coexistence of an electric-polarization vibration with the same frequency

$$\nu \sim v/l\,, \tag{3.23}$$

where v depends on the elastic constants and can formally be expressed as a velocity of sound. Thus $l = 10^{-6}$ cm and $v = 10^5$ cm/s would yield $\nu = 10^{11}$ Hz, but $l = 3 \cdot 10^{-7}$ cm (small protein) with $v = \frac{1}{3} 10^5$ cm/s yields, of course, the same value. The band of polarization waves ω_k would then consist of the interacting vibrations of such units, and also of higher vibrational states of the same unit. It must be emphasized again that the existence of long polarization waves does not require homogeneous material.

A further condition is the absence of unduly high friction, which probably will mean that surfaces facing water should not move. This does not hold, however, for structured water, which is to be considered as an integral part of the particular biological unit. Most likely, these conditions are satisfied by proteins in or near membranes. Thus, if $\Delta\nu$ is the frequency width, we have in this case

$$\nu \sim 10^{11}\ \mathrm{Hz}\,, \qquad \frac{\Delta\nu}{\nu} \ll 1\,, \qquad s > s_0 \tag{3.24}$$

as condition for the excitation of coherent vibrations. Here s is an appropriate average over the rate of energy supply and s_0 is the threshold value for Bose condensation.

It should be emphasized that other frequencies, both higher and lower, are well feasible. Thus larger units such as DNA-protein complexes might

well possess lower frequencies. Higher ones, on the other hand, may be based on a combination of the various rotational and vibrational subgroups of relevant molecules.

4. – The high-polarization model.

Consider [6] a polarizable material, polarization P. Its mean « self-energy » can then be written as $\frac{1}{2}a^2\overline{P^2}$, where the bar indicates an appropriate average and the positive a^2 depends on the polarizability, including size and shape of the material. These definite size and shape imply the elastic energy of the material to be a minimum relative to small deformations. If these are measured in terms of a deformation parameter η, then on deformation the elastic energy increases by $\frac{1}{2}\eta^2\varepsilon$, where $\varepsilon > 0$ depends on the elastic properties. In the case of a sphere, for instance, the deformation could produce an ellipsoid with eccentricity η. Now the polarizability in general is not an extremum at $\eta = 0$, so that a^2 would have to be replaced by $a^2(1 + c\eta)$, where c may be positive or negative. The total contribution to the free energy thus amounts to

$$U = \tfrac{1}{2}a^2(1 + c\eta)\,\overline{P^2} + \tfrac{1}{2}\eta^2\varepsilon\,. \tag{4.1}$$

Minimizing with regard to η at fixed $\overline{P^2}$ yields

$$\eta = -\frac{1}{2}\frac{ca^2}{\varepsilon}\,\overline{P^2}, \tag{4.2}$$

so that

$$U = \frac{1}{2}a^2\,\overline{P^2}\left(1 - \frac{1}{4}\frac{c^2}{\varepsilon}a^2\,\overline{P^2}\right). \tag{4.3}$$

Thus increasing $\overline{P^2}$ ultimately may lead to decreasing U. At this stage, the linear approximation η and the quadratic one in P will probably have become invalid. Higher-order terms may thus reverse this trend and lead to a minimum of U at a certain $\overline{P^2}$. If U at this value is negative, then the material would acquire a spontaneous dipole moment. If U is positive, then a metastable state with spontaneous polarization exists.

These conclusions are of a very general nature. They follows from the shape dependence of the polarizability and require the material to be sufficiently soft, *i.e.* ε should be sufficiently small.

They can be extended to a dynamical model composed of a polarization field $\boldsymbol{P}$ and an elastic (acoustic) field $\boldsymbol{A}$. Let ω_0 be the circular frequency of long-wave polarization waves in the absence of any interaction with the elastic field, then with such an interaction the density of potential energy has the

form

$$W = \tfrac{1}{2}\omega_0^2 \boldsymbol{P}^2 + \left(\tfrac{1}{2}v^2(\operatorname{div}\boldsymbol{A})^2 + c\boldsymbol{P}^2 \operatorname{div}\boldsymbol{A}\right)(1 - d^2\boldsymbol{P}^2)\,, \tag{4.4}$$

where v is the velocity of sound, d^2P^2 is an *ad hoc* term introduced to achieve stabilization for high P^2.

Note that, for long waves,

$$\varrho = \operatorname{div}\boldsymbol{A} \tag{4.5}$$

may be introduced as change in density. The static case requires minimization of (4.4) with regard to ϱ and P^2, so that for homogeneous displacements

$$\varrho = -\frac{cP^2}{v^2}\,, \tag{4.6}$$

yielding

$$W = \frac{1}{2}\omega_0^2 P^2 - \frac{1}{2}\frac{c^2}{v^2}(1 - d^2P^2)\,. \tag{4.7}$$

Apart from the different notation, this result corresponds to (4.3) except that the d^2P^2 term leads to a stabilization at high P^2.

Now we can introduce the density of kinetic energy

$$K = \tfrac{1}{2}(\dot{\boldsymbol{P}}^2 + \dot{\boldsymbol{A}}^2) \tag{4.8}$$

and hence from the Lagrangian density L obtain the equations of motion in the usual way,

$$\delta\int L\,\mathrm{d}^3\boldsymbol{r}\,\mathrm{d}t = \delta\int (K - W)\,\mathrm{d}^3\boldsymbol{r}\,\mathrm{d}t = 0\,, \tag{4.9}$$

where $\boldsymbol{r}$ is the space vector and t the time. The assumption that acoustic phonons are absent then permits the elimination of the $\boldsymbol{A}$-field and yields an equation of motion of the form

$$\ddot{\boldsymbol{P}} + \frac{\partial V(P)}{\partial \boldsymbol{P}} = 0\,. \tag{4.10}$$

The effective potential $V(P)$ has the form

$$V(P) = A + BP^2 + CP^4\,, \tag{4.11}$$

apparently a harmless anharmonic oscillator. This is misleading, however, for the coefficients A, B, C depend on the mean square amplitude

$$p = \overline{P^2}\,, \tag{4.12}$$

such that

$$A = \frac{1}{2}\frac{c^2}{v^2}p^2, \tag{4.13}$$

$$B = \frac{1}{2}\left(\omega_0^2 - 2\frac{c^2}{v^2}p - \frac{d^2c^2}{v^2}p^2\right), \tag{4.14}$$

$$C = \frac{c^2d^2}{v^2}p\,. \tag{4.15}$$

For small p clearly the CP^4 term is negligible and B becomes $\omega_0^2/2$, so that an harmonic oscillation with frequency ω_0 is obtained.

For increasing p, however, the magnitude of B decreases, *i.e.* the mode softens. For appropriate values of the parameters it is then found that a metastable state with permanent polarization $\pm P_0$ exists and that the system has two branches of oscillation: one around $P = 0$ and another with limited amplitude around $\pm P_0$. In this case, the effective potential $V(P)$ has a maximum at $P = 0$ and two minima at $P = \pm P_0$. The value of P_0 is found by minimizing (4.7), which yields

$$P_0^2 = p_0 = \frac{1}{3d^2}\left(1 + \sqrt{1-\lambda}\right), \tag{4.16}$$

where

$$\lambda = \frac{3v^2d^2\omega_0^2}{c^2}. \tag{4.17}$$

Thus $\lambda < 1$ is a necessary condition for the existence of a P_0, and it can be shown that this state is metastable, *i.e.* has positive energy when

$$\frac{3}{4} < \lambda < 1\,. \tag{4.18}$$

For $\lambda < \frac{3}{4}$ its energy is negative, *i.e.* it becomes the (ferroelectric) ground state.

We notice again that small v^2, *i.e.* softness of the material, favours the establishment of a polarized state. We also note that in many ferroelectric materials the effective potential has a minimum at $P = 0$ at high temperatures. The mode softens as we decrease the temperature towards the Curie point and below we find two minima at $P \pm P_0$.

In our case $V(P)$ has its minimum at $P = 0$ at low energies, but supply of energy may establish the quasi-ferroelectric state with minima at $P = \pm P_0$. Thus here as in the case of sect. **3** a long-range order—coherent vibrations, or permanent polarization—is established through the supply of energy.

Clearly actual biological materials will behave in a much more complex way than discussed in this simple model. We should recall in this case that structured water and various ions are to be considered as an integral part of the system.

This implies considerably higher nonlinear effects than discussed above. While these will have to be treated case by case, it seems that they usually will contribute to the stabilization of our metastable state. Thus a polarized molecule will shift mobile ions so that its energy is lowered; in turn, this will, of course, affect the magnitude of polarization.

5. – General consequences.

The two types of excitation discussed in sect. **3** and **4** can have far-reaching biological consequences. First of all, it should be noticed that they require supply of energy, *i.e.* activation. Both can then lead to forces which were absent before activation, and to storage and transport of energy.

Cell water contains considerable amounts of small ions which tend to screen any static electric charges at short range. Hence a system activated to a high dipole moment (sect. **4**), *e.g.* a protein, may exert considerable forces in water-free regions, but will be screened for long-range interaction. This screening does not hold, however, for an oscillating field at frequencies of the order of 10^{11} Hz (sect. **3**). Thus, if the coherently excited vibration represents an oscillating giant dipole, then long-range forces may be activated. They will be shown to be frequency selective.

Consider two oscillating giant dipoles at distance R with frequencies ω_e and ω_s, respectively, in the absence of interaction [4, 7]. At sufficient distance, their interaction is essentially dipolar, since other interactions decrease much more rapidly with R. The combined system then has two normal modes ω_+, ω_-, where

$$\omega_\pm^2 = \tfrac{1}{2}(\omega_e^2 + \omega_s^2)\left(1 \pm (q^2 + Q^2)^{\frac{1}{2}}\right). \tag{5.1}$$

Here (by assuming $\omega_e^2 \geqslant \omega_s^2$)

$$q = \frac{\omega_e^2 - \omega_s^2}{\omega_e^2 + \omega_s^2} > 0 \tag{5.2}$$

and Q essentially is the ratio of the interaction energy between the two giant dipoles and their internal potential energies.

It thus depends on the distance as $1/R^3$.

In the particular case that the giant-dipole oscillations in the two molecules consist in correlated oscillations of, respectively, z_e and z_s elastically bound ions of charge e and of equal mass M, then

$$Q = \gamma e^2 (z_e z_s)^{\frac{1}{2}} / M R^3 (\omega_e^2 + \omega_s^2), \tag{5.3}$$

where γ is a numerical constant.

Clearly the expression for ω_-^2 holds only when $\omega_-^2 > 0$; at short distance thus mode softening occurs.

The interaction energy I is defined as the difference of the free energy F at distance R from its value at $R \to \infty$. Now consider the case that the state with the lower frequency has been coherently excited (n_- quanta) so that the contribution of ω_+ to F can be neglected. Then

$$I = F(R) - F(\infty) = n_- \hbar(\omega_-(R) - \omega_-(\infty)) = \tag{5.4}$$

$$= \frac{1}{\sqrt{2}} n_- \hbar(\omega_e^2 + \omega_s^2)^{\frac{1}{2}} \left\{(1 - \sqrt{q^2 + Q^2})^{\frac{1}{2}} - (1 - q)^{\frac{1}{2}}\right\}.$$

This energy is always negative, *i.e.* attractive. Furthermore, if $\omega_e \neq \omega_s$, then at sufficient distance $Q^2 < q^2$ and we find, after development,

$$I = -\frac{\hbar n_-}{4\sqrt{2}} (\omega_e^2 + \omega_s^2)^{\frac{1}{2}} \frac{Q^2}{q(1-q)^{\frac{1}{2}}} \propto -\frac{1}{R^6}. \tag{5.5}$$

If, however, $\omega_e^2 \simeq \omega_s^2$ and $Q^2 \gg q^2$, $q^2 \ll 1$, then

$$I \simeq -\frac{\hbar n_-}{\sqrt{2}} (\omega_e^2 + \omega_s^2)^{\frac{1}{2}} Q = -\frac{\hbar n_-}{\sqrt{2}} \frac{\gamma e^2 (z_e z_s)^{\frac{1}{2}}}{M(\omega_e^2 + \omega_s^2)^{\frac{1}{2}}} \frac{1}{R^3} \propto -\frac{1}{R^3}. \tag{5.6}$$

In this case of near resonance, the always attractive interaction has very long range. Roughly, if

$$U = \hbar n_- \omega_e \tag{5.7}$$

is the energy of excitation, using $\omega_e \simeq \omega_s$, we have

$$I \simeq -U \frac{e^2 (z_e z_s)^{\frac{1}{2}}}{M \omega_e^2 R^3}. \tag{5.8}$$

While thus activation of coherent vibrations may lead to long-range, frequency-selective interaction, activation of the polar state leads to short-range nonselective forces which might facilitate transport of ions within the particular material, or across it.

Long-range selective forces may lead to selective transportation of molecules like enzymes and thus initiate a particular trend of chemical events. It might also be relevant for the transportation of larger units.

It has been suggested that these properties are an essential feature in the high catalytic power of enzymes [4, 6, 8]. Thus the high internal fields arising from the strongly polarized state could reduce activation energies, while excitation of coherent modes could be used for the transport of energy.

Independently [9], the same conclusion has been reached from an analysis of a great variety of enzymatic reactions.

In [10] it has been suggested that the selective long-range interaction (sect. **3**) may be responsible for the attraction of a homologous pair of chromosomes in meiosis—an interaction which according to [11] has till then remained « a great mystery ».

A number of other possible consequences has been discussed elsewhere [1, 2, 4, 12].

Selective long-range interaction in conjunction with the existence of highly polar metastable states may also be decisive for the establishment of the well-known low-frequency electric vibrations in the brain, whose biological significance is only now being explored [13]. It has been shown that the long-range coherent interaction (based on the 10^{11} Hz coherent vibrations) could in an enzyme-substrate system lead to collective enzymatic reactions resulting in slowly varying (periodic) excitation and de-excitation of the system of enzymes from the only slightly polar ground state into the highly polar metastable state. Thus a slow chemical oscillation is connected with a corresponding electric vibration. The strong electric interaction between the highly polar states in conjunction with the considerable damping of electric currents then imposes on them a limit cycle condition, which recently has been investigated [14]. This model requires large regions to oscillate coherently as has actually been found [15].

6. – Experimental evidence.

In active biological systems experimental evidence for the existence of coherent electric oscillations in the 10^{11} Hz region would require methods which have not yet been fully developed (cf., however, Note added in proofs). Evidence does exist, however, that radiation in this frequency region may have very strong effects on such systems. The most comprehensive report on such effects has been presented in a meeting of the USSR Academy of Science [16]. Properties of a great variety of objects have been found to be strongly influenced by coherent electromagnetic waves in the $(5 \div 7)$ mm (vacuum) wave-length region. Almost all experimental results have in common:

i) That they very strongly depend on frequency in a region in which the optical constants have the appearance of broad bands; often the biological effects showed typical resonance character.

ii) The dependence on intensity exhibits a threshold below which no biological effects occur and above which they are nearly independent of intensity.

To interpret these results [17], we assume that these biological effects are initiated by a first step which is sensitive to a radiation followed by a chain of other possibly very complicated effects which are insensitive to it. The

first step might be connected with a long-range interaction arising from coherent excitation of a molecule, *e.g.* an enzyme which could be attracted to its substrate. Let the time required for this be $t_1(s)$, where s represents the radiative supply of energy to it. According to sect. **3**, excitation of the coherent mode requires a threshold s_0. With growing s ($> s_0$) the time $t_1(s)$ will decrease, so that the total time for the whole process

$$t = t_1(s) + t_2 \to t_2 \tag{6.1}$$

will approach the s-independent time t_2 required for the s-independent chain of further biological steps. Thus in these experiments we find

$$\nu \sim \frac{1}{2} 10^{11}\,\mathrm{Hz}\,, \qquad \frac{\Delta\nu}{\nu} \ll 1\,, \qquad s > s_0\,, \tag{6.2}$$

as predicted in (3.24).

We note that the resonancelike frequency dependence in a region in which the optical constants show little frequency dependence is the best proof that we are not dealing with thermal effects.

The magnitude of s_0 has been estimated theoretically [18] from calculations using methods of laser physics and found of the order 0.04 mW/cm², which agrees with some of the experimental results (*).

The Russian experiments range from various effects on bacteria, growth of yeast cultures, influence on human and animal cells to effects on various genetic properties. Depending on the type of process stimulated by the radiation, a certain biological activity might be increased or decreased. Independently, on a smaller scale and without extensive frequency resolution, biological effects near $0.7 \cdot 10^{11}$ Hz and $1.4 \cdot 10^{11}$ Hz have been found by other investigators [19, 20].

In some of the Russian experiments [21], the biological resonance has been found as narrow as $2 \cdot 10^8$ Hz. This can be understood on the assumption that the only broadening of the excited giant-dipole vibration arises from its electric coupling to the thin layers of structured water whose dielectric absorption lies in this region [22, 23]. The remarkable absence of other frictional effects reminds one, of course, of the various suggestions of the existence of room temperature superconductive regions in biological systems. So far no definite experimental proof for their existence has been found. These narrow resonances are most likely to occur when the level which is excited by the radiation is identical with the level which is strongly populated in Bose condensation,

(*) It should be mentioned that in [18] the presentation of a quadratic Hamiltonian is misleading, though the results are correct. For, as required, the authors do use the influence of a heat bath which in terms of a Hamiltonian would require at least 4th order terms to obtain their eq. (8*a*)—as can be seen from our eq. (3.2) which is responsible for the occurrence of a chemical potential μ.

i.e. when in (3.21) this is the level K_0. Broader bands must be expected when this is not the case.

In experiments of this type, one must be prepared for unexpected non-linear effects of the imposed radiation. Thus, within the range in which the radiation is absorbed, in the first place coherent excitation of some molecules is expected. If the same type of molecules is present at greater depth, then they in turn may become excited by transfer of the excitation energy from the outside molecules. Thus the excitation may be expected to move beyond the penetration depth of the imposed radiation.

Now, in turning to the excitation of the strongly polar metastable state, one would, of course, attempt to excite this state by bringing the molecule in question into a strong electric field. It should be of interest in this context that investigations on nerve membranes have led to the conclusion that the electric field in the membrane makes it nearly impermeable to Na^+ and K^+ ions, whereas the absence of the field leads to a much higher permeability presumably through proteins dissolved in the membrane; it is suggestive to attribute these two different properties to the two states in the proteins.

More direct evidence for conformational changes of certain molecules in membranes due to changes of the electric field is available [24].

A number of other possibilities has been discussed in [4], including results found with the electret method [25] and with laser-induced stimulation of enzyme activity at very high dilution [26].

Thus, while experiments available at present seem to support the theoretical predictions, in a general way much greater refinement of the experimental methods is desirable to obtain more specific experimental results. This in turn should stimulate theoretical investigations of a more specific nature.

7. – Cancer.

The problem of cancer is one of the control of cell division. While bacterial cells under appropriate conditions keep on dividing, cells in a fully grown differentiated tissue or organ divide relatively rarely, though they divide more rapidly in growing tissues and organs. We shall assume that a controlling agency belonging to the whole organ, *i.e.* a collective property, exists and that individual cells respond to it.

The nuclei of these cells contain a DNA-protein complex, whose detailed structure is not well known, though it seems reasonable to assume that its spacial arrangement differs in different organs and depends on the type of differentiation that has taken place. Now we shall assume that these materials are capable of electric vibrations which can be excited coherently in the sense of our model (cf. sect. **3**). Thus metabolic energy would be used to excite these vibrations at a well-defined frequency (or frequencies) which would be char-

acteristic for the particular organ or tissue. Note that experiments discussed in sect. **6** confirm the existence of such vibrations in some biological materials.

This then implies that a coherent electric vibration carried collectively by the individual nuclei extends through the whole tissue (organ), and that the vibrations of each individual cell respond to it and are thus held in the appropriate phase, such that the total electric energy is a minimum. In general, this vibrating polarization field will form a complicated spacial pattern.

Now, preceding cell division, the DNA-protein complex of the nucleus of a cell must unfold and may then be expected to change some of its physical characteristics, such that it looses its resonance with the collective mode. This implies that its energy of interaction with the collective mode is increased, *i.e.* energy has to be spent to perform this change (apart from changes in the internal energy). This energy may not always be available and, as a consequence, the interaction of an individual cell with the collective mode acts as an inhibition on cell division. The strength of this inhibition depends on the magnitude of excitation of the collective mode. This excitation may well depend on the size of the tissue because of the long range of the interactions as discussed in sect. **5**. The possibility of a size-dependent inhibition on cell division thus exists.

A situation thus prevails which in some respects corresponds to order in physical systems described in terms of an order parameter as, for instance, in lasers (cf. also [27]). In our case, the collective vibration represents the order parameter. It is built up of the contributions of the relevant vibrations of the individual cells, and they in turn obey the instruction of the order parameter, namely not to divide.

Now assume that changes have taken place in a particular cell such that its frequency is altered. This cell no longer contributes to the coherent collective vibration, whose intensity will thus be slightly reduced, nor does this cell obey the instruction of the coherent mode, *i.e.* the inhibition to divide disappears. The slight reduction of intensity of the coherent mode due to the change of property of a single cell should have a minute effect on other cells. A critical intensity should exist, however, below which the control of cell division no longer is effective. It is feasible, therefore, that something like a phase transition takes place from the controlled to the uncontrolled state once a critical number of cells has been affected.

Such an approach requires that—while the controlling vibration is strong enough—a cell in the tissue will be induced to transfer into the correctly vibrating state provided it possesses such a state, for only thus can the system exhibit stability.

This may have two types of consequences:

i) Differentiation may be expected to consist in rearrangements of the DNA-protein complex, such that it obtains relevant frequencies assumed

absent before differentiation. Under the influence of the coherently excited mode(s) of the particular tissue (organ), this transition is energetically favourable and hence is stimulated [28]. The differentiated cell then gets subjected to the control.

ii) Mutations which lead to alterations in the relevant frequency(cies) are energetically less favourable than those which do not alter them and which thus still respond to the control and are, therefore, statistically less likely.

Thus, in our model, the sufficiently excited collective vibration exerts an influence on individual cells which inhibits both cell division and « unfavourable » mutations (which would not react to control) and, at the same time, tends to bring cells which do not respond to the control into states in which they do react to it.

If by external influences some cells are changed such that their relevant frequencies are altered, then *a*) these cells no longer respond to the control and *b*) the strength of the controlling vibration is reduced. Both these effects are rectified by the above-mentioned stabilizing influence provided they are not too large. If, however, the number of altered cells is so large that the excitation of the collective mode is reduced below a critical value, then the control becomes ineffective. As a consequence, all cells divide more frequently, and simultaneously « unfavourable » mutations (having « wrong » frequencies) become possible, *i.e.* a phase transition from order to disorder (cancer) has taken place.

It should be noted here that the frequencies of the changed cells will differ from each other; thus no collective mode with sharp frequency will arise: the ordered state has a sharp frequency, the disordered one a broad frequency band.

Which are the external influences that can lead to a change in frequencies? It has been shown [2] that supply of unattached electrons could have such an influence. Such electrons might occur in the conduction band of various molecules. They are then capable of long-wave electric vibrations, plasma waves, whose frequencies increase proportional to the square root of the electron density. These vibrations will interact with the coherent vibrations of our system and under appropriate circumstances can significantly alter the frequency.

Such electrons might be supplied by excitation through ultraviolet light. Or they might be supplied by foreign molecules which are electron donors. A change of frequency might, of course, also arise from an invasion by foreign molecules into the relevant DNA-protein complex.

Other possibilities may arise, however, if we assume that the frequency sharpness found in some of the experiments discussed in sect. **6** is a feature of the proposed collective vibrations. It has been previously indicated that electric coupling with structured water would account for the measured frequency width indicating absence of all other types of friction. Some structural

changes might be sufficient to eliminate this property and with it the effectiveness of control. The affected cells could then still have frequencies in the region of the nonaffected one, but exhibit greater width.

The model described here can without difficulty be presented in mathematical form by using well-known techniques developed, *e.g.*, in connection with laser physics [29]. Even without doing so, it will be noticed that the term « collective mode » might be given a different physical interpretation from the one suggested here, provided individual cells have selective properties which enable them to react to the collective mode and which could be destroyed by external influences.

If the interpretation of the collective mode as a vibration is correct, then one would wish to know the order of magnitude of relevant frequencies. The 10^{11} Hz region is, of course, a first candidate both for theoretical and experimental reasons (sect. **3** and sect. **6**). It might well be, however, that quite different frequency regions are also of importance, and it will in the first place be a task for experiments to find these frequencies.

Most desirable would, of course, be experiments which measure the excitations of the biological material *in vivo*. For frequencies near or above 10^{12} Hz laser Raman effect would be suitable. For frequencies near or below 10^{11} Hz, this becomes inefficient because of the broadening of the primary line through water. Thus the frequencies for which selective biological effects have been discovered (sect. **6**) are beyond the available range. New methods (corresponding to Brillouin scattering) are being developed, but must still be adapted for use in biological materials.

It should be emphasized in this context that vibrations in this frequency range may very sensitively depend on details of the arrangements and on the conformation of relevant molecules including structured water and ions, which must both be considered an integral part of the system. Investigations on molecules, *e.g.* proteins, outside their natural surroundings may but need not give relevant results. Something similar holds of the measurement of absorption of electromagnetic waves by the biological material. We have mentioned above that the excited collective mode might have a complicated spacial pattern. This will probably not be optically active. Optically active modes are, however, to be expected in the neighbourhood of each « relevant » frequency; they would show up as a band rather than as the relatively sharp line we expect from the collective mode, except in cases in which the spacial pattern is homogeneous in the range of penetration of the electromagnetic wave—as might hold for thin layers of the material. It should be emphasized in this context that some of the experiments mentioned in sect. **6** exhibit extremely sharp frequency response for biological effects, whereas broad bands are found in optical measurements. Nevertheless changes found in the optical spectra would certainly indicate major changes and could thus provide valuable information.

One might in particular wish to measure the changes arising from the excitation of electrons into the conduction bands discussed above. It must be pointed out in this context that plasma waves are not optically active except through the influence of surfaces. The same is likely to hold of the collective mode with which they interact. For optical detection, therefore, one would use biological samples which are small compared with the wavelength of the electromagnetic radiation with the relevant frequency. When using a thin layer, then the electric vector should have a component perpendicular to the surface. Internal surfaces may, however, also be effective which to some extent may relax these conditions.

Finally biological effects of the electromagnetic radiation are most likely to provide significant results. If we can learn from the experiments mentioned in sect. **6**, then we might expect sharp frequency resonances in the biological response. As a consequence, a tunable source will be required.

In the first place, we would expect the naturally excited mode to be in its most favourable state. External excitation of modes with the same frequency, though probably with different spacial pattern, would thus interfere with the efficiency of the collective controlling mode. In a grown material this would then increase the rate of cell division, while in a growing system it should decrease it.

Finally, when under the influence of external agencies a sufficient number of cells have undergone changes in frequency to the extent that the controlling collective mode has become ineffective, then a stronger externally imposed vibration of the correct frequency (though possibly with different spacial pattern) might still be able to restore the original order. This, of course, would be the most startling influence of electromagnetic waves on biological systems.

Note added in proofs on 27/9/1977.

Conclusive evidence for the metabolic excitation of vibrational states in bacteria has now been obtained by Raman effect. WEBB and STONEHAM (*Phys. Lett.*, **60** A, 267 (1977)) show that, while resting bacteria exhibit only bands in the (100÷200) cm^{-1} region, active bacteria show strong lines. Moreover (S. J. WEBB, M. E. STONEHAM and H. FRÖHLICH: *Phys. Lett.*, A, in print), the intensity ratio of anti-Stokes to Stokes lines in thermal equilibrium should be about 0.5, whereas it is found close to 1.0, evidence for strong nonthermal excitation.

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Research Report

MICROTUBULE IONIC CONDUCTION AND ITS IMPLICATIONS FOR HIGHER COGNITIVE FUNCTIONS

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The neuronal cytoskeleton has been hypothesized to play a role in higher cognitive functions including learning, memory and consciousness. Experimental evidence suggests that both microtubules and actin filaments act as biological electrical wires that can transmit and amplify electric signals via the flow of condensed ion clouds. The potential transmission of electrical signals via the cytoskeleton is of extreme importance to the electrical activity of neurons in general. In this regard, the unique structure, geometry and electrostatics of microtubules are discussed with the expected impact on their specific functions within the neuron. Electric circuit models of ionic flow along microtubules are discussed in the context of experimental data, and the specific importance of both the tubulin C-terminal tail regions, and the nano-pore openings lining the microtubule wall is elucidated. Overall,

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these recent results suggest that ions, condensed around the surface of the major filaments of the cytoskeleton, flow along and through microtubules in the presence of potential differences, thus acting as transmission lines propagating intracellular signals in a given cell. The significance of this conductance to the functioning of the electrically active neuron, and to higher cognitive function is also discussed.

Keywords: Cytoskeleton; ionic conduction; cognitive function.

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1. Introduction

The cytoskeleton is a web of filamentous protein polymers that define the structure of all eukaryotic cells. Composed of microtubules (MTs), actin filaments (AFs), and intermediate filaments (IFs), such as neurofilaments and keratin, the cytoskeleton provides architectural support for the cell, giving rise to its name, as well as facilitating cargo transport via motor proteins. Additionally, MTs perform specific functions relevant to cell motility, when composed as part of cilia or flagella, and to cell division, when forming the mitotic spindle. MTs have been discovered to play a role in the communication between the cell exterior and the cell nucleus, although the exact molecular mechanism of this process is not yet fully understood [1–3]. In recent years, it has been proposed that the AF and MT elements act as transmission lines to support the propagation of electrical signals within the cell [4–7].

Within axons and dendrites, the neuronal cytoskeleton displays a dense concentration of MTs stabilized by MT associated proteins (MAPs), specifically MAP-tau and MAP2, in polarized (axon), and mixed-polarity (dendrite) orientation. This is linked to the AF structures lining the walls of axons with specific MT architectures characterizing the presynaptic terminal [8]. In dendrites, the MT network connects to AF structures which enter into the postsynaptic density forming a highly elaborate framework. The IFs known as neurofilaments are the most abundant cytoskeletal components of large myelinated axons, although less is known about them compared to MTs and AFs. As such, due to the unique structure and function of the cytoskeleton in neurons, AFs and MTs play a major role in neuronal development, e.g., differentiation and synaptic generation, and neuronal plasticity.

The role of the neuronal cytoskeleton does not appear to be simply one of architecture or transport either. The functioning of the $GABA_A$ receptor, a key target for the action of anesthetics, appears to rely on the microtubular cytoskeleton as its destruction has been shown to cause a 78% rundown in receptor currents [9]. Additionally, hypoxia-induced cytoskeletal re-ordering has been shown to be responsible for suppression of fast sodium channels in brainstem respiratory neurons [10], while drug-induced modifications of MTs altered the activity of calcium-dependent inactivation of calcium channels in isolated rat hippocampal neurons [11–14]. These results suggest that the neuronal cytoskeleton may mediate ion channel responses.

The specialized neuronal cytoskeletal architecture and function, and the inherently electric nature of neurons, point to the importance of the potential function of the cytoskeleton as a transmitter of electric signals. It is this potential electric

signaling function of the MT cytoskeleton that will be explored within the present article. Accordingly, the remainder of this article will be presented in the following manner: Sec. 2 discusses the involvement of the MT cytoskeleton in higher cognitive functions of learning, memory and consciousness, Sec. 3 will present the current experimental evidence for ionic conduction and theoretical models of these phenomena, Sec. 4 describes the link between cytoskeletal networks, the neuronal membrane, and neuron function, Sec. 5 discusses the particular importance of the C-terminal region and the enzymatic regulation of intracellular signaling, and finally Sec. 6 provides discussion and conclusions.

2. The Microtubule Cytoskeleton in Learning, Memory and Consciousness

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While there is, as yet, no direct experimental confirmation that the role of the MT cytoskeleton includes electrical signal conduction in neurons, there is a large amount of experimental evidence supporting a role for MTs in higher cognitive function and by extension, neuronal signaling. The role of the neuronal cytoskeleton in learning and memory has been previously reviewed [4, 5, 15]. While in most tissues MTs exhibit growth, catastrophe (shrinkage) and tread-milling movement, they are less dynamic in neurons, largely because of the stabilizing effect of MAPs. However, rearrangement of the MT cytoskeleton does take place during learning. Learning has been shown to correlate with an increase in MTs [16], and to be impaired by the MT polymerization inhibitor colchicine [17–20], indicating that learning relies on dynamic MTs.

Correlations have been shown between learning and an increase in MAP2 [21], a MT cross-linking and stabilizing protein found mainly in dendrites [22, 23]. The converse is also true. Cerebral hypoperfusion, a condition which impairs learning, correlates with decreased MAP2 [24], and mice that exhibit learning and memory deficits also expresses less cortical MAP2 [25]. However, fear conditioning has been shown to produce significant changes in MAP2 immuno-histochemical staining [26–28], suggesting MAP2 breakdown during learning. This apparent contradiction is clarified when taken in the context of memory consolidation, the conversion of short-term memory into long-term memory. To reorganize the MT matrix in memory formation, the stabilizing protein MAP2 must first be broken down before reorganization can occur, and then reproduced to stabilize the new matrix. In fact, during memory consolidation, tubulin capping proteins increase their binding to other proteins [29], which is accompanied by MAP2 expression alterations [29, 30].

MAP-tau, another MT stabilizing protein found mainly in axons [22, 23], is also associated with learning and memory. Mice with increased levels of tau showed correlating improved memory before tau aggregation, and memory impairment after aggregation [31], and tau excess produced measurable impairment in memory in fruit fly and mice models [32, 33]. This suggests that an overabundance of tau causes excessive stabilization of the MT matrix inhibiting the dynamic reorganization

required in learning and memory. Indeed, a well-known hallmark of Alzheimer's disease, a disease of memory deterioration, is the accumulation of hyper-phosphorylated tau, unbound from MTs, into neurofibrillary tangles which results in neuronal degeneration and malformed neurites [34], suggesting that memory depends on proper functioning of the MT cytoskeleton.

Research on the molecular mechanisms of memory identifies Ca^{2+}/calmodulin-dependent protein kinase II (CaMKII) as crucial to long-term potentiation (LTP) contributing to learning and memory [35, 36]. Phosphorylation of MAP2 by CaMKII is fundamental to the potentiated synaptic response [37], and experiments indicate increased concentrations of dendritic mRNA for MAP2 and CaMKII during the intermediate phase of LTP [38]. CaMKII also directly phosphorylates the MT subunit protein tubulin on both its α and β monomer in the C-terminal region [39]. Additionally, learning via passive avoidance training induces changes in coexpressed MAP2 and protein kinase C (PKC) in rat models [40], and transgenic mice expressing altered MAP2 show decreased MAP2 phosphorylation by cAMP-dependent kinase (PKA) along with impaired contextual learning [41]. Such enzymatic events may be a triggering point for the MT matrix reorganization required for learning and memory formation.

While consciousness itself is hard to define, it is generally agreed upon that general anesthetics result in unconsciousness, making them a natural probe for functional sites of consciousness. Hallmarks of general anesthesia include hypnosis (unconsciousness) and amnesia (the prevention of memory formation), and it has been argued that hypnosis and amnesia are one and the same [42, 43]. The involvement of the cytoskeleton in memory has already been discussed above.

In recent years, evidence has accumulated for the interaction of anesthetics with neuronal MTs. Photo-labeling and gel electrophoresis have shown that the volatile anesthetic halothane binds directly to both α- and β-tubulin [44]. Differential in-gel electrophoresis investigations of protein expression determined that of 500 detectable proteins in rat cortical and hippocampal neurons, tubulin was altered prominently and was among the 2% of proteins affected by halothane, and among the 1% altered by isoflurane (being one of three proteins altered by both anesthetics) [45]. Additionally, tubulin has been found to be altered by desflurane [46], sevoflurane [47], as well as the neurosteroid anesthetics [48, 49].

The notion that the cytoskeleton plays a role in general anesthesia reaches back almost 40 years when Allison and Nunn first proposed the idea based on the depolymerization and structural alteration of MTs by volatile anesthetics [50]. This hypothesis was largely invalidated when it was found that the concentrations of anesthetics required to depolymerize MTs was far above the clinical dosage [100]. However, in more recent years this idea has been revived by quantum mobility theory [51, 52], which does not necessitate the depolymerization of MTs, but rather relies on hindrance of normal MT function by anesthetics. While the validity of quantum effects in MTs has been highly debated [53–55] this theory has resulted in much work being done in the area of cytoskeletal signaling and information processing in neurons.

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Along the line of this argument, without the need to invoke quantum theory, hindrance of intracellular signaling, through the interaction of anesthetics with MT, could be a contributing factor to the mechanism of general anesthesia, and memory formation.

3. Ionic Signaling via Cytoskeletal Microtubules

The electric properties of biomolecules (and biomolecular structures composed of them), such as tubulin, the constituent protein of MTs, and actin, are of crucial importance to a biophysical understanding of cellular processes. This is all the more important to the understanding of the electrically active neuron which possesses an extremely complex cytoskeletal architecture not seen in the cells of other tissues [8].

MTs have been shown to be affected by external electric fields. It has been demonstrated that pulsed electric fields can align MTs assembled *in vitro* [56], and that AC electric fields can have a pronounced effect on the mitotic spindles in several cancerous cell lines [57], thus indicating the importance of the electrical properties of MTs. These electric properties may involve both intrinsic effects of the protein and ionic contributions from counterions attracted to and concentrated near the MT by its electric field. Unfortunately, conductance measurements on biopolymers are fraught with difficulties due to the structural heterogeneity and instability of these polymers, the liquid state of samples and the dependence of biological systems on environmental factors such as pH, temperature, and ionic concentrations. Nanotechnology advances have improved techniques allowing more precise investigations to take place. Fortunately, despite the inherent difficulties, a number of experimenters have investigated either the intrinsic [58–60] or ionic [61, 62] conductivities of MTs.

MTs are hollow cylindrical structures comprised of the globular protein tubulin. The tubulin molecule is a heterodimer composed of α- and β-tubulin monomers. The α-β complex that assembles into MTs has dimensions of approximately $8\,\text{nm} \times 5\,\text{nm} \times 4\,\text{nm}$ [63] (see Fig. 1(a)). The inner and outer dimensions of an average MT cylinder are 15 nm and 25 nm, respectively, while their length typically varies from less than 200 nm to several micrometers, or more (see Fig. 1(b)). Each dimer possesses a highly flexible C-terminal tail region that can extend up to 4 nm from the dimer surface. Dimers bind together end-to-end (i.e., β to α) in string-like structures known as protofilaments. MTs in human cells typically have 13 protofilaments arranged side by side, generally in parallel arrangement [63], which wrap to form a cylinder, with the dimer C-terminal tails extending outwards from the MT surface. A slight shift between monomers in adjacent protofilaments results in a spiraling three-start left-handed helix of monomers. Two types of lattice structures have been observed [64]: the A-lattice structure possessing rotational symmetry and continuous wrapping, and the B-lattice which possesses a physical discontinuity called a seam running down the length of the MT (see Fig. 1). It is commonly accepted that, *in vivo*, MTs exhibit the B-lattice arrangement [65], however it has been suggested that dendritic MTs may prefer the A-lattice arrangement [52]. Due to the shape and spacing of

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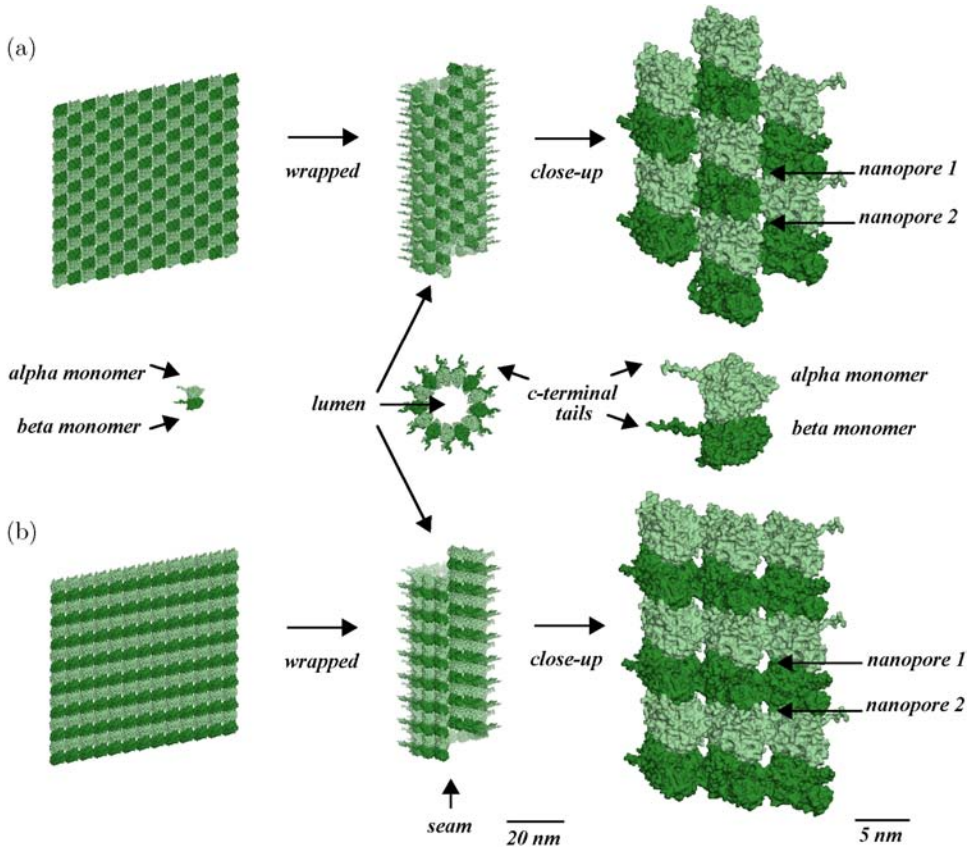

Fig. 1. Microtubule lattice geometries. (a) A-lattice showing perfect helical symmetry. (b) B-lattice showing the physical discontinuity known as a seam. Scale bar before close-up is 20 nm, after close-up is 5 nm.

the tubulin dimers the MT possesses small, nanometer size openings between the outer environment and the inner MT lumen (see Fig. 1). Two distinct types of these nanopores exist for each lattice geometry.

Ionic conductance along MTs relies both on their electrostatic and structural properties. In particular, the surface of the tubulin dimer is largely negative, with a specifically large negative charge located on the C-terminal tail, with the remainder concentrated along the protofilament ridges [66] (see Fig. 2). This results in an overall negatively charged surface for a MT with an average linear charge density of 52.2 electronic charges (e) per dimer, or approximately 85 e/nm [62]. In solution, positive ions are attracted to the MT surface, while negative ions are repelled, creating a condensed ion cloud to counter the MT negative surface charge (see Fig. 3(b)).

In the presence of an applied voltage, the loosely held counterion cloud is free to migrate, creating an ion flow and an associated electrical current, giving rise to ionic conduction (see Fig. 3(c)). Based upon Ohm's law, conductivity is proportional

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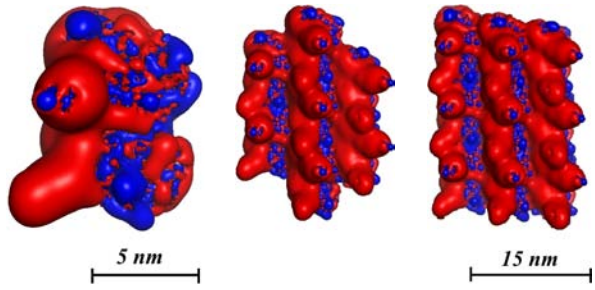

Fig. 2. Electrostatic map: Positive (blue) +0.5 kT/e, Negative (red) −0.5 kT/e. (a) Tubulin dimer. (b) Patch of A-lattice microtubule showing a 7-dimer neighborhood. (c) Patch of B-lattice microtubule showing a 9-dimer neighborhood.

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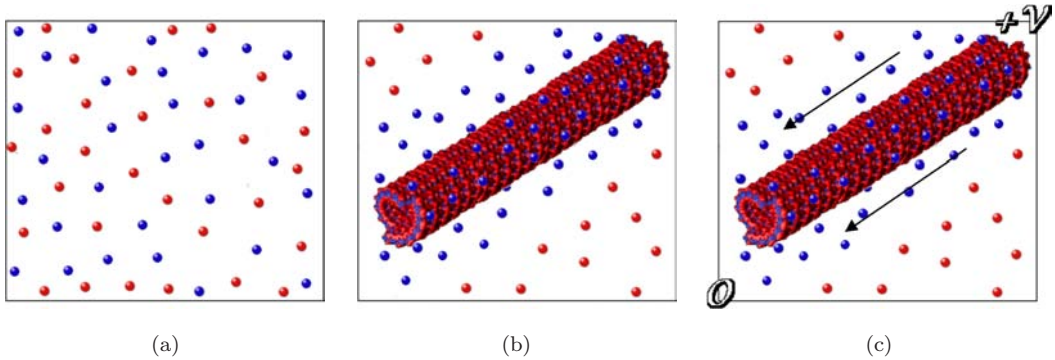

Fig. 3. Cartoon of ion flow along a microtubule. (a) Positive (blue) and negative (red) ions in solution. (b) Positive ions loosely condense around a microtubule due to its negative surface charge (red). (c) In the presence of a potential difference, positive ions flow down the microtubule. (C-terminal tails of the microtubule are not shown.)

to ionic concentration, and so the condensed ionic environments surrounding AFs and MTs are expected to be good conductors in the presence of a voltage source. Consequently, these biofilaments can be thought of as conductive elements in the cells' cytoskeletal circuitry.

In regards to the ionic conductivity, direct measurement attempts of conductivity along MTs in solution were made in filled microchannels and determined to have an upper bound of 90 S/m [61]. Results well within this bound were found by Minoura *et al.* using an electro-orientation method which yielded an MT conductivity of 150 mS/m for intact MTs and 90 mS/m for subtilisin-treated MTs [62]. The enzyme called subtilisin cleaves the C-terminal tail region from MTs. The difference in conductivity due to subtilisin treatment was shown to be roughly proportional to the change in charges per dimer associated with the absence of C-termini. This suggests that counterions on the surface, including those on the negatively charged C-termini, are responsible for the 150 mS/m observed conductivity via the flow of

these condensed ions around the surface of the MT, a value roughly 15 times larger than the conductivity of the surrounding solution (9.5 mS/m).

Priel *et al.* applied a patch-clamp technique to measure electrical currents flowing along MTs following electrical stimulation [7]. They showed that the electrical current generated in solution by a pulse of applied voltage across a 20–50 μm distance is approximately twice the value when MTs are present. Additionally, the group found a maximum MT conductance of 9 nS. These results taken together strongly suggest that, for MTs in solution, the flow of condensed ions around the MT surface provides the observed MT conductivity.

Ionic conductance around seamless, A-lattice MTs was modeled using a theory of resistor networks, where a single MT was represented as a two-dimensional periodic helical lattice of monomer pairs, represented as resistors creating a series of conducting pathways connected in parallel [67]. Furthermore, this model was tested against random walk simulations of ionic conduction along the monomer lattice by considering the relationship between the stochastic random walk process and the theory of electrical networks. Theoretical modeling and computer simulation yielded comparable results.

Another circuit-based investigation used a nonlinear cable-like model of a MT [68], involving resistive, capacitive, and inductive elements in series and in parallel. The model developed considered the MT lumen separated from the rest of the ions in the bulk solution by the counterion condensation cloud, which acts as a dielectric medium providing resistive and capacitive components to the circuit. Additionally, the helical structure of the MT provided an inductive component in analogy to the helical ion flow induced by a solenoid. Solitary wave solutions, of a potentially unstable nature, were predicted qualitatively supporting ionic signal propagation and amplification by MTs.

It must be noted that in the above models, the effect of MAPs was not considered, and neither were the effects of the tubulin C-terminal tails. The charge on the C-terminal tails accounts for approximately 40% of the total MT negative surface charge, with the exact charge being determined by the particular tubulin isotype (or a mixture thereof) comprising the MT. As such, it is expected that positive counterions will condense in larger concentrations around these tails to compensate for their large electrostatic negativity. Fluctuations in the highly flexible C-terminal tail domain must be associated with corresponding movements of the surrounding ion cloud to preserve charge compensation. Thus, the effects of C-terminal tail dynamics are expected to have a significant impact on MT ionic conduction.

Molecular dynamics modeling of the time-dependent conformational states of C-termini interacting with neighboring C-termini was carried out where the MT surface interacts with the C-termini via positively charged surface regions. The inclusion of interactions with adjacent MAPs indicated a potential for C-termini generated electrostatic perturbations of counterions to be transmitted between adjacent MTs via MAP2 [69, 70], forming the basis of a network of electrical elements created by

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the architecture of the cytoskeleton in the cell. More recently, the effects of the C-terminal tails on ion transport along the MT have been specifically investigated [71]. In this case, ionic conductance was modeled in terms of positive ions hopping along a seamless lattice of ionic traps representing the large negative charge of the C-terminal tails along the MT surface. The simulations predicted critical ion concentrations required for conductance to emerge as a percolation threshold. It was shown that this effect can lead to the formation of complex conduction pathways, both of which are highly dependent on the charge of the C-terminal tails, something which is determined by the particular tubulin isoform whose distribution in the body is strongly tissue dependent with, for example, βIII tubulin being largely concentrated in neurons.

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A nonlinear RLC transmission line model of MTs viewed as rings of 13 tubulin dimers coupled together forming an electric circuit has been recently proposed [72]. In this model, the MT, with condensed counterions, was viewed as spatially separated from the rest of the ions in the bulk solution by a dielectric medium composed of an ionic layer depleted of positive ions. This layer provided both a fixed resistive component and a capacitive component that changes with the increasing concentration of counterions due to the flexibility of the C-terminal tail. Inductive components were also considered due to the 13 parallel currents arising from ionic wave flows along each protofilament. Solutions to this model indicated stable and robust anti-kink wave propagation with a velocity value close to experimental findings. It was also shown that even weak electric fields, such as a cellular intrinsic field, could tune both the direction and velocity of this propagation.

The latest research in this area generated an electric circuit model of the MT which incorporates not only the C-terminal tails, but also the nanopores within the MT wall, thus including not only ionic conductance along the outer surface of the MT, but conductance through the MT wall, and in the lumen as well [73]. This latest model effectively includes a 3D representation of the MT with ions flowing both along its surface and across it, in and out of the luminal space. Experimental evidence suggests importance for these nanopores in drug diffusion and MT–protein interactions. For example, the MT-stabilizing MAP tau, with its crucial function in axons, is believed to bind at a location in the MT lumen, suggesting that it passes through the nanopore to reach its binding site. It has been demonstrated that certain drugs (e.g., paclitaxel and epothilone) traverse the nanopores to find their binding sites in the MT lumen [74], or bind in the nanopore itself. Thus, it is assumed that based on their size, the nanopores are quite permeable to ions. Brownian dynamics simulations of ion conductance through B-lattice nanopores indicated a larger conductance traveling into the lumen for positive ions than for ions moving outside the MT, while negligible conductance was found for negative ions in either direction [73]. These results suggest that MTs give rise to complex electric conduction phenomena involving ions, solute molecules and potentially protons.

Finally, several authors have suggested that the flexible C-terminal tails can interact with positively charged residues on the body of tubulin [75, 76]. The formation and disruption of these interactions was proposed to act as a stochastic alternating voltage source on the cations close to outer surface of MT. Simulations including the effects of the asymmetrical conductance and this internal voltage source indicated that an ionic current can travel through the MT wall and propagate down the MT lumen [73]. This current adds directly to the longitudinal current along the outer MT surface arising from an external voltage source, indicating that MT can amplify low-intensity endogenous currents within the cell. Because the voltage source arising from the dynamic behavior of C-terminal tails drives cations into the lumen of the MT all along its outer surface, this effect could be an important factor enabling the dissipation of a potential gradient to be focused towards one end of the MT, regardless of the orientation of the source relative to the MT.

Overall, these results suggest that counterions condensed around the surface of the MT cytoskeleton are allowed to flow in the presence of potential differences, effectively making it a network of biological electrical wires. This may have important implications in the coupling of intracellular signals in all cells. However, as stated earlier, electrically active cells, such as neurons, generate potential differences via the flux of charged ions, and in these instances the mechanism of conduction along MT may be of extreme significance.

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4. Connections Between Microtubule and Actin Networks and the Neuronal Membrane

Both AFs and MTs form networks using various binding proteins. There are over 100 actin-binding proteins implicated in cell types and locations. Of particular relevance are α-actinin and spectrin that promote neural plasticity. The protein α-actinin is known to link AFs to membrane channels, thereby coupling channel activity to the intracellular network. MAPs in neurons are responsible for the structured parallel network observed in the axon, as well as the mixed-polarity arrangement in dendrites. Specifically, MAP-tau is abundant in the axon, creating polarized bundles of MTs to span its length, while in developed neurons MAP2 is present in dendrites where it stabilizes the network of MTs to form a mixed polarity structure. Since MAP2 is relatively heavy and long compared to MAP-tau, the average distance between the MTs in the network developed in the axon is about 20 nm whereas in dendrites it measures 50–90 nm.

The dynamic interactions between both AF and MT cytoskeletons are central to neuronal cell morphology and function. Only recently have clear functional interactions between those cytoskeletal structures become apparent. Association of MT with AF in filopodia appears to guide MT growth along the most efficient path and play a key role in neurite initiation [77]. AF-MT interactions are also important for growth cone turning underlying axonal path-finding mechanisms [78].

Like MTs, AFs are also highly charged polymers, with a negative charge density of about 4 e/nm [79]. They are thinner than MTs with diameters of about 7 nm, they are less rigid by a factor of about 300, are typically shorter, often cross-linked by other proteins into bundles, and are closely associated with neuronal cell membranes [80]. Application of a patch-clamp technique to measure electrical currents flowing along AFs following electrical stimulation found that the charge transferred was consistent with the dynamic behavior of ion flow along the AF longitudinal filament axis and that this flow was doubled by including AFs in the solution [81]. Additionally, osmotic stress applied to a polymerized actin solution revealed a reduction in actin surface charge density with increasing stress and a nonlinear electric response upon application of electric fields, indicating that F-actin in solution may display non-ideal electro-osmotic properties consistent with ionic "cable" behavior [82]. Also, measured electrical signals from the end of an AF after an input voltage was applied indicate that AFs support ionic waves in the form of axial nonlinear currents [79]. Thus, like MTs, AFs are capable of ionic signal conduction and amplification.

In neurons, there is an increased presence of binding proteins capable of interacting with both MTs and F-actin, mediating signaling between both cytoskeletons. For example, MAP2 binds to F-actin efficiently, and induces actin bundle formation. MAP2 has also been found to localize in regions devoid of MTs in neurons [83], in particular in dendritic spines. Since actin bundles enter the spine neck and bind to the postsynaptic density (PSD), and on the other end are connected to the MT network filling the dendritic shaft, it has been hypothesized that a direct regulation of ion channels and thus synaptic strength by AFs and associated cytoskeletal structures controls and helps modify the electrical response of the neuron. In this picture, MTs arranged in networks of mixed polarity, receive signals in the form of electrical perturbations, from synapses via AFs connected to MTs by MAP2 [84], or via direct MT connections to PSD proteins by molecules such as CRIPT [85]. The MT network operates on the propagating waves and modulates the information it carries to control the output measured by remote membrane areas. This intracellular information processing mechanism effectively regulates the cell's electrical response [5].

Since voltage-gated Na^+ and Ca^{2+} channels play an important role in amplifying the excitatory postsynaptic potential (EPSP) responses towards the soma, channel regulation will be important also for dendritic interactions in the vicinity of the synaptic input. Voltage-gated channels may dynamically modify the local input properties — resistance and time constant, which in turn would influence the spatio-temporal integration of EPSPs and inhibitory postsynaptic potentials (IPSPs), thus, promoting highly nonlinear interactions. Hence, the nonlinear properties of dendrites may act as amplification/modulation devices. This control of membrane potential would modify synaptic integration and therefore modulate the whole neural response.

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5. Enzymatic Control of Signal Propagation via Modification of Tubulin C-Termini

Reviews of the specialized role of tubulin isoforms suggest that the evolution of new cell types corresponds to the origin of specific tubulin isoforms [86]. It is interesting to note that of all tissues the brain possesses the largest variety of tubulin isotypes [87]. In particular, this suggests high probability for a very significant role of tubulin isoforms in neural network complexity and its origin [88].

The C-terminal is the most variable region among the many different tissue specific isotypes of tubulin, and, as such, has been suggested to mediate the specific cellular functions of MTs [87, 89]. The importance of the C-termini to the structure and function of MTs has been reviewed extensively [90–92], and includes effects on MT polymerization, and interaction with MAPs and a host of other proteins.

In this regard, it has been suggested that this C-terminal region acts as a chaperone to interactions with other proteins such as MAPs [93]. In fact, it has been shown that MAP tau triggers structural changes in elements on the MT surface [94], and simulations indicate these elements may be the C-termini [90]. The C-termini are also the sites of many of the post-translational modifications of tubulin, including polyglutamylation, polyglycylation, detyrosination/tyrosination, removal of glutamatic acid and phosphorylation of serine and tyrosine residues [95, 96]. These modifications are the result of enzymatic processes [97], and considering the involvement of the C-termini in MT polymerization and interaction with MAPs, are expected to have dramatic dynamic effects on the MT network. For example, modifications that change the electrostatic nature of the C-termini, by adding or removing charged residues, would effect the capacitance of the condensed ionic atmosphere surrounding a MT, as well as the magnitude of the voltage source arising from tail–body interactions. Thus, should the C-termini play a significant role in ionic conduction along MTs, as proposed by the simulations mentioned above, then these enzyme-triggered events would have great effects on ion flow, providing a mechanism for the regulation of intracellular signaling.

6. Discussion and Conclusions

MTs composed of tubulin and some associated proteins are ubiquitous cellular structures of eukaryotic cells important for organizing cytoplasmic properties and cellular movement. Their α- and β-tubulin subunits constitute heterodimers that associate into protofilaments with a strict α/β alternation. Protofilaments are able to form rings and spirals or associate laterally building up both different kinds of sheets and MTs or even macrotubules, depending on the prevailing environmental conditions during the assembly process. The various geometrical possibilities that MTs, AFs and other proteins present in the cytoskeleton give rise to the design of very complex architectures of these filamentous structural elements of neurons. However, their intrinsic electrical conduction and dielectric polarization properties open

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an additional, entirely new spectrum of functional applications within the cellular machinery.

Due to the presence of a large electric dipole in each of the tubulin units (including the layer of bound structured water surrounding the solvent exposed area) and the negatively charged tubulin C-termini, electric fields are expected both to transport suspended MTs into a desired direction and to steer MTs actively driven by motor proteins. Correspondingly, in experiments performed at nearly physiological conditions, (pH 6.8 and about 120 mM ionic strength) individual suspended MTs were found to move steadily towards the positive electrode with an electrophoretic mobility: $2.6 \times 10^{-4}\,\mathrm{cm}^2/\mathrm{Vs}$ [98]. The electrophoretic mobility decreases with ionic strength and pH. The MT movement changed direction to the anode at pH 4.1, indicating a positive net charge has been created on MTs when they were kept at this low value of pH. Interpolation of the isoelectric point gives a value of about pH 4.2 for MTs, which is remarkably lower than that of α- and β-tubulin monomers found to be near pH 5.5. At pH 6.8, external electric fields forced MTs to glide across a kinesin-coated surface into a preferred direction to the anode, in accordance with the negative net charge of the MTs at this pH.

Interestingly for potential nanotechnology applications, the surface of tubulin molecules exposes a defined pattern of amino acid residues (histidines) that provides active sites for nucleation, organization, and binding of metal nano-particles. The doping or decorating of the MT with these nano-particles can be performed by using various processes such as precipitation, solvent evaporation, and vapor phase reactions. The formation of nanometre-size (diameter: 1.5–3.5 nm) immobilized noble metal particles on the MT surface has been experimentally demonstrated [99]. The particles formed have shown definite patterns reflecting the regular arrangement of the α- and β-tubulin subunits within the MT. By further particle growth, a quasi-continuous MT coating can be obtained resulting in Pd nanowires and, furthermore these wires can be assembled into circuits [99] (see Fig. 4). Such an isomolecular organization of nano-particles into functional 2D and 3D structures is a potential method for developing electronic devices with novel biophysical properties. These nano-particles may be deposited randomly or pseudo-randomly within the underlying MT geometry. This results in the formation of a MT with mechanical, electrical and magnetic properties that are different from the original MT. Electrically modified MTs can become useful in applications such as artificial nerves, neural networks and logic circuits, biosensors, neuron-to-neuron couplers, bioisolators and bioswitches. It is feasible to apply the intrinsic properties of MTs, as well as electrically and magnetically modified MTs, to enable the construction of electronic devices and electronic circuits. For example, a four-terminal device can be fabricated from MTs by depositing two parallel gold pads on a substrate, and adding a layer of MAPs between the gold pads to aid in the contact between the gold pads and the MTs. The application of an electric field across the two gold pads will align the MTs in a parallel plane, and allow for a low resistance state. Conversely, the removal or lack of an electric field between the gold pads provides for random

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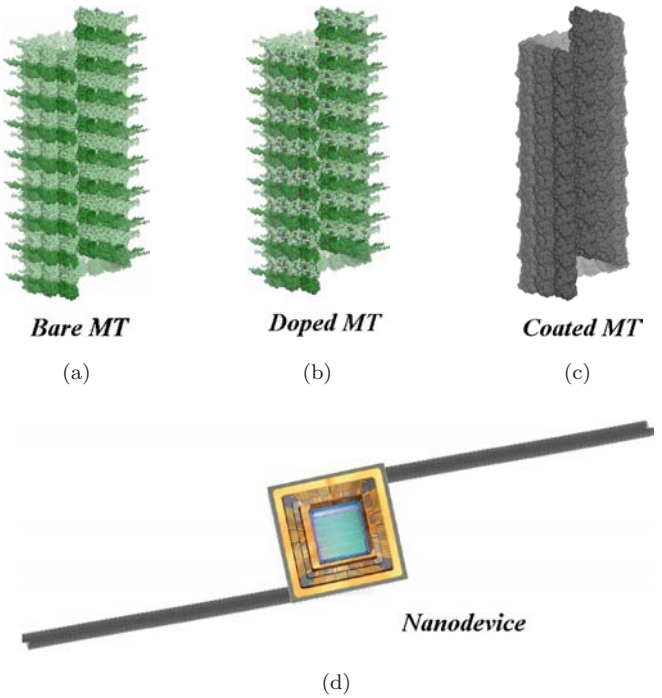

Fig. 4. Nanowires. (a) Bare B-lattice microtubule. (b) B-lattice microtubule doped with silver nano-particles at histidine residues. (c) Silver nano-particle-coated microtubule. (d) Hypothetical nano-device wired with silver coated microtubules.

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orientation of the MTs with respect to each other and to the gold pads. Variations on the configuration of the gold pads provide for inter-changeability between high- and low-resistance states when an electric field is applied. The above four-terminal logic device may also be used to provide electrical isolation of unwanted signals in the presence of an electric field. Noble metals that were used to decorate or coat the MTs for electric field response include gold and palladium. The use of ferromagnetic materials such as iron to decorate MTs may lead to creation of bioengineered structures that are highly responsive to an applied magnetic field.

The conventional thinking ascribes consciousness to the central nervous system only and hence, it is crucial to investigate what makes the MTs there suitable for information processing or at least for aiding the cognitive process. This would be in contrast to the MTs formed in almost every other cell of the body. On the other hand, cell shape determines whether or not external electric fields, such as that of the nerve impulse, may penetrate the cell. This implies a feedback mechanism between MTs and the cell shape. The goal of the studies reviewed in this article is to learn more about the function of tubulin and MTs and to discover novel properties which make them particularly useful within the nervous system. We hope to have shed light on the reason why MTs are important for neuron function and consciousness. We have also concluded that modifying biophysical properties of MTs using nanotechnology

may open a new range of possibilities such as bioelectronic artificial intelligence devices.

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Microtubule as Nanobioelectronic Nonlinear Circuit*

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Abstract: In recent years, the use of biological molecules has offered exciting alternatives to conventional synthetic methods. Specific methods use various biological templates to direct the deposition and patterning of inorganic materials. Here we have established a new electrical model of microtubules as a biological nanoscale circuit based on polyelectrolyte features of cylindrical biopolymers. Our working hypothesis is that microtubules play an active role in sub-cellular computation and signaling via electronic and protonic conductivity and can thus be made useful in hybrid materials that offer novel electronic characteristics. We verify these hypotheses both computationally and analytically through a quantitative model based on the atomic resolution structures of the key functional proteins.

Keywords: Nanobioelectronics, Microtubule, Protein, Nano pore, Nonlinear circuit.

1 Introduction

Nanobioelectronics is a thriving new area of research at the interface between the life sciences and nanotechnology, which deals with structures of dimensions ranging from 1 nm to 100 nm, below the range of lithographic fabrication techniques [1]. Nanobiotechnology aims to exploit biomolecules and the processes carried out by them for the development of novel functional materials and devices [2] and, more speculatively, nanomachines, perhaps nanorobots. These developments open up an exciting possibility of building integrated systems in which electronic and biomolecular components function side–by–side. Biopolymer structures are attractive as templates to form nanoscale architectures for electronics because of their size, geometry and ability to interact with inorganic materials.

Proteins represent fertile territory for nanobioelectronics because they have properties ideal for engineering purposes. They possess sophisticated architectures at nanoscale dimensions, rich chemistry and versatile enzymatic

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activities. Proteins are capable of carrying out complex tasks in cells. We need think only of examples such as the linear motors of the cytoskeleton, voltage–gated ion channels, DNA replication complexes, or the photosynthetic reaction centres [2]. By genetic engineering and/or chemical modification or by using proteins in ways not found in nature, nanobiotechnology can harness the power of proteins to create new components for materials and devices.

Most studies in this area have been performed on DNA molecules [3, 4]. It has been shown that DNA may be used as a template to form silver nanowires connecting gold electrodes [5]. Microtubules (MTs) comprise another interesting type of protein structure that may be a good candidate for designing and manufacturing electronic nanodevices. MTs are cytoskeletal biopolymers shaped as nanotubes. *In vivo* MT cylinders have 13 protofilaments (PFs), as shown in Fig. 1a. The external and internal diameters of MTs are 25 and 15 nm, and the length can reach up to 10–15 μm in a cell. The dimensions of a constituent tubulin dimer (TD) are (4×4×8) nm^3, and there are 13 dimers from 13 PFs considered as an elementary ring of MT. An important point is that every tubulin has a long thin flexible tail (TT) protruding out from the MT cylinder (every dimer has two TT). These TTs have an important role in electric capacitance in our model, as shown Fig. 1b.

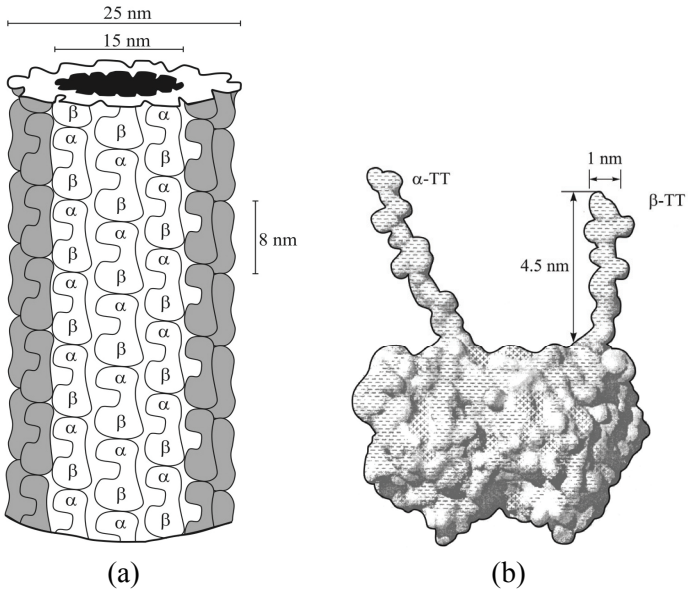


Fig. 1 – (a) *A MT hollow cylinder of* 13 *parallel PFs with denoted characteristic dimensions: outer and inner diameters of* 25 nm *and* 15 nm*, respectively, and TD length of* 8 nm; (b) *The topology of a* TD *with* TT*s whose dimensions are: the length of* 4.5 nm *and diameter of* 1 nm.

manning condensation = counterion condensation

In this paper, we explained the polyelectrolyte character of MT and described the basic components of the MT as a biomolecular nanoscale nonlinear electronic circuit. Then we set up an electrical model of the MT using capacitive and resistive components. Also, we analytically and numerically analyzed the voltage equation.

2 Characterization of the Electric Components of MT

In as much MTs are mostly negatively charged on their outer surface they are true polyelectrolyte polymers. This is the consequence of the fact that numerous amino acids forming tubulin have many negatively charged residues under physiological conditions. As a result, every MT attracts positive counter ions close to its surface creating so called ionic cloud (IC), while negative ions of cytosol are repelled such that a cylindrical depletion area is created around the MT. The width of this depleted layer is called the Bjerrum length, l_B, defined by equating the Coulomb attraction energy of the ions with the thermal energy:

$$e^2 / \left(4\pi\varepsilon_0 \varepsilon l_B\right) = \mathrm{k_B} T. \tag{1}$$

Taking $e = 1.6\times10^{-19}$ C, $\varepsilon_0 = 8.85\times10^{-12}$ F/m, $\varepsilon = 80$ (for cytosol) and $k_B = 1.38\times10^{-23}$ J/K, for the physiological temperature $T = 310$ K we obtain $l_B = 0.67$ nm. The occurrence of Manning condensation around the polymer of radius r also depends on the fact that the salt concentration n in the cytosol should be low enough to satisfy the inequality $l_{Db} >> r$, where the Debye length, $l_{Db,}$ is defined by

$$l_{Db}^{-1} = \left(8\pi n l_B\right)^{1/2}. \tag{2}$$

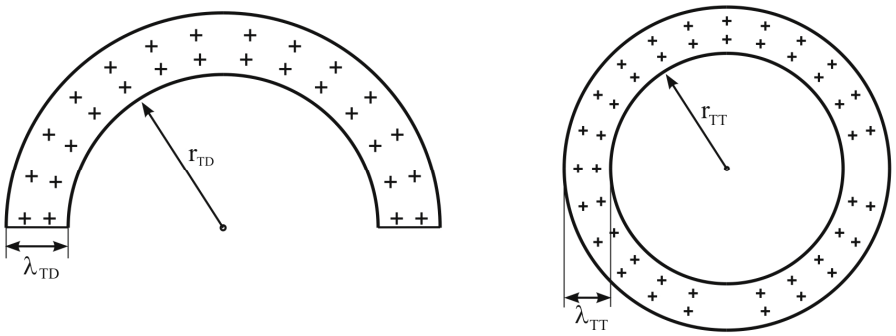


Fig. 2 – *Schematic representation of the counter–ion charge distributions surrounding a* TD (*left panel*) *and a* TT (*right panel*).

EUP = Elementary Unit of Protofilament

Since the radius of a TT is of the order of 0.5 nm and the radius of a TD is of the order of 2.5 nm, while the Debye length for the cytosol is of the order of 10 nm [6], we see that the condition in (2) holds for both radii. Thus, we are now able to reliably estimate the respective condensate thickness λ:

$$\lambda = A\left(rl_{Db}\right)^{1/2}, \quad A<1, \tag{3}$$

where A depends only weakly on the Manning parameter q_0. Taking $A = 1/2$ one finds the corresponding values for the TD (λ_{TD}) and TT (λ_{TT}) as follows: $\lambda_{TD} = 2.5$ nm; $\lambda_{TT} = 1.1$ nm. These values will be used in our calculations of the corresponding capacity and resistivity, see Fig. 2.

2.1 The capacitance of an elementary unit of MT

In an earlier paper [7], a detailed Poisson-Boltzmann approach was used to evaluate the capacitance of an elementary ring of an MT which consists of 13 dimers. Here, we adopt the same expression which reads:

$$C_0 = \frac{2\pi\varepsilon_0\varepsilon l}{\ln\left(1+\dfrac{l_B}{R_{IC}}\right)}, \tag{4}$$

where l stands for the length of a polymer unit and $R_{IC} = r_{TD} + \lambda_{TT}$ for the outer radius of an IC. The other parameters have already been introduced. We first estimate the elementary unit of each MT protofilament (EUP) capacitance contributed by a TD, see Fig. 3. With $l_{TD} = 8$ nm and $R_{IC} = r_{TD} + \lambda_{TT} = 5$ nm, we find for TD (including only the outer surface) $C_{TD} = 1.4\times10^{-16}$ F. Analogously, we can consider an extended TT as a smaller cylinder with the radius $r_{TT} = 0.5$ nm and the thickness of its IC equal to $\lambda_{TT} = 1$ nm. Its extended effective length should be $l_{TT}^{eff} = 4.5\text{ nm} - 2.5\text{ nm} = 2\text{ nm}$ meaning that its part close to the tubulin surface is already embedded. Thus, we now estimate the corresponding capacitance, $C_{TT} = 0.26\times10^{-16}$ F. Accounting for the fact that two TTs are present in each tubulin dimer, we finally obtain $2\times C_{TT} = 0.52\times10^{-16}$ F. The two capacitance values above are considered to correspond to a parallel arrangement with respect to each other, so that the total maximal capacitance of an EUP is readily estimated as:

$$C_0 = C_{TD} + 2\times C_{TT} = 1.92\times10^{-16}\,\text{F}. \tag{5}$$

We have emphasized that TTs capacitance must change with an increasing concentration of condensed cations due to the shrinking of flexible TTs. These changes are slightly different due to the different structures of α and β type TTs. To include this case we introduce the reduced factor of nonlinearity as follows

$$b_0 = \frac{b_\alpha b_\beta}{b_\alpha + b_\beta}, \tag{6}$$

where b_α and b_β stand for the respective TTs. This implies that the charge of an EUP capacitor diminishes with an increased voltage in a nonlinear way $\Delta C_0 = C_0 b_0 v$, $b_0 v << 1$.

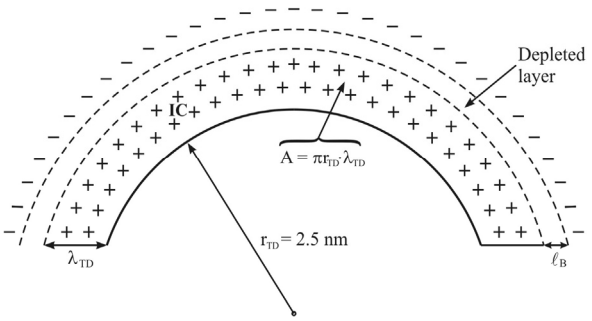


Fig. 3 – *Schematic illustration of the calculation of the MT capacitance.*

Additionally, we account for the tilting movements of TTs under the combined action of thermal fluctuations [8] and a changing voltage due to an incoming ionic wave. Thus, the part of EUP capacitance contributed by TTs should also change by TTs tilt as shown in Fig. 4. The change of the effective length of a TT is an additional factor affecting the capacitance, ΔC_{TT}. We assume that this change can be adequately described by the oscillating function

$$\Delta l_{TT}^{eff} = l_{TT}^{eff} \sin\left[\Omega\left(t - t_0\right)\right] \cong l_{TT}^{eff} \Omega\left(t - t_0\right). \tag{7}$$

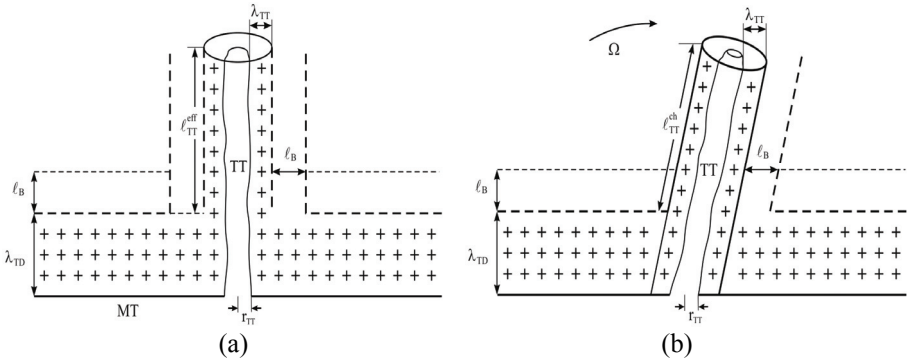


Fig. 4 – *A comparison between*: (a) *an extended* TT;
(b) *a tilting* TT *due to oscillations.*

So that the capacitance changes linearly with a change in the effective TTs length

$$\Delta C_{TT} = C_0 \Gamma_0 \Omega(t - t_0), \tag{8}$$

where the frequency Ω is much lower than the inverse charging time of the EUP capacitor due to the strong viscous damping of the TT tilt, thus justifying the

linearization of the above sine function in (7). In some sense, this effect is similar in character to the thermal ratchet mechanism combined with an asymmetric ionic potential. Including the two aspects of TTs dynamics described above, the charge of EUP can be expressed as follows

$$Q = C_0\left[1-\Gamma_0\Omega(t-t_0)-b_0 v\right]v, \tag{9}$$

where Γ_0 is a dimensionless parameter.

2.2 The conductance of Nano–pores

Between neighboring PFs there are two distinct types of (NPs). NP–1, see Fig. 5 left, is located where an inter–dimer β/α interface of one TD lies next to the inter–dimer β/α interface of the adjacent TD molecule. The so–called NP–2 arises where an intra–dimer interface of one dimer lies next to the intra–dimer interface of an adjacent TD of a neighboring PF, see Fig. 5 right. Freedman et al [8] used HOLE and AMBER programs to estimate the effective radius of the narrowest points within these NPs and found them to be 0.4 nm and 0.47 nm for type–1 and type–2 NPs, respectively.

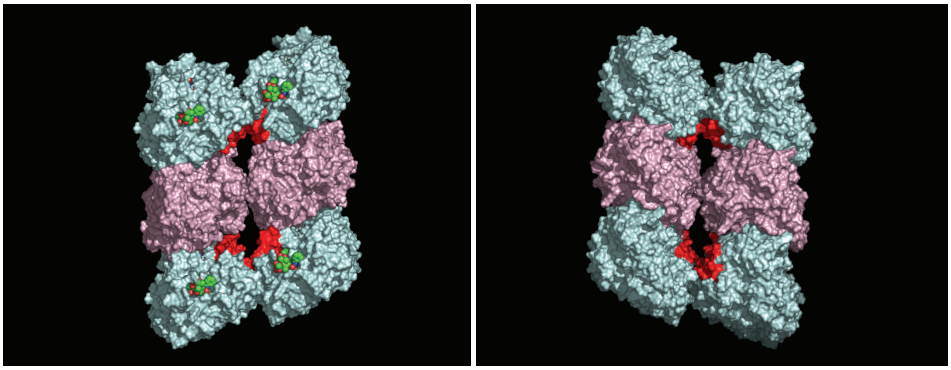


Fig. 5 – *An illustration of the NPs in the A and B MT lattices, respectively.*

Based on the modified version of the Poisson–Boltzmann equation, the conductances of NPs are determined in terms of Brownian dynamics of ions. For the type–1 NP, inner and outer cationic conductances were calculated to be 2.93 nS and 1.22 nS, respectively, while for type–2 NP the respective values of 7.80 nS and 4.98 nS were found.

Finally, we could include the conductance of both NPs to account for the leakage of IC cations into the lumen area. Thus we have

$$G_0 = \sigma_1 + \sigma_2 = (2.93 + 7.8)\text{nS} = 10.7\,\text{nS}. \tag{10}$$

It is expected that NPs exert greater resistance than the volume of IC around EUP.

2.3 The resistance of an elementary unit of MT

Regarding the ohmic resistance, if we ignore ionic current leaks through the depleted layer, the dominant current flows in parallel with the MT axis charging EUP capacitors and partly leaking through NPs. The resistance attributed to this kind of ionic flow can be estimated on the basis of experimental evidence provided by the electro–orientation method performed on MTs *in vitro* [6]. Taking the reported measured value of MT ionic conductivity as $\sigma = (0.15 \pm 0.01)\ \mathrm{Sm}^{-1}$, and adopting a simplifying assumption that the resistivity within an IC patch beyond an EUP is homogeneous, the resistance of an EUP with the length l = 8 nm and the cross–sectional area $A = \pi r_{TD}\lambda_{TD}$, $A = 19.625\ \mathrm{nm}^2$, see Fig. 3, is estimated as:

$$R_0 = \frac{1}{\sigma}\frac{l}{A} = 2.7\times10^9\,\Omega\ . \tag{11}$$

This is obviously an extremely high value of resistance which is most likely an over–estimate. We therefore prefer the computed resistance shown in Table 3 of reference [8], referred to as the outer sheath–outer sheath resistance for complete 13 PFs, namely $R_{13} = 4.75\times10^6\ \Omega$. The resistance for our EUP will be 13 times greater, i.e.,

$$R_0 = 13R_{13} = 6.2\times10^7\,\Omega\ . \tag{12}$$

This value appears to be more consistent with the resistance of two NPs which follows from (10).

2.4 The inductance of an elementary unit of MT

The straightforward calculation performed in [9] revealed that the order of magnitude of the inductance of an elementary ring in MT is of the order of L~10−15 H. This is very small value regarding pertaining reactive resistance. For example, if the frequency is of the order of MHz we have ωL~10−9 Ω for inductive impedance, while capacitive impedance amounts (ωC)–1~109 Ω. Similar discrepancy holds for the energies of corresponding fields

$$\frac{1}{2}Li^2 \ll \frac{1}{2}C_0v^2, \tag{13}$$

since i is of the order of nA and v of the order of several mV. Thus, the role of inductance in this model could be safely ignored.

3 The Model of MT as Nonlinear Circuit

On the basis of above estimations for the resistive components of the MT we are now in the position to establish the corresponding model of MT as nonlinear nanoscale circuit. A typical section scheme is shown in Fig. 6. Applying Kirchhoff's low to ladder of ERs, we have:

$$i_n - i_{n+1} = \frac{\partial Q_n}{\partial t} + G_0 v_n, \qquad v_{n-1} - v_n = R_0 i_n \ . \tag{14}$$

Introducing the characteristic impedance of our system $Z = 1/(C_0\omega)$ and establishing new function $u(x, t)$, $u_n = Z^{1/2} i_n = Z^{-1/2} v_n$, we can safely expand u_n in a continuum approximation using a Taylor series in terms of a small spatial parameter l (the length of a dimer). Then using the travelling–wave form of the function $u(x, t)$ with dimensionless space and time variables (ξ, τ), we get the following voltage equation

$$\begin{aligned} &\left(\frac{ZC_0 s}{T_0} - 2\right)\frac{\partial u}{\partial \tau} + \frac{1}{3}\frac{\partial^3 u}{\partial \xi^3} + ZC_0\Gamma_0\Omega\left(\xi - \xi_0\right)\frac{\partial u}{\partial \xi} + \\ &+ 2\frac{Z^{3/2} b_0 C_0 s}{T_0} u \frac{\partial u}{\partial \xi} + \left(ZG_0 + Z^{-1}R_0 - ZC_0\Gamma_0\Omega\right)u = 0. \end{aligned} \tag{15}$$

Here, $T_0 = R_0 C_0$ is the characteristic charging (discharging) time of an EUP capacitor C_0 through the resistance R_0 and $v_0 = l/T_0$ is the characteristic velocity of spreading the ionic wave. The dimensionless speed, space and time variables are chosen to be respectively:

$$s = \frac{v}{v_0} \le 1, \quad \xi = \frac{x}{l} - \tau, \quad \tau = s\frac{t}{T_0} \ . \tag{16}$$

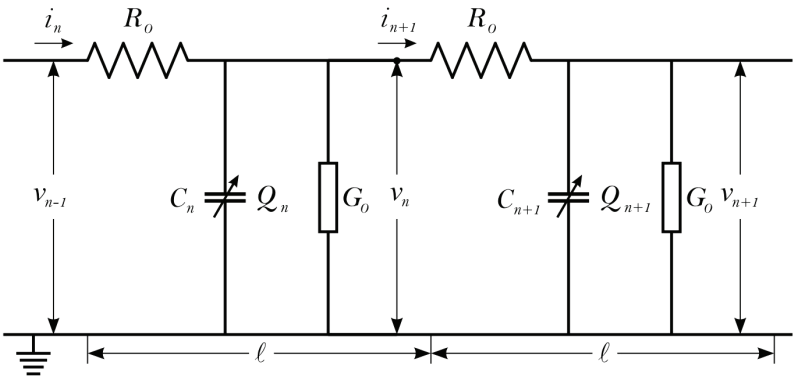


Fig. 6 – *An effective circuit diagram for the n-th* ER *with characteristic elements for Kirchhoff's laws.*

4 Analisis of the Voltage Equation

Imposing that the condition

$$\frac{ZC_0 s}{T_0} > 2 \tag{17}$$

holds, we then estimate the characteristic impedance of EUP as $Z = 1.24\times10^8\ \Omega$ for $s = 1$. For $s = 1$ one obtains the cutoff frequency $\omega_{max} = 4.3\times10^7\ s^{-1}$ or $f_{max} = 6.8\times10^6$ Hz. This indicates that the characteristic frequency matches the order of magnitude of frequency Ω which describes the TTs oscillations. We now establish the compact form of (15) as:

$$\frac{\partial u}{\partial \tau} + \beta\frac{\partial^3 u}{\partial \xi^3} + \alpha u\frac{\partial u}{\partial \xi} + \gamma(\xi)\frac{\partial u}{\partial \xi} + \delta u = 0, \tag{18}$$

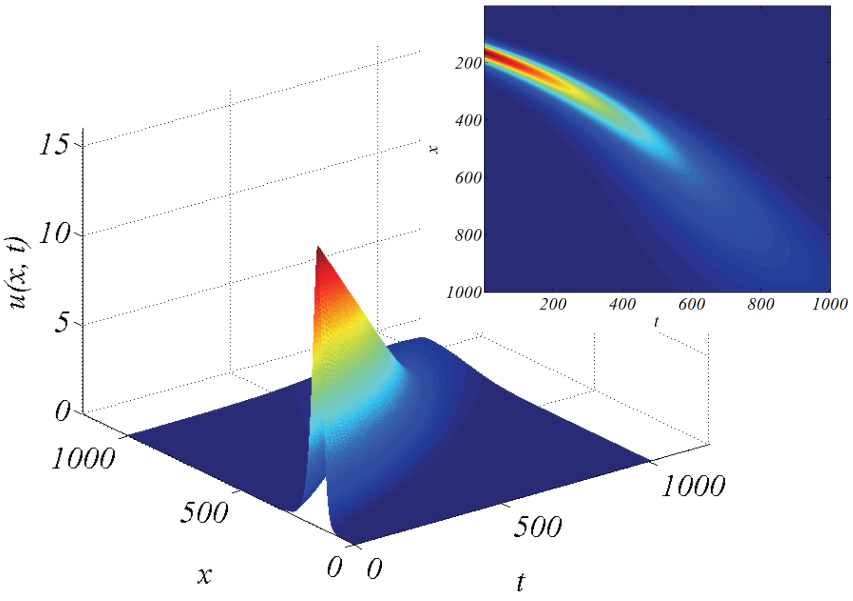
where the abbreviations were introduced as follows:

$$\alpha = \frac{2Z^{3/2}b_0C_0s}{T_0\left(\frac{ZC_0s}{T_0}-2\right)}, \quad \beta = \frac{1}{3\left(\frac{ZC_0s}{T_0}-2\right)},$$
$$\delta = \frac{ZG_0 + Z^{-1}R_0 - ZC_0\Gamma_0\Omega}{\left(\frac{ZC_0s}{T_0}-2\right)}, \quad \gamma(\xi) = \frac{ZC_0\Gamma_0\Omega(\xi-\xi_0)}{\left(\frac{ZC_0s}{T_0}-2\right)} = \gamma_0(\xi-\xi_0). \tag{19}$$

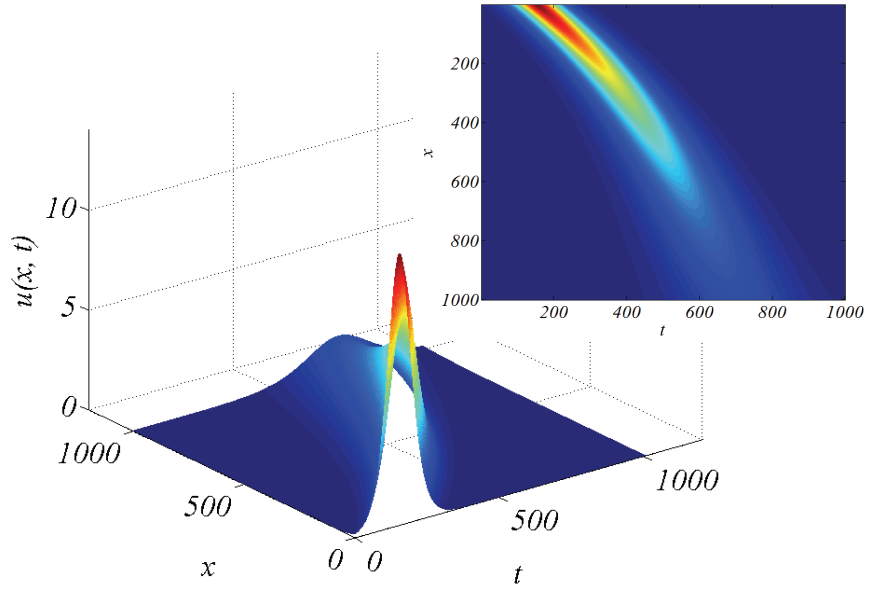
The detailed procedure of solving (18) is given in [10], so that the function $u(\xi, \tau)$ has the following form:

$$u(\xi,\tau) = u_0\exp(-2\gamma_0\tau)\cosh^{-2}\left\{\left[\frac{\alpha u_0}{4\beta}\exp(-2\gamma_0\tau)\right]^{1/2}\cdot\right.$$
$$\left.\cdot\left[\xi-\xi_0\left(1-\exp(\gamma_0\tau)\right)+\frac{\alpha u_0}{3\beta}\left(1-\exp(3\gamma_0\tau)\right)\exp(-2\gamma_0\tau)\right]\right\}. \tag{20}$$

If we use the set of estimated and chosen parameters, we readily see that all dimensionless parameters (19) lie between zero and one ($\beta = 0.33$, $\gamma_0 = 0.17$, $\delta = 0.34$) staying within the same order of magnitude. This suggests that we should inspect numerically the behaviour of the solution of (18) when the parameters α, β, γ_0 and ξ_0 take slightly different values which are done next. Analysing the set of plots in Fig. 7 we can see the competition between nonlinearity α and the dispersion β, as well as the role of inhomogeneity γ_0. In the case represented in Fig. 7b we have balanced all parameters. The soliton solution preserves its width but its amplitude decays rather rapidly so that over the length of about 500 l it becomes negligible. The above view shows the deceleration of the soliton solution along its path. Fig. 7c shows the case with increased nonlinearity ($\alpha = 0.5$). It is remarkable that it exhibits not only a higher localization but also a slower decay of its amplitude. The advantage of this case lies in the fact that the velocity of the soliton solution decreases very slightly.



(a)



(b)

Fig. 7 – *Numerical solutions of u(x, t) for:*
(a) $\alpha = 0.2$, $\beta = 0.33$, $\gamma_0 = 0.17$, $\xi_0 = 0.1$;
(b) $\alpha = 0.1$, $\beta = 0.1$, $\gamma_0 = 0.1$, $\xi_0 = 0.1$.

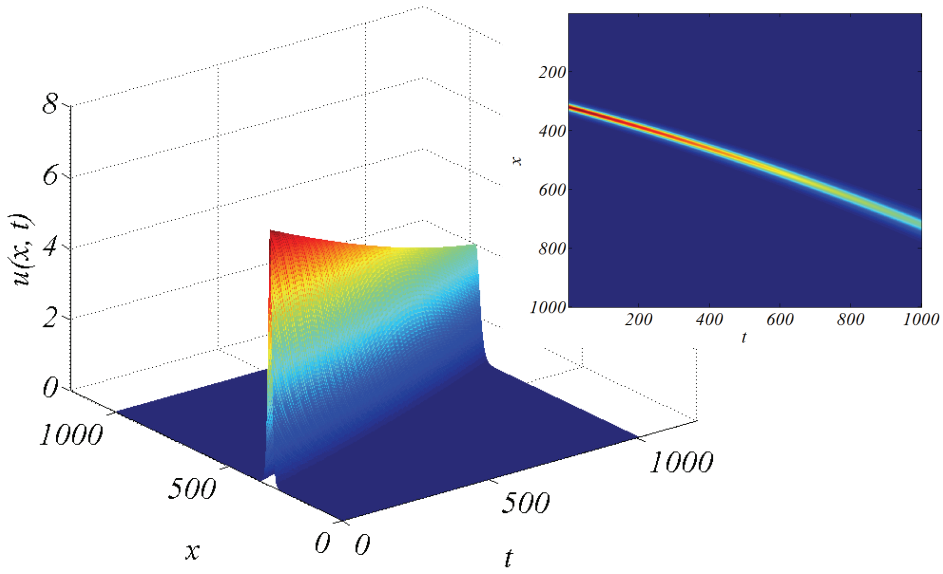


Fig. 7c – *Numerical solutions of u(x, t) for*: α = 0.5, β = 0.1, γ_0 = 0.1, ξ_0 = 0.1.

We estimate its average velocity as follows:

$$\left.\begin{aligned}\Delta x &\approx 400l = 400\times 8\text{nm} = 3.2\mu\text{m},\\ \Delta t &= 1000T_0 = 10^3\times 1.2\times 10^{-8}\,\text{s} = 1.2\times 10^{-5}\,\text{s}\end{aligned}\right\}\ v = \frac{\Delta x}{\Delta t} = 0.26\frac{\text{m}}{\text{s}}.$$

The range of this soliton is 3.2 μm which is of the order of the cell's diameter. Therefore, it appears that the ionic pulse with such parameter values could be efficiently transferred within the cell.

5 Conclusions

In this paper, we theoretically analysed the use of MTs as protein structure for building biomolecular nanoscale nonlinear circuit in the context of the polyelectrolyte character of these cytoskeletal filaments. We have taken into account the role of two aspects of this cylindrical biopolymer, its NPs and very sensitive TTs. Both of them are responsible for the nonlinear character of the overall electrical capacitance of MTs. Our solution, (20), was analyzed numerically for the set of parameters in Fig 7. It is apparent that the solitonic wave loses its energy due to ohmic resistance but it preserves the stable localized form. In the case presented in Fig. 7c with an increased nonlinearity parameter, the solitonic pulse exhibits greater robustness and it progresses with an almost constant velocity and only a slight decay of its initial amplitude. This

demonstrates the role of flexible TTs which could be of decisive importance for the stable localized character of ionic pulses along MTs. The order of magnitude of soliton–like localized pulses arising within the scope of our study ranges from mm/s to a several cm/s which are very reasonable values.

In this study, the particular attention was paid to the role of NPs existing between neighboring dimmers within a MT wall which exhibit properties like ionic channels. These NPs are candidates to explain some properties of MTs resembling to unipolar transistors enabling the rectification and amplification of ionic currents.

Finally, the experimental work of Priel et al [11] demonstrated that the electrical signal amplification resulting from the presence of MTs can be directly attributed to nonlinear ionic currents guided by MTs in close similarity to the framework developed by our present model. In fact, these measurements inspired the authors to consider the role MTs can play as biotransistors taking part in cognitive processes which is another interesting application of nonlinear ionic conductance of MTs.

6 Acknowledgments

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Model of ionic currents through microtubule nanopores and the lumen

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It has been suggested that microtubules and other cytoskeletal filaments may act as electrical transmission lines. An electrical circuit model of the microtubule is constructed incorporating features of its cylindrical structure with nanopores in its walls. This model is used to study how ionic conductance along the lumen is affected by flux through the nanopores, both with and without an external potential applied across its two ends. Based on the results of Brownian dynamics simulations, the nanopores were found to have asymmetric inner and outer conductances, manifested as nonlinear *IV* curves. Our simulations indicate that a combination of this asymmetry and an internal voltage source arising from the motion of the *C*-terminal tails causes cations to be pumped across the microtubule wall and propagate in both directions down the microtubule through the lumen, returning to the bulk solution through its open ends. This effect is demonstrated to add directly to the longitudinal current through the lumen resulting from an external voltage source applied across the two ends of the microtubule. The predicted persistent currents directed through the microtubule wall and along the lumen could be significant in directing the dissipation of weak, endogenous potential gradients toward one end of the microtubule within the cellular environment.

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I. INTRODUCTION

It has been proposed that cytoskeletal elements including actin and microtubules are involved in facilitating the propagation of electrical signals in the cell (see [1,2] for an in-depth discussion). Priel *et al.* [3] experimentally demonstrated that microtubules are excellent conductors of electrical signals and can amplify an electrical current stimulated by a pulse of applied voltage. The current collected 20–50 μm further down the microtubule (MT) was found to be double its value in the buffer solution with no MT present. Likewise, Minoura and Muto [4] found using electro-orientation experiments that the conductivity of the MT was 1.5×10^2 mS/m, which was 15 times higher than that of the surrounding buffer. Similar results have been obtained for electrical signal conduction along actin filaments [5]. Filamentous actin and MTs both share the characteristics of being protein biopolymers with highly negative linear charge densities and thus both structures are surrounded by condensed atmospheres of counterions lining their outer surfaces, which would lead to high conductances along these ionic atmospheres. However, the structure of the MT is somewhat more complex, being cylindrical in shape with a hollow interior and an additional negative linear charge density along its inner surface and having a latticelike wall interspersed with nanopores connecting the inner and outer surfaces. The question arises whether this cylindrical structure, and particularly its decorating nanopores, might be associated with novel bioelectronic properties that would contribute to the MT's proposed role in enhancing electrical signaling.

MTs self-assemble by polymerization of α/β-tubulin protein heterodimers (see supplementary material [6], Fig. S1a). These building blocks are arranged linearly into protofilaments and a ring of 13 protofilaments composes the wall of the cylindrical MT, which is typically several micrometers in length (and up to millimeters under special circumstances) and is 11.4 nm in radius at the midpoint of its wall. Each dimer in the MT has a length of 8.1 nm, a width of about 6.5 nm, and a radial dimension of about 4.6 nm [7]. A wide space, approximately 15 nm in diameter, known as the lumen, makes up the inner core of the cylinder.

The wall of a MT is generally characterized by what is known as a *B*-lattice structure, displaying two distinct types of nanopores (although it should be noted that there are two additional types of nanopores seen in the *A*-lattice structure). One nanopore in the *B*-lattice (nanopore 1) is located in the region where the interdimer β/α interface of one tubulin molecule lies next to the interdimer β/α interface of the adjacent tubulin molecule. A second nanopore (called nanopore 2) arises where the α/β intradimer interface of one tubulin molecule lies next to the α/β intradimer interface of the adjacent tubulin molecule (see supplementary material [6], Figs. S1a and S1b). The effective radius at the narrowest point of the pore is approximately 4.0 Å for the type 1 pore and 4.7 Å for the type 2 pore when calculated by the HOLE program [8] with AMBER radii [9].

Little is known about the biological relevance of the nanopores or of the lumen, which are difficult to probe experimentally. Interestingly, chemotherapy drugs of the taxane family are believed to diffuse through the MT wall via the nanopores to a binding site known to be located in the lumen [10,11]. There is also evidence for binding of ligands within the nanopore; for example, the MT-disrupting agent cyclostreptin has been shown to bind at a site within the MT nanopore [12]. Additionally, the microtubule-associated protein (MAP) tau has been observed binding to a site in the lumen

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[13,14]. Based on their size, the nanopores are expected to be quite permeable to the atmosphere of counterions covering both sides of the MT. The flexible carboxy-terminal tail domains of tubulin, consisting of approximately the last ten residues on α-tubulin and 18 residues on β-tubulin, depending on the tubulin isotype, extend from the outer surface of the MT over the nanopores (see supplementary material [6], Fig. S1a). These tail domains have a high proportion of negatively charged residues; for example, the tail domain of βI tubulin has nine Glu residues and two Asp residues and that of αIV tubulin includes five Glu residues and two Asp residues, although the exact numbers vary slightly with isotype. This requires large numbers of counterions for charge compensation and the fluctuations of the tail domains must be associated with accompanying movements of these counterions.

No other study has yet focused on the motion of the surrounding counterions through the nanopores and around the MT, although this effect is a distinct possibility even just due to its geometrical similarity with ion flows across ion channels in membranes. Here, we intend to examine the flux of these cations under both equilibrium conditions and under a constant perturbative electrostatic potential difference between the two ends of the MT. That is, we have constructed a physical model to calculate electric current through the sides of the MT, in addition to that flowing in the longitudinal direction along a MT. To parameterize this model, we have performed detailed atomic-scale calculations of the conductances of ions through a MT's nanopores using three-dimensional (3D) Brownian dynamics. The results of our simulations will be presented in this paper and will hopefully guide future experimental validation of the predicted effect.

There is evidence in many cellular systems of an important and complex role for MTs in regulating ion-channel gating in the plasma membrane, although the mechanism of this influence has not been explained. For example, an intact MT structure has been shown to be essential for the activity of gamma-aminobutyric acid receptors in hippocampal neurons [15], for the functioning of polycystin-2 calcium channels in the kidney [16,17], and has been associated with a decreased activation threshold of Na^+ currents [18] and increased currents of Na^+ [19] and of Ca^{2+} [20,21] in neurons. On the other hand, in other studies, MT depolymerization has been associated with increased calcium currents in cardiac myocytes and in neurons [22–25]. Studies also show that ion channels can be anchored to MTs through other proteins [15,26]. Thus electrically excitable cells may possess ion channels which, by a somewhat different mechanism, have a similarly important dependence on MTs to that of mechanosensitive channels, which are tethered by other proteins to MT bundles thought to regulate their opening and closing [27,28].

The observations that MTs can regulate ion-channel activity, together with their remarkable conductive properties, and also the fact that both dendrites and axons in neurons are dense in MTs and moreover display distinctly developed cytoskeletal architectures [29] suggest that MTs may play a role in electrical signal conduction in neurons. Various instances suggest that local amplification of ion concentration could be important during the extremely complex process of action potential generation. These include the generation of an axonal action potential by distributed processing of the synaptic activity at distinct dendritic sites [30], the observation of noninactivating behavior in a small percentage of dendritic sodium channels [31], and the reduced voltage firing threshold for channels located in the axonal hillock [32].

While there is as yet no experimental confirmation that this includes electrical signal conduction, an abundance of experimental evidence supports some role of MTs during neuronal signaling. In most tissues, MTs are dynamic, growing and shrinking, but are less so in the neurons, where however rearrangement of MTs occurs during learning. Impaired learning has been found to correlate with decreased MAP2 levels [33–35], an excess of MAP tau [36,37], and can be brought on by colchicine-induced changes in MT dynamics [38–40]. Additionally, memory consolidation is accompanied by increased binding of tubulin to other neuronal proteins [41], by alterations in MAP2 expression [41–46], and by MAP 1B phosphorylation [47].

The remainder of this paper is divided into three sections. In the first, the various elements of the model we develop for ion propagation along MTs are described. Methods used to estimate the necessary parameters are described, including calculations of the electrical conductances of the two major types of pores in the MT wall by Brownian dynamics, and voltage differences arising from the fluctuating *C*-terminal domains. In the next section, we present the parameter set derived for the circuit model and give results. Finally, there is a conclusion section, in which results of our modeling are used to suggest future experimental and computational studies.

II. MODEL DEVELOPMENT

We have developed an electrical circuit model for the conduction of cations along a MT, with various electrical elements, which we describe below. Since anions are scarce in the intracellular environment and in the medium surrounding the negatively charged MT in particular, we make the approximation that potassium cations are the only charge carriers present.

The approximation is also made that the MT has cylindrical symmetry, which allows it to be represented in two dimensions, with the y axis representing the radial direction and the x axis oriented along the axis of the MT. The volume surrounding the MT is divided into four concentric cylindrical shells. Because counterion clouds result in reduced resistivities, two cylindrical shells are chosen to represent the condensed ionic atmospheres adjacent to the inner and outer surfaces, respectively, while two additional shells correspond to outer bulk solution and the luminal volume. In Fig. 1, linear wires are used to depict each of these cylindrical volumes, since in our model, averaging is performed of currents and voltages over units of cylindrical shells extending the length of one tubulin monomer along the MT. The row of tubulin monomers, which has been sketched in Fig. 1, represents the wall of the MT. A set of resistances regulates the numbers of cations allowed to cross between adjacent cylindrical shells in response to a voltage difference across the

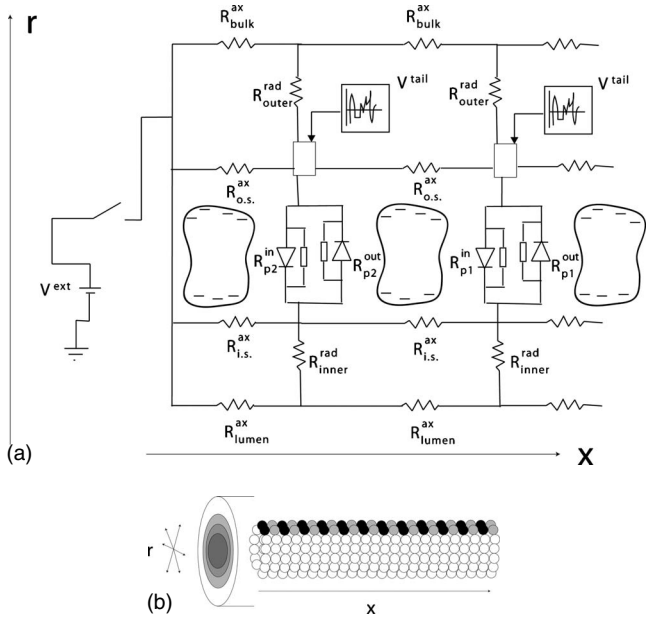

FIG. 1. (a) Two-dimensional electrical circuit model of the microtubule. The vertical direction is radial and the horizontal direction is along the axis of the microtubule. The various electrical resistances, in addition to the voltage sources, are labeled to show the physical placements of these components relative to the microtubule in the model. Abbreviations "o.s." and "i.s." are used to denote the ionic atmospheres close to the outer surface and inner surface, respectively; "ax" stands for axial and "rad" for radial. (b) Schematic representation of a microtubule, depicting the relationship between the molecular structure and the electrical circuit image shown in part (a).

boundary between the two volumes. In the case of motion of cations across the wall of the microtubule, these resistances depend on the permeabilities of the two types of nanopores.

The specific currents that we have chosen to calculate and analyze in the present work are the local currents flowing out of each of the four regions at the right-hand end of the MT. Note that this does not represent the current between the two ends of the MT, as there is also a current flowing at the other end.

Considering that the Bjerrum length, defined as the separation at which the electrostatic interaction energy equals thermal energy, is 7.13 Å at a temperature of 293 K [2], we chose to consider a 10-Å-wide layer near the inner surface and a layer near the outer surface 40-Å wide, which is approximately 10 Å extending beyond the positions of the *C*-termini.

A. Conductances

To determine conductances, we made use of the well-known relation, based on the equivalent of Ohm's law for ionic flows

$$G = \frac{\sigma(c)A}{l} = \frac{\Lambda_0^{K^+} cA}{l}, \tag{1}$$

where c is the cationic concentration, l is the distance in the direction of ionic flow, $\sigma(c)$ is the conductivity at a concentration c of cations, A is the cross-sectional area, and Λ_0 is the proportionality coefficient determining conductivity as a function of ionic concentration, which is approximately equal to 7.5 $(\Omega m)^{-1}\,M^{-1}$ for K^+ [48,49]. Further details of the evaluation of conductances are given in the supplementary material [6].

B. Brownian dynamics

The conductances of the nanopores occurring in the MT lattice limit the flow of ions between the inside and the outside of the MT. We made use of a grand canonical Monte Carlo program (GCMC-BD) created by the Roux group for computing channel conductances by Brownian dynamics simulations [50–53]. This method has been shown to be successful at determining accurate current-voltage (*I-V*) curves for ion channels, especially those with sufficiently wide cross-sectional areas ($>\sim 20$ Å^2) to allow ion motion to be suitably described by Langevin dynamics [53,54]. The nanopores in the MT wall are comparable in area to ion channels to which the program has been successfully applied.

The program uses the Ermak-McCammon equation [55] to generate the dynamics of ions undergoing random solvent collisions. All interactions in the system are being described via a multi-ion potential of mean force (PMF) and thus the equations of motion can be expressed as

$$\dot{\mathbf{r}}_i(t) = -\frac{D_i}{k_B T}\nabla_i W(\mathbf{r}_1, \mathbf{r}_2, \ldots) + \zeta_i(t), \tag{2}$$

where $\boldsymbol{r}_i$ is the position, D_i is the diffusion constant of ion i, and $\zeta_i(t)$ is the Gaussian random noise. The multi-ion PMF, W, is given by [51,52]

$$W(\mathbf{r}_1, \mathbf{r}_2, \ldots) = \sum_{i,j} u_{ij}(|\mathbf{r}_i - \mathbf{r}_j|) + \sum_i U_{core}(\mathbf{r}_i) + \Delta W_{sf}(\mathbf{r}_1, \mathbf{r}_2, \ldots), \tag{3}$$

where u_{ij} is the direct ion-ion interaction, U_{core} is the repulsive potential arising from the excluded volume of the protein, and ΔW_{sf} is the electrostatic potential from the static charge distribution of the protein together with the electric field term V [56,57]. ΔW_{sf} is determined as

$$\Delta W_{sf}(\mathbf{r}_1, \mathbf{r}_2, \ldots) = \sum_i q_i \phi_{sf}(\mathbf{r}_i), \tag{4}$$

where the electrostatic potential ϕ_{sf} is calculated using a modified version of the Poisson-Boltzmann equation, known as the PB-V equation,

$$\nabla[\varepsilon(r)\,\nabla\,\phi(r)] - \bar{\kappa}^2(r)[\phi(r) - V\Theta(r)] = -4\pi\lambda\rho^{prot}(r). \tag{5}$$

Here, Θ is the Heavyside step function equal to 0 for negative values of the z coordinate of the position vector $\mathbf{r} = (x, y, z)$ and 1 for nonnegative z, where the z axis is oriented through the pore and perpendicular to the axis of the MT; ε is the space-dependent dielectric constant, set equal to 1 in the inner and buffer regions; $\bar{\kappa}^2$ is the space-dependent ionic screening factor, set equal to 0 in the inner and buffer

regions; and ρ^{prot} is the charge density of the protein [56,57]. Details of Brownian dynamics calculations are given in the supplemental section.

Some explanation is in order concerning the source of the electric field term V. In situations in which GCMC-BD has been applied in the literature [50–54], the voltage V represents a membrane potential arising from a difference in local ionic composition on either side of the membrane in which the channel is embedded and is accounted for by Eq. (5). Here, V is accounted for by the same formalism, but represents, rather, a potential gradient across the nanopore in the MT wall arising from the fluctuating positions of the *C*-terminal tail domains and the fact that the number of ions screening the charges on these tubulin domains can be either in excess or insufficient. In what follows, this issue will be addressed in greater detail.

C. *C*-terminal tails

The charge of the *C*-terminal β-tubulin tail is generally -12 or -11 elementary charges, depending on the β-tubulin isotype, and that of the α-tubulin (isotype *IV*) tail is -7 charges. Although the remainder of the tubulin dimer is also negatively charged (with a total of ~ 50 negative charges), each tubulin monomer has a positively charged cluster of residues, which may interact with the *C*-terminus of the neighboring monomer [58].

We have performed an investigation of potential tail-body interactions of the *C*-terminal tails with positively charged residues on the surface of the tubulin dimer by molecular-dynamics simulation. Briefly, we first employed a molecular-dynamics simulated annealing approach to determine tail conformations. 15 high-temperature implicitly solvated MT dimers were generated at a temperature of 5000 K and were then slowly cooled to a target temperature of 300 K. Our annealing schedule used steps of 1000 K over periods of 400 ps until the temperature was below 2000 K and then used a 0.8 power-law cooling schedule. The tubulin dimers were then placed into a MT lattice and following equilibration, molecular-dynamics simulations were performed over periods of 10 ns, with periodic boundary conditions at a temperature of 300 K. The details of these simulations will be published separately. However, it was found that *C*-terminal tails can exist in multiple states involving different numbers of salt bridges and that the tails dynamically oscillate between these states as tail-body interactions form and then give way to interactions of tail residues with water and ions, with several such switches occurring within a single nanosecond.

When the negatively charged residues of the tails interact with positively charged residues on the tubulin dimer, fewer cations in the surrounding solvent are required to stabilize the negative charges on the protein, in comparison to when the tail residues are more extended away from the MT surface. There is also expected to be a corresponding decrease in the equilibrium number of surrounding anions, which however are not considered in our model. Since the exact equilibrium number of counterions within the condensed ionic atmosphere surrounding the outer surface of the MT varies as the positions of the *C*-terminal tails undergo thermal fluctuations, the chemical potential oscillates stochastically between being attractive or repulsive for the influx of additional cations. Thus, the motion of the *C*-terminal tails as they alternately attach and detach from the MT surface may be viewed as a stochastic voltage source acting on the surrounding cations.

To estimate the voltage difference corresponding to the variation in the *C*-terminal tail position, we performed calculations of the capacitance of the volume corresponding to the outer counterion sheath. This capacitance was defined as the coefficient of proportionality between the number of K^+ ions in this outer ionic atmosphere and a potential placed between this layer and the outer bulk solution. Ion counts for a range of potential differences were calculated by solving the PB-V Eq. (5) as implemented in the PB-PNP program [51,52]. Further computational details are given in the supplemental section [6]. The resulting curves for each pore type shown in the supplementary material [6], Figs. S2a and S2b, are highly linear and the two have a nearly identical slope corresponding to a capacitance of 1.15×10^{-8} nF.

With a variation of the tail-body interactions, corresponding to different positions of the *C*-terminal tails, the linear plot of equilibrium ion count versus potential is expected to shift up or down. If a single additional cation beyond the equilibrium number is present within the outer counterion sheath after a *C*-terminal tail has switched position to interact with the tubulin body, the line is shifted down by $1e$. This corresponds to a repulsive voltage equal to $1e$ divided by the capacitance, or ~ 14 mV. Likewise, when the *C*-terminal tail switches to a more extended conformation, there is an attractive potential of the same magnitude. In reality, a continuum of many different states is taken on by the *C*-terminal tails. Thus, the thermal fluctuations of the *C*-terminal tails on a ring of 13 monomers in the MT can be modeled as a stochastically alternating battery with a maximum voltage magnitude of $\Delta n^{*}7$ mV, where Δn is the variation in the equilibrium number of counterions in the condensed ionic atmosphere surrounding the ring. The frequency of oscillation was chosen to be 10/ns.

Note that the assertion that a current is driven by the stochastic motion of the *C*-terminal tails in no way represents a violation of energy conservation, as molecular-dynamics simulations clearly demonstrate that thermal energy is sufficient to drive large-scale fluctuations in the positions of these domains.

D. Electrical circuit

A circuit model was created to represent a MT with 20 monomers along its length using the Simulink and Simpowersystems modules of MATLAB R2009A [59]. Resistances, as well as both the external voltage and the stochastic alternating voltage and its frequency, were read as inputs. The program solved for voltages and currents at each segment of the circuit and propagated these in time using a time step of 1 ns. A switch, or circuit breaker, seen in Fig. 1, was implemented to connect the circuit to the external voltage source and opens at time zero to allow current to cross. The sources of time dependence in the model consisted of a stochastic bat-

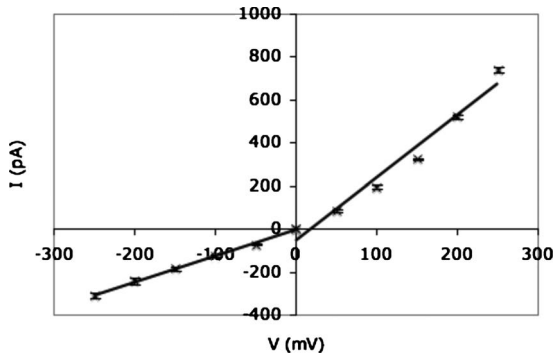

FIG. 2. Current-voltage relation from GCMC-BD simulation for conductance of cations through the type 1 pore. Error bars represent standard deviations over five independent data subsets.

tery representing the motion of the *C*-terminal tails as described above, as well as the integrators used to determine net charge transfer. The random battery generates uniformly distributed numbers over an interval between the positive and negative values of its specified voltage, with a specifiable starting seed. Because the time-dependent current showed large stochastic oscillations, its time integration was carried out to determine the amount of net charge transferred as a function of time at the end of the MT held at lower potential in each of the four cylindrical regions.

III. RESULTS

A. Pore conductances determined by Brownian dynamics

Calculated currents as a function of voltage difference for the conduction of cations across the nanopore of type 1 are plotted in Fig. 2. The *IV* curve showing the conduction of anions through this pore, as well as curves depicting the conduction of positive and negative ions through the nanopore of type 2, is shown in supplementary material [6], Fig. S3. Error bars representing standard deviations, also shown in the figure, were at most 12, 9.0, 32, and 3.7 pA for Fig. 2 and parts (a), (b), and (c), respectively, in Fig. S3 of [6]. Conductances through nanopores were determined from the slopes of linear fits of the *IV* curves shown in the figures. Both types of pores show asymmetric inner and outer conductances of cations. For the type 1 pore, inner and outer cationic conductances were found to be 2.93 and 1.22 nS, respectively. The conductance of cations was found to be somewhat greater for the type 2 pore, with calculated values of 7.80 and 4.98 nS, respectively, for inner and outer conductances. The cylindrical shell-like wall of the MT has a sandwichlike electrostatic potential being negative on the inside, even more negative on the outside, and positive in the center (see supplementary material [6], Fig. S1b). The very negative potential on the outer surface attracts cations into the pore at a positive potential and also makes it difficult for anions to enter the pore at a negative potential.

Both nanopores were found to be almost completely impermeable to anions at negative potentials (43.7 and 11.8 pS for types 1 and 2 pores, respectively). The positive parts of the anion conductance curves in supplementary material [6], Figs. S3a and S3c, are fit well by exponential curves, as shown. Conductances of anions into the pores increase substantially as the voltage is raised above 150 mV, from about 11.8 to 198 pS for the type 2 pore and from 240 pS to 1.10 nS for the type 2 pore, as determined from linear fits of the positive voltage data in the intervals between 0 and 150 mV and between 150 and 250 mV, respectively. The difficulty for anions to enter the pore is most likely partially due to the composition of the constriction zone. As determined by the HOLE program [8], the constriction zone of pore 1 is bounded by α-tubulin residue Thr 130, by Phe 212 of the adjacent β tubulin, and Asp 327 of the other α tubulin. In pore 2, the constriction zone is bounded by residues Thr 94 of α tubulin, Asp 128 of the adjacent β tubulin, and Asp 218 of the other α tubulin. Thus, the negatively charged Asp residues may make it difficult for anions to breach this area and especially so in the type pore 2 where there are two such charged residues.

Asymmetric conductances have been measured in ion channels by GCMC-BD in the past, giving good agreement with experiment [50]. For example, for the OmpF pore, conductances were calculated to be 1.36 and 1.15 nS with positive and negative voltages across the channel, respectively, compared to values of 1.25 and 1.13 nS determined experimentally [53].

To assess the role of the protein charge distribution in contributing to nanopore conductances, calculations were repeated for the type 2 pore, but this time with all protein charges turned off. We found that this led to the cation conductance being reduced by a factor of about 15. The conductance of anions was much larger than for the charged model, being increased by a factor of about 24 at a negative potential and by a factor of about 4 at high positive potentials. These results show that electrostatics has a very significant effect on the conductive properties of the nanopores. We also investigated the effect of a different *C*-terminal tail conformation; with a different position with respect to the MT, the conductance of cations was changed by only 11% and there was very little effect on the conductance of anions. Within our circuit model, a pair of forward-biased diodes was used to model the asymmetric conductance of each MT pore, with one controlling the flow of cations into the MT and the other regulating the outward flux, as is illustrated in Fig. 1.

B. Parameterization of electrical circuit

The repeating unit of two monomers is illustrated in Fig. 1. Values of resistances, which were calculated as described above for use in our model, are given in Table I. The bulk resistance shown in the table was calculated using an outer radius of 50 nm for the entire system. This value was chosen based on the typical distance of 100 nm separating parallel MTs in the cell such as a neuron. Resistance is smallest in the bulk region, followed by the outer condensed ionic atmosphere and the inner condensed ionic atmosphere, with the inner lumen having the smallest conductance.

C. Application of circuit model to dynamics of ionic flow

The time-integrated currents running parallel to the MT axis through each of the four cylindrical regions, which is

TABLE I. Computed resistances in Ohms. Diagonal elements describe the flow of K^+ ions along cylindrical shells running parallel to the microtubule axis in each of four regions. Values corresponding to flow between adjacent volumes are shown as off-diagonal elements, with the region of positive potential given in the first column; upper and lower values represent flow through pore 1 and pore 2, respectively.

	Inner lumen	Inner sheath	Outer sheath	Outer bulk
Inner lumen	1.77×10^{7}	1.07×10^{6}		
Inner sheath	1.07×10^{6}	1.25×10^{7}	6.41×10^{7}	
			1.54×10^{7}	
Outer sheath		2.65×10^{7}	4.75×10^{6}	4.06×10^{6}
		9.86×10^{6}		
Outer bulk			4.06×10^{6}	5.31×10^{5}

along each of the four electrical transmission lines shown in Fig. 1, were measured at the right-hand end of the MT; these are plotted in Fig. 3. A value of 28 mV was used for the stochastic voltage source and 1 mV for the external voltage source applied across the two ends of the MT.

Asymmetric conduction of microtubule pores enhances ionic flow through the lumen. The elements of the circuit model that confer nonlinearity are the asymmetric conductances of the nanopores and the stochastic tail battery. To determine what effect these have on the longitudinal currents, we compared results for the case where there is no internal voltage from the tail movement (see supplementary material [6], Fig. S4), when there is no asymmetry in the conductance in supplementary material [6], Fig. S5, and when there is neither asymmetry nor an internal voltage. The same values corresponding to inward resistances were assigned to the outer resistances to model symmetric conduction.

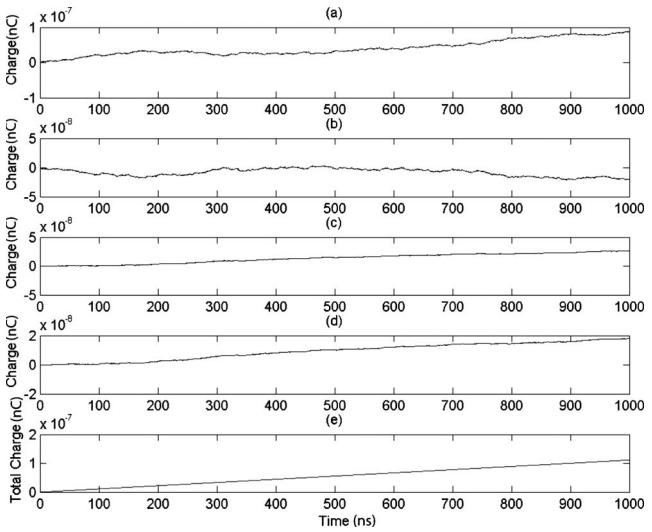

FIG. 3. Time-integrated current (nC) for asymmetric pore conductance plotted against time (ns), with an external voltage of 1 mV and a stochastic voltage of 28 mV. The five panels are, from top to bottom, time-integrated current through (a) the bulk solution outside the microtubule, (b) the outer ionic atmosphere of the microtubule, (c) the inner ionic atmosphere of the microtubule, (d) the rest of the lumen and (e) summed for (a-d).

Without the voltage created by the stochastic motion of the *C*-terminal tails, plots of charge transferred versus time are linear, indicating a constant current. In this case, it makes no difference whether the conductance of the nanopores is symmetric (not shown) or asymmetric, as there is no driving force on the ions in the lateral direction. The net charge transferred in the bulk region after the end of 1000 ns is 1.0×10^{-7} nC, while a net charge of about 1.1×10^{-8} nC flows along the outer ionic atmosphere, which has a smaller area although its conductivity is greater. Only 4×10^{-9} nC is transferred along the inner ionic atmosphere of the MT and even less (3×10^{-9} nC) through the central lumen.

When the *C*-terminal tail fluctuation is accounted for in the model but with symmetric pore conduction, the summed charge transferred along the inner ionic atmosphere and the lumen is 3.1×10^{-10} nC at the end of the 1000 ns time period. Small fluctuations on the time scale of the tail motions are exhibited in the plots, as well as larger time-scale fluctuations, representing an oscillatory exchange of ions between the bulk solution and the condensed atmosphere of counterions near the MT's surface.

When the conductance asymmetry of the pores is included in the model, the total charge transfer in the lumen increases to about 4.4×10^{-8} nC, with 2.6×10^{-8} nC moving along the inner ionic atmosphere and 1.8×10^{-8} nC moving through the inner lumen. It may be noted by comparing values at 500 ns to those at 1000 ns that the average currents in the two luminal regions are well converged by 1000 ns. Thus, the overall charge transferred divided by the total time may be thought of as the average current and the total average current through the lumen predicted by our circuit model is 4.4×10^{-2} nA for Fig. 3. These results demonstrate that the conductance asymmetry of the MT wall allows the random internal battery to pump cations transversely through the nanopores into the lumen, from which they are directed out of one of the open ends of the MT.

Total longitudinal charge transferred along MT is independent of nanopores. In each of Fig. 3 and Figs. S4 and S5 of supplementary material [6], the value of total charge transferred after 1000 ns is identical. Therefore, the summed current along the outer ionic atmosphere and in the bulk, which without the asymmetric conductance or the internal random battery had been 1.1×10^{-1} nA, is reduced by the additional 3.7×10^{-2} nA current which is now shunted through the lumen; that is, the additional luminal current is drawn from the outer ionic atmosphere and the outer bulk solution.

Two-dimensional (2D) spatial-temporal plots reveal that net ionic current flows inward through some pores and outward through others. Net charge transfer was determined as a function of time and position along the MT by measuring time-integrated current outputs loaded from the four electrical wires at positions immediately to the right of each of the 20 tubulin monomers to yield 2D spatial-temporal plots. A random battery voltage of 28 mV was used and external voltages of 0 and 1 mV, as shown in Fig. 4 and Fig. S6 of supplementary material [6], respectively. The average currents at the end of the 1000 steps, given by the overall charge transferred divided by total time, are directed to the right

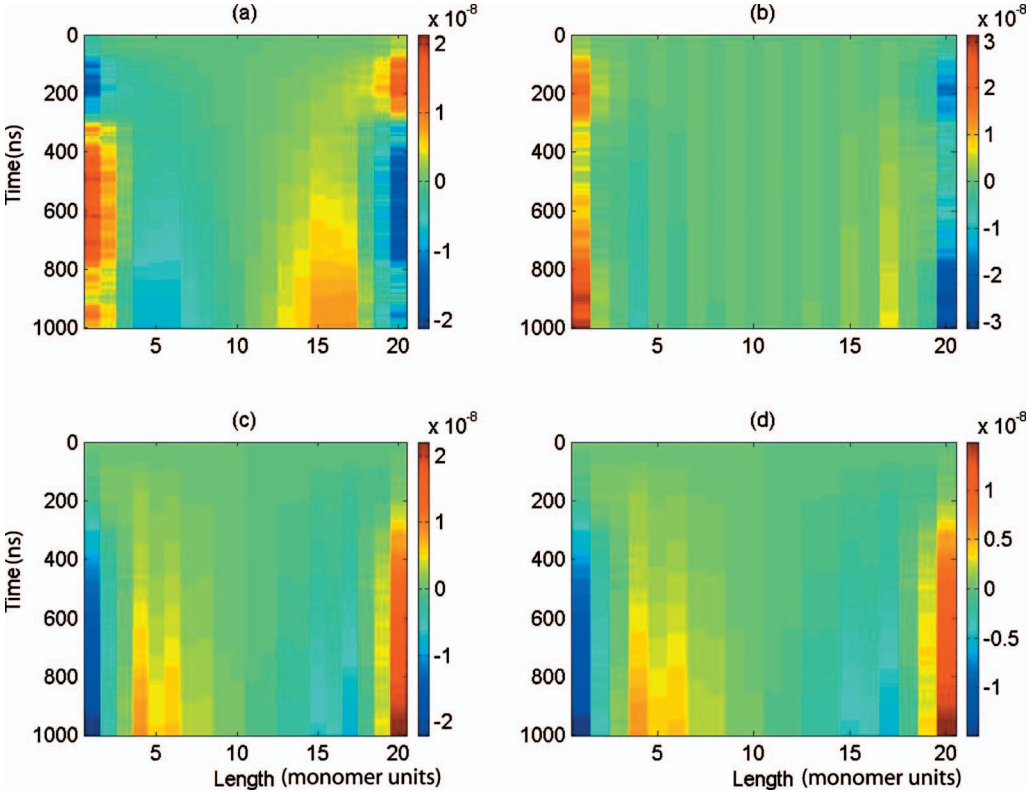

FIG. 4. (Color) Color plot showing integrated current (nC) as a function of time (ns) and tubulin unit through (a) the bulk solution outside the microtubule, (b) the outer ionic atmosphere of the microtubule, (c) the inner ionic atmosphere of the microtubule, and (d) the rest of the lumen. A stochastic voltage of 28 mV was used and an external voltage of 0 mV.

where the area adjacent to the x axis is colored red or yellow and to the left if the coloring is blue. An interesting pattern of currents arises surrounding the MT's wall in the 0 mV case. Close to the ends of the MT, the current flows into the MT and out through the open ends (since, for example, at the left-hand end, the direction of the current is to the right just outside of the microtubule's wall [Fig. 4(b)] and to the left along the wall's inner side [Fig. 4(c)]). On either side of the center, the current follows a loop, so that there are currents flowing away from the center on the outside of the MT and toward the center on the inside. The pattern of currents is symmetric about the center of the MT. The current driven through the wall of the MT leaves either one or the other end of the MT with equal probability. This is to be expected, since no potential gradient between the two ends of a MT means no net current between the two ends. With the 1 mV applied voltage, within the lumen, there is a multidirectional spatial gradient of charge transfer, whereas in the bulk solution at any given time, the direction of charge flow is much more uniform along the length of the MT.

Net current driven through pores adds directly to that induced by the external potential, independent of value of external potential or MT length. In Fig. 5, the current-voltage (IV) curves are shown for total luminal current versus external voltage for values of the random battery equal to 14 and 28 mV, respectively. Simulations were run for values of the external voltage ranging from −100 to 100 mV. Plots were determined to be linear over the entire range of voltages. In the case of symmetric conductance, the IV curve passes through the origin. Asymmetric conductance of the pores shifts the curve toward positive currents. However, the slope is unchanged, showing that the current being driven through the nanopores adds directly to the current induced by the external potential. The effect of varying the length of the MT was also examined using MTs composed of 10 or 30 monomers (see Fig. S7 of supplementary material [6]). The y intercepts of the resulting plots both coincide with that obtained for a length of 20 monomers, although the slope changes as a result of the altered voltage per unit distance. That is, additional tail batteries associated with a longer MT cannot increase the transverse current across the MT wall

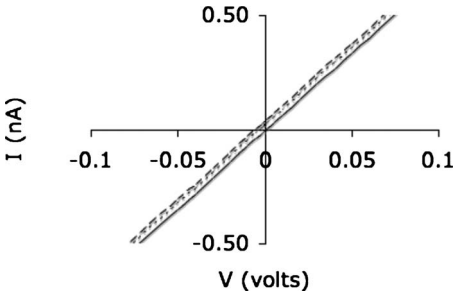

FIG. 5. Current-voltage relations calculated using the circuit model for conductance of cations through the microtubule lumen. A microtubule length of 20 monomers was used. Dotted line corresponds to a stochastic voltage of 14 mV and dashed line to a stochastic voltage of 28 mV. Solid line corresponds to symmetric pore conductance.

and the same net current is pumped across independent of length.

It is important to note that, within our model, the main function served by the thermally driven tails is the pumping of cations from the bulk solution to the lumen. While this continually adds to the population of cations within the lumen, at the same time (in the absence of an external voltage), the added cations leave the lumen mainly from both ends of the microtubule, returning back to the bulk to maintain an equilibrium state. As a result, there is neither a net current into the lumen nor through it in the longitudinal direction, i.e., no charge accumulates within the lumen.

Net current through pores is roughly linear in the value of the tail battery. The positive shifts of the *IV* curves given by the *y* intercepts of the plots in Fig. 5 were 0.021 and 0.037 nA for random battery voltages of 14 and 28 mV, respectively. If the random battery is increased to 140 mV, then the shift is 0.19 mV. Thus the value of the current scales almost linearly with the unknown random voltage.

Luminal current at the end of the microtubule driven by fluctuations of C-terminal tails exceeds that driven by the external battery at small values of the external voltage. At zero or even small negative values of the voltage, a positive current persists at the end of the MT, just as there was a backward flow of charges at the opposite end for small positive values of the voltage. The *x* intercept on each plot in Fig. 5 or Fig. S7 of supplementary material [6] corresponds to not only the external voltage at which forward current vanishes, but also the magnitude of external voltage required to halt the backward flow of ionic current. Most importantly, for positive external voltages smaller than the absolute value of the *x* intercept on the plot, the potential associated with the stochastic fluctuation of *C*-terminal tails is the dominant driving force acting on the cations, whereas as the external potential is increased beyond this point, the stochastic tail voltage, while still adding to the total longitudinal current, contributes less than the external battery to this total. In summary, the addition to the longitudinal current by the current through the MT wall is most important at small values of the external voltage.

D. Theoretical justification

Our computational model demonstrates that the influx of cations through the nanopores results in an addition to the transfer of cations to the more negative end of the MT through the lumen. We now attempt to provide some basic understanding of this effect by using a simple theoretical model.

Consider the *n*th unit of a MT, shown in Fig. 6. Here I_n^b, I_n^{os}, I_n^{is}, and I_n^l are the current in the bulk (*b*), outer surface (os), inner surface (is), and the lumen (*l*), respectively. The random tail batteries $\varepsilon_n(t)$ and $\varepsilon_{n+1}(t)$ are time dependent and could be in different phases, and so in general $\varepsilon_n(t)$ will not equal $\varepsilon_{n+1}(t)$ in both magnitude and sign. The time-dependent resistor blocks $R_n(t)$ and $R_{n+1}(t)$ are determined by the two sets of resistances R_{p1}^{in} and R_{p1}^{out}, and R_{p2}^{in} and R_{p2}^{out}, respectively,

$$R_n(t) = (1/2)[R_{p2}^{\text{in}}(1 + \varepsilon_n/|\varepsilon_n|) + R_{p2}^{\text{out}}(1 - \varepsilon_n/|\varepsilon_n|)], \quad (6)$$

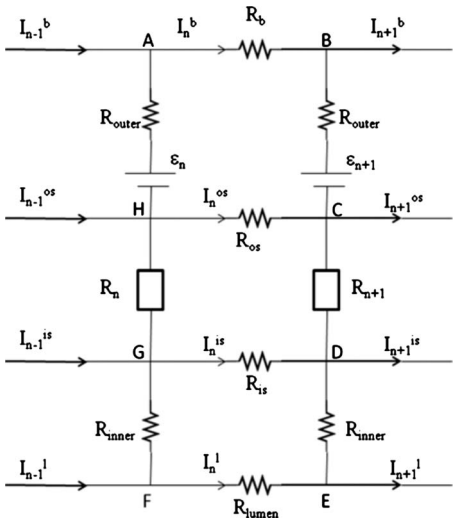

FIG. 6. *n*th unit of the same microtubule model shown in Fig. 1.

$$R_{n+1}(t) = (1/2)[R_{p1}^{\text{in}}(1 + \varepsilon_{n+1}/|\varepsilon_{n+1}|) + R_{p1}^{\text{out}}(1 - \varepsilon_{n+1}/|\varepsilon_{n+1}|)]. \quad (7)$$

The direction of the current in the branch labeled HG(CD) in Fig. 6 for HG (CD) branch depends on whether the tail battery ε_n (ε_{n+1}) is pushing the charge carriers through the nanopores or pulling out from them. As a result, the current flows in or out through different resistors. The term $\varepsilon_n/|\varepsilon_n|$ in the above equation represents the dependence of these resistances on the state of the tail battery. In the symmetric case $R_{p2}^{\text{out}} = R_{p2}^{\text{in}} = R_{p1}^{\text{out}} = R_{p1}^{\text{in}}$, and so $R_n(t) = R_{n+1}(t)$.

To extract some information on the current or the resulting accumulated charges in the lumen, Kirchhoff's rules are applied:

Loop ABCHA

$$\varepsilon_n(t) - \varepsilon_{n+1}(t) - (R_b + 2R_{\text{outer}})I_n^b + R_{\text{outer}}(I_{n-1}^b + I_{n+1}^b) + R_{\text{os}}I_n^{\text{os}} = 0. \quad (8a)$$

Loop HCDGH

$$R_{\text{is}}I_n^{\text{is}} - (R_{\text{os}} + R_n + R_{n+1})I_n^{\text{os}} + R_nI_{n-1}^{\text{os}} + R_{n+1}I_{n+1}^{\text{os}} = 0. \quad (8b)$$

Loop GDEFG

$$R_{\text{lumen}}I_n^l - (R_{\text{is}} + 2R_{\text{inner}})I_n^{\text{is}} + R_{\text{inner}}(I_{n-1}^{\text{is}} + I_{n+1}^{\text{is}}) = 0. \quad (8c)$$

The difference $\Delta I_n^l = I_{n+1}^l - I_n^l$ represents the change in the current through the lumen from the *n*th unit to $(n+1)$th unit, which can be shown by combining Eqs. (8a)–(8c) to satisfy

$$R_{\text{os}}R_{\text{lumen}}\Delta I_n^l = [R_{\text{os}} - \Delta^2 R_n(t)][\Delta^2\varepsilon_n(t) + R_b\Delta I_n^b], \quad (9)$$

where $\Delta F_n = F_{n+1} - F_n$ and $\Delta^2 F_n = F_{n+1} - 2F_n + F_{n-1}$ for any function F_n. For small values of the external battery, one expects that the bulk current does not change significantly and so $\Delta I_n^b \sim 0$. Therefore,

$$R_{\text{os}}R_{\text{lumen}}\Delta I_n^l \sim [R_{\text{os}} - \Delta^2 R_n(t)]\Delta^2\varepsilon_n(t). \quad (10)$$

In the symmetric case $R_{n+2}(t)=R_{n+1}(t)=R_n(t)$ and so $\Delta^2 R_n(t)=0$,

$$R_{\text{lumen}}\Delta I_n^l \sim \Delta^2 \varepsilon_n(t) \tag{11}$$

or

$$R_{\text{lumen}}(d/dt)\Delta Q_n^l \sim \Delta^2 \varepsilon_n(t), \tag{12}$$

where ΔQ_n^l is the change in the accumulated charges in the lumen between units n and $n+1$. Since the tail battery $\varepsilon_n(t)$ is random in nature, its average value over the course of time should be zero. Therefore, $\Delta Q_n^l \sim 0$ and so one would not expect to see a significant change in the average value of the current through the lumen or in the resulting accumulated electric charges. This is what is seen in Fig. S5 of supplementary material [6].

In the asymmetric case, however, $\Delta^2 R_n(t)$ is not zero and so one has

$$R_{\text{os}}R_{\text{lumen}}\Delta Q_n^l \sim -\int dt \Delta^2 R_n(t)\Delta^2 \varepsilon_n(t). \tag{13}$$

It is clear that the above integrand has zeroth- and second-order terms in $\varepsilon_n(t)$,

$$\begin{aligned}\Delta^2 R_n(t)\Delta^2 \varepsilon_n(t) = [R_{p2}^{\text{in}} + R_{p2}^{\text{out}} - R_{p1}^{\text{in}} - R_{p1}^{\text{out}}] + (1/2)(R_{p2}^{\text{in}} - R_{p2}^{\text{out}}) \\ \times(\varepsilon_{n+1}/|\varepsilon_{n+1}| + \varepsilon_{n-1}/|\varepsilon_{n-1}|) \\ -(R_{p1}^{\text{in}} - R_{p1}^{\text{out}})\varepsilon_n/|\varepsilon_n|](\varepsilon_{n+1} - 2\varepsilon_n + \varepsilon_{n-1}).\end{aligned} \tag{14}$$

As a result, one expects the average accumulated charge to be nonzero over the course of time as we move along the MT units, as seen in Fig. 3.

IV. CONCLUSIONS

The key finding of the present work is the determination of the thermally driven currents involving the motion of cations circulating in and around the microtubule. Our model demonstrates how the MT's distinctive geometry, being a cylindrical biopolymer with channel-like nanopores and *C*-terminal tails extending outward from the surface, may enhance the conduction of cations through the lumen, as a result of asymmetry in ion conduction through the pores and fluctuating tail-body interactions. This was demonstrated by comparing the longitudinal current at the end of the microtubule obtained by our model with the results for symmetric pore conduction or where there is no stochastic voltage source from the tail movement.

However, our treatment of the difference in potential between the two ends of the MT as fixed entails that, within this model, the sum of longitudinal currents through all four cylindrical regions is necessarily constant, independent of the presence or absence of conduction through the sides of the MT. The type of situation that is more likely to arise in the cellular environment is a short-lived state of increased potential at one end of the MT, slowly dispersing as ions flow toward the other end. In that case, it may be expected that transferred charges would accumulate at the end with the more negative potential and that current passing through the lumen would not all back-propagate in the opposite direction along the bulk and outer surface, as in the results presented here. That is, the total longitudinal current may increase as a result of the current being focused through the pores into the lumen. In future work, a more realistic computational model allowing for dynamic equilibration, for example, by solution of the Smoluchowski equation, could be used to test this hypothesis. From a biological perspective, this could imply that when a weak potential difference occurs across its two ends, a MT may allow the resulting current to be amplified along its direction, preventing dissipation as random noise throughout the cell. It is possible that within the cellular environment, this may contribute to the opening of voltage-gated ion channels with which the MTs form a terminal connection.

It is also tantalizing to speculate about the role of the most structurally variable element of tubulin, its *C*-terminal tail region, that can be thought of as a tubulin isotype identifier. With major differences across the isotypes in both the length and charge of *C*-termini, their specialized role as an active dynamical element of the electrical circuitry formed by a MT may provide an explanation for tubulin isotype distribution differences among different cell types, e.g., the abundance of βIII tubulin in neurons [60].

Interestingly in this connection, Minoura and Muto [4] found that when the *C*-terminal tail region of the MT was cleaved by subtilisin digestion or when the pH was varied, the effect on conductance was proportional to the decrease in the net number of negative charges, indicating that the high conductivity was primarily due to the polarization response of the condensed atmosphere of counterions to the external voltage source. The volume for which conductance was calculated was 21.6 nm in radius and thus would have included a region of what we have referred to as the outer bulk solution. These observations agree with our results predicting that the stochastic motion of *C*-terminal tails has no effect on overall charge conduction. However, we expect that the "pipelike" conduction of cations through the lumen predicted here would increase the current focused at a narrow region at one end of the MT. Experimental testing of the results of the present model would require a narrowly focused current sensor capable of recording the isolated luminal current. Digestion of *C*-terminal tails by subtilisin or changes in pH or ionic concentrations could then be used to test our predictions of the involvement of *C*-terminal tails by ascertaining how MT conductivity through the lumen varies with the corresponding net negative charge of the dimer. The much-needed experimental validation of our predictions would undoubtedly lead to exciting new opportunities in the development of biological nanoscale electronic applications.

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PROTEINS
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Molecular dynamics modeling of tubulin C-terminal tail interactions with the microtubule surface

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ABSTRACT

Tubulin, an α/β heterodimer, has had most of its 3D structure analyzed; however, the carboxy (C)-termini remain elusive. Importantly, the C-termini play critical roles in regulating microtubule structure and function. They are sites of most of the post-translational modifications of tubulin and interaction sites with molecular motors and microtubule-associated proteins. Simulated annealing was used in our molecular dynamics modeling to predict the interactions of the C-terminal tails with the tubulin dimer. We examined differences in their flexibility, interactions with the body of tubulin, and the existence of structural motifs. We found that the α-tubulin tail interacts with the H11 helix of β-tubulin, and the β-tubulin tail interacts with the H11 helix of α-tubulin. Tail domains and H10/B9 loops interact with each other and compete for interactions with positively-charged residues of the H11 helix on the neighboring monomer. In a simulation in which α-tubulin's H10/B9 loop switches on sub-nanosecond intervals between interactions with the C-terminal tail of α-tubulin and the H11 helix of β-tubulin, the intermediate domain of α-tubulin showed more fluctuations compared to those in the other simulations, indicating that tail domains may cause shifts in the position of this domain. This suggests that C-termini may affect the conformation of the tubulin dimer which may explain their essential function in microtubule formation and effects on ligand binding to microtubules. Our modeling also provides evidence for a disordered-helical/helical double-state system of the T3/H3 region of the microtubule, which could be linked to depolymerization following GTP hydrolysis.

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Key words: protein structure; theory; modeling; computer simulation; conformational changes; dynamics of biomolecules; cytoskeleton; polymerization.

INTRODUCTION

The tubulin protein polymerizes to form microtubules (MTs) which are cylindrical structures consisting typically of 13 protofilaments, each of which is comprised of repeats of α–β tubulin hetero-dimers. The carboxy (C)-terminal tail domains of all tubulin monomers are found on the outer surface of the MT. Both α- and β-tubulin consist of numerous isotypes, differing in amino acid sequence and cellular and tissue distribution.[1] The lengths and sequences of the C-terminal tails vary greatly depending on isotype; human isotypes typically consist of 10–13 residues on α-tubulin, and 18–24 residues on β-tubulin, and have a high proportion of negatively charged residues.[2,3] A great deal of evidence indicates that the C-terminal tails of both α- and β-tubulin play critical roles in regulating MT assembly and function, and in determining the conformation of the tubulin molecule,[4–8] roles whose details are very poorly known. C-terminal tail residues are necessary for the assembly of normal MTs; one recent cloning study has demonstrated that the first two tail residues of β-tubulin are indispensable for the formation of MTs in HeLa cells.[4] Tubulin polymers assembled *in vitro* from tubulin lacking both C-terminal tails (denoted as $\alpha_S\beta_S$ tubulin) display an altered morphology characterized by bundles of twisted filaments, sheets,[5] and hooks.[6,7] Moreover, the presence of alkali metal cations[8] or cleavage of tails[5] allows polymers to form at lower tubulin concentrations, and the reduction of negative charges on C-termini by covalent modification lowers the rate constant for MT disassembly.[9] Tubulin polymers lacking only the β-tubulin tail ($\alpha\beta_S$ tubulin) display a ring con-

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formation, do not require GTP for polymerization, and are temperature insensitive.[10]

It is likely that the effects of the C-termini on MT assembly and tubulin conformation are critically dependent on the conformational interaction between the C-termini and other parts of the tubulin molecule. Nevertheless, the natures of these interactions are almost entirely unknown. Although a series of studies have determined the structure of the rest of the tubulin molecule, the structures of the C-termini have remained largely invisible to crystallographic methods. Thus, a molecular-level picture of the structures of tubulin protofilaments obtained by molecular dynamics (MD) simulations, including both tail dynamics and their interactions with the remainder of the protein, offers an opportunity to provide unique insights into a better comprehension of the biological functions of tail domains and their significance for MTs.

Depending on the isotype, the charge of the C-terminal tail of human β-tubulin is generally between −10 and −12 *e* and that of the α-tubulin tail is −7 or −8 *e*, not including post-translational modifications.[2,3] Although the remainder of the tubulin dimer is also negatively charged overall (the total charge of the dimer is approximately −50 *e*), there are also positively charged regions on the protein surface. By using reductive methylation, residue Lys 394, which together with Arg 390 and His 393 forms a positively-charged pocket on the surface of α-tubulin, has been shown to be essential to MT polymerization.[11,12] Because Lys 394 is less accessible to methylation in the MT polymer compared to when present in the dimer,[12,13] it has been suggested that this residue is involved in electrostatic interactions with the negatively-charged acidic residues in the tail regions.[14] The existence of such a "tail–body" electrostatic interaction was also supported by a study[15] in which the detached α-tubulin tail was computationally docked to the tubulin surface and found to bind at a site near Lys 394. Experimental findings that C-terminal tail cleavage can affect ligands that bind to tubulin, and that this influence seems to be due to electrostatic interactions, also indicate that tail–body interactions contribute to tubulin conformation. This evidence includes a study showing that cleaving the C-terminal tail on α-tubulin lowers the activation energy for colchicine binding to tubulin, and leads to pH independent binding.[16] Another such study showed that the influence of oligoanions on the stabilization of tubulin polymers by vinblastine is eliminated by the cleavage of the C-terminal tails of β-tubulin.[17]

lysine

The objective of this article is to use MD modeling in order to elucidate the issues surrounding the role of C-termini in structural and functional effects on the tubulin dimer and, consequently, in the biological behavior of MTs. The specific modeling tool that we apply to our analysis is simulated annealing in conjunction with MD simulation. Simulated annealing is an enhanced sampling technique in which heating followed by re-cooling of a molecular system can be used to determine its equilibrium structure. Heating allows the system to overcome energy barriers that would otherwise prevent the system from fully sampling configuration space, and slow cooling causes the system to freeze into a minimum energy structure. This method has frequently been applied within Monte Carlo simulations to determine conformational free energy minima[18,19]; more recently simulated annealing has been used within MD simulations,[20] as it is in the present work. Here, this technique is applied to obtain positions of C-terminal tails for the tubulin dimer. Subsequently, simulations are run of protofilaments formed from αβ tubulin, $\alpha\beta_s$ tubulin, and from $\alpha_s\beta_s$ tubulin. In this way, the effects of the tails on protofilament dynamics are investigated computationally.

We analyze MD trajectories to investigate the occurrence of tail–body interactions of the C-terminal tail regions with positively-charged residues on the tubulin surface. Structural motifs intrinsic to the C-terminal tail domains seen in the MD trajectories are also considered. Finally, we analyze the root mean square fluctuations of residues in trajectories with and without tail domains. We show that interactions of the tail domains with exposed residues on the surface of the remainder of the protein may play an important role in MT dynamics by influencing the large-scale conformation of tubulin's intermediate domain.

MATERIAL AND METHODS

Simulated annealing

After heating the tubulin dimer system to 5000 K, each of 12 models was slowly cooled to 300 K over 5.2 ns of simulation time. Out of these 12, three models were selected as starting positions in simulations of two explicitly-solvated tubulin protofilaments. The flowchart in Figure 1 summarizes the simulated annealing/MD simulation procedure adopted in our study to determine tubulin tail conformations, which we now describe in detail.

Our system consisted of a tubulin dimer, composed of human isotypes αIV and βI, modeled as described elsewhere[21]; these have C-terminal tails with lengths of 10 and 18 residues (residues α439–448 and β427–444), and charges of −8 and −12 *e*, respectively. Initial set-up was performed in the *leap* and *antechamber* modules of the AMBER9[22] program package using the AMBER03 force field to parameterize the tubulin protein.[23,24] MD was performed using the sander module of AMBER9.[22] Twelve high temperature implicitly-solvated MT dimers were generated by heating the dimer to 5000 K (chosen to make the tails mobile on a reasonable time scale), and then extracting structures from the simulation every

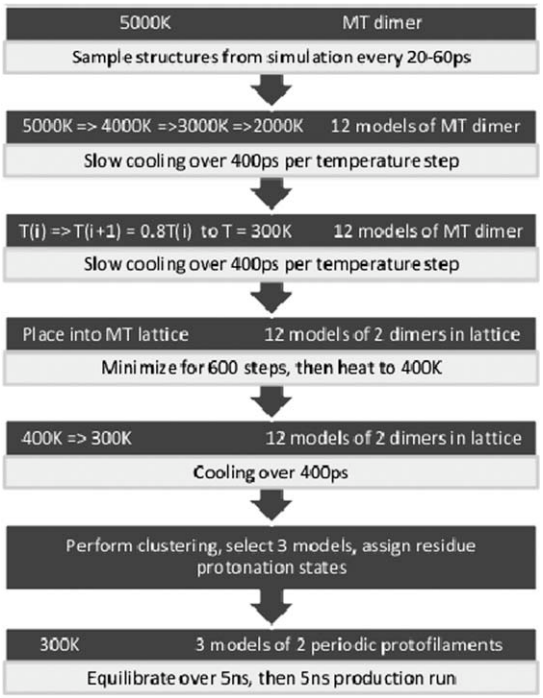

Figure 1

Flowchart illustrating the simulated annealing/MD simulation procedure used in this work to determine tubulin tail conformations.

20–60 ps. These were then slowly cooled to a target temperature of 300 K. Our annealing schedule used temperature reduction steps of 1000 K over 200 ps time periods, with equilibrations being performed at each of these temperatures for a period of another 200 ps, until the temperature was below 2000 K, and then used a 0.8-power law cooling-schedule again with 400 ps spent for each temperature stage. A time step of 1 fs was used throughout all simulations. The modified generalized Born method GB^{OBC} corresponding to model II in Ref. [25] (igb = 5) was employed to represent a 150 m*M* ionic solution and the temperature was maintained by a Langevin thermostat with a collision frequency of 2 ps^{-1}. All bonds involving hydrogens were constrained using the Shake[26] algorithm. A restraint weight of 15 kcal/mol was placed on residues 1–438 of α tubulin and 1–428 of β tubulin. Removal of center-of-mass translational and rotational motion was disabled. The non-bonded cut-off used was 50 Å.

Structures were fit to the original homology model by running simulations for 5000 steps with a restraint of 50 kcal/mol on the α-carbons of all residues not belonging to the C-terminal tail domains, forcing the α-carbons belonging to the main body of the dimer to assume a structure similar to the one determined by electron crystallography experiments. In some cases, the *autosculpt* feature of the Pymol program[27] was employed to slightly adjust the positions of β tubulin tail residues which overlapped with parts of the opposite β tubulin. Tubulin dimers were then placed into a configuration consistent with MT geometry by using the model of Li *et al.*[28] Two adjacent dimers were created as would exist in the MT lattice by using an in-house VMD[29] script, as described elsewhere.[30] These were then relaxed for 300 steps without solvent, and another 300 steps with implicit solvent. Afterwards the structures were heated to 400 K, and then cooled to 300 K over a period of 400 ps for further energy minimization, with the same restraint on the α-carbons belonging to the main body.

The 12 models of the tubulin dimer resulting from our simulated annealing procedure are shown in Figure 2. Structures were clustered in the *ptraj* module of AMBER using the means algorithm[31] to produce three clusters each for α- and β tubulin tails, by comparing positions of alpha-carbons based on root-mean-squared distance (RMSD) after fitting all tubulin dimers, excluding tail residues, to the position of one of the dimers. Clusters of α tubulin tail peptides contained five, four, and three elements, respectively, with mass weighted root-mean-square distances to the centroid of 0.6 Å for each of the three clusters. Clustering the β tubulin tail peptides resulted in one group of six elements, with a mass weighted root-mean-square distance to centroid of 1.4 Å, a second cluster with five elements and a mass weighted root-mean-square distance to centroid of 1.3 Å, and a third cluster with a single element only. Cluster

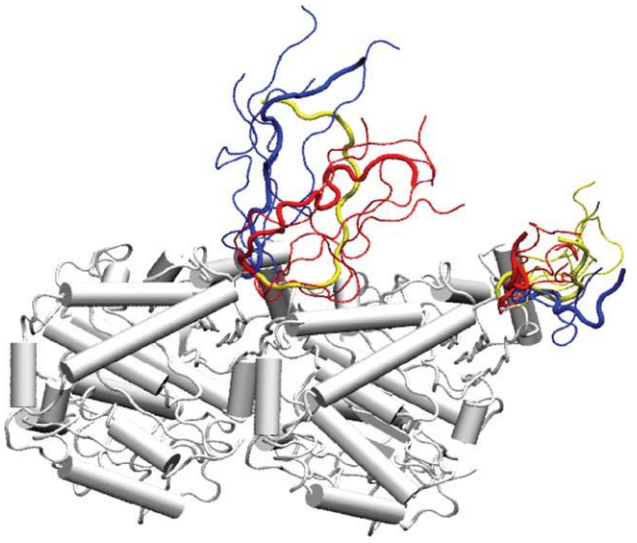

Figure 2

Tubulin dimer, showing three clusters obtained by simulated annealing MD for the C-terminal tail domains, colored in yellow, red, and blue, respectively. Cluster representatives *p1* (yellow), *p2* (red), and *p3* (blue) are depicted as thicker coils compared to other cluster elements. The dimer is viewed from above if positioned in a MT lattice with β-tubulin on the left and α-tubulin on the right.

representatives were chosen as the dimers having β tubulin tails closest to the cluster centroids. Because each of these three dimers fell into a different one of the three clusters corresponding to the α tubulin tails, we chose to retain the original pairing between α- and β tubulin monomers, rather than pairing cluster representatives for β tubulin tails with similarly chosen cluster representatives for α tubulin tails. As a further check on the appropriate number of representatives, hierarchical clustering of both β- and α-tubulin tails was performed using the MMTSB Toolset[32] command *jclust* with the *rmsd* mode, requesting a maximum of 50 structures, which gave two main clusters for the α-tubulin tail and three for the β-tubulin tail.

The three representative dimers, which were used to perform MD simulations, are shown in Figure 2. In the text that follows, we refer to these three models as *p1*, *p2*, and *p3*. In addition, we performed simulations with one or both tail domains removed. The α-tubulin tail position for the $\alpha\beta_S$ model was taken from model *p3*.

MD simulations

The *propka* program[33,34] was used to assign protonation states to Asp, Glu, His, and Lys residues on each of the three models; as a result His 309 was protonated on α-tubulin of model *p1*, His 307 on β-tubulin of models *p2* and *p3*, and α-tubulin residue Asp 46 was protonated on all three models.

The tubulin dimers were then placed into a MT lattice. The positions of GDP at the exchangeable site and GTP at the non-exchangeable site were taken from the PDB file 1TVK[35] and those of the Mg^{2+} ions from the PDB file 1JFF.[36] Parameterization of nucleotide co-factors was taken from reference.[37] The MD simulations included two adjacent protofilaments composed of two tubulin heterodimers; identical tail positions were used on both protofilaments. One hundred three Na ions were added to neutralize the system electrostatically, and following this 80 each of Na^+ and Cl^- were added to bring the ionic concentration to 100 m*M*. The system was placed into a periodic box with dimensions 81.2 Å by 107.5 Å by 152.5 Å, which was filled with pre-equilibrated explicit TIP3P water molecules[38] using the *leap* module of AMBER9. The protofilaments were aligned with the *x*-axis of the periodic box, and by setting the *x*-dimension of the box equal to 81.2 Å, or the length of one tubulin dimer, the periodic boundary conditions were taken advantage of to model the system as two periodic protofilaments. Structures were then converted from AMBER to Gromacs-style coordinate and topology formats using the *amb2gmx* perl script,[39] and simulations were performed with the Gromacs v3.4[40,41] program.

Structures were energy minimized, which also relieved any steric clashes with waters. The entire system was relaxed using 5000 steps of steepest descent (1000 steps), followed by conjugate gradient minimization. Following equilibration for 5 ns with a time step of 2 fs, MD simulation was performed over a period of another 5 ns with the same time step, with periodic boundary conditions at a temperature of 300 K and at constant atmospheric pressure. The Shake algorithm[26] was used to constrain bonds involving hydrogen in all simulations and we used a non-bonded cut-off of 12 Å. PME was employed to calculate long-range electrostatics.

Analysis

Analysis was then performed on the 5 ns production MD runs by using the suite of tools available in the Gromacs program package.[40,41] Only tail domains belonging to the protofilament positioned on the left when viewed from the minus to the plus end was used for analysis. The restriction of our analysis to these tail domains was made because of their physically central position relative to our models, in contrast to the peripheral position of the other set of tail domains, as seen in Figure 5.

For the examination of structural motifs, a matrix of scaled rho (rhosc) values was computed using the Gromacs[40,41] tool *g_rms*. This program compares each structure in trajectory *p1* to each structure in trajectory *p3*, after fitting the structures on top of one another by rotation and translation.[42] The structural similarity value rho is defined as a function of the two structures A and B by:

$$\rho(\mathrm{A},\ \mathrm{B}) = \frac{2\mathrm{RMSD}(\mathrm{A},\ \mathrm{B})}{\sqrt{R_{\mathrm{gyr}}^2(\mathrm{A})R_{\mathrm{gyr}}^2(\mathrm{B}) - \mathrm{RMSD}^2(\mathrm{A},\ \mathrm{B})}},$$

where R_{gyr} is the radius of gyration of a structure, and where RMSD gives the root mean scale deviation between two structures. Scaling the aligned principal moments of inertia of the two structures A and B to values that are all equal and satisfy $R_{\mathrm{gyr}} = 1$ prior to comparing structures, leads to a more universal, size-independent measure of similarity, rhosc, assuming values between 0 and 2. Values of rhosc less than 0.3 were used here as an indication of structural similarity, and the percentage of comparisons for which rhosc < 0.3 represents the percentage of total time for which the conformation of the motif is similar between the two models A and B.

The *g_min_dist* tool was employed to determine minimum distances between charged or aromatic residues belong to tubulin tail domains and other charged or aromatic residues. Residue pairs were included in this analysis if they came closer than 0.3 nm at any point during the simulations.

Average root-mean-square fluctuations (RMSF) were calculated using the *g_rmsf* program, as the averages over the simulation time, $\langle |\Delta R_i|^2 \rangle$, in the square fluctuation of

Table I
Charged and Aromatic Residues Involved in Contacts Between α-Tubulin's C-Terminal Tail Domain, α-Tubulin's H10/B9 Loop, and β-Tubulin's H11 Helix

	αtail-βH11	βH11-αH10/B9loop	αtail-αH10/B9loop
p1	αTyr440,Glu441-βArg390 αAsp442,Glu443-βArg391	βArg390- αAsp345	αTyr440- αAsp345
p2	αTyr440-βArg390 αTyr440-βArg391	βArg391- αTrp346	
p3	αTyr440-βArg390	βArg390- αAsp345 βArg391- αTrp346	αTyr440- αAsp345
$\alpha\beta_S$	αGlu445,Glu447,Glu448-βLys379 αAsp442-βArg390		αTyr440- αAsp345
$\alpha_S\beta_S$		βArg390,Arg391- αTrp346	

the position $\boldsymbol{R}_i$ of the α-carbon corresponding to each residue *i*.

Two programs *do_dssp* and *g_helix* were used in the analysis of helical content. The *do_dssp* program assigns secondary structure by identifying patterns of hydrogen bonding using the DSSP[43] algorithm. On the other hand, *g_helix* determines helicity based upon both hydrogen bonds and ϕ/ψ angles.

RESULTS AND DISCUSSION

Simulated annealing

Following annealing, α-tubulin tails were generally found to point in the longitudinal direction toward where the β-tubulin of the neighboring dimer would be in a MT lattice (see Fig. 2). There was a tendency of the β-tubulin tails to point laterally in the direction of where the right-hand adjacent protofilament would be if the dimer were part of a MT being viewed from the minus to the plus end (i.e., towards the intermediate, taxol-binding domain). However, the β-tubulin tails covered a large range of conformational space (Fig. 2).

By clustering the final endpoints of simulated annealing, and choosing representatives of different clusters (*p1*, *p2*, and *p3*) for subsequent simulation and analysis, we ensured that these three simulations were run from well-separated initial positions. Thus, while each of these simulations on its own does not demonstrate complete time-sampling of MD, a broader picture of the dynamic behavior of the protein system was obtained by considering all three simulations.

Structural motifs

A previous set of replica exchange MD simulations studied peptides composed of the residues forming the C-terminal tails of β-tubulin and identified three transient structural motifs as potential sites for casein kinase-2 (CK-2) phosphorylation and MAP2 binding.[3] We sought to determine, by analysis of our trajectories, whether these structural motifs found for the isolated tail structures, still characterize the tail domains within the context of a MT.

The proposed CK-2 motif, consisting of the residues DATA, occurs at the start of the β tubulin tail and is evolutionarily conserved between tubulin isotypes.[3] The structure of these four residues beginning with residue Asp 427 may be seen in Figure 3. The percentage of total comparisons for which rhosc values are less than 0.3 was found to be 93.6%, indicating a high degree of structural similarity for this sequence and is higher than that found in the original study.

Two motifs on the C-terminal tail, that were previously identified as making up the MAP2 binding site,[44] consist of sequences EEAE (residues 438–441) and EEEA (residues 441–444), respectively, contained in the last seven residues (EEAEEEA) of the β-tubulin tail domain.[3] The respective counts of rhosc values less than 0.3 for these two sequences are 14.7% and 2.8% of total comparisons. This is considerably less than in the previous study and our analysis suggests that these two regions take on little specific structure; however, the bound β-tubulin tail conformers identified below indicate that these residues remain solvent exposed and the possibility of a MAP2 induced structure cannot be ruled out.

Tail–body interactions

We performed analysis on our MD trajectories to investigate the occurrence of tail–body interactions of the C-terminal tail regions with positively-charged residues on the tubulin surface. Tables I and II show the charged residues that, at any point during our simulations, are involved in close contacts of oppositely-charged side-chains, with inter-residue distances of less than 0.3 nm, for each model. Because there were some very noticeable interactions between aromatic residues from the tails and charged residues from the tubulin body and vice versa, these were also included in our analysis. Figures 3 and 4 display models highlighting the tubulin residues involved in tail–body interactions of the β- and α-tubulin tails, respectively, for any one of the four models *p1*, *p2*, *p3*, and $\alpha\beta_S$.

Table II
Charged and Aromatic Residues Involved in Contacts Between β-Tubulin's C-Terminal Tail Domain, β-Tubulin's H10/B9 Loop, and α-Tubulin's H11 Helix

	βtail-αH11	αH11-βH10/B9loop	βtail-βH10/B9loop
p1	βGlu431,Glu432- αHis393	αLys394- βGlu343	βGlu431- βTrp344
	β Glu432- αLys394	αHis393, Lys394- βTrp344	
p2	βGlu432- αLys401		
p3		α Lys401- βTrp344	βAsp427- βTrp344
$\alpha\beta_S$		αArg390,His393,Lys394- βGlu343	
		α Lys401- βTrp344	
$\alpha_S\beta_S$		α Lys401- βTrp344	

We observed that two kinds of tail–body interactions could take place. First, charged Glu and Asp residues of the C-terminal tails can interact with a cluster of positively-charged residues on H11 of the neighboring tubulin dimer. That is, the α-tubulin tail interacts with the H11 helix of β-tubulin, and the β-tubulin tail interacts with the H11 helix of α-tubulin. Figure 5 shows the positions of the H11 domain within the MT lattice, as well as the positions of the other domains that will be discussed below. At the inter-dimer longitudinal interface, negatively-charged residues of the α-tubulin tail interact with residues Lys 379, Arg 390, and Arg 391 on the H11 helix of β-tubulin. Likewise, at the intra-dimer interface, the β-tubulin tail interacts with the positively-charged cluster on α-tubulin containing the residue Lys 394, which has previously been identified as essential for MT polymerization, and His 393; Lys 401 on H11 can also be involved in this type of interaction. The charged tail residues involved are 441–443, 445, and 447–448 on α-tubulin and 431–432 on β-tubulin. We also noted that the ar-

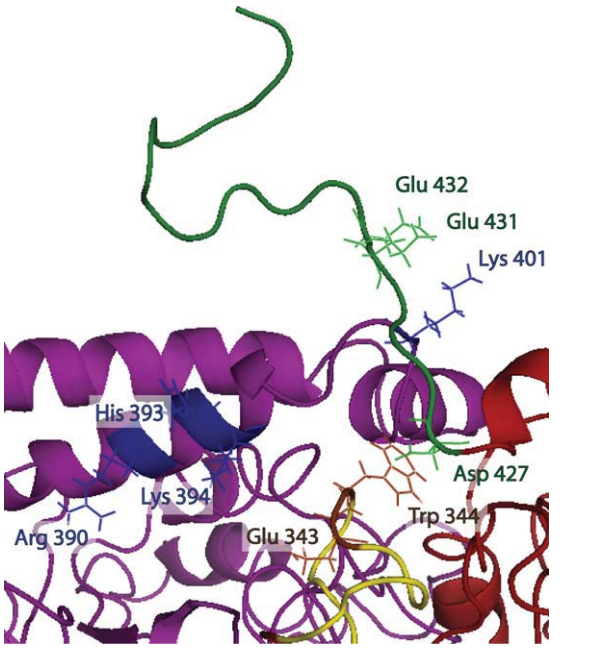

Figure 3
Illustration of the intra-dimer longitudinal interface within a MT between α-tubulin (purple) and β-tubulin (red). Charged and aromatic residues have been labeled and represented in stick mode that are involved in close contacts between the β-tubulin tail, shown in green cartoon, and either the β-tubulin H10/B9 loop, shown in yellow cartoon, or the α-tubulin H11 helix, as well as those involved in interactions between the latter two regions. This figure was created using the Pymol molecular viewing program,[27] using model *p2*.

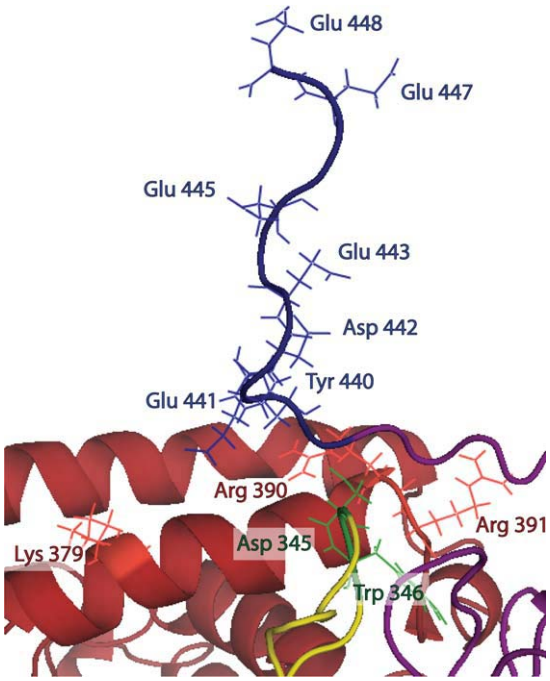

Figure 4
Illustration of the inter-dimer longitudinal interface within a MT between α-tubulin (purple) and β-tubulin (red). Charged and aromatic residues have been labeled and represented in stick mode that are involved in close contacts between the α-tubulin tail, shown in blue cartoon, and either the α-tubulin H10/B9 loop, shown in yellow cartoon, or the β-tubulin H11 helix, as well as those involved in interactions between the latter two regions. This figure was created using the Pymol molecular viewing program,[27] using model *p3*.

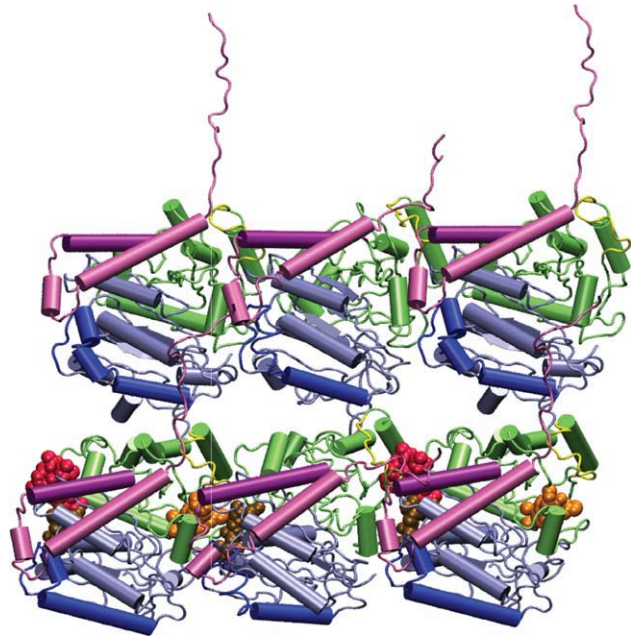

Figure 5

Representation of part of the MT lattice, showing β-α-β tubulin monomers comprising each of two protofilaments oriented with plus ends on the left-hand side. N-terminal domains are colored in blue, intermediate domains in green, and C-terminal domains in purple. The T3/H3 region is shaded in dark blue, the H10/B9 loop in yellow, and the H11 helix in dark purple. The positions of the binding sites of colchicine (orange), vinblastine (red), and nucleotides GDP and GTP (brown) are also illustrated for one of the protofilaments.

omatic residue Tyr 440 on the α-tubulin tail interacts with charged residues Arg 390 and Arg 391 on the H11 helix of β-tubulin by nonpolar side-chain stacking, as well as by interactions between charged Arg side-chains and the polar hydroxyl substituent on the Tyr ring.

The second kind of tail–body interaction observed was between tail residues and residues on the H10/B9 loop of the same tubulin monomer. These contacts are not between two charged residues but rather involve an aromatic residue and a charged residue. In the case of α-tubulin, the tail residue Tyr 440 interacts with Asp 345 of the H10/B9 loop; this contact involves both hydrophobic interactions between the two bulky side chains, and electrostatic attraction between the charged carboxylic acid group of the Asp side-chain and the polar hydroxyl substituent on the Tyr ring. In the case of β-tubulin, tail residues Asp 427 or Glu 431 engage in close nonpolar contacts with Trp 344 of the H10/B9 loop.

Interestingly, the H10/B9 loop can also interact with the positively-charged cluster on the H11 helix of the neighboring dimer, and these interactions are also shown in Tables I and II. In the case of the α H10/B9 loop and the β H11 helix, there can be a salt bridge established between αAsp 345 and residue Arg 390 of H11; also the bulky aromatic side chain of αTrp 346 engages in non-polar interactions with the long side chains of Arg 390 or Arg 391 from β H11. In the case of interaction between β-tubulin's H10/B9 loop and α-tubulin's H11 residues, Glu 343 of the β-tubulin loop can interact with Arg 390, His 393, and/or Lys 394 on α-tubulin's H11. Here also β Trp 344 interacts with residues His 393, Lys 394, and/or Lys 401 from the H11 helix of α-tubulin through both non-polar side-chain interactions and contacts between the aromatic rings of Trp and the positively-charged amine group on Lys. Note that it is likely that these (and possibly some of the other instances of interactions between positively-charged and aromatic residues discussed above) are cation–pi interactions, although our simulations did not employ a polarizable force-field capable of representing the electronic nature of these interactions.

For model *p1*, we plot in Figure 6 how minimum distances involving tail–body interactions of the α-tubulin tail vary throughout the simulation. During the first half of this simulation there is a strong interaction between β-tubulin's helix H11 and the H10/B9 loop of α-tubulin, at the longitudinal inter-dimer interface. During the second half of the simulation these interactions are reduced and instead increased interactions are observed between the α-tubulin tail and both β-tubulin's helix H11 and the H10/B9 loop of α-tubulin. This switch in interactions between the C-terminal tail, the H10/B9 loop, and the H11 helix of the neighboring dimer may be indicative of competing interactions between these three regions.

At the intra-dimer interface, as seen in Supporting Information Figure S3, for the *p1* model there are multiple close contacts between the β-tubulin tail, α-tubulin's helix H11, and the H10/B9 loop of α-tubulin. Although, as may be seen from Tables I and II, many residues or pairs of residues involved in tail–body interactions and in H11-H10/B9loop interactions are shared between models *p1*, *p2*, and *p3*, these models explore different regions in conformational space and thus their trajectories display qualitatively different behaviors. For example, in model *p2*, at the intra-dimer interface, the β-tubulin tail interacts with residue Lys 401 of the H11 helix on α-tubulin, but there is no interaction between αH11 and the H10/B9 loop of β-tubulin. On the other hand, in both the *p3* and $\alpha\beta_S$ simulations, αH11 interacts with β-tubulin's H10/B9 loop but not with the β-tubulin tail. In the $\alpha_S\beta_S$ simulation, without either tail domain present, almost continuous interaction was observed between the H10/B9 loop and the H11 helix of the adjacent monomer, at both monomer interfaces (see Supporting Information Fig. S2).

A physical model was previously created of ion dynamics in and around MTs, based on the application of a random alternating electric force on the surrounding cations by fluctuations in C-terminal tail positions.[45] The continuous breaking and formation of salt bridges give rise to an alternating voltage source on ions as the

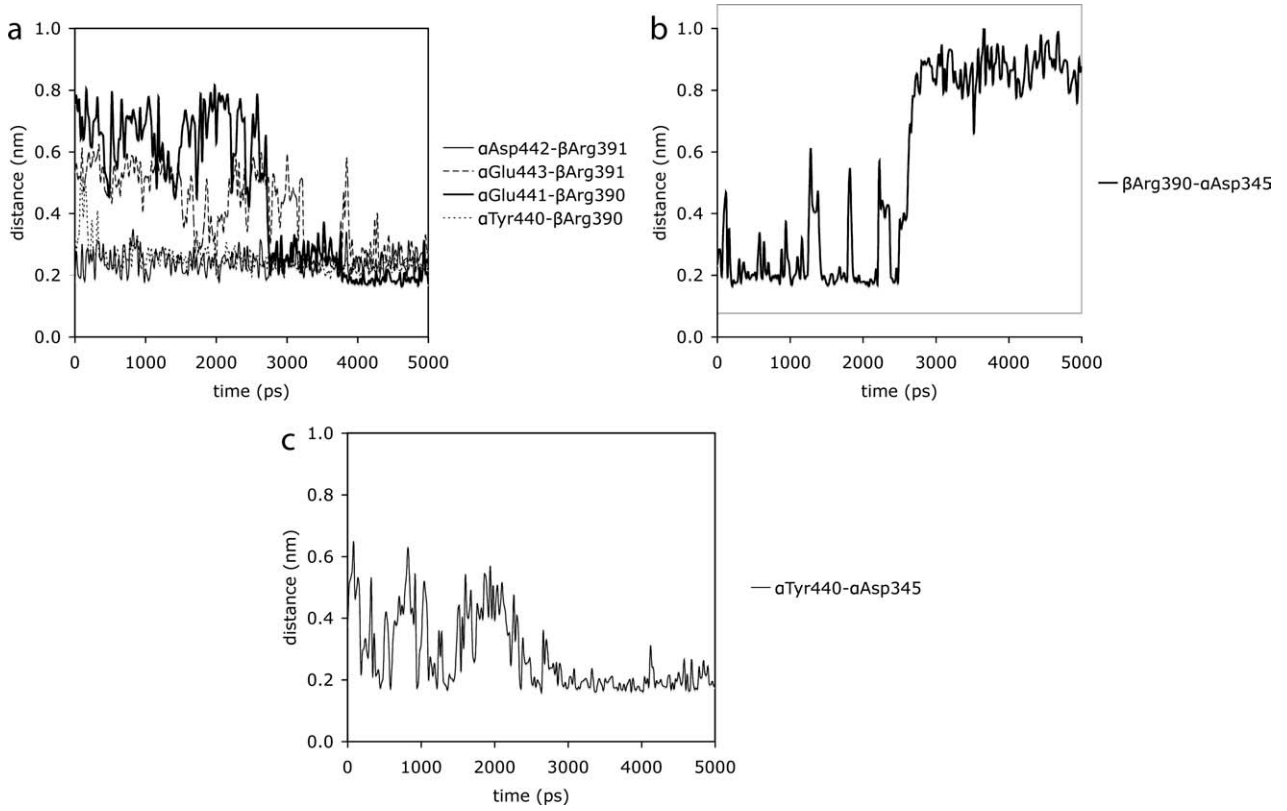

Figure 6

Minimum distances (nm), plotted for the *p1* model as a function of simulation time (ps), between charged or aromatic residues belonging to (**a**) α-tubulin's C-terminal tail domain and β-tubulin's H11 helix; (**b**) β-tubulin's H11 helix and α-tubulin's H10/B9 loop; and (**c**) α-tubulin's C-terminal tail domain and α-tubulin's H10/B9 loop.

tails dynamically oscillate between conformations where they either interact with the tubulin body or extend into more solvated conformations, as seen in Figure 6. This effect observed in our simulations appears to support the hypothesis that local charge reconfiguration due to the motion of C-termini tails can affect ionic flows and hence endow MTs with unique electro-conformational properties.

Structure and flexibility of tubulin body

The average RMSF of each residue of α-tubulin (excluding tail-residues) is plotted as a function of residue index *i* in Figure 7, and that of β-tubulin in Supporting Information Figure S6, corresponding to each of our simulations. The peaks in the RMSF plots show regions of high flexibility, and the highest peaks generally occur at solvent-accessible loops. Tubulin is known to have three main domains, which have been shown to be associated with major correlations in structural fluctuations.[46] These are the N-terminal nucleotide-binding domain, the intermediate taxol-binding domain, and the C-terminal domain, and consist of residues 1–205, 206–381, and 382–448, respectively, on α-tubulin and residues 1–203, 204–371, and 372–444, respectively, on β-tubulin. The last few residues of C-terminal domains show a high degree of flexibility in all models; the loop connecting helices H11 and H12, and H11 itself, may also be seen from the RMSF plots to be quite flexible for the *p3* and $\alpha_S\beta_S$ models. We will now briefly examine the main characteristics of RMSF plots for α- and β-tubulin that we observed first for the N-terminal domain, and then for the intermediate domain; we will also discuss some notable differences between models in RMS fluctuations and their significance.

N-terminal domain (residues α1-205, β1-203)

In the N-terminal domain, peaks are seen in the RMSF plots (Fig. 7 and Supporting Information Fig. S6) at residues α43–63, β45–47 in the region of the H1/B2 loop, which forms lateral contacts with neighboring monomers; at residues α69–74, β66–71 corresponding to the T2 loop; in the region of the H3 helix and T3 loop at residues α95–119, β91–120; and at residues α141–146 in the region of the glycine-rich signature loop, T4. In

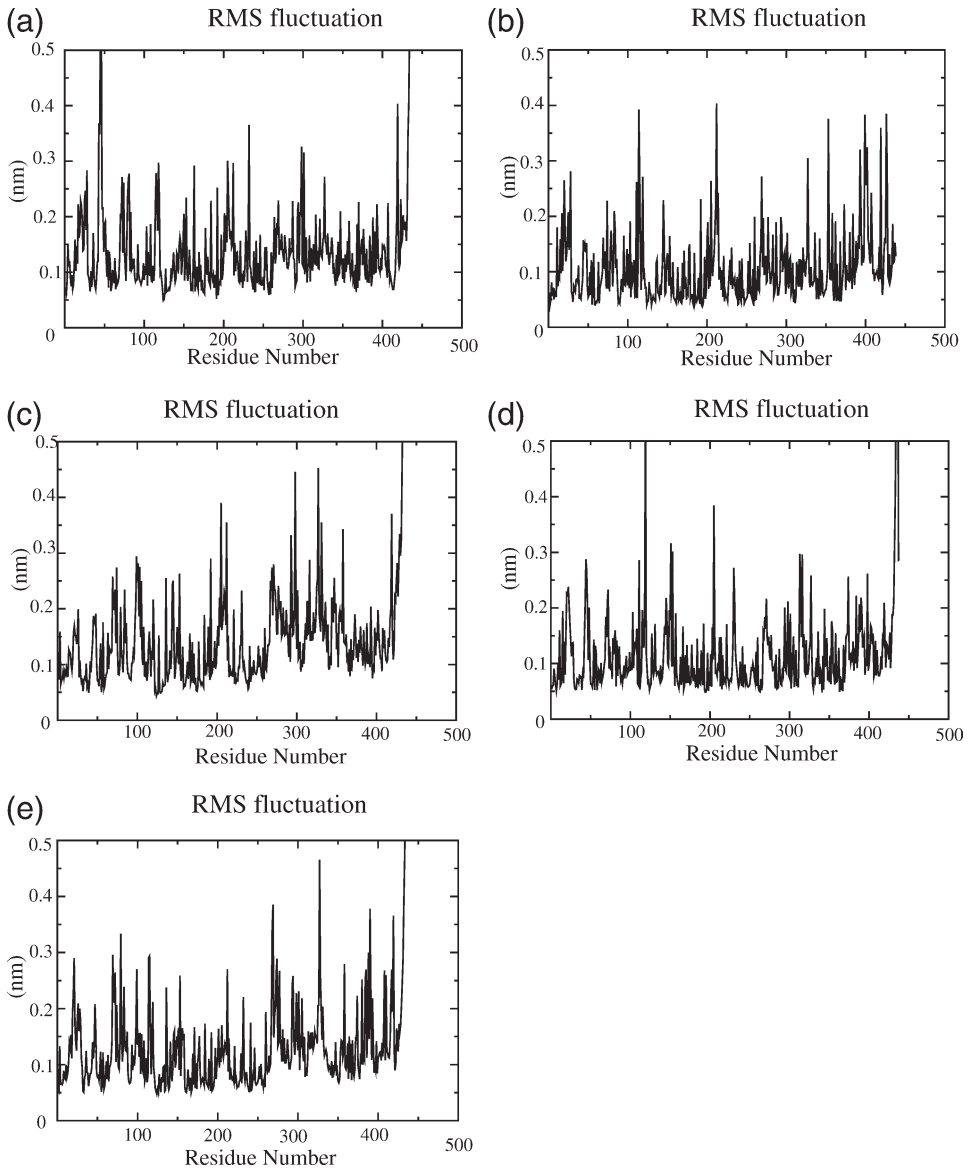

Figure 7

Time-averaged RMSF (nm) of each residue of α-tubulin plotted as a function of residue number for model (**a**) $\alpha\beta_S$, (**b**) $\alpha_S\beta_S$, (**c**) *p1*, (**d**) *p2*, and (**e**) *p3*.

Figure 7 and Supporting Information Figure S6, it may be noted that the peaks corresponding to the T3/H3 region on both α- and β-tubulin are much more pronounced in the RMSF plots for model *p2* than in the other plots. The T3/H3 region (residues α97–128 and β95–126) is located close to the nucleotide binding site, the exchangeable one in the case of β-tubulin, and the non-exchangeable one for α-tubulin. When secondary structure of the N-terminal domain is compared between the simulation trajectories (see Table III and Supporting Information Table S1), it is observed that there is a loss of α-helical structure for the *p2* trajectory compared to for the other trajectories. Upon examination, one noticeable instance of this is a less helical appearance of the H3 helix.

We, therefore, analyzed local secondary structure of the T3 loop and of the H3 helix for each of our models. In Figure 8 and Supporting Information Figure S7, we have plotted the percentage of the total simulation time during which each residue in this region is found in a helical form, calculated by the *g_helix* analysis tool. The plots in Figure 8 show that on average during the simulations the α-tubulin T3/H3 region has lost helicity for both the $\alpha\beta_S$ model and *p2* models, and that the β-tubulin T3/H3 region is much less helical for model *p2* compared to those in the other models.

Table III
Percentages of Residues in α-Tubulin's N-Terminal Domain Forming Various Types of Secondary Structure, Determined Using the do_dssp Gromacs[32,33] Analysis Tool

	$\alpha\beta_S$	$\alpha_S\beta_S$	*p1*	*p2*	*p3*
Structure	36	53	40	43	45
Coil	35	27	33	26	32
β-Sheet	2.2	7.4	1.9	11	6.2
β-Bridge	3.2	2.9	2.4	4.4	1.7
Bend	26	18	25	25	21
Turn	15	21	15	21	17
α-Helix	15	22	21	6.6	20
5-Helix	0.0	0.0	0.0	0.0	0.0
3-Helix	3.6	2.1	1.4	6.4	2.2

The category "Structure" represents all residues classified as α-helix, β-sheet, β-bridge, or turn.

Intermediate domain (residues α206–381, β204–371)

Major peaks seen in the RMSF plots (Fig. 7 and Supporting Information Fig. S6) in tubulin's intermediate domain are those corresponding to the region of longitudinal contacts at residues α200–232, β201–215; the flexible M-loop on β-tubulin at residues 276–284, and the N-loop on α-tubulin at residues 264–288; and the H10/B9 loop at residues α339–350 and β337–348. Greater RMS fluctuations of the intermediate domain of α-tubulin in the *p1* simulation compared to in the other simulations can be seen from the higher peaks on the RMSF plot corresponding to this simulation, and generally increased RMSF values for large parts of this domain. One instance of this is the greater fluctuation seen for model *p1* of residues corresponding to the H10/B9 loop. This is most likely related to the interchange of interactions between the H10/B9 loop, the H11 helix of β-tubulin, and the α-tubulin tail domain, as is observed in Figure 6 for this trajectory. That is, the formation and breaking of salt bridges causes the H10/B9 loop to assume two or more different positions. These interactions may indirectly lead to higher fluctuations of other parts of the intermediate domain of α-tubulin as well. In Figure 6, the sporadic interchange of interactions between the three domains is seen to occur primarily during the first half of the simulation. We thus examined the RMSF plot calculated for the first half of the simulation only [Supporting Information Fig. S8 (a)]. This plot also shows a high degree of fluctuation of residues in tubulin's intermediate domain. However, consistent with the more constant pattern of interactions during the second half of the simulation, the RMSF plot corresponding to this time interval shows much less fluctuation of the intermediate domain [Supporting Information Fig. S8 (b)]. Notice also that models *p2* and $\alpha_S\beta_S$ for which interactions between residues of

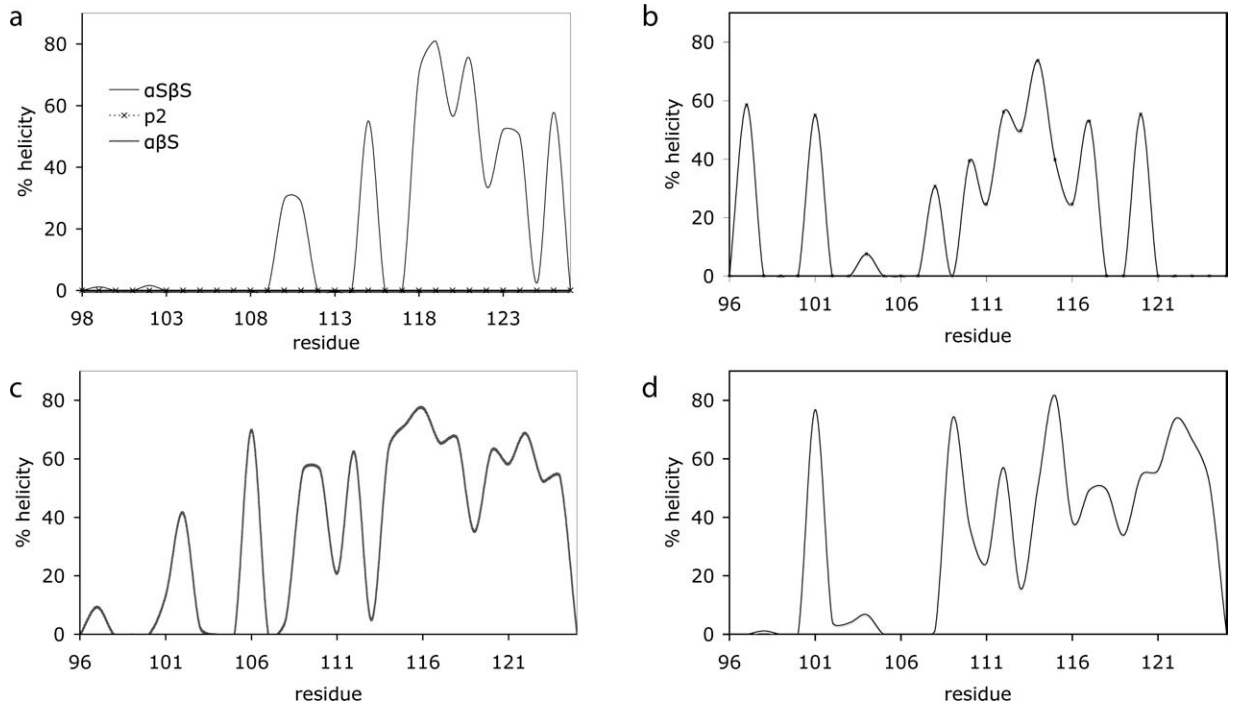

Figure 8
Percentage of total simulation time during which residues are found in a helical form plotted as a function of residue number for (**a**) the T3 loop/H3 helix domain of α-tubulin corresponding to models *p2*, $\alpha\beta_S$, and $\alpha_S\beta_S$; and for the same domain of β-tubulin corresponding to models (**b**) *p2*, (**c**) $\alpha\beta_S$, and (**d**) $\alpha_S\beta_S$.

tail domains and H10/B9 loops are entirely absent show much less fluctuation of residues in the intermediate domain than any of the other three simulations.

Experiments have shown that the polymers assembled from $\alpha_S\beta_S$ tubulin display an increased number of lateral interactions compared to normal MTs, causing junctions to form between polymer walls so that structures resemble hooked MTs with crescent-shaped cross-sections.[6,7] Structural changes in the intermediate domains of normal MTs caused by the disruption of H10/B9 loop-H11 helix interactions by tail domains may possibly lead to changes in positions of key structural elements involved in lateral contacts, which are located in the intermediate domain, such as the M or N loop.

Relevance to drug binding and efficacy

The observations made above, from our simulations, of interactions of the tubulin C-terminal tails with the rest of the protein and our suggestion that changes in the position of tubulin's intermediate domain arising from these interactions may offer an explanation for the role of tail domains in regulating MT assembly, will now be considered in the context of previous work performed by experimentalists.

As has been mentioned in the introduction, experiments have indicated that tail–body interactions within tubulin and tubulin polymers are capable of influencing the binding or efficacy of the tubulin-targeted cancer chemotherapy drugs, colchicine and vinblastine. Colchicine and vinblastine bind to opposite ends of β-tubulin, the former at the intra-dimer interface and the latter on the other end of β-tubulin close to where the N terminus of the α-tubulin tail from the neighboring dimer interfaces with β-tubulin in the context of a protofilament[47] (see Fig. 5). On the other hand, as we will now discuss, colchicine's binding is affected by cleaving the α-tubulin tail,[16] and vinblastine activity by cleaving the β-tubulin tail.[17] In this connection, it is noteworthy to mention that the effect of phomopsin A on the conformational stability of tubulin is also affected by the cleavage of tail domains. Phomopsin A, a non-competitive inhibitor of vinblastine binding[48] is an extremely good inhibitor[49,50] of the spontaneous denaturation[51,52] to which the tubulin dimer is prone. $\alpha_s\beta_s$ tubulin is more resistant to this decay than is undigested αβ tubulin, and the stabilizing effect of phomopsin A is even stronger with $\alpha_s\beta_s$ tubulin than with αβ tubulin.[53]

In experiments performed by Rai and Wolff[17] it was observed that the negative effect that oligo-anions such as sodium triphosphate have on vinblastine-induced polymerization of tubulin is prevented when the C-terminal tail on β-tubulin is removed by subtilisin digestion, although subsequently removing the α-tubulin C-terminal tail to form $\alpha_S\beta_S$ tubulin has no further effect. Note that these polymers are not MTs but are filamentous spirals.[54] Because this negative effect of oligo-anions on vinblastine-induced polymerization of tubulin could be induced in $\alpha\beta_S$ tubulin (recall that $\alpha\beta_S$ tubulin defines dimers lacking only the β-tubulin tail) by increasing the concentration of oligo-anions from 1 m*M* to 10 m*M*, a model was suggested in which there is an interaction favorable to polymer stabilization that is interfered with by both the β-tubulin tail domain and oligo-anions. More specifically, it was proposed that some residues on the tubulin body might interact with a positively-charged residue or region on the tubulin surface, and compete with β-tubulin tail residues for interactions with the same positive cluster.[17] An additive negative effect on stabilization from both this tail–body interaction and the blockage of the positive region by the added anions would then provide an explanation for both how the cleavage of tails allows vinblastine to stabilize tubulin polymers in the presence of a low concentration of anions, and how, when this concentration is increased, vinblastine cannot stabilize even $\alpha\beta_S$ polymers. The interaction between the H10/B9 loop on β tubulin and the positively-charged cluster on H11 of the neighboring α-tubulin, observed in our simulations, along with our observation of interactions between the β-tubulin tail and both of these domains competing with the former interaction, support such a model. From this perspective, the interaction between the H10/B9 loop on β tubulin and positively-charged residues on H11 of the neighboring α tubulin could be strengthened by vinblastine binding, contributing to vinblastine-induced polymer stabilization. On the other hand, the addition of anions would be expected to interfere with this longitudinal interaction. However, without the additive effect of the tail–body interactions, in $\alpha\beta_S$ polymers, vinblastine (albeit while bound to the end of β-tubulin distant from the H10/B9 loop) may be capable of stabilizing either the longitudinal loop–helix interaction and/or the accompanying large-scale effect on the position of the intermediate domain, in the presence of anions. The latter effect can be overridden by adding enough anions.

The effect of vinblastine on tubulin polymerization has been the object of investigations for many years. It turns out that vinblastine binds to MT ends with high affinity but also binds with a much lower affinity to tubulin sites located along the surface of the MT cylinder.[55–57] Approximately, 16–17 high-affinity binding sites located at MT ends have been determined to exist per MT. They are responsible for the well-known kinetic suppression effects of vinblastine, that is, its stabilization of MTs, at low concentrations (sub-micromolar).[58] The binding of vinblastine to the low affinity binding sites on the MT surface is believed to be responsible for the depolymerization of MTs at intermediate ($> 10\ \mu M$) drug concentrations. Finally at drug concentrations of $> 10\ \mu M$, the tubulin polymer stabilized by vinblastine, studied by Rai and Wolff,[17] is an alternate form of the polymer, not the

MT polymer. One possible explanation of the opposite effects of vinblastine on MT polymerization at submicromolar versus intermediate concentrations may be that in the vinblastine-bound polymer, the H11-H10/B9 loop interaction is stronger than normal, which can lead to increased MT stabilization when vinblastine binds to MT tips only; increasing the strengths of these interactions throughout other parts of the MT may lead to destabilization.

In a separate set of experiments, Chakraborty *et al.*[16] found that cleaving the α-tubulin tail domain lowers the activation energy and the on-rate for colchicine binding to the tubulin dimer. In contrast to the case of vinblastine effects on tubulin polymers, the effect is seen in $\alpha_S\beta_S$ tubulin but not $\alpha\beta_S$ tubulin, that is, cleavage of the β-tubulin tail alone has no effect on colchicine binding. Moreover, the binding affinity of colchicine for the tubulin dimer is normally dependent on pH, a factor known to affect tail conformation, but for the $\alpha_S\beta_S$ tubulin dimer, binding is independent of pH. To explain these findings, the authors suggested that although colchicine binds to β-tubulin at a site distant from the α-tubulin tail, tail–body interactions of the α-tubulin tail can cause large-scale conformational changes in the tubulin protein, which both promotes the binding of colchicine and leads to a higher activation energy for binding.[16] In support of this hypothesis it was noted that a 100 m*M* concentration of added NaCl, which would partially screen the electrostatic tail–body interactions, also lowers the activation energy and the on-rate for colchicine binding. It was speculated that the tail–body interaction might involve bending of the α-tubulin tail towards the positively-charged cluster on H11 of α-tubulin, affecting the colchicine binding site and leading to the altered kinetics.[15,16] This was not observed in any of our simulations. However, our MD modeling does support the proposed mechanism by which a tail–body interaction causes a conformational change in α-tubulin by a slightly different explanation, since our results indicate that the α-tubulin tail can interact with the H10/B9 loop, which may enable the intermediate domain to flex out laterally, leading to significant structural changes between normal and $\alpha_S\beta_S$ tubulin. By this line of reasoning, these changes would occur not only on the side of α-tubulin's intermediate domain close to the tail domain, but also extend to the distant side of this intermediate domain, near where colchicine binds to β-tubulin. It is also interesting to speculate whether, conversely, the drug activity of colchicine, which inhibits MT assembly, may involve altering the position of the intermediate domain of α-tubulin.

In summary, the combined results of the experiments from the literature,[16,17] which we have just described, and the simulations reported in the present article seem to indicate that both vinblastine effects on tubulin polymers and colchicine binding to tubulin are affected by shifts in the position of the entire intermediate domain of tubulin compared to that found in $\alpha_S\beta_S$ polymers or tubulin dimer, in the former case that of the β-tubulin monomer and in the latter that of α-tubulin. We propose that this shift is brought on by interactions between the tail domain, the H10/B9 loop, and in the case of vinblastine, the H11 helix of the neighboring monomer, all at the other longitudinal monomer interface from where drug-binding occurs. Experimental confirmation of large-scale conformational flexibility for tubulin's intermediate domain has already been supplied by photo-affinity labeling results demonstrating that cysteines 295, 305, 315, and 316 in the intermediate domain of α tubulin, but distant from the colchicine binding site, are affected by colchicine binding.[59]

CONCLUSIONS

Using computational techniques, we have determined two modes of interaction between the C-terminal tail domains with the main body of tubulin. Specifically, we have found an inter-monomer mode between charged residues on the α(β) tail and β(α)H11 and an intra-monomer mode, pairing aromatic (charged) residues on the α(β) tail with charged (aromatic) residues on the α(β) H10/B9 loop. As one would expect from the heretofore-elusive structures of the C-terminal tails, these interactions are non-specific and transient.

However, we also observe that the binding sites for the tails may bind laterally with each other, that is, the α(β) H10/B9 loop binds to β(α)H11. The result of this competitive interference is to increase flexibility in the intermediate domain with respect to $\alpha_S\beta_S$ tubulin. Such competitive behavior, in which another region of tubulin forms electrostatic interactions with the same positively charged region that the tubulin tail also binds to, has previously been hypothesized based on experimental results implicating tail–body interactions in vinblastine effects on tubulin polymers.[17] Based on the connection that we predict between flexibility of the intermediate domain and tail–body interactions, together with the experimental data showing that the binding or activity of colchicine[16] and vinblastine,[17] respectively, depend on electrostatic interactions of the tubulin tail domains far from the drug binding site, it seems likely that the binding or activity of these two drugs may be linked to the flexibility of tubulin's intermediate domain.

A natural test of the intermediate domain flexibility hypothesis and our proposed intra-monomer binding mode is to mutate residues Glu431 and Glu 432 belonging to the C-terminal tail domain of β-tubulin to alanines and then determine whether the negative effect of oligo-anions on vinblastine-induced polymerization of tubulin is eliminated, just as it is by cleavage of β-tubulin tail domains.[17] The analogous test for the inter-monomer mode is to mutate either of the two residues

Asp α345 on the H10/B9 loop or Tyr α440 on the C-terminal tail, predicted here to form extensive interactions with each other, to alanine, which should lead to pH independent binding of colchicine, in a similar manner that cleavage of the α-tubulin tail domain does.[16]

The C-terminal tails are also involved in the binding of several MAPs essential to the function of MTs. Following Luchko *et al.*,[3] we characterized three structural motifs possibly involved in the CK-2 phosphorylation and MAP2 binding. Falquet *et al.*[60] previously identified a conserved sequence for CK-2, which is found as DATA in the βI-tubulin tail. The DATA motif was found to have a very high similarity among our simulations; at least for the βI tail sequence, this indicates that the structure is persistent, and does not need to be induced for phosphorylation. As this structural motif has now been identified by two independent MD protocols, it should be considered a strong prediction. Contrary to the previous findings, the two proposed MAP2 binding motifs did not maintain similar structures between the models compared. However, since the sequence is experimentally known to be significant[44] and the residues are fully solvated, the possibility that these motifs are induced fit binding sites cannot be ruled out. Rather, further experimental and computational studies are needed to address this question. Some of this type of experimental study has been carried out on α-tubulin tails already, but we have not investigated proposed binding sites from this work.[61]

Beyond the scope of this study are the wide variety of post-translational modifications, particularly effecting the C-terminal tail domain, that tubulin is subject to. These include tyrosinolation/detyrosinolation of α, deglutamylation of α, polyglutamylation of both α and β, polyglycylation of both α and β, and phosphorylation of certain isotypes of β.[1] All of these could well affect the conformation of the tail domains and several also would alter the charges. These modifications may play critical roles in regulating the functions of MTs; for example, detyrosination of α promotes MT stability,[62] while polyglycylation of β is necessary for normal axoneme formation and polyglutamylation of α and/or β appears to mediate binding of MT-associated proteins (MAPs) and the motor protein kinesin to tubulin.[14,63–67] Some of these effects are likely to be indirect, that is, the presence of the modification in the C-terminal could conceivably act as a signal to some other protein or proteins which in turn could mediate the effect, that is, MT stability, axoneme formation, and so forth. Regardless, these modifications are of such scale that it is easy to hypothesize that they cause major conformational changes in the C-termini and that these changes in turn may mediate the ultimate effects of these modifications. It is likely that there exists a complex feedback loop between the probability of modification occurring and the conformational effect of modification. In addition, the binding of the C-termini to MAPs and motors is critical to the function of a variety of MT populations and is likely to also be regulated by conformational interactions involving the C-termini. These conformational effects and interactions and their likely functional outcomes are currently being examined.

Finally, we wish to comment on the lack of helicity of the T3/H3 region that was noted in two of the simulations. The issue of how MTs are destabilized by GTP hydrolysis is currently highly debated.[68–72] Along with MTs, actin filaments comprise one of three major components of the cytoskeleton and these two types of protein filaments have many remarkably similar properties including the utilization of a hydrolyzable nucleotide in polymer dynamics.[73] Experimental and computational studies indicate that in the case of actin polymerization, a loop-to-helix transition of the DNase-I binding (DB) loop is associated with hydrolysis of the nucleotide ATP to ADP and with the simultaneous structural change to a less stable actin polymer, as lateral contacts are lost in the less flexible helical structure.[74,75] Thus, the variability in the helicity of the T3/H3 region seen in our simulations together with the fact that the T3 loop of β-tubulin has been observed to be destabilized upon hydrolysis of the bound GTP to GDP,[68] raise the possibility that upon hydrolysis of the GTP bound to β-tubulin, the T3/H3 region might undergo a loop-to-helix transition similar to that in actin, causing a large-scale shift in the entire MT structure. Experimental testing of this hypothesis could be performed by applying techniques which have previously been used to analyze small changes in protein helical structure, such as Trp fluorescence,[76] FRET (Forster resonance energy transfer) emission spectra,[77,78] or EPR (electron paramagnetic resonance) data.[78] In this way, structural comparisons could be made of the T3/H3 region in MTs bound by different nucleotides or nucleotide-analogues.

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PROTEINS
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Molecular dynamics modeling of tubulin C-terminal tail interactions with the microtubule surface

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ABSTRACT

Tubulin, an α/β heterodimer, has had most of its 3D structure analyzed; however, the carboxy (C)-termini remain elusive. Importantly, the C-termini play critical roles in regulating microtubule structure and function. They are sites of most of the post-translational modifications of tubulin and interaction sites with molecular motors and microtubule-associated proteins. Simulated annealing was used in our molecular dynamics modeling to predict the interactions of the C-terminal tails with the tubulin dimer. We examined differences in their flexibility, interactions with the body of tubulin, and the existence of structural motifs. We found that the α-tubulin tail interacts with the H11 helix of β-tubulin, and the β-tubulin tail interacts with the H11 helix of α-tubulin. Tail domains and H10/B9 loops interact with each other and compete for interactions with positively-charged residues of the H11 helix on the neighboring monomer. In a simulation in which α-tubulin's H10/B9 loop switches on sub-nanosecond intervals between interactions with the C-terminal tail of α-tubulin and the H11 helix of β-tubulin, the intermediate domain of α-tubulin showed more fluctuations compared to those in the other simulations, indicating that tail domains may cause shifts in the position of this domain. This suggests that C-termini may affect the conformation of the tubulin dimer which may explain their essential function in microtubule formation and effects on ligand binding to microtubules. Our modeling also provides evidence for a disordered-helical/helical double-state system of the T3/H3 region of the microtubule, which could be linked to depolymerization following GTP hydrolysis.

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Key words: protein structure; theory; modeling; computer simulation; conformational changes; dynamics of biomolecules; cytoskeleton; polymerization.

INTRODUCTION

The tubulin protein polymerizes to form microtubules (MTs) which are cylindrical structures consisting typically of 13 protofilaments, each of which is comprised of repeats of α–β tubulin hetero-dimers. The carboxy (C)-terminal tail domains of all tubulin monomers are found on the outer surface of the MT. Both α- and β-tubulin consist of numerous isotypes, differing in amino acid sequence and cellular and tissue distribution.[1] The lengths and sequences of the C-terminal tails vary greatly depending on isotype; human isotypes typically consist of 10–13 residues on α-tubulin, and 18–24 residues on β-tubulin, and have a high proportion of negatively charged residues.[2,3] A great deal of evidence indicates that the C-terminal tails of both α- and β-tubulin play critical roles in regulating MT assembly and function, and in determining the conformation of the tubulin molecule,[4–8] roles whose details are very poorly known. C-terminal tail residues are necessary for the assembly of normal MTs; one recent cloning study has demonstrated that the first two tail residues of β-tubulin are indispensable for the formation of MTs in HeLa cells.[4] Tubulin polymers assembled *in vitro* from tubulin lacking both C-terminal tails (denoted as $\alpha_S\beta_S$ tubulin) display an altered morphology characterized by bundles of twisted filaments, sheets,[5] and hooks.[6,7] Moreover, the presence of alkali metal cations[8] or cleavage of tails[5] allows polymers to form at lower tubulin concentrations, and the reduction of negative charges on C-termini by covalent modification lowers the rate constant for MT disassembly.[9] Tubulin polymers lacking only the β-tubulin tail ($\alpha\beta_S$ tubulin) display a ring con-

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formation, do not require GTP for polymerization, and are temperature insensitive.[10]

It is likely that the effects of the C-termini on MT assembly and tubulin conformation are critically dependent on the conformational interaction between the C-termini and other parts of the tubulin molecule. Nevertheless, the natures of these interactions are almost entirely unknown. Although a series of studies have determined the structure of the rest of the tubulin molecule, the structures of the C-termini have remained largely invisible to crystallographic methods. Thus, a molecular-level picture of the structures of tubulin protofilaments obtained by molecular dynamics (MD) simulations, including both tail dynamics and their interactions with the remainder of the protein, offers an opportunity to provide unique insights into a better comprehension of the biological functions of tail domains and their significance for MTs.

Depending on the isotype, the charge of the C-terminal tail of human β-tubulin is generally between −10 and −12 *e* and that of the α-tubulin tail is −7 or −8 *e*, not including post-translational modifications.[2,3] Although the remainder of the tubulin dimer is also negatively charged overall (the total charge of the dimer is approximately −50 *e*), there are also positively charged regions on the protein surface. By using reductive methylation, residue Lys 394, which together with Arg 390 and His 393 forms a positively-charged pocket on the surface of α-tubulin, has been shown to be essential to MT polymerization.[11,12] Because Lys 394 is less accessible to methylation in the MT polymer compared to when present in the dimer,[12,13] it has been suggested that this residue is involved in electrostatic interactions with the negatively-charged acidic residues in the tail regions.[14] The existence of such a "tail–body" electrostatic interaction was also supported by a study[15] in which the detached α-tubulin tail was computationally docked to the tubulin surface and found to bind at a site near Lys 394. Experimental findings that C-terminal tail cleavage can affect ligands that bind to tubulin, and that this influence seems to be due to electrostatic interactions, also indicate that tail–body interactions contribute to tubulin conformation. This evidence includes a study showing that cleaving the C-terminal tail on α-tubulin lowers the activation energy for colchicine binding to tubulin, and leads to pH independent binding.[16] Another such study showed that the influence of oligoanions on the stabilization of tubulin polymers by vinblastine is eliminated by the cleavage of the C-terminal tails of β-tubulin.[17]

The objective of this article is to use MD modeling in order to elucidate the issues surrounding the role of C-termini in structural and functional effects on the tubulin dimer and, consequently, in the biological behavior of MTs. The specific modeling tool that we apply to our analysis is simulated annealing in conjunction with MD simulation. Simulated annealing is an enhanced sampling technique in which heating followed by re-cooling of a molecular system can be used to determine its equilibrium structure. Heating allows the system to overcome energy barriers that would otherwise prevent the system from fully sampling configuration space, and slow cooling causes the system to freeze into a minimum energy structure. This method has frequently been applied within Monte Carlo simulations to determine conformational free energy minima[18,19]; more recently simulated annealing has been used within MD simulations,[20] as it is in the present work. Here, this technique is applied to obtain positions of C-terminal tails for the tubulin dimer. Subsequently, simulations are run of protofilaments formed from αβ tubulin, $\alpha\beta_s$ tubulin, and from $\alpha_s\beta_s$ tubulin. In this way, the effects of the tails on protofilament dynamics are investigated computationally.

We analyze MD trajectories to investigate the occurrence of tail–body interactions of the C-terminal tail regions with positively-charged residues on the tubulin surface. Structural motifs intrinsic to the C-terminal tail domains seen in the MD trajectories are also considered. Finally, we analyze the root mean square fluctuations of residues in trajectories with and without tail domains. We show that interactions of the tail domains with exposed residues on the surface of the remainder of the protein may play an important role in MT dynamics by influencing the large-scale conformation of tubulin's intermediate domain.

MATERIAL AND METHODS

Simulated annealing

After heating the tubulin dimer system to 5000 K, each of 12 models was slowly cooled to 300 K over 5.2 ns of simulation time. Out of these 12, three models were selected as starting positions in simulations of two explicitly-solvated tubulin protofilaments. The flowchart in Figure 1 summarizes the simulated annealing/MD simulation procedure adopted in our study to determine tubulin tail conformations, which we now describe in detail.

Our system consisted of a tubulin dimer, composed of human isotypes αIV and βI, modeled as described elsewhere[21]; these have C-terminal tails with lengths of 10 and 18 residues (residues α439–448 and β427–444), and charges of −8 and −12 *e*, respectively. Initial set-up was performed in the *leap* and *antechamber* modules of the AMBER9[22] program package using the AMBER03 force field to parameterize the tubulin protein.[23,24] MD was performed using the sander module of AMBER9.[22] Twelve high temperature implicitly-solvated MT dimers were generated by heating the dimer to 5000 K (chosen to make the tails mobile on a reasonable time scale), and then extracting structures from the simulation every

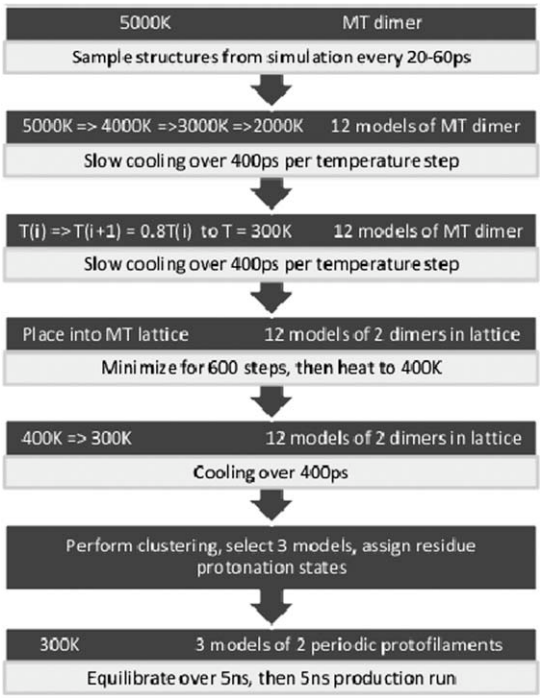

Figure 1

Flowchart illustrating the simulated annealing/MD simulation procedure used in this work to determine tubulin tail conformations.

20–60 ps. These were then slowly cooled to a target temperature of 300 K. Our annealing schedule used temperature reduction steps of 1000 K over 200 ps time periods, with equilibrations being performed at each of these temperatures for a period of another 200 ps, until the temperature was below 2000 K, and then used a 0.8-power law cooling-schedule again with 400 ps spent for each temperature stage. A time step of 1 fs was used throughout all simulations. The modified generalized Born method GB^{OBC} corresponding to model II in Ref. [25] (igb = 5) was employed to represent a 150 m*M* ionic solution and the temperature was maintained by a Langevin thermostat with a collision frequency of 2 ps^{-1}. All bonds involving hydrogens were constrained using the Shake[26] algorithm. A restraint weight of 15 kcal/mol was placed on residues 1–438 of α tubulin and 1–428 of β tubulin. Removal of center-of-mass translational and rotational motion was disabled. The non-bonded cut-off used was 50 Å.

Structures were fit to the original homology model by running simulations for 5000 steps with a restraint of 50 kcal/mol on the α-carbons of all residues not belonging to the C-terminal tail domains, forcing the α-carbons belonging to the main body of the dimer to assume a structure similar to the one determined by electron crystallography experiments. In some cases, the *autosculpt* feature of the Pymol program[27] was employed to slightly adjust the positions of β tubulin tail residues which overlapped with parts of the opposite β tubulin. Tubulin dimers were then placed into a configuration consistent with MT geometry by using the model of Li *et al.*[28] Two adjacent dimers were created as would exist in the MT lattice by using an in-house VMD[29] script, as described elsewhere.[30] These were then relaxed for 300 steps without solvent, and another 300 steps with implicit solvent. Afterwards the structures were heated to 400 K, and then cooled to 300 K over a period of 400 ps for further energy minimization, with the same restraint on the α-carbons belonging to the main body.

The 12 models of the tubulin dimer resulting from our simulated annealing procedure are shown in Figure 2. Structures were clustered in the *ptraj* module of AMBER using the means algorithm[31] to produce three clusters each for α- and β tubulin tails, by comparing positions of alpha-carbons based on root-mean-squared distance (RMSD) after fitting all tubulin dimers, excluding tail residues, to the position of one of the dimers. Clusters of α tubulin tail peptides contained five, four, and three elements, respectively, with mass weighted root-mean-square distances to the centroid of 0.6 Å for each of the three clusters. Clustering the β tubulin tail peptides resulted in one group of six elements, with a mass weighted root-mean-square distance to centroid of 1.4 Å, a second cluster with five elements and a mass weighted root-mean-square distance to centroid of 1.3 Å, and a third cluster with a single element only. Cluster

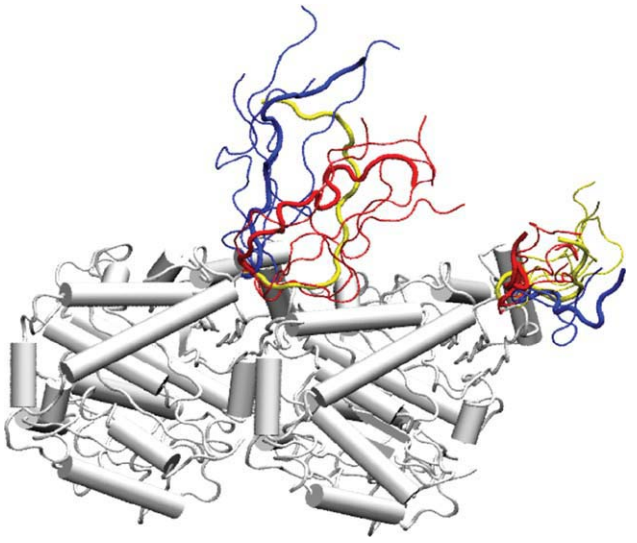

Figure 2

Tubulin dimer, showing three clusters obtained by simulated annealing MD for the C-terminal tail domains, colored in yellow, red, and blue, respectively. Cluster representatives *p1* (yellow), *p2* (red), and *p3* (blue) are depicted as thicker coils compared to other cluster elements. The dimer is viewed from above if positioned in a MT lattice with β-tubulin on the left and α-tubulin on the right.

representatives were chosen as the dimers having β tubulin tails closest to the cluster centroids. Because each of these three dimers fell into a different one of the three clusters corresponding to the α tubulin tails, we chose to retain the original pairing between α- and β tubulin monomers, rather than pairing cluster representatives for β tubulin tails with similarly chosen cluster representatives for α tubulin tails. As a further check on the appropriate number of representatives, hierarchical clustering of both β- and α-tubulin tails was performed using the MMTSB Toolset[32] command *jclust* with the *rmsd* mode, requesting a maximum of 50 structures, which gave two main clusters for the α-tubulin tail and three for the β-tubulin tail.

The three representative dimers, which were used to perform MD simulations, are shown in Figure 2. In the text that follows, we refer to these three models as *p1*, *p2*, and *p3*. In addition, we performed simulations with one or both tail domains removed. The α-tubulin tail position for the $\alpha\beta_S$ model was taken from model *p3*.

MD simulations

The *propka* program[33,34] was used to assign protonation states to Asp, Glu, His, and Lys residues on each of the three models; as a result His 309 was protonated on α-tubulin of model *p1*, His 307 on β-tubulin of models *p2* and *p3*, and α-tubulin residue Asp 46 was protonated on all three models.

The tubulin dimers were then placed into a MT lattice. The positions of GDP at the exchangeable site and GTP at the non-exchangeable site were taken from the PDB file 1TVK[35] and those of the Mg^{2+} ions from the PDB file 1JFF.[36] Parameterization of nucleotide co-factors was taken from reference.[37] The MD simulations included two adjacent protofilaments composed of two tubulin heterodimers; identical tail positions were used on both protofilaments. One hundred three Na ions were added to neutralize the system electrostatically, and following this 80 each of Na^+ and Cl^- were added to bring the ionic concentration to 100 m*M*. The system was placed into a periodic box with dimensions 81.2 Å by 107.5 Å by 152.5 Å, which was filled with pre-equilibrated explicit TIP3P water molecules[38] using the *leap* module of AMBER9. The protofilaments were aligned with the *x*-axis of the periodic box, and by setting the *x*-dimension of the box equal to 81.2 Å, or the length of one tubulin dimer, the periodic boundary conditions were taken advantage of to model the system as two periodic protofilaments. Structures were then converted from AMBER to Gromacs-style coordinate and topology formats using the *amb2gmx* perl script,[39] and simulations were performed with the Gromacs v3.4[40,41] program.

Structures were energy minimized, which also relieved any steric clashes with waters. The entire system was relaxed using 5000 steps of steepest descent (1000 steps), followed by conjugate gradient minimization. Following equilibration for 5 ns with a time step of 2 fs, MD simulation was performed over a period of another 5 ns with the same time step, with periodic boundary conditions at a temperature of 300 K and at constant atmospheric pressure. The Shake algorithm[26] was used to constrain bonds involving hydrogen in all simulations and we used a non-bonded cut-off of 12 Å. PME was employed to calculate long-range electrostatics.

Analysis

Analysis was then performed on the 5 ns production MD runs by using the suite of tools available in the Gromacs program package.[40,41] Only tail domains belonging to the protofilament positioned on the left when viewed from the minus to the plus end was used for analysis. The restriction of our analysis to these tail domains was made because of their physically central position relative to our models, in contrast to the peripheral position of the other set of tail domains, as seen in Figure 5.

For the examination of structural motifs, a matrix of scaled rho (rhosc) values was computed using the Gromacs[40,41] tool *g_rms*. This program compares each structure in trajectory *p1* to each structure in trajectory *p3*, after fitting the structures on top of one another by rotation and translation.[42] The structural similarity value rho is defined as a function of the two structures A and B by:

$$\rho(\mathrm{A},\ \mathrm{B}) = \frac{2\mathrm{RMSD}(\mathrm{A},\ \mathrm{B})}{\sqrt{R_{\mathrm{gyr}}^2(\mathrm{A})R_{\mathrm{gyr}}^2(\mathrm{B}) - \mathrm{RMSD}^2(\mathrm{A},\ \mathrm{B})}},$$

where R_{gyr} is the radius of gyration of a structure, and where RMSD gives the root mean scale deviation between two structures. Scaling the aligned principal moments of inertia of the two structures A and B to values that are all equal and satisfy $R_{\mathrm{gyr}} = 1$ prior to comparing structures, leads to a more universal, size-independent measure of similarity, rhosc, assuming values between 0 and 2. Values of rhosc less than 0.3 were used here as an indication of structural similarity, and the percentage of comparisons for which rhosc < 0.3 represents the percentage of total time for which the conformation of the motif is similar between the two models A and B.

The *g_min_dist* tool was employed to determine minimum distances between charged or aromatic residues belong to tubulin tail domains and other charged or aromatic residues. Residue pairs were included in this analysis if they came closer than 0.3 nm at any point during the simulations.

Average root-mean-square fluctuations (RMSF) were calculated using the *g_rmsf* program, as the averages over the simulation time, $\langle |\Delta R_i|^2 \rangle$, in the square fluctuation of

Table I
Charged and Aromatic Residues Involved in Contacts Between α-Tubulin's C-Terminal Tail Domain, α-Tubulin's H10/B9 Loop, and β-Tubulin's H11 Helix

	αtail-βH11	βH11-αH10/B9loop	αtail-αH10/B9loop
p1	αTyr440,Glu441-βArg390 αAsp442,Glu443-βArg391	βArg390- αAsp345	αTyr440- αAsp345
p2	αTyr440-βArg390 αTyr440-βArg391	βArg391- αTrp346	
p3	αTyr440-βArg390	βArg390- αAsp345 βArg391- αTrp346	αTyr440- αAsp345
$\alpha\beta_S$	αGlu445,Glu447,Glu448-βLys379 αAsp442-βArg390		αTyr440- αAsp345
$\alpha_S\beta_S$		βArg390,Arg391- αTrp346	

the position $\boldsymbol{R}_i$ of the α-carbon corresponding to each residue *i*.

Two programs *do_dssp* and *g_helix* were used in the analysis of helical content. The *do_dssp* program assigns secondary structure by identifying patterns of hydrogen bonding using the DSSP[43] algorithm. On the other hand, *g_helix* determines helicity based upon both hydrogen bonds and ϕ/ψ angles.

RESULTS AND DISCUSSION

Simulated annealing

Following annealing, α-tubulin tails were generally found to point in the longitudinal direction toward where the β-tubulin of the neighboring dimer would be in a MT lattice (see Fig. 2). There was a tendency of the β-tubulin tails to point laterally in the direction of where the right-hand adjacent protofilament would be if the dimer were part of a MT being viewed from the minus to the plus end (i.e., towards the intermediate, taxol-binding domain). However, the β-tubulin tails covered a large range of conformational space (Fig. 2).

By clustering the final endpoints of simulated annealing, and choosing representatives of different clusters (*p1*, *p2*, and *p3*) for subsequent simulation and analysis, we ensured that these three simulations were run from well-separated initial positions. Thus, while each of these simulations on its own does not demonstrate complete time-sampling of MD, a broader picture of the dynamic behavior of the protein system was obtained by considering all three simulations.

Structural motifs

A previous set of replica exchange MD simulations studied peptides composed of the residues forming the C-terminal tails of β-tubulin and identified three transient structural motifs as potential sites for casein kinase-2 (CK-2) phosphorylation and MAP2 binding.[3] We sought to determine, by analysis of our trajectories, whether these structural motifs found for the isolated tail structures, still characterize the tail domains within the context of a MT.

The proposed CK-2 motif, consisting of the residues DATA, occurs at the start of the β tubulin tail and is evolutionarily conserved between tubulin isotypes.[3] The structure of these four residues beginning with residue Asp 427 may be seen in Figure 3. The percentage of total comparisons for which rhosc values are less than 0.3 was found to be 93.6%, indicating a high degree of structural similarity for this sequence and is higher than that found in the original study.

Two motifs on the C-terminal tail, that were previously identified as making up the MAP2 binding site,[44] consist of sequences EEAE (residues 438–441) and EEEA (residues 441–444), respectively, contained in the last seven residues (EEAEEEA) of the β-tubulin tail domain.[3] The respective counts of rhosc values less than 0.3 for these two sequences are 14.7% and 2.8% of total comparisons. This is considerably less than in the previous study and our analysis suggests that these two regions take on little specific structure; however, the bound β-tubulin tail conformers identified below indicate that these residues remain solvent exposed and the possibility of a MAP2 induced structure cannot be ruled out.

Tail–body interactions

We performed analysis on our MD trajectories to investigate the occurrence of tail–body interactions of the C-terminal tail regions with positively-charged residues on the tubulin surface. Tables I and II show the charged residues that, at any point during our simulations, are involved in close contacts of oppositely-charged side-chains, with inter-residue distances of less than 0.3 nm, for each model. Because there were some very noticeable interactions between aromatic residues from the tails and charged residues from the tubulin body and vice versa, these were also included in our analysis. Figures 3 and 4 display models highlighting the tubulin residues involved in tail–body interactions of the β- and α-tubulin tails, respectively, for any one of the four models *p1*, *p2*, *p3*, and $\alpha\beta_S$.

Table II
Charged and Aromatic Residues Involved in Contacts Between β-Tubulin's C-Terminal Tail Domain, β-Tubulin's H10/B9 Loop, and α-Tubulin's H11 Helix

	βtail-αH11	αH11-βH10/B9loop	βtail-βH10/B9loop
p1	βGlu431,Glu432- αHis393	αLys394- βGlu343	βGlu431- βTrp344
	β Glu432- αLys394	αHis393, Lys394- βTrp344	
p2	βGlu432- αLys401		
p3		α Lys401- βTrp344	βAsp427- βTrp344
$\alpha\beta_S$		αArg390,His393,Lys394- βGlu343	
		α Lys401- βTrp344	
$\alpha_S\beta_S$		α Lys401- βTrp344	

We observed that two kinds of tail–body interactions could take place. First, charged Glu and Asp residues of the C-terminal tails can interact with a cluster of positively-charged residues on H11 of the neighboring tubulin dimer. That is, the α-tubulin tail interacts with the H11 helix of β-tubulin, and the β-tubulin tail interacts with the H11 helix of α-tubulin. Figure 5 shows the positions of the H11 domain within the MT lattice, as well as the positions of the other domains that will be discussed below. At the inter-dimer longitudinal interface, negatively-charged residues of the α-tubulin tail interact with residues Lys 379, Arg 390, and Arg 391 on the H11 helix of β-tubulin. Likewise, at the intra-dimer interface, the β-tubulin tail interacts with the positively-charged cluster on α-tubulin containing the residue Lys 394, which has previously been identified as essential for MT polymerization, and His 393; Lys 401 on H11 can also be involved in this type of interaction. The charged tail residues involved are 441–443, 445, and 447–448 on α-tubulin and 431–432 on β-tubulin. We also noted that the ar-

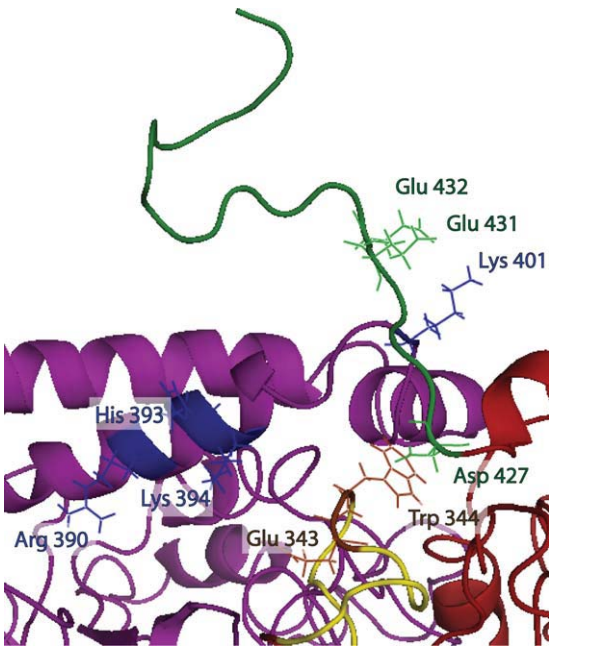

Figure 3
Illustration of the intra-dimer longitudinal interface within a MT between α-tubulin (purple) and β-tubulin (red). Charged and aromatic residues have been labeled and represented in stick mode that are involved in close contacts between the β-tubulin tail, shown in green cartoon, and either the β-tubulin H10/B9 loop, shown in yellow cartoon, or the α-tubulin H11 helix, as well as those involved in interactions between the latter two regions. This figure was created using the Pymol molecular viewing program,[27] using model *p2*.

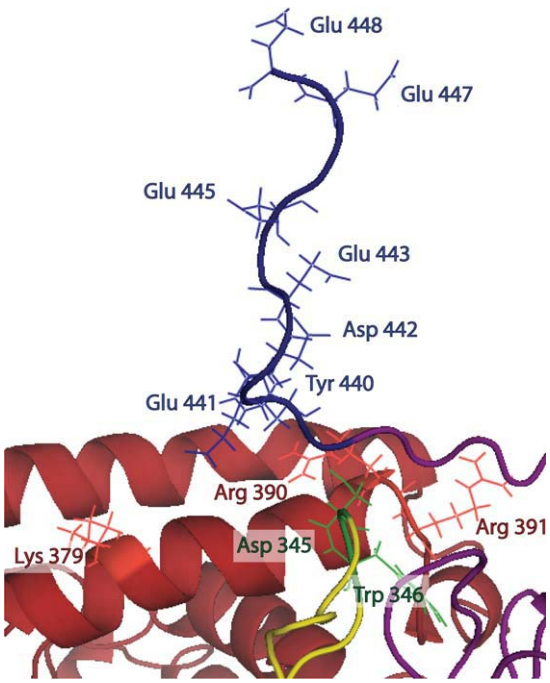

Figure 4
Illustration of the inter-dimer longitudinal interface within a MT between α-tubulin (purple) and β-tubulin (red). Charged and aromatic residues have been labeled and represented in stick mode that are involved in close contacts between the α-tubulin tail, shown in blue cartoon, and either the α-tubulin H10/B9 loop, shown in yellow cartoon, or the β-tubulin H11 helix, as well as those involved in interactions between the latter two regions. This figure was created using the Pymol molecular viewing program,[27] using model *p3*.

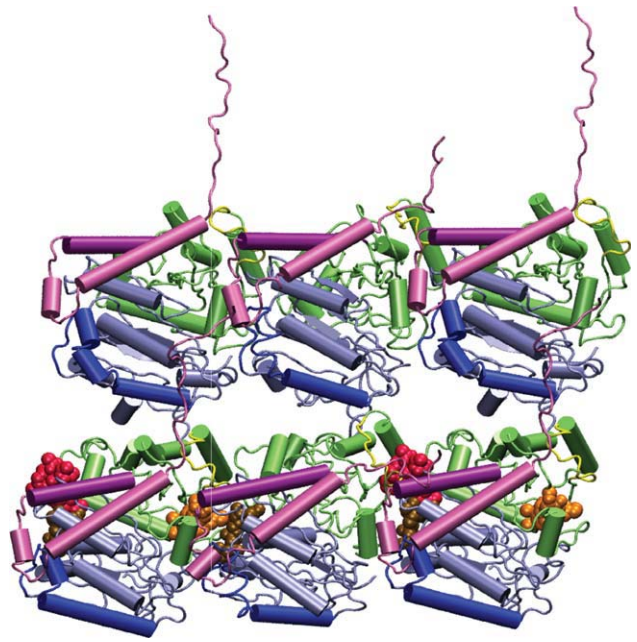

Figure 5

Representation of part of the MT lattice, showing β-α-β tubulin monomers comprising each of two protofilaments oriented with plus ends on the left-hand side. N-terminal domains are colored in blue, intermediate domains in green, and C-terminal domains in purple. The T3/H3 region is shaded in dark blue, the H10/B9 loop in yellow, and the H11 helix in dark purple. The positions of the binding sites of colchicine (orange), vinblastine (red), and nucleotides GDP and GTP (brown) are also illustrated for one of the protofilaments.

omatic residue Tyr 440 on the α-tubulin tail interacts with charged residues Arg 390 and Arg 391 on the H11 helix of β-tubulin by nonpolar side-chain stacking, as well as by interactions between charged Arg side-chains and the polar hydroxyl substituent on the Tyr ring.

The second kind of tail–body interaction observed was between tail residues and residues on the H10/B9 loop of the same tubulin monomer. These contacts are not between two charged residues but rather involve an aromatic residue and a charged residue. In the case of α-tubulin, the tail residue Tyr 440 interacts with Asp 345 of the H10/B9 loop; this contact involves both hydrophobic interactions between the two bulky side chains, and electrostatic attraction between the charged carboxylic acid group of the Asp side-chain and the polar hydroxyl substituent on the Tyr ring. In the case of β-tubulin, tail residues Asp 427 or Glu 431 engage in close nonpolar contacts with Trp 344 of the H10/B9 loop.

Interestingly, the H10/B9 loop can also interact with the positively-charged cluster on the H11 helix of the neighboring dimer, and these interactions are also shown in Tables I and II. In the case of the α H10/B9 loop and the β H11 helix, there can be a salt bridge established between αAsp 345 and residue Arg 390 of H11; also the bulky aromatic side chain of αTrp 346 engages in non-polar interactions with the long side chains of Arg 390 or Arg 391 from β H11. In the case of interaction between β-tubulin's H10/B9 loop and α-tubulin's H11 residues, Glu 343 of the β-tubulin loop can interact with Arg 390, His 393, and/or Lys 394 on α-tubulin's H11. Here also β Trp 344 interacts with residues His 393, Lys 394, and/or Lys 401 from the H11 helix of α-tubulin through both non-polar side-chain interactions and contacts between the aromatic rings of Trp and the positively-charged amine group on Lys. Note that it is likely that these (and possibly some of the other instances of interactions between positively-charged and aromatic residues discussed above) are cation–pi interactions, although our simulations did not employ a polarizable force-field capable of representing the electronic nature of these interactions.

For model *p1*, we plot in Figure 6 how minimum distances involving tail–body interactions of the α-tubulin tail vary throughout the simulation. During the first half of this simulation there is a strong interaction between β-tubulin's helix H11 and the H10/B9 loop of α-tubulin, at the longitudinal inter-dimer interface. During the second half of the simulation these interactions are reduced and instead increased interactions are observed between the α-tubulin tail and both β-tubulin's helix H11 and the H10/B9 loop of α-tubulin. This switch in interactions between the C-terminal tail, the H10/B9 loop, and the H11 helix of the neighboring dimer may be indicative of competing interactions between these three regions.

At the intra-dimer interface, as seen in Supporting Information Figure S3, for the *p1* model there are multiple close contacts between the β-tubulin tail, α-tubulin's helix H11, and the H10/B9 loop of α-tubulin. Although, as may be seen from Tables I and II, many residues or pairs of residues involved in tail–body interactions and in H11-H10/B9loop interactions are shared between models *p1*, *p2*, and *p3*, these models explore different regions in conformational space and thus their trajectories display qualitatively different behaviors. For example, in model *p2*, at the intra-dimer interface, the β-tubulin tail interacts with residue Lys 401 of the H11 helix on α-tubulin, but there is no interaction between αH11 and the H10/B9 loop of β-tubulin. On the other hand, in both the *p3* and $\alpha\beta_S$ simulations, αH11 interacts with β-tubulin's H10/B9 loop but not with the β-tubulin tail. In the $\alpha_S\beta_S$ simulation, without either tail domain present, almost continuous interaction was observed between the H10/B9 loop and the H11 helix of the adjacent monomer, at both monomer interfaces (see Supporting Information Fig. S2).

A physical model was previously created of ion dynamics in and around MTs, based on the application of a random alternating electric force on the surrounding cations by fluctuations in C-terminal tail positions.[45] The continuous breaking and formation of salt bridges give rise to an alternating voltage source on ions as the

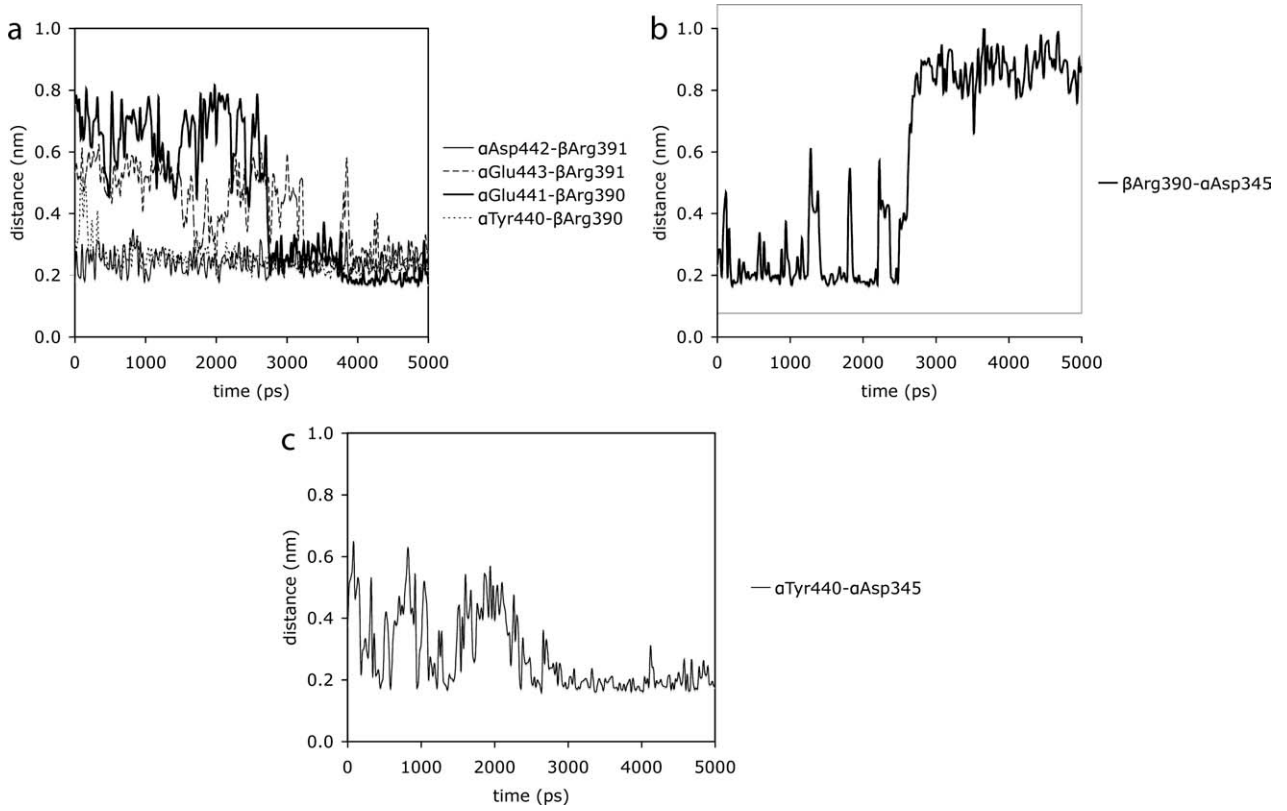

Figure 6

Minimum distances (nm), plotted for the *p1* model as a function of simulation time (ps), between charged or aromatic residues belonging to (**a**) α-tubulin's C-terminal tail domain and β-tubulin's H11 helix; (**b**) β-tubulin's H11 helix and α-tubulin's H10/B9 loop; and (**c**) α-tubulin's C-terminal tail domain and α-tubulin's H10/B9 loop.

tails dynamically oscillate between conformations where they either interact with the tubulin body or extend into more solvated conformations, as seen in Figure 6. This effect observed in our simulations appears to support the hypothesis that local charge reconfiguration due to the motion of C-termini tails can affect ionic flows and hence endow MTs with unique electro-conformational properties.

Structure and flexibility of tubulin body

The average RMSF of each residue of α-tubulin (excluding tail-residues) is plotted as a function of residue index *i* in Figure 7, and that of β-tubulin in Supporting Information Figure S6, corresponding to each of our simulations. The peaks in the RMSF plots show regions of high flexibility, and the highest peaks generally occur at solvent-accessible loops. Tubulin is known to have three main domains, which have been shown to be associated with major correlations in structural fluctuations.[46] These are the N-terminal nucleotide-binding domain, the intermediate taxol-binding domain, and the C-terminal domain, and consist of residues 1–205, 206–381, and 382–448, respectively, on α-tubulin and residues 1–203, 204–371, and 372–444, respectively, on β-tubulin. The last few residues of C-terminal domains show a high degree of flexibility in all models; the loop connecting helices H11 and H12, and H11 itself, may also be seen from the RMSF plots to be quite flexible for the *p3* and $\alpha_S\beta_S$ models. We will now briefly examine the main characteristics of RMSF plots for α- and β-tubulin that we observed first for the N-terminal domain, and then for the intermediate domain; we will also discuss some notable differences between models in RMS fluctuations and their significance.

N-terminal domain (residues α1-205, β1-203)

In the N-terminal domain, peaks are seen in the RMSF plots (Fig. 7 and Supporting Information Fig. S6) at residues α43–63, β45–47 in the region of the H1/B2 loop, which forms lateral contacts with neighboring monomers; at residues α69–74, β66–71 corresponding to the T2 loop; in the region of the H3 helix and T3 loop at residues α95–119, β91–120; and at residues α141–146 in the region of the glycine-rich signature loop, T4. In

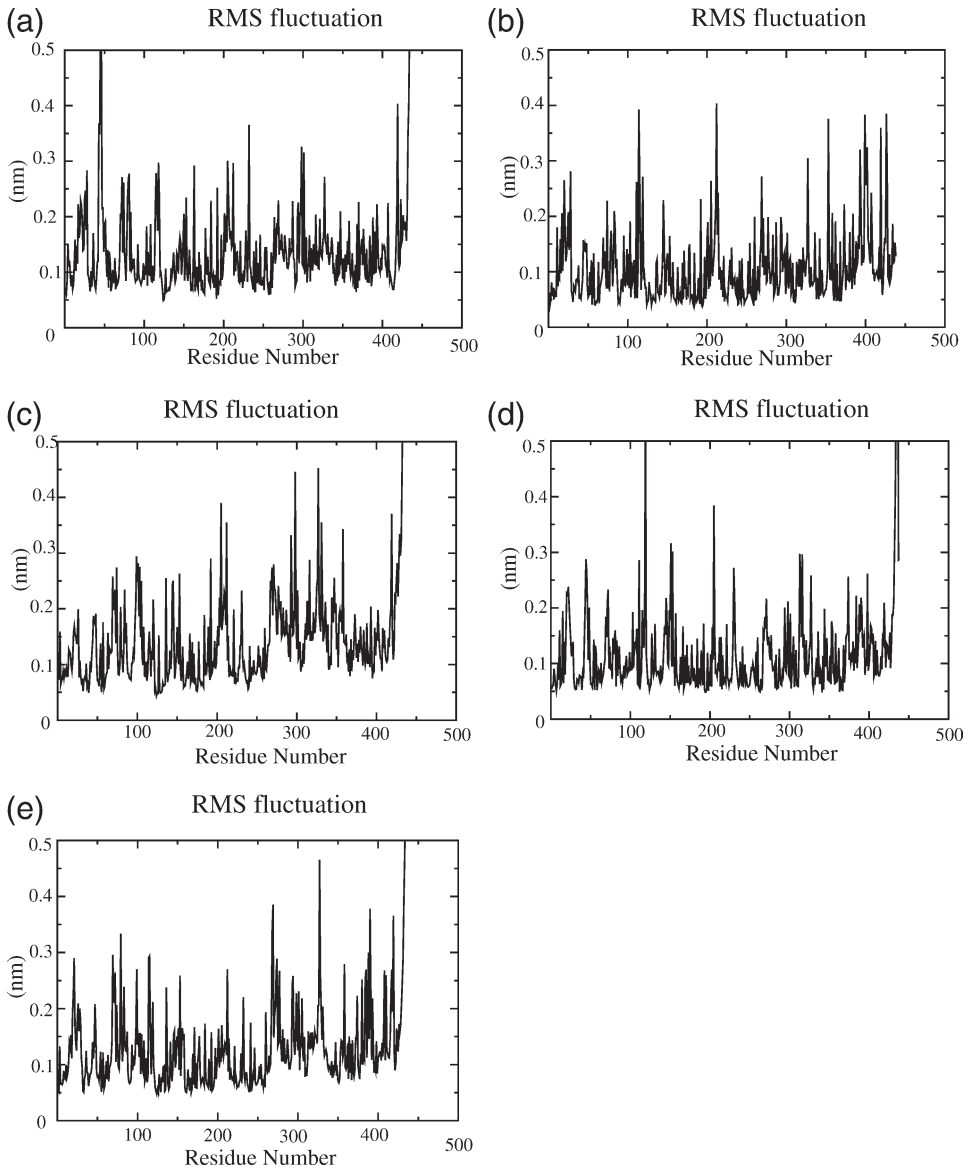

Figure 7

Time-averaged RMSF (nm) of each residue of α-tubulin plotted as a function of residue number for model (**a**) $\alpha\beta_S$, (**b**) $\alpha_S\beta_S$, (**c**) *p1*, (**d**) *p2*, and (**e**) *p3*.

Figure 7 and Supporting Information Figure S6, it may be noted that the peaks corresponding to the T3/H3 region on both α- and β-tubulin are much more pronounced in the RMSF plots for model *p2* than in the other plots. The T3/H3 region (residues α97–128 and β95–126) is located close to the nucleotide binding site, the exchangeable one in the case of β-tubulin, and the non-exchangeable one for α-tubulin. When secondary structure of the N-terminal domain is compared between the simulation trajectories (see Table III and Supporting Information Table S1), it is observed that there is a loss of α-helical structure for the *p2* trajectory compared to for the other trajectories. Upon examination, one noticeable instance of this is a less helical appearance of the H3 helix.

We, therefore, analyzed local secondary structure of the T3 loop and of the H3 helix for each of our models. In Figure 8 and Supporting Information Figure S7, we have plotted the percentage of the total simulation time during which each residue in this region is found in a helical form, calculated by the *g_helix* analysis tool. The plots in Figure 8 show that on average during the simulations the α-tubulin T3/H3 region has lost helicity for both the $\alpha\beta_S$ model and *p2* models, and that the β-tubulin T3/H3 region is much less helical for model *p2* compared to those in the other models.

Table III
Percentages of Residues in α-Tubulin's N-Terminal Domain Forming Various Types of Secondary Structure, Determined Using the do_dssp Gromacs[32,33] Analysis Tool

	$\alpha\beta_S$	$\alpha_S\beta_S$	*p1*	*p2*	*p3*
Structure	36	53	40	43	45
Coil	35	27	33	26	32
β-Sheet	2.2	7.4	1.9	11	6.2
β-Bridge	3.2	2.9	2.4	4.4	1.7
Bend	26	18	25	25	21
Turn	15	21	15	21	17
α-Helix	15	22	21	6.6	20
5-Helix	0.0	0.0	0.0	0.0	0.0
3-Helix	3.6	2.1	1.4	6.4	2.2

The category "Structure" represents all residues classified as α-helix, β-sheet, β-bridge, or turn.

Intermediate domain (residues α206–381, β204–371)

Major peaks seen in the RMSF plots (Fig. 7 and Supporting Information Fig. S6) in tubulin's intermediate domain are those corresponding to the region of longitudinal contacts at residues α200–232, β201–215; the flexible M-loop on β-tubulin at residues 276–284, and the N-loop on α-tubulin at residues 264–288; and the H10/B9 loop at residues α339–350 and β337–348. Greater RMS fluctuations of the intermediate domain of α-tubulin in the *p1* simulation compared to in the other simulations can be seen from the higher peaks on the RMSF plot corresponding to this simulation, and generally increased RMSF values for large parts of this domain. One instance of this is the greater fluctuation seen for model *p1* of residues corresponding to the H10/B9 loop. This is most likely related to the interchange of interactions between the H10/B9 loop, the H11 helix of β-tubulin, and the α-tubulin tail domain, as is observed in Figure 6 for this trajectory. That is, the formation and breaking of salt bridges causes the H10/B9 loop to assume two or more different positions. These interactions may indirectly lead to higher fluctuations of other parts of the intermediate domain of α-tubulin as well. In Figure 6, the sporadic interchange of interactions between the three domains is seen to occur primarily during the first half of the simulation. We thus examined the RMSF plot calculated for the first half of the simulation only [Supporting Information Fig. S8 (a)]. This plot also shows a high degree of fluctuation of residues in tubulin's intermediate domain. However, consistent with the more constant pattern of interactions during the second half of the simulation, the RMSF plot corresponding to this time interval shows much less fluctuation of the intermediate domain [Supporting Information Fig. S8 (b)]. Notice also that models *p2* and $\alpha_S\beta_S$ for which interactions between residues of

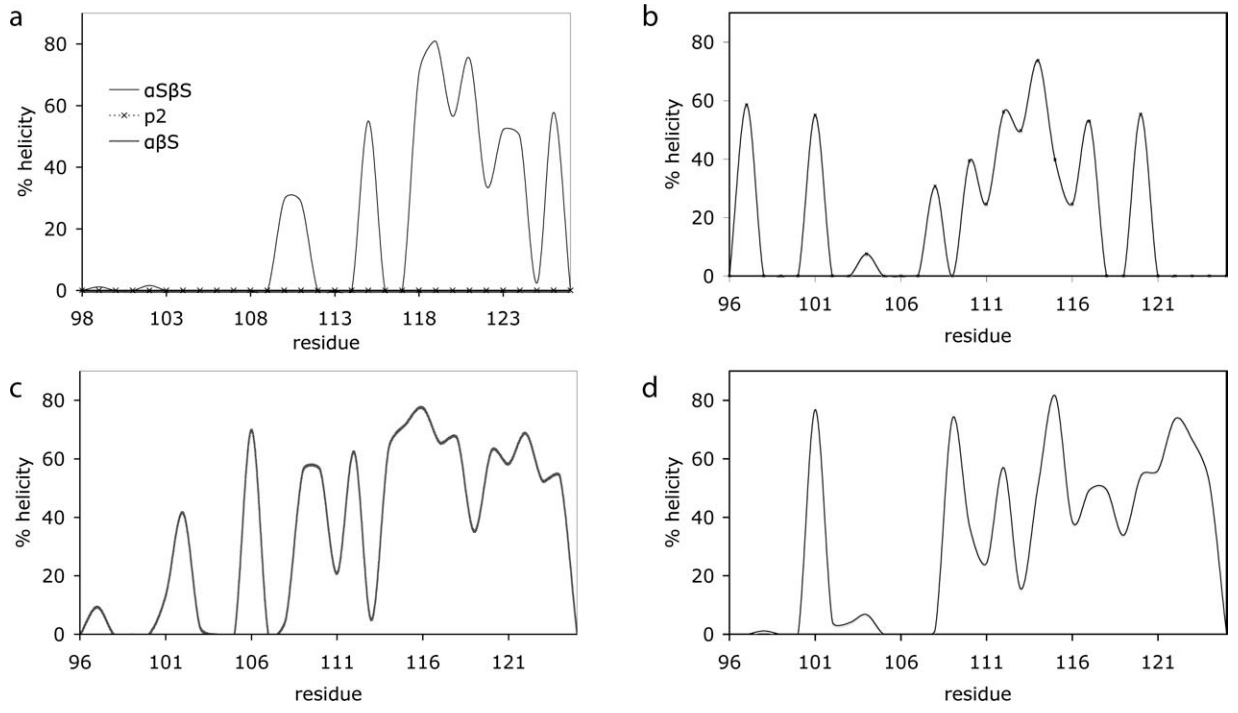

Figure 8
Percentage of total simulation time during which residues are found in a helical form plotted as a function of residue number for (**a**) the T3 loop/H3 helix domain of α-tubulin corresponding to models *p2*, $\alpha\beta_S$, and $\alpha_S\beta_S$; and for the same domain of β-tubulin corresponding to models (**b**) *p2*, (**c**) $\alpha\beta_S$, and (**d**) $\alpha_S\beta_S$.

tail domains and H10/B9 loops are entirely absent show much less fluctuation of residues in the intermediate domain than any of the other three simulations.

Experiments have shown that the polymers assembled from $\alpha_S\beta_S$ tubulin display an increased number of lateral interactions compared to normal MTs, causing junctions to form between polymer walls so that structures resemble hooked MTs with crescent-shaped cross-sections.[6,7] Structural changes in the intermediate domains of normal MTs caused by the disruption of H10/B9 loop-H11 helix interactions by tail domains may possibly lead to changes in positions of key structural elements involved in lateral contacts, which are located in the intermediate domain, such as the M or N loop.

Relevance to drug binding and efficacy

The observations made above, from our simulations, of interactions of the tubulin C-terminal tails with the rest of the protein and our suggestion that changes in the position of tubulin's intermediate domain arising from these interactions may offer an explanation for the role of tail domains in regulating MT assembly, will now be considered in the context of previous work performed by experimentalists.

As has been mentioned in the introduction, experiments have indicated that tail–body interactions within tubulin and tubulin polymers are capable of influencing the binding or efficacy of the tubulin-targeted cancer chemotherapy drugs, colchicine and vinblastine. Colchicine and vinblastine bind to opposite ends of β-tubulin, the former at the intra-dimer interface and the latter on the other end of β-tubulin close to where the N terminus of the α-tubulin tail from the neighboring dimer interfaces with β-tubulin in the context of a protofilament[47] (see Fig. 5). On the other hand, as we will now discuss, colchicine's binding is affected by cleaving the α-tubulin tail,[16] and vinblastine activity by cleaving the β-tubulin tail.[17] In this connection, it is noteworthy to mention that the effect of phomopsin A on the conformational stability of tubulin is also affected by the cleavage of tail domains. Phomopsin A, a non-competitive inhibitor of vinblastine binding[48] is an extremely good inhibitor[49,50] of the spontaneous denaturation[51,52] to which the tubulin dimer is prone. $\alpha_s\beta_s$ tubulin is more resistant to this decay than is undigested αβ tubulin, and the stabilizing effect of phomopsin A is even stronger with $\alpha_s\beta_s$ tubulin than with αβ tubulin.[53]

In experiments performed by Rai and Wolff[17] it was observed that the negative effect that oligo-anions such as sodium triphosphate have on vinblastine-induced polymerization of tubulin is prevented when the C-terminal tail on β-tubulin is removed by subtilisin digestion, although subsequently removing the α-tubulin C-terminal tail to form $\alpha_S\beta_S$ tubulin has no further effect. Note that these polymers are not MTs but are filamentous spirals.[54] Because this negative effect of oligo-anions on vinblastine-induced polymerization of tubulin could be induced in $\alpha\beta_S$ tubulin (recall that $\alpha\beta_S$ tubulin defines dimers lacking only the β-tubulin tail) by increasing the concentration of oligo-anions from 1 m*M* to 10 m*M*, a model was suggested in which there is an interaction favorable to polymer stabilization that is interfered with by both the β-tubulin tail domain and oligo-anions. More specifically, it was proposed that some residues on the tubulin body might interact with a positively-charged residue or region on the tubulin surface, and compete with β-tubulin tail residues for interactions with the same positive cluster.[17] An additive negative effect on stabilization from both this tail–body interaction and the blockage of the positive region by the added anions would then provide an explanation for both how the cleavage of tails allows vinblastine to stabilize tubulin polymers in the presence of a low concentration of anions, and how, when this concentration is increased, vinblastine cannot stabilize even $\alpha\beta_S$ polymers. The interaction between the H10/B9 loop on β tubulin and the positively-charged cluster on H11 of the neighboring α-tubulin, observed in our simulations, along with our observation of interactions between the β-tubulin tail and both of these domains competing with the former interaction, support such a model. From this perspective, the interaction between the H10/B9 loop on β tubulin and positively-charged residues on H11 of the neighboring α tubulin could be strengthened by vinblastine binding, contributing to vinblastine-induced polymer stabilization. On the other hand, the addition of anions would be expected to interfere with this longitudinal interaction. However, without the additive effect of the tail–body interactions, in $\alpha\beta_S$ polymers, vinblastine (albeit while bound to the end of β-tubulin distant from the H10/B9 loop) may be capable of stabilizing either the longitudinal loop–helix interaction and/or the accompanying large-scale effect on the position of the intermediate domain, in the presence of anions. The latter effect can be overridden by adding enough anions.

The effect of vinblastine on tubulin polymerization has been the object of investigations for many years. It turns out that vinblastine binds to MT ends with high affinity but also binds with a much lower affinity to tubulin sites located along the surface of the MT cylinder.[55–57] Approximately, 16–17 high-affinity binding sites located at MT ends have been determined to exist per MT. They are responsible for the well-known kinetic suppression effects of vinblastine, that is, its stabilization of MTs, at low concentrations (sub-micromolar).[58] The binding of vinblastine to the low affinity binding sites on the MT surface is believed to be responsible for the depolymerization of MTs at intermediate ($> 10\ \mu M$) drug concentrations. Finally at drug concentrations of $> 10\ \mu M$, the tubulin polymer stabilized by vinblastine, studied by Rai and Wolff,[17] is an alternate form of the polymer, not the

MT polymer. One possible explanation of the opposite effects of vinblastine on MT polymerization at submicromolar versus intermediate concentrations may be that in the vinblastine-bound polymer, the H11-H10/B9 loop interaction is stronger than normal, which can lead to increased MT stabilization when vinblastine binds to MT tips only; increasing the strengths of these interactions throughout other parts of the MT may lead to destabilization.

In a separate set of experiments, Chakraborty *et al.*[16] found that cleaving the α-tubulin tail domain lowers the activation energy and the on-rate for colchicine binding to the tubulin dimer. In contrast to the case of vinblastine effects on tubulin polymers, the effect is seen in $\alpha_S\beta_S$ tubulin but not $\alpha\beta_S$ tubulin, that is, cleavage of the β-tubulin tail alone has no effect on colchicine binding. Moreover, the binding affinity of colchicine for the tubulin dimer is normally dependent on pH, a factor known to affect tail conformation, but for the $\alpha_S\beta_S$ tubulin dimer, binding is independent of pH. To explain these findings, the authors suggested that although colchicine binds to β-tubulin at a site distant from the α-tubulin tail, tail–body interactions of the α-tubulin tail can cause large-scale conformational changes in the tubulin protein, which both promotes the binding of colchicine and leads to a higher activation energy for binding.[16] In support of this hypothesis it was noted that a 100 m*M* concentration of added NaCl, which would partially screen the electrostatic tail–body interactions, also lowers the activation energy and the on-rate for colchicine binding. It was speculated that the tail–body interaction might involve bending of the α-tubulin tail towards the positively-charged cluster on H11 of α-tubulin, affecting the colchicine binding site and leading to the altered kinetics.[15,16] This was not observed in any of our simulations. However, our MD modeling does support the proposed mechanism by which a tail–body interaction causes a conformational change in α-tubulin by a slightly different explanation, since our results indicate that the α-tubulin tail can interact with the H10/B9 loop, which may enable the intermediate domain to flex out laterally, leading to significant structural changes between normal and $\alpha_S\beta_S$ tubulin. By this line of reasoning, these changes would occur not only on the side of α-tubulin's intermediate domain close to the tail domain, but also extend to the distant side of this intermediate domain, near where colchicine binds to β-tubulin. It is also interesting to speculate whether, conversely, the drug activity of colchicine, which inhibits MT assembly, may involve altering the position of the intermediate domain of α-tubulin.

In summary, the combined results of the experiments from the literature,[16,17] which we have just described, and the simulations reported in the present article seem to indicate that both vinblastine effects on tubulin polymers and colchicine binding to tubulin are affected by shifts in the position of the entire intermediate domain of tubulin compared to that found in $\alpha_S\beta_S$ polymers or tubulin dimer, in the former case that of the β-tubulin monomer and in the latter that of α-tubulin. We propose that this shift is brought on by interactions between the tail domain, the H10/B9 loop, and in the case of vinblastine, the H11 helix of the neighboring monomer, all at the other longitudinal monomer interface from where drug-binding occurs. Experimental confirmation of large-scale conformational flexibility for tubulin's intermediate domain has already been supplied by photo-affinity labeling results demonstrating that cysteines 295, 305, 315, and 316 in the intermediate domain of α tubulin, but distant from the colchicine binding site, are affected by colchicine binding.[59]

CONCLUSIONS

Using computational techniques, we have determined two modes of interaction between the C-terminal tail domains with the main body of tubulin. Specifically, we have found an inter-monomer mode between charged residues on the α(β) tail and β(α)H11 and an intra-monomer mode, pairing aromatic (charged) residues on the α(β) tail with charged (aromatic) residues on the α(β) H10/B9 loop. As one would expect from the heretofore-elusive structures of the C-terminal tails, these interactions are non-specific and transient.

However, we also observe that the binding sites for the tails may bind laterally with each other, that is, the α(β) H10/B9 loop binds to β(α)H11. The result of this competitive interference is to increase flexibility in the intermediate domain with respect to $\alpha_S\beta_S$ tubulin. Such competitive behavior, in which another region of tubulin forms electrostatic interactions with the same positively charged region that the tubulin tail also binds to, has previously been hypothesized based on experimental results implicating tail–body interactions in vinblastine effects on tubulin polymers.[17] Based on the connection that we predict between flexibility of the intermediate domain and tail–body interactions, together with the experimental data showing that the binding or activity of colchicine[16] and vinblastine,[17] respectively, depend on electrostatic interactions of the tubulin tail domains far from the drug binding site, it seems likely that the binding or activity of these two drugs may be linked to the flexibility of tubulin's intermediate domain.

A natural test of the intermediate domain flexibility hypothesis and our proposed intra-monomer binding mode is to mutate residues Glu431 and Glu 432 belonging to the C-terminal tail domain of β-tubulin to alanines and then determine whether the negative effect of oligo-anions on vinblastine-induced polymerization of tubulin is eliminated, just as it is by cleavage of β-tubulin tail domains.[17] The analogous test for the inter-monomer mode is to mutate either of the two residues

Asp α345 on the H10/B9 loop or Tyr α440 on the C-terminal tail, predicted here to form extensive interactions with each other, to alanine, which should lead to pH independent binding of colchicine, in a similar manner that cleavage of the α-tubulin tail domain does.[16]

The C-terminal tails are also involved in the binding of several MAPs essential to the function of MTs. Following Luchko *et al.*,[3] we characterized three structural motifs possibly involved in the CK-2 phosphorylation and MAP2 binding. Falquet *et al.*[60] previously identified a conserved sequence for CK-2, which is found as DATA in the βI-tubulin tail. The DATA motif was found to have a very high similarity among our simulations; at least for the βI tail sequence, this indicates that the structure is persistent, and does not need to be induced for phosphorylation. As this structural motif has now been identified by two independent MD protocols, it should be considered a strong prediction. Contrary to the previous findings, the two proposed MAP2 binding motifs did not maintain similar structures between the models compared. However, since the sequence is experimentally known to be significant[44] and the residues are fully solvated, the possibility that these motifs are induced fit binding sites cannot be ruled out. Rather, further experimental and computational studies are needed to address this question. Some of this type of experimental study has been carried out on α-tubulin tails already, but we have not investigated proposed binding sites from this work.[61]

Beyond the scope of this study are the wide variety of post-translational modifications, particularly effecting the C-terminal tail domain, that tubulin is subject to. These include tyrosinolation/detyrosinolation of α, deglutamylation of α, polyglutamylation of both α and β, polyglycylation of both α and β, and phosphorylation of certain isotypes of β.[1] All of these could well affect the conformation of the tail domains and several also would alter the charges. These modifications may play critical roles in regulating the functions of MTs; for example, detyrosination of α promotes MT stability,[62] while polyglycylation of β is necessary for normal axoneme formation and polyglutamylation of α and/or β appears to mediate binding of MT-associated proteins (MAPs) and the motor protein kinesin to tubulin.[14,63–67] Some of these effects are likely to be indirect, that is, the presence of the modification in the C-terminal could conceivably act as a signal to some other protein or proteins which in turn could mediate the effect, that is, MT stability, axoneme formation, and so forth. Regardless, these modifications are of such scale that it is easy to hypothesize that they cause major conformational changes in the C-termini and that these changes in turn may mediate the ultimate effects of these modifications. It is likely that there exists a complex feedback loop between the probability of modification occurring and the conformational effect of modification. In addition, the binding of the C-termini to MAPs and motors is critical to the function of a variety of MT populations and is likely to also be regulated by conformational interactions involving the C-termini. These conformational effects and interactions and their likely functional outcomes are currently being examined.

Finally, we wish to comment on the lack of helicity of the T3/H3 region that was noted in two of the simulations. The issue of how MTs are destabilized by GTP hydrolysis is currently highly debated.[68–72] Along with MTs, actin filaments comprise one of three major components of the cytoskeleton and these two types of protein filaments have many remarkably similar properties including the utilization of a hydrolyzable nucleotide in polymer dynamics.[73] Experimental and computational studies indicate that in the case of actin polymerization, a loop-to-helix transition of the DNase-I binding (DB) loop is associated with hydrolysis of the nucleotide ATP to ADP and with the simultaneous structural change to a less stable actin polymer, as lateral contacts are lost in the less flexible helical structure.[74,75] Thus, the variability in the helicity of the T3/H3 region seen in our simulations together with the fact that the T3 loop of β-tubulin has been observed to be destabilized upon hydrolysis of the bound GTP to GDP,[68] raise the possibility that upon hydrolysis of the GTP bound to β-tubulin, the T3/H3 region might undergo a loop-to-helix transition similar to that in actin, causing a large-scale shift in the entire MT structure. Experimental testing of this hypothesis could be performed by applying techniques which have previously been used to analyze small changes in protein helical structure, such as Trp fluorescence,[76] FRET (Forster resonance energy transfer) emission spectra,[77,78] or EPR (electron paramagnetic resonance) data.[78] In this way, structural comparisons could be made of the T3/H3 region in MTs bound by different nucleotides or nucleotide-analogues.

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Molecular Dynamics Simulations of Tubulin Structure and Calculations of Electrostatic Properties of Microtubules

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Abstract— We present the results of molecular dynamics computations based on the atomic resolution structure of tubulin. Values of net charge, charge distribution and dipole moment components are obtained for the tubulin heterodimer. Physical consequences of these results are discussed for microtubules in terms of the effects on test charges, test dipoles, and neighboring microtubules. © 2005 Elsevier Ltd. All rights reserved.

1. INTRODUCTION

Microtubules (MTs) are protein filaments of the cytoskeleton [1] with their outer diameter roughly 23 nm, and a hollow interior with a diameter of roughly 15 nm (see Figure 1). Their lengths vary but commonly reach 5–10 μm dimensions. They are composed of 12 to 17 protofilaments when self-assembled in vitro and almost exclusively of 13 protofilaments in vivo. These protofilaments are strongly bound internally and are connected via weaker lateral bonds to form a sheet that is wrapped up into a tube in the nucleation process [2].

MTs are found in nearly all eukaryotic cells and they perform a variety of key cellular functions. In addition, to providing rigidity and structural integrity to a living cell, they serve as tracks for motor protein transport. They also form the core of cilia and flagella which beat in a coordinated manner to either move objects along the cell membrane or to propel the cell through its environment. Perhaps most importantly, microtubules form mitotic spindles that segregate chromosomes during cell division.

In general, there are three types of action modes of chemical antitumor compounds, namely:

(a) DNA targeting compounds that kill the cell by destroying or blocking the use of its genetic material,

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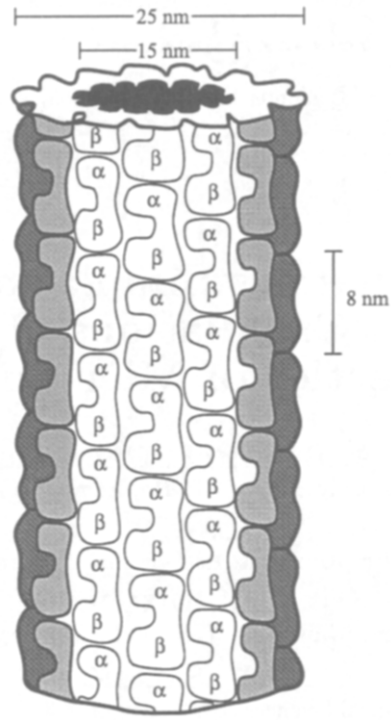


Figure 1. A section of a typical microtubule demonstrating the helical nature of its construction and the hollow interior which is filled with cytoplasm. Each vertical column is known as a protofilament and the typical MT has 13 protofilaments.

(b) compounds which inhibit normal cell functions, and
(c) the so-called spindle poisons which block the mitosis by interfering with the normal behavior of microtubules.

All the above-mentioned chemicals are very toxic. However, one interesting feature of the spindle poison compounds is that they more specifically target fast dividing cells, which is a particular property of cancerous cells. There are two ways by which the spindle poisons can block the mitosis. First, colchicine and vinblastine block the mitosis by preventing the formation of the mitotic spindle. In fact, they inhibit the polymerization of tubulin into microtubules. Second compounds such as taxol and rhazinilam stop the mitosis between the metaphase and the anaphase. They do so by stabilizing the microtubule polymer through binding to tubulin at specific locations as shown in Figure 3. Since the depolymerization is blocked making the microtubules static, the chromosomes cannot migrate toward the poles and cell division cannot be accomplished. It is our belief that analyzing some key physical properties of tubulin and microtubules such as their electrical charge and dipole distributions, we will gain important insights into the mechanism of cell division and possible means of controlling it via sophisticated physical and chemical agents.

The general structure of MTs has been well established experimentally [3,4]. A small difference between the α and β monomers of tubulin allows the existence of several lattice types (see Figure 2). Moving around the MT in a left-handed sense, protofilaments of the A lattice have a vertical shift of 4.92 nm upwards relative to their neighbors. In the B lattice this offset is only 0.92 nm because the α and β monomers have switched positions in alternating filaments. This change results in the development of a structural discontinuity in the B lattice known as a seam [4,5]. In addition, Chrétien *et al.* [6] observed that, the protofilament number need not be conserved along the length of a microtubule leading to the emergence of structural defects in the lattice. Furthermore, Sosa *et al.* [7] showed evidence of more than one seam in microtubules.

The idea that protofilaments have flexible connections, allowing for the presence of defects [8], could explain the anti-parallel alignment of protofilaments in the presence of Zn^{2+} ions. The zinc ion may favor one orientation for hydrophobic bonding. This is also consistent with the

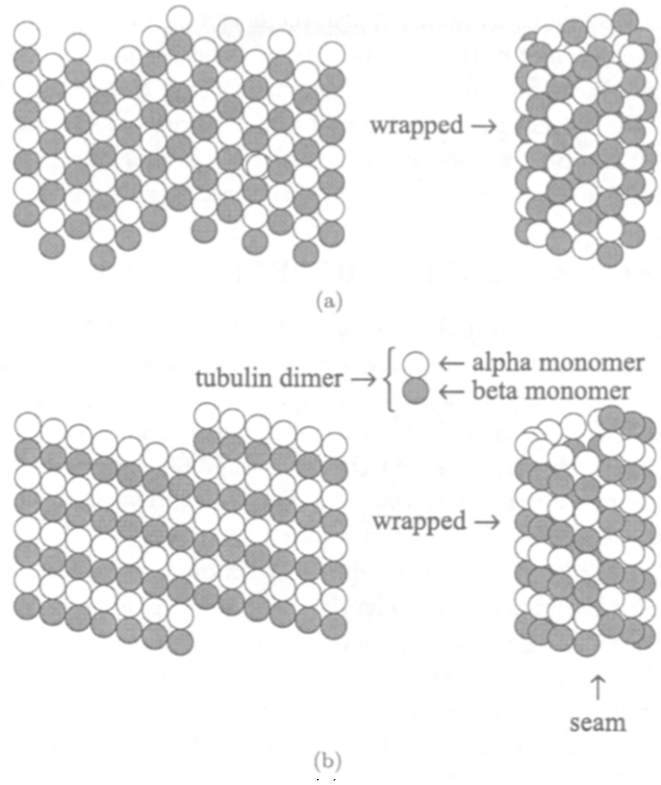


Figure 2. The 13A and 13B MT lattices: (a) in the A lattice, perfect helical spirals are formed. (b) in the B lattice, there is a structural discontinuity known as the seam.

observation of MT assembly in which a seam appears to "zip up" the cylinder behind the assembly edge [2].

In 1998 Nogales *et al.* [9] reported crystallization of tubulin in the presence of zinc ions. Their results were made available through the Protein Data Bank (PDB) [10] (PDB entry: 1TUB) which allowed us to view the three-dimensional atomic resolution structure of tubulin (see Figure 3) using RasMol [11]. Each tubulin monomer is composed of more than 400 amino acids and, in spite of their similarity, slight folding differences can be seen. It is worth stressing that several different versions of both tubulin α and β forms exist an are called isotypes [12] when found in

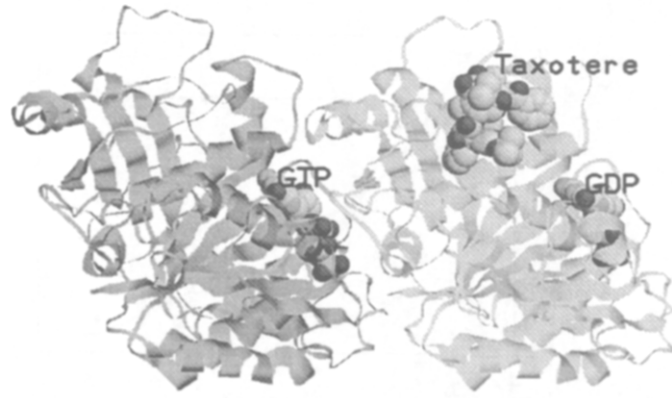


Figure 3. A ribbon diagram of the tubulin molecule produced from the [9] crystallographic data shows the similarity between the α-subunit (up half) and the β-subunit (lower half). The stick outlines near the base of each subunit indicate the location of GTP when bound.

the same organism. In humans at least six distinct α isotypes and seven β isotypes of tubulin have been identified. What is intriguing and, so far, unexplained, is the correlation between localization of certain isotypes and the functions performed by microtubules assembled from them. For example, the β_1 and β_4 are not found in cell nuclei but are present along with β_2 in the mitotic spindle. It has been suggested that, the interaction of tubulin with extrinsic proteins may direct the architecture and organization of MTs according to the isotypes used [13].

2. ELECTROSTATIC MODELLING OF TUBULIN

The method of distributed multipole analysis (DMA) provides a fairly accurate means of calculating the electrostatic field around a biomolecule. Diatomics, triatomics, and tetratomics are described to high precision with the use of monopoles, dipoles and quadrupoles providing an accurate picture of molecular bonding. In the case of α and β tubulin, each monomer is comprised of approximately 450 amino acids, i.e., on the order of 7000 atoms. In the initial part of our calculations, we present results for the electrostatic potential in vacuum surrounding the protein. Later on, we discuss the effects of solvent with ions.

Nogales *et al.* [9] imaged tubulin heterodimers to atomic resolution establishing that, the structures of α and β tubulin are nearly identical and confirming the consensus speculation. A detailed examination shows that, each monomer is formed by a core of two β-sheets that are surrounded by α-helices. The monomer structure is very compact, but can be divided into three functional domains: the amino-terminal domain containing the nucleotide binding region, an intermediate domain containing the taxol binding site, and the carboxy-terminal domain, which probably constitutes the binding surface for motor proteins.

Calculations of the potential energy were done with the aid of a molecular dynamics package, TINKER [14]. This computer program serves as a platform for molecular dynamics simulations and includes the facility to use several force-field parameter sets, some of which are protein specific. The most common of these parameter sets for proteins are AMBER [15] and CHARMM [16]. AMBER was selected over CHARMM on the rationale that it is a more up-to-date parameter set. The overall performance of the program gave us confidence that the results it provided for tubulin were meaningful. It was determined using TINKER that tubulin is quite highly negatively charged at physiological pH and that much of the charge is concentrated on the C-terminus. (Perhaps as much as 40% of the overall charge.)

Figure 4 shows an experimental titration curve for the net charge on the tubulin $\alpha\beta$ heterodimer (with the C-termini) as a function of pH. We have also been able to obtain a detailed map of the electric charge distribution on the surface of the tubulin dimer (see Figure 5). It is clear that, the

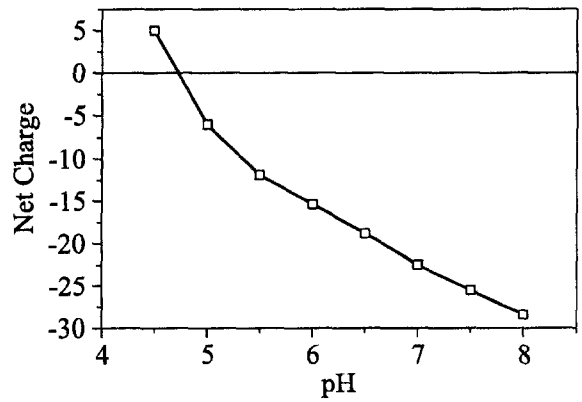


Figure 4. Tubulin titration curve for the tubulin $\alpha\beta$ heterodimer as a function of pH; obtained with no salt and no intra-molecular charge compensation. Figure courtesy of Sackett [17].

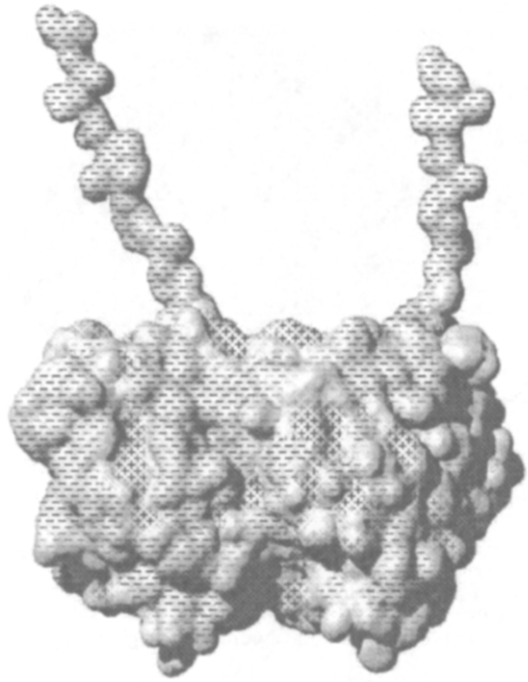


Figure 5. A map of the electric charge distribution on the surface of a tubulin dimer with C-termini tails present. Figure prepared using MOLMOL [18].

C-termini which extend outward carry a significant electric charge. At neutral pH, the negative charge on the carboxy-terminus causes it to remain extended due to the electrostatic repulsion within the tail. Under more acidic conditions, the negative charge of the carboxy-terminal region is reduced by associated hydrogen ions. The effect is to allow the tail to acquire a more compact form by folding (see Figure 6). Although this is probably the largest structural effect which occurs due to changes in the cell's pH, we expect that other structural changes, perhaps the result of post-translational modifications in the process of microtubule assembly, can similarly affect the electrostatics of the tubulin dimer.

In Figure 7, we have constructed a hydrophobicity map showing the location of hydrophobic and hydrophilic regions in the α tubulin monomer. In it, regions of dark minus signs are strongly hydrophobic, the lighter ones less so while increasingly dark plus signs correspond to an increasing level of hydrophylicity.

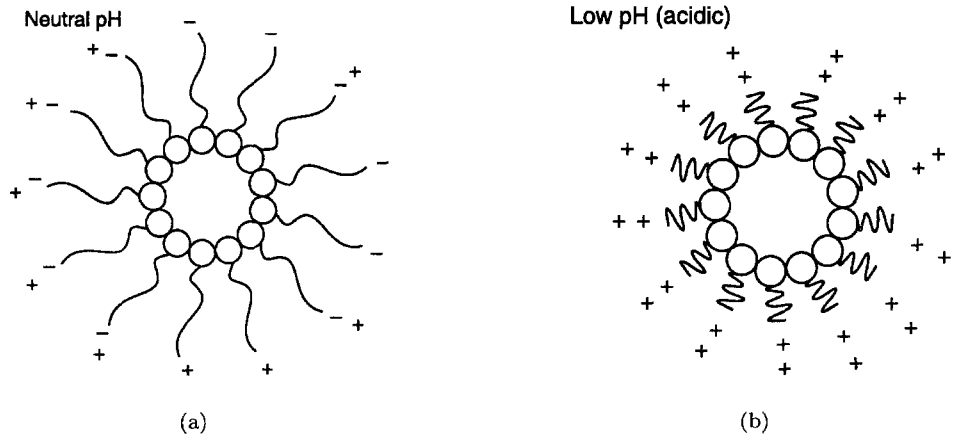


Figure 6. Cross-section of a MT including the carboxy-termini of the tubulin subunits. The folding shown of the carboxy-termini of the tubulin dimer demonstrates the change in the geometry of the molecule with pH. Neutral pH is shown on top, the tail folds at lower pH as the negative charges are screened.

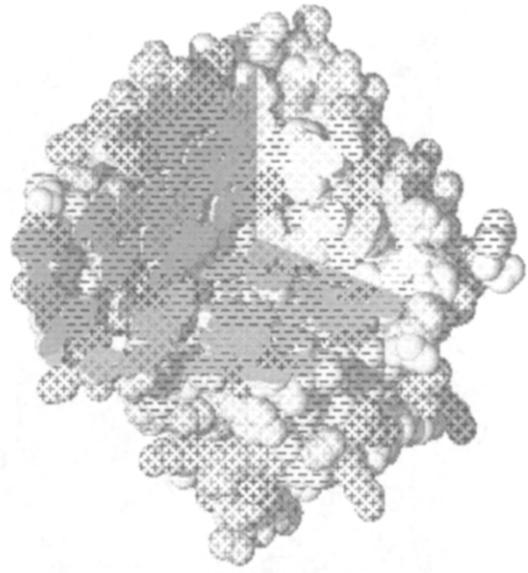
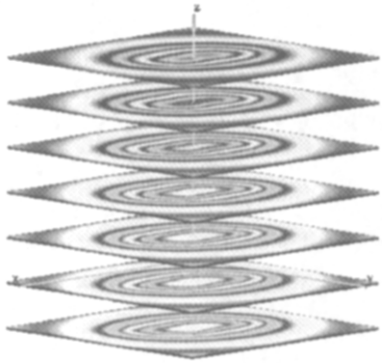


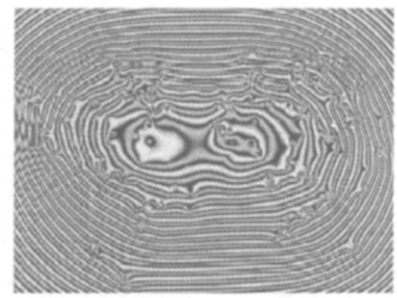
Figure 7. A cut-open view of the α-monomer of tubulin illustrating the location of hydrophobic regions (minus signs) and hydrophilic regions (plus signs).

In Figure 8a, we have imaged the electrostatic potential inside the tubulin dimer. It is interesting to note that, the hydrophobic interior of the protein gives rise to a nicely symmetric electrostatic potential with the different rings indicating equipotential surfaces. While the topography of the potential slices taken in the plane perpendicular to the dimer axis is, for the most part, concentric, towards the top layer of the dimer the sampled cross-sections reveal a double-well structure. This may indicate that charged groups present in this region could execute tunneling motions between the two equivalent energy minima. A close-up view of one such cross-section is given in Figure 8b.

In Table 1, we listed the dipole moment of the dimer without the tails (see Figure 9a). The story here, however, is more complicated. As shown in Figure 9, there are several additional sources of dipoles when tubulin is present in an ionic solution. When two dimers are bound within a protofilament, their positively and negatively charged ends form a double layer with a net dipole moment along the protofilament axis (Figure 9b). Beside each tubulin monomer there is a hydrophobic pocket that may develop a double-well structure (see Figure 8). This can give rise to an internal (switchable) dipole moment due to electronic transitions on this positive



(a) Cross-section slicing perpendicular to its axis.



(b) A double-well potential region close to the top of the $\alpha - \beta$ heterodimer.

Figure 8. Images of the electrostatic potential inside a tubulin dimer.

Table 1. Tubulin's electrostatic properties (tail region excluded)[a].

Tubulin Properties	Dimer	α Monomer
Charge (Electron Charges)	−10	−5
dipole (Debyes)		
overall	1714	566
P_x component	337	115
P_y component	−1669	−554
P_z component	198	−6

[a] The x-direction coincides with the protofilament axis. The α monomer is in the direction of increasing x values relative to the β monomer. The y-axis is oriented radially towards the MT centre and the z-axis is tangential to the MT surface.

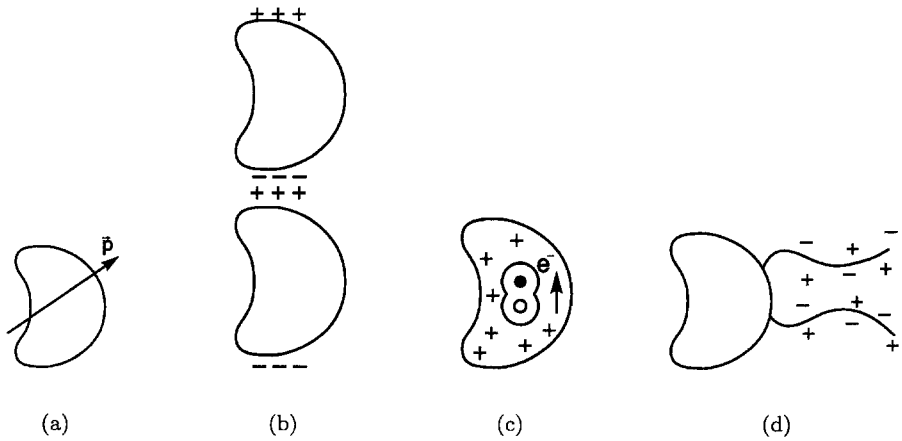


Figure 9. The various contributions to the dipole moment of a tubulin dimer (a) the intrinsic dipole moment of the globular protein, (b) the double layer formed when two dimers are bound in a protofilament, (c) a possible internal dipole created by electronic transitions in the hydrophobic pocket, and (d) a double layer formed by counter ions surrounding the C-termini tails.

background (see Figure 9c). Finally, as is shown in Figure 9d the C-termini which are negatively charged are surrounded by counter ions in solution leading to the formation of double layers. The principal contribution to the dipole moment of a tubulin dimer comes, however, from the location of partial charges on the constituent amino acids.

3. ELECTROSTATIC POTENTIAL AROUND TUBULIN

Having obtained the charge distribution on the tubulin surface, we have attempted to investigate its role in the microtubule lattice formation and its interaction with ions and macromolecules. We have confined our examination largely to the surfaces of tubulin that form the exterior surface and the protofilament-protofilament contacts when assembled into a MT. The first result that may be derived from the electrostatic potential is that there are those regions of the MTs outer surface that are negatively charged and which may attract hydrogen ions.

In calculating the electrostatic potential, 2.0 nm was selected as the cutoff distance for charge, dipole, and van der Waal interactions. The electrostatic potential was calculated for a 12 nm segment of the line, thereby including an additional 2 nm above and below each tubulin molecule. Periodic boundary conditions were then applied in the direction of the protofilament because this is the configuration of the tubulin dimers within a MT. The resulting profiles of the electrostatic potential are shown in Figure 10 and are located about the tubulin dimer as shown in Figure 11. The lateral boundary conditions were not considered in the calculation of the potential.

Consider the profile of the electrostatic potential in Figures 10a and 10b and compare them with the profiles in Figures 10c and 10d. These are left and right sides, respectively, of the

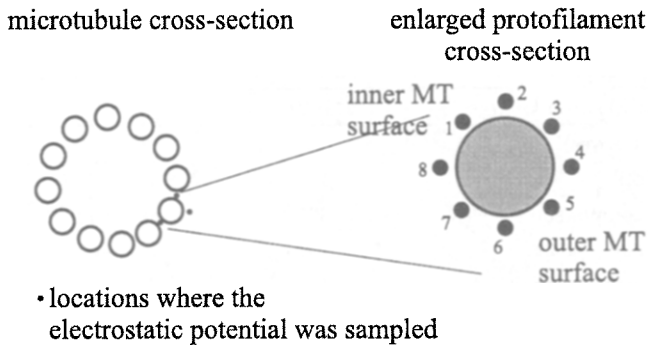


Figure 10. A MT cross-section illustrates where the electrostatic potential was examined along lines parallel to the protofilament axis (perpendicular to the page) in the preceding figure.

a and b are left sides
c and d are right sides

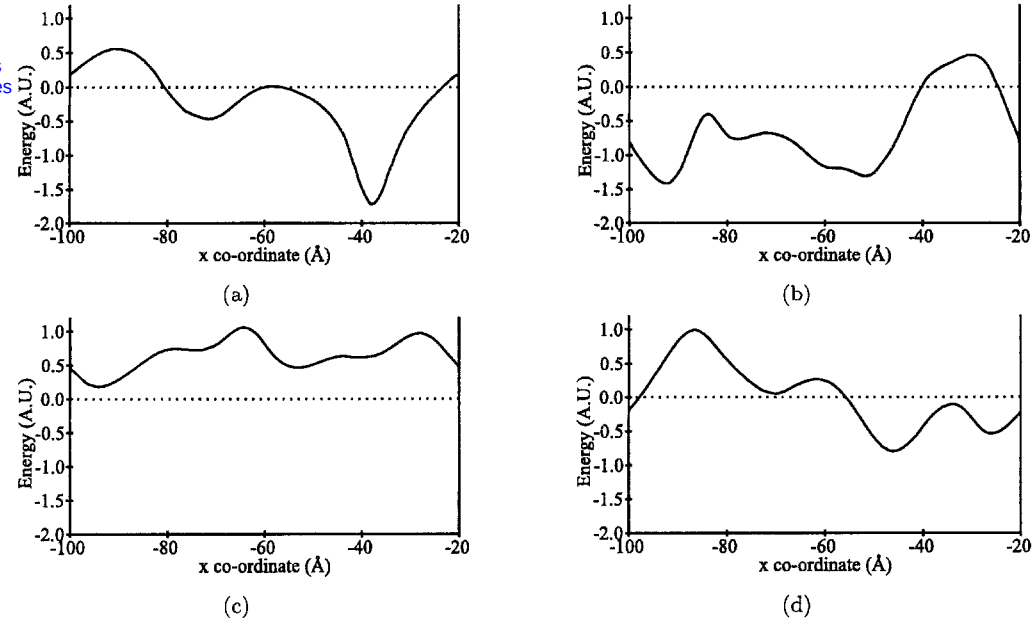


Figure 11. Electrostatic profiles along lines parallel to the protofilament axis. (a) Line 3-MTs interior on the A side of the protofilament-protofilament interface. (b) Line 4-MTs exterior on the A side of the protofilament-protofilament interface. The profile is largely negative indicating the negative surface charge. (c) Line 7-MTs exterior on the B side of the protofilament-protofilament interface. The largely positive surface charge is complementary to the opposite side of the dimer and contributes to protofilament-protofilament binding. (d) Line 8-MTs exterior on the B side of the protofilament-protofilament interface.

tubulin molecule, which interact laterally to hold one protofilament together with neighboring protofilaments. In these figures, each unit of energy represents 14.4 kcal/mol or 0.62 eV. This is roughly the energy available from the hydrolysis of two to three molecules of GTP or just a little more than the hydrolysis of one molecule of ATP. What is interesting is that, the electrostatic potential is largely negative on the left side and positive on the right side. Thus, there is a net electrostatic attraction between tubulin dimers with parallel alignment when their opposite sides face each other. In fact, if the minima in the left side's profile are aligned with the maxima in the electrostatic potential of the right side, we find that the neighboring tubulin dimer will be shifted by 1.4 nm or 5.4 nm which compares reasonably well to the observed 0.9 nm or 4.9 nm offsets that depend on the lattice type. The simple change of a residue on the surface offers the possibility of specifying one shift and locking the resulting MT into either the A or B type lattice. Hence, post-translational modification or more likely the expression of a particular isotype over another could select a specific lattice.

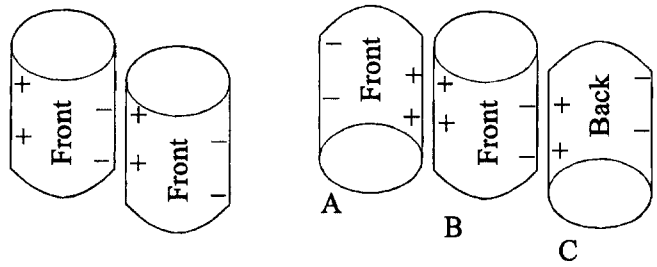


Figure 12. Protofilament-protofilament interaction. At left, tubulin dimers associate such that there is a vertical offset between protofilaments. At right, when the dimer's up-down orientation is reversed, it must also be rotated from front to back since the interaction between A and B is destabilizing but the interaction between B and C is stabilizing.

In the event that, we wish to consider tubulin aggregations such as the zinc(II) ion-induced sheets that were prepared in the tubulin structure determination experiments [9], the protofilaments have an anti-parallel configuration. As a result, proceeding on the premise that electrostatic interactions determine the protofilament-protofilament binding, the zinc(II) ion must work to alter this interaction. Since, the pattern of the electric potential consisting of two electrostatic wells on one side and two electrostatic peaks on the opposite side must be maintained, to explain the efficient binding of the dimers into the lattice of the tubulin sheet or MT, the potential is presumably not tremendously distorted. Since each tubulin dimer will be affected in the same manner, forming anti-parallel protofilaments does require that either the energy profile changes such that each side has a well and peak in the potential or conversely, that there is a small change in the potential which now favors binding where the "front" of the tubulin dimer is presented to an observer along protofilaments with the first orientation while the "back" of the tubulin dimer faces the observer for the protofilaments with the opposite orientation (see Figure 12).

4. ELECTROSTATIC EFFECTS OF MICROTUBULE STRUCTURE

When the previous investigations of the electrostatics of tubulin are brought to bear on the structure of the microtubule, we predict the emergence of a very unusual anti-ferroelectric structure. This is shown in Figure 13 with tubulin's permanent dipoles placed almost perpendicular to the surface of the microtubule cylinder and almost canceling one another due to rotational symmetry. It is tantalizing to speculate that such a dipolar structure may be constructed to facilitate the docking process of motor proteins such as kinesin. Bearing in mind that tubulin is

rotational symmetry = means that each dipole on the opposite ends of the cylindrical structure cancel each other out

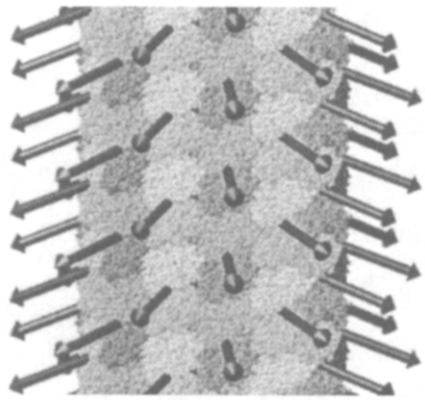


Figure 13. The arrows indicate the orientation of the permanent dipole moments of individual tubulin dimers with respect to the surface of a microtubule. Figure prepared using MOLMOL [18].

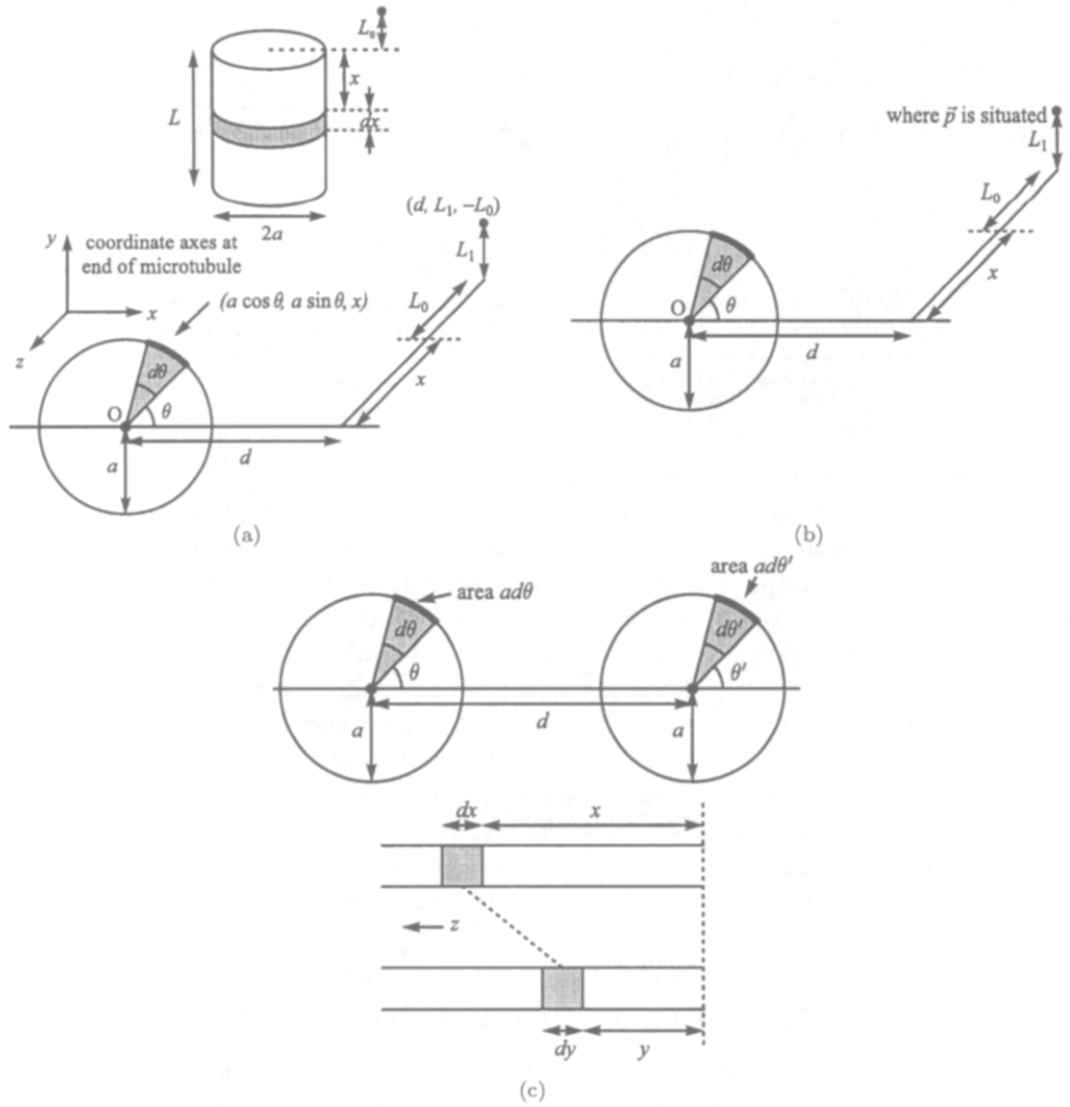


Figure 14. Geometrical arrangements for the calculations of (a) test charge-microtubule interaction; (b) test dipole-microtubule interaction; and (c) the interaction between two parallel microtubules.

both highly charged and possesses a permanent dipole moment we have attempted to estimate the strength of electrostatic effects on,

(a) a test charge,
(b) a test dipole,
(c) another microtubule in the vicinity, and
(d) the dipole-dipole interaction between two microtubules.

Below, we summarize our calculations and Figure 14 illustrates the geometrical details.

A. Microtubule-Charge Interaction

Consider a microtubule with charge per unit area σ on its surface, with a length L, and a diameter $2a$. Suppose a point charge is situated at the points $(d, L_1, -L_0)$ relative to the origin as shown in Figure 14a. We calculate the electric field intensity at the point-charge location assuming the electrical relative permittivity to be ε. Using the formula

$$E = \int_0^L \int_0^{2\pi} \frac{\sigma a \, d\theta \, dx}{4\pi\varepsilon_0\varepsilon(d^2 + a^2 - 2ad\cos\theta - 2aL_1\sin\theta + L_1^2 + (x + L_0)^2)}, \tag{4.1}$$

we assume $d \gg a$ so that $d^2 + a^2 + L_1^2 + (x + L_0)^2$ is much greater than the terms in ad and aL_1. In the angular part of the integration we retain the leading term only and find that

$$E \approx \frac{\sigma a}{2\varepsilon_0 \varepsilon} \frac{1}{\sqrt{d^2 + a^2 + L_1^2}} \left[\tan^{-1} \left(\frac{L + L_0}{\sqrt{d^2 + a^2 + L_1^2}} \right) - \tan^{-1} \left(\frac{L_0}{\sqrt{d^2 + a^2 + L_1^2}} \right) \right]. \tag{4.2}$$

For sufficiently long microtubules

$$E \approx \frac{\sigma a}{2\varepsilon_0 \varepsilon} \frac{1}{\sqrt{d^2 + a^2 + L_1^2}}. \tag{4.3}$$

The results presented for the electrostatic potential in this paper represent "vacuum" results given that, the solvent is not explicitly taken into account. If the surrounding mixture of ions is considered, then the potential due to a point charge does not fall off simply as $1/r$ but instead as $\phi \propto (1/r)e^{-K(r-r_0)}$ where K^{-1} is the Debye length, typically close to 1 nm under physiological conditions [19]. The constant r_0 is the ionic radius. Since we consider locations within several nanometers of the MT surface, they are not completely screened by the ions of the solution.

As an example consider an average length of the microtubule of $L = 5\,\mu$m, $a = 12.5$ nm, and $\sigma = 0.5\,\mathrm{e/nm^2}$ which is consistent with our earlier discussion. With a test charge of $+5$ e located with $L_1 = 0$ and distance 5 nm from the surface we obtain a force of electrostatic attraction of 6 pN in water which is reduced to only 0.5 pN in standard ionic solution with a Debye length of 0.6–1.5 nm depending on the ionic strength. This would indicate that, the maximum distance over which a microtubule can exert an influence on a charged particle is on the order of 5 nm from its surface.

B. Microtubule-Dipole Interaction

Here, we consider a microtubule with a dipole moment per unit area, σ_D, on its surface, each dipole moment being perpendicular to its surface. The test dipole is assumed to be at the point $(d, +L_1, -L_0)$ as we illustrate in Figure 14b. The distance between this latter dipole and a dipole on a small element of area $\sigma_D a\, d\theta\, dx$ at $(a\cos\theta,\, a\sin\theta, x)$ is given by

$$\sqrt{(a\cos\theta - d)^2 + (a\sin\theta - L_1)^2 + (L_0 + x)^2}. \tag{4.4}$$

As a result the test dipole, with dipole moment $\vec{p}$, has a dipolar energy H_E, due to its interaction with the microtubule, given by

$$H_E = \sigma_D \int_0^L \int_0^{2\pi} \frac{a\, d\theta\, dx \left[\hat{n} \cdot \vec{p} - 3\,(\vec{n} \cdot \hat{n})\,(\vec{n} \cdot \vec{p})\right]}{4\pi\varepsilon_0 \varepsilon \left(d^2 + a^2 - 2ad\cos\theta - 2aL_1 \sin\theta + L_1^2 + (x + L_0)^2\right)^{3/2}}, \tag{4.5}$$

where $\hat{n}$ is a unit vector in the direction of the dipole moment of the element of area on the microtubule. That is

$$\hat{n} = \hat{\imath}\cos\theta + \hat{\jmath}\sin\theta. \tag{4.6}$$

Here, $\hat{\imath}$ and $\hat{\jmath}$ are unit vectors along the x and y coordinate axes. On the other hand, $\vec{n}$ is a unit vector along the direction from the surface element to the location of the test dipole. Hence,

$$\vec{n} = \frac{(a\cos\theta - d)\hat{\imath} + (a\sin\theta - L_1)\hat{\jmath} + (L_0 + k)\hat{k}}{\sqrt{(a\cos\theta - d)^2 + (a\sin\theta - L_1)^2 + (L_0 + x)^2}}, \tag{4.7}$$

where $\hat{k}$ is a unit vector along the z coordinate axis parallel to the length of the microtubule. The angular integral in equations (4.5) is evaluated by taking only the leading terms (since on

expansion, using the approximations in Section 4a, terms in $\sin\theta$ and $\cos\theta$ will vanish). Denoting the x, y, and z components of the test dipole by p_x, p_y, and p_z we find

$$\begin{aligned} H_E \approx & \frac{a\sigma_D}{4\pi\varepsilon_0\varepsilon\gamma} \frac{12ad\pi p_x + 12aL_1\pi p_y}{\left(d^2+a^2+L_1^2\right)^2} \\ & \times \left[\sin\left(\tan^{-1}\left(\frac{L+L_0}{\sqrt{d^2+a^2+L_1^2}}\right)\right) - \frac{1}{3}\sin^3\left(\tan^{-1}\left(\frac{L+L_0}{\sqrt{d^2+a^2+L_1^2}}\right)\right)\right. \\ & \left. - \sin\left(\tan^{-1}\left(\frac{L_0}{\sqrt{d^2+a^2+L_1^2}}\right)\right) + \frac{1}{3}\sin^3\left(\tan^{-1}\left(\frac{L_0}{\sqrt{d^2+a^2+L_1^2}}\right)\right)\right] \\ & + a^2\sigma_D 2\pi p_z \left(\left(d^2+a^2+L_1^2+(L+L_0)^2\right)^{-3/2} - \left(d^2+a^2+L_1^2+L^2\right)^{-3/2}\right). \end{aligned} \tag{4.8}$$

Note that, for an infinite cylindrical dipole surface with a uniform value of the dielectric constant in all space, the potential at an arbitrary point in space exterior to the cylinder is zero. However, here we deal with a finite cylinder. More importantly, there is a large difference between the dielectric constant in the medium (close to 80 for H_2O) and inside the protein (in the range of 2 to 4). Since interactions of a test dipole with the layer facing it are through the medium only while those with the layer turned away from it also involve a large portion of the space filled with protein, there will be a non-zero result. For distances comparable to the MT diameter we have introduced an appropriate correction parameter γ which varies between 2 and 3 depending on the distance. We have used the dielectric constant of water for ε.

To get a sense of energies involved in the interaction between a polar molecule in solution and a microtubule a distance of 5 nm away, we consider a 5 μm microtubule whose tubulin dimers possess a dipole moment on the order of 2000 D each, the latter may include an additional contribution from a double layer of water and ions. Assume first that a macromolecule carrying a dipole moment of 200 D along the x axis is situated in the equatorial plane of the cylinder where $L_1 = L_0 = 0$. Using equation (4.8) we obtain the energy of the dipole-dipole interaction, in this case to be a disappointing 0.03 meV which is much less than the energy of thermal fluctuations. However, repeating this calculation for a fictitious tubulin dimer in solution whose 2000 D dipole moment is oriented along the y axis and which is located so that $L_1 = 1\,\mu$m above the plane, everything else remaining the same as before (with $\gamma = 3$), we find that the interaction energy is now just above 2 meV—a value smaller than room temperature thermal fluctuations. Clearly, this indicates a minimal role of the dipolar structure of microtubules in promoting orientational order. This effect may become significant, however, as the assembling molecule approaches the surface for docking both on account of short distances and the decreasing dielectric constant [19]. Notice that, it transpires from equation (4.8) that, the interaction with the z-component of the dipole (which is along the microtubule axis) is very small.

C. Charged Microtubule-Charged Microtubule Interaction

In this section, we consider the force of interaction between the charge distributions, assumed uniform, on two parallel microtubules. We assume there is a charge σ per unit area on each, and consider two elements of area $a\,d\theta\,dx$ and $a\,d\theta'\,dy$, respectively. The square of the distance between the elements (see Figure 14c) is given by $(x-y)^2 + D^2$ where $D^2 = (a\sin\theta - a\sin\theta')^2 + (d - a\cos\theta - a\cos\theta')^2$. The force, F_1, is given by

$$F_1 = \int_0^{2\pi}\int_0^{2\pi}\int_0^{L}\int_0^{L} \frac{\sigma^2 a^2}{4\pi\varepsilon_0\varepsilon\left[(x-y)^2+D^2\right]}\,dx\,dy\,d\theta\,d\theta'. \tag{4.9}$$

Assuming that $d^2 + (x-y)^2$ is very much greater than the other terms in the denominator of equation (4.9) we may expand and retain only leading terms in $\sin\theta, \cos\theta, \sin\theta'$, and $\cos\theta'$. When

integrated over θ and θ' these will vanish giving

$$F_1 \approx \int_0^L \int_0^L \frac{(2\pi)^2\sigma^2 a^2}{4\pi\varepsilon_0\varepsilon\left[(x-y)^2+d^2+2a^2\right]}\,dx\,dy. \tag{4.10}$$

We now change the variables from x, y to $z = x - y$ and $w = x + y$ with a Jacobian of $J = 1/2$. The integral part of equation (4.10), namely

$$I = \int_0^L \int_0^L \frac{dx\,dy}{(x-y)^2+d^2+2a^2}, \tag{4.11}$$

then becomes

$$I = \frac{1}{2}\int_0^L \int_{-w}^{w} \frac{dz\,dw}{z^2+d^2+2a^2} + \frac{1}{2}\int_L^{2L} \int_{w-2L}^{-w+2L} \frac{dz\,dw}{z^2+d^2+2a^2}. \tag{4.12}$$

The integrals in equation (4.12) then become elementary and therefore, F_1 is given by

$$F_1 \approx \frac{2\pi a^2\sigma^2}{\varepsilon_0\varepsilon}\left\{\frac{L}{\sqrt{d^2+2a^2}}\tan^{-1}\left(\frac{L}{\sqrt{d^2+2a^2}}\right) + \ln\left[\cos\left(\tan^{-1}\frac{L}{\sqrt{d^2+2a^2}}\right)\right]\right\}. \tag{4.13}$$

We have estimated this effect by using the same charge density as before and the length of each microtubule to be $5\,\mu$m. The distance between two neighboring microtubules cannot be less than 35 nm measuring it between the centres due to the repulsion between the C-termini. The force of repulsion in equation (4.13) for $d = 40$ nm in an aqueous environment is found to be staggering 0.2×10^6 pN. However, when Debye screening due to ions at a concentration of 150 nM is included the result is reduced to 9 pN. This is a significant number that may explain various ordering effects in microtubule bundles or asters.

D. Dipole-Dipole Interactions Between Two Microtubules

Consider the interaction between the dipole moment densities on two microtubules and suppose on each the dipole moment per unit area is σ_D. We consider an element on one microtubule with coordinates $(a\cos\theta, a\sin\theta, x)$ and an element at $(d + a\cos\theta', a\sin\theta', y)$ on the other. (The positioning of elements in space is the same as in Figure 14c.) Once again d is the distance between the axes of the microtubules which will be assumed parallel and a is their radius. The force between them becomes

$$F_2 = \int_0^L\int_0^L\int_0^{2\pi}\int_0^{2\pi} \frac{\sigma_D^2\left[\hat{n}_x\cdot\hat{n}_y - 3\left(\vec{n}\cdot\hat{n}_x\right)\left(\vec{n}\cdot\hat{n}_y\right)\right]d\theta'\,d\theta\,dy\,dx}{4\pi\varepsilon_0\varepsilon\gamma^2\left[(x-y)^2+\left(a\sin\theta-a\sin\theta'\right)^2+\left(d+a\cos\theta'-a\cos\theta\right)^2\right]^2}, \tag{4.14}$$

where $\vec{n}$ is a unit vector along the line joining the two elements and given by

$$\vec{n} = \frac{\left(d+a\cos\theta'-a\cos\theta\right)\hat{\imath}+\left(a\sin\theta'-a\sin\theta\right)\hat{\jmath}+(y-x)\hat{k}}{\left[\left(d+a\cos\theta'-a\cos\theta\right)^2+\left(a\sin\theta'-a\sin\theta\right)^2+(y-x)^2\right]^2}. \tag{4.15}$$

The unit vector $\hat{n}_x$ denotes the direction of the dipole moment on the element at $(a\cos\theta, a\sin\theta, x)$ so, that $\hat{n}_x = \cos\theta\hat{\imath} + \sin\theta\hat{\jmath}$.

The direction of the dipole moment on the element at $(d + a\cos\theta', a\sin\theta', y)$ is defined by the unit vector

$$\hat{n}_y = \frac{\left(d+a\cos\theta'\right)\hat{\imath}+a\sin\theta'\hat{\jmath}}{\sqrt{\left(d+a\cos\theta'\right)^2+\left(a\sin\theta'\right)^2}}. \tag{4.16}$$

Assuming that, the square in equation (4.14) may be expanded by factoring out $(x-y)^2+d^2+2a^2$, similarly in equation (4.15) and extracting d^2+a^2 in equation (4.16) we have linearized and retained leading terms as we did in Section 4c. This is done by first performing the angular integrals and then dropping all denominators which involve $(x-y)^2+d^2+2a^2$, d^2+a^2, and $[(x-y)^2+d^2+2a^2][d^2+a^2]$ which we assume are small. The net result is that the force, F_2, becomes

$$F_2 \approx \frac{a^2\sigma_D^2}{4\pi\varepsilon_0\varepsilon\gamma^2}\frac{-2ad^2\pi^2}{\sqrt{d^2+a^2}}\int_0^L dx\int_0^L \frac{dy}{\left((x-y)^2+2a^2+d^2\right)^3}. \tag{4.17}$$

The integrals now become tractable if we change variables again to $z=x-y$ and $w=x+y$. We find that, the resultant force is

$$\begin{aligned} F_2 \approx & \frac{a^3\sigma_D^2 d^2\pi}{\varepsilon_0\varepsilon\gamma^2\sqrt{d^2+a^2}\left(d^2+2a^2\right)^2} \\ & \times\left\{\frac{3}{8}\frac{L}{\sqrt{d^2+2a^2}}\tan^{-1}\left(\frac{L}{\sqrt{d^2+2a^2}}\right)+\frac{1}{8}\sin^2\left[\tan^{-1}\left(\frac{L}{\sqrt{d^2+2a^2}}\right)\right]\right\}. \end{aligned} \tag{4.18}$$

Taking as before the polarization density of 2000 D per dimer and a separation between microtubule centres to be 40 nm and using $L=5\,\mu$m as well as $\gamma=3$ results in the attractive force, F_2, between the two microtubules of 330 pN which is very significant. Moving the distance between microtubules to 90 nm (which is the mean separation between axonal microtubules) reduces the attraction force by a factor of 4000 to a mere 0.08 pN. It is also worth emphasizing that a combined action of monopole-monopole and dipole-dipole forces will have a competitive nature with Coulomb repulsion on short distances due to negative charges of the microtubular surfaces and an attractive dipole-dipole interaction that evidently extends over a larger range. Hence, it is expected that an equilibrium distance between neighboring microtubules may be established as a global energy minimum.

5. DISCUSSION AND CONCLUSION

In this paper, we have summarized our calculations regarding the net charges and dipole moments of tubulin and microtubules. These calculations were performed using an atomic resolution structure of tubulin in conjunction with molecular dynamics simulations. We found a large negative charge concentrated on the outer surface of the protein, almost half of which is located on flexible peptide tails attached to the outer face. This charge is undoubtedly instrumental in tubulin-tubulin interactions as well as the interactions of tubulin with motor proteins such as kinesin. Figure 15 shows the results of our force field calculations where two tubulin dimers are placed in each other's neighborhood and allowed to interact electrostatically. The brush strokes represent the direction of the Coulomb force of attraction.

We have also shown that, the microtubule structure, in particular the lateral binding between protofilaments, is consistent with the location of positive and negative segments of the electrostatic potential for optimal binding. It is worth mentioning that, a recent paper [20] showed the electrostatic surface of the whole microtubule following computations involving the Poisson-Boltzmann equation. From these calculations a dramatic difference between the plus and minus ends of a microtubule has been revealed. It is very likely that, this difference leads to the well-known difference is polymerization kinetics involving these two ends. In our paper, we have also considered the role of electrostatics in the interactions between microtubules and other charged or polarized molecules. In particular, in spite of Debye screening, a microtubule can exert a Coulomb force on a charged particle that is up to 5 nm away from its surface. The dipole-dipole forces that have been found are negligible for the most part. However, they can be felt by dipoles that are perpendicular to the microtubule surface and removed from the equatorial plane. When two microtubules are found in the same vicinity, they can exert significant forces of repulsion even

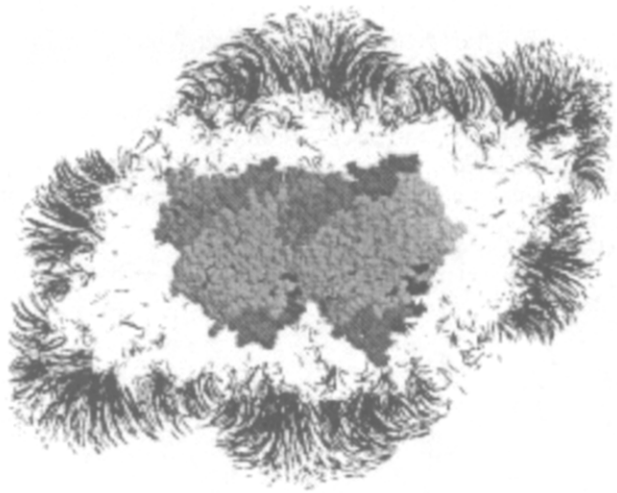


Figure 15. A view of the attractive regions about a tubulin dimer as would be experienced by another dimer. The smallest principal moment of inertia of the dimers is perpendicular the page, the middle one is aligned vertically, the largest principal moment horizontally. See, text for more details. Figure prepared using [11].

in the presence of ionic screening. Since the negatively charged C-termini protrude perpendicularly to the microtubule surface this effect is additionally increased and explains the existence of the so-called "zone of exclusion" [21].

In conclusion, our calculations demonstrate a significant role played by charge and dipole forces in both the formation of microtubules and their interactions with other proteins and possibly drug molecules. The latter observation may lead to a fruitful search for microtubule binding drugs such as derivatives of taxol. In addition, there has already been progress in understanding cell division by the inclusion of electrostatic interaction mechanisms as shown in a recent paper [22].

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Regular Article

Nonlinear ionic pulses along microtubules

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Abstract. Microtubules are cylindrically shaped cytoskeletal biopolymers that are essential for cell motility, cell division and intracellular trafficking. Here, we investigate their polyelectrolyte character that plays a very important role in ionic transport throughout the intra-cellular environment. The model we propose demonstrates an essentially nonlinear behavior of ionic currents which are guided by microtubules. These features are primarily due to the dynamics of tubulin C-terminal tails which are extended out of the surface of the microtubule cylinder. We also demonstrate that the origin of nonlinearity stems from the nonlinear capacitance of each tubulin dimer. This brings about conditions required for the creation and propagation of solitonic ionic waves along the microtubule axis. We conclude that a microtubule plays the role of a biological nonlinear transmission line for ionic currents. These currents might be of particular significance in cell division and possibly also in cognitive processes taking place in nerve cells.

1 Introduction

Microtubules (MTs) are major cytoskeletal protein polymers assembled from the protein called tubulin that plays a number of crucial biological roles in all eukaryotic cells [1]. The role of MTs in cell division has been extensively discussed in the cell biology literature. In this paper, we focus on the importance of MTs in the process of ionic current regulation within biological cells. Many years ago Matsumoto *et al.* [2] revealed that axoplasmic MTs and 260 K proteins in the structure underlying the axolema play a role in generating Na^+ currents in squid giant axons. Another seminar paper [3] considered the importance of Ca^{++} ionic currents in cell division and emphasized that MTs in the mitotic apparatus might be controlled by the local Ca^{++} concentration which should be tuned by nearby endoplasmic reticulum (ER). Spindle MTs appear to be very sensitive to Ca^{++} concentration increases in such a way that MT depolymerisation occurs when the concentration is raised above 1 μM. It was suggested that during anaphase Ca^{++} release from ER activates MT depolymerisation and this facilitates movement of chromosomes to the spindle poles [4]. Here, we postulate that localized Ca^{++} ionic currents along MTs serve as triggers for the onset of their depolymerisation. The synchronously regulated separation of sister chromatid pairs in mitotic spindles has been recently described by Matsson [5,6] using an elegant nonlinear theoretical model. We propose that the process of MT depolymerisation involved in cell division is finely orchestrated by Ca^{++} influx from ER which propagates along MTs as a nonlinear wave.

The structure of MTs is cylindrical and it typically involves 13 parallel protofilaments *in vivo*, see fig. 1. The building block of a MT is a tubulin dimer (TD) that contains approximately 900 amino acid residues comprising some 14000 atoms with a combined mass of 110 kDa (1 Da is the atomic unit of mass, 1 Da = 1.7×10^{-27} kg). The TD is made up of two structurally slightly different monomers called α and β tubulin, respectively. Each TD in the MT has a length of 8 nm along the MT cylinder axis, a width of about 6.5 nm and the radial dimension of 4.6 nm. The inner core of the cylinder, known as the lumen, is approximately 15 nm in diameter. The surface of MTs is mostly characterized by a so-called B-lattice structure [7], see fig. 2.

Between neighboring protofilaments there are two distinct types of nanopores (NP). NP-1, see fig. 2, left panel, is located where an inter-dimer β/α interface of one TD lies next to the inter-dimer β/α interface of the adjacent TD molecule. The so-called NP-2 arises where an intra-dimer interface of one dimer lies next to the intra-dimer interface of an adjacent TD of a neighboring protofilament, see fig. 2, right panel. Freedman *et al.* [8] used HOLE and AMBER programs to estimate the effective radius of the narrowest points within these NPs and found them to be 0.4 nm and 0.47 nm for type-1 and type-2 NPs, respectively.

The biological relevance of NPs in MT is still largely unknown. However, it has been experimentally determined

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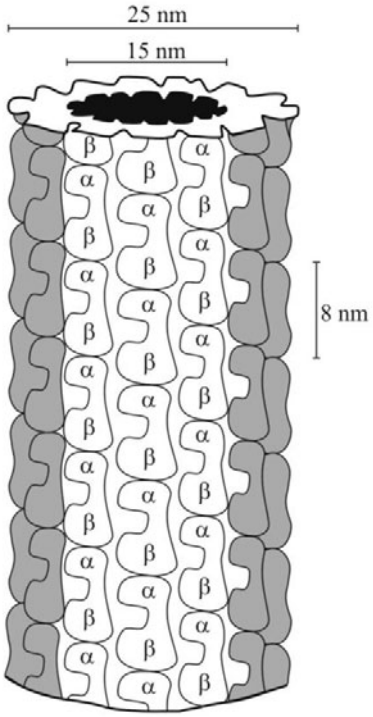

Fig. 1. A MT hollow cylinder of 13 parallel protofilaments with denoted characteristic dimensions: outer and inner diameters of 25 nm and 15 nm, respectively, and tubulin dimer length of 8 nm.

that the molecules of some chemotherapy drugs such as paclitaxel and epothilone diffuse through MT NPs to their corresponding binding sites mostly positioned in the lumen [9]. Thus, it can be safely assumed that these NPs, on the basis of their size, should be quite permeable to the counter-ions from the ionic clouds covering outer and inner sides of the MT wall. This paper will theoretically examine the role of these NPs in the regulation of ionic currents along MTs within the regime comparable to transmission lines used in electric engineering applications. We have earlier made preliminary attempts to evaluate MTs as nonlinear transmission lines. The first stage was a model which disregards the role of NPs but includes the nonlinear electric capacity of MTs [10]. The outcome of this model was the appearance of stable localized kink-like pulses of cations propagating along MTs. The next step was extended by the inclusion of NPs [11]. This approach was based on the simple conical geometry pattern of NPs considered earlier by the group of Eisenberg [12]. This preliminary description of NPs in our model has led to the emergence and propagation of ionic pulses analogous to the celebrated Korteveg de Vries (KdV) solitons and was arrived at using reasonable values of model parameters.

This paper is intended to develop a much more realistic model based on the sophisticated simulations performed by Freedman *et al.* [8]. We briefly state the main results of the simulations reported in [8] which are relevant for the model considered here. Based on the modified version of the Poisson-Boltzmann equation, the conductances of NPs are determined in terms of Brownian dynamics of ions. For the type-1 NP, inner and outer cationic conductances were calculated to be 2.93 nS and 1.22 nS, respectively, while for type-2 NP the respective values of 7.80 nS and 4.98 nS were found. Interestingly, it was demonstrated that both types of NPs are impermeable to anions whose respective conductances are 0.044 nS and 0.012 nS for type-1 and type-2 NPs. Freedman *et al.* [8] also explained in great detail how the constriction zones of NPs, due to the presence of negative ASP and THR residues, prevent the passage of more negative ions through the NPs. Similar asymmetric conductance has been found in various experiments involving ion channels, for example Omp F porin [13].

We now focus on the second, an even more important constituent of MTs in the context of the present model. Each tubulin monomer of the MT lattice has a short C-terminal alpha-helix H12 followed by a highly acidic amino acid sequence projecting out of the MT outer surface which is referred to as a tubulin tail (TT). Geometrically, these TTs are hair-like projections of 4–5 nm length, see fig. 3. Their approximate molecular masses are 5.5 kDa and 6.7 kDa for α and β tubulin, respectively. TTs are of great functional importance because they participate in the binding of proteins through which MT interact with many cellular structures such as motor proteins (kinesin and dynein) and a plethora of microtubule associated proteins (MAPs) [14]. Satarić and Tuszynski [15] developed a model of nonlinear waves created by tilting excitations of TTs emphasizing their possible role in the control of motor protein traffic along MTs. In α tubulin the TT consists of the last 10 residues in their sequence, while in β tubulin the corresponding TT is longer comprising 18 residues. The most common isotype of α tubulin has a TT with 5 Glu and 2 Asp residues, while its β counterpart has 9 Glu and 2 Asp residues. The negative charge on TTs is the reason why they are surrounded by a large number of cations required for overall charge neutrality. The important aspect of the aforementioned modeling [8] is the claim that TTs could contribute to ionic currents flowing along MTs by playing the role of small voltage generators (batteries). This claim is based on their larger time-scale thermal fluctuations which result in an oscillatory transfer of ions between the bulk solution and the cloud of counterions located near the MT surface. Here, we partly rely on the results of the simulations reported in [8] and partly on our earlier transmission line analysis [11], in order to elucidate the possible active role MTs play in ionic signaling throughout cytoskeletal networks.

The paper is organized as follows: In sect. 2 we explain the polyelectrolyte character of MTs. In sect. 3 we describe the basic elements of MTs as a biological nanoscale transmission line. In sect. 4 we analyze a nonlinear differential equation of the modified KdV type for the electric potential accompanying a localized ionic wave along an MT. Then, we solve and analyze the pertaining differential equation which exhibits the common features with spherical and cylindrical solitons arising in ion-acoustic waves in plasma. Section 5 provides conclusions stemming from the results of this paper.

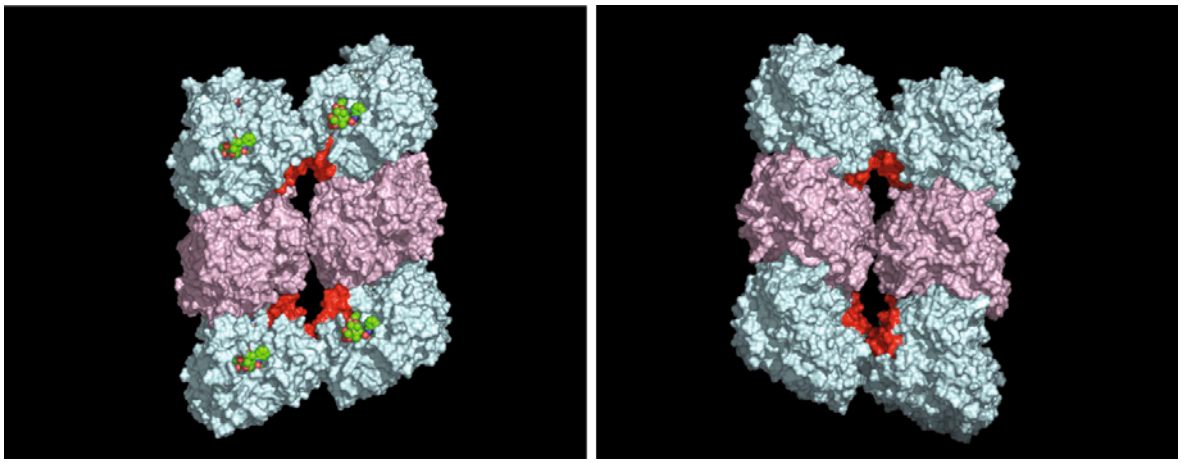

Fig. 2. An illustration of the nanopores in the A and B MT lattices, respectively.

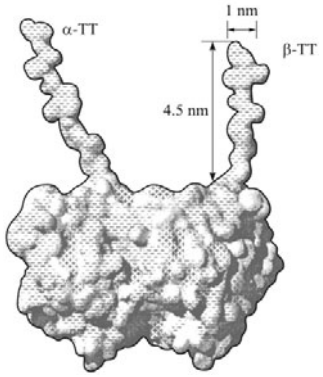

Fig. 3. The topology of a tubulin dimer with TTs whose dimensions are: length of 4.5 nm and diameter of 1 nm. The surface charge distribution is indicated by plus and minus signs.

2 The biophysical basis for the model

Extensive molecular dynamics simulations were performed on the TD structure and MTs [16,17]. Their results indicate the existence of a highly negatively charged solvent exposed surface of the TD with a net electric charge on the order of (20–25)e per monomer, depending on the isotype of tubulin considered. This net charge is distributed preferentially on the outer surface of the MT at an approximate charge ratio of 2:1 between the outer surface and the rest of the protein. Therefore, we conclude that MTs as highly charged polyelectrolytes are expected to attract a fraction of their surrounding counter-ions in the form of a condensed ionic cloud (IC) preferentially localized around the MT surface.

Manning's theory of polyelectrolyte solutions [18] states that ionic condensation occurs along the stretch of a charged polymer, if a sufficiently high linear electric charge density is present on the polymer's surface. On this basis, it can be concluded that the cylindrical sheath of depleted ions outside the ionic cloud surrounding the MT serves as an electrical shield. The "cable-like" behavior of such a structure can be supported by the protein polymer itself and the "condensed" counter-ions, which are physically "bound" to the polymer. The strength of this interaction has been estimated to be such that even under infinite dilution conditions, the counter-ions are still attached to the polymer and do not diffuse out from its vicinity. Although this theory was originally postulated for such polyelectrolytes as DNA, the same applies to highly charged quasi-one-dimensional, or cylindrical polymers such as actin filaments and MTs. These two cases were studied theoretically in more detail in other publications [8,10,19–22] and we adopt this approximation as valid for the case considered in our paper.

Moreover, experimental confirmation of the novel nonlinear amplification of ionic currents by MTs serves as support for this model [23]. Since our model is directly based on the ionic condensation effect around the MT cylinder, we first evaluate the effective charge density around the MT length. The Bjerrum length, denoted l_B, describes the distance beyond which thermal fluctuations are stronger than the electrostatic interactions between charges in solution whose dielectric constant is ε. Specifically, the following equation is satisfied:

$$\frac{e^2}{4\pi\varepsilon_0\varepsilon l_B} = k_B T; \quad \varepsilon_0 = 8.85 \times 10^{-12}\,\mathrm{F/m}. \tag{2.1}$$

At a physiological temperature ($T = 310\,\mathrm{K}$), taking the elementary charge as $e = 1.6 \times 10^{-19}\,\mathrm{C}$, Boltzmann's constant as $k_B = 1.38 \times 10^{-23}\,\mathrm{J/K}$ and the relative dielectric permittivity of the cytosol as $\varepsilon = 80$, one readily obtains

$$l_B = 0.67 \times 10^{-9}\,\mathrm{m} = 0.67\,\mathrm{nm}. \tag{2.2}$$

Accordingly, every MT should attract an IC of positive counter-ions close to its surface and along TTs, while negative ions of the cytosol are repelled away from the MT surface so that a roughly cylindrical ionic depletion area around TDs and TTs emerges. The thickness of such a depleted area is approximately equal to the Bjerrum length.

In addition, counter-ion condensation occurs when the mean distance between two charges, b, is such that

$$S = \frac{l_B}{b} > 1. \tag{2.3}$$

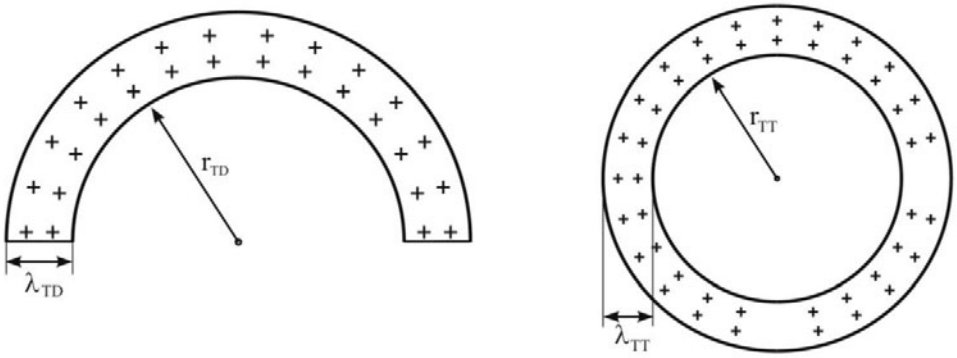

Fig. 4. Schematic representation of the counter-ion charge distributions surrounding a microtubule (left panel) and a C-terminal tail (right panel).

As a reasonable approximation we assume that each β tubulin monomer carries an excess of 24 negative charges in vacuum while each α tubulin monomer carries 22 charges. Moreover, some 30–40% of the net charge is located in areas of tubulin that are not solvent accessible. Therefore, we expect to have 15–18 exposed negative charges per TD, without counting the charges present on the protruding TTs. Since each dimmer is 8 nm long and 13 protofilaments form a MT, we readily find that there is a linear charge of (20–30)e/nm and a linear charge spacing of $b = (0.33\text{–}0.44) \times 10^{-10}$ m which gives S in the range between 16 and 21. As the effective charge q_{eff} is defined as the bare value of net charge divided by S, we estimate it to be $q_{\text{eff}} = (0.43\text{–}0.57)e$/monomer. This quantity has been indirectly measured in experimental assays where MTs were subjected to external electrical fields [23]. However, it should also be noted that this number represents a "far-field" value which could be overly high since it belies additional screening which is achieved by all ions attached to the MT but not all necessarily condensed on it. We conclude that in view of the experimental data, these estimates are acceptable for our continuum model calculations. For this type of model it is important to estimate the thickness of an IC of positive counter-ions around MTs and TTs. This is dictated by the geometry of the surface landscape and by its negative charge distribution.

The occurrence of Manning condensation around the polymer of radius r also depends on the fact that the salt concentration n in the cytosol should be low enough to satisfy the inequality [24]

$$l_{\text{Db}} \gg r, \tag{2.4}$$

where the Debye length l_{Db} is defined by

$$l_{\text{Db}}^{-1} = (8\pi n l_B)^{1/2}. \tag{2.5}$$

Since the radius of a TT is of the order of 0.5 nm and the radius of a TD is of the order of 2.5 nm, while the Debye length for the cytosol is of the order of 10 nm [25], we see that the condition in (2.4) holds for both radii. Thus, we are now able to reliably estimate the respective condensate thickness λ

$$\lambda = A(r l_{\text{Db}})^{1/2}; \quad A < 1, \tag{2.6}$$

where A depends only weakly on the Manning parameter q_0. Taking $A = 1/2$ one finds the corresponding values for the TD (λ_{TD}) and TT (λ_{TT}) as follows:

$$\lambda_{\text{TD}} = 2.5\,\text{nm}; \qquad \lambda_{\text{TT}} = 1.1\,\text{nm}. \tag{2.7}$$

These values will be used in our calculations of the corresponding capacity and resistivity, see fig. 4.

We expect that this IC which is shielded from the cytosol by a depleted layer is an analog of a coaxial cable used in electrical engineering applications, so MTs may effectively act as nonlinear inhomogeneous transmission lines capable of propagating nonlinear dispersive ionic waves within an IC. Independent corroboration of this idea can be found in experimental assays first performed for actin filaments [26] demonstrating that an IC current along a single filament exhibits localized wave patterns similar to solitons in nonlinear electric LC transmission lines. Recently, a similar experiment was conducted on single MTs providing evidence of the amplification of input electrical signals [23]. In our model of IC currents we only consider one of the 13 parallel currents associated with a single protofilament. The elementary unit of this "nanowire" is one tubulin dimer with adjacent NPs (type-1 and type-2) and two associated TTs (α and β). The shape of a tubulin dimer with extended α- and β-TTs is presented in fig. 3, including the charge distribution on its surface. At neutral pH values, the negative charge on each TT causes it to be extended away from the MT surface due to the electrostatic repulsion. When the negative charges of the amino acids within a TT are partly neutralized by counter-ions condensed around, it will cause TTs to contract by folding as shown in fig. 5.

As mentioned earlier, the differences in the structures of α- and β-TTs will cause them to fold differently so that the β-TT is expected to shrink to a greater extent. In addition to these electrostatically generated conformations of TTs, their tilting oscillations due to thermal fluctuations of the environment are also present. Quantitative features of this effect will be presented in the next section. These effects are accompanied by a change in the electrostatic capacity contributed by TTs which inevitably leads

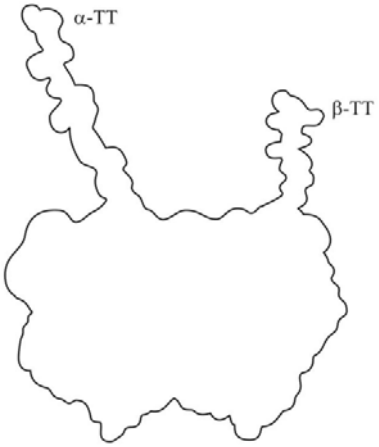

Fig. 5. The shape of TTs, where β-TT is shown to have shrunk due to the surrounding counter-ions, thus diminishing its electrostatic capacity.

to a slightly nonlinear overall capacity of every elementary unit of the ionic current along each filament.

3 The model parameters of an MT as an electric transmission line

Here, we proceed to estimate the electric parameters of an elementary unit of each MT protofilament (EUP). The essential assumption could be that an MT with a condensed IC is separated from the rest of the ions in the bulk solution by a depleted layer which plays the role of a dielectric medium located between the MT and the bulk cytosol. This IC is a conductive medium which dictates the electrical features of MTs. The depleted layer results in the IC and repelled anions playing the roles of "conductive plates" of a cylindrical capacitor. The ions injected from an appropriate cellular compartment are primarily confined to flow along an MT within its IC whose estimated thickness is $\lambda_{\rm TD} = 2.5\,{\rm nm}$.

For example, the opening of inositol (1, 4, 5) triphosphate (IP_3) receptors clustered at specific sites on the ER can lead to large-scale intracellular calcium waves [3,27]. By novel experimental techniques that permit the visualization of subcellular events it has become clear that Ca^{++} dynamics is a highly localized process with respect to these release events and also by the channelling of these waves along cytoskeletal filaments, primarily MTs. We have earlier proposed that such Ca^{++} waves could cause controlled MT depolymerisation during mitosis.

3.1 The capacitance of an EUP

In an earlier paper [10], a detailed Poisson-Boltzmann approach was used to evaluate the capacitance of an elementary ring of an MT which consists of 13 dimers. Here, we

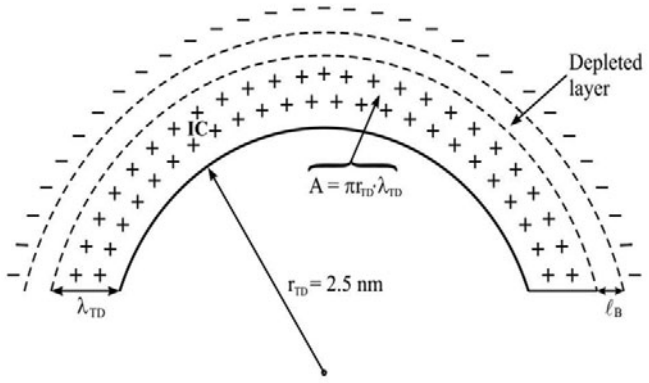

Fig. 6. Schematic illustration of the calculation of the MT capacitance.

adopt the same expression which reads

$$C_0 = \frac{2\pi\varepsilon_0\varepsilon l}{\ln\left(1+\frac{l_B}{R_{\rm IC}}\right)}\,, \tag{3.1}$$

where l stands for the length of a polymer unit and $R_{\rm IC} = r_{\rm TD} + \lambda_{\rm TT}$ for the outer radius of an IC. The other parameters have already been introduced.

We first estimate the EUP capacitance contributed by a TD, see fig. 6. With $l_{\rm TD} = 8\,{\rm nm}$ and $R_{\rm IC} = r_{\rm TD} + \lambda_{\rm TT} = 2.5\,{\rm nm} + 2.5\,{\rm nm} = 5\,{\rm nm}$, we find for TD (including only the outer surface)

$$C_{\rm TD} = \frac{3.14\times 80\times 8.85\times 10^{-12}\times 8\times 10^{-9}}{\ln\left(1+\frac{0.67}{5}\right)}\,{\rm F} = 1.4\times 10^{-16}\,{\rm F}. \tag{3.2}$$

Analogously, we can consider an extended TT as a smaller cylinder with the radius $r_{\rm TT} = 0.5\,{\rm nm}$ and the thickness of its IC equal to $\lambda_{\rm TT} = 1\,{\rm nm}$. Its extended effective length should be $l_{\rm TT}^{\rm eff} = 4.5\,{\rm nm} - 2.5\,{\rm nm} = 2\,{\rm nm}$, meaning that its part close to the tubulin surface is already embedded in the IC accounted for in (3.2). Thus we now estimate the corresponding capacitance as:

$$C_{\rm TT} = \frac{2\times 3.14\times 80\times 8.85\times 10^{-12}\times 2\times 10^{-9}}{\ln\left(1+\frac{0.67}{1.5}\right)}\,{\rm F} = 0.26\times 10^{-16}\,{\rm F}. \tag{3.3}$$

Accounting for the fact that two TTs are present in each tubulin dimer, we finally obtain

$$2\times C_{\rm TT} = 0.52\times 10^{-16}\,{\rm F}. \tag{3.4}$$

The two capacitance values above are considered to correspond to a parallel arrangement with respect to each other, so that the total maximal capacitance of an EUP is readily estimated as

$$C_0 = C_{\rm TD} + 2\times C_{\rm TT} = 1.92\times 10^{-16}\,{\rm F}. \tag{3.5}$$

We have earlier emphasized that TTs capacitance must change with an increasing concentration of condensed

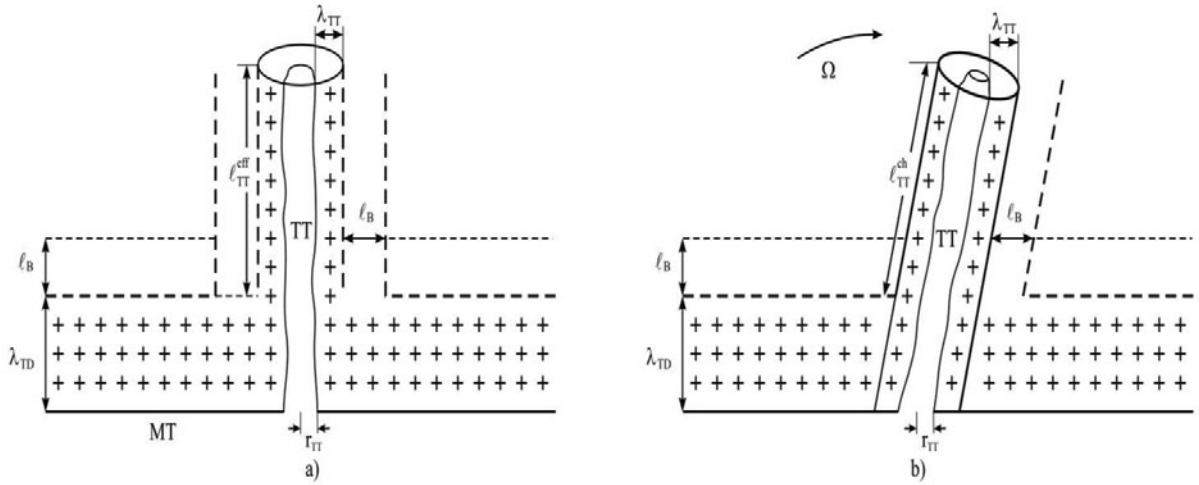

Fig. 7. A comparison between an extended TT (a) and a tilting TT due to oscillations (b).

cations due to the shrinking of flexible TTs. These changes are slightly different due to the different structures of α- and β-type TTs. To include this case we introduce the reduced factor of nonlinearity as follows:

$$b_0 = \frac{b_\alpha b_\beta}{b_\alpha + b_\beta}, \tag{3.6}$$

where b_α and b_β stand for the respective TTs. This implies that the charge of an EUP capacitor diminishes with an increased voltage in a nonlinear way $\Delta C_0 = C_0 b_0 v$, $b_0 v \ll 1$. Additionally, we account for the tilting movements of TTs under the combined action of thermal fluctuations [8] and a changing voltage due to an incoming ionic wave. Thus, the part of EUP capacitance contributed by TTs should also change by TTs tilt as shown in fig. 7. The change of the effective length $l_{\rm TT}^{\rm eff}$ of a TT is an additional factor affecting the capacitance $\Delta C_{\rm TT}$. We assume that this change can be adequately described by the oscillating function

$$\Delta l_{\rm TT}^{\rm eff} = l_{\rm TT}^{\rm eff} \sin[\Omega(t - t_0)] \cong l_{\rm TT}^{\rm eff} \Omega (t - t_0). \tag{3.7}$$

So that the capacitance changes linearly with a change in the effective TTs length $\Delta l_{\rm TT}^{\rm eff}$

$$\Delta C_{\rm TT} = C_0 \Gamma_0 \Omega (t - t_0), \tag{3.8}$$

where the frequency Ω is much lower than the inverse charging time of the EUP capacitor due to the strong viscous damping of the TT tilt, thus justifying the linearization of the above sine function in eq. (3.7). In some sense, this effect is similar in character to the thermal ratchet mechanism combined with an asymmetric ionic potential. Including the two aspects of TTs dynamics described above, the charge of EUP can be expressed as follows

$$Q = C_0[1 - \Gamma_0 \Omega (t - t_0) - b_0 v]v, \tag{3.9}$$

where Γ_0 is a dimensionless parameter.

In the earlier simplified version of the model of ionic currents along MT [10], a proof was given that the order of magnitude of magnetic inductance is so small that its role in these transmission lines can be safely ignored. We accept this statement in the present paper.

3.2 The resistance of an EUP

Regarding the Ohmic resistance, if we ignore ionic current leaks through the depleted layer, the dominant current flows in parallel with the MT axis charging EUP capacitors and partly leaking through NPs. The resistance attributed to this kind of ionic flow can be estimated on the basis of experimental evidence provided by the electro-orientation method performed on MTs *in vitro* [25]. Taking the reported measured value of MT ionic conductivity as

$$\sigma = (0.15 \pm 0.01)\,{\rm S\,m^{-1}}, \tag{3.10}$$

and adopting a simplifying assumption that the resistivity within an IC patch beyond an EUP is homogeneous, the resistance of an EUP with the length $l = 8\,{\rm nm}$ and the cross-sectional area $A = \pi r_{\rm DT} \lambda_{\rm DT} = 3.14 \times 2.5\,{\rm nm} \times 2.5\,{\rm nm} = 19.625\,{\rm nm^2}$, see fig. 6, is estimated as $R_0 = \frac{1}{\sigma}\frac{l}{A} = 2.7 \times 10^9\,\Omega$. This is obviously an extremely high value of resistance which is most likely an over-estimate.

We therefore prefer the computed resistance shown in table 3 of ref. [8], referred to as the outer sheath-outer sheath resistance for complete 13 protofilaments, namely $R_{13} = 4.75 \times 10^6\,\Omega$. The resistance for our EUP will be 13 times greater, *i.e.*

$$R_0 = 6.2 \times 10^7\,\Omega. \tag{3.11}$$

This value appears to be more consistent with the resistance of two NPs which follows from eq. (3.12).

Finally, we could include the conductance of both NPs to account for the leakage of IC cations into the lumen area. Thus we have

$$G_0 = \sigma_1 + \sigma_2 = (2.93 + 7.8)\,{\rm nS} = 10.7\,{\rm nS}; \quad \frac{1}{G_0} = 9.3 \times 10^7. \tag{3.12}$$

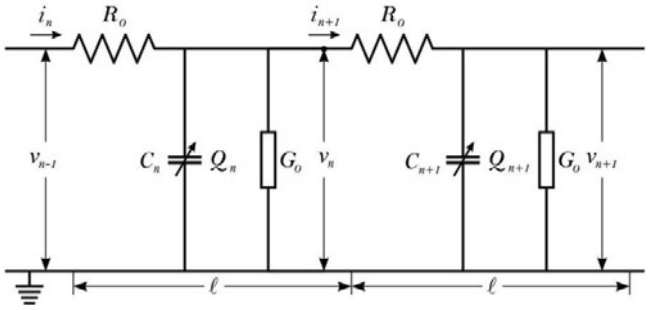

Fig. 8. An effective circuit diagram for the n-th ER with characteristic elements for Kirchhoff's laws.

It is expected that NPs exert greater resistance than the volume of IC around EUP. The above features of EUP electric parameters will lead to biophysical implications of our model. The next section develops the underlying equations governing the ionic flows along MTs.

4 The generalized model of an MT as a nonlinear transmission line

We now construct a simple periodic electric circuit simulating one protofilament of an MT. It consists of a long ladder network composed of a lumped section equal to identical EUPs as represented in fig. 8.

The longitudinal ionic current is represented by the series of resistors with Ohmic resistance R_0 for one EUP, as represented in (3.11). The nonlinear capacity with the charge Q_n for the n-th site of the ladder network is in parallel with the total conductance G_0 of the two TTs expressed by eq. (3.12). This is true if the bulk cytosol and MT lumen are considered to be grounded. This permits the application of Kirchhoff's law as

$$i_n - i_{n+1} = \frac{\partial Q_n}{\partial t} + G_0 v_n, \tag{4.1}$$

$$v_{n-1} - v_n = R_0 i_n. \tag{4.2}$$

Using eq. (3.9) and taking the time derivative results in a nonlinear form of the governing equation since

$$\frac{\partial Q}{\partial t} = C_0 \frac{\partial v_n}{\partial t} - C_0 \Gamma_0 \Omega v_n - C_0 \Gamma_0 \Omega (t - t_0) \frac{\partial v_n}{\partial t} - 2 b_0 C_0 v_n \frac{\partial v_v}{\partial t}. \tag{4.3}$$

Inserting eq. (4.3) into eq. (4.1) we arrive at the explicit system of equations describing the ionic current within IC, and an accompanying voltage

$$i_n - i_{n+1} = C_0 \frac{\partial v_n}{\partial t} - C_0 \Gamma_0 \Omega v_n - C_0 \Gamma_0 \Omega (t - t_0) \frac{\partial v_n}{\partial t} - 2 b_0 C_0 v_n \frac{\partial v_v}{\partial t} + G_0 v_n, \tag{4.4}$$

$$v_{n-1} - v_n = R_0 i_n. \tag{4.5}$$

The next step is to establish the auxiliary function $u(x,t)$ unifying the voltage and its accompanying ionic current as follows:

$$u_n = Z^{1/2} i_n = Z^{-1/2} v_n, \tag{4.6}$$

where the characteristic impedance of EUP is defined in the usual way as

$$Z = \frac{1}{\omega C_0}. \tag{4.7}$$

Since the discrete voltage v_n as well as the associated current i_n change gradually from an EUP to its neighbors, we can safely justify the expansion of u_n in a continuum approximation using a Taylor series with respect to a small spatial parameter l ($l = 8\,\mathrm{nm}$ is the length of an EUP). Then using the travelling-wave form of the function $u(x,t)$ with dimensionless space and time variables (ξ, τ), we eventually get the following inhomogeneous nonlinear partial differential equation (for details see appendix B):

$$\left(\frac{ZC_0 s}{T_0} - 2\right) \frac{\partial u}{\partial \tau} + \frac{1}{3} \frac{\partial^3 u}{\partial \xi^3} + ZC_0 \Gamma_0 \Omega (\xi - \xi_0) \frac{\partial u}{\partial \xi} + 2 \frac{Z^{3/2} b_0 C_0 s}{T_0} u \frac{\partial u}{\partial \xi} + (ZG_0 + Z^{-1} R_0 - ZC_0 \Gamma_0 \Omega) u = 0. \tag{4.8}$$

Here, the characteristic charging (discharging) time of an EUP capacitor C_0 through the resistance R_0, $T_0 = R_0 C_0$, on the basis of eqs. (3.5) and (3.11) yields the value

$$T_0 = 6.2 \times 10^7\,\Omega \times 1.92 \times 10^{-16}\,\mathrm{F} = 1.2 \times 10^{-8}\,\mathrm{s}. \tag{4.9}$$

The characteristic propagation velocity of the ionic wave is defined as follows:

$$v_0 = \frac{8 \times 10^{-9}\,\mathrm{m}}{1.2 \times 10^{-8}\,\mathrm{s}} = 0.67 \frac{\mathrm{m}}{\mathrm{s}}. \tag{4.10}$$

The dimensionless speed, space and time variables are chosen to be, respectively,

$$s = \frac{v}{v_0} \leq 1; \quad \xi = \frac{x}{l} - \tau; \quad \tau = s \frac{t}{T_0}. \tag{4.11}$$

Imposing that the condition

$$\frac{ZC_0 s}{T_0} > 2 \tag{4.12}$$

holds, we then estimate the characteristic impedance of EUP as

$$Z > \frac{2R_0}{s} \quad \text{or} \quad Z > 1.24 \times 10^8\,\Omega \quad \text{for} \quad s = 1. \tag{4.13}$$

For $s = 1$ one obtains the cutoff frequency

$$\omega_{\max} = 4.3 \times 10^7\,\mathrm{s}^{-1} \qquad \text{or} \qquad \nu_{\max} = 6.8 \times 10^6\,\mathrm{Hz}. \tag{4.14}$$

This indicates that the characteristic frequency matches the order of magnitude of frequency Ω which describes

$$W(\xi,\tau)=\frac{u_0}{\cosh^2\left\{\left[\frac{\alpha u_0}{4\beta}\exp(-2\gamma_0\tau)\right]^{1/2}\left[\xi-\xi_0\left(1-\exp(\gamma_0\tau)\right)+\frac{\alpha u_0}{3\beta}\left(1-\exp(3\gamma_0\tau)\right)\exp(-2\gamma_0\tau)\right]\right\}}, \tag{4.28}$$

$$u(\xi,\tau)=\frac{u_0\exp(-2\gamma_0\tau)}{\cosh^2\left\{\left[\frac{\alpha u_0}{4\beta}\exp(-2\gamma_0\tau)\right]^{1/2}\left[\xi-\xi_0\left(1-\exp(\gamma_0\tau)\right)+\frac{\alpha u_0}{3\beta}\left(1-\exp(3\gamma_0\tau)\right)\exp(-2\gamma_0\tau)\right]\right\}}. \tag{4.29}$$

the TTs oscillations. We now establish the compact form of eq. (4.8) as

$$\frac{\partial u}{\partial\tau}+\beta\frac{\partial^3 u}{\partial\xi^3}+\alpha u\frac{\partial u}{\partial\xi}+\gamma(\xi)\frac{\partial u}{\partial\xi}+\delta u=0, \tag{4.15}$$

where the abbreviations were introduced as follows:

$$\left.\begin{aligned}
&\alpha=\frac{2Z^{3/2}b_0C_0s}{T_0\left(\frac{ZC_0s}{T_0}-2\right)};\quad \beta=\frac{1}{3\left(\frac{ZC_0s}{T_0}-2\right)};\\
&\delta=\frac{ZG_0+Z^{-1}R_0-ZC_0\Gamma_0\Omega}{\left(\frac{ZC_0s}{T_0}-2\right)};\\
&\gamma(\xi)=\frac{ZC_0\Gamma_0\Omega(\xi-\xi_0)}{\left(\frac{ZC_0s}{T_0}-2\right)}=\gamma_0(\xi-\xi_0);\\
&\gamma_0=\frac{ZC_0\Gamma_0\Omega}{\left(\frac{ZC_0s}{T_0}-2\right)}.
\end{aligned}\right\} \tag{4.16}$$

We can further perform the following transformation of space-time variables $(\xi,\tau)\to(\rho,\theta)$:

$$\left.\begin{aligned}
&\rho(\xi,\tau)=(\xi-\xi_0)\exp(-\gamma_0\tau),\\
&\theta(\tau)=-\frac{1}{3\gamma_0}\exp(-3\gamma_0\tau),\\
&u(\xi,\tau)=W(\rho,\theta)\exp(-2\gamma_0\tau),
\end{aligned}\right\} \tag{4.17}$$

so that eq. (4.15) takes the simplified form

$$\frac{\partial W}{\partial\theta}+\alpha W\frac{\partial W}{\partial\rho}+\beta\frac{\partial^3 W}{\partial\rho^3}+\frac{\kappa}{\theta}W=0, \tag{4.18}$$

where the new parameter κ reads

$$\kappa=\frac{2}{3}-\frac{\delta}{3\gamma_0}. \tag{4.19}$$

If we assume that the Ohmic losses in this circuit are balanced by the "thermal ratchet" action of TTs, then the last term in eq. (4.15) vanishes so that

$$\frac{2}{3}-\frac{\delta}{3\gamma_0}=0;\quad \delta=2\gamma_0. \tag{4.20}$$

This gives the following condition:

$$ZG_0+Z^{-1}R_0=3ZC_0\Gamma_0\Omega;\quad Z=\frac{1}{\omega C_0}. \tag{4.21}$$

Furthermore, if we impose the simple condition

$$\left(\frac{ZC_0s}{T_0}-2\right)=1,\quad \text{with}\quad s=1,$$

we then obtain the values

$$Z=1.87\times10^8\,\Omega\quad\text{and}\quad\omega=2.7\times10^7\,\mathrm{s}^{-1}. \tag{4.22}$$

Subsequently, equating $\omega=\Omega=2.7\times10^7\,\mathrm{s}^{-1}$ we estimate the parameter Γ_0 directly from eq. (4.21) as

$$\Gamma_0=\frac{G_0}{3C_0\omega}+\frac{C_0R_0\omega}{3}=0.07+0.11=0.18. \tag{4.23}$$

Eventually, the spatial inhomogenity parameter is found to be

$$\Gamma_0\,\Omega=4.8\times10^6\,\mathrm{s}^{-1}, \tag{4.24}$$

which is one order of magnitude smaller than the inverse characteristic time $T_0^{-1}=1.83\times10^7\,\mathrm{s}^{-1}$, thus justifying eq. (3.7). In such a case ($\kappa=0$), eq. (4.18) reduces to the modified nonlinear KdV equation with spatial inhomogeneity which exhibits a single bell-shaped soliton solution subjected to the distortion due to the Ohmic dissipation. In reality, the condition in eq. (4.20) seems to be reasonably met. Otherwise, if $\kappa=1$ or $\kappa=2$, eq. (4.18) reduces to a nonlinear spherical (cylindrical) KdV equation which has already been derived in the context of ionic-acoustic plasma waves [28,29]. In general, by introducing the new transformation [30]

$$F=W\exp\left(\sigma\int\frac{\mathrm{d}\theta}{\theta}\right), \tag{4.25}$$

and substituting to eq. (4.18) we obtain

$$\frac{\partial F}{\partial\theta}=\Lambda(\theta)\cdot F\frac{\partial F}{\partial\xi}+\beta\frac{\partial^3 F}{\partial\xi^3}=0, \tag{4.26}$$

where the new function $\Lambda(\theta)$ reads

$$\Lambda(\theta)=\alpha\exp\left(-\alpha\int\frac{\mathrm{d}\theta}{\theta}\right). \tag{4.27}$$

The solution of (4.18) with $\kappa=0$ is expressed as

see equation (4.28) *above*

so that the function $u(\xi,\tau)$ has the form

see equation (4.29) *above*

If we use the set of estimated and chosen parameters in eqs. (3.5), (3.11), (3.12), (4.21) and (4.24), we readily see that all dimensionless parameters (4.16) lie between zero and one ($\beta = 0.33$, $\gamma_0 = 0.17$, $\delta = 0.34$) staying within the same order of magnitude. This suggests that we should inspect numerically the behavior of the solution of eq. (4.29) when the parameters α, β, γ_0, ξ_0 take slightly different values which is done next.

Analyzing the set of plots in fig. 9 we can see the competition between nonlinearity α and the dispersion β, as well as the role of inhomogenity γ_0. In the case represented in fig. 9b we have balanced all parameters. The soliton solution preserves its width but its amplitude decays rather rapidly so that over the length of about $500l$ it becomes negligible. The above view shows the deceleration of the soliton solution along its path. Figure 9c shows the case with increased nonlinearity ($\alpha = 0.5$). It is remarkable that it exhibits not only a higher localization but also a slower decay of its amplitude. The advantage of this case lies in the fact that the velocity of the soliton solution decreases very slightly. We estimate its average velocity as follows:

$$\Delta x \approx 400l = 400 \times 8\,\mathrm{nm} = 3.2\,\mu\mathrm{m},$$
$$\Delta t = 1000T_0 = 10^3 \times 1.2 \times 10^{-8}\,\mathrm{s} = 1.2 \times 10^{-5}\,\mathrm{s},$$
$$v = \frac{\Delta x}{\Delta t} = 0.26\frac{\mathrm{m}}{\mathrm{s}}\,.$$

The range of this soliton is $3.2\,\mu$m which is of the order of the cell's diameter. Therefore, it appears that the ionic pulse with such parameter values could be efficiently transferred within the cell.

5 Conclusions and discussion

In this paper we have developed a nonlinear model of ionic currents along MTs in the context of the polyelectrolyte character of these cytoskeletal filaments. We have taken into account the role of two aspects of this cylindrical biopolymer, its NPs and very sensitive TTs. Both of them are responsible for the nonlinear character of the overall electrical capacitance of MTs. Segmenting an MT into identical EUPs and estimating the values of corresponding electrical parameters, we were able to establish the difference equations of lumped EUP sections, eq. (4.1) and eq. (4.2). On the basis of a gradual change of voltage along MT protofilaments we imposed a continuum approximation limit leading to the nonlinear KdV type equation in (4.15) with dissipative and inhomogeneity terms, δ and $\gamma(\xi)$, eq. (4.16). This sort of equation was derived earlier within non-uniform plasma applications where ion acoustic solitons were identified, both in the cases of cylindrical and spherical geometry.

Our solution, eq. (4.29), was analysed numerically for the set of parameters in fig. 9. It is apparent that the solitonic wave loses its energy due to Ohmic resistance but it preserves the stable localised form. In the case presented in fig. 9c with an increased nonlinearity parameter, the solitonic pulse exhibits greater robustness and it progresses with an almost constant velocity and only a slight decay of its initial amplitude. This demonstrates the role of flexible TTs which could be of decisive importance for the stable localized character of ionic pulses along MTs. Interestingly, TTs are the structural elements of tubulin with the greatest diversity across cells and biological species which could be of functional significance. Moreover, it was earlier shown that MTs posses ferroelectric properties creating an intrinsic electric field [31]. It is expected that this intrinsic electric field within an MT provides the necessary electrostatic energy which may compensate for the losses due to viscosity and thus enable the near lossles propagation of solitonic pulses over long distances along MTs.

We expect that this type of localized ionic waves could play a fundamental role in many cellular processes. First the process of cell division, in the context described by Hepler [3,4], needs the synchronized depolymerisation of MTs. This can be caused by the localized Ca^{++} waves described by our model which could reach MT plus ends and trigger the onset of massive detachment of TDs from MT tips [32]. If this process were to be based on pure diffusion of Ca^{++} ions through the bulk cytosol, the resulting depolymerisation would be prone to frequent mistakes making this process very unreliable which known not to be the case.

In a very interesting paper Duke [33] explained that in hair bundles (kinocilium) in the inner ear consisting of MT doublets, the myosin motor-driven oscillation are controlled by Ca^{++} ions directed from ion channels along MTs. These ions cause a fraction of myosin motors to detach and thus tune the oscillations of kinocilium as reaction to a corresponding acoustic signal. This mechanism can also be seen in the context of localized Ca^{++} waves described in the present paper.

Finally, the experimental work of Priel *et al.* [23] demonstrated that the electrical signal amplification resulting from the presence of MTs can be directly attributed to nonlinear ionic currents guided by MTs in close similarity to the framework developed by our present model. In fact, these measurements inspired the authors to consider the role MTs can play as bio-transistors taking part in cognitive processes which is another interesting application of nonlinear ionic conductance of MTs.

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Appendix A. Glossary

ER: endoplasmic reticulum
EUP: elementary unit of protofilament
IC: ionic cloud
KdV: Korteveg de Vries
MT: microtubule
NP: nano-pore
TD: tubulin dimer
TT: tubulin tail

ER = elementary ring/ not endoplasmic reticulum

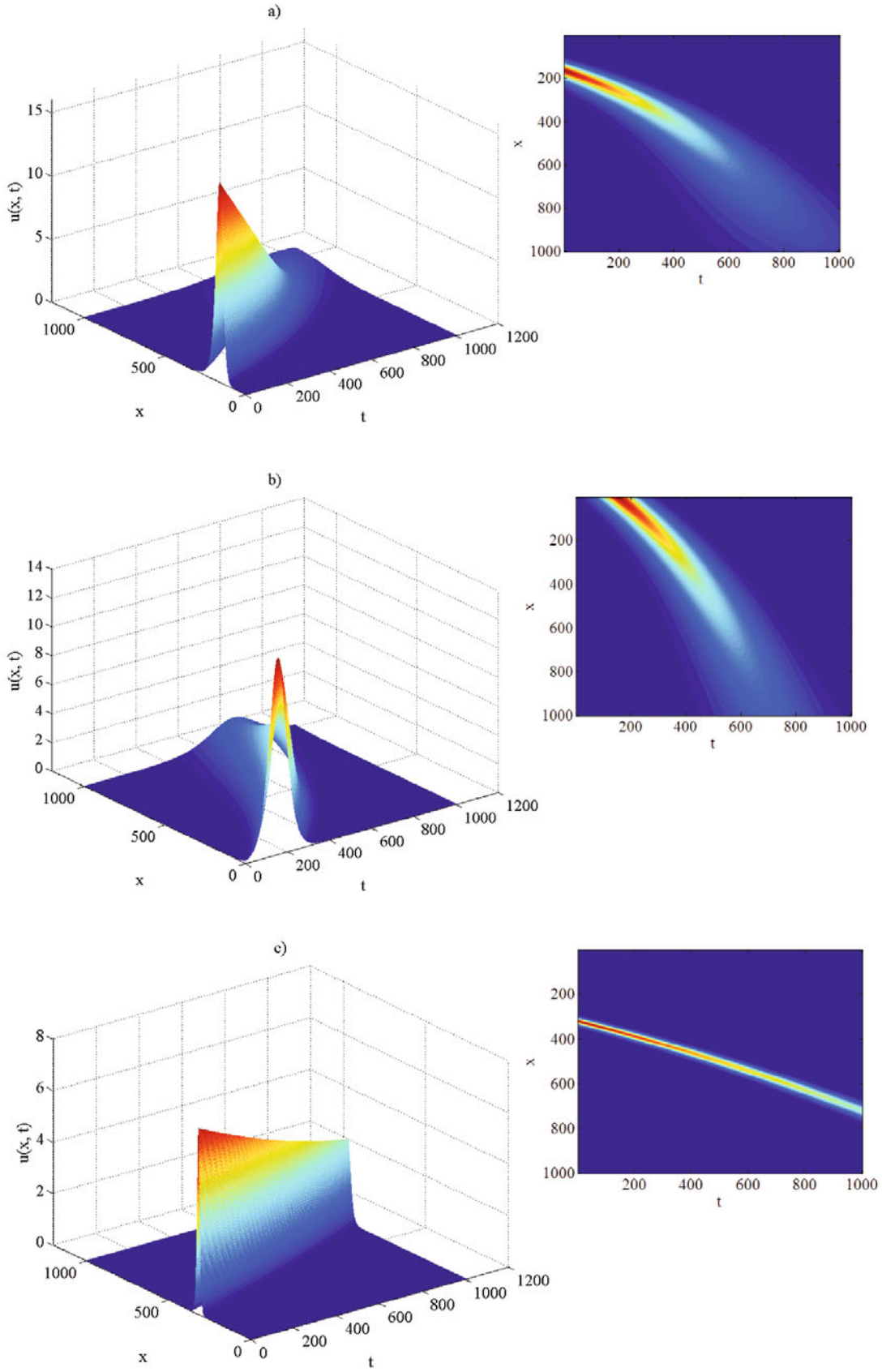

Fig. 9. Numerical solutions of $u(\xi, \tau)$ for: a) $\alpha = 0.2$; $\gamma_0 = 0.17$; $\beta = 0.33$; $\xi_0 = 0.1$. b) $\alpha = 0.1$; $\gamma_0 = 0.1$; $\beta = 0.1$; $\xi_0 = 0.1$. c) $\alpha = 0.5$; $\gamma_0 = 0.1$; $\beta = 0.1$; $\xi_0 = 0.1$.

Appendix B.

Using a Taylor series as follows:

$$u_{n\pm1} = u \pm l\frac{\partial u}{\partial x} + \frac{l^2}{2!}\frac{\partial^2 u}{\partial x^2} \pm \frac{l^3}{3!}\frac{\partial^3 u}{\partial x^3} + \frac{l^4}{4!}\frac{\partial^4 u}{\partial x^4} \pm \frac{l^5}{5!}\frac{\partial^5 u}{\partial x^5} + \ldots, \tag{B.1}$$

and applying on eqs. (4.4) and (4.5) this system now reads

$$-\frac{\partial u}{\partial x} - \frac{l}{2}\frac{\partial^2 u}{\partial x^2} - \frac{l^2}{6}\frac{\partial^3 u}{\partial x^3} - \frac{l^3}{24}\frac{\partial^4 u}{\partial x^4} - \frac{l^4}{120}\frac{\partial^5 u}{\partial x^5} - \frac{ZG_0}{l}u - \frac{ZC_0}{l}\frac{\partial u}{\partial t} + \frac{ZC_0\Gamma_0\Omega}{l}u + \frac{ZC_0\Gamma_0\Omega}{l}(t-t_0)\frac{\partial u}{\partial t} + 2\frac{Z^{3/2}b_0C_0}{l}u\frac{\partial u}{\partial t} = 0, \tag{B.2}$$

$$-\frac{\partial u}{\partial x} + \frac{l}{2}\frac{\partial^2 u}{\partial x^2} - \frac{l^2}{6}\frac{\partial^3 u}{\partial x^3} + \frac{l^3}{24}\frac{\partial^4 u}{\partial x^4} - \frac{l^4}{120}\frac{\partial^5 u}{\partial x^5} - \frac{Z^{-1}R_0}{l}u = 0. \tag{B.3}$$

Adding up the last two equations and retaining the terms up to the order of the third derivative, due to smallness of l^4, we obtain

$$-2\frac{\partial u}{\partial x} - \frac{l^2}{3}\frac{\partial^3 u}{\partial x^3} - \frac{ZC_0}{l}\frac{\partial u}{\partial t} + \frac{ZC_0\Gamma_0\Omega}{l}(t-t_0)\frac{\partial u}{\partial t} + 2\frac{Z^{3/2}b_0C_0}{l}u\frac{\partial u}{\partial t} - \frac{1}{l}(ZG_0 + Z^{-1}R_0 - ZC_0\Gamma_0\Omega)u = 0. \tag{B.4}$$

Then the standard travelling wave with speed v, for the unified function $u(x,t)$, can be employed as follows:

$$\left.\begin{aligned} &u(x,t) = u\left(\frac{x}{l} - v\frac{t}{l}\right) = u\left(\frac{x}{l} - \frac{v}{v_0}v_0\frac{t}{l}\right) = \\ &u\left(\frac{x}{l} - s\frac{t}{T_0}\right) = u(\xi), \\ &\xi = \frac{x}{l} - \tau; \quad \tau = s\frac{t}{T_0}; \quad v_0 = \frac{l}{T_0}; \quad s = \frac{v}{v_0}. \end{aligned}\right\} \tag{B.5}$$

In that respect, the set of the following transformations holds:

$$\frac{\partial u}{\partial x} = \frac{1}{l}\frac{\partial u}{\partial \xi}; \quad \frac{\partial^3 u}{\partial x^3} = \frac{1}{l^3}\frac{\partial^3 u}{\partial \xi^3}; \quad \frac{\partial u}{\partial t} = \frac{s}{T_0}\frac{\partial u}{\partial \tau} = -\frac{s}{T_0}\frac{\partial u}{\partial \xi}. \tag{B.6}$$

Using the transformations given by eqs. (B.5) and (B.6) and retaining the first derivatives of τ and going over to space derivatives of ξ in the inhomogeneous (fourth) term and nonlinear (fifth) term of eq. (B.4), we readily find

$$\left(\frac{ZC_0 s}{T_0} - 2\right)\frac{\partial u}{\partial \tau} + \frac{1}{3}\frac{\partial^3 u}{\partial \xi^3} + ZC_0\Gamma_0\Omega(\xi - \xi_0)\frac{\partial u}{\partial \xi} + 2\frac{Z^{3/2}b_0C_0 s}{T_0}u\frac{\partial u}{\partial \xi} + \left(ZG_0 + Z^{-1}R_0 - ZC_0\Gamma_0\Omega\right)u = 0. \tag{B.7}$$

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ON QUANTUM MECHANICAL ASPECTS OF MICROTUBULES

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We discuss possible quantum mechanical aspects of MicroTubules (MT), based on recent developments in quantum physics. We focus on potential mechanisms for "energy-loss-free" transport along the microtubules, which could be considered as realizations of Fröhlich's ideas on the rôle of solitons for superconductivity and/or biological matter. In particular, by representing the MT arrangements as *cavities*, we present a novel scenario on the formation of macroscopic (or mesoscopic) quantum-coherent states, as a result of the (quantum-electromagnetic) interactions of the MT dimers with the surrounding molecules of the ordered water in the interior of the MT cylinders. Such states decohere due to dissipation through the walls of the MT. Transfer of energy without dissipation, due to such coherent modes, could occur only if the decoherence time is larger than the average time scale required for energy transfer across the cells. We present some generic order of magnitude estimates of the decoherence time in a typical model for MT dynamics. Our conclusion is that the quantum coherent states play a rôle in energy transfer if the dissipation through the walls of the MT cavities is fairly suppressed, corresponding to damping time scales $T_r \geq 10^{-4}$–10^{-5} sec, for moderately large MT networks. We suggest specific experiments to test the above-conjectured quantum nature of the microtubular arrangements inside the cell. These experiments are similar in nature to those in atomic physics, used in the detection of the Rabi-Vacuum coupling between coherent cavity modes and atoms. Our conjecture is that a similar Rabi-Vacuum-splitting phenomenon occurs in the absorption (or emission) spectra of the MT dimers, which would constitute a manifestation of the dimer coupling with the coherent modes in the ordered-water environment (dipole quanta), which emerge due to "super-radiance".

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1. Introduction

MicroTubules (MT) appear to be one of the most fundamental structures of the interior of living cells.[1] These are paracrystalline cytoskeletal structures which seem to play a fundamental rôle for the cell *mitosis*. It is also believed that they play an important rôle for the transfer of electric signals and, more general, of energy in the cell.

In this latter respect it should be mentioned that energy transfer across the cells, without dissipation, had been conjectured to occur in biological matter by Fröhlich,[2] already some time ago. The phenomenon conjectured by Fröhlich was based on his one-dimensional superconductivity model: in one dimensional electron systems with holes, the formation of solitonic structures due to electron–hole pairing results in the transfer of electric current without dissipation. In a similar manner, Fröhlich conjectured that energy in biological matter could be transfered without dissipation, if appropriate solitonic structures are formed inside the cells. This idea has lead theorists to construct various models for the energy transfer across the cell, based on the formation of kink classical solutions.[3]

In the early works no specific microscopic models had been considered.[3] Recently, however, after the identification of the MT as one of the most important structures of the cell, both functionally and structurally, a model for their dynamics has been presented in Ref. 4, in which the formation of solitonic structures, and their rôle in energy transfer across the MT, is discussed in terms of classical physics. In Ref. 5 we have considered the *quantum aspects* of this one-dimensional model, and argued on the consistent quantization of the soliton solutions, as well as the fact that such semiclassical solutions may emerge as a result of "decoherence" due to environmental entanglement, according to recent ideas.[6]

The basic assumption of the model used in Ref. 5 was that the fundamental structures in the MT (more specifically of the brain MT) are Ising spin chains (one-space-dimensional structures). The interaction of each chain with the neighboring chains and the surrounding water environment had been mimicked by suitable potential terms in the one-dimensional Hamiltonian. The model describing the dynamics of such one-dimensional sub-structures was the ferroelectric distortive spin chain model of Ref. 4.

The one-dimensional nature of these fundamental building blocks, then, opens up the way for a mathematical formulation of the chain as a *completely intergrable* field theory model, characterized by an infinite-number of conservation laws. The latter are associated with global excitation modes, completely *delocalised* in the chain space-time. This integrability structure proves sufficient in providing a satisfactory solution to memory coding and capacity.[5] Such features might turn out to be important for a model of the brain as a *quantum computer*.

This formulation allowed the authors of Ref. 5 to employ the so-called Liouville (noncritical) string theory approach to decoherence,[7] which is a formalism to treat such completely integrable spin systems embedded in a "two-dimensional quantum

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gravity environment". At this point we should stress that such an environment does not *necessarily* represent realistic space-time effects. As stressed in Ref. 5, an external stimulus may distort the neighboring space-time of the unpaired spins of the MT dimers, and such a distortion may be *effectively* described by a two-dimensional "metric" in the Liouville picture of MT,[5] which should not be necessarily identified with a relatistic four dimensional space time metric.[a]

The coupling of the MT chain with its "environment" may lead, according to the analysis of Ref. 5, to the formation of *macroscopic quantum coherent solitonic* states. Then, the details of the environment would play a crucial rôle in determining the time scales over which such coherent states could be sustained *before collapsing to classical ground states.*

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Such analyses have yield important results in the past,[6,10–12] concerning the passage from the quantum to classical world. In particular, in Ref. 6 it was suggested that quantum coherent states may appear as a result of decoherence, depending on the nature of the environment.[13] Such states are minimum entropy/uncertainty states, which propagate "almost classically" in time, in the sense that their shape is retained during evolution. Such "almost classical" states are termed "pointer states" by Zurek.[6] In view of the results of Ref. 12, concerning the localization of the wave function in open (stochastic) quantum-mechanical systems, the emergence of pointer states, as a result of decoherence, may be interpreted as implying that the localization process of the state vector had stopped at a stage where it is not complete, but is such that the resulting (minimum-entropy) state is least susceptible to the effects of the environment.

Not all environments admit such pointer states.[13] Here lies one of the advantages of viewing the MT system of chains as a completely integrable Liouville-string theory. The model possesses a pointer basis,[14] which in Ref. 5 has been identified with the quantized (via squeezed-coherent states) soliton solutions of Ref. 4. Another advantage lies on the fact that energy is *conserved* on the average in the model, despite the environmental entanglement. This is for purely stringy reasons pertaining to the Liouville approach. The reader may find details in Ref. 7. Physically this means that such a formalism is capable of describing *energy-loss-free* transport in biological cells, thereby providing a realization of Fröhlich's ideas.[2]

In this article we focus our attention on the *microscopic nature* of such coherent states, and try to understand their emergence by as much "conventional" physics arguments as possible. By "conventional" we mean scenaria based on *quantum electrodynamics*, which are the major interactions expected to dominate at the energy

[a]It is however, possible, that such a distortion involves a sufficient movement of mass so as to create a virtual microscopic singularity in the surrounding space time[8,9,5]; this could then cause collapse of the macroscopic quantum state, leading to *conscious perception.* Such daring assumptions/conjectures have gain some support by the fact that in certain physical models of MT dynamics[5] the estimated time of collapse is of order $\mathcal{O}(1\text{ sec})$, which is in excellent qualitative agreement with a plethora of experimental/observational findings in Neurobiology. In the present work we shall not discuss these issues.

scales of the MT[b] We make an attempt to pick up possible sources of decoherence, consistent with the above-mentioned energy-loss-free transfer scenario in biological cells, and estimate the associated time scales.

The structure of this article is a follows: in Sec. 2 we review briefly the physical one-dimensional model describing MT dynamics, and discuss briefly the mathematical background for obtaining solitonic structures. An important rôle for the existence of the solitonic structures is played by the ordered water in the interior of the MT. The importance of the water in the interior of the MT has been emphasized by S. Hameroff, in connection with information processing, already some time ago.[15,16] In our article we present a scenario for the formation of electric-dipole quantum coherent modes in the water of the MT, inspired by earlier suggestions on "laser-like" behavior of the water.[17–19] Such coherent modes couple with the unpaired electrons of the MT dimers.

In this work we conjecture that such a coupling would result in the so-called Vacuum Field Rabi Splitting (VFRS),[20] a sort of dynamical Stark effect occuring in the emission or absorption spectra of the dimers as a result of the vaccum quantum fluctuations. This phenomenon is characteristic of the behavior of atoms inside quantum electrodynamical cavities, the rôle of which is played in our work by the MT cylindrical structures themselves.

A review of this phenomenon in the atomic physics is given in Sec. 3. The atomic Rabi Splitting is a consequence of the coupling of the atoms with the coherent modes of electromagnetic radiation inside the cavity. It is considered by many as a "proof" of the quantum nature of the electromagnetic radiation.[20] In our work, we consider this phenomenon, if true, as a manifestation of the quantum mechanical nature of the MT arrangement inside the cell.

This is discussed in Sec. 4, where we present a scenario according to which the MT arrangements inside the cell act as fairly isolated *cavities*. The ordered-water molecules in their interior, then, provides an environment, necessary to form coherent quantum modes (dipole quanta),[17–19] whose coupling with the dimers results in VFRS in the respective absorption spectra. Due to dissipation from the cavity walls, there is decoherence of the combined dimer-cavity system coherent state. The resulting decoherence time scales are estimated, for various sources of envirnoments. The main conclusion of our analysis is that in order for the quantum coherent states to play a rôle in dissipationless energy transfer, i.e. the decoherence time to be larger than the scale required for energy transfer in MT — estimated to be of $\mathcal{O}(5 \times 10^{-7}$ sec), for a moderately long MT, in the model of Ref. 4 — the damping time scales of the MT cavities must be $T_r \geq 10^{-4}$–10^{-5} sec.

[b]According to the above discussion, then, quantum gravity entanglement could be, if at all, important for the decoherence of the coherent state formed as a result of the water-dimer coupling. However this would require extreme isolation of the MT system, so that any conventional decoherence due to ordinary environmental entanglement is suppressed. Such isolated MT subsystems have been conjectured to occur in certain parts of brain, in which case quantum gravity plays a rôle in consciousness by inducing a collapse of the coherent-state of the MT dimers.[5,8,9] At present, we do not know whether this is feasible inside the brain.

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Finally, in Sec. 5, we present our conclusions and suggest some experiments to test the above-conjectured quantum mechanical origin of MT arrangements inside the cell. The experiments are inspired by the physics of Rydberg atoms in electromagnetic cavities.[21,22]

2. Review of the Physical Model for Microtubule Dynamics

2.1. *Classical considerations*

MicroTubules (MT) are hollow cylinders comprised of an exterior surface (of cross-section diameter 25 nm) with 13 arrays (protofilaments) of protein dimers called tubulines. The interior of the cylinder (of cross-section diameter 14 nm) contains *ordered water* molecules, which implies the existence of an electric dipole moment and an electric field. The arrangement of the dimers is such that, if one ignores their size, they resemble triangular lattices on the MT surface. Each dimer consists of two hydrophobic protein pockets, and has an unpaired electron. There are two possible positions of the electron, called α and β *conformations.* When the electron is in the β-conformation there is a 29° distortion of the electric dipole moment as compared to the α conformation.

In standard models for the simulation of the MT dynamics, the "physical" degree of freedom — relevant for the description of the energy transfer — is the projection of the electric dipole moment on the longitudinal symmetry axis (x-axis) of the MT cylinder. The 29° distortion of the β-conformation leads to a displacement u_n along the x-axis, which is thus the relevant physical degree of freedom. This way, the effective system is one-dimensional (spatial), and one has the possibility of a quantum integrable system.[5]

Information processing occurs via interactions among the MT protofilament chains. The system may be considered as similar to a model of interacting Ising chains on a trinagular lattice, the latter being defined on the plane stemming from fileting open and flattening the cylindrical surface of MT. Classically, the various dimers can occur in either α or β conformations. Each dimer is influenced by the neighboring dimers resulting in the possibility of a transition. This is the basis for classical information processing, which constitutes the picture of a (classical) cellular automatum.

The *quantum computer character* of the MT network results from the *assumption* that each dimer finds itself in a superposition of α and β conformations.[8] Viewed as a *two-state* quantum mechanical system, the MT tubulin dimers couple to conformational changes with 10^{-9}–10^{-11} sec transitions, corresponding to an angular frequency $\omega \sim \mathcal{O}(10^{10})$–$\mathcal{O}(10^{12})$ Hz. In the present work we assume the upper bound of this frequency range to represent (in order of magnitude) the characteristic frequency of the dimers, viewed as a two-state quantum-mechanical system:

$$\omega_0 \sim \mathcal{O}(10^{12})\ \text{Hz}\,. \tag{1}$$

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The scenario for quantum computation in MT presuposes that there exists a macroscopic coherent state among the various chains. Let us now try to understand its emergence. Let u_n be the displacement field of the nth dimer in a MT chain. The continuous approximation proves sufficient for the study of phenomena associated with energy transfer in biological cells, and this implies that one can make the replacement

$$u_n \to u(x,t) \tag{2}$$

with x a spatial coordinate along the longitudinal symmetry axis of the MT. There is a time variable t due to fluctuations of the displacements $u(x)$ as a result of the dipole oscillations in the dimers. At this stage, t is viewed as a reversible variable.

The effects of the neighboring dimers (including neighboring chains) can be phenomenologically accounted for by an effective potential $V(u)$. In the model of Ref. 4 a double-well potential was used, leading to a classical kink solution for the $u(x,t)$ field. More complicated interactions are allowed in the string picture, as explained in Ref. 5. More generic polynomial potential have also been considered in Ref. 5.

The effects of the surrounding water molecules can be summarized by a viscuous force term that damps out the dimer oscillations,

$$F = -\gamma \partial_t u \tag{3}$$

with γ determined phenomenologically at this stage. This friction should be viewed as an environmental effect, which however does not lead to energy dissipation, as a result of the non-trivial solitonic structure of the ground-state and the non-zero constant force due to the electric field. This is a well known result, directly relevant to energy transfer in biological systems.[3]

In mathematical terms the effective equation of motion for the relevant field degree of freedom $u(x,t)$ reads:

$$u''(\xi) + \rho u'(\xi) = P(u)\,, \tag{4}$$

where $\xi = x - vt$, v is the velocity of the soliton, $\rho \propto \gamma$,[4] and $P(u)$ is a polynomial in u, of a certain degree, stemming from the variations of the potential $V(u)$ describing interactions among the MT chains.[5] In the mathematical literature[23] there has been a classification of solutions of equations of this form. For certain forms of the potential, the solutions include *kink solitons* that may be responsible for dissipation-free energy transfer in biological cells.[c] A typical propagation velocity of the kink

[c]In the "string picture of MT" the various solutions are related[5] by appropriate Renormalization-Scheme changes on the world-sheet of the string (generalizaing appropriately the particle like structures). This is reflected in certain *ambiguities* in the form of the string potential $V(u)$; what seems to be a unique phenomenological potential in point-like theory, determining various forms of soliton solutions, may not have such a unique interpretation in the completely-integrable (noncritical) string picture of MT. This suggests that the point-like field theory soliton solutions, discussed above, constitute only a "low-energy" approximation of more complicated (solitonic) ground states of the string. As discussed in the context of Liouville strings in Ref. 5, such *stringy ground states* share similar properties with their point–like theory counterparts as far as dissipation-free energy transfer is concerned.

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solitons of Ref. 4 is $v \sim 2$ m/sec. This implies that for moderately long microtubules, of length $L \sim 10^{-6}$ m, such kinks transport energy in

$$t_{\rm F} \sim 5 \times 10^{-7} \text{ sec}. \tag{5}$$

This scale is larger than the time scale that Fröhlich had conjectured as corresponding to the frequency of the coherent phonons in biological matter ($t \sim 10^{-11}$–10^{-12} s). In this article, however, we shall keep calling the time (5) the Fröhlich scale, since, upon quantization, the kink soliton solutions of Ref. 4 yield quantum coherent states[5] that are very similar to Fröhlich's phonons.

Therefore, in the above "phenomenological" approach to the MT physics, the importance of the water environment can be seen formally as follows: were it not for the friction term (3) there would be no stable solitonic structures in the ferroelectric distortive model of Ref. 4.[d]

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2.2. *Quantum considerations*

Let us attempt a microscopic analysis of the physics underlying the interaction of the water molecules with the dimers of the MT. Our investigation points towards the fact that, as a result of the ordered structure of the water environment, there appear *collective* coherent modes, which in turn interact with the dimer structures (mainly through the unpaired electrons of the dimers) leading to the formation of a quantum coherent solitonic state that extends over the whole network of MT. According to the idea put forward in Ref. 5, following Ref. 6, such coherent states should be viewed as the result of *decoherence* of the dimer system due to its interaction/coupling with the water environment.

As we shall argue below, such a coupling could be detected by a phenomenon analogous to what is happening in atoms interacting with coherent modes of the electromagnetic radiation in *Cavities*, namely the *Vacuum-Field Rabi Slitting* (VFRS).[20] Our *conjecture* is that the interior of MT, full of ordered water molecules, can be viewed as a *cavity* rather than a *wave guide*; the cavity structure can be formed by "closing" the ends of the MT, and is a way of providing a fairly isolated system which can sustain coherent modes.

This type of environmental entanglement is described in the Liouville framework by the world-sheet renormalization-group β-function, β^u, of the displacement field, $u(x)$, of the dimers, described above. A non-trivial $\beta^u \neq 0$ expresses deviations from *conformal invariance*, as a consequence of environmental entanglement.[5] The β-function corresponds to an energy-dependent environmental coupling, since its magnitude depends on the energy of the field $u(x)$[5]:

$$\beta^u = \mathcal{O}[(E/M_s)^2] \tag{6}$$

[d]In the (noncritical) Liouville string picture such a coupling is responsible for the appearance of the appropriate "dilaton"-like terms in the world-sheet action, which are essential for consistency of the Liouville string.[24,25,5]

where M_s is a string scale, characteristic of the problem. The scale M_s acts as an ultraviolet cutoff for the energies of the low-energy field theory obtained from the "string", and its size depends on the details of the subsystem $u(x,t)$ as well as the environment.

The Master Equation for the time evolution of the density matrix ρ of the displacement field u in the Liouville approach reads[7,5]:

$$\partial_t \rho = i[\rho, H] + i\beta^u[\rho, u]M_s\,, \tag{7}$$

where H is the Hamiltonian of the dimer chain. The decoherence time in this approach is estimated as[5]:

$$t_{\text{decoh/Liouv}} \sim \frac{1}{\mathcal{N}\beta^u M_s} \tag{8}$$

where $\mathcal{N}$ denotes the number of dimers in the MT chain. The time scale (8) is the time over which the solitonic quantum coherent pointer states are formed in the MT networks, as a result of the ordered-water-induced decoherence.

We now remark that, in the approach of viewing the MT as cavities, the main source of dissipation will then be the leakage of photons or other coherent modes from the cavity. Its rate can be assumed *small* for the MT of the brain, otherwise the incoherent mode will dominate. This dissipation constitutes an ordinary environment, which cannot lead to the formation of pointer states, but instead induces eventual collapse of the solitonic states into completely classical ground states.

Hence, according to the above discussion, there are two stages when decoherence plays an important rôle.

(i) The first stage concerns the coupling of the system of dimers with the coherent modes formed in the ordered water. This coupling pertains to a specific environment which induces decoherence in the dimer sub-system, producing quantum coherent states ("pointer states") according to Refs. 6 and 5. The time scale of this decoherence corresponds to the time scale necessary for the formation of the solitonic coherent states of Ref. 5, see Eq. (8) above.

(ii) The second stage refers to the decoherence due to ordinary dissipation through the walls of the (imperfect) MT cavities. Such an ordinary environment does not admit a pointer basis,[13] but causes collapse of the quantum coherent state of stage (i) down to a classical ground state.

In this article, we shall estimate the decoherence time scales of both stages (i) and (ii) and compare them with the Fröhlich scale. Moreover, we shall propose specific experimental tests for the detection of the quantum-mechanical behavior of the MT arrangements inside the cell. An important feature of all these tests is, as we mentioned previously, the Vacuum-Field Rabi Splitting phenomenon, to a brief description of which we now turn for instructive purposes.

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3. On the Rabi Splitting in Atomic Physics and the Quantum Nature of Electromagnetic Radiation

3.1. *Description of the Rabi splitting phenomenon*

In this section we shall recapitulate briefly the VFRS in atomic physics. This phenomenon has been predicted for the emission spectra of atoms inside electromagnetic cavities,[20] in an attempt to understand the *quantized nature* of the electromagnetic radiation. The basic principle underlying the phenomenon is that, in the presence of an interaction among two oscillators in resonance, the frequency degeneracy is *removed* by an amount proportional to the strength of the coupling. In the cavity QED case of Ref. 20, one oscillator consists of a small collection of N atoms, whilst the other is a resonant mode of a high-Q(uality) cavity.[e] Immediately after the suggestion of Ref. 20, a similar phenomenon has been predicted for absorption spectra of atoms in cavities.[27]

By now, the situation has been verified experimentally on a number of occasions.[21] In such experiments one excites the coupled atom-cavity system by a tuneable field probe. The excitation is then found resonant *not* at the "bare" atom or cavity frequencies but at the *split* frequencies of the "dressed" atom-field system. The splitting is enhanced for collections of atoms. For instance, as we shall review below, for a system of N atoms, the split is predicted to be[27]:

$$\text{Rabi splitting} = 2\lambda\sqrt{N}\,, \qquad 2\lambda = \text{Rabi splitting of a single atom}\,. \tag{9}$$

Despite its theoretical prediction by means of quantum mechanical oscillator systems coupled with a *quantized* radiation field mode in a cavity, at present there seems to be still a *debate* on the nature of the phenomenon:

(i) the dominant opinion is that the Rabi splitting is a manifestation of the *quantum nature* of the electromagnetic radiation (cavity field), and is caused as a result of an *entanglement* between the atom and the cavity coherent modes of radiation. It is a sort of Stark effect, but here it occurs in the absence of an external field.[20] This "dynamical Stark effect" is responsible for a splitting of the resonant lines of the atoms by an amount proportional to the collective atomic-dipole amplitude.

(ii) there is however a dual interpretation,[28] which claims that the splitting can be observed in optical cavities as well, and it is simply a result of *classical* wave mechanics inside the cavity, where the atomic sample behaves as a *refractive medium* with a *complex* index, which splits the cavity mode into two components.

Irrespectively of this second classical interpretation, one *cannot deny* the presence of the phenomenon in entangled atom-quantum-coherent mode systems.

[e]The quantity Q is defined as the ratio of the stored-energy to the energy-loss per period,[26] by making the analogy with a damped harmonic oscillator.

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This is the point of view we shall be taking in this work, in connection with our picture of viewing MT filled with ordered water as cavities. We shall try to make specific experimental predictions that could shed light in the formation of quantum coherent states, and their eventual decoherence. As mentioned above, the latter could be due to the interaction of the dimer unpaired spins (playing the rôle of the atoms in the Rabi experiments) with the ordered-water coherent modes (playing the rôle of the cavity fields). Possible scenaria for the origin of such cavity coherent modes will be described below.

Let us first recapitulate briefly the theoretical basis of the Rabi-splitting phenomenon, which will allow the non-expert reader to assess the situation better. We shall present the phenomenon from a point of view that will help us transcribe it directly to the MT case. Consider an atom of a frequency ω_0 in interaction with a single coherent mode of electromagnetic radiation field of frequency ω. The relevant Hamiltonian is:

$$H = \hbar\omega_0 \sum_i S_i^z + \hbar\omega a^\dagger a + \sum_i (\hbar\lambda S_i^+ a + \text{h.c.}), \tag{10}$$

where $a^\dagger$, a are the creation and annihilation cavity radiation field modes, S_i^z, $S_i^\pm$ are the usual spin-1/2 operators, and λ is the atom-field coupling. The atom-field system is not an isolated system, since there is *dissipation* due to the interaction of the system with the surrounding world. An important source of dissipation is the leakage of photons from the cavity at some rate κ. If the rate of dissipation is not too big, then a quantum coherent state can be formed, which would allow the observation of the vacuum-field Rabi oscillations. The density matrix ρ of the atom-field system obeys a Markov-type master equation for the evolution in time t[27]:

$$\partial_t \rho = -\frac{i}{\hbar}[H, \rho] - \kappa(a^\dagger a \rho - 2a\rho a^\dagger + \rho a^\dagger a) \tag{11}$$

The limit $\kappa \ll \lambda\sqrt{N}$ guarantees the possibility of the formation of a quantum coherent state, i.e. this limit describes environments that are weakly coupled to the system, and therefore the decoherence times (see below) are very long. In this limit one can concentrate on the off-diagonal elements of the density matrix, and make the following ("secular") approximation for their evolution[27]:

$$\partial_t \rho_{ij} = -\frac{i}{\hbar}(E_i - E_j)\rho_{ij} - \Gamma_{ij}\rho_{ij}\,, \tag{12}$$

where Γ_{ij} denotes the damping factor, pertaining to the weak coupling of the atom-field system to the environment. The analysis of Ref. 27 pertained to the evaluation of the susceptibility tensor of the system, $\chi_{\alpha\beta}$, which can be calculated by considering its interaction of the system with an external field of frequency Ω. The absorption spectrum is proportional to $\text{Im}\,\chi(\Omega)$. A standard quantum-mechanical

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computation yields the result[27]:

$$\mathrm{Im}\,\chi(\Omega) = \cos^2\theta \frac{\Gamma_-/\pi}{\Gamma_-^2 + \{\Omega - \omega_0 + \Delta/2 - (1/2)(\Delta^2 + 4N\lambda^2)^{1/2}\}^2} + \sin^2\theta \frac{\Gamma_+/\pi}{\Gamma_+^2 + \{\Omega - \omega_0 + \Delta/2 + (1/2)(\Delta^2 + 4N\lambda^2)^{1/2}\}^2} \tag{13}$$

with $\Delta \equiv \omega_0 - \omega$. In the above expression the damping factors $\Gamma_\pm$ represent the damping in the equation of motion for the off-diagonal element of the density matrix $\langle \Psi_0 | \rho | \Psi_\pm^{S,C} \rangle$ where $\Psi_\pm$ are eigenfucntions of H, classified by the eigenvalues of the operators S^2, and $S^z + a^\dagger a \equiv C$, in particular[27]: $S = N/2$, and $C = 1 - N/2$.

The expression (13) summarizes the effect of Rabi-vacuum splitting in absorption spectra of atoms: there is a doublet structure (splitting) of the absorption spectrum with peaks at:

$$\Omega = \omega_0 - \frac{\Delta}{2} \pm \frac{1}{2}(\Delta^2 + 4N\lambda^2)^{1/2}\,. \tag{14}$$

For resonant cavities the splitting occurs with equal weights

$$\Omega = \omega_0 \pm \lambda\sqrt{N}\,. \tag{15}$$

Notice here the *enhancement* in the effect for multi-atom systems $N \gg 1$. This is the "Vacuum Field Rabi Splitting phenomenon", predicted in emission spectra in Ref. 20. As we have already mentioned, the above derivation pertains to absorption spectra, where the situation is formally much simpler.[27] It is also this latter case that is of interest to us for the purposes of this work.

The quantity $2\lambda\sqrt{N}$ is called the "Rabi frequency".[20] From the emission-spectrum theoretical analysis an estimate of λ may be inferrred which involves the matrix element, $\underline{d}$, of atomic electric dipole between the energy states of the two-level atom[20]:

$$\lambda = \frac{E_{\mathrm{vac}}\underline{d}\cdot\underline{\epsilon}}{\hbar} \tag{16}$$

where $\underline{\epsilon}$ is the cavity (radiation) mode polarization, and

$$E_{\mathrm{vac}} \sim \left(\frac{2\pi\hbar\omega_c}{\varepsilon_0 V}\right)^{1/2} \tag{17}$$

is the r.m.s. vacuum field amplitude at the center of the cavity of volume V, and of frequency ω_c, with ε_0 the dielectric constant of the vacuum[f]: $\varepsilon_0 c^2 = 10^7/4\pi$, in M.K.S. units. As mentioned above, there are simple experiments which confirmed this effect,[21] involving beams of Rydberg atoms resonantly coupled to superconducting cavities.

[f]For cavities containing other dielectric media, e.g. water in the case of the MT, ε_0 should be replaced by the dielectric constant ε of the medium.

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The situation which is of interest to us involves atoms that are *near resonance* with the cavity. In this case $\Delta \ll \omega_0$, but such that $\lambda^2 N/|\Delta|^2 \ll 1$; in such a case, formula (14) yields two peaks that are characterized by dispersive frequency shifts $\propto 1/\Delta$:

$$\Omega \simeq \omega_0 \pm \frac{N\lambda^2}{|\Delta|} + \mathcal{O}(\Delta)\,, \tag{18}$$

whilst no energy exchange takes place between atom and cavity mode. This is also the case of interest in experiments using such Rabi couplings to construct Schrödinger's cats in the laboratory, i.e. macro(meso)scopic combinations "measuring apparatus + atoms" to verify decoherence experimentally. The first experiment of this sort, which confirms theoretical expectations, is described in Ref. 29.

Another important issue, which has been used above, and in Ref. 29, is the *dephasing* of the atom as a result of the above-described atom-field Rabi entanglement. To understand better the situation, let us discuss a more generic case, that of a three-state atom, f, e, g, with energies $E_g > E_e > E_f$. Suppose one is interested in the transition $f \to e$ by absorption, in the presence of atoms in interaction with a cavity mode. Calling $D^+_{ef} \equiv |e\rangle\langle f|$, $D^-_{ef} \equiv |f\rangle\langle e| = (D^+_{ef})^\dagger$, we have the effective Hamiltonian for the transition $f \to e$[30]:

$$H^{ef}_{\rm eff} = \hbar\omega_{\rm eff} D^+_{ef} D^-_{ef}\,, \qquad \omega_{\rm eff} = \omega_{ef} + \frac{\lambda^2 n}{\Delta}\,. \tag{19}$$

where the "effective" frequency $\omega_{\rm eff}$ incorporates the dispersive frequency shifts (18) of the Rabi effect, appropriate for near-resonant atom-cavity-field systems; n is the number of cavity photons.[g]

Consider now an experiment to measure, say, the photon number n in the cavity. The relevant probe P can be the above-described three-state atom, in a superposition of e and f states. In this picture the photon number n is an eigenvalue of the cavity signal operator $a^\dagger_{\rm s} a_{\rm s}$; the interaction Hamiltonian between atom and cavity then reads[30]:

$$H_{\rm I} = \frac{\hbar\lambda^2}{\Delta} a^\dagger_{\rm s} a_{\rm s} D^+_{ef} D^-_{ef}\,. \tag{20}$$

The probe observable is the atomic dipole operator:

$$A_{\rm P} = \frac{1}{2i}(D^+_{ef} - D^-_{ef})\,. \tag{21}$$

The Heisenberg evolution of this operator yields

$$i\hbar \frac{d}{dt} A_{\rm P} = [A_{\rm P}, H^{ef}_{\rm eff} + H_{\rm I}] \tag{22}$$

[g]Notice that the $\sqrt{n}$ characteristic scaling law for the Rabi splitting (15) is *also valid* in the case of interaction of a *single* atom with n cavity oscillator quanta (e.g. a coherent cavity mode).

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from which it is easily seen that in a time interval t the phase of the probe is changing by:

$$\Delta\phi = \omega_{ef}t + \frac{\lambda^2 n}{\Delta}t\,. \tag{23}$$

The case of interest for the experiment of Ref. 29 is a two-state atom; the resulting phase shift is then obtained from (23) by setting $\omega_{ef} = 0$. In that case, the phase entanglement due to the atom-field Rabi coupling is

$$\Delta\phi_{\mathrm{R}} = \frac{\lambda^2 n}{\Delta}t \tag{24}$$

for a near resonance atom-field system, with a small detuning Δ. Such a situation is encountered precisely in the experiment of Ref. 29, which will constitute our prototype for the extension of the above results to the MT case. Before doing so it is instructive to review briefly the experiment of Ref. 29, where decoherence of an open mesoscopic system ("Schrödinger's cat") *has been demonstrated.*

3.2. *Rabi splitting and decoherence: Experimenal verification*

The experiment consists of sending a Rubidium atom with two circular Rydberg states e and g, through a microwave cavity storing a small coherent field $|\alpha\rangle$. The coherent cavity mode is mesoscopic in the sense of possessing an average number of photons of order $\mathcal{O}(10)$. The atom-cavity coupling is measured by the Rabi frequency $2\lambda/2\pi = 48$ kHz. The condition for Rabi dispersive shifts (18) is satisfied by having a $\Delta/2\pi \in [70, 800]$ kHz.

The set up of the experiment is as follows: The atom is prepared in the superposition of e, g states, by virtue of a resonant microwave cavity R_1. Then it crosses the cavity C, which is coupled to a reservoir that damps its energy (*dissipation*) at a characteristic time scale $T_r \ll 1.5$ ms. Typical cavities, used for atomic scale experiments such as the above, have dimensions which lie in the mm range or at most cm range. In the cavity C a number of photons varying from 0 to 10 is injected by a pulse source. The field in the cavity relaxes to vacuum — thereby causing *dissipation* through leakeage of photons through the cavity — during a time T_r, before being regenerated for the next atom. The experiment is at an effective temperature of $T = 0.6$ K, which is low enough so as to minimize thermal effects. After leaving C, the atom passes through a second cavity R_2, identical to R_1; one then measures the probability of finding the atom in the state, say, g. This would mean decoherence. The decoherence time is measured for various photon numbers; this helps testing the theoretical predictions that decoherence between two "pointer states" of a quantum superposition occurs at a rate proportional to the square of the distance among the states.[6,11,31,32]

Let us understand this latter point better. The coherent oscillator states, characterizing the cavity modes, constitute such a pointer basis: an oscillator in a coherent state is defined by the average number of oscillator quanta n as: $|\alpha\rangle : |\alpha| = \sqrt{n}$.

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Then, consider the measurement of the above-described experiment, according to which there is only a phase entanglement between the cavity and the atom: the combined atom-cavity (meter) system is originally in the state

$$|\Psi\rangle = |e, \alpha e^{i\phi}\rangle + |g, \alpha e^{-i\phi}\rangle \tag{25}$$

According to (24) the dephasing is atomic-level dependent $\phi \propto \lambda^2 tn/\Delta$. Coupling the oscillator to a reservoir, that damps its energy in a characteristic time scale T_r, produces decoherence. According to the general theory[10,31–32] the latter occurs during a time scale inversely proportional to the square of the distance between the "pointer" states D^2:

$$t_{\text{collapse}} = \frac{2T_r}{D^2} \tag{26}$$

This is easy to verify in a prototype toy model, involving only a phase entanglement, i.e. without energy dissipation. To this end, consider the Hamiltonian:

$$H = \hbar\omega a^\dagger a + a^\dagger a\Gamma \tag{27}$$

with $a^\dagger$, a creation and annihilation operators of a quantum oscillator, and Γ a phase damping. The "pointer" states for this problem are characterized by the eigenvalues n of the number operator $a^\dagger a$ which commutes with the Hamiltonian $[a^\dagger a, H] = 0$, i.e. the pointer basis is $\{|n\rangle\}$. The pertinent Markov master equation for the density matrix, ρ, reads:

$$\partial_t \rho = \frac{\kappa}{2}(2a^\dagger a\rho a^\dagger a - \rho a^\dagger a a^\dagger a - a^\dagger a a^\dagger a\rho) \tag{28}$$

Writing $\rho(t) = \sum_{m,n} \rho_{mn}(t)|n\rangle\langle m|$ it is straightforward to determine the time dependence of $\rho_{nm}(t)$ from (28)

$$\rho_{nm}(t) = e^{-\kappa(n-m)^2 t/2}\rho_{nm}(0) \tag{29}$$

which implies that the coherence between two different ($m \neq n$) pointer (number) states is damped at a typical scale given by (26), where $T_r = 1/\kappa$, and the distance D is given by $n - m$ in the above example. As mentioned previously, this inverse D^2-behavior seems to be a *generic* feature of open (decohering) systems, including open string models.[7,5]

In the set up of Ref. 29 the distance D is given by

$$D = 2\sqrt{n}\sin\phi \simeq 2n^{3/2}\frac{\lambda^2 t}{\Delta} \tag{30}$$

for Rabi couplings 2λ, such that $\lambda^2 tn \ll \Delta$. For mesoscopic systems $n \sim 10$, $D > 1$, and, hence, decoherence occurs over a much shorter time scale than T_r; in

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particular, for $\Delta/2\pi \sim 70$ kHz the decoherence time is $0.24T_r$.[29] This concludes the construction of a Schrödinger's cat, and the associated "measurement process". Notice that the above construction is made in two stages: first it involves an interaction of the atom with the cavity field, which results in a coherent state of the combined "atom-meter", and then *dissipation* is induced by coupling the cavity (measuring apparatus) to the environment, which damps its energy, thereby inducing *decoherence* in the "atom + meter" system. The important point to realize is that the more macroscopic the cavity mode is (i.e. the higher the number of oscillator quanta), the shorter the decoherence time is. This is exactly what was to be expected from the general theory.[6,7,10,11]

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4. Rabi Vacuum Coupling in MT and Energy Transfer in the Cell

4.1. *Microscopic mechanisms for the formation of coherent states in MT*

Above we have sketched the experimental construction of a mesoscopic quantum coherent state (a "Schrödinger's cat" (SC)). The entanglement of the atom with the coherent cavity mode, manifested experimentally by the "vacuum Rabi spliting", leads to a quantum-coherent state for the combined atom-cavity system (SC), comprising of the superposition of the states of the two-level Rydberg atom. Dissipation induced by the leakage of photons in the cavity leads to decoherence of the coherent atom-cavity state, in a time scale given by (26). This time scale depends crucially on the nature of the coupled system, and the nature of the "environment". It is the point of this session to attempt to discuss a similar situation that conjecturally occurs in the systems of MT. We believe that understanding the formation of Schródinger's cats in MT networks will unravel, if true, the mysteries of the brain as a (quantum) computer, which might also be related to the important issue of "conscious perception", as advocated in Refs. 5, 8 and 9.

The first issue concerns the nature of the "cavity-field modes". Our point in this section is to argue that the presence of *ordered water*, which seems to occupy the interior of the microtubules,[16] plays an important rôle in producing coherent modes, which resemble those of the ordinary electromagnetic field in superconducting cavities, discussed above.

Let us first review briefly some suggestions about the rôle of the electric dipole moment of water molecules in producing coherent modes after coupling with the electromagnetic radiation field.[17] Such a coupling implies a "laser-like" behavior. Although it is not clear to us whether such a behavior characterizes ordinary water, in fact we believe it does not due to the strong suppression of such couplings in the disordered ordinary water, however it is quite plausible that such a behavior characterizes the ordered water molecules that exist in the interior of MT.[16] If true, then this electric dipole-quantum radiation coupling will be responsible, according to the analysis of Ref. 17, for the appearance of *collective* quantum coherent modes.

The Hamiltonian used in the theoretical model of Ref. 17 is:

$$H_{\rm ow} = \sum_{j=1}^{M} \left[\frac{1}{2I} L_j^2 + \underline{A} \cdot \underline{d}_{ej} \right] \tag{31}$$

where A is the quantized electromagnetic field in the radiation gauge,[17] M is the number of water molecules, L_j is the total angular momentum vector of a single molecule, I is the associated (average) moment of inertia, and d_{ej} is the electric dipole vector of a single molecule, $|d_{ej}| \sim 2e \otimes d_e$, with $d_e \sim 0.2\,\text{Å}$. As a result of the dipole-radiation interaction in (31) coherent modes emerge, which in Ref. 18 have been interpreted as arising from the quantization of the Goldstone modes responsible for the *spontaneous breaking* of the electric dipole (rotational) symmetry. Such modes are termed "dipole quanta" in Ref. 18.

This kind of mechanism has been applied to microtubules,[19] with the conclusion that such coherent modes cause "super-radiance", i.e. create a specific quantum-mechanical ordering in the water molecules with characteristic times much shorter than those of thermal interaction. In addition, the optical medium inside the internal hollow core of the microtubule is made transparent by the coherent photons themselves.[19] Such phenomena, if observed, could verify the coherent-mode emission from living matter, thereby verifying Fröhlich's ideas.

In our picture of viewing the MT arrangements as cavities, these coherent modes are the quantum coherent "oscillator" modes of Sec. 3, represented by annihilation and creation operators a_c, $a_c^\dagger$, which play the rôle of the cavity modes, if the ordered-water interior of the MT is viewed as an isolated cavity.[h] The rôle of the small collection of atoms, described in the atomic physics analogue above, is played in this picture by the protein dimers of the MT chains. The latter constitute a two-state system due to the α and β conformations, defined by the position on the unpaired spin in the dimer pockets. The presence of unpaired electrons is crucial to such an analogy. The interaction of the dipole-quanta coherent modes with the protein dimers results in an entanglement which we claim is responsible for the emergence of *soliton quantum coherent states*, extending over large scale, e.g. the MT or even the entire MT network.

The issue, we are concerned with here, is whether such coherent states are responsible for *energy-loss-free transport*, as well as for *quantum computations* due to their eventual collapse, as a result of "environmental" entanglement of the entire

[h]This was not the picture envisaged in Ref. 17. However, S. Hameroff, as early as 1974, had conjectured the rôle of MT as "dielectric waveguides" for photons,[15] and in Ref. 19 some detailed mathematical construction of the emergence of coherent modes out of the ordered water are presented. In our work in this article we consider the implications of such coherent modes for the system of dimers, in particular for the formation of kink solitons of Ref. 4. Thus, our approach is different from that in Refs. 17 and 19, where attention has been concentrated only on the properties of the water molecules. We should emphasize that the phenomenon of optical transparency due to super-radiance may co-exist with the formation of kink soliton coherent states along the dimer chains, relevant to the dissipation-free energy transfer along the MT, discussed in the present work.

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"MT dimers + ordered water" system. An explicit construction of such solitonic states has been made in the field-theoretic model for MT dynamics of Ref. 5, based on classical ferroelectric models for the displacement field $u(x,t)$ discussed in Sec. 2.[4] The quantum-mechanical picture described here should be viewed as a simplification of the field-theoretic formalism, which, however, is sufficient for qualitative estimates of the induced decoherence.

4.2. *Decoherence and dissipationless energy transfer*

To study quantitatively the effects of decoherence in MT systems we make the plausible *assumption* that the environmental entanglement of the "ordered-water cavity" (OWC), which is responsible for dissipation, is attributed *entirely* to the leakage of photons (electromagnetic radiation quanta) from the MT interior of volume V ("cavity"). This leakage may occur from the *nodes* of the MT network, if one assumes fairly isolated interia. This leakage will cause decoherence of the coherent state of the "MT dimer-OWC" system. The leakage determines the damping time scale T_r in (26).

The dimers with their two conformational states α, β play the rôle of the collection of N *two-level* Rydberg atoms in the atomic physics analogue described in Sec. 3. If we now make the assumption that the ordered-water dipole-quantum coherent modes couple to the dimers of the MT chains in a way similar to the one leading to a Rabi splitting, described above, then one may assume a coupling λ_0 of order:

$$\lambda_0 \sim \frac{d_{\rm dimer} E_{\rm ow}}{\hbar} \tag{32}$$

where $d_{\rm dimer}$ is the single-dimer electric dipole matrix element, associated wiith the transition from the α to the β conformation, and $E_{\rm ow}$ is a r.m.s. typical value of the amplitude of a coherent dipole-quantum field mode.

Given that each dimer has a mobile charge[1]: $q = 18 \times 2e$, e the electron charge, one may *estimate*

$$d_{\rm dimer} \sim 36 \times \frac{\varepsilon_0}{\varepsilon} \times 1.6 \times 10^{-19} \times 4 \times 10^{-9} \sim 3 \times 10^{-18}\,{\rm Cb} \times {\rm \AA} \tag{33}$$

where we used the fact that a typical distance for the estimate of the electric dipole moment for the "atomic" transition between the α, β conformations is of $\mathcal{O}(4\,{\rm nm})$, i.e. of order of the distance between the two hydrophobic dimer pockets. We also took account of the fact that, as a result of the water environment, the electric charge of the dimers appears to be screened by the relative dielectric constant of the water, $\varepsilon/\varepsilon_0 \sim 80$. We note, however, that the biological environment of the unpaired electric charges in the dimer may lead to further suppression of $d_{\rm dimer}$ (33).

The amplitude of the collective modes of the dipole-quanta may be estimated using the formula (17). In our case the "cavity volume" is:

$$V \sim 5 \times 10^{-22}\,{\rm m}^3 \tag{34}$$

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which is a typical MT volume, for a moderately long MT $L \sim 10^{-6}$ m, considered as an isolated cavity,[i] and ω_c a typical frequency of the dipole quanta collective mode dynamics. To estimate this frequency we assume, following the "super-radiance" model of Ref. 19, that the dominant modes are those with frequencies in the range $\omega_c \sim \epsilon/\hbar$, where ϵ is the energy difference between the two principal energy eigenstates of the water molecule, which are assumed to play the dominant rôle in the interaction with the (quantized) electromagnetic radiation field. For the water molecule:

$$\hbar\omega_c \sim 4\,\text{meV} \tag{35}$$

which yields

$$\omega_c \sim \frac{\epsilon}{\hbar} \sim 6 \times 10^{12}\,\text{s}^{-1} \tag{36}$$

This is of the same order as the characteristic frequency of the dimers (1), implying that the dominant cavity mode and the dimer system are almost in resonance. Note that this is a feature shared by the Atomic Physics systems in Cavities examined in Sec. 3, and thus we may apply the pertinent formalism to our system.

From (17) one obtains for the r.m.s E_{ow} in order of magnitude:

$$E_{\text{ow}} \sim 10^4\ \text{V/m} \tag{37}$$

where we took into account the relative dielectric constant of water $\varepsilon/\varepsilon_0 \sim 80$.

It has not escaped our attention that the electric fields of such order of magnitude can be provided by the electromagnetic interactions of the MT dimer chains, the latter viewed as giant electric dipoles.[4] This may be seen to suggest that the super-radiance coherent modes ω_c, which in our scenario interact with the unpaired electric charges of the dimers and produce the kink solitons along the chains, owe their existence to the (quantized) electromagnetic interactions of the dimers themselves.

We assume that the system of $\mathcal{N}$ MT dimers interacts with a *single* dipole-quantum mode of the ordered water and we ignored interactions among the dimer spins.[j] In our work here we concentrate our attention on the formation of a coherent soliton along a single dimer chain, the interactions of the remaining 12 chains in a protofilament MT cylinder being represented by appropriate interaction terms

[i]Here we consider only a single unit of the MT network. The MT network consists of a large number of such cavities/MT. Our scenario here is to examine soliton formation in a single (isolated) cavity MT, with the only source of dissipation the leakage of photons. This would explain the formation of kinks in a single MT.[4,5] If the entire network of MT is viewed as a cavity, then the volume V is complicated, but in that case the volume can be estimated roughly as $\mathcal{N}_{mt}V$, with V the average volume of each MT, and $\mathcal{N}_{mt}$ the number of MT in the network population. In such a case, the solitonic state extends over the entire network of MT. It is difficult to model such a situation by simple one-dimensional Hamiltonians as in Refs. 4, 5.

[j]More complicated situations, including interactions among the dimers, as well as of the dimers with more than one radiation quanta, which might undoubtedly occur in nature, complicate the above estimate.

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in the effective potential of the chain MT model of Ref. 4. In a moderately long microtubule of length $L \simeq 10^{-6}$ m there are

$$\mathcal{N} = \frac{L}{8} \simeq 10^2 \tag{38}$$

tubulin dimers of average length 8 nm in each chain.

Then, from (32), (33) and (37), the conjectured Rabi-vacuum splitting, describing the entanglement of the interior coherent modes with the dimers can be estimated to be of order

$$\text{Rabi coupling for MT} \equiv \lambda_{\rm MT} = \sqrt{\mathcal{N}}\lambda_0 \sim 3 \times 10^{11}\,{\rm s}^{-1} \tag{39}$$

which is, on average, an order of magnitude smaller than the characteristic frequency of the dimers (1). In this way, the perturbative analysis of Sec. 3, for small Rabi splittings $\lambda << \omega_0$, is valid. Indeed, as can be seen from (1) and (36), the detuning Δ is of order:

$$\frac{\Delta}{\lambda_0} \sim \mathcal{O}(10)\text{–}\mathcal{O}(100) \tag{40}$$

implying that the condition $\lambda_0^2\mathcal{N}/|\Delta|^2 \ll 1$, necessary for the perturbative analysis of Sec. 3, is satisfied. The Rabi frequency (39) corresponds to an energy splitting $\hbar\lambda_{\rm MT} \sim 0.1$ meV for a moderately long MT.

Having estimated the Rabi coupling between the dimers and the ordered-water coherent modes we now proceed to estimate the average time scale necessary for the formation of the pointer coherent states which arise due to this entanglement.[6,5] This is the same as the decoherence time due to the water-dimer coupling. In such a case one may use the Liouville model of MT,[5] discussed in Subsec. 2.2, which is known to possess pointer coherent states.[14] The relevant decoherence time scale is given by (8).

For an accurate estimate of the decoherence time one should have a precise knowledge of the violations of conformal invariance associated with the water-dimer coupling. Unlike the quantum-gravity induced decoherence case, where a microscopic model for the description of the distortions caused in the environment of the chain by the external stimuli is available,[5] at present a precise conformal-field-theory model for the description of the water-dimer coupling is lacking.

However, for our purposes here, it suffices to use our generic approach to the string-theory representation of MT chains, described in Ref. 5, where a good order-of-magnitude estimate for β^u is provided by (6). We, therefore, merely need to have an estimate of the scales E and $M_{\rm s}$ for the problem at hand.

To estimate the "string" scale $M_{\rm s}$ in our case,[k] where only quantum-electromagnetic interactions enter, we notice that in the standard string theory $M_{\rm s}$

[k]In Ref. 5, where scenaria for gravity-induced conscious perception were discussed, the scale $M_{\rm s}$ was taken to be the Planck scale 10^{19} GeV. In that case decoherence was induced by the coupling of the entire network of brain MT to quantum-gravity fluctuations due to the distortion of the surrounding space time, as result of abrupt conformational changes in the dimers, caused by external stimuli.[9,8,5]

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acts as an ultraviolet cut-off in the energy of the effective target-sace field theory, in our case the field theory of the displacement field $u(x,t)$. Since in the ferroelectric chain the closest distance separating two unpaired electrons, which are assumed to have the dominant interactions with the water, is the size of a tubulin dimer $\sim d_{\rm min} = 4\,{\rm nm}$, an order of magnitiude estimate of $M_{\rm s}$ is:

$$M_{\rm s} \sim \frac{\hbar v_0}{d_{\rm min}} \sim 1.5 \times 10^{-4}\,{\rm eV}\,, \tag{41}$$

where we took into account that the Liouville model for MT in Ref. 5 is formally a relativistic string model, but with the rôle of the velocity of light played by the sound velocity $v_0 = 1$ Km/sec in organic biological materials. It is interesting to note that this cut-off in energies is less than the kinetic energies of fast kinks, propagating with the valocity of sound. Indeed such energies are of order $E_0 \sim 10^{-2}\,{\rm eV}$. Such fast kinks have been associated in Ref. 5 with quantum gravitational effects, due to abrupt distortions of space time, and hence their exclusion in the approach of the present work, by considering kinks with energies much less than those, as implied by $M_{\rm s}$, is consistent with our considering only electromagnetic interactions among the dimers and its environment.

The above estimate allows us to get an idea of the typical energy scales of the excitations of the MT systems that dominate the ordered-water-dimer coupling, in the above scenario, and lead to decoherence. As discussed in Ref. 5, the dominant part of the energy of a kink in the model of Ref. 4 is of order 1 eV, and thus much higher than the $M_{\rm s}$ (41). Such scales may play a rôle in the decoherence due to quantum-gravity entanglements,[5] which however is much weaker than the electromagnetic ones considered here. On the other hand, as we shall argue now, a typical energy scale for the dimer displacement field $u(x,t)$, much smaller than $M_{\rm s}$, and therefore consistent with our low-energy approach, is provided by the kinetic energy of the kink, which is estimated to be of order[4]

$$E_{\rm kin} \simeq 5 \times 10^{-8}\,{\rm eV} \tag{42}$$

As we shall argue below, this energy scale may be used for our estimate of the time scale (8), which is necessary for the formation of the coherent state due to the water-dimer coupling. We stresss that (42) should not be considered as a typical energy scale associated with the excitation spectrum of the dimers, pertaining to the Rabi splitting discussed above. It is rather an "effective scale", characterising "friction" effects between the dimers and the water environment. The latter are responsible[5] for decoherence and the eventual formation of the kink coherent states, which are minimum-entropy states,[6] least susceptible to the effects of the water environent.

To justify the above estimate, one should notice that the formation of coherent quantum states through decoherence due to friction[6] is the quantum analogue of the "drift velocity" acquired by a Brownian particle in classical mechanics. In the (non-relativistic) conformal-field-theory setting of Ref. 5 such a friction could be described by the formation of point-like defects on the dimer chains, which could

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be described by appropriate non-relativistic membrane backgrounds in the $(1+1)$-dimensional string theory representation of the MT dynamics. Such membranes are stringy defects, which, for instance, could describe the result of an abrupt conformational change of a given dimer due to its coupling with the water environment. Scattering of stringy excitations $u(x,t)$ off the defect causes "recoil" of the latter, which starts moving with a velocity $v_{\rm d}$. The recoil is due to quantum fluctuations as argued in Ref. 33. Such recoil effects are necessary for energy and momentum conservation in the case of (non-relativistic) heavy membranes, considered in Ref. 33. That model represents a pilot model for discussing decoherence due to water environment in our case.

The important feature to notice is that the "effective mass" of the defect in such a stringy representation turns out to be inversely proprotional to the string-coupling constant $g_{\rm s}$,[33]

$$m_{\rm defect} = (8\sqrt{2}\pi g_{\rm s})^{-1} \tag{43}$$

which, in turn, depends on the vacuum expectation value of the effective "dilaton" field Φ: $g_{\rm s} = e^{-\langle\Phi\rangle}$. In the case at hand, as discussed in Ref. 5, the dilaton field is proprotional to the friction coefficient γ, so that $g_{\rm s} \sim \exp(-\gamma)$ in order of magnitude. Strong friction, therefore, between water molecules and dimers, which may be assumed in our physical model of MT, implies weak string coupling, and hence the non-relativistic approximation for the membranes proves sufficient.

Within the framework of identifying target time with the Liouville mode,[7] then, the analysis of Ref. 33 has shown that the *quantum recoil* degrees of freedom of the membrane defect carry *information* and induce decoherence of the dimer (string) subsystem, corresponding to violations of conformal invariance of order:

$$\beta^{\rm recoil} \sim \mathcal{O}\left(\frac{v_{\rm d}^2}{16\pi g_{\rm s} M_{\rm s}}\right) \tag{44}$$

where $v_{\rm d}^2$ is the recoil velocity of the defect. By energy-momentum conservation, the kinetic energy of the recoiling (non-relativistic) defect is *at most* of the same order as the kinetic energy of the displacement field (42). This would yield a lower bound on the decoherence time (8) estimated on the basis of (44) to be of order:

$$t_{\rm owdecoh} \gtrsim 10^{-10}\,{\rm sec} \tag{45}$$

for a moderately long MT, with $\mathcal{N} = 10^2$ dimers. This is the time scale over which solitonic coherent pointer states in the MT dimer system are formed ("pumped"), according to our scenario. Note that the scale (45) is not far from the original Fröhlich scale,[2] 10^{-11}–10^{-12} sec.

To answer the question whether quantum coherent pointer states are responsible for loss-free energy transport across the MT one should examine the time scale of the decoherence induced by the coupling of the MT to their biological environment as a consequence of dissipation through the walls of the MT cylinders. Such an

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ordinary environmental entanglement has been ignored in the derivation of (45). It is this environment that will induce decoherence and eventual collapse of the pointer states formed by the interaction of the dimers with the coherent modes in the ordered water. Using typical numbers of MT networks, we can estimate this decoherence time in a way similar to the corresponding situation in atomic physics (26) and (30):

$$t_{\rm collapse} = \frac{T_r}{2n\mathcal{N}\sin^2(\mathcal{N}n\lambda_0^2 t/\Delta)} \tag{46}$$

where we took into account that the dominant (dimer)-(dipole quanta) coupling occurs for ordered-water "cavity" modes which are *almost at resonance* with the dimer oscillators (c.f. Eqs. (36) and (1)), slightly detuned by Δ : $\lambda_0/\Delta \ll 1$, c.f. (40).

We also assume that a typical coherent mode of dipole/quanta contains an average of $n = \mathcal{O}(1)$–$\mathcal{O}(10)$ oscillator quanta. The *macroscopic* character of the Schrödinger's cat dimer-dipole-quanta system comes from the N dimers in a MT (or $\mathcal{N}_{\rm mt}\mathcal{N}$ in MT networks).[5]

The time t appearing in (46) represents the "time" of interaction of the dimer system with the dipole quanta. A reasonable estimate of this time scale in our MT case can be obtained by equating it with the average life-time of a coherent dipole-quantum state, which, in the super-radiance model of Ref. 19 can be estimated as

$$t \sim \frac{c\hbar^2 V}{4\pi d_{\rm ej}^2 \epsilon N_{\rm w} L} \tag{47}$$

with $d_{\rm ej}$ the electric dipole moment of a water molecule, L the length of the MT, and $N_{\rm w}$ the number of water molecules in the volume V of the MT. For typical values of the parameters for moderately long MT, $L \sim 10^{-6}$ m, $N_{\rm w} \sim 10^8$, a typical value of t is:

$$t \sim 10^{-4} \text{ sec} \tag{48}$$

We remark at this point that this is considerably larger than the average life-time of a coherent dipole quantum state in the water model of Ref. 17. Indeed in that model, the corresponding life-time is estimated to be:

$$t \sim \frac{2\pi}{\omega_0} \tag{49}$$

where $\omega_0 \sim 1/I$, is a typical frequency of resonating electromagnetic mode in the ordered water. For a typical value of the water molecule moment of intertia[17] this yields a time-scale associated with the coherent interaction $\mathcal{O}(10^{-14}$ sec).

The time scale T_r, over which a cavity MT dissipates its energy, can be identified in our model with the average life-time (47) of a coherent-dipole quantum state:

$$T_r \sim t \sim 10^{-4} \text{ sec} \tag{50}$$

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which leads to a naive estimate of the quality factor for the MT cavities, $Q_{\rm MT} \sim \omega_{\rm c} T_{\rm r} \sim \mathcal{O}(10^8)$. We note, for comparison, that high-quality cavities encountered in Rydberg atom experiments dissipate energy in time scales of $\mathcal{O}(10^{-3})$–$\mathcal{O}(10^{-4})$ sec, and have Q's which are comparable to $Q_{\rm MT}$ above. Thus, it is not unreasonable to expect that conditions like (50), characterizing MT cavities, are met in Nature.

From (46), (39) and (50), one then obtains the following estimate for the collapse time of the kink coherent state of the MT dimers due to dissipation:

$$t_{\rm collapse} \sim \mathcal{O}(10^{-7})\text{–}\mathcal{O}(10^{-6}) \text{ sec} \tag{51}$$

which is larger or equal than the Fröhlich scale (5) required for energy transport across the MT by an average kink soliton in the model of Ref. 4. The result (51), then, implies that Quantum Physics may be responsible for dissipationless energy transfer across the MT.

We close this section by remarking that if the condition $T_{\rm collapse} \gtrsim 5 \times 10^{-7}$ sec is not met, then the above picture, based on mesoscopic coherent states, would be inconsistent with energy-loss free energy transport, since decoherence due to environmental entanglement would occur before energy could be transported across the MT by the preformed quantum soliton. In such a case energy would be transported due to different mechanisms, one of which is the classical solitons scenario of Refs. 4 and 3. However, even in such cases of fast decoherence, the Rabi coupling predicted above could still exist and be subjected to experimental verification.

5. Outlook: Possible Experimental Tests Using Micro-Tubules as Cavities

In the present work we have put forward a conjecture concerning the representation of the MT arrangements inside the cell as *isolated* high-Q(uality) *cavities*. We presented a scenario according to which the presence of the ordered water in the interior of the cylindrical arrangements results in the appearance of electric dipole quantum coherent modes, which couple to the unpaired electrons of the MT dimers via Rabi vacuum field couplings, familiar from the physics of Rydberg atoms in electromagnetic cavities.[20] In quantum optics, such couplings are considered as experimental proof of the quantized nature of the electromagnetic radiation. In our case, therefore, if present, such couplings could indicate the existence of the coherent quantum modes of electric dipole quanta in the ordered water environment of MT, conjectured in Refs. 18 and 17, and used in the present work.

Some generic decoherence time estimates, due to environmental entanglement of the MT cavities, have been given. The conclusion was that only in *fairly isolated* cavities, which we conjecture exist inside the biological cells, decoherence occurs in time scales which are in agreement with the Fröhlich scale ($\sim 5 \times 10^{-7}$ sec) for energy transfer across the MT via the formation of kink solitonic structures. In such a case, *dissipationless energy transfer* might occur in biological systems, an in particular in MT which are the substratum of cells, in much the same way as frictionless electric

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current transport occurs in superconductors, i.e. via *quantum coherent modes* that extend over relatively large spatial regions. A phenomenological analysis indicated that for moderately long MT networks such a situation could be met if the MT cavities dissipate energy in time scales of order $T_r \sim 10^{-4}$–10^{-5} sec. This lower bound is comparable to the corresponding scales of atomic cavities,[21] $\sim 10^{-4}$ sec, which, in turn, implies that the above scenaria, on dissipationless energy transport as a result of the formation of quantum coherent states, have a good chance of being realized at the scales of MT, which are comparable to those in Atomic Physics.

We have conjectured in this work that an indirect verification of such a mechanism would be the experimental detection of the aforementioned Vacuum field Rabi coupling, $\lambda_{\rm MT}$, between the MT dimers and the ordered water quantum coherent modes. This coupling, if present, could be tested experimentally by the same methods used to measure VFRS in atomic physics,[21] i.e. by using the MT themsleves as *cavity environments*, and considering tunable probes to excite the coupled dimer-water system. Such probes could be pulses of (monochromatic) light, for example, passing through the hollow cylinders of the MT. This would be the analogue of an external field in the atomic experiments described above, which would then resonate, not at the bare frequencies of the coherent dipole quanta or dimers, but at the *Rabi splitted* ones, and this would have been exhibited by a double pick in the absorption spectra of the dimers.[21] By using MT of different sizes one could thus check on the characteristic $\sqrt{N}$-enhancement of the Rabi coupling for MT systems with N dimers.

The technical complications that might arise in such experiments are associated with the absence of completely resonant cavities in practice. In fact, from our discussion in this article, one should expect a slight detuning Δ between the cavity mode and the dimer, of frequency ω_0. As discussed in Sec. 3, the detuning produces a split of the vacuum-Rabi doublet into a cavity line $\omega_0 + \lambda^2/\Delta$ and an "atomic line" $\omega_0 - \lambda^2/\Delta$. In atomic physics there are well established experiments[21,22] to detect such splittings. In fact, detection of such lines is considered as a very efficient way of "quantum non-demolition" measurement[30] for small microwave photon numbers. Such atomic physics experiments, therefore, may be used as a guide in performing the corresponding biological experiments involving MT (as cavities), as suggested in the present work.

We believe that such experimental set-ups will help in sheding some light on the issue of the quantum mechanical nature of the MT arrangements, and even on the processes of the transmission of electric signals (stimuli) by the neuronic systems. Clearly much more work needs to be done before even tentative conclusions are reached, concerning the nature of the MT arrangements. However, we believe that the present work constitutes a useful addition to the programme of understanding the nature of the MT arrangements inside the cell, and the associated processes of energy transfer across the cells.

Acknowledgments

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On the capacitive properties of individual microtubules and their meshworks

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ABSTRACT

Microtubules are hollow cylindrical polymers composed of the highly negatively-charged (~23e), high dipole moment (1750 D) protein α, β- tubulin. While the roles of microtubules in chromosomal segregation, macromolecular transport and cell migration are relatively well-understood, studies on the electrical properties of microtubules have only recently gained strong interest. Here, we show that while microtubules at physiological concentrations increase solution capacitance, free tubulin has no appreciable effect. For a particular microtubule concentration, we were able to quantify these effects by determining the capacitance and resistance of a single 20 μm-long microtubule to be 1.86×10^{-12} F and $1.07 \times 10^{12}\Omega$, respectively. Further, we observed a decrease in electrical resistance of solution, with charge transport peaking between 20-60 Hz in the presence of microtubules, consistent with recent findings that microtubules exhibit electric oscillations at such low frequencies. Our results show that in addition to macromolecular transport, microtubules also act as charge-storage devices through counterionic condensation across a broad frequency spectrum. We conclude with a hypothesis of an electrically-tunable cytoskeleton where the dielectric properties of tubulin are polymerization-state dependent.

KEYWORDS: *microtubules, biological nanowires, intracellular charge storage, intracellular ion conduction*

INTRODUCTION

Microtubules (MTs) are cylindrical polymers composed of the heterodimers of protein α, β- tubulin that play a variety of well-recognised intracellular roles, such as maintaining the shape and rigidity of the cell, aiding in positioning and stabilisation of the mitotic spindle for allowing chromosomal segregation, acting as 'rails' for macromolecular transport and forming cilia and flagella for cell movement. Since the tubulin dimer possesses a high negative electric charge of ~23e and a large intrinsic high dipole moment of 1750 D [1-2], MTs have been implicated in electrically-mediated biological roles [3-6]. They have been modelled as nanowires capable of enhancing ionic transport [7-8], and simulated to receive and attenuate electrical oscillations [4, 9-11]. In solution, MTs have been shown to align with applied electric fields [2, 12-16]. Recently, MTs have also been modelled as the primary cellular targets for low-intensity (1-2 V), intermediate-frequency (100-300 kHz) electric fields that inhibit cancer cell proliferation, in particular glioma [17-19]. Indeed, MTs have been reported to decrease buffer solution resistance [12-13], leading to a conductance peak at TTField-like frequencies [20]. While these studies show that MTs are highly sensitive to external electric fields, answers to the questions 'How do MTs effect a solution's capacitance?' and 'What is the capacitance of a single MT?' are still elusive and crucial to the determination of the dielectric properties of living cells. The tubulin concentration in mammalian cells varies in the micromolar range (~10-25 μM) [21-22]. *In vitro*, polymerizing tubulin at such high concentrations can lead to the formation of entangled meshworks, confounding quantification of the individual MT response to electric fields. Electro-rotation, dielectrophoresis and impedance spectroscopy are thus performed using low concentrations of tubulin, in the nanomolar regime, to enable robust observation of individual MTs.

MT formation and stability are known to be optimal in buffers with ionic strength between 80 and 100 mM [23-24]. A background of BRB80 (which consists of 80 mM PIPES, 2 mM $MgCl_2$ and 0.5 mM EGTA), is thus used to study the dynamics and mechanical properties of MTs. To study their electrical properties however, the usage of such high ionic-strength solutions has historically been problematic because any dielectric attenuation caused by MTs is overwhelmed by the noise and high conductivity from the background. In the low-frequency regime (1 Hz-100 kHz), two approaches have thus far been used to estimate the dielectric properties of MTs and tubulin. One is to electrically observe low concentrations of MTs (tubulin concentration in the nanomolar regime) in the presence of low ionic strengths [12-13, 20, 25-26]. Such studies overlook the intrinsic ionic concentration of mammalian cytosol, which varies between 200 to 500 mM depending on the cell type [27-28]. Another approach to electrically interrogate MTs is to dry them: the conductivity of the buffer is nullified by evaporation, leaving polymeric tubulin behind [29-30] In a physiological situation however, MTs are solvated by the highly conductive and noisy cytosol.

In this paper, we report on our efforts overcome the barrier posed by a high ionic strength by performing electrochemical impedance spectroscopy (EIS) on cellular concentrations of tubulin. We have been able to successfully observe differences in impedance using a background of BRB80, which contains within an order of magnitude of physiologically measured ionic concentrations. Surprisingly, we find that MTs increase the solution capacitance of BRB80 whereas free tubulin does not, implicating a difference in electrical properties based only on the morphology of this protein solute. We also report a 'reversal' in the resistive behaviour of MTs compared to BRB80, with a reduction in solution resistance peaking at 20-60 Hz, a finding consistent with recent reports showing that polymerized tubulin quasi-resonantly responds to electric oscillations at ~39 Hz [9-10]. Using an equivalent circuit model for MTs, we experimentally

determine the capacitance and resistance of the MT meshwork to be 1.27×10^{-5} F and 9.74×10^{4} Ω at physiological concentrations of tubulin. Using approximations, we quantify the capacitance and resistance for an individual MT as 1.86×10^{-12} F and 1.07×10^{12}Ω. Our values are in agreement with previous calculations and indicate that the polymerization of tubulin into MTs alters spatial and temporal charge distribution, altering the electrical properties through charge storage in the cell.

RESULTS

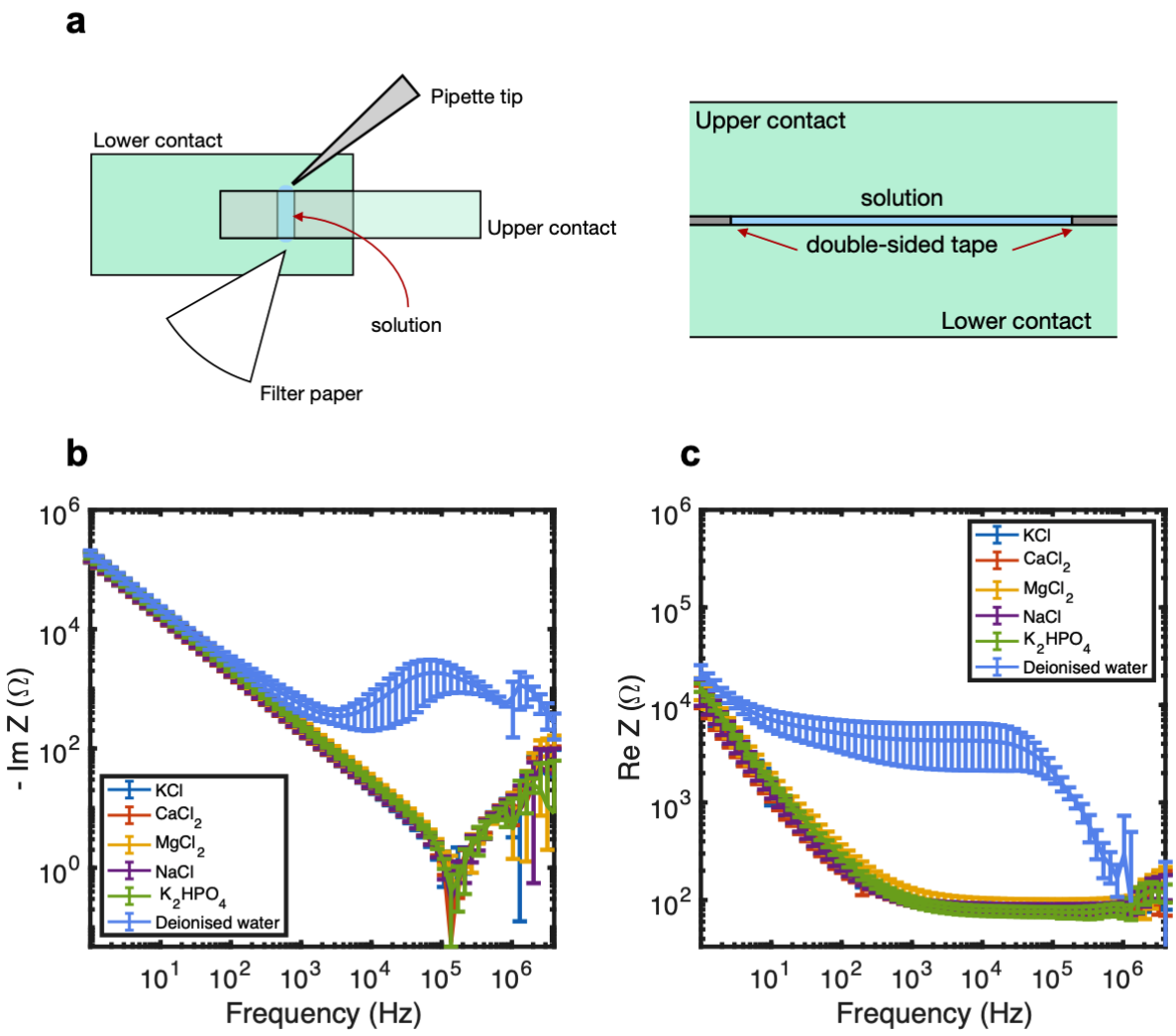

Figure 1. A parallel-plate contact device to measure the impedance properties of MTs compared to tubulin. The operation of the parallel plate device showing (a) top view (left) and side view (right). The upper and lower contacts, double-sided tape and solution are labelled in green, gray and blue, respectively. (b) Imaginary component of impedance for electrolytic solutions at 100 mM and de-ionized water. (c) Real component of impedance for electrolytic solutions at 100 mM and de-ionized water. Data display average values collected between 15 and 21 times. Error bars represent standard deviation. For measurement setup and perfusion protocols, see Materials and Methods.

A parallel-plate contact device can accurately measure dielectric properties of ionic solutions such as the cytosol

To determine the differences in the dielectric properties of solution caused by the presence of MTs, we aimed to create an electrode geometry that would be experimentally robust and easily modelled. We fabricated an FTO-coated parallel-plate contact device (Figure. 1a, Materials and Methods), which allowed EIS using a solution-exchange method.

We started by performing EIS on electrolytes found in the cytosol and observed a decrease in the imaginary component of impedance as a function of decreasing input frequency (Figure. 1b). The total impedance of our system was given by:

$$Z = \mathrm{r_c} + \frac{\mathrm{r_s}}{1+(\mathrm{r_s}\omega\mathrm{C})^2} + j(\omega\mathrm{L_c} - \frac{\mathrm{r_s}^2\omega\mathrm{C}}{1+(\mathrm{r_s}\omega\mathrm{C})^2}) \quad (1)$$

Here, Z is the impedance, ω is the angular frequency (given by $2\pi f$ where f is the input voltage frequency), C is the system capacitance, $\mathrm{L_c}$ is the cable inductance, $\mathrm{r_s}$ and $\mathrm{r_c}$ are the solution and cable resistances respectively. We also observed a decrease in the real component of impedance as a function of decreasing input frequency (Figure 1c). Such a trend is expected from Warburg impedance [32-33] and follows the equation:

$$Z_{complex} = \frac{A_\omega}{\sqrt{\omega}} + \frac{A_\omega}{j\sqrt{\omega}} \quad (2)$$

where $Z_{complex}$ is complex impedance and A_ω is the Warburg coefficient. Our circuit simplifies to the equation below if we ignore the effect of cable inductance ωL_c, at frequencies below 10^5 Hz:

$$Z = r_c - j\frac{1}{\omega C} \quad (3)$$

Our results emulated previous data [34-36] and validated the experimental setup for further analysis.

3.2 MTs increase solution capacitance compared with background, while unpolymerized tubulin does not have a significant effect

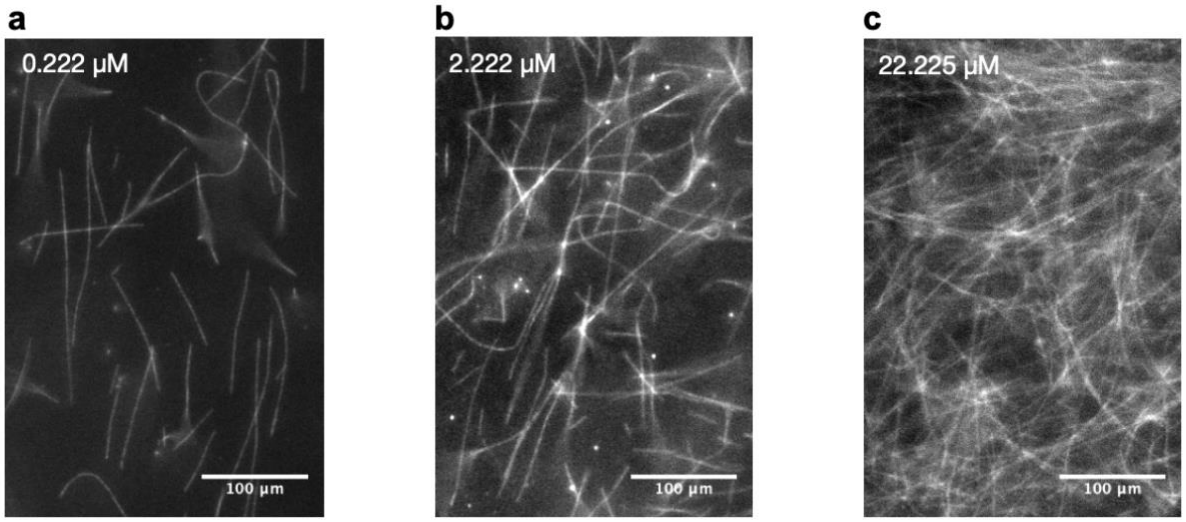

Figure 2. Microtubule imaging at different tubulin concentrations. Polymerization was performed using 45 µM tubulin, and MTs were stabilized with 50 µM paclitaxel, and subsequently diluted to a final concentration of (a) 0.222 µM tubulin (b) 2.222 µM tubulin (c) 22.225 µM tubulin, respectively. For imaging setup and polymerization protocols, see Materials and Methods.

We reconstituted and polymerized fluorescent tubulin from a stock of 45.45 μM tubulin solution (Materials and Methods). MTs were stabilised using 50 μM paclitaxel [37-38] and imaged using an epi-fluorescence microscope. On diluting MT concentration across three orders of magnitude (0.222, 2.222 and 22.225 μM tubulin), we observed that while individual MTs at low concentrations were separated by large distances, those at cellular concentrations formed enmeshed networks reported previously (Figure 2 a, b, c) [39]. Such interconnected MT meshworks are utilized by molecular motors for long-range macromolecular transport [40-41]. Here, their presence demonstrated successful MT polymerization for electrical characterization.

We performed EIS on BRB80, BRB80T (BRB80 supplemented with 50 μM paclitaxel; background for all MT-containing solutions), and MT-containing solutions in increasing order of concentration (Figure S3 a, b). We subtracted impedance values obtained for BRB80T alone from those in MT containing solutions, to determine the MT contribution to impedance. Our results showed that with an increasing MT concentration, the value of imaginary impedance became more negative, resulting in positive impedance differences (Figure 3a, b, c, d). This effect was highest at 1 Hz and decreased with a decreasing input frequency (Figure S4 a, b, c, d, e, f). Experiments with unpolymerized tubulin at the same concentrations were performed using the identical procedure, but using BRB80C (BRB80 was supplemented with 50 μM colchicine) as a background, to prevent MT nucleation [42-43]. Results with tubulin did not show an appreciable deviation at any concentration. Based on the above we can conclude that polymerization of tubulin into MTs alters their ensemble electrical properties, increasing the solution's capacitance on forming MTs and their meshworks. An increase in the solution's capacitance due to MTs has

previously been modelled, [7, 44-45] indicating an increase in charge storage as free tubulin polymerizes.

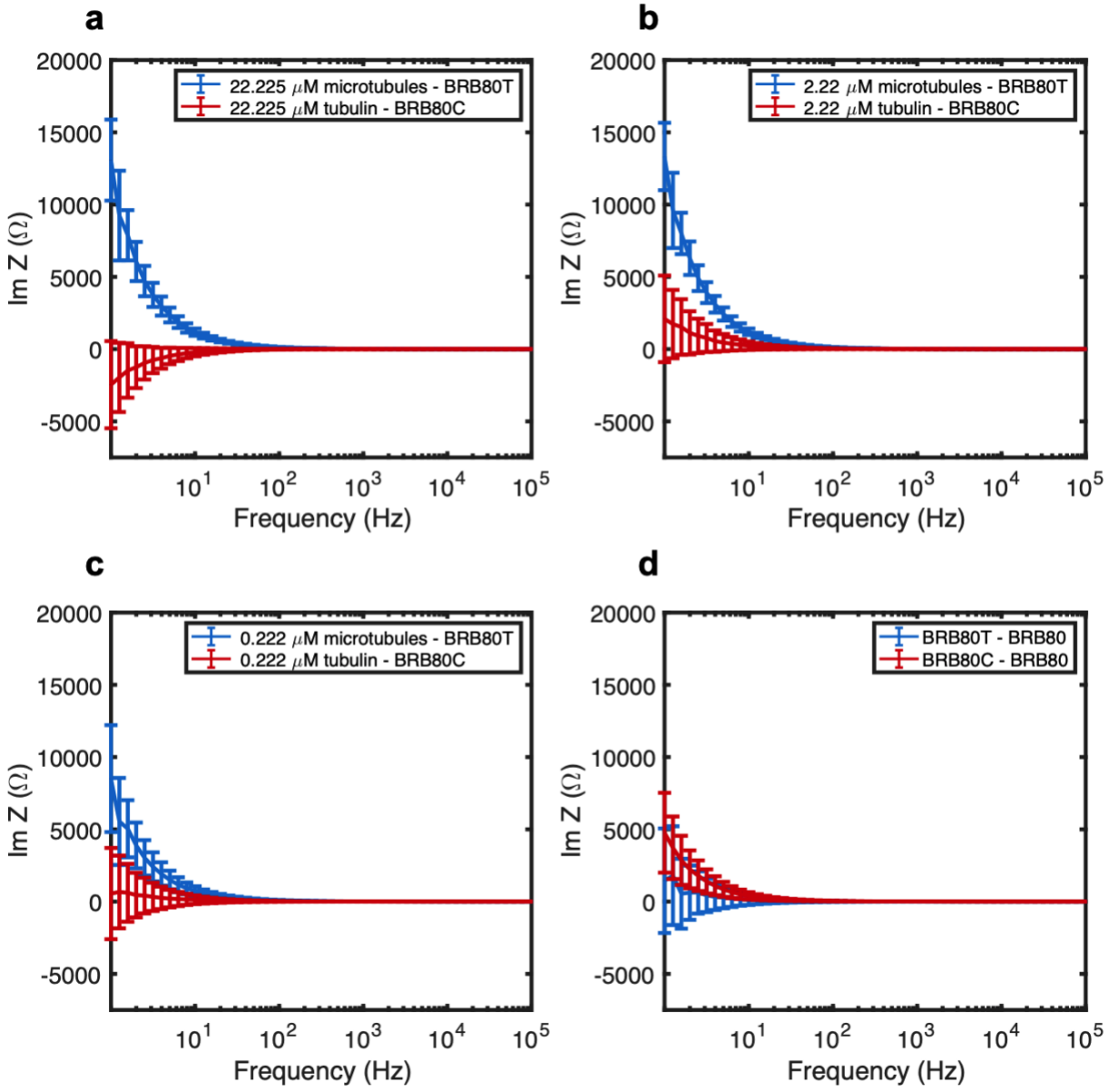

Figure 3. Mean differences in the imaginary component of impedance as a function of decreasing input AC frequency at total tubulin concentrations of (a) 22.225 μM (n = 22 experiments for tubulin, n = 21 for MTs), (b) 2.222 μM (n = 35 experiments for tubulin, n = 49 for MTs) (c) 0.222 μM (n = 35 experiments for tubulin, n = 49 for MTs), (d) comparison of the effects of paclitaxel

(BRB80T) and colchicine (BRB80C, n = 49 experiments for BRB80T, n = 35 for BRB80C, n = 84 experiments for BRB80). Error-bars represent standard deviation.

Microtubules increase solution resistance compared to background with this effect's reversal observed at low frequencies

Next, we looked for differences between MTs and tubulin in the real component of impedance (solution resistance). Previous studies using nanomolar tubulin concentrations and low ionic strengths (1-12 mM) have indicated that MTs enhance charge-transport in solutions [13, 20, 46]. To evaluate if this observation held true at physiologically relevant tubulin concentrations and at higher ionic strengths, we also analyzed the real component of impedance. Addition of both MTs and tubulin generally led to solution led to an increase in resistance (Figure S7 a, b, c, d), with MTs having a higher resistance at low frequencies (1-20 Hz) compared to unpolymerized tubulin. Unexpectedly, a 'reversal' of this behaviour was observed at higher frequencies as MTs began to lower solution resistance compared to tubulin (Figure S5 a, b, c, d, e, f). The reversal took place gradually between 10 and 300 Hz, with a peak between 20 and 60 Hz (Figure 4 a, b, c). Interestingly, within this range, we also found that the addition of MTs lowered solution resistance compared to background buffer BRB80T.

A reversal frequency in resistance between MTs and tubulin has not been reported before and displays the utility of our 'cell-like' approach, because the extent of this reversal decreased with decreasing concentration and was not displayed in the imaginary component of impedance or when BRB80C and BRB80T were compared. It is worth noting that recent findings using this approach showed that MT bundles and tubulin sheets generate electrical signals at ~39 Hz [9-10], and predicted

an increase in solution conductance in the range that we observe, indicating that the conductance behaviour at such frequencies was due to MT- generated electrical oscillations.

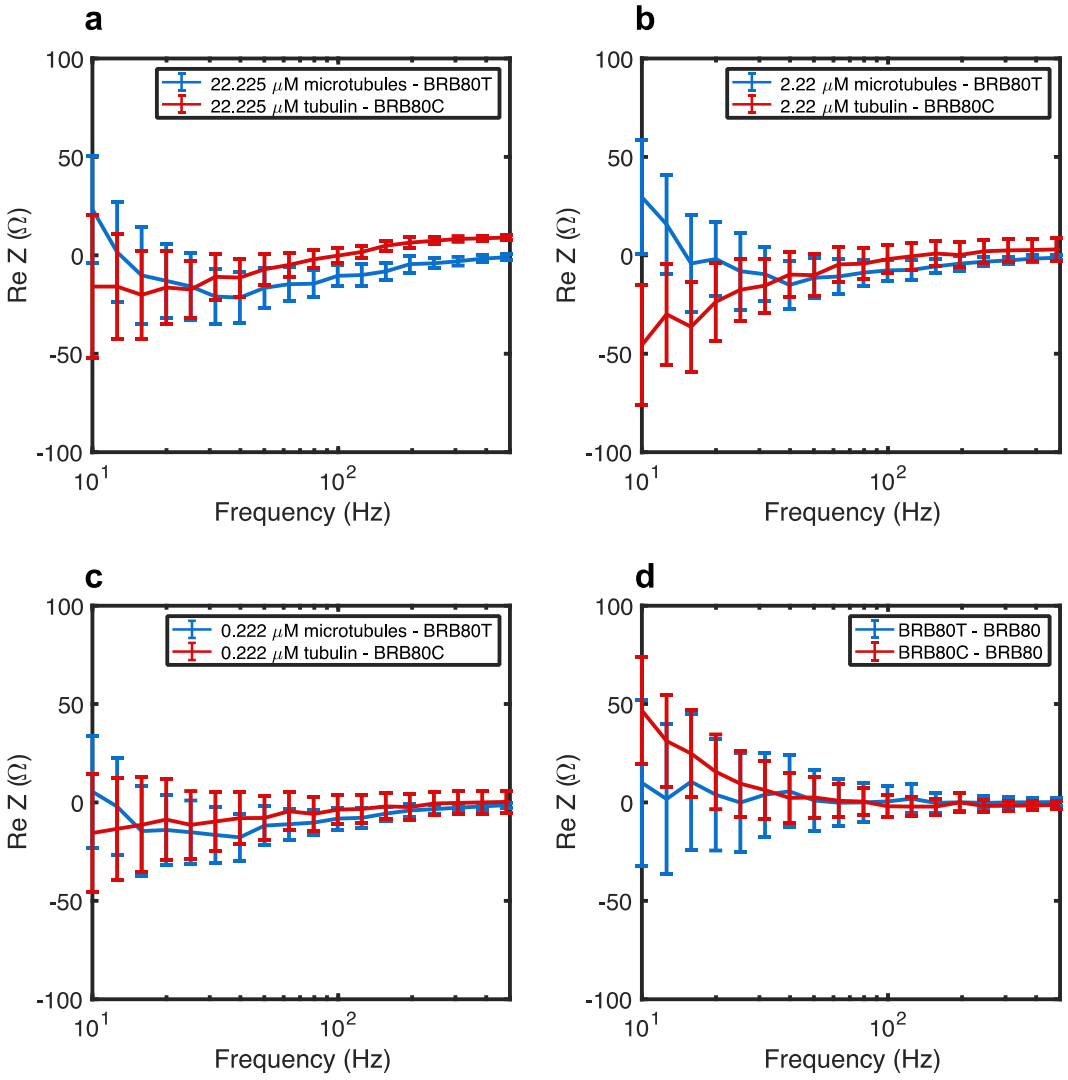

Figure 4. Mean differences in the real component of impedance as a function of decreasing input AC frequency at total tubulin concentrations of (a) 22.225 μM, (b) 2.222 μM, (c) 0.222 μM, (d) comparison of the effect of paclitaxel and colchicine on impedance. Error-bars represent standard deviation.

The microtubule meshwork can be described as an RC circuit in parallel

Using our experiments, our next aim was to quantify the electrical properties of a single MT. Impedance difference curves appeared linear on log-log plots with a slope of approximately negative unity, suggesting that the MT meshwork resulted in the addition of a capacitive element to the solution. We examined several combinations but a parallel RC circuit to represent the entire MT meshwork provided that best fit to observed curves (Figure 5):

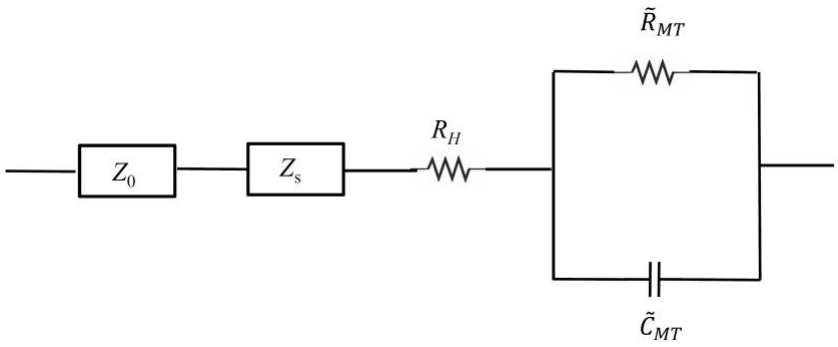

Figure 5. The equivalent electrical circuit model representing the microtubule meshwork as a parallel RC circuit, with meshwork resistance $\tilde{R}_{MT}$ and capacitance $\tilde{C}_{MT}$. The external element has impedance Z_0, while solution has impedance Z_s. R_H is the small constant resistance that is ascribed to small fraction of unpolymerized tubulin that is present in MT containing solutions.

We modelled the impedance caused by external circuit elements and BRB80T as Z_0 and Z_s respectively, as shown in Fig 5. The net impedance of the background BRB80T was thus given by:

$$Z_{\mathrm{buffer}} = Z_0 + Z_\mathrm{s} \tag{4a}$$

Denoting the impedance, resistance and capacitance of the entire MT meshwork by $\tilde{Z}_{\mathrm{MT}}$, $\tilde{R}_{MT}$ and $\tilde{C}_{MT}$ respectively, the impedance for the circuit with MTs is given by:

$$Z_{\mathrm{MT+buffer}} = Z_0 + Z_{\mathrm{s}} + R_{\mathrm{H}} + \tilde{Z}_{\mathrm{MT}} \tag{4b}$$

where,

$$\frac{1}{\tilde{Z}_{\mathrm{MT}}} = \frac{1}{\tilde{R}_{\mathrm{MT}}} + \mathrm{j}\omega\tilde{C}_{\mathrm{MT}}$$

Additionally, the impedance differences between solutions with and without MTs are given by:

$$\Delta Z = Z_{\mathrm{MT+buffer}} - Z_{\mathrm{buffer}} = R_{\mathrm{H}} + \tilde{Z}_{\mathrm{MT}} \tag{5}$$

where

$$\tilde{Z}_{\mathrm{MT}} = \frac{\tilde{R}_{\mathrm{MT}}}{1 + (\omega\tilde{C}_{\mathrm{MT}}\tilde{R}_{\mathrm{MT}})^2} - \mathrm{j}\frac{\omega\tilde{C}_{\mathrm{MT}}\tilde{R}_{\mathrm{MT}}{}^2}{1 + (\omega\tilde{C}_{\mathrm{MT}}\tilde{R}_{\mathrm{MT}})^2} \tag{6}$$

We subsequently fit experimental impedance difference curves shown in Figure 3 and Figure S7 to real and absolute value of imaginary parts of ΔZ using R_{H}, $\tilde{R}_{MT}$ and $\tilde{C}_{MT}$ as our fit parameters. Here, R_{H} is a resistance ascribed to the nominal fraction of unpolymerized tubulin present in MT containing solutions. The fitted curves are displayed in Figure 6 and the optimal fit parameters are listed in Table 1 (see Materials and Methods for details).

Tubulin] (μM)	$\widetilde{\boldsymbol{C}}_{\boldsymbol{MT}}$ **(F)**	$\boldsymbol{\delta\widetilde{C}_{MT}}$ **(F)**	$\widetilde{\boldsymbol{R}}_{\boldsymbol{MT}}$ (Ω)	$\boldsymbol{\delta\widetilde{R}_{MT}}$ (Ω)	$\widetilde{\boldsymbol{R}}_{\boldsymbol{H}}$ (Ω)	$\boldsymbol{\delta\widetilde{R}_{H}}$ (Ω)
22.222	1.27×10^{-5}	1.48×10^{-7}	9.74×10^{4}	1.18×10^{4}	2.12	40.61
2.222	1.25×10^{-5}	1.67×10^{-7}	1.00×10^{5}	1.40×10^{4}	0.61	34.79
0.222	2.01×10^{-5}	3.38×10^{-7}	9.97×10^{4}	2.82×10^{4}	0.41	31.95

Table 1. Fit parameters attained by fitting the real and imaginary components of impedance to equation 6. Fit parameters represent effective capacitance $\tilde{C}_{MT}$, and resistance $\tilde{R}_{MT}$ introduced into

the solution through the addition of the MT meshwork at different concentrations. R_{H} is the small constant resistance that is ascribed to small fraction of unpolymerized tubulin that is present in MT containing solutions. $\delta\tilde{R}_{MT}$, $\delta\tilde{C}_{MT}$ and $\delta\tilde{R}_{H}$ correspond to 95% confidence intervals for the fit parameters. Corresponding graphs are displayed in Figure 6.

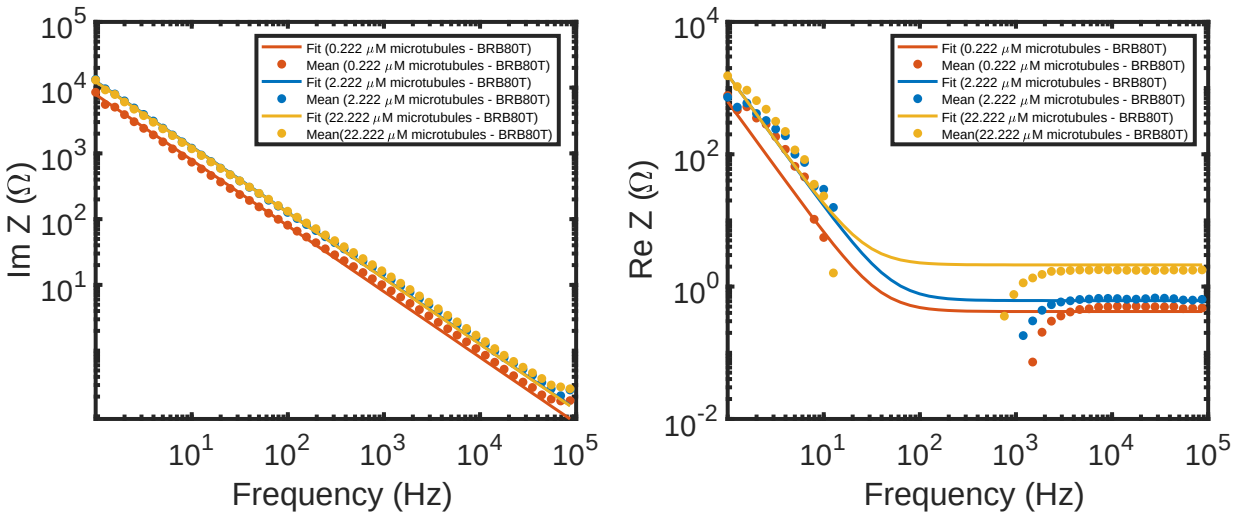

Figure 6. Mean differences of (a) imaginary and (b) real impedance curves for 0.222 μM, 2.222 μM and 22.222 μM, are fitted with the model described in Eq. 5 and Figure 5. Fit parameters and confidence intervals are displayed in Table 1. For detailed fitting methodology, see Materials and Methods.

Experimentally derived values for the capacitance and resistance of a single microtubule match theoretical predictions.

To determine the capacitance and resistance of a single MT from corresponding meshwork mean values, we made two assumptions. (1) The average length of an MT was taken as 20 μm based on visual inspection and (2), MTs were oriented in parallel to one another at 0.222 μM concentration (Figure 2 a). We did not extend the second assumption to concentrations of 2.22 μM and 22.22

μM due to the complex nature of the meshwork, for which a superficial series and/or parallel circuit combination would be inadequate. However, our simple assumptions do provide an upper limit for the capacitance of a single MT. Given the presence of $N = 1.08 \times 10^7$ MTs in our solution volume at the 0.222 μM concentration, the capacitance for a single 20 μm long MT was calculated to be $C_{MT} = \tilde{C}_{MT}/N = 1.86 \times 10^{-12}$ F. We extended our set of assumptions to calculate the resistance of a single 20 μm long MT at 0.222 μM concentration. Here, the total resistance of the MT meshwork would be given by $\tilde{R}_{MT} = R_{MT}/N$ where each MT has resistance R_{MT}, implicating $R_{MT} = \tilde{R}_{MT} \times N = 1.07 \times 10^{12}\Omega$.

To determine if these values were consistent with theoretical predictions, we first calculated the Debye Length λ_D of an MT when present in an ionic concentration $c = 80$ mM to account for an environment consisting of BRB80T.

$$\lambda_D = \frac{\lambda_D^0}{\sqrt{c}} \tag{7}$$

Here, λ_D^0 is a Debye length (calculated to be ~10 nm) in a solution with ionic concentration $c_0 = 1$ mM [13]. Based on the Debye length, the total capacitance of the MT cylinder has been calculated by previous models to be given by [8, 44]:

$$C_{MT,theor} = \frac{2\pi\varepsilon_0\varepsilon l}{\ln\left(\frac{r+\lambda_D}{r}\right)} \tag{8}$$

Substituting the value of λ_D from equation (7) and using the Taylor series expansion for $\ln(1+x)$, we get:

$$C_{MT,theor} = \frac{2\pi\varepsilon_0\varepsilon l}{\ln\left(1 + \frac{\lambda_D^0}{r\sqrt{c}}\right)} \approx \frac{2\pi\varepsilon_0\varepsilon l r\sqrt{c}}{\lambda_D^0} \quad (9)$$

We continued our previous assumption that each microtubule has length $l = 20$ µm, radius r = 12.5 nm and is surrounded by an atmosphere of dielectric constant $\varepsilon = 80$. Placing these values in Equation 9, we calculated the total capacitance of an MT to be 9.8 pF. Interestingly, previous work shows that $\varepsilon < 10$ for tubulin protein [1], enabling an extrapolation of $\varepsilon = 40$ between the protein and the bulk solution [47]. The use of this intermediate value for ε leads to an MT capacitance of 4.9 pF. It is worth noting that both these values are consistent with our experimentally observed value of 1.86 pF per MT.

Fluorescence images (Figure 2 a) showed that MTs orient in the plane of glass surfaces at low concentrations, indicating that resistance to ionic charge transport in solution arose from flow across the porous MT cross section, rather than along their lengths. The lower limit of Ionic current through nanopores present between adjacent tubulin dimers in two-dimensional zinc-induced sheets is previously calculated to be $0.02fA$, at a driving voltage of 5 mV [10]. This leads to a nanopore resistance of $R_{nanopore} = 2.50 \times 10^{14}$ Ω. We note that the conditions of our study vary significantly from those in this reference: Firstly, unlike two-dimensional sheets which present ions with a single resistance, cylindrical microtubules used in our study present both MT entering and MT exiting resistances (Figure 8 a). Secondly, the study utilizing sheets was performed at an ionic concentration of 140 mM, whereas our study was performed at a corresponding concentration of 80 mM. Thirdly and importantly, the resistance calculated above applies to resonant conditions,

with the driving input frequency in the range of 10-100 Hz and would reduce as the input frequency is altered.

We calculated the total resistance for ions entering and exiting transversely through an MT as $R_{MT,enter} = R_{MT,exit} = \frac{R_{nanopore}}{n} = 7.69 \times 10^9\ \Omega$ where n is the total number of nanopores (equal to the number of tubulin dimers) in a 20 μm long MT. We then addressed the first dissimilarity mentioned above, regarding MT entering and exiting impedances, by adding these resistances in series. This leads to total resistance $R_{MT} = R_{MT,enter} + R_{MT,exit} = 1.54 \times 10^{10}\ \Omega$ for ionic movement across the cross section. We also note that theoretically, the total ionic resistance across the cross section has been calculated by [48]:

$$R_{MT,theor} = \frac{\ln\left(1 + \frac{\lambda_D}{r}\right)}{2\pi l\sigma} \approx \frac{\lambda_D}{2\pi l\sigma} \tag{10}$$

Here, the solution conductivity σ is proportional to the ionic concentration c. Additionally, because λ_D depends inversely on the square root of ionic concentration, $R_{MT,theor}$ scales with $\frac{1}{c^{3/2}}$. Thus, to correct for the second dissimilarity arising from the difference in ionic concentration, we multiplied our resistance by a factor of $\mu = \left(\frac{140}{80}\right)^{3/2} = 2.31$. Thus, our computed final value for the resistance across the MT cross section was $\mu \times R_{MT,theor} = 4.71 \times 10^{10}\ \Omega$. We expect the resistance of the MT to be higher in non-resonant conditions (at frequencies >100 Hz), implicating a close match to our experimentally derived value of $1.07 \times 10^{12}\Omega$. As the applied AC voltage oscillates with the frequency it switches electric field direction every half cycle, aiding the movement of in-phase ions across the MT wall, but impeding this flow in out-of-phase ions. Thus, the total resistance is also a function of the input frequency.

Ionic movement to enter the MT cylinder takes place against the potential gradient set up by the tubulin dipole in the MT wall, which could additionally account for the high resistance that we observe.

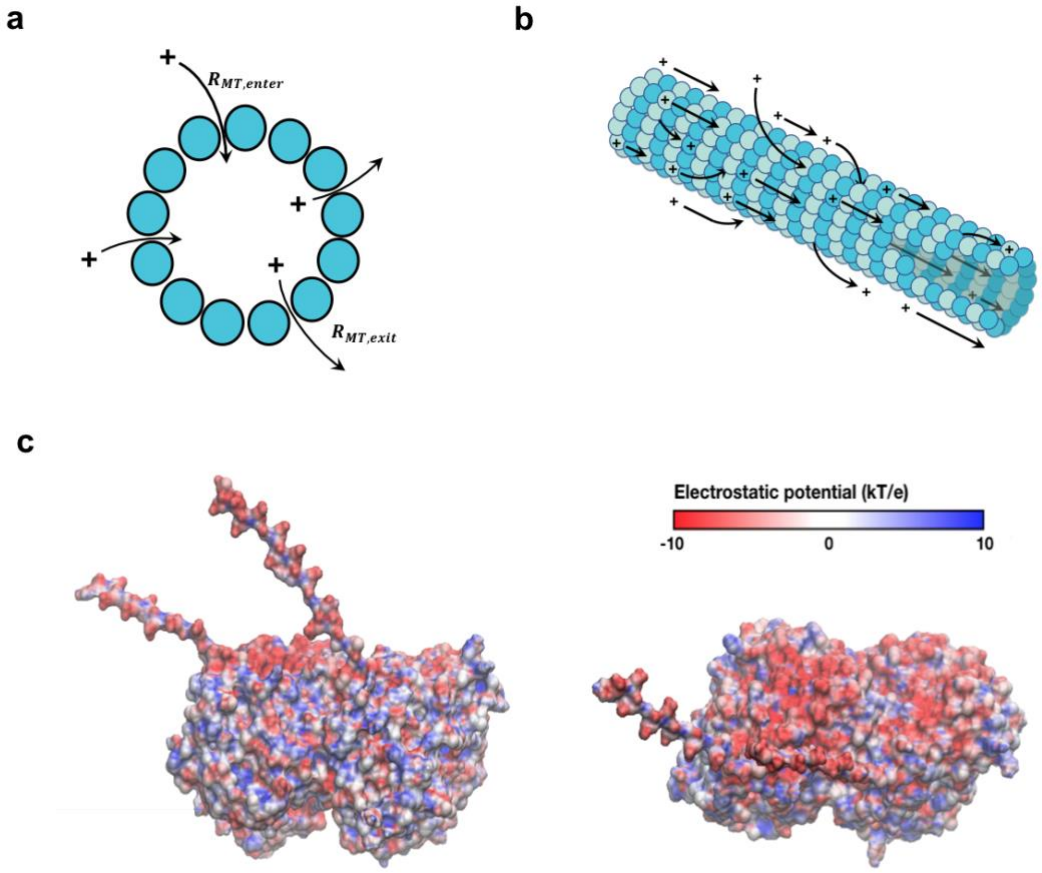

Figure 8. Schematic of charge transport along and across an MT. (a) A representation of charge flow across the MT cross section through nanopores present between adjacent protofilaments. (b) A representation of charge flow through both inner and outer modes along an MT. Arrows depict charge flow via both mechanisms, enabling MT charge storage across a broad spectrum of frequencies, and charge transport at low AC frequencies in the cell. (c) Side view (left) and top view (right) of the tubulin dimer, displaying distribution of electrostatic potential at different

locations. The negatively charged C-termini face towards the solution and contains ~50 % of the total negative charge on a tubulin dimer.

DISCUSSION

Our measurements using a parallel plate contact device reveal interesting electrical properties of MTs at physiological concentrations. Unlike studies exposing MT-containing solutions to non-uniform electric fields [12-14, 20], our device allowed robust quantification of electrical impedance in the presence of spatially uniform electric fields. Our results show that the addition of MTs mimics a parallel RC element placed in series with the high-ionic strength solution, with a nonlinear dependence on the MT number. Unpolymerized tubulin did not alter capacitance significantly, indicating changes in electrical properties of tubulin as it polymerizes.

The physical underpinnings of an increased capacitance

Dense counterion condensation on the MT surface has been previously extensively predicted and simulated due to a variety of reasons [7-8, 44, 49-50]. Firstly, the negative charge of the tubulin dimer attracts counterions in solution, leading to the presence of a double layer and depletion region outside the microtubule surface [7-8, 46, 50]. In addition to this, the charge distribution in the MT protein wall is highly non-uniform, with the outer surface containing approximately four times the charge compared to the inner surface [49]. This asymmetry between the inner and outer electrostatic potentials serves to enhance capacitance and is responsible for the abnormally large dipole moment of the tubulin dimer [1]. The asymmetry also manifests through C-terminal ‘tails’ composed of 10-12 amino-acids, that can extend 4-5 nm outwards from each tubulin monomer. These slender C-termini tails are highly negative, containing about 50 % of the charge of the tubulin dimer [51]. As

they stretch outwards into the solution in a pH and ionic strength dependant manner, they increase the effective area of the tubulin dimer and significantly contribute to the overall MT capacitance [7-8].

Coherent oscillations of these C-terminal tails are modelled to generate solitonic pulses of mobile charge along the outer surface of an MT, creating ionic currents along its' length [7, 44, 52].

Ions from the bulk solution are also modelled to be pumped into the hollow MT lumen through nanopores in its' wall, resulting in charge accumulation inside the cylindrical MT over time [45]. A recent study using molecular dynamics simulations showed that the permeability of the MT lumen was significantly higher for Na^+ and K^+ as opposed to Ca^{2+}, allowing for free movement of selective ions into the MT lumen across its porous surface [49]. To the best of our knowledge, our findings are the first to experimentally quantify this resistance encountered by charge flow across the MT cross section. These results implicate not only ionic movement along the microtubule axis, but also across and inside it, enhancing the modelled roles of MTs as complex subcellular nanowires.

Implications for the cell

Our findings on cell-like complex MT networks at physiological tubulin concentrations indicate a cellular role for MTs as wires that store charge. Neuronal environments where the MT are spontaneously nucleated from free tubulin, such as growth cones, would experience large capacitance changes over short bursts of time. This ability would significantly impact action potentials, that are known to depend strongly on the local charge distributions [53]. Additionally, ionic movement across the MT wall would enhance their roles as attenuators of local cation

distributions. In nonneuronal environments, transient ionic currents around on a MT during mitosis could impact MT dynamics and potentially influence chromosome segregation. Specifically, Ca^{2+} ion storage/flow about an MT would trigger its' depolymerization, whereas waves of Mg^{2+} or lowering in the local pH (increasing H^{+}) would lead to MT stabilization [54-55]. The attraction of Zn^{2+} or Mn^{2+} ions in the vicinity would lead to formation of two-dimensional tubulin polymers [56-57]. Properties of the cytoplasm such as polarizability and relative permittivity would get severely attenuated due to the presence of MTs in the vicinity. Due to the polymerization state of tubulin altering solution capacitance, our findings implicate a temporal evolution of capacitance and ionic flows as the ratio of MTs to free unpolymerized tubulin changes [58-60]. MT lattice defects, that take place when a tubulin dimer is missing in an MT wall [61-62], would cause a large ionic flux to develop at the defect site. Such localised shifts in charge distributions would be most significant at the MT end, where large localized charge fluxes would form due to sudden changes in both ionic resistance and capacitance. Free/polymerized tubulin would thus regulate local and global electrical properties, creating spatially dynamic gradients of charge storage and flux. We envision a cytoskeleton that, in addition to transporting macromolecules, stores and transports ionic signals and electrical information across the cytoplasm (Figure 8 b).

Our findings can be coupled with a vast array of bio-nanodevices that utilize MTs and MAPs (microtubule-associated proteins) for construction of bio-nanotransporters and bio-actuators (Diez 2003; Hess 2018; Li 2018; Kapitein 2018; Yokokawa 2012). In specific conditions, MAP-MT systems are capable of repositioning organelles (Kapitien 2017), directionally transporting microtubules (Yokokawa 2004; Isozaki 200) and even drive their movement within zero-mode waveguides (Yokokawa 2018) and inorganic nanotubes (Li 2014). Charge storage and transport

along MTs can be coupled to mechanical MAP-based movement to develop a wide range of nano-actuators and nano-sensing devices.

Future work

When compared to cells, the rates of MT nucleation and polymerization are significantly lower in BRB80. This difference can be attributed to the absence of MAPs and macromolecular crowding [63-64]. Mammalian cells contain high concentrations of K^+ ions (140-300 mM) [27-28], which, in addition to MAPs and molecular crowding agents, would be included in a future study to attain physiological equivalence. Raw data from complex impedance will be used to derive properties of individual MTs such as the dielectric constant, refractive index, polarizability and Debye relaxation time through the Clausius-Mossotti equation. To improve the present two-contact parallel-plate device, a four-contact device will lower electrode polarization effects and enable quantification of the relative permittivity of MT and tubulin containing solutions. Our device geometry will also be used to perform DC (direct-current) measurements, determine the contribution of MTs to impedance relaxation time and evaluate the voltage dependence of capacitance on MT-containing solutions. Interestingly, this aspect has been discussed previously: the inductance of a single protofilament is calculated to be <1 fH [8]. Further investigation is required to experimentally confirm these predictions.

CONCLUSIONS

We used EIS to compare the complex impedance of MT- and tubulin-containing solutions. A high ionic strength buffer (BRB80) created a high noise, low impedance background, which was countered through the use of physiological concentrations of tubulin. While the presence of MTs

increased solution capacitance, unpolymerized tubulin did not have any appreciable effect. We used an RC circuit to model the MT meshwork across three orders of concentrations, and found at physiological levels, the capacitance and resistance of the meshwork is to be 1.27×10^{-5} F and 9.74×10^{4} Ω . In a study that is the first of its' kind to the best of our knowledge, we determined the capacitance of a single MT to be 1.86×10^{-12} F. We used our lowest, 0.222 μM concentration, for this calculation and assumed that MTs were oriented in parallel to one another. We envision a dual electrical role for MTs in the cell: that of charge storage devices across a broad frequency spectrum (acting as storage locations for ions), and of charge transporters (bionanowires) at specific, low frequencies between 20 and 60 Hz. Our findings also indicate that the electrical properties of tubulin dimers change as they polymerize, revealing the potential impact of MT nucleation and polymerization on the cellular charge distribution. Our work shows, that by storing charge and attenuating local ion distributions, microtubules play a crucial role in governing the bioelectric properties of the cell.

MATERIALS AND METHODS

Tubulin reconstitution

Before reconstituting tubulin solution from lyophilized protein powder, G-PEM buffer was prepared by adding 2.5 μL GTP to 247.5 μL BRB80 (Cytoskeleton Inc; BST01). 180 μL of G-PEM buffer was pipetted into lyophilized tubulin powder (Cytoskeleton Inc; T240). 20 μL of MT cushion buffer (60% glycerol in BRB80) was subsequently added. All steps were performed on ice.

Rhodamine-labelled tubulin solution was prepared by adding 4 μL of G-PEM buffer to labelled tubulin powder (Cytoskeleton Inc; TL590m) and subsequently adding 1 μL of MT cushion buffer.

A final tubulin stock solution of 45.45 μM was thus prepared, and a final labelling ratio of 1:15 was used. Finally, aliquots were snap-frozen and stored at -80 °C.

MT polymerization and stabilization

MT polymerization was performed by incubating 45.45 μM tubulin aliquots in a 37 °C water bath for 30 minutes. BRB80 solution was heated alongside tubulin during the first 15 minutes of polymerization. Subsequently, BRB80 was incubated at room temperature, and paclitaxel solution (Cytoskeleton Inc, TXD01; 2 mM stock) was thawed at room temperature alongside it. After 30 minutes of tubulin polymerization brought to completion, 100 μL of BRB80 was added to 5 μL of 2 mM paclitaxel. For preparing 0.222 μM MTs, 99.5 μL of this solution was added to 0.5 μL tubulin solution. For 2.222 μM MTs, 95 μL of this solution was added to 5 μL tubulin solution. For 22.225 μM MTs, 5 μL of this solution was added to 5 μL of tubulin solution. For preparing BRB80T, 45 μL of this solution was added to 45 μL of BRB80.

For tubulin stabilization, 2 μL of colchicine stock solution (Sigma-Aldrich, C9754; 5 mM in DMSO) was added to 100 μL BRB80. For preparing 0.222 μM tubulin, 99.5 μL of this solution was added to 0.5 μL unpolymerized tubulin solution. For 2.222 μM tubulin, 95 μL of this solution was added to 5 μL unpolymerized tubulin solution. For 22.225 μM tubulin, 5 μL of this solution was added to 5 μL of unpolymerized tubulin solution. For preparing BRB80C, 45 μL of this solution was added to 45 μL of BRB80.

Fluorescence imaging of MTs

Imaging was performed on a Zeiss Examiner.Z1 microscope using a Hamamatsu EMCCD C9100 camera, a Zeiss plan-Apochromat 1.4 NA 63x lens. After pipetting MT solution (2-5 μL) onto a glass slide (VWR 48382-173) a coverslip (VWR 48393-070) was placed on the solution, allowing it to spread. The microscope used an EXFO X-Cite 120 fluorescence source and excitation and

emission filters of 535 nm and 610 nm, respectively. Exposure times between 50 ms and 300 ms were used for imaging to validate the presence of MTs.

Electrode design and device construction

Each 'plate' in the parallel-plate contact device was formed by FTO (Fluorine-doped Tin Oxide)-coated glass slides (Sigma Aldrich, 735140). The slides were cleaved to dimensions of 1.5 mm x 10 mm x 50 mm for the upper contact and 1.5 mm × 27 mm × 50 mm for the lower contact. The cleaving dimensions were set using 3D printed devices that were placed as holders (The Shack, University of Alberta; Figure S1). The slides were sonicated and subjected to Reactive Ion Etching (RIE) using a 5-minute exposure to oxygen plasma (Oxford Instruments, NGP80) to remove surface particulate matter. 70 μm thick double-sided tape was used as a spacer, which formed a chamber of dimensions 3 mm × 1.25 cm × 70 μm. The top electrode was placed using a separate 3D-printed holder device (Figure S1). Once the device was constructed using the above protocol, solution was perfused into the chamber using a pipette and a filter paper for suction (Video S1). We used flat copper electrode clips in a three-electrode configuration to connect to our capacitor device. The counter electrode was connected to the lower electrode, and the working and reference electrodes were connected to the top electrode of our device.

Impedance measurements

Experiments were conducted using Electrochemical Impedance Spectroscopy (EIS) on a Zahner Zennium impedance analyzer. The parallel-plate contact device was placed into the 3D-printed holder for stabilisation (Figure S1). The contacts from the machine were connected to the parallel-plate device using flat-faced copper alligator clips. A three-electrode configuration was used: the counter electrode was attached to the lower contact of the parallel-plate device, whereas the working electrode was attached to the upper contact with the reference electrode orthogonally

clipped onto the clip of the working electrode. Within the Thales Z3.04 environment, the Potentiostat Mode was ON; the stabilisation delay was set to 1 s, the rest potential drift tolerance was set to 250 μV, V_{rms} was set to 5 mV. Solutions were perfused into the experimental chamber using a micropipette tip at one opening, and a filter paper at the other opening for suction, similar to protocols used for TIRF (Total Internal Reflection Fluorescence) microscopy[31]. The frequency range of the EIS measurement was set from 4 MHz to 1 Hz, and data were subsequently collected.

Data analysis

MT and tubulin samples were analyzed using data from five to seven days of experiments. Each day consisted of three to seven solutions for each concentration being tested, with one frequency sweep per solution. Readings of each sweep were saved as a .csv file, and next sample was loaded by solution exchange method. Water was run as the first solution for each day of experiments. BRB80T was run prior to MT solutions, and BRB80C were run prior to the free tubulin containing solutions. MT- and free tubulin-containing solutions were run on separate days, in increasing order of concentration. MATLAB (The Mathworks; Natick, MA) scripts were used for data analysis. Fitting to the real and imaginary components of impedance was performed using the function lsqnonlin. Initial guess values for the MT meshwork resistance and capacitance were 10^5 F and 10^{-5} Ω, respectively, based on visual inspection of raw data. The initial guess values for the nominal series resistor, R_H, were set at 1.78, 0.6 and 0.4 Ω for with tubulin concentrations of 0.222, 2,222 and 22.222 μM respectively. The 95% confidence intervals were determined using the function nlparci. Error propagation was performed assuming no relationship between various days of data collection.

AUTHOR INFORMATION

Author contributions.

AK, JT and KS designed the paper. AK, SP, AB and UM collected the data. AK, SP, AB, KS and JP analysed and validated the data. AK, VR, KS and JT wrote the paper. AK, SP and JP made figures for the paper. KS, JT and JL provided resources for experimentation and analysis.

Notes

The authors declare no competing financial interests.

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Supporting Information for

On the capacitive properties of individual microtubules and their meshworks

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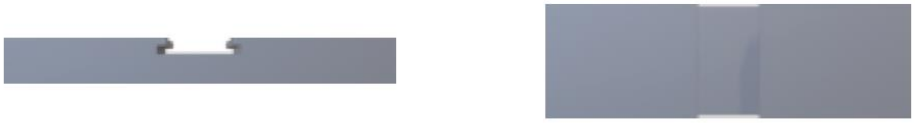

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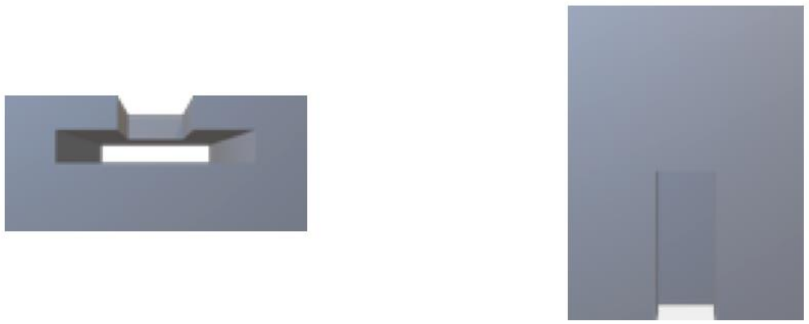

Figure S1. 3D printed holders used to fabricate and align the parallel-plate contact device. (a) Top view (left) and side view (right) of holder for the parallel plate device used to perform impedance measurements. (c) Top view (left) and side view (right) of slider used to position the double-sided tape exactly to fabricate the device. (d) Top view (left) and side view (right) of the holder used to position the upper contact precisely on the lower contact.

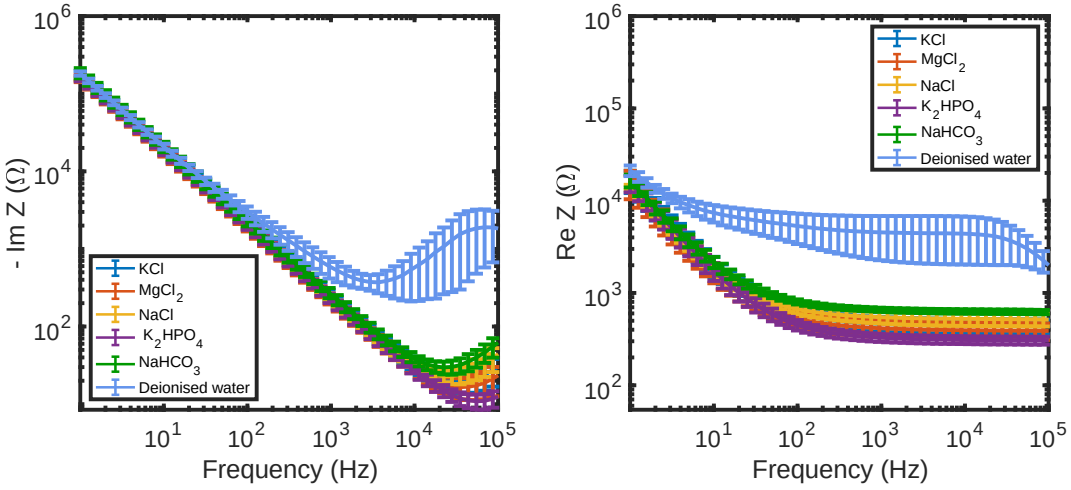

Figure S2. Validation of parallel-plate contact device using 0.5 mM electrolytic solutions (a) Imaginary component of impedance for electrolytic solutions at 0.5 mM and de-ionised water. (b) Real component of impedance for electrolytic solutions at 100 mM and de-ionised water. Data displays average values collected between 15 and 21 times. Error bars represent standard deviation.

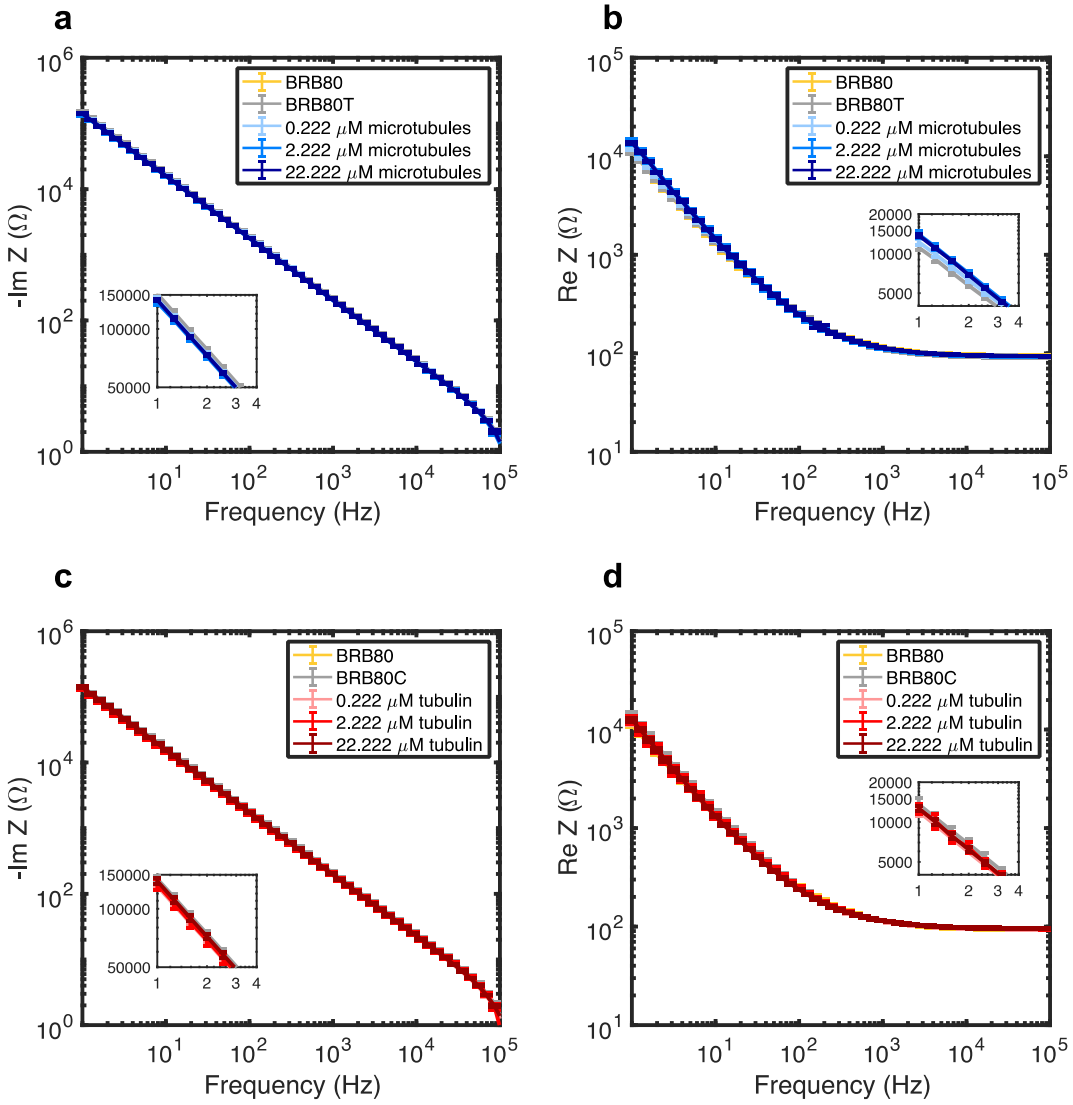

Figure S3. Example of microtubule and tubulin data for the purpose of displaying typical impedance values. Raw values for (a) imaginary and (b) real component of impedance at different

concentrations of microtubules. Raw values for (c) imaginary and (d) real component of impedance at different concentrations of unpolymerized tubulin.

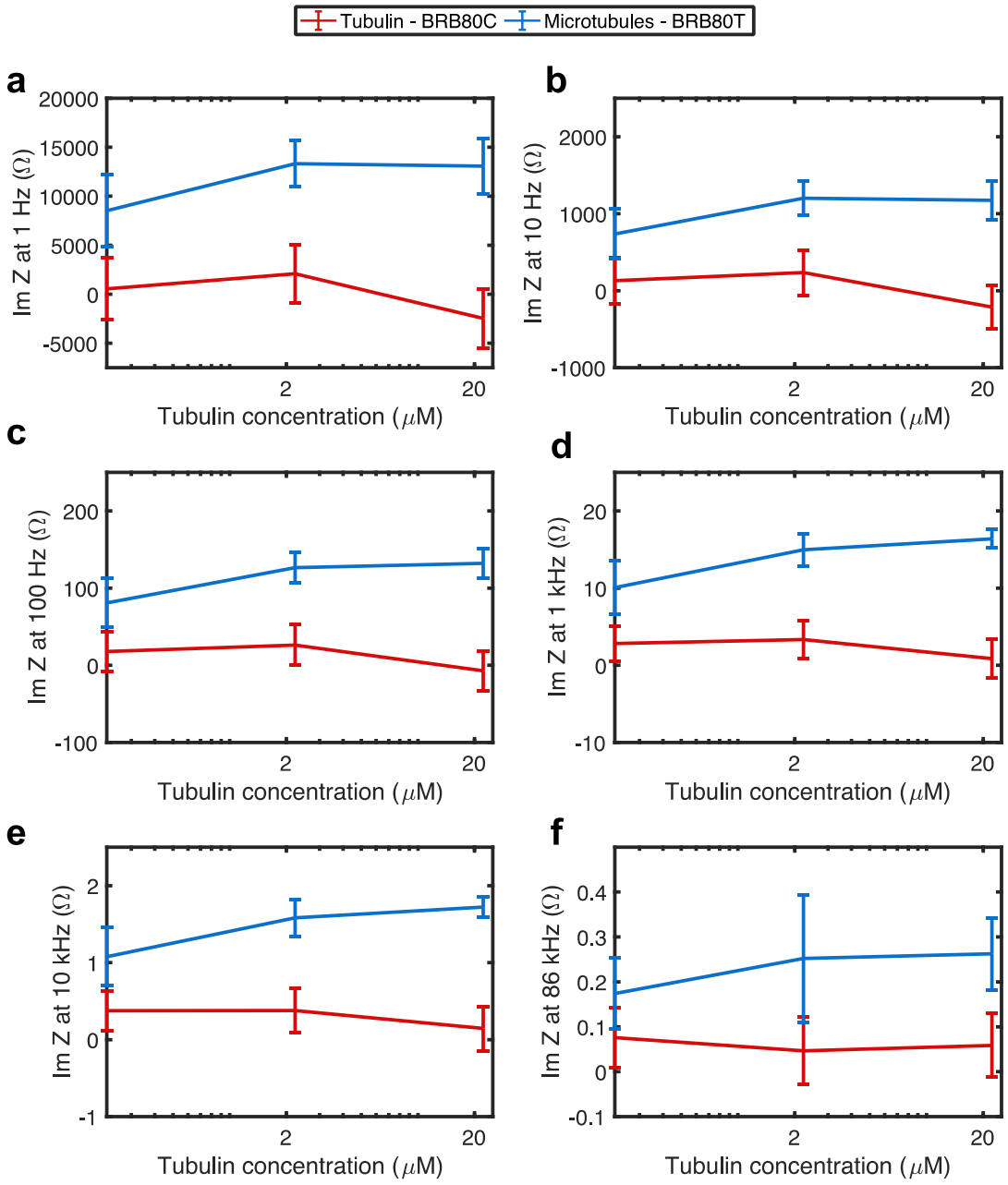

Figure S4. Microtubules increase solution capacitance while unpolymerized tubulin does not. Graphs showing differences in the imaginary component of impedance as a function of tubulin concentration at input AC frequencies of (a) 1Hz, (b) 10 Hz, (c) 100 Hz, (d) 1 kHz, (e) 10 kHz and (f) 86 kHz.

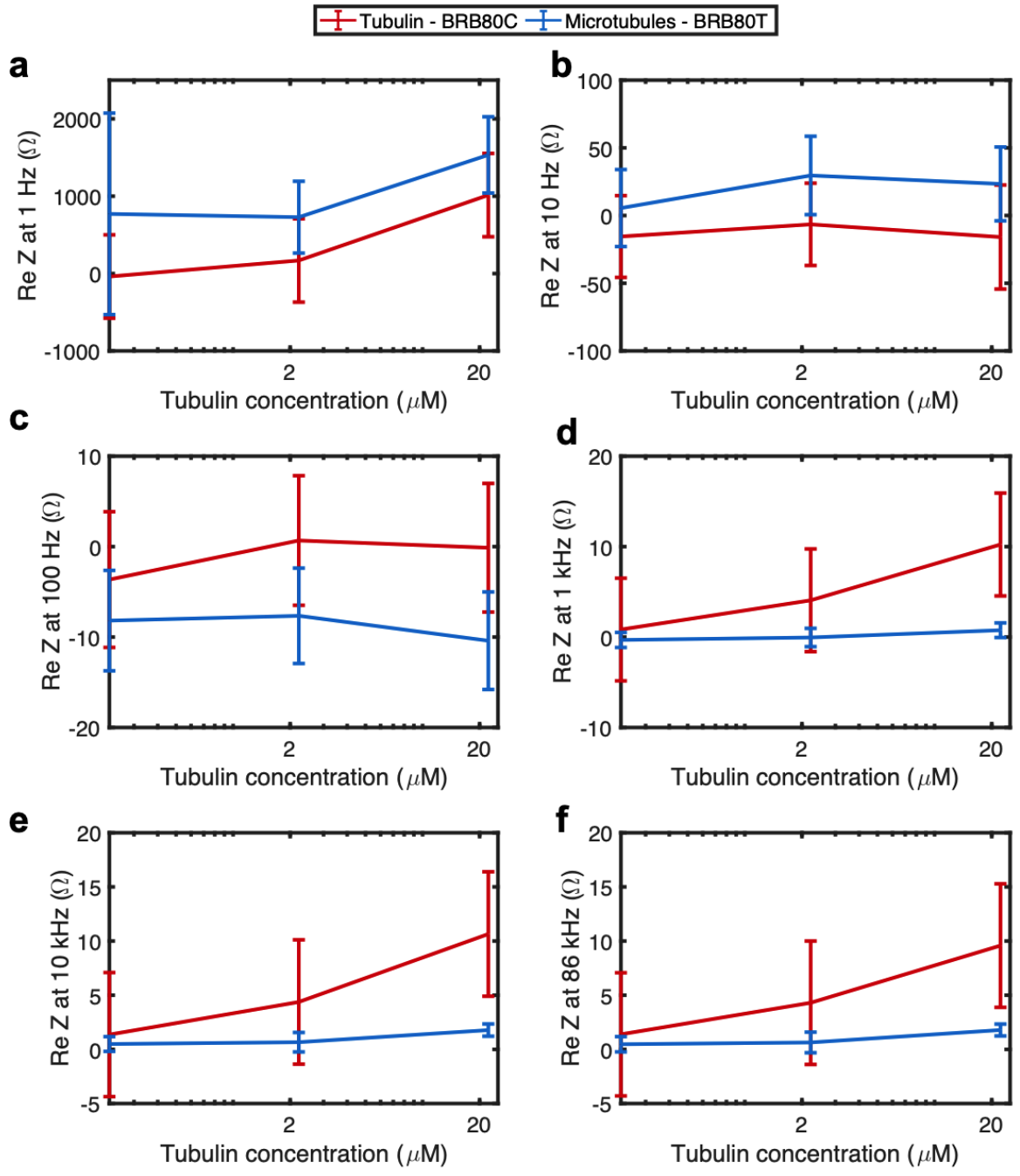

Figure S5. A 'reversal' in the resistive behavior of microtubules is observed between 10 and 100 Hz. Graphs showing differences in the real component of impedance as a function of tubulin concentration at input AC frequencies of (a) 1Hz, (b) 10 Hz, (c) 100 Hz, (d) 1 kHz, (e) 10 kHz and (f) 86 kHz.

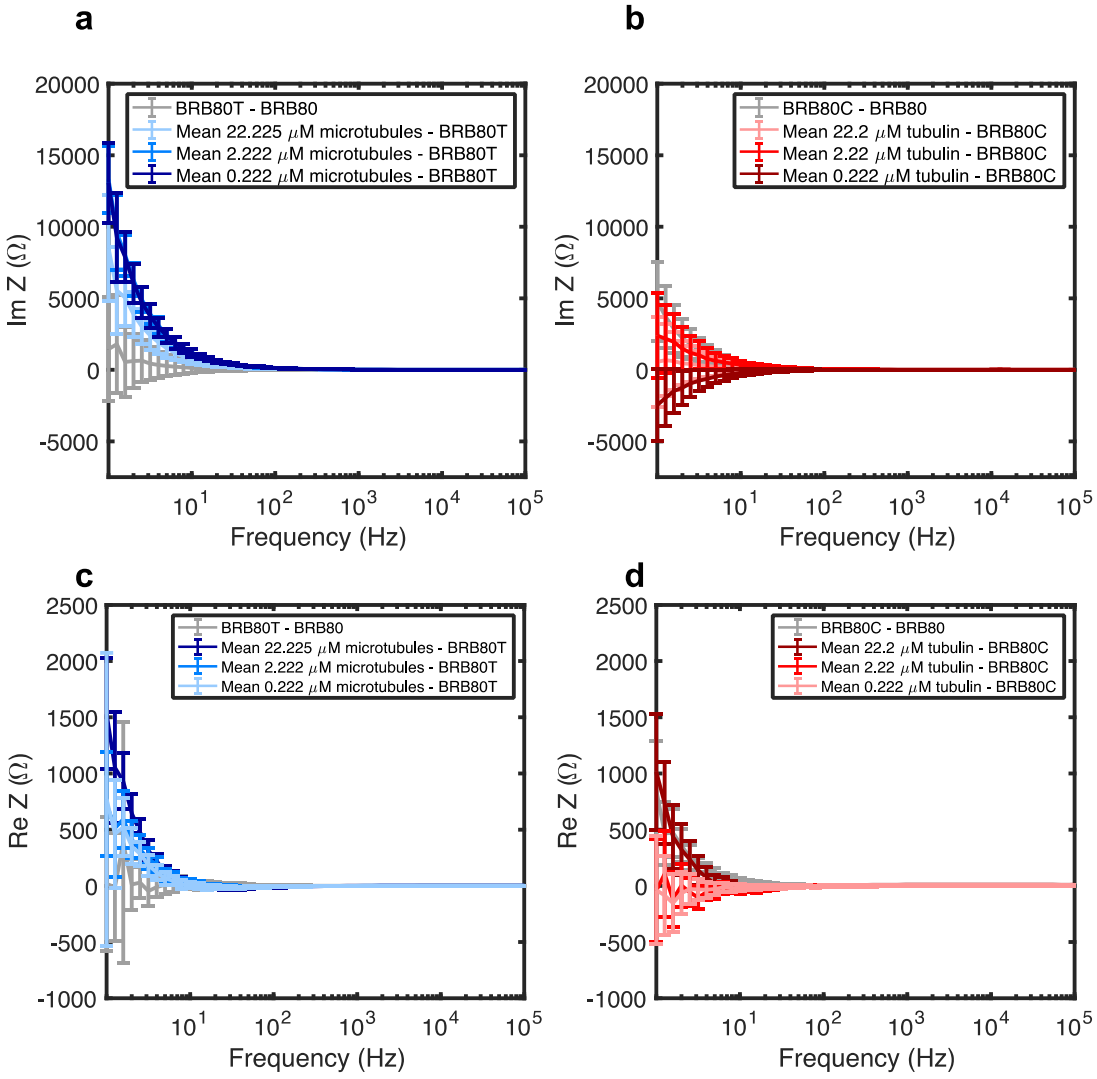

Figure S6. Example of microtubule and tubulin subtraction with backgrounds, to display typical impedance differences. Raw values for (a) imaginary and (b) real component of impedance at different concentrations of microtubules. Raw values for (c) imaginary and (d) real component of impedance at different concentrations of tubulin.

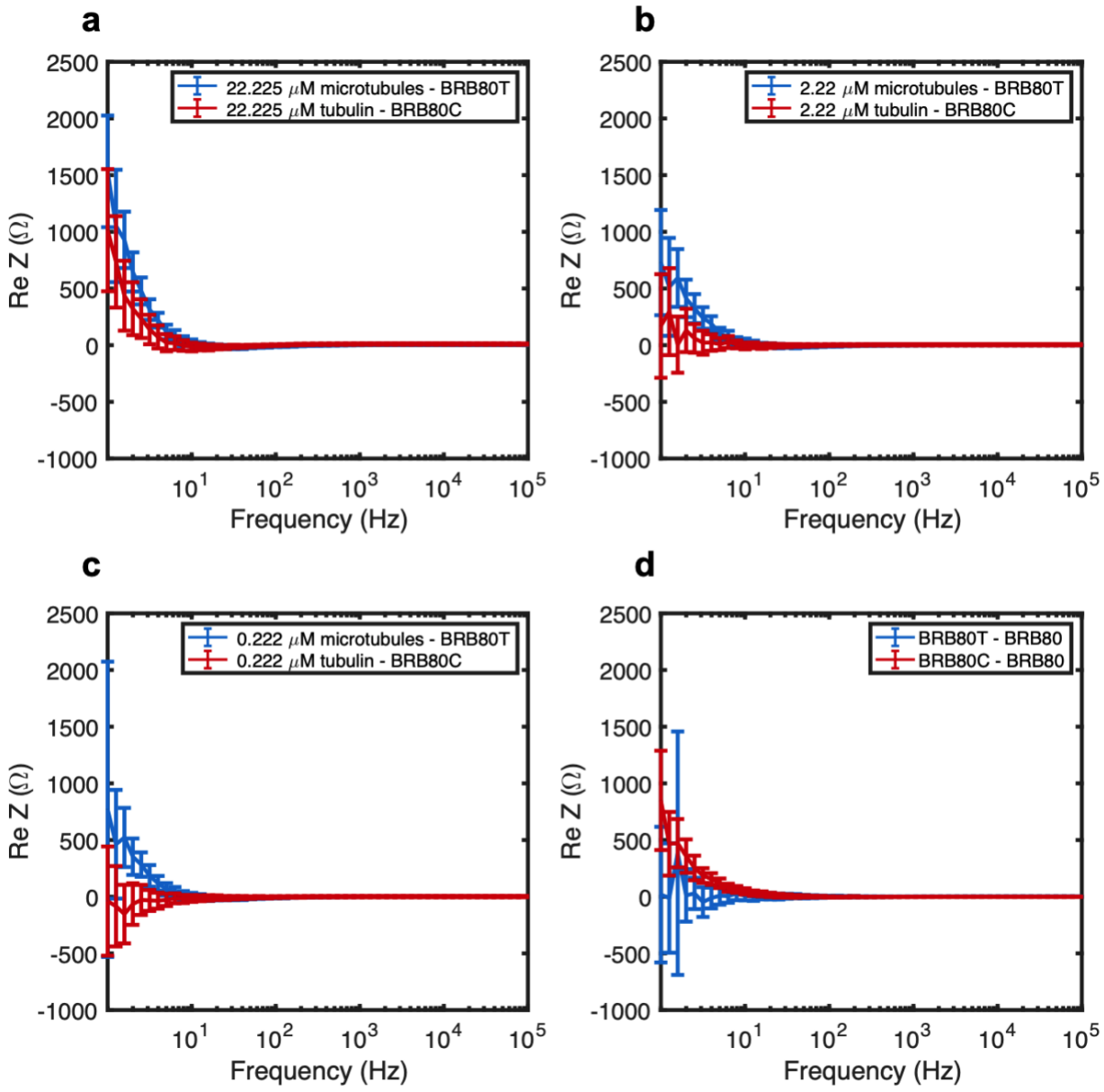

Figure S7. The trend for both microtubules and tubulin displays increasing solution resistance. Graphs showing differences in the real component of impedance as a function of decreasing input AC frequency at total tubulin concentrations of (a) 22.225 μM (n = 22 experiments for tubulin, n = 21 for microtubules), (b) 2.222 μM (n = 35 experiments for tubulin, n = 49 for microtubules) (c) 0.222 μM (n = 35 experiments for tubulin, n = 49 for microtubules), (d) comparison of the effects of paclitaxel (BRB80T) and colchicine (BRB80C, n = 49 experiments for BRB80T, n = 35 for

BRB80C, n = 84 experiments for BRB80). Graphs display average values. Error-bars represent standard deviation.

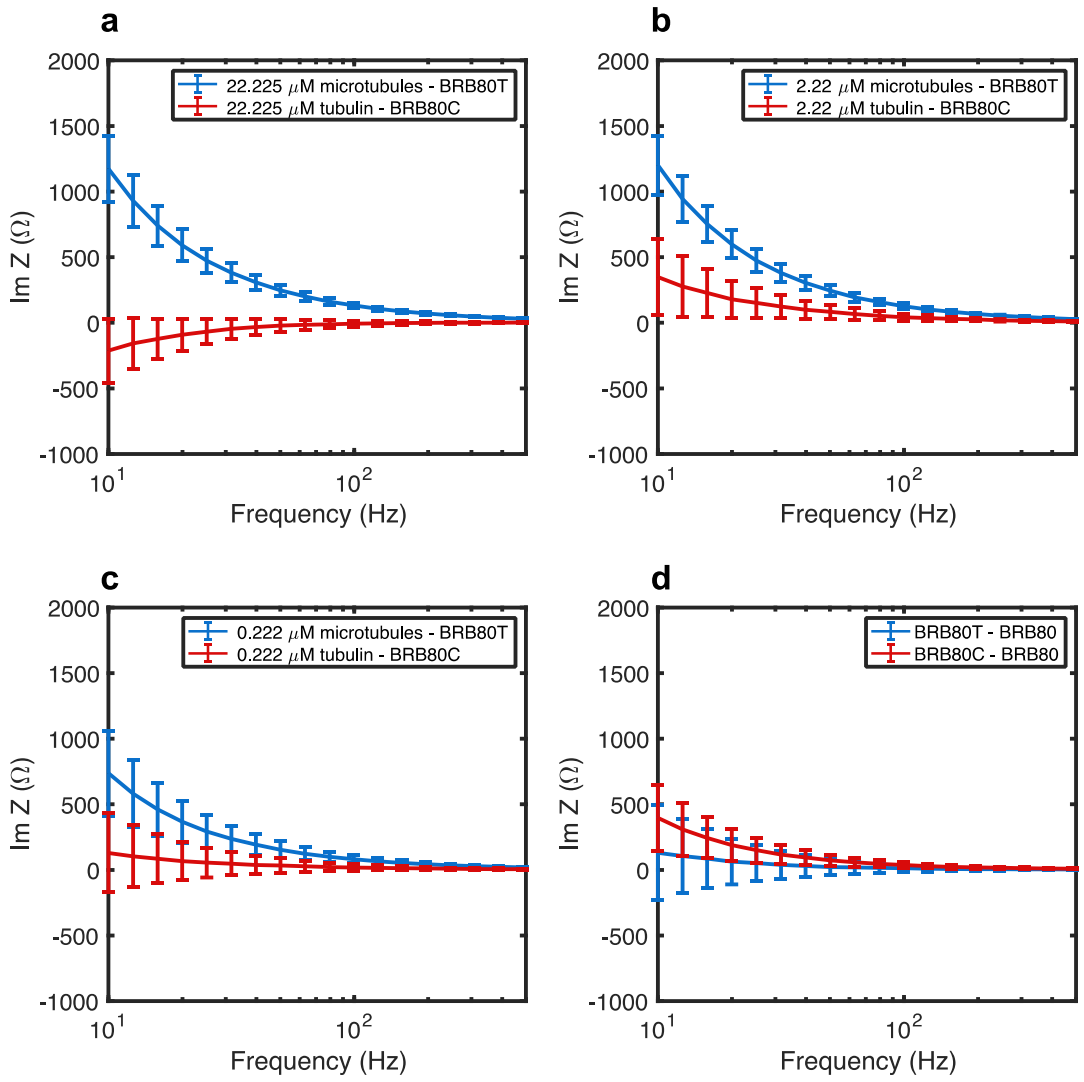

Figure S8. No 'reversal' in the resistive behavior of microtubules is observed between 10 and 100 Hz. Graphs showing differences in the real component of impedance as a function of decreasing input AC frequency at total tubulin concentrations of (a) 22.225 μM, (b) 2.222 μM, (c) 0.222 μM, (d) comparison of the effect of paclitaxel and colchicine on impedance.

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Optical spectrum of a Hubbard chain

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The optical absorption of the one-dimensional Hubbard model is calculated by three different methods. Linear chains of five atoms and rings of five and seven atoms containing four electrons are treated numerically. The infinite-chain problem is solved first in the t-matrix approximation of Lyo and Holstein. It is shown that in this approximation, most of the high-frequency absorption is due to a bound state which lies above the band continuum. Finally, the absorption is evaluated in the memory-function formalism of Götze and Wölfle, which reduces to ordinary perturbation theory at high frequency. The three approaches are in qualitative agreement, and the differences between them can be explained by the nature of the approximations involved. Applications to tetrathiafulvalene-tetracyanoquinodimethane (TTF-TCNQ) and the platinum salt $K_2Pt(CN)_4Br_{0.3}\cdot 3H_2O$ (KCP) are discussed.

I. INTRODUCTION

The goal of this paper is to calculate the optical absorption of a one-dimensional, interacting system of electrons. Our work was motivated by the infrared and optical anomalies recently observed in the quasi-one-dimensional metals tetrathiafulvalene-tetracyanoquinodimethane (TTF-TCNQ) and $K_2Pt(CN)_4Br_{0.3}\cdot 3H_2O$ (KCP). It has been suggested by Torrance *et al.*[1] that the optical spectra of many TCNQ salts could be interpreted in terms of a strongly correlated Hubbard model, in which the long-range part of the Coulomb interaction is neglected. An unexplained anomaly has also been found[2] near the plasma edge of KCP; although it could plausibly be attributed to interband transitions,[3] the possibility that the electron-electron interaction plays a role in these anomalies cannot be discarded. Such experiments point out the need for a reliable theory of optical-absorption processes in correlated one-dimensional systems. In the present paper, we discuss three different methods to calculate the absorption spectrum of the Hubbard model. The first method consists in calculating the spectra of finite chains numerically; this brute-force approach can only be used for relatively short chains. The second method is based on the t-matrix approximation introduced by Lyo and Holstein,[4,5] who used it to calculate the low-frequency conductivity of the Hubbard model. We find that, in this approximation, most of the absorption is due to a bound state which lies above the band continuum, and which was neglected in their calculation (which is justified in the low-frequency limit considered by these authors). Finally, we calculate the optical conductivity in the memory-function approximation of Götze and Wölfle.[6] This approach is exact in the weak-coupling limit, since it reduces to ordinary perturbation theory at high frequency. Our purpose in using this last approach is twofold. First, it provides a test for the validity of the t-matrix approach in the weak-coupling limit. Second, it allows us to check the accuracy of the memory-function approximation when extrapolated to higher values of the coupling. We find that although the result of such an extrapolation has the correct order of magnitude, the memory-function approximation cannot be trusted for couplings which are not very weak. The failure of this approximation is related to that of the first Born approximation in the two-body scattering problem.

Three basic conclusions can be drawn from our results. First, the remarkable agreement between the numerical spectra and the t-matrix results indicates that the optical conductivity is indeed a measure of short-range correlations, which are correctly treated in the t-matrix approximation. This suggests that a similar method could also be used to treat more realistic models of a quasi-one-dimensional conductor. Second, we propose that the validity of the first Born approximation in a one- or two-body problem provides a useful criterion for the applicability of the memory-function approximation to the corresponding many-body problem. Third, we do find an absorption peak at a frequency of order U, the on-site repulsion energy, as suggested by Torrance *et al.* This peak is due to transitions in which the final state contains two electrons sitting at the same site. However, if we use parameters appropriate for TTF-TCNQ and KCP, we find that the optical (and infrared) absorption is quite small; most of the absorption occurs at or near zero frequency. This is especially true for KCP, which has a low carrier density (0.3 hole per Pt atom). Thus, it seems that some of the correlation effects neglected by the Hubbard model must play an important role in the optical anomalies observed in real quasi-one-dimensional conductors. This is hardly surprising, since the Hubbard model assumes (i) one tight-binding band, (ii)

a short-range interaction, and (iii) no electron-phonon interaction. On the other hand, our results do not imply that the correlation effects included in the Hubbard model are negligible; we only show that these effects do not *by themselves* explain the observed anomalies.

In the Hubbard approximation, the electron-electron interaction is represented by a term $\sum_i U\hat{n}_{i\uparrow}\hat{n}_{i\downarrow}$ in the Hamiltonian; U is the repulsion energy experienced by two electrons of opposite spin sitting at the same site i. *A priori*, one expects to see large effects in the conductivity if U is comparable to the half-bandwidth ϵ_0. The magnitude of U has been estimated at ~ 0.3 eV in NMP-TCNQ,[7] which is comparable to the bandwidth in TCNQ compounds. U has been calculated for two electrons on a transition metal atom,[8] and found to be in the range 2–4 eV; this is somewhat less than the estimated bandwidth of KCP. These rough estimates clearly indicate that both KCP and TTF-TCNQ are strongly coupled systems.

The Hamiltonian for the one-dimensional Hubbard model is

$$\hat{H} = -\frac{\epsilon_0}{2}\sum_{i,\sigma}(\hat{a}^\dagger_{i,\sigma}\hat{a}_{1,\sigma} + \hat{a}^\dagger_{i,\sigma}\hat{a}_{i-1,\sigma}) + U\sum_i \hat{n}_{i\uparrow}\hat{n}_{i\downarrow}\,, \quad (1)$$

where $\hat{a}^\dagger_{i\sigma}$ creates an electron with spin σ at site i, $\hat{n}_{i\sigma} = \hat{a}^\dagger_{i\sigma}\hat{a}_{i\sigma}$ and $2\epsilon_0$ is the bandwidth. The two basic parameters of the theory are c, the average number of electrons per site, and $\mu = U/\epsilon_0$, which measures the strength of the interaction. The low-energy excitations of the interacting one-dimensional electron gas have been discussed extensively in the literature. In particular, exactly soluble models yield correlation functions exhibiting power-law singularities in the limit $\omega, q \to 0$.[9] Such results have been used to discredit perturbative treatments, in which the power-law behavior manifests itself through logarithmic (infrared) singularities.[10] Exact methods, however, rely on the Luttinger Hamiltonian,[11] which is characterized by an infinite bandwidth and a linear dispersion relation. While such simplifications of the band structure may not qualitatively affect the low-frequency behavior of the correlation functions, it will be apparent from our results that the finite bandwidth associated with the tight-binding Hamiltonian (1) plays an important role in optical absorption. It is therefore necessary to avoid the simplifications inherent in the Luttinger model, and to use some form of perturbation theory—which raises the specter of logarithmic divergences. There are two reasons, however, why we think that perturbation theory may give better results for the $(q = 0)$ conductivity than for other quantities such as the $2k_F$ electrical susceptibility. First of all, in the high-frequency limit $(\omega \sim \epsilon_0)$ the correlation functions should measure short-range rather than long-range correlations, and one therefore expects that an adequate theory of the optical absorption can be obtained if only short-range correlations are included—which is easily achieved in approximate theories. The second reason is that logarithmic singularities actually disappear from the perturbation-theoretic expression of the conductivity, at least in the simple approximation discussed in this paper. Loosely speaking, the low-energy excitations of the system contribute to normal electron-electron scattering processes, which do not affect the electrical current and do not contribute to the resistivity. The finite-frequency absorption is mainly due to Umklapp processes. In non-half-filled systems, such processes are inelastic and involve high-energy excitations to which a perturbative treatment may be applicable.

We can make the above remarks more precise by looking at the kinematics of electron-electron scattering in one dimension. In a noninteracting system, the crystal momentum k is conserved and there is no optical absorption. In the presence of interactions, a pair of electrons in a state $|k_i, k_i'\rangle$ can be scattered into a state $|k_f, k_f'\rangle$, thereby contributing to the absorption if the initial and final values of the current are different. In the low-frequency limit $\omega \ll \epsilon_0$, one can restrict oneself to elastic collisions. For given transfers of energy ω and crystal momentum q, there are only two possibilities for the initial and final electron states. In the first of these, $k_i' = k_f$ and $k_f' = k_i$; this is a normal process which leaves the total current unchanged. The second possibility is illustrated in Fig. 1. It corresponds to an Umklapp process, in which $k_i' = -\pi/a - k_i$ and $k_f' = \pi/a - k_f$, where a is the lattice constant. In this case, the value of the current does change. However, this process requires that one electron should initially lie in the top half of the band. For systems which are less than half filled $(c < 1)$, this is very unlikely except at high

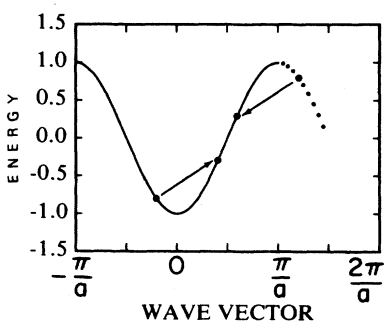

FIG. 1. Energy–wave-vector diagram for a one-dimensional elastic collision. Energy is in units of ϵ_0.

temperature, when the logarithmic divergences of perturbation theory disappear anyway. It follows, then, that for temperatures and frequencies much less than the bandwidth, collisions cannot effectively degrade an electrical current, and the conductivity should look free-carrier-like. A similar reasoning applies to the case $c > 1$.

The above argument breaks down in a strongly correlated system, in which the interaction can excite many electrons out of the Fermi sea. In spite of this, our conclusion about the low-frequency behavior of the conductivity does apply to the limit $U \to \infty$, in which the system becomes equivalent to a noninteracting, spinless fermion gas. In this limit, all of the absorption is concentrated at $\omega = 0$, and the optical absorption vanishes. It is only in the intermediate coupling regime $U \sim \epsilon_0$ that one would expect to see some structure in the low-frequency conductivity.

In Sec. II, we present the general formalism and discuss our numerical spectra. Section III is devoted to the t-matrix approximation. Although this is a low-density approximation, we find that it gives good results even for nearly half-filled systems. In Sec. IV, we discuss the memory function approximation for $\sigma(\omega)$, which reduces to straightforward perturbation theory at high frequency. The good overall agreement between the three approaches gives us confidence that our approximate analysis provides an essentially correct description of the optical spectrum of the infinite Hubbard chain. Of course, our approach breaks down at low frequency, where the detailed structure of the low-lying excitations plays an important role. However, the existence of an exact sum rule for the conductivity of the Hubbard model allows us to indirectly determine the contribution of the zero-frequency absorption peak to the sum rule, which is sufficient for our purposes.

II. FINITE CHAINS AND NUMERICAL RESULTS

This section is devoted to the theory of the optical absorption in a finite chain of N "atoms" or sites. The motivation for the numerical work described later in this section is based on the assumption that the absorption spectra of such finite chains resemble the spectra of infinite chains having the same electron density. There can be no doubt that this is indeed true for large values of N (say, $N = 10^6$). The question is, then, how large an N should be chosen for the assumption to be valid. In principle, this could be checked by calculating the spectra of larger and larger chains, keeping the electron density approximately constant. Unfortunately, the dimension of the Hilbert space grows exponentially with N. Since a knowledge of all the excited states is necessary to describe the spectrum of a chain, one quickly runs into formidable numerical difficulties even for $N \sim 10$. In fact, the numerical results reported here were obtained for only four electrons distributed over up to seven sites. While slightly larger "molecules" could certainly be handled as well, we found that the numerical spectra were in good qualitative agreement with the approximate theories which we developed to treat infinite chains. In particular, all the features of the numerical spectra can be interpreted in terms of physical processes that were included in the approximate treatments. Since the same physical processes are at work in larger chains, it is reasonable to assume that our results reflect the intrinsic properties of the Hubbard model rather than spurious size effects.

For simplicity, we first consider the case of a long chain, in which boundary effects can be neglected. This will enable us to introduce periodic boundary conditions, thereby simplifying the theoretical analysis. In the presence of an oscillating, uniform electric field $Ee^{-i\omega t}$ parallel to the chain, one must add to the Hamiltonian (1) a term representing the coupling to the applied field. This is conveniently done by introducing a vector potential $A(t) = cE/(-i\omega)e^{-i\omega t}$ into the Hamiltonian. Here and in the next two equations, c denotes the speed of light; this is not to be confused with the number of electrons per site, also denoted by c. Some care is needed to preserve gauge invariance in a tight-binding system; if x_i is the position of the ith atom in the chain, then a canonical transformation $\hat{a}'_i = e^{-i(e/c)Ax_i}\hat{a}_i$ should remove the coupling to a static vector potential A. This indicates that in the presence of a time-dependent vector potential $A(t)$, the first term in the Hamiltonian (1) becomes ($\hbar = 1$)

$$\hat{H}'_0 = -\frac{\epsilon_0}{2}\sum_{i,\sigma}(\hat{a}^\dagger_{i,\sigma}\hat{a}_{i+1,\sigma}e^{-ieaA(t)/c} + \hat{a}^\dagger_{i,\sigma}\hat{a}_{i-1,\sigma}e^{ieaA(t)/c}), \tag{2}$$

where a is the lattice constant. Expanding to second order in A, we find that the coupling term is

$$\hat{H}_A = (A/c)\hat{j} + \tfrac{1}{2}(e/c)^2A^2a^2(-\hat{H}_0), \tag{3}$$

where $\hat{H}_0$ is the kinetic energy, i.e., the first term in the Hamiltonian (1) and $\hat{j}$ is the current operator

$$\hat{j} = iea\frac{\epsilon_0}{2}\sum_{i,\sigma}(\hat{a}^\dagger_{i,\sigma}\hat{a}_{i+1,\sigma} - \hat{a}^\dagger_{i,\sigma}\hat{a}_{i-1,\sigma}). \tag{4}$$

We will base our discussion on the fluctuation-dissipation theorem, which relates the conductivity to the equilibrium current-current correlation function χ:

$$\sigma(\omega) = [\chi(\omega) - ne^2/m^*]/i\omega, \tag{5}$$

where $n = cN$ is the number of electrons and m^* is

an effective mass given by[12]

$$1/m^* = -a^2\langle \hat{H}_0 \rangle /n. \tag{6}$$

In the dilute limit, electrons lie near the bottom of the band, and $1/m^* = a^2\epsilon_0$. More generally, in a noninteracting system, one finds $1/m^* = a^2\epsilon_0(\sin\pi c/2)/(\pi c/2)$. The retarded correlation function χ is defined by

$$\chi(\omega) = i\int_0^\infty dt\, e^{i\omega t}\langle[\hat{j}(t),\hat{j}(0)]\rangle\,. \tag{7}$$

In terms of plane wave states, the current $\hat{j}$ is given by

$$\hat{j} = -e\sum_{k,\sigma}\hat{a}^\dagger_{k,\sigma}\hat{a}_{k,\sigma}v_k, \quad v_k = a\epsilon_0\sin ka\,, \tag{8}$$

where k is the (one-dimensional) wave vector, and a is the lattice constant.

Later on in this article, we will use approximations whose validity is restricted to high frequencies. To obtain information about the low-frequency conductivity, we will use the analog of the f-sum rule for a tight-binding system:

$$\frac{2}{\pi}\int_0^\infty \sigma'(\omega)d\omega = ne^2/m^*\,, \tag{9}$$

where σ' denotes the real part of the conductivity and m^* is the effective mass defined in Eq. (6).

Before proving the sum rule, let us consider in some detail the case of a finite chain. The site indices in the Hamiltonian (1) are now restricted by the condition $1 \le i \le N$. If we denote by $|n\rangle$ and E_n the exact eigenstates and energy levels of the many-body Hamiltonian (1), the current-current correlation function (7) becomes

$$\chi(\omega) = \sum_n |\langle 0|\hat{j}|n\rangle|^2\left(\frac{1}{\omega + E_n - E_0 - i\eta} - \frac{1}{\omega - E_n + E_0 + i\eta}\right), \tag{10}$$

where $|0\rangle$ and E_0 denote the ground state and the ground-state energy, respectively. From Eq. (5), the real part of the conductivity is

$$\sigma'(\omega) = \pi\sum_n |\langle 0|\hat{j}|n\rangle|^2\frac{\delta(|\omega| - E_n + E_0)}{E_n - E_0}\,. \tag{11}$$

Thus, the spectrum of a finite chain consists of a series of sharp lines corresponding to transitions from the ground state to the excited states. When N is large, these lines merge into a continuum and σ is a continuous function of the frequency. To compare the spectra of finite and infinite chains, it is convenient to introduce the normalized integrated conductivity $Z(\omega)$, defined by

$$Z(\omega) = \frac{2m^*}{\pi ne^2}\int_0^\omega d\nu\,\sigma'(\nu)\,,$$
$$= \frac{m^*}{n}\sum_{E_n - E_0 < \omega}\frac{|\langle 0|\hat{v}|n\rangle|^2}{E_n - E_0}\,, \tag{12}$$

where $\hat{v}$ is the velocity operator $-\hat{j}/e$. Clearly, the sum rule implies that $Z \to 1$ as $\omega \to \infty$. To prove the sum rule, we observe that the current operator satisfies the equation

$$\hat{j} = ie[\hat{x},\hat{H}], \tag{13}$$

where $\hat{x}$ is the sum of the position operators of the electrons. Therefore,

$$\langle 0|\hat{j}|n\rangle = ie(E_n - E_0)\langle 0|\hat{x}|n\rangle; \tag{14}$$

inserting this into Eq. (11), we find

$$\frac{2}{\pi}\int_0^\infty \sigma'(\omega)\,d\omega = ie\sum_n(\langle 0|\hat{x}|n\rangle\langle n|\hat{j}|0\rangle - \langle 0|\hat{j}|n\rangle\langle n|\hat{x}|0\rangle)\,, \tag{15}$$
$$= -ie\langle 0|[\hat{j},\hat{x}]|0\rangle = -e^2a^2\langle 0|\hat{H}_0|0\rangle,$$

where in the last step we have used the identity [see Eq. (4)]

$$[\hat{j},\hat{x}] = iea^2\frac{\epsilon_0}{2}\sum_{i,\sigma}(\hat{a}^\dagger_{i,\sigma}\hat{a}_{i+1,\sigma} + \hat{a}^\dagger_{i,\sigma}\hat{a}_{i-1,\sigma})\,,$$
$$= -iea^2\hat{H}_0. \tag{16}$$

From the definition (6) of the effective mass, we see that Eq. (16) yields the sum rule (9).

The quantity $Z(\omega)$ was calculated numerically for $N = 5$, $n = 4$, and various values of the interaction constant $\mu = U/\epsilon_0$; the results are illustrated by the dotted lines in Fig. 2. To interpret these spectra, it is instructive to investigate the noninteracting case $\mu = 0$ in some detail. The one-electron wave functions which diagonalize the kinetic energy are easily seen to be

$$\psi_m(k) = \begin{cases}(\frac{2}{5})^{1/2}\sin(\frac{1}{6}\pi km) & (m = 2,4)\\ (\frac{2}{5})^{1/2}\cos(\frac{1}{6}\pi km) & (m = 1,3,5)\end{cases} \tag{17}$$

with corresponding energies $-\epsilon_0\cos\frac{1}{6}\pi m$. Here, k is a site index running from -2 to $+2$. Thus, the ground state of the "molecule" can be represented as in Fig. 3. Since the current operator is odd in k, the only allowed (dipole) transitions from occupied to excited states are $(m = 1) \to (m = 4)$ and $(m = 2) \to (m = 3 \text{ or } 5)$. The corresponding excitation energies are $\Delta E_{14} = \Delta E_{25} = 1.38\epsilon_0$ and $\Delta E_{23} = 0.5\epsilon_0$. These transitions are apparent in the spectrum shown in Fig. 2(a); since these are the only possible transitions, they must exhaust the sum rule (9). The spectrum of an *infinite* noninteracting

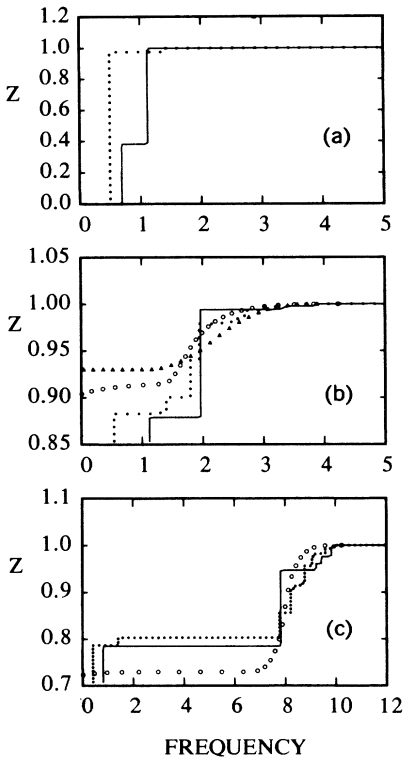

FIG. 2. Normalized, integrated conductivity as defined in Eq. (12) for a Hubbard system with $c = 0.8$ electrons per atom and (a) $U = 0$, (b) $U = \epsilon_0$, (c) $U = 8\epsilon_0$. Continuous curves, five-atom ring; dots, five-atom linear chain; open circles, t-matrix approximation; triangles, memory-function approximation. Frequency is in units of ϵ_0.

chain, on the other hand, would simply be a step function with a discontinuity at $\omega = 0$. Thus, we can look at the finite chain spectrum as a coarse grained approximation to the infinite chain spectrum, with an energy resolution of order $2\epsilon_0/N$. As the interaction parameter μ increases, some of the low-frequency absorption is shifted to higher energy, which would correspond to a peak in the

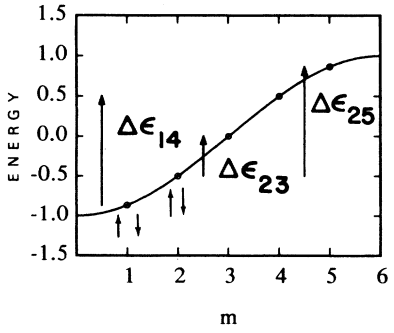

FIG. 3. Ground-state and optical transitions of a noninteracting five-atom linear chain.

real part of the conductivity. This absorption peak occurs at a frequency which increases with μ, and is roughly $U = \mu\epsilon_0$ when $\mu \gg 1$. It must therefore correspond to a transition in which the final state contains two electrons of opposite spin sitting on the same atom. In Sec. III, it will be shown that this transition involves a high-energy "bound state" which is responsible for most of the high-frequency absorption.

The model discussed above has a density of $c = 0.8$ electron per site, which is a nearly half-filled system. In order to increase the number of sites and reduce the value of c, we have also calculated the function $Z(\omega)$ for rings of 5 and 7 sites. The advantage of working with a ring is that periodic boundary conditions can be imposed, and wave-vector conservation ensues. One can then restrict oneself to translation-invariant states, which reduces the dimension of the Hilbert space considerably. Although it is still possible to induce an electric field along the ring by changing the magnetic flux through it, the corresponding "conductivity" is related to the diamagnetic susceptibility of the system and does not satisfy the sum rule (9). Therefore, it is preferable to apply a uniform electric field; the coupling term in the Hamiltonian can again be calculated from Eq. (2), and is given by (c now denotes the speed of light)

$$\hat{H}_{A,r} = \left(\frac{A}{c}\right)\hat{j}_r + \frac{1}{2}\left(\frac{e}{c}\right)^2 A^2 a^2 (-\hat{H}_0) + \frac{\epsilon_0}{4}\left(\frac{e}{c}\right)^2 A^2 a^2 \sum_{k,\sigma} \cos\left(\frac{4\pi k}{N} + 2\psi\right) \times (\hat{a}^\dagger_{k+1,\sigma}\hat{a}_{k,\sigma} + \hat{a}^\dagger_{k,\sigma}\hat{a}_{k+1,\sigma}) , \quad (18)$$

where the current operator is now

$$\hat{j}_r = iea\frac{\epsilon_0}{2^{1/2}} \sum_{k,\sigma} \cos\left(\frac{2\pi k}{N} + \psi\right)(\hat{a}^\dagger_{k+1,\sigma}\hat{a}_{k,\sigma} + \hat{a}^\dagger_{k,\sigma}\hat{a}_{k+1,\sigma}) . \quad (19)$$

In these equations, ψ is a phase angle depending on the orientation of the electric field. The factor $2^{1/2}$ in the last equation provides a convenient normalization. The expectation value of the last term in Eq. (18) in the (translation-invariant) ground state is obviously zero, and the conductivity is again given by Eqs. (5)–(7), in which $\hat{j}$ is replaced by $\hat{j}_r$. The sum rule (9) remains unchanged. Numerical results for rings of five atoms containing four electrons are shown in Fig. 2 (continuous lines), and are in good agreement with the results for the linear chain (dotted lines). In the noninteracting case $\mu = 0$, the spacing between the one-electron energy levels is larger than in the linear case, which results in a poorer energy resolution. The locations

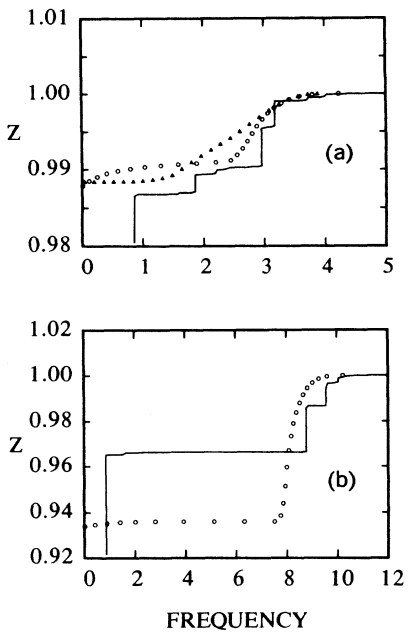

FIG. 4. Normalized, integrated conductivity as defined in Eq. (12) for a Hubbard system with $c = 0.571$ electrons per atom and (a) $U = \epsilon_0$, (b) $U = 8\epsilon_0$. Continuous curves, seven-atom ring; open circles, t-matrix approximation; triangles, memory-function approximation. Frequency is in units of ϵ_0.

and strengths of the main absorption peaks, however, are in good agreement. This indicates that the high-frequency conductivity is relatively insensitive to the boundary conditions, even for $N = 5$. Figure 4 shows numerical results for rings of seven atoms. Note that this yields an electron density $c = 0.571$, which is close to the observed density of TTF-TCNQ.[13] The comparison with the approximate theories of the infinite chain absorption will be carried out in Secs. III and IV.

III. T-MATRIX APPROACH

In the absence of interactions, $\hat{j}$ commutes with the Hamiltonian and the commutator in Eq. (3) vanishes, so that there is no optical absorption and the dc conductivity is infinite. Thus, absorption processes must involve collisions between electrons. In a dilute system, collisions are rare and can be treated "one at a time." Let us therefore consider a system of two electrons having opposite spins and interacting via a Hubbard potential U. If the electrons are in a triplet state, the spatial wave function is antisymmetric and they do not interact. Hence, it will always be assumed that the colliding electrons are in a singlet state. The matrix elements of the two-particle Hamiltonian are

$$\langle k_1, k_2 | \hat{H} | k_1', k_2' \rangle = -\epsilon_0 \delta_{k_1, k_1'} \delta_{k_2, k_2'} (\cos a k_1 + \cos a k_2) + (1/N) \delta_{k_1 + k_2, k_1' + k_2'} U, \tag{20}$$

where $|k_1, k_2\rangle$ and $|k_1', k_2'\rangle$ denote normalized plane wave states. It is convenient to introduce the (dimensionless) "center of mass" and "relative" coordinates $q = \frac{1}{2} a(k_1 + k_2)$ and $\alpha = \frac{1}{2} a(k_1 - k_2)$, so that the kinetic energy of a state $|k_1 k_2\rangle$ takes the simple form $\epsilon_{q,\alpha} = -2\epsilon_0 \cos q \cos\alpha$. If we restrict q and α to the domain $0 < q < \pi$, $-\pi < \alpha < \pi$, q is a conserved quantity. Since we are interested in the singlet state, we limit ourselves to spatially symmetric wave functions. Hence we may further assume that $\alpha > 0$. To find the eigenstates of the two-particle Hamiltonian, we solve the Lippmann-Schwinger equation [a scattering eigenstate with initial quantum numbers q and α is denoted by $|q, \alpha)$, while $|q, \alpha\rangle$ denotes a non-interacting eigenstate].

$$\langle q, \alpha' | q, \alpha) = \delta_{\alpha, \alpha'} + (2/N) U \times \sum_{\alpha'' > 0} \langle q, \alpha'' | q, \alpha) / (\epsilon_{q,\alpha} - \epsilon_{q,\alpha'} - i\eta). \tag{21}$$

This separable equation is immediately solved, with the result that

$$\langle q, \alpha' | q, \alpha) = \delta_{\alpha, \alpha'} + \frac{(1/N) U f(q, \cos\alpha)}{(\epsilon_{q,\alpha} - \epsilon_{q,\alpha'} - i\eta)}, \tag{22}$$

where the "scattering amplitude" $f(q, \cos\alpha)$ is given by

$$1/f(q, \cos\alpha) = 1 + U/(2i\epsilon_0 |\sin\alpha \cos q|). \tag{23}$$

In addition to the scattering states, the Hamiltonian (20) has a bound state $|q, b)$ for arbitrary q. For the repulsive interaction $U > 0$ considered here, the bound state lies *above* the highest band energy $2\epsilon_0$. The ability of a repulsive potential to bind electrons from a tight-binding band was first noted in Ref. 14 in the case of an impurity. The equation for the bound state is analogous to (21) without the inhomogeneous term $\delta_{\alpha,\alpha'}$; the normalized bound-state wave function is found to be

$$\langle q, \alpha | q, b) = \frac{(2U^3/\epsilon_q)^{1/2}}{\epsilon_{q,\alpha} - \epsilon_q}, \tag{24}$$

where the bound state energy ϵ_q is given by

$$\epsilon_q = (U^2 + 4\epsilon_0^2 \cos^2 q)^{1/2}. \tag{25}$$

We now use these results to calculate the correlation function χ for a two-electron system in an eigenstate $|q, \alpha)$, which we will denote by $\chi(q, \alpha, \omega)$. We will then approximate the correlation function for the dilute Hubbard chain by averaging $\chi(q, \alpha, \omega)$ over all the pairs of electrons present in

the system according to

$$\chi(\omega)=\sum_{k_1>k_2}\chi(\tfrac{1}{2}a(k_1+k_2),\tfrac{1}{2}a(k_1-k_2),\omega)n_{k_1,k_2}, \quad (26)$$

where n_{k_1,k_2} is the joint distribution function for a pair of electrons in a singlet state:

$$n_{k_1,k_2}=\langle\hat{S}^{\dagger}_{k_1,k_2}\hat{S}_{k_1,k_2}\rangle,$$

$$\hat{S}_{k_1,k_2}=\frac{1}{2^{1/2}}(\hat{a}_{k_1\uparrow}\hat{a}_{k_2\downarrow}-\hat{a}_{k_1\downarrow}\hat{a}_{k_2\uparrow}). \quad (27)$$

Writing $\chi=\chi'+i\chi''$, one finds from Eq. (7)

$$\chi''(q,\alpha,\omega)=\pi\sum_{\alpha'>0}|(q,\alpha|\hat{j}|q,\alpha')|^2[\delta(\omega+\epsilon_{q,\alpha}-\epsilon_{q,\alpha'})-\delta(\omega-\epsilon_{q,\alpha}+\epsilon_{q,\alpha'})]$$
$$+\pi|(q,\alpha|\hat{j}|q,b)|^2[\delta(\omega+\epsilon_{q,\alpha}-\epsilon_q)-\delta(\omega-\epsilon_{q,\alpha}+\epsilon_q)]. \quad (28)$$

Note that this expression is very similar to that obtained for interband absorption in a semiconductor. The main difference is, of course, that in the present case the optical transitions occur between two- rather than one-particle states. The first term contains the effect of transitions to continuum (scattering) states, and the second term arises from transitions to the bound state. Using the above expressions for the eigenstates of the two-electron Hamiltonian, the matrix elements of the current operator are easily found to be (assuming $\alpha\neq\alpha'$)

$$(q,\alpha|\hat{j}|q,\alpha')=2a(U/N)\tan q f(q,\cos\alpha')f^*(q,\cos\alpha),$$
$$(q,\alpha|\hat{j}|q,b)=(2U^3a^2/\epsilon_q N)^{1/2}f^*(q,\cos\alpha)\tan q. \quad (29)$$

Substituting these matrix elements in Eq. (28) for the correlation function, one finds after some algebra

$$\chi''(q,\alpha,\omega)=2(U/N)a^2\tan^2q|f(q,\cos\alpha)|^2F(q,\alpha,\omega)$$
$$+2\pi(U^3a^2/\epsilon_q N)|f(q,\cos\alpha)|^2\tan^2q$$
$$\times[\delta(\omega+\epsilon_{q,\alpha}-\epsilon_q)-\delta(\omega-\epsilon_{q,\alpha}+\epsilon_q)], \quad (30)$$

where $[\Theta(x)=0$ if $x<0$, 1 if $x>0]$

$$F(q,\alpha,\omega)=\Theta(4\epsilon_0^2\cos^2q-(\omega+\epsilon_{q,\alpha})^2)$$
$$\times\mathrm{Im}f\left(q,\frac{\omega+\epsilon_{q,\alpha}}{2\epsilon_0\cos q}\right)$$
$$-\Theta(4\epsilon_0^2\cos^2q-(\omega-\epsilon_{q,\alpha})^2)$$
$$\times\mathrm{Im}f\left(q,\frac{\epsilon_{q,\alpha}-\omega}{2\epsilon_0\cos q}\right). \quad (31)$$

The first term in the right-hand side of Eq. (30) arises from transitions into continuum two-particle states. This term agrees with the result of Lyo and Holstein.[5] The second term accounts for transitions into the bound state. Since q, α, and k_F are of order c and since $f\sim|\sin\alpha|\sim c$ in the dilute limit, both terms are of order c^4. After integrating over q and α to obtain the real part of the conductivity, these terms are of order c^6 and c^5 respectively (the δ function in the bound state term removes one power of c). This result can be understood by noting that in the dilute limit, electrons sit near the bottom of the band and the dispersion relation is approximately parabolic, leading to momentum conservation and the vanishing of the absorption.

To obtain the current-current correlation function of the system, hence the conductivity, it remains to sum Eq. (31) over the momenta of the incoming particles as specified by Eq. (26). This was done numerically. The open circles in Figs. 2 and 4 show the results for the normalized integrated conductivity $Z(\omega)$, defined in Eq. (12). The corresponding electron densities are $c=\frac{4}{5}$ and $c=\frac{4}{7}$ respectively. In the case $\mu=1$, we have neglected the effect of correlation in the effective mass and in the two-particle distribution function. Thus, we set

$$1/m^*=a^2\epsilon_0\frac{\sin(\pi c/2)}{\pi c/2},$$
$$n_{k_1k_2}=\begin{cases}1 & \text{if } |k_1|,|k_2|<k_F,\\ 0 & \text{otherwise } (\mu \text{ small}).\end{cases} \quad (32)$$

In the strongly interacting case, however, the oscillator strength of the main absorption peak was found to be several times smaller than found numerically for finite chains and rings. This is clearly due to correlation effects, which tend to shift some of the weight of the two-particle distribution function to higher values of the momenta. Since the response function (29) increases rapidly with the reduced momenta q and α, this enhances the absorption. As a crude way to include such effects, we assumed that the particle momenta were uniformly distributed between $-2k_F$ and $+2k_F$.

Thus, we set

$$\frac{1}{m^*} = a^2\epsilon_0 \frac{\sin(\pi c)}{\pi c},$$

$$n_{k_1k_2} = \begin{cases} \frac{1}{4} & \text{if } |k_1|, |k_2| < 2k_F \\ 0 & \text{otherwise } (\mu \text{ large}). \end{cases} \tag{33}$$

This ansatz can be justified as follows. It is easily seen that, in the limit $\mu \to \infty$, the spatial part of the ground-state wave function of the one-dimensional Hubbard model is identical with that of a system of noninteracting, spinless fermions within a factor ± 1. This factor arises because the spinless fermion wave function is odd under the interchange of two spatial coordinates, while the wave function of the Hubbard model is odd under the simultaneous interchange of spatial and spin coordinates. If we concentrate on a pair of neighboring electrons in a singlet state, therefore, the full spatial wave function must behave as a sum of terms of the form $C\left|\sin\frac{1}{2}(k_1-k_2)(x_1-x_2)\right| \times \exp[i\frac{1}{2}(k_1+k_2)(x_1+x_2)]$, where x_1, x_2 denote the coordinates of the two electrons and the two momenta k_1, k_2 are uniformly distributed between $-2k_F$ and $+2k_F$; C is a constant which depends on the remaining coordinates $x_3, \ldots$. Together with the observation that only short-range correlations should be important, we are naturally led to the above ansatz.

With Eqs. (32) and (33), the t-matrix predictions for $Z(\omega)$ are in good agreement with the numerical spectra. The largest discrepancy arises in the case $c = 0.571$, $\mu = 8$ as shown in Fig. 4 (b). We checked that the discrepancy decreases with increasing U. This confirms our interpretation of the main absorption peak in terms of a bound state consisting of two electrons sitting at the same site. Figure 5 shows the dimensionless conductivity σ^*, defined by

$$\sigma^*(\omega) = \frac{2}{\pi}\frac{m^*\epsilon_0}{ne^2}\sigma'(\omega) \tag{34}$$

in the same approximation. Clearly, it is only in the weak coupling limit $\mu \ll 1$ that continuum transitions (open circles) contribute significantly to the absorption; for $\mu \sim 1$ most of the absorption is due to the bound state. It must be noted that the relatively large contribution of continuum transitions to the low-frequency absorption is a spurious effect, due to our neglect of the Pauli principle in the summation over final states in Eq. (28). Restricting the sum over α' to unoccupied states would suppress this contribution. However, this correction is formally of higher order in the electron density c. Since the contribution of continuum transitions to the t-matrix conductivity is small when $\mu \sim 1$, we did not include this refinement in the t-matrix calculation.

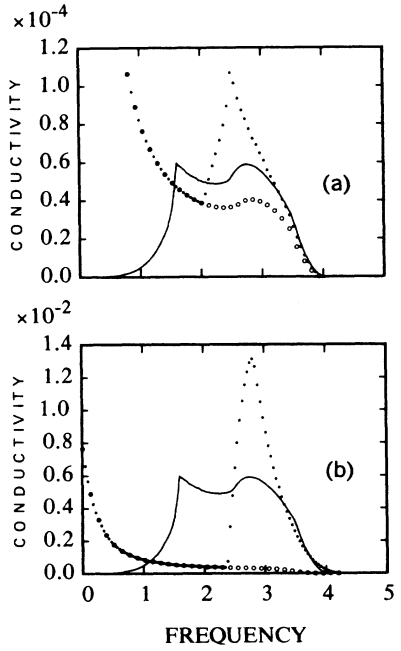

FIG. 5. Dimensionless conductivity [Eq. (34)] calculated in the t-matrix approximation (dotted curves) and in the memory function approximation (continuous curves) for $c = 0.571$ electrons per atom and (a) $U = 0.1\,\epsilon_0$, (b) $U = \epsilon_0$. The open circles represent the contribution of transitions into the continuum to the t-matrix conductivity. Frequency is in units of ϵ_0.

IV. MEMORY-FUNCTION APPROACH

In this section, we will use the memory-function approach of Götze and Wölfle to calculate the conductivity of an infinite Hubbard chain in the weak coupling limit (i.e., $\mu \to 0$). We begin with an outline of the formalism. The memory function $M(\omega)$ is defined in terms of the conductivity by the formula[6]

$$\sigma(\omega) = i(ne^2/m^*)[1/(\omega + M(\omega))]. \tag{35}$$

In particular, $M(\omega) = i/\tau$ in the Drude approximation. Thus, M can be interpreted as a frequency-dependent generalization of the relaxation rate. Just as the conductivity, the memory function is analytic in the upper frequency half-plane. The basic assumption of the theory is that M is a well-behaved function of ω and of the strength of the interaction. One can then expand Eq. (35) at high frequency:

$$\sigma(\omega) = i(ne^2/m^*)[1/\omega - M(\omega)/\omega^2 + \cdots]. \tag{36}$$

Thus, $M(\omega)$ is simply obtained by expanding the conductivity to leading order in the interaction. Extrapolation to all frequencies yields the memory-function approximation. Although we have made no effort toward rigor, it can be show that this approximation is actually the first term of a con-

tinued fraction expansion of the conductivity. In the present context, we view the memory-function approximation as the simplest approximation which satisfies the sum rule (9) and reduces to perturbation theory at high frequency.

From Eqs. (36) and (7), one finds after some algebra that the memory function can be written as

$$M(\omega)=\frac{m^*}{ne^2\omega}\left[\pi(\omega)-\pi(0)\right], \tag{37}$$

where the correlation function π is given by

$$\pi(\omega)=i\int_0^\infty dt\, e^{i\omega t}\langle[\hat{F}(t),\hat{F}(0)]\rangle . \tag{38}$$

Here, the "force operator" $\hat{F}$ is defined according to

$$\hat{F}=-i[\hat{j},\hat{H}]. \tag{39}$$

It is now straightforward to calculate the correlation function (37) to leading order in the interaction U. The force operator (38) is simply

$$2iea\epsilon_0\frac{U}{N}\sum_{k,p,q}\hat{a}^\dagger_{k+q\uparrow}\hat{a}_{k\uparrow}\hat{a}^\dagger_{p-q\downarrow}\hat{a}_{p\downarrow}$$
$$\times\sin\tfrac{1}{2}a(k+p)\{\cos a[\tfrac{1}{2}(k-p)+q]-\cos\tfrac{1}{2}a(k-p)\} \tag{40}$$

so that, to leading order in U, the average in Eq. (38) can be taken in a noninteracting system. The corresponding diagram is shown in Fig. 6. Using the standard rules of perturbation theory, one finds ($M=M'+iM''$)

$$M''(\omega)=\frac{\pi U_a^2}{2\omega\epsilon_0\sin\pi c/2}\int_0^{2\pi/a}\frac{dq}{2\pi}\int_0^\omega\frac{d\Omega}{\pi}\left[2\chi_2(q,\Omega)\chi_0(-q,\omega-\Omega)+2\chi_1(q,\Omega)\chi_1(-q,\omega-\Omega)\right], \tag{41}$$

where

$$\chi_n(q,\omega)=\int_0^{2\pi/a}\frac{dp}{2\pi}\,\delta(\epsilon_p-\epsilon_{p-q}-\omega)$$
$$\times(n_{p-q}-n_p)(v_p-v_{p-q})^n. \tag{42}$$

Here $\epsilon_p=-\epsilon_0\cos ap$ is the electron energy, $v_p=a\epsilon_0\sin ap$ is the velocity, and n_p is the zero-temperature occupation number, $n_p=1$ if $|p|<k_F=\pi c/2a$, $=0$ otherwise. The two-dimensional integral in Eq. (42) is easily evaluated numerically, and the real part M' of the memory function is obtained by using a dispersion relation.

The results for the normalized integrated conductivity $Z(\omega)$ [Eq. (12)] are shown in Figs. 2 and 4, and those for the dimensionless conductivity σ^* [Eq. (34)] in Fig. 5. The absorption predicted by the memory function approximation is of the correct order of magnitude for $\mu=1$. In the weak coupling limit $\mu\ll1$, one would expect the t-matrix and memory function approaches to yield similar results. Figure 5 shows that this is indeed the case for $\mu=0.1$, although (i) the bound-state contribution to the t-matrix is still substantial even though it is formally of order μ^3 [see Eq. (30)], and (ii) the t-matrix absorption is too large at low frequency, as discussed in the previous section. The peak around $\omega=1.6\epsilon_0$ in the memory function approximation curves arises from a singularity in the integrand of the expression (41) for the memory function. This is illustrated in Fig. 7, which shows the domains in the (q,Ω) plane where the susceptibilities $\chi_m(q,\Omega)$ and $\chi_n(-q,\omega-\Omega)$ are nonvanishing [see Eq. (42)].

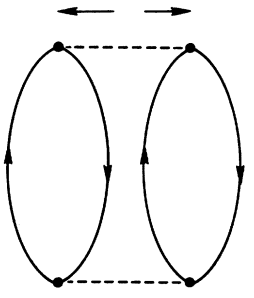

FIG. 6. Feynman diagram for the correlation function π defined in Eq. (38).

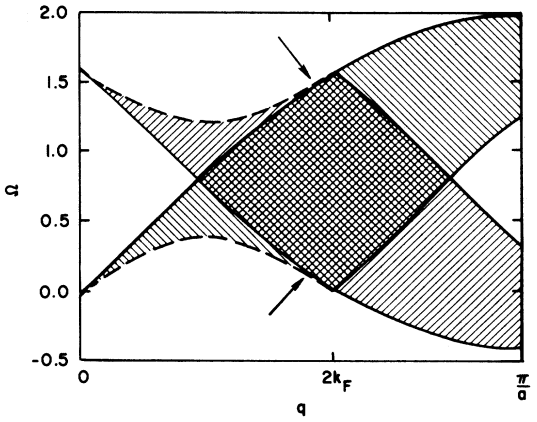

FIG. 7. Domains in which the factors $\chi_m(q,\Omega)$ and $\chi_n(-q,1.58\epsilon_0-\Omega)$ in Eq. (41) are different from zero. The arrows indicate the regions of critical contact, and the dashed lines are the boundaries imposed by the Pauli principle. Ω is in units of ϵ_0.

For certain values of ω, the boundaries of these domains are in critical contact; such values of ω will correspond to singularities in the slope of the memory function, provided that the velocity factor in Eq. (42) does not vanish at the point of critical contact. As discussed in the previous section, our t-matrix approximation neglects the Pauli principle in the summation over final states. The domains shown in Fig. 7 are then enlarged, and the points of critical contact disappear. This explains why the t-matrix curve goes on increasing below the value $\omega = 1.58\epsilon_0$.

Figure 5(b) shows the dimensionless conductivity for $\mu = 1$, and it is clear that the apparent order-of-magnitude agreement between the memory-function approximation and the numerical spectra is due to a fortuitous cancellation of two effects. On the one hand, the memory-function approximation neglects the important bound state contribution. On the other hand, the Born approximation used in the memory-function approximation overestimates the cross-section for scattering into continuum states. In the strong coupling limit, the second effect is stronger and the memory-function approximation breaks down completely.

V. CONCLUSION

The numerical results presented in this paper substantiate the claim made in the introduction that the optical absorption of the one-dimensional Hubbard model with an electron density comparable to that of TTF-TCNQ is small. Thus, for $U = 8\epsilon_0$, we find that the main absorption peak exhausts about 4% of the sum rule. Moreover, within the "energy resolution" $\Delta\omega \sim \epsilon_0$ of our finite chain and ring calculations, we find no low-frequency anomaly in the spectrum. In the case of KCP, we find that the absorption is even smaller; for instance, setting $c = 0.3$ and $U = \epsilon_0$, the t-matrix approximation yields a normalized conductivity $\sigma^*(\omega)$ which is less than 2×10^{-3} for $\omega > 0$. On the positive side, we have shown that the t-matrix approximation is in good agreement with the numerical spectra of finite chains and rings. This indicates that the short-range correlations which are important for the optical conductivity are properly taken into account by that approximation. It also suggests that a similar approximation may be useful in the study of more sophisticated models including both the electron-electron and the electron-phonon interactions.

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ORDERING OF DIPOLES IN DIFFERENT TYPES OF MICROTUBULE LATTICE

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Microtubules (MTs) are the largest protein polymers in the cytoskeleton of eucaryotic cells in which they perform various functions. They are important in cell division, cell movement, they seem to be the devices through which are transferred signals in the nervous system. In this paper we continued to investigate the hypothesis that MTs can be viewed as assemblies of dipoles which are carried by the MT subunits, tubulin heterodimers. These assemblies were studied by means of the two-dimensional Ising model for both the A- and B-type arrangements of the tubulin subunits in a MT and the number of protofilaments 12, 13 and 14. We found that depending on these parameters and the magnitudes and orientations of the dipoles a MT may be at body temperature in an ordered phase or in a phase characterized by a random configuration of dipoles. The type of the ordered phase is determined by the above parameters as well, and it can be ferroelectric, antiferroelectric or ferrielectric. The dipolar ordering also depends on the presence of microtubule associated proteins, assuming that they can locally alter the dipolar interactions by binding to a MT, and external electric fields. The model presented here started by Tuszyński[1–3] can be one of the first steps in the theoretical investigation of the electromagnetic features of MTs and their role in the MT behavior.

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1. Introduction

Microtubules (MTs) are structural elements of eucaryotic cells.[4] They are hollow tubes formed by the polymerization of tubulin molecules. Each tubulin subunit is a dimer composed of two monomers termed α- and β-tubulin.[5–7] The dimer is approximately 8×10^{-9} m long and its width is about 5×10^{-9} m. The molecules of tubulin bound within a MT arrange themselves into protofilaments. The majority of MTs *in vivo* have 13 protofilaments.[8,9] MTs assembled *in vitro* consist mostly of 13 and 14 protofilaments. Changes in the number of protofilaments occur also within a single MT, even though rather infrequently.[9]

Almost all MTs inside the cell form an array around the centrosome.[4] In a non-dividing cell these MTs continuously grow and shrink due to the constant addition and dissociation of tubulin molecules at both MT ends. This phenomenon is known as dynamic instability[10] and its microscopic mechanism is largely unexplained. It is utilized in cell division when MTs growing in random directions from two centro-

somes positioned at the opposite poles of the cell eventually attach to chromosomes in order to pull an equal number of them to each of the cell extremities. In a stable state MTs serve as tracks for the transport of organelles. This transport is mediated by motor proteins which attach to MTs.

Stable MTs form parallel networks in dendrites and axons of neurons. These networks have been compared to parallel information processors that may be basic for the functioning of the nervous system.[11] Nondynamic MTs also form the cores of cilia and flagella that extend from the cell surface and move in a coordinated way. It is conjectured that such motion is a result of the cooperative interactions between the ciliary motor proteins.

The physical pathways of the MT activities briefly outlined above have been a subject of growing theoretical investigation. Several researchers have suggested that the energy freed in the GTP hydrolysis which takes place soon after the addition of each tubulin molecule to a MT end propagates along a MT as a solitary wave.[1,12–14] This wave may locally alter the elastic state of tubulin which may consequently lead to a dissociation of a tubulin dimer from a MT end or may be involved in the coordinated behavior of the motor proteins.

The present paper is focused on the investigation of the dipolar order in MTs. This question has been studied by Tuszyński *et al.* who put forward the hypothesis that some aspects of the MT behavior may be due to the interactions between the permanent dipoles carried by the tubulin molecules.[2] So far Tuszyński and his coworkers have investigated the dipolar ordering only in MTs that are composed of 13 protofilaments and are represented by the A lattice (see the following section).[3] In this work, a similar type of analysis is extended to MTs with both A and B lattices and number of protofilaments 12, 13 and 14.

2. Microtubule Lattice

Neighboring MT protofilaments are longitudinally shifted with respect to each other. Consequently, the tubulin dimers bound in a MT are arranged in a hexagonal manner and a MT can be represented by a hexagonal lattice.

As mentioned earlier, the most common number of protofilaments in a MT *in vivo* is 13 and *in vitro* 13 and 14 where MTs consisting of different numbers of protofilament in the range $p = 12$–17 have been also observed.[15] In all MTs, independent of the number of protofilaments, the tubulin monomers are arranged along a 3-start helix, i.e. each MT consists of three helices of tubulin monomers.

Depending on the geometrical arrangement of the α and β monomers two types of the MT lattice can be defined. They have been termed the A lattice and the B lattice. In MTs that are represented by the A lattice the α and β monomers along a 3-start helix alternate. If the number of protofilaments is odd then the connection between two monomers which are positioned in the last and first protofilament along a 3-start helix is heterologous, i.e. it is an α-β or a β-α connection. Hence, MTs with these A lattices are helically symetric [Fig. 1(a)]. In MTs that possess an A

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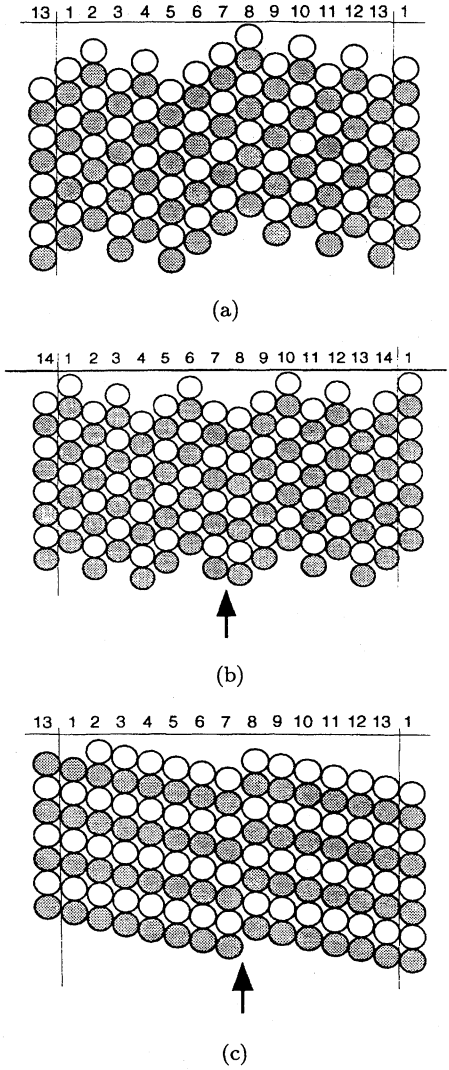

Fig. 1. A two-dimensional schematic view of a MT: (a) A lattice for $p = 13$, (b) A lattice for $p = 14$, (c) B lattice for $p = 13$. The α monomers are white, the β monomers are grey. The seam is indicated with an arrow. [Drawn according to (Martin *et al.*, 1993).]

lattice with an even number of protofilaments this connection is homologous, i.e. it is an α-α or a β-β connection. This means that in such MTs there is a discontinuity between the first and the last protofilament. This discontinuity is called a "seam" [Fig. 1(b)]. A seam is present in all MTs that possess a B lattice. In these MTs each of the three 3-start helices consists of sections that are composed of p monomers of one type. Along one 3-start helix each such section is connected at both ends to sections formed of p monomers of the opposite type. This results in heterologous monomer-monomer contacts along the seam in MTs with a B lattice [Fig. 1(c)].

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To maintain the 3-start helix monomer symmetry of MTs that are composed of different numbers of protofilaments one can adjust the longitudinal shift between two neighboring protofilaments that is also referred to as stagger, as was done in Figs. 1(a) and 1(b). In this way the distances between the monomers and dimers are slightly changed so that the values of the stagger were calculated according to the expression $(4 + 12/p) \times 10^{-9}$ m for the A lattice and $(12/p) \times 10^{-9}$ m for the B lattice[a] (see Fig. 4).

From the consideration above it is obvious that if each monomer in the MT was represented by a site, the A and B type arrangements of tubulin dimers would be represented by the same lattice.[8] The fact that the A and B lattices are equivalent on the level of monomers may suggest that MTs possess only one of them. Recent experimental evidence indicates that the MT lattice is a B lattice.[6,16,17] This result was obtained indirectly by decorating tubulin with motor protein kinesin since due to the very similar structure of α- and β-tubulin current observational techniques cannot satisfactorily detect the difference between the two tubulin monomers.[6,8] However, at present no definite conclusions can be made about the type of the MT lattice.

3. A Model of Dipolar States of Tubulin and their Interactions in a Microtubule

Experimental investigations of Athenstaedt[18] have shown that a number of biological structures including MTs exhibit pyroelectric and piezoelectric behavior. Such properties originate as a result of the elementary dipoles carried by the constituent particles of the structure. This implies that a tubulin dimer carries a permanent dipole moment. However, the charge distribution within the tubulin molecule is not very well known. It seems, though, that at one end of the tubulin heterodimer the density of ions sitting on the amino acid residues of the protein chains is higher than in the rest of the molecule.[19] This attracts charges of the opposite polarity from the MT surroundings and can give rise to a dipole. What would be the charge and the length of the dipole is not precisely clear. Consequently, in the numerical calculations presented here these parameters were adjustable. As will be seen below the adopted value of the magnitude of the dipole moment was not unreasonable.

The model of dipolar interactions utilized in this paper was suggested by Tuszyński *et al.*[1–3] According to this model a MT is viewed as an assembly of dipoles that are placed at sites of a two-dimensional hexagonal MT lattice. In this hexagonal lattice each three nearest sites form identical triangles. Hence the MT lattice is also a triangular lattice. The directions along the sides of a triangular cell determine the three directions of the dipole–dipole interactions in the MT lattice (see Fig. 2).

[a]An A or a B lattice that represents a MT which consists of p protofilaments will be also termed p-A, p-B, where p can be 12, 13 or 14.

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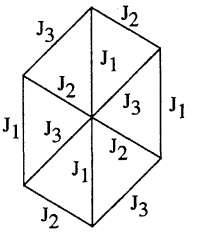

Fig. 2. A section of a MT lattice that represents a dipole and its six nearest neighbors. Each of the three directions along the sides of a triangular cell is associated with one of the energy constants J_1, J_2 and J_3.

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In the model each tubulin dipole interacts only with its six nearest neighbors and it can be in one of two orientational states that are characterized by the same magnitude of the dipole moment. These states were called the α and the β state. (The α and the β state of a tubulin dimer originally refer to two electrostatic states of tubulin that have been adopted in the cellular automata models.[11] These states were associated with a mobile electron that could be localized either more toward the α or more toward the β monomer of the dimer.) Since the longitudinal axis of the tubulin dimers assembled in a MT is parallel to the MT axis we have assumed that in the two dipolar states the projections of the dipole on the MT axis point in opposite directions. This direction was chosen to be positive in the α state and negative in the β state. Hence, the two dipolar states can be also termed the up and down state.

First we considered the simplest case in which the dipole on the tubulin is parallel to the MT axis in both α and β state. If an array of such dipoles is placed at sites of a triangular lattice and it is subjected to an external electric field $\mathbf{E}$ the total potential energy of the whole system is

$$H = -\sum_{j=1}^{N_C}\sum_{i=1}^{N_R}(J_1\sigma_{i,j}\sigma_{i+1,j} + J_2\sigma_{i,j}\sigma_{i-1,j+1} + J_3\sigma_{i,j}\sigma_{i,j+1})$$

$$-\mathbf{p}\cdot\mathbf{E}\sum_{j=1}^{N_C}\sum_{i=1}^{N_R}\sigma_{i,j}\,. \tag{1}$$

In the sum (1) N_R is the number of rows and N_C is the number of columns of the triangular lattice, $\sigma_{i,j}$ is a dipole variable that can have two values $+1$ or -1 which correspond to the two tubulin dipole states $\pm qs$ and J_1, J_2, J_3 are energy constants that characterize the interactions between the nearest dipoles along the three directions in the MT lattice (J_1, J_2, J_3 direction, Fig. 2). As can be seen Eq. (1) represents the Hamiltonian of a two-dimensional Ising model.[20]

The interaction constants J_n were found from the expression

$$W_n^{\uparrow\uparrow} - W_n^{\uparrow\downarrow} = -2J_n\,, \tag{2}$$

where $W_n^{\uparrow\uparrow}$ is the interaction energy between two neighboring dipoles in the nth direction when they both are in the up state and $W_n^{\uparrow\downarrow}$ is the interaction energy between two neighboring dipoles in the nth direction when one dipole is up and the other dipole is down. $W_n^{\uparrow\uparrow}$ and $W_n^{\uparrow\downarrow}$ were calculated using the classical formula for the interaction energy between two dipoles

$$W_n = \frac{1}{4\pi\epsilon_0\epsilon_r} \frac{(\mathbf{p}_1 \cdot \mathbf{p}_2)_n - 3(\mathbf{u} \cdot \mathbf{p}_1)_n(\mathbf{u} \cdot \mathbf{p}_2)_n}{|(\mathbf{x}_1 - \mathbf{x}_2)_n|^3} . \tag{3}$$

Here $\mathbf{p}_1$ and $\mathbf{p}_2$ are the dipole moments of the dipoles whose components with respect to the MT axis in our case have the magnitudes $\pm qs$, where q is the charge of the dipole and s is its length, $\mathbf{u}$ is a unit vector oriented along a line that connects the centers of the two dipoles, $\mathbf{x}_1$, $\mathbf{x}_2$ are the position vectors of the centers of the dipoles, ϵ_0 is the permittivity of vacuum and ϵ_r is the relative permittivity of the medium.

Since the magnitude and the orientation of the tubulin dipole are not very clear we have also considered a more general case in which the dipole in one of the states, for example, β is tilted with respect to the protofilament axis (Fig. 3). Because a MT is a cylinder the orientation of the dipole in the β state is given by two angles. One angle gives the orientation with respect to the MT axis and the other angle with respect to an axis that is perpendicular to the MT surface and passes through the center of the dipole. In the calculations presented here the curvature of the MT was not taken into account. The MT was treated as a sheet and the dipoles were assumed to lie in the plane of the sheet so that the tilting in the β state was given only by one angle θ. As will be seen in Sec. 9 varying θ results in various low temperature configurations of dipoles in the MT as the interaction constants J_n's change [see Eq. (3)]. Including the angle that characterizes the curvature would have the same effect. Hence to show the capabilities of the model it seemed to us sufficient to do calculations using the two-dimensional case at this moment.

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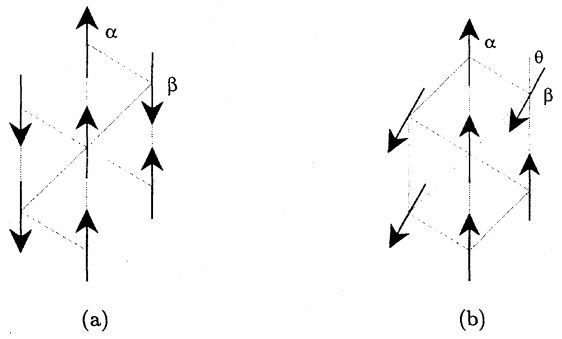

Fig. 3. A dipole and its nearest neighbors in a MT A lattice. (a) In both α and β state the dipole is parallel to the MT axis and oriented in opposite directions as is indicated by arrows. (b) In the β state the dipole is tilted by an angle θ with respect to the negative MT axis.

In the MT lattice in which the dipoles are tilted in the β state and lie in a two-dimensional plane the potential energy due to the interaction between the dipoles can be given by the following expression (Yeomans, 1992)

$$H_t = \frac{1}{4}\sum_{n=1}^{3}\sum_{\langle \mathrm{NN}\rangle} J_{n1}(1+s_k)(1+s_l) + J_{n2}(1-s_k)(1-s_l) + J_{n3}[(1+s_k)(1-s_l)+(1-s_k)(1+s_l)]\,. \tag{4}$$

In the equation above, the sum over n runs over the three sides of each triangular cell and $\langle \mathrm{NN}\rangle$ denotes the sum over the nearest neighbors. The variable s_k can have two values $+1$ or -1 which correspond to the α and β state, respectively. Each direction n of the triangular cell is characterized by three interaction energies: J_{n1} is the interaction energy between two dipoles that are in the nontilted state, J_{n2} is the interaction energy between two dipoles that are both tilted and J_{n3} is the interaction energy between a dipole in the tilted and a dipole in the nontilted state.

The terms in Eq. (4) can be rearranged so that one can be write it in the form

$$H_t = H_{\mathrm{int}} + V_{\mathrm{eff}}\sum_{j=1}^{N_\mathrm{C}}\sum_{i=1}^{N_\mathrm{R}}\sigma_{i,j} + C\,, \tag{5}$$

where H_{int} is the same as the interaction part of (1) but with a plus sign, V_{eff} is an effective potential energy, and C is a constant. The three interaction constants J_n in H_{int} are given by the equations

$$\begin{aligned} J_1 &= \frac{1}{4}(J_{11}+J_{12}-2J_{13})\,,\\ J_2 &= \frac{1}{4}(J_{21}+J_{22}-2J_{23})\,,\\ J_3 &= \frac{1}{4}(J_{31}+J_{32}-2J_{33})\,, \end{aligned} \tag{6}$$

where the interaction energies J_{nm} were calculated from the formula (3). The effective potential energy V_{eff} and the constant term C have the following form

$$V_{\mathrm{eff}} = \frac{1}{2}(J_{11}-J_{12}+J_{21}-J_{22}+J_{31}-J_{32}) \tag{7}$$

and

$$C = \frac{N}{4}\times(J_{11}+J_{12}+2J_{13}+J_{21}+J_{22}+2J_{23}+J_{31}+J_{32}+2J_{33})\,. \tag{8}$$

In the latter equation $N = N_\mathrm{C}\times N_\mathrm{R}$ is the total number of lattice sites.

In Eqs. (1) and (5) toroidal boundary conditions have been used. Due to the helical arrangement of the tubulin dimers a dipole in the last protofilament, σ_{i,N_C},

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is neighboring to the dipole $\sigma_{i+a,1}$, where a depends on the stagger. The formulas (6)–(8) are the same both inside and at the boundaries of the A lattice with an odd number of protofilaments. All other lattices contain a seam at which the distances between the tubulin dimers are different and, consequently, the values of the interaction constants are different from those inside the lattice. This has to be taken into consideration in the three equations above.

It is not difficult to show from Eq. (3) that the values of the interaction constants J_n and the interaction energies J_{nm} depend on the type of the MT lattice, i.e. on the stagger which determines the distances $|(\mathbf{x}_1 - \mathbf{x}_2)_n| = r_n$ (Table 1) and therefore the angles θ_n (Fig. 4), on the orientation of the tubulin dipole when it is tilted and on the parameters q, s and ϵ_r. The dependence on the latter three parameters can be reduced to a dependence on the single parameter $\bar{Q} = q^2 s^2 / \epsilon_r$. In a realistic model, q should have a value of the order of one electronic charge 1.602×10^{-19} C and s should be of the order of several nanometers. As a relative permittivity one can use the value measured for proteins. This value is between 20 and 30 and it is

Table 1. Staggers and distances r_2 and r_3 inside the A and B lattices. The parameter $r_1 = 8 \times 10^{-9}$ m (length of the tubulin dimer).

A LATTICE			
p	12	13	14
stagger (10^{-9} m)	5.00	4.92	4.86
r_2 (10^{-9} m)	5.83	5.87	5.90
r_3 (10^{-9} m)	7.07	7.02	6.97

B LATTICE			
p	12	13	14
stagger (10^{-9} m)	1.00	0.92	0.86
r_2 (10^{-9} m)	8.60	8.67	8.72
r_3 (10^{-9} m)	5.10	5.08	5.07

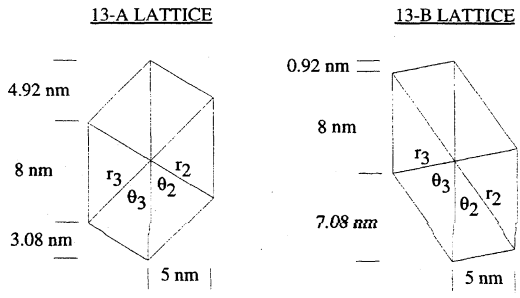

Fig. 4. The geometrical parameters of the 13-A and inside the 13-B lattice. The lateral separation between two neighboring protofilaments was taken as 5×10^{-9} m (Rasmussen *et al.*, 1990).

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the effective relative permitivitty, i.e. such which reflects the response of both the protein and its solvent.[21] We can point out that this value is not so much different from the value used in the previous papers by Tuszyński *et al.* This was $\epsilon_r = 70$ which is the value for free water in the static case (for nonoscillating dipoles) at body temperature.[22]

4. Numerical Procedure: Monte Carlo Method

The numerical calculations presented in this work have been performed on the basis of the classical Monte Carlo method that was realized by means of the Metropolis algorithm.[23] The procedure was used to find the configurations of the interacting system of dipoles in a MT at different temperatures. The configuration is a result of the competition between the potential energy of the dipoles and thermal energy. The interactions with neighboring dipoles and an electric field force a dipole to assume a state with the lowest potential energy (ground state). If the thermal energy is small compared to the potential energy the probability that the dipole will change its state due to a random thermal fluctuation is very small. With increasing thermal energy this probability also increases as the charges of the dipoles become more free to move.

In the algorithm the change of state of a dipole (a flip) happens when the number

$$\exp\left[-\frac{(E_{\text{new}} - E_{\text{old}})}{k_{\text{B}}T}\right], \tag{9}$$

is greater than a random number with a value between 0 and 1. In (9) E_{old} is the energy of a dipole in its old state, E_{new} is the energy of the dipole in its possible new state and k_{B} is Boltzmann's constant. Since in our model each dipole interacts only with its nearest neighbors the two energies depend only on the configuration of these dipoles.

It is obvious that a dipole always flips if $E_{\text{new}} < E_{\text{old}}$ because in that case the argument of the exponential in (9) is a positive number. In the opposite situation the argument is negative and the dipole flips only if (9) is larger than a randomly generated number whose value is between 0 and 1. This means that the probability of flipping a dipole is larger at higher temperatures. Such a procedure ensures that locally configurations of dipoles can form that are not in a ground state. With the increasing temperature the number of these regions increases and eventually at high temperatures the whole array of dipoles in a MT is completely randomized.

In the program used for the calculations shown below the number (9) was calculated and compared to the random number at each lattice site. The lattice was examined column by column. One such examination of the lattice is one Monte Carlo step. The dipolar ordering was calculated at different temperatures in the range 1–2000 K. The temperature step varied from $\Delta T = 1$ K to $\Delta T = 5$ K. In order for the array of dipoles to become uncorrelated with its initial state, at each investigated temperature 1000 Monte Carlo steps were performed and the last 200

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were used to calculate polarization per site (relative polarization). If the simulation was started from the low temperature end we chose as the starting configuration one of the ground states of the system. In the simulations started from the high temperature end the configuration at the beginning of the simulation was a random configuration. The final configuration from the previous Monte Carlo step was used as the initial configuration for the next Monte Carlo step.

Polarization per site was calculated after each Monte Carlo step (starting from the step 801) by dividing the sum of the values of the dipole variables $\sigma_{i,j}$ at all lattice sites by the total number of lattice sites. A dipole tilted in the β state contributed to the sum by $(-\cos\theta)$. The values of relative polarization plotted in figures below are the average of the values calculated in the Monte Carlo steps 801 to 1000. All Monte Carlo simulation shown here were performed for a 100 rows long MT (with the exception of simulations that involved MAPs, see Sec. 8) and $\bar{Q} = 12 \times 10^{-56}$ C^{2}m^2. The polarization curves and data in tables in Secs. 5–8 were obtained assuming that the dipole in the β state is nontilted. The calculations for the case when the dipole in the β state is tilted are discussed in Sec. 9.

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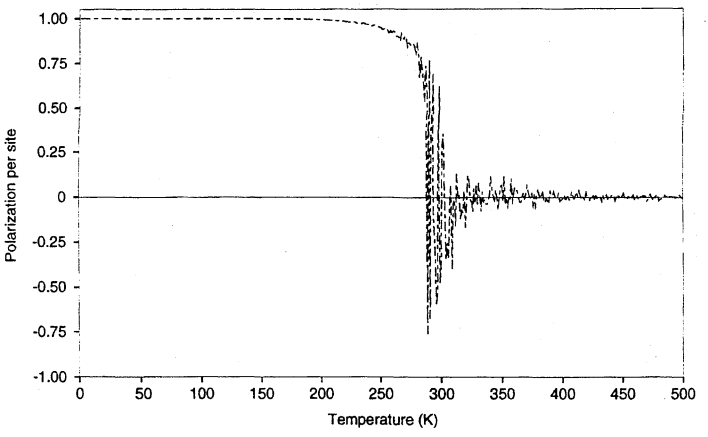
5. Microtubule Represented by the A Lattice

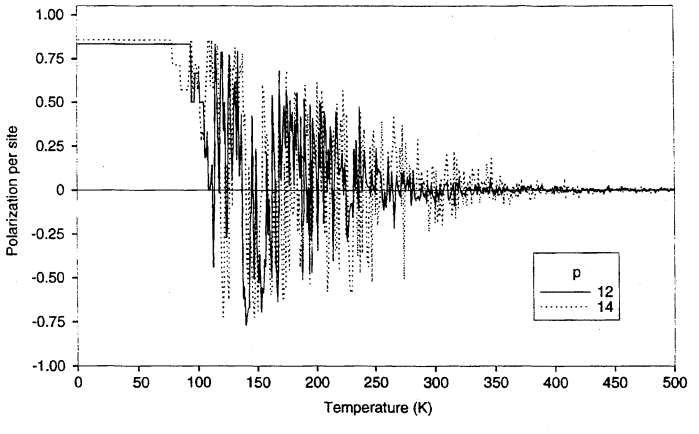
Figure 5(a) is a plot of polarization per site as a function of temperature calculated for a MT represented by the 13-A lattice. As can be seen at low temperatures the MT is in a completely ordered state with all dipoles pointing in the positive direction along the MT axis as was chosen at the beginning of the simulation. At temperatures close to the room temperature (300 K) the system undergoes a transition into a disordered state when the dipoles are oriented randomly and the total polarization is zero. The small fluctuations around zero are due to the thermal fluctuations. In analogy with ferroelectric crystals the two states of the array of dipoles in a MT will be further termed a ferroelectric and a paraelectric phase, respectively.[24]

It is not difficult to explain the character of the curve shown in Fig. 5(a). By examining the values of the corresponding interaction constants that are listed in Table 2 it becomes obvious that all J_n's are of the order of magnitude 10^{-21} J. The transition from the low temperature phase to the high temperature phase takes place when the energy of the thermal fluctiations $k_{\mathrm{B}}T$ is sufficiently large to allow the dipoles to exist in configurations that do not correspond to one of the ground states. This happens when $k_{\mathrm{B}}T$ is comparable to the interaction energy between the dipoles.[25] Since the value of the Boltzmann's constant is 1.38×10^{-23} JK^{-1} the transition occurs at temperatures $T \sim 100$ K.

The considerations above imply that increasing the interaction energies between the dipoles in the MT increases the transition temperature and *vice versa* which we also verified by the Monte Carlo simulations. As was noted in Sec. 3, the magnitudes of the interaction constants J_n depend on the value of the parameter $\bar{Q} = q^2 s^2/\epsilon_{\mathrm{r}}$ and the geometrical configuration of the tubulin dimers in the MT which is given



(a)



(b)

Fig. 5. Polarization per site: (a) 13-A MT, (b) 12-A and 14-A MT.

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by the type of the MT lattice A or B and the number of protofilaments. Changing these parameters changes the transition temperature and, consequently, determines the character of the dipolar order in a MT at body temperature (310 K).

Curves plotted in Fig. 5(b) represent polarization per site calculated for a 14-A and 12-A MT. The figure shows that at low temperatures in the 12- and 14-protofilament MTs not all dipoles point in the same direction — the value of the total relative polarization is less than one. This is due to the presence of the seam. As can be found in Table 2 in the 12-A and 14-A lattices the constant J_{3s} is negative and has a larger magnitude than any other interaction constant. The second most dominant interaction constant is J_1 which is positive and characterizes the dipole–dipole interaction along the longitudinal direction in the MT lattice.

Table 2. The interaction constants in the MT A lattice for $p = 12, 13, 14$. J_1, J_2, J_3 characterize the interaction inside the lattice, J_{2s} and J_{3s} refer to the seam.

p	12	13	14
J_1 (10^{-21} J)	+4.21	+4.21	+4.21
J_2 (10^{-21} J)	−1.12	−0.94	−0.79
J_3 (10^{-21} J)	+1.53	+1.49	+1.45
J_{2s} (10^{-21} J)	+1.67	−0.94	+1.65
J_{3s} (10^{-21} J)	−7.20	+1.49	−7.56

The form of the Hamiltonian (1) is such that when the interaction between two nearest dipoles is characterized by a negative interaction constant, this bond is in the state with the lowest energy if the dipoles point in the opposite directions. On the contrary, interaction between two nearest dipoles characterized by a positive interaction constant requires the dipoles to be aligned. Thus, the dipoles in the two protofilaments along the seam are at low temperatures aligned but in a different direction in each protofilament giving the total relative polarizations $10/12 = 0.833$ and $12/14 = 0.857$ for 12-A and 14-A lattices, respectively.

In a MT that is assembled from 13 protofilaments such a situation does not occur. This MT does not have a seam and the two dominant interaction constants J_1 and J_3 are positive. Consequently, even though the energetically more favorable state along the J_2 direction is when each two nearest dipoles point in the opposite directions, the effect of J_1 and J_3 overcomes the effect of J_2 and all dipoles at low temperatures are in the same state.

It can also be noticed that the interval of temperatures over which the transition from the ordered to the disordered state in the 12-A and 14-A lattices takes place is much larger than in the 13-A lattice. This effect is also caused by the presence of the seam. Since the magnitude of the interaction constant J_{3s} is larger than the magnitudes of the remaining four J_n's, the energy of the thermal fluctuations must be sufficiently large to eventually overcome also the interaction through J_{3s} that forces the dipoles to be ordered along the seam in opposite directions which affects the dipolar ordering in the whole lattice.

Another feature of the array of dipoles with the potential energy (1) is that the state in which all dipoles are oriented in one or the other direction along the MT axis has the same energy. Due to this the ground state of the 13-A lattice is doubly degenerate and the 12-A and 14-A lattices possess a multitude of ground states. For example, all states in which the assembly of dipoles in the MT forms two subclusters, one consisting of protofilaments 1 to $N_T < N_C$ and all dipoles in it are up/down and the other of protofilaments $(N_T + 1)$ to N_C with all dipoles pointing down/up, have the same energy. For instance, the simulation for the 14-A lattice in Fig. 5(b) was started from the configuration in which $N_T = 1$ that is shown in Fig. 6(a).[b]

[b]In all figures below that depict configurations of dipoles are included columns 0 and $(N_C + 1)$.

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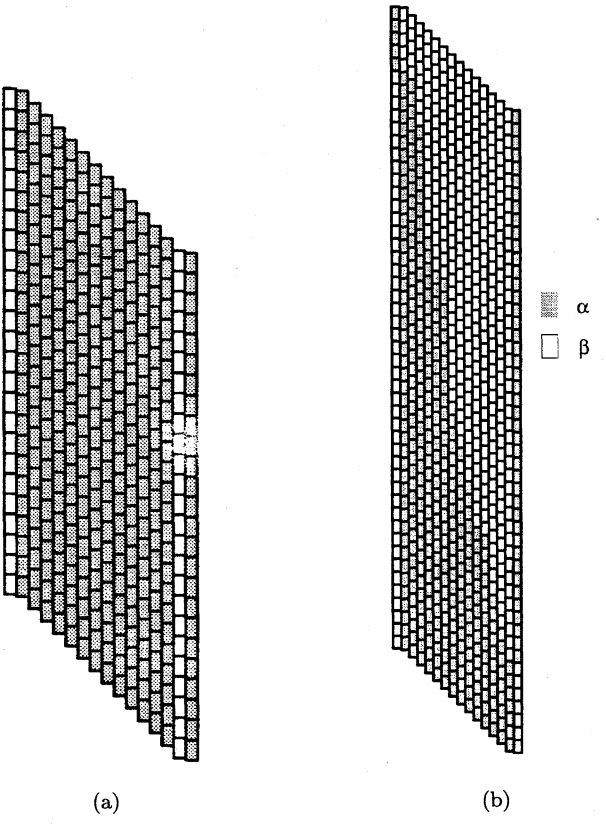

Fig. 6. (a) One of the ground states of a 14-A MT in which in the protofilaments 1 and 14 along the seam the dipoles are aligned in different directions and in the rest of the MT they point up. (b) A configuration of dipoles in a 50 row long section of the 14-A MT at $T = 5$ K. In the whole lattice (100 rows) the two subclusters in which the directions polarization are opposite have almost the same size so that the total relative polarization is almost zero.

The two clusters with different directions of polarization can also have a different form, for example, such as in Fig. 6(b). The figure shows a configuration of dipoles in a section of the 14-A lattice at temperature $T = 5$ K at which the system of dipoles arrived when it was initially completely randomized at $T = 1000$ K. This configuration is not a groundstate because in the 14-A lattice in a groundstate all dipoles in one protofilament have to point in the same direction. However, the total interaction energy is only slightly higher than the energy in a groundstate.

The polarization curve corresponding to the configuration shown in Fig. 6(b) is the solid line in Fig. 7. The dotted line in Fig. 7 represents a simulation that was also started from a random configuration of dipoles but the generated sequence of random numbers used by the Monte Carlo algorithm in this calculation was different. In this case in the ground state all dipoles in protofilaments 1 to 8 pointed up and 9 to 14 down so that the corresponding polarization was $2/14 = 0.143$.

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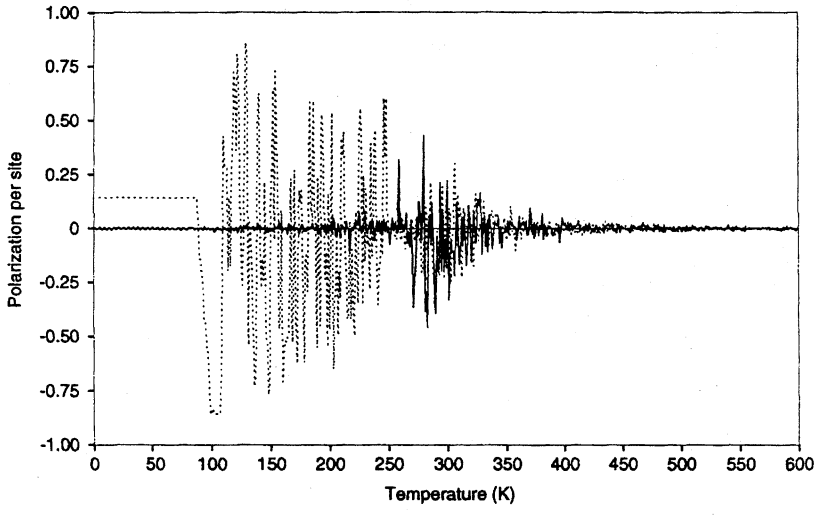

Fig. 7. Polarization curves calculated for a 14-A MT using two different sequences of random numbers. The simulations were started from random configurations of dipoles at $T = 1000$ K.

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We can conclude that the presence of a seam in a MT with the A-lattice introduces a multitude of states that are characterized by different configurations of dipoles but the same energy. The ground state that the regular array of dipoles in a MT reaches in a Monte Carlo simulation depends on the initial configuration of the dipoles and the chosen sequence of random numbers. If the configuration of dipoles in a MT is changed by some external factor (for example, an external electric field) a MT can switch into a different ground state. At low temperatures the MT remains in the ground state for a long time. At high temperatures it is easier for a MT to switch between various states due to the increase in thermal energy. [According to Figs. 5(b) and 7 at temperatures from the range of about 0–100 K the polarization in the 12-A and 14-A lattice has the same value. Thus, a thermal fluctuation by several degrees does not alter the value of polarization. However, at higher temperatures the polarization can fluctuate within 1 K.]

6. Microtubule Represented by the B Lattice

As was explained earlier for any number of protofilaments a MT represented by the B lattice contains a seam along which the values of the interaction constants are different from those inside the MT (Table 3). Because the largest interaction constant in the B lattices is J_3 and the second largest J_1 the ground state of a MT that exhibits a B lattice is antiferroelectric: all dipoles in one protofilament point in the same direction and in each two neighboring protofilaments these directions are opposite. Since J_{2s} and J_{3s} are several times smaller than J_1 and J_3 the dipolar order along the seam will also be determined by the latter interaction constants.

Table 3. The interaction constants in the MT B lattice.

p	12	13	14
J_1 (10^{-21} J)	+4.21	+4.21	+4.21
J_2 (10^{-21} J)	+1.67	+1.66	+1.65
J_3 (10^{-21} J)	−7.20	−7.40	−7.56
J_{2s} (10^{-21} J)	−1.12	−0.94	−0.79
J_{3s} (10^{-21} J)	+1.53	+1.49	+1.45

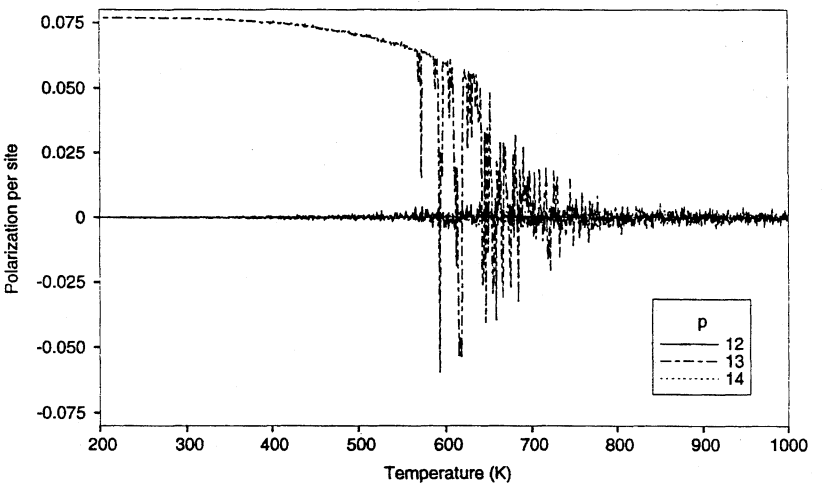

Fig. 8. Polarization per site in 12-B, 13-B and 14-B lattices.

Polarization curves for the B type lattices are shown in Fig. 8. All simulations were started from the low temperature end. The initial configuration was antiferroelectric (Fig. 9) which corresponds to the total relative polarization zero in the 12-B and 14-B and $N_R/(13 \times N_R) = 1/13 = 7.69 \times 10^{-2}$ in the 13-B lattice. As can be seen in Fig. 8 the lattices undergo a transition from the ordered antiferroelectric phase to the disordered paraelectric phase. Because the interaction constants in the B lattice are larger than in the A lattice the transition happens at higher temperatures. Since the two directions of polarization along the MT axis are energetically equivalent the antiferroelectric ground states are doubly degenerate.

7. Effects of an External Electric Field

According to the Ising Hamiltonian (1) the array of dipoles in a MT subjected to an external electric field has the tendency to point in the same direction as the field (to minimize the potential energy due to the interaction with the electric field). Figure 10 shows the effect of an external electric field that points in the positive

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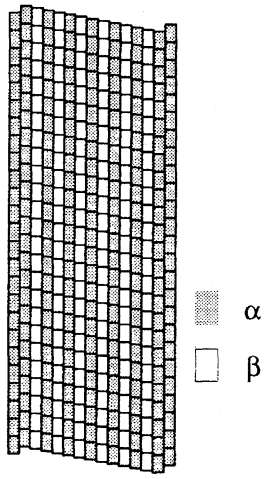

Fig. 9. Antiferroelectric arrangement of dipoles in the 13-B lattice that was used as the initial configuration in the corresponding simulation shown in Fig. 8.

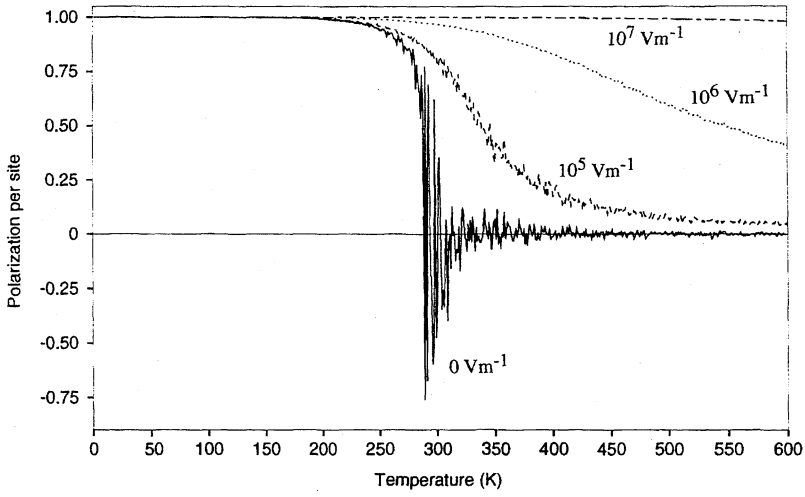

Fig. 10. Polarization curves calculated for a 13-A MT that is subjected to external electric fields 10^5 Vm^{-1}, 10^6 Vm^{-1} and 10^7 Vm^{-1}.

direction along the MT axis on a 13-A MT. To calculate the polarization curves plotted in this figure and also in Figs. 11 and 12 we chose the dipole strength $p = qs = 10^{-27}$ Cm. As can be seen in Fig. 10 the stronger is the electric field the larger is the number of dipoles that at a temperature T point in the positive direction. This is a consequence of adding the electric potential energy to the dipole–dipole interaction energy. Therefore a larger thermal energy is required to overcome the total potential energy.

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The character of the plots in Fig. 10 implies that the electric field starts to be effective at about 10^4–10^5 Vm^{-1}. This is because for these fields the potential energy $-pE$ becomes of the order of magnitude of the dipole–dipole interaction energies (see Sec. 3). For $E = 10^7$ Vm^{-1} up to very high temperatures all dipoles point in the direction of the electric field. It can be noted that Vassilev *et al.*[26] applied electric pulses with amplitudes of the order of magnitude 10^0 to 10^1 Vm^{-1} to assemblies of MTs with the result that the MTs aligned along the electric field lines. According to our model this would mean that the assembly of dipoles in the tested MTs was in an ordered state so that even small electric fields align the MTs along their field lines. The electric fields that exist inside the cell have been reported to be of similar magnitudes than those used by Vassilev *et al.*[27] Hence in our model the electric fields inside the cell wouldn't have an effect on the dipolar ordering in MTs. However, for example, in membranes the electric fields are of the order of 10^6–10^7 Vm^{-1}.

In the 12- and 14-protofilament A lattices the effect of an external electric field is more complex than in the 13-protofilament A lattice due to the presence of the seam. As was already shown in these lattices the diagonal interaction along the seam that is the strongest causes the two protofilaments along the discontinuity to be polarized in opposite directions. Consequently, at lower fields it is still more energetically favorable for the dipoles to be aligned in opposite directions in the two protofilaments along the seam. At the same time even though the electric field is not sufficiently strong to flip the dipoles along the seam it is strong enough to flip the dipoles inside the lattice. Hence, for such electric fields at low temperatures all protofilaments are polarized in the direction of the electric field except for one of the two protofilaments that form the seam. When the electric field is sufficiently large so that also the dipoles along the seam are in the state with the lowest potential energy if they point in the same direction as the external electric field, the whole lattice becomes completely ordered at low temperatures.

The effect of an external electric field on a 14-A MT is shown in Fig. 11. For $0 \leq E \leq 3 \times 10^6$ Vm^{-1} at low temperatures all dipoles point in the same direction (positive) as the electric field with the exception of one of the protofilaments along the seam in which all dipoles point in the opposite direction. At some transition temperature the polarization starts to decrease with the increasing temperature. When $E = 5 \times 10^6$ Vm^{-1} at low temperatures the configuartion of dipoles is the same as for the lower fields. However, as the thermal energy increases with the increasing temperature, in the protofilament along the seam in which the dipoles are oriented in the negative direction some of them can flip up due to random thermal fluctuations while the dipoles in the rest of the MT point in the positive direction. Consequently, the relative polarization is larger than 12/14. After reaching its maximum the polarization decreases as the temperature increases. The electric fields with the magnitude 6×10^6 Vm^{-1} and higher are sufficiently strong so that at low temperatures all dipoles in the MT point in the same direction as the fields.

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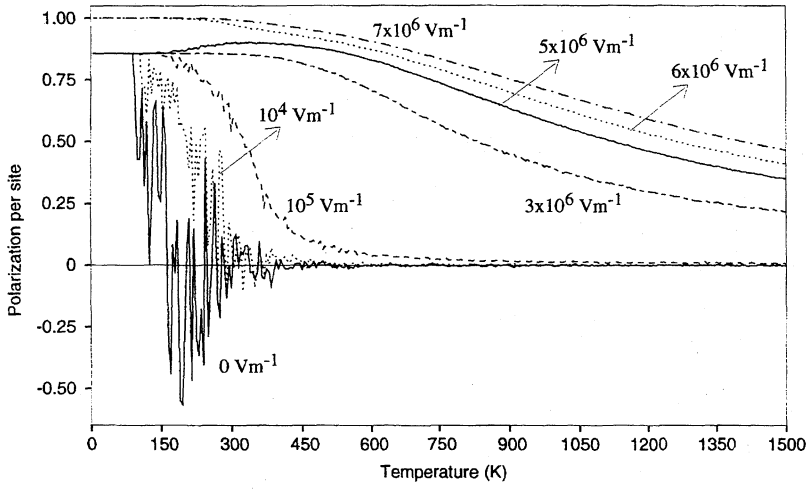

Fig. 11. Polarization per site in a 14-A MT that is subjected to external electric fields. The curves for $E = 3 \times 10^6$ Vm^{-1} and larger fields were calculated starting from random configurations at $T = 2000$ K.

The picture would be very similar in the 12-A lattice since its interaction constants do not substantially differ from those in the 14-A lattice (see Table 2).

The behavior of the array of dipoles in the B lattices under the influence of changing external electric field has a similar character as in the 12-A and 14-A lattices and it is shown for a 13-B lattice in Fig. 12. As was discussed in Sec. 6 the ground state of a MT represented by the B lattice is antiferroelectric because the largest interaction constant is J_3. As can be seen in Fig. 12 when the MT is subjected to lower electric fields, the ground state of the array of dipoles at low temperatures is antiferroelectric. However, as the thermal energy increases with the increasing temperature the random thermal fluctuations allow also the dipoles in the protofilaments with negative polarization to point in the same direction as the electric field. As a result, at fields 5×10^6 Vm^{-1} to 1.1×10^7 Vm^{-1} at higher temperatures the polarization per site increases above 1/13. Once the polarization reaches its maximum given by the combination of the three factors — thermal energy, dipole–dipole interaction energy and the potential energy that is due to the interaction with the electric field — it slowly decreases with the increasing temperature. When the external electric field is larger than a certain value, at low temperatures all dipoles in the lattice point in the direction of the field and at some critical temperature the total polarization starts to decrease. For even larger electric fields the whole lattice would be ordered also at high temperatures.

It can be pointed out that in the 13-B MT due to the odd number of protofilaments the dipoles in the two protofilaments along the seam point in the same direction in the antiferroelectric phase. However, since the values of the interaction constants along the seam are several times smaller than J_3 and J_1 this feature would

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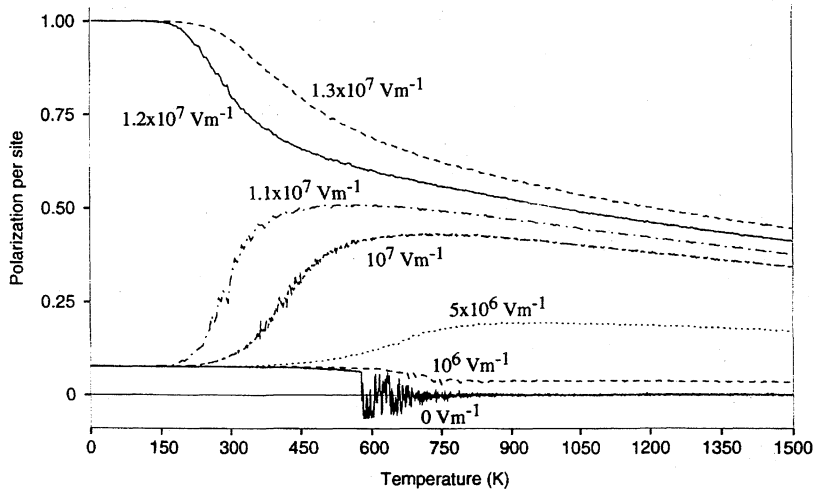

Fig. 12. Polarization per site calculated for a 13-B MT in the presence of external electric fields. The curves for $E = 1.1 \times 10^7$ Vm^{-1} and larger fields were calculated starting from random configurations at $T = 2000$ K.

not introduce a substantial difference in the character of the dipolar order in this lattice. Since the values of the interaction constants in the 12-B and 14-B lattices are not very different from those in the 13-B lattice the array of dipoles in MTs with these lattices when they are subjected to external electric fields with various magnitudes will behave in a similar manner. The difference in the magnitudes of the interaction constants will cause different critical temperatures and at low fieds and low temperatures the polarization in the B-lattices with even number of protofilaments will be zero.

8. Effects of MAPs

Experimental evidence indicates that MAPs can bind to MTs in both regular and irregular way.[8] Since a MT is a helical structure it has been suggested that the regular MAP patterns form superhelices.[8,11] Based on this the influence of MAPs on the dipolar order in the MT lattice was studied here for three different superhelical MAP arrangements. The superhelices were constructed according to[11] and they are shown for a 13-A MT in Fig. 13. In the other types of lattices these patterns do not form superhelices due to the presence of the seam (see Figs. 16, 17 and 19). The ratio of the number of MAPs (MAP ratio) to the total number of lattice sites in the three MAP arrangements is 1/11, 1/22 and 1/48. In the simulations these patterns were imposed on MTs 99, 110 and 96 rows long, respectively. Some simulations for MAP ratio 1/11 were performed for a shorter 44 row long MT.

As can be noticed, in forming the patterns we have assumed that each MAP is attached to one tubulin dimer. However, since most MAPs are several times heavier than tubulin they most likely bind to more than one tubulin heterodimer.[4,28]

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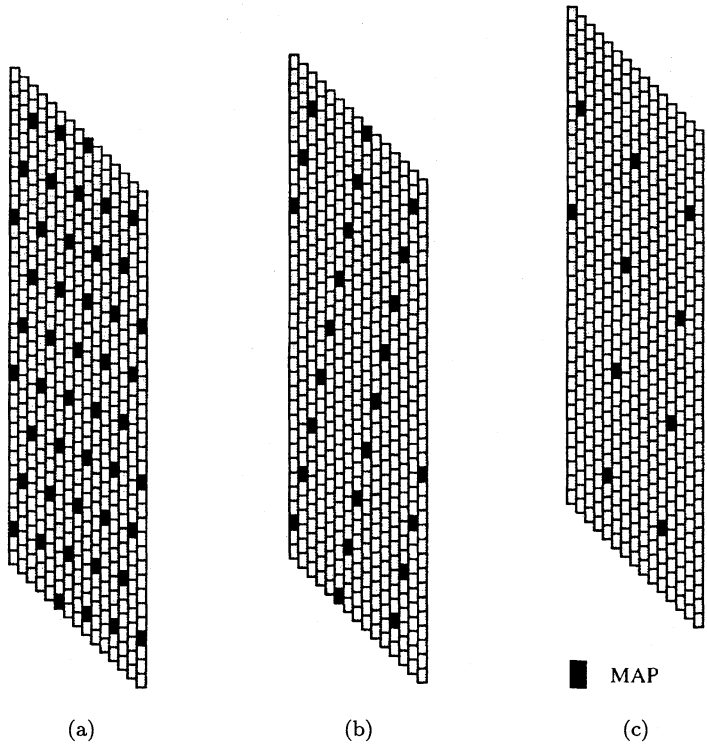

Fig. 13. Three superhelical patterns of MAPs for which the ratios of the number of MAPs to the total number of lattice sites are (a) 1/11, (b) 1/22, (c) 1/48. The patterns are imposed on a 13-protofilament A lattice.

We further assumed that a MAP attached to a tubulin subunit in the MT either prevents the dipole on the tubulin to flip or that it neutralizes the tubulin molecule. Hence, the dipole variable $\sigma_{i,j}$ at the sites of the MT lattice at which MAPs were bound (MAP sites) was set either to one or to zero, respectively.

The polarization curves that represent the dipolar order in a 13-A MT to which MAPs attach according to the three patterns depicted in Fig. 13 are shown in Fig. 14. Figure 14(a) represents a set of calculations in which at the MAP sites $\sigma_{i,j} = 1$. The polarization curves show that the number of tubulin dipoles that point in the same direction, in this case up, is larger in the lattices in which MAPs are present compared to the lattice without MAPs. This behavior is due to the fact that in the 13-A lattice in the state with the lowest energy all dipoles point either up or down (see Sec. 5). If some of the dipoles permanently point up they will force their neighbors also to point up. The more MAPs bind to a MT the greater is at a temperature T the number of dipoles which point in the same direction as the permanent dipoles carried by the tubulins to which the MAPs are attached.

The numerical data obtained for the case when MAPs are represented by $\sigma_{i,j} = 0$ are plotted in Fig. 14(b). The data were normalized with respect to the total number of lattice sites excluding the MAP sites. As can be seen the bound MAPs do not

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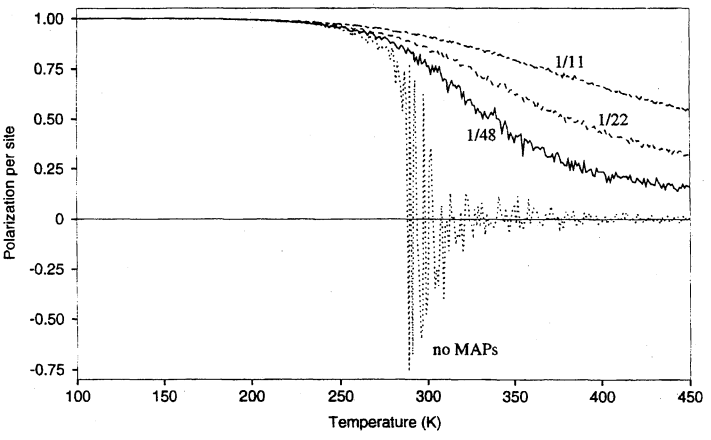
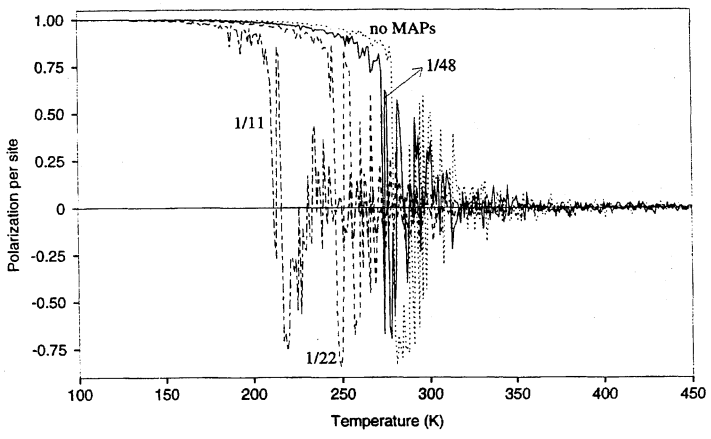


(a)



(b)

Fig. 14. Polarization curves calculated for a 13-A MT in which MAPs bound to the molecules of tubulin form the three patterns shown in Fig. 13. At the MAP sites $\sigma_{i,j}$ is (a) 1, (b) 0.

change the character of the dipolar order that exists in the lattice in which MAPs are not present — the array of dipoles undergoes a transition from the low temperature ferroelectric phase to the high temperature paraelectric phase. However, the MAPs affect the transition temperature. The larger is the MAP ratio, the lower is the temperature of the transition from the ferroelectric to the paraelectric phase. The reason for this behavior is quite obvious. As was discussed in Sec. 5, in the 13-A lattice the dipolar order is determined by the positive interaction constants J_1 and J_3. Along the two directions in the MT lattice characterized by these interaction constants in the configuration x, 1 it requires less energy for the dipole that is up to flip down if $x = 0$ compared to the case when $x = 1$. In the configuration x, -1

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the dipole that is down flips up at less energy cost for $x = 0$ than for $x = -1$. Thus, if in the starting configuration at low temperatures all dipoles in the MT point in the same direction, it requires less energy and, consequently, lower thermal energy k_BT to undergo a transition to the paraelectric phase if some tubulin molecules in the MT are neutral.

The type of the dipolar ordering in the presence of MAPs in a MT A lattice that has an even number of protofilaments is represented in Fig. 15 that shows a simulation for the 14-A lattice in which at the MAP sites $\sigma_{i,j} = 1$. As in the 13-A lattice the number of dipoles that point up increases with the increasing MAP ratio except for at the seam where the dipoles that interact through the large and negative interaction constant J_{3s} are oriented in opposite directions. Because of the chosen MAP patterns and the type of the boundary conditions the MAPs that are placed in protofilaments 1 and 14 happen to be nearest neighbors along the J_{3s} direction (Fig. 16). Hence, the total relative polarization in the low temperature state is slightly higher than in the case when no MAPs are present in the lattice. Another consequence of this arrangement of MAPs along the seam in the 14-A lattice is that in each of the protofilaments 1 and 14 the direction of polarization can be different in the sections between two nearest MAPs. Consequently, this lattice possesses a multitude of groundstates that are characterized by different configurations of dipoles along the seam.

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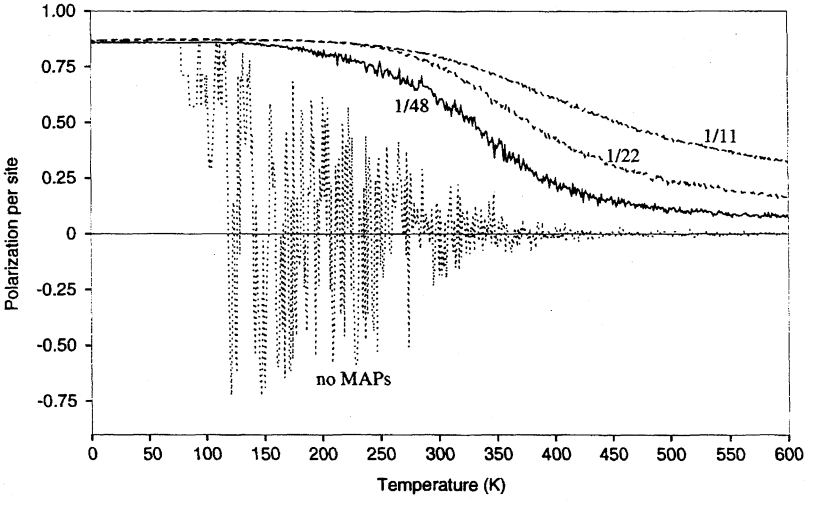

Fig. 15. Polarizations per site in a 14-A MT to which MAPs are bound according to the three patterns shown in Fig. 13. At the MAP sites $\sigma_{i,j} = 1$.

As in the 13-A lattice representing MAPs by $\sigma_{i,j} = 0$ in the 12-A and 14-A lattices does not favor any of their ground states. The dipolar configurations are of a similar type as those found in MTs that do not bind MAPs, however, the

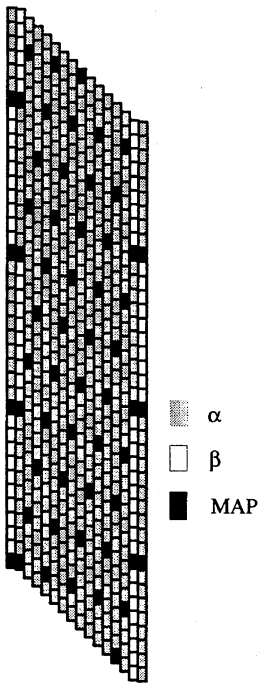

Fig. 16. A section of a 14-A MT that corresponds to the curve for the MAP ratio 1/11 in Fig. 15 at $T = 5$ K.

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presence of MAPs introduces also groundstates of different types. We performed numerical calculations for a 14-A MT for all three MAP ratios starting from random configurations at high temperatures. The configuration at which the array of dipoles arrived at low temperatures most often was of the type shown in Fig. 17(a) for the MAP ratio 1/11. It can be seen that along the seam each two dipoles that interact through J_{3s} point in opposite directions. The assembly of dipoles forms two subclusters in each of which the directions of polarization is different [compare to Fig. 6(b)]. The boundary between the two subclusters forms along one of the diagonal lines that connect MAPs since it consumes less energy to flip a dipole in the neighborhood of a tubulin dimer that does not carry any dipole moment.

As was discussed in Sec. 5 the configuration in Fig. 17(a) is not a ground state of the array of dipoles, but it is close to it. Another low temperature state that is close to the state with the lowest energy was obtained for a MT that bound MAPs in the ratio 1/22. In this state all dipoles inside the MT pointed down and all dipoles neighboring along J_{3s} were oriented in opposite directions, but the polarization of dipoles varied between two neighboring MAPs along the protofilaments 1 and 14. In many simulations performed for lower MAP ratios or shorter MTs the MT lattice reached one of the groundstates. Two such groundstates are shown in Figs. 17(b) and (c). Figure 17(b) depicts a section of a 110 rows long MT that is in a groundstate

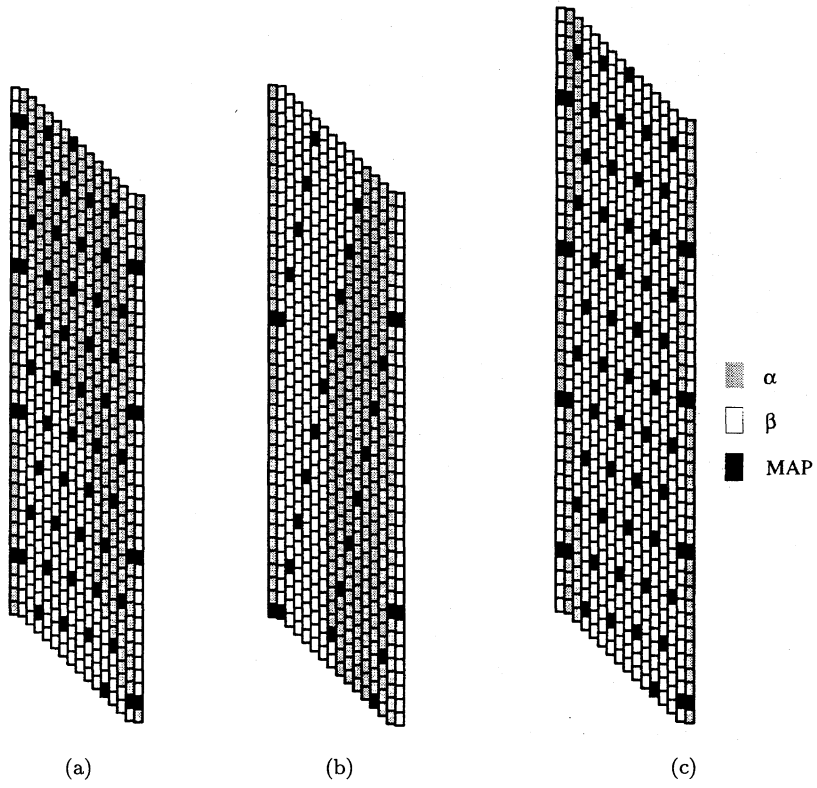

Fig. 17. Three different low temperature dipolar configurations in MTs in the presence of MAPs that are represented by $\sigma_{i,j} = 0$. (a) A section of a MT that is 99 rows long, MAP ratio is 1/11. (b) A section of a 110 row long MT, MAP ratio is 1/22. (c) A 44 row long MT, MAP ratio is 1/11.

for MAP ratio 1/22. Figure 17(c) shows a groundstate in a 44 row long MT to which bind MAPs in the ratio 1/11. The figures imply that there exists a large multitude of groundstates of the types shown. We also found the groundstate configurations that consist of two subclusters, one polarized up and the other down each consisting of 1 up to 13 neighboring protofilaments [see Fig. 6(a)].

Figure 18 shows a set of calculations performed for the 12-B lattice for all three MAP ratios when $\sigma_{i,j} = 1$ at the MAP sites. The simulations were started from the low temperature end. All three lattices are in a ground state or close to a ground state at low temperatures. At $T \sim 200$ K the polarization in the lattice exhibits a maximum, then decreases until it reaches a minimum and eventually increases and at very high temperatures oscillates around the value equal to the MAP ratio. The latter effect cannot be seen because it occurs at temperatures of the order of 10^4 K.

A character of dipolar ordering such as the one represented by Fig. 18 is given by the combination of antiferroelectric interactions, the MAP configuration and thermal energy. At low temperatures at least one of the ground states is such as shown in Fig. 19 for the MAP ratio 1/48. In the background of the antiferroelectric

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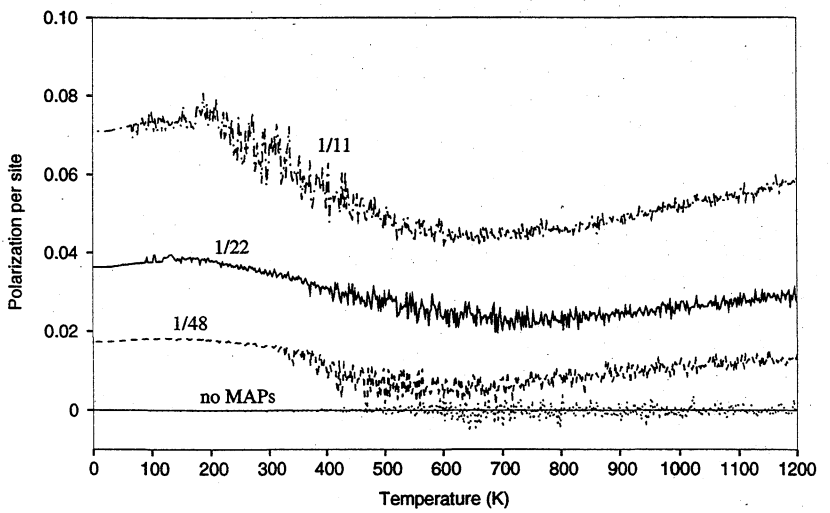

Fig. 18. Polarization curves calculated for the 12-B lattice that contains the three patterns of MAPs. The calculations were started from random configurations at $T = 1500$ K and $\sigma_{i,j} = 1$ at the MAP sites.

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arrangement of dipoles are inserted MAPs which at low temperatures do not have an effect on their surrounding dipoles inside the lattice. Along the seam, however, for the dipoles in the 12th row which are neighboring to MAPs in the 11th row it is more energetically favorable to be down. The energy of each of these dipoles is -0.86×10^{-21} J. The relative polarization that corresponds to this state in the lattice is then $20/(12 \times N_{\mathrm{R}}) = 0.0174$. As the temperature increases the thermal fluctuations allow the dipoles along the seam which are neighboring to MAPs to flip into the energetically unfavorable up state ($+0.86 \times 10^{-21}$ J) since the flip does not require much energy. To flip any other dipoles would require several times more energy. Thus, the polarization increases. It should be realized, though, that this increase is due to flipping of only a few dipoles. After the polarization reaches its maximum it decreases because the thermal energy increases and as a consequence some of the dipoles inside the lattice can flip. From its minimum the polarization slowly increases and eventually reaches a state in which all dipoles in the lattice except for at the MAP sites are oriented at random so that the total polarization is equal to the MAP ratio. This happens at temperatures of about 5×10^4 K at which the thermal energy is very large so that it completely overcomes the influence of the dipole–dipole interactions.

For the MAP ratio 1/48 the ground state shown in Fig. 19 was reached at low temperatures in the simulation shown in Fig. 18. When the MAP ratios were 1/22 and 1/11 the states at low temperatures were close to this groundstate. But even for the highest MAP ratio 1/11 the groundstate above existed in shorter MTs (44 rows). We also found that at low temperatures in lattices that contain a smaller

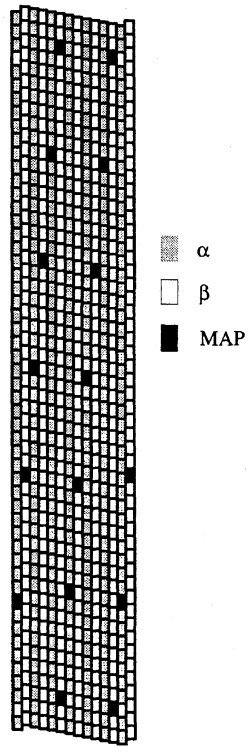

Fig. 19. A ground state in the 12-B lattice that corresponds to $T = 4$ K on the curve for the MAP ratio 1/48 in Fig. 18.

number of MAPs, so that the spaces between two neighboring MAPs along one protofilament are larger, and for $\sigma_{i,j} = 1$ at the MAP sites, along one protofilament can form subsections in which the dipoles are oriented in opposite directions. Most MAPs in the lattice occur in the subsections in which the dipoles are up to minimize the energy of the interaction along the J_1 direction.

In the 13-B and 14-B lattices the effect of MAPs represented by $\sigma_{i,j} = 1$ is similar as in the 12-B lattice. The dipolar order in the MT is a result of the competition between the thermal energy, dipole–dipole interactions and a MAP configuration. Unlike the lattices that consist of an even number of protofilaments in the 13-B lattice to the antiferroelectric ground state can correspond two different values of polarization: $-1/13$ when all dipoles in six rows point up and all dipoles in seven rows point down, or $+1/13$ in the alternative case. The positive branch of polarization is most likely to appear in lattices that contain higher ratios of MAPs which are represented by $\sigma_{i,j} = 1$. When $\sigma_{i,j} = 0$ at the MAP sites, the lattice is equally likely to choose one of the two branches of polarization. This feature of the 13-protofilament B lattice is an additional factor to those listed above that determine the complex dipolar behavior in the lattices that contain MAPs.

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As in the A lattice, if the dipole variable at each MAP site is zero the MAPs do not affect the type of the dipolar ordering in the B lattice, but their presence lowers the transition temperature. In the antiferroelectric phase in the two directions characterized by the large interaction constants J_3 and J_1 it costs less enegy to flip a dipole one of whose nearest neighbors is a MAP. Hence, the transition between the antiferroelectric phase and the paraelectric phase takes place at lower temperatures compared to the lattices in which MAPs are not present. To see this, in Fig. 20 is plotted polarization per site calculated for a 13-protofilament MT which contains MAPs in the ratio 1/11 represented by $\sigma_{i,j} = 0$. The simulation started from the high temperature end and the lattice which contains MAPs chose the negative branch of polarization.

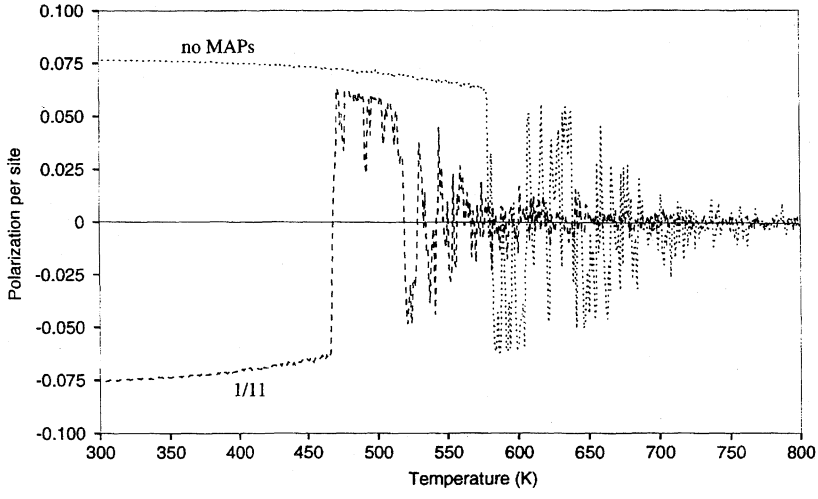

Fig. 20. Relative polarization in the 13-B lattice that contains MAPs represented by $\sigma_{i,j} = 0$ in the ratio 1/11. The calculations were started from a random configuration at $T = 1500$ K.

9. Dipoles Tilted in the β State

To study the dipolar order in lattices in which the dipoles in the β state are tilted by an angle θ with respect to the MT axis we performed calculations for the values of θ in the range $+90°$ to $-90°$ with a step $\pm 10°$ except for the largest angles that were taken to be $\pm 89°$. The sign "+" refers to tilting to the left and "−" refers to tilting to the right with respect to the MT axis. It should be pointed out that tilting to the right and tilting to the left by the same angle are not equivalent in the 12-, 13- and 14-protofilament MT lattices because the distances between two nearest tubulin dimers in the two diagonal directions are not the same (see Table 1, Figs. 3 and 4).

In Fig. 22(a) are plotted curves that represent relative polarization in the 13-A lattice for some "+" values of θ. When $\theta = +10°$ to $+70°$, at low temperatures

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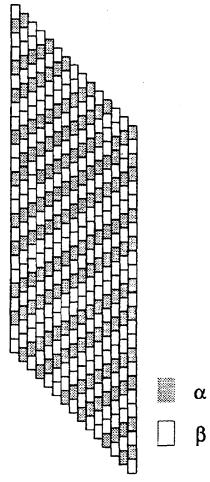

Fig. 21. A section of a MT that corresponds to the curve for $\theta = +89°$ in Fig. 22(a) at $T = 4$ K.

all dipoles are in the tilted state and the value of relative polarization is equal to $(-\cos\theta)$, i.e. it increases with increasing θ. Since $|\cos\theta| < 1$ for $0° < \theta < 90°$, at high temperatures, when due to the thermal energy the dipoles flip into the nontilted state, the polarization per site becomes positive.

The configuration of dipoles for $\theta = +80°$ and $+89°$ at low temperatures is shown in Fig. 21. The dipoles that are in the same state have the tendency to order themselves along the diagonal direction characterized by the constant J_2 and along the J_3 direction the dipole states tend to alternate.

Table 4. The interaction constants in the 13-A lattice for $\theta = -10°$, $-30°$, $\pm 40°$, $+60°$, $+70°$, $\pm 80°$.

θ (°)	+40	+60	+70	+80	−10	−30	−40	−80
J_1 (10^{-21} J)	−3.07	−1.98	−1.43	−0.94	−4.14	−3.54	−3.07	−0.94
J_2 (10^{-21} J)	−3.97	−5.29	−5.46	−5.32	+2.11	+3.76	+4.14	+2.93
J_3 (10^{-21} J)	+1.33	+1.90	+1.92	+1.80	−2.29	−3.58	−3.99	−3.62

Table 4 which contains the values of the interaction constants J_1, J_2 and J_3 for certain angles θ shows why the above types of dipolar order arise. For angles less than $+70°$ the constants J_1 and J_2 are negative and larger in magnitude than the positive constant J_3. Consequently, in the state with the lowest energy all dipoles have the same orientation. Whether they are tilted or nontilted is determined by the second term in the Hamiltonian (5) [the dipolar ordering depends on the difference of energies, see (9)].[c] For $\theta = +70°$ the magnitude of the negative J_1 is smaller

[c]The energy of a dipole due to the interaction with its nearest neighbors can be calculated directly from (4) by summing up the appropriate constants J_{nm}.

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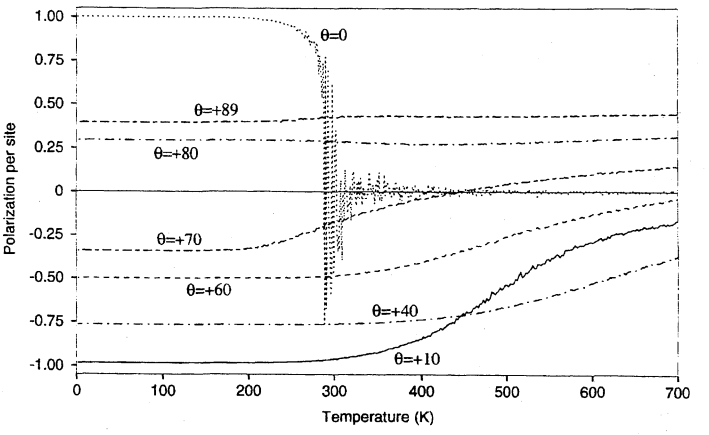
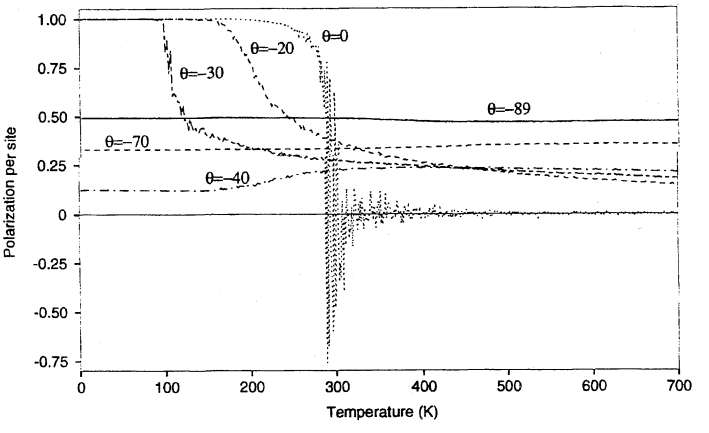
than the positive J_3 but in the ground state all dipoles are nontilted as a result of the combination of J_n's, V_{eff} and thermal energy. For $\theta = +80°$, J_3 is sufficiently large compared to the magnitude of J_1. As a consequence the dipoles mostly are in the same state along the J_2 direction and their states mostly alternate along the J_3 direction. Hence, when θ is large enough and positive the low temperature configuration of dipoles in the 13-A lattice is close to antiferroelectric with respect to the diagonal J_2 direction.



(a)



(b)

Fig. 22. Polarization per site in the 13-A lattice for selected angles θ. The calculations were started from random configurations at $T = 1000$ K. (a) "+" θ's, (b) "−" θ's.

Figure 22(b) represents polarizations per site in the 13-A lattice for negative θ's. At low temperatures for angles up to $-30°$ all dipoles in the MT lattice are nontilted. The ground state for $\theta = -40°$ and larger angles is shown in Fig. 23.

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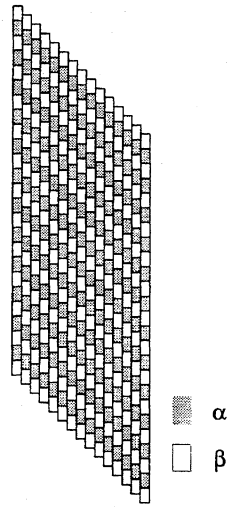

Fig. 23. A section of a MT that corresponds to $T = 4$ K on the curve for $\theta = -40°$ in Fig. 22(b).

The dipoles in the same state form rows along the J_3 direction and the sign of polarization along the J_2 direction alternates. This is a similar kind of order as in the case of $\theta = +80°$ and $+89°$ except for that the types of polarization along the two diagonal directions characterized by the constants J_2 and J_3 are interchanged and this lattice is antiferroelectric with respect to the the J_3 direction (see Table 4).

The dipolar ordering in MTs represented by the A lattices with an even number of protofilaments can be expected to be similar to that in the 13-A lattice. Due to the different number of protofilaments the stagger and therefore the interaction energies in these lattices are slightly different. The configuration of dipoles is also influenced by the presence of the seam. Relative polarization in the 14-A lattice for selected positive θ's is shown in Fig. 24(a). In the low temperature state for $\theta < +40°$ in one of the protofilaments along the seam all dipoles are nontilted and in the rest of the lattice the dipoles are in the tilted state. When θ ranges from $+40°$ to $+70°$ dipoles in the whole lattice are tilted at low temperatures. For angles $+80°$ and $+89°$ in the ground state the configuration of dipoles is antiferroelectric with respect to the direction characterized by J_2 and it is shown in Fig. 25.

In Fig. 24(b) are plotted polarization curves obtained for the 14-A lattice for some negative angles θ. When $-40° < \theta$ the dipoles in the ground state are nontilted except for one of the protofilaments along the seam in which all dipoles are tilted. For $\theta < -40°$ the configuration of dipoles in the ground state is antiferroelectric with respect to the J_3 direction.

To illustrate the influence of tilting of the dipole in the β state on the dipolar ordering in a MT that possesses a B lattice, in Fig. 26 are shown polarizations per site calculated for the 13-B lattice for different angles θ. In agreement with expectations as the tilting of the dipole and therefore the values of the interaction

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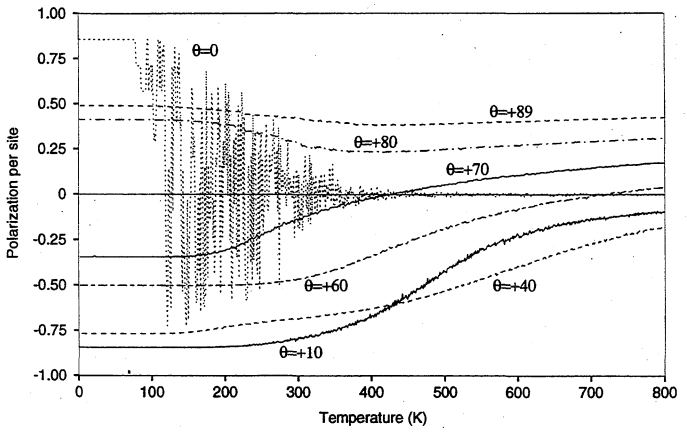
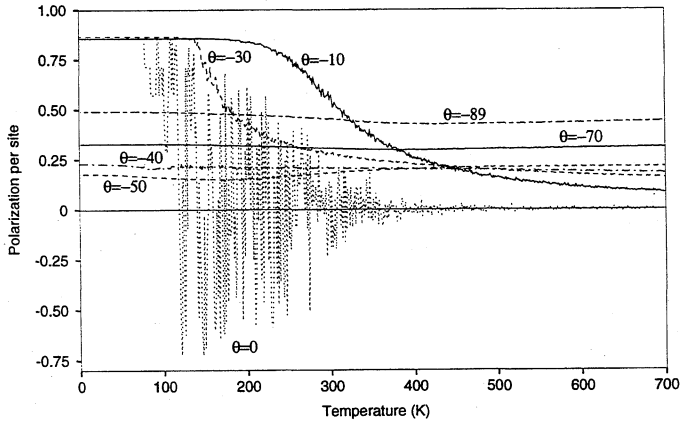


(a)



(b)

Fig. 24. Polarization per site in the 14-A lattice for selected angles θ. The calculations were started from random configurations at $T = 1000$ K. (a) "+" θ's, (b) "−" θ's.

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constants in the lattice change (Table 5), the type of the low temperature configuration of the dipoles is altered. For angles $0 < \theta \leq +50°$ the configuration in the ground state of the 13-B lattice is antiferroelectric. The dipoles are tilted in odd and nontilted in even protofilaments. When θ is in the range 0 to $-30°$, the low temperature configuration is also antiferroelectric but the dipolar states in the odd and even protofilaments are interchanged compared to the previous case. In the ground states that correspond to $+60° \leq \theta \leq +90°$ and $-90° < \theta < -40°$ all dipoles in the lattice are tilted.

The effect of external electric fields and MAPs in a MT lattice in which the dipoles are tilted with respect to the longitudinal axis at least in one of their states can be investigated in a similar way as in the case when the dipole in both of its

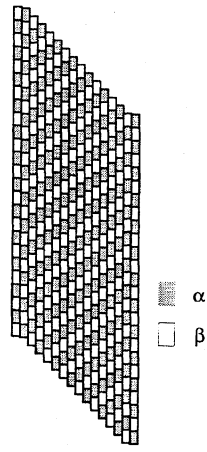

Fig. 25. A section of a MT that corresponds to $T = 4$ K on the curve for $\theta = +89°$ in Fig. 24(a).

Table 5. The interaction constants in the 13-B lattice for $\theta = -10°$, $+20°$, $-30°$, $-40°$, $+50°$, $+60°$, $\pm 80°$.

θ (°)	+20	+50	+60	+80	−10	−30	−40	−80
J_1 (10^{-21} J)	−3.09	−2.54	−1.98	−0.94	−4.14	−3.54	−3.07	−0.94
J_2 (10^{-21} J)	−2.34	−2.60	−2.46	−1.92	−1.23	−0.35	−0.04	+0.78
J_3 (10^{-21} J)	+7.96	+5.47	+4.09	+1.30	+6.41	+3.42	+1.66	−3.78
J_{2s} (10^{-21} J)	−1.67	−4.78	−5.29	−5.32	+2.11	+3.76	+4.14	+2.93
J_{3s} (10^{-21} J)	+0.10	+1.70	+1.90	+1.80	−2.29	−3.58	−3.99	−3.62

states is nontilted. However, since the tilted and nontilted dipolar states are not energetically equivalent the effect is more complex — it matters in which direction point the external electric fields and whether the dipole states on the tubulin dimers to which bind MAPs are tilted or nontilted.

10. Discussion

In this paper the classical two-dimensional Ising model has been used to investigate the properties of an array of dipoles placed at sites of the hexagonal MT lattice at different temperatures. The lattices of both A and B type for the numbers of MT protofilaments 12, 13 and 14 have been studied. The analysis was supported by numerical simulations using the Monte Carlo method. Each dipole, which represented the charge distribution within one tubulin dimer, was assumed to possess two distinct states whose projections on the MT axis were oriented in opposite directions. The two dipolar states were assumed to be (i) both parallel to the MT axis, (ii) the dipole in one of the states was assumed to be tilted with respect to the MT axis in the two-dimensional space.

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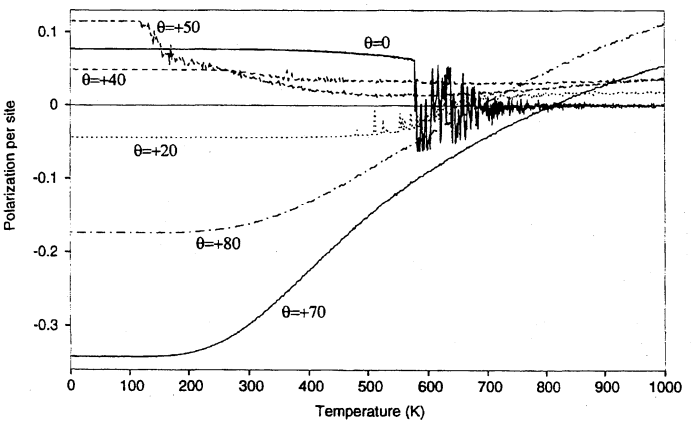
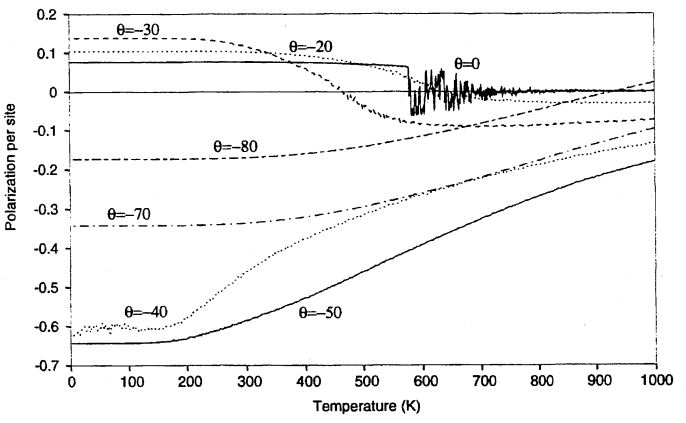


(a)



(b)

Fig. 26. Polarization per site in the 13-B lattice for selected angles θ. The calculations were started from random configurations at $T = 1000$ K. (a) "+" θ's, (b) "–" θ's.

The investigation has shown that a MT may exist in different dielectric phases at different temperatures. At low temperatures the array of dipoles can form regular patterns. The kind of the pattern depends on the type of the lattice, the number of protofilaments and the angle by which the dipole on the tubulin is tilted. With increasing temperature the regular ordering decays due to increasing thermal energy and at sufficiently high temperatures the dipoles are oriented at random. The range of temperatures over which the transition from the ordered phase to the disordered phase takes place depends on the magnitude and orientation of the dipole moment, relative permittivity of the MT and the geometrical dimensions of the MT lattice.

The dipolar ordering is also affected by the presence of external electric fields and binding of MAPs. According to the model the electric field forces the dipoles

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to point in its direction. This means that when a MT is subjected to an electric field it is more likely to be in a ferroelectric phase. The influence of MAPs was studied for three different superhelical patterns that the attached MAPs formed on a MT. We assumed that each MAP binds only to one tubulin dimer and that the associated protein either prevents the dipole on this dimer from flipping or that it neutralizes it. In the first case, naturally, the MAPs enhance the number of dipoles that point in the same direction as the dipole on the MAP tubulin. In the second case the character of the dipolar ordering is the same as in MTs that do not bind any MAPs, however, the presence of neutral tubulin dimers shifts the transition from the ordered to the disordered phase to lower temperatures. If each MAP binds to more than one tubulin molecule these effects can be expected to be greater. In both cases the numerical simulations have shown that the presence of MAPs gives rise to a number of groundstate dipolar configurations that have a more complex character than those found in MTs without MAPs.

The model can be improved by taking into account the curvature of the MT which is a more precise description than the two-dimensional view employed here (see Sec. 3). Then the orientation of the dipole would be determined by two angles in the three-dimensional space. One could also consider the dipole on the tubulin dimer to be tilted in both states by different angles. Introducing these additional parameters would provide more variability to the model. It can be noted that we have made the simplest assumption and took into account only the nearest neighbor interactions between the tubulin dimers. The interactions with the more distant dipoles could be also included but their importance decreases nearly as r^{-3}.

In this work a variety of dipolar configurations has been discussed that can exist in a two-dimensional hexagonal lattice at different temperatures for different parameters of the lattice and different orientation of one of the two dipolar states. The combination of the latter parameters, the dipole length, the charge on the dipole and the permittivity of the MT determines which of these dielectric states might exist in a MT at body temperature. The arrangement of the tubulin heterodimers in a MT is quite well known from experimental measurements. As was discussed in Sec. 3 the charge distribution within a tubulin dimer is not exactly known but it may have dipolar character due to the larger density of charges at one tubulin heterodimer end. Taking the value of the relative permitivity 20 and $q^2s^2/\epsilon_r = 12 \times 10^{-56}$ C ^{2}m^2 gives the dipole strength $qs = 15.5 \times 10^{-28}$ Cm. Then for $s = 10^{-9}$ m the charge on the dipole is about 10 electronic charges and for $s = 2 \times 10^{-9}$ m the q is about 5 electronic charges. This means that when the dipole switches its state quite a few charges have to be moving. These assumptions must hold if the states characterized by the regular patterns of dipoles should exist in MTs (see Sec. 5). It should also be pointed out that *in vivo* only a few MTs do not consist of 13 protofilaments (L. A. Amos and W. B. Amos, 1991).

As has been discussed here[18,19] the tubulin dimers carry permanent dipoles. As was mentioned above the experiments of Vassilev *et al.*[26] suggest that MTs are both electric and magnetic structures. Electromagnetic properties of MTs can be the

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basis of some of the MT functions. For example, an attractive/repulsive electrostatic interaction between the dipole on free tubulin and the electric field produced by the charges on the tubulin dimers assembled in the MT could play a role in the MT assembly/disassembly. Electromagnetic interaction could be responsible for regular patterns formed by MTs such as the mitotic spindle during cell division.[26] The dipolar states existing on the tubulin dimers coupled to tubulin elastic states may give rise to the solitary waves mentioned above. The solitary waves propagate when the dipoles in a MT form regular low temperature arrangements — ferroelectric, antiferroelectric or ferrielectric.[14] Thus according to the investigation presented in this paper the existence of such waves and the possible effects they could have on the MT behavior would depend on the parameters of the MT lattice and the tubulin dipoles. It should be pointed out that the electrostatic properties of the tubulin heterodimer provide means for the MTs to interact with cellular electric fields (including electric fields from other MTs) which may play an important role in controlling the above mentioned biological processes.

In the future using a more precise model of the charge distribution in the tubulin dimer the interactions directly between the charges could be calculated. If the charges can change positions, as we and some other authors[1–3,11,29] assume, then different electrostatic fields arise. By moving while switching their positions the charges create electric currents and consequently magnetic fields. Hence the investigation of MTs in the direction suggested in this paper could provide very useful insights into the electromagnetic nature of these cytoskeletal structures.

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PRODUCTION OF COHERENT STATES IN BIOLOGICAL SYSTEMS

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It is shown that Glauber-type coherent states exist in biological systems as a result of the broken symmetry of the free phonons. The symmetry of the system is reduced by the presence of the interaction terms in the hamiltonian. The coherent states so produced allow for the existence of giant dipoles capable of long-range interactions.

During recent years Fröhlich [1,2] has shown that under certain conditions a phenomenon resembling the well-known Bose–Einstein condensation can occur and produces important biological consequences. One of these consequences is the fact that if the quasiparticles, that have undergone condensation, are characterized by the presence of dipoles then the entire condensate behaves as a giant dipole. If this giant dipole possesses oscillatory character then Fröhlich [3] has shown that long-range (r^{-3}) potentials between two dipoles can occur. Experimentally, Rowlands and co-workers [4] have shown that human red blood cells, in the presence of metabolism and under conditions where transmembrane potentials exist, are capable of exerting long-range forces on each other that result in the production of aggregates commonly referred to as rouleaux. In order to develop a theoretical model for this phenomenon the author of this paper has utilized the idea of a giant dipole on each cell with the concomitant production of a long-range interaction [5,6]. This theory has successfully accounted for Rowlands observation. The theoretical structure is based upon the primary premise regarding the possibility of a dipolar coherent state on each cell. In his original work Fröhlich used a phenomenological rate equation for which a steady-state solution was obtained and recast into a standard Bose–Einstein-type distribution (displaying condensation and coherence) after certain assumptions have been introduced. Such an approach, immediately, became the subject of serious attack by Lifshits [7] and the occurrence of condensation was questioned. Wu and Austin [8] considered the problem from a microscopic point of view and refuted Lifshits contention. The approach adopted by Wu and Austin is in many ways similar to that of Fröhlich since a rate equation is first obtained from the microscopic hamiltonian and then a steady-state solution resembling that of Fröhlich is derived. This derivation has not, however, settled the matter since Yushina [9] repeated the calculations using essentially the Wu–Austin hamiltonian and showed that the specific and desired form of the steady-state solution, leading to Bose–Einstein condensation, does not result. The work of Yushina has been recently attacked by Mills [10–12] who has proved the existence of non-thermal excitations even under conditions where Bose condensation fails. The work presented by Mills provides a firm theoretical foundation for phenomena requiring the existence of polarization waves.

In this paper we adopt the Wu–Austin hamiltonian and proceed via an alternative route that does not depend upon a rate equation. Our approach is a group-theoretic one and is based upon the fact that the interaction terms of the hamiltonian break the phase symmetry of the free hamiltonian. This condition of broken symmetry leads to a translation of the phonon-field operators by a c-number and the production of a coherent state. Since the operators of the phonons of interest are dipolar in nature it follows that the system may be associated with a giant dipole. This resultant dipole is the outcome of the coherent state because a chaotic state would produce a vanishing average value for the dipole moment.

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The Wu–Austin hamiltonian is given by

$$H = H_0 + H_1 \,, \tag{1}$$

where

$$H_0 = \sum_1 \epsilon_1^{\mathrm{a}} a_1^{+} a_1 + \sum_1 \epsilon_1^{\mathrm{b}} b_1^{+} b_1 + \sum_1 \epsilon_1^{\mathrm{P}} P_1^{+} P_1 \,, \tag{2}$$

$$H_1 = \sum_{1,2} (\lambda b_1^{+} a_2 + \lambda^{*} b_1 a_2^{+}) + \sum_{1,2,3} (\chi a_1^{+} a_2 b_3^{+} + \chi^{*} a_1 a_2^{+} b_3) + \sum_{1,2} (\xi P_1 a_2^{+} + \xi^{*} P_1^{+} a_2) \,. \tag{3}$$

Here, the definitions are the same as those used in ref. [8] except the indices are enumerated as numbers and the energies of the phonons are given the symbol ϵ.

If we now make the transformation

$$a_i = \mathrm{e}^{\mathrm{i}\theta} a_i \tag{4}$$

(where θ is some phase) we find that H_0 is invariant to this transformation but H_1 possesses a lower symmetry. Thus the total hamiltonian H is of lower symmetry than H_0 and we do not expect the quantum statistical average $\langle\langle a_l \rangle\rangle$ to vanish with the exact density operator ρ

$$\rho = \mathrm{e}^{-\beta H}/Q \,, \tag{5}$$

where Q is the exact partition function. In the subsequent analysis we adopt the following definitions

$$\langle \text{operator} \rangle \equiv \mathrm{tr}(\mathrm{e}^{-\beta H_0}/Q_0)\,(\text{operator}), \quad \langle\langle \text{operator} \rangle\rangle \equiv \mathrm{tr}(\mathrm{e}^{-\beta H}/Q)\,(\text{operator}) \,. \tag{6,7}$$

Physically, we may view a non-vanishing $\langle\langle a_l \rangle\rangle$ as follows:

(1) We start with a collection of three types of non-interacting phonons: the system phonons, the bath phonons and the pump phonons.

(2) This collection is in complete thermal equilibrium and yields

$$\langle a_l \rangle = 0 \tag{8}$$

(3) When the interactions are switched on, via H_1, a translation of the operators occurs. This translation is given by

$$a_l = a_l + c_l \,, \tag{9}$$

with

$$c_l \equiv \langle\langle a_l \rangle\rangle. \tag{10}$$

The term translation or displacement was used by Glauber [13] because eq. (9) can be induced by a unitary operator

$$D(c_l) \equiv \exp(c_l a_l^{+} - c_l^{*} a_l) \,. \tag{11}$$

In this terminology the vacuum state is converted to an eigenstate of the operator a_l:

$$|c_l\rangle = D(c_l)|0\rangle \,, \tag{12}$$

where

$$a_l |c_l\rangle = c_l |c_l\rangle \,. \tag{13}$$

States such as $|c_l\rangle$ are called *coherent states.* If we assume that the dipole moment μ_l of the phonons are related to the coordinate q_l

$$\mu_l \propto q_l \propto (a_l^+ + a_l) \tag{14}$$

then in the coherent state

$$\langle \mu_l \rangle \neq 0 \,. \tag{15}$$

In this description the macroscopic dipole moments $\langle \mu_l \rangle$ arise quite naturally and the rest of the theoretical arguments, requiring the existence of these dipole moments, can be carried out. The translation phenomenon indicated by eq. (9) has been recognized as a generalized criterion for Bose–Einstein condensation by other workers (see for example Umezawa [14]). The remainder of this paper will be devoted to the explicit computation of c_l.

From eqs. (7) and (6) along with the definition of H we have

$$c_l = Q^{-1} \operatorname{tr}\{\exp[-\beta(H_0 + H_1)]\, a_l\} = \langle \mathcal{S}(\beta) a_l \rangle_{\text{connected}} \,, \tag{16}$$

where the finite temperature expansion of the density operator is utilized and by the retention of only the connected Feynman diagrams the necessity of computing the exact partition function Q is eliminated. This simplification is the result of the well-knonw Wick theorem (see for example Paul [15]). The operator $\mathcal{S}(\beta)$, appearing in eq. (6), is defined by

$$\mathcal{S}(\beta) = \sum_{n=0}^{\infty} \left(\frac{-1}{\hbar}\right)^n \frac{1}{n!} \int_0^{\beta\hbar} \mathrm{d}u_1 \mathrm{d}u_2 \ldots \mathrm{d}u_n \, \mathrm{T} H_1(u_1) H_1(u_2) \ldots H_1(u_n) \,. \tag{17}$$

Here T is the well-known Wick time ordering operator and u_i are imaginary time variables that are defined within the limits $0 \leqslant u_i \leqslant \beta\hbar$. In order to facilitate a diagramatic development it is convenient to introduce an imaginary time variable u in the operator in a_l in eq. (16). This allows us to calculate $c_l(u)$ from which c_l can be obtained by setting $u = 0$ at the end of the computation.

If we examine the interaction hamiltonian given in eq. (3) we notice the presence of six vertices which are displayed in fig. 1 and the meaning of arrows are given in the following rules:

(1) Phonons with operators a_i, a_i^+ are denoted by a single arrow and since these phonons are dipolar in nature they will be called "active" phonons.

(2) Bath phonons (b_i, b_i^+) are given two arrows.

(3) Pump phonons (P_i, P_i^+) are given three arrows.

(4) All diagrams will contain an external condensate point which carries the label l. This point will be designated by a square $\square_l$.

(5) All zero-order Green's (G_{ij}^{a}, G_{ij}^{b} and G_{ij}^{P}) functions will be indicated by thin lines carrying the appropriate numbers of arrows as indicated by rules (1), (2) and (3).

(6) All renormalized Green's functions ($\bar{G}_{ij}^{\mathrm{a}}$, $\bar{G}_{ij}^{\mathrm{b}}$ and $\bar{G}_{ij}^{\mathrm{P}}$)will be indicated by heavy lines with the appropriate numbers of arrows as indicated by rules (1), (2) and (3).

The simplest renormalized diagrams that appear in the present theory are shown in fig. 2. It is our main objective, in this paper, to demonstrate the existence of a nonvanishing field translation given by the topologically dis-

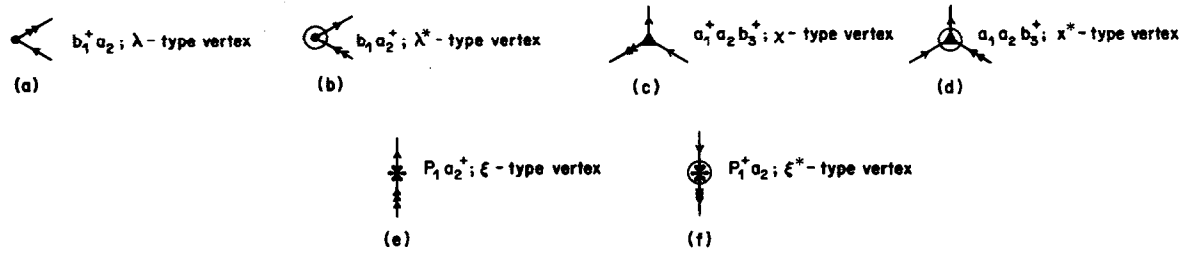

Fig. 1.

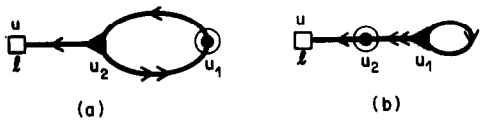

Fig. 2.

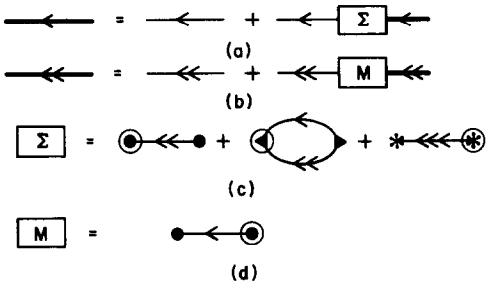

Fig. 3.

tinct diagrams (2a) and (2b). It is sufficient to show a non-vanishing result from either of these diagrams since possible cancellation amongst such topologically distinct diagrams is very unlikely.

For the renormalization procedure we adopt the diagrams displayed in fig. 3, where we use the symbol Σ to designate the self-energies of the active phonons and the symbol M for the self-energies of the bath phonons. The actual computations are done, most conveniently, in a frequency space which is related to the imaginary time space by the following equation:

$$f(u) = \frac{1}{\beta\hbar} \sum_{n=-\infty}^{\infty} f(\omega_n) \exp(-i\omega_n u) , \tag{18}$$

where the discrete frequencies ω_n are given for the Bose operators by

$$\omega_n = 2\pi n/\beta\hbar . \tag{19}$$

The zero-order Green's function (quantities corresponding to the thin lines) for the three kinds of phonons are given by

$$G^s_{ij}(\omega_n) = (i\omega_n - \hbar^{-1}\epsilon^s_i)^{-1}\,\delta_{ij} \quad (s = \text{a, b or P}) . \tag{20}$$

With this definition of the zero-order Green's functions and employing the diagrams of fig. 3 we obtain

$$\bar{G}^a_{i,j}(\omega_m) = \delta_{ij}/[i\omega_m - \hbar^{-1}\epsilon^a_i - \Sigma(\omega_m)] , \quad \bar{G}^b_{i,j}(\omega_m) = \delta_{ij}/[i\omega_m - \hbar^{-1}\epsilon^b_i - M(\omega_m)] . \tag{21,22}$$

The self-energies are given by

$$\Sigma(\omega_m) = \frac{|\lambda|^2}{2\hbar^2} \sum_2 \frac{1}{i\omega_m - \hbar^{-1}\epsilon^b_2} - |\chi|^2 \sum_{2,3} \frac{f(\epsilon^a_2) - f(-\epsilon^b_3)}{i\omega_m + \hbar^{-1}(\epsilon^a_2 + \epsilon^b_3)} + |\xi|^2 \sum_2 \frac{1}{i\omega_m - \hbar^{-1}\epsilon^P_2} , \tag{23}$$

$$M(\omega_m) = \frac{|\lambda|^2}{2\hbar^2} \sum_2 \frac{1}{i\omega_m - \hbar^{-1}\epsilon^a_2} , \quad f(\epsilon^s_d) \equiv [\exp(\beta\epsilon^s_j) - 1]^{-1} . \tag{24,25}$$

If eqs. (21) and (22) are substituted for the renormalized Green's functions in fig. 2b then the field displacement in frequency space becomes

$$c_l(\omega_m) = \frac{\chi\lambda^*}{2\hbar^2[\hbar^{-1}\epsilon^a_l + \Sigma(0)]} \sum_{2,4} \frac{1}{i\omega_m - \hbar^{-1}\epsilon^a_2 - \Sigma(\omega_m)} \frac{1}{i\omega_m - \hbar^{-1}\epsilon^b_4 - M(\omega_m)} . \tag{26}$$

From this equation the static-field displacement can be obtained by first transforming into an imaginary time space v and then allowing $u \to 0$:

$$c_l = \lim_{u\to 0} \frac{1}{\beta\hbar} \sum_m c_l(\omega_m) \exp(-i\omega_m u)$$

$$= \frac{\chi\lambda^*}{2\beta\hbar^3(\hbar^{-1}\epsilon_l^a + \Sigma)} \sum_{2,4,m} \frac{1}{i\omega_m - \hbar^{-1}\epsilon_2^a - \Sigma(\omega_m)} \frac{1}{i\omega_m - \hbar^{-1}\epsilon_4^b - M(\omega_m)} . \tag{27}$$

Here $\Sigma \equiv \Sigma(0)$. The frequency summation over m can be carried out by integrating in the complex plane if the frequency dependence of the self-energies $\Sigma(\omega_m)$ and $M(\omega_m)$ can be neglected. To carry out this process we define the spectral functions for the two self-energies appearing in eq. (27) as follows:

$$\Sigma(z) = \frac{1}{\pi} \int_{-\infty}^{\infty} d\omega' \frac{\Gamma_a(\omega')}{\omega' - z} , \quad M(z) = \frac{1}{\pi} \int_{-\infty}^{\infty} d\omega' \frac{\Gamma_b(\omega')}{\omega' - z} . \tag{28,29}$$

Here, Γ_a and Γ_b are the spectral functions for the two self-energies and in ordinary spectroscopy these quantities play the role of spectroscopic line broadening. The variable z is a complex frequency that represents an analytic continuation of the discrete frequencies ω_m over the entire complex plane. Just above and below the real frequency axes we obtain

$$\Sigma(\omega \pm i\eta) = \frac{1}{\pi} \int_{-\infty}^{\infty} d\omega' \frac{\Gamma_a(\omega')}{\omega' - \omega \mp i\eta} = P\frac{1}{\pi} \int_{-\infty}^{\infty} d\omega' \frac{\Gamma_a(\omega')}{\omega' - \omega} \pm i\Gamma_a(\omega) \equiv S_a(\omega) \pm i\Gamma_a(\omega) . \tag{30}$$

Similarly

$$M(\omega \pm i\eta) \equiv S_b(\omega) \pm i\Gamma_b(\omega) . \tag{31}$$

Here ω is a real frequency and will be treated as a constant in the frequency summation involved in eq. (27). The actual computation of the function $S_s(\omega)$ and $\Gamma_s(\omega)$ (with s = a, b) involves summations shown in eqs. (24) and (25). These summations over the phonon states cannot be carried out exactly and require numerical work. In this paper we do not propose to give actual numerical values for the field translations. It is our purpose to show that such a translation does occur and to display its mathematical form, therefore the actual computation of the frequency shift (S_s) and broadening will not be carried out. If we define the complex frequencies of the dressed phonons by ν_i^a and ν_j^b where

$$\nu_i^a \equiv \hbar^{-1}\epsilon_i^a + S_a \pm i\Gamma_a , \quad \nu_j^b \equiv \hbar^{-1}\epsilon_j^b + S_b \pm i\Gamma_b , \tag{32,33}$$

then the explicit summation over m in eq. (27) yields

$$c_l = \frac{\chi\lambda^*}{2\hbar^2\nu_1^a} \sum_{2,4} \frac{f(\hbar\nu_2^a) - f(\hbar\nu_4^b)}{\nu_4^b - \nu_2^a} . \tag{34}$$

This equation shows that as long as the coupling constants χ and λ^* are present there will always be a finite field translation. The magnitude of this translation is inversely proportional to the frequency of the dressed mode l ($c_l \propto 1/\upsilon_l^a$), this implies that the low-energy modes undergo a relatively more severe distortion or the main contribution to the condensate density arises from small values of l. If we compare our result given by eq. (34) with the result of Mills then at a superficial level they appear contradictory. This is not, however, the case. The main difference arises from the fact that eq. (34) does not explicitly contain the coupling constant ξ (or ξ^*) corresponding to the pump and in Mills work it is precisely the energy pump that produces the non-thermal excitations. Moreover the importance of the pump has been well established by various experimental measurements. The apparent contradic-

tion disappears when it is realized that the present result is a group-theoretical one and is a direct consequence of the symmetry properties of the hamiltonian given by eq. (2). It is easy to see that part of the symmetry breaking is induced in the hamiltonian by the linear coupling with the bath through the $\lambda(\lambda^*)$ dependent terms and the form of these terms is the same as that given by the pump-dependent terms. From a purely mathematical standpoint the inclusions of the symmetry-breaking terms in the hamiltonian have already biased the problems towards coherence. In the treatment based upon the rate equation further physical assumptions (not contained in the hamiltonian or the Heisenberg's equations of motion) are introduced and these give the pump the special significance it finally achieves. These additional physical assumptions are perfectly reasonable and the results obtained are correct, however, the work of this paper is meant to augment the phenomenon of coherence given by Mills and not to negate it.

If the index l is taken to be the phonon propagation vector ($\boldsymbol{k}$) then, in our theory, $\boldsymbol{k}$ is the propagation vector of the dressed phonons and the condensate will be composed, predominantly, of modes with $|\boldsymbol{k}| \to 0$. With this description of the index l we may define the field operators:

$$\psi(r) = \sum_{k} a_k \exp(\mathrm{i}\boldsymbol{k}\cdot r) + \sum_{k} c_k \exp(\mathrm{i}\boldsymbol{k}\cdot r) . \tag{35}$$

The density matrix becomes

$$\rho(r, r) = \langle \psi^+(r)\phi(r') \rangle = \sum_{kk'} \langle a_k^+ a_{k'} \rangle \exp(-\mathrm{i}\boldsymbol{k}\cdot r + \mathrm{i}\boldsymbol{k}'\cdot r') + \sum_{kk'} c_k^* c_{k'} \exp(-\mathrm{i}\boldsymbol{k}\cdot r + \boldsymbol{k}'\cdot r')$$

$$= \sum_{k} n_k \exp[\mathrm{i}\boldsymbol{k}\cdot(r-r')] + \sum_{kk'} c_k^* c_{k'} \exp(-\mathrm{i}\boldsymbol{k}\cdot r + \mathrm{i}\boldsymbol{k}'\cdot r') . \tag{36}$$

The first term of this expression describes the normal (uncondensed) phonons in the system and vanishes when $|r - r'| \to \infty$ due to the oscillatory nature of the exponential and produces no long-range order. The second term clearly does not display this feature and corresponds to the well-known phenomenon of off-diagonal long-range order (ODLRO) that has been observed in other branches of physics [16].

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QED-CAVITY MODEL OF MICROTUBULES IMPLIES DISSIPATIONLESS ENERGY TRANSFER AND BIOLOGICAL QUANTUM TELEPORTATION

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We refine a QED-cavity model of microtubules (MTs), proposed earlier by two of the authors (N.E.M. and D.V.N.), and suggest mechanisms for the formation of biomolecular mesoscopic coherent and/or entangled quantum states, which may avoid decoherence for times comparable to biological characteristic times. This refined model predicts dissipationless energy transfer along such "shielded" macromolecules near room temperatures as well as quantum teleportation of states across MTs and perhaps neurons.

Keywords: Microtubules; cavities; QED; quantum coherence; entanglement; biological quantum computation.

1. Introduction

Observable quantum effects in biological matter such as proteins are naively expected to be strongly suppressed, mainly due to the macroscopic nature of most biological entities as well as the fact that such systems live near room temperature. These conditions normally result in a very fast *collapse* of the pertinent wave-functions to one of the allowed classical states. However, we suggest that under certain circumstances it is in principle possible to obtain the necessary

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isolation against thermal losses and other environmental interactions, so that *meso- and macroscopic* quantum-mechanical coherence, and conceivably entanglement extending over scales that are considerably larger than the atomic scale, may be achieved and maintained for times comparable to the characteristic times for biomolecular and cellular processes.

In particular, we have shown[1] how microtubules (MTs)[2] can be treated as quantum-mechanically isolated (QED) cavities, exhibiting properties analogous to those of electromagnetic cavities routinely used in quantum optics.[3–6] Recently, our speculative model has been supported by some indirect experimental evidence. It has been experimentally shown,[7] that it is possible to maintain partial entanglement of the bulk spin of a macroscopic quantity of Caesium (Cs) atoms ($N = 10^{12}$), *at room temperature*, for a relatively long time (0.5 ms). Note that in this experiment, the large quantity of atoms was of paramount importance in creating and maintaining the entanglement, and even though the gas samples were in constant contact with the environmental heat bath, by using a careful experimental arrangement, Julsgaard *et al.* managed to detect the existence of entanglement for a much longer time than one would intuitively expect. Here, we outline the main features of our model of the quantum mechanical properties of MTs (described in detail in Ref. 1), and we exhibit the relevance of the Julsgaard *et al.* experiment to our model. A direct consequence of our model for MTs as QED cavities is that virtually every experimentally known QED-cavity-based observation may have an analogue in living MTs and we show this analytically with the specific case of intra- and inter-cellular dissipationless energy transfer and quantum *teleportation* of coherent quantum states.

Dissipationless energy would account for minimal energy in cancer

Energy transfer across cells, without dissipation, had been first speculated to occur in biological matter by Fröhlich.[8] The phenomenon conjectured by Fröhlich was based on a one-dimensional superconductivity model: a one-dimensional electron system with holes, where the formation of solitonic structures due to electron-hole pairing results in the transfer of electric current without dissipation. Fröhlich suggested that, if appropriate solitonic configurations are formed inside cells, energy in biological matter could also be transferred without any dissipation (supereefficiently). This idea has lead theorists to construct various models for cellular energy transfer, based on the formation of kink classical solutions.[9]

In these early works no specific microscopic models had been considered.[9] In 1993 Sataric *et al.* constructed a *classical* physics model for microtubule dynamics,[10] in which solitons transfer energy across MTs without dissipation. In the past, we have considered the *quantum aspects* of this one-dimensional model, and developed a framework for consistent quantization of the solitonic solutions.[1] Our work suggested that such semiclassical solutions may emerge as a result of "decoherence" due to environmental interactions, in agreement with ideas in Ref. 11.

The basic assumption used in creating the model of Ref. 1 was that the building blocks of MTs, the tubulin molecule dimers, can be treated as elements of Ising spin chains (one-space-dimensional structures). The interaction of each tubulin chain

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(protofilament) with neighboring chains and the surrounding water environment has been accounted for by suitable potential terms in the one-dimensional Hamiltonian. The model describing the dynamics of such one-dimensional sub-structures was the ferroelectric distortive spin chain model of Ref. 10.

Ferroelecricity is an essential ingredient of the quantum-mechanical mechanism of energy transfer that we propose. We have speculated[12] that the ferroelectric nature of MTs will be that of *hydrated* ferroelectrics, i.e. the ordering of the electric dipole moment of the tubulin molecules will be due to the interaction of the tubulin dimers' electric dipoles with the *ordered water* molecules in the interior and possibly exterior of the microtubular cavities. Ferroelectricity induces a dynamical dielectric "constant" $\varepsilon(\omega)$ which is dependent on the frequency ω of the excitations in the medium. Below a certain frequency, such materials are characterized by almost vanishing dynamical dielectric "constants", which in turn implies that electrostatic interactions inversely proportional to ε will be enhanced, and thus become dominant against thermal losses. In the case of microtubules, the pertinent interactions are of the electric dipole type, scaling as $1/(\varepsilon r^3)$ with the distance r. For ordinary water media, the dielectric constant is of order 80. In the ferroelectric regime however, this ε is diminished significantly. As a result, the electric dipole-dipole interactions may overcome the thermal losses proportional to $k_{\mathrm{B}}T$ at room temperature inside the interior cylindrical region of MT bounded by the dimer walls of thickness of order of a few Angstroms.[1] The situation is depicted in Fig. 1.

It is known experimentally,[13] that in a thin exterior neighborhood of MTs there are areas of atomic thickness, consisting of charged ions, which isolate the MT from thermal losses. This means that the electrostatic interaction overcome thermal agitation. It seems theoretically plausible, albeit yet unverified, that such thermally isolated exterior areas can also operate as *cavity regions*, in a manner similar to the areas interior to MTs. At this point it is unclear whether there exists the necessary coherent dipole quanta in the ionic areas. Further experimental and theoretical (simulation) work needs to be done regarding this issue and this is in progress.

Once such an isolation is provided, one can treat the thin interior regions as electromagnetic cavities in a way similar to that of QED cavities.[a] QED cavities are well-known for their capacity to sustain in their interior coherent modes of electromagnetic radiation. Similarly, one expects that such coherent cavity modes will occur in the thin interior regions of MTs bounded by the protein dimer walls. Indeed, as we discussed in Ref. 1, these modes are provided by the interaction of the electric dipole moments of the *ordered water* molecules in the interior of MT with the quantized electromagnetic radiation.[15,16] Such coherent modes are termed *dipole quanta*. It is the interaction of such cavity modes with the electric dipole excitations of the dimers that leads to the formation of coherent (dipole) states on

[a]Note that the role of MT as *waveguides* has already been proposed by S. Hameroff some time ago.[14] On the other hand, we are interested in isolated regions inside the MT which play the role of *QED cavities*.

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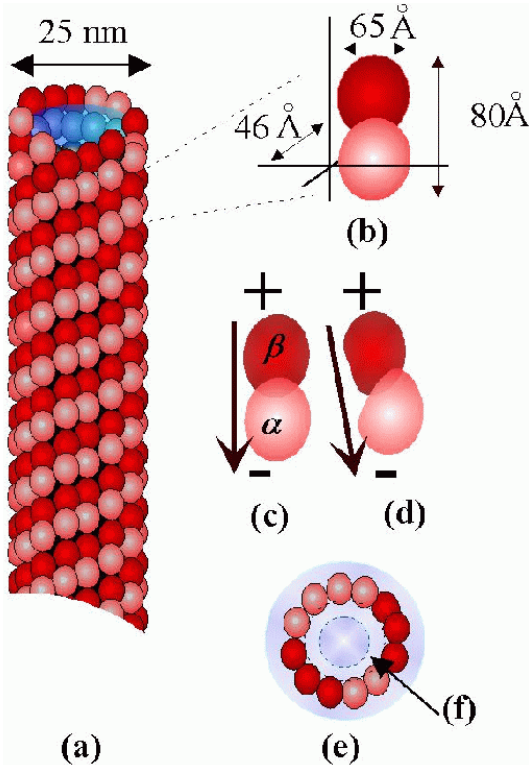

Fig. 1. (a) Typical microtubule made of 13 tubulin protofilaments. Each protofilament is displaced vertically from its neighbor a fifth of the dimer vertical height. (b) Dimensions of the heterodimer as solved by electron crystallography of zinc-induced sheets[19] note that an MT consists of (c) GTP-tubulin and (d) GDP-tubulin. Arrows indicate the direction of the electric dipole moment for the two conformations. (e) A cross section of an MT showing the surrounding water layer. (f) The thin interior layer that is to be treated as a QED cavity.

the tubulin dimer walls of MTs. A review of how this can happen, and what *purely quantum* effects can emerge from the QED nature of MTs, will be the main topic of this communication.

In Sec. 2 we present a concise exposé of our mechanism[1] that justifies the application of quantum physics to the treatment of certain aspects of MT dynamics. We also develop an analogy of our mechanism to the experimental setup used by Julsgaard *et al.* In Sec. 3 we present a straight forward calculation of how quantum teleportation of states can occur in MTs, in direct analogy to the suggested experimental quantum teleportation in optical cavities that has been observed recently.[17,18] We also draw a parallel between certain geometrical features of MTs such as their ordered structure which obeys a potentially information-encoding code and suggest how this can be exploited for (quantum) error-correction and dense coding. Section 4 contains our conclusions and outlook.

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2. Quantum Coherence in Biological Matter?

2.1. *Tubulin, microtubules and coherent states*

Tubulin is a common polar protein found mainly in the cytoskeleton of eukariotic cells which is especially enriched in brain tissue. Many of its properties have been studied both experimentally and theoretically because of its importance in mitosis, its role as the building block of microtubules and its relevance to several diseases including Cancer and Alzheimer's. Measurements[19] have confirmed earlier data suggesting that the tubulin heterodimer has dimensions $46 \times 80 \times 65$ Angstrom [Fig. 1(b)]. Under normal physiological conditions, tubulin is found as a heterodimer, consisting of two nearly identical monomers called the α- and β-tubulin each of molecular weight of about 55 kDalton.[19] MTs are hollow [25 nm-outer diameter, 14 nm inner diameter — see Fig. 1(a)] tubes forming the main component of the cytoskeleton and apart from giving shape and support to the cell, they also play a major role in cellular transport and have been hypothesized to be central in cellular information processing. The interior of the MT, seems to contain *ordered water* molecules,[1] which implies the existence of an electric dipole moment and an electric field. We stress that the intracellular *ordered water* which is full of proteins and other molecules is different from ordinary water in various respects, e.g. as is implied in Ref. 20. It is to be understood that unless otherwise specified, from now on by "tubulin" we refer to the $\alpha\beta$-dimer. Free tubulin can self-assemble into MTs both *in vivo* and *in vitro* where the most common arrangement of the tubulin dimers is such that, if one treats them as points, they resemble triangular lattices on the MT surface. The β-tubulin monomer of the heterodimer can bind guanosine $5'$ triphosphate (GTP) in which case it exists in an energy-rich form that favors polymerization, or it can bind guanosine $5'$ diphosphate (GDP-tubulin) thus being in an energy-poor form (GDP-tubulin) that favors dissociation. The structure of MTs has been the subject of comprehensive study and it transpires that MTs come in a variety of arrangements the predominant of which is a 5-start, period-13 helical tube of dimers which resembles a corn ear [Fig. 1(a)] made out of 13 offset protofilaments. Certain interesting phenomena arise during the (de-)polymerization of tubulin such as length oscillations, treadmilling, etc. are generally referred to as "dynamic instability" and these have been studied extensively[21,22] but are not directly relevant to our analysis at this stage as such phenomena are suppressed in the remarkably stable axonal neural cytoskeleton. It has been shown that the GDP-GTP exchange (hydrolysis) releases approximately 0.42 eV per molecule and is accompanied by a conformational change.[23] This change has been modelled as resulting in a $27°42'$ angle[24] between the original line connecting the centers of the α and β monomers and the new center-to-center line [Fig. 1(d)]. As a result of this change in the geometry of the tubulin molecule, the orientation of the electric dipole moment also changes magnitude and direction. It has been put forward that each dimer has two hydrophobic protein pockets, containing 2×18 unpaired electrons[2] that have at least two possible configurations associated with the GTP and GDP

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states of tubulin, which we will call ↑ and ↓ electron (or equivalently electric dipole moment) conformations respectively.

Using the fact that a typical "distance" for the transition between the ↑ and ↓ conformations is of the order of the distance between the two hydrophobic dimer pockets, i.e. $\mathcal{O}(4\,\text{nm})$, a simplistic estimate of the free tubulin electric dipole moment d can be obtained based on a mobile charge of 36 electrons multiplied by this separation of 4 nm, giving a magnitude of $d = 2.3\times10^{-26}$ C · m (or 700 Debye) while using a more sophisticated molecular simulation, d has been quoted at 1714 Debye.[25] It is evident that an experimentally determined electric dipole moment for the tubulin molecule and its dynamics are important areas of study that have to be undertaken if these studies are to move forward.

If we account for the effect of the water environment that screens the electric charge of the dimers by the relative dielectric constant of the water, which is $\varepsilon/\varepsilon_0 \sim 80$, we arrive at a value of

$$d_{\text{dimer}} \sim 3 \times 10^{-28}\ \text{C} \cdot \text{m} \tag{1}$$

Note that under physiological conditions, the unpaired electric charges in the dimer may lead to even further suppression of d_{dimer} (1).

At physiological pH (= 7.2) MTs are negatively charged[26] due to the presence of a 15-residue carboxyl-terminus "tail" and there have been suggestions that this C-terminus is important in polymerization, protein interactions and perhaps charge conduction.[13] This terminus has not been included in the electron crystallography data of Nogales and Downing[19] so all values concerning the dipole moment are quoted with the understanding that d has been calculated ignoring the effect of the C-terminus. It is also known that at pH 5.6 MTs become neutral. Finally, there have been some preliminary experiments aimed at measuring the electric field around MTs[27–29] indicating that MTs could be ferroelectric, as we have suggested in our model.[1,12] Note that although the "caps" of the MT contain both GTP and GDP tubulin, it is well-known experimentally[24] that the tubulin comprising the "trunk" of the MT is GDP-tubulin incapable of acquiring a phosphate and becoming GTP tubulin. However, this does not preclude electric-dipole moment flip wave propagation down the MT, as a flip at the cap can be propagated without phosphorylation or hyrdolysis but rather via the mechanism suggested below. In view of this, the value of the yet undetermined electric dipole moment direction flip angle θ_{flip} is much smaller than the $27°42'$ value for free tubulin. The ↑ and ↓ states still exist though, but are hard to observe experimentally as they are not associated with a large-scale geometrical mass shift. Note that virtually all of the alternative suggested MT-based "quantum brain" hypotheses today, fail to take this into account and instead wrongly suggest that θ_{flip} is of the order of 27° and that such large distortions occur in the trunk of the polymerized MT.

In standard models for the simulation of MT dynamics,[10] the physical degree of freedom which is relevant for the description of energy transfer is the projection of the electric dipole moment on the longitudinal symmetry axis (x-axis) of the MT

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cylinder. The θ_{flip} distortion of the $\downarrow$-conformation leads to a displacement u_n along the x-axis. This way, the effective system is one-dimensional (spatial), and one has the possibility of being quantum integrable.[1]

It has been suggested for quite some time that information processing via interactions among the MT protofilament chains can be sustained on such a system, if the system is considered as a series of interacting Ising chains on a triangular lattice. For such schemes to work, one must first show that the electromagnetic interaction among the tubulin dimers are strong enough to overcome thermal noise. It is due to this problem that such models for intra-neuronal information processing have been critisized as unphysical.[30] We shall return to this issue later. Classically, the various dimers can only be in the $\uparrow$ and $\downarrow$ conformations. Each dimer is influenced by the neighboring dimers resulting in the possibility of a transition. This is the basis for classical information processing, which constitutes the picture of a (classical) cellular automaton.

If we assume (and there is good theoretical basis for such an assumption[1]) that each dimer can in fact find itself in a superposition of $\uparrow$ and $\downarrow$ states, a *quantum nature* results. Tubulin can then be viewed as a typical *two-state* quantum mechanical system, where the dimers couple to conformational changes with $10^{-9} - 10^{-11}$ sec transitions, corresponding to an angular frequency $\omega \sim \mathcal{O}(10^{10}) - \mathcal{O}(10^{12})$ Hz. In the present work we assume the upper bound of this frequency range (in order of magnitude) to represent the characteristic frequency of the dimers, viewed as a two-state quantum-mechanical system:

$$\omega_0 \sim \mathcal{O}(10^{12}) \text{ Hz} \tag{2}$$

As we shall see below, such a frequency range is not unusual in biology.

Let u_n be the displacement field of the nth dimer in a MT chain. The continuous approximation proves sufficient for the study of phenomena associated with energy transfer in biological cells, and this implies that one can make the replacement

$$u_n \to u(x,t) \tag{3}$$

with x a spatial coordinate along the longitudinal symmetry axis of the MT. There is a time variable t due to fluctuations of the displacements $u(x)$ as a result of the dipole oscillations in the dimers.

The effects of the neighboring dimers (including neighboring chains) can be phenomenologically accounted for by an effective potential $V(u)$. In the model of Ref. 10 a double-well potential was used, leading to a classical kink solution for the $u(x,t)$ field. More complicated interactions are allowed in the picture of Ref. 1, where we have considered more generic polynomial potentials.

The effects of the surrounding water molecules can be accounted for by a viscous force term that damps out the dimer oscillations,

$$F = -\gamma \partial_t u \tag{4}$$

with γ determined phenomenologically at this stage. This friction should be viewed as an environmental effect, which however does not lead to energy dissipation, as

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a result of the nontrivial solitonic structure of the ground-state and the nonzero constant force due to the electric field. This is a well-known result, directly relevant to energy transfer in biological systems.[9]

The effective equation of motion for the relevant field degree of freedom $u(x,t)$ reads:

$$u''(\xi) + \rho u'(\xi) = P(u) \tag{5}$$

where $\xi = x - vt$, v is the velocity of the soliton, $\rho \propto \gamma$ in Ref. 10, and $P(u)$ is a polynomial in u, of a certain degree, stemming from the variations of the potential $V(u)$ describing interactions among the MT chains.[1] In the mathematical literature[31] there has been a classification of solutions of equations of this form. For certain forms of the potential[1] the solutions include *kink solitons* that may be responsible for dissipation-free energy transfer in biological cells[9]:

$$u(x,t) \sim c_1(\tanh[c_2(x-vt)] + c_3) \tag{6}$$

where c_1, c_2, c_3 are constants depending on the parameters of the dimer lattice model. For the form of the potential assumed in the model of Ref. 10 there are solitons of the form

$$u(x,t) = c_1' + \frac{c_2' - c_1'}{1 + e^{c_3'(c_2'-c_1')(x-vt)}}\,,$$

where again $c_i', i = 1, \ldots, 3$ are appropriate constants.

A semiclassical quantization of such solitonic states has been considered in Ref. 1. The result of such a quantization yields a modified soliton equation for the (quantum corrected) field $u_q(x,t)$[32]

$$\partial_t^2 u_q(x,t) - \partial_x^2 u_q(x,t) + \mathcal{M}^{(1)}[u_q(x,t)] = 0 \tag{7}$$

with the notation

$$M^{(n)} = e^{\frac{1}{2}(G(x,x,t) - G_0(x,x))\frac{\partial^2}{\partial z^2}} U^{(n)}(z)|_{z=u_q(x,t)} \quad \text{and} \quad U^{(n)} \equiv d^n U/dz^n\,.$$

The quantity U denotes the potential of the original soliton Hamiltonian, and $G(x,y,t)$ is a bilocal field that describes quantum corrections due to the modified boson field around the soliton. The quantities $M^{(n)}$ carry information about the quantum corrections. For the kink soliton (6) the quantum corrections (7) have been calculated explicitly in Ref. 32, thereby providing us with a concrete example of a large-scale quantum coherent state.

A typical propagation velocity of the kink solitons (e.g. in the model of Ref. 10) is $v \sim 2$ m/sec, although, models with $v \sim 20$ m/sec have also been considered.[33] This implies that, for moderately long microtubules of length $L \sim 10^{-6}$ m, such kinks transport energy without dissipation in

$$t_F \sim 5 \times 10^{-7}\ \text{sec}\,. \tag{8}$$

Energy will be transferred super-efficiently via this mechanism only if the decoherence time is of the order of, or longer than, this time. We shall see in fact that

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indeed such time scales are comparable to, or smaller in magnitude than, the decoherence time scale of the coherent (solitonic) states $u_q(x,t)$. This then implies that fundamental quantum mechanical phenomena may be responsible for frictionless, dissipantionless super-efficient energy (and signal) transfer and/or transduction across microtubular networks in the cell.

2.2. *Microtubules as cavities*

In Ref. 1 we have presented a microscopic analysis of the physics underlying the interaction of the water molecules with the dimers of the MT, which is responsible for providing the friction term (4) in the effective (continuum) description. Below we briefly review this scenario.

As a result of the ordered structure of the water environment in the interior of MTs, there appear *collective* coherent modes, the so-called dipole quanta.[15] These arise from the interaction of the electric dipole moment of the water molecule with the quantized radiation of the electromagnetic field,[16] which may be self-generated in the case of MT arrangements.[1,33] Such coherent modes play the role of "cavity modes" in the quantum optics terminology. These in turn interact with the dimer structures, mainly through the unpaired electrons of the dimers, leading to the formation of a quantum coherent solitonic state that may extend even over the entire MT network. As mentioned above, such states may be identified[1] with semiclassical solutions of the friction equation (5). These coherent, almost classical, states should be viewed as the result of the *decoherence* of the dimer system due to its interaction/coupling with the water environment.[11]

Such a dimer/water coupling can lead to a situation analogous to that of atoms interacting with coherent modes of electromagnetic radiation in *quantum optical cavities*, namely to the so-called *Vacuum-Field Rabi Splitting* (VFRS) effect.[4] VFRS appears in both the emission and absorption spectra of atoms[5] in interaction with a coherent mode of electromagnetic radiation in a cavity. For our purposes below, we shall review the phenomenon by restricting ourselves for definiteness to the absorption spectra case.

Consider a collection of N atoms of characteristic frequency ω_0 inside an electromagnetic cavity. Injecting a pulse of of frequency Ω into the cavity causes a doublet structure (splitting) in the absorption spectrum of the atom-cavity system with peaks at:

$$\Omega = \omega_0 - \frac{\Delta}{2} \pm \frac{1}{2}(\Delta^2 + 4N\lambda^2)^{1/2} \tag{9}$$

where $\Delta = \omega_c - \omega_0$ is the detuning of the cavity mode, of frequency ω_c, compared to the atomic frequency. For resonant cavities the splitting occurs with equal weights

$$\Omega = \omega_0 \pm \lambda\sqrt{N}\,. \tag{10}$$

Notice here the *enhancement* of the effect for multi-atom systems $N \gg 1$. The quantity $2\lambda\sqrt{N}$ is called the "Rabi frequency".[4] From the emission-spectrum

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analysis an estimate of λ can be inferred which involves the matrix element, $\underline{d}$, of atomic electric dipole between the energy states of the two-level atom[4]:

$$\lambda = \frac{E_c \underline{d} \cdot \underline{\epsilon}}{\hbar} \tag{11}$$

where $\underline{\epsilon}$ is the cavity (radiation) mode polarization, and

$$E_c \sim \left(\frac{2\pi\hbar\omega_c}{\varepsilon V}\right)^{1/2} \tag{12}$$

is the r.m.s. vacuum (electric) field amplitude at the center of a cavity of volume V, and of frequency ω_c, with ε the dielectric constant of the medium inside the volume V. In atomic physics, the VFRS effect has been confirmed by experiments involving beams of Rydberg atoms resonantly coupled to superconducting cavities.[6]

In the analogy between the thin cavity regions near the dimer walls of MTs with electromagnetic cavities, the role of atoms in this case is played by the unpaired two-state electrons of the tubulin dimers[1] oscillating with a frequency (2). To estimate the Rabi coupling between cavity modes and dimer oscillations, one should use (11) for the MT case.

We have used some simplified models for the *ordered water* molecules, which yield a frequency of the coherent dipole quanta ("cavity" modes) of order[1]:

$$\omega_c \sim 6 \times 10^{12}\ \mathrm{s}^{-1} \tag{13}$$

Notably this is of the same order of magnitude as the characteristic frequency of the dimers (2), implying that the dominant cavity mode and the dimer system are almost in resonance in our model of Ref. 1. Note that this is a feature shared by atomic physics systems in cavities, and thus we can apply the pertinent formalism to our system. Assuming a relative dielectric constant of water w.r.t to that of vacuum ϵ_0, $\varepsilon/\varepsilon_0 \sim 80$, one obtains from (12) for the case of MT cavities:

$$E_c \sim 10^4\ \mathrm{V/m}\,. \tag{14}$$

Electric fields of such a magnitude can be provided by the electromagnetic interactions of the MT dimer chains, the latter viewed as giant electric dipoles.[10] This suggests that the coherent modes ω_c, which in our scenario interact with the unpaired electric charges of the dimers and produce the kink solitons along the chains, owe their existence to the (quantized) electromagnetic interactions of the dimers themselves.

The Rabi coupling for the MT case then is estimated from (11) to be of order:

$$\text{Rabi coupling for MT} \equiv \lambda_{\mathrm{MT}} = \sqrt{\mathcal{N}}\lambda_0 \sim 3 \times 10^{11}\ \mathrm{s}^{-1} \tag{15}$$

which is, on the average, an order of magnitude smaller than the characteristic frequency of the dimers (2).

In the above analysis, we have assumed that the system of tubulin dimers interacts with a *single* dipole quantum coherent mode of the *ordered water* and hence we ignored dimer-dimer interactions. More complicated cases, involving interactions

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either among the dimers or of the dimers with more than one radiation quantum, which undoubtedly occur *in vivo*, may affect the above estimate.

The presence of such a coupling between water molecules and dimers leads to quantum coherent solitonic states of the electric dipole quanta on the tubulin dimer walls. To estimate the decoherence time we remark that the main source of dissipation (environmental entanglement) comes from the imperfect walls of the cavities, which allow leakage of coherent modes and energy. The time scale, T_r, over which a cavity-MT dissipates its energy, can be identified in our model with the average life-time t_L of a coherent-dipole quantum state, which has been found to be[1]: $T_r \sim t_L \sim 10^{-4}$ sec. This leads to a first-order-approximation estimate of the quality factor for the MT cavities, $Q_{\rm MT} \sim \omega_c T_r \sim \mathcal{O}(10^8)$. We note, for comparison, that high-quality cavities encountered in Rydberg atom experiments dissipate energy in time scale of $\mathcal{O}(10^{-3}) - \mathcal{O}(10^{-4})$ sec, and have Q's which are comparable to $Q_{\rm MT}$ above. The analysis of Ref. 1 then yields the following estimate for the collapse time of the kink coherent state of the MT dimers due to dissipation:

$$t_{\rm collapse} \sim \mathcal{O}(10^{-7}) - \mathcal{O}(10^{-6}) \text{ sec}\,. \tag{16}$$

This is larger than the time scale (8) required for energy transport across the MT by an average kink soliton in some models.[10,33] The result (16), then, implies that quantum physics is relevant as far as dissipationless energy transfer across the MT is concerned.

In view of this specific model, we are therefore in stark *disagreement* with the conclusions of Tegmark,[30] i.e. that only classical physics is relevant for studying the energy and signal transfer in biological matter. Tegmark's conclusions did not take proper account of the possible isolation against environmental interactions, which seems to occur inside certain regions of MTs with appropriate geometry and properties MT.

We would now like to discuss the feasibility of the above, admittedly speculative, ideas by making a brief report on recent progress made by experimentally demonstrating macroscopic quantum entanglement at room temperature in atomic physics.

2.3. *On ordered water in biological systems*

The above scenario may find some support by independent studies of water in biological matter, which we summarize below. Recent experimental spectroscopic studies of resonant intermolecular transfer of vibrational energy in liquid water[34] have established that energy is transfered extremely rapidly and along many water molecules before it dissipates. This energy is in the form of OH-stretch excitations and is thought to be mediated by dipole-dipole interactions in addition to a yet unknown mechanism which speeds up the transfer beyond that predicted by the so-called Förster expression for the energy transfer rate between two OH oscillators, k.

$$k = T_1^{-1} \left(\frac{r_0}{r}\right)^6 \tag{17}$$

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where T_1 is the lifetime of the excited state, r the distance between the oscillators and r_0 the Förster radius. The Förster radius, which is a parameter experimentally determined for each material, characterizes the intermolecular energy transfer and has been determined by Woutersen *et al.*[34] to be $r_0 = 2.1 \pm 0.05$ Å while the typical intermolecular distance (at room temperature) for water is ~ 2.8 Å. It is evident from these data that the energy transfer in pure water will be extremely fast (of order 100 ps) and yet experimentally it is determined to be even faster than that. Woutersen *et al.* speculate that this extremely high rate of resonant energy transfer in liquid water may be a consequence of the proximity of the OH groups in liquid water which causes other, higher-order-uples to also exchange energy. We propose another mechanism to explain the rapidity of the energy transfer, namely kink-soliton propagation. This proposition is based on the phenomenological realization that this is exactly the kind of energy transfer that one would expect to see experimentally as a result of the existence of kink-solitons. It is evident that such a mechanism, regardless of exact origin, is *ideal* for loss-free energy transfer between OH groups located on either different biomolecules or along extended biological structures such as MTs which would be covered (inside and out) with water. Note also that such a mechanism would predict that OH groups in hydrophobic environments would be able to remain in a vibrationally excited state longer than OH groups in hydrophilic environment lending credence to our working assumption that the electrons inside the hydrophobic pockets of the tubulin molecules are sufficiently isolated from thermal noise.

It must be stressed though that such solitonic states in water may not be quantum in origin in the case of microtubules. The 25 nm diameter of the MT is too big a region to allow for quantum effects to be sustained throughout, as we discussed above. Such solitons may be nothing other than the ones conjectured in Ref. 35, which may be responsible for the optical transparency of the water interior of MTs. However, such classical solitons in the bulk of the water interior may co-exist with the quantum coherent states on the dimer walls.[1]

2.4. *Error correction and long-lived quantum entanglement of macroscopic sample at room temperature*

As we have seen above, under appropriate environmental isolation, it is possible to obtain quantum coherence on *macroscopic populations* of tubulin dimers in microtubule systems, which can be sustained for long enough times so that dissipationless energy and signal (information) transfer can occur in a cell.

In a recent article[7] Julsgaard *et al.* describe the macroscopic entanglement of two samples of Cs atoms at room T^0. The entangling mechanism is a pulsed laser beam and although the atoms are far from cold or isolated from the environment, partial entanglement of bulk spin is unambiguously demonstrated for 10^{12} atoms for ~ 0.5 ms. The system's resilience to decoherence is in fact *facilitated* by the existence of a large number of atoms as even though atoms lose the proper spin orientation

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continuously, the *bulk* entanglement is not immediately lost. Quantum informatics, the science that deals with ways to encode, store and retrieve information written in qubits has to offer an alternative way of interpreting the surprising resilience of the Cs atoms by using the idea of "redundancy". Simply stated, information can be stored in such a way that the logical (qu)bits correspond to many physical (qu)bits and thus are resistant to corruption of content. Yet another way of looking at this is given in the work by Kielpinski *et al.*[36] where they have experimentaly demonstrated a decoherence-free quantum memory of one qubit by encoding the qubit into the "decoherence-free subspace" (DFS) of a pair of trapped Berrilium $^9Be^+$ ions. They achieved this by exploiting a "safe-from-noise-area" of the Hilbert space for a *superposition* of two basis states for the ions, thus encoding the qubit in the superposition rather than one of the basis states. By doing this they achieved decoherence times on average an order of magnitude longer.

Both of the above works show that it is possible to use DFS, error correction and high redundancy to both store information and to keep superpositions and entanglements alive for biologically relevant times in macroscopic systems at high temperature.

Thus it may not be entirely inappropriate to imagine that in biological *in vivo* regimes, one has, under certain circumstances, such as specified above, similar entanglement of tubulin/MT arrangements.

3. Possible Implications of Quantum Coherence to the Functioning of Cells

The above raises the question of how such phenomena can affect the functioning of cells. In other words, would the existence of such coherent states and the emergence of quantum mechanical entanglement be somehow useful or beneficial to biological function? Is it then reasonable to propose that in certain cases, natural selection may have favored molecules and cellular structures that exhibited such phenomena? If we accept the notion that according to the laws of quantum physics certain macroscopic arrangements of atoms will exhibit such effects, is it not reasonable then to expect that biomolecules and (by extension) cellular structures and whole cells have "found" a use for such phenomena and have evolved to incorporate them? We stress that at a given instant in time, the different microtubule coherent states participating in a specific bulk entanglement would be almost identical due to the fact that they are related/triggered by a specific "external agent" (e.g. the passing of a specific train of action potentials.) This is of outmost importance since it increases the system's resilience to decoherence (by entangling a large number of nearly identical states), in addition to facilitating "sharp decision making" (i.e. rapid choice among a vast number of very similar states) as explained in Ref. 37 which is presumably a trait favored by natural selection. Here we *digress* to investigate one possible use of such effects by noting a straight-forward application of entanglement to *teleportation* of coherent quantum states across and between cells.

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We define teleportation as the complete transfer of the *coherent state* of an MT *without any direct transfer of mass or energy.* This means that the "receiver" MT finds itself in an identical state to the "sender" MT. We will demonstrate that given the possibility for entangled states, teleportation between microtubule A and microtubule C can happen as follows:

A coherent state in microtubule A (referred to as simply A and designated as $|\Psi(A)\rangle)$ of the (collective) dipole moment(s) being in either of the two classically allowable states with probability amplitude ω_0 and ω_1 can be written as:

$$|\Psi(A)\rangle = \omega_0|0\rangle + \omega_1|1\rangle\,. \tag{18}$$

Step 1: The cell finds itself with *microtubule B and microtubule C* — which can be close together or collinear in an entangled state written as:

$$|\Psi(B,C)\rangle = \frac{1}{\sqrt{2}}(|1_B, 0_C\rangle + |0_B, 1_C\rangle)\,. \tag{19}$$

The combined state of A, B, C can be written as:

$$|\Psi(A,B,C)\rangle = |\Psi(A)\rangle \otimes |\Psi(B,C)\rangle \tag{20}$$

which upon expanding the outer product can be written as:

$$\begin{aligned}|\Psi(A,B,C)\rangle = \frac{1}{\sqrt{2}}(&\omega_0(|0_A, 1_B, 0_C\rangle + |0_A, 0_B, 1_C\rangle)\\ &+ \omega_1(|1_A, 1_B, 0_C\rangle + |1_A, 0_B, 1_C\rangle))\,.\end{aligned} \tag{21}$$

We can also express the combined state $|\Psi(A,B,C)\rangle$ in a different basis, known as the "Bell basis". Instead of $|0\rangle$ and $|1\rangle$, the basis vectors will now be,

$$|\Psi^{\pm}(A,B)\rangle = \frac{1}{\sqrt{2}}(|0_A, 1_B\rangle \pm |1_A, 0_B\rangle) \tag{22}$$

and

$$|\Phi^{\pm}(A,B)\rangle = \frac{1}{\sqrt{2}}(|0_A, 0_B\rangle \pm |1_A, 1_B\rangle)\,. \tag{23}$$

In this new basis, our state of the three microtubules $|\Psi(A,B,C)\rangle$ is written as:

$$\begin{aligned}|\Psi(A,B,C)\rangle = \frac{1}{2}(&|\Psi^+(A,B)\rangle \otimes (\omega_0|0_C\rangle + \omega_1|1_C\rangle) + |\Phi^+(A,B)\rangle \otimes (\omega_0|1_C\rangle\\ &+ \omega_1|0_C\rangle) + |\Psi^-(A,B)\rangle \otimes (\omega_0|0_C\rangle - \omega_1|1_C\rangle)\\ &+ |\Phi^-(A,B)\rangle \otimes (\omega_0|1_C\rangle - \omega_1|0_C\rangle))\,.\end{aligned} \tag{24}$$

This concludes the first step of teleporting the state of MT A to MT C.

Step 2: Notice that so far, MT A has not been manipulated by the cell, i.e. the coherent state of A which we designated as $|\Psi(A)\rangle = \omega_0|0\rangle + \omega_1|1\rangle$ has not been touched. Now the part of the cell containing A and B (let us call it the "sender part") makes a "measurement" — which in our case can be an electromagnetic

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interaction with a passing action potential or the binding of a MAP molecule. If this "measurement" or forced collapse is done in the Bell basis, on $|\Psi^{\pm}(A,B)\rangle$ it will project the state in MT C (!) to:

$$|\Psi^{\pm}(C)\rangle = \langle\Psi^{\pm}(A,B)|\Psi(A,B,C)\rangle = \omega_0|0_C\rangle \pm \omega_1|1_C\rangle\,, \tag{25}$$

similarly

$$|\Phi^{\pm}(A,B)\rangle \to |\Phi^{\pm}(C)\rangle = \omega_0|1_C\rangle \pm \omega_1|0_C\rangle\,. \tag{26}$$

This effectively concludes the teleportation of the state of MT A to MT C with one caveat. There is a probabilistic nature to this process, which means that MT C may receive the exact copy of the state of MT A i.e. $|\Psi^{+}(C)\rangle$ *or* it may receive a state which is a unitary transformation away from the original $|\Psi(A)\rangle$ (one of the other three possibilities: $|\Psi^{-}\rangle$ or $|\Phi^{\pm}\rangle$. MT C can reproduce the state of MT A if there is a "hardwired" condition that when MT C receives $|\Psi^{+}\rangle$ it does nothing further, yet if it receives one of the other three, it performs the correct unitary transformation to obtain the correct state from A. This "hardwired" behavior can be implemented through the use of codes, not unlike the Koruga bioinformation[38] code that MTs follow.

Teleportation is a direct consequence of the existence of entanglement and can be imagined as the basis of intra- and inter-cellular correlation which leads to yoked function (e.g. intracellulary during translation and intercellulary during yoked neuron firing). Experiments to check for such teleportation of states can be designed based on the Surface Plasmon Resonance (SPR) paradigm[39] as applied to sheets of polymerized tubulin immobilized on a dextran-layered gold film.

Note that our use of pure state vectors, $|\Psi\rangle$, to describe the coherent states along a MT arrangement is justifiable since they do not obey the ordinary Schrödinger evolution equation. Instead, they obey the stochastic equations of open systems, of the form discussed in Ref. 40. Nowhere in our proof of teleportation above did we make use of the presice form of the evolution equations. As argued in Ref. 40, by using appropriate stochastic (Langevin type) equations one may recover, for instance, the standard Lindblad form of evolution equations for the corresponding density matrices $\rho = \mathrm{Tr}_{\mathcal{M}}|\Psi\rangle\langle\Psi|$, where $\mathcal{M}$ is an appropriate subset of environmental degrees of freedom, non-accessible to the observer.

4. Conclusions and Outlook

In Ref. 1 we have put forward a comprehensive model conjecture treating certain regions inside MTs as *isolated* high-Q(uality) QED *cavities*. We also presented a scenario according to which the presence of *ordered water* in the interior of MTs results in the appearance of electric dipole quantum coherent modes, which couple to the unpaired electrons of the MT dimers via Rabi vacuum field couplings. The situation is analogous to the physics of Rydberg atoms in electromagnetic cavities.[4] In quantum optics, such couplings may be considered as experimental proof of

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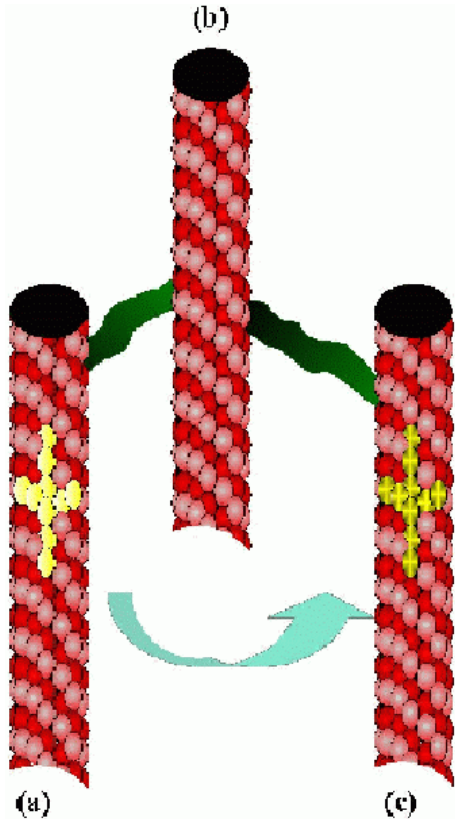

Fig. 2. Schematic of a microtubular quantum teleportation of states. MT (a) sends its state (represented by a cross) to MT (c) without any transfer of mass or energy. Both MT (a) and MT (c) are entangled with MT (b) (entanglement represented by presence of connecting MAPs.

the quantized nature of the electromagnetic radiation. In our case, therefore, if experimentally detected, such couplings would indicate the existence of coherent quantum modes of electric dipole quanta in the *ordered water* environment of MT, as conjectured in Refs. 15 and 16, and used here.

To experimentally verify such a situation, one must first try to detect the emergent ferroelectric properties of MTs, which are predicted by this model and are potentially observable. Measurement of the dipole moment of the tubulin dimers is also an important step. A suggestion along these lines has been put forward in Ref. 12.

In addition, one should verify the aforementioned vacuum field Rabi coupling (VFRS), $\lambda_{\rm MT}$, between the MT dimers and the *ordered water* quantum coherent modes. The existence of this coupling, could be tested experimentally by the same methods used to measure VFRS in atomic physics,[6] i.e. by using the MTs themselves as *cavity environments*, in the way described above, and considering tunable

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probes to excite the coupled dimer-water system. Such probes could be pulses of (monochromatic) light coupling to MTs. This would be the *analogue* of an external field in the atomic experiments mentioned above. The field would then resonate, not at the bare frequencies of the coherent dipole quanta or dimers, but at the *Rabi splitted* ones, leading to a double peak in the absorption spectra of the dimers.[6] By using MTs of different sizes one could thus check on the characteristic $\sqrt{N}$-enhancement of the (resonant) Rabi coupling (10) for MT systems with N dimers.

In the quantum-mechanical scenario for MT dynamics discussed above, as suggested in Ref. 1, a quantum-hologram picture for information processing of MT networks emerges. Further, the existence of solitonic quantum-coherent states along the MT dimer walls implies a role for these biological entities as logic gates.[41] Consider, for instance, a node (junction) of three MTs connected by microtubule associated proteins (MAPs) see Fig. 3. The quantum nature of the coherent states makes the junction interaction *probabilistic*. Therefore at any tube junctions, one is facing a *Probabilistic Boolean Interaction*.[b] The probability for having a solitonic coherent state in a MT branch does depend on its geometric characteristics (such as length). By modulating the length of the tubes and the binding sites of the MAPs one, a bias can be introduced between bit states which can affect the probabilistic final outcomes. This has obvious implications for information processing by MT networks.

Such a binary information system can then provide the basic substrate for quantum information processing inside a (not exclusively neural) cell. In a typical MT network, there may be about 10^{12} tubulin dimers. Such a number is macroscopic, and one is tempted to express doubt as to whether, in realistic biological situations, such macroscopic populations of "particles" can be entangled quantum mechanically, with the entangled state being maintained for a relatively long period of time. It is worth stressing again that in atomic physics the experiments of Ref. 7 have demonstrated *experimentally* the existence of long-lived entangled states of *macroscopic* populations of Cs gas samples, each sample containing 10^{12} atoms. In such experiments entanglement is generated via interaction with pules of light. Thus it is not impossible that *in vivo* one has, under certain circumstances as specified above, similar entanglement of MT coherent quantum states.

If it is experimentally confirmed that treating MTs as QED cavities is a fair approximation to their function, then it may not be so foolhardy to imagine that nature has provided us with the necesasary structures (microtubules) to operate as the basic substrate for *quantum computation* either *in vivo*, e.g. in the way the brain works, or *in vitro*, i.e. it would allow us to construct quantum computers by using microtubules as building blocks, in much the same way as QED cavities in quantum optics are currently being used in successful attempts at implementing qubits.[18]

[b]We would like to thank Dr. Deeph Chana for suggesting this terminology.

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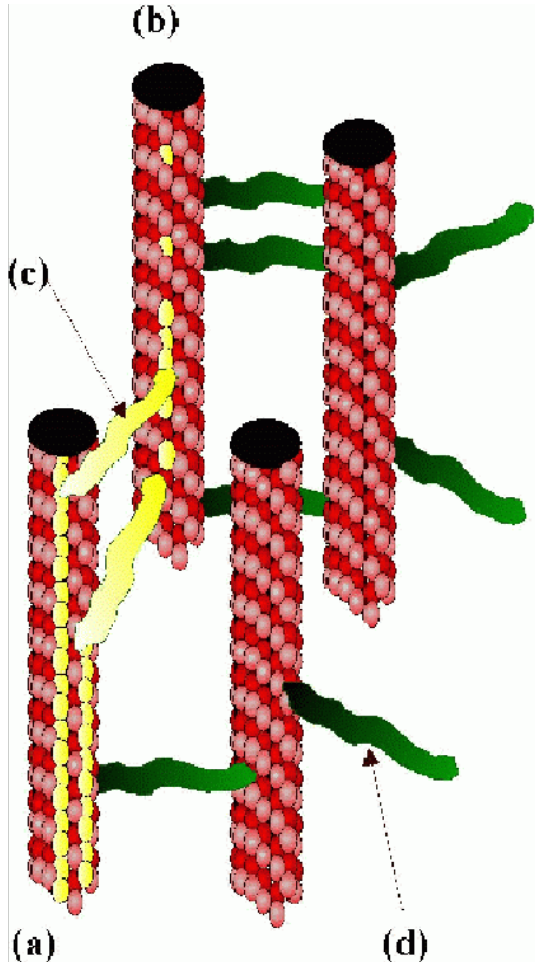

Fig. 3. Example of a microtubular logic gate: A XOR logic gate where "0" is represented by absence of soliton and "1" by presence of soliton. (a) Input MT. (b) Output MT. (c) A MAP transmitting a soliton. (d) A "quiet" MAP. MT (a) has two solitons travelling, encountering two MAPs that transmit both solitons to MT (b). (b) In this hypothetical scenario, the solitons arrive out of phase at MT (b) and cancel each other out. The truth table for XOR reads: $0,0 \to 0; 0,1 \to 1; 1,0 \to 1, 1,1 \to 0$. and in this case is realized by MTs if the MAPs are arranged such that each can transmit a soliton independently but if they both transmit, the solitons cancel out.

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Quantum computation in brain microtubules: Decoherence and biological feasibility

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The Penrose-Hameroff orchestrated objective reduction (orch. OR) model assigns a cognitive role to quantum computations in microtubules within the neurons of the brain. Despite an apparently ''warm, wet, and noisy'' intracellular milieu, the proposal suggests that microtubules avoid environmental decoherence long enough to reach threshold for ''self-collapse'' (objective reduction) by a quantum gravity mechanism put forth by Penrose. The model has been criticized as regards the issue of environmental decoherence, and a recent report by Tegmark finds that microtubules can maintain quantum coherence for only 10^{-13} s, far too short to be neurophysiologically relevant. Here, we critically examine the decoherence mechanisms likely to dominate in a biological setting and find that (1) Tegmark's commentary is not aimed at an existing model in the literature but rather at a hybrid that replaces the superposed protein conformations of the orch. OR theory with a soliton in superposition along the microtubule; (2) recalculation after correcting for differences between the model on which Tegmark bases his calculations and the orch. OR model (superposition separation, charge vs dipole, dielectric constant) lengthens the decoherence time to 10^{-5}–10^{-4} s; (3) decoherence times on this order invalidate the assumptions of the derivation and determine the approximation regime considered by Tegmark to be inappropriate to the orch. OR superposition; (4) Tegmark's formulation yields decoherence times that increase with temperature contrary to well-established physical intuitions and the observed behavior of quantum coherent states; (5) incoherent metabolic energy supplied to the collective dynamics ordering water in the vicinity of microtubules at a rate exceeding that of decoherence can counter decoherence effects (in the same way that lasers avoid decoherence at room temperature); (6) microtubules are surrounded by a Debye layer of counterions, which can screen thermal fluctuations, and by an actin gel that might enhance the ordering of water in bundles of microtubules, further increasing the decoherence-free zone by an order of magnitude and, if the dependence on the distance between environmental ion and superposed state is accurately reflected in Tegmark's calculation, extending decoherence times by three orders of magnitude; (7) topological quantum computation in microtubules may be error correcting, resistant to decoherence; and (8) the decohering effect of radiative scatterers on microtubule quantum states is negligible. These considerations bring microtubule decoherence into a regime in which quantum gravity could interact with neurophysiology.

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I. INTRODUCTION

In the conventional biophysical approach to understanding cognitive processes, it has been generally accepted that the brain can be modeled, according to the principles of classical physics, as a neural network [1–5]. Investigations in this field have delivered successful implementations of learning and memory along lines inspired by neural architectures and these have promoted optimism that a sufficiently complex artificial neural network would, at least in principle, incur no deficit in reproducing the full spectrum and extent of the relevant brain processes involved in human cognition and consciousness.

However, physical effects in the functioning of the nervous system, which lie outside the realm of classical physics, suggest that such simulations may ultimately prove insufficient to the task. One finds ample support for this in an analysis of the sensory organs, the operation of which is quantized at levels varying from the reception of individual photons by the retina [6,7] to thousands of phonon quanta in the auditory system [8]. Of further interest is the argument that synaptic signal transmission has a quantum character [9,10], although the debate on this issue has not been conclusive [11]. We note that using the thermal energy at room temperature in the position-momentum uncertainty relation, and assuming a 1 Å uncertainty for quantal effects, Beck and Eccles [9] concluded that a particle whose mass is just six proton masses would cease to behave quantum mechanically and become classical for all intents and purposes. This seems a serious underestimate, based on the de Broglie wavelength alone. In any case, it is known that quantum modes of behavior exist in much larger structures, such as peptides, DNA, and proteins [12]. For instance, Roitberg *et al.* [13] demonstrated functional protein vibrations that depend on quantum effects centered in two hydrophobic phenylalanine residues, and Tejada *et al.* [14] have evidence to suggest that quantum coherent states exist in the protein ferritin. Finally, new developments in magnetic resonance imaging of the brain demonstrate that induced quantum coherences of proton spins separated by distances ranging from micrometers to 1 mm are sustained for tens of milliseconds and longer [15–17]. While these unentangled quantum couplings are not the type of quantum processes that are likely to prove useful in brain function, they nonetheless demonstrate that mesoscopic quantum coherence can indeed survive in the brain's milieu. Similarly, the aforementioned quantum

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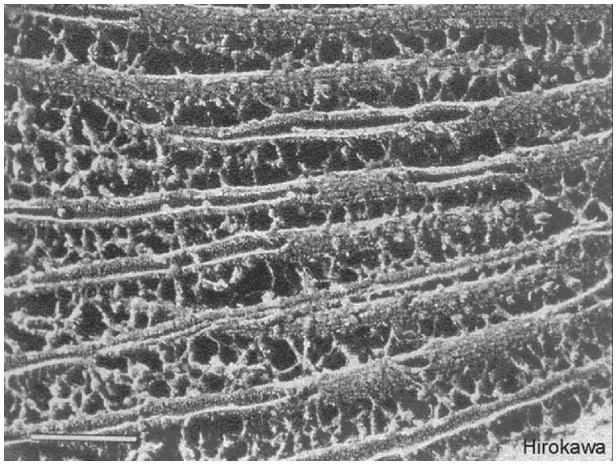

FIG. 1. Immunoelectron micrograph of dendritic microtubules interconnected by microtubule-associated proteins (MAPs). Some microtubules have been sheared, revealing their hollow inner core. Scale bar at lower left represents 100 nm. With permission from Hirokawa [79].

modes in peptides, DNA, and proteins are not the stable, entangled superpositions required for quantum computation, but show that biology can take advantage of quantum modes in clever ways.

The inadequacy of classical treatments is further suggested at the cognitive level, not only as regards longstanding difficulties related to, for instance, accounts of semantics [18], binding [19], and the neural correlate of consciousness, but even in the rather modest goal of reproducing cognitive computational characteristics. Penrose, in particular, has argued that human understanding must involve a noncomputational element [20,21] inaccessible to simulation on classical neural networks and this might be realized through a biological instantiation of quantum computation.[1] Along these lines, Penrose and Hameroff have put forth a specific model [22–24]—orchestrated objective reduction (orch. OR)—positing quantum computation in microtubule protein assemblies in the neurons of the brain.

Microtubules are hollow cylinders whose walls consist of 13 columns (protofilaments) of the protein tubulin arranged in a skewed hexagonal lattice (see Fig. 1). Along with other structures, microtubules comprise the internal scaffolding—the ''cytoskeleton''—in cells including neurons. Determinants of both structure and function, cytoskeletal structures are dynamically active, performing a host of activities instrumental to cellular organization and intelligence [25]. Earlier models (see, for instance, [26–32]) proposed classical information processing among the tubulin ''dimers'' composing microtubules—molecular-level automata regulating real-time cellular behavior. More recently, arguments have been made for quantum computation at the level of individual proteins [33]. In particular, functional protein conformational states are mediated by quantum van der Waals (London) forces [34], the relevance of which is demonstrated by the mechanism of action of the general anesthetic gases that reversibly ablate consciousness. Anesthetics act by disturbing such forces in the hydrophobic pockets of various brain proteins [35,36]. Microtubules are thus poised to mediate between a tubulin-based quantum computation and the classical functioning of neurons.

The Penrose-Hameroff proposal suggests that coherent superpositions of tubulin proteins are inherently unstable and subject to self-collapse under a quantum gravitational criterion (Penrose objective reduction or OR). In the orch. OR model [22,23], the phase of quantum superposition/computation is a preconscious process, and each self-collapse event corresponds to an instantaneous ''moment of conscious experience.'' The mode of collapse, i.e., the fate of proposed superpositions, is a separate issue, the question here being the duration of microtubule quantum states in the face of environmental decoherence. To have an impact on cognitive processes, microtubule superpositions need to survive long enough to interact with the brain's neurophysiological events, typically in the range of milliseconds to hundreds of milliseconds. Calculations indicate that superpositions of these durations would indeed reach self-collapse by quantum gravity in the context of the brain [22,23,37,38]. Quantum coherence that persists, or has influence over such a neurophysiologically relevant time frame, could regulate neural processes, in a manner accounting for the noncomputational element, by ''orchestrating'' state vector reductions to perform quantum computation.

According to the orch. OR model, the neuronal cytoplasm in which microtubules are embedded alternates between phases of (1) isolated quantum superposition/computation (solid, or gelatinous ''gel,'' actin polymerization) and (2) classical states of input/output communication (liquid solution or ''sol''). The input to and output from each OR event evolve as classical microtubule cellular automata—in the form of patterns of tubulin conformational states—to regulate synaptic function, membrane activities, and attachment sites for microtubule-associated proteins (MAPs). Input from synaptic activities may be provided by metabotropic receptors, which interface between membrane synaptic functions and the internal cytoskeleton/microtubules. For example, when the prevalent brain neurotransmitter acetylcholine binds to its postsynaptic receptor, a cascade of activities results in the microtubule-associated protein MAP2 decoupling from microtubules. Woolf [39] has suggested that such decoupling initiates isolation of quantum states in microtubules from the membrane and outside environment. MAPs also interconnect microtubules in bundles, or networks, and are suggested to regulate, or orchestrate, microtubule quantum states prior to OR by their particular attachment sites on the microtubular lattice.

An apparent vulnerability of the orch. OR model and other models of quantum processes relevant to cognition is the question of environmental decoherence. Specifically, can environmental decoherence be avoided and quantum super-

[1]It has been noted that fundamentally analog mechanisms, based on continuous rather than traditional discrete (Turing) computation, might also constitute noncomputation in the relevant sense and equally evade Penrose's argument. However, the essentially discrete nature of exocytosis implies that no such description can be framed in terms of the neurochemical basis of synaptic function.

position sustained long enough for the system to reach threshold for OR? In the original orch. OR formulation, it was assumed that decoherence would have to be avoided for periods long enough to be consistent with membrane electrophysiological events, i.e., 10^{-3}–10^{0} s. However, it has been suggested recently [40] that orch. OR events of far briefer duration may be sufficient. For example, sequences of many orch. OR events in the range of 10^{-7}–10^{-6} s may culminate in electrophysiological events in the 10^{-2}–10^{0} s range.

Recently, Tegmark [41] has responded to this and other models of brain function invoking a quantum element by contending that the relevant degrees of freedom cannot reasonably be sufficiently shielded from environmental, and particularly thermal, influence to maintain quantum superpositions until self-collapse. It is well known that technological quantum devices often require extremely low temperatures to avoid decoherence through environmental interaction. The survival of a delicate quantum coherence in the warm, wet, and noisy milieu of the brain long enough for quantum computation to play a neurophysiological role, therefore, seems unlikely to many observers. Tegmark maintains that orthodox mechanisms of decoherence would destroy a superposed microtubule-associated quantum state on a time scale of the order of 10^{-13} s, much shorter than that associated with events, such as neural firings. In the following, we critically review the assumptions, calculations, and claims that have been made in order to ascertain whether existing treatments accurately reflect the potential for quantum computation in the brain.

II. DECOHERENCE RATES

Tegmark considers in his paper [41] two different scales at which quantum computation might occur in the brain—one involving superpositions of neurons firing and not firing (with calculated decoherence times of 10^{-20} s), and another involving microtubules (calculated decoherence times of 10^{-13} s). We agree that superpositions at the level of neural firing are unlikely, and in fact play no role in the orch. OR or any other contemporary quantum model. In the orch. OR approach, neural firings are entirely classical, though they may be initiated by the outputs of microtubule quantum processes in the neuronal interior. We therefore focus our attention on Tegmark's assertions regarding decoherence times for microtubule-associated quantum superpositions.

Though Tegmark specifically implicates Penrose, his criticisms target neither the Penrose-Hameroff orch. OR model—the only detailed, quantitative model of quantum processes in microtubules with which Penrose is associated—nor any other that is currently or has been under investigation. His remarks appear to be directed against a spuriously quantum version of a classical model, put forth by Sataric *et al.* [29], to treat lossless energy transfer in microtubules in terms of kink solitons traveling along their length.

Tegmark considers a model in which kink soliton solutions, such as those of Sataric *et al.* [29], exist in a quantum superposition of different positions along the microtubule. The actual orch. OR model, on the other hand, is framed in terms of superpositions of the conformational state of a tubulin dimer, as illustrated in Fig. 2, in which superposition separation occurs at the level of each of the protein's atomic nuclei. The separated states are coupled to delocalizable electrons residing in the hydrophobic pocket of the tubulin dimer protein, pointing to a process of conformational change in the dimer controlled by quantum van der Waals (London) forces. There is thus a considerable conceptual disparity between this model and that considered by Tegmark. Nevertheless, it is equally critical to the actual orch. OR model that the mechanisms of decoherence analyzed do not destroy quantum coherence prematurely, before a quantum gravity-induced self-collapse can come into play. Below, we consider both numerical and theoretical concerns that bear on the results presented by Tegmark.

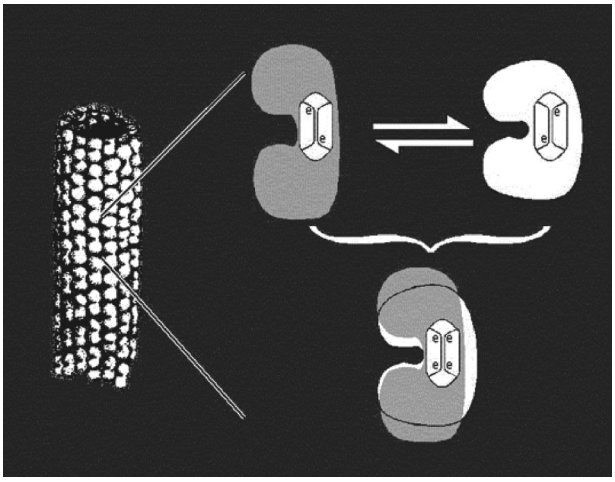

FIG. 2. Each tubulin dimer composing a microtubule can switch between two (or more) conformations coupled to van der Waals (London) forces in the hydrophobic pocket of each. In the orch. OR model, it is posited that tubulin can also exist in a superposition of both conformational states.

In the microtubule case, Tegmark determines the time to decoherence, τ, due to the long-range electromagnetic influence of an environmental ion to be

$$\tau \sim \frac{4\pi\epsilon_0 a^3\sqrt{mkT}}{Nq_e^2 s}, \tag{1}$$

where T is the temperature, m is the mass of the ionic species, a is the distance to the ion from the position of the superposed state, N is the number of elementary charges comprising that state, and s is the maximal ''separation'' between the positions of the tubulin mass in the alternative geometries of the quantum superposition. Since any difference in the mass distributions of superposed matter states will impact upon the underlying space-time geometry, such alternative geometries must presumably be permitted to occur within the superposition.

A. Superposition separation

Superposition occurs not only at the level of a mass distribution separated from itself, but concomitantly at the level of the underlying space-time geometry. According to Pen-

rose's quantum gravitational criterion for objective reduction, superpositions involving different space-time geometries are considered inherently unstable, with the rate of collapse determined by a measure of difference in the geometries. As this measure approaches the order of a Planck length, it becomes problematic to determine a consistent standard by which to match up points in the superposed geometries. Yet if the superposed spaces cannot be resolved into one and the same space, then the different matter states in the superposition must occur in separate spaces and the meaning of superposition in this context becomes obscure. Thus the rate of collapse in Penrose's suggestion for objective reduction must become significant before the measure of difference in superposed space-time geometries grows to the Planck scale. Since gravitational forces are inherently weak, however, the mass distributions of the superposed matter states can be substantially ''separated'' before incurring a large measure of difference in the associated space-time geometries.

Tegmark assumes that this separation s must be at least as large as the diameter of a microtubule, $D=24$ nm, for superpositions spanning many tubulin dimers. This estimate is based on a picture of tubulin dimers literally ''beside themselves'' in superposition. However, in the orch. OR theory, the authors contemplate separation only at the level of the individual atomic nuclei of amino acids comprising the protein.

Hameroff and Penrose [42] surveyed three different levels at which separation related to the protein conformational state of the tubulin dimers might occur: (1) partial separation (10%) of protein spheres, (2) complete separation of atomic nuclei, and (3) complete separation of nucleons. The gravitational self-energy in each instance is taken to be inversely proportional to the decoherence time according to the energy-time uncertainty relation. In the case of protein spheres, the energy E for partial separation is obtained from

$$E=\frac{GM^2s^2}{2r^3}\left(1-\frac{3s}{8r}+\frac{s^3}{80r^3}\right), \tag{2}$$

where M is the monomer mass of 55 kDa, r is the radius of a monomer sphere, and $s=\frac{1}{10}r$ is the superposition separation. For complete separations at the level of either atomic nuclei or nucleons, the contribution to the self-energy determined in separating the mass distributions to a distance of one diameter (the contact position in a spherical approximation of the masses) is of the same order as that determined by increasing the separation further, even to infinity, so the contribution in moving from coincidence to contact is a good order of magnitude estimator of the self-energy for complete separations.

Mass separation of granular arrays of atomic nuclei yields the highest energies of the three cases, and hence the shortest decoherence times, and it is this level that will thus dominate in an orchestrated reduction.[2] Thus mass separation is effected at separations the size of the nucleus, on the order of femtometers, some seven orders of magnitude smaller than Tegmark's estimate.

B. Polarization and charge

In his analysis of the polarization associated with the microtubule, Tegmark defines $p(x)$ to be the average component, in the direction parallel to the microtubule axis, of the electric dipole moment due to the tubulin dimers, a polarization function given in units of (charge) $\times$ (length). The subsequent claim that $-p'(x)$ represents the net charge per unit length along the microtubule, cannot then, on dimensional grounds alone, be well founded. Nevertheless, on this basis, Tegmark integrates over the length of the microtubule across the kink to obtain a net charge that incorrectly bears the units of an electric dipole moment. This, in effect, treats the microtubule as a line of uniform charge rather than a polarized line, and this is how he obtains the magnitude of the *polarization* function by simply summing the *charge* of the ions arrayed around the microtubule at the level of the kinklike propagation. The value of N that figures in his estimate of the decoherence time is then this sum expressed in units of the electron charge q_e. Aside from the dimensional incongruities in this procedure, Tegmark accounts only for the presence of 18 Ca^{2+} ions, bound to the C terminus of the tubulin on each of 13 protofilaments in a cross section of the microtubule. This overlooks the negative charges borne by amino acid side groups and numerous other charges associated with tubulin, all of which attract counterions from the surrounding medium.

Tubulin has been imaged to atomic resolution only within the last two years, following 20 years of difficult work with this protein. Nogales *et al.* published the structure of α- and β-tubulin, co-crystallized in the heterodimeric form [44]. The work establishes that the structures of α and β tubulin are nearly identical and confirms the consensus speculation. A detailed examination shows that each monomer is formed by a core of two β sheets that are surrounded by α helices. The monomer structure is very compact, but can be divided into three functional domains: the amino-terminal domain containing the nucleotide-binding region, an intermediate domain containing the taxol-binding site, and the carboxy-

[2]Estimates of the time to self-collapse (decoherence time) due to such a quantum gravitational mechanism will depend on the number of participating tubulin subunits. For example, calculating energies based on a separation at the level of atomic nuclei, a decoherence time of 500 ms is obtained for 10^9 participating tubulin, or about 10^3 neurons if it is assumed that 10% of the tubulin contained becomes involved (there are roughly 10^7 tubulin per neuron [43]). For a decoherence time of 25 ms (i.e., for coherent 40 Hz oscillations), 2×10^{10} tubulin, or about 20 000 neurons, would be involved. Orch. OR events with a decoherence time of 10^{-6} s would involve roughly 10^{14} tubulin, or about 10^8 neurons (approximately 0.1%–1% of the entire brain). A series of such fast orch. OR events could culminate in membrane depolarization or synaptic transmission.

TABLE I. Calculated values of some electrostatic properties of tubulin.

Tubulin property		Calculated value
Charge		$-10q_e$
Dipole moment		1714 D
Dipole	p_x	337 D
Moment	p_y	-1669 D
Components	p_z	198 D

terminal (C-terminus) domain, which probably constitutes the binding surface for motor proteins [44].

Recently, tubulin's electrostatic properties, including its potential energy surface, were calculated [45] with the aid of the molecular dynamics package TINKER. This computer program serves as a platform for molecular dynamics simulations and includes a facility to use protein-specific force fields. With the C-terminus tail excluded, the electrostatic properties of tubulin are summarized in Table I, following Brown [45].

Since 1 debye$=\frac{1}{3}\times10^{-29}$ C m, we find that the total dipole moment is approximately 5.7×10^{-27} C m, but only a fifth of it is oriented along the protofilament axis.[3]

It turns out that tubulin is quite highly negatively charged at physiological pH, but that much of the charge is concentrated on the C-terminus. This is the one portion of the tubulin dimer which was not imaged by Nogales *et al.* [44] due to its freedom to move following formation of the tubulin sheet. This tail of the molecule extends outward away from the microtubule and into the cytoplasm and has been described by Sackett [46]. At neutral *p*H, the negative charge on the carboxy terminus causes it to remain extended due to the electrostatic repulsion within the tail. Under more acidic conditions, the negative charge of the carboxy-terminal region is reduced by associated hydrogen ions. The effect is to allow the tail to acquire a more compact form by folding.

Any exposed charge in a cytoplasm will be screened by counterions forming a double layer. The screening distance provided by these counterions and water is the Debye length and, in the case of microtubules, its value is typically 0.6–1.0 nm under physiological conditions. Due to the exposure of negatively charged amino acids in the C terminus, a Debye layer is formed as shown in Fig. 3, screening thermal fluctuations due to the stronger Coulomb interactions over distances within the Debye length.

Ionic forces thus tend to cancel over even relatively short distances so that the forces mediating between tubulin and its environment should instead be characterized by dipolar interactions. This suggests that Tegmark's derivation of the decoherence time in Eq. (1) should be replaced with one that

[3]The x direction coincides with the protofilament axis. The α monomer is in the direction of increasing x values, relative to the β monomer. This is opposite to the usual identification of the β monomer as the "plus" end of the microtubule, but all this identifies is whether the microtubule is pointed towards or away from the cell body.

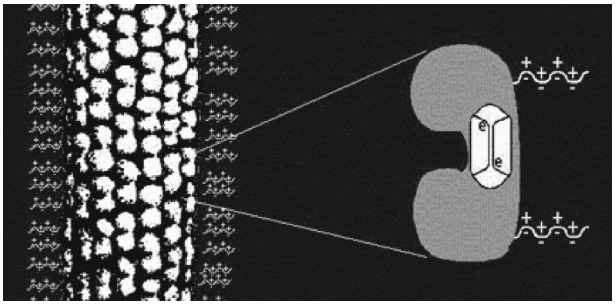

FIG. 3. Negative charges on the C-terminus tail of the tubulin dimer are screened under physiological conditions by counterions forming a Debye layer around the microtubule, as described by Sackett [46].

characterizes tubulin in terms of its electric dipole moment, thus avoiding the need to make a rather arbitrary cut in selecting which charges are to be constitutive of the overall charge of the kinked microtubule and which are to be neglected. Such a modification is accomplished by replacing the Coulomb potential, $V_{\text{Coulomb}}=q^2/4\pi\epsilon_0|\mathbf{r}_1-\mathbf{r}_0|$, describing the interaction of a quantum state of charge q at $\mathbf{r}_0$ and a similarly charged environmental ion at $\mathbf{r}_1$, in favor of a dipole potential, $V_{\text{dipole}}=q\mathbf{p}\cdot(\mathbf{r}_1-\mathbf{r}_0)/4\pi\epsilon_0|\mathbf{r}_1-\mathbf{r}_0|^3$, parametrized by $\mathbf{p}$, the electric dipole moment due to tubulin of the kinked microtubule. The interaction is well approximated, for the purposes of an order-of-magnitude estimate, by this dipole potential in the case that a, the distance between the environmental ion and the superposed state, is greater than the separation of charges in the determination of the electric dipole moment. This separation will not generally be larger than the length of a tubulin dimer, 8 nm, whereas $a=\frac{1}{2}D+n^{-1/3}\approx14$ nm for the same ionic density used by Tegmark, $n=\eta n_{\text{H}_2\text{O}}$ with $\eta\approx2\times10^{-4}$.

As in the Coulomb case of interacting charges, the force resulting from the dipole potential contributes only a phase factor in the evolution of the (reduced) density matrix, traced over the environmental degrees of freedom. These are thus tidal effects that determine the leading contribution to the rate of decoherence. In terms of the vector $\mathbf{a}\equiv\mathbf{r}_1^0-\mathbf{r}_0^0$, between the initial average positions of the environmental ion and the polarized quantum state, these tidal effects are given by the Hessian matrix of second derivatives of the interaction potential

$$\mathbf{M}=\frac{3qp}{4\pi\epsilon_0 a^4}[(5\hat{\mathbf{a}}\hat{\mathbf{a}}^T-\mathbf{I})(\hat{\mathbf{p}}\cdot\hat{\mathbf{a}})-(\hat{\mathbf{a}}\hat{\mathbf{p}}^T+\hat{\mathbf{p}}\hat{\mathbf{a}}^T)]. \tag{3}$$

Under the same assumptions that give rise to Eq. (1), the dipole case yields a decoherence time scale of

$$\tau\sim\frac{4\pi\epsilon_0 a^4\sqrt{mkT}}{3q_e ps}\Omega_{\text{dipole}}, \tag{4}$$

where

$$\Omega_{\text{dipole}}=(5\cos^2\theta\cos^2\varphi-4\cos\theta\cos\varphi\cos\psi+\cos^2\theta+\cos^2\varphi+\cos^2\psi)^{-1/2}$$

is a geometric factor fixed in terms of the angles between **p**, **a**, and **s**,

$$\cos\theta = \hat{\mathbf{a}}\cdot\hat{\mathbf{s}},$$

$$\cos\varphi = \hat{\mathbf{p}}\cdot\hat{\mathbf{a}},$$

$$\cos\psi = \hat{\mathbf{s}}\cdot\hat{\mathbf{p}}.$$

In our calculations, Ω_{dipole} is taken to be of order one.[4]

The calculation of the decoherence time scale in Eq. (4) can be made more realistic by taking into account the dielectric permittivity of tubulin in cytosol, neglected in the original calculation. Since the intracellular medium is primarily water, its dielectric constant can be quite high. The precise value of the permittivity of water is both temperature and frequency dependent but can be as high as $\epsilon \approx 80$ [47,48]. Conservatively estimating the dielectric constant of the surrounding medium by $\epsilon \approx 10$, and using the values given in Table I for the component of tubulin's electric dipole moment along the microtubule axis, together with the correction to the separation s discussed in Sec. II A, yields a decoherence time, $\tau \approx 10^{-5}$–10^{-4} s, that is already eight or nine orders of magnitude longer than that suggested by Tegmark, and sufficient for orch. OR events in the range of 10^{-6} s.

We also wish to point out that Mavromatos and Nanopoulos [38] estimated decoherence times for dipolar excitations in microtubules. Depending on the set of assumptions adopted, the value of τ obtained ranged from as low as 10^{-10} s using a conformal field theory method to as high as 10^{-4} s using a coherent dipole quantum state. For a kink state similar to that discussed by Tegmark, that value is of the order of 10^{-7}–10^{-6} s. Table II summarizes the relevant decoherence time scales.

TABLE II. Relevant decoherence time scales.

Proposed quantum superpositions	Calculated decoherence times
Superposition of neural firing (Tegmark)	10^{-20} s
Soliton superposition (Tegmark)	10^{-13} s
Orch. OR superpositions (corrected)	$10^{-5}-10^{-4}$ s
Coherent dipole state [38]	10^{-4} s
	of magnitude?

C. Dynamical time scales, shielding and error correction

Given the sizeable discrepancy between these estimates and those of Tegmark, it seems reasonable to reevaluate whether the assumptions and conclusions of his calculations are valid. In particular, the derivation requires that the decoherence time scale should fall far short of any relevant dynamical time scale for either the quantum object or the ionic environment, if the noninteracting contribution to the Hamiltonian is to be neglected relative to the interaction contribution. With the substantially modified decoherence times calculated above, this assumption is no longer justified, even by Tegmark's own estimates that place the dynamical time scale for a kinklike excitation traversing a microtubule at $\tau_{\text{dyn}} \approx 5 \times 10^{-7}$ s. The results of the derivation thus fail to be consistent with its assumptions, and Tegmark's formulation of the decoherence time must be rejected as inappropriate to the superposition under consideration. Without this crucial assumption validated, Tegmark's calculation fails to achieve self-consistency in the approximation regime considered, and must be rejected as inappropriate to the superpositions of interest in the orch. OR theory.

Two possible avenues might be envisioned in the framework of the orch. OR theory by which to overcome the influence of decoherence due to scattering and tidal effects, such that decoherence by quantum gravitational effects might play a role. The most obvious solution is to require that the shortest decoherence times be those due to quantum gravity. An equally viable approach, however, is to require that decoherence due to other mechanisms be effectively countered by dynamical processes operating on time scales more rapid than that of the relevant form of decoherence. This is the means by which quantum systems, such as lasers, maintain quantum coherence against thermal disruption at room temperature. The lasing phenomenon is about a supply of energy that is transformed from incoherent to coherent form, with an attendant reduction in entropy and increase in order. The introduction of nonequilibrium conditions impacts decisively on the conclusions that one might derive from calculations of the decoherence rate, which assume that equilibrium conditions hold sway. The significance of nonequilibrium situations in addressing phenomena in living media—decidedly not in equilibrium—cannot be overstated. Whereas the dynamical time scale for lasers is determined by the rate at which the system is pumped by an incoherent source of energy, appropriate dynamical time scales in the microtubule case might be determined by the characteristic rates at which incoherent metabolic energy is provided to processes that might counteract decoherence by scattering, such as actin gelation in sol-gel cycles and the ordering of water, both discussed below. Relevant rates could include that of GTP hydrolysis (known to control the stability of microtubules [49,50]), dephosphorylation of microtubule-associated proteins, and ATP hydrolysis (required for actin polymerization).

The transition between the alternating phases of solution and gelation in cytoplasm depends on the polymerization of actin (see Fig. 4), and the particular character of the actin gel in turn depends on actin cross linking. Of the various cross-linker related types of gels, some are viscoelastic, but others (e.g., those induced by the actin cross-linker avidin) can be

[4]Though Ω_{dipole} increases without bound as the three vectors, $\hat{\mathbf{a}}$, $\hat{\mathbf{p}}$, and $\hat{\mathbf{s}}$, approach mutual orthogonality, randomly oriented vectors rarely come close enough to satisfying this condition to make an order of magnitude difference in the decoherence time.

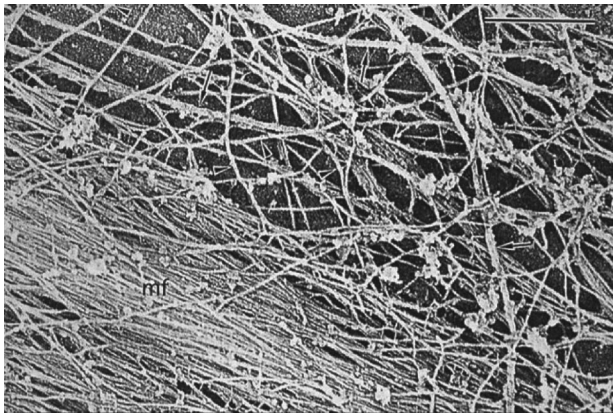

FIG. 4. Immunoelectron micrograph of cytoplasm showing microtubules (arrows), intermediate filaments (arrowheads), and actin microfilaments (mf). A dense actin gel (lower left) completely obscures the microtubules below. Scale bar at upper right represents 500 nm. With permission from Svitkina *et al.* [80].

deformed by an applied force without response [51]. Cycles of actin gelation can be rapid [52], and in neurons, have been shown to correlate with the release of neurotransmitter vesicles from presynaptic axon terminals [53,54]. In dendritic spines, whose synaptic efficacy mediates learning, rapid actin gelation and motility mediate synaptic function, and are sensitive to anesthetics [55–57].

In the orch. OR model, actin gelation encases microtubules during their quantum computation phase. Afterwards, the gel liquifies to an aqueous form suitable for communication between microtubule states and the external environment. Such alternating phases can explain how input from, and output to, the environment can occur without disturbing quantum isolation.

The water within cells is itself not truly liquid, but has been shown to be, to a large extent, ordered [58]. Most of the ordered water in the cell in fact surrounds the cytoskeleton [59]. Neutron diffraction studies indicate several layers of ordered water on such surfaces, with several additional layers of partially ordered water. Tegmark himself allows that the dynamical process of ordering water in the vicinity of the microtubule[5] could protect the quantum system from short-range sources of decoherence, such as the scattering of nearby molecules.

In fact, there is a long history to the hypothesis that macroscopic quantum coherence might be supported biologically by maintaining a supply of energy at a rate exceeding a threshold value [60–62]. The collective effects responsible for the ordering of water arise in the context of a supply of metabolic energy [58]. Empirical evidence indicates that, in the presence of an activation energy approximating the amount required for the formation of a soliton on the microtubule ($\approx$0.3 eV), the surrounding water can be easily brought to an electret state [63,64]. Spontaneous breaking of the dipole rotational symmetry in the interaction of the electric dipole moment of water molecules with the quantized electromagnetic field would then give rise to the dipolar wave quanta that are postulated to mediate collective effects [65–67].

In the gel phase, the water-ordering surfaces of a microtubule are within a few nanometers of actin surfaces that also order water. Thus bundles of microtubules encased in actin gel may be effectively isolated with the decoherence-free zone *a* extending over the radius of the bundle, of the order of hundreds of nanometers. If the *a* dependence of the decoherence time is accurately reflected in the previously corrected version (4) of Tegmark's equation, an order of magnitude increase in the decoherence-free zone results in an increase of three orders of magnitude in the decoherence time for the microtubule bundle.

Technological quantum computing is, in general, feasible because of the use of quantum error correction codes. It has been suggested that error correction may be facilitated by topologies—for instance, toroidal surfaces [68,69]—in which global, topological degrees of freedom are protected from local errors and decoherence. Topological quantum computation and error correction have been suggested in microtubules.[6]

D. Temperature dependence

An examination of the temperature dependence in the formulation of decoherence time casts further doubt on the validity of the reasoning that led Tegmark to claim an ultrarapid decoherence rate due to long-range forces. If the adoption of Eq. (1), even in the modified form, Eq. (4), were justified, it would require that the decoherence time be a monotonically *increasing* function of temperature; decoherence times would tend to grow as the square root of temperature. The apparent implication is that one should expect to find longer-lived quantum coherent states at higher temperatures, contrary to the intuition from thermodynamics, statistical mechanics, and kinetic theory, that increased thermal agitation should have a disruptive effect on the formation and preservation of quantum coherence (at least in thermal equilibrium, the only paradigm considered in [41]).

It might be objected that this intuition proceeds from a consideration of the increased impact of ionic and radiative

[5]While the point is conceded with respect to the water *inside* the microtubule, Tegmark finds it more contentious as regards the water *outside* the microtubule, which ''fills the entire cell volume.'' Geometric considerations aside, the mechanism of ordering is independent of whether the water is inside or outside the microtubule, and is only contended for the water closely approaching the microtubule.

[6]The microtubule lattice features a series of helical winding patterns that repeat on longitudinal protofilaments at 3,5,8,13,21 and higher numbers of subunit dimers (tubulins). These particular winding patterns (whose repeat intervals match the Fibonacci series) define attachment sites of the microtubule-associated proteins (MAPs), and are found in simulations of self-localized phonon excitations in microtubules [70]. They suggest topological global states in microtubules which may be resistant to local decoherence [71]. Penrose [72] has suggested that the Fibonacci patterns on microtubules may be optimal for error correction.

(see subsequent Sec. II E) scatterers at higher temperatures. Since we are investigating tidal influences of long-range forces, influences that are not amenable to formulation in terms of scattering states, one might choose to disregard such intuitions. Noting that it is only the shortest decoherence time, amongst those calculated for competing decohering processes, that will be observable, it might be argued that a counterintuitive temperature dependence in the decoherence rate due to a particular process would remain invisible if this rate were not the fastest. For instance, if the decoherence rate, R_s due to a species of scatterers beat the decoherence rate R_d, due to the tidal influence of distant ions, we would have no physical indication of the temperature dependence in R_d. While this allows us to preserve our physical intuitions, it defeats the purpose of calculating R_d. In [41], R_d is calculated because it is expected to be the dominant rate given that R_s is suppressed by the ordering of water near microtubules.

Nevertheless, it might be contended that R_d dominates decoherence only within a range of temperatures that does not include very low temperatures—where our experience with quantum coherent states argues against the counterintuitive temperature dependence in Eq. (1)—and also does not include very high temperatures, where thermal considerations must eventually reassert themselves over other effects. This could only be maintained, however, if the temperature dependence in R_d were not monotonic and increasing without bound in the low temperature limit (as is Tegmark's). In that case, R_d would eventually overtake R_s as temperature decreases, with the result that we would not observe the longest-lived quantum superpositions at the lowest temperature. Further, it does not seem possible, in order to escape this conclusion, to claim that the temperature dependence in R_s is also counterintuitive since this is precisely the case in which we must expect the strictures of thermodynamics and statistical mechanics to be binding. It should be noted in this context that even the formulas derived in [41] to account for the effect of scattering determine decoherence times that increase with increasing temperature, rather than decrease.

Tegmark's Eq. (1) suggests that low temperatures, at which decohering environmental interactions presumably have the least impact, are deemed most inhospitable to quantum coherence, contrary to experience. Though the equation is formulated to be valid at room temperature, the stated assumptions appear to remain valid in the low temperature limit. Both object and environment should be well localized in this limit and, unless it is imagined that the dynamical time scale goes to zero in the low temperature limit even more rapidly than the decoherence scale—entailing a dynamical rate that increases without bound as absolute zero is approached—the requirement that the decoherence scale must lie well below the dynamical scale is also met. Accounting for the temperature dependence implicit in a, which must decrease to a minimum in the absence of thermal agitation, only exacerbates the counterintuitive trend.

While the equations derived in [41] in the case of ion-ion and ion-water scattering do not exhibit the expected temperature dependence over the range of temperatures in which they are deemed valid, it may be possible to recover the appropriate dependence in the low temperature limit. The approximation regime valid at room temperature will no longer be appropriate in the low temperature limit since the de Broglie wavelength λ of the scatterer is no longer expected to be much smaller than the superposition separation s of the quantum state. Making the alternate approximation, $s \ll \lambda$, which should be valid at sufficiently low temperatures, the ion-water scattering case recovers the intuitive temperature dependence, but the ion-ion scattering case does not, suggesting that the problems in the derivation lie deeper. Even in the ion-water case, adjusting the approximation regime relieves the counterintuitive temperature dependence only at temperatures near absolute zero. Moreover, in the case of interest here—that of decoherence due to tidal influences of long-range forces—there is no analogous approximation to adjust at low temperatures.

Regardless of any deviations from the predicted trend that might mitigate the failure of the low temperature limit in the neighborhood of absolute zero, the functional dependence of the decoherence time, across the range of temperatures in which the calculation is deemed valid (and in particular at room temperature), runs contrary to the expectation that the onset of decoherence should, in general, be more rapid at higher temperatures.

E. Other mechanisms of decoherence

Since any process that irreversibly conveys a flow of information from the system to the environment acts as a source of decoherence, our remarks concerning the threat of decoherence through the tidal effects of long-range forces, while addressing an important source of decoherence, clearly do not exhaust the potential for disruptive environmental influence. In particular, we would like to have some quantitative estimate of the decoherence time associated with ubiquitous radiative scatterers.

A decoherence formalism for apparent wave function collapse due to scattering has been developed in [73], and is summarily reviewed below. The treatment adopts the assumption that the interaction between the system and scatterer can be treated as approximately instantaneous relative to the dynamical time scales of the Hamiltonian governing the evolution of the system. Note that in this approximation, it is these interaction times (this might be given by the transit time of the scatterer through the system), not decoherence times, that are presumed short in comparison to dynamical times, and hence this is not the same approximation whose validity was questioned in Sec. II C.

The change in the density matrix ρ for the system and scatterer together is given in terms of a transition matrix T by $\rho \rightarrow \rho' = T\rho T^{\dagger}$. If T conserves energy and momentum, then the transition matrix element has the form

$$\langle \mathbf{p}'\mathbf{k}' | T | \mathbf{p}\mathbf{k} \rangle = \delta(\mathbf{p}' + \mathbf{k}' - \mathbf{p} - \mathbf{k}) a_{\mathbf{pk}}(\mathbf{p}' - \mathbf{p}), \qquad (5)$$

where $|\mathbf{pk}\rangle$ denotes the state in which the system has momentum $\mathbf{p}$ and the scatterer has momentum $\mathbf{k}$, and $a_{\mathbf{pk}}(\mathbf{p}' - \mathbf{p})$ is the probability amplitude for a momentum transfer to the system of $\mathbf{p}' - \mathbf{p}$. If the velocity of the incident scatterer is much greater than the spread of velocities in the density

matrix (which will generally be the case for the types of scatterers to be considered subsequently), the probability amplitude can be treated as independent of **p**. With the further assumption that the incident particle is in a momentum eigenstate or an incoherent mixture of momentum eigenstates, it is shown in [73] that the effect of such a scattering event on the density matrix is

$$\rho(\mathbf{x},\mathbf{x}') \rightarrow \rho'(\mathbf{x},\mathbf{x}') = \rho(\mathbf{x},\mathbf{x}')\hat{P}(\mathbf{x}'-\mathbf{x}), \tag{6}$$

where $\hat{P}(\mathbf{x}'-\mathbf{x})$ is the Fourier transform of

$$P(\mathbf{p}'-\mathbf{p}) = \int |a_{\mathbf{k}}(\mathbf{p}'-\mathbf{p})|^2 \mu(\mathbf{k}) d^3k, \tag{7}$$

representing an incoherent superposition of plane waves with momentum probability distribution $\mu(\mathbf{k})$. A generic anisotropic spectrum $\mu(\mathbf{k})$ will result in radiation pressure, itself a potential source of decoherence [74]. Forces due to radiation pressure, however, are in general significantly weaker than those due to scattering and are unlikely to yield a faster rate of decoherence. Consideration is thus limited to an isotropic spectrum,

$$\mu(k) = \lambda_0 \nu(\lambda_0 k)/4\pi k^2, \tag{8}$$

where $k = |\mathbf{k}|$, λ_0 is a characteristic wavelength and ν is a probability distribution over the real line. In particular, we will be interested in the Planck distribution

$$\nu(x) = \frac{1}{2\zeta(3)} \frac{x^2}{e^x - 1} \tag{9}$$

for radiative scatterers in ambient visible light, in the flux of thermal radiation from our surroundings, and in the cosmic microwave background radiation, spectra corresponding to blackbody temperatures of 5800 K, 300 K, and 2.7 K, respectively.

With a flux Φ of scatterers and a total cross section σ the incidence of scattering events is governed by a Poisson distribution with intensity, $\Lambda = \sigma\Phi$. If it were scattering processes alone that evolved the density matrix, it would thus be subject to

$$\dot{\rho}(\mathbf{x},\mathbf{x}',t) = -\Lambda(1-\hat{P}(\mathbf{s}))\rho(\mathbf{x},\mathbf{x}',t), \tag{10}$$

where we have defined $\mathbf{s} = \mathbf{x}' - \mathbf{x}$. If the time scale given by Λ^{-1} is short compared to all the dynamical time scales in the system Hamiltonian, this is approximately the case that prevails. The solution to Eq. (10),

$$\rho(\mathbf{x},\mathbf{x}',t) = \rho(\mathbf{x},\mathbf{x}',t_0) e^{-\Lambda t(1-\hat{P}(\mathbf{s}))}, \tag{11}$$

will then adequately capture the behavior of the density matrix for the short times over which the usual Schrödinger evolution can be neglected.

In the isotropic case, the mean of the distribution, $\hat{P}(\mathbf{s})$, vanishes and the covariance matrix is proportional to the identity matrix, $S_{ij} = v\,\delta_{ij}$ with the constant of proportionality given by the variance v. This allows us to make the following Gaussian approximation in the solution above:

$$\rho(\mathbf{x},\mathbf{x}',t) \approx \rho(\mathbf{x},\mathbf{x}',t_0)\exp[-\Lambda t(1-e^{-s^2/2\lambda_{\text{eff}}^2})], \tag{12}$$

where $s = |\mathbf{s}|$ is the superposition separation of Sec. II A, and $\lambda_{\text{eff}} = 1/\sqrt{v}$ is given by

$$\lambda_{\text{eff}} = \frac{\lambda_0}{\sqrt{-g''(0)\langle x^2\rangle}}. \tag{13}$$

Here, $\langle x^2\rangle = 4!\zeta(5)/2!\zeta(3)$, denotes the expectation value of x^2 over the Planck distribution $\nu(x)$, and the function

$$g(x) = \frac{\sin x}{x} \int_0^{2\pi} \int_{-1}^{1} e^{\imath x u} f(\arccos u, \varphi) du d\varphi \tag{14}$$

results from the Fourier integral in $\hat{P}(\mathbf{s})$ over the momentum probability distribution in Eq. (8), such that

$$\hat{P}(\mathbf{s}) = \int_0^\infty g\left(\frac{ws}{\lambda_0}\right) \nu(w) dw. \tag{15}$$

The differential cross section $f(\theta,\varphi)$ in Eq. (14) has been normalized so as to integrate to 1. In the case of *s*-wave scattering, appropriate for treating opaque systems much larger than the wavelength of the scattering photons, $f(\theta,\varphi) = 1/4\pi$, so $g(x) = \sin^2 x/x^2$ and $g''(0) = -2/3$. According to Eq. (13), one then calculates $\lambda_{\text{eff}} = 0.381\lambda_0$. In the case of photons scattering from dielectric spheres much smaller than the wavelength of the photons, the appropriate normalized differential cross section [75] is $f(\theta,\varphi) = 3(1 + \cos^2\theta)/16\pi$, which determines

$$g(x) = \frac{3\sin x}{2x^3}\left(\cos x + (x^2-1)\frac{\sin x}{x}\right), \tag{16}$$

and $g''(0) = -11/15$. The effective wavelength in this case is then $\lambda_{\text{eff}} = 0.363\lambda_0$.

In the near-diagonal approximation appropriate when $s \ll \lambda_{\text{eff}}$, the solution in Eq. (12) reduces to

$$\rho(\mathbf{x},\mathbf{x}',t) \approx \rho(\mathbf{x},\mathbf{x}',t_0)\exp(-\Lambda s^2 t/2\lambda_{\text{eff}}^2), \tag{17}$$

which identifies the decoherence time as

$$\tau \sim \frac{\lambda_{\text{eff}}^2}{\Lambda s^2}. \tag{18}$$

Table III summarizes estimates, adapted from [73], for Λ, Φ, and λ_{eff} for each of the scattering sources under consideration, and gives values of τ calculated on the basis of Eq. (18). For Λ and τ the results are expressed over a range extending from the case of a single electron in superposition to that of a mote 10 μm in size. The cross section σ for interaction with an electron is estimated at roughly 5×10^{-29} m^2, and for interaction with a 10 μm mote at 5×10^{-10} m^2. Even in the case of a superposed system as

TABLE III. Characteristic values, adapted from [73], for several scattering processes. Decoherence times are quoted in the near-diagonal approximation $\tau=\lambda_{\mathrm{eff}}^2/\Lambda s^2$ appropriate to the superposition separations, s, given in Sec. II A.

		Ambient visible light	300 K photons	Cosmic microwaves
Λ (s^{-1})	Electron	5×10^{-8}	5×10^{-6}	5×10^{-12}
	10 μm mote	5×10^{11}	5×10^{13}	5×10^{7}
Φ (m^{-2} s^{-1})		10^{21}	10^{23}	10^{17}
λ_{eff} (m)		9×10^{-7}	2×10^{-5}	2×10^{-3}
τ (s)	Electron	10^{25}	10^{26}	10^{36}
	10 μm mote	10^{6}	10^{7}	10^{17}

large as 10 μm, decoherence times due to radiative scatterers are substantially longer than those postulated to result from the effects of quantum gravity in the orch. OR theory, a result attributable to the fact that the superposition separation is so small. The decohering effect of radiative scatterers thus does not appear to pose a significant threat to the proposed mechanism.

III. OUTLOOK

Among the many features of brain function not currently understood, subjectivity remains the most problematic, motivating both classical and quantum explanations. Quantum approaches, in particular the orch. OR model in microtubules, are potentially able to account for features like the unified nature of conscious experience and subjectivity [76–78], but are based on speculative quantum level activities and appear on the surface to suffer from a decoherence problem.

The mechanisms of decoherence discussed here, however, fail to rule out biologically instantiated quantum coherence of the sort envisioned in the orch. OR hypothesis, particularly when proposed biological mechanisms of coherent pumping, screening, and quantum error correction are considered. When appropriately revised, both theoretically and numerically, decoherence times due to the tidal influence of Coulomb forces appear to be in line with the relevant dynamical times, in the range 10^{-5}–10^{-4} s, in accord with biological phenomena. These revisions place the microtubule decoherence time in a range invalidating Tegmark's assumption that the decoherence time scale is much shorter than relevant physiological dynamical time scales, and suggest that the approximation scheme used is inappropriate to the superposition under consideration. Further, the temperature dependence of Tegmark's formulation fails to confirm, in the case of both tidal influences of Coulomb forces and decoherence due to scattering, the intuition that increased thermal disruption should have a deleterious effect on the formation and maintenance of quantum coherent states. These intuitions are expected to be particularly robust in thermal equilibrium, which is the context that Tegmark considers. Consideration of other mechanisms of decoherence, such as that due to radiative scattering, indicate that these play no role in the determination of the decoherence rate. Thus the issue of organized quantum processes in the brain remains open, and subject to experimental verification, an indication that there is cause for optimism that some of the enigmatic features of the cognitive processes occurring in consciousness might yet be understood in a quantum theoretical framework.

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Quantum Information Processes in Protein Microtubules of Brain Neurons

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Abstract— **We study biologically 'orchestrated' coherent quantum processes in collections of protein microtubules of brain neurons, which correlate with, and regulate, neuronal synaptic and membrane activity. In this situation the continuous Schrodinger evolution of each such process terminates in accordance with the specific Diosi–Penrose (DP) scheme of 'objective reduction' ('OR') of the quantum state. This orchestrated OR activity ('Orch OR') is taken to result in moments of conscious awareness and/or choice. We analyze Orch OR in light of advances and developments in quantum physics, computational neuroscience and quantum biology. Much attention is also devoted to the 'beat frequencies' of faster microtubule vibrations as a possible source of the observed electroencephalographic ('EEG') correlates of consciousness.**

Keywords— **quantum information processes, quantum consciousness, protein microtubules of brain neurons.**

I. INTRODUCTION

All living cells of plants, animals and humans continuously emit ultraweak biophotons (ultraweak electromagnetic waves) in the optical range of the spectrum, which is associated with their physiological states and can be measured using special equipment. Neural cells also continuously emit biophotons. The intensity of biophotons is in direct correlation with neural activity, cerebral energy metabolism, EEG activity, cerebral blood flow and oxidative processes [1, 2]. According to Van Wijk et al [3], there are significant correlations between the fluctuations in biophoton emission and fluctuations in the strength of electrical alpha wave production in the brain. Some unpublished observations suggest that the state of the biophoton field of a person may be connected to the state of the brain as measured by the EEG (e.g., degree of synchronization and coherence) [4]. Certain meditative states characterized by a high degree of coherence in the EEG may well be accompanied by a high degree of coherence in the biophoton field [4], although measurements correlating the coherence of the biophoton field and the EEG readings have not been made yet. In this situation it is very important to study the interaction of biomolecules with biophotons of brain neurons and to quantitatively investigate possible connections between the EEG and the biophoton production.

There is no doubt that EEG waves are deeply involved with the basic functioning of the brain. The EEG waves associated with two distant neurons are strongly correlated and this supports the view that EEG waves are related to the properties of the brain as a coherent quantum system.

In modern optogenetics much attention is devoted to the possibility to manipulate neurons with light [5-7]. In this situation spatially-precise neuronal control and monitor the activities of individual brain neurons are realized with the help of optogenetic actuators (channelrhodopsin, halorhodopsin, archaerhodopsin) and temporally-precise recordings can be made with the help of optogenetic sensors for calcium (Aequorin, Cameleon, GCaMP), chloride (Clomeleon) or membrane voltage (Mermaid).

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At present much attention is devoted to the nature of consciousness and mechanism by which it occurs in the brain [8]. In this situation consciousness depends on biologically 'orchestrated' coherent quantum processes in collections of microtubules within brain neurons [9]. It is very important to unify micro and macro levels: quantum-statistical properties of quantized fields insight MTs and EEG signals [10], because superradiant optical computing in networks of microtubules may provide a basis for biomolecular cognition and a substrate for consciousness [11], fMRI and SPECT (single-photon emission computed tomography) are also very important tools for investigation of cooperative processes in protein microtubules of brain neurons. There has been a strong recent interest in applying quantum theory in cognitive science [12]. It should be noted that special evaluation of altered states of consciousness (ASC), induction of ASC by different physical methods: electrical stimulation (direct and alternating current), magnetic stimulation, sound (binaural) stimulation compared with placebo in healthy volunteers will give more insight about nature of consciousness from quantum biophysical point of view [13-15].

II. Fluctuation Function of Protein Microtubules

At present much attention is paid to the coupling between fluctuation function $S_z(t)$ of protein MTs and EEG signal, influence of coherence in biophoton field (Fig. 1) on coherence in neuronal system and correlations between the fluctuations in biophoton emission and fluctuations in the strength of electrical alpha wave production in the brain [10]

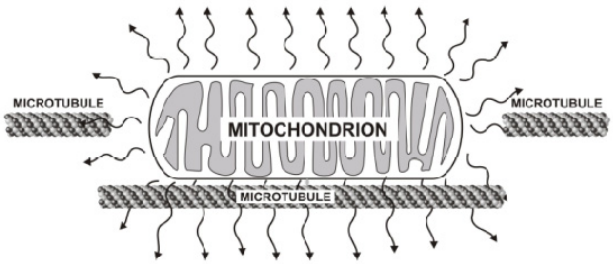

Fig. 1 Representation of biophotons produced by mitochondria and the interaction of biophotons with microtubules

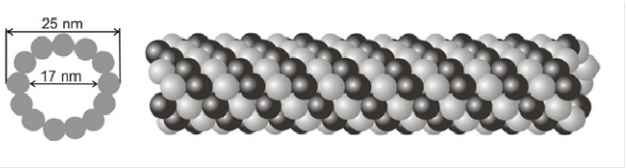

Fig. 2 MTs are hollow cylinders composed of protein units called tubulin. The inner diameter of an MT is 17 nm and the outer diameter is 25 nm. The lengths of MTs vary widely from nanometers to micrometers. MTs have been considered to act as QED-cavities [16, 17]

It should be emphasized that MTs have been considered as optical cavities (Fig. 2) [16] with quantum properties [17], capable of supporting only a single mode [18] or perhaps a few widely spaced (in the frequency domain) modes. Rahnama et al [10] is successfully applied a fully quantum mechanical formalism of the Jaynes-Cummings model (JCM) [19-21]. It is noticed that MTs are biological hollow cylinders with a 17 nm inner diameter and a 25 nm outer diameter (Fig. 2) [22], composed of units called tubulin dimers, each of which has the dimensions $4\text{nm}\times 8\text{nm}\times 6\text{nm}$ [16]. Tubulin can be viewed as a typical two-state quantum mechanical system, where the dimers couple to conformational changes with 10^{-9} - 10^{-11} sec transitions due to electron transitions in hydrophobic pockets, corresponding to an angular frequency in the range $\omega_0 \sim O(10^{10}) - O(10^{12}) Hz$ [16]. Using a first-order-approximation estimate of the quality factor for the MT cavities (i.e. Q_{MT}), it has been found that $Q_{MT} \sim O(10^8)$ [16]. High-quality cavities encountered in Rydberg atom experiments dissipate energy on time scales of $O(10^{-3}) - O(10^{-4})$ sec and have Q's which are comparable to Q_{MT} [16]. Frequencies of visible light are on the order of THz and Wang et al [23] have detected visible light in the brain as biophotons. Also, transition frequencies in tubulins are on the order of THz [16]. In order to get more physical insight about Quantum Brain Dynamics it is necessary to

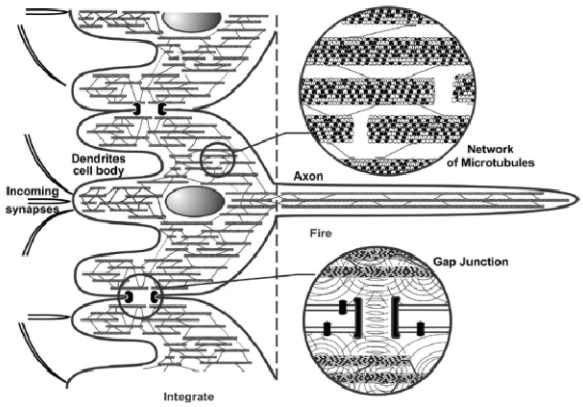

Fig. 3 An 'integrate-and-fire' brain neuron, and portions of other such neurons are shown schematically with internal microtubules. In dendrites and cell body/soma (left) involved in integration, microtubules are interrupted and of mixed polarity, interconnected by microtubule-associated proteins (MAPs) in recursive networks (upper circle, right). Dendritic–somatic integration (with contribution from microtubule processes) can trigger axonal firings to the next synapse. Microtubules in axons are unipolar and continuous. Gap junctions synchronize dendritic membranes, and may enable entanglement and collective integration among microtubules in adjacent neurons (lower circle right). In Orch OR, microtubule quantum computations occur during dendritic/somatic integration, and the selected results regulate axonal firings which control behavior.

carefully examine properties of a single protein microtubule [24] from quantum-mechanical point of view. This will give new possibilities for realization and manipulation of quantum information processes with the help of entanglement and non-local canals in the ensemble of microtubules of brain neurons (Fig. 3).

Let us investigate quantum brain dynamics and analyze fluctuation function of MTs in the case, in which MT has three energy states: ground state $|g\rangle$, first excited $|e_1\rangle$ and second excited $|e_2\rangle$ biomolecular states. In this situation it is assumed that at the initial moment t = 0 MT is situated in the first excited state $|e_1\rangle$ and quantized electromagnetic field in Holstein–Primakoff SU(1,1) coherent state [19, 20]

$$|\psi(0)\rangle_{biophotons} = (1-|\xi|^2)^k \sum_{m=0}^{\infty} \left\{ \frac{\Gamma(m+2k)}{m!\Gamma(2k)} \right\}^{1/2} \xi^m |m,k\rangle \tag{1}$$

where $|m,k\rangle$ are the eigenvectors of the Casimir operator

$$C = K_0^2 - \frac{K_+K_- + K_-K_+}{2} = k(k-1)I \tag{2}$$

k is the so-called Bargmann index and $\xi = |\xi| \exp(i\varphi)$ $(0 \le |\xi| \le 1)$. In the present model k=½ and SU(1,1) CS can be written through oscillator number states

$$|\psi(0)\rangle_{biophotons} = (1-|\xi|^2)^{1/2} \sum_{n=0}^{\infty} \xi^n |n\rangle = \sum_{n=0}^{\infty} Q_n |n\rangle \tag{3}$$

In this case the initial state – vector of the coupled MT - field system takes the following form

$$|\psi(t=0)\rangle = |\xi\rangle \otimes |e_1\rangle = \sum_{n=0}^{\infty} Q_n |e_1, n\rangle \tag{4}$$

Exact analytical solution for state-vector of the MT - field system can be obtained by using Schrodinger equation:

$$\begin{aligned} |\psi(t)\rangle = & \sum_{n=0}^{\infty} e^{-i\omega_0\{S_z+n\}t} Q_n \{\cos(\sqrt{n^2+\alpha^2(n+1)^2}\,\tau)|e_1,n\rangle \\ & - \frac{i\alpha(n+1)}{\sqrt{n^2+\alpha^2(n+1)^2}} \sin(\sqrt{n^2+\alpha^2(n+1)^2}\,\tau)|g,n+1\rangle \\ & - \frac{in}{\sqrt{n^2+\alpha^2(n+1)^2}} \sin(\sqrt{n^2+\alpha^2(n+1)^2}\,\tau)|e_2,n-1\rangle\} \end{aligned} \tag{5}$$

where dimensionless time is $\tau = \lambda_1 t$, $\alpha = \lambda_2 / \lambda_1$, λ_1 and λ_2 are MT - field coupling constants [10].

By using solution (5) one can obtain exact analytical solution for fluctuation function $S_z(t)$ in the following manner

$$\begin{aligned} S_z(t) = & -\left(1-|\xi|^2\right) \sum_{n=0}^{\infty} \left(|\xi|^2\right)^n \\ & \times \sin^2(\sqrt{n^2+\alpha^2(n+1)^2}\,\tau) \frac{n^2-\alpha^2(n+1)^2}{n^2+\alpha^2(n+1)^2} \end{aligned} \tag{6}$$

In figure 4 we show $S_z(t)$ as a function of time for cases $|\xi|$=0.6 and 0.9, α = 0.1 and 0.01. In this situation it is observed that the higher the initial mean biophoton number n(t = 0)=$|\xi|^2/(1-|\xi|^2)$, the higher $S_z(t)$. There are also some jumps in the fluctuation function, in which a strong energy exchange between the MTs and field subsystems takes place.

It should be noted that present JCM with intensity-dependent coupling [19, 20] in the best way describes quantum vibrations of MTs of brain neurons and gives more insight about faster microtubule vibrations as a possible source of the observed EEG correlates of consciousness from quantum biophysical point of view.

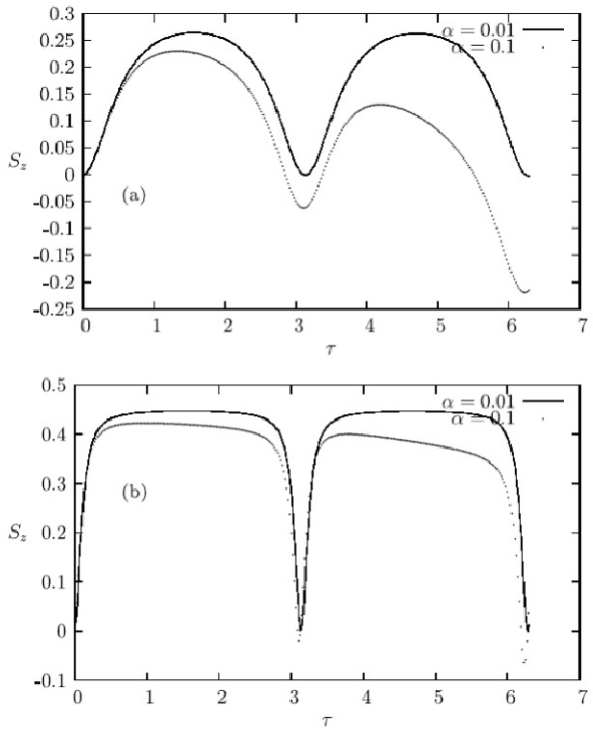

Fig. 4 Sz(t) as a function of the dimensionless time τ for cases α = 0.1 and 0.01: (a) |ξ|=0.6, (b) |ξ|=0.9.

III. Quantum Entanglement and Non-local Models of Quantum Information Processes in Microtubules of Brain Neurons

Quantum entanglement is, according to Erwin Schrodinger in 1935, the essence of quantum physics and inspires fundamental questions about the principles of nature. By testing the entanglement of particles we are able to ask fundamental questions about realism and locality in nature. Local realism imposes certain constraints in statistical correlations of measurements on multi-particle systems. Quantum mechanics, however, predicts that entangled systems have much stronger than classical correlations that are independent of the distance between the particles and are not explicable with classical physics.

In non-local models of quantum mechanics quantum system behaves as a whole no matter how far separated its components. In other words, correlations between electrons or photons, composing the whole quantum system exist at distances much larger than the dimensions of composed parts. It should be noted that experimental demonstration of quantum correlations over more than 10 km was realized by Tittel et al [25]. Moreover, quantum entanglement distribution over 100-kilometre free-space channels is successfully demonstrated by Yin et al [26]. It should be emphasized that quantum entanglement takes place also in living species, in which biophotons are generated by mitochondria [10]. In this situation each living cell is giving off, or resonating, a biophoton field of coherent energy. If each cell is emitting this field, then the whole living system is, in effect, a resonating field-a ubiquitous nonlocal field. And since biophotons are the entities through which the living system communicates, there is near-instantaneous intercommunication throughout, which is the basis for coherent biological organization - referred to as quantum coherence. In this case much attention is devoted to the timeless approaches and quantum vacuum [27, 28]. Capacity for evolution rests not on aggressive struggle and rivalry but on the capacity for communication and cooperation. In this sense the built-in capacity for species evolution is not based on the individual but rather living systems that are interlinked within a coherent whole: Living systems are thus neither the subjects alone, nor objects isolated, but both subjects and objects in a mutually communicating universe of meaning. Experiments with non-local signal transfer between seeds, plant organisms and bio-hybrid systems confirm this point [29-32]. Just as the cells in an organism take on different tasks for the whole, different populations enfold information not only for themselves, but for all other organisms, expanding the consciousness of the whole, while at the same time becoming more and more aware of this collective consciousness.

Because protein microtubules of brain neurons are quantum optical devices [9-11], [16, 17] it is instructive to study the generation of squeezing and entanglement in microtubules to obtain more insight about nature of consciousness from quantum biophysical point of view. In this situation non-local correlations takes place between different microtubules of brain neurons and therefore quantum information in the brain may be transmitted through non-local canals (https://www.youtube.com/watch?v=0fu_k8_tae0).

IV. Conclusions

In the present paper the possibility of quantum information processes in MTs of brain neurons is investigated. Much attention is devoted to the quantum-statistical properties of protein microtubules. It is shown that fluctuation function of MTs $S_z(t)$ has the tendency towards oscillations but exact periodicity of $S_z(t)$ oscillations is violated. It should be noted that exact periodicity of $S_z(t)$ oscillations takes place when one of the dipole moment matrix transition elements between neighbouring biomolecular levels tends to zero. In this situation fluctuation function determines number of generated biophotons in protein microtubules and is connected with oscillations of EEG signal. Moreover, entanglement and non-local correlations exist between biophotons and tubulin structures of different MTs of brain neurons, situated at distances much larger than their own dimensions.

Conflict of Interest

The authors declare that they have no conflict of interest.

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Quantum optical coherence in cytoskeletal microtubules: implications for brain function

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Abstract

'Laser-like,' long-range coherent quantum phenomena may occur biologically within cytoskeletal microtubules. This paper presents a theoretical prediction of the occurrence in biological media of the phenomena which we term 'superradiance' and 'self-induced transparency'. Interactions between the electric dipole field of water molecules confined within the hollow core of microtubules and the quantized electromagnetic radiation field are considered, and microtubules are theorized to play the roles of non-linear coherent optical devices. Superradiance is a specific quantum mechanical ordering phenomenon with characteristic times much shorter than those of thermal interaction. Consequently, optical signalling (and computation) in microtubules would be free from both thermal noise and loss. Superradiant optical computing in networks of microtubules and other cytoskeletal structures may provide a basis for biomolecular cognition and a substrate for consciousness.

Key words: Quantum theory; Quantum coherence; Photon coherence; Neural holography; Microtubules; Spontaneous symmetry breaking; Water molecules; Photon signalling network; Consciousness

1. Introduction

Several features of brain activity have prompted connections to quantum physics. One feature is that the physical correlates of consciousness, presumably patterns composed of many states of many entities, are distributed throughout the brain. Despite this extension in space, consciousness has a 'unity' that is difficult to explain by classical means (Marshall, 1989). Non-local quantum coherence may provide the unity necessary to break the impasse known as the 'binding problem' in neurophysiological terms (Crick and Koch, 1990; Singer, 1993).

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look for the binding problem in neuroscience

Another feature, as argued by Penrose (1989), is that many brain functions are non-algorithmic and non-computational, and thus inexplicable by serial and/or parallel neural processing. Quantum coherence (in microtubules specifically, according to Penrose (1994)) can account for these functions. Other links between consciousness and quantum theory include probabilistic synaptic functions (Eccles, 1986; Beck and Eccles, 1992), holography (Pribram, 1966, 1971, 1991; Gabor, 1968), and general application of the uncertainty principle (Bohm, 1951; Bohr, 1958).

In the mid 1960s the advent of optical holography heralded a new departure in understanding the relationship between brain, memory and perception. The resistance of aspects of memory and of perception to relatively extensive brain damage indicated that memory storage and perceptual processing are distributed procedures. Before the engineering instantiations of Gabor's holographic equations were available, it was difficult to imagine what such a distributed procedure would be like.

Over the next decades computer programs inspired by holography were developed that reflected some of the associative characteristics of the parallel distributed processes that constitute the essence of holographic optical procedures. These 'connectionist', 'neural network' procedures made it possible to stimulate in vitro many perceptual and memory processes and to explore the extent and limitations of the simulations.

Meanwhile, it was recognised that an unmodified holographic metaphor was an inappropriate model for brain processing because its spread function is unlimited (infinite). Data from neurophysiological experiments showed that the cortical function modelled by holography is better represented by a sinusoid limited by a Gaussian envelope — a mathematical formulation put forward by Gabor (1946) to measure the maximum efficiency with which a telephone message could be sent across the Atlantic cable. Gabor (1948, 1968) used the same equations used by Heisenberg to describe units in microphysics and, therefore, called the unit of communication a 'quantum of information'.

A major interest confronting brain science is, therefore, to probe the possible mechanisms that lead to the processing of quanta of information by the brain.

A specific quantum field theoretical approach was originated by Ricciardi and Umezawa (1967). Without reference to the structural constituents of brain cells, they formulated a relationship between quantum physics and memory. They emphasized that each neuron specifies a spatially distributed system with quantum mechanical degrees of freedom whose physical properties can be understood by quantum field theory. Quantum field theory is a framework of quantum physics capable of describing fundamental processes of elementary particle physics as well as condensed matter physics (Umezawa, 1993). Ricciardi and Umezawa (1967) suggested that memory is a quantum vacuum state that violates an original dynamical symmetry property of the full complement of the spatially-distributed quantum mechanical degrees of freedom in each brain cell.

The introduction of quantum theory in brain science is perhaps analogous to the recent advent of the new field, quantum cosmology. The fundamental equations of cosmology, recast as quantum theoretical operator expressions acting on a 'wavefunction of the universe' permit an interpretation in which the universe is born in imaginary time. A quantum tunnelling process would then allow the universe to evolve into real time, according to a proposal by Hawking. Similarly, the model advanced by Ricciardi and Umezawa (1967) relies on the quantum theoretical concept of a vacuum state which violates a dynamical symmetry property. While the marriage of quantum theory and cosmology is uneasy so long as quantum theory remains uninterpretable outside the context of a classical environment, the union of brain science with quantum concepts is consistent with currently accepted interpretations. Memory of an external stimulus is imprinted as an ordering of the vacuum state (i.e. the lowest energy state or the ground state). If this ordered vacuum state no longer manifests the original dynamical symmetry property, the Nambu–Goldstone theorem suggests creation of long-range correlation

waves with zero energy requirement which participate in the system's quantum dynamics (Ricciardi and Umezawa, 1967).

Such a long-range correlation wave can be seen as a Bose quantum ('Goldstone boson' or 'massless boson'). Ricciardi and Umezawa were perhaps the first to suggest the existence of a spatially distributed system with the full range of quantum mechanical degrees of freedom in brain cells, and also introduced quantum ordering and spontaneous symmetry breaking as tools to investigate the biological system. Slightly later, Fröhlich (1968) proposed a similar idea for biological cells in general but with more explicit materialistic formulation. Focusing on a thin layer just beneath the cell membrane, Fröhlich theorized that energy can be stored without thermal loss in this two-dimensional region in coherent, dipolar propagating waves. Biological molecules with dipolar vibrational activity adjacent to cell membranes may thus manifest a globally coherent mode of dynamics so that the thin layer could be seen as a biological superconducting medium or a biological plasma effectively isolated from thermal environments. Fröhlich waves are predicted to appear in a frequency region between 10^{11} and 10^{12} s^{-1} called the Fröhlich frequency. It can be concluded that energy supplied to biological cells with magnitude equal to the 'Fröhlich frequency' times the Planck constant may not be completely thermalized but stored in a highly ordered fashion.

Experimental evidence for Fröhlich excitations in biological systems include observation of GHz-range phonons in proteins (Genberg et al., 1991), sharp-resonant non-thermal effects of microwave irradiation on living cells (Grundler and Keilmann, 1983), GHz-induced activation of microtubule pinocytosis in rat brain (Neubauer et al., 1990), and resonance Raman detection of Fröhlich frequency energy (Genzel et al., 1976; Webb et al., 1977; Webb and Stoneham, 1977; Webb, 1980).

Fröhlich waves need not be restricted to thin layers adjacent to cell membranes. Dipolar oscillations maintained by hydrogen bonds and non-localized electrons trapped in hydrophobic regions of protein molecules may manifest a collective mode considered as a Fröhlich wave propagating in a one-dimensional medium. Davydov described dipolar solitary waves propagating along alpha-helices within proteins. This mode of solitary wave propagation, known in quantum field theory to carry energy and information without loss due to thermalization, is called the Davydov soliton or dipolar soliton (Davydov, 1979).

About 10 years after the original idea of 1967, Umezawa together with two colleagues, Takahashi and Stuart, proposed a refined physical model of a spatially-distributed system with fully quantum mechanical degrees of freedom, not only within each brain cell but also among brain cells (Stuart et al., 1978, 1979). The system extended among brain cells is thought to be composed of two kinds of spatially distributed degrees of freedom subject to quantum dynamics. As a consequence of spontaneous symmetry breaking, these two modes are distinguished as the 'corticon' field and an exchange boson field which interact as a quantum dynamical system — a theoretical model of non-local memory storage and recall processes. More recently, Jibu and Yasue presented a physically realistic picture of the system of corticon and exchange boson fields, and proposed to call the new quantum field theoretical framework 'Quantum Brain Dynamics' ('QBD') (Jibu and Yasue, 1992a,b, 1993a,b). They describe corticon and exchange boson fields as dipolar vibrational fields distributed along protein filaments of the cytoskeleton and extracellular matrices and associated water fields. Marshall (1989) proposed that Fröhlich's 'pumped phonons' result in a Bose–Einstein condensate in brain which is the substrate for consciousness.

Inspired by application of quantum theoretical methods to brain and other biological cells initiated by Ricciardi and Umezawa and by Fröhlich in the 1960s, other scientists began to view brain functioning from the microscopic view of quantum physics during the 1980s. Several groups focused especially on networks of filamentous proteins which dynamically organize the interiors of living cells: the cytoskeleton (Hameroff and Watt, 1982; Del Giudice et al., 1983). Among cytoskeletal filamentous polymers (actin, interme-

Such a condition may be present in the isolated cores of microtubules surrounded by ordered water molecules

diate filaments, etc.), microtubules are the most central to cellular organization and information processing. Consisting of skewed hexagonal lattices of subunit protein dimers called tubulin, microtubules self-assemble into hollow cylinders 25 nm across, adapt, transport and govern cellular activities. Their hollow geometry and periodic lattice structure have suggested several quantum field aspects. Hameroff (1987) applied Fröhlich's coherent excitations to tubulin subunits in microtubules as the basis of computational cellular automata. Rasmussen et al. (1990) showed that networks of such microtubule automata are capable of learning. Considering the layer of ordered water outside and inside microtubules, Del Giudice et al. (1986) proposed that the formation of microtubules' cylindrical structure from 'tubulin' subunits may be understood by the concept of self-focussing of electromagnetic energy by ordered water. Like the Meissner effect for superconducting media, electromagnetic energy would be confined inside filamentous regions around which the tubulin subunits gather. Del Giudice et al. (1986) showed that this self-focussing should result in filamentous beams of radius 15 nm, precisely the inner diameter of microtubules (see Figs 1–4).

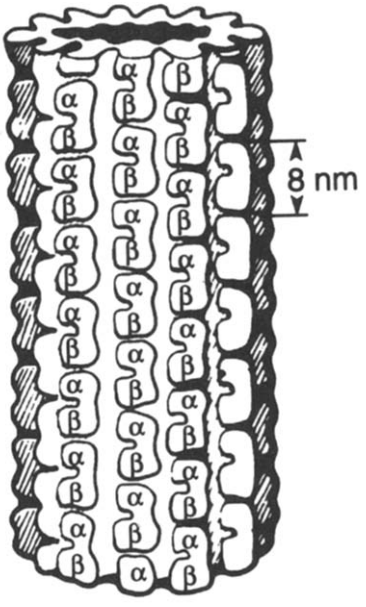

Fig. 2. Microtubule (MT) structure from X-ray crystallography (Amos and Klug, 1974). Tubulin subunits are 8-nm dimers comprised of α and β monomers.

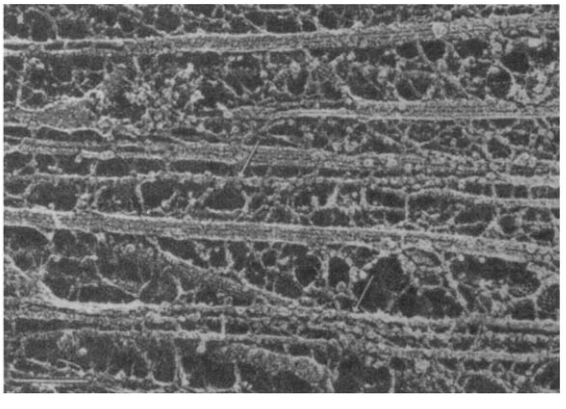

Fig. 1. Electron micrograph of quick frozen, deep, etched neuronal microtubules (MTs) and neurofilaments polymerized with microtubule associated proteins (MAPs). Arrows point at neurofilaments; larger diameter structures are microtubules. Scale bar, 100 nm. (With permission from Hirokawa, 1991).

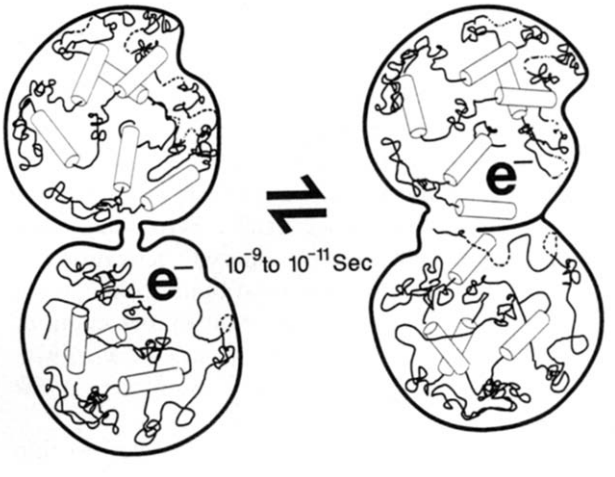

Fig. 3. The tubulin dimer protein has two states in which a quantum event (electron mobility, quasi-particle phonon) within a hydrophobic region couples to conformational change with 10^{-9}–10^{-11}-s transitions. Cylinders within protein are α-helical regions. (Original drawing by Djuro Koruga).

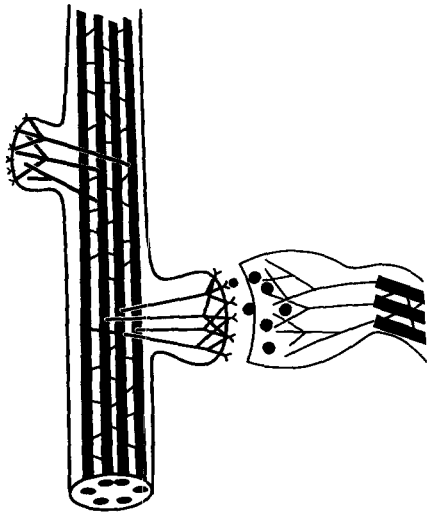

Fig. 4. Schematic drawing of a synapse showing underlying cytoskeletal structures. Right: axon terminal with MT-MAP network connecting with synapsin filaments involved in release of neurotransmitter vesicles (circles) into synaptic cleft. Left: ventrical dendrite with MT-MAP network, two dendritic spines with cytoskeletal structures connected to receptors in synaptic membrane. (Adapted from Hirokawa, 1991).

A waveguide is a structure that guides waves, such as electromagnetic waves or sound, with minimal loss of energy by restricting expansion to one dimension or two.

Another perspective was proposed by Hameroff (1974) in which microtubules were thought to act like 'dielectric waveguides' for photons, that is, quantum dynamical modes of an electromagnetic wave. Indeed, living tissue transmits light more readily than non-living material; in experiments with mammalian brains, temporal poles and hippocampus manifest the maximum light penetration (Hameroff, 1987). This evidence led Hameroff to consider microtubules as waveguides for photons. Furthermore, he proposed that cytoplasmic interference of coherent sources from and among multiple microtubules may lead to a holographic information processing mechanisms. Holographic patterns stored in cytoplasmic sol-gel states may become 'hardwired' by calcium-induced assembly of finer components of the cytoskeleton such as actin, actin-binding proteins and the micro-trabecular lattice. Lechleiter et al. (1991) imaged coherent cytoplasmic spiral waves of calcium induced by acetylcholine binding to membrane receptors of living cells. Coherent sources of either charge carriers or conformational waves in microtubules suggest a revival of holographic brain theory of Pribram and others (Pribram, 1966, 1971, 1991; Gabor, 1968; Margenau, 1984; Eccles, 1986; Marshal, 1989). Our current view is that coherence manifests at the molecular, quantum level rather than (or in addition to) coherence at the level of neural firing (Singer, 1993).

Hameroff suggested in very general terms two key concepts in understanding cytoskeletal brain activity from the point of view of quantum physics: microtubules acting as waveguides for photons and as holographic information processors. Holography is an optical phenomenon related to interference of coherent electromagnetic waves or photons. The microtubule may be seen as a waveguide for photons, and Fröhlich's theory suggests coherent excitations in microtubules. Further, the periodic lattice structure of microtubules may provide periodically arrayed 'slits' (spaces between dimers) through which photons may pass. No proof of coherent photon generation or emission necessary for optical holography has been found in microtubules or any microscopic biological structure. However, self-focused photons and self-trapped, non-thermalizing wave entities should be difficult to detect (see Section 4). In any case, no thorough theoretical investigation of these concepts has occurred from the standpoint of physics except the water laser (Del Giudice et al., 1988).

In the present paper, we propose a new quantum theoretical framework suggesting microtubules possess ordered and systematic properties yielding cooperative quantum dynamics called superradiance and self-induced transparency. Specifically, the quantum dynamical system of water molecules and the quantized electromagnetic field confined inside the hollow microtubule core can manifest a specific collective dynamic called superradiance by which the microtubule can transform any incoherent, thermal and disordered molecular, electromagnetic or atomic energy into coherent photons inside the microtubule. Furthermore, it will also be shown that such coherent photons created by superradiance penetrate perfectly along the internal hollow core of the microtubule as if

the optical medium inside it were made 'transparent' by the propagating photons themselves. This is a quantum theoretical phenomenon called self-induced transparency.

Superradiance and self-induced transparency in cytoskeletal microtubules can lead to 'optical' neural holography. Neurons (and other cells) may contain microscopic coherent optical supercomputers with enormous capacity like the fictional but famous HAL 9000 of '2001: A Space Odyssey' with holographic memory manipulation (Clarke, 1968).

This paper is a result of the long-range cooperation among physicists and brain scientists providing several by-products:

1. Superradiance and self-induced transparency of microtubules may be understood to predict theoretically the existence of a peculiar phenomenon: weak coherent photon emission from living matter.
2. Quantum coherence among microtubules spaced hundreds of micrometers apart (which in turn can have quantum coherence with other microtubules hundreds of micrometers away, etc.) can solve the unity and binding problems which plague classical theories of consciousness.
3. A source of coherence at the biomolecular level can lead to quantum interference and holographic information representation at the sub-cellular level.
4. General anesthesia may be explained by blockade of quantum level events which affect collective cooperative macro-level events.

2. Superradiance in microtubules

Let us look at a typical microtubule in the cytoskeletal structure of brain cells or general biological cells. It is a hollow cylinder about 25 nm in diameter whose wall is a polymerized array of protein subunits ('tubulins'). Its length may range from tens of nanometers to micrometers, and possibly further to meters in nerve axons of large animals. For the purpose of simplifying the physical formulation, we consider the microtubule as a hollow cylinder with radius r_{MT} and length l_{MT}. Their real values would be $r_{MT} \approx 12$ and $l_{MT} \approx 10^2$–10^3 in nanometers. We denote the spatial region inside the microtubule cylinder by V and restrict our discussion to quantum mechanical dynamics taking place in this spatial region V.

Let us introduce a Cartesian system of coordinates O_{xyz} with the xy-plane attached parallel to one of the two ends of the microtubule cylinder so that the origin O coincides with the centre of the end cap. The z-axis lies naturally along the longitudinal center axis of the microtubule cylinder. Then, any position in the region V can be labelled by giving its coordinates $\boldsymbol{r} = (x,y,z)$.

The spatial region V inside the microtubule is not empty, but is likely to be filled with water molecules. Of course, there may be other molecules, although their population is relatively small. We consider the ideal case in which the existence of molecules other than those of water can be neglected. However, below we will consider the more realistic case in which we have impurities among water molecules such as anesthetic molecules. It is most likely that the density of water confined inside the microtubule cylinder remains almost constant. Therefore, we may be allowed to fix the total number of water molecules inside the region V, say N.

Let us take a look at a typical water molecule, say the jth water molecule. Here, j running from 1 to N denotes the fictitious number labelling the N water molecules in question. Its position is given by coordinates $\boldsymbol{r}^j = (x^j,y^j,z^j)$. From a physical point of view, a water molecule has a constant electric dipole, and so it can be seen as a quantum mechanical spinning top with an electric dipole moment. The average moment of inertia and electric dipole moment of a water molecule are estimated to be $I = 2m_p d^2$ with $d \approx 0.82$ Å and $\mu = 2e_p P$ with $P \approx 0.2$ Å, respectively. Here, m_p denotes the proton mass and e_p the proton charge.

Due to the electric dipole moment μ, the water molecule interacts strongly with the quantized electromagnetic field in the spatial region V. Although the water molecule has many energy eigenstates as a quantum mechanical spinning top and so it can exchange energy between the quantized electromagnetic field in many different

values, we restrict our discussion to the most likely case in which only the two principal energy eigenstates take part in the energy exchange. This coincides with the conventional two-level approximation in describing the energy exchange between atoms and the quantized electromagnetic field in laser theory.

Then, one sees immediately that the quantum dynamics of the jth water molecule can be well described by a spin variable $s^j = 1/2\sigma$, where $\sigma = (\sigma_x,\sigma_y,\sigma_z)$ and the σ_x are Pauli spin matrices denoting the three components of the angular momentum for spin 1/2. Let ϵ be the energy difference between the two principal energy eigenstates of the water molecule. Its real value is $\epsilon \approx 200$ cm^{-1} (Franks, 1972). Then, the Hamiltonian governing the quantum dynamics of the jth water molecule is given by ϵs^j_z, and so the total Hamiltonian for N water molecules becomes

$$H_{WM} = \epsilon \sum_{j=1}^{N} s_z^j \qquad (1)$$

s = spin
e = energy

So the Hamiltonian for the water molecules becomes sum of the respective energies of the spin of the individual water molecules

Two energy eigenvalues of the former Hamiltonian are $-1/2\epsilon$ and $1/2\epsilon$ reflecting the fact that only the two principal energy eigenstates with energy difference ϵ are taken into account.

Now, let us consider the quantized electromagnetic field in the spatial region V. It is convenient to describe the quantized electromagnetic field in terms of an electric field operator $\boldsymbol{E} = \boldsymbol{E}(\boldsymbol{r},t)$. Let us assume for simplicity that the electric field is linearly polarized, obtaining $\boldsymbol{E} = \boldsymbol{e}E$, where $\boldsymbol{e}$ is a constant vector of unit length pointing in the direction of linear polarization. Then, the quantized electromagnetic field in question comes to be well described by a scalar electric field $E = E(\boldsymbol{r},t)$ governed by the usual Hamiltonian

r = (x,y,z).

$$H_{EM} = \frac{1}{2}\int_V E^2 d^3r \qquad (2)$$

Next, we take the interaction between the quantized electromagnetic field and the totality of water molecules by which they can exchange energy in terms of creation and annihilation of photons. Let us divide the electric field operator into positive and negative frequency parts

$$E = E^+ + E^- \qquad (3)$$

Then, the interaction Hamiltonian of the quantized electromagnetic field and the totality of water molecules becomes

$$H_I = -\mu \sum_{j=1}^{N} \{E^-(\boldsymbol{r}^j,t)s_-^j + s_+^j E^+(\boldsymbol{r}^j,t)\} \qquad (4)$$

where

$$s_\pm^j = s_x^j \pm i s_y^j \qquad (5)$$

The total Hamiltonian governing the quantum dynamics of the electromagnetic field, the dipolar vibrational field of water molecules, and their interaction is given by

$$H = H_{EM} + H_{WM} + H_I. \qquad (6)$$

Since the spatial region V inside the microtubule cylinder may be considered as a cavity for the electromagnetic wave, it is convenient to introduce the normal mode expansion of the electric field operator $E = E^+ + E^-$, obtaining

$$E^\pm(\boldsymbol{r},t) = \sum_{\boldsymbol{k}} E^\pm_{\boldsymbol{k}}(t) e^{\pm i(\boldsymbol{k}\cdot\boldsymbol{r} - \omega_k t)} \qquad (7)$$

Here, ω_k denotes the proper angular frequency of the normal mode with wave vector $\boldsymbol{k}$. We are mainly interested in the ordered collective behavior among the water molecules and the quantized electromagnetic field in the cavity region V. Let us introduce hence collective dynamical variables $S^\pm_{\boldsymbol{k}}(t)$ and S for water molecules by

$$S^\pm_{\boldsymbol{k}}(t) = \sum_{j=1}^{N} s^j_\pm(t) e^{\pm i(\boldsymbol{k}\cdot\boldsymbol{r}^j - \omega_k t)} \qquad (8)$$

and

$$S = \sum_{j=1}^{N} s_z^j \qquad (9)$$

Then, the total Hamiltonian (Eq. 6) becomes

$$H = H_{EM} + \epsilon S - \mu \sum_k (E_k^- S_k^- + S_k^+ E_k^+) \qquad (10)$$

It seems worthwhile to note here that this total Hamiltonian for the system of N water molecules and the quantized electromagnetic field in the region V inside the microtubule cylinder is essentially of the same form as not only Dicke's Hamiltonian for the laser system but also that of Stuart et al. for Quantum Brain Dynamics (Dicke, 1954; Stuart et al., 1979). Therefore, it can be expected that each microtubule in the cytoskeletal structure of brain cells manifests not only the memory printing and recalling mechanism in QBD but also a laser-like coherent optical activity. The former seems not so surprising because Jibu and Yasue (1992a, 1993a,b) developed a physical picture of QBD in terms of water molecules and protein filaments. Hence, we will not discuss this in the present paper. On the other hand, the latter seems highly surprising because it may open a completely new picture of the fundamental process of brain functioning, drastically different from the conventional one. Namely, in addition to the usual pathway conduction of neural impulses in terms of transmembrane ionic diffusions among nerve cells, the brain system may possibly use another but much more microscopic and elaborated fundamental physical process in terms of coherent photon emission and transfer in the microtubule just like the optical computer with lasers, optical fibers and other optical devices. It is this very surprising possibility we are going to discuss throughout the present paper.

However, it is difficult to look for the possibility of realizing laser devices in brain cells even though the microtubule may manifest the same quantum dynamical behavior, in terms of water molecules and the electromagnetic field, as is governed by the Hamiltonian of a typical laser system. This is because we need initially to pump up the majority of water molecules to the higher energy eigenstate by certain incoherent but high intensity light. In actual artificial laser devices, for example, the initial pumping is due to xenon flash lamps. In other words, the laser system cannot emit coherent photons without some pumping mechanism. We can hardly expect the existence of such a pumping light in the brain. We are, therefore, forced to look for another possibility with the total Hamiltonian (Eq. 10) leading to a different mechanism of coherent photon emission without pumping light.

Fortunately, a physical inspection of the form of the total Hamiltonian (Eq. 10) reveals that it manifests a dynamical symmetry property not evident in the ground state and so the resulting quantum dynamics is known to involve certain long-range order creating phenomena due to spontaneous symmetry breaking (Stuart et al., 1979). The spatial dimension of this long-range order, that is, the coherence length l_c can be estimated to be inversely proportional to the energy difference ϵ, obtaining $l_c \approx$ hundreds of micrometers. Among those long-range order creating phenomena we may find a specific one in which the collective dynamics of the majority of water molecules inside the microtubule cylinder V can give rise to cooperative spontaneous emission of photons without any pumping light. Any incoherent and disordered energy distribution among the water molecules due to the macroscopic thermal dynamics of the polymerized array of protein subunits (i.e. tubulins) forming the wall of microtubule cylinder can be gathered collectively into coherent and ordered dynamics ready to emit coherent photons cooperatively. This laser-like process of coherent photon emission without pumping light was first introduced by Dicke (1954) and called superradiance.

Let us investigate the superradiance in the microtubule cylinder V starting from the total Hamiltonian (Eq. 10). We assume for simplicity that only one normal mode with a specific wave vector, say $\boldsymbol{k}_0$, has a proper angular frequency $\omega_{\boldsymbol{k}_0}$ resonating to the energy difference ϵ between the two principal energy eigenstates. Namely we have

$$\epsilon = \hbar \omega_{\boldsymbol{k}_0} \qquad (11)$$

and all the other normal modes are neglected. In the conventional laser theory, this is known as a single mode laser. What we analyse below may hence be considered as a single mode superradiance in the microtubule.

Since we have only one normal mode with wave

vector k_0, we may omit all the wave vector indices of the dynamical variables. Then, the total Hamiltonian (Eq. 10) becomes

$$H = H_{EM} + \epsilon S - \mu(E^- S^- + S^+ E^+). \tag{12}$$

The corresponding Heisenberg equations of motion for the three collective dynamical variables, S and $S^\pm$, for water molecules and the two variables, $E^\pm$, for the quantized electromagnetic field are given by:

$$\frac{dS}{dt} = -i\frac{\mu}{\hbar}(E^- S^- - S^+ E^+) \tag{13}$$

$$\frac{dS^\pm}{dt} = \pm i\frac{2\mu}{\hbar} S E^\mp \pm i\frac{\epsilon}{\hbar} S^\pm \tag{14}$$

and

$$\frac{dE^\pm}{dt} = \pm i\frac{2\pi\epsilon\mu}{\hbar V} S^\mp \tag{15}$$

Because of the short length of the microtubule cylinder — $l_{MT} \approx 10^2$–10^3 nm — the pulse mode propagating along the microtubule cylinder in the direction of the z-axis stays in the cavity region V only for a short transit time $t_{MT} \equiv l_{MT}/c$, where c stands for the speed of light. As this transit time of the pulse mode is much shorter than the characteristic time of thermal interaction due to the disordered environment, the system of water molecules and quantized electromagnetic field in this single mode superradiance is free from thermal loss and can be considered as a closed system well-described by the Heisenberg equations of motion (Eqs. 13–15). Furthermore, the time derivative of the dynamical variables $E^\pm$ of the quantized electromagnetic field can be approximated by $E^\pm/l_{MT}$ in the case of a pulse mode propagating along the longitudinal axis of the microtubule. Eq. 15 then yields

$$E^\pm = \pm i\frac{2\pi\epsilon\mu l_{MT}}{\hbar V} S^\mp \tag{16}$$

This means that a pulse mode of the quantized electromagnetic field in the microtubule cylinder cavity follows the collective dynamics of water molecules inside the microtubule. In other words, once a collective mode with long-range order is created in the dynamics of water molecules due to spontaneous symmetry breaking, coherent emission of pulse modes of the quantized electromagnetic field — that is, photons — follows. This is the mechanism of superradiance.

The last question is whether such a collective mode can be realized in the dynamics of water molecules starting from incoherent and disordered initial conditions. Notice that such an incoherent and disordered initial dynamical configuration of water molecules is due to the interaction between water molecules and thermally disordered states of the conformational dynamics of tubulins. The onset of this collective mode can be seen by rewriting the three Heisenberg equations for the collective variables of water molecules (Eqs. 13 and 14) by substituting Eq. 16. Namely, we have

$$\frac{dS^\pm}{dt} = \beta S S^\pm \pm i\epsilon S^\pm \tag{17}$$

and

$$\frac{dS}{dt} = -\beta S^+ S^- \tag{18}$$

where

$$\beta = \frac{4\pi\epsilon\mu^2 l_{MT}}{\hbar^2 V}.$$

These are coupled non-linear differential equations for non-commuting operators S and $S^\pm$ subject to certain commutation relations, and it is not so easy to find their solutions. However, if we regard these equations as ordinary coupled non-linear differential equations for classical (i.e. commuting) dynamical variables, we can find special solutions with respect to any incoherent and disordered initial conditions. They are known as semi-classical approximations of the quantum

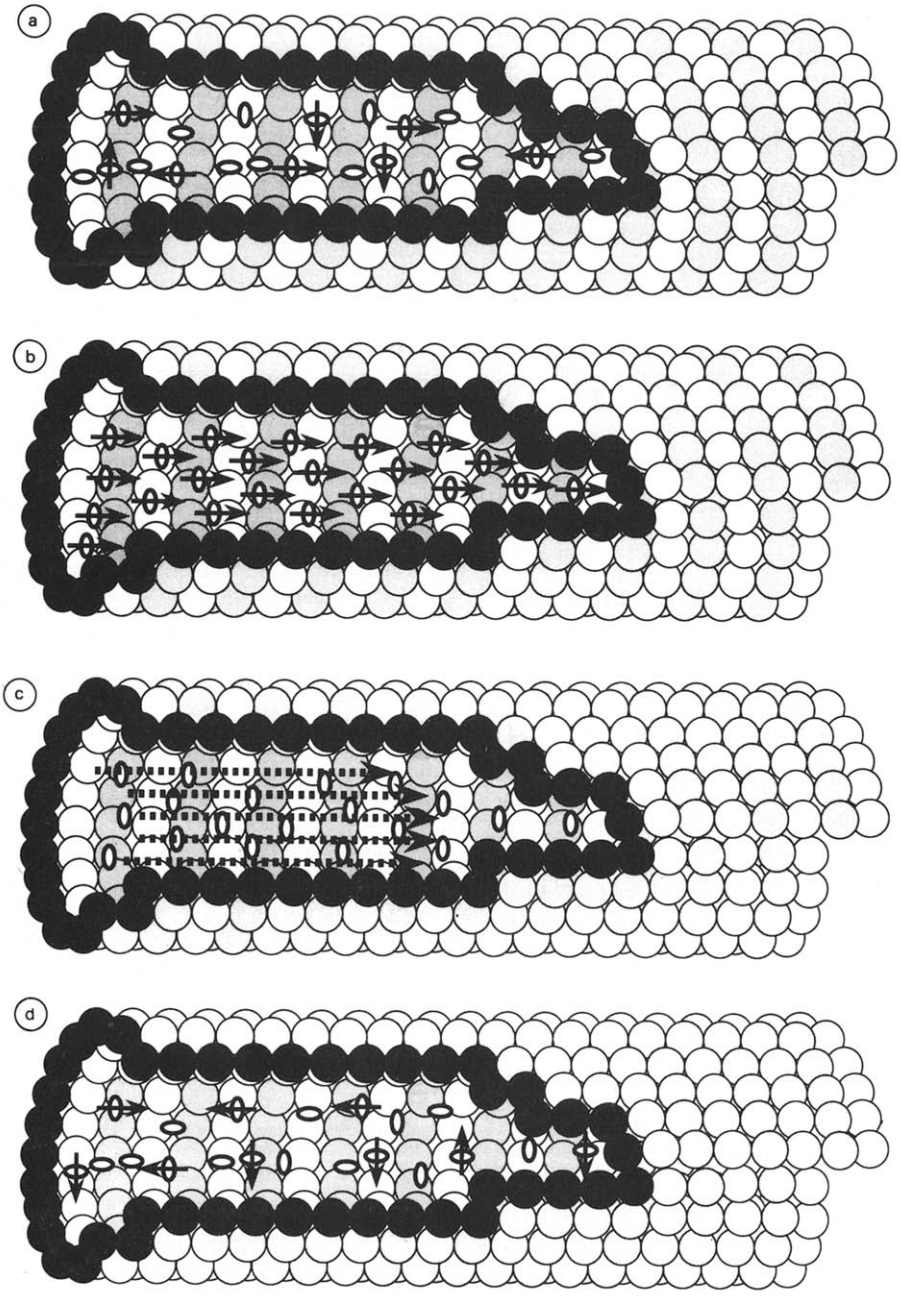

Fig. 5. A schematic representation of the process of superradiance in a microtubule. Each oval without an arrow stands for water molecule in the lowest rotational energy state. Each oval with an arrow stands for a water molecule in the first excited rotational energy state. The process is cyclic (a → b → c → d → a → b), and so on. (a) Initial state of the system of water molecules in a microtubule. Energy gain due to the thermal fluctuation of tubulins increases the number of water molecules in the first excited rotatinal energy state. (b) A collective mode of the system of water molecules in rotationally excited states. A long-range coherence is achieved inside a microtubule by means of spontaneous symmetry breaking. (c) A collective mode of the system of water molecules in rotationally excited states loses its energy collectively, and creates coherent photons in the quantized electromagnetic field inside a microtubule. (d) Water molecules, having lost their first excited rotational energies by superradiance, start again to gain energy from the thermal fluctuation of tubulins, and the system of water molecules recover the initial state (a).

dynamical system of superradiance (Agarwal, 1971). A straightforward calculation yields that the intensity of coherent photon emission in the microtubule cylinder due to superradiance can be given in this approximation by

$$I = \frac{\hbar^2}{(4t_R\mu)^2}\mathrm{sech}^2\frac{t - t_0}{2t_R} \tag{19}$$

where

$$t_R = \frac{c\hbar^2 V}{4\pi\mu^2\epsilon N l_{MT}}$$

and $t_0 = t_R \ln 2N$ denotes the life time and delay of the superradiance, respectively. Notice that the intensity of superradiance is proportional to N^2 and its delay time is inversely proportional to N. These facts are characteristic to the long-range order creating process involving N water molecules (see Fig. 5).

We have found that the quantum collective dynamics of water molecules and a quantized electromagnetic field inside the microtubule cylinder manifests the long-range cooperative phenomenon of superradiance in which collective excitation of water molecules can be induced by incoherent and disordered perturbations due to the macroscopic thermal dynamics of protein molecules forming the wall of the microtubule cylinder. This fact ensures that each microtubule in the cytoskeletal structure of brain cells, that is, neurons and astrocytes, may play an important role in the optical information processing regimen of brain function as a superradiant device which converts the macroscopic disordered dynamics of water molecules and protein molecules into the long-range ordered dynamics of water molecules and a quantized electromagnetic field involving a pulse mode emission of coherent photons. In other words, each microtubule is a coherent optical encoder in a dense microscopic optical computing network in the cytoplasm of each brain cell, if such a network is realized in actual cytoplasmic structure. This last point is far from evident and deserves to be discussed in the following section.

3. Self-induced transparency in microtubules

We have shown the possibility of a completely new mechanism of fundamental brain functioning in terms of coherent photon emission by superradiance in microtubules. Unlike a laser, superradiance is a specific quantum mechanical ordering process with a characteristic time much shorter than that of thermal interaction. Therefore, microtubules may be thought of as ideal optical encoders providing a physical interface between (1) the conventional macroscopic system of classical, disordered and incoherent neural dynamics in terms of transmembrane ionic diffusions as well as thermally perturbed molecular vibrations and (2) the as yet unknown microscopic optical computing network system of ordered and coherent quantum dynamics free from thermal noise and loss. As Feynman (1985) proposed, the optical computing network is the most realizable quantum mechanical computer among many possibilities such as a superconducting computer. However, it is not clear whether the pulse mode coherent photons created in the microtubule cylinder by superradiance can be safely transmitted, preserving its long-range coherence. It seems most likely that even coherent photons emitted by superradiance will lose immediately their coherence and long-range order due to the noisy thermal environment. In any case, we absolutely require some quantum dynamical mechanism to maintain the coherent transmission of photons in the as yet unknown microscopic optical computing network in brain cells. In this section, we will show that it is again the microtubule in the cytoskeletal structure of brain cells which provides us with such a mechanism.

Let us suppose that the pulse mode coherent photons are created in a small segment of the microtubule cylinder by superradiance. Then, those photons propagate along the longitudinal axis of the microtubule cylinder, that is, the z-axis. If the region V inside the microtubule cylinder were maintained at vacuum, they would transmit through the region just as they do along a waveguide. However, the region is filled up with water molecules, and it is not evident at all whether the pulse mode coherent photons can be

safely transmitted through the region without absorption or loss of coherence.

For the purpose of describing the transmission of pulse mode coherent photons along the region inside the microtubule cylinder, it seems convenient to work with a semi-classical approximation in which the long-range ordered dynamics of water molecules is described by quantum mechanics but the electromagnetic field is described classically. This approximation becomes reliable when the intensity of the pulse mode coherent photons is large, and this is indeed the case for superradiance emission (Feynman et al., 1957).

Let us consider the Maxwell equation for the scalar electric field $E = E(z,t)$, standing for the pulse mode coherent photons propagating along the z-axis coupled to the collective dynamical variables $S^{\pm}$ of water molecules inside the microtubule cylinder,

$$\frac{\partial E^{\pm}}{\partial z} + \frac{\partial E^{\pm}}{\partial t} = \mp i \frac{2\pi\epsilon\mu}{\hbar V} S^{\mp} \tag{20}$$

This equation is valid under the condition that the collective dynamical variables $S^{\pm}$ are slowly varying, that is,

$$\frac{\partial S^{\pm}}{\partial t} \ll i\omega S^{\pm} \tag{21}$$

Then, taking the expectation of those quantum mechanical variables in this Maxwell equation and the Heisenberg equations of motion (Eqs. 13 and 14), eliminating all the expectation values of those variables referring to water molecules, and introducing new variables for the scalar electric field by

$$\theta^{\pm}(z,t) = \frac{2\mu}{\hbar} \int_{-\infty}^{t} E^{\pm}(z,u)\mathrm{d}u \tag{22}$$

we can obtain dynamical equations for the electromagnetic field in the region V inside the microtubule cylinder

$$\frac{\partial^2 \theta^{\pm}}{\partial\tau\partial\zeta} = -\sin\theta^{\pm} \tag{23}$$

Here,

$$\tau = \sqrt{\frac{2\pi\epsilon\mu^2 N}{\hbar^2 V}} \left(t - \frac{z}{c}\right)$$

and

$$\zeta = \sqrt{\frac{2\pi\epsilon\mu^2 N}{\hbar^2 V}} \frac{z}{c}$$

This is a typical non-linear partial differential equation called the Sine–Gordon equation, and several exact solutions are obtained by means of the inverse scattering method of soliton equations (Ablowitz et al., 1974).

The most interesting solution of the Sine–Gordon equation (Eq. 23) gives rise to an explicit form of the time evolution of the scalar electric field E in the region V inside the microtubule cylinder

$$E = \sqrt{\frac{2\pi\epsilon N v_0}{V(c - v_0)}} \operatorname{sech} \sqrt{\frac{2\pi\epsilon\mu^2 N v_0}{\hbar^2 V(c - v_0)}} \left(t - \frac{z}{v_0}\right) \tag{24}$$

This is nothing but a soliton solution and tells us that the pulse mode photons propagate along the dielectric waveguide of the microtubule cylinder filled up with water molecules with a certain constant speed v_0 less than the speed of light in vacuum c. It is important to see that the pulse form of the soliton solution is kept unchanged due to the non-linearity of the Sine–Gordon equation.

We have found that microtubules play the role of dielectric waveguides and that pulse mode coherent photons propagate through them as if they were perfectly transparent. This phenomenon is termed self-induced transparency and known to be a typical non-linear effect in quantum optics (McCall and Hahn, 1967). The microtubule may be an ideal microscopic optical device for use as a perfectly transparent pathway for pulse mode photons, free from thermal noise and loss. Combined with superradiance, this self-induced transparency of the microtubule allows us to conclude that the brain is essentially a dense assembly of

microscopic and elaborated optical computing networks of microtubules in the cytoskeletal structure of brain cells. Coherent photon emission and transfer in each microtubule are ensured by superradiance and self-induced transparency characteristic to long-range ordering phenomena in quantum dynamics.

Both superradiance and self-induced transparency may be essential to cooperative physical activity of multiple microtubules in cytoskeletal structure in single cells and among many cells. Thus biological systems in general and the brain in particular may utilise a quantum level holographic information system (Hameroff, 1987; Pribram, 1991).

4. Implications of brain function

We have developed a theoretical framework for the occurrence of quantum coherence of photons in microtubules and their associated water. Fundamental involvement of microtubules and the cytoskeleton in orchestration and control of 'real time' cell functions suggests that such quantum coherence, if it exists, could play a major role in cellular communication, signalling and information processing (Conrad, 1988).

Several aspects of quantum coherence in microtubules may be considered:

1. Superradiance and self-induced transparency occurring in ordered water within the hollow core of cylindrical microtubules behaving as waveguides will result in coherent photons. This coherence, estimated to be capable of superposition of states among microtubules spatially distributed over hundreds of micrometers, which in turn are in superposition with other microtubules hundreds of micrometers away in other directions and so on, could account for a coupling of microtubule dynamics over wide areas. This in turn could account for a unity of thought and consciousness. For example, Insinna (1992) has attributed to quantum coherence in microtubules the phenomenon of Jungian synchronicity.
2. Coherent interference leading to holographic information representation would result from quantum optical effects. At the cytoskeletal level, this could occur via calcium induced sol-gel variations causing polymerization of actin or other cytoskeletal filaments resulting in a holographic representation of short-term memory (Hameroff, 1987). The holographic paradigm suggests coherence and holographic interference across multiple scales (Pribram, 1991). At a more 'macro' level, coherence of neuronal firing among widely dispersed neurons in the 40-Hz range may be related to quantum coherence of microtubules within coherently firing neurons (Singer, 1993).
3. Sequelae of quantum events in individual microtubule subunits, acoustoconformational vibrations could propagate as signals through cytoskeletal networks and membranes and extracellular matrices. Such proposed signals (in the context of the cellular automata theory) have been shown theoretically capable of learning, logic and adaptation (Rasmussen et al., 1990; Lahoz-Beltra et al., 1993).

We have described a complex energy/information field originating in cytoskeleton and associated water and cooperatively pervading other biostructures. 'Unifying field theories' have a long and unsuccessful history in biology and biophysics and connote a derisive term: 'vitalism'. We introduce a neo-vitalistic, quantum coherent holographic field based on microtubule-derived Goldstone bosons as a medium of biological information, communication and representation of external world. In its largest 'brain-wide' scale, this should constitute consciousness.

Fueled by biochemical energy from ATP and GTP hydrolysis and protein phosphorylation, microtubule quantum coherence and self-focussing should be sensitive to general anesthesia. Anesthetic gas molecules reversibly inhibit consciousness by weak, van der Waals binding in hydrophobic regions of proteins (Franks and Lieb, 1982). If quantum optical coherence in microtubules is essential for consciousness, anesthetics must somehow inhibit it. One possible way is by membrane inhibition which could disconnect and isolate cytoskeletal dynamics from the external

milieu. Alternatively, anesthetics may directly alter cytoskeletal quantum coherence. General anesthetics do bind to microtubules and at high enough concentrations cause depolymerization (Allison and Nunn, 1968). At concentrations of anesthesia just sufficient for ablation of consciousness, inhibition of tubulin conformational dynamics could have profound effects. Wulf and Featherstone (1957) showed that anesthetic binding within protein hydrophobic regions altered protein-water binding at the protein surface. Thus, slight anesthetic effects on tubulin conformational dynamics and cooperative water binding would have significant effects on quantum coherence. Anesthesia is also known to be reversed by pressure (Halsey, 1976). By condensing a non-linear optical medium, pressure increases quantum cooperative coherent phenomena. Thus our model is consistent with the mechanism of general anesthesia.

5. Summary

This paper presents a theoretical framework for quantum optical coherence in cytoskeletal microtubules. Specifically, the quantum dynamical system of water molecules and the quantized electromagnetic field confined inside the hollow microtubule core manifests a specific collective dynamics called superradiance by which coherent photons are created inside the microtubule. Furthermore, such coherent photons created by superradiance penetrate perfectly along the internal hollow core of the microtubule as if the optical medium inside it were made transparent by the propagating photons themselves. This quantum effect is called self-induced transparency. Such effects can account for cellular signalling, holographic information processing and unity of consciousness.

6. Acknowledgements

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Relationship between Fröhlich and Davydov models of biological order

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Two physical models to explain the unusual behavior of biological systems put forward by Fröhlich and Davydov are reviewed. The Davydov Hamiltonian is transformed into normal coordinates and the Fröhlich Hamiltonian is canonically transformed into an equivalent form within the Hartree-Fock approximation. A model Hamiltonian capable of relating both theories is proposed. The resulting analysis employs only one type of second-quantized operators which describe the effect on the active dipolar modes of vibration in biological cells. An equation of motion is derived for the creation and annihilation operators that are used. When the phenomenon of Bose condensation is taken into account, the field translation variable is shown to obey a nonlinear Schrödinger equation. Its solutions are provided and their significance is discussed. Some of the solutions are shown to correspond to properties already examined by Fröhlich and Davydov. New solutions are found and interpreted as interfaces between either an ordered and a disordered phase or two ordered phases existing in the biosystem. The relationship between the two theories of biological order is demonstrated and discussed.

I. THE FRÖHLICH MODEL

Fröhlich presented, in a series of papers,[1-7] a quantum theory of long-range coherence in biological cells predicting the existence of coherent elastic vibrations in the microwave region (10^{11}–10^{12} Hz). The effect presupposes

(i) the existence of a large transmembrane potential difference,

(ii) a continuous supply of metabolic energy within a rate of supply exceeding a critical value, and

(iii) internal organization of the biosystem resulting from a spatial arrangement and a sufficient strength for the coupling constants between the component parts.

The biosystem is schematically divided into three interacting subsystems (Fig. 1):

(i) an oscillating segment of giant dipoles located along the chain formed by macromolecules,

(ii) a heat bath (containing all water and suspended ions), and

(iii) an external source of energy supply.

Following Fröhlich,[7] Wu and Austin[8-11] introduced the following Hamiltonian to adequately describe the above situation:

$$H=\sum_i \omega_i a_i^\dagger a_i+\sum_i \Omega_i b_i^\dagger b_i+\sum_i \theta_i P_i^\dagger P_i + \tfrac{1}{2}\sum_{i,j,k}(\chi a_i^\dagger a_j b_k^\dagger+\chi^* a_j a_i^\dagger b_k) + \sum_{i,j}(\lambda b_i a_j^\dagger+\lambda^* b_i^\dagger a_j)+\sum_{i,j}(\xi P_i a_j^\dagger+\xi^* P_i^\dagger a_j)\,, \quad (1)$$

where $(a_i^\dagger,a_i)$, $(b_i^\dagger,b_i)$, and $(P_i^\dagger,P_i)$ are, respectively, the cell, heat-bath, and energy-pump creation and annihilation (boson-type) operators.

Provided that the oscillating dipoles are within a narrow band of resonance frequencies ($\omega_{\min}\le\omega_i\le\omega_{\max}$) and that the coupling constants χ, λ, and ξ are large enough, the following effects are expected to occur.

(i) The existence of a strong, long-range (~ 1 μm) attractive forces between the dipoles. The effective Fröhlich potential between two interacting dipoles which vibrate with frequencies $\omega_1=\omega_2=\bar{\omega}$ is given by[4]

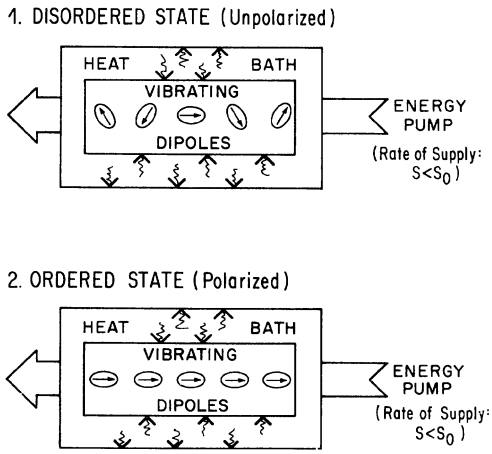

FIG. 1. Illustration of the Fröhlich model.

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$$U(r)=-\frac{E}{r^3}+\frac{F}{r^6}\ , \tag{2}$$

where F is the London—van der Waals coefficient and E is the long-range constant given by[4]

$$E=\frac{\hbar e^2 Z\,|\,\bar{A}\,|\,\gamma}{4\bar{\omega}M}\left[\frac{1}{\epsilon'(\omega_+)}-\frac{1}{\epsilon'(\omega_-)}\right], \tag{3}$$

where M is the mass of a dipole, r is the lattice spacing, e is the electron charge, Z is the number of elementary charges on each dipole, and $\bar{A}$ is an angle constant. The quantities $\omega_\pm$ are the new dipole vibration frequencies which arise in the system as a result of the interactions

$$\omega_\pm=\left[\tfrac{1}{2}(\omega_1^2+\omega_2^2)\pm\left[\tfrac{1}{4}(\omega_1^2-\omega_2^2)+\frac{\beta_0^4}{(\epsilon'_\pm)^2}\right]^{1/2}\right]^{1/2}, \tag{4}$$

where $\beta_0=\gamma^2e^2\sqrt{Z_1Z_1}/Mr^3$ and γ is a coupling constant. Paul *et al.*[12,13] presented a detailed derivation of the dielectric constant ϵ using the Wu-Austin Hamiltonian, Eq. (1), and obtained

$$\epsilon'(\omega)=1+\frac{4\pi^2}{kT}\rho d^2 f(\omega_{ij}\,n_i)\,|\,\chi\,|^2\ , \tag{5}$$

where ρ is the density of the substance, d is the average dipole moment, and f is a complicated function of the eigenfrequencies ω_i and occupation numbers n_i. Chatterjee *et al.*[14] applied this result to cholesterol and found $\chi\sim5\times10^{-3}$.

(ii) Bose condensation of the vibrational modes ω_i to one with the lowest allowed frequency ω_0. This can be viewed as an energy redistribution in a coherent state with $\omega=\omega_0$, which is a polarized, metastable state and, consequently, has a nonzero dielectric polarization: $\langle\vec{P}\rangle\neq\vec{0}$. Further, Paul[15] showed that the nonzero polarization is a manifestation of symmetry breaking associated with Bose condensations. He also demonstrated using Eq. (1) that $\langle\langle a_l\rangle\rangle\neq0$ and $\langle\langle a_l^\dagger\rangle\rangle\neq0$, where $\langle\langle\ \rangle\rangle$ is the average employing the Hamiltonian H of Eq. (1). Since $\vec{d}_l\propto x_l\propto(a_l^\dagger+a_l)$, $\langle\langle\vec{d}_l\rangle\rangle\neq\vec{0}$, where $\vec{d}_l$ is the dipole moment connected with the lth mode.

Experimental evidence in support of Fröhlich's conjecture about the existence of nonthermal excitations was found in Raman scattering studies of *Escherischia coli*[16,17] and *Chlorella pyrenoidosa*.[18] An indirect effect of increased yeast growth upon irradiation by specific frequencies of millimeter waves[19,20] seems to indicate the importance of the Fröhlich-type excitations in growth control. Moreover, the interactions of human erythrocytes leading to the effect of rouleau formation can be understood using the Fröhlich theory.[21–26] In fact, Paul *et al.*[12] examined this phenomenon in detail and found good consistency with the theoretical model.

One question that has not been addressed in the studies involving the Fröhlich model is the propagation of order (polarization) throughout the system. However, a seemingly unrelated theory of quantum interactions in biological systems has been proposed by Davydov specifically to deal with this problem. We shall review the Davydov model in Sec. II.

II. THE DAVYDOV MODEL

Davydov and his associates, in a series of papers[27–32] and a book,[33] presented a model to explain the almost lossless energy transfer in quasi-one-dimensional biological substances such as protein and peptide chains and deoxyribonucleic acid (DNA). As an example, consider a peptide chain formed by H—N—C=O groups. Each peptide group has a constant electric dipole moment $d\sim0.4$ debye. The neighboring groups interact through a dipole-dipole potential, Eq. (2), whose coefficients differ from those given by Fröhlich since the dipoles involved have no freedom of oscillations. The Davydov constant of attraction is

$$E=(3\cos^2\theta-1)d^2\ , \tag{6}$$

where θ is the angle between the dipole $\vec{d}$ and the chain axis. The constant E reaches its maximum $2d^2$ when $\vec{d}$ is parallel to the axis. The excitation energy required for an electron to overcome the potential barrier is ~0.4 eV and electron transfer is facilitated by the presence of hydrogen bonds between the neighboring peptide groups. Furthermore, Davydov assumed strong coupling between the internal C=O vibrations and the elastic deformations of the chain.

The total Hamiltonian is written in the Heitler-London approximation as[27,28]

$$H=\tfrac{1}{2}\sum_n(M\dot{x}_n^2+\omega\rho_n^2)+\sum_n\left[(\epsilon-D_n)B_n^\dagger B_n-U(B_{n+1}^\dagger B_n+B_{n+1}B_n^\dagger)\right], \tag{7}$$

where

$$\rho_n=r-|\,x_n-x_{n-1}\,|\ , \tag{8}$$

$$D_n=D\left[1+\frac{\beta}{r}\rho_n\right], \tag{9}$$

and D,β are positive parameters of the theory, and x_n is the x coordinate of the nth site of the chain. The electronic excitation operators B_n and $B_n^\dagger$ obey the Bose-

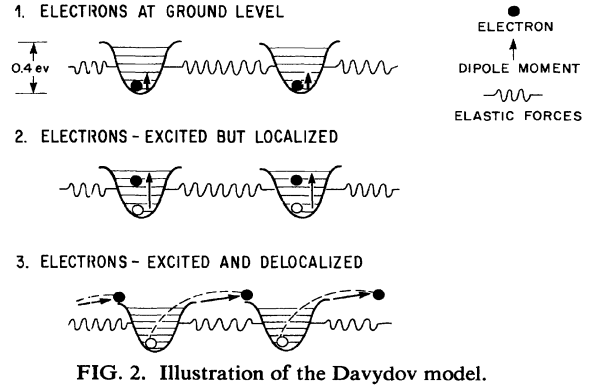

FIG. 2. Illustration of the Davydov model.

Einstein commutation relations.

Next, Davydov derived a nonlinear Schrödinger equation for the amplitude of the collective, electronic excitation $a(t)$ by using the Hamiltonian given above and the corresponding time-dependent Schrödinger equation for the wave function $|\phi(t)\rangle$:

$$i\dot{a}(t)-\lambda a(t)+U\nabla^2 a(t)+G_v|a(t)|^2 a(t)=0 . \tag{10}$$

Here, $|\phi(t)\rangle=\sum_n a_n(t)B_n^\dagger|0\rangle$ is a wave function of the total system and the remaining coefficients are

$$\lambda=\epsilon+T+U_H-D-2U$$

and

$$G_v=\frac{\beta^2 D^2}{\omega r^2[1-(v/V)^2]}$$

with U_H representing the potential energy and T the kinetic energy of the lattice vibrations. The solutions of Eq. (10) are solitary waves of permanent profile which propagate along the chain with constant velocity v (v is less than the velocity of sound V).

Davydov's model was further elaborated by Scott[34] and applied to study the properties of the green algae: *Chlorella pyrenoidosa*.[35] The contraction mechanism of tranversely strained animal muscles[36] was also elucidated with the use of this model. Moreover, it has recently been suggested[37] that this is applicable to the internal pulse in blood vessels.

We shall next explore the relationship between the Fröhlich and Davydov models. In order to compare the respective Hamiltonians [Eqs. (1) and (7)], certain transformations will be made to represent them in a compatible fashion. As a first step, we write the Davydov Hamiltonian using normal coordinates.

III. THE TRANSFORMATION OF DAVYDOV'S HAMILTONIAN TO NORMAL COORDINATES

The most apparent difference between the Wu-Austin and Davydov Hamiltonians is that the former employs normal coordinates whereas the latter is written in position coordinates. The transition from one set of coordinates to the other reflects the change from a localized to a collective picture of the excitations in the system. Davydov's Hamiltonian is transformed in the following steps.

(i) The potential energy of elastic vibrations is rewritten as

$$U_H=\frac{Nwr^2}{2}-wr\sum_{n=1}^{N}|x_n-x_{n-1}| +w\sum_{n=1}^{N}x_n^2-w\sum_{n=1}^{N}x_n x_{n-1} . \tag{11}$$

Since this represents a harmonic approximation, the length of the chain L is constant:

$$L=\sum_{n=1}^{N}|x_n-x_{n-1}| . \tag{12}$$

Using periodic boundary conditions the total energy of elastic vibrations may be written as

$$H_{\rm el}=\sum_{n=1}^{N}\left[\frac{p_n^2}{2M}+\sum_{n'=1}^{N}V_{nn'}x_n x_{n'}\right]+V_0 , \tag{13}$$

where p_n denotes the momentum operator at the nth site and

$$V_{nn'}=\begin{cases}2w & \text{if } n'=n\\ -w & \text{if } n'=n\pm1\\ 0 & \text{otherwise .}\end{cases} \tag{14}$$

The constant energy shift V_0 is

$$V_0=NrW\left|\frac{r}{2}-\frac{L}{N}\right| . \tag{15}$$

Following a standard procedure,[38] the variables x_n and p_n are Fourier transformed to obtain

$$H_{\rm el}=\sum_q\left[\frac{1}{2M}p_q p_q^\dagger+\tfrac{1}{2}M\Omega_q^2 x_q x_q^\dagger\right]+V_0 . \tag{16}$$

Here, x_q and p_q are the Fourier transforms of x_n and p_n, respectively, and $\Omega_q=2(W/M)^{1/2}\sin(q/2)$. Finally, defining the phonon annihilation operator as

$$b_q\equiv(2M\Omega_q)^{-1/2}(M\Omega_q x_q+ip_q^\dagger) \tag{17}$$

and the creation operator $b_q^\dagger$ as the Hermitian conjugate of b_q, a fully diagonalized, second-quantized operator for $H_{\rm el}$ is obtained:

$$H_{\rm el}=\sum_q\Omega_q(b_q^\dagger b_q+\tfrac{1}{2})+V_0 . \tag{18}$$

(ii) The diagonal part of the electron excitation energy

$$H_{\rm ex}=\sum_n[\epsilon-D(1+\beta)]B_n^\dagger B_n \tag{19}$$

can be transformed into normal coordinates by introducing the operators a_q and $a_q^\dagger$ as the Fourier transforms of the operators B_n and $B_n^\dagger$, respectively:

$$H_{\rm ex}=\sum_q\bar{\omega}_q a_q^\dagger a_q , \tag{20}$$

where $\bar{\omega}_q=\epsilon-D(1+\beta)$. Note that $\bar{\omega}_q$ is independent of q.

(iii) The off-diagonal part of the electronic excitation energy

$$H_{\rm off}=-\sum_n U(B_{n+1}^\dagger B_n+B_{n+1}B_n^\dagger) \tag{21}$$

is transformed in the same manner as $H_{\rm ex}$ into

$$H_{\rm off}=-\sum_q U(2\cos q\, a_q^\dagger a_q+1) . \tag{22}$$

Now, $H_{\rm ex}$ and $H_{\rm off}$ are combined to form the total electronic contribution to the Hamiltonian:

$$H_{\rm exc}=H_{\rm ex}+H_{\rm off}=\sum_q(\omega_q a_q^\dagger a_q-U) , \tag{23}$$

where the new energy eigenvalues are $\omega_q=\epsilon-D(1+\beta)-2U\cos q$. Note that ω_q is an even

function of q and $(\partial\omega_q/\partial q)_{q=0}=0$.

(iv) The remaining term of Eq. (7) represents the interactions between lattice vibrations and electronic excitations:

$$H_{\rm int}=\sum_n D\frac{\beta}{r}\,|x_n-x_{n-1}|\,B_n^\dagger B_n\ , \tag{24}$$

and, with the use of the Fourier transformations introduced before,

$$H_{\rm int}=\sum_{p,q,r}\chi_{pqr}(b^\dagger_{-p}+b_p)a_q^\dagger a_r\ , \tag{25}$$

where the coupling constant is

$$\chi_{pqr}=\frac{2D\beta}{r}\left[\frac{w}{N^2M^3}\right]^{1/4}\delta(q,r)[\sin(q/2)]^{3/2}\ .$$

Finally, we write the entire Davydov Hamiltonian, Eq. (7), in normal coordinates as

$$H=H_0+\sum_q\Omega_q b_q^\dagger b_q+\sum_q\omega_q a_q^\dagger a_q+\sum_{p,q,r}\chi_{pqr}(b^\dagger_{-p}+b_p)a_q^\dagger a_r\ , \tag{26}$$

where

$$H_0=N\left[rw\left|\frac{r}{2}-\frac{L}{w}\right|-U\right]+\tfrac{1}{2}\sum_q\Omega_q\ , \tag{27}$$

$$\Omega_q=2\sqrt{w}/M\sin(q/2)\ , \tag{28}$$

$$\omega_q=\epsilon-D(1+\beta)-2U\cos q\ , \tag{29}$$

$$\chi_{pqr}=\frac{2D\beta}{r}\left[\frac{w}{N^2M^3}\right]^{1/4}[\sin(q/2)]^{3/2}\delta(q,r)\ . \tag{30}$$

In Sec. IV we transform the Wu-Austin Hamiltonian, Eq. (1), to a form which is essentially identical to Eq. (26). As a result, we will be able to connect the two models by a single Hamiltonian.

IV. THE CANONICAL TRANSFORMATION OF THE WU-AUSTIN HAMILTONIAN

By comparing the normal coordinate representation of the Davydov Hamiltonian, Eq. (26), with the Wu-Austin Hamiltonian, Eq. (1), many similarities may be seen. However, the main differences that the Wu-Austin Hamiltonian involves are the interactions of the cell phonons with the energy pump (i.e., the terms proportional to $P_i a_j^\dagger$ and $P_i^\dagger a_j$) as well as first-order interactions with the heat bath (i.e., the terms proportional to $b_i a_j^\dagger$ and $b_i^\dagger a_j$). Fortunately, this problem may be eliminated by a canonical transformation of the Wu-Austin Hamiltonian and obtaining an equivalent operator in a form identical to the Davydov Hamiltonian.

Assuming that the *second-order* cell-phonon–heat-bath interactions are sufficiently small perturbations, we neglect them for the moment and rewrite Eq. (1) as

$$H=H_0+H_I\ , \tag{31}$$

where H_0 is the unperturbed part of the Hamiltonian

$$H_0=\sum_i\omega_i a_i^\dagger a_i+\sum_i\Omega_i b_i^\dagger b_i+\sum_i\theta_i p_i^\dagger p_i \tag{32}$$

and H_I contains the *first-order* perturbation terms:

$$H_I=\sum_{i,j}(\lambda b_i a_j^\dagger+\lambda^* b_i^\dagger a_j)+\sum_{i,j}(\xi P_i a_j^\dagger+\xi^* P_i^\dagger a_j)\ . \tag{33}$$

In order to effectively *decouple* the cell operators $a_i^\dagger, a_i$ from the heat-bath operators $b_i^\dagger, b_i$ and the energy-pump operators $P_i^\dagger, P_i$, we use the following unitary transformation:

$$\tilde{H}=e^{-S}He^{+S} \tag{34}$$

with the matrix S defined as

$$S=\sum_{i,j}(\mu b_i^\dagger a_j+\upsilon b_i a_j^\dagger)+\sum_{i,j}(\Delta P_i a_j^\dagger+\sigma P_i^\dagger a_j)\ . \tag{35}$$

Applying the Baker-Hausdorff formula

$$e^{-S}He^{+S}=H_0+H_I+[H_0,S]+[H_I,S]+\tfrac{1}{2}[[H_0,S],S]+\tfrac{1}{2}[[H_I,S],S]+\cdots \tag{36}$$

together with the requirement

$$H_I=-[H_0,S] \tag{37}$$

yields the following values of the transformation coefficients:

$$\mu=\frac{\lambda}{\Omega_i-\omega_j},\quad \upsilon=\frac{\lambda^*}{\omega_j-\Omega_i}\ , \tag{38}$$

$$\Delta=\frac{\xi}{\theta_i-\omega_j},\quad \sigma=\frac{\xi^*}{\omega_j-\theta_i}\ . \tag{39}$$

Therefore, $\sigma=-\Delta^*$ and $\upsilon=-\mu^*$. Consequently, the series expansion, Eq. (36), is evaluated up to the fifth term as

$$\tilde{H}=H_0+\Delta H\ , \tag{40}$$

where ΔH is the term representing the net effect of the first-order perturbations

$$\Delta H=\tfrac{1}{2}\sum_{i,j}\left\{\sum_k\left[|\xi|^2\left[\frac{1}{\omega_i-\theta_k}+\frac{1}{\omega_j-\theta_k}\right]+|\lambda|^2\left[\frac{1}{\omega_i-\Omega_k}+\frac{1}{\omega_k-\Omega_k}\right]\right]\right\}a_i^\dagger a_j$$

$$+\tfrac{1}{2}\sum_{i,j}\left\{\sum_k\left[|\xi|^2\left[\frac{1}{\theta_i-\omega_k}+\frac{1}{\theta_j-\omega_k}\right]\right]\right\}P_i^\dagger P_j+\tfrac{1}{2}\sum_{i,j}\left\{\sum_k\left[|\xi|^2\left[\frac{1}{\Omega_j-\omega_k}+\frac{1}{\Omega_j-\omega_k}\right]\right]\right\}b_i^\dagger b_j\ . \tag{41}$$

Although Eq. (41) contains off-diagonal terms, ΔH remains Hermitian. Moreover, ΔH preserves the total number of quasiparticles in the system. The sixth term in the expansion, Eq. (36), reintroduces the coupling between the cell phonons and the heat bath and the energy pump. However, the new coupling coefficients are proportional to $|\xi|^2\xi$ and $|\lambda|^2\lambda$ this time. Repeating the canonical transformation procedure removes these couplings beyond the fifth term in the new expansion according to Eq. (36). consequently, the operator ΔH after the second transformation still preserves the initial form given by Eq. (41) save for the energy constants which are supplemented by corrections proportional to $|\xi|^4$ and $|\lambda|^4$. Carrying out this procedure up to infinity brings the following form of $\tilde{H}$:

$$\tilde{H}=\sum_{i,j}\omega_{ij}a_i^\dagger a_j+\sum_{i,j}\Omega_{ij}b_i^\dagger b_j+\sum_{i,j}\theta_{ij}P_i^\dagger P_j\ , \tag{42}$$

where

$$\omega_{ij}=\omega_i\delta_{ij}+\tfrac{1}{2}\sum_k\left[|\xi|^2\left(\frac{1}{\omega_i-\theta_k}+\frac{1}{\omega_j-\theta_k}\right)+|\lambda|^2\left(\frac{1}{\omega_i-\Omega_k}+\frac{1}{\omega_j-\Omega_k}\right)\right]+\Sigma_{n\geq 2}(|\xi|^{2n})+\Sigma_{n\geq 2}(|\lambda|^{2n})\ , \tag{43}$$

where $\Sigma_{n\geq 2}(|\xi|^{2n})$ represents a series in $|\xi|^{2n}$, with $n=2,3,\ldots,\infty$ and $\Sigma_{n\geq 2}(|\lambda|^{2n})$ a series in $|\lambda|^{2n}$, with $n=2,3,\ldots,\infty$;

$$\Omega_{ij}=\Omega_i\delta_{ij}+\tfrac{1}{2}\sum_k|\lambda|^2\left(\frac{1}{\Omega_i-\omega_k}+\frac{1}{\Omega_j-\omega_k}\right)+\Sigma_{n\geq 2}(|\lambda|^{2n})\ ; \tag{44}$$

$$\theta_{ij}=\theta_i\delta_{ij}+\tfrac{1}{2}\sum_k|\xi|^2\left(\frac{1}{\theta_i-\omega_k}+\frac{1}{\theta_j-\omega_k}\right)+\Sigma_{n\geq 2}(|\xi|^{2n})\ . \tag{45}$$

The applicability of this procedure rests entirely on the detailed energy structure and the strength of the coupling constants which determine the radii of convergence of the series involved in Eqs. (43)–(45) denoted: ξ_1,λ_1 and ξ_2 and λ_2, respectively. All the energy constants converge if $|\xi|<\min(\xi_1,\xi_2)$ and if $|\lambda|<\min(\lambda_1,\lambda_2)$. Since the heat and energy reservoirs are assumed much larger than the cell, we approximate $\Omega_{ij}\simeq\Omega_i\delta_{ij}$ and $\theta_{ij}\simeq\theta_i\delta_{ij}$. Furthermore, in the process of quantum averaging the off-diagonal terms (those with $i\neq j$) will disappear and the entire Wu-Austin Hamiltonian may be finally presented as

$$H=H_0'+\sum_i\omega_i a_i^\dagger a_i+\sum_i\Omega_i b_i^\dagger b_i+\tfrac{1}{2}\sum_{i,j,k}(\chi a_i^\dagger a_j b_k^\dagger+\chi^* a_j a_i^\dagger b_k)\ , \tag{46}$$

where $H_0'=H_0+\sum_i\theta_i P_i^\dagger P_i$ since the energy-pump operator commutes with all the remaining operators in Eq. (46). The new cell energy constant is $\bar{\omega}_i=\omega_{ij}\delta_{ij}$ where ω_{ij} is that of Eq. (43). This Hamiltonian, Eq. (46), contains the second-order cell-phonon—heat-bath interactions as the only coupling terms. The effect of all first-order interactions is incorporated into the new energy constant $\bar{\omega}_i$.

It is concluded that, provided the interactions between the subsystems are sufficiently small, the Wu-Austin Hamiltonian can be successfully and accurately transformed into an effective operator which is exactly of the same form as the Davydov Hamiltonian, Eq. (26). Therefore, Eq. (46) can be considered a prototype Hamiltonian embracing both theories. Before deriving equations of motion for the operators involved in this Hamiltonian, another canonical transformation will be applied in order to clarify further the role of the remaining interaction term.

V. THE CANONICAL TRANSFORMATION OF THE MODEL HAMILTONIAN

We have seen thus far that the two seemingly unrelated theories of biological order can, indeed, be described using the same Hamiltonian

$$H=H_0+H_I\ , \tag{47}$$

where

$$H_0=H_0'+\sum_i\bar{\omega}_i a_i^\dagger a_i+\sum_i\Omega_i b_i^\dagger b_i \tag{48}$$

and

$$H_I=\tfrac{1}{2}\sum_{i,j,k}(\chi b_{-k}^\dagger+\chi^* b_{+k})a_i^\dagger a_j\ . \tag{49}$$

We have assumed that the indices i, j, and k refer to conserved quantities. Equation (47) can be regarded as a model Hamiltonian for the phenomena described by both theories. It employs two types of creation and annihilation operators: $\{a_i^\dagger,a_i\}$ and $\{b_i^\dagger,b_i\}$. The first type of operators refers to the active modes of the cell: dipole vibrations in the Fröhlich model or electronic excitations in the Davydov model. The second type of operators belongs to those of the thermal environment as displayed by the heat bath in the Fröhlich model or the elastic lattice vibrations in the Davydov model. Both types of operators are coupled through a second-order interaction term, Eq. (49), and it is assumed that the heat reservoir has a much larger energetic capacity than the cell. Consequently, only the cell will be noticeably affected by the coupling. The particular (dispersion) properties of the energy constants ω_i and Ω_i and of the coupling constant χ remain a characteristic feature of each model. In general, χ is assumed to be a function of i and j since the third index k is determined from a conservation law: $Q(i)-Q(j)-q(k)=0$, where Q and q refer to a conserved quantity (e.g., momentum) of the cell and heat-bath quasiparticle, respectively.

In order to see the role of the interaction term, Eq. (49), better, the model Hamiltonian, Eq. (47), is replaced by its

equivalent, in much the same way as was applied to the Wu-Austin Hamiltonian in Sec. IV. Choosing the transformation matrix S as

$$S=\sum_{i,j,k}(Ab^{\dagger}_{-k}+Bb_{+k})a_i^{\dagger}a_j \tag{50}$$

and using a Baker-Hausdorff expansion, Eq. (36), the transformation coefficients A and B are found as

$$A=-\frac{\chi_{ij}}{(\omega_i-\omega_j)+\Omega_{-k}},\quad B=-\frac{\chi^*_{ij}}{(\omega_i-\omega_j)-\Omega_k}\ . \tag{51}$$

The first nonvanishing correction term in the series, Eq. (36), is

$$\Delta H_I=\sum_{i,j,k}\Delta_{ijk}a_i^{\dagger}a_j a_k^{\dagger}a_{i-j+k}\ , \tag{52}$$

where

$$\Delta_{ijk}=|\chi_i|^2\frac{\Omega_{j-k}}{(\omega_j-\omega_k)^2-\Omega^2_{j-k}}\ .$$

This is referred to as the Hartree-Fock (or the mean-field) approximation in which the effect of the external environment is averaged out to form a homogeneous background. The coupling between both modes of excitation is reintroduced into the Hamiltonian in the sixth-order term of the expansion, Eq. (36). The same procedure may be reapplied to remove this coupling and, as a result, correction terms involving an increasing number of second-quantized operators will be added to Eq. (52). However, on the assumption that the coupling constant χ is, relatively, of small magnitude, the higher-order terms are neglected as they are proportional to $|\chi|^{2n}$ with $n=2,3,\ldots$.

We can now separate the model Hamiltonian H into two commuting (noninteracting) parts H' and H'',

$$H=H'+H''\ , \tag{53}$$

where H' is an equivalent operator representing the combined effect of the surroundings on the active modes of the cell. Using the more conventional wave number indices,

$$H'=\sum_k W_K a_k^{\dagger}a_k+\sum_{k,k',q}\Delta_{kk'q}a^{\dagger}_{k'+q}a^{\dagger}_{k-q}a_k a_{k'}\ , \tag{54}$$

where

$$W_k=\bar{\omega}_k-\left[\frac{|\chi_0|^2}{\Omega_0}N\right], \tag{55}$$

N is the total number of cell phonons, and $\Delta_{kk'q}$ is the effective coupling constant,

$$\Delta_{kk'q}=|\chi_{kk'}|^2\frac{\Omega_q}{(\omega_k-\omega_{k'})^2-(\Omega_q)^2}\ . \tag{56}$$

H'' describes the state of the external environment,

$$H''=H'_0+\sum_q\Omega_q b_q^{\dagger}b_q\ . \tag{57}$$

Since $[H',H'']=0$, H'' can be henceforth neglected in the discussion. H' contains all the relevant information about the perturbation effects on the biosystem.

VI. THE EQUATION OF MOTION

Having derived the model Hamiltonian, Eq. (53), which combines both theories of biological order, we shall investigate the ensuing time-dependent phenomena. In general this is a very complicated problem. Fortunately, however, the Hamiltonian presents certain features that can be used to advantage. It is easy to see that, if in each of the creation and annihilation operators in Eq. (53) we make the following transformations:

$$a_k\rightarrow e^{i\theta_k}a_k \tag{58}$$

$$a_k^{\dagger}\rightarrow e^{-i\theta_k}a_k^{\dagger}\ , \tag{59}$$

(where θ_k is some arbitrary phase factor), the Hamiltonian H' does *not* remain invariant. It has been shown by Umezawa and co-workers[39] that such a lack of invariance signifies the phenomenon of symmetry breaking. Paul[15] has shown that this phenomenon is accompanied by a field translation of the operators

$$a_k\rightarrow a_k+c_k,\ a_k^{\dagger}\rightarrow a_k^{\dagger}+c_k^*\ .$$

Here, the quantities c_k and c_k^* are not operators but c-numbers. The occupation number states upon which the operators a_k and $a_k^{\dagger}$ act are transformed into the Glauber coherent states. Thus

$$|n_1,n_2,\ldots\rangle\rightarrow|c_1,c_2,\ldots\rangle\equiv|\{c\}\rangle\ .$$

The $|\{c\}\rangle$ are the Glauber coherent states and are eigenstates of the annihilation operators

$$a_k|\{c\}\rangle=c_k|\{c\}\rangle\ . \tag{60}$$

The adjoint of this equation yields

$$\langle\{c\}|a_k^{\dagger}=c_k^*\langle\{c\}|\ . \tag{61}$$

The Hamiltonian for the system within the coherent-state description is given by the expectation of H' [as given by Eq. (54)]. We will adopt the symbol $\mathscr{H}'$ to designate this expectation value:

$$\mathscr{H}'=\frac{\langle\{c\}|H'|\{c\}\rangle}{\langle\{c\}|\{c\}\rangle}\equiv\mathscr{H}'(\{c\},\{c^*\})\ . \tag{62}$$

This can be readily calculated by substituting Eq. (54) in Eq. (62) and employing Eqs. (60) and (61). We now have

$$\mathscr{H}'(\{c\},\{c^*\})=\sum_k W_k c_k^*c_k+\sum_{k,k',q}\Delta_{kk'q}c^*_{k'+q}c^*_{k-q}c_k c_{k'}\ . \tag{63}$$

In this representation the Hamiltonian is no longer an operator but a c number depending upon the field-translation parameters. The equations of motion for the quantities $\{c\}$ and $\{c^*\}$ can be obtained from Hamilton's equations

$$i\dot{c}_l=\frac{\partial\mathscr{H}'}{\partial c_l^*},\quad -i\dot{c}^*_l=\frac{\partial\mathscr{H}'}{\partial c_l}\ . \tag{64}$$

By inserting $\mathscr{H}'$ from Eq. (62) and carrying out the differentiation we get, along with its complex conjugate,

$$i\dot{c}_l=W_l c_l+\sum_{k,k'}Y_{kk'}c^{\dagger}_{k+k'+l}c_k c_{k'}\ , \tag{65}$$

where

$$Y_{kk'}=\Delta_{k,k',l-k'}+\Delta_{k,k',k-l} \ . \tag{66}$$

Similarly, an equation of motion for c_l^* can be derived.

It is easy to see that within the coherent-state description the expectation value of the annihilation and creation operators do not vanish, but propagate throughout the system without loss of energy. Before demonstrating these solutions (in Sec. III) it should be mentioned that if only one of the following conditions is met, ordinary waves exhibiting diffusion will result instead of solitons.

$$\frac{\langle\{c\}|a_k|\{c\}\rangle}{\langle\{c\}|\{c\}\rangle}=c_k^* \ , \tag{67}$$

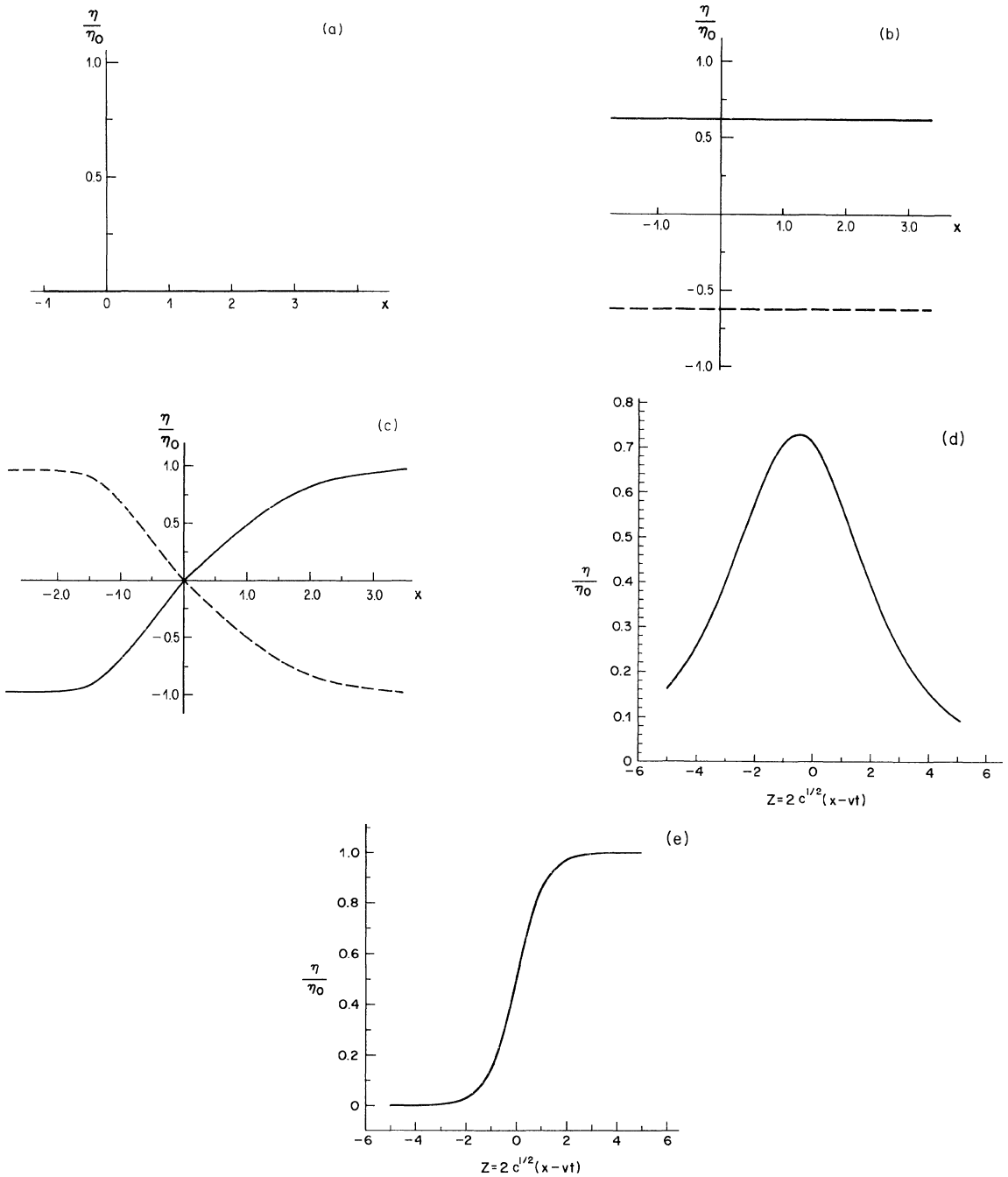

FIG. 3. (a) Solution 1 (static) representing the disordered phase. (b) Solution 2 (static) representing the two possible ordered phases. (c) Solution 3 (static) representing domain walls between the two possible ordered phases. (d) Solution 4 (dynamic) representing the kink-antikink couple. (e) Solution 5 (dynamic) representing the moving kink connecting the ordered and disordered phases.

(i) The third- or a higher-order term in the expansion of W_k in a series of k is significant and cannot be neglected.

(ii) The first-order term W_1 does not vanish, as, for example, in the case of acoustic phonons where $W(k) \propto \sin(k/2)$.

(iii) The Bose condensation does not occur and the averages over $|0\rangle$, and not $|0\rangle\rangle$, must be taken. Then a linear differential equation in the form of a diffusion equation will be obtained:

$$i\dot{\eta} = W_0\eta + W_2\eta_{xx} \,. \tag{68}$$

The amplitudes of the solutions to (68) decay exponentially with x in direct proportionality to the diffusion constant W_2.

VII. SOLUTIONS

The present equation, Eq. (65), is a special case of a nonlinear Schrödinger equation and is solved following the detailed derivation[40] for a general case. The parameters in Eq. (65) have a physical interpretation which depends on the particular model used. In both cases, however, the nonlinearity parameter Y is proportional to the strength of the cell-heat reservoir coupling: $|\chi|^2$. In the Davydov model the diffusion constant W_2 equals the dipole-dipole potential U and the linear coefficient $W_0 = \epsilon - D(1+B) - 2U - (|\chi_0|^2/\Omega)N$.

All solutions of Eq. (65) can be divided into two groups: (i) static (or time-independent) and (ii) dynamic (or time-dependent). The static solutions follow.

(1) $\eta_1(x) \equiv 0$. It exists when $W_0 \geq 0$ and the cell—heat-bath coupling increases the value of W_0 since Ω_k are negative. This solution represents a disordered, unpolarized phase since the field translation is proportional to the net polarization. When $\eta_1(x) = 0$, there is neither Bose condensation nor a net dielectric polarization.

(2) A solution corresponding to the homogeneous, ordered phase whose amplitude is

$$\eta_2(x,t) = \pm[(-W_0)/T]^{1/2} \tag{69}$$

and it can only exist when $W_0 \leq 0$ which is facilitated by the presence of a sufficiently strong coupling:

$$|\chi|^2 \geq \frac{\omega_0}{N}\Omega_0 \,.$$

This solution represents an ordered, polarized phase existing when Bose condensation takes place. Solutions (1) and (2) were, in fact, the subject of investigations by Fröhlich and they describe the time-independent aspect of the problem. Note that we use the terminology of phase transitions since the field translation η can be treated as a time- and space-dependent order parameter. Moreover, since without the presence of external electric fields breaking the symmetry manifested by the appearance of "$\pm$" in Eq. (69) two ordered phases are allowed, there is an additional solution connecting the two phases. It has the form of a domain wall and is given by

(3) an interface solution,

$$\eta_3(x,t) = \eta_0 \tanh\left[\frac{x}{\sqrt{2}\xi_c(T)}\right] , \tag{70}$$

where $\eta_0 = |\eta_2|$ and $\xi_x(T)$ is the so-called correlation length which equals $\xi_c(T) = -W_2/W_0$. Thus, it is necessary that $W_2 > 0$ and $W_0 < 0$ for this solution to exist.

The dynamic solutions are solitons.

(4) A kink-antikink pair corresponding to the Davydov soliton:

$$\eta_4(x,t) = \exp[i\delta_c(\Delta x_c - v_c t)]\{[(1+A)/4C]\sinh[x_1 \pm 2\sqrt{C}(x-vt)] + [(1-A)/4C]\cosh[x_1 \pm 2\sqrt{C}(x-vt) - D/2C]\}^{-1/2} \,, \tag{71}$$

where δ_c, x_c, and v_c are the carrier width, phase shift, and velocity, respectively, and x_1 is the envelope phase shift. The following symbols have also been introduced:

$$C = \frac{-W_0}{W_2} + \tfrac{1}{2}\left[\frac{v}{W_2}\right]^2 , \tag{72}$$

$$A = -\left[\frac{T}{W_2}\right]^2 \leq 0 \,, \tag{73}$$

$$D = \frac{-T}{W_2} \,. \tag{74}$$

This solution can be interpreted as a solitary wave of polarization-soliton propagating through the system at a constant velocity v without loss of energy. Both directions of motion are allowed and the width of this soliton is $\delta_e = 2\sqrt{C}$ which vanishes as $W_0 \to 0$ or $W_2 \to \infty$.

(5) This solution is a moving order-disorder phase wall given by

$$\eta_5(x,t) = \exp[i\delta_c(\Delta x_c - v_c t)] \times \{(1/2C)\exp[x_2 \pm 2\sqrt{C}(x-vt)] - D/2C\}^{-1/2} \,. \tag{75}$$

It exists only when $A = 0$ (i.e., if $T = 0$ or $W_2 = \infty$) and $C > 0$. This solution separates two neighboring regions, one of which is disordered (solution 1) and the other of which is ordered (solution 2) and therefore exists at the transition point where $W_c = 0$.

Neither the static domain wall (solution 3) nor the dynamic domain wall (solution 5) have been found before and they seem to play important roles in these phenomena. Moreover, the remaining solutions which are divided into two distinct groups appear to describe different aspects of the same phenomenon. The static solutions described first by Fröhlich are manifestations of the Bose condensation in the system and the dynamic solutions are polarization waves propagating across the system as soli-

tons in a lossless manner. Therefore, the two seemingly different models of biological order are not only mathematically equivalent but also physically complementary. We have shown that they can be unified into a general, complete theory.

VIII. CONCLUSIONS

Using a series of Fourier and canonical transformations, we have demonstrated the relationship between the two seemingly independent theories of biological order put forward by Fröhlich and Davydov. Within the bounds of the approximations used, the two models are mathematically equivalent as they yield identical effective Hamiltonians. A model Hamiltonian encompassing both models is constructed which yields an equation of motion for the ground-state expectation value of the creation and annihilation operators involved, called a field translation. Under the conditions for Bose condensation the Fourier transform of the field-translation variable is shown to obey a nonlinear Schrödinger equation (NLSE) whose solutions are well known. The set of NLSE solutions can be classified as static (time independent) or dynamic (time dependent). The first group was the subject of Fröhlich's studies whereas the second was discussed by Davydov. Thus, the two models seem to complement each other physically. We have found two new solutions: a standing interface between two ordered phases whose amplitudes are of opposite sign and a moving interface between an ordered and a disordered phase. It is concluded that the field translation associated with the presence of Bose condensation can be considered as an order parameter in biological systems. In the absence of Bose condensation or when the dispersion function for the cell phonons does not meet certain restrictions, the resultant equation of motion does not admit solitons, but ordinary waves which diffuse and decay.

Based on our findings, we expect both static and dynamic effects to occur simultaneously and they are interpreted as two aspects of the same phenomenon. Thus, Bose condensation of membrane-vibrational modes should be accompanied by the solitonic propagation of a polarization wave. Conversely, the solition transport of bond energy along the peptide chain should be accompanied by Bose condensation of lattice vibrations. We note that the existence of solitons and Bose condensation have been shown to be related[41,42] also elsewhere, which supports our conclusions.

The main difference between the Davydov and Fröhlich models is one of emphasis on different manifestations of the same problem. Fröhlich's model concentrates on time-independent effects (stable states) and leads to conditions for establishing order in the system whereas Davydov's model investigates time-dependent effects and highlights propagation of this order throughout the system. Furthermore, due to the differences in the particulars of the systems under study, the energy constants and coupling coefficients appear to have different strengths and dispersion properties. Notably, Fröhlich assumed the dipoles to vibrate at high frequencies and his approach implies[43] that the dipole-dipole energy is less than kT. Davydov, on the other hand, considers the case when the dipoles are statically aligned and their interaction energy far exceeds kT.[44] Both models, however, arrive at an attractive long-range effective potential which is proportional to r^{-3}. Since the active objects are of different dimensions in these theories (Fröhlich, membranes; Davydov, electrons), it is not surprising to find that the two potentials differ significantly in strength. Paul *et al.*[18] calculated, using the Wu-Austin Hamiltonian, that $E \simeq -2.3\times10^{-36}$ J m^3 for human erythrocytes whereas for the peptide group $E \simeq -2.0\times10^{-60}$ J m^3 using the Davydov expression. It would be necessary to investigate the dispersion properties of the coefficients appearing in the Wu-Austin Hamiltonian, without which it would be difficult to argue that solitons do exist in biological membranes.

We note in closing that Fröhlich[43] himself predicted the significance of electronic excitations in biological cells although he did not relate them explicitly to the model of membrane vibrations. Having related Fröhlich's theory to Davydov's theory of electron transport it seems natural to explore the possibilities of membranes and electrons attaining similar frequencies. Such as analysis could quantify Fröhlich's speculations about the role of free electrons in destroying the growth control mechanisms.[43]

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SELECTIVE LONG RANGE DISPERSION FORCES BETWEEN LARGE SYSTEMS

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Two large molecular systems capable of giant dipole vibrations of frequencies ω_1 and ω_2 respectively may exert selective very strong long range forces upon each. The selectivity refers to frequency resonance ($\omega_1 \approx \omega_2$). Further conditions are (i) the dielectric constant of the medium between them shows strong dispersion in the relevant frequency region or (ii) coherent vibrations are excited by pumping. The latter is always attractive, the former may be either attractive or repulsive.

The interaction between two large systems (e.g. giant molecules) at a distance R, larger than their linear dimensions L arises principally from the interaction between their giant dipole vibrations (= zero wave number polarisation waves). Let ω_1 and ω_2 respectively ($\omega_1 \geqslant \omega_2$) be the frequencies of these vibrations in the separated systems. Then through their interaction at finite distance two frequencies ω_+ , ω_- of the joint system arise,

$$\omega_\pm^2 = \tfrac{1}{2}(\omega_1^2 + \omega_2^2) \pm (\Delta^2 + \beta^4/\epsilon_\pm^2)^{\frac{1}{2}},$$
$$\Delta = \tfrac{1}{2}(\omega_1^2 + \omega_2^2) \qquad (1)$$

where $\epsilon_\pm$ is the longitudinal dielectric constant at frequency $\omega_\pm$ of the medium separating the systems and

$$\beta^2 = e^2 |A| (z_1 z_2)^{\frac{1}{2}} / MR^3. \qquad (2)$$

Here the two systems are considered to contain z_1 and z_2 elastically bound particles of mass M and charge e. $|A|$ is a positive constant of magnitude unity which depends on various angles. The frequencies ω_1 and ω_2 depend on the macroscopic shape of the systems and for simplicity will be assumed non degenerate. Generalisations to more complicated systems are straight forward. Clearly the frequency separation increases with the size of the systems and for large systems may reach considerable values such that in a sispersion region of the intermediate medium ϵ_+ may differ considerably from ϵ_-. The result is a resonance-like interaction energy between the systems which is linear in β^2 (i.e. $\propto 1/R^3$) if $\epsilon_+ \neq \epsilon_-$ but $\omega_1 \approx \omega_2$. For $\epsilon_+ = \epsilon_-$ on the other hand the second order London interaction ($\propto 1/R^6$) is obtained which is not frequency slective.

The interaction energy I is defined as the difference of the free energy F at distance R from its value at $R \to \infty$. Thus in the extreme quantum case (I_q) when $KT/\hbar$ is small compared with ω_-,

$$I_q/kT = \hbar((\omega_+ + \omega_-) - (\omega_1 + \omega_2))/2kT \qquad (3)$$

whereas in the extreme classical case

$$I_c/kT = \log \omega_+ \omega_- / \omega_1 \omega_2. \qquad (4)$$

To discuss the result consider first the case of weak coupling defined as

$$(\Delta^2 + \beta^4/\epsilon_\pm^2)^{\frac{1}{2}} - \Delta \ll \omega_2^2 \qquad (5)$$

Systematic development near resonance then yields

$$I_q/kT = \frac{\hbar}{4kT}\left\{\Delta\left(\frac{1}{\omega_2} - \frac{1}{\omega_1}\right) + \left(\frac{\Delta^2}{\omega_1^2} + \frac{\beta^4}{\omega_1^2\epsilon_+^2}\right)^{\frac{1}{2}} - \left(\frac{\Delta^2}{\omega_2^2} + \frac{\beta^4}{\omega_2^2\epsilon_-^2}\right)^{\frac{1}{2}}\right\} \qquad (6)$$

Hence at resonance, $\omega_1 = \omega_2 = \omega$, $\Delta = 0$,

$$\frac{I_{q\,res}}{kT} = \frac{\hbar\omega}{4kT}\frac{\beta^2}{\omega^2}\left(\frac{1}{\epsilon_+} - \frac{1}{\epsilon_-}\right), \quad \omega_1 = \omega_2. \qquad (7)$$

If $\hbar\omega/4kT$ is large, $I_q/kT > 1$ may well arise although (5) requires $\beta^2/\omega^2\epsilon < 1$. The result in the classical case is similar to (7) except for the factor $\hbar\omega/kT$ so that $I_c < kT$. Outside the resonance region the usual London result $\propto 1/R^6$ obtains; eq. (6) must then be taken to the next order.

If the weak coupling condition (5) is not fulfilled

then according to (1) considerable softening of the mode ω_- may arise. The case $\omega_-^2 < 0$ is, of course, well known from the theory of mode softening in ferroelectrics and will not be considered here. Small but positive ω_-^2 implies that the classical limit only need be considered. According to (4) it may lead to strong attractive interaction but is not sensitive to resonance conditions.

Our main result thus indicates the possibility of strong, frequency selective, long range forces between large systems provided the longitudinal dielectric constant $\epsilon(\omega)$ of the intermediate medium shows appropriate dispersion in the relevant frequency region; the sign of $\partial\epsilon/\partial\omega$ decides whether these forces are attractive or repulsive.

Selective long range forces which are always attractive are obtained if the system is pumped as has been suggested to occur in biological systems through their metabolism [1]. It has been shown then that the lowest frequency in a band of polarisation waves should be very strongly excited. Now at infinite separation of our two systems the frequency of the polarisation waves with zero wave number (responsible for long range interaction) is not in general the lowest one but according to (1), ω_- is depressed at closer approach and this may become the lowest frequency at a distance below a critical one. The interaction then is always negative, (n_- = number of quanta in ω_-)

$$I = (n_- + \tfrac{1}{2})\hbar(\omega_- - \omega_2) \approx$$
$$-\hbar(n_- + \tfrac{1}{2})((\Delta^2 + \beta^4/\epsilon^2)^{\frac{1}{2}} - \Delta)/2\omega_2 \qquad (8)$$

Thus we have found that a system capable of giant dipole vibrations may attract another strongly, and selectively in frequency. The range of this interaction is much larger than that of the more usual forces. At closer distance, however, the behavior of the individual atoms composing the large systems should become more relevant and the highly specific nature of the London forces as demonstrated by Jehle et al. [2] as well as other chemical interaction may become of principal importance. The two aspects are complementary and should be decisive in many aspects of selective and specific interaction of large systems [3,4].

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Self-focusing and Ponderomotive Forces of Coherent Electric Waves: A Mechanism for Cytoskeleton Formation and Dynamics

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Coherent electric waves have been predicted to exist in biological systems.(1) We try to link the consequences of their propagation in the cell medium to some peculiar features of cell architecture and organization such as cytoskeleton formation and dynamics.(2) This will give a consistency argument in favour of the existence of coherent electric waves in living systems.(3)

The general ponderomotive force produced by an electric field $\vec{E}$ and a magnetic field $\vec{H}$ acting on a dielectric with polarization $\vec{P}$ and magnetization $\vec{M}$ is given by(4)

$$\vec{f} = \rho\vec{E} + \frac{1}{c}\dot{\vec{j}} \times \vec{H} + (\vec{P}\cdot\nabla)\vec{E} + (\vec{M}\cdot\nabla)\vec{H} + \frac{1}{c}\frac{\partial\vec{P}}{\partial t} \times \vec{H} - \frac{1}{c}\frac{\partial\vec{M}}{\partial t} \times \vec{E} \qquad (1)$$

In a first approach we will drop the terms depending on the magnetization $\vec{M}$ (which is expected to be important only in a specialized class of cells) and we will also omit the linear term arising from the "true" current source $\dot{\vec{j}}$. Therefore

$$\vec{f} = \rho\vec{E} + (\vec{P}\cdot\nabla)\vec{E} + \frac{1}{c}\frac{\partial\vec{P}}{\partial t} \times \vec{H} \qquad (1')$$

In eq. (1') we have the usual linear term $\rho\vec{E}$, the nonlinear term, called the "gradient force" $(\vec{P}\cdot\nabla)\vec{E}$, and the term of "radiation pressure" $\frac{1}{c}\frac{\partial\vec{P}}{\partial t} \times \vec{H}$. Since coherent waves can bring very large values of $\vec{E}$ (of the order of 10^7 V/m), the nonlinear terms of eq. (1') are expected to dominate on the linear one. Consequently, we will drop the first term in the right hand side of eq. (1'). The gradient term can be written as(5)

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$$(\vec{P}\cdot\nabla)\vec{E} = -\ \text{grad}\ p_o(\rho,T) + \frac{1}{8\pi}\ \text{grad}\ E^2\rho\left(\frac{\partial\varepsilon}{\partial\rho}\right)_T + \qquad (2)$$

$$-\frac{1}{8\pi}\ E^2\ \text{grad}\ \varepsilon$$

where ε is the dielectric permittivity of the medium and $p_o(\rho,T)$ is the pressure in the medium in the absence of an external field, at the same volume and temperature T. The pressure gradient term, although relevant in many applications,can be disregarded in a microscopic approach. Moreover, we will use an approximation valid for gaseous media and dilute solutions (as the cytoplasm can be considered in a rough approximation):

$$\rho\frac{\partial\varepsilon}{\partial\rho} \sim (\varepsilon - 1) \qquad (3)$$

After trivial manipulations one gets from eq. (1')

$$\vec{f} = \frac{1}{8\pi}(\varepsilon-1)\nabla E^2 + \frac{1}{c}\frac{\partial\vec{P}}{\partial t} \times \vec{H} \qquad (4)$$

The first term in the right hand side of eq. (4) shows that a high power, strong directional electric wave

$$\vec{E}(\vec{r},t) = \frac{1}{2}\vec{A}(\vec{r},t)\ \exp\left[i(\omega t - \vec{k}\cdot\vec{r})\right] + \text{c.c.}$$

can produce condensation or rarefaction in a medium according to the sign of the coefficient of ∇E^2. This coefficient is a function both of the frequency ω of the beam and of the typical resonance frequencies ω_{ok} of the dielectric. It has been shown[6] that

$$\frac{1}{8\pi}(\varepsilon-1)\nabla E^2 = C\sum_k\left[\frac{\omega_{ok}^2 - \omega^2}{(\omega_{ok}^2 - \omega^2)^2+\Gamma_k^2}\right]\nabla E^2 \qquad (5)$$

where C is some positive constant and Γ_k is the damping of the k-th oscillator. This gradient force is important when the incoming frequency ω approaches ω_o, one of the frequencies of the medium. An attractive (repulsive) force in the direction transverse to the propagation axis of the beam will then be present when $\omega_o > \omega\ (\omega_o < \omega)$.

The gradient force produces then a waveguide in the medium along the intense and directional electric beam, where a condensation of matter takes place. This process eliminates the geometric and diffraction divergences of the beam, leading to a very efficient propagation within the waveguide ("self-focusing").

In order to discuss this kind of propagation, the gradient force can be expressed in terms of the properties of the refractive index of the nonlinear medium[7]. On the basis of the Kerr properties of protein solutions[8], we will consider a "cubic"

medium, i.e. $n = n_o + n_2|A|^2$ ($\varepsilon = \varepsilon_o + \varepsilon_2|A|^2$).

From the Clausius-Mossotti equation for an isotropic medium

$$\frac{n^2-1}{n^2+2} = \frac{4}{3}\pi\rho\alpha \tag{6}$$

where α is the polarizability, we can express the variation Δn (at the lowest order) induced by an electric field $\vec{E}$ as

$$\begin{aligned} \Delta n &= (\Delta n)_\rho + (\Delta n)_\alpha \\ (\Delta n)_\rho &= \frac{1}{2} K_\rho \lambda |A|^2 \\ (\Delta n)_\alpha &= \frac{1}{2} K_\alpha \lambda |A|^2 \\ \lambda &= \frac{2\pi}{n}\frac{c}{\omega} \end{aligned} \tag{7}$$

K_ρ, K_α are the electrostrictive and ac Kerr constants respectively. $(\Delta n)_\rho$ is the variation of the refractive index due to electrostriction; $(\Delta n)_\alpha$ is the variation of the refractive index arising from ac Kerr contribution. It should be noticed[9] that only in the case of non-polar molecules it is possible to approximate the ac Kerr constant K_α with the contribution coming from the induced dipole moments $(K_1)_{dc}$ to the dc Kerr constant; this contribution is always positive whereas the dc Kerr constant K_{dc} can also be negative. For polar molecules with large dipole moments, K_α is in general smaller than the absolute value of the dc Kerr constant. Again we find as it appeared from eq. 2, that the propagation of an electric wave in a nonlinear dielectric is governed by an electrostriction term and by a molecular orientational one(Kerr term). From eqs. (2) and (7) the Kerr contribution appears to depend on ω and can be strongly enhanced when ω approaches a resonant frequency of the medium.

As far as the propagation of the beam is concerned, it can be shown[7] that eq.(7) implies the "self-focusing" of the beam, i.e. the beam shrinks and, after travelling in the medium a critical length

$$R_{nl} = \frac{a}{2}\left(\frac{n_o}{n_2|A|^2}\right)^{\frac{1}{2}}$$

(a being the radius of the cross-section of the beam), gets confined in a thin waveguide ("self-trapping") provided that its power be greater than a critical value

$$P_{cr} = (1.22)^2\lambda^2 c/256\, n_2$$

The above discussion compels us to conclude that coherent electric waves form in the medium a network of waveguides which in turn may attract "selectively" molecules in solution which have appropriate vibrational frequencies. Self-focusing strengthens very much the role of the gradient force by shrinking the beam and steepening its profile. Therefore molecules will experience stronger forces when the conditions

for the beam self-trapping are met. An increase of concentration and an alignment of the solute molecules are then generated around the beam, resulting in a dynamic self-association, which is a requirement for polymerization. It has been recently reported[10] that polymerization can be actually induced by ordering the monomers.

At this point, let us turn to the radiation pressure term in eq.(4). The direction in which this force is exerted depends upon the longitudinal or transversal character of the time dependent electric field : a longitudinal wave produces a force transversal to the direction of propagation whereas in the transversal case the force is directed along the beam axis. As a consequence, in the first case the gradient force will change its magnitude; in the second case, molecules will also be pushed along the beam axis. The interplay between gradient forces and radiation pressure gives rise to an "optical mechanics"as shown by the experiments on laser induced"levitation" performed by Ashkin[11] and on laser isotope separation[12]: biological systems seem to be an excellent lab for such an"optical mecanics".The radiation pressure squeezes then the molecules, selected and attracted by the gradient force, in the waveguide and transports them along it.This squeezing enhances the probability that monomers ordered by the gradient force could polymerize. The contribution of the radiation pressure term to the assembly of the monomers could account for the dynamics of polymerization: the polymer can lose monomers at the end of the self-trapped beam while acquiring them at the opposite end.

The cytoskeleton consists of three different networks of filaments built up with protein (mainly actin and tubulin) polymers. These filaments exhibit a labile character, and their formation and dynamics strictly depend on a number of factors such as cell shape,timing, environment [13,14]. Moreover, a "treadmilling" behaviour has been observed[15]: monomers are assembled at one end of the filament and disassembled at the opposite end, giving rise to a unidirectional flow of matter. These features, insofar unexplained, can thus be accounted for by the ponderomotive forces generated by coherent electric waves propagating in form of filaments in the nonlinear medium.

An evidence for the general character of the proposed model is provided by the experiments on blood clotting[16]; filaments - the so-called contractils - appear when erythrocytes interact among themselves "à la Fröhlich" and an appropriate macromolecule is present in the suspension.

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Solitary Excitons in One-Dimensional Molecular Chains

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The change in time of the state of intramolecular excitation in an one-dimensional molecular chain is investigated, where its deformability is taken into account. It is shown that the excitation which is accompanied by a local deformation of the chain, can move uniformly along the chain, the size of the region under excitation remaining constant. These excitations in one-dimensional molecular chains can be called *particle-like excitons* or *solitary excitons.*

Es werden die zeitlichen Änderungen des intramolekularen Anregungszustandes in einer ein-dimensionalen molekularen Kette untersucht, wobei deren Deformierbarkeit berücksichtigt wird. Es wird gezeigt, daß die Anregung der Kette, die durch eine lokale Deformation begleitet ist, sich gleichförmig längs der Kette bewegen kann, wobei die Größe des angeregten Bereichs konstant bleibt. Diese Anregungen in ein-dimensionalen Molekülketten können *partikel-ähnliche Exzitonen* oder *solitäre Exzitonen* genannt werden.

1. One-Dimensional Molecular Chain

The stationary states of the electron excitation taking crystal lattice deformation into account were considered in [1, 2]. The change of such a state in time is of certain interest. The present paper is devoted to this problem.

Consider a one-dimensional infinite molecular chain. The Hamiltonian of the electron excitation in the Heitler-London approximation has the form (see [3]):

$$H = \sum_n [\varepsilon + D(n)]\, B_n^+ B_n + \sum_{l,n}{}' M_{ln}(r_l - r_n)\, B_n^+ B_n\,, \tag{1}$$

where ε is the excitation energy of an isolated molecule;

$$D(n) = \sum_{l(\neq n)} D_{ln}(r_l - r_n)\,,$$

$D_{ln}(r_l - r_n)$ is the change in the interaction energy between molecules l and n, when the molecule n is excited; $M_{ln}(r_l - r_n)$ are the matrix elements of the resonance interaction; r_n is the coordinate along the chain of the molecule n; B_n^+, B are the creation and annihilation operators of the electronic excitation over the molecule n.

The kinetic energy T of the molecules of the chain is

$$T = \frac{m}{2} \sum_n \left(\frac{\partial r_n}{\partial t}\right)^2, \tag{2}$$

where m is the mass of the molecule.

In the nearest-neighbour approximation the potential energy of the chain is

$$U = \sum_n u(R_n)\,, \tag{3}$$

where $R_n = r_{n+1} - r_n > 0$ are the intermolecular distances;

$$u(R_n) = \frac{C}{R_n^s} - \frac{A}{R_n^q},$$

$C, A, s, q > 0$ are the constants defining the interaction of unexcited molecules; $s > q$.

In the ground state $|0\rangle$ (without intramolecular excitations) the chain possesses periodicity and the intermolecular distances obtained from the condition of a minimum of potential energy (3) are equal to

$$R = \left(\frac{s\,C}{q\,A}\right)^{1/(s-q)}. \tag{4}$$

In the state $|\psi\rangle$ when the intramolecular excitation is present, the intermolecular distances R_n are different from R and the potential energy of the chain changes by the value ΔU, as compared to the energy of the chain in the ground state. In the harmonic approximation

$$\Delta U = \sum_n [u(R_n) - u(R)] \approx w \sum \varrho_n^2, \tag{5}$$

where

$$R_n = R - \varrho_n, \qquad |\varrho_n| < R, \tag{6}$$

$$w = \frac{1}{2}\left[\frac{\partial^2 u}{\partial R_n^2}\right]_{R_n = R}. \tag{7}$$

Taking into account (6), the formula (2) for the kinetic energy of the molecules can be transformed to

$$T = \sum_n \left\{\frac{m}{2}\left[\sum_l \frac{\partial \varrho_{l+n}}{\partial t}\right]^2\right\}. \tag{8}$$

Then the total energy E of the deformation of the chain is written as

$$E = T + \Delta U = \sum_n \left\{\frac{m}{2}\left[\sum_{l=n}^{\infty} \frac{\partial \varrho_l}{\partial t}\right]^2 + w\,\varrho_n^2\right\}. \tag{9}$$

2. The Nearest-Neighbour Approximation

In the nearest-neighbour approximation the operator (1) takes on the form

$$H = \sum_n \{[\varepsilon + D(R_n)]\, B_n^+ B_n + M(R_n)\,[B_n^+ B_{n+1} + B_{n+1}^+ B_n]\}. \tag{10}$$

In (10) we adopt

$$D_{n-1,n}(R_{n-1}) + D_{n+1,n}(R_n) \approx D(R_n),$$

$$M_{n,n+1}(R_n) \approx M_{n+1,n}(R_n) = M(R_n).$$

Let

$$D(R_n) = -\frac{\alpha}{R_n^p}, \tag{11}$$

where $\alpha, p > 0$ are the parameters of theory.

In the case of a dipole–dipole resonance excitation (and only this excitation is considered below)

$$M(R_n) = \frac{b}{R_n^3}. \tag{12}$$

We assume that the dipole moment of transition, $\boldsymbol{d}$ is oriented along the chain. Then in (12) $b = -2d^2$. Suppose that the following inequality is valid for distances close to R:

$$D > M\,, \tag{13}$$

where $D \equiv -D(R) > 0$; $M \equiv -M(R) > 0$. Using (6), (11), and (12), $D(R_n)$ and $M(R_n)$ in (10) can be expanded near the point R. In $D(R_n)$ we keep only the term linear in ϱ_n, and in $M(R_n)$ the zero term, i.e. we set $D(R_n) \approx \approx -(1 + p\,\varrho_n/R)\,D$, $M(R_n) \approx -M$. Then (10) takes the form

$$H = \sum_n \{[\varepsilon + (1 + p\,\varrho_n/R)\,D]\,B_n^+\,B_n - M\,[B_n^+\,B_{n+1} + B_{n+1}^+\,B_n]\}\,. \tag{14}$$

The operator $\mathcal{H}$ of the excitation energy of the chain with its deformation energy taken into account reads

$$\mathcal{H} = H + E\,. \tag{15}$$

The solutions of the Schrödinger equation

$$i\,\hbar\,\frac{\partial}{\partial t}\,|\Psi(t)\rangle = \mathcal{H}\,|\Psi(t)\rangle \tag{16}$$

are looked for in the form

$$|\Psi(t)\rangle = \sum_n a_n(t)\,B_n^+|0\rangle \tag{17}$$

provided

$$\langle\Psi(t)|\Psi\rangle = \sum_n |a_n(t)|^2 = 1\,. \tag{18}$$

The operator (15) is the function of a set of the values of ϱ_n. Moreover, these values are dependent on the state $|\Psi(t)\rangle$.

In the region of the chain where the density of the probability of finding the excitation $|a_n(t)|^2$ is great, one should expect greater deformation of the chain which is defined by the values of ϱ_n. From the condition of a minimum of the total excitation energy $\langle\Psi(t)|\Psi(t)\rangle$ at every fixed moment of time t, and the fact that $\partial\varrho_n/\partial t$ is independent of ϱ_n, we find

$$\varrho_n = \frac{p\,D}{2\,w\,R}\,|a_n(t)|^2\,. \tag{19}$$

Substituting (19) into (14), we obtain

$$H = \sum_n \{[\varepsilon - D]\,B_n^+\,B_n - M\,[B_n^+\,B_{n+1} + B_{n+1}^+\,B_n] - G\,|a_n(t)|^2\,B_n^+\,B_n\}\,, \tag{20}$$

where $G = p^2\,D^2/2\,w\,R^2$ is the parameter of nonlinearity. Further, according to (9) and (19)

$$E = \frac{1}{2}\sum_n \left\{F\left[\sum_{l=n}\frac{\partial}{\partial t}\,|a_l(t)|^2\right]^2 + G\,|a_n(t)|^4\right\}, \tag{21}$$

where

$$F = \frac{m\,G}{2\,w}.$$

Now the equation (16) takes the form

$$\sum_n \left\{i\,\hbar\,\frac{\partial a_n(t)}{\partial t} + e\,a_n(t) - M\,[a_{n+1}(t) + a_{n-1}(t)] - G\,|a_n(t)|^2\,a_n(t)\right\} B_n^+|0\rangle = 0\,, \tag{22}$$

where

$$e = \varepsilon - D + E \,.$$

Using (22) and the fact that the system $B_n^+|0\rangle$ is complete and orthogonal, we obtain the equation for $a_n(t)$

$$i\hbar \frac{\partial a_n(t)}{\partial t} + M\,[a_{n+1}(t) + a_{n-1}(t)] - e\,a_n(t) + G\,|a_n(t)|^2\,a_n(t) = 0\,. \tag{23}$$

In (23) we go over to a long-wave approximation and replace the differences by the derivatives. Thus

$$a_n(t) \to \varphi(\xi, t)\,, \qquad \varphi(n, t) = a_n(t)\,, \tag{24}$$

$$\varphi(\xi + 1, t) + \varphi(\xi - 1, t) \approx 2\,\varphi(\xi, t) + \frac{\partial^2 \varphi(\xi, t)}{\partial \xi^2}\,, \tag{25}$$

where $\xi = r/R$ is the dimensionless coordinate. Then the differential-difference equation (23) transforms to

$$i\hbar \frac{\partial \varphi}{\partial t} + M \frac{\partial^2 \varphi}{\partial \xi^2} - \lambda\,\varphi + G|\varphi|^2\,\varphi = 0\,, \tag{26}$$

where

$$\lambda \equiv \varepsilon - D - 2\,M + \frac{1}{2}\int_{-\infty}^{\infty} \mathrm{d}\xi \left\{ F\left[\int_{\xi}^{\infty} \mathrm{d}\eta \frac{\partial}{\partial t}\,|\varphi(\eta, t)|^2 \right]^2 + G\,|\varphi(\xi, t)|^2 \right\}, \tag{27}$$

3. The Particle-Like Excitation

The normalized solution of the nonlinear equation (26) has the form of a solitary wave

$$\varphi_v(\xi, t) = \frac{\sqrt{\frac{\mu}{2}} \exp\left\{ i \left[\frac{\hbar\, v}{2\,M} (\xi - \xi_0) - \Omega\, t \right] \right\}}{\cosh\left[\mu\,(\xi - \xi_0 - v\,t)\right]}\,, \tag{28}$$

where

$$\mu = \frac{G}{4\,M}\,,$$

$$\hbar\,\Omega = \varepsilon - D - 2\,M - \frac{G^2}{48\,M} + \frac{m^*\,R^2\,v^2}{2}\,, \tag{29}$$

$$m^* = m_0^* + m_{\mathrm{d}}^* = \frac{\hbar^2 + (G\,F/6)}{2\,M\,R^2}\,; \tag{29a}$$

$m_0^* = \hbar^2/2\,M\,R^2$ is the effective mass of the excitation in the chain with rigidly fixed molecules; $m_{\mathrm{d}}^* = m\,G^2/24\,M\,w\,R^2$ is the addition to the effective mass which is caused by a local deformation.

According to (28), the excitation is localized in the neighbourhood $\xi = \xi_0 + v\,t$. Consequently, the parameter defines the velocity $V = v\,R$ of the excitation motion along the chain. To every value of v there corresponds an excitation energy of the chain, $\hbar\,\Omega$. The excitation localized near the point ξ_0 corresponds to the value $v = 0$. Its energy $\varepsilon - D - 2\,M - G^2/48\,M$ is less by the value $G^2/48\,M$ than the excitation energy of the chain with rigidly fixed molecules.

Using (19), we find that in the excitation region the equilibrium distances between molecules decrease by the value

$$\varrho(\xi, t) = \frac{\sqrt{\frac{G}{2w}}\frac{\mu}{2}}{\cosh^2[\mu(\xi - \xi_0 - vt)]}. \tag{30}$$

Thus, an appreciable change in the distances between the molecules, which is proportional to the parameter μ, spreads over the region

$$\Delta\xi = \frac{\pi}{\mu} = \frac{4\pi M}{G} \tag{31}$$

around the point $\xi_0 + vt$. This region travels with constant speed along the chain. The average distance between the molecules in the excitation region is

$$\langle R_n \rangle \approx R - 0.3\mu\sqrt{\frac{G}{2w}}. \tag{32}$$

Therefore, when the inertia and deformability of a chain is taken into account, the motion of the excitation is accompanied by a local distortion of the chain and the velocity V of an excitation decreases, as the effective mass (29a) increases.

Unlike the states of motion of free particles described by wave packets, the excitation state corresponding to the function (28) does not "spread" in the course of time. These excitations in one-dimensional molecular chains can be called *particle-like excitons*, or solitary excitons.

The long-wave approximation used by us requires that the inequality $\Delta\xi > 1$, i.e. $\mu = G/4M < \pi$ is satisfied. On the other hand, for the particle-like excitons the following inequality should be valid:

$$\Delta\xi \ll N \quad \text{i. e.} \quad \mu > \frac{\pi}{N},$$

where N is the total number of molecules in the chain; $N \gg 1$.

At the very small values of the parameter μ ($\mu \ll 1$) the expression (25) can be transformed to the form of a simple harmonic wave

$$\varphi_v(\xi, t) \approx \sqrt{\frac{\mu}{2}} \exp\left\{i\left[\frac{\hbar v}{2M}(\xi - \xi_0) - \Omega t\right]\right\}.$$

Consequently, in this limiting case the excitation is practically nonlocalized.

Excitations such as (28) can arise in the chain under the influence of an external perturbation $\Phi(\xi, t)$ which satisfies the inequality

$$\int \Phi(\xi, t)\, \varphi_v^*(\xi, t)\, d\xi \neq 0.$$

We have not taken into account the processes of relaxation of the arising excitation. Phenomenologically the relaxation processes can be taken into account by replacing the excitation energy (26) by the complex value $\hbar\Omega - i\hbar\gamma$. Then an exponentially decreasing factor $\exp(-\gamma t)$ appears in the function (28) determining the average life-time $(1/\gamma)$ of the excitation.

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Solitonic Ionic Currents Along Microtubules

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Microtubules are very important cytoskeletal structures implicated in different cellular activities. We should mention the cell division and traffic of organelles (mitochondria, vesicles and other cargos) by kinesin and dynein motor proteins. Microtubules are also playing important roles in higher neuronal functions including mechanisms of consciousness and memory. We here investigate the circumstances and the conditions enabling microtubules to act as nonlinear electrical transmission lines for ions flow along their cylinders. We established a model in which each tubulin dimmer protein is an electric element with a capacitive, resistive and negative incrementally resistive property due to polyelectrolyte nature of microtubules in cytosol. The particular attention was paid to the role of nano-pores existing between neighboring dimmers within a MT wall which exhibit properties like ionic channels. These nano-pores are candidates to explain some properties of microtubules resembling to unipolar transistors enabling the rectification and amplification of ionic currents.

Keywords: Cytoskeleton, Microtubule, Polyelectrolyte, Nonlinear Transmission Line, Ionic Waves, Soliton, Nano-Pore.

1. INTRODUCTION

The cytoskeleton consists of three mayor types of thin rod-like filaments that span the cytoplasm: actin-based filaments (i.e., microfilaments), tubulin-based filaments (i.e., microtubules-MTs) and intermediate filaments (e.g., neurofilaments-keratin). There are at list three well-studied mechanical functions of the cytoskeleton *in vivo*, providing the mechanical strength of the cell, segregating the chromosomes and participating in the cellular traffic.[1]

MTs are long and rigid tubulin polymers. They are the key protein filaments that form several μm long hollow cylinders typically $d = 25$ nm in outer diameter, Figure 1, and exhibit dynamically unstable polymerization kinetics.

MTs are formed as ferroelectric polymers whose building block is a protein called tubulin. The hollow cylindrical wall of a MT is made up of parallel chains or protofilaments of tubulin-hetero-dimer. Each dimer, 8 nm in length, is composed of two units, a α-tubulin and a β-tubulin monomer. Each MT is generally made up of 13 protofilaments *in vivo*. Every tubulin monomer of the MT cylinder has a C-terminal helix, H12, followed by a highly acidic amino-acid sequence protruding out of the MT outer surface and called C-terminus tail (TT). The TTs sequences have length $l_{TT} = 4{,}5$ nm when completely outstretched. The TTs are being essential for MT interactions whit MT associated proteins and motor proteins, (kinesin and dynein).[2,3] More details about TTs structures and properties can be found in Ref. [4]. The shape of tubulin dimer with pertaining TTs is presented in Figure 2, including the detailed charge distributions on the tubulin surface calculated by team lead by Tuszynski.[5]

At neutral pH the negative charge on TT causes it to remain extended due to electrostatic repulsion. Under more acidic conditions the negative charges of pertaining amino acids within TT are partially neutralized by counter-ions condensed on them. This effect allows the β-TTs to be compacted by folding, as shown in Figure 3.

This is the largest structural effect in MTs and this fact has the great importance in establishing of our model with respect to the variable effective electrical capacitance of MTs. The detailed charge distribution on the MT surface, calculated in Ref. [5], enabled the determination of an electrostatic potential around a MT. It exhibits non-uniformity along the MT radius with a peak and a trough corresponding to every of 13 protofilaments and the area between them populated with mentioned nano-pores. These findings play an essential role in presented modeling of MT as a "coaxial cable" for intracellular communication.

In as much MTs are mostly negatively charged on their outer surface they are true polyelectrolyte polymers. This is the consequence of the fact that numerous amino acids forming tubulin have many negatively charged residues under physiological conditions, as was already stressed for TTs. Thus the conditions for application of Manning's theory[6] are provided enabling "condensation" of positive

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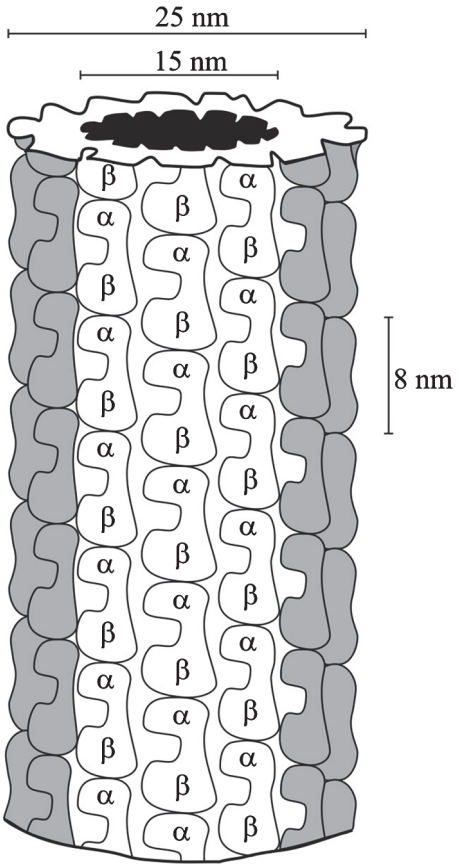

Fig. 1. A MT hollow cylinder of 13 parallel protofilaments with denoted characteristic dimensions: outer and inner diameters of 25 nm and 15 nm and tubulin dimer length of 8 nm.

counter-ions near the negative surface of MT landscape. As a result, every MT attracts positive counter-ions in narrow shell (ionic cloud-IC) near its surface, while negative ions of the cytosol are being repelled such that a roughly

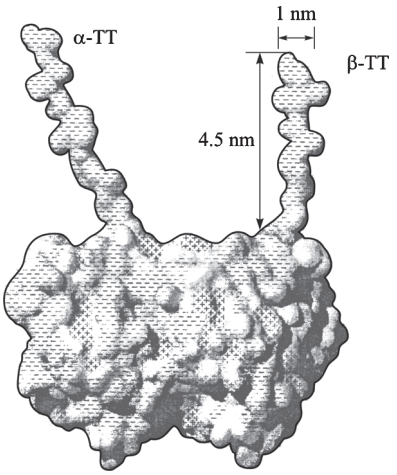

Fig. 2. The landscape of a tubulin dimer with dimensions of TTs the length of 4.5 nm and diameter of 1 nm, and surface charge distribution according to Tuszynski et al.[5]

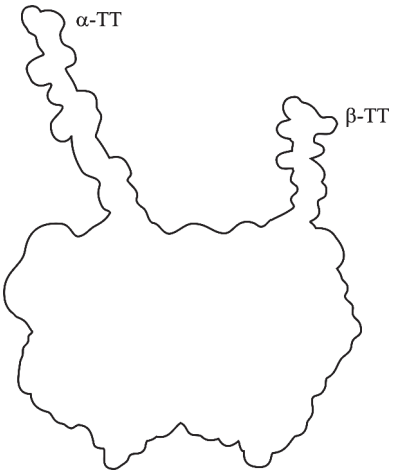

Fig. 3. The shape of TTs, where β-TT is being shrank, thus diminishing its capacity.

cylindrical depletion area around a MT is being created, Figure 4. The thickness of this depleted area is defined by so-called Bjerrum length, which is the distance from the MT surrounding IC at which the Coulomb energy of the screened surface charges equals the thermal energy

$$\frac{e^2}{4\pi\varepsilon_o\varepsilon l_B} = k_B T \tag{1}$$

Here e is the charge of an electron, ε_0 the permittivity of vacuum, ε the relative permittivity of cytosol and k_B is Boltzmann's constant. At a physiological temperature of $T = 310$ K, and taking $\varepsilon = 80$, it is clear from Eq. (1) that we have

$$l_B = 6.7 \times 10^{-10} \text{ m} \tag{2}$$

On the other hand, the thickness of the IC of positive counter-ions is dictated by the surface landscape and by its negative charge distributions. In the case of actin filament, Pollard and Cooper[6] estimated that the radius of IC is of the order of 8 nm, while the filament radius is 2.5 nm, giving IC thickness of 5.5 nm. From the best of our knowledge the evidences for MTs IC thickness are still missing but we anticipate that the corresponding value should be of the order of 2 nm since the surface of MTs is pretty less negative than in actin filament. Significant ionic movements within this bound IC are therefore allowed along the length of the MT provided that it is shielded from the bulk solution by the depletion layer of thickness l_B.

Bearing in mind the IC geometry around the MT we see that effectively MTs may act as biological "electrical wires" which can be modeled as nonlinear inhomogeneous transmission lines that are able to propagate nonlinear dispersive ionic waves. First attempts to elaborate the cytoskeleton as a network of nonlinear polyelectrolyte transmission lines are made by Tusyznski et al.,[7] for actin filaments, and later further elaborated by Sataric et al.[8] Similar approach was applied for MTs in Ref. [4]. Here

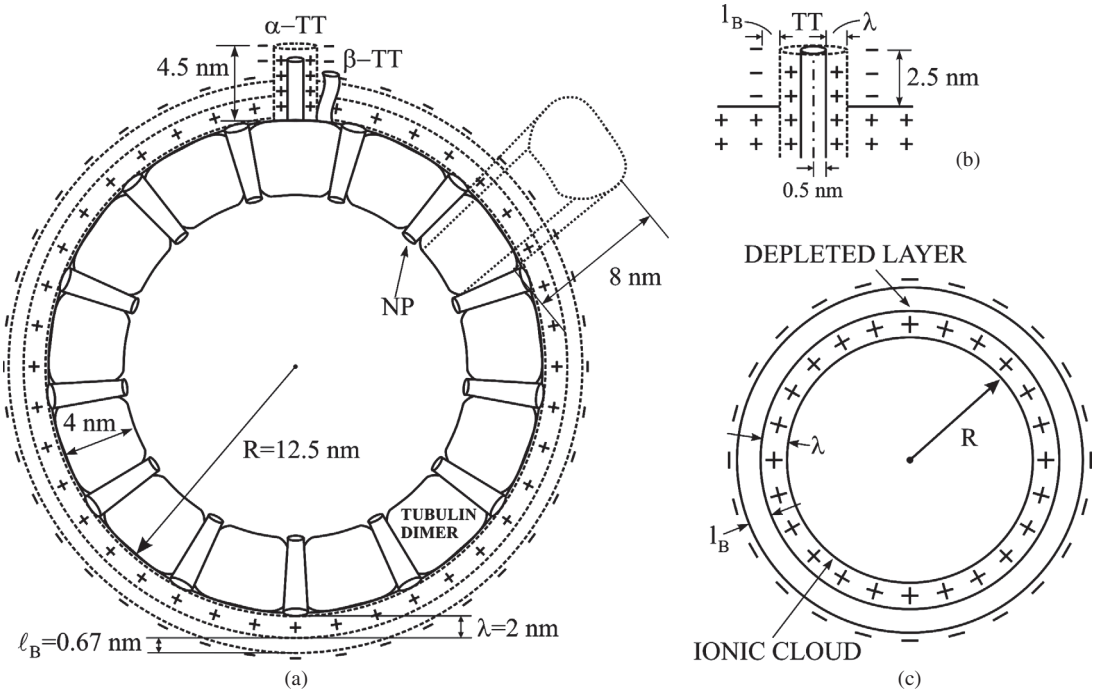

Fig. 4. The geometry of a MT with ionic cloud and depleted layer clearly depicted, including corresponding dimensions: the thickness of IC $\lambda = 2$ nm and Bjerrum length $l_B = 0.67$ nm. The top part of (a) shows α and β TTs.

we further amend the model introduced in Ref. [4] paying particular attention on the role of nano-pores (NPs) embedded in MT wall. In that respect we partially relied on the results published by the group lead by Eisenberg,[9, 10] which emphasize the particular role of calcium ions in dynamics of NPs.

Some fifteen years ago, Lin and Cantiello[11] performed an assay demonstrating that IC waves along a single actin filament were highly nonlinear in nature exhibiting the wave patterns remarkably similar to recorded solitary waveforms from various experimental studies on electrically stimulated nonlinear transmission lines.[12] Recently Tuszynski and Cantiello[13] conducted similar experiment on single MT which even manifested transistor-like amplification of electrical signals. It is our objective in this article to amend our earlier model[3, 4] in order to explain these features of MTs and their possible role in information processing within neurons.

The paper is organized in the following way:

In the Section 2 we reexamined the estimations of elementary capacitance, inductance and resistance components of elementary MT ring and elaborated the role NPs.

The Section 3 is dedicated to the modeling of each MT as a nonlinear C–R–G transmission line and evaluating KdV- like equation and its solution.

Section 4 deals with the influence of viscous dissipation and intrinsic electrical field on the dynamics of ionic solitons along MTs.

The Last section offers the conclusions regarding this interesting physical mechanism in terms of the cell's electrical activity provided by the MTs and their interactions with ions from MT lumen.

2. THE ELEMENTS OF A MT AS AN ELECTRIC TRANSMISSION LINE

The basic assumption is that a MT with condensed counter-ions is separated from the rest of the ions in the bulk solution by a depleted layer which plays the role of a dielectric medium located between MTs and the cytosol solution. This IC is electro-conductive medium responsible for electrical properties of MT. The depleted layer provides that IC and repelled ions represent the "conductive plates" of cylindric capacitor. The ions injected from appropriate cell's compartment are confined to flow parallel with MTs within the IC of the supposed thickness $\lambda = 2$ nm. We here pay particular attention to the case where calcium ions are injected within IC.

2.1. The Capacitance of a MT Elementary Ring

The elementary ring (ER) consists of 13 dimers, Figure 4(a), forming the ring of radius of 12.5 nm and of length $l = 8$ nm. On the basis of the Poisson-Boltzmann equation the detailed derivation given in Ref. [4] yielding the basic capacity of such cylindric ER as follows

$$C_0 = \frac{2\pi\varepsilon_0\varepsilon l}{\ln(1 + (l_B/R))} \tag{3}$$

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where $l = 8 \times 10^{-9}$ m is length of an ER, (length of one tubulin dimer), $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m, $\varepsilon = 80$, $l_B = 0.67 \times 10^{-9}$ m. In contrast to,[4] we here take $R = (d/2 + \lambda) = (12.5 + 2)$ nm leading to elementary capacity:

$$C_0^{(1)} = \frac{6.8 \times 8.85 \times 10^{-12} \times 80 \times 8 \times 10^{-9}}{\ln(1 + (0.67/14.5))} = 0.8 \times 10^{-15} \text{ F} \tag{4}$$

In the same way, we consider an extended TT as a smaller cylinder with the length $l_{TT} = 4.5$ nm and the radius $R_{TT} = 0.5$ nm giving $R = R_{TT} + \lambda = 2.5$ nm. From formula (3) we calculate an additional capacity $C_0^{(2)}$ for $n = 13 \times 2 = 26$ TTs, contained in the same ER, composed of 13 dimers, see Figure 4(a), top position, and Figure 4(b).

The length of each TT capacitor is now $l_{eff} = (4{,}5\text{-}2)$ nm $= 2.5$ nm, since 2 nm are embedded in part of already calculated capacitor $C_0^{(1)}$

$$C_0^{(2)} = \frac{26 \times 6.28 \times 8.85 \times 10^{-12} \times 80 \times 2.5 \times 10^{-9}}{\ln(1 + (0.67/2.5))} = 1 \times 10^{-15} \text{ F} \tag{5}$$

Two above capacities are being connected in parallel, so the total maximal capacity of the ER is now estimated to be

$$C_0 = C_0^{(1)} + C_0^{(2)} = 1.8 \times 10^{-15} \text{ F} \tag{6}$$

It was already mentioned that the increase of negative voltage should increase the concentration of positive counter-ions in IC. These ions additionally condense on the surface of a MT and on its TTs. This brings about β-TTs to shrink diminishing their effective length $l_{eff} = 4.5$ nm, and thus the capacity $C_0^{(2)}$. It is reasonable to assume that this change of capacity is linear with respect to the applied voltage. This circumstance is of fundamental importance for establishing of an adequate model of MTs as nonlinear transmission lines for ionic currents. It will be elaborated in next Section in details.

2.2. The Effective Inductance of an ER in MT

The straightforward calculation performed in Ref. [4] revealed that the order of magnitude of the inductance of an ER in MT is of the order of $L \sim 10^{-15}$ H. This is very small value regarding pertaining reactive resistance. For example, if the frequency ω is of the order of MHz we have $\omega L \sim 10^{-9}$ Ω while capacitive impedance amounts $1/(\omega C) \sim 10^{9}$ Ω. Similar discrepancy holds for the energies of corresponding fields

$$\frac{1}{2} L i^2 \ll \frac{1}{2} C_0 v^2 \tag{7}$$

since i is of the order of nA and v of the order of several mV. Thus, the role of inductance in this model could be safely ignored. This circumstance removes the discrepancy between estimated wave velocity obtained for ionic current along actin filament[14] and the experimental evidences.

2.3. The Ohmic Resistance of an ER in MT

We here safely rely on recently published experimental findings, regarding electro-orientation method, performed by MTs *in vitro*.[15] This assay provided the value of MT conductivity as follows:

$$\sigma = (0.15 \pm 0.01) \frac{S}{m} \tag{8}$$

If we now take that the resistivity within an IC around a MT (with the thickness λ) is roughly homogeneous, the resistance of the ER with length $l = 8 \times 10^{-9}$, and the area $A = 2\ \pi R \lambda$, see Figures 4(a, c), is estimated to be of the order of:

$$R = \frac{1}{\sigma} \frac{l}{2\pi R \lambda} = \frac{8 \times 10^{-9}}{0.15 \times 2\pi \times 12.5 \times 10^{-9} \times 2 \times 10^{-9}} \ \Omega = 0.34 \times 10^{9}\ \Omega \tag{9}$$

2.4. The Nano-Pores in MT

The structural design of the MT is remarkable in terms of its biological functions. By cryoelectron microscopy, the group led by Nogales,[16] performed an assay which clearly revealed that between tubulin dimers within MT wall the conical NPs of about 1.7 nm in diameter exist, Figure 5. These NPs are characterized by the presence of the α-helix, H6, which can move down in MT structure increasing the effective size of the holes enhancing the diffusion of small ligands and ions into the MT lumen and vice versa. It could be expected that this movement may be controlled by physiological conditions.

We believe that NPs in MT are not just tubes that allow diffusion across it, but more sophisticated molecular machines able to respond on the changes in their environment and undergo precisely timed conformational change in order to be selective and to tune electric currents and signals confined within considered IC.

For comparison, it is well known that some ionic channels in cell membrane are being gated by ATP hydrolysis.[17] We could speculate that in the case of NPs in MTs the hydrolysis of GTP ligand binding process could cause the mentioned movement of H6, thus gating them. Such closed NPs are being open after an activation signal (in this case electrical) passes by. We here pay attention

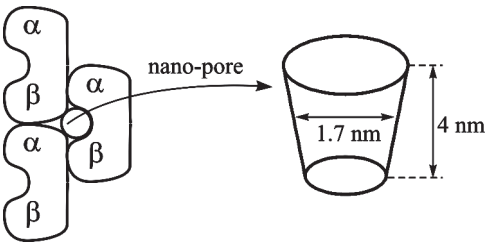

Fig. 5. The sketch of a NP with approximates shape and dimensions, the conical shape follows from MT geometry.

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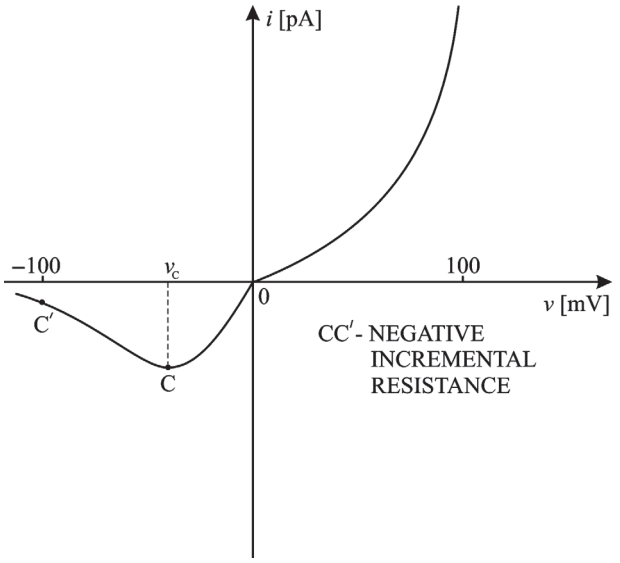

Fig. 6. The qualitative shape of current–voltage characteristics for NPs in the context of Eisenberg's group's work.[9,10] The segment CC′ shows the negative incremental resistance.

on the possible role of calcium ions which are good candidates for being injected from endoplasmic reticulum, and thus transported along MTs.

The group lead by Eisenberg[9,10] examined an artificial single asymmetric conical ionic NP-system whose rectifying properties for ionic current are changed dramatically by addition of a small concentration of calcium ions. They suggest that binding and unbinding of calcium ions to the pore walls produces pumping of potassium ions, by a flashing-rachet mechanism. This simplified case could suggest that we envisage NPs in MT as possible candidates for this kind of calcium-induced voltage gating. Nevertheless NPs in MTs are much more complex because the potential across the NP is a dynamic variable, hence the conductive properties of the pore must vary accordingly.

We could start this segment by the fact that voltage–current functions for mentioned artificial NPs with calcium-gating exhibit negative incremental resistance for a specific domain of sufficient negative voltage across NPs, Figure 6, section C–C′.[9,10] The threshold voltage v_c depends on calcium concentration and on overall characteristics of NPs. But we will later include the effects of nonlinear dynamical variations of NPs conductivity within MT.

3. THE MODEL OF MT AS A NONLINEAR TRANSMISSION LINE

On the basis of above estimations for the resistive components of the ER of a MT we are now in the position to establish the corresponding model. We consider a long transmission line (or ladder networks) composed of lumped sections equal to ERs. A typical section scheme is shown in Figure 7.

The longitudinal current is represented by the series of negligible inductance L_n and ohmic resistance R_n for one ER, as being estimated in previous section. The nonlinear capacity charged with Q_n is in parallel with total conductance of all NPs, G_n contained in an ER of the length $l = 8$ nm. This is because we consider the bulk cytosol and MT lumen as grounded. Applying Kirchoff's low to ladder of ERs, envisaged to represent coupled electrical circuits, we have

$$i_n - i_{n+1} = \frac{\partial Q_n}{\partial t} + G_n v_n \tag{10}$$

$$v_{n-1} - v_n = R_n i_n \tag{11}$$

We already stressed that the ER's capacitance C_0 must change with an increasing concentration of counter-ions due to the shrinking of flexible β-TTs. This implies that the charge on this capacitor diminishes with increased voltage in a nonlinear way as follows:

$$Q_n = C_0(1 - bv_n)v_n \tag{12}$$

where b is expected to be small enough ($bv_n \ll 1$) since the change of the area of a β-TT is small compared with total

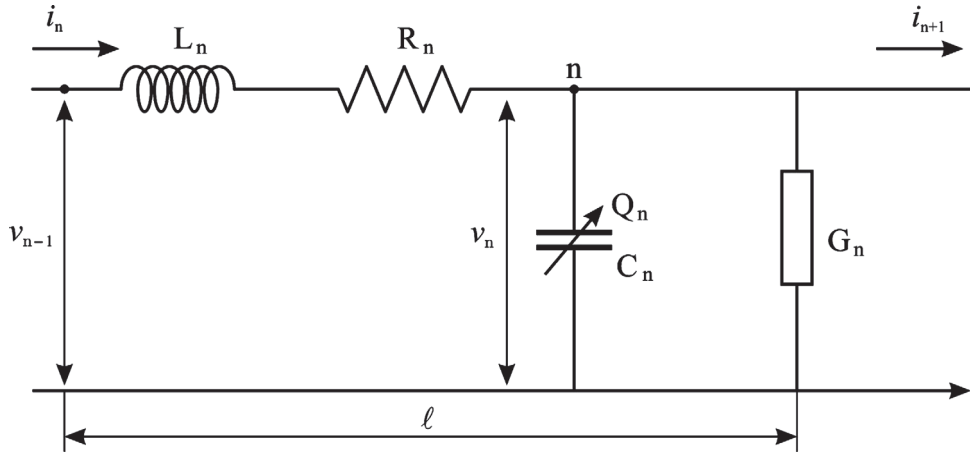

Fig. 7. An effective circuit diagram for the n-th ER with characteristic elements for Kirchoff's laws.

outer surface of the corresponding tubulin dimmer plus surface of α-TT. Now, we pay attention on the role of NPs in the context of negative incremental resistance (conductance) discovered by Eisenberg's group.[9, 10] We upgraded this idea by choosing the conductance of NPs to have the form dependent of voltage

$$G_n = -G_0\left(1-\beta\frac{dv_n}{dt}\right) \tag{13}$$

thus bringing an additional nonlinearity of system.

Replacing (12) and (13) in Eq. (10), one gets

$$i_n - i_{n+1} = -G_0 v_n + C_0\frac{\partial v_n}{\partial t} + \beta G_0 v_n\frac{\partial v_n}{\partial t} - 2bC_0 v_n\frac{\partial v_n}{\partial t} \tag{14}$$

$$v_{n-1} - v_n = R_n i_n \tag{15}$$

We now introduce the characteristic impedance of our system

$$Z = \frac{1}{C_0\omega} \tag{16}$$

and establish new function $u(x, t)$, unifying voltage and current as follows

$$u_n = Z^{1/2} i_n = Z^{-1/2} v_n \tag{17}$$

Bearing in mind that voltage v_n and current i_n change gradually from ER_n to its neighbors, we can safely expand u_n in a continuum approximation using a Taylor serie in terms of a small spatial parameter l (the length of a dimmer):

$$u_{n\pm1} = u \pm l\left(\frac{\partial u}{\partial x}\right) + \frac{l^2}{2!}\left(\frac{\partial^2 u}{\partial x^2}\right) \pm \frac{l^3}{3!}\left(\frac{\partial^3 u}{\partial x^3}\right) + \frac{l^4}{4!}\left(\frac{\partial^4 u}{\partial x^4}\right) \pm \ldots \tag{18}$$

By using Eqs. (16–18), system of Eqs. (14) and (15) now reads

$$\frac{\partial u}{\partial x} + \frac{l}{2}\frac{\partial^2 u}{\partial x^2} + \frac{l^2}{6}\frac{\partial^3 u}{\partial x^3} + \frac{l^3}{24}\frac{\partial^4 u}{\partial x^4} + \frac{l^4}{120}\frac{\partial^5 u}{\partial x^5} - G_0 Z u + \frac{ZC_0}{l}\frac{\partial u}{\partial t} + \frac{\beta Z^{3/2} G_0}{l} u\frac{\partial u}{\partial t} - \frac{2bZ^{3/2}C_0}{l} u\frac{\partial u}{\partial t} = 0 \tag{19}$$

$$\frac{\partial u}{\partial x} - \frac{l}{2}\frac{\partial^2 u}{\partial x^2} + \frac{l^2}{6}\frac{\partial^3 u}{\partial x^3} - \frac{l^3}{24}\frac{\partial^4 u}{\partial x^4} + \frac{l^4}{120}\frac{\partial^5 u}{\partial x^5} - \frac{RZ^{-1}}{l} = 0$$

Adding last two equations and neglecting terms beyond the order of l^5, we get

$$2\frac{\partial u}{\partial x} + \frac{l^2}{3}\frac{\partial^3 u}{\partial x^3} + \frac{1}{l}\left(RZ^{-1} - G_0 Z\right)u + \frac{ZC_0}{l}\frac{\partial u}{\partial t} + \left(\frac{\beta Z^{3/2} G_0}{l} - \frac{2bZ^{3/2}C_0}{l}\right)u\frac{\partial u}{\partial t} = 0 \tag{20}$$

We proceed by imposing the condition that third term in above equation should be zero

$$(RZ^{-1} - G_0 Z) = 0 \tag{21}$$

This condition enables that the ohmic loss could be balanced by fresh ions injected from NPs since these act as ionic pumps in suitable voltage regime.

It is now the important step to go over to dimensionless variables. In that respect, we introduce the characteristic scales of time and length. The characteristic time of charging ER-capacitor C_0 is $\tau = RC_0$. Taking the values of Eqs. (6) and (9), we have

$$\tau = 0.34\times10^{9}\times1.8\times10^{-15}\ \text{s} = 0.6\times10^{-6}\ \text{s} \tag{22}$$

The characteristic velocity of spreading the ionic wave is $v_0 = l/\tau$, which yields

$$v_0 = \frac{8\times10^{-9}}{0.6\times10^{-6}} = 13.3\ \frac{\text{mm}}{\text{s}} \tag{23}$$

Now we will use the standard travelling-wave form for unified function $u(x, t)$:

$$u(x,t) = u\left(\frac{x}{l} - v\frac{t}{l}\right) = u\left(\frac{x}{l} - \frac{v}{v_0}v_0\frac{t}{l}\right) = u\left(\frac{x}{l} - s\frac{t}{\tau}\right);\quad s = \frac{v}{v_0} \tag{24}$$

Introducing joined space-time dimensionless variable

$$\xi = \frac{x}{l} - s\frac{t}{\tau} \tag{25}$$

we have

$$\frac{\partial u}{\partial x} = \frac{1}{l}\frac{du}{d\xi};\quad \frac{\partial^3 u}{\partial x^3} = \frac{1}{l^3}\frac{d^3 u}{d\xi^3};\quad \frac{\partial u}{\partial t} = -\frac{s}{\tau}\frac{du}{d\xi} \tag{26}$$

It brings Eq. (20) into the form of ordinary differential equation, as follows

$$\frac{d^3 u}{d\xi^3} - 3\left(\frac{ZC_0 s}{\tau} - 2\right)\frac{du}{d\xi} + 3\frac{(2Z^{3/2}bC_0 - Z^{3/2}\beta G_0)s}{\tau} u\frac{du}{d\xi} = 0 \tag{27}$$

If we go back to the definition $u = Z^{1/2} i$, and then introduce the dimensionless current $W = i/i_0$, with i_0 being the peak value, we have

$$u = Z^{1/2} i_0 W = Z^{1/2} i \tag{28}$$

Inserting Eq. (28) into (27) yields

$$\frac{d^3 W}{d\xi^3} - 3\left(\frac{ZC_0 s}{\tau} - 2\right)\frac{dW}{d\xi} + 3\frac{i_0 s}{\tau}(2Z^2 bC_0 - Z^2\beta G_0)W\frac{dW}{d\xi} = 0 \tag{29}$$

Introducing the abbreviations

$$A = 3\left(\frac{ZC_0 s}{\tau} - 2\right); \quad B = 3\frac{i_0 s}{\tau} Z^2 (2bC_0 - \beta G_0) \quad (30)$$

we have

$$\frac{d^3W}{d\xi^3} - A\frac{dW}{d\xi} + BW\frac{dW}{d\xi} = 0 \quad (31)$$

After two successive integrations with the appropriate boundary conditions as follows

$$W_\infty = \left(\frac{dW}{d\xi}\right)_\infty = 0$$

one gets

$$\frac{dW}{d\xi} = \left(\frac{B}{3}\right)^{1/2} (DW^2 - 2W^3)^{1/2}; \quad D = \frac{3A}{B} \quad (32)$$

By introducing the new variable y

$$W = \frac{1}{2} D \cos h^2(y) \quad (33)$$

we finally obtain

$$W = \left[\frac{3(s\sigma - 2)}{s\sigma i_0 Z(2b - \beta(G_0/C_0))}\right] \times \sec h^2\left\{\frac{1}{2}\left[\frac{3}{2}(s\sigma - 2)\right]^{1/2}\left(\frac{x}{l} - s\frac{t}{\tau}\right)\right\} \quad (34)$$

with

$$\sigma = \frac{ZC_0}{T} \quad (35)$$

Let us estimate the dimensionless parameter σ. Starting from $Z = 1/C_0\omega$ we get $\sigma = 1/\tau\omega$. If we choose the moderate frequency of range $\omega = 10^4$ s^{-1}, which is based on the fact that the characteristic oscillations of ionic currents in Ref. [13] are of the order of kHz. Combining it with Eq. (22), it readily follows

$$\sigma = \frac{1}{6 \times 10^{-7} \times 10^4} = 1.67 \times 10^2 \quad (36a)$$

Accordingly the pertinent impedance reads

$$Z = \frac{1}{C_0\omega} = 5.56 \times 10^{10}\ \Omega \quad (36b)$$

Taking from the definition $i = i_0 W$, the IC current now has the form:

$$i(x,t) = \left[\frac{3(s\sigma - 2)}{s\sigma Z(2b - \beta(G_0/C_0))}\right] \sec h^2\left\{\frac{1}{2}\left[\frac{3}{2}(s\sigma - 2)\right]^{1/2}\left(\frac{x}{l} - s\frac{t}{\tau}\right)\right\} \quad (37)$$

This solution we will call ionic cloud soliton (ICS). It is apparent that down-limit of it velocity follows from the necessary condition

$$s\sigma - 2 > 0 \quad (38)$$

yielding

$$s \geq 1.2 \times 10^{-2} \quad (39)$$

Relying on Eq. (23) we now have the inequality for the speed of ICS

$$v > 0.16\ \frac{\text{mm}}{\text{s}} \quad (40)$$

Interestingly that the upper limit of ICS speed also exists on the basis of condition that its minimal width d can not be smaller of the length of one ER:

$$d = \frac{l}{1/2[(3/2)(s\sigma - 2)]^{1/2}} \geq l; \quad s\sigma \leq \frac{14}{3} \quad (41)$$

It gives

$$s \leq 2.8 \times 10^{-2} \text{ or } v \leq 0.34\ \frac{\text{mm}}{\text{s}} \quad (42)$$

Thus we get a pretty narrow window of possible solitonic speeds

$$(0.16 \leq v \leq 0.34)\ \frac{\text{mm}}{\text{s}} \quad (43)$$

In the other hand, using the balance condition, Eq. (21) we may estimate the negative conductance of NPs contained in an ER:

$$G_0 = \frac{R}{Z^2} = \frac{3.4 \times 10^8 \Omega}{5.6^2 \times 10^{20} \Omega^2} = 1{,}1 \times 10^{-13}\ \text{Si} \quad (44)$$

The order of magnitude of corresponding current is

$$i_{\text{NP}}^{(13)} = vG_0 = 0.1\ \text{V} \times 1{,}1 \times 10^{-13}\ \text{Si} = 1,1 \times 10^{-14}\ \text{A} \quad (45)$$

If the conductance nonlinearity parameter (βG_0) is much smaller than capacitance one, (bC_0) the ICS has very small amplitude of the order of 0.3 pA. But in the case where the conductance nonlinearity is getting competitive with capacitance nonlinearity, the ICS amplitude increases accordingly.

Thus, if we take $\sigma s = 2.5$, and assuming small margin of nonlinearities as follows

$$2b - \beta\frac{G_0}{C_0} = 0.1$$

which with $Z = 5.56 \times 10^{10}\ \Omega$, Eq. (37), yields:

$$i_0 = 27\ \text{pA} \quad (46)$$

If we compare it with experimental evidence from,[13] where the collected currents were in the range (2–6) nA, we could infer that dozens of ICS constituting the train-like pulses are involved in measured currents along MT.

Interestingly, the nonlinearities in NP conductance and TTs capacity have the opposite signs. This is the consequence of the fact that pumping of positive ions (potassium) by NPs from MT lumen causes diminishing of TTs capacity, and vice versa. The opposite signs of two nonlinearities lead to increase of amplitude of ICS, but do not affect its width and speed. We stress here that this bell-shaped ICS qualitatively differs from the localized wave obtain in Ref. [4] which was anti-kink-like.

We have shown a couple of numerical illustrations of ICS for different values of $s\sigma$ as represented in Figure 8.

We here emphasize that ICSs of our model exhibit the typical properties of KdV solitons in fluid dynamics. Their velocities are bellow the characteristic velocity v_o and increasing the velocity leads to increase of their amplitude and decrease of corresponding width (higher localisation).

Eventually, we could estimate the charge carried by one ICS with $s\sigma = 3$. Starting from $Q_n = C_0(1 - bv_n)v_n$ and using continuum approximation, the total charge is

$$Q = \frac{C_0}{l}\int_{-\infty}^{+\infty}(v - bv^2)\,dx; \quad v(x,t) = Z \cdot i(x,t) \tag{47}$$

or explicitly:

$$Q = \frac{C_0}{l}\left[i_o Z\right]\int_{-\infty}^{+\infty}\operatorname{sec h}^2\left\{\frac{1}{2}\left[\frac{3}{2}\times 1\right]^{1/2}\left(\frac{x}{l} - s\frac{t}{\tau}\right)\right\}dx$$
$$-\beta\frac{C_0}{l}[i_0 Z]^2\int_{-\infty}^{+\infty}\operatorname{sec h}^4\left\{\frac{1}{2}\left[\frac{3}{2}\times 1\right]^{1/2}\left(\frac{x}{l} - s\frac{t}{\tau}\right)\right\}dx$$

The second addend could be safely ignored, as very small. Using Eq. (47) with already known parameters C_0, Z and $i_0 = 27$ pA, yields

$$Q = 2.7\times 10^{-15}\ \text{C} = 1.7\times 10^4\ \text{e} \tag{48}$$

One could also assess the energy carried by an ICS. The basic assumption is that the energy loss due to the ohmic resistance is being compensated through fresh supply of ions by NPs. Under such circumstances it is clear that the energy contained in ER capacitors should be preserved. This energy amounts:

$$E = \frac{Q^2}{2C_0} = \frac{(2.7\times 10^{-15}\ \text{C})^2}{2\times 1.8\times 10^{-15}\ \text{F}}$$
$$= 2\times 10^{-15}\ \text{J} = 1.7\times 10^4\ \text{eV} \tag{49}$$

Taking the charge of $Q = 1.7\times 10^4$ e, Eq. (3.39), to be attributed to monovalent ions, the energy per single ion involved in ICS is of the order of 0.8 eV which is one thirty times greater of thermal energy (25 meV).

4. THE INFLUNCE OF VISCOUS DISSIPATION AND INTRINSIC ELECTRIC FIELDS ON THE DYNAMICS OF ICSs ALONG MTs

In the absence of dissipation the ICS, Eq. (46), is expected to move at a constant speed s. But, since ICS behaves like

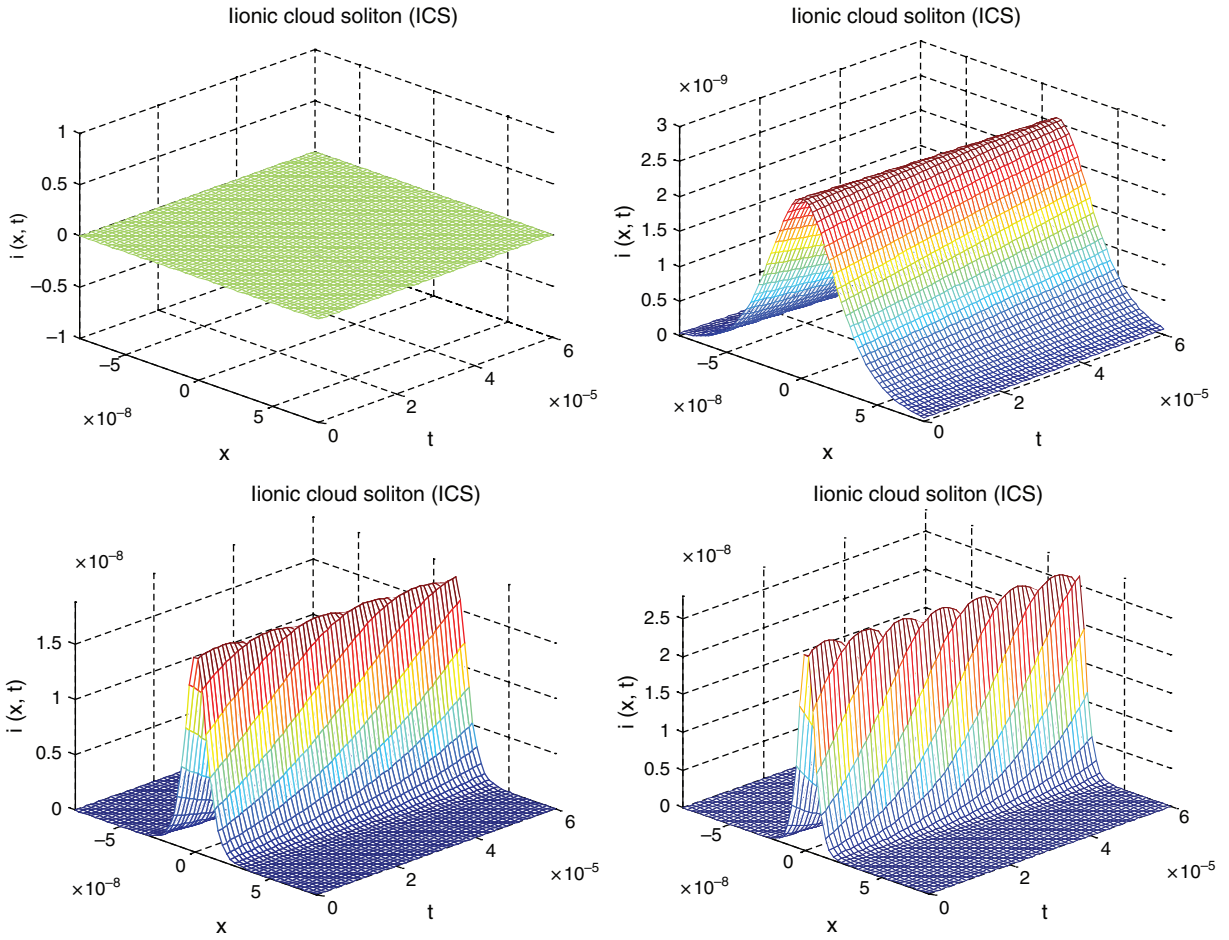

Fig. 8. Numerical solutions: (a) for $s\sigma = 2$ (the smearing limit), (b) for $s\sigma = 2.1$, (c) for $s\sigma = 3$ and (d) for $s\sigma = 4$.

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a "particle" in a viscous fluid, it is evident that the role of viscous cytosol is to decrease its speed.

The rate of deceleration could be calculated on the basis of Energy method.[18, 19] This rate appears to follow exponential decay tending to the down-limit ($s\sigma = 2$), Eq. (38), when ICS is being totally smeared out.

In the other hand, since ICS is being subjected under the action of intrinsic electrical field of MT[20, 21] it should be accelerated.

The most interesting case is when these two opposite tendencies are mutually balanced. The viscous damping is proportional to the speed of an ICS:

$$F_{\text{vis}} = -kv \tag{50}$$

where k depends on dynamical viscosity of cytosol and on the geometry of ICS, while the force of electric field simply reads

$$F_{\text{if}} = Q \cdot E_i \tag{51}$$

with E_i being the strength of intrinsic electrical field along MT.

Equations (50) and (51) we can calculate the value of parameter k on the basis of estimated $E_i = 10^4$ V,[21] and taking $v = 0.2$ mm/s, and Q from Eq. (48), we get

$$k = \frac{Q \cdot E_i}{v} = \frac{2.7 \times 10^{-15} \times 10^4}{0.2 \times 10^{-3}} = 1.35 \times 10^{-7} \ \frac{\text{kg}}{\text{s}} \tag{52}$$

Obviously the strength of intrinsic electric field could be an important mechanism for tuning dynamics of ICS.

5. CONCLUSION AND DISCUSSION

We here farther improved the former model[3, 4] for ionic currents propagating in narrow layer along MTs. This model is established in terms of polyelectrolyte concept applied to the molecular structure and geometry of MT, and taking into account its interaction with solvent ions. As the consequence, the MT cylinder with flexible TTs, surrounded by counter-ions from cytosol, is considered as an electrical transmission line, with R, C and G elements estimated on the basis of approximate MT geometry and other experimental evidences available. The essentially important nonlinearity of model pertains to its capacitance arising from structure and dynamics of β-TTs and from dynamic conductance of NPs.

We here paid the particular attention to the possible role of NPs acting as ionic pumps thus creating the negative incremental resistance in the considered circuit.

As a result, the nonlinear second order differential equation for ionic current along MT was evaluated. The travelling-wave solution of Eq. (48) is localized ICS which exhibits the features of celebrated Korteveg de Wries soliton from fluid dynamics. It is important to emphasize the qualitative difference of this bell-shaped ICS compared with anti-kink solution obtained in former approach.[4] It appears that the role of NPs exerts some kind of self-focusing effect on IC increasing its amplitude.

Interestingly, this ICSs have pretty narrow window of possible velocities with the order of magnitude being a fraction of (mm/s), based on the assessed capacity and ohmic resistance of an ER of such transmission line.

We estimated the order of magnitude of pertaining current along a single MT, together with the charge of ICS and its energy. These parameters have pretty reasonable values, being comparable with still scarce experimental data.

Let us especially pay attention on the interesting experimental assay performed by the group led by Tuszynski.[13] The measured values for the ionic current along the single MT were in the range (2–6) nA. Moreover they stated that the electrical amplification by MTs is equivalent to their ability to act as the bimolecular living transistors. In that respect the NP current of pumped ions, Eq. (45), in our model is expected to play the control role as well as the base current does play in bipolar transistor.

It is worth to mention the experimental results regarding intracellular calcium dynamics.[22] Calcium ions released from endoplasmic reticulum could be candidate for propagation along MT in the context of present model. The calcium puffs launched from above compartment could have enough energy (proper velocity) to form ICS. This localized propagation of ions is being controllable by cellular electric fields and goes faster than the pure diffusion through cytosol. Experimental speed of diffusive calcium waves, according to Ref. [22], is of the order of 12 (μm/s), which is one order of magnitude bellow the ICS window, Eq. (43).

We finally considered biophysically realistic circumstances where the damping effects are being balanced by energy supply provided by existing cell's intrinsic electrical fields.

Our results could encourage experimentalists to conduct more subtle assays in an attempt to ellucidate this important aspect of cellular activities.

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9

Collective Properties of Biological Systems

Solitons and Coherent Electric Waves in a Quantum Field Theoretical Approach

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ABSTRACT: We present a dynamical scheme for biological systems. We use methods and techniques of quantum field theory since our analysis is at a microscopic molecular level. Davydov solitons on biomolecular chains and coherent electric dipole waves are described as collective dynamical modes. Electric polarization waves predicted by Fröhlich are identified with the Goldstone massless modes of the theory with spontaneous breakdown of the dipole-rotational symmetry. Self-organization, dissipativity, and stability of biological systems appear as observable manifestations of the microscopic quantum dynamics.

1. Introduction

Biological systems are collective systems indeed. They seem to be satisfactorily described when the list of their components is supplemented by the prescription of their location inside the system and the timing of their operation. Biologists refer to such a timing as a "cell clock"; what is now

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required is the engine responsible for such a dynamical ordering. Dynamical mechanisms that could give account for collective behavior have been proposed in the frame of different approaches.

H. Fröhlich[1] has introduced, in a framework of far-from-equilibrium processes, coherent electric polarization waves as the physical agent able to control the working of distant and separate parts of the system, making them cooperative. On the other hand, it has been shown that biomolecules are able to host on their own chains, in a conservative framework, highly nonlinear subdynamics which give rise to deep structural and conformational changes.

Very interesting subdynamics have been investigated by A. S. Davydov,[2,3] who has shown that metabolic reactions can result in strictly confined "bumps" traveling along biomolecular chains and carrying energy in an efficient way a long distance away from the point of release.

Our aim is to provide a systematic approach to the dynamics of a biological system[4,5] by using the widely accepted tool for exploring the collective properties of a system made from a large number of elementary microscopic components, namely, quantum field theory (QFT). We choose quantum field theory instead of classical field theory since quantum theory has proved to be the most adequate tool for describing atoms, molecules, and their interactions, and we wish actually to carry out an investigation of the biological system at a molecular level. Moreover, we choose quantum field theory instead of quantum mechanics because we are not interested in singling out just a few degrees of freedom of the system; on the contrary we would like to get a comprehensive description of the dynamics of the whole system (collective behavior).

The QFT approach could be sketched as follows. In a preliminary stage the microscopic components of the system are enumerated; the state of minimum energy is called the ground state or the vacuum. It is possible that more than one ground state exists for the same system, each of them being characterized by different macroscopic properties. In such a case different regimes or phases will correspond to different vacua. Subsequently, the microscopic components are linked to field operators whose dynamical evolution is obtained through equations of motion derived from a Lagrangian. The usual prescription is to choose this Lagrangian as symmetric as possible in order to get the simplest equations of motion while that part of the dynamics which is less symmetric is taken into account via the vacuum.

Where the vacuum does not exhibit the same invariance of the Lagrangian, one says that a spontaneous breakdown of the symmetry occurs. General theorems have been derived which elucidate the dynamical consequences on the whole system of the interplay between a symmetric Lagrangian and a nonsymmetric vacuum. The Goldstone theorem[6] is a

very important one; it states that massless modes (usually called Goldstone bosons) appear when the Lagrangian is invariant, while the vacuum is noninvariant under a certain group of transformations. Notice that the invariance or noninvariance of the Lagrangian and of the vacuum under a transformation compels us to take into account the whole of the system degrees of freedom so that the Goldstone modes are really collective modes. A massless mode actually describes a long-range interaction among the system components and such an interaction establishes a correlation (i.e., order) among the components. The noninvariance of the vacuum under a certain transformation thus implies that this vacuum is an ordered state, in which absorption and emission of quanta of the Goldstone field, undergoing the transformation of noninvariance, take place. The mathematical structure of the theorem makes evident that the above-mentioned boson quanta exist in a condensed form and their condensation onto the ground state corresponds to a sort of "printing" of the symmetry pattern on the vacuum.

It is interesting to compare the collective properties arising from a microscopic theory with the macroscopic predictions obtained at a thermodynamic level in the framework of different approaches. The ordering of biological systems by coherent electric waves as proposed by Fröhlich could be understood, for instance, as the phenomenological manifestation of the breakdown of some symmetry at a microscopic level.[4,5,7] In this case the fact that the condensation, i.e. the emission and absorption, of Goldstone boson quanta does not require any energy expense is the microscopic counterpart of the thermodynamic dissipativity condition, which is a necessary condition[1] for the appearance of these waves. Dissipativity is a very general and important feature of living systems as far-from-equilibrium systems.[8] If we consider the Goldstone modes not as particles but as waves, the condensation of quanta producing an ordered state would lead us to take into account coherent waves. Coherent electric polarization waves require then, in the above framework, a vacuum with a nonzero value for the relevant field, namely, for the polarization field. As a matter of fact, water around biomolecules exhibits a nonzero polarization, i.e., an electret.[9,10]* Zero frequency modes have been also observed, which point to the existence of massless modes.[11]

Nonlinear dynamical effects on one-dimensional systems such as biomolecular chains, e.g., Davydov solitons, can also be analyzed in terms of boson condensation.[12–14] In the Fröhlich wave case, the condensation

* We define the electret state of a dielectric material, in agreement with the definition of Refs. 9 and 10, as a metastable state characterized by a stored polarization. Its decay time must be assumed long with respect to the characteristic time of experiments performed on the material.

induced by a symmetry breakdown occurs on large regions homogeneously, whereas in one-dimensional molecular systems one has quite localized condensations producing solitons. QFT can provide a unified description of such physically different situations. It is also interesting to note that an analysis by means of boson condensation can be performed in a variety of systems in many-body physics.[12] Thus a QFT approach has the appealing feature of extending methods and mathematics currently used in condensed matter physics to living matter.

In the following sections we present such a unified treatment. First we sketch qualitatively the interplay between Davydov and Fröhlich mechanisms (Section 2). Then we present an account of both of them and we discuss the transition from one regime to the other (Section 3). Finally we discuss the phenomenological implications of the theoretical scheme and the available supporting experimental evidence (Section 4).

2. *Solitons, Electrets, and Fröhlich Waves*

A very appealing schematization features a biological system as a set of electric dipoles, a dielectric with peculiar properties. Both water molecules—which account for a large amount of the global weight—and the macromolecules having a role in the biochemical machinery (DNA, RNA, proteins, and so on) have relevant dipole values or exhibit atomic subgroups with large dipole values. The starting point of the Fröhlich approach, for instance, is just to treat living matter as a dielectric, where the correlations among different parts are established by electric polarization waves. On the other hand the dynamics of a one-dimensional set of dipoles gives rise to vibrational solitons on biomolecular chains as proposed by Davydov.

Let us first discuss these two mechanisms separately. Biological systems are considered by Fröhlich as three-dimensional sets of dipoles. It is well known that the basic dipole–dipole interaction is rotationally invariant, so that we expect that the Lagrangian describing the whole system—whatever its detailed mathematical form is—be rotationally invariant too. On the other hand an electret (not a ferroelectret) has been detected for almost all the biomolecules.[9,10] Furthermore, this electret does not belong to the bare molecular structure but to its "dressing," the surrounding hydration water. Some sort of molecular dynamics produces a nonzero electric polarization in the surroundings, thus breaking the rotational symmetry of the dipole dynamics. It is possible consequently to introduce the formalism and theorems concerning spontaneously broken symmetries and their dynamical rearrangement (see Section 3).[15] In a first rough treatment one could use the Goldstone theorem, which predicts the

appearance of massless bosons. A more adequate treatment would consider the electric field as a gauge field; a massive electric quantum excitation would then appear. It is known that the mass of a quantum of a field is inversely proportional to its propagation range; consequently, while a rigorously massless mode would correspond to an infinite propagation range (infinite size of the system), a nonzero mass corresponds to the observed finite range correlations (finite size of the system). A further improvement would concern the introduction of finite temperature effects which will allow thermal excitations. The problem will then arise of the interaction of coherent electric waves with thermal effects.

Let us now consider the one-dimensional dynamics occurring on the macromolecular chains. Davydov has shown[2,3,16] that a vibrational soliton—namely, a localized excitation strictly bound to a localized deformation—occurs on one-dimensional chains made by weakly bound monomers under two conditions:

i. the initial excitation is localized, i.e., energy is supplied to the system at small sized sites;
ii. the coupling between monomers exhibits a positive anharmonicity.

Under the above conditions solitons are created and travel quite slowly on the chain exhibiting a remarkable stability when their lifetime is compared with the one of the original excitation. Solitons have been therefore proposed as the long-range carriers of the energy produced by metabolic reactions and stored on the molecules.[17] For this reason they were used in a model of muscle contraction by Davydov.

Apart from the storage of energy, many additional and partially unexplored consequences arise on the biomolecules because of solitons. Conformational and structural changes may occur as well as changes in the vibrational frequency patterns. The binding of hydration water molecules to biomolecules is also affected. Davydov has recently pointed out[18] that a soliton on the α-helix of a protein behaves as a mobile trap for loosely bound electrons, giving rise to an electrosoliton which has the features of a supercurrent. In other words, once a vibrational soliton is produced when a certain amount of energy is supplied, an electric current is obtained without further expense of energy. It is reasonable to assume that such a supercurrent, via its associated fields, might induce a preferred direction (an orientation) among water molecules surrounding the biomolecule. Actually, it is a fact[9] that the activation energy that one must supply to a biomolecule in order that an electret may appear around it is just of the same order of magnitude (0.2–0.4 eV) as the energy required for the production of a soliton on an α-helix protein through the ATP hydrolysis.

The above-discussed mechanisms are particular ones. It is possible to integrate them in a provisional, unified scheme giving a partial qualitative

picture of cell operation.[19] The metabolic reaction, governed by nutrition and membrane activity, produces localized excitations at selected points of selected molecules (DNA). Such excitations propagate in soliton form along the chains storing energy on them; the rate of accumulation is controlled by the difference between the number of solitons produced per unit time (metabolic reaction rate) and the number of solitons decaying in unit time. This rate is linked to the chain elasticity, a time-dependent parameter. We have described what we call the energy charge phase of the cell cycle. Simultaneously, a water electret is created around the molecule because of the soliton accumulation. When the electret reaches a threshold corresponding to the breakdown of the rotational symmetry of the dipole set (representing a finite-sized cell), then a different dynamical regime is switched on, where electric polarization waves must appear according to the previous discussion on the dynamical rearrangement of symmetry. These waves, propagating in the system, are instrumental in its biochemical organization, as discussed elsewhere.[19,20]

3. *Boson Condensation and Collective Modes*

Quantum field theory provides a two-level description for a system:

a. the dynamical level, where basic microscopic fields, say $\psi(x)$, are introduced, which are called the Heisenberg fields and obey the dynamic equations

$$\Lambda(\partial)\,\psi(x) = J[\psi(x)] \tag{3.1}$$

b. the phenomenological level, where observable quantities are expressed in terms of physical fields, say $\phi(x)$, which are called also asymptotic or quasiparticle fields and satisfy free field equations

$$K(\partial)\,\phi(x) = 0 \tag{3.2}$$

In Eqs. (3.1) and (3.2), $\Lambda(\partial)$ and $K(\partial)$ are kinetic differential operators, x stands for $(\mathbf{x}, t)$, and $J[\psi(x)]$ is a nonlinear functional of $\psi(x)$.

As a next step, the set of the physical states, or Fock space, of the system is introduced. It is denoted by $\{|0\rangle\}$ and contains all the states obtained by repeated application of the "quasiparticle" creation operators on the ground state, or "vacuum," $|0\rangle$. The "quasiparticle" creation operators are functions of the fields $\phi(x)$.

At the first level of description the "forces" acting among the system components appear through the functional $J[\psi(x)]$, while, at the second level, the fields are considered free, because only fairly independent quan-

tities are observable. In a scattering process, for instance, the particles are observed only before and after the collision. In a complex system either the whole compound system or the correlations among the constituents—namely, the free particles acting as messengers (phonons and so on)—are observed. These "free" phenomenological fields are of course the output of the inner dynamics described at the first level and such relationship is provided by the "dynamical map"[12,15]:

$$\langle a|\ \psi(x)\ |b\rangle = \langle a|\ \Psi[\phi(x)]\ |b\rangle \tag{3.3}$$

Equation (3.3) is the bridge between the deep underworld where the "forces" are at work and the apparent world where observations are performed. In Eq. (3.3) $|a\rangle$ and $|b\rangle$ are wave-packet states, which belong to the Fock space $\{|0\rangle\}$ and $\Psi[\phi(x)]$ is a functional of (normally ordered) products of the $\phi(x)$ fields.

Let us now investigate how a symmetry at the dynamical level would affect the phenomenological description. Consider a transformation which leaves Eq. (3.1) invariant, namely, let g be an element of the symmetry group of Eq. (3.1):

$$\psi'(x) = g\psi(x) \tag{3.4}$$

$$\Lambda(\partial)\,\psi'(x) = J[\psi'(x)] \tag{3.5}$$

The dynamical map (3.3) would then select a field $\phi'(x)$ corresponding to $\psi'(x)$, defining a transformation h which would give $\phi'(x)$ as a function of the field $\phi(x)$ corresponding to the original $\psi(x)$ via the dynamical map

$$\langle a|\ g\psi(x)\ |b\rangle = \langle a|\ \Psi[h\phi(x)]\ |b\rangle \tag{3.6}$$

Usually the functional $\Psi[\phi(x)]$ is highly nonlinear, so that h is expected to be different from g. However, since g leaves Eq. (3.1) invariant (i.e., it corresponds to a physical process which requires no expense of energy), h leaves in turn Eq. (3.2) invariant

$$\phi'(x) = h\phi(x) \tag{3.7}$$

$$K(\partial)\,\phi'(x) = 0 \tag{3.8}$$

If the transformations g form a group G, then the transformations h form a group H, too.

We are interested in the situation $H \neq G$, namely, when the dynamical map (3.6) rearranges the symmetry transformations so that the system acquires a symmetry at the phenomenological level, which is quite different from the one at the microscopic dynamical level. In this case we say that a dynamical rearrangement of the symmetry occurs.

We are going to explore the consequences of the dynamical rearrangement of the symmetry in the two cases of the Davydov vibrational solitons on macromolecules and of the Fröhlich electric waves in the whole biological system. In the first case we deal with an explicit form of the equations, so that a complete treatment is possible, while in the second case we limit ourselves to the consideration of the dynamical rearrangement of the symmetry and of the consequent appearance of massless modes.

Let us consider first the soliton dynamics on the biomolecular chains.

3.1. Davydov Soliton Regime

A biomolecular chain can be schematized as a one-dimensional chain of weakly bound monomers of mass M, whose electric dipole moment is aligned along the chain. Consequently the internal dipole quantum number, giving the spatial orientation of the dipole, can be taken as frozen; the electric dipole, which would be described by a spinor in the usual three dimensions, is described in this one-dimensional case by a scalar.

In a first approximation the molecule length is assumed infinite and corrections due to the finite length are introduced as a second step. The intramolecular excitation brought by an external supply of energy (e.g., the output of a chemical reaction) is described by a complex Heisenberg boson field $\psi(\xi, t)$, solution of a nonlinear Schrödinger equation[2,3,16]:

$$\left(i\hbar\frac{\partial}{\partial t} - \Lambda + J\frac{\partial^2}{\partial \xi^2}\right)\psi(\xi, t) = -G\psi^+(\xi, t)\,\psi(\xi, t)\,\psi(\xi, t) \tag{3.9}$$

The right-hand side of Eq. (3.9) is understood to be normally ordered. Λ is a constant energy term which may depend on the free exciton energy, on the static interaction energy, and on the deformation energy. J is the resonance interaction energy between neighboring monomers. The quasiparticle free equation (3.2) becomes in this case

$$\left(i\hbar\frac{\partial}{\partial t} - \Lambda + J\frac{\partial^2}{\partial \xi^2}\right)\phi(\xi, t) = 0 \tag{3.10}$$

The operator $\phi(\xi, t)$ is the free exciton field. As a first step we are going to show that the c-number Davydov soliton[2,3,16]

$$\chi_c(\xi, t) = \left(\frac{\mu}{2}\right)^{1/2} \frac{\exp i\left[\dfrac{\hbar v}{2J}(\xi - \xi_0) - E_v\dfrac{t}{\hbar}\right]}{\cosh \mu(\xi - \xi_0 - vt)} \tag{3.11}$$

$$\int |\chi_c(\xi, t)|^2\, d\xi = 1 \tag{3.12}$$

is just a convenient vacuum expectation value for $\psi(\xi, t)$ in the "classical" limit. In Eq. (3.11) $\hbar$ is the Planck constant and

$$E_v = \Lambda + \frac{\hbar v^2}{4J} - J\mu^2, \qquad \mu = \frac{G}{4J} \tag{3.13}$$

Let us assume a transformation B such that

$$\psi(\xi, t) \to \psi'(\xi, t) = B\psi(\xi, t) \tag{3.14}$$

$$\chi_c(\xi, t) = \langle 0|\, \psi'(\xi, t)|0\rangle\,|_{\hbar=0} = \langle 0|\, B\psi(\xi, t)\,|0\rangle\,|_{\hbar=0} \tag{3.15}$$

which leaves Eq. (3.9) invariant. The dynamical map (3.3) then becomes

$$\langle 0|\, B\psi(\xi, t)\,|0\rangle = \langle 0|\, \psi[\beta\phi(\xi, t)]\,|0\rangle \tag{3.16}$$

which defines β as an invariant transformation for Eq. (3.10). Let us require also that β be a canonical transformation, namely, β preserves the canonical commutation relations

$$\begin{aligned}
&[\phi(\xi, t), \dot{\phi}(\xi', t)] = [\phi'(\xi, t), \dot{\phi}'(\xi', t)] = i\delta(\xi - \xi') \\
&[\phi(\xi, t), \phi(\xi', t)] = [\phi'(\xi, t), \phi'(\xi', t)] = 0 \\
&[\phi(\xi, t), \phi(\xi', t)] = [\phi'(\xi, t), \phi'(\xi', t)] = 0
\end{aligned} \tag{3.17}$$

where

$$\phi'(\xi, t) = \beta\phi(\xi, t) \tag{3.18}$$

We consider now the general class of canonical transformations

$$\phi'(\xi, t) = \beta\phi(\xi, t) = \phi(\xi, t) + f(\xi, t) \tag{3.19}$$

where the field $\phi(\xi, t)$ is translated by a c-number function $f(\xi, t)$. The transformation (3.19) would leave Eq. (3.10) invariant, provided that $f(\xi, t)$ is a (c-number) solution of the same equation, namely,

$$f(\xi, t) = ce^{-\mu(\xi - vt)} \exp\left[i\left(\frac{\hbar v}{2J}\xi - \frac{E_v t}{\hbar}\right)\right] \tag{3.20}$$

Let us show that the transformation (3.19) on the fields ϕ, which corresponds to the "translation" (3.20), induces through the dynamical map just the transformation (3.14) on the fields ψ, so that Eq. (3.15) is satisfied. The generator of (3.19) in the $v = 0$ frame of reference is

$$D = -\int_{-\infty}^{\infty} d\xi[g(\xi, t)\,\dot{\phi}(\xi, t) - \dot{g}(\xi, t)\,\phi(\xi, t)] \tag{3.21}$$

where

$$g(\xi, t) \doteq \theta(\xi)\, f_0(\xi, t)$$
$$f_0(\xi, t) = f(\xi, t)\,|_{v=0} \tag{3.22}$$

$\theta(\xi)$ is the usual step function: $\theta(\xi) = 0$ for $\xi < 0$ and $\theta(\xi) = 1$ for $\xi > 0$. It is possible to read Eq. (3.19) as

$$\phi'(\xi, t) = \beta\phi(\xi, t) = \lim_{g \to f_0} e^{-iD}\phi(\xi, t)\, e^{iD} \qquad \text{at} \quad v = 0 \tag{3.23}$$

and

$$\langle 0|\, e^{-iD}\phi(\xi, t)\, e^{iD}\, |0\rangle\,|_{g=f_0} = \langle f_0|\, \phi(\xi, t)\, |f_0\rangle = f_0(\xi, t) \tag{3.24}$$

The state $|f_0\rangle$ is defined as

$$|f_0\rangle \equiv e^{iD}\, |0\rangle\,|_{g=f_0}, \qquad \langle f_0\,|\,f_0\rangle = 1 \tag{3.25}$$

We introduce now the annihilation operator $a_{\bar{k}}$ of a quantum of the field $\phi(\xi, t)$ whose momentum is $\bar{k}$; let us also consider the c-number function $\alpha(\bar{k})$, the Fourier transform of a function which coincides with $f_0(\xi, t)$ in a suitable ξ domain; then

$$a_{\bar{k}}\, |f_0\rangle = \alpha(\bar{k})\, |f_0\rangle \tag{3.26}$$

i.e., $|f_0\rangle$ is a coherent state for $\phi(\xi, t)$, namely, any functional of ϕ acts on $|f_0\rangle$ as a c-number.

The dynamical map (3.3) then becomes

$$\langle f_0|\, \psi(\xi, t)\, |f_0\rangle = \Psi[f_0(\xi, t)] \tag{3.27}$$

$$\langle f_0|\, J[\psi(\xi, t)]\, |f_0\rangle = J[\Psi[f_0(\xi, t)]] \tag{3.28}$$

A boost to the original frame of reference and the substitution of Eqs. (3.27) and (3.28) into Eq. (3.9) produce at last

$$\left(i\hbar\frac{\partial}{\partial t} - \Lambda + J\frac{\partial^2}{\partial\xi^2}\right)\psi[f] = -G\,|\psi[f]|^2\,\psi[f] \tag{3.29}$$

Equation (3.29) is a c-number equation which coincides with the one proposed by Davydov [see Eq. (2.10) of Ref. 2]. The solution of (3.29) is, then, just the soliton (3.11)

$$\Psi[f(\xi, t)] = \chi_c(\xi, t) \tag{3.30}$$

which allows us to determine the constant c in Eq. (3.20):

$$c = (2\mu)^{1/2} e^{\mu\xi_0} \exp\left(-i\frac{\hbar v}{2J}\xi_0\right) \tag{3.31}$$

In conclusion from the above formalism we can deduce that the Davydov soliton is the macroscopic envelope of the localized boson condensation of the excitation quanta $\phi(\xi, t)$ induced by the β transformation (3.19).[12–14] The vanishing of the coefficient G of the nonlinear term of Eq. (3.29), via Eq. (3.13), implies the disappearance of the soliton, which then looks like a necessary consequence of the nonlinearities of the dynamics.

As a further step, we are going now to show how the QFT approach could recover all the relevant Davydov results starting from a restricted set of assumptions.[4]

We started actually from the nonlinear Schrödinger equation (3.9) governing the $\psi(\xi, t)$ field only. A deformation (phonon) field has not, so far, been independently introduced. But the density probability ρ^f of ψ in the soliton state $|f\rangle$

$$\rho^f(\xi, t) = \gamma \langle f | \psi^+(\xi, t)\psi(\xi, t) | f \rangle = \gamma |\chi_c(\xi, t)|^2 \tag{3.32}$$

or [see Eq. (3.11)]

$$\rho^f(\xi, t) = \gamma \frac{\mu}{2} \frac{1}{\cosh^2 \mu(\xi - \xi_0 - vt)} \tag{3.33}$$

is a solution of the (phonon) equation

$$\left(\frac{\partial^2}{\partial t^2} - v^2 \frac{\partial^2}{\partial \xi^2}\right) \rho^f(\xi, t) = 0 \tag{3.34}$$

The phonon equation is then a direct consequence of the self-interacting intramolecular excitation field ψ, which is the only independent physical object. As a matter of fact, the ψ-excitation field dynamics induces conformational changes in the chain lattice which are described by a phonon field.

It is useful to recast Eq. (3.34) in the more familiar form of the Davydov Eq. (2.56) of Ref. 2. Let us define

$$\begin{aligned} v_{ac} &= \left(\frac{W}{M}\right)^{1/2} \\ v^2 &= v_{ac}^2 - v_{ac}^2(1 - s^2) \\ s &= \frac{v}{v_{ac}} \ll 1 \end{aligned} \tag{3.35}$$

where w is the chain elasticity coefficient and v_{ac} is the sound velocity on the chain in absence of solitons. Equation (3.34) is consequently transformed into

$$\left(\frac{\partial^2}{\partial t^2} - v_{ac}^2 \frac{\partial^2}{\partial \xi^2}\right) \beta^f(\xi, t) - \frac{2K}{M} \frac{\partial}{\partial \xi} |\chi_c(\xi, t)|^2 = 0 \tag{3.36}$$

where

$$K = 1/2w(1 - s^2)\,\gamma \tag{3.37}$$

$$-\frac{\partial}{\partial \xi} \beta^f(\xi, t) = \rho^f(\xi, t) \tag{3.38}$$

Equation (3.36) is just the Davydov equation (2.56) of Ref. 2. Moreover Eq. (3.29) can be transformed as follows:

$$\left[i\hbar \frac{\partial}{\partial t} - \Lambda + J \frac{\partial^2}{\partial \xi^2} - 2K \frac{\partial \beta^f(\xi, t)}{\partial \xi} \right] \chi_c(\xi, t) = 0 \tag{3.39}$$

where

$$\mu = \frac{K^2}{w(1 - s^2)\,J} \tag{3.40}$$

Equation (3.39) coincides with Eq. (2.5a) of Ref. 2. We conclude that the present QFT approach is equivalent to the Davydov analysis. All the results of the latter theory can be recovered in the QFT framework, such as, for instance, the soliton stability, the soliton–photon interaction and the attraction of electrons by the self-generating potential wells leading to quasisuperconducting properties of the molecular chains, hosting solitons. The chains considered above were assumed to be infinitely long; let us now drop this assumption. This change affects mainly the stability of the soliton. The soliton is stable when the local deformation $\beta^f(\xi, t)$ [or $\rho^f(\xi, t)$] is stable and vice versa [see Eqs. (3.36) and (3.39)]. The conditions for the stability of β^f have been investigated in Ref. 14. For the sake of conciseness, we do not report here such an analysis (see also Ref. 4), but we give the conclusion. In a finite chain the soliton state has a nonzero transition amplitude with any state of n free excitation quanta, so that its lifetime has to be finite. The longer the chain, the longer the lifetime.

Very important features of solitons, such as their rate of creation and decay, their density on the chain, the effects of external perturbations (for instance, changes of the soliton density and decay rate induced by variations of the physical and chemical structure of the chain) are yet a matter of investigation.

All the above analysis shows that an incoherent energy supply can be transformed into a coherent but localized form (the soliton) through the

inner nonlinear dynamics occurring on the molecules constituting the biological system. The next step is the dynamical appearance of self-organization in the system without further expense of energy; energy stored in soliton form during the "charge" or Davydov regime is completely released outward during the "discharge" or Fröhlich regine.

3.2. Fröhlich Regime

It has been experimentally found that water surrounding biomolecules can be brought in an electret state by low electric fields at low frequency.

We have said in Section 2 that the electret activation energy is of the same order of magnitude as the energy required for the production of a soliton on a biomolecule. The numerical coincidence suggests that the soliton might induce the observed electret while traveling on the chain. How it could happen is at present under investigation; we limit ourselves here to suggesting that the soliton could be a trigger for the water polarization. As a consequence the dipole quantum number carried by the dipole excitation ψ field would no longer be frozen since the water electret has transformed the previously one-dimensional system into a three-dimensional one. Let us describe the dipole quantum number by the spinor:

$$\psi(x) = \begin{pmatrix} \psi\uparrow(x) \\ \psi\downarrow(x) \end{pmatrix} \tag{3.41}$$

where $x \equiv (\vec{x}, t)$. The Lagrangian $\mathscr{L}$ of the whole biological system is very complicated, but it is safe to assume its invariance under the rotation of the dipole degree of freedom of the doublet field (3.4). More technically, we assume its invariance under the $SU(2)$ group of transformations

$$\psi(x) \to \psi'(x) = e^{i\theta_i \sigma_i} \psi(x), \qquad i = 1, 2, 3 \tag{3.42}$$

θ_i are real parameters and σ_i are the Pauli matrices. Let us now get as much information as possible by using symmetry arguments only.

The path integral formalism of QFT appears to be the most suitable one. We introduce[20,21,22] the generating functional

$$\begin{aligned} W[J, j, n] = \frac{1}{N} \int [d\psi][d\psi^+] \exp \Big\{ i \int d^4x \\ \times [\mathscr{L}[\psi(x)] + J^+(x)\,\psi(x) + \psi^+(x)\,J(x) \\ + j^+(x)\,D_\psi^{(-)}(x) + D_\psi^+(x)\,j(x) + D_\psi^{(3)}(x)\,n(x) - i \in D_\psi^3(x)] \Big\} \end{aligned} \tag{3.43}$$

$D_\psi^{(i)}(x)(i=+,-,3)$ is the dipole density operator, whose explicit form is not needed here, but whose transformation law under $SU(2)$ group is

$$D_\psi^{(i)} \to D_\psi^{(i)}(x) - \lambda_j \varepsilon_{ijk} D_\psi^{(k)}(x) \qquad (i,j,k=1,2,3) \tag{3.44}$$

Notice that $D_\psi^{(\pm)}(x) = D_\psi^{(1)}(x) \pm i D_\psi^{(2)}(x)$ The number N in Eq. (3.43) is

$$N = \int [d\psi][d\psi^+] \exp\{i\, d^4x [\mathscr{L}[\psi(x)] - i\varepsilon D_\psi^{(3)}(x)]\} \tag{3.45}$$

The fields ψ and ψ^+ and the sources J and J^+ anticommute, while j and j^+ commute. The ε term in (3.43) takes care, as usual, of the vacuum non-vanishing polarization:

$$\langle 0|\, D_\psi^{(3)}(x)\, |0\rangle = P \neq 0 \tag{3.46}$$

The parameter ε has to be put to zero only *after* all the mathematical manipulations have been performed. Equation (3.46) is referred to as the spontaneous breakdown of the $SU(2)$ dipole rotational symmetry.

The ground state does not share the Lagrangian full invariance, but it is invariant only under rotations around the third axis (i.e., the direction of the electret). This group of residual invariance transformations is a subgroup of the original $SU(2)$ symmetry group. The nonvanishing electric polarization (3.46) is the trigger of the Fröhlich regime.

In Refs. 20, 21, 22 a detailed mathematical treatment is given which illustrates the consequences of the condition (3.46) on the formalism derived from the generating functional (3.43). Here we sketch only the main points and the conclusions. The first point is the validity, in the absence of long-range fields (gauge fields), of the Goldstone theorem. Two massless boson modes $P(x)$ and $P^+(x)$ must appear under the condition (3.46). The set of "phenomenological" fields [namely, the solution of Eq. (3.2)] in our case is then the field $\phi(x)$ of the free excitation quanta and the Goldstone fields $P(x)$ and $P^+(x)$:

$$\Lambda(\partial)\, \phi(x) = 0 \tag{3.47}$$

$$\left[i\frac{\partial}{\partial t} + \omega(\partial) \right] P(x) = 0 \tag{3.48}$$

The form of the differential operators $\Lambda(\partial)$ and $\omega(\partial)$ depends on the details of the dynamics and cannot be given here. Of course, more excitation fields can be added, but here we are interested *in* the Goldstone modes only. In particular we leave for further research the effects of the finite temperature, which imply the consideration of thermal excitations.

Let us concentrate on the Goldstone modes. They are the necessary

consequence, through the Goldstone theorem, of the nonvanishing vacuum polarization (water electret). The interaction of these modes with the "matter" field ϕ is embodied in the scattering matrix S:

$$S(\phi, \phi^+, P, P^+) = \langle :\exp[-iA(\phi, \phi^+, P, P^+)]: \rangle \tag{3.49}$$

while the dynamical map (3.3) is expressed in a path integral formalism as

$$SD^{(i)}(x, \phi, \phi^+, P, P^+) = \langle D_\psi^{(i)}(x) : \exp[-iA(\phi, \phi^+, P, P^+)]: \rangle \tag{3.50}$$

and

$$\begin{aligned} A(\phi, \phi^+, P, P^+) = \int d^4x [&\rho^{-1/2} P(x)\, K(\vec{\partial})\, D_\psi^{(-)}(x) \\ &+ \rho^{-1/2} D_\psi^{(+)}(x)\, K(-\overleftarrow{\partial})\, P^+(x) \\ &+ z^{-1/2} \phi^+(x)\, \Lambda(\vec{\partial})\, \psi(x) \\ &+ z^{-1/2} \psi^+(x)\, \Lambda(-\overleftarrow{\partial})\, \phi(x)] \end{aligned} \tag{3.51}$$

ρ and z are the wave function renormalization constants, $\vec{\partial}$ and $\overleftarrow{\partial}$ denote space-time derivatives operating on the right and left side, respectively, and

$$K(\partial) = -\left[i\frac{\partial}{\partial t} + \omega(\partial) \right] \tag{3.52}$$

By standard techniques it is possible to get the rules of transformation of the fields ϕ and $P(P^+)$ when the field $\psi(x)$ undergoes an $SU(2)$ transformation (3.42):

(a) case $\theta_2 = \theta_3 = 0$:

$$\begin{aligned} &\phi(x) \to \phi'(x) = \phi(x) \\ &P(x) \to P'(x) = P(x) + i\theta_1 \left(\frac{P}{2}\right)^{1/2} \\ &P^+(x) \to P'^+(x) = P^+(x) - i\theta_1 \left(\frac{P}{2}\right)^{1/2} \end{aligned} \tag{3.53}$$

(b) case $\theta_1 = \theta_3 = 0$:

$$\begin{aligned} &\phi(x) \to \phi'(x) = \phi(x) \\ &P(x) \to P'(x) = P(x) - \theta_2 \left(\frac{P}{2}\right)^{1/2} \\ &P^+(x) \to P'^+(x) = P^+(x) - \theta_2 \left(\frac{P}{2}\right)^{1/2} \end{aligned} \tag{3.54}$$

(c) case $\theta_1 = \theta_2 = 0$:

$$\begin{aligned} \phi(x) &\to \phi'(x) = e^{(1/2)i\theta_3\sigma_3}\phi(x) \\ P(x) &\to P'(x) = e^{-(1/2)i\theta_3}P(x) \\ P^+(x) &\to P'^+(x) = e^{+(1/2)i\theta_3}P^+(x) \end{aligned} \tag{3.55}$$

The generators of the above three transformations are, respectively, in the space of physical states,

$$\begin{aligned} D^{(1)} &= \left(\frac{P}{2}\right)^{1/2} \int d^3x [P(x) f(x) + P^+(x) f^*(x)] \\ D^{(2)} &= -i\left(\frac{P}{2}\right)^{1/2} \int d^3x [P(x) f(x) - P^+(x) f^*(x)] \\ D^{(3)} &= \int d^3x \left[\phi^+(x)\frac{\sigma_3}{2}\phi(x) - P^+(x) P(x)\right] \end{aligned} \tag{3.56}$$

The function $f(x)$ is a square-integrable c-number solution of the free field equation (3.48):

$$K(\partial) f(x) = 0 \tag{3.57}$$

and makes the integrals (3.56) convergent. The condition (3.57) makes also the generators (3.56) time independent, so that the corresponding transformations can be physically implemented without any expense of energy.

In other words the transformations (3.53)–(3.55) generated by the operators (3.56) are invariant transformations of the free-field "phenomenological" equations (3.47) and (3.48) for any choice of $f(x)$. Notice that the $P(x)$ field "translations" in Eqs. (3.53) and (3.54) are understood as the limit for $f(x) \to 1$ of the transformation generated by $D^{(1)}$ and $D^{(2)}$, namely,

$$P(x) \to P'(x) = \lim_{f(x)\to 1} \left[P(x) + i\theta_1 f(x)\left(\frac{P}{2}\right)^{1/2}\right] \tag{3.58}$$

The transformations (3.53)–(3.55) do not belong to the original $SU(2)$ group of symmetry [see Eq. (3.42)] but to the so-called $E(2)$ group—the group of the transformations of the Euclidean two-dimensional space into itself. The $E(2)$ group is the group "contraction" of $SU(2)$. We have shown explicitly that the group of symmetry at the level of observable quantities is

different from the group of symmetry at the level of the basic fields. This difference vanishes when the vacuum net polarization P vanishes.

Let us show now that the vacuum electric polarization is induced by a coherent homogeneous condensation of the massless dipole-wave quanta $P(x)$ in the vacuum $|0\rangle$, which is controlled in turn by the field "translation" (3.53) and (3.54). We get indeed, by using Eq. (3.56),

$$\langle 0|\, e^{-iD^{(2)}} e^{-iD^{(1)}} P^{+}(x)\, P(x)\, e^{iD^{(1)}} e^{iD^{(2)}} |0\rangle = P\,|f(x)|^2 \xrightarrow[f\to 1]{} P \tag{3.59}$$

We define

$$|f\rangle = e^{iD^{(1)}} e^{iD^{(2)}} |0\rangle \tag{3.60}$$

Then Eq. (3.59) becomes

$$\langle f|\, P^{+}(x)\, P(x)\, |f\rangle = P\,|f(x)|^2 \xrightarrow[f\to 1]{} P \tag{3.61}$$

By introducing the annihilation operator $P_{\bar{K}}$ of a quantum of the field $P(x)$ with momentum $\bar{K}$ and taking the Fourier amplitude $f_{\bar{K}}$ of $f(x)$, we get

$$P_{\bar{K}}\, |f\rangle = P^{1/2} f_{\bar{K}}\, |f\rangle \tag{3.62}$$

which shows that $|f\rangle$ is a coherent state for the $P(x)$ field. In conclusion, a coherent homogeneous (i.e., not space dependent) condensation of dipole-wave quanta $P(x)$ is responsible for the ordered state of polarization. No expense of energy is required since $D^{(1)}$ and $D^{(2)}$ are time independent. We can then identify the dipole-wave Goldstone fields $P(x)$ and $P^{+}(x)$ with the coherent electric polarization waves predicted by Fröhlich.

A relevant feature of this theory is the so-called "low-energy theorem," analogous to corresponding theorems which hold in ferromagnetism and in particle physics. The invariance of the "phenomenological" field equations under the transformations (3.53)–(3.55) requires a corresponding invariance of the S matrix. In the case of an infinite volume [namely, $f(x) \equiv 1$], this requirement is shown to imply

$$S(\phi(x), \phi^{+}(x), P(x), P^{+}(x)) = S(\phi(x), \phi^{+}(x), \partial P(x), \partial P^{+}(x)) \tag{3.63}$$

The S matrix, then, would be independent of $P(x)$ in the zero-momentum limit; in the infinite volume case, the Fröhlich regime would be stable against external perturbations exciting soft (zero momentum) modes.

In realistic systems, however, the volume is finite and surface effects become important. For a finite volume V it has been shown[4] that

$$\begin{aligned}\langle 0|\, D_{\Psi}^{(3)}(x)\,|0\rangle &= \lim_{\varepsilon\to 0} i\varepsilon \int_{-1/\eta}^{1/\eta} d^3x \int \frac{d^3p}{2\pi} e^{i\vec{p}\cdot\vec{x}}\rho(p) \\ &\quad \times \left(\frac{1}{\omega(p)-i\varepsilon a}-\frac{1}{\omega(p)+i\varepsilon a}\right) \\ &= \lim_{\varepsilon\to 0} i\varepsilon \int d^3p\, \delta_\eta(p)\,\rho(p)\left(\frac{1}{\omega(p)-i\varepsilon a}-\frac{1}{\omega(p)+i\varepsilon a}\right) \\ &= \lim_{\varepsilon\to 0} i\varepsilon\rho(\eta)\left(\frac{1}{\omega(p=\eta)+i\varepsilon a}-\frac{1}{\omega(p=\eta)-i\varepsilon a}\right) \qquad \text{at } \eta\sim 0\end{aligned} \tag{3.64}$$

where $\eta = V^{-1/3}$, $\delta_\eta(p)$ approaches $\delta(p)$ as $\eta\to 0$ and the following relation has been used:

$$\lim_{\eta\to 0}\int d^3p\,\delta_\eta(p)\,f(p) = f(0) = \lim_{\eta\to 0}\int d^3p\,\delta(p-\eta)\,f(p)$$

The function $\omega(p)$ is the energy of a dipole-wave quantum of momentum p; for $V\to\infty$ $(\eta\to 0)$ $\omega(p=0)=0$. Then, as $V\to\infty$, or, alternatively "far from the boundaries" of the system, $\omega(p=\eta)\to 0$ and Eq. (3.64) reduce to Eq. (3.46) [notice that $\rho(0)=p/2$]. Near the boundaries (η different from zero), $\omega(p=\eta)$ behaves as an "effective mass" of the quantum of the $P(x)$ field.

Consequently the range of the correlation mediated by $P(x)$ and the domain of the coherence will extend over a distance η^{-1}. Finite (but long) range correlation forces are then expected in realistic (finite volume) systems. A similar conclusion can be drawn from Eq. (3.58) when an appropriate cut-off function $f_\eta(x)$ is introduced. In conclusion a relation between the size of the system and the not exactly zero mass of the P-field quantum is obtained. As announced in Section 2 a more correct treatment, which considers the electric field as a gauge field, would lead to a dynamically constructed (nonzero) mass of the quanta, producing thus a finite volume for the biological system.

Let us now look at the form the low-energy theorem assumes for $\eta\neq 0$. The field equation becomes

$$K_\eta(\partial)\,P(x) = 0 \tag{3.65}$$

The operator $K_\eta(\partial)$ depends upon the effective mass. Then the transformation rule of $P(x)$ changes to

$$P(x) \to P(x) + f_\eta(x)\,\text{const} \tag{3.66}$$

where $f_\eta(x)$ is prescribed to be a c-number solution of Eq. (3.65). In Eq. (3.63) $\partial P(x)$ must be replaced by $K_\eta(\partial)\,P(x)$ in order to have an S matrix invariant under the $E(2)$ group transformations. Thus an energy threshold, equal to the effective mass of the quantum, appears, through $K_\eta(\partial)$, in the S matrix. The consequence is that external perturbations whose energy is below threshold cannot perturb the system. Since the effective mass increases as $1/\eta$, the smaller the system the higher the threshold. The existence of an energy threshold matches a corresponding feature of the Fröhlich theory.

4. *Phenomenological Evidence*

In the above dynamical scheme a crucial role is played by solitons, water electrets, and coherent electric waves. In this section we present some experimental findings which provide evidence for the existence of such collective modes in biological systems.

4.1. *Solitons on Macromolecules*

Support for solitons on different kinds of molecules has been accumulated in these last years; α-helix proteins have been extensively investigated.[3] It has been recognized that the hydrogen-bond filaments along α-helix chains support a pathway for solitons in the nonlinear dynamics of the chains. It has been shown that a simple positive anharmonicity in the interaction among nearest-neighbour monomers is the required nonlinearity in the chain dynamics (C=O bonds in α-helix).[27,28]

A measure of the dipole–dipole interaction energy between two adjacent monomers and of the anharmonicity of the chain can be obtained by an inspection of the bandwidth of the absorption band of the monomer excitation, that forms the exciton. For α-helix proteins in which inhomogeneous broadening has been eliminated by ordering processes of the samples, the bandwidth of the C=O absorption at 1660 cm^{-1} [29] gives evidence of the excitonlike collectivization of the vibrational C=O excitation along the chain. This is a prerequisite for the existence of Davydov solitons on the chain.[2]

In recent years, it has been suggested that Davydov solitons could actually occur on a number of one-dimensional chains of biological

interest. Raman experiments on green algae *Chlorella P.*[30] suggested the possible formation of Davydov solitons on carotenoid chains.[31] Infrared absorption studies of acetanilide, a model system for one-dimensional chains of H bonds, have been recently interpreted in terms of Davydov-like solitons, involving hydrogen-bonded amide groups.[28,32] Of particular interest is the possible existence of solitons on nucleic acids.[19,33–35] Whether Davydov solitons could actually occur on DNA, has been examined.[19] The stretching vibration of the P–O bond in the phosphodiester backbone of the double helix, being directed along the axis of the chain, satisfies the symmetry and polarity requirements for a collectivization of the excitation along the chain. However, only DNA in B-form displays the proper anharmonicity for the formation of a vibrational P–O exciton. This can be seen from the width of the P–O Raman bond at 830 cm^{-1} [36,37] For DNA in A-form the band of the P–O stretching is shifted at 807 cm^{-1} and narrowed in a sharper line, showing that the P–O excitation remains localized on individual P–O groups and does not spread along the chain in exciton form. A prerequisite for the appearance of Davydov solitons on DNA is then fulfilled only for B helix structures. Different biological functions are associated to B (resting) and A (active) DNA; a soliton dynamics could have a major role in DNA structural and functional transitions. Indeed, B → A,[35] B → Z[34] transitions as well as the opening of DNA helices[33] have been described in terms of soliton dynamics.

4.2. Water in Biological Systems

Water in biological systems exhibits peculiar properties. Only a percentage of total water appears in the form of usual liquid water.[38,39] A relevant part is made up of water molecules which are bound to different sites of macromolecules in the form of hydration water, while a sizable amount, although not fixed to any definite molecular site, is strongly affected by macromolecular fields; this kind of water has been termed "vicinal" water.[39] "Vicinal" water has been extensively investigated in the last decade and it exhibits noticeable properties.[40] It does not have a unique freezing temperature but an interval ranging from -70 to $-50°C$; it is a very bad solvent for electrolytes, but nonelectrolytes have the same solubility properties in it as in usual water. Its viscosity is enhanced and its NMR response is anomalous. The whole of these physical properties point to the existence of some dynamic correlation among its constituents.

It is interesting to notice that most of the "vicinal" water surrounds the elements of the cell cytoskeleton.[39] It is not unlikely that the coherent electric waves discussed in this paper are responsible for such a correlation.[20] Bound water also exhibits some kind of collective behavior;

actually it has been shown that bound water produces electrets around many macromolecules. Activation energies of about 0.3–0.4 eV have been detected and relaxation times up to 10^5 s have been measured, suggesting the existence of zero-frequency modes of oscillation.[9,10] A zero-frequency mode, i.e., in our theoretical framework a massless mode, appears when dynamical rearrangements of symmetry are at work. Since the electret activation energy is of the same order of magnitude as the energy required for a Davydov soliton, we think that the electret is the result, via a spontaneous symmetry breakdown, of the soliton regime occurring on biomolecules.

It is worthwhile noticing that hysteresis has been observed in all cases[9,10]; in our model then the electret would appear different in subsequent cycles (alternance of Davydov and Fröhlich regimes) during the cell lifetime, giving rise to a nonstationary evolution instead of a periodic one (on a time scale comparable with the lifetime of the biological system under consideration).

4.3. Coherent Electric Waves in Living Matter

Up to now, different kinds of evidence can be quoted, which support the Fröhlich proposal of electric waves in living matter. First of all, we mention those resonance experiments where external electromagnetic waves of selected frequencies interact with biological targets inducing macroscopic nonthermal consequences. There is a growing literature[41–43] about microwave interaction with different types of cells. It has been observed that at given frequencies the effects occur only when a threshold is overcome and are independent from different intensities of irradiation.

A similar phenomenon, as far as the biological effects are concerned, has been observed in the visible region for a long time,[44] but only recently has satisfactory phenomenological evidence been achieved.[45] In this case, astonishingly, there is only one frequency in the visible domain (at 632.8 nm, the wavelength of He–Ne lasers) which is the same for different types of cells (for instance *E. coli* and He–La). Moreover, active cells emit electric and electromagnetic radiation on a very large frequency spectrum. Measurements have been performed at frequencies lower than 100 GHz[46,47] and in the optical range.[48] This last type of emission, which could be of coherent type, has been attributed to the dynamics of DNA in the cell.

Further evidence for coherent electric waves comes from Raman experiments on living cells.[49] These experiments show a time-dependent pattern of lines in IR and IR regions. The connections between Raman spectroscopic data and the underlying (spatial and temporal) order in living matter is discussed elsewhere.[50–52]

A different kind of support for Fröhlich ideas has been brought by experiments on rouleaux formation in erythrocyte suspensions.[53] These experiments show that selective long-range interactions appear among metabolically active red blood cells. The connection between the experimental outcome and Fröhlich theory has been recently discussed.[20,54]

In conclusion, phenomenological evidence supports the main elements of our QFT approach. We feel that the task of getting a satisfactory dynamical description of living systems is not out of reach.

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Solitons and Energy Transfer Along Protein Molecules

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It is shown that the energy released in the hydrolysis of ATP molecules can be transferred in the form of vibration solitons along α-helical protein molecules. The vibration solitons are collective excitations travelling along a chain of successively arranged peptide groups and corresponding to amide I vibrations. The exceptional stability of solitons in one-dimensional structures can account for the small probability of their energy transforming into that of disordered heat motion.

1. Introduction

Many biological phenomena are known to be associated with spatial energy transfer along protein molecules. To elucidate the mechanism for this energy transfer at the molecular level is one of the most urgent problems in bioenergetics.

It is now established that the energy (about 0·43 eV or 10 kcal/mol) released in the hydrolysis of adenosine triphosphate molecules (ATP) is the universal energy unit used in all biological phenomena. This energy is only 20 times greater than the average energy of heat motion at physiologic temperatures. It is insufficient to excite the electronic states of molecules. Therefore the importance of vibration excitations (with an energy less than 0·4 eV) of separate groups of atoms in protein molecules has been repeatedly indicated.

The opponents of this viewpoint have frequently mentioned the fact that the lifetime (with respect to transformation into heat) of such excitations is very small (of the order of 10^{-12}–10^{-13} s). At the same time it is usually forgotten that when the molecular system is organized in a certain manner the lifetime of excitations of some of its degrees of freedom can increase by many orders. For example, in condensed luminescent molecular systems (in particular, in crystalline anthracene) the electronic excitations of a system of conjugated π-electrons give off almost all their energy by radiation during the time (10^{-8}–10^{-9} s) exceeding considerably the time of thermal relaxation.

In addition to the fact that the energy released in the hydrolysis of ATP molecules is insufficient to excite electronic states, the excitation of the latter is also disadvantageous because their lifetime relative to radiation is considerably smaller than the corresponding lifetime of vibration states. For example, the lifetime (relative to radiation) of valence vibrations C=O (amide I with an energy of 0·21 eV) in peptide groups of proteins is equal to about 0·01 s. It exceeds the lifetime of electronic excitations of the same group ($\sim$ 4·4 eV of energy) by a factor of $(4{\cdot}4/0{\cdot}21)^4 = 1{\cdot}9 \times 10^5$.

In alpha-helical protein molecules the peptide groups (H—N—C=O) are situated along the molecule in three chains. In each of these chains the neighbouring peptide groups are held together by hydrogen bonding between atomic H and O. Just these bonds provide for a relatively rigid structure of the α-helix.

Amide I vibrations are quite characteristic, i.e. they are weakly connected with other normal vibrations of protein molecules. This fact is confirmed by a very small difference in the amide I vibration frequencies in proteins of different kinds.

The collective excitations of a chain of successively arranged peptide groups which correspond to amide I vibrations are described by "solitary waves"—the vibration solitons. They have a particle-like character, propagate rapidly along the chain and have big lifespan, because they interact with other degrees of freedom of the molecule much more weakly than separate amide I vibrations.

The solitons were first observed as solitary, rapidly moving water loops in narrow channels filled with water. The "tsunami" covering huge distances in the oceans is also identified with solitons by some people. Numerous investigations (Hasegawa & Tappert, 1973; Scott, Chu & McLaughlin, 1973; Toda, 1975) showed that a remarkable property of the solitons described by the same non-linear equation as vibration solitons is their exceptional stability for outer disturbances.

The formation of stable excitations such as vibration solitons in the α-helical portions of protein molecules guarantees the high efficiency of the transfer of energy, released in the hydrolysis of ATP molecules, by protein. The exceptional stability of solitons in one-dimensional structures can account for the small probability of their energy transforming into that of disordered heat motion.

2. Solitons in One-Dimensional Chains

The dipole–dipole interaction between the adjacent peptide groups situated along a hydrogen-bonded chain in the α-helical protein molecules is an order of magnitude higher than a similar interaction between the

nearest groups of neighbouring chains. Therefore the excitations of the three chains may thus be considered independently. For the sake of simplicity we shall further investigate the excited states of only one of these chains.

Consider an infinite chain of weakly bound peptide group with mass M situated along axis x at distance R. Assume that the internal excitations of peptide groups are characterized by an energy ε (equal to 0·21 eV) and an electric dipole moment d directed along the chain. To investigate the collective excitations of this chain we shall use the results obtained by the author and Kislukha (1976).

Let x_n be a co-ordinate characterizing the position of the nth group and B^+ the creation operator for an amide I vibration excitation.

The excited state of such a chain is defined by the wave function

$$\psi(t) = \sum_n a_n(t) B_n^+ |0\rangle \tag{1}$$

with the normalized condition

$$\langle\psi(t)|\psi(t)\rangle = \sum_n |a_n(t)|^2 = 1. \tag{2}$$

Here $|0\rangle$ is the function of the ground state. The function (1) must satisfy the Schrodinger equation

$$i\hbar \frac{\partial \psi}{\partial t} = H\psi.$$

Since the effective mass M of the peptide group is large, we treat their motion classically, and the amide I vibration excitations terms of quantum mechanics.

The Hamiltonian of the chain (provided the interaction between adjacent groups only is taken into account) can then be written as

$$H = T + U + \sum_n [(\varepsilon - D_n) B_n^+ B_n - J(B_{n+1}^+ B_n + B_{n+1} B_n^+)] \tag{3}$$

where the function $-D_n$ characterizes the change in the static interaction (the energy of the hydrogen bond) of the nth group with neighbouring groups when it is excited. Making allowance for the interaction with nearest neighbours we have

$$D_n = \mathscr{D}_n(|x_{n+1} - x_n|) + \mathscr{D}_n(|x_{n-1} - x_n|) \approx \left(1 + \frac{\beta}{R}\rho_n\right) D$$

where

$$\rho_n = R - |x_n - x_{n-1}|$$

is the decrease in equilibrium distances between peptide groups; $D = 2\mathscr{D}(R)$ and β are the positive parameters of a theory. $J = 2d^2/R^3$ is the operator

of the resonance interaction determining the transition of an excitation from one group to another;

$$T+U=\tfrac{1}{2}\sum_n\left[M\left(\frac{\partial x_n}{\partial t}\right)^2+w\rho_n^2\right]$$

are the kinetic and the potential energies of the displacement of peptide groups; w is the elasticity constant of the chain.

The set of functions $a_n(t)$ and $\rho_n(t)$ determine the distribution of excitations of peptide groups in the chain and the relative changes in the equilibrium distances between them. They obey the set of equations

$$i\hbar\frac{\partial a_n}{\partial t}=\left[\varepsilon+T+U-\left(1+\frac{\beta}{R}\rho_n\right)D\right]a_n-J(a_{n+1}-a_{n-1}), \tag{4}$$

$$\frac{\partial^2\rho_n}{\partial t^2}=-\frac{w}{M}(2\rho_n-\rho_{n+1}-\rho_{n-1})+\frac{\beta D}{MR}(2|a_n|^2-|a_{n+1}|^2-|a_{n-1}|^2). \tag{5}$$

Equations (4) and (5) define the collective excitations and deformations of the chain.

We are interested only in excitations distributed over some region of the chain. Therefore we can apply the continual model. At this approximation we can replace discrete functions $a_n(t)$ and $\rho_n(t)$ by the continuous functions $a(\xi, t)$ and $\rho(\xi, t)$ of the dimensionless co-ordinate $\xi = x/R$. Using the expression

$$f(\xi\pm1, t)\approx\left(1\pm\frac{\partial}{\partial\xi}+\frac{1}{2}\frac{\partial^2}{\partial\xi^2}\right)f(\xi, t),$$

we can transform equations (4) and (5) to

$$\left\{i\hbar\frac{\partial}{\partial t}-\left[\lambda-\frac{\beta D}{R}\rho(\xi, t)\right]+J\frac{\partial^2}{\partial\xi^2}\right\}a(\xi, t)=0, \tag{6}$$

$$\left[\frac{\partial^2}{\partial t^2}-v_a^2\frac{\partial^2}{\partial\xi^2}\right]\rho(\xi, t)+\frac{\beta D}{MR}\frac{\partial^2}{\partial\xi^2}|a(\xi, t)|^2=0, \tag{7}$$

where

$$\lambda\equiv\varepsilon+T+U-D-2J,$$

$$T+U\equiv\frac{M}{2}\int_{-\infty}^{\infty}d\xi\left[\left(\int_{-\infty}^{\xi}d\eta\frac{\partial}{\partial t}\rho(\eta, t)\right)^2+v_a^2|a(\xi, t)|^4\right], \tag{8}$$

$v_a^2\equiv w/M$, therefore $V_a=v_aR$ is the longitudinal sound velocity in the chain.

We shall look for solution of the set of equations (6) and (7) in the form of excitations propagating along the chain with velocity $V=vR$. For this

we set

$$\rho(\xi, t) = \rho(\xi - vt), \qquad a(\xi, t) = \Phi(\xi - vt) \exp\left[i\theta(\xi, t)\right].$$

We then obtain from equation (7)

$$\rho(\xi, t) = \frac{\beta D |a(\xi, t)|^2}{MR(v_a^2 - v^2)}. \tag{9}$$

Substituting this value in equations (6) and (8) we transform them to

$$\left\{ i\hbar \frac{\partial}{\partial t} - \lambda + J \frac{\partial^2}{\partial \xi^2} + G_v |a(\xi, t)|^2 \right\} a(\xi, t) = 0, \tag{10}$$

$$T + U = \frac{G_v}{2(v_a^2 - v^2)} \int_{-\infty}^{\infty} d\xi \left[\left(\int_{-\infty}^{\xi} d\eta \frac{\partial}{\partial t} |a(\eta, t)|^2 \right)^2 + v_a^2 |a(\xi, t)|^4 \right], \tag{11}$$

where

$$G_v = \frac{G_0}{1 - (V/V_a)^2}, \qquad G_0 = \frac{\beta^2 D^2}{wR^2}. \tag{12}$$

The exact normalized solution of the non-linear Schrodinger equation (10) is called soliton. It has the form

$$a_v(\xi, t) = \frac{\sqrt{\mu_v/2} \exp\left\{ i \left[\frac{\hbar v}{2J} (\xi - \xi_0) - \Omega_v t \right] \right\}}{\cosh\left[\mu_v(\xi - \xi_0 - v)\right]}, \tag{13}$$

where

$$\mu_v = G_v/4J; \tag{14}$$

ξ_0 is the arbitrary constant due to the translational invariance of equations (6) and (7). Each value $V = vR$ corresponds to the soliton energy

$$E_V \equiv \hbar\Omega_V = \hbar\Omega_0 + \frac{\hbar^2 V^2}{2JR^2} + \frac{G_0^2(z^2 + \frac{3}{2}z^4 - \frac{1}{2}z^6)}{24J(1 - z^2)^3}, \tag{15}$$

where $z \equiv V/V_a$ and

$$\hbar\Omega_0 = \varepsilon - D - 2J - G_0^2/48J \tag{16}$$

is the internal energy of the soliton.

It follows from the expression (15) that when the soliton velocity approaches the longitudinal sound velocity V_a the soliton energy tends to infinity. At all finite values of the soliton energy its velocity is therefore always less than V_a. The soliton energy thus cannot transform into the energy of longitudinal sound waves. This is one of the reasons of the great stability of the solitons.

At small velocities of the soliton when $V \ll V_a$ is valid, the soliton energy (15) transforms to a simple expression

$$E_V = \hbar\Omega_0 + \tfrac{1}{2}m^*V^2, \tag{17}$$

where

$$m^* = \frac{1}{2J}\left[\frac{\hbar^2}{R^2} + \frac{G_0^2}{12V_a^2}\right] \tag{18}$$

is effective mass of the soliton. In the same approximation the parameter

$$\mu_v \approx \frac{G_0}{4J}[1+(V/V_a)^2] \approx \frac{G_0}{4J} = \mu_0$$

is not dependent upon the soliton velocity.

Substituting equation (13) into equation (9) we find the distribution of the change in relative distances between the peptide groups. At small velocities of the soliton we shall get

$$\rho(\xi, t) = \frac{\sqrt{G_0^3/w}}{8J}\,\mathrm{sech}^2\,[\mu_0(\xi - \xi_0 - vt)]. \tag{19}$$

According to equation (19), the maximal relative changes in distances between peptide groups at $\xi = \xi_0 + vt$ moves along the chain with the velocity $V = vR$. The deformation moving along the chain is accompanied by the internal excitation of the peptide groups. The region of the chain embraced by the excitation and deformation is defined by

$$R\Delta\xi \approx \frac{\pi R}{\mu} = \frac{4\pi JwR^3}{\beta^2 D^2}. \tag{20}$$

3. Excitons in One-Dimensional Chains

In an absolutely rigid chain $w = \infty$, so that the non-linearity parameter (12) is equal to zero and equation (10) reduces to linear Schrodinger equation which has solutions in the form of plane waves. In this case an excitation is uniformly distributed along the whole chain. The excitations of this type is called excitons. The internal energy of the exciton is

$$E_0 = \varepsilon - D - 2J. \tag{21}$$

The effective mass of the exciton is

$$m_0^* = \hbar^2/2R^2J\,. \tag{22}$$

At the final value w the non-linear equation (10) has also solution in the form

of plane wave (exciton)

$$a(\xi, t) = A \exp\{i[k\xi R - \omega(k)t]\} \tag{23}$$

which characterizes the uniform distribution of an exciton along the whole chain. To each wave (23) there corresponds the energy

$$\hbar\omega(k) = E_0 + \hbar^2 k^2/2m_0^*.$$

The internal energy exciton E_0 and an effective mass m_0^* are determined by expressions (21) and (22). Therefore the internal energy of the exciton exceeds the internal energy (16) of the soliton by the value $G_0^2/48J$. Thus the excitations (23) in one-dimensional molecular systems are unstable.

In order to satisfy the normalization condition (2), we must set up in a chain of infinite length a wave packet with values of k in some narrow interval $k_0 - \Delta k \leqslant k \leqslant k_0 + \Delta k$, using the solution (23). The excitation described by such a wave packet propagates with the group velocity $V_g = \hbar k_0/m_0^*$. In this state the energy and the momentum of the exciton have not well-defined values. The region of the chain embraced by the excitation increases, because these wave packets are known to "spread". If the excitation was initially localized on segment $(\Delta x)_0 = 2\pi/\Delta k$ this segment increases eventually, so that in the time t

$$(\Delta x)_t \approx \sqrt{(\Delta x)_0^2 + \hbar^2 t^2/[m_0^*(\Delta x)_0]^2}.$$

4. The Stability of Solitons

A remarkable property of the vibration solitons at small velocities is that the distribution of excitation and deformation characterized by the function (19) remains completely unchanged in motion. The value of the maximum, the shape and the width (20) are independent of the velocity (energy) of a soliton and are defined only by the properties of the chain, i.e. the quantities J, G_0, R. When the soliton slows down, its velocity changes, leaving the shape and the size of the excitation region unchanged. The stability property of solitons enables us to regard them as particle-like formations.

The stability of solitons is connected with the competition of two processes: the spreading of excitation region, determined by the resonance interaction J, and the tendency towards excitation localization which is due to the non-linear terms of equation (10) proportional to the parameter (12). In protein molecules the formation of solitons corresponding to the vibration amide I is favoured by the one-dimensional arrangement of peptide groups, the relative softness of hydrogen bonds between peptide groups (the small w) and the orientation of electric dipole moments of vibrations along the chain. Only if the electric dipole moments in the excitation region are arranged

in this manner the distances between peptide group decrease, thus promoting excitation localization. In very rigid chains the ratio G_0/J is very small and the formation of solitons is impossible.

Solitons can be formed only under a strong coupling of the internal vibration with a local deformation of the chain. This coupling is proportional to the non-linearity parameter $G_0 = \beta^2 D^2/wR^2$. This parameter is big enough in comparatively soft chains of the peptide groups in protein molecules.

The condition of the formation solitons in α-helical protein molecules resembles the condition of the super conductivity formation in a metal. The superconductivity state due to the interaction of the electrons of a metal with phonons (the shifts of ions). Good conductors (silver, copper, gold) do not turn into a superconductive state. Superconductive states arise only in the metals which have a strong enough electron–phonon interaction. The strong electron–phonon interaction leading to a large resistivity in the normal state of metal makes the formation a superconducting state without any resistivity.

The above-considered solitons transfer with a very great efficiency the energy of excitation between different portions of the protein molecule. The energy transfer is accompanied by the deformation moving along the protein molecule. These solitons have a longitudinal polarization, since the electric dipole moment is directed along the chain.

The formation of solitons can occur only under the action of local excitations. These types of excitations also include the chemical energy released in the hydrolysis of ATP molecules which often attach themselves to certain portions of protein molecules (for example, to the heads of myosin molecules in the muscles). The vibration solitons cannot be excited by the infrared radiation of suitable frequency ($\sim 1660\ \text{cm}^{-1}$), since the wave length of this radiation is very large ($\sim 6 \times 10^4$ Å), and the photon localization is impossible with an accuracy higher than the wave length of radiation.

The concept of solitons which occur in myosin molecules was used by the present author† (Davydov, 1973, 1974, 1975) to elucidate the contraction mechanism of transversely striated muscles at the molecular level. Only the kinetic energy of solitons is actively used in muscle contraction, since the sliding of thin filaments relative to thick ones in the sarcomeres is due to the movement of solitons in myosin molecules contained in thick filaments.

† In these papers it was assumed that the distortion of the chain follows in an inert manner the movement of excitation. The notion of inert motion of peptide groups is valid only if the soliton velocity is much less than the longitudinal sound velocity in chain of the peptide groups.

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Solitons in a Nonlinear Inhomogeneous Transmission Line

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Nondestructive propagation of solitons is described in a nonlinear, electrical transmission line having loss and inhomogeneity. According to its nonlinear property expressed in terms of the [P+1]-th power, the transmission line demands a certain balance between the magnitude of loss or activity and inhomogeneity to preserve a unique soliton form of $\mathrm{sech}^{2/P}$ with exponentially varying amplitude. The adiabatic amplification effect for waves propagating in the line is considered by estimating their conserved quantities. The theoretical anticipation is confirmed by several numerical calculations. It is also shown that, for $P \geq 4$, it seems that solitons never travel stably in the real nonlinear transmission line.

§1. Introduction

In this decade, various transmission line models for solitons and related phenomena have been discussed in many articles.[1,2] The recent advance is, in particular, some discovery of the transmission line models which can be shown to have the exact N-interacting soliton solution. For the first time Hirota and Suzuki[3] showed that a certain nonlinear network is an electrical model analog to the well known Toda's exponential lattice[4] in dynamics. Later, other different networks which could be treated analytically were investigated by Hirota and Satsuma,[5] and others.[6-8] However, as a solvable model, each section of such ladder networks has to contain both nonlinear inductor and capacitor which indicate peculiar characteristics of nonlinear mathematical functions.

In parallel with above theoretical investigations another conventional approach has been performed to get a good description of nonlinear phenomena in the realistic nonlinear transmission lines. Noguchi[9] applied the perturbation method[10] to analyze an electrical ladder network of the low-pass filter, and found to explain the comparatively low frequency nonlinear waves such as video solitons in the line. According to this perturbation method, the low frequency waves in the lossless transmission line with [P+1]-th power nonlinearity can be well approximated in the P-th nonlinear K-dV equation.[9,11] In other words, the low frequency nonlinear waves are described by the fundamentals of both linearized dispersion relation and nonlinear property of the transmission line.

The main aim of this article is to extend this method to a case of lossy and inhomogeneous nonlinear transmission lines, and to discuss some interesting consequences. We can find several studies on this problem, *e.g.*, a lossless but linearly inhomogeneous case,[12] a randomly inhomogeneous case,[13] and a nonlinear transmission line with small dissipations.[14] In these examples, however, the describing equations could not have any analytical solutions as puls-like waves, and so experimental or numerical treatments were used in their estimations.

We restrict to an exactly solvable case of a lossy and inhomogeneous transmission line with [P+1]-th power nonlinearity. In the next section, we introduce a model of the nonlinear transmission line, and derive an approximate equation which describes the comparatively low frequency nonlinear waves. We also discuss on the particular case of the approximate equation calles the P-th nonlinear K-dV equation with loss and inhomogeneity. Section 3 is the most interesting one in which

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we consider about the properties of soliton solutions, and use numerical calculations to compare with the corresponding theoretical results. Section 4 is devoted to more careful discussions on the mechanism of the amplification or damping of video solitons. The last section is a conclusion.

§2. Derivation of a Modified Type of the K-dV Equation

We consider an infinitely long transmission line (or ladder networks) composed of lumped sections; a typical section is shown in Fig. 1. If $\Phi_n(i_n)$ is the nonlinear flux per unit length Δ at stage n, the flux of the section is $\Delta\Phi_n(i_n)$ webers. Similarly the resistance of the section is ΔR_n ohms, the nonlinear charge is $\Delta Q_n(v_n)$ coulombs, the leakage conductance is ΔG_n mohs.

The fundamental equations derived from Fig. 1 are

$$i_n - i_{n+1} = \Delta G_n v_n + \Delta \frac{\partial Q_n}{\partial t}, \tag{1}$$

$$v_{n-1} - v_n = \Delta R_n i_n + \Delta \frac{\partial \Phi_n}{\partial t}. \tag{2}$$

We express each parameters of the line as follows:

$$\Phi_n = L_0[1 - \Gamma_L(n, t) - \alpha_L i_n^{\mathrm{P}}/(\mathrm{P}+1)]i_n, \tag{3}$$

$$Q_n = C_0[1 - \Gamma_C(n, t) - \alpha_C v_n^{\mathrm{P}}/(\mathrm{P}+1)]v_n, \tag{4}$$

$$R_n = R_0[1 - \Gamma_R(n, t)], \tag{5}$$

$$G_n = G_0[1 - \Gamma_G(n, t)], \tag{6}$$

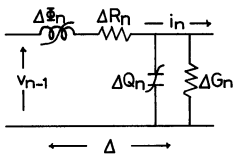

Fig. 1. A nonlinear transmission line, (a unit section).

$$\Gamma_L/\Gamma_{L_0} = \Gamma_C/\Gamma_{C_0} = \Gamma_R/\Gamma_{R_0} = \Gamma_G/\Gamma_{G_0} = \Gamma(n, t). \tag{7}$$

Where $\Gamma(n, t)$ describes an inhomogeneity of the line. We assume that $\Gamma(n, t)$ is a slowly varying function with respect both n and t and so,

$$\begin{aligned}\Gamma(n, t) &= \Gamma(n\Delta - (L_0C_0)^{-1/2}t) = \Gamma(\xi) \\ &= \Gamma(\xi_0 + (\xi - \xi_0)) \\ &\simeq \Gamma(\xi_0) + (\xi - \xi_0)\left.\frac{\partial \Gamma}{\partial \xi}\right|_{\xi=\xi_0},\end{aligned} \tag{8}$$

and the magnitudes of dissipations (R_0/G_0) are assumed to be both small. The nonlinear inhomogeneous transmission line shown here slightly differs from the previous line[15] which has the frequency dispersion even in the limit of distributed network $(\Delta \to 0)$. In the examination of the present transmission line we must remember that the line is constructed from lumped parameters $(\Delta = 1)$. So we consider the continuum approximation of eqs. (1) and (2). This is a reasonable approximation for the long wavelength (or low frequency) waves comparing with the length of a section of the line. By using

$$f_{n\pm1}(t) = \left[f(x, t) \pm \Delta \frac{\partial f}{\partial x} + \frac{\Delta^2}{2}\frac{\partial^2 f}{\partial x^2} \pm \frac{\Delta^3}{6}\frac{\partial^3 f}{\partial x^3} + \frac{\Delta^4}{24}\frac{\partial^4 f}{\partial x^4} \pm \cdots \right]_{x=n\Delta}, \tag{9}$$

rewriting i_n, v_n, i_{n+1} and v_{n-1} in eqs. (1) and (2), we obtain a set of nonlinear partial differential equation for $v(x, t)$ and $i(x, t)$. From the Ohm's law, if we can assume

$$v \simeq \left[\frac{\partial \Phi}{\partial i}\Big/\frac{\partial Q}{\partial v}\right]^{1/2} i \simeq \left(\frac{L_0}{C_0}\right)^{1/2} i = Zi, \tag{10}$$

and introduce $u(x, t)$ as the right progressing waves;

$$u = Z^{1/2} i = Z^{-1/2} v, \tag{11}$$

the next two equations are derived as,

$$\frac{\partial u}{\partial x} + \frac{\Delta}{2}\frac{\partial^2 u}{\partial x^2} + \frac{\Delta^2}{6}\frac{\partial^3 u}{\partial x^3} + \frac{\Delta^3}{24}\frac{\partial^4 u}{\partial x^4} + \frac{\Delta^4}{120}\frac{\partial^5 u}{\partial x^5} + \cdots + RZu + Z\frac{\partial Q(Z^{1/2}u)}{\partial t} = 0, \tag{12}$$

$$\frac{\partial u}{\partial x} - \frac{\Delta}{2}\frac{\partial^2 u}{\partial x^2} + \frac{\Delta^2}{6}\frac{\partial^3 u}{\partial x^3} - \frac{\Delta^3}{24}\frac{\partial^4 u}{\partial x^4} + \frac{\Delta^4}{120}\frac{\partial^5 u}{\partial x^5} - \cdots + GZ^{-1}u + Z^{-1}\frac{\partial \Phi(Z^{-1/2}u)}{\partial t} = 0. \tag{13}$$

Adding eqs. (12) and (13), and neglecting small terms,

$$2\frac{\partial u}{\partial x}+\frac{\Delta^2}{3}\frac{\partial^3 u}{\partial x^3}+(GZ+RZ^{-1})u+Z\frac{\partial Q}{\partial t}+Z^{-1}\frac{\partial \Phi}{\partial t}=0. \tag{14}$$

In eq. (14), we introduce a coordinate system moving to the right with the velocity of an infinitesimally small amplitude wave along the line,

$$\xi=x-(L_0C_0)^{-1/2}t, \quad \tau=(L_0C_0)^{-1/2}t, \tag{15}$$

and using the simple relation; $\partial/\partial\tau \simeq -\partial/\partial\xi$, we get the following modified type of the P-th nonlinear K-dV equation having loss and nonuniformity terms,

$$\frac{\partial u}{\partial \tau}+\alpha u^{\mathrm{P}}\frac{\partial u}{\partial \xi}+\beta\frac{\partial^3 u}{\partial \xi^3}+\gamma\frac{\partial u}{\partial \xi}+\delta u=0, \tag{16}$$

where

$$\alpha=\alpha_C Z^{\mathrm{P}/2}+\alpha_L Z^{-\mathrm{P}/2}, \quad \beta=\frac{\Delta^2}{6}=\frac{1}{6},$$

$$\gamma=\Gamma_C+\Gamma_L=(\gamma_{C_0}+\gamma_{L_0})\Gamma(\xi)=\gamma_0\Gamma(\xi), \quad \delta=\gamma_0+(GZ+RZ^{-1})/2. \tag{17}$$

The above eq. (16) with eqs. (17) seems to describe well the comparatively long wavelength waves in the line. Hereafter, we examine a simpler case of which the previous assumption (10) is completely valid;

$$\gamma_{C_0}=\gamma_{L_0}=\gamma_0/2, \quad \alpha_L Z^{-\mathrm{P}/2}=\alpha_C Z^{\mathrm{P}/2}=\alpha_0/2, \quad \text{and} \quad G=R=d_0/(Z+Z^{-1})=\text{const.} \tag{18}$$

The coefficients of eq. (16), eqs. (17) are then,

$$\alpha=\alpha_0, \quad \beta=1/6, \quad \gamma=\gamma_0(\xi-\xi_0), \quad \text{and} \quad \delta=\gamma_0+d_0/2. \tag{19}$$

Besides the above critical case, even in the below cases,

$$|\alpha_C Z^{\mathrm{P}/2}-\alpha_L Z^{-\mathrm{P}/2}|, \quad |\gamma_{C_0}-\gamma_{L_0}|, \quad |\gamma_{G_0}Z+\gamma_{R_0}Z^{-1}|=O(\varepsilon), \tag{20}$$

the assumption (10) and so eq. (16) with eqs. (17) may still hold sufficiently well.[15]

Here, we introduce the following dependent variable transformations;

$$U(\zeta,\theta)=\exp(2\gamma_0\tau/\mathrm{P})u(\xi,\tau),$$
$$\theta=\exp(-3\gamma_0\tau)/(-3\gamma_0),$$
$$\zeta=\exp(-\gamma_0\tau)(\xi-\xi_0), \tag{21}$$

and rewrite eq. (16) to the P-th nonlinear K-dV equation with an additional term,

$$\frac{\partial U}{\partial \theta}+\alpha U^{\mathrm{P}}\frac{\partial U}{\partial \zeta}+\beta\frac{\partial^3 U}{\partial \zeta^3}+\sigma\theta^{-1}U=0, \tag{22}$$

where $\sigma=2/3\mathrm{P}-\delta/3\gamma_0$. Figure 2 indicates the conditions in which the last term of eq. (22) disappears ($\sigma=0$) for the various nonlinear property P of the line. In such cases, eq. (22) reduces to the usual P-th nonlinear K-dV equation which has the single soliton solution to each natural number P. We discuss in detail on the property of these soliton solutions in the next section.

In the final of this section, we denote some other solvable case without the condition $\sigma=0$. If $\sigma=1$ ($\sigma=2$), eq. (22) reduces to the P-th nonlinear spherical (cylindrical) K-dV equation. For P=1, such spherical (cylindrical) soliton like pulses have been already found as ion-acoustic plasma waves.[16,17] It is interesting that the behavior of three dimensional

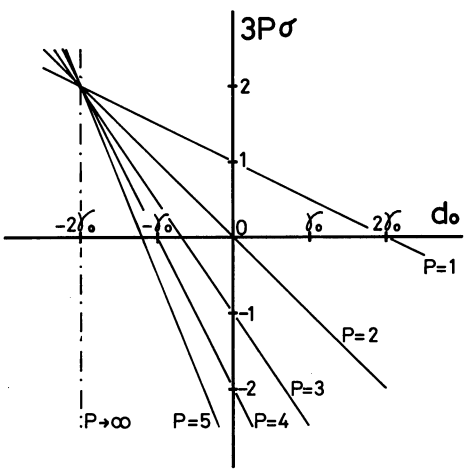

Fig. 2. Schematic description of conditions $\sigma=0$.

nonlinear waves can be simulated in the one-dimensional nonlinear transmission line with loss and nonuniformity.

In general, by introducing the transformation;[18]

$$U' = U \exp\left(\sigma \int \theta^{-1}\, d\theta\right), \tag{23}$$

and substituting to eq. (22), then

$$\frac{\partial U'}{\partial \theta} + \alpha'(\theta) U'^{P} \frac{\partial U'}{\partial \zeta} + \beta \frac{\partial^3 U'}{\partial \zeta^3} = 0, \tag{24}$$

where

$$\alpha'(\theta) = \alpha \exp\left(-P\alpha \int \theta^{-1}\, d\theta\right).$$

The asymptotic solutions of the above variable-coefficient K-dV equation (24) for P=1 and 2 have been discussed in detail,[19,20] so we can obtain a little more information on the solutions of eq. (24), and so of eq. (16).

§3. Properties of Solitons

In this section, we show the single soliton solution of the P-th nonlinear K-dV equation with loss and nonuniformity terms (16). The single soliton solution is derived from that of the usual K-dV equation (22), through the explicit transformations (21), under the condition of

$$\sigma = 2/3P - \delta/3\gamma_0 = 0, \tag{25}$$

as

$$u(\xi, \tau) = u_s(\tau)\, \mathrm{sech}^{2/P} \times \left[W_s^{-1}(\tau) \left\{ \xi - \xi_0 (1 - e^{\gamma_0 \tau}) + \frac{V_s(\tau)}{3\gamma_0} (1 - e^{3\gamma_0 \tau}) \right\} \right], \tag{26}$$

where

$$u_s(\tau) = u_0 \exp(-2\gamma_0 \tau / P),$$
$$V_s(\tau) = 2\alpha u_s(\tau)^P / (P+1)(P+2),$$
$$W_s(\tau) = [P^2 V_s(\tau) / 4\beta]^{-1/2}.$$

The amplitude dependent velocity of the single soliton (26) is

$$\frac{d\xi}{d\tau} = -\frac{\partial \phi}{\partial \tau} \Big/ \frac{\partial \phi}{\partial \xi} \gamma_0 (\xi - \xi_0) + V_s(\tau) \exp(-2\gamma_0 \tau), \tag{27}$$

where $\phi(\xi, \tau)$ is the phase of $\mathrm{sech}^{2/P}[\phi]$ in eq. (26). This velocity becomes a constant value $V_s(\tau)|_{\gamma_0 - 0}$ to each natural number P. For any P, if $\gamma_0 > 0$ ($\gamma_0 < 0$), then the single soliton comes to broaden (compress) its own width during the propagation, since the front (tail) part of the body moves faster than tail (front). In Fig. 3, 4, and 5, we show typical results of numerical simulation of eq. (16) for P=1, 2, and 3, respectively. We used the single soliton-form (26) as the initial wave profile, and other experimental conditions were denoted in the corresponding figure captions. As is already shown in eq. (25), to preserve a unique soliton form without emitting any oscillatory waves, the following relations must be satisfied between the magnitudes of inhomogeneity and loss or activity;

$$\gamma_0 = (GZ + RZ^{-1})/2 \quad \text{for } P=1,$$
$$GZ + RZ^{-1} = 0 \text{ and any } \gamma_0 \quad \text{for } P=2,$$
$$\gamma_0 = -(GZ + RZ^{-1})/(2 - 4/P) \quad \text{for } P \geq 3.$$

It is evaluated that, for the cases of $P \geq 2$, the amplitudes of solitons are grown ($\gamma_0 < 0$) by the inhomogeneity even in the transmission

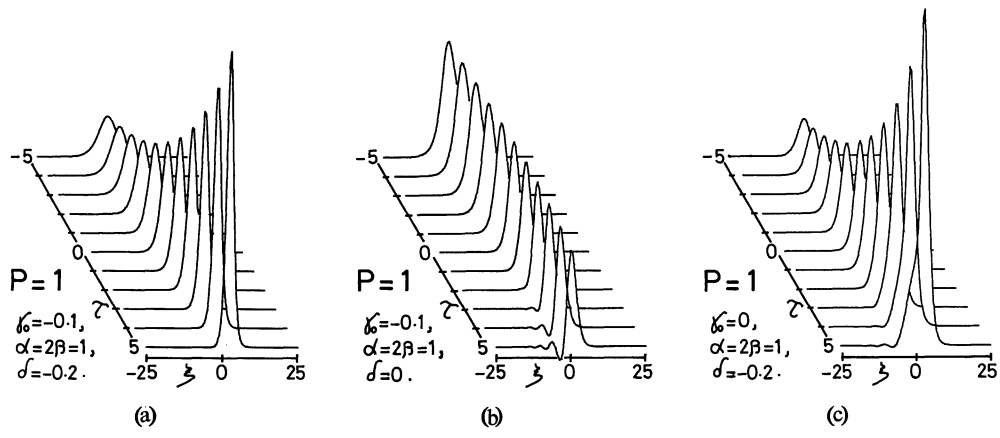

Fig. 3. Amplification and damping of one-soliton for P=1; (a) $\sigma=0$, ($\gamma_0=-0.1$, $\delta=-0.2$), (b) $\sigma\neq 0$, ($\gamma_0=-0.1$, $\delta=0$), (c) $\sigma\neq 0$, ($\gamma_0=0$, $\delta=-0.2$).

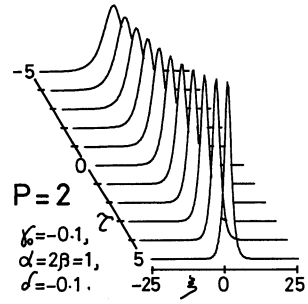

Fig. 4. Amplification and damping of one-soliton for P=2.

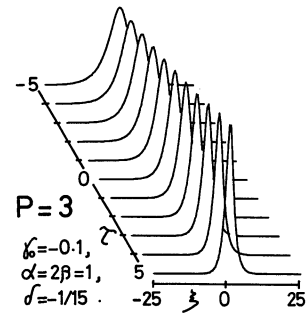

Fig. 5. Amplification and damping of one-soliton for P=3.

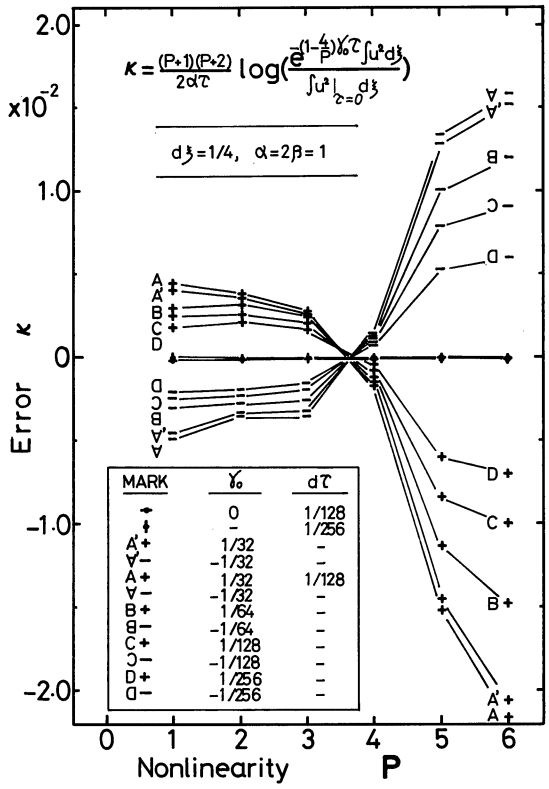

Fig. 6. An estimation of the errors due to instability of one-soliton.

line having real dissipations.

We examine such adiabatic amplification effect by a gradual inhomogeneity of parameters. As will be shown in the next section, the P-th nonlinear K-dV equation possesses at least three polynomial conserved quantities. The instantaneous energy of waves propagating in the line is proportional to u^2, because of the assumption (11). By using the result (eq. (32)) of the next section, the total energy of the single soliton (26) is derived in the moving frame as,

$$E_P(\tau)=\int_{-\infty}^{\infty}\left(\frac{u^2}{2}\right)d\xi=\int\frac{1}{2}u_s^2\,\mathrm{sech}^{4/P}[\phi]d\xi$$
$$=\frac{A_P}{2}u_s^2W_s, \tag{28}$$

where

$$A_P=\int_{-\infty}^{\infty}\mathrm{sech}^{4/P}\xi'\,d\xi', \quad A_1=4/3,$$
$$A_2=2, \quad A_3\simeq 2.587, \quad \text{and} \quad A_4=\pi.$$

The amplitude u_0 and time dependences of the above total energy E_p are calculated as,

$$E_P(\tau)\propto u_0^{2-P}\exp[(1-4/P)\gamma_0\tau]. \tag{29}$$

We note that, for P>4, eq. (29) shows the total energy of soliton comes to depend negatively on its amplitude, and the time dependence of the energy also changes. The total energy of the single soliton comes to decrease with the passage of time even for $\gamma_0<0$ against the growth of the amplitude of soliton. For P=4, the energy of the soliton becomes a constant value with respect to both amplitude and time. Consequently, it seems that such anomalous dependences of the total energy for the occasions of P≥4, (i.e., $\partial E_p/\partial u_0\leq 0$) never permit the corresponding single soliton to propagate stably in the nonlinear transmission line. We conclude this section by showing a brief confirmation on the above conjecture in Fig. 6. The figure indicates that the obvious difference between two groups, P≤3 and P>4 in the range of stable propagation of the single solitons.

§4. Adiabatic Amplification and Conserved Quantities

It is well known that the usual K-dV equation and the modified K-dV equation (in this paper,

they correspond to the cases P=1 and 2 of eq. (22) with $\sigma=0$, respectively) have the N-soliton solutions[21,22] which denote the time and space evolution of N-multiple interacting solitons. So, we can easily obtain the N-soliton solutions of eq. (16) by using already known N-soliton for P=1 and 2 through our transformations (21).

The N-soliton solutions $U(\zeta, \theta)$ for P=1 and 2 are described as follows:[21-24]

$$U^{\mathrm{P}}=\frac{2\beta(\mathrm{P}+1)(\mathrm{P}+2)}{\alpha \mathrm{P}^2}\frac{\partial^2}{\partial\zeta^2}[\log w], \quad w=\det|H|,$$

$$H_{ij}=\delta_{ij}+\left[\frac{2(k_ik_j)^{1/2}}{k_i+k_j}\right]\exp\left(\frac{\zeta_i+\zeta_j}{2}\right),$$

$$\zeta_m=k_m\zeta-\beta k_m^3\theta-\theta_m, \quad m=1, 2, 3, \cdots, N, \quad (30)$$

where δ_{ij} is the Kronecker's delta and θ_m constants. It should be noted again that the description (30) is valid only for P=1 and 2. As an example, in Fig. 7, we show the configuration of 3-interacting solitons for P=2. Although, we can see the effect of amplification or damping in Fig. 7, from the condition (25), in this case the nonlinear transmission line may not have any activities nor losses. Such adiabatic (or generally, parametric) amplification or damping is understood by estimating the following conserved quantities

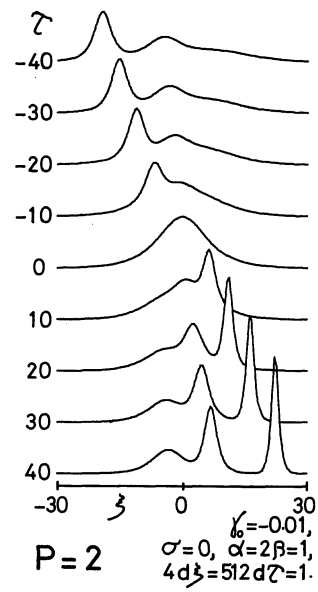

Fig. 7. Three interacting solitons for P=2, ($\sigma=0$).

(or constants) of motion. To any natural number P, the first three conserved quantities are obtained as:

$$I_1=\int_{-\infty}^{\infty} U\,\mathrm{d}\zeta=\mathrm{e}^{(2/\mathrm{P}-1)\gamma_0\tau}\int u\,\mathrm{d}\xi, \quad (31)$$

$$I_2=\int_{-\infty}^{\infty}\frac{U^2}{2}\,\mathrm{d}\zeta=\mathrm{e}^{(4/\mathrm{P}-1)\gamma_0\tau}\int\frac{u^2}{2}\,\mathrm{d}\xi, \quad (32)$$

$$I_3=\int_{-\infty}^{\infty}\frac{1}{\mathrm{P}+1}\left[U^{\mathrm{P}+2}-\frac{(\mathrm{P}+1)(\mathrm{P}+2)\beta}{2\alpha}\left(\frac{\partial U}{\partial\zeta}\right)^2\right]\mathrm{d}\zeta$$

$$=\mathrm{e}^{(4/\mathrm{P}+1)\gamma_0\tau}\int\frac{1}{\mathrm{P}+1}\left[u^{\mathrm{P}+2}-\frac{(\mathrm{P}+1)(\mathrm{P}+2)\beta}{2\alpha}\left(\frac{\partial u}{\partial\xi}\right)^2\right]\mathrm{d}\xi. \quad (33)$$

These quantities are considered to hold as the conservation of momentum, energy, and another higher order constant of motion for each P. Only for P=1 and 2, further higher order (infinite) conserved quantities have been derived.[23,24]

We show the next higher order constant of motion as an example of the use of the transformetions (21),

$$I_4=\int_{-\infty}^{\infty}\frac{1}{2(\mathrm{P}+1)}\left[U^{2(\mathrm{P}+1)}-\frac{2(\mathrm{P}+1)(2\mathrm{P}+1)\beta}{\alpha}U^{\mathrm{P}}\left(\frac{\partial U}{\partial\zeta}\right)^2+\frac{6(\mathrm{P}+1)(2\mathrm{P}+1)\beta^2}{5\alpha^2}\left(\frac{\partial^2 U}{\partial\zeta^2}\right)^2\right]\mathrm{d}\zeta$$

$$=\mathrm{e}^{(4/\mathrm{P}+3)\gamma_0\tau}\int\frac{1}{2(\mathrm{P}+1)}\left[u^{2(\mathrm{P}+1)}-\frac{2(\mathrm{P}+1)(2\mathrm{P}+1)\beta}{\alpha}u^{\mathrm{P}}\left(\frac{\partial u}{\partial\xi}\right)^2+\frac{6(\mathrm{P}+1)(2\mathrm{P}+1)\beta^2}{5\alpha^2}\left(\frac{\partial^2 u}{\partial\xi^2}\right)^2\right]\mathrm{d}\xi, \quad (34)$$

for P=1 and 2. Since the interaction of 3-solitons for P=2 is known to be described by the fifth conserved quantity itself, the existence of the corresponding integral, which depends only on time as $\exp(-7\gamma_0\tau)$, is indirectly indicated by the fact that famous particle-like interaction of solitons can still hold as shown in Fig. 7.

§5. Concluding Remarks

In this paper, it is shown that the P-th power nonlinear transmission lines with a certain

loss and inhomogeneity can still support the single solitons to each natural number P, and especially to P=1 and 2, even the exact N-multi interacting solitons. We have obtained a generalization of the result of the previous report[15] in which a special case as eq. (20) in this article had been treated. The theoretical conditions of balancing between the magnitudes of loss and inhomogeneity to each nonlinearity P show that the nondestructive amplification or damping of solitons without radiations may seem to be done in the P-th power nonlinear transmission lines with loss or activity and inhomogeneity.

Although the stability analysis on the propagating single solitons shown in the last half of section 3 is considered not to be completed, a numerical result briefly indicates that, in the transmission line with P-th ($P \geq 4$) nonlinearity, we can not only amplify nor reduce the amplitude of soliton nondestructively but even see the stable propagation of the single soliton. The similar method for the stability analysis of the single soliton have been already reported by several authors,[25,26] and we can see their results for the K-dV equation and a different nonlinear evolution equation bears very well resemblance to ours.

It is not difficult to extend our result for the nonlinearity P=1 to a shallow water waves,[27] a nonlinear lattice waves,[28] and a few types of nonlinear plasma waves.[29] The Alfven waves[30] in a nonuniform plasma may be also described by our modified type of the K-dV equation (16) with the nonlinearity P=2.

As well as the amplification or damping of solitons, it is also interesting to consider nonlinear periodical waves such as the cnoidal waves in inhomogeneous media. The well known recurrence phenomenon brought up by Fermi, Pasta, and Ulam[31] has been also observed in nonlinear transmission lines having quadratic (P=1)[32,33] and cubic (P=2)[32] nonlinearity. In conjunction with the problem of thermalization, the recurrence like phenomenon in the nonlinear inhomogeneous media is considered to be worth studying.

As a final remark, to realize the nondestructive amplification or damping of amplitude of solitons in the transmission line having the nonlinearity P=2, we can use a coupling between high and low frequency waves propagating together in the line. The method and result will be appeared in the next paper.

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Solitons in One-Dimensional Molecular Chains

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A theory of collective excitations in one-dimensional chains of weakly bound molecules which correspond to intramolecular excitations is developed. It is shown that these excitations are described by solitary waves (solitons) propagating (without "smearing") along the chain together with local deformation. The theory is developed without applying the assumption of inertialess motion of molecules. It is shown that the ultimate velocity of solitons is the velocity of sound waves corresponding to longitudinal vibrations of molecules. The problem of soliton stability is discussed.

Es wird eine Theorie kollektiver Anregungen in eindimensionalen Ketten von schwach gebundenen Molekülen entwickelt, die intramolekularen Anregungen entsprechen. Es wird gezeigt, daß diese Anregungen beschrieben werden durch solitäre Wellen (Solitonen), die sich (ohne „Verschmierung") in Richtung der Kette ausbreiten, zusammen mit einer lokalen Deformation. Die Theorie wird ohne die Annahme einer trägheitslosen Bewegung der Moleküle entwickelt. Es wird gezeigt, daß die Grenzgeschwindigkeit der Solitonen die Geschwindigkeit von Schallwellen ist, die den longitudinalen Schwingungen der Moleküle entsprechen. Das Problem der Solitonenstabilität wird diskutiert.

1. Introduction

A special type of collective excitations which has been discussed by the authors in [1] and called particle-like excitations or solitary excitons is possible in one-dimensional molecular chains which consist of periodically repeated molecules (or groups of strongly bound atoms) interacting by relatively weak forces. In what follows these excitations will be referred to as solitons. Particular cases of these systems are α-helical protein molecules in which the peptide groups made up of four atoms (HNCO) bound by chemical forces are arranged along the molecule in the form of three chains. In each chain the neighbouring peptide groups are linked by hydrogen bonds which provide for a rigid structure of α-helix.

The collective excitations of a one-dimensional molecular chain which correspond to intramolecular electronic or vibrational excitations with an electric moment directed along the chain are of a very specific nature different from ordinary excitons in three-dimensional crystals. In protein molecules these excitations include excitations (based on valent Amide I vibrations) with an energy of 0.21 eV and an electric dipole moment of 0.3 Debye directed along the axis of the molecule.

The solitons (solitary excitons) considered in [1] are excitations localized in some region of a chain; they are connected with varying distance between the molecules and propagate with a certain velocity along the chain. The theories of electronic excitations and vibrational amide I excitations of α-helical protein

molecules, developed in [1] and [2], respectively, assumed that the distortion of the chain follows in an inertialess manner the motion of the excitation. This assumption imposes some restrictions on the excitation velocity. In the present paper we develop a theory avoiding the concept of the inertialess motion of the molecules. We show that the ultimate velocity of a soliton in a molecular chain is the velocity of sound waves corresponding to molecular vibrations along the chain. The problem of the soliton stability is also discussed.

2. The Basic Equations and Their Solution in the Form of Solitons

Consider an infinite chain of weakly bound molecules (or groups of atoms) with mass M, which are at distance R. We also assume that the internal excitations of molecules (electronic or vibrational ones) in the chain are characterized by an energy ε and an electric dipole moment $\boldsymbol{d}$ directed along the chain. When the molecule becomes excited the forces of its interaction with neighbouring molecules are changed, this resulting in a change in the equilibrium distances between the molecules. Besides, since the molecules are identical, there arise some additional resonance interactions causing the excitation transfer from one of the molecules to another. The internal excitations of the molecules and their motion about equilibrium positions are inseparably linked.

In a one-dimensional chain it is sufficient to take into account only the interaction between the neighbouring molecules. In this approximation the collective excitations of the chain which correspond to an intramolecular excitation with an energy ε may be described by the Hamiltonian

$$H = T + U + \sum_n [(\varepsilon - D_n) B_n^+ B_n - J(B_{n+1}^+ B_n + B_{n+1} B_n^+)] , \qquad (1)$$

here the subscript n labels the molecules occupying the positions x_n in the chain; B_n^+, B_n are the creation and the annihilation operators of an intramolecular excitation. They satisfy the commutation relations

$$[B_n, B_m^+] = \delta_{nm} , \qquad [B_n, B_m] = 0 .$$

The function $-D_n$ characterizes the change in the static interaction of the n-th molecule with the neighbouring molecules when it is excited. If ϱ_n is defined by the expression

$$x_n - x_{n-1} = R - \varrho_n , \qquad (2)$$

then (allowing for the interaction with the nearest neighbours)

$$D_n = \mathcal{D}_n(|x_{n+1} - x_n|) + \mathcal{D}(|x_{n-1} - x_n|) \approx \left(1 + \frac{\beta}{R} \varrho_n\right) D , \qquad (3)$$

where $D \equiv 2\mathcal{D}(R)$ and β are the positive parameters of the theory. The resonance interaction determining the transition of an intramolecular excitation from one molecule to another is characterized by the quantity $-J$ which is defined by the electric dipole moment using the equality

$$J = 2\boldsymbol{d}^2 R^{-3} . \qquad (4)$$

The first two terms in (1) correspond to the potential and the kinetic energy of the longitudinal displacements of molecules. The potential energy is chosen in the harmonic approximation:

$$U = \tfrac{1}{2} w \sum_n \varrho_n^2 . \qquad (5)$$

When the equation $\dot{x}_n = -\sum_{-\infty<l\leq n} \dot{\varrho}_l$ (5a) is taken into account the kinetic energy can be written as

$$T = \tfrac{1}{2} M \sum_n \dot{x}_n^2 = \tfrac{1}{2} M \sum_n \Big(\sum_{-\infty<l\leq n} \dot{\varrho}_l \Big)^2 . \tag{6}$$

Dotted symbols in (6) and in subsequent expressions mean a derivative with respect to time.

The collective excitations of interest are described by the wave function

$$|\psi\rangle = \sum_n a(t)\, B_n^+ |0\rangle . \tag{7}$$

The coefficients $a_n(t)$ satisfy the condition

$$\sum_n |a_n(t)|^2 = 1 \tag{7a}$$

and characterize the distribution of excitation along the chain of molecules. These coefficients satisfy a set of equations which can be obtained from the Schrödinger equation

$$i\hbar \frac{\partial}{\partial t} |\psi\rangle = H\, |\psi\rangle . \tag{8}$$

Using the explicit form of the operator H and (8), we obtain the equation

$$\sum_n \left\{ i\hbar \frac{\partial a_n}{\partial t} - \left[\varepsilon + T + U - \left(1 + \frac{\beta}{R} \varrho_n \right) D \right] a_n - J(a_{n+1} + a_{n-1}) \right\} B_n^+ |0\rangle = 0 .$$

Since the functions $B^+ |0\rangle$ corresponding to different values of n are orthogonal, we can use this equation to obtain the following set of equations:

$$i\hbar \frac{\partial a_n}{\partial t} = \left[\varepsilon + T + U - \left(1 + \frac{\beta}{R} \varrho_n \right) D \right] a_n - J(a_{n+1} + a_{n-1}) . \tag{9}$$

It is now convenient to introduce the functional of the variables ϱ_n, $\dot{\varrho}_n$ and a_n

$$F \equiv \langle \psi |\, H\, |\psi\rangle =$$
$$= \sum_n \left\{ \left[\varepsilon + T + U - \left(1 + \frac{\beta}{R} \varrho_n \right) D \right] a_n^* a_n - J a_n^* (a_{n+1} + a_{n-1}) \right\} . \tag{10}$$

Following Toda [3], we associate the displacements with canonically conjugated variables

$$s_n = \frac{\partial T}{\partial \dot{\varrho}_n} = -M \sum_{l\geq n} \dot{x}_l .$$

This expression yields the equation

$$\dot{x}_n = \frac{1}{M} (s_n - s_{n-1}) ,$$

which enables the kinetic energy of the molecular motion to be expressed in terms of the variables

$$T = \frac{1}{2M} \sum_n (s_n - s_{n-1})^2 . \tag{11}$$

Thus at fixed values of the coefficients $a_n(t)$ the function F is the Hamiltonian function with respect to the molecular displacements ϱ_n and their

canonical conjugates s_n. We can therefore write the Hamilton equations

$$\dot{\varrho}_n = \frac{\partial F}{\partial s_n} = \frac{1}{M}(2s_n - s_{n+1} - s_{n-1}),$$

$$\dot{s}_n = -\frac{\partial F}{\partial \varrho_n} = -w\varrho_n + \frac{\beta D}{R}|a_n(t)|^2 .$$

Eliminating s_n from these equations we find

$$\ddot{\varrho}_n = -\frac{w}{M}(2\varrho_n - \varrho_{n+1} - \varrho_{n-1}) + \frac{\beta D}{MR}(2|a_n|^2 - |a_{n+1}|^2 - |a_{n-1}|^2) . \quad (12)$$

The set of equations (9) and (12) defines the collective excitations of a chain. If we are interested only in excitations distributed over some region of a chain, we can apply the continuum model. Let us introduce a dimensionless variable $\xi = x/R$ and continuous functions $\varrho(\xi, t)$ and $a(\xi, t)$ such that

$$\varrho(n, t) = \varrho_n(t) , \qquad a(n, t) = a_n(t) .$$

Using the expressions

$$\varrho(\xi \pm 1, t) \approx \varrho(\xi) \pm \frac{\partial \varrho(\xi, t)}{\partial \xi} + \frac{1}{2}\frac{\partial^2 \varrho(\xi, t)}{\partial \xi^2},$$

$$a(\xi \pm 1, t) \approx a(\xi) \pm \frac{\partial a(\xi, t)}{\partial \xi} + \frac{1}{2}\frac{\partial^2 a(\xi, t)}{\partial \xi^2},$$

$$|a(\xi \pm 1, t)|^2 \approx |a(\xi, t)|^2 \pm \frac{\partial}{\partial \xi}|a(\xi, t)|^2 + \frac{1}{2}\frac{\partial^2}{\partial \xi^2}|a(\xi, t)|^2,$$

we can transform (9) and (12) to

$$i\hbar \frac{\partial a(\xi, t)}{\partial t} - \left[\lambda - \frac{\beta D}{R}\varrho(\xi, t)\right] a(\xi, t) + J\frac{\partial^2 a(\xi, t)}{\partial \xi^2} = 0 , \quad (13)$$

$$\frac{\partial^2 \varrho(\xi, t)}{\partial t^2} - v_a^2 \frac{\partial^2 \varrho(\xi, t)}{\partial \xi^2} + \frac{\beta D}{MR}\frac{\partial^2}{\partial \xi^2}|a(\xi, t)|^2 = 0 , \quad (14)$$

where

$$\lambda \equiv \varepsilon + T + U - D - 2J , \quad (15)$$

$$T + U = \frac{M}{2}\int_{-\infty}^{\infty} d\xi \left[\left(\int_{-\infty}^{\xi} d\eta \frac{\partial}{\partial t}|\varrho(\eta, t)|^2\right)^2 + v_a^2 |a(\xi, t)|^4\right], \quad (16)$$

$v_a = \sqrt{w/R}$, $v_a R = V_a$ is the longitudinal sound velocity in the chain.

We shall look for solutions of the set of (13), (14) in the form of excitations propagating along the chain with the velocity $V = vR$. In other words, we set

$$\varrho(\xi, t) = \varrho(\xi - vt) , \qquad a(\xi, t) = \Phi(\xi - vt)\, e^{i\theta(\xi, t)} . \quad (17)$$

We then obtain from (14)

$$\varrho(\xi, t) = \frac{\beta D |a(\xi, t)|^2}{MR(v_a^2 - v^2)} . \quad (18)$$

Substituting the right-hand side of (18) into (13) and (16), one obtains

$$i\hbar \frac{\partial a(\xi, t)}{\partial t} - \lambda a(\xi, t) + J \frac{\partial^2 a(\xi, t)}{\partial \xi^2} + G_\nu |a(\xi, t)|^2 a(\xi, t) = 0 , \tag{19}$$

$$T + U = \frac{G_\nu}{2(\nu_a^2 - \nu^2)} \int_{-\infty}^{\infty} \mathrm{d}\xi \left[\left(\int_{-\infty}^{\xi} \mathrm{d}\eta \frac{\partial}{\partial t} |a(\eta, t)|^2 \right)^2 + \nu_a^2 |a(\xi, t)|^4 \right], \tag{20}$$

$$G_\nu \equiv \frac{\beta^2 D^2}{M R^2 (\nu_a^2 - \nu^2)} = \frac{G_0}{1 - V^2/V_a^2}, \tag{21}$$

$$G_0 \equiv \beta^2 D^2 / M V_a^2 . \tag{22}$$

The exact solution of the nonlinear equation (19), normalized by

$$\int_{-\infty}^{\infty} |a(\xi, t)|^2 \, \mathrm{d}\xi = 1$$

has the form

$$a_\nu(\xi, t) = \frac{\sqrt{\frac{\mu_\nu}{2}} \exp\left\{ i \left[\frac{\hbar \nu}{2J} (\xi - \xi_0) - \Omega_\nu t \right] \right\}}{\cosh\left[\mu_\nu (\xi - \xi_0 - \nu t) \right]}, \tag{23}$$

where ξ_0 is an arbitrary constant due to the translational invariance of (13) and (14). The other quantities contained in (23) are defined by

$$\mu_\nu = G_\nu / 4J , \tag{24}$$

$$\hbar \Omega_\nu = \lambda + (\hbar^2 \nu^2 - \tfrac{1}{4} G_\nu^2)/4J . \tag{25}$$

When (23) is taken into account the expression (20) takes the form

$$T + U = \frac{G_\nu^2 (\nu^2 + \nu_a^2)}{24J(\nu_a^2 - \nu^2)} .$$

Substituting this into (15), we obtain the final expression for the energy of a soliton described by the function (23), when the velocity of its motion along the chain is fixed $V = \nu R$

$$\hbar \Omega_\nu = \hbar \Omega_0 + \frac{\hbar^2 V^2}{4JR^2} + \frac{G_0^2 (Z^2 + \frac{3}{2} Z^4 - \frac{1}{2} Z^6)}{24J(1 - Z^2)^3}, \tag{26}$$

where

$$Z \equiv V/V_a , \qquad \hbar \Omega_0 = \varepsilon - D - 2J - G_0^2/48J \tag{27}$$

is the internal energy of the soliton.

It follows from the expression (26) that when the soliton velocity approaches the longitudinal sound velocity V_a, the soliton energy tends to infinity. At all finite values of the soliton energy its velocity is therefore less than V_a. The soliton energy thus cannot be transformed into the energy of the longitudinal waves.

At small velocities of the soliton ($V \ll V_a$) the soliton energy (26) takes the form

$$\hbar \Omega_\nu = \hbar \Omega_0 + \frac{\hbar^2 V^2}{2R^2 J} + \frac{G_0^2 V^2}{24 V_a^2 J} \left[1 - \frac{9}{2} (V/V_a)^2 + \cdots \right].$$

In the same approximation the parameter

$$\mu_v \approx \frac{G_0}{4J}[1 + (V/V_a)^2] \approx \frac{G_0}{4J} = \mu_0 \tag{28}$$

depends but very weakly upon the soliton velocity.

Substituting (23) into (18) we find the distribution of the changes in the relative distances between the molecules

$$\varrho(\xi, t) = \frac{\sqrt{G_0^3/w}}{8J(1 - V^2/V_a^2)^2} \operatorname{sech}[\mu_v(\xi - \xi_0 - vt)] . \tag{29}$$

According to (29), the maximal deviation

$$\varrho_{\max} = \frac{\sqrt{G_0^3/w}}{8J(1 - V^2/V_a^2)^2} \tag{30}$$

obtained at $\xi = \xi_0 + vt$ moves along the chain with the velocity $V = vR$. The deformation moving along the chain is accompanied by the internal excitation. The region of the chain embraced by the internal excitation and the deformation is defined by

$$R\,\Delta\xi \approx \frac{\pi R}{\mu_v} = 4\pi J R/G_v . \tag{31}$$

The applicability condition of the continuum approximation is associated with the validity of the inequality $\Delta\xi = 4\pi J/G_v \gg 1$.

3. The Stability of Solitons

A remarkable property of solitons is that once the velocity is fixed the distribution of excitation and deformation described by the functions (23) and (29) remains unchanged during the soliton motion. At $V \ll V_a$ we have $G_v \approx G_0$, so that the size and shape of the soliton are both independent of its velocity, i.e. the soliton slows down without changing its shape.

In order to study the stability of solitons we consider the expression (10) at zero velocity ($v = 0$). In this case $a(\xi, t)$ represents a state with a simple time dependence, i.e.

$$a(\xi, t) = \Phi(\xi) \exp(-i\Omega_0 t) . \tag{32}$$

Using the continuum approximation in (10) and taking (32) into account, we obtain

$$F_0\{\Phi\} = \varepsilon - D - 2J - \int_{-\infty}^{\infty} \Phi(\xi)\left[J\frac{\partial^2\Phi(\xi)}{\partial\xi^2} + \frac{1}{2}G_0\Phi^3(\xi)\right] d\xi . \tag{33}$$

The expression (33) can be regarded as a functional of $\Phi(\xi)$; its absolute minimum defines the soliton energy at zero velocity, i.e.

$$(\hbar\Omega)_{\min} = \min F_0\{\Phi\} . \tag{33a}$$

If $\Phi(\xi)$ is taken to be a normalized function depending on a single parameter μ

$$\Phi(\xi) = \sqrt{\mu/2} \operatorname{sech}[\mu(\xi - \xi_0)] , \tag{34}$$

we obtain

$$F_0(\mu) = \varepsilon - D - 2J + \tfrac{1}{3}\left[J\mu^2 - \tfrac{1}{2}G_0\mu\right]. \tag{35}$$

The minimum (35) corresponds to the value $\mu_0 = G_0/4J$, and the energy

$$(\hbar\Omega)_{\min} = \varepsilon - D - 2J - G_0^2/48J \tag{36}$$

is exactly the same as that given by (27). The functions different from (34), or the values $\mu \neq \mu_0$ correspond to higher energies. In particular, the value of the functional (35) at $\mu = 0$ corresponds to nonlocalized excitations — the excitons. Their energy (at zero velocity) is equal to $\varepsilon - D - 2J$ thus exceeding the soliton energy by the value $G_0^2/48J$. These excitations are unstable in one-dimensional molecular systems.

If the electric dipole moment d is directed perpendicular to the chain, the parameter J which characterizes the resonance interaction will have a negative value ($\approx -d^2/R^3$). In this case the functional (35) has no minimum, the continuum approximation is not valid and there are no stable excitations such as the soliton. The excitation tends to be completely localized on a single molecule.

In conclusion we investigate the stability of the solitons for external disturbances. As a disturbance we consider a sound wave

$$A\cos\omega\,(t - \xi/v_a) \tag{37}$$

assuming the frequency ω to be low enough so that the sound wavelength exceeds considerably the size of an exciton,

$$\frac{2\pi v_a R}{\omega} \gg R\,\Delta\xi = 4\pi J R/G_0\,. \tag{38}$$

As mentioned above, at finite values of the soliton energy its velocity is much less than that of longitudinal sound waves. Therefore when investigating the soliton–longitudinal sound wave interaction the solitons may be regarded as unmoving.

To take into account the role of the exciton–sound wave interaction we investigate the solutions of (13) and (14) using

$$\varrho(\xi, t) = \varrho_0(\xi) + A\cos\omega(t - \xi/v_a)\,, \tag{39}$$

Here

$$\varrho_0(\xi) = \frac{G_0}{8J}\sqrt{\frac{G_0}{w}}\,\mathrm{sech}^2\,(\mu_0\xi)\,, \qquad \mu_0 = G_0/4J$$

is the solution for a free soliton at rest. In this case (14) is exactly satisfied by the function

$$a(\xi, t) = \Phi(\xi)\exp\left[-i\vartheta(t)\right], \tag{40}$$

if

$$\Phi(\xi) = \sqrt{\mu_0/2}\,\mathrm{sech}\,\mu_0\xi\,. \tag{40a}$$

At $|\xi| > \Delta\xi$ equation (13) is satisfied approximately, the function (40) and its derivatives tending rapidly to zero. In the region $|\xi| \leqq \Delta\xi$, under the condition (38) the sound wave has the same phase. Equation (13) thus reduces to

$$\frac{\partial\vartheta(t)}{\partial t} + \frac{\beta DA}{\hbar R}\cos\omega t - \Omega_0 = 0\,. \tag{41}$$

The soliton energy is defined by the integral

$$E = i\hbar \int a^*(\xi, t) \frac{\partial a(\xi, t)}{\partial t} \, d\xi \, .$$

Therefore, when the expressions (40) and (41) are taken into account we have

$$E = \hbar \frac{\partial \theta(t)}{\partial t} = \hbar \Omega_0 - \frac{\beta D A}{R} \cos \omega t \, . \tag{42}$$

The average value of soliton energy (when it is averaged over the period of a sound wave) is thus equal to the energy $\hbar \Omega_0$ of a free soliton. According to (40), the sound wave does not change the shape of a soliton, defined by the function $\Phi^2(\xi) = (\mu_0/2) \operatorname{sech}^2 (\mu_0 \xi)$.

Thus the excitations of a molecular chain in the form of solitons have great stability in retaining the shape, magnitude, and the size of the region occupied by the excitation. This fact has been used by one of the authors [2] to account for the great efficiency of the transfer of energy released in the hydrolysis of ATP molecules in living organisms along the α-helical protein molecules and to elucidate the contraction mechanism of transversely striated animal muscles at the molecular level [4].

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Soliton solution for nonlinear partial differential equations by cosine-function method

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Abstract

In this Letter, we established a traveling wave solution by using Cosine-function algorithm for nonlinear partial differential equations. The method is used to obtain the exact solutions for five different types of nonlinear partial differential equations such as, general equal width wave equation (GEWE), general regularized long wave equation (GRLW), general Korteweg–de Vries equation (GKdV), general improved Korteweg–de Vries equation (GIKdV), and Coupled equal width wave equations (CEWE), which are the important soliton equations.
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1. Introduction

The study of numerical methods for the solution of nonlinear partial differential equations has enjoyed an intense period of activity over the last 40 years from both theoretical and practical points of view. Improvements in numerical techniques, together with the rapid advances in computer technology, have meant that many of the partial differential equations arising from engineering and scientific applications, which were previously intractable, can now, be routinely solved [1]. Finite difference methods approximate the differential operators and hence, solve the difference equations. In finite element method the continuous domain is represented as a collection of a finite number N of subdomains known as elements. The collection of elements is called the finite element mesh. For time dependent problems, the differential equations are approximated by the finite element method to obtain a set of ordinary differential equations in time. These differential equations are solved approximately by finite difference methods or other methods. In all finite difference and finite elements it is necessary to have a boundary and initial conditions. But the Adomian decomposition method, which has been developed by George Adomian (Solving Frontier Problem of Physics: The Decomposition Method, Kluwer Academic Publishers, Boston, MA, 1994) depends only on the initial conditions to obtain solution in series form which almost converges to the exact solutions of the problem. In recently years, some other ansatz methods have been developed, such as, the tanh method [2,3], the extended tanh-function method [4,5], the modified extended tanh-function method [6–11], variational iteration method [12–17], the generalized hyperbolic-function [18,19], the separation of variables method [20,21] and the sine–cosine method [22,23].

The aim of this Letter is extended the cosine function-method to find the exact solutions of some important nonlinear partial differential equations such as the general equal width wave (GEW) equation, the general Regularized long wave (GRLW) equation,

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the general Korteweg–de Vries (GKdV) equation, the general improved Korteweg–de Vries (GIKdV) equation and the coupled equal width wave equations, which are the five of important soliton equations.

2. The cosine function method

Consider the nonlinear partial differential equation in the form

$$F(u, u_t, u_x, u_{xx}, u_{xxt}, \ldots) = 0, \tag{1}$$

where $u(x,t)$ is the solution of nonlinear partial differential equation Eq. (1). We use the transformations,

$$u(x,t) = f(\xi), \tag{2}$$

where $\xi = x - ct$. This enables us to use the following changes:

$$\frac{\partial}{\partial t}(\cdot) = -c\frac{d}{d\xi}(\cdot), \qquad \frac{\partial}{\partial x}(\cdot) = \frac{d}{d\xi}(\cdot), \qquad \frac{\partial^2}{\partial x^2}(\cdot) = \frac{d^2}{d\xi^2}(\cdot), \quad \ldots. \tag{3}$$

Using Eq. (3) to transfer the nonlinear partial differential equation Eq. (1) to nonlinear ordinary differential equation

$$G(f, f', f'', f''', \ldots) = 0. \tag{4}$$

The solution of Eq. (4) can be expressed in the form:

$$f(\xi) = \lambda \cos^{\beta}(\mu\xi), \qquad |\xi| \leqslant \frac{\pi}{2\mu}, \tag{5}$$

where λ, β and μ are unknown parameters which will be determined. Then we have:

$$\begin{aligned} f' &= \frac{df(\xi)}{d\xi} = -\lambda\beta\mu\cos^{\beta-1}(\mu\xi)\sin(\mu\xi), \\ f'' &= \frac{d^2 f(\xi)}{d\xi^2} = -\lambda\beta\mu^2\cos^{\beta}(\mu\xi) + \lambda\mu^2\beta(\beta-1)\cos^{\beta-2}(\mu\xi) - \lambda\mu^2\beta(\beta-1)\cos^{\beta}(\mu\xi). \end{aligned} \tag{6}$$

Substituting Eq. (6) into the nonlinear ordinary differential equation Eq. (4) gives a trigonometric equation of $\cos^{\alpha}(\mu\xi)$ terms. To determine the parameters first balancing the exponents of each pair of cosine to determine α. Then we collect all terms with the same power in $\cos^{\beta}(\mu\xi)$ and put to zero their coefficients to get a system of algebraic equations among the unknowns β, λ and μ. Now, the problem is reduced to a system of algebraic equations that can be solved to obtain the unknown parameters β, λ and μ. Hence, the solution considered in Eq. (5) is obtained. The above analysis yields the following theorem:

Theorem. *The exact analytical solution of the nonlinear partial differential equations Eq. (1) can be determined in the form Eq. (5) where all constants found from the algebraic equations after its solutions.*

3. Applications

In order to illustrate the effectiveness of the proposed method five examples in mathematical are chosen as follows

3.1. Test problem

3.1.1. The general equal width wave equation (GEWE)

Consider the following problem: Find functions $u(x,t)$ satisfying the general equal width wave equation (GEWE) [24–27] in the form,

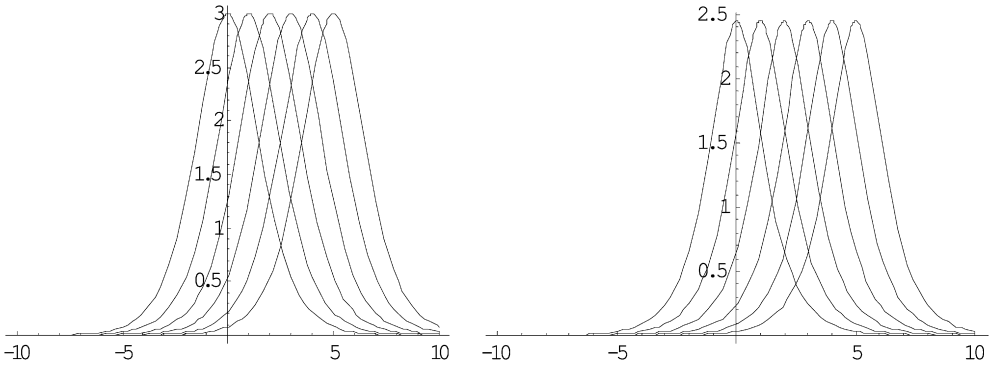
$$u_t + \varepsilon u^p u_x - \nu u_{xxt} = 0. \tag{7}$$

By using the wave variable $\xi = x - ct$ and $u(x,t) = f(\xi)$ then GEWE equation becomes

$$-c\frac{df(\xi)}{d\xi} + \varepsilon f^p(\xi)\frac{df(\xi)}{d\xi} + c\nu\frac{d^3 f(\xi)}{d\xi^3} = 0. \tag{8}$$

Integrating Eq. (8) gives

$$-cf(\xi) + \frac{\varepsilon}{p+1}\big(f(\xi)\big)^{p+1} + c\nu\frac{d^2 f(\xi)}{d\xi^2} = 0. \tag{9}$$



(a) (b)

Fig. 1. (a) Soliton solution for EWE; (b) soliton solution for MEWE.

Table 1
The conservative quantities the EWE, MEWE and GEWE equations with increase time

T / p	I_1			I_2		
	1	2	3	1	2	3
0	12	7.6953	6.04182	28.8	16	11.4365
1	12	7.6953	6.04182	28.8	16	11.4365

Substituting Eq. (6) into (9) gives:

$$-c\lambda\cos^{\beta}(\mu\xi)+\frac{\varepsilon\lambda^{p+1}}{p+1}\cos^{(p+1)\beta}(\mu\xi)-c\nu\lambda\beta\mu^2\cos^{\beta}(\mu\xi)+\nu c\lambda\beta(\beta-1)\mu^2\cos^{\beta-2}(\mu\xi)-c\nu\lambda\beta(\beta-1)\mu^2\cos^{\beta}(\mu\xi)=0. \tag{10}$$

The last equation is satisfied only if the following system of algebraic equations holds:

$$\begin{aligned}&-c\lambda-c\lambda\nu\beta\mu^2-c\lambda\mu^2\nu\beta(\beta-1)=0, \qquad (p+1)\beta=\beta-2,\\ &\frac{\varepsilon\lambda^{p+1}}{p+1}+c\lambda\mu^2\nu\beta(\beta-1)=0,\end{aligned} \tag{11}$$

using MATHEMATICA package software for solving the system Eq. (11) we obtain:

$$\beta=\frac{-2}{p}, \qquad \mu=\pm\frac{ip}{2\sqrt{\nu}} \quad \text{and} \quad \lambda=2^{-1/p}\left(\frac{c(2+3p+p^2)}{\varepsilon}\right)^{1/p}, \tag{12}$$

substituting Eq. (12) into Eq. (5) we obtain the exact soliton solution of the GEWE equation in the form

$$u(x,t)=2^{-1/p}\left(\frac{c(2+3p+p^2)}{\varepsilon}\right)^{1/p}\cos^{-2/p}\left(\frac{\pm ip}{2\sqrt{\nu}}(x-ct)\right). \tag{13}$$

As a special case when $p=1$ gives the soliton solution of the EWE equation as in [24], when $p=2$ gives the soliton solution of the MEWE equation as in [29] and also we can get the soliton solution of the GEWE equation as in [30]. Fig. 1 shows the soliton solution for the EWE, MEWE and GEWE equations with increase time. Also, we can calculate the conservative quantities when $p=1$ and 2 as given in Table 1. It is to be noted that the obtained results are quite good with that obtained [28–30].

3.1.2. The general regularized long wave (GRLW) equation

We consider the general regularized long wave equation for long waves in shallow water written as [31–35]:

$$u_t+u_x+\varepsilon u^p u_x-\nu u_{xxt}=0. \tag{14}$$

Using the transformation $u(x,t)=f(\xi)$ where $\xi=x-ct$, then the GRLW equation becomes

$$-c\frac{df(\xi)}{d\xi}+\frac{df(\xi)}{d\xi}+\varepsilon f^p(\xi)\frac{df(\xi)}{d\xi}+c\nu\frac{d^3f(\xi)}{d\xi^3}=0. \tag{15}$$

Integrating Eq. (15) gives

$$(1-c)f(\xi)+\frac{\varepsilon}{p+1}\big(f(\xi)\big)^{p+1}+c\nu\frac{d^2f(\xi)}{d\xi^2}=0. \tag{16}$$

Substituting Eq. (6) into Eq. (16) becomes

$$(1-c)\lambda\cos^{\beta}(\mu\xi)+\frac{\varepsilon\lambda^{p+1}}{p+1}\cos^{(p+1)\beta}(\mu\xi)-c\nu\lambda\beta\mu^2\cos^{\beta}(\mu\xi)$$
$$+c\nu\lambda\beta(\beta-1)\mu^2\cos^{\beta-2}(\mu\xi)-c\nu\lambda\beta(\beta-1)\mu^2\cos^{\beta}(\mu\xi)=0. \tag{17}$$

By equating the exponents and the coefficients of each pair of the cosine function we obtain the following system of algebraic equations:

$$(1-c)\lambda-c\nu\lambda\beta\mu^2-c\nu\lambda\mu^2\beta(\beta-1)=0,\qquad (p+1)\beta=\beta-2,$$
$$\frac{\varepsilon\lambda^{p+1}}{p+1}+c\nu\lambda\mu^2\beta(\beta-1)=0. \tag{18}$$

By using MATHEMATICA package software for solving the system Eq. (18) we get:

$$\beta=\frac{-2}{p},\qquad \mu=\pm\frac{\sqrt{1-c}\,p}{2\sqrt{\nu c}}\quad\text{and}\quad \lambda=2^{-1/p}\left(\frac{-2+2c-3p+3cp-p^2+cp^2}{\varepsilon}\right)^{1/p}. \tag{19}$$

Then by substituting Eq. (19) into Eq. (5) then, the exact soliton solution of the GRLW equation can be written in the form:

$$u(x,t)=2^{-1/p}\left(\frac{-2+2c-3p+3cp-p^2+cp^2}{\varepsilon}\right)^{1/p}\cos^{-2/p}\left(\pm\frac{\sqrt{1-c}\,p}{2\sqrt{\nu c}}(x-ct)\right). \tag{20}$$

3.1.3. The GKdV and GIKdV equations

The general Korteweg–de Vries (GKdV) equation [36,37], for long waves in shallow water has the form:

$$u_t+\varepsilon u^p u_x+\gamma u_{xxx}=0. \tag{21}$$

In the same manner to solve Eq. (21) by cosine-function method, we obtain the following system of equations as follows:

$$-c\lambda-\gamma\lambda\beta\mu^2-\gamma\lambda\mu^2\beta(\beta-1)=0,\qquad (p+1)\beta=\beta-2,$$
$$\frac{\varepsilon\lambda^{p+1}}{p+1}+\gamma\mu^2\beta(\beta-1)=0. \tag{22}$$

This can be solved using MATHEMATICA package software to get

$$\beta=\frac{-2}{p},\qquad \mu=\pm\frac{i\sqrt{c}\,p}{2\sqrt{\gamma}},\qquad \lambda=2^{-1/p}\left(\frac{c(2+3p+p^2)}{\varepsilon}\right)^{1/p}. \tag{23}$$

So the exact soliton solution of GKdV equation can be written in the form

$$u(x,t)=2^{-1/p}\left(\frac{c(2+3p+p^2)}{\varepsilon}\right)^{1/p}\cos^{-2/p}\left(\pm\frac{i\sqrt{c}\,p}{2\sqrt{\gamma}}(x-ct)\right). \tag{24}$$

In a similar way, when we consider the general improved Korteweg–de Vries (GIKdV) equation in the form [27,37]

$$u_t+\varepsilon u^p u_x+\gamma u_{xxx}-\nu u_{xxt}=0. \tag{25}$$

The exact soliton solution of this soliton equation can be deduced to the form:

$$u(x,t)=2^{-1/p}\left(\frac{c(2+3p+p^2)}{\varepsilon}\right)^{1/p}\cos^{-2/p}\left(\pm\frac{i\sqrt{c}\,p}{2\sqrt{\gamma+c\nu}}(x-ct)\right). \tag{26}$$

3.1.4. Coupled equal width wave equation (CEWE)

In this case we consider the coupled equal width wave equations, in the normalized form

$$u_t+uu_x-u_{xxt}+vv_x=0,\qquad v_t+vv_x-v_{xxt}=0, \tag{27}$$

where the subscripts t and x denoting to the differentiation with respect to time and space respectively, the boundary conditions $u,u',u''\to c_1$ and $v,v',v''\to c_2$ as $x\to\pm\infty$ such that c_1, c_2 are arbitrary constants. Suppose $u(x,t)=f(\xi)$ and $v(x,t)=g(\xi)$ then the coupled EWE becomes

$$-c\frac{df(\xi)}{d\xi}+f(\xi)\frac{df(\xi)}{d\xi}+c\frac{d^3f(\xi)}{d\xi^3}+g(\xi)\frac{dg(\xi)}{d\xi}=0,$$

$$-c\frac{dg(\xi)}{d\xi} + g(\xi)\frac{dg(\xi)}{d\xi} + c\frac{d^3g(\xi)}{d\xi^3} = 0. \tag{28}$$

We define the two functions

$$f(\xi) = \lambda_1 \cos^{\beta_1}(\mu_1\xi), \qquad g(\xi) = \lambda_2 \cos^{\beta_2}(\mu_2\xi). \tag{29}$$

Substituting Eq. (29) into Eq. (28) we obtain a system of algebraic equations

$$\begin{aligned}
&-c\lambda_1 - c\lambda_1\beta_1\mu_1^2 - c\lambda_1\mu_1\beta_1(\beta_1-1) = 0, \qquad \beta_1 + 2\beta_2 + 2 = 0,\\
&\frac{\lambda_1^2}{2} + \frac{\lambda_2^2}{2} + c\lambda_1\mu_1^2\beta_1(\beta_1-1) = 0, \qquad -c\lambda_2 - c\lambda_2\beta_2\mu_2^2 - c\lambda_2\mu_2\beta_2(\beta_2-1) = 0, \qquad \beta_2 + 2 = 0,\\
&\frac{\lambda_2^2}{2} + c\lambda_2\mu_2^2\beta_2(\beta_2-1) = 0.
\end{aligned} \tag{30}$$

Solving the system of Eq. (30), with the aid of MATHEMATICA, we have

$$\begin{aligned}
&\lambda_1 = (0.5 - 2.95804i)c, \qquad \beta_1 = 2, \qquad \mu_1 = \pm 0.5i,\\
&\lambda_2 = 3c, \qquad \beta_2 = -2, \qquad \mu_2 = \pm 0.5i.
\end{aligned} \tag{31}$$

So we obtain the exact soliton solution of the CEWE equation in the form

$$\begin{aligned}
&u(x,t) = (0.5 - 2.95804i)c\cosh^2\big[0.5(x-ct)\big],\\
&v(x,t) = 3c\operatorname{sech}^2\big[0.5(x-ct)\big].
\end{aligned} \tag{32}$$

4. Conclusion

In this Letter, the cosine-function method has been successfully applied to find the solution for five nonlinear partial differential equations such as GEWE, GRLW, GKdV, GIKdV and CEWE equations. The cosine-function method is used to find a new exact solution. Thus, we can say that the proposed method can be extended to solve the problems of nonlinear partial differential equations which arising in the theory of solitons and other areas.

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6

Some Aspects of Charge Transfer in Biological Systems

Felix Gutmann

ABSTRACT: The following modes of biological charge transfer are discussed: via solitons, as electronic conductance as hydrated proteins, via surface charge transfer complexes, and by protonic charge transfer complexes. Electron transfer involving the cytochromes is treated in the light of electrochemical measurements including voltammetry. A hypothesis for the lateral, surface conductance of biological membranes is proposed, based on electron donor–acceptor surface interactions giving rise to polaritons.

1. Introduction

In a way, all biological charge transfer is electrochemical in nature. Thus, this field is so huge that the following discussion will have to be confined to a few selected aspects. Many relevant topics are treated by the other contributors to this volume.

Regular and periodic organization of molecules is essential for the living state, and fundamental to its processes are structured instabilities which are reflected in the solid state by, e.g., charge transfer complexes. Many *in vivo* properties are difficult to explain by classical chemical processes in solution but appear to fit solid state physical processes in cells. Solid state events involving conduction are evident in animate aggregations and may well be an essential characteristic of life, which may be an electromagnetic phenomenon. A growing body of reviews and texts[1] is

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available to support these views. Chelate metal complexes of purines, pyrimidines, and their nucleosides have been reviewed in detail.[2]

For charge transfer to occur, free charges must either be initially, inherently present, or injected from outside, or else be generated within the system by means of an injection of energy.

When a molecule is excited by a sufficiently energetic event, an electron can be ejected. The resulting molecular cation polarizes electrons from its immediate environment in a time of about 10^{-16} s, and this polarization energy stabilizes the final state of the system. Frohlich's excitation may play here

When two opposite charges in a medium of relative permittivity ε_r approach one another, the Coulombic energy of attraction between them equals kT at a critical distance R_c so that[3]

$$R_c = e^2/4\varepsilon_r\varepsilon_0 kT$$

or (1)

$$R_c = 55.9/e_r \text{ nm}$$

A limit therefore exists for the quantity of charge of one kind which can be imposed without resulting in recombination.[3] This limits the amounts to about 10^{17} electronic charges/cm^3, while there are about 10^{21} molecules/cm^3. Electric neutrality need not be preserved in organic media. In an organic layer of about 5 nm the total of such uncompensated space charge is approximately 10^{-10} C/cm^2, which, if it were uniformly distributed, would yield a rather small electric field of about 10^3 V cm^{-1}. Potential differences of about 5 mV across 5 nm, i.e., 10^4 V cm^{-1}, do not represent a serious problem for space charge limitation of current flow through biological systems typically several nanometers thick. It also follows that only one molecule in several thousand can be charged if compensating charges are absent. Thus, modeling a biological system, a localized representation based on individual charged species is preferable to a representation of a charge density distributed over all the molecules.[3]

If the distance between M^+ and M^- is less than R_c, they exist as an ion pair, i.e., in a charge transfer state. If M^+ and M^- are locked into original lattice states then the distance can have only certain values. From the energies for linear polyacenes, determined accurately by Silinsh,[4] it appears that, as the number of rings increased,* charge separation in the

* Silinsh estimated[4] the energy (E_{CT}) of the charge transfer state of pentacene to be about one order of magnitude smaller than that of naphthalene. As the number of conjugated rings increases it becomes easier to separate the charges in a CT state; the photoinduced M^+M^- thermalized CT state of pentacene has, under some conditions, charge separation distances as large as 12.5 nm.[5]

charge transfer state is facilitated. This could be significant for biological systems containing highly conjugated molecules such as chlorophyll.[3]

Electrochemical methods have been applied to the study of a variety of biological charge transfer problems. Thus, to give just a few examples, voltammetry has been used to study tissue,[6] zeta potentials have been related to cystic fibrosis,[7] an electrophoretic mobility test of macrophage for malignancy has been suggested[8] and polarography has been applied, *inter alia*, to the study of carcinogenicity of several chemical compounds by the Czech school.[9]

2. *Solitons*[73]

The important role played by solitons* in biological transfer remains to be clarified: solitons are said[11] to arise from localized disordered regions on, say, a membrane surface in the presence of the transmembrane electric field; the local perturbation then tends to spread and to move leading to changes in the orientation of the lipid membrane molecules. The soliton energy is considerably below the "energy band gap" of a polypeptide chain but may initiate proton transfer[12] in a hydrogen bonded chain in the presence of an electrostatic field.

Solitons may result in energy transporting along biological macromolecules. Their genesis and propagation as phonons in macromolecules establish important system properties in these molecules. Electromagnetic energy may be converted to soliton conducted energy as a transductive step. These solitons propagate quite slowly along the molecule, exhibiting threshold and possibly "windowed" characteristics in relation to the exciting energy. In transductive coupling of exciting events on cell membranes, they thus offer a vehicle for highly selective relationships in the electrochemical environment. In the transmembrane coupling of cell surface events to specific intracellular enzyme systems or organelles, either at internal membrane sites or via the cytoskeletal system to submembrane regions, solitons may also play a major role.

Binding of drug molecules to helical protein molecules[67] may inhibit soliton propagation. Further studies along these lines may allow "tailoring" of new drugs with optimal therapeutic capabilities. Their actions may be modulated by concurrent exposure to selected electromagnetic fields. Similar selective modulations of tissue functions by modification of the

* A soliton[10] is a localized collective excited state likely to arise in nonlinear dispersive media. The soliton is capable of migrating over relatively long distances with very little energy dissipation. It may be electrically charged or may be neutral such as a kink in a polymeric chain, or a domain wall.

electromagnetic environment may be seen in wound healing, tissue regeneration, and cancer therapy. These studies in biological interactions with the electromagnetic environment may yield important new knowledge about biophysical bases of circadian and other biological rhythms, with important therapeutic implications in endrocrine and immune disorders.

The solitons are thought[68] to induce simultaneously a displacement of the vibrational energy levels of the macromolecular chain and a rearrangement of the aqueous environment, until some sort of resonance is attained between these two processes, and a transition between charge and discharge regimes occurs. The result is a region of highly structured water and what often is called a Fröhlich regime, characterized by the existence of coherent electrical oscillations having frequencies of the order of 10–100 gHz, and discussed by Fröhlich himself in Chapter 8 of this volume. In other words, the metabolic energy reaches definite, localized sites within the macromolecular chain, e.g., DNA or protein. The energy causes a local deformation which travels along the chain as a soliton, causing the two effects mentioned above. Eventually, the soliton energy is transferred into the aqueous phase as a Fröhlich polarization wave. Maintained by the continuing supply of metabolic energy, this wave then plays a major part in affecting the cell organization.

One main problem still remains, viz., the isolation and characterization of the solitons as appearing in biological systems.

3. *Conduction and Biological Structure*

Much of what follows is based on discussions by Pethig and coworkers.[13] (See his contribution to this volume, Chapter 7.)

Proteins have the general structure of polypeptide chains with –C–C–N– as a repeat unit; see Figure 1.

Each peptide unit lies in a plane because it consists of a delocalized system of π electrons associated with π orbitals of the C and O atoms together with the lone electron-pair orbital of the N atom. Such an electron resonance structure is sufficient to produce significant diamagnetic anisotropy in the protein.[14] Some two dozen amino-acid residues make up polypeptides, but only glycine and proline have a first atom in the side chain R which is not a carbon atom. Thus, many features of regularity are present which may cooperate in forming energy bands, particularly in the extended arrays of α helices and β-pleated sheets. In fact, spectra of DNA

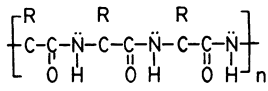

FIGURE 1. General structure of proteins.

and RNA are not basically different from those of their constituent groups because randomization of the constituents washes out all features of the electronic spectrum. Tong[15] suggests that this is due to the presence of geometry-independent lines and allowed energy bands. Altieri and Krizan,[16] using a self-consistent method, show term energies of reasonable magnitude to exist if DNA and related models are characterized as intrinsic semiconductors with a band energy gap of approximately 2 eV. Many proteins have a semiconduction activation energy of about this value.[17]

Whether charge carriers can migrate freely through proteins depends greatly on these energy bands. Coherent electronic motion can occur if broad bands of electronic states are available for the ground states (valency bands) and the excited states (conduction bands). If the bands of the extended states are less than about 2 kT, or if the energy states are localized, charge transport takes place by activated hopping or by tunneling processes.[13]

It appears that all proteins in the natural "pure" state are insulators. The valence band of the extended states is completely occupied by electrons, and the band gap is so large that at physiological temperatures the possibility of promoting electrons across the gap is negligible. Then, calculations[13] indicate that the energy band widths are greater for atomic interactions along the polypeptide backbone chains (–C–C–N–C–C–N–) than those arising from the hydrogen-bonded network interactions which stabilize the protein's tertiary structure. Thus, water molecules in a gramicidine A channel form a linear structure along the channel axis and containing about seven molecules.[96] The majority of semiconduction experiments have involved soluble proteins, probably because these are easily purified and crystallized. However, the structural proteins, those which support the main functional elements, are usually discarded in purification processes, as residues. Experimental evidence has existed for some time for electronic conduction in sea-animal integuments, e.g., crab shells.[12] In order to function, such structural proteins will generally be complexed with other molecules in a manner analogous to dopants and thereby facilitate charge transport.

4. Hydration and Charge Transfer

4.1. Proteins

In their natural state many proteins are bound into hydrophobic lipid matrices. The energetics of charge separation and mobility are governed largely by the dielectric constant of the surrounding medium. Calculations[13] have shown that the internal structures have an effective

high-frequency relative permittivity of about 2.6, which value does not increase much upon hydration. Even in the presence of a fully extended hydrogen-bonded network, conduction along the backbone is more likely, but the particular properties of a protein will depend on its precise conformation, environment, and interactions with other molecules. Thus, it has been shown[19] that the measured conductances and permittivities of bovine serum albumin and lysozyme are determined by the degree of hydration as well as of the NaCl concentration in the salt solution used as the medium. The electrical behavior is governed by protein–water interactions; the conductance is due to a proton site-to-site hopping process.[19]

Considerable differences in the resistivity between that of ordinary proteins and that of electron transfer proteins have been reported[20,21] and also between ferrocytochrome-*c* and ferricytochrome-*c*. This is attributed to the electronic state of the central metal atom.[20] The anhydrous cytochromes exhibit resistivities many orders of magnitude higher; thus anhydrous ferricytochrome c_3 has a room temperature of $4.1 \times 10^{12}\ \Omega$ cm and the ferrocytochrome c_3 a value of 1.6×10^{10}; these data are even more remarkable because the ferri compound is said[24] not to follow an Arrhenius type relationship in its temperature dependence while the ferro compound is reported to yield a negative value for its thermal activation energy. Cytochromes will be discussed in more detail in the next section.

Another interesting example for severe hydration effects in relation to charge transfer is the purple membrane of *Halobacterium halobium*: it acts as a photoelectret, producing at least semipermanent electric charges upon illumination; this activity has been shown to be greatly affected by hydration.[95]

Water in proteins has been determined from hydration isotherms obtained gravimetrically and the more recent quartz crystal resonator techniques.[39] The steady state conductivity of protein samples increases rapidly with water absorbed. The weight percentage, m, relates to the conductivity according to the Spivey equation,[40]

$$\lambda = \lambda_D \exp(\alpha m) \tag{2}$$

where λ_D is the dry-state conductivity and α a constant. For m of about 5 wt %, a change occurs such that $\alpha = 1.3$ for $m < 5$ wt % and $\alpha = 0.9$ for $m > 5$ wt %. This may be explained by assuming that at 5 wt % nearly all the protein primary sorption sites are occupied by water molecules and the population of secondary hydration sites begins. The frequency f_m of maximum loss of dielectric dispersion is also found[13] to increase with increasing hydration following the relationship

$$\tau = \tau_0 \exp(-\beta m) \tag{3}$$

where τ is the relaxation time and β a constant. Again at 5 wt % the value of the constant changes.[13] Although f_m changes, the area under the dielectric loss factors ε'' vs. $\log(f)$ curve does not change with hydration, suggesting[13] that the observed dielectric dispersions are not directly related to the relaxation of water dipoles.

4.2. Cytochromes

Cytochrome-c_3 is specially interesting because this gobular hemoprotein (molecular weight about 12,500) is a defined chemical entity and known[22] to be the electron carrier in the respiratory chain of, *inter alia*, *Desulfovibrio vulgaris*. The electron transfer rate in its reversible electrochemical electrode[98] redox reaction has the enormous value of 0.1 cm sec^{-1}. Its resistivity in the ferro- as well as the ferri-form is shown in Figure 2. The conductivity was ohmic and appears to be electronic in nature[20]; values are said[20] to be well reproducible. The low-temperature branch of ferrocytochrome-*c* exhibits the usual Arrhenius form, but the high-temperature region shows a most unusual behavior, as seen from Figure 2. The resistivities of these two forms are reported[20] to differ by 11 orders of magnitude, with the reduced form having the extraordinarily low value of 57 Ω cm at the transition temperature T_M, a value comparable to that of germanium. It is remarkable that even anhydrous films of tetrahemoprotein cytochrome C_3 exhibit[23] good conductivity at 292 K.

The polypeptide chain within the cytochrome molecule has virtually no conductivity at all, while the complete hemoprotein exhibits a conductivity of the order of 10^{-9}–$10^{-11}(\Omega\,\text{cm})^{-1}$, suggesting a considerable contribution of the heme units to the overall electron transfer. Since the porphyrin subunit is about 13 Å diameter, sufficient delocalization of π electrons should be expected[20] to impart conductivity. It has been

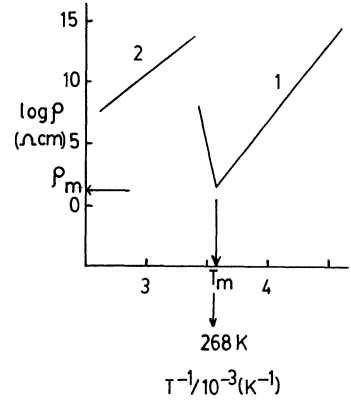

FIGURE 2. Temperature dependence of the resistivity of (1) ferrocytochrome-c_3 and (2) ferricytochrome-c_3.[20]

hypothesized that some part of the hemes in cytochrome-c_3 are exposed to the protein surface so that electrons may tunnel through a peptide residue.[19] The heme–heme interaction in the cytochrome c_3 molecule was inferred from NMR and Mössbauer spectroscopy. Cytochrome c_3 undergoes a reversible oxidoreduction by catalytic action of hydrogenase.[26]

Table 1 collects the electrical conductivity data of several protein solids.[27] The ferri-forms of both cytochromes have a similar resistivity in magnitude at room temperature. On the other hand, very drastic differences appeared in the resistivity of the ferro-form.

The contribution of heme to the electrical conduction of biological materials is clear because the polypeptide chain itself has a resistivity 10^{14} Ω cm. On the other hand, the resistivity of one hemoprotein is about 10^9–10^{11} Ω cm.

Taking into account the size of porphyrin, 1.3 nm in diameter, the distance is small enough to cause the overlapping of π electrons. Electrochemical measurement of the reduction of cytochrome c_3 on a mercury electrode also suggests[31] that some parts of hemes are exposed to the protein surface and/or the electrons tunnel through a peptide residue.

Rapid direct electron transfer between cytochrome c and a gold electrode takes place in the presence of 4,4′-bipyridyl and 1,2-bis(4-pyridyl)ethylene, which form an adsorbed layer on the electrode surface thus facilitating the electron-transfer reaction.[32] There are some strik-

TABLE 1. Electrical Conductivity of Several Protein Solids[a]

Substance	Temperature range (°C)	E in $E/2kT$ (eV)	Resistivity (Ω cm) at temperature indicated (°C)
Cytochrome c oxidized[b]	—	2.7	6.1×10^{16}(30)
Cytochrome c oxidized[b]	50 to 85	1.2	3.1×10^{11}(30)
Cytochrome c reduced[b]	10 to 60	1.2	3.1×10^{9}(30)
Cytochrome c_3 oxidized[b]	20 to 70	3.3	2.3×10^{12}(30)
Cytochrome c_3 reduced[b]	−40 to −5	7.7(?)	5.7×10^{1}(−5)
Ferricytochrome c	—	1.2	6.5×10^{10}(55)
Ferrocytochrome c	—	1.2	6.5×10^{8}(55)
Lysozyme	—	—	$>10^{14}$(30)
Myoglobin	—	0.31	3.6×10^{10}(30)
Ribonuclease	—	—	$>5 \times 10^{14}$(30)
Trypsin	—	—	$>10^{14}$(30)

[a] After Nakahara, Kimura, and Inokuchi.[30]
[b] All samples in the condensed form measured in surface cells with gold electrodes.

ing analogies between the reaction of cytochrome *c* at this electrode and its reaction with cytochrome oxidase with respect to the effect of chemical modification of the cytochrome *c* lysine residues and the effect of poly-L-lysine on the electrode reaction. The results suggest that cytochrome *c* binds to the 4,4′-bipyridyl-modified gold electrode surface prior to electron transfer in a manner similar to its interaction with the oxidase.

The electron transfer reaction between cytochrome *c* and cytochrome oxidase proceeds via a protein complex in which the ε-amino groups of the cytochrome *c* lysine residues are believed[33] to play an imporant role. Chemical modification of these lysines is well known[34] to affect the cytochrome *c*–oxidase electron-transfer reaction. The importance of lysine residues in the protein–protein interaction is further illustrated by the effect[35] of poly-L-lysine, a competitive inhibitor of the cytochrome *c*–oxidase reaction which acts by binding to the oxidase.[36] Both *N*-acetimidyllysyl and *N*-guanidinyllysyl horse heart cytochrome *c* are electroactive at a gold electrode in the presence of 4,4′-bipyridyl or 1,2-bis(4-pyridyl)ethylene, giving rise to quasireversible diffusion-controlled dc and ac voltammograms indistinguishable from those (Figure 3) of the native protein,[37] with a half-wave potential, $E_{1/2}r = 0.25$ V vs. NHE. Similarly[36] both are enzymatically active[37] in the cytochrome–oxidase system. The enzymatically inactive *N*-trifluoroacetyl and *N*-maleyl derivatives are both electroinactive.[32]

Poly-L-lysine inhibits the electrode reaction of native horse heart cytochrome *c*, as shown by dc voltammetry (Figure 3), again analogous to its inhibiting effect on the cytochrome *c*–oxidase reaction. The effect on the ac cyclic voltammetry peak current, i_p(ac), is more marked. The varation with poly-L-lysine concentration is consistent with adsorption of poly-L-lysine onto the electrode surface, decreasing the effective free electrode area.

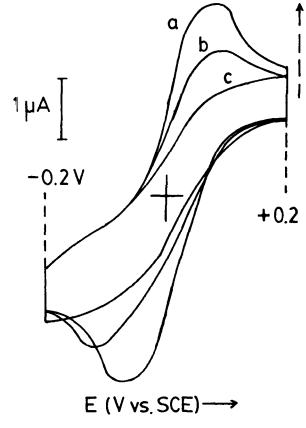

FIGURE 3. dc Voltammogram of horse heart ferricytochrome *c*, 0.4 nM in 0.1 *M* $NaClO_4$, 0.02 *M* phosphate buffer, pH 7, saturated solution of 1,2-bis(4-pyridyl)ethylene in the potential range from +0.2 to −0.2 V vs. SCE with poly-L-lysine: (a) 0 mg ml^{-1}, (b) 1 mg ml^{-1}, (c) 1.5 mg ml^{-1}, dc potential scan rate, 100 mV s^{-1}. After Eddowes *et al.*[32]

The electron-transfer reaction of the native protein has been studied[32] by ac impedance measurements. The heterogeneous electron-transfer rate was found to be as fast as that determined[38] for the ferri/ferrocyanide couple ($0.3–3.0 \times 10^{-4}$ msec^{-1}).

Double-potential-step chronocoulometry, rotating disk, and ac rotating ring disk indicate[32] that the native cytochrome binds to the 4,4′-bipyridyl-modified gold electrode surface prior to electron transfer. Thus it appears that 4,4′-bipyridyl acts by forming a suitable surface at the electrode–solution interface to which the cytochrome can bind. The binding of the protein to the electrode may therefore be similar in kind to that observed in the cytochrome *c*–oxidase reaction, involving the ε-amino groups of lysine residues in the vicinity of the cytochrome *c* heme crevice. The similarities to the cytochrome *c*–oxidase reaction may result from a preferred orientation of the cytochrome when bound at the electrode surface or to cytochrome oxidase such that the exposed heme edge is adjacent to the electrode surface or the cytochrome oxidase, thereby enabling rapid electron transfer to occur.

Hydrated melanin exhibits a reversible switching behavior[41]; upon application of an electric field above a certain threshold value this substance changes reversibly and reproducibly from a state of high resistivity to one of low resistivity. The value of the critical switching field depends on the source, nature, and mode of preparation of the pigment; only hydrated specimens show the effect. Cope proposes[42] that the high-resistivity form may be considered as an emulsion of electrons in an electron-poor matrix; copious electron injection by virtue of the switching field then inverts the emulsion yielding electron-poor islands in an electron-rich matrix. This should affect the redox potential, and this has indeed been observed.[43] The model is further supported by the negative photoconductivity reported[44] for melanin: this effect, viz., a drop in conductivity upon illumination, is rare in organic materials; it was first reported[45] in 1965 for chlorpromazine. It is thought to be associated with the presence of centers capable of existing in two different valency, or electronic, states.

5. *Charge Transfer and Adsorption. Surface Effects*

5.1. *Introduction*

Surface effects in aqueous media always involve electric double layers characterized by more or less—depending on location—immobilized charges and by a firmly held layer of structured water exhibiting a permittivity considerably below that of its bulk value. There is an extensive literature on this subject[46]; see also the discussion in Pethig's article in this volume.

An adsorbate behaves as an electron donor if the energy of its lowest unoccupied level evaluated in its adsorbed state is situated at or above the Fermi level; the adsorbate, conversely, acts as an acceptor if the highest occupied level of the system substrate plus adsorbate is at or below the Fermi level. Surface complexes are often dimerlike with their energy states well localized within the energy gap, and resulting in a depletion of states in the vicinity of the Fermi levels; any excess charge tends to accumulate on the surface complex,[47] raising the carrier hopping probability. Some surface complexes consist of an adatom plus its nearest neighbor on the surface, again a dimerlike structure.[49] The first stage of an adsorption process involves a charge transfer, donor–acceptor, interaction between the adsorbent and the adsorbate,[48] resulting in an increased effective ionization potential, or electron affinity, of the adsorbed donor or acceptor, and a rise in the local energy levels of the substrate. This stage is followed[48] by electron, or hole, transfer into stabilized vacant energy levels of the adsorbate. Surface excitons and polaritons are likely to arise on surfaces,[50] their binding energy is reported to be of the order of 0.1 eV.[51]

The topic of surface CTC is basic to the theories of catalysis and is usually treated in that context.[52]

Surface CTC of chlorpromazine (CPZ) has been shown to be involved[53] in hypothermia and sedation of mice; *in vitro* surface adducts between chlorpromazine and collagen have been demonstrated, as well as with lignocain and with phenytoin.[54] The existence of such CPZ surface adducts has been directly confirmed by electroreflectance measurements.[94]

If an acceptor is incorporated into a cationic micelle, then the rate of reactions involving hydrated electrons is increased by a factor of about 60 times that prevailing at outside negatively charged anionic micelles.[55]

Since counterions interact with, and are attracted by the, say, hydrophillic heads of the colloidal particles, micellar assemblages of such particles should be catalytically active. This, in general, has been found to be the case,[56] and should be a rewarding field for the study of micellar CTC as well as electrocatalysis. Formation of micellar CTC is also reported to result in considerably shifted ion exchange equilibria.[57]

5.2. Membranes

5.2.1. The Transmembrane Conductance

This topic has been extensively studied and many reviews and treatises dealing with it are available.[59] In this volume, it is discussed in contributions by Pethig (Chapter 7) and by Habib and Bockris (Chapter 3).

5.2.2. *Membrane Surfaces*

The surface of a biological membrane exhibits electron donating as well as electron accepting centers, though the surface as a whole is not, in general, electrically neutral, exhibiting the well-known membrane potential. We now wish to suggest that interactions with an external molecular electron acceptor or donor produces a conformational change of the complementary region on the membrane surface, perhaps initiated by Poole–Frenkel, thermally assisted field emission of carriers. The resulting transient voltage pulse may then be reversed by an opposing membrane potential which transfers the carrier back and thus restores the initial steric configuration on the membrane surface.

This hypothesis requires a finite, nonzero, conductance in the surface plane of the membrane because the electron, or hole, initially transferred forms a movable excited surface state, i.e., a polariton.[60] This may well give rise to a soliton, given a supply of energy at the proper locale.

No direct contact between donor and acceptor is envisaged; charge transfer is expected to occur once the donor approaches the acceptor, or vice versa, within tunneling distance, say 20 Å. The donor–acceptor charge transfer thus yields a pattern of surface charges and of sterically oriented proteins. Moreover, the highly nonuniform, inhomogeneous local electrical fields in the vicinity of the charges produce large, local, mechanical forces which are liable to lead to the opening or closing of ionically conducting transmembrane channels. One could then still further hypothesize that it is this surface charge pattern which is involved in the recognition of "self" and the disturbance of which plays a part in uncontrolled growth, viz., carcinogenesis. It may be significant in this context that of 289 carcinogens listed in the latest, Tenth, Collective Index of Chemical Abstracts, all but one—benzene—are electron acceptors or have pronounced electron accepting sites in their molecular structure.

The transmembrane, transcellular transient mentioned has indeed been observed,[61] e.g., in the hyphae of the water mould *Achyla bisexualis* as well as in other fungi.

The electron involved in the donor–acceptor interaction need not originate from the actual surface layer but may well come from the interior of the bilipid membrane, for example, from intramembrane proteins. The electron escape depth in organic materials is comparable to intermolecular distances.[62]

Alternatively, it may arise from an "activating ion" which enters into a surface complex with an adsorbed enzyme; it has been shown[97] that the stability of such complexes is greatly affected by the electric double layer structure of the membrane surface. As the electrostatic screening is

increased, the complex tends to become less stable and the enzyme activity drops.[97]

It is notable that all condensed polycyclic aromatics with a planar molecular structure are carcinogenic—and all have the same and relatively high value of polarization energy, viz., 1.7 eV and thus are capable of acting as efficient electron donors as well as acceptors.[63] Whether a given protein is capable of insertion into the membrane is chiefly determined by hydrophobic forces, but the final orientation of the protein depends on electrostatic interactions such as the forces arising from surface charge asymmetries—which are a feature of the hypothesis proposed above—and from local electrochemical events.[64] Any changes in the electrical surface pattern are likely to propagate to adjacent cell membranes by a process akin to epitaxy, as has been demonstrated for inorganic layers by Distler[65] for distances of up to about 230 Å. A photoelectret process is said[65] to be involved.

6. *Proton Transfer Complexes*

The electron transfer involved in the formation of a charge transfer complex may in suitable systems be coupled to a proton transfer,[69] resulting in proton transfer complexes. This holds, especially for surface reactions, e.g., at electrodes, and a fortiori in nonaqueous media, because the proton affinity of water is so very high, viz., about 8.9 eV.[70]

The proton affinity A_p is defined as the enthalpy of the reaction

$$X + H^+ \rightarrow XH^+ \tag{4}$$

In the gas phase, where there are energy contributions from solvation or polarization, A_p may be measured by means of ion cyclotron spectroscopy.[80] Proton affinities are often given in kJ/mole; 1 kJ/mole equals 0.0104 eV; 1 kcal/mole equals 0.0436 eV. Exchange repulsion energies do not enter into proton affinities.[72] The "relative protonicity" of solvents, akin to the donicity concept for electrons, is discussed by Bayless *et al.*[74]

Hydroxy-dinitro pyridines, e.g., act as electron acceptors and/or as proton donors to, e.g., napthalene derivatives.[71]

The field of protonic CTC has been opened up by Matsunaga and his school,[69] and has been reviewed by Morokuma.[72] This author points to the close linkage existing in such adducts between proton and concurrent electron transfer; one does not occur without the other. The proton complex is said[72] to involve an electron transfer from, e.g., an amine to the

proton. There is evidence that the electron transfer in some cases may be catalyzed by the presence of a proton and thus enhanced above its thermally controlled rate.[75]

Proton transfer may, sometimes, be really nothing more than a case of conventional hydrogen bonding, but in many cases the simultaneous transfer of an electron and a proton produces a new different type of adduct; then the difference between mere hydrogen bonding; and complex formation may indeed be dramatic, as pointed out by Arnett and Mitchell.[76] There is no correlation between the heats of protonation and of hydrogen bonding.[76]

Ion pair and hydrogen bonded complexes[77] are beyond the scope of this paper. Here, only a few representative examples of protonic complexes can be discussed.

The 1:1 and 1:2 complexes between oxalic acid and ε amino acids also involve a proton transfer from the carboxyl group of the oxalix acid to the carboxyl ion of the amino acid; there is a similar interaction in the 1:1 complex between malonic acid and glycine.[78] These adducts may well involve a great deal of proton delocalization resulting in the formation of proton energy bands rather than energy levels. Proton tunneling is a well-established fact,[79] and even halogen ions have been shown to be able to tunnel between two suitable energy levels. Ionic charge transfer complexes between benzyl radicals and halide ions have also been reported.[80]

The average time of residence of the proton at a temporary equilibrium site is of the order of one vibrational period of the OH group and of the average lifetime of H_3O^+, viz., about 2.5×10^{-13} s; since the dielectric relaxation time of water is about 10^{-11} s, it appears that dielectric relaxation and frictional processes are not involved in proton transfer.[81] However, in what appears to be protonic charge transfer complexes between, e.g., aliphatic amines and phenols, a considerable increase in viscosity is reported.[82]

Intramolecular proton transfer in electronically excited molecules has been reviewed[83]; in salicyclic acid esters it is said to lead to deexcitation of the excited electron[89]; the free energy change involved in that proton transfer is reported as about 0.13–0.22 eV.[84]

The effect has been shown to accompany at least some cases of electronic excitation[85] in organic molecules, such as aromatic compounds. These effects have hardly been touched upon as far as they affect biologically important transitions, as they are bound to do. This is an important and rather new field which should be explored.

Electron excitation causes only a very small change in the enthalpy of hydrogen bonding; a value of about 0.0044 eV is reported.[86]

Bacteriorhodopsin is a pigment found as a single protein component of the purple membrane of *Halobacterium halobium* and similar extreme

halophiles.[87] The purple membrane converts light energy by translocating protons across the membrane to generate an electrochemical potential. A review[88] of this subject has been published. The naturally recurring crystalline structure has been determined with electron microscopy at 7 Å resolution. The protein field is folded into seven α-helical chains, all of which span the hydrophobic core of the membrane.[88] The chromophore has been located in a lysine residue in a sequence of the second helical chain from the amino terminal group[89]; cf. discussions of methyl-glyoxal-protein complexes. The chromophere's double bond chain makes an angle of about 20° to the plane of the membrane,[90] with its ionone ring close to the center of the membrane.[91]

The role of bacterio-rhodopsin in the proton pumping activity of halobacteria has been further investigated by Bagyinka *et al.*,[92] while the light transduction via the pigmented bilipid membranes, of the purple membrane of *H. halobium*, has been studied by Ti Tien.[93]

Acknowledgments

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SPECIFICITY OF THE FRÖHLICH INTERACTION OF ERYTHROCYTES

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The aggregation of mixtures of red blood cells from different species of mammals provides evidence for the specificity of the Fröhlich quantum mechanical interaction between living cell membranes.

We have shown [1,2] that there is an ultra-long-range ($\sim\mu$m) attraction between human erythrocytes and that this interaction is in accord with Fröhlich's theory [3,4] of quantum-mechanical coherent vibrations in the membranes, organelles and macromolecules of living cells.

A Fröhlich interaction should be specific; that is attractive, repulsive or null, depending on the relative frequencies or phases of the vibrations in the interacting entities and on the dispersion, with frequency, of the dielectric constant of the surroundings [4]. We have adduced [2,5] the experiments of Sewchand and Canham [6] as evidence of specificity in the erythrocyte interaction. Mixtures of erythrocytes from different species will aggregate into rouleaux and Sewchand and Canham [6] showed that the ordering of the cells in the rouleaux is not random. Cells from the same species adhere to each other significantly more frequently than expected from the results of passive aggregation experiments and from theory [7]. Sewchand and Canham studied ten pairs of different species and the only exception was the combination of erythrocytes from rats and mice. Rats and mice are of the same biological family and their exception suggests that we are dealing with a living rather than a physicochemical phenomenon [2].

The flaw in the argument for specificity is that colliding cells may come apart again. When human erythrocytes, executing brownian movement in their own plasma, touch they adhere, and, on the time scale of the slow movement, quickly slide together face to face to form rouleaux [8]. If, in the interspecies experiments [6], this occurred with like cells but, on the contrary, unlike cells had a small probability of coming apart after contact, the results of Sewchand and Canham [6] would have a trivial explanation. Accordingly we have used our aggregation technique [2,5] on mixtures of cells. Plasma cannot be used as the suspending medium because of immunological incompatibility. The cells were suspended in phosphate-buffered-saline [5] to which had been added a rouleau-forming agent, polyvinylpyrrolidone MW 360,000 [9]. The mixture was placed in a sealed haemacytometer chamber and the aggregation process was recorded by time lapse cinémicroscopy [1,2,5,9].

We carried out four experiments with dog–human mixtures and ten experiments with cat–human mixtures, both of which are known [6] to give non-random rouleaux. In each experiment about 200 cells were in view, of which about 40 pairs would collide and 10–15 of these pairs would come from the different species. Careful scrutiny of the fourteen films revealed no case of cells, once in contact, coming apart again. The non-randomness of the arrangement of the cells in the resulting rouleaux was obvious in the cat–human experiments because of the marked difference in the diameter of the cells from these two species [6].

We also analysed the experiments by our previously described method [2,5] which measures deviations from brownian movement. Deviations are expressed as multiples of a rate constant (interaction coefficient, Ξ [2,5]). $\Xi = 1$ if the motion is brownian. $\Xi > 1$ im-

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Table 1
Interaction coefficients Ξ, for explanation see text.

Cat–human mixture	Cat only or human only
4.19	6.59
3.64	5.01
4.03	5.26
3.22	7.92
6.32	6.13
	4.20
	5.20
mean 4.28 ± 1.1 SD	mean 5.76 ± 1.1 SD

plies an attraction. With a solution of polyvinylpyrrolidone rather than plasma [1,2,5] as the suspending solution, cells tend to interact variably with the surface of the haemacytometer and so can lead to inconsistent values for the drag coefficients [2,5]. This led us to reject all the dog–human experiments and five of the cat–human experiments for quantitative assessment. The interaction coefficients from the remaining five experiments were compared with those from seven experiments, four of which employed cat cells only and three of which employed human cells only (table 1). There is a difference between the two series which is highly significant by the "t" test for the differences between averages [10]. The figures suggest a reduced but positive interaction between the two species, but the variability in the figures is too great to draw firmer conclusions. These experiments will not be repeated by the current method because it has now been superseded by quasi-elastic-light scattering which is much quicker and less prone to artefact. Our published results [2,5] have been confirmed qualitatively and quantitatively by this method (O.G. Fritz Jr., private communication).

Although quantitatively these experiments were a disappointment because of technical difficulties, the observation that cells never come apart after initial contact makes Sewchand and Canham's experiments [6] on the non-randomness of the order in mixed rouleaux into firm evidence for the specificity of the interaction between erythrocytes from different species.

Fröhlich [4] and Webb [11] have reviewed the evidence for the presence of vibrations of the predicted frequency [4] in micro organisms and in cells in tissue culture. Pohl [12,13] has detected an attraction by dividing cells for inorganic particles of high dielectric constant and we have shown that a *specific* interaction exists between erythrocytes which is dependent on a membrane potential and a supply of metabolic energy [1,2,5]. Over a limited range of organisms all the major predictions of Fröhlich's theory of coherent excitations in biological systems are confirmed.

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Statistical-Mechanical Theory of Irreversible Processes. I.

General Theory and Simple Applications to Magnetic and Conduction Problems

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A general type of fluctuation-dissipation theorem is discussed to show that the physical quantities such as complex susceptibility of magnetic or electric polarization and complex conductivity for electric conduction are rigorously expressed in terms of time-fluctuation of dynamical variables associated with such irreversible processes. This is a generalization of statistical mechanics which affords exact formulation as the basis of calculation of such irreversible quantities from atomistic theory. The general formalism of this statistical-mechanical theory is examined in detail. The response, relaxation, and correlation functions are defined in quantummechanical way and their relations are investigated. The formalism is illustrated by simple examples of magnetic and conduction problems. Certain sum rules are discussed for these examples. Finally it is pointed out that this theory may be looked as a generalization of the Einstein relation.

§ 1. Introduction

The principal purpose of the present paper is to develop a general scheme for the calculation of kinetic coefficient, or admittance for external forces such as electric or magnetic susceptibility for alternating field, electric conductivity, heat conductivity and so on. The most common method traditionally employed to calculate such non-equilibrium quantities is to set up the so-called kinetic equation or transport equation for the molecular distribution functions. This will be solved for stationary or periodic conditions. It is however to be remineded that the kinetic equation is itself an approximation and cannot be derived without a certain condition which is rather strict and is not often satisfied.[1] For weakly interacting particles, it may be justified if the nature of interaction is appropriate. But certainly it is not always true, and we may not be able even to write down any equation of that sort.

On the other hand, the recent development of the thermodynamic of irreversible processes[2] indicates the possibility of constructing a general statistical-mechanical theory at least for those irreversible processes which are not far apart from thermal equilibrium. In fact, there have been a number of attempts in this direction[3]. A remarkable progress has been made by Kirkwood[4] who showed that the friction constant, for instance, of a Brownian particle is determined by the correlation of forces acting on the particle. This may be looked as a particular case of the so-called dissipation-fluctuation theorem. As was discussed from a general point of view by Callen, Welton,[5] Takahasi[6] and others, the general admittance for an external disturbance is related to the expectation of the square of Fourier component of certain physical quantity which fluctuates in time in thermal equilibrium. Thus it is shown that, as far as linear responses are concerned, the admittance is reduced to the calculation of time-fluctuations in equilibrium.

This idea was used by the present author as the basis of the theory of magnetic resonance problem. In a previous paper[7] the author developed a quantum-mechanical theory of this and has shown a certain method of practical application. A straightforward application of this theory was made by Nakano[8] for the conduction problem. He showed that the simplest approximation of the fundamental expression of conductivity gives the well-known Grüneisen's formula.

A generalized form of Nyquist's theorem can be easily derived by application of perturbation theory either classical or quantum-

mechanical. The complete conductivity tensor for a given frequency of applied electric field can be rigorously expressed in terms of electric current components fluctuating spontaneously in equilibrium state[9]. We emphasize here that this expression is quite rigorous though admittedly somewhat abstract. It is however of great value, because the traditional theory of conductivity has not been aware of existence of such rigorous expression (although the Heisenberg-Kramers dispersion formula is an equivalent). The abstract nature of the formula may be compared to that of general theory of statistical mechanics. The statistical mechanics can give a rigorous expression for the equation of state in terms of partition function, but its calculation is not necessarily easy.

This theory of electric conduction has been actively discussed in last two years in Japan since the appearance of our paper on magnetic resonance[7] and its application to conduction problem by Nakano. Feynman also discussed this when he visited Japan in the summer of 1955. Also it has been discussed by Lax, Luttinger, Kohn and possibly by some others in the United States. Recently Mori[20] has published a paper on the same problem.

In the present paper, which will be the first part of a series of papers devoted to the development of statistical mechanics of irreversible processes along this line and to its applications, we shall examine in detail the implications of this new method.

In § 2, we shall treat the response of a system to an external force by simple perturbation method. In order to make the idea clearer we first discuss the classical case. Quantum-mechanical formulation is quite in parallel with this classical theory. However, we notice a difficult problem of quantum-mechanical observation which might disturb the natural development of the motion. It will be pointed out that the logical basis of the whole theory is closely related to the ergodic properties of the system and the physical quantities we are concerned with. This is discussed in § 3 in connection with the definition of relaxation functions which describe the relaxation of the system after removal of external disturbance.

In § 4, the correlation functions are introduced which describe the time correlation of two quantities at different time points. The relation between the correlation function and the relaxation function is derived. This is somewhat complicated in quantum-mechanical case.

§ 5 is devoted to the application of general argument to simple examples of magnetic and conduction phenomena. This is mostly to illustrate the general idea. The symmetry of the relaxation and response functions is discussed in § 6. The Onsager's reciprocity is seen most clearly by this formalism. The complete form of fluctuation-dissipation theorem will be given in § 7 taking the example of conductivity tensor.

As an interesting application of the theory, some general rules for the admittance is discussed in § 8. The most general sum rule is mentioned with regard to the symmetric part of conductivity tensor. The integration of this over the whole range of frequency must be constant irrespective of the presence of interactions or magnetic field and at any temperature. Another type of sum rule is derived for the antisymmetric part of conductivity. An interesting sum rule for the magnetic problem is that the integrated absorption intensity of circular wave by magnetic media is directly related to the gyromagnetic ratio of magnetic particles.

In § 9, it will be pointed out that the theory may be looked as a generalization of Einstein relation which relates the mobility and the diffusion constant. The general nature of Einstein relation is thus most clearly revealed.

In the present paper we confined ourselves to such type of external disturbance which can be expressed definitely as an additional term in the Hamiltonian of the system. Thus the heat flow, or mass flow phenomena under the gradient of temperature or chemical potential is logically out of the frame of this treatment. This problem has been treated recently by Mori.[10] We shall, however, discuss this in a forthcoming paper from a somewhat different point of view.

We also have applied the present theory to the galvanomagnetic effect at high magnetic field[11]. This is an example for which the traditional transport equation has to be modified. Our theory will be reported in another paper.

§ 2. Response Function and Admittance of an Isolated System

Let us consider an isolated system, the Hamiltonian of which is denoted by $\mathscr{H}$. The dynamical motion of the system determined by $\mathscr{H}$ is called the "*natural motion*" of the system. We suppose that an external force $F(t)$ is applied to the system, the effect of which is represented by the perturbation energy,

$$\mathscr{H}'(t) = -AF(t) . \tag{2.1}$$

The motion of the system is perturbed by this force, but the perturbation is small if the force is weak. We confine ourselves to weak perturbation and ask for the response of the system in the linear approximation. The response is observed through the change $\Delta B(t)$ of a physical quantity B. The problem is now to express $\Delta B(t)$ in terms of the natural motion of the system.

First, we shall treat this in classical mechanics. We conceive a statistical ensemble which is represented by a distribution function $f(p, q)$ in the phase space. The *natural motion* is described by the equation of motion,

$$\frac{\partial f}{\partial t} = -\sum\left(\frac{\partial f}{\partial q}\frac{\partial \mathscr{H}}{\partial p} - \frac{\partial f}{\partial p}\frac{\partial \mathscr{H}}{\partial q}\right) = (\mathscr{H}, f), \tag{2.2}$$

where p and q mean the set of the canonical moments and coordinates and the bracket means the Poisson bracket,

$$(A, B) = \sum\left(\frac{\partial A}{\partial q}\frac{\partial B}{\partial p} - \frac{\partial A}{\partial p}\frac{\partial B}{\partial q}\right). \tag{2.3}$$

We assume that the distribution function is given by f at $t=-\infty$, that is, at the infinite past. It is assumed to be in equilibrium, that is, $(\mathscr{H} f)=0$. The perturbation (2.1) is inserted adiabatically at $t=-\infty(F(-\infty)=0)$. The distribution function $f'(t)$ obeys the equation

$$\frac{\partial f'(t)}{\partial t} = (\mathscr{H}, f') + (\mathscr{H}'(t), f) . \tag{2.4}$$

Since we take the linear approximation, we put

$$f'(t) = f + \Delta f$$

and replace (2.4) by

$$\frac{\partial \Delta f}{\partial t} = (\mathscr{H}, \Delta f) - F(t)(A, f) . \tag{2.5}$$

The solution is easily found to be

$$\Delta f(t) = -\int_{-\infty}^{t} e^{i(t-t')L}(A, f)\, F(t')\, dt' \tag{2.6}$$

where the operator L is defined by the operation[4]

$$iLg \equiv (\mathscr{H}, g) . \tag{2.7}$$

Therefore, the change $\Delta B(t)$ of a dynamical quantity B is statistically given by

$$\begin{aligned}\Delta B(t) &= \int \Delta f(t) \cdot B(p, q)\, d\Gamma \\ &= -\int d\Gamma \int_{-\infty}^{t} \left\{e^{i(t-t')L}(A, f)\right\} \cdot F(t') B\, dt' \\ &= -\int d\Gamma \int_{-\infty}^{t} (A, f) B(t-t') F(t')\, dt' ,\end{aligned} \tag{2.8}$$

$d\Gamma$ being the volume element of the phase space. The last expression is obtained from the second remembering that the transformation $\exp(iLt)$ is the natural motion which conserves the measure of the phase space and that $B(t)$ means the dynamical motion of the phase function $B(p, q)$ which follows the equation

$$\dot{B}(p, q) = (B, \mathscr{H}) .$$

This corresponds to the Heisenberg equation of motion in quantum-mechanics. Eq. (2.8) means that the response $\Delta B(t)$ is a superposition of the effects of pulses $F(t')\, dt'$, $-\infty < t' < t$. The response for an unit pulse will be called the *response function* or the *after-effect function* $\phi_{AB}(t)$. From Eq. (2.8) we find at once that it is given by

$$\phi_{BA}(t) = -\int d\Gamma (A, f)\, B(t) \tag{2.9}$$

which describes the response ΔB at the time t after the pulse. The response $\Delta B(t)$, (2.8), is written as

$$\Delta B(t) = \int_{-\infty}^{t} \phi_{BA}(t-t')\, F(t')\, dt' . \tag{2.10}$$

The above consideration applies even when the initial distribution is very sharp, as far the perturbation is weak enough. But our aim is rather to apply this for a macroscopic system, in which case the statistical average itself acquires a realistic meaning. If A and B are both macroscopic quantities, we may think the ensemble average $\Delta B(t)$ is what we actually observe, because a macroscopic system can be conceived to consist of smaller systems so that observed value of ΔB is a sum of a number of components and has only

a very small relative fluctuation.

Let us now turn to quantum mechanics. The distribution function in phase space is now replaced by the density matrix ρ. The initial ensemble which represents statistically the initial state of the system is specified by the density matrix ρ satisfying $[\mathscr{H}, \rho]=0$, whereas the motion of the ensemble under the perturbation (2.1) is represented by $\rho'(t)$, which obeys the equation,

$$\frac{d}{dt}\rho'(t)=\frac{1}{i\hbar}[\mathscr{H}+\mathscr{H}'(t), \rho'(t)] . \quad (2.11)$$

With the initial condition

$$\rho'(-\infty)=\rho ,$$

we expand $\rho'(t)$ as

$$\rho'(t)=\rho+\Delta\rho(t) .$$

The same procedure as we used for (2.5) leads us at once to

$$\Delta\rho(t)=-\frac{1}{i\hbar}\int_{-\infty}^{t} \exp(-i(t-t')\mathscr{H}/\hbar)[A, \rho] \times\exp(i(t-t')\mathscr{H}/\hbar) F(t')\,dt' . \quad (2.12)$$

Incidentally, we note that it is sometimes very convenient to introduce the operator $a^{\times}$ operating on another operator b by following definition, i.e.,

$$a^{\times}b=[a, b] , \quad (2.13)$$

for which one find the rule,

$$e^{a^{\times}}b=e^{a}be^{-a} . \quad (2.14)$$

With this notation, Eq. (2.11) and (2.12) can be written as

$$\dot{\rho}'(t)=\frac{1}{i\hbar}(\mathscr{H}^{\times}+\mathscr{H}'(t)^{\times})\rho'(t) \quad (2.15)$$

$$\Delta\rho(t)=-\frac{1}{i\hbar}\int_{-\infty}^{t} \exp(-i(t-t')\mathscr{H}^{\times}/\hbar)\times \times[A, \rho]F(t')dt' \quad (2.16)$$

This will make clear the similarity of Eqs. (2.6) and (2.12).

The response $\Delta B(t)$ of the quantity B is statistically

$$\begin{aligned}\Delta B(t)&=\mathrm{Tr}\,\Delta\rho(t)B\\ &=-\frac{1}{i\hbar}\mathrm{Tr}\int_{-\infty}^{t}\exp(-i(t-t')\mathscr{H}/\hbar)\\ &\quad\times[A, \rho]\exp(i(t-t')\mathscr{H}/\hbar)BF(t')\,dt'\\ &=-\frac{1}{i\hbar}\mathrm{Tr}\int_{-\infty}^{t}[A, \rho]B(t-t')F(t')\,dt'\end{aligned} \quad (2.17)$$

where $B(t)$ is the Heisenberg representation of B following the equation

$$\dot{B}(t)=\frac{1}{i\hbar}[B(t), \mathscr{H}], \quad B(0)=B . \quad (2.18)$$

Eq. (2.17) corresponds to Eq. (2.8)

The response function or the after-effect function is now

$$\phi_{BA}(t)=-\frac{1}{i\hbar}\mathrm{Tr}\,[A, \rho]\,B(t) \quad (2.19)$$

corresponding to Eq. (2.9). We naturally have Eq. (2.10) for (2.17) with the use of response function (2.19). If the force is periodic, i.e.,

$$F(t)=F_0\cos\omega t ,$$

the response $\Delta B(t)$ will be conveniently written as

$$\Delta B(t)=\mathrm{Re}\,\chi_{BA}(\omega)\,F_0\,e^{i\omega t} \quad (2.20)$$

with the complex admittance $\chi_{BA}(\omega)$, for which we have from Eq. (2.10)

$$\chi_{BA}(\omega)=\int_0^{\infty}\phi_{BA}(t)\,e^{-i\omega t}\,dt , \quad (2.21)$$

or more exactly

$$\chi_{BA}(\omega)=\lim_{\varepsilon\to 0+}\int_0^{\infty}\phi_{BA}(t)\,e^{-i\omega t-\varepsilon t}\,dt .$$

We summarize the above by the theorem:

Theorem 1. *The linear response of a physical quantity B to an external force $F(t)$ is represented by Eq. (2.10) as the superposition of after-effects. The after-effect function or the response function is given by*

$$\phi_{BA}(t)=\int d\Gamma(f, A)\,B(t) \quad \text{(classical)} \quad (2.22a)$$

$$=\frac{1}{i\hbar}\mathrm{Tr}\,[\rho, A]\,B(t) \quad \text{(quantal)} . \quad (2.22b)$$

The admittance $\chi_{BA}(\omega)$ is given by

$$\chi_{BA}(\omega)=\lim_{\varepsilon\to 0+}\int_0^{\infty}e^{-i\omega t-\varepsilon t}\,dt\int d\Gamma(f, A)\,B(t)$$

(classical) (2.23a)

$$=\lim_{\varepsilon\to 0+}\int_0^{\infty}e^{-i\omega t-\varepsilon t}\,dt\,\frac{1}{i\hbar}\mathrm{Tr}\,[\rho, A]\,B(t) .$$

(quantal) (2.23b)

Eqs. (2.22) may be written as

$$\phi_{BA}(t)=\int d\Gamma f(A, B(t)) \quad \text{(classical)} \quad (2.24a)$$

$$=\frac{1}{i\hbar}\mathrm{Tr}\,\rho[A, B(t)] \quad \text{(quantal)} \quad (2.24b)$$

which are sometimes more convenient. One must however be careful about the boundary conditions of f or ρ, which may, however,

be replaced by a suitable potential under certain circumstances.

As we mentioned before, there is no difficulty in classical cases to interprete the response thus obtained as the average which approximates closely the actual observation for large systems. In quantum-mechanical cases, however, we must note that there exists a certain difficulty. ΔB as given by Eq. (2.17) is the quantum-mechanical expectation of the first order change of B at the time t. This means that we prepare at $t=-\infty$ an ensemble of systems which corresponds to the statistical operator ρ. At a certain time t, we pick up a number of systems and observe B. The average of the observed values will be $\Delta B(t)$. We shall not follow these systems after this observation. At a later time point t', we choose different systems which have not been observed before. This is the way to follow the time change $\Delta B(t)$ to avoid the quantum-mechanical disturbance of observation process.

For the most of practical applications, we are rather interested in a continuous observation of a particular system, which is however macroscopic. The use of ensemble average for the actual observation will be justified by the same argument we made for the classical case. But the quantum-mechanical disturbance of observation is another thing. We may, however, expect for most of macroscopic quantities which we observe this disturbance can be neglected. This is a condition for macroscopic observations. Unfortunately, exact conditions for the existence of such quantities are not known.

Thus we shall rather introduce without proof the assumption that the identification of the statistically calculated $\Delta B(t)$ with the actully observed time variation is legitimate at least for those macroscopic quantities we shall be concerned with. We should however make reservation that this assumption may not be necessary for the identification of the admittance calculated by Eq. (2.23b) with what will be actually observed, because the observation of admittance, for example, the refractivity and absorption in optical problems, is not the observation of the response function itself. As a matter of fact, Eq. (2.23b) is equivalent to the well known dispersion formula of Heisenberg and Kramers[12] plus the absorption calculated by time-dependent perturbation method.[7] The dispersion formula represents the oscillation of expectation value of dipole moments if the formula is applied to optical problems. This is not directly observed. We observe the refraction by measuring the number of photons passed through the material. The logical relation between this and that is not clear enough, but the most common apologize is the use of correspondence principle. Any further analysis of this delicate problem is not our main purpose in this paper so we shall leave it for a future occasion.

Before concluding this section, we remark that the method employed here is also applicable to higher approximations than the linear. For this purpose we note Eq. (2.5) can be used for successive approximation, namely

$$\frac{\partial \Delta_k f}{\partial t}=(\mathscr{H}, \Delta_k f)-F(t')(A, \Delta_{k-1} f)\,, \tag{2.25}$$

where $\Delta_k f$ means the k-th term of the expansion

$$f'(t)=f+\Delta_1 f+\Delta_2 f+\cdots\cdots\,. \tag{2.26}$$

The solution of Eq. (2.25) has the same from as Eq. (2.6), f being replaced by $\Delta_{k-1} f(t')$. Therefore we obtain the complete solution of Eq. (2.4) as

$$\begin{aligned} f'(t)&=f+\sum_{k=1}^{\infty}(-)^k\int_{-\infty}^{t}\int_{-\infty}^{t_1}\cdots\int_{-\infty}^{t_{k-1}} e^{iL(t-t_1)}\Big(A, e^{iL(t_1-t_2)}(A, \cdots e^{iL(t_{k-1}-t_k)}(A, f))\cdots\Big) \\ &\qquad\times F(t_1)\,F(t_2)\cdots F(t_k)\,dt_1\cdots dt_k \\ &=f+\sum_{k=1}^{\infty}(-)^k\int_{-\infty}^{t}\int_{-\infty}^{t_1}\cdots\int_{-\infty}^{t_{k-1}} e^{iLt}\Big(A(t_1), (A(t_2), \cdots (A(t_k), f)\cdots\Big)F(t_1)\cdots F(t_k)dt_1\cdots dt_k\,. \end{aligned} \tag{2.27}$$

where $A(t)$ is defined by

$$A(t)=A(p_t, q_t)\,, \quad p_{t=0}=p\,, \quad q_{t=0}=q\,.$$

The same expression is obtained for quantum-mechanical case, namely

$$\rho'(t)=\rho+\sum_{k=1}^{\infty}\left(-\frac{1}{i\hbar}\right)^k\int_{-\infty}^{t}\int_{-\infty}^{t_1}\cdots\int_{-\infty}^{t_{k-1}}\exp\left(-it\mathscr{H}^{\times}/\hbar\right)A(t_1)^{\times}A(t_2)^{\times}\cdots A(t_k)^{\times}\rho F(t_1)\cdots F(t_k)\,dt_1\cdots dt_k \tag{2.28}$$

where the cross means the commutator operation with the operation on its right hand side (see Eqs. (2.13) and (2.14)). These solutions allow us to write down the expression of responses in higher approximation. For instance, the second order change of B at the time t can be written as

$$\Delta_2 B(t)=\left(\frac{1}{i\hbar}\right)^2\mathrm{Tr}\int_{-\infty}^{t}\int_{-\infty}^{t_1}[A(t_1-t_2)[A,\,\rho]]\,B(t_1-t_2)F(t_1)F(t_2)dt_1dt_2 \tag{2.29}$$

The corresponding classical formula is similar so that it is omitted here.

§ 3. Relaxation Function and Other Useful Formulae for Canonical Ensemble

In the following we consider quantum-mechanical problems, because the classical limit is easily obtained from quantum-mechanical formulae by replacing $[\ ,\ ]/i\hbar$ by Poission bracket or by making $\hbar\to 0$.

First let us observe the behavior of the response function $\phi_{BA}(t)$ in the limit $t\to\infty$. It may approach to a limiting value or may not approach to any definite value. If it does, we must have

$$\lim_{t\to\infty}\phi_{BA}(t)=0 \text{ (if the limit exists).} \tag{3.1}$$

This is evident from Abel's theorem

$$\lim_{t\to\infty}\phi_{BA}(t)=\lim_{\varepsilon\to 0+}\varepsilon\int_0^{\infty}\phi_{BA}(t)\,e^{-\varepsilon t}\,dt \tag{3.2}$$

which holds if the left hand side exists. The right hand side is zero unless $\phi_{BA}(t)$ has a finite Fourier-amplitude at zero frequency. If $\phi_{BA}(t)$ has finite Fourier components for a set of discrete frequencies, $\phi_{BA}(t)$ will oscillate indefinitely and does not have any limit. But, for calculation of admittance we have the convergence factor $e^{-\varepsilon t}$ so that we may even in this case regard the relation (3.1) to hold. If $\phi_{BA}(t)$ is expressed by a Fourier integral with continuous Fourier spectrum, Eq. (3.1) is generally true. This corresponds to actual cases of large systems, because the complicated interaction within the system is always enough to split the energy levels into a structure so fine that the observation can not discriminate. The error in the observation of frequency certainly introduces an average which allows us to replace the originally discrete Fourier spectrum by continuous spectrum. Classically this corresponds to ellimination of long Poincaré cycles.

If the perturbation is applied continuously from $t=-\infty$ up to $t=0$ and is cut off at $t=0$, then the response ΔB will relax following the formula

$$\Delta B(t)=\int_{-\infty}^{0}\phi_{BA}(t-t')dt'F=F\int_t^{\infty}\phi_{BA}(t')dt',\qquad t>0\,. \tag{3.3}$$

Therefore, the function

$$\Phi_{BA}(t)=\lim_{\varepsilon\to 0+}\int_t^{\infty}\phi_{BA}(t')\,e^{-\varepsilon t'}\,dt' \tag{3.4}$$

describes the relaxation of ΔB after removal of the outer disturbance. This will be called the relaxation function.

Here, for later reference we note the theorem:

Theorem 2. *The admittance $\chi_{BA}(\omega)$ can be calculated by the following formulae*;

$$\begin{aligned}\chi_{BA}(\omega)&=\lim_{\varepsilon\to 0+}\int_0^{\infty}\phi_{BA}(t)\,e^{-i\omega t-\varepsilon t}\,dt\\&=\lim_{\varepsilon\to 0+}\frac{1}{i\omega+\varepsilon}\left\{\dot{\phi}_{BA}(0)+\int_0^{\infty}\phi_{BA}(t)\,e^{-i\omega t-\varepsilon t}dt\right\}\\&=\Phi_{BA}(0)-i\omega\int_0^{\infty}\Phi_{BA}(t)\,e^{-i\omega t}\,dt\,.\end{aligned} \tag{3.5}$$

These transformations are rather evident.

The expression of $\phi_{BA}(t)$ has already been given by Eq. (2.22). We note here that other functions $\dot{\phi}_{BA}(t)$ and $\Phi_{BA}(t)$ have convenient transform if the statistical ensemble is assumed to be *canonical*. For this purpose, we observe the identity

$$\begin{aligned}&[A,\exp(-\beta\mathscr{H})]\\&=\exp(-\beta\mathscr{H})\int_0^{\beta}\exp(\lambda\mathscr{H})[\mathscr{H},A]\\&\qquad\times\exp(-\lambda\mathscr{H})\,d\lambda\end{aligned}$$

$$=\frac{\hbar}{i}\exp(-\beta\mathscr{H})\int_0^S \exp(\lambda\mathscr{H})\dot{A} \times\exp(-\lambda\mathscr{H})\,d\lambda$$

$$=\frac{\hbar}{i}\exp(-\beta\mathscr{H})\int_0^\beta \dot{A}(-i\hbar\lambda)\,d\lambda \tag{3.6}$$

or

$$[\rho_\beta, A]=i\hbar\int_0^\beta \rho_\beta\dot{A}(-i\hbar\lambda)\,d\lambda \tag{3.7}$$

which is easily seen by writing the expression in matrix form in the representation diagonalizing $\mathscr{H}$.

Let us now assume that the system we observe is statistically represented by the canonical density matrix,

$$\rho=\exp(-\beta(\mathscr{H}-\Psi)),\quad \beta=1/kT,$$
$$\exp(-\beta\Psi)=\mathrm{Tr}\exp(-\beta\mathscr{H})\,. \tag{3.8}$$

By Eq. (2.22b) we have

$$\phi_{BA}(t)=\frac{1}{i\hbar}\mathrm{Tr}[\rho, A]B(t)$$
$$=\int_0^\beta \mathrm{Tr}\,\rho\dot{A}(-i\hbar\lambda)B(t)\,d\lambda$$
$$=-\int_0^\beta \mathrm{Tr}\,\rho A(-i\hbar\lambda)\dot{B}(t)\,d\lambda \tag{3.9}$$

and also

$$\dot{\phi}_{BA}(t)=\frac{1}{i\hbar}\mathrm{Tr}[\rho, A]\dot{B}(t)$$
$$=\frac{1}{i\hbar}\mathrm{Tr}\,\rho[A, \dot{B}(t)]=\frac{-1}{i\hbar}\mathrm{Tr}\,\rho[\dot{A}, B(t)]$$
$$=\int_0^\beta \mathrm{Tr}\,\rho\dot{A}(-i\hbar\lambda)\dot{B}(t)\,d\lambda \tag{3.10}$$

One may use any of these expressions depending on the nature of the quantities A and B. This will be illustrated later by examples.

With the aid of Eq. (3.9), the relaxation is transformed into

$$\Phi_{BA}(t)=\frac{i}{\hbar}\int_t^\infty \mathrm{Tr}\,\rho[B(t'), A]\,dt' \tag{3.11a}$$
$$=\int_0^\beta \mathrm{Tr}\,\rho A(-i\hbar\lambda)B(t)d\lambda-\beta\mathrm{Tr}\rho A^0B^0 \tag{3.11b}$$

where A^0 and B^0 are the diagonal parts of A B with respect to $\mathscr{H}$. The second term on the right hand side of Eq. (3.11) is the limit

$$\lim_{t\to\infty}\int_0^\beta \mathrm{Tr}\,\rho A(-i\hbar\lambda)B(t)\,d\lambda$$
$$=\lim_{t\to\infty}\frac{1}{t}\int_0^t dt\int_0^\beta \mathrm{Tr}\,\rho A(-i\hbar\lambda)B(t)\,d\lambda$$
$$=\beta\,\mathrm{Tr}\,\rho A^0B^0 \tag{3.12}$$

because this limit process selects the zero-frequency component of $B(t)$. The limit of the first expression, if it exists, must be equal to the last.

We may also define the function

$$\Phi^e{}_{BA}(t)=\int_0^\beta \mathrm{Tr}\,\rho A(-i\hbar\lambda)(B-B(t)) \tag{3.13}$$

which we shall call the *excitation* function for convenience, because this describes the increase of $\Delta B(t)$ when a constant force is applied from $t=0$. Thus

$$\lim_{t\to\infty}\Phi^e{}_{BA}(t)=\Phi_{BA}(0)\,. \tag{3.14}$$

In particular, for $t=0$, we have from Eq. (3.11) and Eq. (3.5)

$$\chi_{BA}(0)=\Phi_{BA}(0)=\Phi^e{}_{BA}(\infty)$$
$$=\int_0^\beta \mathrm{Tr}\,\rho A(-i\hbar\lambda)Bd\lambda-\beta\,\mathrm{Tr}\,\rho A^0B^0$$
$$=\int_0^\beta \mathrm{Tr}\,\rho(A(-i\hbar\lambda)-A^0)(B-B^0)d\lambda \tag{3.15}$$

where $\chi_{BA}(0)$ is the static admittance.

Eq. (3.15) is a little embarrassing, since the isothermal admittance, $\chi_{BA}{}^T$, is

$$\chi_{BA}{}^T=\int_0^\beta \mathrm{Tr}\,\rho(A(-i\hbar\lambda)-\overline{A})(B-\overline{B})\,d\lambda \tag{3.16}$$

where $\overline{A}$ and $\overline{B}$ and the equilibrium expectations of A and B in thermal equilibrium with $F=0$.

Eq. (3.16) is obtained by using the expansion

$$\exp\{-\beta(\mathscr{H}-FA\}$$
$$=\exp(-\beta\mathscr{H})\{1+\int_0^\beta A(-i\hbar\lambda)d\lambda F+0(F^2)\}$$

for the expression

$$\Delta B=\frac{\mathrm{Tr}\exp\{-\beta(\mathscr{H}-FA)\}B}{\mathrm{Tr}\exp\{-\beta(\mathscr{H}-FA)\}}-\frac{\mathrm{Tr}\exp(-\beta\mathscr{H})B}{\mathrm{Tr}\exp(-\beta\mathscr{H})}.$$

$\chi_{BA}{}^T$ is defined by

$$\chi_{BA}{}^T=\Delta B/F\,.$$

Expressions (3.15) and (3.16) are different unless

$$\mathrm{Tr}\,\rho A^0B^0=\overline{A}\,\overline{B}=\mathrm{Tr}\,\rho A\cdot\mathrm{Tr}\,\rho B\,. \tag{3.17}$$

This will not however hold without certain restrictions imposed on A and B. In fact, $\chi_{BA}(0)$ as given by Eq. (3.15) is the static admittance of an isolated system on which the force F is inserted adiabatically. On the other hand $\chi_{BA}{}^T$, Eq. (3.16), is isothermal,

that gives the response when the system is in thermal contact with the heat reservoir. Thus they need not be equal.

But, there are situations in which the difference is quite negligible. For instance, suppose a magnetic system is magnetized by external field. If the magnetic energy is only a very small fraction of the total energy, the magnetization process treated by Eq. (3.15) is practically isothermal. This means that, if the heat capacities associated with A and B are only a small fraction of total heat capacity, the equality (3.17) must be practically satisfied.

It is noted here that this is concerned with the ergodic property of the system. Khinchin[13] has shown that the ergodicity for a quantity A, namely

$$\langle A\rangle=\lim_{t\to\infty}\frac{1}{t}\int_0^t A(t)\,dt \tag{3.18}$$

must hold if the autocorrelation of A satisfies the relation

$$\lim_{t\to\infty}\langle A(0)A(t)\rangle=\langle A\rangle^2\,. \tag{3.19}$$

Here $\langle A\rangle$ means the phase average of a phase function $A(p, q)$ over a microcanonical ensemble. The ergodicity of A, (3.18), means that the time average of A starting from any point P on the ergodic surface approaches to $\langle A\rangle$ independent of the initial point P. This will hold on the assumption of (3.19) for almost all of P except a set of measure zero. This Khintchin's theorem can be reversed. It can be proved that the correlation function $\langle A(0)A(t)\rangle$ satisfies Eq. (3.19) if (3.18) holds uniformly.

In classical limit, (3.11b) in replaced by

$$\Phi_{BA}(t)=\beta\{\langle A(0)B(t)\rangle-\langle A^0B^0\rangle\}\,, \tag{3.20}$$

where $\langle\ \rangle$ means the phase average, and A^0 means the invariant part of $A(p_t, q_t)$ with respect to the natural motion. Eq. (3.16) becomes classically

$$\chi_{BA}{}^T=\beta\langle(A-\langle A\rangle)(B-\langle B\rangle)\rangle\,. \tag{3.21}$$

In general we must expect

$$\begin{aligned}\Phi_{BA}(0)&=\Phi_{BA}{}^e(\infty)=\chi_{BA}(0)\\&=\beta\langle(A-\mathrm{A}^0)(B-B^0)\rangle\fallingdotseq\chi_{BA}{}^T\,.\end{aligned} \tag{3.22}$$

The left hand side is the static admittance for an isolated system. This means that

$$\lim_{t\to\infty}\langle A(0)(B(t)\rangle\fallingdotseq\langle A\rangle\langle B\rangle \tag{3.23}$$

and hence that the ergodicity (3.18) does not hold. This is quite true as one would easily understand by the example of magnetization process of an ideal magnetic system. The isothermal and adiabatic susceptibilities are different, the later being zero.

However, our aim is to apply the whole theory to realistic systems which contain various interactions. As we have mentioned before, we should expect that

$$\chi_{BA}(0)=\chi_{BA}{}^T \tag{3.24}$$

will hold if the total system is large enough compared to the degrees of freedom associated with the observed quantities A and B.

For a very large system, the canonical distribution is almost equivalent to a microcanonical distribution, the relative fluctuation being very small. The ensemble average in Eqs. (3.20), etc., can be regarded as that over an ergodic surface. Therefore, Eq. (3.24) must hold if the quantities A, B, and $A(0)B(t)$ are ergodic, namely

$$\begin{aligned}\langle A\rangle&=\lim_{t\to\infty}\frac{1}{t}\int_0^t A(t)\,dt\,,\\\langle B\rangle&=\lim_{t\to\infty}\frac{1}{t}\int_0^t B(t)\,dt\,,\\\langle A(0)\,B(t)\rangle&=\lim_{t'\to\infty}\frac{1}{t'}\int_0^{t'}\langle A(t')B(t'+t)\rangle\,dt'\end{aligned} \tag{3.25}$$

Here the right hand sides of the equations are primarily determined by the location of the initial point of time average, but they do not depend on that. That is the assertion of ergodicity.

If the requirement of ergodicity, (3.25), is fulfilled, we shall have

$$\begin{aligned}&\lim_{t\to\infty}\langle A(0)\,B(t)\rangle\\&\quad=\lim_{t\to\infty}\lim_{t'\to\infty}\frac{1}{t'}\int_0^{t'}\langle A(t')B(t'+t)\rangle\,dt'\\&\quad=\lim_{t\to\infty}\frac{1}{tt'}\int_0^t\int_0^{t'}\langle A(t')\,B(t'+t)\rangle\,dt'dt\\&\quad=\langle\langle A\rangle\langle B\rangle\rangle=\langle A\rangle\langle B\rangle\,.\end{aligned} \tag{3.26}$$

Thus Eq. (3.22) holds.

The above arguments apply to quantum-mechanical cases with small modifications that the function $\int_0^\beta\langle A(-i\hbar\lambda)\,B(t)\rangle\,d\lambda$ enters in place of the correlation function.

Let us consider a magnetic system for illustration. The system, M, of the magnetic units are embedded in another large system,

R, with a certain interaction between M and R. The total system S we consider is $M+R$. If the heat capacity of R is large compared with M, R is a sort of heat bath for M. If we treat only M, assuming a canonical distribution and ignore the interaction between M and R, the relaxation function calculated by (3.11) describes only the internal process inside the subsystem M. The magnetic susceptibility, for instance, obtained from this, is adiabatic. For this canonical distribution, the ergodicity such as (3.25) will not hold.

If we consider the interaction and suppose, for instance, that it induces a Markoffian process for the transition of M among its various states, the ergodicity (3.25) will become true; the Markoffian assumption makes it easy to treat the relaxation process. The Markoffian assumption may be adequate if the interaction is weak and satisfies certain conditions which may be required. If such conditions are not satisfied, we are forced to deal directly with the combined system $M+R$. The relaxation function or the response function has to be treated with the complete Hamiltonian for $M+R$. In this case, the ergodicity (3.25) may be approximately true if the subsystem M is small compared to R. A rigorous formulation of mathematical proof of this is not easy, so that we merely anticipate this.

§ 4. Correlation Functions

We have seen in the previous section that the relations function $\Phi_{BA}(t)$ in classical case for canonical ensemble is essentially the correlation function of A and B; namely

$$\Phi_{BA}(t)=\beta\langle A(0)B(t)\rangle-\beta\langle A^0B^0\rangle \quad \text{(classical)} \tag{4.1}$$

where the bracket means the phase average over the canonical distribution. Or, the excitation function is

$$\Phi_{BA^e}(t)=\beta\langle A(0)(B-B(t))\rangle\,. \tag{4.2}$$

Other functions, for example, the response functions, are also proportional to certain correlation functions; e.g.

$$\phi_{BA}(t)=-\beta\langle A(0)\dot{B}(t)\rangle\,. \tag{4.3}$$

These are also valid for the microcanonical ensemble provided that the system in question is large.

Now the question arises how the correlation functions are defined in quantum-mechanics. We may define this by

$$\Psi_{BA}(t)=\mathrm{Tr}\,\rho\{A(0)B(t)\}=\mathrm{Tr}\,\rho\{A(t_1)B(t_1+t)\} \tag{4.4}$$

where ρ is the equilibrium density matrix, either canonical or microcanonical. The bracket means the symmetrized product,

$$\{AB\}=\frac{1}{2}(AB+BA)\,.$$

We may call this the time correlation of B and A at the time interval t, although it is not the correlation of two sets of actually observed values of B and A at different time points. The disturbance due to the observation is not taken into account. If the nature of the system and of A and B is such that this disturbance is not serious, which will be true for macrosopic systems and macroscopic observations, the function (4.4) describes the actual correlation. Since ρ is equilibrium density matrix, the time variation of A and B makes a stochastic time series which is stationary in time. For the sake of brevity, we rewrite the relaxation function as

$$\Phi_{BA}(t)=\int_0^\beta \mathrm{Tr}\,\rho(A(-i\hbar\lambda)-A^0)(B-B^0)\,d\lambda \tag{4.5}$$

and in the following we shall deal with quantities A and B which satisfy the condition

$$\mathrm{Tr}\,\rho A=\mathrm{Tr}\,\rho B=0\,, \tag{4.6}$$

which is achieved merely by substracting the invariant part from each of the quantities.

We can show easily that the relaxation function and the correlation function are mutually connected by the following theorem:

Theorem 3. *Let the Fourier transforms of* $\Phi_{BA}(t)$ *and* $\Psi_{BA}(t)$ *be*

$$f_{BA}(\omega)=\frac{1}{2\pi}\int_{-\infty}^{\infty}\Phi_{BA}(t)\,e^{-i\omega t}\,dt\,,$$

$$g_{BA}(\omega)=\frac{1}{2\pi}\int_{-\infty}^{\infty}\Psi_{BA}(t)\,e^{-i\omega t}\,dt\,, \tag{4.7}$$

then we have

$$g_{BA}(\omega)=E_\beta(\omega)f_{BA}(\omega) \tag{4.8}$$

where $E_\beta(\omega)$ *is the average energy of the oscillator with the frequency* ω *at temperature* $T=1/k\beta$,

$$E_\beta(\omega)=\frac{\hbar\omega}{2}\coth\frac{\beta\hbar\omega}{2}\,. \tag{4.9}$$

Eq. (4.8) *is equivalent to*

$$\Phi_{BA}(t)=\int_{-\infty}^{\infty}\Gamma(t-t')\Psi_{BA}(t')\,dt' \tag{4.10}$$

where $\Gamma(t)$ *is defined*

$$\Gamma(t)=\frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{e^{i\omega t}}{E_\beta(\omega)}\,d\omega=\frac{2}{\hbar\pi}\log\coth\frac{\pi}{2\beta\hbar}|t|\,, \tag{4.11}$$

or to

$$\Psi_{BA}(t)=E_\beta\left(\frac{d}{idt}\right)\Phi_{BA}(t)\,. \tag{4.12}$$

The easiest way of the proof is to write first the functions $\Phi_{BA}(t)$ and $\Psi_{BA}(t)$ as double series over the quantum states of $\mathscr{H}$ and rearrange the double sum. An alternative method is to assume a function-theoretical conditions, i.e.,

$$\lim_{Ret\to\pm\infty}\mathrm{Tr}\,\rho AB(t)=0\,, \tag{4.13}$$

and that $\mathrm{Tr}\,\rho AB(t)$ is analytic in the domain

$$0\leqq\mathrm{Im}\,t\leqq\beta\hbar\,.$$

The condition (4.13) is fulfilled if one extends the meaning of the limit process as was discussed in § 2 and § 3. The second condition is equivalent to the existence $\mathrm{Tr}\,\rho AB$ and $\mathrm{Tr}\,\rho BA$.

Now Eq. (4.8) is derived as follows. By definition we can write

$$\begin{aligned}f_{BA}(\omega)&=\frac{1}{2\pi}\int_{-\infty}^{\infty}\int_0^\beta\mathrm{Tr}\,\rho AB(t+i\hbar\lambda)\exp(-i\omega t)d\lambda dt\\&=\frac{1}{2\pi}\int_{-\infty}^{\infty}\int_0^\beta\mathrm{Tr}\,\rho AB(t+i\hbar\lambda)\\&\qquad\times\exp\{-i\omega(t+i\hbar\lambda)-\lambda\hbar\omega\}\,d\lambda dt\\&=\frac{1}{2\pi}\int_0^\beta d\lambda\exp(-\lambda\hbar\omega)\\&\qquad\times\int_{-\infty+i\hbar\lambda}^{\infty+i\hbar\lambda}\mathrm{Tr}\,\rho AB(t)\exp(-i\omega t)\,dt\,.\end{aligned}$$

Thus interchanging the order of integration and shifting the path of integration, we obtain by the above-mentioned assumptions

$$f_{BA}(\omega)=\frac{1-\exp(-\beta\hbar\omega)}{\hbar\omega}\times\frac{1}{2\pi}\int_{-\infty}^{\infty}\mathrm{Tr}\,\rho AB(t)\exp(i\omega t)\,dt\,. \tag{4.14}$$

Similarly we can show that

$$\int_{-\infty}^{\infty}\mathrm{Tr}\,\rho AB(t)e^{-i\omega t}\,dt=\exp(\beta\hbar\omega)\int_{-\infty}^{\infty}\mathrm{Tr}\,\rho B(t)\,A\exp(-i\omega t)\,dt \tag{4.15}$$

by shifting the integration path to $-i\hbar\beta-\infty$ to $-i\hbar\beta+\infty$. In the above transformations, we took ρ as canonical given by Eq. (3.8).

Eqs. (4.15) gives

$$\int_{-\infty}^{\infty}\mathrm{Tr}\,\rho AB(t)e^{-i\omega t}\,dt=\frac{2}{1+\exp(-\beta\hbar\omega)}g_{BA}(\omega)\,,$$

so that Eq. (4.14) leads Eq. (4.8).

Eq. (4.10) is easily obtained from (4.8). The kernel $\Gamma(t)$, (4.11) is calculated using the partial fraction expansion of $\tanh z/z$. We note further that it has the property

$$\lim_{\hbar\to0}\Gamma(t)=\beta\delta(t)\,, \tag{4.16}$$

which in naturally to be expected.

§ 5. Simple Examples

It will be adequate to insert here simple examples to illustrate the general idea developed in the previous sections. The author has applied this in a previous paper[7] to the problem of magnetic resonance absorption. An uniform magnetic field $\boldsymbol{H}(t)$ is applied to a magnetic system, the total magnetic moment of which is denoted by $\boldsymbol{M}$. The perturbation energy due to $\boldsymbol{H}(t)$ is now

$$\mathscr{H}'(t)=-\boldsymbol{M}\boldsymbol{H}(t)\,. \tag{5.1}$$

The natural motion of the magnetic moment in the absence of the external field is represented by $\boldsymbol{M}(t)$. Then, the response function $\phi_{\mu\nu}(t)$ for the magnetization in μ-direction when the external field $\boldsymbol{H}(t)$ lies in ν-direction, is by Eq. (2.22).

$$\begin{aligned}\phi_{\mu\nu}(t)&=\frac{i}{\hbar}\langle[M_\mu(t),\,M_\nu]\rangle\\&=\int_0^\beta\langle\dot{M}_\nu(-i\hbar\lambda)M_\mu(t)\rangle\,d\lambda\,,\end{aligned} \tag{5.2}$$

and the relaxation function is, by Eq. (3.11),

$$\Phi_{\mu\nu}(t)=\int_0^\beta\langle(M_\nu(-i\hbar\lambda)-M_\nu{}^0)(M_\mu(t)-M_\mu{}^0)\rangle\,d\lambda \tag{5.3}$$

where $M_\nu{}^0$ and $M_\mu{}^0$ mean the diagonal parts of M_ν and M_μ with respect to the unperturbed Hamiltonian $\mathscr{H}$.

If we consider a system with unit volume, the admittance becomes the magnetic susceptibility. Thus the complex magnetic susceptibility tensor can be expressed either by $\phi_{\mu\nu}(t)$ or $\Phi_{\mu\nu}(t)$. The convenient expression is, by Eq. (3.5)

$$\chi_{\mu\nu}(\omega)=\Phi_{\mu\nu}(0)-i\omega\int_0^\infty\Phi_{\mu\nu}(t)\,e^{-i\omega t}\,dt\,. \tag{5.4}$$

The static susceptibility is, in particular,

$$\chi_{\mu\nu}(0)=\int_0^\beta \langle (M_\nu(-i\hbar\lambda)-M_\nu^0)(M_\mu-M_\mu^0)\rangle\, d\lambda \tag{5.5}$$

which is the susceptibility for the isolated system and is not necessarily equal to the isothermal susceptibility,

$$\chi_{\mu\nu}{}^T=\int_0^\beta \langle M_\nu(-i\hbar\lambda)M_\mu\rangle d\lambda-\beta\langle M_\nu\rangle\langle M_\mu\rangle. \tag{5.6}$$

In the expression for $\chi_{\mu\nu}(0)$, the diagonal parts of M_ν and M_μ are substracted. This corresponds to the fact the magnetization in an isolated system proceeds adiabatically, the occupation probabilities of the levels remaining unchanged. On the other hand they are changed in isothermal process for $\chi_{\mu\nu}{}^T$. The difference becomes smaller if the environment is taken into account in greater extent.

In the classical limit, Eq. (5.5) becomes

$$\chi_{\mu\nu}(0)=\frac{1}{kT}\langle (M_\nu-M_\nu^0)(M_\mu-M_\mu^0)\rangle\,. \tag{5.7}$$

while Eq. (5.6) becomes

$$\chi_{\mu\nu}{}^T=\frac{1}{kT}\langle\, (M_\nu-\langle M_\nu\rangle)(M_\mu-\langle M_\mu\rangle)\,\rangle\,. \tag{5.8}$$

The last expression is first noticed by Kirkwood[14] for the classical theory of dielectric polarization. Eq. (5.7) is its extension to adiabatic susceptibility, and Eq. (5.4) to general complex susceptibility for non-equilibrium states. It is evident that the similar formulae are obtained for dielectric polarizations.

As a second example we shall consider the electric conductivity. For an electric $\boldsymbol{E}(t)$, the perturbation energy is

$$\mathscr{H}'(t)=-\sum_i e_i \boldsymbol{r}_i \boldsymbol{E}(t) \tag{5.7}$$

where e_i is the charge of the i-th particle in the system and $\boldsymbol{r}_i$ its position vector. The response of current in μ-direction when a pulse of electric field is applied in ν-direction at $t=0$ is, by Eq. (2.22) or Eq. (3.9)

$$\begin{aligned}\phi_{\mu\nu}(t)&=\frac{1}{i\hbar}\mathrm{Tr}\,[\rho,\ \sum_i e_i x_{i\nu}]\sum e_i\,\dot{x}_{i\mu}(t)\\ &=\int_0^\beta \langle\, J_\nu(-i\hbar\lambda)\, J_\mu(t)\,\rangle\, d\lambda\end{aligned} \tag{5.10}$$

where

$$J_\mu=\sum e_i\,\dot{x}_{i\mu}$$

is the total current in the system.

If we take the volume of the system as unity, the conductivity tensor $\sigma_{\mu\nu}(\omega)$ for periodic field is given by

$$\sigma_{\mu\nu}(\omega)=\int_0^\infty e^{-i\omega t}\,dt\int_0^\beta d\lambda\,\langle\, J_\nu(-i\hbar\lambda)\, J_\mu(t)\,\rangle\,. \tag{5.11}$$

This is an exact expression for the conductivity. In particular, for the static conductivity we have

$$\sigma_{\mu\nu}=\int_0^\infty\int_0^\beta d\lambda\,\langle\, J_\nu(-i\hbar\lambda)\, J_\mu(t)\,\rangle\,. \tag{5.12}$$

This may be expressed in another form,

$$\sigma_{\mu\nu}=\lim_{s\to 0+}\frac{1}{s}\left\{\phi_{\mu\nu}(0)+\int_0^\infty \dot{\phi}_{\mu\nu}(t)\,e^{-st}\,dt\right\} \tag{5.13}$$

where

$$\dot{\phi}_{\mu\nu}(t)=-\frac{1}{i\hbar}\,\mathrm{Tr}\,\rho[J_\nu,\, J_\mu(t)]\,. \tag{5.14}$$

The above expressions of conductivity tensor are, as a matter of fact, what one would obtain by simple application of dispersion formula of Heisenberg and Kramers. But they are more convenient and much clearer for physical interpretation. Nakano[8] has shown that the Grüneisen formula is obtained from (5.11) as the first approximation. We emphasize again that the formulae given in the above are exact although they are admittedly rather abstract and have to be approximated in some way to deduce useful answers for assumed physical models. We can show that the usual method of employing Boltzmann-Bloch type transport equations makes a certain approximation to calculate these exact expressions. But the transport equation can not always be justified since the Markoffian assumption involved is only valid under rather strict conditions for the scattering processes.[1] The exact theory will certainly give a starting point of actual calculations when the traditional methods are not justified. A good example of this is provided by the problem of electronic conduction in strong magnetic field.[11] This will be treated in a forthcoming paper.

§ 6. Symmetry Relations

We note here some symmetry properties of the response and relaxation functions and those of the admittance. Let us consider for convenience the relaxtion function defined by Eq. (3.11), for which we have the theorem:

Theorem 4. *The relaxation function $\Phi_{BA}(t)$ has the following properties;*

1) *$\Phi_{BA}(t)$ is real* (6.1)

2) $\Phi_{BA}(-t)=\Phi_{AB}(t)$ (6.2)

3) *reciprocity law: if a static magnetic field* $\boldsymbol{H}$ *is present, the reversal of the direction of* $\boldsymbol{H}$ *results in*

$$\Phi_{BA}(t, \boldsymbol{H})=\varepsilon_A\varepsilon_B\Phi_{BA}(-t, -\boldsymbol{H}) =\varepsilon_A\varepsilon_B\Phi_{AB}(t, -\boldsymbol{H}) \quad (6.3)$$

where ε_A *or* ε_B *is* $+1$ *or* -1 *according to the quantity* A *or* B *is even or odd with respect to the reversal of time.*

The proof is simple. The complex conjugate of $\Phi_{BA}(t)$ is calculated as follows.

$$\begin{aligned}\overline{\Phi}_{BA}(t)&=\int_0^\beta \mathrm{Tr}\,\rho B(t)\,A(i\hbar\lambda)\,d\lambda\\ &=\int_0^\beta \mathrm{Tr}\,\rho B(t-i\hbar\lambda)\,A\,d\lambda\\ &=\int_0^\beta \mathrm{Tr}\,\rho B(t-i\hbar(\beta-\lambda))\,A\,d\lambda\\ &=\int_0^\beta \mathrm{Tr}\,\rho AB(t+i\hbar\lambda)d\lambda=\Phi_{BA}(t)\,.\end{aligned}$$

The transformation from the second to the third line is simply the change of integration variable from λ to $\beta-\lambda$.

Similarly, we may proceed as

$$\begin{aligned}\Phi_{BA}(-t)&=\int_0^\beta \mathrm{Tr}\,\rho AB(-t+i\hbar\lambda)\,d\lambda\\ &=\int_0^\beta \mathrm{Tr}\,\rho A(t-i\hbar\lambda)B\,d\lambda\\ &=\int_0^\beta \mathrm{Tr}\,\rho BA(t+i\hbar\lambda)\,d\lambda=\Phi_{BA}(t)\,.\end{aligned}$$

Eq. (6.3) follows from the fact that the simultaneous reversal of t and $\boldsymbol{H}$ changes the q-representation of the Hamiltonian and the wave function to their complex conjugates. A corollary of the theorem is the symmetry of other functions such as $\phi_{BA}(t)=-\dot{\Phi}_{BA}(t)$. This is clear and so is omitted here.

The second corollary is the symmetry of admittance. We write down here the symmetry of the quantity defined by

$$\sigma_{BA}(\omega)=\int_0^\infty \Phi_{BA}(t)\,e^{-i\omega t}\,dt \quad (6.4)$$

for which we have

$$\mathrm{Re}\,\sigma_{BA}(\omega)=\mathrm{Re}\,\sigma_{BA}(-\omega)\,, \quad (6.5)$$

$$\mathrm{Im}\,\sigma_{BA}(\omega)=-\mathrm{Im}\,\sigma_{BA}(-\omega)\,, \quad (6.6)$$

$$\sigma_{BA}(\omega, -\boldsymbol{H})=\varepsilon_A\varepsilon_B\sigma_{AB}(\omega, \boldsymbol{H})\,. \quad (6.7)$$

Eq. (6.7) is the well-known Onsager's relation.[15] The derivation here made is very simple.

As examples, we note the symmetry of magnetic susceptibility tensor and that of the conductivity tensor. The susceptibility tensor has the symmetry

$$\begin{aligned}\mathrm{Re}\,\chi_{\mu\nu}(\omega, \boldsymbol{H})&=\mathrm{Re}\,\chi_{\mu\nu}(-\omega, \boldsymbol{H})\\ &=\mathrm{Re}\,\chi_{\nu\mu}(\omega, -\boldsymbol{H})\,,\\ \mathrm{Im}\,\chi_{\mu\nu}(\omega, \boldsymbol{H})&=-\mathrm{Im}\,\chi_{\mu\nu}(-\omega, \boldsymbol{H})\\ &=\mathrm{Im}\chi_{\nu\mu}(\omega, -\boldsymbol{H})\,.\end{aligned} \quad (6.8)$$

For the conductivity tensor we have the same relations

$$\begin{aligned}\mathrm{Re}\,\sigma_{\mu\nu}(\omega, \boldsymbol{H})&=\mathrm{Re}\,\sigma_{\mu\nu}(-\omega, \boldsymbol{H})\\ &=\mathrm{Re}\,\sigma_{\nu\mu}(\omega, -\boldsymbol{H})\,,\\ \mathrm{Im}\,\sigma_{\mu\nu}(\omega, \boldsymbol{H})&=-\mathrm{Im}\,\sigma_{\mu\nu}(-\omega, \boldsymbol{H})\\ &=\mathrm{Im}\,\sigma_{\nu\mu}(\omega, -\boldsymbol{H})\,.\end{aligned} \quad (6.9)$$

Thus, for the change ω to $-\omega$, the real part is even while the imaginary part is odd. For the reversal of $\boldsymbol{H}$, the symmetric part is even and the antisymmetric part is odd.

§ 7. Fluctuation-Dissipation Theorems

We have shown in § 4 that the response and relaxation functions can be expressed by correlation functions. Thus the admittance can also be expressed in terms of correlation functions. This is rather trivial for classical cases, but in quantum mechanics it becomes rather complicated. The relation between the admittance and the Fourier components correlation functions is usually called the fluctuation-dissipation theorem. Thus our aim here is to give this theorem in its complete form.

We treat as a typical example the case of electric conduction. Let us rewrite Eq. (5.11) in terms of the Fourier component of response function $\phi_{\mu\nu}(t)$, i.e.,

$$f_{\mu\nu}(\omega)=\frac{1}{2\pi}\int_{-\infty}^{\infty}\phi_{\mu\nu}(t)\,e^{-i\omega t}\,dt\,. \quad (7.1)$$

Eq. (5.11) is transformed as

$$\begin{aligned}\sigma_{\mu\nu}(\omega)&=\int_0^\infty \phi_{\mu\nu}(t)\,e^{-i\omega t}\,dt\\ &=\int_0^\infty dt\int_{-\infty}^{\infty}d\omega'\,f_{\mu\nu}(\omega')\,e^{i\omega' t-i\omega t}\\ &=\pi f_{\mu\nu}(\omega)+i\int_{-\infty}^{\infty}\frac{f_{\mu\nu}(\omega')}{\omega'-\omega}\,d\omega'\,.\end{aligned} \quad (7.2)$$

Thus we obtain at once

$$\mathrm{Re}\,\sigma^s{}_{\mu\nu}(\omega)=\pi f^s{}_{\mu\nu}(\omega)\equiv\frac{\pi}{2}\left\{f_{\mu\nu}(\omega)+f_{\nu\mu}(\omega)\right\}$$

$$\mathrm{Im}\,\sigma^{a}_{\mu\nu}(\omega)=\frac{\pi}{2}f^{a}_{\mu\nu}(\omega)\equiv\frac{\pi}{2i}\left\{f_{\mu\nu}(\omega)-f_{\nu\mu}(\omega)\right\} \tag{7.3}$$

$$\mathrm{Im}\,\sigma^{s}_{\mu\nu}(\omega)=\int_{-\infty}^{\infty}\frac{f^{s}_{\mu\nu}(\omega')}{\omega'-\omega}d\omega'$$

$$\mathrm{Re}\,\sigma^{a}_{\mu\nu}(\omega)=i\int_{-\infty}^{\infty}\frac{f^{a}_{\mu\nu}(\omega')}{\omega'-\omega}d\omega' , \tag{7.4}$$

where the superscript s or a means the symmetric or the anti-symmetric part of the tensor. By the theorem 4 one has the symmetry

$$\overline{f_{\mu\nu}}(\omega)=f_{\nu\mu}(\omega) , \tag{7.5}$$

which shows $f^{s}_{\mu\nu}(\omega)$ is real while $f^{a}_{\mu\nu}(\omega)$ is purely imaginary. Eqs. (7.3) and (7.4) give the well-known Kramers-Kronig relation. Note that it holds for each of the symmetric and antisymmetric parts separately.

Now let us introduce the correlation function of current components,

$$\Psi_{\mu\nu}(t)=\mathrm{Tr}\,\rho\{j_{\nu}(0)\,j_{\mu}(t)\} , \tag{7.6}$$

which describes the correlation of spontaneous current fluctuating in equilibrium. We can show the following theorem:

Theorem 5. *The conductivity tensor $\sigma_{\mu\nu}(\omega)$ is connected with correlation function of current components, (7.6), by the following relations:*

$$E_{\beta}(\omega)\,\mathrm{Re}\,\sigma^{s}_{\mu\nu}(\omega)=\pi g^{s}_{\mu\nu}(\omega)=\int_{0}^{\infty}\Psi^{s}_{\mu\nu}(t)\cos\omega t\,dt \tag{7.7a}$$

$$E_{\beta}(\omega)\,\mathrm{Im}\,\sigma^{a}_{\mu\nu}(\omega)=\frac{\pi}{i}g^{a}_{\mu\nu}(\omega)=-\int_{0}^{\infty}\Psi^{a}_{\mu\nu}(t)\sin\omega t\,dt \tag{7.7b}$$

$$\Psi^{s}_{\mu\nu}(t)=\frac{2}{\pi}\int_{0}^{\infty}E_{\beta}(\omega)\,\mathrm{Re}\,\sigma^{s}_{\mu\nu}(\omega)\cos\omega t\,d\omega \tag{7.8a}$$

$$\Psi^{a}_{\mu\nu}(t)=-\frac{2}{\pi}\int_{0}^{\infty}E_{\beta}(\omega)\,\mathrm{Im}\,\sigma^{a}_{\mu\nu}(\omega)\sin\omega t\,d\omega , \tag{7.8b}$$

and

$$\mathrm{Im}\,\sigma^{s}_{\mu\nu}(\omega)=\int_{-\infty}^{\infty}\frac{1}{E_{\beta}(\omega')}\cdot\frac{g^{s}_{\mu\nu}(\omega')}{\omega'-\omega}d\omega'$$
$$=-2\int_{0}^{\infty}\Gamma(\tau)\cos\omega\tau d\tau\int_{\tau}^{\infty}\Psi^{s}_{\mu\nu}(t)\sin\omega t\,dt$$
$$-2\int_{0}^{\infty}\Gamma(\tau)\sin\omega\tau d\tau\int_{0}^{\tau}\Psi^{s}_{\mu\nu}(t)\cos\omega t\,dt \tag{7.9a}$$

$$\mathrm{Re}\,\sigma^{a}_{\mu\nu}(\omega)=-\frac{1}{i}\int_{-\infty}^{\infty}\frac{1}{E_{\beta}(\omega')}\frac{g^{a}_{\mu\nu}(\omega')}{\omega'-\omega}d\omega'$$
$$=2\int_{0}^{\infty}\Gamma(\tau)\cos\omega\tau d\tau\int_{\tau}^{\infty}\Psi^{a}_{\mu\nu}(t)\cos\omega t\,dt$$
$$-2\int_{0}^{\infty}\Gamma(\tau)\sin\omega\tau d\tau\int_{0}^{\tau}\Psi^{a}_{\mu\nu}(t)\sin\omega t\,dt \tag{7.9b}$$

$$\Psi^{s}_{\mu\nu}(t)=\frac{2}{\pi}\int_{0}^{\infty}\mathrm{Im}\,\sigma^{s}_{\mu\nu}(\omega)\,\mathrm{Re}\,\Delta(\omega,t)\,d\omega \tag{7.10a}$$

$$\Psi^{a}_{\mu\nu}(t)=\frac{2}{\pi}\int_{0}^{\infty}\mathrm{Re}\,\sigma^{a}_{\mu\nu}(\omega)\,\mathrm{Im}\,\Delta(\omega,t)\,d\omega , \tag{7.10b}$$

where the kernel $\Delta(\omega, t)$ is defined by

$$\Delta(\omega,t)=\frac{1}{\pi}\int_{-\infty}^{\infty}\frac{du}{u-\omega}E_{\beta}(u)e^{iut} . \tag{7.11}$$

These expressions reduce, in classical limit, to

$$\mathrm{Re}\,\sigma_{\mu\nu}(\omega)=\int_{0}^{\infty}\Psi_{\mu\nu}(t)\cos\omega t\,dt$$
$$\mathrm{Im}\,\sigma_{\mu\nu}(\omega)=-\int_{0}^{\infty}\Psi_{\mu\nu}(t)\sin\omega t\,dt \tag{7.12}$$

First we note that the symmetry of $\Psi_{\mu\nu}(t)$ is the same as that we have for $\phi_{\mu\nu}(t)$ (Theorem 4). Eqs. (7.7) and the first parts of (7.9a) and (7.9b) are obvious from Eq. (4.8). Eq. (7.8a, b) is simply the inverse of Eq. (7.7a, b). We need some calculation to derive (7.9) and (7.10) but the details are omitted here. Only we note for the convenience of reference that the function (7.11) can be transformed to

$$\mathrm{Re}\,\Delta(\omega,t)=-E_{\beta}(\omega)\sin\omega t\cdot\mathrm{sign}\,t+\frac{1}{\pi\beta}\sum_{n=1}^{\infty}\frac{(n\pi)^{2}\exp(-2n\pi|t|/\beta\hbar)}{(n\pi)^{2}+(\beta\hbar\omega)^{2}/4} \tag{7.13a}$$

$$\mathrm{Im}\,\Delta(\omega,t)=E_{\beta}(\omega)\cos\omega t\cdot\mathrm{sign}\,t-\frac{1}{\pi\beta}\sum_{n=1}^{\infty}\frac{(n\pi/2)\beta\hbar\omega\exp(-2n\pi|t|/\beta\hbar)}{(n\pi)^{2}+(\beta\hbar\omega)^{2}/4}\mathrm{sign}\,t . \tag{7.13b}$$

An useful corollary of the theorem is the expression of static conductivity $\sigma_{\mu\nu}(0)$. We have the formulae;

$$\sigma^{s}_{\mu\nu}(0)=\beta\int_{0}^{\infty}\Psi^{s}_{\mu\nu}(t)\,dt\equiv\frac{\beta}{2}\int_{-\infty}^{\infty}\Psi^{s}_{\mu\nu}(t)\,dt \tag{7.14}$$

$$\sigma^{a}_{\mu\nu}(0)=2\int\Gamma(\tau)\,d\tau\int_{\tau}^{\infty}\Psi^{a}_{\mu\nu}(t)\,dt$$
$$=2\int_{0}^{\infty}\Psi^{a}_{\mu\nu}(t)\int_{0}^{t}\Gamma(\tau)\,d\tau . \tag{7.15}$$

Thus the symmetric part of the static conductivity is calculated either as the integration of the response function or as that of the correlation function. The situation is more complicated for the antisymmetric part. If one wishes to use the correlation function instead of the response function, one has to be careful about a quantum effect which appears as $\Gamma(\tau)$ in Eq. (7.15).

Another remark is that, even at finite frequencies, the symmetric part is easily obtained if the correlation function is known. This is, of particular interest, for instance, in the case of magnetic problem[7]. For a linearly polarized radiation, the absorption is determined by the imaginary part of susceptibility tensor, for which we have from Eq. (5.4)

$$\chi''_{xx}(\omega)=-\mathrm{Im}\,\chi_{xx}=\frac{\omega}{2E_\beta(\omega)}\int_{-\infty}^{\infty}\langle\{M_x(0)M_x(t)\}\rangle\cos\omega t\,dt\ . \tag{7.16}$$

The corresponding expression for conduction is (7.7a) itself, i.e.,

$$\sigma_{xx}(\omega)=\frac{1}{2E_\beta(\omega)}\int_{-\infty}^{\infty}\langle\{J_x(0)J_x(t)\}\rangle\cos\omega t\,dt \tag{7.17}$$

The last equation is the well-known Niquist's theorem[16]. If the fluctuating current is expanded into the Fourier components, Eq. (7.17) reduces to

$$\sigma_{xx}(\omega)=\frac{\pi}{E_\beta(\omega)}\langle|J_x(\omega)|^2\rangle\ . \tag{7.18}$$

§ 8. Sum Rules

It is interesting that the exact expressions of admittance derived in the above give proof of certain general laws. The simplest example of this is the well-known sum-rule of oscillator strength. This is generalized in the following way.

We consider the expression (6.4), from which we obtain

$$\frac{1}{\pi}\int_{-\infty}^{\infty}\sigma_{BA}(\omega)d\omega=2\int_0^{\infty}\Phi_{BA}(t)\delta(t)dt=\Phi_{BA}(0). \tag{8.1}$$

and

$$\lim_{\omega\to\infty}i\omega\,\sigma_{BA}(\omega)=\Phi_{BA}(0) \tag{8.2}$$

by Abel's theorem if the left hand side does exist.

Let us now apply this for electric conduction. By Eq. (5.10) one can write

$$\phi_{\mu\nu}(t)=\frac{1}{i\hbar}\mathrm{Tr}\,\rho\left[\sum e_i x_{i\nu},\ \sum e_j\dot{x}_{j\mu}(t)\right] \tag{8.3}$$

If the Hamiltonian $\mathscr{H}$ has the general form

$$\mathscr{H}=\sum_i\left[p_{i\nu}-\frac{e_i}{c}A_\nu(\boldsymbol{r}_i)\right]^2\Big/2m_i+V(\boldsymbol{r}_1,\cdots\boldsymbol{r}_N),$$

$\boldsymbol{A}$ being the vector potential we have the commutation rule

$$[x_{i\nu},\ \dot{x}_{j\mu}]=\frac{i\hbar}{m_i}\delta_{ij}\,\delta_{\mu\nu} \tag{8.4}$$

which gives

$$\phi_{\mu\nu}(0)=\sum_{i=1}^{N}\frac{e_i^2}{m_i}\delta_{\mu\nu}=\sum_r\frac{n_r e_r^2}{m_r} \tag{8.5}$$

where r means the species of charge carriers, n_r being the number of carriers of r-th kind.

Eq. (8.3) is derived from Eq. (5.10) by changing the order in the trace operation. One might doubt whether this is justfied or not, pointing out that the operators $x_{i\nu}$ may be singular if one uses the Born-Karman boundary conditions. As a matter of fact this will not affect the validity of the proof, because the commutators are actually regular operators. It might be worth while to see what one has in classical case. Classically we may proceed as (see Eq. (2.22a))

$$\begin{aligned}\phi_{\mu\nu}(t)&=\int d\Gamma(f,\ \sum e_i x_{i\nu})\sum e_j\dot{x}_{j\mu}(t)\\&=-\int d\Gamma\sum_i e_i\frac{\partial f}{\partial p_{i\nu}}\cdot\sum e_j\dot{x}_{j\mu}(t)\\&=\int d\Gamma\, f\sum_i\sum_j e_i e_j\frac{\partial}{\partial p_{i\nu}}\dot{x}_{j\mu}(t)\ .\end{aligned} \tag{8.6}$$

Thus we have

$$\begin{aligned}\phi_{\mu\nu}(0)&=\int d\Gamma f\sum_i\sum_j e_i e_j\frac{\partial}{\partial p_{i\nu}}\cdot\frac{1}{m_j}\left\{p_{j\mu}-\frac{e_j}{c}A_\mu(\boldsymbol{r}_j)\right\}\\&=\int d\Gamma f\sum_i e^2{}_i/m_i\,\delta_{\mu\nu}\end{aligned} \tag{8.7}$$

This is exactly the same as Eq. (8.5). This does not prove directly the validity of (8.3), but will be enough to make it convincing. As as proof one may introduce a suitable potential to avoid the difficulty of singular operators $x_{i\nu}$, and make it vanish if one wants.

Thus we have proved the sum rule

$$\frac{2}{\pi}\int_0^{\infty}\mathrm{Re}\,\sigma^s{}_{\mu\nu}(\omega)d\omega=\sum_r\frac{e^2{}_r n_r}{m_r}\delta_{\mu\nu} \tag{8.8}$$

and

$$\lim_{\omega\to\infty}\omega\,\mathrm{Im}\,\sigma^{s}_{\mu\nu}=-\sum_{r}\frac{e^2_r n_r}{m_r}\delta_{\mu\nu} \tag{8.9}$$

which hold for any system irrespective of the interaction of particles, the temperature, the statistics and even in the presence of magnetic field. This is the most general form of the sum rule.

For a system of electrons, Eqs. (8.8) and (8.9) are written as

$$\frac{2}{\pi}\int_0^\infty \mathrm{Re}\,\sigma^{s}_{\mu\nu}(\omega)\,d\omega=\frac{ne^2}{m}\delta_{\mu\nu} \tag{8.10}$$

$$\mathrm{Im}\,\sigma^{s}_{\mu\nu}(\omega)\to-\frac{e^2n}{m\omega}\delta_{\mu\nu}\quad(\omega\to\infty)\,. \tag{8.11}$$

Here we should remember that m is the true mass of electrons. The integration of $\sigma(\omega)$ has to be carried over all range of frequencies.

If one considers electrons in a crystal and confines himself to the electrons in a particular band neglecting interband transitions, the sum rule has to be modified to

$$\begin{aligned}\frac{2}{\pi}\int_0^\infty &\mathrm{Re}\,\sigma^{s}_{\mu\nu}(\omega)\,d\omega\\ &=-\lim_{\omega\to\infty}\omega\,\mathrm{Im}\,\sigma^{s}_{\mu\nu}(\omega)\\ &=e^2\,\mathrm{Tr}\,\{\rho\cdot\partial^2E(p)/\partial p_\nu\partial p_\mu\}\,.\end{aligned} \tag{8.12}$$

This holds if the electron system is described by the Hamiltonian

$$\mathscr{H}=\sum_i E(p_i)+V(r_1,\cdots r_N) \tag{8.13}$$

where $E(p)$ is the energy of an electron with the crystal moments p. Also one has to omit the interband elements of the potential V. The theorem (8.12) is wider than that usually given in text books only for the oscillator strength connecting one-electron states.

Another interesting result is obtained from the second expression of (3.5) when applied to conduction problem. This give

$$\begin{aligned}&\frac{1}{\pi}\int_{-\infty}^{\infty}\{i\sigma_{\mu\nu}(\omega)\,\omega-\phi_{\mu\nu}(0)\}d\omega\\ &=-\lim_{\omega\to\infty}\omega^2\left\{\sigma_{\mu\nu}(\omega)-\frac{\phi_{\mu\nu}(0)}{i\omega}\right\}=\dot{\phi}_{\mu\nu}(0)\end{aligned} \tag{8.14}$$

The tensor

$$\dot{\phi}_{\mu\nu}(0)=-\frac{i}{\hbar}\,\mathrm{Tr}\,\rho\,[J_\mu,J_\nu] \tag{8.15}$$

is antisymmetric. Therefore Eq. (8.14) is rewritten as

$$\begin{aligned}&-\frac{1}{\pi}\int_{-\infty}^{\infty}\mathrm{Im}\,\sigma^{a}_{xy}(\omega)\,\omega d\omega\\ &=\lim_{\omega\to\infty}\omega^2\sigma^{a}_{xy}(\omega)=-\sum_r\frac{n_re_r^3H_z}{m_r^2c}\,.\end{aligned} \tag{8.16}$$

Both sides of this equation are identically zero unless the system is in a magnetic field (or rotating, in which case e_rH_z/m_rc must be replaced by the angular frequency of the rotation). This is seen from the fact that the velocity of charged particles in magnetic field is

$$\boldsymbol{v}=\frac{1}{m}\left(\boldsymbol{p}-\frac{e}{c}\boldsymbol{A}\right)$$

and the commutation rule is now

$$[v_x,v_y]=-\frac{eH_z}{m^2c}\frac{\hbar}{i}\,. \tag{8.17}$$

This gives

$$\frac{i}{\hbar}\,\mathrm{Tr}\,\rho\,[J_x,J_y]=\sum n_re_r^3\,H_z/m_r^2c\,, \tag{8.18}$$

which is the last expression in Eq. (8.16).

In particular, for electrons we have

$$\begin{aligned}\frac{2}{\pi}\int_0^\infty\mathrm{Im}\,\sigma^{a}_{\mu\nu}(\omega)\omega d\omega&=-\lim_{\omega\to\infty}\omega^2\sigma^{a}_{\mu\nu}(\omega)\\ &=\frac{ne^3}{m^2c}H_z\,.\end{aligned} \tag{8.19}$$

This is another type of sum-rule which holds quite generally, but is not usually remarked. Eq. (8.19) may also be written as

$$\sum_l\mathrm{Im}\langle k|J_x|l\rangle\langle l|J_y|k\rangle=-\frac{ne^3}{2m^2c}H_z \tag{8.20}$$

for a system of n electrons.

Similar arguments can also be made for tensors of other kinds. For instance, the magnetic susceptibility tensor satisfies the rules,

$$-\frac{1}{\pi}\int_{-\infty}^{\infty}\mathrm{Im}\,\chi^{s}_{\mu\nu}(\omega)\,\frac{d\omega}{\omega}=\chi^{s}_{\mu\nu}(0)\,, \tag{8.21}$$

$$\lim_{\omega\to\infty}\mathrm{Re}\,\chi^{s}_{\mu\nu}(\omega)=0\,, \tag{8.22}$$

$$\begin{aligned}&\frac{1}{\pi}\int_{-\infty}^{\infty}\mathrm{Re}\,\chi^{a}_{\mu\nu}(\omega)\,d\omega\\ &=-\lim_{\omega\to\infty}\omega\,\mathrm{Im}\,\chi^{a}_{\mu\nu}(\omega)=\frac{i}{\hbar}\,\mathrm{Tr}\,\rho[M_\mu,M_\nu]\,.\end{aligned} \tag{8.23}$$

The relation (8.21) is well-known. This states that the properly weighted integration of absorption intensity for linearly polarized radiation is equal to the static susceptibility. The third relation, (8.23), requires more at-

tention.

If M_μ is written as

$$M_\mu=\sum_i \gamma_i I_{i\mu} ,$$

where I_i is the angular momentum associated with the magnetic moment, we have

$$\frac{i}{\hbar}[M_x, M_y]=-\sum \gamma_i^2 I_{iz} .$$

Therefore Eq. (8.23) can be written as

$$\frac{1}{2}\int_0^\infty \mathrm{Re}\,\chi^a{}_{xy}(\omega)\,d\omega = -\lim_{\omega\to\infty}\omega\,\mathrm{Im}\,\chi^a{}_{xy}(\omega)=-\sum\gamma_i^2\overline{I}_{iz}=-\sum\gamma_r\overline{M}_{rz} \tag{8.24}$$

where $\overline{M}_{rz}$ means the magnetization due to r-th component in the system.

For circularly polarized radiation, the rate of energy absorption per unit volume is

$$Q=\frac{\omega}{2}H_1^2\{\chi'_{xy}-\chi'_{yx}+\chi''_{xx}+\chi''_{yy}\}$$

H_1 being the amplitude of the magnetic field of radiation. Thus, $\mathrm{Re}\,\chi^a{}_{xy}(\omega)$ is related to the absorption intensity of circular waves. The total integration of the absorption intensity is directly connected with the gyromagnetic ratio γ. This may not be of much value if γ can be measured accurately by sharp resonance. But one has to remember that (8.24) is generally valid even in the presence of strong interaction between magnetic moments. Thus the effects related to the absorption of circular wave may be used for the determination of γ, for instance, for electronic magnetic moments in ferromagnets or paramagnets. This sort of experiments may be considered to corresponds in a sense to those of gyromagnetic effects.

§ 9. Einstein Relations

We note here that the expression of conductivity given by Eq. (5.12) or (7.14) is the most general form of Einstein relation, which connects the conductivity or the mobility with the diffusion constant.

To see this, let us take the simplest example of charged particles which are moving independently. By Eq. (7.14), the conductivity is

$$\sigma^s{}_{\mu\nu}=\frac{ne^2}{kT}\int_0^\infty \langle v_\nu(0)\,v_\mu(t)\rangle\,dt , \tag{9.1}$$

if we regard the system to be classical. If we admit

$$\lim_{t\to\infty}\langle v_\nu(0)v_\mu(t)\rangle=\langle v_\nu\rangle\langle v_\mu\rangle=0$$

we can transform the right hand side of Eq. (9.1) as

$$\begin{aligned}&\int_0^\infty\langle v_\nu(0)v_\mu(t)\rangle\,dt\\ &=\lim_{T\to\infty}\frac{1}{2T}\int_0^T\int_0^T\langle v_\nu(t')v_\mu(t)\rangle\,dt\,dt'\\ &=\lim_{T\to\infty}\frac{1}{2T}\langle(x_\nu(T)-x_\nu(0))(x_\mu(T)-x_\mu(0))\rangle\\ &=D_{\mu\nu} .\end{aligned} \tag{9.2}$$

where $D_{\mu\nu}$ is the diffusion constant defined by the third expression of Eq. (9.2). Eqs. (9.2) and (9.1) give at once

$$\sigma^s{}_{\mu\nu}=e^2nD_{\mu\nu}/kT \tag{9.3}$$

which is the well known Einstein relation.

For a system of interacting particles, the exact form of Einstein relation is

$$\sigma^s{}_{\mu\nu}=e^2\overline{\Delta n^2}D_{\mu\nu}/kT \tag{9.4a}$$

$$=e^2D_{\mu\nu}\Big/\left(\frac{\partial\mu}{\partial n}\right)_T \tag{9.4b}$$

where $D_{\mu\nu}$ is the diffusion constant in isothermal conditions, $\overline{\Delta n^2}$ the average fluctuation of particle density, and μ the chemical potential of the particle. This relation gives, for the diffusion constant,

$$\begin{aligned}D^s{}_{\mu\nu}&=\left(\frac{\partial\mu}{\partial n}\right)_T\int_0^\infty dt\int_0^\beta \mathrm{Tr}\,\rho I_\nu(-i\hbar)I_\mu(t)d\lambda\\ &=\left(\frac{\partial\mu}{\partial n}\right)_T\frac{1}{2}\int_{-\infty}^\infty dt\,\mathrm{Tr}\,\rho\{I_\nu I_\mu(t)\}\end{aligned} \tag{9.5}$$

where

$$I_\nu=\sum_i v_{i\nu}$$

is the total particle flow per unit volume. Eq. (9.5) can be used for the calculation of diffusion constant in gases and also condensed phases. This will be discussed further in a forthcoming paper.

It might be worth while to note that Eq. (9.4) indicates at once how the factor $1/kT$ appearing in the expression of electric conductivity, Eq. (7.14) cancels out for degenerate Fermi particles. In this case, $\overline{\Delta n^2}$ is proportional to T, or in other words $(\partial\mu/\partial n)_T$ is a constant of the order of μ_0/n, μ_0 being the Fermi-level at the absolute zero.

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The evolution of the structure of tubulin and its potential consequences for the role and function of microtubules in cells and embryos

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ABSTRACT This paper discusses the results of homology modeling and resulting calculation of key structural and physical properties for close to 300 tubulin sequences, including α, β, γ, δ and ε -tubulins. The basis for our calculations was the structure of the tubulin dimer published several years ago by Nogales *et al.* (1998), later refined to 3.5 Å resolution by Löwe *et al.* (2001). While, it appears that the α, β and γ-tubulins segregate into distinct structural families, we have found several differences in the physical properties within each group. Each of the α, β and γ- tubulin groups exhibit major differences in their net electric charge, dipole moments and dipole vector orientations. These properties could influence functional characteristics such as microtubule stability and assembly kinetics, due to their effects on the strength of protein-protein interactions. In addition to the general structural trends between tubulin isoforms, we have observed that the carboxy-termini of α and β-tubulin exists in at least two stable configurations, either projecting away from the tubulin (or microtubule) surface, or collapsed onto the surface. In the latter case, the carboxy-termini form a lattice distinctly different from that of the well-known A and B lattices formed by the tubulin subunits. However, this C-terminal lattice is indistinguishable from the lattice formed when the microtubule-associated protein tau binds to the microtubule surface. Finally, we have discussed how tubulin sequence diversity arose in evolution giving rise to its particular phylogeny and how it may be used in cell- and tissue-specific expression including embryonal development.

KEY WORDS: *tubulin, microtubule, isoform, homology model, dipole*

Introduction

Microtubules (MTs) are cylindrical protein filaments found in all eukaryotes and are critically involved in a variety of cellular processes including cellular motility, cytoplasmic transport and mitosis (Hyams, 1994). Their component protein, tubulin forms a heterodimer that is composed of two homologous polypeptides, designated α and β tubulin (Postingl, 1981; Krauhs, 1981). Each individual tubulin heterodimer is 8 nm in length and interacts laterally and longitudinally to form protofilaments and then MTs (Downing and Nogales, 1998). In addition to microtubules, whose protofilament numbers vary betweens 8 and 17, under special experimental conditions tubulin can also give rise to polymorphic assemblies that include sheets, macrotubes, ribbons and several additional, exotic structures (Unger, 1990).

Microtubules undergo cycles of rapid growth and disassembly in a process known as dynamic instability which has been extensively studied both *in vivo* and *in vitro* (Mitchison and Kirschner, 1984). This instability appears to be critical for proper MT function, especially during mitosis (Kirschner and Schulze, 1986). The assembled tubulin dimer hydrolyzes a bound guanosine triphosphate (GTP) at its exchangeable nucleotide binding site to GDP; the kinetics of this process in β tubulin is critical in regulating dynamic instability by affecting the loss of the "lateral cap" (the top tubulin layer in the growing MT that stabilizes the entire structure).

In addition to forming MTs, tubulin interacts with a large number of ancillary proteins. Some of these, such as tektin, may play structural roles, while others like *tau* or MAP2 are known as microtubule-associated proteins (MAPs) are involved in MT assembly, stability and may also mediate interactions with other

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proteins. Yet another subset of proteins, such as kinesin and dynein, are motor proteins that move cargoes, such as cellular vesicles, along MTs as part of active intracellular transport (Kikkawa, 2001; Wang and Sheetz, 2000).

An interesting feature of tubulin is its presence as multiple isoforms in eukaryotic organisms and its numerous post-translational modifications, such as phosphorylation, acetylation, detyrosination and polyglutamylation (Banerjee, 2002; Luduena, 1998). In addition to the ubiquitous α and β-tubulin, MTs often require the presence of an additional tubulin protein,'γ-tubulin, for correct assembly as a templating protein (Erickson, 2000; Luduena, 1998). Two additional tubulins, designated δ and ε, have also been characterized and although their role in MT assembly remains uncertain (Vaughan, 2000), models of their function in MT assembly have been proposed (Inclan and Nogales, 2001).

At the molecular level, the role of tubulin is extremely complex and seems to be related to the structural variations observed between α and β isoforms (Richards, 2000). The existence and distribution of numerous homologous forms of α and β-tubulin provides a link to the structure of tubulin and the polymerization and stability properties of MTs. It has been demonstrated that

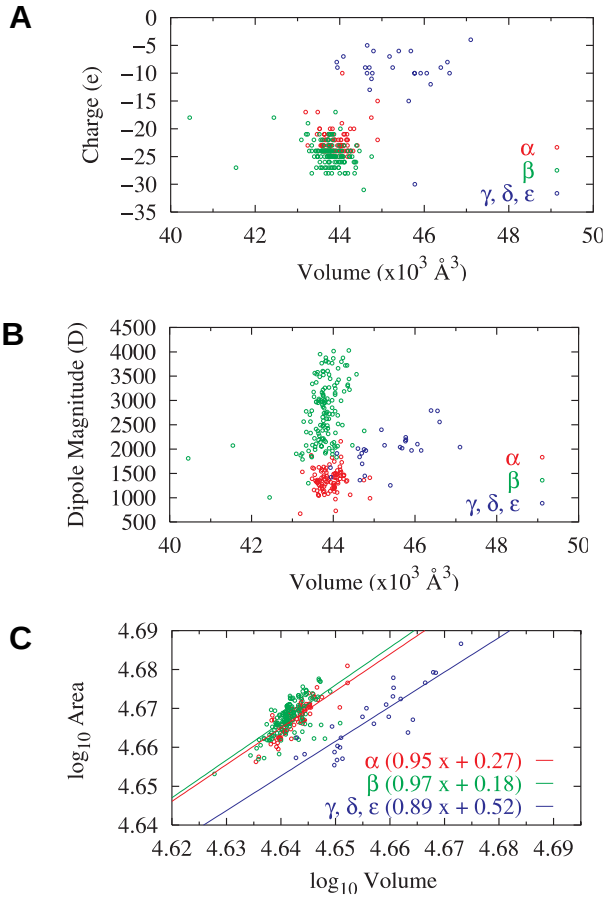

Fig. 1. Illustration of the distribution of physical properties of the monomer models. *Each ring corresponds to one model and is color-coded to show three groups within the library. Three pairs of dependencies are shown:* **(A)** *charge vs. volume,* **(B)** *dipole moment magnitude vs. volume and* **(C)** *log surface area vs. log volume.*

apparently small overall differences in the structures and therefore binding energies and chemical affinities between different tubulins may translate into significant deviations in the growth and catastrophe rates for MTs. The three-dimensional structure of bovine brain tubulin has been determined crystallographically, resulting in several atomic structures that are available in the Protein Data Bank, entries 1TUB (Nogales, 1998) and 1JFF (Lowe, 2001) being typical examples. Using these structures, it is possible to computationally predict the structures of related tubulin sequences with some degree of accuracy. We have created homology models of nearly 300 different tubulins, representing α and β-tubulins, as well as γ, δ and ε-tubulins from animals, plants, fungi and protists. For most of the resulting tubulin structures, we have been able to estimate the magnitudes and orientations of their dipole moments, charge distributions, surface areas and volumes. The magnitudes and orientations of the dipople moment of the tubulin dimer may play a role in MT assembly and stability due to their contribution to the total energy via dipole-dipole interactions. In addition, we have been able to generate plausible conformations for the carboxy-terminal region of each tubulin monomer. The C-termini of α- and β-tubulin were not resolved in the original crystallographic structures of tubulin due to their flexibility and possibly sample inhomogeneity (Nogales, 1998). The importance of the carboxy-termini is highlighted by the fact that they are the site of most of tubulin's post-translational modifications, that they bind to MAPs and that most of the differences among tubulin isoforms cluster here. We have observed that the carboxy-termini can either project out to a considerable distance from the MT or collapse onto the surface. Our results raise the possibility that the conformational transitions in the C-termini of tubulin may play a significant role in kinesin and dynein movement and that they may have a novel role in ion transport along MTs, especially in the axoplasm of neurons.

Tubulin's diversity

Gene orthologs are homologous proteins with amino acid sequences that have diverged due to mutations accumulated since their separation by speciation events (Fitch, 2000). The variations that occur can be neutral: implying that they are irrelevant to the process of natural selection, crucial: meaning that they adapt the function of a protein to a given selective pressure, or can be somewhere in between. In general, a weak but statistically significant correlation is found between this sequence variability and a protein's solvent accessible surface area (Rodionov and Blundell, 1998). Interestingly, an apparent conservation of the three-dimensional pattern of conserved amino acids has been observed in several families of structurally homologous proteins that points to the existence of a folding nucleus (Mirny and Shakhnovich, 1999). Previous reports agree that buried residues within a protein tend to evolve more slowly, but most of these analyses have been restricted to globular enzymes. Interestingly, the mapping of evolutionary rates onto the sequence of the α/β tubulin dimer demonstrate the opposite pattern (Roger, 1996). The relative rates of evolution at sites within the core of the tubulin proteins are faster than the rates of sites at the surface. The large number of intermolecular interactions between tubulins and other proteins apparently restrains the sites at the protein surface, relative to the constraints imposed by side-chain packing

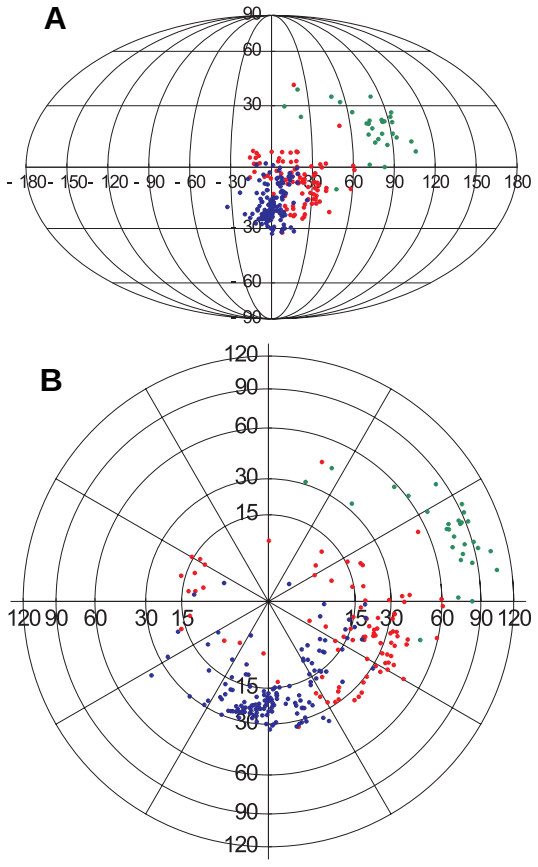

Fig. 2. Distribution of center-of-mass dipole moment orientations within the monomer library. Color-coding is as in Fig. 1. *The center of each plot corresponds to a dipole moment normal to a cylindrical microtubule.* **(A)** *An equal area (Mollweide) projection with the vertical axis representing angle with the protofilament axis and the horizontal axis angle with the protofilament-radial plane.* **(B)** *A polar projection plot, using a nonlinear radial scale for the angle for the normal position; the vertical axis corresponding to the microtubule axis.*

within the core of the proteins. This is consistent with the common assumption that co-evolution of many interacting proteins leads to a high degree of sequence constraint within the proteins where these interactions occur.

Raff *et al.* (1997) emphasized overwhelming evidence in favor of the idea that FtsZ is a homolog of tubulin, the ubiquitous eukaryotic cytoskeletal protein involved in many essential cellular processes including mitosis. Despite only limited primary sequence homology centered on a GTP-binding motif termed the 'tubulin signature sequence', the recently solved crystal structures of FtsZ and tubulin show extensive structural similarity throughout the proteins. In addition, FtsZ, like tubulin, binds and hydrolyzes GTP and assembles into protofilaments that have structures similar to those within MT's. This assembly is GTP-dependent and disassembly occurs when the GTP supply is exhausted, suggesting that FtsZ polymers, like MTs, are dynamically unstable. Tubulin and FtsZ share a common fold of two domains connected by a central helix (Nogales, 1998). Structure-based sequence alignment shows that common residues localize in the nucleotide-binding site and regions that interact with the nucleotide of the next subunit in the protofilament, suggesting that tubulin and FtsZ use similar contacts to form filaments. Surfaces that would make lateral interactions between protofilaments or interact with motor proteins are, however, different.

Most eukaryotic organisms express multiple tubulin genes, with each distinct tubulin isoform having an essential function. It has been shown, that in the testis of *Drosophila* and the touch neurons of *Caenorhabditis elegans*, not every tubulin molecule is interchangeable. For example, the *mec7* and *mec12* tubulins are required for assembly of specialized MTs that contain 15 protofilament. It has been demonstrated that γ tubulin is required for the initiation of MT assembly. Additionally, four new members of the tubulin family have recently been identified (Dutcher, 2001). The first, δ-tubulin was identified as a flagellar assembly mutant in the green algae *Chlamydomonas reinhardti* and has also been found in *Chlamydomonas Euplotes*, human, rat, *Xenopus*, mouse, dog and Trypanosomids. Second, η-tubulin was discovered due to a mutation in *Paramecium* that results in a basal body duplication defect. The ε and ζ-tubulins, were discovered using bioinformatics techniques, however their cellular functions have yet to be determined. ε-tubulin is present in *Chlamydomonas*, human, mouse, rat, *Xenopus* and Trypanosomids. ζ-tubulin has been found in *Trypanosoma* and *Leishmania* and as an expressed sequence tag in *Xenopus laevis*. Interesingly, the δ, ε, ζ and η-tubulins do not appear to have a ubiquitous distribution throughout all eukaryotic organisms, as is the case for the α and β tubulins. The genomes of *Saccharomyces cerevisiae*, *Schizosaccharomyces pombe*, *Aradbidopsis thaliana* and *C. elegans* do not have clear homologs to the δ, ε, ζ and η-tubulins. The absence of these tubulins from fungi and plants can be correlated with the lack of centrioles and basal bodies. The high degree of divergence among the new tubulins in both the same and different families suggests that their functions are not as constrained by their primary sequences as those of the α, β and γ-tubulins. The elucidation of the biological function of these new members of the tubulin superfamily will hopefully resolve their roles within the cell.

Early sequence comparisons for tubulin gave valuable insights into the function of α and β-tubulin (Little, 1981; Little and Seehaus, 1988; Burns, 1991; Burns and Surridge, 1994). Alignments between the orthologs and paralogs of α and β-tubulin are unambiguous due to the highly conserved amino acid sequences between these two proteins. Recently, Kuchnir-Fygenson *et al.* (2004) analyzed the homology within the tubulin family using multiple sequence alignment techniques. They compared members of the tubulin super-family from human, *Drosophila*, *Chlamydomonas*, *Trypanosoma* and *Paramecium*. Within this subset, it was shown that all the α-tubulins are 89–95% similar to one another while the β-tubulins were only 88–94% similar. It was also demonstrated that the γ-tubulins have similarities ranging from 72% to 94%, while the three δ-tubulins were only 47–57% similar. The two complete sequences for the ε-tubulin are only 58% similar. A portion of the ε-tubulin gene from *Chlamydomonas* is 79% similar to the human gene over the 87 amino acids within the region that is most conserved between humans and *Trypanosoma*. The predicted *Xenopus* sequences for ζ and η-tubulin, 89 and 97 amino acids long, are 60% and 61% similar to the *Trypanosoma* and *Paramecium* sequences, respectively. Among the 21 full-length tubulin sequences, only 7 amino acids are

conserved and include the the GDP/GTP-binding site, an aspartate within helix 7 and a proline near the M-loop that interacts with the H3 helix within a tubulin from the adjacent protofilament within the MT.

By examining sequence alignments between α- and β-tubulin, Kuchnir-Fygenson *et al.* (2004) identified residues that differ significantly in variability. Most of these residues were shown to be clustered around the nucleotide-binding pocket, where the greatest functional differences between the two types of tubulin exist. The remaining residues associated with large differences in variability are found in the N-terminal loop between helix (H) 1 and beta sheet (S) 2. The statistical distribution of residue variability in both α and β-tubulins is strongly peaked at low values, with >50% of residues scoring in the bottom 10% of the variability range. In the structure-based alignment, the correlation coefficient for variability between the tubulins is R(α/β)= 0.42. This alignment involves two gaps in β-tubulin: a small one in the disordered N-terminal loop (β, 39–40) and a larger one in the loop between S9 and S10 (β, 362–365). Three distinct regions are misaligned to accommodate multiple residues with large differences in variability between the paralogs. These misaligned regions include a total of approximately 100 residues located in helices H1 and H4, sheets S4, S5 and S9, two turns (T4 and T5) and the disordered N-terminal loop (L1). Among these, there are 40 positions that appear to be of particular functional importance where homologous residues differ in variability by more than a standard deviation from the mean. Half of these are clustered around the nucleotide-binding pocket, four are clustered around the taxol binding site on β-tubulin and one participates in lateral binding between MT protofilaments. Of the 20 residues near the nucleotide-binding pocket, 16 are more variable in the «N-site», which binds GTP without catalyzing its hydrolysis and less variable in the «E-site», which hydrolyzes GTP as dimers assemble into protofilaments. Of the four residues that are less variable in the N-site, the tyrosine residue at position 172 in α-tubulin is particularly interesting as a target for directed mutagenesis because it interacts directly with the nucleotide. Because analogous differences in variability were so plausibly connected with functional differences in the other misaligned regions, Kuchnir-Fygenson (2004) predicted that these residues in the N-terminal loop have a role in making the biochemical functions of α and β-tubulin distinct. It is suspected that the functional distinction is related to GTP hydrolysis and that the tenuously structured glycines within β-tubulin are involved in a hydrolysis-driven conformational change that eventually results in a catastrophe event.

We conclude this introductory survey on tubulin diversity stating that while it is tantalizing to speculate about a direct relationship between tubulin structure and function, the precise molecular basis of the properties of tubulin is still not well understood. This is in part because tubulin's highly flexible C-terminal conformation (Keskin, 2002) makes it difficult to crystallize this region.

Tubulin phylogeny

Phylogenetics is the science of estimating and analyzing evolutionary relationships. Phylogenetic relationships among micro-organisms are especially difficult to discern. DNA, RNA and proteins can be considered to be 'information molecules' since they retain a record of an organism's evolutionary history. A logical approach then is to compare nucleic acid or protein sequences from different organisms and estimate the evolutionary relationships based on the degree of homology between the sequences. The nucleotide or amino acid differences within a gene can therefore reflect the evolutionary distance between two organisms. Closely related organisms will exhibit fewer sequence differences than distantly related organisms. Through the molecular approach in determining phylogenetic relationships, the differences are readily quantifiable as sequences from different organisms can be compared and the number of differences computed. These data are often expressed in the form of phylogenetic trees, where the positions and lengths of the "branches" depict the relatedness of sequences between organisms. Interestingly, phylogenetic trees that are produced from different gene sequences may produce different topologies. Possible reasons for these inconsistencies are: (a) disparities in evolutionary rates among lineages, (b) uneven taxonomic sampling, (c) single explosive radiation of major eukaryotic taxa, (d) horizontal DNA transfer between species and (e) gene paralogies (i.e., duplications) and gene conversion. The first two phenomena result in a long-branch attraction artifact in which many slowly evolving sequences cluster to the exclusion of a few rapidly evolving sequences. The long branches that are far apart in the lower portion of the eukaryotic branch may be a result of the experimental procedure. In addition, events like horizontal DNA transfer and gene duplications complicate the analysis of molecular phylogenetic data. Some of these problems are resolved by combining data into consensus trees, for example, by combining protein data from actin, α-tubulin and β-tubulin (Roger, 1996).

Wilson and Borisy (1997) stated that although highly conserved, the variable regions within α and β-tubulins show divergence from other α and β-tubulins within the same species, but can show conservation between different species. Such conservation raises the question of whether the diversity in tubulin sequence can mediate the diversity in MT organization. Recent studies probing the function of β-tubulin isoforms in axonemes of insects suggest that tubulin structure, through interactions with extrinsic proteins, can direct the architecture and supra-molecular organization of MTs. The tubulin gene family includes, at least, three highly conserved subfamilies, α and β and γ-tubulin, that

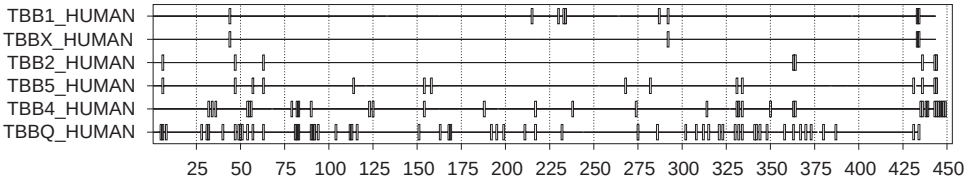

Fig. 3. Comparison amongst human -tubulin sequences. *Each of the horizontal lines represents one sequence and has boxes at horizontal locations corresponding to residues at which the sequence differs from a consensus sequence. Note the presence of a hypervariable region, beginning after residue 425.*

arose from a series of gene duplications early in eukaryotic evolution (Edlind *et al.*, 1996; Keeling and Doolittle, 1996). Currently, α and β-tubulin genes have the widest taxonomic representation and have been identified in early diverging eukaryotic lineages (Burns, 1996; Edlind, 1996; Keeling and Doolittle, 1996). Cleveland *et al.* (1980) concluded that the human genome contains approximately 14 copies of the α and β-tubulin genes.

Adoutte *et al.* (1984) produced a survey of the electrophoretic behavior of the tubulins of 23 species (mostly protists) as well as their reactivity towards 4 anti-tubulin antibodies. This allowed a rough evaluation of evolutionary relatedness between the various groups of protists analyzed. The results indicated a striking agreement between a number of published phylogenies. Gaertig *et al.* (1993) cloned and sequenced the two β-tubulin genes of the ciliated protozoan *Tetrahymena thermophila*. These two genes encode identical 443 amino acid peptides which are 99.7% identical to the β-tubulin proteins of *T. pyriformis* and 95% identical to the human β1-tubulin. *T. thermophila* contains only one α-tubulin gene, implying that all of the diverse MT structures in this unicellular organism must be formed from a single α and a single β-tubulin heterodimer. Additionally, Gaertig *et al.* (1993) also carried out a phylogenetic analysis of 84 complete β-tubulin peptide sequences. This analysis supports the following two hypotheses regarding β-tubulin evolution and function: (a) that multifunctional β-tubulins are under greater evolutionary constraint than β-tubulins present in specialized cells or in cells with very few MT-related functions, which can evolve rapidly; and (b) that cells which form axonemes maintain a homogeneous population of tubulins. Luduena (1998) reviewed the distribution and possible functional significance of the various forms of tubulin. In analyzing the differences among the tubulin isoforms that are encoded by different genes, it was shown that some appear to have no functional significance, while some increase the overall adaptability of the organism to environmental challenges and yet others appear to perform specific functions including formation of particular organelles and interactions with specific proteins (Keeling, 1998).

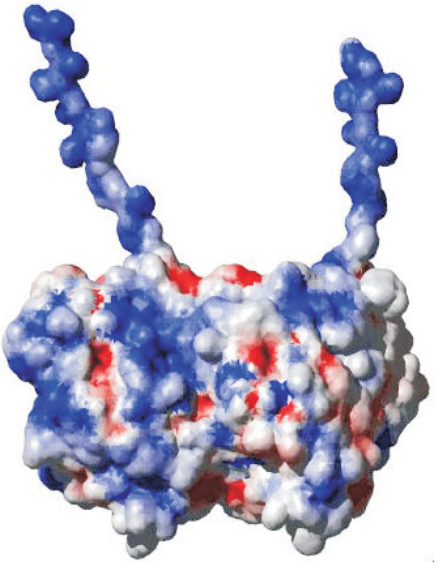

Fig. 4. A map of the electric potential on the surface of a tubulin dimer with C-termini tails. *Red regions represent positive charge, while blue regions indicate negative charge. The intensity of coloring indicates the local surface strength and polarity of the field. Figure prepared using MolMol (Koradi* et al.*, 1996).*

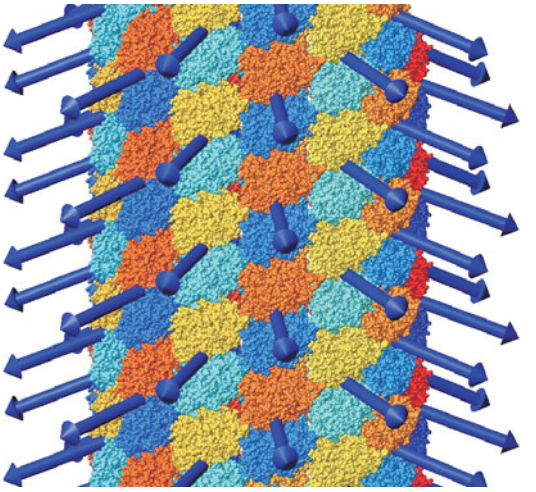

Fig. 5. Dipole orientation of dimers in a microtubule. *The arrows indicate the orientation of the permanent dipole moment with respect to the centers-of-mass of individual tubulin-tubulin dimers with respect to the surface of a microtubule. Note that dipole moments point into the dimers. Figure prepared using MolMol (Koradi, 1996).*

Keeling and Doolittle (1996) sequenced genes from a number of protozoa and constructed phylogenetic trees of α and β-tubulins. These trees were consistent with each other, but inconsistent with other molecular phylogenies. The agreement between α and β-tubulin trees could arise only by the co-adaptation of one molecule to variants of the other as a result of their intimate steric association in MTs. Therefore, these trees may not provide independent support for the phylogenetic results. Relationships between the protist tubulins were also examined by constructing trees of all three types. These trees were found to be of limited value for determining the position of the root within each subfamily because of the great interfamily distances, but they confirmed the classification of all known genes into three monophyletic subfamilies. Divergent genes from *C. elegans* and *S. cerevisiae* representing the novel classes of δ and ε-tubulin were found to be specifically related to γ-tubulins from animals and fungi respectively and therefore are seen as rapidly evolving orthologs of γ-tubulin.

Edgcomb *et al.* (2001) elucidated the relationships among jakobid flagellates and other early-diverging eukaryotic lineages by characterizing α and β-tubulin genes from four jakobids. Tubulin gene phylogenies were in general agreement with mitochondrial gene phylogenies and ultrastructural data. The final representation of major eukaryotic groups was pared down to the alignments of 42 α and 39 β-tubulin sequences that were amenable to more rigorous phylogenetic analyses. A variety of traditional eukaryotic groups that are well established on the basis of morphology or molecular phylogeny were also recovered in either α or β-tubulin trees. In addition, several «higher- order» eukaryotic groupings were observed, such as fungi plus microsporidia, animals plus fungi and ciliates plus apicomplexa (alveolates). Unlike rRNA trees, tubulin phylogenies do not show many inde-

pendent deep-branching lineages leading to a «crown» radiation of eukaryotes but, instead, a deep split occurs within eukaryotes separating two robust groupings, a plant-protist «superclade» and an animals-plus-fungi group, with several protist lineages intervening. It is likely that the extreme conservation of both tubulin genes leaves few phylogenetically informative sites, especially for groups in the plant-protist superclade (Keeling, 1999; 2000).

Homology modeling

The presence of multiple forms of tubulin provides us with the opportunity to create a model library from which we can estimate physical characteristics for a large number of tubulin sequences. Once three-dimensional protein structures are determined, it becomes possible to use techniques, such as homology modeling to predict structures of proteins with similar sequences (Chothia and Lesk, 1986). Homology modeling utilizes several structural motifs from template proteins and pieces them together to form the final model. A scoring function then assesses both the sequence identity between the target sequence and template and the overall quality of the template that is being considered. The scores are ranked and the fold with the best score is assumed to be the one adopted by the target sequence. A computer program uses alignment of the sequences with known related structures to obtain spatial restraints that the output structure must satisfy. Additional restraints derived from statistical studies of representative protein and chemical structures are also used to ensure a physically plausible result. Missing regions are predicted by simulated annealing of a molecular mechanics model. Since the 3D structures of tubulin obtained from crystallography lack the extreme carboxy-termini, we used this capability to create structure within this region of the protein. Our experience with molecular mechanics models of these regions suggests that the residues are too mobile for such structures to be anything more than one of many possible conformations. It is our approxiation that any models with gross errors will be insignificant when comparisons are made amongst several models.

We have performed homology modeling on a set tubulin sequences, containing α and β-tubulins from animals, plants, fungi and protists in the Uniprot Knowledgebase. As an initial step, the Swiss-Prot database Release 40.3 of 08-Nov-2002 (Boeckmann, 2003) (available at http://www.expasy.org/sprot/) was searched for tubulin amino acid sequences. A search using the keyword «tubulin» was manually filtered to separate actual tubulin sequences from those of other tubulin-related proteins. This provided some 290 sequences, representing a wide range of species. Of these, 27 sequences are annotated as being fragmentary leaving 263 complete tubulin monomer sequences. Of particular interest were the 15 human sequences obtained in this process. We believe that this core library is large enough to adequately sample the diversity of tubulin variation, therefore no attempt to systematically add to this set from other data sources has been made, although select sequences of interest from other sources have also been studied.

Appendix A summarizes all the tubulin sequences used in this study. The table names the source organism and for each α, β, γ, δ and ε, gives the name used in the database. It is important to relate the biochemical data encapsulated by the amino acid sequence to the biologically relevant information presented in this table. Fragmentary sequences were processed like all the others, however the results from such sequences must be reviewed before accepting data from them and were filtered out when appropriate in the data sets presented here. In some circumstances it may be appropriate to replace missing regions with a standard sequence to estimate results for these fragments.

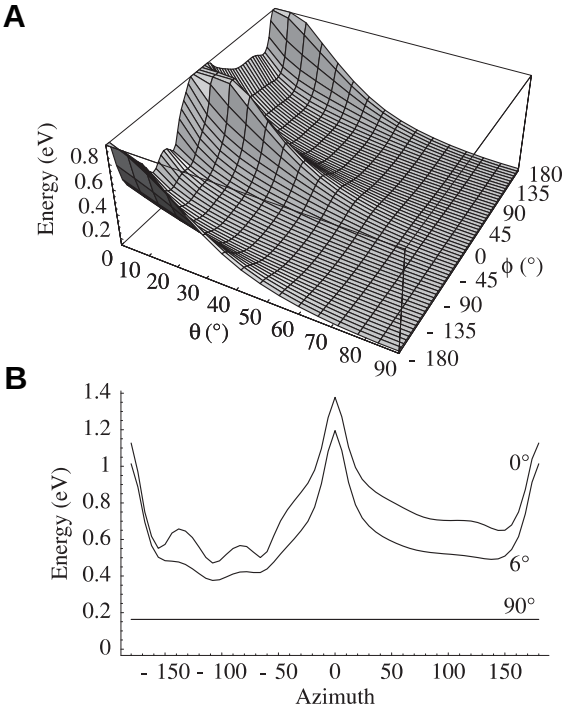

Fig. 6. Graph of energy level versus position of rigid C-terminal tail in a simplified electrostatic model. (A) *Energy of system as function of azimuth (ϕ) and declination (θ).* **(B)** *The energy levels from part (A) for declinations of 0°, 6° and 90° from top to bottom respectively. Note energy minima for sticking straight out (θ = 90°) and for lying on the surface (θ = 0°). See Priel* et al. *(2005).*

Currently, a total of three structures, providing a total of four templates are available in the Brookhaven Protein Databank upon which we can construct homology models of tubulin. The first and most obvious choice, was the initial Nogales structure of the α/β tubulin heterodimer (PDB identifier 1TUB) (Nogales, 1998). However, this structure was solved at a relatively low resolution and contains numerous omissions and misalignments due to difficulties in density fitting. A refinement of the 1TUB structure, PDB identifier 1JFF, was subsequently deposited with a slightly better resolution at 3.5 Å (Lowe, 2001). Along with increased resolution, the 1JFF structure also addressed a number of misalignment errors within the 1TUB structure and therefore makes a better choice for a homology modeling template. However, the 1JFF structure contains a large, nine-residue gap within the α tubulin M loop (serine 277 to leucine 286). Fortunately, this gap was not present in the 1TUB structure, providing us with a template structure for this region. Shortly after we began modeling the tubulin isoforms, a new structure of tubulin complexed with stathmin was released (PDB identifier 1SA0) (Ravelli, 2004). This structure was solved at 3.58 Å and contained most of the a

loop that was not present in the 1JFF structure. The presence of the complete missing a loop in 1TUB and partial loop in 1SAO provided us with adequate templates for modeling this section of the α-tubulins. A fourth template structure from which to build the tubulin homology models was available to us. This structure was that of the *E. coli* cell division protein 1FSZ (Lowe and Amos, 1998). Unfortunately, these five template structures are not entirely unique, the 1TUB and 1JFF being largely the same models.

When generating models of the α tubulin monomers, only the FtsZ and the two α-structures were input to Modeller as templates (Sanchez and Sali, 2000). Similarly, when modeling β-monomers only the FtsZ and the two β structures were used. In other cases all five structures were input. Models were also constructed without including the FtsZ structure, a putative prokaryotic homologue of tubulin (Erickson, 1997; 1998). These were judged to be insignificantly different. After constructing several test models we determined that the presence of extended loops within 1FSZ, that were absent from all of the tubulin sequences tended to alter the alignment scores and produce alignment anomalies between the tubulin alignments. It was therefore decided to eliminate the FtSZ structure from our templates and because of the high sequence homology between the β and α-tubulins we chose to use only 1JFF as our template.

Results

With the resulting library of structural tubulin models, various computational estimates of physical properties of the different tubulins may be made. These include the volume, surface area, net charge and dipole moments. All volumes and surface areas reported herein are calculated using a Connolly solvent accessible surface. Similarly, all dipole moments reported are with respect to the protein center-of-mass or alternatively a geometric mean of the model co-ordinates which is expected to be a good approximation of the former. We performed these calculations on the model structures, typically using analysis tools within the Gromacs (Lindahl, 2001) molecular dynamics package (version 3.1.4). Because only a single δ and a single ε sequence are present and these somewhat resemble γ, it is convenient to consider three groups: α, β and γ\δ\ε. We show some scatter plots of these three groups below in Figs. 1 and 2.

We also analyzed the properties of the C-terminal projection. We first needed to define this region. We used Clustal W (version 1.82) (Thompson, 1994) in order to obtain a multiple sequence alignment amongst the peptides. Such an alignment is illustrated in Fig. 3 which highlights variations in amino acid sequences. The multiple alignment then allows rapid identification of corresponding residues in all of sequences. Because this alignment is done amongst different sequences and with different parameters than the alignments performed with Modeller it may be that slight differences in the alignments have occurred. Studying these may be instructive in understanding some of the uncertainties associated with the Modeller outputs. However, we have ignored this possibility and defined residue 436 to be the first in the tail. This corresponds well to the beginning of the hypervariable region and the last contact of the backbone with the globular structure in 1TUB and 1JFF. From this definition, we can then readily read and analyze the C-terminal projection sequences for each tubulin species.

The library of tubulin structures including α, β and γ from animals, plants, fungi and protists, as well as δ and ε from humans, were analyzed by molecular mechanics to determine their net charges, dipole moment components, dipole orientations, volumes, surface areas and the lengths and charges of their C-termini. The results of our computations in this regard are shown in Appendix B.

Fig. 1A shows a scatter diagram of the net charge/volume ratios of the different tubulins. This plot is striking in that the net charge on the β-tubulins is by far the greatest ranging between 17 and 32 elementary charges (e) depending on the particular β tubulin with an average value in this case at approximately 25 e. Next come the α tubulins with net charges varying between 10 and 25 elementary charges. Finally, the γ, δ and ε tubulins have a much lower value of the net charge; in some cases being very close to zero. There appears to be little if any correlation between the size of a protein and its charge, although the γ-tubulins are noticeably larger in volume. Further, it should be kept in mind, that the charge on a tubulin dimer will be neutralized in solution due to the presence of counter-ions which almost completely screen the net charge. This was experimentally determined for tubulin by the application of an external electric field; the resulting value of an unscreened charge of approximately 0.2 e per monomer was found (Stracke, 2002). What is, however, of great interest in connection with polymerization of tubulin into MT's and with protein-protein and drug-protein binding is the actual distribution of charges on the surface of tubulin. Fig. 4 illustrates this for the Downing-Nogales structure with plus signs indicating the regions of positive charge and minus signs negatively charged locations. This figure shows C-termini in two very upright positions. Each of the different tubulins will show differences in this regard but lack of space prevents us from showing in detail where the differences occur for specific monomers.

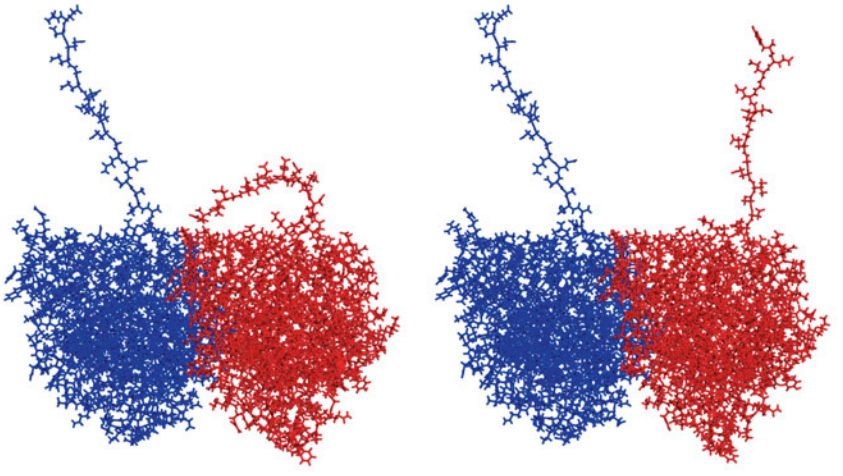

Fig. 7. Examples of C-termini in up and down positions. *Although only the C-terminus on an α - monomer is shown in a lowered state, both C-termini can be in either state.*

Fig. 1C shows the logarithm of surface area against the logarithm of volume for the different tubulins. Note the high level of correlation between the tubulin family and the surface-volume functional dependence. It is worth noting that the γ, δ and ε-tubulins are significantly more compact than the α tubulins, which are slightly more compact than the β-tubulins. We also show a best-fit line to these scatter plots for each of the three families. Note that the α and β families have a very similar slope with a value close to the unity that is indicative of cylindrical symmetry in the overall geometry. The γ, δ and ε-tubulins are characterized by a lower value of the slope indicating a greater amount of spherical symmetry for which surface-to-volume dependence scales with a power of 2/3.

As seen in Fig. 1B, α-tubulins have relatively low dipole moments about their centers-of-mass ranging between 1,000 and

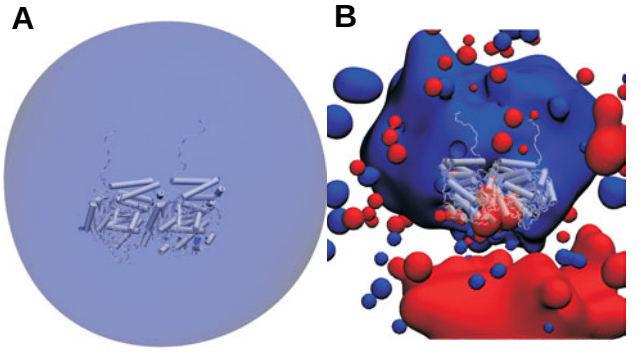

Fig. 8. A tubulin dimer in vacuum (A) *is seen principally as a monomer from the point of view of electrostatic interactions;* **(B)** *it is mainly dipolar when surrounded by counter ions and water. Figure prepared with VMD (Humphrey, 1996).*

2,000 debye, while the β-tubulins are very high in this regard with the corresponding values ranging between 1,000 and 4,000 debye and with the average value close to 3,000 debye. The γ-, δ- and ε-tubulins have dipole moments whose values are in between those two ranges. There is clearly some correlation between the magnitude of the dipole moment and volume for β- and γ-tubulins while the α tubulins are so compact that no correlation is readily apparent. In Fig. 2 we have illustrated the important aspect of dipole organization for tubulin, namely its orientation. Fig. 2A shows a Mollweide projection of dipole orientation in tubulin. The center of the diagram corresponds to the orientation perpendicular to the cylinder surface in a MT. The northerly direction represents the direction along the protofilament towards the plus-end while the southerly direction points towards the minus-end. Points along the horizontal axis are oriented tangentially to the MT surface. We conclude from this diagram and its magnification around the center region in Fig. 2 that both α- and β-tubulins orient their dipole moments in a direction that is close to being perpendicular to the MT surface. Notably, the same cannot be said about the dipole orientation for γ- tubulin, whose dipole orientation is largely tangential to the cylindrical surface of the MT formed out of its subunits.

The C-terminal tails (residues beyond the H-12 helix) display a largely random configuration, in stark contrast to the rest of the monomer. This is due to the inherent flexibility of this region of the sequence and the *in vacuo* simulation techniques employed by Modeler. While the variety of configurations was not unexpected they were included for completeness and do not detract from the rest of the model. Furthermore, they reflect the conformational flexibility of these regions. When tubulin is modeled as a dimer or monomer in solution, implicit or explicit, that simulates physiological conditions we see a variety of conformations for the C-terminal tails. While conformations that point away from the MT dominate (as would be entropically expected) the tails will also occasionally make contact with what would be the surface of the MT. Generally speaking, the C-terminal tails sweep out a large volume of space over several nanoseconds. Our models show that only α- and β-tubulins have C-terminal tails that project outwards from the tubulin, due to their high negative charges. Fig. 6 shows the energy levels of different orientations of the C-termini in a toy model with rigid tails and simplified electrostatics and suggests that there is relatively little energetic difference between the tails projecting straight outward from the rest of the tubulin and lying on the surface of tubulin in certain energy minima. While other surface states are quite strongly disfavored, the surface minima occur in positions where their negative charges form electrostatic bonds with certain positively charged residues on the tubulin surface. This variation is illustrated by Fig. 7. In a more realistic model this would need to be extended to consider different C-terminal structures and surface charges. This aspect is still underway and it will be extended to investigate the effects of post-translational modifications such as phosphorylation and polyglutamation on the equilibrium C-terminal states.

Discussion

The structures of α and β-tubulins are known to be quite similar, being nearly indistinguishable at approximately 6 Å (Li, 2002) despite only a 40% amino acid homology. Since the sequences within the α or β-tubulin family are more similar to each other than to those sequences belonging to the other families of tubulins, it is reasonable to believe that any given sequence should produce a structure very similar to another member of a given family. Further support for this comes from the published structures of Nogales (1998) and Lowe (2001) which are of a porcine sequence, but which were fit to structural data from an inhomogeneous bovine sample. Accordingly, by substituting appropriate amino acid side chains and properly adjusting other residues to accommodate insertions and deletions in the sequence, crystallographic structures can be used as a framework to produce model structures with different sequences with a high degree of confidence. Indeed, we have demonstrated that homology modeling produced a set of models that are essentially indistinguishable from one another. Therefore, homology modeling may not be an appropriate test, as it attempts to move the resulting models towards the templates. The universal convergence that we see here only proves that the structure can accommodate all of the tubulin sequences that we have investigated. While homology modeling or threading may be the best way to generate a large number of models, additional methods are required to test the stability of the resulting conformations obtained from the homology modeling. Having said that, it is still of interest to compare the results of our calculations and find possible consequences of the differences in the physical characteristics between various tubulin isoforms.

The dipole moment of tubulin could play a role in MT assembly and in other processes such as interactions with neighboring proteins and ions, espcially when the dipole moments can be ordered leading to collective effects (Brown and Tuszynski, 1999) or even dynamical propagating states (Trpisova and Tuszynski, 1997). This could be instrumental in the docking process of molecules to tubulin and in the proper steric configuration of a tubulin dimer as it approaches a MT for binding. An isolated (in vacuum) tubulin dimer has an electric field dominated by its net charge as seen in the nearly spherical isopotential surface in Fig. 8A. In contrast, a dimer, shown in Fig.8B, surrounded by water molecules and counter-ions, as is physiologically relevant, has an isopotential surface with two lobes much like the dumbbell shape of a mathematical dipole. In a MT, the individual dimers have near parallel dipole moments relative to their nearest neighbors; this is illustrated in Fig. 5. The greater the individual dipole of each of its units, the less stable the MT will generally be since these near parallel dipole-dipole interactions are repulsive, tending to push apart the MT. Note that the strength of the interaction potential is proportional to the square of the dipole moment; hence MT structures formed from tubulin units with larger dipole moments should be more prone to undergo disassembly catastrophes (Trpisova and Tuszynski, 1997) compared to those MTs that contain low dipole moment tubulins. For organisms that express more than one type of tubulin isoform in the same cell, one can conceive that MT dynamic behavior could be regulated by altering the relative amounts of the different isoforms according to their dipole moments. However, these effects are moderated by local interactions between two dimers, which produces a more complicated electrostatic field than the assembly of dipoles described here would suggest; for details see (Baker, 2001).

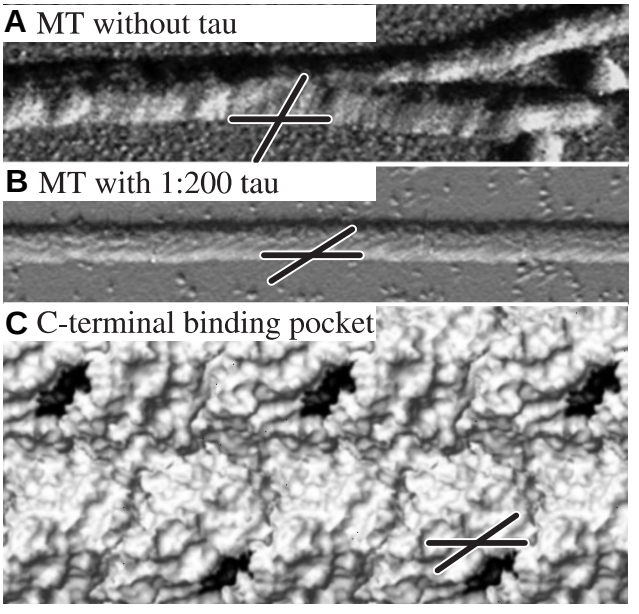

Fig. 9. Microtubule surface patterns show different angular structures. *STM observations of MTs* **(A)** *without tau MAPs and* **(B)** *with tau as reported by (Makrides, 2003).* **(C)** *A view of a microtubule surface with a C-terminal binding site observed in molecular dynamics simulations highlighted. The three angle markers are at 65°, 32.5° and 35°, respectively.*

In terms of surface/volume ratios, α and β-tubulins are the least compact, while γ, δ and ε are the most compact. There is abundant evidence that both α and β have variable conformations. This is attested to by their interaction with drugs and is consistent with the dynamic instability of MTs. In contrast, there is no evidence of γ, δ and ε participating in dynamic instability nor is there any theoretical reason to imagine such flexibility. It is reasonable to postulate that a less compact structure may have a more flexible conformation.

Isoform composition has a demonstrable effect on MT assembly kinetics (Panda, 1994; Banerjee and Kasmala, 1998). This could be due to changes in the electrostatics of tubulin, in particular, its net charge. This property although significantly screened by counter-ions does affect MT assembly by influencing dimer-dimer interactions over relatively short distances (approximately 5 nm) as well as the kinetics of assembly. These short-range interactions have been recently studied by (Sept, 2003) by calculating the energy of protofilament-protofilament interactions. These authors concluded from their work that the two types of MT lattices (A and B lattices differ in the relative positions of the protofilaments relative to each other) correspond to the local energy minima. However, the matters can be significantly more complicated since not only does tubulin exist as multiple isoforms, but the protein can also undergo various post-translational modifications, such as phosphorylation, acetylation, detyrosination and polyglutamylation (Banerjee, 2002; Luduena, 1998).

Our models predict that the C-terminal tails of α and β can readily adopt the two extreme conformations: either projecting outwards from the tubulin (and the MT surface) or lying on the surface, albeit such that their charged residues can form electrostatic bonds with complimentary charges on the surface. The state of the C-terminus (upright, down, or in intermediate states; see Fig. 7) is easily influenced by local ion concentrations including pH. This conformational complexity has many implications. First, a projecting C-terminus could play a major role in signaling. The fact that tubulin isoforms differ markedly in their C-termini (Sackett, 1995) suggests that specific sequences may mediate the functional roles of the isoforms. These sequences would be readily available for interactions with other proteins in a projecting C-terminus (Sarkar, 2001). Second, the C-termini are the sites of many of the post-translational modifications of tubulin, i.e. polyglutamylation, polyglycylation, detyrosination/tyrosination, removal of the penultimate glutamatic acid and phosphorylation of serine and tyrosine (Redeker, 1998). It is known that the C-termini are essential to normal MT function (Duan, 2002); a projecting C-terminus would be easily accessible to enzymes that effect these modifications and the modification could influence the likelihood of the C-terminus changing conformation. In addition, if the modification plays a role in signaling, then the signal would be readily available in a projecting C-terminus, as mentioned above. Third, projecting C-termini would automatically create spacing between MT's. It is known that MT's are never closely packed and are surrounded by what is sometimes referred to as an «exclusion zone» (Dustin, 1984). This is a region of space around them that is strongly unfavorable to the presence of other MT's in the vicinity. Although MAP's likely play a role in establishing such spacing, electrostatic repulsion among C-terminal ends might also influence this as well. The C-termini are the major sites of binding of the MAP's to tubulin. A projecting C-terminus may

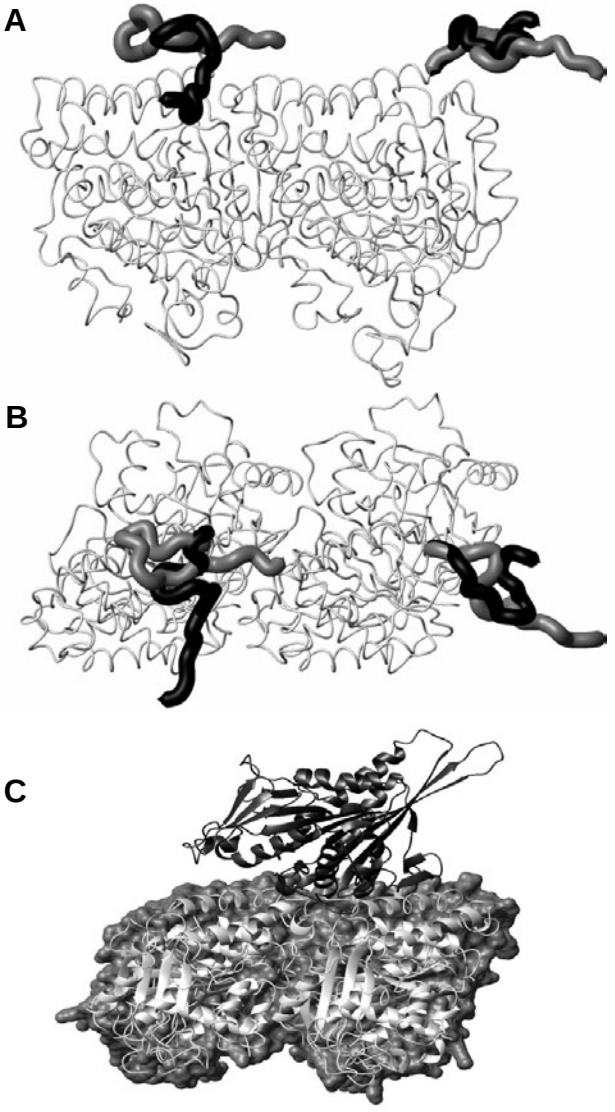

Fig. 10. Microtubule surface decorations. *Two states of the C-terminal tail are shown:* **(A)** *view tangential to the microtubule surface,* **(B)** *at right angles looking in from the outside of a microtubule. For comparison,* **(C)** *the tubulin-kinesin motor domain complex. This figure is from PDB file 1IA0 (Kikkawa, 2001) and shows two monomers from adjacent dimers with an -monomer on the left and a -monomer on the right.*

facilitate MAP binding and, conversely, MAP binding could influence the conformation of the C-terminus. Evidence for this is provided by the work of Makrides *et al*. (2003) who showed that when the MAP tau binds to MT's, it triggers a structural change on the MT surface whereby a structural element, presumably tau, lies on the surface of the MT forming a lattice whose alignment angle is much more acute than that of the lattice of tubulin subunits. This lattice is apparently superimposed on top of the normal MT (A or B) lattice. The orientations of the C-termini when they are lying on the surface of the MT form nearly the same kind of pattern that Makrides *et al*. (2003) observed, a striking confirmation of the potential accuracy of our modeling. The observed angles are listed in the caption of Fig. 9. These results raise the possibility that the orientation of the C-termini of the α and β-subunits determines the arrangement of tau molecules on the MT.

In addition, the state of the C-termini could mediate how motor proteins such as kinesin or dynein bind to and move on MT's. Our models show that kinesin binds preferentially to upright C-termini and not to C-termini lying on the surface of the MT (see Fig. 10). Very minor changes in the local ionic environment or the pH could halt the processive motion of a two-headed kinesin (Wriggers and Schulten, 1998) by collapsing the C-termini. One can postulate that the proportion of C-termini that are in the upright conformation in a given portion of the MT could determine the actual rate of kinesin movement. It is likely that such arguments could apply to other motor proteins as well. One might imagine that the very fine coordination of movements that occurs in processes such as mitosis could be influenced or even caused by the conformational state of the C-termini in particular areas of the MT. Finally, one can speculate that the C-termini could collapse in waves that could simultaneously couple to a wave of ions that could polarize or depolarize a membrane. This could be a form of MT signaling that has not yet been considered in the literature. A quantitative model of ionic wave transmission coupled to coordinated motion of the C-termini of dendritic MT's has been recently developed by Priel *et al*. (2005).

The role and function of MTs in embryos has been discussed in depth by Gordon (1999). Based on important studies of the multi-gene families of tubulin (Lewis and Cowan, 1990), it can be concluded that all tubulins in the same organism appear able to participate in all major functions. Nonetheless, some tubulins are produced primarily in certain tissues at specific stages of development. Interestingly, tissue-specific tubulins are often expressed at lower evolutionary levels in some other tissues. To complicate matters more, nearly identical tubulins may have entirely different patterns of expression. However, some tubulins differ in their ability to participate in dynamic equilibrium phenomena, their role in MT nucleation as well as sperm formation. Particularly intriguing is the fact that tubulins can even differ from one part of the cell in an embryo to another and during mitosis. This is suggestive of a specialized role of tubulin isoform in cell splitter effects. In particular, the role of tubulin isoforms was outlined by Gordon (1999) concluding with the following observations:

1. Different tubulin isoforms are involved in cortical rotation.
2. Tubulins in nematodes may vary between cell types.
3. The origin of tubulin isoforms coincides with the origin of new cell types and specifically with their cell state splitters (Kube-Granderath and Schliwa, 1998).
4. Some tubulin isoforms nevertheless occur in multiple tissues (Lu, 1998).
5. There is a highly probable and very intriguing role of tubulin isoforms in brain complexity and its origin.
6. Tubulin isoforms could be used to distinguish otherwise histologically equivalent cell types.

Two hypotheses regarding multiple tubulin gene sequences have been advanced, both of which may have some factual basis. First, individual tubulin genes may encode functionally divergent polypeptides conferring unique properties to the final MT that is polymerized by the cell. Alternatively, multiple polypeptides may themselves be functionally equivalent but represent the products of duplicated genes that have evolved to possess different regulatory sequences for activation of transcription during alternative

programs of differentiation.

While this paper attempted to shed light on the gross biophysical features distinguishing between different tubulins, we have been able to relate such differences only to the presence of the major sub-families. Very subtle differences in the amino-acid composition within each sub-family while still elude our ability to relate structure and function precisely, may contain significant changes in tubulin's functional domains. Our quest to uncover these relationships is still ongoing.

Acknowledgments

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Appendix A

Tubulin sequences

Achlya klebsiana	β TBB_ACHKL
Acremonium coenophialum	β TBB_ACRCO
Ajellomyces capsulata (Histoplasma capsulatum)	α TBA_AJECA; β TBB_AJECA
Anemia phyllitidis (Fern)	α TBA1_ANEPH, TBA2_ANEPH; β TBB1_ANEPH, TBB2_ANEPH, TBB3_ANEPH; γ TBG_ANEPH
Arabidopsis thaliana (Mouse-ear cress)	α TBA1_ARATH, TBA2_ARATH, TBA3_ARATH, TBA6_ARATH; β TBB1_ARATH, TBB2_ARATH, TBB4_ARATH, TBB5_ARATH, TBB6_ARATH, TBB7_ARATH, TBB8_ARATH, TBB9_ARATH; γ TBG2_ARATH
Aspergillus flavus	β TBB_ASPFL
Aspergillus parasiticus	β TBB_ASPPA
Avena sativa (Oat)	α TBA_AVESA; β TBB1_AVESA
Babesia bovis	β TBB_BABBO
Blepharisma japonicum	α TBA_BLEJA
Bombyx mori (Silk moth)	α TBA_BOMMO; β TBB_BOMMO
Botrytis cinerea (Botryotinia fuckeliana)	β TBB_BOTCI
Brugia pahangi	β TBB1_BRUPA
Caenorhabditis briggsae	β TBB7_CAEBR
Caenorhabditis elegans	α TBA2_CAEEL, TBA8_CAEEL; β TBB2_CAEEL, TBB4_CAEEL, TBB7_CAEEL; γ TBG_CAEEL
Candida albicans (Yeast)	α TBA_CANAL; β TBB_CANAL; γ TBG_CANAL
Cephalosporium acremonium (Acremonium chrysogenum)	β TBB_CEPAC
Chlamydomonas incerta	β TBB_CHLIN
Chlamydomonas reinhardtii	α TBA1_CHLRE, TBA2_CHLRE; β TBB_CHLRE; γ TBG_CHLRE
Chlorella vulgaris	α TBA_CHLVU
Chondrus crispus (Carragheen)	β TBB1_CHOCR
Cicer arietinum (Chickpea) (Garbanzo)	β TBB_CICAR
Cochliobolus heterostrophus (Drechslera maydis)	γ TBG_COCHE
Colletotrichum gloeosporioides (Anthracnose fungus) (Glomerella cingulata)	β TBB1_COLGL, TBB2_COLGL
Colletotrichum graminicola (Anthracnose fungus) (Glomerella graminicola)	β TBB1_COLGR, TBB2_COLGR
Cyanophora paradoxa	β TBB1_CYAPA
Daucus carota (Carrot)	β TBB1_DAUCA, TBB2_DAUCA
Dictyostelium discoideum (Slime mold)	α TBA_DICDI; β TBB_DICDI
Drosophila erecta (Fruit fly)	β TBB2_DROER
Drosophila melanogaster (Fruit fly)	α TBA1_DROME, TBA2_DROME, TBA3_DROME, TBA4_DROME; β TBB2_DROME, TBB3_DROME; γ TBG2_DROME
Ectocarpus variabilis	β TBB5_ECTVR, TBB6_ECTVR
Eimeria tenella	β TBB_EIMTE
Eleusine indica (Goosegrass)	α TBA1_ELEIN, TBA2_ELEIN, TBA3_ELEIN; β TBB1_ELEIN, TBB2_ELEIN, TBB3_ELEIN, TBB4_ELEIN
Emericella nidulans (Aspergillus nidulans)	α TBA1_EMENI, TBA2_EMENI; β TBB1_EMENI, TBB2_EMENI; γ TBG_EMENI
Entamoeba histolytica	α TBA1_ENTHI; γ TBG_ENTHI
Epichloe typhina	β TBB_EPITY
Erysiphe graminis (subsp. hordei) (Grass mildew)	β TBB_ERYGR
Erysiphe pisi (Pea powdery mildew)	β TBB2_ERYPI
Euglena gracilis	α TBA_EUGGR; β TBB_EUGGR
Euplotes aediculatus	γ TBG_EUPAE
Euplotes crassus	β TBB_EUPCR; γ TBG2_EUPCR
Euplotes focardii	β TBB_EUPFO
Euplotes octocarinatus	α TBA_EUPOC; β TBB_EUPOC; γ TBG2_EUPOC
Euplotes vannus	α TBA_EUPVA
Gadus morhua (Atlantic cod)	β TBB1_GADMO
Gallus gallus (Chicken)	α TBA1_CHICK, TBA2_CHICK, TBA3_CHICK, TBA4_CHICK, TBA5_CHICK, TBA8_CHICK; β TBB1_CHICK, TBB2_CHICK, TBB3_CHICK, TBB4_CHICK, TBB5_CHICK, TBB6_CHICK, TBB7_CHICK
Geotrichum candidum (Oospora lactis)	β TBB1_GEOCN, TBB2_GEOCN
Giardia lamblia (Giardia intestinalis)	β TBB_GIALA
Gibberella fujikuroi (Bakanae and foot rot disease fungus) (Fusarium moniliforme)	β TBB_GIBFU
Glycine max (Soybean)	β TBB1_SOYBN, TBB2_SOYBN, TBB3_SOYBN
Haemonchus contortus (Barber pole worm)	α TBA_HAECO
Haliotis discus (Abalone)	β TBB_HALDI
Homarus americanus (American lobster)	α TBA1_HOMAM, TBA2_HOMAM, TBA3_HOMAM; β TBB1_HOMAM, TBB2_HOMAM
Homo sapiens (Human)	α TBA1_HUMAN, TBA2_HUMAN, TBA4_HUMAN, TBA6_HUMAN, TBA8_HUMAN; β TBB1_HUMAN, TBB2_HUMAN, TBB4_HUMAN, TBB5_HUMAN, TBBQ_HUMAN, TBBX_HUMAN; γ TBG1_HUMAN, TBG2_HUMAN; δ TBD_HUMAN; ε TBE_HUMAN
Hordeum vulgare (Barley)	α TBA1_HORVU, TBA2_HORVU, TBA3_HORVU; β TBB_HORVU
Leishmania mexicana	β TBB_LEIME
Leptomonas seymouri	α TBA_LEPSE
Lupinus albus (White lupine)	β TBB1_LUPAL, TBB2_LUPAL
Lymnaea stagnalis (Great pond snail)	β TBB_LYMST
Lytechinus pictus (Painted sea urchin)	α TBA_LYTPI; β TBB_LYTPI
Manduca sexta (Tobacco hawkmoth) (Tobacco hornworm)	β TBB1_MANSE
Mus musculus (Mouse)	α TBA1_MOUSE, TBA2_MOUSE, TBA3_MOUSE, TBA6_MOUSE, TBA8_MOUSE; γ TBG1_MOUSE, TBG2_MOUSE
Mycosphaerella graminicola (Septoria tritici)	α TBA_MYCGR
Mycosphaerella pini (Dothistroma pini)	β TBB_MYCPJ
Naegleria gruberi	α TBA_NAEGR; β TBB_NAEGR
Neurospora crassa	α TBA1_NEUCR, TBA2_NEUCR; β TBB_NEUCR; γ TBG_NEUCR
Notophthalmus viridescens (Eastern newt) (Triturus viridescens)	α TBA_NOTVI
Notothenia coriiceps neglecta (Black rockcod) (Yellowbelly rockcod)	β TBB1_NOTCO
Octopus dofleini (Giant octopus)	α TBA_OCTDO; β TBB_OCTDO
Octopus vulgaris (Octopus)	α TBA_OCTVU

Onchocerca gibsoni	β TBB_ONCGI
Oncorhynchus keta (Chum salmon)	α TBA_ONCKE
Oncorhynchus mykiss (Rainbow trout) (Salmo gairdneri)	α TBAT_ONCMY
Oryza sativa (Rice)	α TBA1_ORYSA; β TBB1_ORYSA, TBB2_ORYSA, TBB3_ORYSA; γ TBG2_ORYSA
Oxytricha granulifera	α TBA_OXYGR
Paracentrotus lividus (Common sea urchin)	α TBA1_PARLI; β TBB_PARLI
Paramecium tetraurelia	β TBB1_PARTE
Patella vulgata (Common limpet)	α TBA2_PATVU
Pelvetia fastigiata	α TBA1_PELFA, TBA2_PELFA
Penicillium digitatum	β TBB_PENDI
Pestalotiopsis microspora	β TBB_PESMI
Phaeosphaeria nodorum (Septoria nodorum)	β TBB_PHANO
Physarum polycephalum (Slime mold)	α TBAD_PHYPO, TBAE_PHYPO, TBAN_PHYPO; β TBB1_PHYPO, TBB2_PHYPO
Physcomitrella patens (Moss)	γ TBG_PHYPA
Phytophthora cinnamomi	β TBB_PHYCI
Picea abies (Norway spruce) (Picea excelsa)	α TBA_PICAB
Pisum sativum (Garden pea)	α TBA1_PEA; β TBB1_PEA, TBB2_PEA, TBB3_PEA
Plasmodium berghei yoelii	α TBA_PLAYO
Plasmodium falciparum	α TBA_PLAFK; β TBB_PLAFA, TBB_PLAFK; γ TBG_PLAFO
Plasmodium falciparum (isolate K1 / Thailand)	α TBA_PLAFK; β TBB_PLAFK
Plasmodium falciparum (isolate NF54)	γ TBG_PLAFO
Pleurotus sajor-caju (Oyster mushroom)	β TBB_PLESA
Pneumocystis carinii	α TBA1_PNECA, TBAA_PNECA; β TBB_PNECA
Polytomella agilis	β TBB_POLAG
Porphyra purpurea	β TBB1_PORPU, TBB2_PORPU, TBB3_PORPU, TBB4_PORPU
Prunus dulcis (Almond) (Prunus amygdalus)	α TBA_PRUDU
Pseudopleuronectes americanus (Winter flounder) (Pleuronectes americanus)	β TBB_PSEAM
Rattus norvegicus (Rat)	β TBB1_RAT
Reticulomyxa filosa	γ TBG_RETFI
Rhynchosporium secalis	β TBB_RHYSE
Saccharomyces cerevisiae (Baker's yeast)	α TBA1_YEAST, TBA3_YEAST; β TBB_YEAST; γ TBG_YEAST
Schizophyllum commune (Bracket fungus)	α TBAA_SCHCO, TBAB_SCHCO; β TBB_SCHCO
Schizosaccharomyces japonicus	γ TBG_SCHJP
Schizosaccharomyces pombe (Fission yeast)	α TBA1_SCHPO, TBA2_SCHPO; β TBB_SCHPO; γ TBG_SCHPO
Solanum tuberosum (Potato)	β TBB1_SOLTU, TBB2_SOLTU
Sordaria macrospora	α TBA_SORMA
Strongylocentrotus purpuratus (Purple sea urchin)	β TBB_STRPU
Stylonychia lemnae α TBA1_STYLE, TBA2_STYLE;	β TBB_STYLE
Sus scrofa (Pig) α TBA_PIG;	β TBB_PIG
Tetrahymena pyriformis	α TBA_TETPY; β TBB_TETPY
Tetrahymena thermophila	α TBA_TETTH; β TBB_TETTH
Thalassiosira weissflogii (Marine diatom)	β TBB_THAWE
Torpedo marmorata (Marbled electric ray)	α TBA_TORMA
Toxoplasma gondii α TBA_TOXGO;	β TBB_TOXGO
Trichoderma viride	β TBB1_TRIVI,'TBB2_TRIVI
Triticum aestivum (Wheat)	α TBA_WHEAT; β TBB1_WHEAT, TBB2_WHEAT,—TBB3_WHEAT, TBB4_WHEAT, TBB5_WHEAT
Trypanosoma brucei rhodesiense	α TBA_TRYBR; β'TBB_TRYBR
Trypanosoma cruzi	α TBA_TRYCR; β TBB_TRYCR
Ustilago violacea (Smut fungus) (Microbotryum violaceum)	γ TBG_USTVI
Venturia inaequalis	β TBB_VENIN
Volvox carteri	α TBA1_VOLCA; β TBB1_VOLCA
Xenopus laevis (African clawed frog)	α TBA_XENLA; β'TBB2_XENLA, TBB4_XENLA; γ TBG_XENLA
Zea mays (Maize)	α TBA1_MAIZE,'TBA2_MAIZE, TBA3_MAIZE, TBA4_MAIZE, TBA5_MAIZE, TBA6_MAIZE; β TBB1_MAIZE, TBB2_MAIZE, TBB3_MAIZE, TBB4_MAIZE, TBB5_MAIZE, TBB6_MAIZE, TBB7_MAIZE,'TBB8_MAIZE; γ TBG1_MAIZE, TBG2_MAIZE, TBG3_MAIZE

Appendix B

Summary of computation results

Each of the 290 Swiss-Prot Sequences is listed below with some results. Fragmentary sequences are marked with an initial “F”. Charges are reported in fundamental charge (electronic) units.

	Name	Net		Volume	Area	C-Terminal Tail		
		|M|	Charge	(Å³)	(Å²)	Charge	Size	Sequence
	TBA1_ANEPH	1.344	-22	43.723	46.120	-8	16	GAESGEGEEGDEGEEY
	TBA1_ARATH	1.574	-24	43.726	46.097	-9	15	GGEGAEDDDEEGDEY
F	TBA1_CHICK	1.904	-21	40.490	43.082	-8	16	GVDSVEGEGEEEGEEY
	TBA1_CHLRE	1.042	-21	43.643	45.934	-6	16	GAESAEGAGEGEGEEY
	TBA1_DROME	1.225	-22	44.031	46.824	-6	15	GMDSGDGEGEGAEEY
	TBA1_ELEIN	1.407	-24	43.861	46.749	-9	16	GAEFDEGEEGDEGDEY
	TBA1_EMENI	1.467	-24	44.070	46.434	-7	14	ASDSLEEEGEEVEY
	TBA1_ENTHI	730	-10	44.061	46.461	-3	14	ALDNTIEGESMTAQ
	TBA1_HOMAM	1.502	-22	44.167	46.824	-7	16	GMDSADGEDIEGGDEY
	TBA1_HORVU	1.321	-23	43.591	45.827	-8	15	GAEGADDEGDEGDDY
	TBA1_HUMAN	1.449	-24	44.250	47.174	-8	16	GVDSVEGEGEEEGEEY
	TBA1_MAIZE	1.323	-24	43.835	46.652	-9	16	GAEFDEGEDGDEGDEY
	TBA1_MOUSE	1.470	-24	44.263	47.102	-8	16	GVDSVEGEGEEEGEEY
	TBA1_NEUCR	970	-19	44.052	46.358	-5	13	AGDYNDVDVDAEY
	TBA1_ORYSA	1.625	-24	43.648	45.940	-8	15	GAEGADDENDDGEDY
	TBA1_PARLI	2.160	-25	44.184	46.804	-9	17	GVDSADAEGEEEEGDEY
	TBA1_PEA	1.344	-23	43.568	45.724	-8	17	GAESGDGDDDGLGEEEY
	TBA1_PELFA	1.681	-24	43.907	46.568	-9	18	GAETADGDGEEEEFGEEY
	TBA1_PNECA	1.352	-20	44.335	47.012	-6	14	GQDSIEGEIMEEEY
	TBA1_SCHPO	1.411	-22	44.895	47.968	-7	16	GQDSMDNEMYEADEEY
	TBA1_STYLE	1.531	-23	43.243	45.451	-7	15	GIEIVEGEGEEEGME
	TBA1_VOLCA	1.063	-21	43.630	45.981	-6	16	GAESAEGAGEGEGEEY
	TBA1_YEAST	1.083	-22	43.874	46.462	-5	11	GADSYAEEEEF
F	TBA2_ANEPH	1.640	-15	35.461	37.487	-6	13	AAEGVDEPEGDDY
	TBA2_ARATH	1.632	-23	43.766	46.803	-9	15	GAEGGDDEDDEGEEY
	TBA2_CAEEL	1.373	-22	43.891	46.319	-7	15	GADSNEGGEEEGEEY
	TBA2_CHICK	1.140	-25	43.774	46.365	-6	14	ATDLFEDENEAGDS
	TBA2_CHLRE	1.146	-22	43.601	45.661	-6	16	GAESAEGAGEGEGEEY
	TBA2_DROME	1.300	-21	44.117	46.892	-6	14	GIDSTTELGEDEEY
	TBA2_ELEIN	1.444	-21	43.843	45.941	-8	12	GAEVEEDDEEEY
	TBA2_EMENI	1.570	-21	44.173	46.890	-6	14	AGDSLDMEGEEAEY
	TBA2_HOMAM	1.387	-20	44.252	47.078	-8	16	GIDTADGEDDEEANDY
	TBA2_HORVU	1.542	-24	43.706	46.254	-9	16	GAEFDDGEDGDEGDEY
	TBA2_HUMAN	1.494	-23	44.046	46.631	-7	15	GVDSVEAEAEEGEEY
	TBA2_MAIZE	1.353	-24	43.670	46.060	-9	16	GAEFDEGEEGDDGDEY
	TBA2_MOUSE	1.526	-24	44.189	46.902	-8	16	GVDSVEGEGEEEGEEY
	TBA2_NEUCR	1.189	-22	43.970	46.398	-6	14	AADSMEGEDVEAEY
	TBA2_PATVU	1.727	-24	44.206	46.802	-8	17	GVDSVEGEGEEEGGEEY
	TBA2_PELFA	1.823	-25	43.972	46.730	-9	18	GAETAEGEGEEEDFGEEY
	TBA2_SCHPO	1.057	-23	44.414	47.084	-6	14	GQDSMEVDYMEEEY
	TBA2_STYLE	1.868	-21	43.463	45.795	-7	15	GIETAEGEGEEEGME
	TBA3_ARATH	1.455	-23	43.768	46.341	-9	15	GAEGGDDEEDEGEDY
F	TBA3_CHICK	1.425	-11	31.862	34.077	-3	12	GRNSADGGEFEE
	TBA3_DROME	1.126	-22	44.025	46.744	-6	15	GMDSGDGEGEGAEEY
	TBA3_ELEIN	1.341	-24	43.623	45.927	-8	15	GAEGADDEGDEGEDY
	TBA3_HOMAM	1.742	-24	44.024	46.425	-8	16	GVDSADAEGEEEGEEY
	TBA3_HORVU	1.229	-24	43.774	46.615	-9	16	GAEFDEGEDGDEGDEY
	TBA3_MAIZE	1.600	-20	43.523	45.861	-8	15	AAEGGSDDGDEEEEY
	TBA3_MOUSE	1.494	-23	44.046	46.631	-7	15	GVDSVEAEAEEGEEY
	TBA3_YEAST	1.380	-20	43.773	46.394	-3	9	GADSYAEEF
F	TBA4_CHICK	1.816	-19	31.763	34.085	-7	12	GTDSFEDENDEE
	TBA4_DROME	1.362	-18	44.750	46.802	-8	16	GLDNAEEGGDEDFDEF
	TBA4_HUMAN	1.486	-24	44.006	46.802	-8	13	GIDSYEDEDEGEE
F	TBA4_MAIZE	786	-13	5.653	6.442	-9	15	AEFDEGEDGDEGDEY
	TBA5_CHICK	1.423	-24	44.001	46.787	-8	13	GLDSYEDEEEGEE
	TBA5_MAIZE	1.259	-24	43.664	46.181	-8	15	GAEGADDEGDEGDDY
	TBA6_ARATH	1.472	-23	43.549	45.981	-9	15	GAEGGDDEDDEGEEY
	TBA6_HUMAN	1.230	-23	44.020	46.936	-7	14	GADSADGEDEGEEY
	TBA6_MAIZE	1.304	-24	43.616	45.962	-8	15	GAEGADDEGDEGDDY
	TBA6_MOUSE	1.262	-23	44.005	46.878	-7	14	GADSAEGDDEGEEY
	TBA8_CAEEL	1.483	-21	44.092	46.452	-6	14	SRDTADLEEENDEF
F	TBA8_CHICK	1.420	-17	31.941	34.148	-7	14	GTDSMDGEDEGEEY
	TBA8_HUMAN	1.326	-24	44.109	46.847	-7	14	GTDSFEEENEGEEF
	TBA8_MOUSE	1.109	-23	44.094	46.772	-7	14	GTDSFEEENEGEEF
F	TBA_AJECA	800	-11	40.916	42.810	0	0	
F	TBAA_PNECA	701	0	21.164	22.926	0	0	
	TBAA_SCHCO	1.160	-20	43.529	46.458	-6	14	GMDSADAEEEAAEY
	TBA_AVESA	672	-17	43.193	45.318	-3	12	GAEGADDEVTRG
F	TBA_BLEJA	135	-17	4.939	5.727	-8	18	GIETAEAEGEEEGYGEEL
	TBA_BOMMO	1.189	-23	44.003	46.588	-6	15	GMDSAEGEGEGAEEY
	TBAB_SCHCO	1.389	-17	43.480	46.447	-6	13	GTDSADAEEEGEY
	TBA_CANAL	1.610	-20	43.827	46.383	-6	12	GTDSFPEEEEEY
	TBA_CHLVU	1.279	-23	43.800	46.512	-8	16	GAESAEADGEDEGEEY
	TBA_DICDI	988	-15	44.898	47.487	-6	13	SASTEGEEQEEEY
	TBAD_PHYPO	1.513	-22	43.833	46.203	-6	15	GAESSEAGGDEEGEY

	Name	Net		Volume	Area	C-Terminal Tail		
		|M|	Charge	(Å^3)	(Å^2)	Charge	Size	Sequence
	TBA_PLAFK	1.639	-22	44.160	46.869	-8	18	GIESNEAEGEDEGYEADY
F	TBA_PLAYO	1.414	-12	19.400	20.787	-7	15	GIETNDGEGEDEGYE
	TBA_PRUDU	1.362	-23	43.612	46.258	-8	15	GAESAEGEDDEGDDY
	TBA_SORMA	1.057	-23	43.781	46.691	-6	14	AADSMEGEEVEAEY
	TBA_TETPY	1.189	-21	43.728	46.142	-6	14	GIETAEGEGEEEGY
	TBA_TETTH	1.116	-21	43.758	46.335	-6	14	GIETAEGEGEEEGY
	TBAT_ONCMY	1.411	-23	44.043	46.641	-7	15	GVDSVEGEAEEGEEY
	TBA_TORMA	1.334	-24	44.319	47.358	-8	16	GVDSVEGEGEEEGEEY
	TBA_TOXGO	1.562	-23	44.098	46.709	-8	18	GIETAEGEGEEEGYGDEY
	TBA_TRYBR	1.334	-24	43.868	46.477	-8	16	GAESADMDGEEDVEEY
	TBA_TRYCR	1.425	-25	43.758	46.172	-8	16	GAESADMEGEEDVEEY
	TBA_WHEAT	1.561	-24	43.805	46.562	-9	16	GAEFDEGEDGDEGDEY
	TBA_XENLA	1.313	-23	43.943	46.479	-7	14	GADSADAEDEGEEY
F	TBB1_ANEPH	1.688	-21	43.331	45.950	-8	18	QDATAEREGEYEEDYDEA
	TBB1_ARATH	3.129	-27	43.752	46.147	-12	21	QDATADEEDEYDEEEEQVYES
	TBB1_AVESA	3.182	-25	38.101	41.157	-13	23	QDATADEEGEYEDEEEDLQAEDM
	TBB1_BRUPA	1.961	-26	43.981	46.705	-11	23	QDATADEEGDLQEGESEYIEQEE
	TBB1_CHICK	2.149	-25	43.815	46.865	-11	20	QDATADEQGEFEEEGEEDEA
	TBB1_CHOCR	1.895	-27	43.978	45.919	-13	26	GEATADGVEGYEEEGYENDHPEDDEE
	TBB1_COLGL	3.581	-24	43.617	45.527	-13	22	QDAGMDDDEAEEAYEEEEPVEE
	TBB1_COLGR	2.349	-24	43.341	45.418	-12	22	QDAGMDDEYGEEYEDEAPAEEE
	TBB1_CYAPA	2.061	-25	43.704	46.639	-11	22	QDATAEEEGEGDEEEAEGEAAA
	TBB1_DAUCA	3.080	-17	31.338	33.360	-12	22	QDATADEEEYYEDEEEEAQGM
	TBB1_ELEIN	3.102	-26	43.750	46.610	-12	20	QDATAEDEEEYEDEEEEMAA
	TBB1_EMENI	2.478	-23	43.751	46.675	-11	22	QDASISEGEEEYAEEEIMEGEE
	TBB1_GADMO	1.950	-25	43.818	47.122	-11	20	QDATAEEEGEFEEEGEEELA
	TBB1_GEOCN	1.914	-24	43.809	46.274	-11	22	QNATVDDEDMEYEDELPLEDEM
	TBB1_HOMAM	3.186	-24	44.266	45.948	-11	22	QEATADDEAEFEEEGEVEGEYA
	TBB1_HUMAN	2.504	-25	43.765	46.854	-11	19	QDATAEEEEDFGEEAEEEA
	TBB1_LUPAL	3.531	-25	43.898	46.734	-12	22	QDATADEDGYEYEDEEEIGEEA
	TBB1_MAIZE	2.840	-25	43.777	46.781	-12	21	QDATADEEGEYEDEEEGDLQD
	TBB1_MANSE	1.831	-25	44.083	46.838	-12	22	QEATADEDAEFDEEQEQEIEDN
	TBB1_NOTCO	3.004	-25	43.698	46.443	-11	21	QDATAEEEGEFEEEGEYEDGA
	TBB1_ORYSA	2.216	-25	43.757	46.832	-12	19	QDATADEEYEDEEEEAEAE
	TBB1_PARTE	2.083	-24	43.491	46.266	-9	17	QDATAEEEGEFEEEGEQ
F	TBB1_PEA	3.760	-26	44.209	46.988	-13	25	QDATADEDEYGEEEGDEEEYGQHDI
	TBB1_PHYPO	1.794	-23			-9	42	QDATIDDEEGGEEEEGGAEEEARQRKHYVIDYVPSVCVILIR
F	TBB1_PORPU	2.434	-28	43.887	47.046	-11	32	EAATVEGEEEEDAYAEGAVVNGDQSYEDQYAA
	TBB1_RAT	3.037	-25	43.856	46.824	-11	20	QDATADEQGEFEEEEGEDEA
	TBB1_SOLTU	3.446	-26	43.921	45.964	-12	23	QDATADDEEEYDDEAADDHQYES
	TBB1_SOYBN	2.056	-22	43.717	46.392	-10	20	QDATAVDDHEDEDEDEAMAA
	TBB1_TRIVI	1.906	-21	43.239	45.387	-10	19	QDASADDGEEYEEDAPMEE
	TBB1_VOLCA	2.371	-24	43.623	46.596	-10	18	QDASAEEEGEFEGEEEEN
	TBB1_WHEAT	2.331	-25	44.053	47.377	-12	20	QDATADEEGEYEEEEELEQE
	TBB2_ANEPH	1.809	-18	40.451	43.580	-8	18	QDATAEPEGXYEEDYDEA
	TBB2_ARATH	4.027	-27	44.380	47.533	-14	25	QDATADEEGDYEDEEEGEYQQEEEY
	TBB2_CAEEL	3.151	-24	44.042	46.516	-10	25	QEATAEDDVDGYAEGEAGETYESEQ
	TBB2_CHICK	2.944	-24	43.791	46.642	-11	20	QDATADEQGEFEEEGEEDEA
	TBB2_COLGL	2.737	-24	43.772	46.801	-13	22	QDAGVDEEEEEYEEEAPLEEEV
	TBB2_COLGR	2.964	-24	43.776	46.726	-13	22	QDAGVDEEEEEYEDDAPLEEEV
	TBB2_DAUCA	1.849	-25	43.469	46.734	-9	19	QDATAEEDDYDDGEGSTGD
	TBB2_DROER	3.591	-25	43.757	46.469	-12	21	QEATADEEGEFDEDEEGGGDE
	TBB2_DROME	3.031	-26	43.647	46.257	-12	21	QEATADEEGEFDEDEEGGGDE
	TBB2_ELEIN	2.516	-26	44.115	47.287	-13	23	QDATADEDGEYEDELDGQEEEDM
	TBB2_EMENI	2.477	-22	43.740	46.549	-10	24	QEATVSDGEGAYDAEEGEAYEQEE
	TBB2_ERYPI	2.621	-22	43.844	46.800	-11	22	QDASISEGEEEYEEEQQLENEE
	TBB2_GEOCN	2.995	-28	44.193	46.317	-12	26	QNAGVDEDEELMDHEEYADEGVEDFN
	TBB2_HOMAM	2.078	-24	44.467	45.943	-12	22	QEATADDEAEFEEEGEVEGEYD
	TBB2_HUMAN	2.311	-25	43.874	47.064	-11	20	QDATAEEEGEFEEEAEEEVA
	TBB2_LUPAL	4.016	-26	44.007	46.759	-13	23	QDATADEDGYEYEDEEEVGEEDA
	TBB2_MAIZE	2.952	-23	43.628	46.573	-10	19	QDATADEEADYEEEEAAAE
	TBB2_ORYSA	2.060	-25	44.025	47.077	-12	22	QDATADEEGEYEDEEQQEADDM
	TBB2_PEA	3.660	-28	44.120	47.264	-14	24	QDATAEEDEYEEEEEDYHQEHDEM
	TBB2_PHYPO	3.796	-24	44.198	47.050	-12	29	QEASVDDEAMEDDAEAEGGAGQNEAVEEF
	TBB2_PORPU	2.073	-27	41.546	44.677	-13	23	QDATAEEEGEYDEDEDDEGGDYA
	TBB2_SOLTU	2.778	-26	44.047	46.135	-12	24	QDATADDEEEYDDEAADDHHQYES
	TBB2_SOYBN	3.324	-26	44.355	47.559	-14	24	QDATADEDEYEEEEEEEEFAQHDM
	TBB2_TRIVI	2.655	-24	43.739	46.060	-12	21	QEAGIDEEEEYEDEAPMEAEE
	TBB2_WHEAT	3.606	-27	43.865	46.565	-13	22	QDATSDEEGEYEDEDQEPEEDM
	TBB2_XENLA	2.320	-24	43.640	46.526	-10	18	QDATADEQGEFEEEEDEA
F	TBB3_ANEPH	2.202	-9	24.028	25.945	-8	18	QDASAEPXXEQEEDYEEA
	TBB3_CHICK	3.524	-26	43.756	46.491	-12	20	QDATAEEEGEFEEEAEEEAE
	TBB3_DROME	3.220	-23	44.396	46.320	-10	23	QEATADDEFDPEVNQEEVEGDCI
	TBB3_ELEIN	3.004	-27	43.974	47.142	-13	21	QDATAEEEEEYDDDEEEEVAA
	TBB3_MAIZE	3.339	-25	43.486	46.041	-12	20	QDATAEEYDEEEQDGEEEHD
	TBB3_ORYSA	3.258	-27	43.797	46.374	-13	22	QDATADDEEEDYGDEEEDEVAA
F	TBB3_PEA	3.520	-27	43.323	46.649	-14	24	QDATADEEGEYEDEEEEEPEHGYE
	TBB3_PORPU	3.074	-26	43.530	46.185	-13	20	QEASADDEADEFDEEEGDEE
F	TBB3_SOYBN	1.142	-9	40.339	43.199	0	0	
	TBB3_WHEAT	3.487	-28	43.671	46.343	-13	20	QDATADEEEEYDEEEEEEAA
	TBB4_ARATH	2.867	-25	43.751	46.536	-11	19	QDATAGEEEYEEEEEEYET
	TBB4_CAEEL	2.493	-24	43.684	46.649	-10	19	QEATADDEGEFDEHDQDVE
	TBB4_CHICK	3.244	-24	44.049	46.491	-11	24	QDATAEEEGEMYEDDEEESEQGAK
	TBB4_ELEIN	2.724	-25	43.906	46.994	-12	21	QDATADEEGDYEDEDEALHDE
	TBB4_HUMAN	2.494	-25	44.223	47.074	-11	25	QDATAEEEGEMYEDDEEESEAQGPK
	TBB4_MAIZE	2.580	-24	43.757	46.284	-11	20	QDATAEEYEEEEHDGEEEHA

	Name	Net	Volume	Area	C-Terminal Tail		
		|M|	Charge	(Å^3)	(Å^2)	Charge	Size	Sequence
	TBB8_ARATH	3.721	-25	44.295	47.021	-13	24	QDATADEEEGYEYEEDEVEVQEEQ
	TBB8_MAIZE	2.389	-25	43.822	47.070	-11	20	QDATADEEAEYEDEEAIQDE
	TBB9_ARATH	2.221	-27	43.541	46.710	-12	19	QDATVGEEEYEEDEEEEEA
	TBB_ACHKL	3.371	-27	43.582	46.093	-12	21	QDATAEEEGEFDEDEEMDEMM
	TBB_ACRCO	3.306	-25	44.030	47.131	-13	23	QDAGIDEEEEEYEEEAPVDEPLE
	TBB_AJECA	2.016	-17	43.900	46.112	-6	19	QDASISEGEDEYFDYAWAM
	TBB_ASPFL	3.877	-23	43.818	46.347	-12	23	QDASISEGEEEYLEEEEPLEHEE
	TBB_ASPPA	3.766	-23	43.974	46.808	-12	23	QDASISEGEEEYLEEEEPLEHEE
	TBB_BABBO	1.296	-22	43.390	46.335	-7	16	QEATIDDDADDMVNDY
	TBB_BOMMO	3.873	-25	44.236	47.190	-14	25	QDATIDQEFEDEEEVEEQNDDSDEQ
	TBB_BOTCI	2.912	-22	43.734	46.687	-11	22	QDASISEGEEEYEEEVPIEGEE
	TBB_CANAL	2.998	-27	43.650	45.896	-11	24	QEASIDEEELEYADEIPLEDAAME
	TBB_CEPAC	3.089	-24	43.756	46.590	-13	22	QDAGIDEEEEEYEEELPLEGEE
	TBB_CHLIN	2.565	-24	43.441	46.047	-10	18	QDASAEEEGEFEGEEEEA
	TBB_CHLRE	2.107	-24	43.581	46.513	-10	18	QDASAEEEGEFEGEEEEA
	TBB_CICAR	3.368	-26	44.094	46.346	-13	22	QDAIAEEEDEYEEEGEEQYDEQ
	TBB_DICDI	2.376	-25	44.756	46.369	-12	25	SNQETEEDGGEYQEEHEEHEEQAEN
	TBB_EIMTE	2.763	-24	43.849	46.691	-11	24	QDATAEEEGEFDEEEGVMDAEGAA
	TBB_EPITY	2.713	-24	43.982	47.088	-13	23	QDAGIDEEEEEYEEEAPVDEPLE
	TBB_ERYGR	3.174	-21	43.521	46.022	-10	21	QEASISEGEEEYPEEVSNEEE
	TBB_EUGGR	2.075	-28	43.339	45.941	-11	20	QDATVEEEGEFDEEEDVEQY
	TBB_EUPCR	2.969	-26	43.733	46.318	-11	21	QDATAEEEGEYVEDEDEMDGM
	TBB_EUPFO	2.381	-23	43.737	46.622	-11	19	QDATAEEEGEFDDEEEMDV
	TBB_EUPOC	2.789	-25	43.474	46.099	-10	19	QDATAEEEGEMDEEEGAME
	TBB_GIALA	2.850	-24	43.973	47.134	-12	21	QEAGVDEGEEFEEEEDFGDEQ
	TBB_GIBFU	3.764	-24	43.718	46.451	-13	21	QDAGIDEEEEEYEEELPEGEE
F	TBB_HALDI	1.426	-5	33.790	36.441	0	0	
	TBB_HORVU	3.816	-27	43.828	46.711	-13	22	QDATADEEGEYEDEDQEAEDDM
	TBB_LEIME	2.182	-25	43.641	46.103	-10	18	QDATVEEEGEFDEEEEAY
F	TBB_LYMST	1.499	-16	11.211	12.653	-12	24	QDATAEDEGEFDEEEAEGEGQEYA
F	TBB_LYTPI	3.329	-13	17.813	19.768	-12	21	QDATAEEEGEFDEEEGDEEAA
	TBB_MYCPJ	3.541	-22	43.590	46.322	-11	22	QEASVSEGEEEYDEEAPLEGEE
	TBB_NAEGR	3.441	-26	44.385	47.524	-12	26	QDATAEEEGEFDENEGAEGEEQPADY
	TBB_NEUCR	3.951	-24	43.679	46.401	-13	22	QDAGVDEEEEEYEEEAPLEGEE
	TBB_OCTDO	2.060	-23	44.106	46.618	-8	22	QEARSTDSDEYDNEEYYNQQEE
	TBB_ONCGI	2.425	-21	43.866	46.451	-8	19	QDASADDELNETIEQAETE
	TBB_PARLI	3.785	-26	43.883	46.679	-13	22	QDATAEEEGEFDEEEEGDEEAA
	TBB_PENDI	2.715	-21	43.815	46.891	-10	22	QEASVSEGEEEYLAEDIVDEEV
	TBB_PESMI	2.968	-24	43.723	46.551	-13	21	QDAGVDEEEEEYEEEPLPEDE
	TBB_PHANO	2.991	-21	43.778	46.470	-11	22	QEASISEGEEEYDEEAPLEAEE
	TBB_PHYCI	2.274	-24	43.659	46.213	-9	19	QDGTAEEEGEFDEDEEWMR
	TBB_PIG	2.481	-25	43.854	47.042	-11	20	QDATADEQGEFEEEGEEDEA
	TBB_PLAFA	2.213	-28	43.732	46.620	-11	20	QDATAEEEGEFEEEEGDVEA
	TBB_PLAFK	2.142	-27	43.685	46.607	-11	20	QDATAEEEGEFEEEEGDVEA
	TBB_PLESA	3.119	-25	44.047	46.981	-12	21	QDATADEEEGEYEEEPAEEEQ
	TBB_PNECA	1.903	-22	43.093	45.489	-9	17	EIAGVDEEVELDDEIET
	TBB_POLAG	2.708	-24	43.429	45.899	-10	18	QDASAEEEGEFGEEEEEN
	TBB_PSEAM	2.177	-25	43.784	46.877	-11	20	QDATAEEEGEGEEEGDEEVA
	TBBQ_HUMAN	1.005	-18	42.440	44.994	-3	10	QDATAEGEGV
	TBB_RHYSE	3.601	-21	43.740	46.423	-11	22	QDASISEGEEEYEEEAPMEPEE
	TBB_SCHCO	2.109	-25	43.970	46.855	-11	20	QDATVEEEGEYEEEVIEDQE
	TBB_SCHPO	2.331	-27	43.447	45.914	-11	23	QEAGIDEGDEDYEIEEEKEPLEY
F	TBB_STRPU	3.410	-18	29.276	31.526	-13	22	QDATAEEEGEFDEEEEGDEEAA
	TBB_STYLE	2.563	-24	43.277	45.763	-10	17	QDATAEDEEEMDEEQME
	TBB_TETPY	2.046	-24	43.558	46.340	-10	18	QDATAEEEGEFEEEEGEN
	TBB_TETTH	1.987	-25	43.553	46.384	-10	18	QDATAEEEGEFEEEEGEN
	TBB_THAWE	2.776	-23	43.338	45.963	-10	18	QDATADEEGEFDEDEMEG
	TBB_TOXGO	2.957	-27	43.888	46.653	-12	24	QDATAEEEGEFDEEEGEMGAEEGA
	TBB_TRYBR	1.836	-24	43.476	45.924	-9	17	QDATIEEEGEFDEEEQY
	TBB_TRYCR	1.960	-25	43.397	45.909	-9	17	QDATIEEEGEFDEEEQY
	TBB_VENIN	3.561	-22	43.550	46.257	-11	22	QEASVSEGEEEYDEEAPLEGEE
	TBBX_HUMAN	2.968	-24	43.586	46.372	-11	19	QDATAEEEEDFGEEAEEEA
	TBB_YEAST	3.539	-31	44.569	47.246	-13	32	QEATVEDDEEVDENGDFGAPQNQDEPITENFE
	TBD_HUMAN	1.360	-5	44.651	45.580	0	0	
	TBE_HUMAN	855	-6			0	18	DATKNMPVQDLPRLSIAM
	TBG1_HUMAN	1.838	-10	44.645	45.231	-1	13	TRPDYISWGTQEQ
	TBG1_MAIZE	2.100	-10	46.059	46.111	-4	28	ESPDYIKWGMEDPGEANVVAALDSKLVV
	TBG1_MOUSE	1.995	-11	44.752	45.707	-1	13	TRPDYISWGTQEQ
	TBG2_ARATH	2.558	-10	46.599	47.774	-4	33	ESPDYIKWGMEDPGQLMTGEGNASGVADPKLAF
	TBG2_DROME	1.969	-6	44.800	45.402	-1	18	TQIDYPQWSPAVEASKAG
	TBG2_EUPCR	2.039	-15	45.633	46.882	-4	29	EKMDYINRGKDDEDMDYDPRAPPNFRPIE
	TBG2_EUPOC	2.242	-10	45.772	47.629	-5	29	EKMDYINWGSDDDDMQFDPREPPKFSNIQ
	TBG2_HUMAN	1.697	-13	44.707	45.747	-2	13	TQPDYISWGTQEQ
	TBG2_MAIZE	1.981	-12			-5	28	ESPDYIKWGMEDPGEANVAADLDSKLVV
	TBG2_MOUSE	1.447	-10	44.771	45.967	-1	13	TRPDYISWGTQEQ
	TBG2_ORYSA	1.972	-12	46.151	46.463	-4	28	ESPDYIKWGMEDAGEANVAAALDSKLVV
F	TBG3_MAIZE	2.091	-9	41.587	42.200	-4	28	ESPDYIKWGMEDPGEANVVAALDSKLVV
	TBG_ANEPH	2.793	-9	46.392	47.491	-3	30	ESADYIKWGMEDRSKTLSADGTMDLSLPSS
	TBG_CAEEL	1.420	-9	43.945	45.973	1	10	VQKDYLTRGL
	TBG_CANAL	2.711	-23			-13	38	KEITYLDDDDEDDLEDGDGGGGGNGNGYNNIDDADMGI
	TBG_CHLRE	2.019	-6	45.685	46.544	-1	13	ESADYIQRQMMAS
F	TBG_COCHE	1.096	-2	26.055	27.658	0	0	
	TBG_EMENI	2.009	-9	44.603	46.275	-4	17	ERENYLDPDAGKDEVGV
	TBG_ENTHI	1.255	-6	45.398	46.350	0	26	ESIEYSHYSYPIHKYLKGIAKEKIAE
	TBG_EUPAE	2.163	-10	45.767	47.109	-5	29	EKMDYINWGNDDDDMQFDPREPPKFSNIQ
	TBG_NEUCR	2.077	-9	45.255	46.778	-4	23	EDANYLNPELGENASADTDKRMA

Name		Net	Volume	Area	C-Terminal Tail		
	|M|	Charge	(Å^3)	(Å^2)	Charge	Size	Sequence
TBG_PHYPA	2.785	-8	46.549	47.782	-2	34	ESADYIKWGMEDRGKQVSGEGNTSGTVDSRVGAS
TBG_PLAFO	2.397	-7	45.179	46.542	-2	11	ERDDYFTNTYI
TBG_RETFI	2.039	-4	47.100	48.599	1	44	QTSDYINWGMKQQQSQISQKESSSLANENGNGANNKPGKSAMAL
TBG_SCHJP	1.912	-7	44.087	45.524	-2	8	EQPEYLSM
TBG_SCHPO	1.618	-8	43.930	45.423	-2	8	EDPNYLSL
TBG_USTVI	1.970	-10	45.915	47.039	-2	30	EKADYIDYGAGPGYVKGEDRREKGREAVEG
TBG_XENLA	1.913	-9	44.698	45.368	0	13	TRPDYISWGTQDK
TBG_YEAST	2.193	-30	45.778	47.349	-13	33	EQDSYLDDVLVDDENMVGELEEDLDADGDHKLV

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The Free Energy for Hydrolysis of a Microtubule-Bound Nucleotide Triphosphate Is Near Zero: All of the Free Energy for Hydrolysis Is Stored in the Microtubule Lattice

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Abstract. The standard free energy for hydrolysis of the GTP analogue guanylyl-(a,b)-methylene-diphosphonate (GMPCPP), which is −5.18 kcal in solution, was found to be −3.79 kcal in tubulin dimers, and only −0.90 kcal in tubulin subunits in microtubules. The near-zero change in standard free energy for GMPCPP hydrolysis in the microtubule indicates that the majority of the free energy potentially available from this reaction is stored in the microtubule lattice; this energy is available to do work, as in chromosome movement. The equilibrium constants described here were obtained from video microscopy measurements of the kinetics of assembly and disassembly of GMPCPP-microtubules and GMPCP-microtubules. It was possible to study GMPCPP-microtubules since GMPCPP is not hydrolyzed during assembly. Microtubules containing GMPCP were obtained by assembly of high concentrations of tubulin-GMPCP subunits, as well as by treating tubulin-GMPCPP-microtubules in sodium (but not potassium) Pipes buffer with glycerol, which reduced the half-time for GMPCPP hydrolysis from >10 h to ~10 min. The rate for tubulin-GMPCPP and tubulin-GMPCP subunit dissociation from microtubule ends were found to be about 0.65 and 128 s^{-1}, respectively. The much faster rate for tubulin-GMPCP subunit dissociation provides direct evidence that microtubule dynamics can be regulated by nucleotide triphosphate hydrolysis.

THE hallmark for energy-transducing systems such as myosin (Bagshaw and Trentham, 1973), ion-transporting ATPases (Taniguchi, and Post, 1975; Pickart and Jencks, 1984), the mitochondrial ATPase (Grubmeyer et al., 1982) and chloroplast coupling factor (Feldman and Sigman, 1982) is a near-zero free energy for hydrolysis of bound nucleotide triphosphate (NTP)[1]. Negligible free energy is released during NTP hydrolysis since this is stored in the protein conformation until useful work can be done. We have used the hydrolyzable GTP analogue GMPCPP, which contains a methylene linkage between the alpha and beta phosphates, to determine the free energy for hydrolysis of microtubule-bound NTP. We found that the majority of this free energy is stored in the microtubule lattice, so that the standard free energy for hydrolysis is near zero. This stored energy can do work, as in the NTP-independent movement of chromosomes in a reaction coupled to microtubule disassembly (Koshland et al., 1988; Coue et al., 1991). The retention of the energy of NTP hydrolysis by microtubules parallels the behavior of several energy-transducing systems.

In addition to these equilibrium studies, we have characterized the dynamics of microtubules formed with GMPCPP. The recent observation that this substance is not hydrolyzed in microtubules (Hyman et al., 1992) suggested its use in determining the role of the gamma phosphate moiety in microtubule dynamics. Our rationale was that if conditions could be found for forming microtubules containing GMPCP, it would be possible to determine the kinetic behavior of microtubules with a cognate pair of bound nucleotide tri- and diphosphates. Such a comparison was not possible with GTP analogues containing a nonhydrolyzable linkage between the beta and gamma phosphates (Weisenberg and Deery, 1976; Penningroth and Kirschner, 1978; Arai and Kaziro, 1976; Karr and Purich, 1978).

We found two routes for forming GMPCP-microtubules: by treating GMPCPP-microtubules with glycerol in sodium (but not potassium) Pipes buffer, which induces rapid hydrolysis of bound GMPCPP and stabilizes the resulting polymer; by assembling high concentrations of tubulin-GMPCP subunits. Our observation that subunits containing GMPCP dissociate much faster than subunits containing GMPCPP provides direct evidence that microtubule dynamics can be regulated by NTP hydrolysis. This conclusion was previously inferred from the requirement for GTP for dynamic instability behavior.

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1. *Abbreviation used in this paper*: NTP, nucleotide triphosphate.

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Materials and Methods

Materials

Pig and beef brain tubulin was purified by two cycles of thermal-induced assembly and disassembly, followed by chromatography on phosphocellulose (Weingarten et al., 1975). For some experiments the protein was further purified by a glutamate-induced polymerization step (Voter and Erickson, 1984). The small amount of glutamate that contaminated the resultant protein was removed by chromatography on Sephadex G-25. Some studies with calf brain tubulin were with protein provided by T. J. Mitchison (University of California, San Francisco, CA). We found no significant difference in the properties of beef and pig tubulin.

The amount of bound and free guanine nucleotide contained in purified tubulin was determined by an isotope dilution procedure. Microtubules were assembled with 0.5 mM ATP, a trace amount of [^{3}H]GDP, 0.17 U/ml of recombinant nucleoside diphosphate kinase (2500 U/mg, Nm23-1 expressed in *Escherichia coli* and purified essentially as described by Randazzo et al., 1991) and varying amounts of added GDP; polymer was isolated with an Airfuge (see below). Since the amount of label that is incorporated from radioactive nucleotide into polymer is proportional to the specific activity of the nucleotide, and the specific activity is inversely proportional to the nucleotide concentration, the endogenous guanine nucleotide contained in purified tubulin (E) is described by the relationship

$$\frac{\text{CPM in pellet (GDP added = 0)}}{\text{CPM in pellet (GDP added = N)}} = \frac{E + N}{E} \quad (1)$$

N in this equation corresponds to the concentration of added GDP. We found that the amount of radiolabel incorporated into polymer was reduced by two when N was equal to the tubulin concentration. This result indicates that the tubulin used here contained ~1.0 mol exchangeable guanine nucleotide/mol tubulin; the nucleotide was 30% GTP and 70% GDP.

[^{3}H]Guanylyl-(a,b)-methylene-diphosphonate ([^{3}H]GMPCPP) was obtained in 1984 from ICN Biomedicals, Inc. (Costa Mesa, CA); GMPCPP was also obtained in 1984 from ICN, or was synthesized as described (Hyman et al., 1984). The material obtained from ICN was repurified by chromatography on a 0.7 × 10 cm column containing Waters Protein-Pak DEAE 40HR, developing with a 50 ml 0.05–0.225 M triethylammonium bicarbonate gradient. Gamma-[^{32}P]GMPCPP was synthesized from GMPCP (provided by T. J. Mitchison or synthesized in this laboratory), using ATP-32 and nucleoside diphosphate kinase, as described previously (Hyman et al., 1992).

In preparing tubulin-GMPCP for studies of assembly it was necessary to deplete tubulin subunits of endogenous GTP and GDP. Removal of GTP was necessary since assembly might otherwise result from reaction with this nucleotide. Endogenous GDP could inhibit binding of GMPCP, which was found (see below) to have lower affinity for the E-site. Endogenous nucleotide was depleted by addition of alkaline phosphatase (P4252, 5 U/ml, 60 min, 10°; Sigma), including 1 mM GMPCP, which is not hydrolyzed by alkaline phosphatase, to stabilize the tubulin as the E-site nucleotide was hydrolyzed. To determine the extent of hydrolysis of endogenous GDP and GTP the pool of nucleotide was radiolabeled by a 10-min reaction with a trace amount of [^{3}H]GTP and nucleoside diphosphate kinase (0.7 μg/120 μl) before alkaline phosphatase addition. The nucleoside diphosphate kinase catalyzes isotopic equilibrium between GDP and GTP so that hydrolysis of these can be determined by measuring the radiolabel in GTP, GDP, GMP, and guanosine. Samples were quenched with perchloric acid and analyzed with HPLC. After a 60-min incubation only 0.1% of the initial E-site GTP and 21% of the initial GDP remained. The relatively slow digestion resulted because, as described previously for the reaction of added pyruvate kinase with E-site nucleotide (Brylawski and Caplow, 1983), the tubulin repeatedly recaptures most of the dissociated nucleotide before it can react with alkaline phosphatase. We did not extend the alkaline phosphatase digestion time to hydrolyze the 21% remaining GDP because we were concerned that this might result in loss of activity in the tubulin. Also, any observed assembly with tubulin-GMPCP subunits could not be attributed to tubulin-GDP subunits, since this material does not form microtubules under these conditions. In our calculation of the rate constant for microtubule assembly with GMPCP we have corrected for the presence of 21% tubulin-GDP.

Isolation and Characterization of GMPCPP-Microtubules

All reactions were in BRB80 buffer (80 mM Na Pipes, 1 mM $MgCl_2$, 1 mM EGTA, pH 6.8), at 37°, except for rates measured with video microscopy, where the temperature was 22–24°. Microtubules were made by two methods, in both the tubulin was incubated with GMPCPP for 20 min at 0° before a 30 min incubation at 37°. "5:1 microtubules" were generated with an ~92% yield of protein by assembly with 200 μM GMPCPP and 40 μM tubulin. The amount of GDP in this polymer was determined by including a trace amount of [^{3}H]GDP and [^{3}H]GTP in the assembly reaction; the specific activity of the added label was calculated from the amount of exchangeable GDP or GTP (see Eq. 1). "High-GMPCPP Tubes" were generated by an initial assembly at 25° with 1 mM GMPCPP and 10 μM tubulin, with a second assembly reaction using 10 μM tubulin (obtained by disassembling the initial pellet for 45 min in buffer at 0°) and 50 μM radiolabeled GMPCPP. The amount of GDP in these microtubules was determined by including a trace amount of alpha-[^{32}P]GTP and alpha-[^{32}P]GDP in the second assembly reaction.

Microtubules were isolated using an Airfuge (3–10 min, 37°, 30 psi), or by layering a 50-μl aliquot over a 2-ml cushion, made with 5% Ficoll 400 in BRB80 buffer, with centrifugation in a Beckman TLA100.3 rotor (100,000 rpm, 37°). After the supernatant was removed by suction and the pellet carefully rinsed with buffer, the portion of the tube near the pellet was wiped dry with a Kimwipe. The pellet was dissolved in 0.1 M NaOH for measurement of protein. For nucleotide analysis the pellet was treated with 1.4 M $HClO_4$ for 10 min at 0°, after which the perchloric acid was neutralized with K_2HPO_4. Precipitated salt was removed by a 30-s centrifugation in a microfuge and the nucleotide composition in the supernatant was determined using a SynChrompak AX100 column (Synchrom Inc., Lafayette, IN). GMPCP and GMPCPP eluted at 4.5 and 7.5 min (1 ml/min) when the column was developed with 0.15 M sodium phosphate buffer and 1.2 M sodium acetate, pH 7.3.

The rate of hydrolysis of GMPCPP in microtubules was determined by HPLC analysis of perchloric acid-quenched aliquots of resuspended polymer containing tritium-labeled nucleotide. Also, the rate was determined by measuring release of [^{32}P]Pi (Carlier, et al., 1987) from microtubules that had been assembled with gamma-labeled [^{32}P]GMPCPP.

Video Assay of Assembly and Disassembly at Microtubule Ends

Microtubule dynamics were analyzed using video microscopy, as described previously (Walker et al., 1988). Briefly, microtubules were assembled at the ends of sea urchin axonemes (kindly provided by E. D. Salmon) with tubulin subunits in BRB80 buffer. Disassembly was induced by displacement of subunits with buffer, using the flow cell described by Vale (1991), or, less frequently, with the cell described by Berg and Block (1984).

The rate of assembly of tubulin-GMPCPP was determined with 4 μM tubulin and 1 mM GMPCPP. It was not possible to vary significantly the tubulin subunit concentration since with 3 μM tubulin there was no nucleation at axoneme ends in 10 min and with 6 μM tubulin there was extensive nucleation of free microtubules that obscured observations of axoneme-associated tubes; formation of free microtubules also decreases the tubulin subunit concentration by an unknown amount. It is noted that the requirement for 3 μM tubulin for nucleation is not related to the critical concentration for assembly, which is much lower. Rather, as reported previously (Walker et al., 1988), there is a significant lag before axonemes are elongated with tubulin subunits and this lag was excessively long when the tubulin concentration was below 3 μM.

Tubulin-GMPCP subunits were assembled at axoneme ends by reacting 90–100 μM tubulin with 1 mM GMPCP and alkaline phosphatase. The lag for axoneme elongation was especially significant with tubulin-GMPCP, so that it was necessary to incubate the flow cell containing the reaction mixture for ~5 min at 37° (in a humidified chamber) before viewing samples at room temperature. Less frequently, axonemes were first incubated for about 5 min with 7.5 μM tubulin and 1 mM GMPCPP to generate short stubs at axoneme ends. After the tubulin-GMPCPP subunits had been washed out with buffer the stubs were elongated with the 100 μM tubulin, 1 mM GMPCP, alkaline phosphatase mixture. Electron microscopic images of elongated axonemes were obtained with samples that had been diluted 100-fold into BRB buffer containing 1% glutaraldehyde and pelleted on an electron microscope grid using a Beckman EM90 rotor.

Assays of the Rate of Disassembly of GMPCPP-Microtubules

The rate of disassembly of [^{32}P]GMPCPP-microtubules was determined by sedimenting polymer with an Airfuge (37°, 10 min, 30 psi) after a 30-fold dilution into BRB80 buffer. The radioactive nucleotide contained in the supernatant was measured. Either the microtubules were not fully sedi-

mented, or disassembly continued during a portion of the time required for centrifugation, so that an only 80% yield was obtained with microtubules that were centrifuged immediately after dilution.

The kinetics for dilution-induced disassembly were also measured with high-GMPCPP tubes containing gamma-[^{32}P]GMPCPP, from the rate of release of [^{32}P]Pi formed in the reactions in Eqs. 2 and 3:

$$[^{32}\text{P}]\text{GMPCPP-Microtubule} \xrightarrow{\text{200-fold dilution}} [^{32}\text{P}]\text{GMPCPP-tubulin subunits} \quad (2)$$

$$[^{32}\text{P}]\text{GMPCPP-tubulin subunit} \rightarrow [^{32}\text{P}]\text{GMPCPP} \xrightarrow{\text{alkaline phosphatase}} [^{32}\text{P}]\text{Pi} \quad (3)$$

With alkaline phosphatase at 1 U/ml the cleavage of GMPCPP was rapid (0.56 s^{-1}), so that the observed rate of Pi release closely tracked the rate of disassembly. The molecular rate constant for disassembly was obtained by multiplying the % microtubule disassembled/s times the mean number of tubulin subunits/microtubule. The microtubule length distribution was determined by diluting an aliquot from the reaction mixture 3,000-fold into 1% glutaraldehyde, before the microtubules were sedimented with an Airfuge EM90 rotor onto an electron microscope grid. Lengths were determined by measuring >300 microtubules.

Equilibrium Constants for Nucleotide Binding

The equilibrium constant for GMPCP and for GMPCPP binding to tubulin was determined by competition experiments in which these nucleotides displaced bound [^{3}H]GTP from the tubulin E-site. Tubulin-[^{3}H]GTP subunits were prepared by adding a trace amount of [^{3}H]GTP to the 14.2 μM tubulin that eluted in the void volume when a mixture of 40 μM tubulin and 200 μM GTP was chromatographed on a Sephadex G-25 column. It was necessary to determine the amount of GTP in this mixture, since this is required in calculating the nucleotide specific activity, used in deriving the stoichiometry for nucleotide binding. The amount of GTP was determined by adding a trace amount of [^{3}H]GTP with and without 10 μM GTP before the 14.2 μM tubulin was rechromatographed on Sephadex G-25. Since 10 μM added GTP reduced the amount of radioactive nucleotide that eluted with the protein by 42%, it was concluded that the 14.2 μM tubulin contained 13.8 μM GTP (see Eq. 1).

The K_d values for GMPCPP and GMPCP were determined by incubating these nucleotides with 14.2 μM tubulin-[^{3}H]GTP for 5 min before a 40-μl aliquot was chromatographed on a 1-ml Sephadex G-25 column. In a control reaction without added GMPCPP or GMPCP the tubulin eluted with 0.57 mol[^{3}H]GTP/mol tubulin; however, since 20% of E-site GTP dissociates from tubulin during chromatography under these conditions (see Fig. 1 in Zeeberg, 1980), the E-site stoichiometry was 0.57/0.8 = 0.71.

Results

Nucleotide Composition of Microtubules Assembled with GMPCPP

We were puzzled by the discrepancy between results showing that either GMPCPP (Hyman et al., 1992) or GMPCP (Sandoval et al., 1977) is found in microtubules assembled with GMPCPP. We reinvestigated this problem using [^{3}H]GMPCPP, rather than the gamma-[^{32}P]GMPCPP used by Hyman et al. (1992), so that both GMPCPP and GMPCP could be tracked by the tritium in the guanine moiety.

Table I. Nucleotide Composition of Isolated Microtubules

Isolation Procedure	Pelleted protein	Pelleted [^{3}H]	Pelleted [^{32}P]
	%	%	%
Cushion	85	28	2.7
No cushion	100	30	20

Microtubules were assembled with tubulin (51 μM) and gamma-[^{32}P], [^{3}H]GMPCPP (150 μM) and the resultant polymer was isolated by centrifugation in an Airfuge, or by centrifugation through a cushion containing 25% glycerol.

Microtubules assembled with [^{32}P], [^{3}H]GMPCPP and isolated through a glycerol cushion were found to contain very little radioactive phosphorus (Table I). The 10/1 ratio of [^{3}H] and [^{32}P] indicated that ~90% of the GMPCPP that had been incorporated was hydrolyzed in the pelleted microtubules. In contrast, microtubules that were isolated without using a glycerol cushion contained nearly equal amounts of [^{3}H] and [^{32}P] (Table I). The possibility that the presence of [^{32}P] in the product isolated without glycerol resulted from subunits containing [^{3}H]GMPCP and [^{32}P]Pi was ruled out by HPLC analysis of bound nucleotide: 83% of the tritium was in [^{3}H]GMPCP and 17% in [^{3}H]GMPCPP.

Rate of Hydrolysis of GMPCPP in Microtubules

To determine the basis for the dramatic difference in the extent of GMPCPP hydrolysis in microtubules isolated with or without glycerol (Table I), the rate of nucleotide hydrolysis was determined in microtubules that had been assembled with [^{3}H]GMPCPP and isolated with a Ficoll cushion. Hydrolysis was slow: the fraction of bound tritium as [^{3}H]GMPCP increased from 19.3% in the initially isolated polymer to 30.9% in 2 h and to 59% in 18 h. The rate constant calculated from results during the first 2 h of incubation is 2.1 × 10^{-5} s^{-1}; this is twice the value calculated from the longer incubation, where protein denaturation may have influenced the rate. The rate of release of [^{32}P]Pi from microtubule bound gamma-[^{32}P]GMPCPP was linear with time and an observed 2.4% release of label in 40 min corresponds to a rate constant equal to 1 × 10^{-5} s^{-1} (19 h half-life).

To confirm that glycerol affected the rate of hydrolysis of GMPCPP in microtubules parallel analyses were performed with glycerol added to microtubules in solution. The half-time of GMPCPP hydrolysis was reduced to ~10 min (Fig. 1). To test whether other commonly used chemicals affect the hydrolysis rate, we examined GMPCPP hydrolysis in the

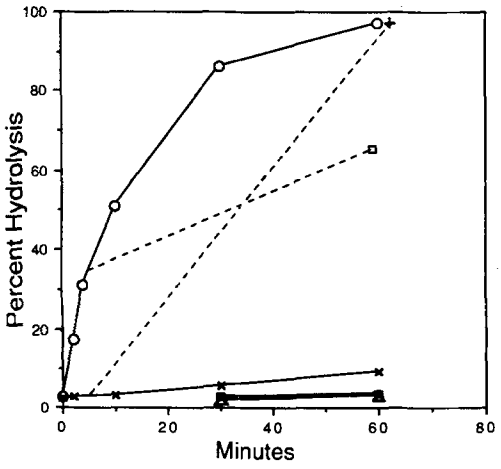


Figure 1. Effect of glycerol on the rate of hydrolysis of GMPCPP in microtubules. The rate of release of [^{32}P]Pi was determined when [^{32}P]GMPCPP-microtubules were diluted into buffer containing alkaline phosphatase and Na- (○) or K-Pipes (×) in glycerol, or in Na- (■) or K-Pipes (△) in the absence of glycerol. The dotted lines describe the reaction when GMPCPP-microtubules in either Na or K-Pipes and glycerol for 5 min were diluted 10-fold into glycerol buffer containing Na-Pipes (+) or K-Pipes (□). The rate in 1 M Na glutamate in Na-Pipes was indistinguishable from that in Na-Pipes and glycerol.

presence of glutamate or following substitution of Na-Pipes buffer with K-Pipes. The former was chosen because it is commonly used in tubulin purification; the latter was used because this had been used in a previous study (Hyman et al., 1992) which found intact GMPCPP. Like glycerol, glutamate stimulated hydrolysis, reducing the half-time to ~10 min. Most surprisingly, the switch from sodium to potassium cations strongly slowed hydrolysis, yielding a rate similar to that reported earlier (Hyman et al., 1992). The rate was also mildly stimulated by glycerol in Li-Pipes buffer: hydrolysis was increased from about 2% to 14% by glycerol, when gamma-[^{32}P]GMPCPP-microtubules that had been assembled in Li-Pipes buffer were incubated for 30 min. These cation effects are reversible, since a buffer change by a 10-fold dilution reduced hydrolysis in K-Pipes and increased it in Na-Pipes (Fig. 1).

Since both glycerol (Solomon, et al., 1973) and glutamate (Hamel and Lin, 1981) stabilize microtubules we examined whether taxol would also affect the hydrolysis rate. Isolated GMPCPP-microtubules were reacted with 25 μM taxol and the [^{3}H]GMPCPP remaining was analyzed by HPLC after a 5-, 45-, and 90-min incubation. This revealed that hydrolysis was essentially unaffected by taxol, as 82–84% of GMPCPP remained intact even at the latest time point.

To determine whether the presence of adjacent tubulin-GDP subunits could simulate the effect of glycerol on the rate of GMPCPP hydrolysis, microtubules were assembled under conditions where GMPCP and GMPCPP constituted only ~5% of the total bound nucleotide. Adjacent tubulin-GDP subunits did not simulate the effect of glycerol since there was no enhanced hydrolysis of GMPCPP in microtubules under these conditions (Table II).

Microtubules Containing GMPCPP Are Very Stable

The rate of microtubule disassembly upon dilution was sensitive to relatively small differences in the degree of substitution with analogue (Fig. 2). The sedimentation assay used here was not suited for determining the molecular rate constant for subunit dissociation, since in 30 min there was only a small increase in radiolabel that was not pelleted, over a relatively high (20%) background (lower curve in Fig. 2). To surmount this technical problem we assayed the rate of release of [^{32}P]Pi from [^{32}P]GMPCPP-microtubules after the polymer had been diluted into buffer containing alkaline phosphatase (see Eqs. 2 and 3). This revealed an initial rate

Table II. Effect of Coassembly of GMPCPP and GTP on GMPCPP Hydrolysis

GTP in assembly	μM Analogue in pellet*	% GMPCPP Hydrolyzed
50 μM	0.64 (3.9%)	25
200 μM	.54 (3.3%)	15
400 μM	.46 (2.8%)	6

Microtubules were assembled by a 30-min incubation of 40 μM tubulin, a GTP regenerating system (250 μM ATP and nucleoside diphosphate kinase) and 16 μM [^{3}H]GMPCPP and isolated by centrifugation in an Airfuge. A Ficoll cushion could not be used here since microtubules containing GDP are not stable to dilution in Ficoll. It was determined that 10.4 μM GDP was in microtubules, from measurement of the fraction of the label pelleted in an identical reaction with label in [^{3}H]GTP.

* The amount of nucleotide is expressed as concentration, after normalizing the volume of the resuspended pellet to the initial reaction volume. The value in parenthesis gives percent total [^{3}H]GMPCPP in the pellet; this exceeded the 0.5–.8% nonspecific entrapment of nucleotide under these conditions.

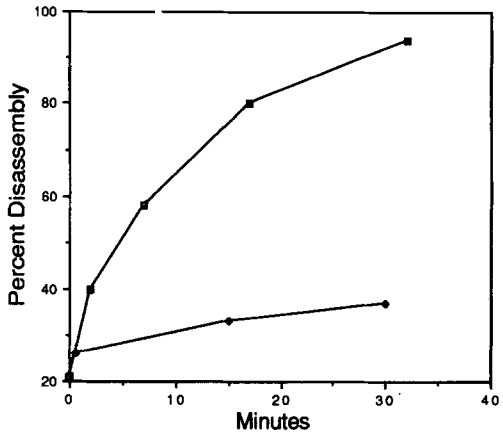


Figure 2. Rate of dilution-induced disassembly of GMPCPP-microtubules measured with a sedimentation assay. Rates with High GMPCPP Tubes (●) and 5:1 microtubules (□) are shown. High GMPCPP Tubes were generated by two cycles of assembly with GMPCPP; these contained 0.82 mol analogue/mol tubulin and <0.07 mol GDP/mol tubulin. 5:1 Tubes were assembled in a single cycle with GMPCPP/tubulin equal to 5. These contained 0.07 mol GMPCP, 0.46 mol GMPCPP, 0.05 mol GDP (from coassembly with tubulin-GDP subunits), and 0.11 mol GDP (from coassembly with tubulin-GTP subunits) per mol tubulin in polymer; this corresponds to 0.69 guanine nucleotide/tubulin, with 77% substitution with analogue.

of 1.7%/min (Fig. 3), and with a mean microtubule length equal to 2.8 μm, the molecular rate constant for subunit dissociation is 1.3 s^{-1}. Since the disassembly rates are about equal at the two ends (see below) the molecular rate constant is 0.65 s^{-1}. The rate of subunit dissociation was not significantly influenced by the identity of the monovalent cation: when microtubules assembled in Na-Pipes were diluted 100-fold the disassembly rate was 0.48%/min when dilution was in Na-Pipes and 0.33%/min in K-Pipes.

To confirm that microtubules were stabilized by GMPCPP, video measurements were made on individual microtu-

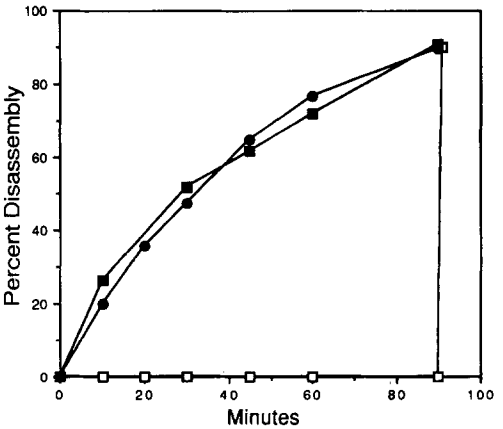


Figure 3. Rate of dilution-induced disassembly of GMPCPP-microtubules. Results from experiments run on consecutive days (●,■) and a reaction where alkaline phosphatase was not included in the diluent (□) are shown. In the latter reaction alkaline phosphatase was added at 90 min, and the quantitative release of Pi demonstrated that the microtubules had disassembled fully.

bules. High GMPCPP microtubules were very stable at both ends (Table III, Fig. 2). Subunit dissociation was extremely slow. Assuming that loss of fewer than 600 subunits (~2 mm on the video screen) would go undetected during a 30-min observation yielded a maximal rate constant for subunit loss of 0.4 s^{-1}, in agreement with the bulk measurements above (Fig. 3). About 15% of the microtubules were observed to disassemble at a rate of about 2 s^{-1}; we have no explanation for this faster rate. Microtubules assembled with a 5:1 GMPCPP/tubulin ratio so that they were 77% substituted with analogue (see legend to Fig. 2) had reduced stability. After dilution to remove tubulin subunits the majority of microtubules disassembled relatively slowly at one end (Table III), while a small fraction of these microtubules did not disassemble at a measurable rate.

To characterize further GMPCPP-microtubules, the rate of elongation from axonemes was determined for 4 μM tubulin-GMPCPP. This revealed a rate of 6.1 (n = 12, SD 1.8) s^{-1}, with no significant difference in the rate at the two microtubule ends. It was not possible to determine the elongation rate at widely varying concentrations of tubulin-GMPCPP because, as described in Materials and Methods, there is massive nucleation of microtubules that are not at axoneme ends when the tubulin concentration exceeds ~6 μM. Thus, the rate constant for elongation with tubulin-GMPCPP could not be determined from a linear plot of the concentration dependence of the rate. Instead, the observed rate with 4 μM tubulin, corrected for the rate for tubulin-GMPCPP subunit dissociation (0.65 s^{-1}, the average for the two ends from results in Fig. 3), was divided by the 4-μM subunit concentration used for assembly. The so-derived second-order rate constant was 1.36×10^6 M^{-1} s^{-1} (at 23°); the rate constant at 37° was reported to be 5×10^6 M^{-1} s^{-1} (Hyman et al., 1992). For comparison, rates equal to 1.2–5.3 $\times$ 10^6 M^{-1} s^{-1} have been reported for the reaction of tubulin-GTP subunits (Mitchison and Kirschner, 1984; Walker et al., 1988).

Table III. Kinetics for Disassembly of GMPCPP-microtubules Measured with Video Microscopy

Polymerization conditions	Glycerol treatment*	Rate end 1	Rate end 2
		s^{-1}	s^{-1}
High-GMPCPP Tubes	–	0.4‡ (41)	2.1 (11, 2)
High GMPCPP Tubes	+	146 (4, 39)	5.5 (6, 3)
5:1 Tubes‡	–	.5§ (19, .4)	10 (15, 7)

Rates were measured after microtubules assembled at axoneme ends were diluted with buffer. Values in parenthesis are number of microtubules analyzed and standard error. Results were from studies where both ends of an axoneme could be observed; however, numerous observations of microtubules on axonemes that could not be viewed at both ends confirmed the results reported here. The properties of High GMPCPP Tubes and 5:1 Tubes are described in the legend for Fig. 2.

* Microtubules that had been elongated at the end of axonemes were exposed to 60% glycerol for 40 min, after which the glycerol was displaced by excess buffer to induce disassembly.

‡ This is an upper limit for the rate since 16 of the 18 microtubules analyzed did not measurably shorten in 25 min of observation. The rate was calculated by assuming that a 2-mm shortening on the video screen would not be detected during the period of observation.

§ There was no measurable disassembly for 15 of 19 microtubules examined. An upper limit to the rate was calculated by assuming that a 1-mm change in length would not be detected in the 10 min of observation.

Stability of GMPCP-microtubules

The stability of GMPCP-microtubules was determined with microtubules that had been assembled with GMPCPP and then treated with glycerol to induce hydrolysis. On subsequent dilution these microtubules disassembled very rapidly, but only at one end (Table III). Microtubules containing GMPCP were also generated by assembling tubulin-GMPCP subunits at axoneme ends. This reaction could be induced with high concentrations of tubulin (91 μM) and GMPCP, using alkaline phosphatase to cleave excess endogenous nucleotides. The possibility that assembly resulted from reaction with residual tubulin-GTP subunits was ruled out since only 0.1% of the initial E-site GTP was found to survive the alkaline phosphatase treatment. Evidence that microtubules were formed was obtained by electron microscopy (Fig. 4). The previous failure to form microtubules in a reaction of purified tubulin with GMPCP (Sandoval, 1977) probably resulted from use of low protein concentrations.

Video microscopy was used to measure the elongation rate with GMPCP. A wide range of rates, much wider than that attributable to experimental error, was observed for elongation of a single axoneme end. For example, two microtubules at one axoneme end grew at rates of 5.7 s^{-1}, while another at the same end grew at 1.6 s^{-1}. A similarly wide range of rates of microtubule assembly (Gildersleeve et al., 1992; Toso et al., 1993) and disassembly (O'Brien et al., 1990) with GTP has also not been accounted for. Also, in agreement with previous reports (Gildersleeve et al., 1992; Toso et al., 1993), we occasionally observed periods during which microtubules temporarily stopped growing, as well as variable rates for both assembly and disassembly of a single microtubule. When results from experiments on three successive days were plotted the assembly rates clustered in two ranges, with rates equal to 16 s^{-1} (n = 16, SD 4) and 8 s^{-1} (n = 11, SD 1.7); two microtubules assembled at 1 s^{-1}. Since assembly with tubulin-GTP occurs more rapidly at the microtubule plus end we assume that the 16 s^{-1} rate is for reaction at this end.

The disassembly rate was more difficult to measure with GMPCP-microtubules since these were out of focus during the flow of buffer and moved violently for some time after the flow was stopped. Also, microtubules were frequently lost by shear and since disassembly is relatively rapid, it was not possible to select a new field or otherwise optimize observations after the flow of buffer. Rates again mainly fell into two categories: 128 s^{-1} (n = 12, SD 23) and 63 s^{-1} (n = 12, SD 15); six other microtubules had a mean rate of 26 (SD 6). Since the ratio of the assembly to disassembly rate is equal to the equilibrium constant for tubulin-GMPCP subunit addition, and this ratio must be equal at the two ends for a reaction where there is no NTP hydrolysis (Wegner, 1976), the rapid and slow assembly rates were coupled with the corresponding disassembly rates to calculate the second-order rate constant for assembly and the equilibrium constant for subunit addition (Table IV). The appropriateness of this coupling of rapid and slow rates was indicated by analysis of the rates for microtubules where both assembly and disassembly could be observed. For example, 12 out of 17 microtubules were found to have assembly and disassembly rates that were both either in the rapid or slow category. Generally, microtubules at opposite microtubule ends were

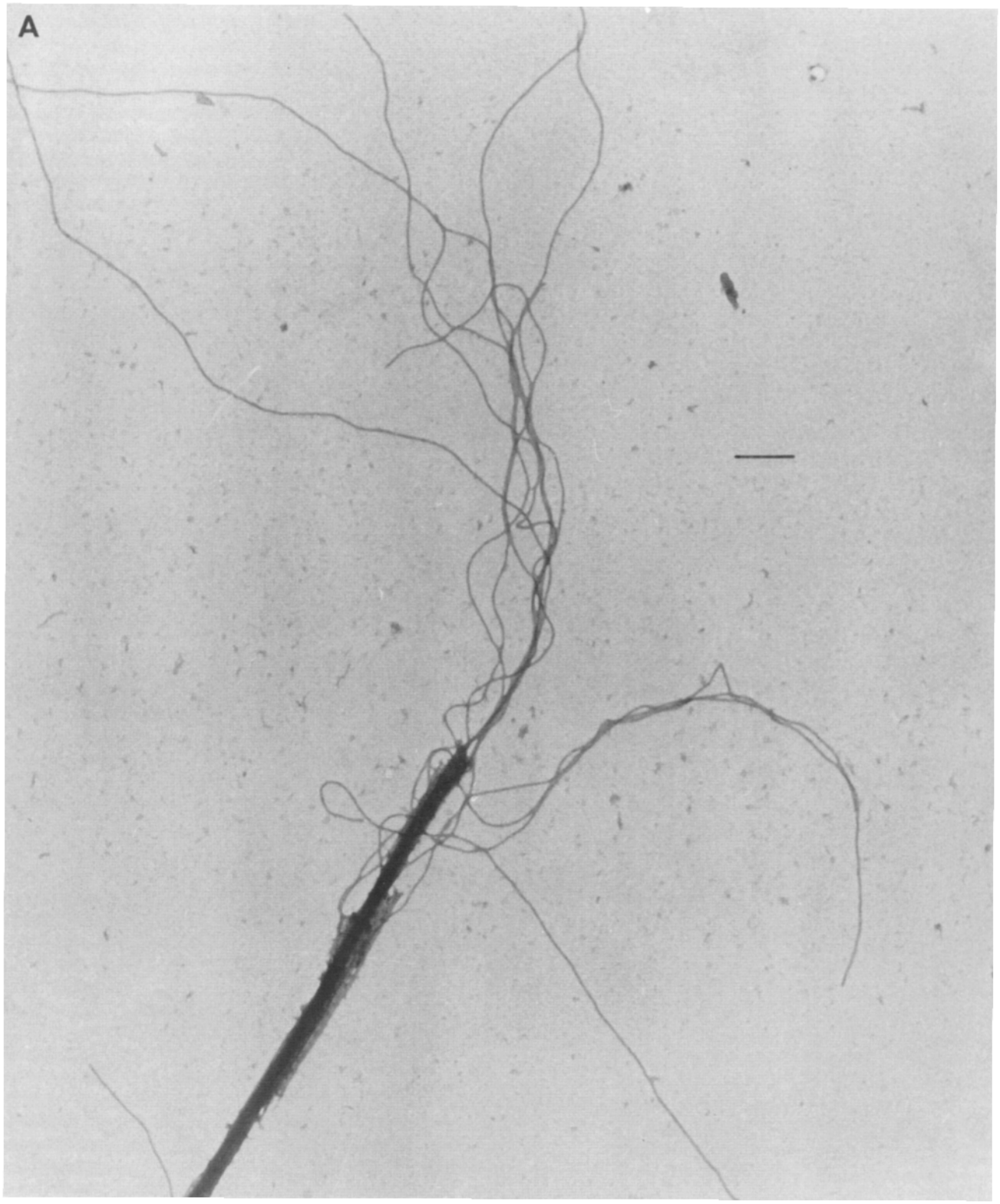


Figure 4. Structure of microtubule assembled with GMPCP. Bars, 0.5 μm.

either in the fast or slow range for assembly or disassembly. Deviation from the rule that either rapid or slow assembly and disassembly must occur at a given end may have resulted because some of the microtubules originated from axonemes with opposite polarity that were stuck together, or because microtubules grew from defects in the axoneme structure and then folded back on the axoneme so that their site of origin was mistaken. The frayed character of the axonemes is apparent in Fig. 4.

The second-order rate constant for assembly with tubulin-

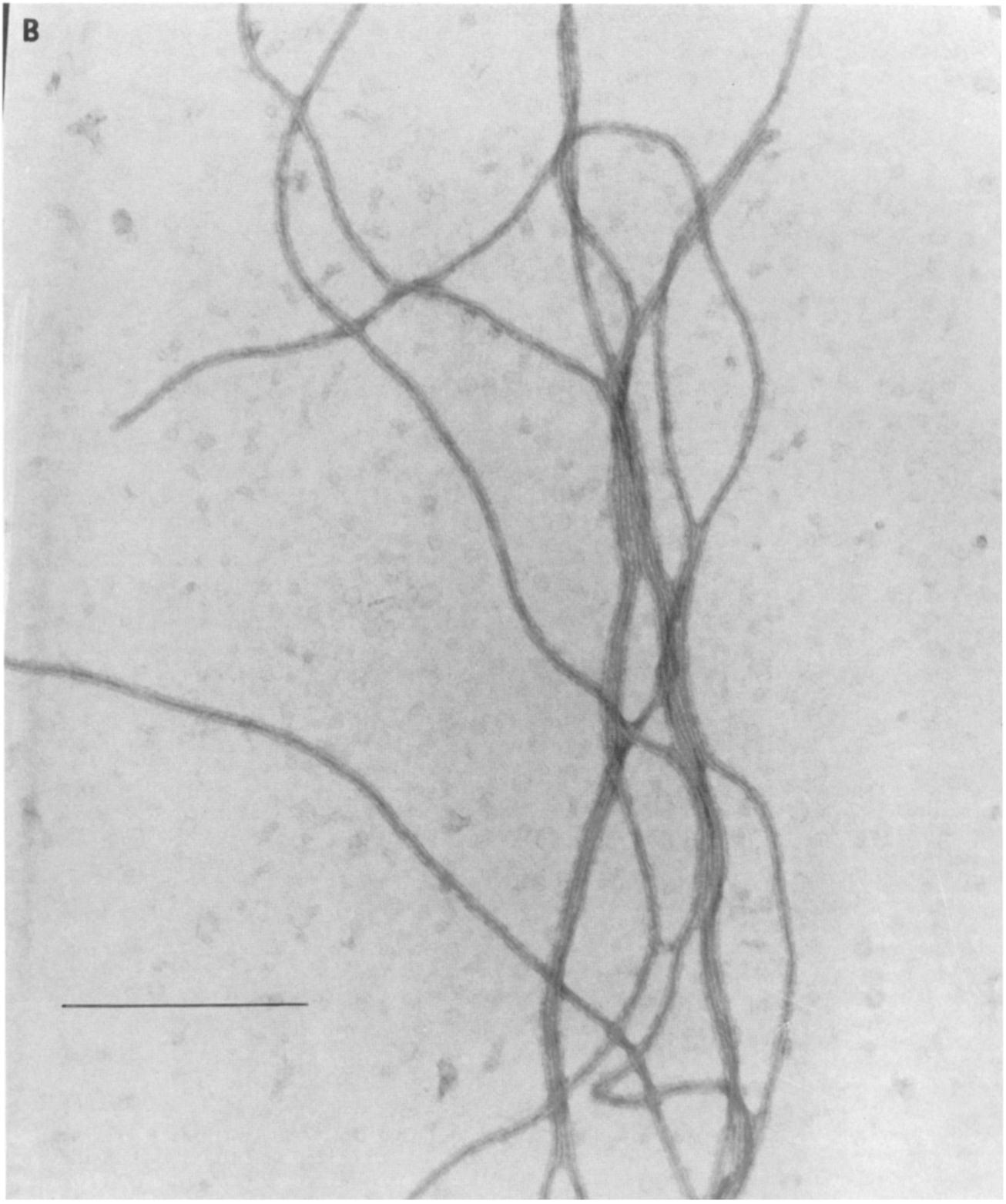


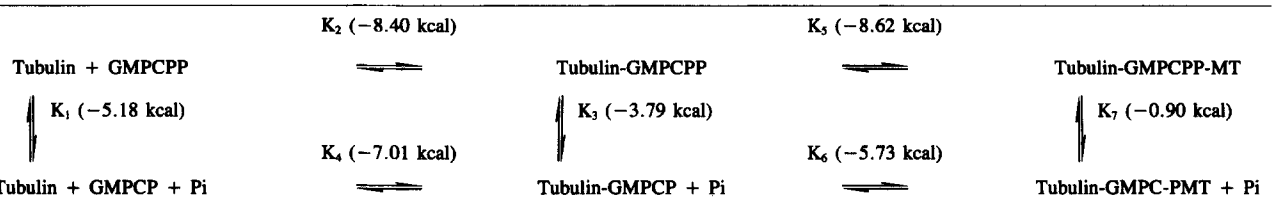
Figure 4.

GMPCP was determined by dividing the sum of the observed elongation and disassembly rates by the tubulin-GMPCP subunit concentration (73 μM, after correcting for the 21% occupancy of the E-site with GDP). The calculated second order-rate constants were 2×10^6 M^{-1} s^{-1} at one end and 1×10^6 M^{-1} s^{-1} at the other end.

Equilibrium for GMPCP and GMPCPP Binding to Tubulin Subunits

The binding affinity of GMPCPP and GMPCP to tubulin subunits was measured by their ability to displace GTP (Table V). The affinity is quite high, with K_d equal to 7.25 μM

Table IV. Equilibrium Constants for Reactions of GMPCPP and GMPCP with Tubulin



Parameter	K_{eq}	How determined
K_1	6.30×10^3	Assumed to be identical to AMPCPP*
K_2	1.45×10^6	See Table V
K_4	1.38×10^5	See Table V
K_3	3.50×10^2	$K_1 K_4 = K_2 K_3$
K_5	2.09×10^6‡	Rate constant assembly/rate constant disassembly§
K_6	1.58×10^4	See K_5‖
K_7	2.65	$K_5 K_7 = K_3 K_6$

* Milner-White and Rycroft, 1983. Since this value was determined by comparison with ATP, we use a free energy for ATP hydrolysis equal to −8.8 kcal/mol (Jencks, 1968), corrected for pH 6.8.
‡ This is equal to 5×10^7, using the rate constants derived by Hyman et al. (1992). The difference is mainly in the rate for subunit dissociation, which was found to be 0.1 s^{-1} for both ends, compared to 1.35 s^{-1} found here. A larger value for K_5 would make the free energy change for K_7 more positive, reinforcing the conclusions we have derived.
§ k(tubulin-GMPCPP subunit addition)/k (tubulin-GMPCPP subunit dissociation) = 1.36×10^6 $m^{-1}/0.65$ s^{-1}. The subunit dissociation rate was derived from the 1.3 s^{-1} measured for both ends, taking into account the fact that the rate must be equal at the two ends. This equality is required because the rate of addition of tubulin-GMPCPP subunits was the same at the two ends, and because K_5, which is equal to the ratio of the rate constants, must be identical at the two microtubule ends when there is no NTP hydrolysis in the reaction (Wegner, 1976).
‖ As with K_5, with rates equal to 2.0×10^6 M^{-1} s^{-1} and 128 s^{-1}, or 1.0×10^6 M^{-1} s^{-1} and 63 s^{-1}.

with GMPCP and 0.69 μM with GMPCPP (at 23°); this corresponds to a free energy for binding equal to −7.01 and −8.04 kcal, respectively.

Discussion

Regulation of Microtubule Dynamics by Controlled Hydrolysis of a GTP Analogue

Although the GTP-cap model for microtubule dynamic instability is generally accepted and even made its way into textbooks, there is conflicting evidence for the presence of tubulin-GTP subunits in microtubules (reviewed by Caplow, 1992). The most convincing evidence that GTP hydrolysis modulates microtubule dynamics has come from comparison of the behavior of microtubules containing nonhydrolyzable GTP analogues and GDP.

The GTP analogue GMPCPP, which is susceptible to hydrolysis because the methylene bridge links the alpha and beta phosphate residues, is unique in binding relatively tightly to the tubulin E-site and in being more effective than GTP in promoting assembly (Sandoval et al., 1977; Hyman et al., 1992).

Table V. Equilibrium Constant for GMPCPP and GMPCP Binding to Tubulin

Nucleotide	% cpm bound	K_d
		(μM)*
1 mM GMPCP	74.5-80.2	4.7, 6.5
2.5 mM GMPCP	66.7-71.0	8.0, 9.8
1 mM GMPCPP	29.6	0.67
2.5 mM GMPCPP	14.6-15.1	0.71, .69

* The equilibrium constant was determined from the relationship: eluted [Tubulin-GMPCPP/Tubulin-GTP] = K_{GTP} $(GMPCPP)_{free}/K_{GMPCPP}$ $(GTP)_{free}$, where K_{GTP} is equal to 2.2×10^{-8} (Zeeberg and Caplow, 1979).

It is of special interest that the rate of tubulin subunit dissociation from microtubules is dramatically increased when the gamma phosphate moiety of a bound NTP has been lost by hydrolysis. The more than 100-fold greater rate for dissociation of tubulin-GMPCP subunits compared to tubulin-GMPCPP subunits provides direct evidence that microtubule dynamics can be regulated by NTP hydrolysis. This conclusion was previously derived from considering the requirement for GTP for dynamic instability, as well as from comparison of the properties of microtubules assembled with nonhydrolyzable GTP analogues that were presumed to exactly mimic GTP, with the properties of GDP-microtubules.

Free Energy for Hydrolysis of Microtubule-bound GMPCPP

The equilibrium constant for hydrolysis of GMPCPP bound to the E-site of tubulin subunits and microtubules was determined from experimental measurements of five of the equilibrium constants in Scheme I (Table IV). The most striking result is the decrease in the free energy for GMPCPP hydrolysis from −5.18 kcal in solution, to −3.79 kcal in subunits, to −0.90 kcal in microtubules. The near zero free energy for hydrolysis of microtubule-bound GMPCPP is of singular interest.

It must be noted that the products for hydrolysis of microtubule-bound GMPCPP (K_7 in Table IV) are the binary complex of GMPCP and tubulin subunits in the microtubule, and free Pi at a 1 M standard state. To estimate the free energy change to form the ternary complex contain-

ing both GMPCP and Pi in the microtubule requires correcting the −0.9 kcal value for K_7 for the free energy expended for Pi dissociation at the 1 M standard state concentration; this depends on the K_d for Pi dissociation. Pi binding to GDP-subunits in microtubules is weak (Carlier, et al., 1988; Caplow et al., 1989; Melki et al., 1990; Stewart et al., 1990; Trinczek et al., 1993) and a K_d equal to 25 mM was previously estimated from the effect of Pi on the disassembly rate in glycerol buffer (Carlier, et al., 1988). Based on this value for K_d, the free energy expended for Pi dissociation at the 1 M standard state concentration is +2.18 kcal ($-RT\ln 1/K_d$). Therefore, the standard free energy for forming the GMPCP-Pi-microtubule ternary complex would be +1.28 kcal (i.e., 2.18 − 0.9), corresponding to an equilibrium constant equal to 0.114. For comparison, the free energy change is −1.3 kcal for conversion of the myosin-ATP to the myosin-ADP-Pi ternary complex (Bagshaw and Trentham, 1973).

It is emphasized that the observed free energy change for NTP hydrolysis under physiological conditions will be more negative than +2.18 kcal, since the position of the equilibrium depends on the Pi concentration. With Pi buffered in cells at 1 mM, the apparent free energy change for NTP hydrolysis in the microtubule would be 4.08 kcal more negative ($-RT\ln 0.001$) than the +1.28 kcal; ie., equal to −2.8 kcal.

The relatively small free energy change for NTP hydrolysis (even at physiological Pi concentrations) indicates that much of the free energy potentially available from this reaction is stored in the microtubule lattice, presumably as a repulsive force between subunits. This repulsive force is manifested in the enormous rate for dissociation of subunits containing GMPCP or GDP. The energy stored in the microtubule lattice can do work, by coupling disassembly to vesicle or chromosome movement (Koshland et al., 1988; Coue et al., 1991). Finally, it is noted that calculation of the free energy for hydrolysis of microtubule-bound GTP is not possible since the equilibrium constant for addition of tubulin-GDP subunits to the microtubule (K_6 in Table IV) is not available.

Glycerol and Glutamate-Induced Hydrolysis

Our observation that glycerol induces hydrolysis of microtubule bound GMPCPP in Na Pipes buffer means that glycerol binds to microtubules. Although this would appear to be a reasonable conclusion, since glycerol is used to stabilize microtubules (Solomon, et al., 1973), the effect of glycerol has primarily been taken to involve destabilization of tubulin subunits. That is, results from studies of the effect of glycerol on the apparent specific volume of tubulin (Na and Timasheff, 1981) were interpreted as proving that glycerol is preferentially excluded from the solvation sphere of tubulin dimers. This increase in the chemical potential of tubulin (and glycerol) is presumably relieved by a shift of tubulin from dimer to microtubule. According to this model, the dramatic reduction by glycerol of the rate of dilution-induced disassembly (Caplow et al., 1986; O'Brien and Erickson, 1989) results because glycerol destabilizes the transition state for disassembly, since this transition state resembles tubulin subunits. However, equally likely is the possibility that glycerol binds to and stabilizes microtubules.

As to the mechanism through which glycerol and glutamate accelerate hydrolysis of GMPCPP in microtubules, we propose that interaction of these compounds with the microtubule induces a protein conformation change. It is suggested GMPCPP is ordinarily stable because the different stereochemistry of the methylene linkage between the alpha and beta phosphate groups alters the position of the gamma phosphate, so it is remote from the residues responsible for catalysis. A protein conformation change may move the phosphate group to the active site for catalysis. The different rates of glycerol-induced hydrolysis with Na and K-Pipes is unaccounted for.

It is noted that GMPCP-microtubules formed by glycerol-induced hydrolysis in GMPCPP-microtubules did not have identical kinetic properties as microtubules formed with GMPCP. The disassembly rates at the two ends were 146 s^{-1} rate and 5 s^{-1} after glycerol treatment (Table III), compared to 128 s^{-1} and 63 s^{-1} with microtubules that had been assembled with tubulin-GMPCP. This difference may result from the ~17% of the GMPCPP that remained after treating GMPCPP-microtubules with glycerol (see discussion of results in Table I). We suggest that the slower rate at one end with glycerol-treated microtubules results because the residual sticky tubulin-GMPCPP subunits that escape hydrolysis slow down the rate. The effect of these tubulin-GMPCPP subunits may be different at the two ends because the unhydrolyzed nucleotide interfaces with the solution at the plus end and with the core of the microtubule at the minus end (Mitchison, 1993).

We are grateful to Harold Erickson for helpful comments on the manuscript and to Tim Mitchison, who used his tubulin and our reagents to repeat our study of the effect of glycerol on GMPCPP hydrolysis. These experiments allowed us to identify the role of sodium and potassium in hydrolysis of GMPCPP in microtubules. Don Cleveland's editorial assistance is greatly appreciated.

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THE HUBBARD MODEL: SOME RIGOROUS RESULTS AND OPEN PROBLEMS

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Dedicated to Gian Fausto Dell'Antonio on his sixtieth birthday

Abstract

The Hubbard model of interacting electrons, like the Ising model of spin-spin interactions, is the simplest possible model displaying many "real world" features, but it is much more difficult to analyze qualitatively than the Ising model. After a third of a century of research, we are still not sure about many of its basic properties. This mini-review will explore what is known rigorously about the model and it will attempt to describe some open problems that are possibly within the range of rigorous mathematical analysis.

0. Introduction

The Hubbard model is to the problem of electron correlations as the Ising model is to the problem of spin-spin interactions; it is the simplest possible model displaying many "real world" features. It is, however, much more difficult to analyze qualitatively than the Ising model. After a third of a century of research, we are still not sure about many of its basic features. It is believed, for example, to have something to do with high temperature superconductivity, and it would be nice to validate this.

One of the most intriguing questions concerns magnetism — particularly the magnetic properties of the ground state. The familiar models of interacting spins, such as the Ising or Heisenberg models, posit an underlying spin Hamiltonian that ultimately comes from a model of itinerant electrons. The mystery here is that neither the electronic kinetic nor the electronic potential energies favor ferromagnetism, but together they sometimes do so. Ferromagnetic spin models, for example, are popular, but the truth is that the antiferromagnetic preference of the kinetic energy usually seems to dominate. In fact the only known examples of saturated itinerant electron ferromagnetism come from the

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Hubbard model in a special limit (infinite repulsion and one hole) devised by Nagaoka [NY] or from the Hubbard model with finite repulsion but on special lattices devised by Mielke and Tasaki [Mi1, TH2, MT] for which the kinetic energy spectrum is macroscopically degenerate. No doubt, further study of the Hubbard model will eventually lead to a better understanding of the way in which the Pauli exclusion principle leads to magnetism.

These notes will explore what is known rigorously about the model and it will attempt to describe some open problems that are possibly within the range of rigorous mathematical analysis. The notes are not a complete review in the sense that every contribution is covered, but an attempt has been made to mention, at least, most topics that can be treated rigorously. I apologize to those authors whose works have been omitted and I hope this will be attributed to my ignorance rather than intent. Given more space I would have liked to discuss the closely related Falicov-Kimball model, for which many rigorous results are known, but which is not a Hubbard model because it does not have $SU(2)$ symmetry. Another interesting chapter would be the infinite dimensional Hubbard model—pioneered by Metzner and Vollhardt [MV]—and which opens exciting mathematical and physical avenues. It has led to a large literature, but much remains to be added in the way of mathematical rigor.

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1. Definition of the Model

Only the original short-range Hubbard model will be considered here. Like the Ising model, the Hubbard model is defined on a graph, i.e., a collection of vertices or sites (denoted by Λ and whose number is $|\Lambda|$) and (unordered) edges or bonds connecting certain distinct pairs of vertices. The word graph instead of lattice is used to avoid any possible implication of translation invariance because most of the results stated here do not depend on such invariance. One is given a **hopping matrix** T, with elements t_{xy}, with x and $y \in \Lambda$, and we assume, as a convention, that $t_{xy} = 0$ if x, y are not connected by an edge. Note that $t_{xx} = 0$. t_{xy} might be complex, signaling the presence of a magnetic field, the line integral of whose vector potential from x to y (thought of now as points in $\mathbf{R}^3$) is $\arg(t_{xy})$. However, T is always self adjoint, $t_{xy} = t^*_{yx}$, with $*$ denoting complex conjugate.

The **bipartite** graphs form an important sub-class; here $\Lambda = A \cup B$, with A and B disjoint, and such that there is no edge between x, y if $x \in A$ and $y \in A$ or if $x \in B$ and $y \in B$. The square lattice is bipartite, the triangular is not.

Electrons, i.e., spin $\frac{1}{2}$ fermions, move on Λ with kinetic energy given in second quantized form by $K = K_\uparrow + K_\downarrow$ with

$$K_\sigma = -\sum_{x,y\in\Lambda} t_{xy} c^\dagger_{x\sigma} c_{y\sigma}. \tag{1.1}$$

Here $\sigma = \pm 1$ denote the two spin states $\uparrow$ and $\downarrow$ while $c^\dagger_{x\sigma}$ is the creation operator for an electron at x with spin σ. We have $c^\dagger_{x\sigma}c_{y\tau} + c_{y\tau}c^\dagger_{x\sigma} = \delta_{xy}\delta_{\sigma\tau}$ and $c_{x\sigma}c_{y\tau} + c_{y\tau}c_{x\sigma} = 0$. The number operator is defined by $n_{x\sigma} = c^\dagger_{x\sigma}c_{x\sigma}$ and has eigenvalues 0 and 1. The total number of each spin species

$$N_\sigma = \sum_{x\in\Lambda} n_{x\sigma}$$

is a conserved quantity. The total particle number is $N = N_\uparrow + N_\downarrow$ which satisfies $0 \leq N \leq 2|\Lambda|$. The **half-filled band**, $N = |\Lambda|$ is especially important and especially amenable to analysis.

In the physics literature it is often assumed that $t_{xy} = \text{constant} = t > 0$ on all edges of Λ, in which case T is a discrete version of the Laplacian, but without the diagonal terms. This assumption will not generally be made here.

At each site x there is also given a number U_x which governs the on-site electron-electron interaction at x. We usually assume all $U_x \geq 0$ (repulsive case) or all $U_x \leq 0$ (attractive case). The total potential energy is[†]

$$W = \sum_{x\in\Lambda} U_x(n_{x\uparrow} - \tfrac{1}{2})(n_{x\downarrow} - \tfrac{1}{2}). \tag{1.2}$$

The total Hubbard Hamiltonian of our system is then

$$H = K + W, \tag{1.3}$$

and most of the results discussed here are about the ground state of this H. In fact, many of the results do not require a point interaction as in (1.2); instead, terms like $U_{xy}(n_{x\uparrow} - \frac{1}{2})(n_{y\downarrow} - \frac{1}{2})$, with the matrix U_{xy} being positive semidefinite, are also allowed.

The interaction in (1.2) includes a one-body term $-\frac{1}{2}\sum_x U_x(n_{x\uparrow} + n_{x\downarrow})$ plus a trivial constant term $\frac{1}{4}\sum_x U_x$. If U_x is independent of x, as is normally assumed in almost all papers on the subject, then this one-body term is trivially a constant proportional to the fixed particle number. It is interesting to consider $U_x \neq$ constant, and the one-body term is included in (1.2) in order to be able to exploit **hole-particle symmetry**: The unitary transformation that maps $c_{x\downarrow} \to c^\dagger_{x\downarrow}$ and $c^\dagger_{x\downarrow} \to c_{x\downarrow}$ (but $c_{x\uparrow} \to c_{x\uparrow}$) is the hole-particle transformation on the down-spins and it maps $(n_{x\downarrow} - \frac{1}{2})$ into $-(n_{x\downarrow} - \frac{1}{2})$. Thus, this transformation maps the repulsive W into the attractive W and vice versa, but it changes the down-spin number from $N_\downarrow$ to $|\Lambda| - N_\downarrow$.

This hole-particle transformation may or may not map $K_\downarrow$ into $K_\downarrow$. It does map $K_\downarrow$ into $\widehat{K}_\downarrow$, defined by $\widehat{t}_{xy} = -t^*_{yx}$.

[†] The usual formulation is $n_\uparrow n_\downarrow$ instead of our $(n_\uparrow - \frac{1}{2})(n_\downarrow - \frac{1}{2})$. If U_x varies with x the two formulations are obviously inequivalent. The formulation here is not without a physical foundation because a neutral atom (with $n_\uparrow + n_\downarrow = 1$) is locally the most stable configuration and adding or removing an electron produces a net local charge that raises the energy roughly equally.

In the special, but important case that T is real and Λ is bipartite we can make a further unitary transformation that will take $\widehat{K}_\downarrow$ into $K_\downarrow$ and $n_{x\downarrow}$ into $n_{x\downarrow}$. This unitary transformation maps $c_{x\downarrow}$ into $(-1)^x c_{x\downarrow}$ (and, of course, $c_{x\uparrow}$ into $c_{x\uparrow}$), where $(-1)^x$ denotes the function on the vertices of Λ which is $+1$ for $x \in A$ and -1 for $x \in B$. Effectively, t_{xy} is mapped into $(-1)^x(-1)^y t_{xy} = -t_{xy}$ for the down-spins. Henceforth, the hole-particle transformation on bipartite lattice is always meant to include this additional unitary $(-1)^x$, so that $K_\downarrow$ is mapped into $K_\downarrow$.

Thus, in the special real, bipartite case we can analyze the repulsive case by analyzing the attractive case. But we have to remember that $N_\downarrow \leftrightarrow |\Lambda| - N_\downarrow$ so that the particle number for one is related to the magnetization of the other. This is a non-trivial distinction and is similar to the relation between the ferromagnetic and antiferromagnetic Ising models on a bipartite graph.

The hole-particle transformation also induces a conservation law in the real, bipartite case that has no classical analogue. Like any electron system this model has an $SU(2)$, i.e., angular momentum, invariance. The generators are

$$J^3 = \tfrac{1}{2}(N_\uparrow - N_\downarrow), \qquad J^+ = \sum_{x\in\Lambda} c^\dagger_{x\uparrow} c_{x\downarrow}, \qquad J^- = (J^+)^\dagger. \tag{1.4}$$

However, the Hamiltonian H in (1.3) is the unitary transform (described above) of another H' (with U_x replaced by $-U_x$) which also has an $SU(2)$ symmetry with generators given by (1.4) in the *transformed* basis. By transforming back we obtain a different set of $SU(2)$ generators

$$\widehat{J}^3 = \tfrac{1}{2}(N_\uparrow + N_\downarrow - |\Lambda|), \qquad \widehat{J}^+ = \sum_{x\in\Lambda} (-1)^x c^\dagger_{x\uparrow} c^\dagger_{x\downarrow}, \qquad \widehat{J}^- = (\widehat{J}^+)^\dagger \tag{1.5}$$

that also commute with H and with the J's in (1.4). We call this the **pseudospin**. This special model thus has an $SU(2)\times SU(2)$ symmetry group,[§] but it must be emphasized that real T is *essential* (zero magnetic field). These operators change the quantum numbers $N_\uparrow$ and $N_\downarrow$. The operators $J^\pm$ change J^3 by one unit while $\widehat{J}^\pm$ change $\widehat{J}^3$ by one unit. The four operators together thus permit us to move around inside rectangles in $(N_\uparrow, N_\downarrow)$ space whose four vertices have the form $(n,m), (m,n), (|\Lambda|-n, |\Lambda|-m), (|\Lambda|-m, |\Lambda|-n)$. This means that to each eigenstate for $N_\uparrow = n$, $N_\downarrow = m$ there is a corresponding eigenstate with the same energy at each point in the rectangle (there are also additional eigenstates, of course). Thus, the usual operators $J^\pm$ permit us to infer all the states of our Hamiltonian from knowledge of the states on the line $N_\uparrow - N_\downarrow = 0$ or the line $N_\uparrow - N_\downarrow = 1$. The pseudospin

[§] Actually, the group is $SO(4) = SU(2)\times SU(2)/Z_2$, as pointed out by Yang and Zhang [YZ]. The reason is that the two operators $w = -\mathbf{1}$ in each $SU(2)$ corresponds to only *one* operator on our Hilbert space, i.e., $w \otimes \mathbf{1} = \mathbf{1} \otimes w$. This reduction to $SO(4)$ coincides with the observation that in every state the spin and pseudospin are either both integral or both half-integral.

operators $\widehat{J}^{\pm}$ permit us to infer everything from knowledge of the line $N_\uparrow + N_\downarrow = |\Lambda|$ (half-filled band) or the line $N_\uparrow + N_\downarrow = |\Lambda| + 1$. But we repeat that this property of $\widehat{J}^{\pm}$ holds *only* for the real, bipartite case.

We note, for future use, that the hole-particle transformation (*without* $(-1)^x$) applied to *both* spins (namely $c_{x\sigma} \leftrightarrow c^\dagger_{x\sigma}$). does not preserve $\vec{J}$. Indeed, $J^3 \to -J^3, J^\pm \to -J^\mp$, but $\mathbf{J}^2 \to \mathbf{J}^2$.

The Hubbard model describes — in the simplest possible fashion — an interacting fermion system. It can be viewed this way, as a toy model, or it can be viewed, a bit more realistically in the repulsive case, as a serious model of π-electrons hopping between localized Wannier orbitals in some molecule such as benzene (with $|\Lambda| = 6$); the half-filled band, $N = |\Lambda|$, is then especially important because it corresponds to neutrality. The ultra-short range interaction is supposed to mimic a highly screened Coulomb potential. From the latter viewpoint it was known first in the chemistry literature as the Pariser-Parr [PP]-Pople [PJ] model; molecules having a bipartite structure are called "alternant molecules". It was a decade later that Hubbard [HJ], Gutzwiller [GMC] and Kanamori [KJ] realized its importance for bulk matter.

2. One-Dimensional Exactly Solvable Model

In 1968 the ground state of H was solved for the translation invariant one-dimensional ring by Lieb and Wu [LW] using the extension of the "Bethe ansatz" technique [BH, LL1] to fermions [MJ, FL1, YC, GM]. Shortly thereafter, Ovchinnikov [OA] used these results to calculate the elementary excitation spectrum at half filling. Recently, Essler and Korepin [ES] obtained a new and illuminating derivation. Coll [CC] extended Ovchinnikov's results to arbitrary filling, as did Woynarovich [WF]. Takahashi [TM] evaluated the magnetism curve at half filling and Shiba [SH] evaluated the magnetic susceptibility for all filling. Koma [KO] has formulated a sequence of approximations to the positive temperature free energy and correlation length (obtained by using the Trotter product formula for the partition function) which converge to the exact answer and such that each approximation can be calculated using the Bethe ansatz without any assumptions, such as the "string hypothesis".

These results were obtained in the thermodynamic limit, in which sums could be replaced by integrals. Woynarovich and Eckle [WE] evaluated the asymptotics of finite size effects on the ground state energy. For small chains the general "Bethe ansatz" solution, while correct, is too complicated for numerical evaluation. Heilmann and Lieb [HL] undertook to evaluate *all* the energy levels for *all* $U > 0$ for the benzene molecule ($N = 6, |\Lambda| = 6$). To our surprise we found many instances of both level crossings and of permanent degeneracy — as a function of U — and which were not accounted for on the basis of the known invariance groups (spin, pseudospin and symmetries of the hexagon). This means that the system has *non-abelian* symmetry groups and these *are dependent on* U (i.e., the group operations commute with H but *not* with K and W separately; a

well known example of this phenomenon, for the hydrogen atom, is the Runge-Lenz vector whose definition depends on the value of the electron's charge).

What are these invariants? Sixteen years later Shastry [SB] (unaware of [HL]) found many invariants and also a two-dimensional classical statistical mechanics vertex model whose transfer matrix commutes with our H. In fact he found a whole commuting family of such transfer matrices — which means that the one-dimensional model can be called "integrable". It is not clear whether all invariants of H are of Shastry's form. A few years later, Grosse [GH] (motivated by [HL]) published another derivation of some of the invariants.

Another interesting question is whether all the eigenstates of H are of the "Bethe ansatz" form. This was answered in the negative by Essler, Korepin and Schoutens [EKS] who went on to demonstrate that when the ring is bipartite (i.e., even length) the $SU(2) \times SU(2)$ generators J^+ and $\widehat{J}^+$ save the day. The lowest weight states (the ones that are annihilated by J^- and $\widehat{J}^-$) are claimed to be *all* of the "Bethe ansatz" type. All the remaining states are then obtained by application of J^+, and $\widehat{J}^+$.

Many of the "Bethe Ansatz" results rely on a "string" hypothesis. Moreover, some assertions in [EKS], notably the linear independence of the solutions, rely on an appeal to some properties of Shastry's invariants that have not been verified. It is desirable to put these matters on a more rigorous basis.

The literature about this one-dimensional solution is vast and the above remarks do not reflect everything that is known about the subject.

3. Magnetism

A. One-Dimension.

It is convenient, now, to take an open chain instead of a ring. Then, by a general theorem of Lieb and Mattis [LM1], whose proof also works for lattice systems with nearest neighbors hopping and a *completely arbitrary* many-body potential, $E^N(S)$, the ground state energy of H, as a function of total spin $S \leq \frac{1}{2}N$ and for N particles, satisfies

$$E^N(\tfrac{1}{2}N) > E^N(\tfrac{1}{2}N-1) > E^N(\tfrac{1}{2}N-2) > \ldots > E^N(0) \text{ or } E^N(\tfrac{1}{2}). \tag{3.1}$$

The numbers U_x are now totally arbitrary and not necessarily of one sign. [Note: The original theorem [LM1] uses a Perron-Frobenius positivity argument, which a-priori requires $t_{xy} > 0$. However, any T can be reduced to this form in one-dimension by the simple unitary gauge transformation $c_{x\sigma} \to \exp[-i\theta(x)]c_{x\sigma}$, which carries t_{xy} into $\widehat{t}_{xy} := t_{xy}\exp[i\theta(x) - i\theta(y)]$, where $\theta(x)$ is an arbitrary real number, selected so that $\widehat{t}_{xy} > 0$ for all x and y.] Since we can then take T to be real, our H is then the (down-spin) hole-particle transform of some other $\widehat{H}$, in which U_x is replaced by $-U_x$. Assuming N to be even, (3.1) says that the ground state of H has $N_\uparrow = N_\downarrow = N/2$. The transformed $\widehat{H}$ system has $\widehat{N}_\uparrow = N_\uparrow$, $\widehat{N}_\downarrow = |\Lambda| - N_\downarrow$, whence $\widehat{S}^3 = \frac{1}{2}(\widehat{N}_\uparrow - \widehat{N}_\downarrow)$ is the predetermined

number $\frac{1}{2}(N - |\Lambda|)$. The $\widehat{H}$ system also satisfies (3.1) since it is one-dimensional, and the spin of the ground state of the $\widehat{H}$ system is therefore

$$\widehat{S} = \tfrac{1}{2}\big|N - |\Lambda|\big|.$$

This number, $\widehat{S}$, is *the pseudospin of our ground state* — a curious result whose physical significance is not entirely obvious. (Note the logic here. Both systems, H and $\widehat{H}$ must be in their respective ground states, consistent with the given conditions on each; for H it is $N_\uparrow + N_\downarrow = N$ and for $\widehat{H}$ it is $N_\uparrow - N_\downarrow = N - |\Lambda|$.)

This theorem was extended many years later by Aizenman and Lieb [AL] to positive temperatures. The main theorem, applicable to any many-body potential in a one-dimensional system, expresses the fact that the free energy is a monotone increasing function of the spin. This is done in terms of total spin S or 3-component s, as follows; neither inequality implies the other.

Classify the eigenstates by the 3-component of spin, $J^3 = s = \frac{1}{2}(N_\uparrow - N_\downarrow)$, and by S, the total spin angular momentum (recall $\mathbf{J}^2 = S(S+1)$). For a given N and inverse temperature β, let $Z_3(s) = \mathrm{Tr}_s e^{-\beta H}$ be the partition function in which only states of a given s value are included. Likewise, let $Z_J(S) = \mathrm{Tr}_S e^{-\beta H}$ be the partition function for a given S. We have the relation $Z_3(S) - Z_3(S+1) = (2S+1)^{-1} Z_J(S)$, which is obvious from the theory of angular momentum. Correspondingly, we have the combinatorial quantities $Y_3(s) := \binom{N}{\frac{N}{2}+s}$ and $Y_J(S) := (2S+1)[Y_3(S) - Y_3(S+1)]$, which are essentially the partition functions of free particles with $H = 0$, and which serve to normalize the Z's. The two theorems are then that

$$\frac{Z_3(s)}{Y_3(s)} \quad \textit{and} \quad \frac{Z_J(S)}{Y_J(S)}$$

are *both strictly monotone decreasing functions of their arguments* (s or S). A corollary of this is that the magnetization is less than its value in the **atomic limit** (or what I prefer to call the pure paramagnetic value). I.e., for all β and magnetic field h

$$M(\beta, h) = \frac{1}{\beta}\frac{\mathrm{d}}{\mathrm{d}h}\ln Z(h) < N\tanh(\beta h). \tag{3.2}$$

Here $Z(h) = \sum_{s=-N/2}^{N/2} Z_3(s)\exp[2\beta s h]$ is the total partition function (recall that the g-factor of an electron is 2).

It is noteworthy that all this holds for *completely arbitrary*, Hermitian t_{xy} and (real) U_x. Indeed, it holds even if we add an arbitrary real one body potential

$$V = \sum_{x\in\Lambda} V_x(n_{x\uparrow} + n_{x\downarrow}).$$

An amusing fact about one-dimension concerns the $U = +\infty$ case. For an open chain the ground state is highly degenerate—indeed, it can have any value of S. For a closed

chain the situation is quite different and the S of the ground state depends on the sign of the t_{xy}'s and whether N is even or odd; S can be $N/2$ in some cases [AL]. The situation is discussed in detail by Mielke [Mi2] who finds that the "average" S is $\sqrt{N}$. Some interesting facts about the closed one-dimensional chain with $U = 0$, but with a magnetic field, can be found in [LL2]. The $U \neq 0$ case is discussed by Fujimoto and Kawakami [FK1].

B. Half-Filled Band.

The hole-particle symmetry notwithstanding, the repulsive case (all $U_x \geq 0$) and the attractive case (all $U_x \leq 0$) are quite different, even for a bipartite lattice. The physical spin of one is the pseudospin of the other.

In the limit $U \to +\infty$ (by which I mean all $U_x \to +\infty$), the energy and wave functions have nice limits. The electrons become hard-core particles. When $N = |\Lambda|$ we just have one electron per site and, since motion is impossible, each electron can be, independently, spin-up or spin-down. This gives us the atomic limit whose partition function is Y_3 or Y_J given above. First order perturbation theory in t/U vanishes, but in second order we have to diagonalize our H among all the $2^{|\Lambda|}$ degenerate ground states just described. This yields an effective Hamiltonian H' that can be written in terms of the three Pauli spin operators $\mathbf{S}$ at each site. It turns out [AP] that for any graph

$$H' = \sum_{x,y \in \Lambda} J_{xy}(\mathbf{S}_x \cdot \mathbf{S}_y - \tfrac{1}{4}) \tag{3.3}$$

with $J_{xy} = |t_{xy}|^2(U_x^{-1} + U_y^{-1})$. This is the spin 1/2 antiferromagnetic Heisenberg Hamiltonian and it is known [LE1, LM2] that its ground state has total spin

$$S = \big||A| - |B|\big| \tag{3.4}$$

on a bipartite graph. It is also known [DLS, KLS] that in the **translation invariant case** (i.e., our graph is a D-dimensional hypercube with periodic boundary conditions and $t_{xy} = \text{constant} = t$ and $U_{xy} = \text{constant} = U$) there is long range order when $D \geq 3$.

The obvious question is whether the results just stated (i.e., (3.4) and the long range order) hold non-perturbatively in the repulsive case. One would also guess that in the attractive case the total spin should be zero in the ground state because when $U_x = -\infty$ for all x the ground state consists simply of bound pairs of electrons sitting on selected sites. These questions about the spin are answered in the following [LE1].

Theorem 1: *Assume t_{xy} is real for all $x, y \in \Lambda$. If $U_x < 0$ for all x, the ground state on any connected graph is unique and has spin $S = 0$* for any even electron number, N, not just $N = |\Lambda|$. *If $U_x > 0$ for all x, if Λ is connected and bipartite and if $N = |\Lambda|$ is even, the ground state is unique (except for the $(2S+1)$-fold degeneracy) and has spin $S = \frac{1}{2}\big||A| - |B|\big|$.*

We can easily have $|A| - |B|$ of the order of $|\Lambda|$ itself. As an example, take a square lattice and add a site at the center of each bond of this square lattice. The original sites

are then B sites and the new sites are A sites. Then $|A| = 2|B|$ and the ground state has a bulk magnetization per site of 1/3. This is really more like ferrimagnetism than ferromagnetism but, in any case, it is one of the few examples known in which the system has a bulk magnetization without an external magnetic field.

There is an interesting corollary of this theorem if Λ is bipartite and if $|A| \geq |B|$. Suppose we ask for the absolute minimum energy, without fixing N. Starting with $U_x > 0$ we find that an optimum N is $N = |\Lambda|$ (by the $SU(2) \times SU(2)$ symmetry mentioned in Sec. 1). But $J^3 = (N_\uparrow - N_\downarrow)/2$ can be anywhere in the interval $(|B|-|A|)/2$ to $(|A|-|B|)/2$ since $S = (|A|-|B|)/2$. Now using hole-particle symmetry to obtain the $U_x < 0$ model, we find that the optimum N is any integer satisfying $2|B| \leq N \leq 2|A|$. (E.g., starting with $N_\uparrow = |A|, N_\downarrow = |B|$, the transformed values are $N_\uparrow = |A|, N_\downarrow = |\Lambda| - |B|$, which yields $N = 2|A|$.) Thus, *there can be a large degeneracy in the attractive case!* (For $U = 0$ this is easily seen from the remark that T has at least $|A| - |B|$ zero eigenvalues.)

This theorem was extended to positive temperature by Kubo and Kishi [KK] who found upper bounds on certain two-point functions. They discuss only the translation invariant case on a hypercubic lattice with $U_x = U =$ constant, but their method easily extends to the general case. For $U < 0$ they bound the spin susceptibility at wave vector q by

$$\chi_q \leq \frac{1}{4|U|} \tag{3.5}$$

for all temperatures and all filling fraction $N/|\Lambda|$ (more precisely, they use the grand canonical ensemble and prove (3.5) for all chemical potentials μ). This result precludes magnetic long range order. In the repulsive case $U > 0$, and with μ adjusted to the half-filled band case $N = |\Lambda|$, namely $\mu = 0$, they bound the charge susceptibility as

$$(\delta\widehat{n}_q, \delta\widehat{n}_{-q}) \leq (\beta U)^{-1} \tag{3.6}$$

and the pairing susceptibility as

$$(\widehat{p}_q, \widehat{p}_{-q}) \leq (\beta U)^{-1}. \tag{3.7}$$

Here (A, B) is the Duhamel two-point function $(A, B) = \int_0^1 \mathrm{Tr}[A^\dagger e^{t\beta H} B e^{(1-t)\beta H}]\, dt$ and $\delta\widehat{n}_q = \widehat{n}_q - \langle\widehat{n}_q\rangle$ and $p_x = c^\dagger_{x\uparrow} c_{x\downarrow}$ and $\widehat{\ }$ denotes spatial Fourier transform. Charge long range order is precluded by (3.6) while Cooper pairing is precluded by (3.7).

Although I am restricting this review to the Hubbard model, I cannot resist the temptation to mention that Theorem 1 has recently been extended [FL2] to another model—the Holstein model—in which electrons interact with a quantized phonon field instead of with each other. Again, the finite system ground state for any even number of electrons is unique and has zero spin. The method of proof of Theorem 1 has also been used by Ueda, Tsunetsugu and Sigrist [UTS] to show that the periodic Anderson model at half-filling has a singlet ground state.

This close connection between the half-filled band, repulsive Hubbard model and the antiferromagnetic Heisenberg model points to the first of our spin problems. The antiferromagnetic Heisenberg model on a hypercubic lattice, in the thermodynamic limit, has no long range order (LRO) at positive temperature in dimensions $D = 1$ or 2. This is a consequence of the Hohenberg-Mermin-Wagner theorem. The same is true for the Hubbard model as first shown by Walker and Ruijgrok [WR], then by Ghosh [GD]. Later, Koma and Tasaki [KT]proved it by a different method—that of McBryan and Spencer [MS]. As far as the ground state is concerned, the Heisenberg model has LRO for $D = 2$ and spin 1 or more per site [KLS] (i.e. $|\mathbf{S}_x|^2 = S(S+1)$ and $S \geq 1$) and it is believed to have LRO also when $S = 1/2$, which is the case of interest for us. The $D = 1$ case is believed to have no LRO in the ground state. For $D \geq 3$ there is LRO in the ground state for all $S \geq 1/2$ [KLS] and at positive temperature [DLS, KLS] for $S \geq 1$ (presumably, also for all $S \geq 1/2$). The obvious conjecture is the following.

Problem 1: *Prove that there is antiferromagnetic LRO in the half-filled band, repulsive Hubbard model on the hypercubic lattice (with $t_{xy} = 1$ for $|x - y| = 1$ and $U_x =$ positive, finite constant) in the ground state when $D = 2$ and for positive temperature when $D \geq 3$. For which values of N will the attractive model have LRO?*

The Falicov-Kimball [FK2] model poses an analogous problem that *can be solved* affirmatively. In this model, one kinetic energy term, $K_\downarrow$, is omitted from the Hamiltonian. Although the down spins are not dynamic their locations are left arbitrary. After "integrating out" the movable particles (up-spins), an effective interaction among the fixed particles (down-spins) is left. This, then, is a classical lattice gas with a complicated interaction. It resembles an Ising (not Heisenberg) model when U is large and it can be shown [KL] to have long range order at low temperatures (and no long range order at high temperatures) in two or more dimensions—as does the Ising model.

C. The Surprising Hole.

In the previous subsection we considered the half-filled band and showed a strong tendency to antiferromagnetism. The only important requirement on T was that it was real (i.e., no magnetic field acts on the orbital motion). Nagaoka [NY] made a surprising discovery about the case $U = +\infty$, but *with one hole* (i.e., $N = |\Lambda| - 1$). Thouless [TD] had a similar result a bit earlier (cf. note 7 in [NY] and the discussion on p. 47 in [LE2]) but there is little doubt that Nagaoka's presentation of this particular result is clearer, more precise and applicable to certain non-bipartite lattices such as *bcc* and *fcc*. The Thouless approach uses a Perron-Frobenius argument that appears to be restricted to bipartite lattices; for such lattices it is true that fermions behave like bosons when $U = +\infty$ and there is one hole. Nagaoka, on the other hand seems to require a regular Bravais lattice, but this is not really necessary provided *all* t_{xy} are nonpositive. The fully general result with a considerably simplified proof, was given by Tasaki [TH1].

It is usually assumed in the condensed matter literature that the t_{xy}'s are nonnegative, but there seems to be neither a compelling reason for this assumption nor many examples

in which it can be verified (S. Kivelson, private communication). (For a bipartite lattice, however, one can have either sign with the help of the unitary operator $(-1)^x$ applied to *both* spin $\uparrow$ and $\downarrow$.) Notice that $t_{xy} > 0$ puts the lowest kinetic energy at one point, namely $k = 0$, in Fourier space. The condition $t_{xy} < 0$ puts it at the edge of the Brillouin zone, and thus the lowest kinetic energy can be highly degenerate. The physical intuition is then quite different in the two cases—a fact that should not be lost sight of because of the hole-particle symmetry that holds for bipartite graphs. The proof in [TH1] shows that negative, not positive t_{xy} is the *natural mathematical assumption* for this theorem. If this upsets anyone's physical proclivity, that is a pity.

Theorem 2: *If $N = |\Lambda| - 1$, if $U_x = +\infty$ for all $x \epsilon \Lambda$, and if $t_{xy} \leq 0$ for all x, y, then the ground state has total spin $S = N/2$. This state is unique up to the trivial $(N+1)$–fold degeneracy if Λ satisfies a certain connectivity condition [TH1].*

The connectivity condition mentioned above is not stringent and it holds for all regular lattices in dimension greater than one (see also [AL]). Essentially it means that there are loops that permit nontrivial permutations of the particles.

In the case that Λ is completely translation invariant, i.e., that all vertices of Λ are equivalent, as is the case on a hypercubic lattice on a torus, a different proof of the theorem, very similar to Nagaoka's, was given by Tian [TG1] and by Trugman [TS].

The obvious next question to ask is this: *If there is more than one hole $(N < |\Lambda| - 1)$ and $U = +\infty$, is the ground state totally ferromagnetic, i.e., is $S = N/2$?* There can be no simple general theorem because numerical calculation on small systems show that the answer seems to be "no" and, at the same time, no simple pattern seems to emerge. Yet there are a few theoretical results, as follows.

(i) The Nagaoka-Tian-Trugman method can be generalized for $\Lambda =$ the D-dimensional hypercubic lattice with periodic boundary conditions and with $t_{xy} = t =$ nonpositive constant to show [TG2, TG3, TS, SQT] that the completely magnetized state energy, $E(S = N/2; \Lambda)$, when compared to the actual ground state energy, $E(\Lambda)$, satisfies

$$\lim_{\Lambda \to \infty} E(S = N/2; \Lambda) - E(\Lambda) = 0 \tag{3.8}$$

in the thermodynamic limit, $\Lambda \to \infty$, when the number of holes $N_h = |\Lambda| - N$ is not too large. The best result is by Shen, Qiu and Tian [SQT], which gives (3.8) when $N_h < |\Lambda|^\alpha$ with $0 < \alpha < 2/(D+2)$. Note that we do not divide by $|\Lambda|$ in (3.8), which thus truly represents the vanishing of a gap. The proof here is elegant and simple. However, one would expect (3.8) to hold as long as $N_h/|\Lambda| \to 0$ as $|\Lambda| \to 0$.

(ii) If there are many holes, $N_h/|\Lambda| > 0.49$ for the $D = 2$ or $N_h/|\Lambda| > 0.32$ for the $D = 3$ hypercubic lattices, and $\Lambda \to \infty$ as in (i) then

$$E(S = N/2; \Lambda) - E(S = N/2 - 1; \Lambda) \not\to 0. \tag{3.9}$$

There really is an instability of the $S = N/2$ state with respect to one spin flip. This was proved by Shastry, Krishnamurthy and Anderson [SKA]; see also [SA1]. The estimate was

improved to $N_h/|\Lambda| > 0.29$ by von der Linden and Edwards [LvE]; Hanisch and Müller-Hartmann [HM] simplified the calculation (but not the estimate of 0.29).

(iii) Several authors [DW, FRDS, SA1, TB] were able to prove, for a translation invariant hypercubic lattice model, that when there are two holes the ground state energies satisfy $E(S = N/2-1; \Lambda) < E(S = N/2; \Lambda)$. However, assumptions have to be made about the relative lengths of the sides. Sütő [SA1] extended this to $2, \ldots, 6$ holes for a bcc lattice.

(iv) Sütő [SA2] shows, as expected, that the energy splitting needed for demagnetization is, in any case, negligible. He proves that $M(\beta, h, \rho)$, the magnetization per site in field h at density $\rho = N/|\Lambda|$ satisfies (in the thermodynamic limit) $M(\beta, h, \rho) \to \tanh(\beta h)$ as $\rho \to 1$.

The results in (ii) and (iii) are achieved with a variational calculation. The value of $E(S = N/2; \Lambda)$ is easy to find exactly because it equals the energy of spinless electrons, i.e.,

$$E(S = N/2; \Lambda) = \sum_{j=1}^{N} \lambda_j(T) \tag{3.10}$$

for any Λ and hopping matrix, T, and in which $\lambda_1(T) \leq \lambda_2(T) \leq \cdots$ are the eigenvalues of T. Thus, the hard problem is to find a good variational function with $S = N/2 - 1$, and this appears to be extraordinarily difficult. Why? No one seems to know! And why is it so difficult to treat $S = N/2 - 2$?

These results, (i)–(iii), show that one cannot expect $S = N/2$ except when $N = |\Lambda|-1$, but one can ask the following.

Problem 2: *With $U = +\infty$, for which $\rho := N/|\Lambda|$ is it true that some (if there is more than one) ground state has $S/|\Lambda| > 0$ in the thermodynamic limit $\Lambda \to \infty$?*

This brings us to two more open problems about the $U = +\infty$ case; the first is a corollary of the second. We take a large Λ and N particles and suppose that the thermodynamic limit $\Lambda \to \infty$ with $\rho = N/|\Lambda|$ fixed is well defined. We set $S =$ spin of the ground state (the maximum such spin if there is more than one ground state) and we set $S_{\max} = N/2$. We also assume that $t_{xy} \geq 0$ for all x and y. Then

Problem 3: *Prove or disprove that*

$$\lim_{\rho \to 0} \lim_{\Lambda \to \infty} \quad S/S_{\max} = 0. \tag{3.11}$$

Problem 4: *Does there exist some number $\rho_c > 0$ such that*

$$\lim_{\Lambda \to \infty} \; S/S_{\max} = 0 \;\; \textit{for all } \rho < \rho_c \,? \tag{3.12}$$

The requirement that $t_{xy} \geq 0$ is important. As we shall see in Sect. D, Mielke's work shows that there can be nice, periodic lattices in any dimension (such as the kagome lattice in two-dimensions) for which $S = S_{\max}$ for all $\rho < \rho_c$ with $\rho_c > 0$, thereby contradicting (3.11) and (3.12). To achieve this, however, one needs $t_{xy} \leq 0$. Perhaps (3.11) and (3.12)

hold in the case $t_{xy} \leq 0$ if we replace "maximum such spin" by "average of such spins", because Mielke's and Tasaki's examples have highly degenerate ground states with spins ranging from 0 to $N/2$.

Closely related in spirit to Theorem 2, but with an interesting, different proof, is the result of Chakravarty, Chayes and Kivelson [CCK]. They start with a half-filled band, $N = |\Lambda|$, and U large. Then they add or subtract one or two particles, and define $E_n := E(N = |\Lambda| + n)$ for $-2 \leq n \leq +2$. They then define

$$\Delta_e = 2E_1 - E_2 - E_0$$

$$\Delta_h = 2E_{-1} - E_{-2} - E_0. \tag{3.13}$$

The interpretation of Δ_e is as a 2-particle binding energy, while that of Δ_h is as a 2-hole binding energy. The picture of Δ_e, for example, is that given two very large systems at half-filling, and given two extra electrons, is it energetically favorable to add the two electrons to one system ($\Delta_e > 0$), or is it favorable to add one electron to each system ($\Delta_e < 0$)? The former, $\Delta_e > 0$, connotes **pair binding**.

It is pointed out in [CCK] that $\Delta_e \leq 0$ and $\Delta_h \leq 0$ when $U = 0$, but they quote numerical studies showing that $\Delta_e > 0$ and $\Delta_h > 0$ for some U and some Λ. They prove, however, that in the limit $U \to \infty$, $\Delta_e \leq 0$ if all t_{xy} are nonnegative and $\Delta_h \leq 0$ if all t_{xy} are nonpositive. For a bipartite graph the sign does not matter (as long as all t_{xy} are positive or all are negative), and thus $\Delta_e \leq 0$ and $\Delta_h \leq 0$ in this case. They interpret this result to mean that the numerical positive binding results are only an intermediate U phenomenon but, strictly speaking, the (unlikely) alternatives $\Delta_e = 0$ and $\Delta_h = 0$ in their theorems would first have to be eliminated.

D. Another Path to Ferromagnetism.

The one-hole, $U = +\infty$ model is not the only one known to have *saturated* ferromagnetism (i.e., $S = N/2$). Mielke [Mi1] and later Tasaki [TH2] and then both [MT] found interesting, but very special models with this property.

The basic idea is to find a graph Λ and a hopping matrix T such that the lowest eigenvalue of T (call it λ_0) is highly degenerate; in fact we want the degeneracy to be at least N and, to be interesting, we want that to be of the order of $|\Lambda|$. Let N_0 denote the degeneracy of this lowest eigenvalue and denote the space of these eigenfunctions by $\mathcal{H}_0$. If $U_x \geq 0$ for all $x \in \Lambda$, it is easy to see that if $N = N_0$ then

(i) There is a ground state having $S = N_0/2$.

(ii) The state is simply a determinant formed from the N_0 vectors in $\mathcal{H}_0$ and its energy is $N_0\lambda_0$.

(iii) The ground state is unique if certain additional conditions (known to be optimal) are met [Mi1].

If $N < N_0$ the ground state manifold will contain at least one state with $S = N/2$, but perhaps others as well (see [Mi1]).

A comparison with Theorem 1 is interesting, but it is not clear whether or not it is misleading. Note that the $S = \big||A| - |B|\big|/2$ result there for a bipartite graph and a half-filled band was somehow related to the $\big||A| - |B|\big|$-fold degeneracy of the zero-mode of T. Thus, a common feature is degeneracy, and it is often said that itinerant ferromagnetism is associated with atomic or kinetic energy degeneracy. But there are also important differences:

(i) The spin in Theorem 1, while it might be proportional to $|\Lambda|$, is not $N/2 = |\Lambda|/2$.

(ii) No fine tuning of T or of Λ was needed for Theorem 1. All that was needed was the bipartite structure, the positivity of the U_x's and the reality of the t_{xy}'s.

Mielke's way of achieving the degeneracy N_0 is to start with some graph G and then to set $\Lambda = L(G)$, the **line graph of G**, which is defined as follows. Make a mark in the center of every edge of G; those marks will be the sites of $L(G)$. Two sites are connected by an edge in $L(G)$ if the two edges of G on which they reside have a G-site in common. Note that $L(G)$ is never bipartite, except for the trivial case of a ring. A well known example of a line graph is the kagome lattice. (Incidentally, kagome is not a person—it is a pattern of woven bamboo.)

The hopping matrix T is defined to be $-t < 0$ on every edge of $\Lambda = L(G)$. Not only is there a restriction on the magnitude of t_{xy} but we see, once again, that negative t_{xy} is the natural sign—as it is for Theorem 2.

Of course, as Mielke and Tasaki note, and as is also true in Theorem 2, one can convert positive t_{xy} into negative t_{xy} by a hole-particle transformation on both spin $\uparrow$ and spin $\downarrow$. This does not alter the interaction or the total spin, S, but it changes N into $N' = 2|\Lambda| - N$. For positive, t_{xy}, then, we can transform to the negative t_{xy} situation and conclude that $S = N'/2$ when $N' \leq N_0$. This translates into $S = |\Lambda| - N/2$ when $N \geq 2|\Lambda| - N_0$ (and not $\leq$). This construction would yield saturated ferromagnetism, $S = N/2$, only at $N = |\Lambda|$ (half-filled band), provided it were possible to achieve $N_0 \geq |\Lambda|$; this is clearly impossible (unless $T = 0$), so the positive t_{xy} choice does *not* yield the desired saturated ferromagnetism. However, it is still possible to have unsaturated ferromagnetism, i.e. $1 > 2S/N \neq 0$ in the thermodynamic limit. By taking $N = 2|\Lambda| - N_0$ and $S = |\Lambda| - N/2$ we have $2S/N = N_0/(2|\Lambda| - N_0)$. (One *can* reasonably argue, however, that the condition $S = |\Lambda| - N/2$ *is indeed* saturated ferromagnetism because this value of the spin is the maximum possible one when $N > |\Lambda|$, given the constraints of the sytem and given the Pauli principle. The system is constrained by allowing only a limited number of states for the electrons, i.e., two per site. I leave the semantic resolution to the reader.)

The homology of G determines $\mathcal{H}_0$ in a simple way. Pick any closed, self avoiding path in G of even length. This path corresponds, in an obvious way, to a closed path, P, of the same length in $L(G)$. If, now, we take the vector $\phi(x) = 0$ for x not a vertex of P and $\phi(x) =$ alternately $+1$ and -1 as we traverse the vertices of P, we see at once that ϕ is an eigenvector with eigenvalue $-2t$. It is also not hard to prove that $-2t$ cannot be improved, i.e., $\lambda_0 = -2t$. Moreover, this construction yields all of $\mathcal{H}_0$.

In [TH2, MT] essentially the same result as Mielke's (with similar requirements on T) is achieved, but with certain decorated lattices with next nearest neighbor hoppings. The ferromagnetism of [TH2, MT] is shown to be stable under small change of the electron density, by using a "grand canonical ensemble" with a fixed electron density. It is also proved that there is a transition to a paramagnetic phase as one decreases the electron density. Thus (3.11) and (3.12) (with S replaced by its grand canonical average) are proved for these special graphs.

In all cases, the eigenvectors in $\mathcal{H}_0$ can all be taken to have compact support, i.e., each $\phi(x)$ vanishes except on a finite set of sites of Λ, and each such set is independent of Λ once Λ is large enough to include the set. This property leads to the result [MT] that the effective Hamiltonian, in a suitable subspace of states that includes the ground states, can be written exactly as a Heisenberg Hamiltonian.

4. The Flux Phase Problem

Very little seems to be known rigorously about the effect of a magnetic field on the orbital motion, that is, if we set

$$t_{xy} = |t_{xy}| \exp[i\theta_{xy}], \qquad \theta_{xy} = \int_x^y A \cdot ds \tag{4.1}$$

for some vector potential A. The spin of the ground state might well change. The known proofs of Theorems 1 and 2 fail when $A \neq 0$. If, indeed, the spin of the ground state changes then we have a new kind of "magnetic field—spin" interaction, brought about by the Pauli exclusion principle. Indeed, something similar is discussed in the one-dimensional context in [FK1].

The energy certainly does change, and it is by now well known that when $U = 0$ the zero flux state does *not* give the lowest ground state energy. When Λ is a square lattice and when $N = |\Lambda|$, it is conjectured that the *maximum* (!) possible flux, namely π in each square ($t_{12}t_{23}t_{34}t_{41} = -1$ around a square), is optimum. (For a survey of this question and for some rigorous results about it see [LL2].) The same conjecture has been made for $U \neq 0$.

Problem 5: *Solve the flux phase problem for $U \neq 0$ (or even for $U = 0$) for a half-filled band on a square lattice.*

The moral of this story is that the Pauli exclusion principle can really upset our ideas about diamagnetism. For one solitary electron, the imposition of a magnetic field raises the ground state energy. For many electrons it can and does lower the energy. It is trivially true, for example, that when $U = +\infty$ and $N = |\Lambda|$, or $U =$ anything and $N = 2|\Lambda|$, a magnetic field has absolutely no effect on the energy.

5. Uniform Density Theorem

The hole-particle transformation has remarkable consequences for the Hubbard model on a bipartite graph with a half-filled band, $N = |\Lambda|$. The ideas given below are well known to chemists—less so to physicists. They go back to Coulson and Rushbrooke [CR] for the $U = 0$ case, and to MacLachlan [MA] who generalized them to many interacting models—including the Hubbard model as a special case. A simplified proof together with an extension to models involving explicity spin-spin interactions (such as the $t-J$ model), and to the Falicov-Kimball model, is given in [LLM].

The results apply equally to three cases: (a) The canonical Gibbs state with $N = |\Lambda|$; (b) The grand-canonical Gibbs state with zero chemical potential; (c) The ground state with $N = |\Lambda|$ and which is defined, in case of degeneracy, to be the $\beta \to \infty$ limit of the canonical Gibbs state. The one-body density matrix $\rho_\sigma(x, y)$ is the expectation value of $c^\dagger_{x\sigma} c_{y\sigma}$ in the state in question.

Theorem 4: *For a half-filled band on a bipartite lattice, the one-body density matrix for each of the above states satisfies (for each* $\sigma =\uparrow$ or $\downarrow$*).*

$$\rho_\sigma(x, y) = \tfrac{1}{2}\delta_{xy} \qquad \textit{if } x, y \epsilon A \textit{ or } x, y \epsilon B. \tag{5.1}$$

If $x\epsilon A$ and $y\epsilon B$, nothing simple can be said. Note that the theorem does *not* require T to be real, and is thus one of the few theorems that applies to complex t_{xy}. However, a true, physical magnetic field would also act on the electron spins and thereby vitiate the hole-particle symmetry needed for the proof.

The complex case is a bit subtle, for it uses more than just a hole-particle transformation (call it W). It also utilizes the nonlinear antiunitary map J that maps a vector ψ, considered as a polynomial in the $c^\dagger_{x\sigma}$'s applied to the vacuum, into the vector ψ^* corresponding to the polynomial with complex conjugate coefficients. While J is nonlinear, JKJ is linear when K is any linear operator, and $TrJKJ = (TrK)^*$. The antiunitary $Y = JW$ satisfies $Y = JW = WJ$ (and hence $Y^2 = \mathbf{1}$) and $Yc_{x\sigma}Y = Wc_{x\sigma}W$. Most important is the invariance of the Hamiltonian, $YHY = H$, which replaces the hole particle invariance, $WHW = H$, which fails for complex T.

What is the significance of this result? It seems to contradict the conjecture in Problem 1 that there can be antiferromagnetic, i.e., staggered, LRO. Such an ordering can occur only in the thermodynamic limit and it has the property that for every state with ordering $\uparrow\downarrow\uparrow\downarrow\cdots$ there is a state with ordering $\downarrow\uparrow\downarrow\uparrow\cdots$. The point about Theorem 4, applied to the thermodynamic limit, is that for every state with ordering there is an equally good state (obtained by changing boundary conditions) with the opposite ordering, and on the average each site will have the same density for each spin value. In other words, *there is no way to adjust the potentials* U_x *or the hopping matrix* t_{xy} *in a clever way so as to enhance the occupation of certain sites* — in a manner independent of boundary conditions. This stability is remarkable and, although it is not true for real materials, the theorem hints

at some kind of remnant stability that might transcend the overly idealized assumptions needed for its proof.

Problem 6: *Is there any residue of (5.1) when* $N \neq |\Lambda|$?

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The molecular theory of polyelectrolyte solutions with applications to the electrostatic properties of polynucleotides

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I. Introduction

Although the importance of the polyelectrolyte character of DNA has been recognized for some time (Felsenfeld & Miles 1967), few of the implications have been explored, primarily because of a lag in translating the breakthroughs in polyelectrolyte theory of the last decade into a form that is well adapted to the analysis of the specialized problems of biophysical chemistry. Perhaps an analogous situation existed in the field of protein chemistry during the period after the formulation and confirmation of the Debye–Hückel theory of ionic solutions but before Scatchard's incorporation of the theory into his analysis of the binding properties of proteins. An achievement for polynucleotide solutions parallel to Scatchard's was recently presented by Record, Lohman, & de Haseth (1976) and further developed and reviewed by Record, Anderson & Lohman (1978). Record and his colleagues have succeeded in the self-consistent incorporation of the central result of the theory of polyelectrolyte solutions – counterion condensation – into the thermodynamic description of multiligate binding. A general conclusion of major importance arising in a systematic way from their analysis is that the release of counterions is a substantial contribution to the driving force for the binding of cationic substances to polynucleotides. A limitation of the thermodynamic approach, however, is a sharply reduced capacity for numerical prediction.

Indeed, polyelectrolyte theory has been a successful example of statistical mechanical methodology, whereby assumptions are made on the molecular level, a molecular model is constructed on the basis of these assumptions, and quantitative information about the behaviour of the model is then derived from well-known computational rules. In optimal cases – and polyelectrolytes present such a case – the parameters of the model are independently known, and then the results of the computation provide *a priori* numerical predictions that may be directly compared with the results of measurement. Good agreement strengthens the belief that the model is an accurate representation of reality; the experimentally observed behaviour has been successfully explained in terms of intermolecular forces and molecular movements. Systematic discrepancy between prediction and observation may also be of value, in so far as specific aspects of physical reality that may have been omitted from the model may be revealed. In this paper I have attempted to extend the theory of polyelectrolyte solutions to cases of

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interest in polynucleotide chemistry. Its statistical mechanical character, and hence its full predictive and explanatory capability, is retained.

I shall begin with a detailed discussion of the ionic state of an extended polynucleotide structure in an aqueous environment of simple counterions, such as the alkali metal and alkaline earth cations. I shall then develop the concept that in such an environment the polynucleotide is 'potentiated' by its counterions for the spontaneous occurrence of any process that results in the relief of the electrostatic stress on the polynucleotide set up by the repulsive interactions among the phosphate groups. Binding of multivalent cationic substances is only one process of this type. Others include transitions to conformation of lower charge density – the DNA helix–coil transition being the prototypical example – and the tendency of DNA to straighten spontaneously from a locally bent configuration (a thermodynamic consequence of its very long persistence length).

To varying degrees of detail, but always on a molecular, as distinct from thermodynamic, level, I will be able to discuss the binding to polynucleotides of cationic substances as diverse as sodium and magnesium ions, spermine, polylysine, and *lac* repressor. At this stage perhaps it will be forgiven if the variation in mathematical detail is inversely related to the structural complexity of the cation. The analysis of ionic effects on conformational transitions includes those induced by strong electric fields as well as by temperature increases. I have also treated the local bending of DNA as a differential conformational transition to a state of higher charge density. The result suggests that DNA would not be a stiff molecule (on the scale of its persistence length) in the absence of the phosphate repulsions. An obvious corollary is that DNA will collapse, or condense, if sufficient charge neutralization by bound cationic substances occurs.

II. The molecular basis of polyelectrolyte behaviour

A. *Counterion condensation – a directly observable property of polyelectrolytes in solution*

Except for cases involving binding to the bases, the interaction of extended polynucleotide configurations with small ions is governed by the general laws of polyelectrolyte solutions. Indeed, the commonly heard dogma that a polyelectrolyte solution of low normality belongs

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nevertheless, because of the high counterion concentration in the immediate vicinity of the polymeric chain, to the realm of concentrated salt solutions – about which little or nothing of a general nature can be quantified – has turned out not to be true. All particular effects in the locally concentrated region are subsumed by the general phenomenon of counterion condensation, which, together with ionic screening interactions at long range, familiar from the classical theory of dilute small-electrolyte solutions, provides a framework for polyelectrolyte solutions (Manning, 1969, 1972*a*, 1974, 1977*a*, 1978; Bailey, 1973; Iwasa, 1977) capable of predicting with remarkable accuracy a large variety of measurable properties. The concept of counterion condensation is a radical departure from the ideas that have traditionally dominated the biochemical literature in this field. When first introduced as the basis for polyelectrolyte theory (Manning, 1969), its status as a physically sharp phenomenon, as opposed to a convenient mathematical device, was uncertain. Experimental progress has been rapid, however, and has established condensation as a fundamental physical property of polyelectrolyte solutions. It has become easy, therefore, to avoid all possibility of investing the definition of counterion condensation with an abstract quality, simply by describing its measurement.

Consider therefore the following ultrasonic absorption experiment (Zana *et al.* 1971). NaOH is added to an aqueous solution of carboxymethylcellulose (CMC), initially in its completely protonated form. No differential ultrasonic absorption (with respect to titration with tetramethylammonium hydroxide) can be detected until the titration has proceeded to a critical extent of neutralization; whereupon, with continuing titration, absorption increases up to complete neutralization of the polyacid. Addition of more Na^+ in the form of its chloride salt has no further effect on the ultrasonic absorption. The explanation is without ambiguity (Zana *et al.* 1971). In the initial stages of the titration the Na^+ counterions are not in close interaction with the CMC polyion. At a critical degree of neutralization, however, further addition of OH^- results not only in chemical neutralization of carboxyl protons but also their replacement near the CMC chain by Na^+ counterions added simultaneously. The volume fluctuations due to interpenetration of the electrostricted hydration layers surrounding both the Na^+ ion and the CMC polyion are detected by the ultrasonic probe. Sodium ions continue to 'condense' on the polyacid as added until the titration end-point is reached. All carboxyl groups are now deprotonated, but

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the charge fraction of the polymer – that is, the charge per carboxylate group – is less than unity if one counts the condensed Na^+ ions as part of the polymeric molecular unit. The charge fraction is then equal to α_{crit}, the critical degree of neutralization for the onset of differential ultrasonic absorption. Further addition of Na^+ as its chloride salt does not lead to increased Na^+ binding; the charge fraction is stable to increases of ionic strength. Conceptually similar experiments have been reviewed elsewhere (Manning, 1977*a*).

For the case of DNA volume effects are so small (an important point that will be expanded in subsequent discussion) that direct 'tracking' of the counterion itself appears to be necessary. Sodium-23 NMR linewidth measurements have established that the charge fraction of DNA in an environment of excess Na^+ is invariant to variations of ionic strength at least up to 0·5 M (Anderson, Record & Hart, 1978). Under these conditions there remain many electrostatically uncompensated phosphate groups, as demonstrated by NMR-monitored titration of tetrabutylammonium DNA by Na^+ ions; association of Na^+ with the DNA polyion is indicated up to $75 \pm 10\%$ compensation of the phosphate charge but no further (Anderson *et al.* 1978). NMR charge-fraction experiments on a variety of other polyelectrolytes have been reviewed (Manning, 1977*a*).

Similar results are obtained for the interaction of Mg^{2+} with DNA. Dialysis-monitored titration of DNA by Mg^{2+} in excess 1 : 1 salt yielded about 85–86% maximum DNA charge neutralization by bound Mg^{2+} (Skerjanc & Strauss, 1968; Bhat, 1974). Over a limited range of added counterion, a minimum non-zero charge fraction independent of counterion concentration has also been confirmed by direct NMR measurement for the cobalt and manganese polyphosphate systems (Spegt & Weill, 1976).

Thus, as a matter of physical observation, polyelectrolytes in general, and DNA in particular, in an environment containing only one species of small cation like Na^+ and Mg^{2+}, possess a saturated charge fraction less than unity but well above zero. The data reveal a further, more restrictive generalization; in each case the value of the charge fraction depends only on the valence of the counterion and the *axial charge density* of the polyelectrolyte. The charge fractions of fully deprotonated sodium polyacrylate (carboxylate groups), sodium polystyrenesulphonate and sodium polyphosphate are equal (Manning, 1975; Kielman, van der Hoeven & Leyte, 1976). If b is the average axial charge spacing

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(along the helical axis for DNA, along the contour axis for more flexible polymers) and if a dimensionless structural parameter ξ is defined as proportional to the charge density,

$$\xi = q^2/\epsilon kTb, \tag{1}$$

where q is the protonic charge, ϵ the bulk dielectric constant of solvent, k Boltzmann's constant, and T the Kelvin temperature, then a quantitative formulation of the physical observation is that the saturated charge fraction is close to $(N\xi)^{-1}$, where N is the valence (without sign) of the counterion. For water at 25 °C, $\xi = (7{\cdot}1/b)$ if b is expressed in Å (the temperature dependence is small and may be neglected for most purposes). The B form of DNA has step height 3·4 Å between base pairs, i.e. between phosphate pairs, so its value of b is 1·7 Å and $\xi_{\mathrm{DNA}} = 4{\cdot}2$. Then the charge fractions of DNA in aqueous Na^+ and Mg^{2+} environments are, respectively, 0·24 and 0·12.

The operational definition of counterion condensation can therefore be formulated thus: *the mode of binding of counterions of valence N to a polyelectrolyte is called condensation if, in an environment containing only the single counterion species, the charge fraction of the polyelectrolyte equals the constant value* $(N\xi)^{-1}$ *over a broad concentration range.*

B. *Nature of the condensed state; delocalized binding*

The extent to which condensed counterions are dehydrated and localized near discrete groups, that is, the 'tightness' of binding, may be expected to vary both with the nature of the charged groups and the counterion species (Strauss, 1974). At one extreme the counterion may be fully dehydrated and directly coordinated to several neighbouring charged groups. About half of the Mn^{2+} and Co^{2+} ions condensed on the polyphosphate chain are bound in this fashion (Spegt & Weill, 1976). At the other extreme the condensed counterion is said to be 'delocalized'. It retains the fully hydrated state characteristic of a pure aqueous environment while moving in an unrestricted and random way along the length of the polyion chain. If, for example, about half of the Mn^{2+} and Co^{2+} population condensed on polyphosphate is, as mentioned, completely dehydrated, the other half has been shown to retain at least a complete inner hydration layer (Spegt & Weill, 1976). NMR relaxation measurements have demonstrated the nearly unrestricted movement of possibly the entire population of Na^+ and Rb^+ condensed on DNA (Reuben, Shporer & Gabbay, 1975). The extremely small

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specificities observed in differential dialysis experiments among alkali metal ions bound to DNA (J. Span & U. P. Strauss, unpublished) indicates the delocalized binding mode for condensed Li^+ and K^+ as well. Relaxation kinetic studies on the binding of Mg^{2+} and Ca^{2+} to poly A and poly C reveal diffusion control of the binding process, a result consistent with delocalized movement of these alkaline earth cations when bound and inconsistent with tight dehydrated binding to localized sites on the polynucleotide (Pörschke, 1976*a*). There is no reason to anticipate dissimilar behaviour for DNA, and the very small dilatometric effects exhibited by the binding of Mg^{2+} to DNA (in contrast to much larger effects for a number of synthetic polyelectrolytes) is indeed direct evidence that Mg^{2+} binds to DNA with almost no dehydration (Skerjanc & Strauss, 1968).

That alkali metal and alkaline earth cations condensed on extended polynucleotide conformations are almost fully hydrated and delocalized on the polyion surface is in fact their *expected* behaviour. Each fixed phosphate group is monovalent and rigidly held (Olson, 1975) at a great distance, about 7 Å, from its nearest-neighbour phosphate groups. Singly charged phosphate monomers do not interact strongly with cations. Activity coefficients of aqueous MH_2PO_4 solutions decrease (i.e. the ionic interactions become stronger) in the order of decreasing *hydrated* radius of the alkali metal ion M^+ (Li^+ interacts more weakly than K^+) (Robinson & Stokes, 1959). No ion-pair formation between Mg^{2+} and $H_2PO_4^-$, monovalent glucose 1-phosphate, or monovalent glycerol 2-phosphate could be detected by sensitive electrochemical measurements (Greenwald, Redish & Kibrick, 1940; Clarke, Cusworth & Datta, 1954). It should be realized that, even if ion-pairing with the monomer is present to a slight extent, delocalization on the polymer would still be likely to occur because of the long-range overlap of coulomb fields. Tight localized binding (site binding) is expected to occur for a polyelectrolyte with monovalent charged groups only if the chain is sufficiently flexible or the nearest-neighbour charge spacing is sufficiently close that two or more groups can co-operate in holding the cation, i.e. when the 'monomer' is effectively multivalent (Strauss & Siegel, 1963). Extended polynucleotide conformations do not satisfy this requirement (Skerjanc & Strauss, 1968; Olson, 1975). Significant binding of alkali metal or alkaline earth cations to the bases in aqueous solution has been ruled out by differential dialysis studies (Shapiro, Stannard & Felsenfeld, 1969*b*).

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The qualitative difference, *as a matter of physical observation*, between polyelectrolyte–counterion binding and association reactions of oppositely charged small ions cannot be overemphasized. The latter is governed by mass action. The *extent* of ion pairing and the *tightness* of the complex are inseparable; the greater the association constant, the greater will be both the extent of reaction (number of ion pairs) and the tightness (stability) of each pair. For a polyelectrolyte in an environment of a single counterion species, however, the extent and tightness of binding are uncorrelated. Polyphosphate and DNA, which possess, roughly, about the same axial charge densities ($\xi_{PP} = 2{\cdot}8$, $\xi_{DNA} = 4{\cdot}2$), bind about the same number of Na^+ ions in accordance with the condensation principle; but condensed Na^+ ions are bound loosely to DNA with little or no dehydration and localization, while Na^+ ions condensed on polyphosphate are extensively dehydrated (Strauss, 1974), although probably still not strongly localized (Kielman *et al.* 1976). In accordance with its higher valence, more Mg^{2+} than Na^+ is bound on an equivalent basis to the synthetic polyelectrolyte PVS (with weakly binding sulphonate groups that are comparable to the monovalent phosphates on polynucleotides), but Na^+ is actually the more *tightly* bound of the two if extent of dehydration per equivalent is used as the criterion (Hen & Strauss, 1974).

Of course if the affinity of a counterionic species for sites on the polyion contains an appreciable covalent, or other non-ionic, component, then the situation is different. The binding of protons to nucleotide bases or to carboxylate groups on polyacrylate, for example, does not conform to the condensation requirement, but rather obeys a mass-action law with an association constant dependent on the extent of binding. There is also a formal similarity to a mass-action binding distribution when a multivalent species (Mg^{2+} or small oligolysines, for example) equilibrates by purely electrostatic interaction with DNA in the presence of excess univalent salt, although important qualitative differences remain. This situation, which is of the greatest practical interest for the biologist, will be discussed subsequently.

C. *Molecular theory of counterion condensation; the counterion diffusion potential*

It has been proved (Manning, 1977*a*, 1978) that condensation of counterions is a simple consequence of the spatial correlation of fixed charge centres along a single axis characteristic of all linear polyelectro-

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lytes (even flexible chains can be expected to be fairly 'straight' over distances of the order of a Debye length because of electrostatic repulsion among the fixed charges). Thus, condensation is a manifestation of long-range electrostatic forces, and its general occurrence regardless of the specific nature of the polyelectrolyte charges is thereby explained.

To investigate the consequences of long-range overlap of the spherical coulomb fields generated by each charged group (assumed univalent) on the polyion, the actual spatial distribution of groups with average axial spacing b may be replaced by a regular linear array of fixed univalent point charges of magnitude q and spacing b (Manning & Zimm, 1965; Manning, 1977*a*). Let θ_N be the number of associated counterions per fixed charge, with the understanding that all counterions M^{N+} in the system have the same valence N. Then the *effective* charge q_{net} on each element of the linear array is lowered from q by the factor $1-N\theta_N$,

$$q_{\text{net}}/q = 1-N\theta_N. \tag{2}$$

Each pair of effective charges is then assumed to contribute a repulsive term to the overall free energy of the form

$$(q_{\text{net}}^2/\epsilon)\,(|i-j|\,b)^{-1}\exp(-\kappa|i-j|\,b), \tag{3}$$

a Debye–Hückel screened coulomb potential between the ith and jth charges in the array, $|i-j|\,b$ being their fixed distance of separation. The Debye screening parameter κ is proportional to the square root of ionic strength (if simple electrolyte is present in excess over polyelectrolyte):

$$\kappa^2 = (8\pi)\times 10^{-3}L_{\text{AV}}(q^2/\epsilon kT)\,I, \tag{4}$$

where L_{AV} is Avogadro's number and I is the ionic strength. For excess 1 : 1 salt in water at 25 °C,

$$\kappa = (3{\cdot}29\times 10^7)\,c_1^{\frac{1}{2}}\ [\text{cm}^{-1}], \tag{5}$$

where c_1 is the molarity of excess univalent counterion (ionic strength of the salt). The theoretical extent of association θ_N will therefore be dictated by the deviation of the actual electrostatic free energy from the Debye–Hückel approximation to it, or, again, the departure from diffuse Debye–Hückel screening interactions (ion atmosphere) is theoretically attributed entirely to counterion association. Thus determined, the theoretical extent of counterion association satisfies the requirements for counterion condensation (see Section IIA).

The total free energy of the system is the sum of contributions G_i from various sources, of which screened repulsions of the form of equation

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(3) are one. In general I shall define

$$\bar{g}_i = G_i/n_P RT, \tag{6}$$

where n_P is the total number of moles of charged groups on the polyions (number of moles of phosphate groups in a polynucleotide system) and R is the gas constant. I shall also assume that

$$I \gg c_P, \tag{7}$$

where c_P is the polyion equivalent concentration (molarity of phosphate groups for a polynucleotide), so that equations (4) and (5) may be used.

The work of charging the phosphate groups against their mutual electrostatic repulsion is obtained by summing equation (3) over all pairs (i, j) of effective charges in the linear array,

$$\bar{g}_{\mathrm{el}} = -(1 - N\theta_N)^2 \, \xi \ln (1 - e^{-\kappa b}), \tag{8}$$

where ξ has been defined by equation (1). Note that the summation, the mathematical device for making the 'overlap' of coulomb fields a quantitative concept, has drastically transformed the qualitative nature of the individual spherical screened coulomb fields; the exponential screening, equation (3), has become logarithmic.

Other contributions to the free energy that depend on θ_N are given by the mixing of free M^{N+} cations, bound M^{N+} cations, and solvent molecules. Lumped together, these terms can be written as

$$\bar{g}_{\mathrm{mix}} = \theta_N \ln (10^3 \theta_N \, V_P^{-1}/c_N), \tag{9}$$

where terms independent of θ_N have been omitted. The argument of the logarithm in equation (9) is the ratio of the local concentration of bound M^{N+} cations to their molarity c_N in free solution. The quantity V_P is the volume of the region surrounding the polynucleotide within which M^{N+} cations are said to be 'bound'. The units of V_P are cm^3/mole P, so that the local concentration $10^3 \theta_N \, V_P^{-1}$ is expressed in molarity units, and the argument of the logarithm is dimensionless. The assertion that there are no other θ_N-dependent contributions to the free energy is equivalent to the assumption that the interaction of bound and free M^{N+} ions with the solvent is the same, and that the 'bound' M^{N+} ions may translate freely within the volume V_P centred along the long axis of the polynucleotide. This assumption is the natural way to incorporate the discussion in II B into a model.

Turning to the qualitative features of the dependence of $\bar{g}_{\mathrm{el}}$ and

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$\bar{g}_{\mathrm{mix}}$ on θ_N, one sees that $\bar{g}_{\mathrm{el}}$ is positive and decreases as θ_N increases, the expected behaviour for the contribution of phosphate–phosphate repulsions, which decrease in magnitude as more counterions bind and effectively lower the phosphate charge. This term alone, then, would lead to complete binding, $N\theta_N = 1$, or minimum electrostatic work of charging the polynucleotide, as the equilibrium state.

However, increased binding of M^{N+} cations lowers the entropy of the system, as expressed by $\bar{g}_{\mathrm{mix}}$, which increases as θ_N increases if, as will be seen to be true for cases of interest here, the local concentration of bound M^{N+} is greater than the concentration of free M^{N+},

$$10^3\theta_N V_P^{-1} > c_N. \tag{10}$$

By itself, $\bar{g}_{\mathrm{mix}}$ would lead to $\theta_N = 0$ as the equilibrium state; that is, complete dissociation. The actual equilibrium state, determined from minimization of the sum $\bar{g}_{\mathrm{el}}+\bar{g}_{\mathrm{mix}}$, is a value of $N\theta_N$ intermediate between 0 and 1, a state of incomplete dissociation and incomplete binding.

The discussion thus far has held no surprises; *any* system of interacting molecules possessing binding capacity behaves similarly, the equilibrium state of incomplete binding representing the balance between maximization of entropy by dissociation and minimization of energy by binding. Polyelectrolyte binding is qualitatively distinctive, however. For a typical binding system dissociation tends to completion as the concentration tends to zero, because the entropy increase becomes infinite while the energy increase is essentially independent of concentration and remains finite. But for the polyelectrolyte case it can be verified from equations (8) and (4) that $\bar{g}_{\mathrm{el}}$ increases without bound as c_N tends to zero. Moreover, the concentration variation is logarithmic, just as for $\bar{g}_{\mathrm{mix}}$. The consequence is that θ_N does not tend to zero with c_N but remains positive. The implications of this distinctive behaviour are far-reaching. (It should be noted that the validity of equation (8) requires neglect of end-effects, an excellent approximation for ionic strengths of practical interest and *poly*nucleotides, as opposed to *oligo*nucleotides.)

To proceed quantitatively, the operation of minimization of $\bar{g}_{\mathrm{el}}+\bar{g}_{\mathrm{mix}}$ with respect to θ_N is performed, with the result that θ_N satisfies the following transcendental relation at the free energy minimum,

$$1+\ln\left(10^3\theta_N V_P^{-1}/c_N\right) = -2N\xi(1-N\theta_N)\ln\left(1-e^{-\kappa b}\right). \tag{11}$$

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In the limit $c_N \to 0$ the only value of θ_N that satisfies the equation is

$$N\theta_N = 1 - (N\xi)^{-1} \tag{12}$$

because only for this value will the right-hand side produce a term $-\ln c_N$ to cancel exactly the singular $-\ln c_N$ term on the left-hand side (see equation (4)). The value of V_P that corresponds to equation (12) in the low-concentration limit is

$$V_P = 4\pi e L_{\Delta V} N' \nu^{-1}(\nu + \nu')(\xi - N^{-1}) b^3, \tag{13}$$

where e is the base of natural logarithms, ν and ν' are the number of counterions and coions, respectively, in the formula for the small electrolyte that is present in excess and hence determines the ionic strength, and N and N' are the valences (without sign) of the counterion and coion, respectively. Note that the only polyelectrolyte parameter on which V_P depends is the axial charge spacing b. Equation (13) is obtained by substituting equation (12) into (11) along with (4) for κ, while linearizing the exponential in accord with the stipulation $c_N \to 0$. For a 1 : 1 salt

$$\theta_1 = 1 - \xi^{-1} \tag{14}$$

and

$$V_P = 41{\cdot}1(\xi - 1) b^3, \tag{15}$$

where V_P has the units cm^3/mole P if b is expressed in ångströms. Clearly, equation (12) and (13) are physically meaningful only for polyelectrolytes of structures such that $\xi \geqslant N^{-1}$, a condition satisfied by all known polynucleotide conformations. Thus, a polyelectrolyte molecule with $\xi > 1$, immersed in 1 : 1 salt of vanishingly small bulk concentration c_1, is nevertheless surrounded by a local concentration c_1^{loc} of counterions

$$c_1^{loc} = 10^3 \theta_1 V_P^{-1} = 24{\cdot}3(\xi b^3)^{-1}, \tag{16}$$

where c_1^{loc} has molarity units if b is expressed in ångströms. Table 1 lists these local concentrations of bound univalent counterions for several polynucleotide structures. Also listed in Table 1 are the values Δx of the thickness of cylindrical shells of volume V_P about the polynucleotide; to calculate Δx it was assumed that the double- and triple-stranded structures are cylinders of radius 10 Å, while the single-stranded structures all have 5 Å radii. As an example from Table 1 the charge fraction of native DNA in excess 1 : 1 salt is 0·24 (76% of the phosphate charge is compensated by bound M^+), and the bound cations

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TABLE 1. *Parameters for bound univalent cations in the limit* $c_1 \to 0$

Polynucleotide	b (Å)†	ξ	θ_1	V_P (cm³/mole P)	Δx (Å)	c_1^{loc} (M)
Poly A·2 poly U	1·0	6·9	0·86	252	5	3·52
Poly A·poly U	1·6	4·5	0·78	589	7	1·32
Native DNA	1·7	4·2	0·76	646	7	1·18
Poly A	3·1	2·3	0·57	1592	12	0·35
Single-stranded DNA	4·0	1·8	0·44	2104	12	0·21
Poly U	4·5	1·6	0·38	2247	12	0·17

† Record, Woodbury & Lohman (1976), pH 7.

translate freely within about 7 Å of the DNA surface (of course, there is rapid exchange between ions in the bound region and bulk solution), where their local concentration is about 1·2 M, even though their bulk concentration c_1 may approach zero. *The local concentration is not governed by the law of mass action; it does not tend to zero with the bulk concentration* c_1.

It is now natural to wonder whether an approximate ionic strength c_1 can be specified below which, for practical purposes, the system has reached the mathematical limit $c_1 \to 0$. That is, over what range of ionic strengths c_1 are the parameters in Table 1 practically applicable? It may be verified that if the condition $\kappa b \ll 1$ is fulfilled, so that the exponential term on the right hand side of equation (11) can be linearized, then equations (14) and (15) for θ_1 and V_P satisfy the free-energy minimization condition equation (11). Thus, for concentrations c_1 such that the Debye screening length κ^{-1} is appreciably larger than the axial phosphate spacing b, Table 1 is applicable. For double- and triple-stranded structures, this requirement is not very restrictive, since the corresponding values of b are so small. For example, κ^{-1} equals 3·0 Å at 1 M according to equation (5), which may be compared to the value $b = 1{\cdot}7$ Å for native DNA. It is expected, then, that Table 1 is applicable to native DNA for ionic strengths in 1 : 1 salt approaching 1 M. For the single-stranded structures anticipation of the applicability of Table 1 would be restricted to concentrations below about 0·1 M, for which $\kappa^{-1} = 9{\cdot}6$ Å.

An equivalent and most appealing criterion for the applicability of Table 1 is $c_1 \ll c_1^{loc}$. It may in fact be verified from equations (5) and (16) that $c_1/c_1^{loc} = 2{\cdot}7\kappa^2 b^2$ (use of the general expressions rather than

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TABLE 2. *Number of bound counterions per phosphate (θ_1) as a function of ionic strength*

$\xi = 4{\cdot}2$		$\xi = 1{\cdot}5$	
c_1 (M)	θ_1	c_1 (M)	θ_1
0·0001	0·76	0·0001	0·33
0·001	0·76	0·001	0·33
0·01	0·76	0·01	0·35
0·1	0·77	0·05	0·39
0·5	0·80	0·10	0·42
1·0	0·83		

the numerical ones identifies the factor 2·7 as e, the base of natural logarithms), so that the conditions $c_1 \ll c_1^{loc}$ and $\kappa b \ll 1$ are indeed equivalent. Thus, Table 1 applies for ionic strengths approaching the listed values of the local concentrations of bound counterions.

In Table 2 these qualitative remarks are validated. Equation (15) for V_P is used in (11) with $N = 1$ and θ_1 is then evaluated by numerical iteration as a function of c_1. For $\xi = 4{\cdot}2$ (native DNA) Table 1 indicates that c_1^{loc} equals 1·2 M, while Table 2 shows that θ_1 retains its limiting value 0·76 up to $c_1 = 0{\cdot}1$ M and does not deviate much from this value even at $c_1 = 1$ M. For $\xi = 1{\cdot}5$ (~poly U) Table 1 lists c_1^{loc} as about 0·2 M, and Table 2 shows that θ_1 remains at its zero-concentration limit up to $c_1 = 0{\cdot}01$ M and is not immoderately different for concentrations five times higher.

The significance of the trend in θ_1 as a function of c_1 exhibited in Table 2 is obscure at the present time. The theory contains inconsistencies that become significant at the same concentrations at which θ_1 begins to deviate from its limiting value (Manning, 1977*a*). From the experimental side no variations of polyelectrolyte charge fractions with ionic strength have been reported; the charge fraction of DNA in NaCl is invariant up to 0·5 M, the highest ionic strength investigated by Anderson *et al.* (1978). The point to emphasize from Table 2 at present is the theoretical *invariance* of θ_1 up to ionic strengths c_1 approaching c_1^{loc}, not the variation of θ_1 at higher ionic strengths.

I can now formulate an evocative description of the ionic state of an extended polynucleotide structure in a solution environment of excess 1 : 1 salt of concentration (ionic strength) $c_1 \ll c_1^{loc}$. The structure is 'potentiated', in the sense that a large entropy increase $nR \ln (c_1^{loc}/c_1)$

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would accompany the diffusion of n M^+ cations from the bound region of locally high concentration to the lower concentration in free solution. This statement is implicit in equation (9), since $c_1^{loc} = 10^3\theta_1 V_P^{-1}$. The process of local diffusion of M^+ into bulk solution does not occur spontaneously only because its presence at high concentration near the polynucleotide is required to offset the repulsive electrostatic free energy of interaction among the phosphate groups, as described by equation (8). However, if the system can avail itself of another mechanism for relieving the stress of phosphate–phosphate repulsion, then M^+ will diffuse spontaneously into free solution as the alternate mechanism is implemented. Conversely and equivalently, the counterion diffusion potential $RT \ln (c_1/c_1^{loc})$ constitutes a thermodynamic driving force for the spontaneous implementation of the alternate mechanism of phosphate charge compensation. As the ionic strength c_1 approaches c_1^{loc}, or 1·2 M for DNA, the polyelectrolyte is no longer potentiated in this sense, and, to the extent that a given process consists of the relief of long-range electrostatic stress, that process will have lost its ability to occur spontaneously.

III. Binding of metal ions in the aqueous DNA/MCl_N system

A. *Charge fraction of DNA and mode of binding*

This subject has already been covered in Section II. The theoretical charge fraction r of DNA in excess MCl_N (fraction of phosphate charge uncompensated by bound M^{N+}) is given by (12):

$$r = 1 - N\theta_N = (N\xi)^{-1} \tag{17}$$

or 0·24 and 0·12 for $N = 1$ and $N = 2$, respectively. These values have been closely confirmed by direct measurement for Na^+ and Mg^{2+} in DNA solutions over wide ranges of ionic strength, and (17) has also been confirmed for Mn^{2+} and Co^{2+} in polyphosphate solutions; see Section II for references. There exists no reason at the present time to doubt the essential validity of (17) for all small univalent and divalent cations except those that may interact strongly with the nucleotide bases in native DNA, like Ag^+ and Cu^{2+}.

Good experimental evidence exists that Na^+, Rb^+, Mg^{2+} and Ca^{2+} are bound loosely to DNA, to the molecule as a whole rather than to discrete sites on it, and are in a state of essentially complete hydration

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and free translational and rotational mobility (see Section II for references). This information has been incorporated into the theoretical model leading to the successful equation (17). The similarity in behaviour among all the alkali metal ions, as well as the nearly identical binding constants to native DNA of Mg^{2+}, Ca^{2+}, Mn^{2+} and Zn^{2+} (Daune, 1970; Clement, Sturm & Daune, 1973) suggests that *all* small cations, except those that can actually penetrate the native DNA structure and enter into strong interaction with the bases, are more or less associated with DNA in this delocalized mode. The burden of proof has now clearly shifted to those who would support a contrary position. As I shall point out in Section III F, the undisputed existence of small specific ionic effects does not constitute such proof.

B. *The free energy of binding*

Binding of M^{N+} cations to DNA in the system DNA/MCl_N/H_2O can be formulated as a chemical reaction (Manning, 1978),

$$M^{N+}\text{ (free)} \rightarrow M^{N+}\text{ (bound)}. \tag{18}$$

Note carefully that this reaction does *not* imply that free M^{N+} combines with a 'site' to form a cation–site complex; it states simply that M^{N+} is transferred from free solution into the spatial region V_P, where it is defined as bound. Application of the standard thermodynamic treatment of chemical reactions to equation (18) is fully equivalent to the treatment in II C and yields the same results as (12) and (13). The integral free energy of reaction (18) – that is, the free energy when DNA is combined with its equilibrium number of counterions (and hence has charge fraction r given by (17)), relative to the non-equilibrium state of complete dissociation ($r = 1$) – is $\Delta\bar{g} = -5{\cdot}3$ for $N = 1$ at ionic strength 0·1 M. That is, condensation of Na^+ ions at 0·1 M stabilizes DNA by 3·2 kcal/mole P at 25 °C. At 10^{-3} M the corresponding stabilization is twice as great.

C. *Colligative properties*

Since, as described in II, the free energy of a polyelectrolyte solution is theoretically calculable, expressions may also be obtained for all the observables associated with classical equilibrium electrochemistry such as activity coefficients, osmotic coefficients, Donnan effects and heats of solution and mixing (Manning, 1972*a*, 1974; Devore & Manning, 1974; Boyd, Wilson & Manning, 1976; Manning, 1976*a*). As an example let

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TABLE 3. *Correlation of activity coefficient data for MgDNA/$MgCl_2$/H_2O*

X	c_P	$(\gamma_{Mg}/\gamma^0_{Mg})_{obs}$†	$(\gamma_{Mg}/\gamma^0_{Mg})_{calc}$‡
1·2	$1{\cdot}0\times10^{-2}$	0·47	0·50
1·4	$6{\cdot}1\times10^{-4}$	0·36	0·46
2·5	$2{\cdot}6\times10^{-3}$	0·35	0·34
3·1	$1{\cdot}0\times10^{-2}$	0·29	0·30
3·1	$6{\cdot}2\times10^{-4}$	0·25	0·30
6·1	$2{\cdot}6\times10^{-3}$	0·21	0·21
6·2	$1{\cdot}0\times10^{-2}$	0·20	0·20
7·7	$6{\cdot}4\times10^{-4}$	0·16	0·18
12·2	$2{\cdot}6\times10^{-3}$	0·14	0·15
15·4	$6{\cdot}5\times10^{-4}$	0·13	0·13
∞	5×10^{-4} to 1×10^{-2}	$0{\cdot}075\pm0{\cdot}05$	0·072

† Data from Lyons & Kotin (1965).
‡ Calculated from equation (19) of text.

us consider the activity coefficient γ_{Mg} of the Mg^{2+} ion in the ternary solution MgDNA/$MgCl_2$/H_2O. Theoretical activity coefficients are easily obtained, since the number of associated counterions, which do not contribute to the apparent activity, is known, while unassociated counterions contribute to simple Debye–Hückel screening interactions. The theoretical formula for γ_{Mg} is (Kwak, O'Brien & MacLean, 1975),

$$\ln(\gamma_{Mg}/\gamma^0_{Mg}) = -\tfrac{1}{4}\xi^{-1}X(\tfrac{1}{2}\xi^{-1}X+\tfrac{3}{2})^{-1} + \ln[(\tfrac{1}{2}\xi^{-1}X+1)(X+1)^{-1}]. \quad (19)$$

In this formula γ^0_{Mg} is the activity coefficient of Mg^{2+} in the binary solution $MgCl_2$/H_2O at the same $MgCl_2$ concentration as in the polyelectrolyte solution, X is the ratio of polyelectrolyte equivalent concentration (normality) to $MgCl_2$ equivalent concentration, and ξ is the charge density parameter defined by equation (1) and equal to 4·2 for DNA in an aqueous environment.

Table 3 presents a comparison of observed values of γ_{Mg} with values calculated from equation (19). X has been defined above, c_P is the DNA normality (molarity of phosphate groups) and $MgCl_2$ normalities are equal to $c_P X^{-1}$. The last entry in the table, $X=\infty$, refers to a pure MgDNA solution in the absence of added $MgCl_2$. Note the remarkably good agreement between observed and theoretical values of $\gamma_{Mg}/\gamma^0_{Mg}$ over the entire composition range, $1 < X < \infty$ and $MgCl_2$ normalities up to 0·01 N. Underlying the predicted values is the concentration-

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invariant theoretical charge fraction $(2\xi)^{-1} = 0{\cdot}12$; Table 3 adds to the evidence for its physical reality.

D. *Transport properties*

It is an easy matter to calculate the electrophoretic mobility U expected for the polyelectrolyte model under discussion, namely a linear array of effective point charges of magnitude $N^{-1}\xi^{-1}q$ and spacing b. The starting point is Kirkwood's theory of the translational friction coefficient for macromolecules composed of identical subunits (Bloomfield, Crothers & Tinoco, 1974). It need only be recognized that the hydrodynamic interaction between each pair of subunits (in this case, point charges) is attenuated, not only by the familiar inverse dependence on distance but also by the oppositely charged Debye–Hückel atmosphere that surrounds each fixed charge and moves in the opposite direction, dragging the solvent with it, under the action of an applied electric field. The quantitative expression of this added attenuation is the appearance of a Debye screening factor $\exp(-\kappa b|i-j|)$ in the hydrodynamic interaction between fixed charges i and j. The summation over all pairs is then exactly the same operation as summing the Debye–Hückel pair potentials of equation (3), and the final result is

$$U = N^{-1}(\epsilon kT/900\pi\eta q)\,|\ln \kappa b|, \tag{20}$$

where η is the viscosity and all other quantities have been previously defined. At 0 °C in water the numerical value of the factor in parentheses is $1{\cdot}37 \times 10^{-4}$; the units of U are standard, $cm^2\ sec^{-1}\ V^{-1}$.

Mobilities predicted from equation (20) are compared in Table 4 with values measured for DNA ($b = 1{\cdot}7 \times 10^{-8}$ cm) in excess $MgCl_2$ ($N = 2$) at 0 °C. It should be noted that, although prediction consistently overestimates observation, the ratio of calculated to observed values is almost constant over a range of $MgCl_2$ normalities n_{MgCl_2} that cover more than two orders of magnitude. The data are therefore in reasonable agreement with the predicted linear dependence on the logarithm of ionic strength, and the discrepancy in absolute values can probably be ascribed to the obviously oversimplified description of the hydrodynamic particle as a linear array of points. Nevertheless, the *electrostatic* contribution to the mobility is apparently, as anticipated, accurately contained in this model. The theoretical mobility explicitly rests on a concentration-invariant charge fraction $(N\xi)^{-1}$, which therefore is supported by the mobility data in Table 4.

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TABLE 4. *Comparison of predicted and observed mobilities*

n_{MgCl_2}	κ^{-1} (Å)	$10^4 U_{obs}$†	$10^4 U_{calc}$†	U_{calc}/U_{obs}
$5{\cdot}6 \times 10^{-4}$	106·2	1·6 ± 0·2	2·83	1·77
$1{\cdot}0 \times 10^{-3}$	79·5	1·5 ± 0·2	2·63	1·75
$5{\cdot}2 \times 10^{-3}$	34·9	1·00	2·07	2·07
$1{\cdot}1 \times 10^{-2}$	24·0	0·86	1·81	2·10
$3{\cdot}2 \times 10^{-2}$	14·1	0·76	1·45	1·91
$6{\cdot}2 \times 10^{-2}$	10·1	0·60	1·22	2·03
$1{\cdot}9 \times 10^{-1}$	5·8	0·45	0·84	1·87

† Data from Papon & Strauss (unpublished).
‡ Calculated from equation (20) of text.

E. *The dissociation–field effect*

The extent of counterion condensation, $N^{-1}(1 - N^{-1}\xi^{-1})$ moles condensed counterion of valence N per polyelectrolyte equivalent, is invariant to changes of ionic strength. Application of a strong electric field E, however, can cause partial dissociation of the condensed counterions, thereby increasing the charge fraction of the polyelectrolyte. The dissociation–field effect is well known in the area of partially associated simple electrolyte solutions, since the linear increase with E of the conductance of such solutions is accepted as a direct measure of an increased dissociation constant (Onsager, 1934; Onsager & Liu, 1965). The conductance of polyelectrolyte solutions also increases linearly with E, but it is not clear at present that the field–dissociation effect is the dominant factor in this behaviour. It is conceivable that charge fractions can be directly measured in the presence of a field, as they have been in its absence, but no attempt in this direction has yet been undertaken. Therefore the field–dissociation effect for polyelectrolytes cannot yet be said to have been directly observed. The great current interest in the effect is due rather to observations that sufficiently strong electric fields can induce conformational changes in ordered polynucleotide structures (Pörschke, 1976*b*; Revzin & Neumann, 1974).

The theory for polyelectrolytes (Manning, 1977*c*) is surprisingly easy, in view of the almost legendary difficulty of Onsager's 1934 paper on small electrolytes. At zero field the contribution to the free energy from condensed counterions is, aside from the effective decreased charge on the polyion, an ideal mixing entropy. When the field E (in e.s.u.) is turned on, the free energy is supplemented by a work term proportional

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to $NqaE$, where Nq is the charge of a counterion and a is a distance. That is, in order to remain condensed, a counterion must do work against the field which tends to tear it from the polyion. The key to the derivation is the unambiguous knowledge of a, which is simply the cylindrical radius of the free volume V_P determined by equation (13). The average position of a counterion, if condensed, is on the cylindrical axis of the free volume; to remain there, the counterion must work against the field over the distance a, past which it is no longer condensed. Values of a have been tabulated as a function of ξ (Manning, 1977*a*); for DNA, $\xi = 4{\cdot}2$ and $a = 14{\cdot}2$ Å for univalent counterions.

With this added work term condensed counterions have higher energy; hence at the free energy minimum there will be fewer condensed counterions than for the field-free case. Define a degree of dissociation α by

$$\theta_N(E)/\theta_N(0) = 1-\alpha, \tag{21}$$

where $\theta_N(E)$ is the fraction of a condensed counterion per polyelectrolyte charge at field E; $\theta_N(0)$, the corresponding field-free quantity, has already been found and is given by equation (12). The relation between α and E at the free energy minimum is

$$\bar{E} = \alpha[1-2(N\xi-1)\ln \kappa b], \tag{22}$$

where

$$\bar{E} = (\tfrac{1}{4}\pi)(NqaE/kT). \tag{23}$$

For an aqueous polyelectrolyte solution at 25 °C containing excess 1 : 1 salt of ionic strength 10^{-2} M, a very high applied field of 402 kV/cm produces only 10% dissociation ($\alpha = 0{\cdot}1$) if $\xi = 4$ (the approximate value for DNA). At the higher ionic strength 10^{-1} M, dissociation requires less work because the dissociated ions enter a more effective screening atmosphere; the field corresponding to $\alpha = 0{\cdot}1$, however, is still 262 kV/cm for $\xi = 4$. Although the extent of counterion condensation is the *same* at 0·01 and 0·1 M in the absence of a field, it may be said that the condensed counterions are held more loosely at the higher ionic strength, in the sense that the field required to tear some of them away is then not as high as in the more dilute solution. In Section VD the theory is applied to the effect of a strong field on the melting of DNA.

F. *Specific ionic effects*

To a slight extent, DNA in aqueous solution has been observed to

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discriminate among cations of the alkali metal series, preferentially selecting Li^+ over Na^+ and K^+ in differential dialysis experiments (Span & Strauss, unpublished) and exhibiting higher T_m's in Li^+ solution relative to Na^+ and K^+ (Gruenwedel, Hsu & Lu, 1971). Since these ions are unable to discriminate between A·T-rich and G·C-rich DNA (Shapiro *et al.* 1969*b*) there is no reason to suspect groove binding, and the almost unrestricted movement of bound Na^+ and Rb^+ (Reuben *et al.* 1975) suggests that varying degrees of 'tightness' of localized binding to phosphate groups is not an appropriate explanation. We have seen that the local concentration of condensed univalent cations in the immediate neighbourhood of DNA is about 1·2 M, regardless of the bulk ionic strength. Moreover, Li^+ is more highly hydrated (i.e. binds *water* more tightly) than Na^+ or K^+ (Robinson & Stokes, 1959). Given the sensitivity of DNA structure to solvent effects, it would be surprising if DNA in a local environment of 1 M aqueous M^+ did *not* exhibit at least small differential effects depending on the species M^+. By the same token circular dichroism variations upon addition of Mg^{2+} (Clement *et al.* 1973) are perhaps also not an unexpected consequence of changes in local solvent properties, since Mg^{2+} binds *water* extremely tightly (Eigen, 1963).

For one such effect a quantitative estimate can be made. If it is assumed that the electrophoretic mobility of DNA is inversely proportional to the *local* solvent viscosity (Abramson, Moyer & Gorin, 1942), then one would predict that mobilities at any ionic strength in aqueous MCl should vary in inverse proportion to the viscosities of 1 M-MCl (the Cl^- ion contributes little to the ionic effect on viscosity). On this basis DNA mobilities would be expected to vary in the proportion 1·23 : 1·13 : 1·00 in aqueous K^+, Na^+ and Li^+, respectively, at 0 °C. The observed variation (Ross & Scruggs, 1964) is 1·37 : 1·16 : 1·00.

A recent development of possibly great importance is the discovery that half of the Na^+ counterions are coordinated to $O_{(2)}$ of uracil and thymine in *crystalline* ApU and pTpT, respectively (Seeman *et al.* 1976; Camerman, Fawcett & Camerman, 1976). The $O_{(2)}$ atom of thymine and uracil is a carbonyl oxygen exposed in the minor groove of double helical structures and not involved in base pair hydrogen binding. Although all current evidence *in solution* points to the absence of groove binding for Na^+ (presumably because its electric field intensity is more easily compensated by complete hydration in solution than in even a relatively highly hydrated crystal), the $O_{(2)}$ atom could be an

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important factor in the apparent ability of the A · T base pair to bind more water in solution than the G · C pair (Tunis & Hearst, 1968) as well as in the small selective affinity of the tetramethyl- and tetraethylammonium cations for A · T-rich regions of DNA relative to G · C-rich regions (Shapiro *et al.* 1969; Melchior & von Hippel, 1973).

Another interesting result is the charge reversal of DNA in aqueous $MgCl_2$ if the concentration is in excess of 1 N-$MgCl_2$ (Papon & Strauss, unpublished). DNA then behaves like a cation in an electrophoresis experiment. It would be of great interest to know if a conformational change, possibly to a compact form, is associated with this phenomenon. The constant anionic charge fraction 0·12 mentioned repeatedly in previous sections, of course, refers to an apparently much simpler situation at lower Mg^{2+} concentrations.

IV. Theory of delocalized binding of N-valent cationic substances to polynucleotides in aqueous excess 1 : 1 salt

A. *Introduction*

The subject of this section is a statistical mechanical analysis of binding constants obtained under typical experimental conditions. The goal is the *a priori* prediction of binding constants, properly defined, under the assumption that the bound cationic substance is delocalized (free translation near the polynucleotide surface). Comparison with measured values will then provide suggestive information about the molecular structure of the cation/polynucleotide complex. The central role of the counterion diffusion potential will be stressed, and the reader may wish to refer to the last paragraph of Section II C before proceeding.

B. *One-variable theory*

1. *Minimization of the free energy.* I shall begin with a discussion of binding relations obtained by treating the free energy as a function of the single variable θ_N, the number of M^{N+} cations bound per phosphate group. As in Section II C, the excess salt condition, equation (7), is assumed. In addition the inequality

$$c_1 \gg c_N, \tag{24}$$

where c_N is the molarity of *free* N-valent cations and c_1 the molarity of supporting 1 : 1 salt, proves useful without imposing a severe restriction

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on applications to commonly employed binding conditions. It allows us to retain equation (5) for the Debye screening constant κ, since c_1 is then effectively equal to the ionic strength I. It also motivates the modelling specification that V_P as given by equation (15) for 1 : 1 salts, defines the region of binding for both univalent and N-valent cations. The numerical values of V_P for several polynucleotides may be found in Table 1.

The one-variable approach turns out to require two separate cases, low binding levels (small θ_N) and high levels (θ_N approaching saturation). The two-variable theory developed subsequently enables one to avoid this separation and also to assess the approximations inherent in one-variable minimization. On the other hand the one-variable theory will provide a simple and attractive interpretation for the otherwise physically obscure two-variable results.

For low θ_N Table 1 motivates the assumption that θ_1, the number of bound M^+ cations per phosphate, is fixed at the value

$$\theta_1 = 1 - \xi^{-1} - N\theta_N, \tag{25}$$

that is, the binding of one M^{N+} ion results in the release of N bound M^+ ions, so that the charge fraction of the polynucleotide (fraction of phosphate charge uncompensated by bound cations) remains invariant at $1 - \theta_1 - N\theta_N = \xi^{-1}$. It is this restriction on θ_1 that allows reduction of the problem to the minimization of the free energy with respect to the single variable θ_N. From equation (25) it follows that the calculation for low θ_N is mathematically applicable to the range

$$0 \leqslant \theta_N \leqslant N^{-1}(1 - \xi^{-1}). \tag{26}$$

With the notation of equation (6) the electrostatic free energy required to charge the phosphate groups is

$$\bar{g}_{el} = -\xi^{-1} \ln (1 - e^{-\kappa b}), \tag{27}$$

obtained by replacing the charge fraction $1 - N\theta_N$ in equation (8) by ξ^{-1}, in accordance with the discussion of equation (25). This contribution to the free energy is independent of θ_N; hence the electrostatic repulsion does not determine, even partially, the extent of binding of the M^{N+} ions, at least in the binding range (26) and subject to the constraint equation (25).

The only other contributions to the free energy of the system are

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assumed to be from the ideal mixing of the various species,

$$\bar{g}_{\mathrm{mix}} = N\theta_N \ln (10^{-3}c_1) + \theta_1 \ln (\theta_1/V_P) + (c_N/c_P) \ln (10^{-3}c_N) + \theta_N \ln (\theta_N/V_P) + \theta_N, \quad (28)$$

where terms independent of θ_N have not been included. The first term on the right-hand side of equation (28) is the free energy of mixing of the free univalent cations M^+, with recognition that $N\theta_N$ M^+ ions per phosphate have been freed from the bound state if the extent of binding of the N-valent molecule is θ_N. The second term is the mixing contribution of the θ_1 M^+ ions remaining bound; θ_1 is given by equation (25) as a function of θ_N. The third term is contributed by free M^{N+} ions of concentration c_N, which satisfies the condition $dc_N/d\theta_N = -c_P$ if the *total* amount of M^{N+} ions is specified as fixed (corresponding to the typical experiment of adding a given amount of M^{N+} to the system, measuring the amount bound or free, and obtaining the amount free or bound by difference). The fourth term is the free energy of mixing of bound M^{N+} ions, assumed delocalized just like the univalent ions. The fifth term combines the mixing entropy of solvent with a higher-order term from the mixing of free M^+. Further detail for the case $N = 2$ may be found in Manning (1977*b*). Minimization of $\bar{g}_{\mathrm{mix}}$ as a function of θ_N yields the equilibrium binding relation

$$\log K_N = \log (V_P/1000e) + N \log [(1000e/V_P)(1 - \xi^{-1} - N\theta_N) c_1^{-1}], \quad (29)$$

where

$$K_N = \theta_N/c_N. \quad (30)$$

The quantity K_N is proportional to the equilibrium constant for the transfer of an M^{N+} ion from free solution to the 'bound' region V_P, and, as such, serves as the 'association constant' for the binding of M^{N+}. Recall also that K_N as defined by equation (30) is the ordinate in the usual Scatchard plot and is a directly accessible experimental quantity. The ionic strength dependence of K_N is given by the coefficient

$$d \log K_N / d \log c_1 = -N, \quad (31)$$

while, as a function of the extent of prior binding θ_N, K_N is a monotone decreasing Nth order polynomial with a single zero at $\theta_N = N^{-1}(1 - \xi^{-1})$.

In Fig. 1 K_2 is represented by a Scatchard plot; the dashed curve for $\theta_2 \leqslant 0{\cdot}38$, a number corresponding to $\xi_{\mathrm{DNA}} = 4{\cdot}2$, is equation (29) in the range specified by (26). The additional information in Fig. 1 and the other figures to which reference is made at this point pertain to calcula-

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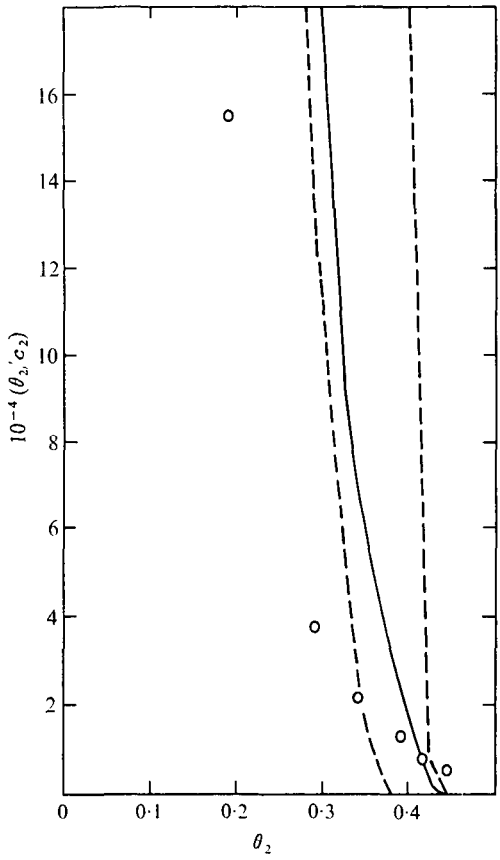

Fig. 1. Scatchard representation of divalent ion binding. Dashed curve for low θ_2, equation (29); dashed curve for high θ_2, equation (36); solid curve, equations (53) and (54). Data points from Clement *et al.* (1973). Ionic strength, 0·001 M.

tions and data that are discussed subsequently. In each of these figures only the dashed curves for $\theta_2 \leqslant 0{\cdot}38$ are germane to the present case. Fig. 2 gives a logarithmic representation of K_2. Fig. 3 shows the ionic strength dependence of K_2, equation (31). Fig. 4 shows how the bound fraction of M^+ cations decreases as M^{2+} is progressively bound, in this case following equation (25). Finally, Fig. 5 gives the total fraction of the phosphate charge compensated by bound cations of both species M^+ and M^{2+} as a function of θ_2; for the present case, as mentioned in connection with equation (25), this number is given by $1-\xi^{-1}$ (equal to 0·76 for DNA) for $\theta_2 \leqslant \frac{1}{2}(1-\xi^{-1})$.

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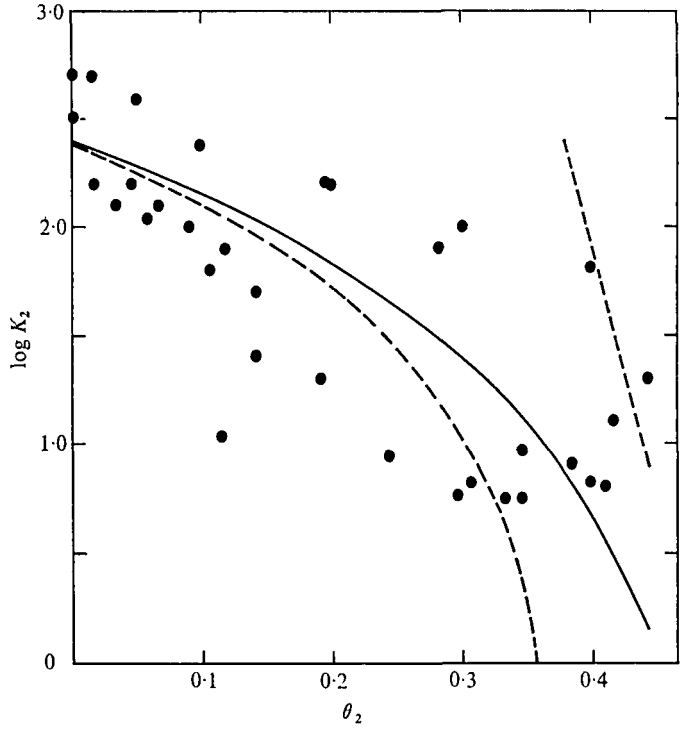

Fig. 2. Logarithmic representation of divalent ion binding. Curves as for Fig. 1; data from Archer *et al.* (1972), Clement *et al.* (1973), Bhat (1974), Skerjanc & Strauss (1968), and Zubay & Doty (1958). Curves are drawn for 0·1 M ionic strength; points represent ionic strengths in the range 0·001–0·2 M but are normalized to 0·1 M according to procedure described in text.

For the binding region

$$\theta_N \geqslant N^{-1}(1-\xi^{-1}) \tag{32}$$

a separate calculation is required. In this region the constraint that reduces the problem to one variable is

$$\theta_1 = 0, \tag{33}$$

that is, all bound M^+ cations have been displaced by M^{N+}. Note that equation (33) is continuous with equation (25) at the mutually included point $\theta_N = N^{-1}(1-\xi^{-1})$. Now

$$\bar{g}_{\mathrm{el}} = -(1-N\theta_N)^2\,\xi \ln (1-\mathrm{e}^{-\kappa b}), \tag{34}$$

in analogy to equation (8), while

$$\bar{g}_{\mathrm{mix}} = (c_N/c_P)\ln (10^{-3}c_N)+\theta_N \ln (\theta_N/V_P)+\theta_N, \tag{35}$$

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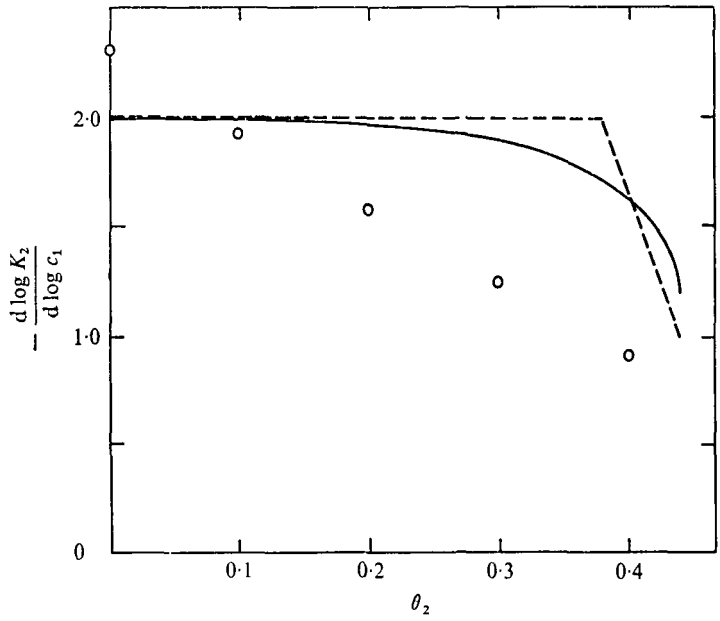

Fig. 3. Ionic strength dependence of divalent ion binding. Dashed curve for low θ_2, equation (31); dashed curve for high θ_2 (after break), equation (37); solid curve computed from equations (53) and (54). Data from Krakauer (1974) based on measurements in the range 0·01–0·1 M.

since only the mixing of free M^{N+}, bound M^{N+}, and solvent depend on θ_N. Minimization of $\bar{g}_{el}+\bar{g}_{mix}$ with respect to θ_N, with recognition that $dc_N/d_{\theta N} = -c_P$, produces the following relation for the value of θ_N at equilibrium:

$$\log K_N = \log (V_P/1000e) - 2N\xi(1-N\theta_N) \log \kappa b, \qquad (36)$$

where K_N is defined by equation (30) and κ is conveniently given by equation (5). The approximation $1-e^{-\kappa b} \approx \kappa b$ has been incorporated into (36) because (35), with which (33) is continuous, is motivated by the behaviour at low concentration illustrated in Tables 1 and 2. For the ionic strength dependence of K_N, it follows from (5) and (36) that

$$d \log K_N/d \log c_1 = -N\xi(1-N\theta_N), \qquad (37)$$

a relation that preserves continuity with (31) at $\theta_N = N^{-1}(1-\xi^{-1})$. As a function of θ_N, K_N decreases exponentially like $\exp [2N^2\xi \ln (\kappa b)\, \theta_N]$.

These relations are illustrated for K_2 and DNA by the dashed curves in Figs. 1–5 corresponding to $\theta_2 \geqslant 0{\cdot}38$. In Fig. 4, of course, the dashed

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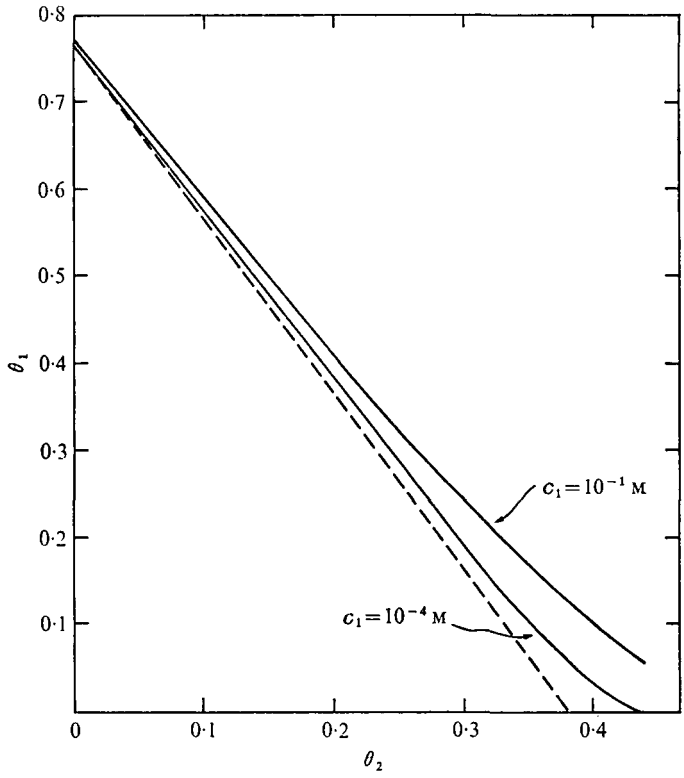

Fig. 4. Fractional number (per phosphate) of bound univalent counterions as a function of fractional number of bound divalent ions. Dashed curve, equation (25). Solid curves computed from equations (53) and (54) for the indicated ionic strengths.

curve in this range coincides with the abscissa, in accordance with (33). An important feature of the figures not yet mentioned is the existence of a maximum value $\theta_N^{\max}$. Due to the constraint imposed by inequality (24), $\theta_N^{\max}$ cannot be obtained from (36), which yields a positive value of K_N for any value of θ_N. However, as M^{N+} continues to be added to the system, so that inequality (24) becomes invalid and then reversed, the case that eventually presents itself is that of the polynucleotide in an aqueous environment of M^{N+} cations with negligible amounts of M^+. This situation is the same as that discussed in Sections II and III. According to (17), then,

$$\theta_N^{\max} = N^{-1}(1 - N^{-1}\xi^{-1}) \tag{38}$$

and the binding range to which (36) is restricted is defined by inequality

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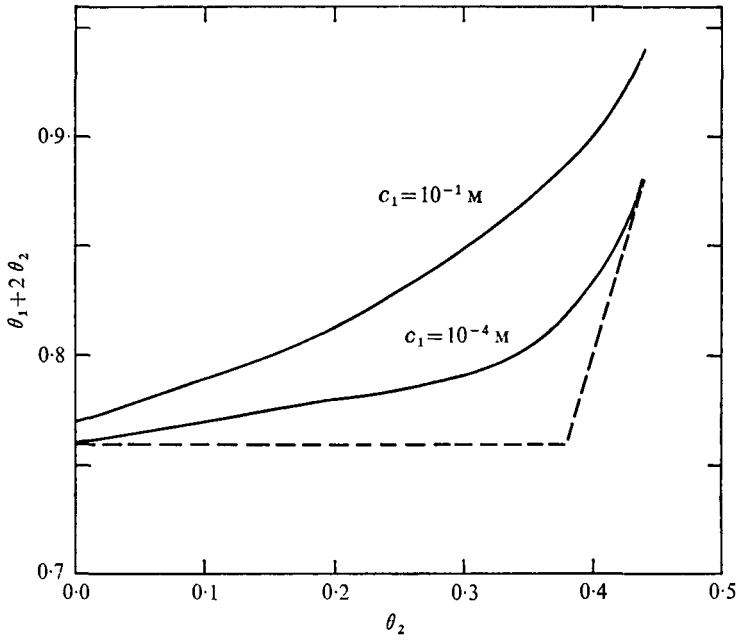

Fig. 5. Fraction of phosphate charge compensated by the combined charge of bound univalent and divalent ions, i.e. $\theta_1+2\theta_2$. Dashed curve for low θ_2 from equation (25); dashed curve for high θ_2 from equation (33) (i.e. this curve is merely $2\theta_2$ *v.* θ_2); solid curves computed from equations (53) and (54) for the indicated ionic strengths.

(32) and (38). For DNA, $\theta_2^{\max} = 0{\cdot}44$, a value that has been closely confirmed by measurement for Mg^{2+}, as discussed in Sections II and III. Recall from III F that this value is not an absolute upper limit; rather, it may represent a saturation plateau that is eventually exceeded, but not before concentrations of about 1 M free Mg^{2+}_f have been attained and factors other than electrostatic may become dominant.

2. *Chemical model.* Additional insights may be obtained by formulating the binding process as a chemical reaction,

$$M^{N+}\,(\text{free}) \rightarrow M^{N+}\,(\text{bound}), \tag{39}$$

and applying to it the standard thermodynamic analysis of chemical reactions. The reaction affinity (Kirkwood & Oppenheim, 1961) $\widetilde{\Delta}\mu_N$ for (39) is

$$\widetilde{\Delta}\mu_N = \mu_{N,b} - \mu_{N,f}, \tag{40}$$

where $\mu_{N,b}$ and $\mu_{N,f}$ are the chemical potentials of bound and free M^{N+}, respectively. By appropriate differentiation of the free energy of

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the system for low binding levels, (28), the required chemical potentials, in units of RT, may be obtained as

$$\mu_{N,f} = \ln 10^{-3} c_N, \tag{41}$$

$$\mu_{N,b} = -(N-1) + N \ln [10^{-3} c_1 V_P (1 - \xi^{-1} - N\theta_N)^{-1}] + \ln (\theta_N / V_P). \tag{42}$$

Note that the local and free concentrations of univalent M^+ are 'hidden variables' relative to reaction (39) and appear in the 'standard state' component of $\mu_{N,b}$ (first two terms in r.h.s. of (42)). From (40), (41), (42), (16) and (25) one has

$$\tilde{\Delta}\mu_N = \tilde{\Delta}\mu_N^0 + \ln (c_N^{\infty} / c_N), \tag{43}$$

where

$$\tilde{\Delta}\mu_N^0 = -(N-1) + N \ln (c_1 / c_1^{\mathrm{loc}}). \tag{44}$$

In (43) the local molarity c_N^{loc} of bound M^{N+} is given in analogy to (16),

$$c_N^{\mathrm{loc}} = 10^3 \theta_N / V_P. \tag{45}$$

It is easily verified that the equilibrium condition $\tilde{\Delta}\mu_N = 0$ yields the equilibrium binding relation, (29).

The equilibrium constant for reaction (39) is conventionally defined by

$$\bar{K}_N = (c_N^{\mathrm{loc}} / c_N)_{\mathrm{equil}} = \exp (-\tilde{\Delta}\mu^0), \tag{46}$$

where the 'bar' on $\bar{K}_N$ distinguishes it from the quantity K_N of (30), to which it is proportional. Then, with (44),

$$\bar{K}_N = e^{N-1} (c_1^{\mathrm{loc}} / c_1)^N. \tag{47}$$

The association constant $\bar{K}_N$ for binding of M^{N+} is large if the concentration c_1 of free M^+ (ionic strength) is appreciably less than the local concentration c_1^{loc} of bound M^+. This conclusion is in complete agreement with the concept of the 'potentiated' polyelectrolyte discussed in II C; reaction (39) is driven by the diffusional entropy of the 'hidden' process $NM^+(c_1^{\mathrm{loc}}) \rightarrow NM^+(c_1)$. The constant contribution $(N-1)$ to $\ln \bar{K}_N$ arises from the osmotic inactivity of the bound cations, which, although free to move on the surface of the polyion, travel with the polyion and hence do not independently lower the activity of the solvent.

Note that, when $N = 1$, the equilibrium condition $\tilde{\Delta}\mu = 0$ reduces to

$$c_N^{\mathrm{loc}} / c_N = c_1^{\mathrm{loc}} / c_1 \quad (N = 1), \tag{48}$$

a relation that underscores the complete absence of specific effects in the model; selectivity is on the basis of valence alone.

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The chemical thermodynamical model allows a systematic calculation of the free energy ΔG stabilizing the M^{N+}/polynucleotide complex,

$$\Delta G/RT = \int_0^{\lambda_{eq}} \tilde{\Delta}\mu_N \, d\lambda. \tag{49}$$

In this equation ΔG is the free energy of the system at equilibrium relative to its free energy at the moment before commencement of reaction (39), when all M^{N+} ions are present and mixed uniformly but none are yet bound. The quantity λ is the progress variable, or degree of advancement of the reaction, and λ_{eq} is its value after the reaction has proceeded to equilibrium (Kirkwood & Oppenheim, 1961). For reaction (39), λ is the number of moles of bound M^{N+}

$$\lambda = \theta_N n_P, \tag{50}$$

where n_P is the number of moles of phosphate groups in the system. After the integration in (49) is carried through, extensive cancellation may be achieved by application of the relation $\tilde{\Delta}\mu(\lambda_{eq}) = 0$ (equivalent to equation (29)), and the final result is

$$\begin{aligned}\Delta\bar{g} = {} & \theta_N^t \ln [1-(\theta_N/\theta_N^t)] \\ & + N\theta_N + (1-\xi^{-1}) \ln [(1-\xi^{-1}-N\theta_N)/(1-\xi^{-1})],\end{aligned} \tag{51}$$

where it is understood that θ_N is the equilibrium value determined by (29) and (30), θ_N^t is the total number of moles of M^{N+} present in the system per mole phosphate, and $\bar{g}$ was defined by (6).

To illustrate this relation, suppose that the phosphate (base) molarity is 10^{-4} M at an ionic strength $c_1 = 0{\cdot}1$ M. For the equilibrium extent of binding $\theta_2 = 0{\cdot}25$ (50% phosphate charge compensation), (29) yields $\theta_2^t = 87$ or a total divalent cation molarity $8{\cdot}7 \times 10^{-3}$ M. One then calculates the value $\Delta\bar{g} = -0{\cdot}56$ from (51) for the B DNA value $\xi = 4{\cdot}2$. At 25 °C, then, the loss of 0·33 kcal free energy/mole phosphate accompanies the binding of 0·25 mole divalent cation/mole phosphate under these conditions. An interesting comparison is that, at the same ionic strength in the absence of divalent metal ion, formation of the bound layer of univalent cations is stabilized by the much larger amount 3·2 kcal/mole phosphate, as discussed in III B.

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C. *Two-variable theory*†

The constraints on θ_1, equation (25) for low binding levels and (33) for high, may be relaxed in a straightforward manner. The electrostatic free energy is taken as

$$\bar{g}_{\mathrm{el}} = -(1-\theta_1-N\theta_N)^2\,\xi \ln (1-\mathrm{e}^{-\kappa b}), \tag{52}$$

that is, both θ_1 and θ_N are variable and the net charge density after binding of univalent and N-valent cations is reduced from the stoichiometric value based on the phosphates alone by the factor $(1-\theta_1-N\theta_N)$. The mixing free energy is identical to equation (28), except that the coefficient $N\theta_N$ in the first term of the r.h.s. of (28) must be replaced by $-\theta_1$ to account for variable θ_1. Minimization of $\bar{g}_{\mathrm{el}}+\bar{g}_{\mathrm{mix}}$ with respect both to θ_1 and θ_N leads to the following two equations for the equilibrium values of θ_1 and θ_N:

$$1+\ln (10^3\theta_1\, V_P^{-1}/c_1) = -2\xi(1-\theta_1-N\theta_N)\ln (1-\mathrm{e}^{-\kappa b}), \tag{53}$$

$$\ln (\theta_N/c_N) = \ln (V_P/10000\mathrm{e})+N\ln [(10000/V_P)\, c_1^{-1}\theta_1]. \tag{54}$$

It is instructive to use these equations first to regain the results of the one-variable theory. Following the analysis of (11), one sees that (25) is indeed an approximate solution of (53) for small θ_N. Substitution of (25) into (54) yields a binding relation for M^{N+} identical to (29). Thus, the one-variable result for low levels of binding, (25) and (29), is an approximate solution of (53) and (54) for small θ_N.

For values of θ_N near saturation, an approximate solution of (53) and (54) is obtained as follows. First, note that

$$V_P = 10000\mathrm{e}\,(1-\xi^{-1})\,(\kappa b)^2\, c_1^{-1} \tag{55}$$

as a consequence of (11) with $N = 1$ and (14). Equation (55) is simply another form of (15). Now, anticipating that θ_1 is small near saturation of M^{N+} binding, substitute $\theta_1 = 0$ into the r.h.s. of (53), along with the approximation $1-\mathrm{e}^{-\kappa b} \approx \kappa b$. With (55) the result is an improved approximation for θ_1,

$$\theta_1 \approx (1-\xi^{-1})\,(\kappa b)^{2[1-\xi(1-N\theta_N)]}. \tag{56}$$

Substitution of this expression for θ_1 into (54), followed by a second application of (55), yields a result identical to (36), the prediction of one-variable theory for high levels of binding. Thus, (33) and (36)

† This section represents work done in collaboration with Nancy de Marky.

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constitute an approximate solution of equations (53) and (54) for high binding levels of M^{N+}.

Equations (53) and (54) can be solved exactly by numerical iteration. It is convenient to fix θ_N, obtain θ_1 from equation (53) by iteration, and then substitute θ_1 into (54) to get $K_N = \theta_N/c_N$ corresponding to the fixed value for θ_N. (An nth iterate for θ_1 may be obtained by substituting the $(n-1)$st into either the left- or right-hand side of (53); the scheme that converges depends on the value of θ_N. For small θ_N, a convenient zeroth-order trial for θ_1 is obtained from (25); for large θ_N, an appropriate starting point is $\theta_1 = 0$.)

The numerical results for DNA ($\xi = 4{\cdot}2$, $V_P = 646$ cm^3; see Table 1) are presented as the full curves in Figs. 1–5, the meanings of which were discussed above in connexion with the one-variable theory (dashed lines). At this point it can be remarked that the one-variable theory has provided useful approximations. Equation (29) is fairly accurate over a broad part of the low-binding end of the range (and nearly exact at $\theta_N = 0$), and (36) is reasonably good right at saturation (it is appreciably better at saturation than indicated by Fig. 2 if ionic strengths lower than 0·1 M are considered). More important than the degree of numerical accuracy of the approximations, since (53) and (54) are easily solved exactly, is the physical insight provided by the one-variable theory and discussed in detail above. The exact solution of (53) and (54) validates that discussion and sets limits to its applicability.

D. *Divalent metal ion binding*

1. *Summary of results for* $N = 2$. A summary of the main theoretical results obtained here must begin with the definition of the association constant K_2 (see equation (30)) as proportional to the equilibrium constant for the transfer of a divalent metal ion from free solution to a local volume region surrounding the polynucleotide, within which the metal cations move freely. For a duplex helix the 'association region' or 'free volume' V_P extends about 7 Å out from the surface of the helix (see Table 1). Thus defined, log K_2 is a linear function of the logarithm of the ionic strength (i.e. concentration of univalent cations from the excess 1 : 1 supporting electrolyte) with slope nearly equal to -2 over most of the binding range but dropping off rapidly to -1 as saturation is approached.

The association is anticooperative; K_2 decreases as the binding level increases. Over most of the binding range the functional dependence of

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K_2 on extent of prior binding θ_2 is intrinsically quadratic (see equation (29)), becoming exponential only at the extremity near saturation. The origin of the anticooperativity is entropic; c_1^{loc} is decreased when divalent cation binds, in accordance with (25), so that the driving force decreases with θ_2; see (47). The repulsive interactions among bound divalent metal ions usually invoked to account for anticooperative behaviour are assumed here to contribute negligibly to the free energy and are not present in the model.

Equation (47), valid over most of the binding range, states that nearly the entire thermodynamic force that drives divalent cation binding is entropic, consisting of the positive concentration difference between bound but delocalized univalent counterions and univalent counterions free in the solution bulk. Binding of a divalent ion is accompanied by the net flow of two bound univalent counterions along this gradient, out into free solution. Only near the saturation end of the binding range does the decrease of electrostatic free energy (predominately still an entropy term, incidentally, since ϵT is nearly independent of temperature for water) become a source of binding free energy.

Substitution of the numerical values $\xi_{\text{DNA}} = 4{\cdot}2$ and $V_P^{\text{DNA}} = 646\ \text{cm}^3$ from Table 1 into equation (29) gives a good numerical approximation for K_2^{DNA} for θ_2 less than about 0·30,

$$\log K_2^{\text{DNA}} = 0{\cdot}39 + 2\log(1 - 2{\cdot}63\theta_2) - 2\log c_1. \qquad (57)$$

The exact equations (53) and (54) may be solved numerically, however, over the entire binding range.

2. *Comparison of theory with Mg^{2+} binding data.* Table 5 presents a comparison of values of $-\text{p}K_2$ calculated from (57) at $\theta_2 = 0$ (where (57) is essentially exact) with published measurements of $-\text{p}K_{\text{Mg}}^0$, the 'intrinsic' Mg^{2+} binding constant extrapolated to $\theta_{\text{Mg}} = 0$. The experimental values in the table are not necessarily the numerical values given for the quantity called 'K' in the cited source; rather, the experimental values in the table conform to the definition of K given in (30), extrapolated to $\theta_{\text{Mg}} = 0$. The agreement is quite good, especially if it is remembered that half a pK unit corresponds to a free energy discrepancy of only 0·7 kcal at 25 °C.

In Fig. 1 the Mg^{2+} binding data of Clement *et al.* (1973) at 10^{-3} M ionic strength clearly show the predicted upward curvature of a Scatchard plot. In Fig. 2 data from a variety of sources have all been 'normalized' to 0·1 M ionic strength by applying the theoretical curve

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TABLE 5. *Comparison of theoretical and experimental values of* K^0_{Mg}

c_1 (M)	log K^0_{Mg} (th)	log K^0_{Mg} (exp)
0·2	1·8	2·2†
0·2	1·8	1·8‡
0·1	2·4	2·6§
0·05	3·0	3·3‖
0·01	4·4	4·9§
0·001	6·4	5·8¶

† From Skerjanc & Strauss (1968); $M^+ = (CH_3)_4N^+$.
‡ From Zubay & Doty (1958); $M^+ = Na^+$.
§ From Archer, Craney & Krakauer (1972); $M^+ = Na^+$, 'DNA' = poly A·poly U.
‖ From Shapiro, Stannard & Felsenfeld (1969); for Ca^{2+}, $M^+ = (CH_3)_4N^+$.
¶ From Clement, Sturm & Daune (1973); $M^+ = Na^+$.

of Fig. 3 to the original data (ranging in ionic strengths from 10^{-3} to 0·2 M). The resulting comparison shows that the ionic strength dependence of K_2 is about right (otherwise, many of the points would lie off the graph) and that the dependence on θ_2 is also given reasonably well. In Fig. 3 Krakauer's curve-fitting expression (1974) for poly A·poly U was used to obtain the points, which accounts for their regularity. The overall variation of slope is about the same for both theory and data points; the detailed curvature as a function of θ_2 cannot be tested on the basis of currently available data. Equation (47) predicts that the primary driving force for non-specific binding of divalent metal ions is the local counterion diffusion potential. If $c_1 \approx c_1^{loc}$, the potential disappears, and binding becomes very weak. Since $c_1^{loc} \approx 1$ M for DNA, the precise prediction is that the association constant for Mg or Ca ion binding should still be appreciable at 0·1 M but should fall to very low values at ionic strengths in the neighbourhood of 1 M. Fig. 2 indicates that K_2 is about 10^2 at 0·1 M. Data are not available for ionic strengths higher than 0·2 M, for which K_{Mg} was measured to be 59 (Zubay & Doty, 1958), but the theoretical prediction

$$\mathrm{d} \log K_2 / \mathrm{d} \log c_1 = -2,$$

which is more or less in line with observation, suggests strongly that measured values of K_2 at 1 M will be found to be of order unity, if indeed such small values can be measured at all.

3. *Conclusion.* From the reasonably successful comparison of theory and experiment for Mg^{2+} binding I conclude that the molecular model on which the theory is based may be quite realistic; in particular, Mg^{2+}

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(and probably many other divalent metal cations) binds to polynucleotides in the delocalized mode. The immediate environment of a bound Mg^{2+} ion is very nearly the same as in free solution; the bound state is characterized by essentially complete hydration and free translational and rotational mobility. In view of the independent evidence outlined in Section II consistent with the same conclusion, the case for delocalized binding of Mg^{2+} is a strong one.

E. *Binding of small cations with complex structure*

1. *Introduction.* It is reasonable to wonder how well equation (29) may work for cationic substances with relatively small values of N, but with 'structure', typically due to discretely protonated nitrogen atoms on the same molecule. One would certainly worry about the naïve use of (29) for any but the lowest values of θ_N, since excluded volume effects among bound cations have not been considered. However, the few binding data that do exist are fortunately concerned with the 'intrinsic' binding at $\theta_N = 0$. For DNA substitution of the values $\xi = 4{\cdot}2$, $V_P = 646$ cm^3 from Table 1 and $\theta_N = 0$ into (29) yields

$$\log K_N = -0{\cdot}62 + 0{\cdot}51N - N \log c_1, \tag{58}$$

and the equation may be easily compared with available data, provided the latter are used in a form consistent with the definition, (30), of K_N.

2. *Oligolysine.* Record, Lohman & de Haseth (1976) have reanalysed the data of Latt & Sober (1967) for the binding of a series of oligolysines with charge state $+N$, $N = 3$ to 8. The experimental results in NaCl could be adequately represented by

$$\log K_N = (-0{\cdot}62 + 1{\cdot}46) + (0{\cdot}51 - 0{\cdot}68)\, N - 0{\cdot}9N \log c_1, \tag{59}$$

where the first two terms have been written as sums to facilitate comparison with equation (58). The polynucleotide used was poly A · poly U, the structure of which is close enough to DNA (see Table 1) to allow direct comparison of (59) with (58). (The reader, if worried by this statement, may change the numerical values $-0{\cdot}62$ and $0{\cdot}51$ in (58) and (59) to $-0{\cdot}66$ and $0{\cdot}55$, respectively, to correspond exactly to the specification for poly A · poly U in Table 1.) The form of (59) is correctly predicted by (58). Log K_N is linear in both N and log c_1. As a function of N it has a non-zero intercept that is independent of c_1 and a slope that varies linearly with log c_1. As a function of log c_1 both the intercept and slope vary linearly with N.

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Interpretation of the numerical value of the intercept at $N = 0$ is made difficult by the presence of an extra lysine bearing an ϵ-DNP group on each member of the series used in the binding experiments. Equation (58) reflects only ionic and ideal mixing effects; the non-zero intercept is due to the translational entropy of the oligolysine molecule, as a unit, within the free volume that defines the bound state. As suggested by Latt & Sober (1967), the DNP group may contribute significantly to the binding, and its contribution would then show up in the intercept, providing thereby a satisfactory explanation for the discrepancy of 1·46 pK units between measured and predicted intercepts. It should be stressed, however, that observation of a non-zero intercept does not imply by itself a specific interaction independent of the ionic contribution, since (58) possesses a non-zero intercept but is derived exclusively from nonspecific ionic theory.

The linear dependence on N originates theoretically from the difference in ideal entropies of mixing of N M^+ ions in the condensed state relative to bulk solution (that is, from the counterion diffusion potential), since N condensed univalent cations diffuse into bulk solution as one N-valent cation binds. There is reasonable agreement between the theoretical slope $-N$ as a function of log c_1 and the measured result $-0{\cdot}9N$.

The concentration-independent second term of (59) is 0·68 pK units lower, on an electrostatically equivalent basis (i.e. divide by N), than the corresponding theoretical term in (58). That is, per equivalent bound, this contribution to the binding free energy is observed to be 0·93 kcal *less* than predicted (at 25 °C). I shall discuss below the possible significance of this fact.

Unpublished data of Lohman and Record for the binding of pentalysine (charge state $N = 5$) to DNA at pH 6 can be represented by (Record, personal communication)

$$\log K_5 = -0{\cdot}62 + (0{\cdot}51 - 0{\cdot}57)(5) - 5{\cdot}3 \log c_1, \qquad (60)$$

where, again, the form facilitates comparison with (58). The observed slope $-5{\cdot}3$ of the log c_1 dependence compares favourably with the predicted value -5 from (58). Having, in writing (60), simply *assumed* that the form of (58) applies with the value $-0{\cdot}62$ as the correct result of extrapolation of log K_N to $N = 0$ (the pentalysine used by Lohman and Record does not have the extra ϵ-DNP lys group of the Latt–Sober compounds), I now note that the apparent discrepancy in the

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second term is 0·57 pK units per equivalent; at room temperature, the observed binding free energy corresponding to this term is 0·78 kcal/(bound equiv.) less than predicted, a value quite similar to that obtained above for the Latt–Sober series. See Record *et al.* (1978) for a discussion of these data from another point of view.

3. *Spermine.* Spermine is a polyamine with charge state $N = 4$. For the ionic strength $c_1 = 0{\cdot}15$ M-NaCl, an observed value of K_4 for binding to DNA may easily be extracted from the data of Shapiro *et al.* (1969*b*). The value is $\log K_4 = 2{\cdot}89$, or

$$\log K_4 = -0{\cdot}62 + (0{\cdot}51 - 0{\cdot}46)(4) - (4)\log(0{\cdot}15). \tag{61}$$

Comparison with (58) yields an apparent discrepancy in the second term of 0·46 pK units; the observed binding free energy is 0·63 kcal/(bound equiv.) less than predicted at 25 °C.

4. *Lys-trp-lys.* Data for the binding of this tripeptide (charge state $N = 2$) to poly A are available (Brun, Toulmé & Hélène, 1975) and may be represented by (Record, Lohman & de Haseth, 1976)†

$$\log K_2 = -1{\cdot}8 - 2{\cdot}4 \log c_1. \tag{62}$$

Parameters for poly A are listed in Table 1; the corresponding numerical form of the theoretical equation (29) at $\theta_2 = 0$ is

$$\log K_N = -0{\cdot}23 - 0{\cdot}02N - N \log c_1. \tag{63}$$

Accordingly, the empirical equation (62) can be recast into the form

$$\log K_2 = -0{\cdot}23 - (0{\cdot}02 + 0{\cdot}77)(2) - 2{\cdot}4 \log c_1. \tag{64}$$

The observed coefficient $-2{\cdot}4$ of the $\log c_1$ dependence is in reasonable agreement with the value -2 predicted from (63). The discrepancy of 0·77 pK units in the second term corresponds to an observed binding free energy lower than predicted by the amount 1·0 kcal/(bound equiv.) at 25 °C. Record *et al.* (1978) have also analysed these data within the context of their binding theory.

5. *Discussion and conclusion.* It is striking that substances with charge states as diverse as lys-trp-lys and octalysine and with structures as distinct as spermine and oligolysine should all have observed binding free energies lower than predicted by 0·6–1·0 kcal/bound equiv. It

† Equation (62) represents the stability constant of the complex denoted as 'A' by Brun *et al.* and conceived by them as not involving interaction of the trp residue with the adenine bases. The analysis given here makes no contact with the question of whether or not A is in dynamic equilibrium with a stacked complex B.

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should first be realized that, whatever the biological importance of small binding energies, the molecular theoretician cannot help but be pleased by such small discrepancies, of the order of random thermal energy, between observation and *a priori* calculation. It is fair to conclude that the model of delocalized binding on which the theory rests may be realistic for these cationic substances. The observation that predicted binding constants *overestimate* measured values strengthens this conclusion, since localization of the cation in potential energy wells on the polynucleotide could be expected to require sources of binding free energy *in addition to* those considered in the model. The difference between observation and an *underestimated* prediction could have been consistently attributed to localization; the overestimate actually found does not encourage this line of reasoning. At the present time, therefore, I believe that interpretations of the binding of these substances in terms of ion-pair formation, stabilized by hydrogen bonds, are not warranted. Binding to the bases has been ruled out for spermine and tetralysine by the differential dialysis measurements of Shapiro, Stannard & Felsenfeld (1969).

The molecular origin of the discrepancy between theory and experiment could be a conformational change induced in the cation while in the bound state. Schematically, one may speculate as follows. When free in solution octalysine tends to project its charged lysine residues on alternate sides of the molecule. In this conformation, not all eight residues can fit simultaneously into the bound region V_P, which is only 7 Å thick, and octalysine would bind like a cation of valence lower than eight. However, if all lysine residues are brought to the same side of the molecule, octalysine can bind like an octavalent cation. Since octavalent cations preferentially displace cations of lower valence (for the same reason that Mg^{2+} displaces Na^+ ions), the latter conformer will be the bound form. The process of binding can be thermodynamically described as the sequential transformation from a '*trans*' to a '*cis*' form in solution followed by binding of the *cis* form. The first step involves bringing like charges closer together (and possibly other strains) and hence costs free energy. The observed binding constant is therefore smaller than what would be expected of a structureless octavalent cation. The structure of spermine suggests less conformational energy expenditure per equivalent than for oligolysine, and the corresponding discrepancy between theory and experiment is in fact the least of all the substances considered. The bulky tryptophan residue may impose

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more severe restrictions on the spatial arrangement of the lysine residues in lys-trp-lys; the corresponding discrepancy is indeed the greatest of those considered.

F. *Binding of high-molecular-weight cationic substances*

1. *Introduction.* Complexes of DNA with a surprisingly large variety of cationic substances are observed to dissociate when the ionic strength is raised to about 1 M. The simplest example, the Mg/DNA complex, has already been discussed in Section IV D. Additionally, the empirical formulas in the preceding section for the binding of oligolysine show association constants that drop off to values of the order unity at 1 M ionic strength. The phenomenon is not restricted to small cations; polylysine/DNA complexes (Leng & Felsenfeld, 1966; Shapiro, Leng & Felsenfeld, 1969) and nucleohistones in chromatin (Weintraub, Palter & van Lente, 1975) dissociate at about 1 M as well. An ionic strength of 0·6 M is sufficient to elute DNA from DEAE columns (Hirsh & Schleif, 1976). Extrapolation of binding data for the complex formed from *lac* repressor and non-operator DNA (de Haseth, Lohman & Record, 1977; Record *et al.* 1978) suggests that the binding constant is of order unity at about 0·5 M ionic strength even though it is of order 10^6 at 0·1 M.

It is likely that such close phenomenological similarity has its molecular origin in structural features possessed in common by all these systems. In fact the above behaviour would be *predicted* under the assumption that all these binding processes are predominately driven by the counterion diffusion potential, the local tendency of counterions to diffuse into free solution from a region of high concentration, which for DNA has the value 1·2 M.

In the absence of clear evidence to the contrary, therefore, it does not seem necessary to invoke a specific, geometrically complementary, structure for each of the complexes, nor is it necessary to conceive of cationic groups as having entered into hydrogen-bonded ion pairs with the phosphate oxygen atoms. The dominant driving force for complexation is brought into play simultaneously with penetration of cationic groups into the free volume V_P containing a locally high concentration of univalent counterions. The resulting complex is 'loose'; the cationic substance (or positively charged region of a protein) is free to slide along the DNA surface with a layer or two of water molecules intervening. Of course, the sliding motion of a protein (a histone, for

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example) can be prevented by purely mechanical means if the DNA helix is wrapped around it.

In the following paragraphs I shall discuss what appear to be implications of this description for three cases of interest: polylysine-DNA, repressor-DNA, and repressor-operator.

2. *The polylysine/DNA complex.* The possibility that polylysine forms a loose *non-specific* complex with DNA in the sense of the preceding paragraph is supported by at least three lines of evidence. As discussed above, the complex dissociates at about 1 M ionic strength, implying stabilization by realization of the non-specific counterion diffusion potential (tendency of counterions to diffuse into free solution from a *loosely* bound state). By direct calorimetry, Ross & Shapiro (1974) showed that the binding of polylysine to DNA is accompanied by only very small enthalpy changes; it is primarily entropy-driven, as would be true for a process driven by local counterion diffusion. (Ross & Shapiro also determined that the same is true for the binding of Mg^{2+}, spermine, and tetralysine.) Finally, Mirzabekov & Melnikova (1974) measured the effect of polylysine binding on methylation of the $N_{(7)}$ atom of guanine in the major groove of DNA and $N_{(3)}$ of adenine in the minor groove. Polylysine had no effect on the methylation reaction in either groove, in contrast to significant interference effects found for a variety of other binding substances. The conclusion suggested by the authors was that polylysine does not make contacts in either groove when bound to DNA. Indeed, it should be asked whether their data do not definitively rule out models which have the entire polylysine backbone buried in a groove with the cationic residues 'sticking out' and attached to the phosphates.

The indisputable fact remains that polylysine can quantitatively precipitate A·T-rich DNA from a mixture of A·T-rich and G·C-rich DNA (Shapiro, Leng & Felsenfeld, 1969). It may be, however, that this phenomenon has its molecular basis entirely in the base-specific properties of DNA itself, and not in any specific properties of the complex. If A·T-rich DNA were inherently more flexible than G·C-rich DNA, then even the loose, nonspecific binding of polylysine to the former could more easily induce folding of the double helix in the vicinity of the bound polylysine molecule. In effect, then, polylysine may bind to a more completely 'doubled over' region of DNA if the latter is rich in A·T base pairs, hence to a region of somewhat higher charge density than would be available in G·C-rich DNA. Even a 5°′

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differential increase in charge density would be enormously magnified, first because, according to (16), the local concentration c_1^{loc} of univalent counterions would be relatively increased by the factor $(1{\cdot}05)^2 = 1{\cdot}10$, and then because (47) implies that the ratio of affinities for a 50-residue polylysine bound to A·T-rich relative to G·C-rich DNA would be $(1{\cdot}10)^{50}$, or two orders of magnitude. Even a 1% effect in the charge densities translates into an affinity ratio of 3 for a 50-residue polylysine and 7 for 100 residues. Concerning the hypothesis that A·T-rich DNA may be less stiff than its G·C-rich counterpart, there is no evidence one way or the other as far as I know; nevertheless, in view of the detailed analysis in Section VI of electrostatic effects on DNA flexibility, it may be a reasonable conjecture.

3. *The complex of lac repressor with non-operator DNA.* As mentioned in paragraph 1 of this section, the affinity of *lac* repressor for non-operator DNA extrapolates to very low values at ionic strengths of the order of 1 M, a behaviour consistent with a loose and mobile complex. In addition, no volume change on binding could be detected by Revzin & von Hippel (1977), the likely implication being that the cationic groups on repressor (the binding region of repressor, for both operator and non-operator DNA, is known to be cationic), do not approach DNA sufficiently closely to release water of hydration from either repressor or DNA.

Methylation protection experiments (Kolchinsky *et al.* 1976) indicate, in agreement with other evidence (Lin & Riggs, 1972; Richmond & Steitz, 1976), that repressor makes no contacts in the major groove of non-operator DNA. Repressor does protect the minor groove against methylation, but the measure of protection is small, about 15%, a figure that can be contrasted to the 78% effect obtained with ethidium bromide.

These considerations are all consistent with a counterion-potentiated complex (de Haseth *et al.* 1977) perturbed by interactions in the minor groove that are too weak to prevent either substantial repressor mobility when bound or complete dissociation when the ionic strength is raised. It is reasonable to speculate that the weak interactions may involve contacts between cationic residues in the binding region of repressor with thymine $O_{(2)}$ atoms located in the minor groove (see Section III F).

4. *Repressor/operator complex.* The repressor/operator complex *is* specific. Under physiological conditions its binding affinity is some eight orders of magnitude higher than the affinity of repressor for non-operator DNA (Record, de Haseth & Lohman, 1977; Revzin &

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von Hippel, 1977). Its binding constant does not extrapolate to a small value at 1 M salt (Record *et al.* 1977). However, the same region of repressor that is responsible for its DNA binding capability is also necessary for binding to the operator segment (Bourgeois & Pfahl, 1976). Moreover, the ionic strength dependencies of the binding to operator and to non-specific DNA, although significantly different, are similar both in form and magnitude (Record *et al.* 1977). It can therefore be concluded (Record *et al.* 1977) that the stability conferred on the repressor/DNA complex by the counterion diffusion potential is largely transferred to the repressor/operator complex, and that the enhanced stability of the latter is due to superposition on the counterion potential of a rather strong interaction in the grooves.

Methylation experiments (Gilbert, Maxam & Mirzabekov, 1975) show that both major and minor grooves on the operator are protected when repressor is bound. Taking into account their result that repressor interacts weakly with only the minor groove in non-operator DNA, Kolchinsky *et al.* (1976) have proposed that when repressor initially binds to a non-operator region of operator-containing DNA, it proceeds to wander around, poking into the minor groove until it roughly recognizes the operator, whereupon it clamps down firmly on the operator, surrounding it and interacting strongly in both grooves. To supplement this attractive picture I offer a conjecture which may be extreme but that is nonetheless consistent with available data. If, as suggested in paragraph 3, the occasional probing by repressor of the non-operator minor groove is accomplished by means of weak contacts of one or two cationic residues with thymine $O_{(2)}$ atoms, it is then possible that, upon encountering a sterically specific sequence of thymine $O_{(2)}$ atoms in the minor groove of the operator, the repressor falls into a deep potential well created by entropic release of one or two water molecules from each of several 'partially ionic pairs' formed from cationic residues and thymine $O_{(2)}$'s. The repressor could be pictured as swinging around to cover the major groove, without actually forming stabilizing contacts there, although that could happen also. This conjecture is consistent with the known facts that the 'extra' stability of the repressor/operator complex, over and above that of the non-operator DNA/repressor complex, is also entropic (Riggs, Bourgeois & Cohn, 1970), and that under conditions conducive to dehydration thymine and uracil $O_{(2)}$ atoms are substantially anionic in character (Section III F).

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V. Ionic effects on the DNA helix–coil transition

A. *Electrostatic free energy and entropy of transition*

The coil form of DNA (separated strands) has a lower charge density than the native duplex helical form. It is expected, therefore, that the electrostatic free energy in the coil phase is lower than that in the helix phase. To proceed quantitatively, define $\Delta\bar{g}$ as the difference 'coil minus helix' of the free energies per mole phosphate in units of RT, and let $\Delta\bar{g}^0$ refer to the non-electrostatic contribution. The prescription given in Section II C for calculating the free energy of a polyelectrolyte solution characterized by any value of b (axial charge spacing) is now applied to both coil and helix forms of DNA, and the difference is taken. The result for DNA in an aqueous solution of excess 1 : 1 salt is

$$\Delta(\bar{g}-\bar{g}^0) = \Delta[(1-\xi^{-1})\ln(c_1^{\infty}/c_1)-\xi^{-1}\ln(\kappa b)], \tag{65}$$

where all quantities have been defined in Section II. The numerical result may be obtained by reference to Table 1 and equation (5),

$$\Delta\bar{g} = \Delta\bar{g}^0 - 1{\cdot}11 + 0{\cdot}36\log c_1, \tag{66}$$

where the coefficient of $\log c_1$ is the numerical value of $\frac{1}{2}(\Delta\xi^{-1})\ln 10$. The ionic contribution to the enthalpy change, which is contained in the slight temperature dependence of the product of absolute temperature and bulk dielectric constant of water, is less than 0·1 kcal per mole phosphate and may be neglected for practical purposes. Equation (66) implies therefore that the ionic contribution to the entropy change is about 3 entropy units per mole phosphate for ionic strengths in the neighbourhood of 0·1 M. For comparison, the total entropy change from all sources is about 11 or 12 entropy units per mole base.

B. *Dependence of T_m on ionic strength*

The dependence of T_m on ionic strength is implicit in the defining relation

$$\Delta\bar{g}(T_m, c_1) = 0. \tag{67}$$

Application of the Gibbs–Helmholtz identity to both sides of this equation, together with equation (65), gives (Manning, 1972*b*)

$$\mathrm{d}T_m/\mathrm{d}\log c_1 = \tfrac{1}{2}(\Delta\xi^{-1})\,(2{\cdot}30RT_m^2/\Delta h^0), \tag{68}$$

where Δh^0 is the enthalpy change *per mole phosphate* at the melting temperature. The value of $\Delta\xi^{-1}$ from Table 1 is 0·32. It is an empirical fact that the quantity $2{\cdot}30RT_m^2/\Delta h^0$ is nearly independent of T_m. Values

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determined from three separate calorimetric studies are 115°, 129° and 156° (Record, Woodbury & Lohman, 1976). The corresponding theoretical values of $dT_m/d \log c_1$ from equation (68) are 18°, 21° and 25° respectively. A recent experimental value of $dT_m/d \log c_1$ is 19·5° (Record, 1975). Agreement between theory and experiment is close; the 'discrepancy' noted in a previous publication (Manning, 1972*b*) was due to use of an incorrect value for b for the coil form.

A few words should be devoted at this point to the appropriate values of b. For the duplex helix the value 1·7 Å given in Table 1 is merely the familiar 3·4 Å step height of the B form divided by the two phosphate groups per step. For the separated strand the value $b = 4{\cdot}0$ Å in Table 1 was chosen as that which provides the best average agreement of (68) with measurement (Record, Woodbury & Lohman, 1976; Manning, 1976*a*). However, closely similar values of b for the coil have also been obtained by independent means (Manning, 1976*a*). The value 3·8 Å was determined by low-angle X-ray scattering on poly C in solution at elevated temperature (Gulik, Inoue & Luzzati, 1970). An identical value was deduced from an *a priori* conformational energy analysis, assuming equal proportions of C(3′)-*endo* and C(2′)-*endo* residues and the absence of all-*trans* arrangements of the phosphodiester link (Olson & Manning, 1976). Therefore equation (68) can either be viewed as a means of converting calorimetric and melting temperature measurements into a numerical value of the axial charge spacing for a single strand, or as an independent theoretical equation which accurately predicts the ionic strength dependence of T_m from calorimetric data and known values of the charge spacings of helix and coil.

C. *Role of the counterion diffusion potential*

The coil form of DNA (separated strands) has a lower charge density than the native duplex helical form. Therefore the structural transition native → coil is a mechanism for relief of phosphate–phosphate repulsive stress. At temperatures greater than T_m this mechanism for compensation of phosphate repulsion becomes available to the system as an alternative to counterion binding. Therefore, in accordance with the discussion at the end of Section II C, the potential for local counterion diffusion is realized and accompanies the spontaneous transition as the ionic contribution to the overall driving force, identifiable quantitatively as the first term on the r.h.s. of equation (65).

The second term in (65) is also electrostatic but acts in the opposite

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direction to stabilize the helix. Its physical origin is in the fact that the net charge per phosphate, $-\xi^{-1}q$, is lower for the helix than for the coil, so that screened electrostatic repulsion among pairs of phosphates is *greater* for the coil form. The relation $(\kappa b)^2 = e^{-1}(c_1/c_1^{1oo})$ mentioned in IIC may be used to show numerically that this 'counter-driving force' is considerably smaller than the counterion diffusion term. Interestingly, however, the magnitude of the ionic-strength dependence (the log c_1 term) contributed by this effect is fully 50% as large as that from counterion diffusion.

A qualitative prediction following from this discussion is that the ionic-strength dependence of T_m should level off as c_1 approaches c_1^{1oo}, or about 1 M. Precisely this behaviour is in fact commonly observed (Schildkraut & Lifson, 1965). The related consequences of ionic strengths of about 1 M for multivalent cation binding should be recalled (see Section IVF, 1).

D. *Effect of strong electric fields on T_m*

The dissociation-field theory described in Section IIIE may be used to calculate the effect of an electric field on the stability of the DNA double helix (Manning, 1977*c*). Let $\bar{g}(0)$ be the free energy, minimized as in Section IIC, of a polyelectrolyte solution in the absence of an electric field (in units of RT per polyelectrolyte equivalent); the corresponding quantity in the presence of a field $\boldsymbol{E}$ is designated by $\bar{g}(\boldsymbol{E})$. The explicit free energy calculation gives, to first order in $\boldsymbol{E}$,

$$\bar{g}(\boldsymbol{E}) - \bar{g}(0) = (1 - \xi^{-1})\,\bar{\bar{E}}, \tag{69}$$

where $\bar{\bar{E}}$ has been defined by equation (23) and all counterions are assumed univalent. The physical meaning of this equation is transparent; the number of condensed counterions is $1 - \xi^{-1}$ per polyion charge, and each of these ions has the additional energy $\bar{\bar{E}}$ (in units of kT) when the field is applied. The actual dissociation of some of the condensed counterions enters only as a negligible second-order effect. It is understood that the field is 'pulsed' for a time sufficiently short to eliminate orientational effects.

If this relation is applied both to double helical DNA and to its separated strands,

$$\Delta\bar{g}(\boldsymbol{E}) - \Delta\bar{g}(0) = \Delta[(1 - \xi^{-1})\,\bar{\bar{E}}]. \tag{70}$$

The quantity $\bar{\bar{E}}$ is slightly different for single strands and double helix

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(since the parameter a in equation (23) depends on ξ), but the dominant effect is simply that $1-\xi^{-1}$ is larger for the double helix relative to separated strands (the right-hand side of equation (70) is negative). Intact DNA binds the greater number of cations and hence suffers the greater increase of energy when the field is turned on ($\Delta\bar{g}(E) < \Delta\bar{g}(0)$). The free energy of the double helix becomes relatively higher, and the tendency for strand separation is enhanced. The T_m is lowered by the field. Upon insertion of the appropriate numerical values equation (70) becomes

$$\Delta\bar{g}(E)-\Delta\bar{g}(0) = -0{\cdot}94\times 10^{-3}\,E, \tag{71}$$

if E is in kV/cm.

Now if strand separation is actually to occur, $\Delta\bar{g}(E)$ must be reduced by the field to the value zero. Field-induced strand separation is a *threshold* phenomenon that does not occur for fields below a critical, or threshold, value. Let E_{th} denote the threshold field. Then, from (71)

$$E_{\text{th}} = 1060\Delta\bar{g}(0). \tag{72}$$

It follows immediately from equation (66) that E_{th} is linear in $\log c_1$ with

$$\mathrm{d}E_{\text{th}}/\mathrm{d}\,\log c_1 = 350\ \text{kV/cm}. \tag{73}$$

Field-induced melting has not been unambiguously observed. The predicted effect is small; experiments would have to be done at an ionic strength such that the T_m in the absence of the field is only a few degrees higher than the temperature of the apparatus. Melting might then be observed at a field of order 50 kV/cm (Manning, 1977*c*). At significantly higher ionic strengths (higher T_m) equation (73) suggests that the highest attainable fields might not be sufficient to induce melting (i.e. to lower the T_m to the ambient temperature). Detailed measurements on the non-cooperative unstacking of poly C, however, have revealed a threshold field that varies linearly with the logarithm of the ionic strength (Pörschke, 1976*b*). It is not clear at the moment whether it is easier to extend the theory to non-cooperative melting or to do the experiment for DNA.

E. *Effect of pH on T_m; average axial phosphate spacing of single strand DNA as a function of pH*

Titration of the bases with acid or alkali affects the polyelectrolyte behaviour only by changing the value of ξ (Record, Woodbury & Lohman, 1976). Heretofore, ξ was considered as being built up by the

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negative phosphate charges only; the bases were tacitly assumed to be untitrated, and (68), for example, applies only to the pH band about pH 7 for which the bases are in fact uncharged, at least if Table 1 is to be used for the numerical values of ξ. Protonation of bases decreases the value of ξ; deprotonation augments the negative phosphate charge, and consequently ξ is increased.

For a given pH, let the extent of titration per nucleotide be k_h and k_c for duplex DNA and a coiled single strand, respectively. Then (Record, Woodbury & Lohman, 1976)

$$\xi_h' = \xi_h(1 \mp k_h), \tag{74}$$

where ξ_h' is the value of ξ to be used in polyelectrolyte calculations, ξ_h is its value based only on the phosphate charges (equal to 4·2), and the upper and lower sign refer, respectively, to acid and alkaline titration. An analogous expression holds for the separated strands. The number of condensed M^+ cations per nucleotide is $1-\xi_h'^{-1}$, while the screening free energy per mole nucleotide given by equation (8) for the neutral pH case with $N = 1$ turns out to be unmodified because of the distinction that must now be made between moles of nucleotides and polyion equivalents (Record, Woodbury & Lohman, 1976). The result for the ionic strength dependence of T_m, analogous to equation (68), is

$$\mathrm{d}T_m/\mathrm{d}\log c_1 = (2{\cdot}30RT_m^2/\Delta h^0)\,(\tfrac{1}{2}\Delta\xi^{-1} \pm \Delta k). \tag{75}$$

Note that if the bases are untitrated ($k_h = k_c = 0$), or if there is no *differential* titration ($\Delta k = 0$), then equation (75) reduces to equation (68). The situation is somewhat more complicated than indicated by equation (75) since acid titration can in practice lower ξ' to the extent that it becomes less than unity, so that no M^+ cations condense (recall that the charge fraction is ξ'^{-1}) and other expressions apply for the free energy (Manning, 1969). The details are presented in full by Record, Woodbury & Lohman (1976).

The extents of titration k_c and k_h are experimentally accessible at each pH, as is $\mathrm{d}T_m/\mathrm{d}\log c_1$ and Δh^0. It may be assumed that $\xi_h = 4{\cdot}2$ regardless of pH because of the double helical rigidity. However, it is possible that the average axial nucleotide spacing on the single strand may increase with the total charge placed on the chain. Hence, ξ_c is regarded as the sole adjustable parameter in equation (75) and the values of b_c in Table 6 are those required to bring (75) into agreement with data (Record, Woodbury & Lohman, 1976). It is seen from the table that the

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TABLE 6. *Variation of single-strand nucleotide spacing (b_c) with pH*†

pH	ξ_c	b_c (Å)
11·3	1·14	6·3
11·0	1·19	6·0
10·8	1·27	5·7
9·0	1·49	4·8
7·0	1·72	4·1
4·8	2·50	2·9
4·5	2·78	2·6
3·7	4·00	1·8
3·2	4·17	1·7

† Reproduced with permission from Record, Woodbury & Lohman (1976).

axial nucleotide spacing (contour length divided by number of bases) does indeed increase with pH, reaching the value characteristic of a fully extended chain at the highest pH. Apparently the all-*trans* conformation, which may be excluded at neutral pH (Olson & Manning, 1976) becomes dominant upon sufficient deprotonation of the bases with consequent strong close-range electrostatic repulsion. Finally, the value at the lowest pH suggests that the single strand has 'folded over' on itself and adopted a spacing similar to that of the double helix (recall that $b_h = 1{\cdot}7$ Å and $\xi_h = 4{\cdot}2$).

F. *Ionic effects of strandedness on T_m*

The four transitions of the (poly A, poly U) system provide a nice test of the theory (Record, Woodbury & Lohman, 1976). Only two of the four transitions are thermodynamically independent and may be chosen as

(1) poly A·poly U → poly A + poly U,
(2) poly A·2 poly U → poly A + 2 poly U.

It is easy to show (Record, Woodbury & Lohman, 1976; de Marky, unpublished) that the corresponding theoretical equations for

$$\mathrm{d}T_m/\mathrm{d}\log c_1$$

have the same form as (68),

$$\mathrm{d}T_m/\mathrm{d}\log c_1 = \tfrac{1}{2}(\Delta_i \xi^{-1})\,(2{\cdot}30RT_m^2/\Delta h^0), \tag{76}$$

but with $\Delta_i \xi^{-1}$ given by the number-averaged differences,

$$\Delta_1 \xi^{-1} = \tfrac{1}{2}\xi_{\mathrm{A}}^{-1} + \tfrac{1}{2}\xi_{\mathrm{U}}^{-1} - \xi_{\mathrm{A{\cdot}U}}^{-1} \tag{77}$$

$$\Delta_2 \xi^{-1} = \tfrac{1}{3}\xi_{\mathrm{A}}^{-1} + \tfrac{2}{3}\xi_{\mathrm{U}}^{-1} + \xi_{\mathrm{A{\cdot}2\,U}}^{-1}, \tag{78}$$

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where the subscripts '1' and '2' refer to the transition number. If it is assumed that $\xi_A = \xi_U$, then $\Delta_i \xi^{-1}$ collapses to the quantity $\Delta\xi^{-1}$ of equation (68). The values of $\xi_{A\cdot U}$ and $\xi_{A\cdot 2U}$ are known from X-ray data: $\xi_{A\cdot U} = 4{\cdot}6$, $\xi_{A\cdot 2U} = 6{\cdot}8$ (Record, Woodbury & Lohman, 1976). The required calorimetric and T_m data are available (Krakauer & Sturtevant, 1968) and have been tabulated in convenient form by Record, Woodbury & Lohman (1976).

Let us assume first that $\xi_A = \xi_U = 1{\cdot}9$, the numerical value for unstacked (high-temperature) poly C (see Section VB). The values of $dT_m/d \log [Na^+]$ predicted from (76) are then 20° for transition (1) and 24° for transition (2); the observed values are 19·6° and 26° respectively. For the K^+ salt the $\Delta h^0/T_m^2$ data are slightly different. Predicted values of $dT_m/d \log [K^+]$ are 21° and 28° for transitions (1) and (2), respectively, while the corresponding measured values are 17·4° and 27°.

Equation (76) may be used in another way (Record, Woodbury & Lohman, 1976). Written for both transitions (1) and (2), two equations in the two unknowns ξ_A and ξ_U are obtained, all other quantities being given from the available data. The values $\xi_A = 2{\cdot}3$ and $\xi_U = 1{\cdot}6$ are obtained from the Na^+ data; the K^+ data yield $\xi_A = 3{\cdot}1$ and $\xi_U = 1{\cdot}6$. It is not unreasonable that the axial phosphate spacing should be greater for unstacked poly U than for poly A, which is partially stacked even at typical melting temperatures. The axial spacing b for poly U corresponding to $\xi = 1{\cdot}6$ (obtained in both Na^+ and K^+ salts) is 4·5 Å, which could be slightly high (Manning, 1976*a*). The axial spacings for poly A corresponding to the ξ values 2·3 (Na^+) and 3·1 (K^+) are 3·1 Å and 2·3 Å. The former value is essentially the same as that which characterizes the fully stacked low-temperature form of poly A (Stannard & Felsenfeld, 1975), but the latter value is much too small to be rationalized. Using a similar method, Pless & Ts'O (1977) have determined the axial charge spacings for poly dA and dT_8 as 3·3 and 4·9 Å, respectively.

G. *Effect of Mg^{2+} on T_m*

Section IVB outlined a method for obtaining the free energy of a polynucleotide solution in which Mg^{2+} ions are present along with excess 1 : 1 salt. Therefore it is possible to calculate $\Delta\bar{g}$ and T_m for the DNA melting transition under these conditions (deMarky & Manning, in preparation). Theory and measurements are compared in Fig. 6. The

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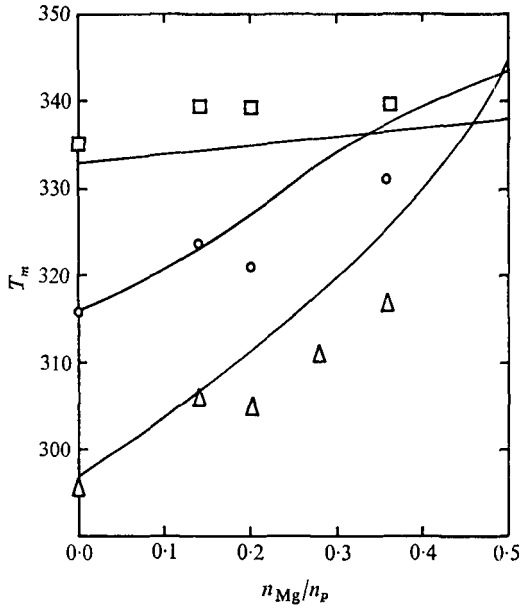

Fig. 6. Melting temperature of DNA as a function of added Mg^{2+} at three different ionic strengths, 10^{-2}, 10^{-3} and 10^{-4} M, from top to bottom at left. The theory (solid curves) has been adjusted to match the data at the single point of zero Mg^{2+} and 10^{-3} M (middle curve at origin). Data from Record (1975).

abscissa in the figure is equivalents of total added Mg^{2+} per mole phosphate, so that the points and curves refer to low binding levels of Mg^{2+}. The data points are taken from Record (1975). The single theoretical point at 10^{-3} M-NaCl and $n_{Mg} = 0$ is forced to coincide with the measured value (since T_m reflects a non-ionic contribution in addition to ionic effects), but all other points on the curves are determined from *a priori* theory with the sole parameters b_h and b_c, both of which are known independently. It is especially to be noted that the near-independence of T_m on added Mg at 10^{-2} M-NaCl, both as observed and predicted by theory, does not mean that Mg^{2+} ions are not bound at this ionic strength. The theoretical and measured binding constants in Table 5 show that Mg remains strongly bound at 10^{-2} M, at least to the double helix, and theoretical values of K_{Mg} for the single strand are also appreciable (an order of magnitude lower). Therefore the invariance of T_m at this ionic strength reflects a delicate differential balance between Mg^{2+} binding to the two DNA forms, and the success-

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ful prediction correspondingly suggests a high degree of theoretical accuracy.

One of the more interesting phenomena associated with Mg^{2+} binding is the cross-over of the curves seen at the high end of the added Mg^{2+} range in Fig. 6. The data do exhibit such a cross-over (Record, 1975; Dove & Davidson, 1962), but at somewhat higher values of added Mg^{2+}. Beyond the region of cross-over addition of 1 : 1 salt (increased ionic strength) results in destabilization of the double helix for a fixed value of total added Mg^{2+}. The explanation of this phenomenon (Manning, 1972*c*; deMarky & Manning, 1975) is particularly simple for levels of added Mg^{2+} sufficiently high that its binding is essentially saturated (polynucleotide charge fraction equal to $\frac{1}{2}\xi^{-1}$) for both double helix and single strand over a range of ionic strengths. In Section C it was pointed out that melting in the absence of Mg^{2+} is accompanied by release of condensed M^+ ions (increased entropy) and an increase of repulsive electrostatic free energy. The former effect dominates, and, since it destabilizes the double helix but decreases in magnitude as the ionic strength increases, the double helix becomes more stable at higher ionic strengths. At saturating levels of Mg^{2+}, however, no M^+ cations are condensed, having been entirely replaced by Mg^{2+}. The melting transition is consequently unaccompanied by release of condensed M^+, since there are none to be released. The only remaining effect dependent on ionic strength is the increase of electrostatic free energy, which stabilizes the double helix but decreases in magnitude with increasing ionic strength. Therefore an increased ionic strength destabilizes the double helix.

The effect of Mg^{2+} on T_m has also been discussed by Krakauer (1974) and by Record *et al.* (1978).

H. *Ionic effects on the renaturation rate*

The rate of association of separated but complementary polynucleotide strands to form double helixes is strongly dependent on ionic strength (Wetmur, 1976). Measured values of the apparent second-order rate constant k_2 (Ross & Sturtevant, 1960; Studier, 1969) can be fitted by the empirical relation (Manning, 1976*b*)

$$\mathrm{d}\log k_2/\mathrm{d}\log c_1 \approx 3{\cdot}6. \tag{79}$$

The theory (Manning, 1976*b*) is based on the assumption that the rate-determining step is a pre-equilibrium between separated strands

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and a non-bonded precursor to the nucleated species (Wetmur & Davidson, 1968). The problem then reduces to the calculation of $\Delta\bar{g}$ for the process of bringing together two separated strands, each characterized by the charge-density parameter ξ_c, to form an aligned but non-bonded double-stranded configuration of charge density $2\xi_c$ and of length Q base pairs. The theoretical prediction is then easily demonstrated (Manning, 1976*b*) to be

$$\mathrm{d}\log k_2/\mathrm{d}\log c_1 = Q/2\xi_c. \tag{80}$$

Comparison of (79) and (80) with the value $\xi_c = 1{\cdot}9$ from Table 1 yields the value $Q = 14$, which is comparable to the persistence length of an unstacked single strand.

The effect of added Mg^{2+} has also been treated, and the observed second-order dependence of the association rate on Mg^{2+} concentration (Felsenfeld & Rich, 1957) has been demonstrated to be consistent with the value of Q given above (Manning, 1976*b*).

I. *The mistake of adopting the Debye–Hückel approximation*

A frequently used procedure to calculate the ionic contributions to the free energy of a closely interacting array of charge centres is pair-wise summation of Debye–Hückel potentials. Although everyone is aware that some sort of approximation is thereby being made, just how bad this approximation really is may not be sufficiently appreciated. The helix–coil transition provides a good example.

Suppose pair-wise summation of Debye–Hückel potentials is performed for an infinitely long, linear array of charges with spacing b, and that the difference is then taken for two values of b. The result, which should immediately be compared with (65), is

$$\Delta(\bar{g}-\bar{g}^0) = -\Delta(\xi \ln \kappa b). \tag{81}$$

The comparison shows clearly that, by ignoring the physical reality of bound counterions and their degrees of freedom, the Debye–Hückel procedure has completely missed the molecular phenomena underlying the transition! Another qualitative error is the direct variation with ξ in equation (81); the correct variation is inverse.

Substitution of numerical values from Table 1 into equation (81) yields

$$\Delta\bar{g} = \Delta\bar{g}^0 - 2{\cdot}93 + 2{\cdot}76 \log c_1, \tag{82}$$

which may be compared with (66). Although (82) does give the correct

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form of the ionic-strength dependence, the slope is an order of magnitude too high. At 0·1 M and 25 °C the electrostatic free energy (sum of the last two terms) predicted by equation (82) is $-3{\cdot}37$ kcal/mole P, about four times the value $-0{\cdot}87$ kcal/mole P from the more accurate (66). Similarly, the slope of T_m *v.* log c_1 is about an order of magnitude too high in the Debye–Hückel approximation.

Within the context of such gross inaccuracy, use of exact coordinates for the charge centres can provide only the illusion of computational sophistication.

VI. Electrostatic contribution to DNA stiffness

A. *Harrington's measurements of persistence length*

Recent measurements (Harrington, 1978) of high-molecular-weight DNA persistence lengths in aqueous NaCl over a range of nearly three magnitudes of ionic strength provide the possibility of new insight into local DNA structure. Harrington's data are portrayed in Fig. 7 in the form of a graph of persistence length a (in ångström units) against log c_1, the logarithm of ionic strength. The measurements cover the range 0·005 M $\leqslant c_1 \leqslant$ 2·0 M. The solid line with greater numerical slope is the least-squares linear best fit based on the four points of lowest ionic strength. The equation of this line is

$$a = 221 - 559 \log c_1 \quad \text{(based on } c_1 \leqslant 0{\cdot}2 \text{ M)}. \tag{83}$$

The other straight line in Fig. 7 is the corresponding fit based on the five most dilute concentrations; its equation is

$$a = 317 - 504 \log c_1 \quad \text{(based on } c_1 \leqslant 0{\cdot}6 \text{ M)}. \tag{84}$$

Although there are not enough points to determine unambiguously whether a is an inherently linear function of log c_1 at the low end of the ionic strength range, the fit is reasonable. Moreover, Fig. 7 is strongly reminiscent of observed plots of T_m *v.* log c_1, which also begin to level off from steep linearity at ionic strengths between 0·2 and 0·6 M (Schildkraut & Lifson, 1965). As discussed in Section V, this type of dependence is related to the counterion diffusion potential. I have accordingly sought for a conformational change, closely linked to the persistence length, which may be driven by the counterion diffusion potential.

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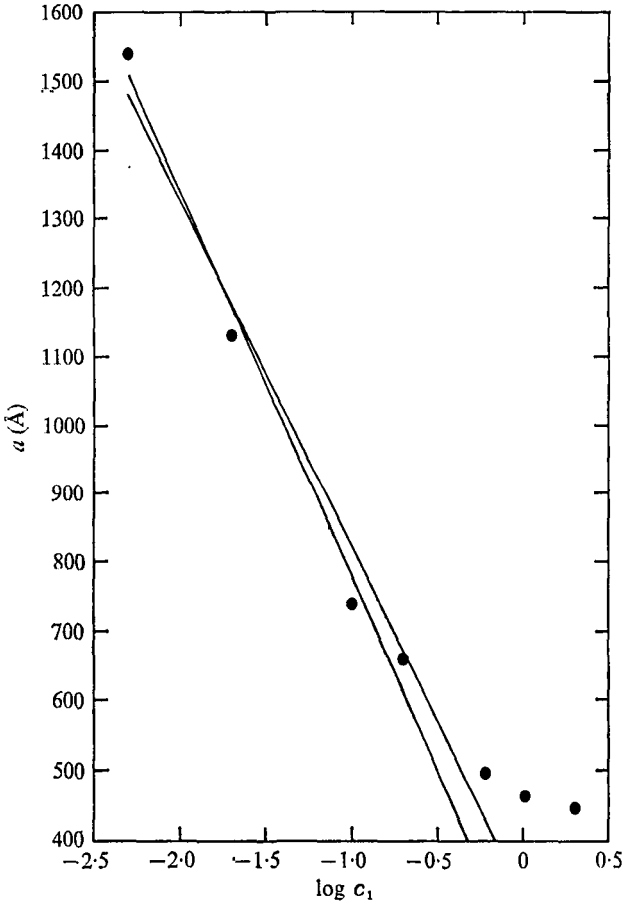

Fig. 7. Harrington's persistence length data (1978) in logarithmic representation. Solid lines are linear regressions as described in text.

B. *Polyelectrolyte theory of the persistence length*

Suppose that on a scale of length l high-molecular-weight DNA behaves like a rigid rod with elastic bending capability. Then a positive free energy ΔG must accompany the bending of a mole of segments of length l through a small angle θ (see Fig. 8). The persistence length is related to ΔG by the formula (Bloomfield *et al.* 1974)

$$\Delta G = (RTa/2l)\,\theta^2. \tag{85}$$

It is the spontaneous process bent → straight, with negative free energy

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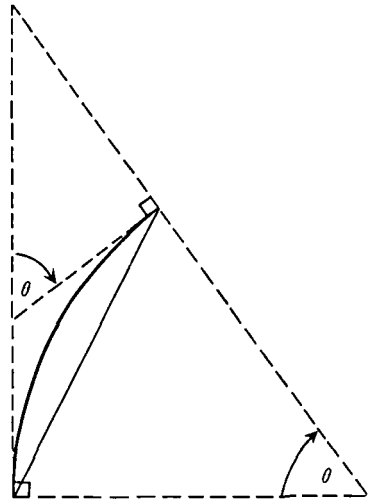

Fig. 8. Geometrical construction used to derive equation (93).

$-\Delta G$, that I presently believe, on the basis of the analysis to follow, to be primarily driven by local counterion diffusion.

With reference to the geometrical construction in Fig. 8, where the heavy circular arc is a segment of DNA of length l bent through an angle θ, let us first note the relation between l, θ, and the end-to-end distance of the arc, denoted by $l(\theta)$ and represented in the figure by the thin solid line:

$$l(\theta) = \sqrt{2}\, l\theta^{-1}\sqrt{(1-\cos\theta)}. \tag{86}$$

Expansion of $\cos\theta$ for small angles θ yields

$$\Delta l = -\tfrac{1}{24} l\theta^2, \tag{87}$$

where

$$\Delta l \equiv l(\theta) - l. \tag{88}$$

When $\theta = 0$, the segment is straight, its end-to-end distance $l(0)$ is the same as its contour length l, and $\Delta l = 0$.

Now denote by n_l the number of phosphates in the segment. This quantity is independent of θ. Define an effective charge spacing $b(\theta)$ of the bent segment by

$$b(\theta) = l(\theta)/n_l. \tag{89}$$

Then, corresponding to equations (87) and (88),

$$\Delta b = b(\theta) - b = -\tfrac{1}{24} b\theta^2, \tag{90}$$

where $b = l/n_l$, the usual axial charge spacing (1·7 Å from Table 1).

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The effective charge spacing $b(\theta)$ in the bent conformation is less than that of the straight conformer. The corresponding formula for $\xi(\theta)$, the charge density parameter generalized from equation (1) as proportional to the reciprocal of $b(\theta)$, is

$$\Delta\xi = \xi(\theta) - \xi = \tfrac{1}{24}\xi\theta^2. \tag{91}$$

The charge density increases when the segment is bent. The electrostatic contribution to ΔG will therefore be positive.

Turn now to equation (65), which gives a formula for $\Delta\bar{g}$, the free energy difference per mole phosphate, in units of RT, between two forms of DNA with charge densities ξ_{final} and ξ_{initial}. Identify the former with $\xi(\theta)$ and the latter with ξ. Expand $\Delta\bar{g}$ in powers of θ, noting that c_1^{loc} is a function of ξ through V_P as well as directly, and retain the lowest-order term only. The result is

$$\Delta\bar{g} = \Delta\bar{g}^0 + (\theta^2/24)\,\xi^{-1}\,[2\xi - 1 + \ln\,(c_1^{\text{loc}}/c_1) + \ln\,\kappa b], \tag{92}$$

where $\Delta\bar{g}^0$, the non-electrostatic contribution, is also assumed proportional to θ^2. In this form it appears that the large positive contribution 2ξ is independent of the counterion diffusion potential. However, it should be remembered that equation (92) is simply a differential form of equation (65); the term 2ξ originates in the expansion of

$$(1 - \xi^{-1})\,\ln\,(c_1^{\text{loc}}/c_1).$$

Substitution of (92) into (85) yields

$$a = a_0 + (l^2/12b)\,\xi^{-1}[2\xi - 1 + \ln\,(c_1^{\text{loc}}/c_1) + \ln\,\kappa b), \tag{93}$$

where a_0 is the contribution to the persistence length originating from the nonelectrostatic free energy $\Delta\bar{g}^0$ of bending. With the numerical values for duplex DNA in Table 1,

$$a = a_0 + 0{\cdot}0815l^2 - 0{\cdot}0134l^2 \log c_1, \tag{94}$$

where a and l are in ångströms.

Comparison of (94) with either of the empirical equations (83) or (84) shows identical forms if l is independent of ionic strength. If the slope of the $\log c_1$ dependence in (83) is compared with that of (94), the numerical value of l emerges as

$$l = 204\ \text{Å}. \tag{95}$$

If (84) is used instead, a similar value 194 Å is found for l. If (95) is now substituted back into (94), the latter becomes

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$$a = a_0 + 3392 - 559 \log c_1. \tag{96}$$

The discussion to follow is based on (95) and (96), which in turn follow from comparison of (94) with (83); if (84) had been used instead, nothing substantive would require change.

C. *Discussion of the numerical results*

To assess the theory properly it is necessary to have some feeling for the numbers involved. The increment in b corresponding to a small bending angle θ, given by (90), is quite plausible. With $b = 1{\cdot}7$ Å and θ taken as one radian, Δb is about a tenth of an ångström, a value that could easily be the consequence of a slight structural distortion of the charge distribution. The value of about 200 Å (about 60 base pairs) for l is equally consistent. Recall that l is the length of a DNA segment which must retain rigid mechanical properties over the range of ionic strengths, $c_1 \lesssim 1$ M, represented by the empirical (83) or (84). Reference to Fig. 7 shows that the persistence length is maintained at a value of about 450 Å even at very high ionic strengths. Therefore a segment l somewhat less than half this length does behave nearly like a rigid rod (Bloomfield *et al.* 1974).

At 0·1 M ionic strength (96) predicts that the polyelectrolyte contribution to the persistence length (sum of last two terms) is 3950 Å, whereas the observed persistence length at 0·1 M from Fig. 7 is only 750 Å. *The net effect of other contributions is to render DNA more flexible.* In more general quantitative terms, the value of a_0 calculated by comparison of (96) and (83) is negative,

$$a_0 = -3171 \text{ Å}. \tag{97}$$

To see if this number is physically sensible, let us translate it into an equivalent free energy by applying (85) with ΔG and a replaced by ΔG_0 and a_0, respectively. At 25 °C, and with a_0 and l given by (97) and (95), respectively, ΔG^0 has the value $-4{\cdot}6$ kcal for an angle θ of 1 radian. Bending of a segment of length l would be a spontaneous process if polyelectrolyte effects are ignored, driven by the free energy $-4{\cdot}6$ kcal/(radian)^2. Since the segment l contains $(204/3{\cdot}4) = 60$ base pairs, this free energy is a mere -77 cal/mole base pair, and it is easy to imagine an increased entropy of base motion (increased breathing) when DNA is locally bent as a possible source of this free energy.

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D. *Implications for DNA condensation and nucleosome formation*

According to the above analysis, removal of the polyelectrolyte effect, or counterion diffusion potential, at ionic strengths for which (83) and (96) are valid (a range that includes physiological conditions) would have as a consequence the spontaneous bending, or folding, of the DNA molecule. Mathematically, removal of the counterion diffusion potential is accomplished by setting the valence of a phosphate group equal to zero; the last two terms in (96) are then replaced by zero, and a reduces to the negative quantity a_0. Physically, 'removal' of the polyelectrolyte effect at 0·1 M salt, for example, occurs at sufficiently high levels of multivalent cation binding. For example, even bound loosely, a polylysine molecule, or possibly two acting simultaneously, can completely neutralize the phosphate charge in its vicinity, neutralization being understood in the long-range sense of zero net charge. Or saturated binding of trivalent spermidine, which produces

$$1-(3\xi)^{-1} = 92\%$$

phosphate charge neutralization, might be enough. The prediction is that DNA should then spontaneously fold, or 'condense', and that is exactly what happens (Gosule & Schellman, 1976). If the histone core of a nucleosome possesses a sufficiently high positive surface charge density, the possibility is also raised that a nucleosome is formed by the spontaneous wrapping of DNA around the protein core, driven solely by the non-specific counterion diffusion potential.

It is interesting that quite different considerations have led Riemer & Bloomfield (1978) to identify electrostatic repulsion as the primary resistance to DNA packaging in phage heads.

E. *Further discussion*

The procedural simplicity and appealing results obtained here should not be allowed to obscure a number of questions that can be raised. The extent to which the theoretical conclusions may depend on the accuracy of the data is not clear. Persistence length measurements are not straightforward. The dependence of Harrington's values on ionic strength is considerably steeper than that indicated by previous estimates (Harrington, 1978). His methods, however, do appear to be considerably more direct than those used previously. At any rate the experimental data require careful assessment, confirmation, and extension to include more data points at low ionic strength.

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An apparent difficulty with the theoretical interpretation is illustrated by the behaviour of poly A, which is rigid when its bases are stacked at low temperatures (Stannard & Felsenfeld, 1975), but becomes flexible as the bases unstack in response to a temperature increase at a fixed ionic strength. The charge densities of unstacked and stacked single strands are similar (Olson & Manning, 1976). Stiffness, therefore, appears to be correlated with base stacking rather than with electrostatic repulsion. However, these two aspects of polynucleotide structure are not necessarily independent. A stacked conformation may provide the steric constraints that render obligatory the accompaniment of bending by the quasi-linear charge compression assumed in the theory; bending of an unstacked single strand may not require so severe an increase in electrostatic stress.

The behaviour of the persistence length at high ionic strengths, outside the range of validity of (96), is interesting and may also be a source of concern. An important distinction must be drawn here between the polyelectrolyte effect, or counterion diffusion potential, and electrostatic effects in general. By definition, mathematical reduction of the phosphate valence to zero abolishes all electrostatic effects along with the counterion diffusion potential. Sufficient cationic binding accomplishes the same thing. However, augmentation of the ionic strength to high levels may not. Although the counterion diffusion potential disappears at high ionic strength, nearest-neighbour electrostatic repulsion may survive because of the finite size of the screening ions. Indeed, the nearest-neighbour phosphate–phosphate distance of about 6–7 Å is not much larger, if at all, than current estimates of the effective hydrated diameter of the Na^+ ion. Under these conditions (96) would be replaced by

$$a = a_0 + a_{nn,\mathrm{el}}, \tag{98}$$

where a_0 would have about the same value as given by (97) and the nearest-neighbour electrostatic phosphate–phosphate repulsive contribution $a_{nn,\mathrm{el}}$ is the only survival of the last two terms of (96). Both terms on the r.h.s. of (98) would be expected to depend slightly on ionic strength indirectly through solvent effects, but I shall assume ionic-strength independence as a first approximation. Fig. 7 shows that $a \approx 450$ Å at high ionic strength. Hence, with (97), an estimate for $_{nn,\mathrm{el}}$ is 3600 Å; translated into a free energy of bending by means of (85), this value is equivalent to a bending free energy of 5·2 kcal/(radian)^2 for the segment of length l, or only 87 cal/mole base pair/

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(radian)^2. As a consistency check, I note that unscreened coulomb repulsions between each of the two phosphates in a base pair and their respective nearest neighbours at distance about 7 Å (which happens also to be about equal to the Bjerrum length) gives a free energy of bending of about $(2RT/24) = 50$ cal. This rough estimate assumes applicability of the bulk dielectric constant and totally unscreened phosphate charges; the factor $(24)^{-1}$ appears in (90).

Let us recapitulate. Even at high ionic strength segments of length $l \approx 200$ Å maintain their rigidity, since a is still observed to be more than twice as great. The rigidity under these conditions cannot be attributed to the counterion diffusion potential, which emerges only at lower ionic strength as the cause of increased segmental rigidity; a 200 Å segment becomes *harder* to bend as the ionic strength decreases, but it is still stiff even at high ionic strength, at least up to 2 M. The residual stiffness at high ionic strength can be attributed to surviving nearest-neighbour repulsions among the phosphates. If the latter could also be eliminated at high ionic strength, then DNA would become locally flexible in accord with what would then be the negative value a_0 of the persistence length. Recall again that at lower ionic strength, including physiological, *all* electrostatic effects can be eliminated by binding of multivalent cationic substances, leaving a_0 as the only contribution and thus rendering DNA locally flexible.

During the course of fifteen years of experience in polyelectrolyte theory I have accumulated many debts to many people. Special thanks are due to Bruno Zimm for initial motivation and guidance; to U. P. Strauss for sustained interest and encouragement; to Wilma Olson for whatever rudimentary knowledge of structure I may possess; to Alfred Holtzer, Tom Record and the late Lars Onsager for particular insights that led to new advances; and to Will McClure and Gary Felsenfeld for recent valuable discussions.

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The Role of Solitons in the Energy and Electron Transfer in One-Dimensional Molecular Systems

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Abstract

Theory of motion of quasi-particles, electrons and a vibrational excitation along molecular chains is developed without using the adiabatic approximation and the perturbation theory. It is shown that the motion of quasi-particles is accompanied by a local chain deformation. The deceleration of a soliton by external dissipative forces is studied. The effect of thermal vibrations on the soliton motion is analysed.

1. Introduction

When studing biological phenomena on a molecular level it was established that many processes are associated with an electron transfer along protein molecules from donor to acceptor molecules or with a transfer of energy released under hydrolysis of adenosine triphosphate (ATP) molecules.

The energy of ATP molecule hydrolysis equal to 0.54 eV is only 20 times greater than the average energy to thermal vibrations. Many researchs suggested that the energy of the ATP hydrolysis moves along protein molecules as vibrational oscillations of atoms C = O contained in peptide groups of protein molecules. However, at standard conditions the life time of such vibrations is only 10^{-13}s. Therefore many doubts have been expressed concerning the possibility of an effective transfer of vibrational energy along protein molecules.

Great difficulties are also experienced in the explanation of the effective transfer from donor to acceptor molecules. The experimental data show that such a transfer is possible the distance of the order of 70 A^{o}. The electron transfer to such great distances cannot be explained by the usual theory of quantum mechanical tunneling.

One of possible explanations of the electron transfer at large distances is connected with the assumption that the transfer process is facilitated by protein molecules to which a donor and acceptor of electrons are attached. The donor-acceptor model of the electron transfer through protein molecules was developed in [1-2]. In these papers a protein molecule was modelling by a quasi-one-dimensional molecular system which containes one extra electron passed into a molecule from a donor. In these and in most other papers in which the motion of quasi-particles (electrons, excitons etc.) was studied in periodic structures an adiabatic approximation was used. When using such an approximation one assumes that molecules slightly displace with respect to stable equilibrium positions. The interaction of a quasi-particle with such displacements is considered as a perturbation decelerating its motion.

Recently in our institute the study has been made of the motion of quasi-particles - electrons and vibrational excitation - along molecular chains without using the adiabatic approximation and the perturbation theory. Giving the perturbation theory up leads to a description of quasi-particle motion by means of nonlinear equations. Their solution shows [3-9] that the motion of a quasi-particle is accompanied by a local deformation of a chain, i.e. a displacement of equilibrium positions. The simultaneous motion of a quasi-particle and a local deformation is described by a wave function in the form of a solitary wave-soliton - which is a solution of a nonlinear Schrodinger equation.

Solitons transport a dipole vibrational excitation or an electron with a local decrease of mutual distances between peptide groups. These solitons always travel with a velocity less than that of a sound, therefore, they do not expend energy to generate phonons. The energy expenditure is required to break the electron coupling or vibrational excitations of peptide groups with a local deformation of a molecule. Thus, the motion of an electron or a vibrational excitation of peptide groups along a protein molecule holds without energy loss and resembles superconductivity and superfluidity phenomena.

Below we consider the basic results of theoretical investigations [3-9].

2. Solitons in a One-Dimensional Molecular System at Very Low Temperatures

We consider a one-dimensional molecular chain with repetitive peptide groups as a simplest protein molecule. As a result of a resonant interaction of vibrational excitations of peptide groups, in such a chain the collective states appear which may be described by the Hamiltonian

$$H = H_e + H_{ph} + H_{int} \tag{2.1}$$

where

$$H_e = \sum_n \left(\mathcal{E}_o - \frac{\hbar^2}{ma^2}\right) B_n^+ B_n + \\ + \frac{\hbar^2}{2ma^2} B_n^+ \left(B_{n-1} + B_{n+1}\right) \tag{2.2}$$

is the Hamiltonian of collective excitations - excitons in the chain with peptide groups fixed rigidly in the equilibrium positions (at the points na); excitons are described by plane waves. a is the distance between the nearest points, m is the effective exciton mass; $\mathcal{E}_o$ is the energy of the exciton band bottom; B_n^+ and B_n are the creation and annihilation operators of the vibrational excitation on the group of the number n.

$$H_{ph} = \frac{1}{2} \sum_n \left(\frac{1}{M} \hat{p}_n^2 + \varkappa u_n^2\right) \tag{2.3}$$

is the operator of longitudinal displacements u_n in peptide groups of the mass M from equilibrium positions, $\varkappa$ is the viscosity coefficient, $\hat{p}_n$ is the momentum operator.

$$H_{int} = \chi \sum_n \left(u_{n+1} - u_{n-1}\right) B_n^+ B_n \tag{2.4}$$

is the operator of interaction of vibrational excitations with displacements of peptide groups. χ is the parameter of this interaction.

The operator (2.1) may also describe the motion of an extra electron in a molecular chain. In this case the operator (2.2) will characterize the extra electron motion in repetitive potential

wells caused by constant electric dipole moments of peptide groups. Then m is the effective electron mass, $\mathcal{E}_0$ is the energy of the conductivity band bottom *).

Let us determine the Hamiltonian function in our system by the equality

$$\mathcal{H} = \langle \psi(t) | H | \psi(t) \rangle$$

where the function

$$\psi(t) = \sum_n \varphi_n(t) U(t) B_n^+ |0\rangle , \quad \sum_n |\varphi_n|^2 = 1 \tag{2.5}$$

contains the unitary operator of displacements

$$U(t) = \exp\left\{ -\frac{i}{\hbar} \sum_n \left[\beta_n(t) \hat{p}_n - \pi_n(t) u_n \right] \right\} . \tag{2.6}$$

The Hamiltonian function $\mathcal{H}$ depends on $\varphi_n(t)$, $\beta_n(t)$ and on their canonically conjugate $i\hbar\varphi_n^*(t)$ and $\pi_n(t)$. In this case $|\varphi_n(t)|^2$ determines the possibility of vibrational excitation of the group n; $\beta_n(t)$ characterizes the displacement of equilibrium positions of peptide groups.

Using the Hamiltonian function $\mathcal{H}$ we may find the Hamiltonian equation

$$i\hbar \frac{\partial \varphi_n}{\partial t} = \frac{\partial \mathcal{H}}{\partial \varphi_n^*} , \quad \frac{\partial \beta_n}{\partial t} = \frac{\partial \mathcal{H}}{\partial \pi_n} , \quad \frac{\partial \pi_n}{\partial t} = -\frac{\partial \mathcal{H}}{\partial \beta_n} .$$

In the continuum approximation these equations are reduced to two equations

$$\left[i\hbar \frac{\partial}{\partial t} - \mathcal{E}_0 + \frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + 2\varkappa \rho(x,t) \right] \varphi(x,t) = 0, \tag{2.7}$$

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*) Below we shall speak only about the motion of vibrational excitation. However all the results equally refer to the electron motion. It is sufficient to replace the word exciton by the word electron.

$$\left(\frac{\partial^2}{\partial t^2} - V_{aq}^2 \frac{\partial^2}{\partial x^2}\right)\rho(x,t) + \frac{2\varkappa a^2}{M}\frac{\partial^2}{\partial x^2}|\varphi(x,t)|^2 = 0 \qquad (2.8)$$

where $V_{aq} = a\sqrt{\frac{\varkappa}{M}}$ is the velocity of the longitudinal sound in the chain

$$\rho(x,t) = -a\,\frac{\partial\beta(x,t)}{\partial x} \qquad (2.9)$$

is the relative decrease in distances between neighbouring peptide groups.

Owing to translational invariance of the chain we may seek solution of the set of equations (2.7) and (2.8) in the form of excitations spreading with a constant velocity V. Assuming

$$\rho(x,t) = \rho(\xi) \quad , \quad |\varphi(x,t)|^2 = f(\xi)$$

where

$$\xi \equiv x - x_0 - Vt \qquad (2.10)$$

we find the equation

$$\rho(\xi) = \frac{2\varkappa}{\varkappa a(1-S^2)}|\varphi(x,t)|^2 \quad , \quad S \equiv V/V_{aq} \qquad (2.11)$$

from Eq.(2.8) integrating twice over the variable ξ. Substituting (2.11) into the equation (2.8) we obtain the nonlinear Schrodinger equation

$$\left(i\hbar\frac{\partial}{\partial t} - \mathcal{E}_0 + \frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + G(S)|\varphi(x,t)|^2\right)\varphi(x,t) = 0 \qquad (2.12)$$

with the nonlinearity parameter

$$G(S) = \frac{G(0)}{1-S^2} \qquad G(0) \equiv 4\varkappa^2/\varkappa a \,. \qquad (2.13)$$

At $S > 1$ the value $G(S) < 0$, therefore the solution normalized by the condition

$$\frac{1}{a}\int |\varphi(x,t)|^2 dx = 1 \tag{2.14}$$

is absent in an infinite chain. In a chain of a finite length L $(L \gg a)$ the solution of Eq.(2.12) is expressed by a plane wave

$$\varphi(x,t) = L^{-1/2} \exp\{i[kx - \omega(k)t]\} \tag{2.15}$$

and corresponds to the energy

$$\hbar\omega(k) = \mathcal{E}_0 + \frac{a}{L}|G(s)| + \frac{\hbar^2 k^2}{2m}, \quad G(s) < 0, \tag{2.15a}$$

where $k = mV/\hbar$. Thus, at $s > 1$ the collective excitations in the chain are excitons.

With velocities V less than that of longitudinal sound V_{aq} the value $G(s) > 0$. In this case the equation (2.12) has the following solution normalized by the condition (2.14)

$$\varphi(x,t) = \frac{\sqrt{a\mu}\, \exp\{i[k\xi - (\omega - kV)t]\}}{\sqrt{2}\, \cosh(\mu\xi)}, \tag{2.16}$$

$$\mu = \frac{amG(s)}{2\hbar^2} = \frac{2amJ^2}{\mathcal{J}\hbar^2(1-s^2)}, \tag{2.17}$$

$$\hbar\omega = \mathcal{E}_0 + \frac{\hbar^2(k^2 - \mu^2)}{2m}, \quad k = mV/\hbar. \tag{2.18}$$

The excitations described by the function (2.16) are called solitons. The value $\Delta x = \pi/\mu$ characterizes the region of the excitation localization (the soliton size). The continuum approximation used is justified when the inequality $\Delta x \gg a$ is satisfied. With fixed parameters of the chain ($\mathcal{J}$, J, m) this inequality imposes a restriction on the soliton motion velocity.

According to (2.11) the decrease in the distances between peptide groups is determined by the expression

$$\rho(\xi) = \frac{a\mu\chi}{(1-s^2)\,2e\cosh^2(\mu\xi)}. \tag{2.19}$$

The displacement of equilibrium positions of molecules is expressed by the function

$$\beta(\xi) = \frac{\chi}{2e(1-s^2)}\left[1-\tanh(\mu\xi)\right]. \tag{2.19a}$$

The soliton energy travelling with velocity V involves also the energy of local deformation. It is determined by the expression

$$E(V) = \mathcal{E}_0 + \frac{1}{2}mV^2 - \frac{4ma^2\chi^4(1-3s^2)}{3e^2\hbar^2(1-s^2)^2}$$

At $s \ll 1$ this expression is transformed to the form

$$E(V) = \mathcal{E}_0 + \frac{1}{2}m_{sol}V^2 - \frac{4ma^2\chi^4}{3e^2\hbar^2}, \tag{2.20}$$

where

$$m_{sol} = m\left(1 + \frac{8M\chi^4}{3e^3}\right). \tag{2.21}$$

In the chain of the finite length L the equation (2.12) permits also solutions of the exciton type described by plane waves (2.15) with the energy

$$\hbar\omega(k) = \mathcal{E}_0 - \frac{aG(s)}{L} + \frac{\hbar^2k^2}{2m}, \quad k = mV/\hbar. \tag{2.22}$$

At $a \ll L$ the inequality

$$\frac{aG(s)}{L} \ll \frac{4ma^2\chi^4}{3e^2\hbar^2} \tag{2.23}$$

is usually satisfied, therefore, the internal soliton energy (at $V = 0$) is less than the sum of the internal exciton and deformation energy. Thus, the exciton excitations are unstable.

The exciton motion is not practically accompanied by the chain deformation. The exciton transport only the vibrational excitation energy. The soliton motion corresponds to the vibrational excitation motion in conjunction with a local deformation. Therefore, the effective soliton mass (m_{sol}) is greater than the effective exciton mass (m). This difference is especially great in soft chains (small $\mathcal{J}$) and at a strong coupling of a vibrational excitation with deformation (large χ). Travelling with a velocity less than that of a sound the solitons expend no energy to generate phonons.

The excitons are excited by light in protein molecules. The soliton excitation is forbidden by the Frank-Condon principle according to which an optical transfer should not be accompanied by displacement of heavy particles. The solitons may be excited only under local external influences for example, under chemical reactions occuring with molecules attached to the end of protein molecule.

The simple model of a quasi-one-dimensional periodic structure under consideration is referred to each of three chains formed in an α-helical protein molecule by hydrogen bonds between peptide groups. In studying the electron motion each chain acts independently. When vibrational excitations of the dipole type are investigated an additional resonance interaction appears between the peptide groups contained in different chains. Therefore, all three chains should be considered as a whole. This problem was solved in [4].

In the paper of Hymen, Mc Layhlin and Scott [14] the collective excitations in alpha-helical protein molecules of the finite length ($L = 200a$) were computed without making use of the continuum approximation. The conditions of soliton excitation and spreading in such molecules were elucidated. Authors of [14] say that numerical studies of dynamic model for α-helix protein confirm the prediction in [4] of soliton formation.

3. Deceleration of Solitons by external dissipative forces

The possibility of making use of the motion of soliton to explain some phenomena in physics and biology is inevitably connected with studying the physical conditions leading to nonlinearity and dispersion. The properties of solitons are manifested under their interaction with external fields and other degrees of freedom of the system. Mathematical study of these interactions is usually carried out by the perturbation theory methods. Different variants of the perturbation theory for solitons in one-dimensional systems were developed by Karpman and Maslov [10], Bishop and Krumhanse et al [11], Kinner and Mc Laughlin [12] and by other investigators.

In this section we give the results of investigations [8] concerning the influence of external effects on solitons described by the set of equations (2.7), (2.8). The nonlinearity of the collective excitations described by these equations is determined by the interaction of an exciton (electron) with a local chain deformation generated by this exciton. Below we shall consider only the solitons moving with velocity V significantly less then that of sound waves V_{aq}.

If a molecular chain is in a certain medium, then a moving soliton will be acted upon by a friction force and it will be decelerated. The friction forces decelerate the displacements of peptide groups. We consider a case when the friction forces are proportional to the velocity of displacements $\sim \gamma \frac{\partial \beta}{\partial t}$. In this case the equations (2.8) determining the relative displacement $\rho(x,t)$ are transformed to the form (see [8]):

$$\left(\frac{\partial^2}{\partial t^2} - V_{aq}^2 \frac{\partial^2}{\partial x^2}\right) \rho(x,t) + \frac{2\chi a V_{aq}^2}{\varepsilon} \frac{\partial^2}{\partial x^2} |\varphi(x,t)|^2 = -\varepsilon\gamma \frac{\partial \rho(x,t)}{\partial t} . \qquad (3.1)$$

For convenience, a small positive parameter $\varepsilon \ll 1$ is introduced into the deceleration coefficient $\varepsilon\gamma$. We introduce the transformation

$$\varphi(x,t) = \Phi(x,t) \exp\left(-\varepsilon_0 t/\hbar\right)$$

then according to (2.7) the function $\Phi(x,t)$ will satisfy the equation

$$\left[i\hbar \frac{\partial}{\partial t} + \frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + 2\varkappa \rho(x,t) \right] \Phi(x,t) = 0 . \qquad (3.2)$$

In order to solve the set of equations (3.1), (3.2) we use the method of slowly varying coefficients introduced by Van der-Pole and developed in detail by Bogoliubov and Mitropolskii [13] .

Assume

$$\Phi(x,t) = \Phi_0(x,t) + \varepsilon \Phi_1(x,t) , \qquad (3.3)$$

$$\rho(x,t) = \rho_0(x,t) + \varepsilon \rho_1(x,t) , \qquad (3.4)$$

where

$$\Phi_0(x,t) = \frac{\sqrt{a\mu}\, \exp\{ i [k(x-\xi) + \Theta] \}}{\sqrt{2}\, \cosh [\mu(x-\xi)]} \qquad (3.5)$$

$$\xi = x_0 + \frac{\hbar k}{m} t , \qquad \mu = \frac{2am\varkappa^2}{\partial e \hbar^2} , \qquad (3.5a)$$

$$\rho_0(x,t) = \frac{2\varkappa}{\partial e} |\Phi(x,t)|^2 . \qquad (3.6)$$

The function (3.5) describes the soliton travelling with velocity $V = \hbar k/m$. The value

$$x = \xi \equiv x_0 + \frac{\hbar k}{m} t \qquad (3.7)$$

determines the position of the soliton centre at the time moment t :

$$\Theta = \hbar (k^2 + \mu^2) t / 2m \qquad (3.8)$$

is the phase of the wave function (3.5).

The presence of friction results in gradual changing the parameters k and ω of the function (3.5) and in a small changing the form of the function (3.5) which is taken into account by the term $\varepsilon\Phi_1$ in (3.3). We express these changes by the unknown functions $f(k)$ and $\varphi(k)$ using the equations

$$\frac{dk}{dt} = \varepsilon f(k), \tag{3.9}$$

$$\frac{d\omega}{dt} = \frac{\hbar^2}{2m}(k^2+\mu^2) + \varepsilon\varphi(k), \tag{3.9a}$$

$$\frac{d^2\xi}{dt^2} = \varepsilon\hbar f(k)/m\,. \tag{3.9b}$$

Substituting (3.4) into Eq.(3.1) and taking into account (3.5) and (3.6) and a smooth dependence $\rho_1(x,t)$ on time we obtain

$$\rho_1(x,t) = -\frac{a\chi\hbar}{2emV_{aq}^2}\{\gamma k^2 + f(k)\}\tanh[\mu(x-\xi)]\,. \tag{3.10}$$

Using (3.3) and (3.4) and the equalities

$$\frac{\partial\Phi_0}{\partial k} = i(x-\xi)\Phi_0\,, \qquad \frac{\partial\Phi_0}{\partial\omega} = i\Phi_0$$

we may transform the equation (3.2) to the form

$$\left[i\hbar\frac{\partial}{\partial t} + \frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + \frac{8\chi^2}{2e}|\Phi_0|^2\right]\Phi_1 + \frac{4\chi^2}{2e}\Phi_0^2\Phi_1^* = S(\Phi_0)\,, \tag{3.11}$$

where

$$S(\Phi_0) \equiv \left[(x-\xi)f(k) + \varphi(k) - 2\chi\rho_1(x,t)\right]\Phi_0\,. \tag{3.12}$$

The unknown functions $f(k)$ and $\varphi(k)$ introduced into (3.9) are determined provided that the function Φ_1 in (3.11) has no secular terms increasing quadratically with time. One may show [8] that

such secular terms contain multipliers $\frac{\partial \Phi_0}{\partial \xi}$ and $\frac{\partial \Phi_0}{\partial \omega}$. Therefore, the condition that Φ_1 involves no secular terms is reduced to the requirement of orthogonality of the functions $\frac{\partial \Phi_0}{\partial \xi}$ and $\frac{\partial \Phi_0}{\partial \omega}$ in the right-hand side of (3.11) i.e. to the fulfillment of the equalities

$$\int S^*(\Phi_0) \frac{\partial \Phi_0}{\partial \xi} dx = \int S^*(\Phi_0) \frac{\partial \Phi_0}{\partial \omega} dx = 0. \tag{3.13}$$

When these equalities are satisfied only virtual exciton states contribute into $\Phi_1(x,t)$. They cause the soliton from distortion and the so called "tail outgrowing" [10].

Since the energy of exciton states exceeds that of a soliton by the value $ma^2\varkappa^4/\varkappa^2\hbar^2$, then at large $\varkappa$ and small $\varkappa$ the value $B_1(x,t)$ will be very small for all times t less than $k_0/\varepsilon f(k_0)$. Therefore, the explicit form of the function $B_1(x,t)$ is not calculated.

The inequalities (3.13) yield the explicit expressions of the unknown functions

$$f(k) = \frac{4m\varkappa}{\hbar} \int \rho_1(x,t) |\Phi_0(x,t)|^2 \tanh[m(x-\xi)] dx,$$
$$\varphi(k) = \frac{2\varkappa}{a\hbar} \int \rho_1(x,t) |\Phi_0(x,t)|^2 dx. \tag{3.14}$$

Substituting ρ_1 and Φ_0 into (3.14) we obtain

$$\varphi(k) = 0, \qquad f(k) = -\frac{\gamma D k}{1+D} \tag{3.15}$$

where

$$D = \frac{8a^2\varkappa^4}{3\varkappa^2\hbar^2 V_{aq}^2} = \frac{m_{sol} - m}{m} > 0. \tag{3.16}$$

Substituting (3.15) into (3.9) and integrating over time we obtain the exponential law of decreasing $k(t)$ and hence of decreasing

soliton velocity

$$V(t) = V_0 \exp(-\alpha t) \tag{3.17}$$

where the damping coefficient due to overcoming the friction forces

$$\alpha = \varepsilon\gamma \frac{m_{sol} - m}{m_{sol}}. \tag{3.17a}$$

If the friction force is proportional to the square of velocity and has an opposite direction, i.e. is proportional to $\varepsilon\lambda\left(\frac{\partial\beta}{\partial t}\right)^2$, then when the soliton moves along the positive direction of the axis x the equation (2.8) is replaced by the equation

$$\left(\frac{\partial^2}{\partial t^2} - V_{aq}^2 \frac{\partial^2}{\partial x^2}\right)\rho(x,t) + \frac{2\chi V_{aq}^2}{ae}\frac{\partial^2}{\partial x^2}|\varphi(x,t)|^2 = -\varepsilon\lambda\frac{V}{a}\frac{\partial}{\partial t}[\rho(x,t)]^2.$$

Assuming

$$\Phi(x,t) = \Phi_0(x,t) + \varepsilon\Phi_1^1(x,t),$$

$$\rho(x,t) = \rho_0(x,t) + \varepsilon\rho_1(x,t),$$

where $\Phi_0(x,t)$ and $\rho_0(x,t)$ are determined by the expressions (3.5) and (3.6), respectively, we obtain

$$\rho_1^1(x,t) = -\frac{a\chi\hbar f(k)}{m a e V_{aq}^2}\tanh[\mu(x-\xi)] - \frac{\lambda V^2 a \chi^2 \mu}{a e^2 V_{aq}^2}\left\{\tanh[\mu(x-\xi)] - \frac{1}{3}\tanh^2[\mu(x-\xi)]\right\}. \tag{3.18}$$

The unkown functions of Eqs (3.9) are now determined by the expressions

$$\varphi(k) = 0, \quad f(k) = \frac{4\mu\chi}{\hbar}\int\rho_1^1(x,t)\tanh[\mu(x-\xi)]\,|\Phi_0|^2\,dx. \tag{3.19}$$

Substituting (3.18) into (3.19) we obtain the final expression

$$f(k) = -\frac{\lambda A}{1+D} V^2 \tag{3.20}$$

where D is determined by the formula (3.16) and

$$A = \frac{16 a \mu^2 \chi^3}{15 e^2 \hbar V_{aq}^2} .$$

Using (3.20) we transform the equation (3.9b) to the form

$$\frac{dV}{dt} = -\delta V^2 , \qquad \delta = \frac{\varepsilon \lambda 16 a \mu^2 \chi^3}{15 m_{sol} e^2 V_{aq}^2} . \tag{3.21}$$

Integrating (3.21) we obtain the final expression of the law of the soliton velocity change in the presence of friction proportional to the square of velocity

$$V(t) = \frac{V_0}{1+V_0 \delta t} \approx V_0 (1 - V_0 \delta t) . \tag{3.22}$$

Thus, with friction proportional to the square of velocity, the soliton velocity decreases with time by the law close to that of linear with the proportionality coefficient δV_0^2 .

4. Motion of a soliton (electron) in the presence of external nonhomogeneous fields

The presence of external and small local nonmonogeneities in the arrangement of molecules may be taken into account by introducing in (3.2) the small potential energy $\varepsilon U(x)$ depending on the coordinate x . In this case the set of equations determining the soliton motion with velocity $V \ll V_{aq}$ takes the form

$$\left[i\hbar \frac{\partial}{\partial t} + \frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + 2\chi \rho(x,t) \right] \Phi(x) = \varepsilon U(x) \Phi(x) , \tag{4.1}$$

$$\left[\frac{\partial^2}{\partial t^2} - V_{aq}^2 \frac{\partial^2}{\partial x^2} \right] \rho(x,t) + \frac{2\chi}{e} V_{aq}^2 \frac{\partial^2}{\partial x^2} |\Phi(x,t)|^2 = 0 . \tag{4.2}$$

We may seek the solution of this set of equation in the first approximation in the form

$$\Phi(x,t) = \Phi_0(x,t) + \varepsilon \Phi_1(x,t), \tag{4.3}$$

$$\rho(x,t) = \rho_0(x,t) + \varepsilon \rho_1(x,t), \tag{4.4}$$

where

$$\rho_0(x,t) = \frac{2\alpha}{\partial e} |\Phi_0(x,t)|^2. \tag{4.5}$$

The function Φ_0 has the form (3.5) with parameters k, ξ, ω which vary with time according to Eqs (3.9).

Substituting (4.4) into Eqs (4.1) and (4.2) we obtain

$$\left[i\hbar\frac{\partial}{\partial t} + \frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + \frac{4\alpha^2}{\partial e}|\Phi(x,t)|^2\right]\Phi(x,t) = \varepsilon\left[U(x) + \rho_1(x,t)\right]\Phi(x,t), \tag{4.6}$$

$$\left(\frac{\partial^2}{\partial t^2} - V_{aq}^2\frac{\partial^2}{\partial x^2}\right)\rho_1(x,t) = -\frac{1}{\varepsilon}\frac{\partial \rho_0(x,t)}{\partial \xi}\frac{d^2\xi}{dt^2}. \tag{4.7}$$

Using the value (3.9b) and a weak dependence $\rho_1(x,t)$ on time we obtain (at $S \ll 1$) the equation

$$\frac{\partial^2 \rho_1}{\partial x^2} = \frac{2a\hbar\alpha\mu^2}{m\partial e V_{aq}^2} f(k)\frac{\sinh[\mu(x-\xi)]}{\cosh^3[\mu(x-\xi)]}.$$

After integration over x we have

$$\rho_1(x,t) = -\frac{a\alpha\hbar}{m\partial e V_{aq}^2} f(k)\tanh[\mu(x-\xi)]. \tag{4.8}$$

To obtain the equation determining the correction $\Phi_1(x,t)$ to the function $\Phi_0(x,t)$, we substitute (4.3) into (4.6). Then we obtain

$$\left(i\hbar\frac{\partial}{\partial t}+\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}+\frac{8\chi^2}{\varepsilon}|\Phi_0(x,t)|^2\right)\Phi_1(x,t)+$$
$$+\frac{4\chi^2}{\varepsilon}\Phi_0^2(x,t)\,\Phi_1^*(x,t)=S(\Phi_0), \tag{4.9}$$

where
$$S(\Phi_0)\equiv[U(x)-2\chi\rho_1]\Phi_0-$$
$$-i\hbar\left[f(k)\frac{\partial\Phi_0}{\partial k}+\varphi(k)\frac{\partial\Phi_0}{\partial\xi}\right]. \tag{4.10}$$

The condition that Φ_1 involves no secular terms is now reduced to the equality

$$\int S^*(\Phi_0)\frac{\partial\Phi_0}{\partial\xi}\,dx=0,$$

from which taking into account (4.8) we obtain two equations

$$\varphi(k)=-\frac{1}{a\hbar}\int U(x)\,|\Phi_0|^2\,dx, \tag{4.11}$$

$$f(k)=-\frac{2\mu}{(1+D)\hbar}\int|\Phi_0(x,t)|^2\,U(x)\tanh[\mu(x-\xi)]\,dx, \tag{4.12}$$

where D is determined by the expression (3.16).

Substituting the value $f(k)$ obtained into Eq.(3.9b) and taking account the equality $m_{sol}=m(1+D)$ we find the equation

$$m_{sol}\frac{d^2\xi}{dt^2}=F \tag{4.13}$$

determining the motion of the soliton centre.

The equation (4.13) resembles the Newton equation for a particle with a soliton mass moving under the influence of the force

$$F \equiv -2\mu\varepsilon \int U(x) |\Phi_0(x,t)|^2 \tanh[\mu(x-\xi)]\, dx. \quad (4.14)$$

Hence, the soliton motion is effected upon by the potential field $U(x)$ over the whole region where $|\Phi_0|^2 \neq 0$.

It is essential that the equation (4.13) involves an effective soliton mass rather than an exciton (electron) mass which enters directly in the nonlinear equation (3.2). This effect is due to the fact that in the molecular chains under consideration the nonlinearity is caused by the interaction of an exciton (electron) with the chain deformation (a phonon subsystem). The perturbation acts not only on an exciton (electrons) but also on a phonon subsystem which leads to the appearance of an additional term $\varepsilon \rho_1(x,t)$ in the right-hand side of Eq. (4.6).

In a number of papers when studying the role of the perturbation $\varepsilon U(x)$ a nonlinear equation of the form

$$\left(i\hbar\frac{\partial}{\partial t} + \frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + G|\Phi(x,t)|^2\right)\Phi(x,t) = \varepsilon U(x)\Phi(x) \quad (4.15)$$

was considered without studying the nature of nonlinearity. In this case we obtain for the function $f(k)$ the value

$$f(k) = -\frac{2m}{\hbar}\int U(x)|\Phi_0(x,t)|^2 \tanh[\mu(x-\xi)]\, dx$$

which leads to the equation

$$m\frac{d^2\xi}{dt^2} = F$$

for a quasi-particle with a mass m oppearing in the equation (4.15).

Thus, in studying the dynamical behaviour of solitons described by a Schrodinger equation it is very important to know physical factors causing the equation nonlinearity.

5. The motion of a soliton in a one-dimensional molecular chain with allowance for thermal vibrations

In the above sections the assumption was made that the chain was

at a absolute zero temperature. It is interesting to investigate the influence of thermal vibrations of molecules on the soliton properties. This problem was considered in Ref. [9] . Below the basic results obtained in that paper are given.

To analize the role of thermal motion it is necessary to express the displacements u_n in the peptide groups through the operators of creation b_q^+ and annihilation b_q of the phonons with wave numbers q

$$u_n = \sum_q \left(\frac{\hbar}{2MN\Omega_q}\right)^{1/2} (b_q + b_{-q}^+) \exp(iqna) \tag{5.1}$$

where $N(\gg 1)$ is the number of molecules in the chain

$$\Omega_q = |q| V_{aq} , \qquad V_{aq} = a\sqrt{2e/M} . \tag{5.2}$$

With temperature distinct from zero we should pass to the statistical description of quantum states of the system. The Hamiltonian function averaged by means of the density matrix of phonon states

$$\rho_{\nu\nu} = \frac{\langle\nu|\exp(-H_{ph}/\Theta)|\nu\rangle}{\sum_\nu \langle\nu|\exp(-H_{ph}/\Theta)|\nu\rangle} , \qquad \Theta = kT, \tag{5.3}$$

takes the form

$$\mathcal{H} = \sum_n \Big\{ (\varepsilon_0 - \frac{\hbar^2}{ma^2}) \varphi_n^* \varphi_n - \frac{\hbar^2 e^{-\bar{W}_n}}{2ma^2} \varphi_n^* (\varphi_{n-1} + \varphi_{n+1}) -$$

$$- \frac{1}{\sqrt{N}} \sum_q F(q) |\varphi_n|^2 (\beta_{qn} + \beta_{-q,n}) + \sum_q \hbar\Omega_q [\bar{\nu}_q + |\beta_{qn}|^2] \Big\} , \tag{5.4}$$

where

$$\bar{W}_n \approx \frac{a^2}{2} \sum_q (2\bar{\nu}_q + 1) q^2 |\beta_{qn}|^2 ,$$

$$F(q) \approx i 2a\varkappa \frac{q}{|q|} \left(\frac{\hbar|q|}{2MV_{aq}}\right)^{1/2} , \tag{5.5}$$

$$\bar{\nu}_q = \left[\exp\left(\frac{\hbar\Omega_q}{\Theta}\right) - 1\right]^{-1}.$$

$\bar{\nu}_q$ is the average number of phonons in the state q ; $\beta_{qn}(t)$ determine the average values of displacements so that

$$\langle u_n(t)\rangle = -\sum_q \left(\frac{\hbar}{2MN\Omega_q}\right)^{1/2} \left[\beta_{qn}(t) + \beta^*_{-q,n}(t)\right].$$

The function (5.4) depends on φ_n , β_{qn} on canonically conjugate generalized momenta $i\hbar\varphi^*_n$ and $i\hbar\beta^*_{qn}$. Using this function we find the Hamiltonian equations which in the continuum approximation (for excitations moving with a velocity $V < V_{aq}$) are reduced to the following approximate equations

$$\left[i\hbar\frac{\partial}{\partial t} - \mathcal{E}_0 + \frac{\hbar^2}{2m(\Theta)}\frac{\partial^2}{\partial x^2} + G(\Theta)\,|\varphi(x,t)|^2\right]\varphi(x,t) = 0, \tag{5.6}$$

$$\rho(x,t) = -\frac{2a\chi\,|\varphi(x,t)|^2}{M(1-s^2)V_{aq}^2}, \qquad s = V/V_{aq}. \tag{5.7}$$

The Schrodinger equation (5.6) involves the parameters depending on temperature Θ .

The nonlinearity parameter is determined by the expression

$$G(\Theta) = \left[1 - \tfrac{1}{4}B\,F(\Theta)\right]G \tag{5.8}$$

where

$$G = 4a^2\chi^2 / M(1-s^2)V_{aq}^2, \tag{5.9}$$

$$B = \pi a^4\chi^2 / M(1-s^2)\hbar V_{aq}^3. \tag{5.10}$$

$$F(\Theta)=\frac{a^2}{\pi^2}\int_0^{\pi/a}\left(1+2\frac{\bar{J}}{q}\right)q\,dq=\begin{cases}\frac{1}{2}+\frac{2\Theta}{\hbar\Omega_0} & \text{if } \frac{\hbar\Omega_0}{\Theta}\ll 1;\ \Omega_0=\pi V_{aq}/a;\\ \frac{1}{2}+\frac{\pi^2\Theta^2}{6\hbar\Omega_0^2} & \text{if } \frac{\hbar\Omega_0}{\Theta}\gg 1.\end{cases} \tag{5.11}$$

The effective exciton (electron) mass increases with increasing temperature by the exponential law

$$m(\Theta)=m\exp\left(\frac{1}{4}\mu_0^2 a^2 B F(\Theta)\right) \tag{5.12}$$

where

$$\mu_0=4ma^3\chi^2/(2MV_{aq}^2\hbar^2(1-s^2)) \tag{5.13}$$

The increase in the effective exciton mass is due to the decrease in the resonance interaction between peptide groups caused by the phase coherence distortion of vibrational excitations in the neighbouring groups.

The nonlinearity parameter (5.8) decreases with increasing temperature. The nonlinearity parameter decrease is associated with the effective decrease in the connection of an exciton with the chain deformation. With sufficiently high temperature the parameter $G(\Theta)$ may become negative. Then the equation (5.6) will have no localized (soliton) solutions. Unfortunately,this conclusion is only qualitatively meaningful since the theory is valid only when the inequality

$$BF(\Theta)<1 \tag{5.14}$$

is satisfied.

With the inequality $S<1$ satisfied, the increase in the velocity of the excitation propagation according to (5.12), (5.10) and (5.13) leads to the increase in the values B and μ_0 causing the exciton mass increase since it acts as increasing temperature. The change of the nonlinearity parameter (5.8) is more complex. At $V\to V_{aq}$, according to (5.9), the value $G(\Theta)$ increases whill the multiplier $\left[1-\frac{1}{4}BF(\Theta)\right]$ decreases.

Solving the equation (5.6) we find that the probability of excita-

tion distribution along the chain is determined by the function

$$|\varphi(x,t)|^2 = \frac{a\mu(\Theta)}{2\cosh^2\{\mu(\Theta)[x-x_0-Vt]\}} \tag{5.15}$$

where

$$\mu(\Theta) = a\, m(\Theta) G(\Theta)/2\hbar^2. \tag{5.16}$$

In accordance with (5.7) the decrease in equilibrium distances between neighbouring molecules is determined by the function

$$\rho(x,t) = -\frac{2a^3\varkappa\mu(\Theta)}{MV_{aq}^2\cosh^2\{\mu(\Theta)[x-x_0-Vt]\}} \tag{5.17}$$

The energy of a soliton travveling with velocity $V \ll V_{aq}$ is proportional to the square of velocity

$$E_{sol}(V) = E_{sol}(0) + \frac{1}{2} m_{sol} V^2, \tag{5.18}$$

with the soliton mass

$$m_{sol} = m(\Theta)\left[1 + \frac{a^2 G^2(\Theta)}{6\hbar^2 V_{aq}^2}\right] \tag{5.19}$$

and the energy at rest

$$E_{sol}(0) = \mathcal{E}_0 - m(\Theta) a^2 G^2(\Theta)/12\hbar^2. \tag{5.20}$$

The soliton size is determined by the value

$$l \cong \frac{\pi}{\mu(\Theta)} = 2\pi\hbar^2/a\, m(\Theta) G(\Theta). \tag{5.21}$$

We substitute the values (5.8) and (5.12) into this expression. Then when the continuum approximation $a\mu_0 \ll 1$ and inequality (5.14) are satisfied

$$l \approx 2\pi\hbar^2/(ma[1-\tfrac{1}{4}BF(\Theta)]G). \tag{5.22}$$

Thus, with increasing temperature the soliton size increases and its properties more and more approximate the exciton properties. The fact whether a soliton transforms into an axciton continuously or discontinuously can be clarified only on the basis of a new theory valid in the region $V \approx V_{aq}$.

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The Theory of Contraction of Proteins under their Excitation

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This paper develops the quantum theory of contraction of α-helical proteins under the excitation of their peptide groups, which form three parallel chains of hydrogen bonds along the molecule. In the region embraced by an excitation, the pitch of the helix decreases, and the "contracted" region of the helix moves along the molecule at a rate proportional to the energy of resonance interaction between peptide groups. The movement of the excitation region from the head of the myosin molecule to its tail results in movement of the myosin filaments along the actin ones in the myofibril of the muscle, and thus leads to a decrease in length of the muscle fibre.

1. Introduction

As yet, there is no complete understanding of the mechanism for muscle contraction in vertebrates or invertebrates. Apparently it is an established fact that the energy required for the contraction of muscle fibres is released when adenosine triphosphate (ATP) transforms into adenosine diphosphate (ADP) and phosphate. Such a transformation occurs in the presence of Ca^{2+} ions, which trigger an enzyme action of the protein molecules of the muscles. The mechanism converting this chemical energy into a mechanical motion is still not completely understood. In the present paper we propose a new mechanism responsible for the contraction of protein molecules when they are excited by the energy released in the hydrolysis of ATP.

2. Current Concepts of the Mechanism for Muscle Contraction

As is well known, long polymer protein molecules (polypeptides) are the chief contractile elements of muscles and other biological objects that con-

vert chemical energy into mechanical motion. The striated muscles of vertebrate animals and insects have received a very detailed study (see, for example, the review of Volkenstein (1970)). These muscles are bundles of fibres each of which is made up of several thousands of densely packed filaments called myofibrils. The myofibril is the functional unit of the muscle contractile system.

Each myofibril is divided into separate strictly repeating segments, or sarcomeres, built up of an interpenetrating system of thin (actin) and thick (myosin) protein filaments. The thick filaments are situated in the centre part of the sarcomeres and consist of several hundred myosin molecules. These molecules are built up of two identical polypeptide chains coiled up in the form of a helix. In turn, each of these chains has the form of an α-helix in the long (tail) portion of the molecule. In the head portion the molecule has two projections ("heads"). Separate myosin molecules in thick filaments are oriented in opposite directions from the centre.

The thin filaments of the sarcomere start from the Z-plates (which separate the sarcomeres from each other) and run to the centre of the sarcomere, partially penetrating into the spaces between thick filaments. They are made up of double helices of globular G-actin molecules. Each filament contains about 400 molecules.

Many hypotheses have been proposed to account for the machinery of contraction. Some of these hypotheses were based on the properties of polyelectrolytes which lead to the folding and unfolding of protein chains when the charges of their radicals are changed. These hypotheses differed from each other in the assumptions concerning the character of the distribution and the change in electric charges along the protein molecules (Meyer, 1929; Kuhn, Toth & Ebner, 1962; Wohlisch, 1940; Botts, Morales & Cellular, 1951; Risemann & Kirkwood, 1948). No direct evidence is available for the rearrangement of molecules during the contraction of muscle fibres.

Hypotheses have often been advanced (Astbury, 1950; Huxley, A., 1957; Ooswa *et al.*, 1961) to account for the contraction and relaxation of muscles by postulating the structural transformation of myosin and actin molecules.

A number of authors (Pryor, 1950; Kuhn, 1960; Flory, 1956; Hill, 1968) have associated muscle contraction with phase transitions of the protein molecules. But the bulk of X-ray evidence on living contracting muscle does not support the hypothesis that there is any large scale rearrangement in the filaments of muscle (see review paper by J. Hanson (1968)).

Using electron-microscope data, Huxley H. & Hanson (1954) and Huxley & Nidergerke (1954) established that the contraction of myofibrils was connected with the fact that thin actin filaments move along the channels between the thick myosin filaments without overall changes in length. These

data served as a basis in constructing the presently widely accepted "sliding model" for the mechanism of muscle contraction and relaxation.

A phenomenological theory of the sliding model has been developed by Davies (1963). Davies' theory runs into a number of difficulties. It overestimates the role of electrostatic repulsion, regarding it as causing the myosin molecule's head to change from an α-helical to a stretched, disordered configuration.

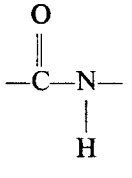
The distance between the G-actin monomers in the actin filament is about 55 Å. Consequently, every miniature contraction cycle should, according to Davies' theory, move the actin filament ~55 Å farther, so that the molecule's head should shorten and stretch by about 70 Å during each cycle.

If the negative electric charge at the base of the myosin molecule's head so strongly repels the ATP^- complex, attached to the head, that the latter stretches, then the same charge will not allow the sarcoplasmic ATP to approach the ADP (which formed at the head after the ATP broke down) to replace it or charge it to ATP.

Davies' theory does not make it clear why the myosin molecule has, besides its head, a very long (~1300 Å) α-helical tail that takes no active part in the contraction. It is unlikely that nature is so wasteful.

3. Collective Excitations of Peptide Groups in α-Helical Protein Molecules

The contractile proteins, like all other proteins, are synthesized by a linear polymerization of various amino acids. Polymerization results in a periodically repeating sequence of peptide groups



(1)

The formation of hydrogen bonds between the oxygen and hydrogen atoms of the peptide groups of the first and fourth, second and fifth, etc., amino acid residues stimulates the polypeptide chains of amino acid residues to fold up into α-helices. Thus there forms an α-helix 10·1 Å in diameter, containing three chains of hydrogen bonds whose directions almost coincide with that of the helix axis.

As the peptide groups are regularly positioned, collective (exciton) excitations forming broad bands correspond to electric dipole excitations. Moffitt (1956), using the theory of molecular crystal spectra developed by

Davydov (1951, 1971), developed the theory of collective excitations of a type which correspond in energy range (5–7 eV) to the electronic excitations of peptide groups. In calculations it was assumed that the peptide groups were fixed in the positions which they occupy in a non-excited molecule.

As this author has shown (1969), if the deformability of a crystal lattice is taken into account one is led to the fact that an exciton excitation in a one-dimensional crystal is accompanied by a local crystal deformation. We shall consider this effect for the case where α-helical molecules are subject to excitations of less than the 0·48 eV of energy released during ATP hydrolysis. This energy range corresponds to the energy of 0·23 eV of the valence vibration C=O in the peptide group. The electric dipole moment of this vibration is directed along the hydrogen bond. The dipole–dipole interaction between adjacent peptide groups placed along the chain of hydrogen bonds is an order of magnitude higher than the similar interaction between the nearest groups of adjacent chains, so that the excitations of the peptide groups of the three hydrogen-bond chains may be treated independently. For the sake of simplicity we shall consider only one of them in what follows.

Let $\mathbf{r}_n$ be a radius-vector determining the position of the nth peptide group (see Fig. 1). The number n takes the values $1, 2, \ldots, N$ ($N \sim 200$–300). Then

$$R_n = |\mathbf{r}_{n+1} - \mathbf{r}_n| \tag{2}$$

is the distance between two adjacent peptide groups. Suppose that the potential interaction energy of these groups when in the ground state is

$$U_0 = \sum_n u_0(R_n), \tag{3}$$

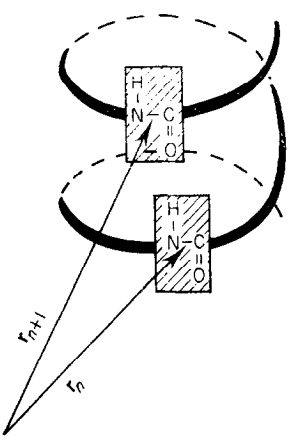


FIG. 1.

where

$$u_0(R) = C/R^s - A/R^3, \tag{4}$$

The quantities C, A and s are the parameters of the theory. They depend on the properties of peptide groups themselves and the environment in which they happen to be (cytoplasm, radicals, etc.). To the minimum of potential energy, equation (3), there correspond the equilibrium distances

$$R_0 = (sC/3A)^{\sigma}, \qquad \sigma = (s-3)^{-1}. \tag{5}$$

The value of R_0 is of the order of 5·4 Å.

Suppose that the peptide group in a protein molecule can go into an excited state with excitation energy ε, and that an electric dipole transition moment directed along the molecule corresponds to this excitation.

Let us introduce the operators of creation B_n^+ and annihilation B_n of an excitation on group n. Then for fixed values of R_n the operator for the excitation energy of the whole chain will have the form

$$H = \sum_n \{[\varepsilon + D(R_n)]B_n^+ B_n + M(R_n)[B_n B_{n+1}^+ + B_n^+ B_{n+1}]\}, \tag{6}$$

where

$$D(R) = -\alpha/R^3 \tag{7}$$

is the change in static interaction energy of two adjacent groups when one of them goes into an excited state:

$$M(R) = -\gamma/R^3 \tag{8}$$

is the energy of resonance dipole–dipole interactions between adjacent groups. $\gamma = 2\mathbf{d}$; $\mathbf{d}$ is the electric moment of the dipole excitation.

To simplify our calculation we take periodic boundary conditions. The $R_N = |\mathbf{r}_1 - \mathbf{r}_N|$ and the sums over n in all formulae are extended over the values from 1 to N.

If the ground state of the chain is $|0>$, then at fixed R_n the excited states are characterized by the functions

$$\Psi = \sum_n a_n B_n^+ |0\rangle \tag{9}$$

which are defined by the sets of complex numbers a_n satisfying the condition

$$\sum_n a_n^* a_n = 1. \tag{10}$$

To every set of numbers a_n and R_n there corresponds the excitation energy

$$E(\{a_n\}, \{R_n\}) = \sum_n \{[\varepsilon + D(R_n)]a_n^* a_n + M(R_n)[a_n a_{n+1}^* + a_n^* a_{n+1}]\}. \tag{11}$$

For a fixed set of numbers $\{a_n\}$, the expression (11) is dependent on the distance R_n and is the potential energy, additional to that in equation (3), for peptide groups.

Hence the total potential energy takes the form

$$U(\{a_n\}, \{R_n\}) = \varepsilon + \sum_n \{u_0(R_n) + D(R_n)a_n^* a_n + M(R_n)[a_n a_{n+1}^* + a_n^* a_{n+1}]\}. \quad (12)$$

Our task is to find sets a_n and R_n corresponding to an absolute minimum of the energy given in equation (12), at a fixed number of simultaneous elementary excitations—that is, when the condition (10) is satisfied.

We obtain the minimization of the energy, (formula (12)), in two stages. We first find the values $R_n = \rho_n$ that correspond to the minimum of formula for a fixed set of values of the a_n. These values of R_n determine new equilibrium distances between the peptide groups in an excited chain:

$$\rho_n = R_0\{1 + \kappa a_n^* a_n + \eta[a_n a_{n+1}^* + a_n^* a_{n+1}]\}^{-\sigma}, \quad (13)$$

where

$$\kappa = \frac{\alpha}{A}, \qquad \eta = \frac{\gamma}{A} > 0 \qquad (\eta < \kappa < 1). \quad (14)$$

To find the coefficients a_n which minimize formula (12), we insert the values (13) and keep the terms of order not higher than κ, η, κ^2. As a result we obtain the functional

$$U(\{a_n\}) = Nw_0 + \varepsilon - (3\sigma+1)\kappa v w_0 - \Lambda \sum_n (a_n a_{n+1}^* + a_n^* a_{n+1}) - G \sum_n (a_n^* a_n)^2, \quad (15)$$

where

$$w_0 = -u_0(R_0) = \frac{A}{\sigma S R_0^3}, \qquad G = \tfrac{3}{2}(\sigma+1)\sigma\kappa^2 w_0, \qquad \Lambda = \frac{(3\sigma+1)\gamma}{\sigma S R_0^3} \quad (16)$$

are parameters independent of n. With our choice of boundary conditions the potential energy (equation (12)) is therefore invariant under the substitution of $n+m$ for n, where m is an integer. Consequently, the energy states of the chain may be classified by means of the wave numbers

$$k = \frac{2\pi}{N\rho_0} j, \quad (j = 0, \pm 1, \pm \ldots N/2), \quad (17)$$

where ρ_0 is the average distance between peptide groups in the region of excitation. Considering only small values of k (long wave approximation), one can go over in equation (12) to a continuous model of the chain.

4. Collective Excitations in Long Wave Approximation

To go over to a continuous model of the chain in the case of long wave excitation we perform the substitution

$$1 = \sum_n a_n^* a_n \Rightarrow \int a^*(x)a(x)\,dx, \quad (18)$$

$$\varepsilon - \Lambda \sum_n (a_n a_{n+1}^* + a_n^* a_{n+1}) \Rightarrow (\varepsilon - 2\Lambda) + \int a^*(x) \left(- \frac{\hbar^2}{2m^* \rho_0^2} \frac{\mathrm{d}^2}{\mathrm{d}x^2} \right) a(x)\, \mathrm{d}x, \quad (19)$$

where

$$m^* = \hbar^2/(2\Lambda\rho_0^2) \quad (20)$$

is the effective mass of a free exciton. Then equation (15) reduces to the form

$$U(k, \{a(x)\}) = (\varepsilon - 2\Lambda) - (3\sigma + 1)\kappa w_0 - N w_0 + \\ + \int a^*(x) \left[- \frac{\hbar^2}{2m^* \rho_0^2} \frac{d^2}{\mathrm{d}x^2} \right] a(x)\, \mathrm{d}x - G \int [a^*(x) a(x)]^2\, \mathrm{d}x. \quad (21)$$

With $G = 0$, to the extremum of the functional (21) correspond the values $a(x)$ with the form of plane waves

$$a(x) = \sqrt{\frac{1}{N}} \exp(ikx\rho_0). \quad (22)$$

With $G \neq 0$ we represent $a(x)$ in the form of modulated plane waves

$$a(x) = \varphi(x) \exp(ikx\rho_0), \quad (23)$$

where $\varphi(x)$ is a real function to be defined from the condition that formula (21) be a minimum, subject to the condition that the equality (18) be satisfied.

Substituting equation (23) into equation (21) we find

$$U(k, \varphi(x)) = -N w_0 + (\varepsilon - 2\Lambda) - (3\sigma + 1)\kappa w_0 + \\ + \frac{\hbar^2 k}{2m^* \rho_0^2} - \frac{\hbar^2}{2m^* \rho_0^2} \int \varphi(x) \frac{\mathrm{d}^2 \varphi(x)}{\mathrm{d}x^2}\, \mathrm{d}x - G \int \varphi^4(x)\, \mathrm{d}x. \quad (24)$$

Varying the functional (24) under the additional condition (18), we derive a non-linear equation

$$\frac{\hbar^2}{2m^* \rho_0^2} \frac{\mathrm{d}^2 \varphi(x)}{\mathrm{d}x^2} + 2G\varphi^3(x) - \chi\varphi(x) = 0 \quad (25)$$

to define $\varphi(x)$. Its solution that corresponds to the minimum of equation (24) at fixed $k \approx 0$, has the form

$$\varphi(x) = \sqrt{\mu/2}\, \mathrm{ch}^{-1}(\mu\xi), \quad (26)$$

where

$$\xi = x - x_0, \qquad \mu \equiv \frac{2G}{\Delta E}, \qquad \chi \equiv \frac{G^2}{\Delta E}, \quad (27)$$

and x_0 is the point on the chain where the maximum value of $\varphi(x)$ is concentrated. $\Delta E \approx 4\Lambda$ is the energy bandwidth of free excitons.

In the long wave approximation, we can put, in expression (13),

$$a_n a_{n+1}^* + a_n^* a_{n+1} \approx 2a_n^* a_n.$$

Going to a continuous model and making use of the values (26) we then obtain

$$\rho(\xi) \approx R_0 \left[1 - \frac{\sigma(\kappa+2\eta)G}{\Delta E . \mathrm{ch}^2(\mu\xi)}\right]. \tag{28}$$

Consequently, a maximum shortening of the distance between peptide groups takes place at the point $x = x_0$. The distance decreases by the amount

$$\Delta\rho_{\max} = \sigma(\kappa+2\eta)GR_0/\Delta E. \tag{29}$$

A substantial decrease in the distances between peptide groups should be observed for the length of the chain which contains $\Delta\xi$ peptide groups, where

$$\Delta\xi = \pi\Delta E/2G. \tag{30}$$

In this region the mean distance between peptide groups becomes

$$\rho_0 = R_0[1 - 0{\cdot}6\sigma(\kappa+2\eta)G/\Delta E]. \tag{31}$$

The total length of the chain in the excited state is determined by the expression

$$L \approx \left(1 - \frac{\sigma(\kappa+2\eta)}{N}\right) \approx NR_0. \tag{32}$$

Substituting equation (26) into equation (24) we find the excitation energy (above the ground-state energy—Nw_0) for a fixed value k

$$E(k) = \left[\varepsilon - \frac{\Delta E}{2} - (3\sigma+1)\kappa w_0\right] + \frac{\hbar^2k^2}{2m^*} - \frac{G^2}{3\Delta E}. \tag{33}$$

Thus, the excitation energy of a chain of resonantly interacting peptide groups of an α-helix is much less than the excitation energy ε of groups of isolated amino acid residues contained in the same cytoplasm.

Using equation (33) it is possible to calculate a group velocity of movement of the excitation along the α-helical molecule

$$V_g = \frac{1}{\hbar}\,\mathrm{grad}_k\, E(k) = \frac{\hbar k}{m^*} = \frac{k}{\hbar}\Delta E\rho_0^2. \tag{34}$$

It follows from equation (34) that the velocity of movement of the contracted region along the molecule is proportional to the energy ΔE of the resonance interaction between peptide groups, and proportional to the quasi-momentum of the excitation. Parallel with the solution (26), equation (25) allows other solutions as well. However, at fixed k they do not correspond to the energy minimum of equation (24). For example, the solutions $\varphi = \sqrt{1/N}$ and $\chi = 2G/N$, which reduce equation (23) to plane waves, correspond to the excitation energy

$$E(k) = \left[\varepsilon - \frac{\Delta E}{2} - (3\sigma+1)\kappa W_0\right] + \frac{\hbar^2k^2}{2m^*} - \frac{G}{N}.$$

Excitations of such a type are uniformly distributed along the whole length of the molecule.

We have considered the collective longitudinal excitations of one chain of peptide groups of an α-helical protein molecule. The excitation energy of this chain is less than 0·23 eV provided that it corresponds to the excitation of the valence vibration C=O. Therefore it appears likely that the disintegration energy of one ATP molecule is sufficient to excite one, two, and even all three chains formed by hydrogen bonds in the molecule.

To simplify calculations we have considered periodic boundary conditions, taking no account of excitation relaxation (attenuation) processes. When relaxation processes are taken into account, an excitation that arises at one end of a long molecule will spread to its other end at a decreasing rate, that is, it will give up a portion of its energy as it travels. At the same time the probability grows that the remaining portion of the excitation energy will all turn into heat before reaching the end of a molecule.

5. Excitation of Myosin Molecules and the "Sliding Model"

A quantum theory of the conformational change of a certain region of α-helical protein molecules under the excitation of their peptide groups has been developed in sections 3 and 4.† In the myofibrils of muscles such excitations may be generated by the disintegration energy of ATP molecules when, in the presence of Ca^{2+}, they interact with the active ATPase centres of the heads of the myosin molecules. The resulting collective excitation leads to shortening of the distances between peptide groups in some region of every hydrogen-bonded chain in the myosin molecule. As a result, in this region the helical mysoin molecule thickens and bends, getting into a closer contact with the surrounding actin filaments.

The resulting collective excitation should travel from the head, where the ATPase centre is located, to the tail of the molecule. Since the frictional resistance (to motion) of the displaced twisted section of myosin filament relative to the surrounding actin filaments is considerably elevated, the movement of this "contracted" region from the head of the molecule to its tail will make the whole myosin filament move along actin filaments in the direction from the tail to the head (see Fig. 2). Such a "sliding" of a myosin filament over actin filaments will result in a slight shortening of the length of the I-discs of the sarcomere. The subsequent excitations that arise in the region

† This conformation corresponds in some respects to the possibilities discussed in the short paper by Volkenstein (1972). In that paper he introduced the notion "conformon" for excitation of the system of the long-wave phonons, electrons and local deformations of macromolecules.

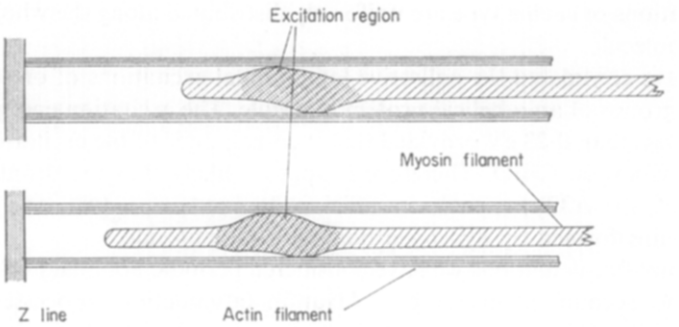


FIG. 2.

of the head (where the ATP is hydrolyzed in the presence of Ca^{2+}) and spread along myosin filaments to the sarcomere centre will lead to further movement of actin filaments with respect to myosin filaments. This movement resembles the movement of a snake ("concertina progression" (Gans, 1970)) in a straight narrow channel.

In the theory proposed here (and agreeing with the experimentally established "sliding model") the whole long helical protein molecule (and not only the head section as in Davies' theory) is the working element of the muscle. The second feature of the theory is that the "contracted" region spreads along the whole molecule, although an excitation initially arises in a separate part where the ATP energy is released.

So the main points of our theory are that: (1) the energy of hydrolysis of ATP is used in muscle to excite the carboxyl group at a peptide bond in the head of a myosin molecule; (2) this excitation is accompanied by a local distortion and contraction of the myosin molecule without gross overall changes in its length (see equation (32)); (3) the excitation with its attendant local contraction propagates down the molecule from head to tail because of resonance interaction between excited and non-excited peptide groups; and (4) this leads to a net movement of the myosin filament relative to the neighbouring actin filaments.

6. Some Numerical Estimates

The electric dipole moment of the vibration C=O in a peptide group with frequency $\sim 1660\,\mathrm{cm}^{-1}$, is equal to $\mathbf{d} \approx 0{\cdot}29$ Debye (Chirgadze & Rashevskaja, 1969). Therefore the radiation lifetime of the excited state with respect to spontaneous emission is

$$\tau = \frac{3\hbar}{4d^2}\left(\frac{c}{\omega}\right)^3 \approx 10^{-1}\ \mathrm{sec}; \qquad \Delta E \approx 2 \times 10^{-15}\ \mathrm{erg}.$$

Excitons with kinetic energy equal $0{\cdot}07\,\Delta E$ have the wave number $k = 0{\cdot}5/R_0$. They have the group velocity $Vg = \hbar km^*/$, $m^* = 2\hbar^2/\Delta E\rho^2$, (Davydov, 1971, Ch. II, sec. 2), which is equal about 30 m/sec. When the deformability of a chain is taken into account, motion of the excitation is accompanied by a local distortion of the chain and this group velocity will decrease, as the effective mass m^* will increase. For quantitative description of this effect one must develop the theory in the next approximation which can take into account the motion of the peptide group. The velocity of the motion of the peptide group is not connected with the velocity of an exciton. The bigger the mass of a peptide group the smaller will be the role of deformation. If the mass of a peptide group is M_0, the increase in the effective mass will be

$$m^* \Rightarrow m^*(1 - m^*B)^{-1},$$

where B is proportional $1/M_0$.

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The Vibrational Structure of Davydov Solitons

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Abstract

Internal oscillations of an alpha-helix soliton are studied numerically on a modified version of Davydov's equations and analytically on an approximating system of coupled nonlinear Schrödinger equations. The numerical results are used to interpret lines in recently measured laser-Raman spectra of metabolically active *E. coli*. Agreement between measured and calculated lines appears to be within the uncertainties of computation and measurement.

1. Introduction

The idea that bond energy could be transported along one-dimensional molecular chains (such as alpha-helix protein molecules) by solitary waves (or "solitons") was first proposed by Davydov and Kislukha in 1973 [1]. Since that time Davydov and his co-workers have published a number of papers that discuss various soliton parameters including lifetime, damping, energy, and wave structure [1]. The physical notion of Davydov's alpha-helix soliton can be understood rather easily with reference to Fig. 1. The energy to be transported is stored as "stretching" quanta of the amide-I bond (i.e., $C=O$) which has an absorption peak at 1653 cm^{-1} corresponding to a quantum unit of 0.205 eV. Naïvely one would expect such energy to disperse rapidly ($\sim 10^{-12}$ s) via dipole–dipole interactions throughout all the amide-I bonds. But if certain nonlinearities of the system are taken into consideration, it is seen that localized bond energy generates longitudinal sound which in turn acts as a potential well to trap the bond energy and prevent its dispersion. Thus the Davydov soliton can be viewed as a local region of conformational change which is mobile.

The most important nonlinearity in the problem is a parameter (χ_1) that gives the change in energy of an amide-I bond (newton-meters) per unit extension of its associated hydrogen bond (meters). In 1979 a theoretical study of an analytically tractable approximation to Davydov's equations (the nonlinear Schrödinger equation) showed that there is a threshold level for χ_1 below which solitons could not be expected to form, and this conclusion was supported by numerical integration of Davydov's equations [3]. Estimates by both Davydov and myself indicated for a real alpha-helix protein that χ_1 was about the right order of magnitude to support soliton formation.

Recently Kuprievich and Kudritskaja [4] have obtained χ_1 from SCF calculations on formamide dimer as 0.34×10^{-10} newtons. Using a form of Davydov's equations that have been extended to improve accuracy, I have recently found exactly the same value of χ_1 as the threshold for soliton formation [5]. Thus there is now substantial numerical evidence that Davydov solitons can form on real alpha-helix protein.

During the course of this study [5] some cursory observations were made of an "interspine oscillation" in bond energy between the three spines (etc., $\cdots$H–N–C=O $\cdots$ H–N–C=O $\cdots$, etc.) that run longitudinally along the helix. (One of these three spines is cross-hatched in Fig. 1). After the manuscript for [5] was completed, I became aware of Webb's recent review of laser-Raman spectra measured on metabolically active cells [6]. Of particular interest was a spectrum (see Fig. 2) taken from *E. coli* and consisting of nine definite lines which could be assigned rather closely to simple sums and differences of two predetermined frequencies: (i) The frequency of interaction of the soliton with unit cells of the alpha-helix, and (ii) The frequency of the above mentioned interspine oscillation [7]. Thus laser-Raman spectroscopy (LRS) may provide direct experimental confirmation that Davydov solitons play a functional role in metabolic processes. If so, LRS could become even more valuable as a probe into the secret details of life processes.

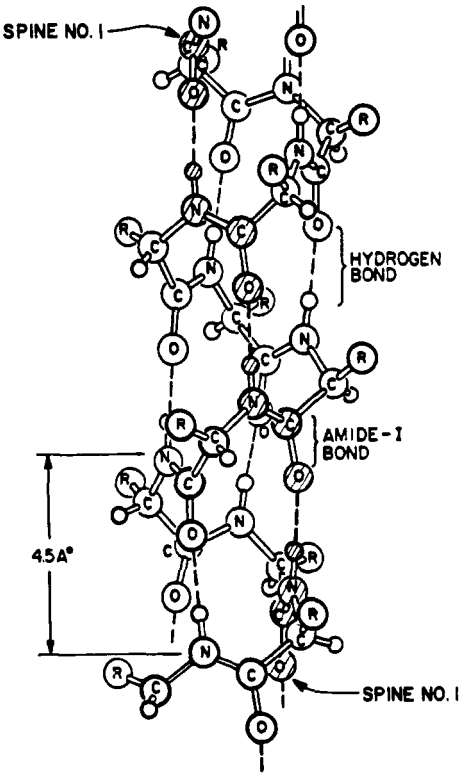

Fig. 1. Alpha-helix (redrawn from [2] with permission of the author). The atoms of a single "spine" are cross-hatched.

The aim of this paper is to provide the basis for a critical evaluation of the above hypothesis (that Fig. 2 displays the vibrational structure of an alpha-helix soliton). The following section lists the numerical and analytical models to be studied beginning with that developed in [5] and proceeding toward simpler and more tractable approximations. Section 3 discusses the linear problem of interspine interaction below the level for soliton formation. This is important not only because the linear

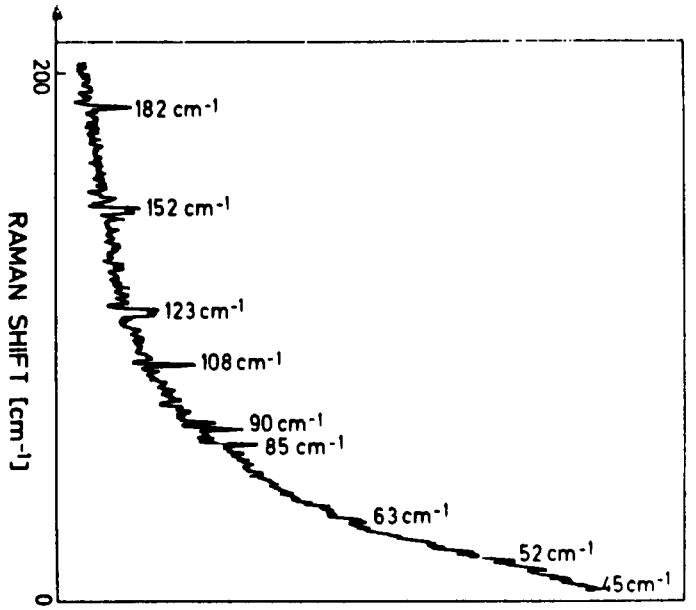

Fig. 2. Laser-Raman spectrum from metabolically active *E. coli* (from [6] with permission of the author).

problem suggests ways to analyze the nonlinear problem above threshold, but also because it is necessary to understand the dynamic differences above and below threshold in order to consider whether Fig. 2 represents the spectrum of a soliton or mere linear vibrations of the protein. In Section 4 the simplest representation of the nonlinear problem is considered: Three coupled nonlinear Schrödinger (NLS) equations. At this level of approximation, a family of exact soliton solutions is obtained which display the salient features of the solitary wave observed on the original numerical model. A detailed description of this numerical solitary wave is presented in Section V and discussed in the context of the analytical solution developed in the previous section. The differences between nonlinear system dynamics above threshold and linear dynamics below threshold are also developed. A critical discussion of the differences between the spectrum of Fig. 2 and my numerical prediction of it is presented in Section VI. In particular, it is shown that the best possible numerical results differ only slightly from those obtained from the numerical model without any adjustment of the physical parameters. The residual differences between predicted and measured spectra must be attributed to errors in measurement. Section 7 provides a concluding discussion. The analytical results of Section 4 are readily extended from three to an arbitrary number of interacting NLS equations; this extension is noted in an appendix.

2. Alpha-helix models

In this section the several numerical and analytical models to be considered are listed in order of decreasing complexity.

2.1. *Modified Davydov (MD) equations*

The following equations were developed in [5] to describe the coupled propagation of bond energy probability density, $A_{n\alpha}$, and longitudinal displacement of an amide-I bond, $B_{n\alpha}$.

$$\mathrm{i}\frac{\mathrm{d}A_{n\alpha}}{\mathrm{d}\tau} + J(A_{n+1,\alpha} - 2A_{n\alpha} + A_{n-1,\alpha}) - k_1(B_{n+1,\alpha} - B_{n-1,\alpha})A_{n\alpha} = LF_L + \{k_2[B_{n-1,\alpha}A_{n-1,\alpha} - B_{n+1,\alpha}A_{n+1,\alpha} + B_{n\alpha}(A_{n+1,\alpha} - A_{n-1,\alpha})] - 2JA_{n\alpha} + NF_N + PF_P + QF_Q + RF_R + SF_S + TF_T + UF_U + VF_V + XF_X + ZF_Z\} \quad \text{(MD1)}$$

$$\frac{\mathrm{d}^2B_{n\alpha}}{\mathrm{d}\tau^2} - (B_{n+1,\alpha} - 2B_{n\alpha} + B_{n-1,\alpha}) = k_3(|A_{n+1,\alpha}|^2 - |A_{n-1,\alpha}|^2) + \{k_4[A^*_{n\alpha}(A_{n+1,\alpha} - A_{n-1,\alpha}) + A_{n\alpha}(A^*_{n+1,\alpha} - A^*_{n-1,\alpha})]\} \quad \text{(MD2)}$$

In these equations the $\{J, L, \ldots, Z\}$ are dipole–dipole interaction energies between pairs of amide-I bonds. They have the normalized values given in Table I. The subscript n counts unit cells along the helix while α specifies one of the three spines (see Fig. 1). The symbol F_L is shorthand for the dipole–dipole interactions between laterally adjacent amide-I bonds; thus:

$$\mathrm{i}\dot{A}_{n1} = L(A_{n2} + A_{n-1,3}) + \text{etc.}$$
$$\mathrm{i}\dot{A}_{n2} = L(A_{n3} + A_{n1}) + \text{etc.}$$
$$\mathrm{i}\dot{A}_{n3} = L(A_{n2} + A_{n+1,1}) + \text{etc.} \quad \text{(1a, b, c)}$$

Similar definitions of the symbols $F_N, \ldots, F_Z$ are given in [5].

The constants $k_1, \ldots, k_4$ represent nonlinear coupling between bond energy dispersion and longitudinal sound. They are

$$k_1 = \frac{10^{-11}}{\hbar}\sqrt{\frac{M}{w}}\chi_1$$
$$k_2 = \frac{10^{-11}}{\hbar}\sqrt{\frac{M}{w}}\chi_2$$
$$k_3 = \frac{10^{11}}{w}\chi_1$$
$$k_4 = \frac{10^{11}}{w}\chi_2 \quad \text{(2a, b, c, d)}$$

where M is one third of the mass of a unit cell, w is the linear force constant of a hydrogen bond, χ_1 (as previously noted) is the change in amide-I bond energy per unit extension of an adjacent hydrogen bond, and χ_2 is the change in the longitudinal (nearest neighbor) dipole–dipole coupling term, J, per unit of differential displacement.

The time scale, τ, measures laboratory time in units of

Table I

Term	Energy (computer units)
J	0.145
L	0.231
N	0.073
P	0.034
Q	0.019
R	0.012
S	0.0089
T	0.0073
U	0.0037
V	0.0030
X	0.0022
Z	0.0017

$$\sqrt{\frac{M}{w}} = 0.99 \times 10^{-13}\,\mathrm{s} \tag{3}$$

and $B_{n\alpha}$ measures longitudinal displacement in units of 10^{-11} m.

It is important to emphasize (see [5] for details) that there are no adjustable parameters in the MD system.

2.2. *Coupled Zakharov (CZ) equations*

The MD system can be simplified by noting [5] that $k_2 \ll k_1$ and $k_4 \ll k_3$ and also from Table I that J and L are substantially larger than the other dipole–dipole interaction terms. Furthermore the term "$-2JA_{n\alpha}$" in MD1 can be removed by scaling. Thus the $\{\ \}$ terms can be neglected in both MD1 and MD2. Also the discrete variable, n, can be approximated as a continuous variable, ξ, by writing

$$B_{n+1,\alpha} - B_{n-1,\alpha} \doteq 2\frac{\partial \tilde{B}_\alpha}{\partial \xi}$$

$$|A_{n+1,\alpha}|^2 - |A_{n-1,\alpha}|^2 \doteq 2\frac{\partial |\tilde{A}_\alpha|^2}{\partial \xi}$$

Finally it is convenient to define

$$\tilde{\rho}_\alpha \equiv \frac{\partial \tilde{B}_\alpha}{\partial \xi} \tag{4}$$

whereupon the MD system reduces to

$$\mathrm{i}\frac{\partial \tilde{A}_\alpha}{\partial \tau} + J\frac{\partial^2 \tilde{A}_\alpha}{\partial \xi^2} - 2k_1\tilde{\rho}_\alpha\tilde{A}_\alpha \doteq L(\tilde{A}_{\alpha+1} + \tilde{A}_{\alpha-1}) \tag{CZ1}$$

$$\frac{\partial^2 \tilde{\rho}_\alpha}{\partial \tau^2} - \frac{\partial^2 \tilde{\rho}_\alpha}{\partial \xi^2} \doteq 2k_3\frac{\partial^2 |\tilde{A}_\alpha|^2}{\partial \xi^2} \tag{CZ2}$$

where $\alpha = 1, 2, 3$. Writing

$$\tilde{A}_\alpha = A_\alpha/4k_1k_3J$$

$$\tilde{\rho}_\alpha = \rho_\alpha/2k_1J$$

$$\xi = Jx$$

$$\tau = Jt$$

puts CZ into the normalized form*

$$\mathrm{i}A_{\alpha,t} + A_{\alpha,xx} - \rho_\alpha A_\alpha \doteq \epsilon(A_{\alpha+1} + A_{\alpha-1}) \tag{CZ1'}$$

$$\rho_{\alpha,tt} - \rho_{\alpha,xx} \doteq (|A_\alpha|^2)_{xx} \tag{CZ2'}$$

where

$$\epsilon = JL \tag{5}$$

2.3. *Coupled nonlinear Schrödinger (CNLS) equations*

For the solitary wave solutions that arise in this study, the velocities observed are small compared with unity (longitudinal sound speed). Thus the first term in (CZ2') can be neglected, leading to the coupled system of nonlinear Schrödinger equations.

$$\mathrm{i}A_{1,t} + A_{1,xx} + |A_1|^2A_1 \doteq \epsilon(A_2 + A_3)$$

$$\mathrm{i}A_{2,t} + A_{2,xx} + |A_2|^2A_2 \doteq \epsilon(A_1 + A_3)$$

$$\mathrm{i}A_{3,t} + A_{3,xx} + |A_3|^2A_3 \doteq \epsilon(A_1 + A_2) \tag{CNLS}$$

* From here on a subscript notation is used to indicate partial differentiation.

2.4. *Coupled nonlinear Schrödinger equations with no bending (CNLSNB)*

A slightly different form of coupled nonlinear Schrödinger equations can be obtained from CZ' under the assumption that the dynamics introduces no bending of the alpha-helix. This assumption is equivalent to requiring

$$\rho_1 = \rho_2 = \rho_3 = \tfrac{1}{3}(|A_1|^2 + |A_2|^2 + |A_3|^2) \tag{6}$$

whereupon the three coupled nonlinear Schrödinger equations can be written in vector form as

$$\mathrm{i}\bar{A}_t + \bar{A}_{xx} + \tfrac{1}{3}(\bar{A}^*\cdot\bar{A})\bar{A} = Z\bar{A} \tag{CNLSNB}$$

where $\bar{A} = \mathrm{col}\,(A_1, A_2, A_3)$ and

$$Z = \begin{bmatrix} 0 & \epsilon & \epsilon \\ \epsilon & 0 & \epsilon \\ \epsilon & \epsilon & 0 \end{bmatrix}. \tag{7}$$

This "no bending" assumption is not unreasonable physically since the valence structure of the helix might be expected to inhibit differences in the ρ_α's, but CNLSNB is primarily interesting because, as we shall see in Section 4, it is completely integrable.

2.5. *Coupled linear Schrödinger (CLS) equations*

As the amplitudes of $\bar{A}$ approach zero, both CNLS and CNLSNB become the coupled linear Schrödinger equations which can be written in vector form as

$$\mathrm{i}\bar{A}_t + \bar{A}_{xx} = Z\bar{A} \tag{CLS}$$

3. Linear analysis below threshold for soliton formation

A general solution of CLS can be written as a generalized sum of elementary solutions of the form $\bar{A}\exp(\mathrm{i}\omega t + \mathrm{i}kx)$ whereupon CLS requires

$$[Z + (\omega + k^2)I]\bar{A} = 0 \tag{8}$$

From the work of Ferrers and of Garbieri [8]

$$\det[Z + (\omega + k^2)I] = (\omega + k^2 - \epsilon)^2(\omega + k^2 + 2\epsilon) \tag{9}$$

Thus eq. (8) has a *symmetric* eigenvector

$$\bar{A}_s = \begin{bmatrix} A_s \\ A_s \\ A_s \end{bmatrix}$$

at $\omega + k^2 = -2\epsilon$, and two linearly independent *unsymmetric* eigenvectors at $\omega + k^2 = \epsilon$ which can be written as

$$\bar{A}_{u1} = \begin{bmatrix} A_{u1} \\ A_{u1}\mathrm{e}^{2\pi\mathrm{i}/3} \\ A_{u1}\mathrm{e}^{-2\pi\mathrm{i}/3} \end{bmatrix} \quad \text{and} \quad \bar{A}_{u2} = \begin{bmatrix} A_{u2} \\ A_{u2}\mathrm{e}^{-2\pi\mathrm{i}/3} \\ A_{u2}\mathrm{e}^{2\pi\mathrm{i}/3} \end{bmatrix}$$

A general solution of CLS is then

$$\bar{A} = \begin{bmatrix} A_s \\ A_s \\ A_s \end{bmatrix} \int_{-\infty}^{\infty} F(k)\exp\mathrm{i}[kx - (k^2 + 2\epsilon)t]\,\mathrm{d}k$$

$$+\begin{bmatrix} A_{u1} \\ A_{u1}e^{2\pi i/3} \\ A_{u1}e^{-2\pi i/3} \end{bmatrix} \int_{-\infty}^{\infty} G(k) \exp i[kx-(k^2-\epsilon)t]\,dk$$

$$+\begin{bmatrix} A_{u2} \\ A_{u2}e^{-2\pi i/3} \\ A_{u2}e^{2\pi i/3} \end{bmatrix} \int_{-\infty}^{\infty} H(k) \exp i[kx-(k^2-\epsilon)t]\,dk \quad (10)$$

In the long wavelength limit ($k \to 0$) interaction of the symmetric mode with either of the unsymmetric modes will generate sinusoidal terms of the form $\exp(3\epsilon i t)$. In general, however, initial conditions will require the solution to be distributed over a band of k values which in turn implies nonperiodic behavior in time.

4. Coupled mode solitons on CNLS and CNLSNB

In this section we seek solitary wave (or "soliton") solutions of coupled nonlinear Schrödinger equations. Beginning with CNLS and introducing the transformation

$$\bar{A} = \bar{\phi} e^{i\omega t} \quad (11)$$

where $\bar{\phi} = \text{col}(\phi_1, \phi_2, \phi_3)$ one finds the condition

$$\det[Z + \omega I] = 0 \quad (12)$$

Equation (12) has a single root at $\omega = -2\epsilon$ corresponding to a *symmetric* solitary wave

$$\bar{A}_s = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \phi e^{-2i\epsilon t} \quad (13)$$

where

$$i\phi_t + \phi_{xx} + |\phi|^2\phi = 0 \quad (14)$$

Equation (12) also has a double root at $\omega = \epsilon$ which correspond to two unsymmetric solitary waves

$$\bar{A}_{u1} = \begin{bmatrix} 1 \\ e^{2\pi i/3} \\ e^{-2\pi i/3} \end{bmatrix} \phi e^{i\epsilon t} \quad \text{and} \quad \bar{A}_{u2} = \begin{bmatrix} 1 \\ e^{-2\pi i/3} \\ e^{2\pi i/3} \end{bmatrix} \phi e^{i\epsilon t} \quad (15a, b)$$

where, again, ϕ is a soliton solution of eq. (14).

It is of particular interest to consider mixed solitary waves with symmetric and unsymmetric components for these display pulsations in the magnitudes of the spine amplitudes. For example taking

$$\bar{A} = a\bar{A}_{u1} + \bar{A}_s = \begin{bmatrix} a e^{i\epsilon t} + e^{-2i\epsilon t} \\ a e^{i\epsilon t + 2\pi i/3} + e^{-2i\epsilon t} \\ a e^{i\epsilon t - 2\pi i/3} + e^{-2i\epsilon t} \end{bmatrix} \phi_m \quad (16)$$

leads to the requirements that ϕ_m must satisfy the three pde's:

$$i\phi_{m,t} + \phi_{m,xx} + |\phi_m|^2\phi_m[1 + a^2 + 2a\cos 3\epsilon t] = 0$$

$$i\phi_{m,t} + \phi_{m,xx} + |\phi_m|^2\phi_m[1 + a^2 + 2a\cos(3\epsilon t + 2\pi/3)] = 0$$

$$i\phi_{m,t} + \phi_{m,xx} + |\phi_m|^2\phi_m[1 + a^2 + 2a\cos(3\epsilon t - 2\pi/3)] = 0 \quad (17a, b, c)$$

Thus this mixed mode satisfies CNLS only approximately for $|a| \ll 1$ or $|a| \gg 1$.

If, however, the mixed mode (16) is substituted into CNLSNB, eqs. (17) become

$$i\phi_{m,t} + \phi_{m,xx} + (1 + a^2)|\phi_m|^2\phi_m = 0 \quad (18)$$

so

$$\phi_m = \frac{A_0}{\sqrt{1+a^2}} \text{sech}\left(\frac{A_0(x-st)}{\sqrt{2}}\right) \exp i\left(\frac{s}{2}x + \frac{A_0^2}{2}t\right) \quad (19)$$

A somewhat more compact analysis of CNLSNB proceeds by introducing the transformation

$$\bar{A} = M(t)\bar{\phi} \quad (20)$$

with additional constraints

$$M^T M^* = 3I \quad (21)$$

and

$$iM_t = ZM \quad (22)$$

Under eq. (20) CNLSNB reduces to

$$i\bar{\phi}_t + \bar{\phi}_{xx} + (\bar{\phi}^* \cdot \bar{\phi})\bar{\phi} = 0 \quad (23)$$

Assuming the columns of M to have time dependence as $\exp(i\omega t)$, eq. (22) implies

$$[Z + \omega I]M = 0 \quad (24)$$

which, with eq. (12), leads to

$$M = \begin{bmatrix} 1 & 1 & e^{-3i\epsilon t} \\ e^{2\pi i/3} & e^{-2\pi i/3} & e^{-3i\epsilon t} \\ e^{-2\pi i/3} & e^{2\pi i/3} & e^{-3i\epsilon t} \end{bmatrix} e^{i\epsilon t} \quad (25)$$

Manakov [9] has shown that eq. (23) is completely integrable. Thus the "mixed solitary wave" given by eqs. (16) and (19) [or from eq. (23) by taking $\bar{\phi} = \text{col}(a\phi_m, 0, \phi_m)$] is a true soliton of CNLSNB. The squared amplitudes (i.e., bond energies) of its three components are:

$$|A_1|^2 = \left[\frac{A_0^2}{1+a^2}\text{sech}^2\ \frac{A_0(x-st)}{\sqrt{2}}\right][1 + a^2 + 2a\cos 3\epsilon t]$$

$$|A_2|^2 = \left[\frac{A_0^2}{1+a^2}\text{sech}^2\ \frac{A_0(x-st)}{\sqrt{2}}\right] \times [1 + a^2 + 2a\cos(3\epsilon t + 2\pi/3)]$$

$$|A_3|^2 = \left[\frac{A_0^2}{1+a^2}\text{sech}^2\ \frac{A_0(x-st)}{\sqrt{2}}\right] \times [1 + a^2 + 2a\cos(3\epsilon t - 2\pi/3)] \quad (26a, b, c)$$

Following Gibbons et al. [10] it is straightforward to show that the Hamiltonian associated with CNLSNB is

$$H = \int_{-\infty}^{\infty} \left\{ \sum_{j=1}^{3} |A_{j,x}|^2 - \tfrac{1}{6}\left[\sum_j |A_j|^2\right]^2 + \epsilon(A_1^*A_2 + A_1A_2^* + A_1^*A_3 + A_1A_3^* + A_2^*A_3 + A_2A_3^*) \right\} dx \quad (27)$$

while the "mass"

$$m = \tfrac{1}{2}\int_{-\infty}^{\infty} [|A_1|^2 + |A_2|^2 + |A_3|^2]\,dx \quad (28)$$

Thus the energy of the soliton is

$$H = \tfrac{1}{2}ms^2 - \frac{m^3}{54} + 2m\epsilon\left(\frac{2-a^2}{1+a^2}\right) \tag{29}$$

Physical significance is readily assigned to the terms in eq. (29): the first term is kinetic energy, the second is binding energy, and the third is interspine interaction energy. It is energetically favorable for the mixed mode to add more of the unsymmetric mode by increasing a.

5. Numerical results on the modified Davydov equations

In this section we turn to numerical results obtained from integration of the MD equations. As in [5] the boundary conditions are

$$A_{n\alpha} \text{ and } B_{n\alpha} = 0 \text{ for: } n = 0, -1, -2, -3,$$

$$n = n_{\max+1}, n_{\max+2}, n_{\max+3}, \text{ and all } \tau$$

The initial condition (at $\tau = 0$) is

$$A_{11} = 1 \quad \text{and} \quad A_{12} = 1$$

$$\text{other } A_{n\alpha} = 0$$

$$\text{all } B_{n\alpha} = 0$$

With $n_{\max} = 200$ and $\chi_1 = 0.3 \times 10^{-10}$ newtons (i.e., just below threshold), Figure 3 shows the peak amplitudes on each of the three spines as a function of τ. Figure 4 shows a corresponding results for $\chi_1 = 0.4 \times 10^{-10}$ newtons (just above threshold). A well developed interspine oscillation is seen in Fig. 4 with a period (in computer units) of

$$T_1 = 21.93 \pm 0.806 \tag{30}$$

where the "±" refers to r.m.s. deviations in a series of measurements. The time dependence in Fig. 4 is qualitatively similar to that derived in eq. (26), but from eq. (5) and Table I, $\epsilon = 0.033$ which implies a period $(3\epsilon)^{-1} = 10$. This disagreement is not unexpected considering the approximations made in going from MD through Z and CNLS to CNLSNB.

The speed of the solitary wave under the conditions of Fig. 4 is

$$s = 0.3733 \pm 0.0189. \tag{31}$$

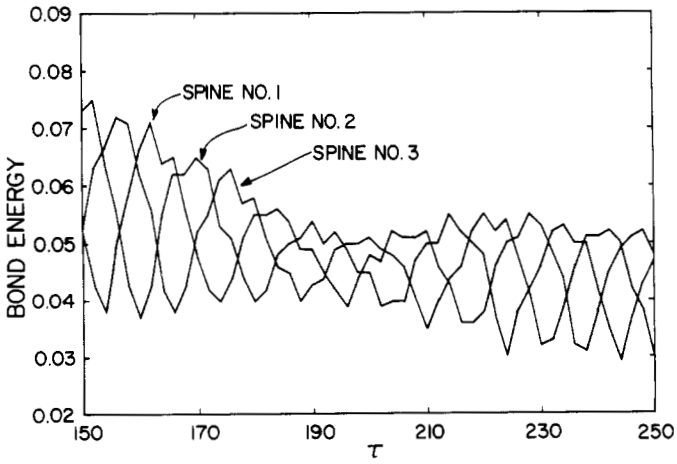

Fig. 3. Peak bond energy on each spine as a function of computer time with the nonlinear parameter $\chi_1 = 0.3 \times 10^{-10}$ newtons which is *below threshold* for soliton formation.

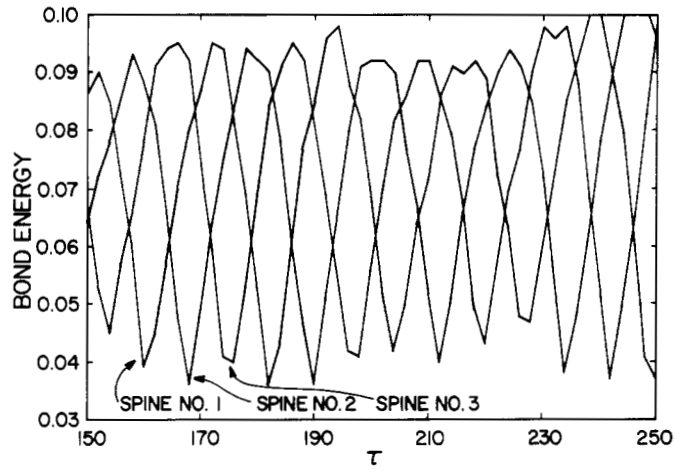

Fig. 4. Same as Fig. 3 but $\chi_1 = 0.4 \times 10^{-10}$ newtons which is *above threshold* for soliton formation.

Assuming the bond energies on each spine to be described by eq. (26), the mixing parameter, a, is

$$a = 0.22 \quad \text{at } \tau = 170$$

$$a = 0.26 \quad \text{at } \tau = 412$$

Thus there is a rather small tendency for the unsymmetric mode component to increase with τ.

As Fig. 3 shows, the situation below threshold is quite different. Although there is some tendency toward an oscillation of period T_1, the bond amplitude variations are much smaller and the frequency is less uniform. Further appreciation of the dynamic differences below and above threshold can be had by comparing the spatial shapes of the waves in Figs. 5 and 6. Not only is the oscillation frequency more variable below threshold, but a smaller fraction of the total bond energy participates in it.

6. Interpretation of laser-Raman spectral measurements as internal vibrations of alpha-helix solitons

The interspine oscillation shown in Fig. 4 has a spectral energy

$$E_1 = \frac{\sqrt{\omega/M}}{cT_1} \text{ cm}^{-1} \tag{32}$$

where numerical values for $\sqrt{\omega/M}$ and T_1 are given in eqs. (3) and (30) respectively, and c is the speed of light. Thus

$$E_1 = 15.35(1 \pm 0.0368)\,\text{cm}^{-1} \tag{33}$$

A second oscillation is generated with a period equal to the time the soliton takes to pass by a unit cell. This is s^{-1} in com-

Table II

Measured (cm^{-1})	Assignments
45	$2E_1$
52	$3E_1$
63	$3E_1$
85	$5E_1$
90	$E_2 - 2E_1$
108	$E_2 - E_1$
123	E_2
152	$E_2 + 2E_1$
182	$E_2 + 3E_1$

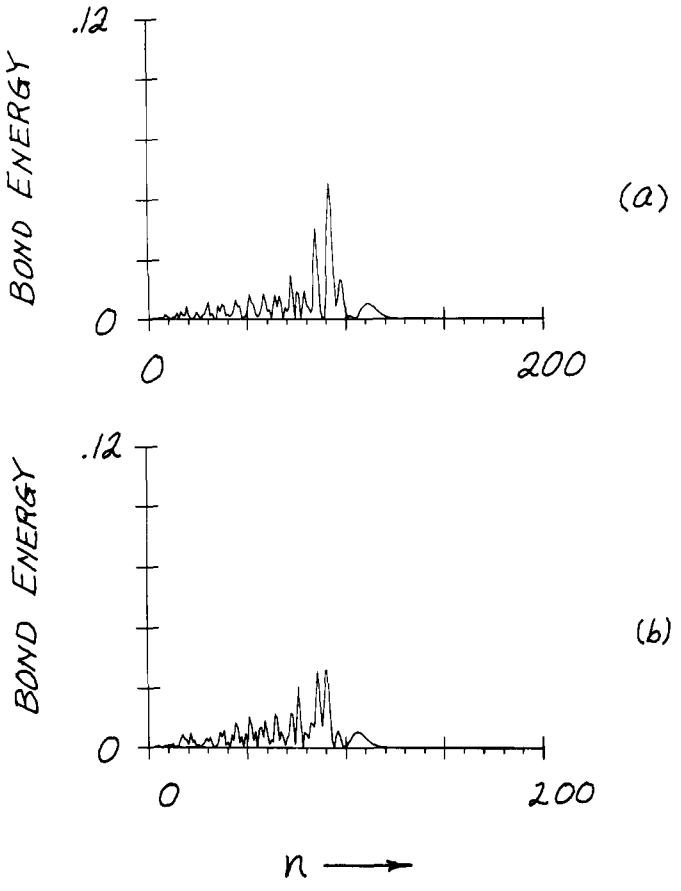

Fig. 5. Spatial wave form on spin No. 2 *below threshold* ($\chi_1 = 0.3 \times 10^{-10}$ newtons). (a) Maximum amplitude at $\tau = 234$. (b) Minimum amplitude at $\tau = 224$.

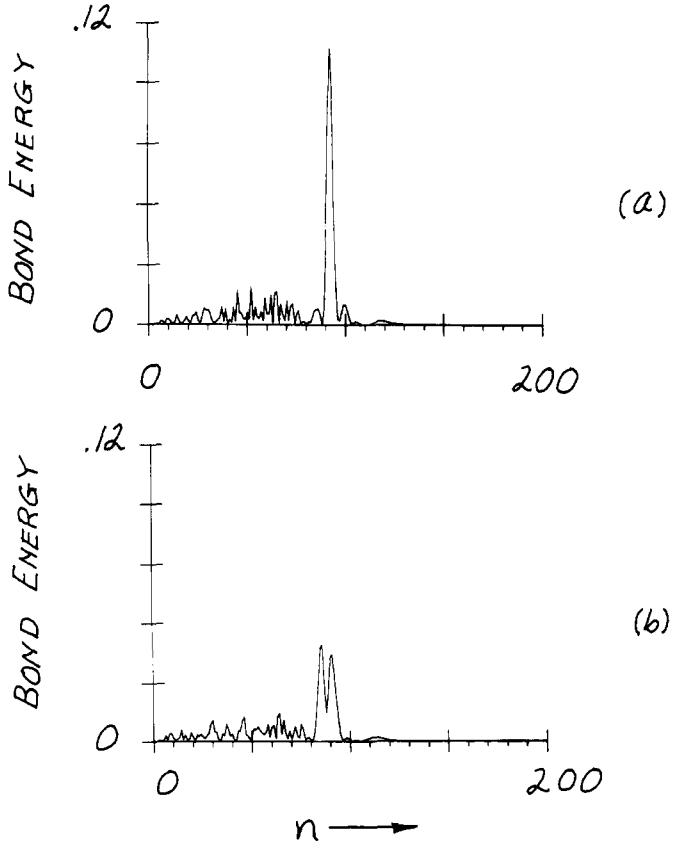

Fig. 6. Spatial waveform on spine No. 2 *above threshold* ($\chi_1 = 0.4 \times 10^{-10}$ newtons). (a) Maximum amplitude at $\tau = 246$. (b) Minimum amplitude at $\tau = 234$.

puter units, where s is given in eq. (31), so the corresponding spectral energy is

$$E_2 = 125.7(1 \pm 0.0506)\,\mathrm{cm}^{-1} \tag{34}$$

In [7] the spectral lines obtained by Webb from metabolically active *E. coli* [6] (see Fig. 2) were tentatively assigned as simple sums and differences of E_1 and E_2 as shown in Table II. With these assignments and the numerical values for E_1 and E_2 given in eqs. (33) and (34), the relation between calculated and measured line energies is plotted in Fig. 7. Broadly speaking, the differences between measurements and calculations can be attributed to errors in the numerical model (MD) and the measurement errors. With the line assignments as in Table II, the best that the numerical model can do is a least squares fit which (on integer values) is at

$$E_1 = 17\,\mathrm{cm}^{-1} \quad \text{and} \quad E_2 = 124\,\mathrm{cm}^{-1} \tag{35}$$

Assuming the three parameters in MD to be such that the fundamental energies are given by eq. (35) rather than eqs. (33) and (34) (a 10% and a 1.3% change respectively), the relation between measured and calculated lines is plotted again in Fig. 8. There are three possible explanations for the differences between measurements and calculations in Fig. 8: (i) The theory presented here is wrong, (ii) The line assignments are incorrect, and (iii) Measurement errors. Only the first line (measured at $45\,\mathrm{cm}^{-1}$) lies significantly more than a spectral bandwidth from agreement. From Fig. 2 it is seen that this line rests high on the background shoulder (of Rayleigh scattering) which may introduce additional errors into the measurement.

Further support for the hypothesis presented here is provided by the data in Fig. 9. This is a histogram of 125 lines obtained from ten experiments done by S. J. Webb and M. E. Stoneham between 1975 and 1979 [11]. With the assignments as indicated, a least squares fit to these data requires

$$E_1 = 16.8\,\mathrm{cm}^{-1} \quad \text{and} \quad E_2 = 126.0\,\mathrm{cm}^{-1} \tag{36}$$

Spectral lines calculated from these frequencies are also indicated on Fig. 9.

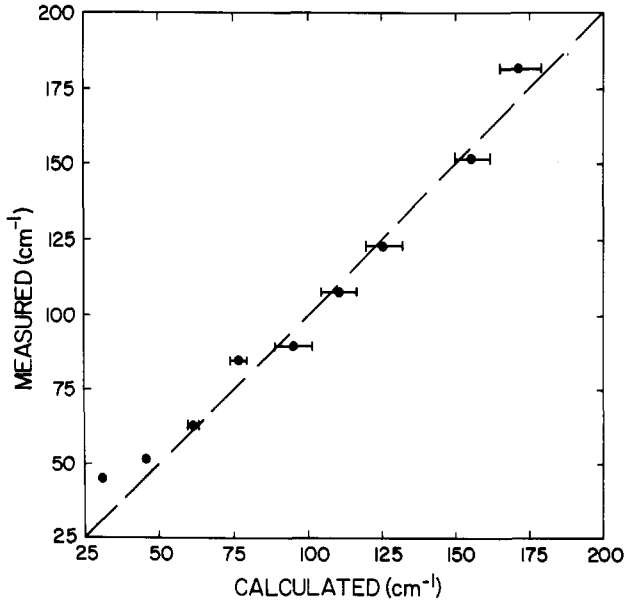

Fig. 7. Comparison laser-Raman lines measured in Fig. 2 with calculations according to the assignments in Table II from E_1 in eq. (33) and E_2 in eq. (34).

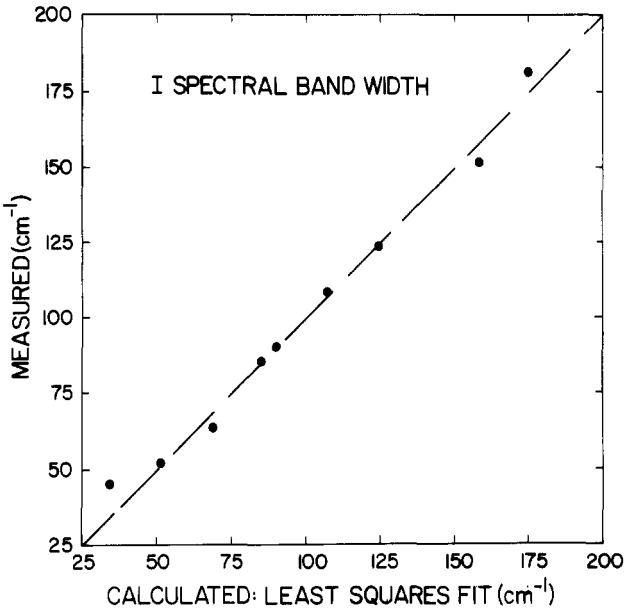

Fig. 8. Same as Fig. 7 but E_1 and E_2 are the "least-squares-fit" given in eq. (35).

7. Concluding discussion

The central question in this paper is to decide whether the laser-Raman spectrum displayed in Fig. 2 can be assigned to internal oscillations of alpha-helix solitons. According to Webb [6] these spectra are characterized by the following facts: (i) At 300 K a Raman spectrum is observed only when the cells are metabolically active, (ii) The intensity ratios of Stokes to anti-Stokes lines indicate that the Raman active states are produced *in vivo* by non-thermal means, and (iii) Spectral lines below 300 cm^{-1} shift to lower wave numbers as the cells progress through their life cycles. Although broad laser-Raman spectra have been observed on the alpha-helix proteins in aqueous solution and in crystalline form [12–14], such an explanation is unlikely for the lines in Fig. 2.

A convenient summary of the foregoing results is provided by Table III.

Since there was no adjustment of parameters in MD prior to the calculations of row (1), the agreement displayed in this table is quite remarkable. There is little difficulty in supposing the MD parameters to be slightly changed to obtain the fundamental energies listed in row (2) or row (3). Since E_2 is related to the frequency at which a soliton interferes with the alpha-helix periodicity, it could be expected to change with the soliton speed under varying metabolic conditions.

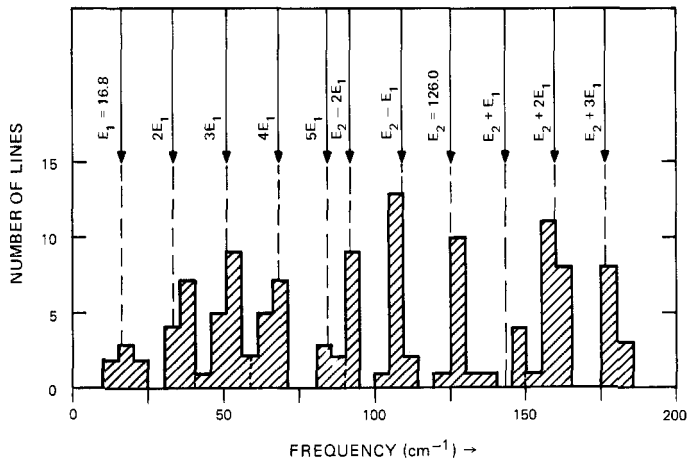

Fig. 9. Histogram of lines from ten laser-Raman experiments on metabolically active *E. coli*. Calculated lines are from the least squares fit values of E_1 and E_2 given in eq. (36). Data from Webb, S. J. (private communication).

Table III

Item	$E_1(cm^{-1})$	$E_2(cm^{-1})$
(1) Calculation of component frequencies from MD.	15.4 ± 3.7%	125.7 ± 5.1%
(2) Least squares fit (over integer values) to lines in Fig. 2.	17	124
(3) Least squares fit to data in Fig. 9.	16.8	126.0

Figure 8 shows the agreement between the lines measured from the spectrum of Fig. 2 and those calculated from the energies listed in the second row of Table III. Figure 9 shows the agreement between data from 125 lines measured during ten separate experiments and those calculated from the energies listed in the third row of Table III. The differences between measured and calculated lines shown in Figs. 8 and 9 must be attributed either to measurement errors or theoretical misconceptions [15]. Noting that [11, 16]: (i) The spectral band width of the measurements is 5 cm^{-1}; (ii) The lines often move several cm^{-1} during the time of measurement (~ 100 minutes); and (iii) Low frequency lines are difficult to resolve because of Rayleigh scattering (see Fig. 2), I conclude that Figs. 8 and 9 show reasonable agreement between measurements and calculations.

The reader may not agree, but certainly the evidence is strong enough to encourage a vigorous search for corroborative experiments.

Appendix

Here we consider a generalization of CNLSNB from three to n interacting channels. Thus

$$i\bar{A}_t + \bar{A}_{xx} + \frac{1}{n}(\bar{A}^* \cdot \bar{A})\bar{A} = Z\bar{A} \tag{A-1}$$

where $\bar{A} = \mathrm{col}\,(A_1, \ldots, A_n)$. If the channels are spaced close together compared with the decay constant for lateral interaction, it is reasonable to assume

$$Z = \begin{bmatrix} 0 & \epsilon & \epsilon & \ldots & \epsilon \\ \epsilon & 0 & \epsilon & \ldots & \epsilon \\ . & . & . & . & . \\ \epsilon & . & . & \epsilon & 0 \end{bmatrix}_{n \times n} \tag{A-2}$$

The transformation (20) with constraints

$$M^T M^* = nI \tag{A-3}$$

and

$$iM_t = ZM \tag{A-4}$$

reduces (A-1) to

$$i\bar{\phi}_t + \bar{\phi}_{xx} + (\bar{\phi}^* \cdot \bar{\phi})\bar{\phi} = 0 \tag{A-5}$$

where $\bar{\phi} = \mathrm{col}\,(\phi_1, \ldots, \phi_n)$.

From (A-4) the time dependence of columns of M is as $\exp(i\omega t)$ where ω satisfies $\det[Z + \omega I] = 0$. The general Ferrers–Garbieri formula for this determinant gives [8]

$$(\omega - \epsilon)^{n-1}[\omega + (n-1)\epsilon] = 0 \tag{A-6}$$

Thus one column of M has time dependence $\exp[i(n-1)\epsilon t]$ and has equal coefficients. The remaining columns have time dependence $\exp(i\epsilon t)$ and are on an orthogonal set in the null space of $Z + \epsilon I$.

Manakov [9] has given Lax operators [17] for eq. (A-5). Thus it is completely integrable using the inverse scattering transform method [18] and it has soliton solutions which, upon transformation by eq. (20), can be interpreted as solutions of eq. (A-1).

If the n channels are not spaced closely with respect to the decay constant for lateral interaction, the non-diagonal elements in Z will not be equal and the Ferrers–Garbieri formula cannot be used to factor $\det[Z + \omega I]$. For four channels arranged at the corners of a square, however,

$$Z = \begin{bmatrix} 0 & \epsilon_1 & \epsilon_1 & \epsilon_2 \\ \epsilon_1 & 0 & \epsilon_2 & \epsilon_1 \\ \epsilon_1 & \epsilon_2 & 0 & \epsilon_1 \\ \epsilon_2 & \epsilon_1 & \epsilon_1 & 0 \end{bmatrix} \tag{A-11}$$

and

$$\det[Z + \omega I] = (\omega - \epsilon_2)^2(\omega + \epsilon_2 - 2\epsilon_1)(\omega + \epsilon_2 + 2\epsilon_1) \tag{A-12}$$

Acknowledgements

It is a pleasure to thank Professor S. J. Webb for correspondence about the accuracy of laser-Raman measurements and for providing the data that is summarized in Fig. 9.

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Theory of Long-range Coherence in Biological Systems. I. The Anomalous Behaviour of Human Erythrocytes

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In this paper, an explicit expression of the interaction potential has been obtained, based on the Fröhlich model of long-range coherence in biological cells. These theoretical expressions are used to correlate with the experimental data obtained from the light scattering and the Brownian motion for the red blood cells (human erythrocytes). The necessary conditions are derived for the formation of rouleaux in human erythrocytes in the presence of long range interactions. The kinetics of rouleaux formation in terms of the properties of the diffusion coefficient has been investigated. It is concluded that both types of experimental data can be interpreted satisfactorily by this model.

1. Introduction

One of the remarkable features of the biological systems is the existence of an extremely high degree of order. Since any form of order in nature must eventually be the result of forces acting amongst the subsystems, a great deal of speculation regarding the origin of these forces has taken place. In a series of papers Fröhlich (1968*a*,*b*, 1970, 1973, 1977) has investigated the origin of these forces from a quantum mechanical standpoint. A brief description of his model will be given later in this paper. Recently some experimental evidence supporting Fröhlich's model has begun to emerge and we shall focus our attention on two different types of experiments for which the theoretical analysis will be carried out.

These two types of experiments are as follows: (1) Rowlands, Sewchand & Enns (1982) have conducted a series of experiments related to the Brownian motion of the human red blood cells (erythrocytes) and observed an increased rate of rouleaux formation which they correlated with the long range forces of Fröhlich. These forces may be either attractive or

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repulsive depending upon specific experimental conditions. The erythrocytes tend to form clusters referred to as rouleaux. The important feature about this phenomenon of rouleaux formation, is that the rate of process cannot be explained on the basis of pure Brownian motion of the erythrocytes. It is necessary to postulate the existence of long range forces of Fröhlich.

(2) In an entirely different experiment, involving quasi-elastic light scattering from suspensions of erythrocytes, one of us has observed the appearance of anomalous diffusion coefficients in a similar condition to the rouleaux formation. It is not at all surprising that the process of rouleaux formation is itself diffusion controlled and may be the consequence of anomalous diffusion. The experimental details of the light scattering experiments will be published elsewhere.

For the purpose of the theoretical investigations developed hereafter, we will adopt the following conjectures: (a) there is evidence of the unusual behaviour of biological systems for two types of experiments; (b) both experimental observations can be explained on the basis of a long range force of Fröhlich operating between the erythrocytes.

The "specific conditions" required for either rouleaux formation or anomalous diffusion are the following: (a) a constant supply of energy is needed which can be provided from the metabolic energy of the cell; (b) the existence of a transmembrance potential is required to polarize the molecules of the system.

If either of these two conditions is not satisfied, then the usual behaviour of the erythrocytes vanishes. Consequently, the interparticle potential must be sensitive to the two factors mentioned above. The details of the theoretical investigations are as follows.

2. Theory

It is convenient to consider the theoretical aspects of this problem in three stages: (1) the formulation of the mathematical form of the long range potential; (2) the development of a theory that suitably incorporates the potential into the kinetics of rouleaux formation; and (3) the inclusion of a long range potential into the formula for the diffusion coefficient of the erythrocytes. This will provide important information about the anomalous diffusion properties.

We now subsequently discuss the three stages outlined above:

THE LONG RANGE POTENTIAL

Fröhlich (1972) showed that the coupled frequency $\omega_{\pm}$ of two giant dipolar molecules separated by a distance r and possessing frequencies ω_1

and ω_2 is given by

$$\omega_{\pm}^2 = \tfrac{1}{2}(\omega_1^2 + \omega_2^2) \pm \sqrt{\tfrac{1}{4}(\omega_1^2 - \omega_2^2) + \frac{\beta^4}{\varepsilon_{\pm}'^2}} \tag{1}$$

where

$$\beta^2 = \frac{\gamma^2 e^2 (Z_1 Z_2)^{1/2}}{Mr^3} \tag{2}$$

and γ is a positive quantity of the order of unity which depends upon various angles, e denotes the electronic charge, M is the mass, Z_1 and Z_2 are the numbers of elastically bound particles in the two molecules and $\varepsilon_{\pm}'$ represents the real part of the frequency dependent dielectric constant determined at $\omega = \omega_+$ and $\omega = \omega_-$, respectively, that is:

$$\varepsilon_{\pm}' = \varepsilon'(\omega_{\pm}). \tag{3}$$

If the two dipolar molecules are identical, then under conditions of frequency resonance ($\omega_1 = \omega_2$), an interaction potential $U(r)$ exists between them expressed by

$$U(r) = \frac{\hbar}{4\bar{\omega}} \frac{e^2 Z r}{M} \left[\frac{1}{\varepsilon'(\omega_+)} - \frac{1}{\varepsilon'(\omega_-)} \right] \frac{1}{r^3} \tag{4}$$

where $\hbar$ is Planck's constant, $Z = Z_1 = Z_2$, $\bar{\omega} = \omega_1 = \omega_2$ reflecting the identical nature of the two molecules. If $\varepsilon'(\omega_+)$ turns out to equal $\varepsilon'(\omega_-)$, then the interaction potential $U(r)$ of (4) vanishes and equation (1) is reduced to a London–Van der Waals type with a much shorter range since the potential is now proportional to r^{-6}. It, therefore, follows that the long range Fröhlich potential is dependent upon the frequency of the coupled oscillators through the real part of the dielectric constant. Since the phenomena of rouleaux formation and anomalous diffusion are manifestations of the potential $U(r)$, it is clear that the quantity $\varepsilon'(\omega)$ plays a cardinal role here. In the second paper of this series (Paul, Chatterjee & Tuszyński, 1983) we have presented a detailed derivation of $\varepsilon'(\omega)$. Here we will simply apply the final equation for it which is:

$$\varepsilon'(\omega) = 1 + \frac{4\pi^2}{VkT} \sum_{5,6} \mu_{5,6}^2 (n_6 - n_5) \frac{(a_{5,6}^2 + b_{5,6}^2 - a_{5,6}\omega)}{(\omega - a_{5,6})^2 + b_{5,6}^2}. \tag{5}$$

The summation indices are expressed numerically, thus 5 and 6 should be regarded as indices not numbers., V is the volume of the suspension of erythrocytes and T is the absolute temperature. Also:

$$\mu_{5,6} = |\langle 5|\boldsymbol{\mu}|6\rangle| \tag{6}$$

where $\boldsymbol{\mu}$ is a dipole moment operator and

$$b_{5,6}^2 = \frac{|\chi|^4}{\pi^2 \hbar^4 (n_6 - n_5)^2} \Gamma_{5,6}^2(\hbar^{-1}\omega_{5,6}) \tag{7}$$

here χ is a phonon-heat bath coupling constant, $\Gamma_{5,6}$ the line width determined under the resonance conditions and n is the occupation number of the phonon state i. Finally,

$$a_{5,6} = \hbar^{-1}\omega_{5,6} \tag{8}$$

where $\omega_{5,6}$ is the difference in energy between the excited (5) and ground (6) states or $\omega_{5,6} = \omega_5 - \omega_6$. The details will be given in our second paper.

With the dielectric constant given by equation (5) and the frequencies $\omega_\pm$ by equation (1), the Fröhlich potential can now be fully defined.

So far we have obtained the form of the interaction between two giant dipoles but the relationship with the etythrocytes has not been discussed yet. A typical erythrocyte is an extremely complex object. Most important, however, is the existence of a very large electrical field across the cell membrane. This field produces intense polarization of the cell membrane resulting in the generation of the giant dipoles as envisaged in Fröhlich's theory. Furthermore, the cell membranes are not stationary objects but display various modes of vibrations which provide us with the features required from a long-range potential. In our model, we assume that the giant dipoles are located at the centre of mass of each erythrocyte. This is, of course, an assumption since the dipoles are actually located on the membranes with different orientations. The exact orientations and numbers of the dipoles are not available to us at the present time forcing us to adopt the simplified model and allow experiment to decide the validity of the assumptions. It is worth pointing out, perhaps, that since our interest lies with the long-range forces between the erythrocytes, the exact cellular details may not be of any significance.

Having obtained the analytical form of the potential acting between the erythrocytes, let us now turn our attention to the nature (attractive or repulsive) of the forces. The frequency ω appearing in equation (5) is a positive quantity and from the discussion given in section 6 of our subsequent paper we see that $\omega_5 > \omega_6$ or that

$$\omega_5 - \omega_6 > 0 \tag{9}$$

which ensures the positive nature of $a_{5,6}$ defined by equation (8). The quantities $b_{5,6}^2$ and $\mu_{5,6}^2$ are positive definite by virtue of their definitions.

From equation (1) the frequency ω_- is lower than ω_+, i.e. $\omega_+ > \omega_-$. Let us define a gap Δ as

$$\omega_+ = \omega_- + \Delta. \tag{10}$$

In order to compute the sign of the Fröhlich potential, it is necessary to calculate the relative magnitudes of the dielectric constants ε'_+ and ε'_-. These two quantities, when substituted into (4), determine the sign of $U(r)$. To evaluate the difference $\varepsilon'_+ - \varepsilon'_-$ we employ equation (5) and get:

$$\varepsilon'(\omega_+) - \varepsilon'(\omega_-) = -\frac{4\pi^2}{VkT} \sum_{5,6} \mu_{5,6}^2 \left[\left(d_{5,6} + \frac{a_{5,6}}{B_{5,6} + C_{5,6}} \right) \right] (n_6 - n_5) \tag{11}$$

where we have adopted the following notation

$$B_{5,6} = (\omega_- - a_{5,6})^2 + b_{5,6}^2 > 0. \tag{12}$$

The above is a positive quantity by virtue of the positive definiteness of the addands. We also used:

$$C_{5,6} = \Delta^2 + 2(\omega_- - a_{5,6})\Delta. \tag{13}$$

Unfortunately, no *a priori* statement about the sign of this term can be made. In a special case of a large enough gap Δ such that:

$$\Delta^2 > 2|(\omega_- - a_{5,6})|\Delta \tag{14}$$

the $C_{5,6}$ is positive:

$$C_{5,6} > 0. \tag{15}$$

We assume that the prevalent conditions allow the inequality (14) to hold thus assuring inequality (15).

Let us define

$$A_{5,6} \equiv a_{5,6}^2 + b_{5,6}^2 - a_{5,6}\omega_-. \tag{16}$$

From this definition and the inequality (15) it follows that

$$\frac{A_{5,6}}{B_{5,6} + C_{5,6}} < \frac{A_{5,6}}{B_{5,6}}. \tag{17}$$

A positive quantity $d_{5,6}$ has been introduced as

$$d_{5,6} = \frac{A_{5,6}}{B_{5,6}} - \frac{A_{5,6}}{B_{5,6} + C_{5,6}} > 0. \tag{18}$$

The right hand side of equation (11) now appears as a sum of terms, each of which contains two factors. One of these factors appearing in the

bracket is positive hence the overall sign is determined by the difference in population numbers $(n_6 - n_5)$. Recalling the fact that the state 5 corresponds to higher energy than that of the state 6 (equation (9)) we deduce that two situations are possible.

(a) If the population inversion occurs $(n_5 > n_6)$, then

$$\frac{1}{\varepsilon'_+} - \frac{1}{\varepsilon'} < 0 \tag{19}$$

and the overall sign of $U(r)$ becomes negative, and the erythrocytes experience an attractive force. An attractive force operating amongst the erythrocytes is, therefore, a signal of population inversion and holds under non-equilibrium conditions.

(b) Under equilibrium conditions $(n_5 < n_6)$ the reverse of inequality (19) must hold true:

$$\frac{1}{\varepsilon'_+} - \frac{1}{\varepsilon'} > 0. \tag{20}$$

This inequality leads to a repulsive force amongst the erythrocytes.

In the above discussion we have shown that the population inversion is of great importance in determining the nature of $U(r)$. We must, now, seek for the possible underlying mechanism causing such an inversion. It is not hard to see that metabolism must play an essential role in supplying the required energy and we are thereby led to an understanding of the prerequisite (a) mentioned in Section 1. Moreover, the reasons for the prerequisite (b) can also be found from the equations we have already developed. If, in fact, the transmembrane field is switched off, then the force causing polarizations and the appearance of the dipole moment μ has been abolished and the Fröhlich potential vanishes.

Having gathered all necessary information about the potential $U(r)$ we proceed to discuss the next stage, i.e. the kinetics of the rouleaux formation.

THE KINETICS OF ROULEAUX FORMATION

In the absence of an interaction potential the process of rouleaux formation is solely controlled by the random Brownian motion of the erythrocytes. From a mathematical standpoint this problem can be formulated in much the same way as the kinetics of diffusion controlled chemical reactions (Rowlands *et al.*, 1982). In this approach one solves the diffusion equation and calculates the flux to a spherical surface. The boundary conditions required here correspond to a zero density of erythrocytes within the spherical surface and a bulk concentration at large distances. Such an

approach has been applied to the phenomenon of colloidal coagulation by Smoluchowski and well-documented literature on the subject is available (see for example Chandrasekhar, 1943). In the event that long range forces are operative between the erythrocytes, however, it is essential to go beyond the diffusion equation. An adequate treatment for diffusion controlled chemical reactions involving interacting particles has been developed by Debye (1942) and is well suited to the present problem. While it is not our intention to present all the details it seems worthwhile to introduce the basic mathematical steps as they may contribute to a better understanding of the physical principles involved. The analysis is most conveniently introduced in the following steps.

(a) Consider two volume elements dV_A and dV_B located by the vectors $\mathbf{r}_A$ and $\mathbf{r}_B$, respectively, as shown in Fig. 1.

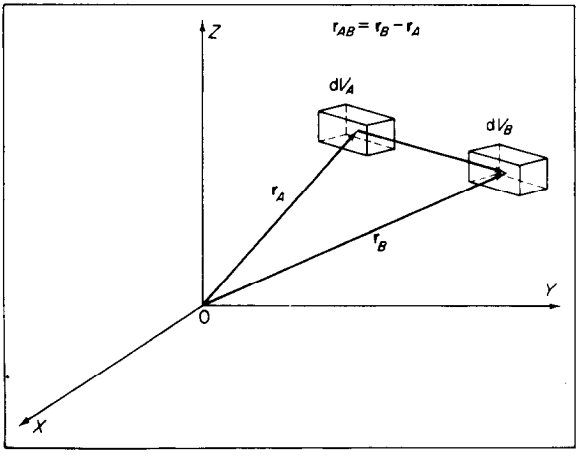

FIG. 1. Volume elements and coordinate systems.

(b) Let us introduce the following definitions. n_{AB} is the probability of finding an erythrocyte in dV_B given that dV_A already contains one erythrocyte. $N_A(\mathbf{r}_A)$ is the probability density of finding an erythrocyte in the vicinity of $\mathbf{r}_A$. N_{AB} is the joint probability of finding an erythrocyte in dV_A and dV_B.

By interchanging the labels A and B on n_{AB} and N_{AB} and by considering $N_B(\mathbf{r}_B)$ instead of $N_A(\mathbf{r}_A)$, one can interchange the physical role played by the two volume elements dV_A and dV_B.

From the above definitions it follows by straightforward physical arguments that the following equations must be true:

$$N_{AB} = N_A n_{AB} \tag{21}$$

$$N_{BA} = N_B n_{BA} \tag{22}$$

$$N_{AB} = N_{BA}. \tag{23}$$

(c) In completely general terms the functions introduced above must have the following spatial dependence:

$$N_{AB} = N_{AB}(\mathbf{r}_A, \mathbf{r}_B, \mathbf{r}_{AB}) \tag{24}$$

$$N_{BA} = N_{BA}(\mathbf{r}_B, \mathbf{r}_A, \mathbf{r}_{BA}) \tag{25}$$

and in the limit of infinite separation between the two volume elements:

$$\lim_{|\mathbf{r}_{AB}|\to\infty} N_{AB}(\mathbf{r}_A, \mathbf{r}_B, \mathbf{r}_{AB}) = N_A(\mathbf{r}_A) N_B(\mathbf{r}_B). \tag{26}$$

With no external gradients or fields present the probabilities defined above cannot depend upon the individual vectors $\mathbf{r}_A$ and $\mathbf{r}_B$ but upon the relative vector $\mathbf{r}_{AB}$ only:

$$N_{AB} = N_{AB}(\mathbf{r}_{AB}). \tag{27}$$

Consequently,

$$\lim_{|\mathbf{r}_{AB}|\to\infty} N_{AB}(\mathbf{r}_{AB}) = N_A N_B. \tag{28}$$

Clearly the erythrocytes are not stationary objects, hence in addition to the spatial dependence, each of the functions above must display time dependence as well.

(d) The joint probability function N_{AB} will, in general, satisfy a continuity equation:

$$-\frac{\partial}{\partial t} N_{AB} = \nabla_A \cdot \mathbf{J}_A + \nabla_B \cdot \mathbf{J}_B \tag{29}$$

where ∇_i, are the gradient operators with respect to $\mathbf{r}_i$ and $\mathbf{r}_i$ are fluxes defined as

$$\mathbf{J}_i \equiv \mathbf{v}_i N_{AB} \tag{30}$$

$$\mathbf{v}_i \equiv \frac{\mathrm{d}\mathbf{r}_i}{\mathrm{d}t}. \tag{31}$$

(e) The fluxes must be selected in keeping with the physical requirements of the problem. In the present case the following two constituent fluxes are employed. First, a diffusional flux governed by Fick's law and involving

a diffusion coefficient D_i, and

$$\mathbf{J}_i^{\text{diff}} = -D_i \nabla_i N_{AB}. \tag{32}$$

Second, a flux caused by a force $\mathbf{K}_{AB}$ acting upon the erythrocyte located at $\mathbf{r}_A$, given that a second erythrocyte is located at $\mathbf{r}_B$. This flux is defined by the following set of equations:

$$\mathbf{J}_A^k = \mathbf{v}_A^k N_{AB} \tag{33}$$

$$\mathbf{v}_A^k = \frac{1}{\gamma_A} \mathbf{K}_{AB}. \tag{34}$$

Here, γ_A is the frictional constant of the erythrocyte at $\mathbf{N}_A$.

Assuming that the forces $\mathbf{K}_{AB}$ and $\mathbf{K}_{BA}$ are due to a potential function U, such that

$$\mathbf{K}_{AB} = -\nabla_A U \tag{35}$$

$$\mathbf{K}_{BA} = -\nabla_B U \tag{36}$$

and inserting the total flux $(\mathbf{J}_i = \mathbf{J}_i^{\text{diff}} + \mathbf{J}_i^k)$ along with the appropriate definitions, from equations (32)–(36) into equation (29), we obtain a modified diffusion equation:

$$\frac{\partial}{\partial t} N_{AB} = D_A \nabla^2 N_{AB} + \frac{1}{\gamma_A} N_{AB} \nabla_A^2 U + \frac{1}{\gamma_A} (\nabla_A U)(\nabla_A N_{AB})$$

$$+ D_B \nabla^2 N_{AB} + \frac{1}{\gamma_B} N_{AB} \nabla_B^2 U + \frac{1}{\gamma_B} (\nabla_B U)(\nabla_B N_{AB}). \tag{37}$$

(f) In order to solve this equation with the aim of applying it to the present problem we first define a set of variables $\mathbf{R}$ and $\mathbf{r}$ as follows:

$$\mathbf{R} \equiv \frac{D_A \mathbf{r}_A + D_B \mathbf{r}_B}{D_A + D_B} \tag{38}$$

$$\mathbf{r} = \mathbf{r}_A - \mathbf{r}_B. \tag{39}$$

Then, we make the following assumptions that U and N_{AB} do not depend upon $\mathbf{R}$ and that spherical symmetry prevails.

Under these conditions a very simple equation results from equation (36).

$$-\frac{\partial N_{AB}}{\partial t} = \nabla_r \cdot \mathbf{J}_{AB}^r \tag{40}$$

with

$$J_{AB}^r = -(D_A + D_B) \frac{\partial}{\partial r} N_{AB} - \left(\frac{1}{\gamma_A} + \frac{1}{\gamma_B}\right) \frac{\partial U}{\partial r} N_{AB}. \tag{41}$$

The J'_{AB} represents the flux of erythrocytes along the vector $\mathbf{r}_{AB}$ where the origin of this vector has been placed at the origin of the coordinate system. By definition the flux is the current (I) per unit area. Thus, if the total surface area through which the erythrocytes move towards the origin is $4\pi r^2$, then

$$J'_{AB} = \frac{I}{4\pi r^2}. \tag{42}$$

By equating the right hand sides of equation (41) and (42) we obtain

$$\frac{dN_{AB}}{dr} + \left(\frac{1}{D_A + D_B}\right)\left(\frac{1}{\gamma_A} + \frac{1}{\gamma_B}\right)\frac{dU}{dr} N_{AB} = \frac{-I}{4\pi(D_A + D_B)r^2}. \tag{43}$$

If we assume that the erythrocytes "react" upon arrival at a critical spherical surface of radius r_0, then equation (43) can be integrated to yield

$$N_{AB}(r) = -\frac{I}{4\pi(D_A + D_B)} \exp\left[-\left(\frac{1}{D_A + D_B}\right)\left(\frac{1}{\gamma_A} + \frac{1}{\gamma_B}\right)U(r)\right]$$
$$\times \int_{r_0}^{r} dr' \frac{e(1/(D_A + D_B))}{r'^2}\left(\frac{1}{\gamma_A} + \frac{1}{\gamma_B}\right)U(r'). \tag{44}$$

(g) From equation (44) we require the current I. Not knowing the analytical form of $N_{AB}(r)$ we cannot do this computation in quite general terms. When $r \to \infty$ then equation (28) is true and if U vanishes asymptotically then we obtain:

$$N_A N_B = \frac{-I}{4\pi(D_A + D_B)} \cdot \frac{1}{r_0^*} \tag{45}$$

where

$$\frac{1}{r_0^*} \equiv \int_{r_0}^{\infty} dr \frac{\exp\left[(1/(D_A + D_B))(1/\gamma_A + 1/\gamma_B)U(r)\right]}{r^2}. \tag{46}$$

Solving equation (45) for I we have

$$-I = 4\pi(D_A + D_B) r_0^* N_A N_B \tag{47}$$

which is just the rate of "reaction" (rouleaux formation).

(h) Assuming that the erythrocytes A and B are identical we obtain the rate of erythrocyte depletion in the form of a simple second order kinetic equation

$$-\frac{dN}{dt} = 8\pi D r_0^* N^2 \tag{48}$$

which upon integration becomes

$$\frac{1}{N(t)} - \frac{1}{N(0)} = \frac{4kT}{3\eta}\left(\frac{r_0^*}{r}\right)t. \tag{49}$$

Here the Einstein definition of the diffusion coefficient along with the Stoke's expression for the friction coefficient have been adopted:

$$D = \frac{kT}{\gamma} \tag{50}$$

$$\gamma = 6\pi\eta r_0. \tag{51}$$

In the experimental work carried out by Rowlands *et al.* (1982), the kinetics of rouleaux formation has been studied by introducing an *ad hoc* parameter Z in the rate constant of a pure diffusion controlled process. This parameter is intended to account for the existence of an interparticle potential but the dependence of A upon the potential or even the form of the potential itself has not been explicitly stated. The expression used by these workers (Rowlands *et al.*, 1982) has the following form:

$$\frac{1}{N(t)} - \frac{1}{N(0)} = \frac{2kT}{3\eta}\Xi t \tag{52}$$

where $\Xi = Z/r_0$. Comparing equation (52) with equation (49) we notice that our parameter r_0^* plays precisely the same role as Z. The advantage, however, of using the generalized model employed by us lies in the fact that r_0^* is explicitly given in terms of Fröhlich's potential and the properties of this potential in the biological context have already been discussed.

(i) In order to link our theory with the experiments on rouleaux formation we must evaluate the integral in equation (46) which can be written as:

$$L(\alpha, r_0) = \int_{r_0}^{\infty} \mathrm{d}r \frac{e^{-\alpha/r^3}}{r^2} \tag{53}$$

where

$$\alpha = \frac{1}{(D_A + D_B)}\left(\frac{1}{\gamma_A} + \frac{1}{\gamma_B}\right)A \tag{54}$$

and

$$U(r) = -\frac{A}{r^3}. \tag{55}$$

Since this integral cannot be solved analytically, we have to resort to the numerical integration instead. We treat α (or alternatively A) as an

unknown parameter we can vary and we calculate the corresponding values of the constant Ξ according to:

$$\Xi = \frac{1}{r_0 L(\alpha, r_0)} \tag{56}$$

We have used $r_0 \cong 4$ μm as indicated by Rowlands *et al.* (1982). The results of our sample calculations for selected attractive potentials are presented in Table 1.

TABLE 1

Selected values of the parameter Ξ and the corresponding coefficients α and A.

α (10^{-9} cm^3)	A (10^{-24} ergs cm^3)	Ξ
$0{\cdot}166 \times 10^{-3}$	0·007	1·11
0·011	0·444	1·16
0·021	0·848	1·21
0·029	1·172	1·25
0·422	17·049	2·66
0·521	21·048	2·91
0·770	31·108	3·45
0·914	36·926	3·73

It is not difficult to see that the integral of equation (53) gives the values of Ξ which are always less than 1 if the Fröhlich potential is repulsive ($A < 0$) and that Ξ tends to 1 as A approaches 0. On the other hand, it is the attractive potential ($A > 0$) that provides $\Xi \geq 1$. The Ξ increases with A starting at $\Xi = 1$ for $A = 0$. In their experiments, Rowlands *et al.* (1982) used a particle concentration of 5×10^{-3} v/v and the value of Ξ they arrived at was 3·0. As seen from Table 1, $\Xi = 3.0$ corresponds to $A \cong 23 \times 10^{-24}$ erg cm^3 or $\alpha \cong 0{\cdot}57 \times 10^{-9}$ cm^3.

From our analysis employing a generalized diffusion equation and the Fröhlich potential we are able to explain the phenomenon of rouleaux formation and the dependence of the phenomenon upon the cell metabolism. Furthermore, by carrying out a numerical integration and using Rowlands' experiments data, the value of the constant in the Fröhlich potential has been obtained. Finally, the reduction of the rate of rouleaux formation to that expected under Brownian motion only by a lowering of the pH as observed by Rowlands *et al.* (1982) is explained by the destruction of the transmembrane field and the attending loss of polarization.

THE PHENOMENON OF ANOMALOUS DIFFUSION

Before we describe the experiments showing anomalous diffusion, some of the qualitative aspects of the diffusion coefficient in the presence of long range potentials must be introduced.

Pusey & Tough (1982) showed, that to a first order approximation, a non-stochastic interparticle potential $U(r)$ yields the following value of the mean square displacement (with the angular bracket denoting the average value):

$$\langle \Delta r^2(t) \rangle = 2D_0 t - \frac{D_0 t^2}{\gamma} \left\langle \frac{\partial^2 U(r)}{\partial r^2} \right\rangle \tag{56}$$

where D_0 is the diffusion coefficient of a Brownian particle in the absence of the potential. It is now possible to introduce a modified diffusion coefficient as follows

$$D = D_0 \left[1 - \frac{1}{2} \frac{t}{\gamma} \left\langle \frac{\partial^2 U(r)}{\partial r^2} \right\rangle \right]. \tag{57}$$

In applying this formula it is necessary to obtain the dependence of the potential $U(r)$ (in our case the Fröhlich potential) upon the particle density ρ. To achieve this objective let us examine equation (5). It is not hard to see that the value of summations must depend upon the total number of phonons N and therefore we write

$$\sum_{5,6} \mu_{5,6}^2 (n_6 - n_5) \frac{(a_{5,6}^2 + b_{5,6}^2 - a_{5,6}\omega)}{(\omega - a_{5,6})^2 + b_{5,6}^2} \cong NL \tag{58}$$

where L contains all the remaining quantities apart from N. With this approximation:

$$\varepsilon'(\omega) \cong 1 + \frac{4\pi^2}{kT} L\rho. \tag{59}$$

Consequently, the constant A in equation (55) to the first order approximation in the series expansion assumes the form

$$A = \frac{B}{\rho} + \ldots \tag{60}$$

where B is a parameter containing a collection of constants. Thus, the modified diffusion coefficient becomes

$$D = D_0 \left[1 + \frac{6t}{\gamma} \left\langle \frac{B}{\rho r^5} \right\rangle \right]. \tag{61}$$

It is now clear that the attractive potential increases the value of D as compared to D_0 and *vice versa*.

In order to extract the density dependence of D we apply the first order of the virial expansion, i.e.:

$$<r^n> \cong \rho^{-n/3} \tag{62}$$

and arrive at

$$\left\langle \frac{B}{\rho r^5} \right\rangle \cong B\rho^{2/3}. \tag{63}$$

We conclude that in keeping with our expectations, the coefficient D increases with the density ρ. This, in fact will account for the numerical differences between the Rowlands' results and the light scattering results.

To measure the diffusion coefficient of erythrocytes suspended in various solutions, the method of quasi-elastic light scattering has been employed. The 488 mm line of an Ar ion laser has been used and the scattered light has been detected through a pair of collimating pinholes by a photomultiplier. Then, the autocorrelation function plotted against time has given the diffusion coefficient as a slope.

The solution used is a phosphate buffered saline (PBS) at various pH to which human serum albumin (HSA) has been added. The additional macromolecules are polyvinylpyrrolidone (PVP) at several molecular weights. The PVP at the molecular weight of 360 000 causes an abnormally high diffusion coefficient when the blood substance is metabolically active and a transmembrane field is present. These are the conditions under which an increased rate of rouleaux formation was also observed by Rowlands *et al.* (1981). The only major difference between the two experiments is the particle concentration. In the light scattering experiments the concentration was $1{\cdot}8 \times 10^{-4}$ v/v, that is almost 28 times lower than that in Rowlands' experiments. Rowlands *et al.* (1981) obtained $\Xi = 3{\cdot}0$ and in a comparable case, our result was:

$$D/D_0 = 1{\cdot}56. \tag{64}$$

Table 2 shows all the results of measurements of the diffusion coefficient (in 10^{-10} cm^2/sec) in various situations.

TABLE 2

The values of diffusion coefficient D in 10^{-10} cm^2/sec *for various solutions*

Contents	PBS		PBS + PVP		
State	pH 7·4	pH 6·3	pH 7·4	pH 6·3	pH 7·4†
Metabolically active	15·9	16·5	23·5	19·5	21·4
Metabolically depleted	15·4	14·8	17·5	19·0	18·1
Metabolically repleted	—	16·2	22·2	19·3	20·6

† Restoration of the pH of the suspension at 7·4 after a measurement had been taken at pH = 6·3.

In spite of the seeming discrepancy between the results of the measurements in two different types of experiments (Rowlands *et al.* (1981) $\Xi = 3{\cdot}0$, light scattering $D/D_0 = 1{\cdot}56$), these numbers are in good agreement when the density corrections are made. Another correction we have to make is the dimensionality factor. Rowlands *et al.*'s experiments (1981, 1982) treated the observed solution as a two-dimensional fluid whereas the light scattering method accounts for all three dimensions. Therefore, the diffusion constant used by Rowlands after Swift & Friedlander (1964) has to be multiplied by a factor of two:

$$K' = \frac{4\pi kT}{3\eta}\Xi. \tag{65}$$

To compare the light scattering diffusion constant with the corresponding result of Rowlands (1981) we therefore require that:

$$2 \cdot \Xi - 1 = \frac{D}{D_0} - 1. \tag{66}$$

However, when the density variation is accounted for we expect that:

$$\frac{2\Xi - 1}{D/D_0 - 1} = \left(\frac{\rho_R}{\rho_D}\right)^{2/3} \tag{67}$$

where ρ_R denotes the density in Rowlands' experiment (1981) and ρ_D is the density in the diffusion experiment. Substituting the relevant numbers we obtain a remarkable agreement as the left hand side produces 8·9 and the right hand side gives 9·2 which indicates the compatibility of the two methods.

3. Conclusions

In this paper we have discussed and extended some important aspects of the Fröhlich theory of the long range coherence in biological systems. We have adopted the assumptions of the Fröhlich model as our basis and using our derivations supplemented the missing expressions for the dielectric constants. The Fröhlich long range potential has, therefore, been explicitly defined mathematically.

We have discussed the necessary conditions for this potential to become attractive. All relevant algebraic expressions for the dielectric constants, frequencies and coupling coefficients have been given. We have found that the population inversion of the cell's energy levels is possible if only the energy supply is maintained above a certain threshold level and if the phonon-heat bath coupling constant is large enough.

We have given a brief review of the diffusion process for particles with appreciable mutual interactions. We have stated the formula for the rate constant in such processes. Combining this result with the analytical expression for the Fröhlich potential we have obtained information about the diffusion constant in rouleaux formation. Our numerical estimates of the rate constant for the rouleaux formation provided us with the values of the Fröhlich potential parameter and a threshold condition for the appearance of an increased rate of rouleaux formation.

We have carried out an analysis of the density dependence of our diffusion constant and concluded that to a first order approximation, D is proportional to $\rho^{2/3}$.

We have reported some experimental work done by one of us on the diffusion coefficient of human erythrocytes with the use of quasi-elastic light scattering technique. The experimental findings reveal a marked difference in the diffusion coefficient between the metabolically active and metabolically depleted cells. The process of metabolic depletion appears to be reversible. The presence of a transmembrane potential has proven to be another crucial factor in the cell behaviour which agrees with the theoretical predictions.

Finally, we have applied the theoretically derived expressions to interpret both types of experiments. We have compared the light scattering results with the corresponding results reported by Rowlands *et al.* (1982) and attempted to explain a seemingly paradoxical difference in the values of diffusion coefficient. When the differences in densities of the solutions and the dimensionality are taken into account, both results agree to within an astonishing 5%.

We conclude that the two experimental results can be interpreted by the same theoretical model. Furthermore, the proposed theory seems to provide a satisfactory explanation of the process of rouleaux formation in both cases.

We hope that applications of our model may well exceed the scope we have touched upon in this work.

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ARTICLE

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Transitions in microtubule C-termini conformations as a possible dendritic signaling phenomenon

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Abstract We model the dynamical states of the C-termini of tubulin dimers that comprise neuronal microtubules. We use molecular dynamics and other computational tools to explore the time-dependent behavior of conformational states of a C-terminus of tubulin within a microtubule and assume that each C-terminus interacts via screened Coulomb forces with the surface of a tubulin dimer, with neighboring C-termini and also with any adjacent microtubule-associated protein 2 (MAP2). Each C-terminus can either bind to the tubulin surface via one of the several positively charged regions or can be allowed to explore the space available in the solution surrounding the dimer. We find that the preferential orientation of each C-terminus is away from the tubulin surface but binding to the surface may also take place, albeit at a lower probability. The results of our model suggest that perturbations generated by the C-termini interactions with counterions surrounding a MAP2 may propagate over distances greater than those between adjacent microtubules. Thus, the MAP2 structure is able to act as a kind of biological wire (or a cable) transmitting local electrostatic perturbations resulting in ionic concentration gradients from one microtubule to another. We briefly discuss the implications the current dynamic modeling may have on synaptic activation and potentiation.

Keywords Neuron · Synapse · MAP · Ionic wave · Protein filament · Electrostatic interaction

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Introduction

Microtubules (MTs) have long been known to play key roles in the functioning of eukaryotic cells. Cell division, for example, is accomplished by the dynamic reorganization of the cytoskeleton and the segregation of the chromosomes by mitotic spindles that are composed of MTs (Alberts et al. 2002). Whereas all living cells contain MTs in neurons they facilitate processes and interactions that extend over large (even macroscopic) distances. Elongated structural elements, for example the dendrites of large pyramidal cells of cerebral cortex, provide an environment where interactions that are electromagnetic in nature may propagate along the long axis of MTs aligned within them (Brown and Tuszynski 1999).

Recently, it has become apparent that neurons utilize MTs in cognitive processing. Both kinesin and microtubule-associated protein 2 (MAP2) were implicated in learning and memory (Wong et al. 2002; Khuchua et al. 2003; Woolf et al. 1999). Dendritic MTs, in particular, were implicated in these processes, and it is highly probable that precisely coordinated transport of critical proteins and mRNAs to the postsynaptic density via kinesin along MT tracks in dendrites is necessary for learning, as well as for long-term potentiation (LTP) (Kiebler and DesGroseillers 2000; Steward and Schuman 2001). As shown schematically in Fig. 1, presynaptic axon terminals make synaptic contacts onto postsynaptic spines. Within the shaft of the dendrite, MTs act as tracks for kinesin's processive movement (Alberts et al. 2002); thus enabling them to transport proteins such as receptors. MTs also transport mRNA that translates to proteins critical to synaptic activity. In general, actin participates in the dynamic changes of the dendrite and the formation of the spines. As for the interaction between the actin and MT as well as the possible role of actin in the signal processing, this is an interesting possibility but it is out of the scope of this paper and will be discussed elsewhere. Our computa-

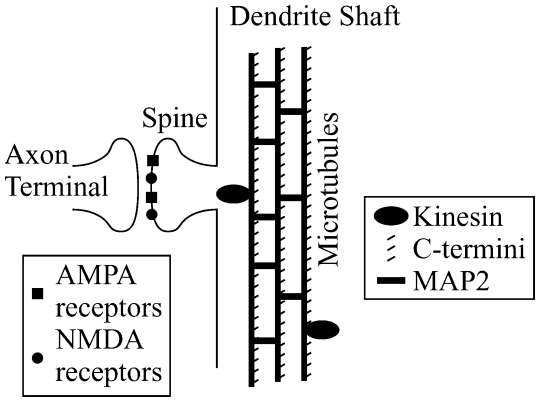

Fig. 1 Diagram of C-termini on the tubulin dimers of microtubules in relationship to incoming synapses. Axon terminals typically input to spines, many of which contain AMPA and NMDA glutamate receptors. Kinesin motors transport cargo along microtubules. C-termini states affect the transport of kinesin along microtubules and this affects transport of materials to active synapses

tional analyses reported in the present paper focus on the C-termini of those MTs whose biophysical properties have a significant influence on the transport of material to activated synapses; as one example, cytoskeletal signal transduction and processing, and, as another, synapses undergoing LTP.

When Nogales et al. (1998) determined a 3D-structure of tubulin, the C-termini could not be resolved because the C-terminus region is highly flexible and lies outside the globular domain of tubulin. However, molecular dynamics enables one to model conformation dynamics of the C-termini using their known amino acid sequences in conjunction with the available crystallographic structures of tubulin. We show some relevant results in the next section.

The C-termini of tubulin are located on the outside of the MT and hence they interact strongly with other proteins, such as MAP2 and kinesin (Sackett 1995). Indeed MAP2 binds to helices 11 and 12 near the C-termini of tubulin in MTs (Al-Bassam et al. 2002). While precise nature of the bond is not known at present, it is suspected to be ionic due to oppositely charged regions of tubulin and MAP2 in the binding areas. To the best of our knowledge, a single MAP2 binds one tubulin dimer. The processivity of kinesin appears to critically involve an interaction with the C-termini of tubulin, although MAP2 may temporarily uncouple to avoid interfering with the movement of kinesin along the MT (Thorn et al. 2000). Removal of the C-terminus of tubulin profoundly affects kinesin processivity; in particular cleavage by subtilisin results in a fourfold decrease of the rate of kinesin transport (Wang and Sheetz 2000).

In an effort to gain an insight into the role played by C-termini in the functioning of dendrites, we have developed a quantitative computational model based on the currently available biophysical and biochemical data regarding some key macromolecular structures involved (tubulin, its C-termini, and MAP2). Our model also accounts for the presence of ions in the surrounding aqueous environment. We start our modeling with the analysis of the possible conformations of the individual C-termini and calculate the resulting electric potential on the outside of the dimer. To elucidate the dynamic properties of C-termini and their interactions with MAP2 we study: (1) the relatively long-lived configurations of C-termini, (2) the interactions between C-termini and tubulin, (3) the interactions between nearest-neighbor C-termini, and (4) interactions between C-termini and MAP2 where we propose a model of wave propagation involving counterions that are weakly bound to MAP2.

Structural and electrostatic properties of tubulin including C-termini

In humans, tubulin exists in dozens of homologous isoforms some of which are preferentially expressed in the human brain (Lu et al. 1998). A 3D structure of tubulin was determined by Nogales et al. (1998) and later refined by Löwe et al. (2001) but several amino acids including some at the C-termini of the protein were not resolved, and hence these amino acid sequences were omitted from the (Protein Data Bank) PDB data files for tubulin (1TUB and 1JFF). However, we have created structure files that include the C-termini whose amino acid sequences correspond to those known for individual human tubulin isotypes. We have also performed computational experiments at physiological temperatures intended to find equilibrium conformations of tubulin's C-termini and these are shown later in the paper. The tubulin structure was a combination of 1JFF with a consistent set of structural elements from 1TUB for some regions absent in 1JFF. However, we must acknowledge the existence of serious limitations in the accuracy of computational studies of a protein as large and complex as tubulin.

An important physical property of proteins is their surface electric charge distributions, as these affect their interactions with other proteins and biomolecules. Fig. 2 illustrates the results of our visualizations based on the PDB file 1JFF. It shows a coarse-grained representation of the electric charge distribution on the tubulin dimer surface. The intention here is to show where charges of different signs are located, not necessarily how densely they are distributed at a given point. The fading of the font used represents the depth of view meant to express the 3D structure of the protein. This figure contains regions including the two C-termini that were not crystallographically resolved, but added here using relevant amino acid sequences. These C-terminal chains' conformations are purely artifacts of the process by which they were added and should not be interpreted as corresponding to any real conformation. The figure and the potential calculation were in part produced using MolMol (Koradi et al. 1996).

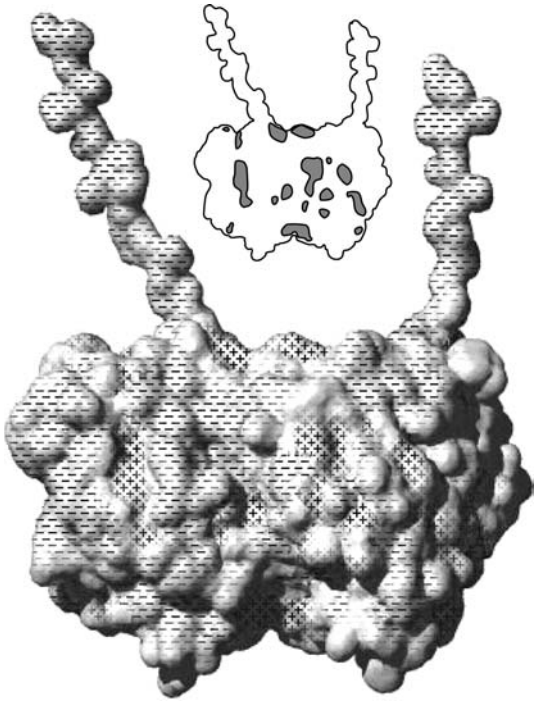

Fig. 2 A map of the electric charge distribution on the surface of a tubulin dimer with C-termini tails present (prepared in part with MolMol (Koradi et al. 1996)). *Pluses* represent the areas of positive charge while *minuses* represent the location of negative ones. The *symbols* do not correspond one-to-one to elementary charge values, only to the presence of charge of a given type

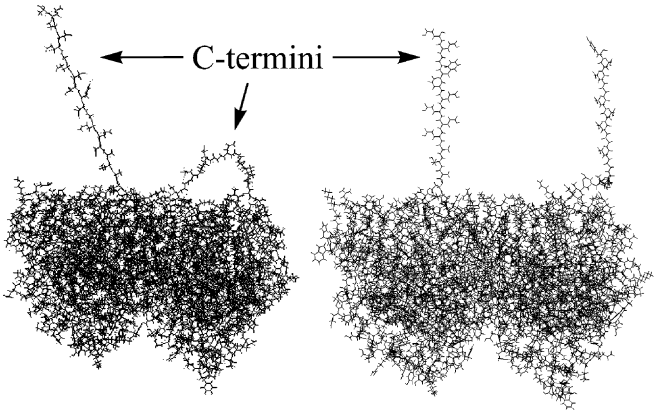

Fig. 3 A reconstruction of the surface of a tubulin dimer with two C-termini attached (based on the PDB and C-terminus sequence. The two states of the C-terminus shown are the result of MD simulation of this system

Frohlich Oscillations

The C-termini of tubulin are strongly negatively charged (having a net charge of up to 11 electrons each) and interact electrostatically with several other charged objects in their vicinity, namely: (a) the surface of the tubulin dimer (which is generally negatively charged, with as many as 16 negative charges per tubulin monomer, but which has a positively charged region(s) on its exposed surface that can bind a C-terminus), (b) with neighboring C-termini and (c) with adjacent proteins such as kinesin or MAP2.

Through the computer simulations that are later described in detail, we have found that the C-termini are likely to exist in two major conformational states: (a) a rather flexible conformation pointing away from the MT surface and subject to thermal fluctuations, or (b) a state in which the C-termini bind to the MT surface. Fig. 3 shows a tubulin dimer where two C-termini configurations are obtained from simulations (not described in this paper); in one of them, both C-termini point upright away from the surface while the other dimer shows one upright C-termini and one bound. It is feasible that bound conformations of C-termini form a regular arrangement on the surface of a MT. Interestingly, a recent paper by Makrides et al. (2003) reports that MAP (tau) binding to neuronal MTs produces a collective change in the MT lattice structure. Their result obtained using AFM (atomic force microscopy) techniques is consistent with our prediction of the existence of a collective state of the C-termini that are bound to the tubulin surface. However, we deliberately avoided any relation to conformational changes of the dimers themselves. This is because the time scale for the collective motions of the C-termini is much shorter than that for the bulk of the protein's conformational motions. Besides, dimer conformational changes (if any) are much smaller with respect to the degrees of freedom available for the C-termini. While this section has dealt with the preliminary application of molecular dynamics simulations to individual tubulin monomers and dimers, the next sections intend to explore the possibility of interactions among the many C-termini of a microtubule as well as between the C-termini and MAPs. Our intention is to examine whether: (a) collective conformational states of these structural elements may be generated, and (b) whether these states can exhibit nontrivial dynamics, e.g. wave propagation. As will be elaborated in the discussion, such collective states may be reversibly switched on and off; this might be of significance to dendritic information processing and axonal trafficking. However, due to the complexity and computational demands of this problem, we are now faced with a need to simplify the modeling effort.

Modeling the C-termini as charged rods

Based on the assumption that C-termini dynamics plays an important role in the functioning of a dendrite, we develop a simplified model of a C-terminus and its interactions with neighboring structures. Analysis of the model indicates the existence of two metastable states that were observed in the full MD simulation performed on a single dimer, and allows further investigation of the interactions with MAP2.

The proposed model of the C-termini microtubule network is schematically illustrated in Fig. 4 where the tubulin dimer is considered to be the basic unit. Each

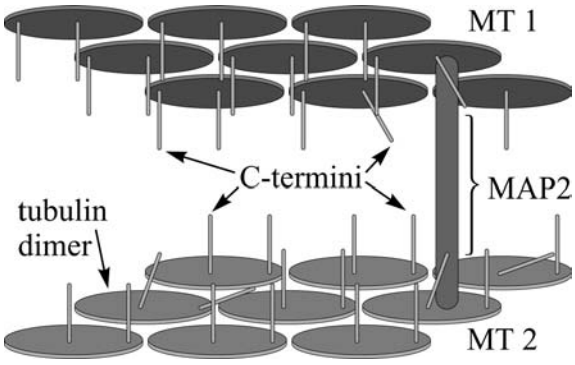

Fig. 4 A schematic diagram of portions of the two microtubules connected by a MAP2. Each portion shows the outer face of a tubulin dimer covered by two C-termini. Each C-terminus can be in one of two metastable states, perpendicular or parallel to the dimer surface

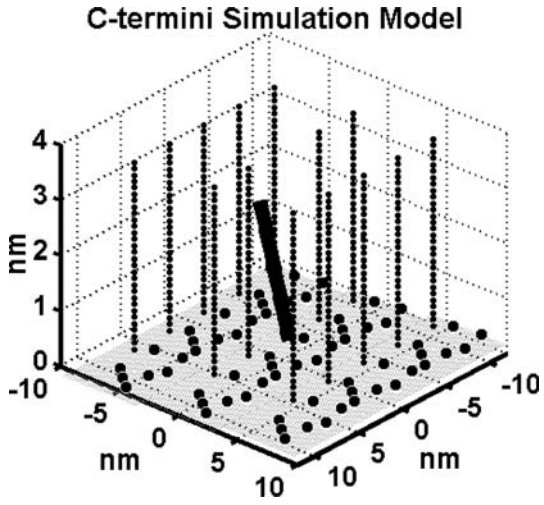

Fig. 5 A portion of the microtubule composed of a few tubulin dimers with their C-termini used as a model for the static simulation. Also shown are a few of the positively charged regions (*circles*) on the surface of a tubulin dimer. The test C-terminus is slanted (*bold*). (Axes are scaled in [nm])

dimer has two C-termini that may either extend outward from the surface of the protofilament or bind to it. The major goal here is to explore how MAP2 binding affects the states of the C-termini, which in turn may affect other sub-cellular processes, the processive motion of kinesin motor proteins transporting mRNA as they walk on the MT.

The most responsive structural elements of the system (i.e. elastic and electric degrees of freedom) can be best understood in terms of the conformational states of the C-termini. Each state of the unbound C-terminus is assumed to evolve such as to minimize the overall interaction energy of the system. Note that, while the exposed surface of the dimer is, on average, highly negatively charged, it exhibits local regions of positive charge that attract the C-termini as thermal fluctuations (or directed fields) may cause them to bend into and bind in what we call a 'down' state. The energy difference between the two metastable states is relatively small, on the order of few k_BT at room temperature, allowing for transitions between the two main conformational states to be thermally activated. We elaborate on this below.

In this preliminary analysis, we map the interaction energy of a C-terminus with its nearest neighbors to investigate its static properties; dynamic properties will be investigated later. Here, for simplicity, we model the C-terminus as a charged rod (of negligible width) that is attached at one end to the surface of the tubulin dimer, and can move otherwise above the surface as a rigid body (see Fig. 5). We describe its location in terms of spherical polar coordinates as: (θ, ϕ) (azimuth, elevation). The main contributions to the interaction energy of the test C-terminus are: (1) its nearest neighbors' C-termini, (2) the negative surface of tubulin dimer below and (3) the positive regions on the surface. Hence, we can describe the total energy of the ith test C-terminus as follows:

$$E_i^{\mathrm{tot}} = \sum_{j \in nn} E_{i,j}^{\mathrm{cc}} + E_i^{\mathrm{cd}} + E_i^{\mathrm{cp}} \qquad (1)$$

Debye length = charge carrier's net electrostatic effect in a solution

The different contributions to the interaction energy are now described explicitly. We use in our calculations the concept of Debye screening due to the presence of ions in a solution (Daune 1999). The electrostatic interaction energy between two C-termini labeled (i, j) can be approximated as:

$$E_{i,j}^{\mathrm{cc}}(\theta_{i,j}, \phi_{i,j}) = K\lambda^2 \int_0^L \int_0^L \mathrm{d}l_i \mathrm{d}l_j \frac{e^{-(D(\cdot)/\lambda_{\mathrm{D}})}}{D(\cdot)\varepsilon_{\mathrm{r}}(D(\cdot))}$$

$$\begin{aligned} D(l_{i,j}, \theta_{i,j}, \phi_{i,j})^2 &= \left(l_i \cos\phi_i \cos\theta_i + x_i - l_j \cos\phi_j \cos\theta_j - x_j\right)^2 \\ &+ \left(l_i \cos\phi_i \sin\theta_i + y_i - l_j \cos\phi_j \sin\theta_j - y_j\right)^2 \\ &+ \left(l_i \sin\phi_i - l_j \sin\phi_j\right)^2 \end{aligned} \qquad (2)$$

where ($x_{i(j)}$, $y_{i(j)}$) is the (fixed) binding location of the C-termini to the surface, $\theta_{i,j}$ is an abbreviation for (θ_i, θ_j), $D(\cdot)$ stands for the distance function, L is the length of the C-terminus, λ is the linear charge density, $K = (4\pi\varepsilon_0)^{-1}$, ϵ_{r} is the relative dielectric constant (which for short ranges in an aqueous solution is 1–80), and λ_{D} is the Debye screening length that depends on the ions and their concentrations in the surrounding solution.

The tubulin dimer has a complex surface both geometrically and electrostatically. For simplicity, we model it as a 2D rectangular area with a certain average charge density σ with several localized positively charged regions whose locations are based on the calculations described above, see Fig. 2. Considering the dimer and one of its C-termini, with the center of the dimer at the origin, the negative surface contribution to the interaction energy is given by:

$$E_i^{\mathrm{cd}}(\theta,\phi) = K\lambda\sigma \int_0^L \int_{-W/2}^{W/2} \int_{-H/2}^{H/2} \mathrm{d}l\,\mathrm{d}w\,\mathrm{d}h \frac{e^{-(D(\bullet)/\lambda_\mathrm{D})}}{D(\bullet)\varepsilon_\mathrm{r}(D(\bullet))}$$

$$D(l,w,h,\theta,\phi)^2 = (l\cos\phi\cos\theta - h - x_0)^2 + (l\cos\phi\sin\theta - w - y_0)^2 + (l\sin\phi + \Delta)^2 \quad (3)$$

where (x_0, y_0) is the binding location to the surface of the C-terminus, H, and W are the height and width of the dimer, respectively, and Δ a is the distance in the z-direction separating the C-terminus and the dimer surface. Similarly, the contribution due to positively charged regions (points) is given by:

$$E_i^{\mathrm{cp}}(\theta,\phi) = K\lambda q_\mathrm{e} \sum_{j=1}^{N_\mathrm{p}} \int_0^L \mathrm{d}l \frac{e^{-(D_j(\bullet)/\lambda_\mathrm{D})}}{D_j(\bullet)\varepsilon_\mathrm{r}(D_j(\bullet))}$$

$$D_j(l,\theta,\phi)^2 = (l\cos\phi\cos\theta - x_j - x_0)^2 + (l\cos\phi\sin\theta - y_j - y_0)^2 + (l\sin\phi + \delta)^2 \quad (4)$$

where N_p is the number of positive charges on the dimer surface, δ is a minimum z-displacement of a positive region with respect to the surface (the residue is, to some extent, exposed) and q_e is the electron charge.

Based on the above contributions, we calculated numerically the interaction energy surface as experienced by a C-terminus. The main assumptions behind this analysis are:

– The empirical estimate of the relative dielectric constant ε_r (Mallik et al. 2002) is linearly approximated by:

$$\varepsilon_r(d) = \begin{cases} (1 + 79 \cdot d)/1.7 & \text{if } d \leqslant 1.7\,\mathrm{nm} \\ 80 & \text{if } d > 1.7\,\mathrm{nm} \end{cases} \quad (5)$$

– Further effects of the ionic solution are taken into account only via the inclusion of the Debye–Huckel factors (Daune, 1999).

The governing equations were evaluated on a $\theta - \phi$ grid defining the inclination of the C-terminus. Fig. 5 depicts the simulation setup where a test C-terminus is rotating with two degrees of freedom (the base is fixed), θ, ϕ denoting the azimuth and elevation, respectively, and the rest of the environment is frozen, i.e. neighboring C-termini, and the dimer's surface, a negatively charged area with specific positive regions. The model parameters used for the numerical evaluation are given as follows. First, we have used the geometrical dimensions of the tubulin dimer as: $H = 8$ nm, $W = 6$ nm. Then, we assumed that approximately 2/3 of the net average charge of tubulin is exposed on the outer surface, which results in the surface charge density of: $\sigma = 12\mathrm{e}^{-}\cdot(4\ \mathrm{nm})^{-2}$. Next, a reasonable value of the linear charge density for the C-termini: $\lambda = 10\mathrm{e}^{-}/L$ (where L is the C-terminus length, $L = 3.5$ nm). The values of the displacement parameters were estimated at $\Delta = 4$ Å, $\delta = 1$ Å.

Figure 6 depicts the interaction energy for several elevation cuts (angles are given in the figure) at azimuth angles, whereas Fig. 7 shows a full surface plot. It transpires that the 'up-state' has the lowest energy. (It corresponds to the C-terminus being perpendicular to the tubulin's surface). However, the cone-angle created by the constraint $E - E_0 < 50$ meV (where $E_0 = E$ ($\phi = 90°$)) is about 40°. This means that the C-termini can move readily within this cone due to thermal fluctuations ($k_\mathrm{B}T$ is approximately 25 meV at physiological temperatures). But an important result is the existence of

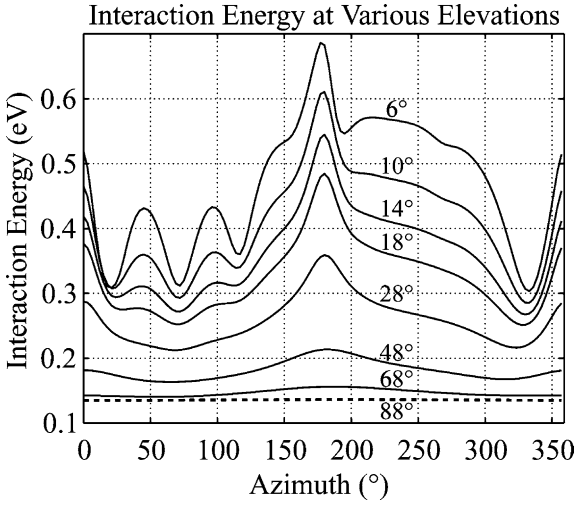

Fig. 6 Evaluation of (1) for the interaction energy between a test C-terminus and the environment vs. the azimuthal angle; each line in the figure describes an elevation cut; values are given

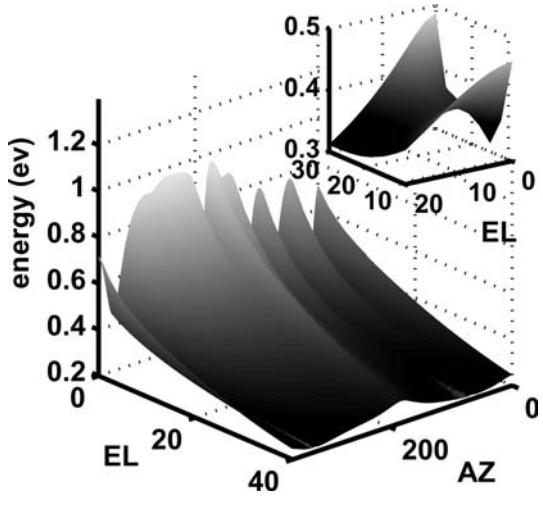

Fig. 7 Energy surface of the interaction between the test C-terminus and the environment. The grayscale is in [eV]. The *inset* shows a zoom of a saddle point on the surface

a local minima associated with the 'down-state' which is only 100 meV higher than the straight-up state, see Fig. 7. A saddle point between the two states is ~160 meV above the 'up-state'. What we deduce here is that, at least under the assumptions of the model, there are two metastable states with a moderate energy barrier separating them. The inset in Fig. 7 shows a close-up near the saddle point.

Finally, we wish to discuss ion condensation that may be expected to take place on MAPs and the C-termini of MTs due to their strong negative charge exposure to the aqueous environment. Many years ago, Manning (1978) postulated an elegant theory that polyelectrolytes may "condense" neutralizing ions from their surroundings. According to Manning's hypothesis, counterions condense along the polymer if a sufficiently high linear charge density is present on the polymer's surface. Thus, a linear polymer may be surrounded by counterions from the saline solution such that co-ions of the salt solution are repelled so that an ion depleted volume is created. The sum of surface charges and associated counterions depends on the counterion's valences and Bjerrum lengths. In general, depending on the charge density and the geometry of the charged surface exposed to ionic solutions (quasi-linear, cylindrical, spherical, etc) some of the ions condense on the surface binding very strongly to the opposite charges on the surface, while a number of ions remain fairly free to move about in the solution still being attracted to the surface. The proportion of the condensed ions to the surface charge can be evaluated for a given situation. The Bjerrum parameter is a phenomenological property of the ion's ability to compensate for the surface charges, and depends on temperature and the solvent. In 1926, Bjerrum developed a modification of Debye–Hückel ion-pair formation theory provided that the ions are small, of high valence, and that the dielectric constant of the solvent is small. The Bjerrum length is the distance at which the Coulomb energy of the screened charges equals $k_B T$, the thermal energy, creating an equilibrium from which the charges do not preferentially move. The cylindrical volume of ionic depletion outside the ion cloud surrounding the polymer serves as an electrical shield. Thus, the 'wire-like' behavior of such a structure is supported by the polymer itself, and the adsorbed counterions. For all practical purposes, these are bound to the polymer. The strength of this interaction is such that even under infinite dilution, counterions are still attracted to the polymer and do not diffuse away from the polymer. Although originally postulated for such polyelectrolytes as DNA, this theory applies to highly charged one-dimensional polymers such as the C-termini of microtubules or MAPs. In recent experiments (Stracke et al. 2002), it was demonstrated under the influence of externally applied electric fields that microtubules drift in the field direction, indicating that not all the protein charge is compensated by bound counterions.

Detailed description of the C-terminal model

In this section, we extend our model in order to simulate the dynamic behavior of a C-terminus. To enable exhaustive simulations, we model the C-terminus as a sequence of beads with flexible connections, instead of the rigid rod, in what is known as the bead-spring model, a common method for simulating polymers. The simulation takes into account the following types of interactions:

Electric field of the dimer

Since the tubulin surface is complicated we used a dense grid that models its charge distribution. The outer surface can be characterized as mainly negative with a few small positive pockets (by visual inspection of our earlier calculations we estimated the existence of 7 such pockets per dimer). The computation of the interaction between the C-terminus and the dimer's surface is highly intensive if done for each update of the position. For example, if we use a 2Å resolution, the number of grid points is (8/0.2)*(6/0.2) = 1,200, assuming a 6×8 nm^2 surface area. If we use a 10-bead model, then the number of point-to-point interactions becomes 12,000. A reasonable number of updates is 10^5–10^6, so this type of interaction becomes a serious bottleneck. To overcome this problem, we assumed that the surface is static and evaluated the field it exerts in a box of (6×8×6) nm^3 volume whose base is the surface. This can be done even in a finer grid than described earlier. Once we have the (static) field vector at the box grid points, we apply real-time linear interpolation for each update to obtain the field at the position of each bead. We assumed a negative charge of $-12e^-$ per tubulin monomer surface, and $+7e^-$ in positive pockets per dimer, hence the net negative contribution to the surface charge distribution is taken to be (−12–7/2) per monomer.

The bead-spring model

A C-terminus has a molecular weight of ~2 kD and is arranged as a long flexible tail. Instead of trying to model its entire secondary structure, we use an approximation where 1–2 amino-acids are modeled by a bead with an equivalent mass while the charge is distributed uniformly among the beads. To maintain the chain, we use the following interactions:

- Coulomb interactions between the electric charges
- The Lennard–Jones potential: Short range forces between all beads whose distance is shorter than a cut-off. This force is responsible for the strong short-range repulsion. The attractive part of this potential is truncated. The equation used is given below:

$$F_{\mathrm{LJ}} = 24(1/r)\left[2(\sigma/r)^{12} - (\sigma/r)^{6}\right] \quad \text{if } r \leqslant R_{\mathrm{c}} \tag{6a}$$

$$F_{\mathrm{LJ}} = 0 \quad \text{if } r > R_{\mathrm{c}} \tag{6b}$$

– FENE: Finite-extensible-nonlinear-elastic springs forces that enforce bond connectivity along the chain. This force is calculated between adjacent beads only, referred to as nearest neighbors (nn).

$$F_{\mathrm{fene}}(r) = \frac{KR_0^2 r}{\left(R_0^2 - r^2\right)} \quad \text{if } r < R_0 \tag{7}$$

Note that a singularity exists at $r = R_0$.

– Angular forces: These are the two types of forces that restrict the angle between every three adjacent beads and the planar angle involving every four adjacent beads, i.e. harmonic angular potential and harmonic dihedral potential. The harmonic cosine potential: (for simplicity, we give here the potential and not the force):

$$U_{\mathrm{angle}} = K_{\theta_a}(\cos(\theta) - \cos(\theta_0))^2 \tag{8}$$

where θ is the angle between every three adjacent beads.

The harmonic dihedral potential:

$$U_{\mathrm{dihedral}} = K_{\theta_d}(1 + \cos(n\varphi - \varphi_0)) \tag{9}$$

where for the quartet $(ijkl)$ we define $\varphi = (ijk)\angle(jkl)$, the angle between the plane spanned by (ijk) and the plane spanned by (jkl).

The other two contributions to the equation of motion are:

– Friction forces

$$F_{\mathrm{f}} = -m\Gamma v, \tag{10}$$

where v is the velocity vector and Γ is the friction coefficient.

– Random forces representing thermal fluctuations denoted by $W(t)$ and satisfying

$$\langle W(t)\rangle = 0, \quad \langle W_i(t)\, W_j(t')\rangle = 6k_{\mathrm{B}}Tm\Gamma\delta_{\mathrm{ij}}\delta(t - t') \tag{11}$$

This combination of friction and noise terms is described in the fluctuation–dissipation theorem and often referred to as Langevin dynamics. The initial conditions adopted in our simulations had C-termini in a straight conformation normal to the surface. We used 600 runs that lasted 500 ps each and they took up 400 ps for relaxation and 100 ps for stabilization of the various resultant conformations that were then statistically analyzed. A detailed description of the model parameters chosen for our simulations is given in Appendix A. Note that the external force is taken in the x-direction with a random projection [–0.2..0.2] in the y-direction. What our simulation indicates here is the ability of an ionic wave to trigger a corresponding wave of C-termini state changes from their upright to downward orientations (see Figs. 8, 9). In the next section, we address the related question whether or not an ionic wave can propagate along a MAP2 that interconnects two neighboring MTs.

In Figs. 9 and 10 we have presented the results of our calculations of the energy minimized positions of the individual beads representing the amino acids of the C-termini in two equivalent forms. The distributions shown in Fig. 10 (top) describe the minimal energy positions of beads representing the constituent residues of a C-terminus ranging from three to the end for each run after relaxation. The energy minimized value can come from each of the beads at positions from 3 to 11. The first bead is permanently connected to the surface and hence has no freedom to move. The surface is taken

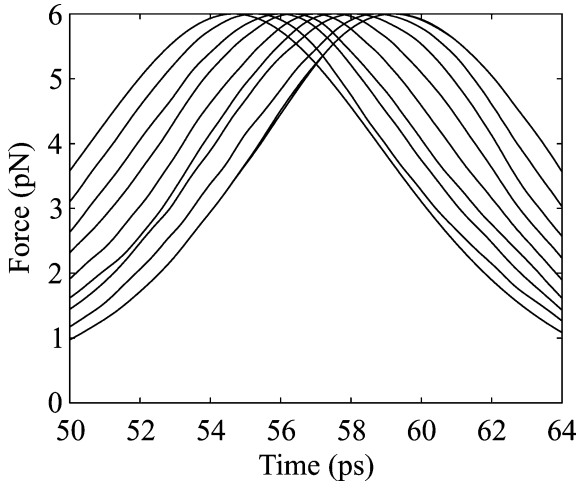

Fig. 8 Plot of the function $F(z(i), t) = \mathrm{amp}\ \mathrm{sech}\ (b(z(i) - vt))^2$ where $z(i)$ is the z-position of the ith bead

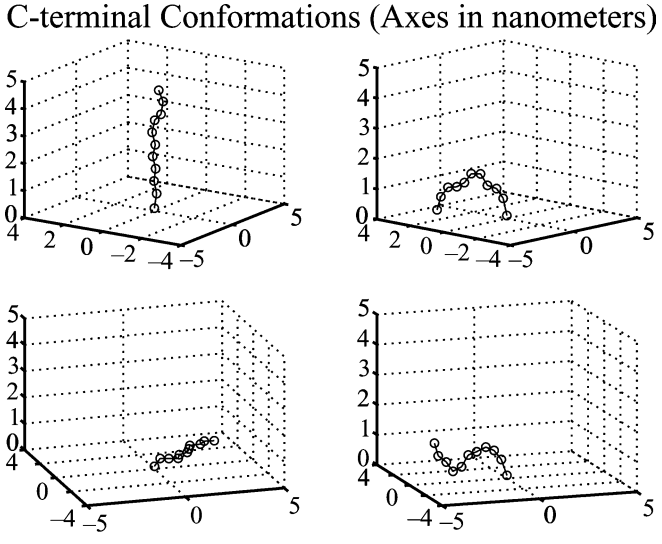

Fig. 9 Four C-termini conformations observed with the flexible model

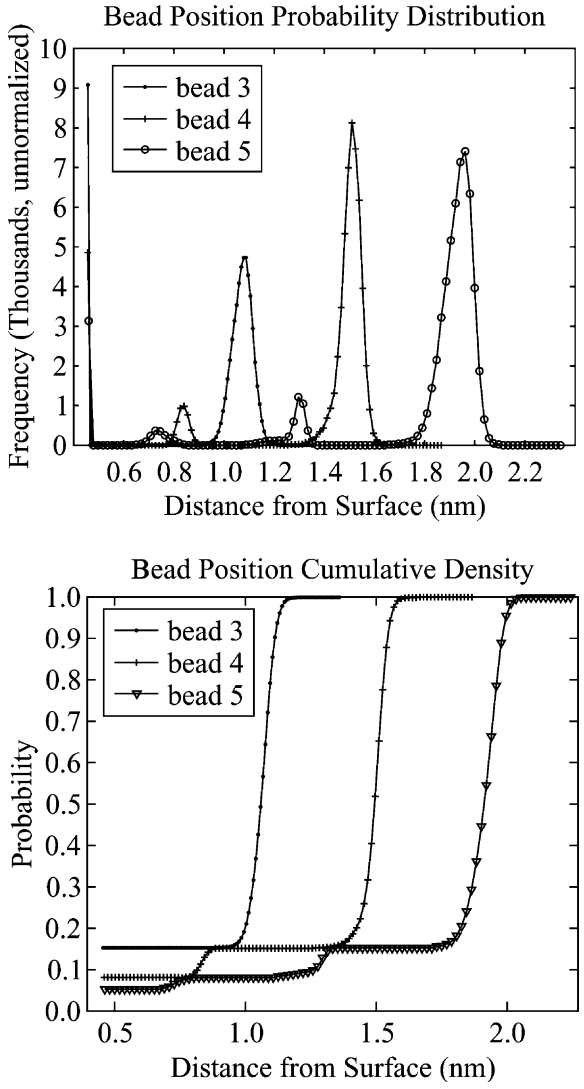

Fig. 10 Top: The histograms shown describe the minimal energy positions of beads representing a C-terminus ranging from 3 to 5 obtained at the end of each run after relaxation. The energy minimized value can come from each of the beads at positions from 3 to 11. The first bead is permanently connected to the surface and hence has no freedom to move. The surface is taken at $z = 0.45$ nm for technical reasons. **Bottom:** The cumulative sum of the probabilities corresponding to the same physical situation as described in Fig. 9 is just another way to present the results. It is shown that the probability of the down position which includes all cases of full or partial attachment is ~15%

to be at the vertical coordinate $z = 0.45$ nm for technical reasons. Figure 10 (bottom) shows the cumulative sum of the probabilities corresponding to the data shown in Fig. 9. It is shown that the probability of the down position which includes all cases of full or partial attachment is ~15%.

Model of an ionic wave propagation along MAP2

The next step in our investigation focuses on the characteristics of MAP2 and its ability to function as a 'wave-guide' that transfers the conformational change in a C-terminus state to an adjacent MT. Following the description of the MAP2 and its electrostatic properties, we describe a simplified model of the interaction between MAP2 and its ionic environment via counterions and their role in propagating the perturbation exerted by an adjacent C-terminus.

The 3D structure of MAP2 is unknown. However, MAP2 binding with MTs was investigated. The binding region is located at the C-terminus and the bond between MAP2 and a MT appears to be electrostatic in nature. Pedrotti et al. (1994) showed that MAP2 binds to MTs in a concentration-dependent manner. Once bound, a high NaCl concentration is not effective in reducing the concentration of MAP2 bound to MTs. During binding, $MgCl_2$ is ten times as effective as NaCl at decreasing MAP2 binding to MTs. The authors concluded that MAP2 may "screen" electrostatic repulsion between tubulin dimers.

Our estimates of charge on the surface of MAP2 are based on a published sequence (Tokuraku et al. 1999) in the Swiss-Prot database (entry P11137: MAP2_HUMAN). We found that the binding domain is composed of roughly the first 418 amino acids so we take the remaining 1,410 amino acids of MAP2b to constitute the projecting domain. Assigning a single positive charge to each Lys and Arg residue, snd and a negative ones to Asp and Glu with His being given an average value of $+1/2$ due to its isoelectric point and leaving the remaining residues neutral, we take the total negative charge of the projecting domain to be $-150\ e^-$ and its charge distribution to be very close to uniform.

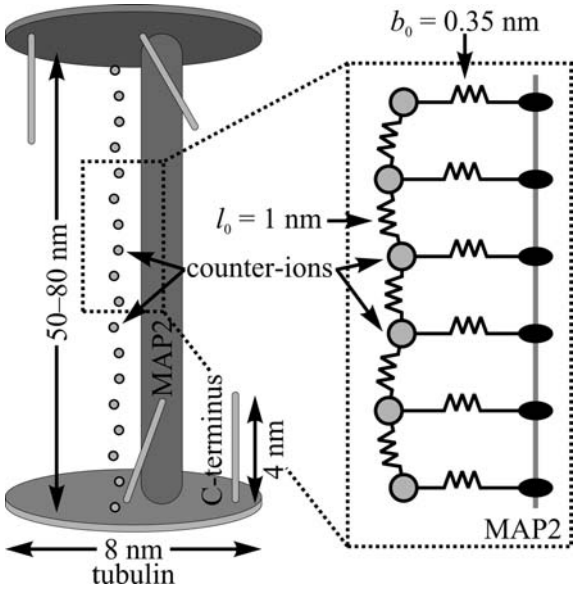

Fig. 11 A schematic model of the MAP2 interacting with positive counterions (*circles*). Also seen are two dimers at both ends of the MAP2 and a slanted C-terminus. The zoom region depicts the springs model used to describe the interactions between the MAP2 and the counterions

We are now able to extend our model to include MAP2 as a supporting structure (backbone) to which counterions are attached. We are aware of the still unresolved controversy regarding the characteristics of the binding between the projecting domain of MAP2 and an adjacent MT. While it is still to be determined if a single MAP2 makes direct physical contact with a MT, it is apparent to us that it at least reaches the immediate vicinity of a given MT. However, our model's results presented below are unaffected by the type of binding present as long as the distance between these two structures is on the order of 1 nm. To simplify the model we assume that the counterion attracting sites are equidistant and arranged in a straight chain along the MAP2. We envisaged that the counterions move only in a plane perpendicular to a cylinder representing the MAP2 (see Fig. 11). We show that a perturbation applied to the counterions at one end of the MAP2 will drive them out of equilibrium and initiate a wave that travels along the MAP2. Subsequently, we derive the equations of motion and investigate the properties of the traveling wave that emerges.

To facilitate the implementation of the model, we use the harmonic approximation and represent the chain as a 2D-spring system in which each counterion is strongly bound to a MAP2 site and weakly bound to its two nearest-neighbor counterions. The main justification for taking only the nearest-neighbor interaction into account is the strong screening effect imposed by the solution (see the previous section for discussion). In addition, we assume external forces of two kinds; the damping viscous force due to temperature fluctuations and a perturbation exerted by a C-terminus while changing its state (details given below). An additional constraint is provided by reflecting boundaries located approximately 0.2 nm from the MAP2 (representing a repelling force).

Our main focus in the following investigation is the characterization of the propagating perturbation along the MAP2. The results reported below were obtained from MD simulations of this model. The model Hamiltonian can be written as follows:

$$H = \frac{M}{2}\sum_i \dot{\mathbf{r}}_i^2 + \frac{1}{2}K^b \sum_i \left(|\mathbf{r}_i - \mathbf{b}_i| - b_0\right)^2 + \frac{1}{2}K^c \sum_i \left(|\mathbf{r}_i - \mathbf{r}_{i-1}| - l_0\right)^2 + W(\mathbf{r}) \tag{12}$$

where we have assumed a harmonic potential for the electrostatic interaction with K^b and K^c denoting the force constants of the MAP2–counterion and counterion–counterion interactions, respectively, $\mathbf{r}_i$ is the position of the counterion, $\mathbf{b}_i$ is the ith MAP2 binding site, b_0, l_0 are the equilibrium chain distances, M is the counterion's mass and $W(\mathbf{r})$ describes the site-dependent potential (which for the purpose of this calculation is ignored). Using the following notation, we obtain

$$\begin{aligned} \mathbf{u}_i^{\mathrm{CM}} &= \mathbf{r}_i - \mathbf{b}_i, \quad u_i^{\mathrm{CM}} = ||\mathbf{u}_i^{\mathrm{CM}}||, \quad \hat{\mathbf{u}}_i^{\mathrm{CM}} = \frac{\mathbf{u}_i^{\mathrm{CM}}}{u_i^{\mathrm{CM}}}, \\ \mathbf{u}_i^{\mathrm{CC}} &= \mathbf{r}_i - \mathbf{r}_{i-1}, \quad u_i^{\mathrm{CC}} = ||\mathbf{u}_i^{\mathrm{CC}}||, \quad \hat{\mathbf{u}}_i^{\mathrm{CC}} = \frac{\mathbf{u}_i^{\mathrm{CC}}}{u_i^{\mathrm{CC}}}, \end{aligned} \tag{13}$$

where the symbols CM and CC label 'counterion–MAP' and 'counterion–counterion' interactions, respectively. The corresponding equations of motion for the counterions are given as follows:

$$\begin{aligned} M\ddot{\mathbf{r}}_i = &-K^b\left(u_i^{\mathrm{CM}} - b_0\right)\hat{\mathbf{u}}_i^{\mathrm{CM}} \\ &- K^C\left[\left(u_i^{\mathrm{CC}} - l_0\right)\hat{\mathbf{u}}_i^{\mathrm{CC}} - \left(u_{i+1}^{\mathrm{CC}} - l_0\right)\hat{\mathbf{u}}_{i+1}^{\mathrm{CC}}\right] - \gamma\dot{\mathbf{r}}_i + \mathbf{f}_i \end{aligned} \tag{14}$$

where damping and the external force, $\mathbf{f}$, have been explicitly introduced. In order to solve these equations, we applied the velocity-Verlet algorithm for time integration (Swope et al. 1982).

Let us now describe the rationale and assumptions underlying the choice of parameters in the model, $K^{b(c)}$, b_0, l_0, M and γ. The equilibrium distance between counterions is dictated by our assumption about the average distance between binding sites. As discussed above, from the analysis of the residues of MAP2 we infer that there are approximately 150 electronic charges. Assuming that MAP2 is a 50-nm rod-like structure, we obtain a charge distribution of about $3e^-$ per nanometer along the MAP2 axis (length estimation is obtained from average measurements of the MAP2). Since our model actually represents a two-dimensional structure perpendicular to the MAP2 surface, we assume the average charge separation to be $l_0 = 1$ nm. The bond length for typical counterions in ionic solution is 2.5–4 Å; in our simulations we therefore used $b_0 = 3.5$ Å. The representative mass of the counterion chosen here is an average between sodium and calcium ions, $M = 30$ amu, or approximately 5×10^{-26} kg. The damping factor γ has units of amu ps^{-1}; in our simulations we have used $\gamma = 1$ amu ps^{-1} which is typical in molecular dynamics simulations. The MAP2-counterion bond force constant takes a typical value of $K_b = 100$ kcal mol^{-1} Å^{-2} and the counterion–counterion force constant is on the order of $K_c = 0.3$ kcal mol^{-1} Å^{-2}. The former estimate is comparable to the strength and distance involved in the formation of a monovalent ion's bond with a protein. The choice of the second force constant reflects a weak Coulomb interaction between adjacent counterions. At any rate, it is the ratio of these two constants that is more important in our calculations than their absolute values.

The interaction between the MAP2 and the C-terminus actually includes a layer of counterions surrounding each of these structures. In our model, the C-terminus is located 0.4 nm from the MAP2-base and is initialized at a given value of (θ, ϕ). It then falls at a certain angular velocity until it reaches the tubulin surface. We apply MD updating steps to the counterions only, since the C-terminus in this case merely serves as a

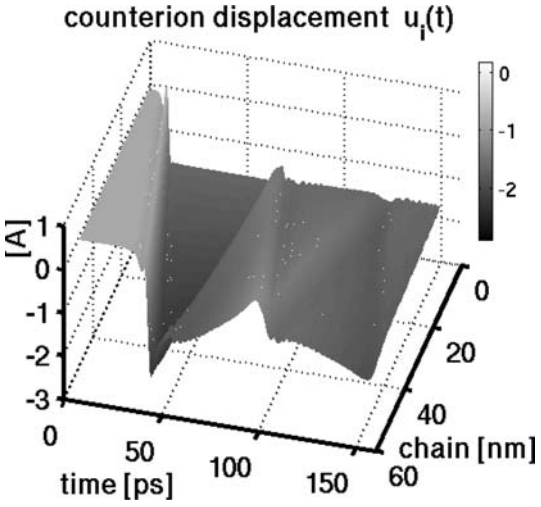

Fig. 12 Results of the MAP2 model simulation describing the counterions displacement (the chain-axis denotes the discrete indices of the counterions). The grayscale is given in [Å]

source of perturbation. In parallel, we evaluate the properties of the wave generated by this perturbation. A typical initial orientation is $\theta = 180°$, $\phi = 50°$ and the C-terminus collapses over a characteristic time of 10 ps.

The following figures show several properties of the wave propagation along the chain of $N = 50$ counterions. Figure 12 depicts the counterion displacement parallel to the MAP2, u_i, where i denotes the ith counterion. The counterions near the MAP2-base are attracted by the C-termini and the displacement is effectively negative. The wave propagates along the MAP2 and then is reflected back from the other end; this process may re-occur a few times. The amplitude of the reflected wave is attenuated and depends on the value of the damping factor γ. The asymmetry between the two edges is mainly an artifact of the experimental setup in which the C-terminus is located only at one end; hence the other end has fewer

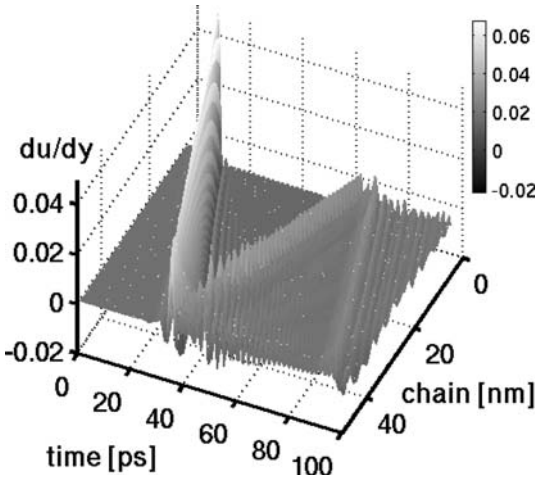

Fig. 14 Spatial derivative of the counterions displacement evaluated using (15b)

constraints and exhibits more fluctuations. The next two figures, Fig. 13 and Fig. 14, present the first derivatives of the displacement with respect to the time and space, respectively. More specifically, the temporal and the spatial derivatives are estimated from the simulation updating steps as follows:

$$\frac{\mathrm{d}u_i(t)}{\mathrm{d}t} \approx \frac{u_i(t+\Delta t) - u_i(t)}{\Delta t} \tag{15a}$$

$$\frac{\mathrm{d}u_i(t)}{\mathrm{d}y} \approx \frac{u_{i+1}(t) - u_i(t)}{\Delta y_i} \tag{15b}$$

where Δt is the simulation time step, Δy_i is the distance between two adjacent counterions and the spatial derivative is obtained only in the y-direction (parallel to the MAP2 axis).

The profiles of the displacements and their derivatives reveal the following interesting result. The projection of

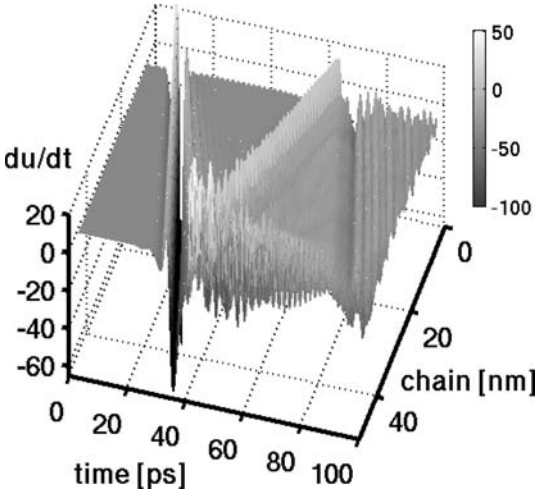

Fig. 13 Time derivative of the counterions displacement evaluated using (15a). The grayscale is given in units of [Å ps^{-1}]

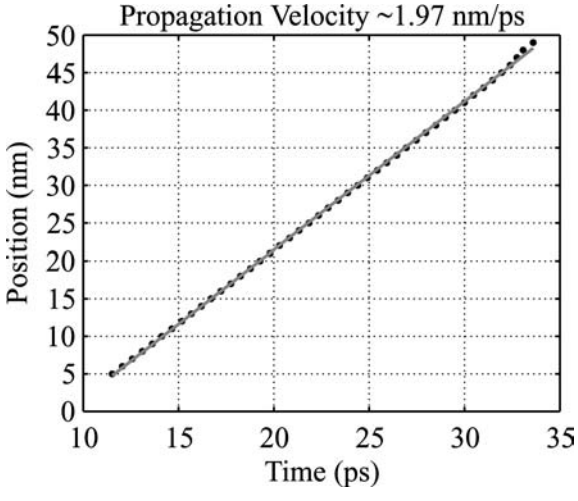

Fig. 15 Evaluation of the phase velocity of the wave propagation along the MAP2 using (16)

the initial perturbation along the MAP2 propagates almost as a "kink", mainly due to the large K_b/K_c ratio; only after the second reflection can one observe a significant spread of the packet. Determination of the phase velocity is done by evaluating the following quantity for all sites:

$$\{t_i: u_i(t) = u_0,\ u_i(0) = 0\} \tag{16}$$

i.e. we find the set of time values $\{t_i\}$ for which the displacement of the ith counterion is u_0 for the first time. We measure the first passage time for all sites from which the propagation velocity is evaluated. Figure 15 shows the results for the above parameters with a velocity estimation of $v_{ph} \approx 2$ nm ps^{-1}. Finally, it is worth mentioning that the velocity scales with the square root of K_c. More details of this analysis will be published separately.

Summary and discussion

In this paper, we have modeled static and dynamic properties of the C-termini of tubulin dimers that comprise neuronal microtubules. We have used various computational tools including molecular dynamics in order to explore the dynamic behavior of conformational states of a C-terminus of tubulin within a given microtubule and assumed that each C-terminus interacts via screened Coulomb forces with the surface of a tubulin dimer, with neighboring C-termini and also adjacent MAP2s. Each C-terminus was shown to either bind to the tubulin surface via one of the several regions we identified as positively charged or to explore the space available in the solution surrounding the dimer fluctuating above the tubulin surface with higher probabilities for conformations with a larger inclination angle. We found that the preferential orientation of each C-terminus is away from the tubulin surface but binding to the surface may also take place at a considerably lower probability. The static properties of the C-terminus were studied by analyzing the energy surface of a system composed of a test C-terminus modeled by a charged rod interacting with the tubulin dimers below and the neighboring C-termini. This analysis confirmed the hypothesis of two major states where the volume described by cone-angles above 50 degrees has energy differences of ~2KT at room temperature; we also found a clear barrier of ~6KT between the two major types of states. We further evaluated the statistical properties of the two major states by studying the dynamic properties of a polymer-like bead-spring model of the C-terminus. The results indicate that the system has indeed two major states with a strong bias towards the stretched-up state. In other words, the system tends to be mostly in the 'up-state' unless driven towards one of the surface binding sites where the probability of attraction increases. The statistical bias favoring the up-state versus the down state of C termini may be tipped depending on the actual physiological conditions, in particular the presence of local polarizing fields and/or screening effects. This may be affected by many processes, for example the effect of actin polymerization in the vicinity of a given microtubule or even by ordering of water molecules surrounding the MT surface.

Finally, we studied the properties of ionic wave propagation from one microtubule to another along MAP2. The trigger for this wave may be a near by C-terminus changing state or ionic flux, e.g. from an adjacent ion-channel. Under the noncondensed counterions assumption, we derived the equation of motion for a one-dimensional chain of counterions along the MAP2 main axis and evaluated the properties of such a wave. It was found that the wave may propagate at a speed of ~2 nm/ps (=2,000 m/s) with a small amplitude attenuation. The results of our model suggest that perturbations generated by the C-termini interactions with counterions surrounding MAP2 may propagate over distances greater than those between adjacent microtubules. Interestingly, a recent paper by Ripoll et al. (2004) has proposed an important integrative role for ion waves guided by protein polymers in providing a new means of signal transduction in eukaryotic cells. Our model offers a quantitative description of such a possibility in neurons.

Below, we briefly discuss the implications the current dynamic modeling may have on synaptic activation and potentiation. The results of the molecular dynamics modeling performed here raise the possibility of transmitting electrostatic perturbations collectively among neighboring C-termini and from C-termini located on one MT to those C-termini on another MT via a MAP2 that physically interconnects MT's in dendrites. Collective conformational states of C-termini and a transmission of perturbations among the C-termini was shown to be a viable mode of behavior in dendritic microtubules. This may suggest the possibility of a hitherto unexplored information processing mechanism operating at a subneuronal level is clear. For example, MAP2 and kinesin bind near to the C-termini on tubulin and the conformational states of C-termini affect this binding (Al-Bassam et al. 2002; Thorn et al. 2000). Hence these conformational states of the C-termini must at some point be taken into account in order to understand neural processing that depends on transport of synaptic proteins inside of neurons (see Fig. 1). In this connection, Kim and Lisman (2001) have shown that inhibition of MT motor proteins reduces an AMPA receptor-mediated response in hippocampal slice. This means that a labile pool of AMPA-receptors depends on MT dynamics, and MT motors determine the amplitudes of excitatory postsynaptic currents (EPSC).

Furthermore, a walking kinesin carries with it a protein or an mRNA molecule. Since kinesin binds to a MT on a C-terminus, as it steps on it, it brings the C-terminus to the MT surface and makes it ineffective in binding for the next kinesin over a period of time that it takes the C-terminus to unbind and protrude outside. From this, we can hypothesize that long stretches of

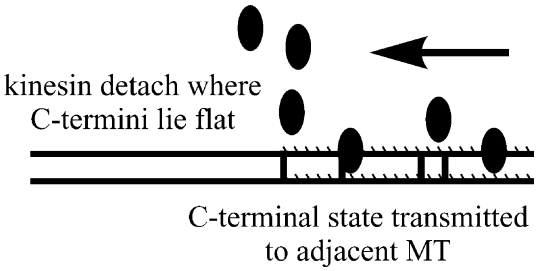

Fig. 16 C-termini that stand upright will increase kinesin processivity, whereas C-termini that lie flat will favor kinesin detachment. This could serve as a mechanism for releasing cargo near to activated synapses. The state of C-termini can be transmitted from one MT to a neighboring MT, in theory. Significant transport toward, and detachment near, active synapses could occur

C-termini in the upright position are more efficient at transporting kinesin and kinesin cargo while C-termini that lie flat may lead to a detachment of kinesin from the MT's (see Fig. 16). Thus, in considering the trafficking of many kinesins, collective conformational effects of C-termini become more and more important. While this is outside the scope of this paper, we may speculate even further and put forth a hypothesis that dendritic processing leading to axonal spikes at the axon hillock (an event not completely explained by membrane signaling) may be regulated or at least affected by the dynamic states of the C-termini in essence leading to a complex dynamic system within a neuron

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Appendix A

The choice of model parameters

- The basic units in our simulation were selected as follows:
  - Mass: 1 amu = 1 D = 1.66×10^{-27} kg
  - Time: 1 ps (1×10^{-12} s)
  - Charge: 1 e^- (1.6×10^{-19} C)
  - Length: 1 nm (hence, velocity is 1 nm/ps)
  - Energy: kJ/mol
  - Force: kJ/(mol · nm)
- The mass of the molecular chain of beads is estimated as 2 kD (summing up the amino acids of the C-terminus of a given human tubulin isotype), or 2000 unit masses. For simplicity we assume identical beads, i.e. for N-beads model the weight of each bead is 2 kD/N. We actually take N = 11, however the first bead is totally attached to the surface.
- The net charge of the C-terminus is assumed to be −10e, again taken from the same isotype of tubulin.
- The nominal length of the C-terminus is taken to be $L = 4.5$ nm.
- The Lennard–Jones parameters are:
  - $\epsilon = 1$ (however recall the truncation)
  - $\sigma = 0.45$ (in nm units)
  - $R_c = 2.0^{1/6}\sigma$
- FENE parameters:
  - $K = 20$–30
  - $R_c = 1.5\sigma$
- Harmonic angle force parameters:
  - $K_{\theta_a} = 40$ (the force constant)
  - $\theta_0 = 135°$ or $3\pi/4$
- Dihedral harmonic angle force parameters:
  - $K_{\theta_d} = 4$ (the force constant)
  - $\phi_0 = 0°$ (or π)
  - $n = 2$ multiplication factor
- The friction coefficient used: $\Gamma = 0.03$
- The noise was taken from a (3D) uniform distribution $[-W,...,W]$ $W = 20$;

Note that the friction coefficient, Γ, enters the equation of motion as part of the fluctuation-dissipation terms in the Langevin dynamics. We use the random and friction forces to represent a virtual solution without explicitly taking into account the particles in the solution. The actual parameters of the simulation are determined from the (room) temperature, the Lennard–Jones constants and the bead mass. The friction is determined as a function of the system's time unit and the time steps of the numerical integration. In our case, we were interested in maintaining the higher vibrational frequencies, so the decay time was kept longer (determined from friction). The amplitude of the stochastic force then follows the Langevin constraints.

External force representation

The external force is simulated as a traveling localized wave with a certain amplitude, as follows: $F(z(i), t) =$ amp $\mathrm{sech}(b(z(i) - vt))^2$ where $z(i)$ is the z-position of the ith bead

Figure 8 shows the actual force on each one of the beads. The parameters of the wave were taken to be:

- $v = 0.8$ [nm/ps]
- amp = 6 [kJ mol^{-1} nm^{-1}]
- $b = 0.25$

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Transmission of the Quantum Interaction of Erythrocytes

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Abstract

The previously demonstrated long-range quantum mechanical interaction between human erythrocytes suspended in plasma also occurs with artificial media. Certain macromolecules dissolved in phosphate-buffered saline will transmit the interaction provided their concentration is above a minimum. Extended macromolecules transmit the interaction, whereas a compact macromolecule (albumin) does not.

Index Entries: Erythrocytes, quantum interaction between; quantum mechanics, and erythrocyte interactions; macromolecules, and quantum erythrocyte interactions; ultra-long-range force, between erythrocytes.

Introduction

We have shown that human erythrocytes interact when less than a few micrometers apart (*1–3*) and that this interaction has features in common with Fröhlich's (*4, 5*) theory of ultra-long-range coherent quantum mechanical interactions in biological systems. The electric field of the membrane potential is about ten million volts per meter, which polarizes the molecules constituting the membrane. The polarized molecules interact, giving rise to modes of vibration that Fröhlich (*5*) calculates to have frequencies of the order of 10^{11} Hz, at which frequency quantum mechanical properties become significant. In the very strong electric field, conformational change in membrane molecules and the readjustment of counterions will give rise

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to metastable states of polarization that are analogous to the states of "population inversion" in the photon laser, though within the membrane the quantum particles are phonons. The theory asserts that if the membrane is supplied with metabolic energy above a minimum level, membrane phonons will trigger into a laser-like coherent mode of vibration that can give rise to forces between membranes, organelles, and macromolecules extending over a range of several micrometers (*5*).

We made measurements of the Brownian movement of human erythrocytes in blood that showed that the movement is no longer random when two erythrocytes are closer than three cell radii (*2, 3*). The interaction disappears when the cells are killed and it disappears *reversibly* when the membrane potential is abolished or when the cells are metabolically depleted. Moreover, there is some evidence from experiments on interspecies erythrocyte rouleau formation (*6*) that the interaction is specific (*2, 3*). At low frequencies, ionic screening would limit the range of a polar-wave interaction to a few nanometers, but Fröhlich demonstrates (*5*) that at 10^{11} Hz ionic screening is insignificant up to distances of micrometers. Pohl (*7, 8*) has shown that dividing cells will attract $BaTiO_3$ particles (dielectric constant, 4000) more readily than $BaSO_4$ particles (dielectric constant 11). He names the process "microdielectrophoresis" (*9*), and attributes the interaction to cellular electromagnetic vibrations of the Fröhlich type, but at a lower frequency. With erythrocytes (murine) he found no interaction. These experiments of Pohl (*7*) were done with cells suspended in artificial media of very low salt concentration, which enhances electromagnetic transmission. Our experiments were done with the erythrocytes suspended in their own plasma. When the cells were suspended in Ringer-Locke solution or phosphate-buffered saline (high salt content similar to that of plasma) there was no interaction. In the experiments described in this paper, we have tested the hypothesis that macromolecules are necessary to transmit the quantum interaction of human erythrocytes. The implication it that this interaction is not a direct polar wave attraction of the type demonstrated by Pohl (*7, 8*).

Methods

The methods and the Brownian theory (*10*) employed in the calculations have been fully described (*3*). The observed process is Brownian movement of erythrocytes, on which the Fröhlich interaction is superimposed. Provided there is an adequate concentration of certain macromolecules in the suspending medium, red cells that come into contact form themselves into easily recognized columns of cells called rouleaux. The macromolecules under study are polyvinylpyrrolidone (PVP-360, average molecular weight 360,000; Sigma Chemicals Co., St. Louis, MO), dextran (Dextran 70, molecular weight 60,000–90,000, ICM, Cleveland, OH), human fibrinogen prepared by the Cohn fractionation method, and human serum albumin (Connaught Laboratories, Willowdale, Ontario). With the exception of albumin, the macromolecules are known rouleau-inducing agents. The macromolecules were made into solutions of various concentrations in phosphate-buffered saline (PBS: composition 150 m*M* NaCl, 7.3 m*M* Na_2HPO_4, 2.8 m*M*

NaH_2PO_4, 2.2 m*M* $CaCl_2$, 4.6 m*M* $MgCl_2$, 310 mosmol, pH 7.4 at 22°C). The viscosity of the solutions was measured with a cone and plate viscometer (Synchro-Lectric, Brookfield, Stoughton, MA). At the start of an experiment the solution was filtered through a 0.3 μm Millipore filter and human blood from a finger prick was added to give a cell concentration (v/v) of ~0.2%. The suspension was stirred and quickly transferred to a siliconized (Dow Corning, Midland, MI: Sylgard 184 or R4-3117) hemacytometer chamber (Spencer Bright-Line, American Optical Corp., Buffalo, NY). The chamber was sealed with silicone grease to prevent evaporation and photographed with a time-lapse ciné-camera on an Ortholux microscope (Wild-Leitz, Rockleigh, NJ) at about 60× magnification. An exposure was taken every 15 s for 1 h.

Analysis of Results

The observed process is an aggregation of cells undergoing Brownian movement on which is superimposed a specific interaction at a distance. The cells sediment to the floor of the chamber and if the motion is purely Brownian the kinetics of rouleau-formation is given (*3*, *10*) by

$$\frac{1}{N_t} - \frac{1}{N_o} = \frac{2kT}{\eta} t \qquad [1]$$

where k is Boltzmann's constant, T is the absolute temperature, and η is the viscosity of the solution. The developed ciné-film was projected with a Photo-optical Data Analyser (Model 224A, L-W International, Woodland Hills, CA) and in the earliest frame at which the cells had settled the total number of cells and rouleaux (N_o) was counted in an area defined by the hemacytometer grid. The count was repeated (N_t) every fifth frame (75 s) and the regression of N_t^{-1} on t was computed. Correlation coefficients averaged 0.98.

Equation [1] is derived on the assumption that the diffusion coefficient is given by the Stokes-Einstein relation

$$D = \frac{kT}{6\pi\eta r} \qquad [2]$$

where the denominator is the Stokes' drag coefficient

$$\mu = 6\pi\eta r \qquad [3]$$

We have done an extensive series of measurements on the drag coefficients under the conditions of our experiments (*2*, *3*, *11*,). These demonstrate (*3*) that, in plasma, the experimental value is 1.78 ± 0.06 (SEM) times greater than the theoretical value, and they imply that cells in contact with a surface experience an added hydrodynamic drag (*11*). Accordingly, the rate constant in Eq. [1] has to be divided by the ratio of the experimentally determined to the theoretical drag coefficient. The drag coefficient is measured in each experiment from the Brownian movement of isolated cells (*2*, *3*) and for each experiment a Brownian motion rate

constant is calculated. This is the factor $2kT/\eta$ in Eq. [1] corrected for the additional hydrodynamic drag. For a viscosity of 1.7×10^{-3} Pas, this rate constant has a value of $2.6 \times 10^{-18} m^3 s^{-1}$. For convenience we use an interaction coefficient Ξ which we define to have a value of unity when representing the Brownian motion rate constant. Experiments on dead cells and microspheres (*2, 3*) demonstrate values for Ξ that do not differ significantly from unity. It is shown in reference (*3*) that if r is the erythrocyte radius, then $Z = \Xi r$ where Z is the radius of the zone of attraction around the erythrocyte.

Results

The results are given in Table 1. Concentrations of macromolecules are given in column 1 and the viscosities of the suspending media in column 2. The measured drag coefficients in column 3 are used to calculate the Brownian motion rate constant as explained under Methods. The interaction coefficients (Ξ column 4) are expressed as multiples of this Brownian motion rate constant. If $\Xi > 1$ an attractive interaction is implied and $\Xi < 1$ implies a repulsion. It will be seen that with

TABLE 1
Results

Concentration of macromolecule, g/100 mL	Viscosity of suspension, $\eta \times 10^3$ Pas	Drag coefficient $\mu \times 10^7$ Nsm^{-1}	Interaction coefficient, Ξ	No. of experiments
Polyvinylpyrrrolidone				
0.05	1.06	3.05	0.67	2
0.1	1.18	3.30	1.75	2
0.2	1.39	2.47	2.92	2
0.4	1.85	3.01	1.94	2
0.6	2.23	4.46	2.98	2
Dextran				
<1.0	1 to 1.13	2.75 ± 0.60[a]	0.23 ± 0.11[a]	12
2.0	1.51	2.38	1.66	2
4.0	2.11	2.84	1.60	2
6.0	2.87	5.27	3.10	2
Fibrinogen				
0.1	1.70	3.98	0.16	1
0.3	1.97	2.50	3.74	2
0.4	2.00	2.42	4.35	2
Albumin				
4.0	1.24	2.48	0.10	2
6.0	1.45	3.73	0.18	2
8.0	1.63	3.08	0.18	2

[a]Standard deviation.

PVP concentration ≥ 0.1 g%, Dextran ≥ 2.0 g%, and fibrinogen ≥ 0.3 g%, Ξ is greater than one. The concentrations of fibrinogen that showed an interaction were in the physiological range, whereas albumin at physiological concentrations gave no indication of an attraction.

In the previously described experiments (*3*) that were all done in plasma, rouleaux formed even when the cells were metabolically depleted or with a zero membrane potential. The counts of (N_o) and (N_t) were unequivocal, although the magnification and resolution used was low to get sufficient cells in the field of view for statistical analysis. With low concentrations of PVP, dextran, and fibrinogen, and with all concentrations of albumin, rouleaux do not form, but a few cells stick together. These are identified by running the film quickly back and forth to check that they indeed remain together though human error may occur because the counting process is long and tedious. For instance, we did pairs of experiments at six different concentrations of dextran between 0.1 and 1.0 g%, but the values of Ξ were distributed randomly. Hence the presentation of these results in the table as a mean and standard deviation. It is possible that there is a threshold concentration above which the quantum interaction appears as a step function. This hypothesis could only be tested by many more experiments, work that will not be done until automated methods of film analysis are available.

The drag coefficients should, of course, change uniformly as the viscosity and the density of the suspending medium increases, but this was not so. In early experiments with plasma we encountered difficulty in getting uniform drag coefficients, but the problem was overcome (*3, 11*). Further work on surface interaction when employing solutions other than plasma should improve the accuracy of the results. Nevertheless, it is important to remember that the drag coefficient was *measured* in each experiment and that the interaction coefficient is a measure of any deviation from the *observed* Brownian motion of isolated cells.

Discussion

Rowlands et al. (*1–3*) have shown that normal human red cells in their own plasma form rouleaux significantly faster than Brownian motion predicts. Varying the membrane potential and the cellular ATP content shows that the interaction is compatible with the major predictions of Fröhlich's theory (*4, 5*) of coherent excitations in biological systems. Cells suspended in phosphate-buffered saline do not interact. The present results show that extended macromolecules must be present in sufficient concentration in the suspending medium, a point that was not predicted by Fröhlich's theory in its present form. PVP and dextran are electrically neutral and fibrinogen carries a net negative charge. Albumin, a compact globular molecule, does not transmit the interaction even when its concentration is at or above the upper limit of normal in plasma. This suggests that the secondary or tertiary structure of the macromolecule determines whether or not it will transmit the Fröhlich interaction.

Human erythrocytes, in addition to having a membrane potential, carry a negative surface charge (*12, 13*) that keeps the membranes apart when the cells are sus-

pended in isotonic solutions. The surface charge is neutralized by counterions up to very short distances of approach, quite invisible in our microscope, of necessity used at low magnification. Brownian movement can overcome a small repulsion and two such erythrocytes may stick, but if they do not, Brownian motion will cause them to separate. This process is monitored by running the ciné-film back and forth (see Results). Nevertheless, with the experiments performed as at present, firm conclusions cannot be drawn from the magnitude of values of the interaction coefficient which are substantially less than unity as exemplified by the results for low concentrations of dextran.

In the plasma experiments (*2, 3*) there was a considerable spread in the values for Ξ for normal cells. This is not unexpected. There is a considerable spread in the area and volume of the erythrocytes in a single sample of human blood (*14*) and an even larger spread in the membrane permeability for individual cells to certain nonelectrolytes (*15*). Therefore, the Fröhlich force is unlikely to be identical between every pair of erythrocytes. And so the variations in the values of Ξ for PVP $\geqslant 0.1$ g% probably reflect the variations noted in the plasma experiments (*3*).

In spite of the reservations discussed above, it can be concluded that the quantum mechanical interaction between human erythrocytes can only be transmitted in the presence of a sufficient concentration of extended macromolecules. Pohl (*7, 8*) has demonstrated an attractive force toward mitotic cells in suspending media devoid of macromolecules, but only if the medium has a very low ionic strength. He did not find an effect with murine erythrocytes.

Our experiments were done at physiological ionic strengths and indeed the original experiments were done in plasma itself (*1–3*), so there is no incompatibility between Pohl's findings and our own. Fröhlich has reviewed (*5*) the evidence for the occurrence of vibrations of the predicted frequency in microorganisms and in cells in tissue culture. Pohl's work and ours demonstrate the predicted ultra-long-range interaction and so all the major features of Fröhlich's theory of coherent excitations in biological systems have been confirmed.

Acknowledgments

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PLOS ONE

RESEARCH ARTICLE

Tubulin's response to external electric fields by molecular dynamics simulations

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Abstract

Tubulin heterodimers are the building blocks of microtubules and disruption of their dynamics is exploited in the treatment of cancer. Electric fields at certain frequencies and magnitudes are believed to do the same. Here, the tubulin dimer's response to external electric fields was determined by atomistic simulation. External fields from 50 to 750 kV/cm, applied for 10 ns, caused significant conformational rearrangements that were dependent upon the field's directionality. Charged and flexible regions, including the α:H1-B2 loop, β:M-loop, and C-termini, were susceptible. Closer inspection of the α:H1-B2 loop in lower strength fields revealed that these effects were consistent and proportional to field strength, and the findings indicate that external electric fields modulate the stability of microtubules through conformational changes to key loops involved in lateral contacts. We also find evidence that tubulin's curvature and elongation are affected, and external electric fields may bias tubulin towards depolymerization.

Introduction

α- and β- tubulin heterodimers spontaneously assemble end to end to form protofilaments, and the helical arrangement of 13 protofilaments constitutes microtubules that are central to cellular rigidity, division, motility, and trafficking of intracellular proteins [1–4]. As the driving force for sister chromatid segregation in mitosis, microtubules have long been targeted by chemotherapies using pharmacological agents that stabilize MTs such as paclitaxel, and those that destabilize them such as vincristine, and vinblastine [5,6]. More recently, microtubules have become the target of a novel treatment modality: electric fields. Alternating electric fields at 2.5 V/cm with a frequency of 100–300 kHz, known as Tumor Treating Fields (TTFields), disrupt microtubules *in vitro*. While no definitive mechanistic model of their action has been established, a putative explanation claims that this is due to the microtubules' large intrinsic charge and dipole [7–10] which interacts with electric fields and their gradients. In a major development in the area of cancer therapy, Optune®, a device for administering the fields to patients, was recently approved by the FDA to treat glioblastoma [11–13].

Compared to TTFields, external electric fields (EEFs) of much greater strengths but shorter timescales can also eliminate cancer cells *in vitro* and *in vivo*. Nanosecond pulsed electric fields

(nsPEFs)—static EEFs of tens of kV/cm that are applied repeatedly over nanosecond durations [14,15]—have demonstrated anti-cancer potential against Jurkat and HL-60 cells *in vitro* at 300 kV/cm [15], papillomas and squamous cell carcinoma *in vivo* at 40 kV/cm [16], and many other cancer models [17–23]. Several mechanisms have been proposed to explain nsPEFs' effects, but their creation of nanopores, which allow an influx of Ca^{2+} from extracellular and intracellular sources, and reduction of mitochondrial membrane potential have received the most attention [14,24,25].

More recent investigations have demonstrated that nsPEFs affect the cytoskeleton. Nanosecond pulsed electric fields cause a breakdown of actin filaments with concomitant cell rounding [26–29] and 44 kV/cm pulses induce microtubule clearance in U87 human glioblastoma cells within minutes, all without observable Ca^{2+} influx and/or osmotic swelling [30]. These findings of microtubule breakdown [30], in conjunction with those of Kirson et al. [31] in TTFields, suggest that EEFs may destabilize microtubules directly in addition to various indirect effects that have been proposed [8].

Other investigations of tubulin in EEFs have found that microtubules can be aligned with an applied electric field [32,33] and that EEFs reduce the Young's modulus of a tubulin heterodimer [34]. Despite these earlier findings, the precise atomic-level details of a single dimer's response to EEFs are unknown. Therefore, we performed an investigation of these effects through Molecular Dynamics, applying EEFs along multiple directions (Fig 1) at time scales and field strengths consistent with nsPEFs (36–38), the results of which are important for understanding macroscopic observations like nsPEF-induced depolymerization. Notably, the α:H1-B2 loop and β:M-loop, that are integral to lateral contacts between protofilaments, are especially susceptible to the influence of electric fields, and the bend angle and elongation of tubulin are affected, which may accelerate the depolymerization process.

Results and discussion

Tubulin's structure and dynamics are affected by EEFs

Structural change as measured by root mean standard deviation. We first investigated large-scale EEF-induced changes to tubulin's root mean standard deviation (RMSD) and dipole magnitude. By both measurements, a 750 kV/cm EEF caused significant structural changes in all directions tested (Fig 2). Most of the change in RMSD was in loops and turns. As expected, we found that rigid structures like alpha helices and beta sheets remained relatively fixed, with an RMSD for alpha carbons close to 1 Å, whereas other structural motifs were more sensitive to the EEFs and exhibited an RMSD of 5–6 Å. By comparison, the same flexible motifs showed an RMSD of around 2–3 Å in the dimer unexposed to an EEF. These more-flexible residues were classified as, in order of prevalence, coils, hydrogen bonded turns, and 3_{10} helices (Fig 2B and Fig 2E). The relaxation time was less than 30 ns, when the EEF was applied along the dimer's transverse axis (Fig 2A), but lasted longer when applied along the longitudinal axis (Fig 2D). Microtubules clearance is observable in nsPEFs applied at 10 Hz, or with 100 ms between their application, which means that there is likely sufficient time for total relaxation in a single dimer between pulses.

Increase of dipole moment by EEFs. In all but the negative transverse direction, or opposite the dimer's dipole, the EEF doubled the magnitude of tubulin's dipole moment (Fig 2C and Fig 2F). Increases in dipoles were longer lasting when the EEF was applied along the longitudinal versus transverse axis, a finding similar to our results in RMSD (Fig 2C and Fig 2F). This effect is explainable by coupling of charged residues with the EEF, in which the dimers' natural charge distributions were exaggerated (in all but the negative transverse simulation). Atomistic simulations of insulin and lysozyme in EEFs have demonstrated that protein dipoles

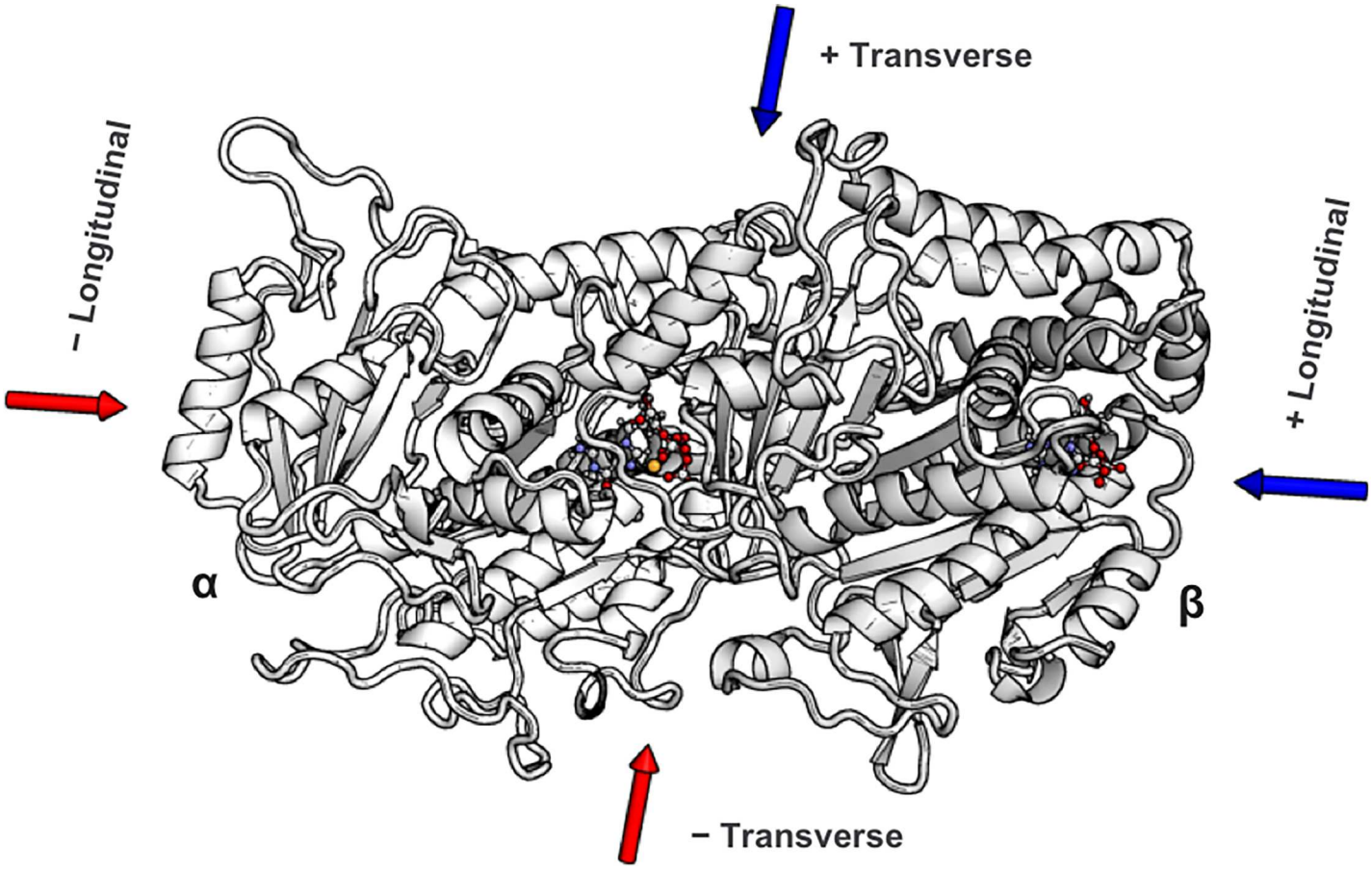

Fig 1. Directionality of the applied electric fields. Electric fields were applied along the direction of the dipole (transverse axis) and along the vector between the beta and alpha monomers' center of mass (longitudinal axis), both in reference to the equilibrated dimer's initial position.

https://doi.org/10.1371/journal.pone.0202141.g001

are affected by the localized rearrangement of charged sidechains and loops [35–37]. Additionally, Ojeda-May et al [38] and Budi et al [36] found that EEFs can convert beta sheets to alpha helices or destabilize alpha helices, respectively, but we observed neither in the 750 kV/cm EEFs studied here (S1 Fig).

Rearrangement is greatest in the α:H1-B2 Loop, β:H6-H7 Loop, and C-termini

Charged, flexible regions undergo large per-residue displacement. To gain a better understanding of EEFs' effects on the tubulin dimer, we superimposed final structures, after 10 ns of EEF exposure in opposite directions, and calculated displacement on a residue-by-residue basis. In flexible regions, structural rearrangement was dependent upon the residue's root mean square fluctuation (RMSF) (S2 Fig) and charge; positive and negative residues move towards the EEF's anode and cathode, respectively. There was overlap in the affected regions within dimers in the transverse and longitudinal EEFs. After exposure to EEFs along the dimer's transverse axis (Fig 3B), displacement was greatest in the α:H1-B2 loop (Fig 3D), α:H4-B5 loop, β:H6-H7 loop (Fig 3C), β:M-loop, and C-termini of both monomers (Fig 3E). Similarly, the longitudinal EEFs greatly affected the α:H1-B2 loop (Fig 4C), β:H6-H7 loop, β:M-loop (Fig 4E), and both C-termini (Fig 4D).

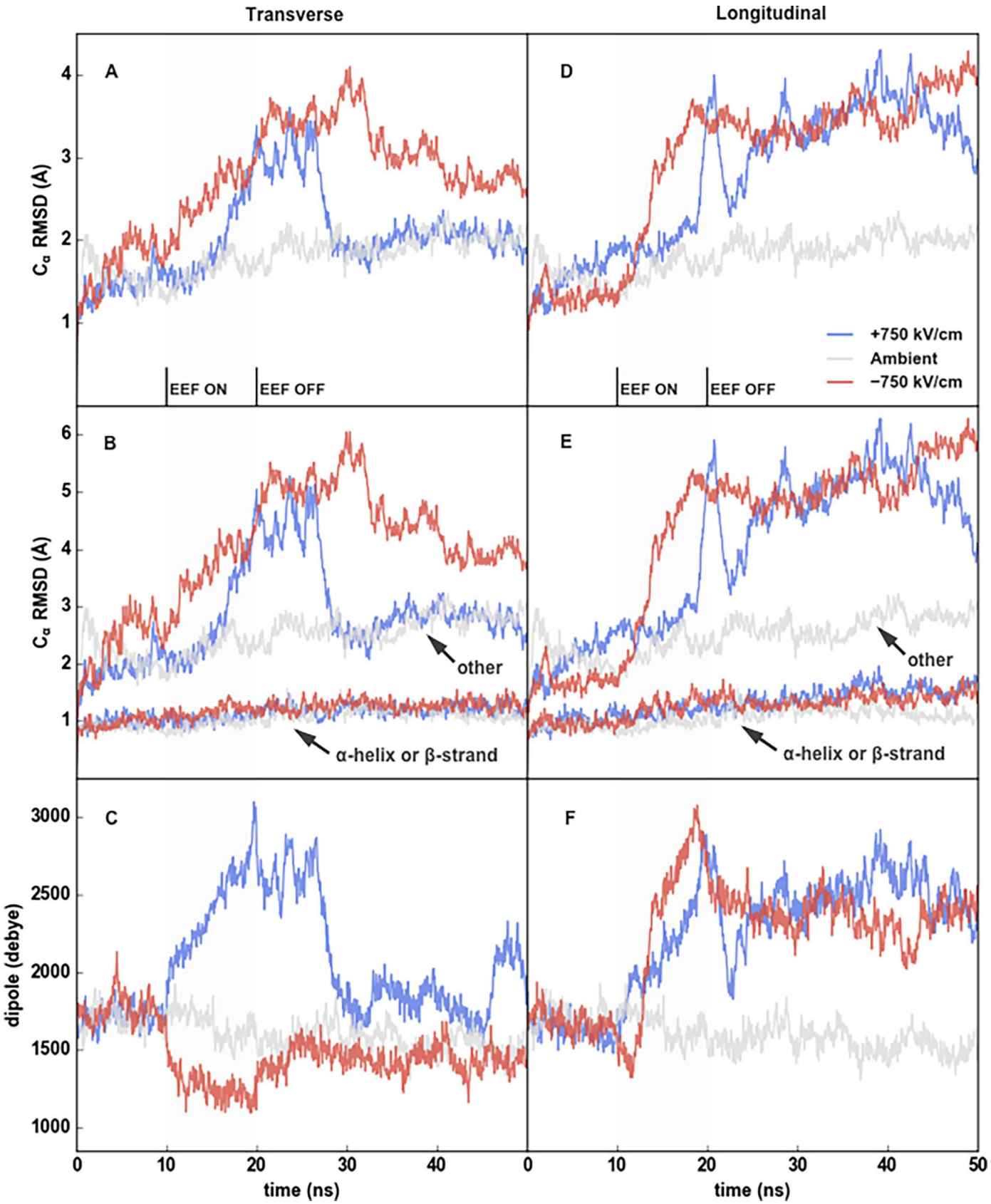
Transverse
Longitudinal
A
D
B
E
C
F
C_α RMSD (A)
dipole (debye)
time (ns)
EEF ON
EEF OFF
+750 kV/cm
Ambient
−750 kV/cm
other
α-helix or β-strand

Fig 2. Structural rearrangement in the presence of a strong EEF. Cα RMSD of a tubulin dimer in the transverse (a) and longitudinal (d) directions. Cα RMSD again but with delineation of displacement in alpha helices and beta sheets versus other structures in the transverse (b) and longitudinal (e) directions. Total dimer dipole magnitudes for the transverse (c) and longitudinal (f) directions.

https://doi.org/10.1371/journal.pone.0202141.g002

C-termini exposed to EEFs showed large movements relative to those that were unexposed, but the extent of their rearrangement was dependent on the field's direction. While the C-termini of the α-monomer and β-monomer shifted inwards in the positive and negative longitudinal EEF, respectively, neither were significantly displaced when the EEF was applied along the opposite, outward direction (Fig 4B). This, plus other research indicating that there are frequent interactions between each monomer's C-terminus and the other's H11 [39], suggests that the C-termini take an "inward" trajectory from a position on the surface of the dimer. In opposition to this movement is the attraction between the negative C-termini and the dimer's positively charged surface. Our equilibrated structure contained salt bridges from α:311K to α:447E and β:292K to β:445E (S3 Fig); the C-termini would necessarily need to escape their attraction to the dimer's positive surface along an inward low-energy path and the EEFs' direction influences whether they reach such an extended state.

EEF-induced modulation of C-termini would have biological relevance because of effects on the dynamics of kinesin [40], dynesin [41], and other microtubule-associated proteins, all of which directly interact with the C-termini of microtubules. In fact, post-translational modifications and removal of the C-termini have already been implicated in changes to motor protein velocity and processivity [41,42]. The interaction between the C-termini and EEFs may be further complicated by post-translational modifications. One such modification is polyglutamylation, which is common to cancer cells, and would result in the addition of negatively charged glutamate amino acids to the already negatively charged C-termini [43] making the latter protrude even more from the mostly negatively charged surface of microtubules. Further investigations of the C-termini's response to EEFs are needed to characterize their transition between contracted and extended states [44] and whether such changes affect the dynamics of microtubule-associated proteins.

Per-residue fluctuations are augmented in EEFs. We also investigated how EEFs affect residue fluctuations. We calculated change in RMSF as $\Delta RMSF = RMSF_{EEF} - RMSF_{AMB}$ where $RMSF_{EEF}$ represents fluctuations of each residue in the dimer exposed to an EEF and $RMSF_{AMB}$ denotes fluctuations in dimer unexposed to an EEF. Therefore, a residue with a large positive $\Delta RMSF$ experienced a significant increase in its fluctuation in an external electric field (relative to the same residue in the dimer unexposed to an external electric field); conversely, a residue with a large negative $\Delta RMSF$ was found to have stabilized by the field. The results (Fig 5 and S4 Fig) further demonstrate the interaction between EEFs and the α:H1-B2 loop, α:H10-B9 loop, M-loop, and C-termini. For these regions, the EEFs' influence on $|\Delta RMSF|$ was consistent, regardless of the direction in which the EEF was applied, but whether $\Delta RMSF$ was positive or negative was variable. For example, there was an increase in fluctuation for the M-loop in both the positive transverse and, especially, negative longitudinal directions. But the same loop showed reduced fluctuation in the negative transverse and positive longitudinal directions. The directional dependence was most apparent in the α:H1-B2 loop which, for three of the four directions tested, experienced the greatest increase in fluctuation among all internal structures, yet was rigidified in the positive transverse EEF.

EEF perturbation results are consistent at lower magnitudes. Given its notable susceptibility to 750 kV/cm EEFs, measurable by both displacement and $\Delta RMSF$, we examined the robustness of α:H1-B2 loop's response to EEFs of lower strengths. We ran 10 ns simulations of EEFs applied along the positive and negative transverse directions at 50, 100, and 200 kV/cm

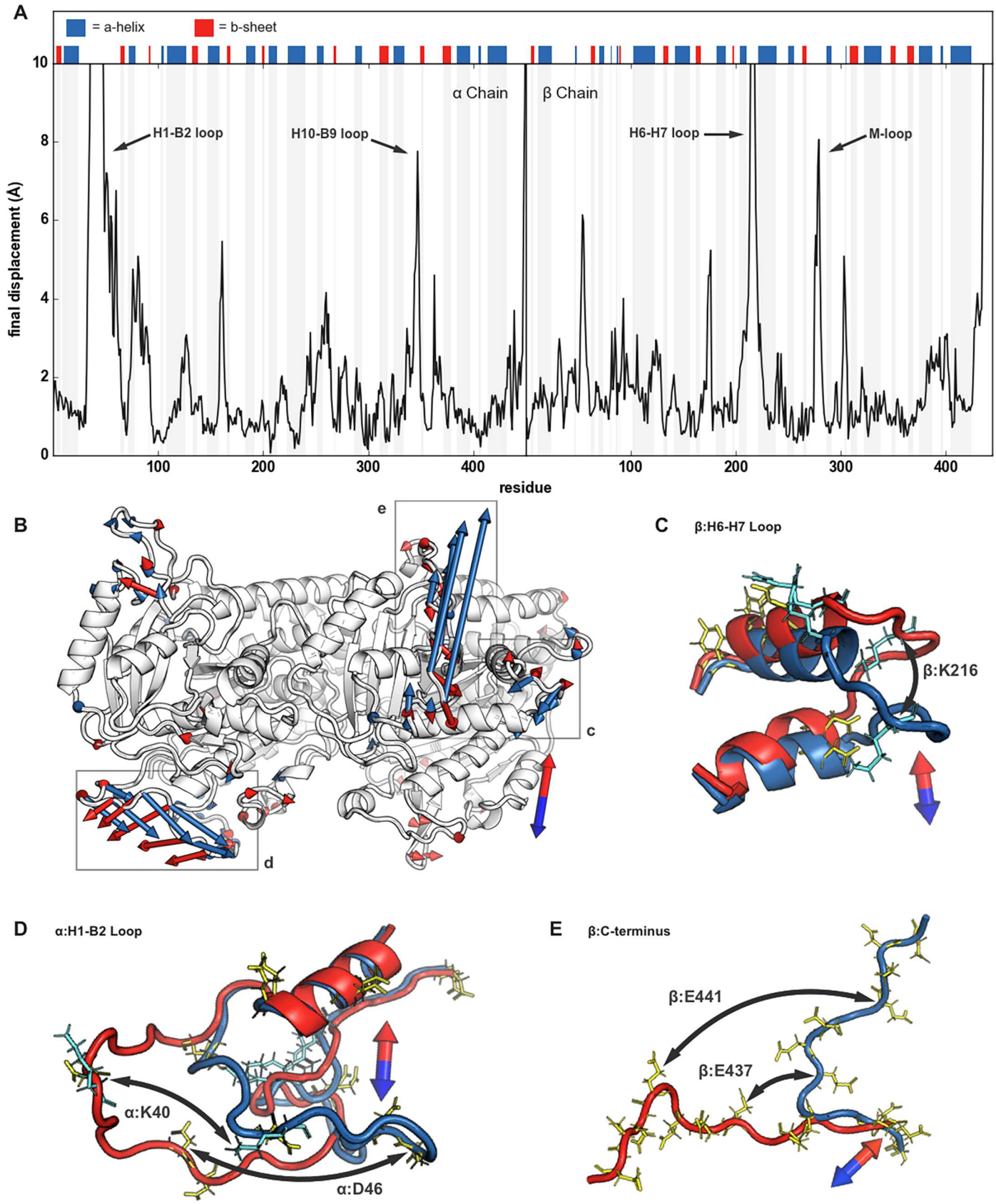
A
= a-helix
= b-sheet
α Chain
β Chain
H1-B2 loop
H10-B9 loop
H6-H7 loop
M-loop
final displacement (Å)
residue
B
e
c
d
C
β:H6-H7 Loop
β:K216
D
α:H1-B2 Loop
α:K40
α:D46
E
β:C-terminus
β:E441
β:E437

Fig 3. Displacement induced by transverse electric fields. (A) Displacement distance (Å) of Cα atoms between final tubulin structures after 10 ns of 750 kV/cm in the positive and negative transverse directions (after superimposition of structures). (B) Displacements exceeding 2 angstroms are visualized by vectors from the residues in the initial, equilibrated structure (white) to the final structure after application along the negative direction (red arrows) and the positive direction (blue arrows). For clarity, every other displacement vector is shown. (C, D, E) Final positions of the dimer are visualized after 750 kV/cm in the negative (red) and positive (blue) transverse directions for the β:H6-H7 Loop, α:H1-B2 Loop, and β:C-terminus. Positively and negatively charged residue side chains are represented by light-blue and yellow sticks, respectively.

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and then visualized the average positions of the loop over the last 5 ns (Fig 6B). We also characterized the loop's movement using measurements of two charged amino acids—greater distances between α:40K and α:H3 (Fig 6C) and α:47D and α:H3 (Fig 6D) correspond to outward movements of the first and second half of the α:H1-B2 loop, respectively. Inspection of the α:H1-B2 loop's response to lower strength EEFs revealed a change in conformation that was proportional to field strength. The greatest effect was observed in the movements of α:40K.

As a positively charged residue, α:40K was more frequently in an "outward" conformation in a positive transversal field and vice versa, even at 50 kV/cm (Fig 6C). Conversely, the negatively charged α:Asp47 was more likely to be in an "inward," closer to the dimer conformation in a positive transverse EEF. Given the field magnitudes and exposure duration tested here, the results indicate that nsPEFs induce rearrangement in the α:H1-B2 loop. The findings are relevant to a mechanistic explanation of the breakdown of microtubules under the influence of EEFs.

A tubulin dimer in a microtubule is held in place through loop-loop interactions. A dimer within a microtubule has both longitudinal and lateral contacts with its own and adjacent protofilaments, respectively [4,45,46]. In the energetically favorable B-lattice confirmation, lateral contacts between the α monomers of two protofilaments include hydrogen bonds between the H1-B2 loop on one side and the M-loop on the other [47,48]. Similarly, lateral contacts between β monomers include hydrogen bonds between the M-loop of one dimer and the H1-B2 loop of the other [2,4,47]. Given that these regions are susceptible to EEFs, we conjecture that this induced rearrangement may promote microtubule catastrophe.

Tubulin's essential dynamics are affected by EEFs

Beyond the per-residue effects of EEFs, we asked whether the lower-frequency motions of tubulin were also affected. Specifically, intradimer curvature and elongation are two modes that are biologically relevant for the polymerization and depolymerization of microtubules [49,50]. These two essential modes in microtubule dynamics were calculated over the last 5 ns of exposure to 750 kV/cm EEFs for all field directions (Fig 7). The negative transverse EEF promoted the greatest change in curvature, with the mean bend angle at 5.9˚ and 8.2˚ for the unexposed and negative transverse EEF, respectively (Fig 7B). Furthermore, EEFs in all four directions promoted elongation, though the greatest increases were in the longitudinal EEFs (Fig 7D).

The curvature of tubulin has long been linked to the dynamics of microtubules. The straight conformation is found in zinc-induced protofilament sheets [52] and are characteristic of stable microtubules [2], while the curved conformation is found in depolymerization peels, which are sometimes referred to in the literature as "ram's horns" [4,53]. A recent model, based on structures produced with cryo-electron microscopy, proposed that the curved dimer conformation is an energetically preferred state that alleviates the strain of elongation produced during the depolymerization process [4]. Given our findings that EEFs can bias tubulin dimers towards bent (Fig 7B) and elongated conformations (Fig 7D), we posit that modulation of essential dynamics is another means by which nsPEFs affect microtubule stability.

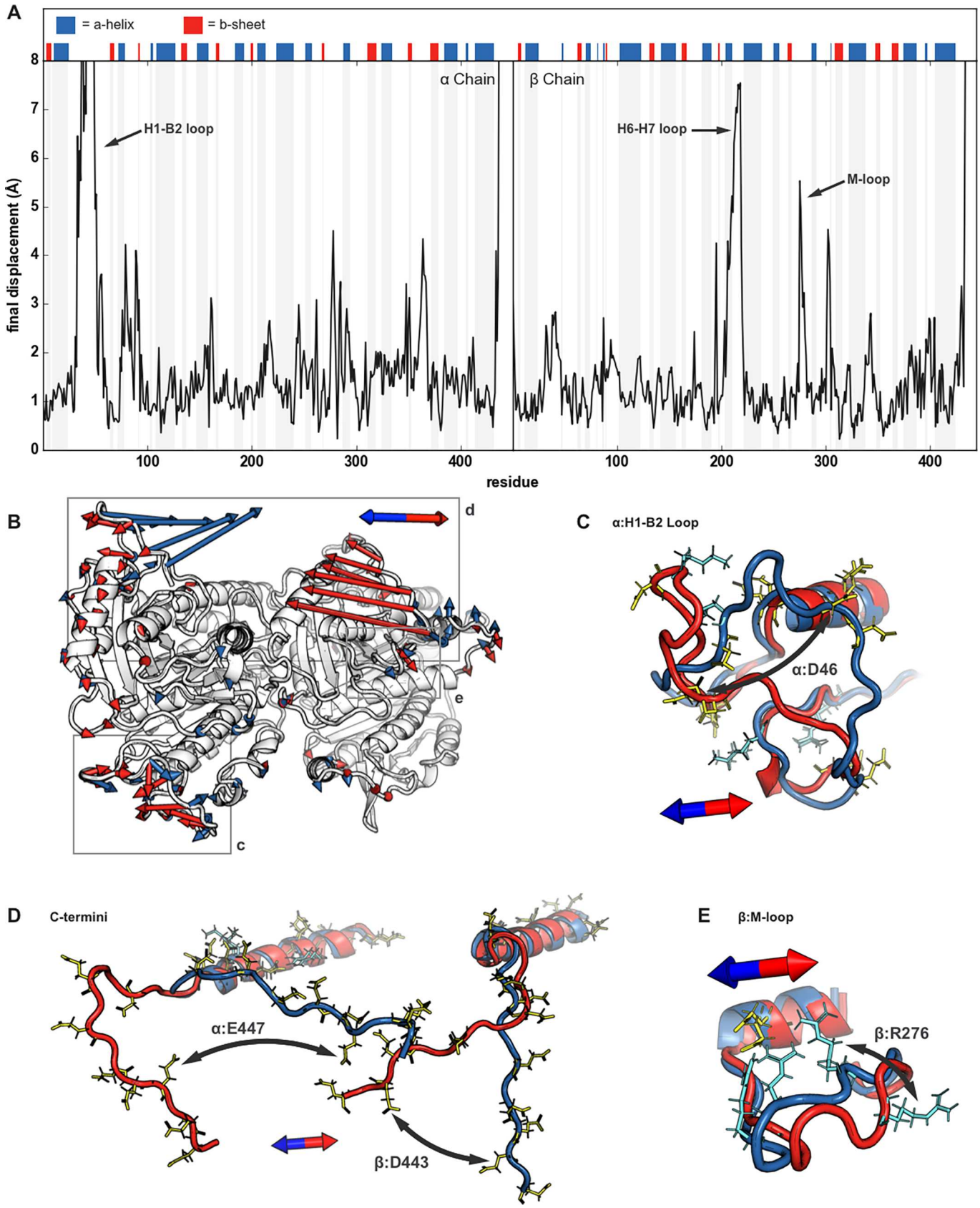
A
= a-helix
= b-sheet
α Chain
β Chain
H1-B2 loop
H6-H7 loop
M-loop
final displacement (Å)
residue
B
d
e
c
C
α:H1-B2 Loop
α:D46
D
C-termini
α:E447
β:D443
E
β:M-loop
β:R276

Fig 4. Displacement induced by longitudinal electric fields. (A) Analogous to Fig 3, but comparing displacement between structures after application of longitudinal EEFs. (B) Displacement vectors are shown from every other Cα, for clarity. (C, D, E) Final positions are visualized for the heterodimer after 750 kV/cm in the negative (red) and positive (blue) longitudinal directions for the α:H1-B2 Loop, C-termini, and M-loop.

https://doi.org/10.1371/journal.pone.0202141.g004

While our model investigates the effects of EEFs on a single tubulin dimer, other groups have proposed that nsPEFs' effects on the cytoskeleton are due to electrophoresis of the large, negatively charged microtubules [30]. In other words, microtubule breakdown in nsPEFs stems from macro-scale forces, because an unaligned microtubule experiences a lateral electrophoretic force in electric fields [54], and the breakdown of loop-loop interactions occurs secondarily. We did observe an increase in the mean squared displacements of dimers that were proportional to the EEFs' magnitude (S5 Fig), but insight into large-scale, electrophoretic breakdown of microtubules would require either a macroscopic model of microtubule networks or an *in vitro* experiment on microtubule buckling in a range of external electric fields. Relatedly, buckling during the alignment process appears to be modulated by loop-loop interactions between adjacent protofilaments, as evidenced by microtubule response to acetylation of α:40K [55]—so EEF-induced changes in loop-orientation may affect microtubules' plasticity and resistance to buckling at the level of the dimer. Future experiments may further this work by increasing tubulin's number—by simulating either a full or partial cross-section of a microtubule with lateral contacts [48]—or by investigating the effects of hydrolysis of GTP on the dimer's response to EEFs.

Conclusions

In this paper, we have reported on our computer simulations of the effects of externally applied alternative electric fields on the structure and dynamics of tubulin dimers. This was motivated by several experimental observations concluding that microtubules become unstable when exposed to sufficiently strong electric fields. In our simulations, we have observed displacement in the charged and flexible regions of tubulin, including the α:H1-B2 loop, β:M-loop, and C-termini. The changes to the α:H1-B2 loop were persistent at lower magnitude EEFs, and there were also detectable perturbations to tubulin's curvature and elongation. Based on these observations, we propose a mechanistic explanation of the microtubule breakdown in EEFs, which is most likely to involve promotion of depolymerization through weakened lateral bonds and increased dimer strain. Finally, we have found significant effects of the field's orientation on the conformational rearrangements of tubulin dimers, an observation that should encourage an experimental validation.

Materials and methods

We created a 3D model of a tubulin heterodimer by modifying a crystal structure of tubulin bound to Taxol: PDB ID: 1JFF [56]. Preparing this structure for computer simulations, we removed Taxol, added missing residues α:1, α:35–60, α:440–451, β:1, and β:428–445, and energy minimized using Modeller [57].

Note that the residue indices referenced here are ungapped and therefore differ from those reported by Löwe et al [56]. The final structure included the dimer's C-termini and consisted of 451 residues in α-tubulin plus GTP and one Mg^{2+} atom, both of which are bound to α-tubulin, and 445 residues in β-tubulin plus GDP, which is bound to β-tubulin. We included missing hydrogen atoms and delta protonated histidine residues with psfgen [58]. We then solvated the dimer in TI3P3 water atoms in a rectangular box with 20 Å of padding along each axis using VMD's Solvate plugin. This box sizing was chosen with consideration of C-terimini movements to prevent interaction with the dimer's image. To neutralize the dimer and

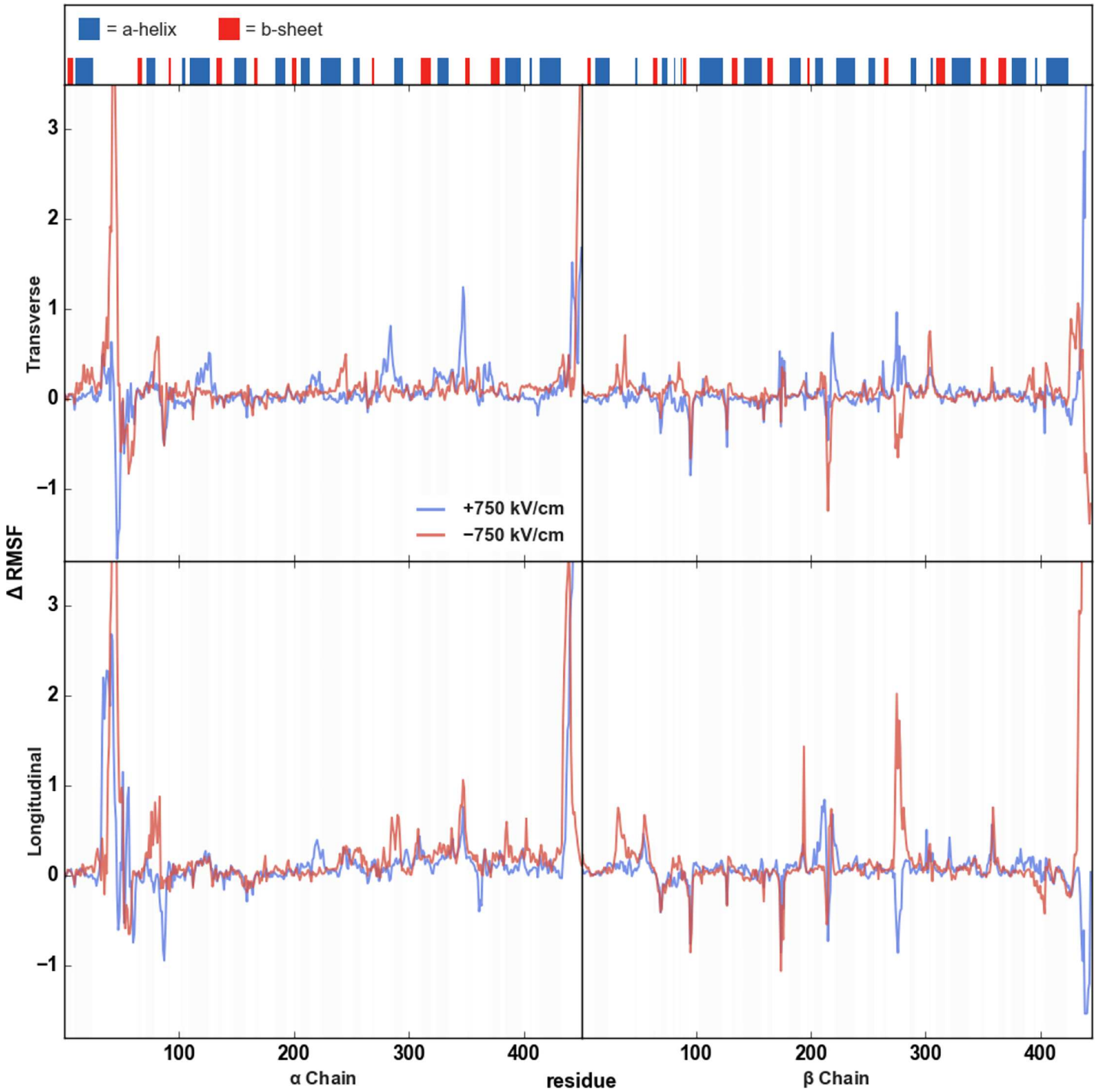

Fig 5. Change in per-residue RMSF. Changes in residue fluctuations for a tubulin dimer in a 750 kV/cm external electric field compared to a tubulin dimer not exposed to an external electric field ($\Delta RMSF = RMSF_{EEF} - RMSF_{AMB}$). The positive transverse (blue) and negative transverse $\Delta RMSF$ plots are along the top row, while the positive longitudinal (blue) and negative longitudinal (red) $\Delta RMSF$ plots are along the bottom.

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produce a NaCl molarity of 150mM, we added 185 Chloride and 240 Sodium ions *via* VMD's Autoionize plugin; ultimately, the periodic cell was 110 x 150 x 110 Å along the x, y, and z axes, respectively, and contained 164,493 water atoms.

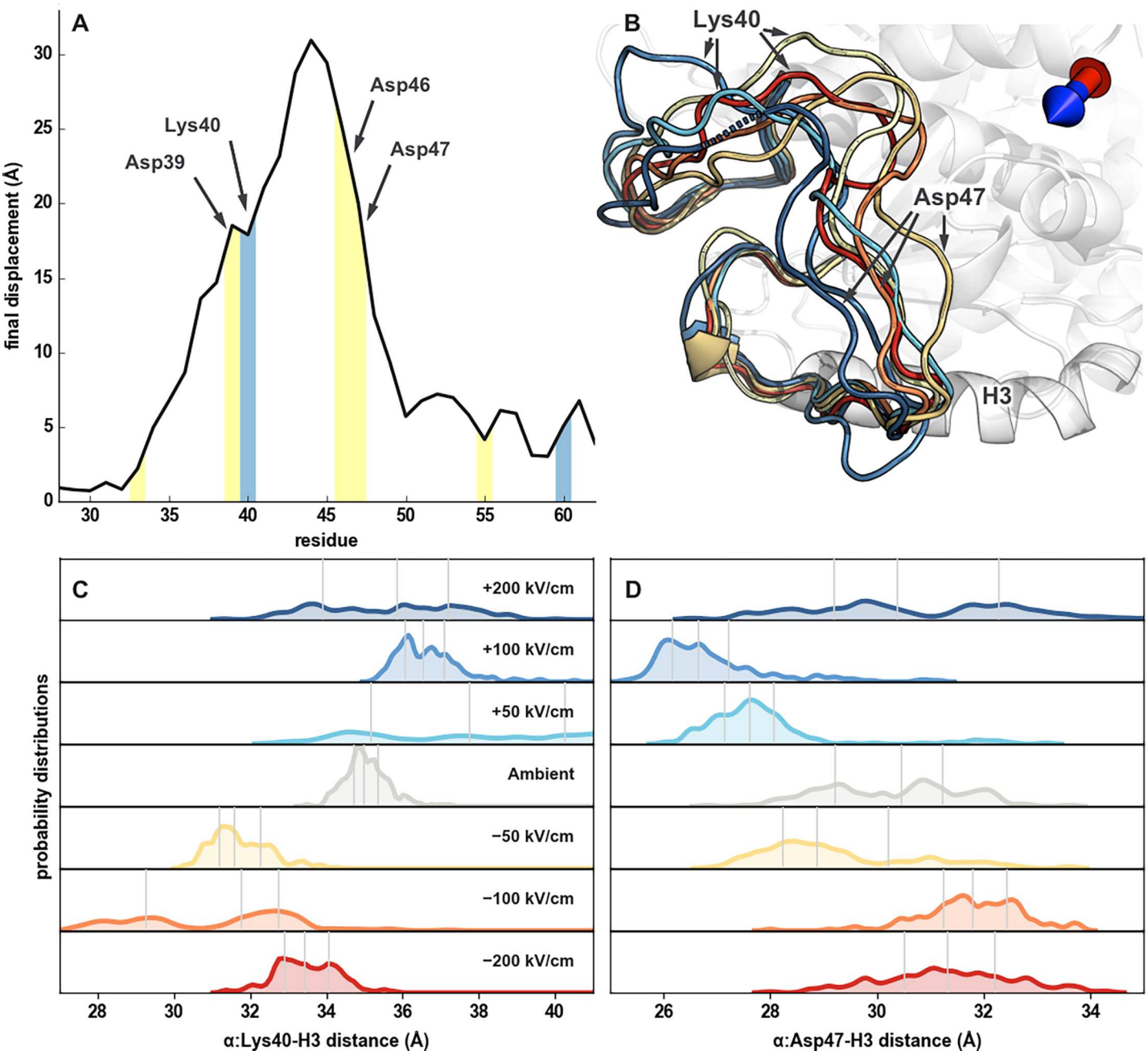

Fig 6. α:H1-B2 Loop's response to transverse external electric fields. (A) Final per-residue displacement, after application a 750 kV/cm EEF along the transverse axis, between a dimer exposed to a positive and negative transverse direction. Negatively and positively charged residues are highlighted with yellow and light-blue, respectively. (B) Average positions of the α:H1-B2 loop, during the final 5 ns of EEF application, along the transverse direction. Loop colors of dark blue, blue, light blue, yellow, light orange, orange, and red correspond to +200, +100, +50, 0, -50, -100, and -200 kV/cm field strengths, respectively. The positive and negative EEF directions are indicated with a blue and red arrowhead, respectively. (C, D) Probability distributions for the distance between α:40K and α:H3 and between α:47D and α:H3, using snapshots from the final 5 ns of EEF exposure, correspond to outward movement of the first and second half of the α:H1-B2 loop, respectively. Quartiles for distance measurements are indicated with vertical gray lines.

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Long-range electrostatics effects were treated with Particle-Mesh Ewald sum and a 12 Å cutoff [59]. We applied rigid bonds to hydrogen bonds, used a 2 femtosecond timestep, and saved frames to trajectories every 25 ps. We energy minimized the solvated structure for 25 ps,

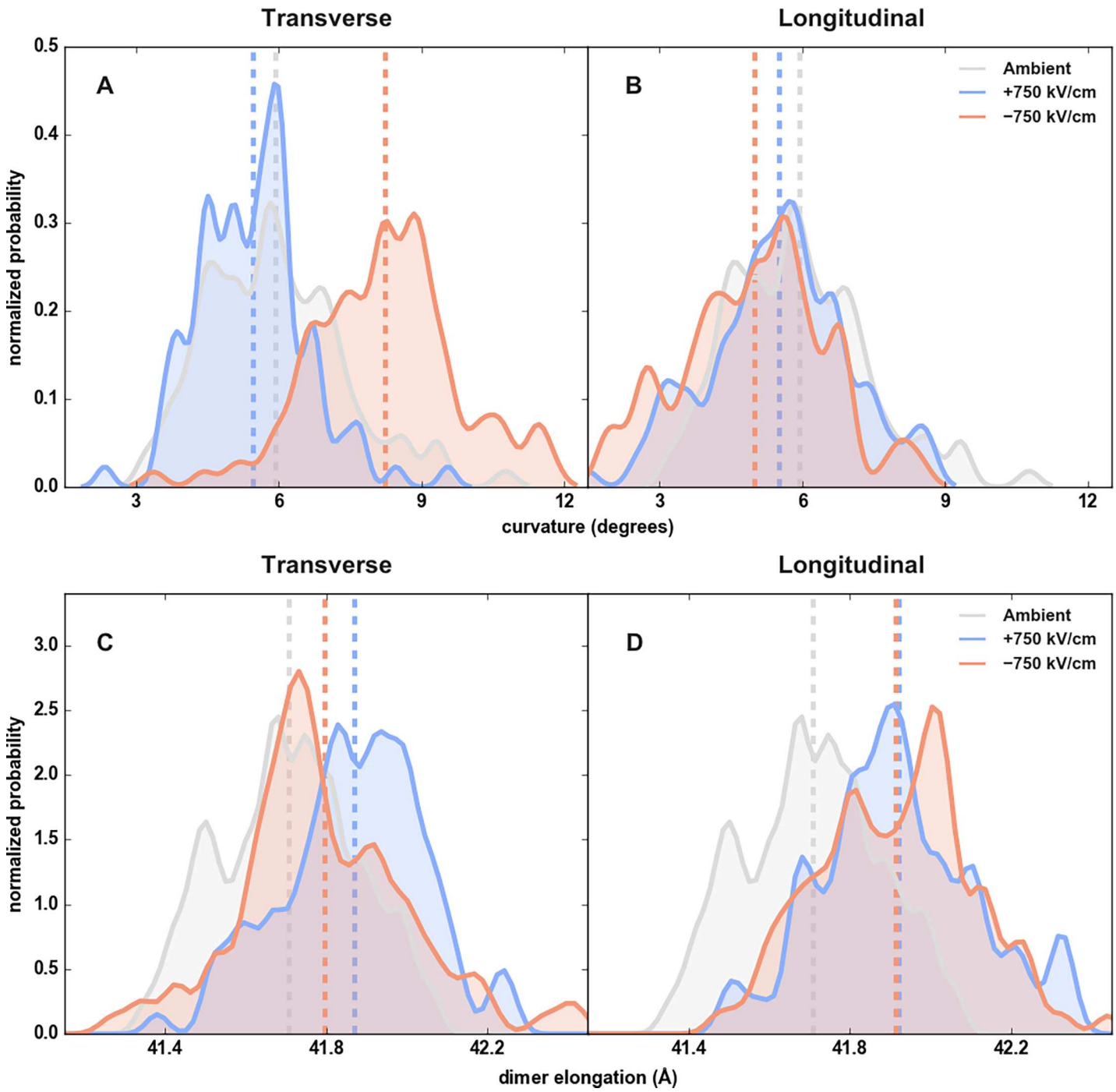

Fig 7. Tubulin essential dynamics in an external electric field. (A, B) Normalized probability distributions of tubulin's curvature and dimer elongation for the last 5 nanoseconds of EEF application. Curvature was calculated by the method presented in [51]. Elongation was measured as the distance between the center of mass of each monomer's alpha helices. Mean curvature and compaction are shown by dashed lines.

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equilibrated it for 500 ps in an NVT ensemble, then equilibrated it for 1 nanosecond in an NPT ensemble. Because of 1JFF's unfavorable initial conformation, we performed extensive additional energy minimization of the dimer by running it for an additional 100 ns in an NPT ensemble until convergence was definitive in RMSD. All simulations were run with the

CHARMM36 force field [60]. Notably, CHARMM36 is a non-polarizable force field in which dipoles are measured purely by the distribution of fixed charges. In future studies, electronic polarization by the EEF and or other modalities could be accounted for with a polarizable force field like Drude-2013 or AMOEBA-2013 [61–63]. NAMD 2.10 [64] was used to run the simulations on clusters of 64 nodes, each containing 16 cores, connected by Infiniband in a BlueGene/Q cluster. We ran equilibration and production simulations at 310˚K and coupled them to a Nosé-Hoover-Langevin piston for pressure control and used Langevin dynamics for temperature control.

We simulated static electric fields with the External Electric Field module of NAMD and applied them along four directions that we define as the positive and negative of the transverse and longitudinal axes, respectively (Fig 1). The positive transverse direction was that of the dimer's dipole moment at the start of the production runs while the negative transverse direction was its negative. The positive longitudinal direction was along the vector formed from the beta to the alpha monomers' center of mass at the beginning of the production runs; here again, the negative longitudinal vector was its negative. In addition to the control run, we ran simulations of EEFs at 750 kV/cm along each of the four major directions; we chose this initial magnitude based on other Molecular Dynamics studies of proteins in EEFs found in the literature [35,65,66]. We performed each production run for 50 ns and applied the EEFs between the 10 and 20 ns time points (using a 10 ns duration that is typical of nsPEFs [30]). Additionally, we simulated lower strength fields at 200, 100, and 50 kV/cm along the two transverse directions for 10 ns using the positions and velocities of the ambient production run at the ten-nanosecond mark for the first frame.

Analysis was performed with modules and scripting in VMD [58]. Per-residue displacement, referred to here as "final displacement," was calculated as the distance between alpha carbons after superimposing heterodimers exposed to two EEFs of opposite directionality, which is loosely based on the concept developed by Hekstra et al [67]. Tubulin's intradimer curvature (or bend angle) was calculated by the method described by Pecquer et al [51] and characterized by Peng et al [53]. Briefly, we calculated curvature as the intersection angle between the two best fit lines—created with the atoms of α:H7 and β:H7—from the Taxol bound tubulin dimer, PDB ID: 1JFF, and the dimer under study, after superimposing the two structures by the atoms of their α:H7s. We measured dimer elongation as the distance between the centers of mass from all alpha helices atoms in the α and β monomers, using atoms classified as belonging to alpha helices in the initial equilibrated structure.

Supporting information

S1 Fig. Secondary residue classifications in a 750 kV/cm EEF. (A, B) The total number of residues that are classified as alpha helices or beta sheets during the application of 750 kV/cm EEFs of four directions: positive transverse, negative transverse, positive longitudinal, and negative longitudinal.
(TIFF)

S2 Fig. Final displacement in a transverse external electric field and RMSF. Final displacement, measured as the distance between residues after 10 ns exposure to a positive and negative transverse EEF (black). Cα RMSF per residue of a tubulin heterodimer, from the 10 to 20 ns time frame, without exposure to an EEF (blue).
(TIFF)

S3 Fig. Salt bridges between the C-termini and the dimer surface. Representative salt bridges experienced by the C-termini when in a contracted state (lying along the dimer's surface).

A distance cutoff of 3.2 Å and the positions of the initial frame, after equilibration, were used to identify possible salt bridges.
(TIFF)

S4 Fig. Visualization of change in per-residue RMSF. Same as Fig 4 but with ΔRMSF of each residue mapped onto the structure. Red and blue indicate an increase and decrease in RMSF, respectively. Field directions are indicated with arrows.
(TIFF)

S5 Fig. Mean standard displacement of tubulin in a transverse external electric field. Mean standard displacement ($Å^2$) of tubulin from the start to the end of the application of a transverse EEF.
(TIFF)

Author Contributions

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Supervision: Jack A. Tuszynski.

Validation: Jordane Preto.

Writing – original draft: Joshua J. Timmons.

Writing – review & editing: Jordane Preto, Jack A. Tuszynski.

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Two Fermions in a Double Well: Exploring a Fundamental Building Block of the Hubbard Model

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We have prepared two ultracold fermionic atoms in an isolated double-well potential and obtained full control over the quantum state of this system. In particular, we can independently control the interaction strength between the particles, their tunneling rate between the wells and the tilt of the potential. By introducing repulsive (attractive) interparticle interactions we have realized the two-particle analog of a Mott-insulating (charge-density-wave) state. We have also spectroscopically observed how second-order tunneling affects the energy of the system. This work realizes the first step of a bottom-up approach to deterministically create a single-site addressable realization of a ground-state Fermi-Hubbard system.

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In the presence of strong correlations, the understanding of quantum many-body systems can be exceedingly difficult. One way to simplify the description of such systems is to use a discrete model where the motion of the particles is restricted to hopping between the sites of a lattice. The paradigmatic example for this approach is the Hubbard model, which reduces the physics of a quantum many-body system to tunneling of particles between adjacent sites and interactions between particles occupying the same site. While this model captures essential properties of electrons in a crystalline solid and provides a microscopic explanation for the existence of Mott-insulating and antiferromagnetic phases, many questions about this Hamiltonian — such as whether it can explain d-wave superfluidity — are still unanswered [1].

A promising approach to answer these questions is to use ultracold atoms trapped in periodic potentials as quantum simulators of the Hubbard model [2–8]. Such experiments have been performed both in large- and small-scale systems. Degenerate gases loaded into optical lattices have been used to observe the transition to the bosonic [9, 10] and fermionic Mott insulator [3, 4]. The first observation of second-order tunneling was achieved in a small-scale system by studying the tunneling dynamics of bosonic atoms in an array of separated double wells [11, 12]. In a recent experiment, these two regimes have been connected by splitting a fermionic Mott insulator into individual double wells. In this way, the strength of the antiferromagnetic correlations in the many-body system could be determined by measuring the fraction of double wells with two atoms in the spin-singlet configuration [13]. But despite the observation of antiferromagnetic correlations[13, 14] current experiments using fermionic atoms have so far failed to reach temperatures below the critical temperature of spin ordering [15, 16].

Recently, new experimental techniques have been developed which allow for the deterministic preparation of few-particle systems in the ground state of a single potential well [17–19]. This makes it feasible to use ultracold atoms to study many-body physics in a bottom-up approach, i.e. to start from the fundamental building block of the system and watch how many-body effects emerge as one gradually increases the system's size [20]. Here we report on the realization of the fundamental building block of the Fermi-Hubbard model at half filling, which consists of one $|\uparrow\rangle$ and one $|\downarrow\rangle$ particle in a spin-singlet configuration in a double-well potential.

By starting from this two-site realization of the Hubbard model, we can test our building block in a regime where the model can still be easily solved. In the Hubbard regime, the spatial wave function $|\Psi\rangle$ of this two-particle system can be written in the basis $\{\,|LL\rangle\,,|LR\rangle\,,|RL\rangle\,,|RR\rangle\,\}$. These basis states are all possible combinations of the localized single-particle states $|L\rangle$ and $|R\rangle$ of one particle in the ground state of either the left or the right well. In this basis, the spatial part of the Hamiltonian is

$$H = \begin{pmatrix} U+2\Delta & -J & -J & 0 \\ -J & 0 & 0 & -J \\ -J & 0 & 0 & -J \\ 0 & -J & -J & U-2\Delta \end{pmatrix}, \qquad (1)$$

with the tunneling matrix element J, the on-site interaction energy U and the energy tilt 2Δ between the wells. Diagonalizing this Hamiltonian leads to three eigenstates ($|a\rangle$, $|b\rangle$ and $|c\rangle$) which are symmetric and one eigenstate ($|d\rangle$) which is antisymmetric with respect to particle exchange (Fig 1b).

To prepare our double-well system, we start with two ^{6}Li atoms in different hyperfine states in the motional ground state of a single optical microtrap [17] (Sec. I of the Supplemental Material [21]). We then slowly ramp on a second potential well and thereby deform our single trap into a double-well potential. During this process, we keep the coupling between the wells negligible and thus initialize the system in state ($|LL\rangle$) where both atoms reside in the ground state of the left well. This state is the

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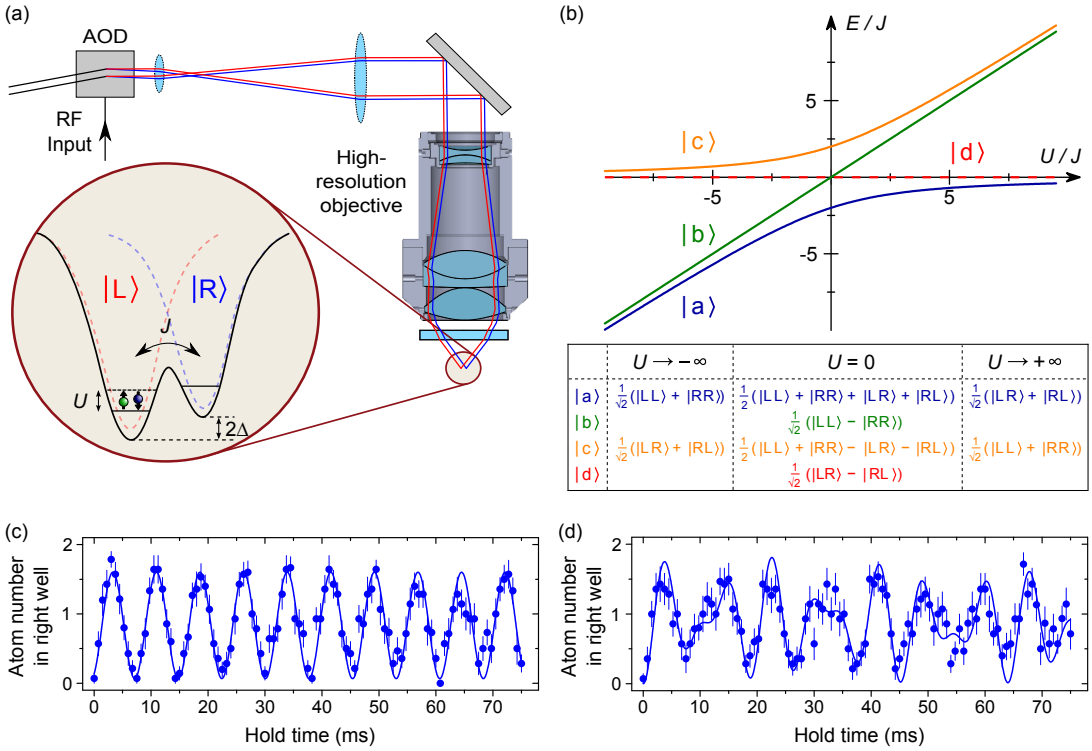

FIG. 1. Experimental realization and eigenenergies of the two-site Hubbard model. (a) Experimental setup: The double-well potential is created by focusing two laser beams with a high-resolution objective. By independently controlling the intensity and position of the two laser beams with an acousto-optic deflector (AOD) we can tune the tunnel coupling J and the tilt Δ between the two wells. (b) Energies of the four lowest two-particle eigenstates in a symmetric double-well potential as a function of the on-site interaction energy U. (c, d) Tunneling of two particles in the double well. The data show the time evolution of the particle number in the right well after initializing the system with both particles in the left well and abruptly switching on the tunnel coupling J between the two sites. For $U \approx 0$ and $\Delta \approx 0$ (c), we can extract the value of the tunnel coupling by fitting the data with a damped sine wave. For intermediate interaction strength ($U \approx J$) (d), we observe correlated tunneling of the two particles, which shows good agreement with the prediction from the Hubbard model (solid line). The error bars denote the 1σ statistical uncertainty.

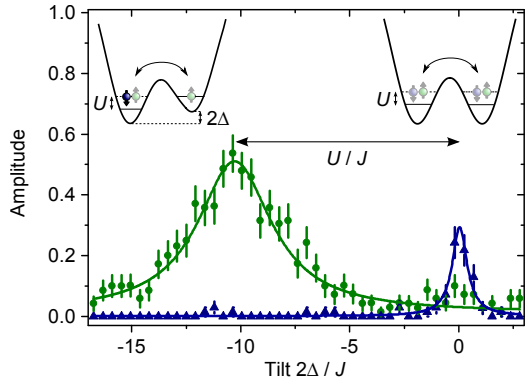

FIG. 2. Single-particle and pair tunneling as a function of the tilt Δ at an interaction energy of $U/J = 10.05 \pm 0.19$. The data points show the time-averaged probability of finding a single particle (green circles) or a pair of particles (blue triangles) in the right well after initializing the system with two atoms in the left well and switching on the tunneling. Pair tunneling is resonant in a symmetric double well, while conditional single-particle tunneling occurs for a tilt of $-2\Delta = U$. The error bars denote the 1σ statistical uncertainty.

starting point for all our measurements. We can prepare it with a fidelity of more than 90 %. The predominant error is that there is only one atom in the trapping potential, while the probability to start with three atoms is $\lesssim 1\,\%$.

Since the spatial wave function of the initial state is symmetric with respect to particle exchange, the spin wave function of the two fermions is in a singlet configuration. In the experiments presented here, we do not couple the position and the spin of the particles and therefore restrict the spatial wave function of the system to the symmetric eigenstates $|a\rangle$, $|b\rangle$ and $|c\rangle$.

In a first set of experiments, we study the tunneling dynamics of the two particles in our double-well potential to characterize the Hubbard parameters J, U, and Δ of our system. To do this, we initialize the system in state $|LL\rangle$ and abruptly reduce the height of the potential barrier, which allows the atoms to tunnel between the wells. To observe the resulting dynamics, we let the system evolve for different durations and then freeze the spatial distribution of the atoms by quickly increasing the barrier height. We then count the number of atoms in one of the wells by recapturing them into a magneto-optical

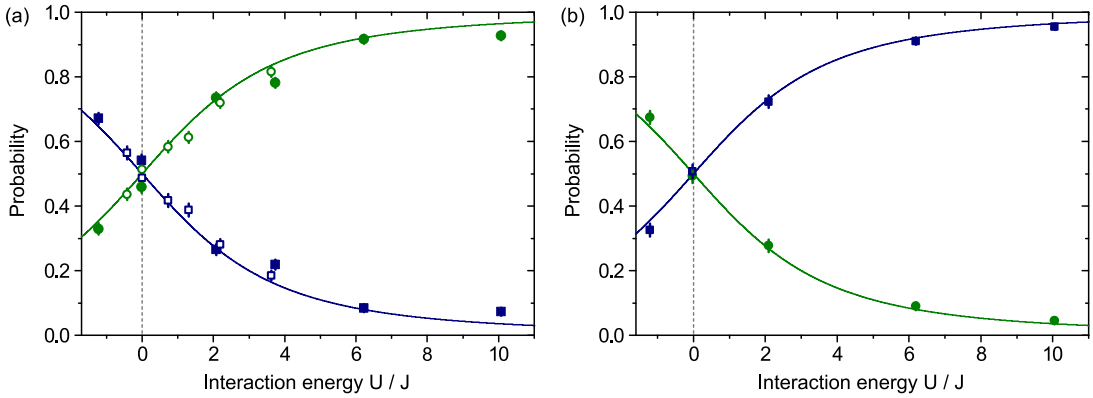

FIG. 3. Occupation statistics as a function of interaction strength. The relative probabilities of measuring both particles in the same well (P_2, blue squares) or in different wells (P_1, green circles) are shown as a function of the on-site interaction energy U. Open (filled) symbols indicate a tunnel coupling of $J/h \simeq 142\,\mathrm{Hz}$ ($J/h \simeq 67\,\mathrm{Hz}$), the solid lines show the prediction of the Hubbard model. (a) For the ground state $|a\rangle$, double occupancy is suppressed for increasing repulsive interactions. This indicates the crossover from a metallic to a Mott-insulating regime. For attractive interactions, double occupancy is enhanced, which we interpret as the onset of a charge-density-wave regime. (b) For the excited state $|c\rangle$, we observe the crossover to the charge-density-wave regime for strong repulsive interactions. For both measurements, the data have been corrected for the effect of the finite fidelities of preparation and detection (Sec. VII and Fig. S5 of [21]). The error bars denote the 1σ statistical uncertainty.

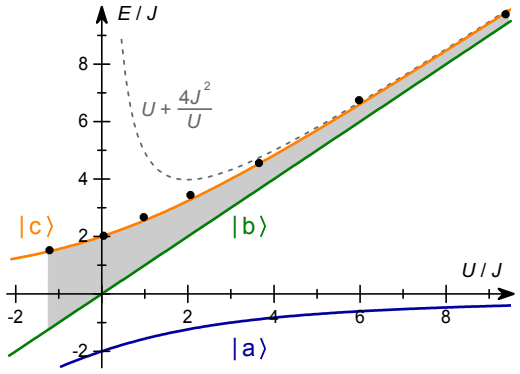

FIG. 4. Transition from first-order to second-order tunneling. To directly measure the influence of the tunnel coupling on the energy of the system, we determine the energy difference $E_{\mathrm{bc}} = E_{\mathrm{c}} - E_{\mathrm{b}}$ (shaded region) between state $|c\rangle$ (orange) and state $|b\rangle$ (green) using trap modulation spectroscopy. To compare our results to the eigenenergies of the Hubbard model (Fig. 1b), we plot the sum of E_{bc} and the energy of state $|b\rangle$ ($E_{\mathrm{b}} = U$). The error bars for the 1σ statistical uncertainties (Sec. VIII of [21]) are smaller than the symbols of the data points. For $U = 0$, we observe that $E_{bc} \approx 2J$, which is consistent with single-particle tunneling. For increasing repulsion, first-order tunneling is suppressed and E_{bc} converges to the superexchange energy $4J^2/U$ (see dashed line).

trap and measuring their fluorescence [28] [17] (Sec. III of [21]).

For a noninteracting system and a small tilt of the double well ($|\Delta| \lesssim J$), we observe long-lived tunneling oscillations whose frequencies we can set by tuning the barrier height (Sec. V.A, Fig. S1 and Fig. S2(a) of [21]). As an example, a typical oscillation for $\Delta \simeq 0$ with a frequency of $2J/h \simeq 134\,\mathrm{Hz}$ is shown in figure 1c. To calibrate Δ, we measure the oscillation frequency for different tilts and find good agreement with the effective coupling strength $J_{\mathrm{eff}} = \sqrt{J^2 + \Delta^2}$ of a two-level system (Sec. V.B and Fig. S2(b) of [21]).

For an interacting system, the interaction energy creates an effective detuning for the tunneling of a single particle. In the limit of strong interactions and a symmetric double well ($U \gg J$ and $\Delta = 0$), the atoms can therefore only tunnel as pairs [29]. However, we can restore single-particle tunneling by setting a tilt that compensates for the interaction shift [11]. This allows us to calibrate the on-site interaction energy U by measuring the strength of single-particle tunneling as a function of the tilt Δ for different interaction strengths (Fig. 2). We find good agreement with a calculation of the interaction energy of two particles in the single well [25] (Sec. V.C and Fig. S3 of [21]).

Using the parameters determined from these measurements, the two-site Hubbard Hamiltonian (Eq. 1) fully describes the oscillations of two interacting particles in our double well (solid line in Fig. 1d).

To use our system as a fundamental building block of the Hubbard model, we must be able to prepare the particles in the ground state of the symmetric double well. To achieve this, we first initialize the system with both atoms in the left well and a tilt $\Delta \ll 0$. Then, we adiabatically change the tilt to bring the system into the ground state of the symmetric double well [30] (Sec. VI and Fig. S4 of [21]) [31].

To show that our fundamental building block already contains the physics that is responsible for the formation of ordered phases in a many-body system, we first measure the influence of the interaction energy U on the distribu-

tion of the two particles between the wells. We therefore determine the probabilities $P_1 = |\langle\Psi|LR\rangle|^2 + |\langle\Psi|RL\rangle|^2$ of finding the two particles on different sites (single occupancy) and $P_2 = |\langle\Psi|LL\rangle|^2 + |\langle\Psi|RR\rangle|^2$ of finding both particles on the same site (double occupancy) by measuring the probabilities of having 0, 1 or 2 atoms in one of the wells (Sec. VII of [21]).

In a noninteracting system ($U = 0$), the spatial wave function of two particles in the ground state is an equal superposition of the basis states $|LL\rangle$, $|LR\rangle$, $|RL\rangle$, and $|RR\rangle$. This leads to equal probabilities P_1 and P_2 which we observe in our measurements (Fig. 3a).

In a system with strong repulsive on-site interactions, it is energetically unfavorable to have two atoms occupying the same site. The ground state of our system then becomes a two-particle analog to a Mott-insulating state which we observe as a reduction of double occupancy. For attractive interactions we observe an increase in double occupancy that marks the onset of a paired state. We interpret this state as the two-particle limit of the cold gas analog of a charge-density-wave state as described in [32]. We perform these measurements for two different barrier heights. For the larger barrier height ($J/h \simeq 67\,\mathrm{Hz}$), we find good agreement with the prediction of the Hubbard model (Fig. 3a). For the smaller barrier height ($J/h \simeq 142\,\mathrm{Hz}$), we observe a small deviation from the theoretical expectations which might indicate that our system is approaching the limits of the Hubbard approximation.

Since in our isolated, two-particle system excited states are stable against relaxation, we can also prepare the system in the highest-energy eigenstate $|c\rangle$ (Fig. S4 of [21]). When measuring the occupation statistics of this state as a function of interaction strength, we find that the number of doubly occupied sites is enhanced for repulsive interactions (Fig. 3b). This allows us to study the charge-density wave regime in a system with repulsive interactions.

In our final measurement, we directly probe the effect of the tunnel coupling on the energy of the system by trap modulation spectroscopy (Sec. VIII of [21]). For this, we initialize the system in state $|c\rangle$ and measure the energy difference to state $|b\rangle$ as a function of interaction strength. For $U = 0$, the atoms are delocalized over the wells by single-particle tunneling. This leads to a change in the kinetic energy of the system that is proportional to the tunnel coupling J (Fig. 4). As we increase the interactions, the system enters the insulating regime and first-order tunneling is suppressed. However, second-order tunneling is still possible and we observe a crossover to a new energy scale given by the superexchange energy $4J^2/U$. This energy is directly responsible for the appearance of spin order in the ground state of the Hubbard model [33].

By combining a series of isolated double wells we can realize a dimerized lattice [13] where each dimer contains two fermions in a spin-singlet configuration with high fidelity. By adiabatically lowering the barriers between the double wells, we can then prepare a low-entropy state in a homogeneous lattice. Since our systems can be prepared with arbitrary filling factors they are also ideally suited to study the effects of hole doping. Additionally, the tunability of our potential allows us to explore finite-size lattices with arbitrary geometries [34] and introduce controlled disorder into our system. Finally, our experiments provide a starting point for scalable quantum computation with neutral atoms [35–38].

ACKNOWLEDGMENTS

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SUPPLEMENTAL MATERIAL

I. PREPARATION OF TWO FERMIONS IN A MICROTRAP

We start our experiments by preparing two ^{6}Li atoms in the $|F = 1/2, m_F = +1/2\rangle$ and $|F = 1/2, m_F = -1/2\rangle$ hyperfine states – labeled $|\uparrow\rangle$ and $|\downarrow\rangle$ – in the motional ground state of an optical microtrap. To do this we follow the procedure established in Serwane et al.[17]. We first prepare a two-component Fermi gas consisting of $N_{|\uparrow\rangle} \approx N_{|\downarrow\rangle} \approx 2.5 \times 10^4$ atoms in a crossed-beam optical dipole trap at a temperature of $T \approx 250\,\mathrm{nK}$. This dipole trap acts as a reservoir from which we load about 1000 atoms into our microtrap, which is created by focusing a far red-detuned laser beam ($\lambda = 1064\,\mathrm{nm}$) with a custom-designed high-resolution objective (NA = 0.55)[22]. We then apply a magnetic field gradient to deform the trapping potential until only a single bound state remains in the microtrap. This spilling technique allows us to prepare systems of one $|\uparrow\rangle$ and one $|\downarrow\rangle$ atom in the ground state of the trap with a typical fidelity above 90%.

To characterize our microtrap, we have measured its trapping frequencies using modulation spectroscopy. We find trap frequencies of $\omega_{\mathrm{ax}} \approx 2\pi \times 2.5\,\mathrm{kHz}$ and $\omega_{\mathrm{rad}} \approx 2\pi \times 16.5\,\mathrm{kHz}$ at an optical power of $P \approx 393\,\mu\mathrm{W}$, which is in good agreement with our expectations for a cigar-shaped dipole trap with a waist of $w_0 \approx 1.66\,\mu m$ and an aspect ratio $\eta \approx 7$.

II. CREATION OF THE DOUBLE-WELL POTENTIAL

To form a double-well potential, we create two microtraps which are partially overlapping. We achieve this by using an acousto-optical deflector (AOD) to create two separated laser beams that are incident upon the objective under slightly different angles, which the objective translates into a spatial separation of the two foci in the focal plane. To control the spacing and the tilt of the resulting double-well potential we tune the relative angle and intensity of the two trapping beams. We do this by changing the frequencies and amplitudes of two RF-signals driving the AOD. To obtain a tunnel coupling which is on the order of 100 Hz at our typical light powers of $200\,\mu\mathrm{W}$, we set the frequencies to values of 32 MHz and 38 MHz which corresponds to a spacing of the double well of $d \approx 2\,\mu\mathrm{m}$. To tune the strength of the tunnel coupling during our experiments, we change the overall depth of the potential which we control by actively stabilizing the overall light power P_{tot} of the trapping beams. At these trap parameters, the individual wells have a depth of approximately $h \times 30\,\mathrm{kHz}$ and the height of the potential barrier between them is approximately $h \times 5.4\,\mathrm{kHz}$. To compare our double well to optical lattice experiments, we can define an effective lattice wavelength $\lambda_{\mathrm{eff}} = 2d$ and calculate the recoil energy $E_{\mathrm{rec}} = h^2/2m\lambda_{\mathrm{eff}}^2 \approx h \times 2.1\,\mathrm{kHz}$, where h is the Planck constant and m is the mass of ^{6}Li.

III. SITE-SELECTIVE ATOM NUMBER DETECTION

Since our experiment allows us to independently address the individual wells of our potential, we can perform site-selective atom-number detection by switching off one of the wells and measuring the number of atoms in the remaining well.

To do this, we first decouple the wells by rapidly increasing the height of the potential barrier within 2 ms. This ramp is fast compared to the time scale of the tunnel coupling but slow enough to avoid heating of the atoms into higher trap levels. Then we slowly turn off one of the wells while keeping the overall light power constant. During this process, the atoms in this well are adiabatically transferred into excited states of the remaining well. Therefore, we use the spilling technique introduced above to remove all atoms from these highly excited states.

To count the number of atoms in the remaining well, we release the atoms from the microtrap, recapture them in a magneto-optical trap and measure their fluorescence with a CCD camera (Andor iXon DV887) for an exposure time of 1 second [17].

IV. THE TWO-SITE HUBBARD MODEL

To experimentally realize the Hubbard model, our double-well system has to fulfill the tight-binding approximation and the single-band approximation.

In a lattice potential, the wavefunction of a single particle can be described as a superposition of localized wavefunctions centered on the individual sites of the lattice. In the limit of tight binding, the overlap between wavefunctions located on different sites is small and they can be approximated by the eigenstates of the individual wells. In this case, the tunnel coupling J between adjacent sites is much smaller than the lowest excitation energy on the individual wells. Since in our double well, the tunneling constant J is approximately $h \times 100\,\mathrm{Hz}$ and the axial trap frequency ω_{ax} in each well is about $2\pi \times 1\,\mathrm{kHz}$, the tight-binding approximation is well fulfilled in our system.

In the single-band approximation, one only considers the motion of the particles in the lowest Bloch band, which is composed of the ground states of the individual wells. To fulfill this approximation, all energy scales in the system have to be much smaller than the band gap to the first excited band. In our double-well system, we initialize both atoms in the ground state of one well (see Sec. I) and limit the interaction energy U to values smaller than the lowest excitation energy of the single wells during all our experiments. For all but the largest values of the interaction energy we estimate the wavefunction overlap

between two interacting particles localized in a single potential well and the respective non-interacting ground state to be above 97 %. For the largest interaction energies of $U \approx h \times 650\,\mathrm{Hz}$ for the dataset with $J = 67\,\mathrm{Hz}$ ($U \approx h \times 400\,\mathrm{Hz}$ for $J = 142\,\mathrm{Hz}$) the overlap decreases to $\approx 92\,\%$ ($\approx 94\,\%$).

Using these two approximations, the spatial wavefunction of a single particle in our double-well system can be described as a superposition of the ground-state wavefunctions of a particle in the left $|L\rangle$ or the right $|R\rangle$ well. This results in a two-level system where the states $|L\rangle$ and $|R\rangle$ are coupled with a tunnel coupling J.

V. CALIBRATION OF THE HUBBARD PARAMETERS

We calibrate the Hubbard parameters of our double well by observing the tunneling dynamics in our system. To determine J and Δ, we study the tunneling of two non-interacting particles between the wells (Sect. V A and Sect. V B). We then determine the on-site interaction energy U for different strengths of the interparticle interaction (Sect. V C).

A. Tunneling oscillations

To prepare the initial state for our tunneling experiments, we start from two non-interacting atoms in the ground state of a single microtrap (see Sec. I). Then we slowly ramp on a second trapping beam (see Sec. II) and thereby change the shape of the potential from a single Gaussian well to a double-well potential. During this ramp, we linearly increase the overall power P_{tot} to keep the depth of the initial well constant. At any time in this process, P_{tot} is large enough to prevent the atoms from tunneling between the wells.

To start the tunneling dynamics, we lower the height of the barrier between the two wells within 2 ms. Since this ramp is fast compared to the time scale of the tunnel coupling we can treat it as a sudden switch-on of the tunneling. Yet, this ramp is slow enough to avoid exciting the atoms into higher bands. We then let the system evolve for different hold times, ramp up the barrier to decouple the wells and finally measure the number of atoms in one of the wells, using the detection scheme described above.

As an example, the tunneling oscillation of two non-interacting atoms in a symmetric double-well at an overall power of $P_{\mathrm{tot}} = 131\,\mu\mathrm{W}$ is shown in Fig. 1. To determine the value of the tunnel coupling J we fit the observed oscillation with a damped sine wave and find $J/h = (142.9 \pm 0.1)\,\mathrm{Hz}$ and a damping time of $(83 \pm 9)\,\mathrm{ms}$. We believe that the main cause for the observed damping is dephasing due to long-term drifts of the tilt of the double-well potential.

To calibrate J for different barrier heights, we perform these measurements for different values of the overall light power P_{tot} (Fig. 2a). For our experiments, we use light powers of $P_{\mathrm{tot}} = 131\,\mu\mathrm{W}$ and $P_{\mathrm{tot}} = 186\,\mu\mathrm{W}$ which result in tunnel couplings of $J/h = (142.0 \pm 0.5)\,\mathrm{Hz}$ and $J/h = (67.3 \pm 0.5)\,\mathrm{Hz}$.

B. Calibration of the tilt

Tunneling between two states that are separated by a potential barrier is only possible if their energy difference Δ is not much larger than their tunnel coupling J. To achieve this for our double-well system, we control the relative depth of the two wells by changing the relative power of the RF signals driving the AOD (see Sec. II).

To calibrate the tilt Δ, we measure the oscillation frequency for various values of the relative RF power (Fig. 2b)[23]. We fit the measured frequencies with the effective coupling $2\,J_{\mathrm{eff}} = 2\sqrt{J^2 + \Delta^2}$ expected for a two-level system. We find good agreement to the data and therefore conclude that our non-interacting system can be described by the two-site Hubbard model (see Sec. IV). Using this fit, we calibrate the tilt Δ in units of the tunnel coupling J and extract the relative RF power where the double well is balanced ($\Delta = 0$).

To observe coherent tunneling in our double-well system, we have to stabilize the relative depth of the two wells with a precision better than J. For our typical trap depth of 30 kHz and tunnel couplings of $J/h \approx 100\,\mathrm{Hz}$ (at an overall light power of $P_{\mathrm{tot}} \approx 200\,\mu\mathrm{W}$), this corresponds to a relative stability of the intensities of the two beams that is on the order of 10^{-4}.

However, switching on the second well leads to heating of the AOD which results in a slow drift of Δ on the order of $h \times 300\,\mathrm{Hz}$ over several seconds. Since these drifts are the same in every experimental cycle, we can compensate them by applying an exponential ramp to the relative RF power. Using this technique, our stability is currently limited by slow drifts of the tilt which are on the order of 10 Hz to 20 Hz per day.

C. Calibration of the on-site interaction

To control the on-site interaction energy U between two particles that occupy the same well, we change the s-wave scattering length a between the atoms with a magnetic Feshbach resonance. To calibrate the interaction energy as a function of the magnetic offset field, we study the tunneling of two interacting particles.

For a system with two particles in one well, the energy is shifted with respect to the non-interacting system by the interaction energy U. If the two particles tunnel together, this interaction energy is conserved and consequently pair tunneling is resonant in a symmetric double well at all interaction strengths. However, if a single particle tunnels to the other well the particles can no longer interact

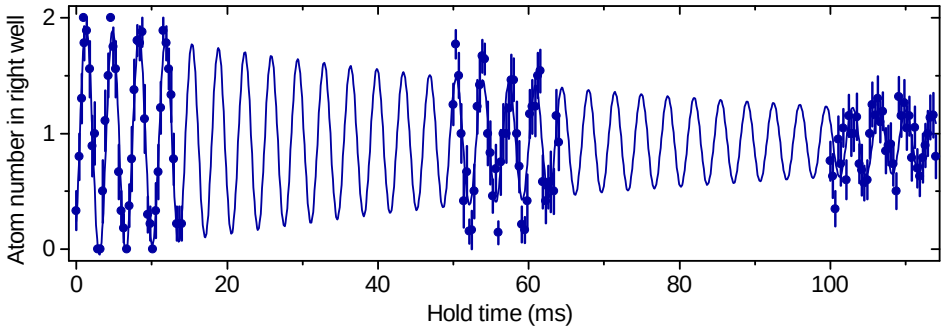


FIG. 1. Tunneling of two non-interacting particles in a symmetric double-well potential. Time evolution of the atom number in the right well for a system with two particles in the left well at $t = 0$. Each data point is the average of 15 measurements. The solid line shows a damped sinusoidal fit to the data from which we determine the tunnel coupling J. The error bars denote the 1σ statistical uncertainties.

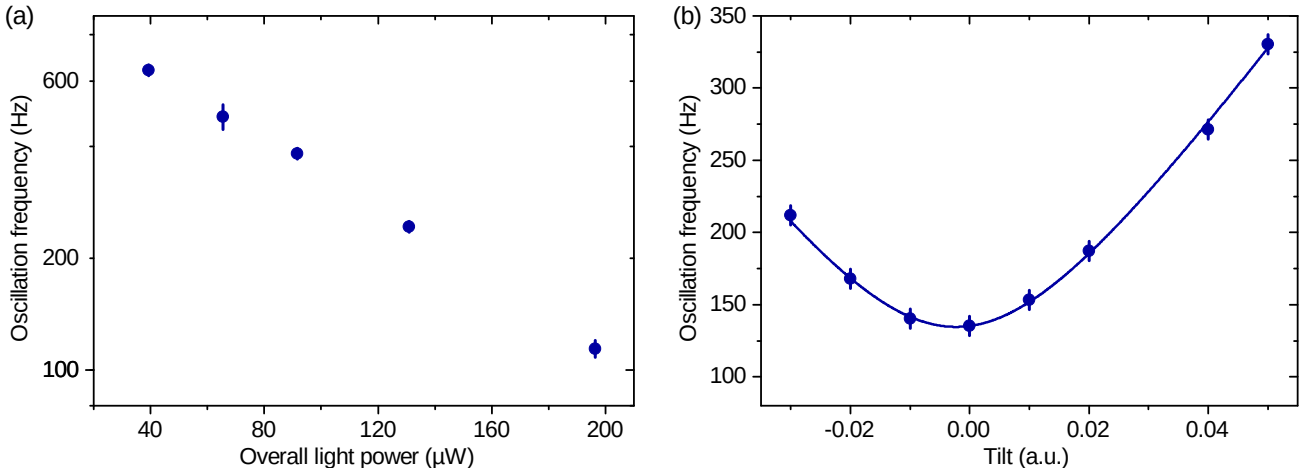


FIG. 2. Oscillation frequencies for different parameters of the double well. (a) Oscillation frequency of two non-interacting particles in a symmetric double well ($\Delta \approx 0$) as a function of the overall light power in the trap beams. A decrease of the total light power lowers the barrier height and thereby leads to a faster tunneling of the atoms between the wells. (b) Oscillation frequency as a function of the relative RF power of the two signals that drive the AOD. The solid line shows a fit with the effective tunnel coupling $2\,J_{\mathrm{eff}} = 2\sqrt{J^2 + \Delta^2}$ expected for a two-level system. From the fit, we extract the relative RF power where the double well is balanced. The error bars denote the 1σ statistical uncertainties.

and hence single-particle tunneling is effectively detuned by the interaction energy U. We can compensate this effective detuning by a tilt of $2\Delta = -U$ between the two wells[11]. This allows us to determine the on-site interaction energy by measuring the value of the tilt at which single-particle tunneling is restored.

To perform this calibration, we measure the amplitudes of single-particle and pair tunneling as a function of tilt. To do this we initialize the system with two interacting atoms in the left well, let them tunnel between the wells for different hold times and measure the number of atoms in the right well. From these measurements, we extract the probabilities of finding 1 or 2 atoms in the right well, average over all hold times and plot them as a function of the tilt (Fig. 2 of main text). We observe that the tilt at which single-particle tunneling is resonant shifts with the interaction strength whereas the pair-tunneling resonance remains at $\Delta = 0$. By fitting Lorentzians to the resonance peaks and measuring their separation, we can therefore determine the value of the on-site interaction U for different magnetic fields (Fig. 3a).

Since the value of the on-site interaction energy depends on the depth of the confining potential, we perform the calibration of U for the two potential depths (see Sec. V C) we use in our experiments. To check whether the measured interaction energies are affected by off-site interactions we compare our results to a calculation of the interaction energy of two particles in a cigar-shaped potential (Fig. 3b)[25, 26]. Within the errors we find good agreement with our data and therefore conclude that the effects of off-site interactions in our experiments are small. The good agreement between the measured and calculated interaction energies allows us to use the results from this calculation as the calibration for U for all our measurements.

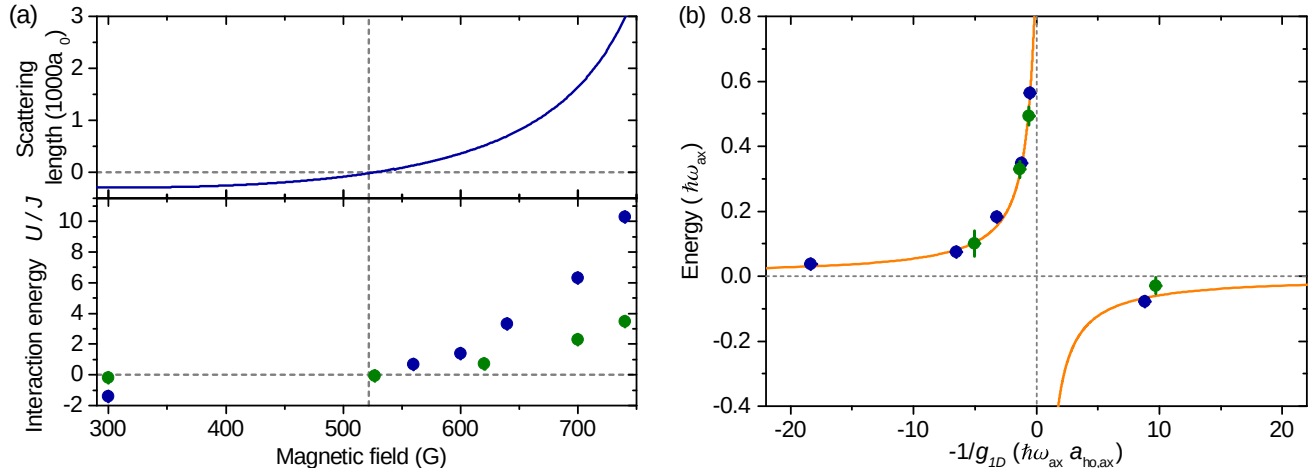


FIG. 3. Measurement of the on-site interaction energy U for two different tunnel couplings. (a) Interaction energy U in units of the tunnel coupling J as a function of the magnetic offset field. The green (blue) dots correspond to a tunnel coupling of $J = h \times (142.0 \pm 0.5)$ Hz ($J = h \times (67.3 \pm 0.5)$ Hz). (b) Interaction energy U in units of the harmonic oscillator energy $\hbar\omega_{\rm ax}$ plotted as a function of the one-dimensional interaction strength $g_{\rm 1D}$ [24] where $a_{\rm ho,ax} = \sqrt{2\hbar/(m\omega_{\rm ax})}$ denotes the harmonic oscillator length in axial direction and m is the mass of a ^{6}Li atom. The data shows good agreement with a calculations of the interaction energy of two particles in a single quasi-1D potential well (solid line)[25]. The error bars denote the 1σ statistical uncertainties.

VI. PREPARATION OF EIGENSTATES OF THE SYMMETRIC DOUBLE WELL

To prepare two atoms in an eigenstate of the symmetric double well, we start with two non-interacting atoms in the ground state of a single well and adiabatically change the shape of the potential to a double well. To prepare the system in the ground state $|a\rangle$, we initialize the system in state $|LL\rangle$, but keep the left well much deeper than the right one. Then, we lower the potential barrier between the wells and thereby set the tunnel coupling J to a value of about $h \times 100$ Hz. By setting a tilt of $|\Delta| \approx \frac{1}{2}\hbar\omega_{ax}$, we make sure that tunneling is far off-resonant ($|\Delta| \gg J$) and the initial state $|LL\rangle$ has almost complete overlap with the ground state of the tilted double well (Fig. 4). By adiabatically reducing the tilt to $\Delta = 0$ within 100 ms we then bring the system into the ground state of the symmetric double well.

Using the same technique, we can also prepare the atoms in the highest energy state $|c\rangle$. To do this, we start again in state $|LL\rangle$, but this time we set the right well to be much deeper than the left one ($\Delta \approx +\frac{1}{2}\hbar\omega_{ax}$). Lowering the barrier then transforms the initial state $|LL\rangle$ into state $|c\rangle$ (Fig. 4).

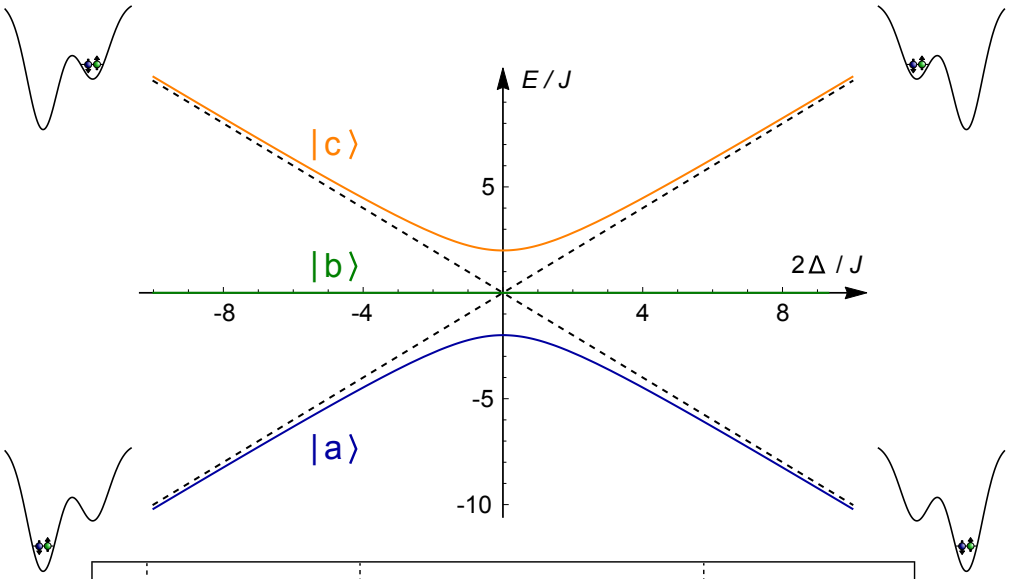
After preparing the non-interacting eigenstate of the double-well system, we adiabatically introduce interparticle interactions by ramping the magnetic field from the zero-crossing of the scattering length at 527 G to values between 300 G and 740 G over 60 ms. This corresponds to scattering lengths between $-288\,a_0$ and $2974\,a_0$ and interaction energies between $U/h \approx -1.3\,J/h \approx -87$ Hz and $U/h \approx 10.1\,J/h \approx 680$ Hz, where a_0 is the Bohr radius (Fig. 3).

Similar to the experimental sequence presented here, one can also start from an interacting system with large tilt and both particles in the deeper well and then adiabatically go to the ground state of the symmetric double well [27].

To confirm the adiabaticity of the ramps in tilt and in interaction strength, we drive the system from state $|LL\rangle$ to the final configuration, hold the system for different durations and then reverse both ramps. To check for excitations, we measure the mean atom number in the ground state both with and without the ramps. Within the statistical uncertainty of our measurement we could not detect any heating caused by the ramps. From these measurements, we determine a fidelity of $91.6 \pm 0.7\%$ of preparing the system in the ground state of the symmetric double well.

VII. NUMBER STATISTICS AND PREPARATION FIDELITY CORRECTION

To observe the transition from the metallic to the insulating regime, we measure the probability of single and double occupancy in the symmetric double well as a function of interaction strength (Fig. 3 of main text). Hence, we first prepare the system either in state $|a\rangle$ or state $|c\rangle$ at different interaction strengths (see Sec. VI). Then, we quickly ramp up the potential barrier to decouple the wells and measure the atom number in either the left or the right well. We repeat this measurement 300 times for each well and count how often we detect zero atoms (N_0), one atom (N_1), or two atoms (N_2). The probabilities of measuring 0, 1, or 2 atoms follow as: $a'_i = N_i/(N)$, where $i \in 0, 1, 2$ and N is the total number of measurements in



	Δ → − ∞	Δ = 0	Δ → + ∞
|a⟩	|LL⟩	$\frac{1}{2}$(|LL⟩ + |RR⟩ + |LR⟩ + |RL⟩)	|RR⟩
|b⟩	$\frac{1}{\sqrt{2}}$(|RL⟩ + |LR⟩)	$\frac{1}{\sqrt{2}}$(|LL⟩ − |RR⟩)	$\frac{1}{\sqrt{2}}$(|RL⟩ + |LR⟩)
|c⟩	|RR⟩	$\frac{1}{2}$(|LL⟩ + |RR⟩ − |LR⟩ − |RL⟩)	|LL⟩

FIG. 4. Eigenenergies and eigenstates in a tilted double well. Energies of states $|a\rangle$, $|b\rangle$, and $|c\rangle$ in a non-interacting system as a function of the tilt Δ between the wells. Depending on the sign of the initial tilt, state $|LL\rangle$ can be used to prepare either state $|a\rangle$ or state $|c\rangle$.

both wells.

In our measurement, we have finite fidelities both for the preparation of the initial state and the detection of the final atom number. Both the preparation fidelity and the detection fidelity reduce the measured atom numbers as compared to an ideal measurement. In a symmetric double well the influence of these processes on the measured probabilities cannot be distinguished. We therefore combine the preparation and detection fidelity into an overall fidelity p which we define as the mean atom number per well. We determine p by averaging over all individual measurements.

Due to the finite fidelity $p < 1$ of our experiment, the measured probabilities a_0', a_1', and a_2' deviate from the probabilities a_0, a_1, and a_2 of an ideal experiment with $p = 1$:

$$a_2' = a_2 p^2 \tag{2}$$
$$a_1' = a_1 p + 2a_2 p(1-p) \tag{3}$$
$$a_0' = a_0 + a_1(1-p) + a_2(1-p)^2 \tag{4}$$

These three equations can be inverted to calculate the ideal probabilities:

$$a_2 = a_2'/p^2 \tag{5}$$
$$a_1 = a_1'/p - 2(1-p)a_2'/p^2 \tag{6}$$
$$a_0 = a_0' - a_1'(1-p)/p + a_2'(1-p)^2/p^2 \tag{7}$$

Using these corrected probabilities, we define single occupancy as the probability of measuring a single atom ($P_1 = a_1$) and double occupancy as the sum of the probabilities of measuring either zero or two atoms ($P_2 = a_0 + a_2$). For comparison, the results for single occupancy and double occupancy in the ground state $|a\rangle$ using both the corrected and the non-corrected probabilities are shown in Fig. 5. To verify that our measurement of the occupation statistics is insensitive to small drifts of the tilt, we performed the measurement for state $|c\rangle$ at different tilts in a range of ± 30 Hz around $\Delta = 0$. We observe that while the tilt affects the probabilities a_0 and a_2 of finding zero or two particles in the individual wells there is no significant change in the double occupancy $a_0 + a_2$.

VIII. TRAP MODULATION SPECTROSCOPY

To directly observe the effect of the tunnel coupling on the energy of our double-well system, we measure the energy difference $E_{bc} = E_c - E_b$ between state $|c\rangle$ and state $|b\rangle$ by trap modulation spectroscopy. We start by initializing state $|c\rangle$ with a tunnel coupling of $J = h \times 67.3$ Hz and interaction energies between $U = -1.3\,J$ and $U = 10.1\,J$, as described above. Then we perform a sinusoidal modulation of the total trapping power with frequencies between $f = 30$ Hz and $f = 300$ Hz for a

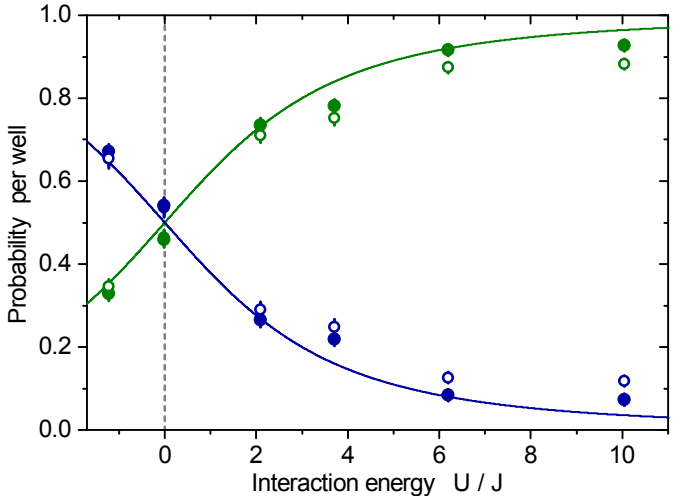


FIG. 5. Corrected and non-corrected occupation statistics. Single occupancy (green) and double occupancy (blue) in state $|a\rangle$ as a function of the on-site interaction energy U at a tunnel coupling of $J/h \simeq 67\,\mathrm{Hz}$. Open symbols correspond to the occupation statistics which were directly calculated from the measured probabilities of finding 0, 1, or 2 atoms in one of the wells. For the filled symbols, these probabilities were corrected for the finite experimental fidelities in the preparation and detection processes before calculating the occupation statistics. The solid lines show the prediction of the Hubbard model. The error bars denote the 1σ statistical uncertainties.

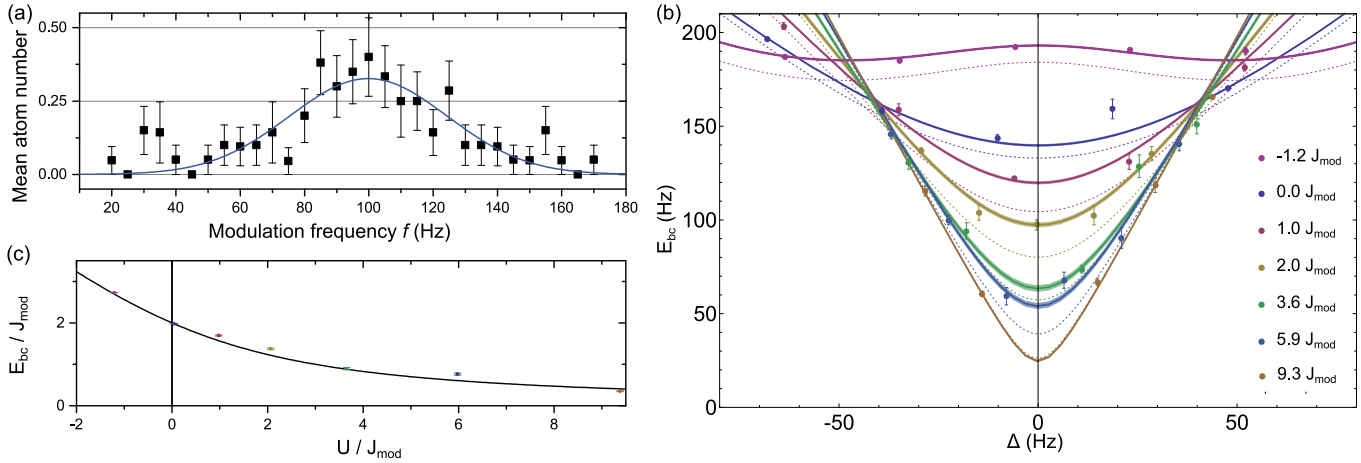


FIG. 6. Energy difference E_{bc} between states $|c\rangle$ and $|b\rangle$. (a) Mean atom number in the right well at different modulation frequencies for $U = 5.9 J_{\mathrm{mod}}$ and $\Delta/h = -22\,\mathrm{Hz}$. The resonance frequency is extracted from a Gaussian fit to the peak (blue curve). (b) Resonance frequency of the transition between states $|c\rangle$ and $|b\rangle$ as a function of the tilt Δ for different interaction energies. The predictions by the Hubbard model without free parameters are shown as dashed lines. The solid curves are fits to the data using the Hubbard model with J as a free parameter, the shaded areas are the respective 1σ confidence bands for the mean predictions. (c) Energy difference E_{bc} of the balanced double well plotted against the interaction energy U. The errors correspond to the width of the respective confidence bands at $\Delta = 0$ in the right panel.

duration of 200 ms. The atoms are transferred resonantly to state $|b\rangle$ if the modulation frequency matches the energy difference E_{bc}. To minimize the distortion of the energy spectrum, we modulate the tunnel coupling only with a small amplitude of $0.11\,J$. To determine whether the atoms have been transferred to state $|b\rangle$, we adiabatically turn off the interactions, ramp to a large tilt $\Delta \gg 0$ and measure the number of atoms in the right well. If the system has remained in state $|c\rangle$ during the modulation, both atoms are then in the left well. However, if the system has been transferred to state $|b\rangle$, there is a large probability to observe a single atom in the right well for $\Delta \gg 0$ (Fig. 4). We plot the mean atom number in the right well as a function of the modulation frequency and extract the resonance frequency with a Gaussian fit (Fig. 6a).

To measure the effects of second-order tunneling, we have to control the tilt better than $4J^2/U$, which is approximately 30 Hz at $U \approx 10J$. This is comparable to the residual drifts in our calibration of the tilt Δ over the

course of several days (see Sec. V B). We solve this problem by measuring the resonance frequency for different tilts around $\Delta = 0$ for each of the interaction energies. We determine the shift $\Delta_0 = \Delta' - \Delta$ between our set value Δ' and the tilt Δ in the experiment for each data set by fitting a Rabi function $f_\mathrm{R} = 2\sqrt{J^2 + (\Delta' - \Delta_0)^2}$ with Δ_0 and J as free parameters. We then center our data by subtracting the respective Δ_0.

For a non-interacting system, the energy difference E_{bc} is given by the transition frequency $f_\mathrm{R}(\Delta = 0)$. For an interacting system, we can still use this fit to center our data, but it does not yield the correct value for E_{bc}. Hence, we compare our measurements to a numeric solution of the Hubbard model with fixed parameters U and J as calibrated from the tunneling measurements (dashed curves in Fig. 6b). We observe resonance frequencies which are systematically higher than predicted. Therefore, we fit the data with a numeric Hubbard model with J' as a free parameter (solid lines). We obtain $E_{bc}(\Delta = 0)$ for the balanced double-well system at different values of U from the fitted curves (Fig. 6c and Fig. 4 of the main text).

One possible cause for the deviation of the fixed-parameter Hubbard model from our data is that the modulation of the trapping potential results in a modified tunnel coupling. Therefore, we scale E_{bc} and U with the modified tunnel coupling $J_\mathrm{mod} = h \times (70.7 \pm 0.3)\,\mathrm{Hz}$ which we obtain from the fit at $U = 0$.